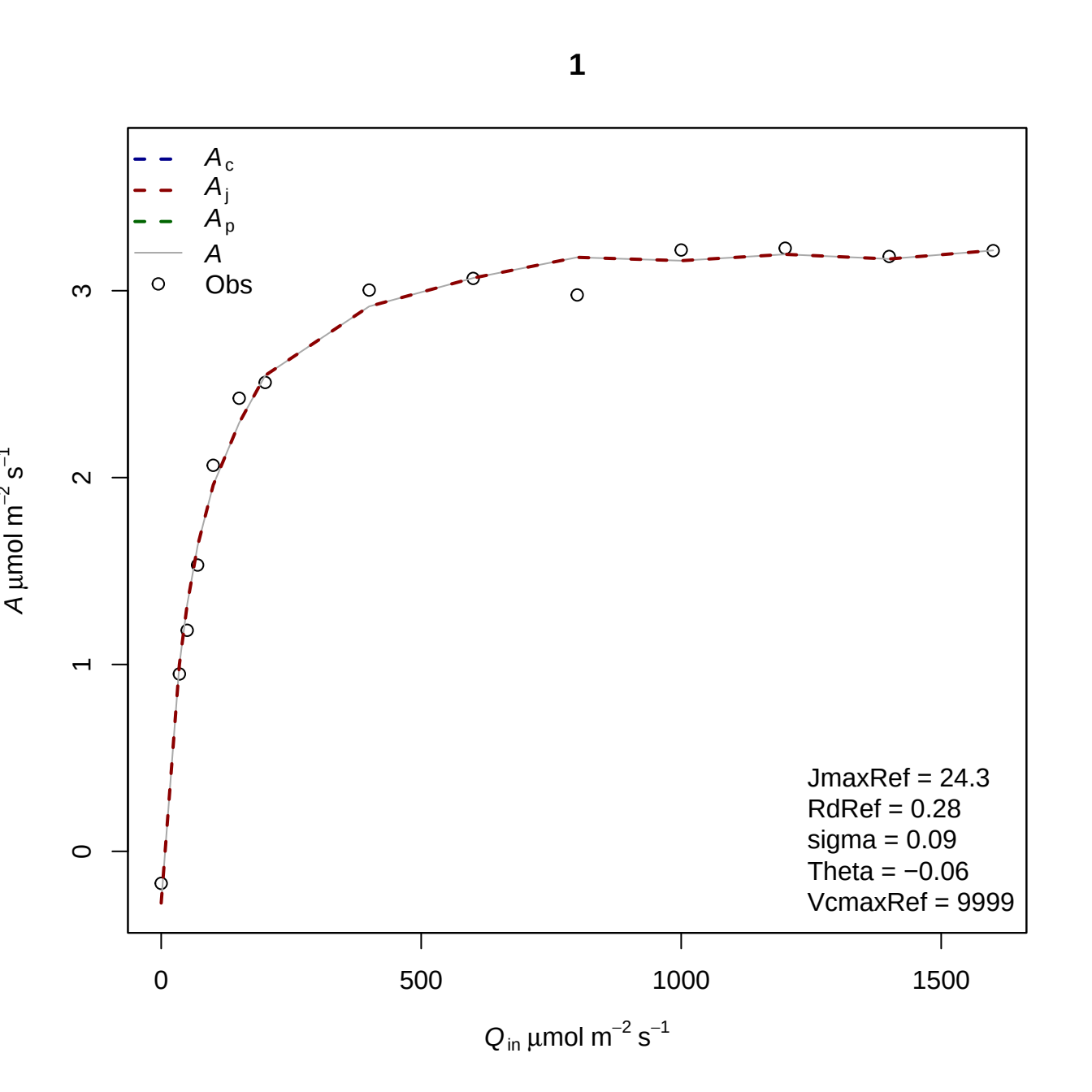
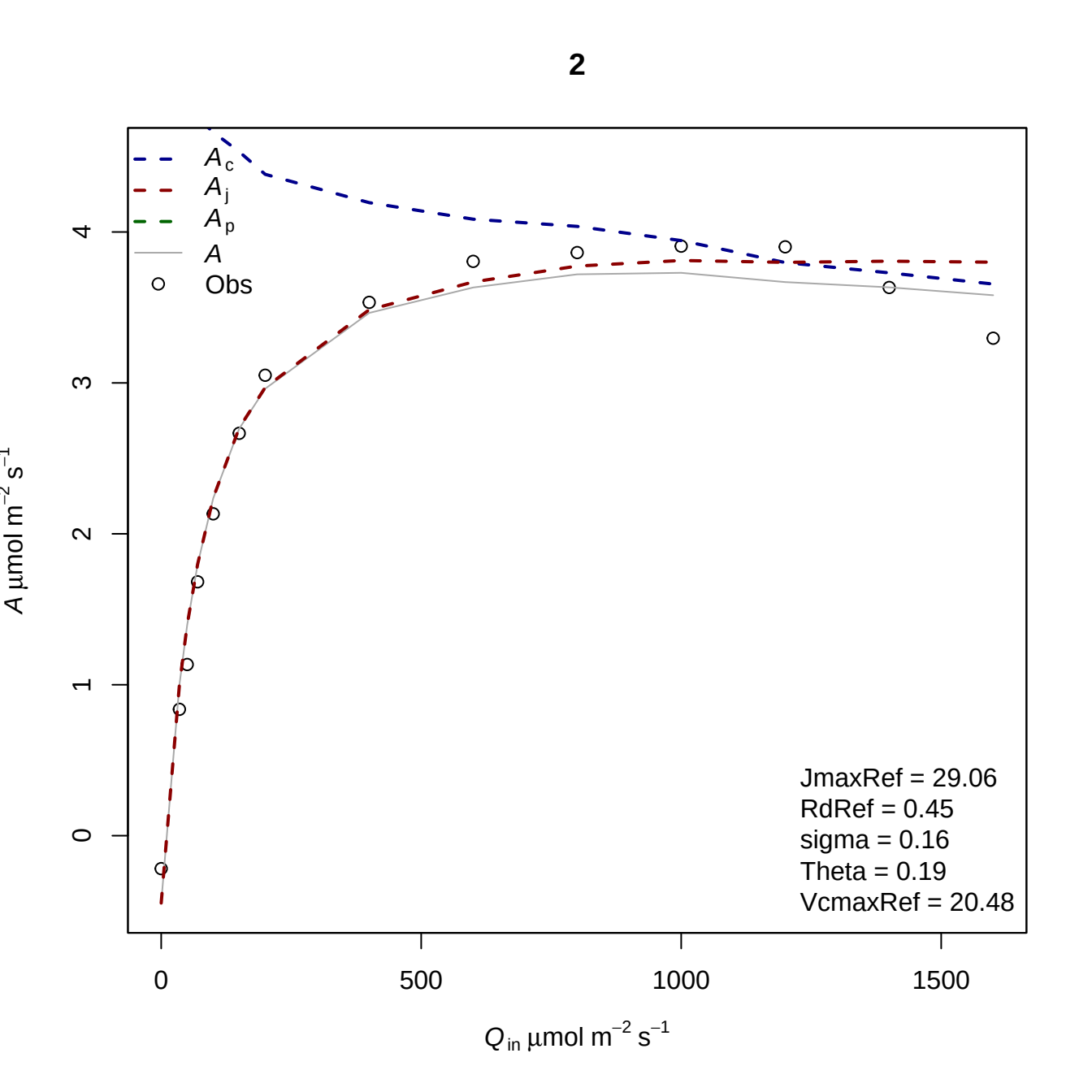
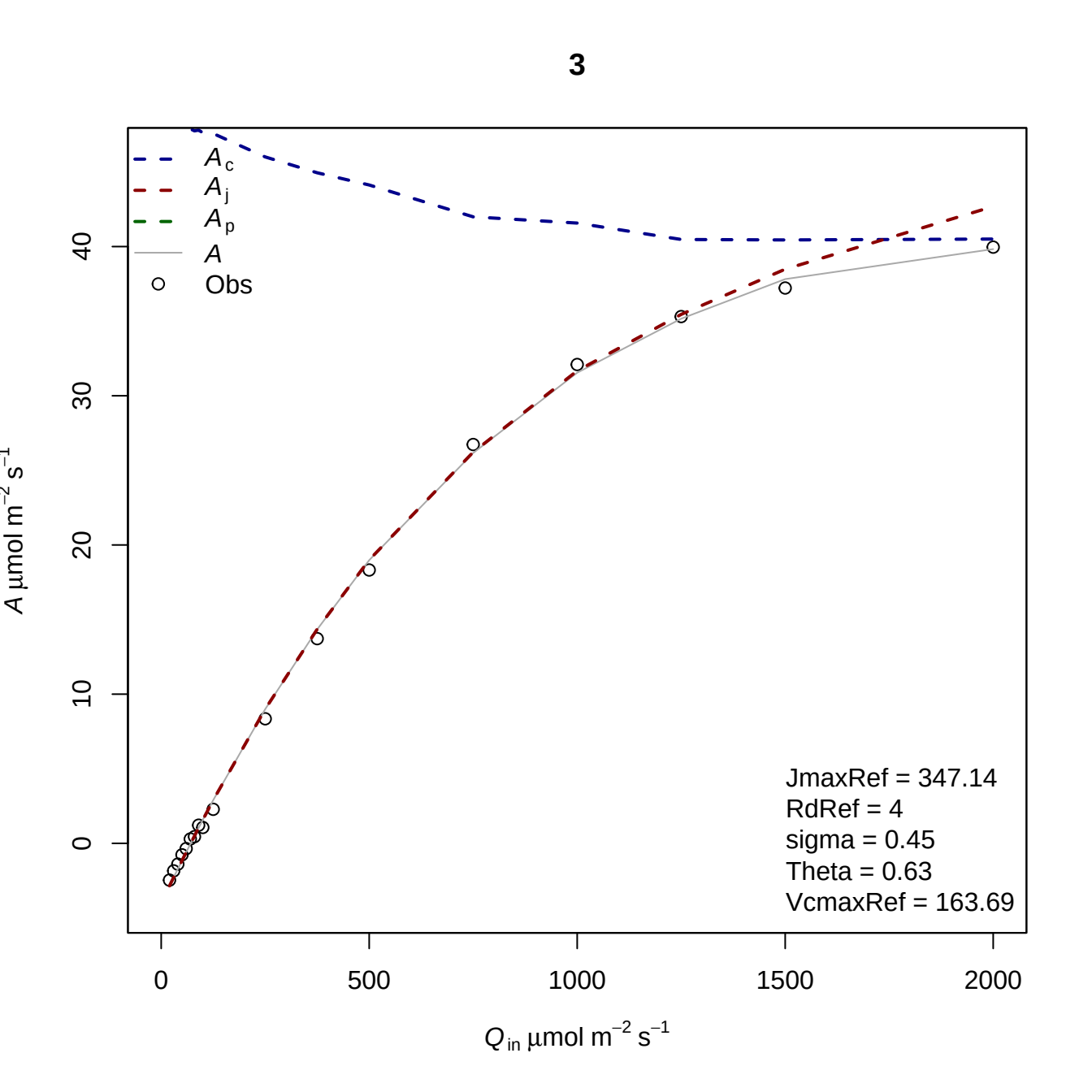
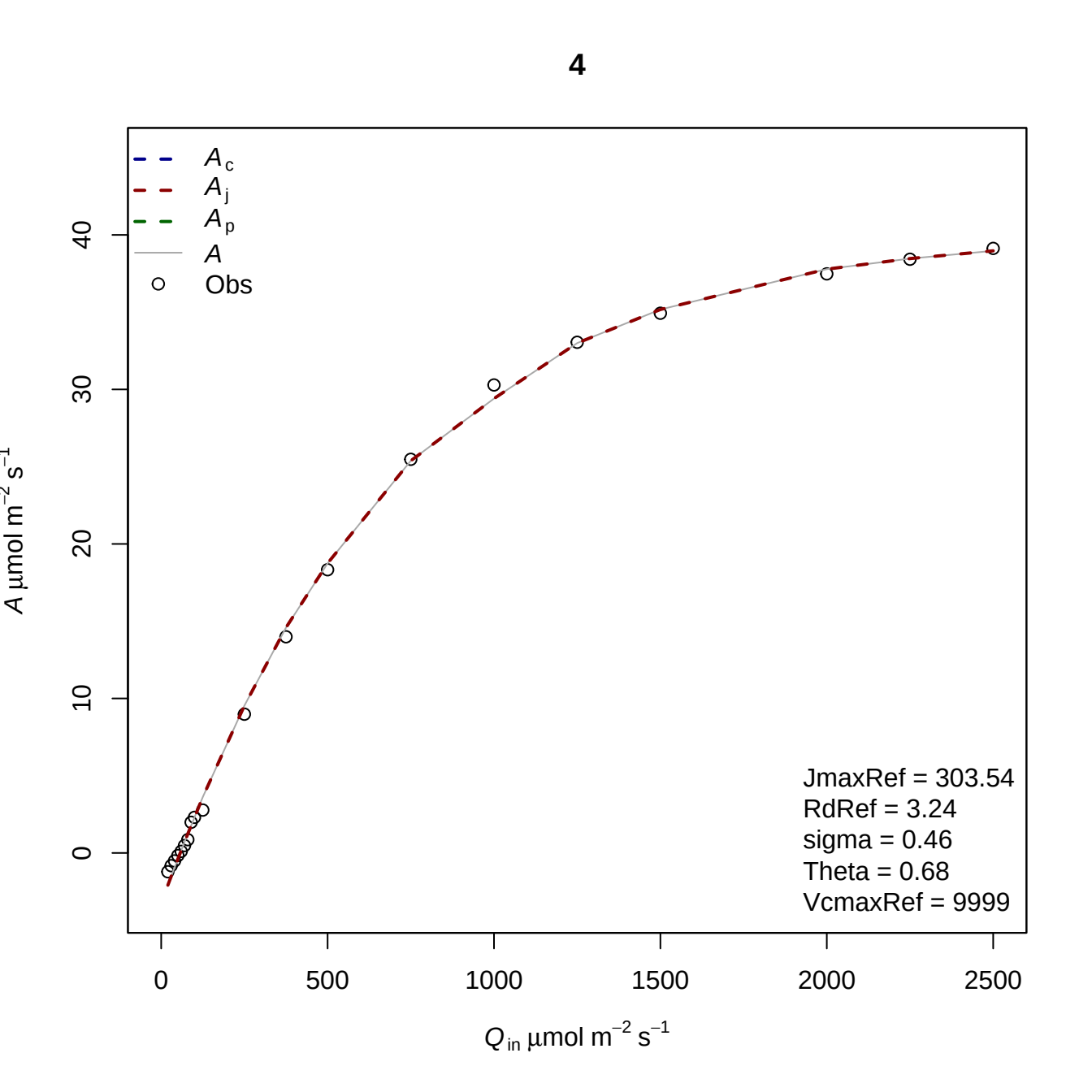
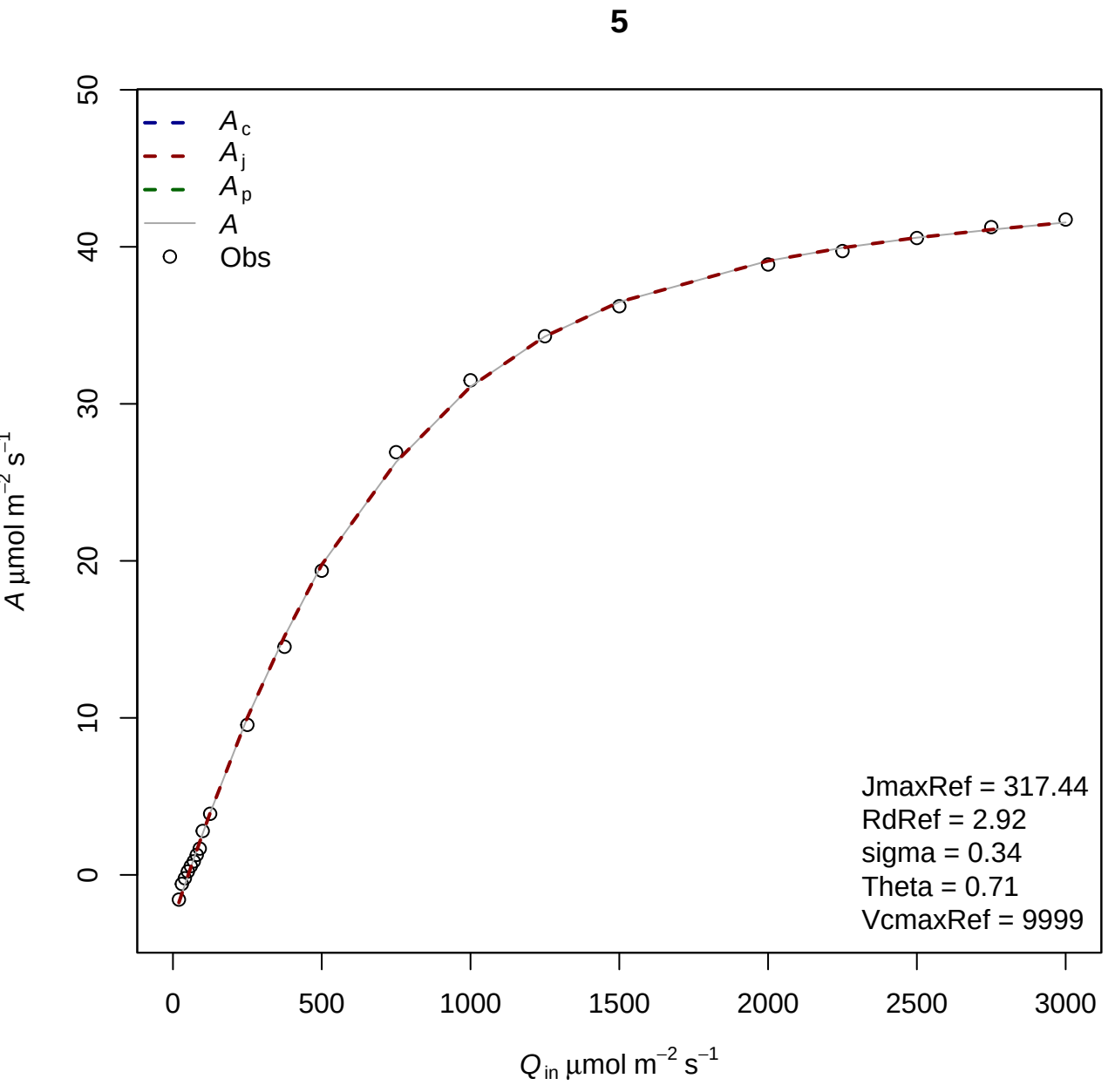
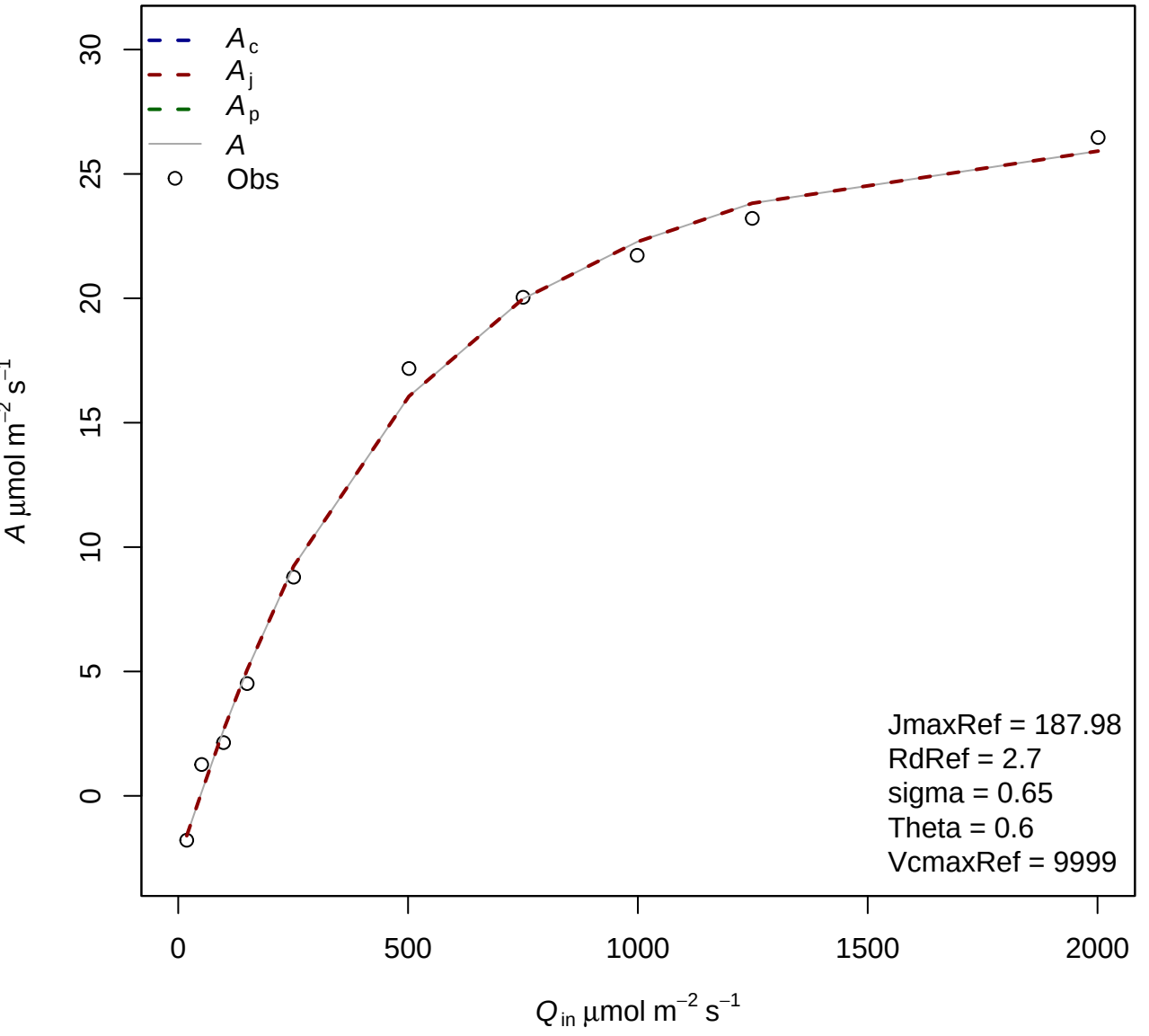

1
A_c
A_j
A_p
A
Obs
JmaxRef = 24.3
RdRef = 0.28
sigma = 0.09
Theta = −0.06
VcmaxRef = 9999
0
1
2
3
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

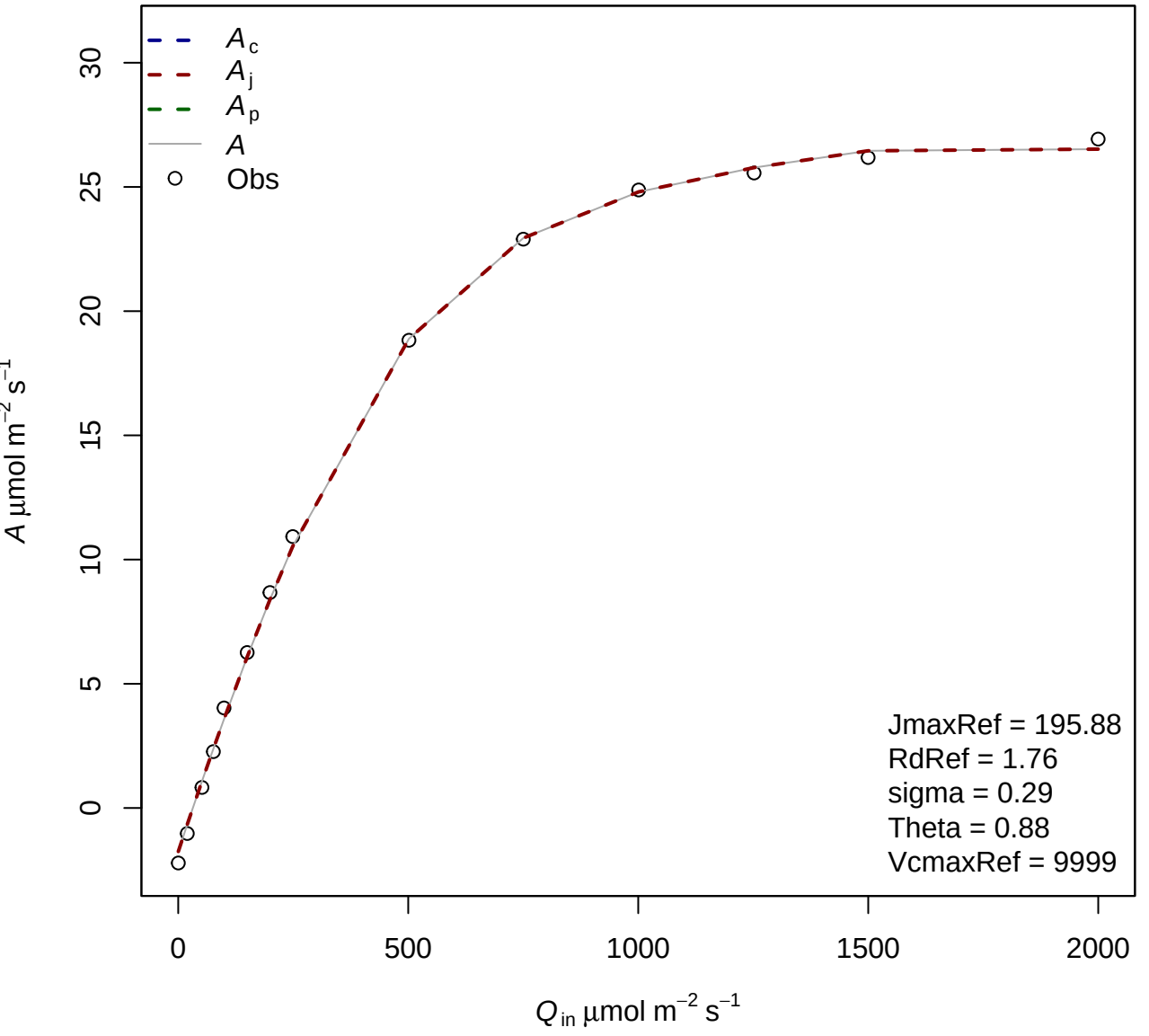

2
A_c
A_j
A_p
A
Obs
JmaxRef = 29.06
RdRef = 0.45
sigma = 0.16
Theta = 0.19
VcmaxRef = 20.48
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

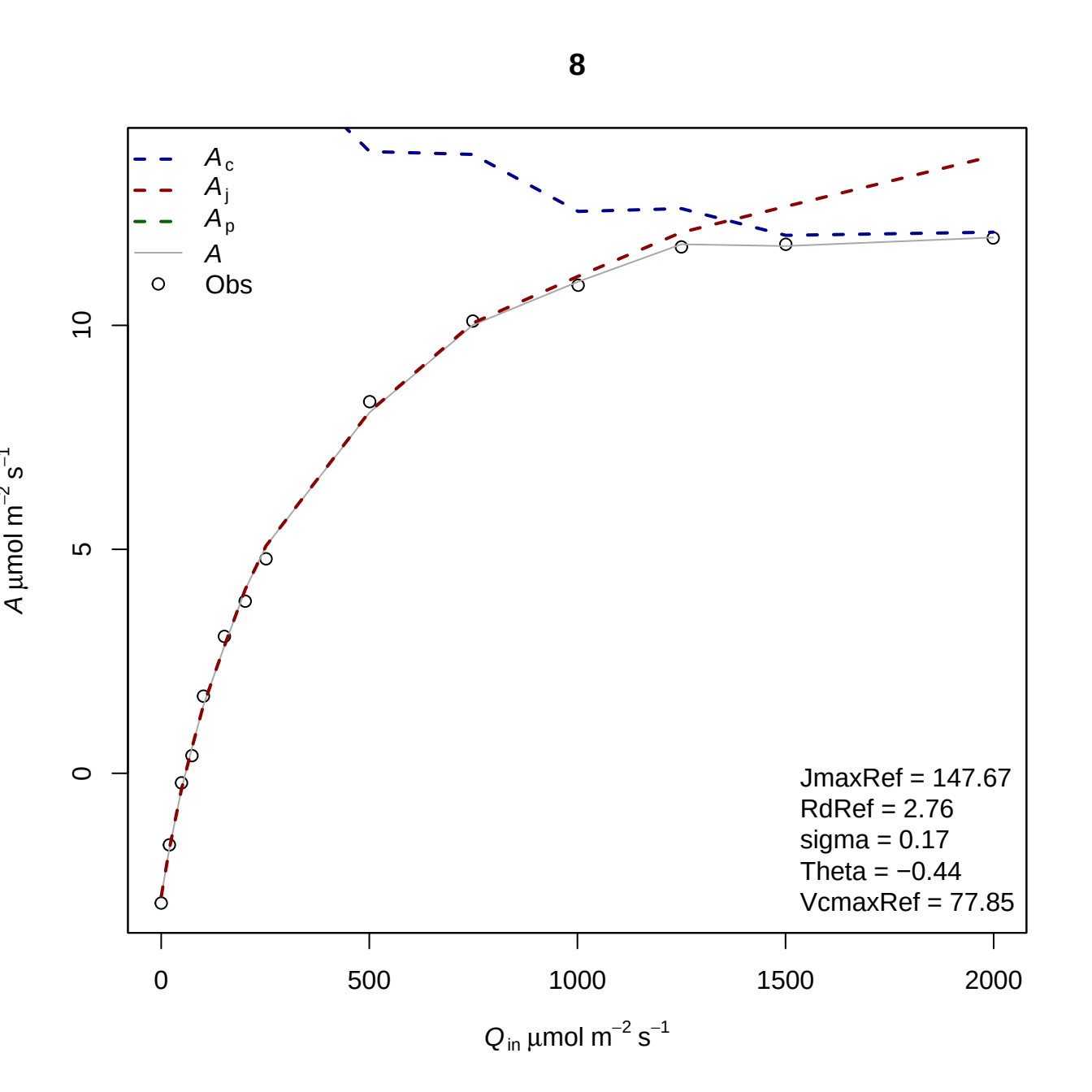

3
A_c
A_j
A_p
A
Obs
JmaxRef = 347.14
RdRef = 4
sigma = 0.45
Theta = 0.63
VcmaxRef = 163.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

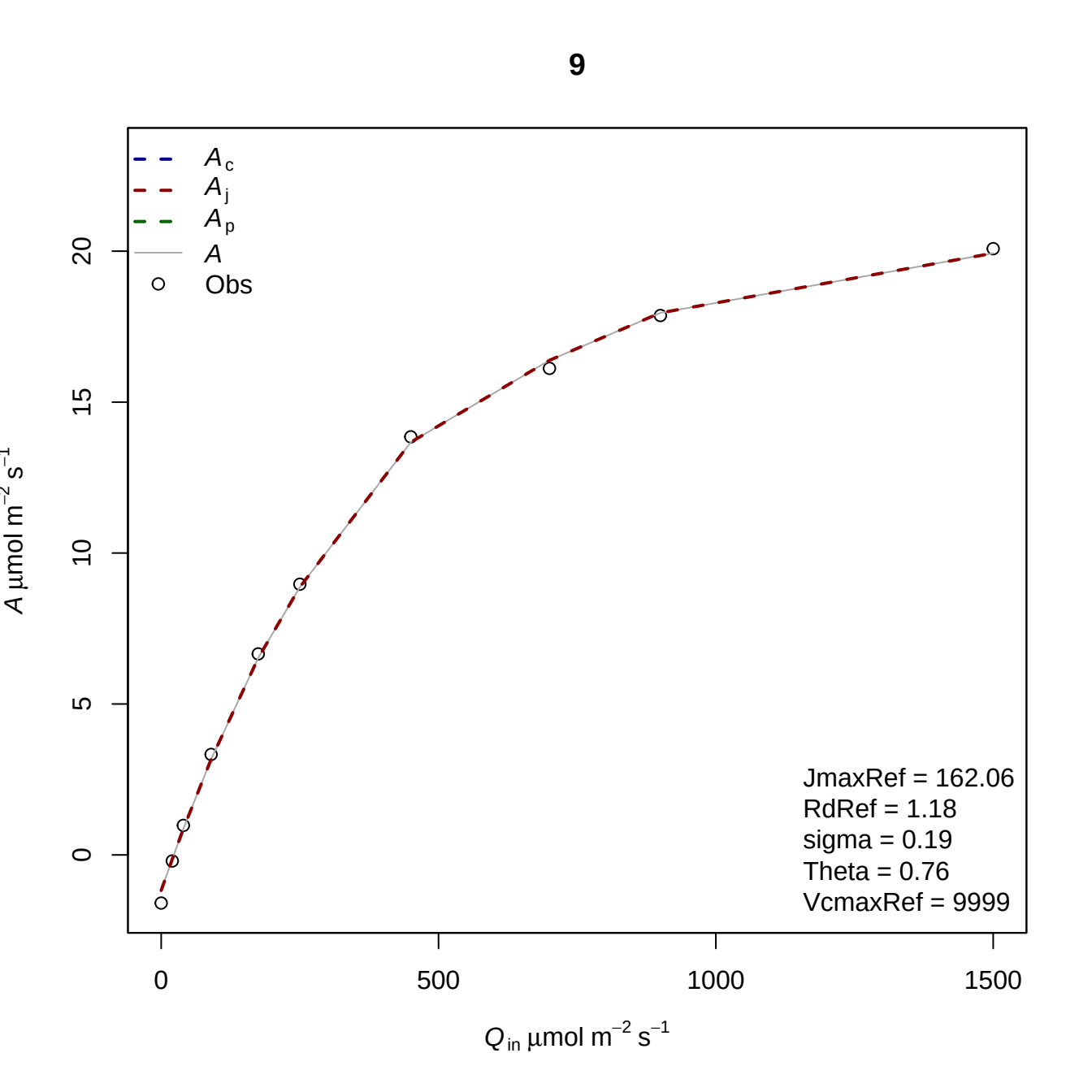

4
A_c
A_j
A_p
A
Obs
JmaxRef = 303.54
RdRef = 3.24
sigma = 0.46
Theta = 0.68
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000
2500

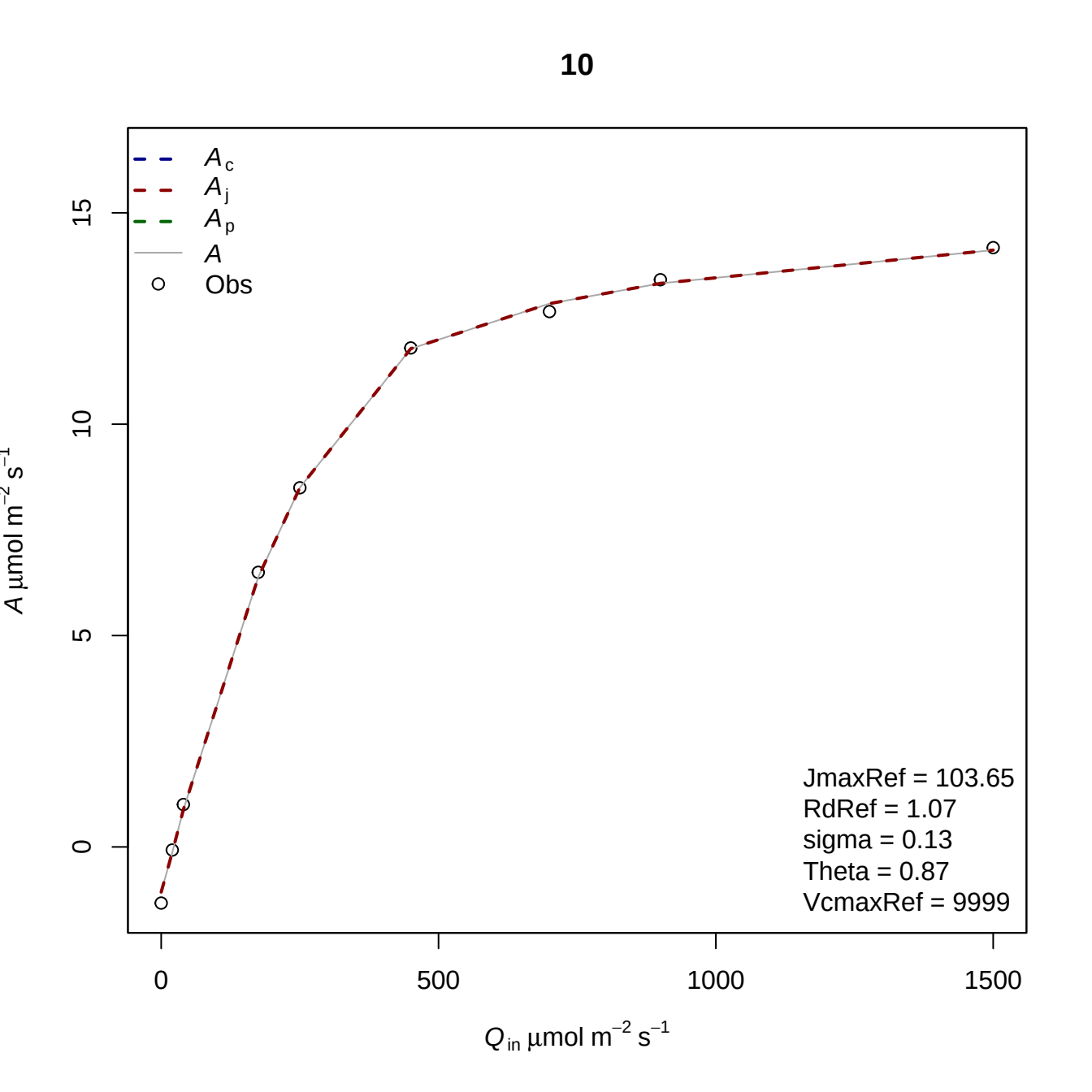

5
A_c
A_j
A_p
A
Obs
JmaxRef = 317.44
RdRef = 2.92
sigma = 0.34
Theta = 0.71
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
50
0
500
1000
1500
2000
2500
3000

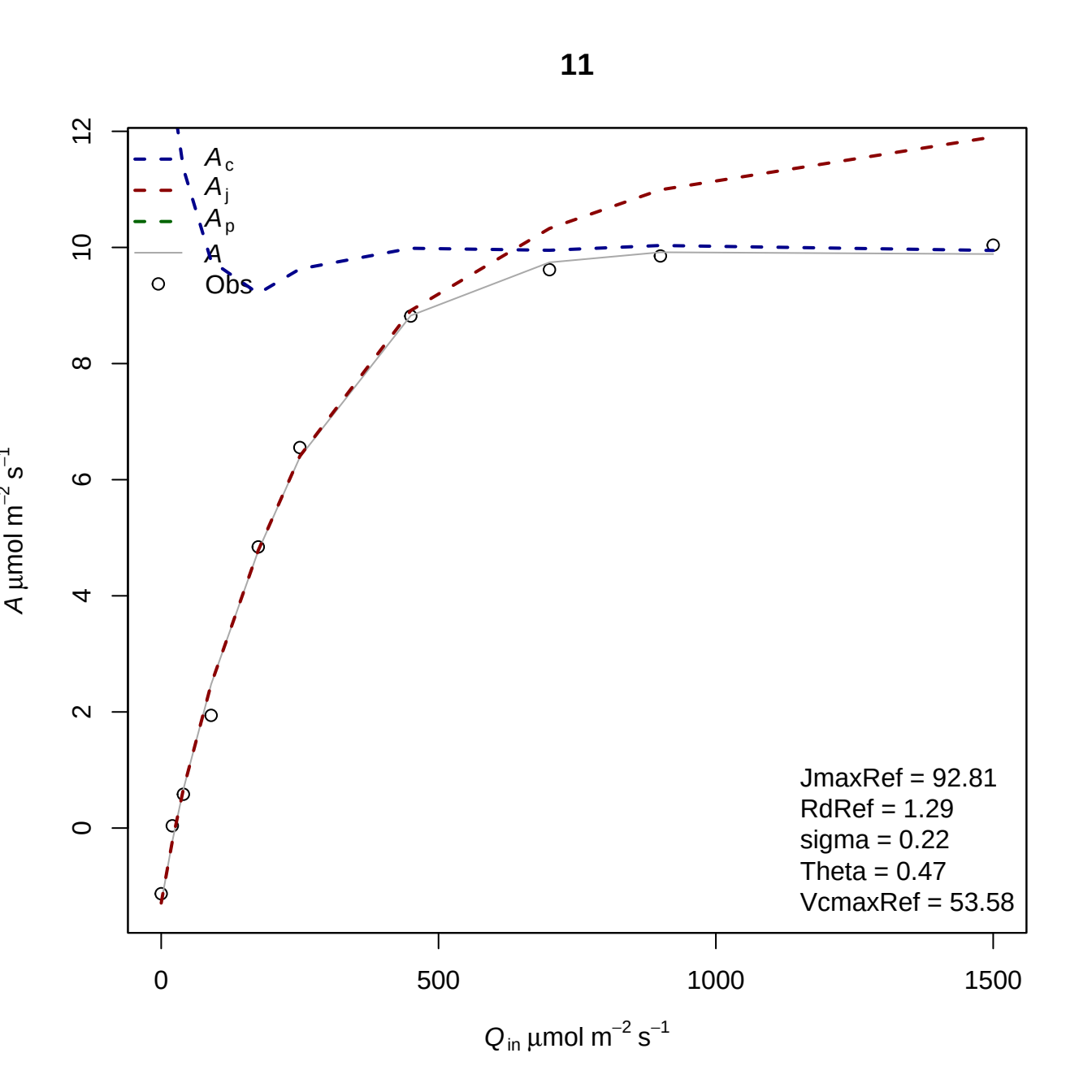

A_c
A_j
A_p
A
Obs
JmaxRef = 187.98
RdRef = 2.7
sigma = 0.65
Theta = 0.6
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

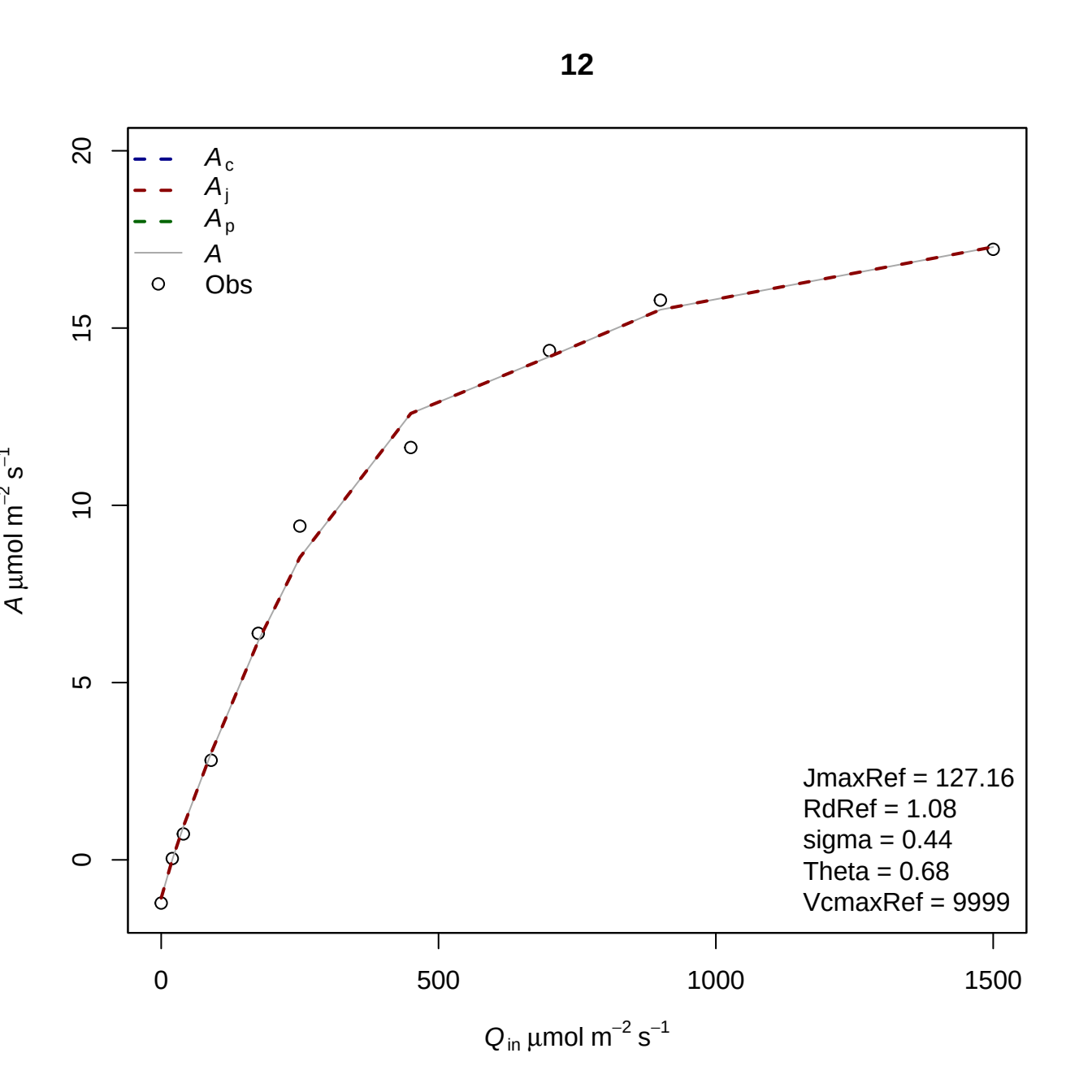

A_c
A_j
A_p
A
Obs
JmaxRef = 195.88
RdRef = 1.76
sigma = 0.29
Theta = 0.88
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

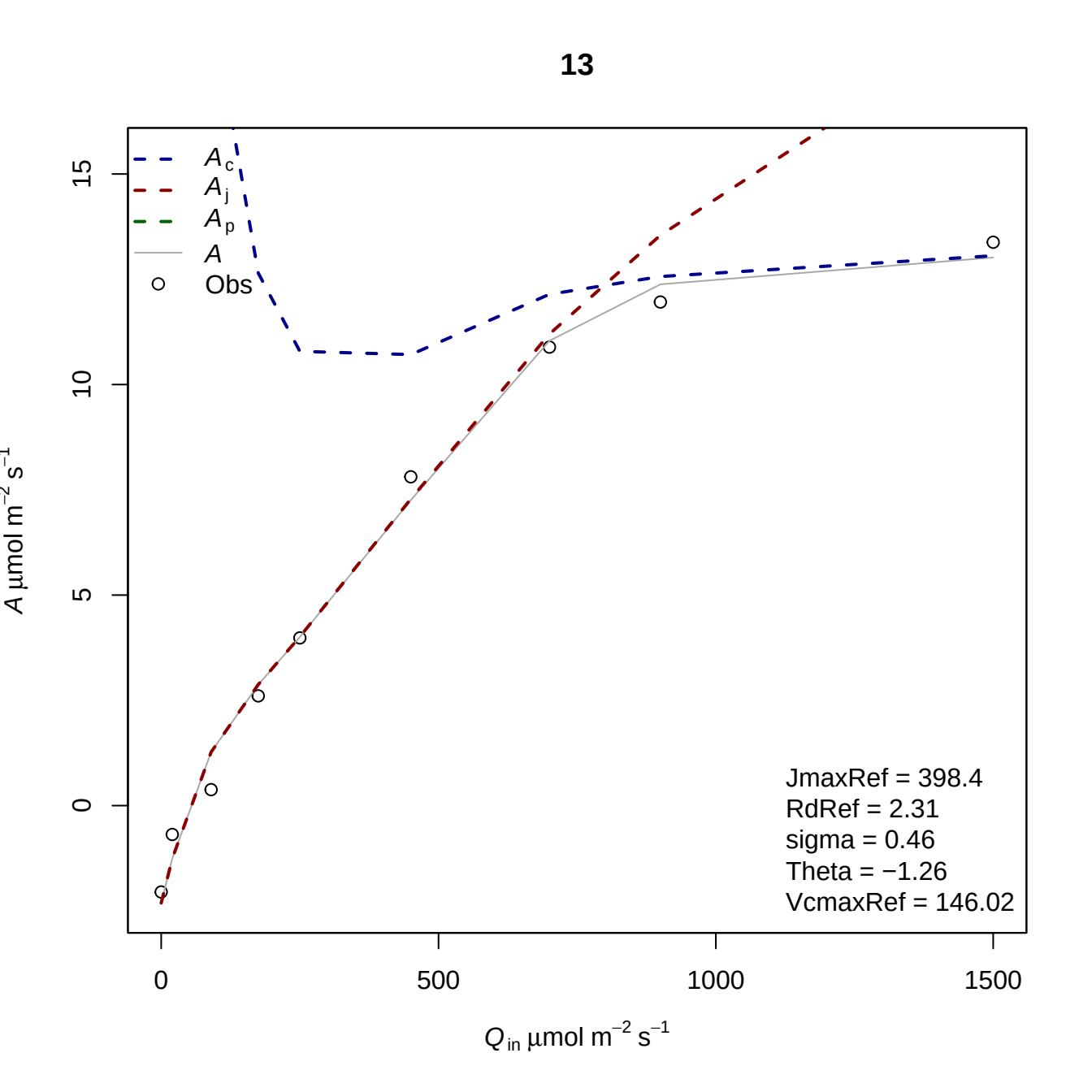

8
A_c
A_j
A_p
A
Obs
JmaxRef = 147.67
RdRef = 2.76
sigma = 0.17
Theta = −0.44
VcmaxRef = 77.85
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

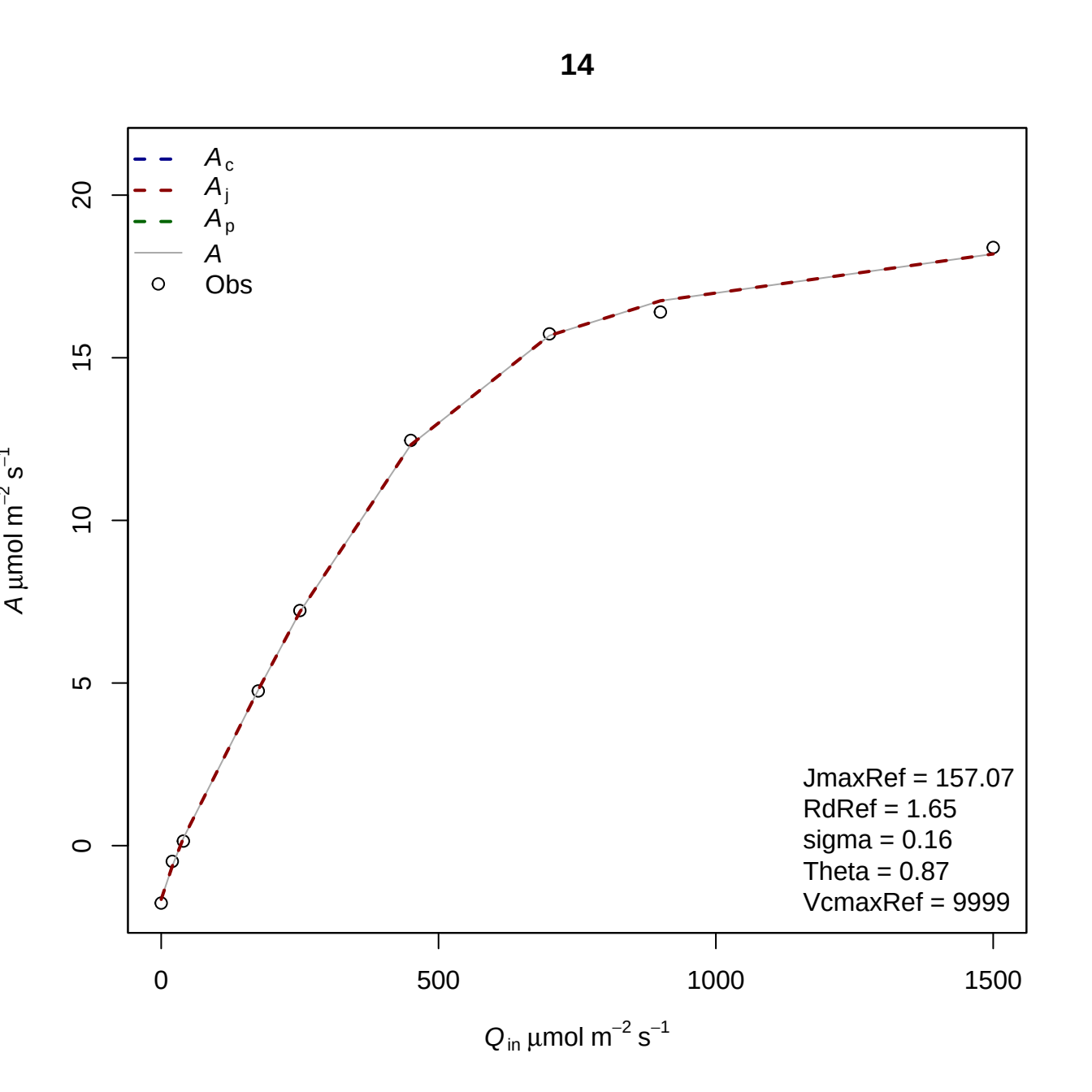

9
A_c
A_j
A_p
A
Obs
JmaxRef = 162.06
RdRef = 1.18
sigma = 0.19
Theta = 0.76
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

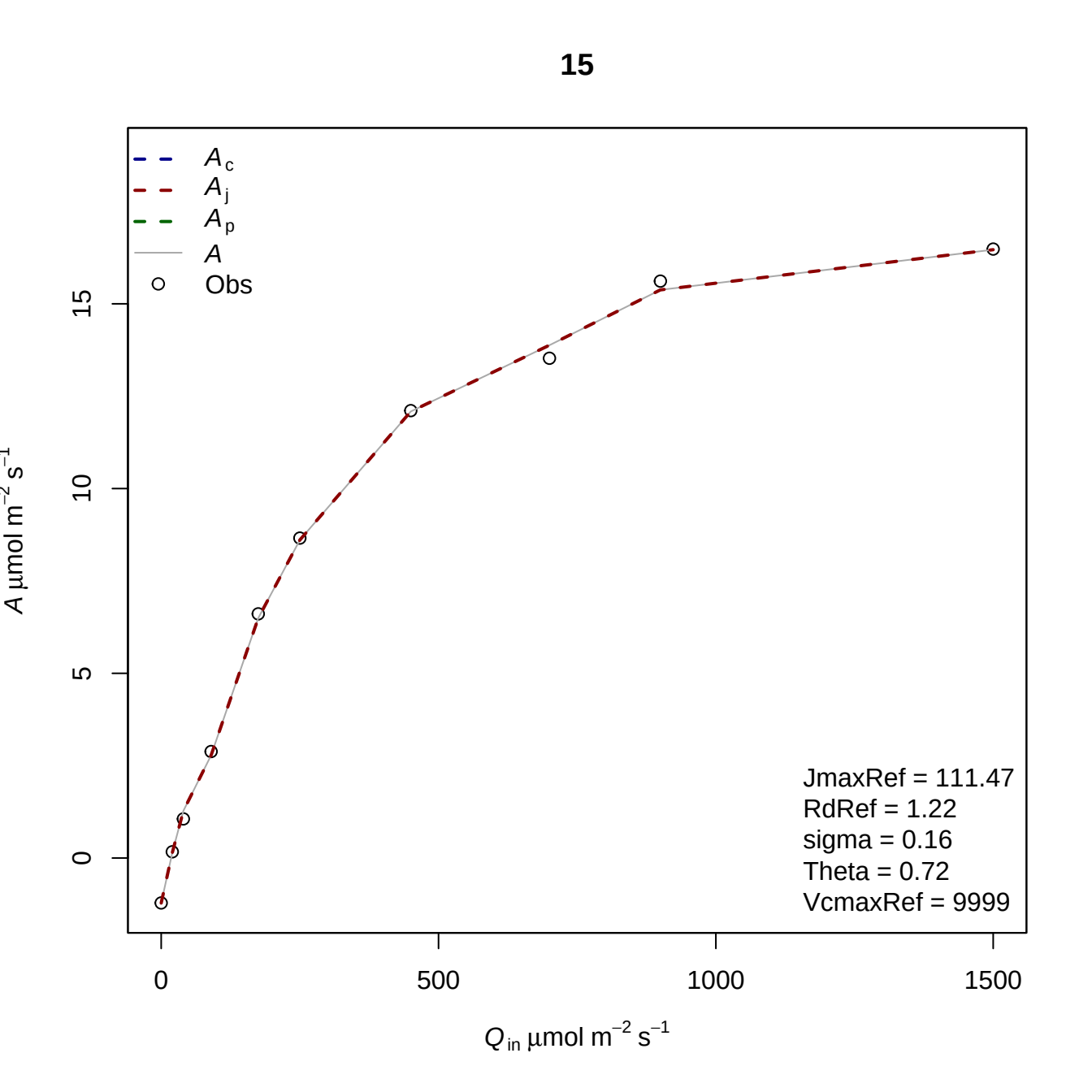

10
A_c
A_j
A_p
A
Obs
JmaxRef = 103.65
RdRef = 1.07
sigma = 0.13
Theta = 0.87
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

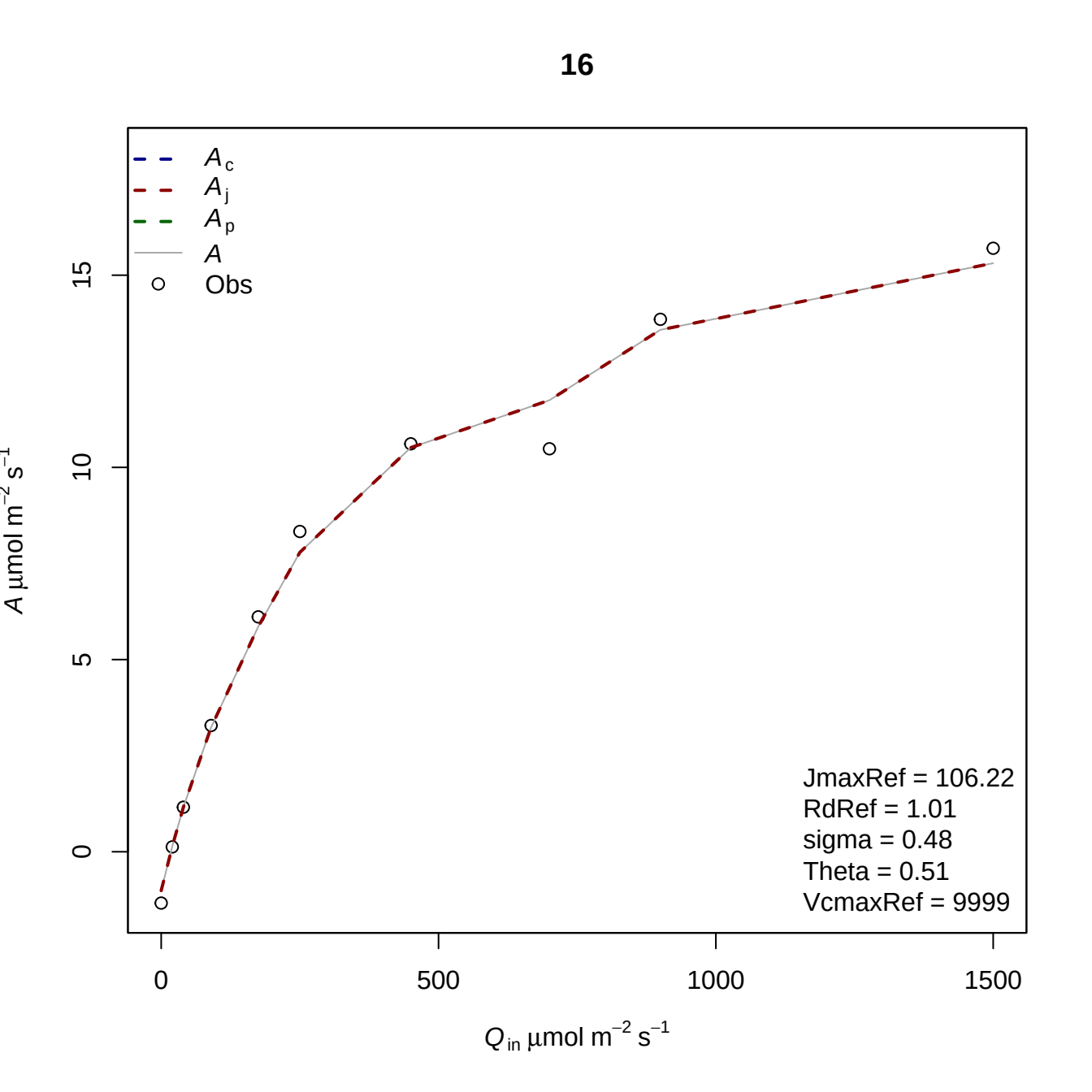

11
A_c
A_j
A_p
A
Obs
JmaxRef = 92.81
RdRef = 1.29
sigma = 0.22
Theta = 0.47
VcmaxRef = 53.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

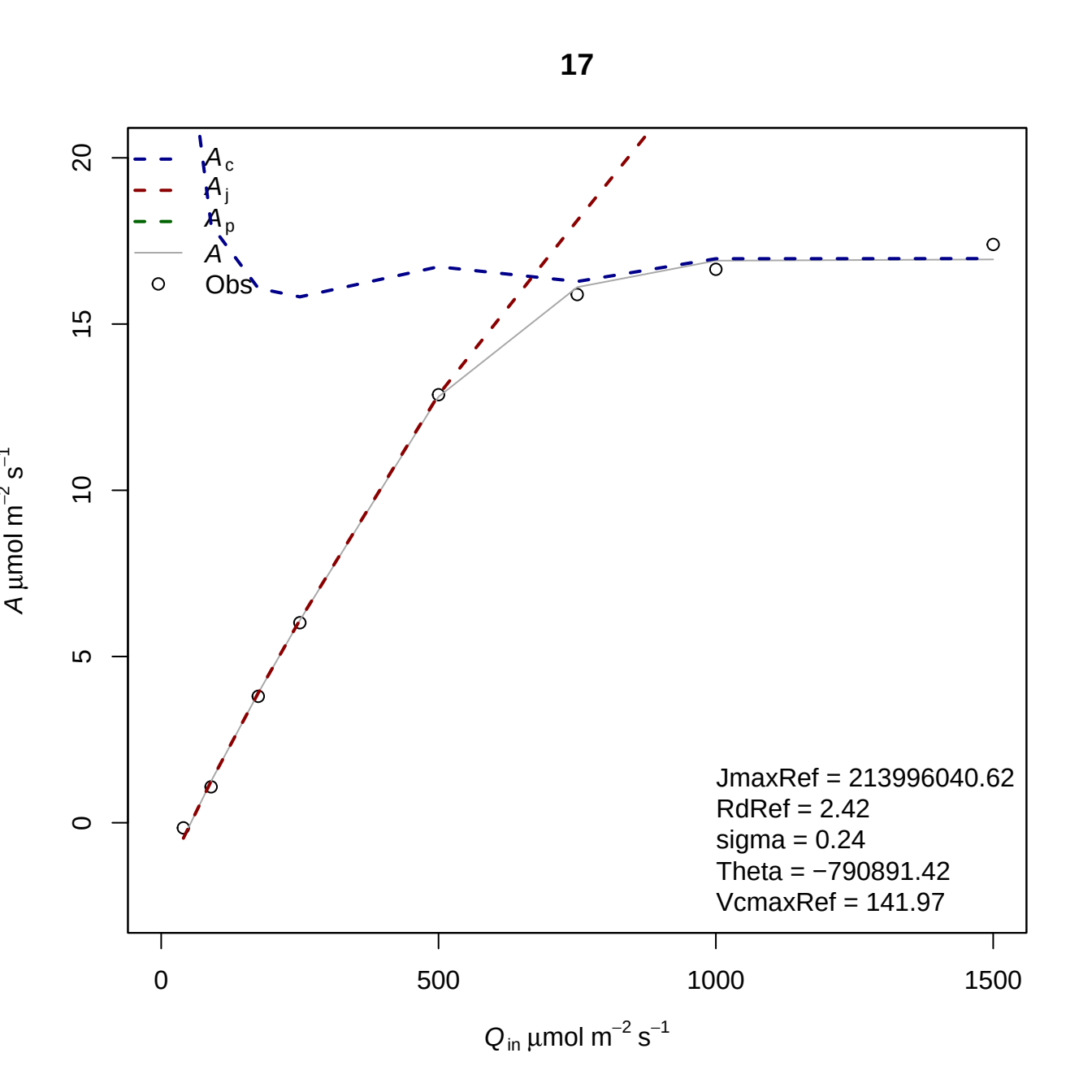

12
A_c
A_j
A_p
A
Obs
JmaxRef = 127.16
RdRef = 1.08
sigma = 0.44
Theta = 0.68
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

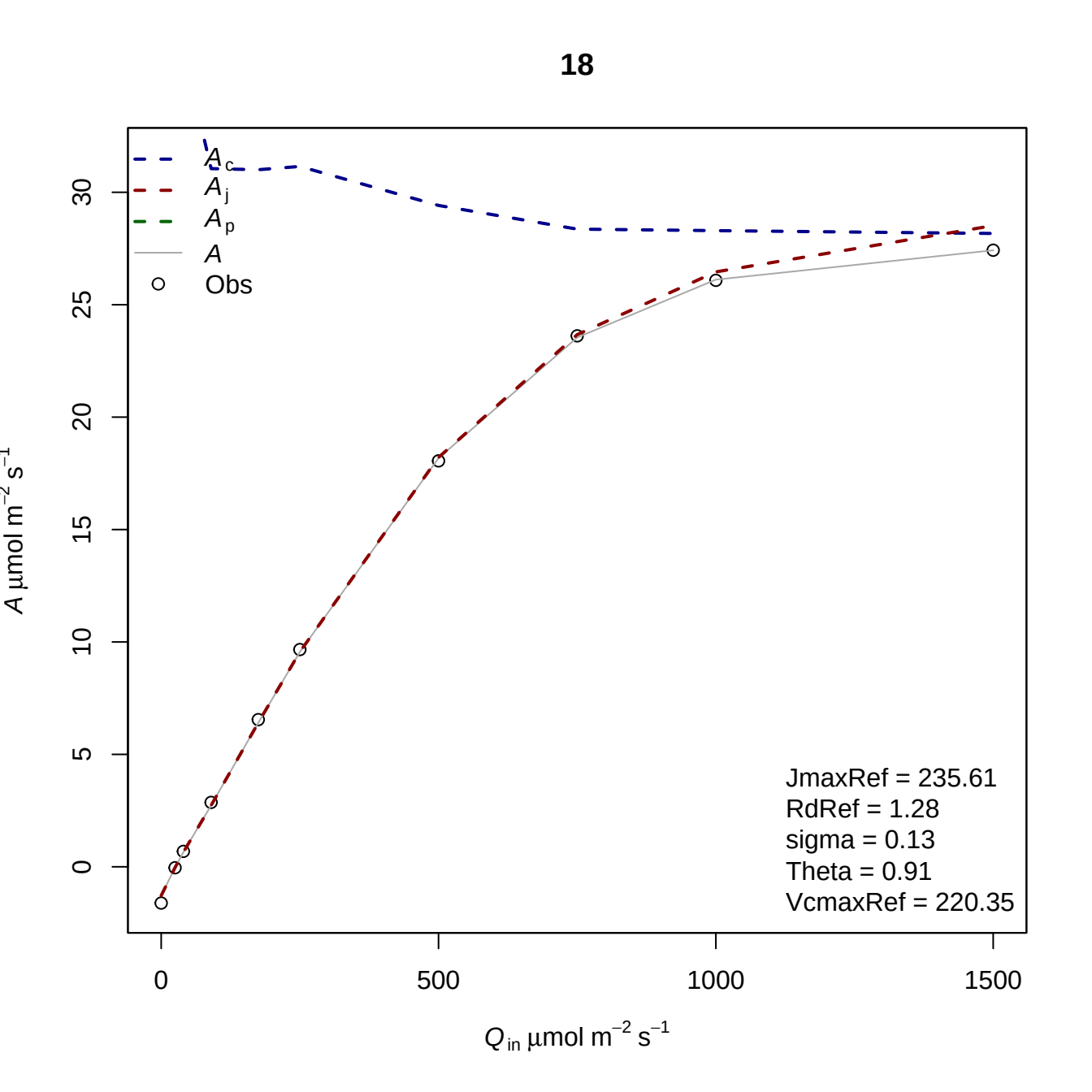

13
A_c
A_j
A_p
A
Obs
JmaxRef = 398.4
RdRef = 2.31
sigma = 0.46
Theta = −1.26
VcmaxRef = 146.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

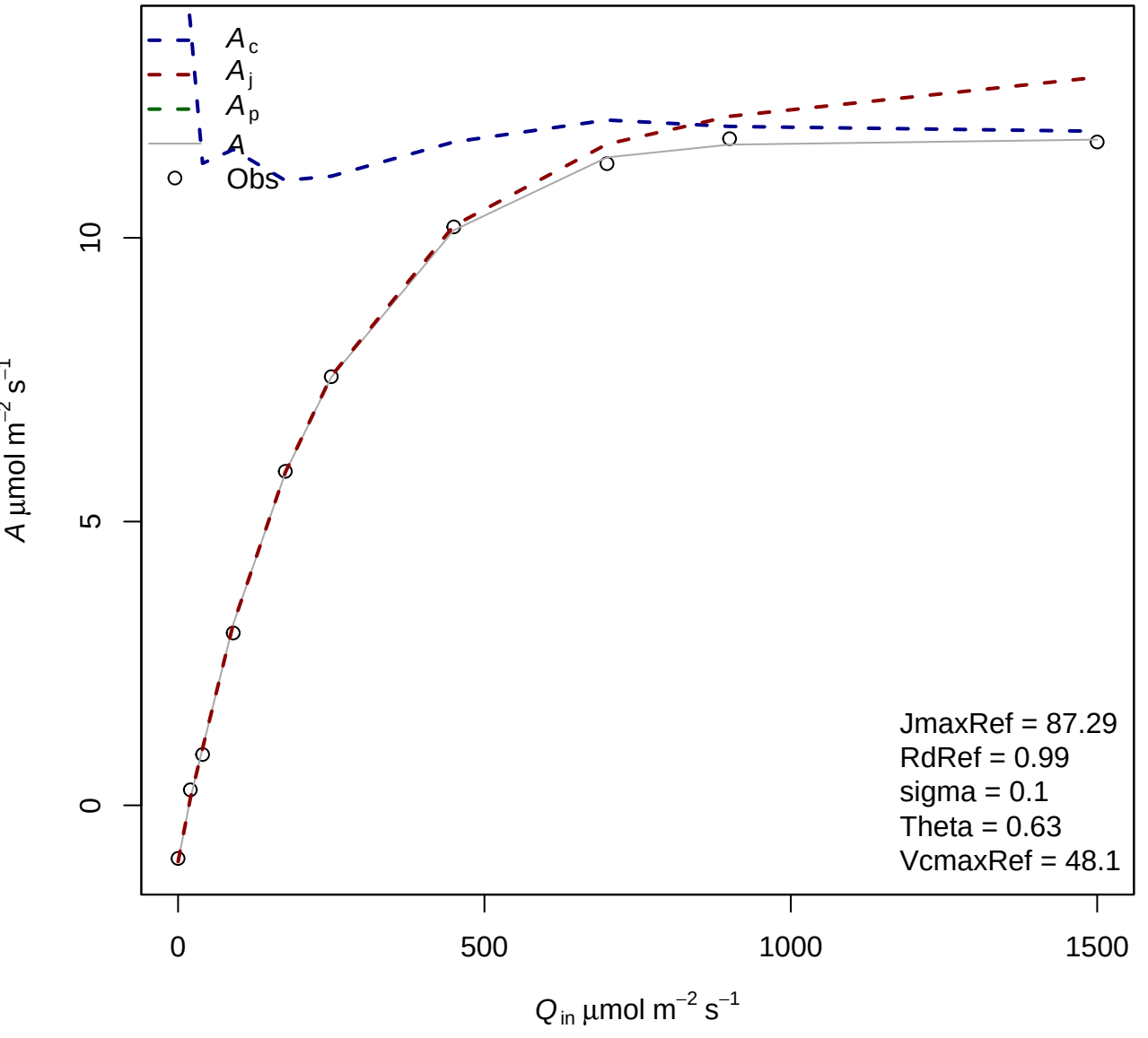

14
A_c
A_j
A_p
A
Obs
JmaxRef = 157.07
RdRef = 1.65
sigma = 0.16
Theta = 0.87
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

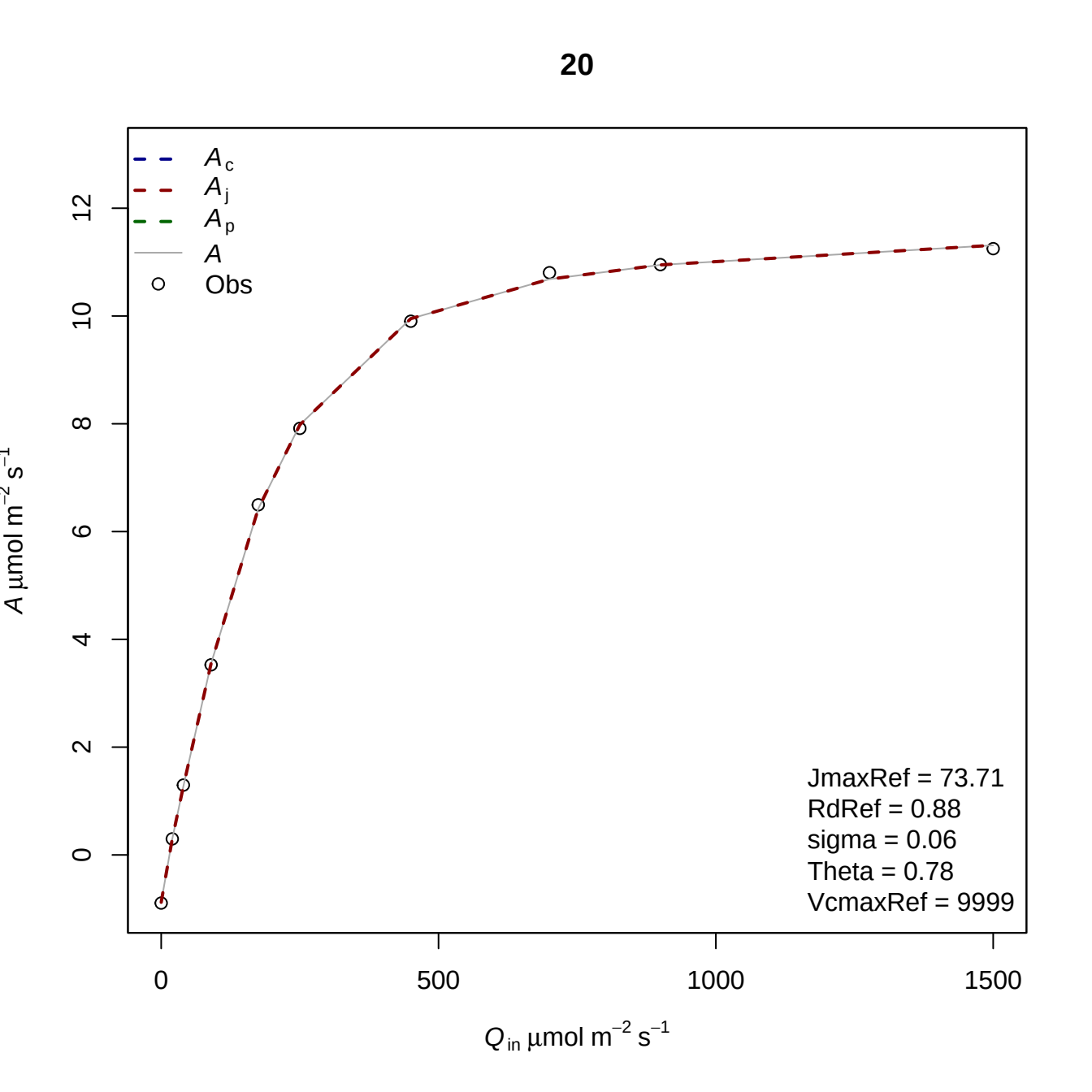

15
A_c
A_j
A_p
A
Obs
JmaxRef = 111.47
RdRef = 1.22
sigma = 0.16
Theta = 0.72
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

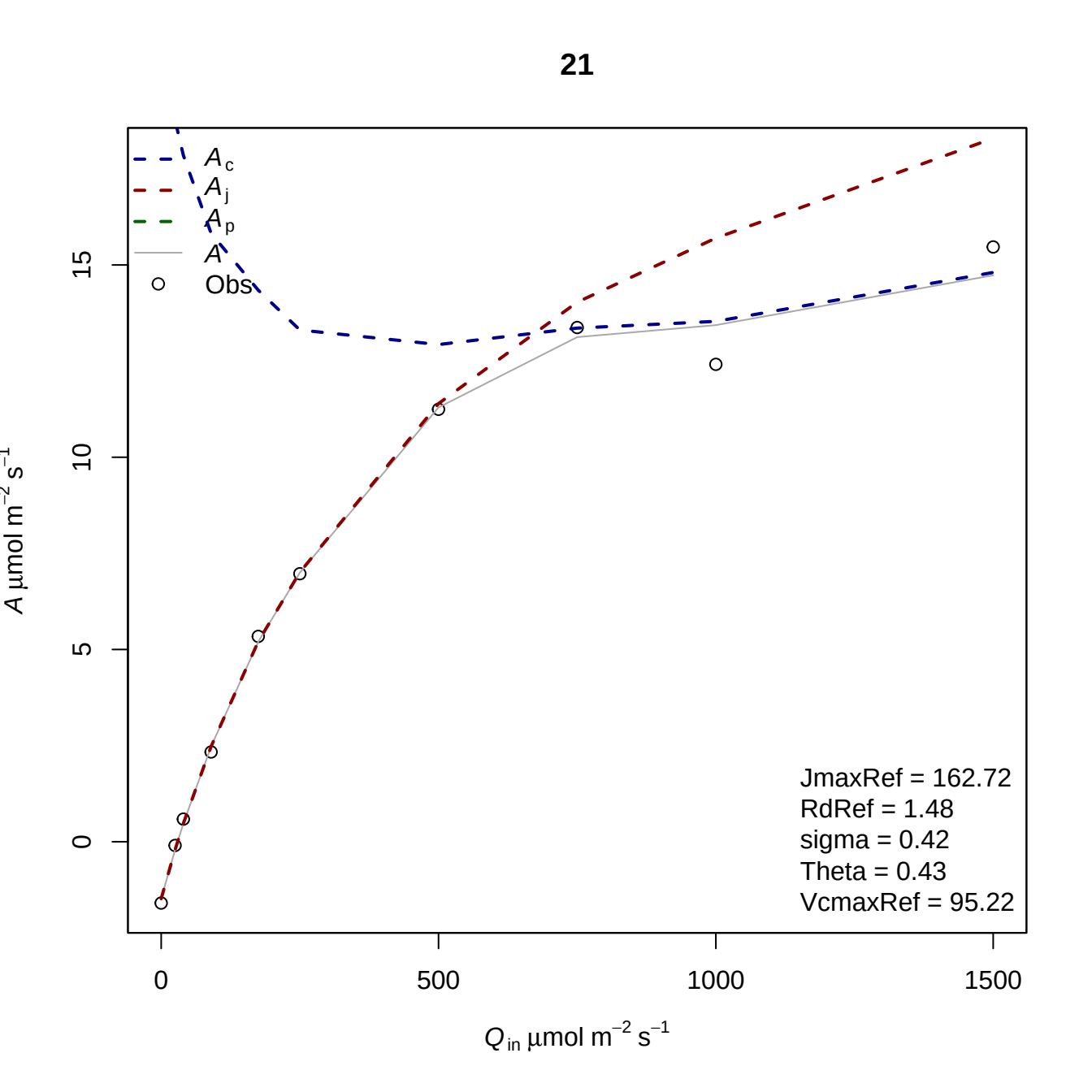

16
A_c
A_j
A_p
A
Obs
JmaxRef = 106.22
RdRef = 1.01
sigma = 0.48
Theta = 0.51
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

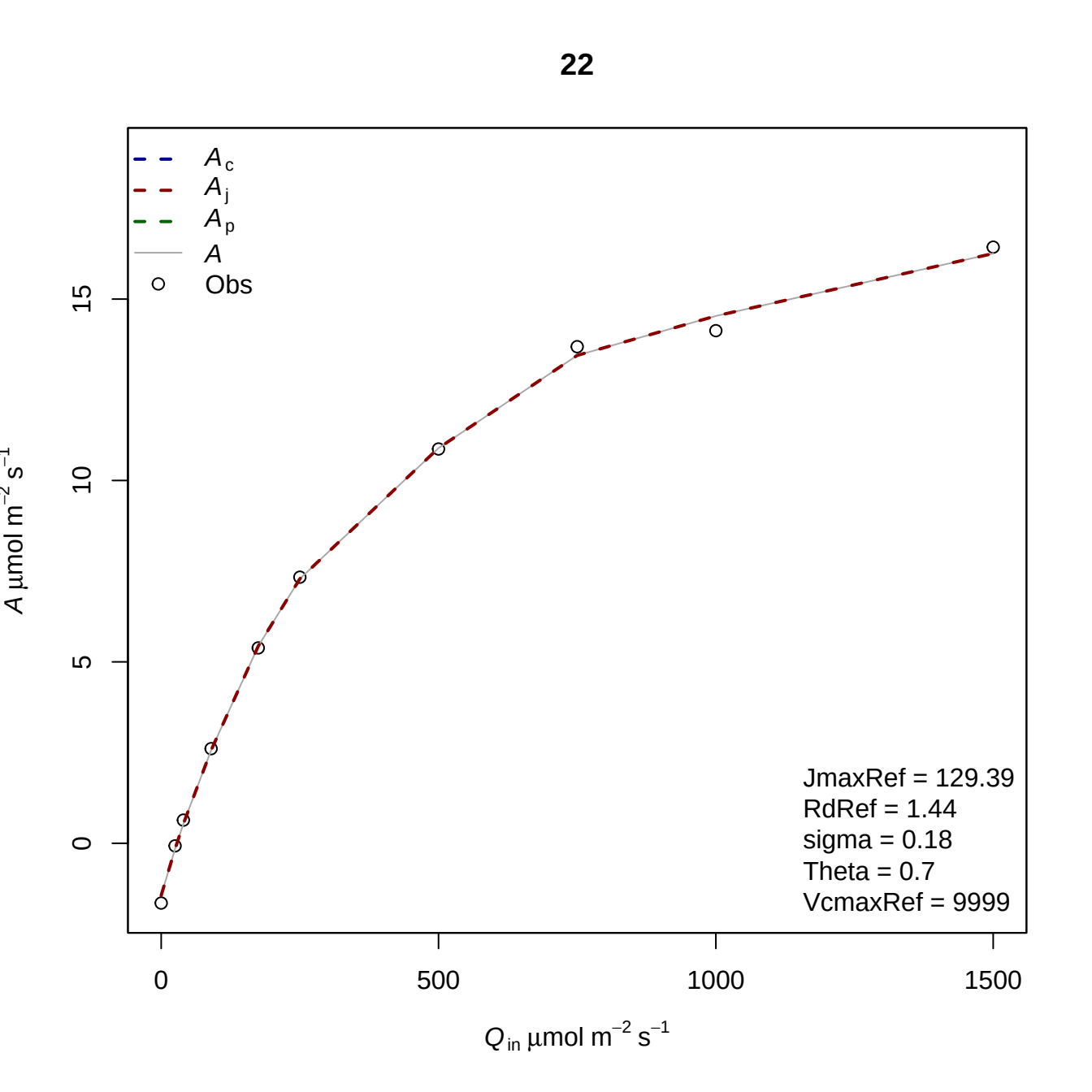

17
A_c
A_j
A_p
A
Obs
JmaxRef = 213996040.62
RdRef = 2.42
sigma = 0.24
Theta = −790891.42
VcmaxRef = 141.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

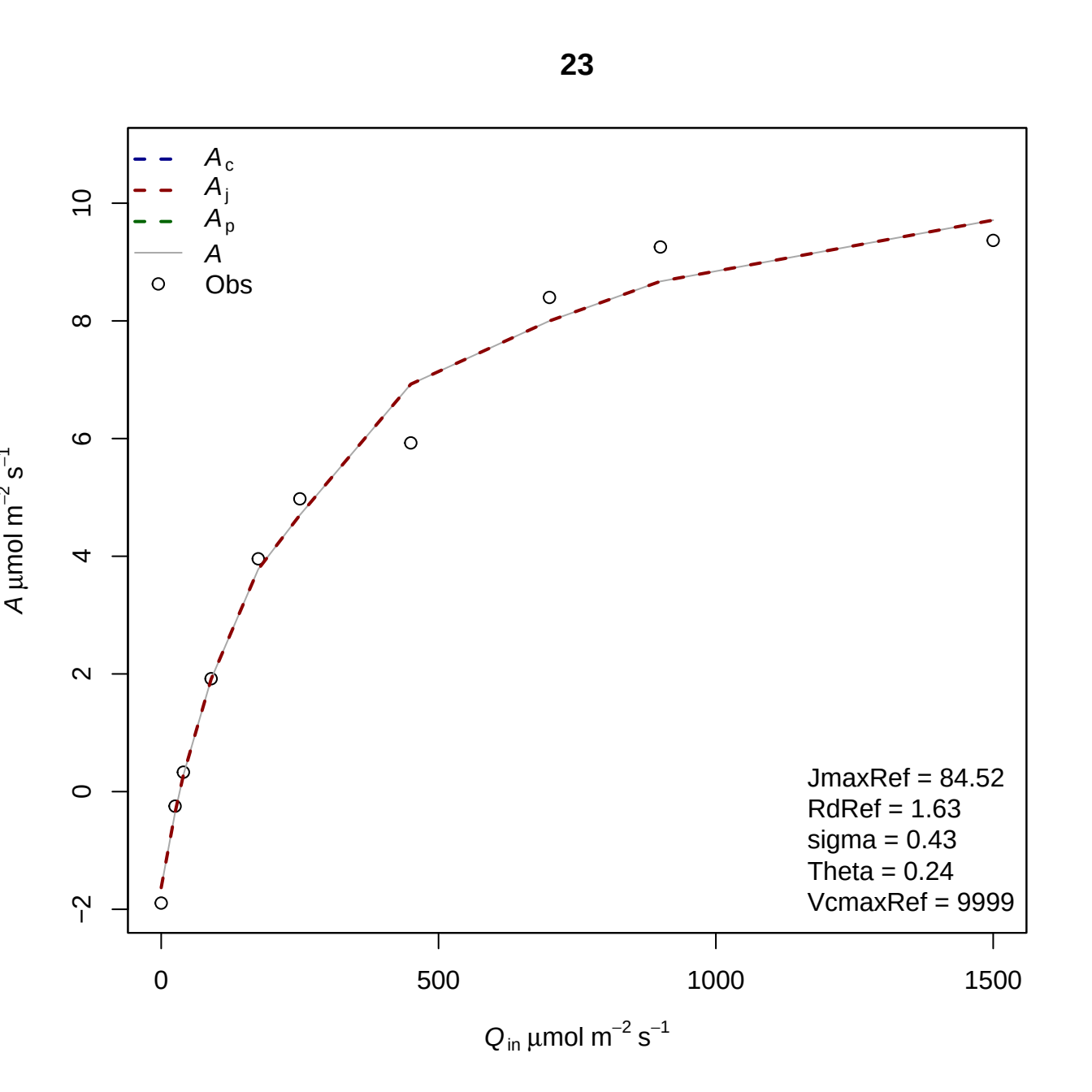

18
A_c
A_j
A_p
A
Obs
JmaxRef = 235.61
RdRef = 1.28
sigma = 0.13
Theta = 0.91
VcmaxRef = 220.35
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

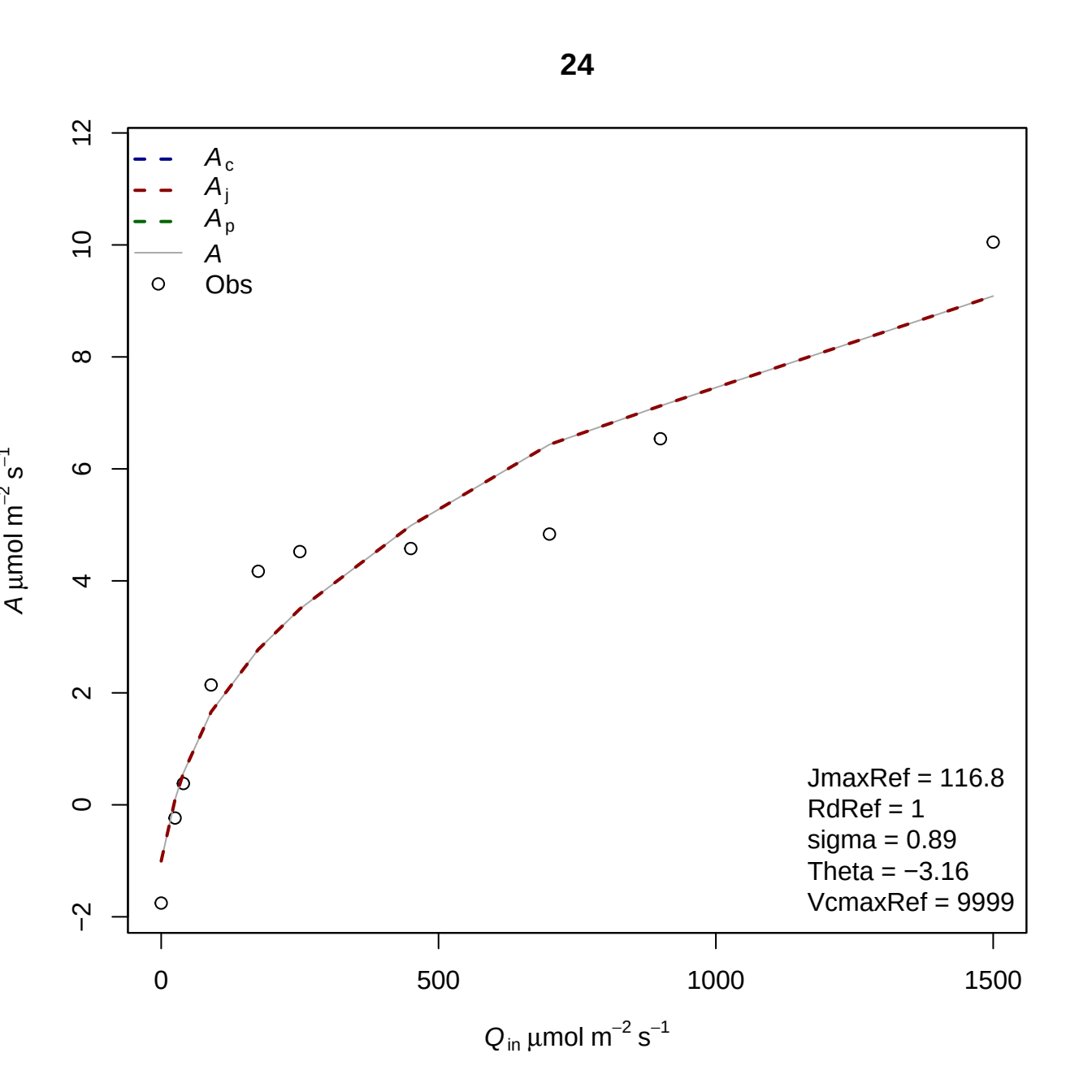

A_c
A_j
A_p
A
Obs
JmaxRef = 87.29
RdRef = 0.99
sigma = 0.1
Theta = 0.63
VcmaxRef = 48.1
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

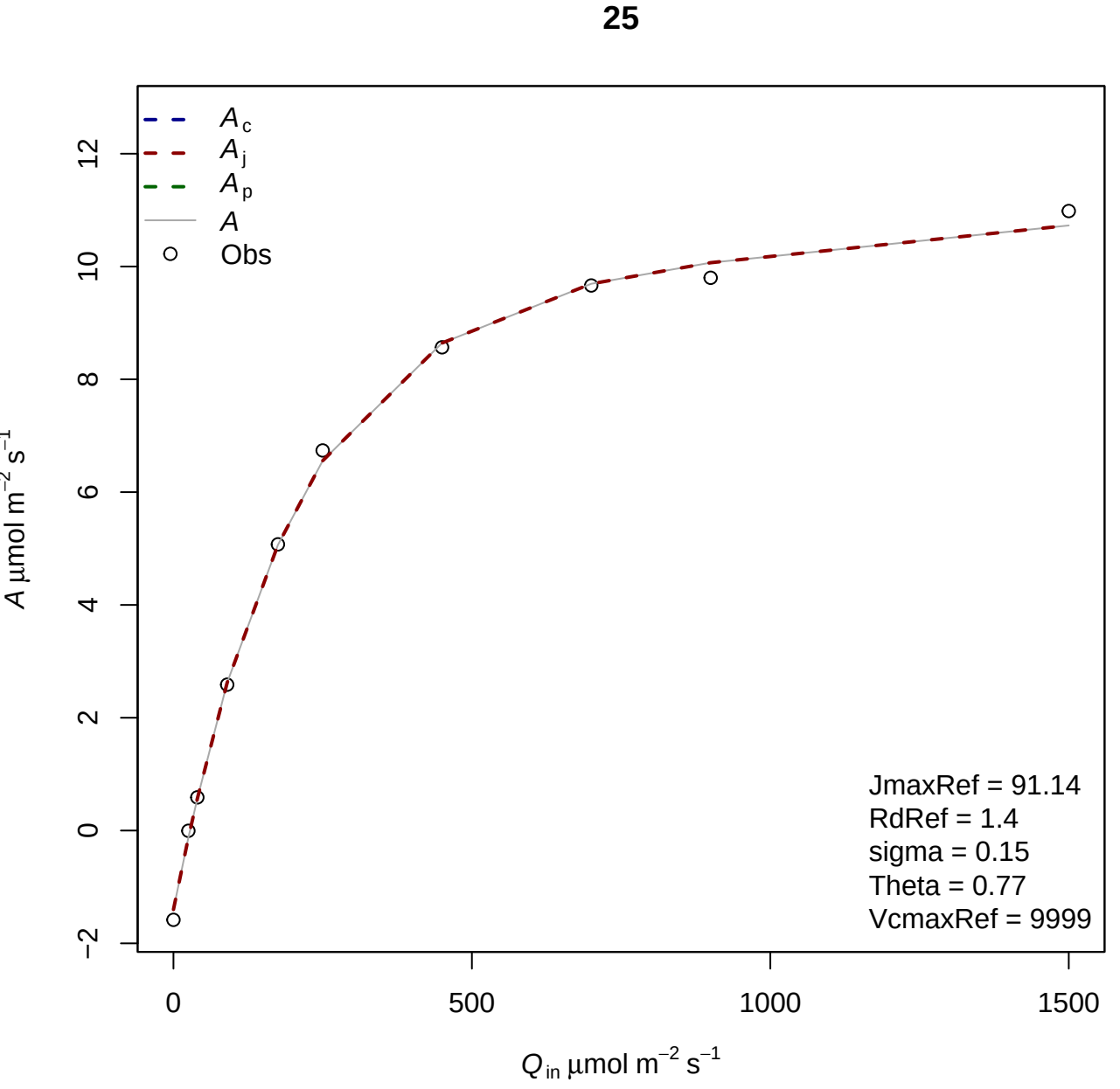

20
A_c
A_j
A_p
A
Obs
JmaxRef = 73.71
RdRef = 0.88
sigma = 0.06
Theta = 0.78
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

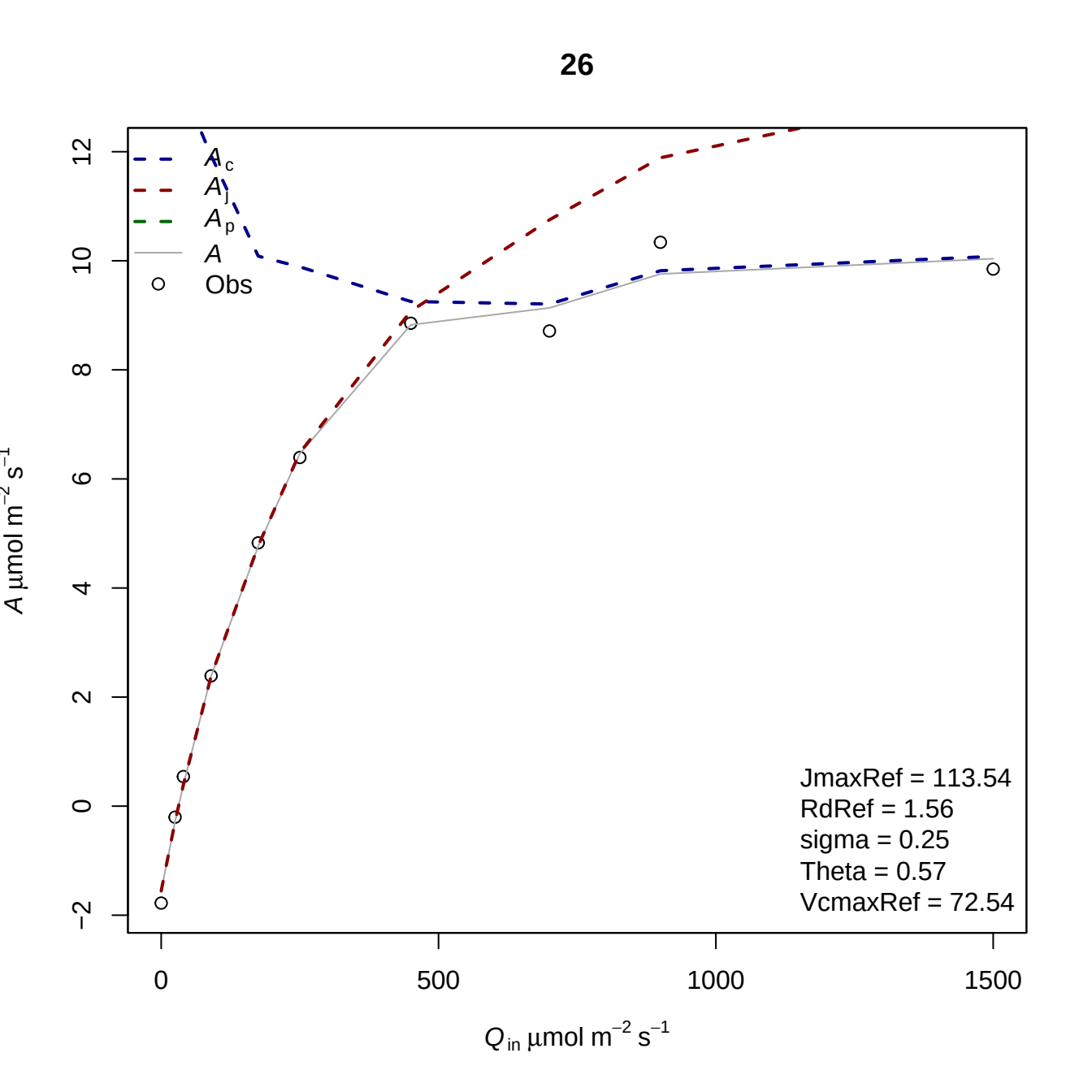

21
A_c
A_j
A_p
A
Obs
JmaxRef = 162.72
RdRef = 1.48
sigma = 0.42
Theta = 0.43
VcmaxRef = 95.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

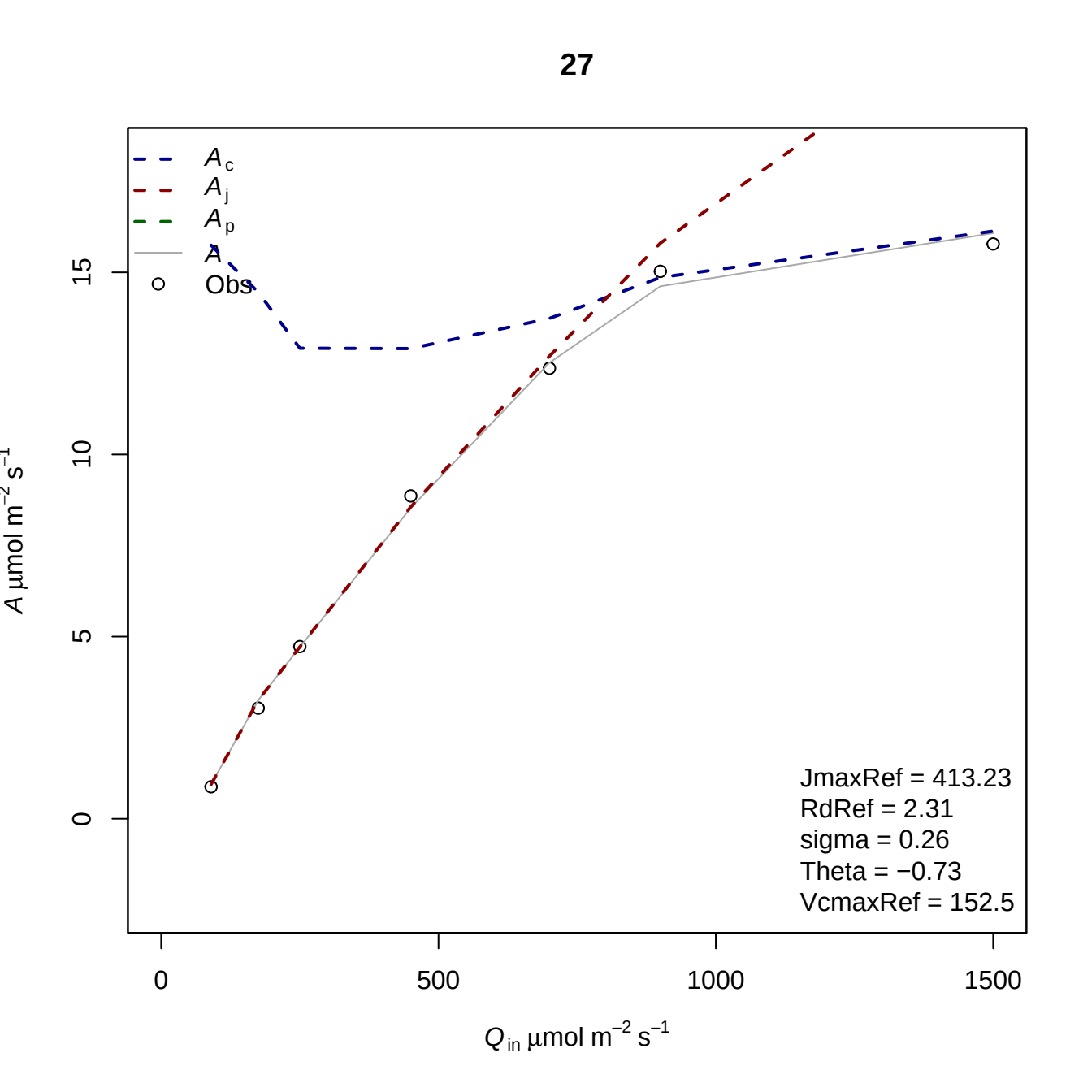

22
A_c
A_j
A_p
A
Obs
JmaxRef = 129.39
RdRef = 1.44
sigma = 0.18
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

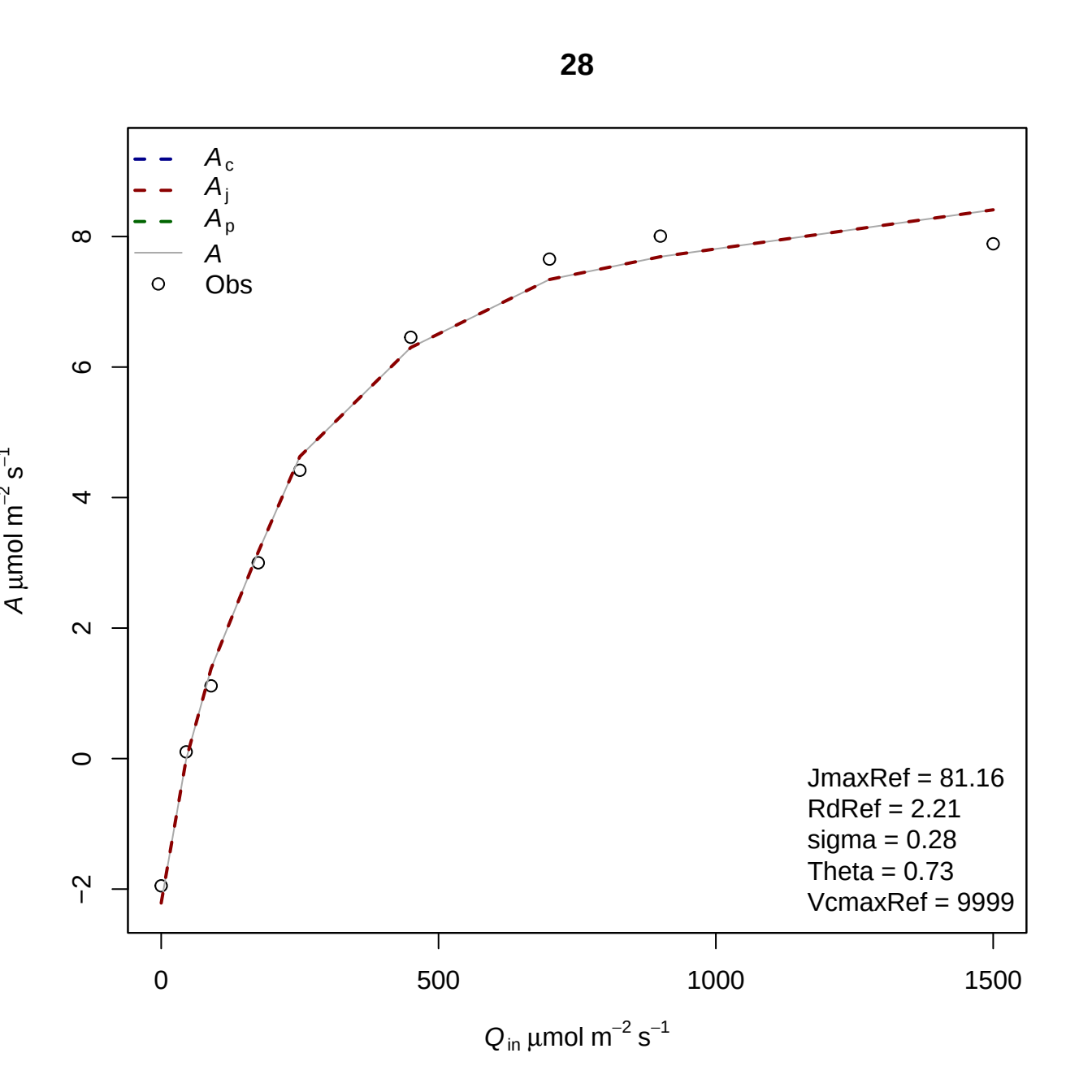

23
A_c
A_j
A_p
A
Obs
JmaxRef = 84.52
RdRef = 1.63
sigma = 0.43
Theta = 0.24
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

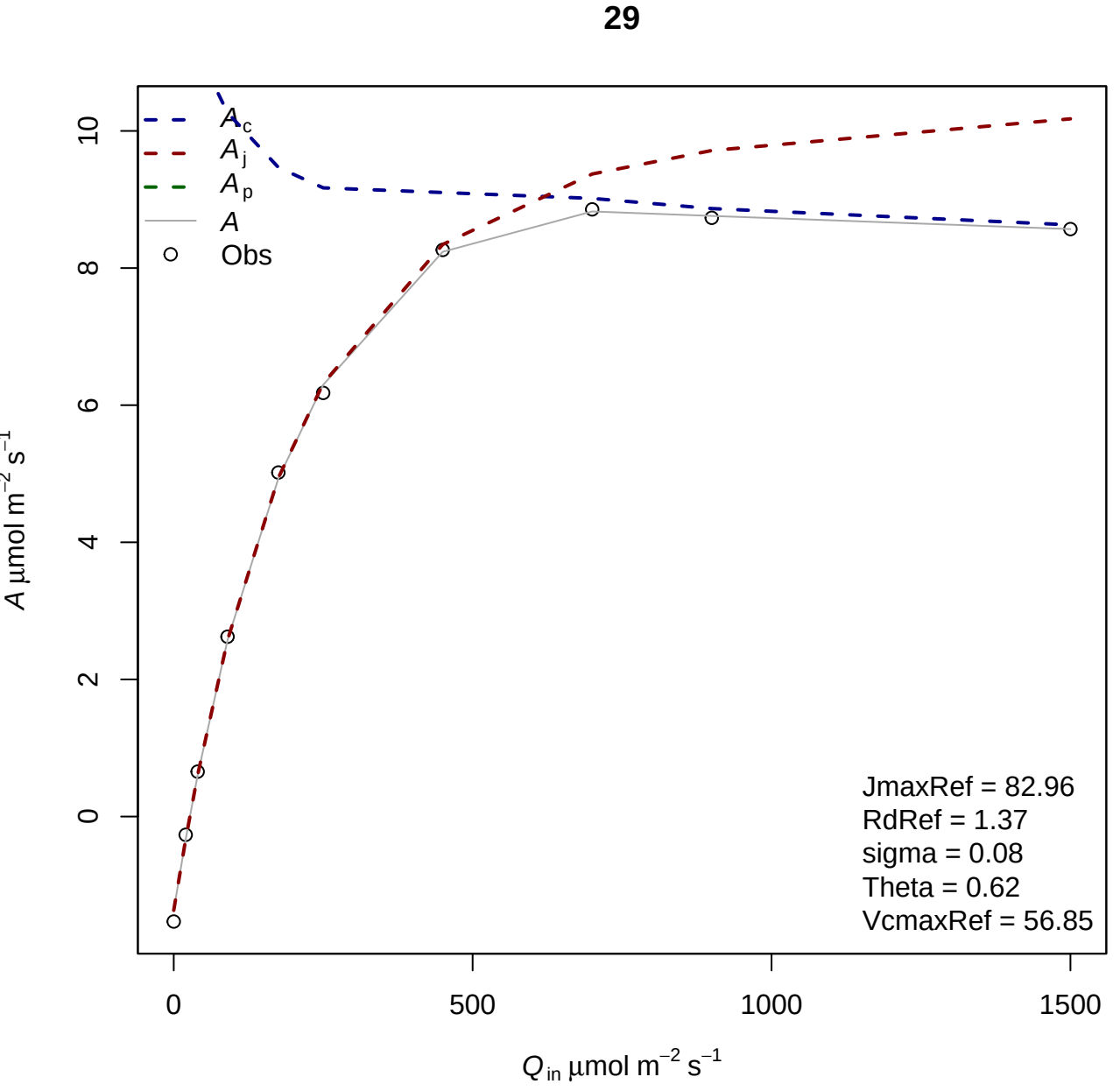

24
A_c
A_j
A_p
A
Obs
JmaxRef = 116.8
RdRef = 1
sigma = 0.89
Theta = −3.16
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

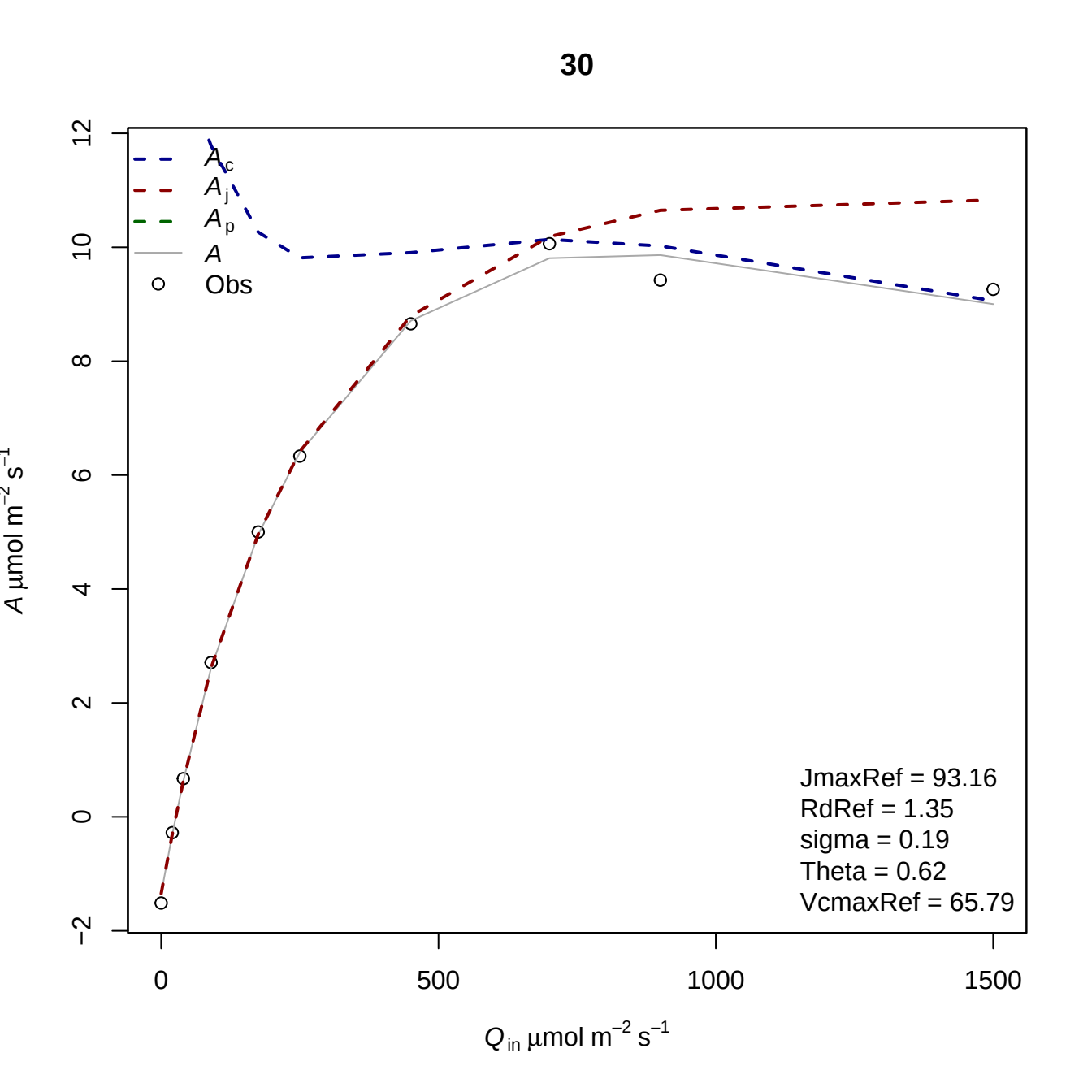

25
A_c
A_j
A_p
A
Obs
JmaxRef = 91.14
RdRef = 1.4
sigma = 0.15
Theta = 0.77
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

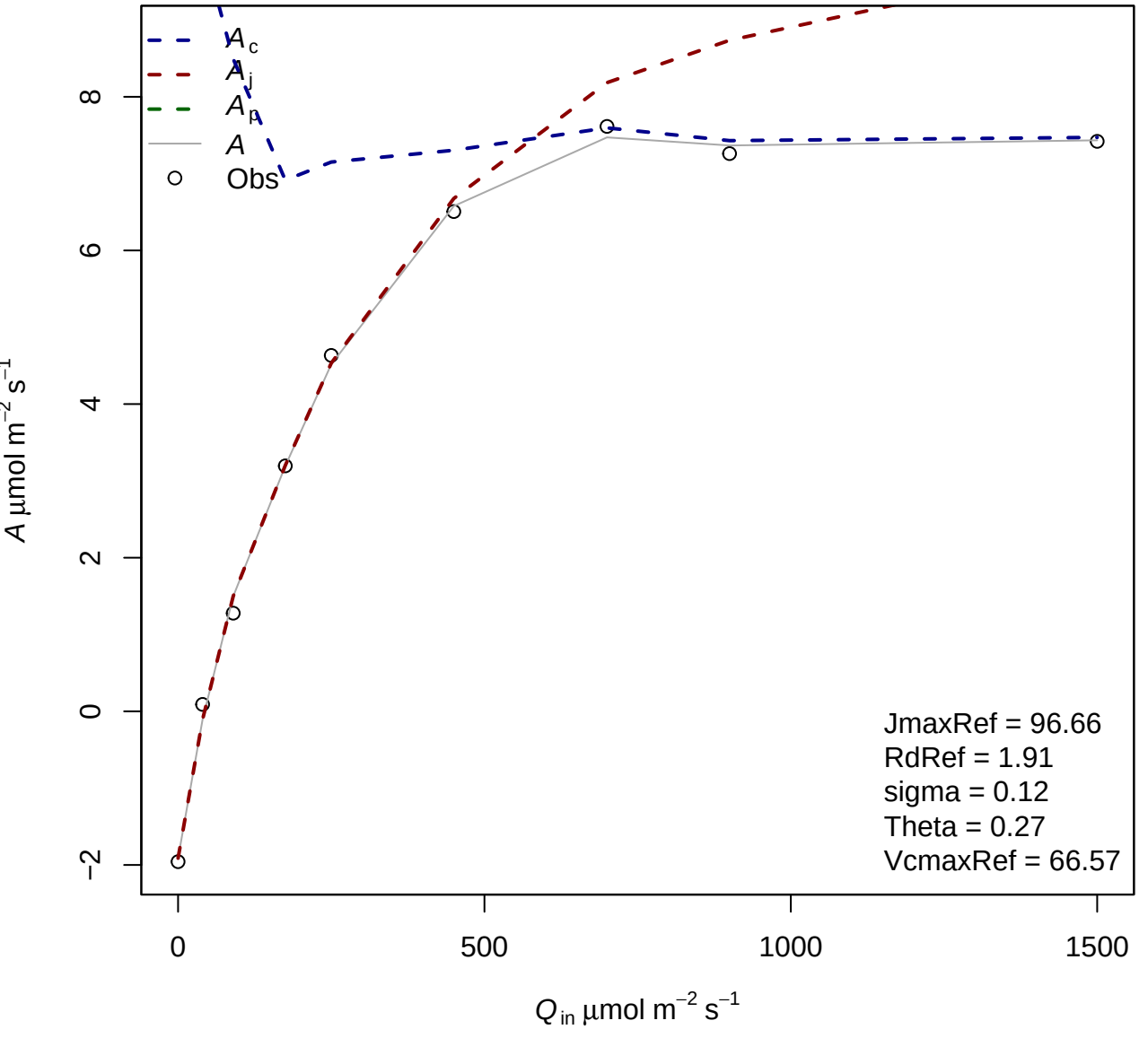

26
A_c
A_j
A_p
A
Obs
JmaxRef = 113.54
RdRef = 1.56
sigma = 0.25
Theta = 0.57
VcmaxRef = 72.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

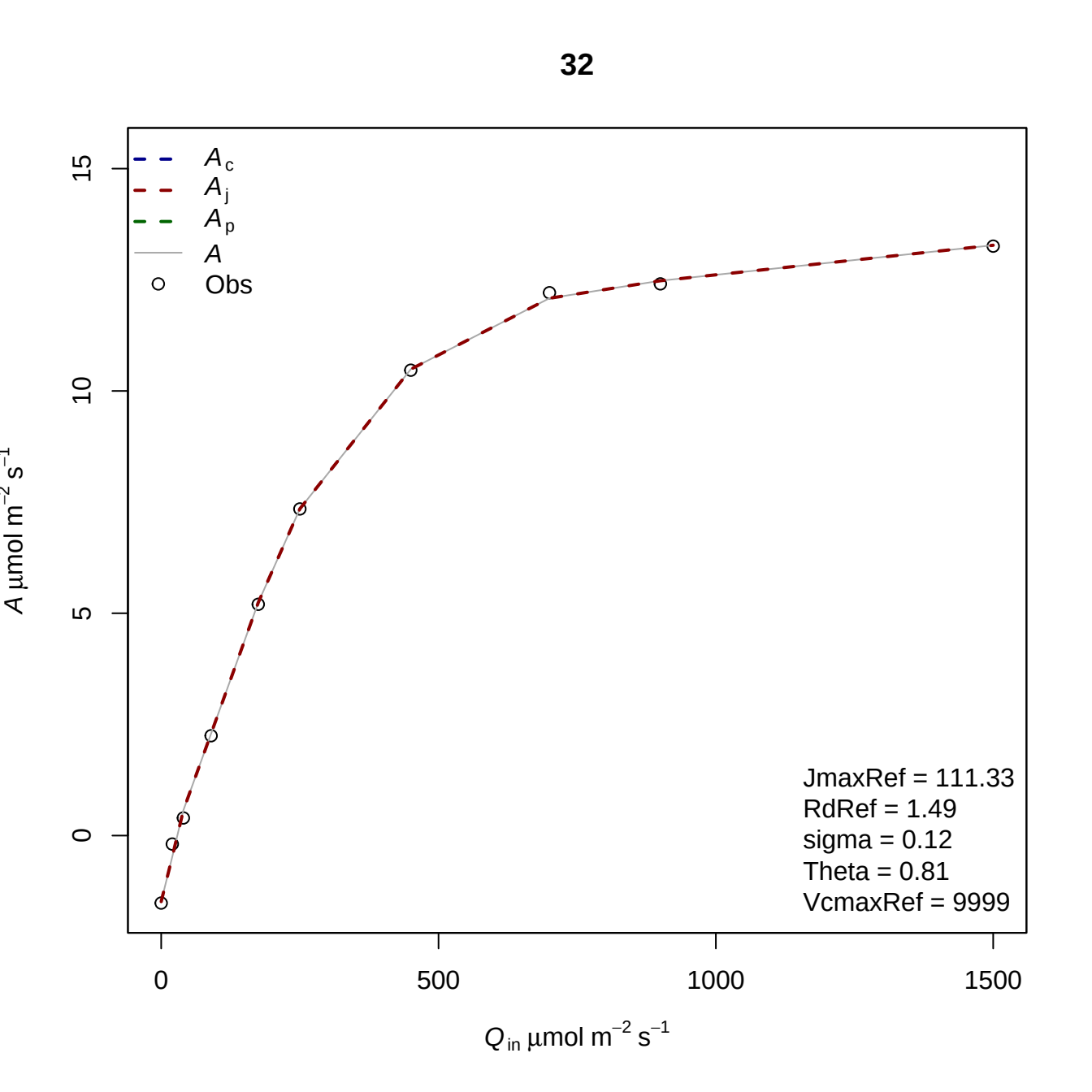

27
A_c
A_j
A_p
A
Obs
JmaxRef = 413.23
RdRef = 2.31
sigma = 0.26
Theta = −0.73
VcmaxRef = 152.5
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

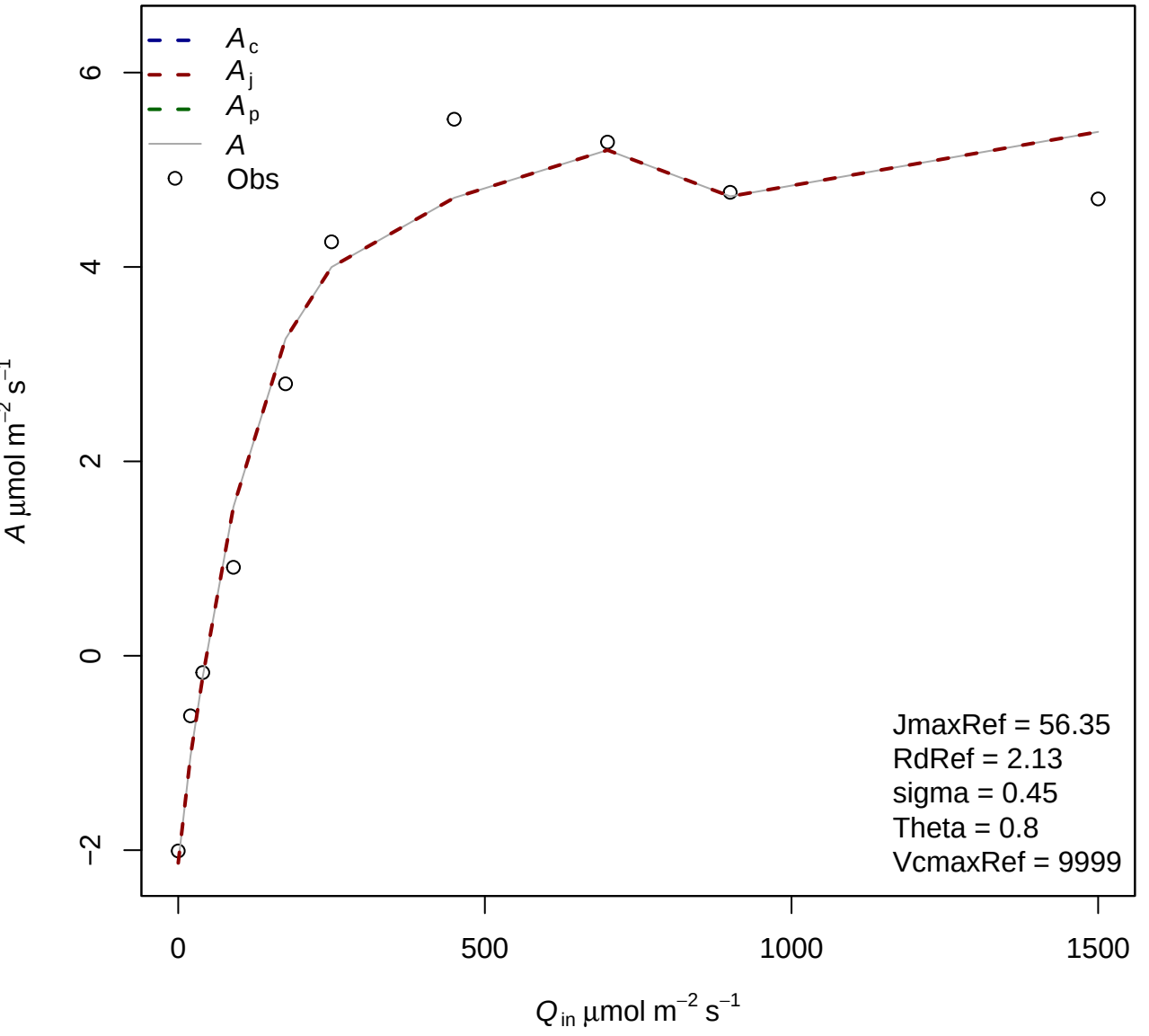

28
A_c
A_j
A_p
A
Obs
JmaxRef = 81.16
RdRef = 2.21
sigma = 0.28
Theta = 0.73
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

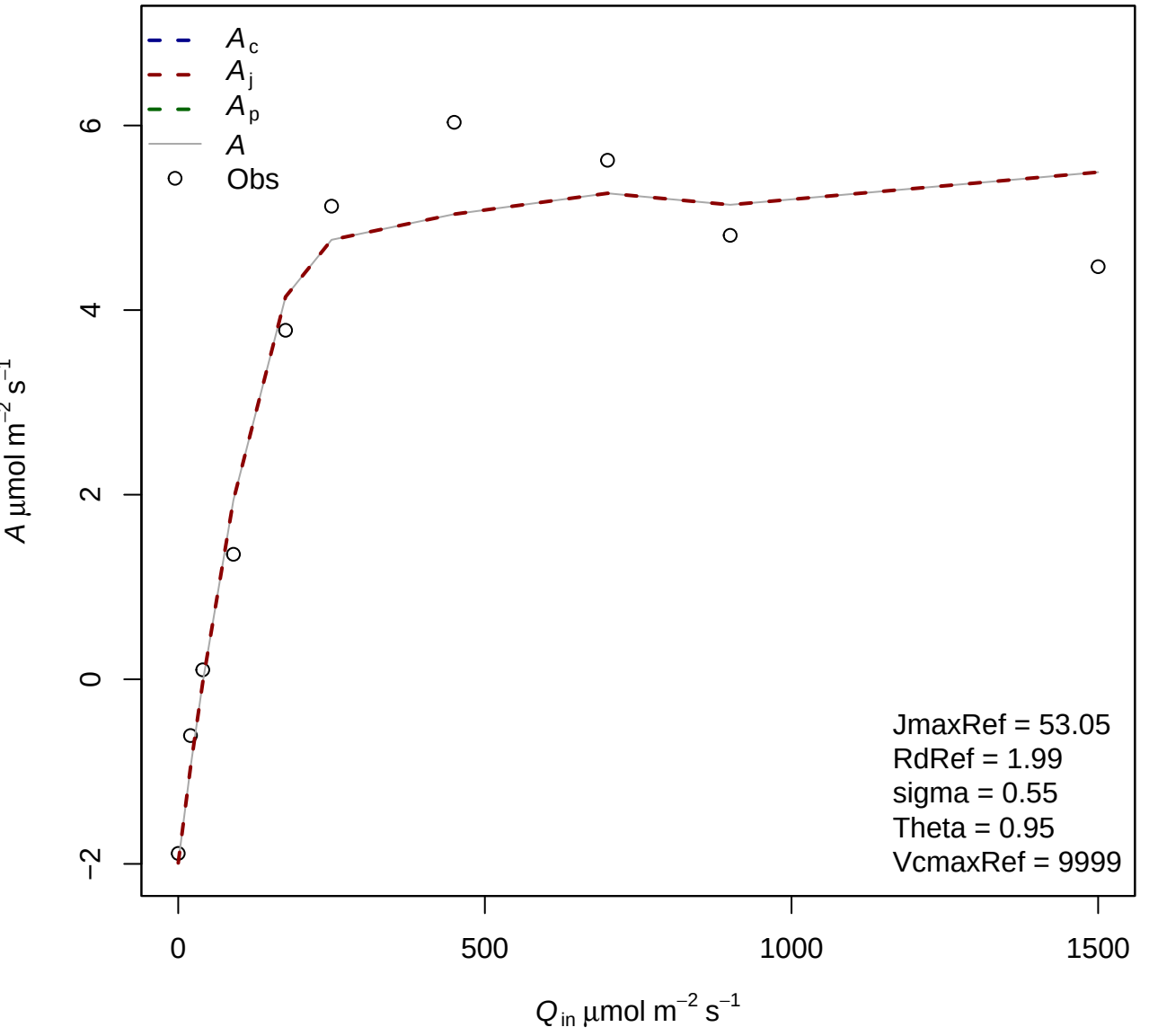

29
A_c
A_j
A_p
A
Obs
JmaxRef = 82.96
RdRef = 1.37
sigma = 0.08
Theta = 0.62
VcmaxRef = 56.85
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

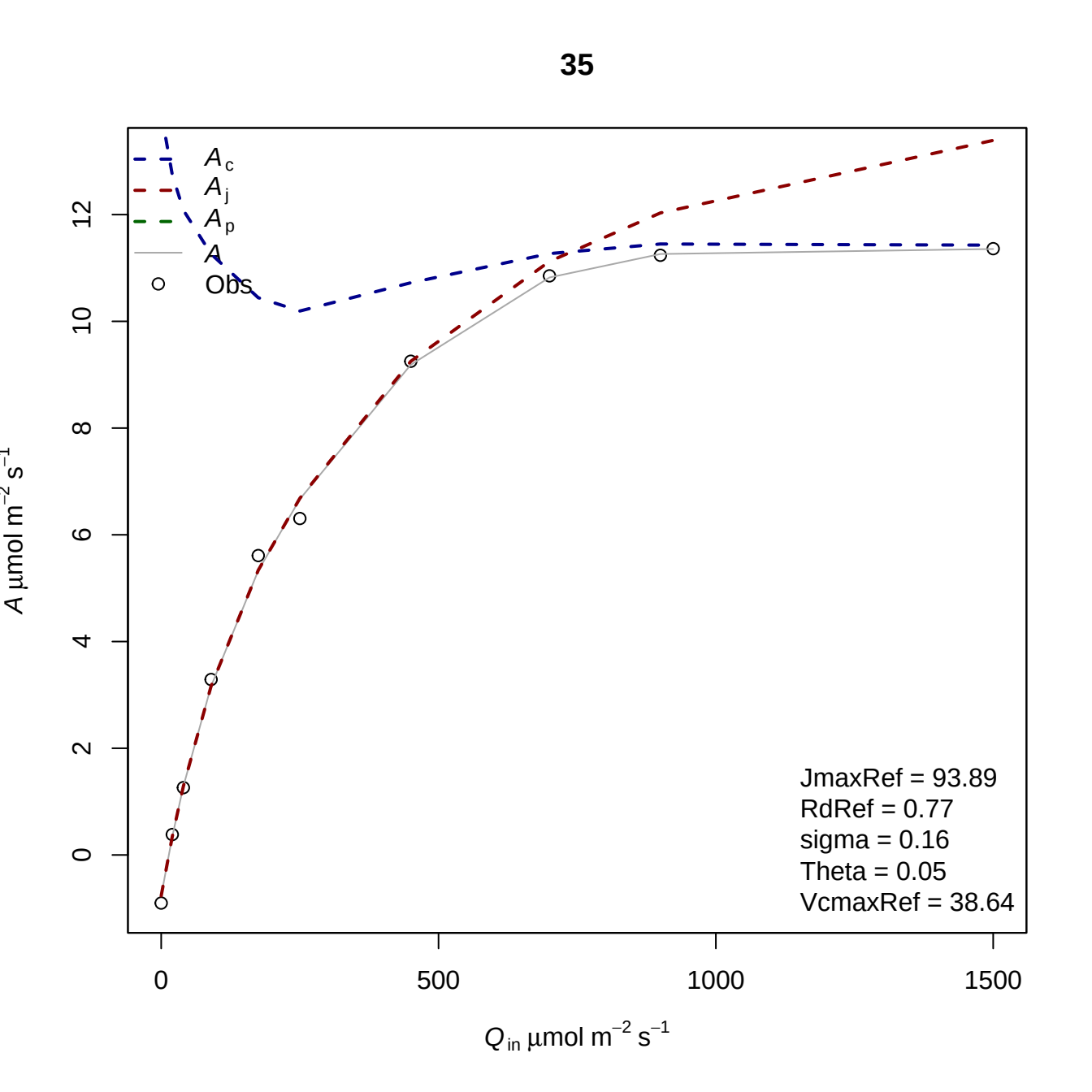

30
A_c
A_j
A_p
A
Obs
JmaxRef = 93.16
RdRef = 1.35
sigma = 0.19
Theta = 0.62
VcmaxRef = 65.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

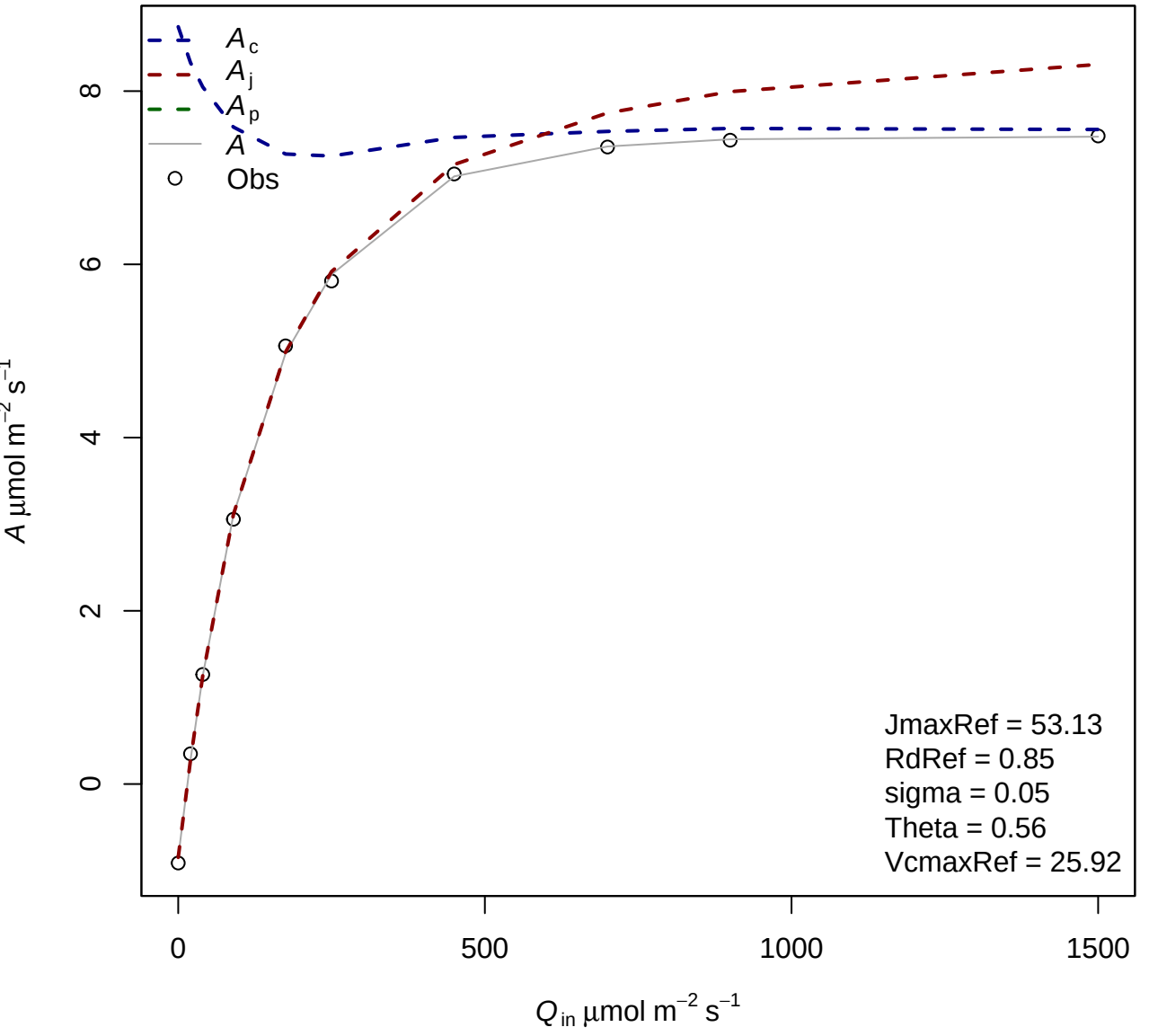

A_c
A_j
A_p
A
Obs
JmaxRef = 96.66
RdRef = 1.91
sigma = 0.12
Theta = 0.27
VcmaxRef = 66.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

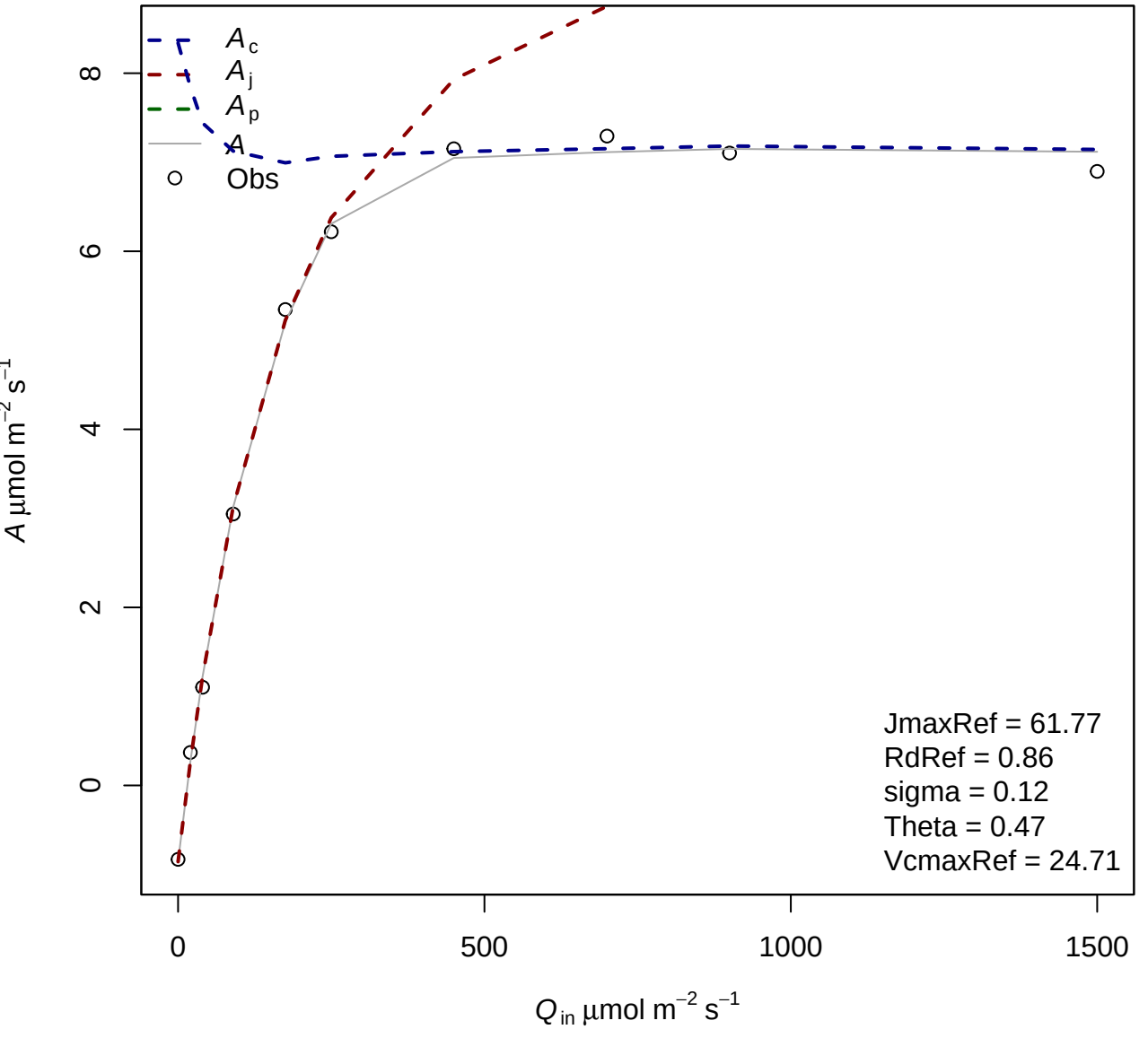

32
A_c
A_j
A_p
A
Obs
JmaxRef = 111.33
RdRef = 1.49
sigma = 0.12
Theta = 0.81
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

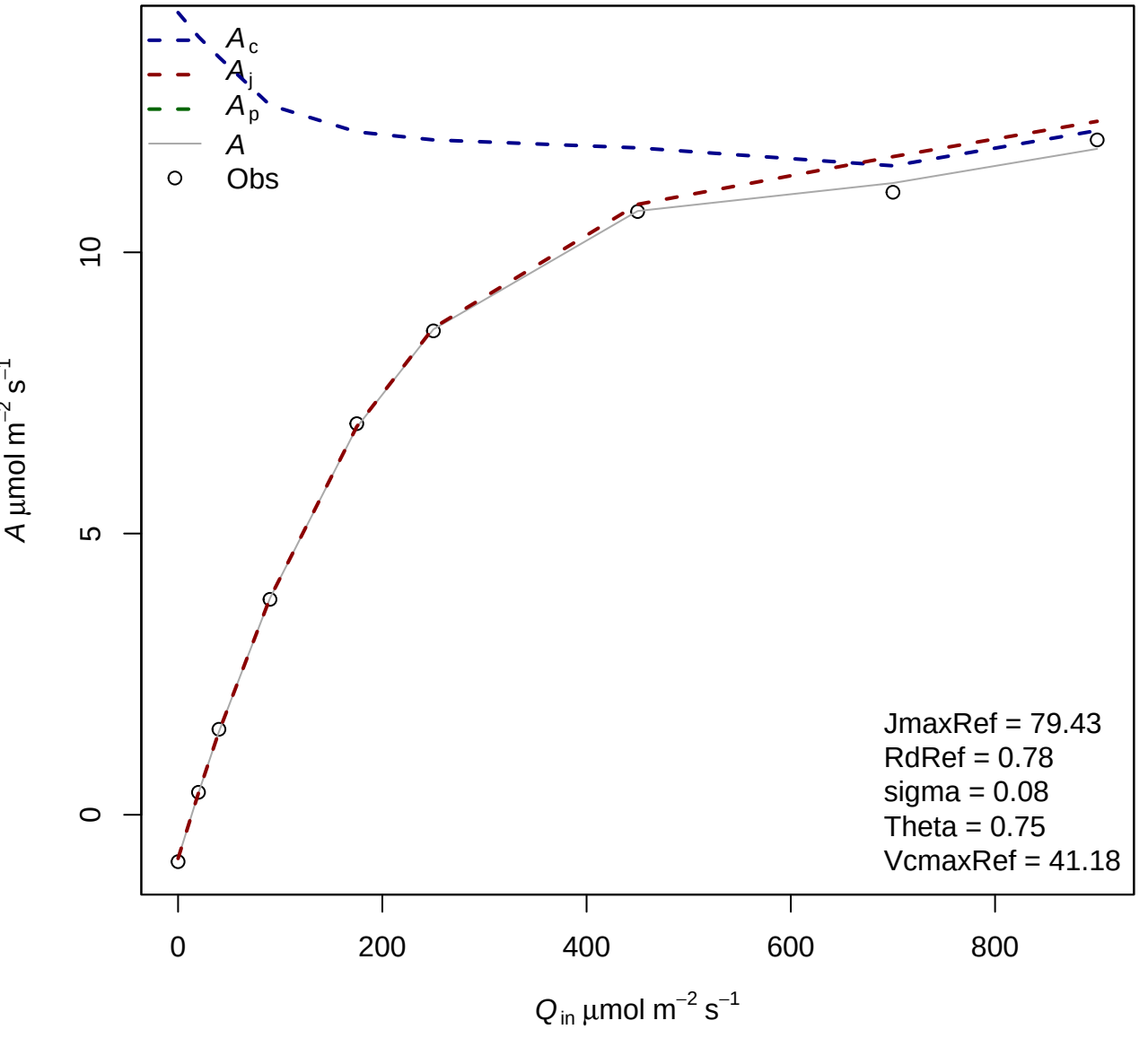

A_c
A_j
A_p
A
Obs
JmaxRef = 56.35
RdRef = 2.13
sigma = 0.45
Theta = 0.8
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

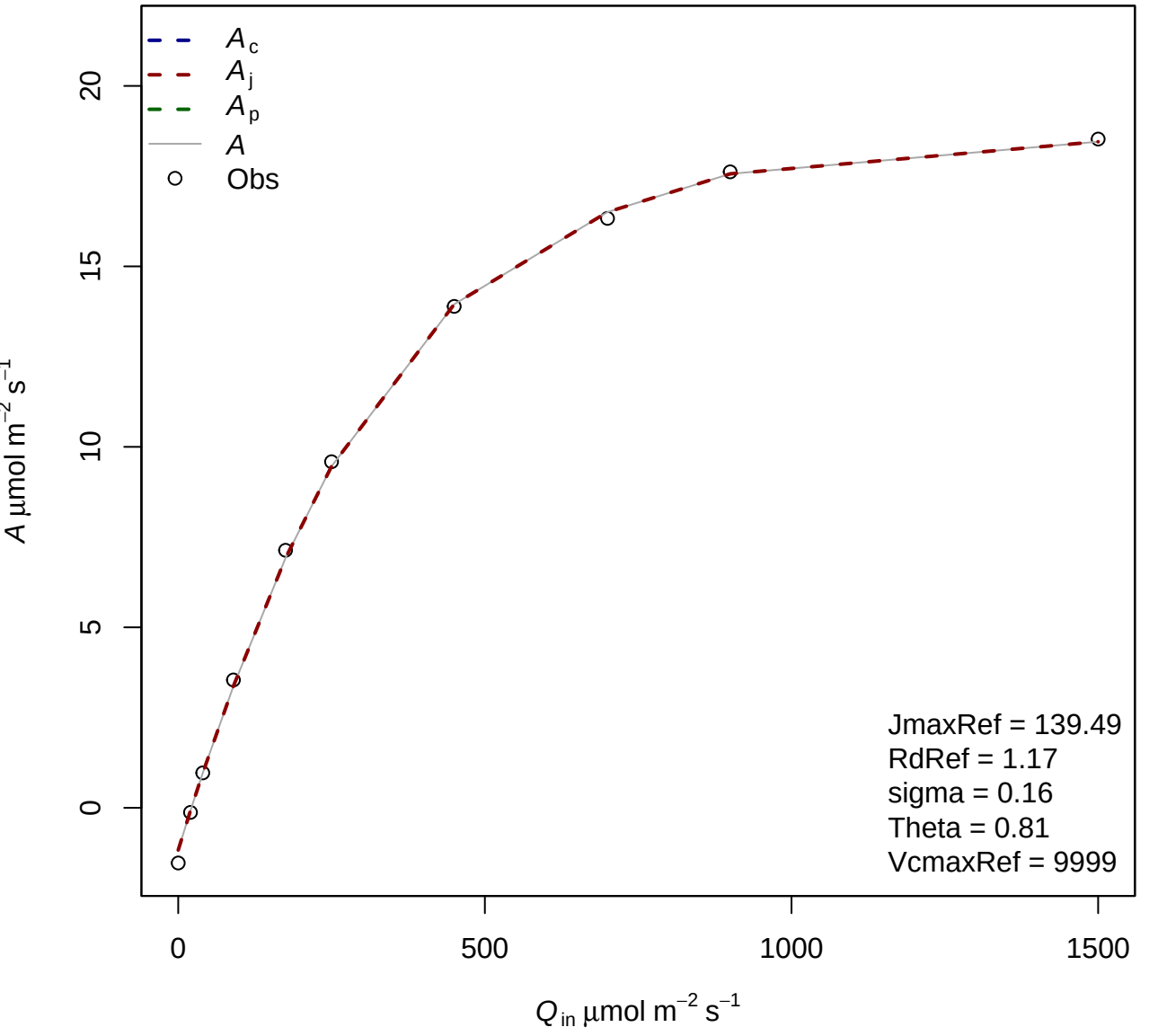

A_c
A_j
A_p
A
Obs
JmaxRef = 53.05
RdRef = 1.99
sigma = 0.55
Theta = 0.95
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

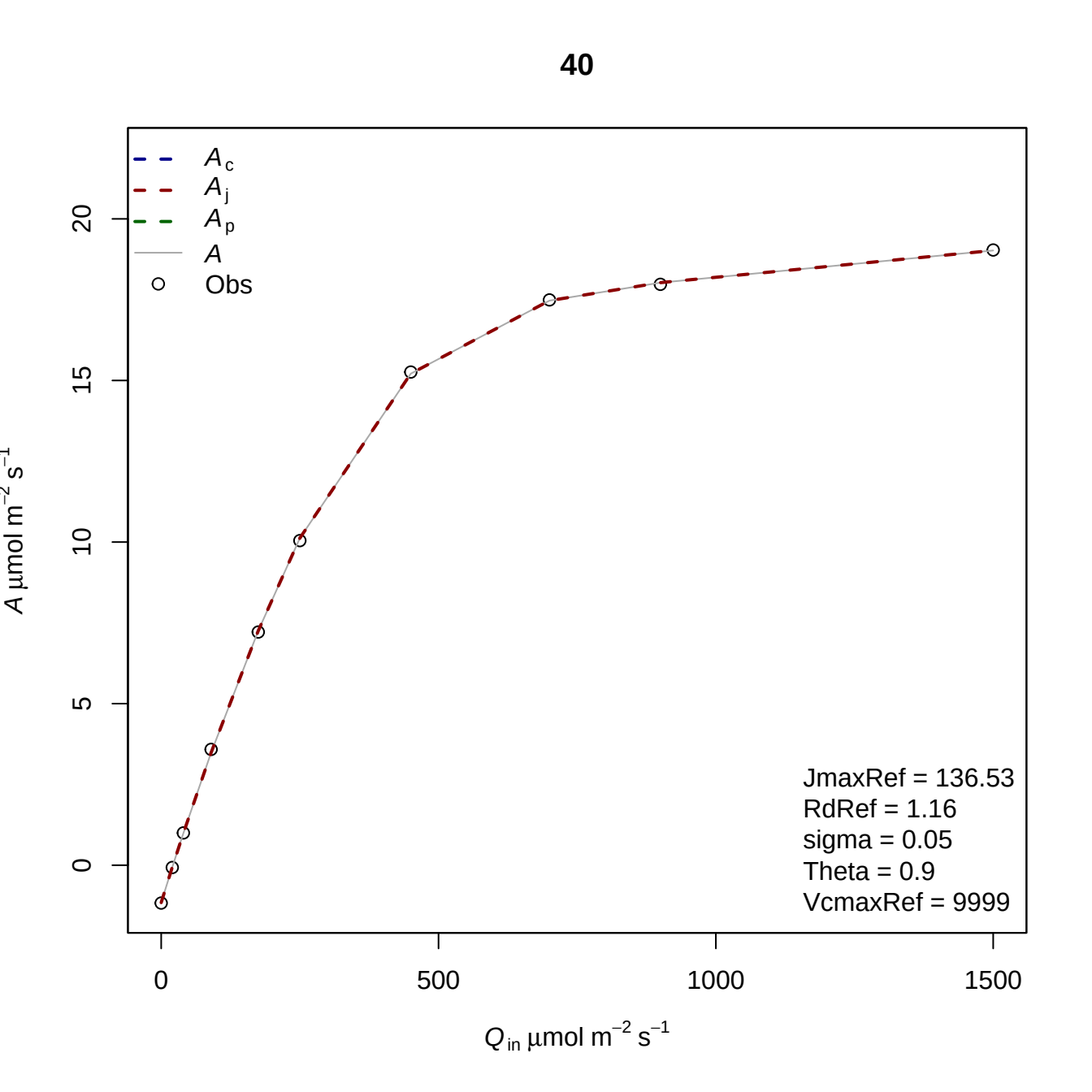

35
A_c
A_j
A_p
A
Obs
JmaxRef = 93.89
RdRef = 0.77
sigma = 0.16
Theta = 0.05
VcmaxRef = 38.64
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

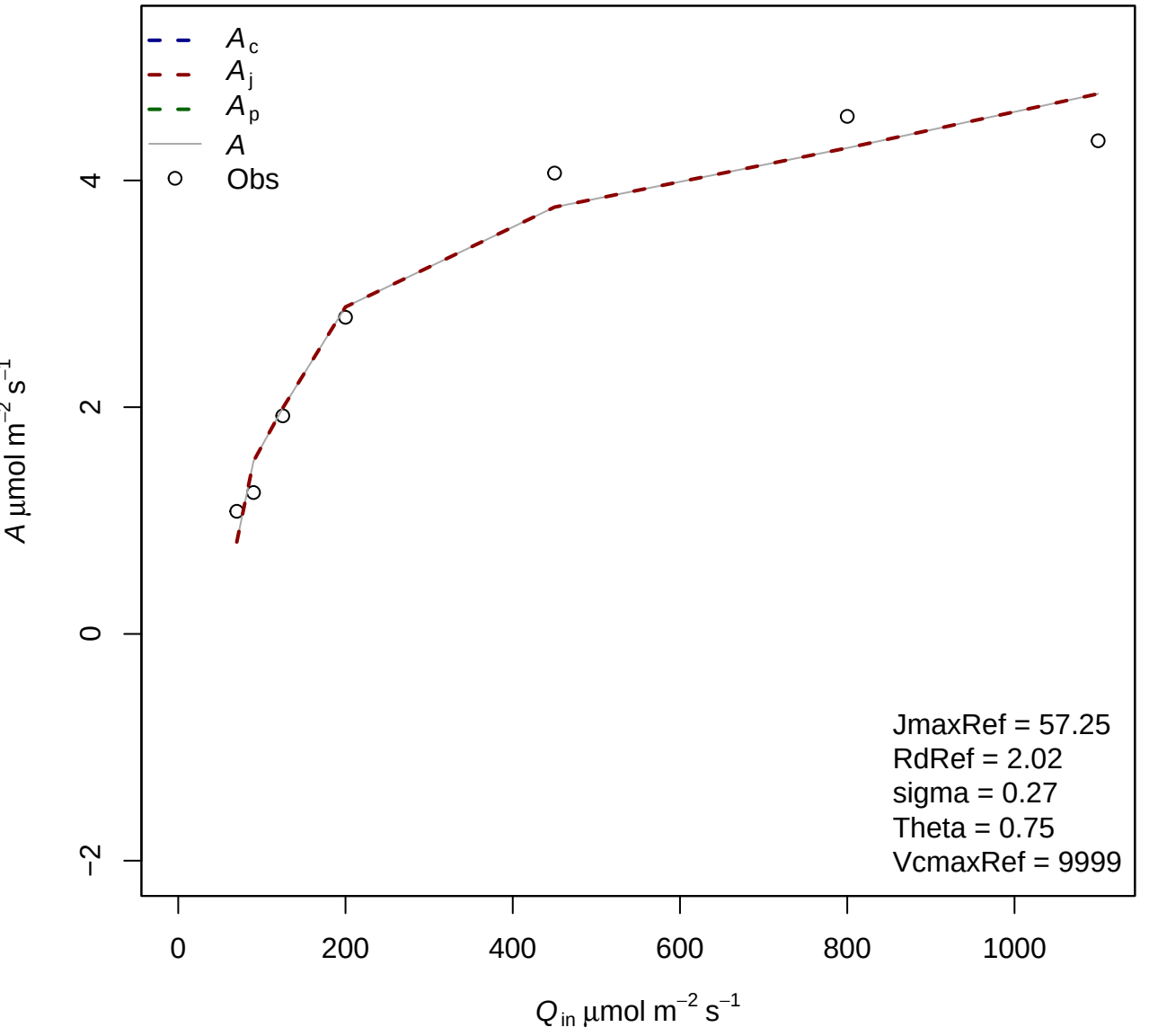

A_c
A_j
A_p
A
Obs
JmaxRef = 53.13
RdRef = 0.85
sigma = 0.05
Theta = 0.56
VcmaxRef = 25.92
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

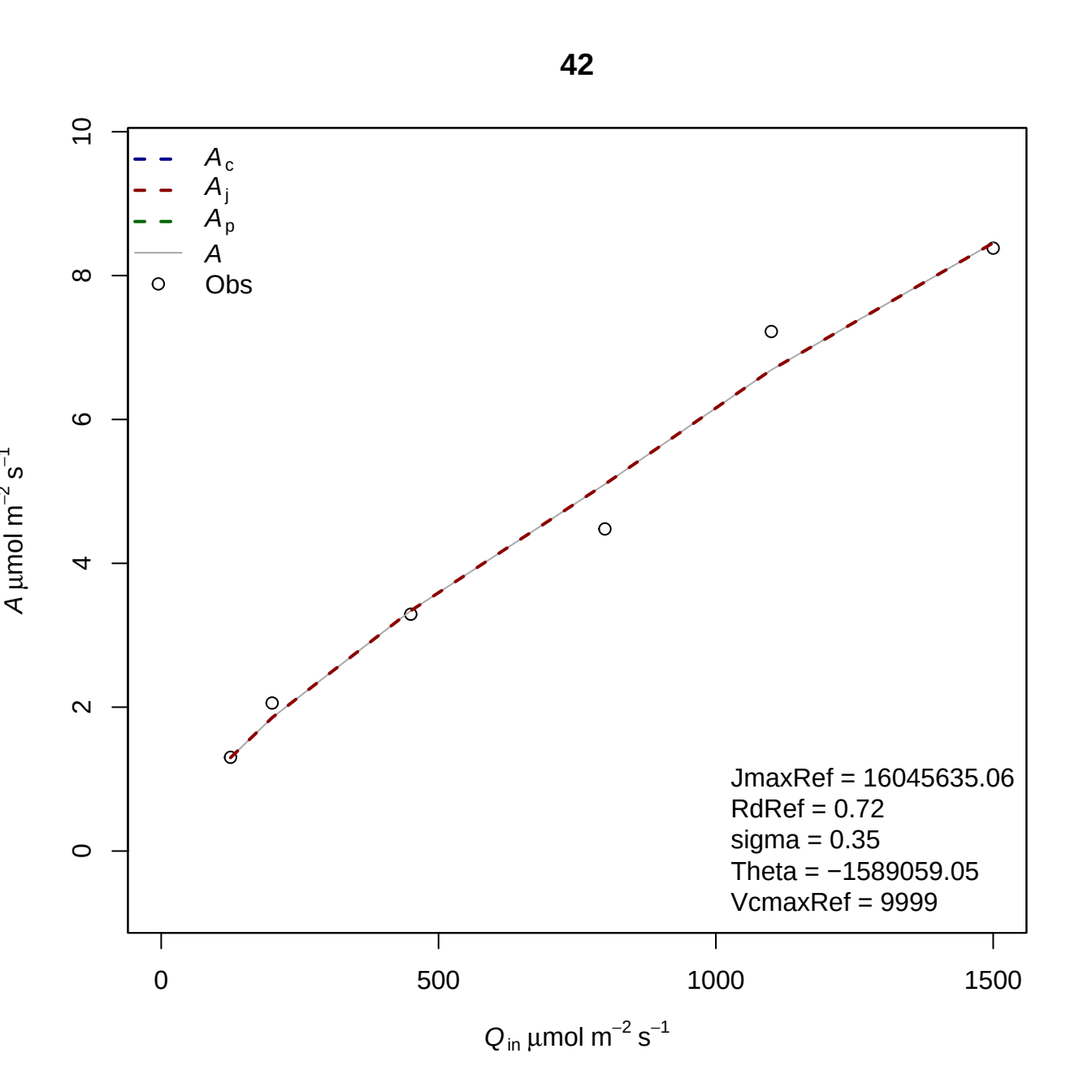

A_c
A_j
A_p
A
Obs
JmaxRef = 61.77
RdRef = 0.86
sigma = 0.12
Theta = 0.47
VcmaxRef = 24.71
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

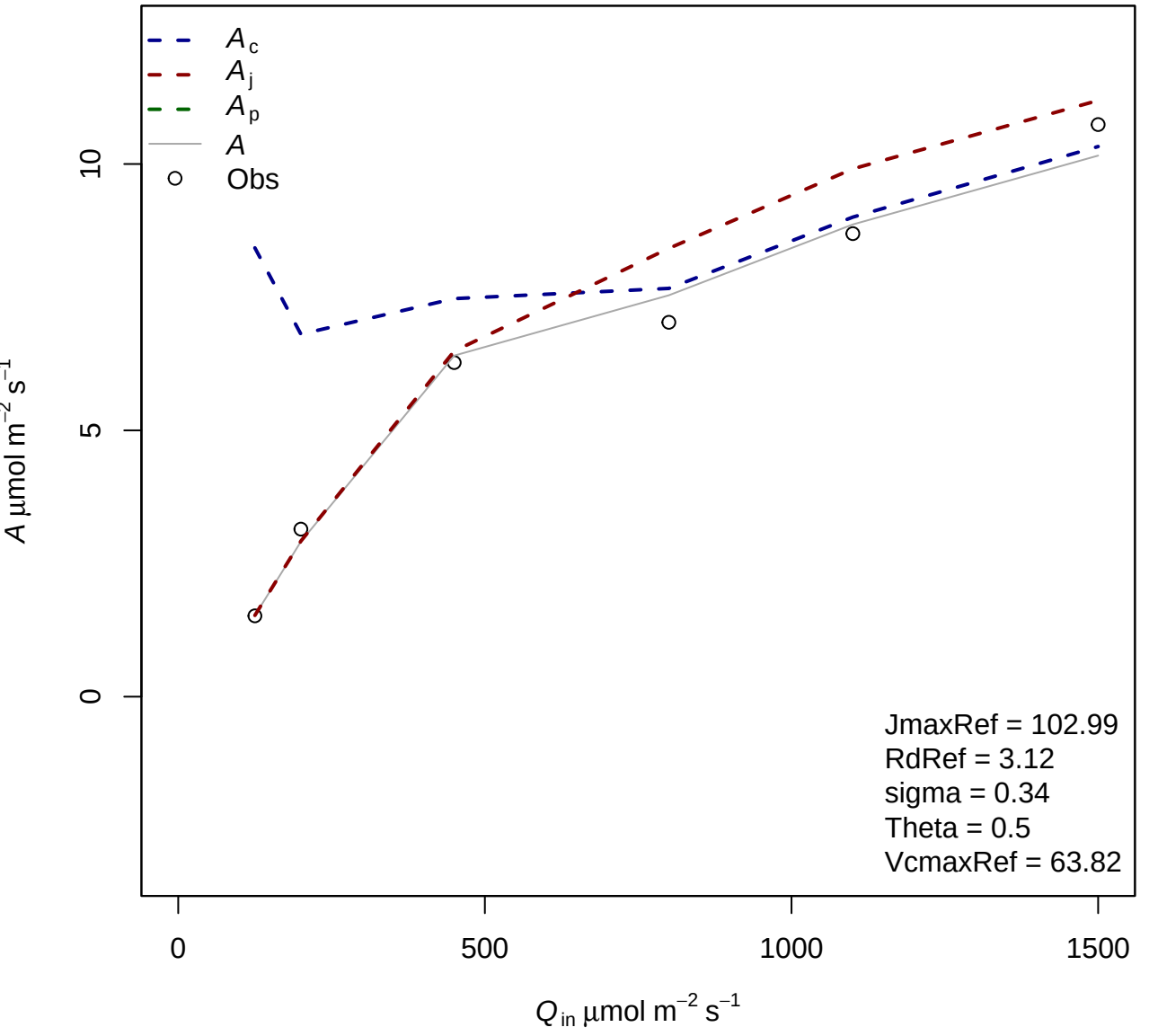

A_c
A_j
A_p
A
Obs
JmaxRef = 79.43
RdRef = 0.78
sigma = 0.08
Theta = 0.75
VcmaxRef = 41.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800

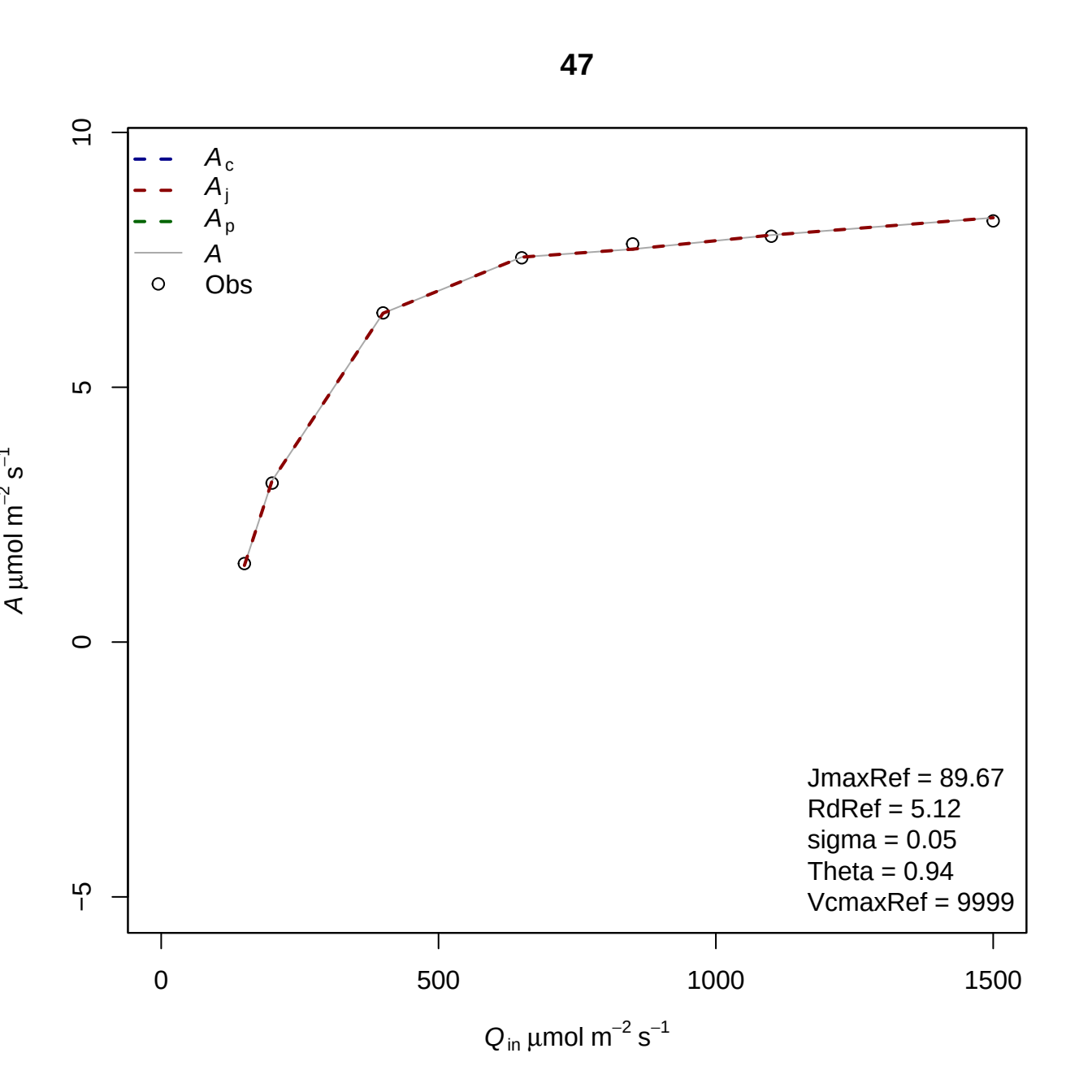

A_c
A_j
A_p
A
Obs
JmaxRef = 139.49
RdRef = 1.17
sigma = 0.16
Theta = 0.81
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

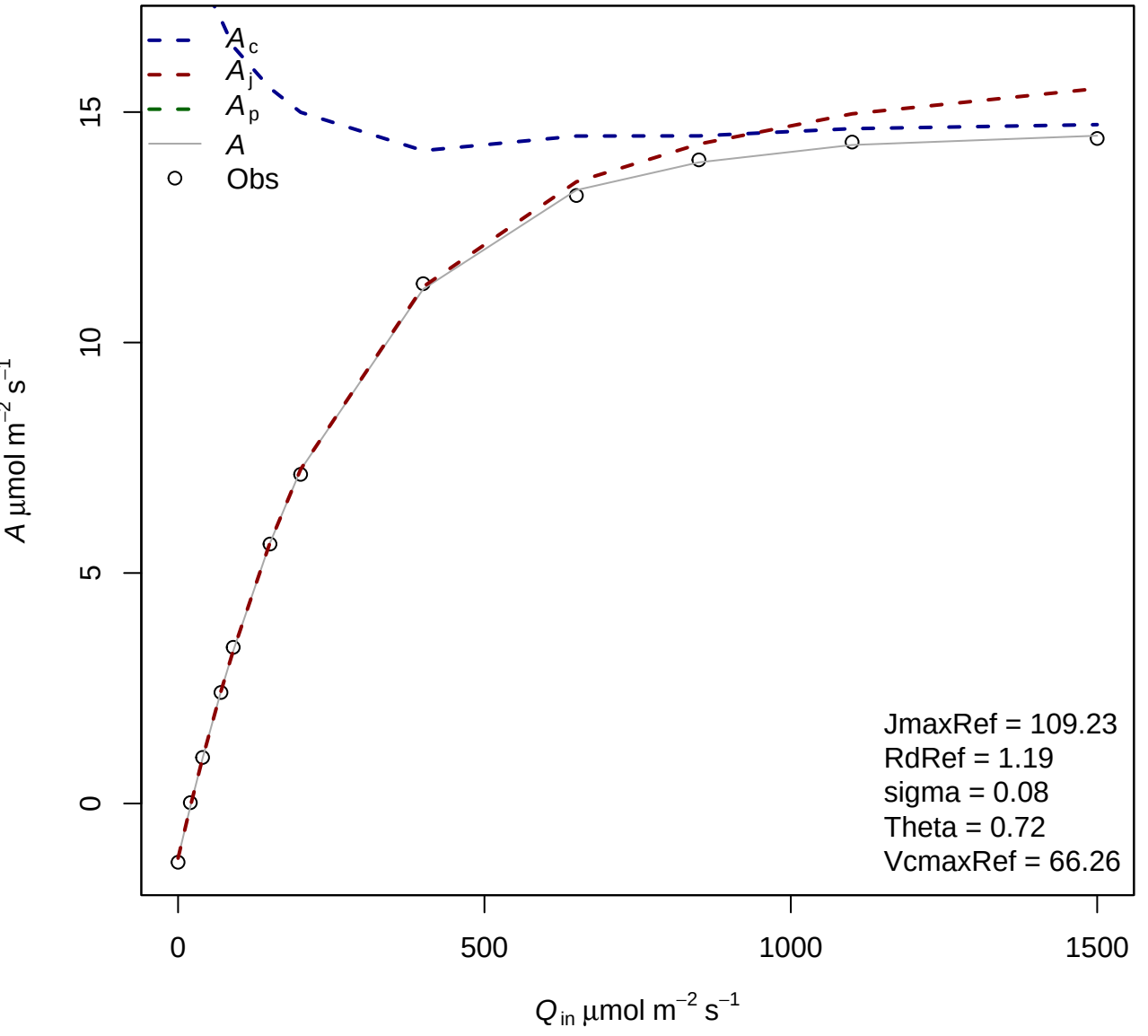

40
A_c
A_j
A_p
A
Obs
JmaxRef = 136.53
RdRef = 1.16
sigma = 0.05
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

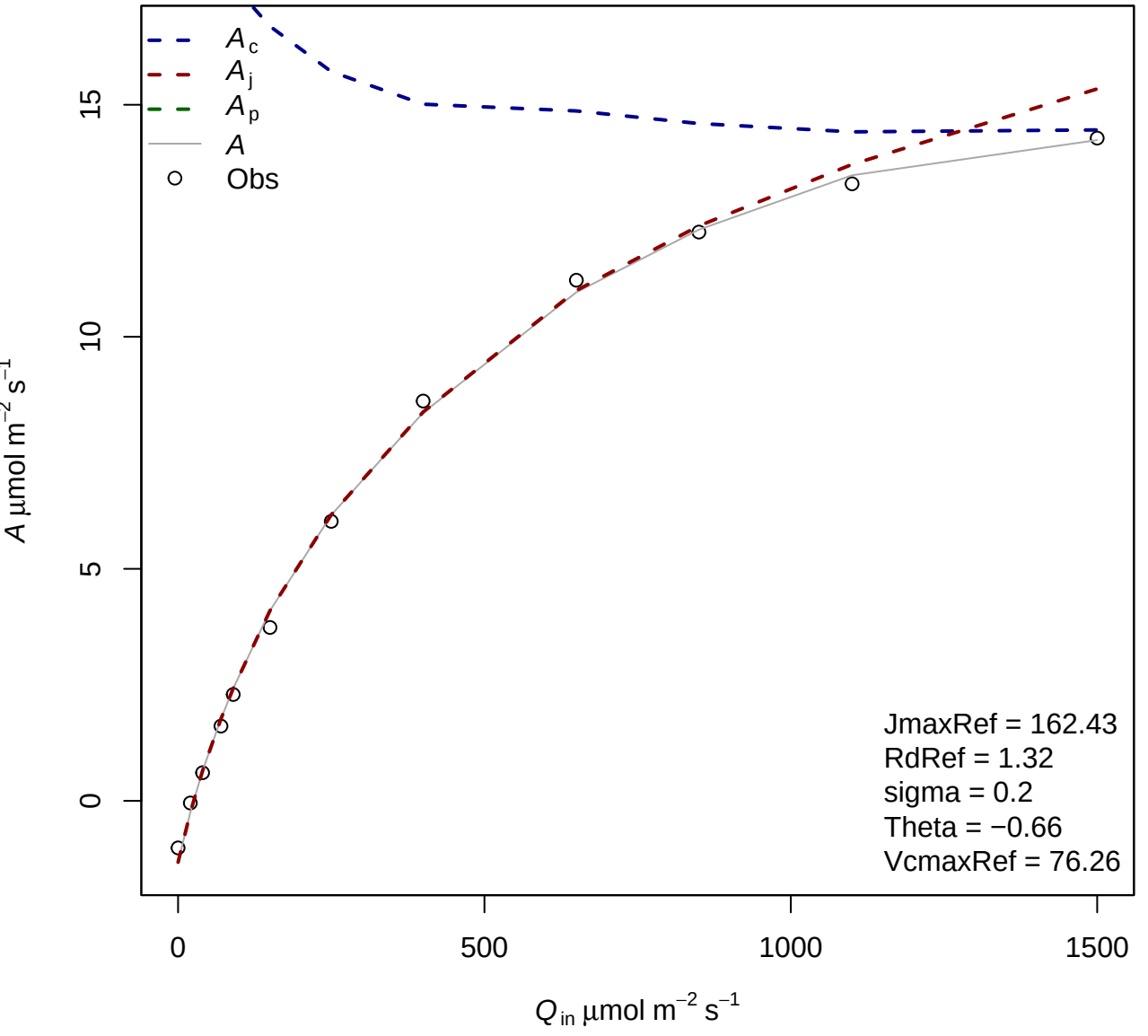

A_c
A_j
A_p
A
Obs
JmaxRef = 57.25
RdRef = 2.02
sigma = 0.27
Theta = 0.75
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

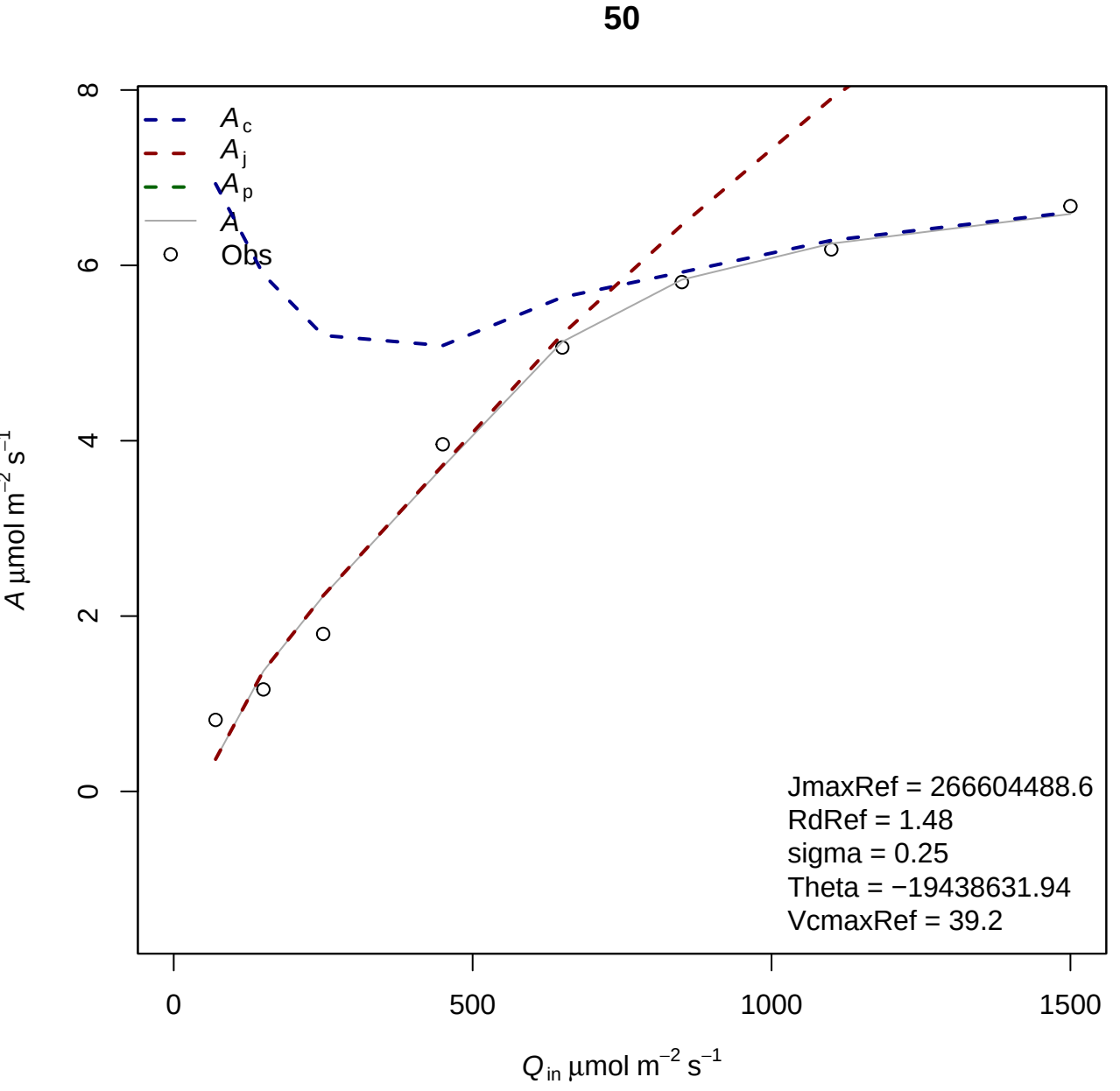

42
A_c
A_j
A_p
A
Obs
JmaxRef = 16045635.06
RdRef = 0.72
sigma = 0.35
Theta = −1589059.05
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

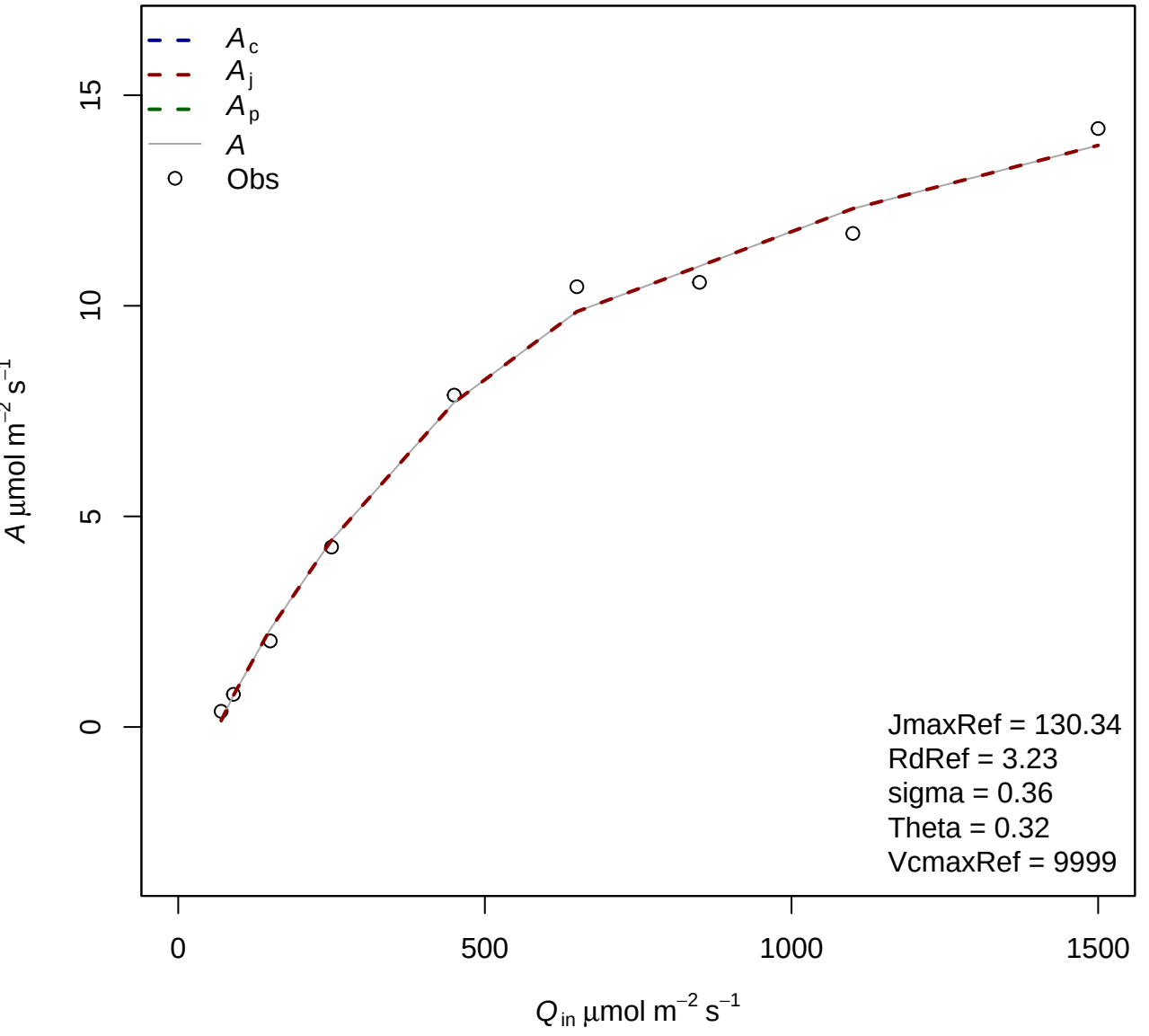

A_c
A_j
A_p
A
Obs
JmaxRef = 102.99
RdRef = 3.12
sigma = 0.34
Theta = 0.5
VcmaxRef = 63.82
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

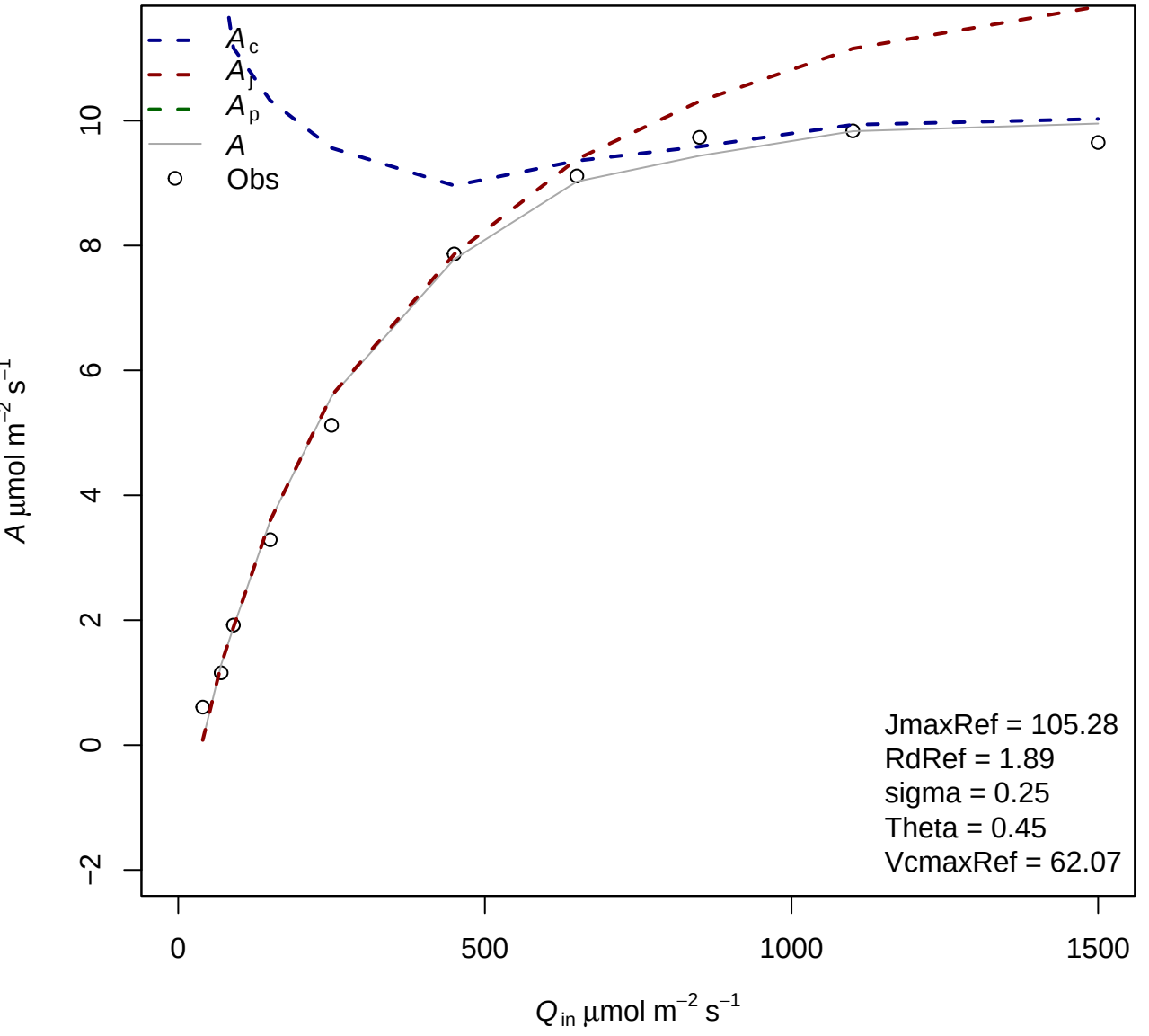

47
A_c
A_j
A_p
A
Obs
10
5
0
−5
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 89.67
RdRef = 5.12
sigma = 0.05
Theta = 0.94
VcmaxRef = 9999

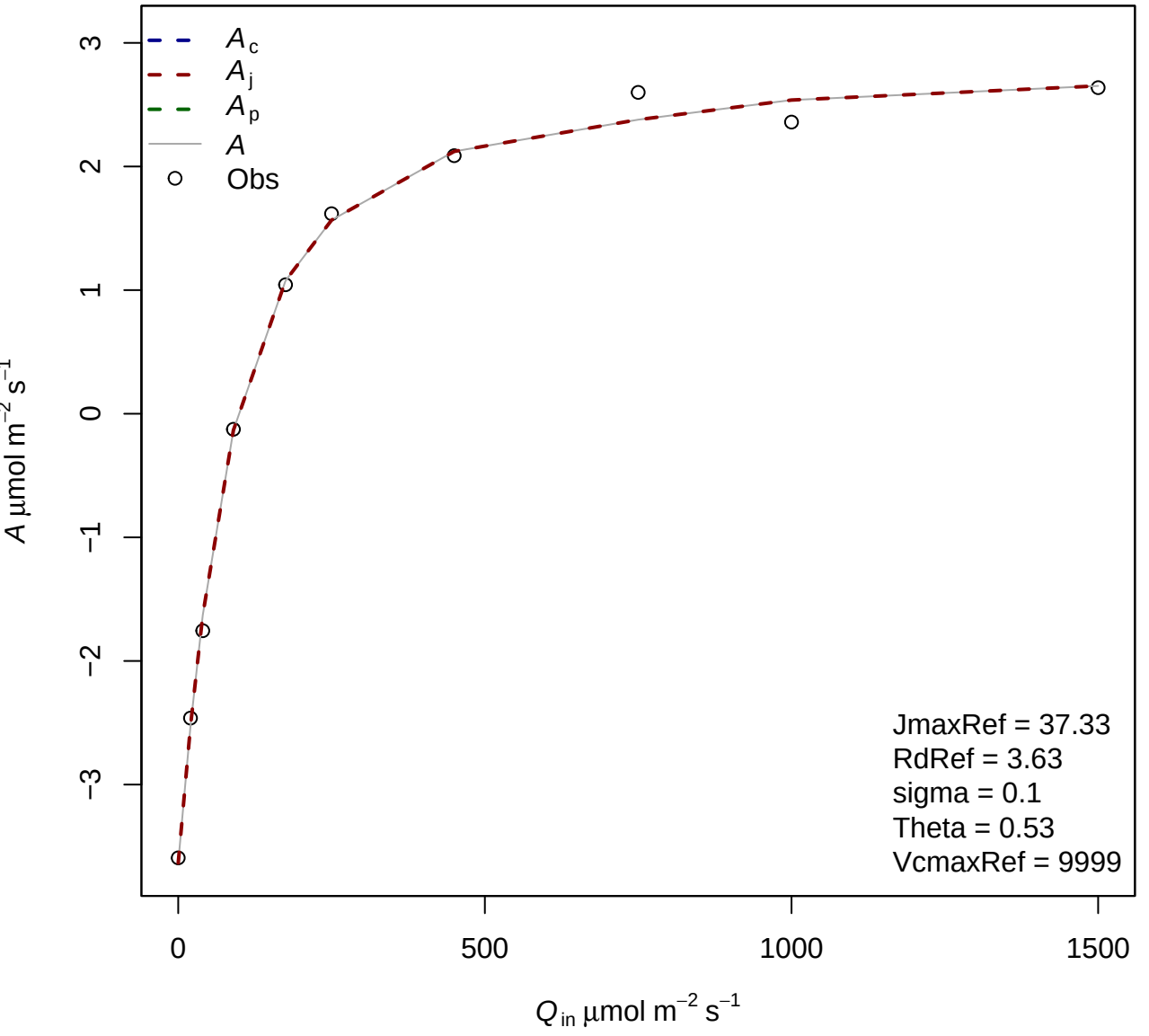

A_c
A_j
A_p
A
Obs
JmaxRef = 109.23
RdRef = 1.19
sigma = 0.08
Theta = 0.72
VcmaxRef = 66.26
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

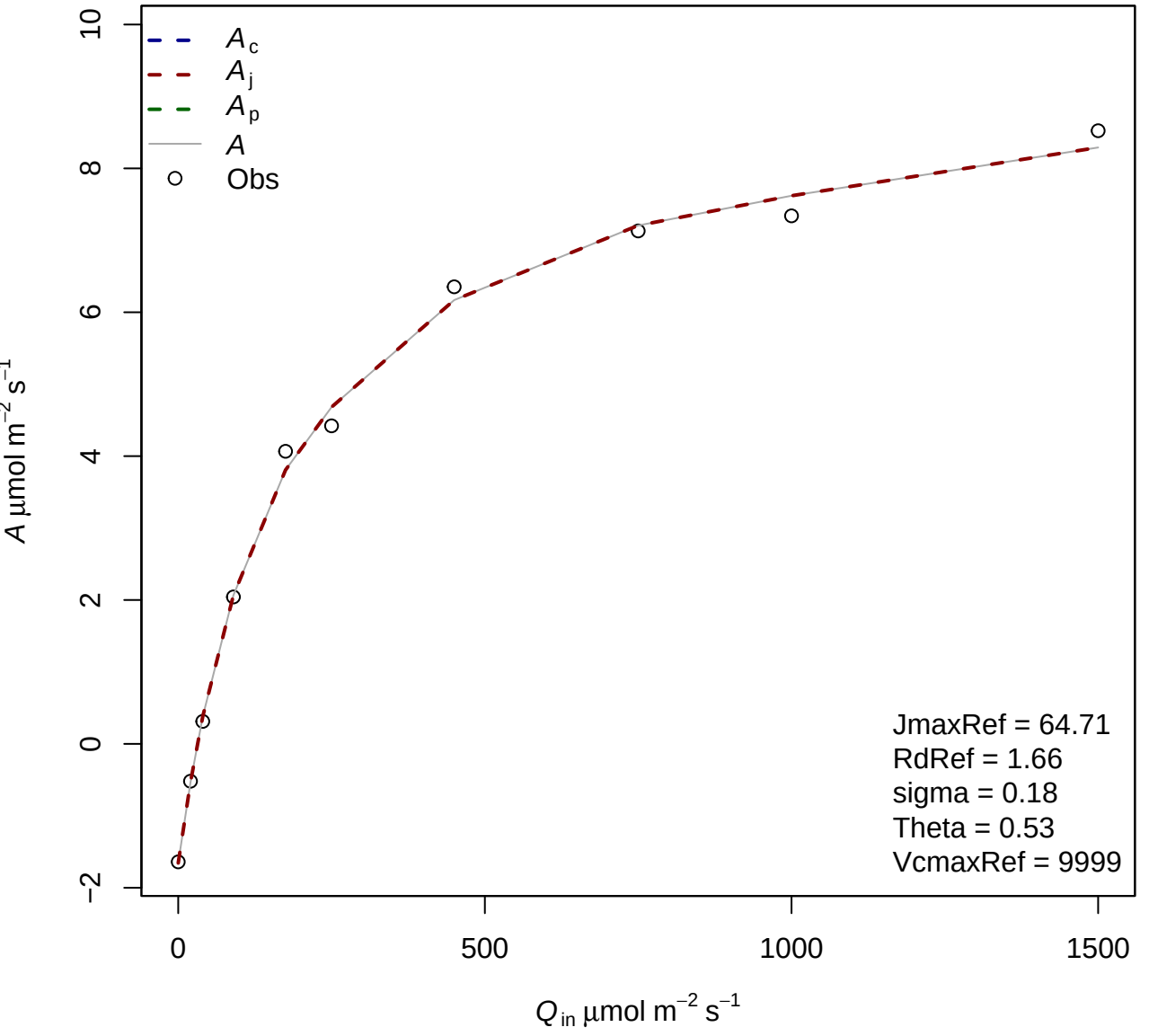

A_c
A_j
A_p
A
Obs
JmaxRef = 162.43
RdRef = 1.32
sigma = 0.2
Theta = −0.66
VcmaxRef = 76.26
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

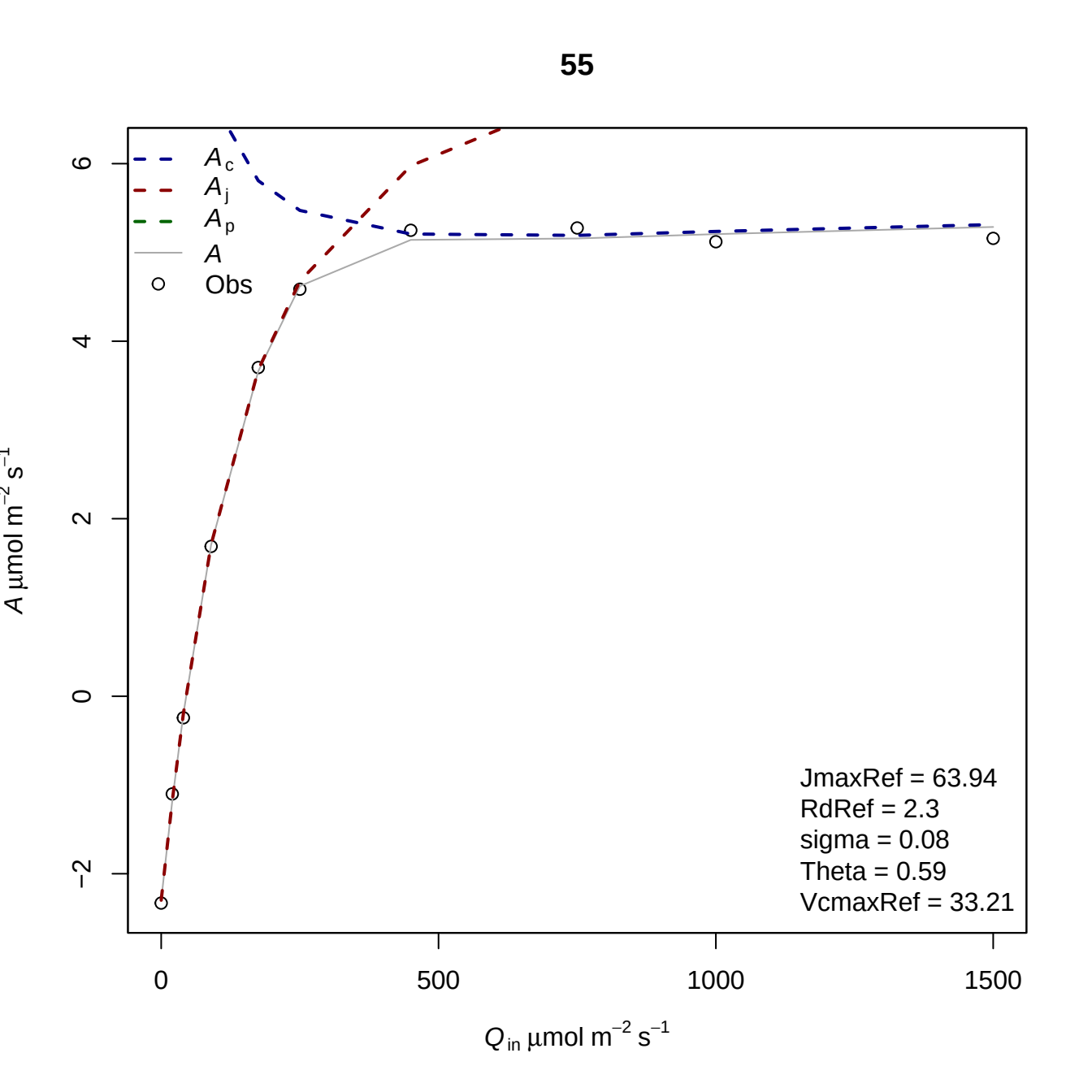

50
A_c
A_j
A_p
A
Obs
JmaxRef = 266604488.6
RdRef = 1.48
sigma = 0.25
Theta = −19438631.94
VcmaxRef = 39.2
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

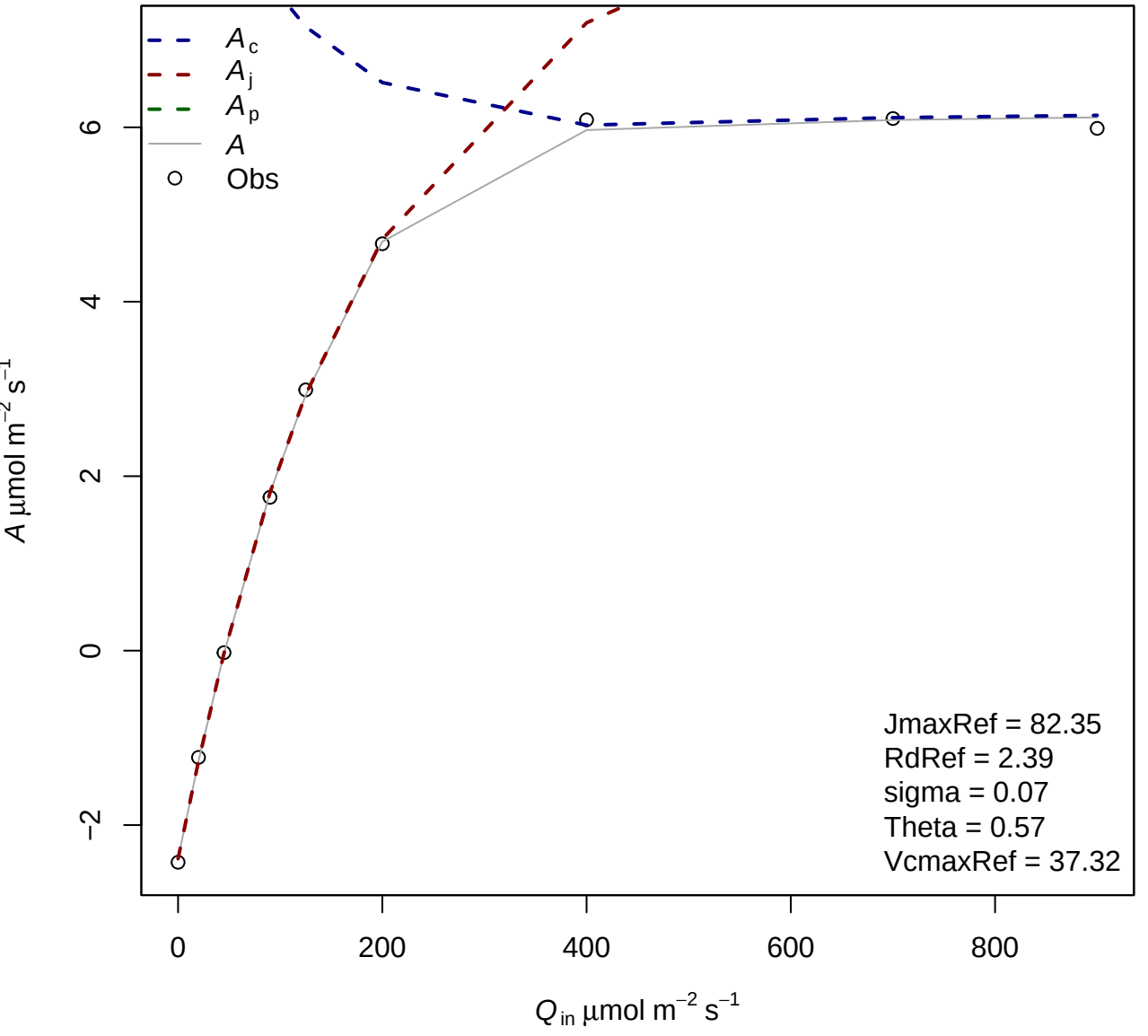

A_c
A_j
A_p
A
Obs
JmaxRef = 130.34
RdRef = 3.23
sigma = 0.36
Theta = 0.32
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

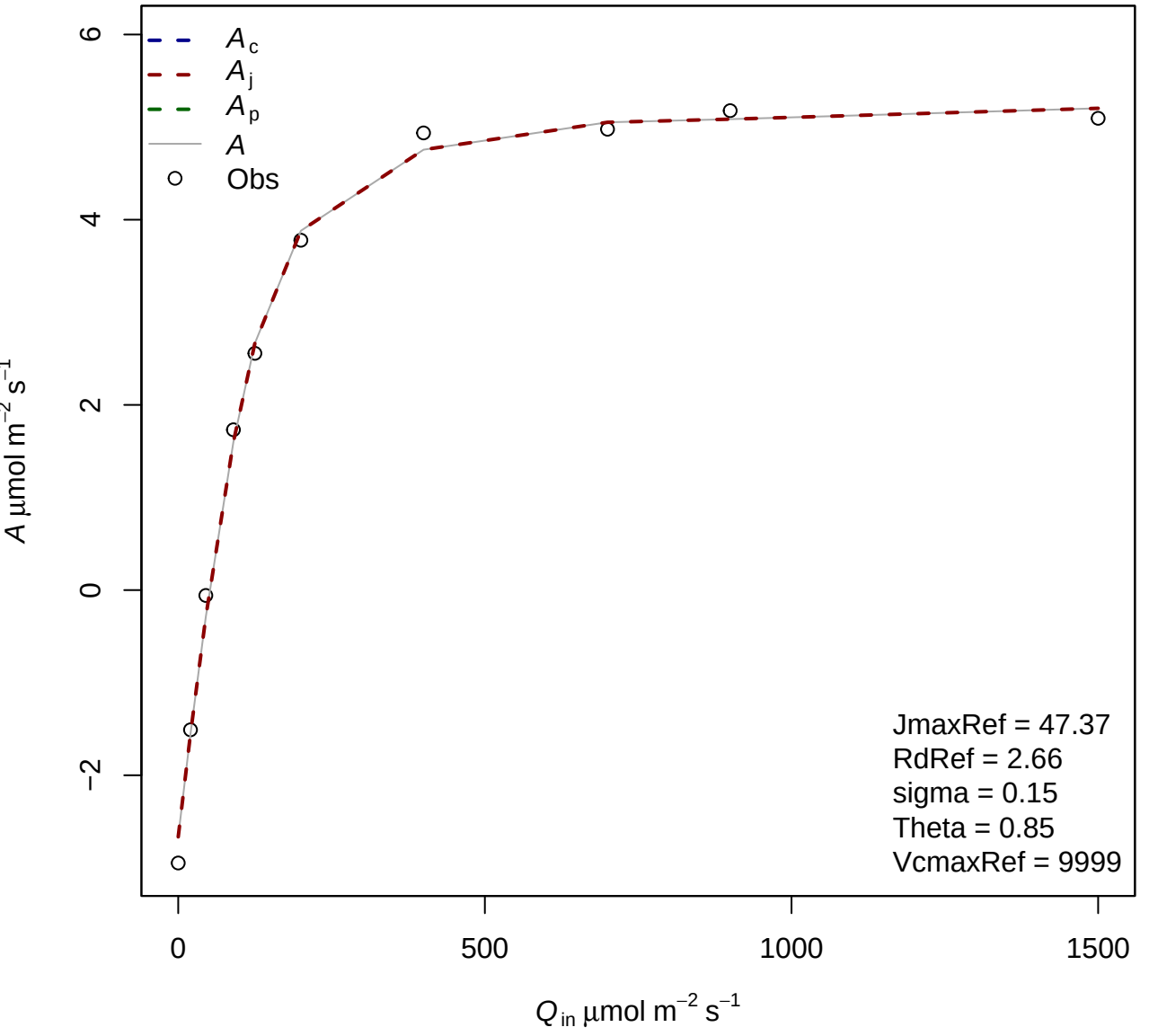

A_c
A_j
A_p
A
Obs
JmaxRef = 105.28
RdRef = 1.89
sigma = 0.25
Theta = 0.45
VcmaxRef = 62.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

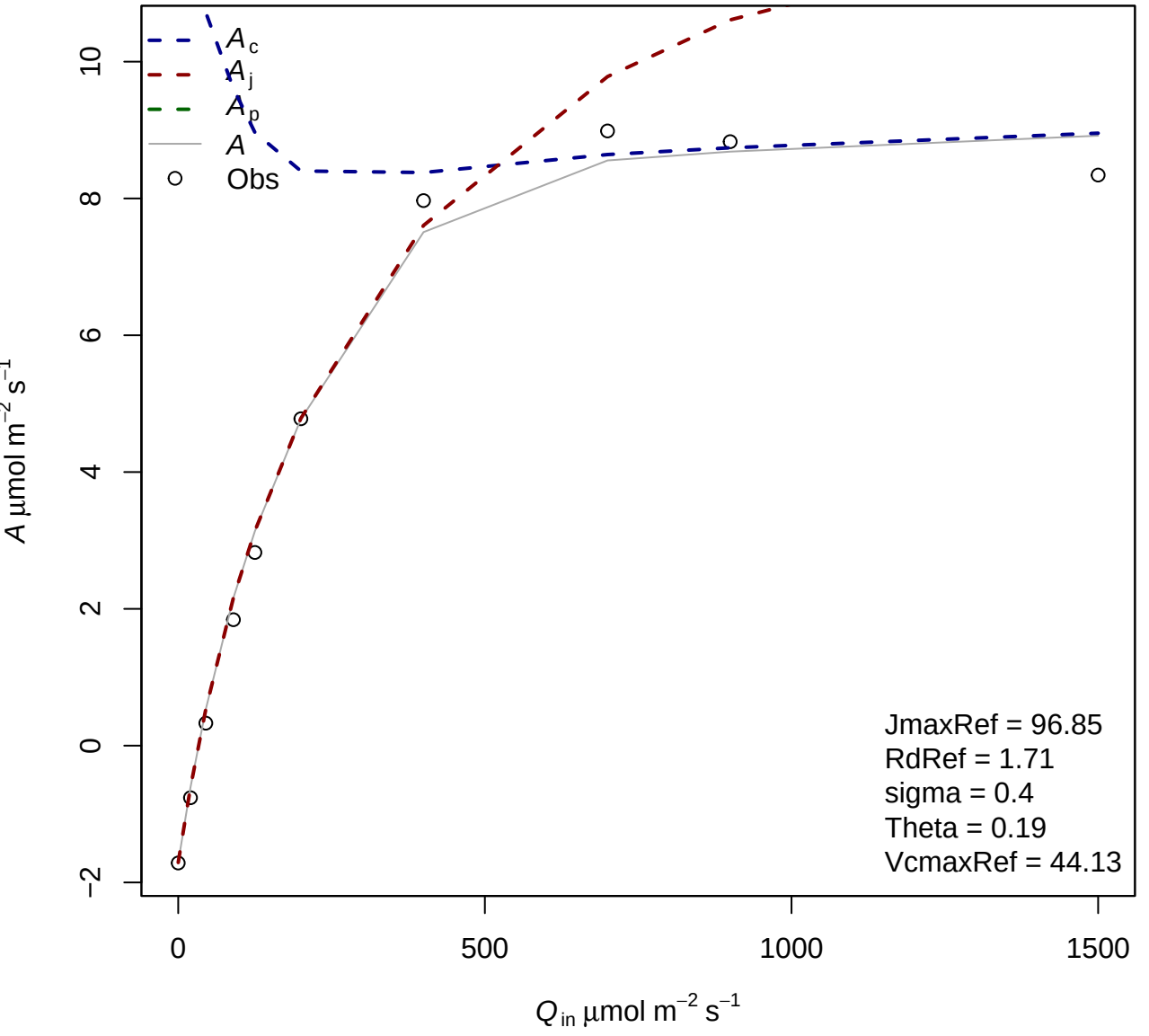

A_c
A_j
A_p
A
Obs
JmaxRef = 37.33
RdRef = 3.63
sigma = 0.1
Theta = 0.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
−3
0
500
1000
1500

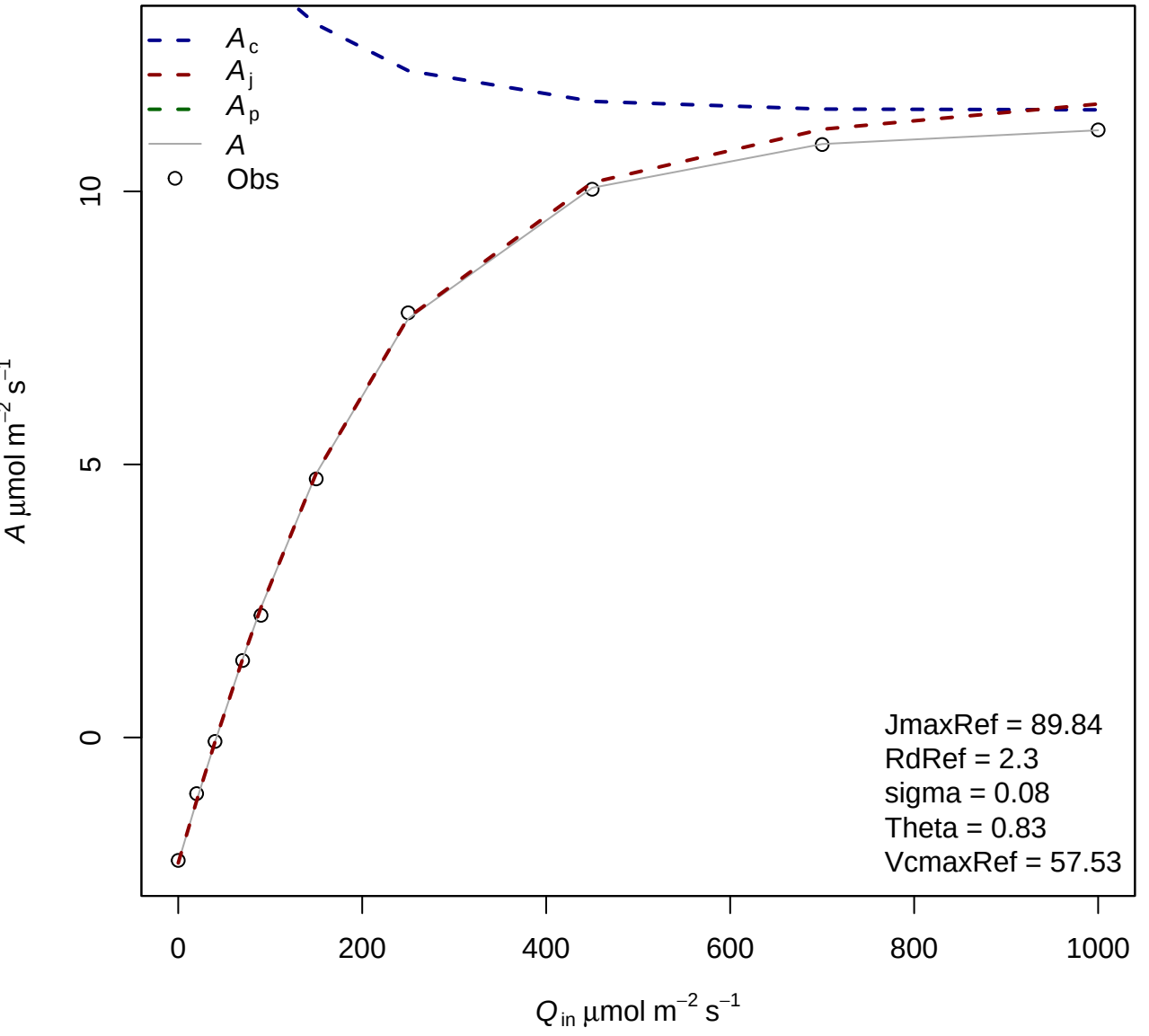

A_c
A_j
A_p
A
Obs
JmaxRef = 64.71
RdRef = 1.66
sigma = 0.18
Theta = 0.53
VcmaxRef = 9999
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

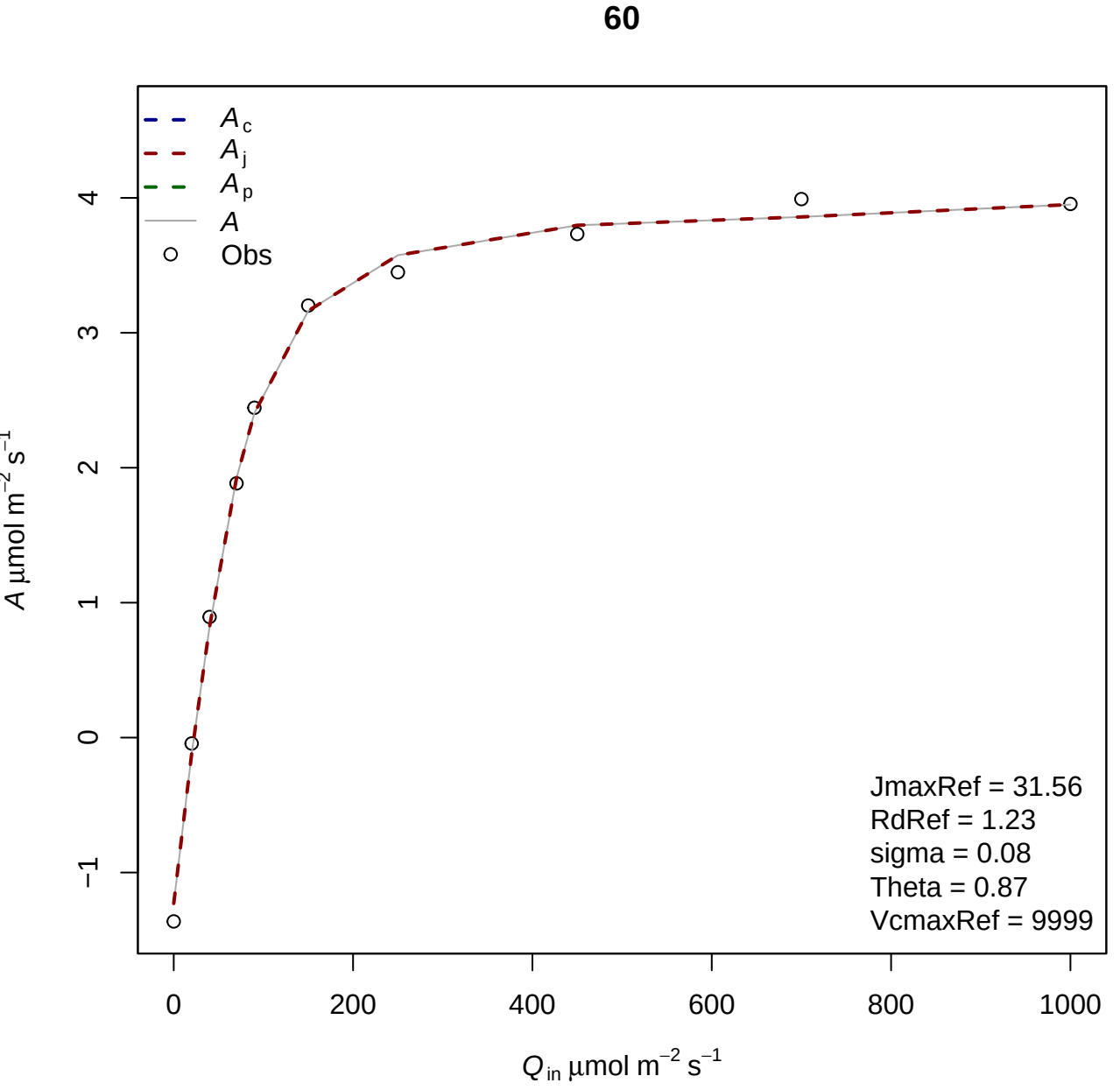

55
A_c
A_j
A_p
A
Obs
JmaxRef = 63.94
RdRef = 2.3
sigma = 0.08
Theta = 0.59
VcmaxRef = 33.21
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

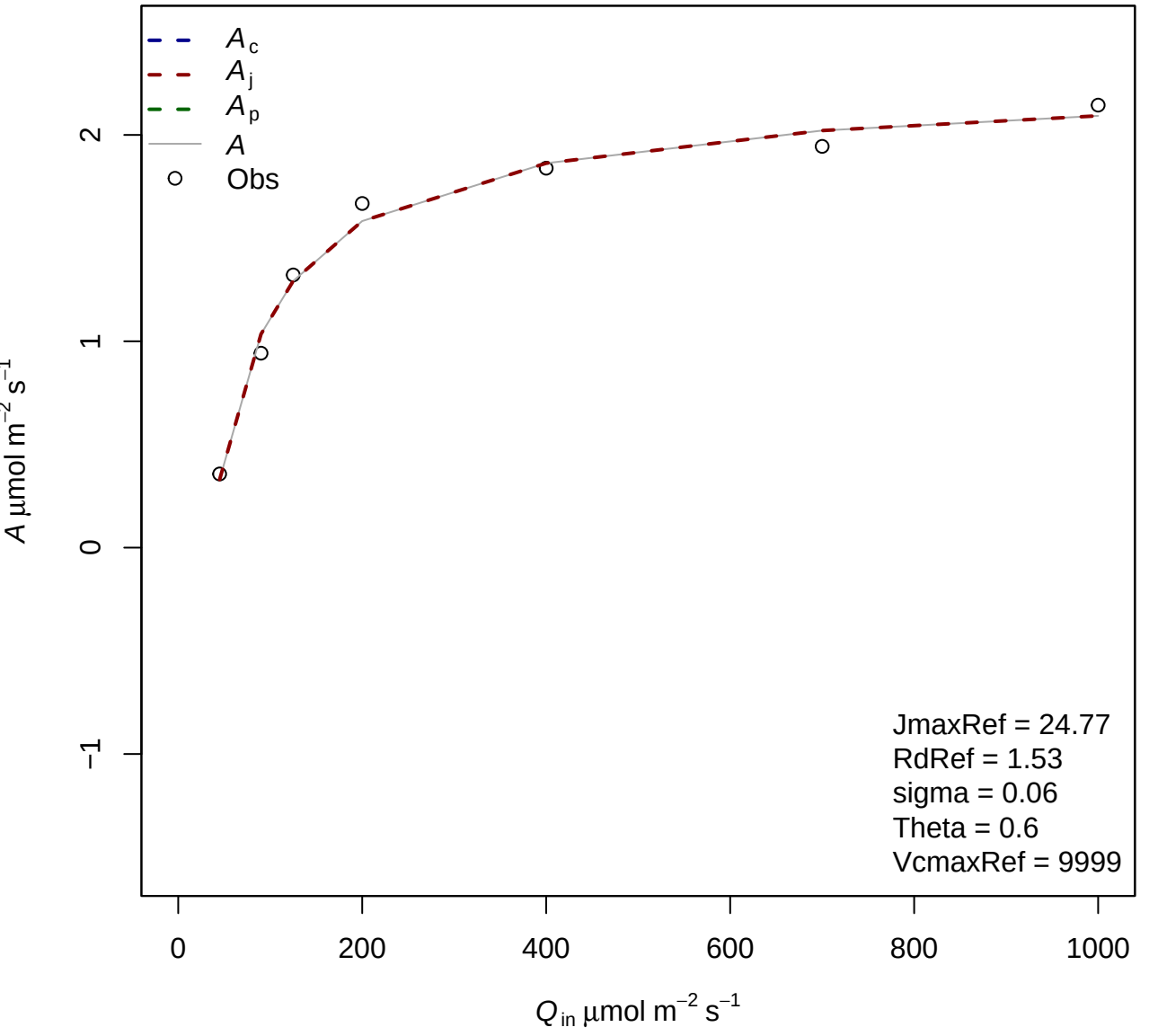

A_c
A_j
A_p
A
Obs
JmaxRef = 82.35
RdRef = 2.39
sigma = 0.07
Theta = 0.57
VcmaxRef = 37.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800

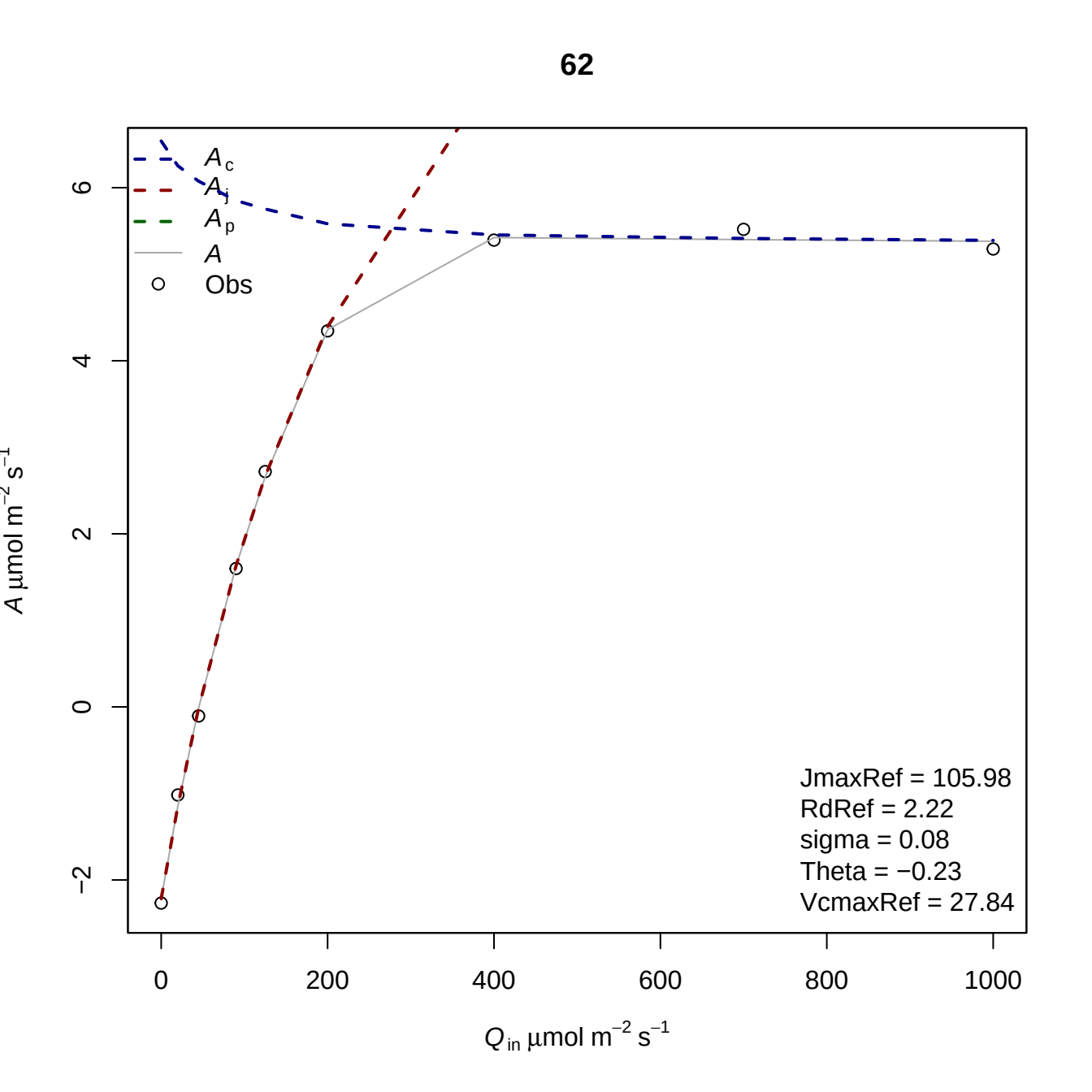

A_c
A_j
A_p
A
Obs
JmaxRef = 47.37
RdRef = 2.66
sigma = 0.15
Theta = 0.85
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

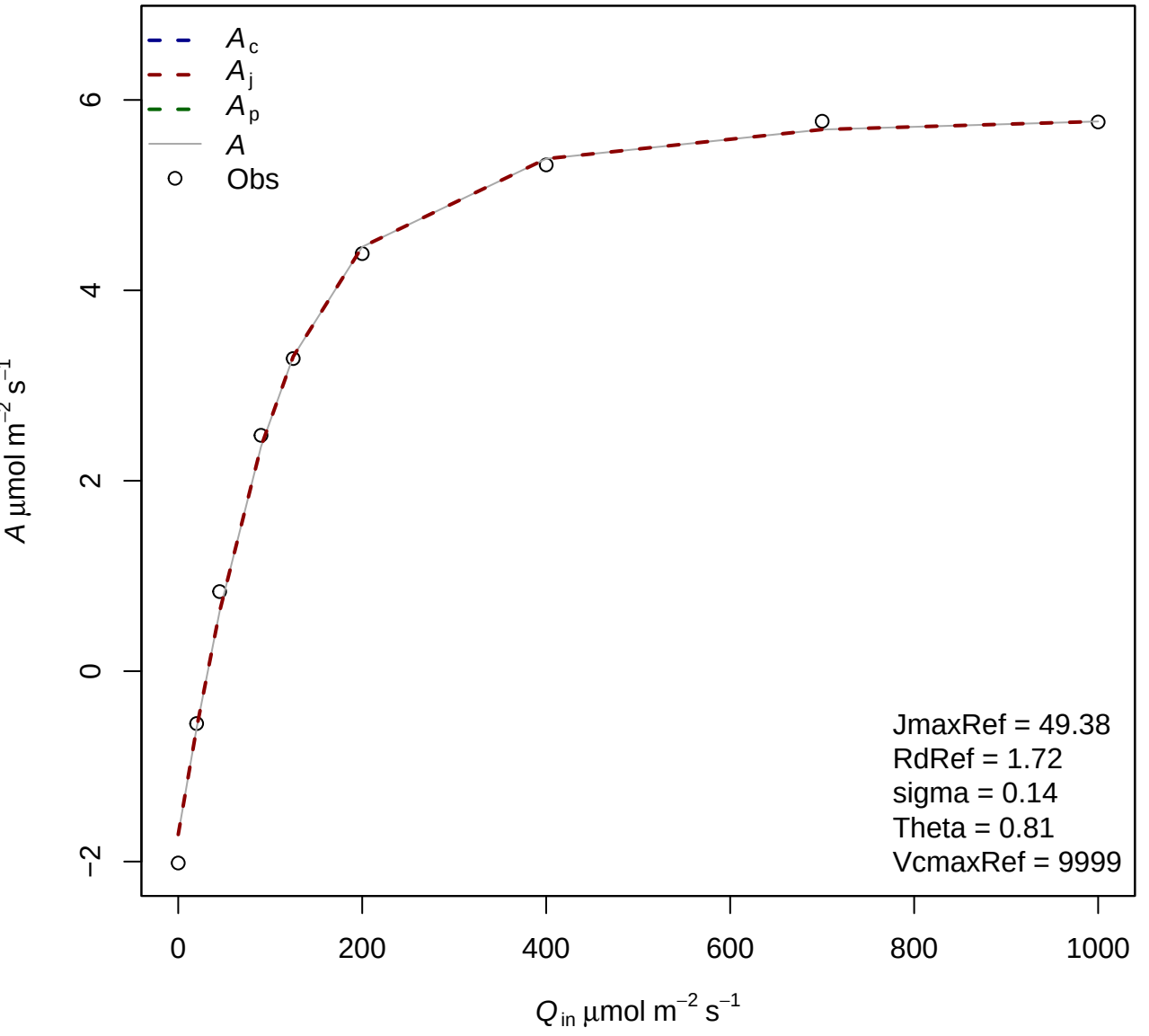

A_c
A_j
A_p
A
Obs
JmaxRef = 96.85
RdRef = 1.71
sigma = 0.4
Theta = 0.19
VcmaxRef = 44.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

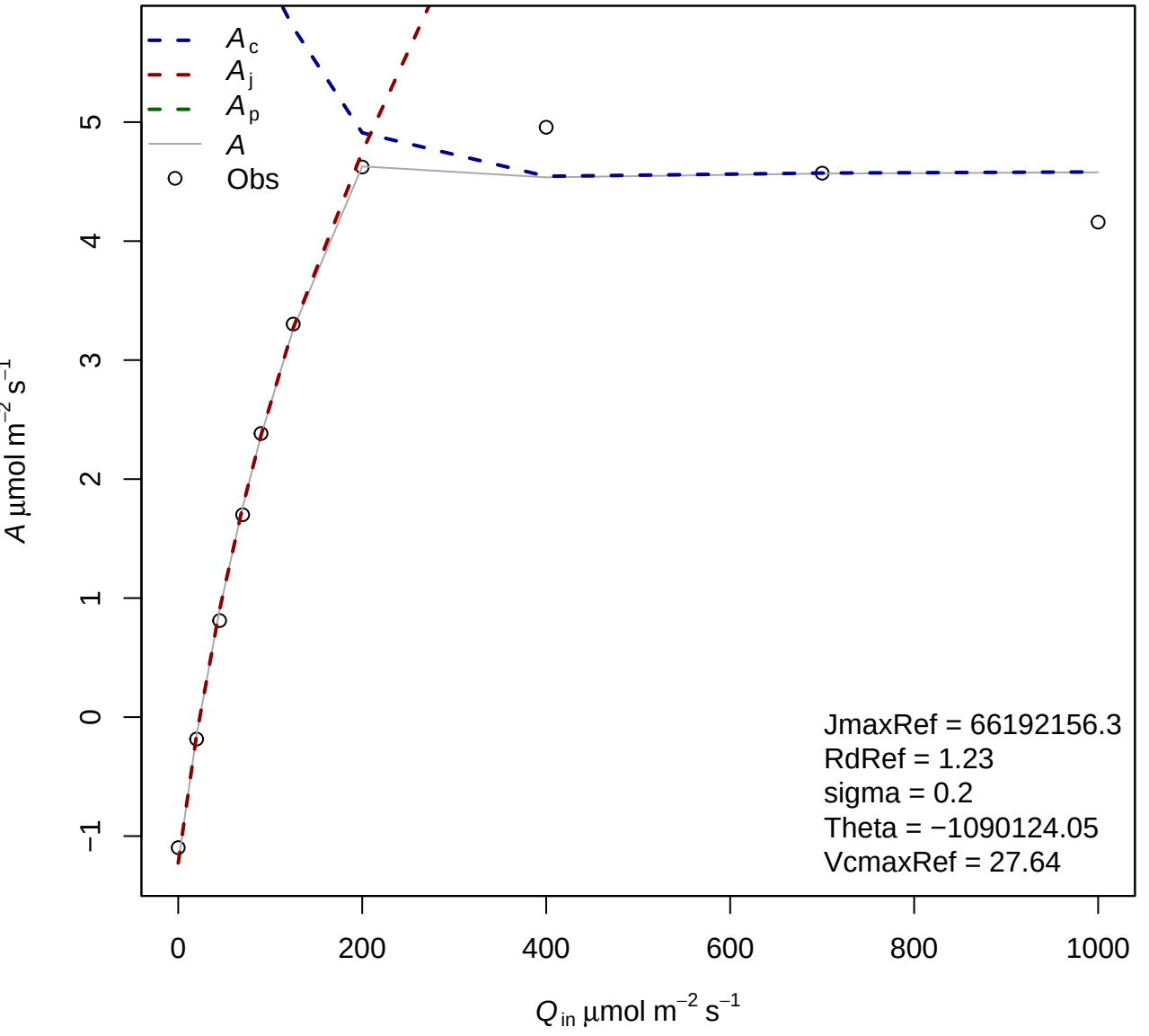

A_c
A_j
A_p
A
Obs
JmaxRef = 89.84
RdRef = 2.3
sigma = 0.08
Theta = 0.83
VcmaxRef = 57.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000

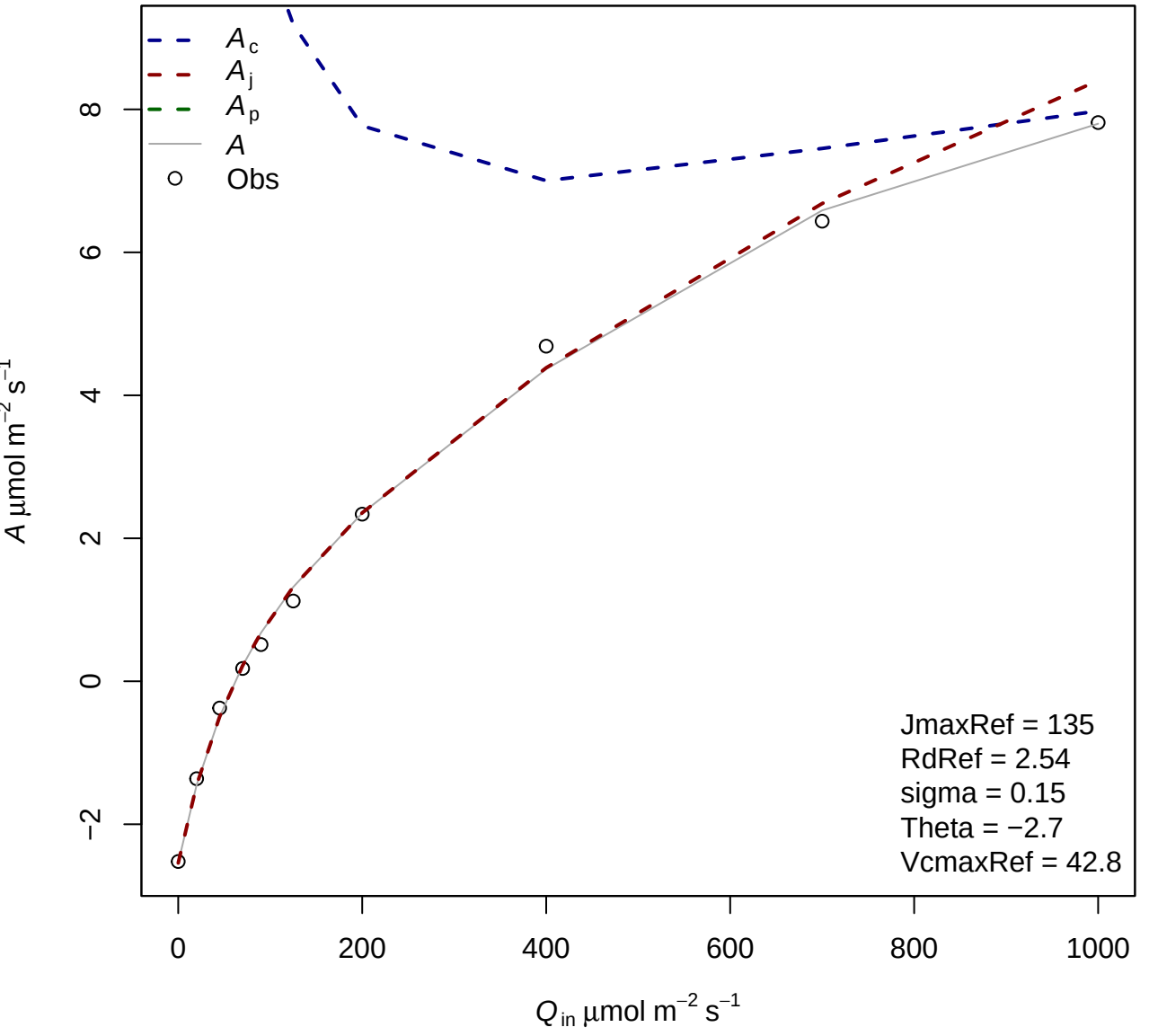

60
A_c
A_j
A_p
A
Obs
JmaxRef = 31.56
RdRef = 1.23
sigma = 0.08
Theta = 0.87
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

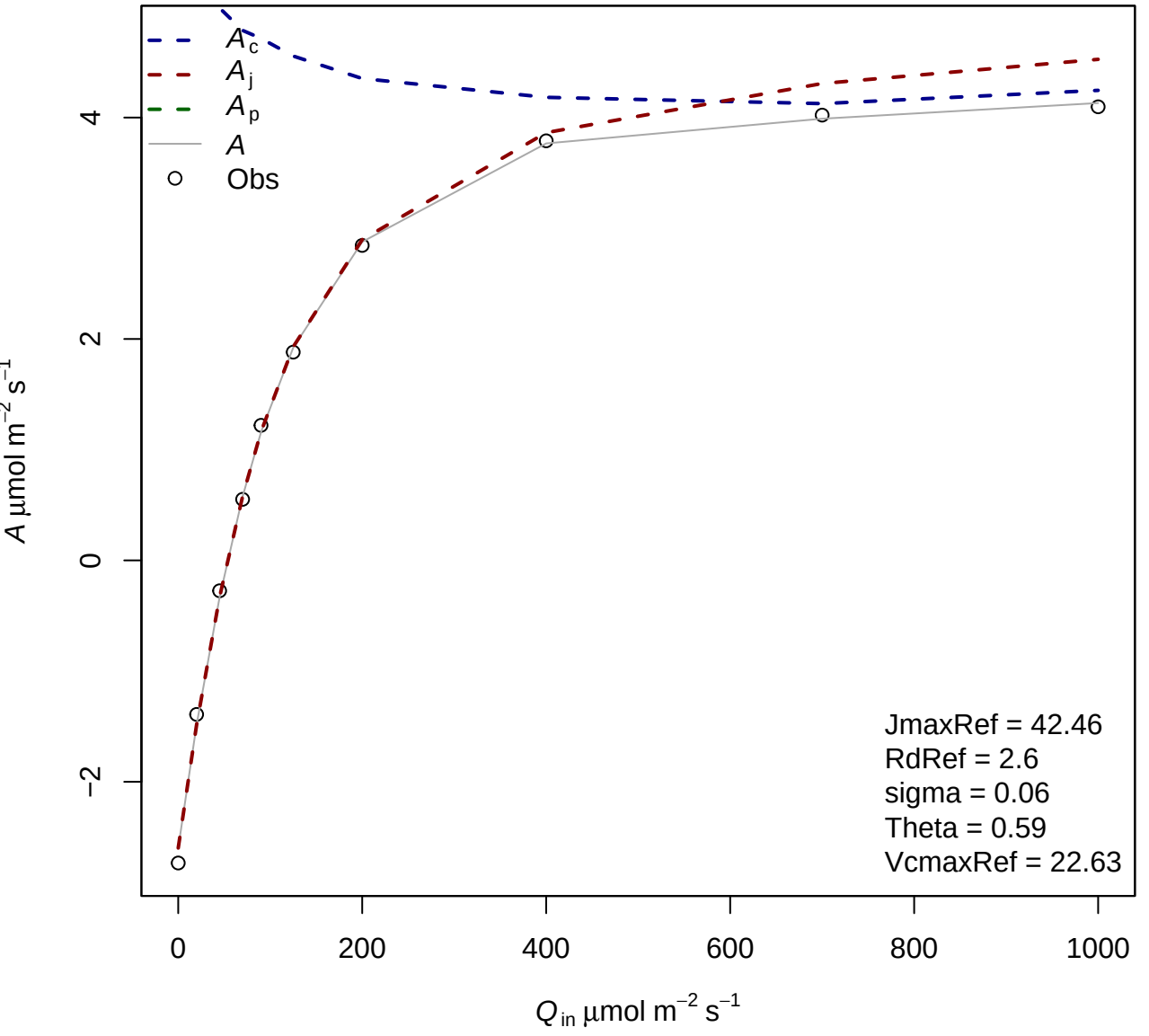

A_c
A_j
A_p
A
Obs
JmaxRef = 24.77
RdRef = 1.53
sigma = 0.06
Theta = 0.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
200
400
600
800
1000

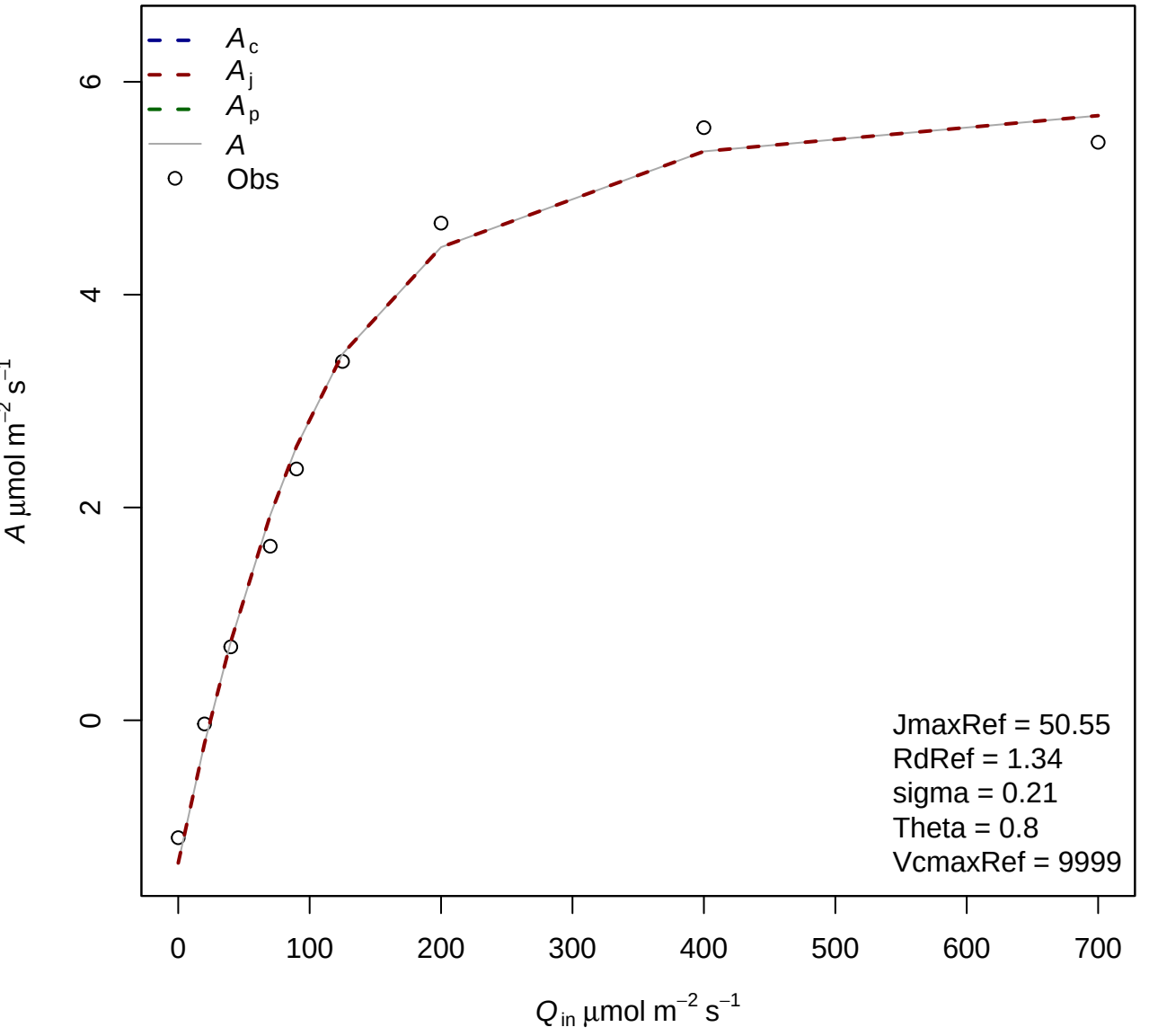

62
A_c
A_j
A_p
A
Obs
JmaxRef = 105.98
RdRef = 2.22
sigma = 0.08
Theta = −0.23
VcmaxRef = 27.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

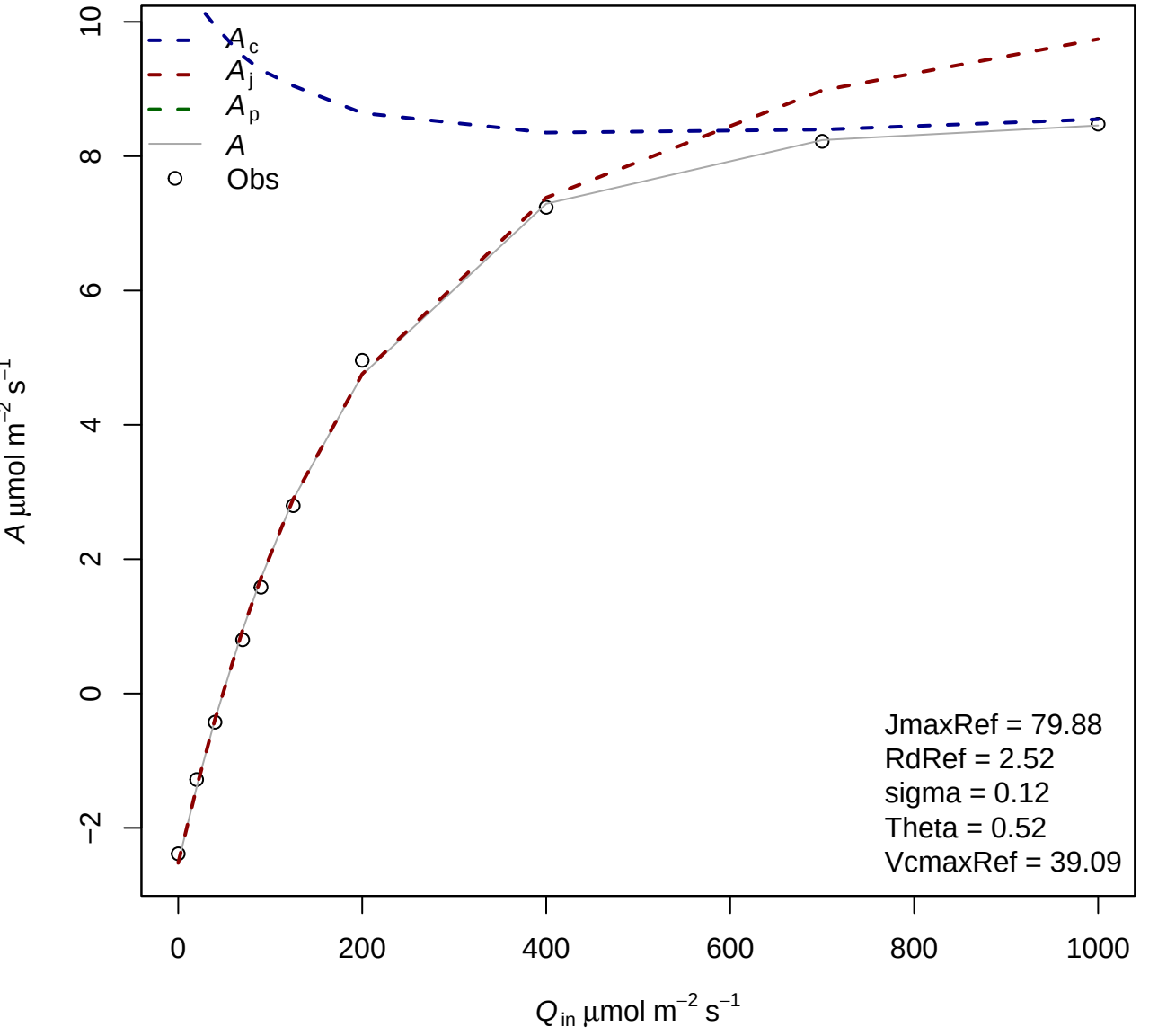

A_c
A_j
A_p
A
Obs
JmaxRef = 49.38
RdRef = 1.72
sigma = 0.14
Theta = 0.81
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

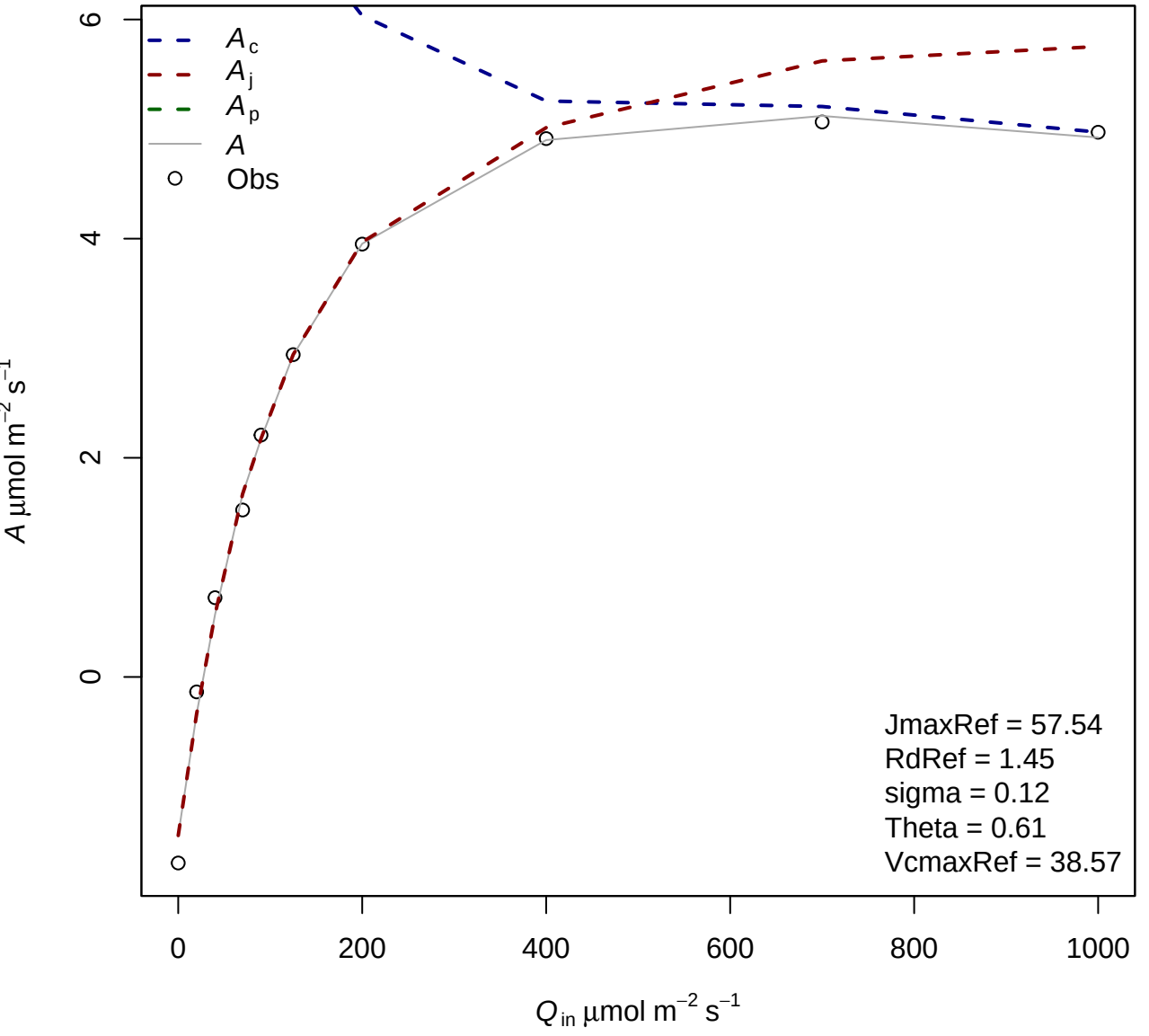

A_c
A_j
A_p
A
Obs
JmaxRef = 66192156.3
RdRef = 1.23
sigma = 0.2
Theta = −1090124.05
VcmaxRef = 27.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000

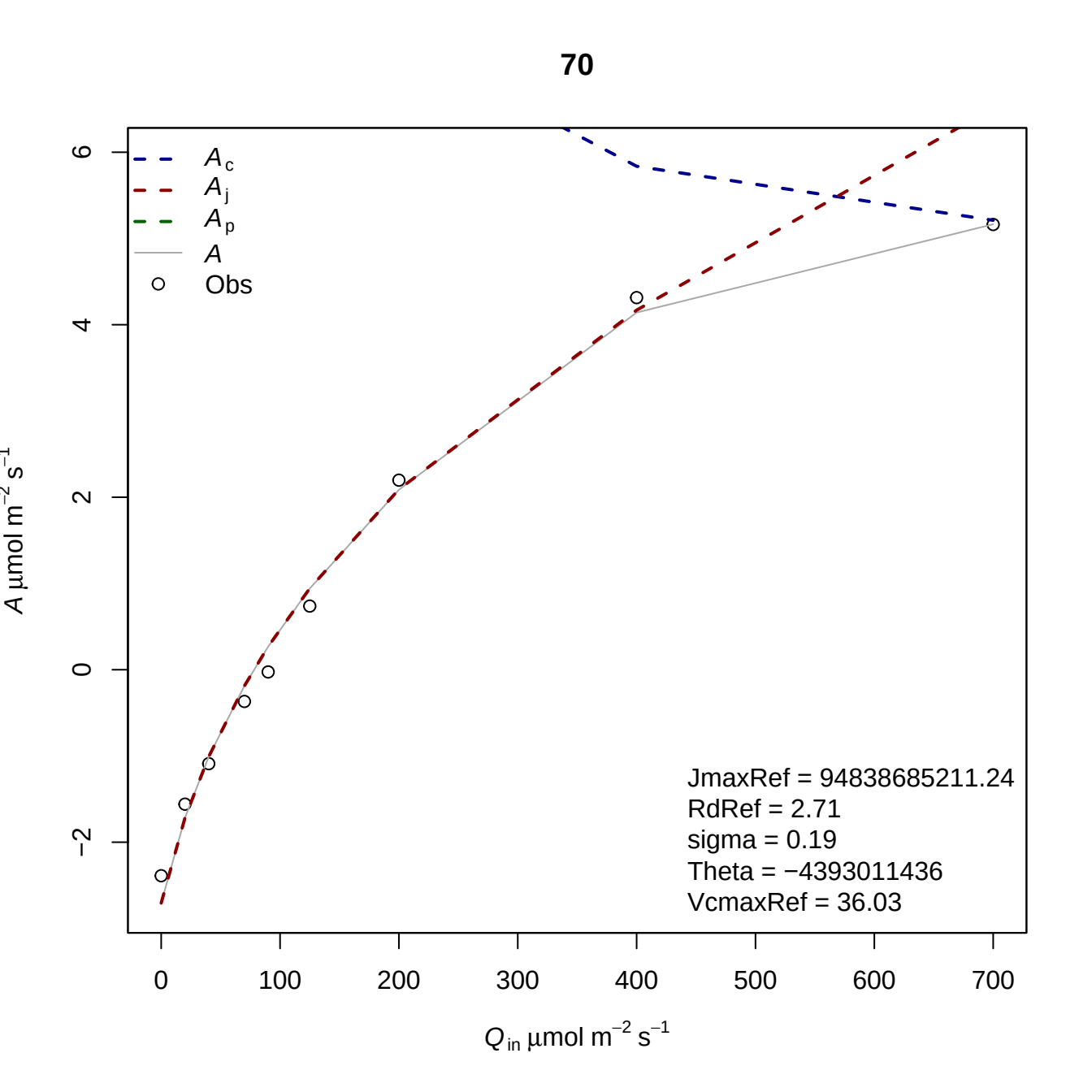

A_c
A_j
A_p
A
Obs
JmaxRef = 135
RdRef = 2.54
sigma = 0.15
Theta = −2.7
VcmaxRef = 42.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000

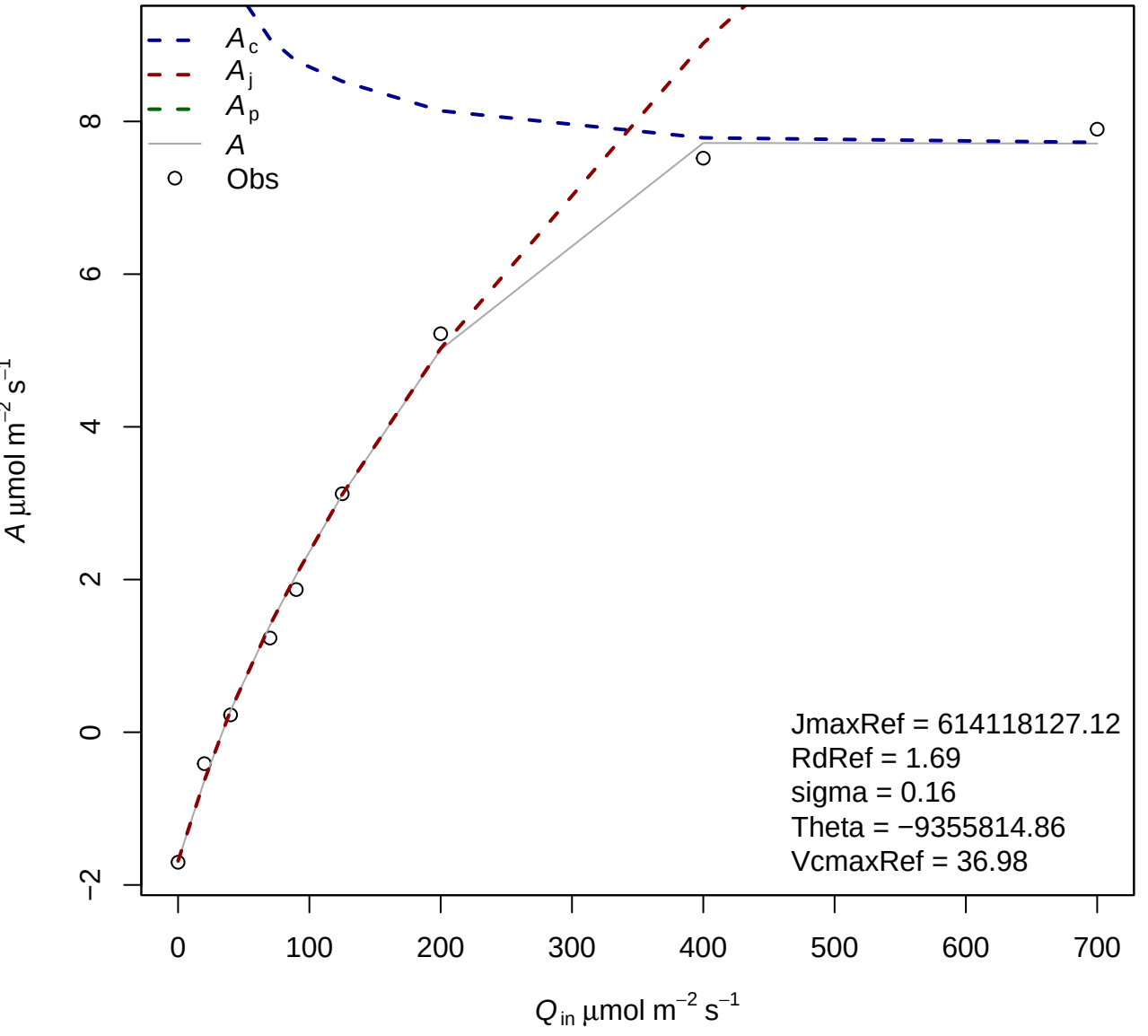

A_c
A_j
A_p
A
Obs
JmaxRef = 42.46
RdRef = 2.6
sigma = 0.06
Theta = 0.59
VcmaxRef = 22.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

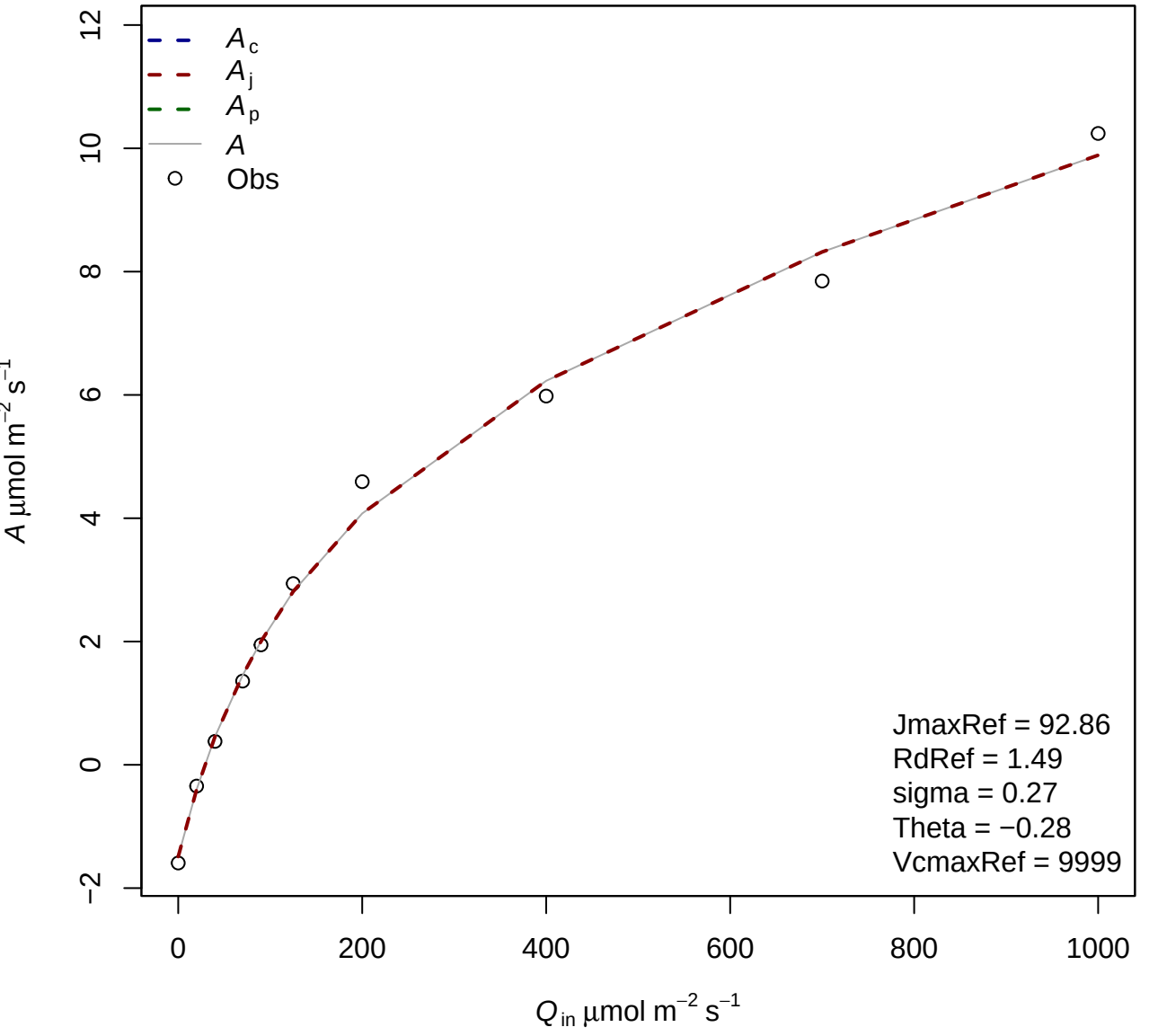

A_c
A_j
A_p
A
Obs
JmaxRef = 50.55
RdRef = 1.34
sigma = 0.21
Theta = 0.8
VcmaxRef = 9999
0
2
4
6
0
100
200
300
400
500
600
700
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

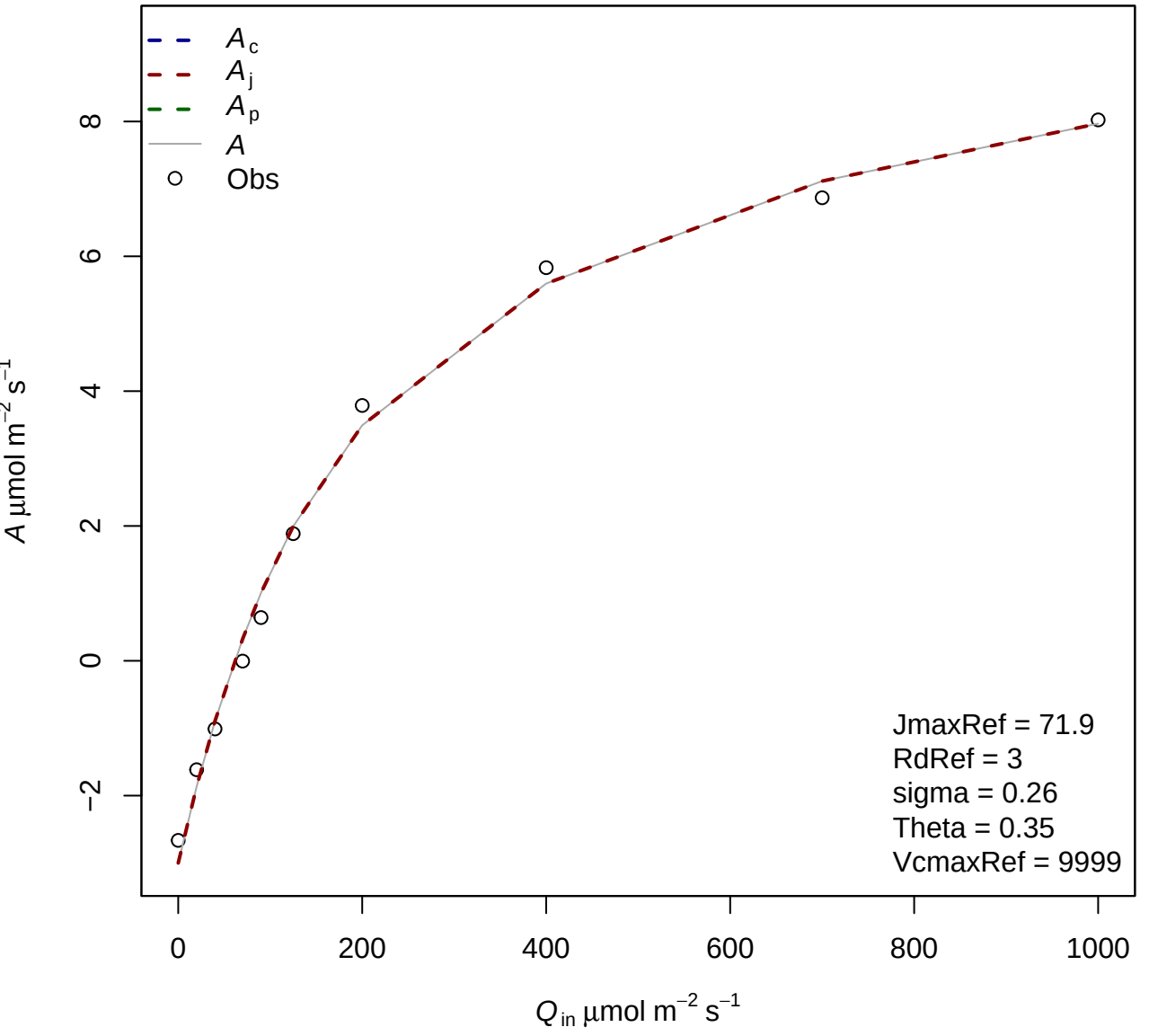

A_c
A_j
A_p
A
Obs
JmaxRef = 79.88
RdRef = 2.52
sigma = 0.12
Theta = 0.52
VcmaxRef = 39.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
200
400
600
800
1000

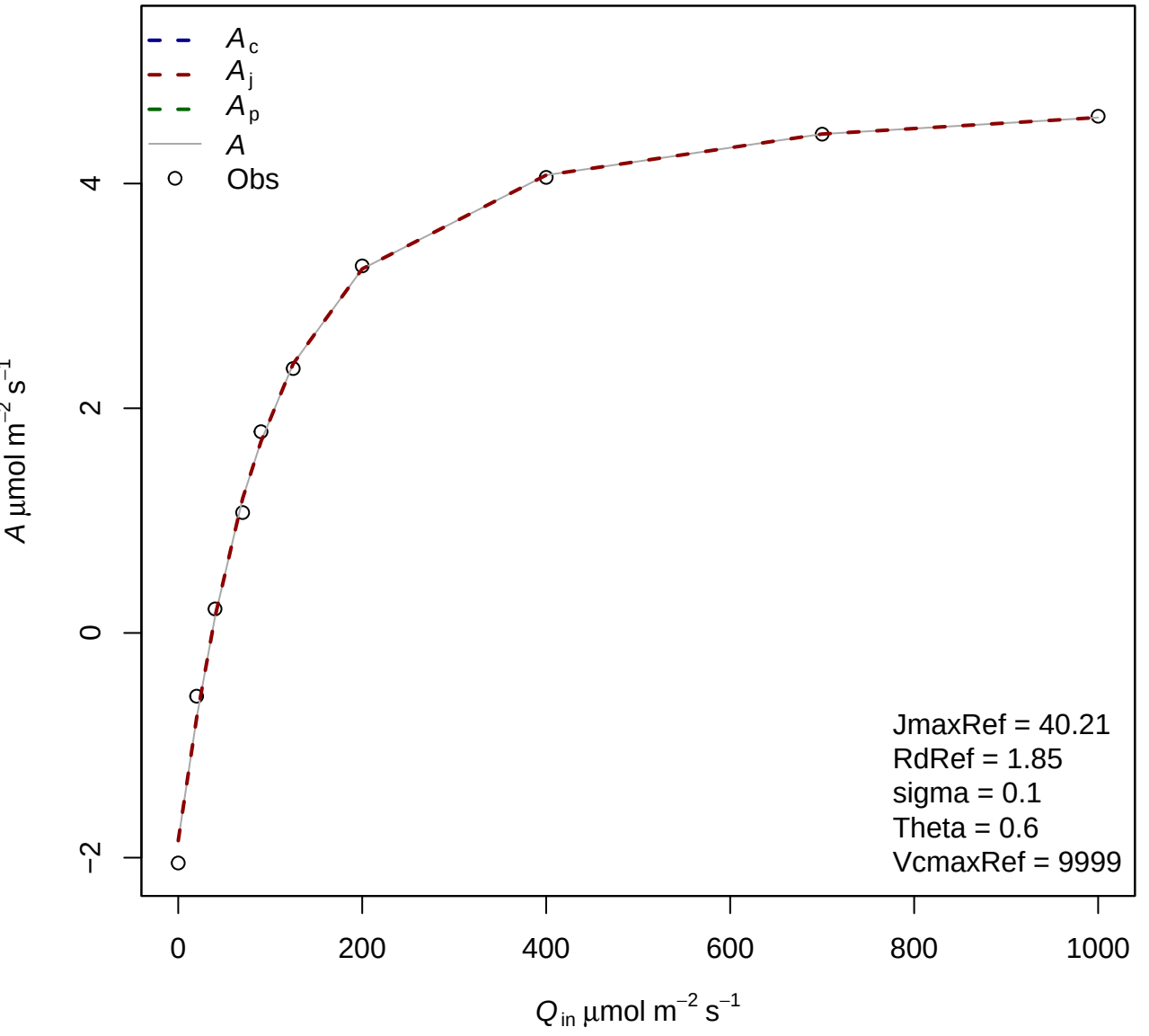

A_c
A_j
A_p
A
Obs
JmaxRef = 57.54
RdRef = 1.45
sigma = 0.12
Theta = 0.61
VcmaxRef = 38.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

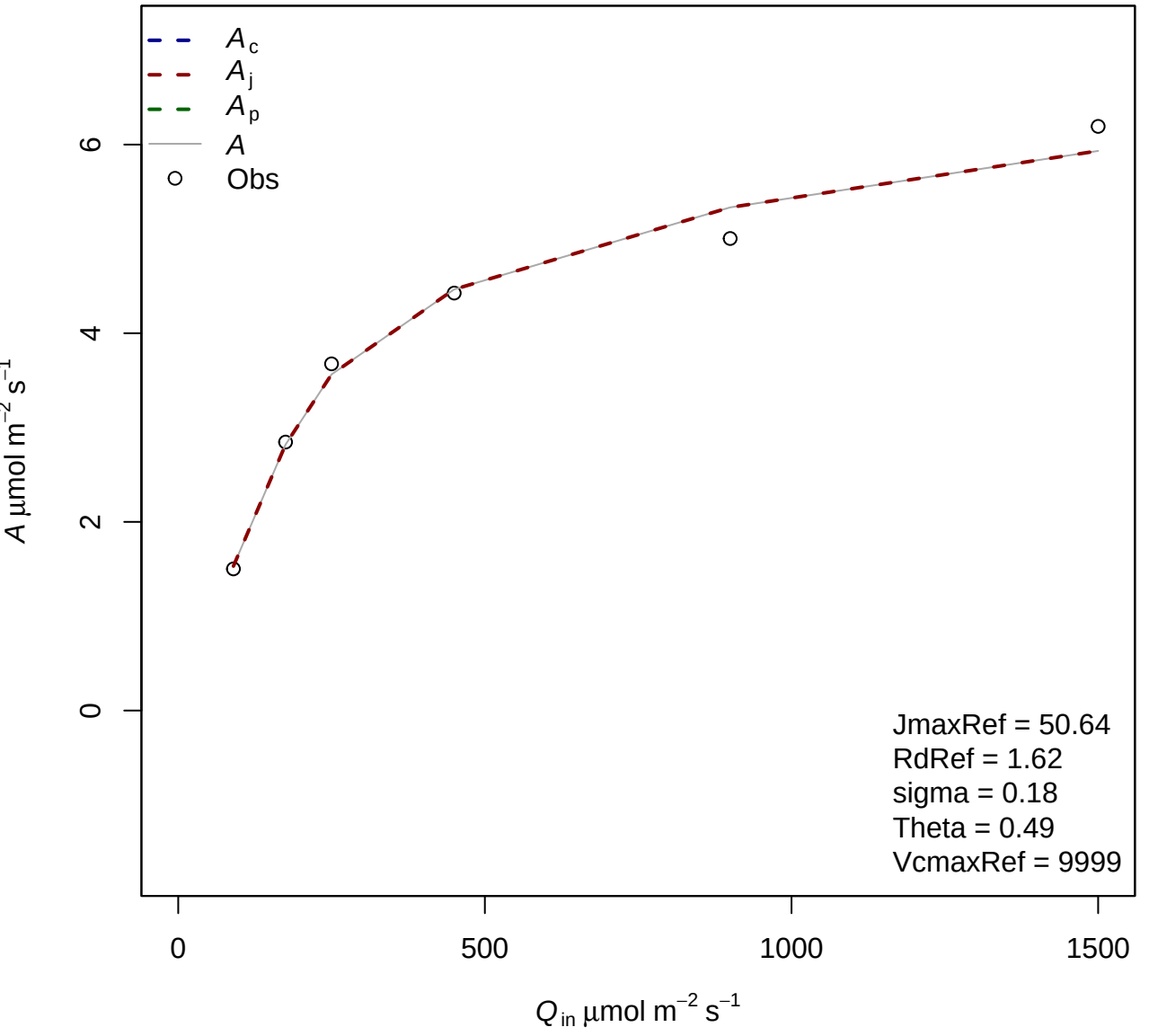

70
A_c
A_j
A_p
A
Obs
JmaxRef = 94838685211.24
RdRef = 2.71
sigma = 0.19
Theta = −4393011436
VcmaxRef = 36.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
100
200
300
400
500
600
700

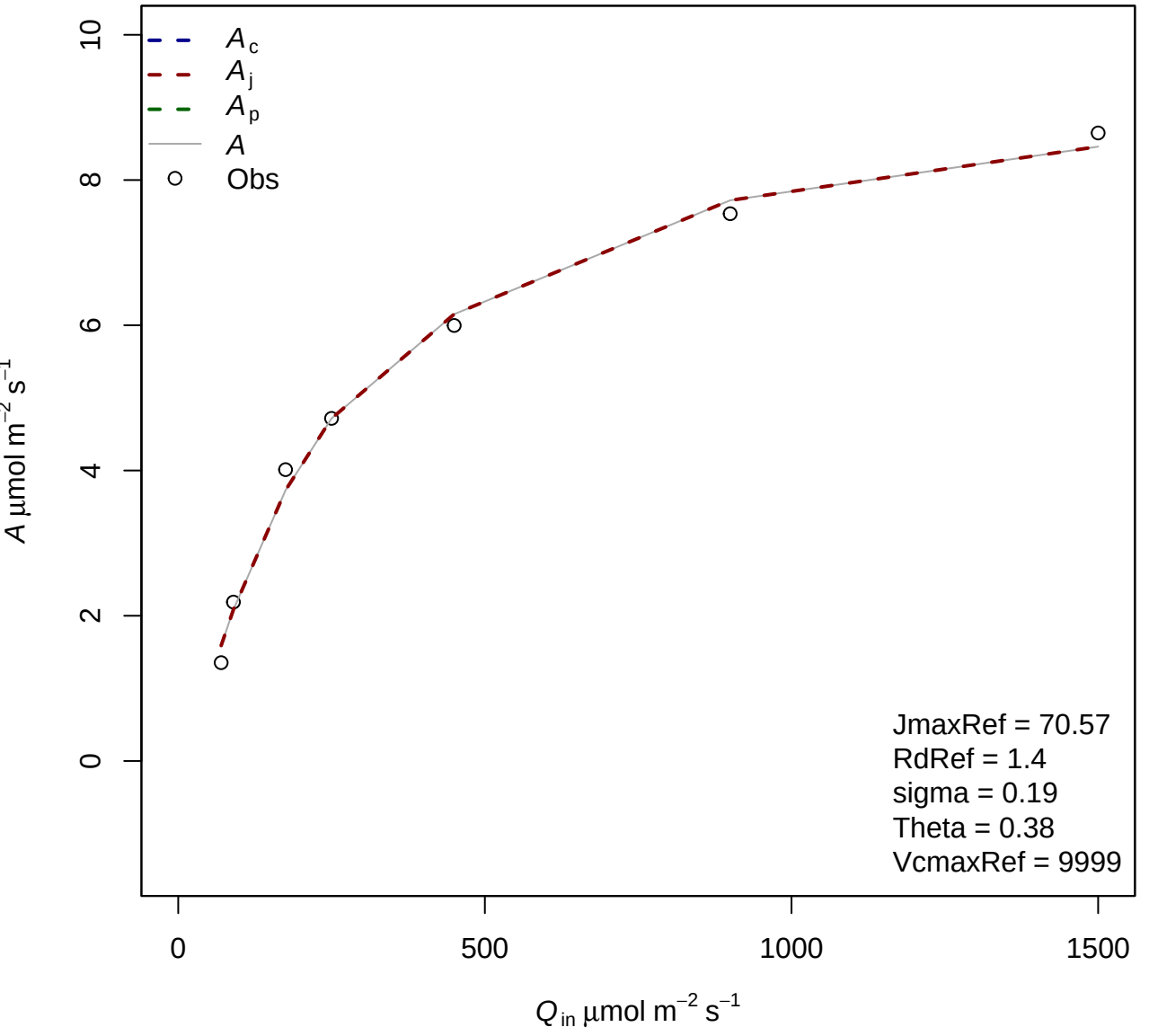

A_c
A_j
A_p
A
Obs
JmaxRef = 614118127.12
RdRef = 1.69
sigma = 0.16
Theta = −9355814.86
VcmaxRef = 36.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
100
200
300
400
500
600
700

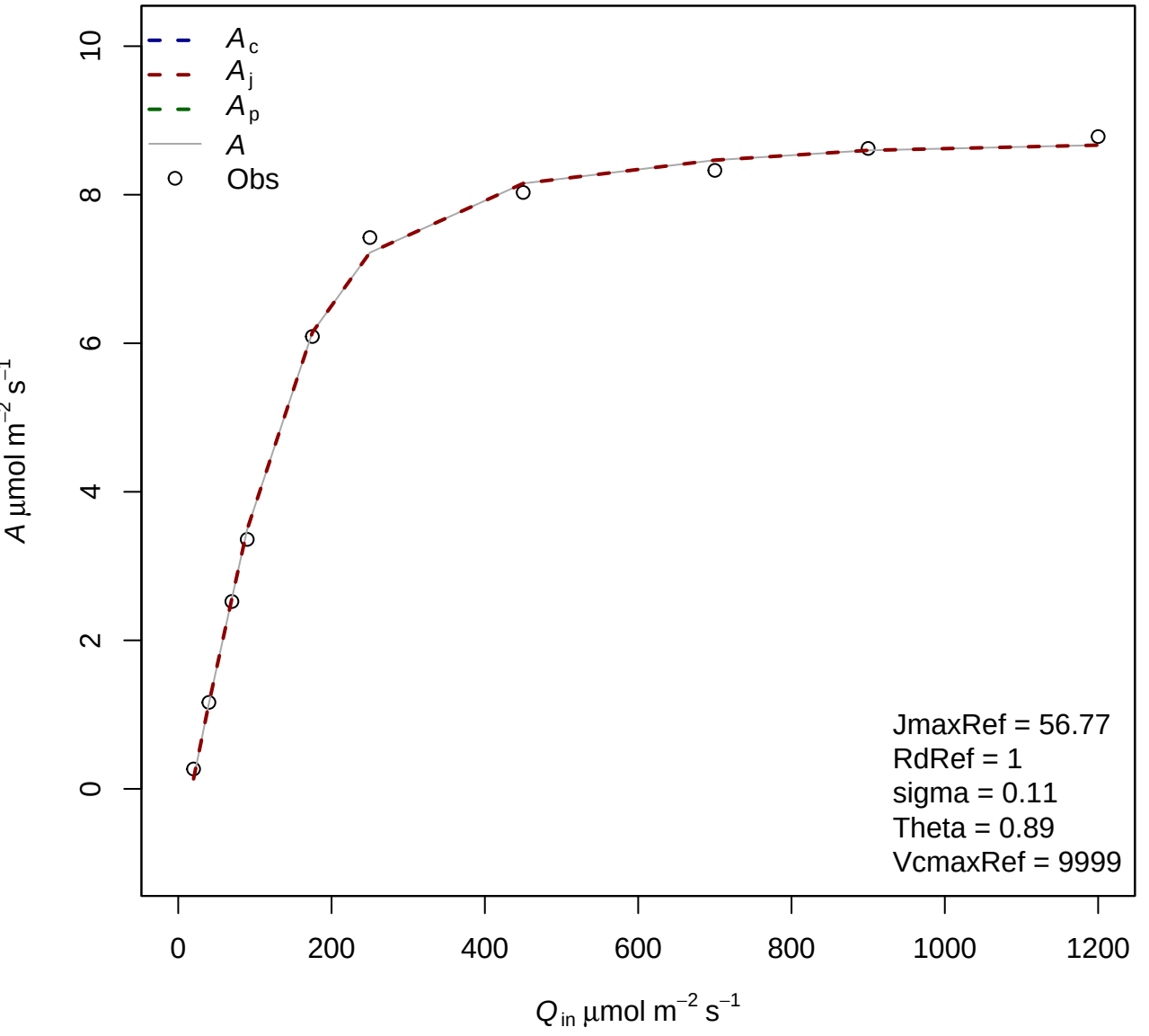

A_c
A_j
A_p
A
Obs
JmaxRef = 92.86
RdRef = 1.49
sigma = 0.27
Theta = −0.28
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000

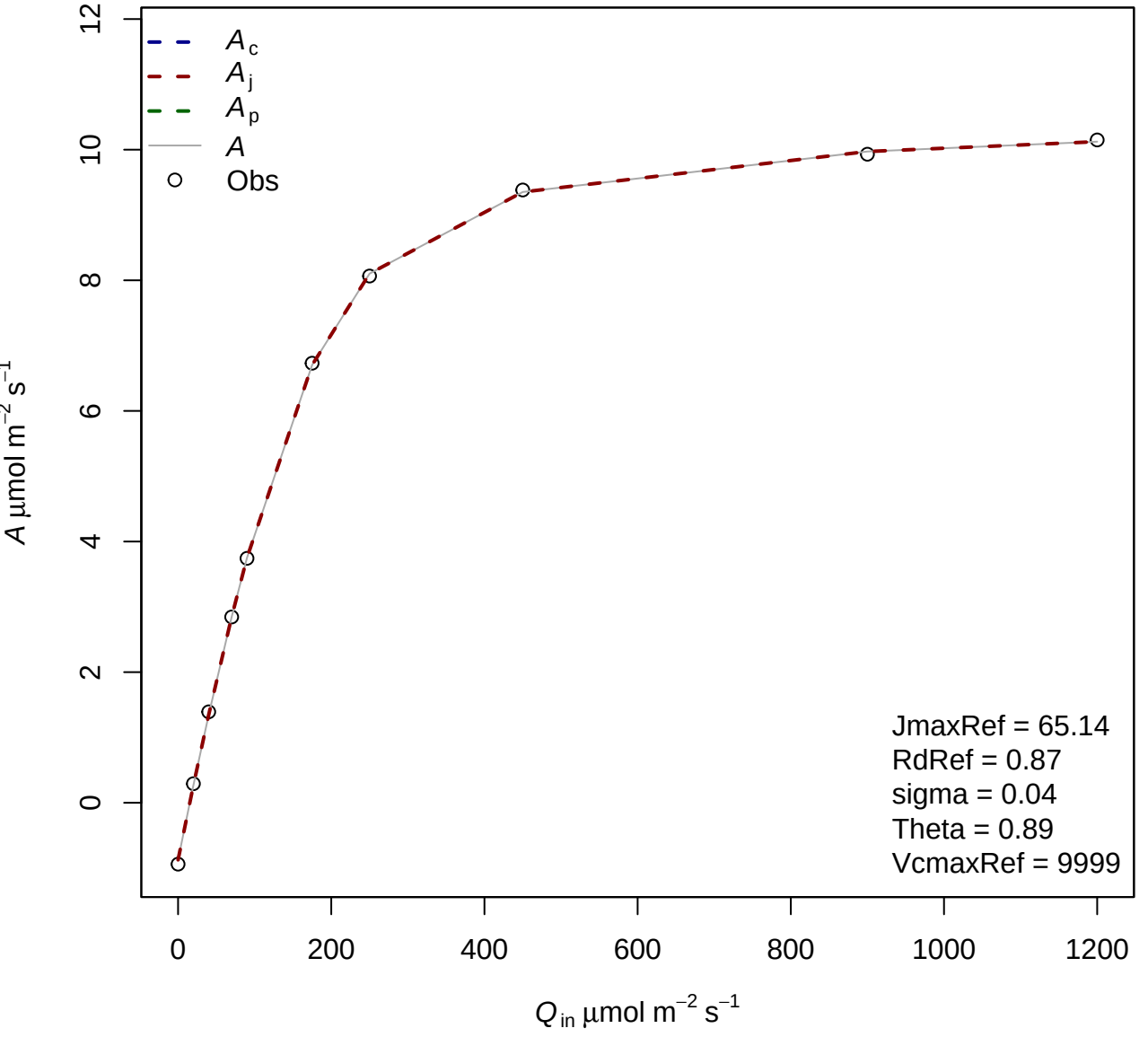

A_c
A_j
A_p
A
Obs
JmaxRef = 71.9
RdRef = 3
sigma = 0.26
Theta = 0.35
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000

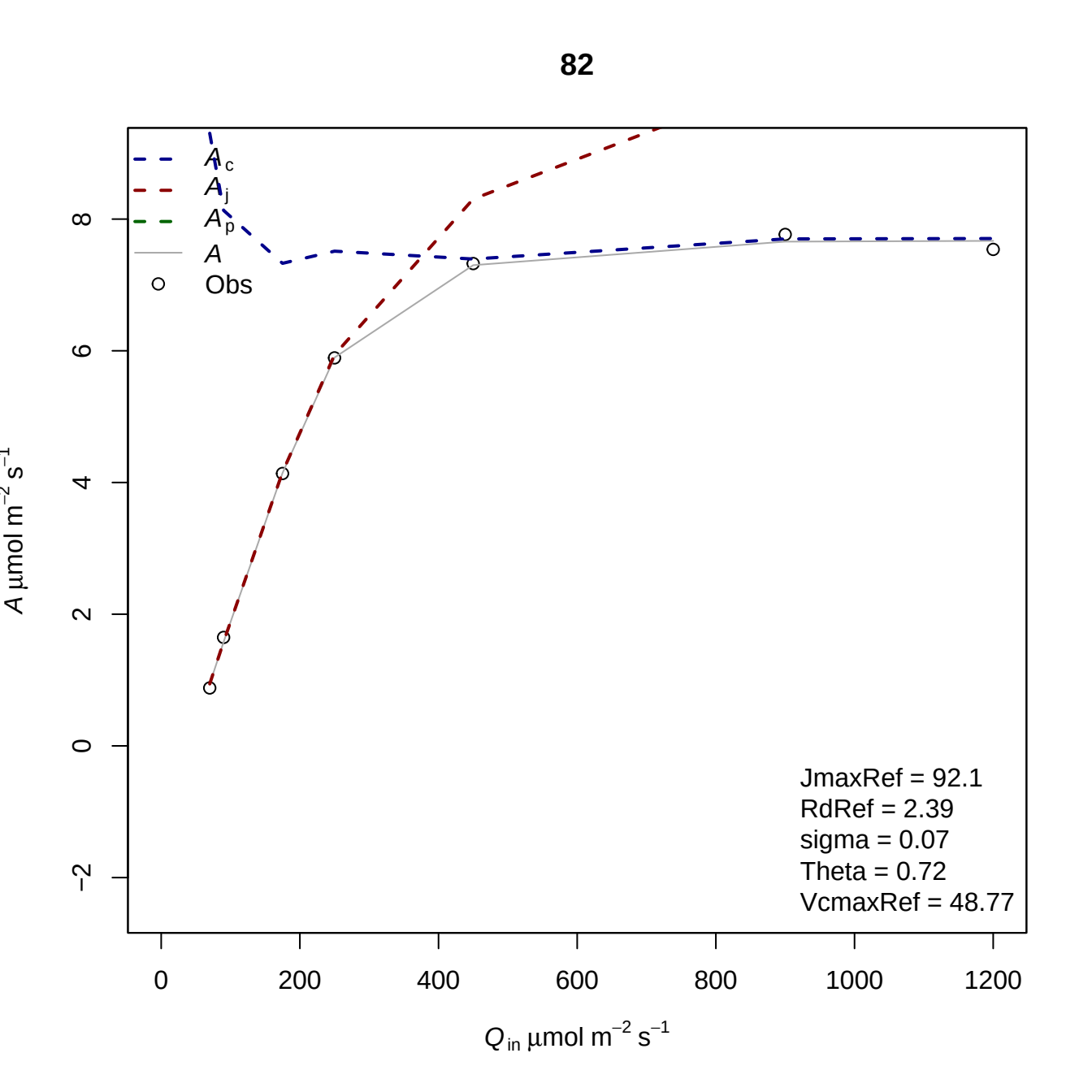

A_c
A_j
A_p
A
Obs
JmaxRef = 40.21
RdRef = 1.85
sigma = 0.1
Theta = 0.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

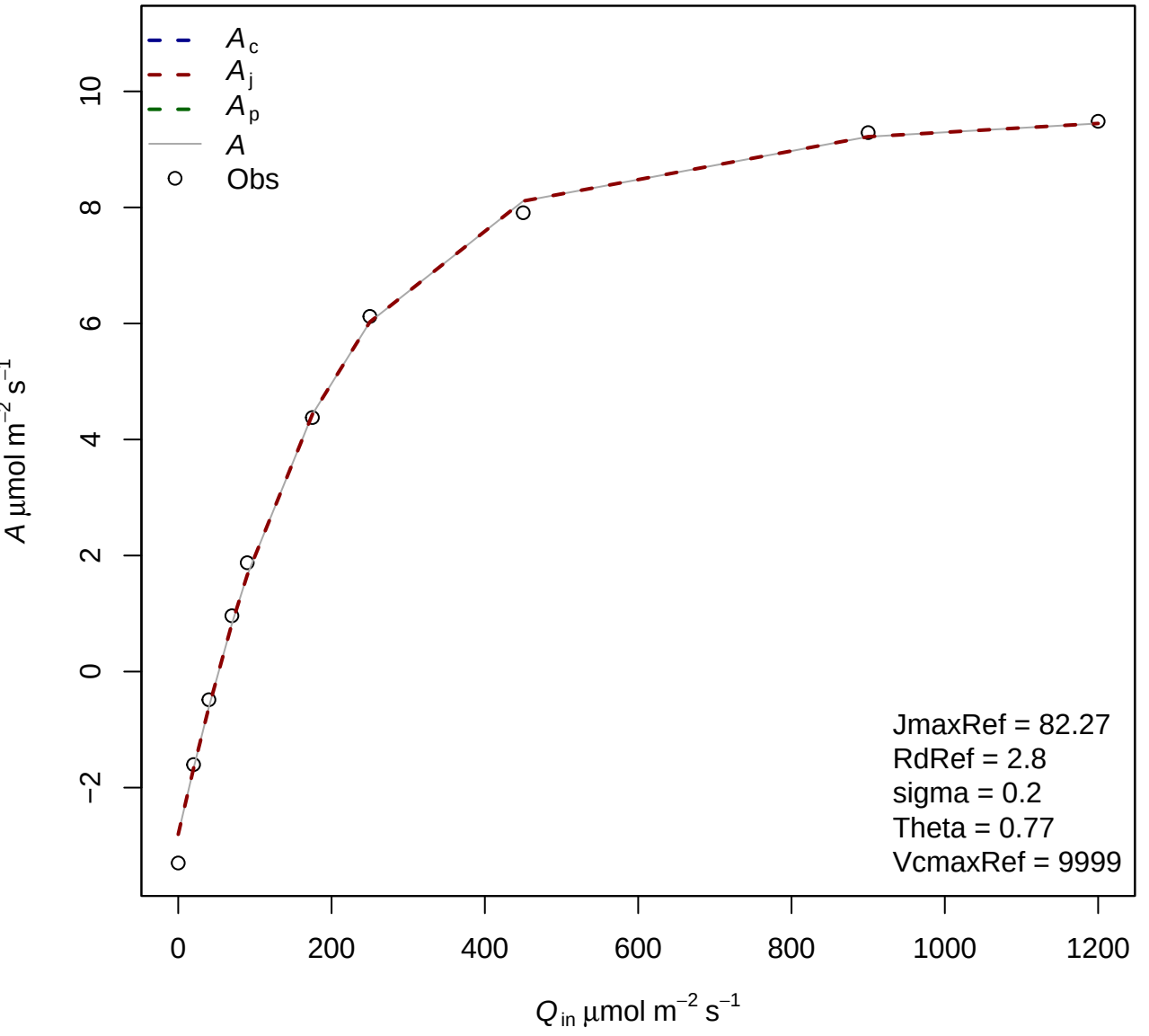

A_c
A_j
A_p
A
Obs
JmaxRef = 50.64
RdRef = 1.62
sigma = 0.18
Theta = 0.49
VcmaxRef = 9999
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

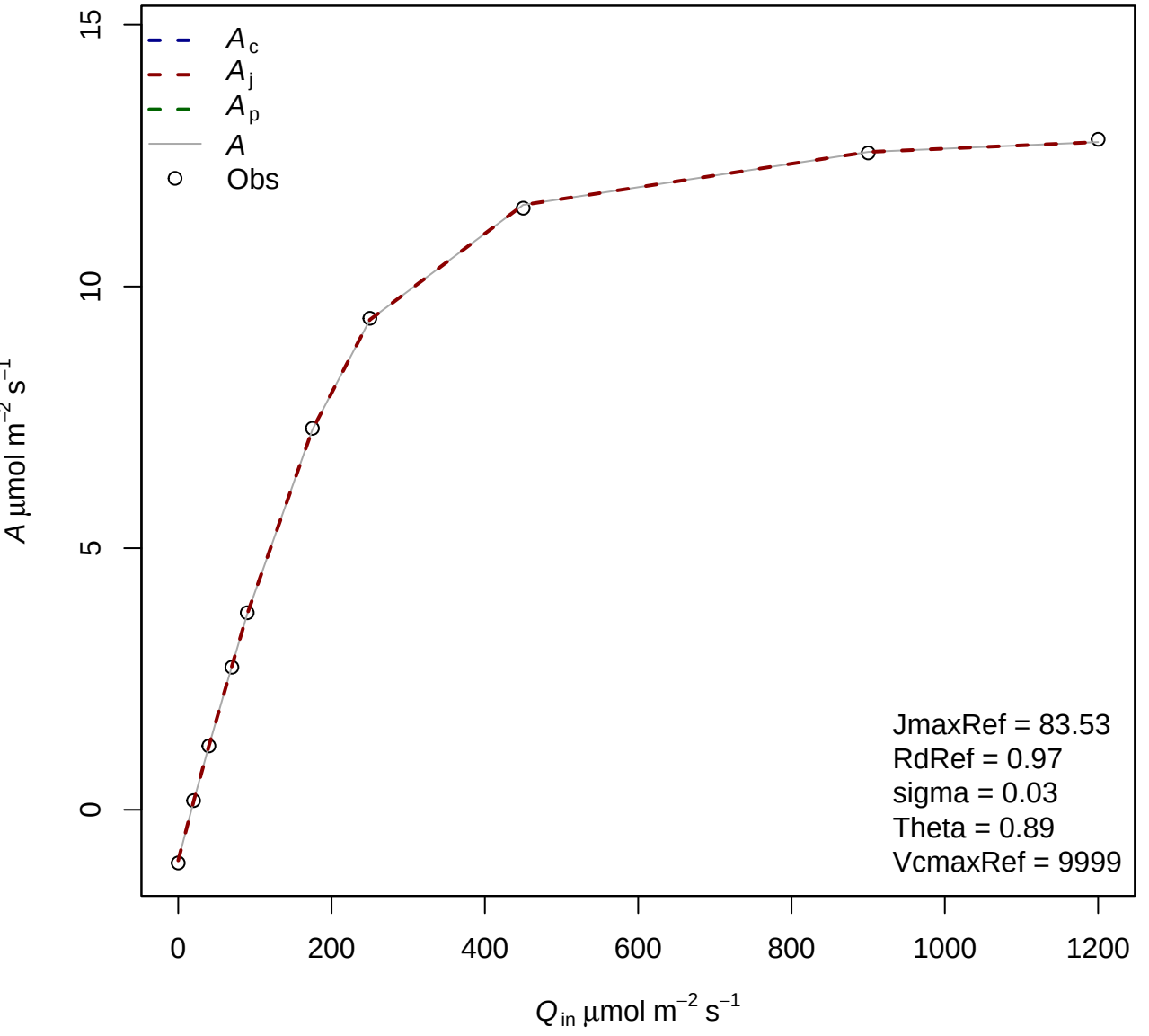

A_c
A_j
A_p
A
Obs
JmaxRef = 70.57
RdRef = 1.4
sigma = 0.19
Theta = 0.38
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

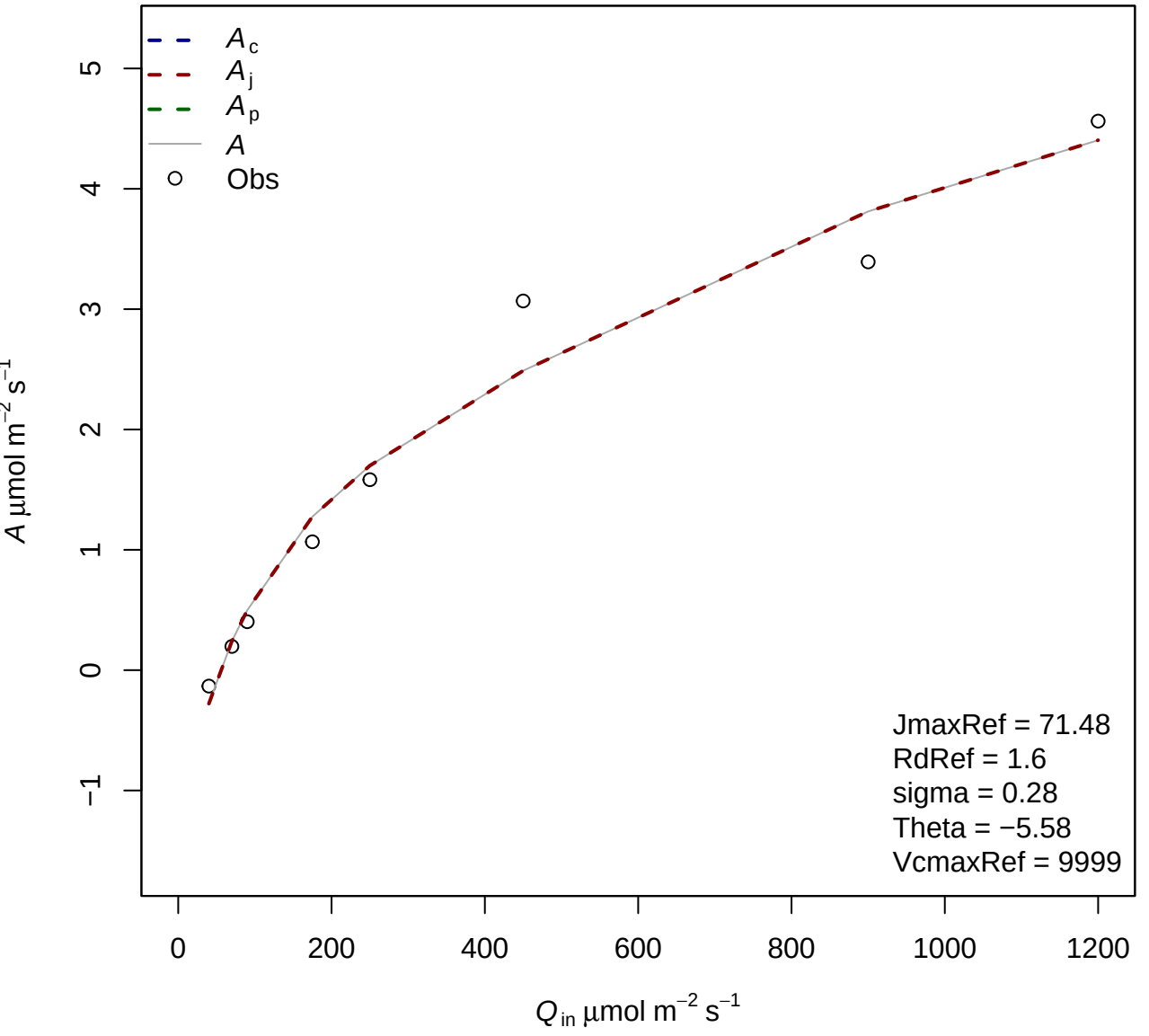

A_c
A_j
A_p
A
Obs
JmaxRef = 56.77
RdRef = 1
sigma = 0.11
Theta = 0.89
VcmaxRef = 9999
0
2
4
6
8
10
0
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

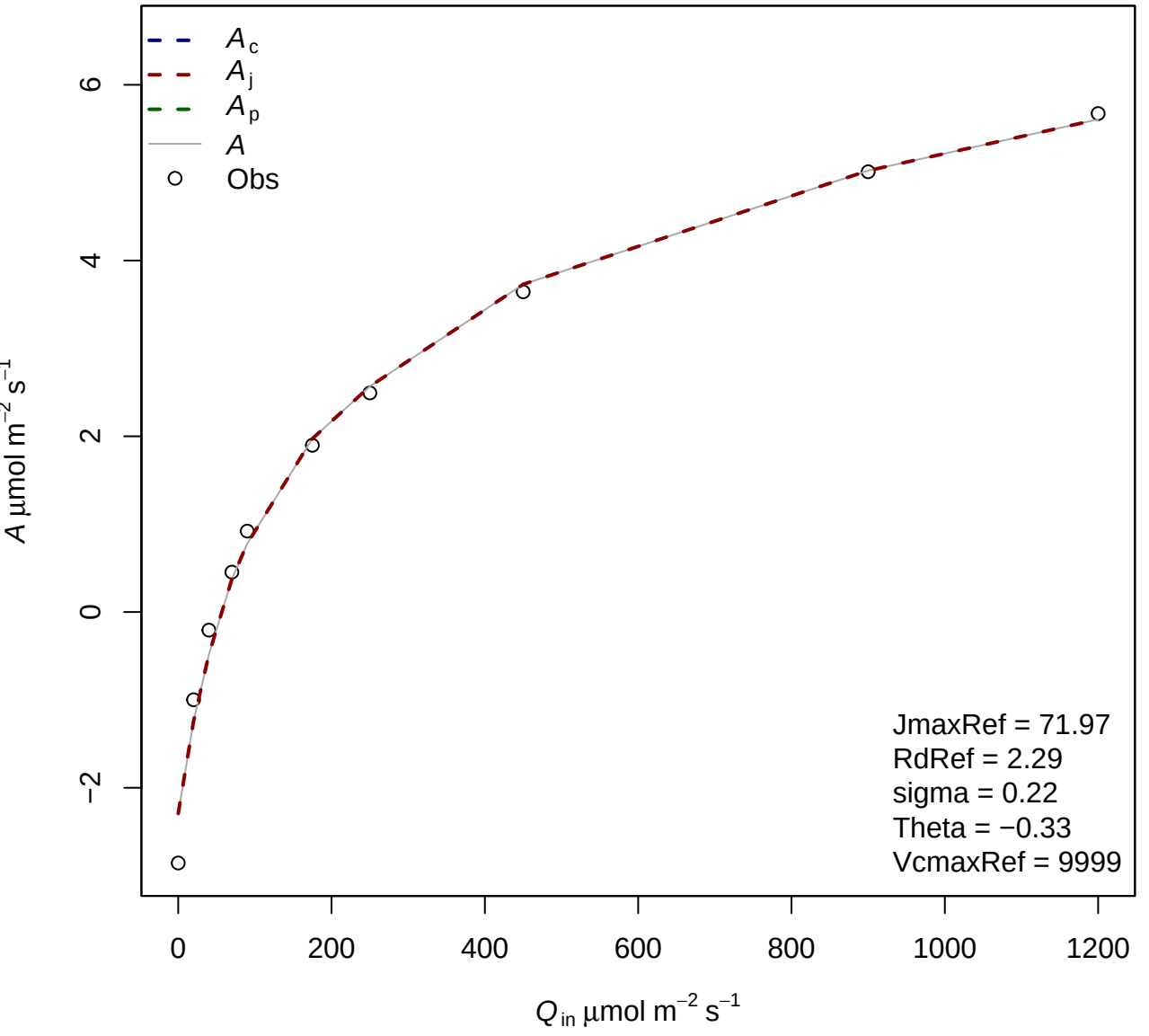

A_c
A_j
A_p
A
Obs
JmaxRef = 65.14
RdRef = 0.87
sigma = 0.04
Theta = 0.89
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

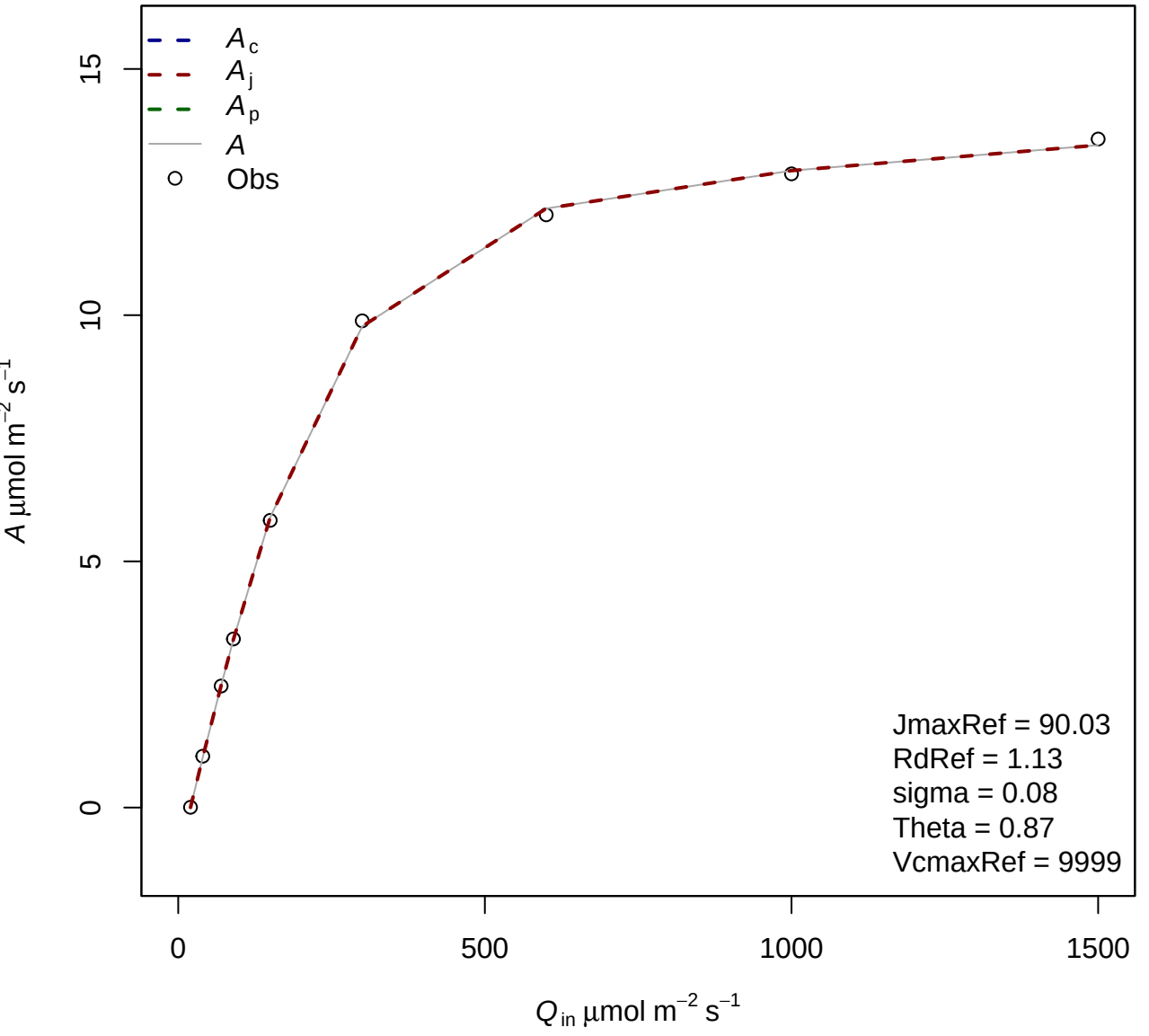

82
A_c
A_j
A_p
A
Obs
JmaxRef = 92.1
RdRef = 2.39
sigma = 0.07
Theta = 0.72
VcmaxRef = 48.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200

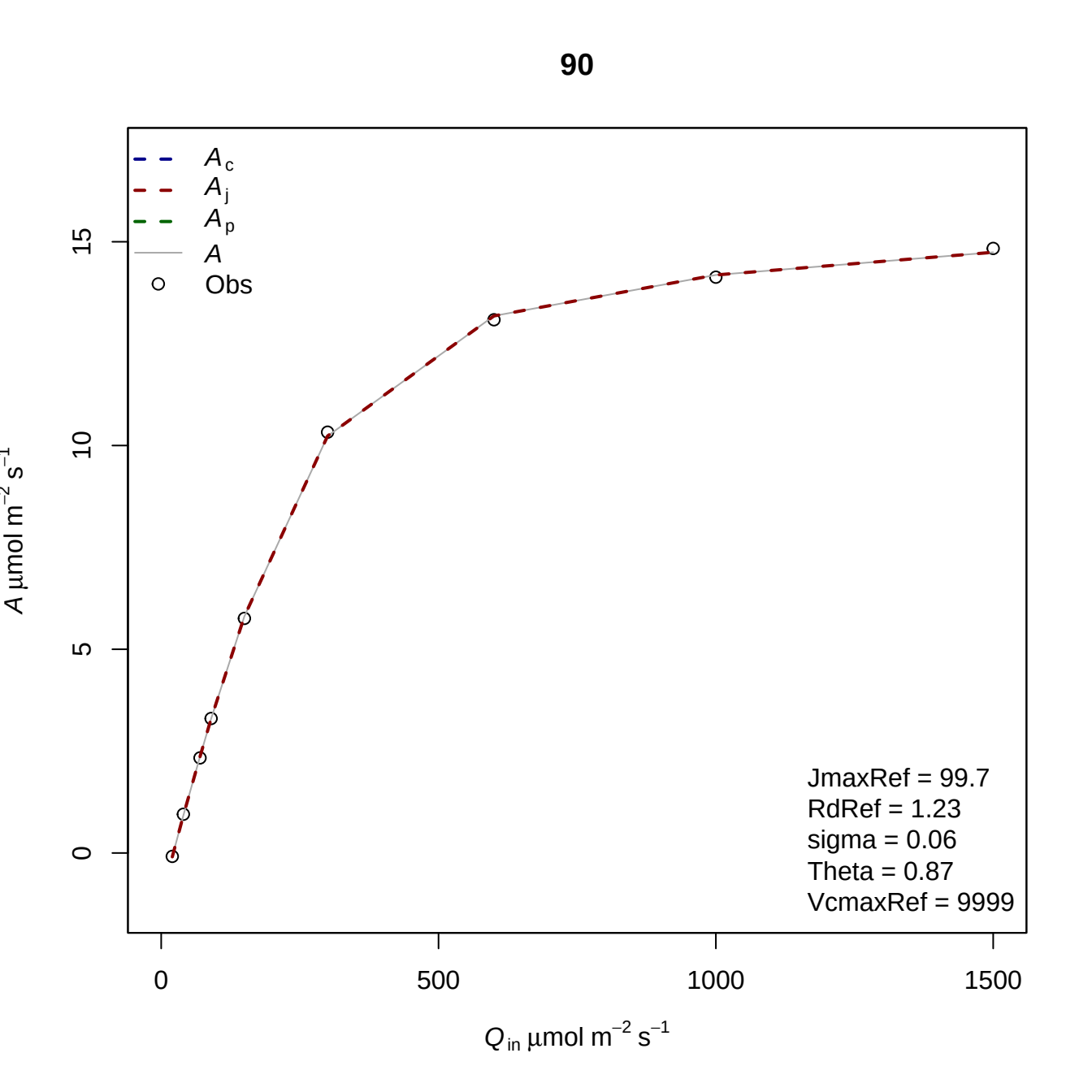

A_c
A_j
A_p
A
Obs
JmaxRef = 82.27
RdRef = 2.8
sigma = 0.2
Theta = 0.77
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200

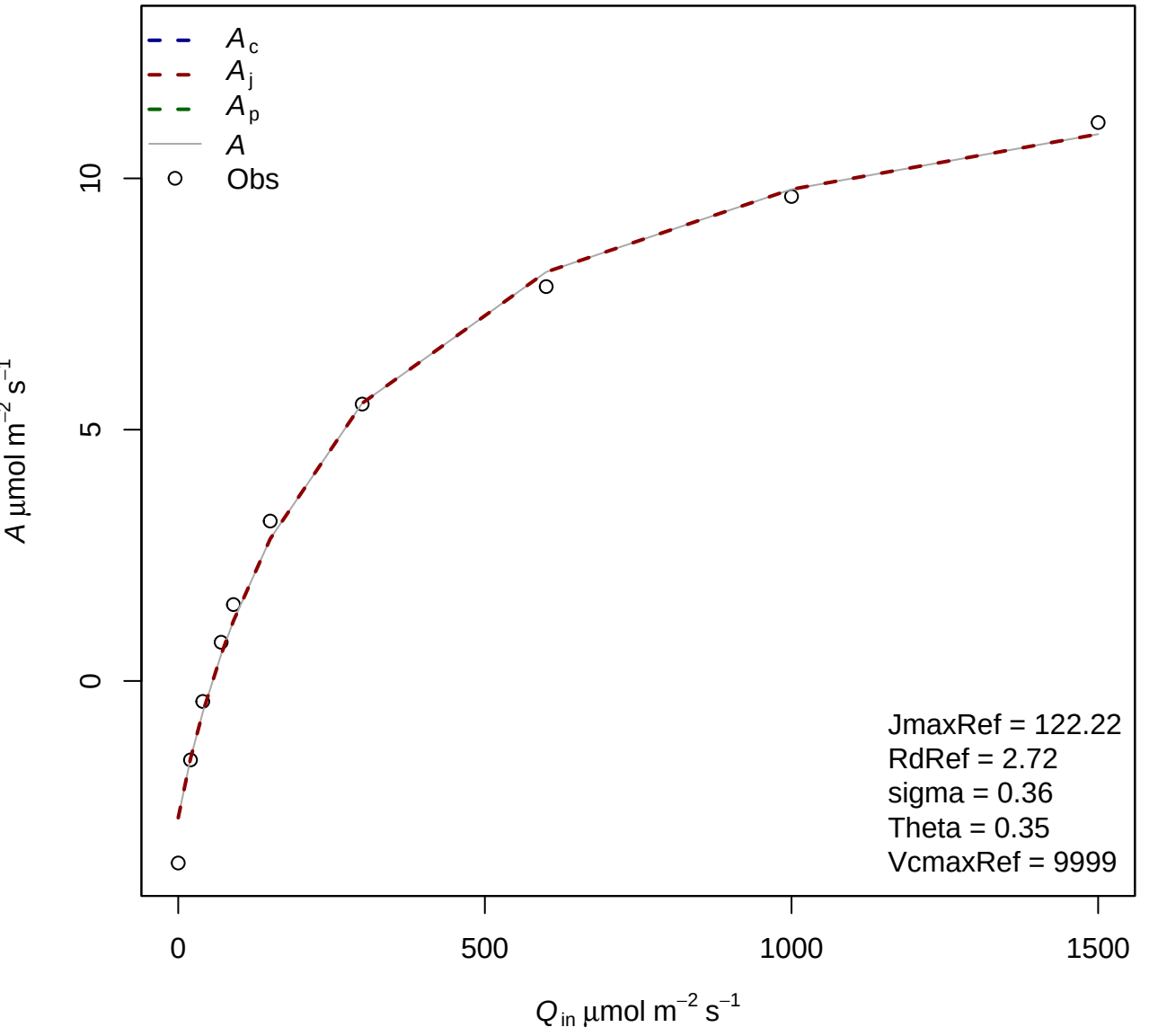

A_c
A_j
A_p
A
Obs
JmaxRef = 83.53
RdRef = 0.97
sigma = 0.03
Theta = 0.89
VcmaxRef = 9999
0
5
10
15
0
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

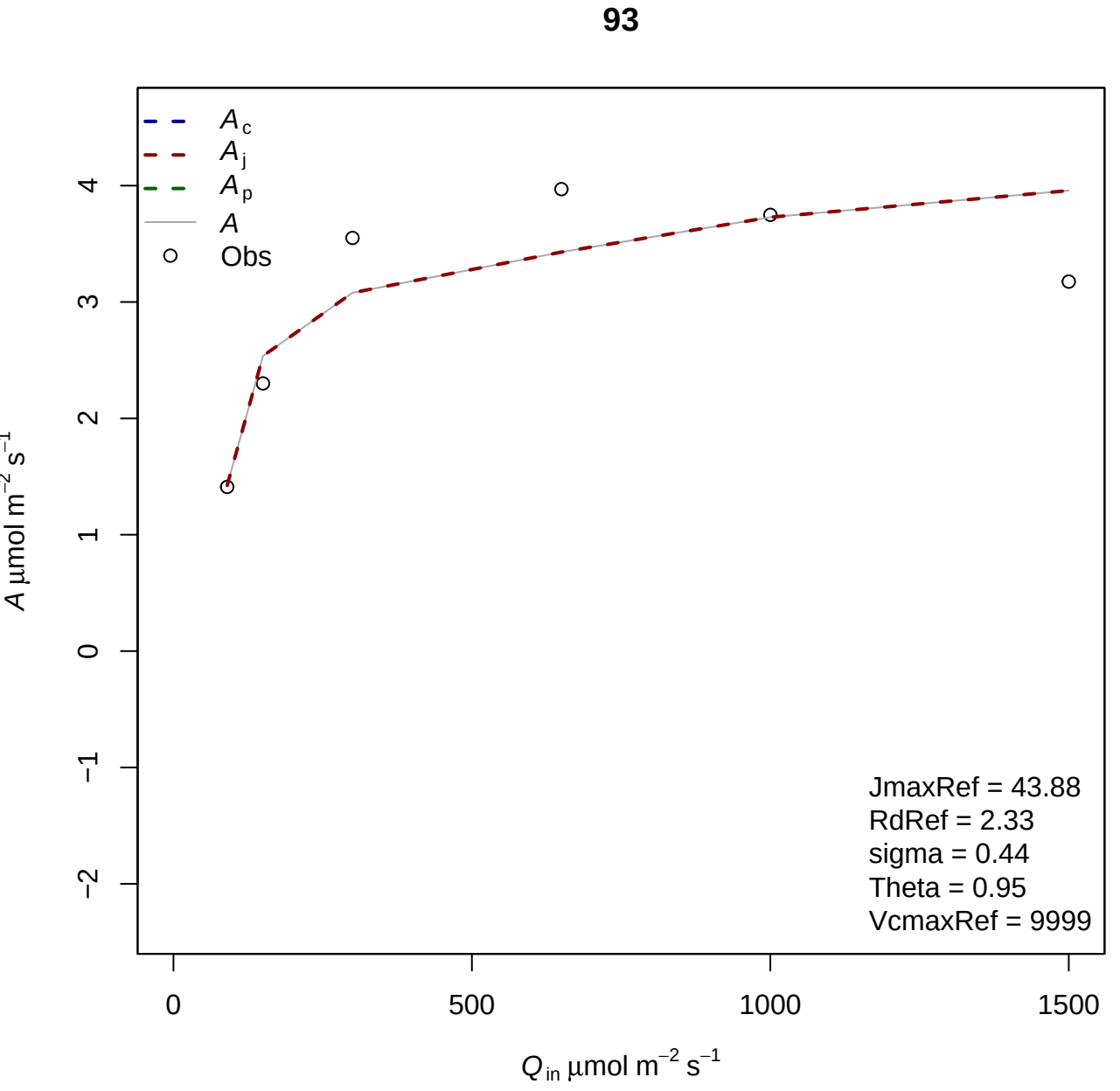

A_c
A_j
A_p
A
Obs
JmaxRef = 71.48
RdRef = 1.6
sigma = 0.28
Theta = −5.58
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000
1200

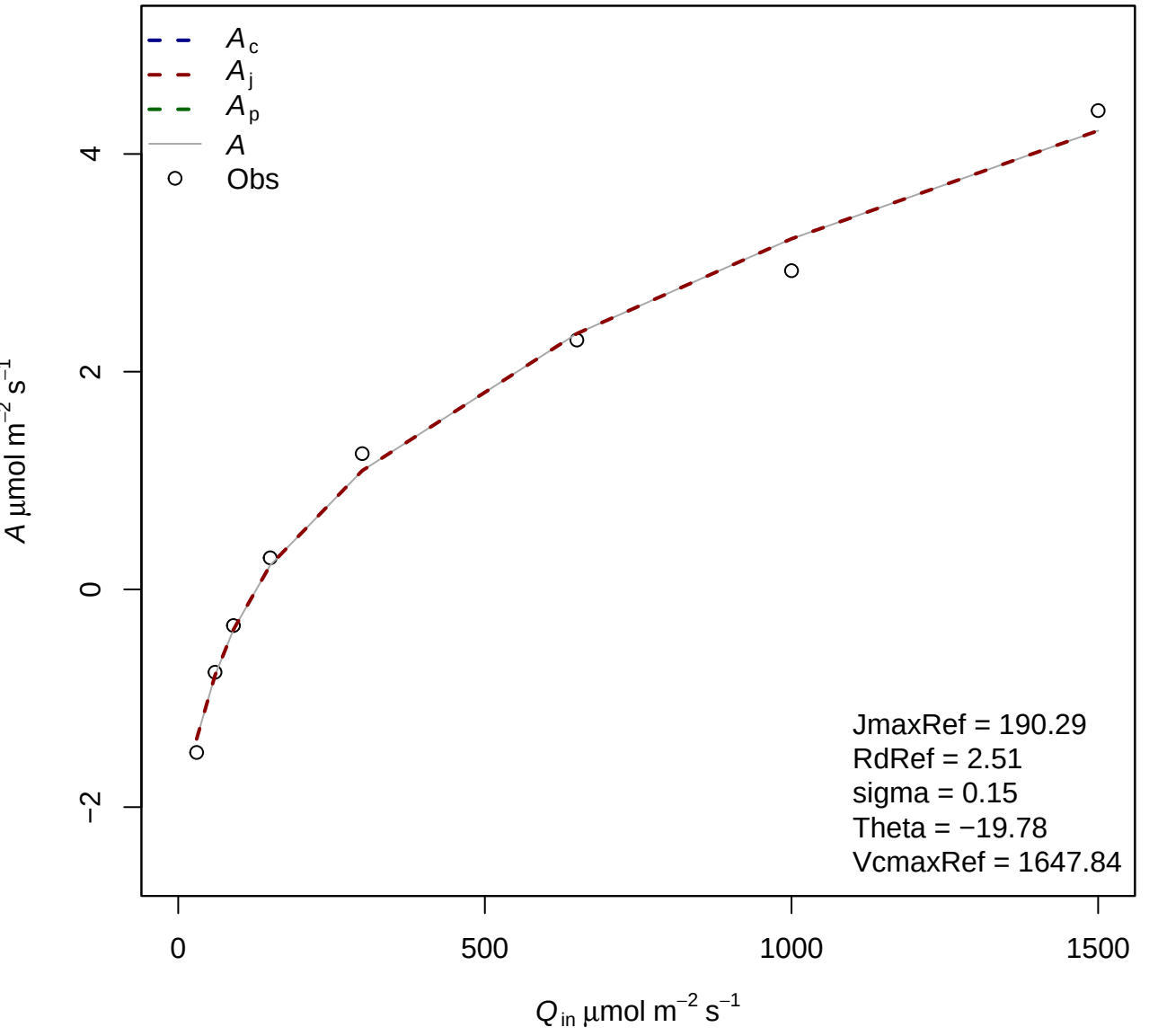

A_c
A_j
A_p
A
Obs
JmaxRef = 71.97
RdRef = 2.29
sigma = 0.22
Theta = −0.33
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

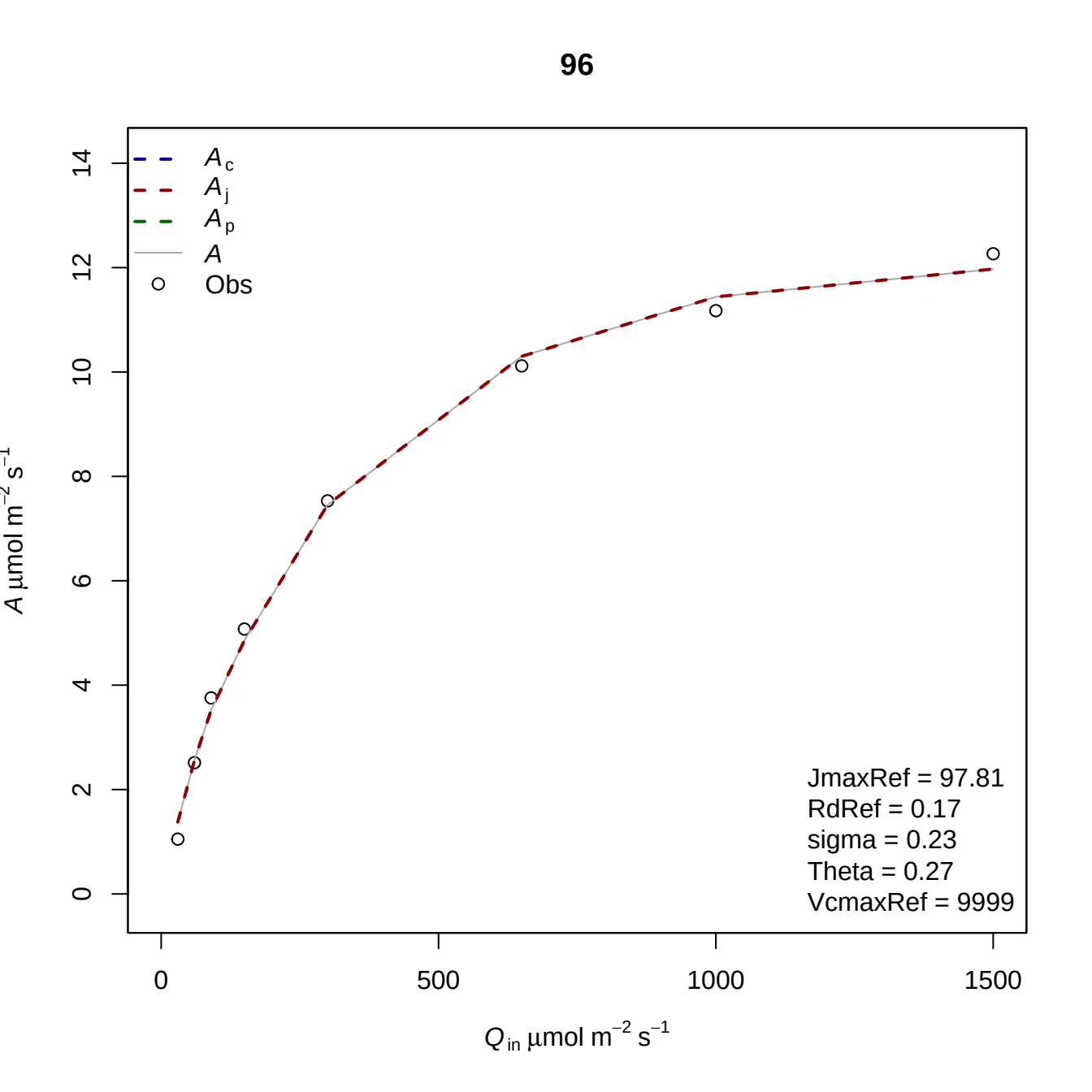

A_c
A_j
A_p
A
Obs
JmaxRef = 90.03
RdRef = 1.13
sigma = 0.08
Theta = 0.87
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

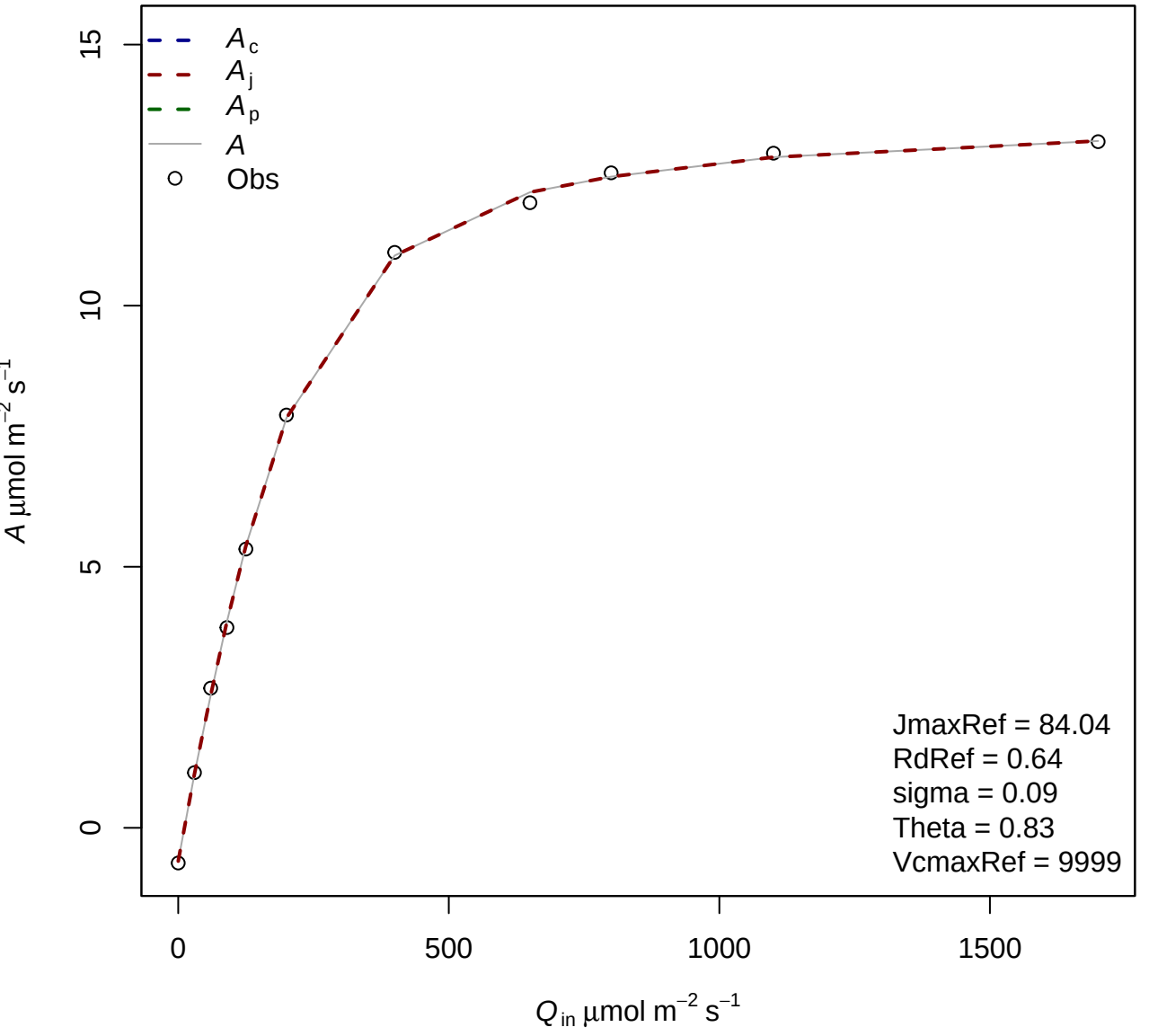

90
A_c
A_j
A_p
A
Obs
JmaxRef = 99.7
RdRef = 1.23
sigma = 0.06
Theta = 0.87
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

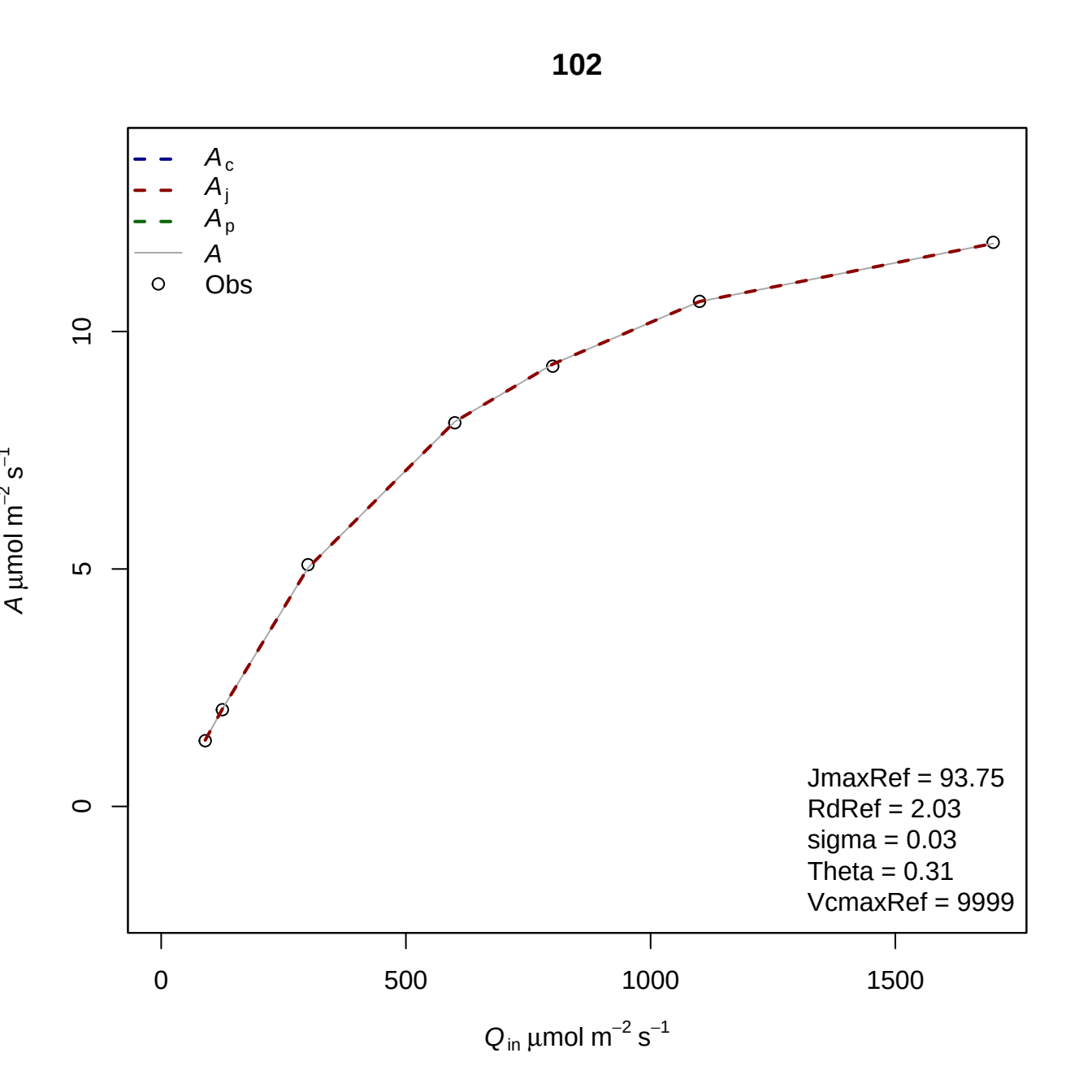

A_c
A_j
A_p
A
Obs
JmaxRef = 122.22
RdRef = 2.72
sigma = 0.36
Theta = 0.35
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

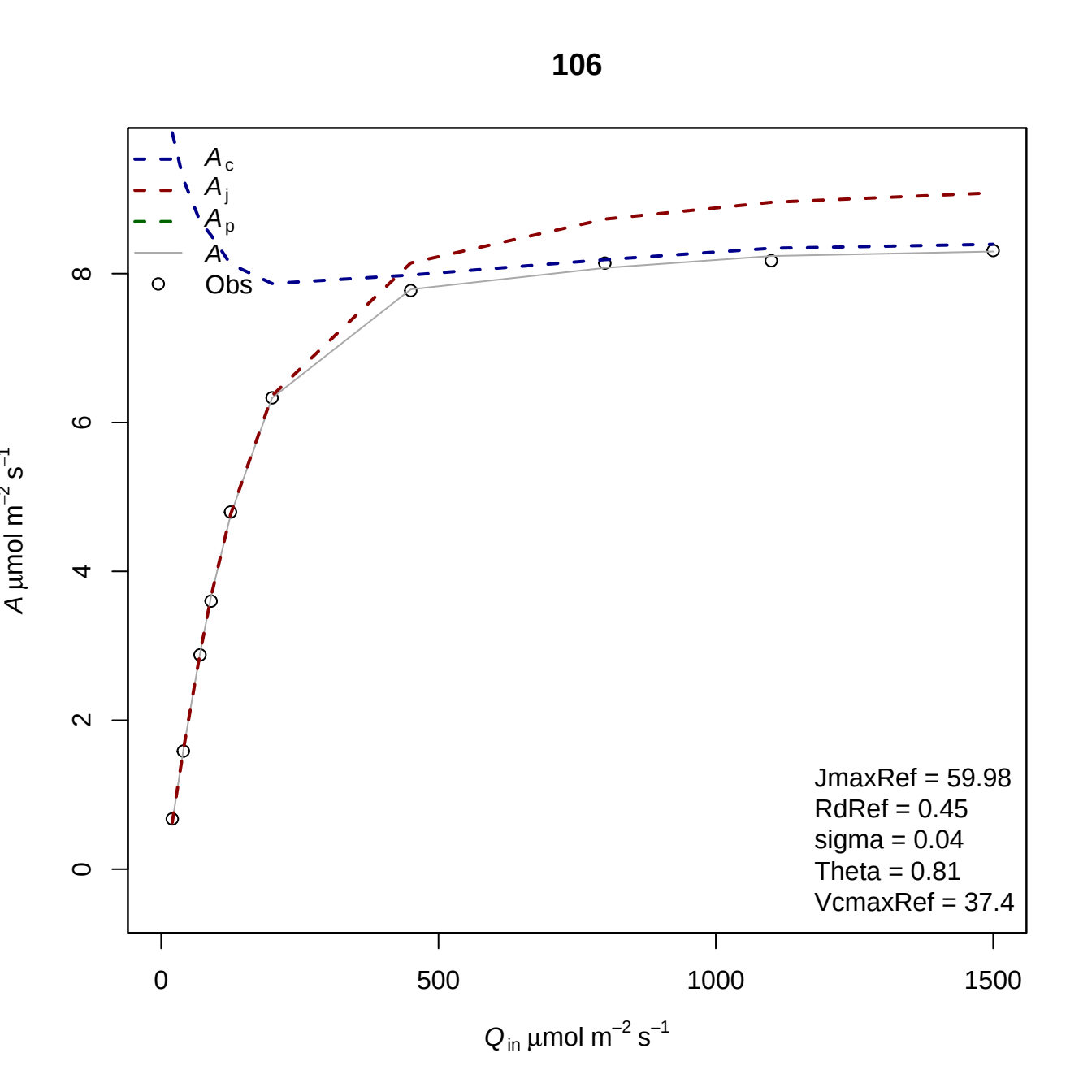

93
A_c
A_j
A_p
A
Obs
JmaxRef = 43.88
RdRef = 2.33
sigma = 0.44
Theta = 0.95
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500

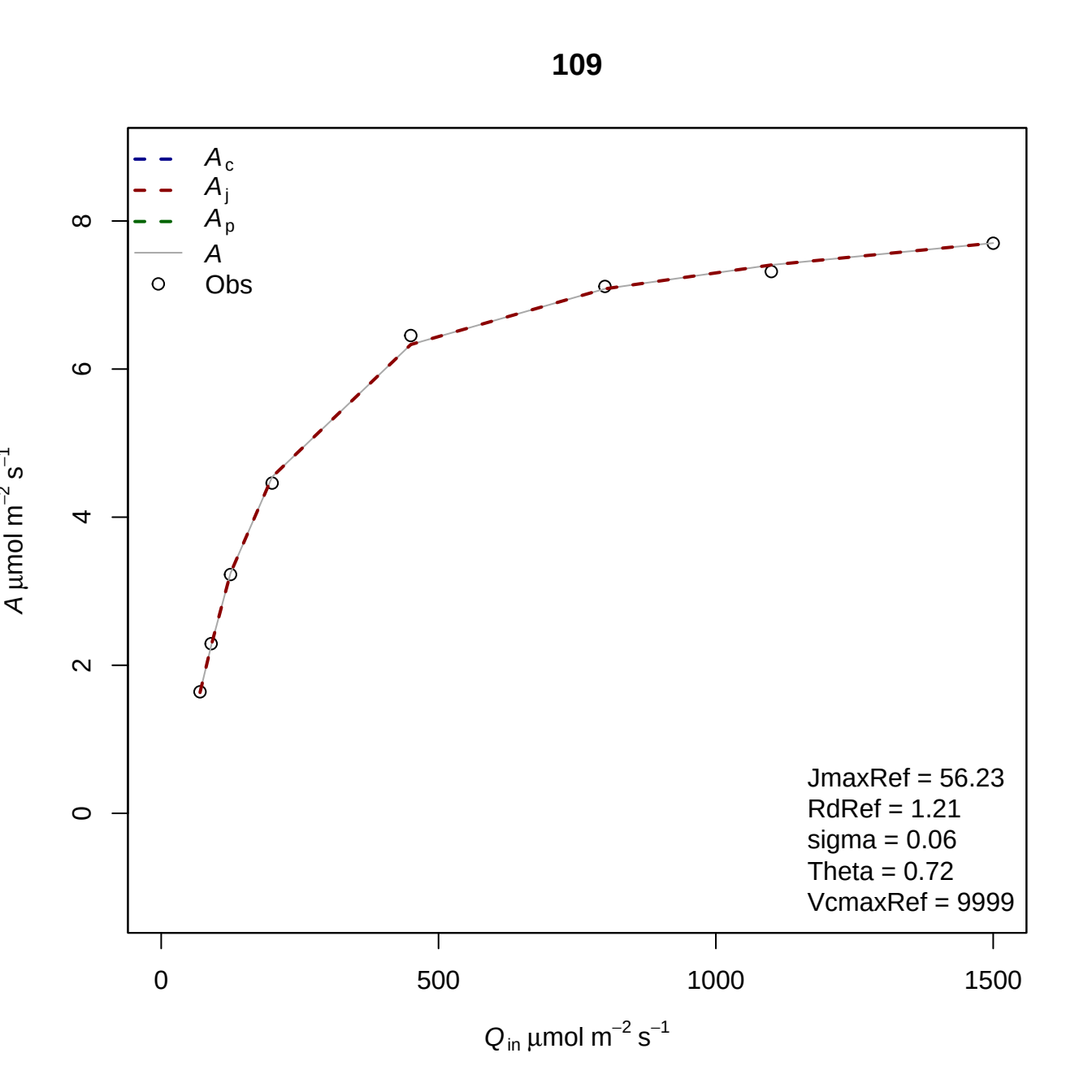

A_c
A_j
A_p
A
Obs
JmaxRef = 190.29
RdRef = 2.51
sigma = 0.15
Theta = −19.78
VcmaxRef = 1647.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

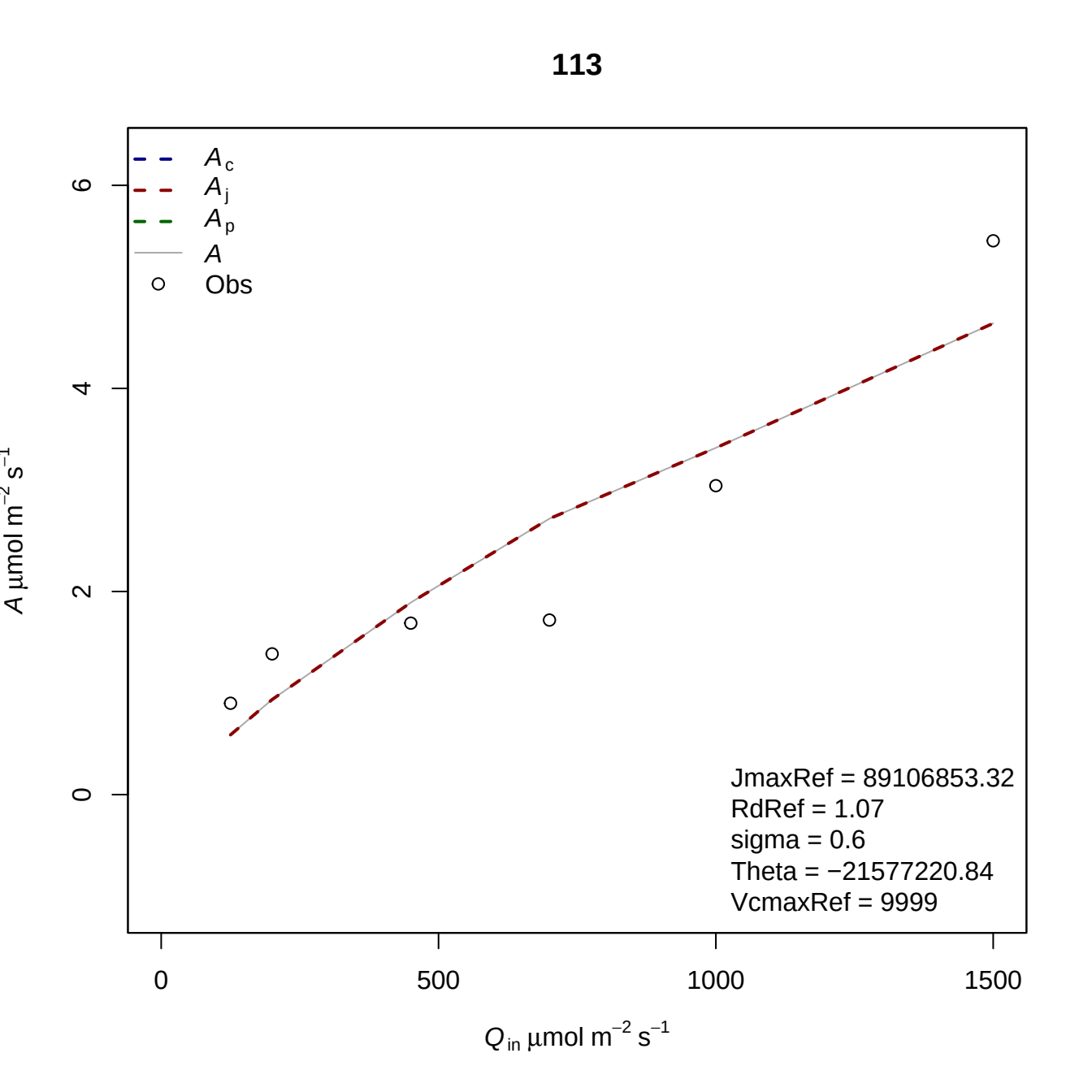

96
A_c
A_j
A_p
A
Obs
JmaxRef = 97.81
RdRef = 0.17
sigma = 0.23
Theta = 0.27
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

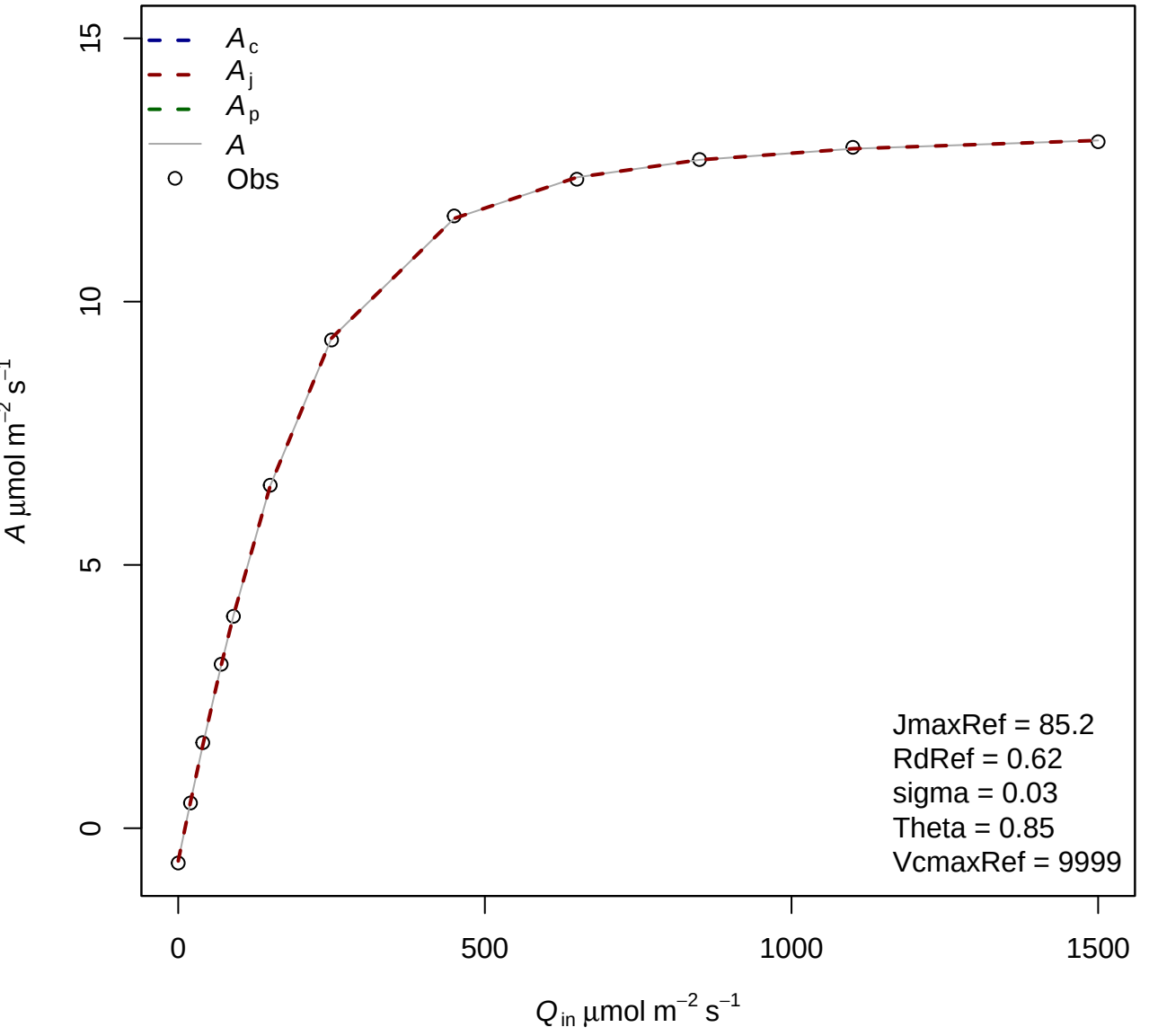

A_c
A_j
A_p
A
Obs
15
10
5
0
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 84.04
RdRef = 0.64
sigma = 0.09
Theta = 0.83
VcmaxRef = 9999

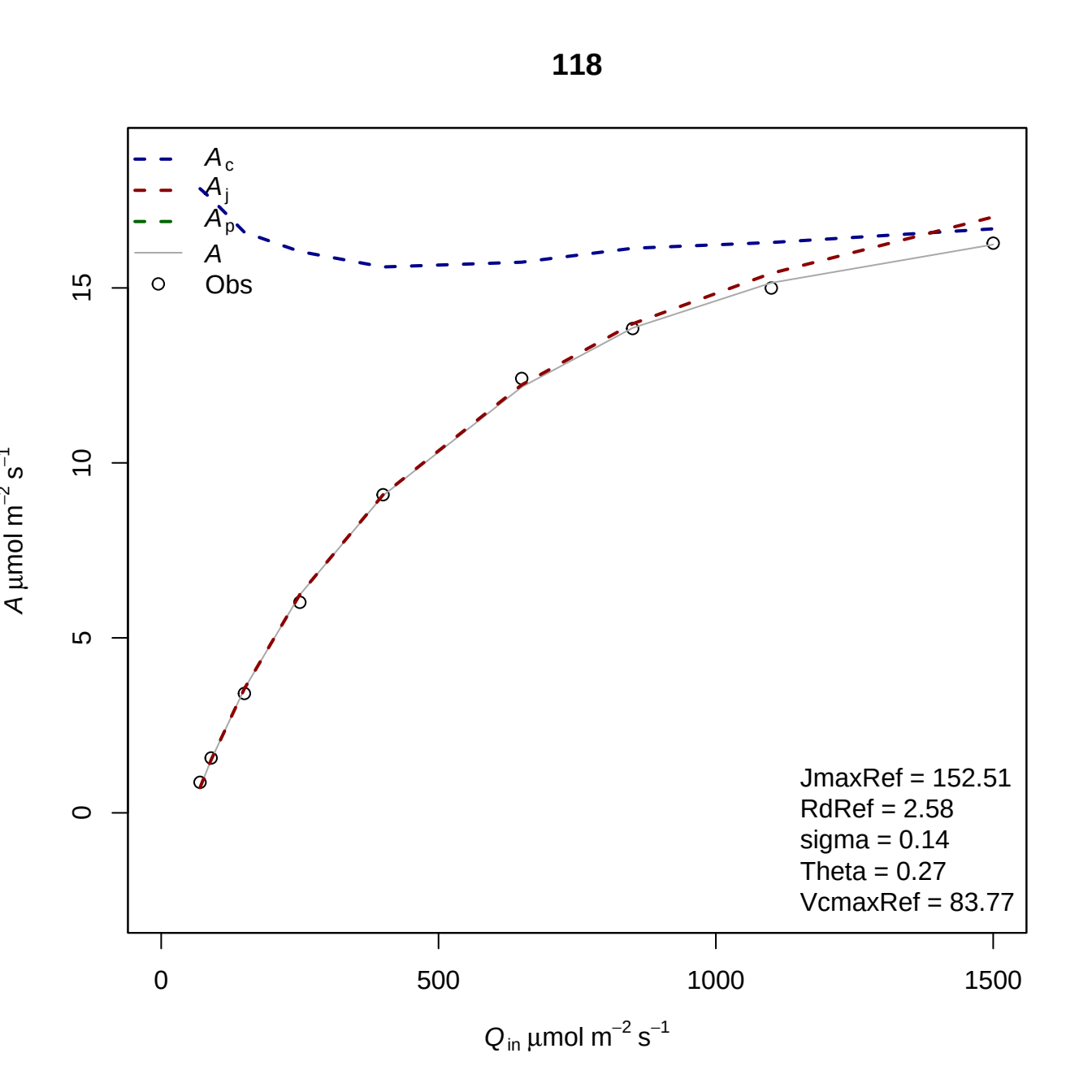

102
A_c
A_j
A_p
A
Obs
JmaxRef = 93.75
RdRef = 2.03
sigma = 0.03
Theta = 0.31
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

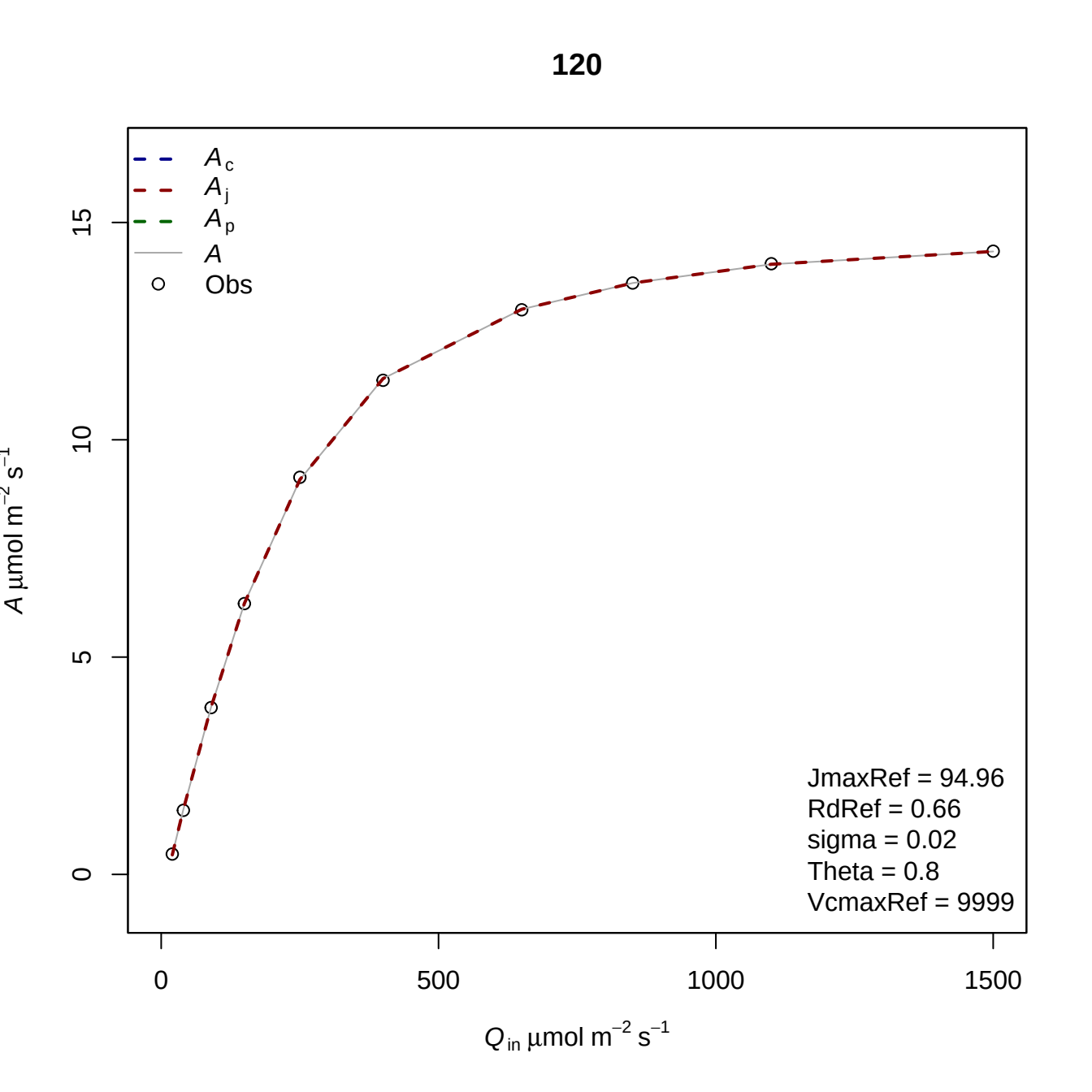

106
A_c
A_j
A_p
A
Obs
JmaxRef = 59.98
RdRef = 0.45
sigma = 0.04
Theta = 0.81
VcmaxRef = 37.4
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

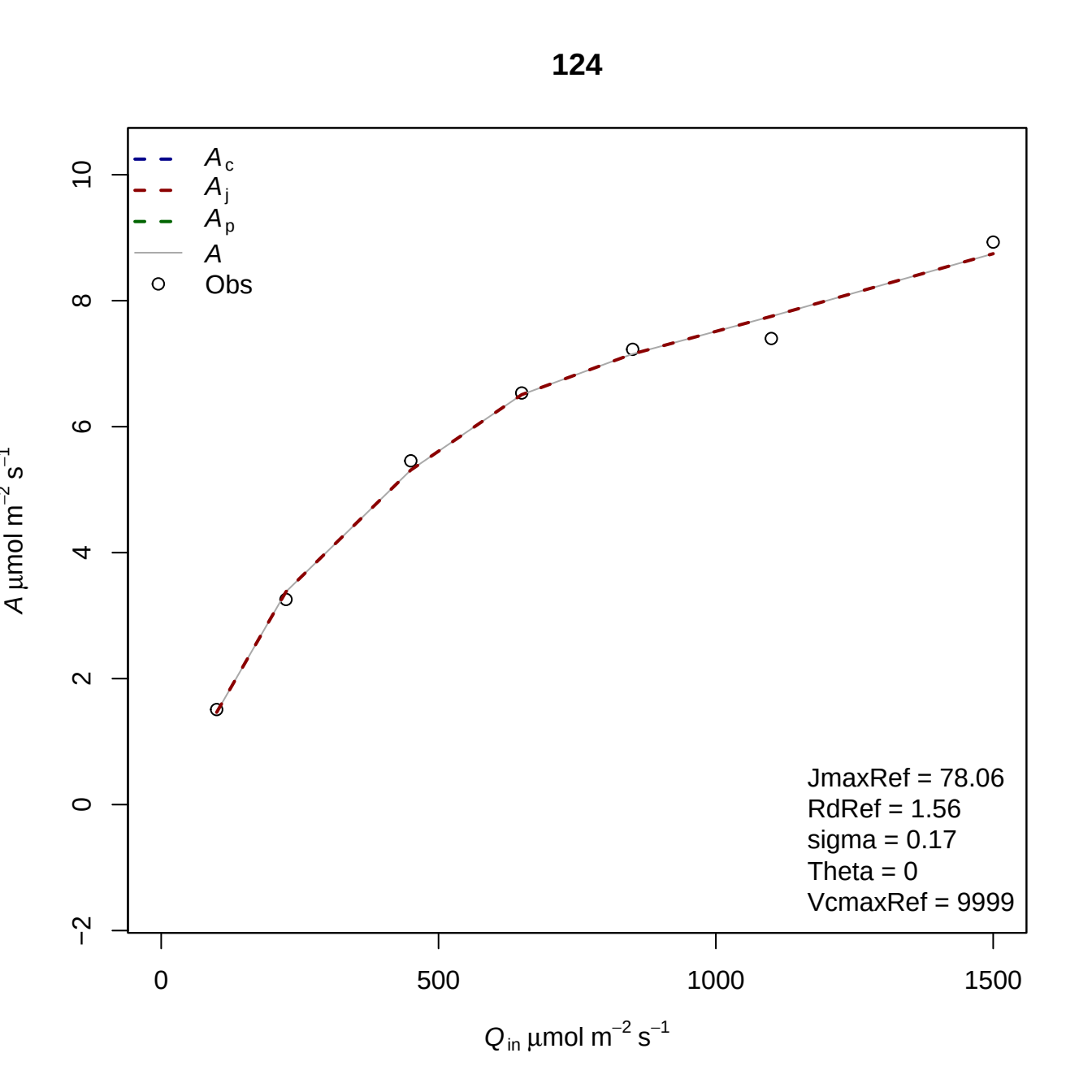

109
A_c
A_j
A_p
A
Obs
JmaxRef = 56.23
RdRef = 1.21
sigma = 0.06
Theta = 0.72
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

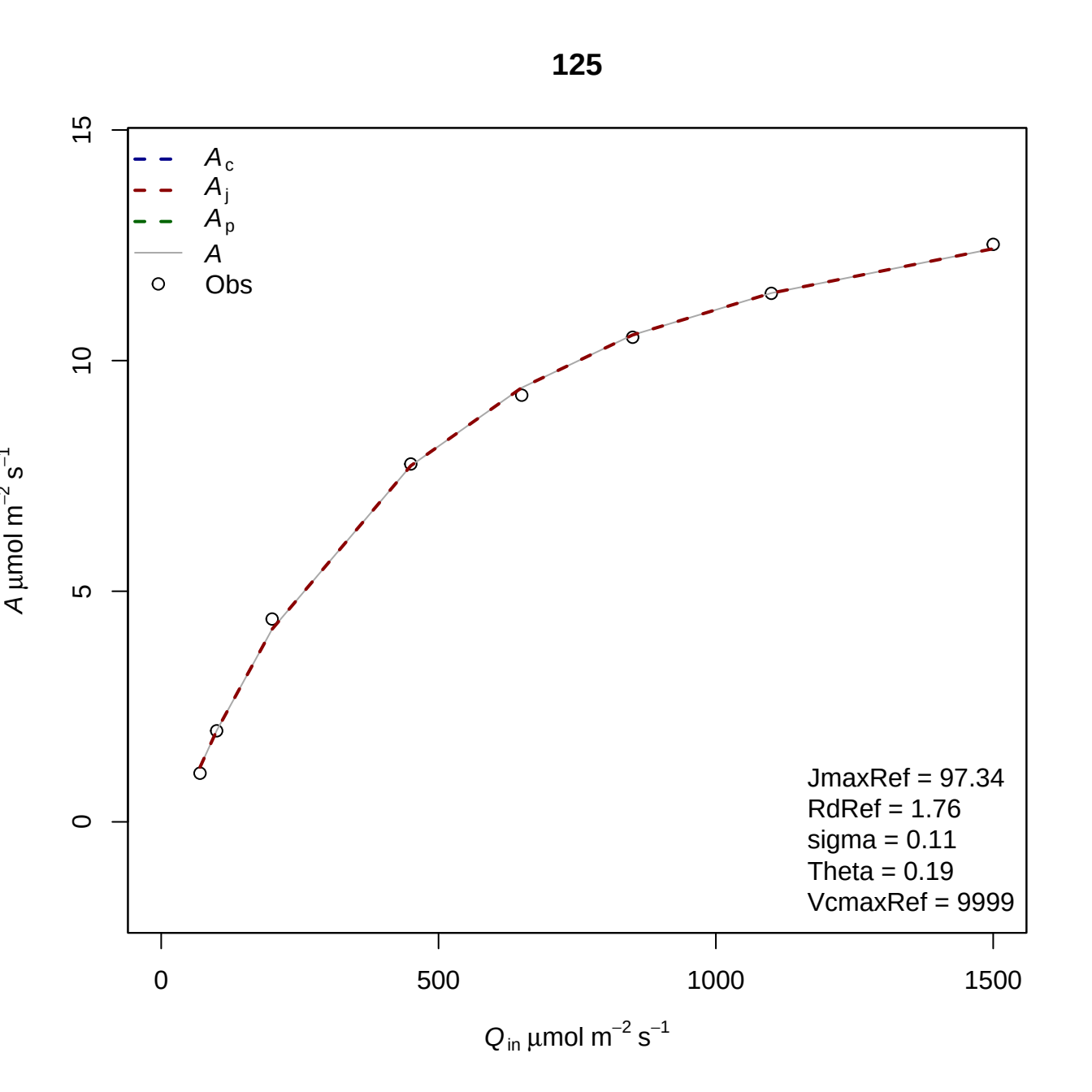

113
A_c
A_j
A_p
A
Obs
JmaxRef = 89106853.32
RdRef = 1.07
sigma = 0.6
Theta = −21577220.84
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

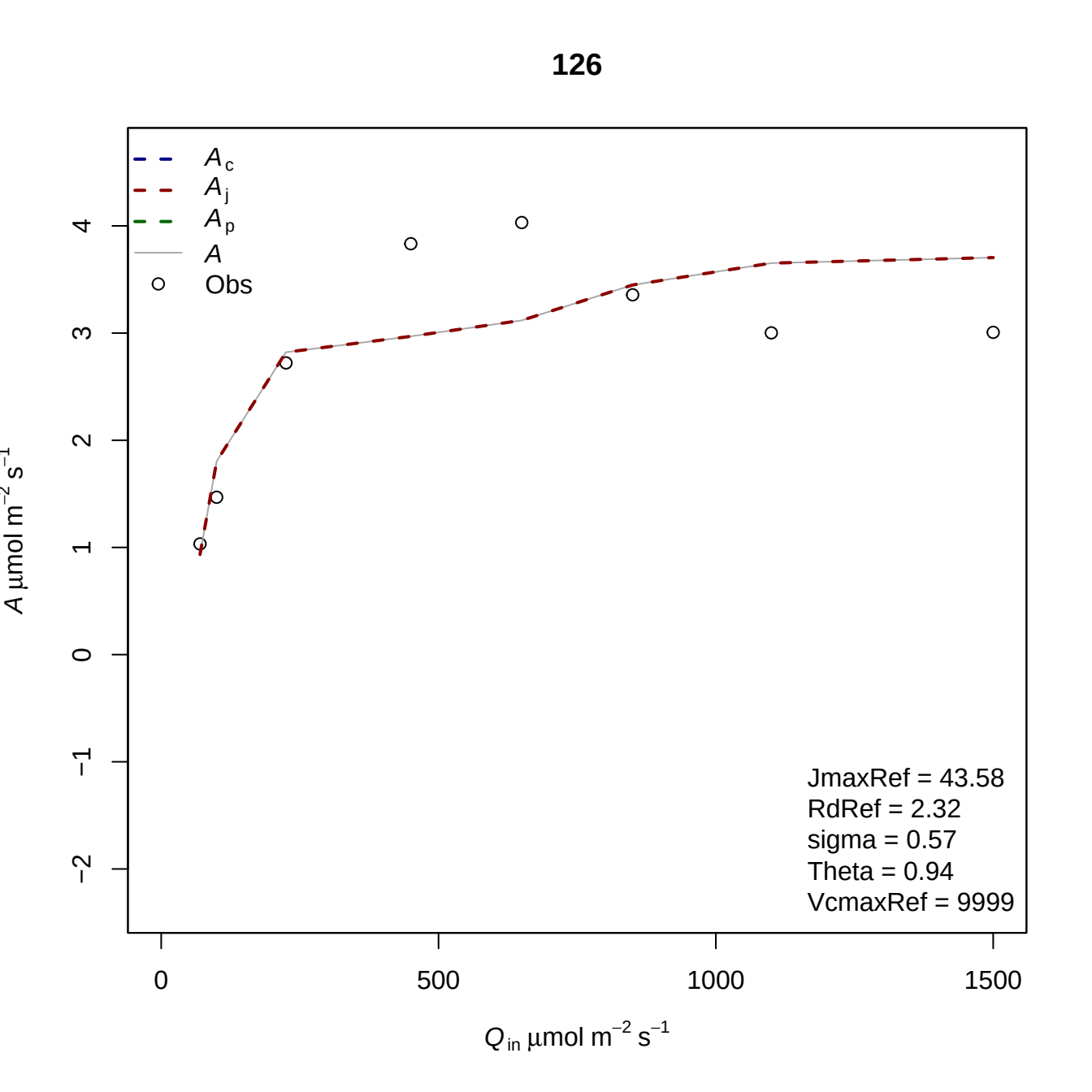

A_c
A_j
A_p
A
Obs
JmaxRef = 85.2
RdRef = 0.62
sigma = 0.03
Theta = 0.85
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

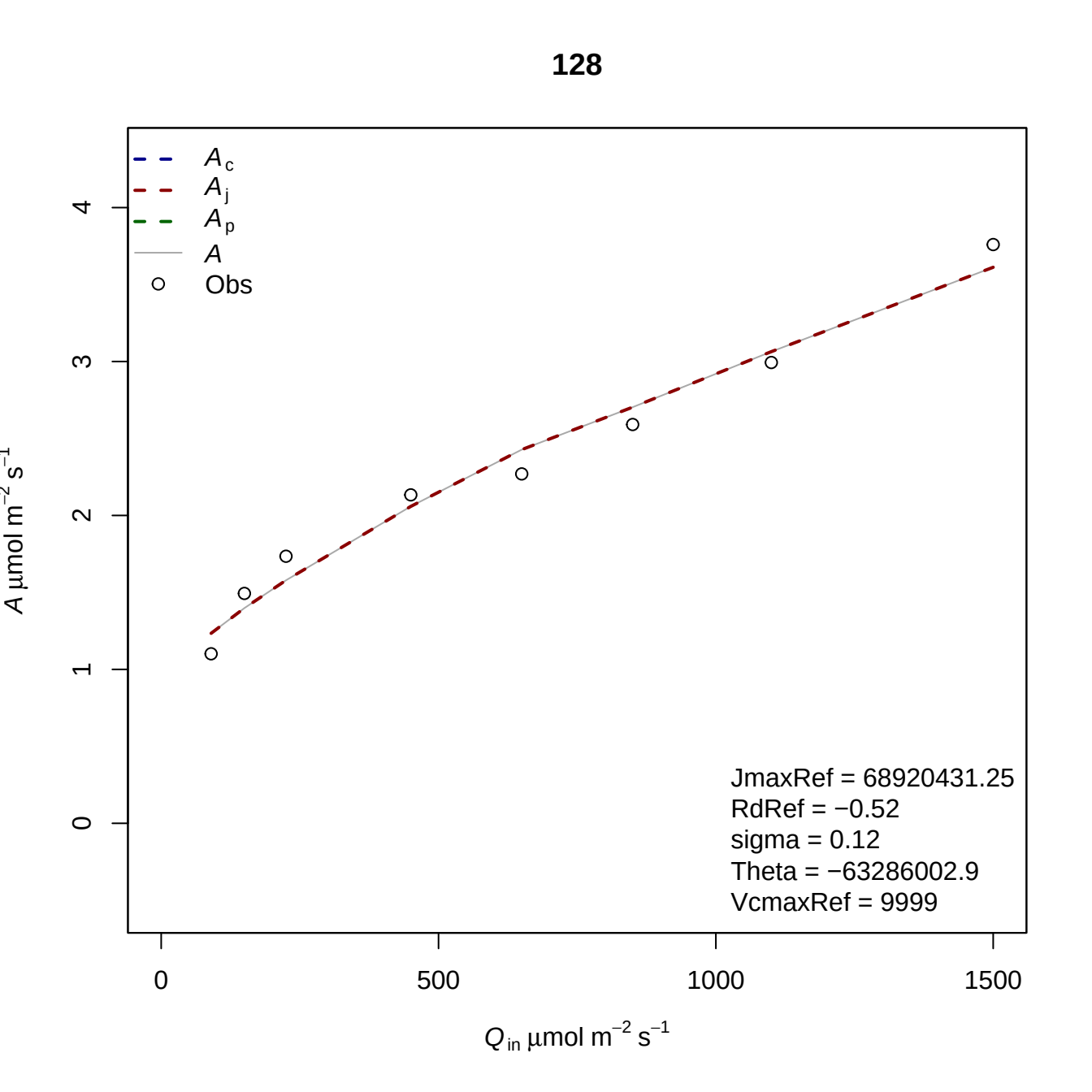

118
A_c
A_j
A_p
A
Obs
JmaxRef = 152.51
RdRef = 2.58
sigma = 0.14
Theta = 0.27
VcmaxRef = 83.77
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

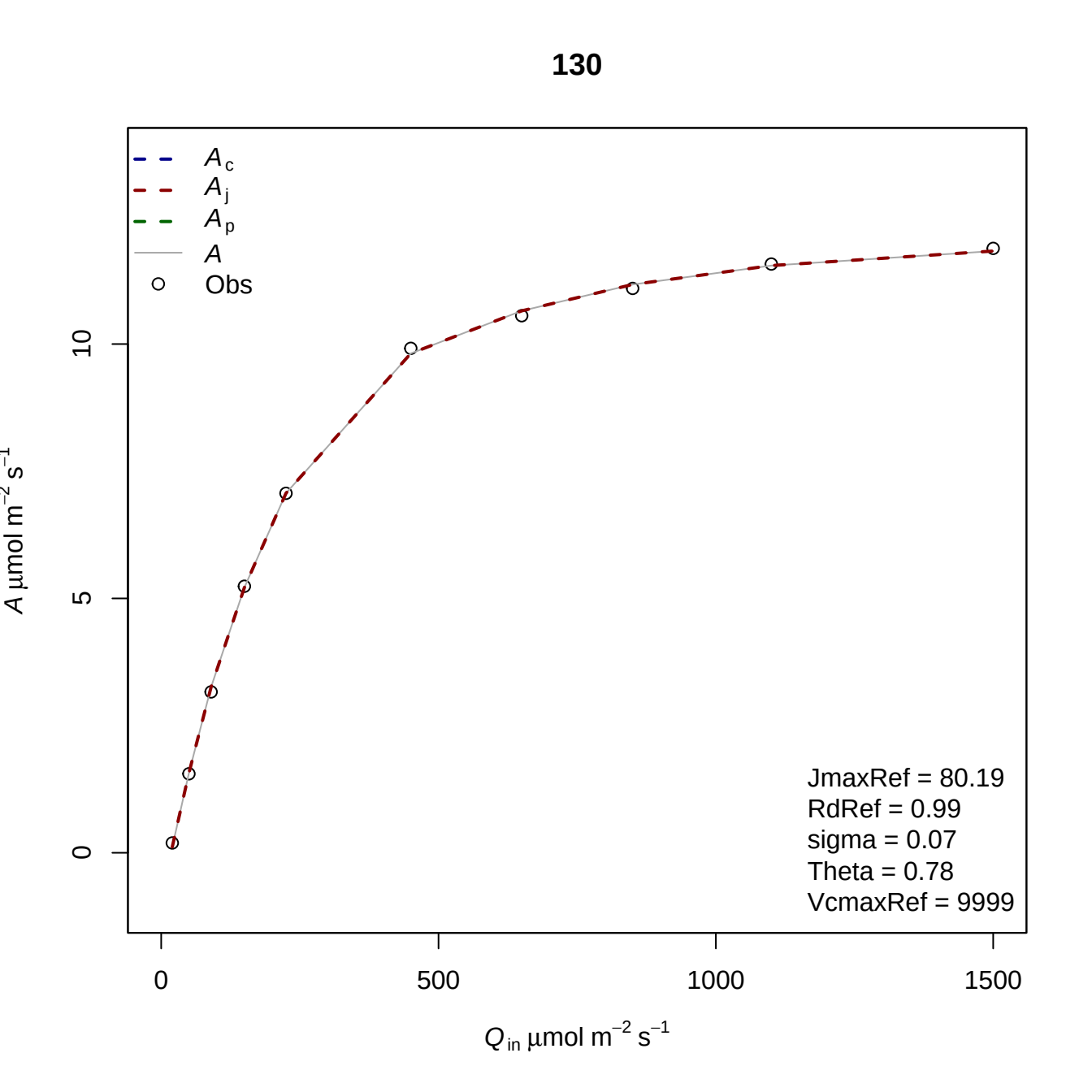

120
A_c
A_j
A_p
A
Obs
JmaxRef = 94.96
RdRef = 0.66
sigma = 0.02
Theta = 0.8
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

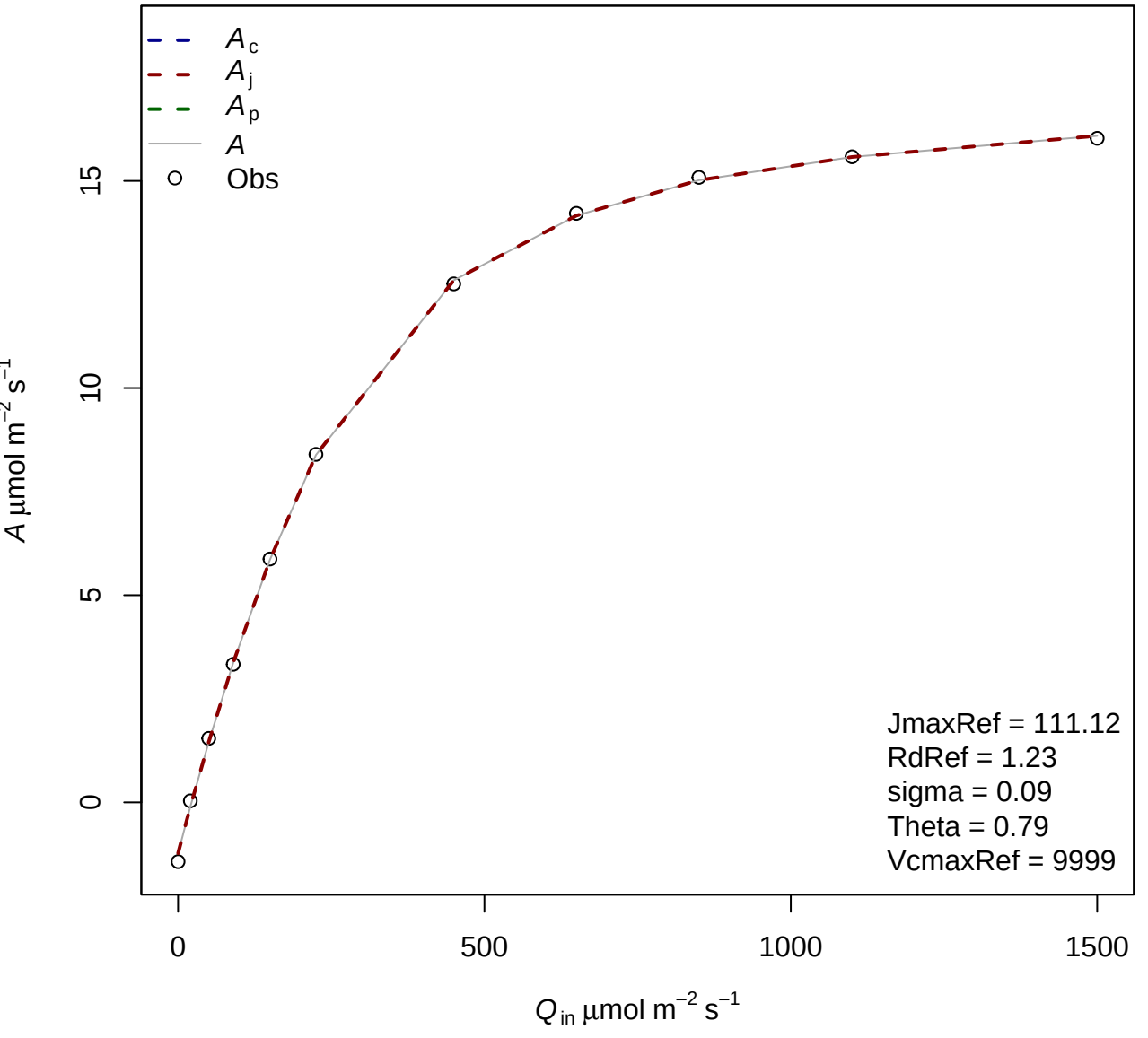

124
A_c
A_j
A_p
A
Obs
JmaxRef = 78.06
RdRef = 1.56
sigma = 0.17
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

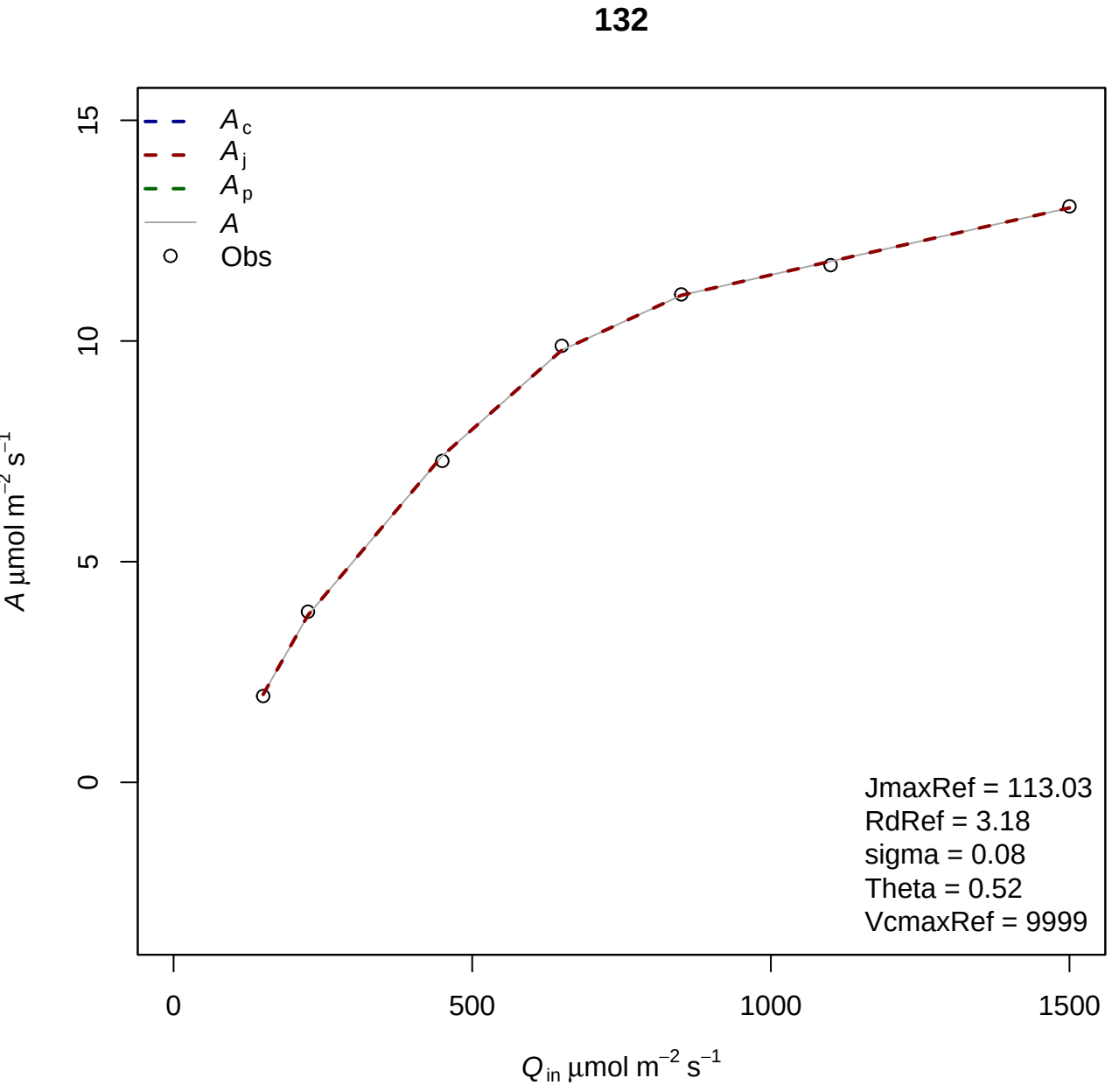

125
A_c
A_j
A_p
A
Obs
JmaxRef = 97.34
RdRef = 1.76
sigma = 0.11
Theta = 0.19
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

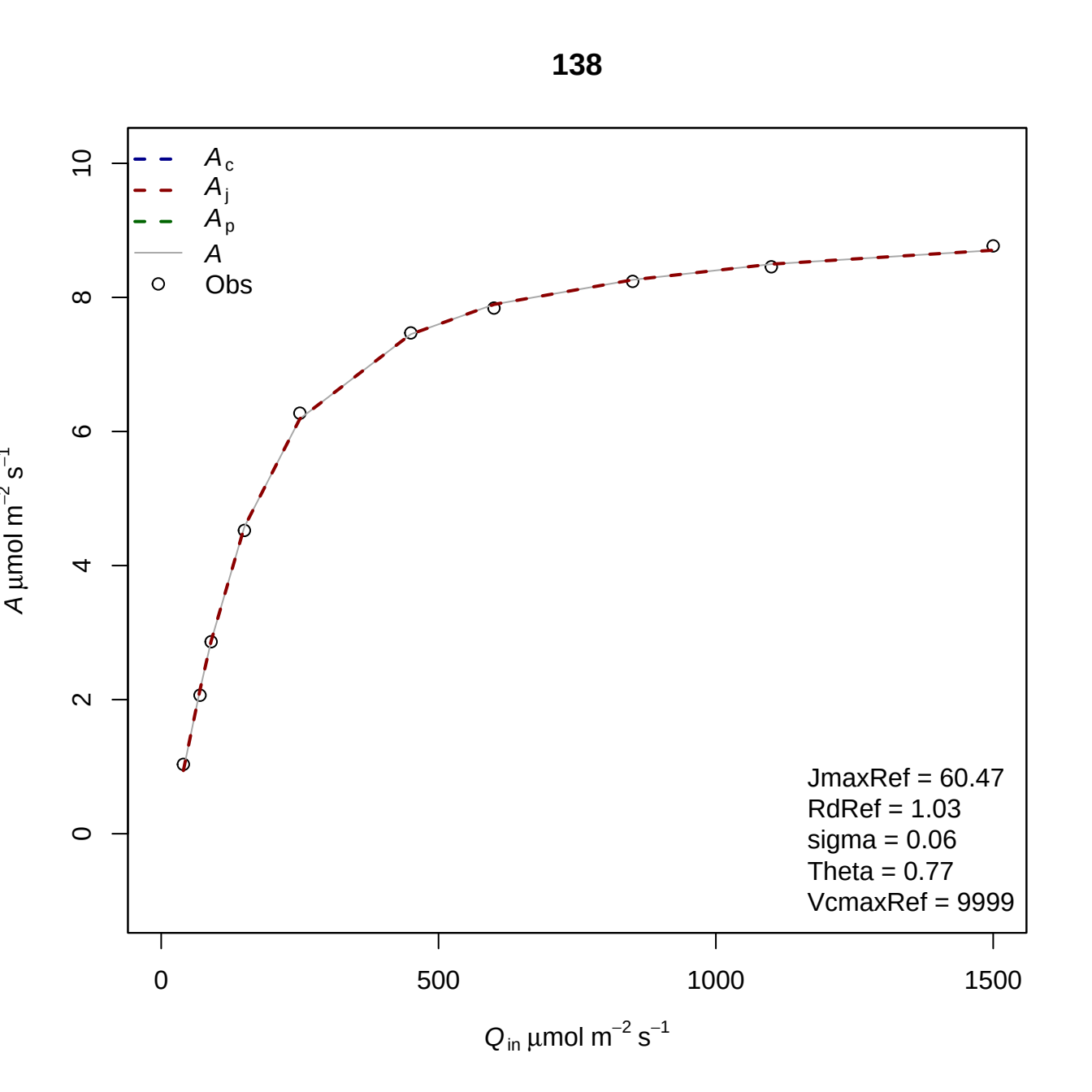

126
A_c
A_j
A_p
A
Obs
JmaxRef = 43.58
RdRef = 2.32
sigma = 0.57
Theta = 0.94
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500

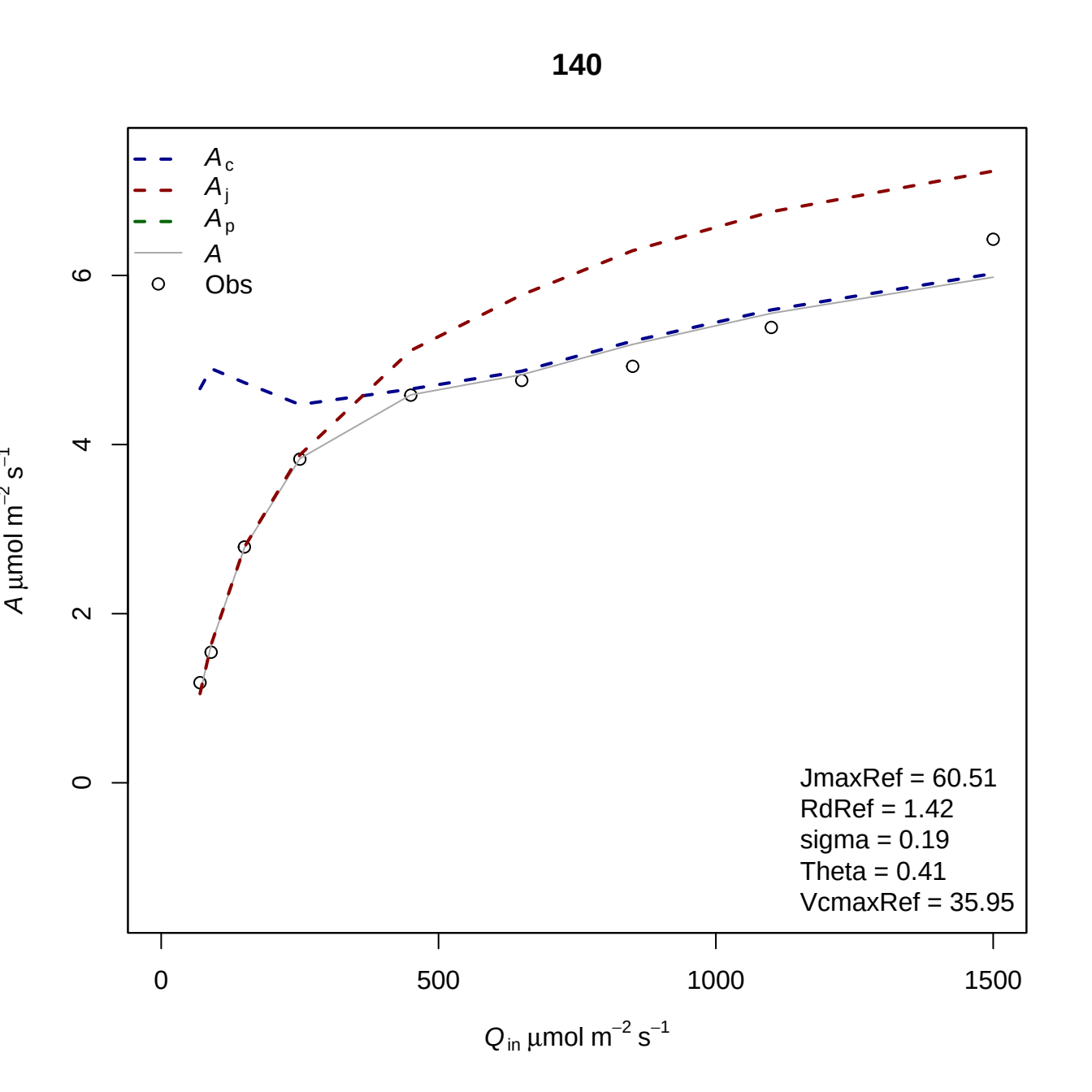

128
A_c
A_j
A_p
A
Obs
JmaxRef = 68920431.25
RdRef = −0.52
sigma = 0.12
Theta = −63286002.9
VcmaxRef = 9999
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

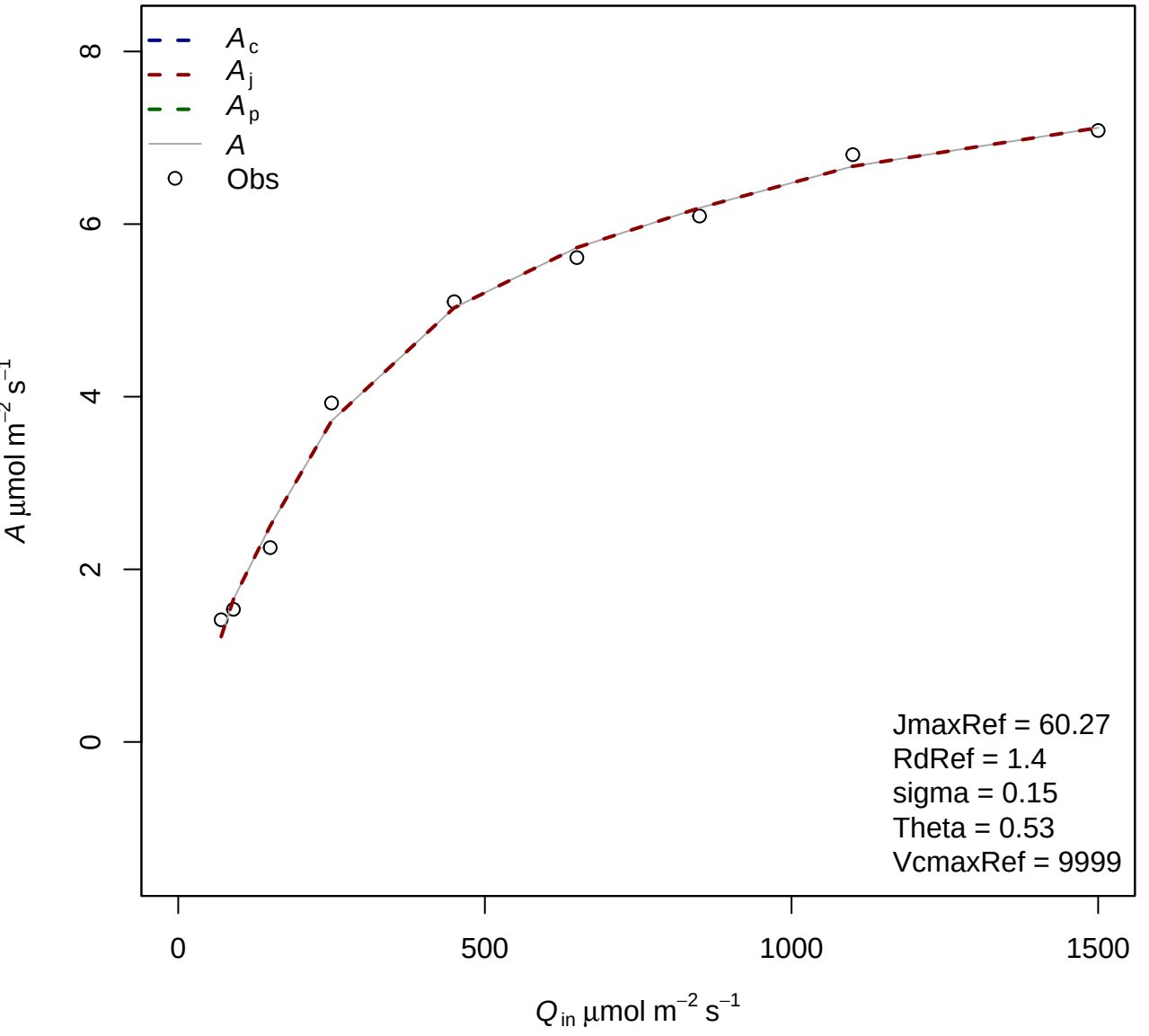

130
A_c
A_j
A_p
A
Obs
JmaxRef = 80.19
RdRef = 0.99
sigma = 0.07
Theta = 0.78
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

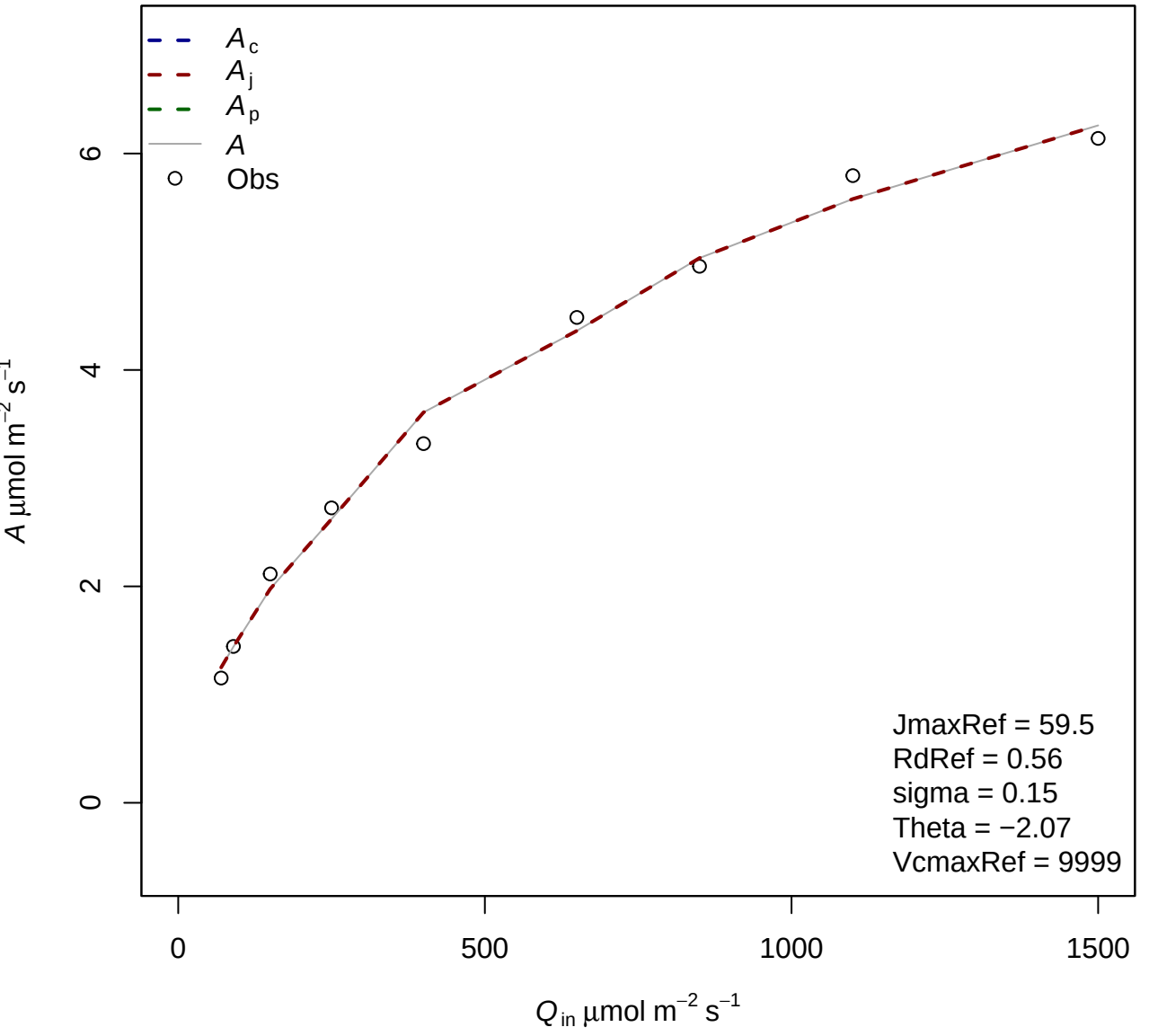

A_c
A_j
A_p
A
Obs
JmaxRef = 111.12
RdRef = 1.23
sigma = 0.09
Theta = 0.79
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

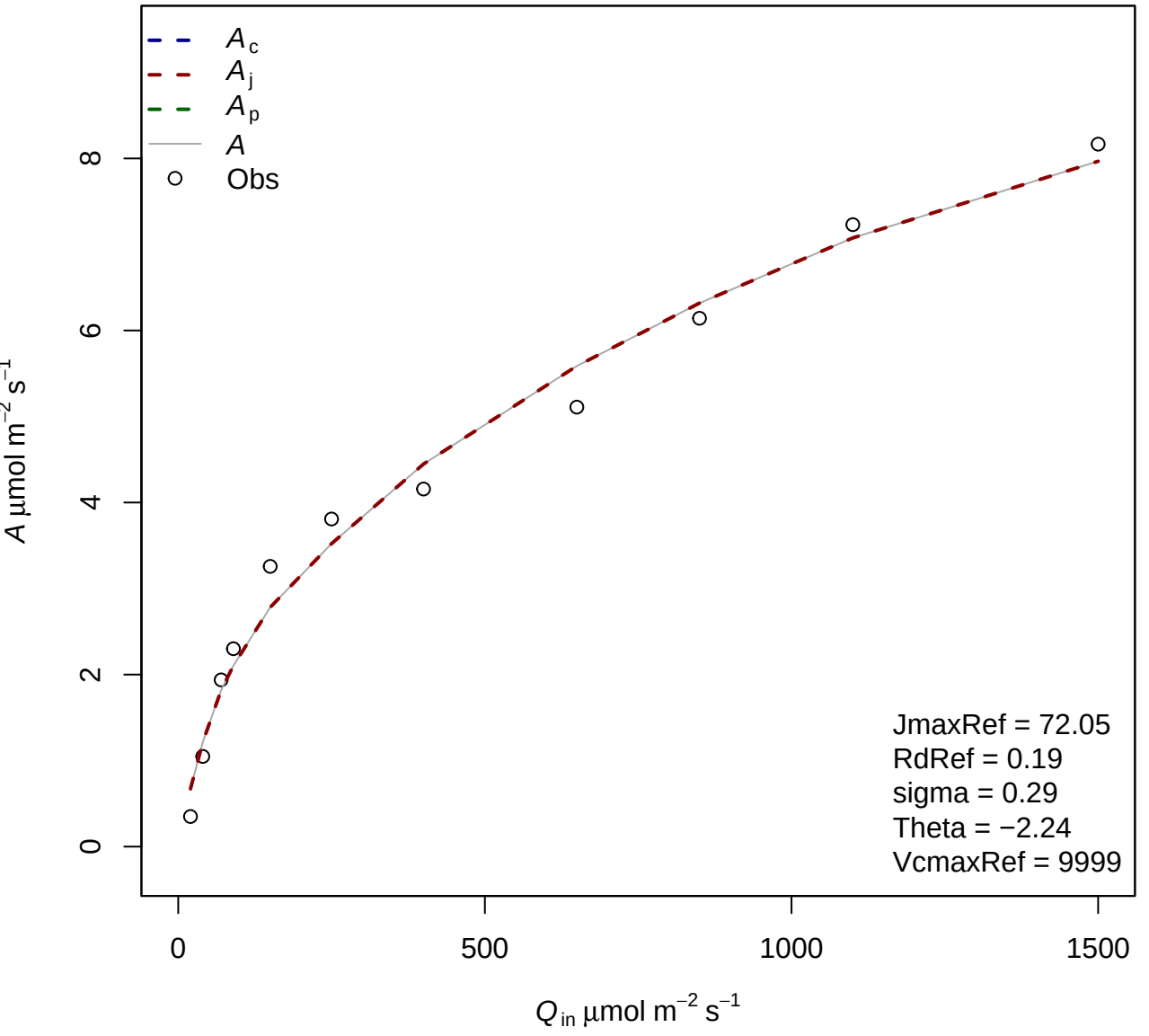

132
A_c
A_j
A_p
A
Obs
15
10
5
0
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 113.03
RdRef = 3.18
sigma = 0.08
Theta = 0.52
VcmaxRef = 9999

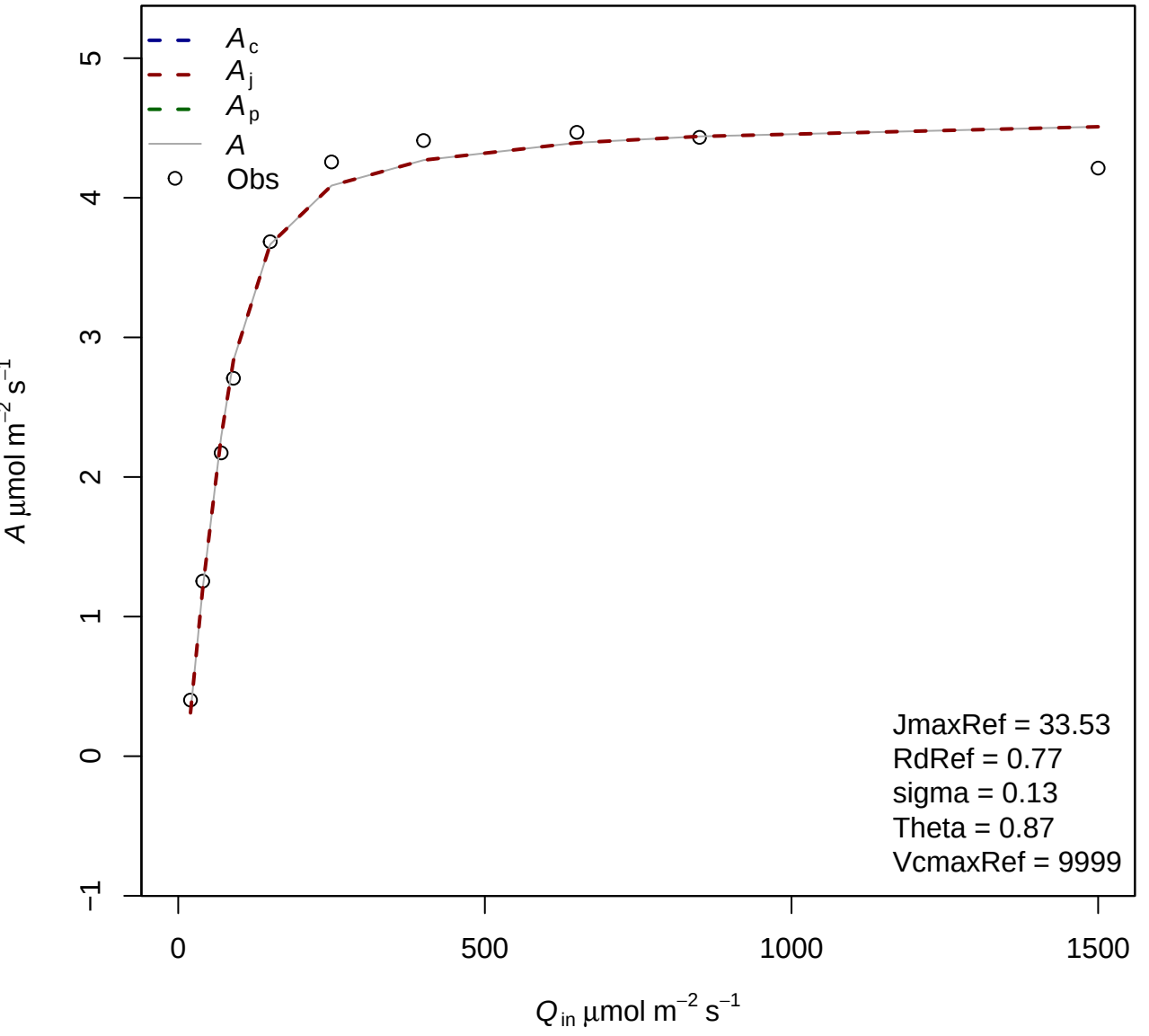

138
A_c
A_j
A_p
A
Obs
JmaxRef = 60.47
RdRef = 1.03
sigma = 0.06
Theta = 0.77
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

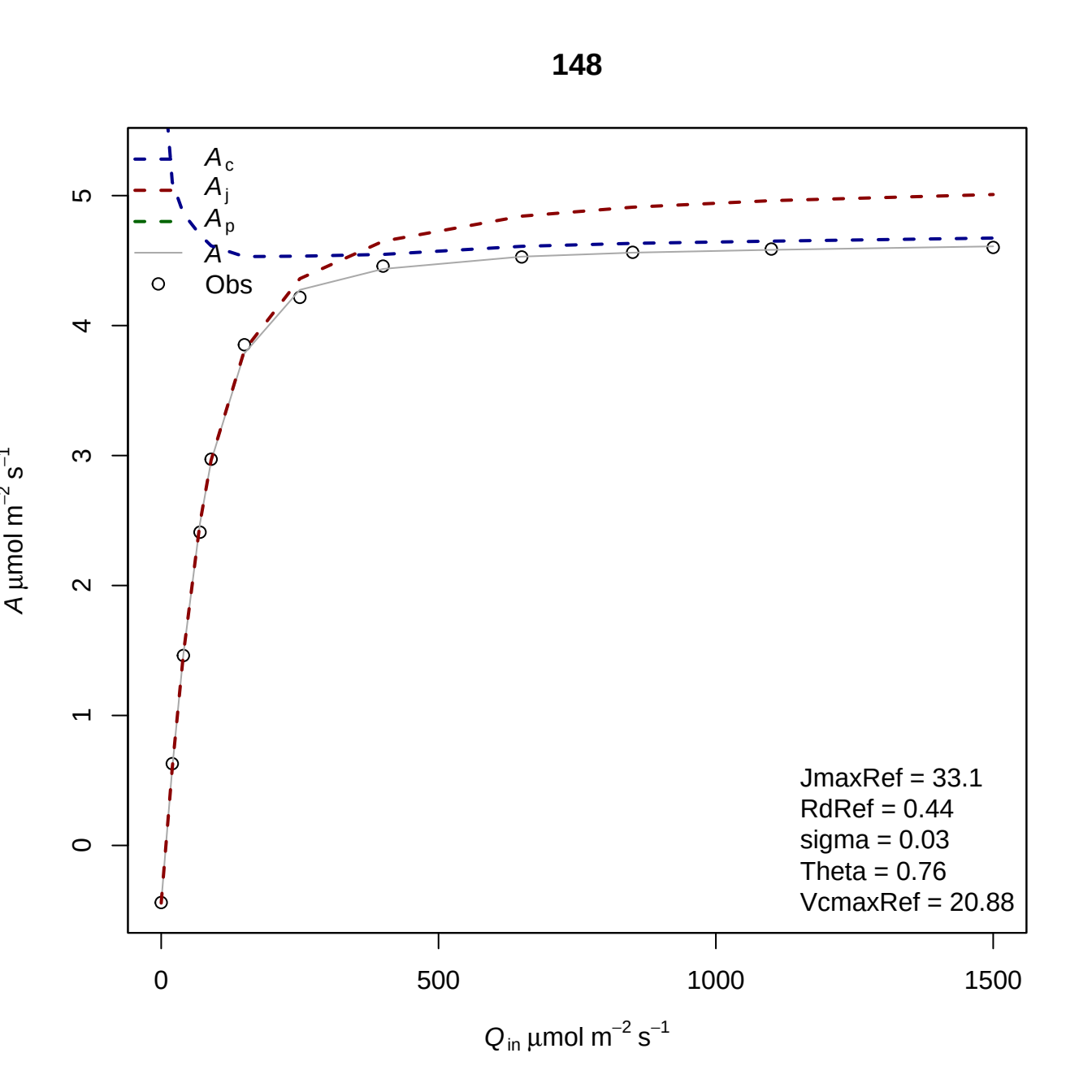

140
A_c
A_j
A_p
A
Obs
JmaxRef = 60.51
RdRef = 1.42
sigma = 0.19
Theta = 0.41
VcmaxRef = 35.95
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

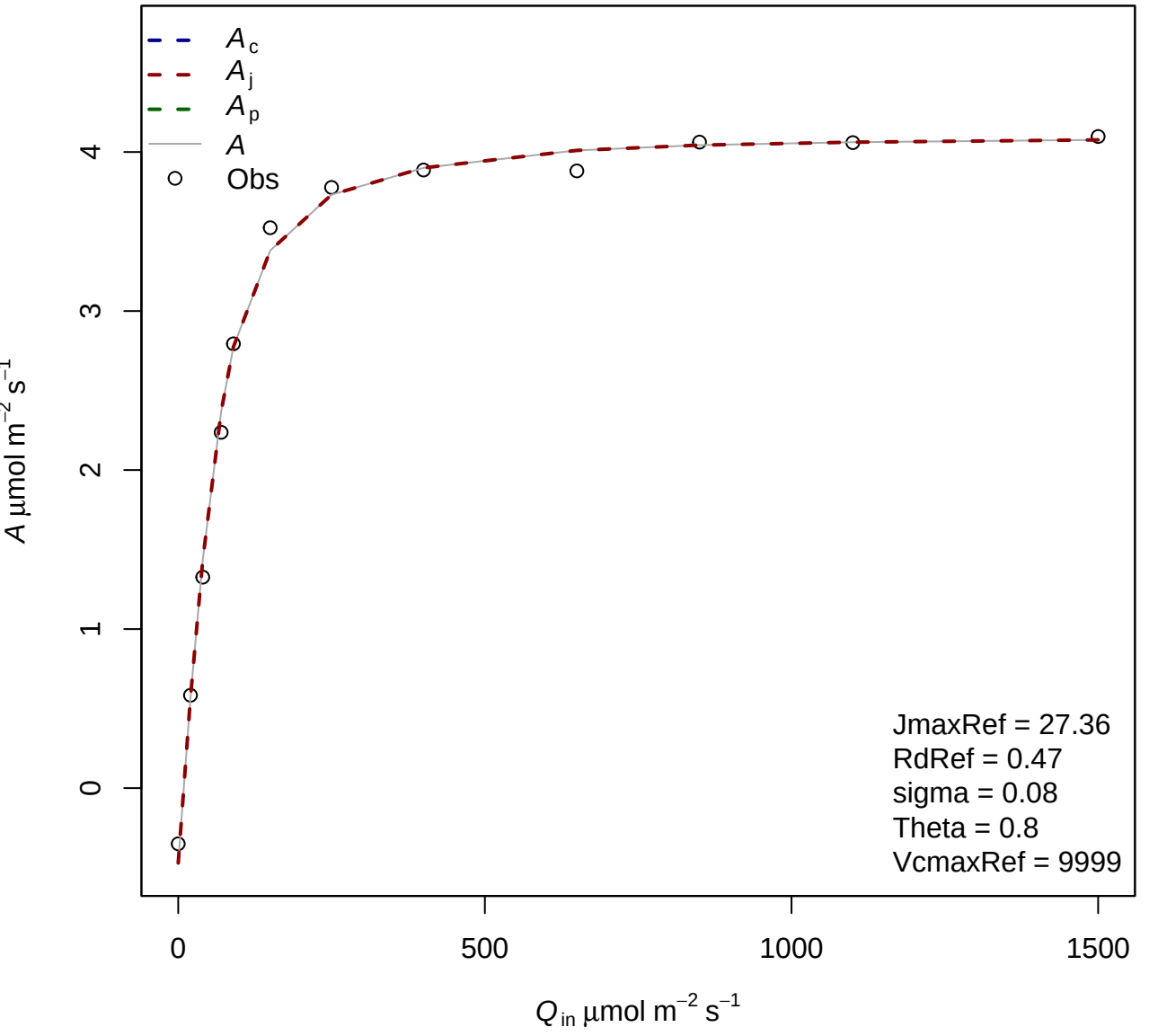

A_c
A_j
A_p
A
Obs
JmaxRef = 60.27
RdRef = 1.4
sigma = 0.15
Theta = 0.53
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

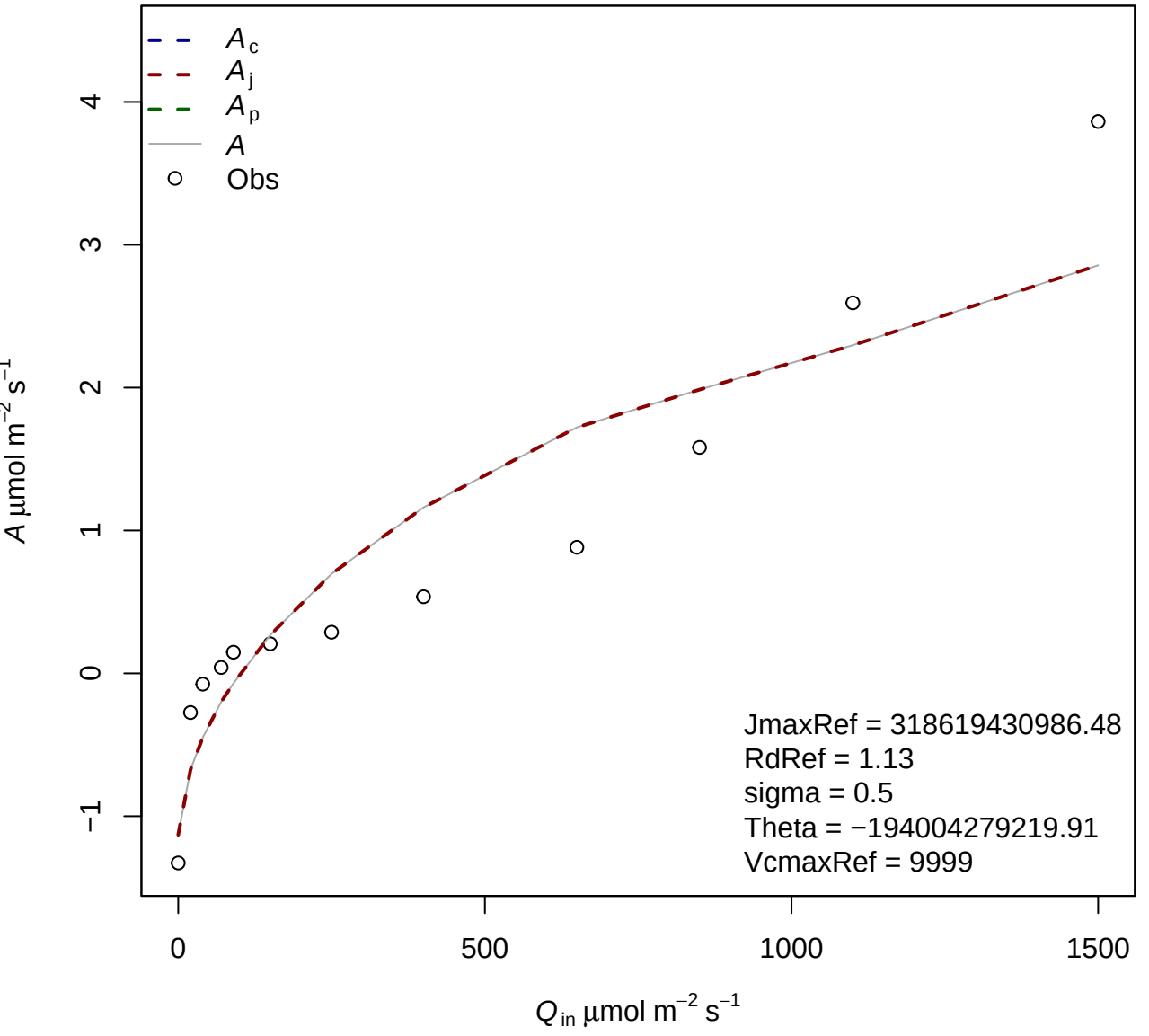

A_c
A_j
A_p
A
Obs
JmaxRef = 59.5
RdRef = 0.56
sigma = 0.15
Theta = −2.07
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

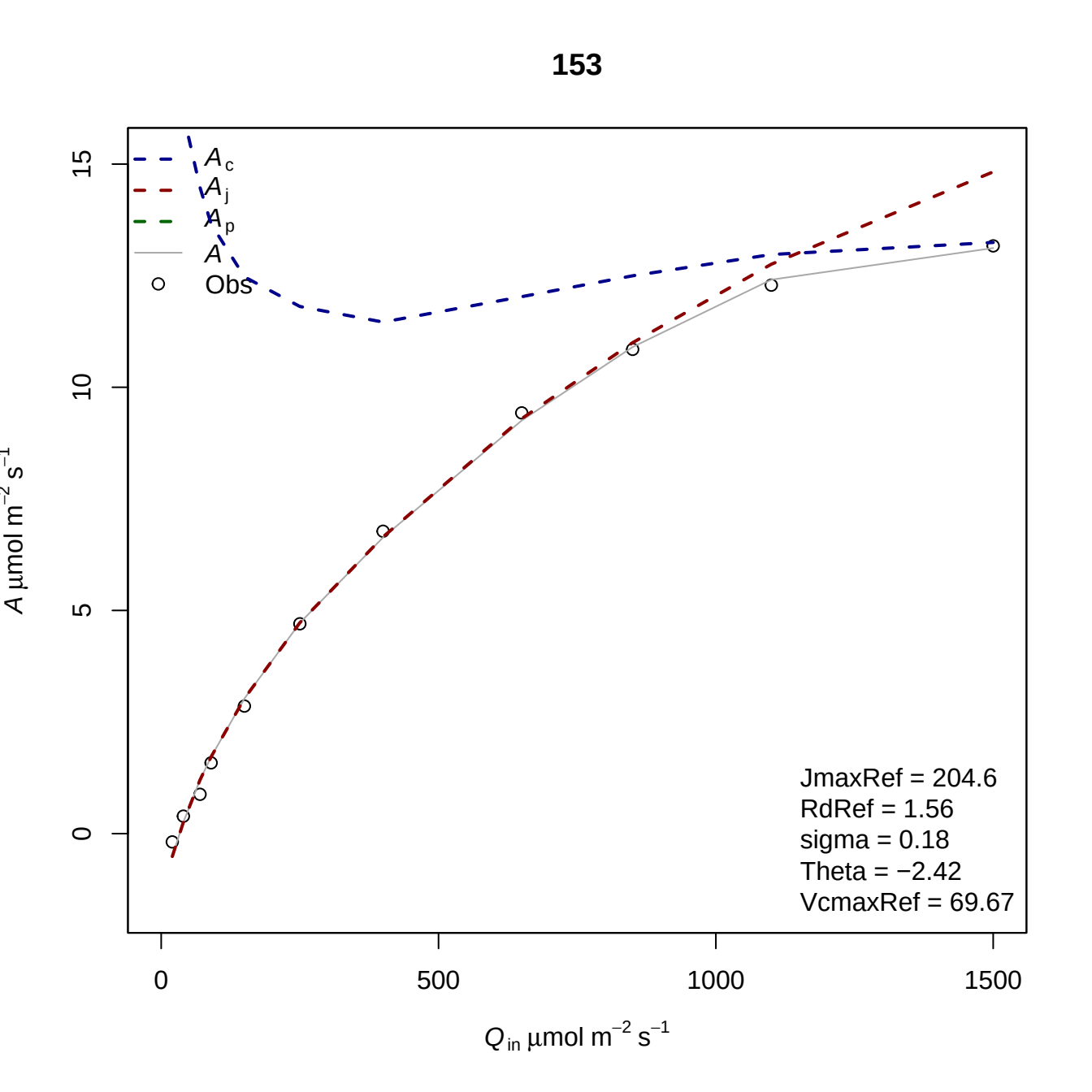

A_c
A_j
A_p
A
Obs
JmaxRef = 72.05
RdRef = 0.19
sigma = 0.29
Theta = −2.24
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

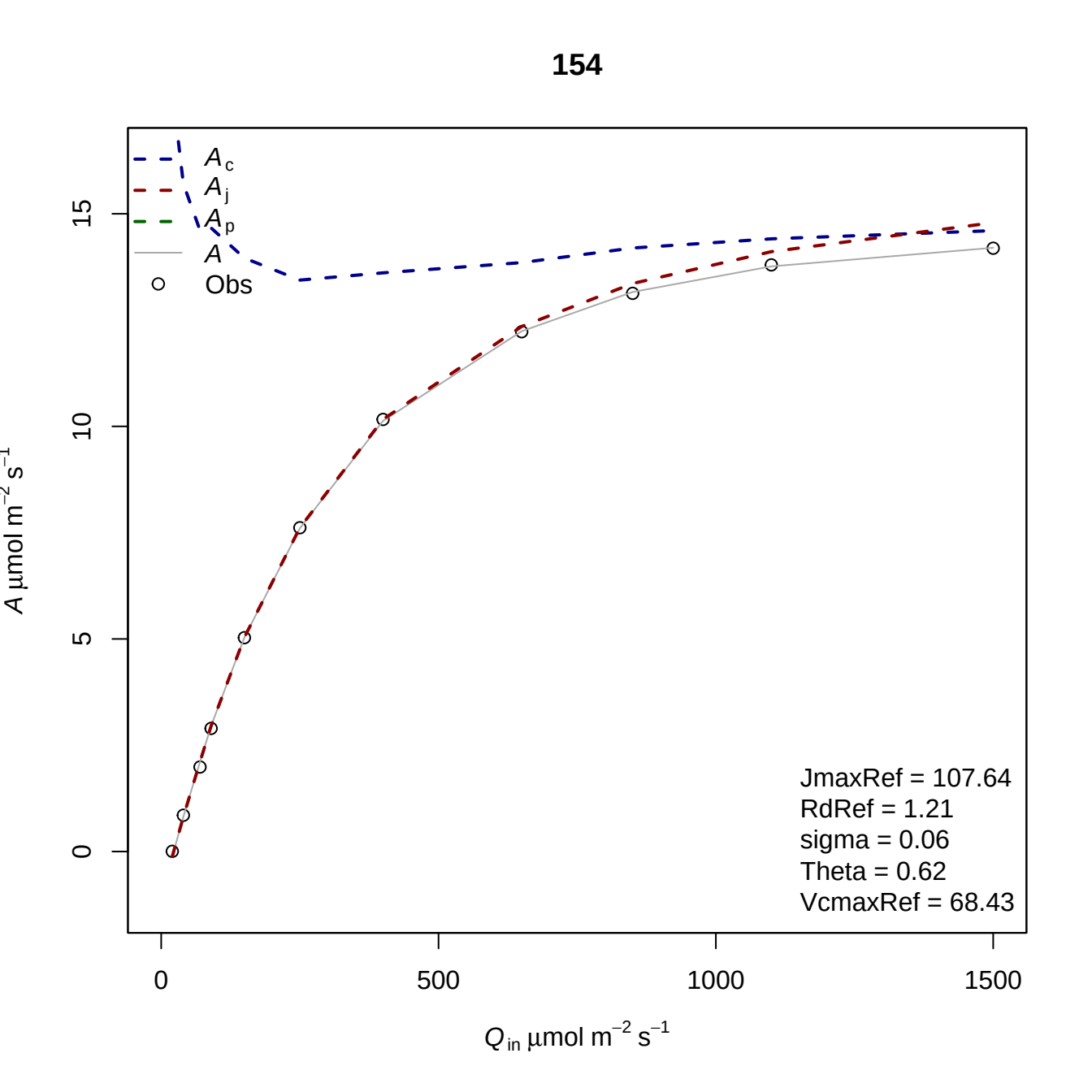

A_c
A_j
A_p
A
Obs
JmaxRef = 33.53
RdRef = 0.77
sigma = 0.13
Theta = 0.87
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

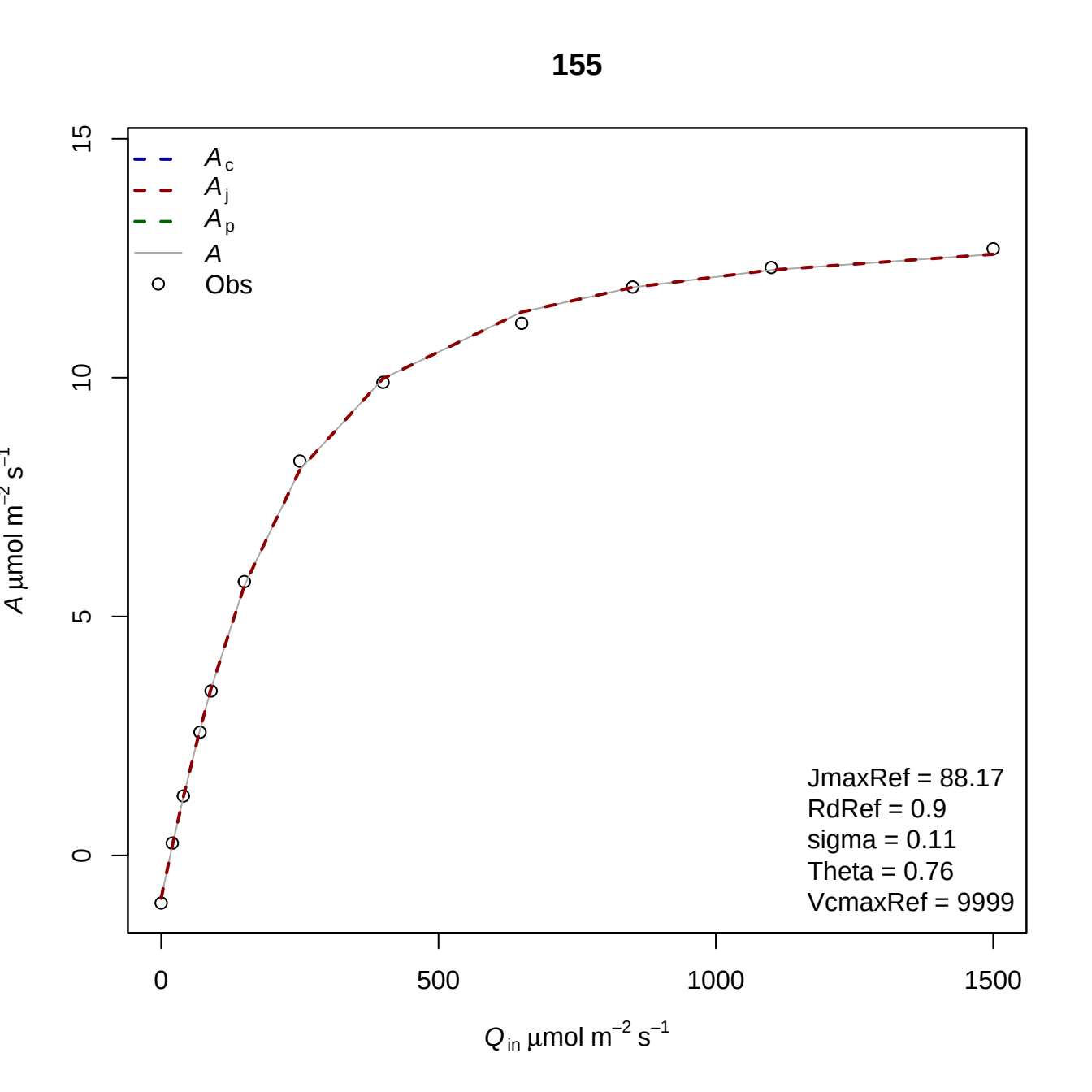

148
A_c
A_j
A_p
A
Obs
JmaxRef = 33.1
RdRef = 0.44
sigma = 0.03
Theta = 0.76
VcmaxRef = 20.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

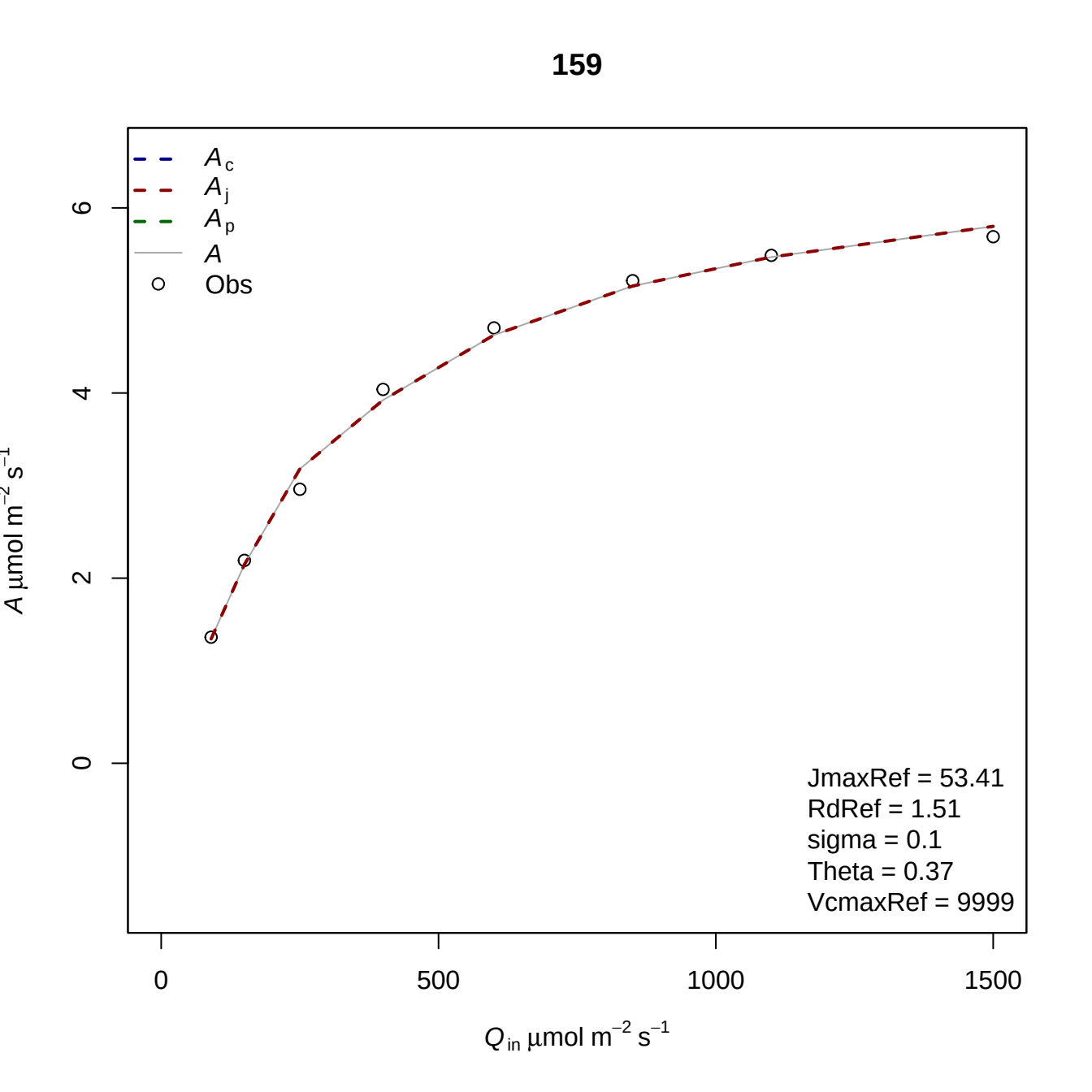

A_c
A_j
A_p
A
Obs
JmaxRef = 27.36
RdRef = 0.47
sigma = 0.08
Theta = 0.8
VcmaxRef = 9999
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

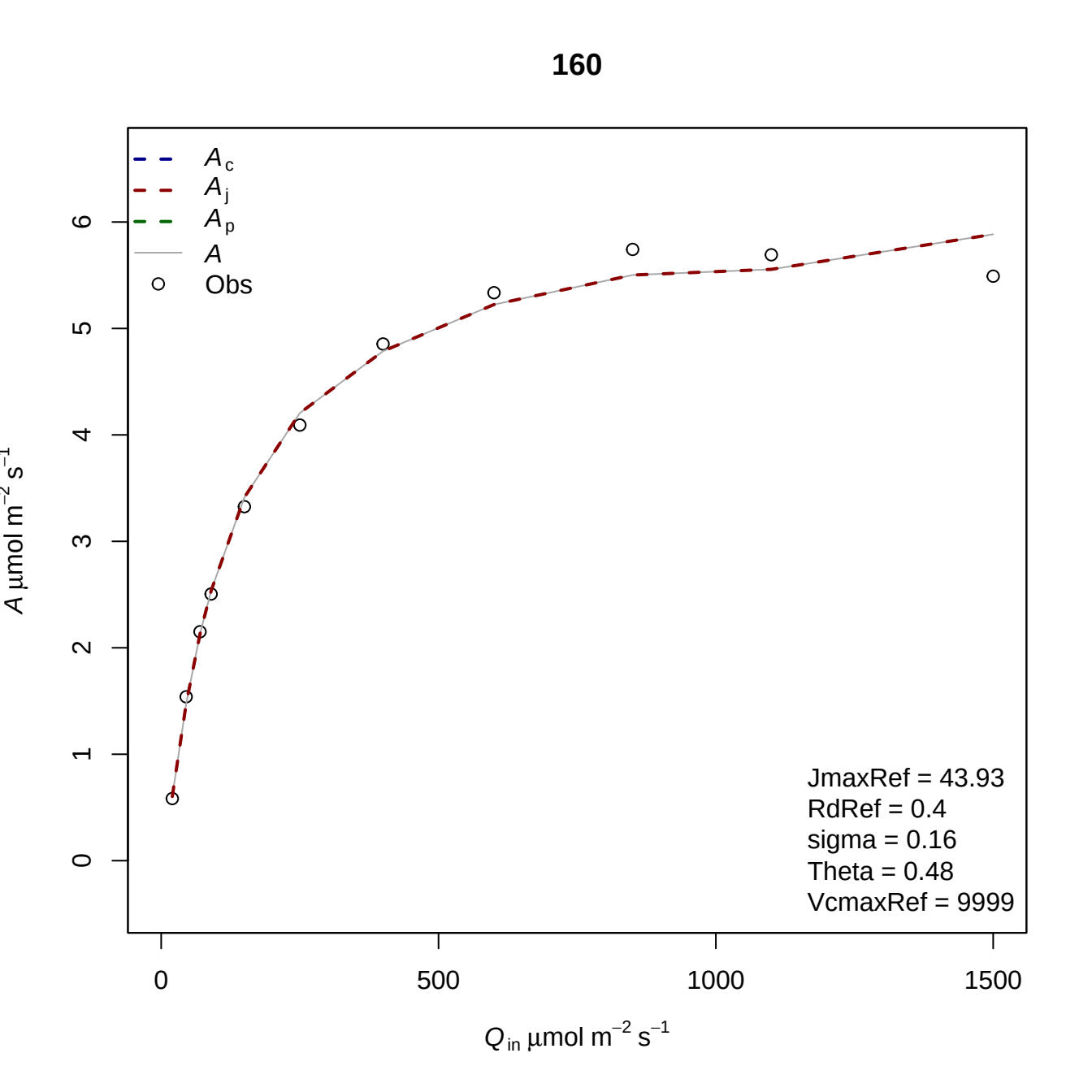

A_c
A_j
A_p
A
Obs
JmaxRef = 318619430986.48
RdRef = 1.13
sigma = 0.5
Theta = −194004279219.91
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

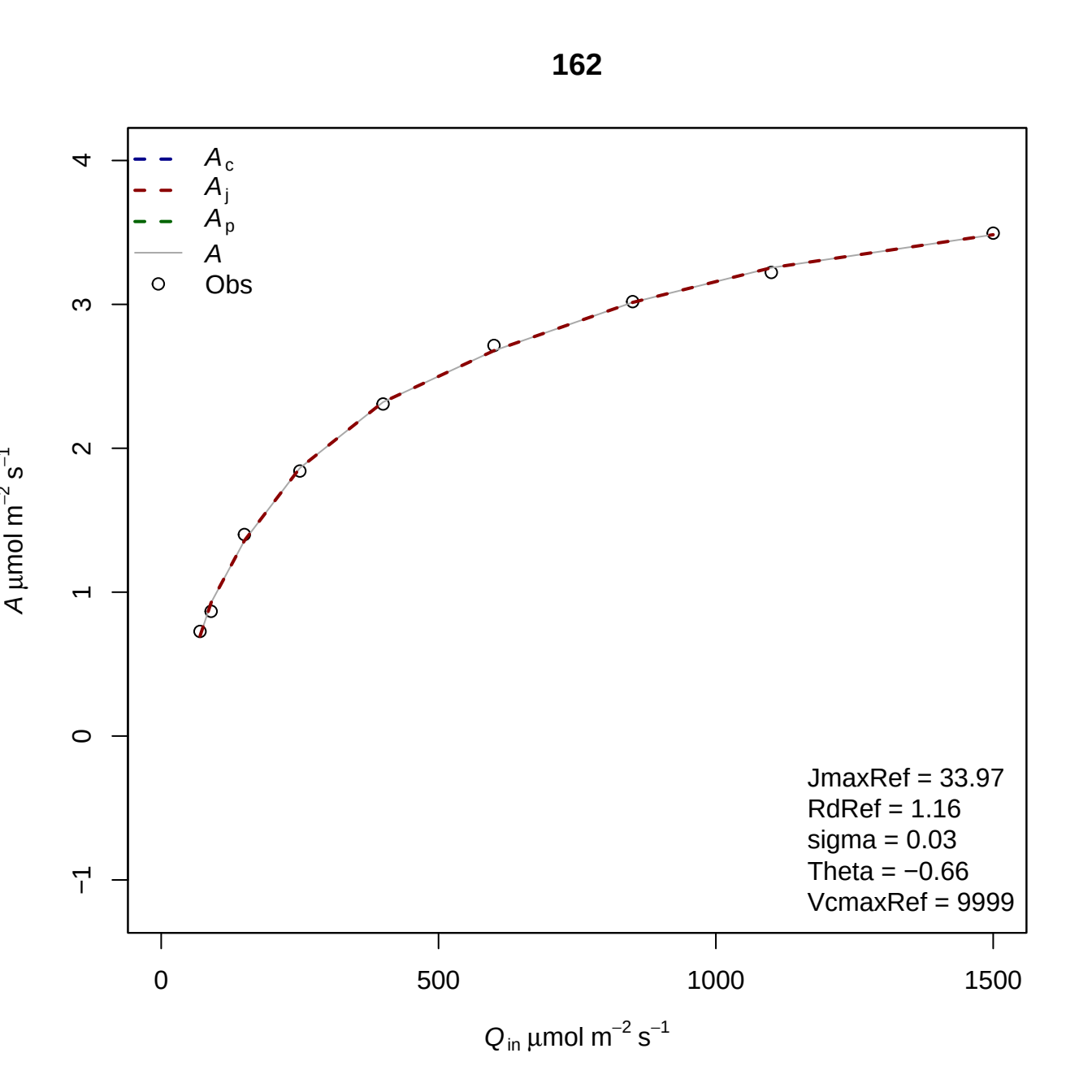

153
A_c
A_j
A_p
A
Obs
JmaxRef = 204.6
RdRef = 1.56
sigma = 0.18
Theta = −2.42
VcmaxRef = 69.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

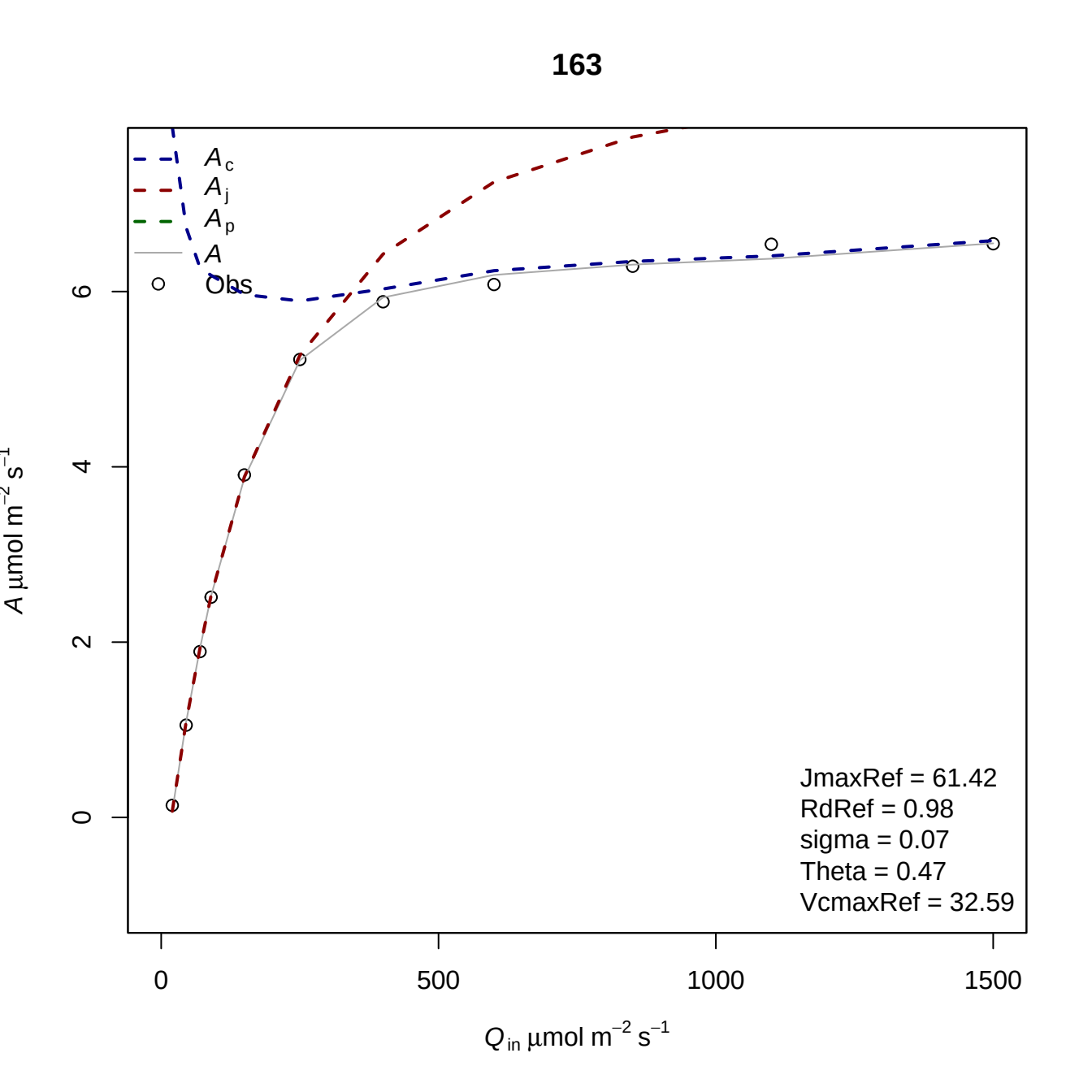

154
A_c
A_j
A_p
A
Obs
JmaxRef = 107.64
RdRef = 1.21
sigma = 0.06
Theta = 0.62
VcmaxRef = 68.43
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

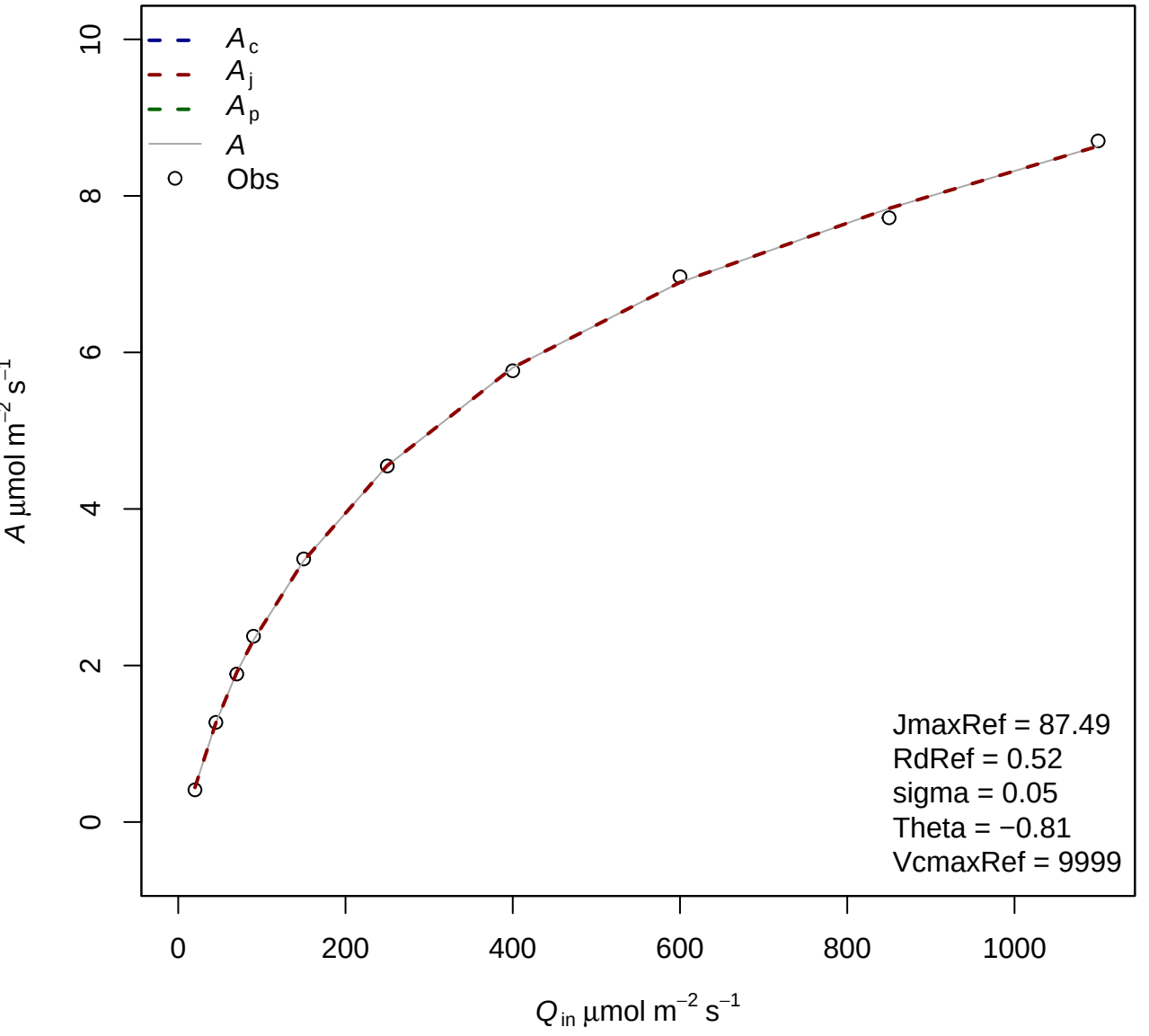

155
A_c
A_j
A_p
A
Obs
JmaxRef = 88.17
RdRef = 0.9
sigma = 0.11
Theta = 0.76
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

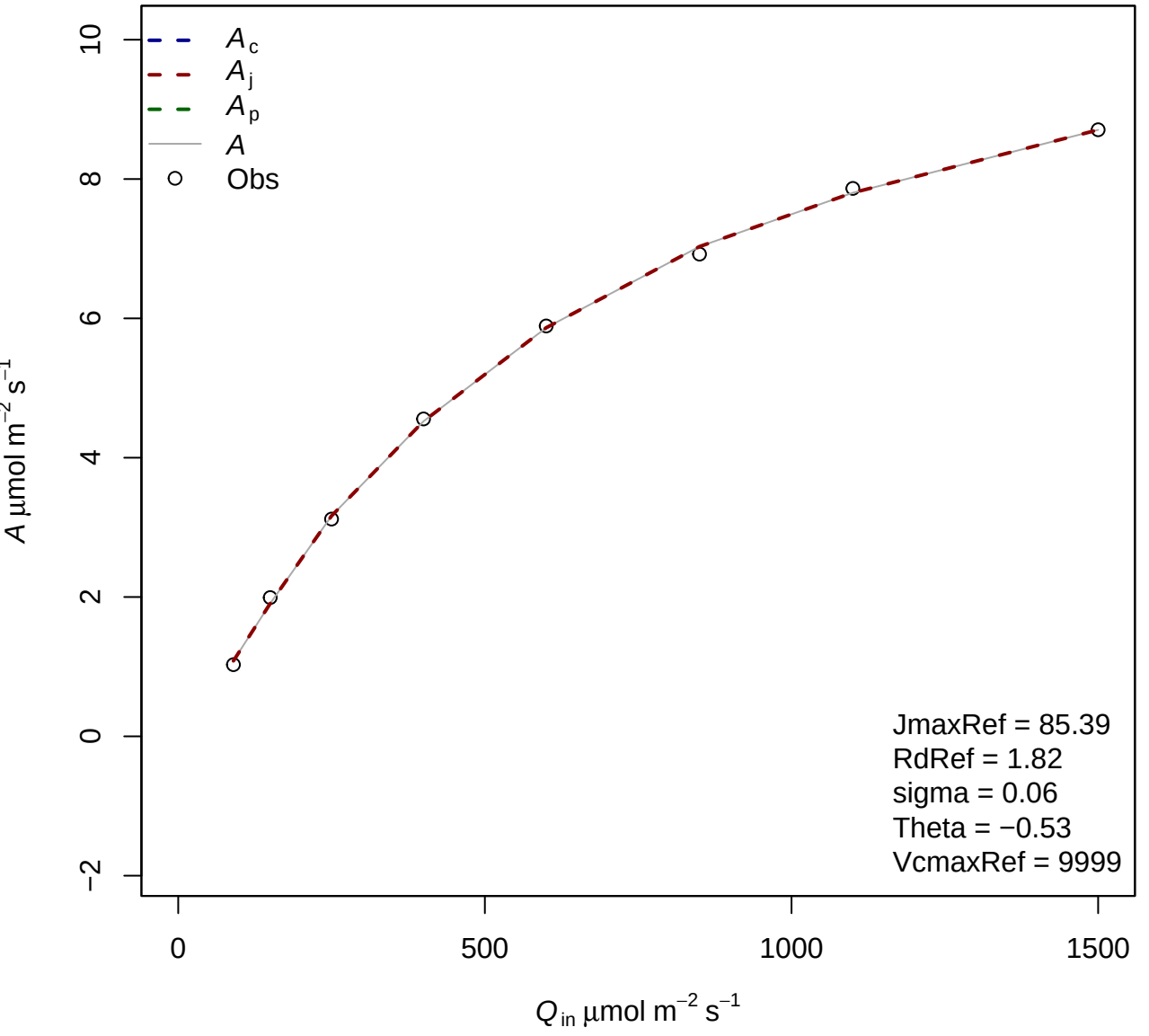

159
A_c
A_j
A_p
A
Obs
JmaxRef = 53.41
RdRef = 1.51
sigma = 0.1
Theta = 0.37
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

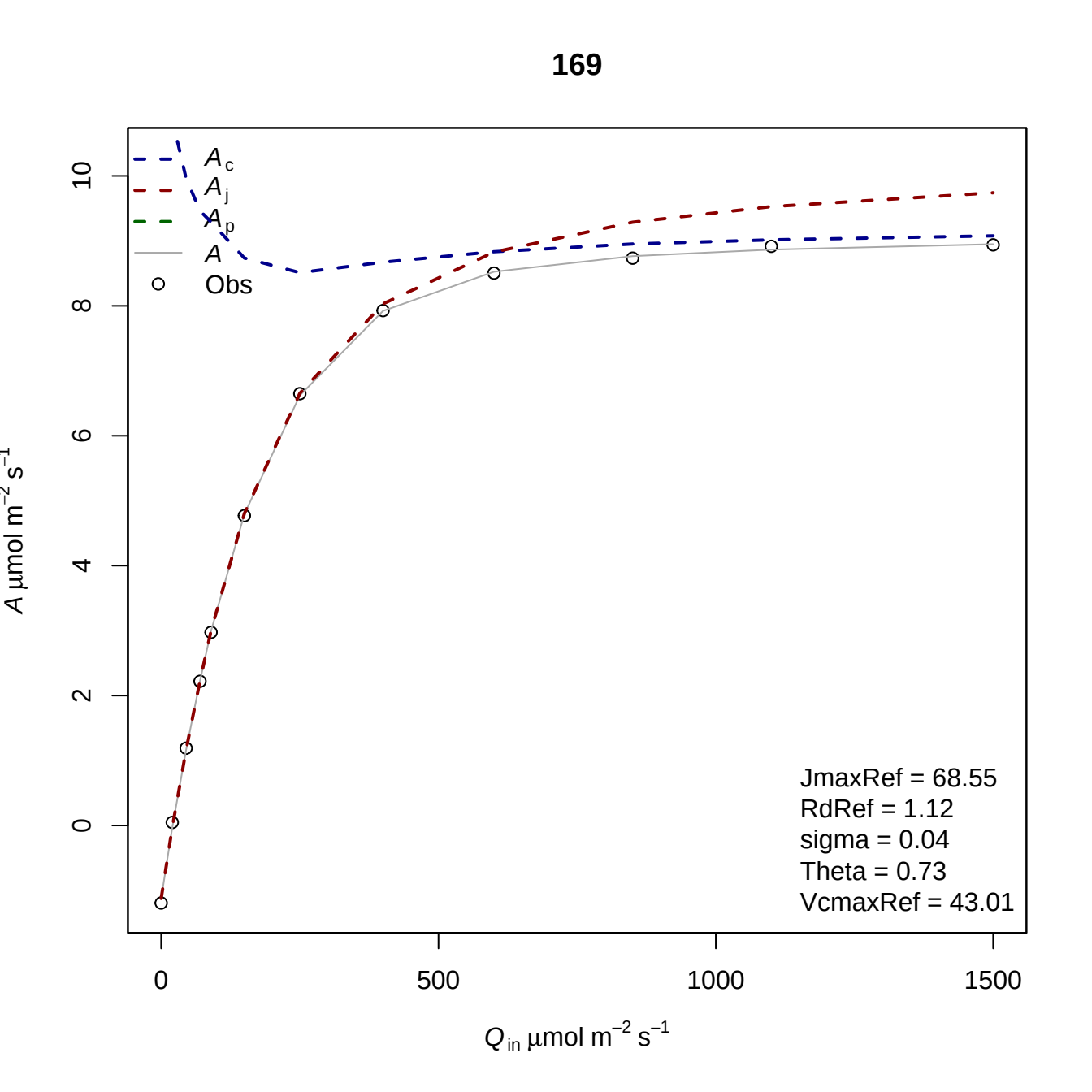

160
A_c
A_j
A_p
A
Obs
JmaxRef = 43.93
RdRef = 0.4
sigma = 0.16
Theta = 0.48
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

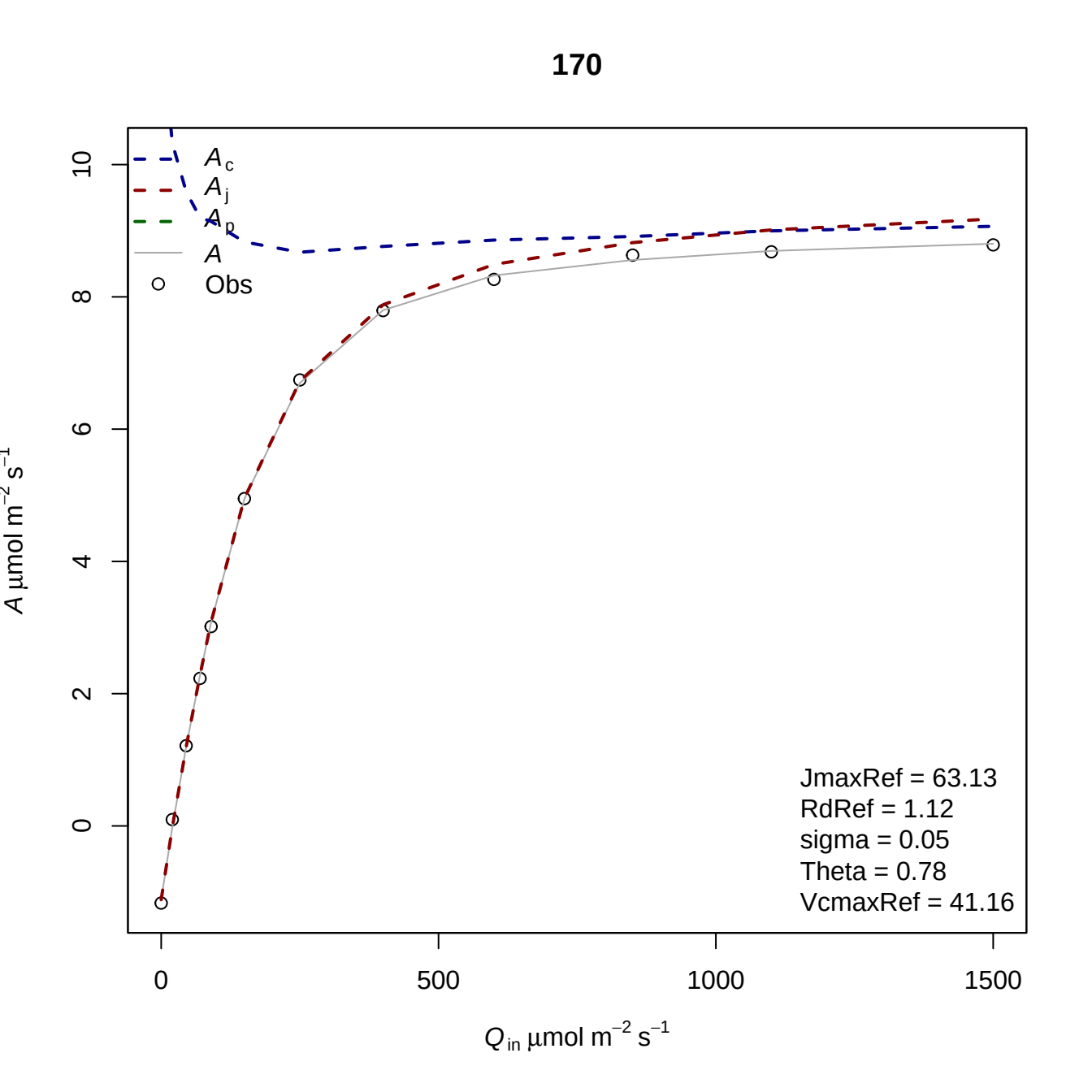

162
A_c
A_j
A_p
A
Obs
JmaxRef = 33.97
RdRef = 1.16
sigma = 0.03
Theta = −0.66
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
500
1000
1500

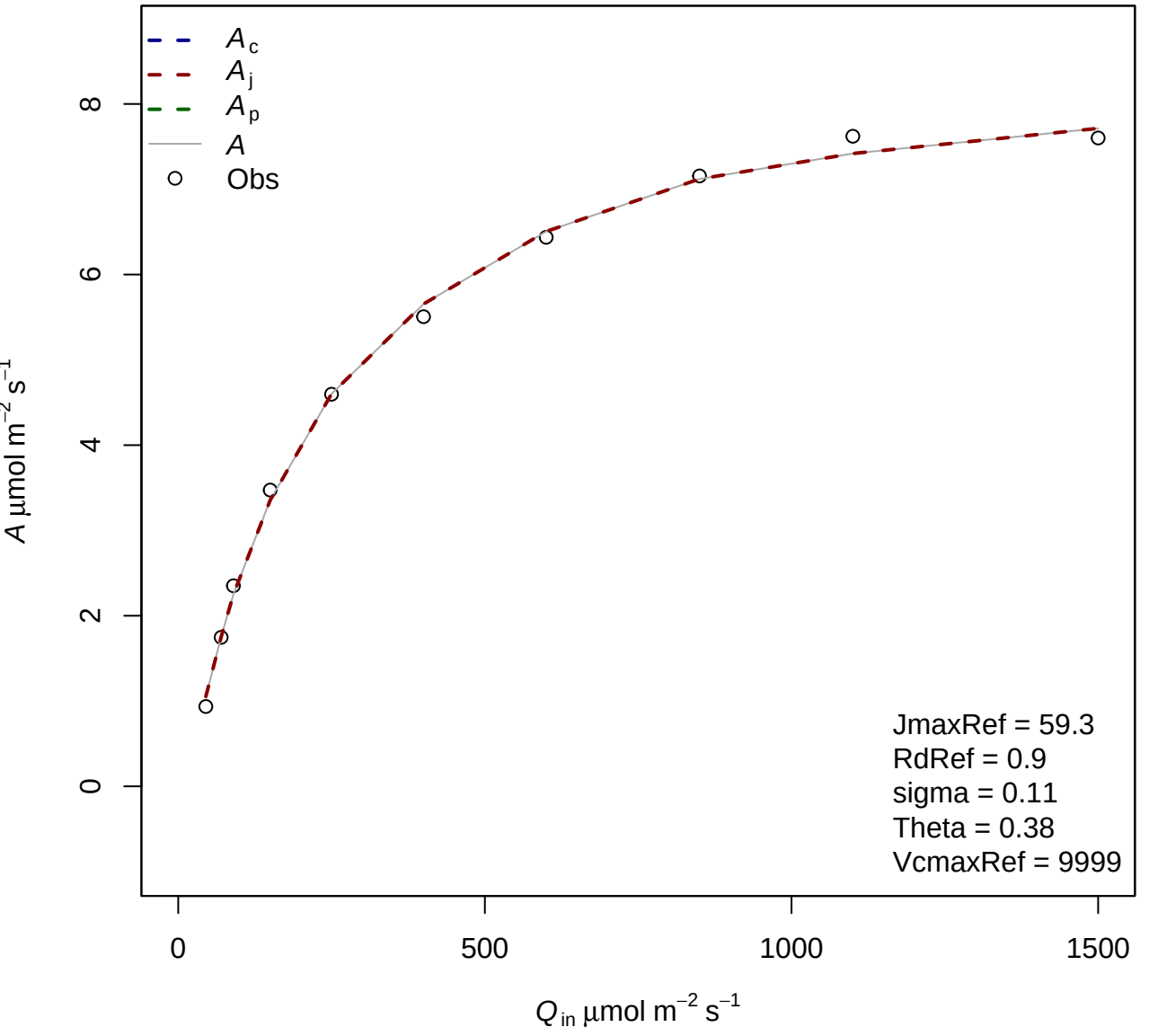

163
A_c
A_j
A_p
A
Obs
JmaxRef = 61.42
RdRef = 0.98
sigma = 0.07
Theta = 0.47
VcmaxRef = 32.59
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

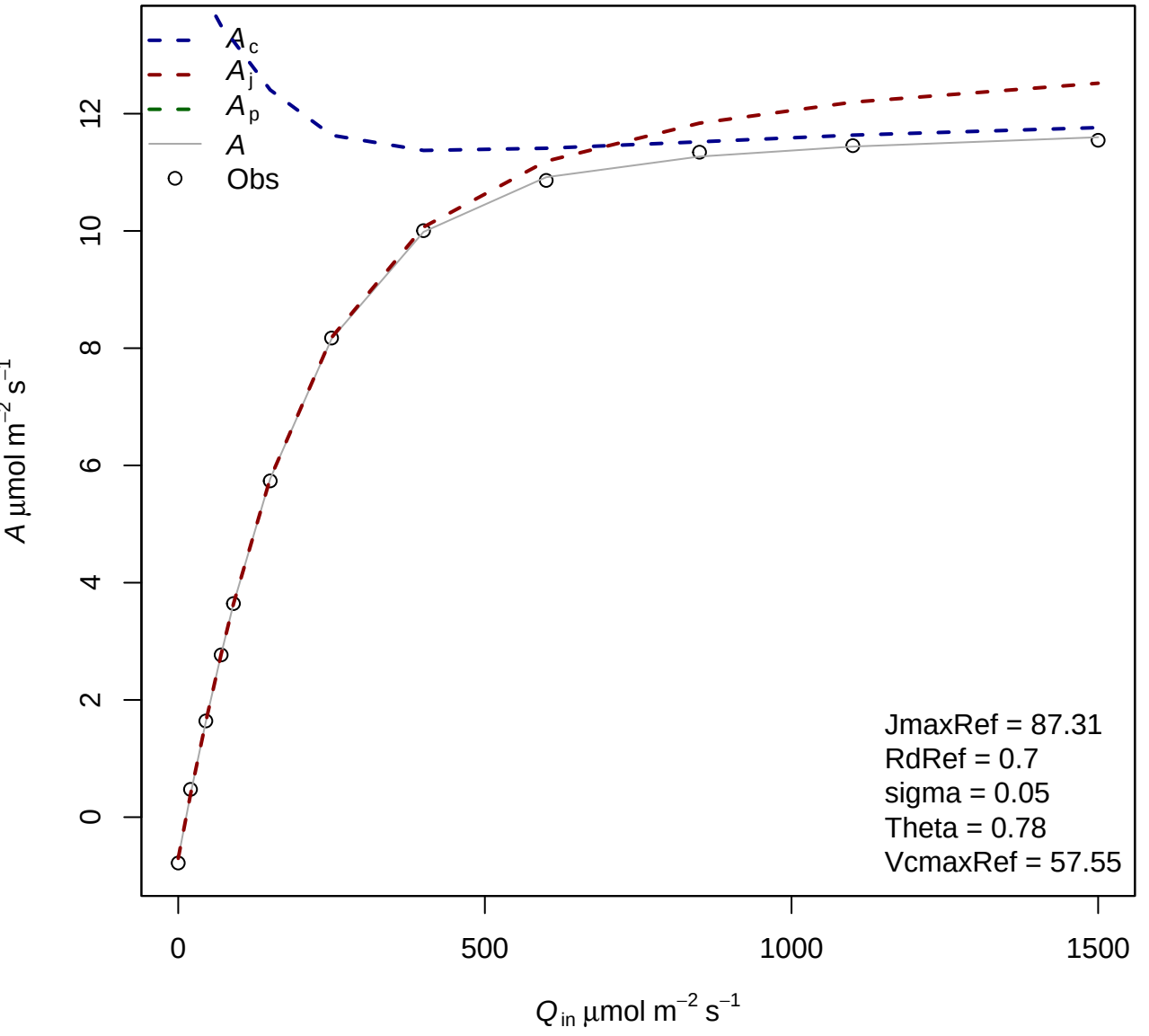

A_c
A_j
A_p
A
Obs
JmaxRef = 87.49
RdRef = 0.52
sigma = 0.05
Theta = −0.81
VcmaxRef = 9999
0
2
4
6
8
10
0
200
400
600
800
1000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

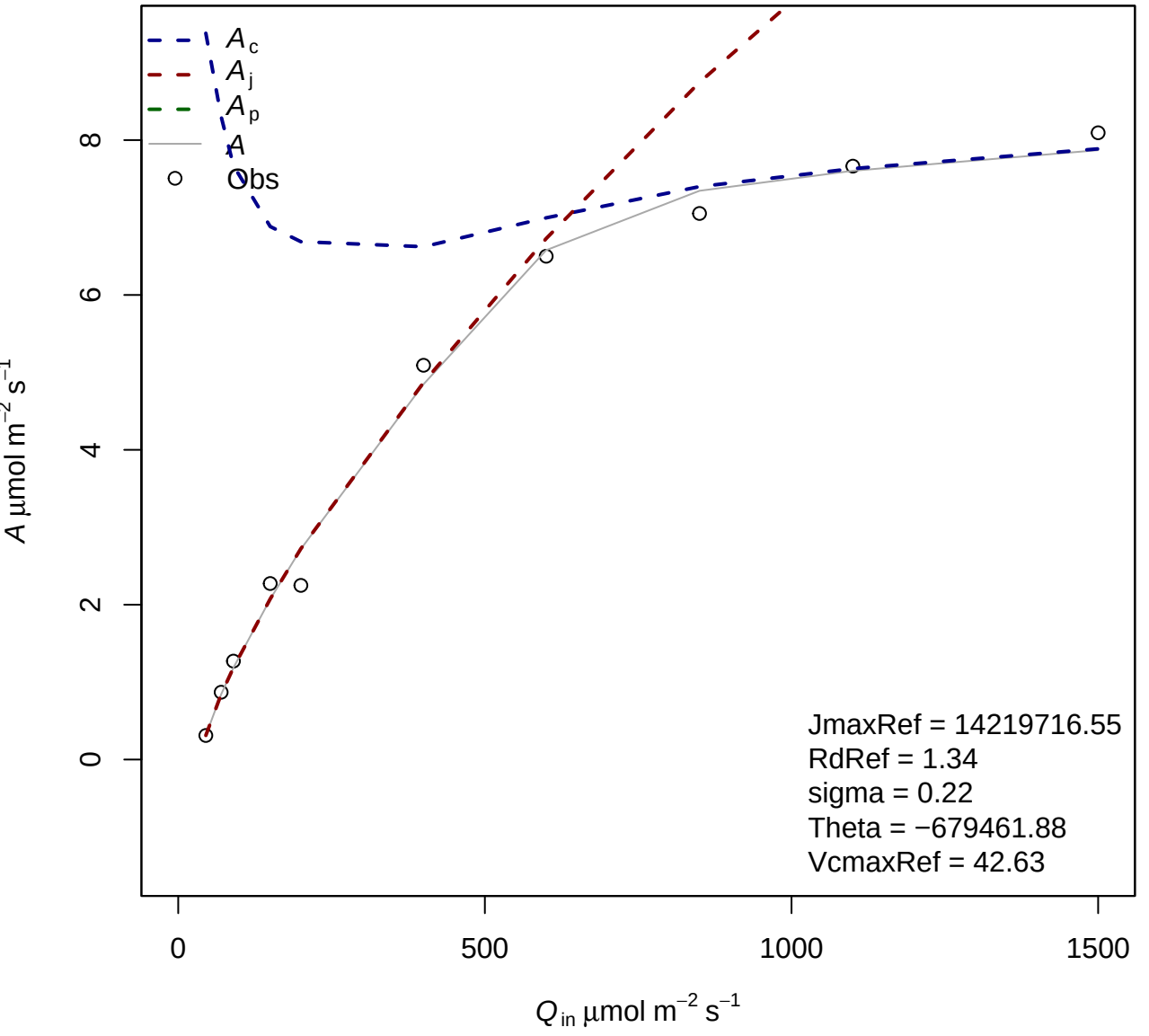

A_c
A_j
A_p
A
Obs
JmaxRef = 85.39
RdRef = 1.82
sigma = 0.06
Theta = −0.53
VcmaxRef = 9999
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

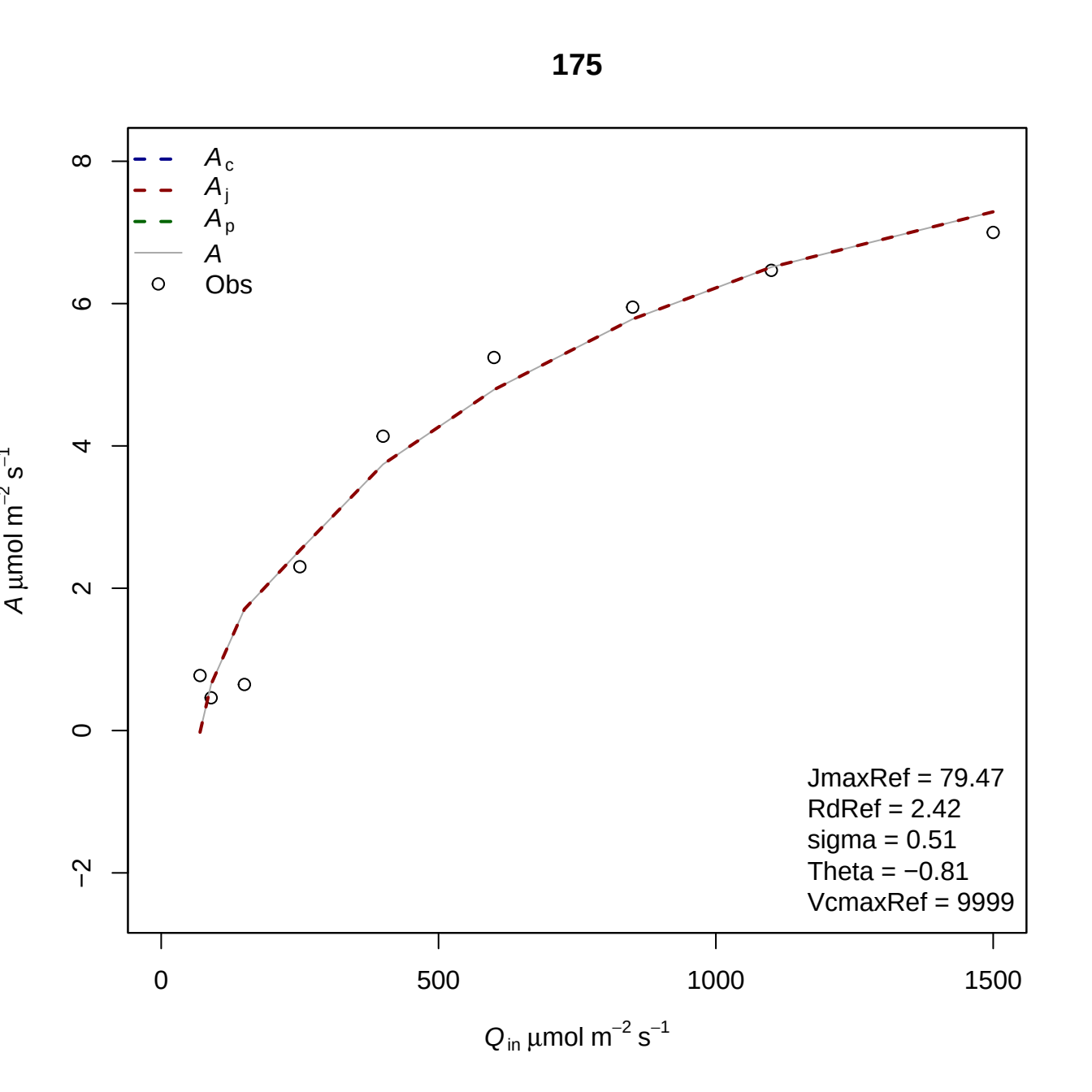

169
A_c
A_j
A_p
A
Obs
JmaxRef = 68.55
RdRef = 1.12
sigma = 0.04
Theta = 0.73
VcmaxRef = 43.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

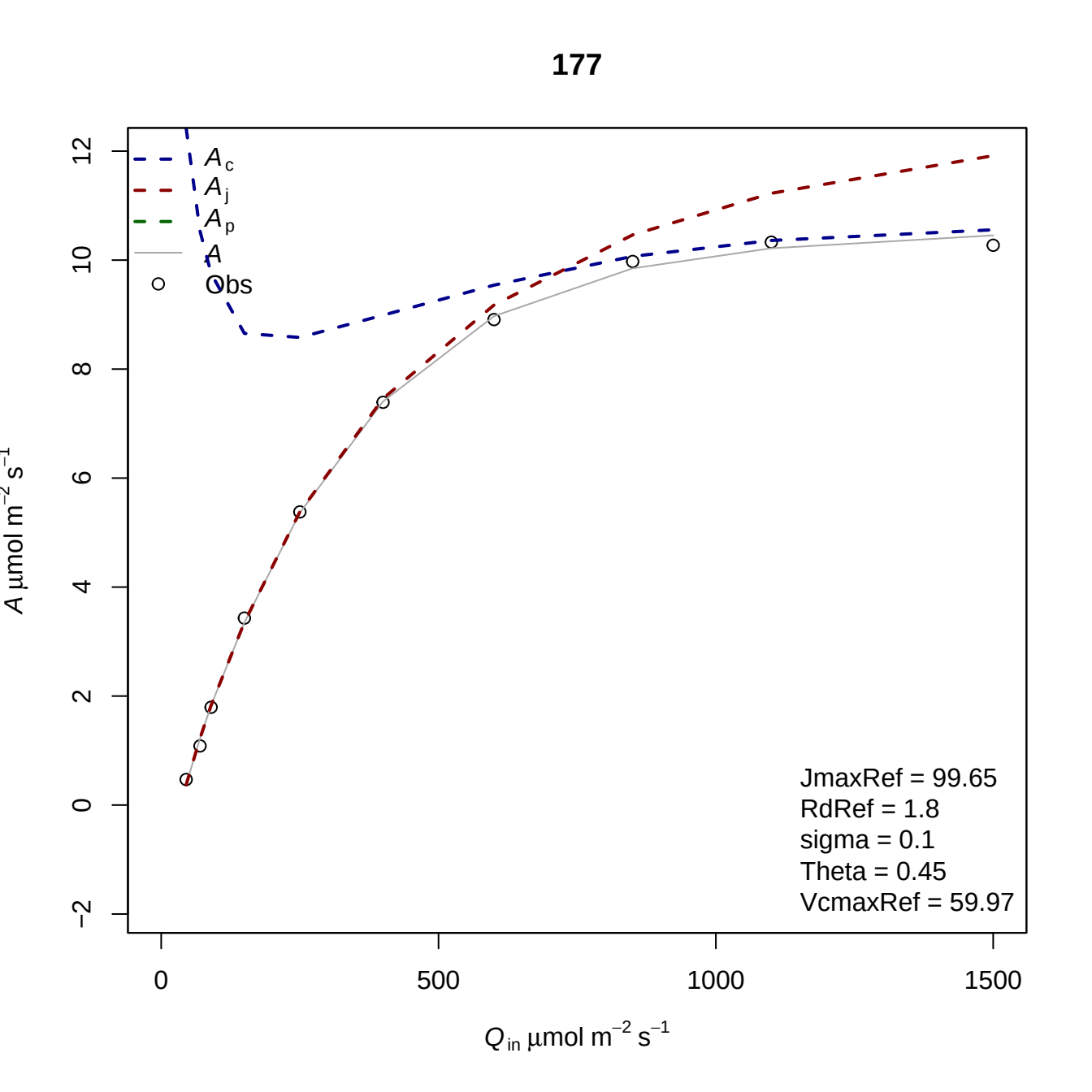

170
A_c
A_j
A_p
A
Obs
JmaxRef = 63.13
RdRef = 1.12
sigma = 0.05
Theta = 0.78
VcmaxRef = 41.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

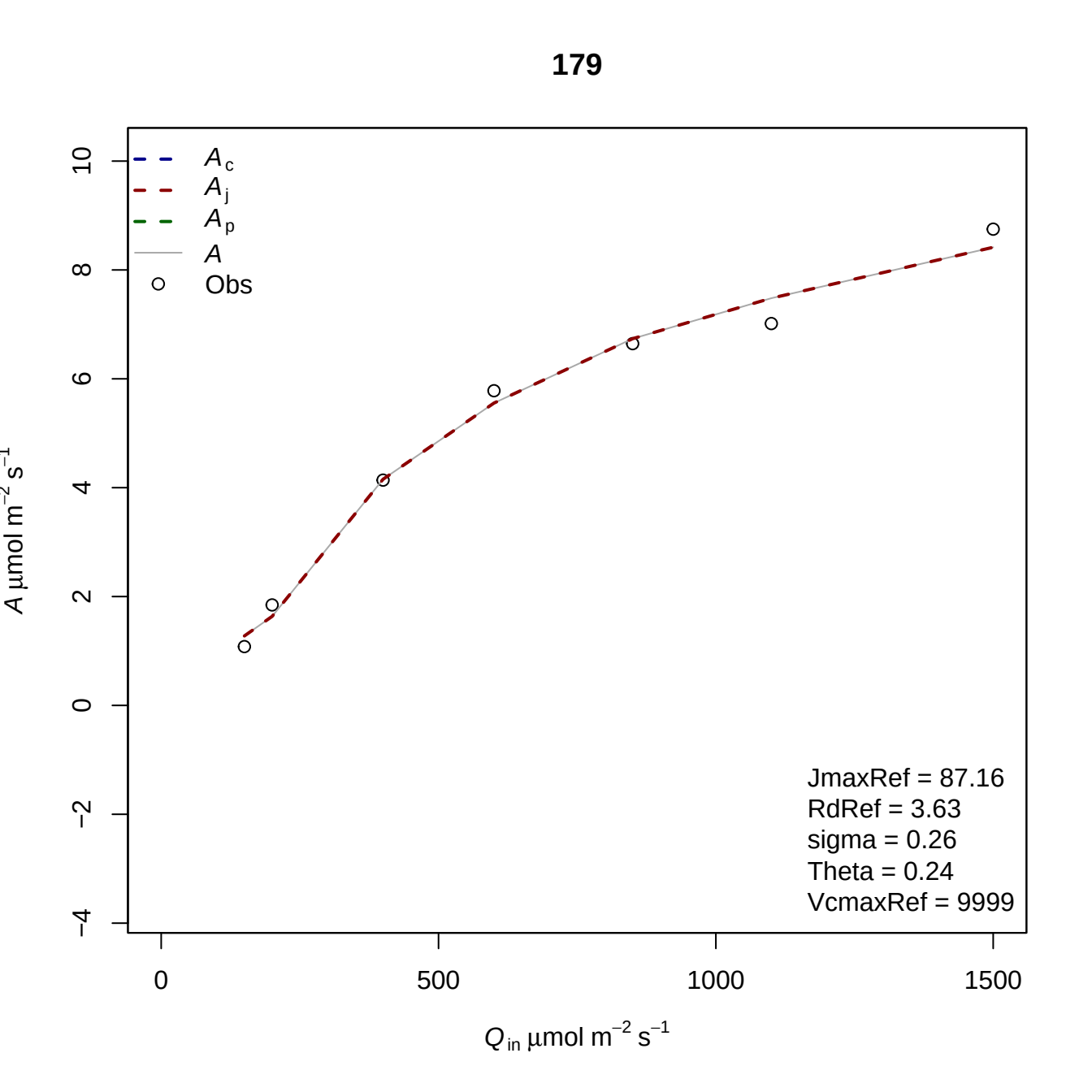

A_c
A_j
A_p
A
Obs
JmaxRef = 59.3
RdRef = 0.9
sigma = 0.11
Theta = 0.38
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

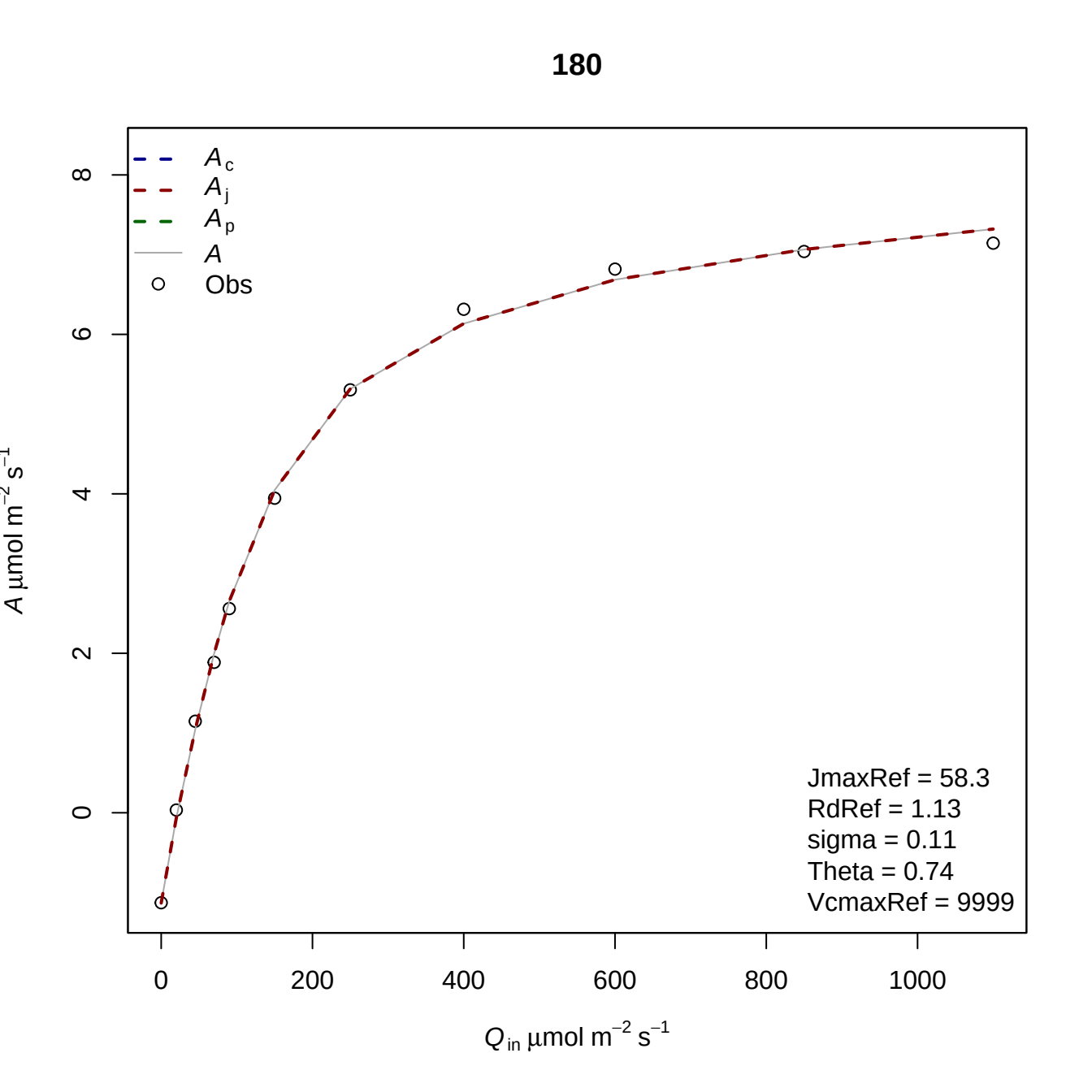

A_c
A_j
A_p
A
Obs
JmaxRef = 87.31
RdRef = 0.7
sigma = 0.05
Theta = 0.78
VcmaxRef = 57.55
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

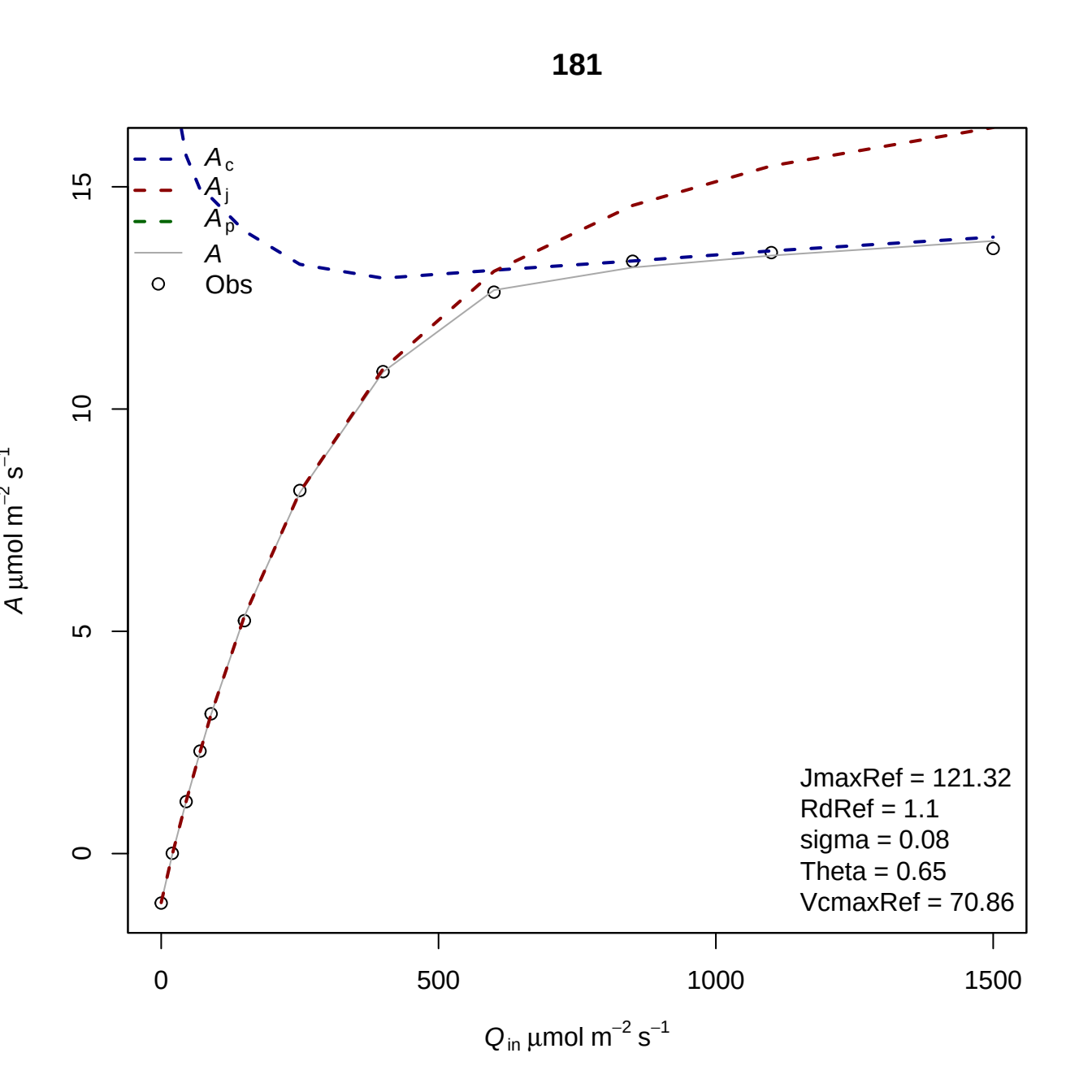

A_c
A_j
A_p
A
Obs
JmaxRef = 14219716.55
RdRef = 1.34
sigma = 0.22
Theta = −679461.88
VcmaxRef = 42.63
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

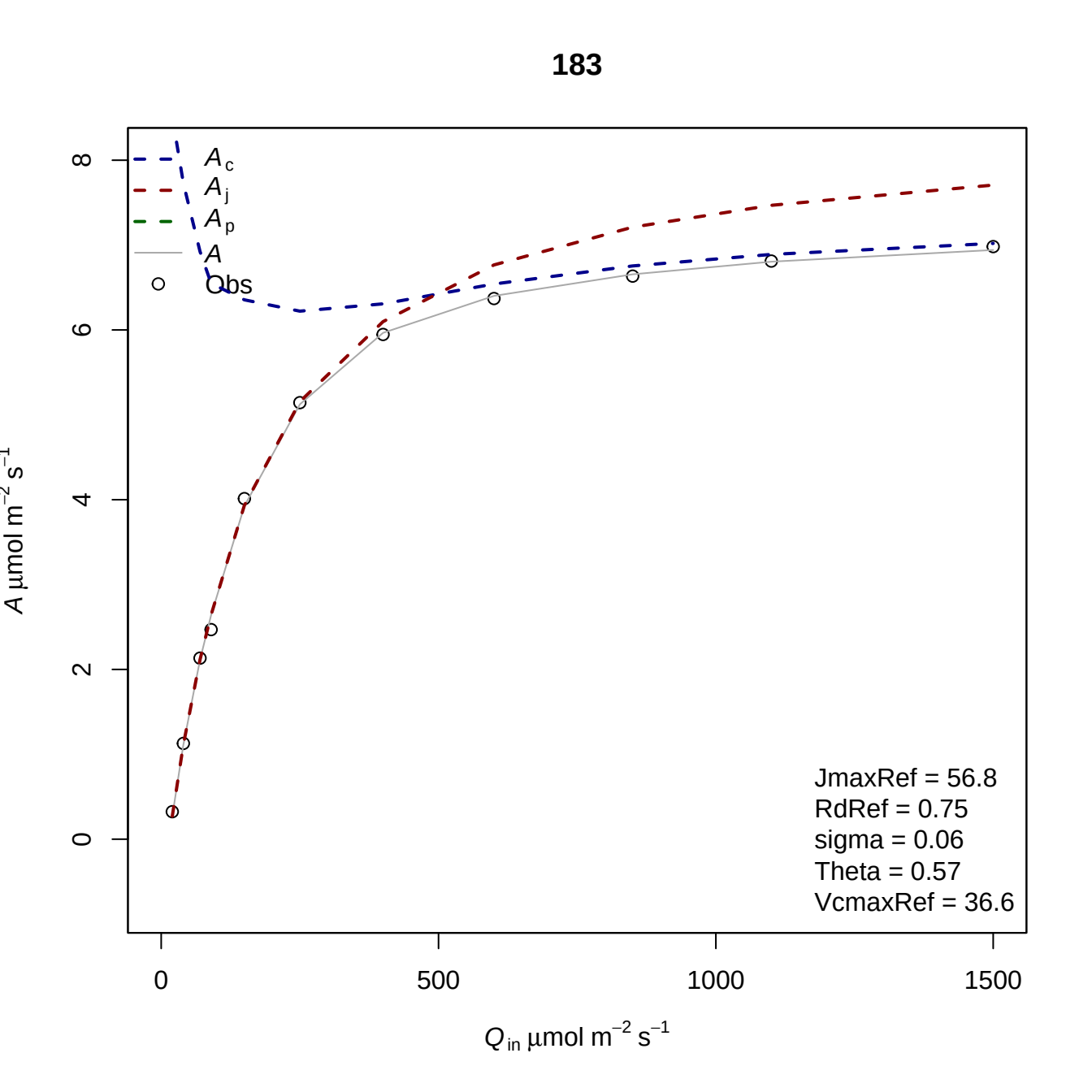

175
A_c
A_j
A_p
A
Obs
JmaxRef = 79.47
RdRef = 2.42
sigma = 0.51
Theta = −0.81
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

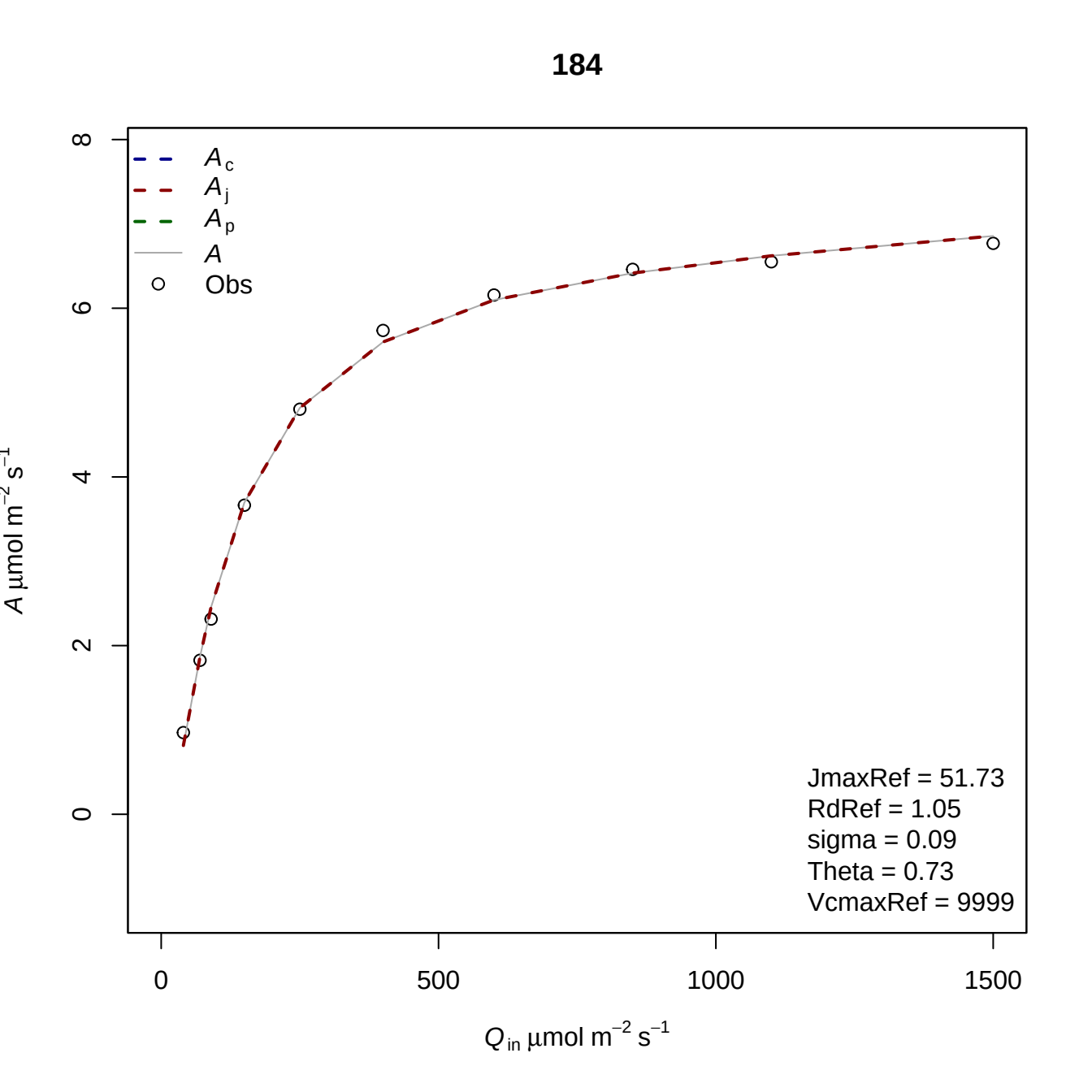

177
A_c
A_j
A_p
A
Obs
JmaxRef = 99.65
RdRef = 1.8
sigma = 0.1
Theta = 0.45
VcmaxRef = 59.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

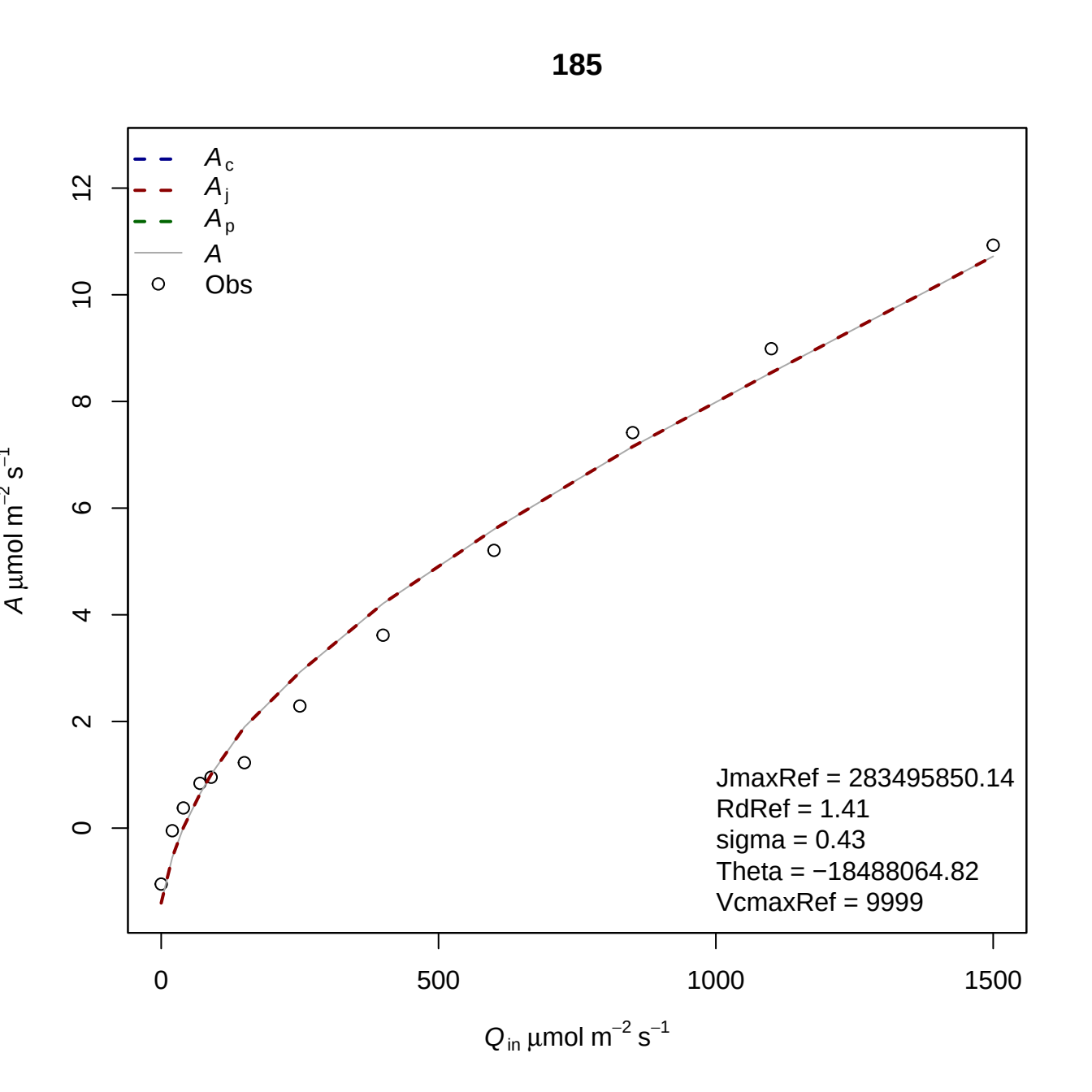

179
A_c
A_j
A_p
A
Obs
JmaxRef = 87.16
RdRef = 3.63
sigma = 0.26
Theta = 0.24
VcmaxRef = 9999
10
8
6
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

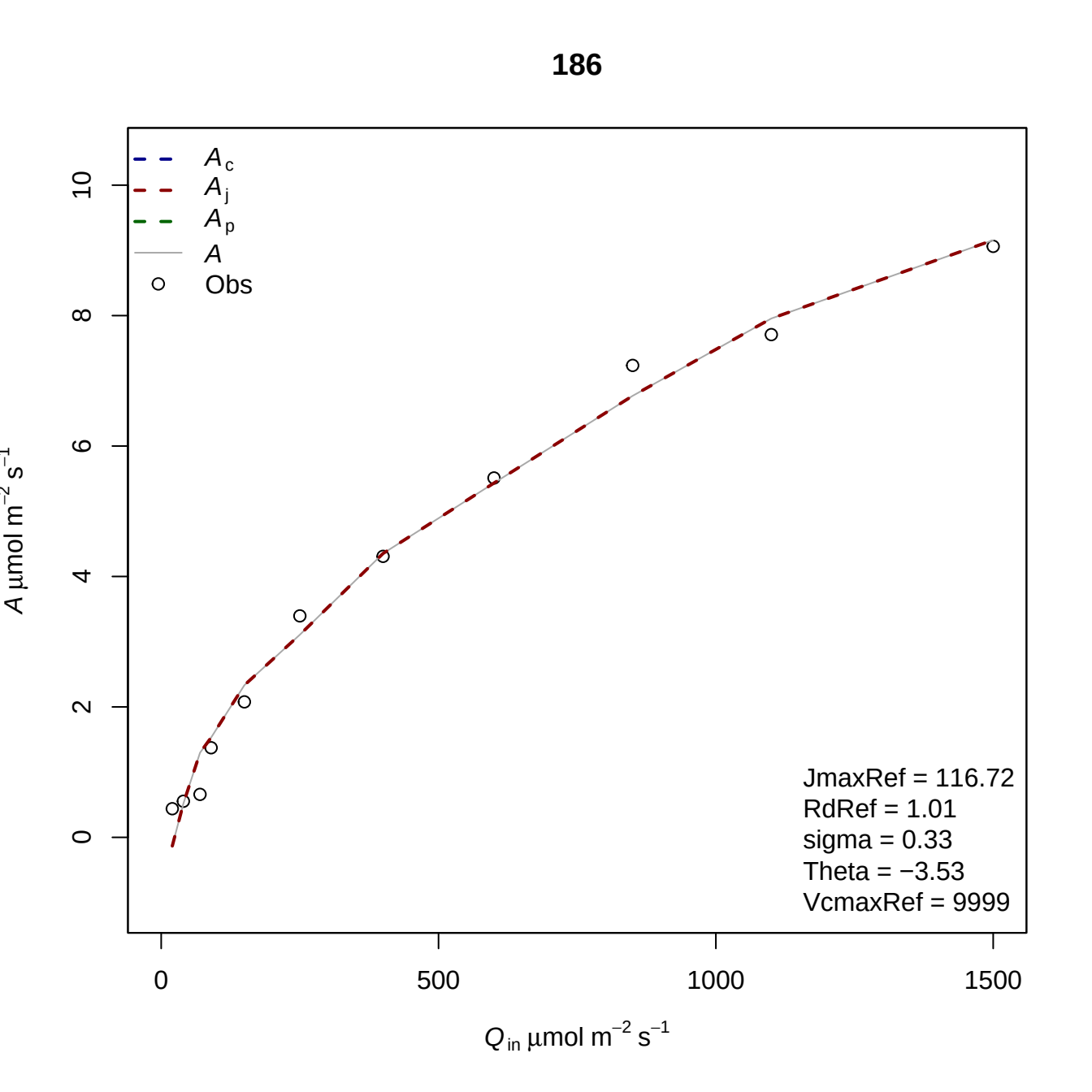

180
A_c
A_j
A_p
A
Obs
JmaxRef = 58.3
RdRef = 1.13
sigma = 0.11
Theta = 0.74
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
Q_{in} µmol m^{-2} s^{-1}

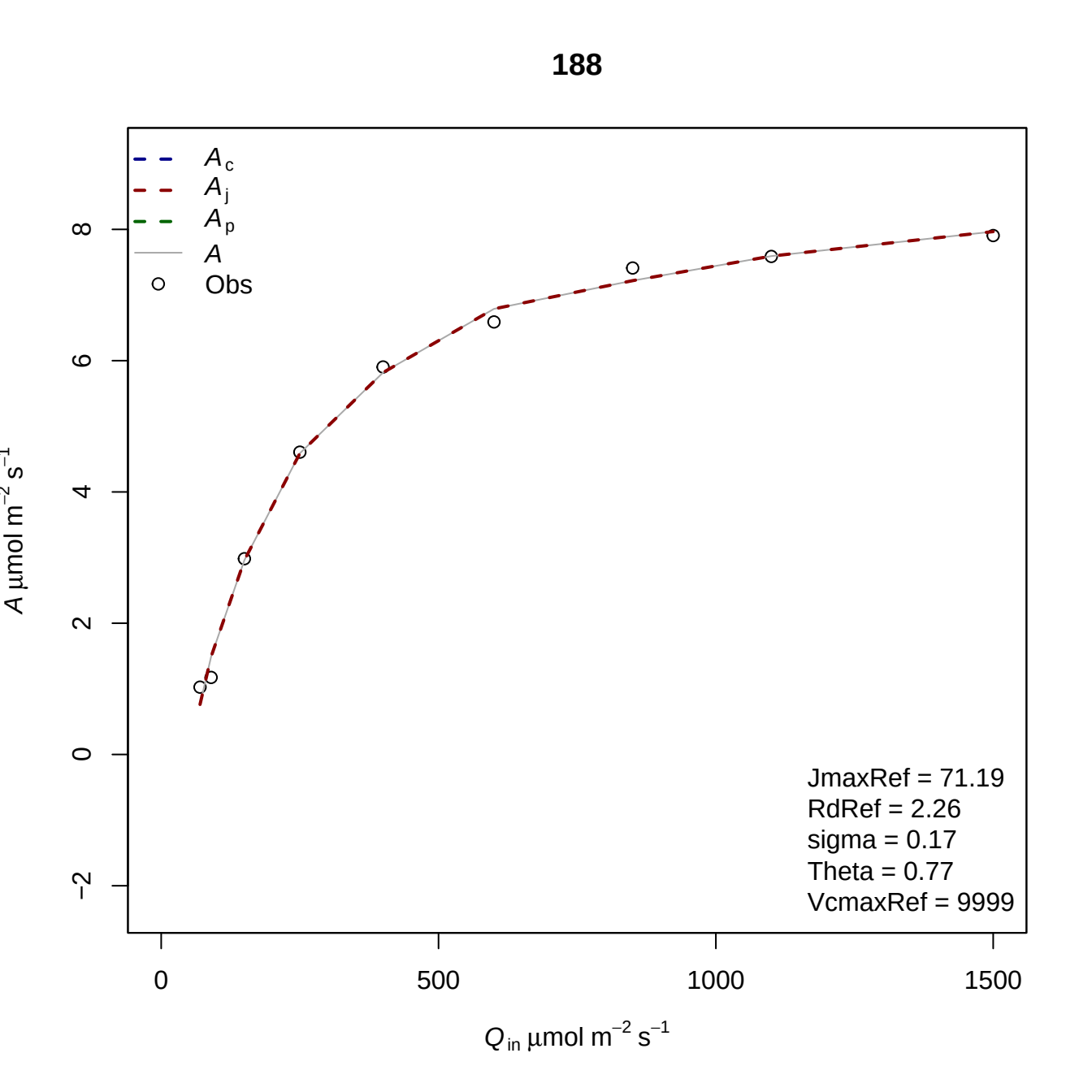

181
A_c
A_j
A_p
A
Obs
JmaxRef = 121.32
RdRef = 1.1
sigma = 0.08
Theta = 0.65
VcmaxRef = 70.86
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

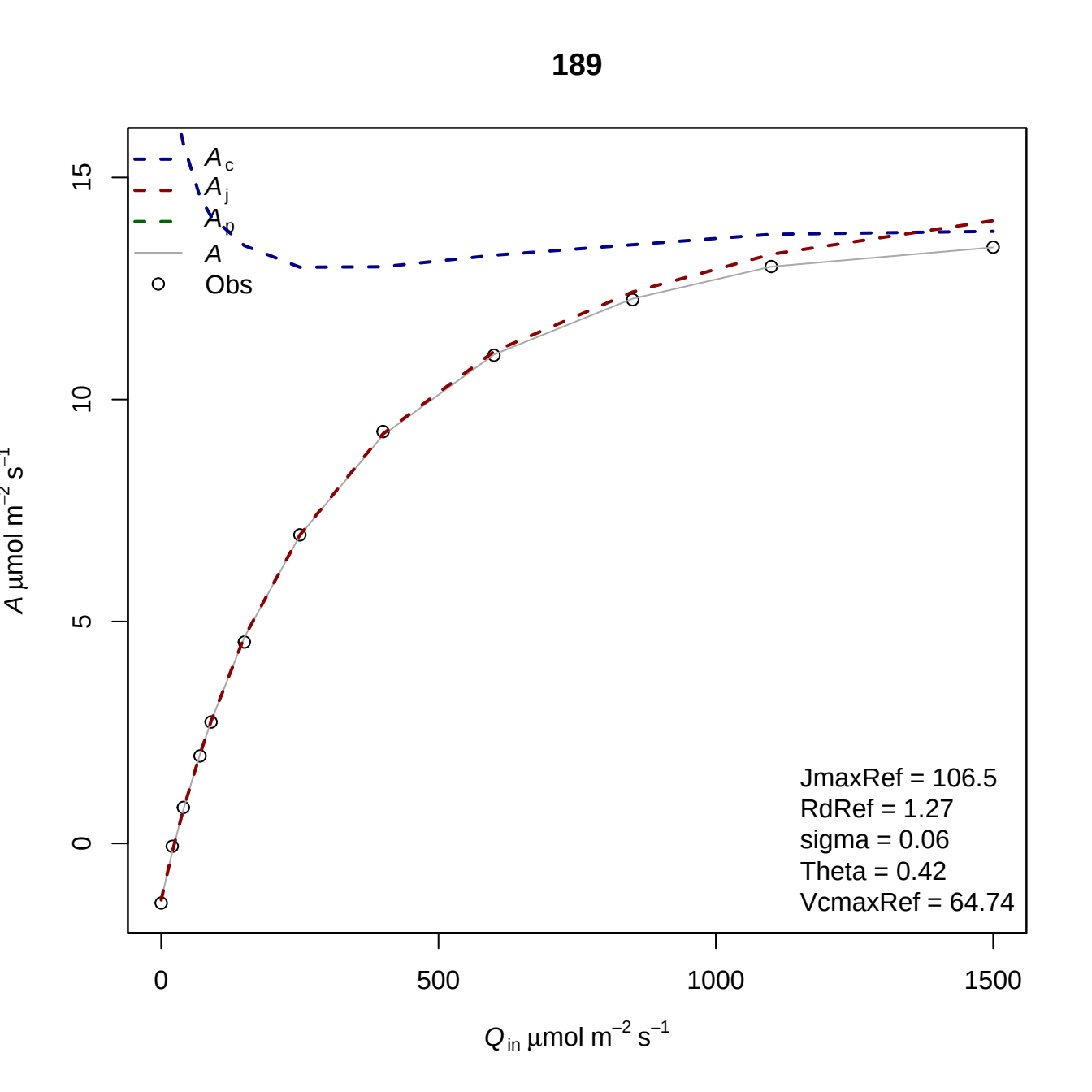

183
A_c
A_j
A_p
A
Obs
JmaxRef = 56.8
RdRef = 0.75
sigma = 0.06
Theta = 0.57
VcmaxRef = 36.6
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

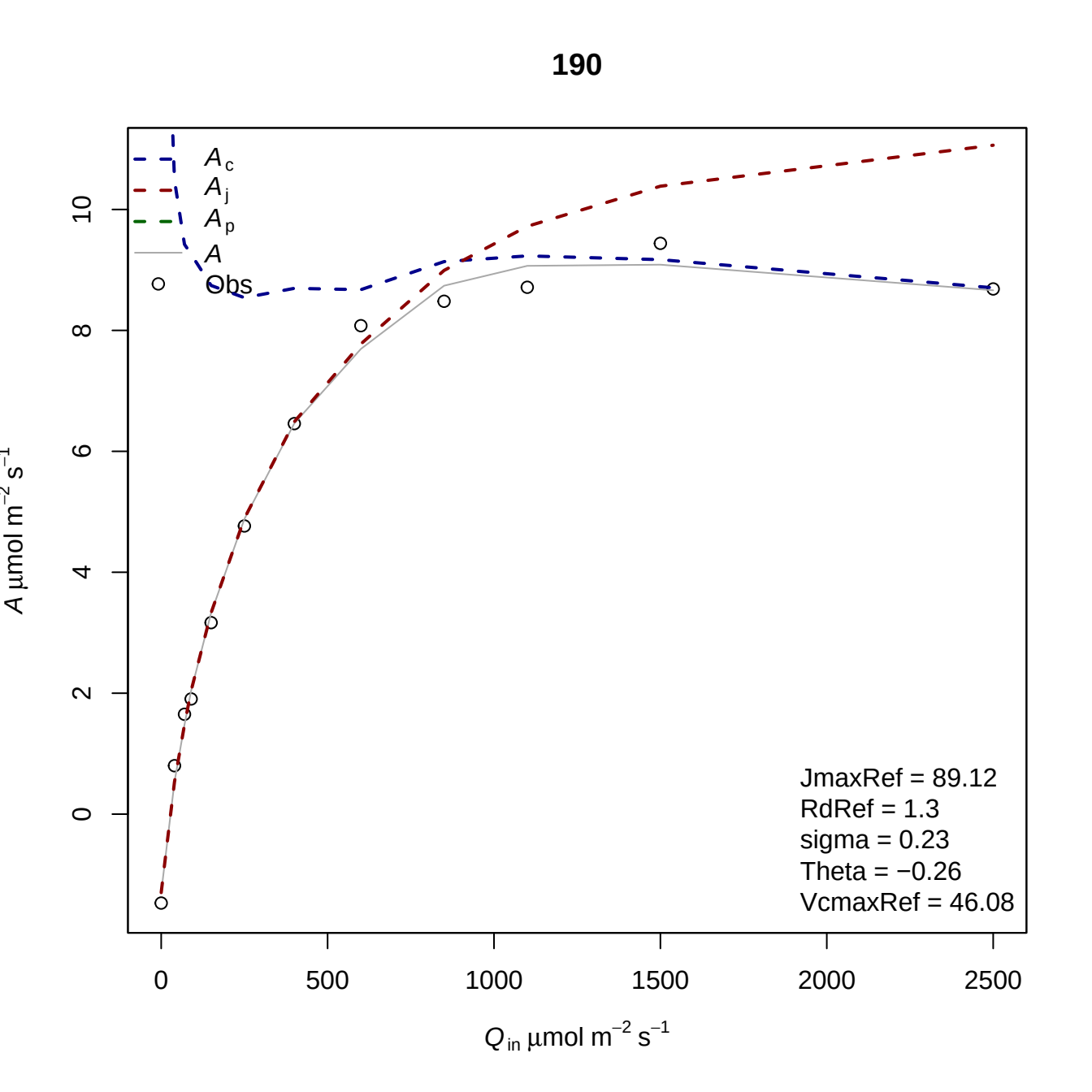

184
A_c
A_j
A_p
A
Obs
JmaxRef = 51.73
RdRef = 1.05
sigma = 0.09
Theta = 0.73
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

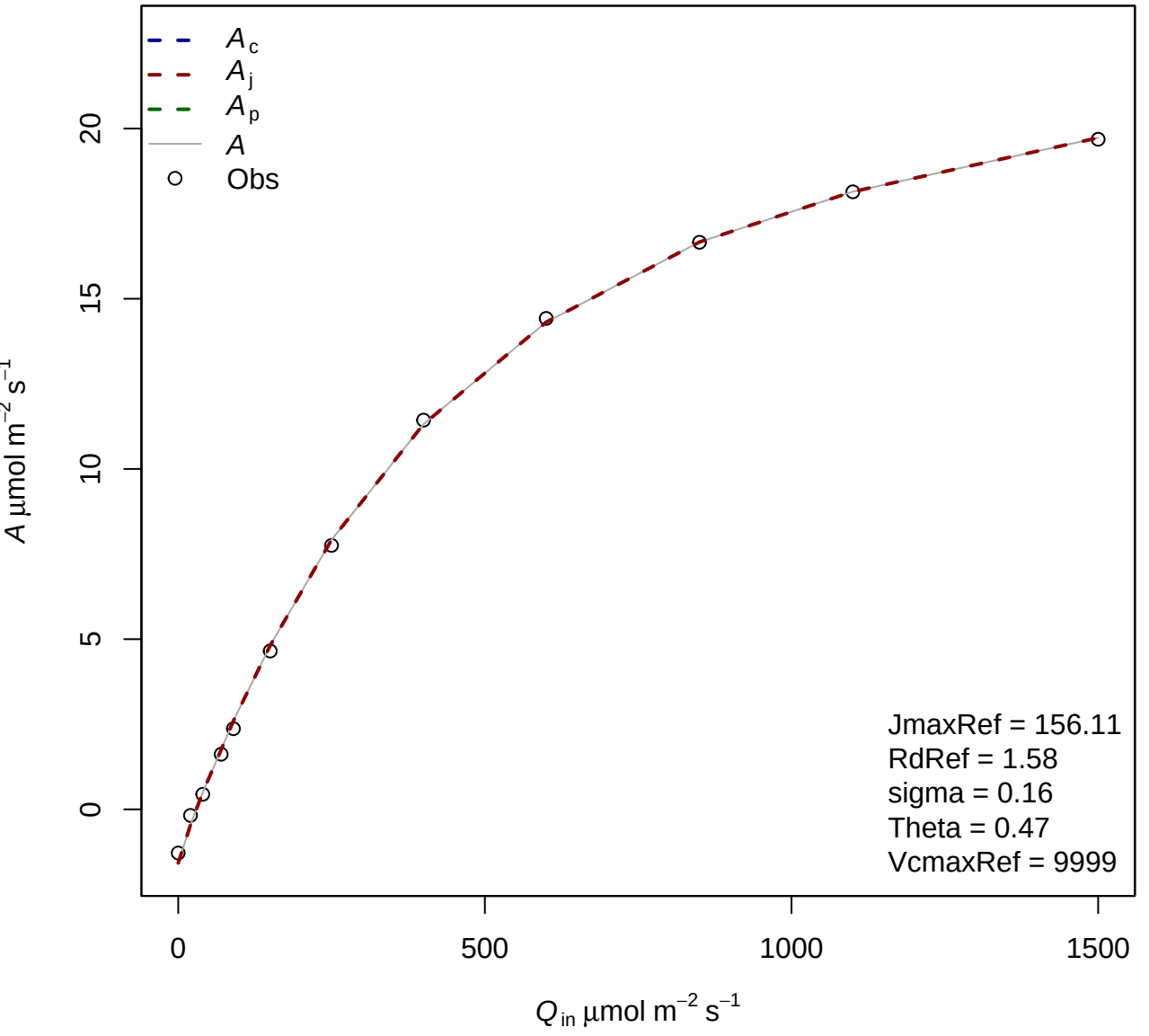

185
A_c
A_j
A_p
A
Obs
JmaxRef = 283495850.14
RdRef = 1.41
sigma = 0.43
Theta = −18488064.82
VcmaxRef = 9999
0
2
4
6
8
10
12
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

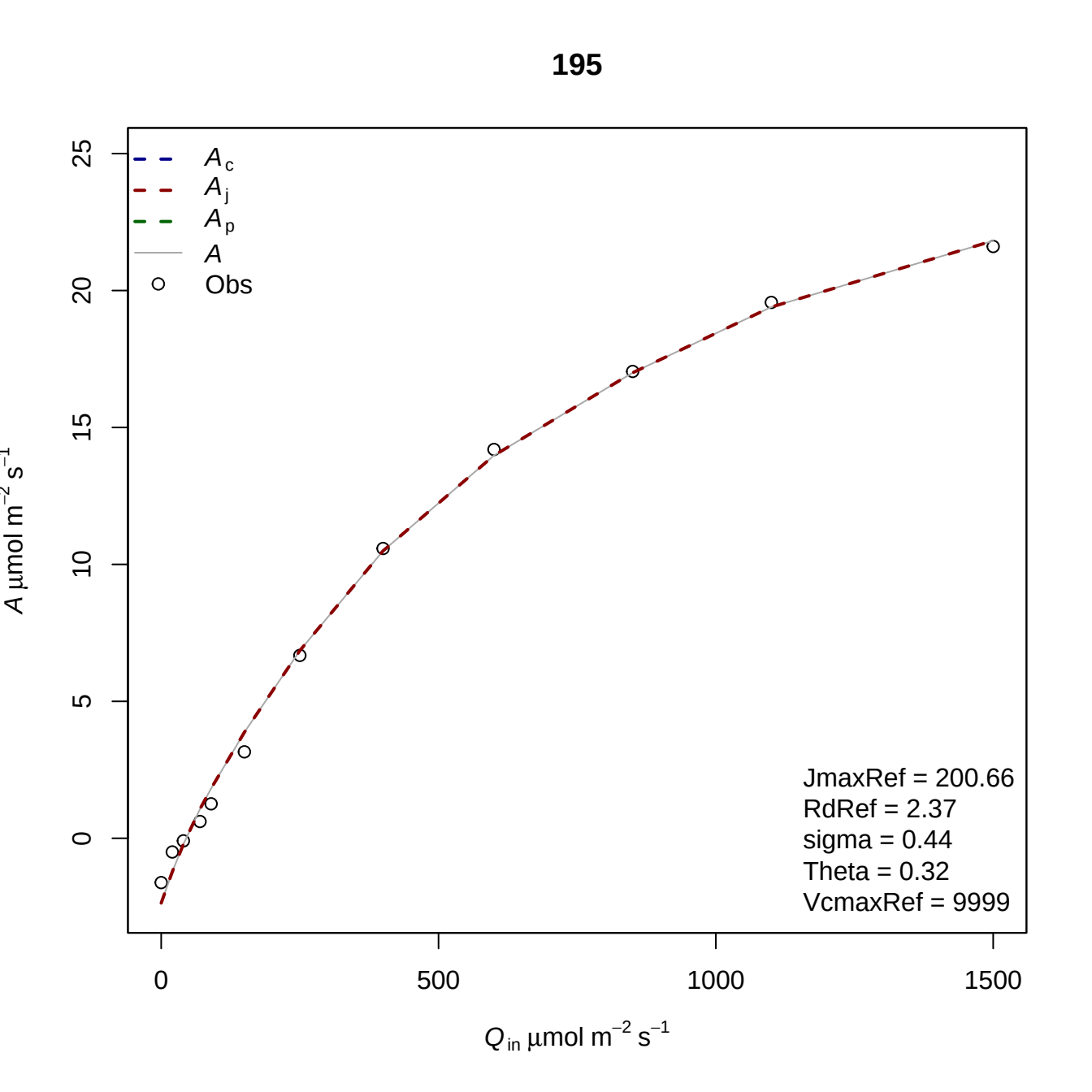

186
A_c
A_j
A_p
A
Obs
JmaxRef = 116.72
RdRef = 1.01
sigma = 0.33
Theta = −3.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

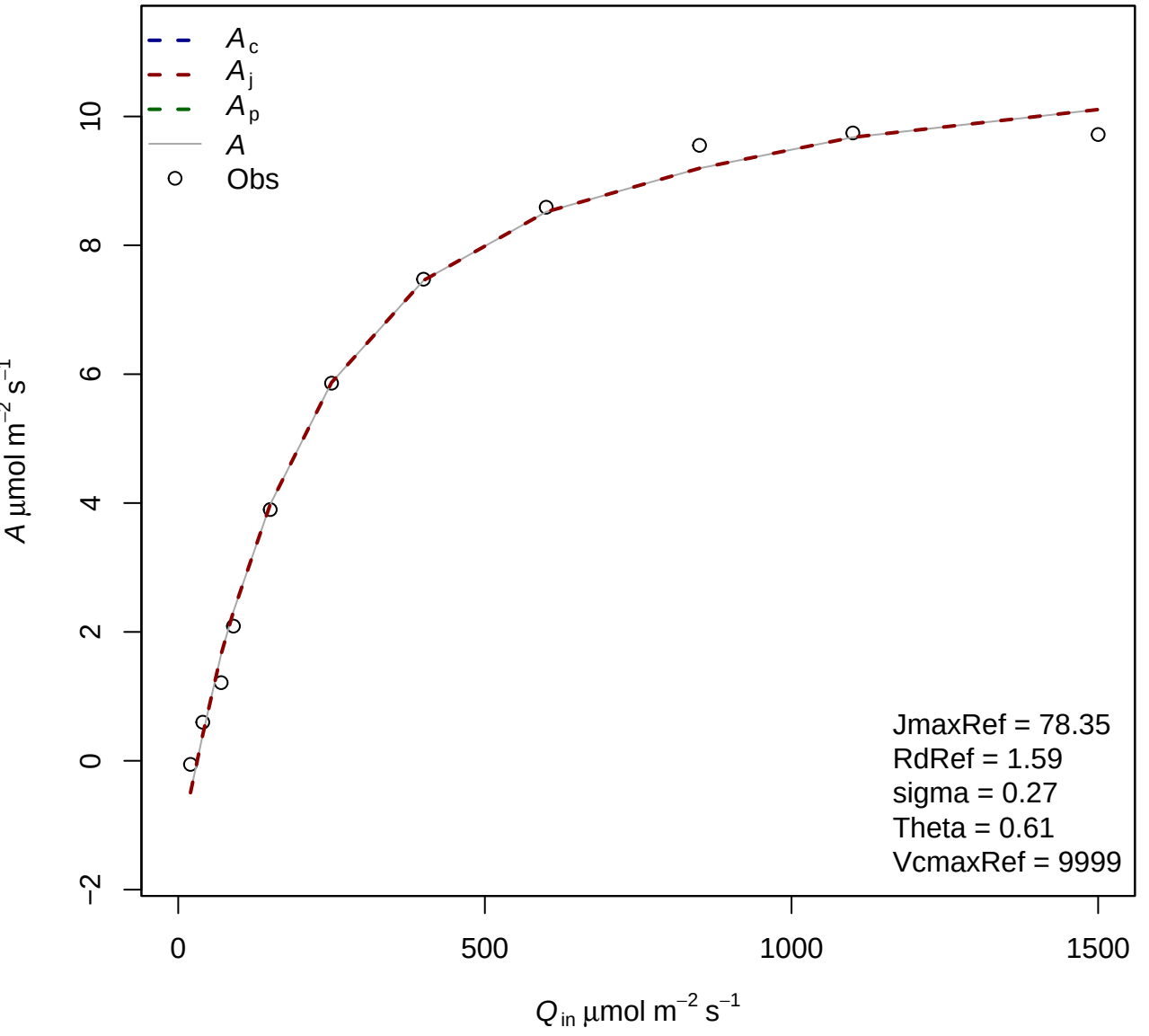

188
A_c
A_j
A_p
A
Obs
JmaxRef = 71.19
RdRef = 2.26
sigma = 0.17
Theta = 0.77
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

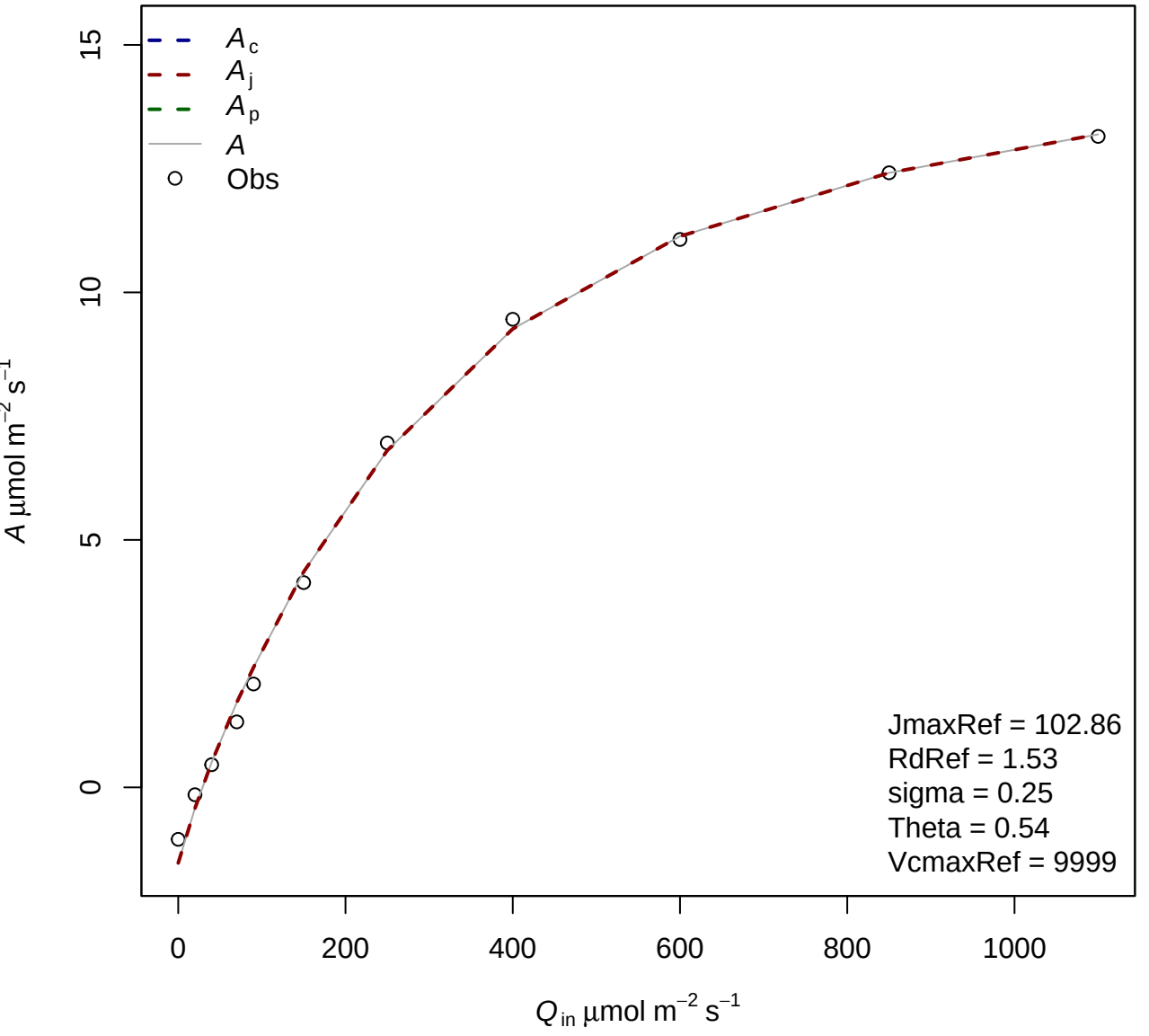

189
A_c
A_j
A_p
A
Obs
JmaxRef = 106.5
RdRef = 1.27
sigma = 0.06
Theta = 0.42
VcmaxRef = 64.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

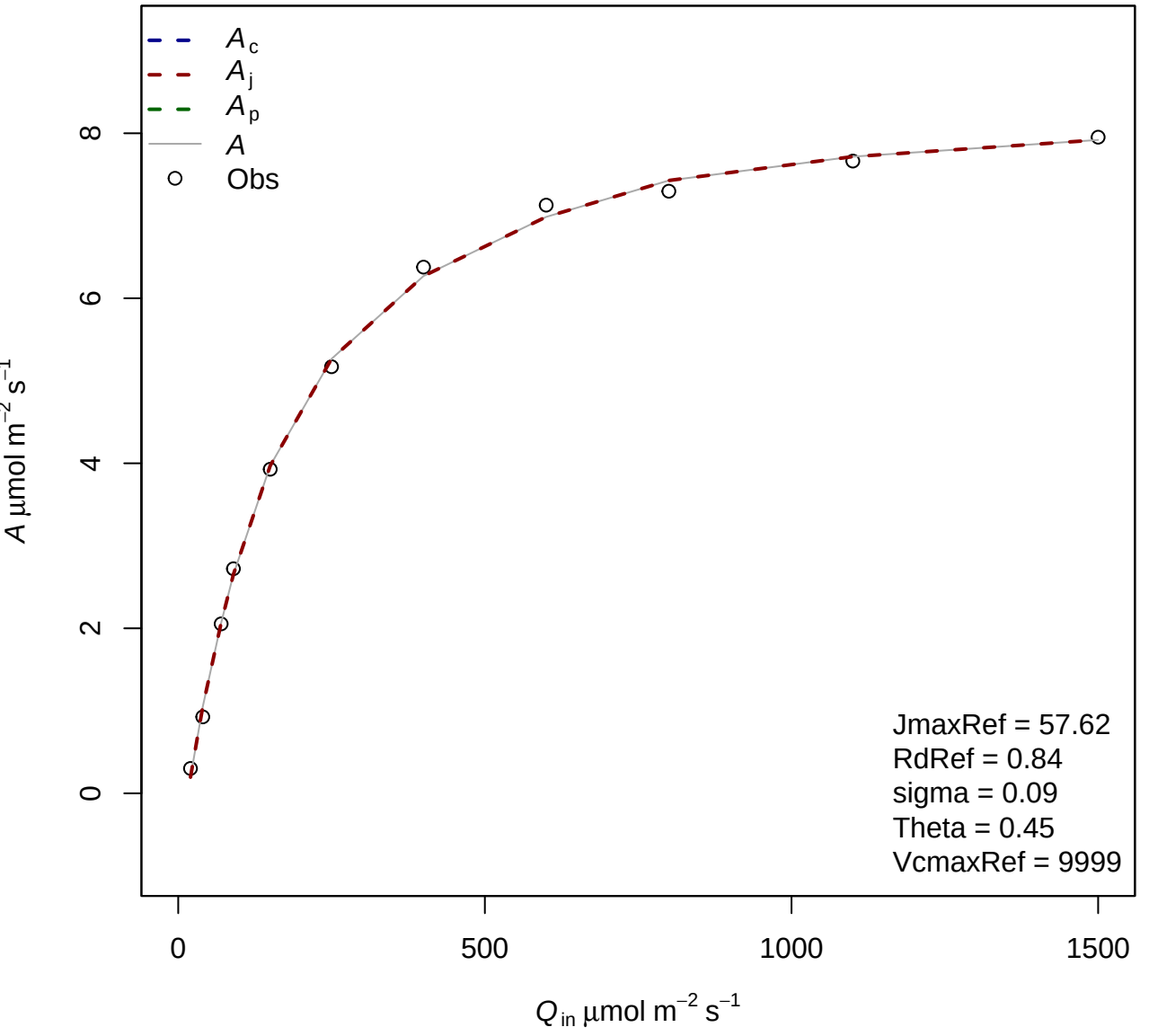

190
A_c
A_j
A_p
A
Obs
JmaxRef = 89.12
RdRef = 1.3
sigma = 0.23
Theta = −0.26
VcmaxRef = 46.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000
2500

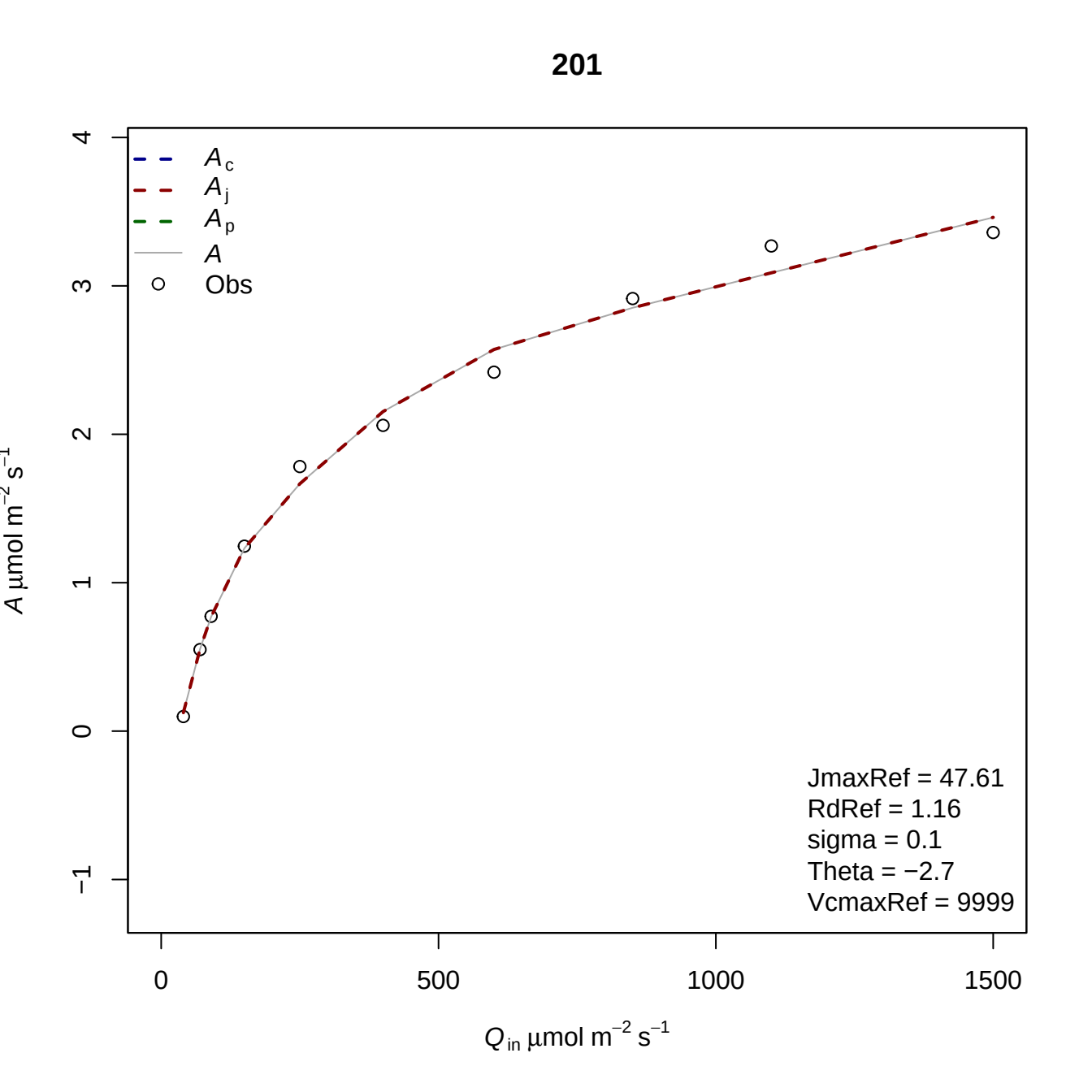

A_c
A_j
A_p
A
Obs
JmaxRef = 156.11
RdRef = 1.58
sigma = 0.16
Theta = 0.47
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

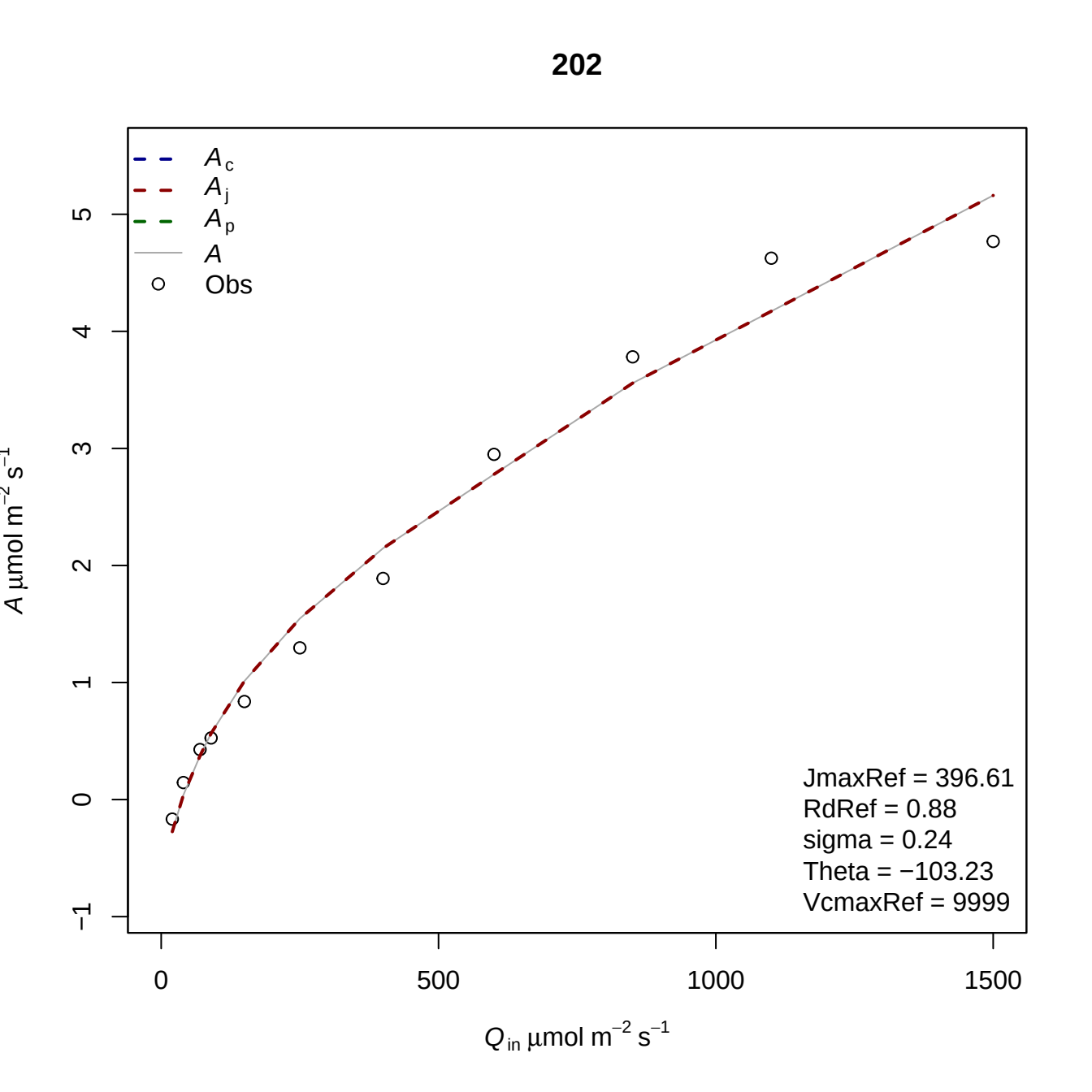

195
A_c
A_j
A_p
A
Obs
JmaxRef = 200.66
RdRef = 2.37
sigma = 0.44
Theta = 0.32
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

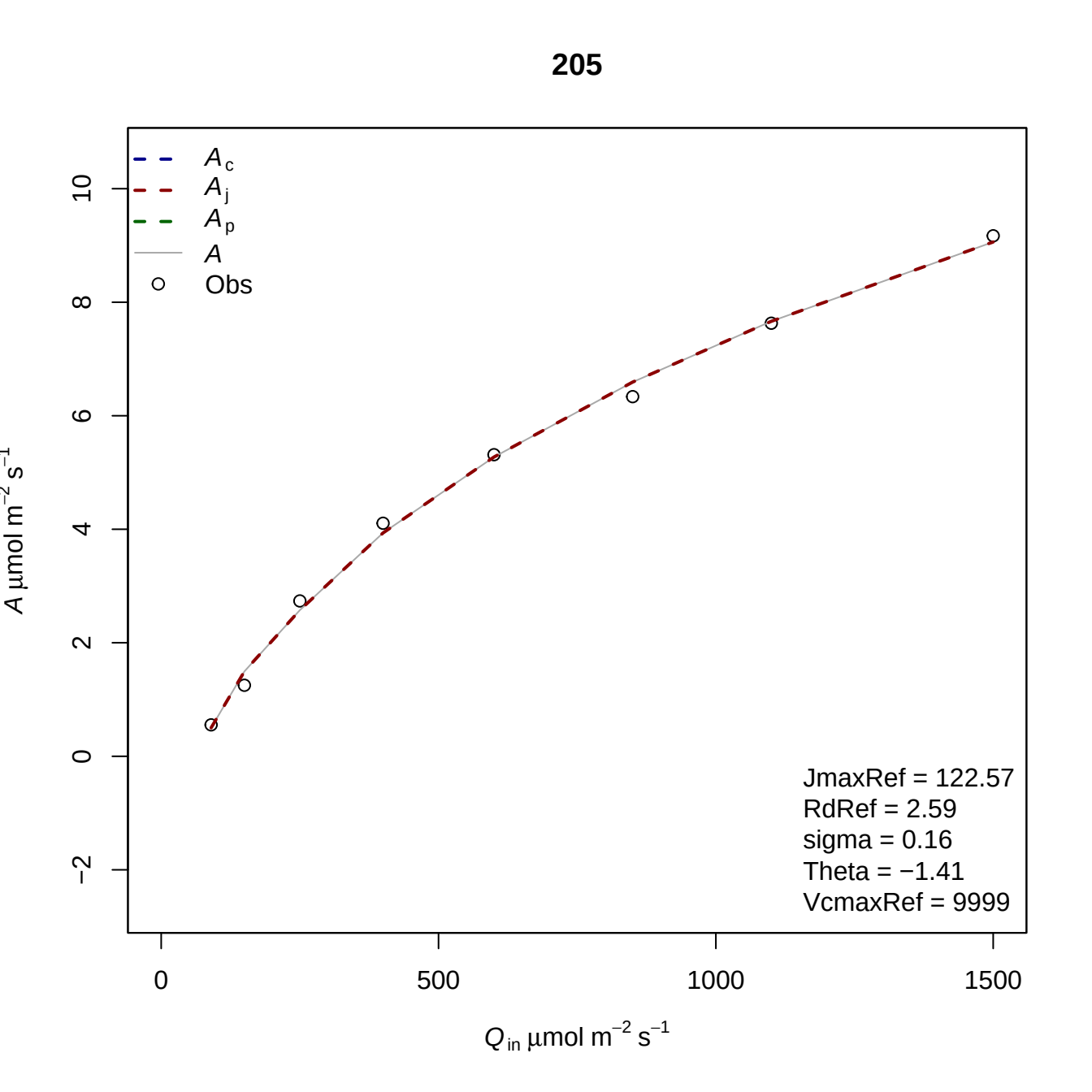

A_c
A_j
A_p
A
Obs
JmaxRef = 78.35
RdRef = 1.59
sigma = 0.27
Theta = 0.61
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

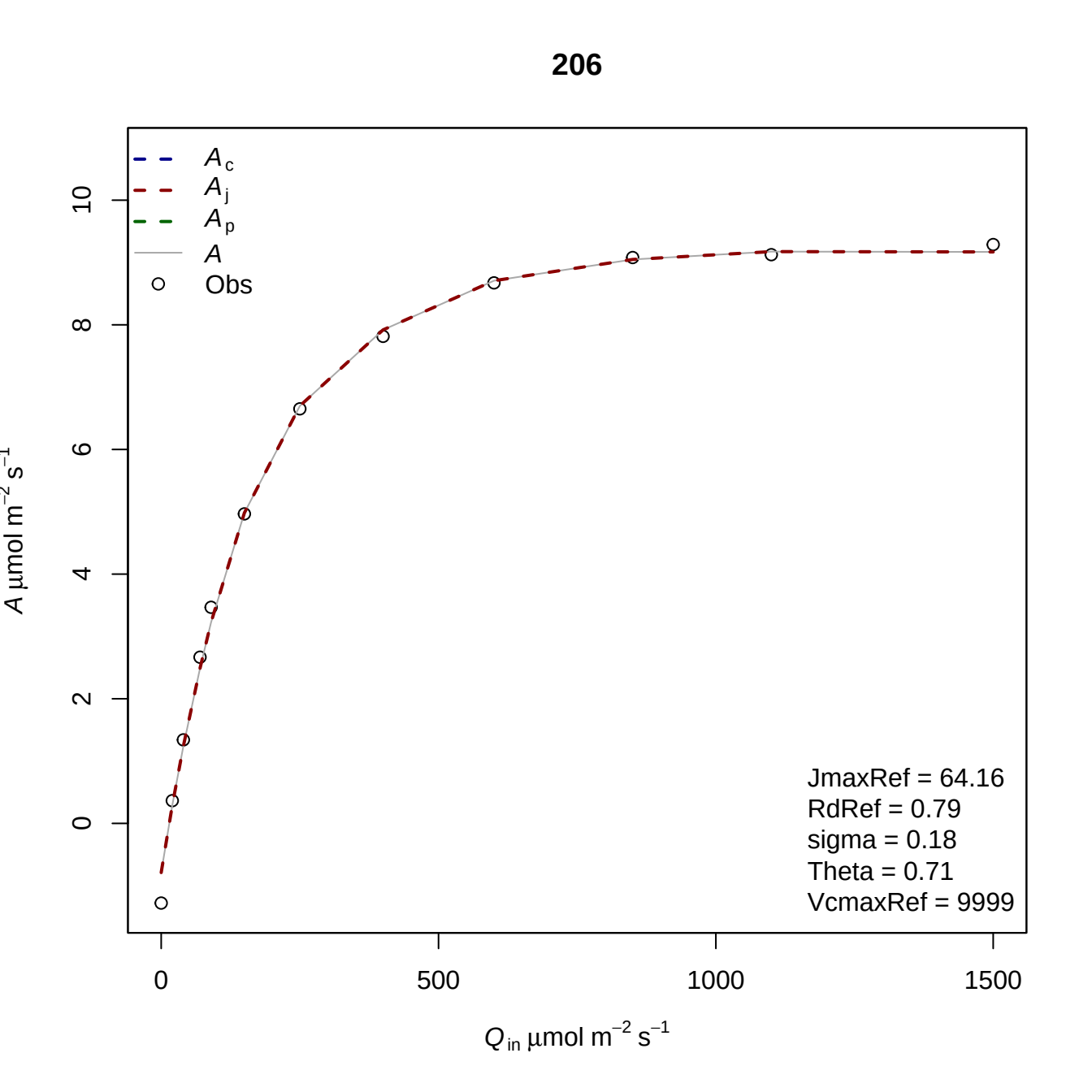

A_c
A_j
A_p
A
Obs
JmaxRef = 102.86
RdRef = 1.53
sigma = 0.25
Theta = 0.54
VcmaxRef = 9999
0
5
10
15
0
200
400
600
800
1000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

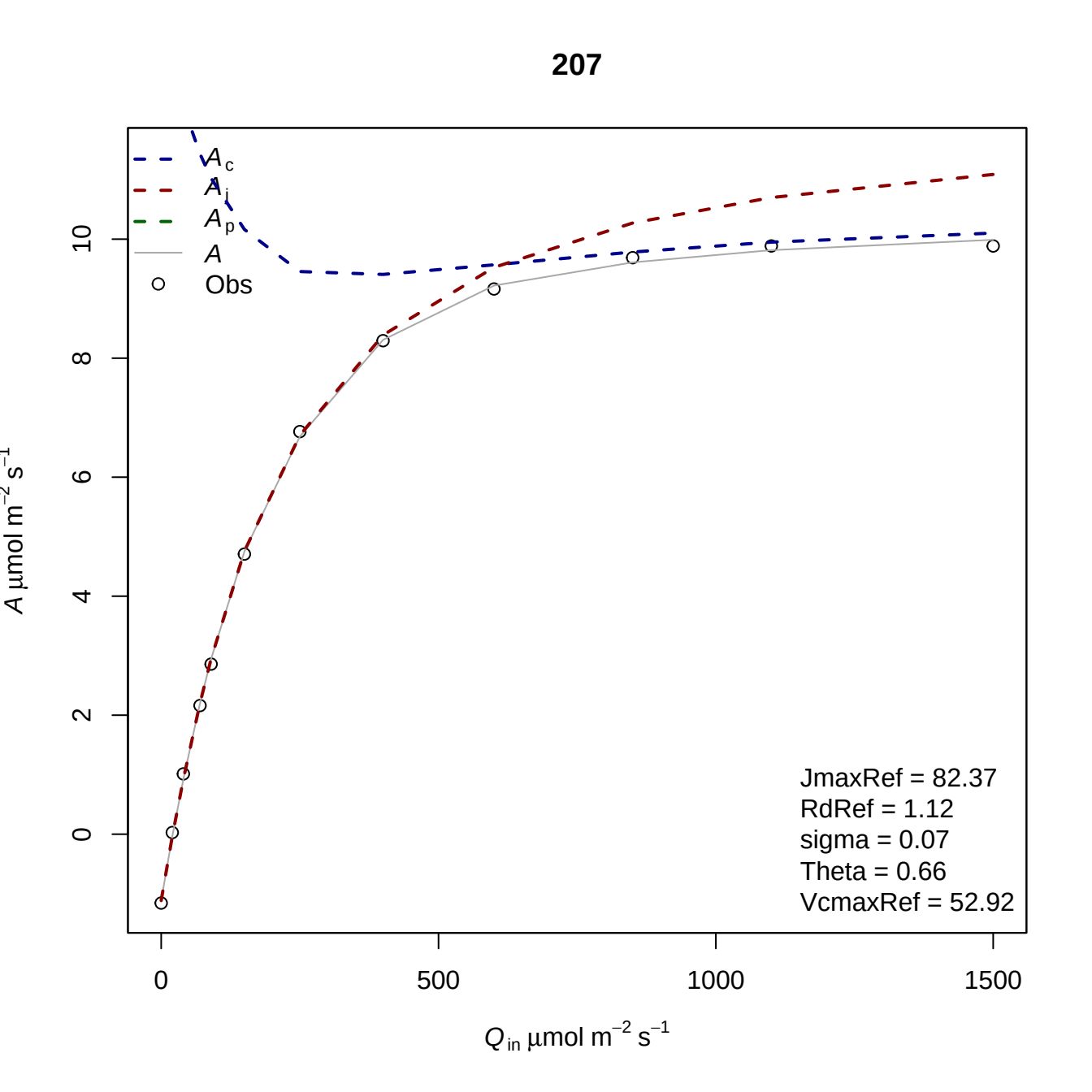

A_c
A_j
A_p
A
Obs
JmaxRef = 57.62
RdRef = 0.84
sigma = 0.09
Theta = 0.45
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

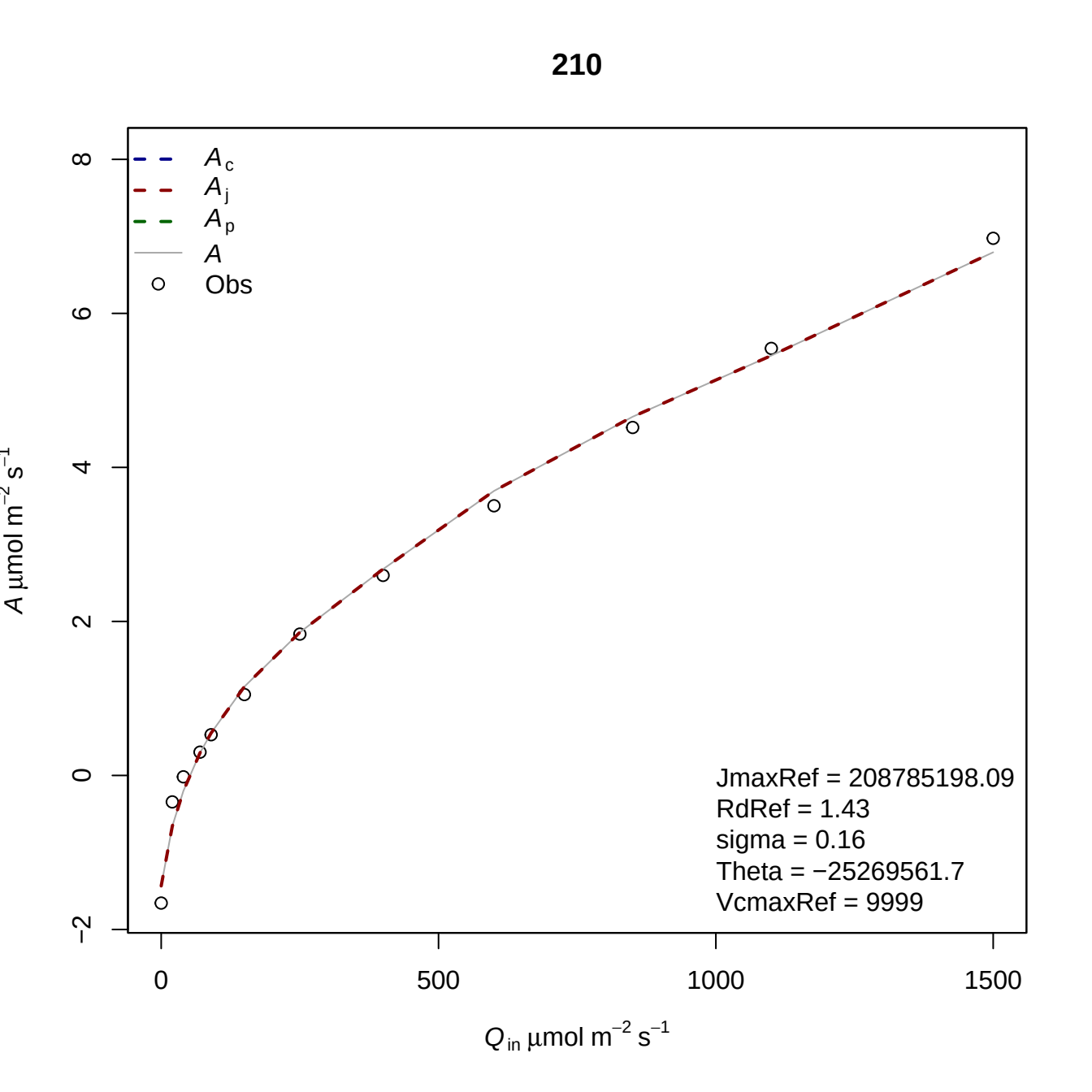

201
A_c
A_j
A_p
A
Obs
JmaxRef = 47.61
RdRef = 1.16
sigma = 0.1
Theta = −2.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
500
1000
1500

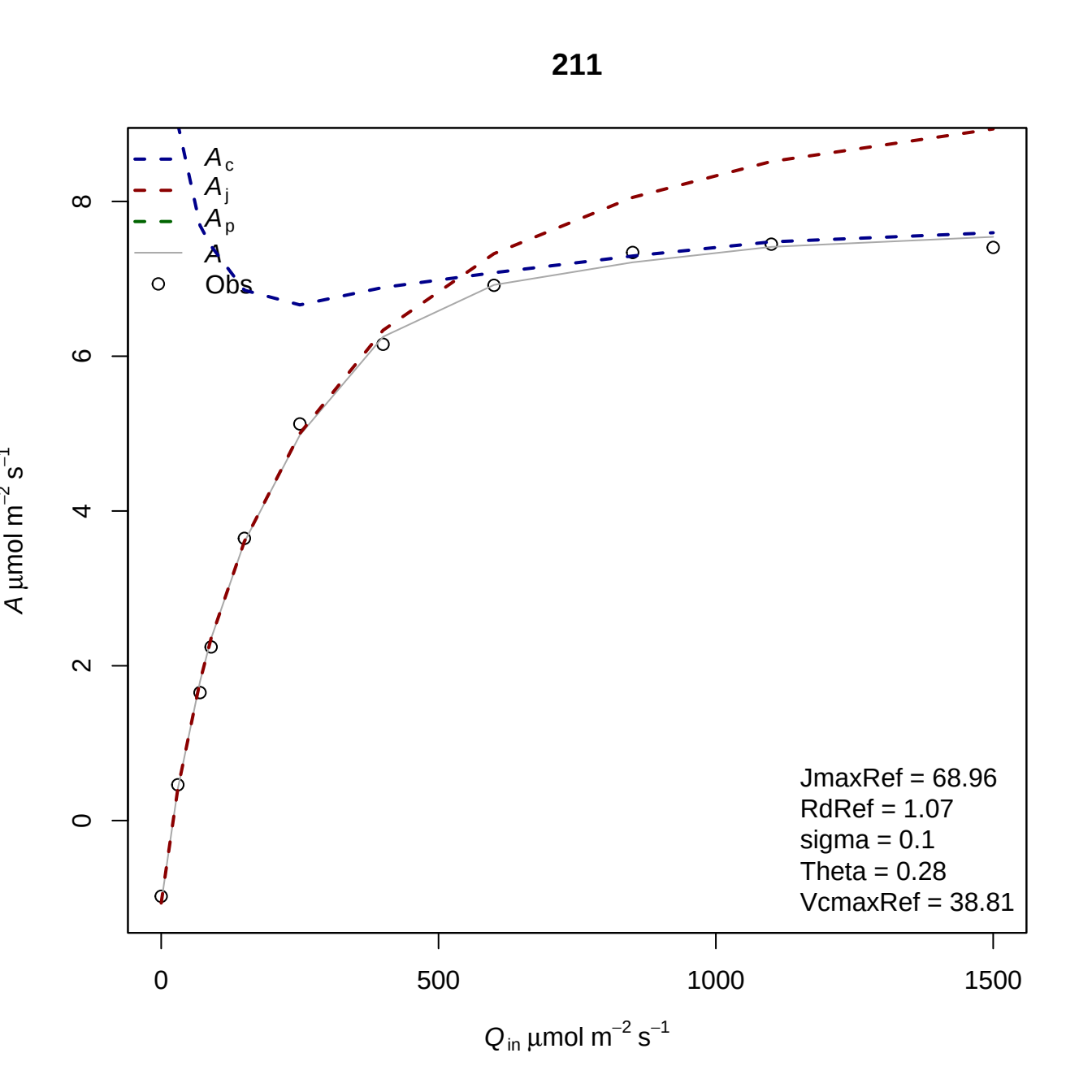

202
A_c
A_j
A_p
A
Obs
JmaxRef = 396.61
RdRef = 0.88
sigma = 0.24
Theta = −103.23
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

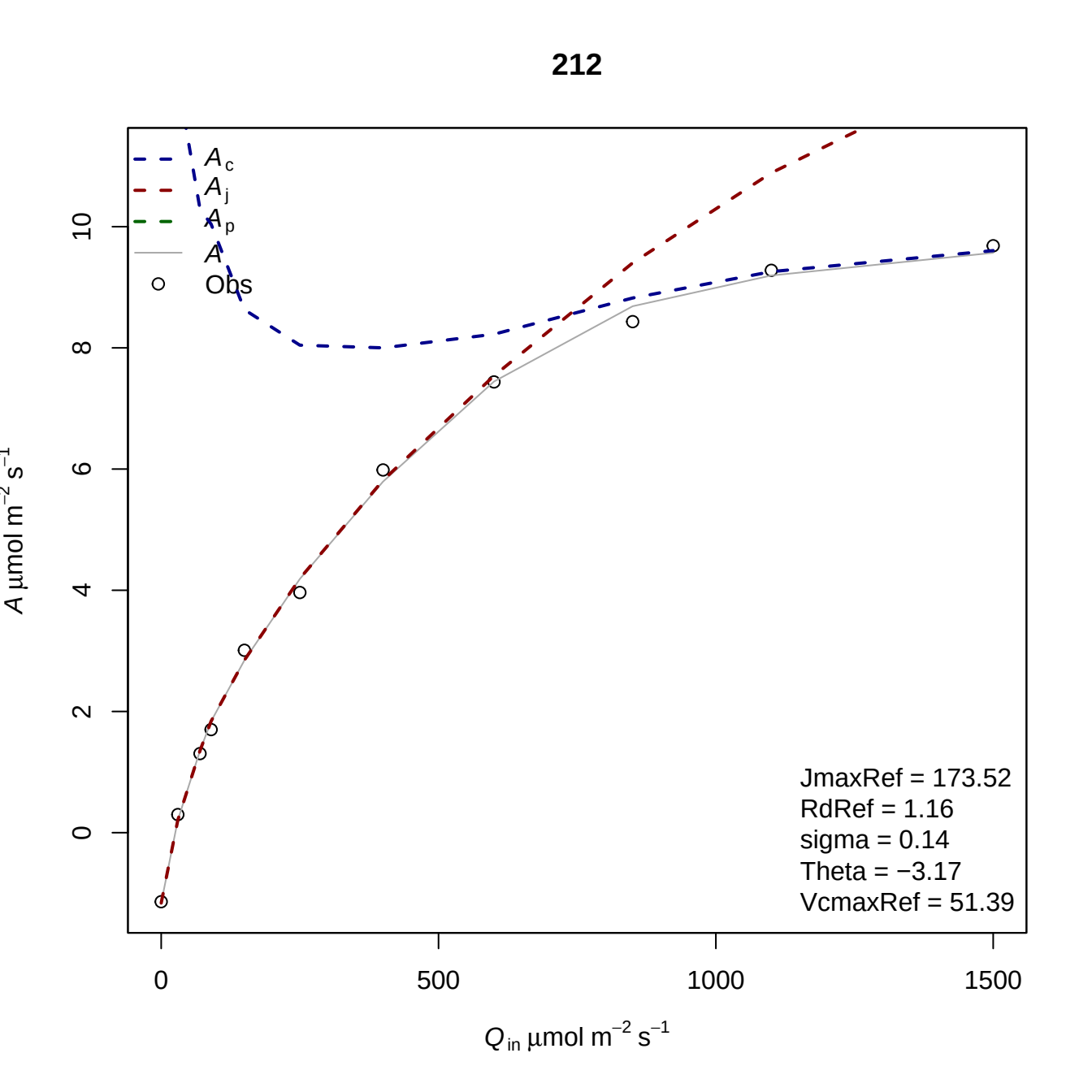

205
A_c
A_j
A_p
A
Obs
JmaxRef = 122.57
RdRef = 2.59
sigma = 0.16
Theta = −1.41
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

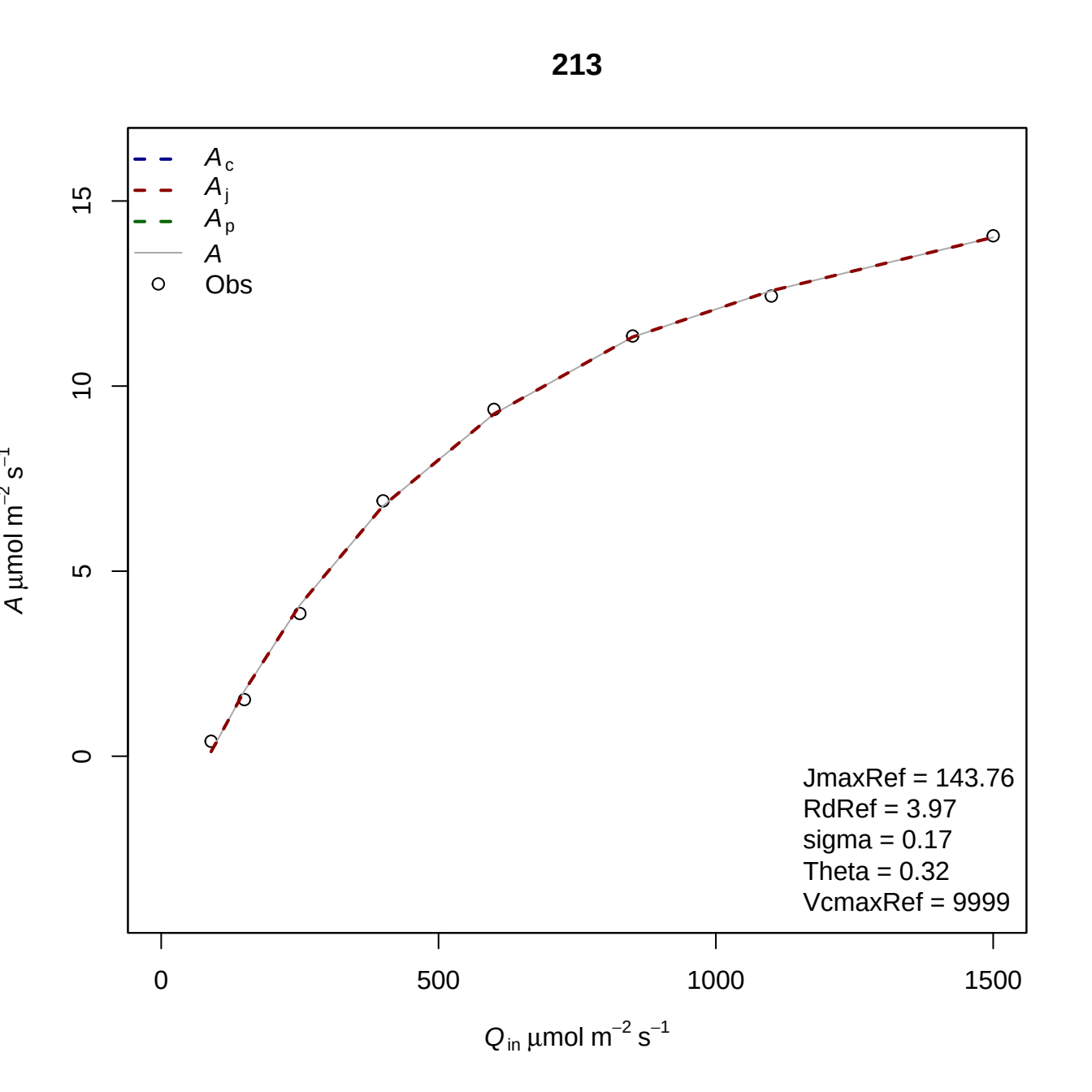

206
A_c
A_j
A_p
A
Obs
JmaxRef = 64.16
RdRef = 0.79
sigma = 0.18
Theta = 0.71
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

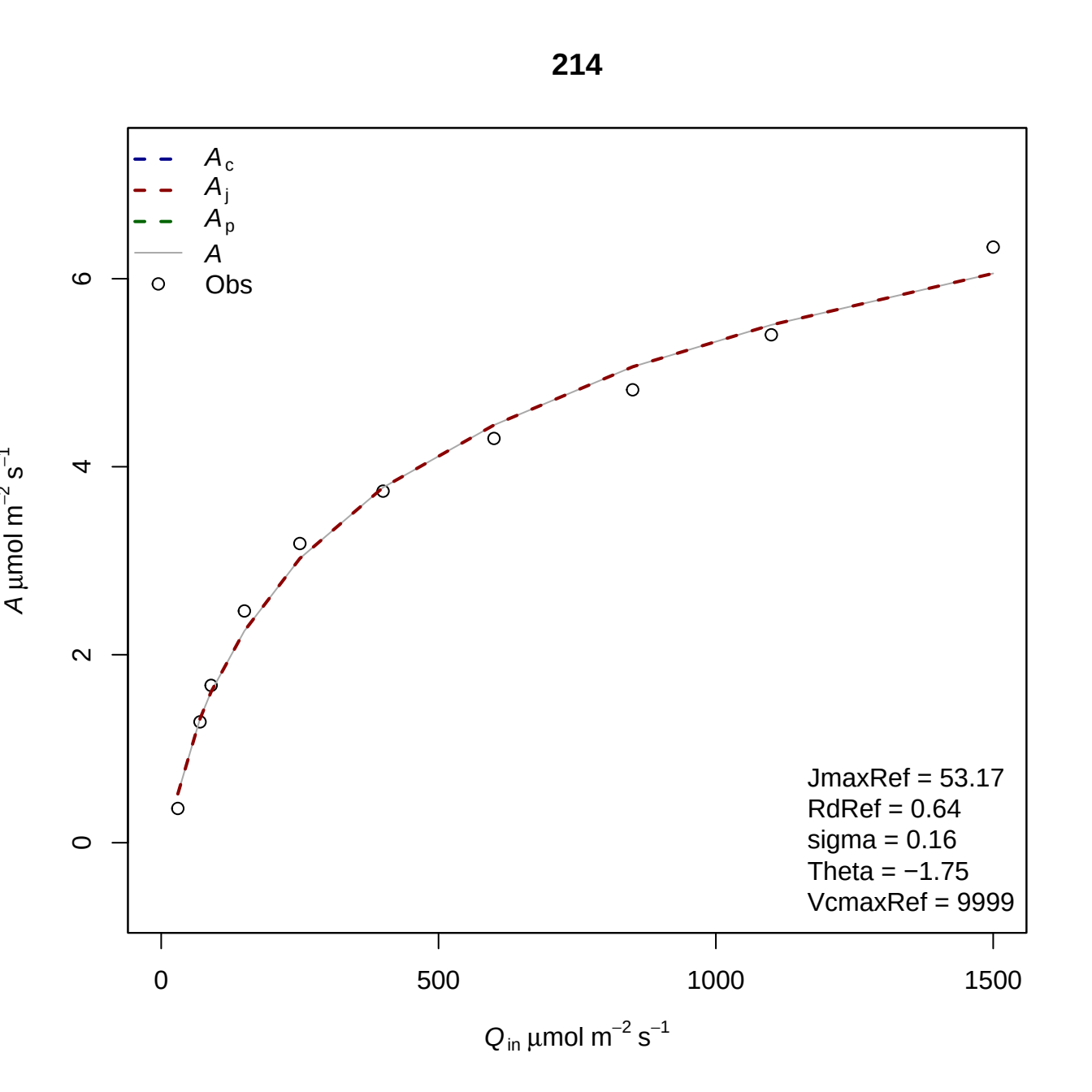

207
A_c
A_j
A_p
A
Obs
JmaxRef = 82.37
RdRef = 1.12
sigma = 0.07
Theta = 0.66
VcmaxRef = 52.92
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

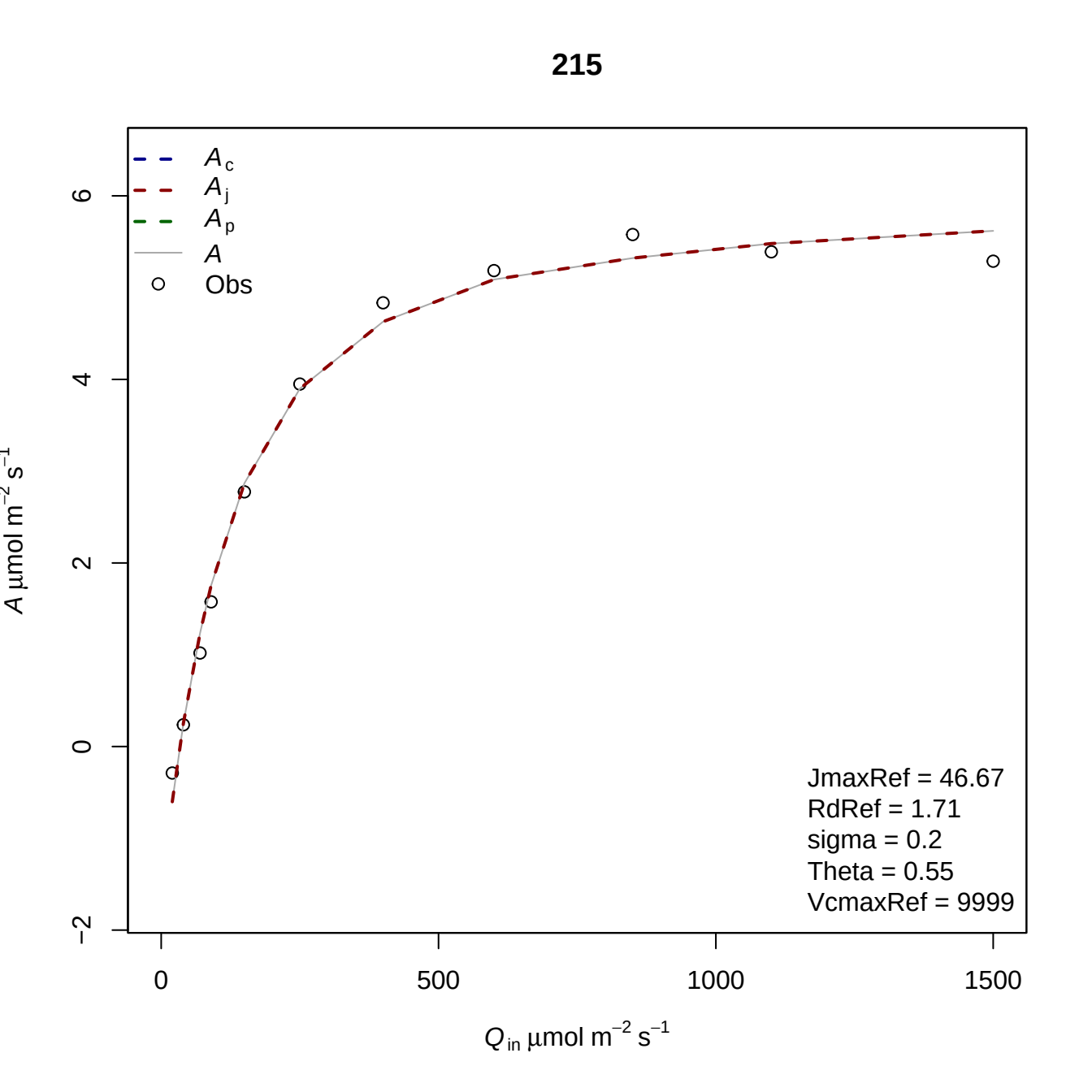

210
A_c
A_j
A_p
A
Obs
JmaxRef = 208785198.09
RdRef = 1.43
sigma = 0.16
Theta = −25269561.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

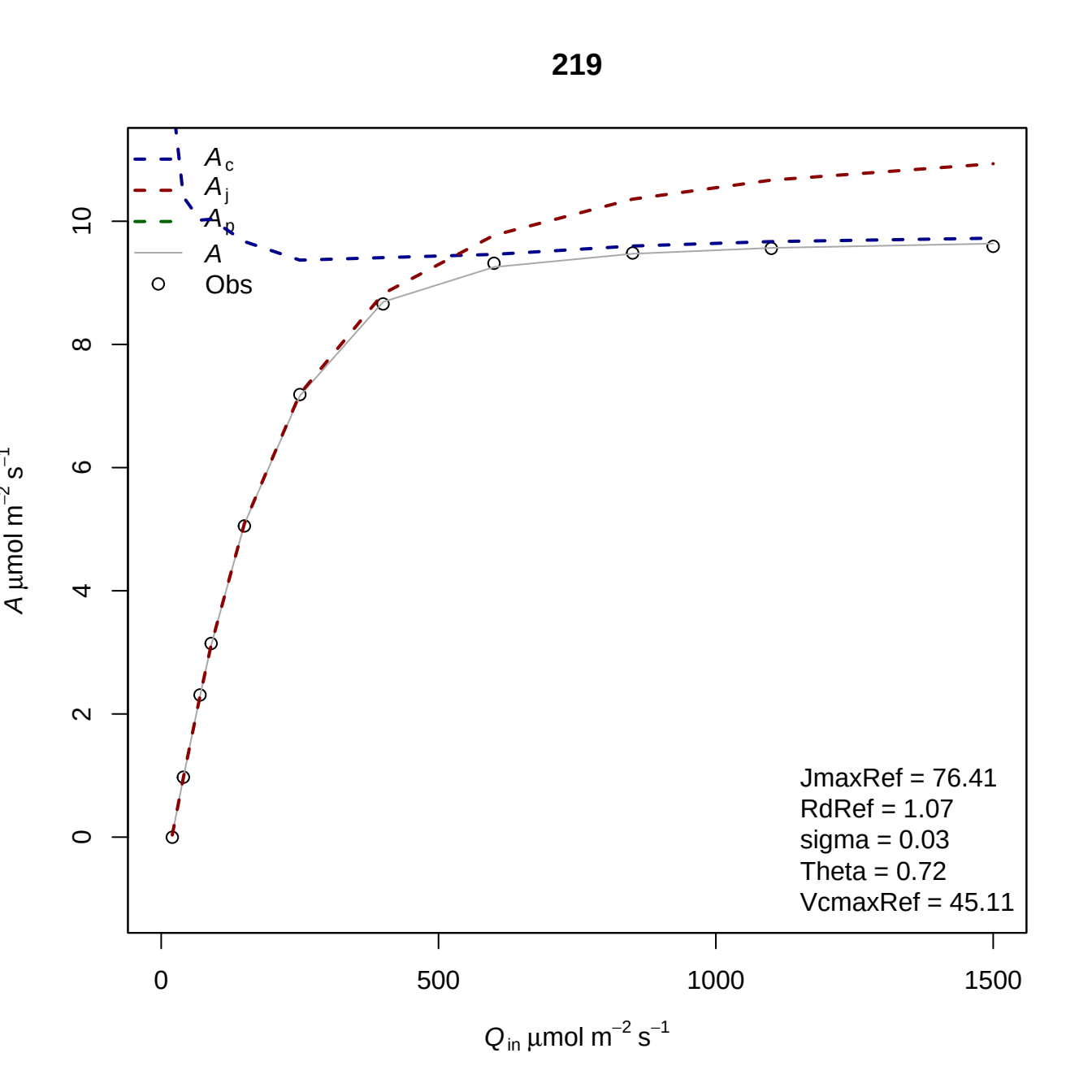

211
A_c
A_j
A_p
A
Obs
JmaxRef = 68.96
RdRef = 1.07
sigma = 0.1
Theta = 0.28
VcmaxRef = 38.81
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

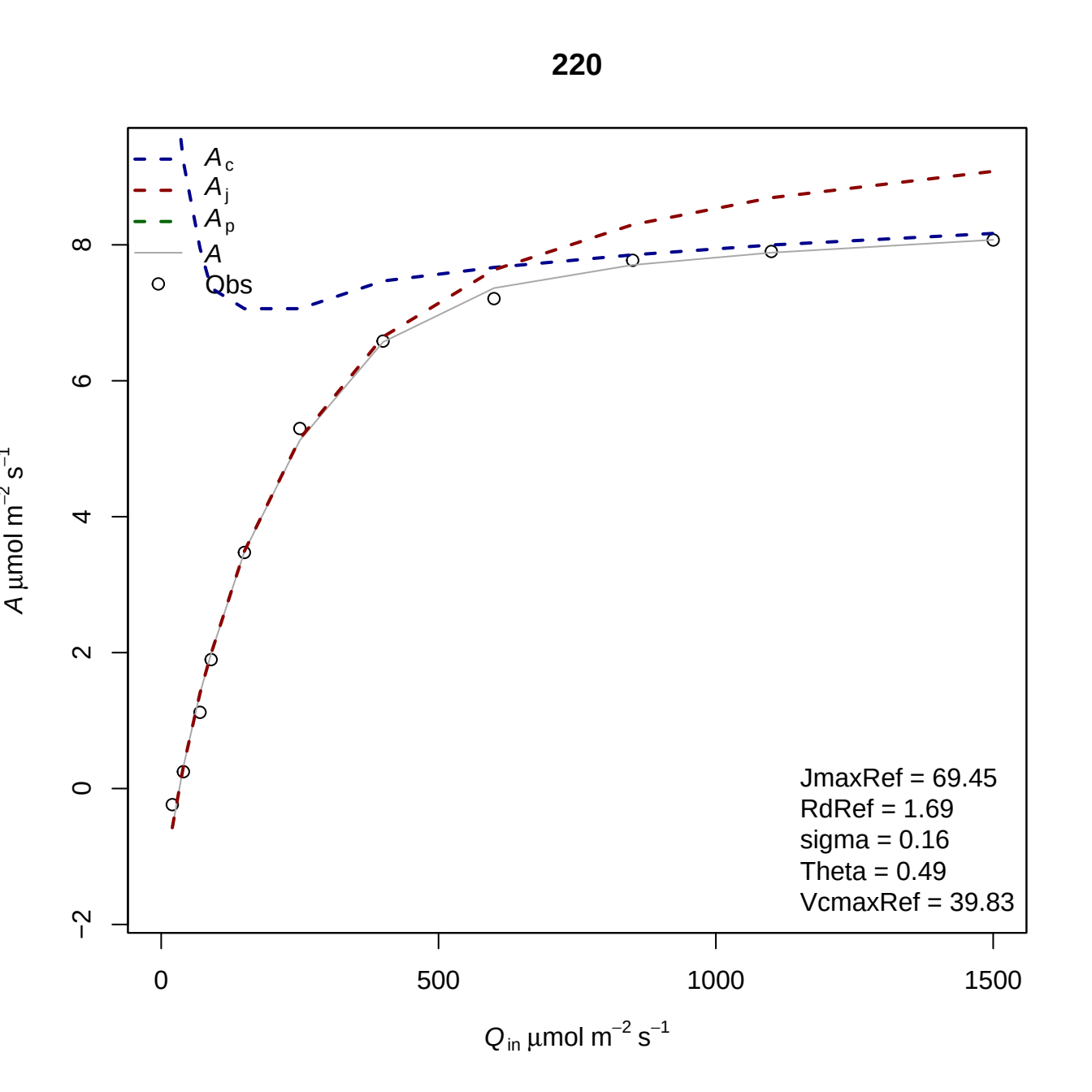

212
A_c
A_j
A_p
A
Obs
JmaxRef = 173.52
RdRef = 1.16
sigma = 0.14
Theta = −3.17
VcmaxRef = 51.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

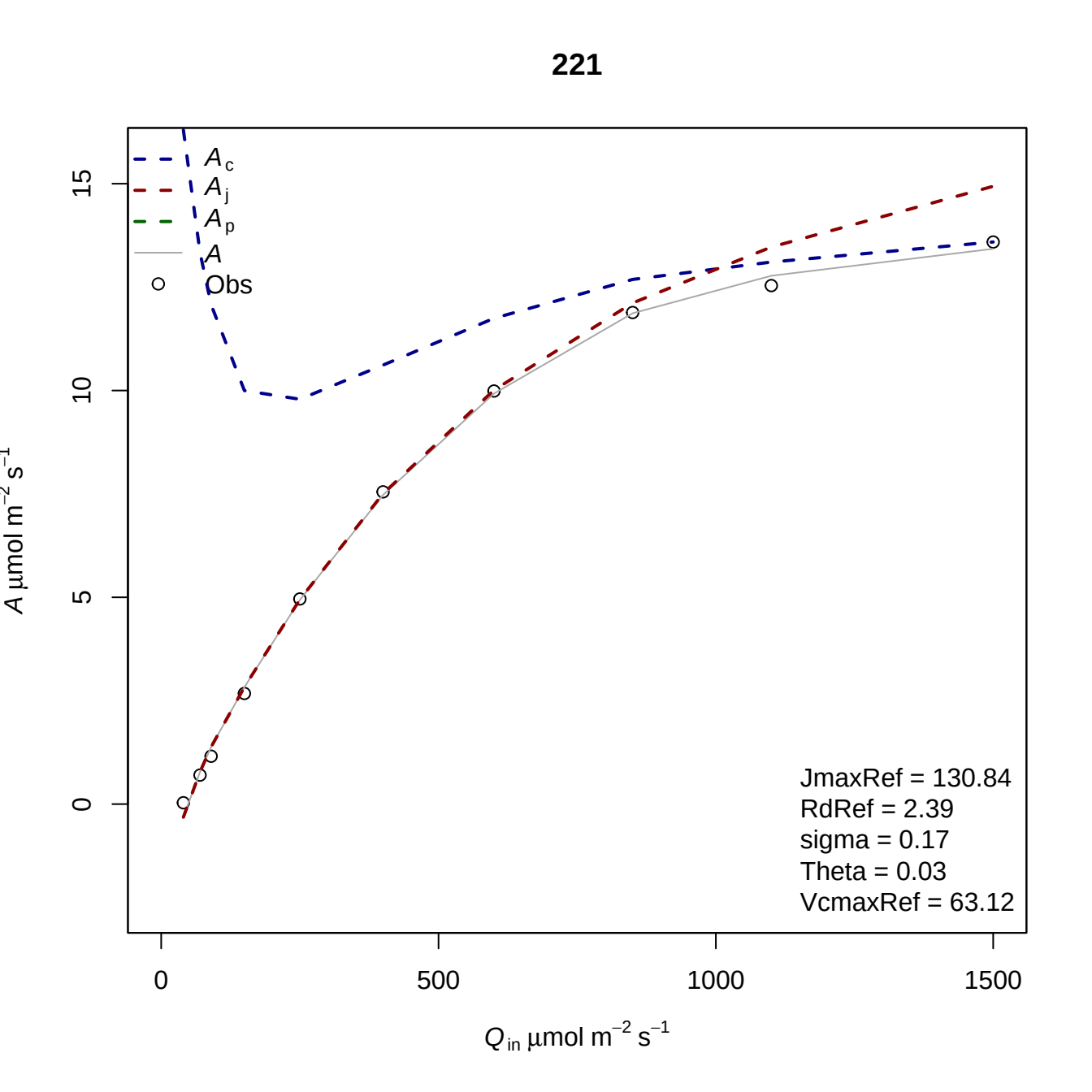

213
A_c
A_j
A_p
A
Obs
JmaxRef = 143.76
RdRef = 3.97
sigma = 0.17
Theta = 0.32
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

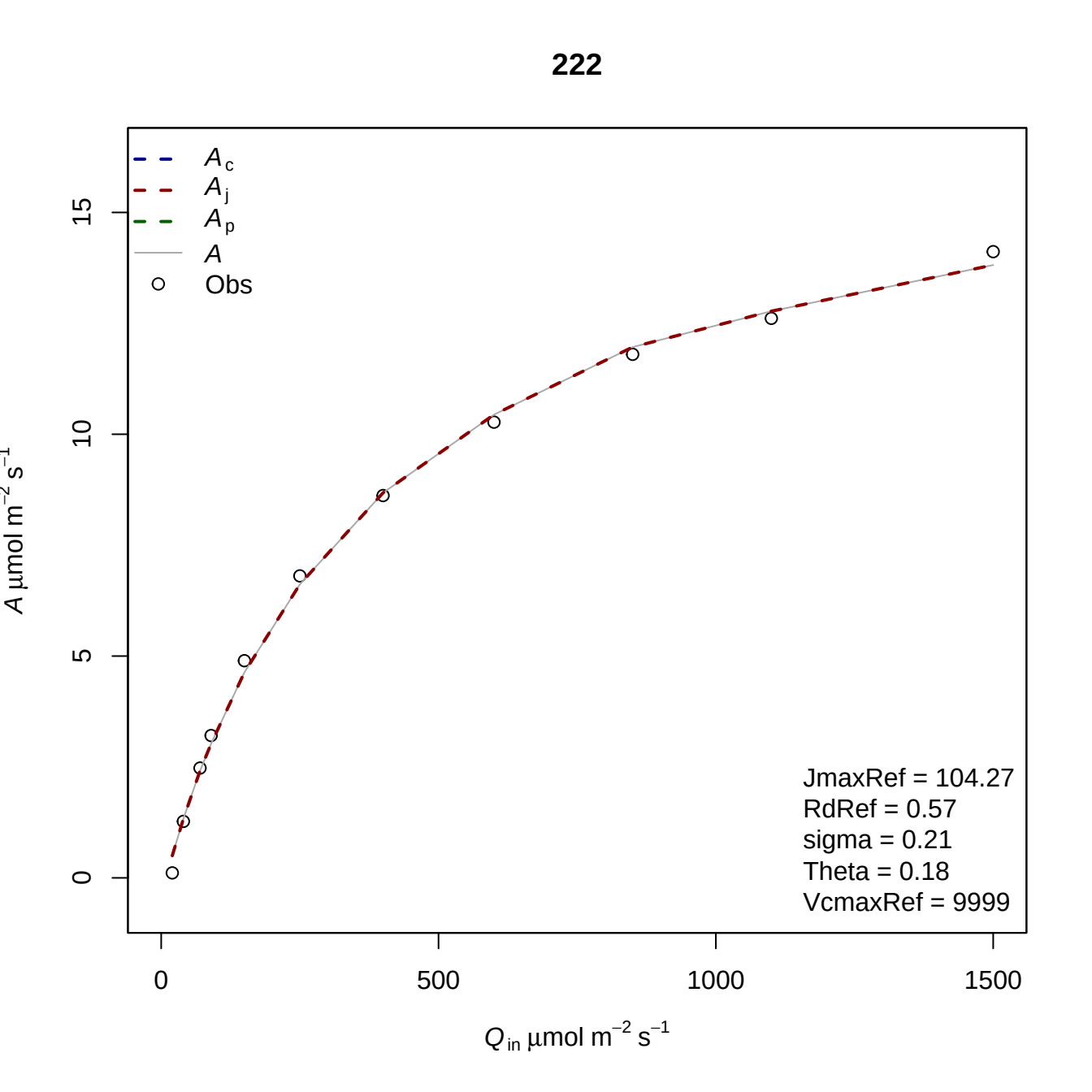

214
A_c
A_j
A_p
A
Obs
JmaxRef = 53.17
RdRef = 0.64
sigma = 0.16
Theta = −1.75
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

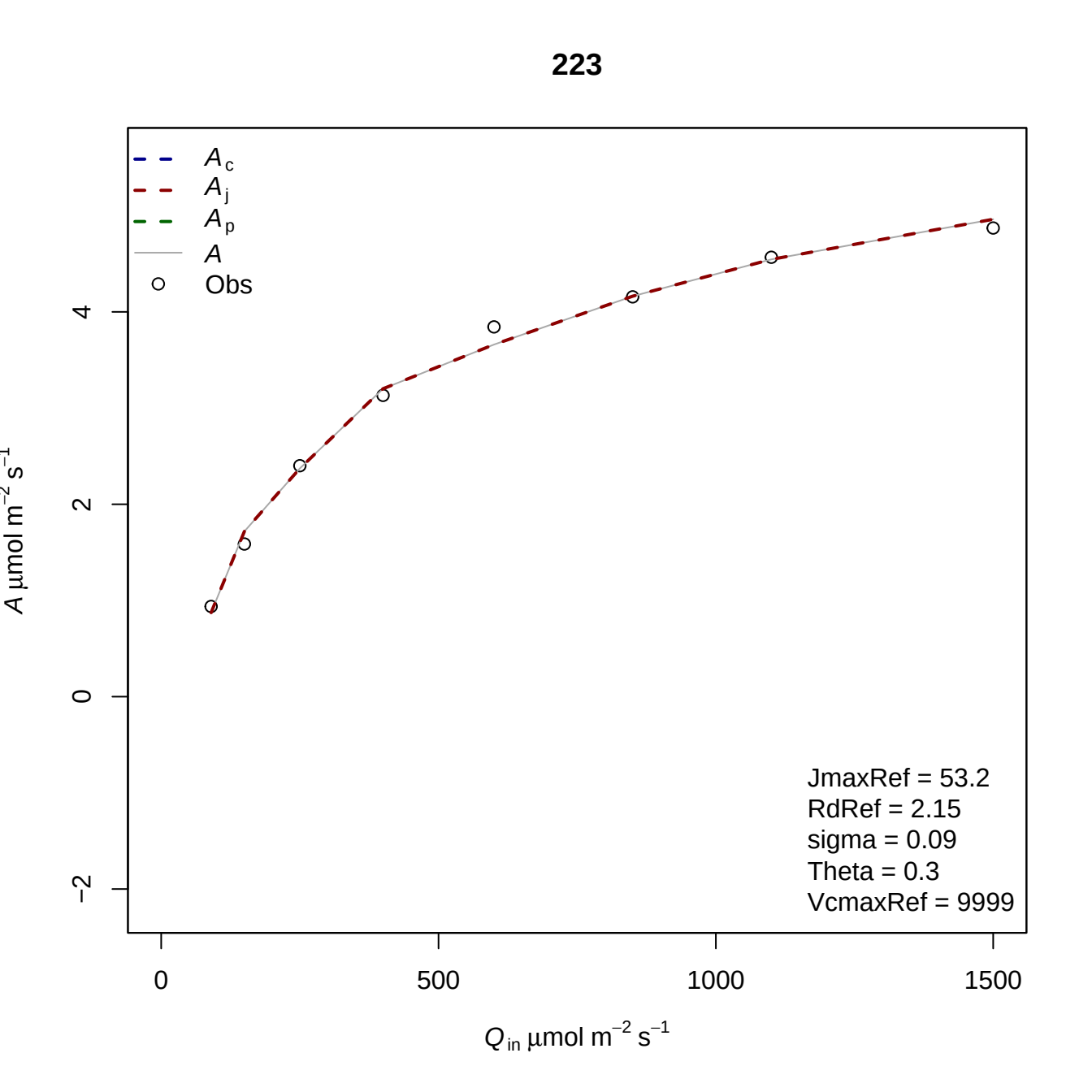

215
A_c
A_j
A_p
A
Obs
JmaxRef = 46.67
RdRef = 1.71
sigma = 0.2
Theta = 0.55
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

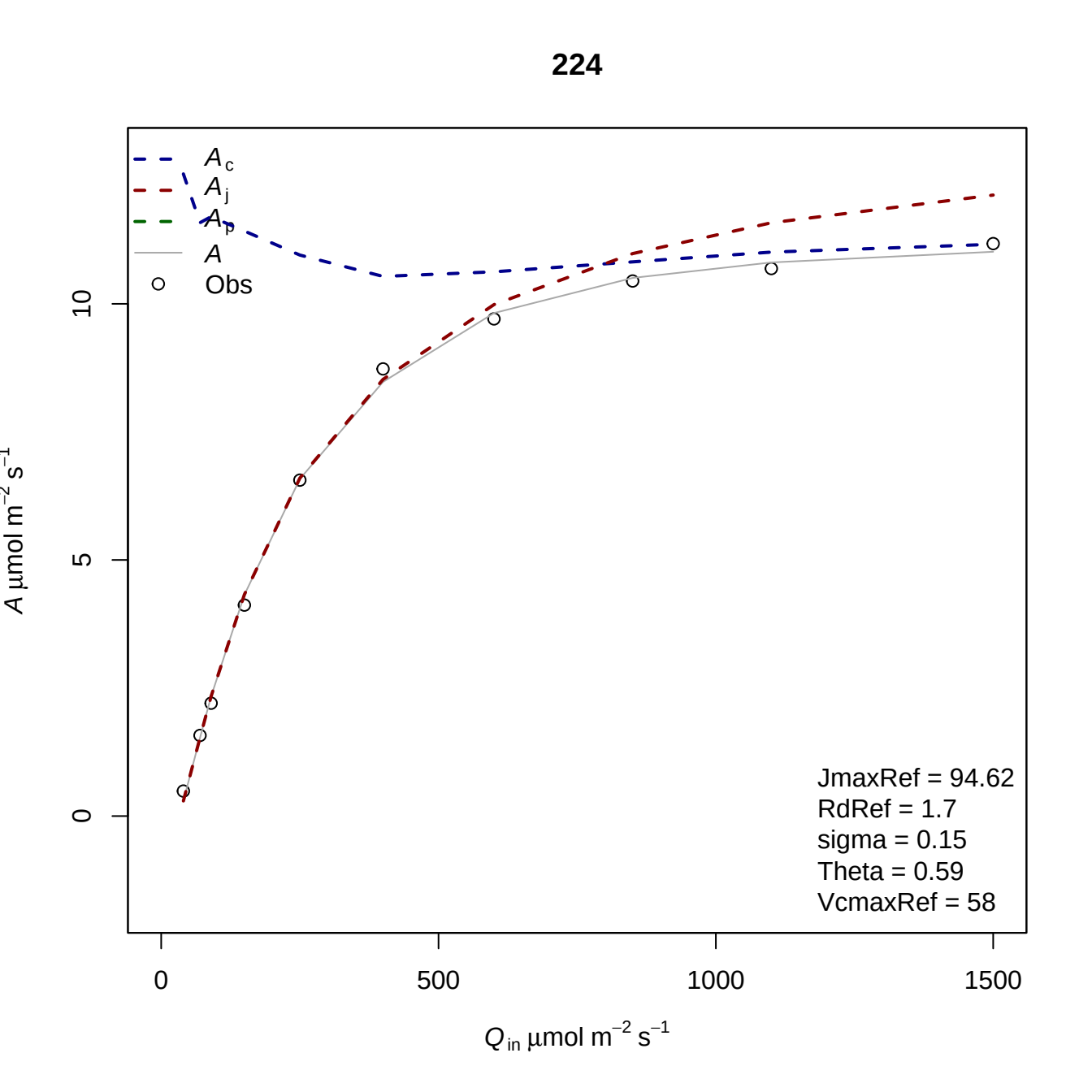

219
A_c
A_j
A_p
A
Obs
JmaxRef = 76.41
RdRef = 1.07
sigma = 0.03
Theta = 0.72
VcmaxRef = 45.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

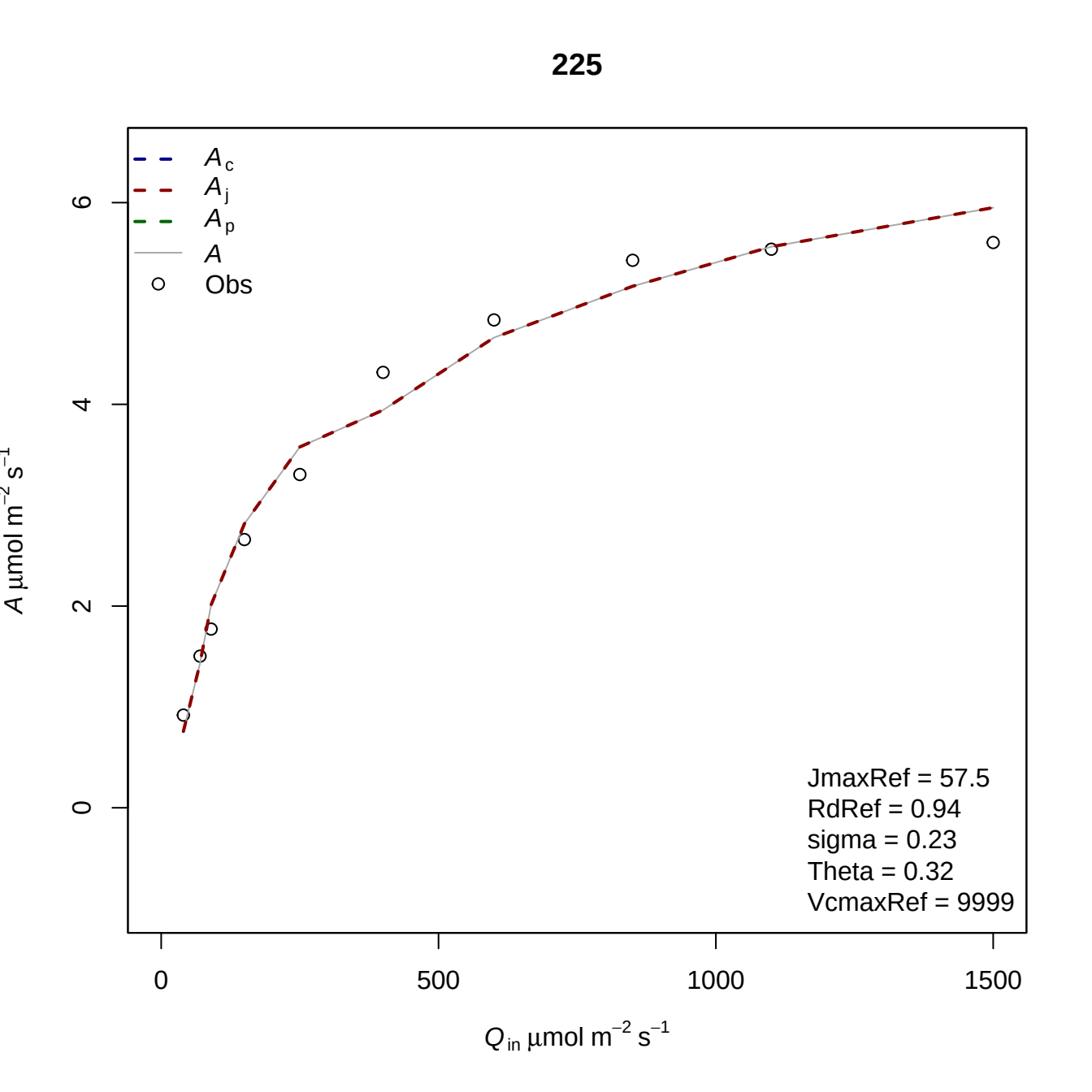

220
A_c
A_j
A_p
A
Qbs
JmaxRef = 69.45
RdRef = 1.69
sigma = 0.16
Theta = 0.49
VcmaxRef = 39.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

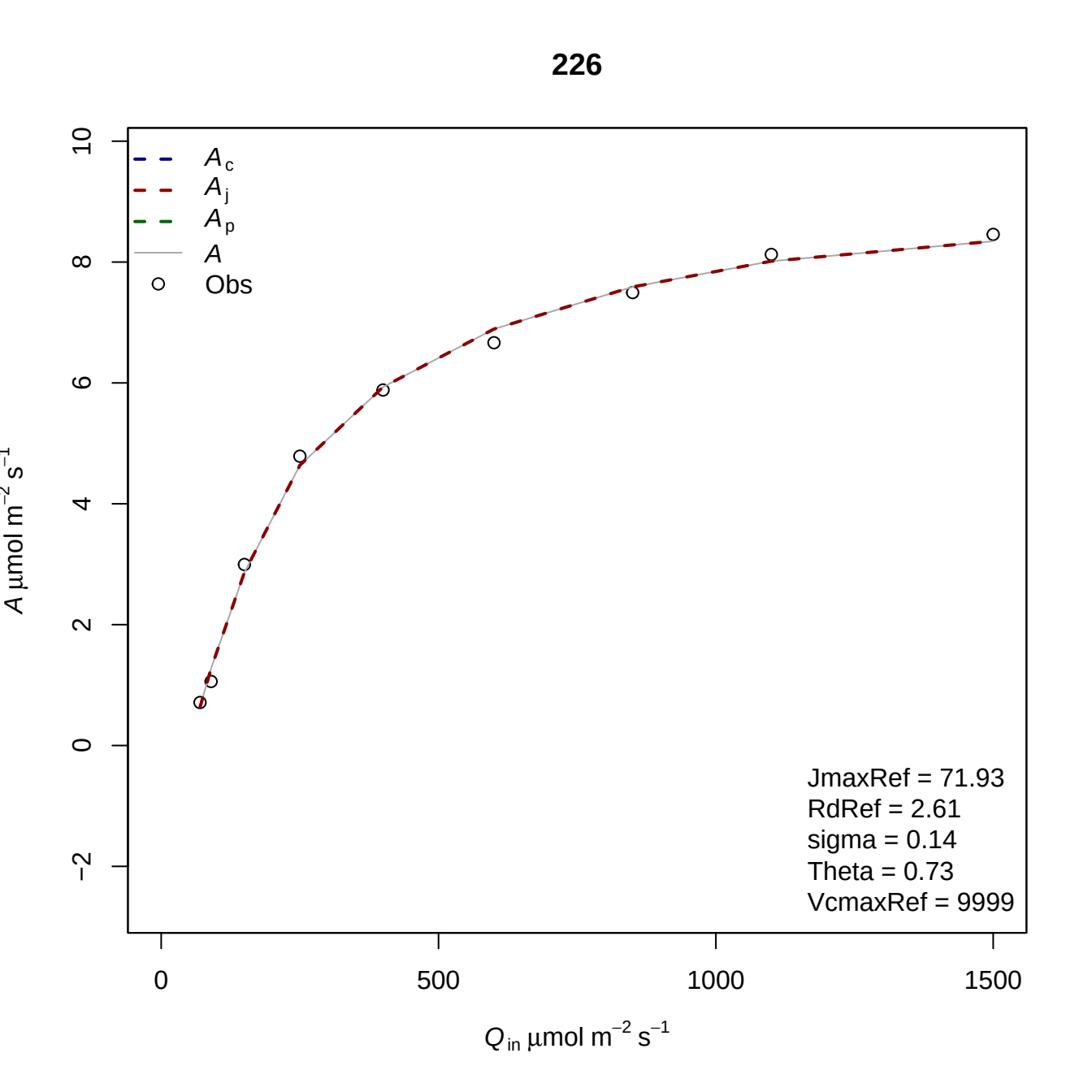

221
A_c
A_j
A_p
A
Obs
JmaxRef = 130.84
RdRef = 2.39
sigma = 0.17
Theta = 0.03
VcmaxRef = 63.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

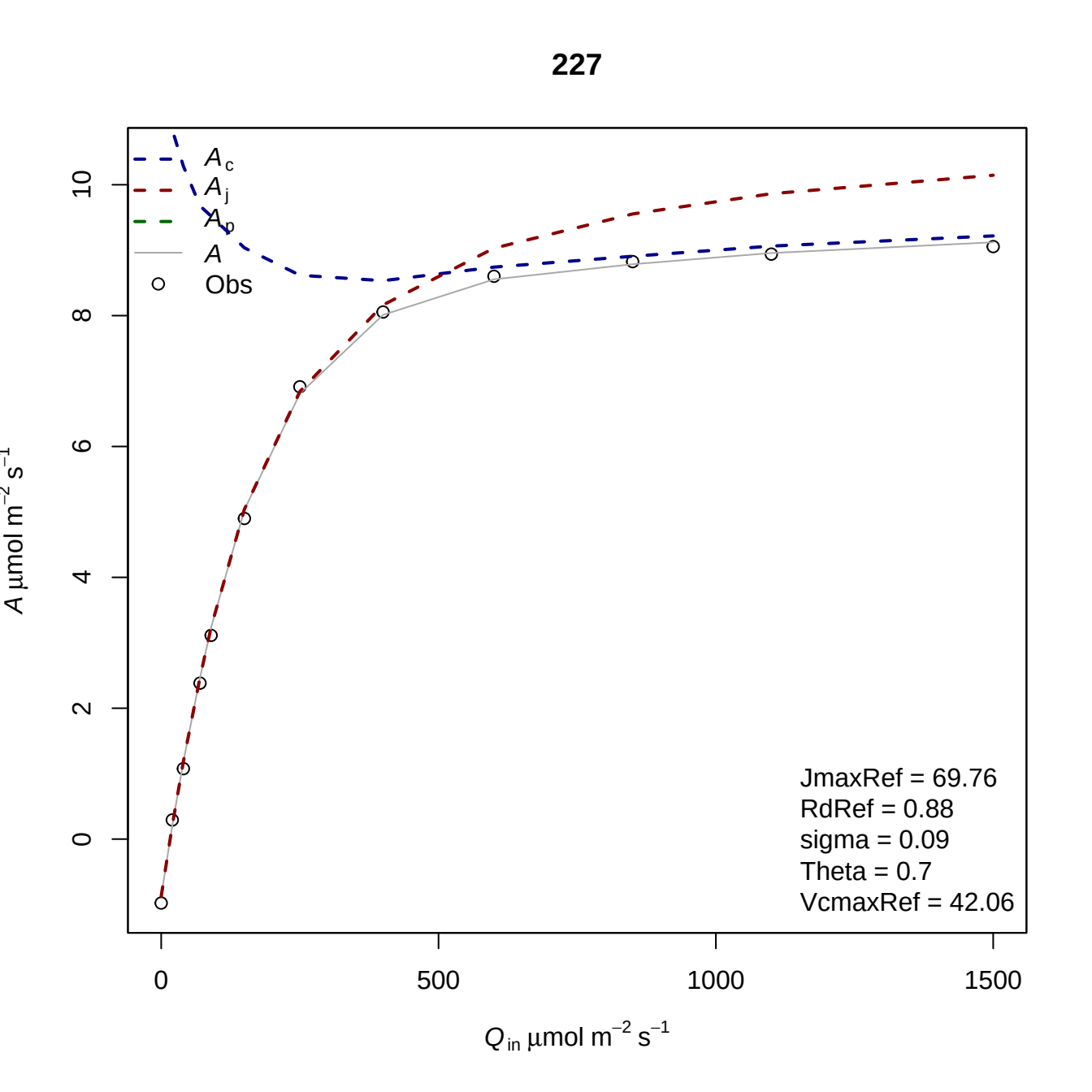

222
A_c
A_j
A_p
A
Obs
JmaxRef = 104.27
RdRef = 0.57
sigma = 0.21
Theta = 0.18
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

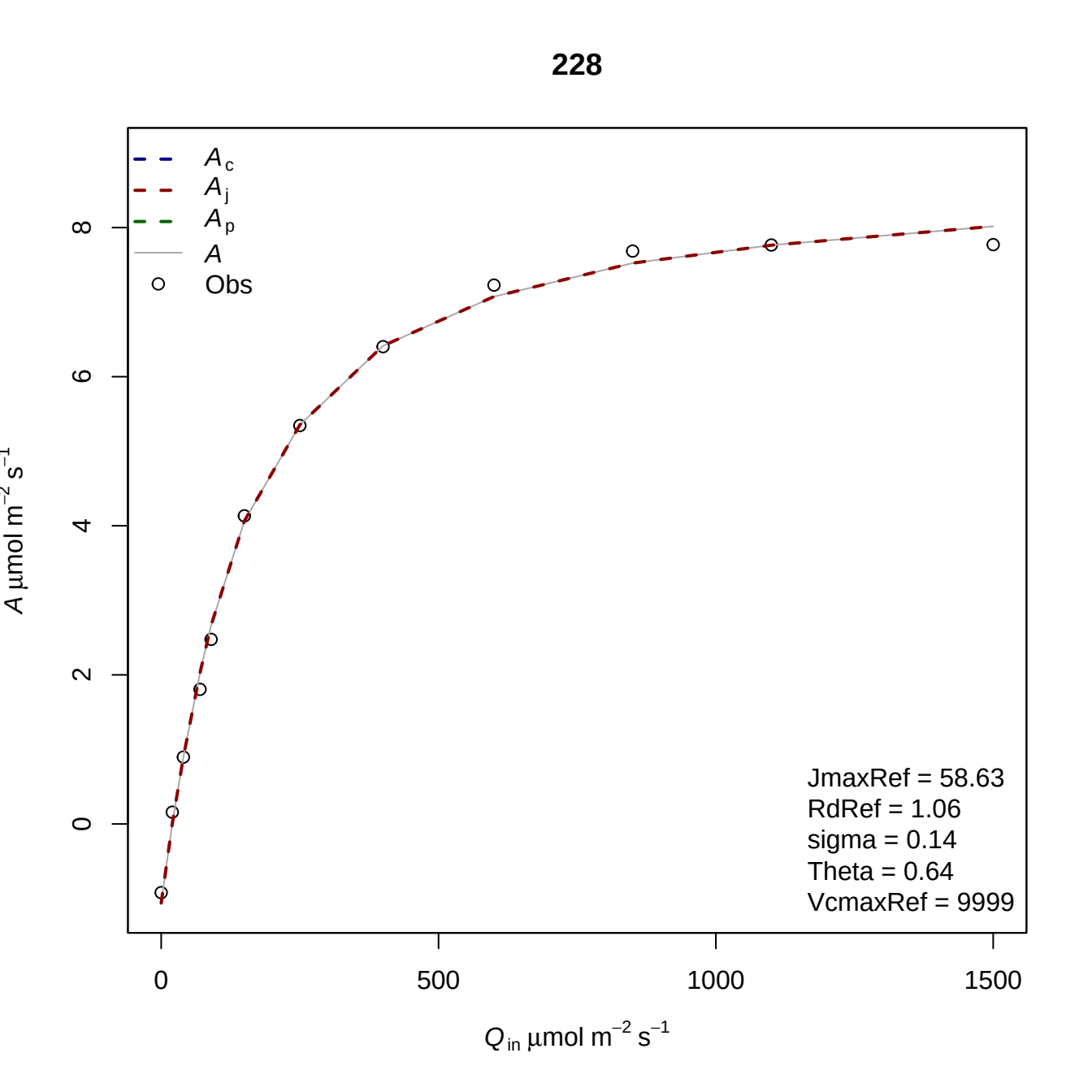

223
A_c
A_j
A_p
A
Obs
JmaxRef = 53.2
RdRef = 2.15
sigma = 0.09
Theta = 0.3
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

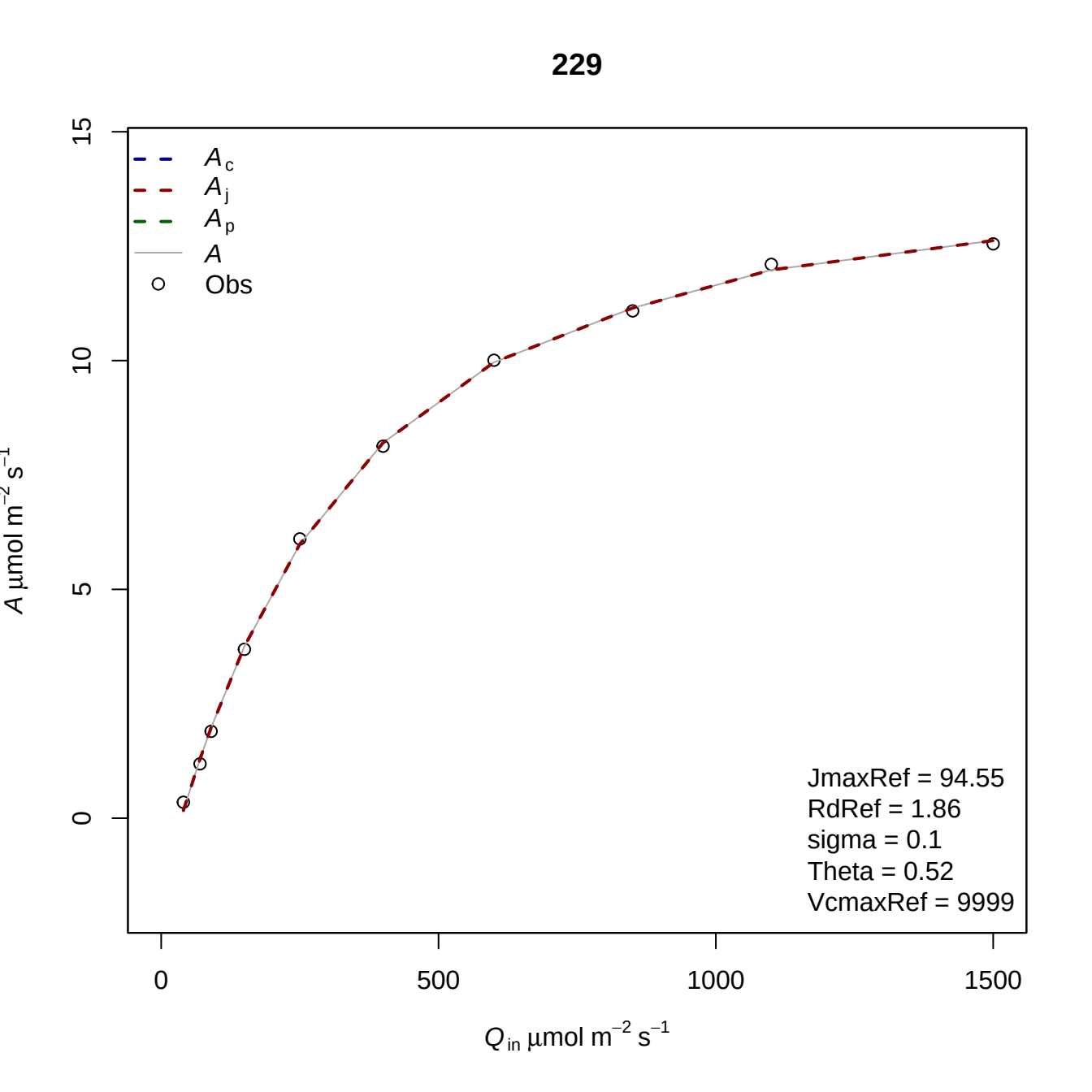

224
A_c
A_j
A_p
A
Obs
JmaxRef = 94.62
RdRef = 1.7
sigma = 0.15
Theta = 0.59
VcmaxRef = 58
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

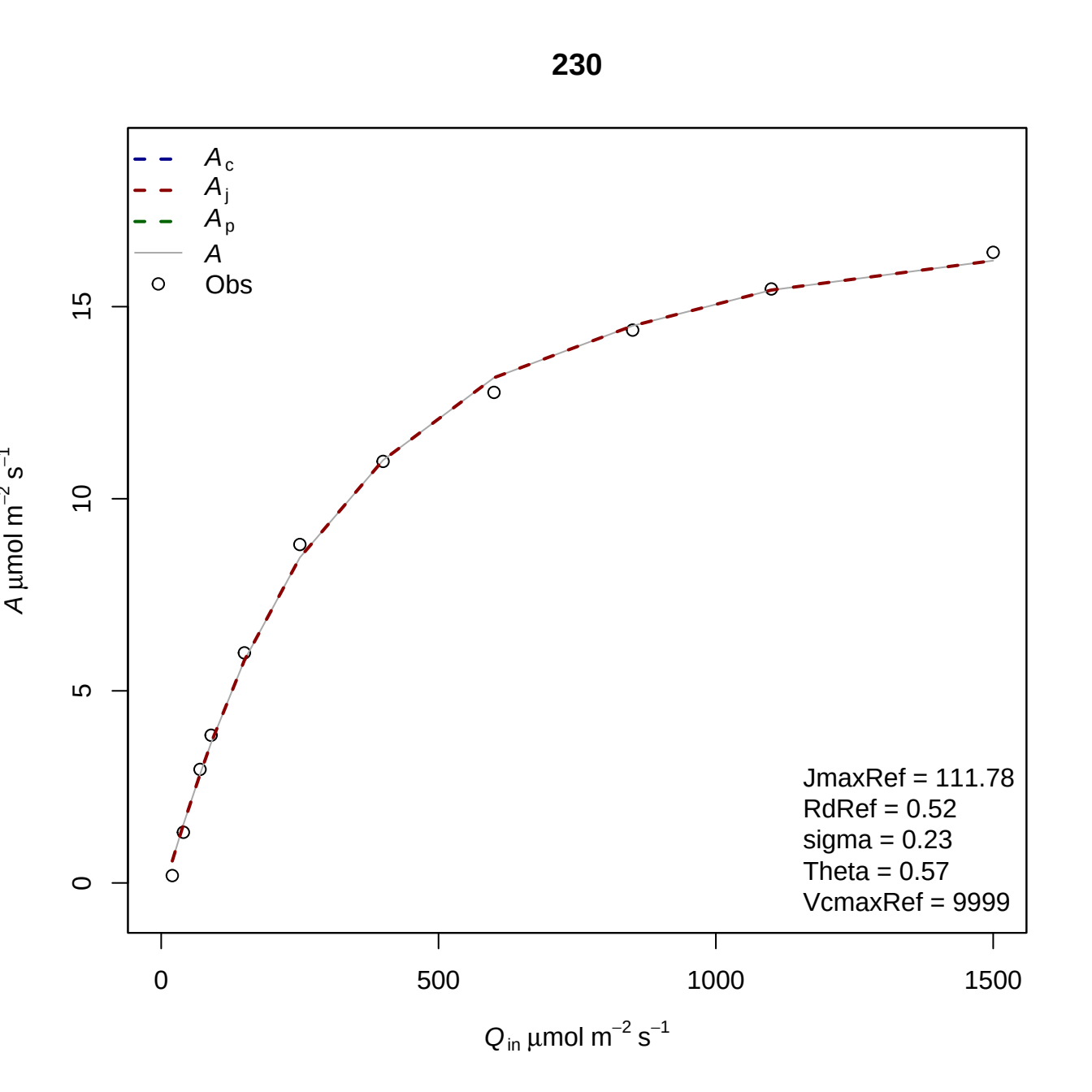

225
A_c
A_j
A_p
A
Obs
JmaxRef = 57.5
RdRef = 0.94
sigma = 0.23
Theta = 0.32
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

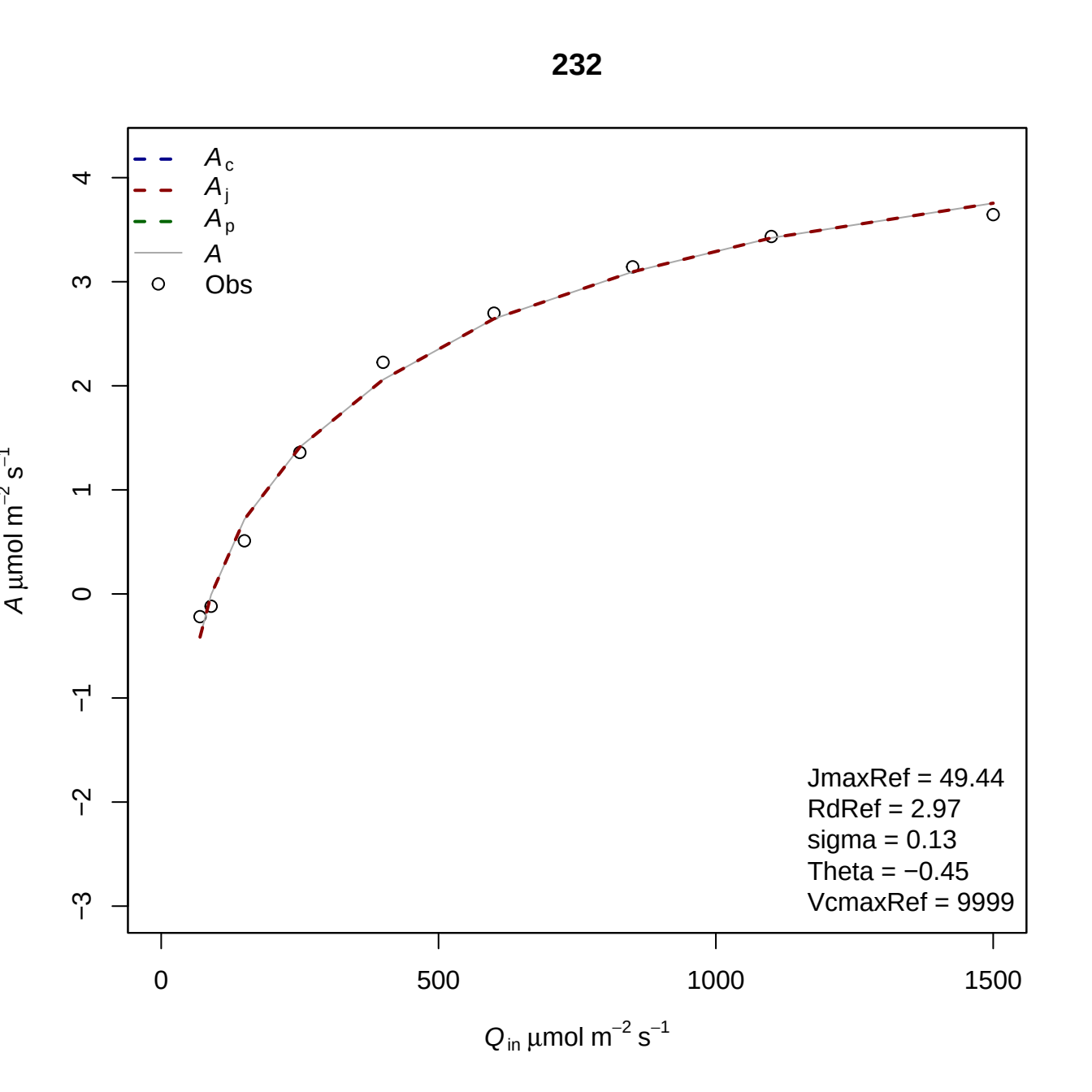

226
A_c
A_j
A_p
A
Obs
JmaxRef = 71.93
RdRef = 2.61
sigma = 0.14
Theta = 0.73
VcmaxRef = 9999
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

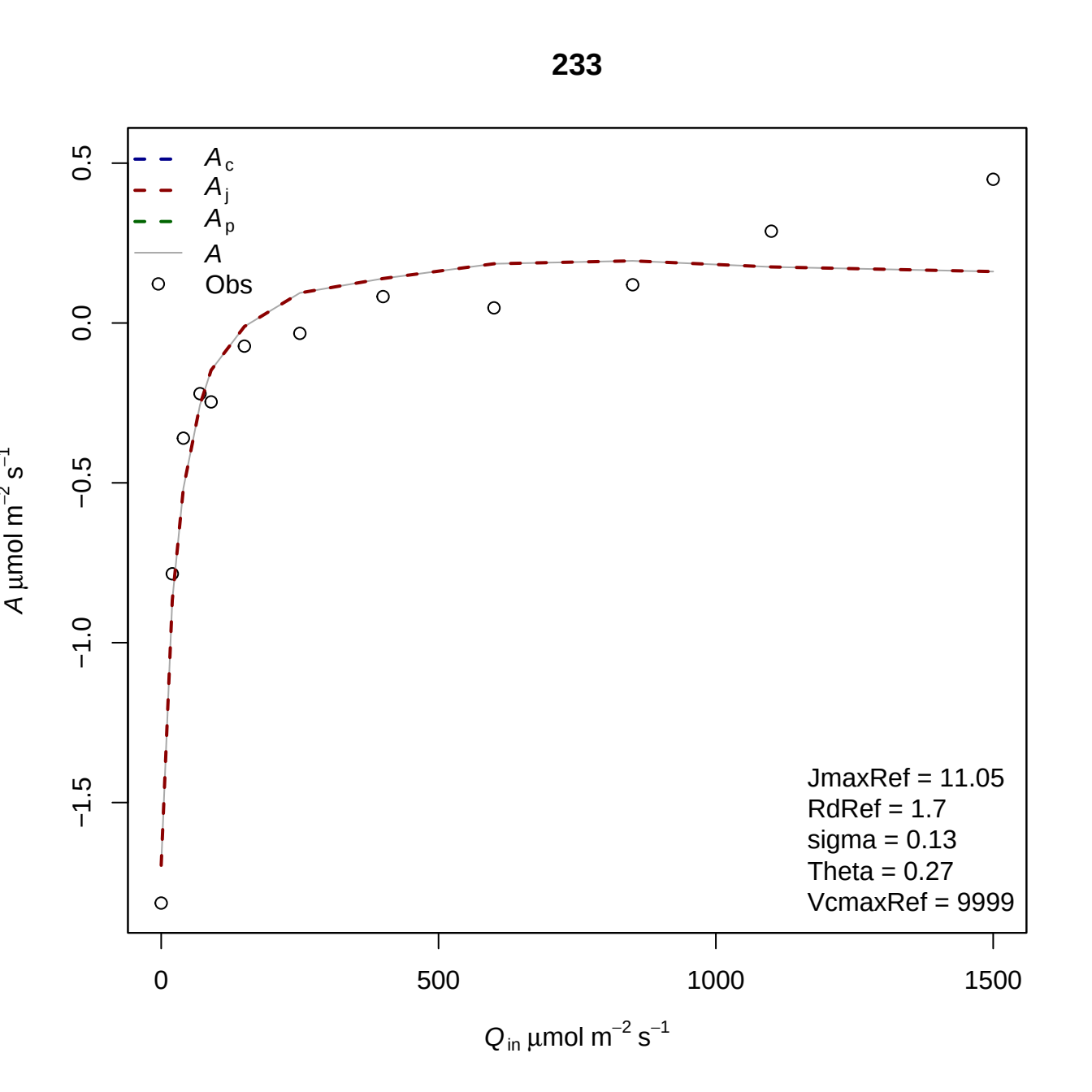

227
A_c
A_j
A_p
A
Obs
JmaxRef = 69.76
RdRef = 0.88
sigma = 0.09
Theta = 0.7
VcmaxRef = 42.06
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

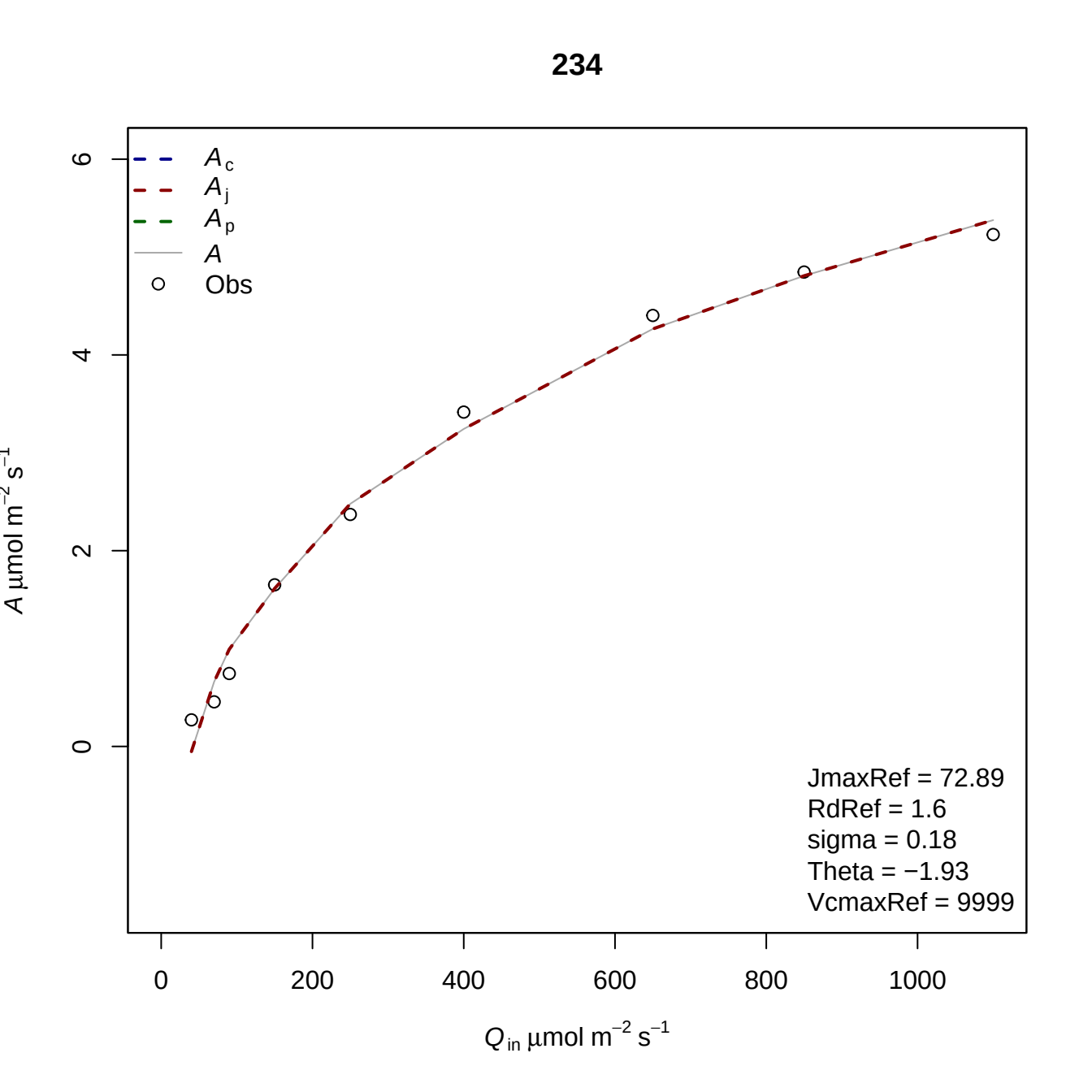

228
A_c
A_j
A_p
A
Obs
JmaxRef = 58.63
RdRef = 1.06
sigma = 0.14
Theta = 0.64
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

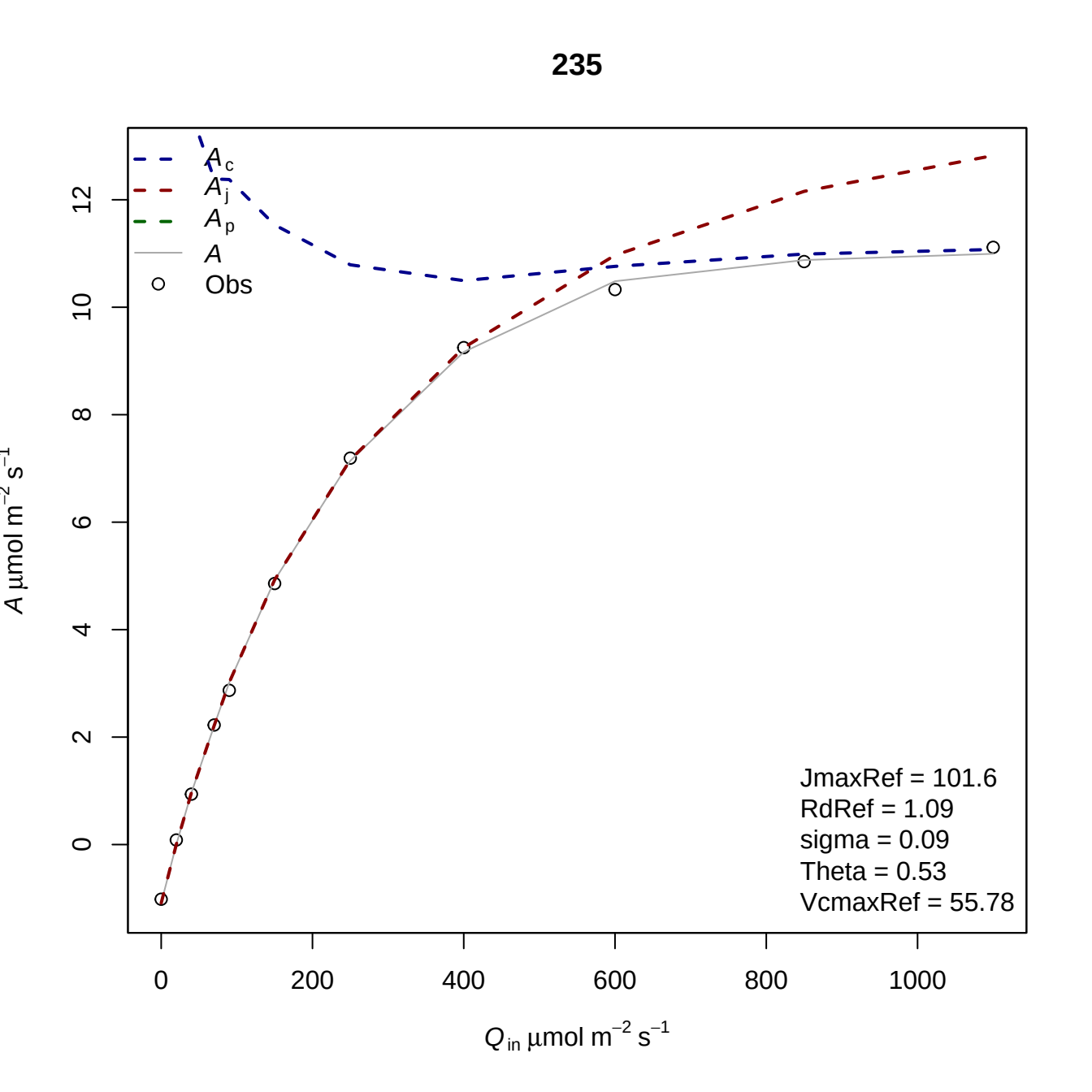

229
A_c
A_j
A_p
A
Obs
JmaxRef = 94.55
RdRef = 1.86
sigma = 0.1
Theta = 0.52
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

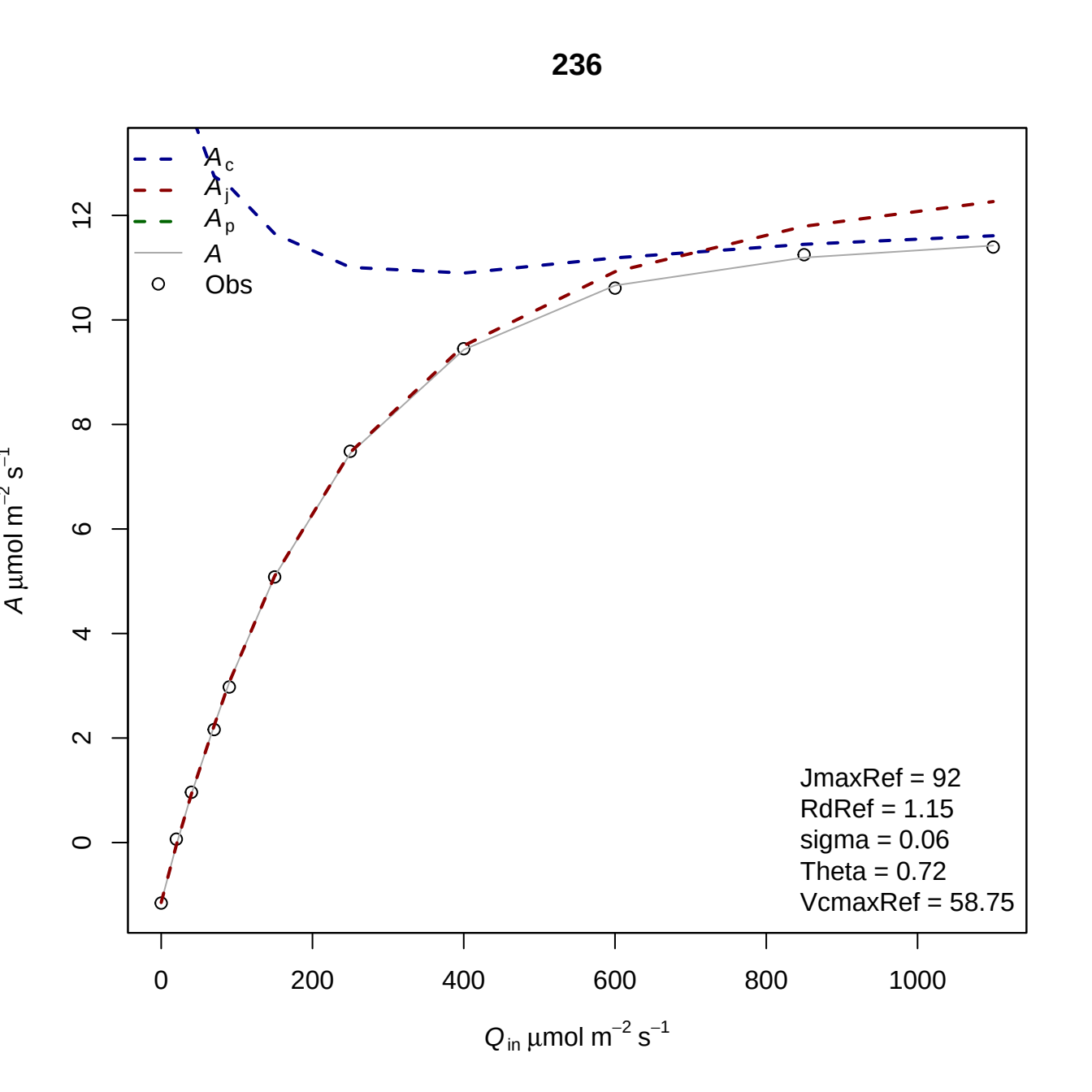

230
A_c
A_j
A_p
A
Obs
JmaxRef = 111.78
RdRef = 0.52
sigma = 0.23
Theta = 0.57
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

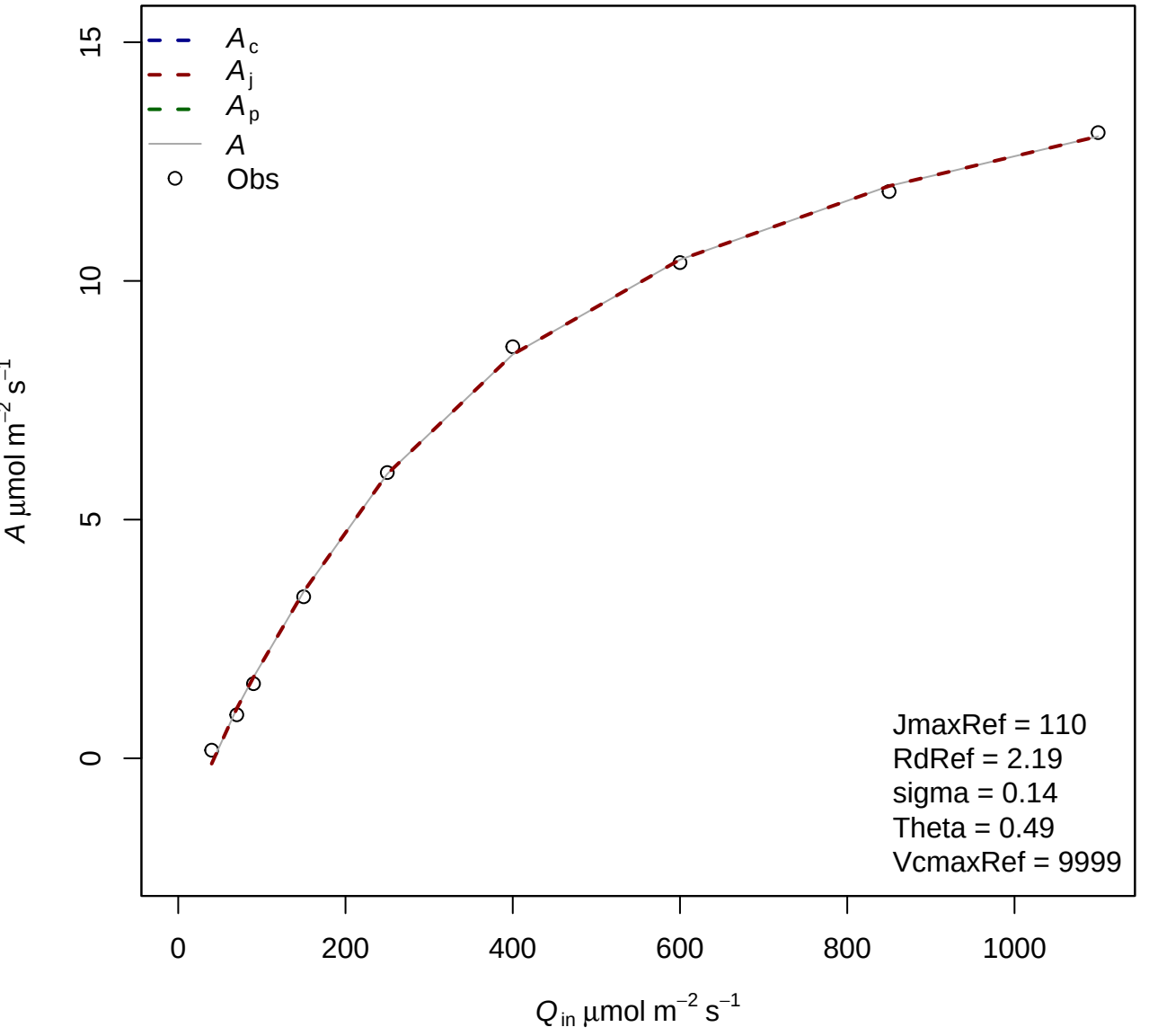

232
A_c
A_j
A_p
A
Obs
JmaxRef = 49.44
RdRef = 2.97
sigma = 0.13
Theta = −0.45
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
−3
0
500
1000
1500

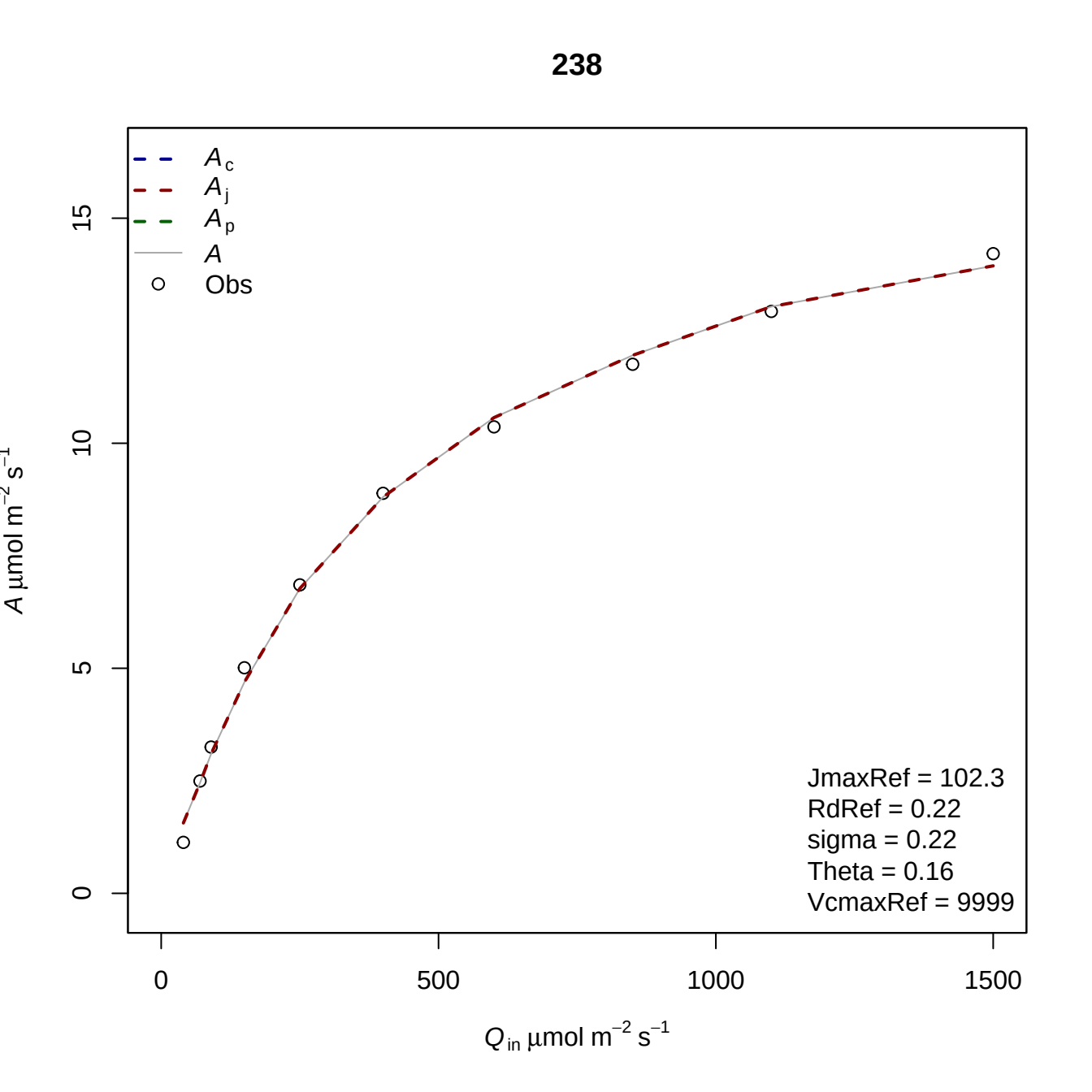

233
A_c
A_j
A_p
A
Obs
JmaxRef = 11.05
RdRef = 1.7
sigma = 0.13
Theta = 0.27
VcmaxRef = 9999
0.5
0.0
−0.5
−1.0
−1.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

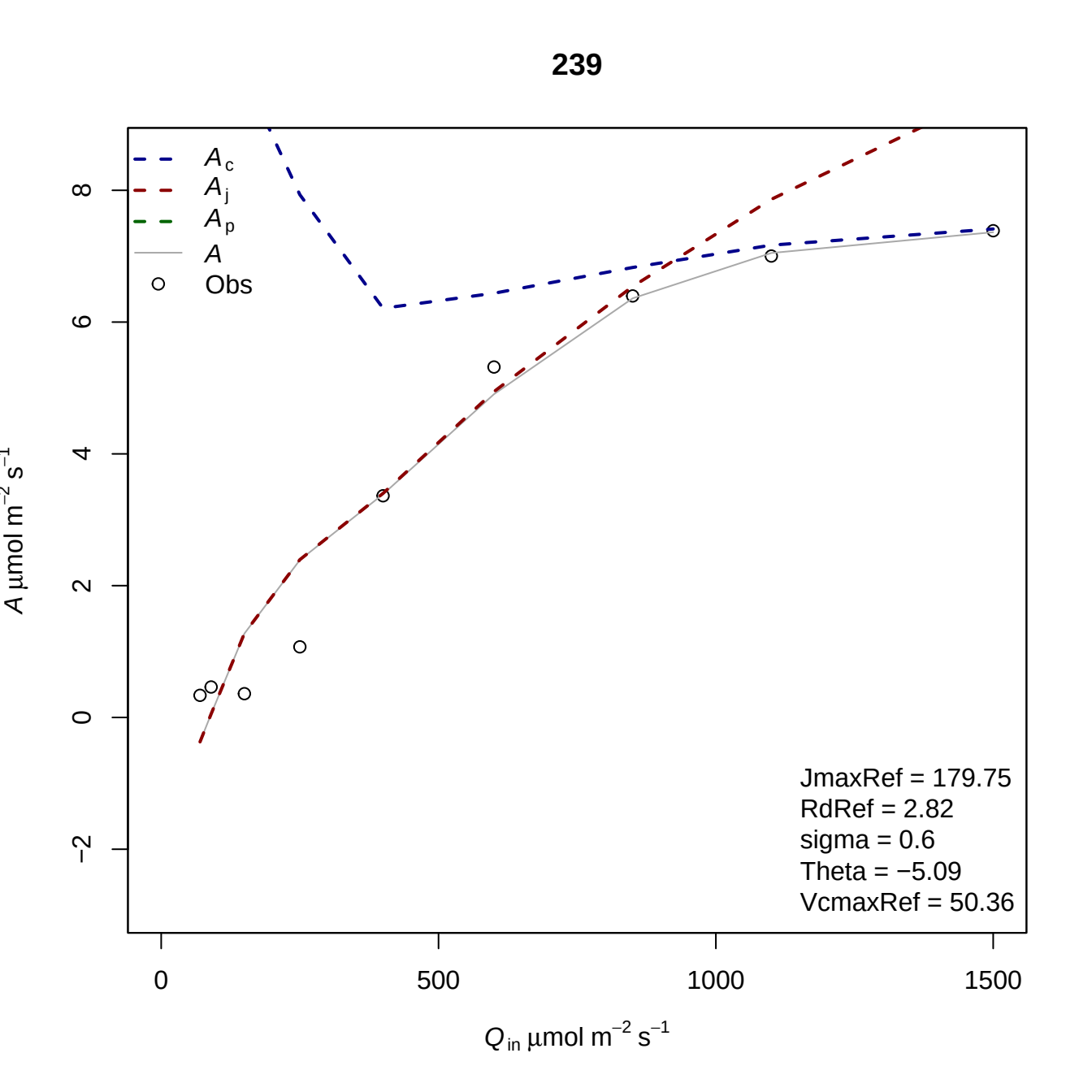

234
A_c
A_j
A_p
A
Obs
JmaxRef = 72.89
RdRef = 1.6
sigma = 0.18
Theta = −1.93
VcmaxRef = 9999
0
2
4
6
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
Q_{in} µmol m^{-2} s^{-1}

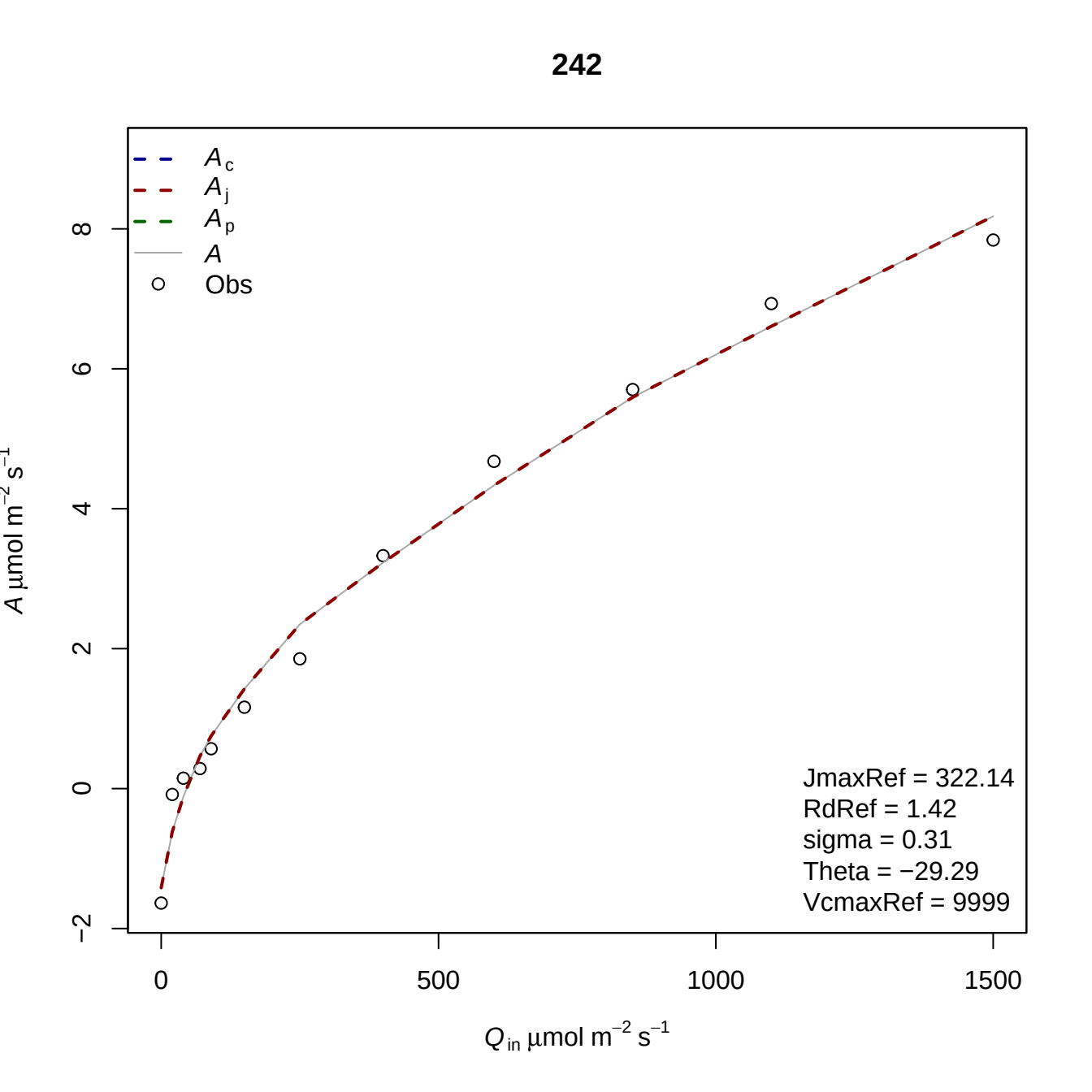

235
A_c
A_j
A_p
A
Obs
JmaxRef = 101.6
RdRef = 1.09
sigma = 0.09
Theta = 0.53
VcmaxRef = 55.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000

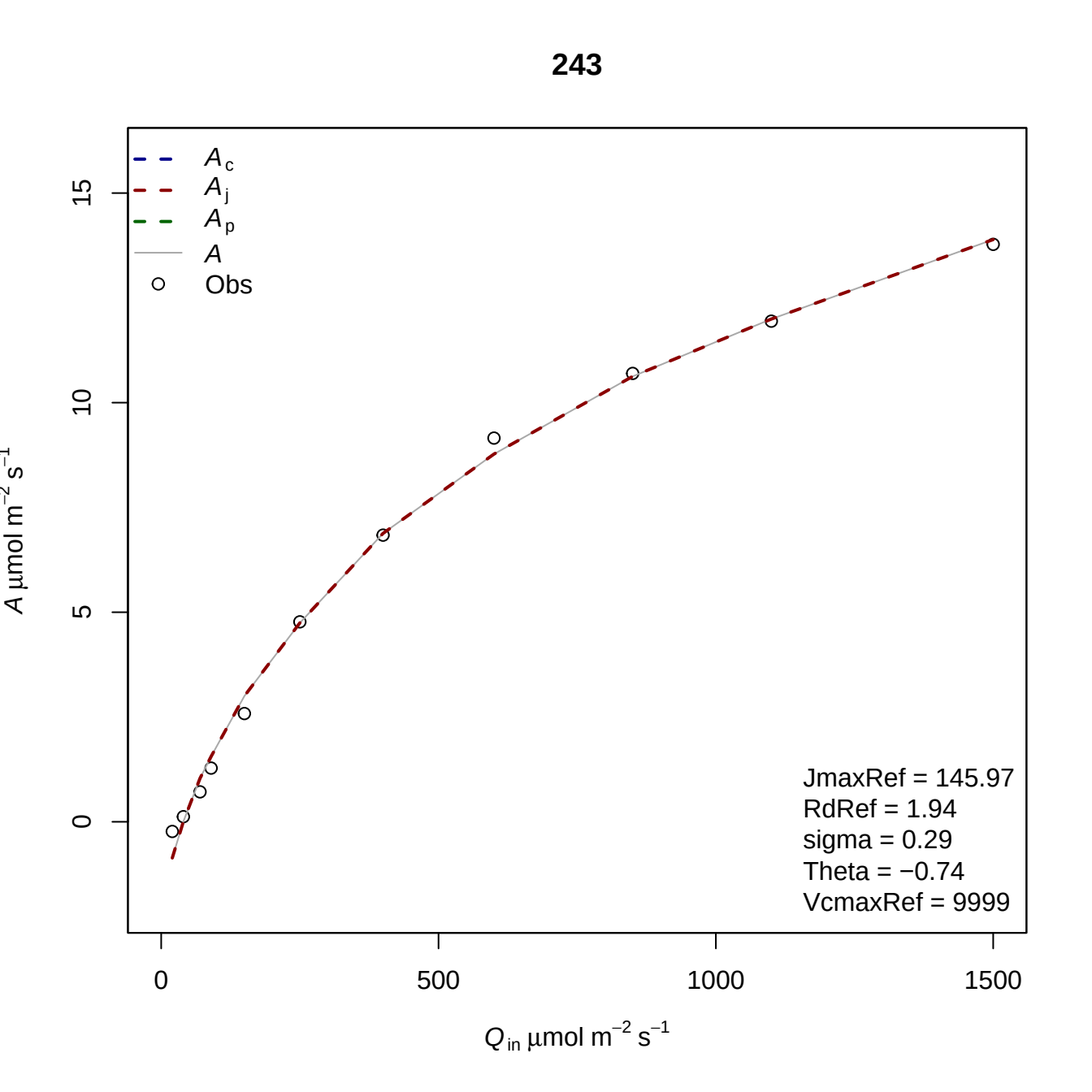

236
A_c
A_j
A_p
A
Obs
JmaxRef = 92
RdRef = 1.15
sigma = 0.06
Theta = 0.72
VcmaxRef = 58.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000

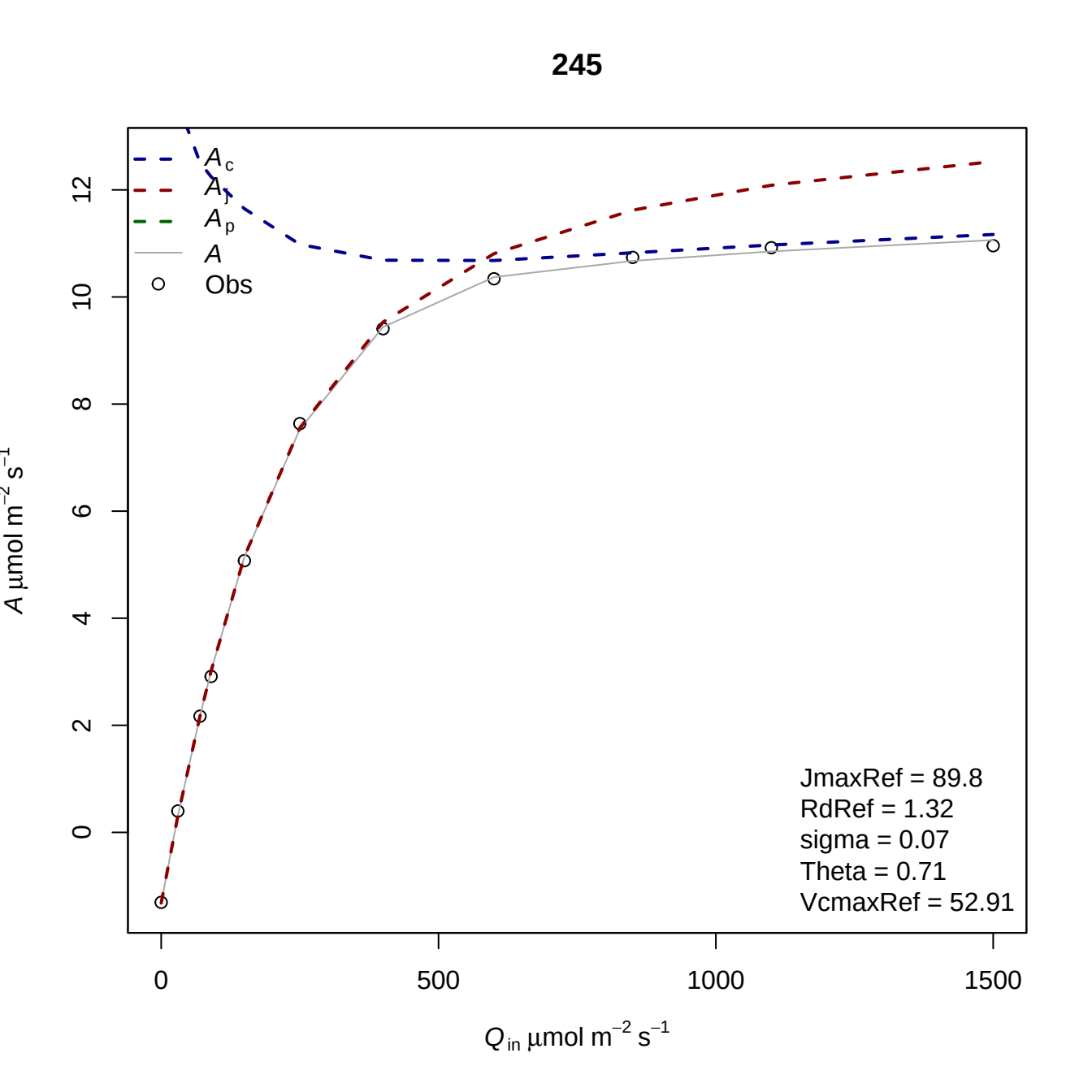

A_c
A_j
A_p
A
Obs
JmaxRef = 110
RdRef = 2.19
sigma = 0.14
Theta = 0.49
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
Q_{in} µmol m^{-2} s^{-1}

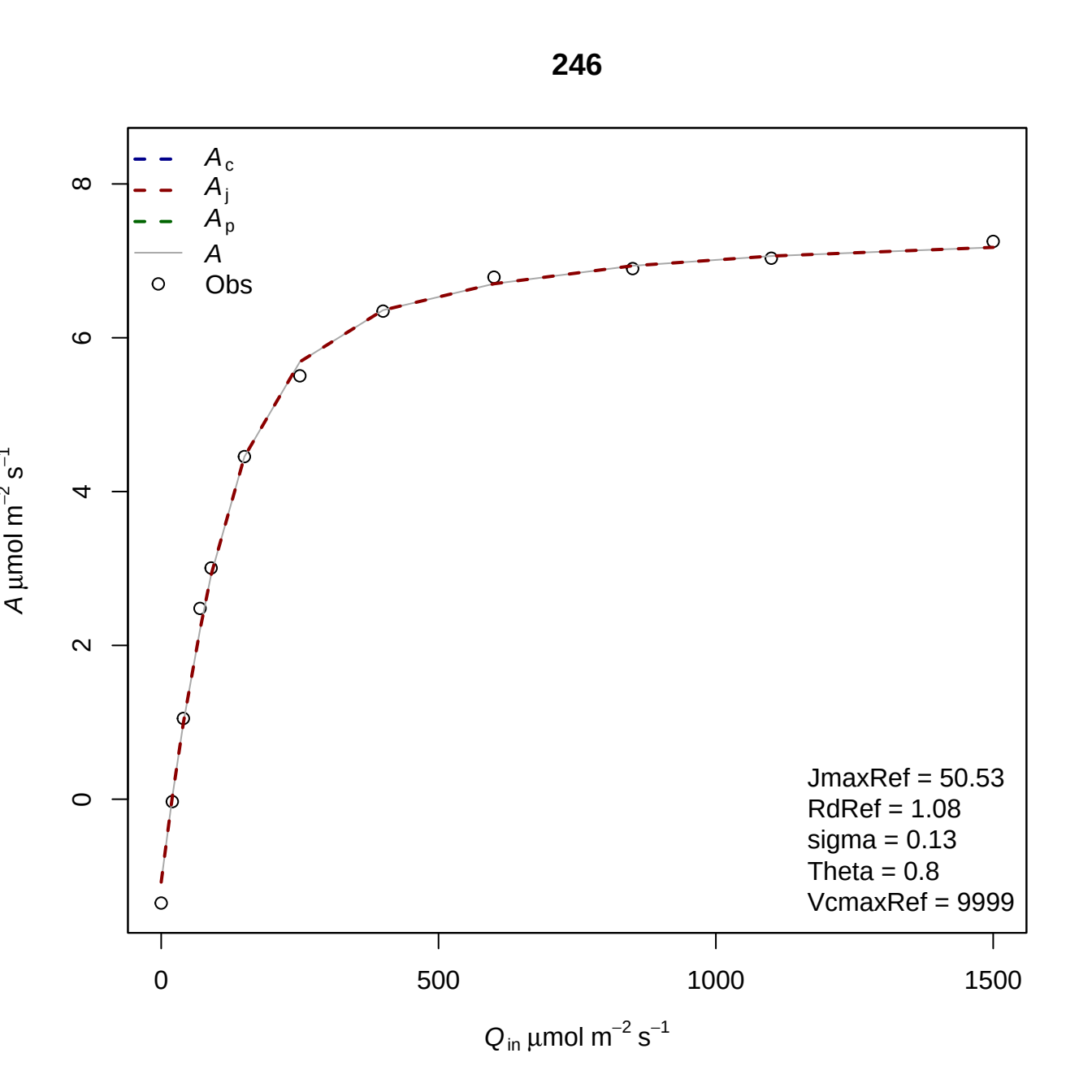

238
A_c
A_j
A_p
A
Obs
JmaxRef = 102.3
RdRef = 0.22
sigma = 0.22
Theta = 0.16
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

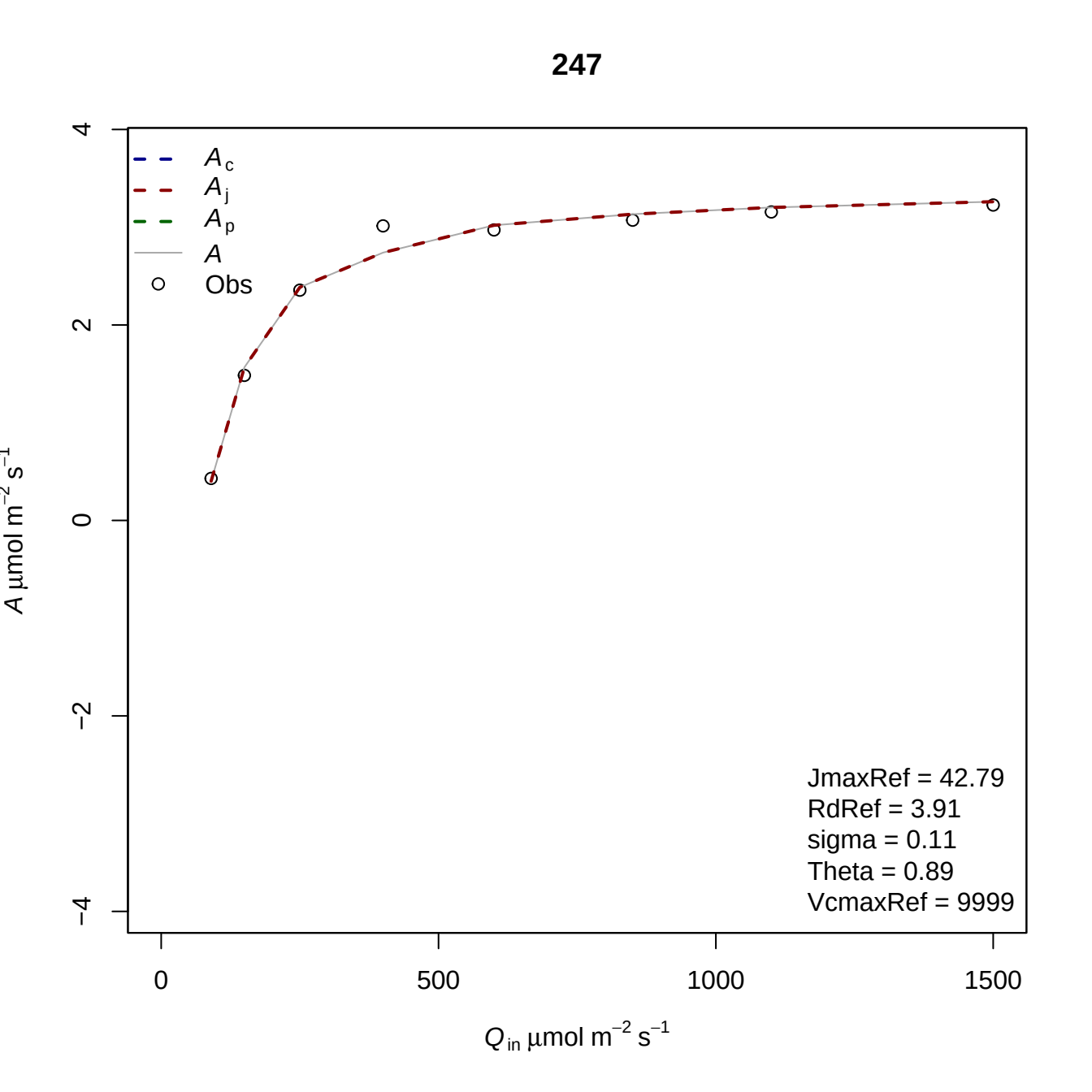

239
A_c
A_j
A_p
A
Obs
JmaxRef = 179.75
RdRef = 2.82
sigma = 0.6
Theta = −5.09
VcmaxRef = 50.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

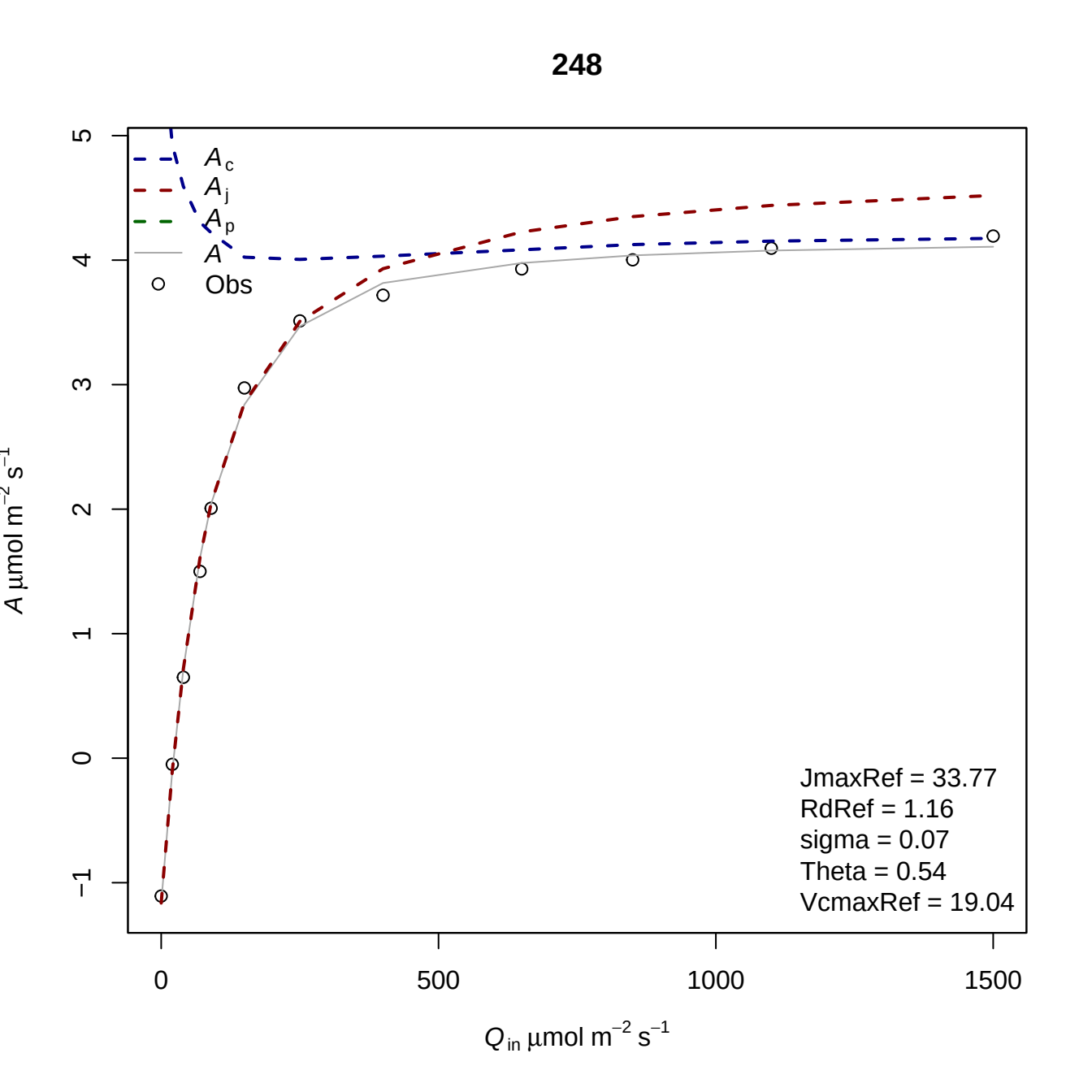

242
A_c
A_j
A_p
A
Obs
JmaxRef = 322.14
RdRef = 1.42
sigma = 0.31
Theta = −29.29
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

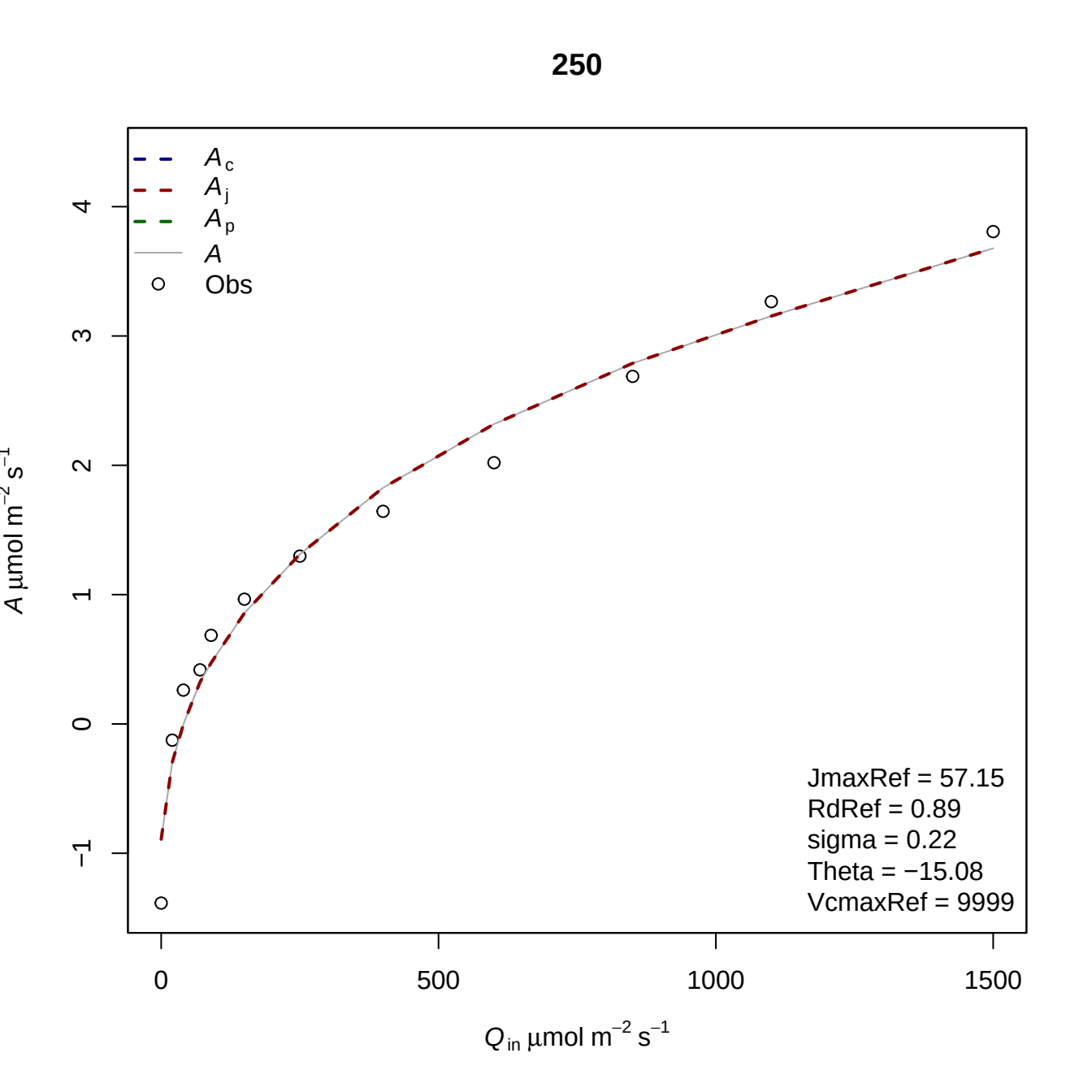

243
A_c
A_j
A_p
A
Obs
JmaxRef = 145.97
RdRef = 1.94
sigma = 0.29
Theta = −0.74
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

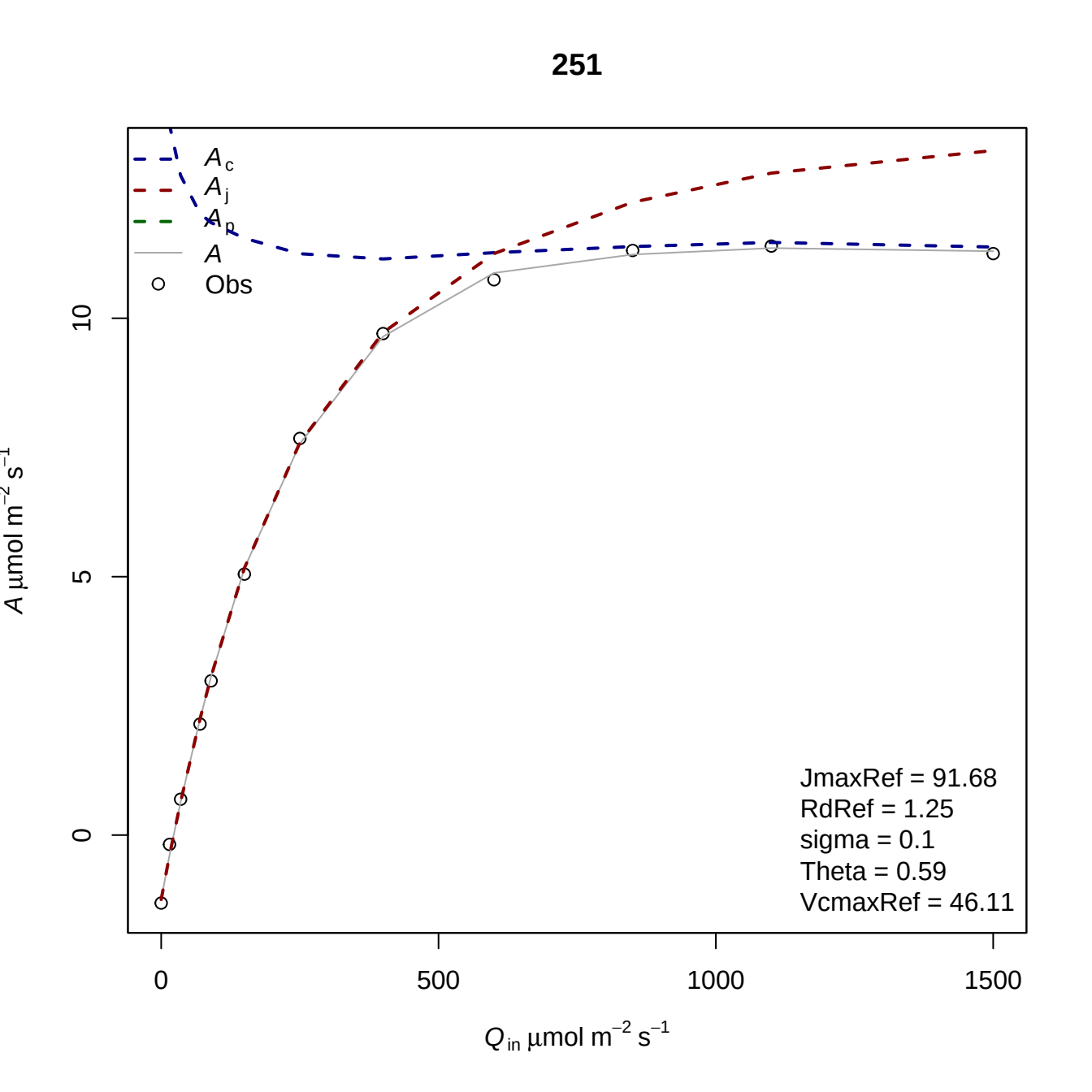

245
A_c
A_j
A_p
A
Obs
JmaxRef = 89.8
RdRef = 1.32
sigma = 0.07
Theta = 0.71
VcmaxRef = 52.91
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

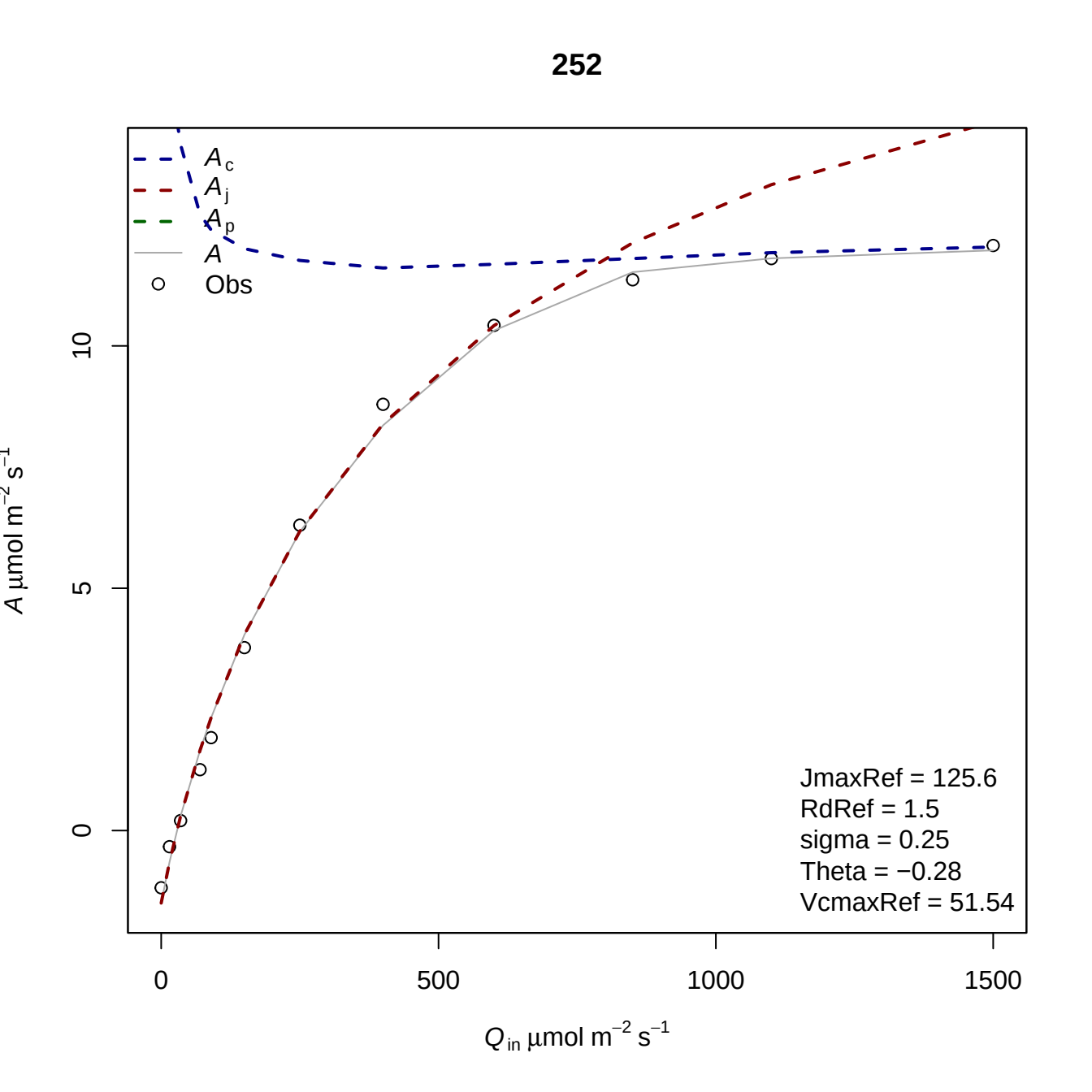

246
A_c
A_j
A_p
A
Obs
JmaxRef = 50.53
RdRef = 1.08
sigma = 0.13
Theta = 0.8
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

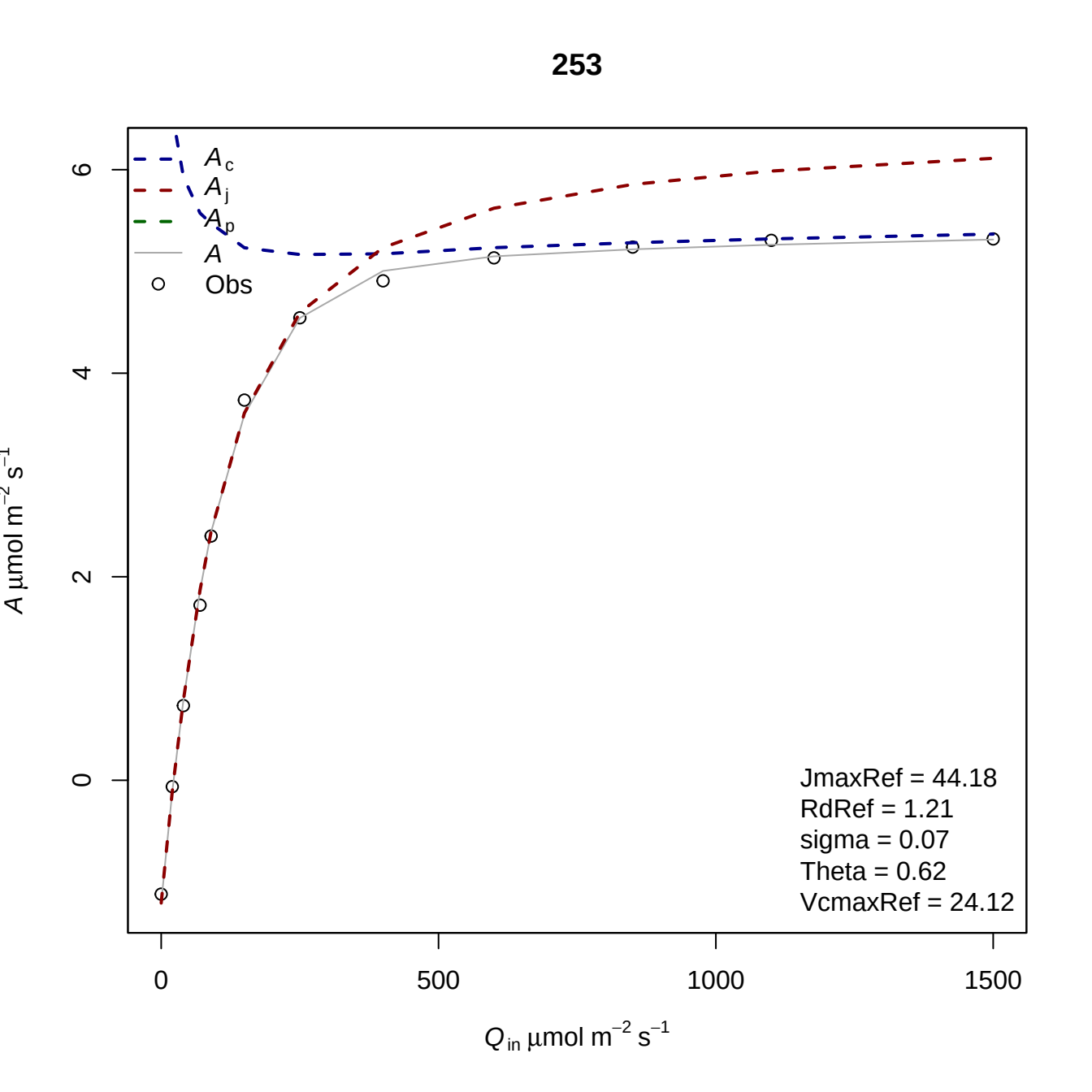

247
A_c
A_j
A_p
A
Obs
JmaxRef = 42.79
RdRef = 3.91
sigma = 0.11
Theta = 0.89
VcmaxRef = 9999
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

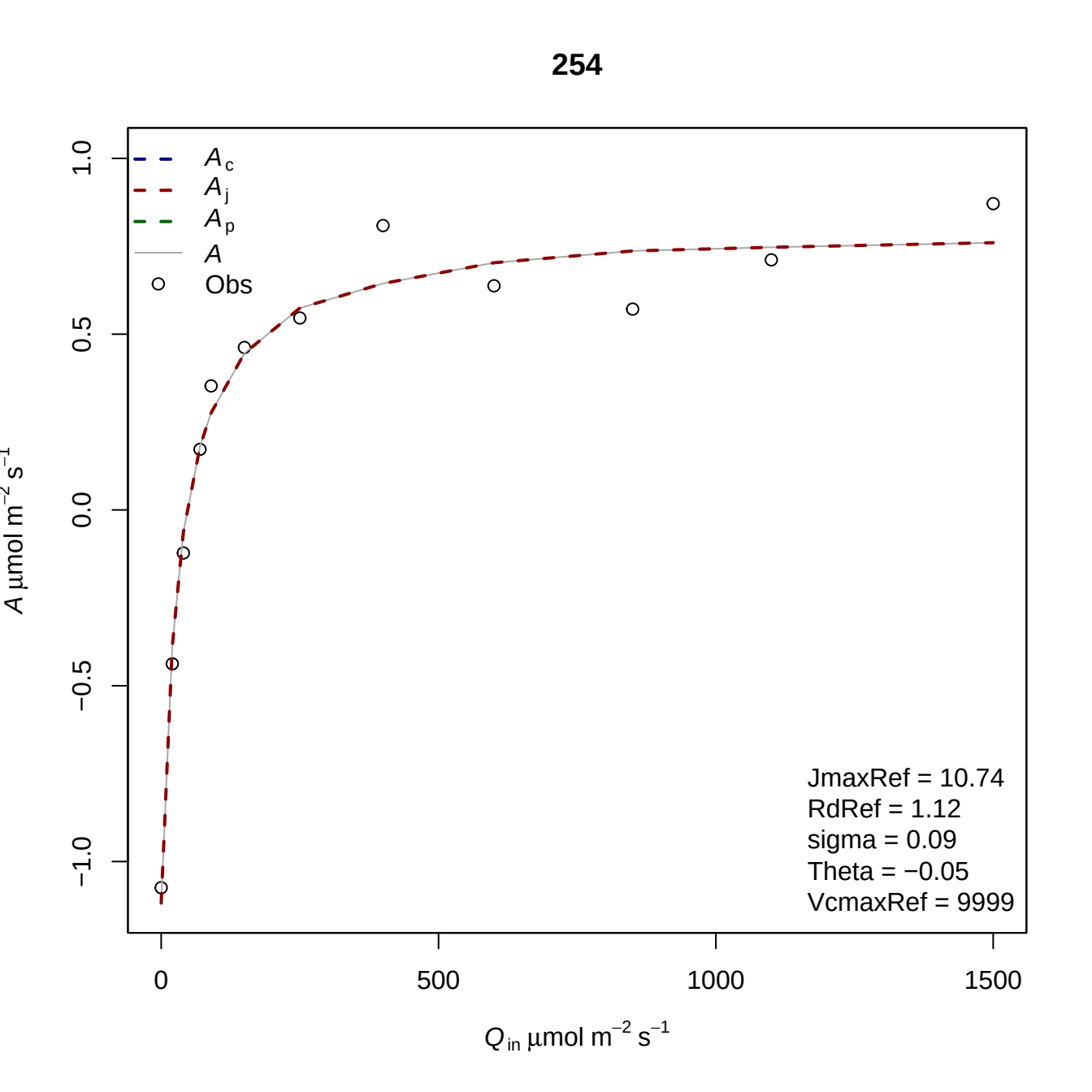

248
A_c
A_j
A_p
A
Obs
JmaxRef = 33.77
RdRef = 1.16
sigma = 0.07
Theta = 0.54
VcmaxRef = 19.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
0
500
1000
1500

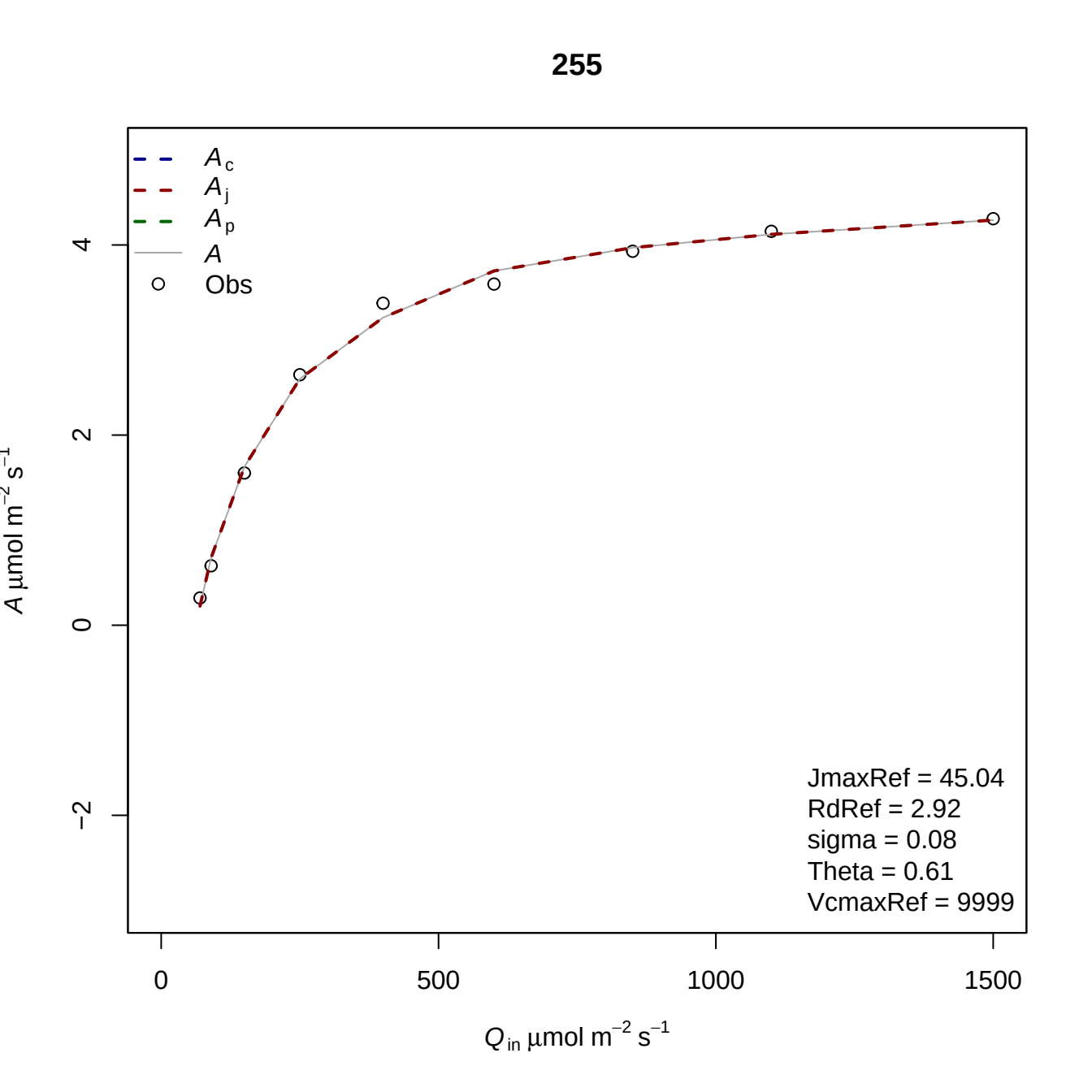

250
A_c
A_j
A_p
A
Obs
JmaxRef = 57.15
RdRef = 0.89
sigma = 0.22
Theta = −15.08
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

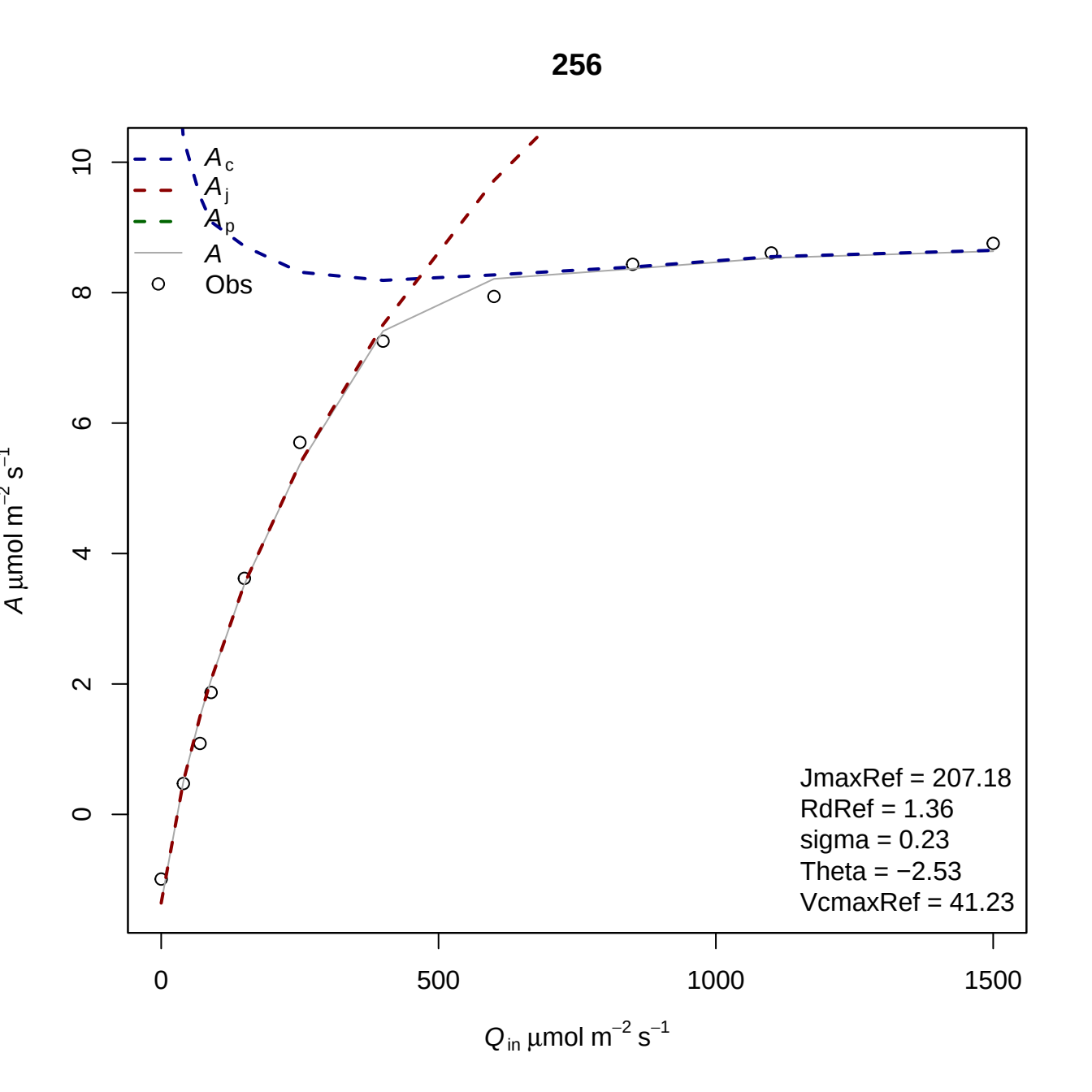

251
A_c
A_j
A_p
A
Obs
JmaxRef = 91.68
RdRef = 1.25
sigma = 0.1
Theta = 0.59
VcmaxRef = 46.11
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

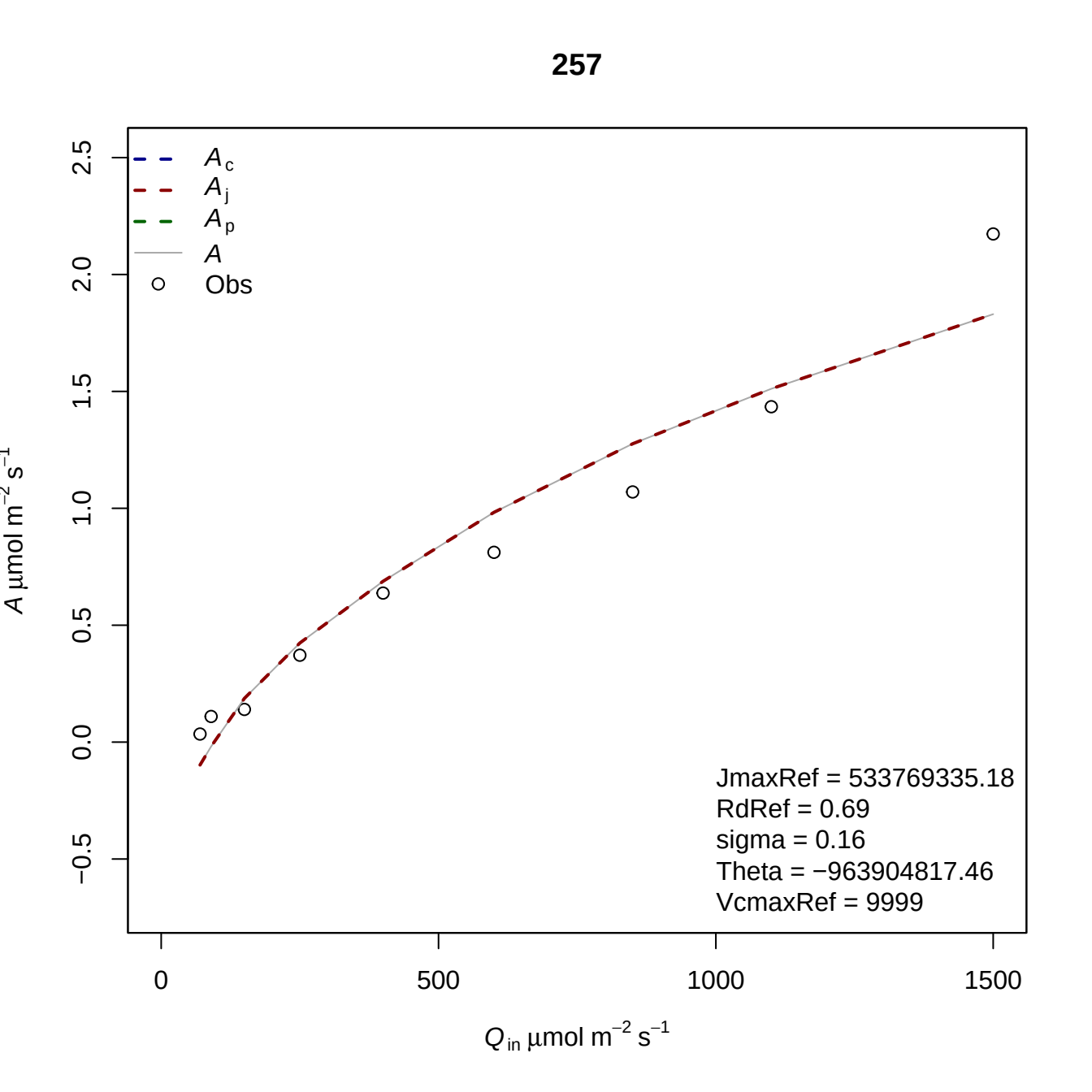

252
A_c
A_j
A_p
A
Obs
JmaxRef = 125.6
RdRef = 1.5
sigma = 0.25
Theta = −0.28
VcmaxRef = 51.54
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

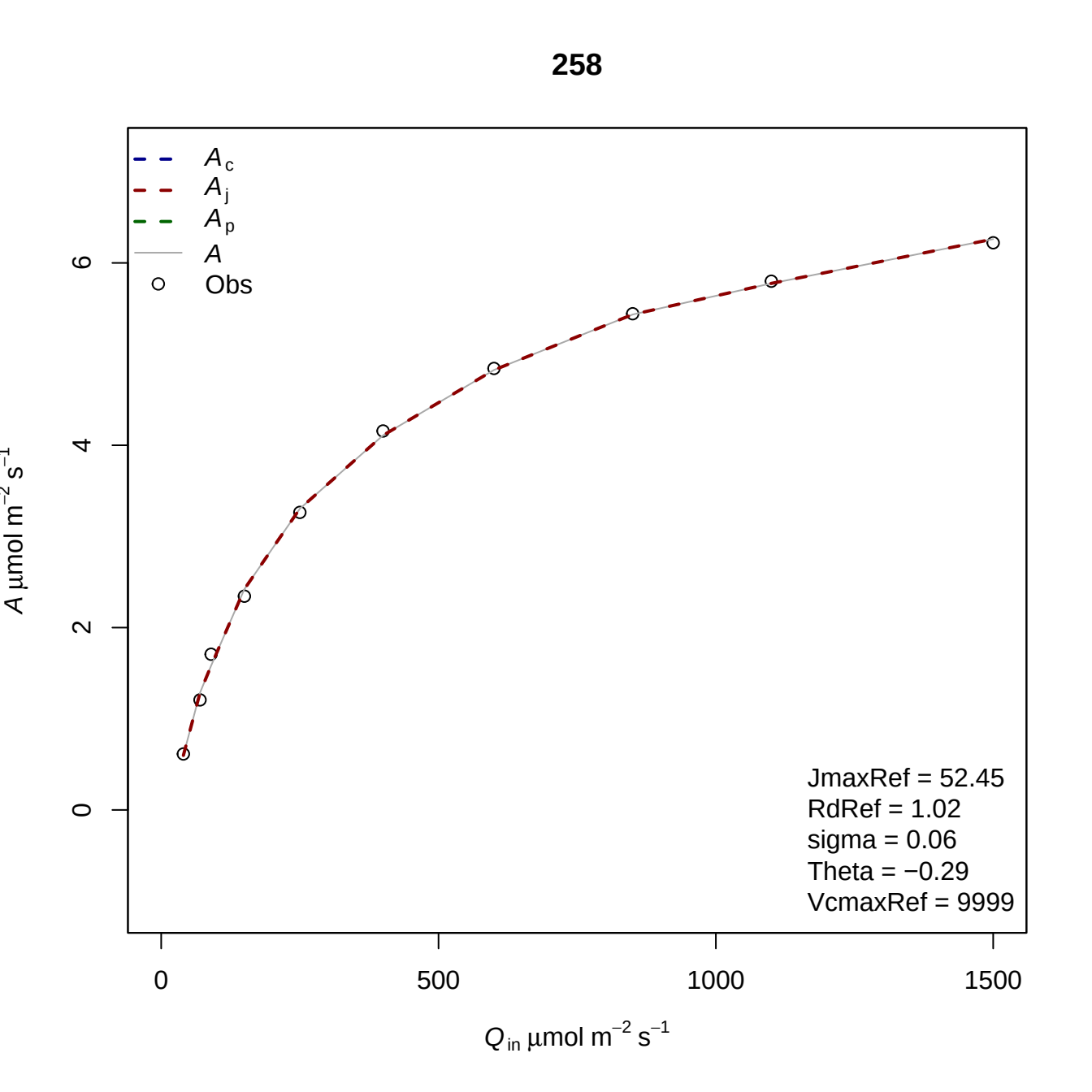

253
A_c
A_j
A_p
A
Obs
JmaxRef = 44.18
RdRef = 1.21
sigma = 0.07
Theta = 0.62
VcmaxRef = 24.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

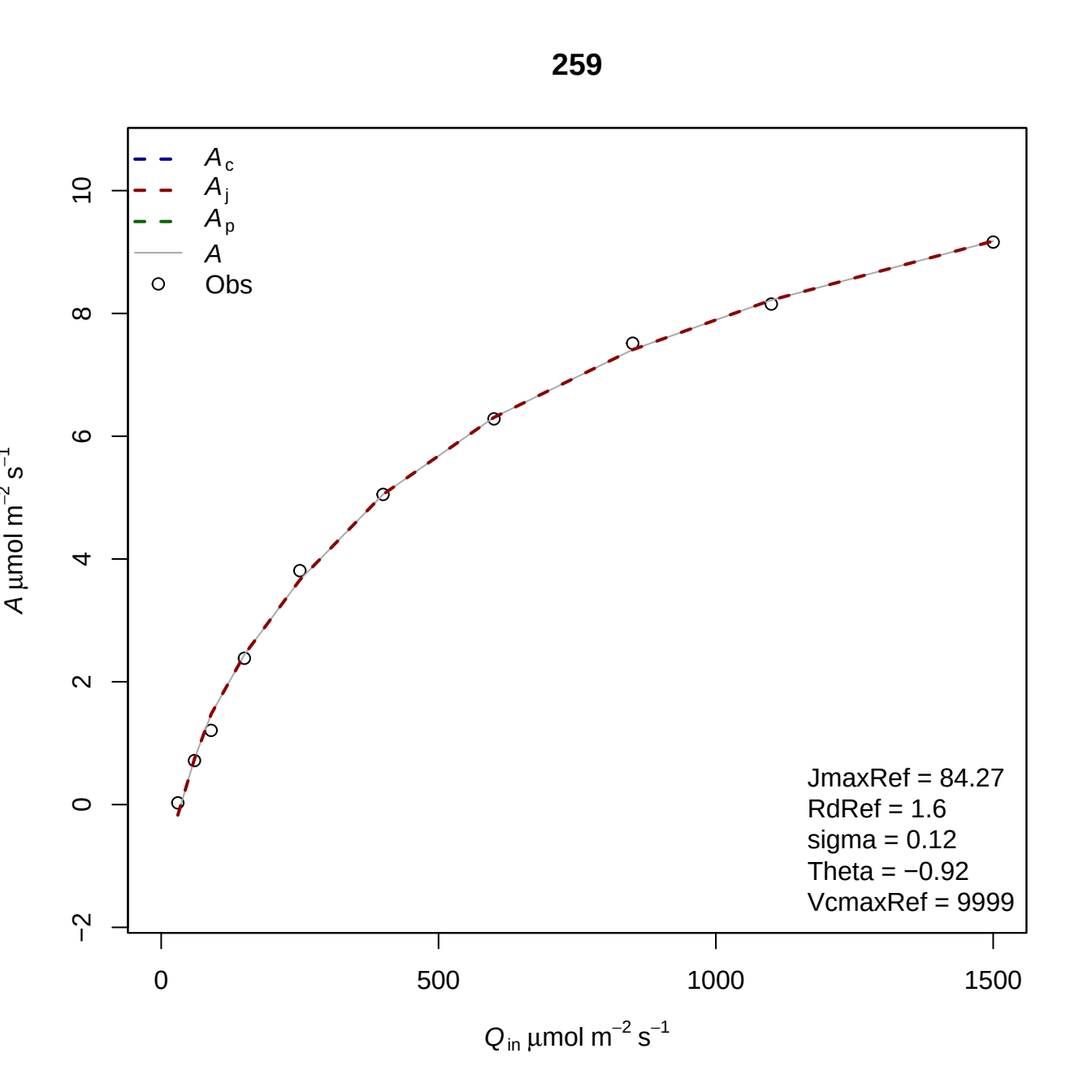

254
A_c
A_j
A_p
A
Obs
JmaxRef = 10.74
RdRef = 1.12
sigma = 0.09
Theta = −0.05
VcmaxRef = 9999
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

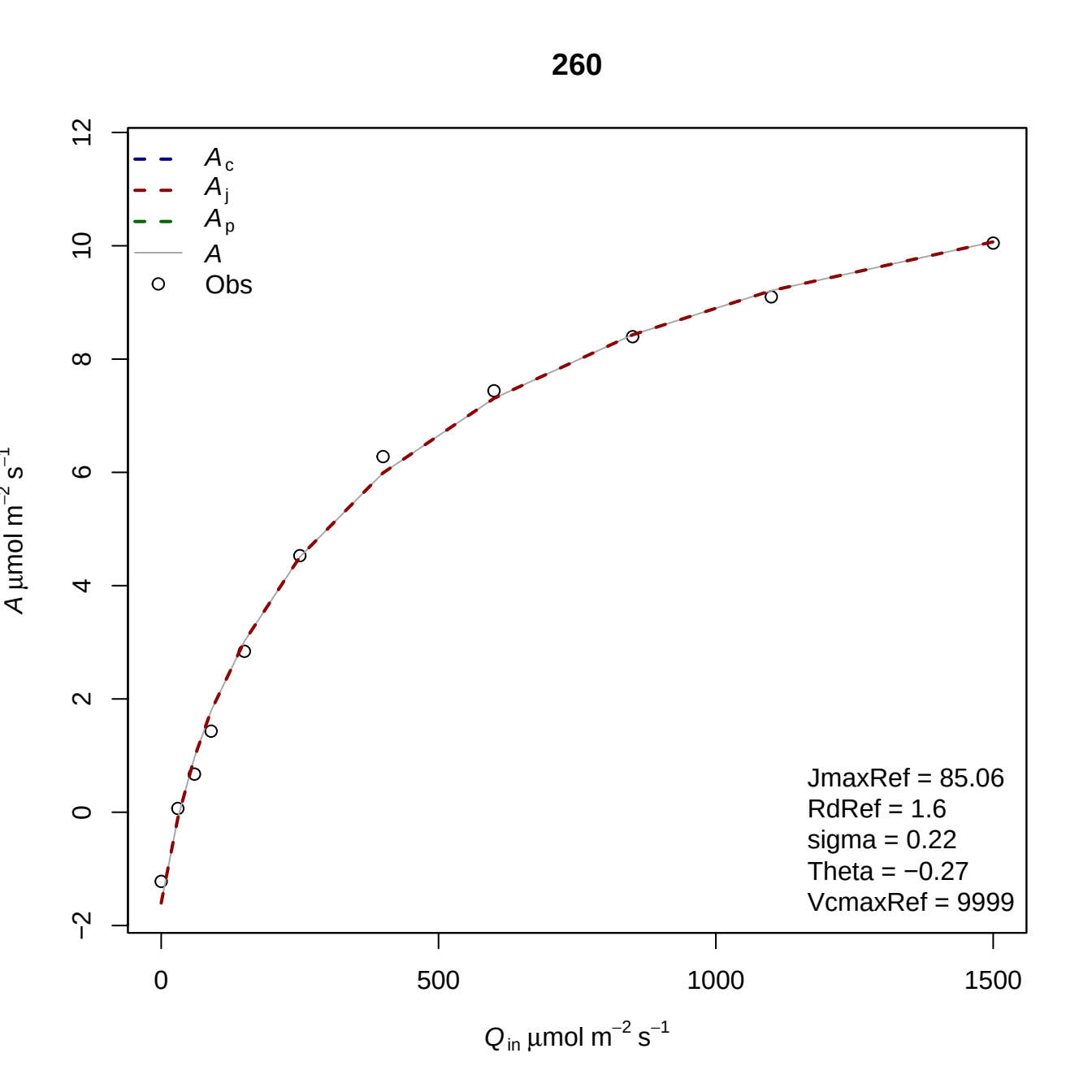

255
A_c
A_j
A_p
A
Obs
JmaxRef = 45.04
RdRef = 2.92
sigma = 0.08
Theta = 0.61
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

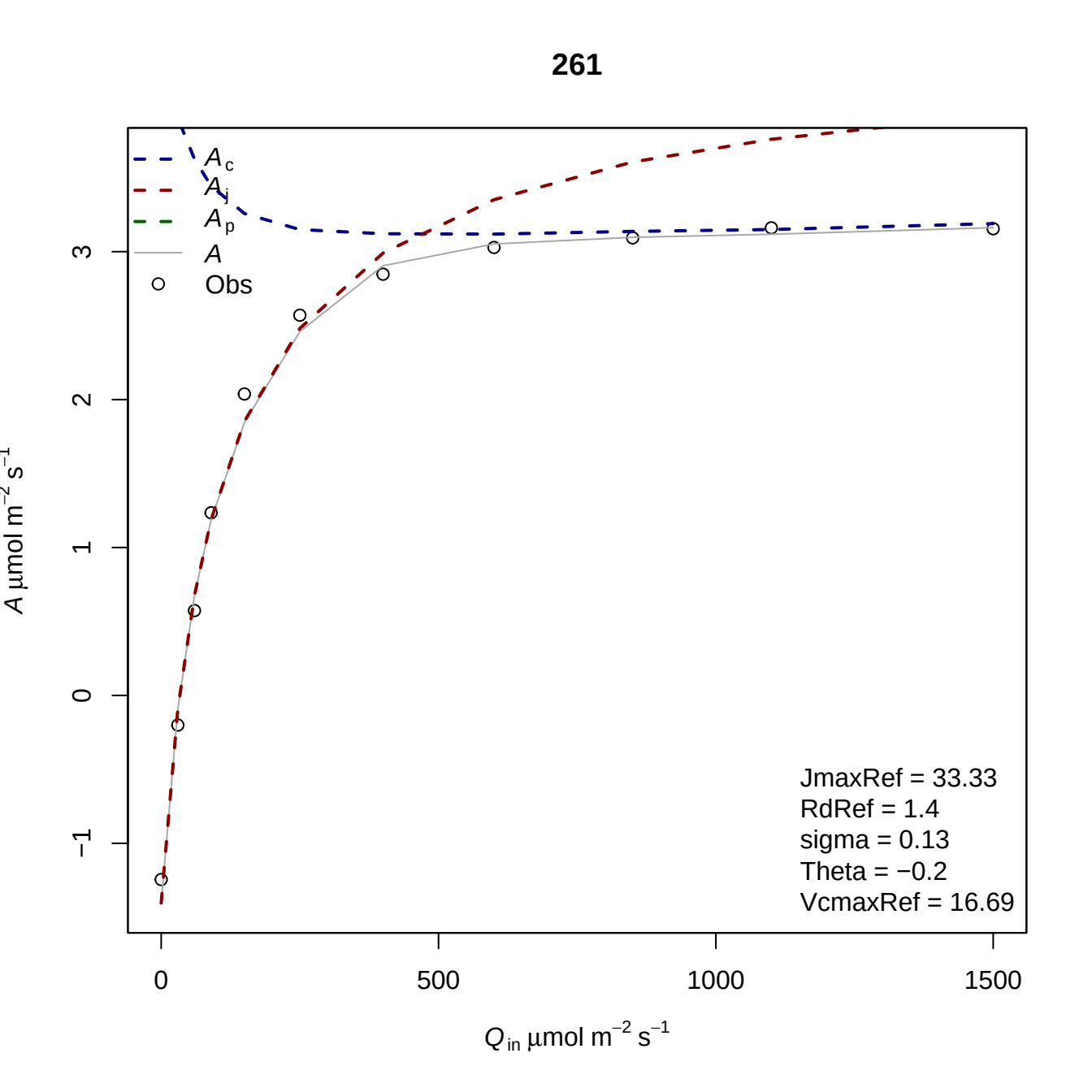

256
A_c
A_j
A_p
A
Obs
JmaxRef = 207.18
RdRef = 1.36
sigma = 0.23
Theta = −2.53
VcmaxRef = 41.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

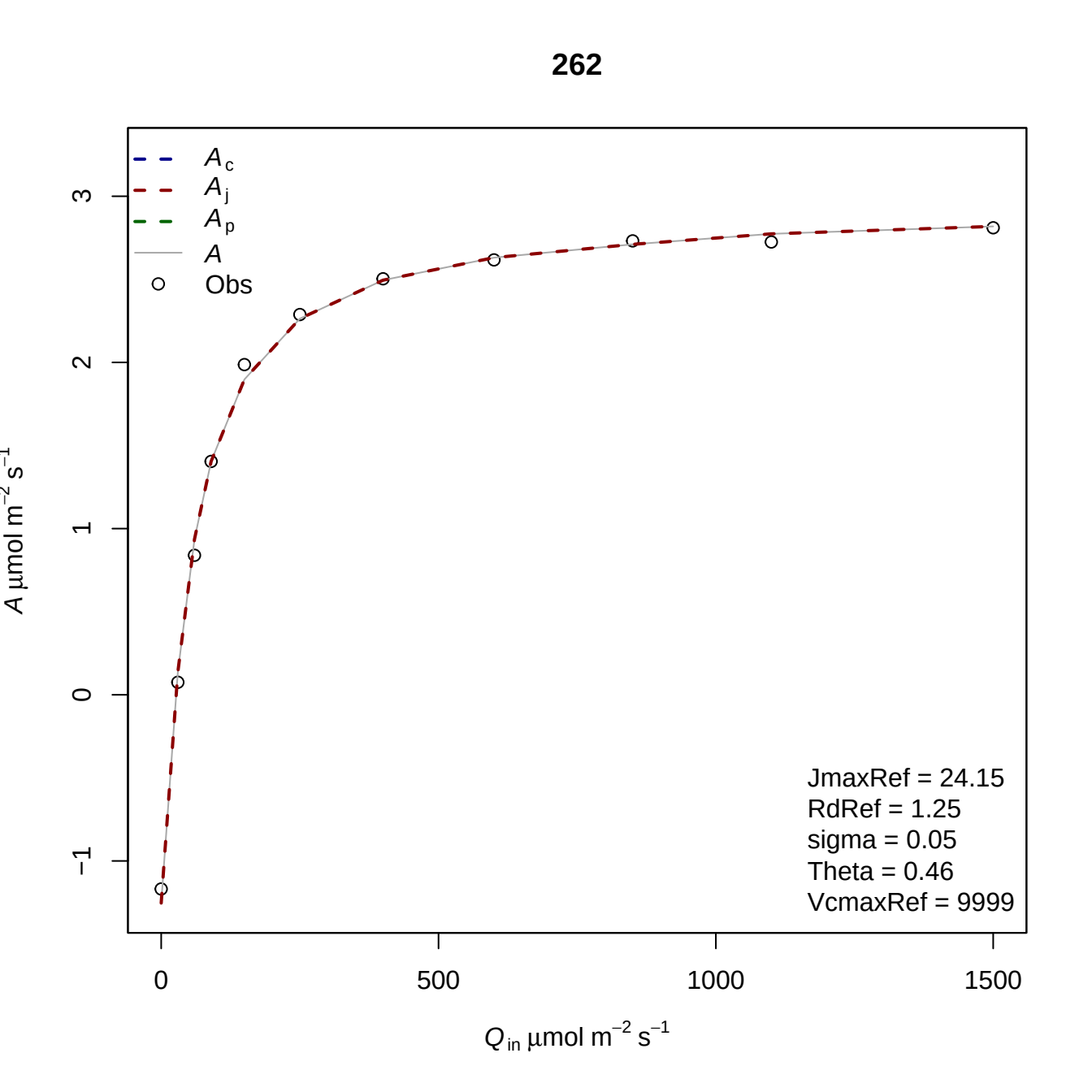

257
A_c
A_j
A_p
A
Obs
JmaxRef = 533769335.18
RdRef = 0.69
sigma = 0.16
Theta = −963904817.46
VcmaxRef = 9999
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

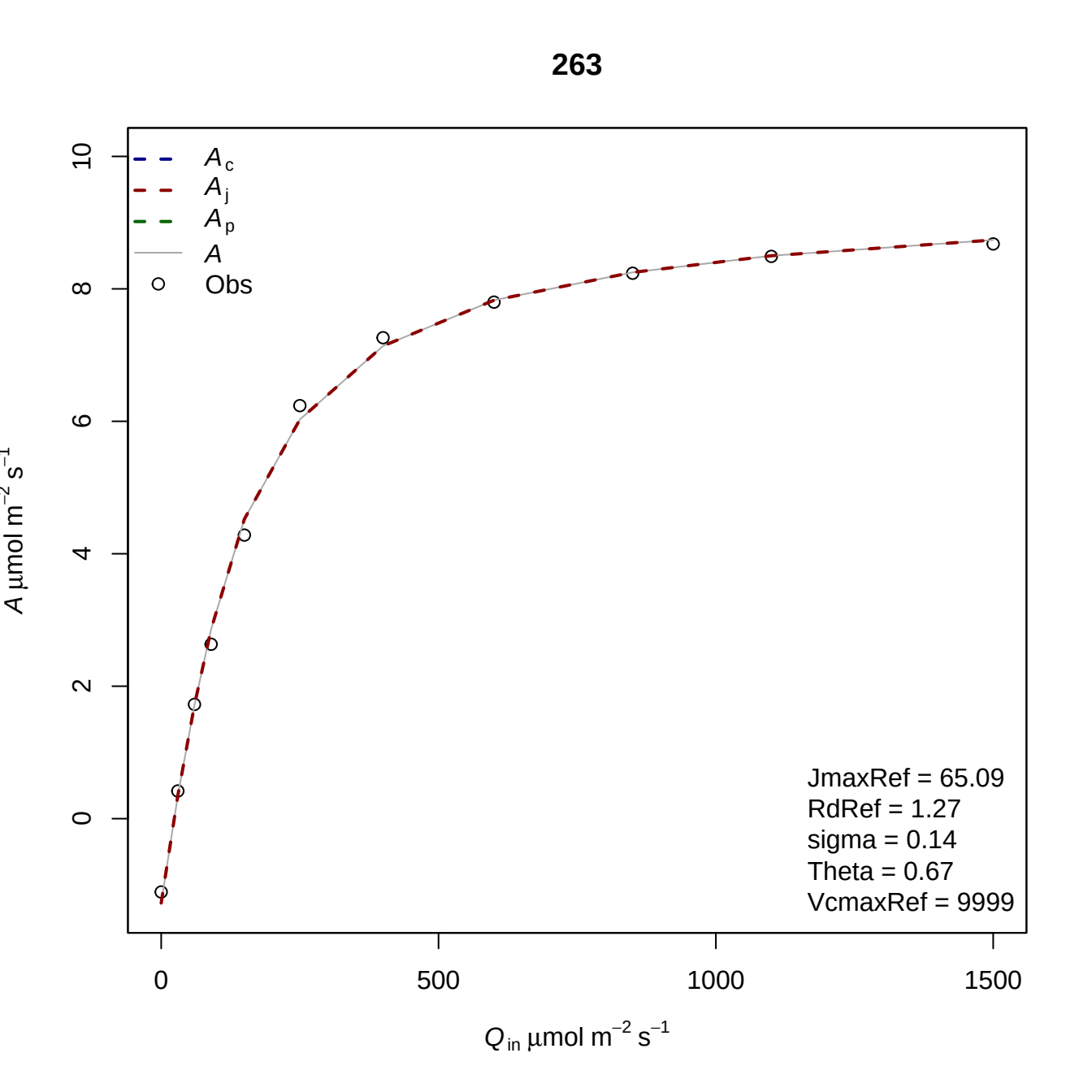

258
A_c
A_j
A_p
A
Obs
JmaxRef = 52.45
RdRef = 1.02
sigma = 0.06
Theta = −0.29
VcmaxRef = 9999
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

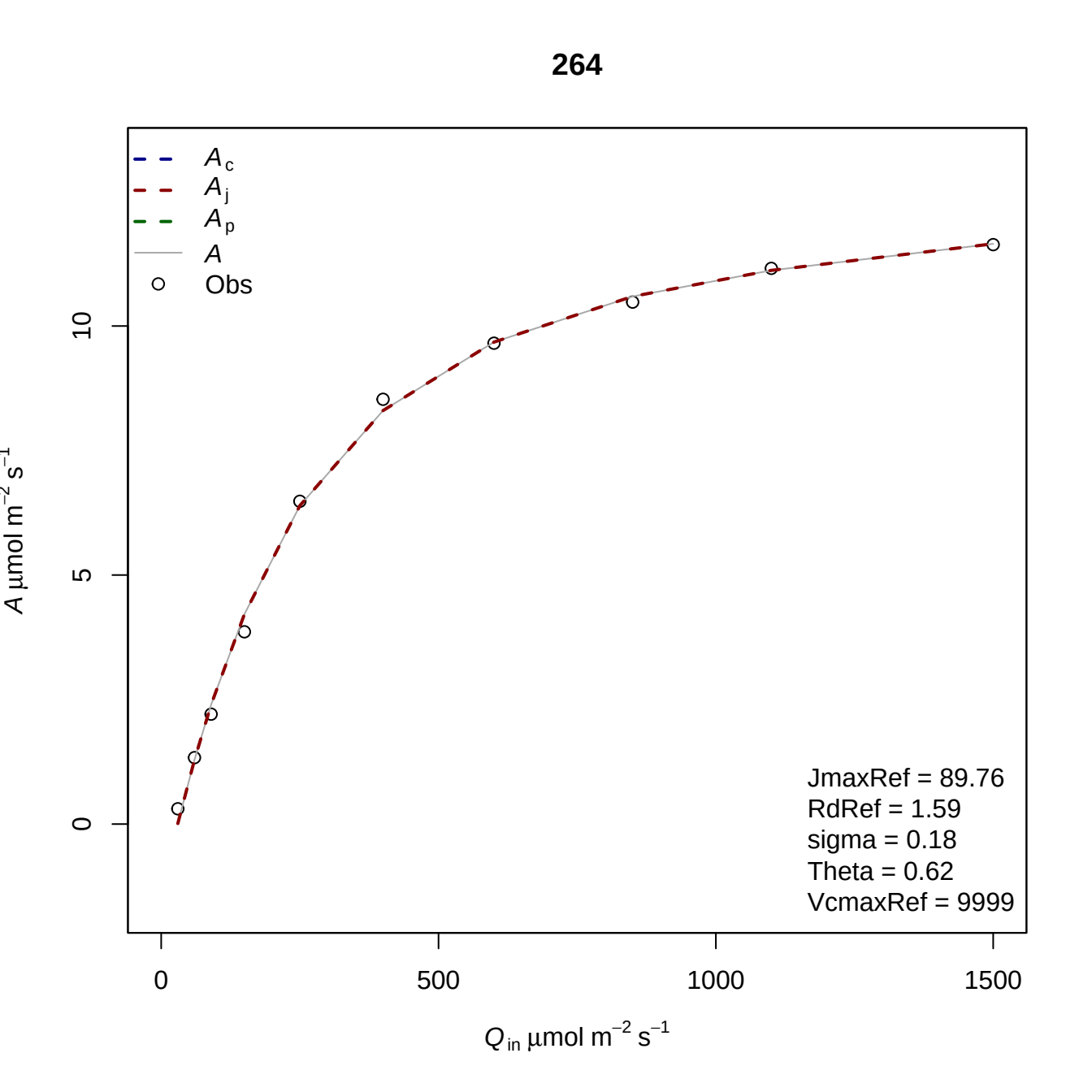

259
A_c
A_j
A_p
A
Obs
JmaxRef = 84.27
RdRef = 1.6
sigma = 0.12
Theta = −0.92
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

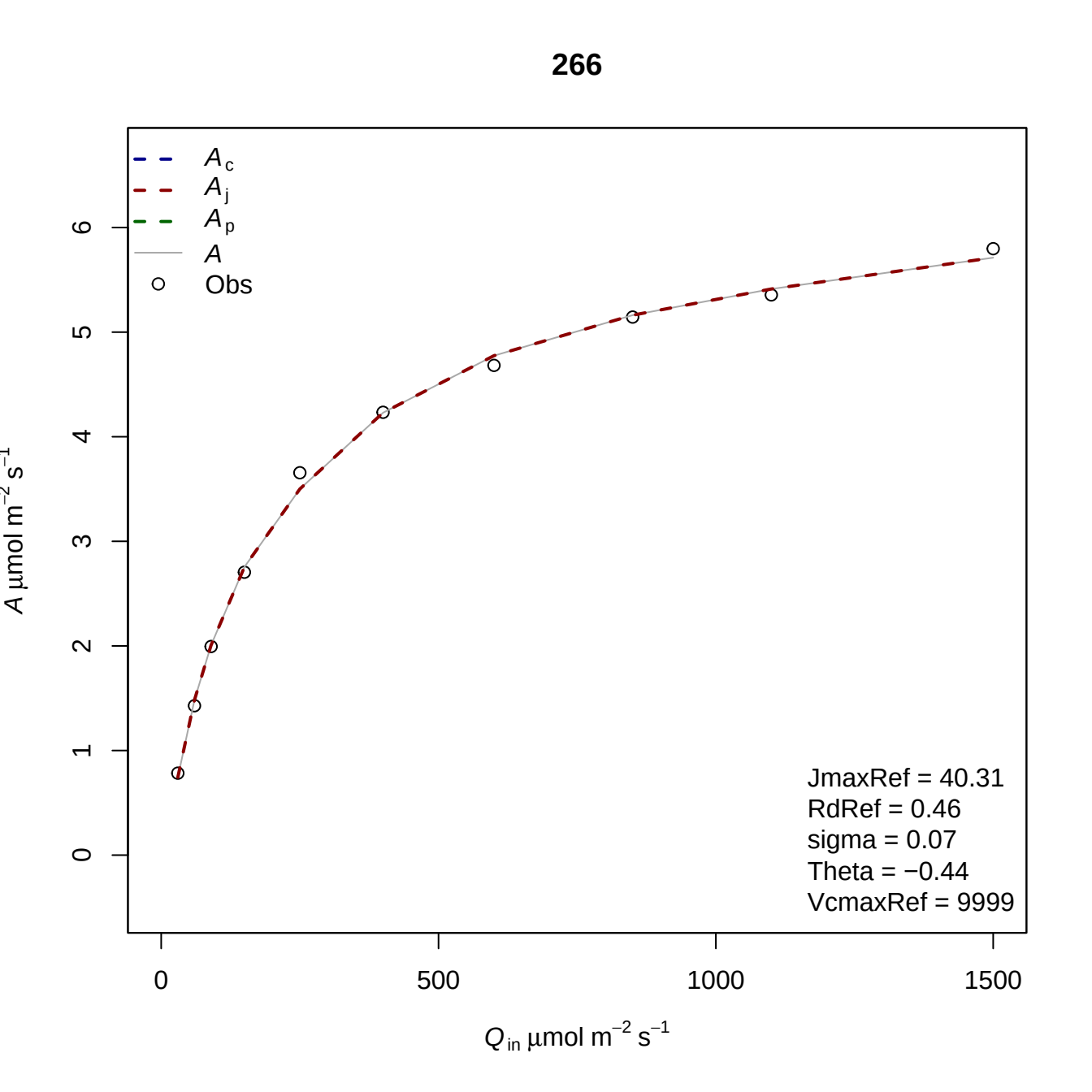

260
A_c
A_j
A_p
A
Obs
JmaxRef = 85.06
RdRef = 1.6
sigma = 0.22
Theta = −0.27
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

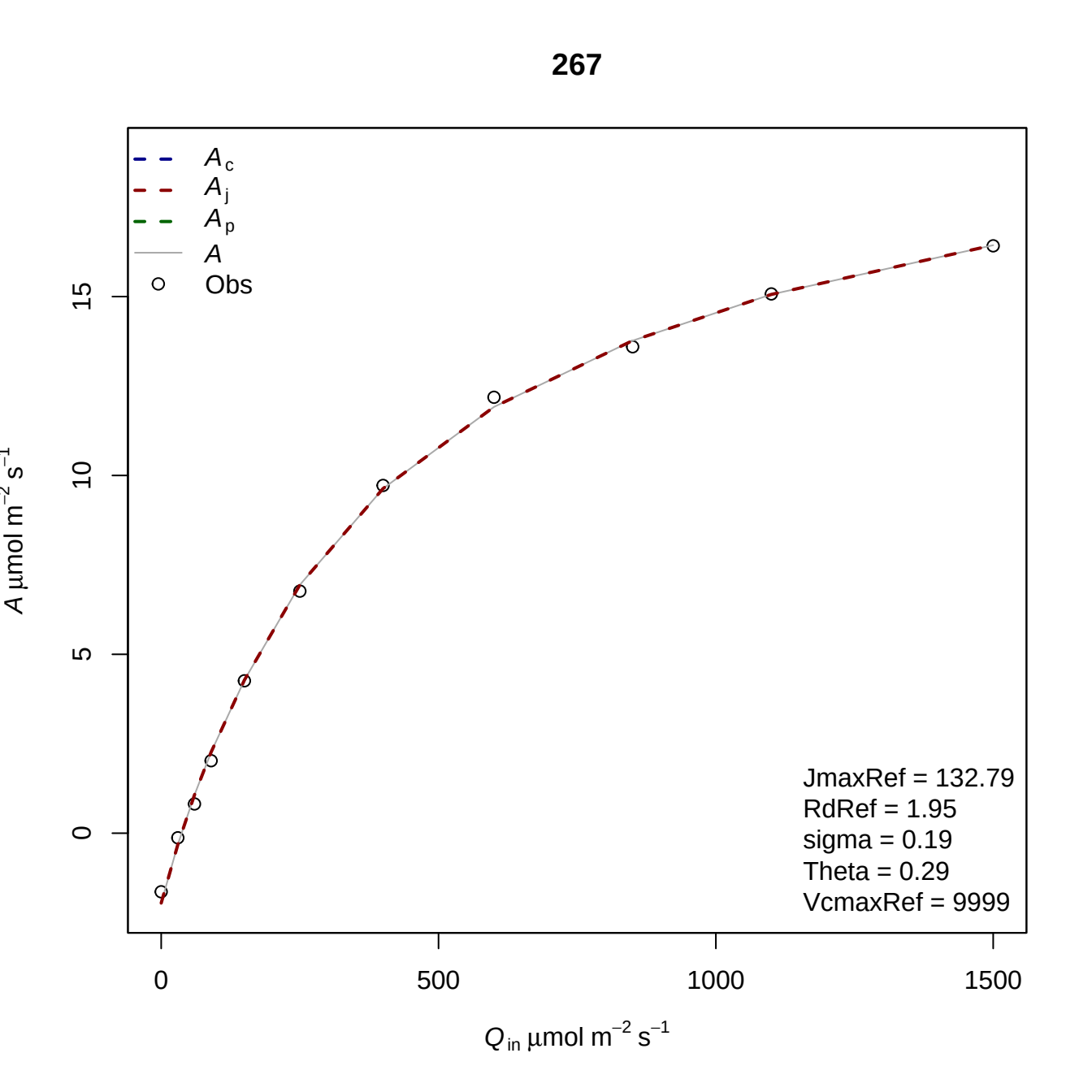

261
A_c
A_j
A_p
A
Obs
JmaxRef = 33.33
RdRef = 1.4
sigma = 0.13
Theta = −0.2
VcmaxRef = 16.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

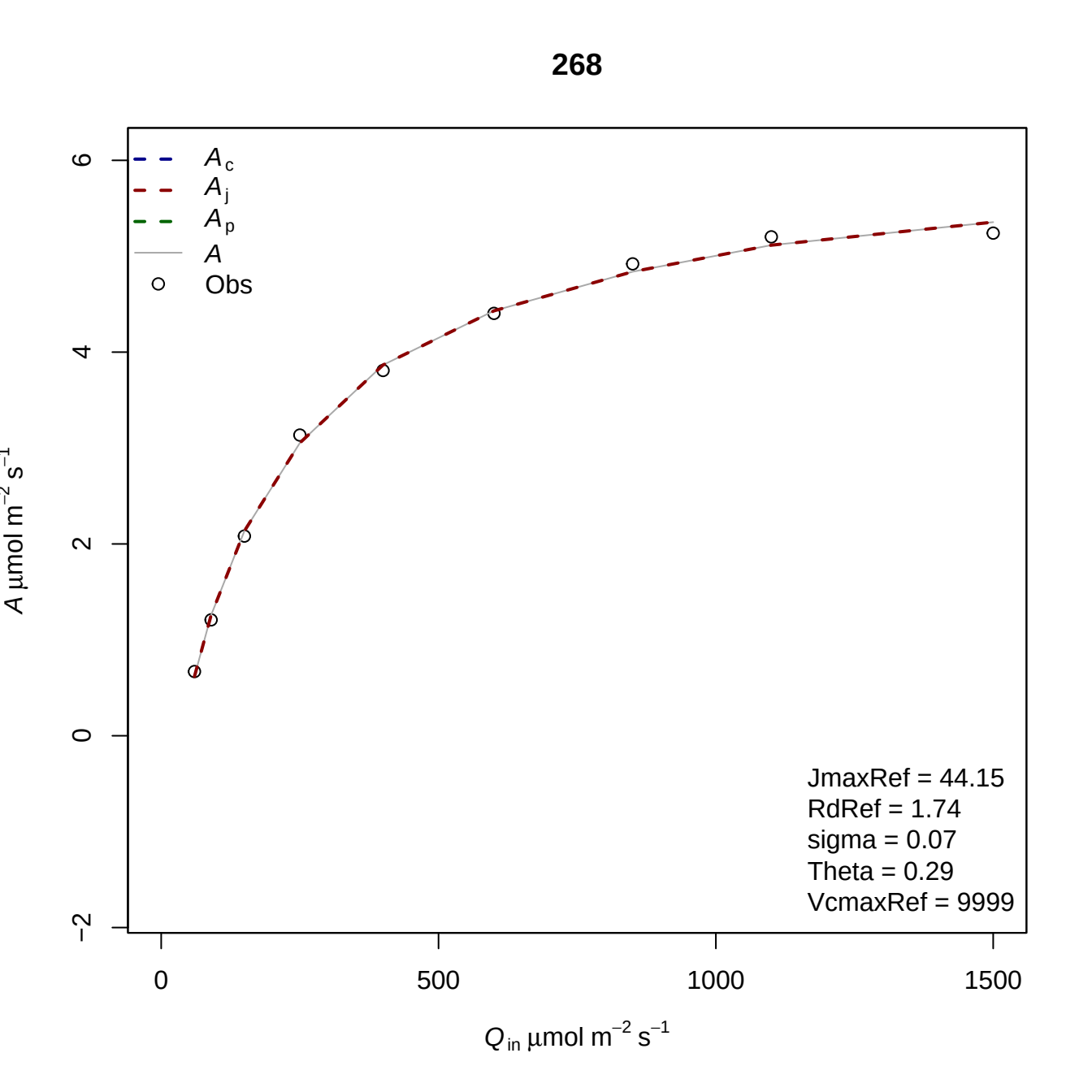

262
A_c
A_j
A_p
A
Obs
JmaxRef = 24.15
RdRef = 1.25
sigma = 0.05
Theta = 0.46
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

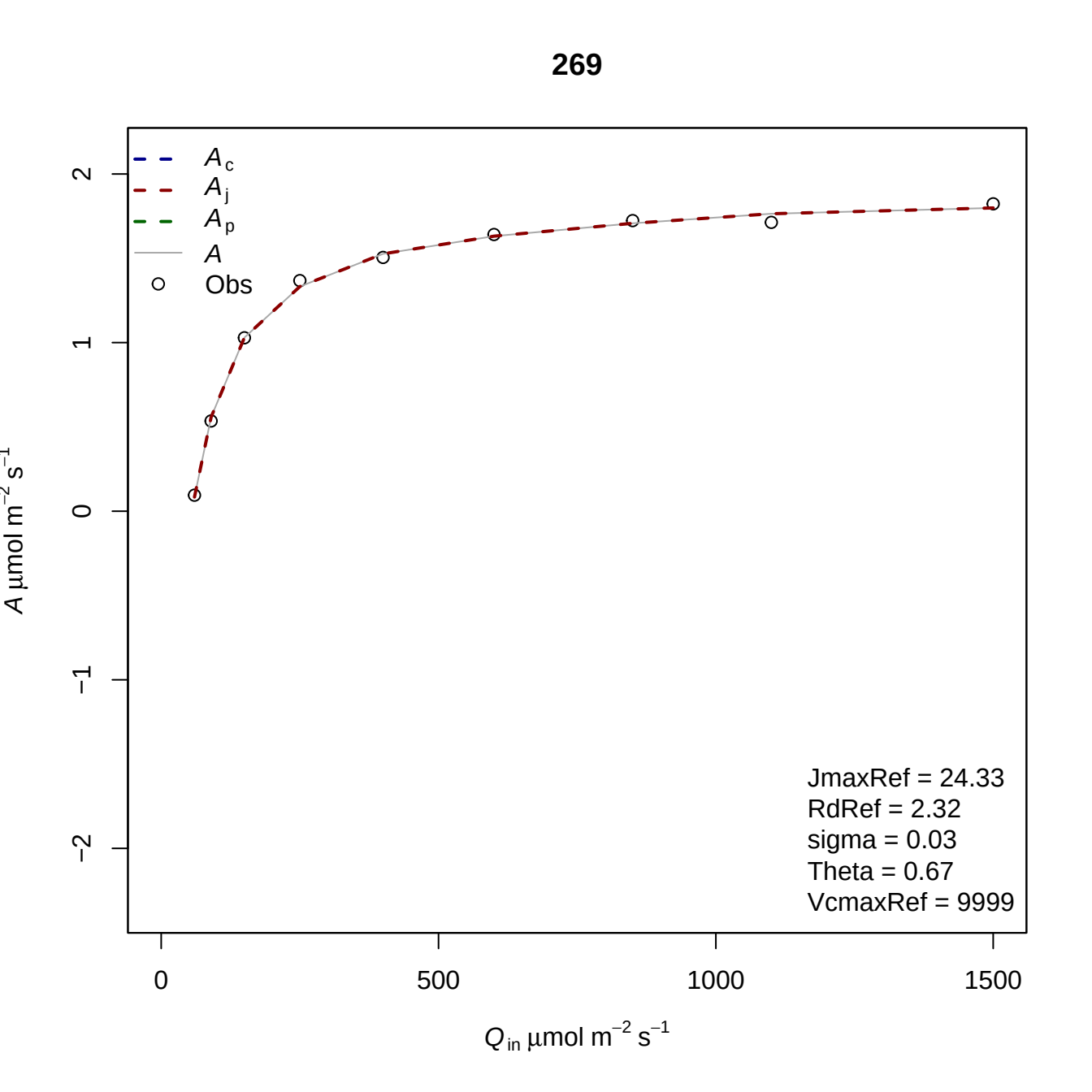

263
A_c
A_j
A_p
A
Obs
JmaxRef = 65.09
RdRef = 1.27
sigma = 0.14
Theta = 0.67
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

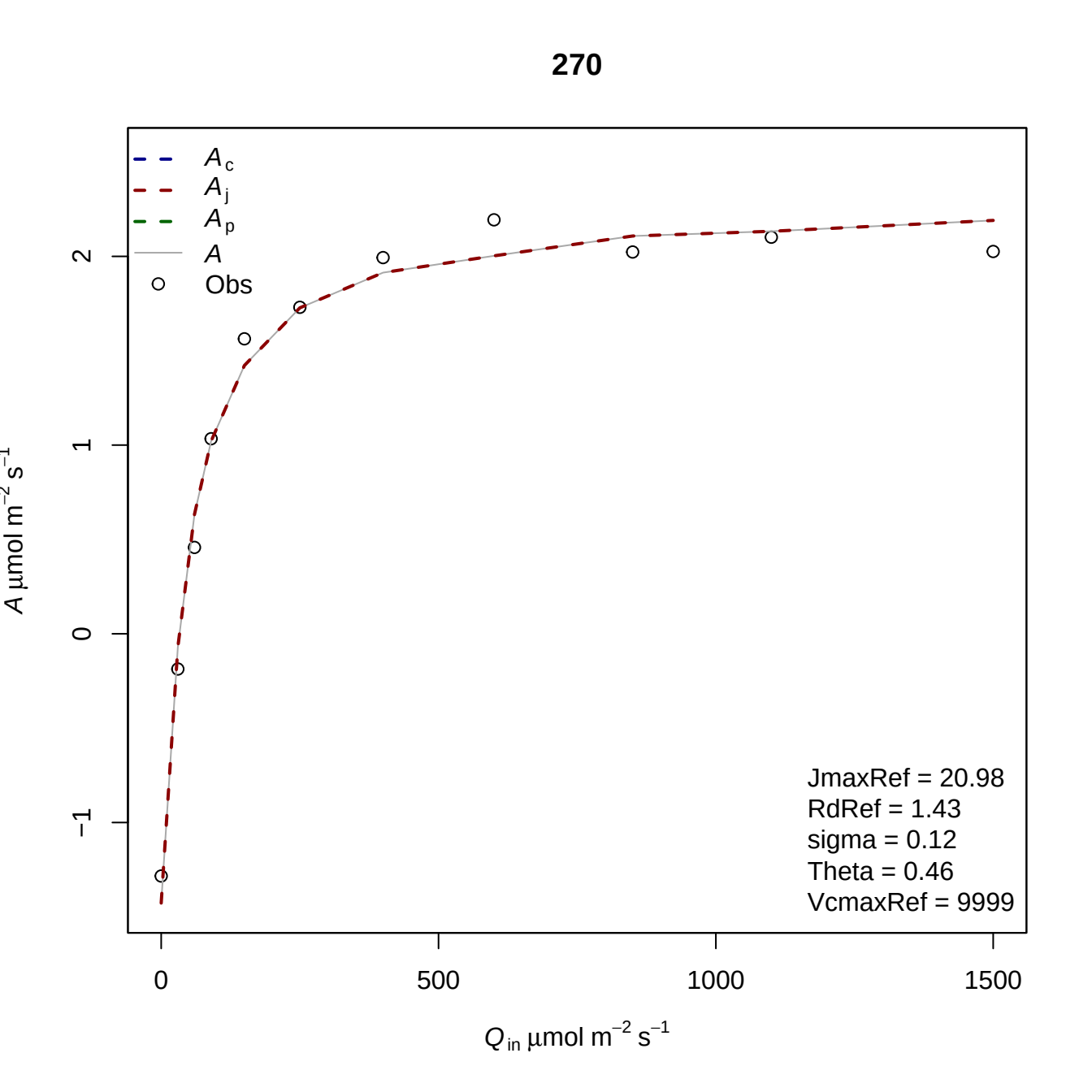

264
A_c
A_j
A_p
A
Obs
JmaxRef = 89.76
RdRef = 1.59
sigma = 0.18
Theta = 0.62
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

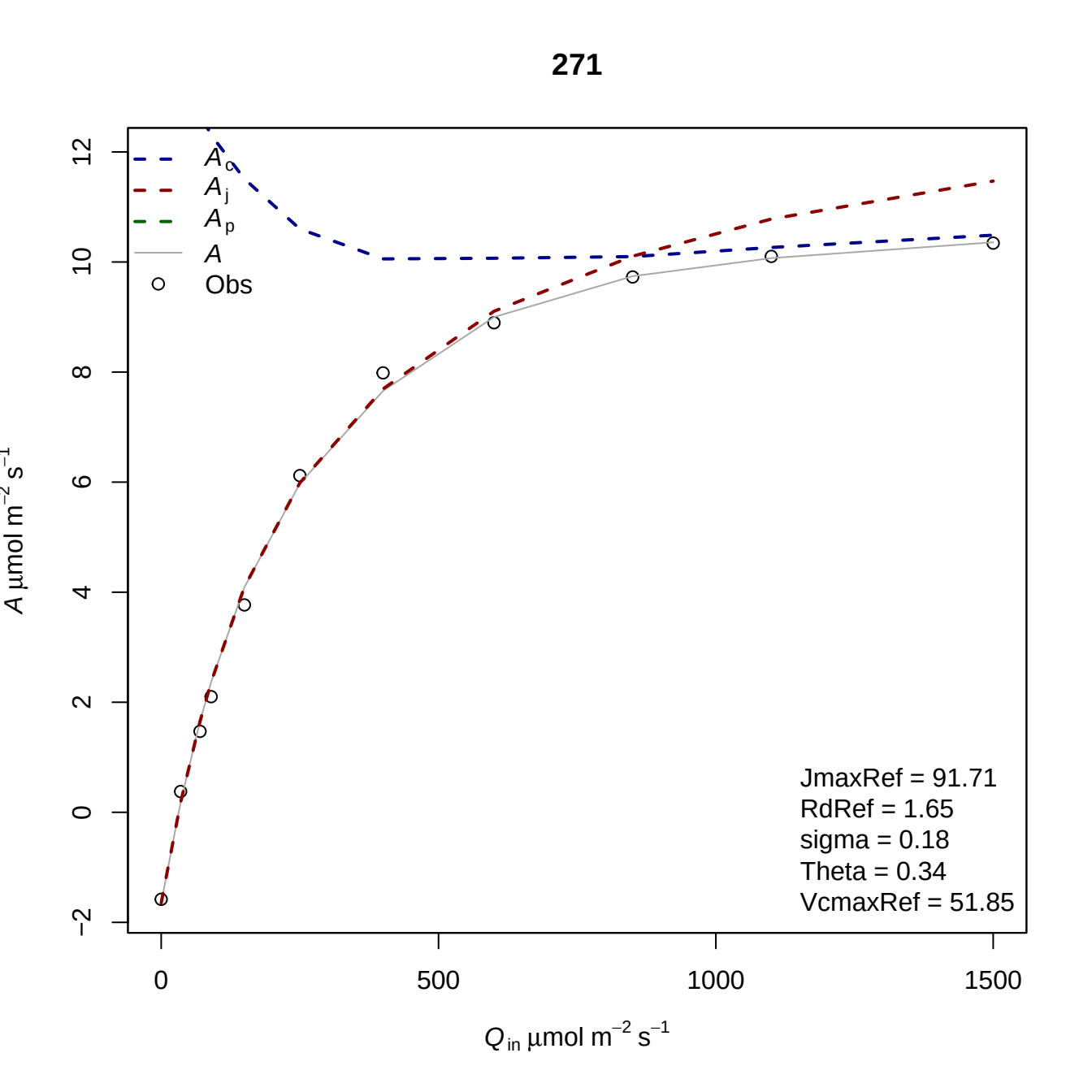

266
A_c
A_j
A_p
A
Obs
JmaxRef = 40.31
RdRef = 0.46
sigma = 0.07
Theta = −0.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

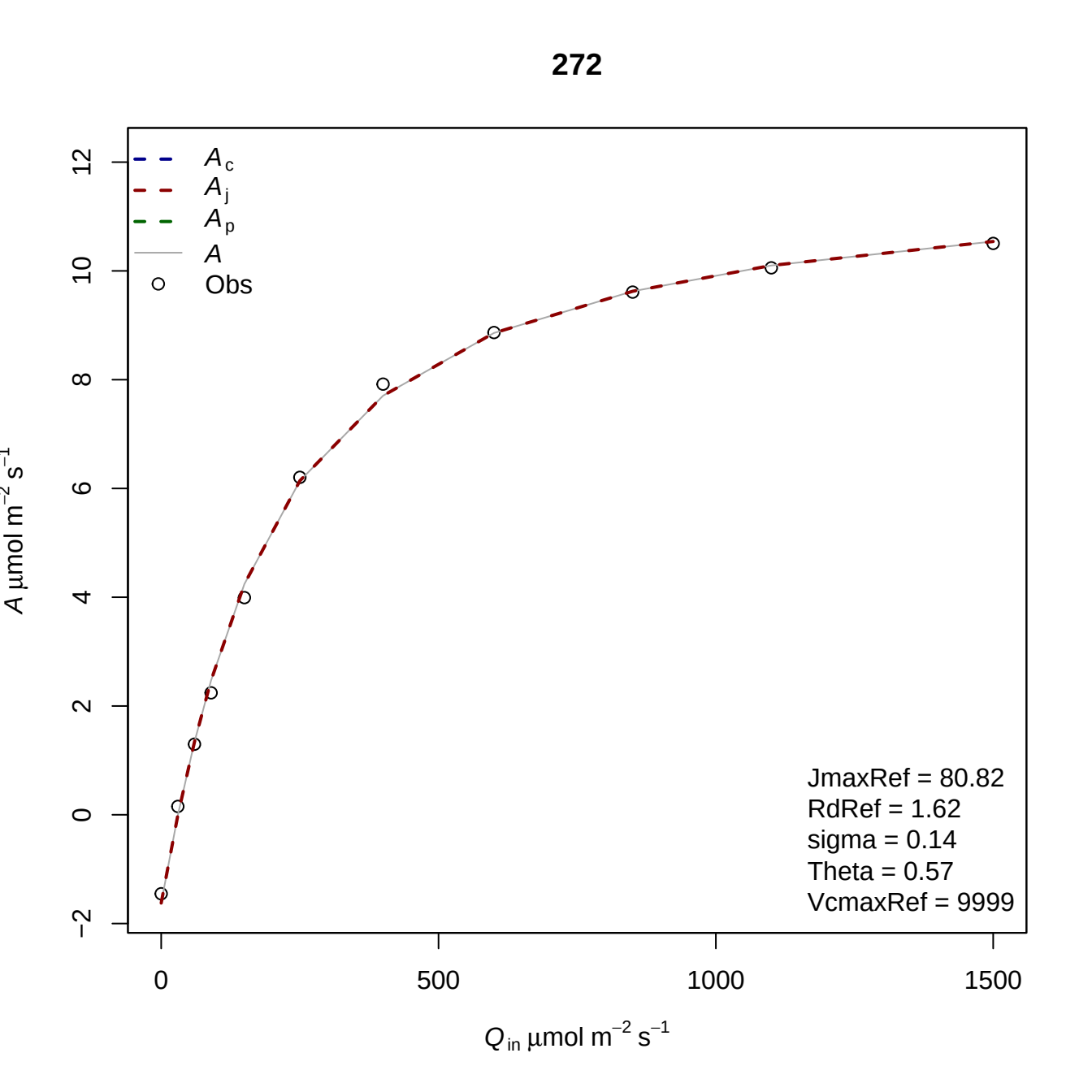

267
A_c
A_j
A_p
A
Obs
JmaxRef = 132.79
RdRef = 1.95
sigma = 0.19
Theta = 0.29
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

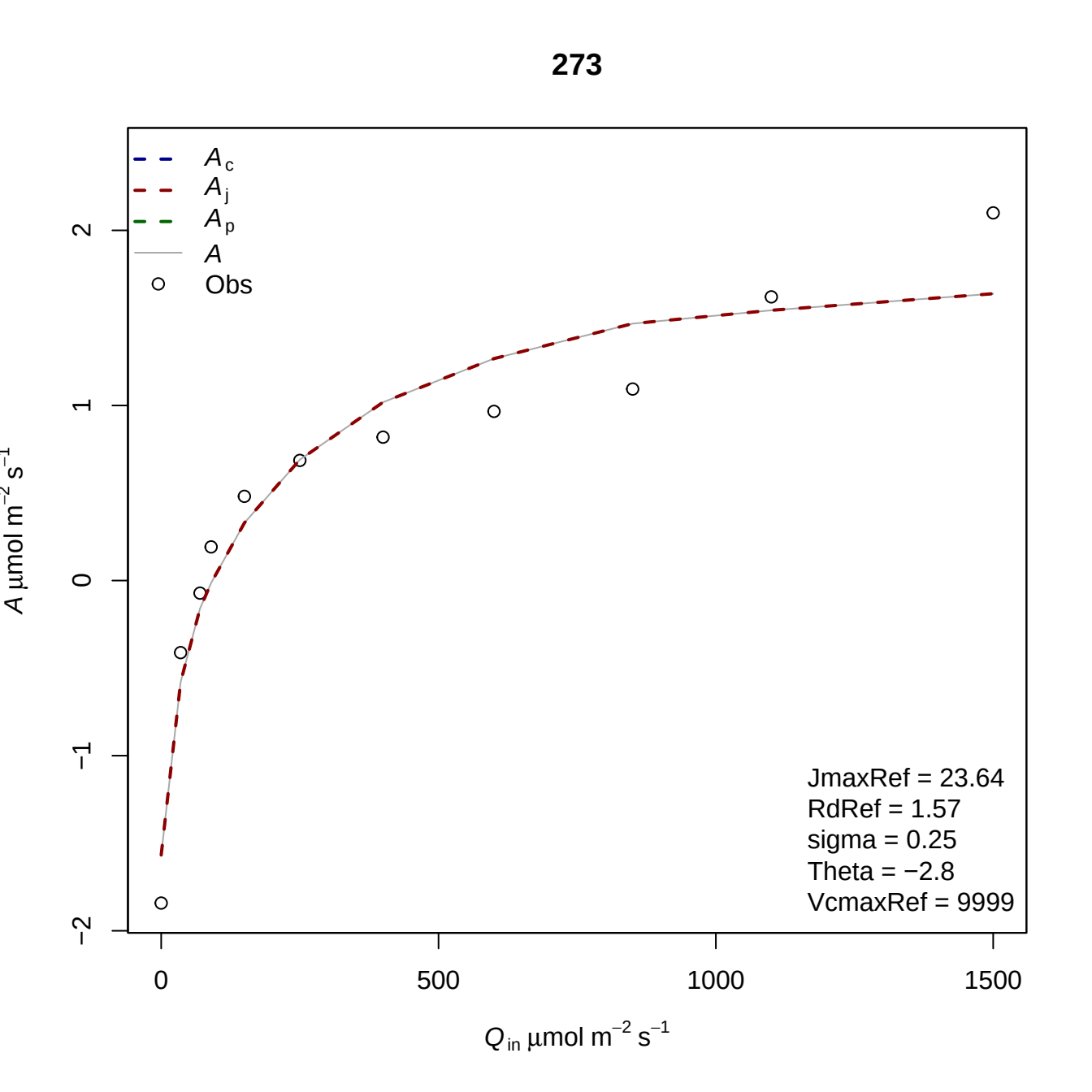

268
A_c
A_j
A_p
A
Obs
JmaxRef = 44.15
RdRef = 1.74
sigma = 0.07
Theta = 0.29
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

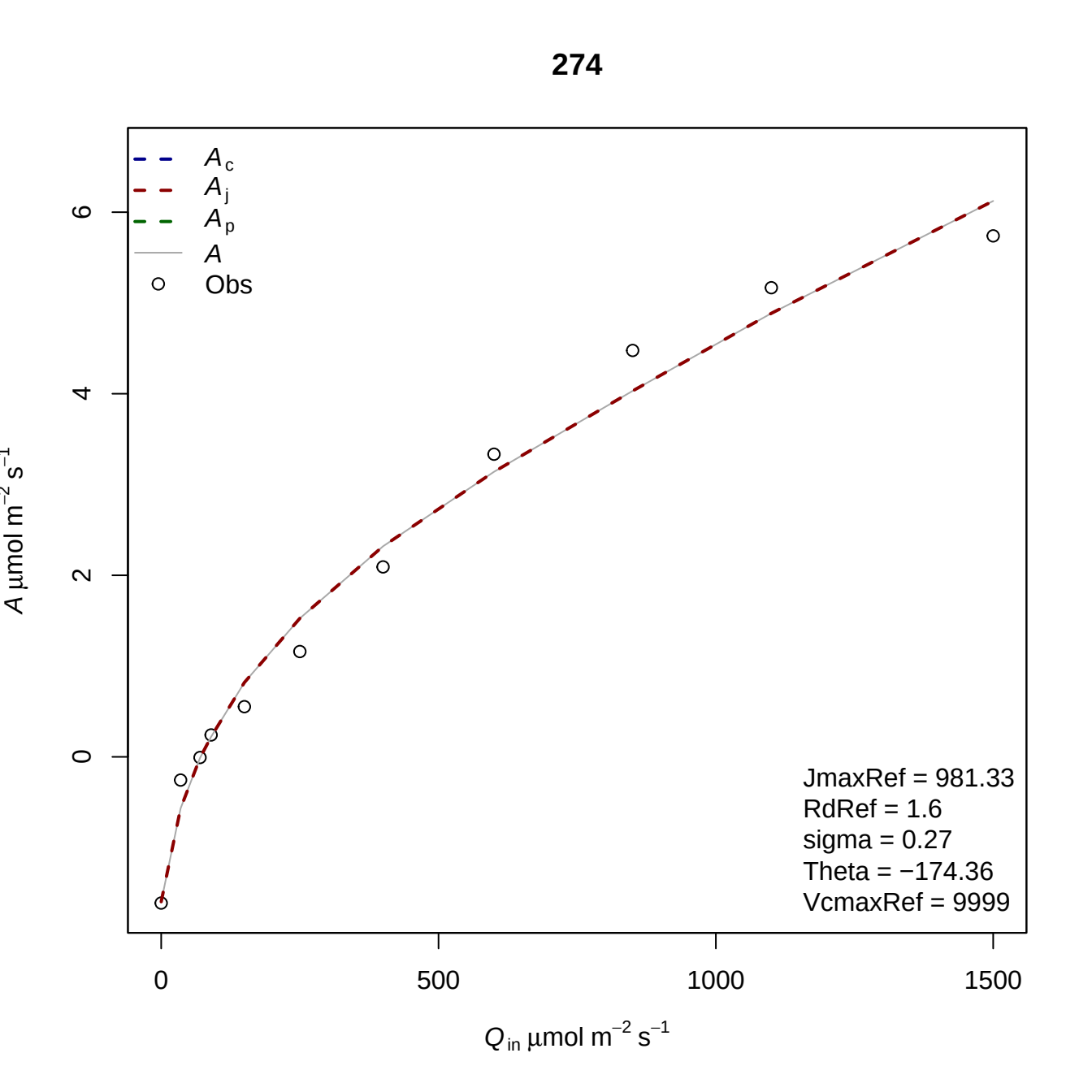

269
A_c
A_j
A_p
A
Obs
JmaxRef = 24.33
RdRef = 2.32
sigma = 0.03
Theta = 0.67
VcmaxRef = 9999
2
1
0
−1
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

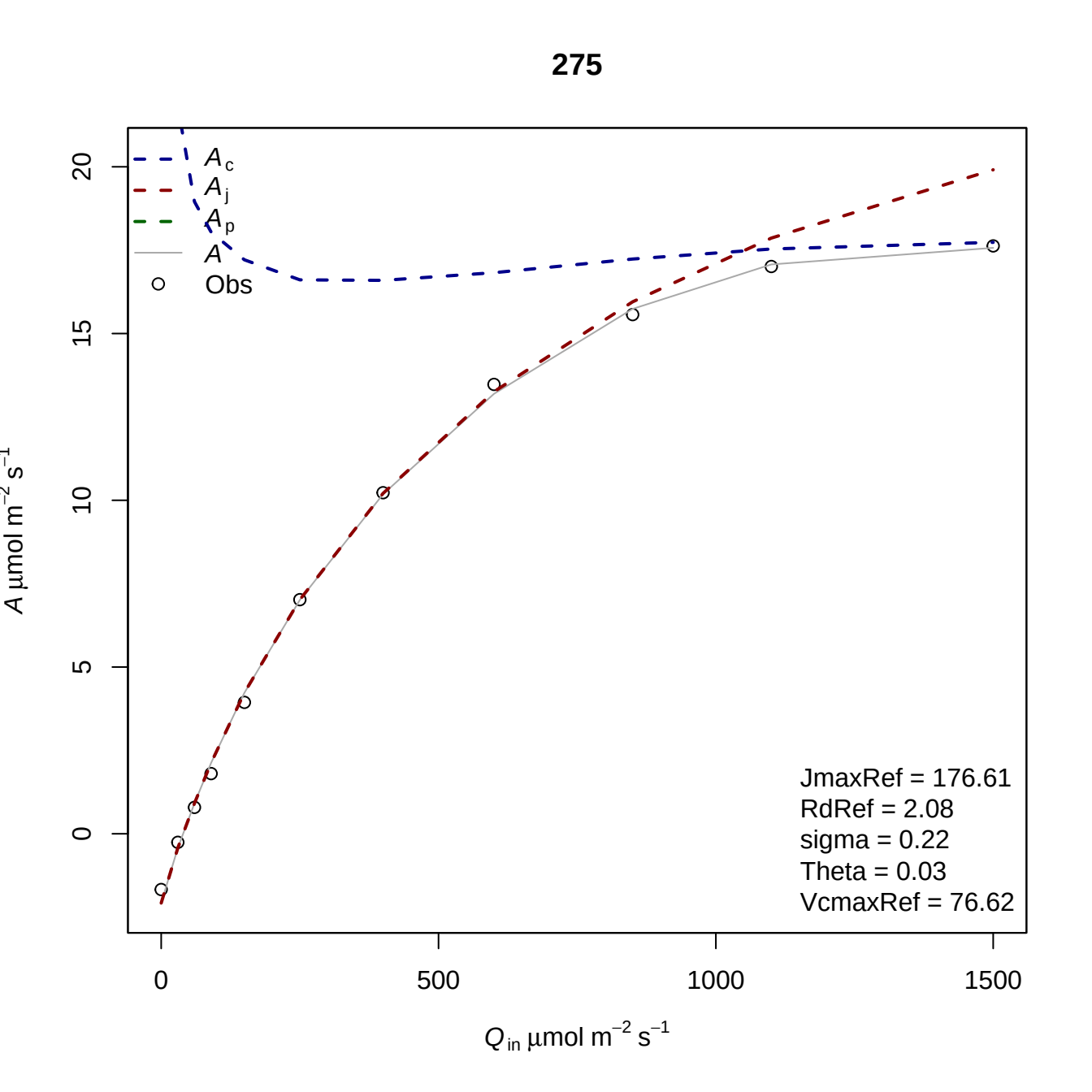

270
A_c
A_j
A_p
A
Obs
JmaxRef = 20.98
RdRef = 1.43
sigma = 0.12
Theta = 0.46
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

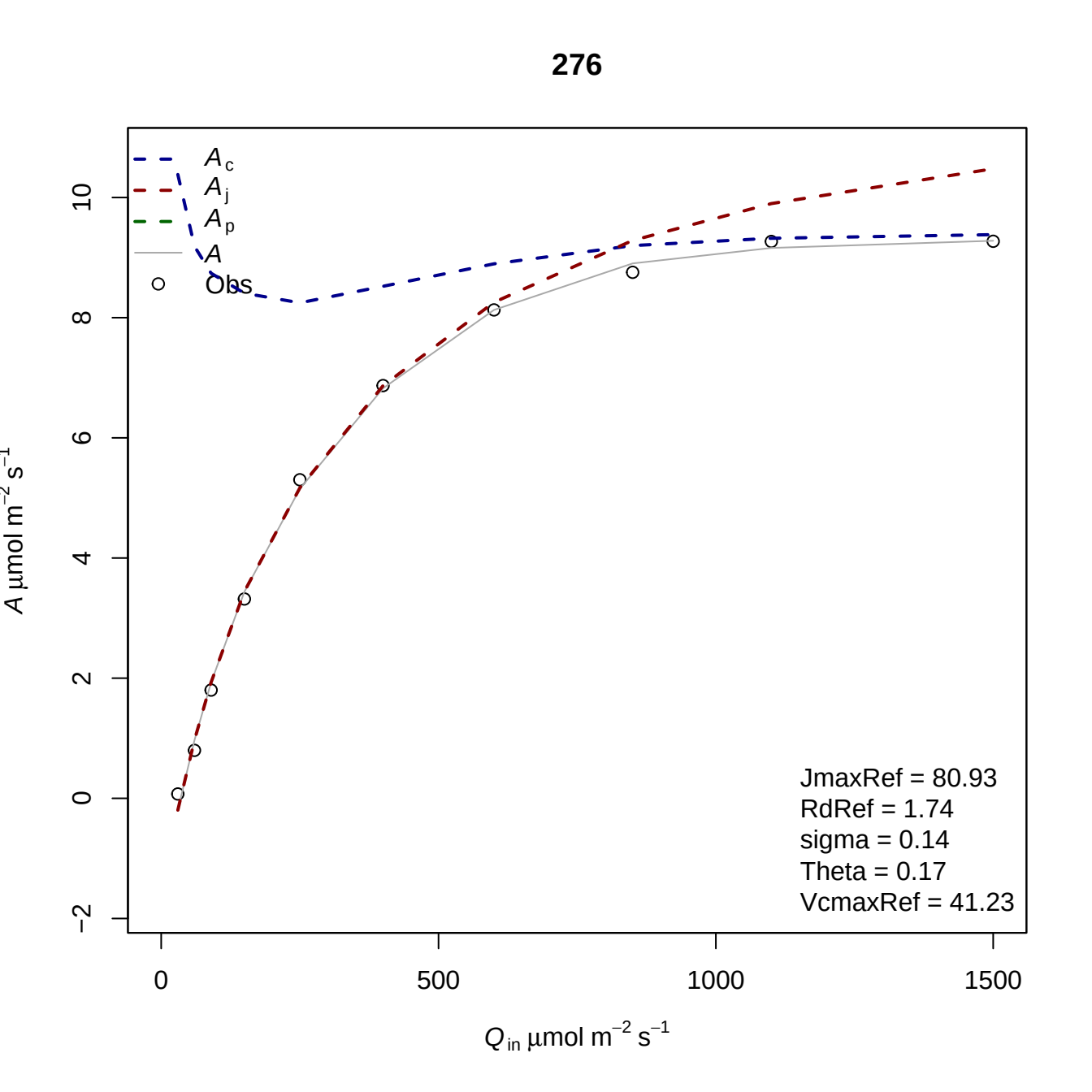

271
A_c
A_j
A_p
A
Obs
JmaxRef = 91.71
RdRef = 1.65
sigma = 0.18
Theta = 0.34
VcmaxRef = 51.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

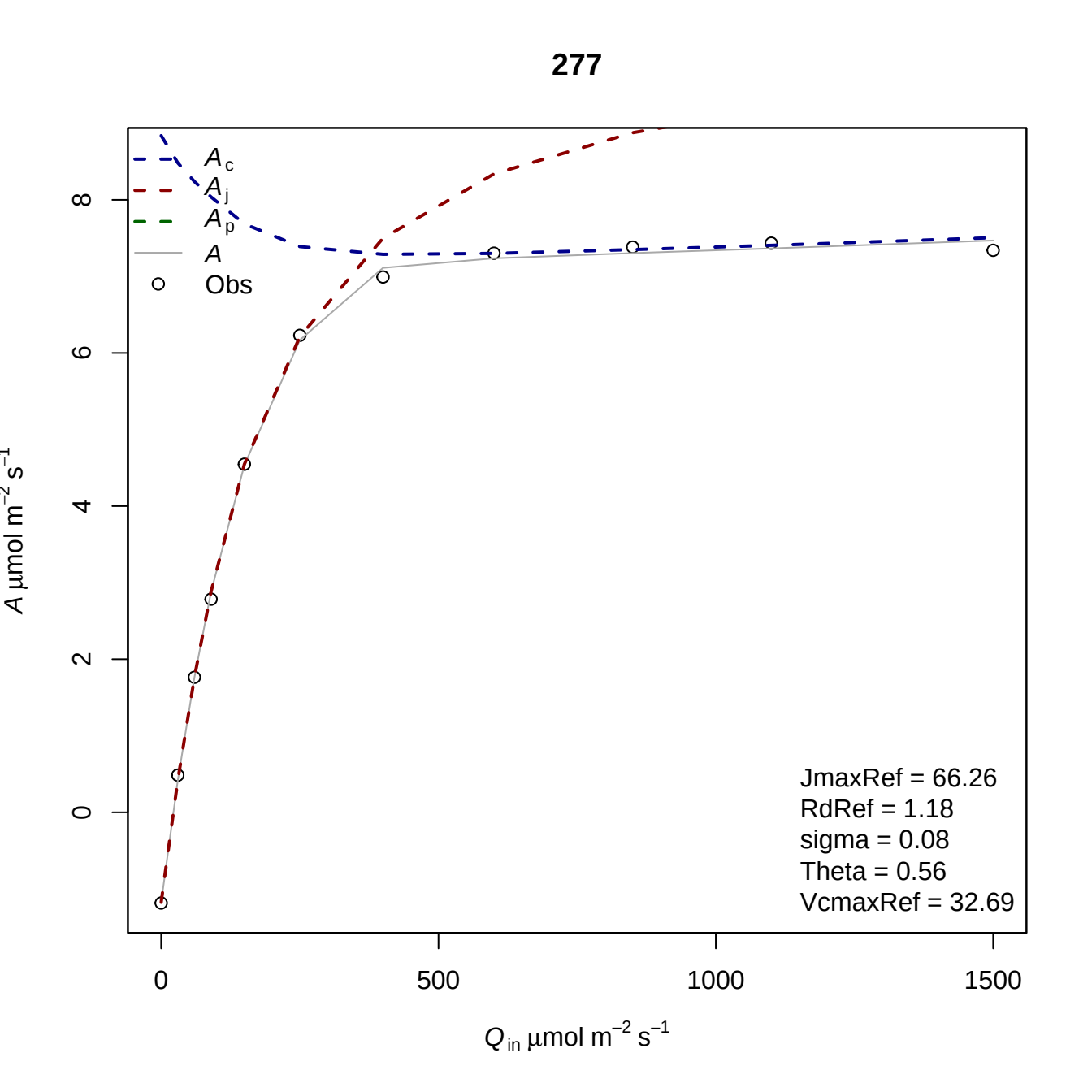

272
A_c
A_j
A_p
A
Obs
JmaxRef = 80.82
RdRef = 1.62
sigma = 0.14
Theta = 0.57
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

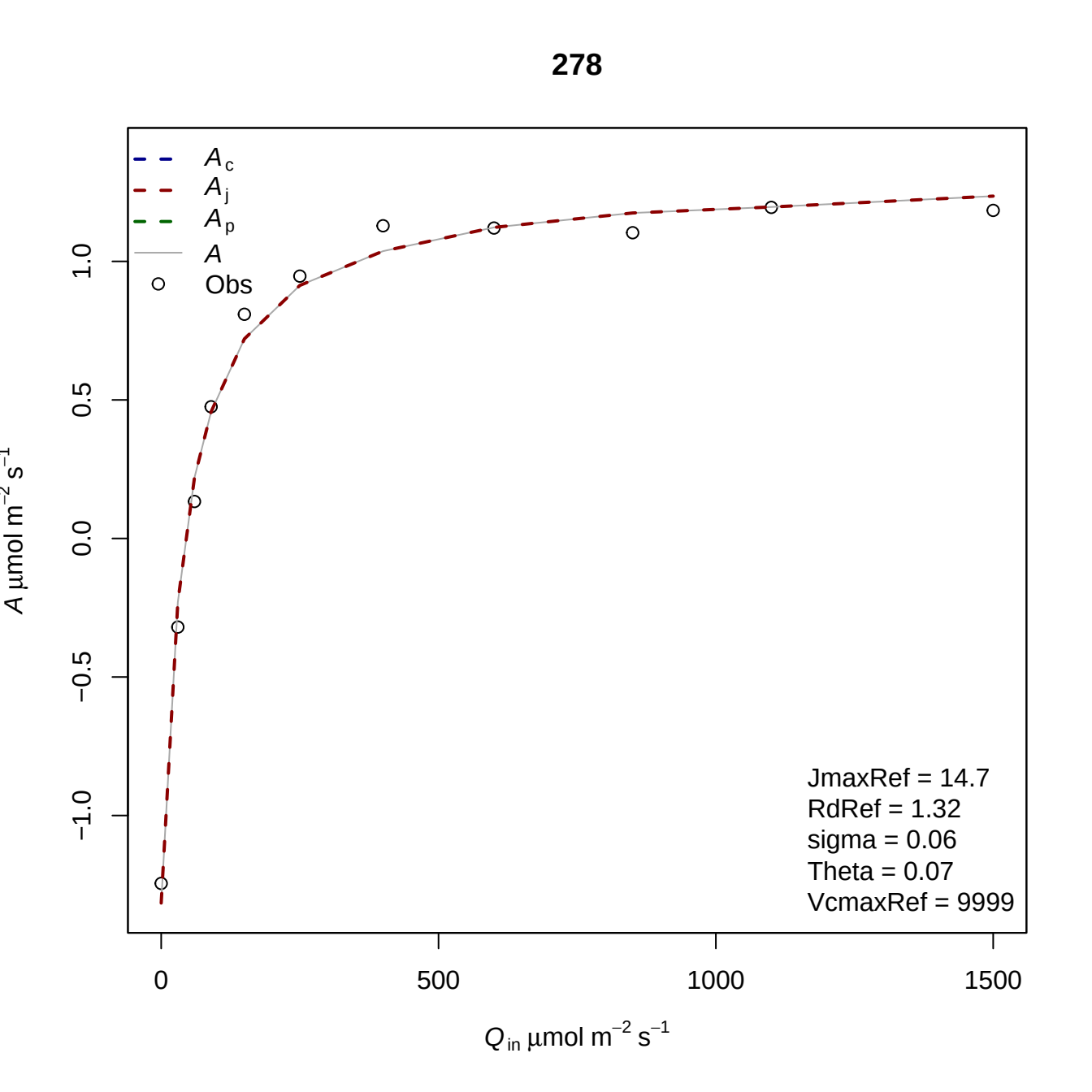

273
A_c
A_j
A_p
A
Obs
JmaxRef = 23.64
RdRef = 1.57
sigma = 0.25
Theta = −2.8
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
0
500
1000
1500

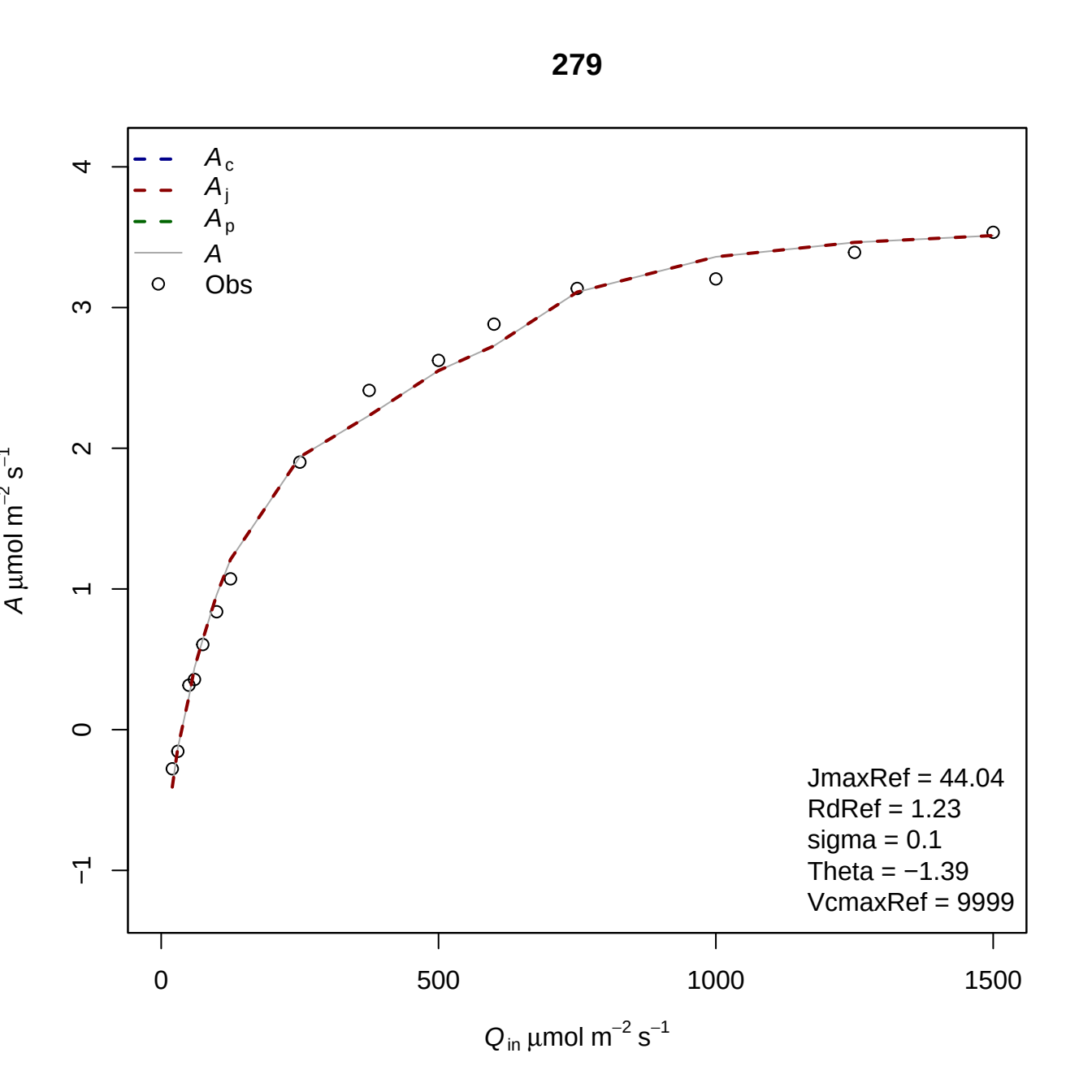

274
A_c
A_j
A_p
A
Obs
JmaxRef = 981.33
RdRef = 1.6
sigma = 0.27
Theta = −174.36
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

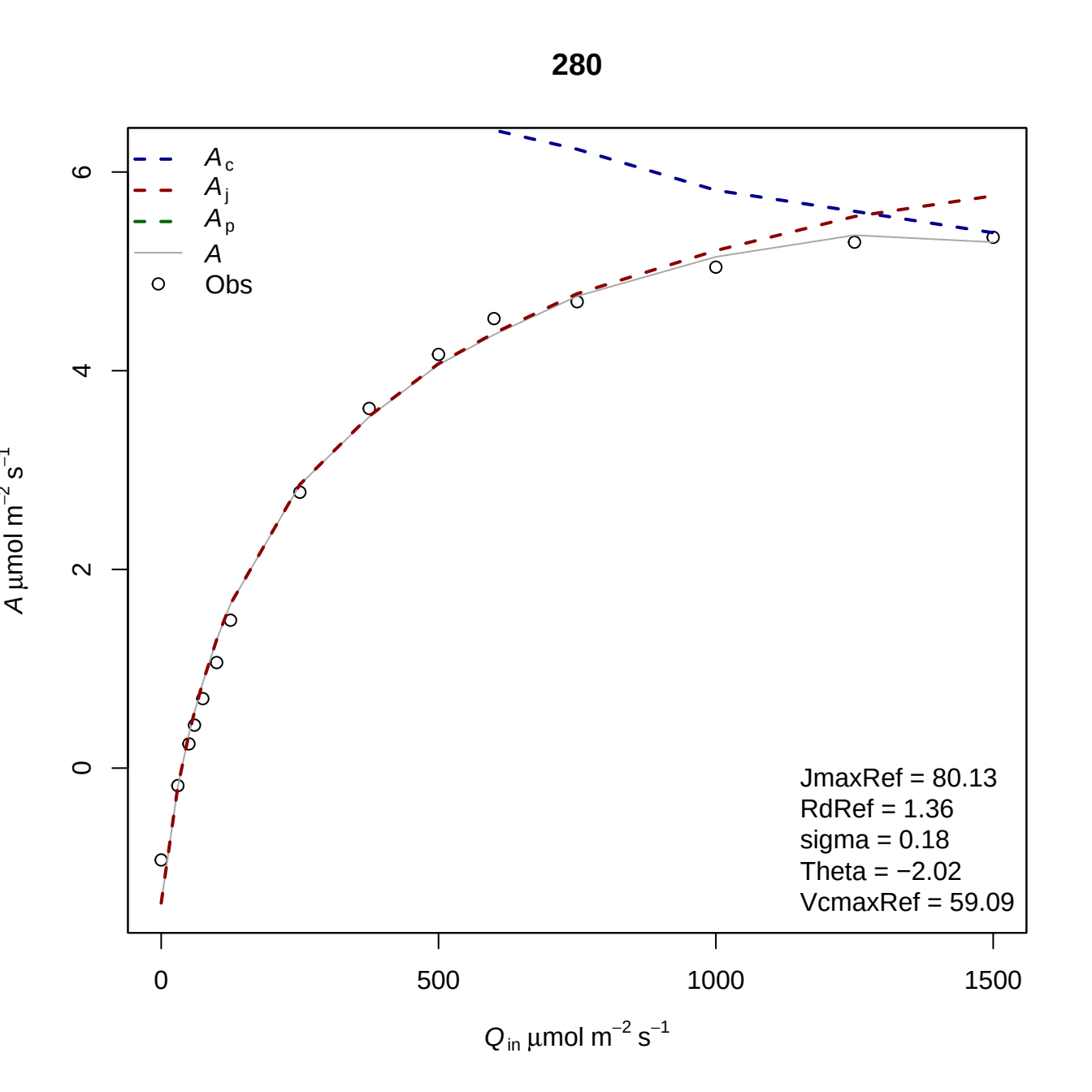

275
A_c
A_j
A_p
A
Obs
JmaxRef = 176.61
RdRef = 2.08
sigma = 0.22
Theta = 0.03
VcmaxRef = 76.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

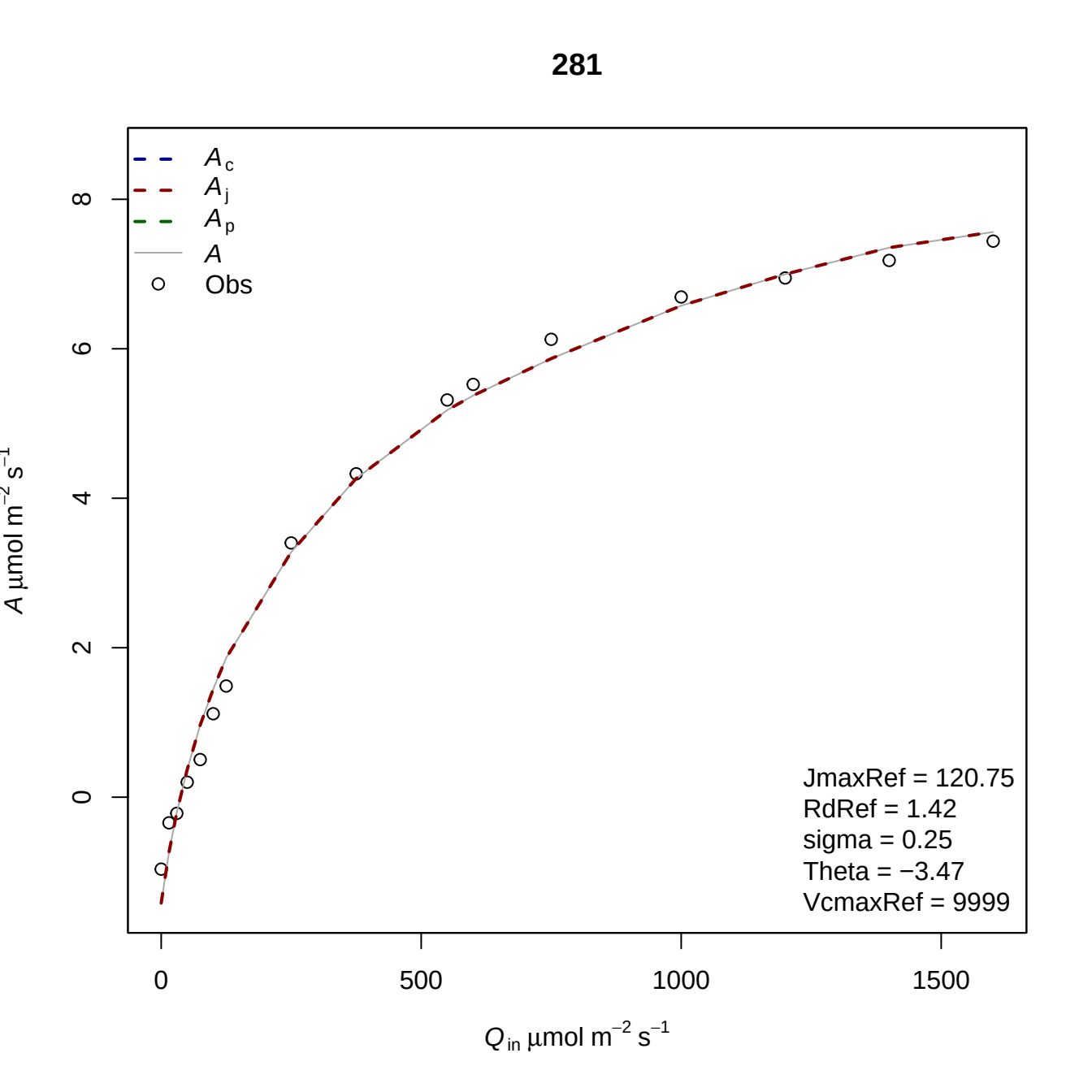

276
A_c
A_j
A_p
A
Obs
JmaxRef = 80.93
RdRef = 1.74
sigma = 0.14
Theta = 0.17
VcmaxRef = 41.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

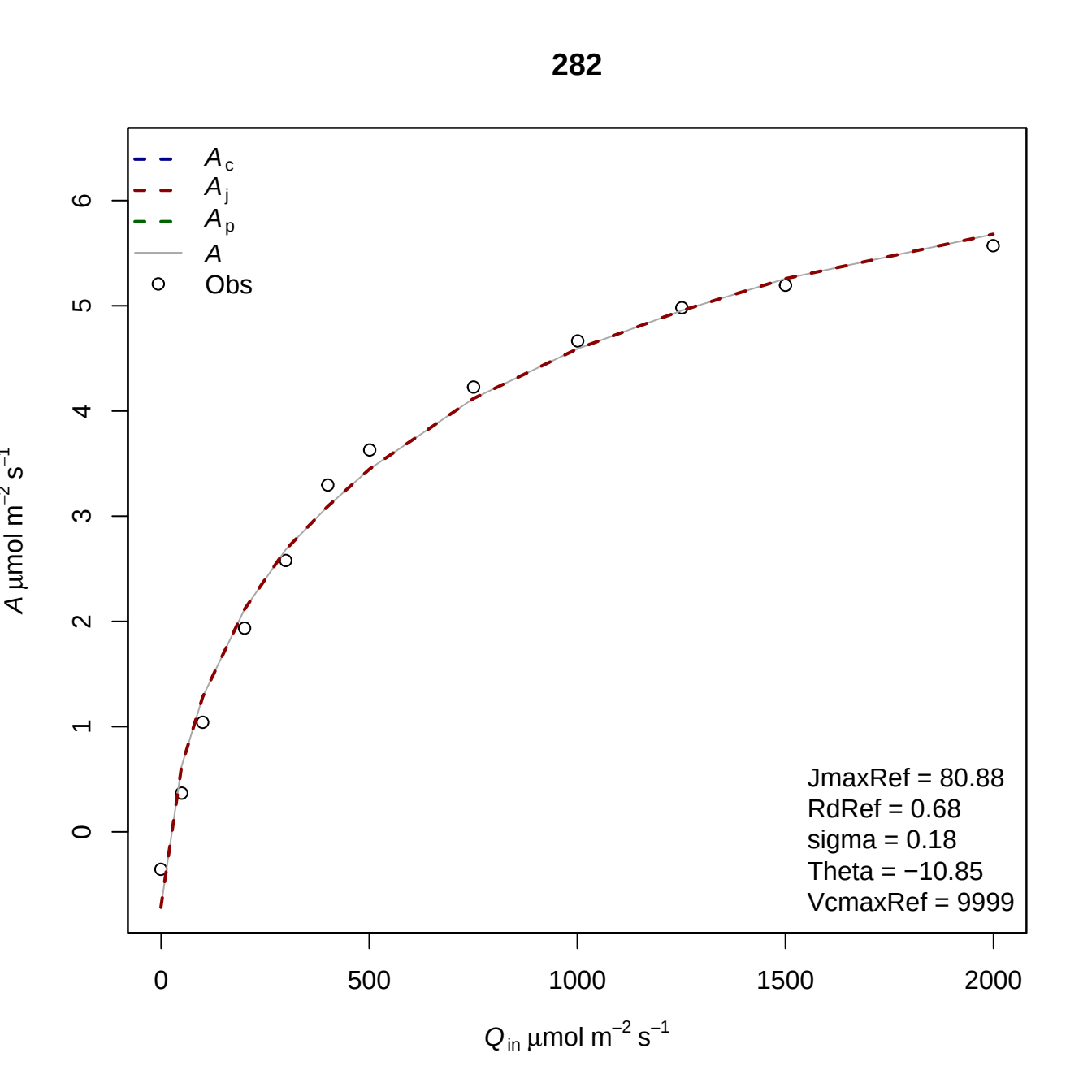

277
A_c
A_j
A_p
A
Obs
JmaxRef = 66.26
RdRef = 1.18
sigma = 0.08
Theta = 0.56
VcmaxRef = 32.69
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

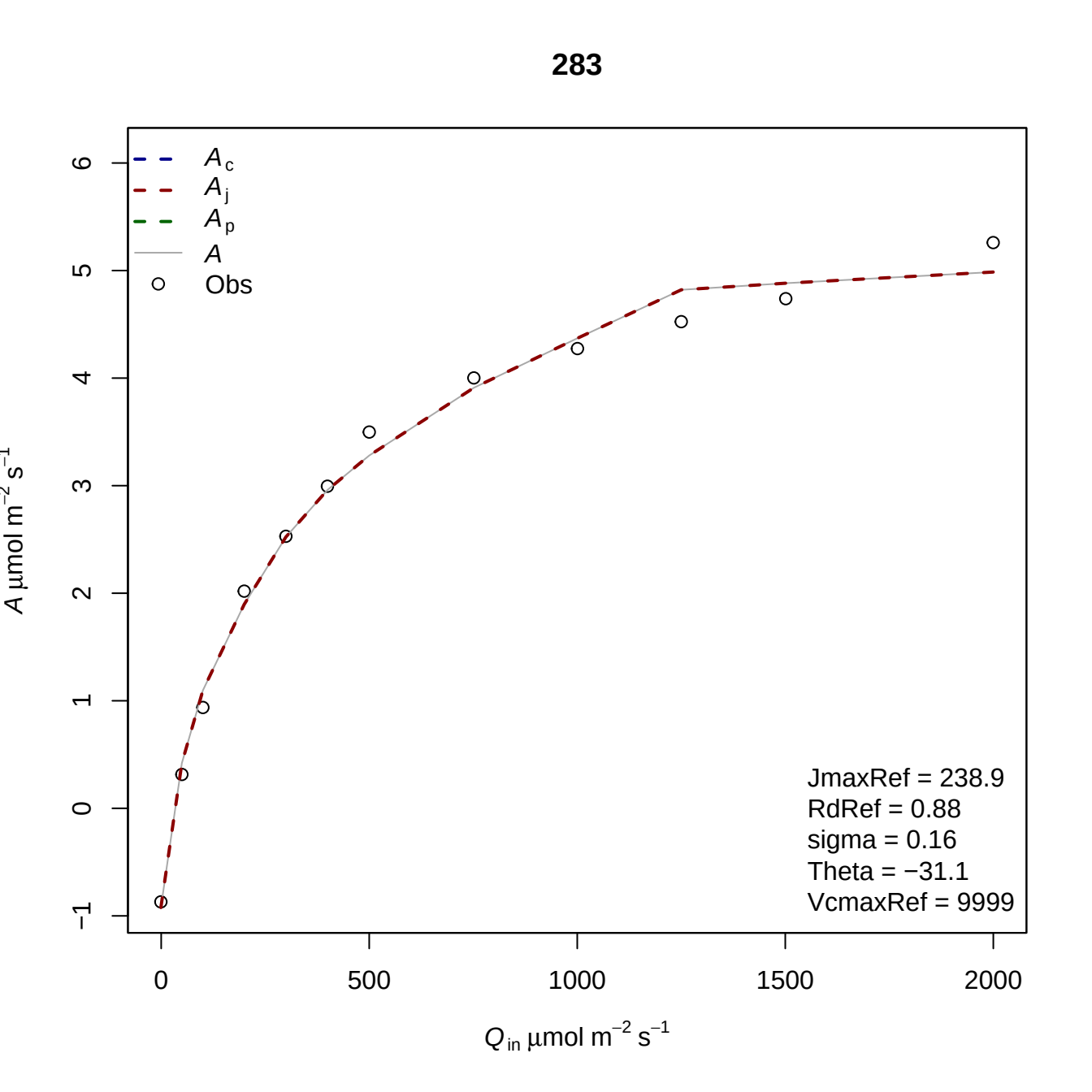

278
A_c
A_j
A_p
A
Obs
JmaxRef = 14.7
RdRef = 1.32
sigma = 0.06
Theta = 0.07
VcmaxRef = 9999
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

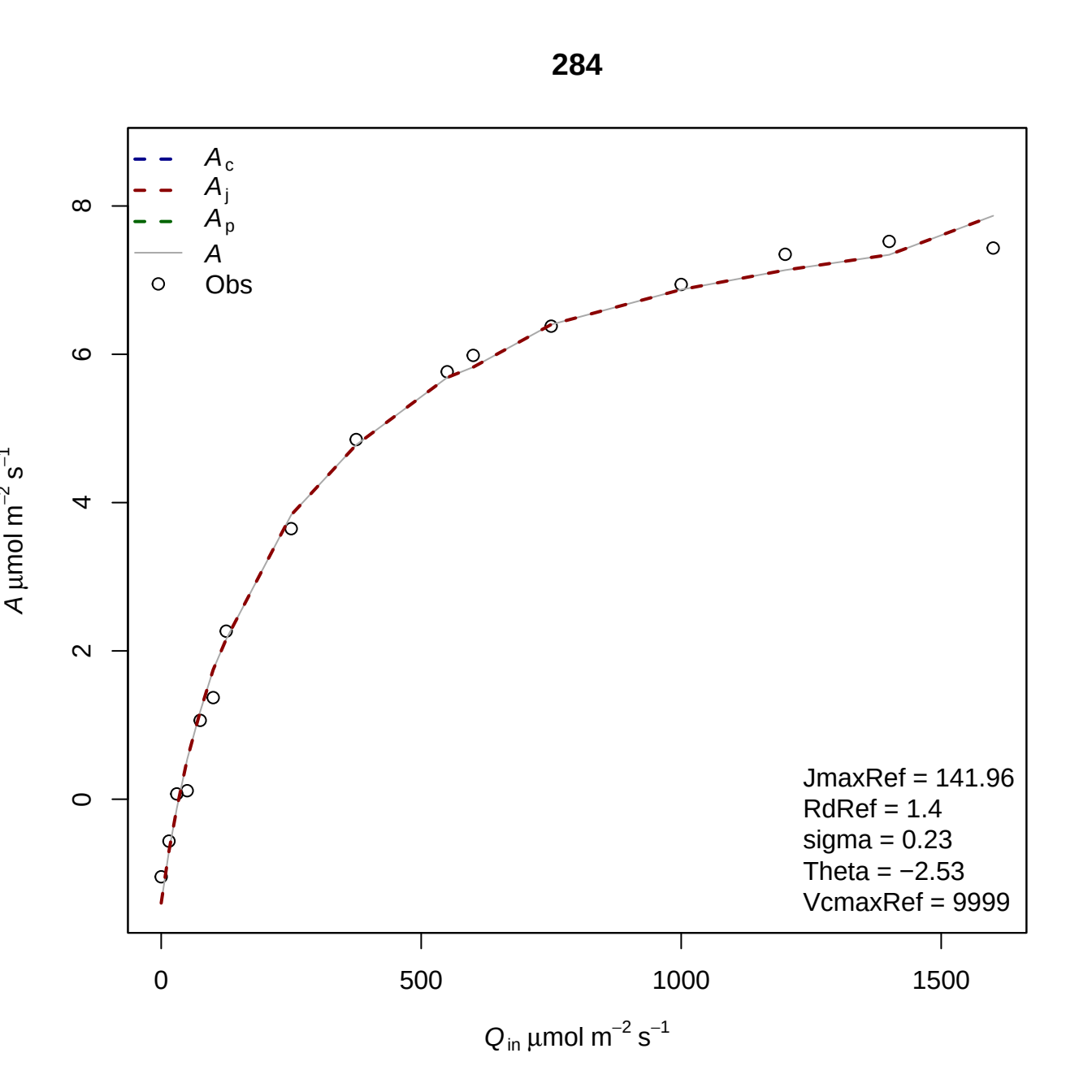

279
A_c
A_j
A_p
A
Obs
JmaxRef = 44.04
RdRef = 1.23
sigma = 0.1
Theta = −1.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

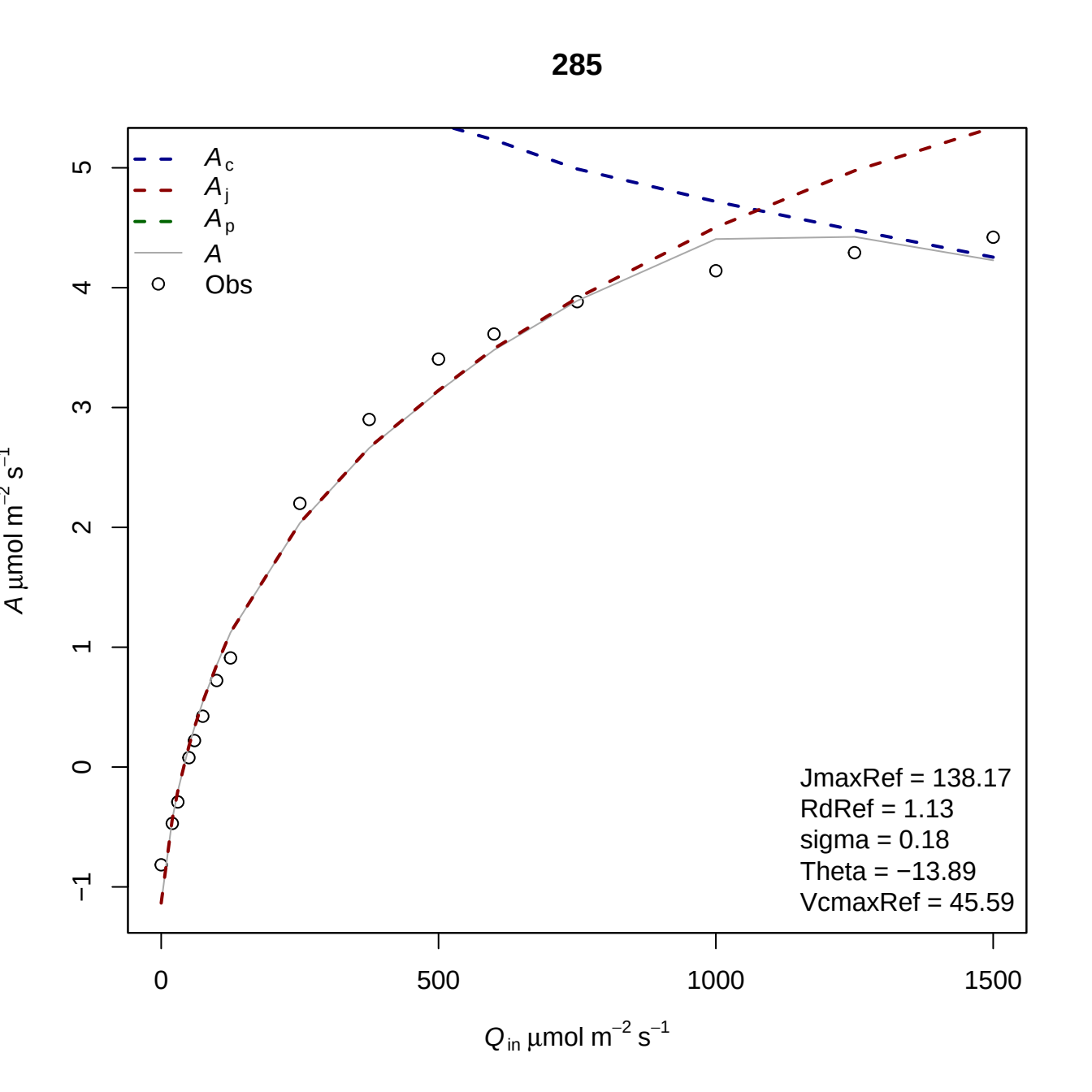

280
A_c
A_j
A_p
A
Obs
JmaxRef = 80.13
RdRef = 1.36
sigma = 0.18
Theta = −2.02
VcmaxRef = 59.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

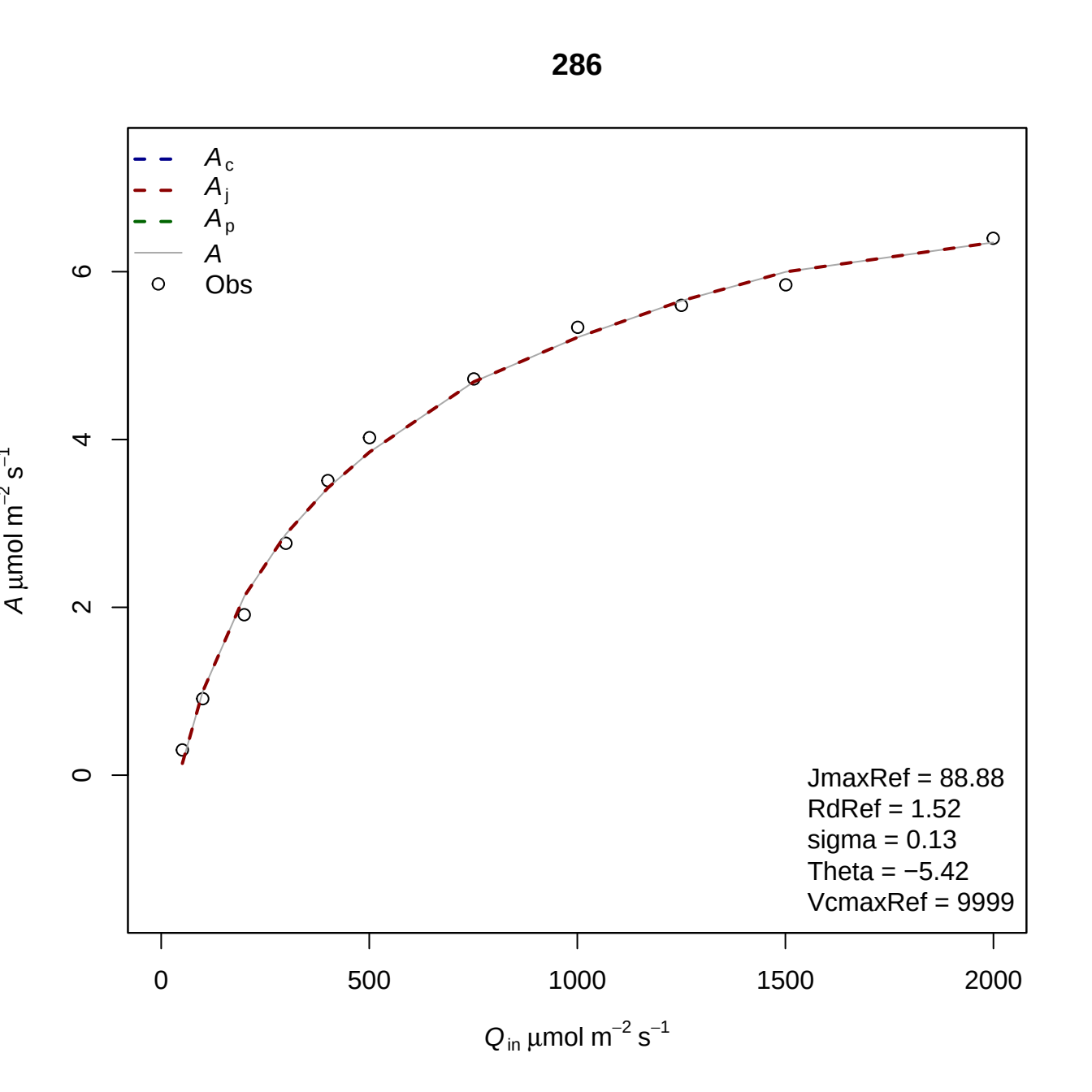

281
A_c
A_j
A_p
A
Obs
JmaxRef = 120.75
RdRef = 1.42
sigma = 0.25
Theta = −3.47
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

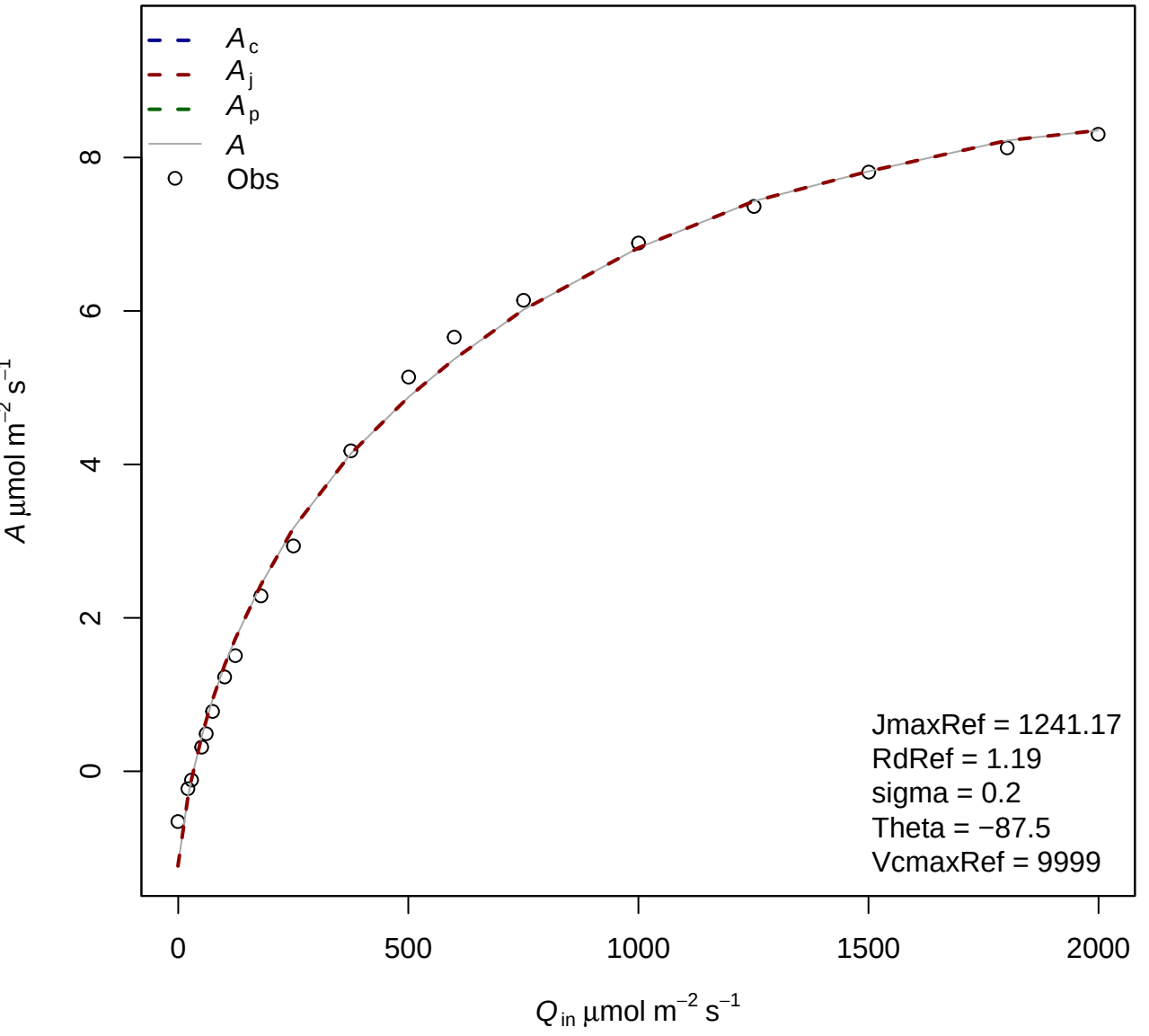

282
A_c
A_j
A_p
A
Obs
JmaxRef = 80.88
RdRef = 0.68
sigma = 0.18
Theta = −10.85
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500
2000

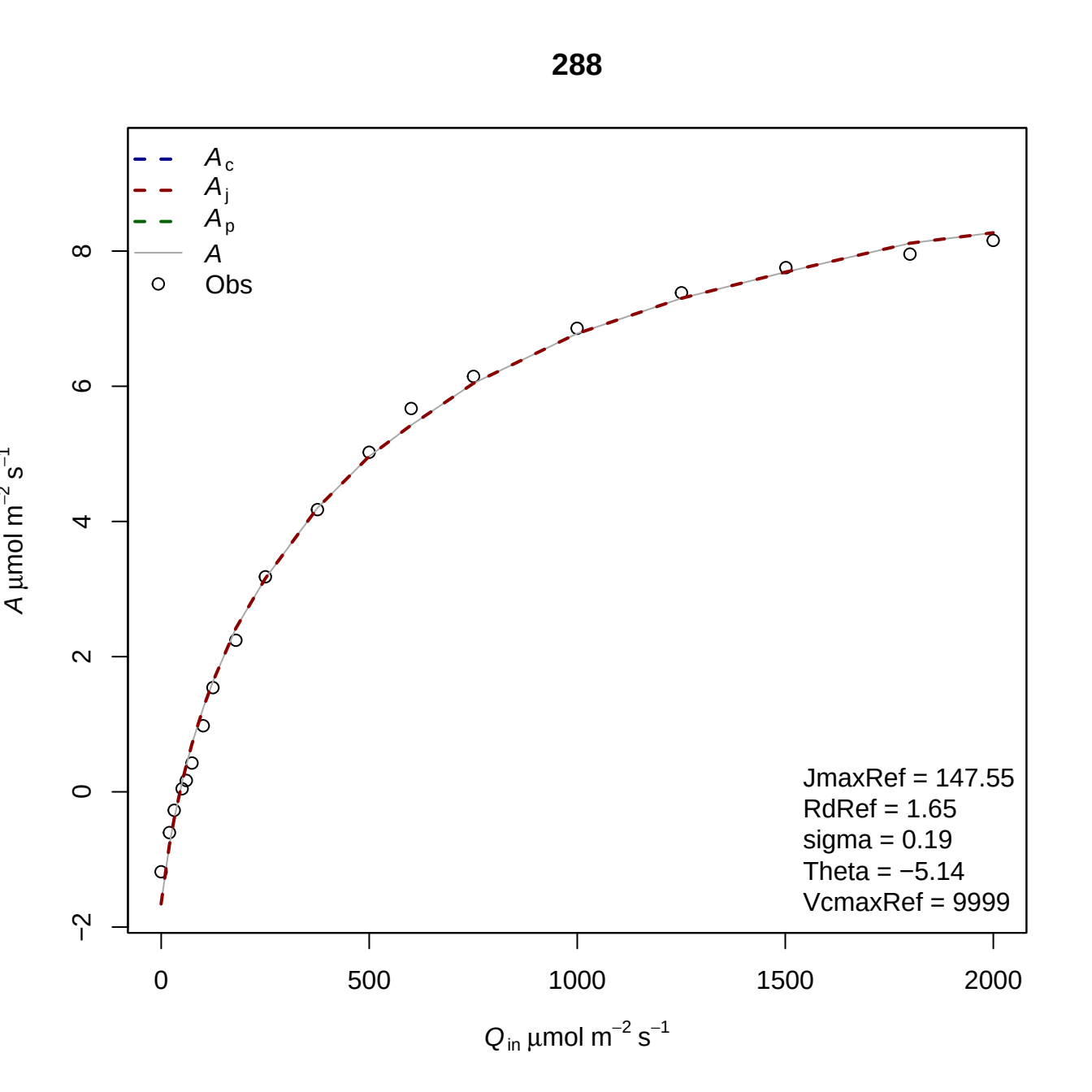

283
A_c
A_j
A_p
A
Obs
JmaxRef = 238.9
RdRef = 0.88
sigma = 0.16
Theta = −31.1
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

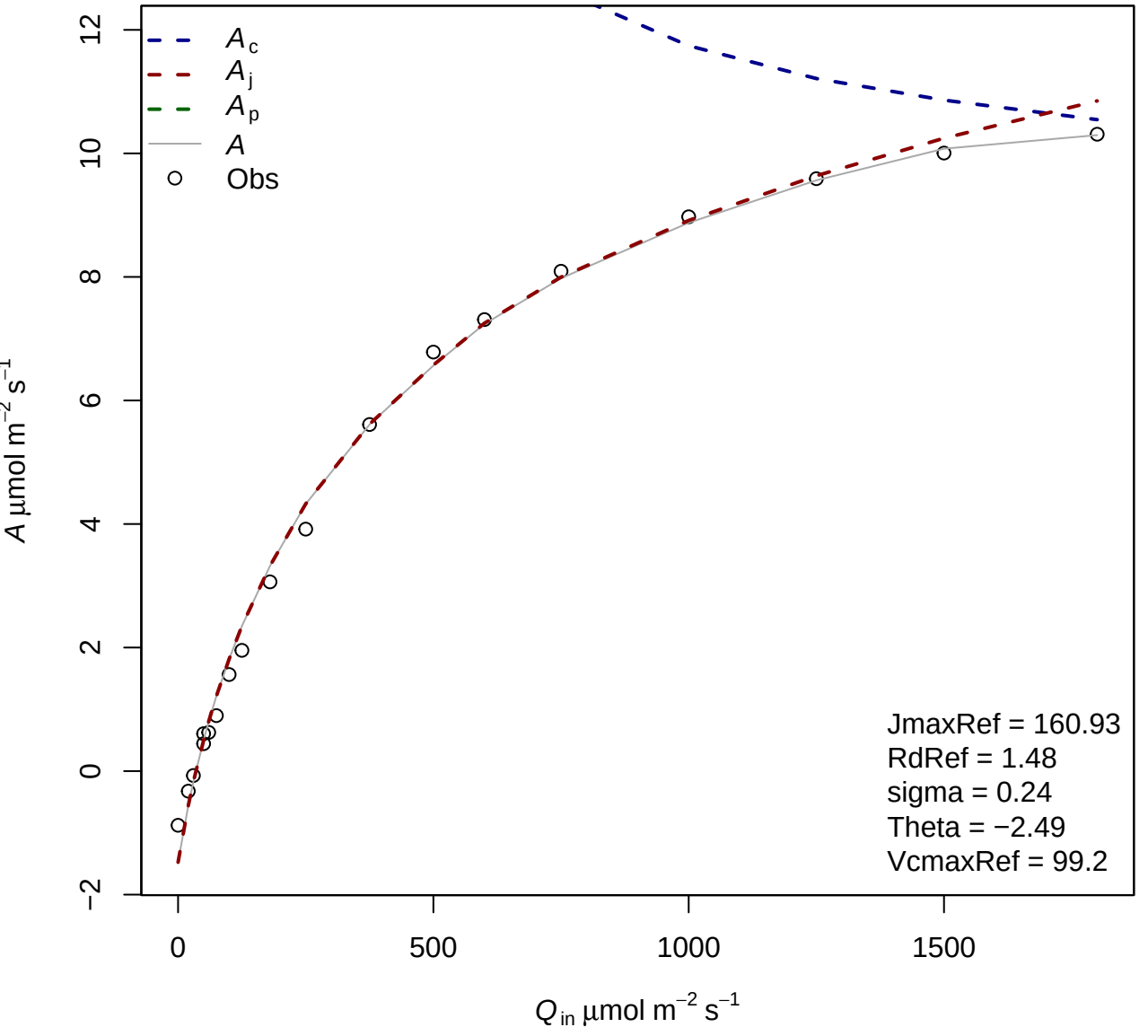

284
A_c
A_j
A_p
A
Obs
JmaxRef = 141.96
RdRef = 1.4
sigma = 0.23
Theta = −2.53
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

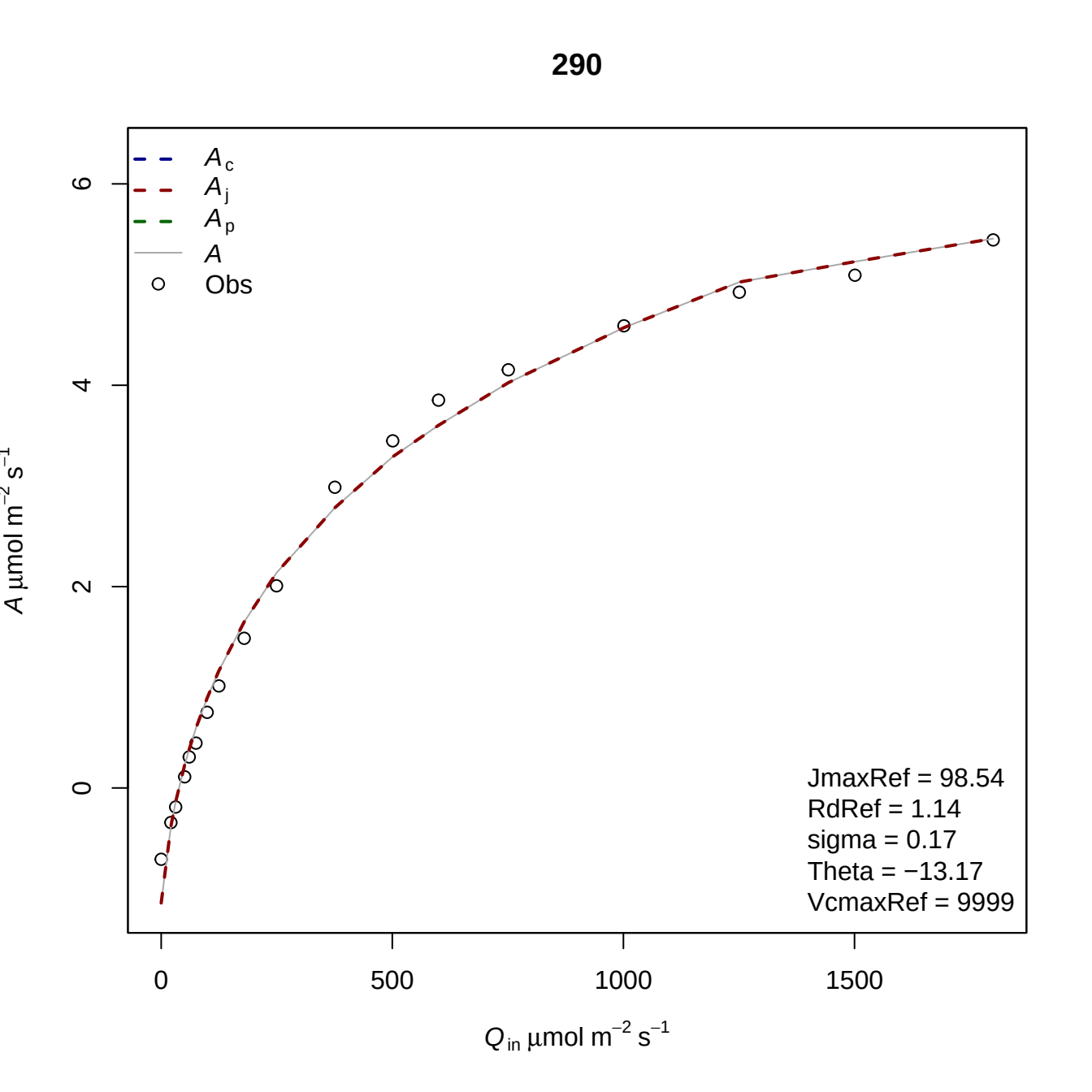

285
A_c
A_j
A_p
A
Obs
JmaxRef = 138.17
RdRef = 1.13
sigma = 0.18
Theta = −13.89
VcmaxRef = 45.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

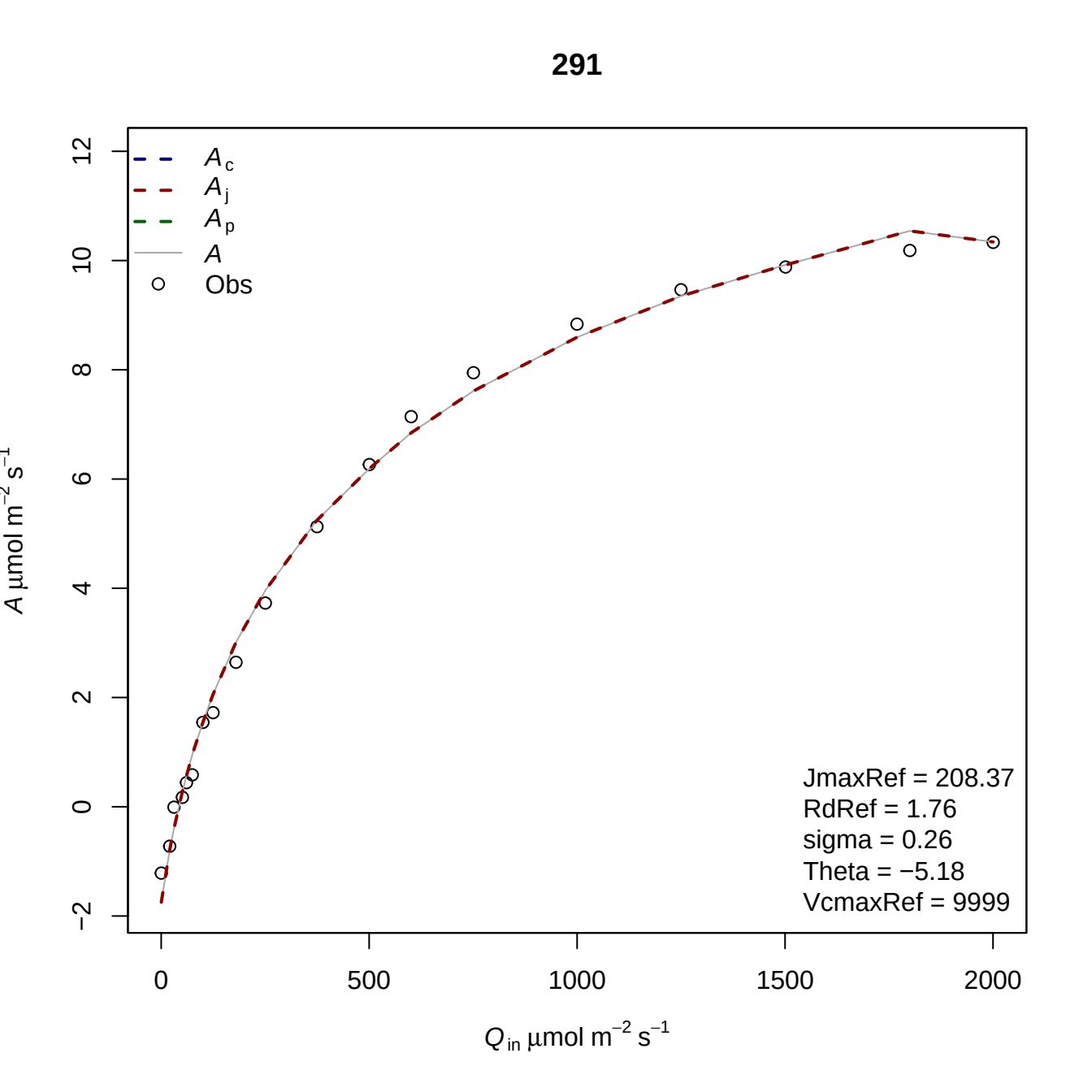

286
A_c
A_j
A_p
A
Obs
JmaxRef = 88.88
RdRef = 1.52
sigma = 0.13
Theta = −5.42
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

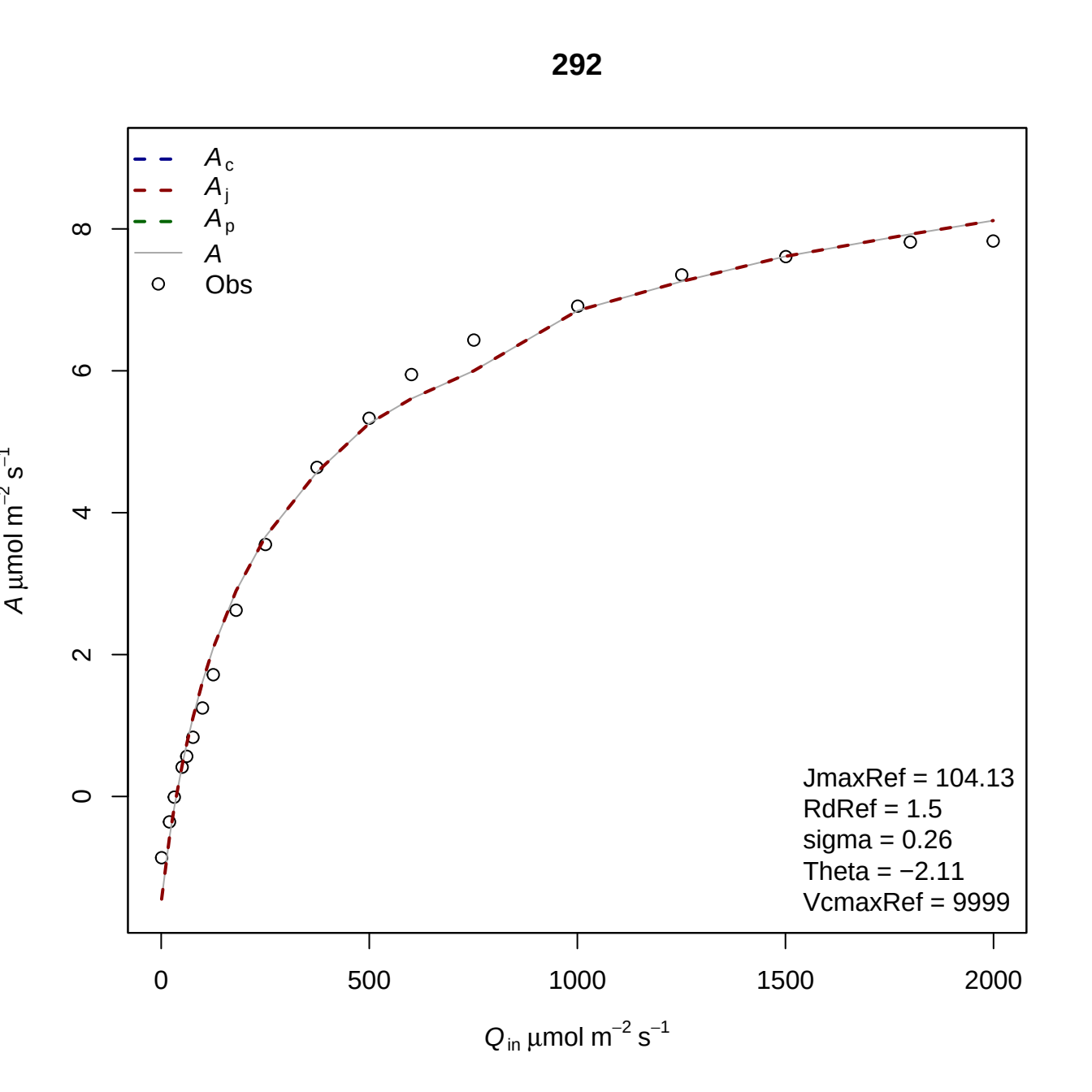

A_c
A_j
A_p
A
Obs
JmaxRef = 1241.17
RdRef = 1.19
sigma = 0.2
Theta = −87.5
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

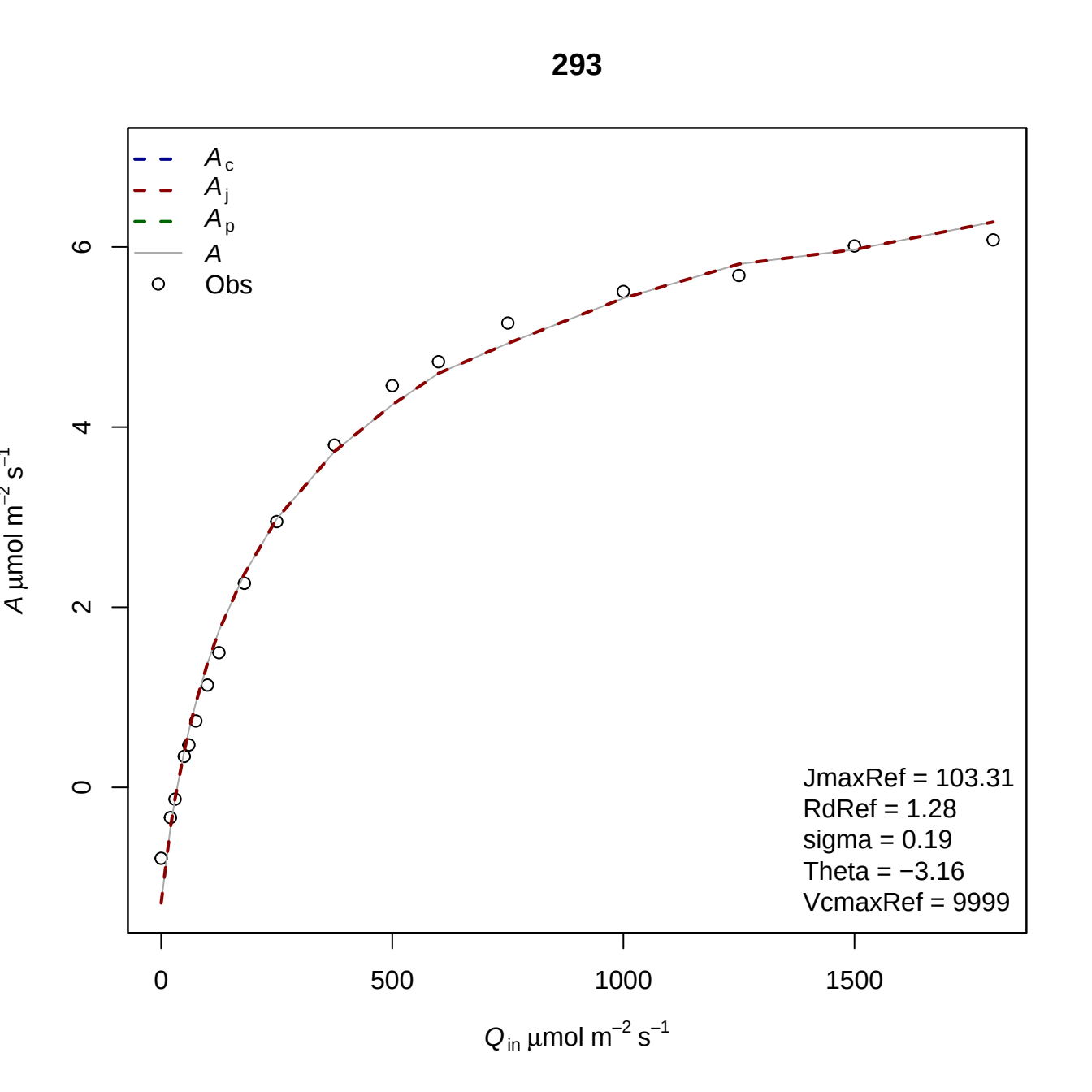

288
A_c
A_j
A_p
A
Obs
JmaxRef = 147.55
RdRef = 1.65
sigma = 0.19
Theta = −5.14
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

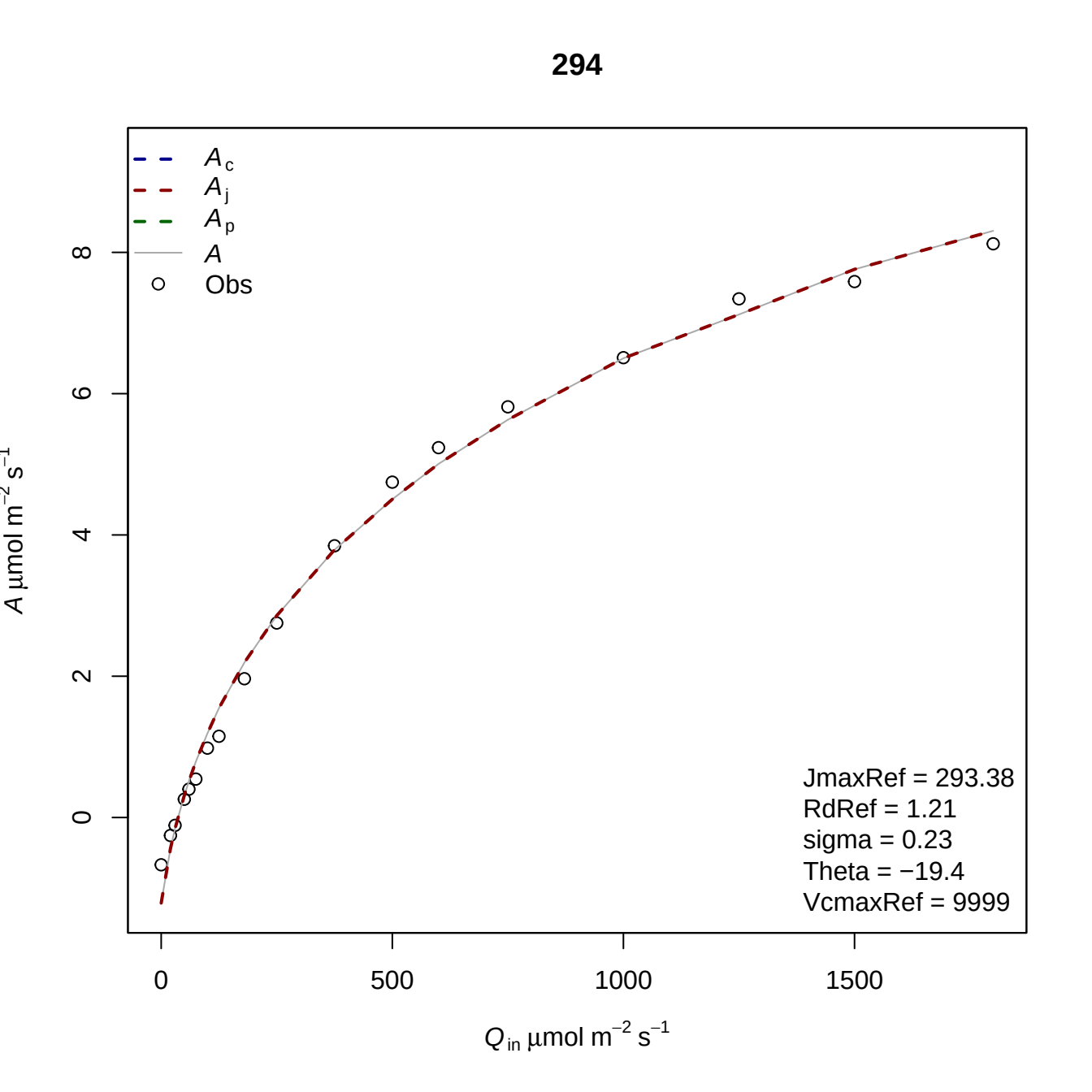

A_c
A_j
A_p
A
Obs
JmaxRef = 160.93
RdRef = 1.48
sigma = 0.24
Theta = −2.49
VcmaxRef = 99.2
12
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

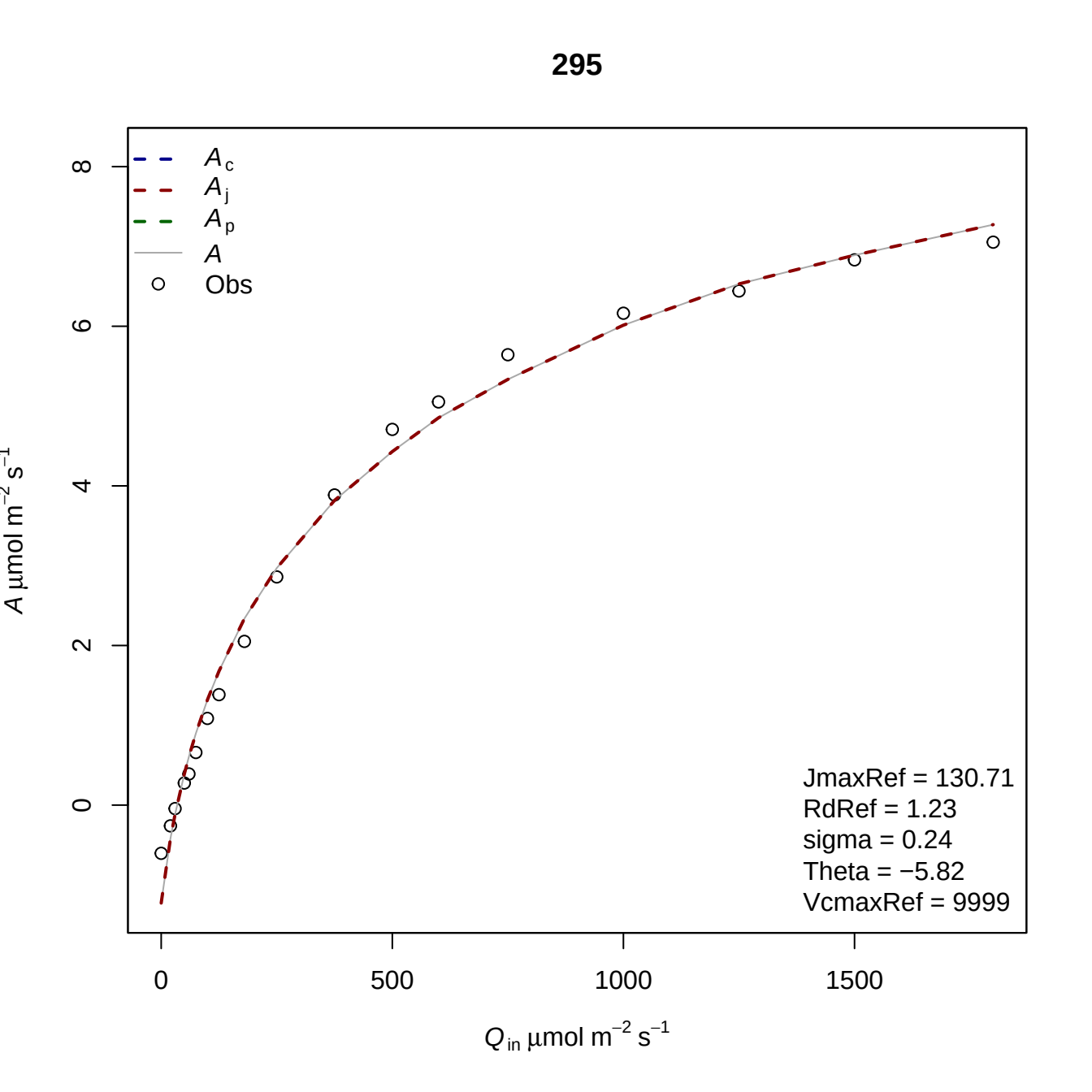

290
A_c
A_j
A_p
A
Obs
JmaxRef = 98.54
RdRef = 1.14
sigma = 0.17
Theta = −13.17
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

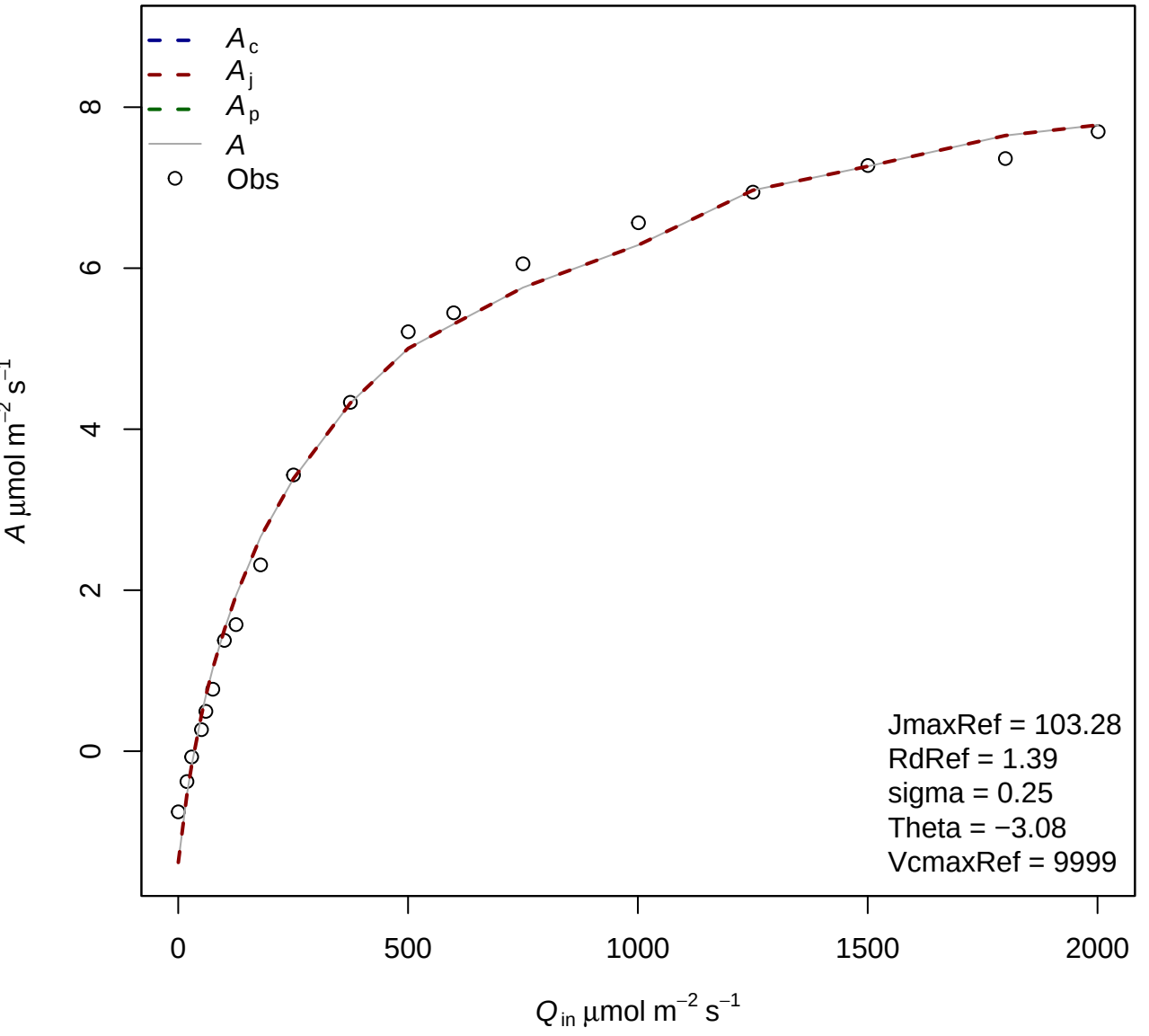

291
A_c
A_j
A_p
A
Obs
JmaxRef = 208.37
RdRef = 1.76
sigma = 0.26
Theta = −5.18
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

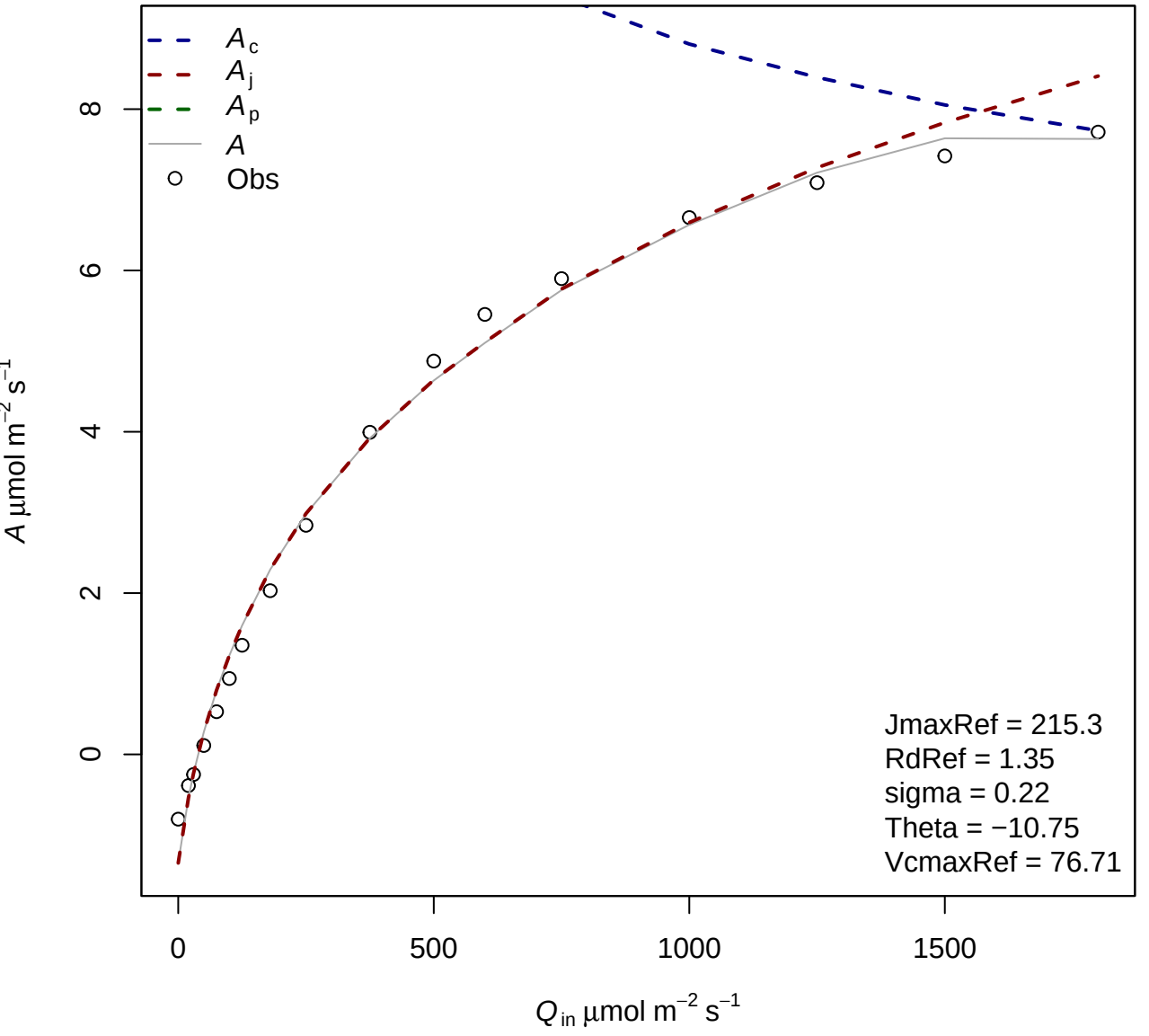

292
A_c
A_j
A_p
A
Obs
JmaxRef = 104.13
RdRef = 1.5
sigma = 0.26
Theta = −2.11
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

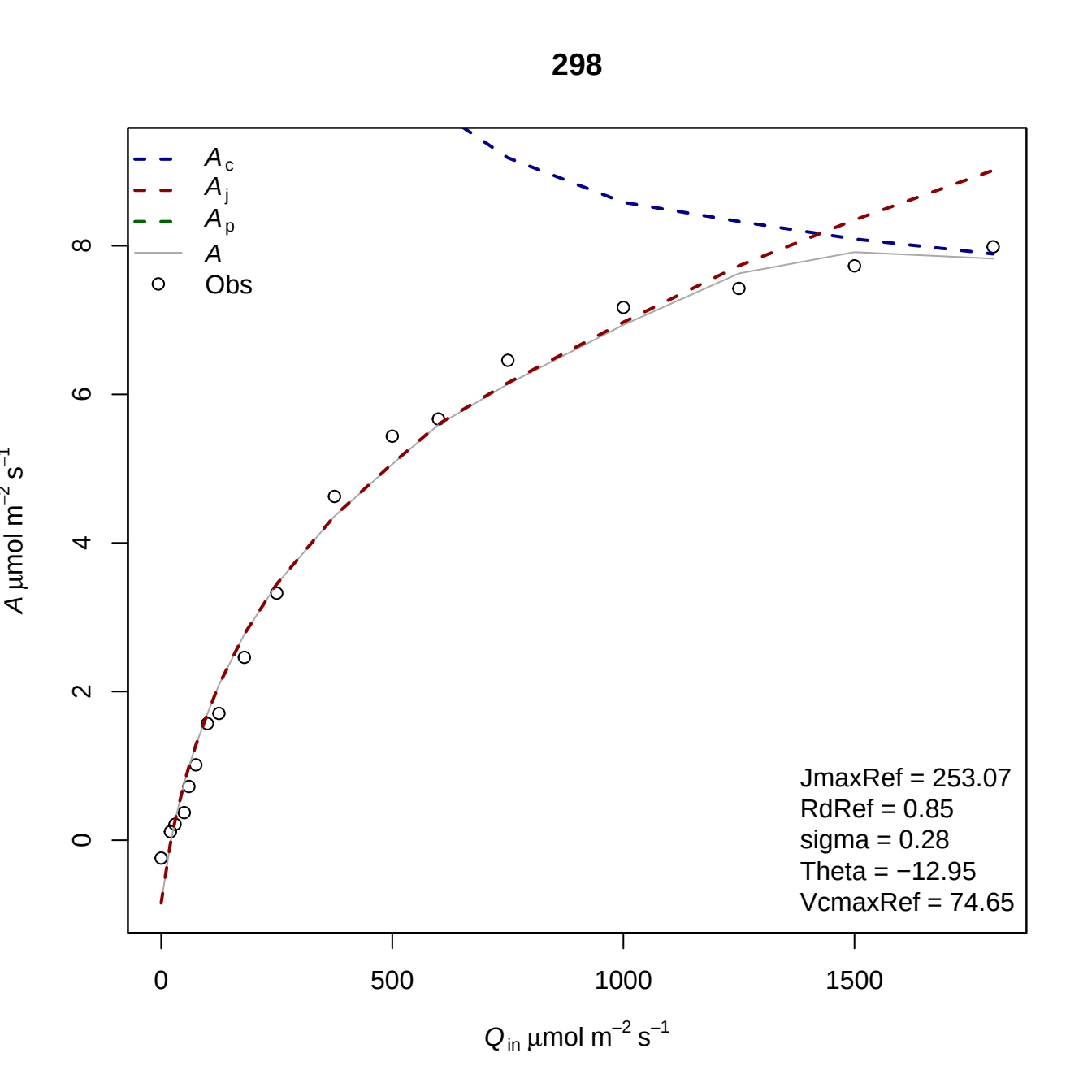

293
A_c
A_j
A_p
A
Obs
JmaxRef = 103.31
RdRef = 1.28
sigma = 0.19
Theta = −3.16
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

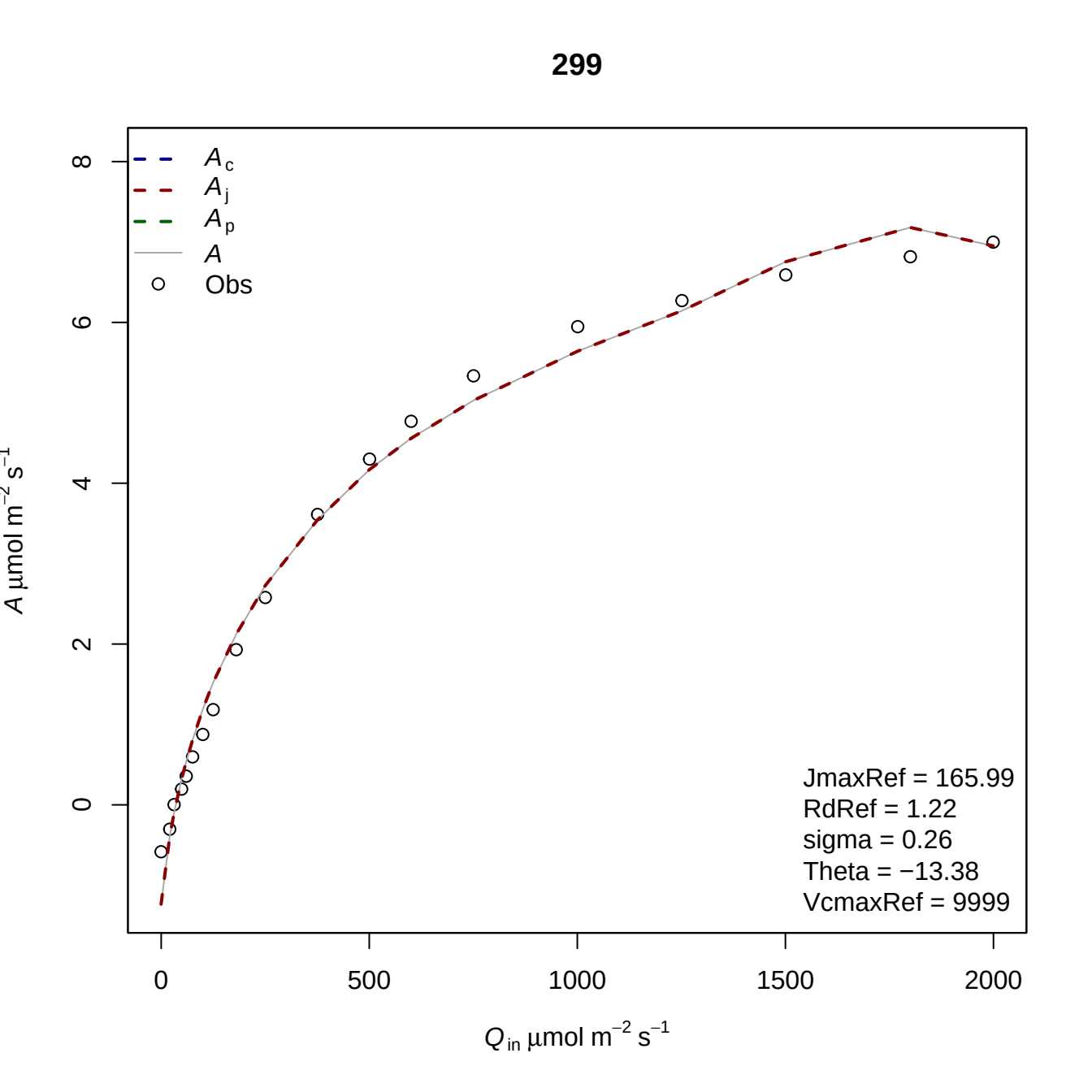

294
A_c
A_j
A_p
A
Obs
JmaxRef = 293.38
RdRef = 1.21
sigma = 0.23
Theta = −19.4
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

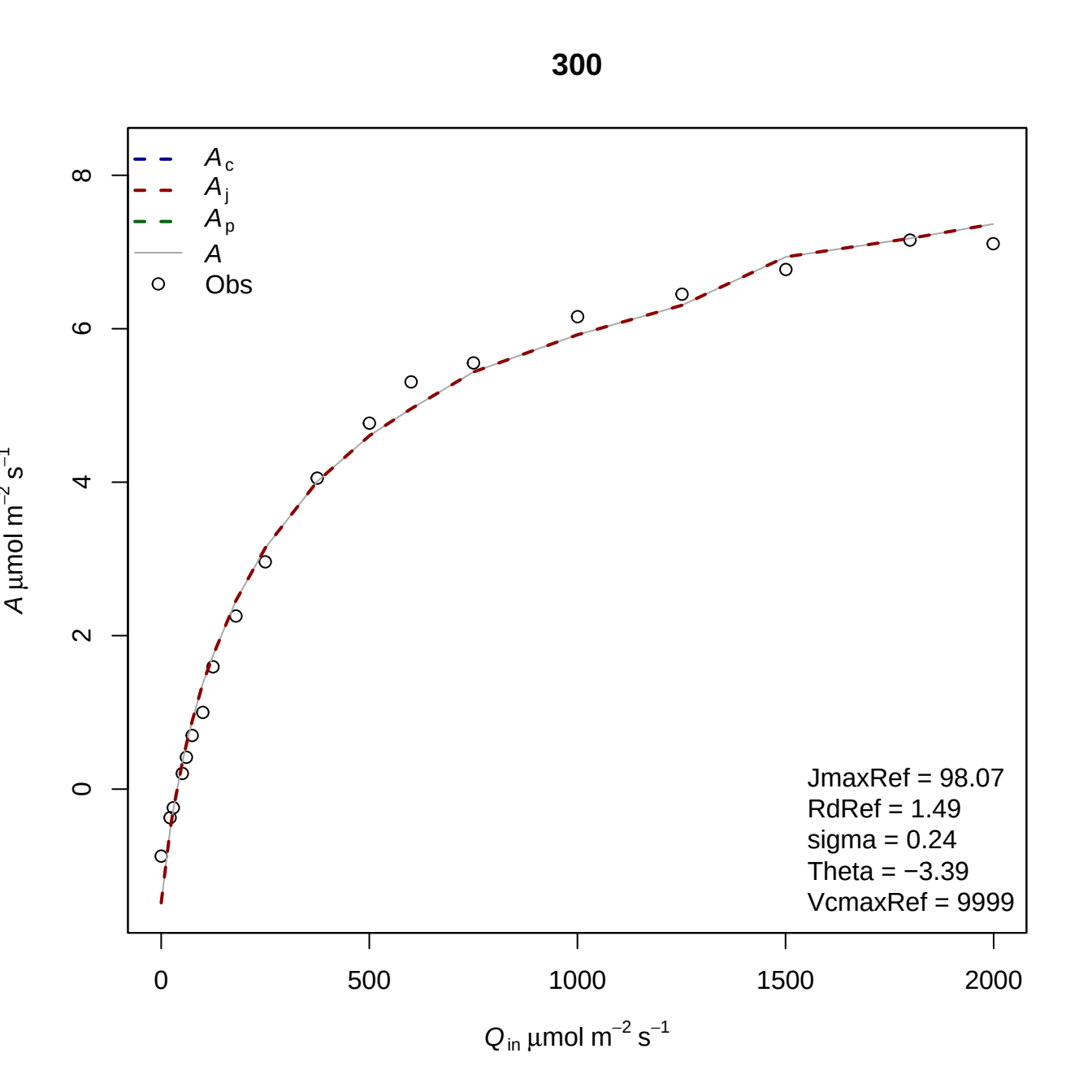

295
A_c
A_j
A_p
A
Obs
JmaxRef = 130.71
RdRef = 1.23
sigma = 0.24
Theta = −5.82
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

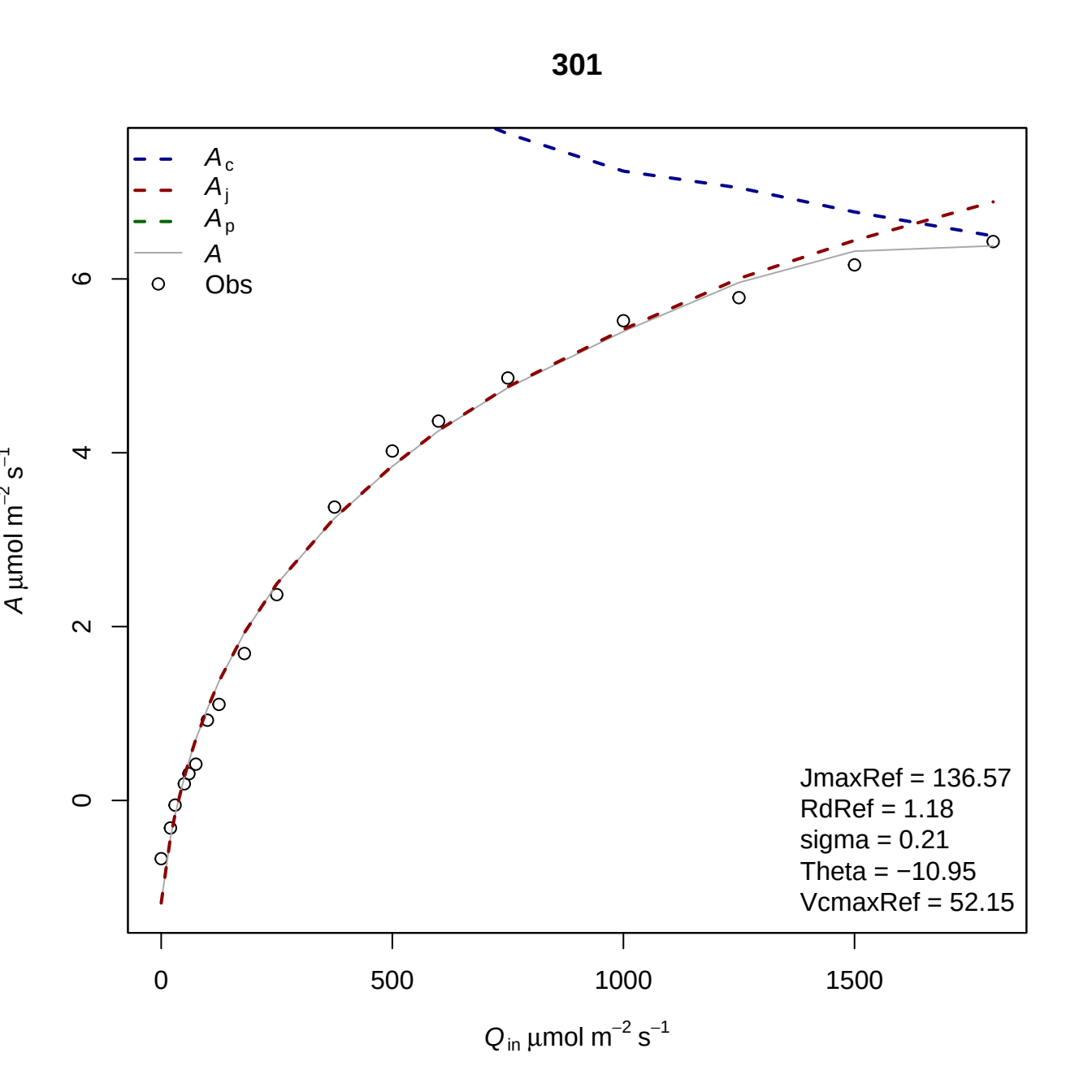

A_c
A_j
A_p
A
Obs
JmaxRef = 103.28
RdRef = 1.39
sigma = 0.25
Theta = −3.08
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

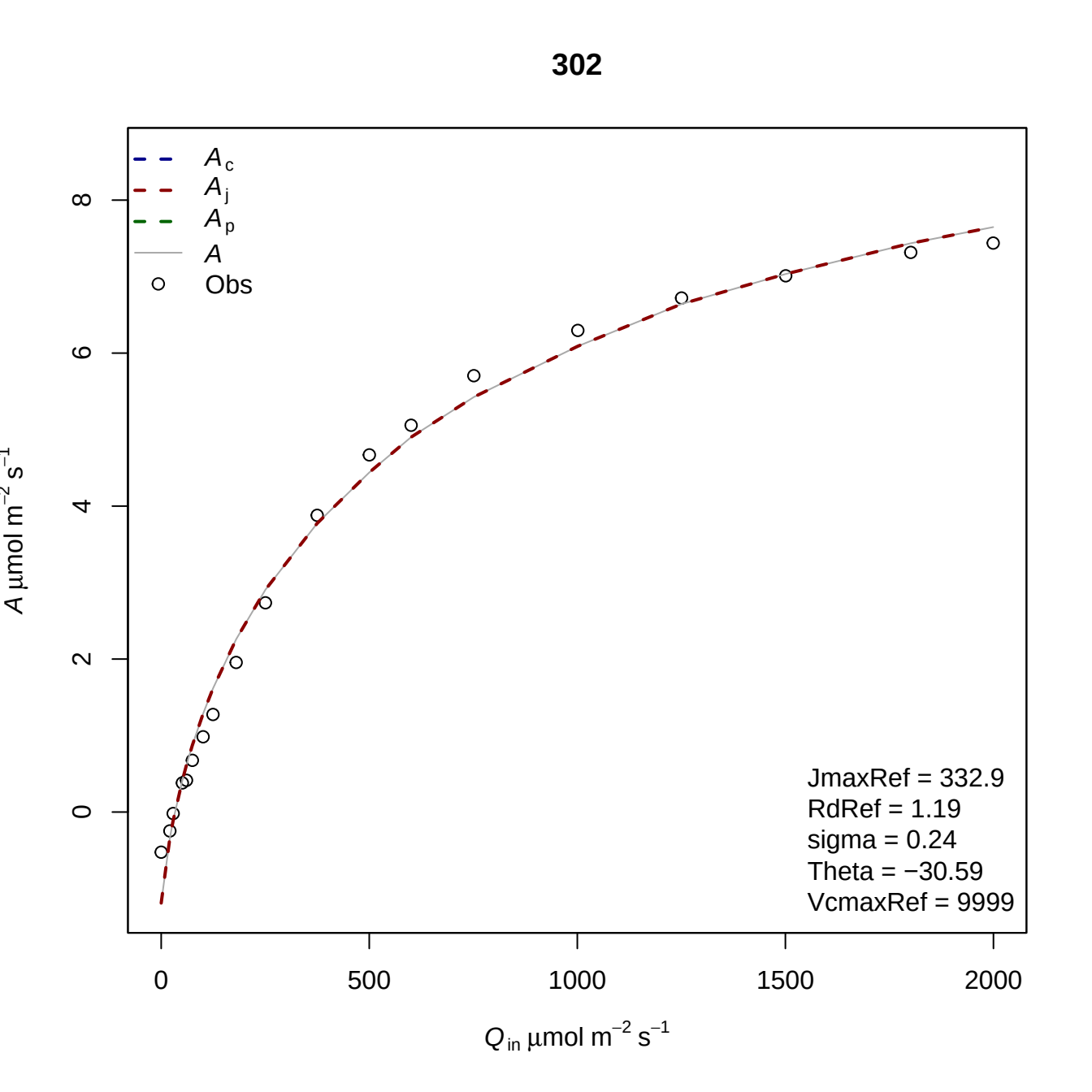

A_c
A_j
A_p
A
Obs
JmaxRef = 215.3
RdRef = 1.35
sigma = 0.22
Theta = −10.75
VcmaxRef = 76.71
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

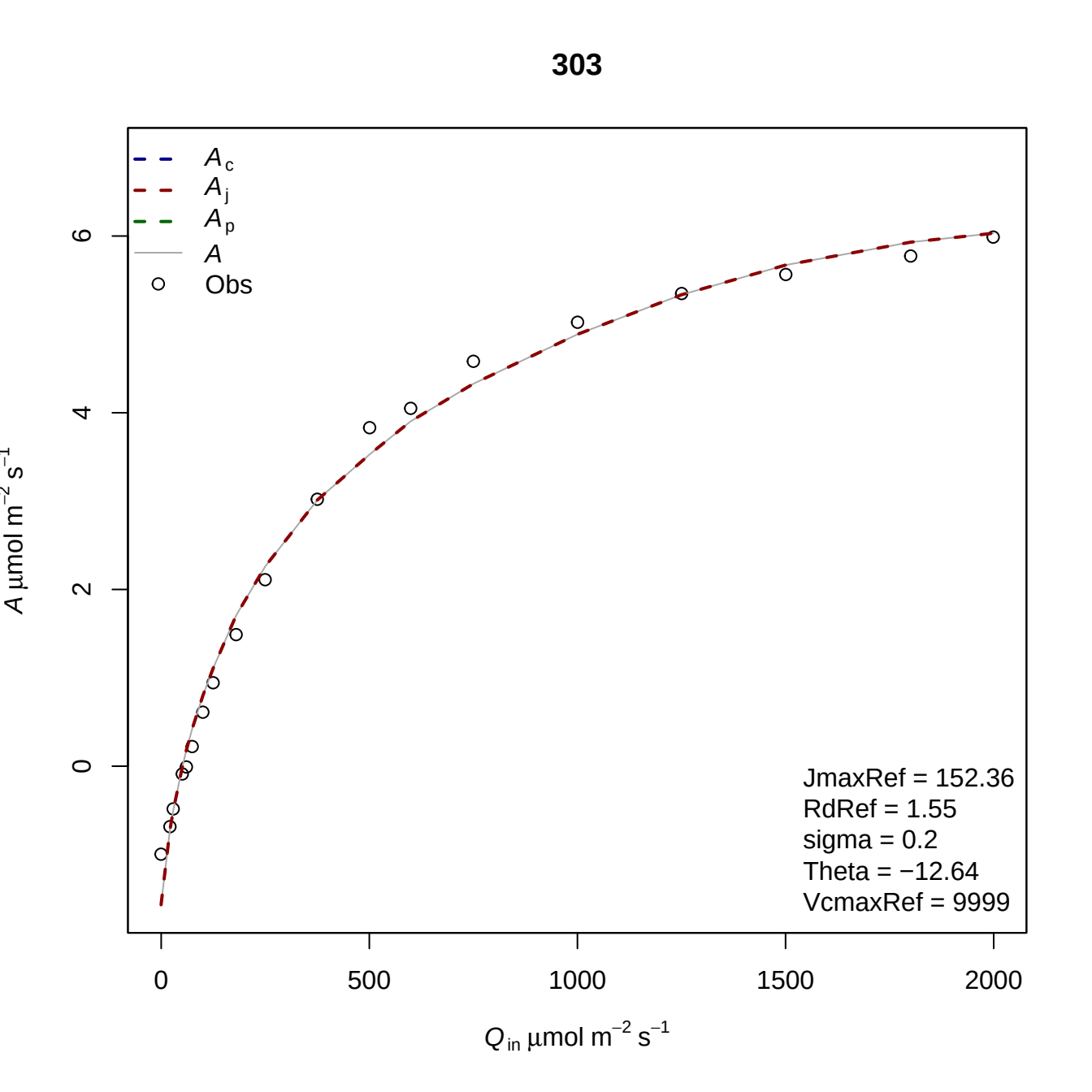

298
A_c
A_j
A_p
A
Obs
JmaxRef = 253.07
RdRef = 0.85
sigma = 0.28
Theta = −12.95
VcmaxRef = 74.65
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

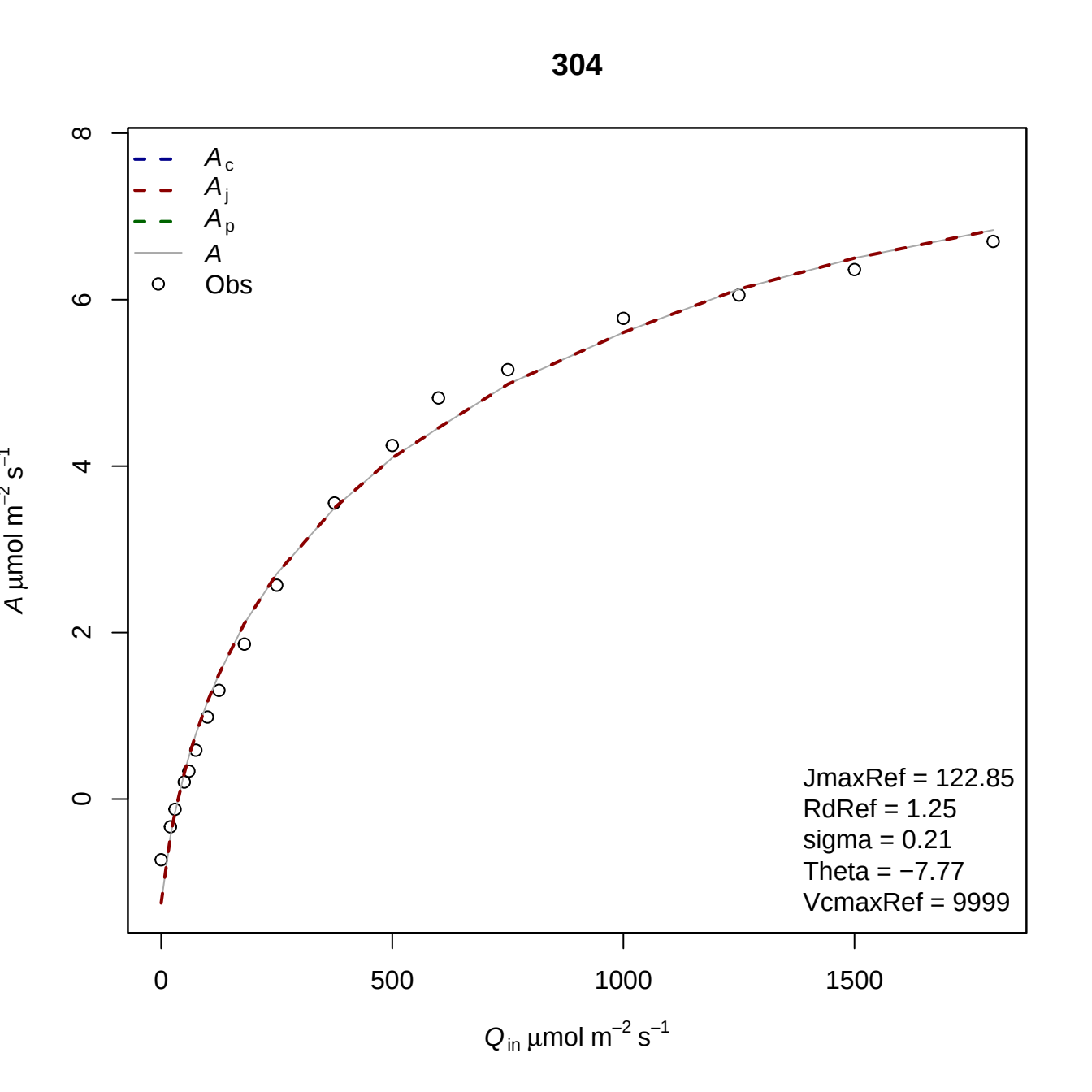

299
A_c
A_j
A_p
A
Obs
JmaxRef = 165.99
RdRef = 1.22
sigma = 0.26
Theta = −13.38
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

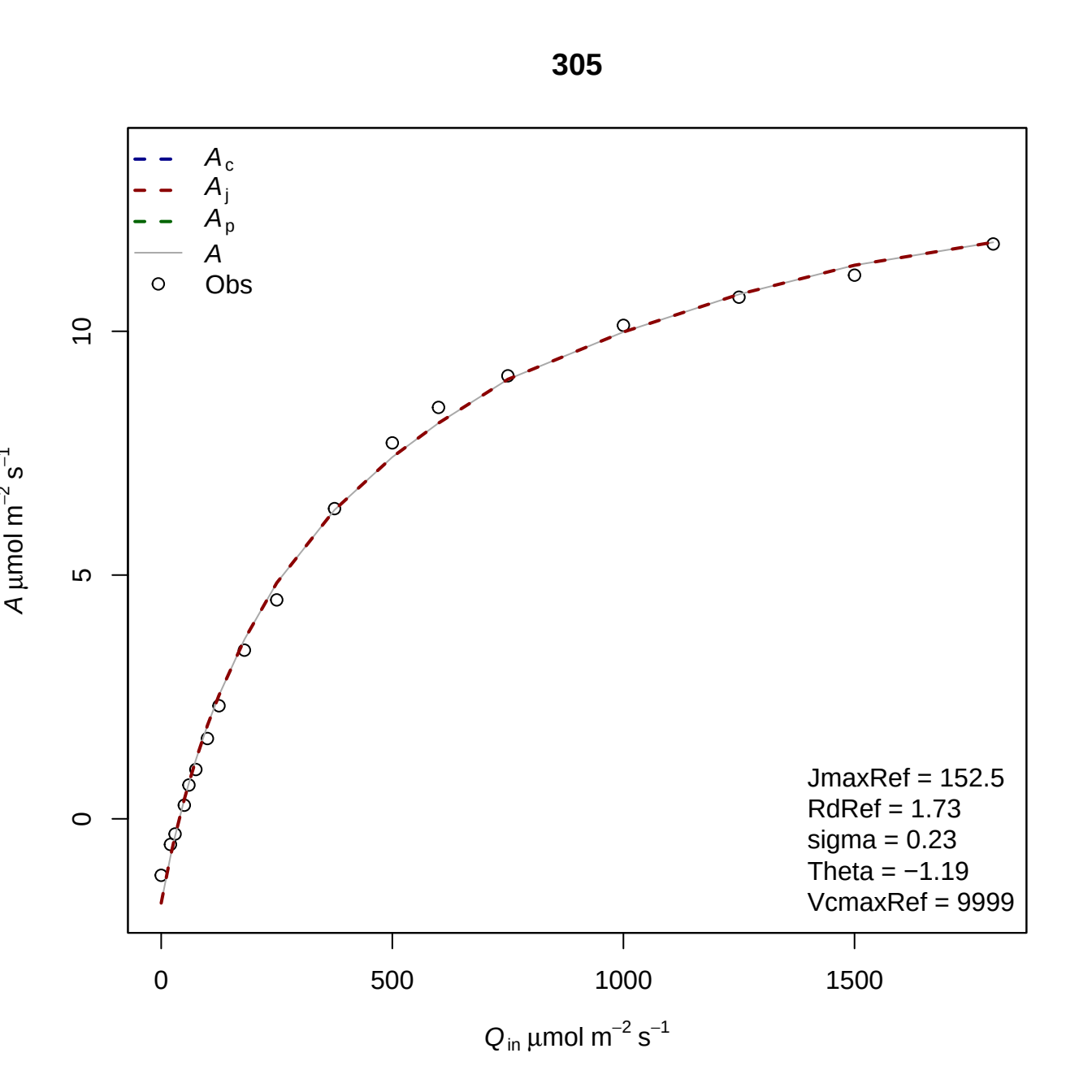

300
A_c
A_j
A_p
A
Obs
JmaxRef = 98.07
RdRef = 1.49
sigma = 0.24
Theta = −3.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

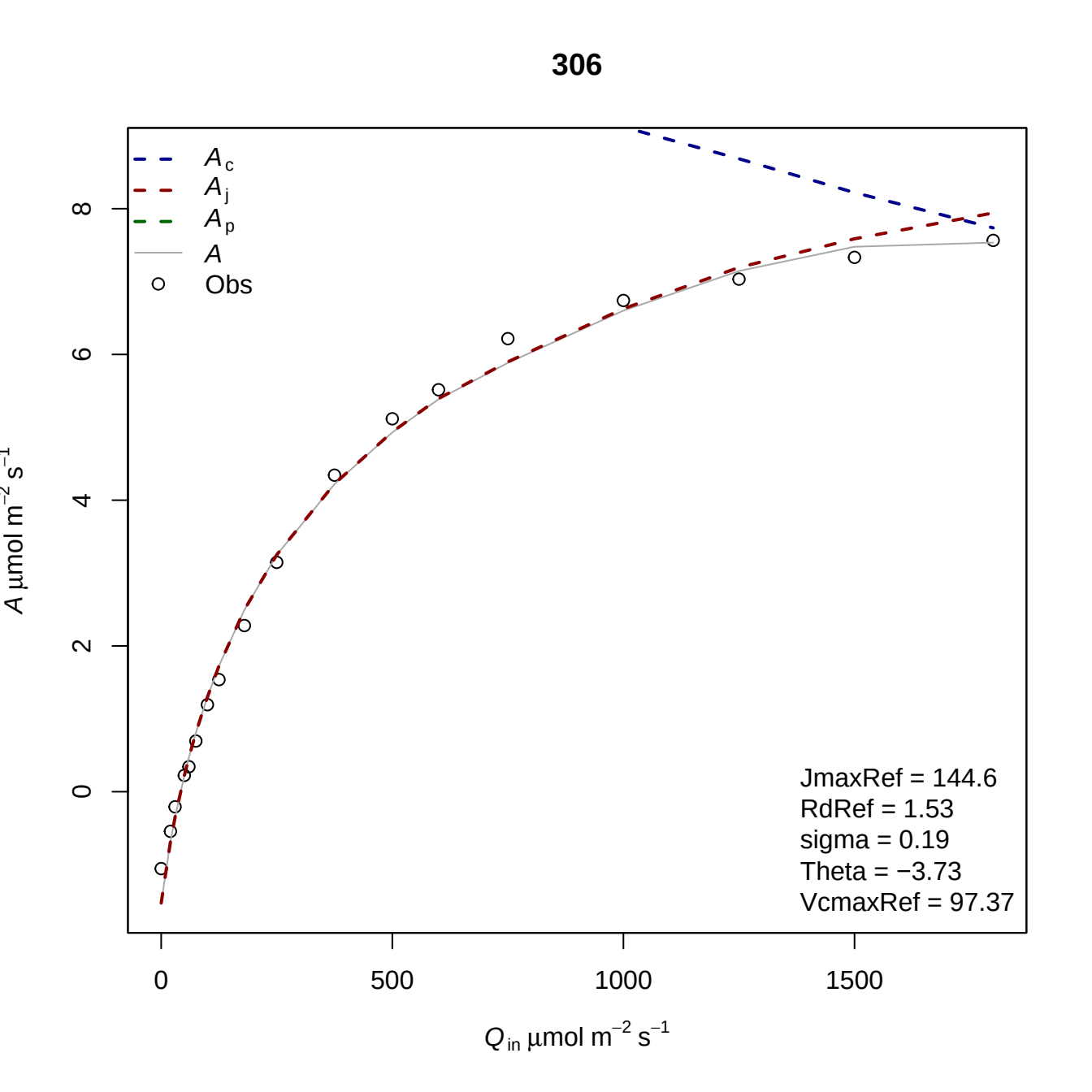

301
A_c
A_j
A_p
A
Obs
JmaxRef = 136.57
RdRef = 1.18
sigma = 0.21
Theta = −10.95
VcmaxRef = 52.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

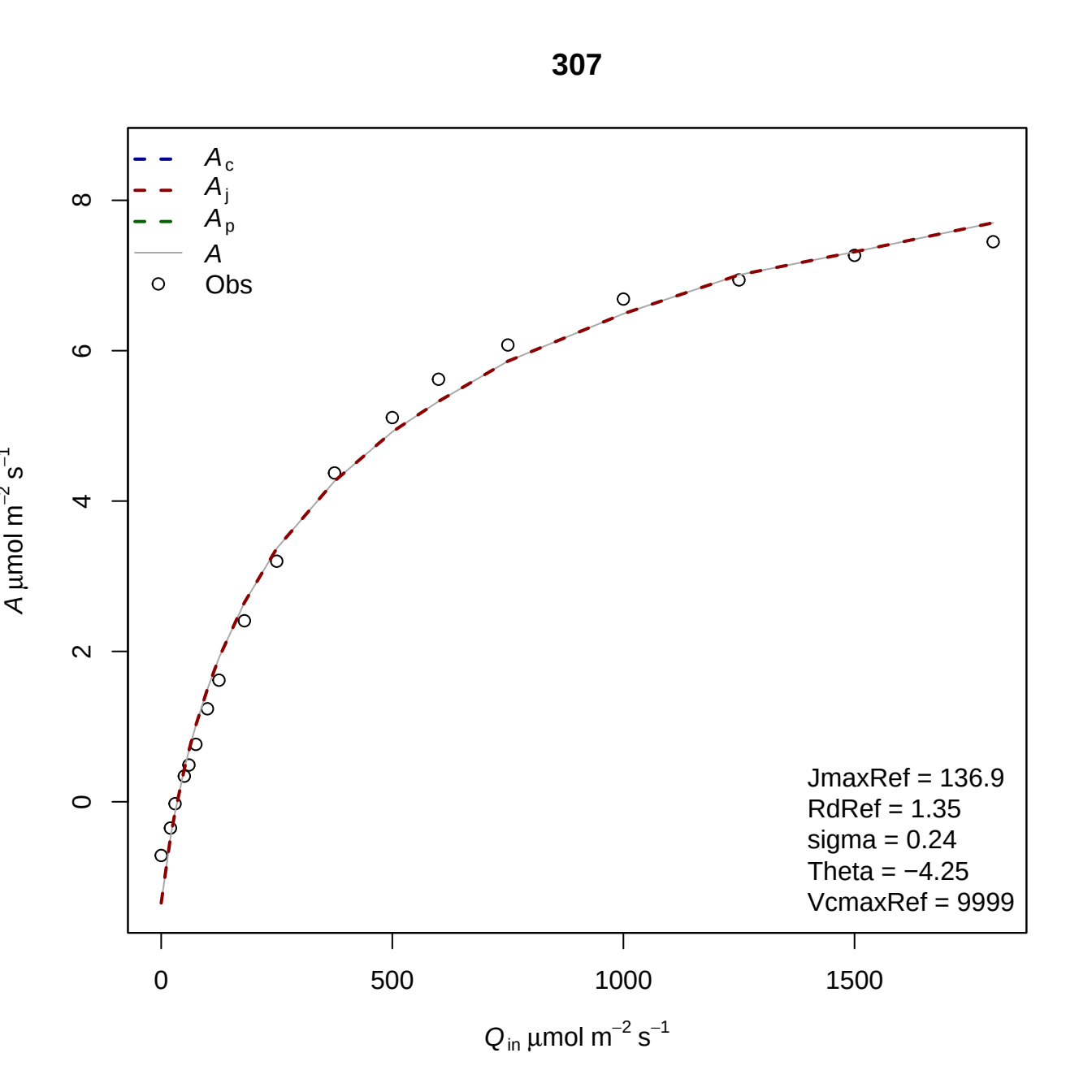

302
A_c
A_j
A_p
A
Obs
JmaxRef = 332.9
RdRef = 1.19
sigma = 0.24
Theta = −30.59
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

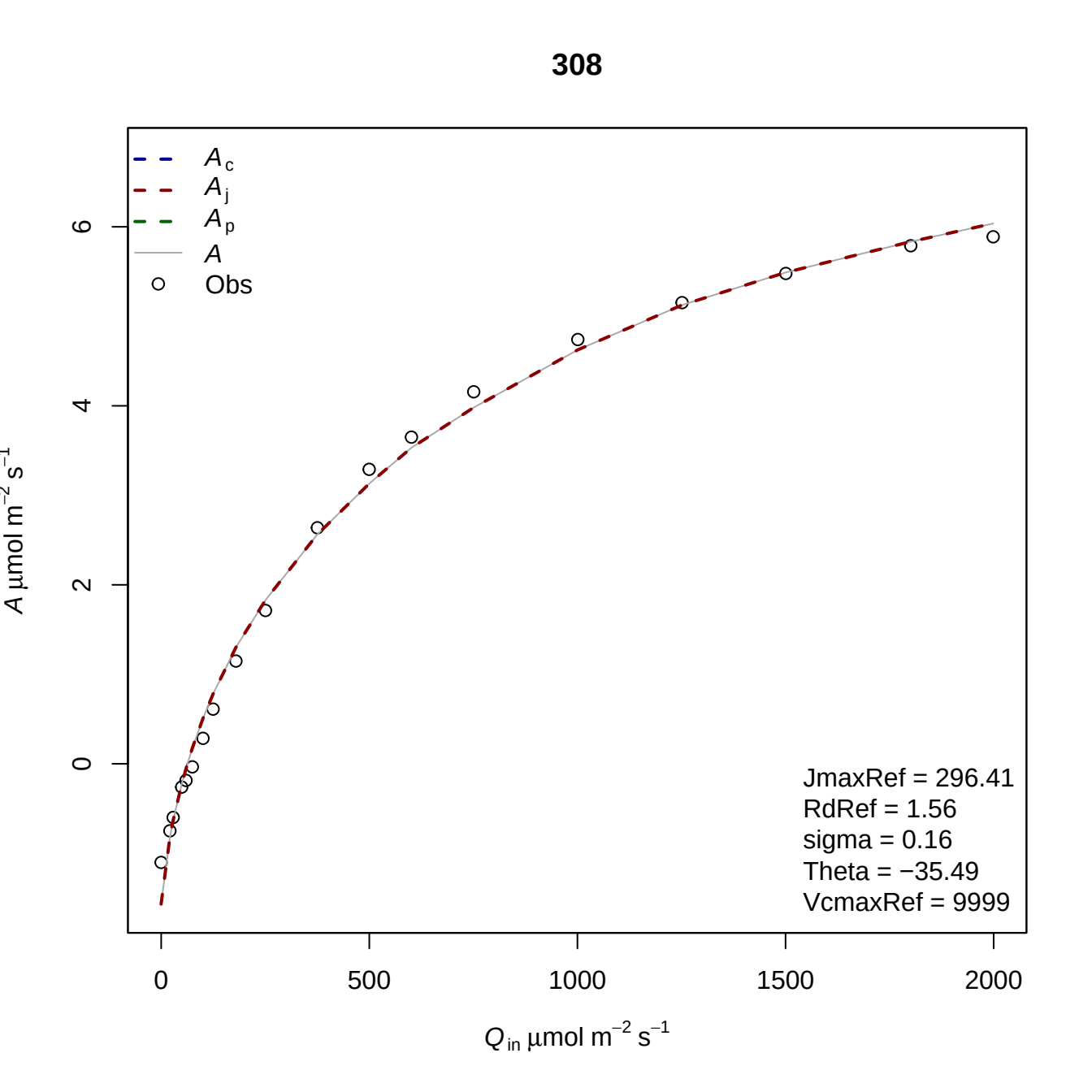

303
A_c
A_j
A_p
A
Obs
JmaxRef = 152.36
RdRef = 1.55
sigma = 0.2
Theta = −12.64
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

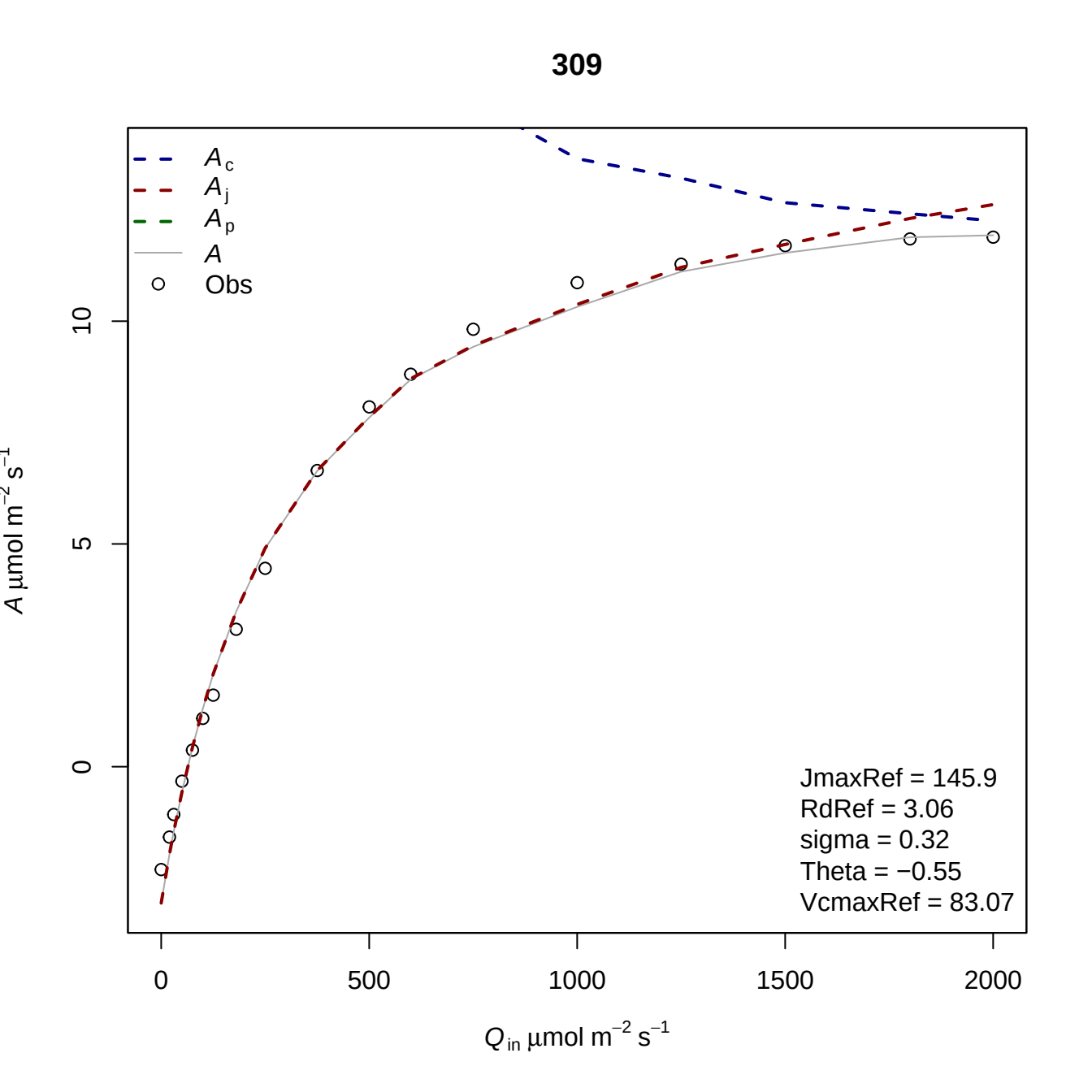

304
A_c
A_j
A_p
A
Obs
JmaxRef = 122.85
RdRef = 1.25
sigma = 0.21
Theta = −7.77
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

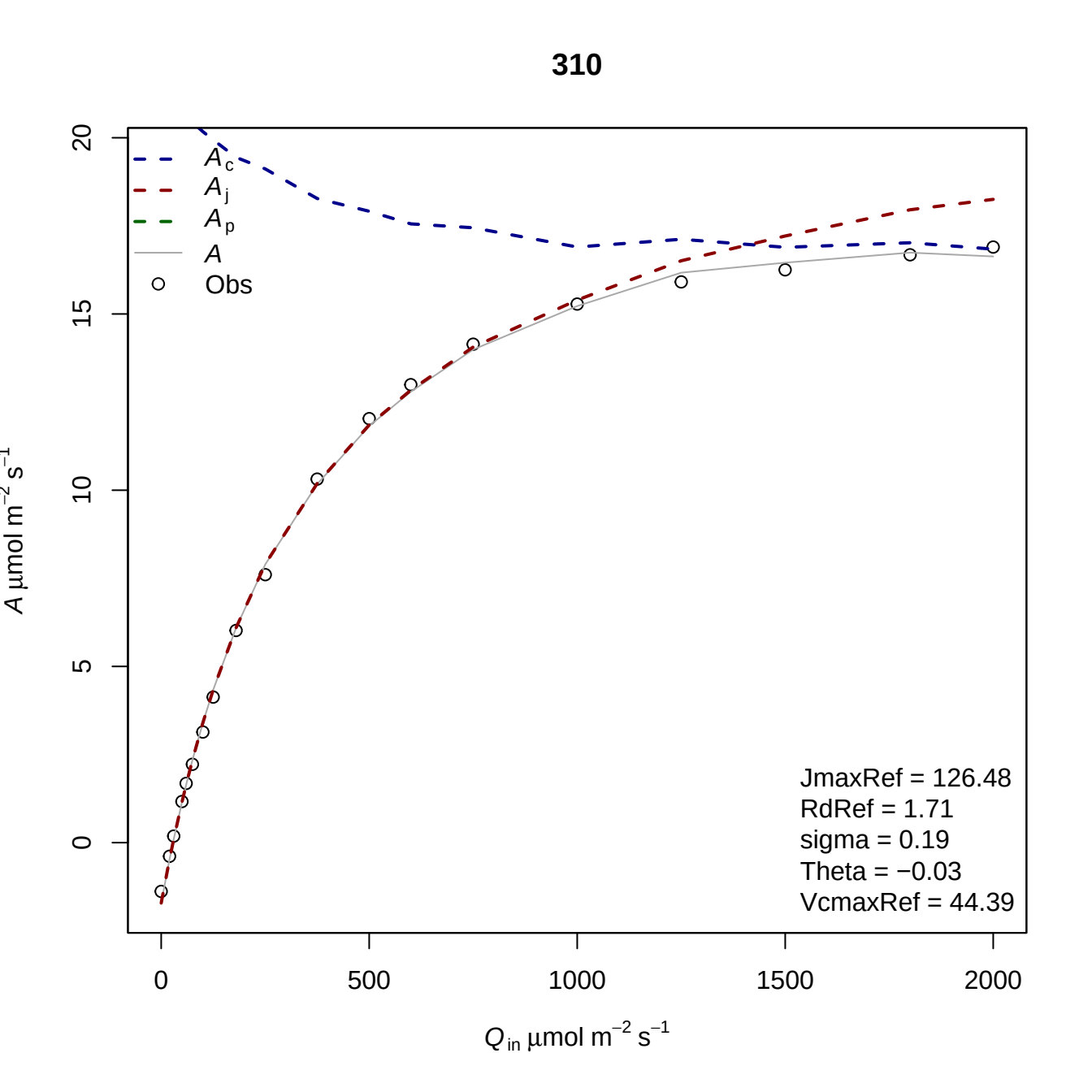

305
A_c
A_j
A_p
A
Obs
JmaxRef = 152.5
RdRef = 1.73
sigma = 0.23
Theta = −1.19
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

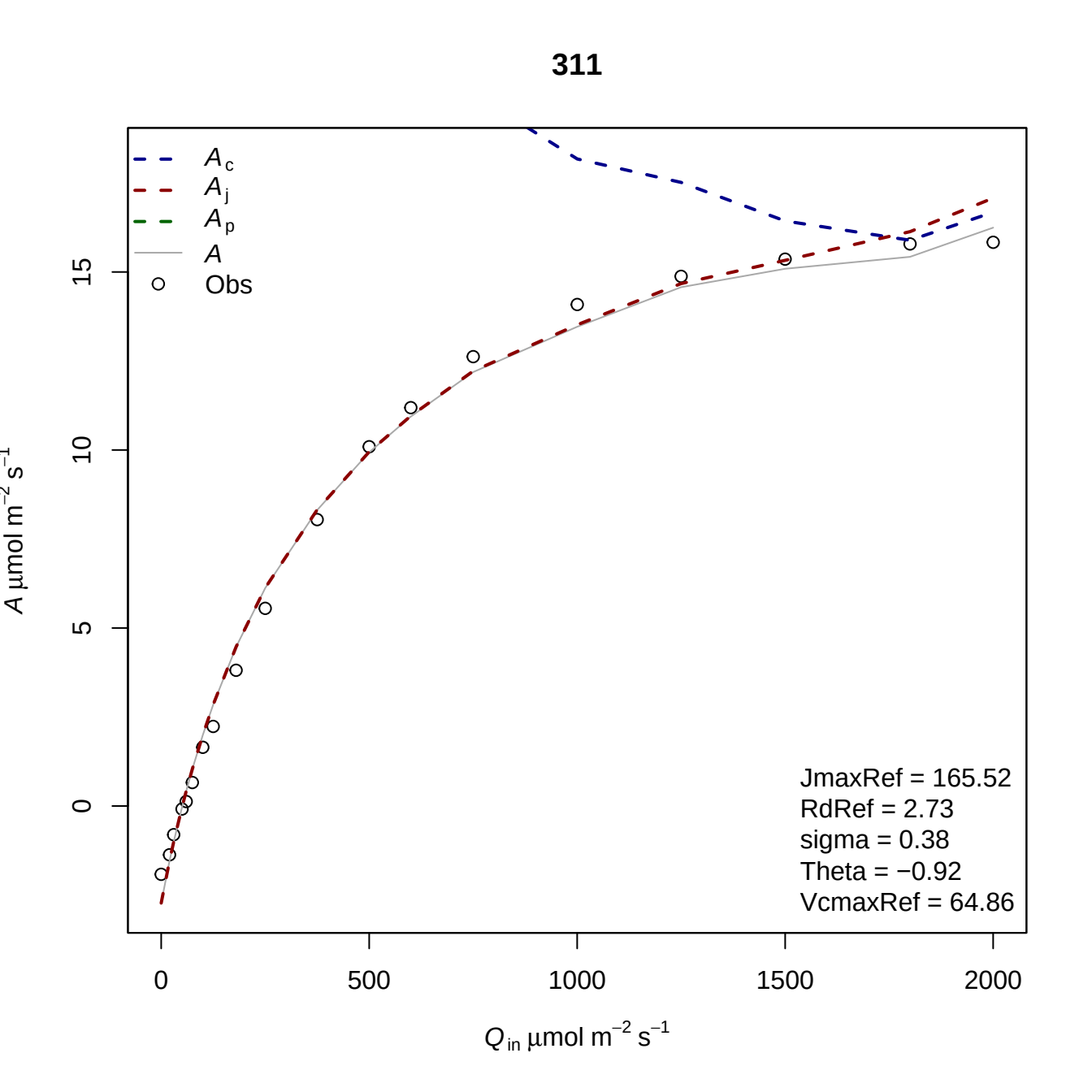

306
A_c
A_j
A_p
A
Obs
JmaxRef = 144.6
RdRef = 1.53
sigma = 0.19
Theta = −3.73
VcmaxRef = 97.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

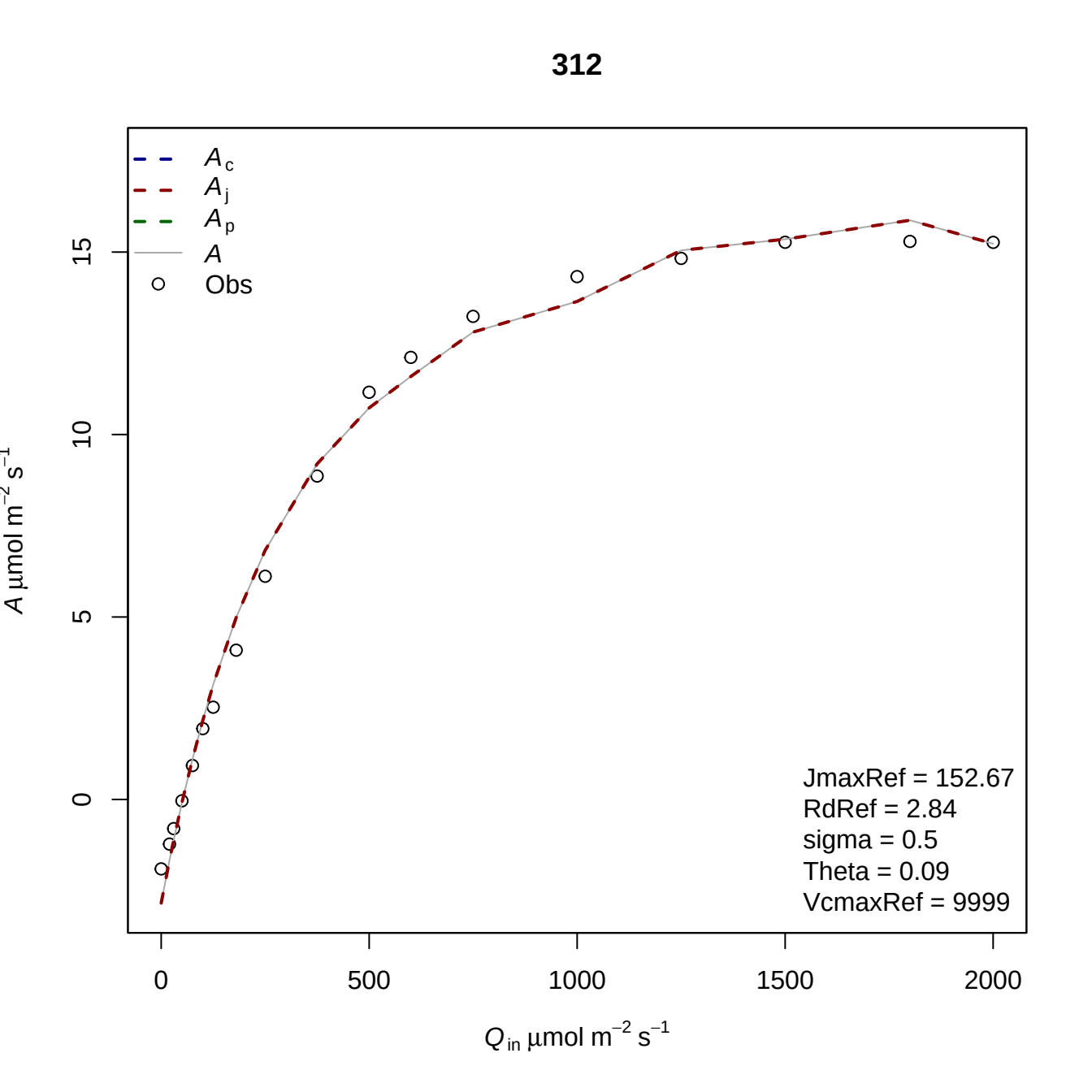

307
A_c
A_j
A_p
A
Obs
JmaxRef = 136.9
RdRef = 1.35
sigma = 0.24
Theta = −4.25
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

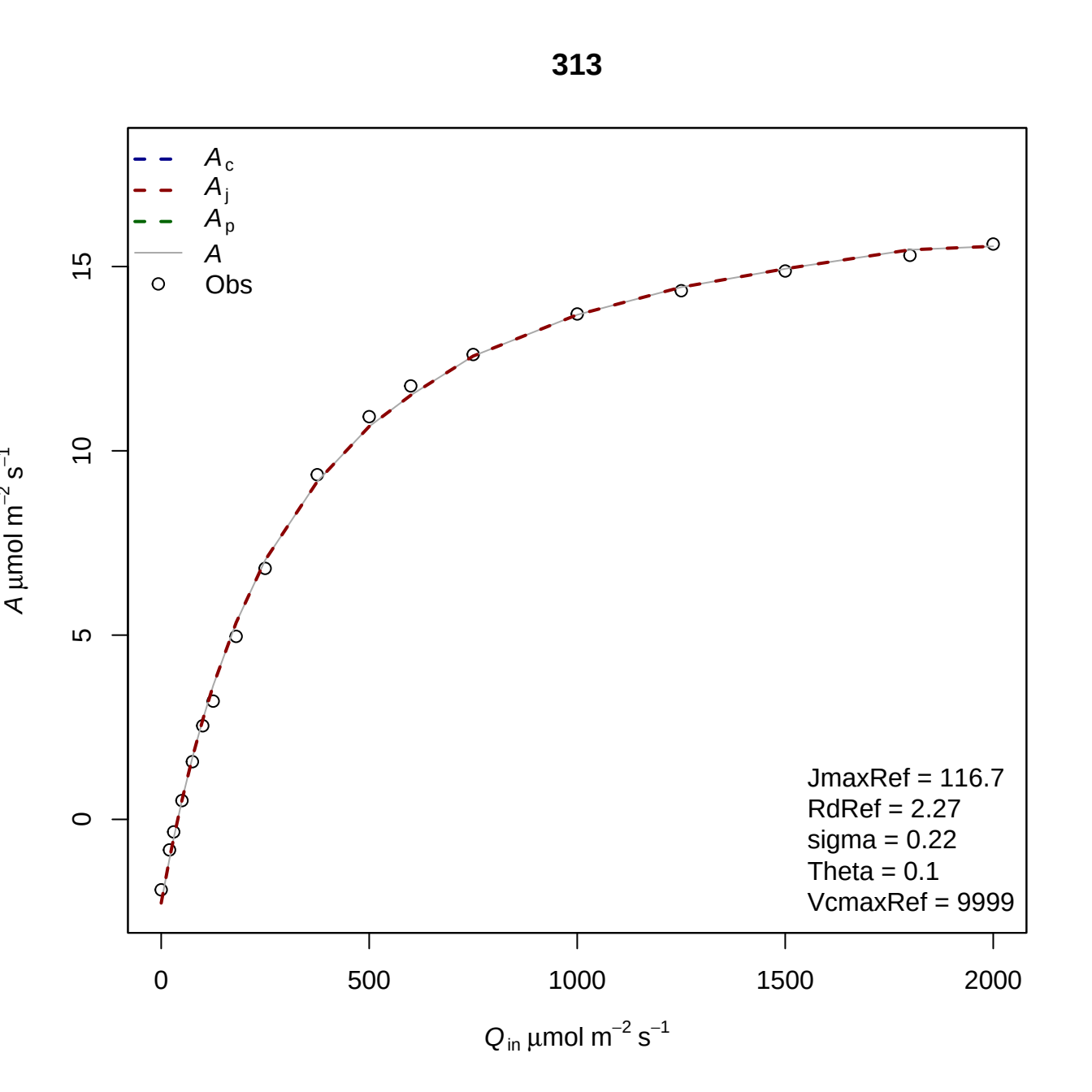

308
A_c
A_j
A_p
A
Obs
JmaxRef = 296.41
RdRef = 1.56
sigma = 0.16
Theta = −35.49
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

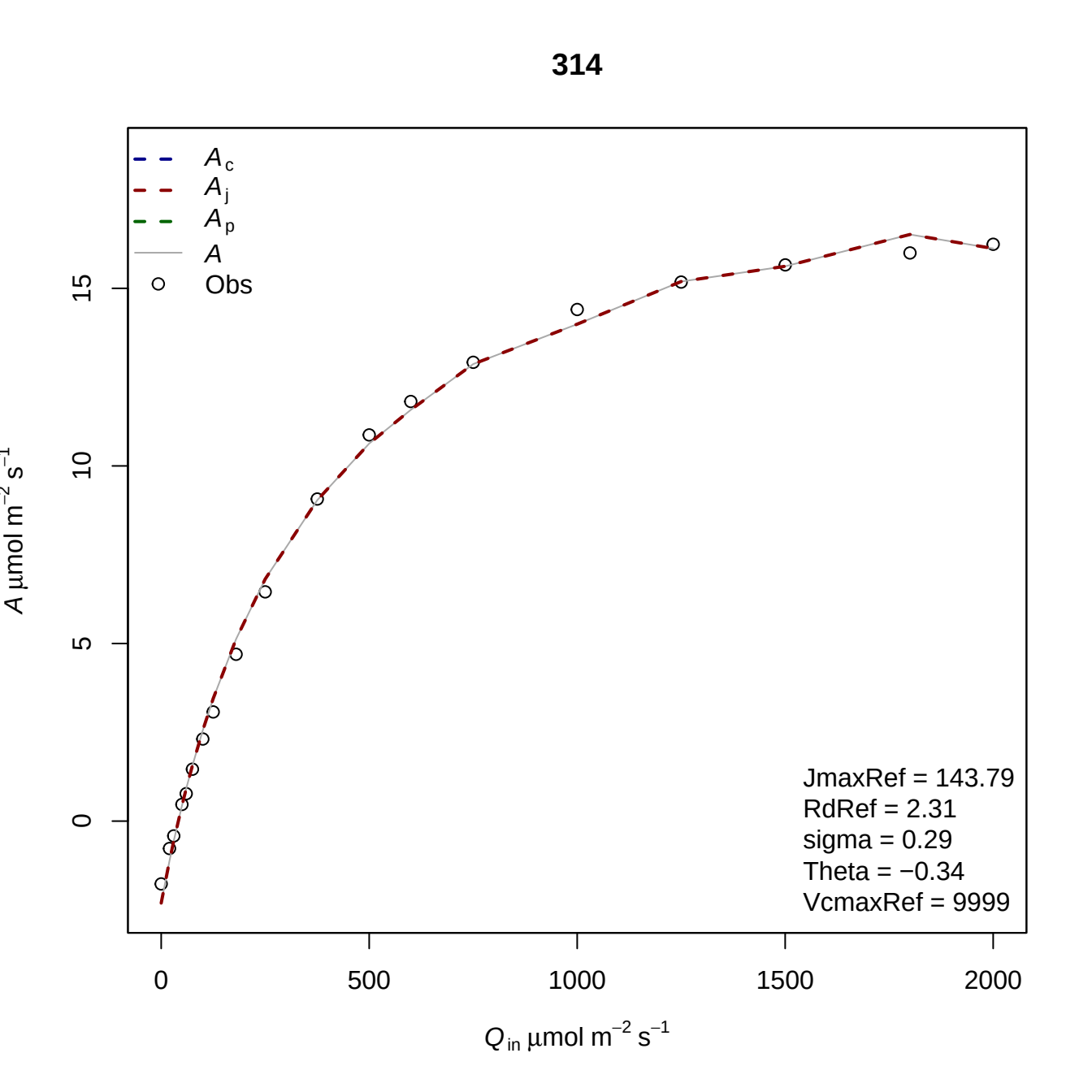

309
A_c
A_j
A_p
A
Obs
JmaxRef = 145.9
RdRef = 3.06
sigma = 0.32
Theta = −0.55
VcmaxRef = 83.07
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

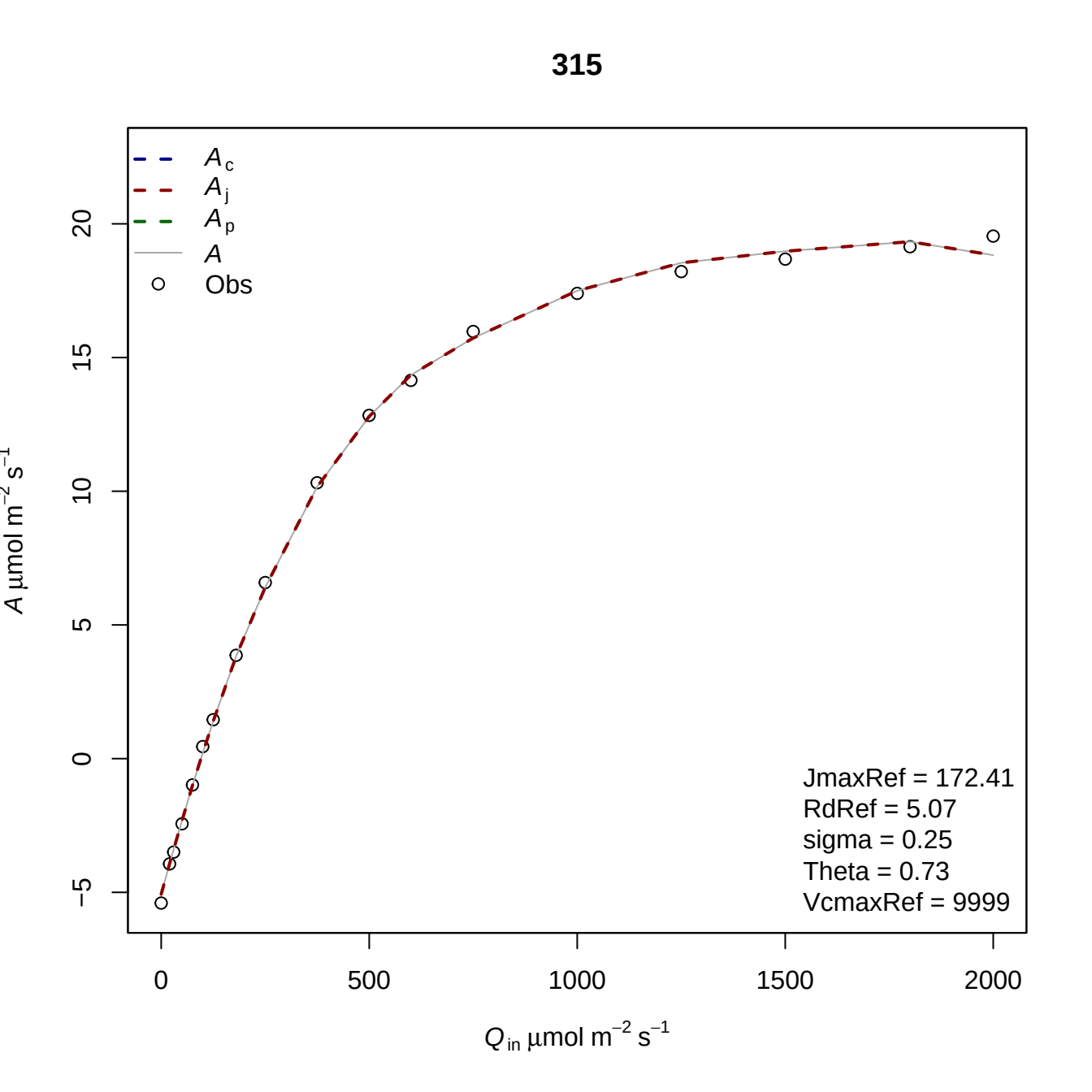

310
A_c
A_j
A_p
A
Obs
JmaxRef = 126.48
RdRef = 1.71
sigma = 0.19
Theta = −0.03
VcmaxRef = 44.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

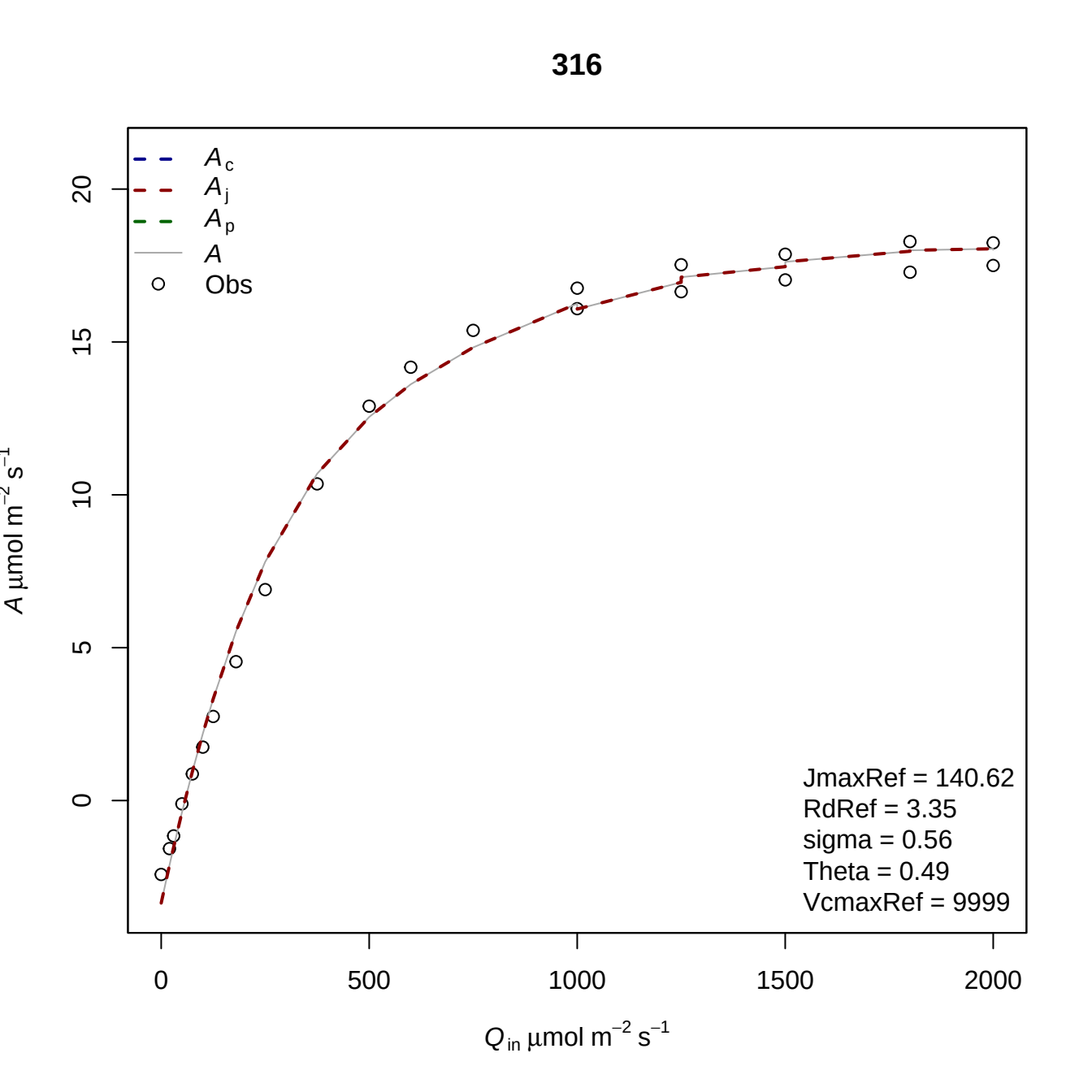

311
A_c
A_j
A_p
A
Obs
JmaxRef = 165.52
RdRef = 2.73
sigma = 0.38
Theta = −0.92
VcmaxRef = 64.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

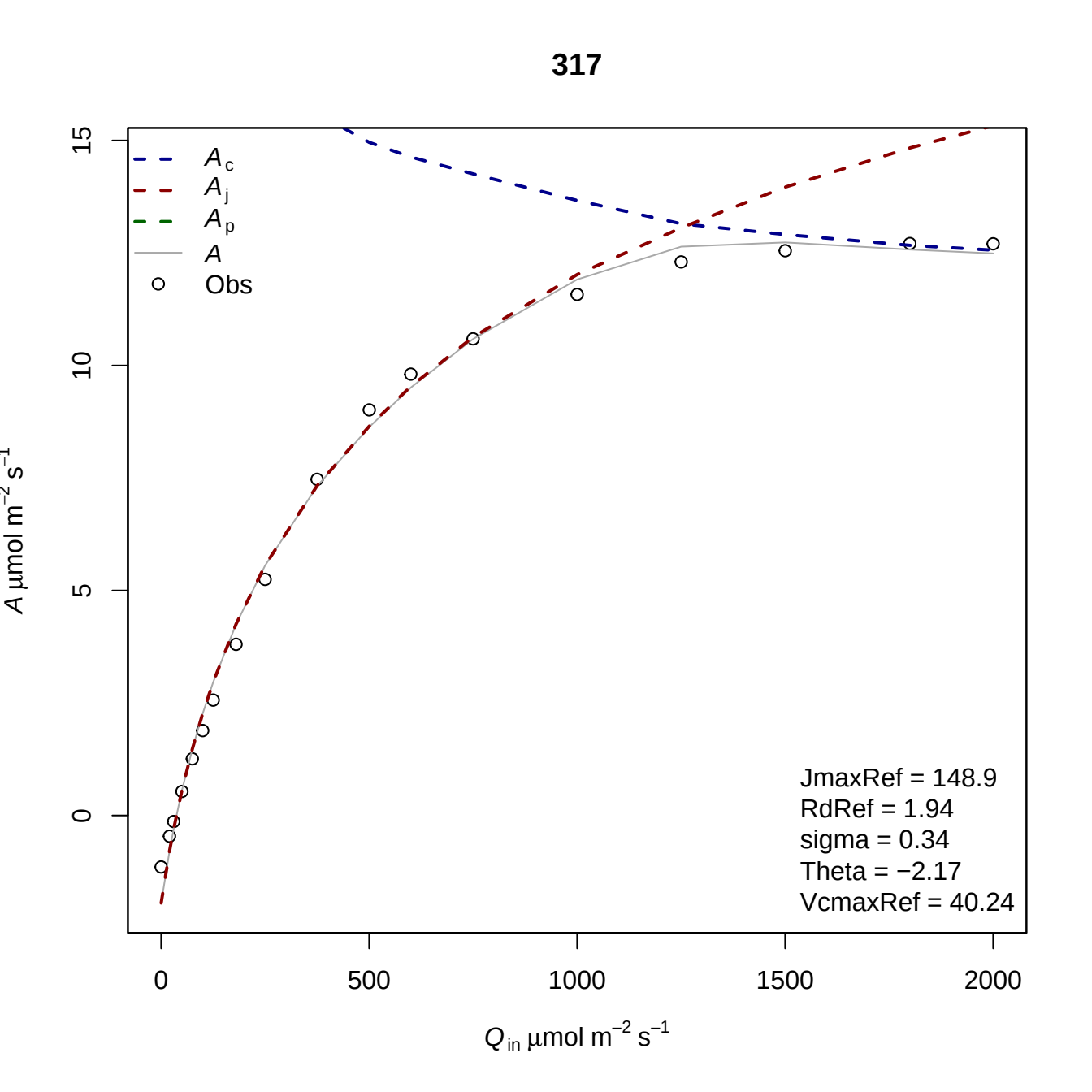

312
A_c
A_j
A_p
A
Obs
JmaxRef = 152.67
RdRef = 2.84
sigma = 0.5
Theta = 0.09
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

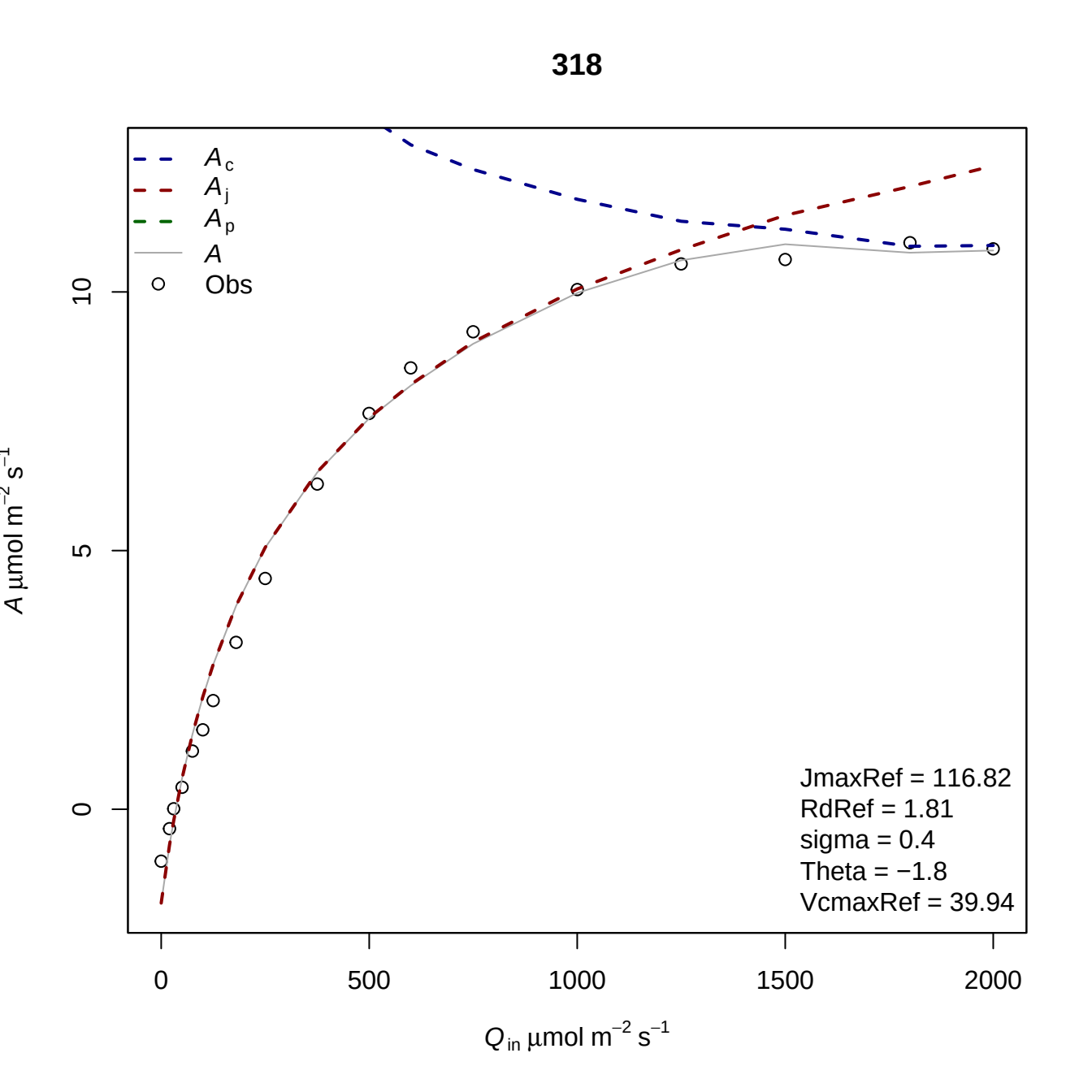

313
A_c
A_j
A_p
A
Obs
JmaxRef = 116.7
RdRef = 2.27
sigma = 0.22
Theta = 0.1
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

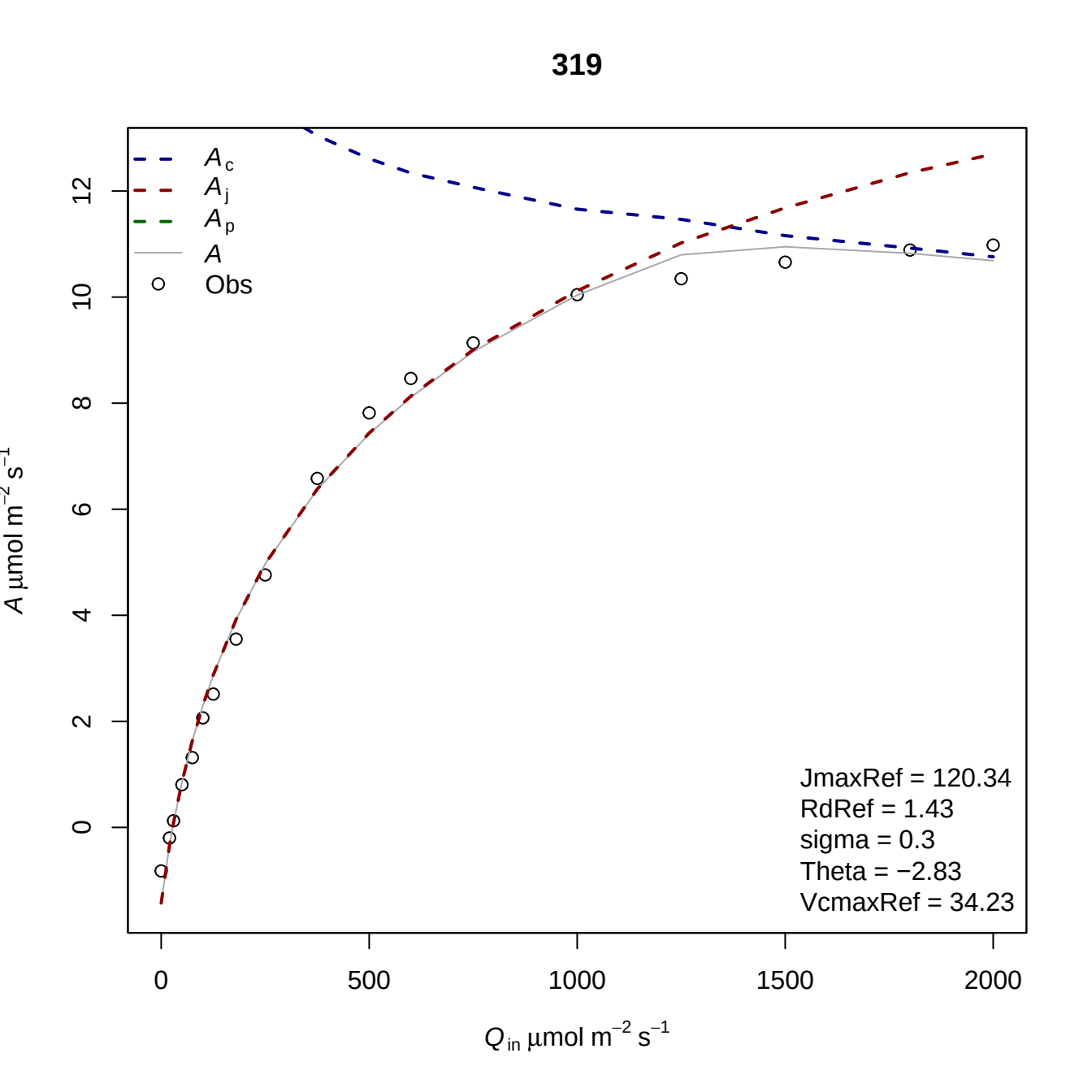

314
A_c
A_j
A_p
A
Obs
JmaxRef = 143.79
RdRef = 2.31
sigma = 0.29
Theta = −0.34
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

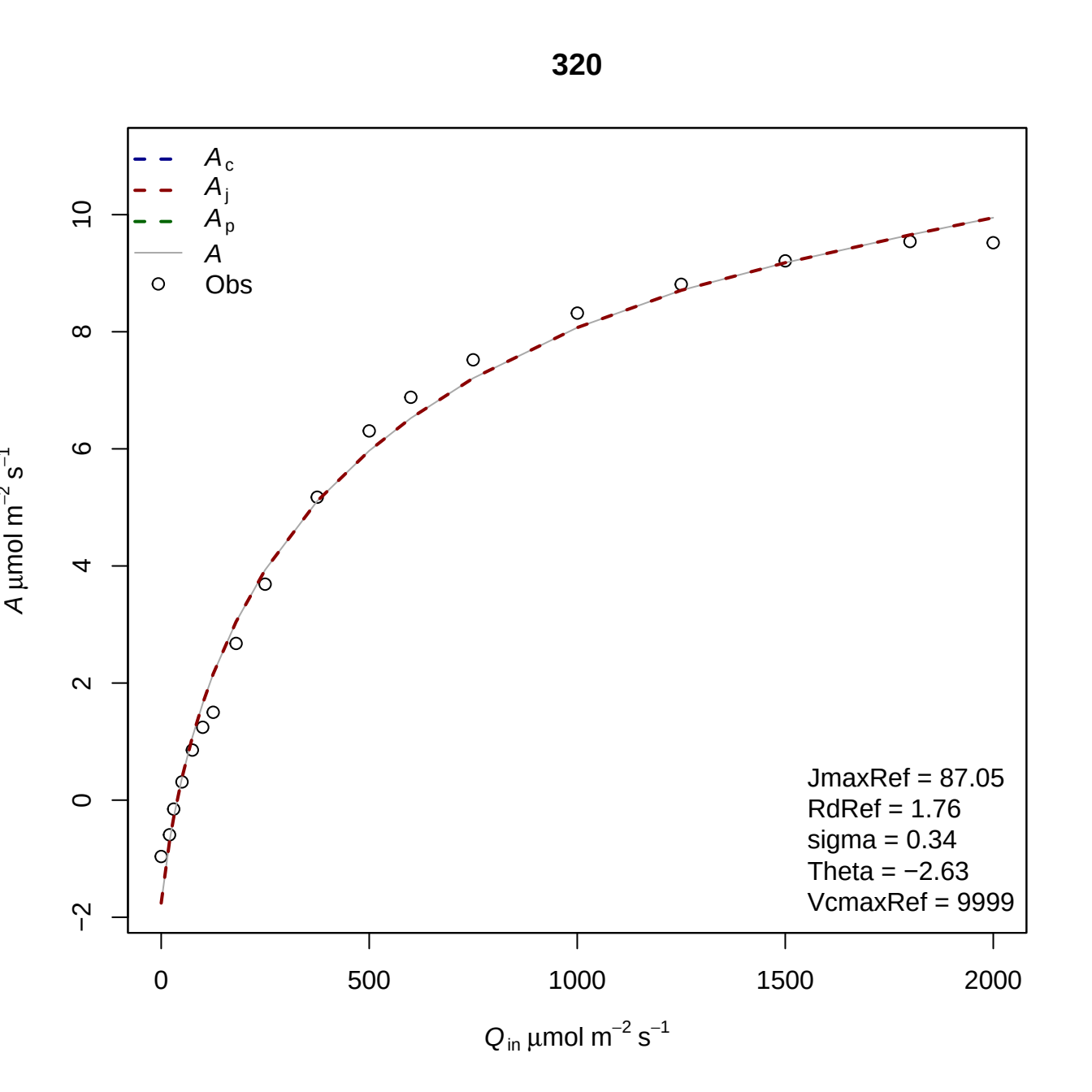

315
A_c
A_j
A_p
A
Obs
JmaxRef = 172.41
RdRef = 5.07
sigma = 0.25
Theta = 0.73
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

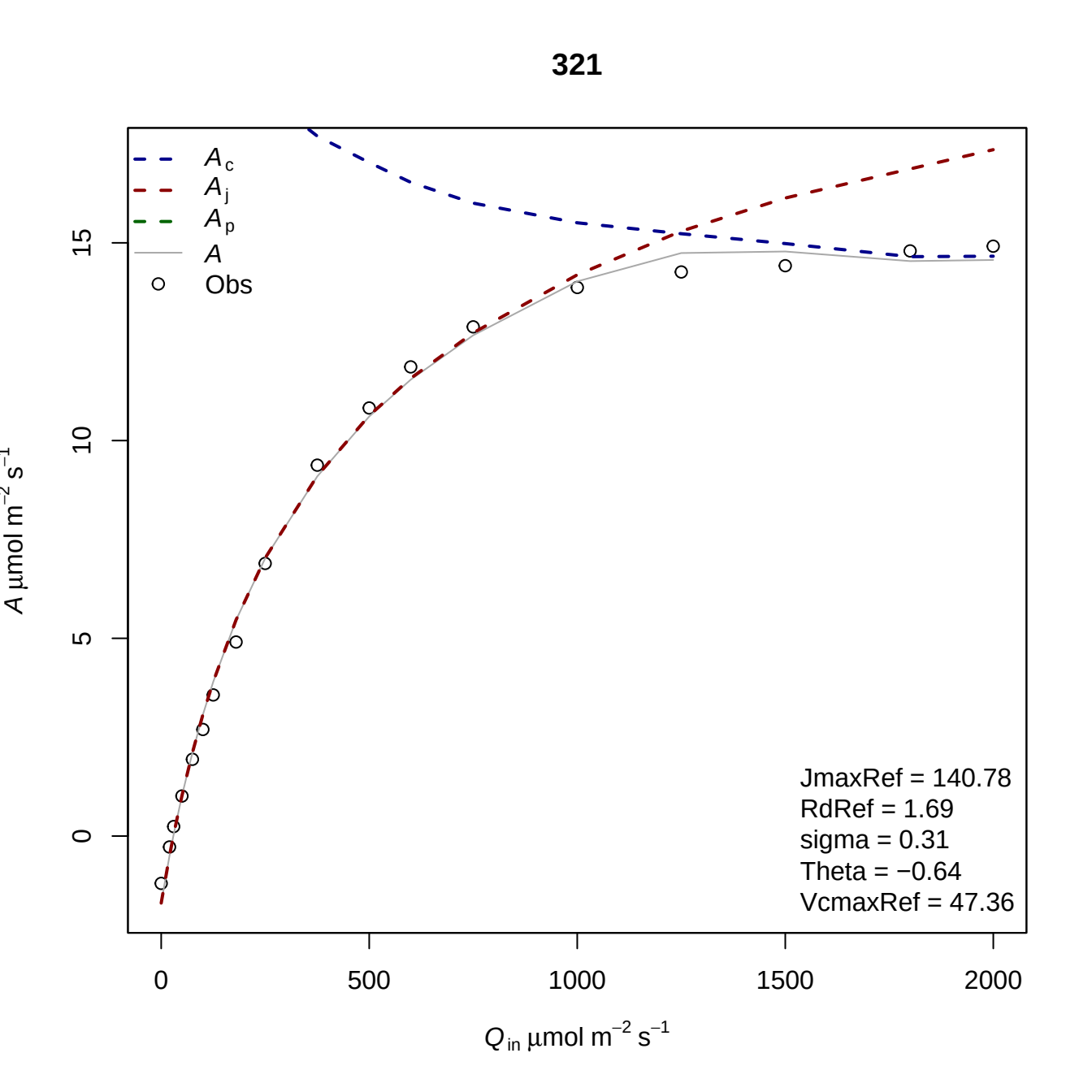

316
A_c
A_j
A_p
A
Obs
JmaxRef = 140.62
RdRef = 3.35
sigma = 0.56
Theta = 0.49
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

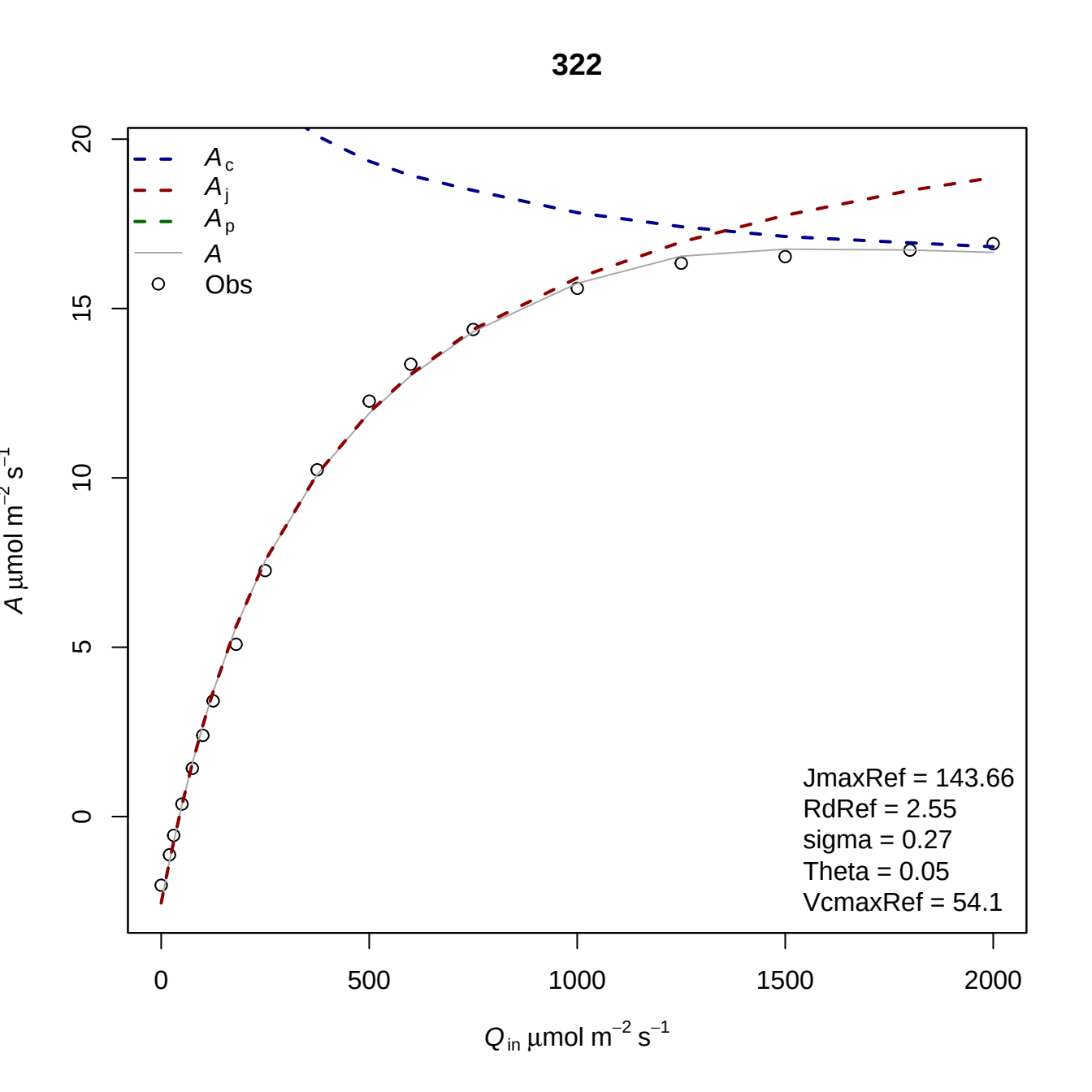

317
A_c
A_j
A_p
A
Obs
JmaxRef = 148.9
RdRef = 1.94
sigma = 0.34
Theta = −2.17
VcmaxRef = 40.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

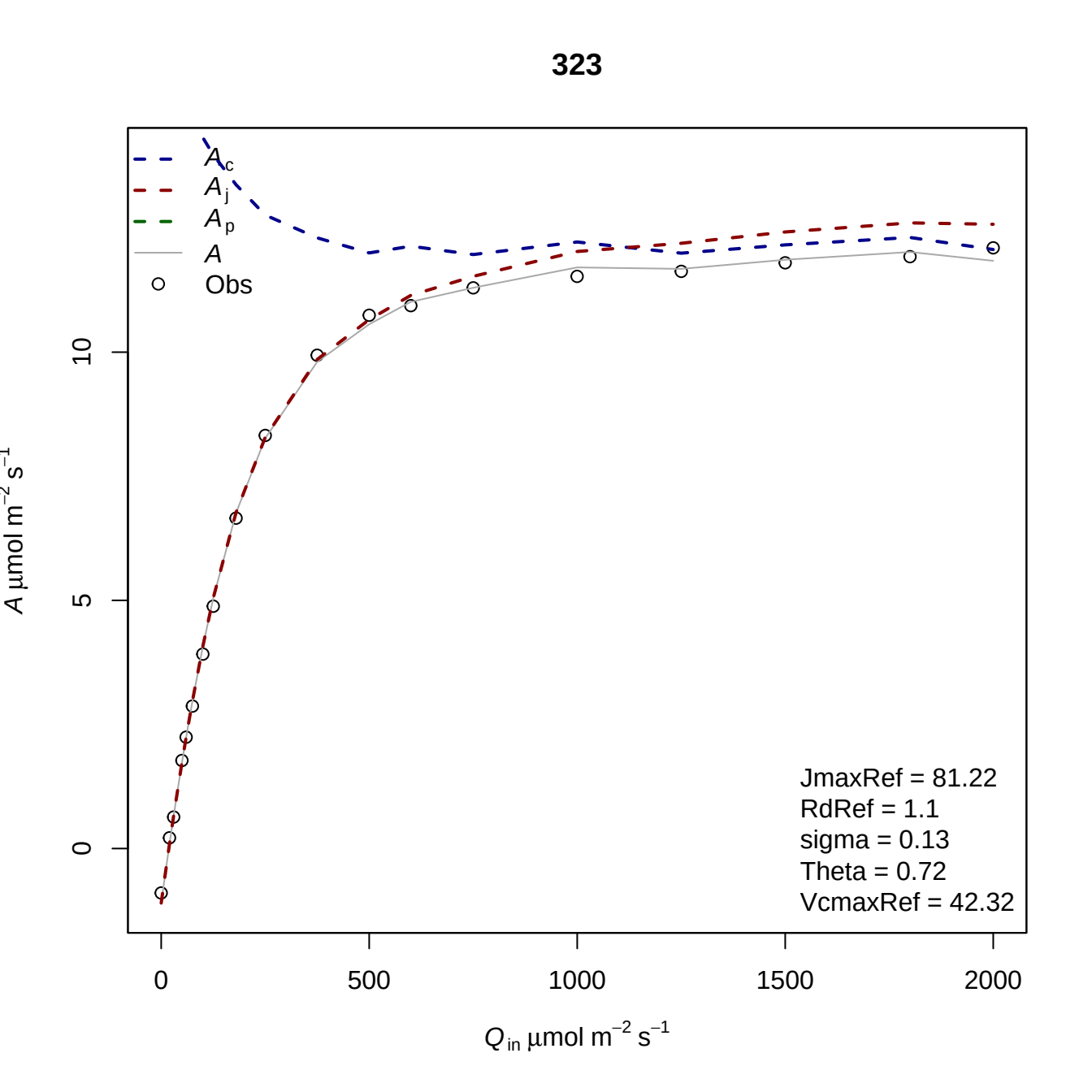

318
A_c
A_j
A_p
A
Obs
JmaxRef = 116.82
RdRef = 1.81
sigma = 0.4
Theta = −1.8
VcmaxRef = 39.94
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

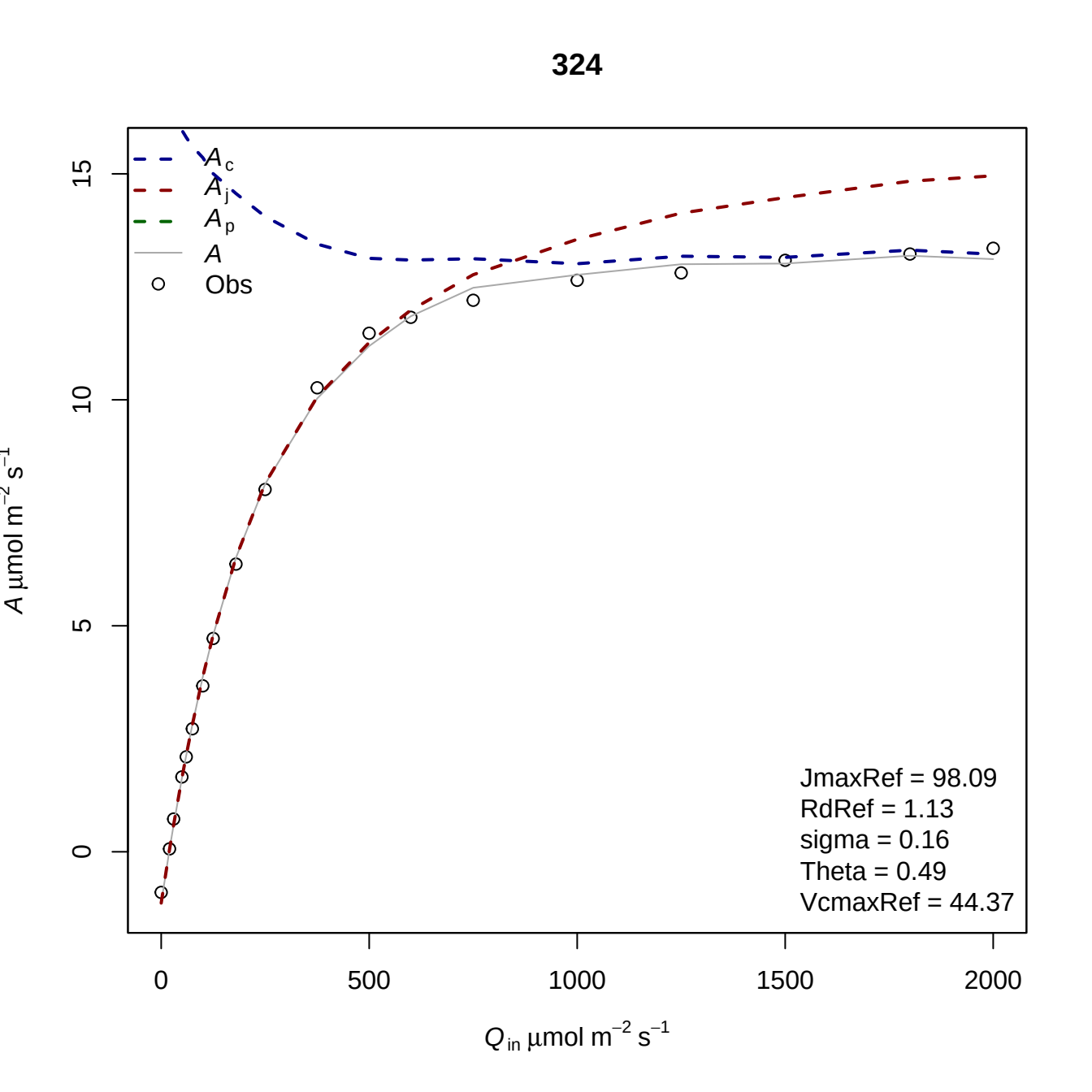

319
A_c
A_j
A_p
A
Obs
JmaxRef = 120.34
RdRef = 1.43
sigma = 0.3
Theta = −2.83
VcmaxRef = 34.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

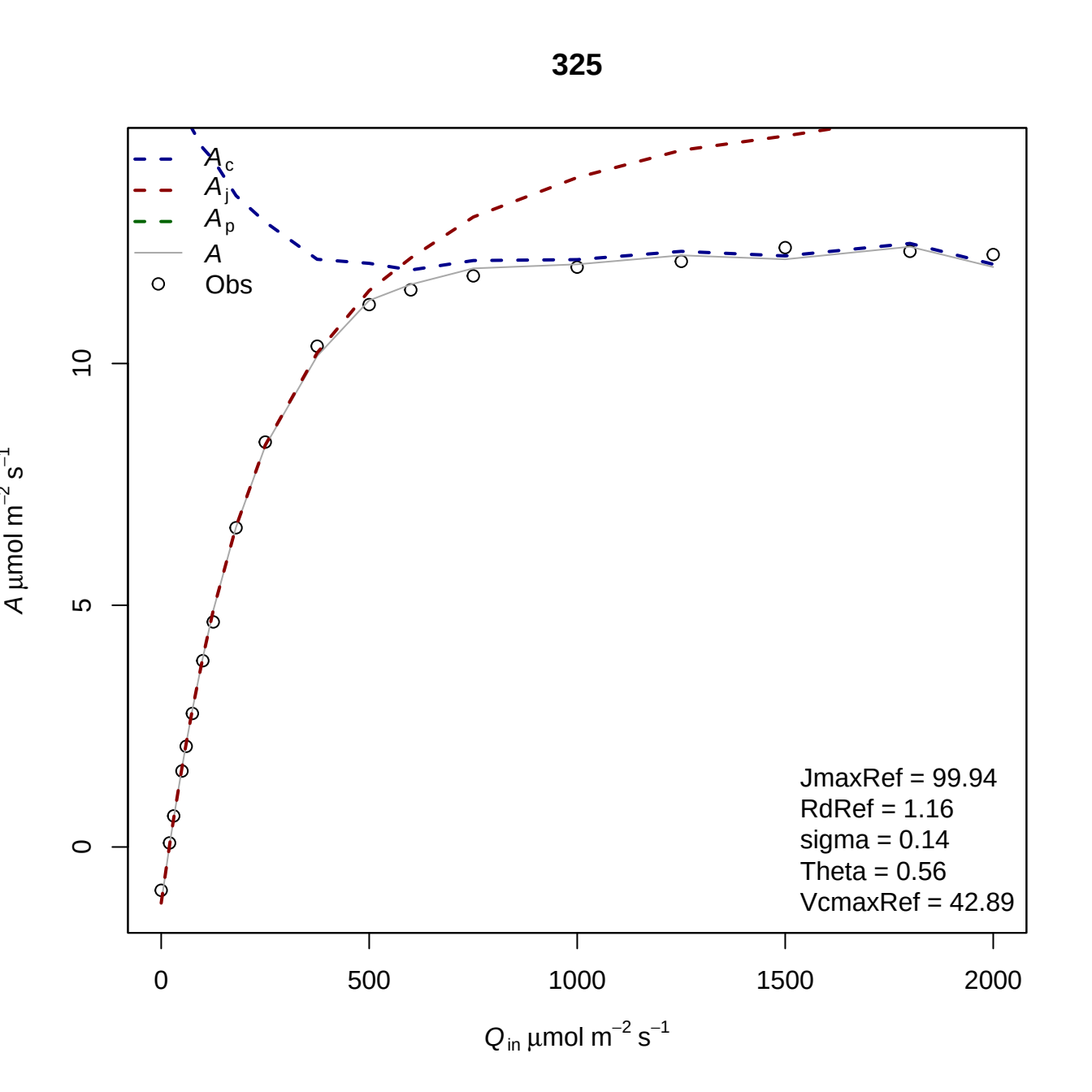

320
A_c
A_j
A_p
A
Obs
JmaxRef = 87.05
RdRef = 1.76
sigma = 0.34
Theta = −2.63
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

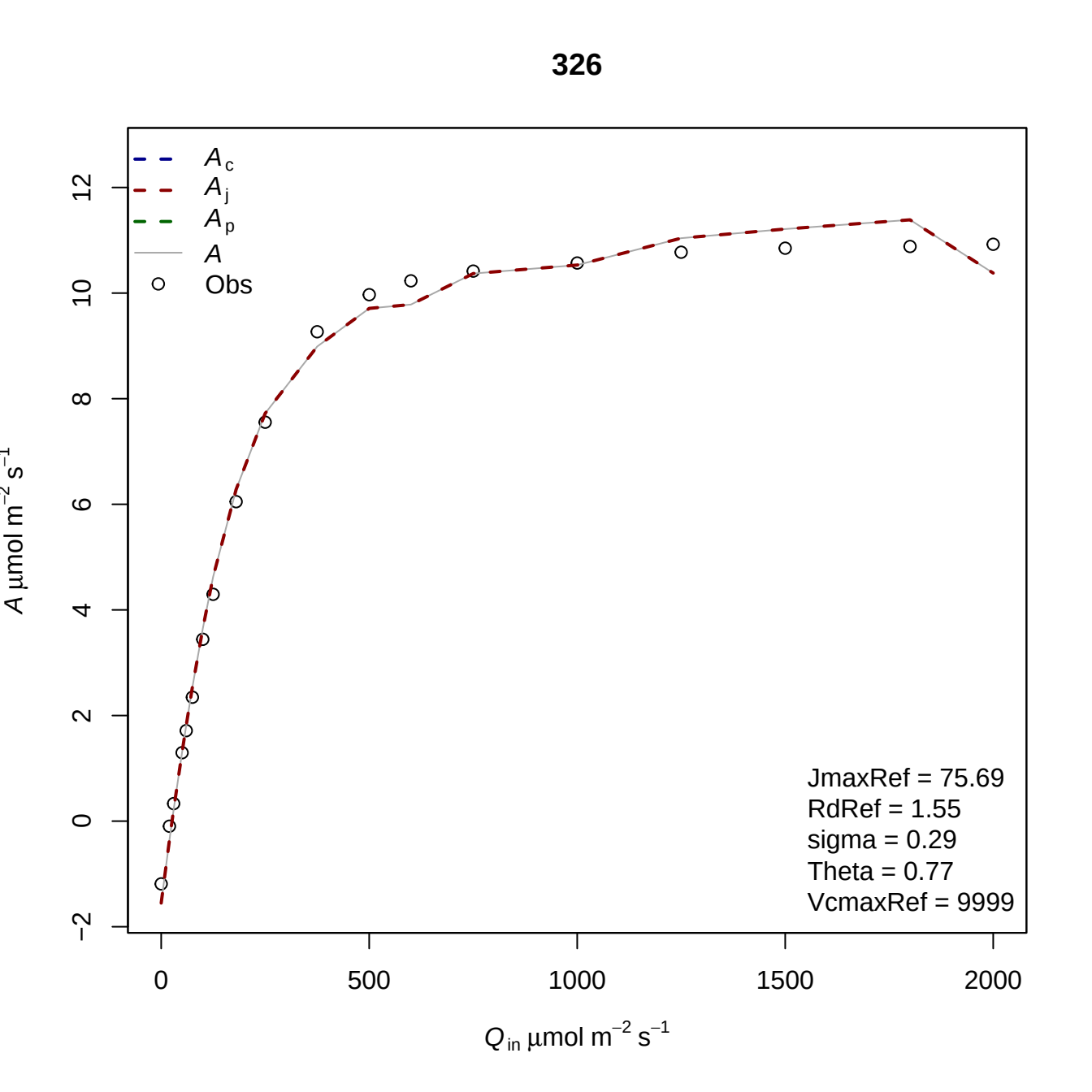

321
A_c
A_j
A_p
A
Obs
JmaxRef = 140.78
RdRef = 1.69
sigma = 0.31
Theta = −0.64
VcmaxRef = 47.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

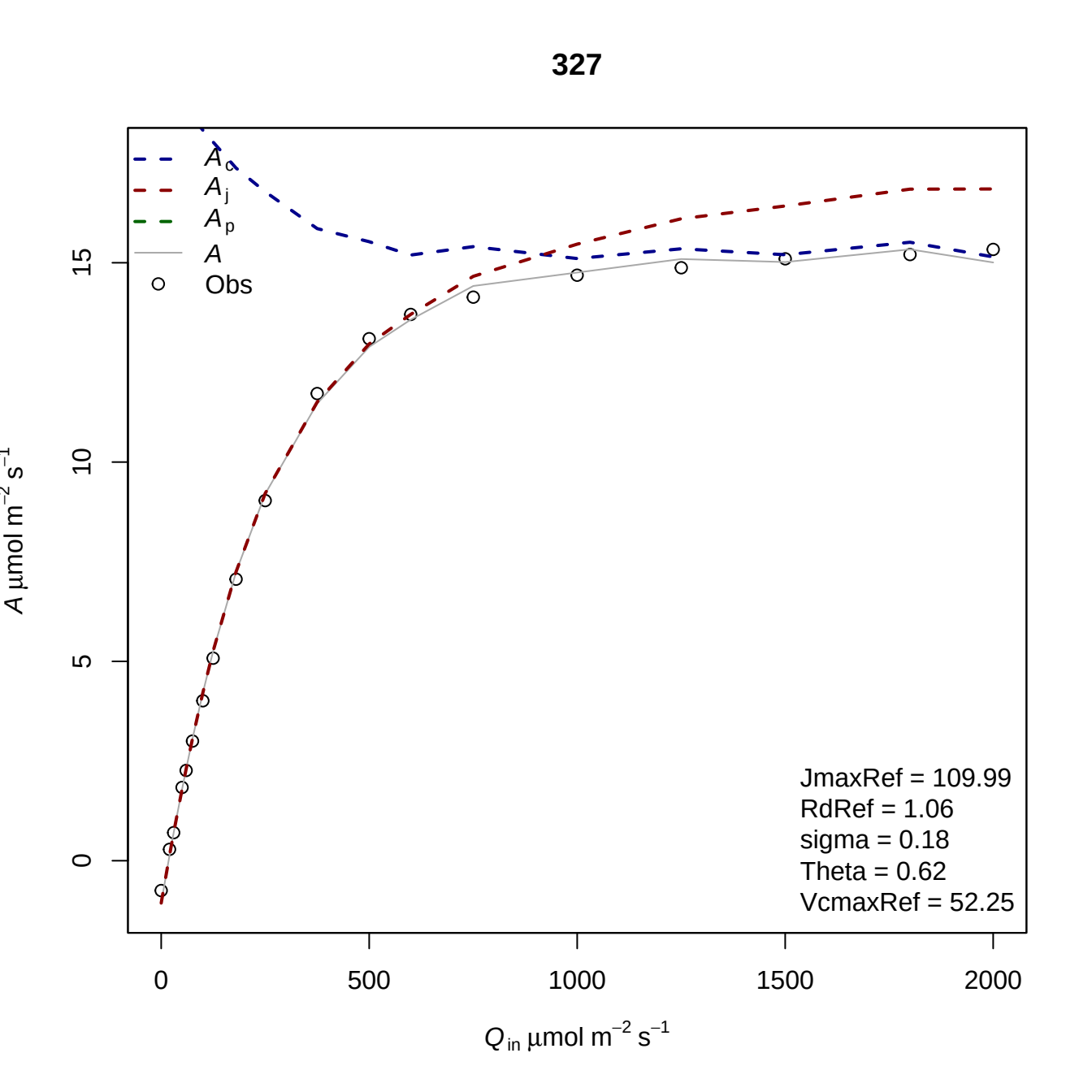

322
A_c
A_j
A_p
A
Obs
JmaxRef = 143.66
RdRef = 2.55
sigma = 0.27
Theta = 0.05
VcmaxRef = 54.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

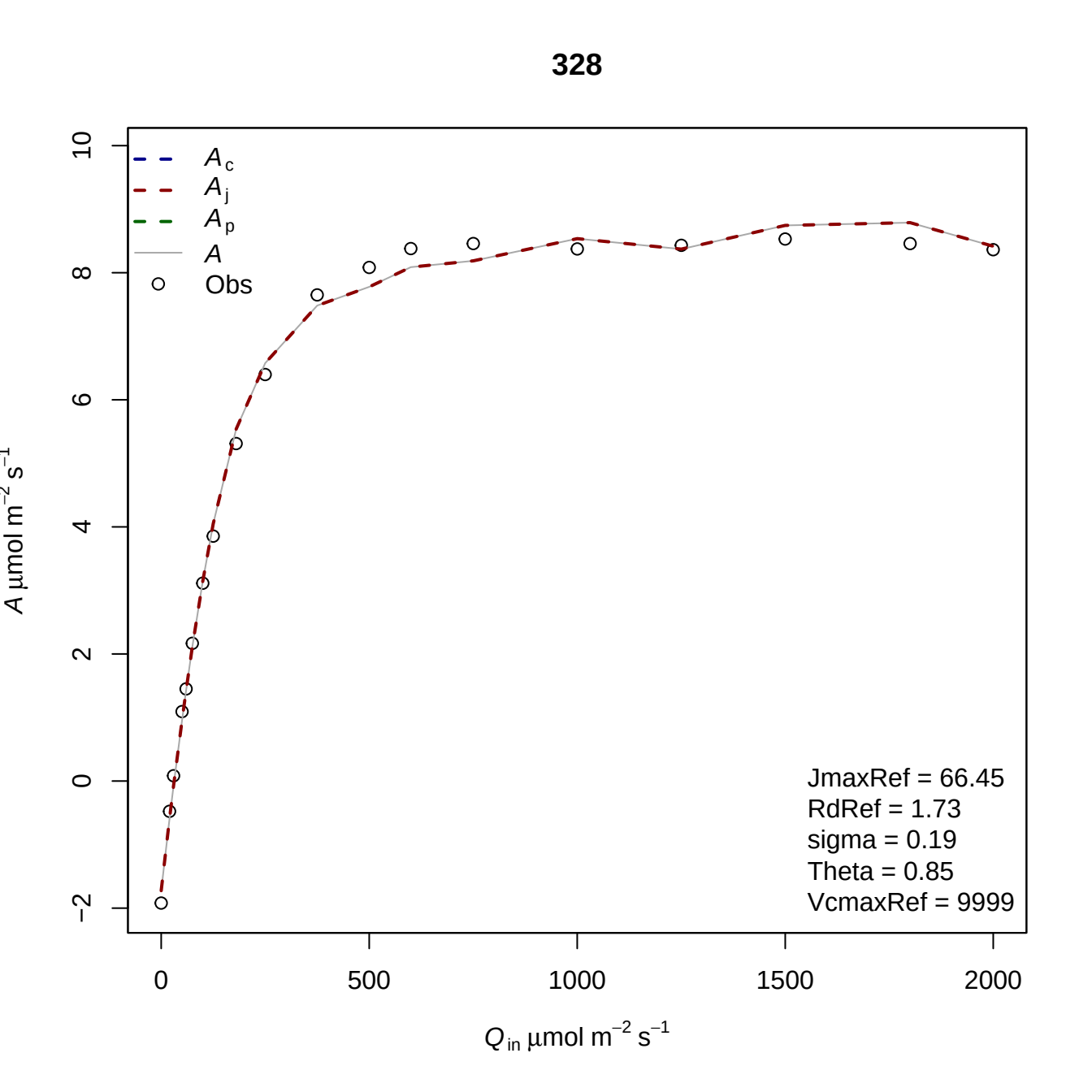

323
A_c
A_j
A_p
A
Obs
JmaxRef = 81.22
RdRef = 1.1
sigma = 0.13
Theta = 0.72
VcmaxRef = 42.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

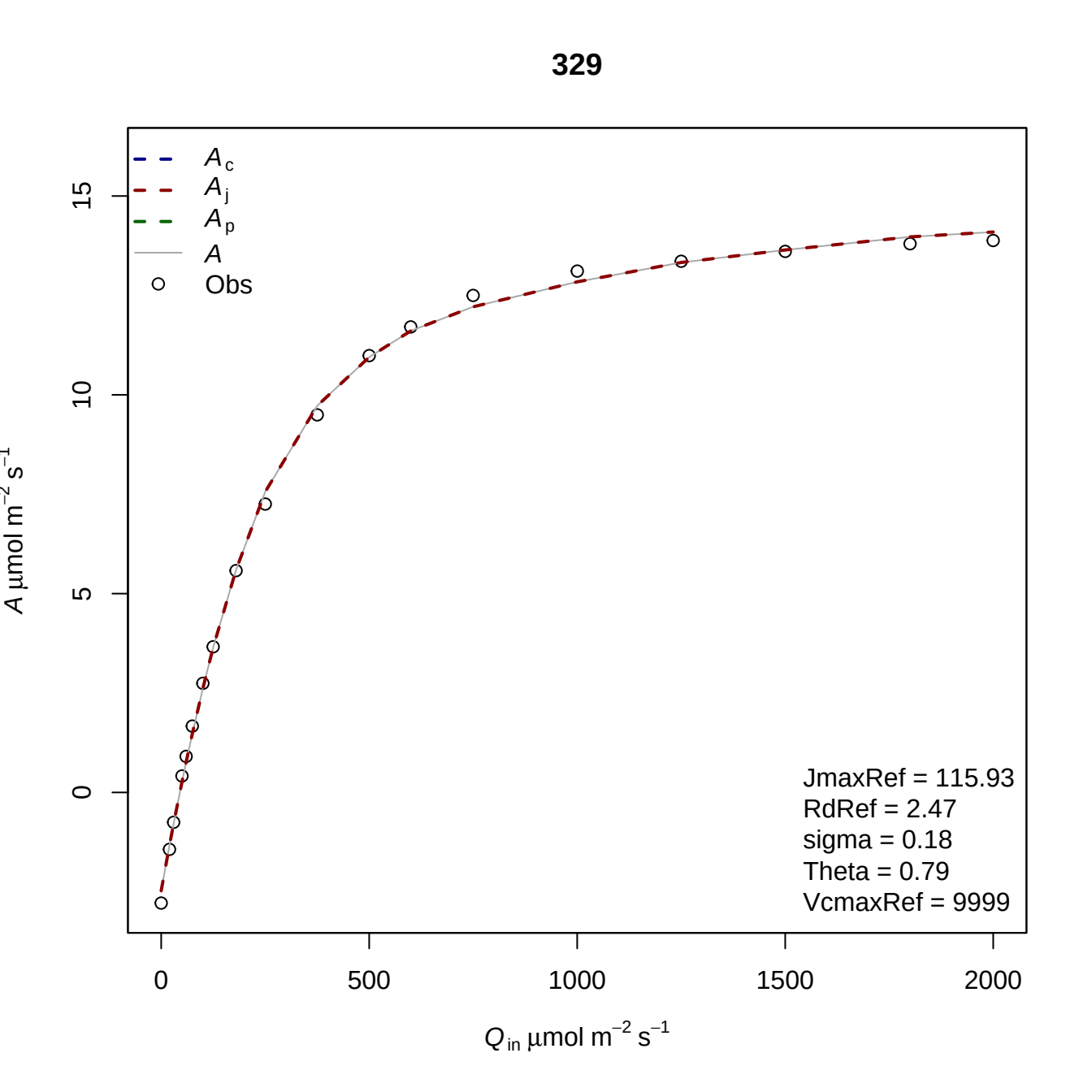

324
A_c
A_j
A_p
A
Obs
JmaxRef = 98.09
RdRef = 1.13
sigma = 0.16
Theta = 0.49
VcmaxRef = 44.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

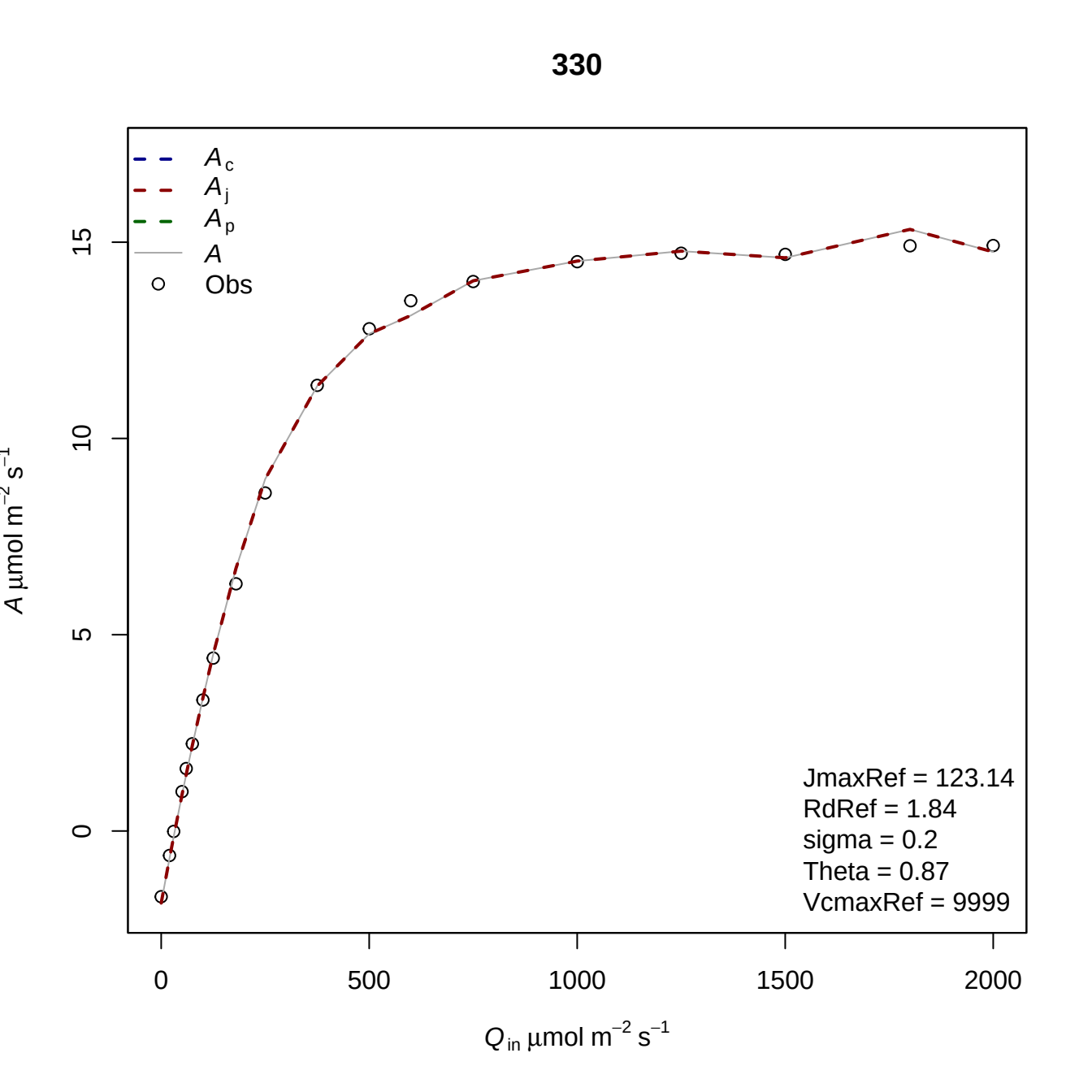

325
A_c
A_j
A_p
A
Obs
JmaxRef = 99.94
RdRef = 1.16
sigma = 0.14
Theta = 0.56
VcmaxRef = 42.89
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

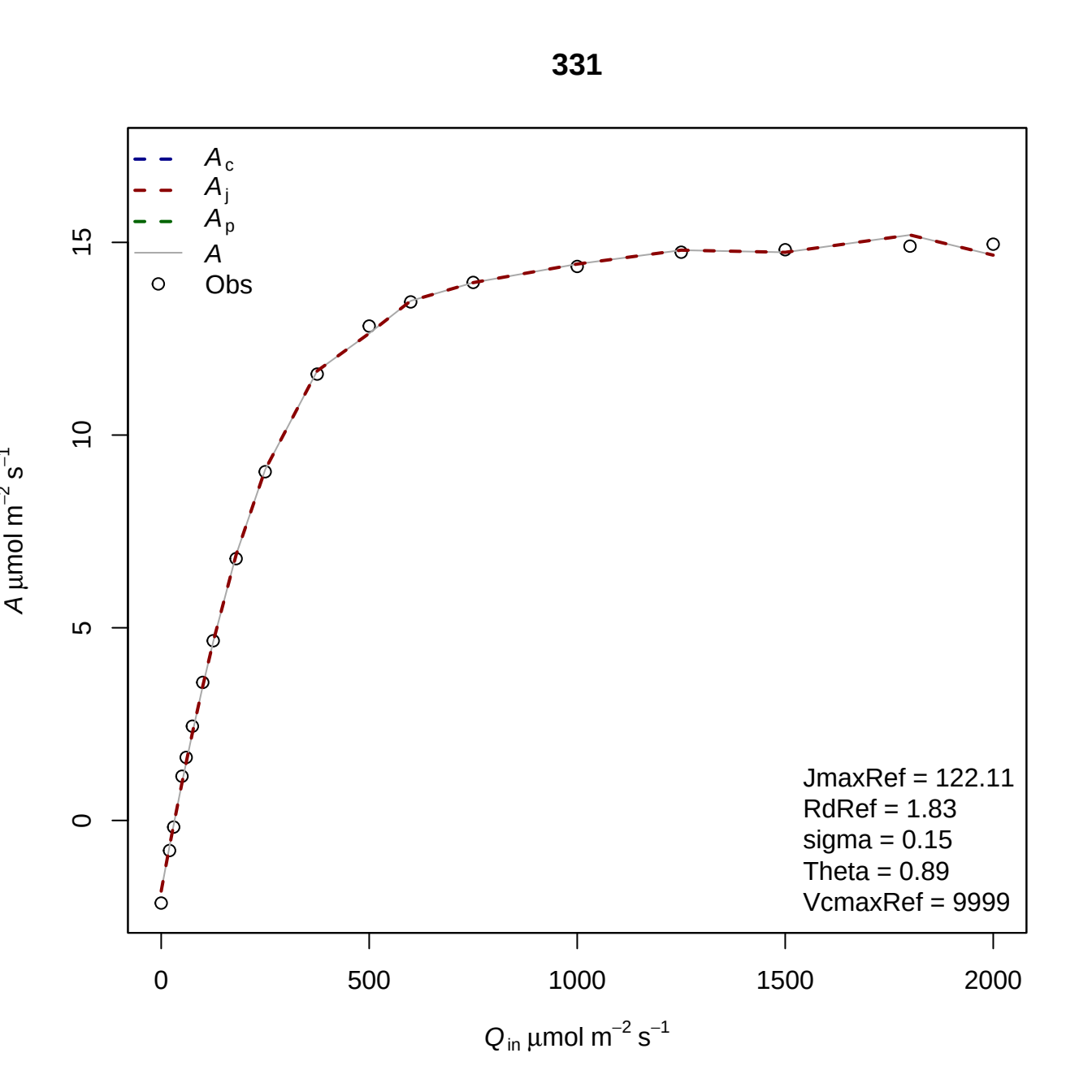

326
A_c
A_j
A_p
A
Obs
JmaxRef = 75.69
RdRef = 1.55
sigma = 0.29
Theta = 0.77
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

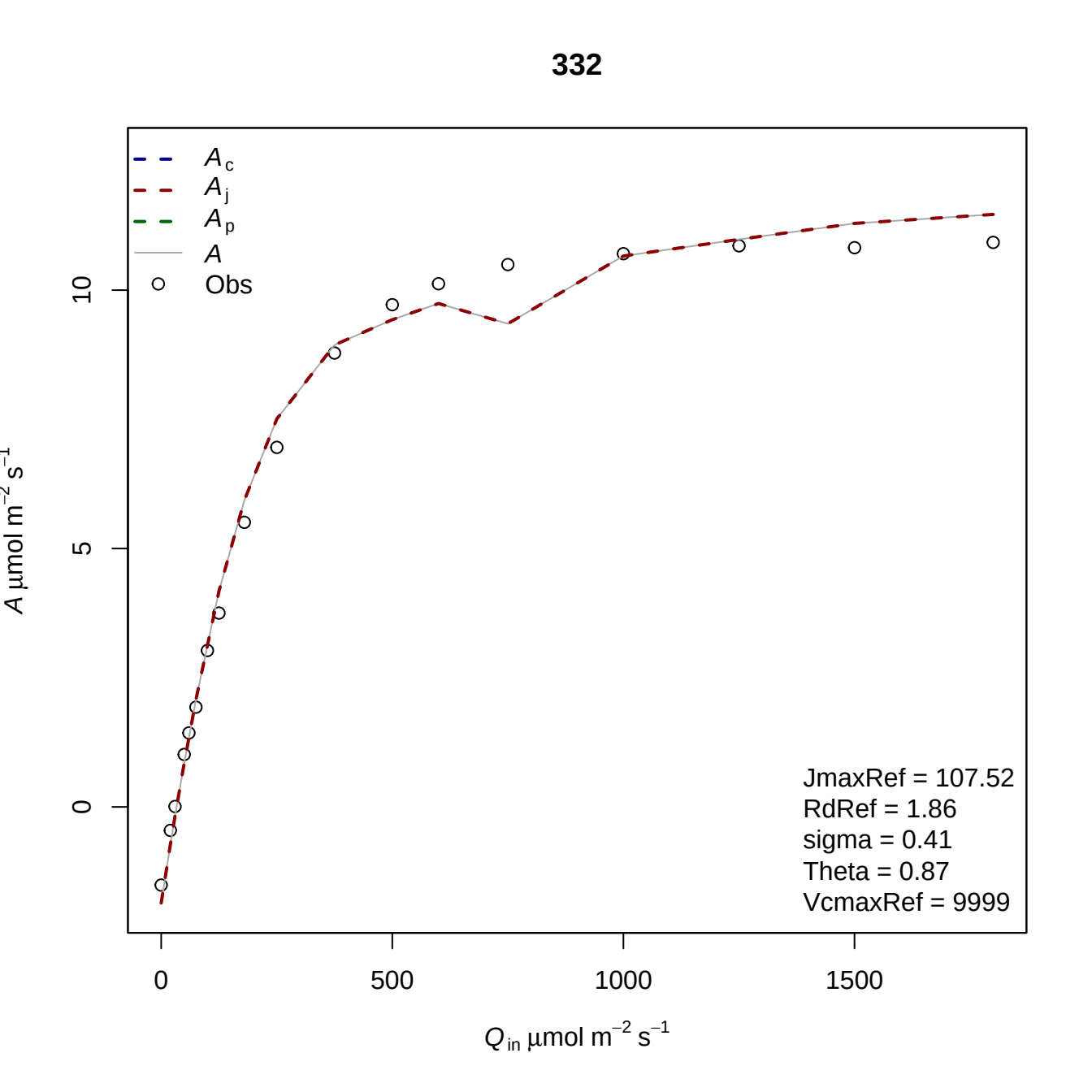

327
A_c
A_j
A_p
A
Obs
JmaxRef = 109.99
RdRef = 1.06
sigma = 0.18
Theta = 0.62
VcmaxRef = 52.25
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

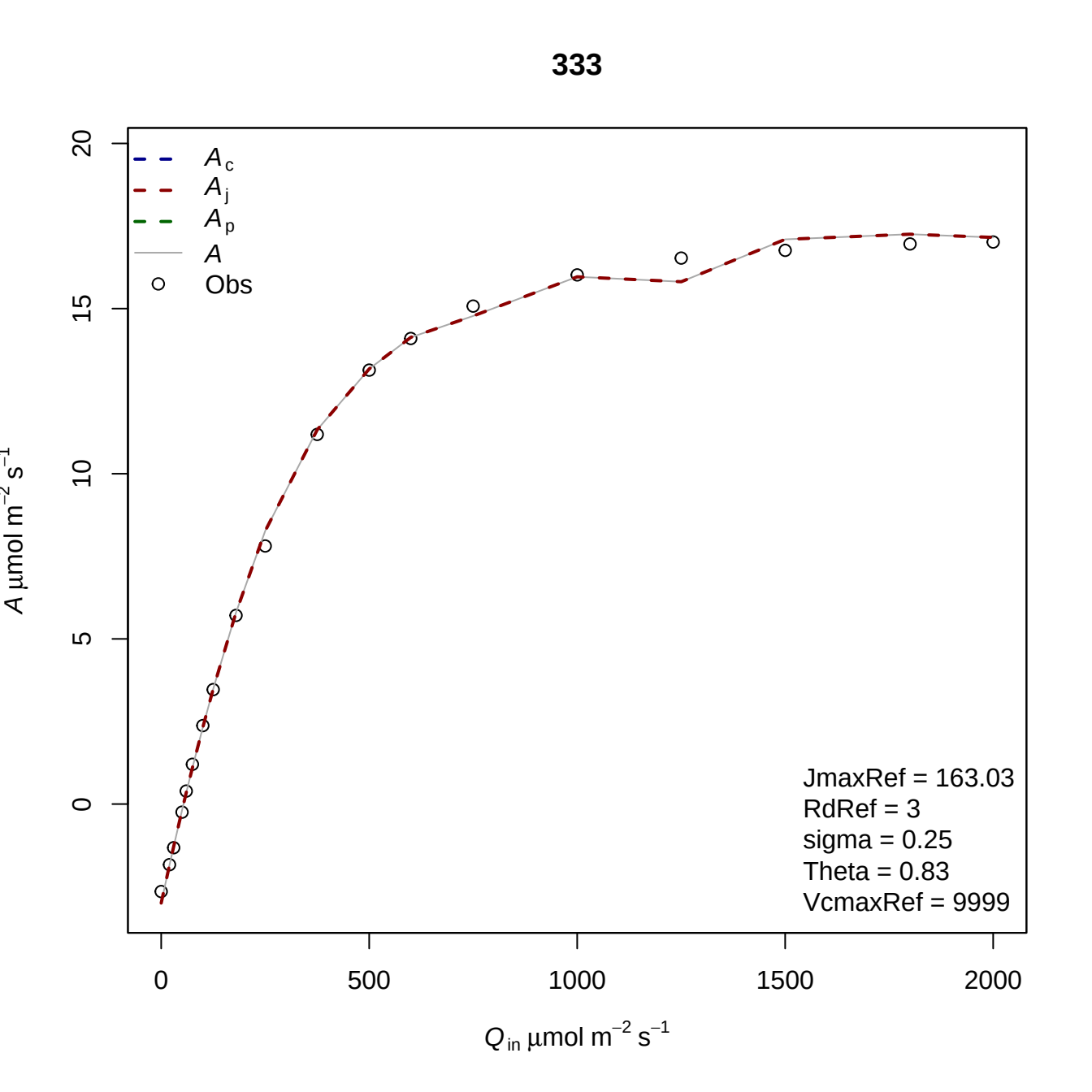

328
A_c
A_j
A_p
A
Obs
JmaxRef = 66.45
RdRef = 1.73
sigma = 0.19
Theta = 0.85
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

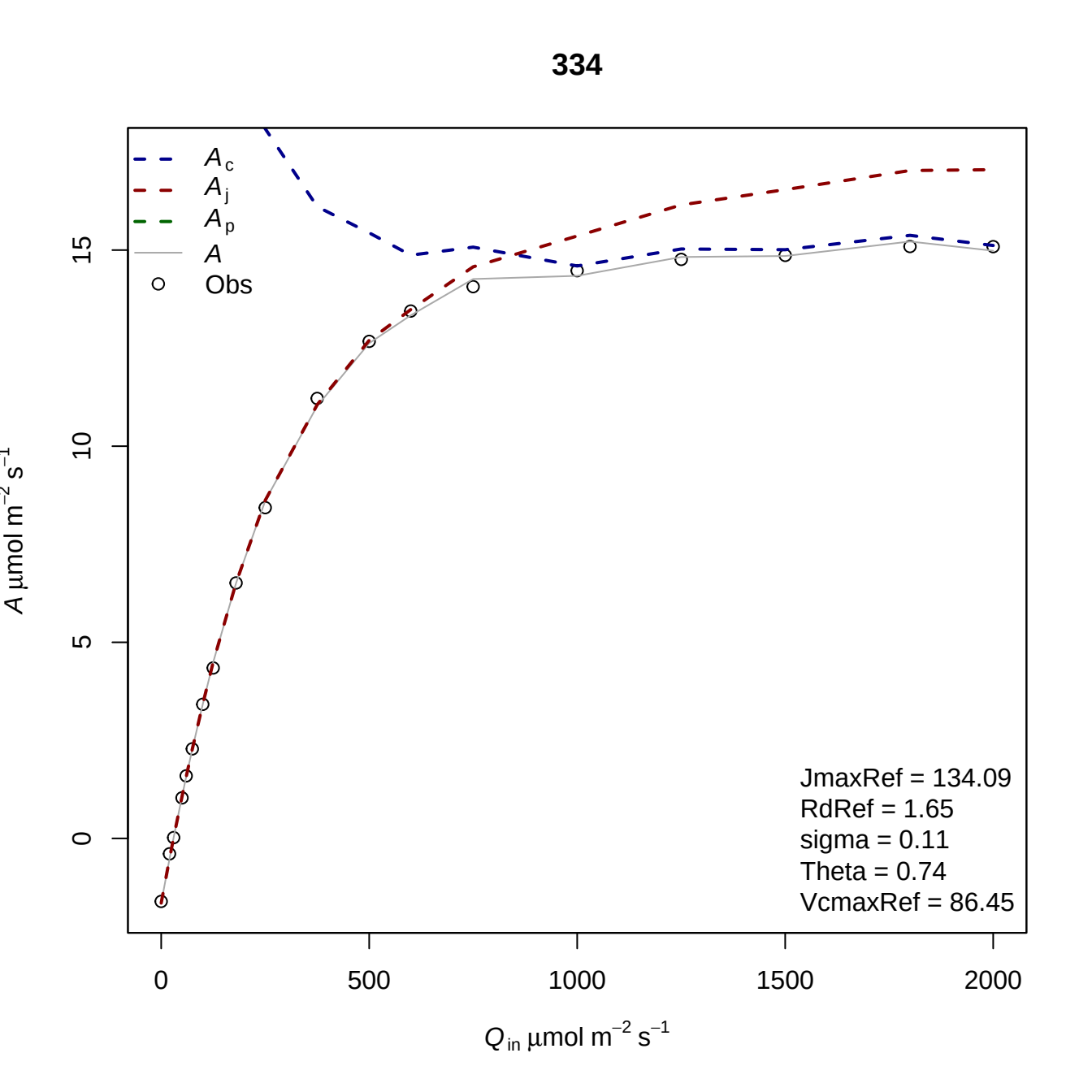

329
A_c
A_j
A_p
A
Obs
JmaxRef = 115.93
RdRef = 2.47
sigma = 0.18
Theta = 0.79
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

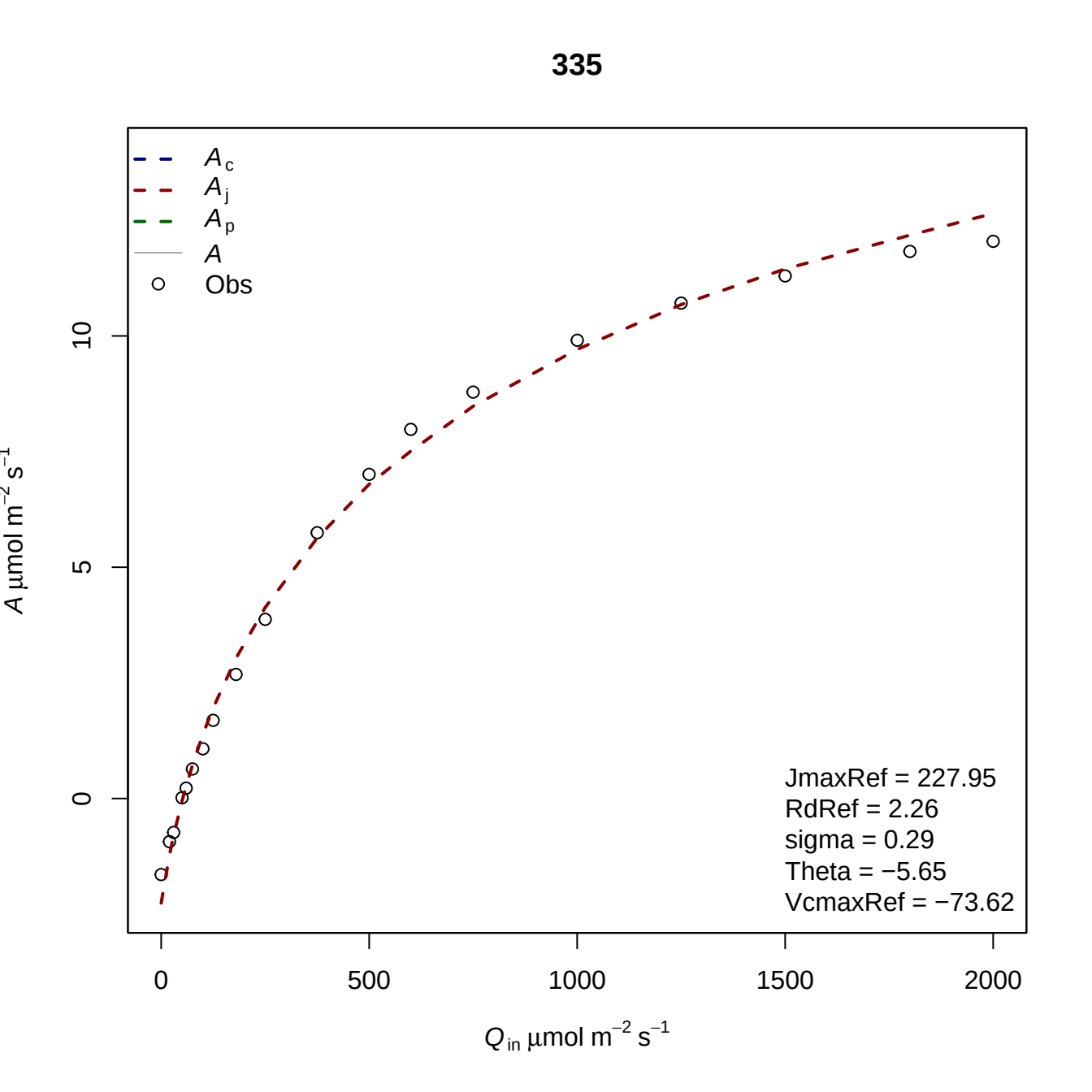

330
A_c
A_j
A_p
A
Obs
JmaxRef = 123.14
RdRef = 1.84
sigma = 0.2
Theta = 0.87
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

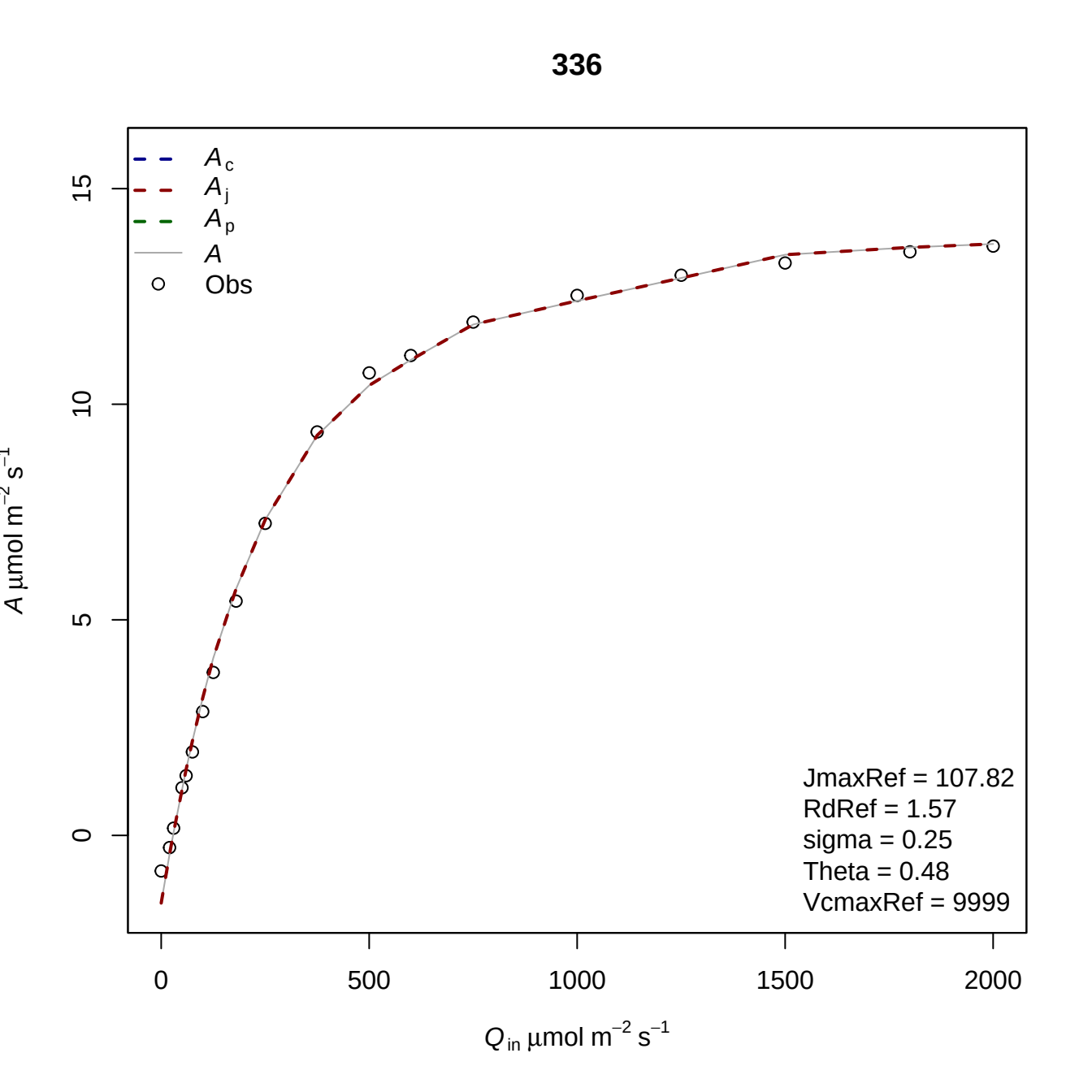

331
A_c
A_j
A_p
A
Obs
JmaxRef = 122.11
RdRef = 1.83
sigma = 0.15
Theta = 0.89
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

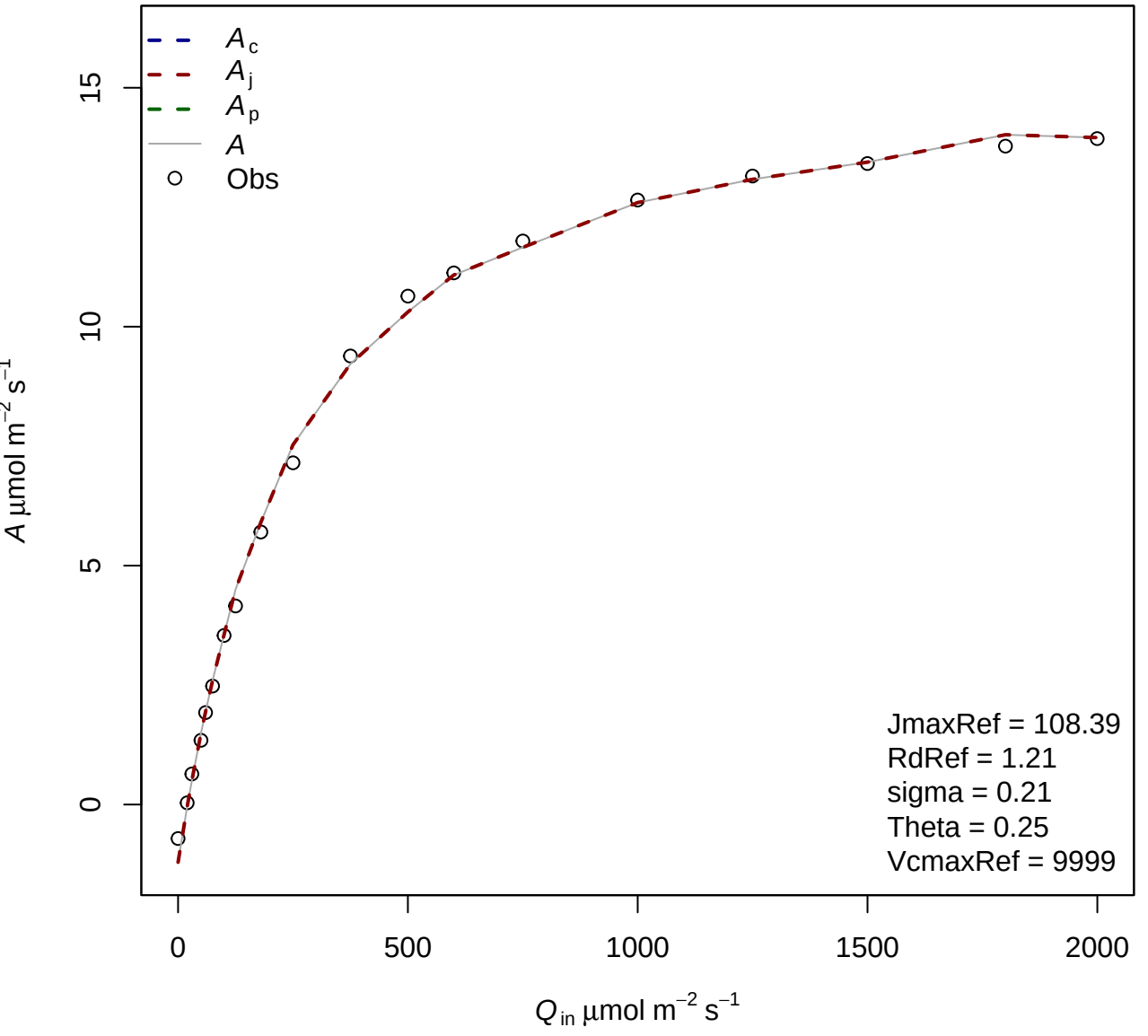

332
A_c
A_j
A_p
A
Obs
JmaxRef = 107.52
RdRef = 1.86
sigma = 0.41
Theta = 0.87
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

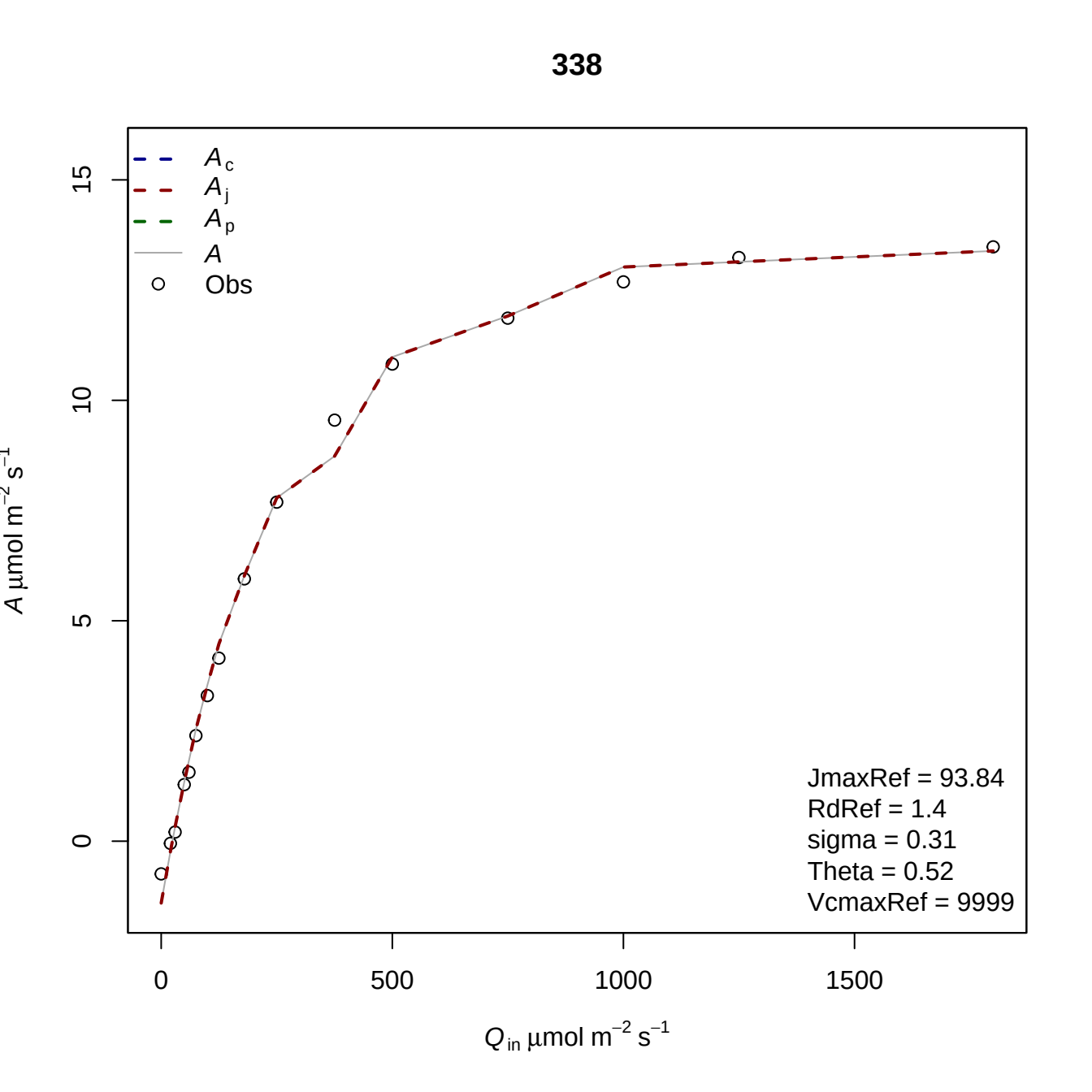

333
A_c
A_j
A_p
A
Obs
JmaxRef = 163.03
RdRef = 3
sigma = 0.25
Theta = 0.83
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

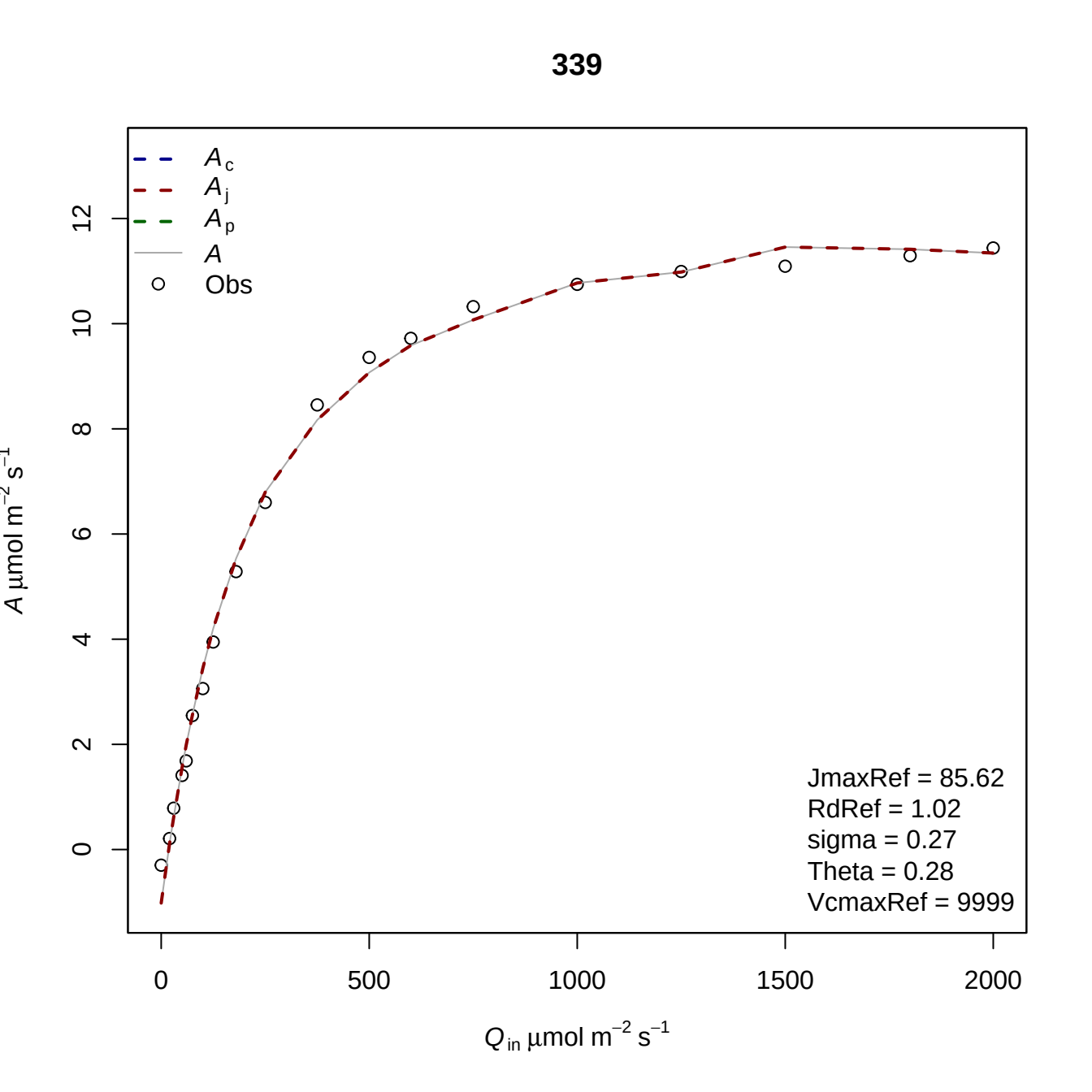

334
A_c
A_j
A_p
A
Obs
JmaxRef = 134.09
RdRef = 1.65
sigma = 0.11
Theta = 0.74
VcmaxRef = 86.45
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

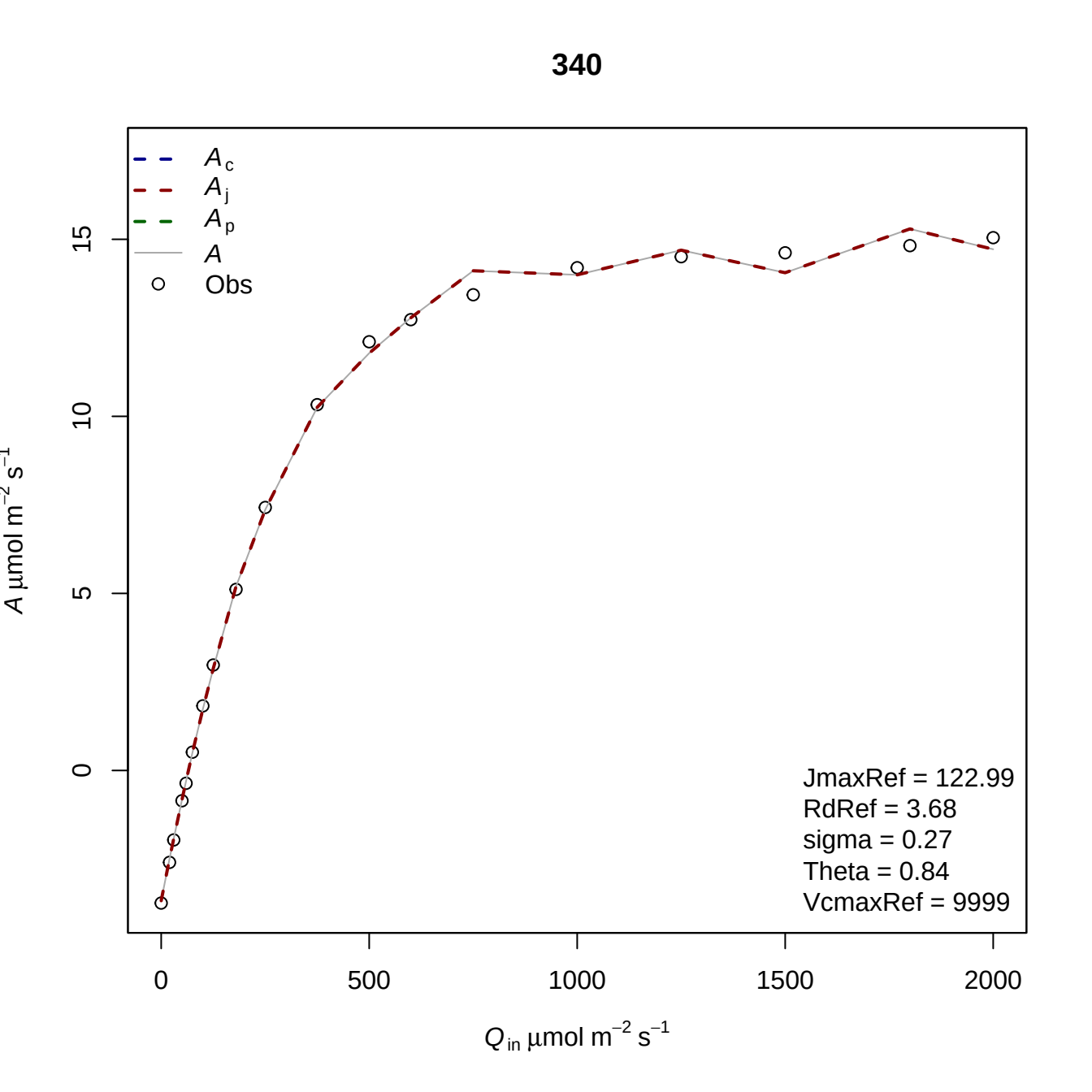

335
A_c
A_j
A_p
A
Obs
JmaxRef = 227.95
RdRef = 2.26
sigma = 0.29
Theta = −5.65
VcmaxRef = −73.62
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

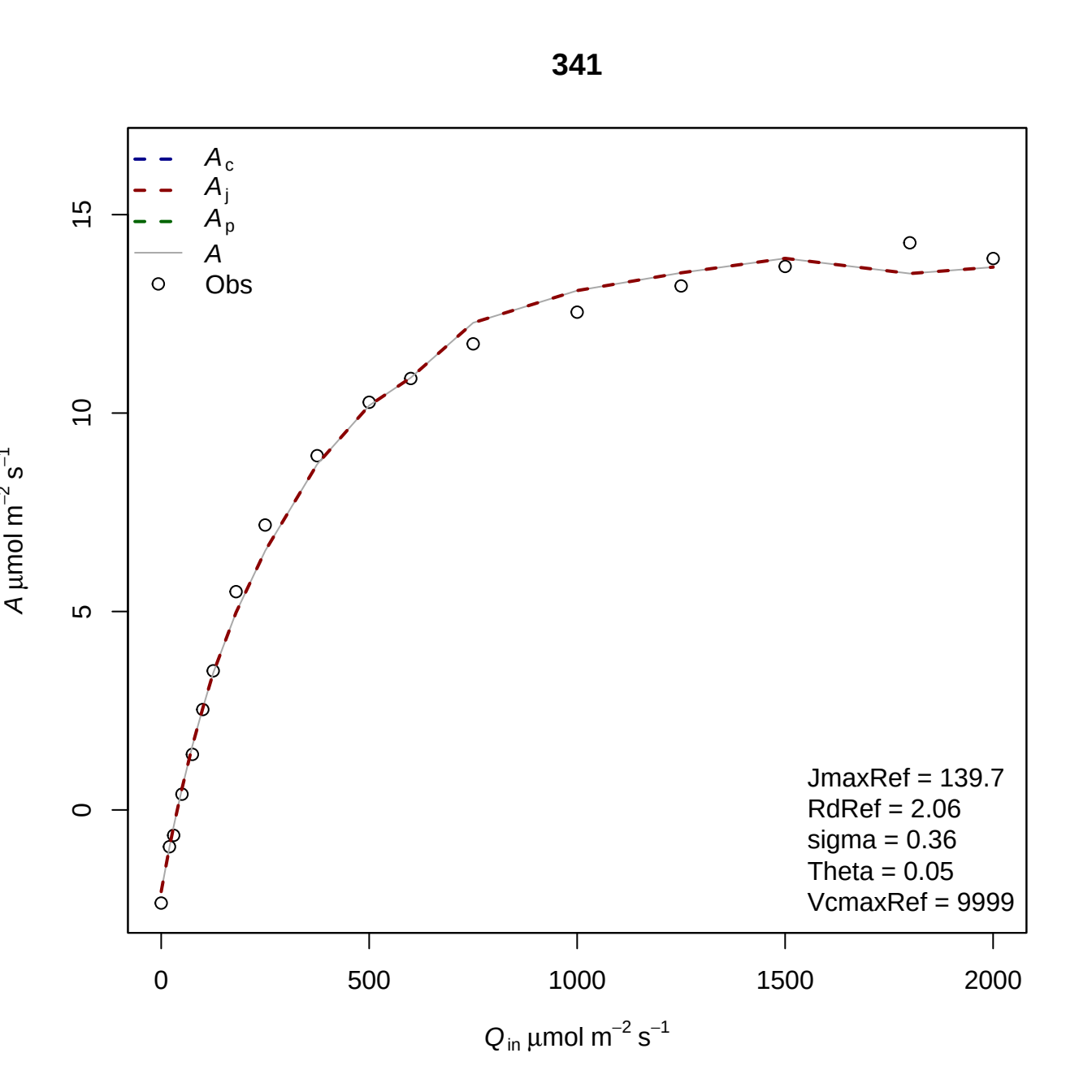

336
A_c
A_j
A_p
A
Obs
JmaxRef = 107.82
RdRef = 1.57
sigma = 0.25
Theta = 0.48
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

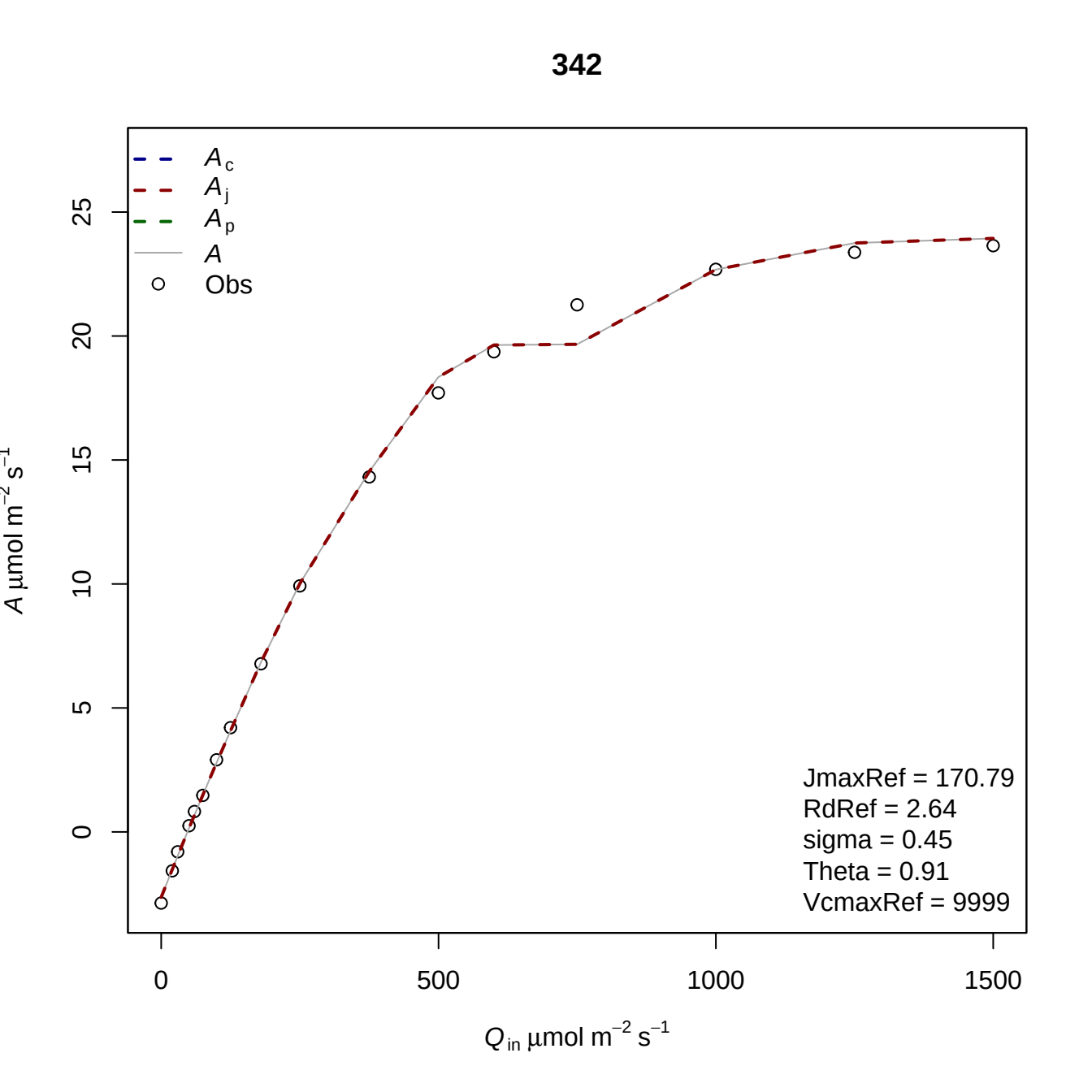

A_c
A_j
A_p
A
Obs
JmaxRef = 108.39
RdRef = 1.21
sigma = 0.21
Theta = 0.25
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

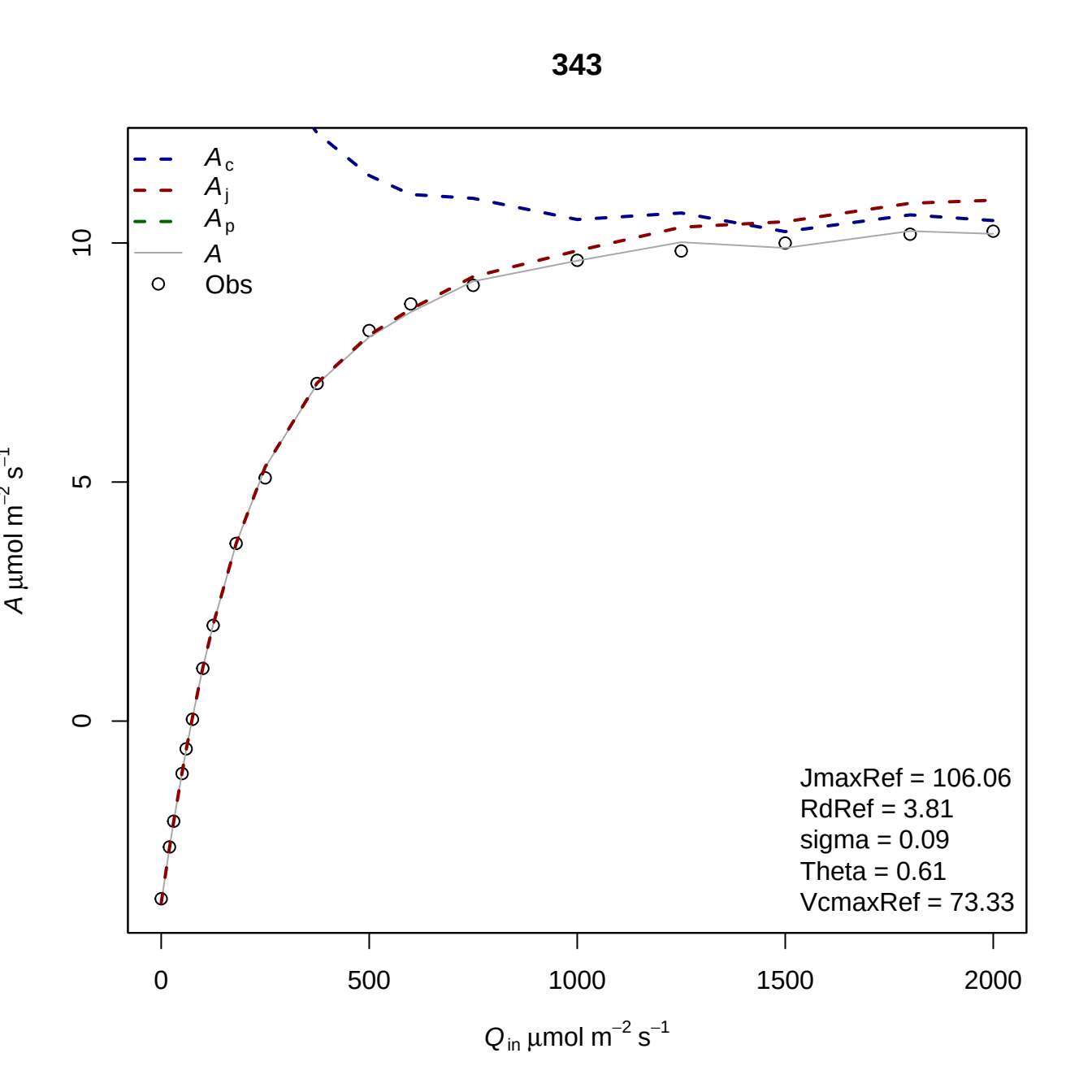

338
A_c
A_j
A_p
A
Obs
JmaxRef = 93.84
RdRef = 1.4
sigma = 0.31
Theta = 0.52
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

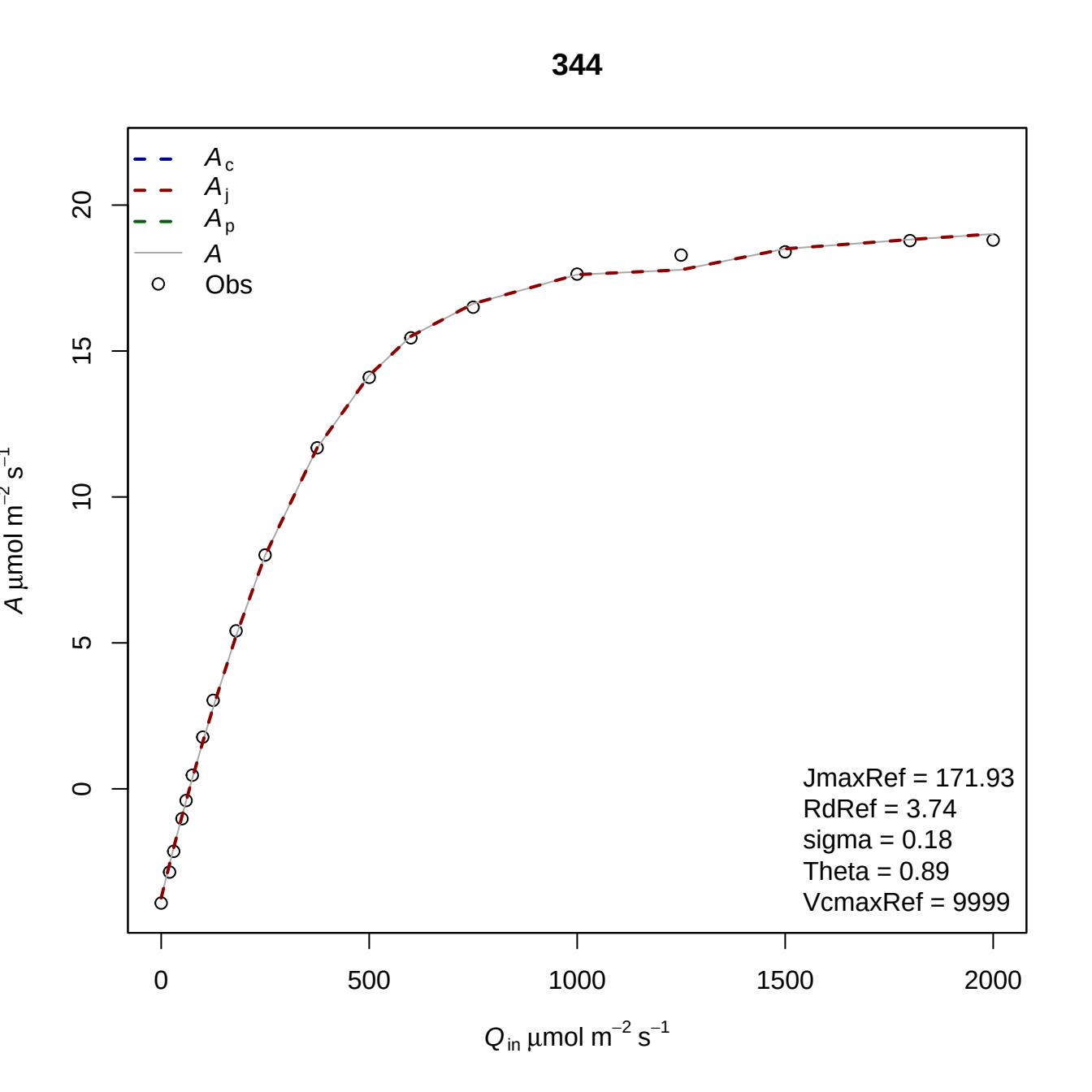

339
A_c
A_j
A_p
A
Obs
JmaxRef = 85.62
RdRef = 1.02
sigma = 0.27
Theta = 0.28
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

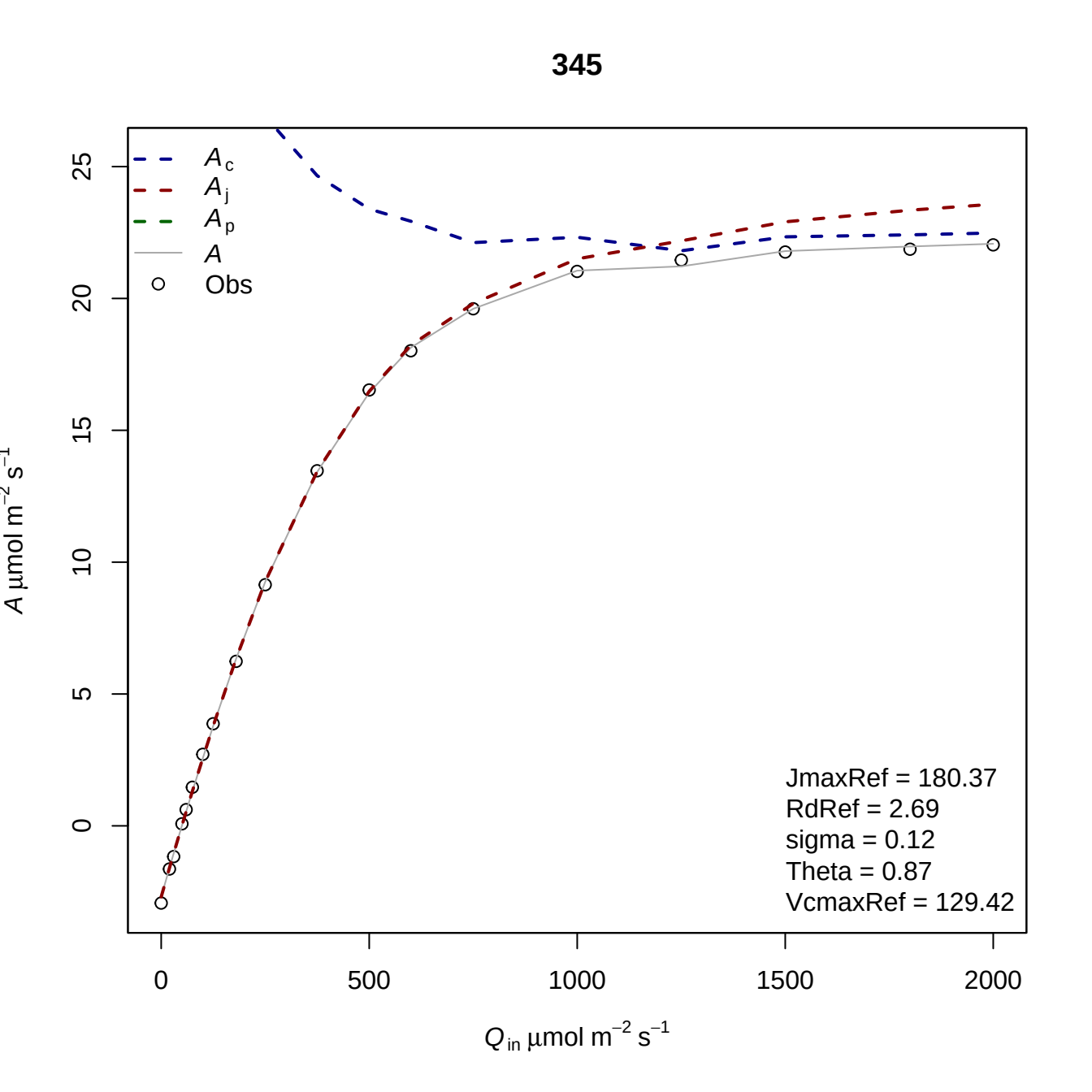

340
A_c
A_j
A_p
A
Obs
JmaxRef = 122.99
RdRef = 3.68
sigma = 0.27
Theta = 0.84
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

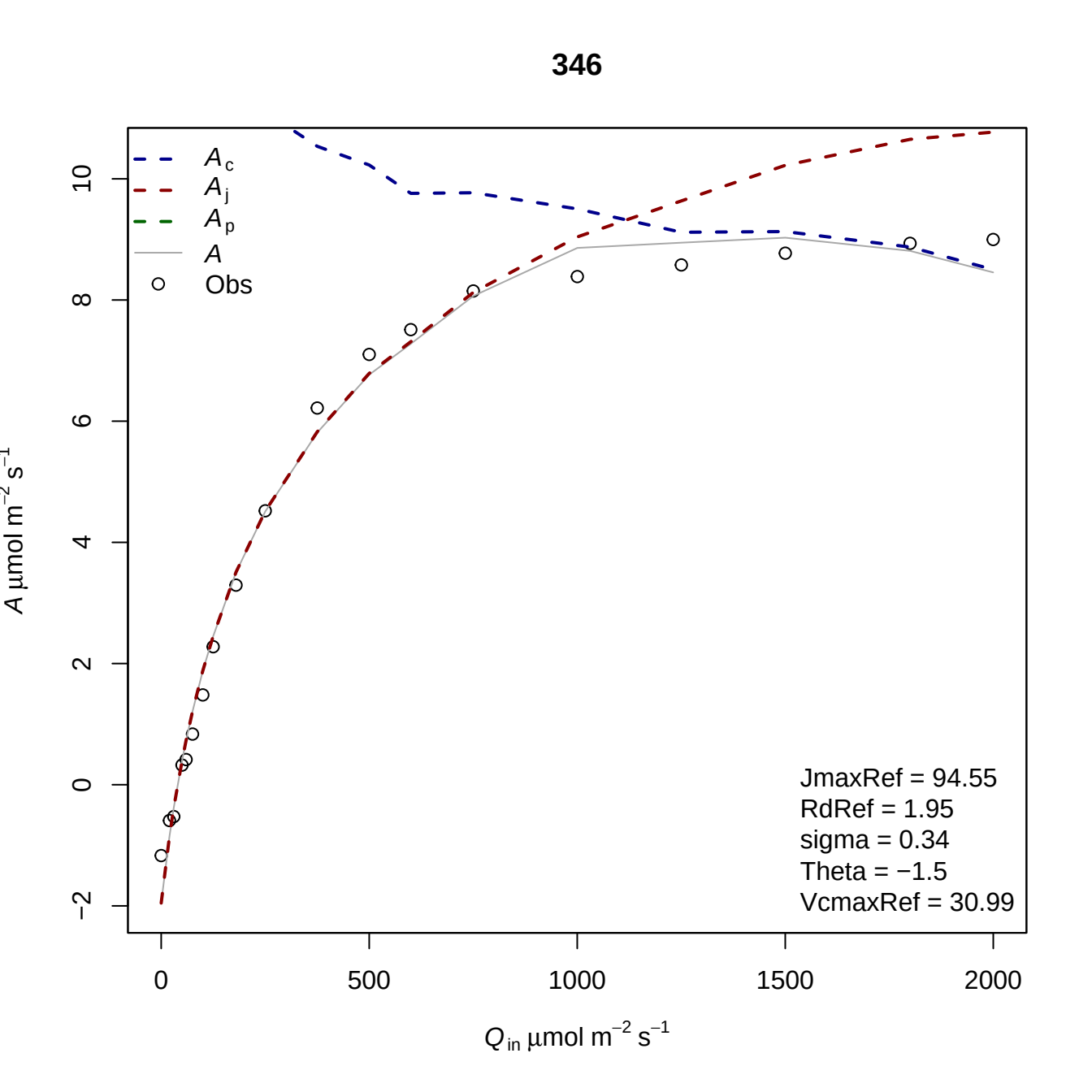

341
A_c
A_j
A_p
A
Obs
JmaxRef = 139.7
RdRef = 2.06
sigma = 0.36
Theta = 0.05
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

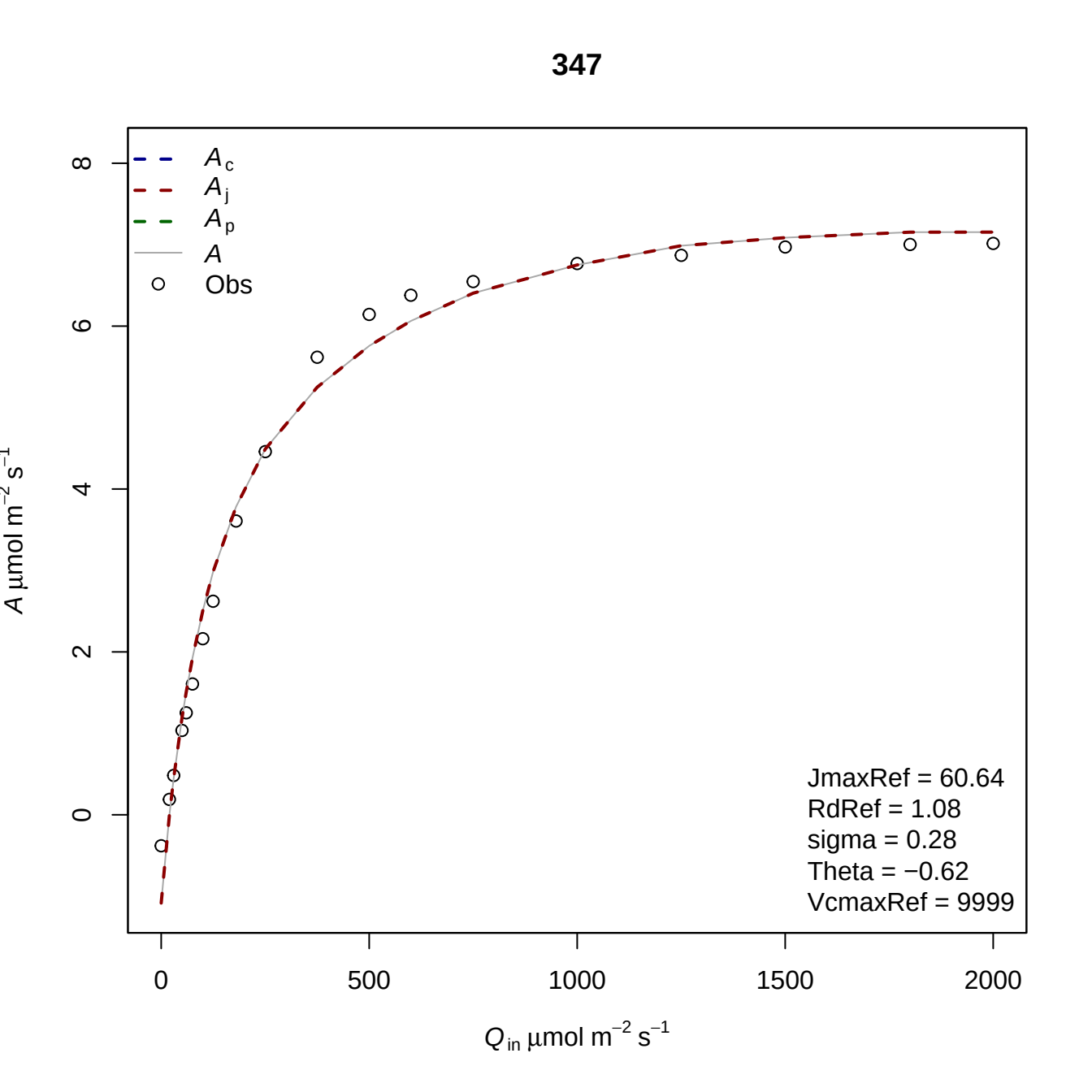

342
A_c
A_j
A_p
A
Obs
JmaxRef = 170.79
RdRef = 2.64
sigma = 0.45
Theta = 0.91
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

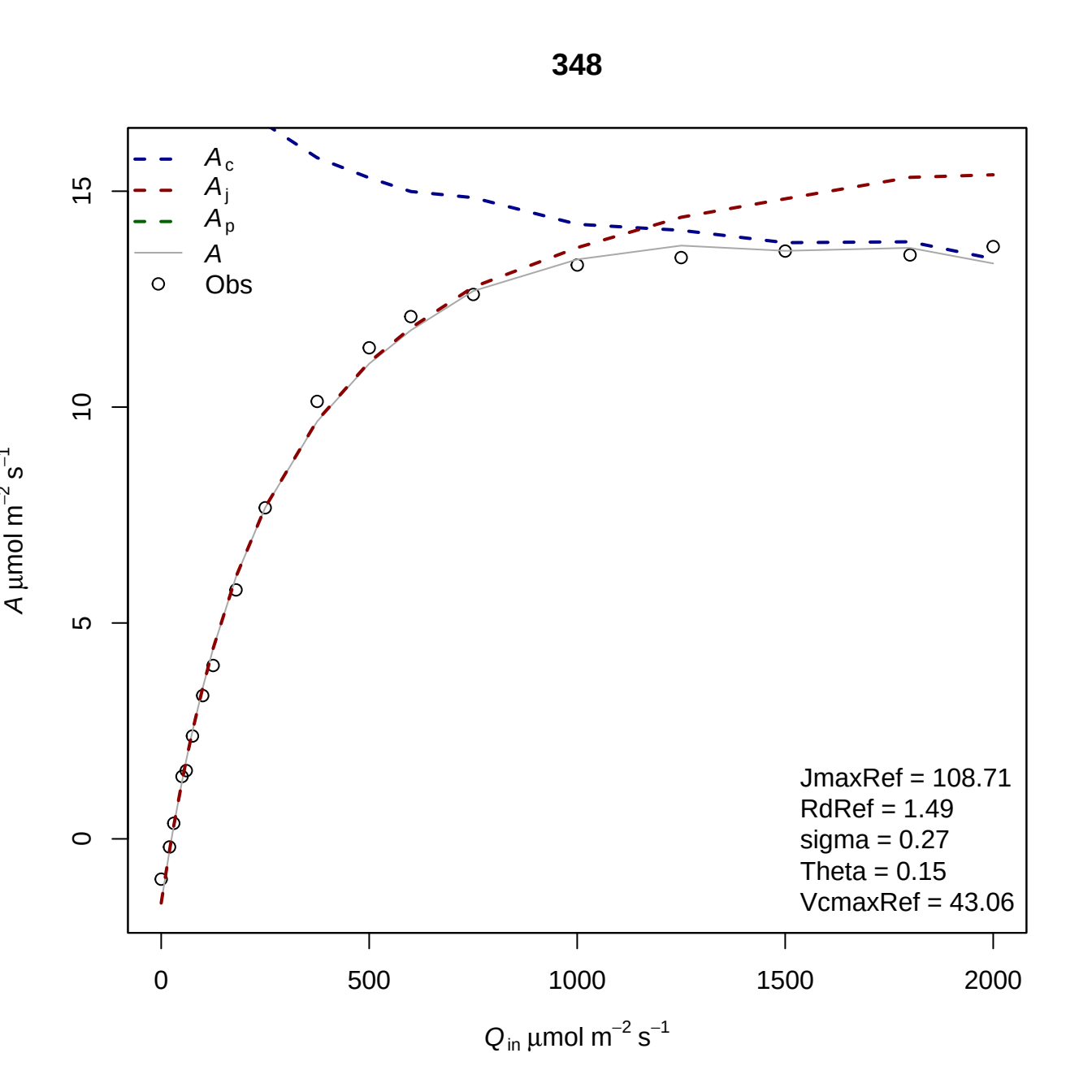

343
A_c
A_j
A_p
A
Obs
JmaxRef = 106.06
RdRef = 3.81
sigma = 0.09
Theta = 0.61
VcmaxRef = 73.33
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

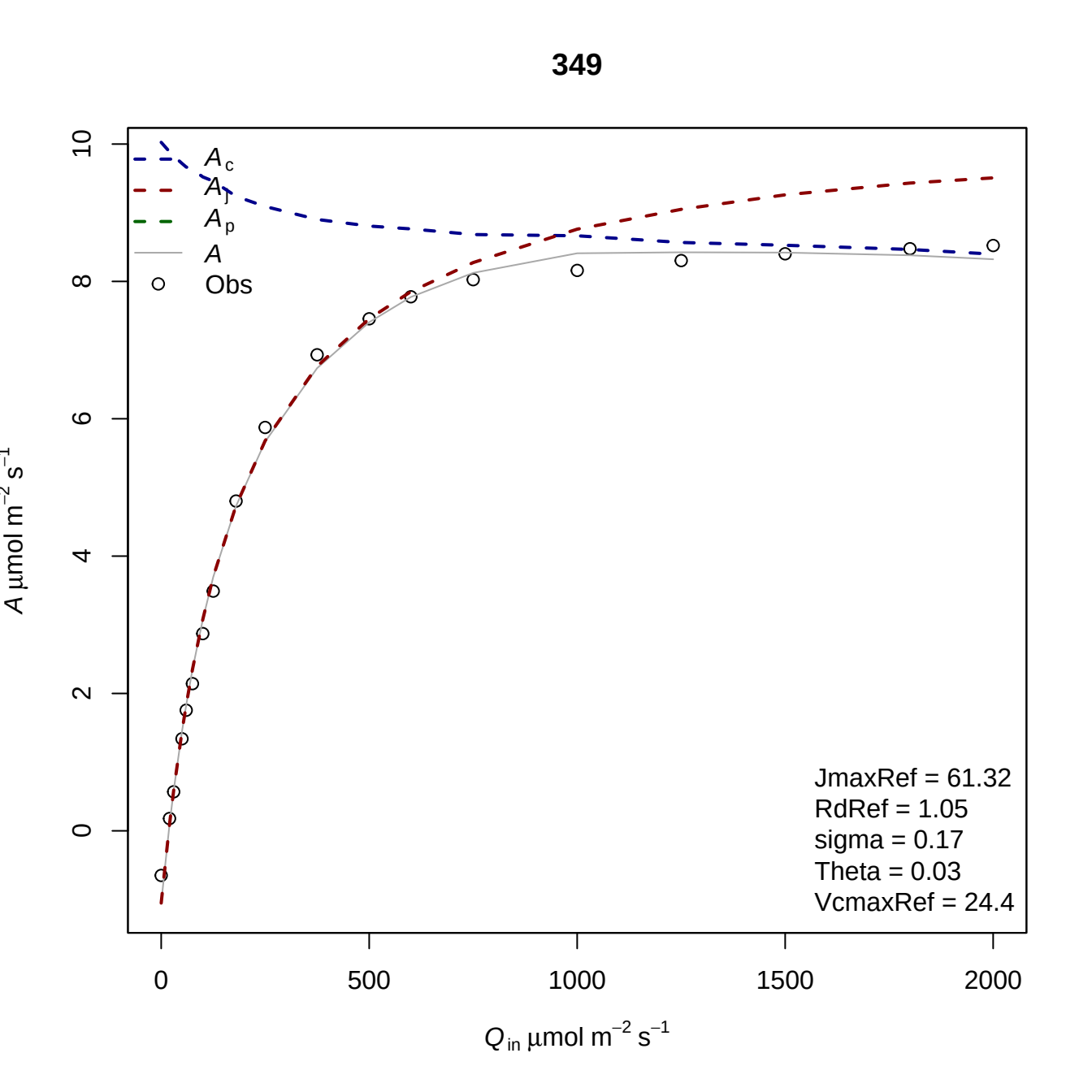

344
A_c
A_j
A_p
A
Obs
JmaxRef = 171.93
RdRef = 3.74
sigma = 0.18
Theta = 0.89
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

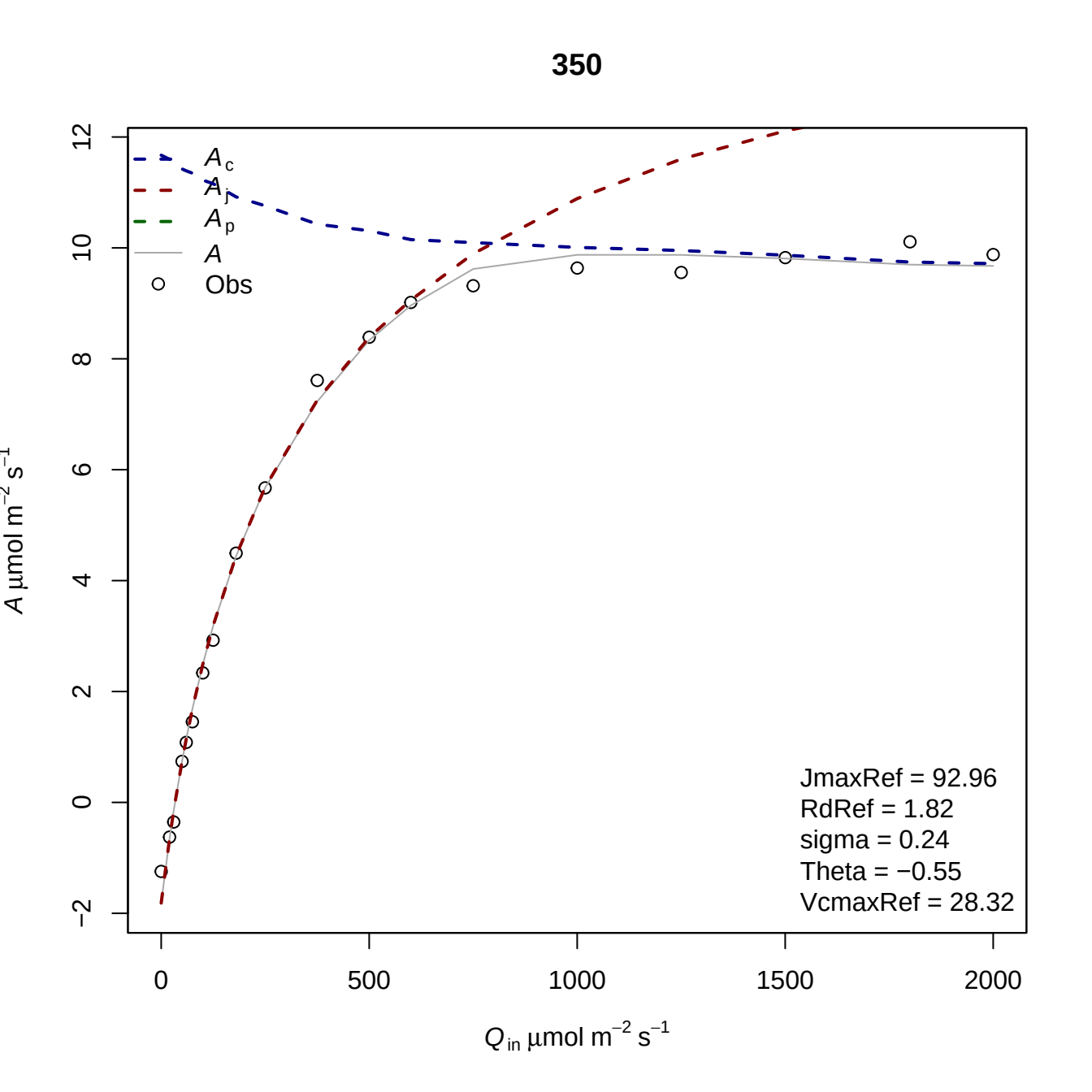

345
A_c
A_j
A_p
A
Obs
JmaxRef = 180.37
RdRef = 2.69
sigma = 0.12
Theta = 0.87
VcmaxRef = 129.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

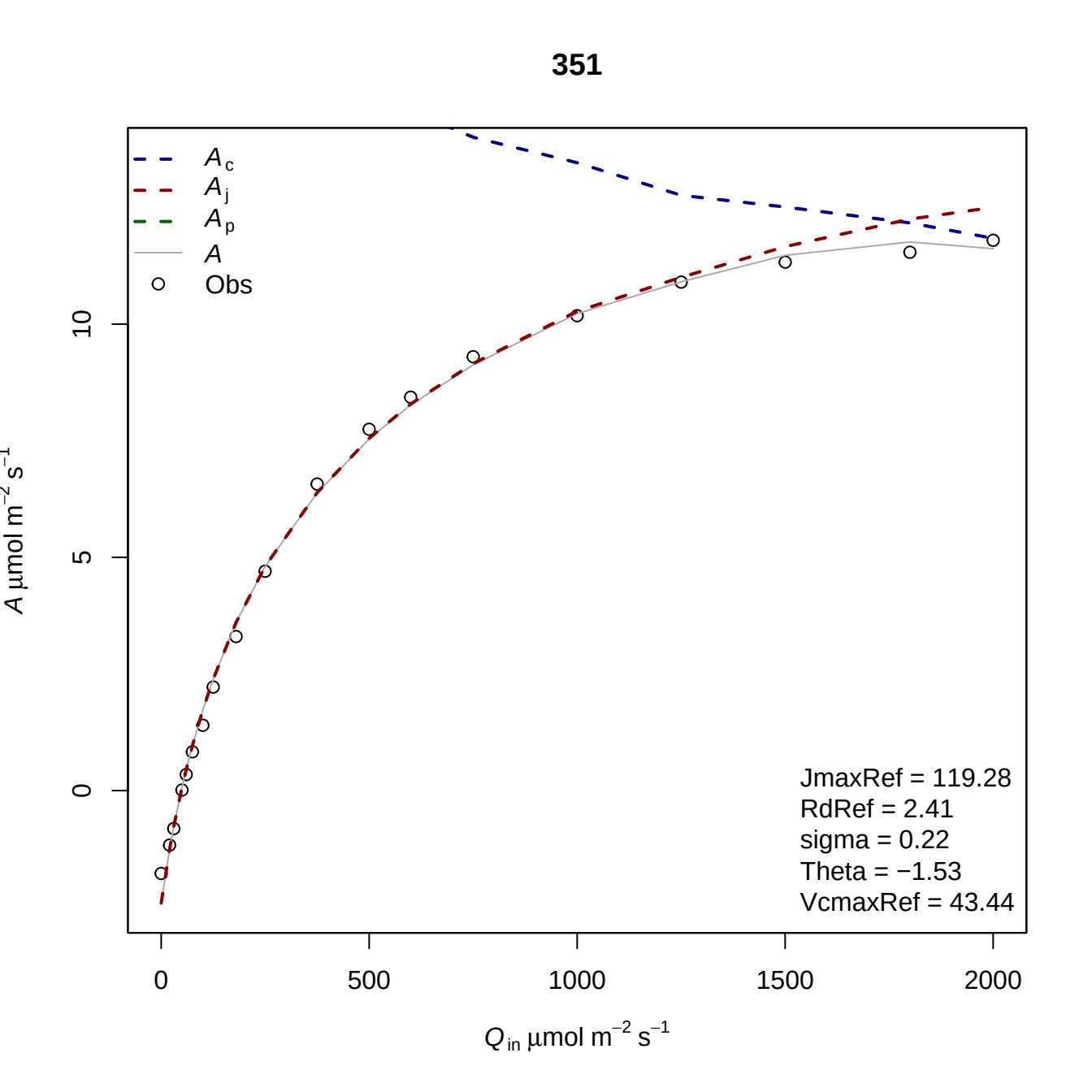

346
A_c
A_j
A_p
A
Obs
JmaxRef = 94.55
RdRef = 1.95
sigma = 0.34
Theta = −1.5
VcmaxRef = 30.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

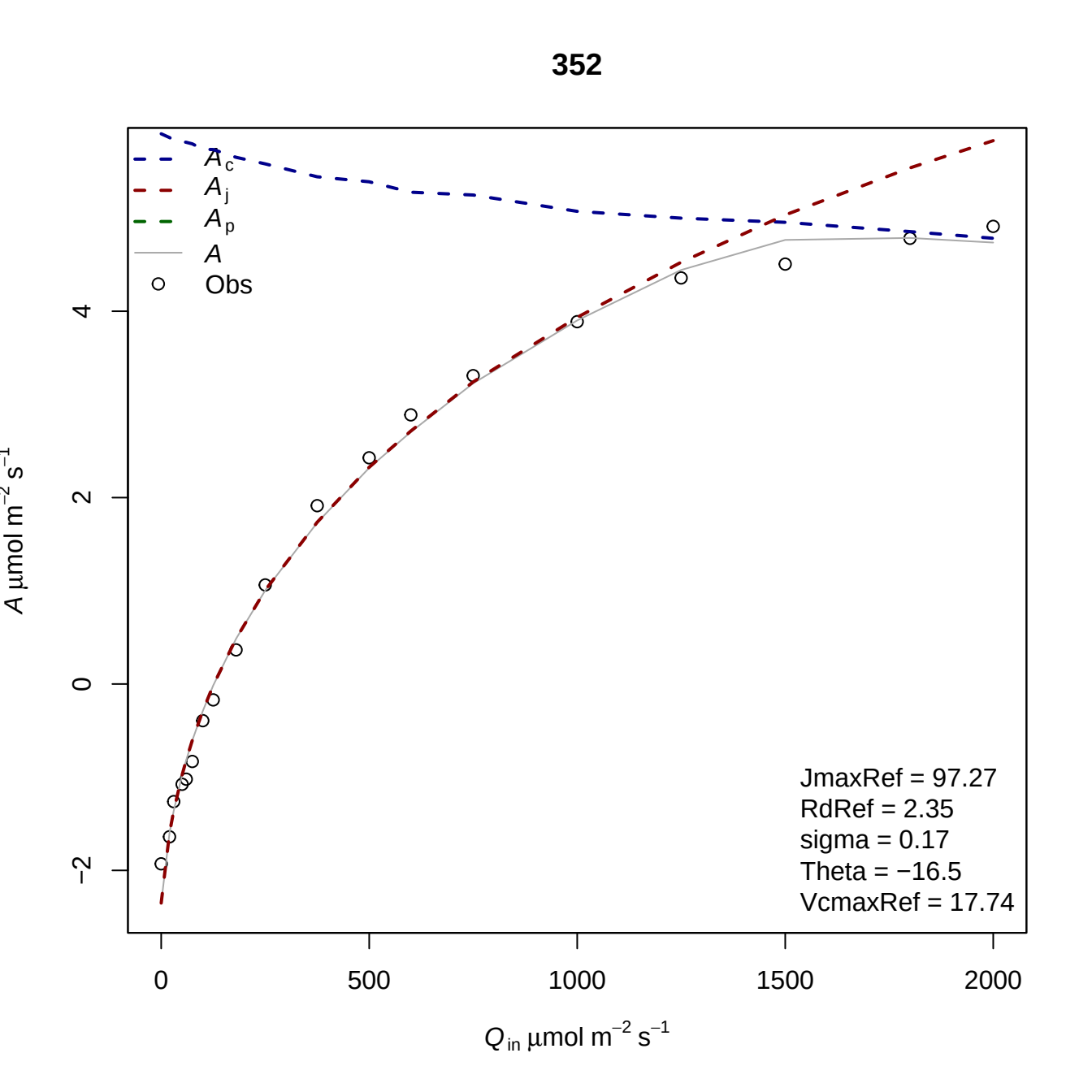

347
A_c
A_j
A_p
A
Obs
JmaxRef = 60.64
RdRef = 1.08
sigma = 0.28
Theta = −0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

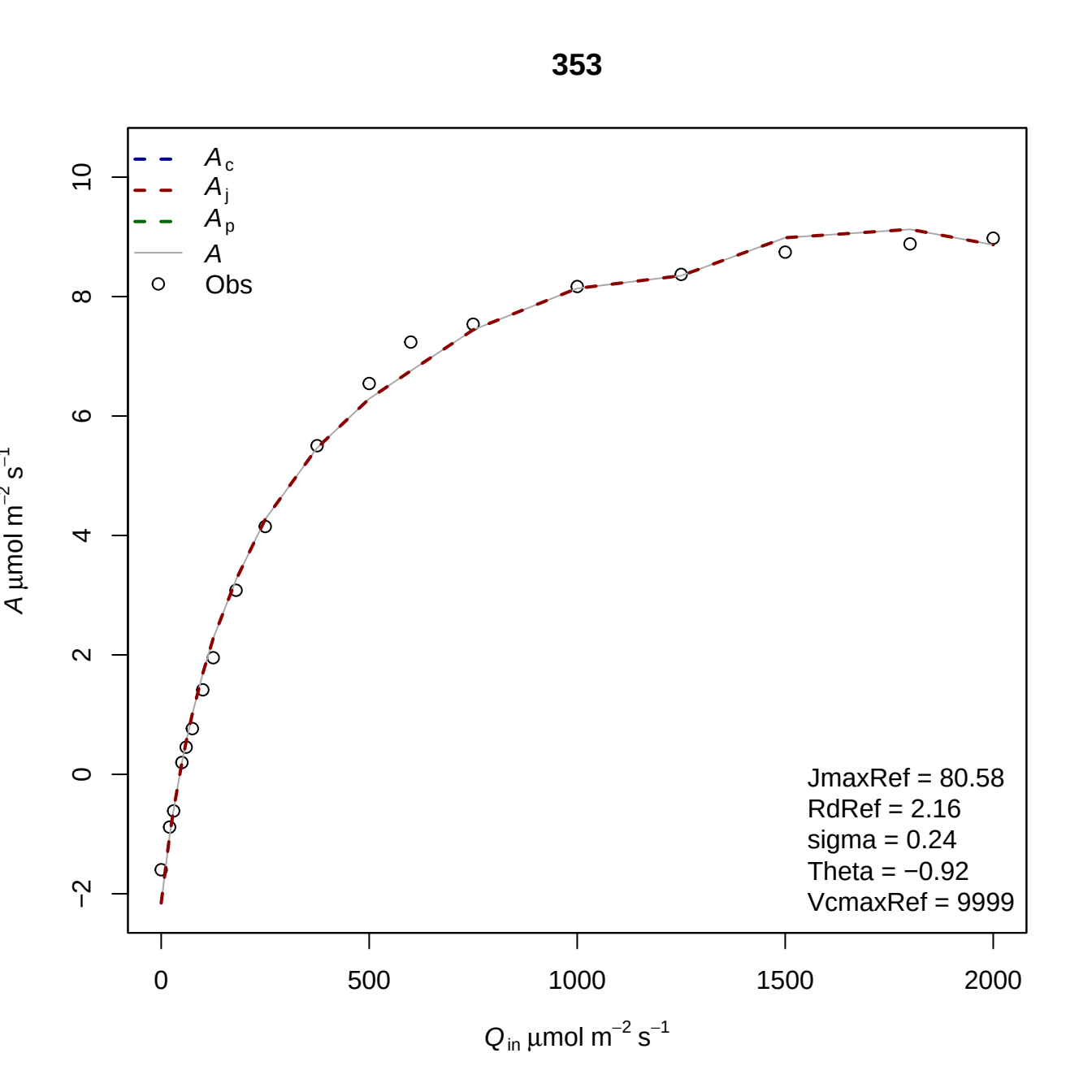

348
A_c
A_j
A_p
A
Obs
JmaxRef = 108.71
RdRef = 1.49
sigma = 0.27
Theta = 0.15
VcmaxRef = 43.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

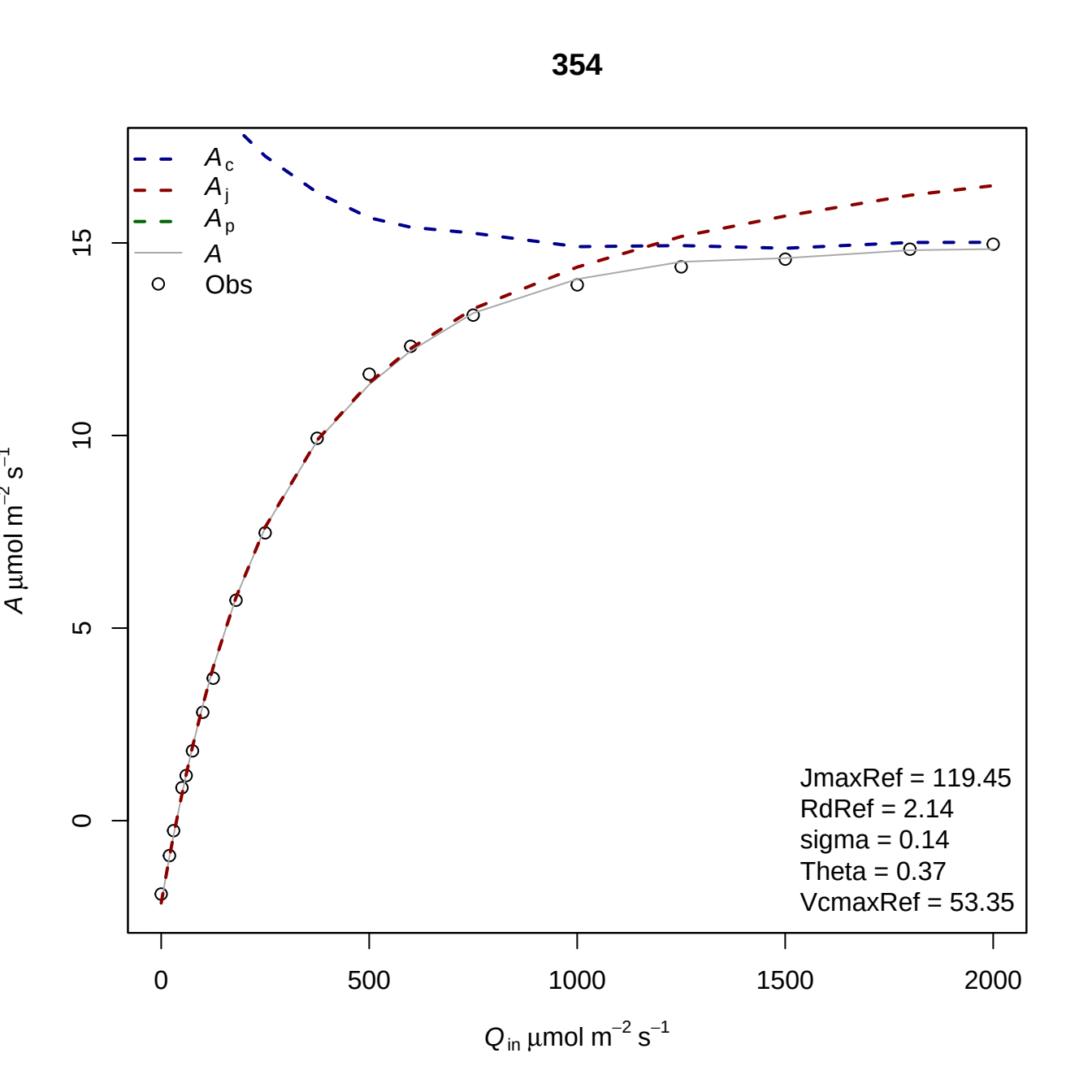

349
A_c
A_j
A_p
A
Obs
JmaxRef = 61.32
RdRef = 1.05
sigma = 0.17
Theta = 0.03
VcmaxRef = 24.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

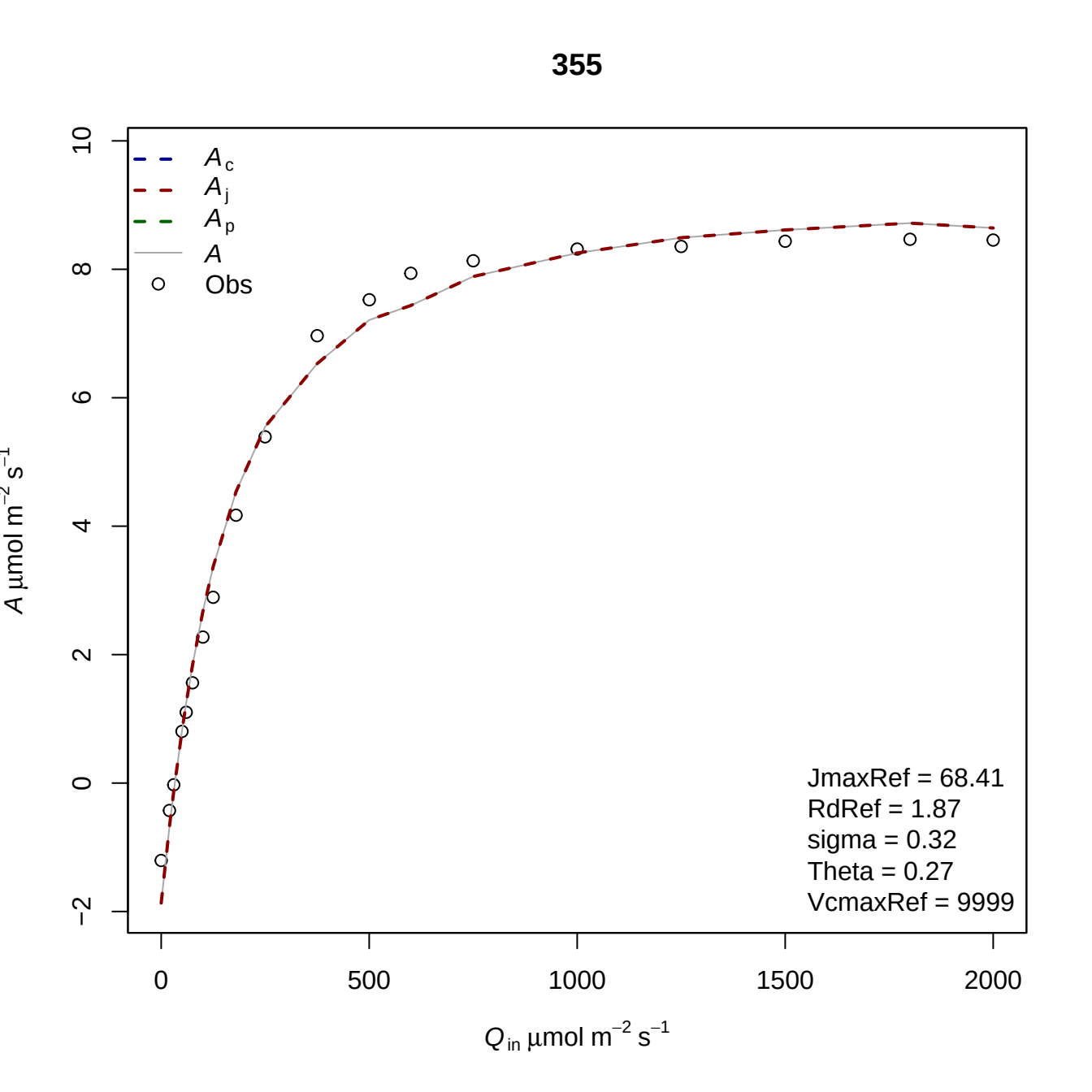

350
A_c
A_j
A_p
A
Obs
JmaxRef = 92.96
RdRef = 1.82
sigma = 0.24
Theta = −0.55
VcmaxRef = 28.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

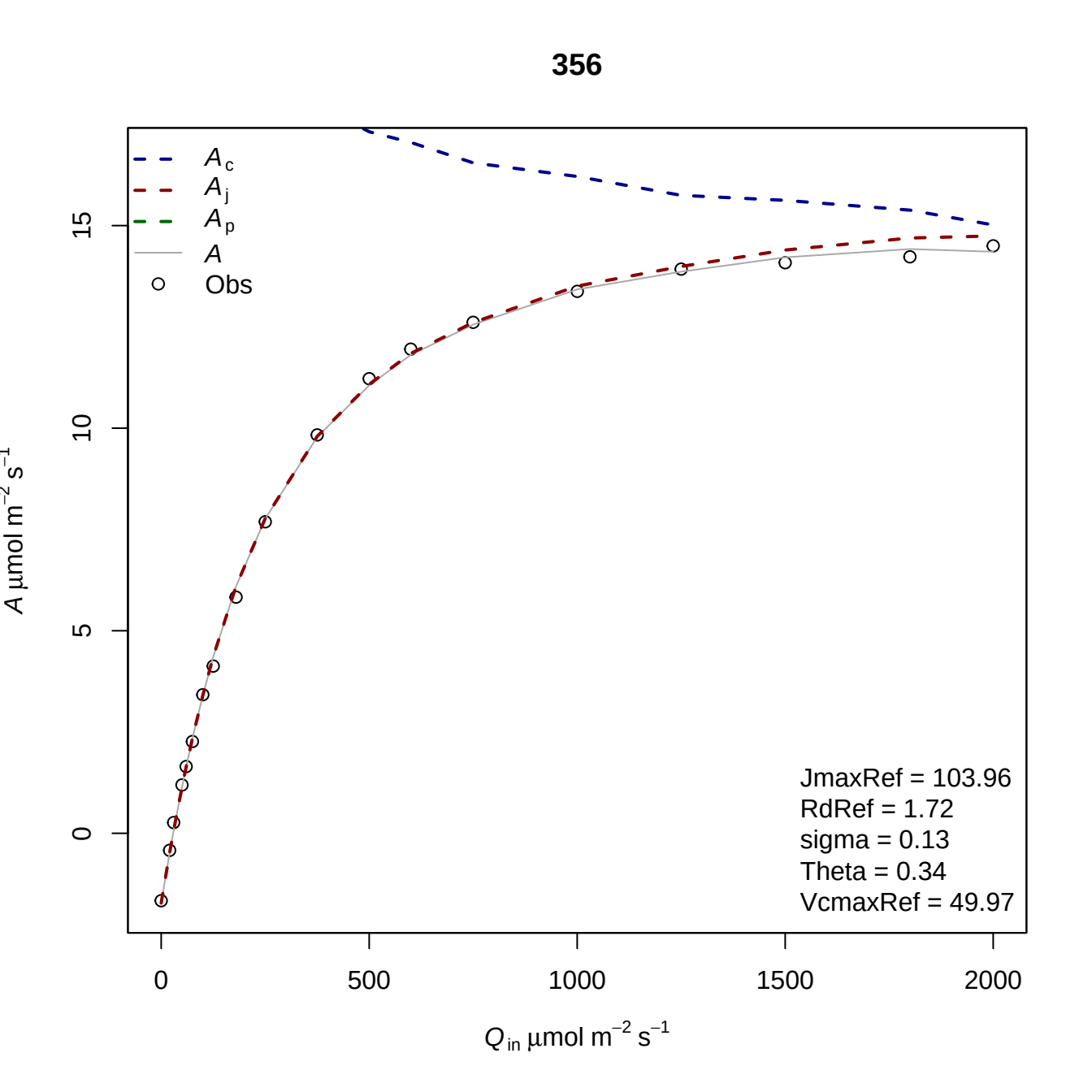

351
A_c
A_j
A_p
A
Obs
JmaxRef = 119.28
RdRef = 2.41
sigma = 0.22
Theta = −1.53
VcmaxRef = 43.44
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

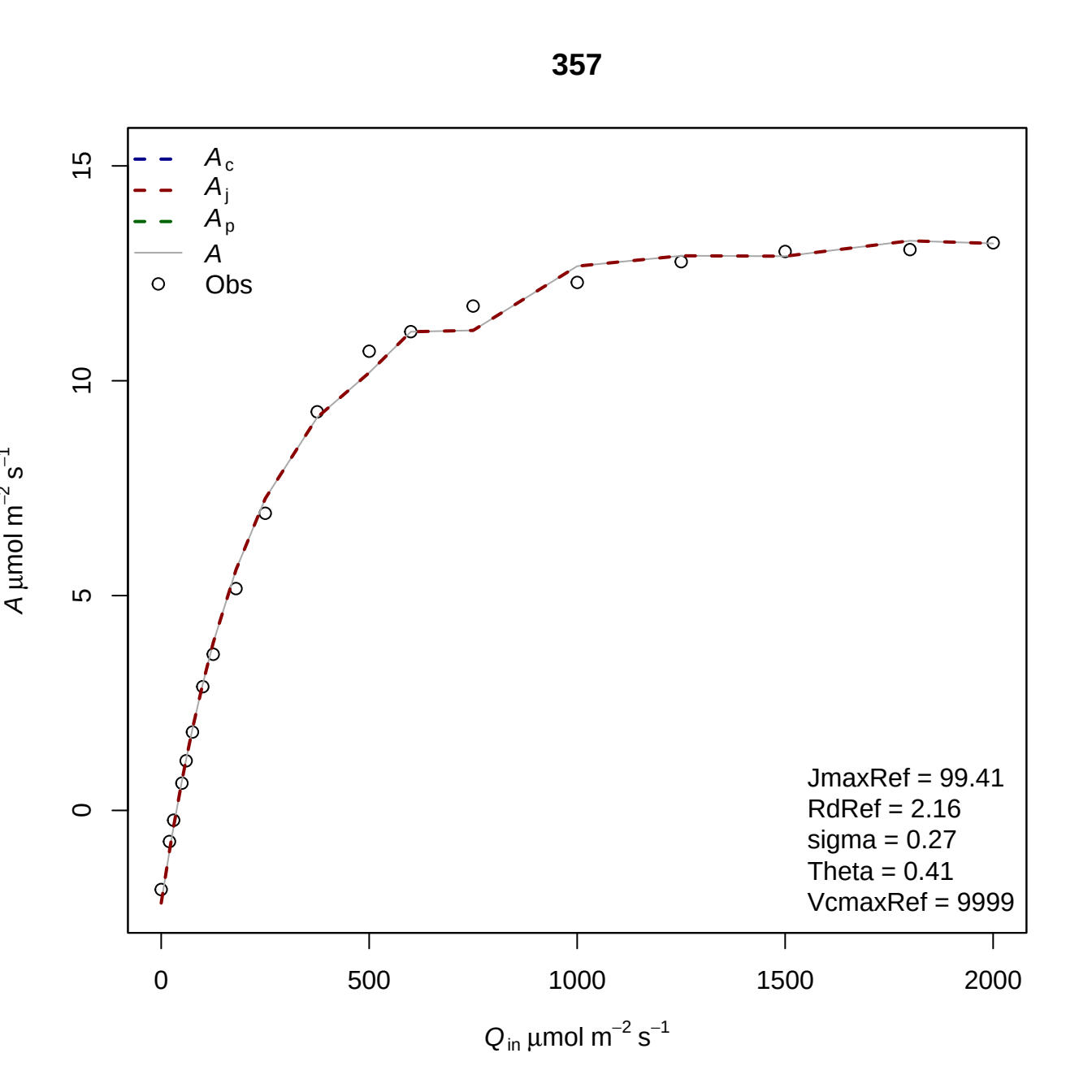

352
A_c
A_j
A_p
A
Obs
JmaxRef = 97.27
RdRef = 2.35
sigma = 0.17
Theta = −16.5
VcmaxRef = 17.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

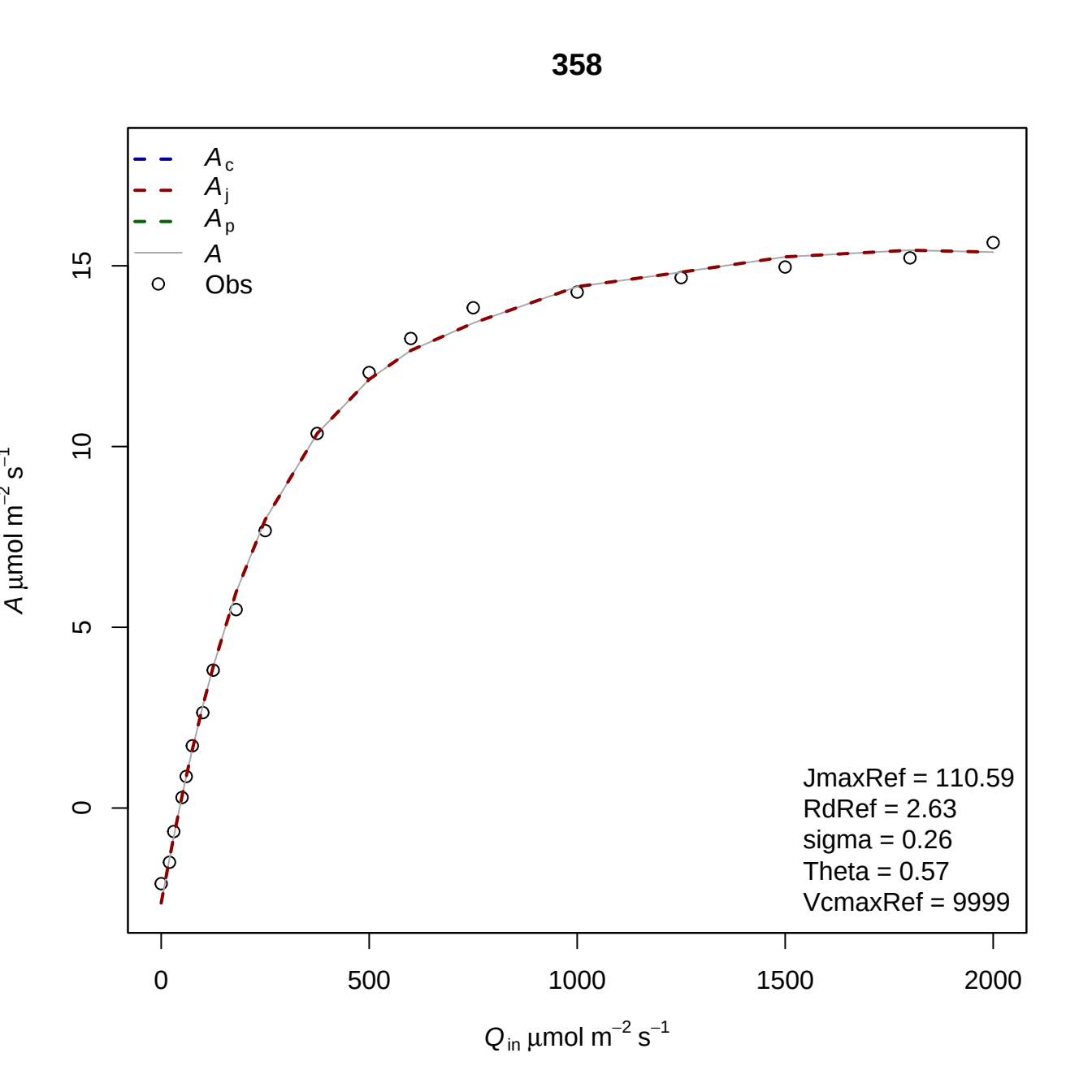

353
A_c
A_j
A_p
A
Obs
JmaxRef = 80.58
RdRef = 2.16
sigma = 0.24
Theta = −0.92
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

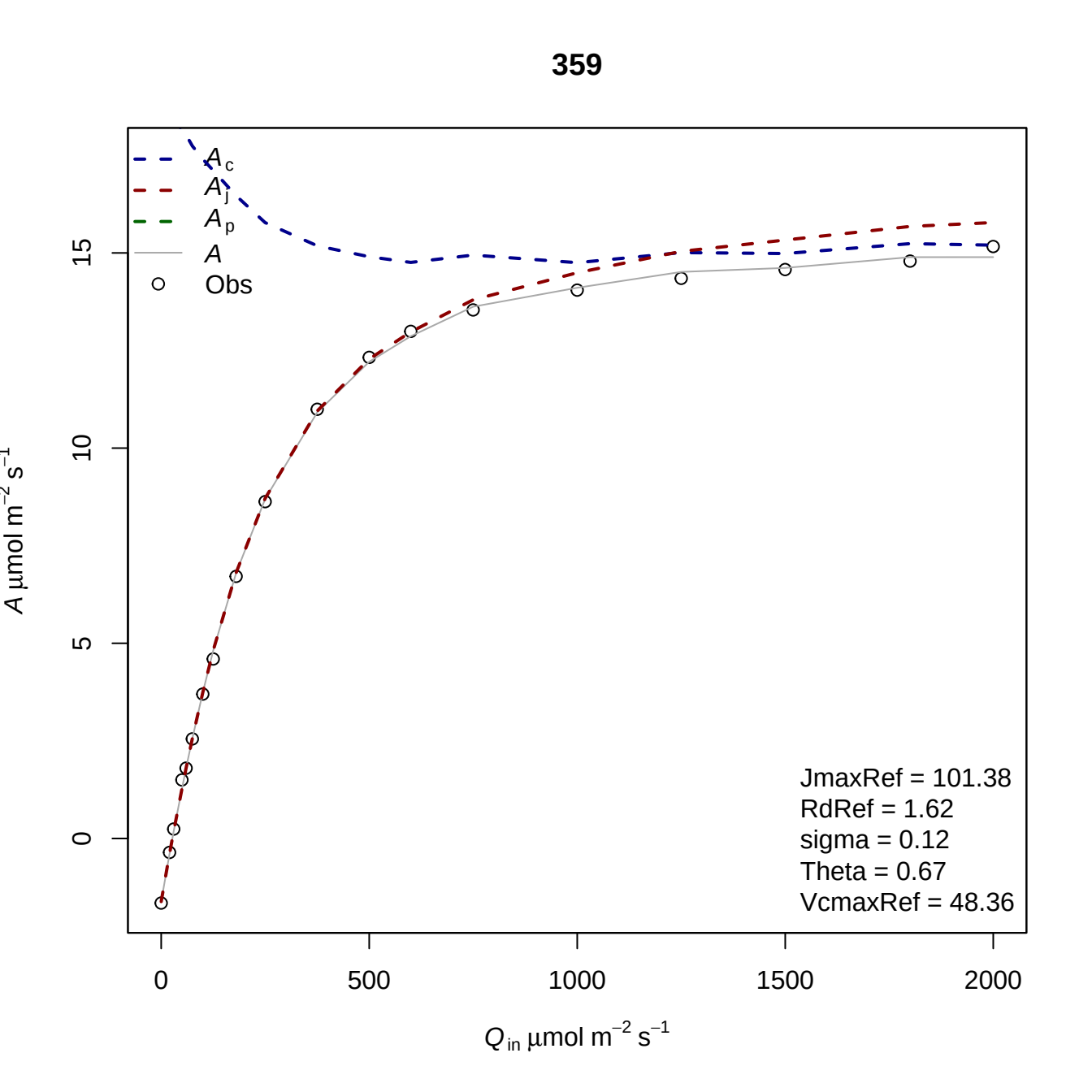

354
A_c
A_j
A_p
A
Obs
JmaxRef = 119.45
RdRef = 2.14
sigma = 0.14
Theta = 0.37
VcmaxRef = 53.35
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

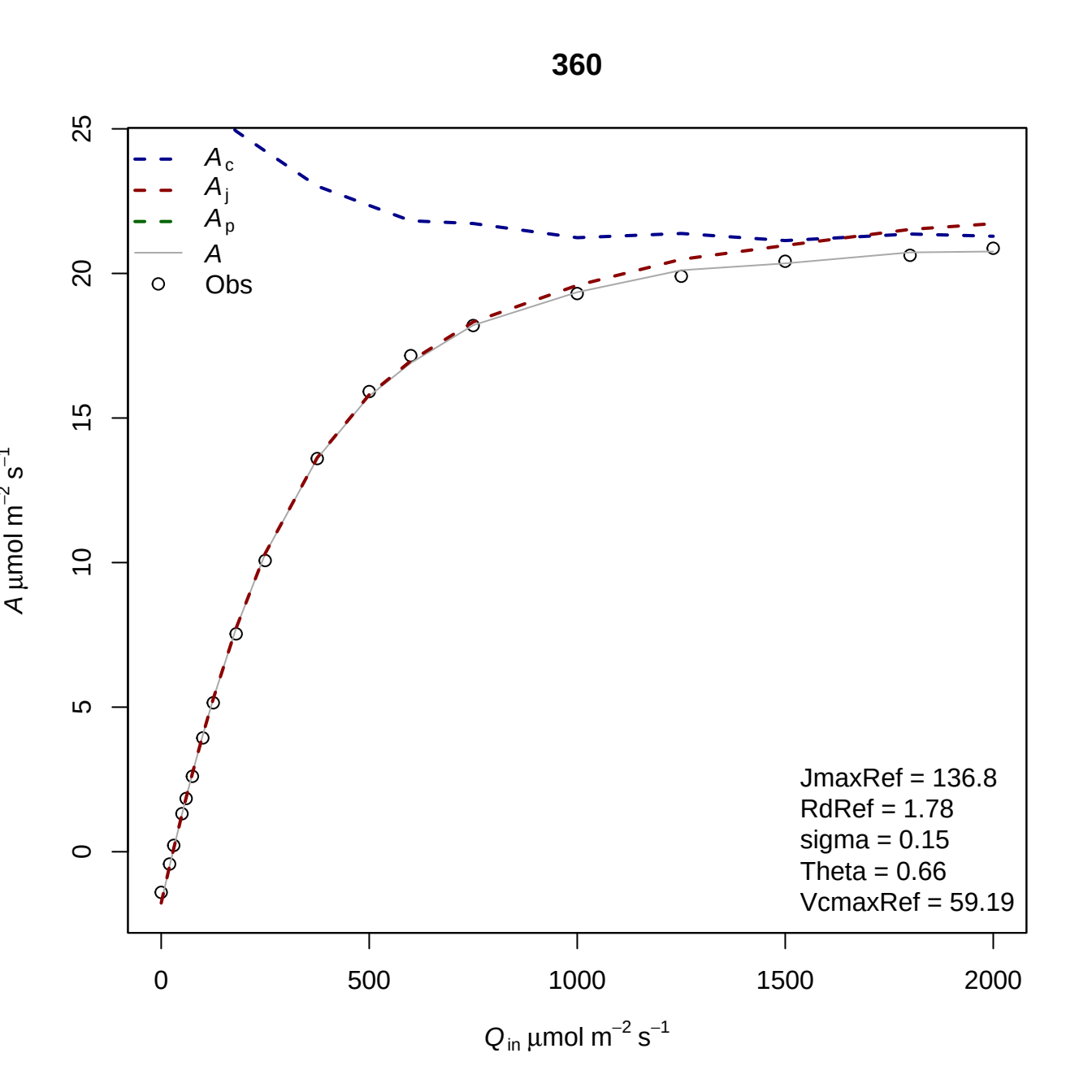

355
A_c
A_j
A_p
A
Obs
JmaxRef = 68.41
RdRef = 1.87
sigma = 0.32
Theta = 0.27
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

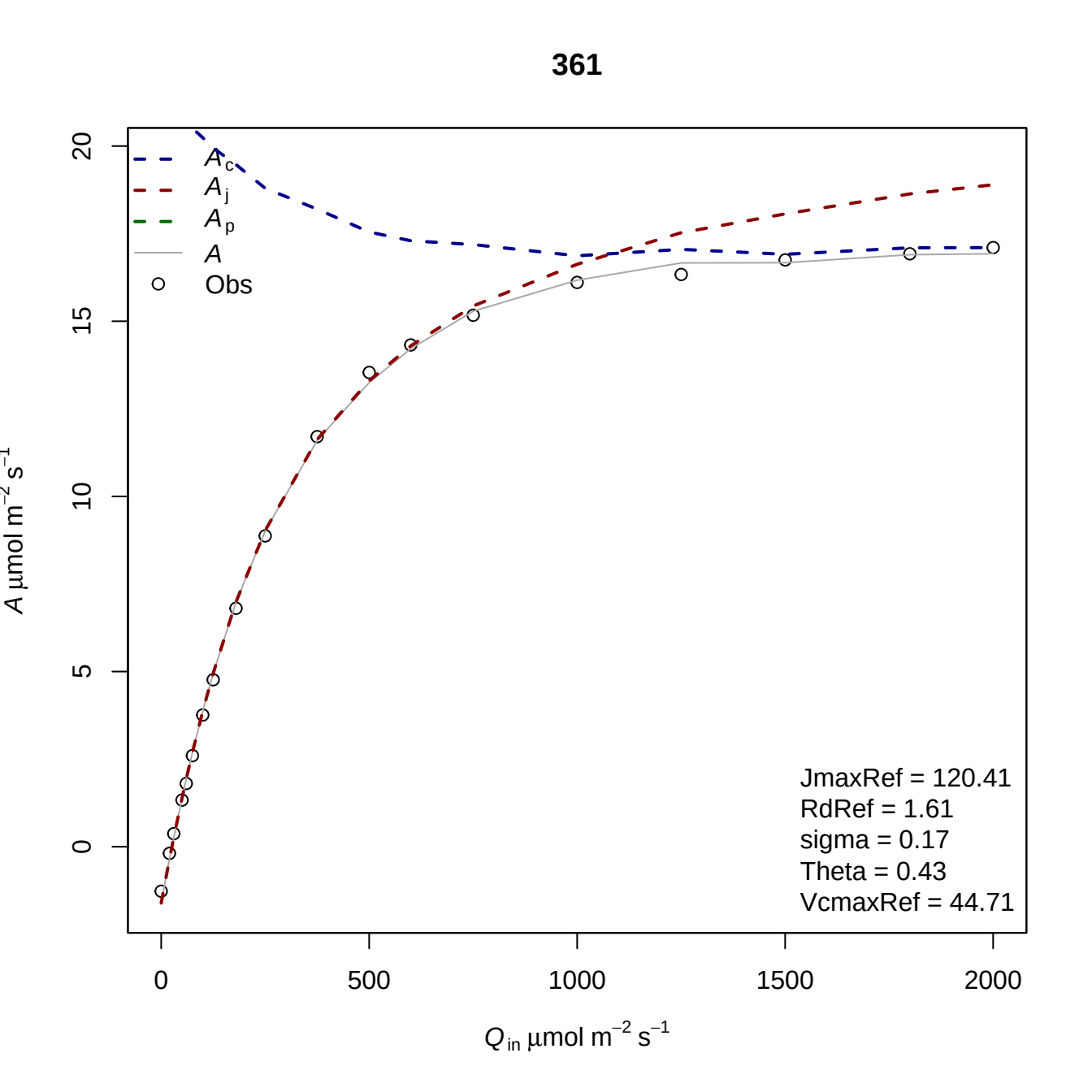

356
A_c
A_j
A_p
A
Obs
JmaxRef = 103.96
RdRef = 1.72
sigma = 0.13
Theta = 0.34
VcmaxRef = 49.97
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

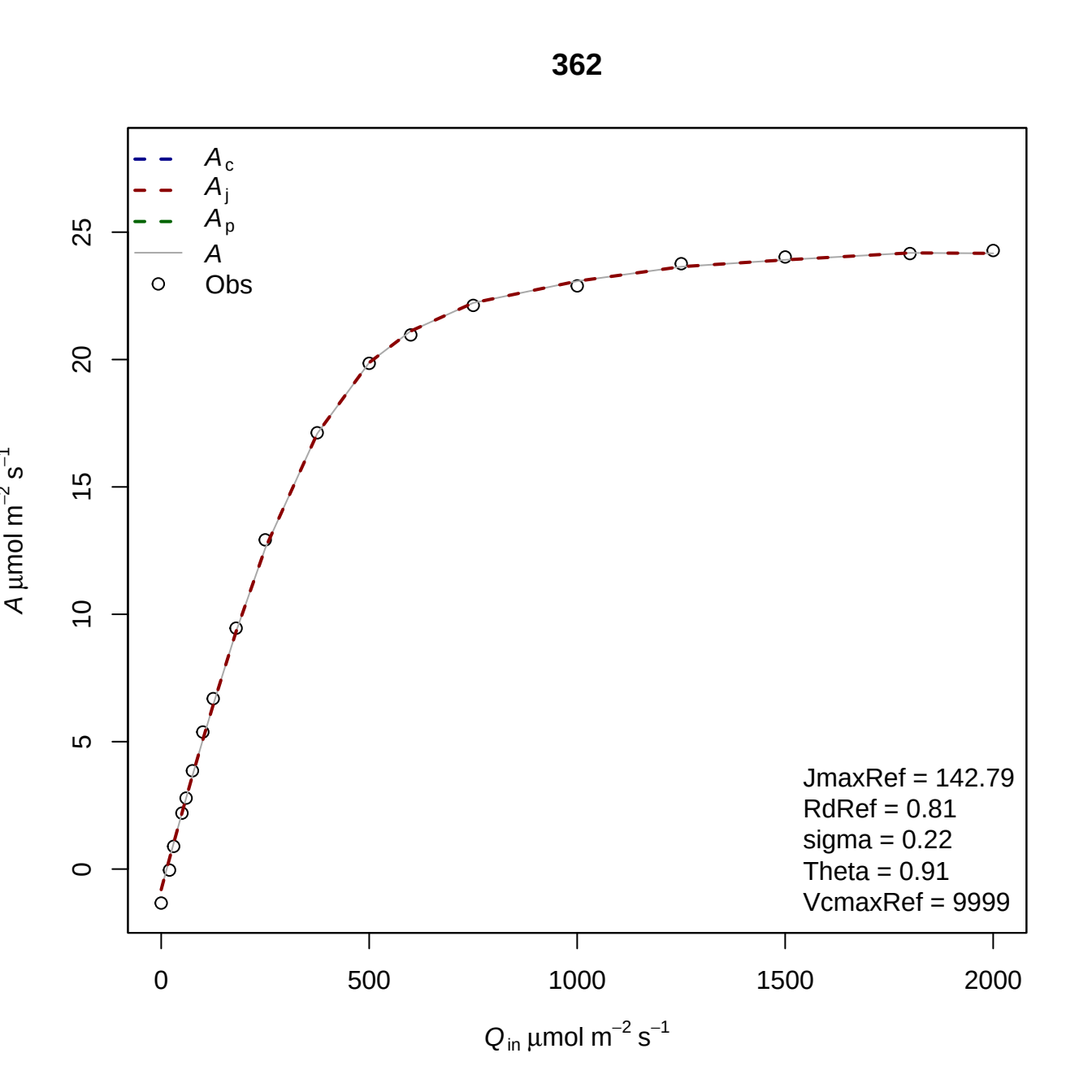

357
A_c
A_j
A_p
A
Obs
JmaxRef = 99.41
RdRef = 2.16
sigma = 0.27
Theta = 0.41
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

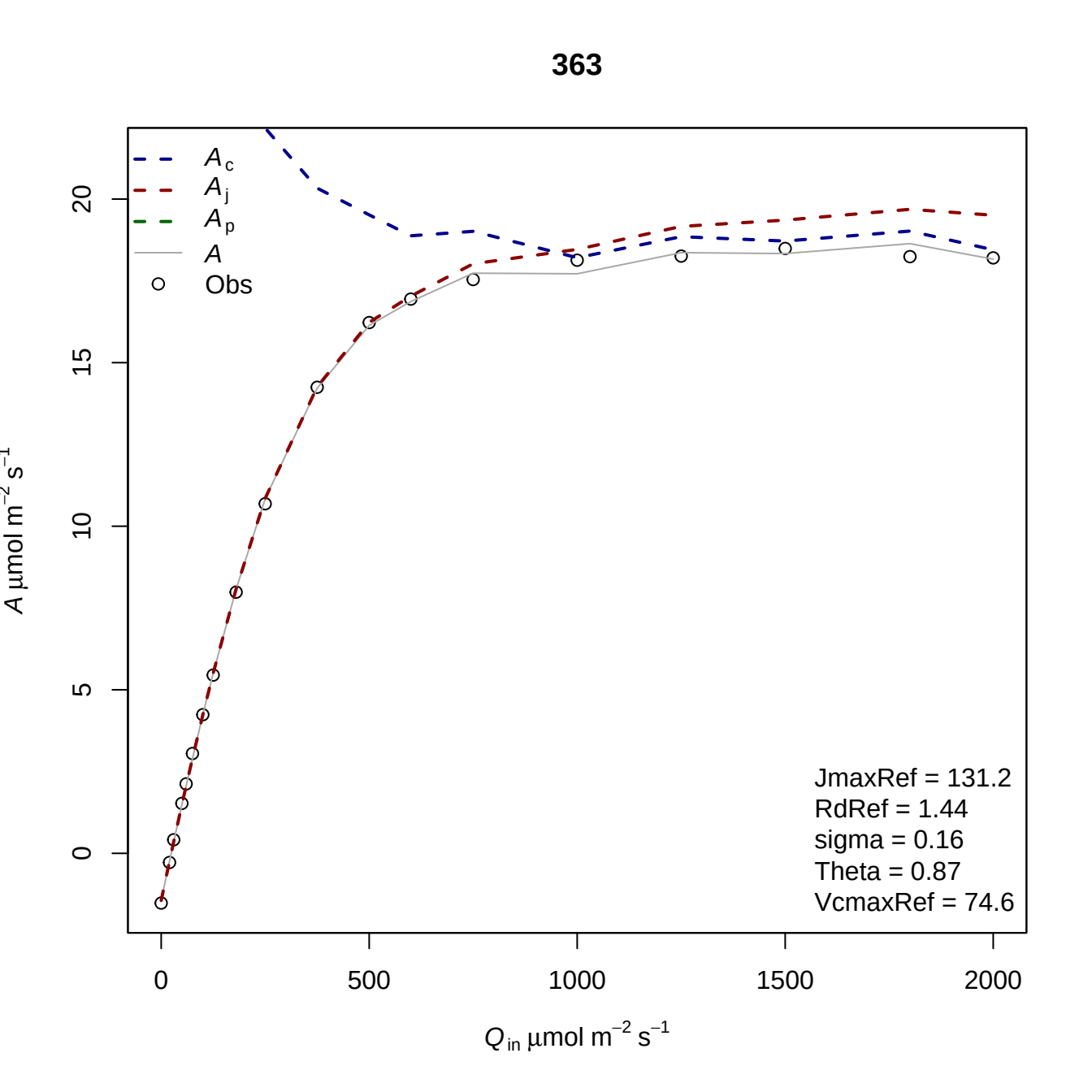

358
A_c
A_j
A_p
A
Obs
JmaxRef = 110.59
RdRef = 2.63
sigma = 0.26
Theta = 0.57
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

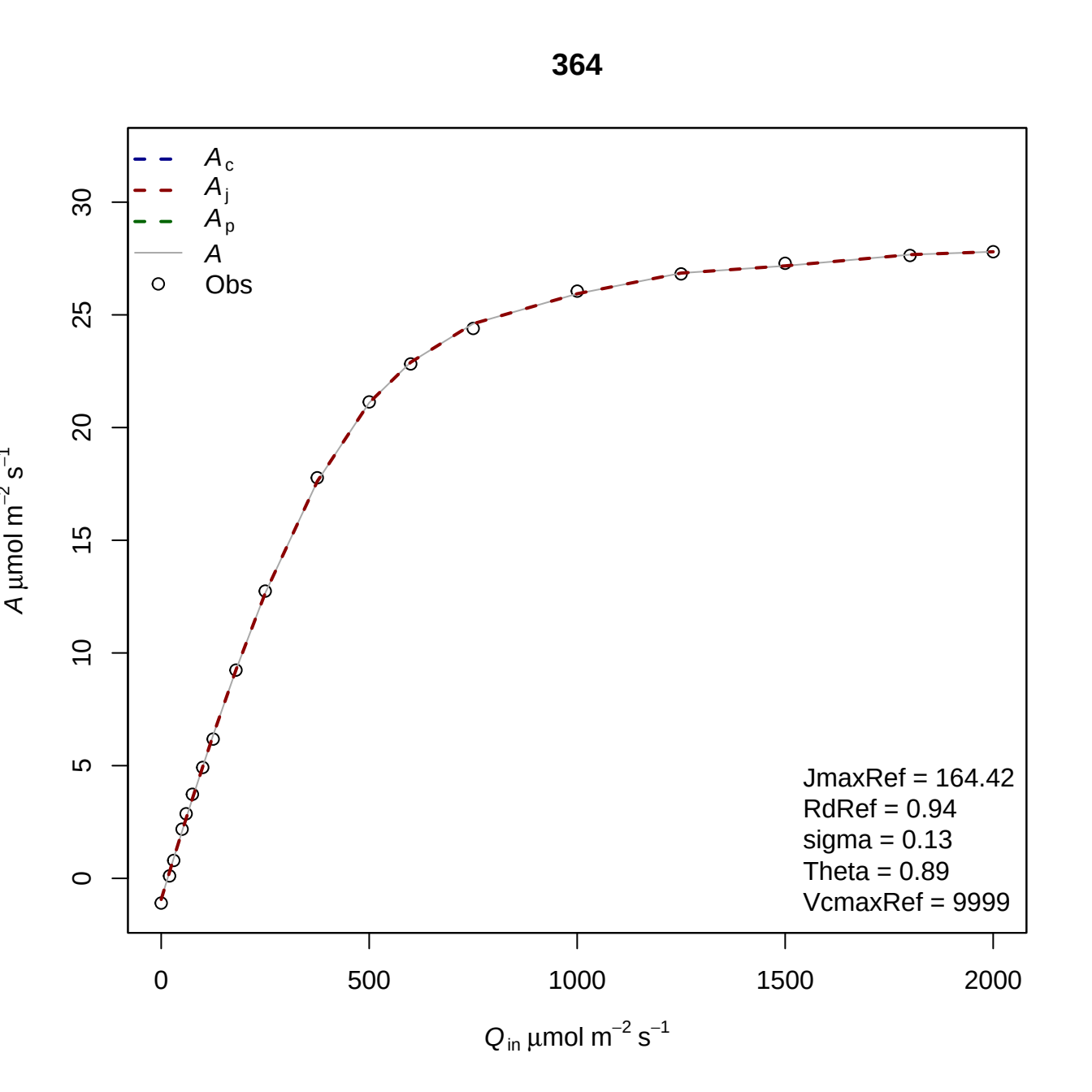

359
A_c
A_j
A_p
A
Obs
JmaxRef = 101.38
RdRef = 1.62
sigma = 0.12
Theta = 0.67
VcmaxRef = 48.36
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

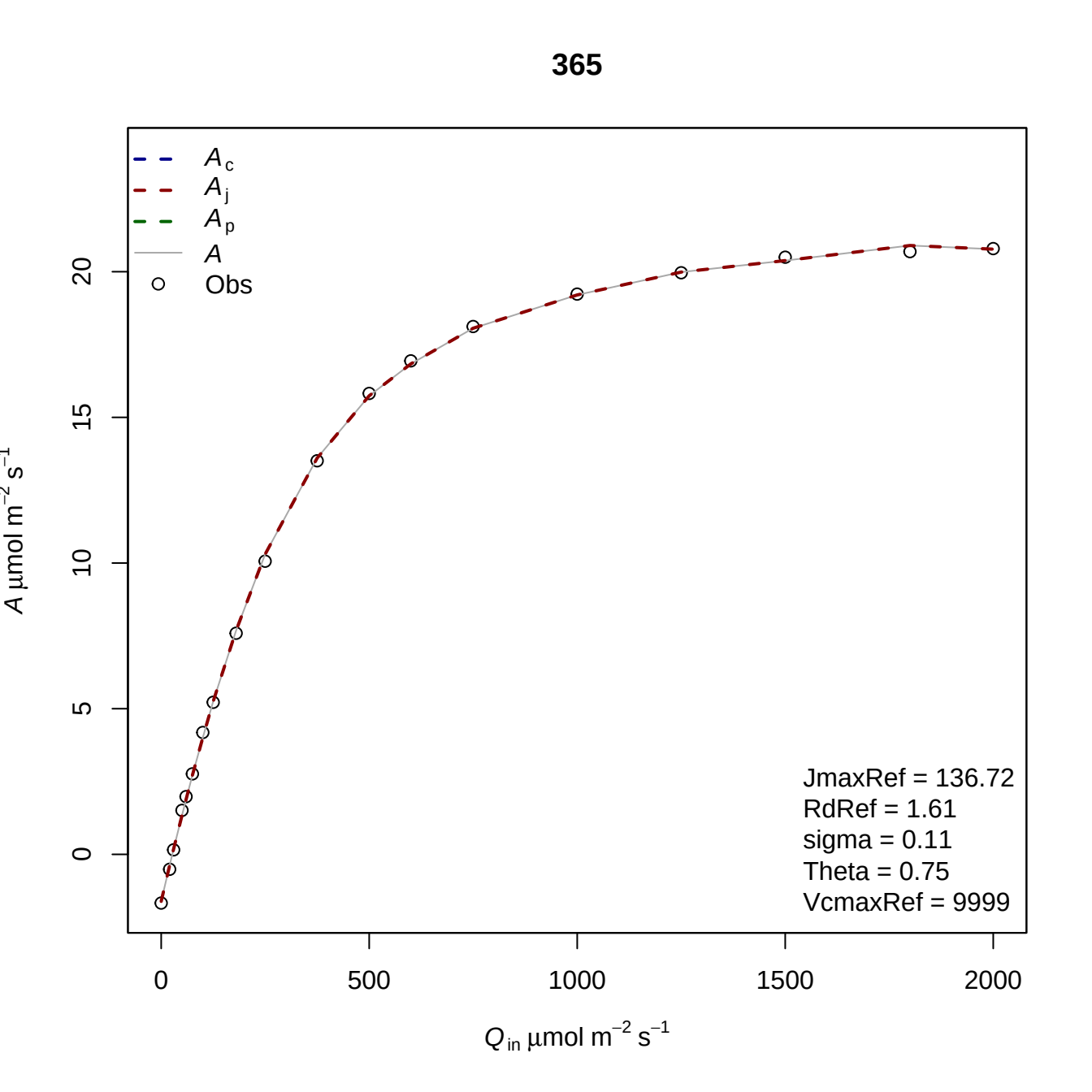

360
A_c
A_j
A_p
A
Obs
JmaxRef = 136.8
RdRef = 1.78
sigma = 0.15
Theta = 0.66
VcmaxRef = 59.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

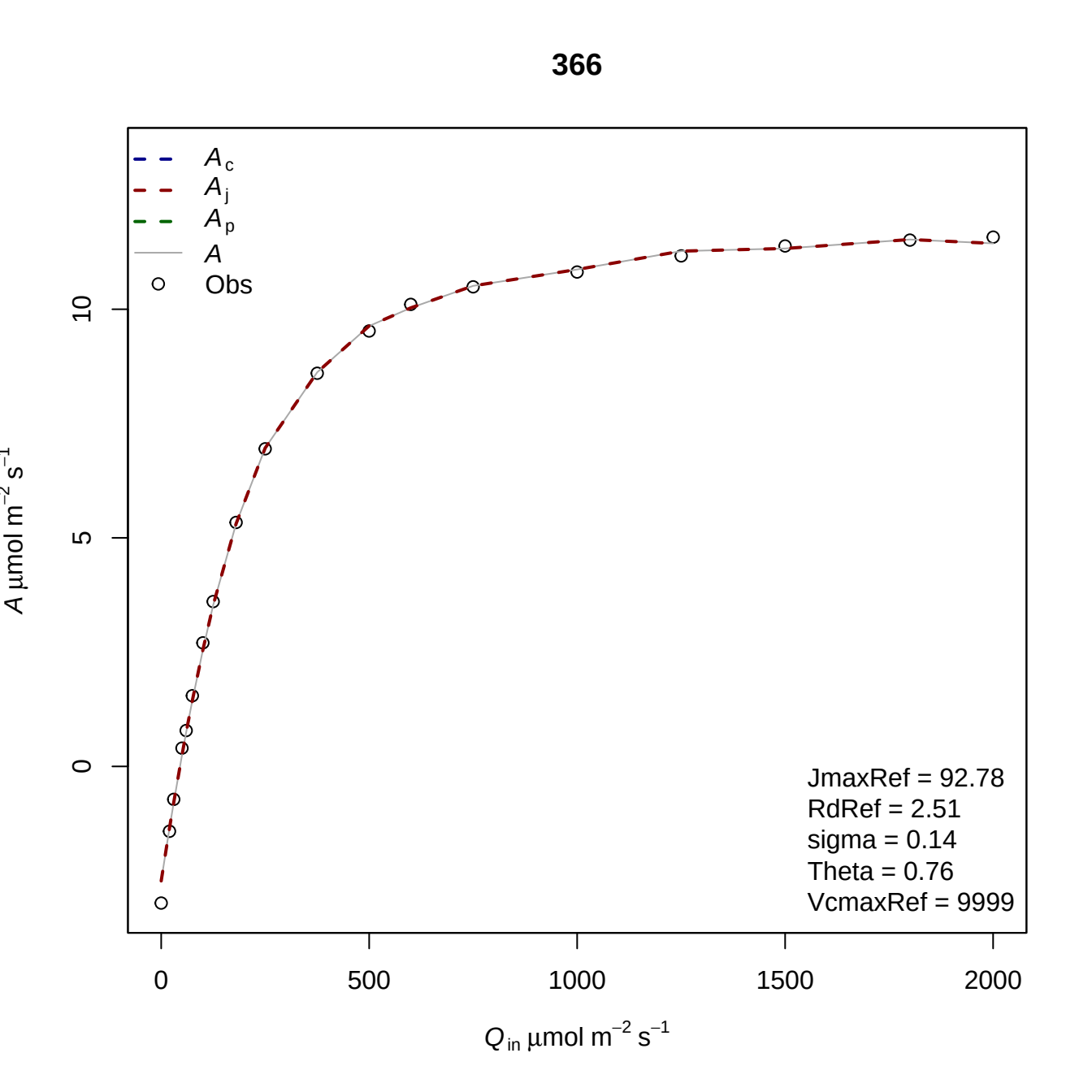

361
A_c
A_j
A_p
A
Obs
JmaxRef = 120.41
RdRef = 1.61
sigma = 0.17
Theta = 0.43
VcmaxRef = 44.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

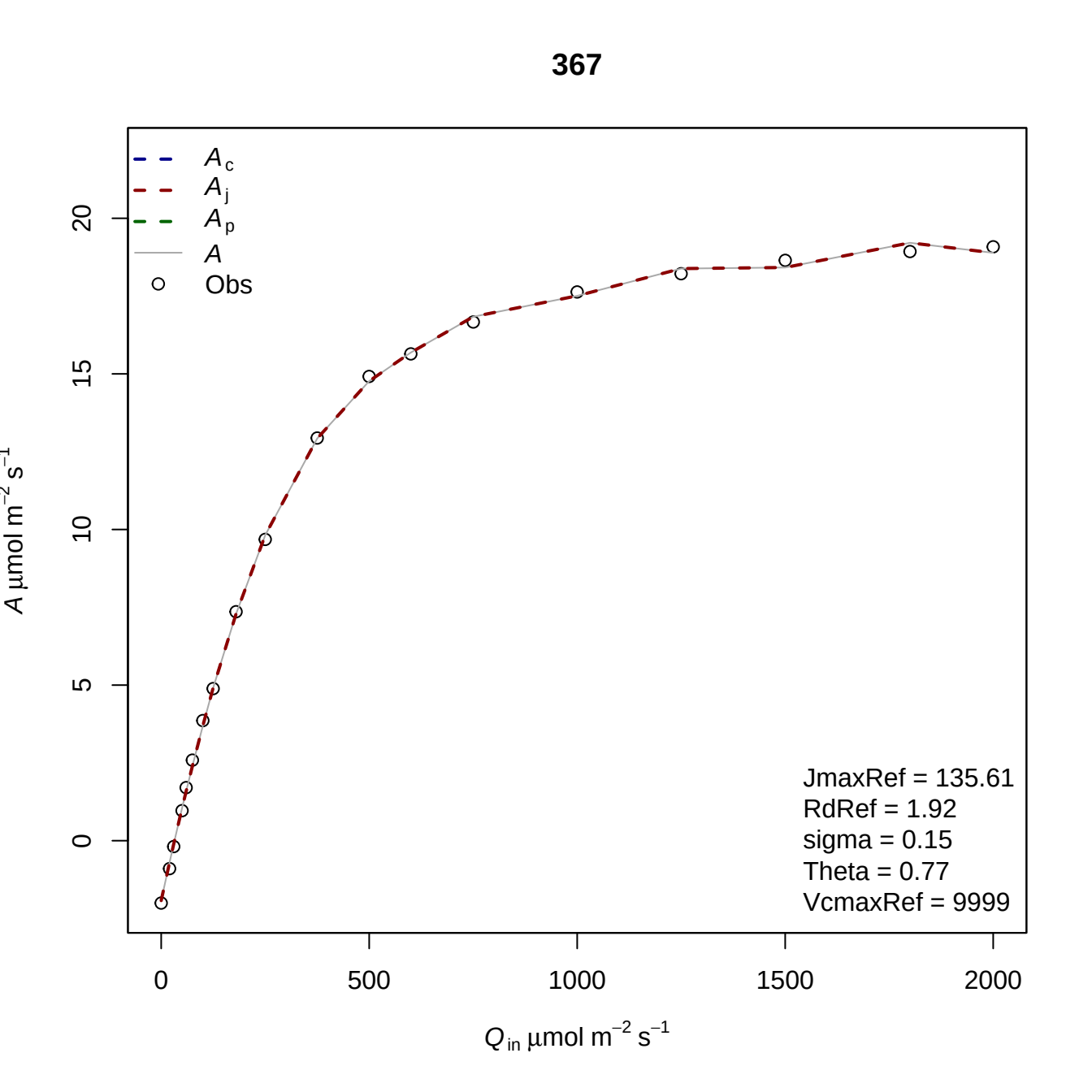

362
A_c
A_j
A_p
A
Obs
JmaxRef = 142.79
RdRef = 0.81
sigma = 0.22
Theta = 0.91
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

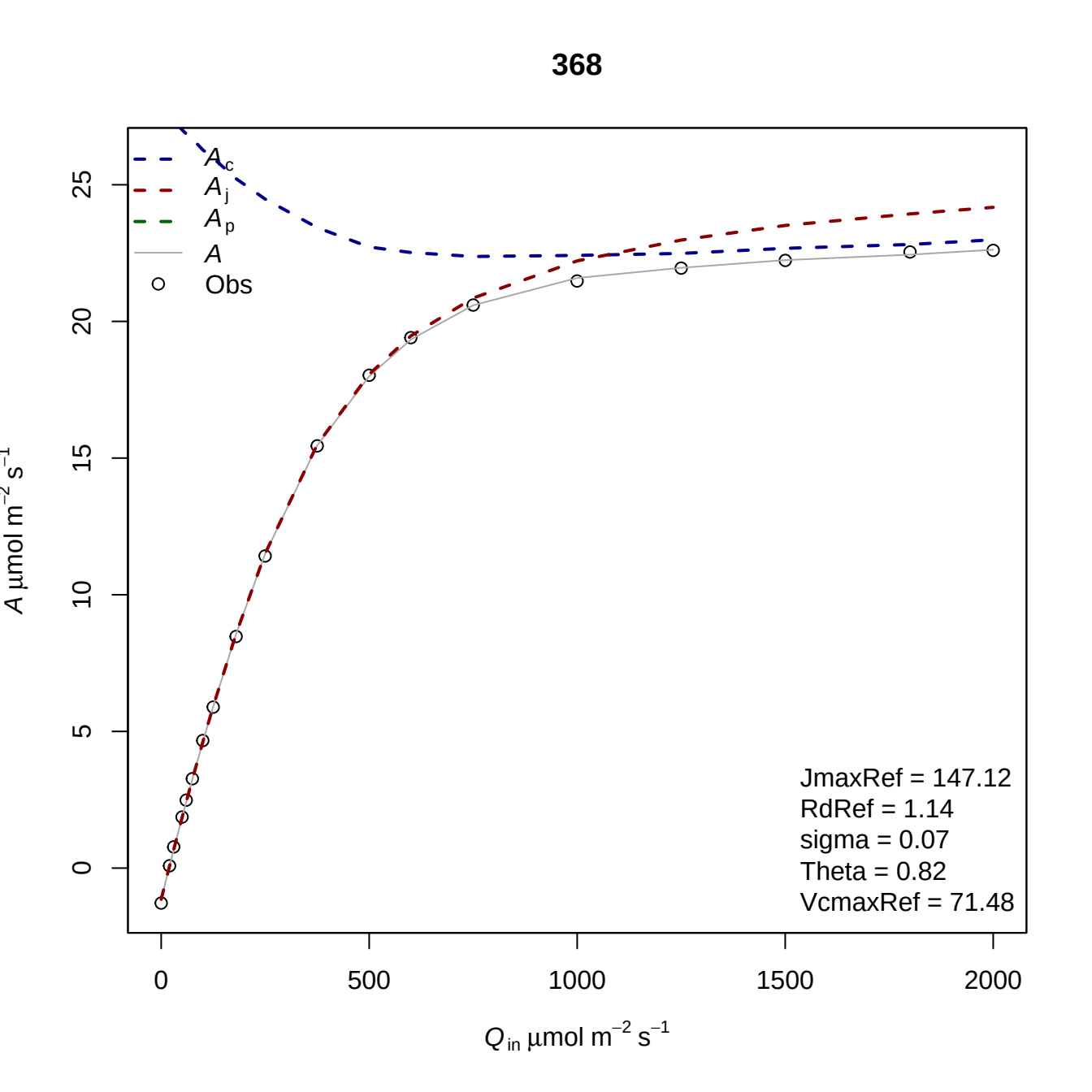

363
A_c
A_j
A_p
A
Obs
JmaxRef = 131.2
RdRef = 1.44
sigma = 0.16
Theta = 0.87
VcmaxRef = 74.6
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

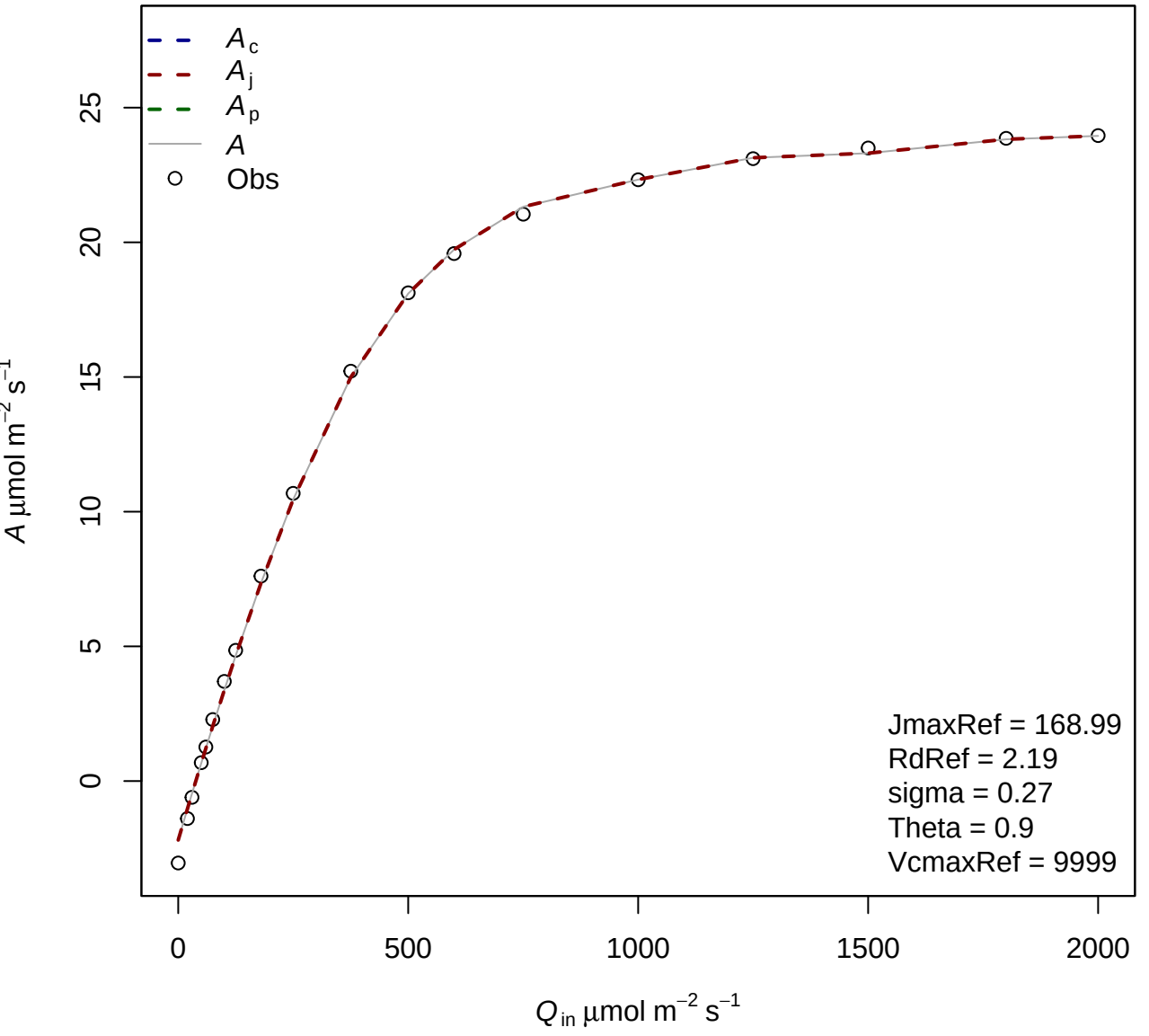

364
A_c
A_j
A_p
A
Obs
JmaxRef = 164.42
RdRef = 0.94
sigma = 0.13
Theta = 0.89
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

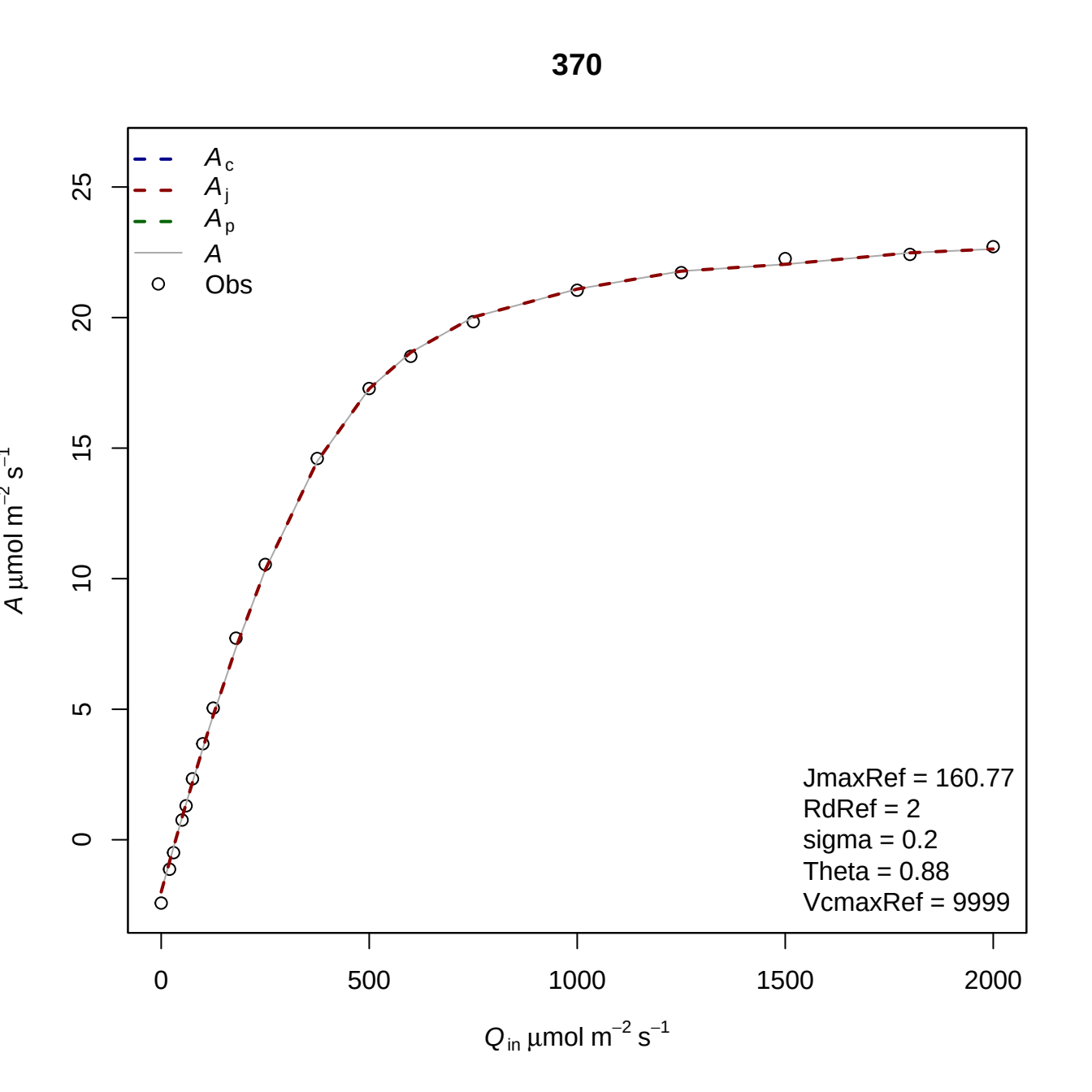

365
A_c
A_j
A_p
A
Obs
JmaxRef = 136.72
RdRef = 1.61
sigma = 0.11
Theta = 0.75
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

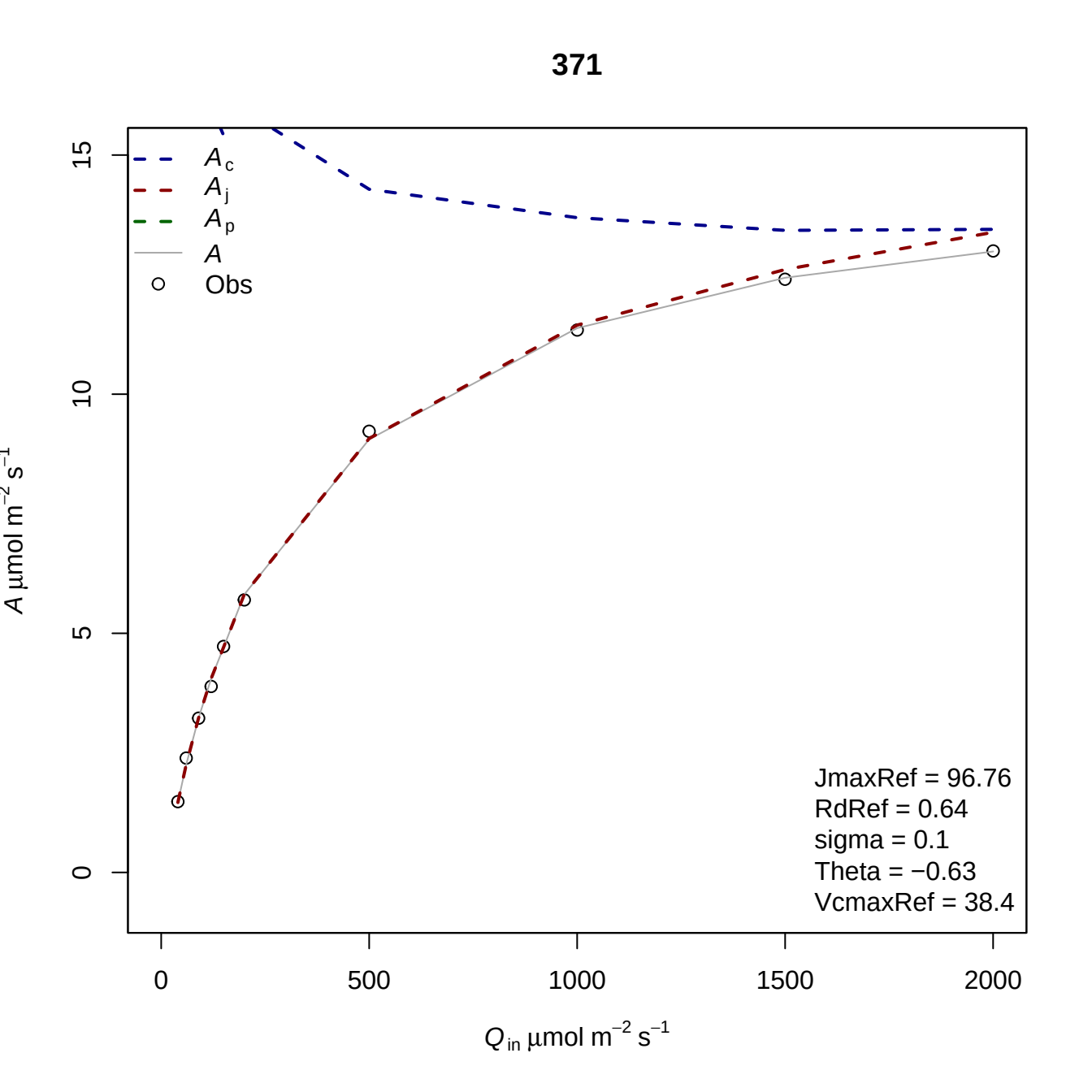

366
A_c
A_j
A_p
A
Obs
JmaxRef = 92.78
RdRef = 2.51
sigma = 0.14
Theta = 0.76
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

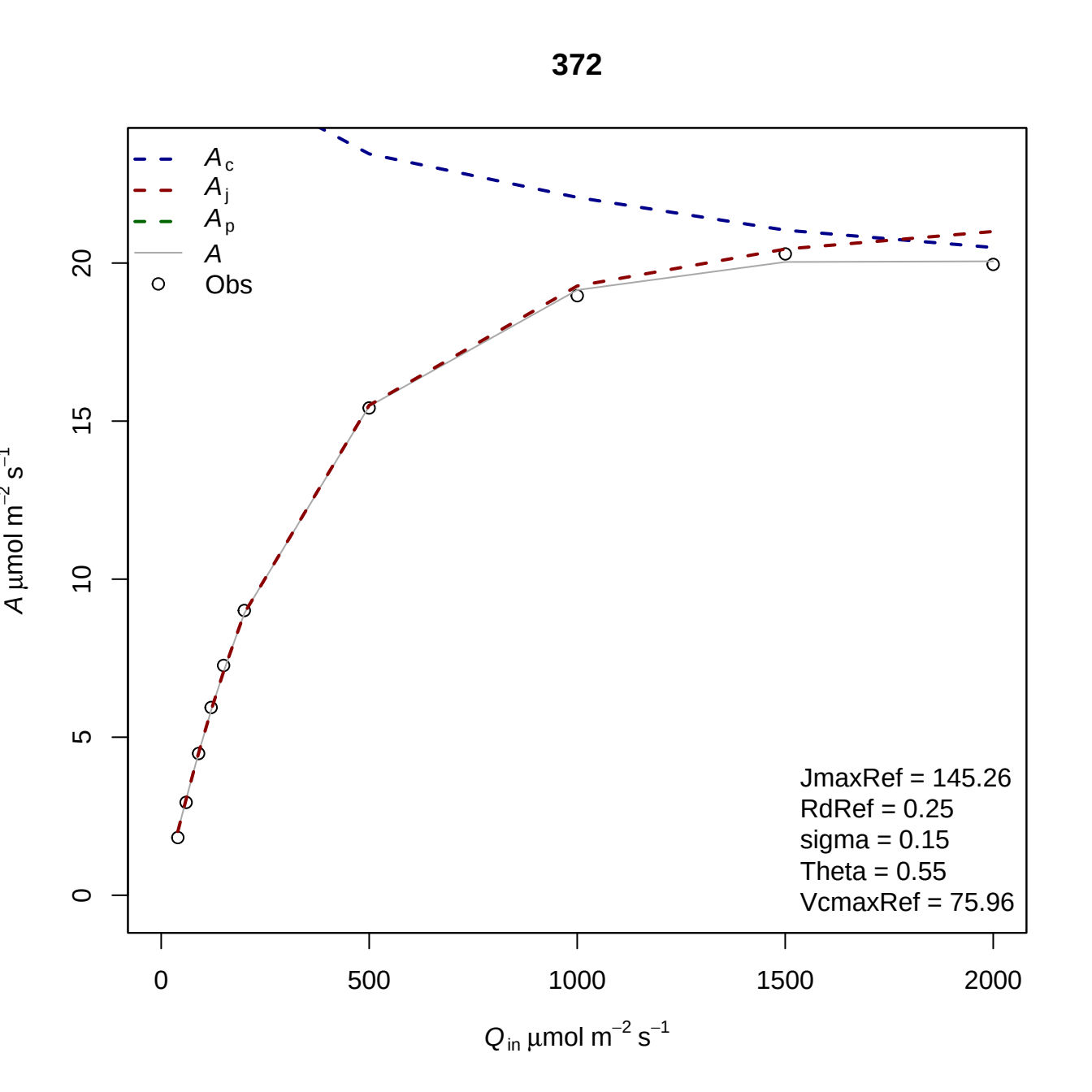

367
A_c
A_j
A_p
A
Obs
JmaxRef = 135.61
RdRef = 1.92
sigma = 0.15
Theta = 0.77
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

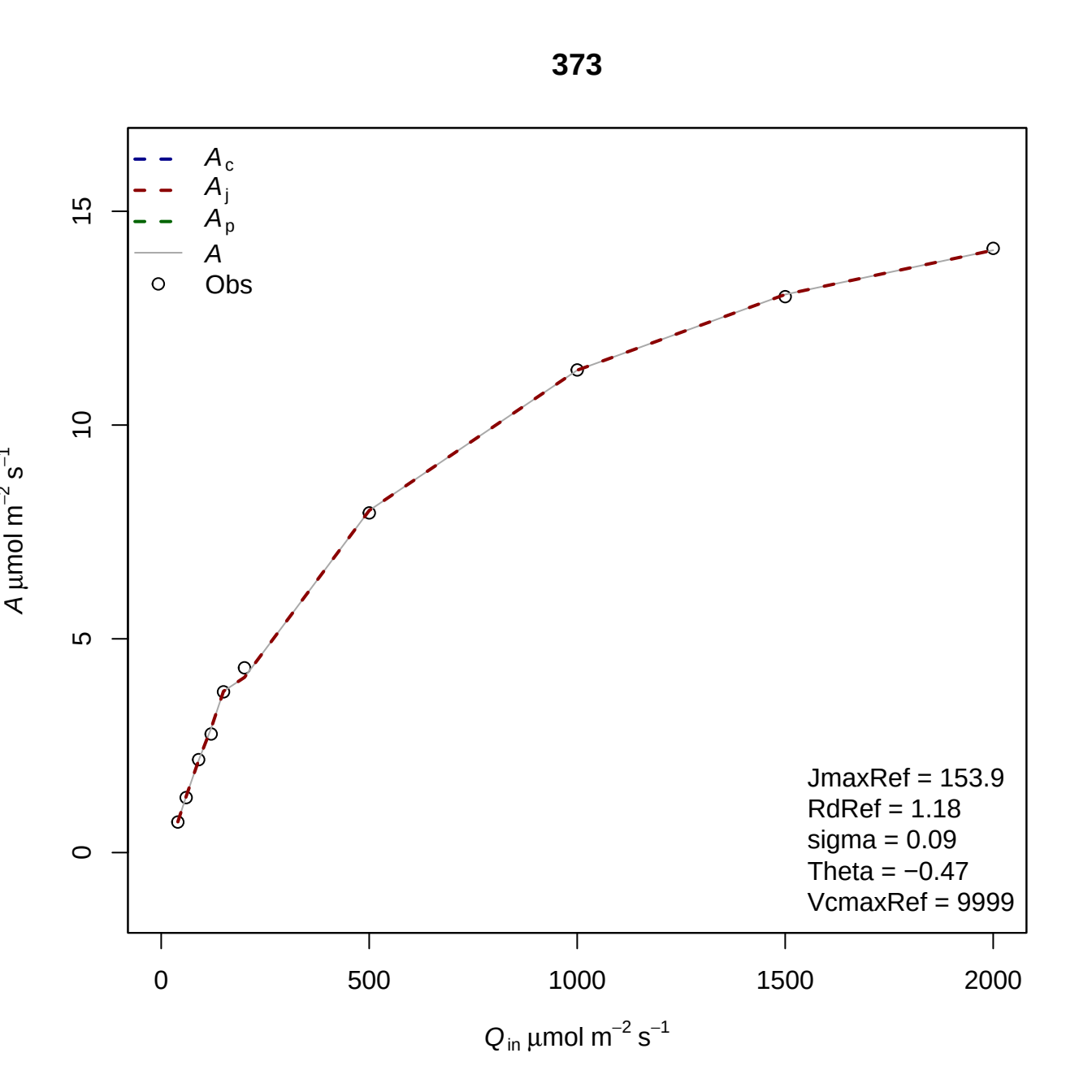

368
A_c
A_j
A_p
A
Obs
JmaxRef = 147.12
RdRef = 1.14
sigma = 0.07
Theta = 0.82
VcmaxRef = 71.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

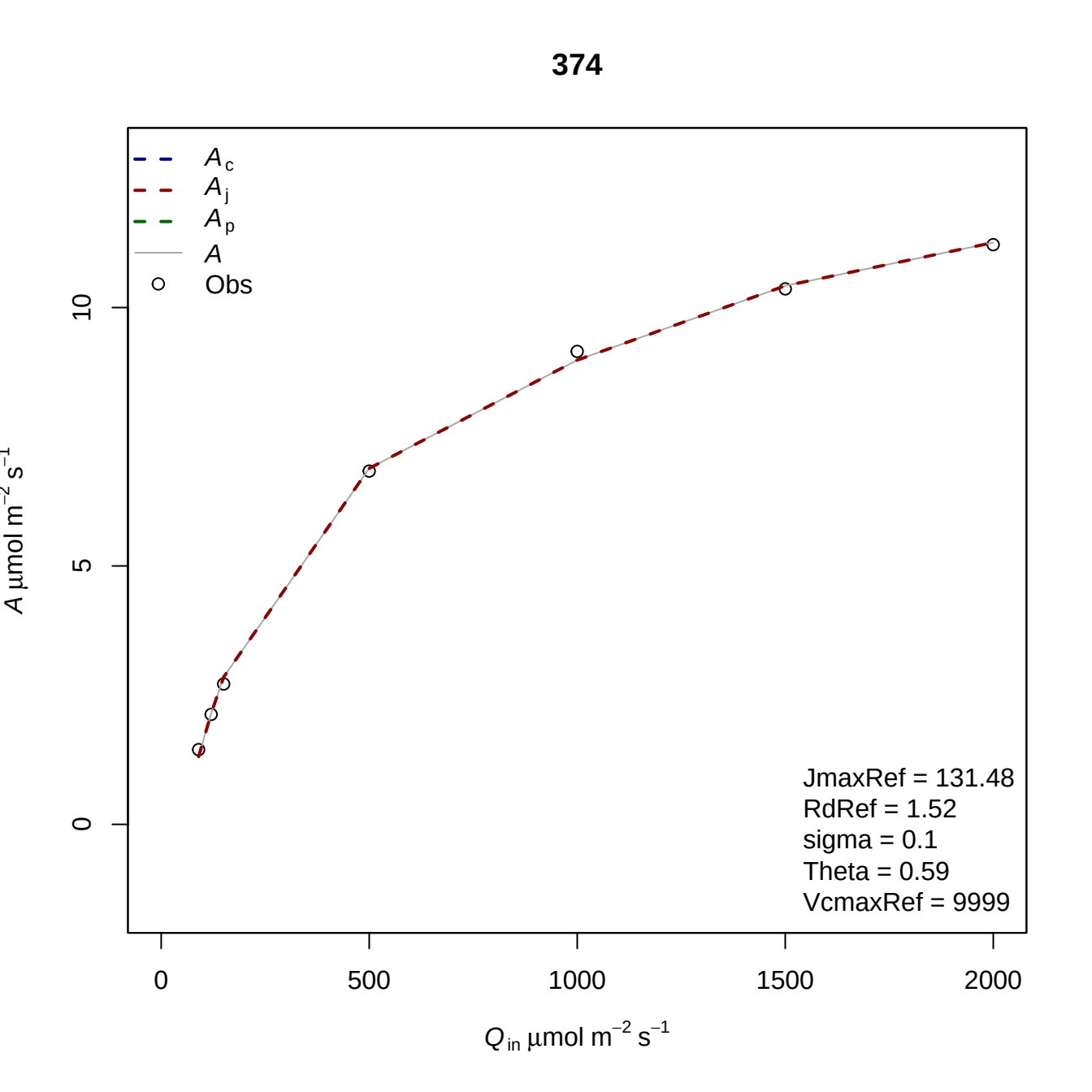

A_c
A_j
A_p
A
Obs
JmaxRef = 168.99
RdRef = 2.19
sigma = 0.27
Theta = 0.9
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

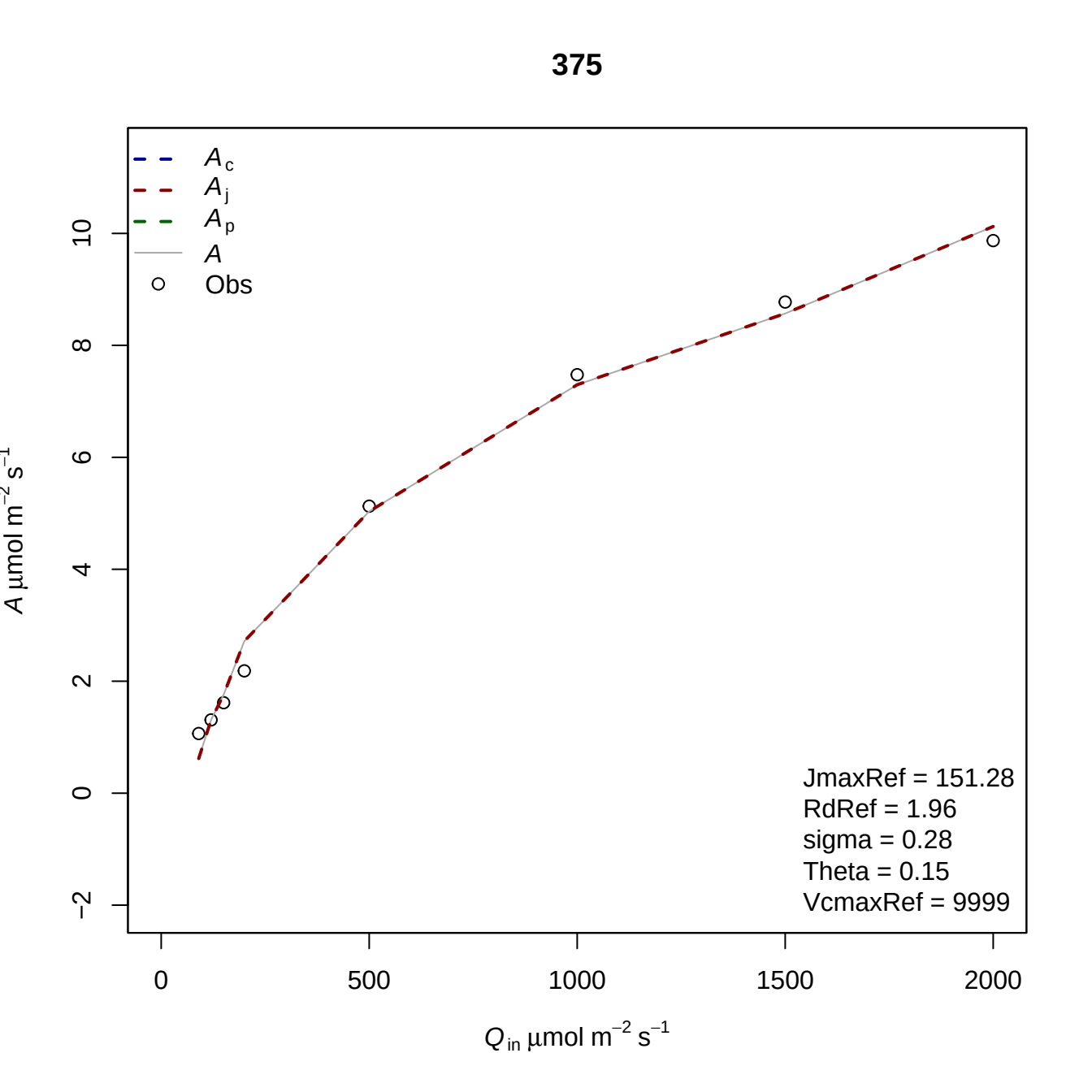

370
A_c
A_j
A_p
A
Obs
JmaxRef = 160.77
RdRef = 2
sigma = 0.2
Theta = 0.88
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

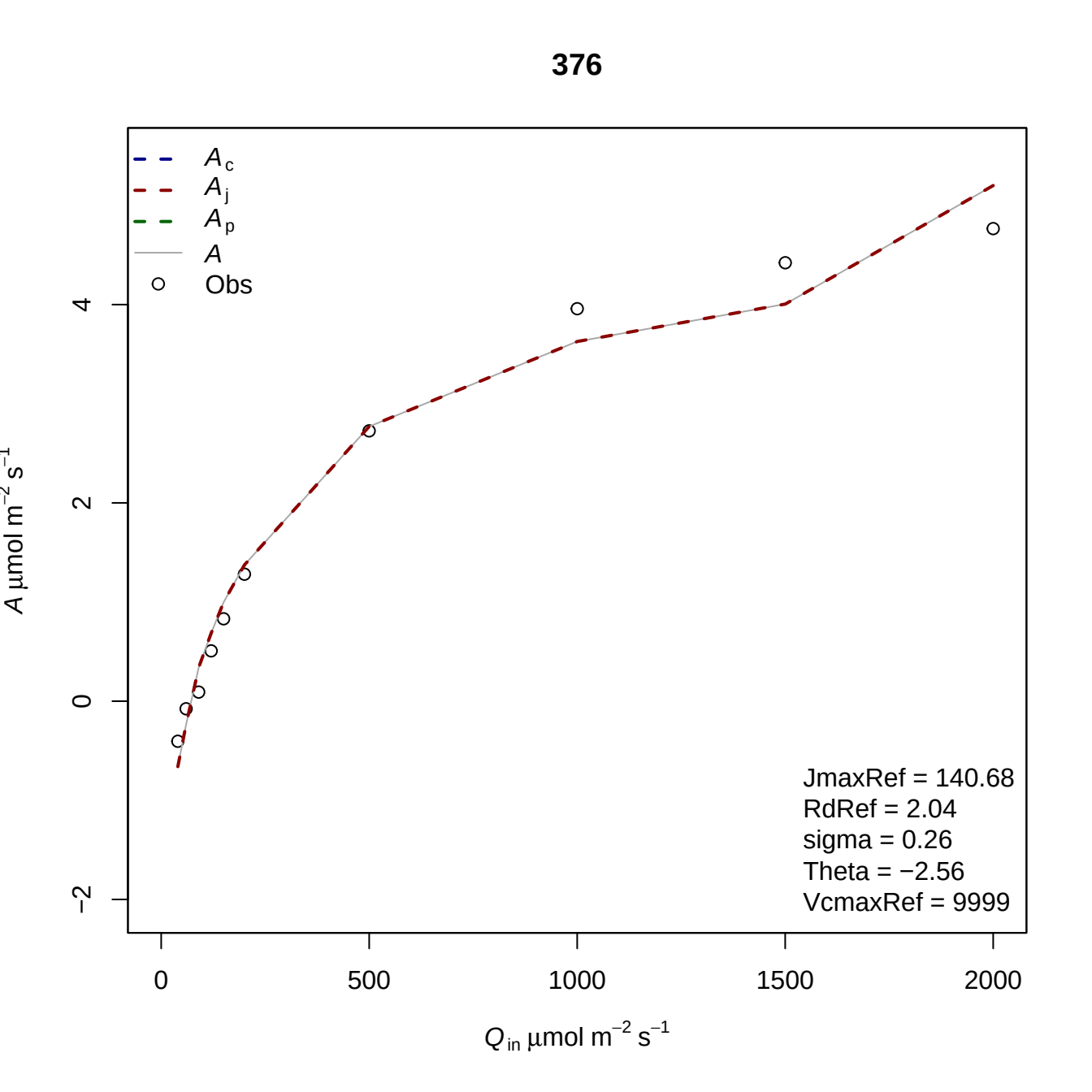

371
A_c
A_j
A_p
A
Obs
JmaxRef = 96.76
RdRef = 0.64
sigma = 0.1
Theta = −0.63
VcmaxRef = 38.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

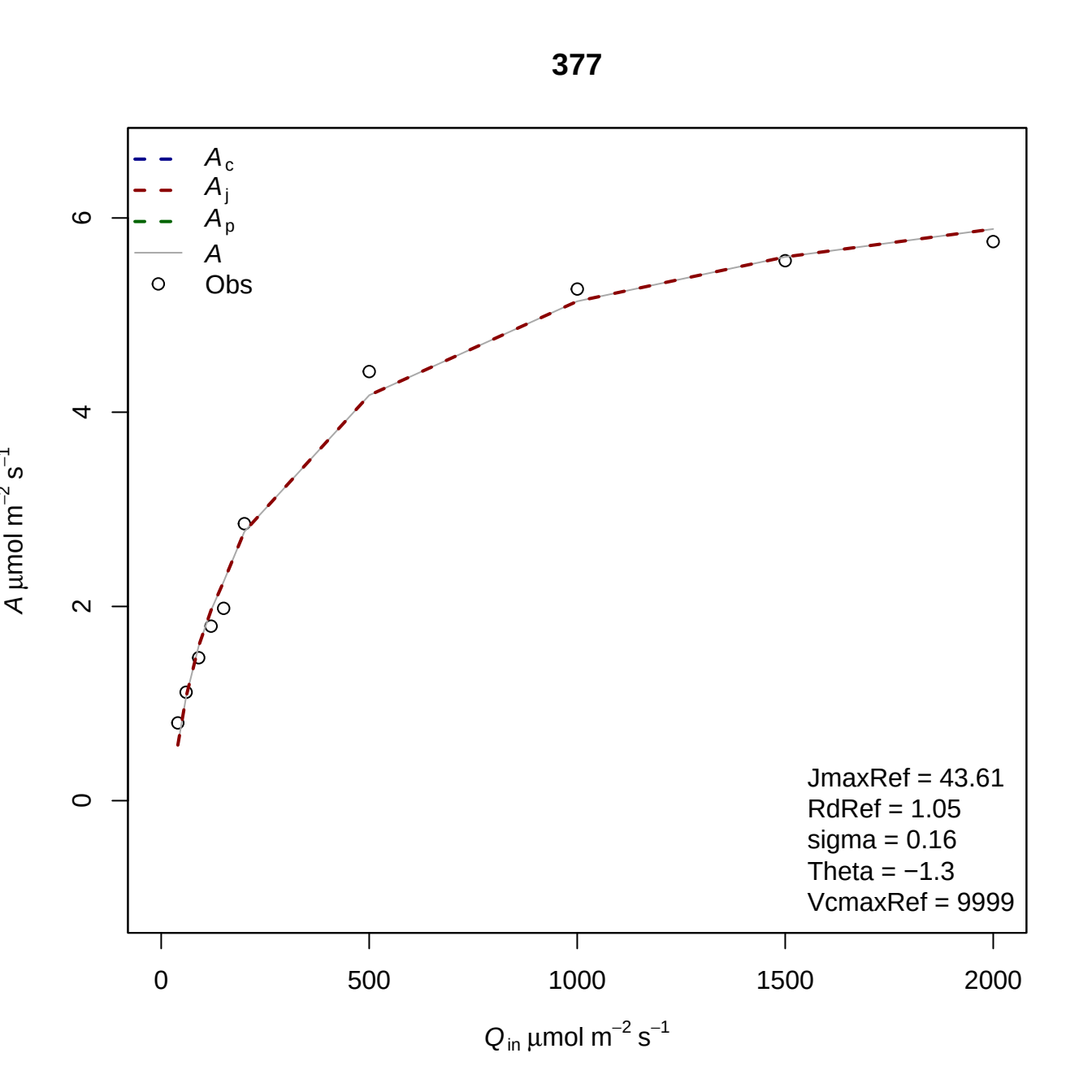

372
A_c
A_j
A_p
A
Obs
JmaxRef = 145.26
RdRef = 0.25
sigma = 0.15
Theta = 0.55
VcmaxRef = 75.96
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

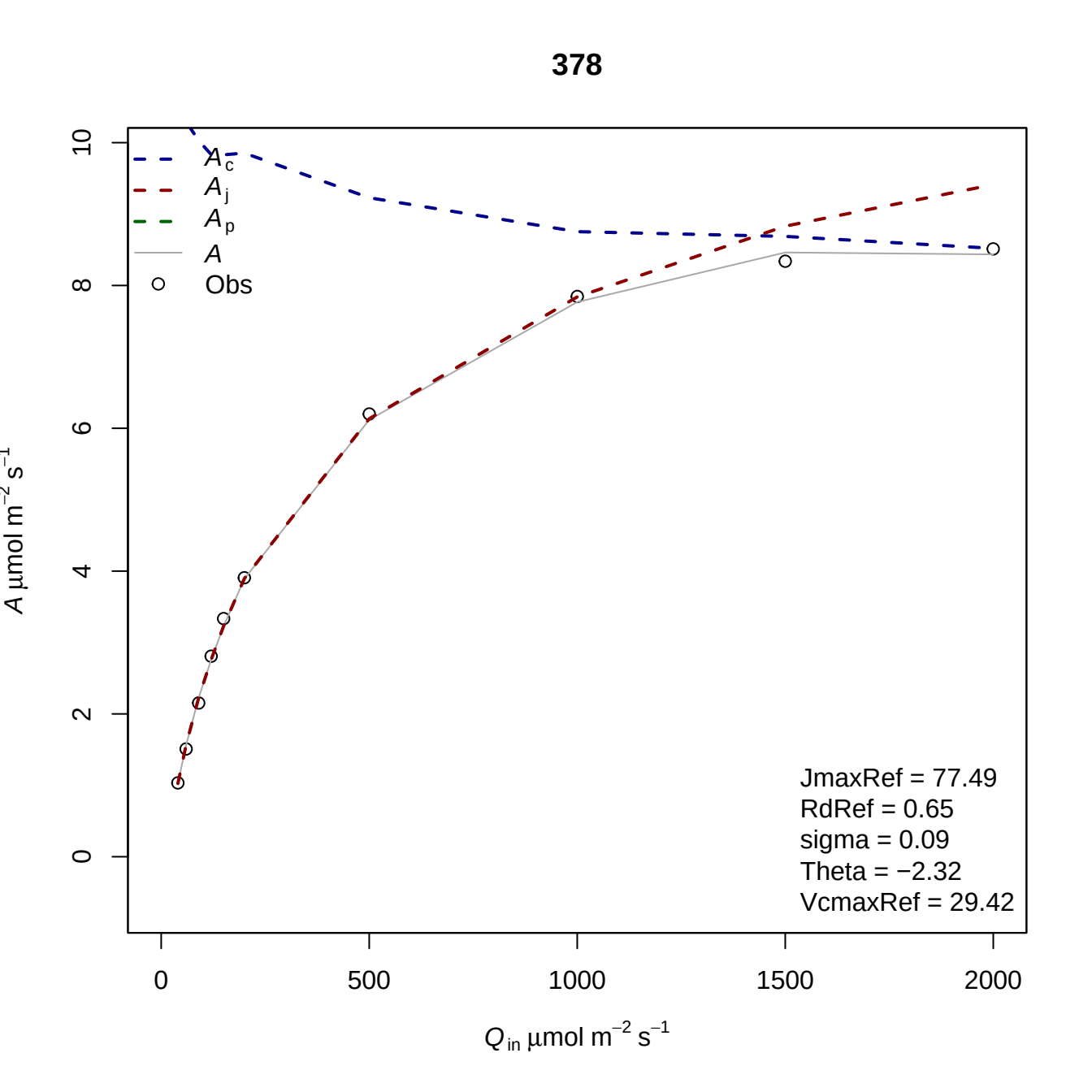

373
A_c
A_j
A_p
A
Obs
JmaxRef = 153.9
RdRef = 1.18
sigma = 0.09
Theta = −0.47
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

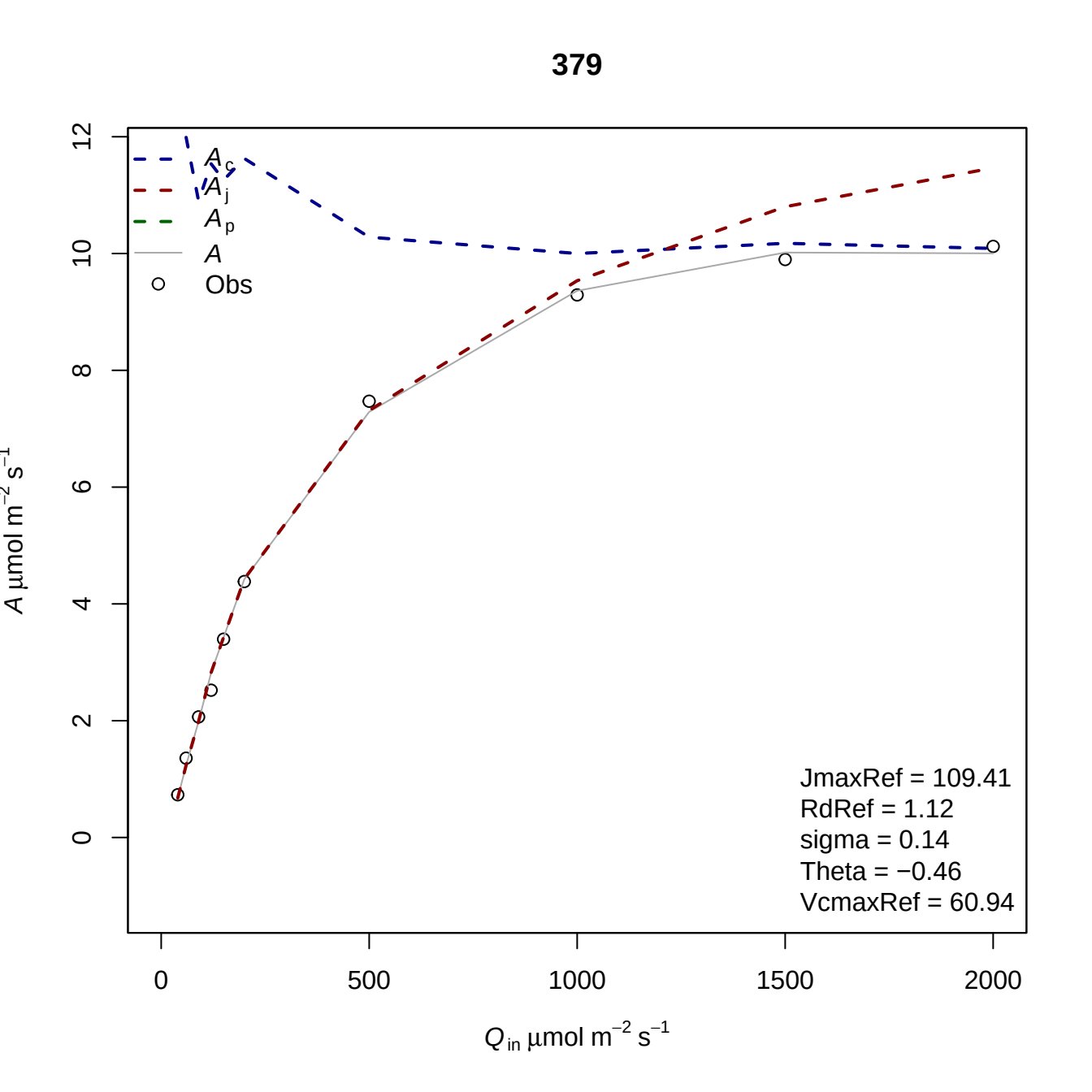

374
A_c
A_j
A_p
A
Obs
JmaxRef = 131.48
RdRef = 1.52
sigma = 0.1
Theta = 0.59
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

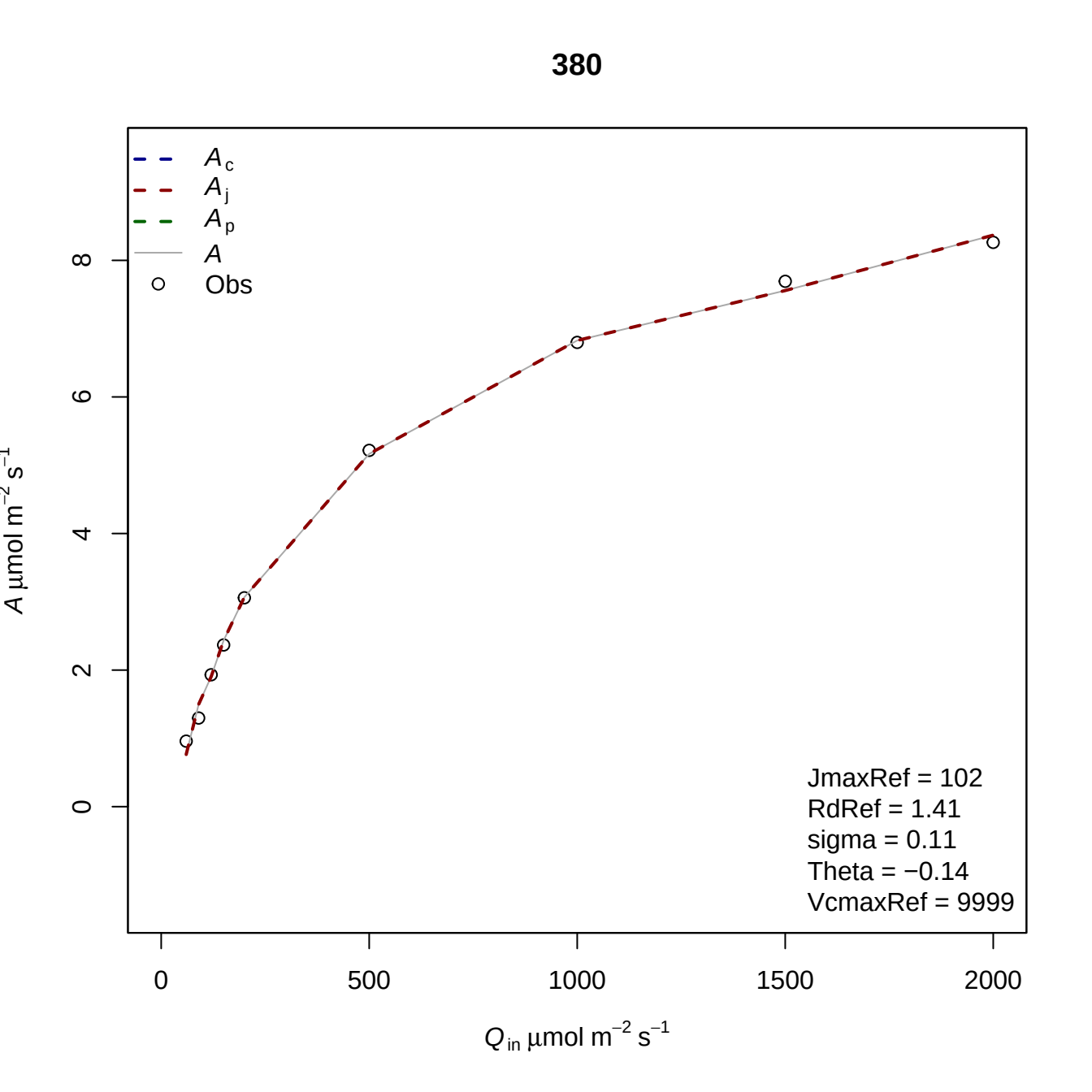

375
A_c
A_j
A_p
A
Obs
JmaxRef = 151.28
RdRef = 1.96
sigma = 0.28
Theta = 0.15
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

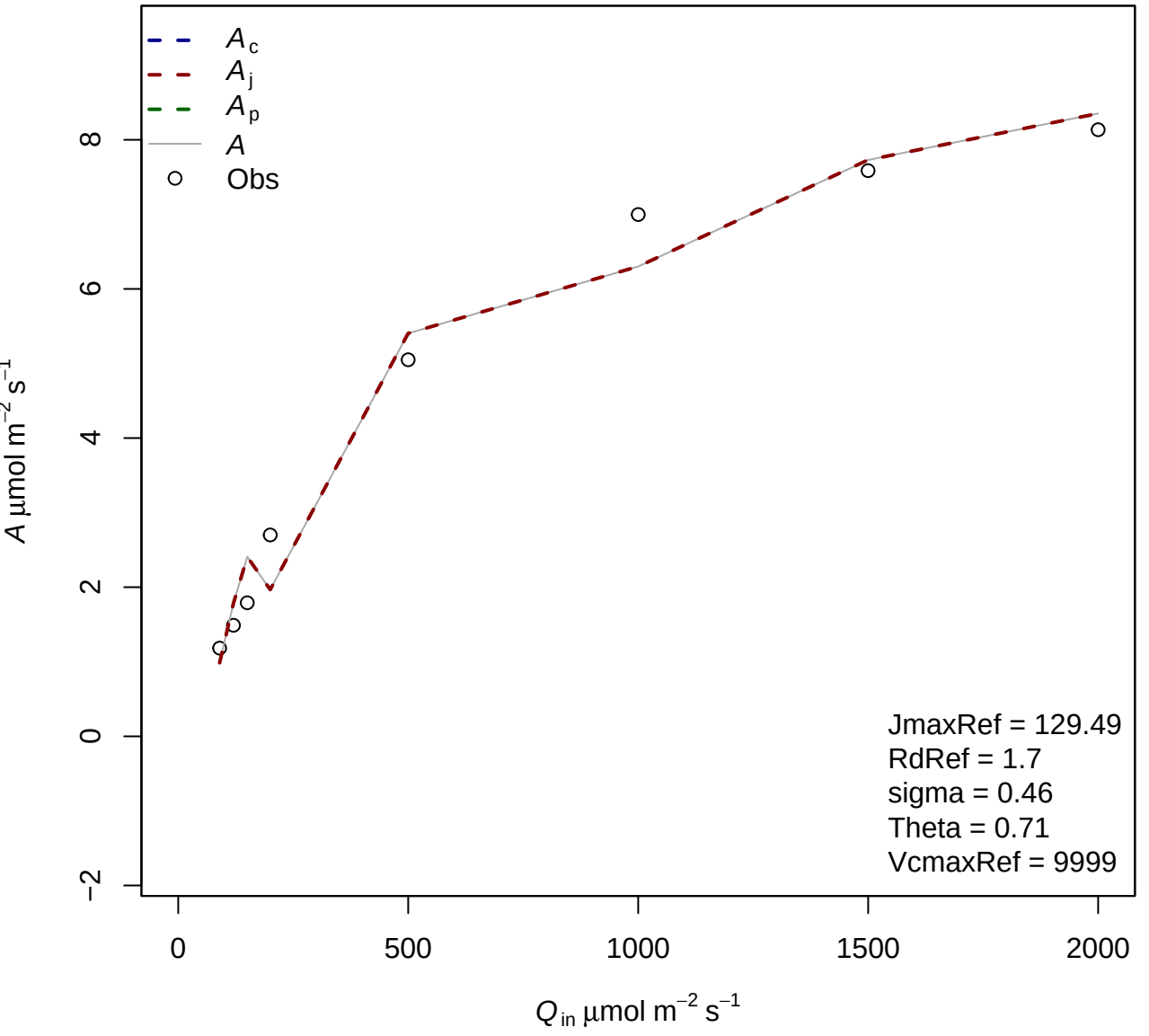

376
A_c
A_j
A_p
A
Obs
JmaxRef = 140.68
RdRef = 2.04
sigma = 0.26
Theta = −2.56
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

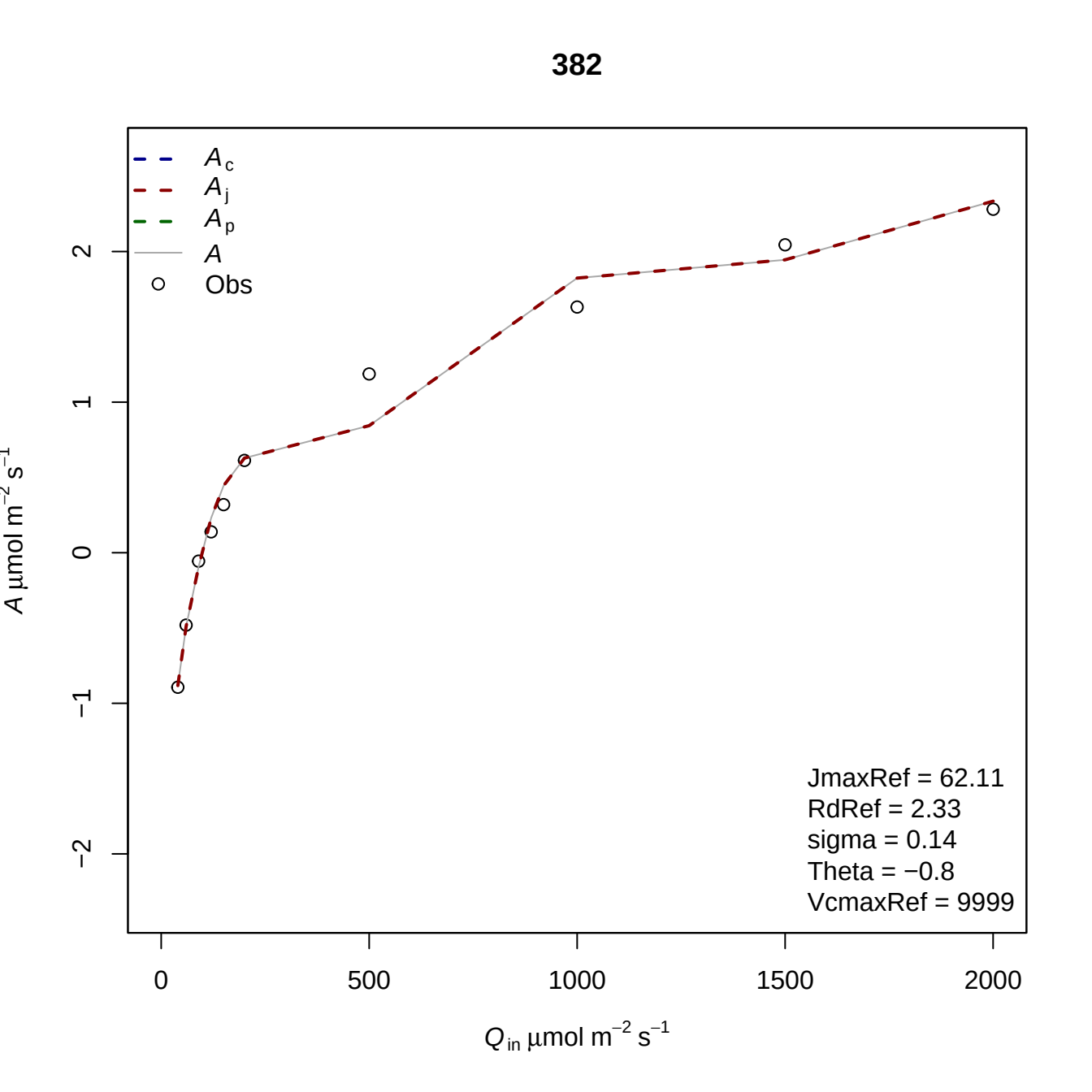

377
A_c
A_j
A_p
A
Obs
JmaxRef = 43.61
RdRef = 1.05
sigma = 0.16
Theta = −1.3
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

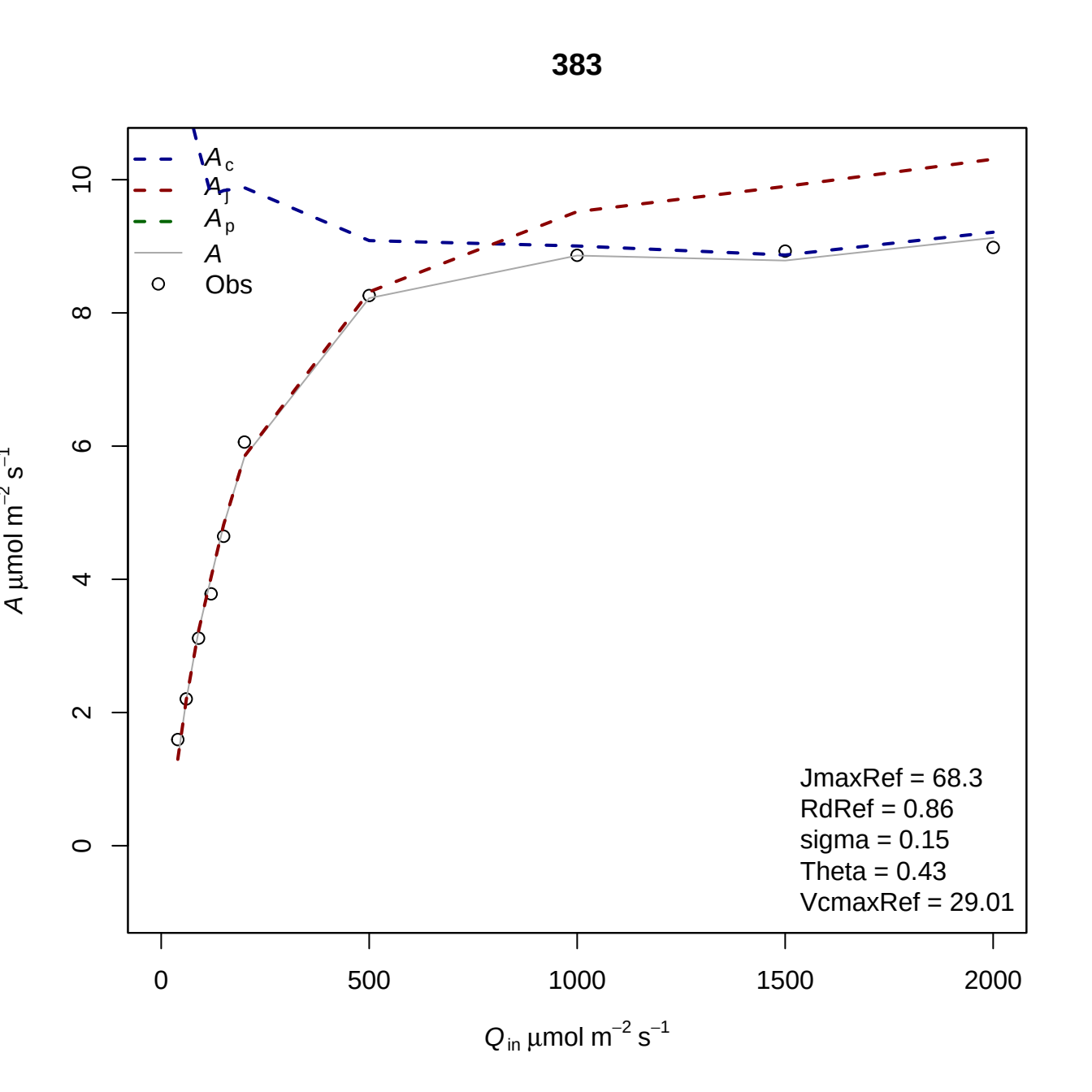

378
A_c
A_j
A_p
A
Obs
JmaxRef = 77.49
RdRef = 0.65
sigma = 0.09
Theta = −2.32
VcmaxRef = 29.42
0
2
4
6
8
10
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

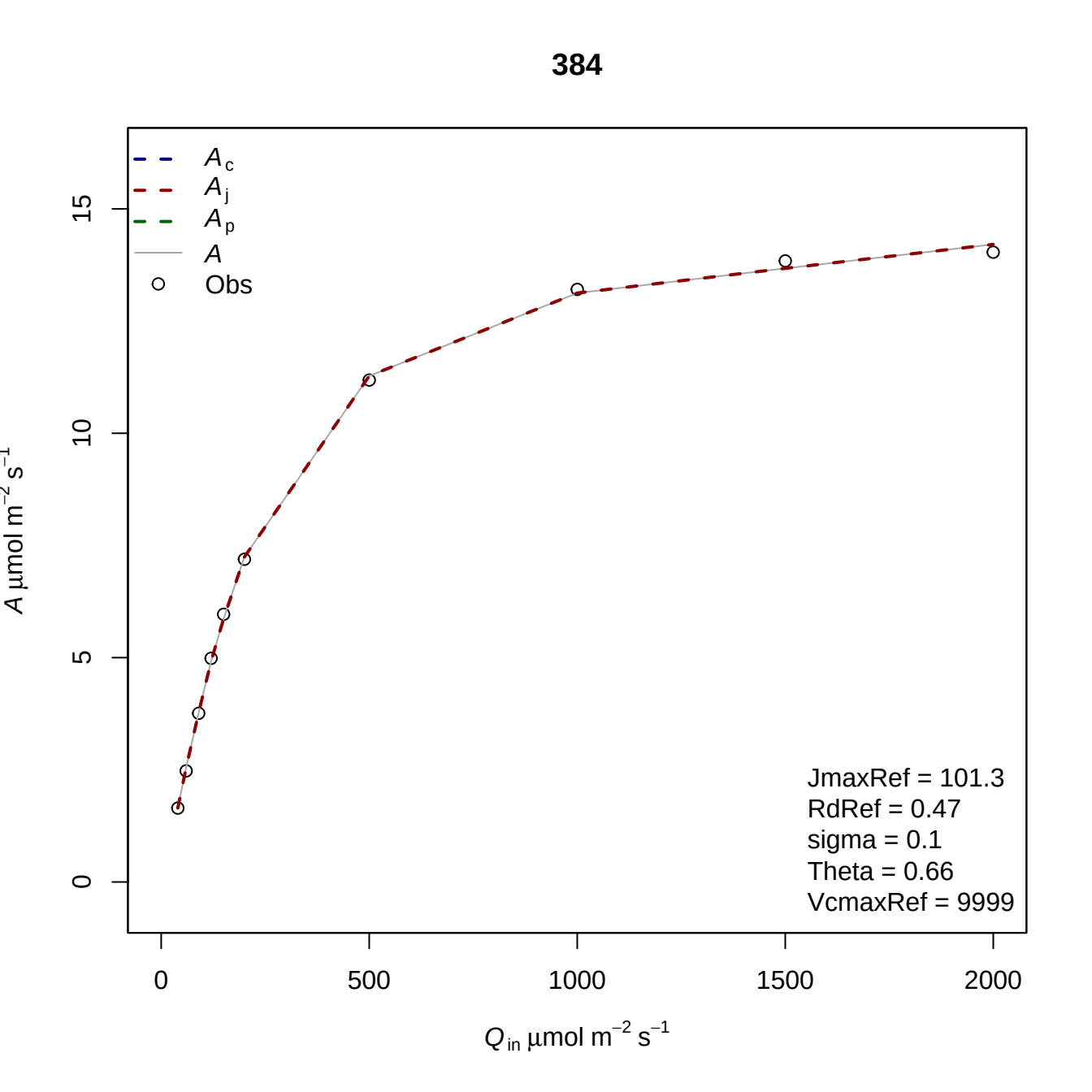

379
A_c
A_j
A_p
A
Obs
JmaxRef = 109.41
RdRef = 1.12
sigma = 0.14
Theta = −0.46
VcmaxRef = 60.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

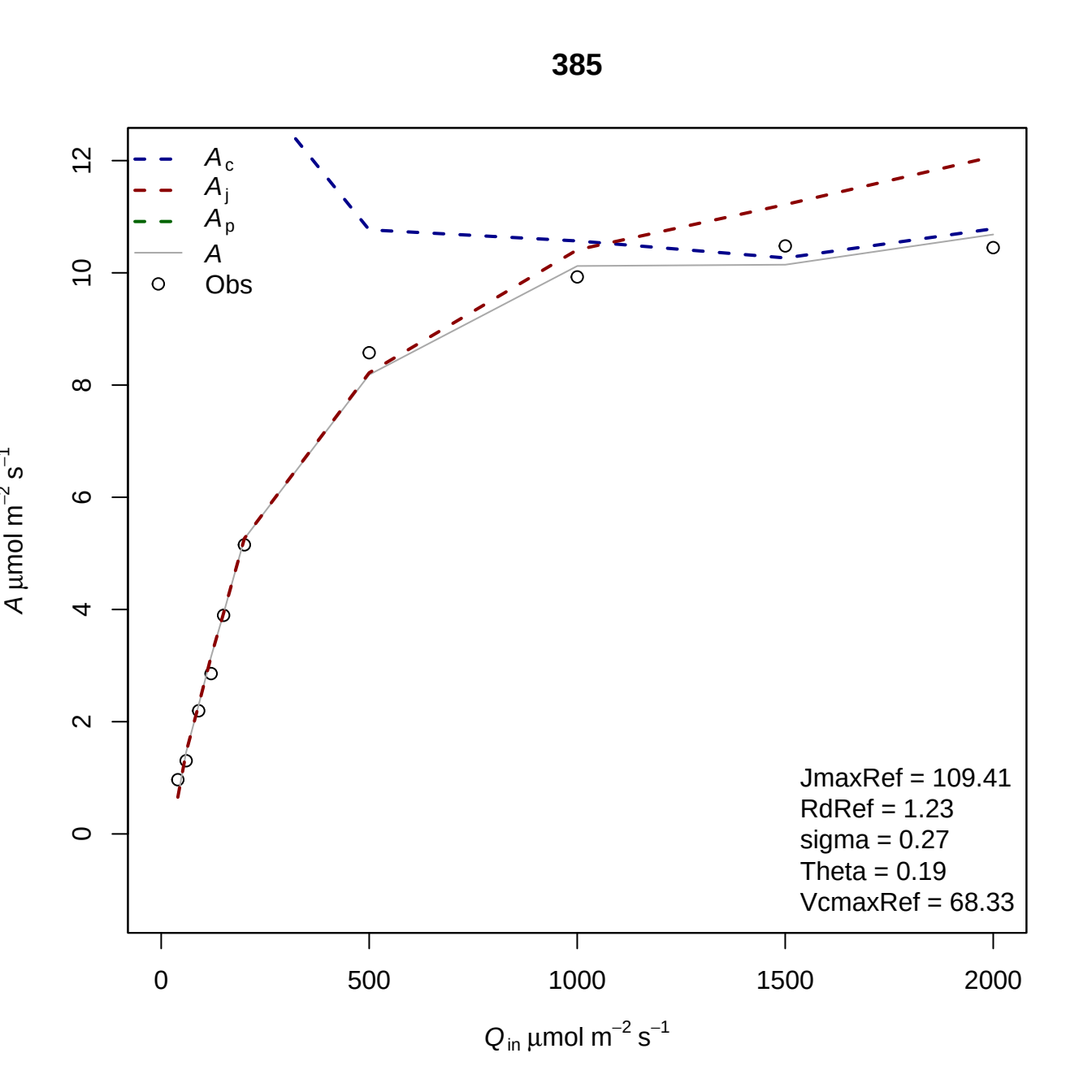

380
A_c
A_j
A_p
A
Obs
JmaxRef = 102
RdRef = 1.41
sigma = 0.11
Theta = −0.14
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

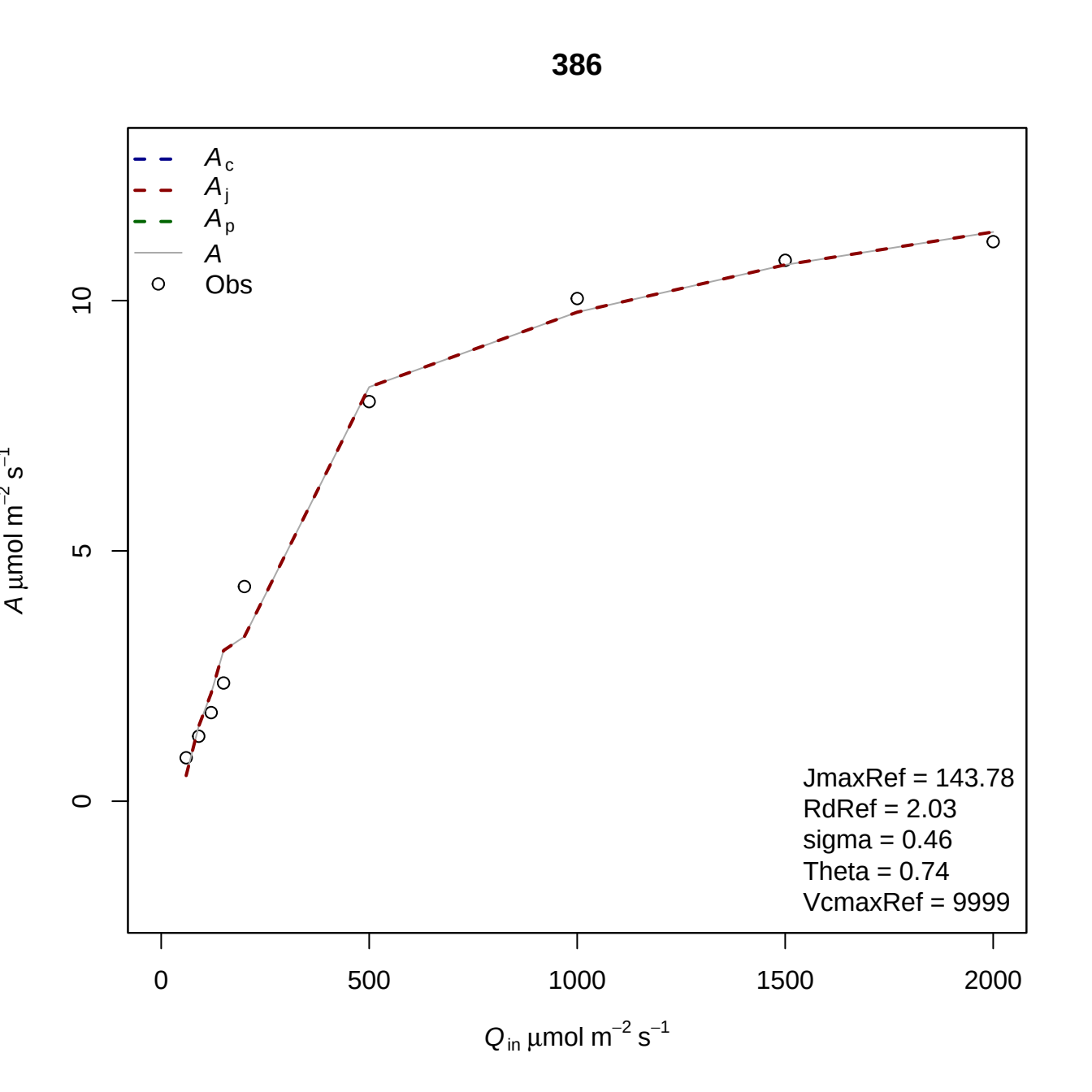

A_c
A_j
A_p
A
Obs
JmaxRef = 129.49
RdRef = 1.7
sigma = 0.46
Theta = 0.71
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

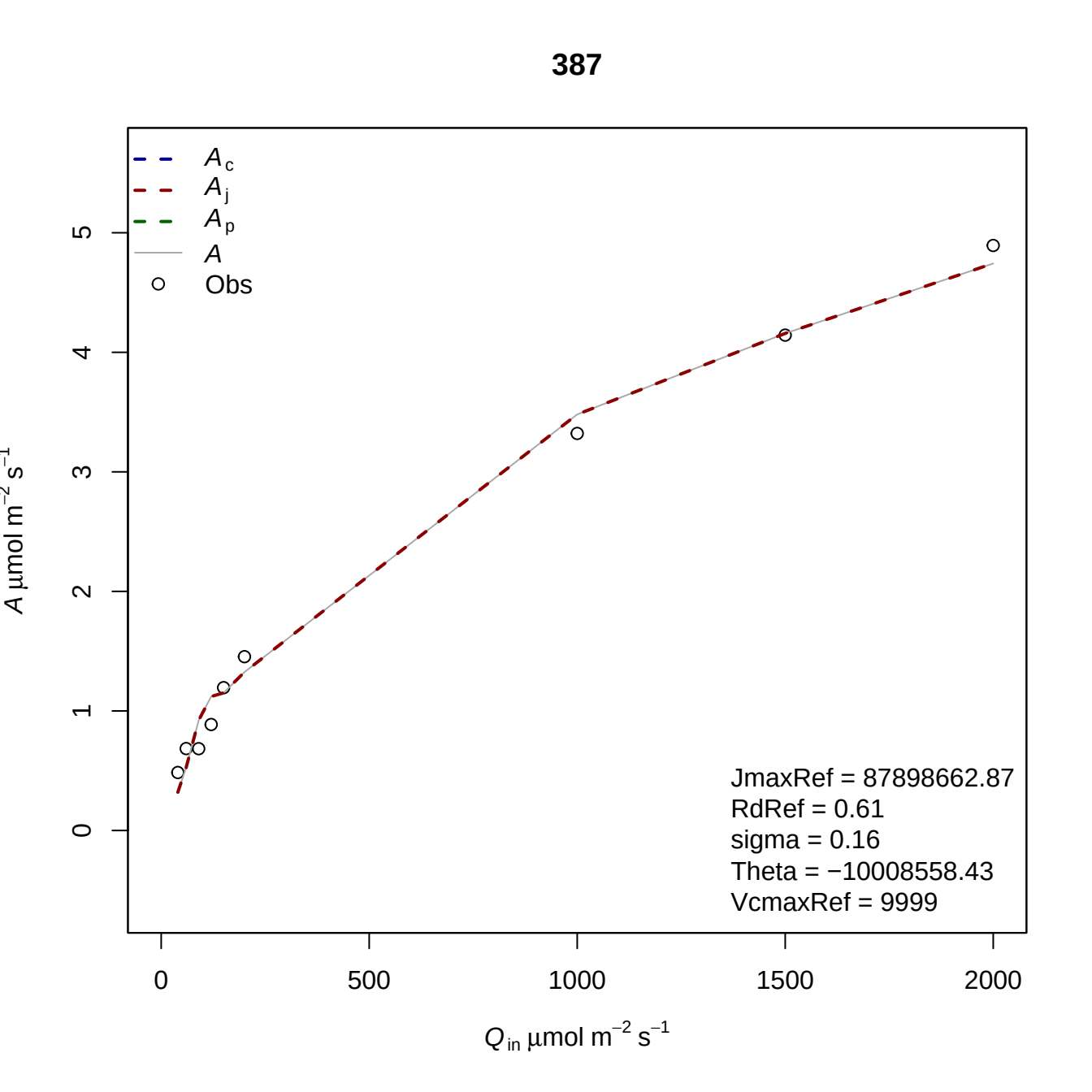

382
A_c
A_j
A_p
A
Obs
JmaxRef = 62.11
RdRef = 2.33
sigma = 0.14
Theta = −0.8
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
0
500
1000
1500
2000

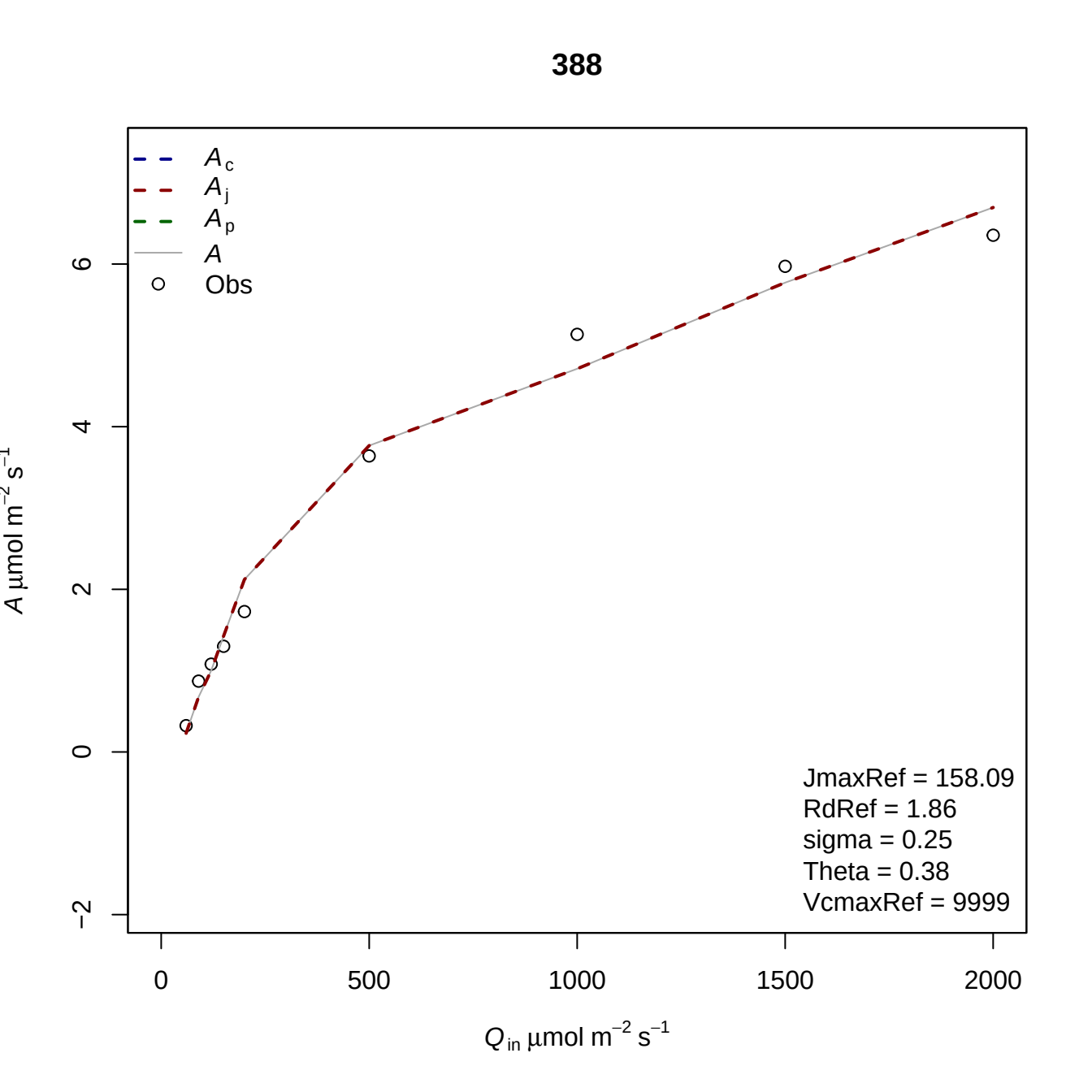

383
A_c
A_J
A_p
A
Obs
JmaxRef = 68.3
RdRef = 0.86
sigma = 0.15
Theta = 0.43
VcmaxRef = 29.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

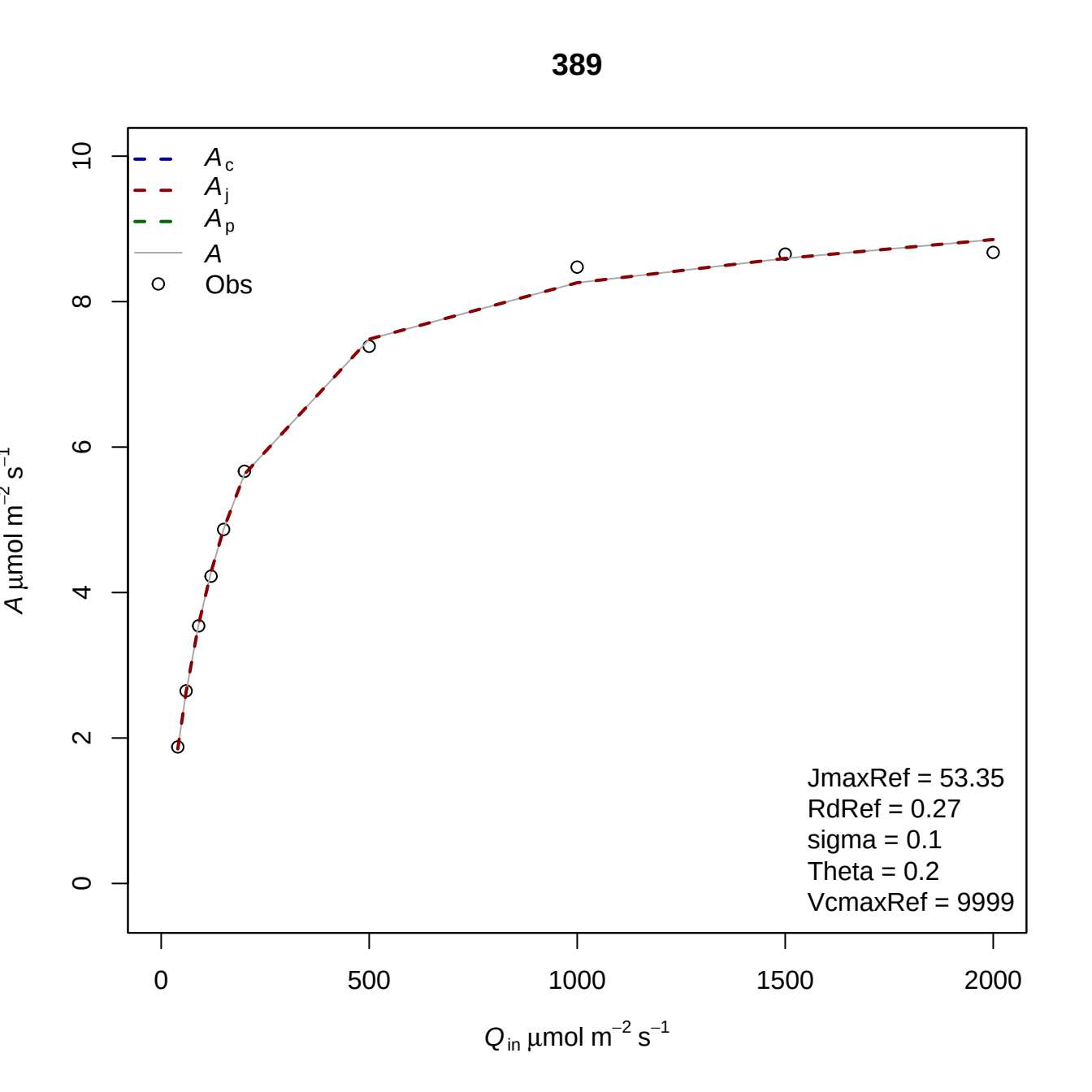

384
A_c
A_j
A_p
A
Obs
JmaxRef = 101.3
RdRef = 0.47
sigma = 0.1
Theta = 0.66
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

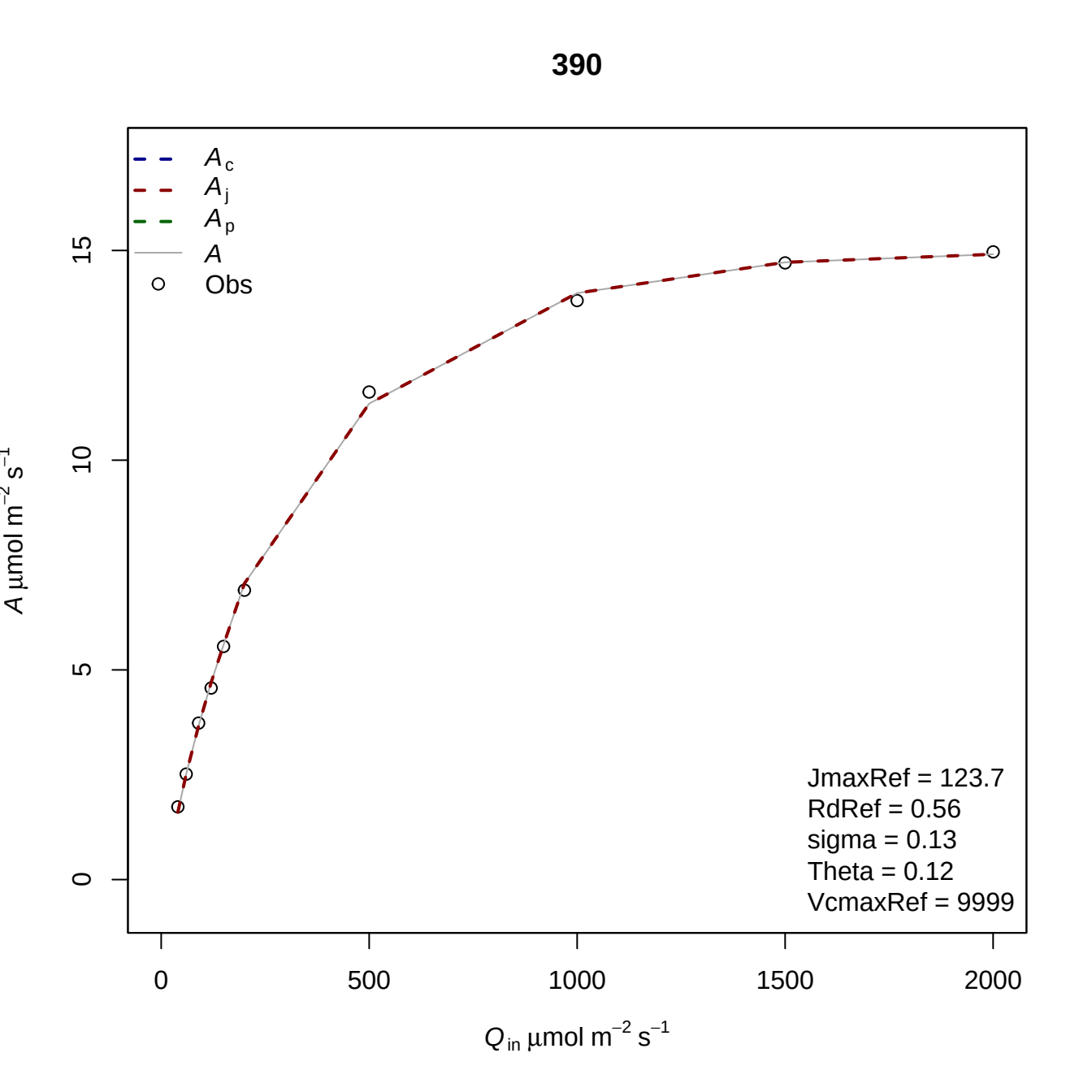

385
A_c
A_j
A_p
A
Obs
JmaxRef = 109.41
RdRef = 1.23
sigma = 0.27
Theta = 0.19
VcmaxRef = 68.33
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

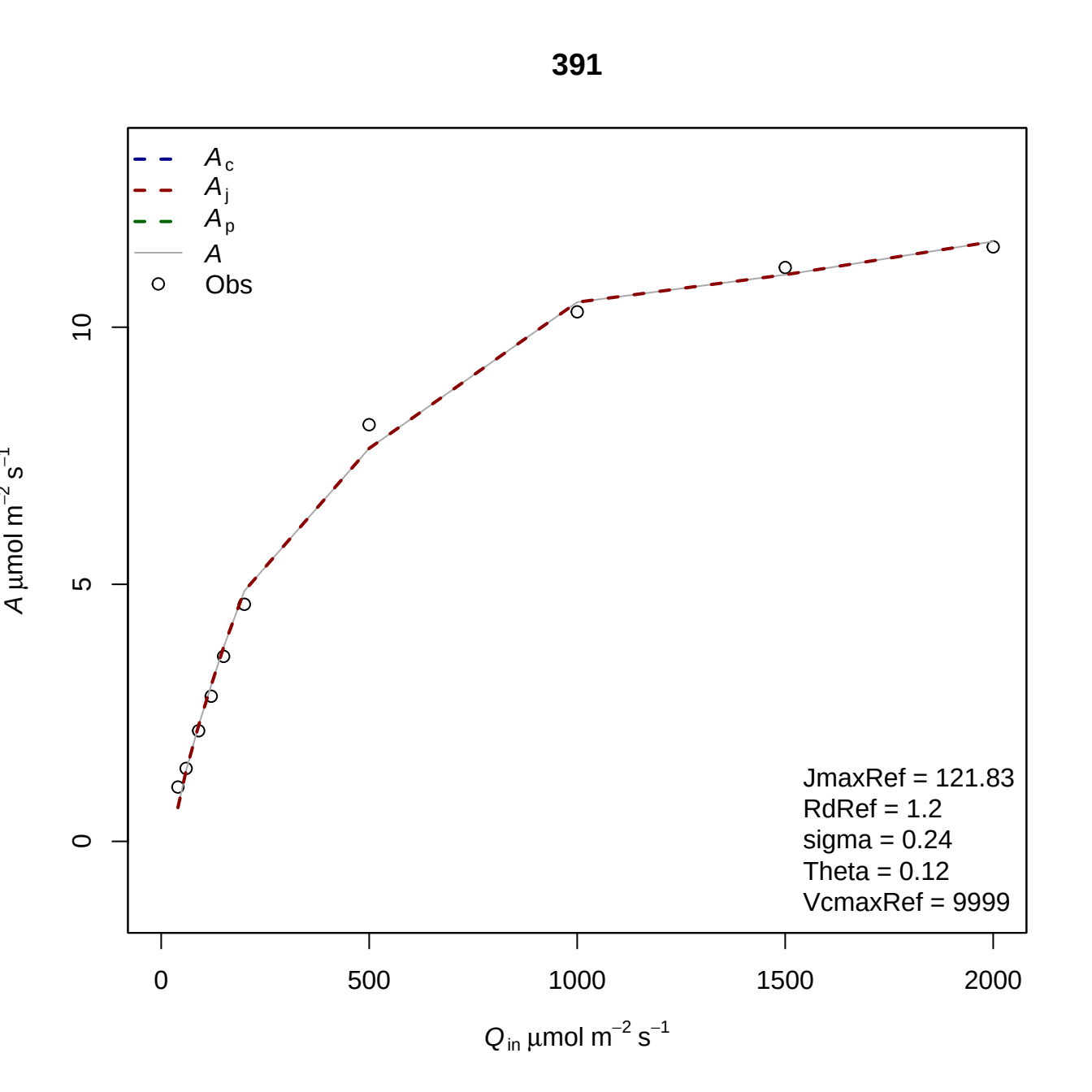

386
A_c
A_j
A_p
A
Obs
JmaxRef = 143.78
RdRef = 2.03
sigma = 0.46
Theta = 0.74
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

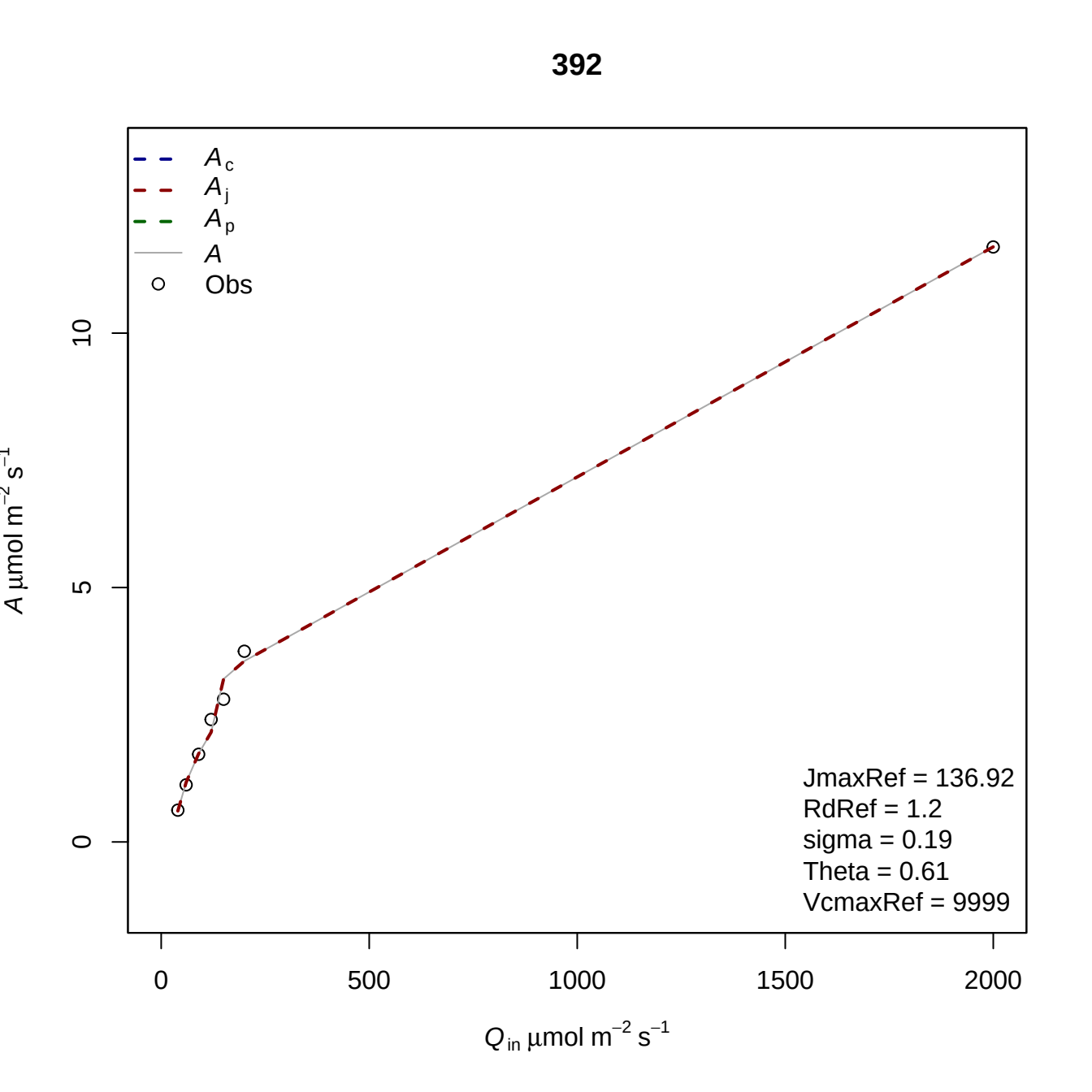

387
A_c
A_j
A_p
A
Obs
JmaxRef = 87898662.87
RdRef = 0.61
sigma = 0.16
Theta = −10008558.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500
2000

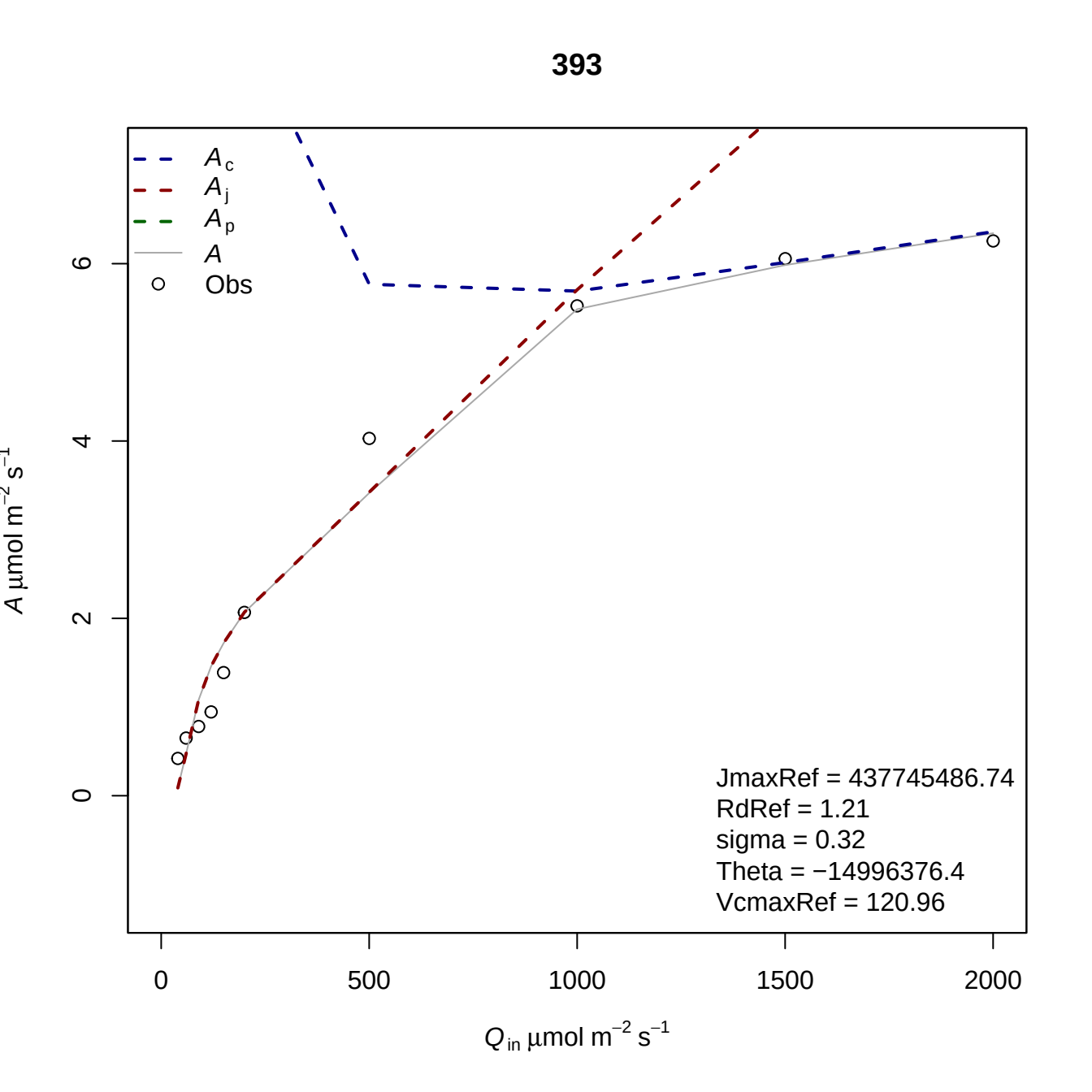

388
A_c
A_j
A_p
A
Obs
JmaxRef = 158.09
RdRef = 1.86
sigma = 0.25
Theta = 0.38
VcmaxRef = 9999
−2
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

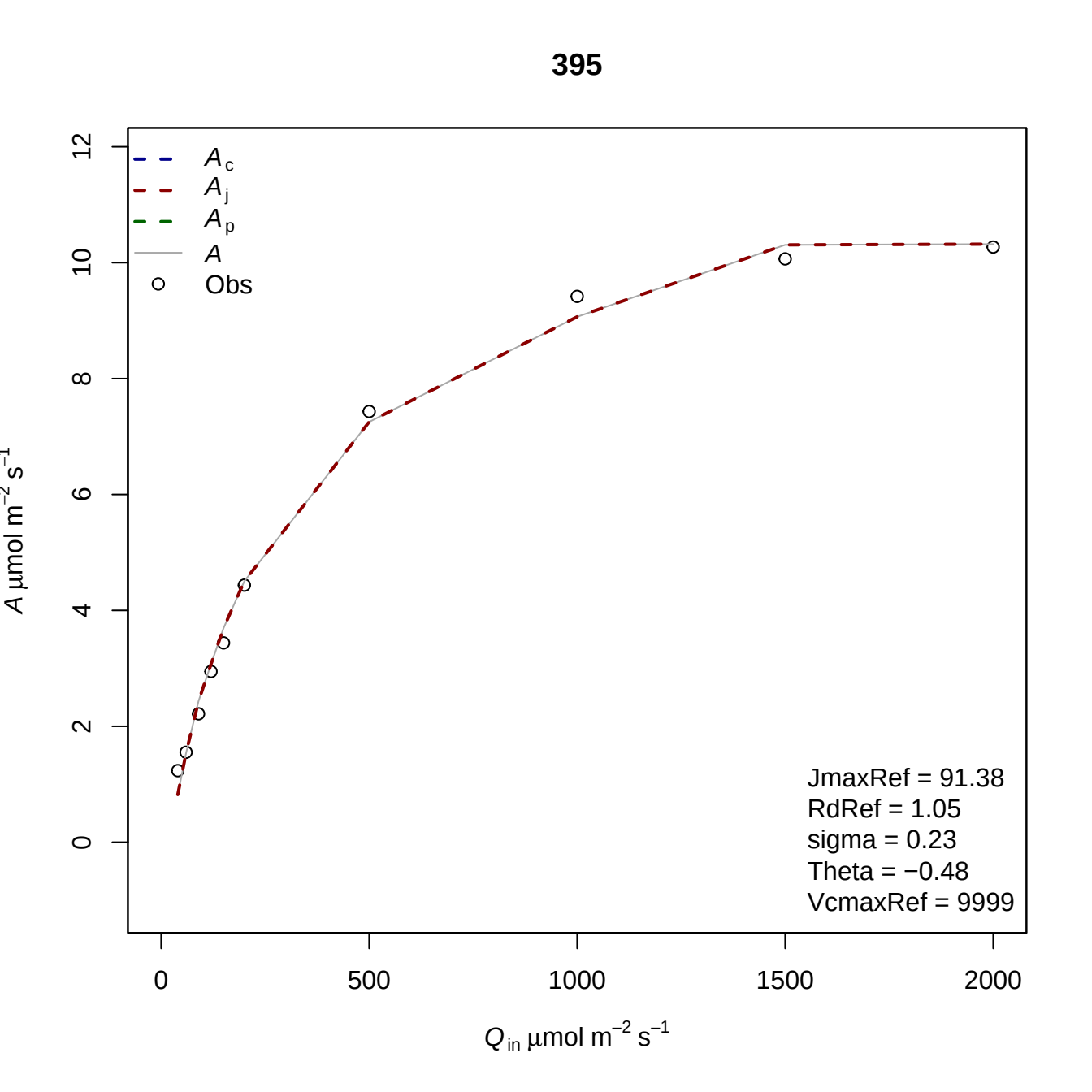

389
A_c
A_j
A_p
A
Obs
JmaxRef = 53.35
RdRef = 0.27
sigma = 0.1
Theta = 0.2
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

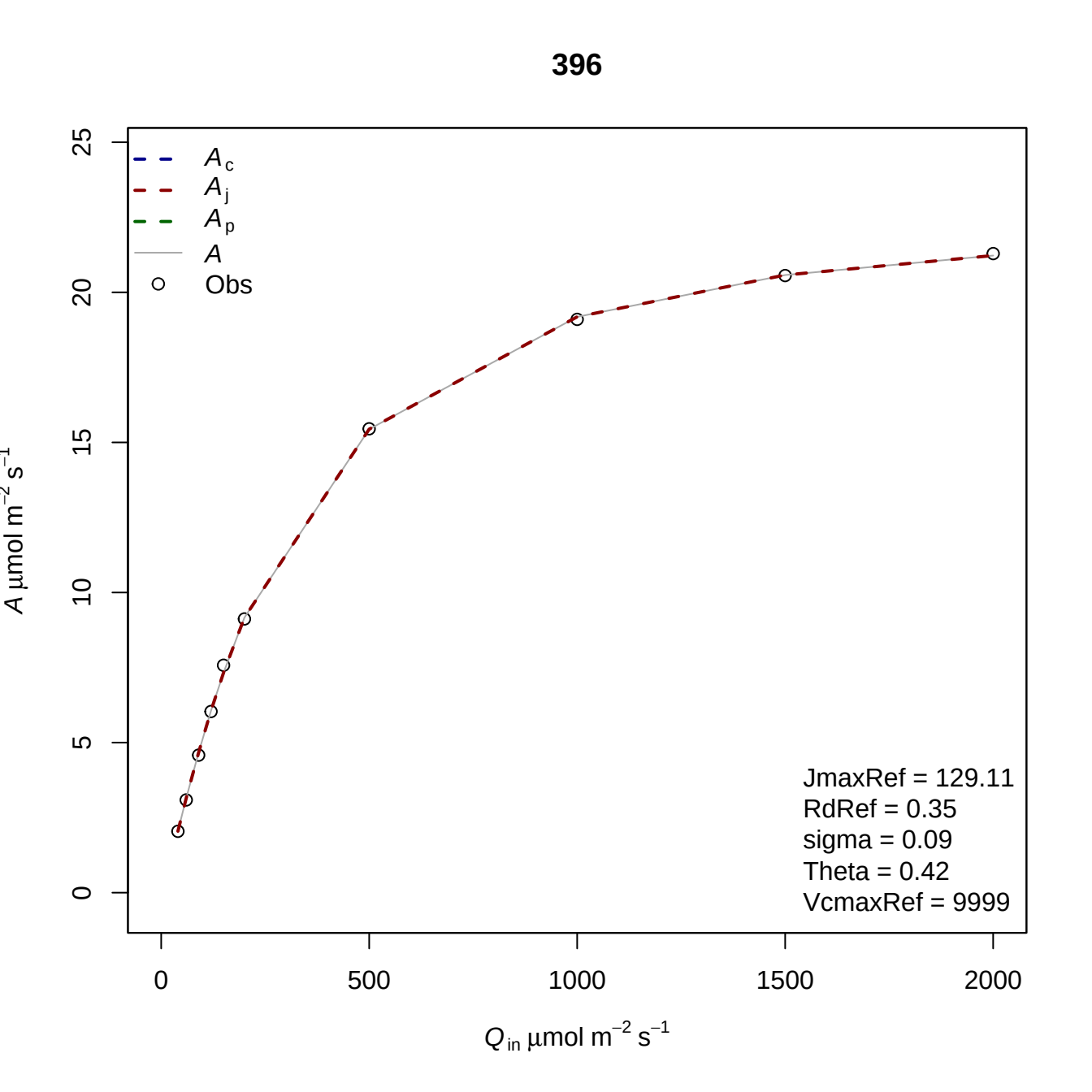

390
A_c
A_j
A_p
A
Obs
JmaxRef = 123.7
RdRef = 0.56
sigma = 0.13
Theta = 0.12
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

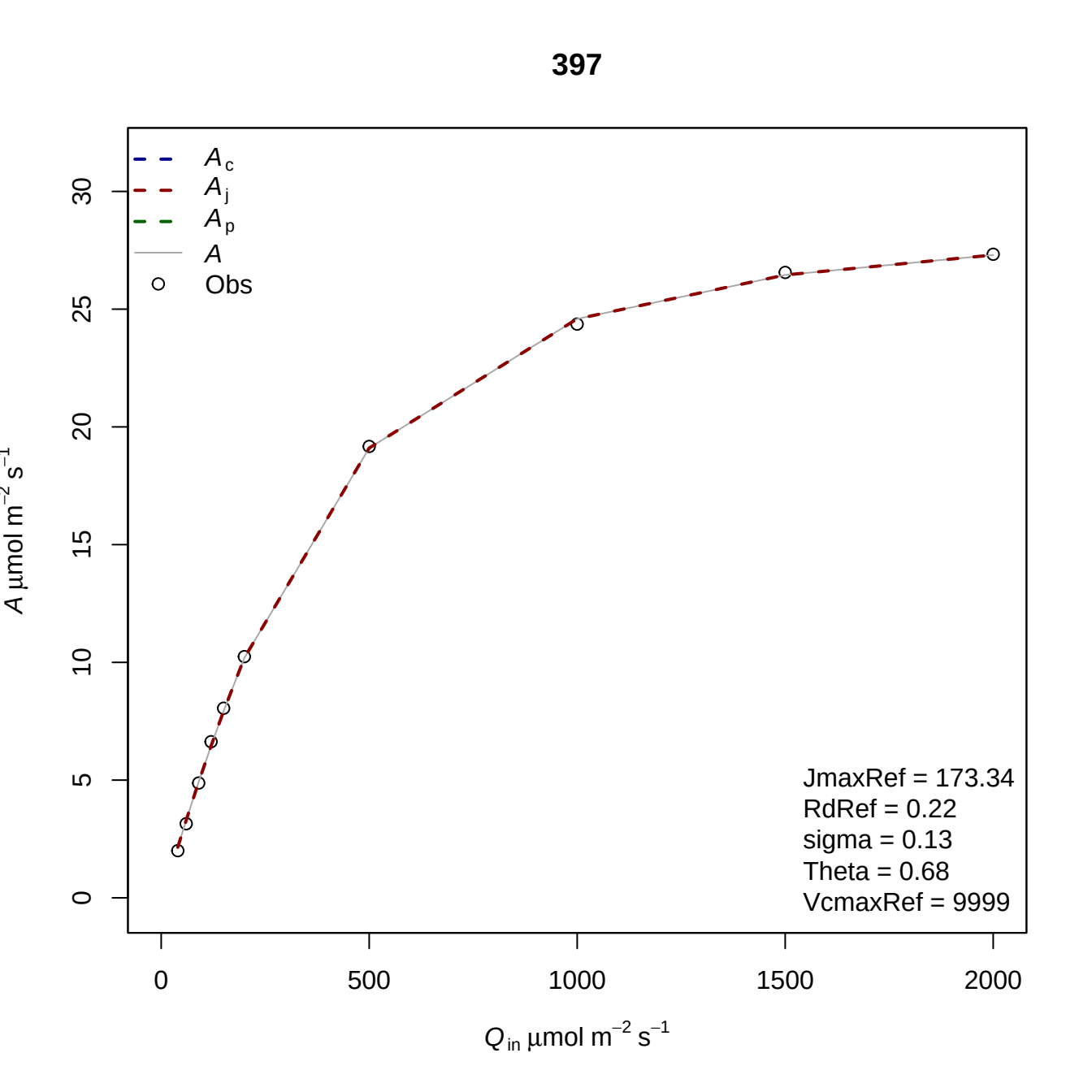

391
A_c
A_j
A_p
A
Obs
JmaxRef = 121.83
RdRef = 1.2
sigma = 0.24
Theta = 0.12
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

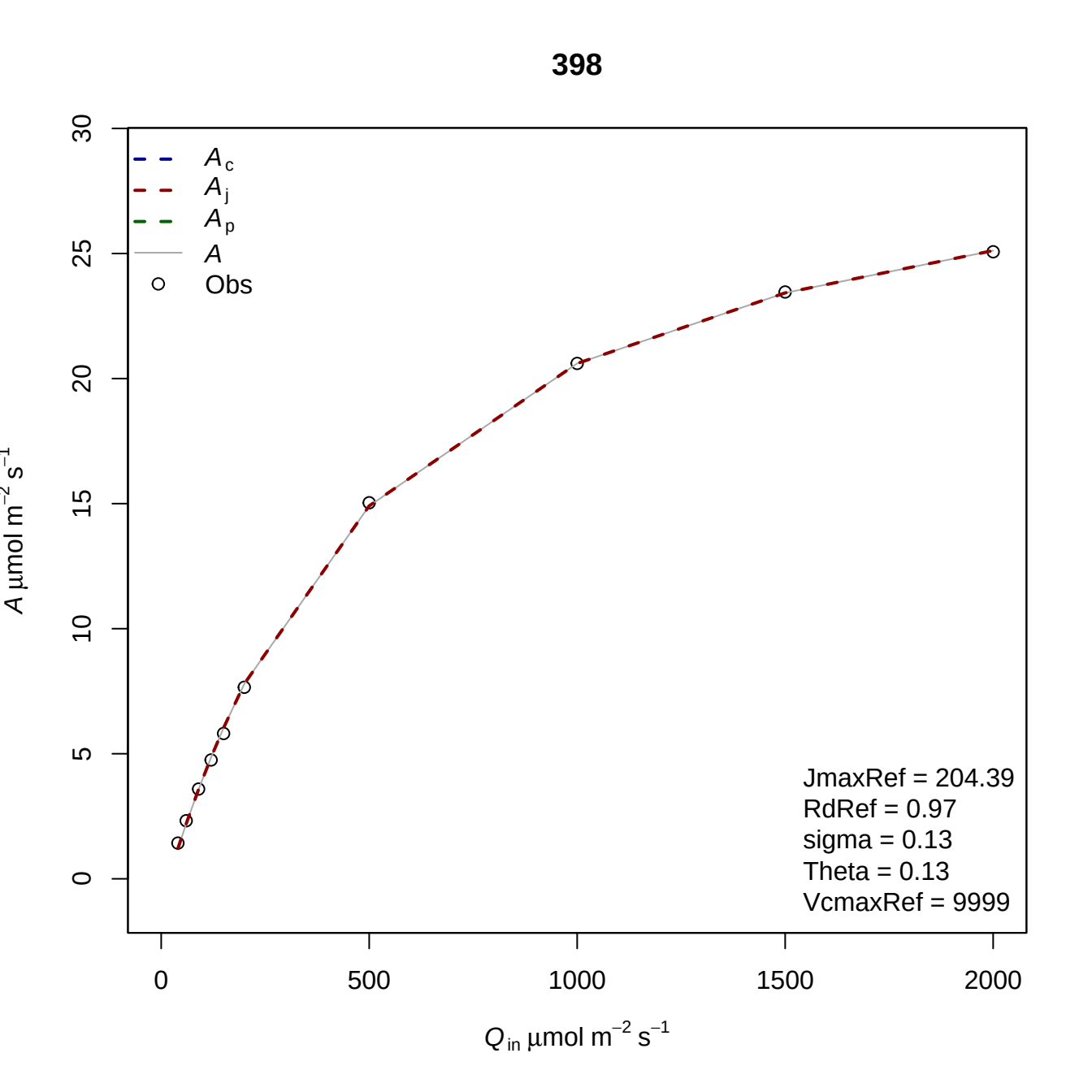

392
A_c
A_j
A_p
A
Obs
JmaxRef = 136.92
RdRef = 1.2
sigma = 0.19
Theta = 0.61
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

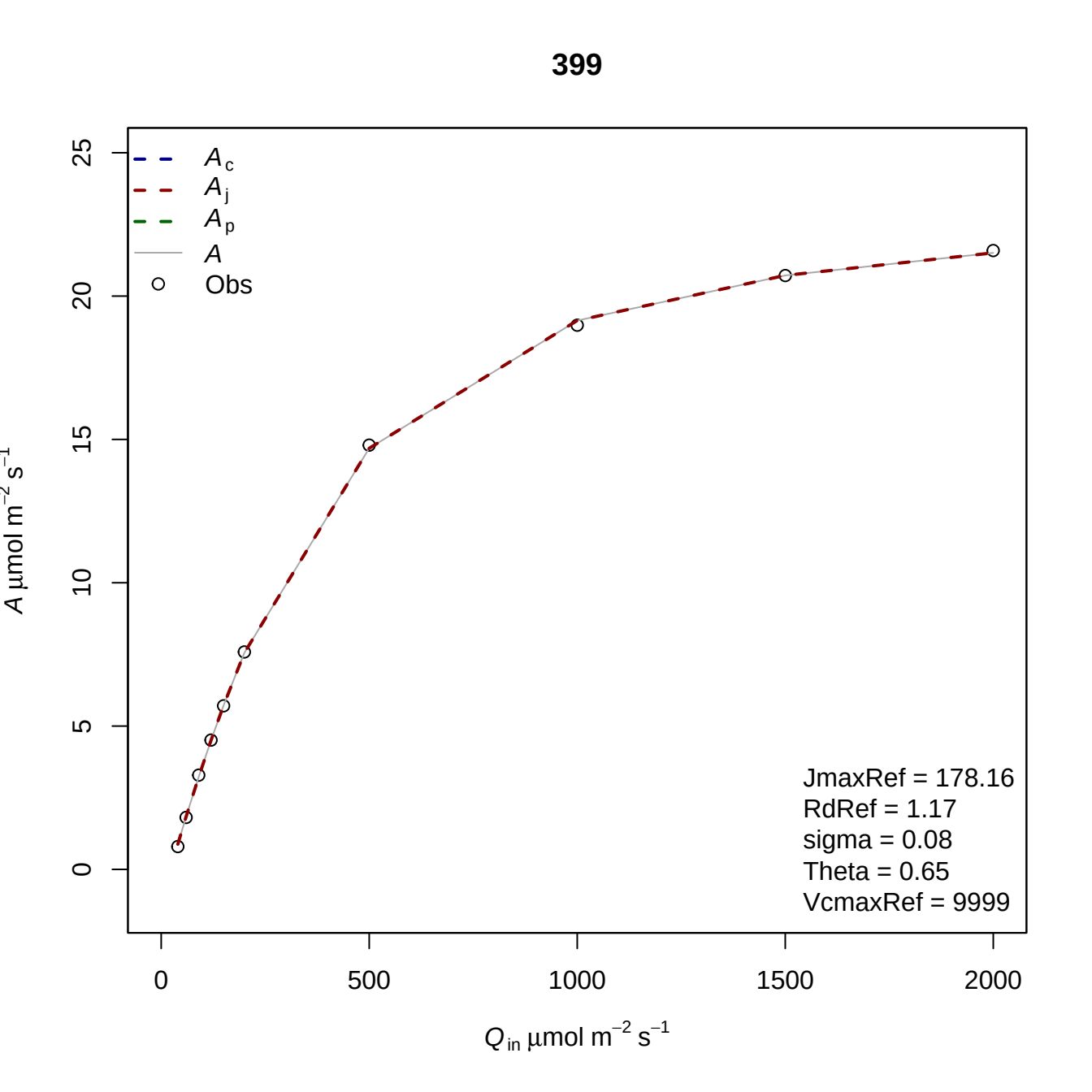

393
A_c
A_j
A_p
A
Obs
JmaxRef = 437745486.74
RdRef = 1.21
sigma = 0.32
Theta = −14996376.4
VcmaxRef = 120.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

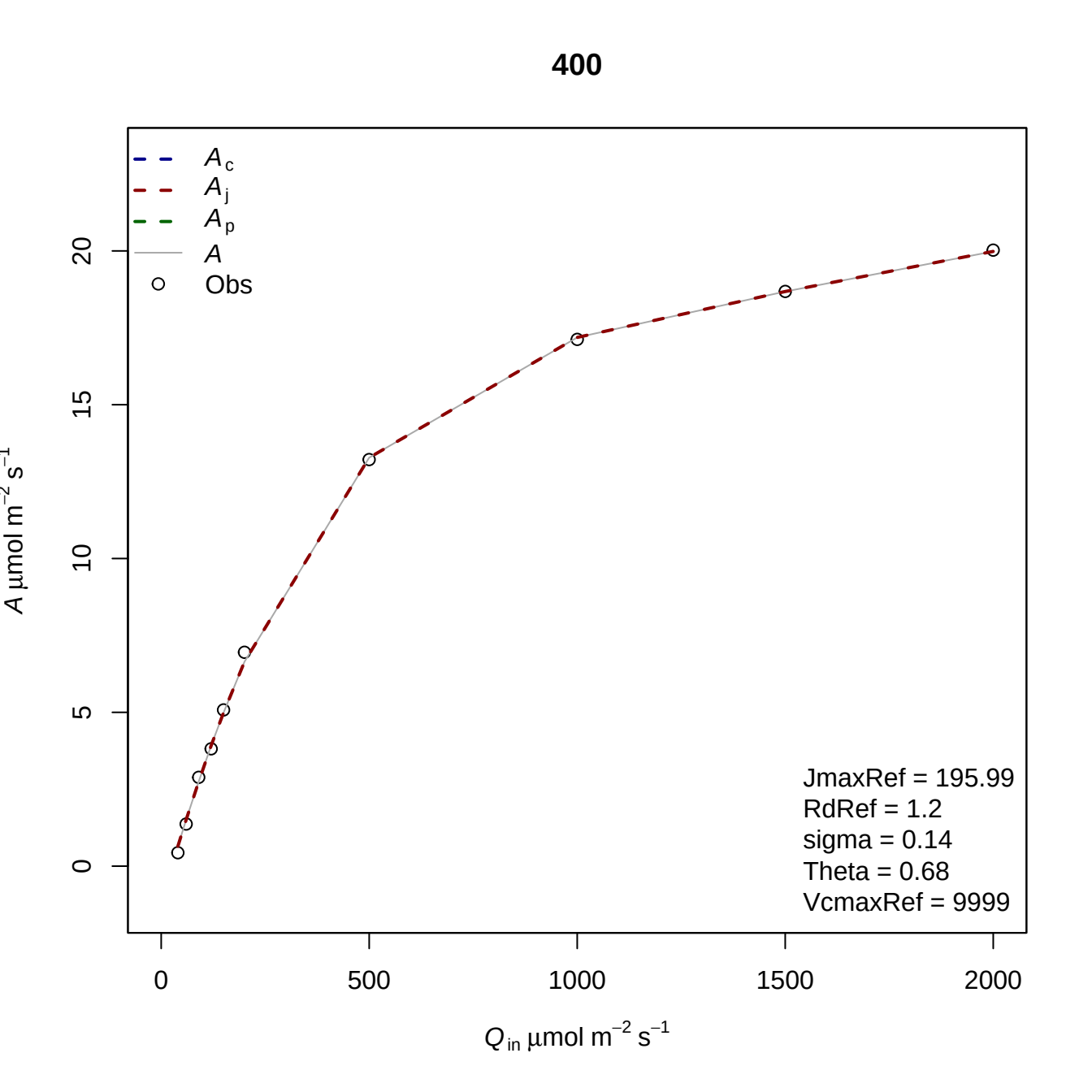

395
A_c
A_j
A_p
A
Obs
JmaxRef = 91.38
RdRef = 1.05
sigma = 0.23
Theta = −0.48
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

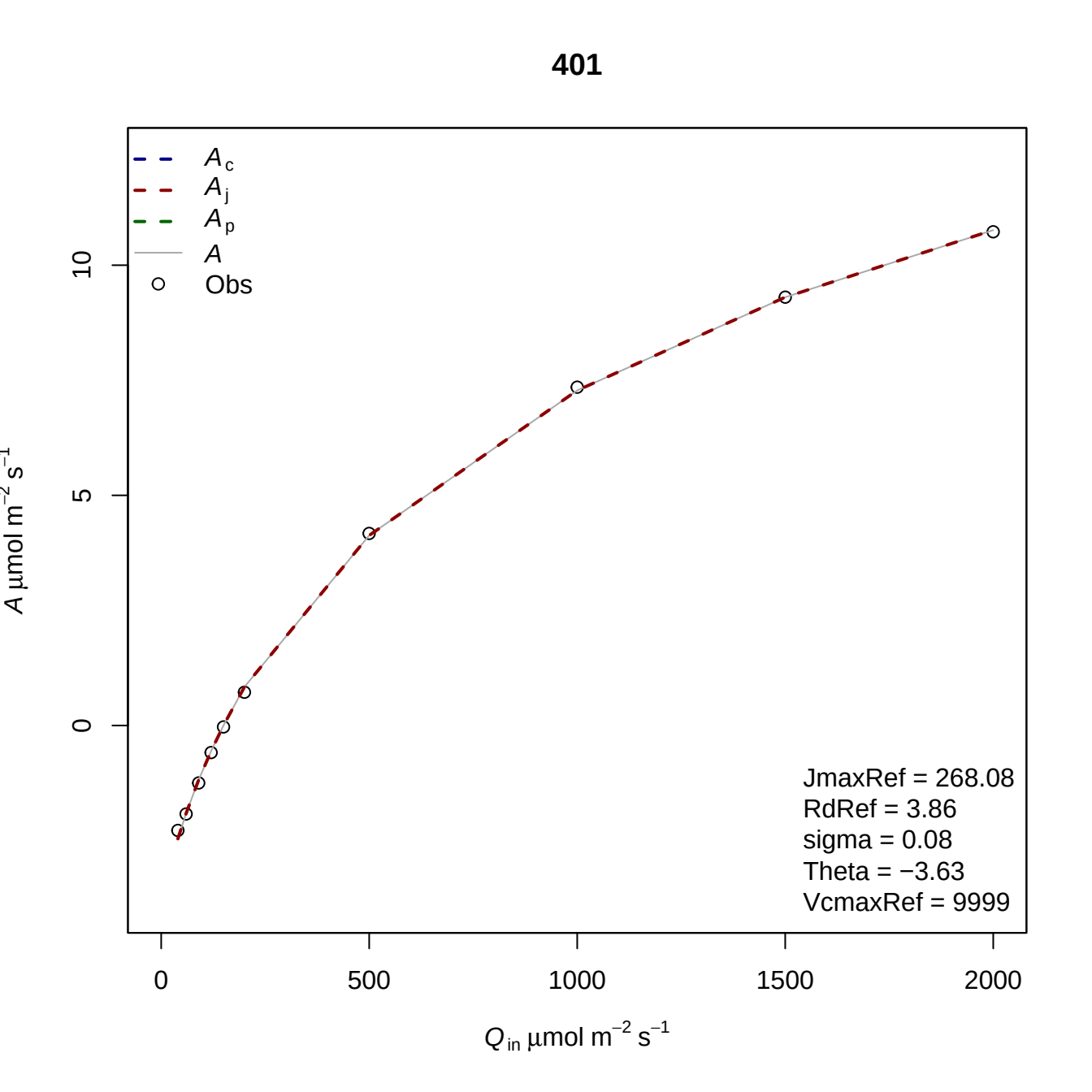

396
A_c
A_j
A_p
A
Obs
JmaxRef = 129.11
RdRef = 0.35
sigma = 0.09
Theta = 0.42
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

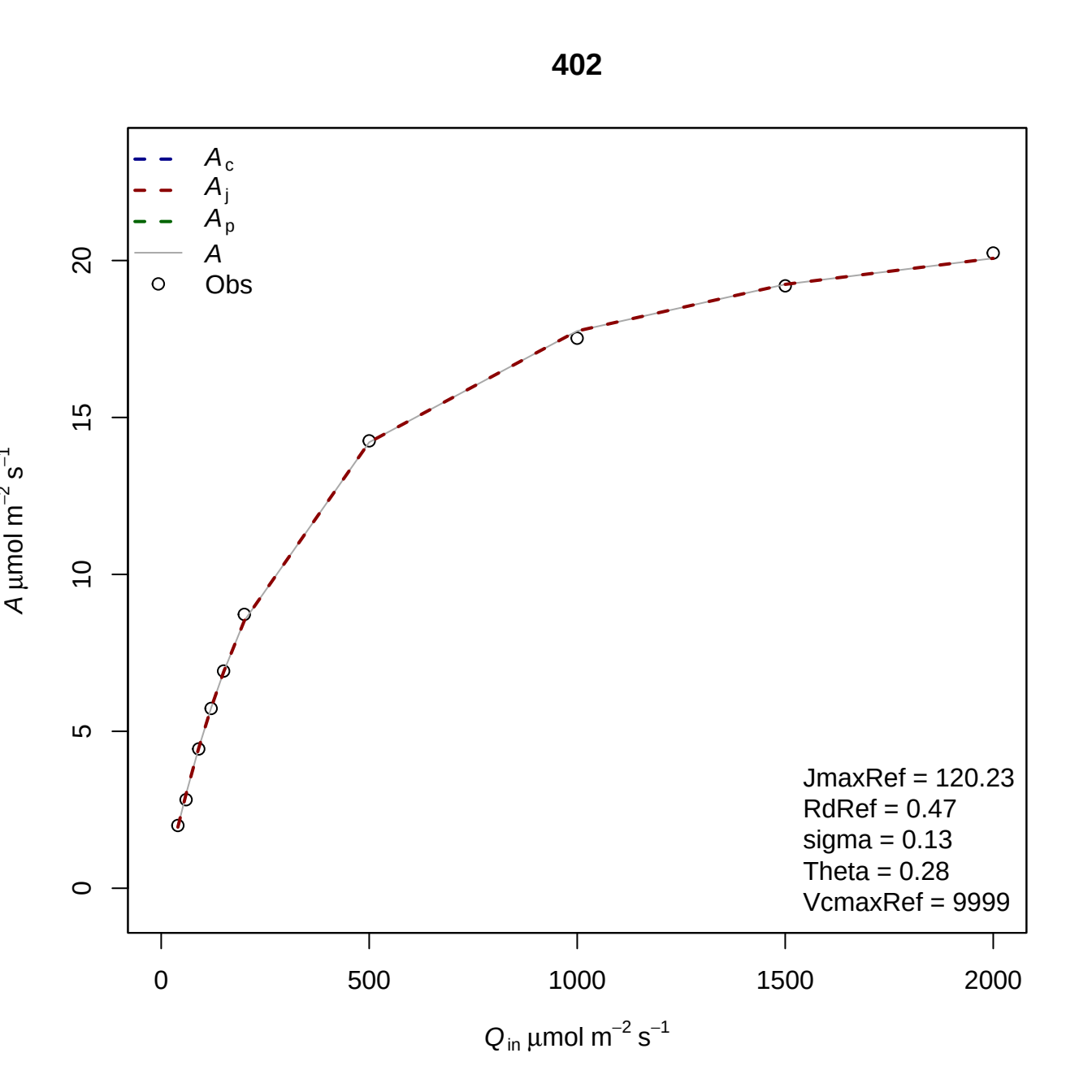

397
A_c
A_j
A_p
A
Obs
JmaxRef = 173.34
RdRef = 0.22
sigma = 0.13
Theta = 0.68
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

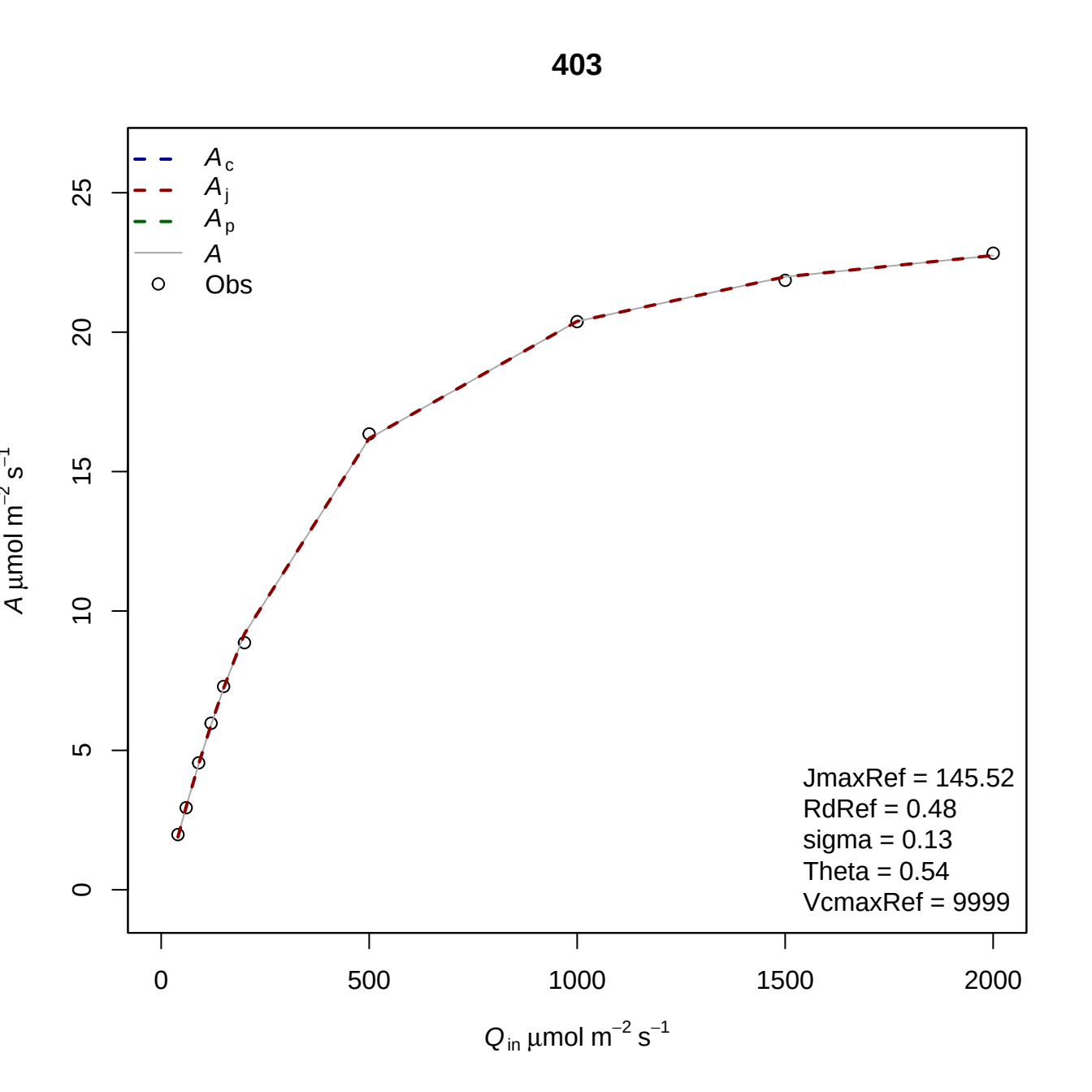

398
A_c
A_j
A_p
A
Obs
JmaxRef = 204.39
RdRef = 0.97
sigma = 0.13
Theta = 0.13
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

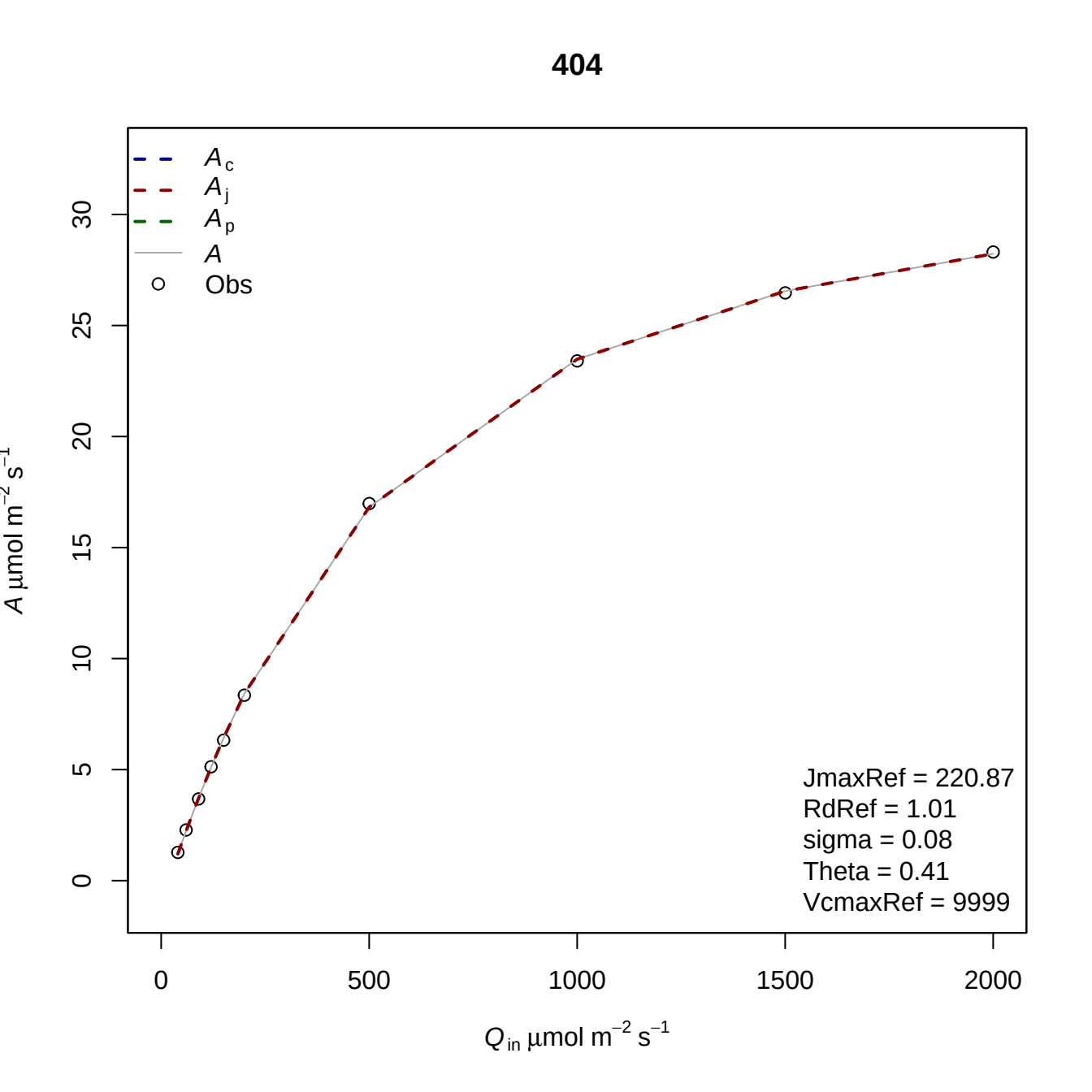

399
A_c
A_j
A_p
A
Obs
JmaxRef = 178.16
RdRef = 1.17
sigma = 0.08
Theta = 0.65
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

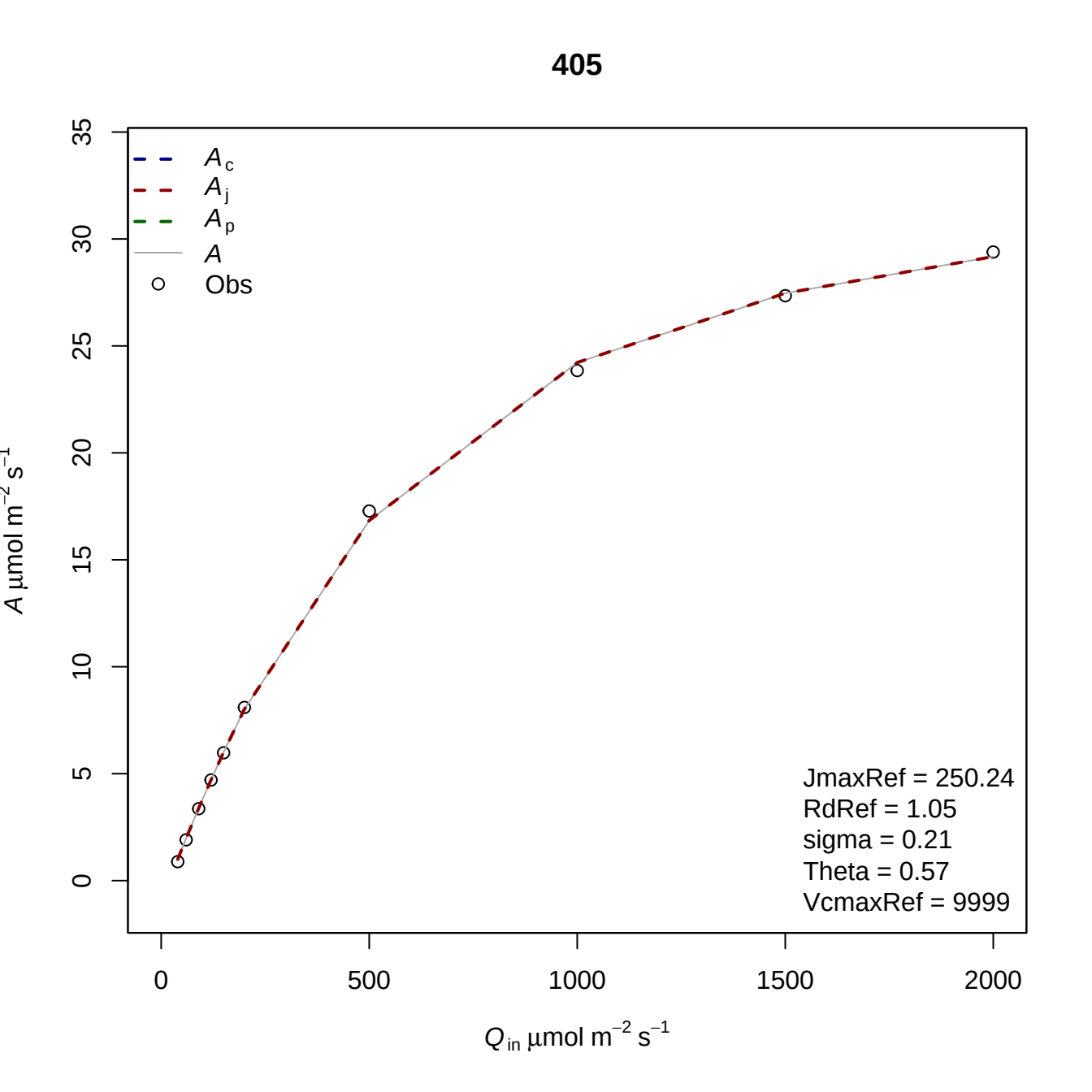

400
A_c
A_j
A_p
A
Obs
JmaxRef = 195.99
RdRef = 1.2
sigma = 0.14
Theta = 0.68
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

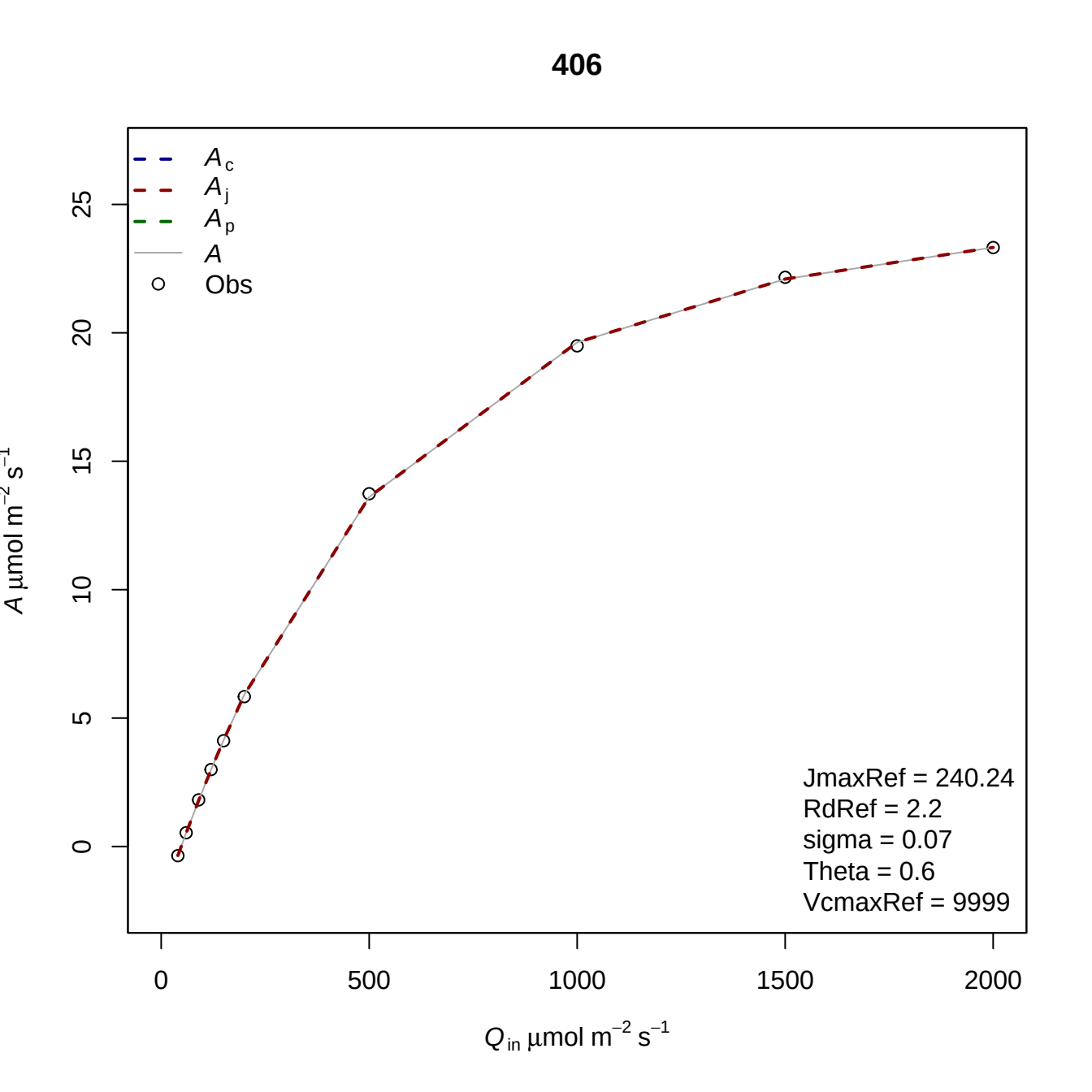

401
A_c
A_j
A_p
A
Obs
JmaxRef = 268.08
RdRef = 3.86
sigma = 0.08
Theta = −3.63
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

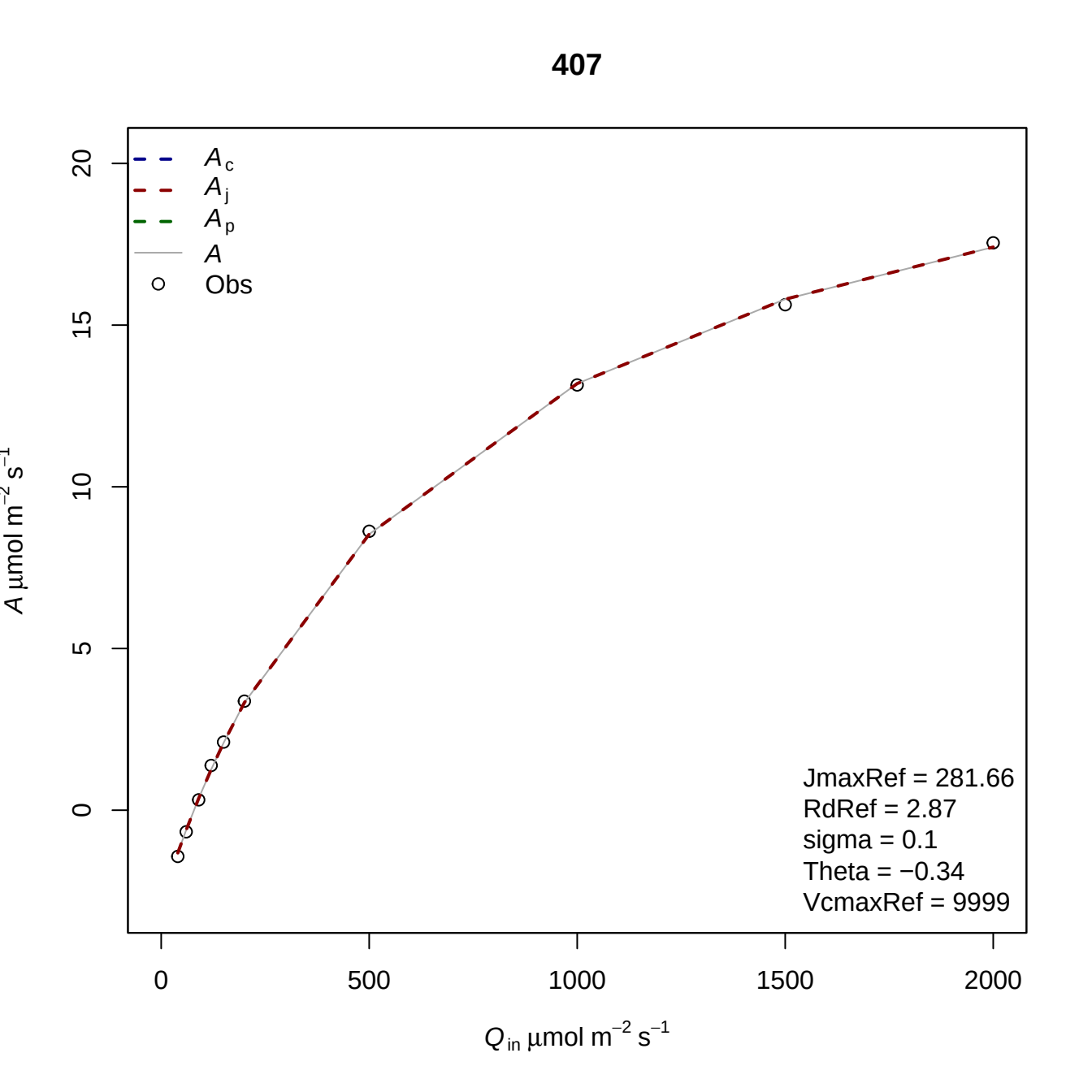

402
A_c
A_j
A_p
A
Obs
JmaxRef = 120.23
RdRef = 0.47
sigma = 0.13
Theta = 0.28
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

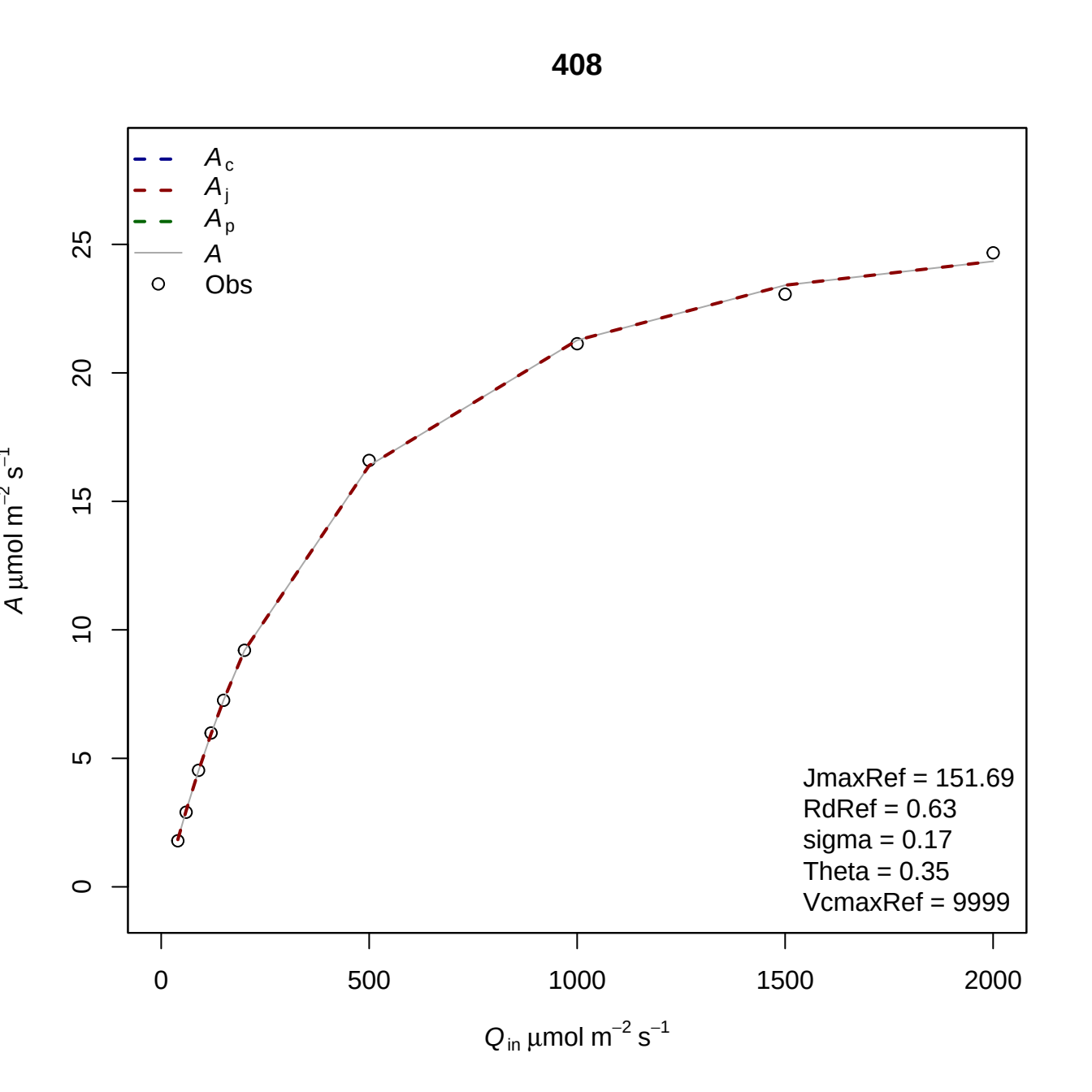

403
A_c
A_j
A_p
A
Obs
JmaxRef = 145.52
RdRef = 0.48
sigma = 0.13
Theta = 0.54
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

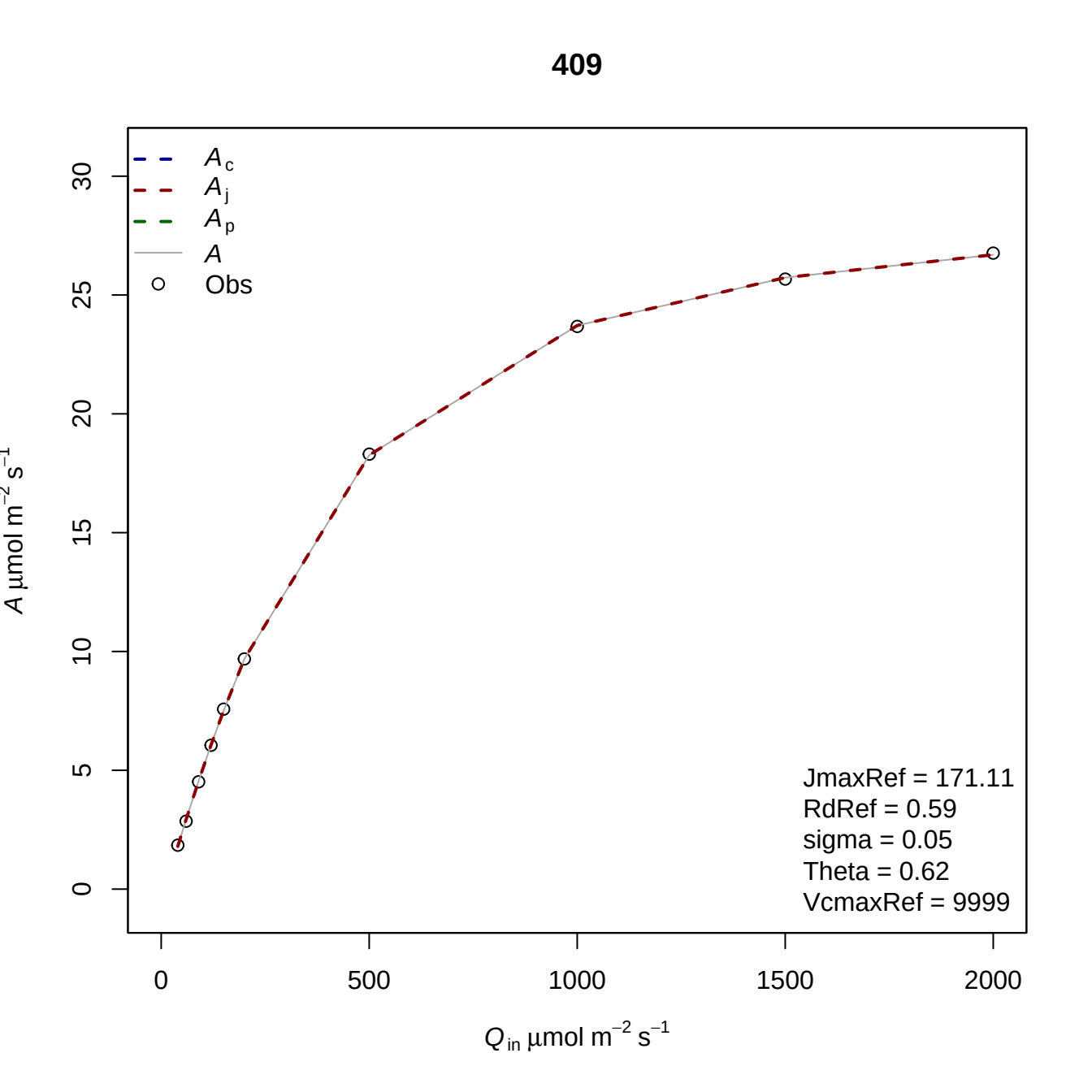

404
A_c
A_j
A_p
A
Obs
JmaxRef = 220.87
RdRef = 1.01
sigma = 0.08
Theta = 0.41
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

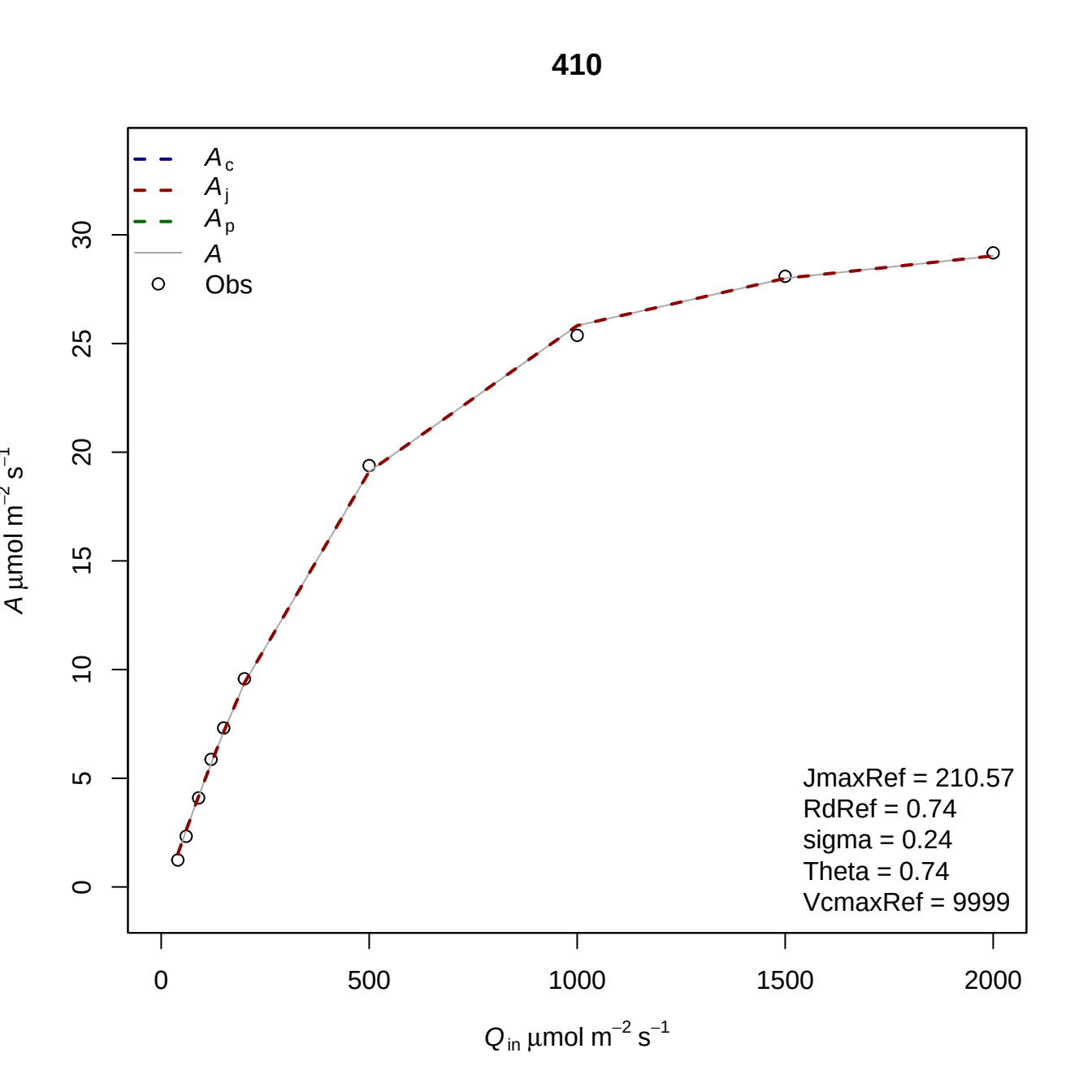

405
A_c
A_j
A_p
A
Obs
JmaxRef = 250.24
RdRef = 1.05
sigma = 0.21
Theta = 0.57
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

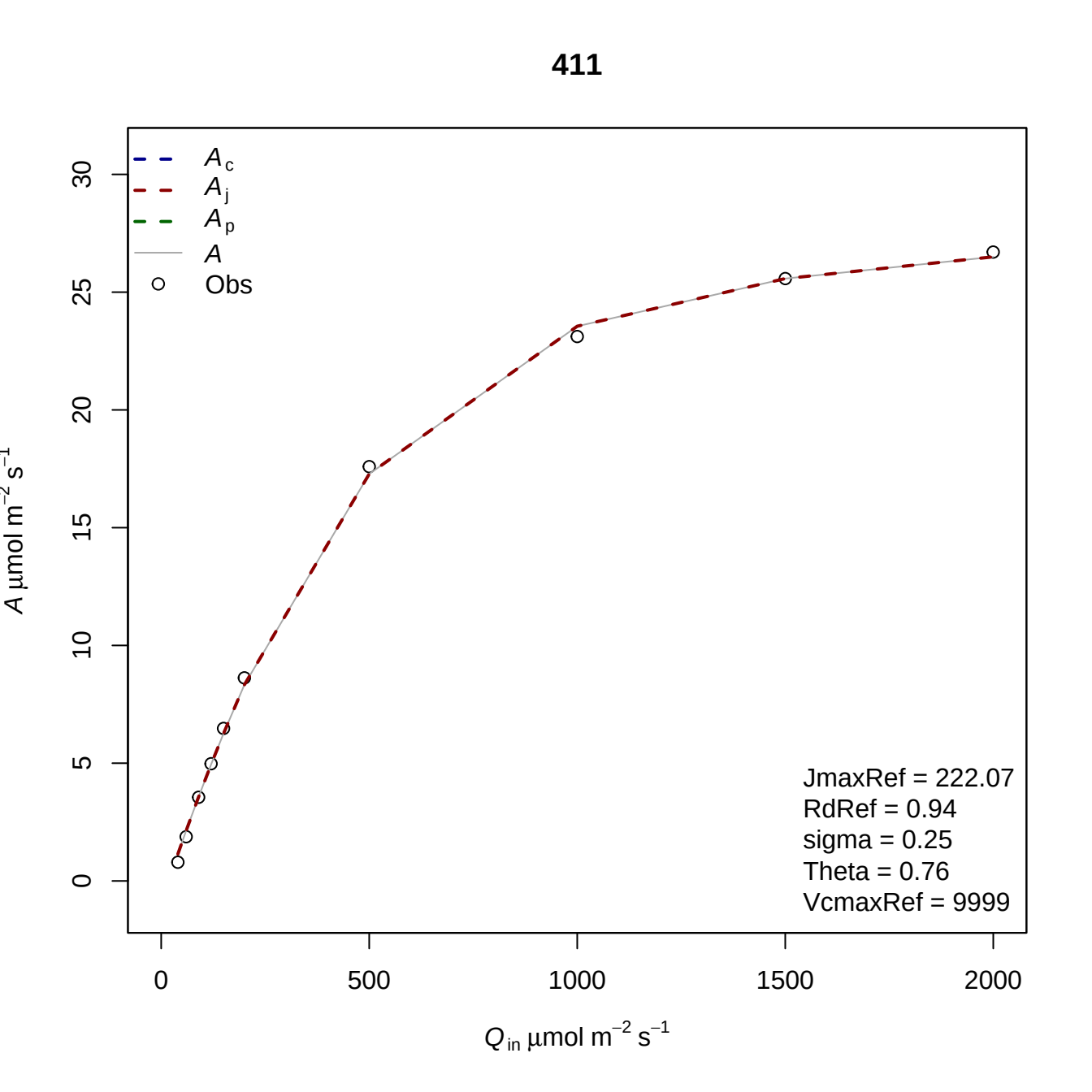

406
A_c
A_j
A_p
A
Obs
JmaxRef = 240.24
RdRef = 2.2
sigma = 0.07
Theta = 0.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

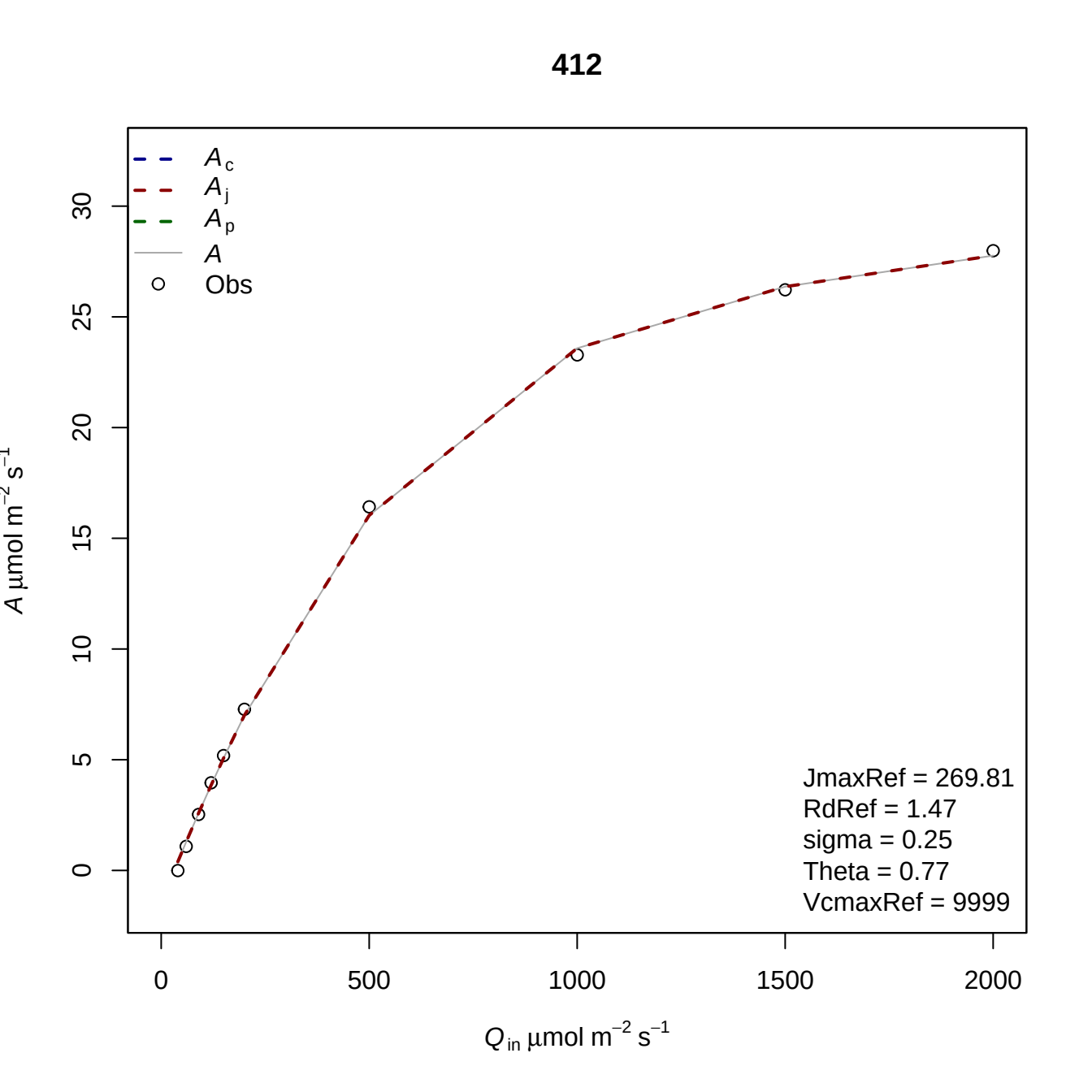

407
A_c
A_j
A_p
A
Obs
JmaxRef = 281.66
RdRef = 2.87
sigma = 0.1
Theta = −0.34
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

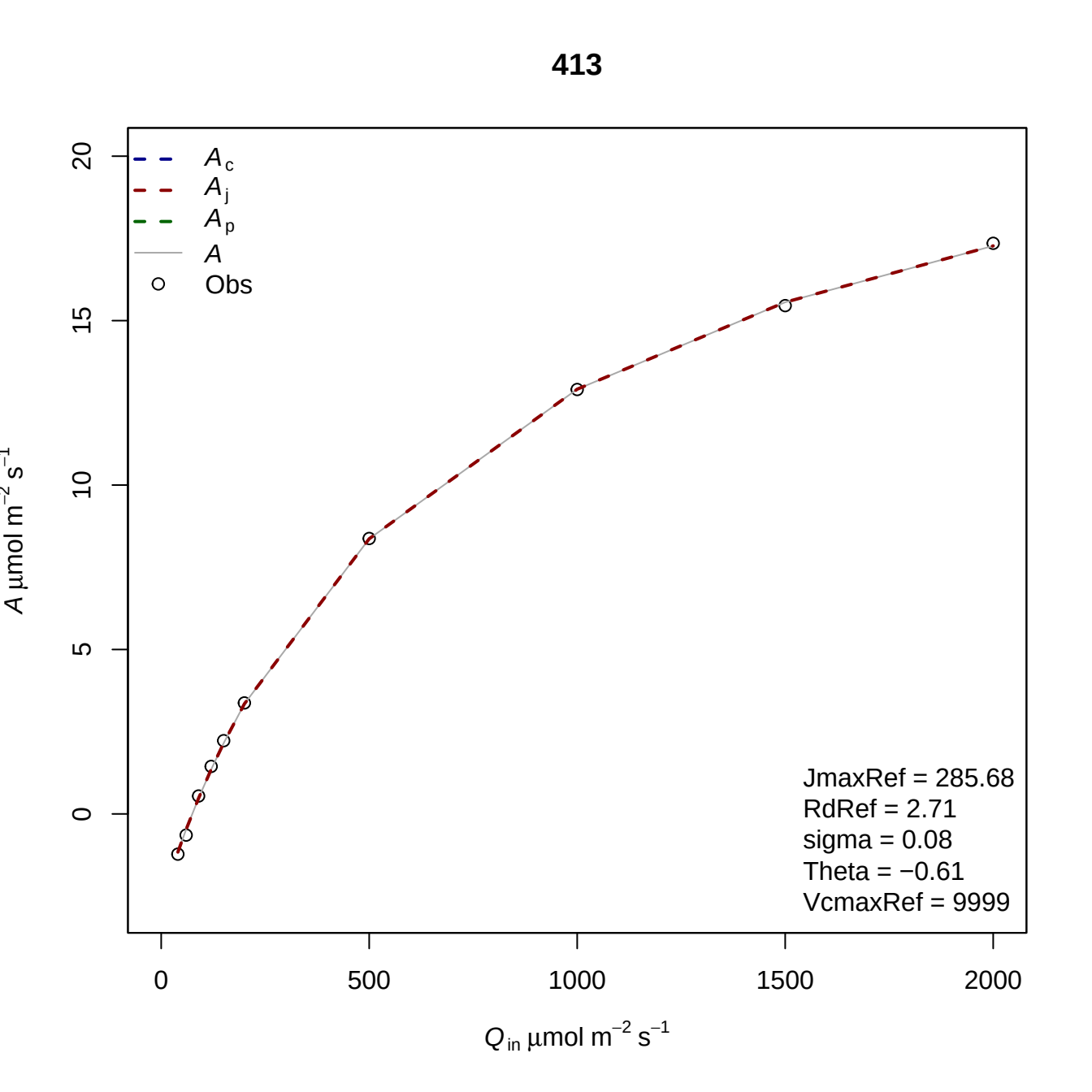

408
A_c
A_j
A_p
A
Obs
JmaxRef = 151.69
RdRef = 0.63
sigma = 0.17
Theta = 0.35
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

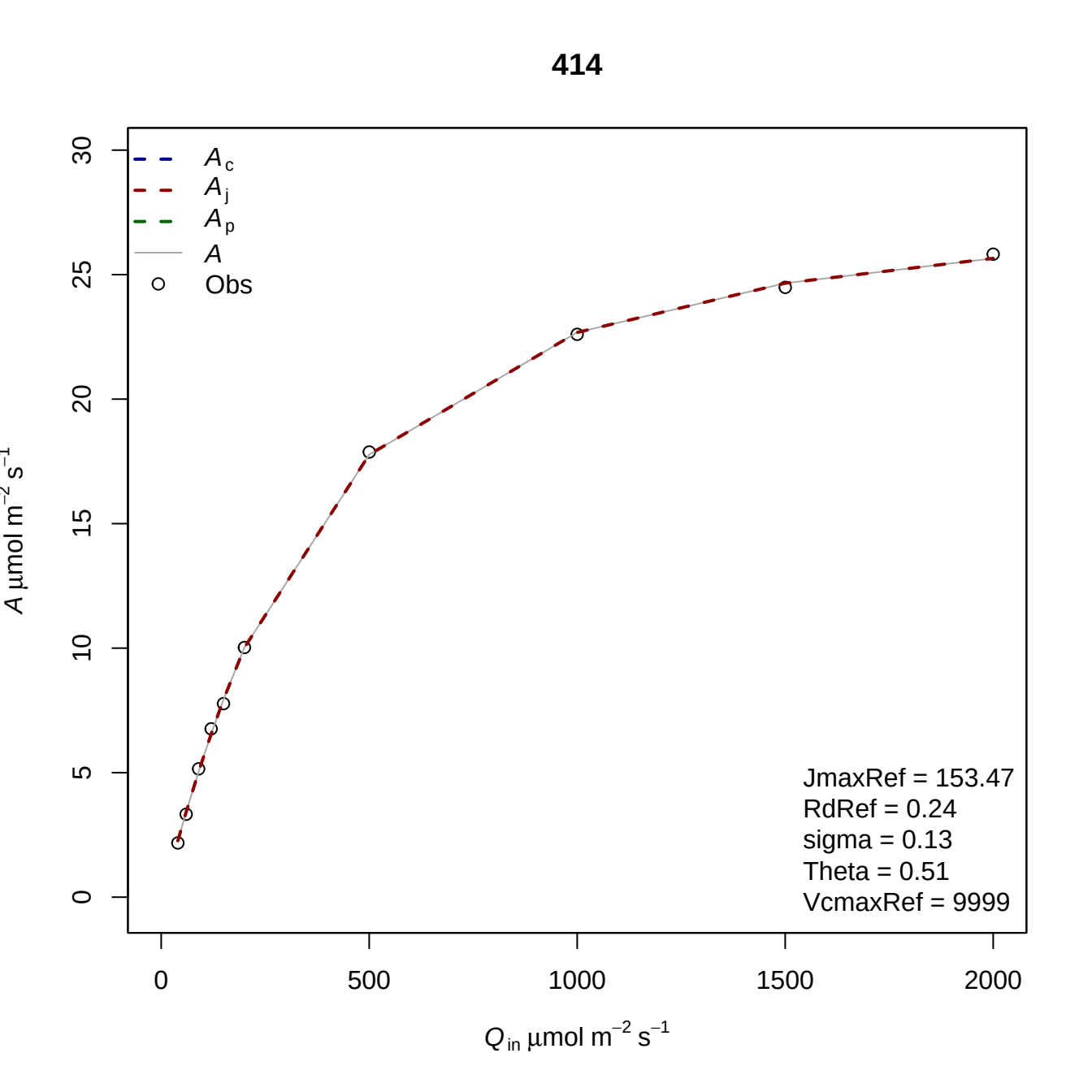

409
A_c
A_j
A_p
A
Obs
JmaxRef = 171.11
RdRef = 0.59
sigma = 0.05
Theta = 0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

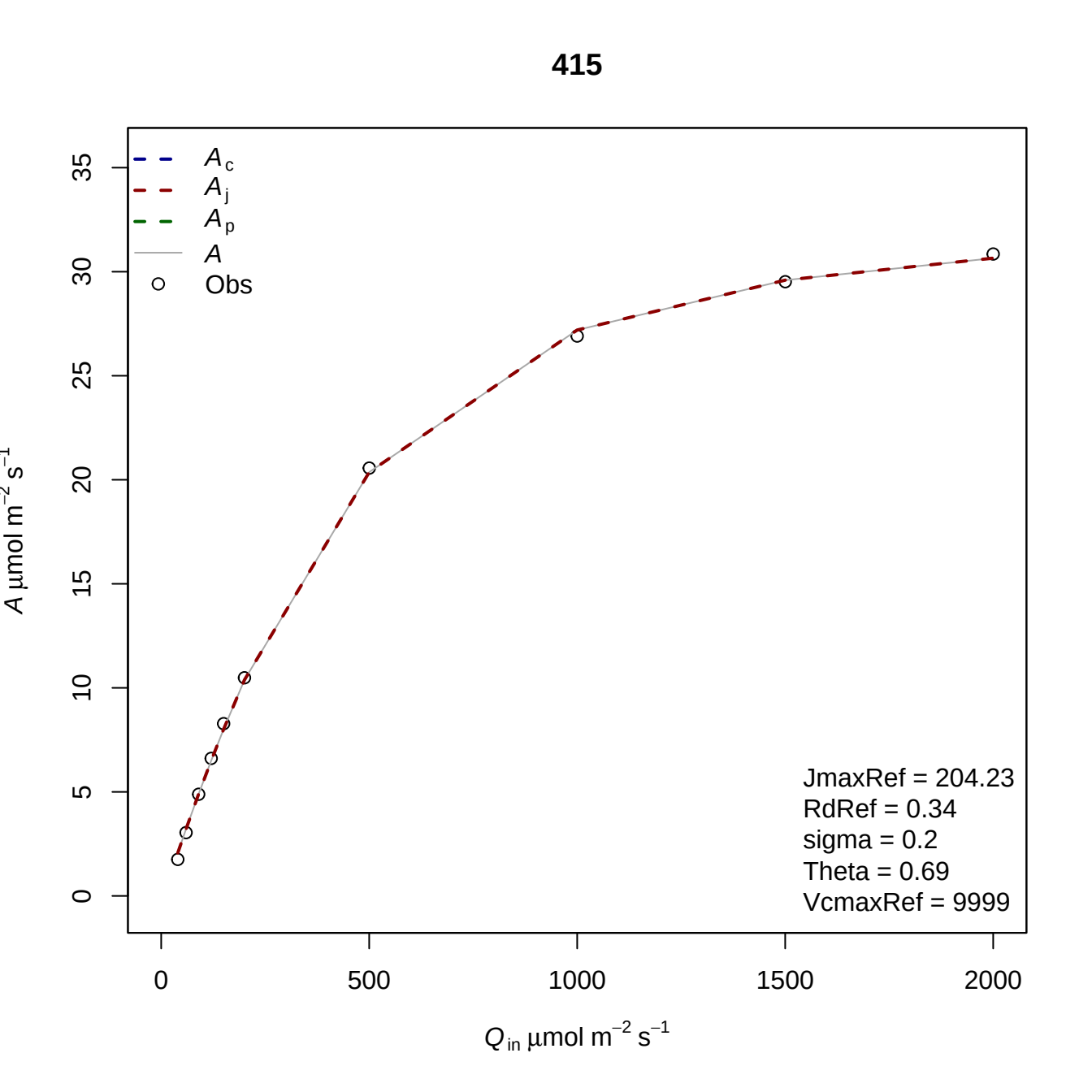

410
A_c
A_j
A_p
A
Obs
JmaxRef = 210.57
RdRef = 0.74
sigma = 0.24
Theta = 0.74
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

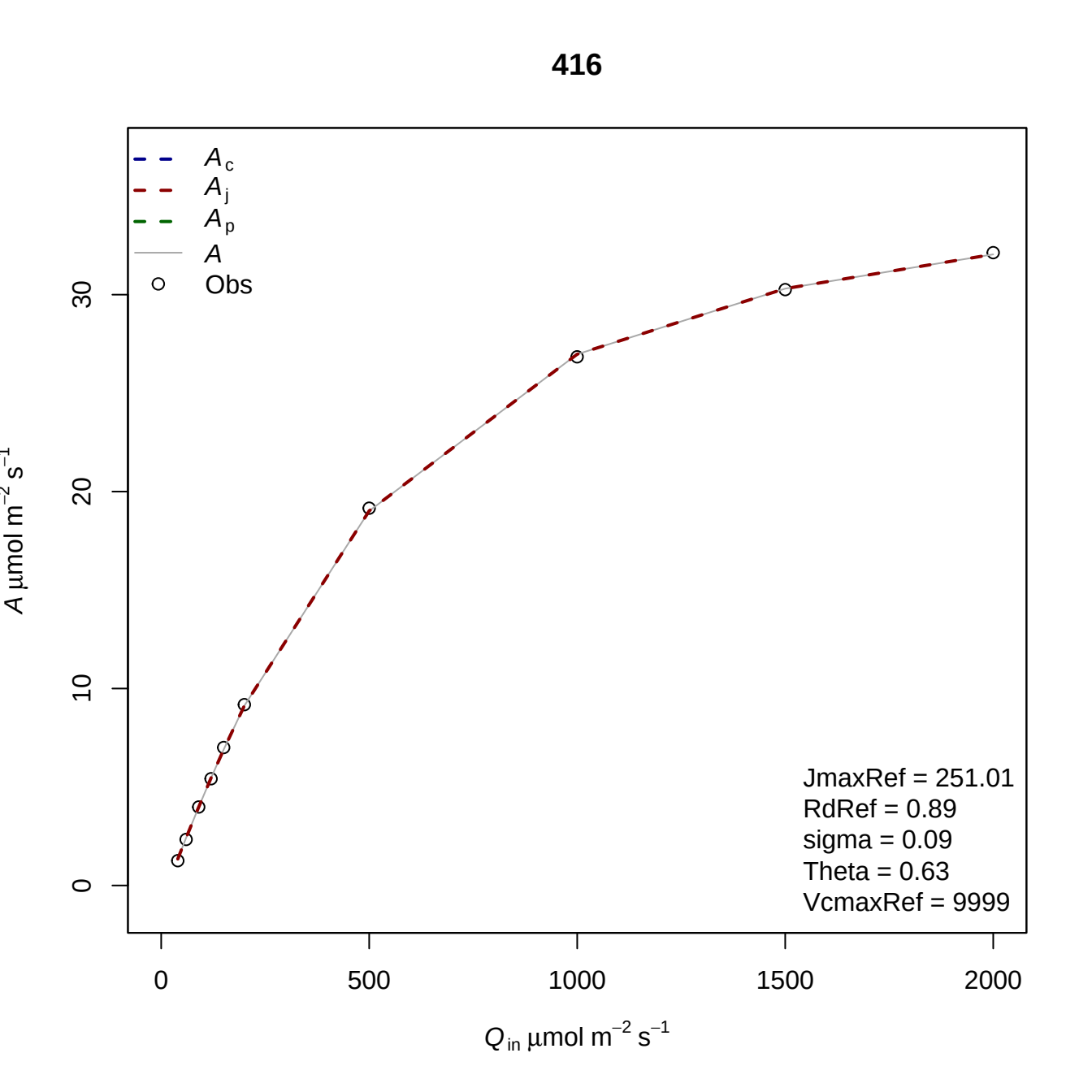

411
A_c
A_j
A_p
A
Obs
JmaxRef = 222.07
RdRef = 0.94
sigma = 0.25
Theta = 0.76
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

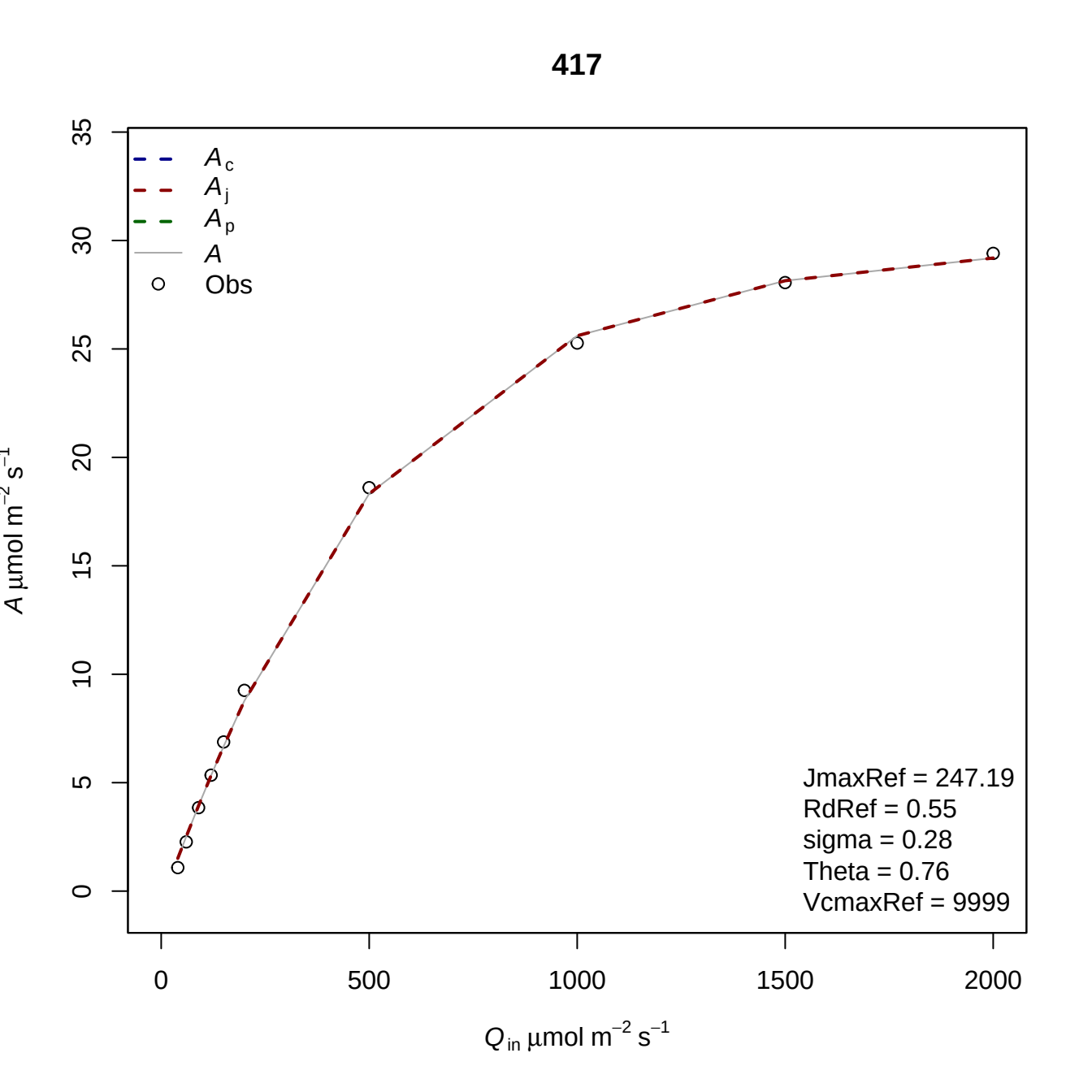

412
A_c
A_j
A_p
A
Obs
JmaxRef = 269.81
RdRef = 1.47
sigma = 0.25
Theta = 0.77
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

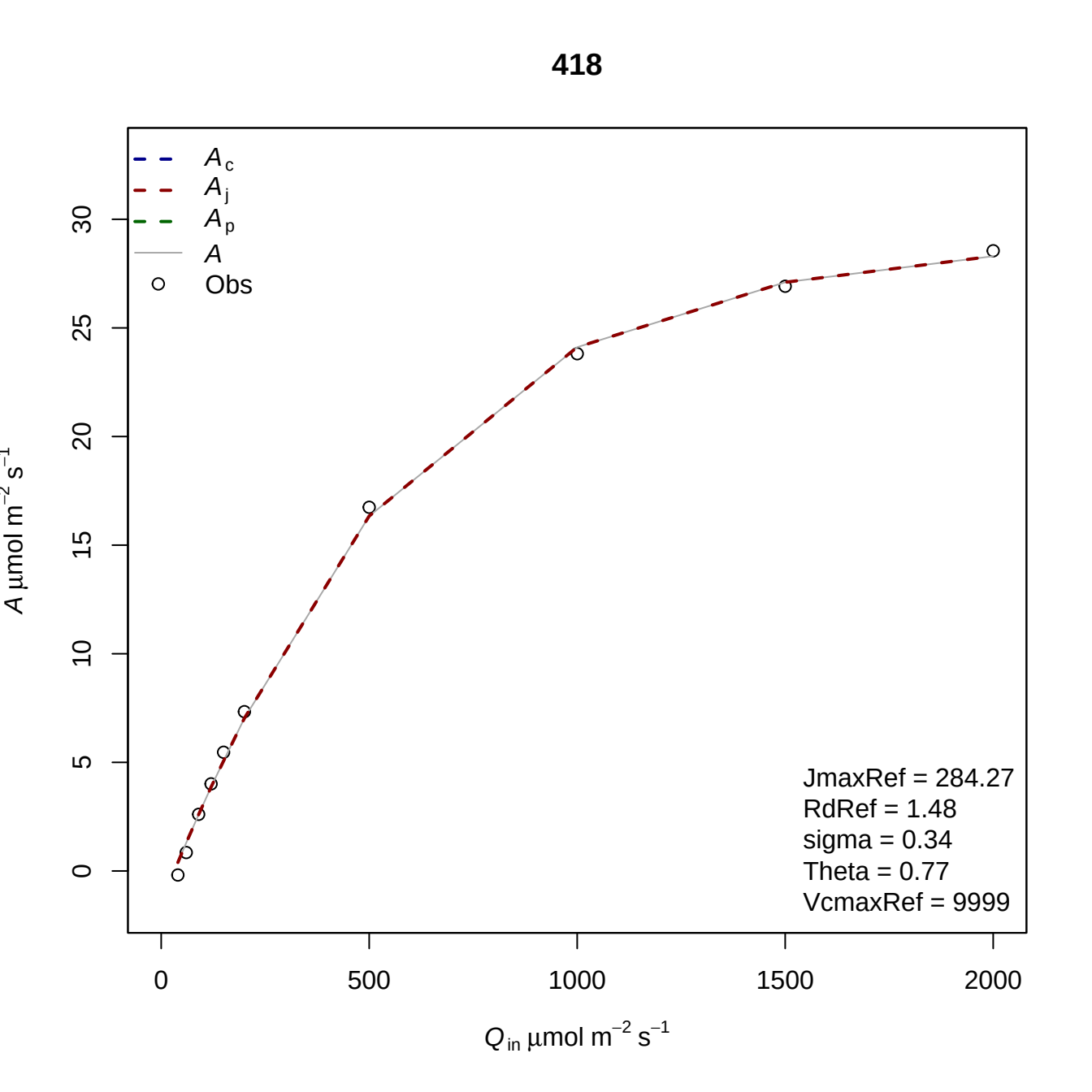

413
A_c
A_j
A_p
A
Obs
JmaxRef = 285.68
RdRef = 2.71
sigma = 0.08
Theta = −0.61
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

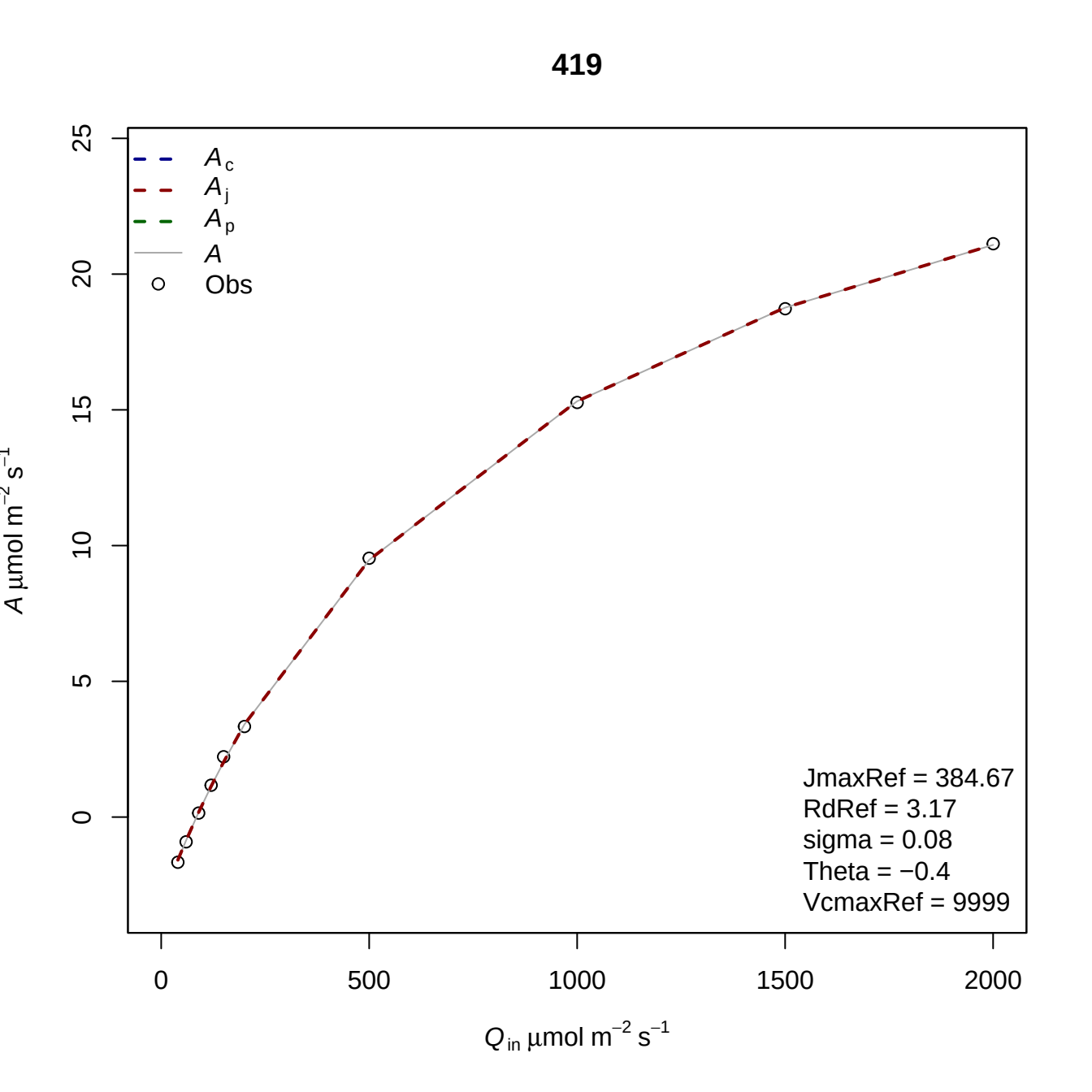

414
A_c
A_j
A_p
A
Obs
JmaxRef = 153.47
RdRef = 0.24
sigma = 0.13
Theta = 0.51
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

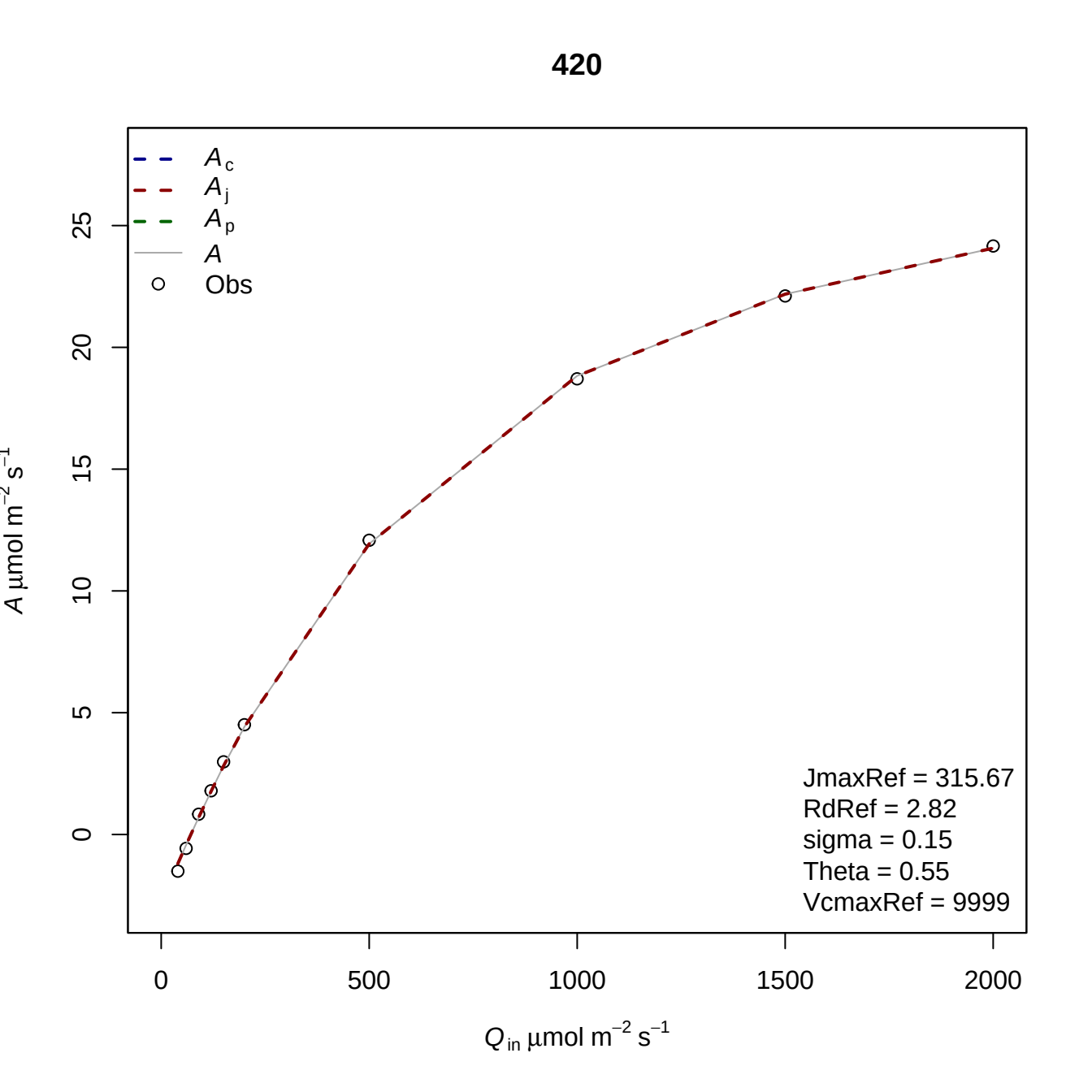

415
A_c
A_j
A_p
A
Obs
JmaxRef = 204.23
RdRef = 0.34
sigma = 0.2
Theta = 0.69
VcmaxRef = 9999
0
5
10
15
20
25
30
35
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

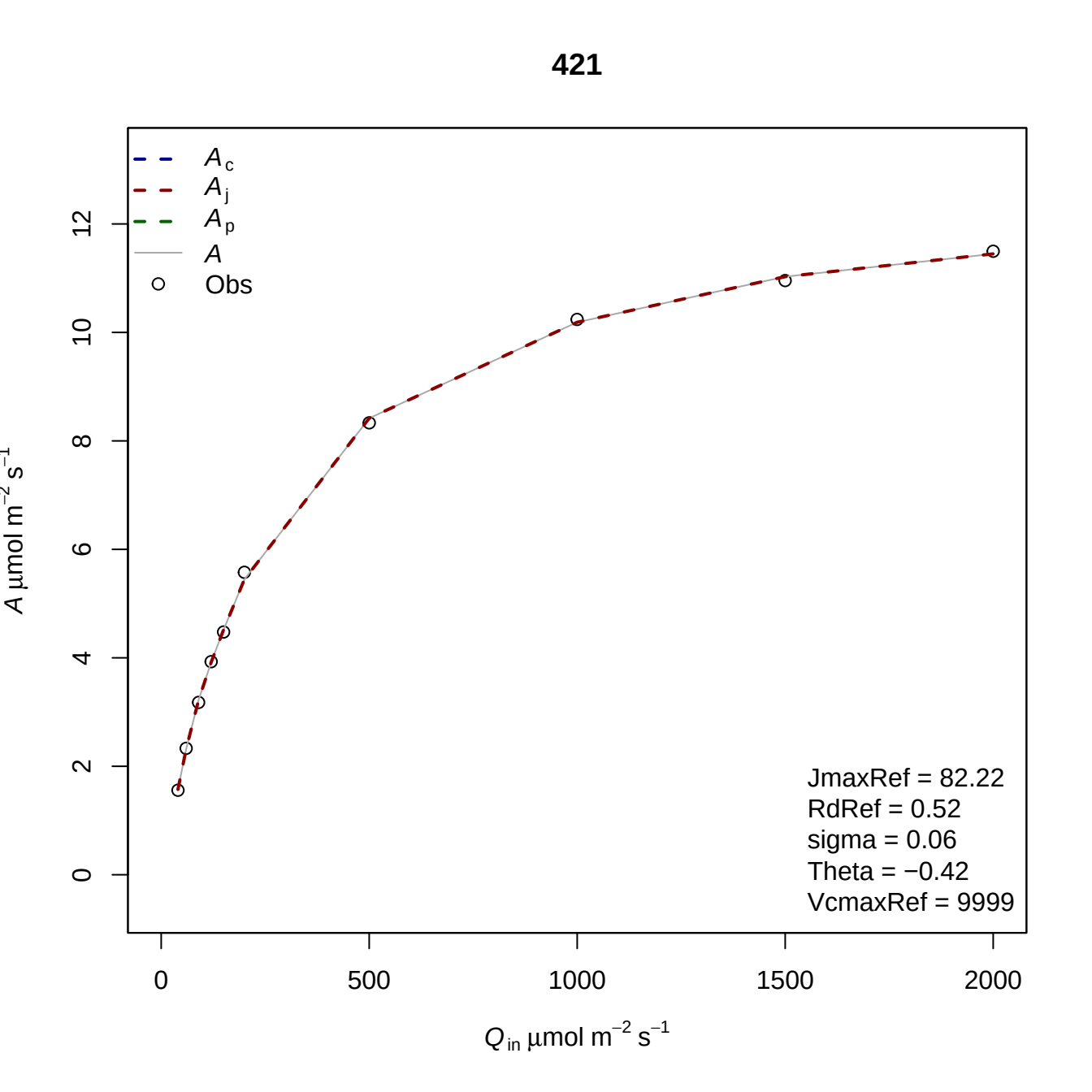

416
A_c
A_j
A_p
A
Obs
JmaxRef = 251.01
RdRef = 0.89
sigma = 0.09
Theta = 0.63
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

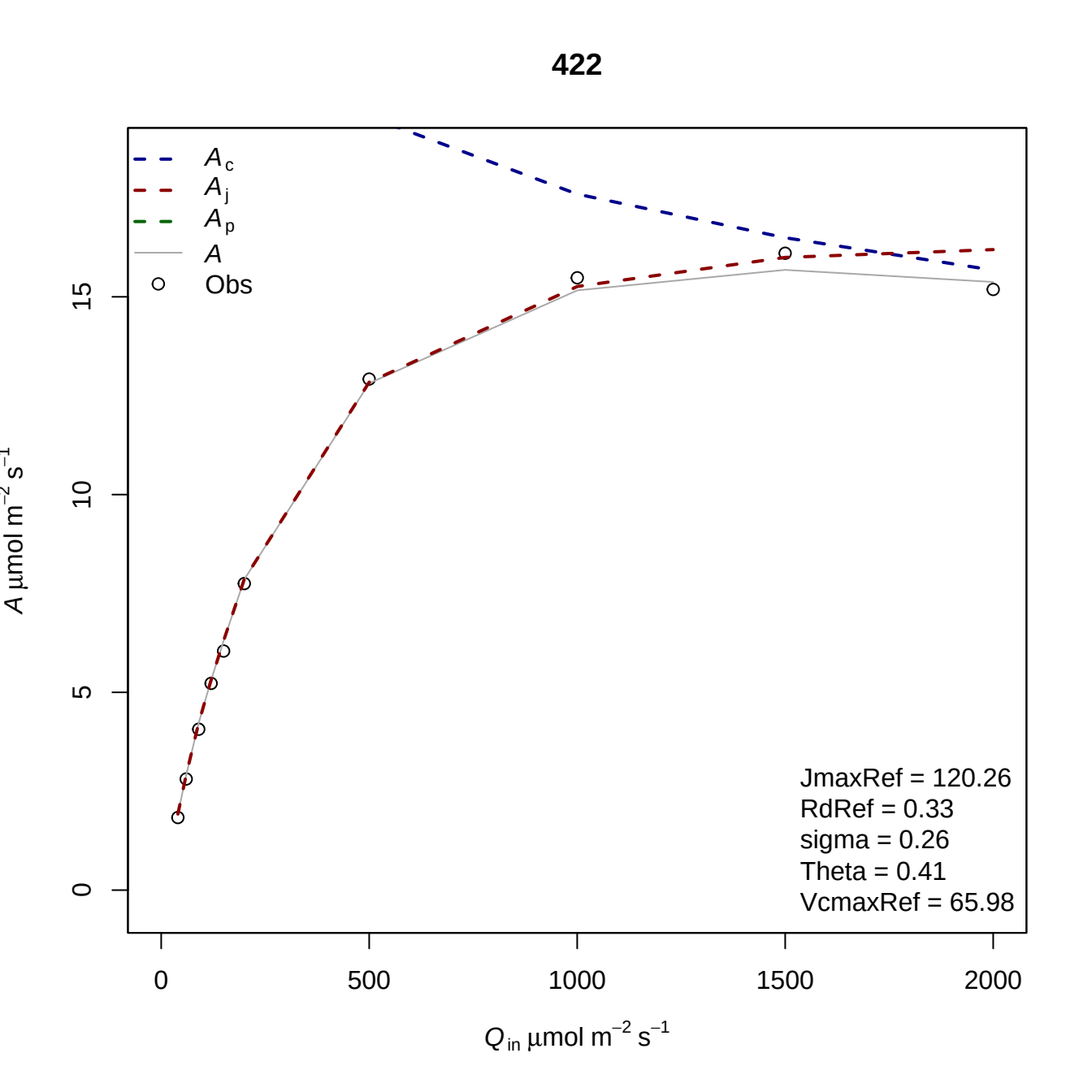

417
A_c
A_j
A_p
A
Obs
JmaxRef = 247.19
RdRef = 0.55
sigma = 0.28
Theta = 0.76
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

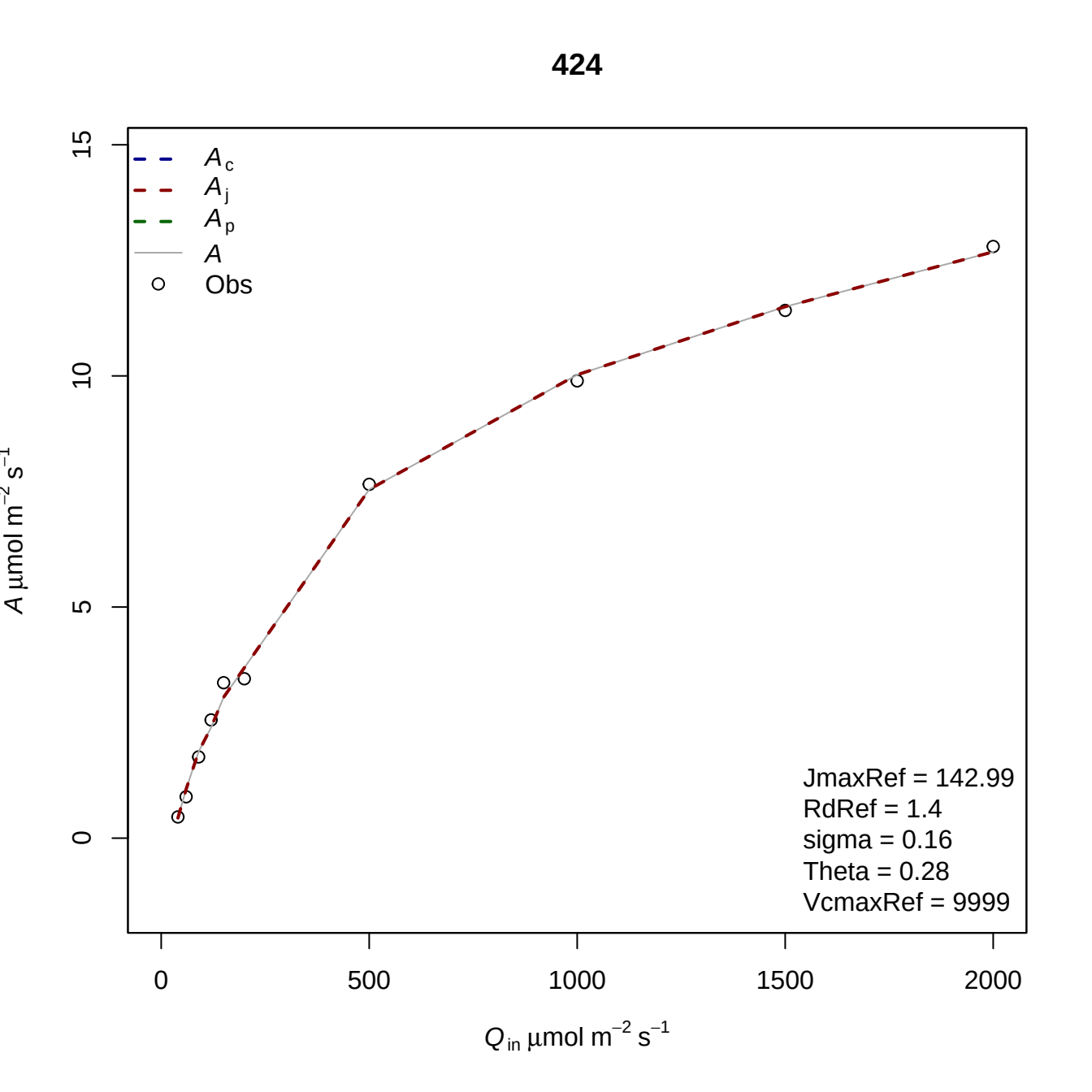

418
A_c
A_j
A_p
A
Obs
JmaxRef = 284.27
RdRef = 1.48
sigma = 0.34
Theta = 0.77
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

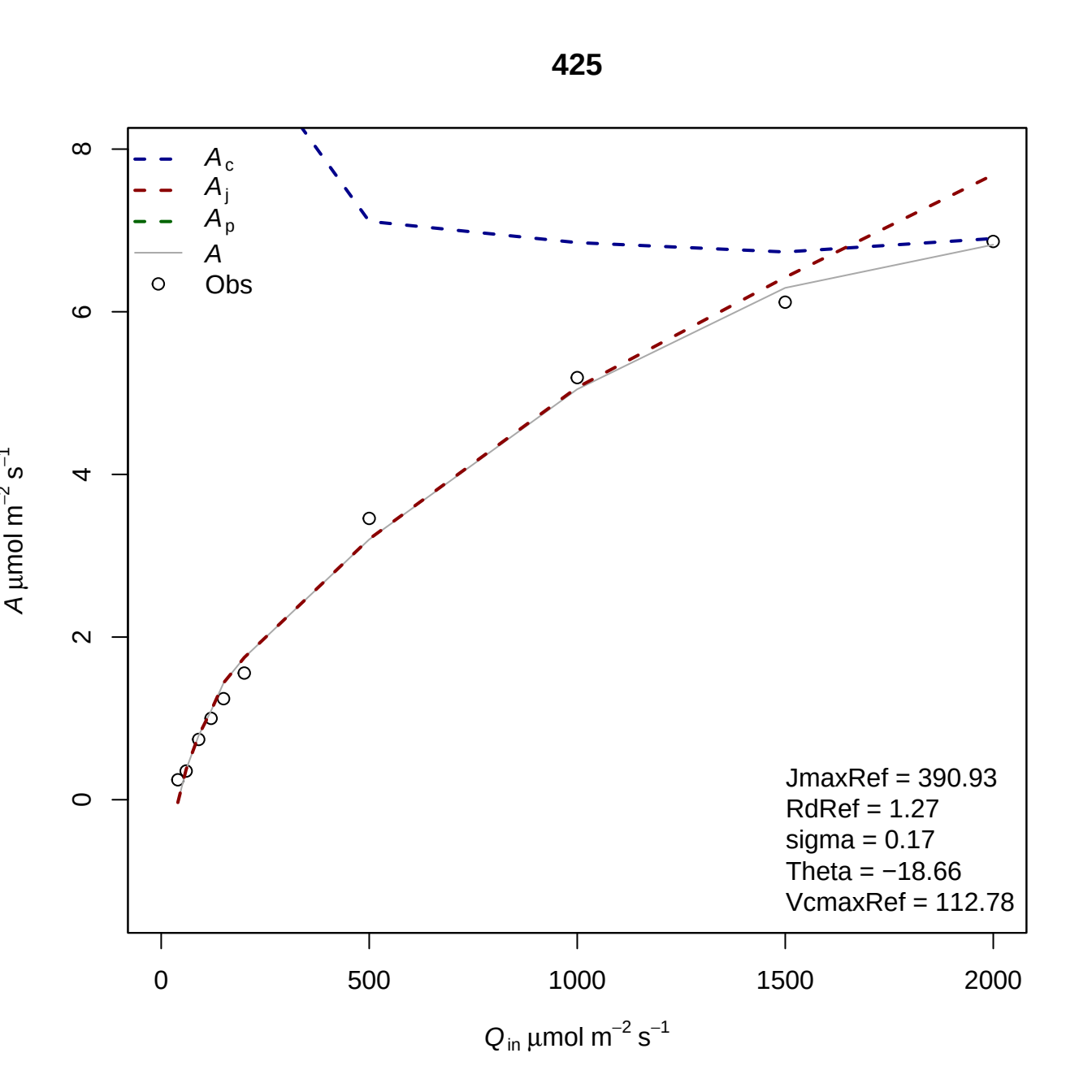

419
A_c
A_j
A_p
A
Obs
JmaxRef = 384.67
RdRef = 3.17
sigma = 0.08
Theta = −0.4
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

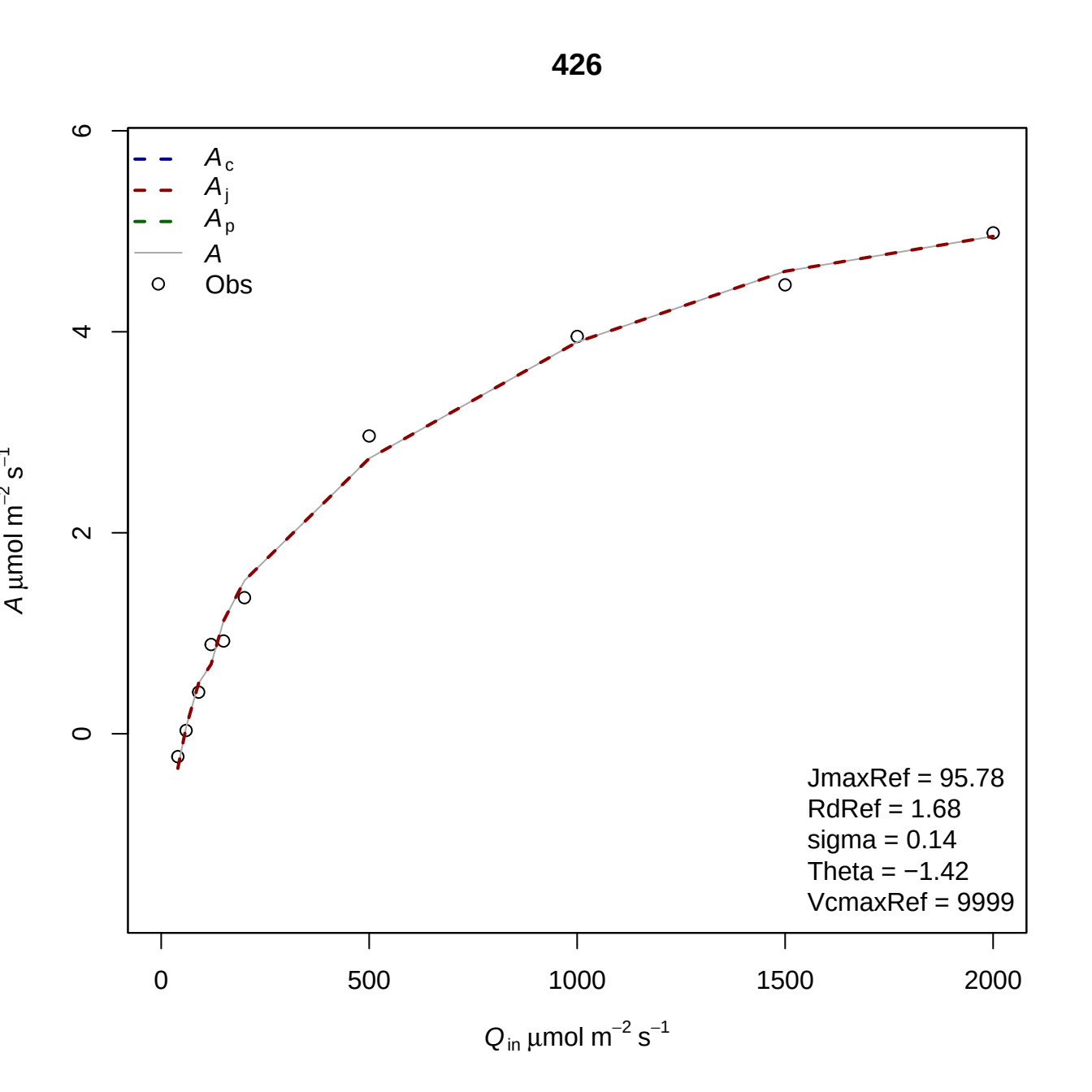

420
A_c
A_j
A_p
A
Obs
JmaxRef = 315.67
RdRef = 2.82
sigma = 0.15
Theta = 0.55
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

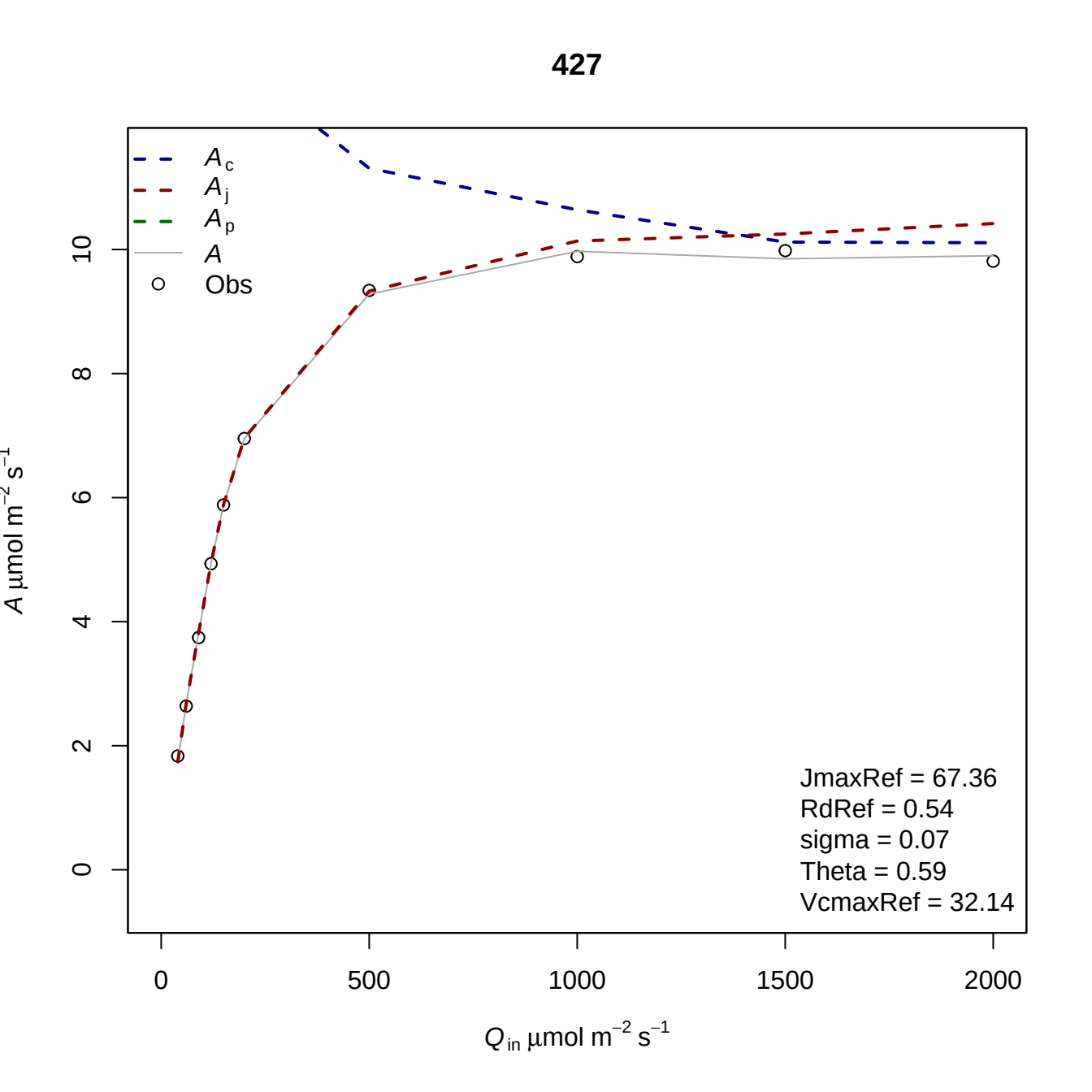

421
A_c
A_j
A_p
A
Obs
JmaxRef = 82.22
RdRef = 0.52
sigma = 0.06
Theta = −0.42
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

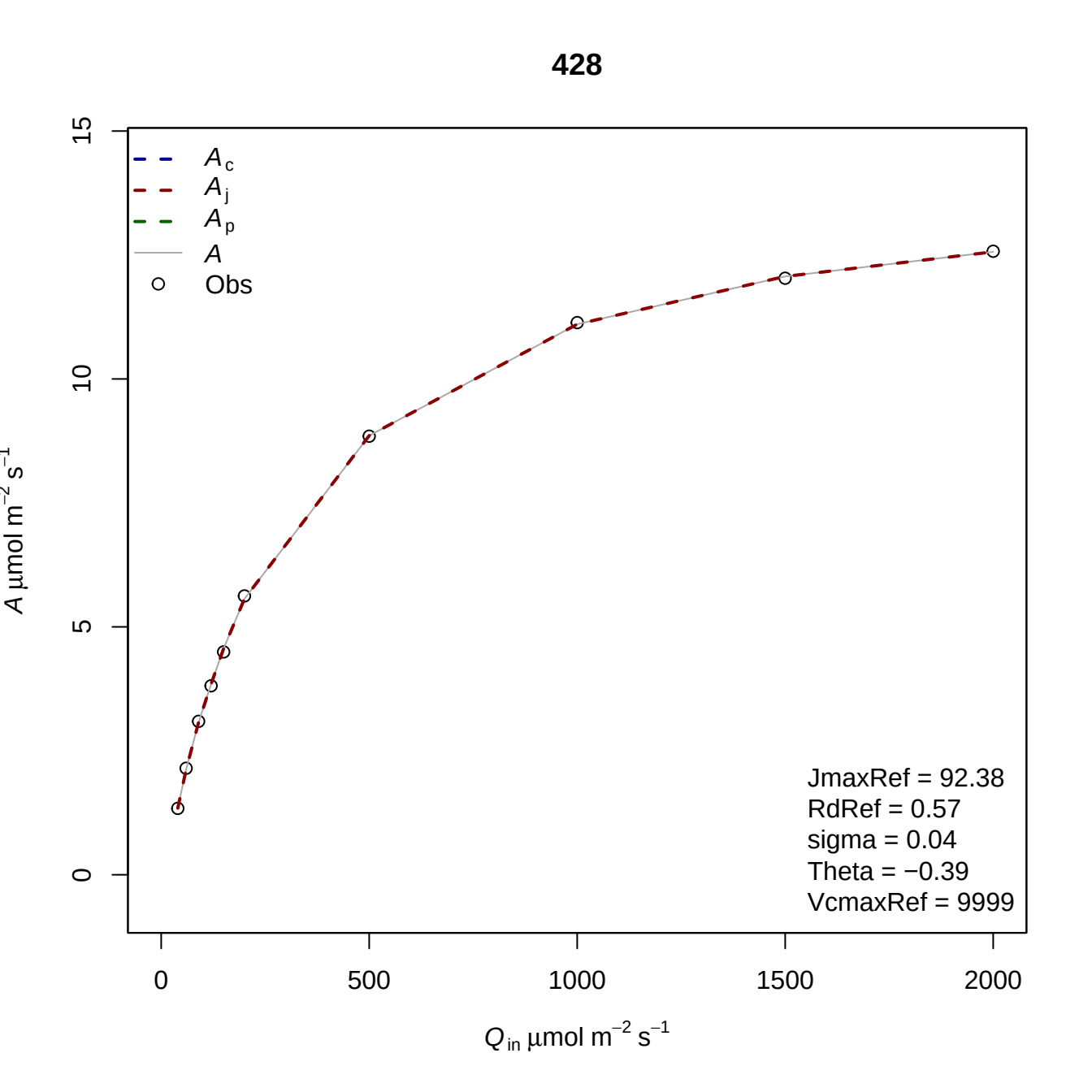

422
A_c
A_j
A_p
A
Obs
JmaxRef = 120.26
RdRef = 0.33
sigma = 0.26
Theta = 0.41
VcmaxRef = 65.98
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

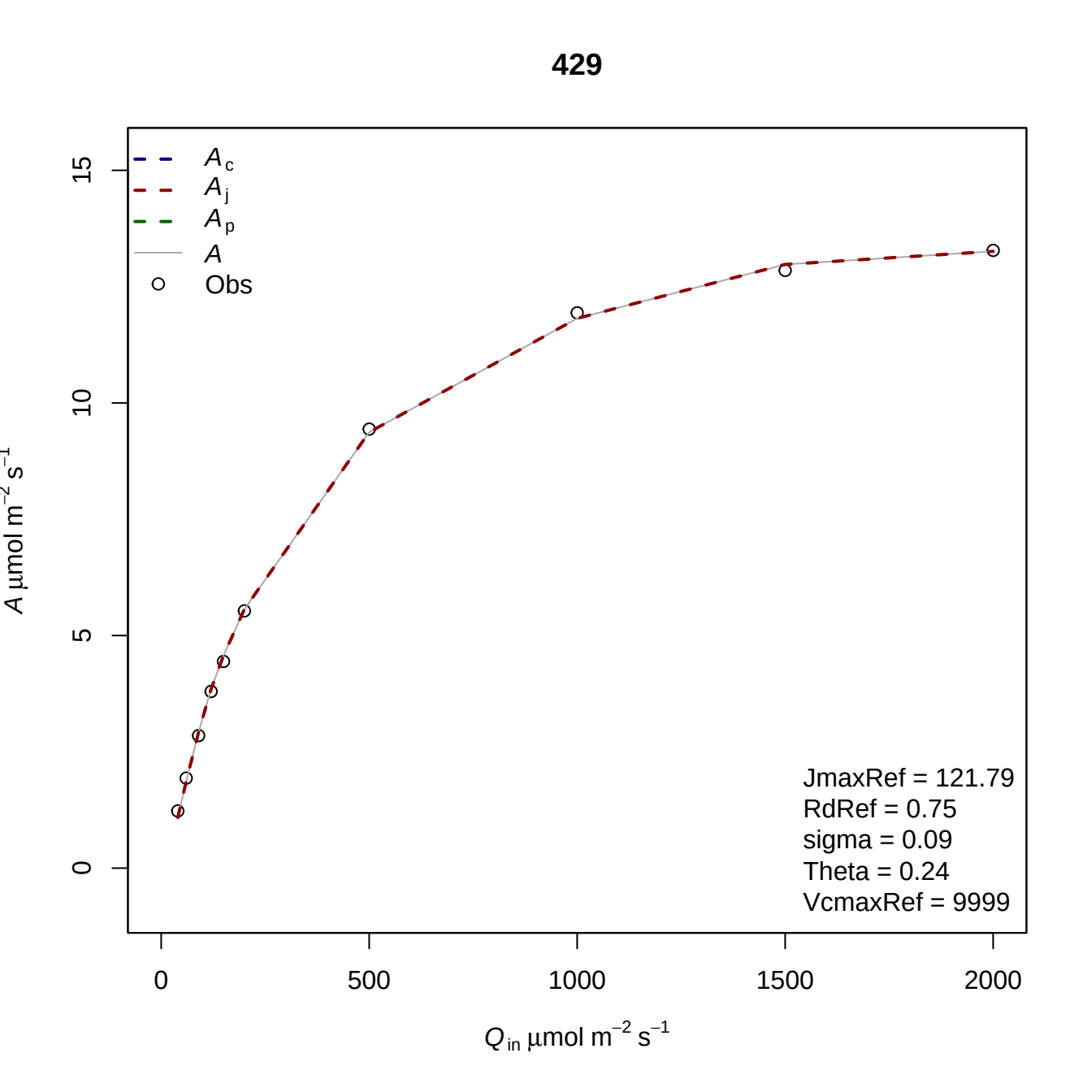

424
A_c
A_j
A_p
A
Obs
JmaxRef = 142.99
RdRef = 1.4
sigma = 0.16
Theta = 0.28
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

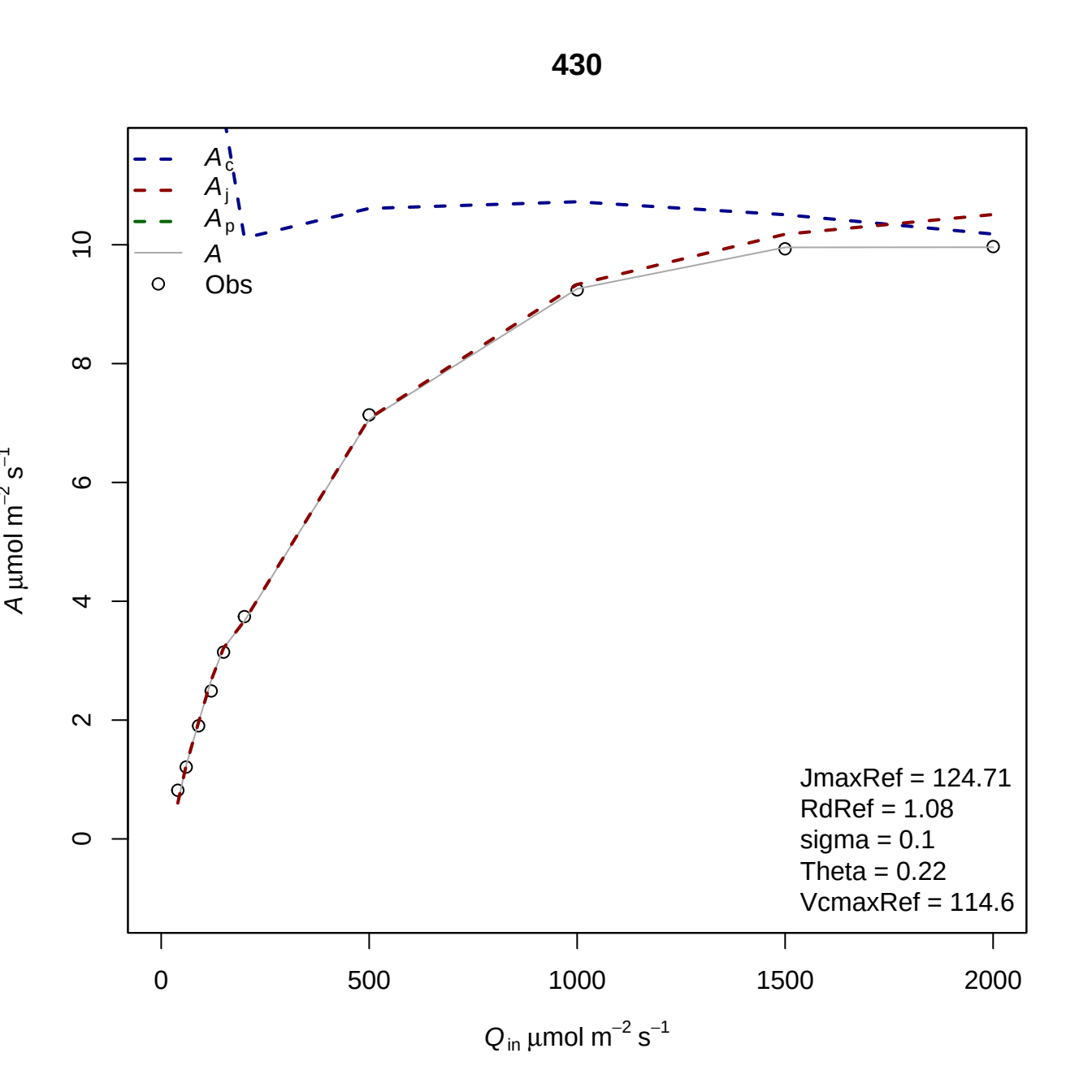

425
A_c
A_j
A_p
A
Obs
JmaxRef = 390.93
RdRef = 1.27
sigma = 0.17
Theta = −18.66
VcmaxRef = 112.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

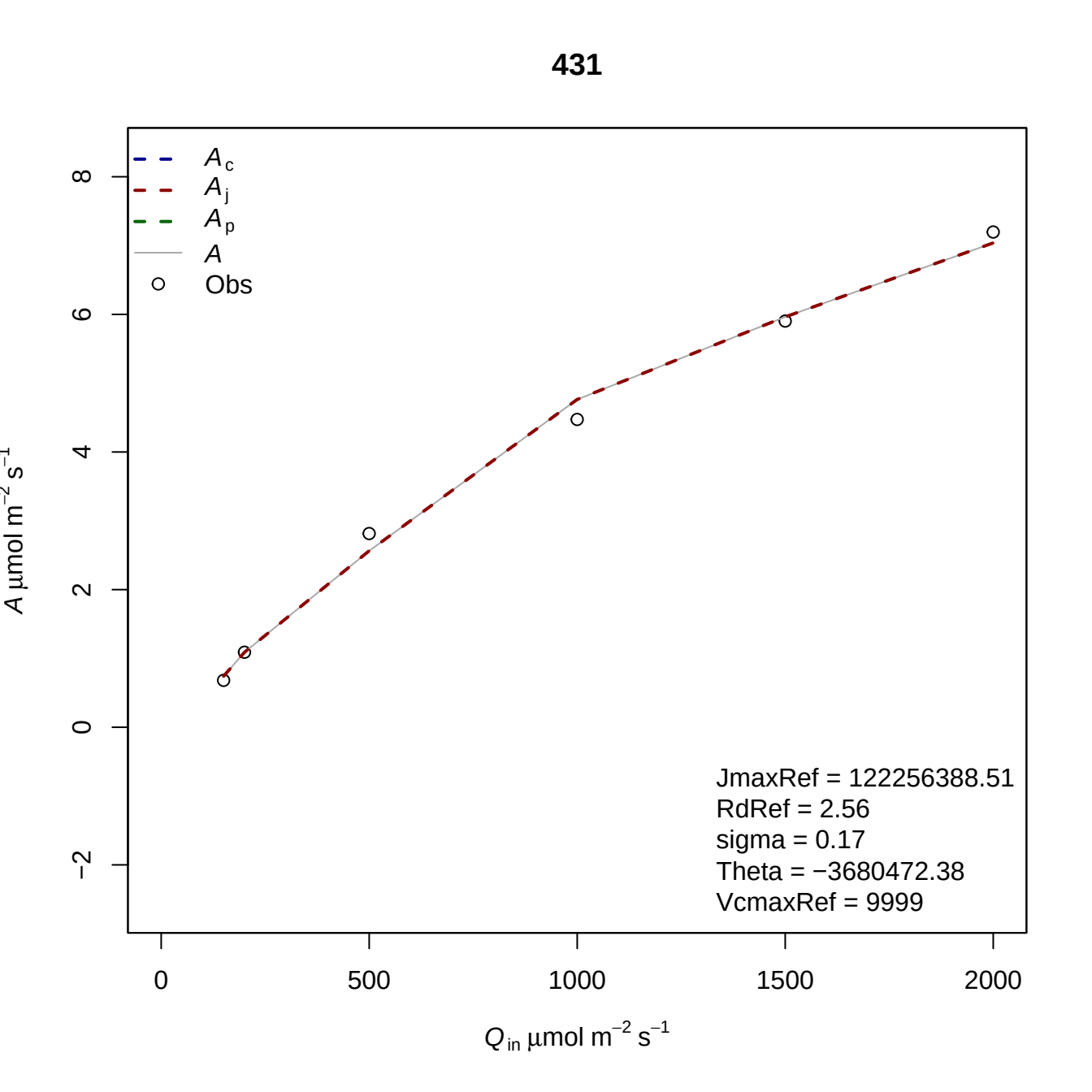

426
A_c
A_j
A_p
A
Obs
JmaxRef = 95.78
RdRef = 1.68
sigma = 0.14
Theta = −1.42
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

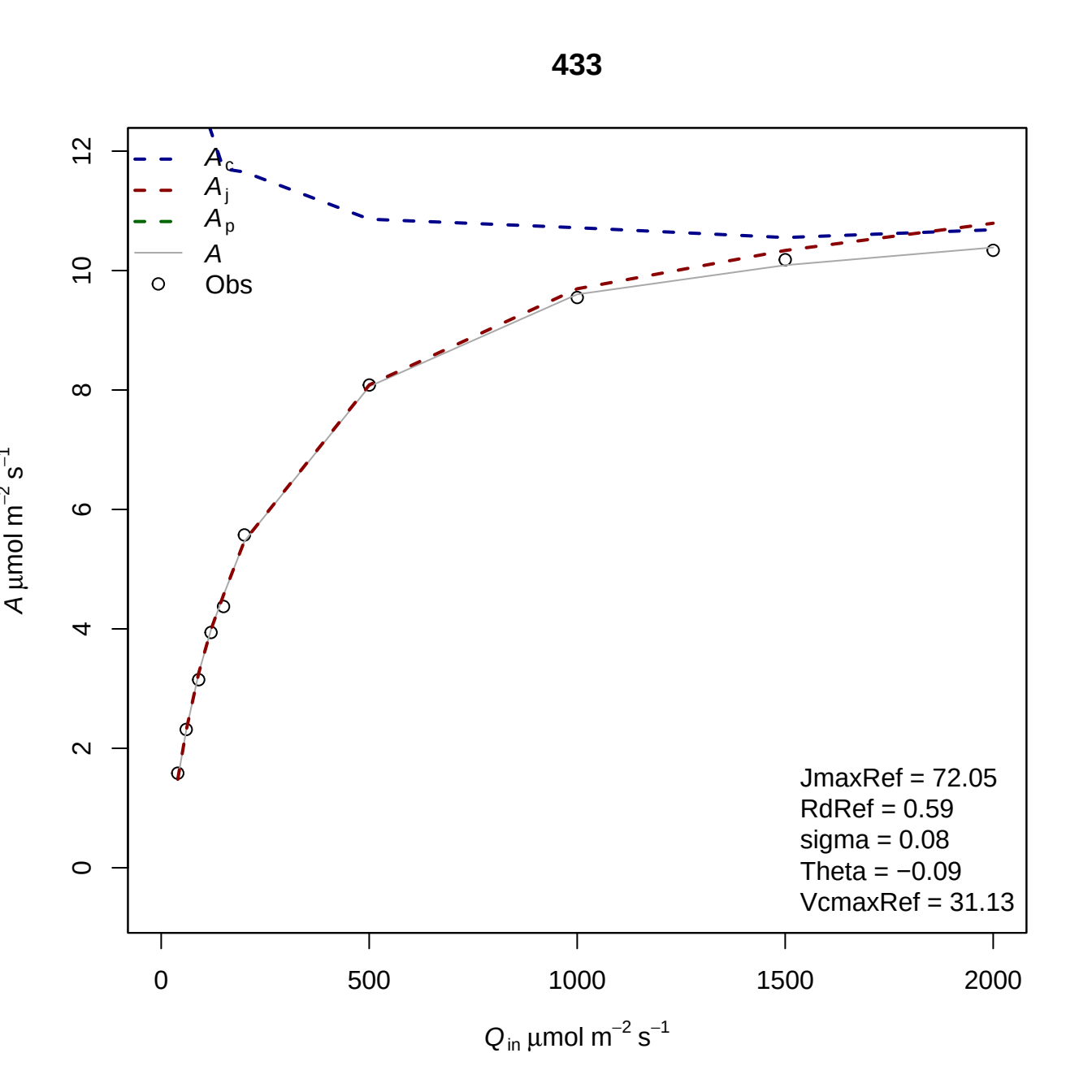

427
A_c
A_j
A_p
A
Obs
JmaxRef = 67.36
RdRef = 0.54
sigma = 0.07
Theta = 0.59
VcmaxRef = 32.14
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

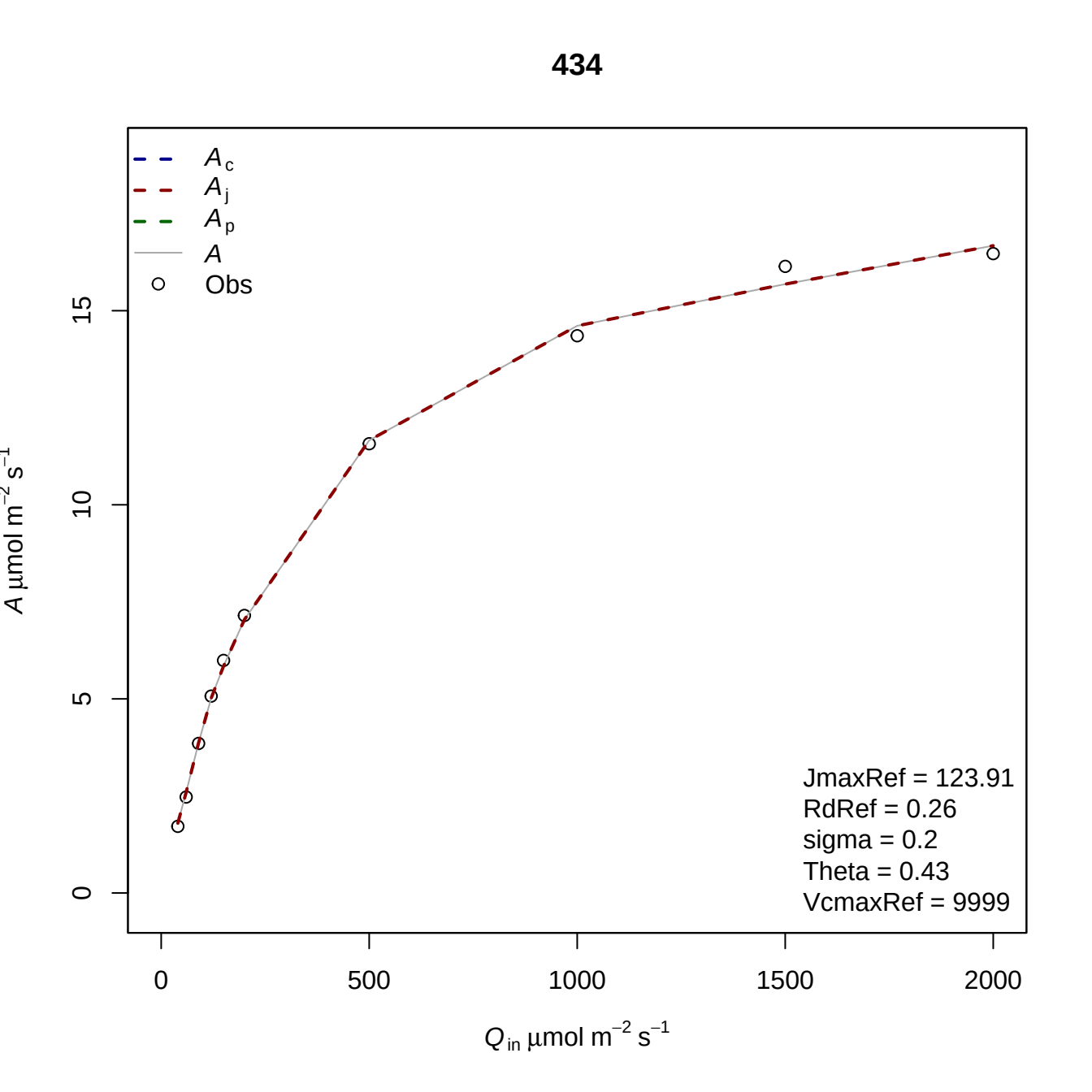

428
A_c
A_j
A_p
A
Obs
JmaxRef = 92.38
RdRef = 0.57
sigma = 0.04
Theta = −0.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

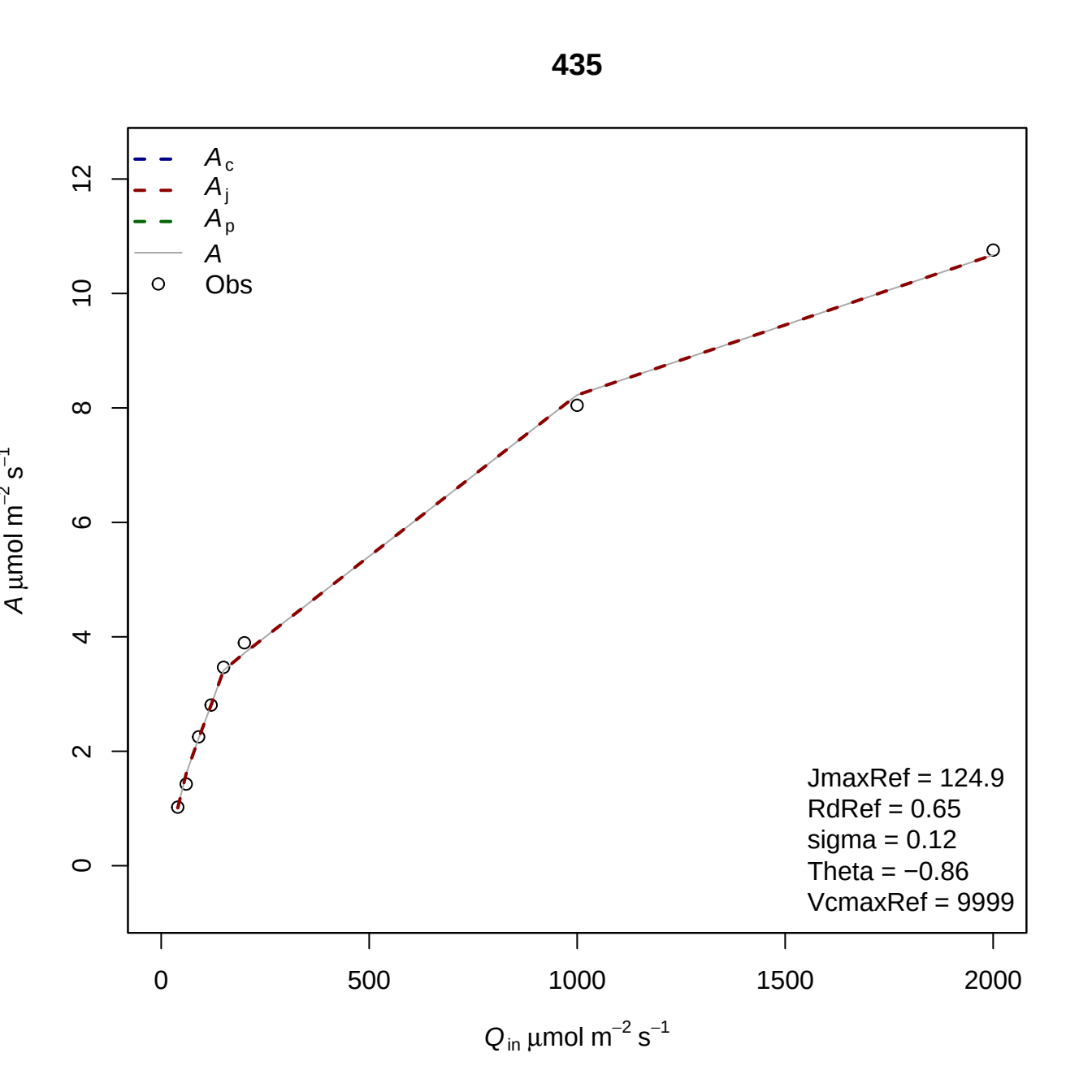

429
A_c
A_j
A_p
A
Obs
JmaxRef = 121.79
RdRef = 0.75
sigma = 0.09
Theta = 0.24
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

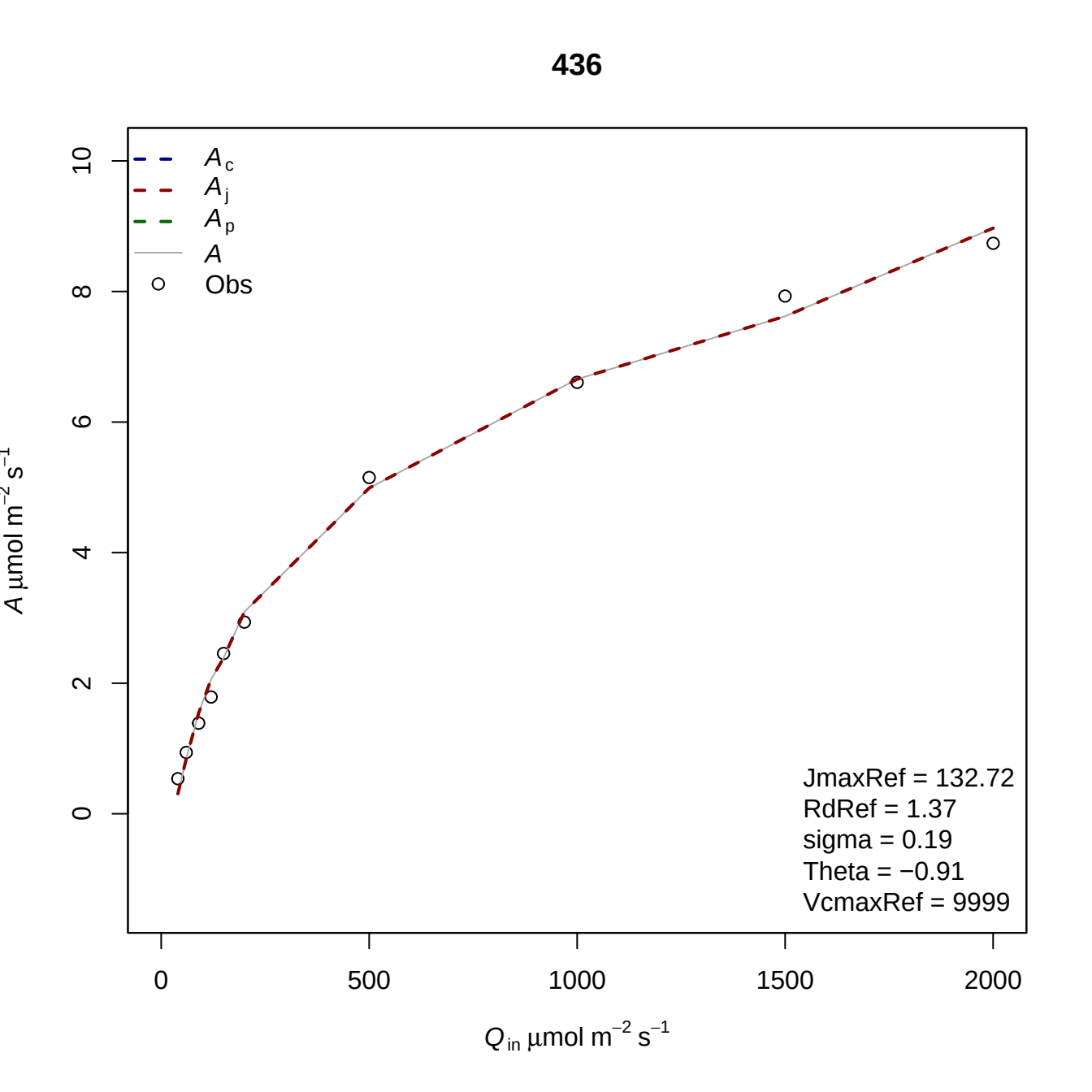

430
A_c
A_j
A_p
A
Obs
JmaxRef = 124.71
RdRef = 1.08
sigma = 0.1
Theta = 0.22
VcmaxRef = 114.6
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

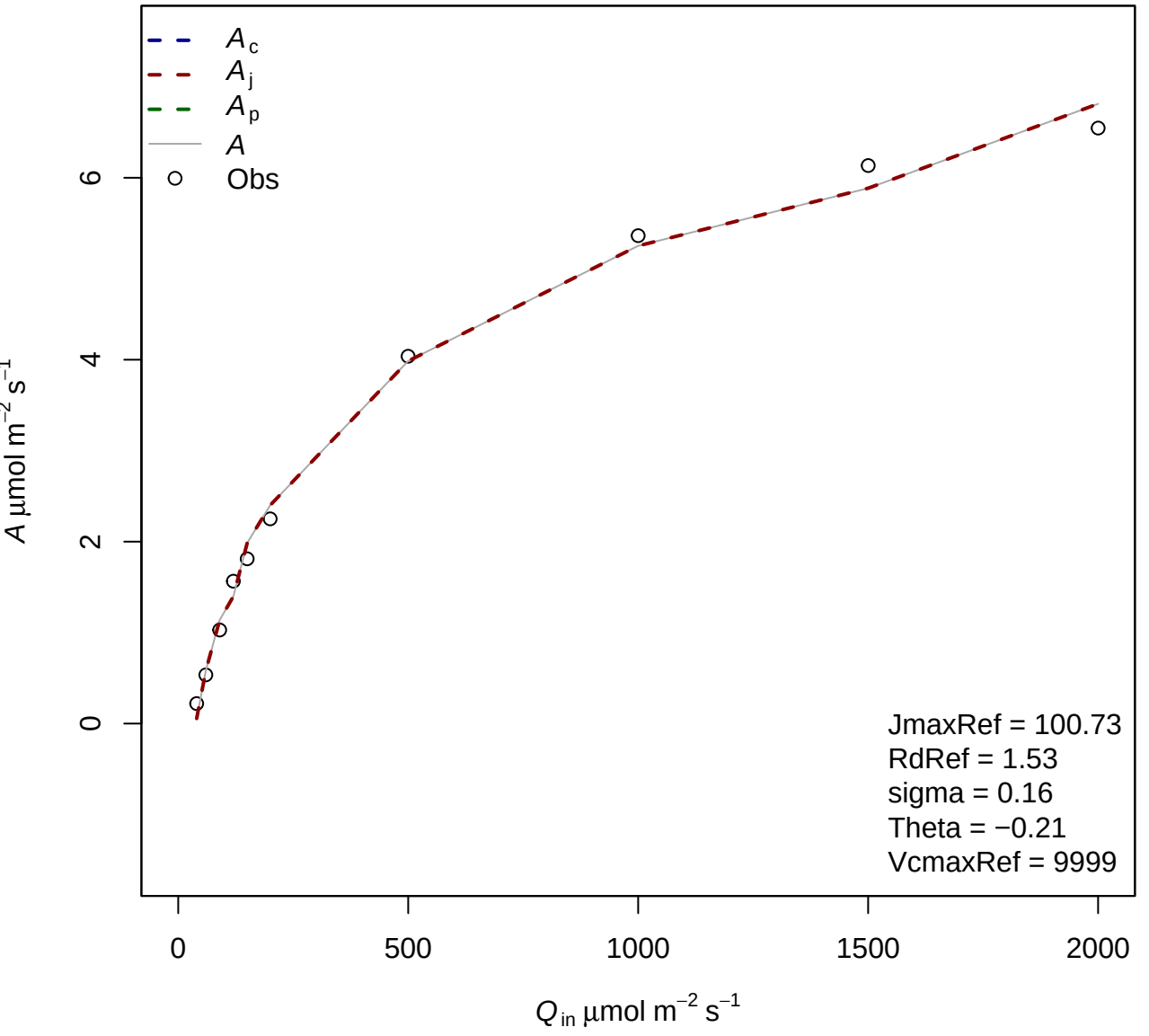

431
A_c
A_j
A_p
A
Obs
JmaxRef = 122256388.51
RdRef = 2.56
sigma = 0.17
Theta = −3680472.38
VcmaxRef = 9999
8
6
4
2
0
−2
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

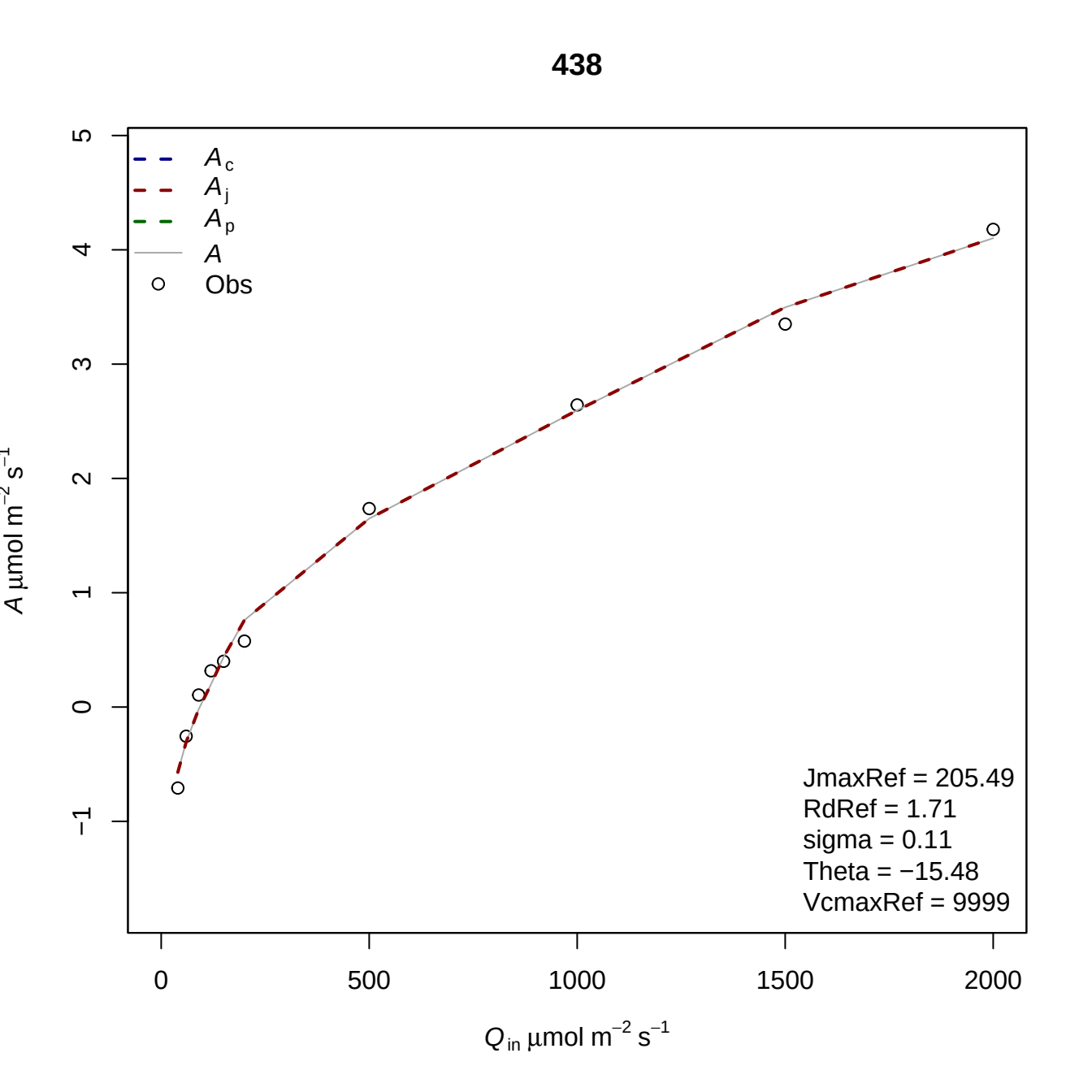

433
A_c
A_j
A_p
A
Obs
JmaxRef = 72.05
RdRef = 0.59
sigma = 0.08
Theta = −0.09
VcmaxRef = 31.13
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

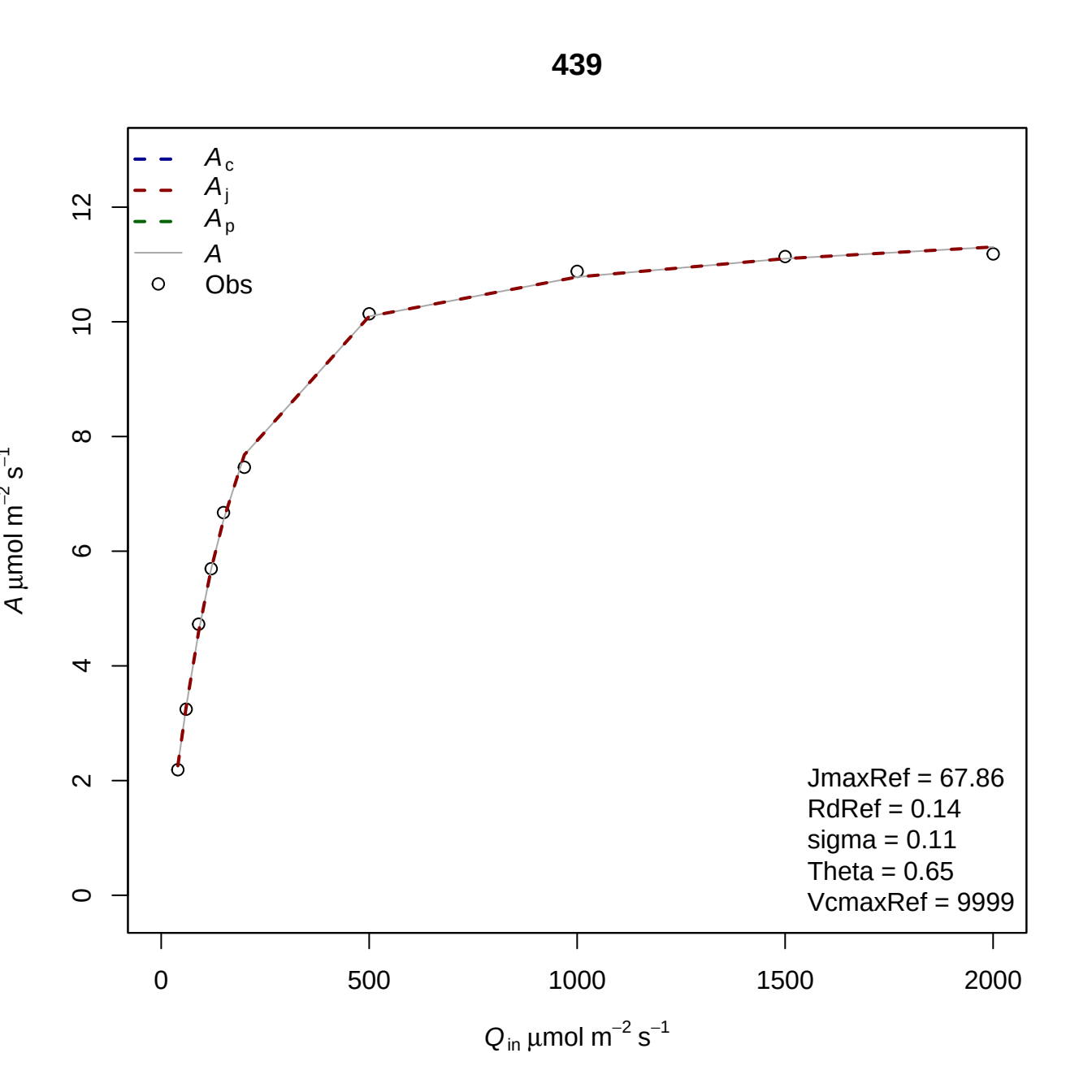

434
A_c
A_j
A_p
A
Obs
JmaxRef = 123.91
RdRef = 0.26
sigma = 0.2
Theta = 0.43
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

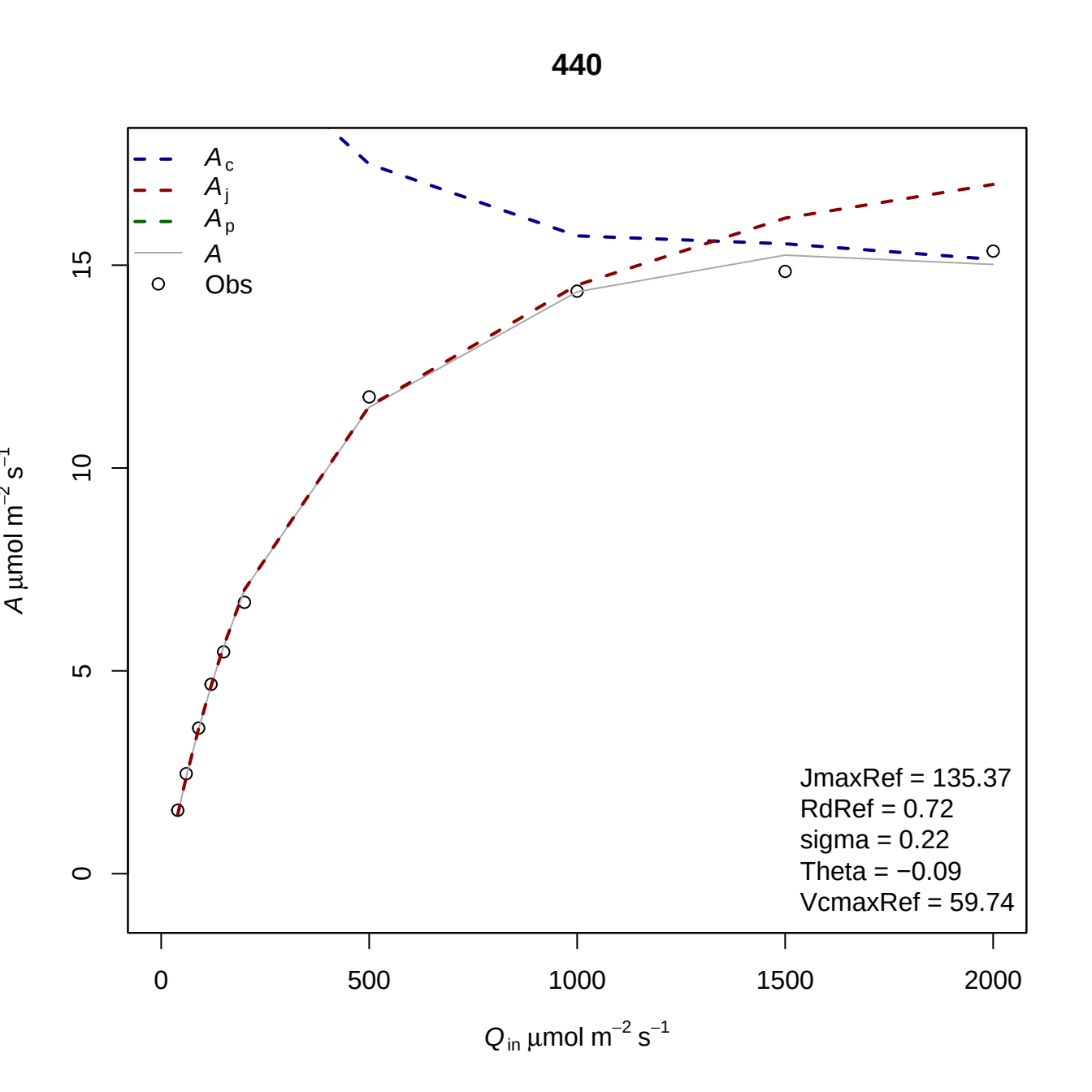

435
A_c
A_j
A_p
A
Obs
JmaxRef = 124.9
RdRef = 0.65
sigma = 0.12
Theta = −0.86
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

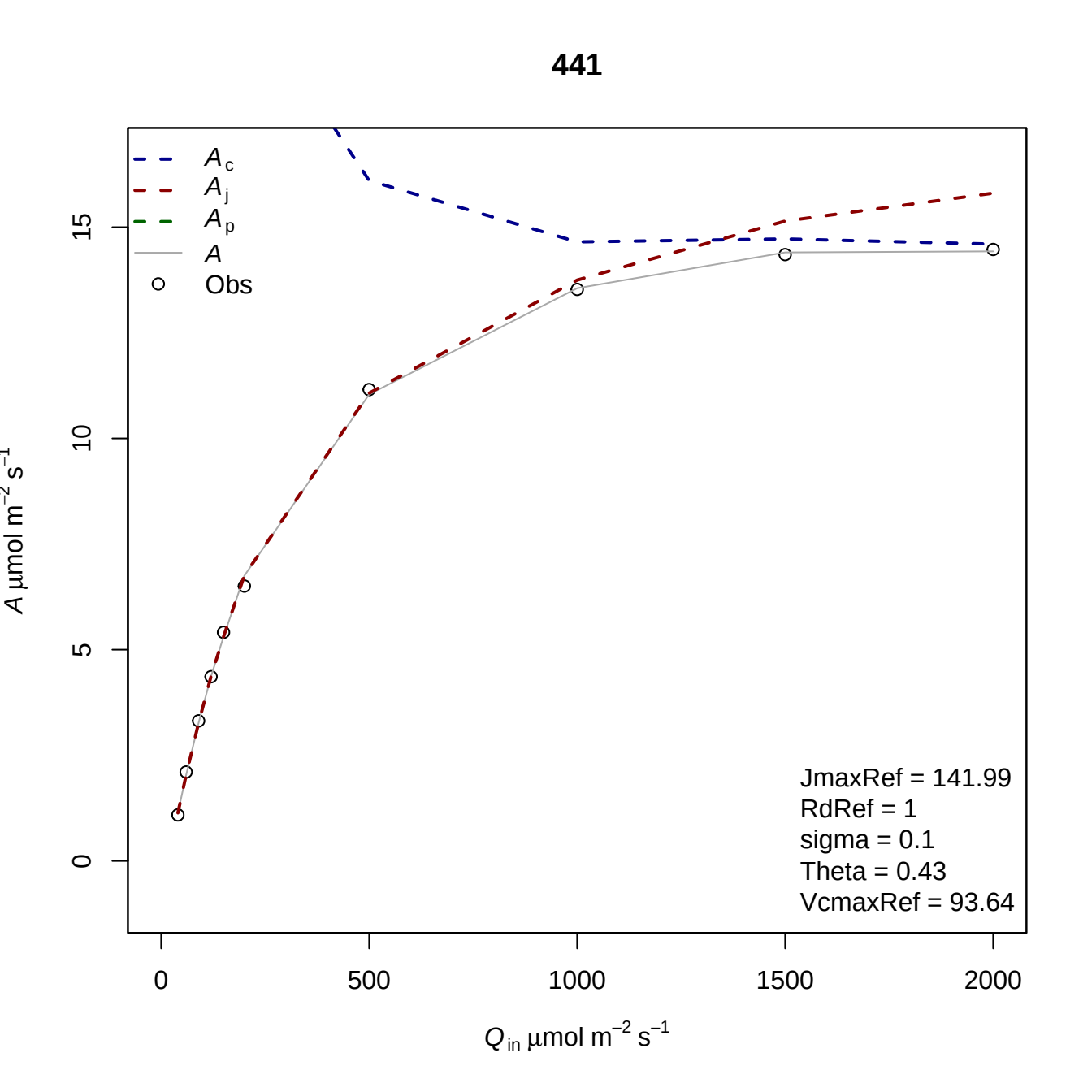

436
A_c
A_j
A_p
A
Obs
JmaxRef = 132.72
RdRef = 1.37
sigma = 0.19
Theta = −0.91
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

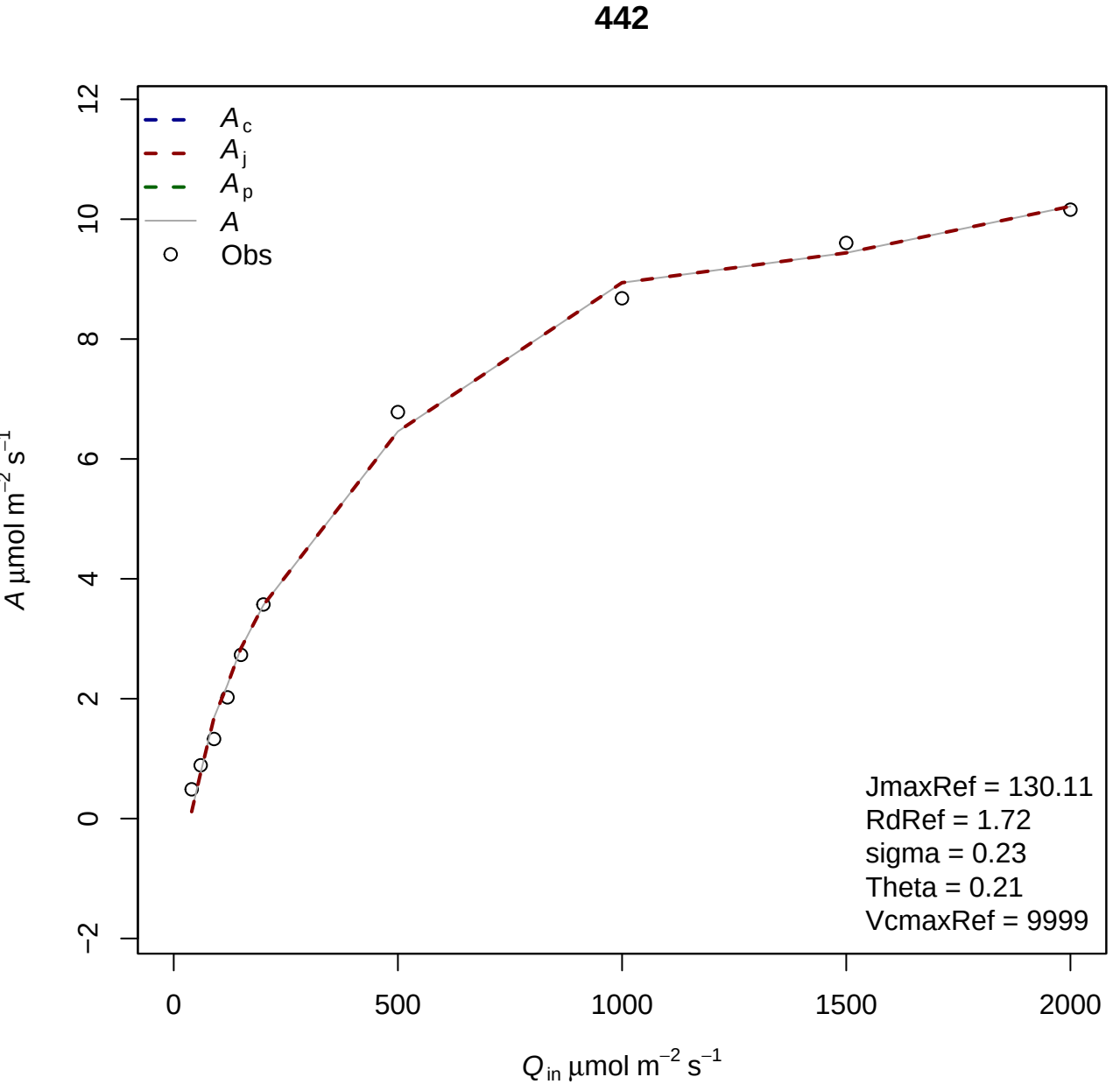

A_c
A_j
A_p
A
Obs
JmaxRef = 100.73
RdRef = 1.53
sigma = 0.16
Theta = −0.21
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

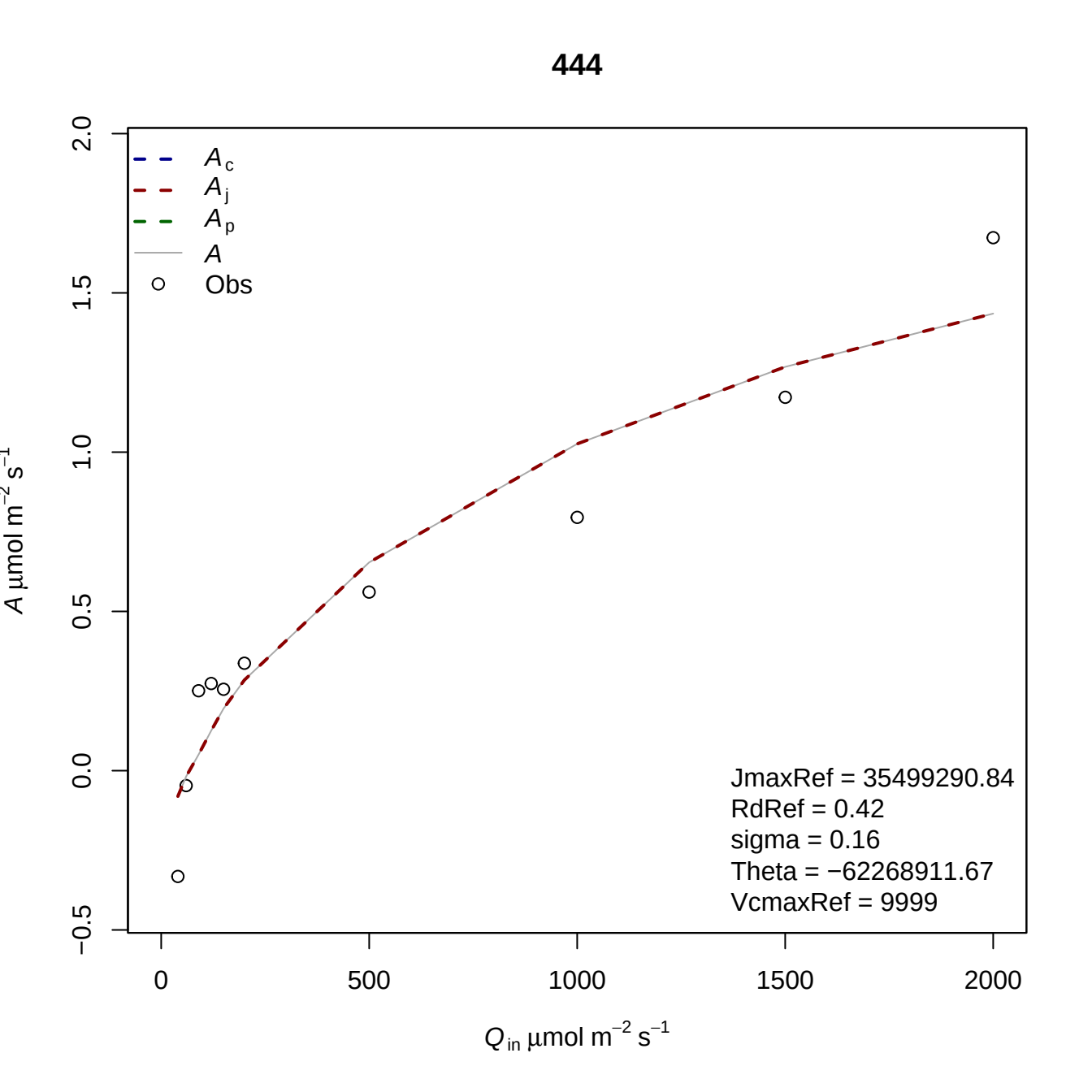

438
A_c
A_j
A_p
A
Obs
JmaxRef = 205.49
RdRef = 1.71
sigma = 0.11
Theta = −15.48
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

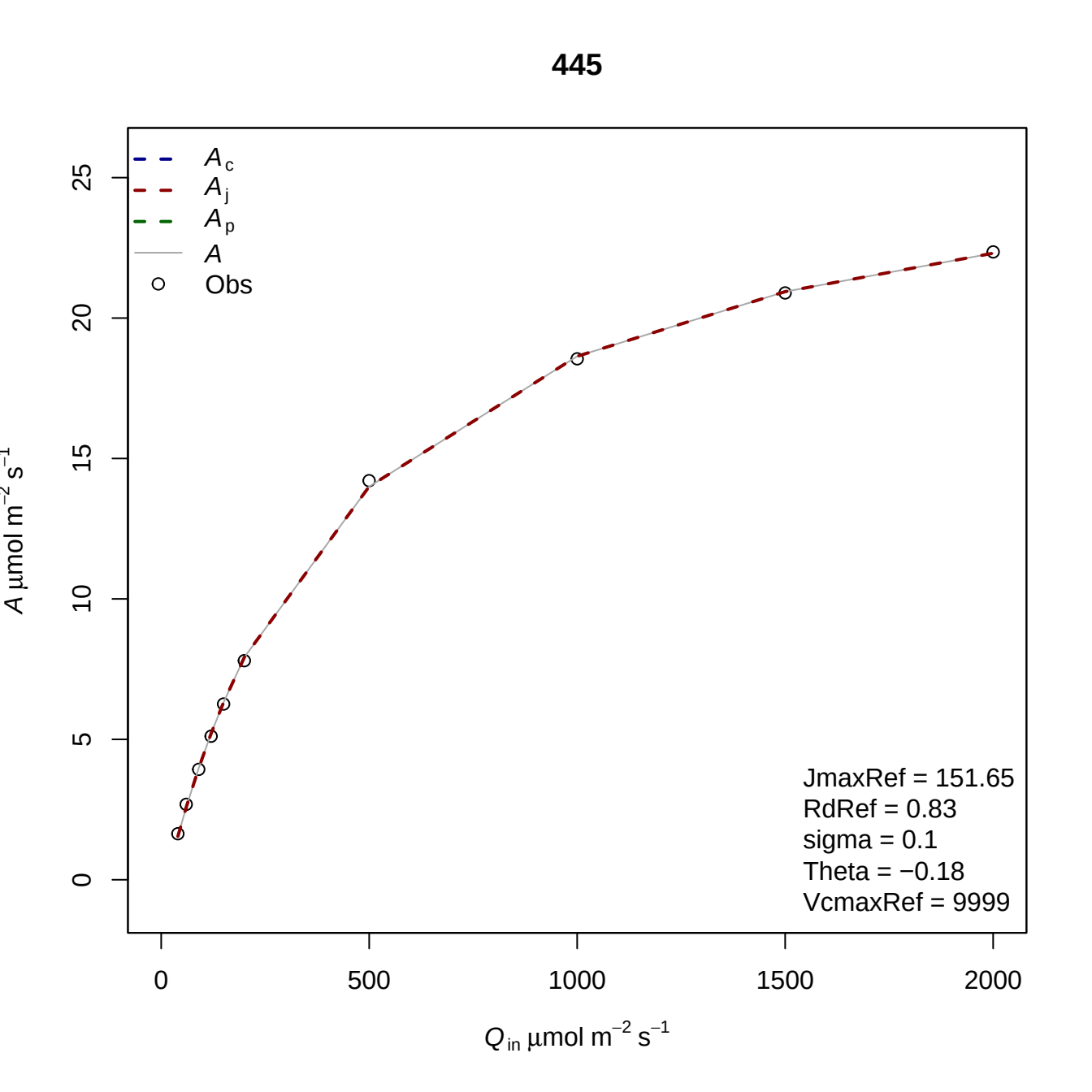

439
A_c
A_j
A_p
A
Obs
JmaxRef = 67.86
RdRef = 0.14
sigma = 0.11
Theta = 0.65
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

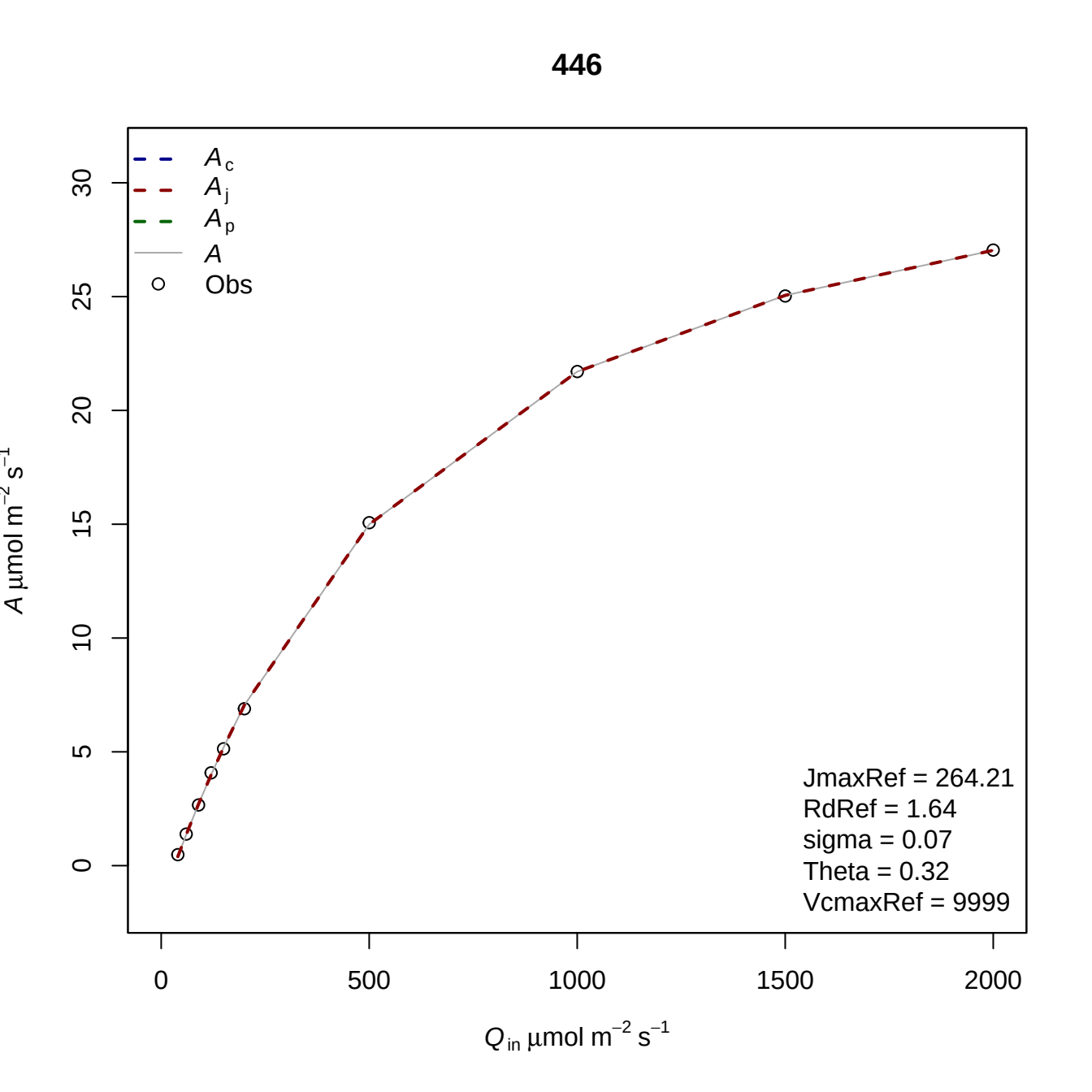

440
A_c
A_j
A_p
A
Obs
JmaxRef = 135.37
RdRef = 0.72
sigma = 0.22
Theta = −0.09
VcmaxRef = 59.74
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

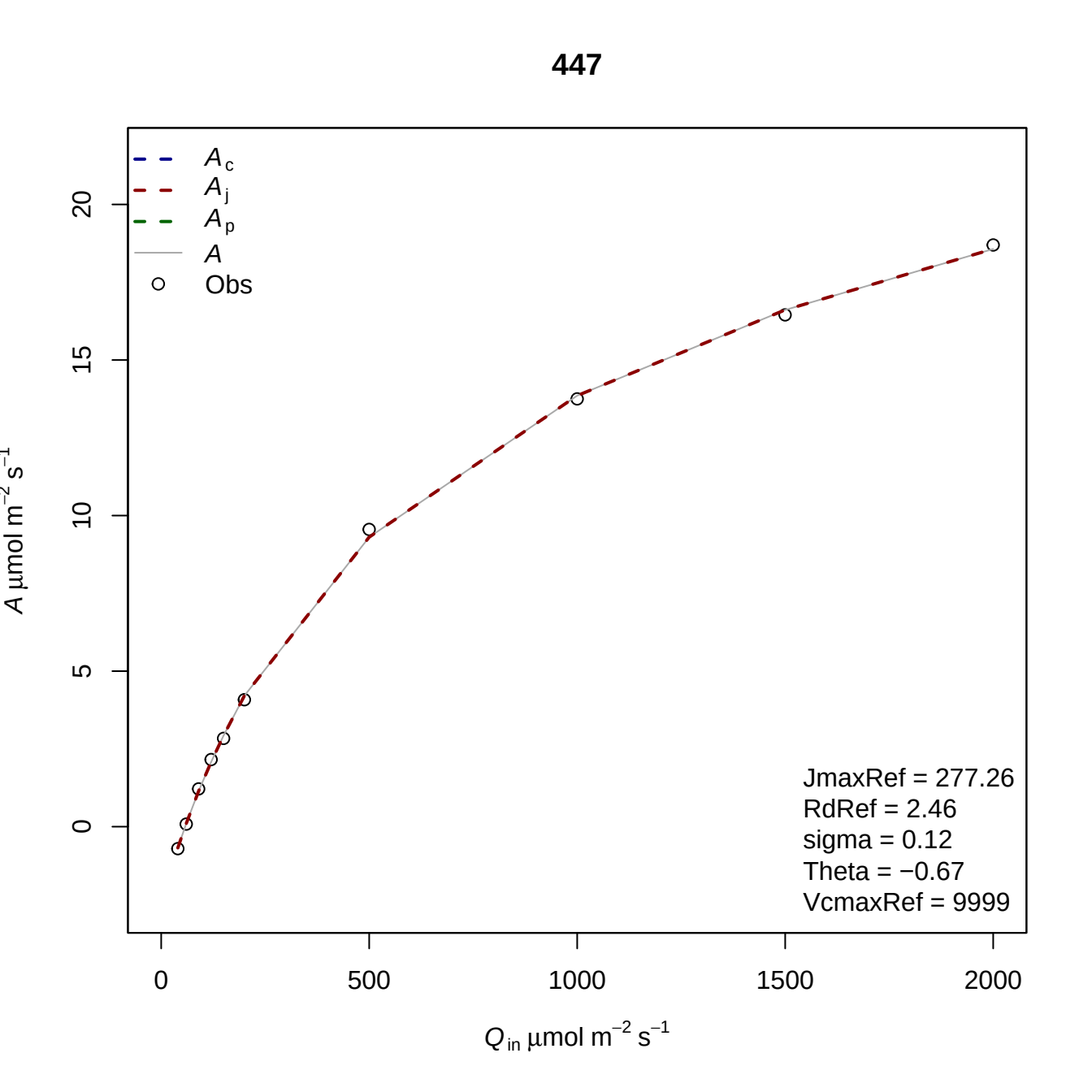

441
A_c
A_j
A_p
A
Obs
JmaxRef = 141.99
RdRef = 1
sigma = 0.1
Theta = 0.43
VcmaxRef = 93.64
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

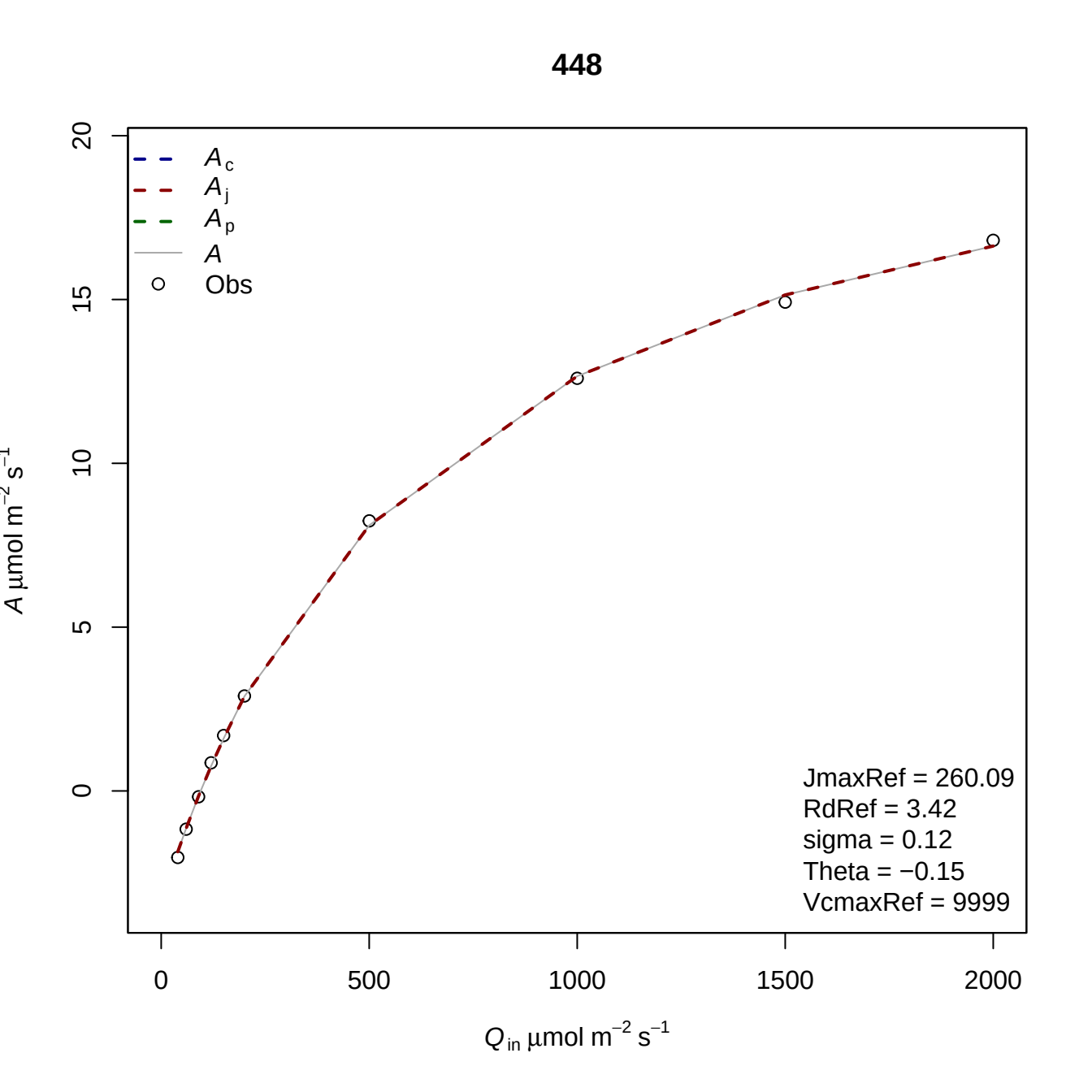

442
A_c
A_j
A_p
A
Obs
JmaxRef = 130.11
RdRef = 1.72
sigma = 0.23
Theta = 0.21
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

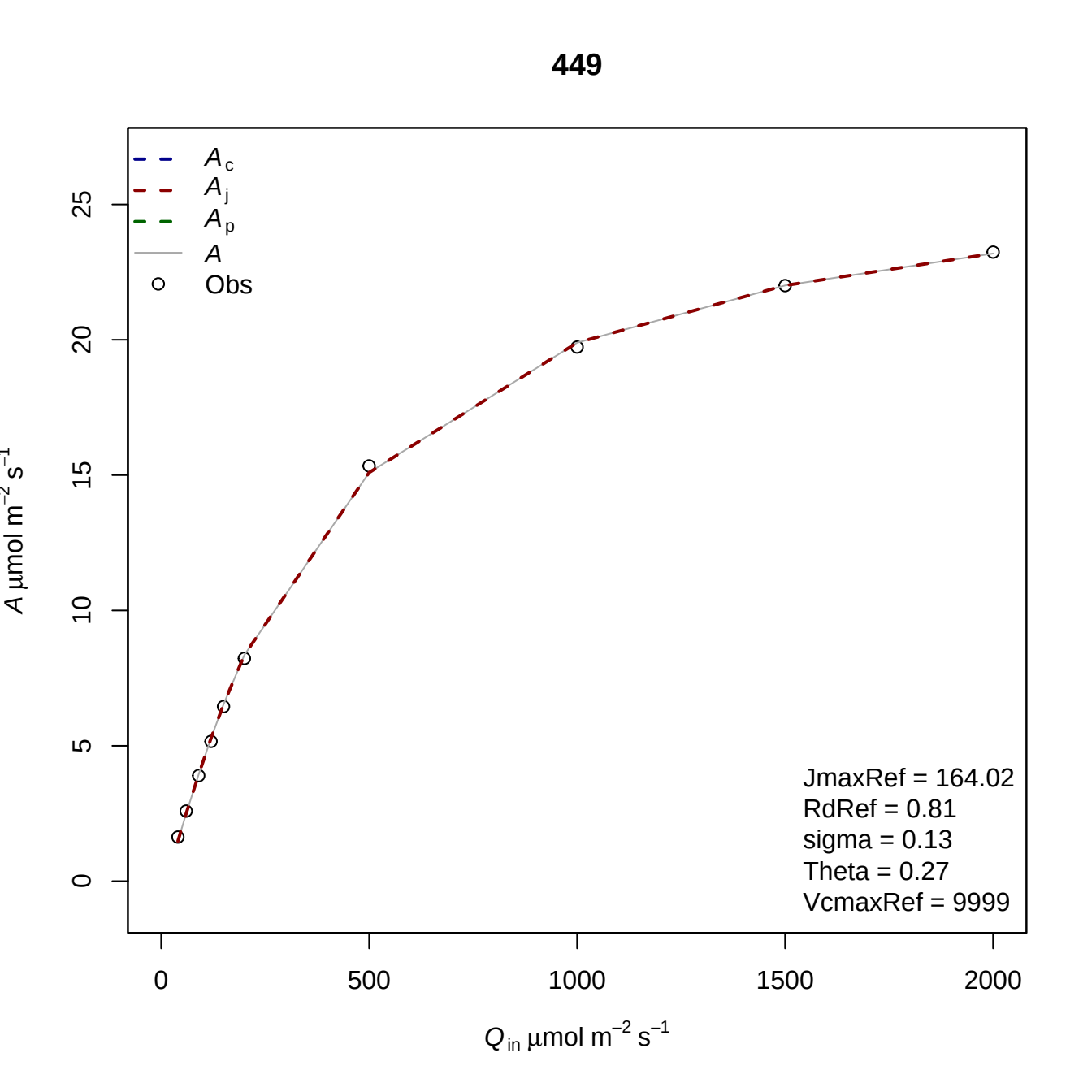

444
A_c
A_j
A_p
A
Obs
JmaxRef = 35499290.84
RdRef = 0.42
sigma = 0.16
Theta = −62268911.67
VcmaxRef = 9999
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

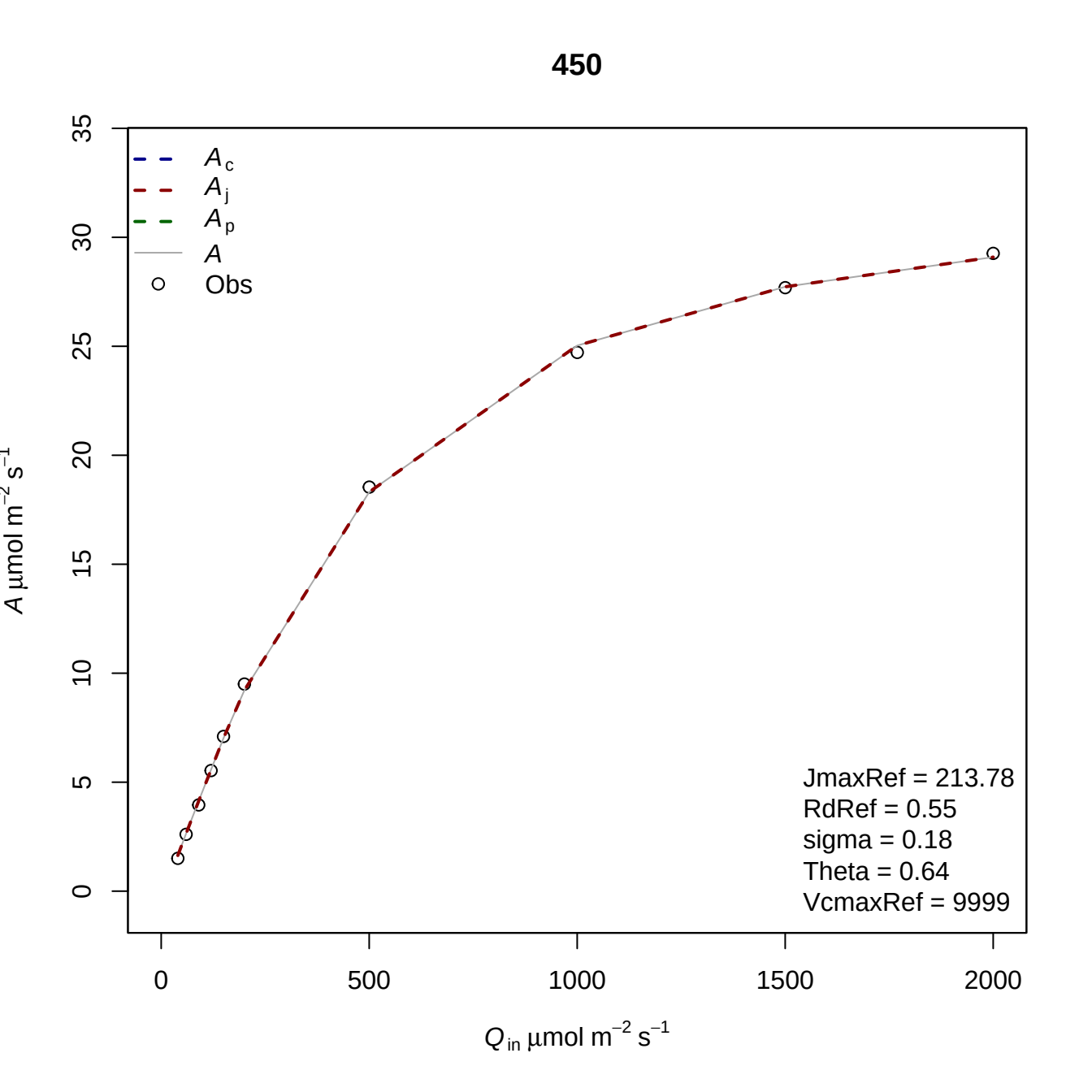

445
A_c
A_j
A_p
A
Obs
JmaxRef = 151.65
RdRef = 0.83
sigma = 0.1
Theta = −0.18
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

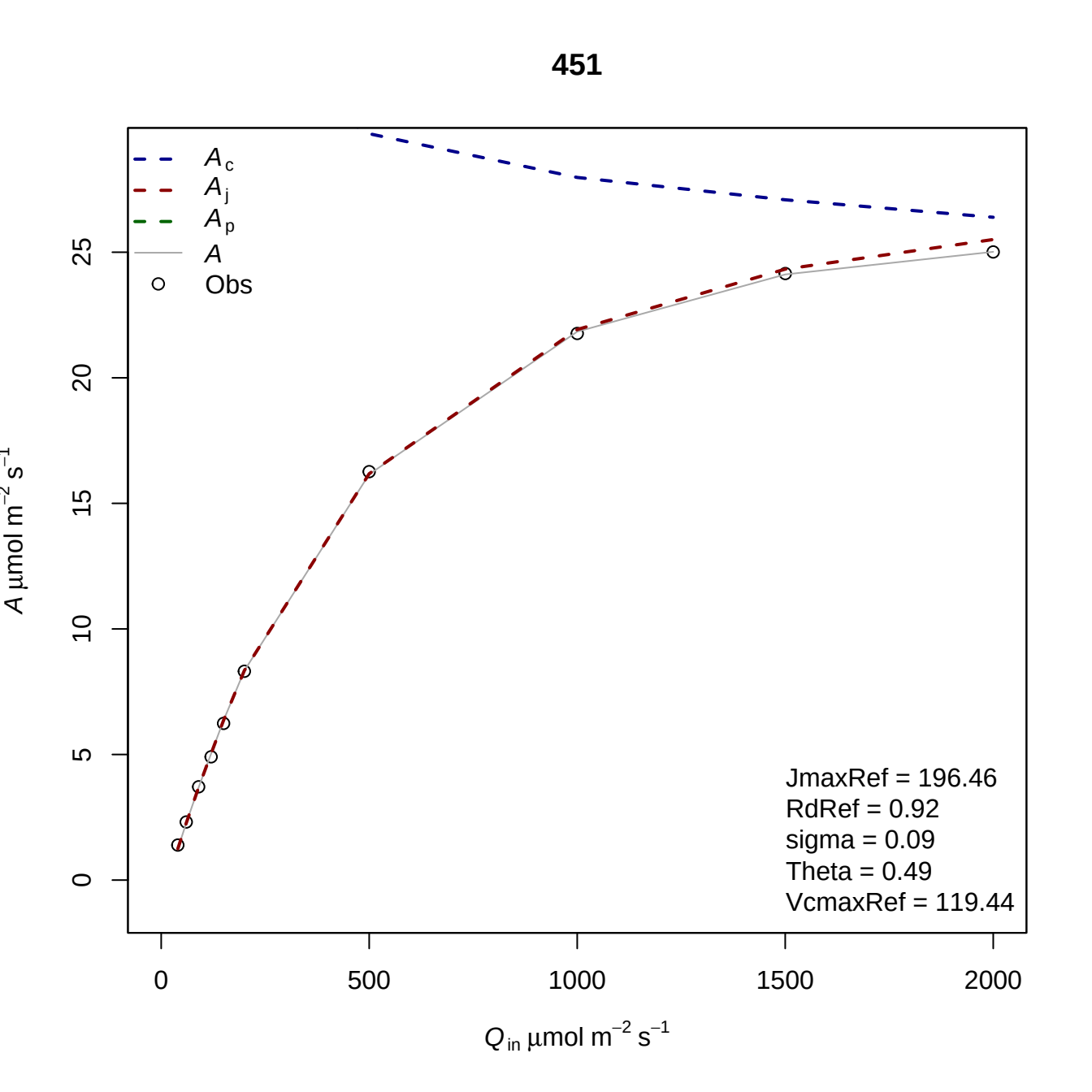

446
A_c
A_j
A_p
A
Obs
JmaxRef = 264.21
RdRef = 1.64
sigma = 0.07
Theta = 0.32
VcmaxRef = 9999
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

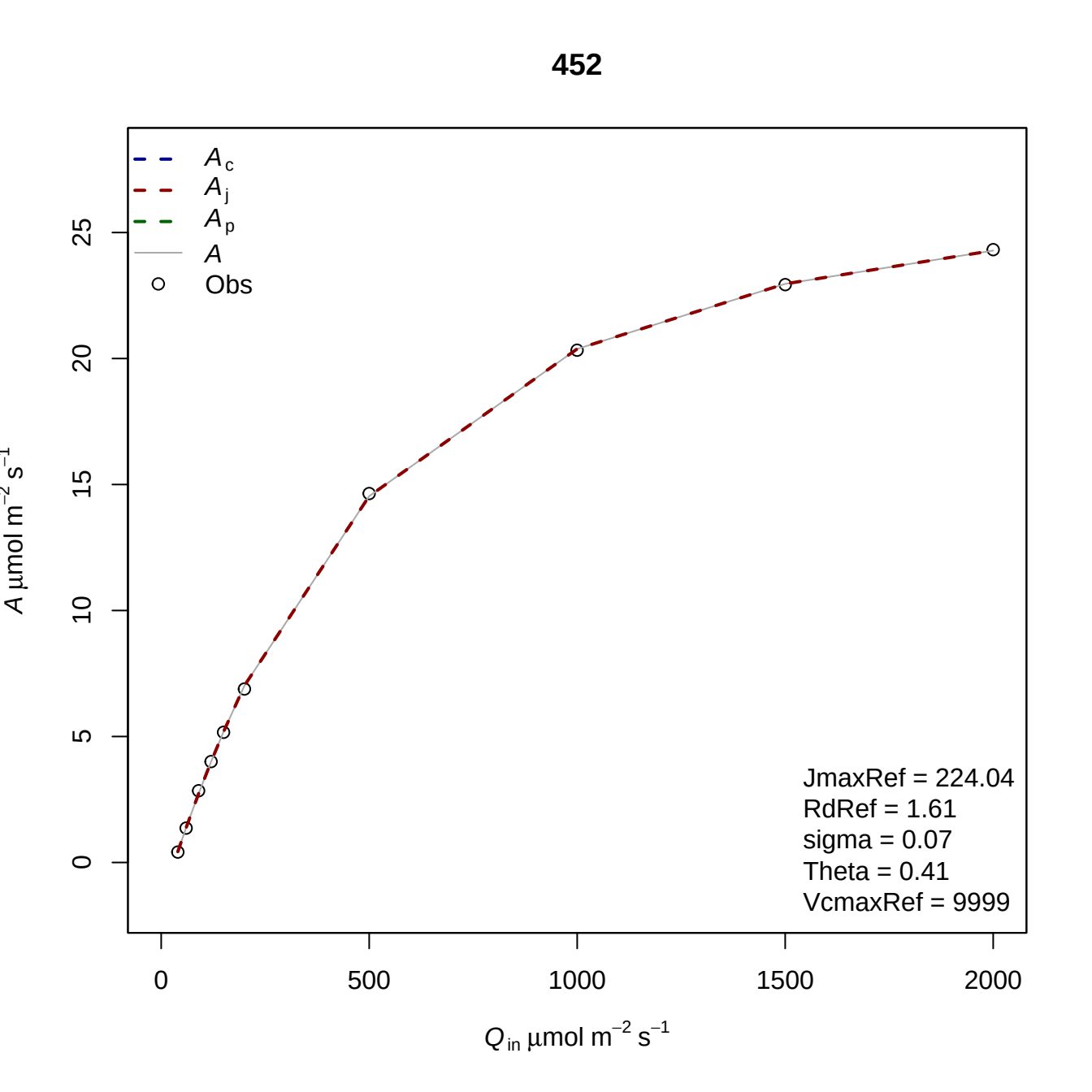

447
A_c
A_j
A_p
A
Obs
JmaxRef = 277.26
RdRef = 2.46
sigma = 0.12
Theta = −0.67
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

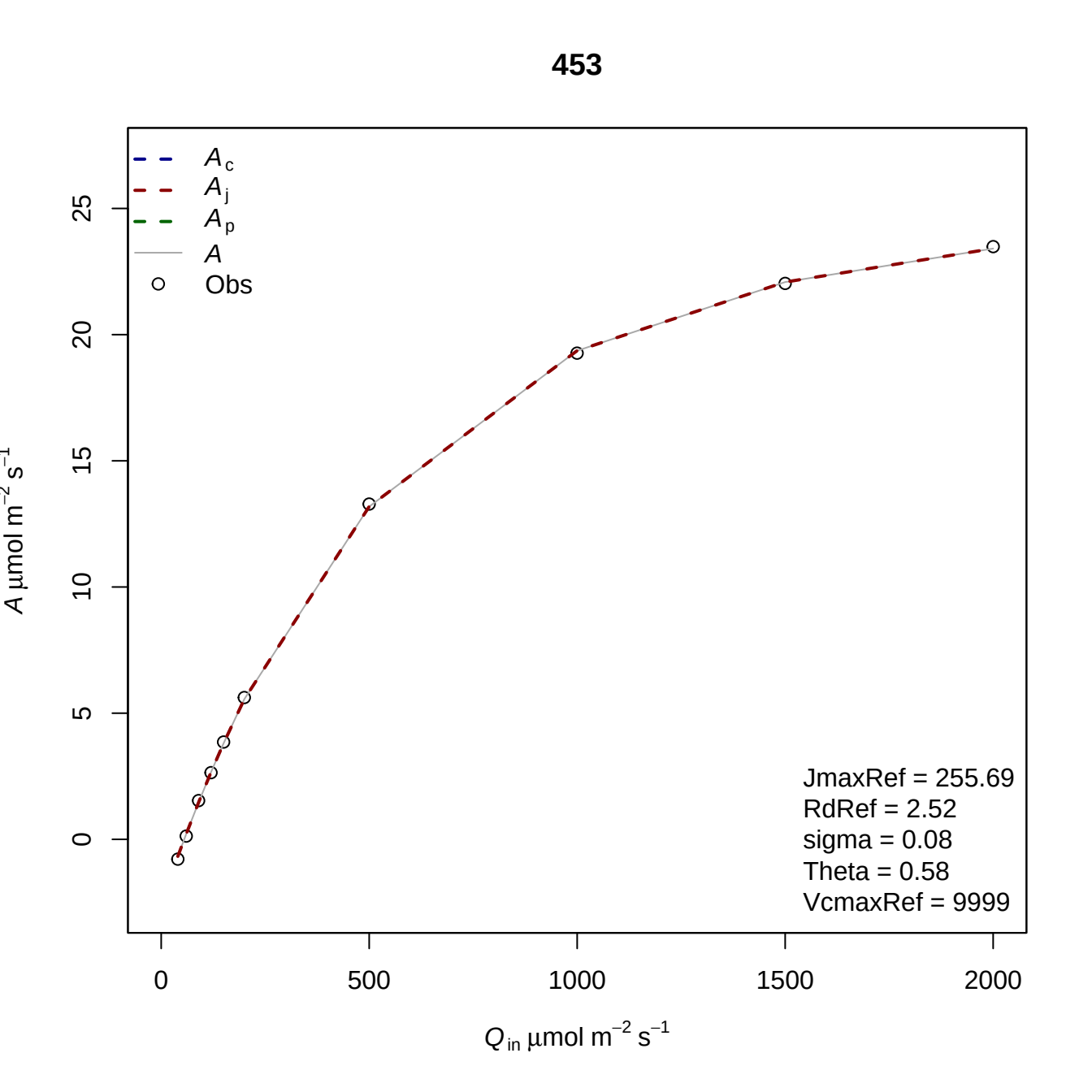

448
A_c
A_j
A_p
A
Obs
JmaxRef = 260.09
RdRef = 3.42
sigma = 0.12
Theta = −0.15
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

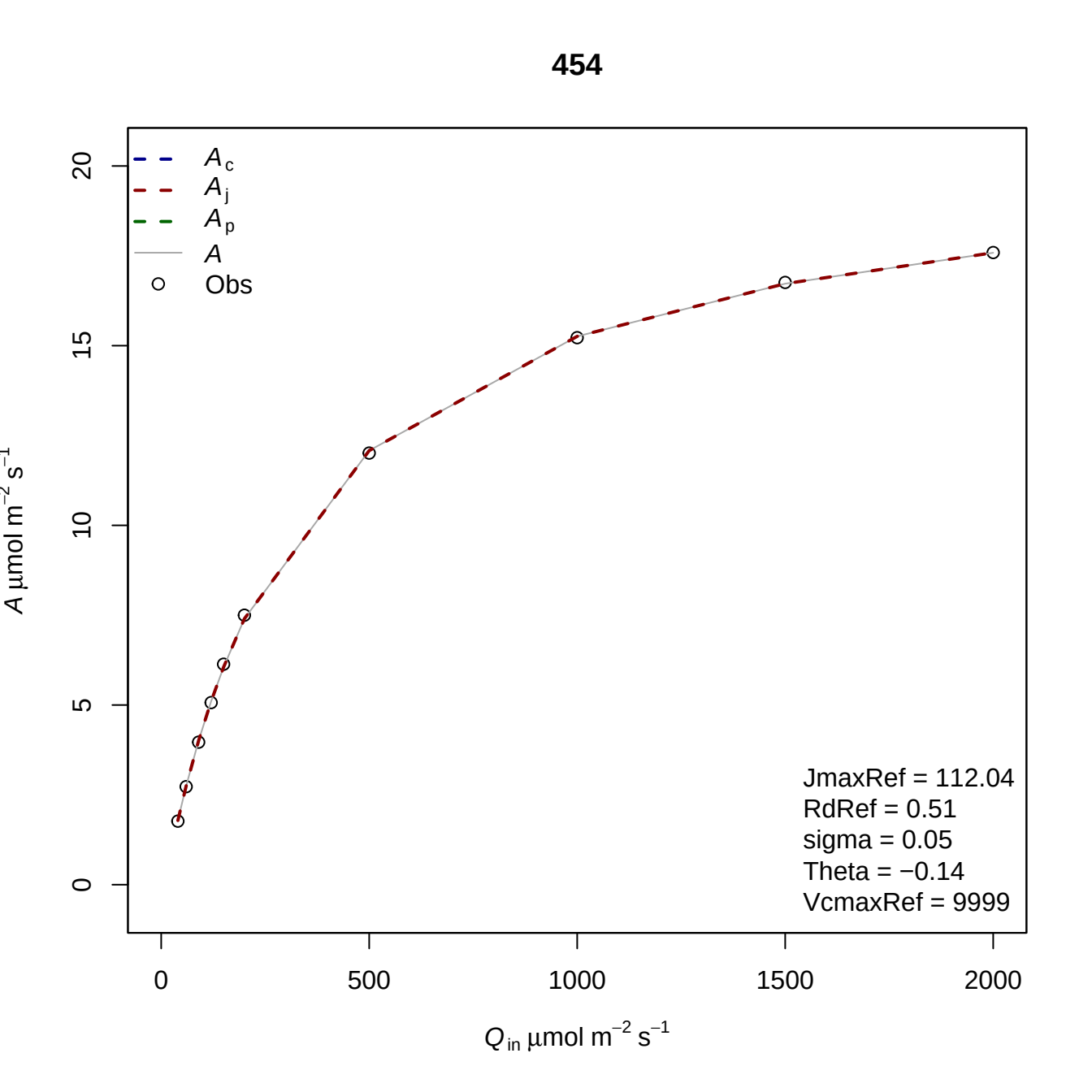

449
A_c
A_j
A_p
A
Obs
JmaxRef = 164.02
RdRef = 0.81
sigma = 0.13
Theta = 0.27
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

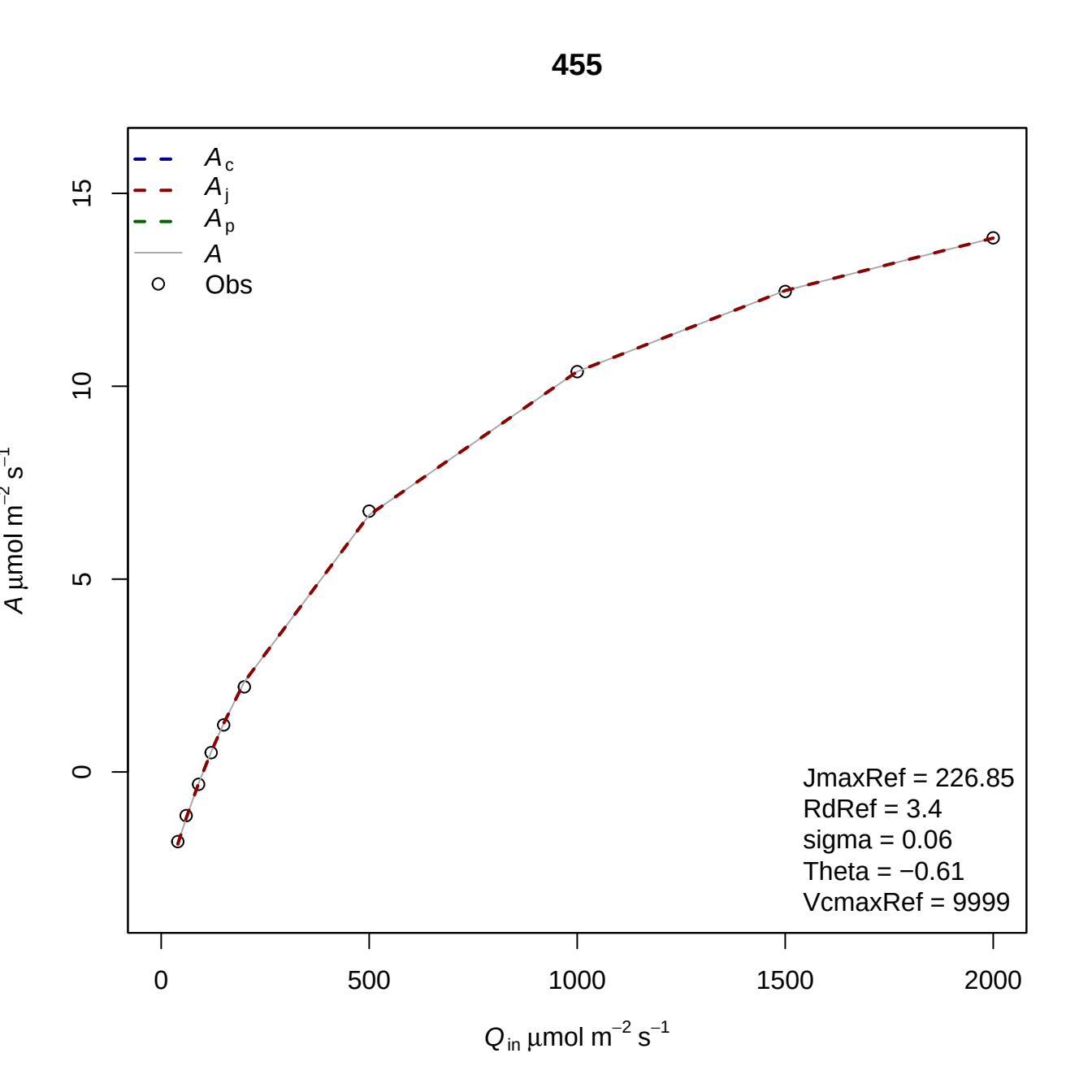

450
A_c
A_j
A_p
A
Obs
JmaxRef = 213.78
RdRef = 0.55
sigma = 0.18
Theta = 0.64
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

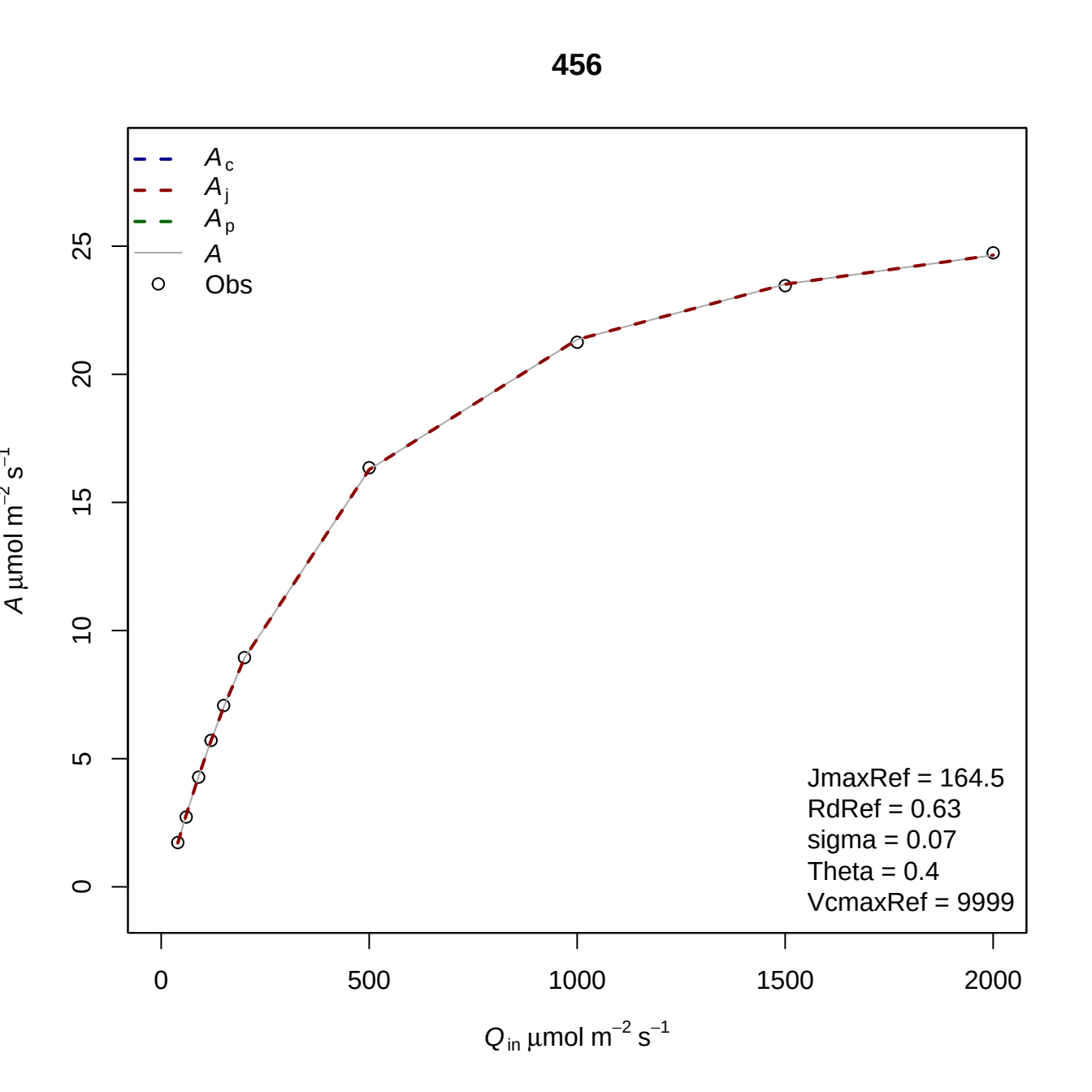

451
A_c
A_j
A_p
A
Obs
JmaxRef = 196.46
RdRef = 0.92
sigma = 0.09
Theta = 0.49
VcmaxRef = 119.44
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

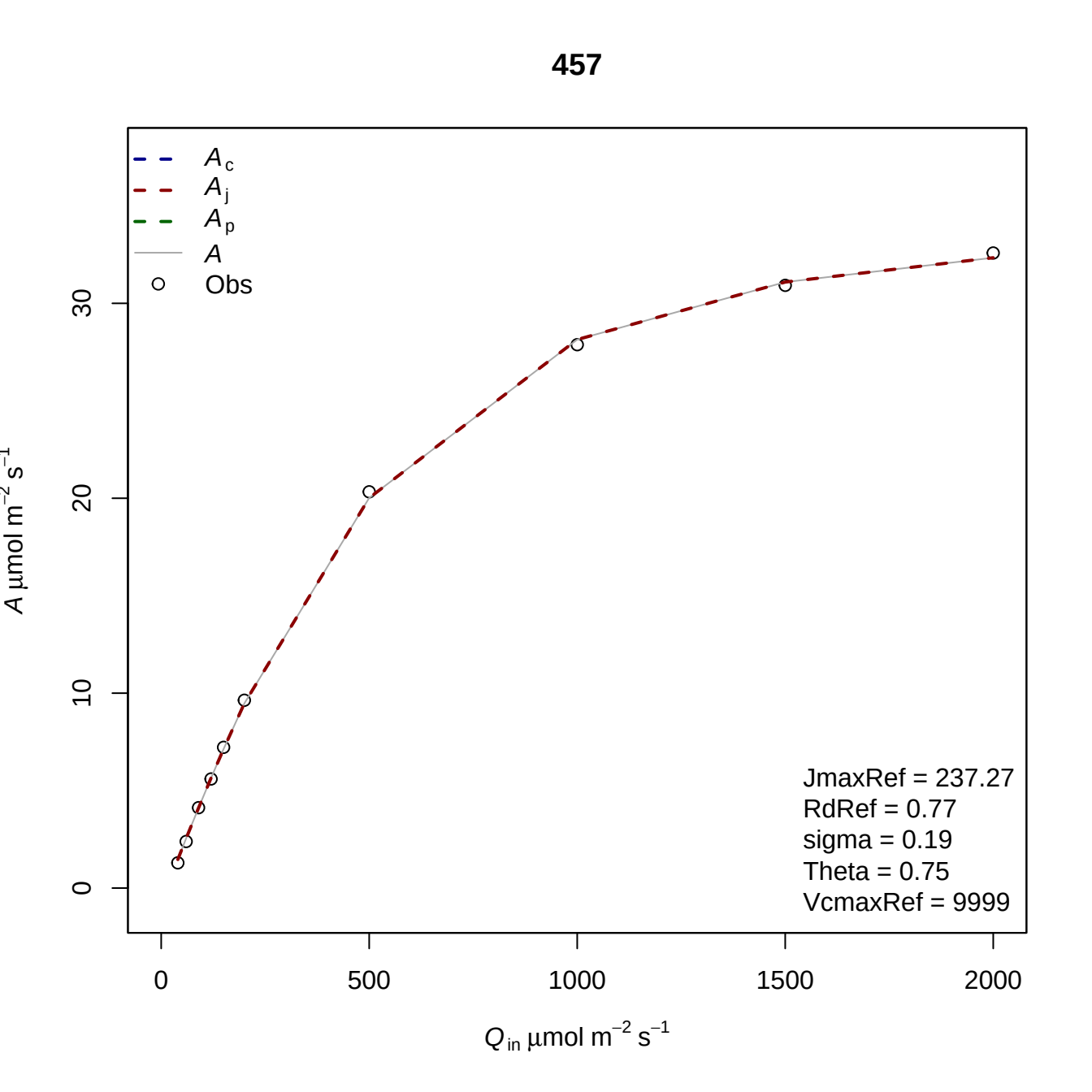

452
A_c
A_j
A_p
A
Obs
JmaxRef = 224.04
RdRef = 1.61
sigma = 0.07
Theta = 0.41
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

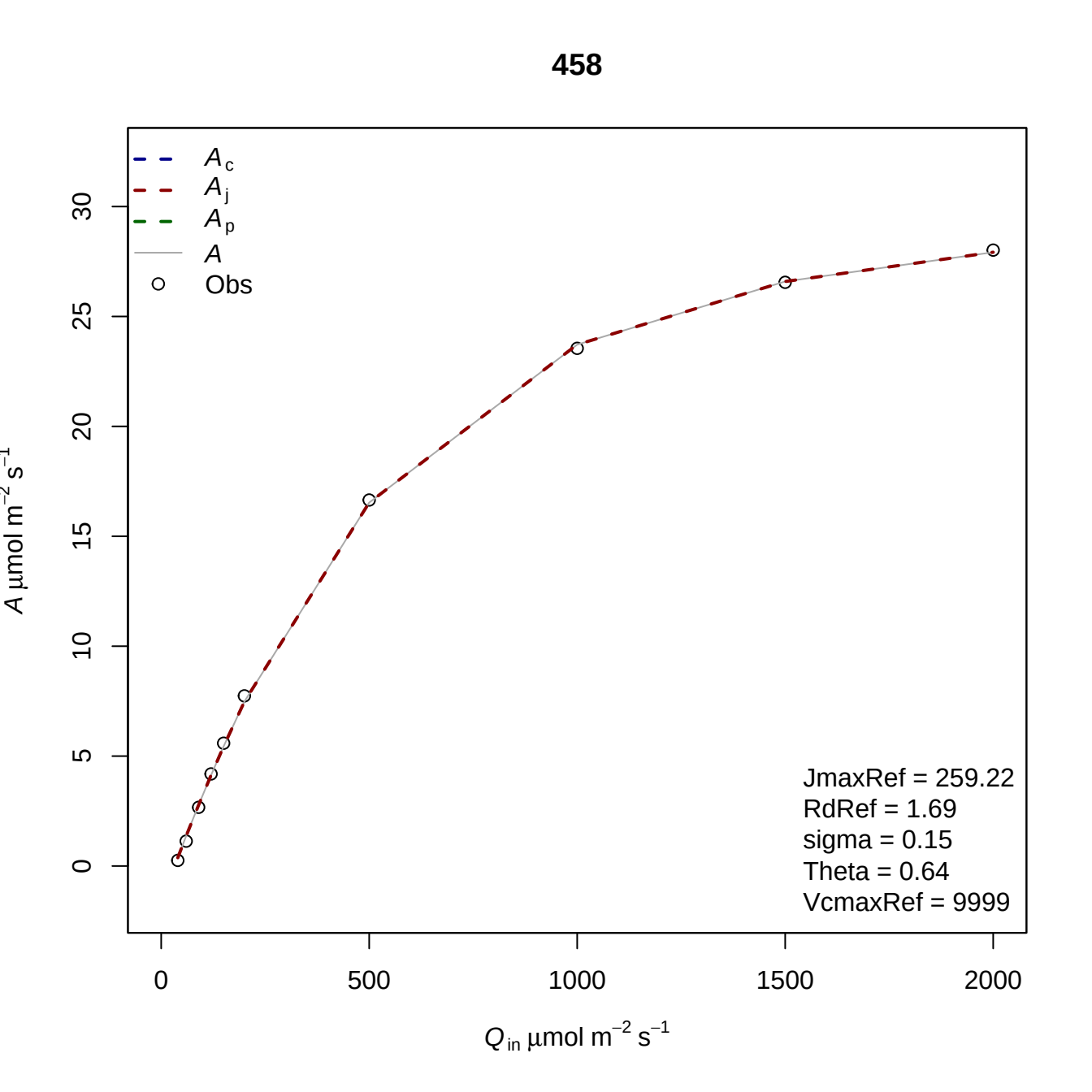

453
A_c
A_j
A_p
A
Obs
JmaxRef = 255.69
RdRef = 2.52
sigma = 0.08
Theta = 0.58
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

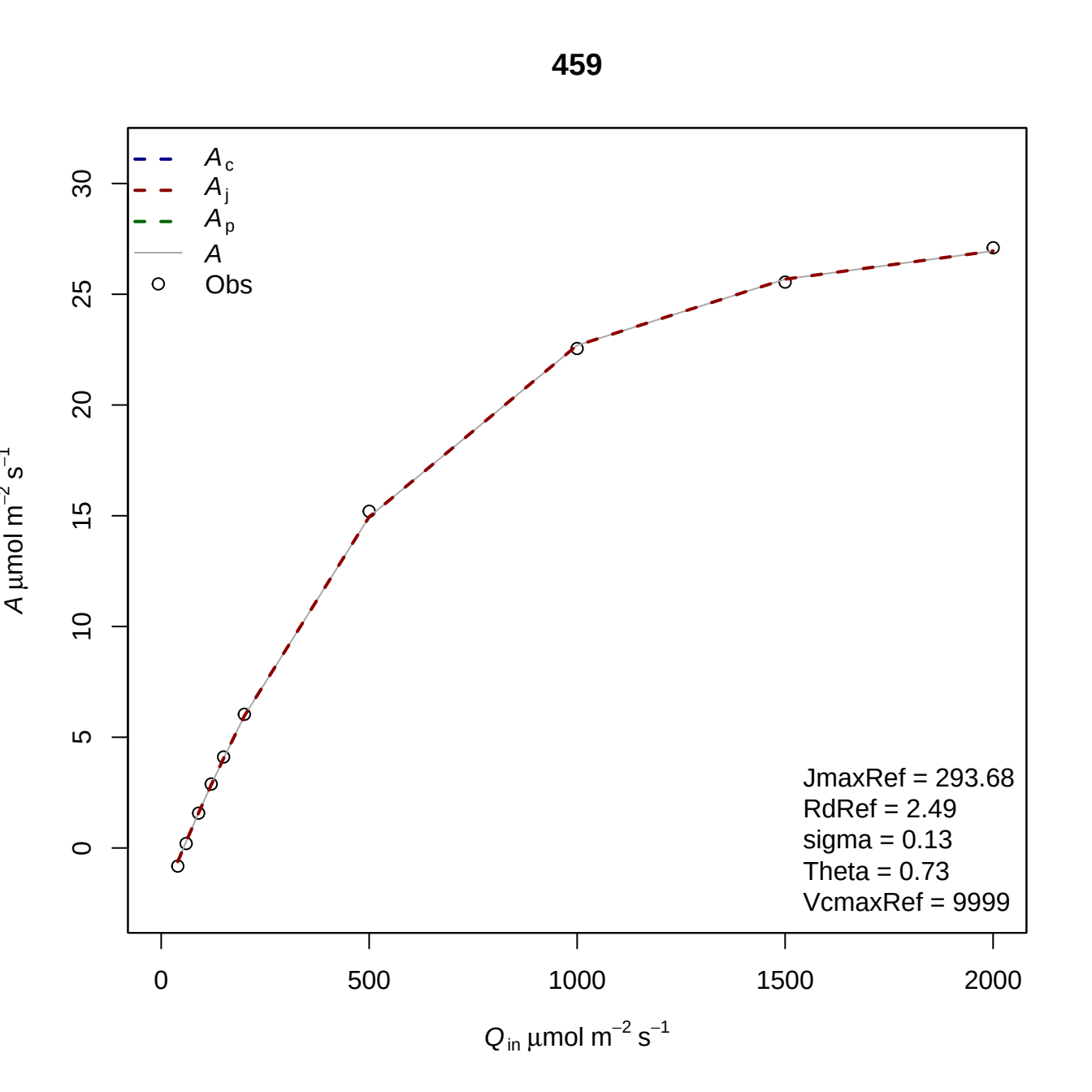

454
A_c
A_j
A_p
A
Obs
JmaxRef = 112.04
RdRef = 0.51
sigma = 0.05
Theta = −0.14
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

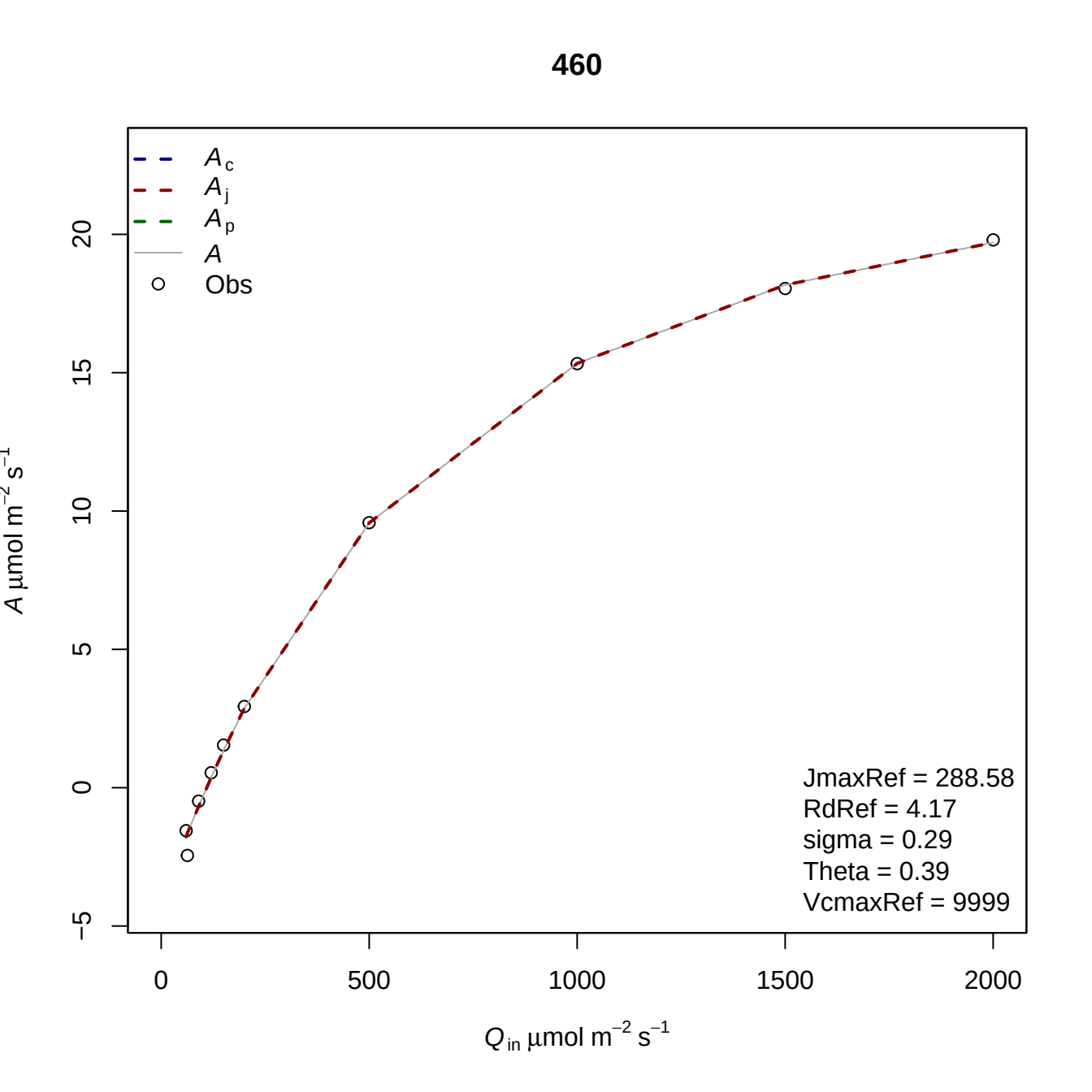

455
A_c
A_j
A_p
A
Obs
JmaxRef = 226.85
RdRef = 3.4
sigma = 0.06
Theta = −0.61
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

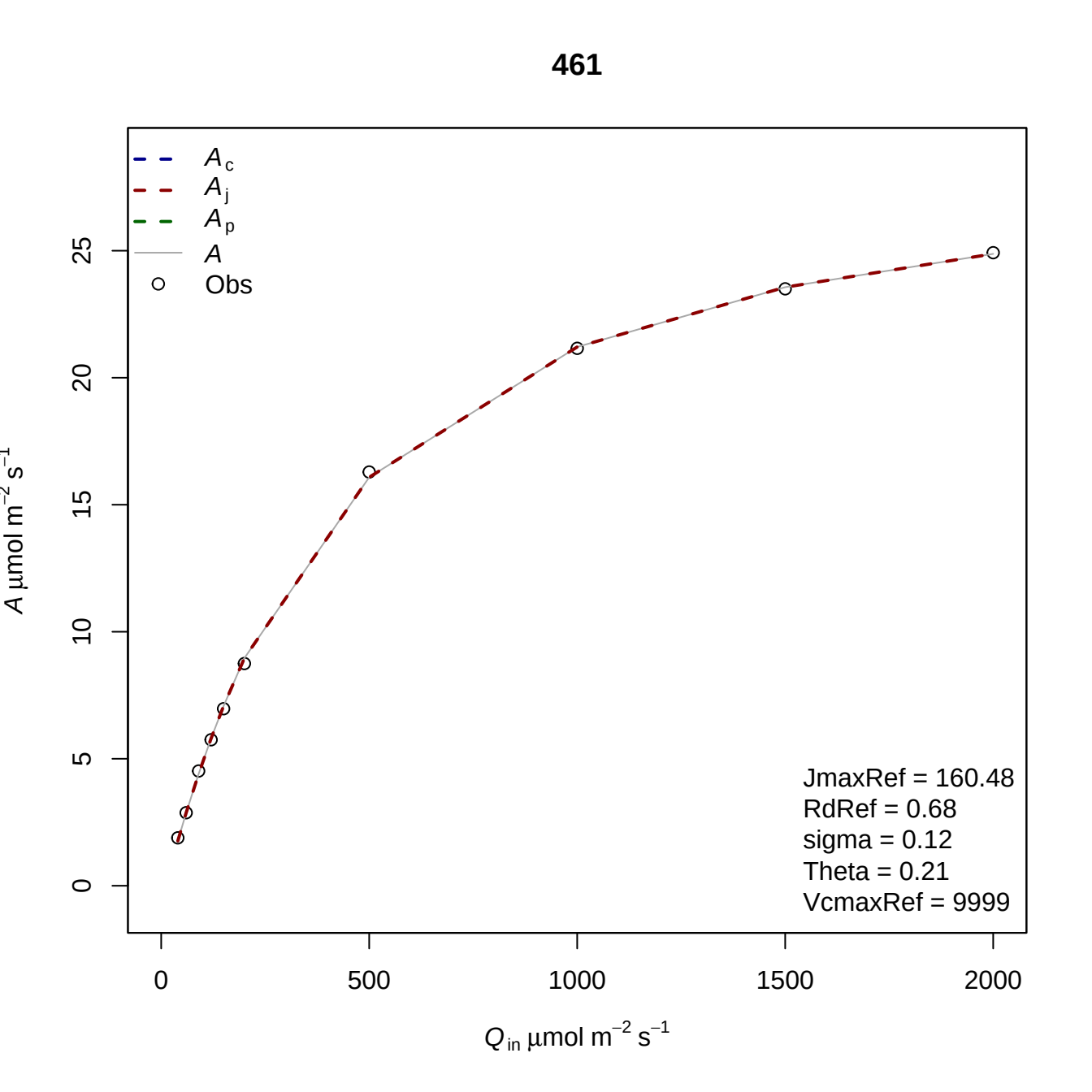

456
A_c
A_j
A_p
A
Obs
JmaxRef = 164.5
RdRef = 0.63
sigma = 0.07
Theta = 0.4
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

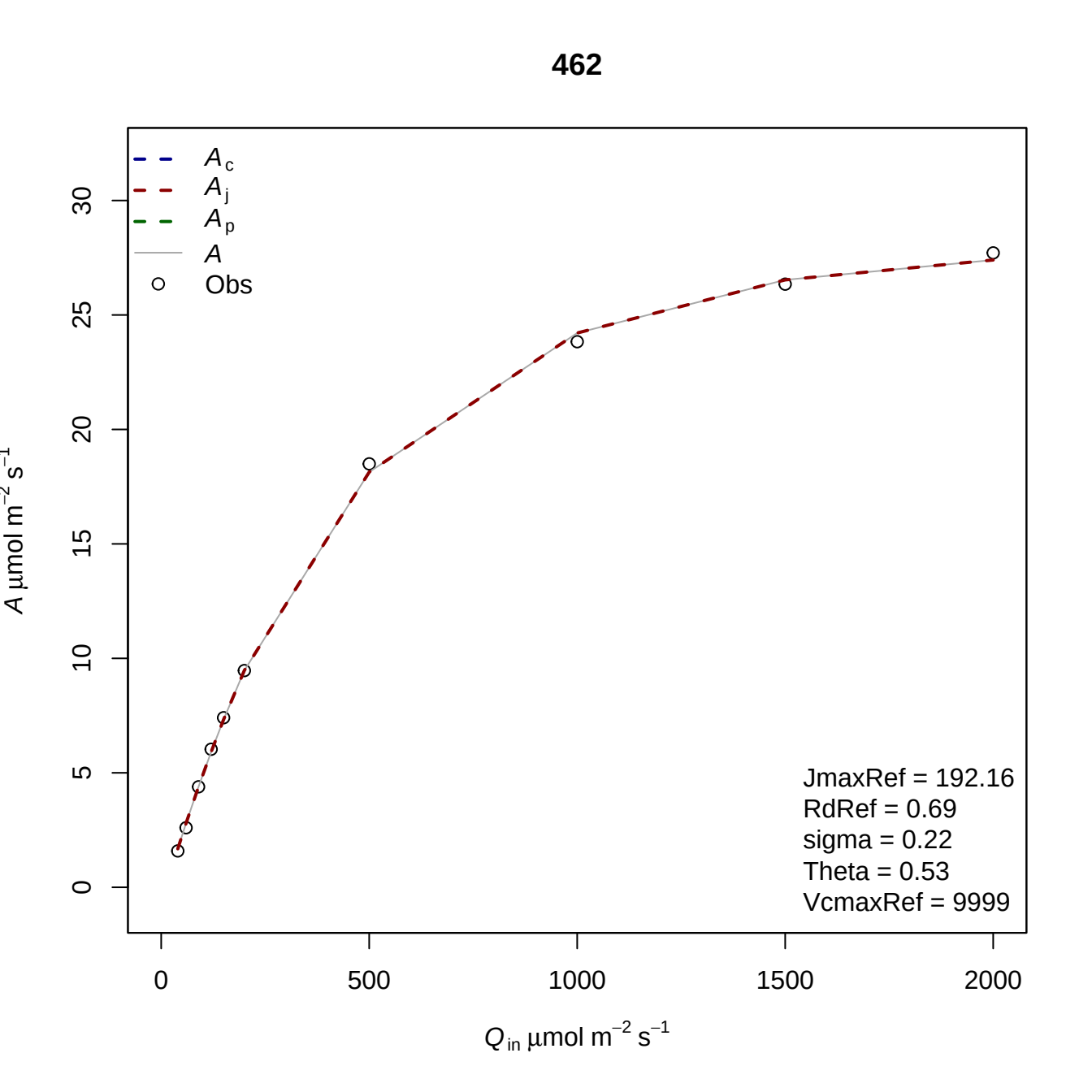

457
A_c
A_j
A_p
A
Obs
JmaxRef = 237.27
RdRef = 0.77
sigma = 0.19
Theta = 0.75
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

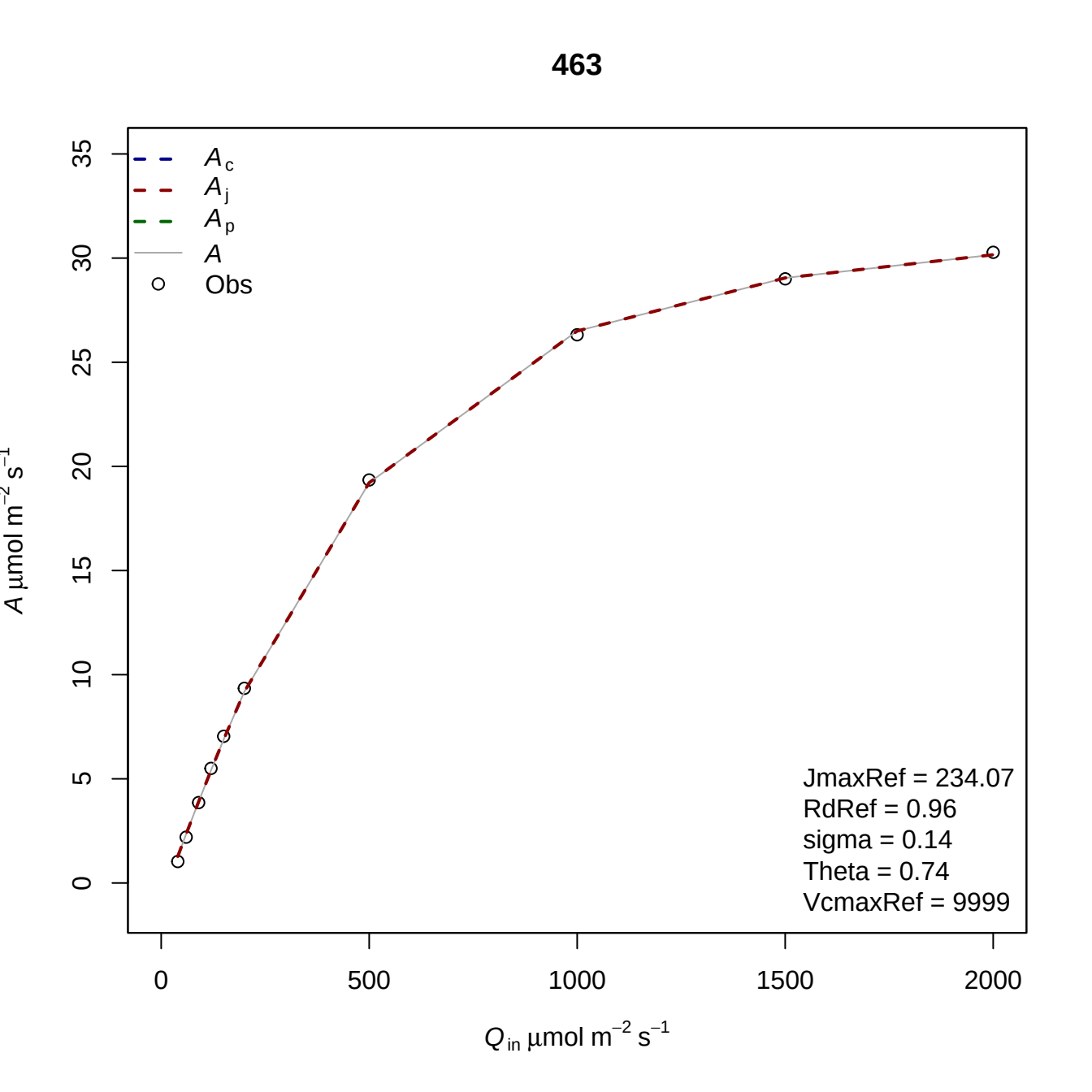

458
A_c
A_j
A_p
A
Obs
JmaxRef = 259.22
RdRef = 1.69
sigma = 0.15
Theta = 0.64
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

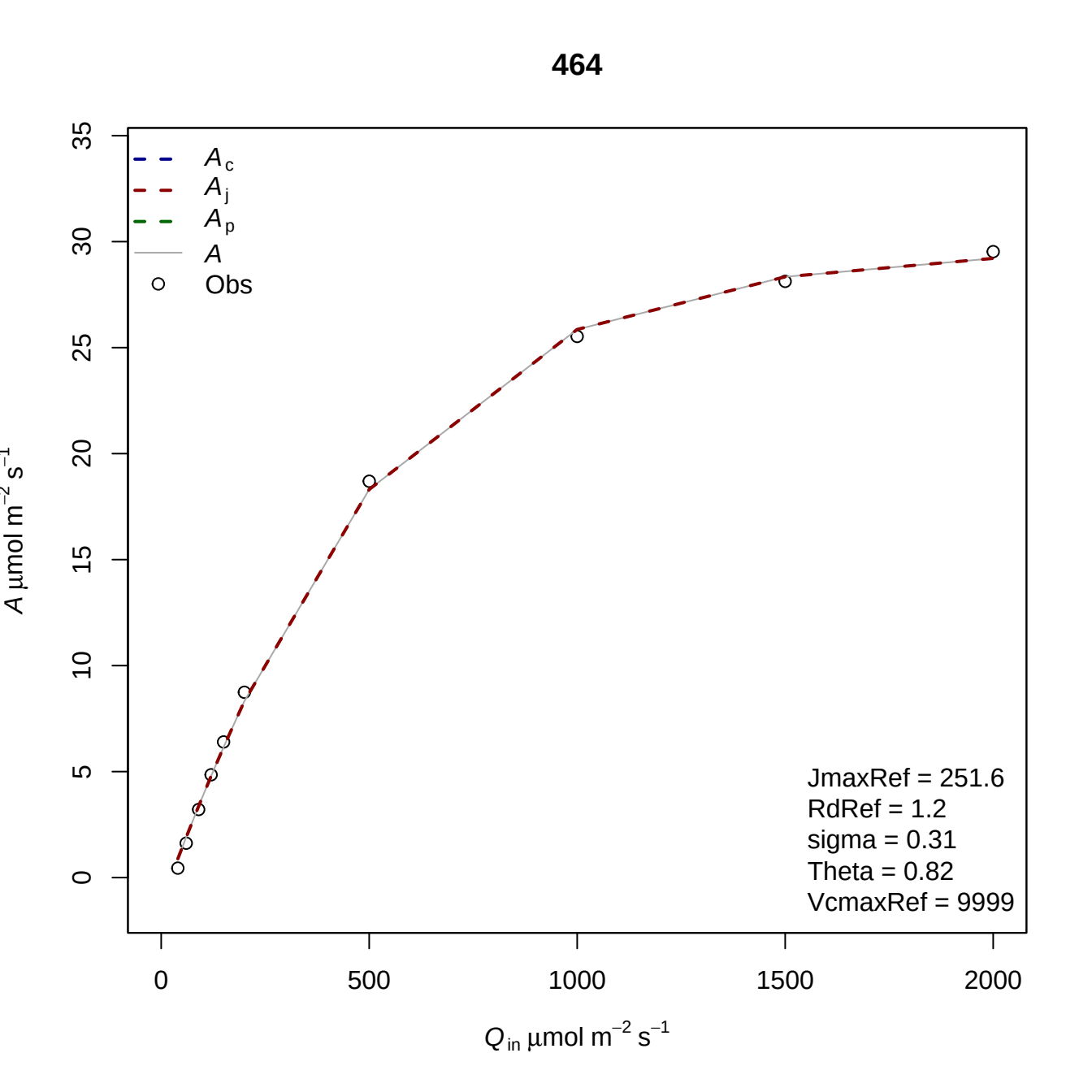

459
A_c
A_j
A_p
A
Obs
JmaxRef = 293.68
RdRef = 2.49
sigma = 0.13
Theta = 0.73
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

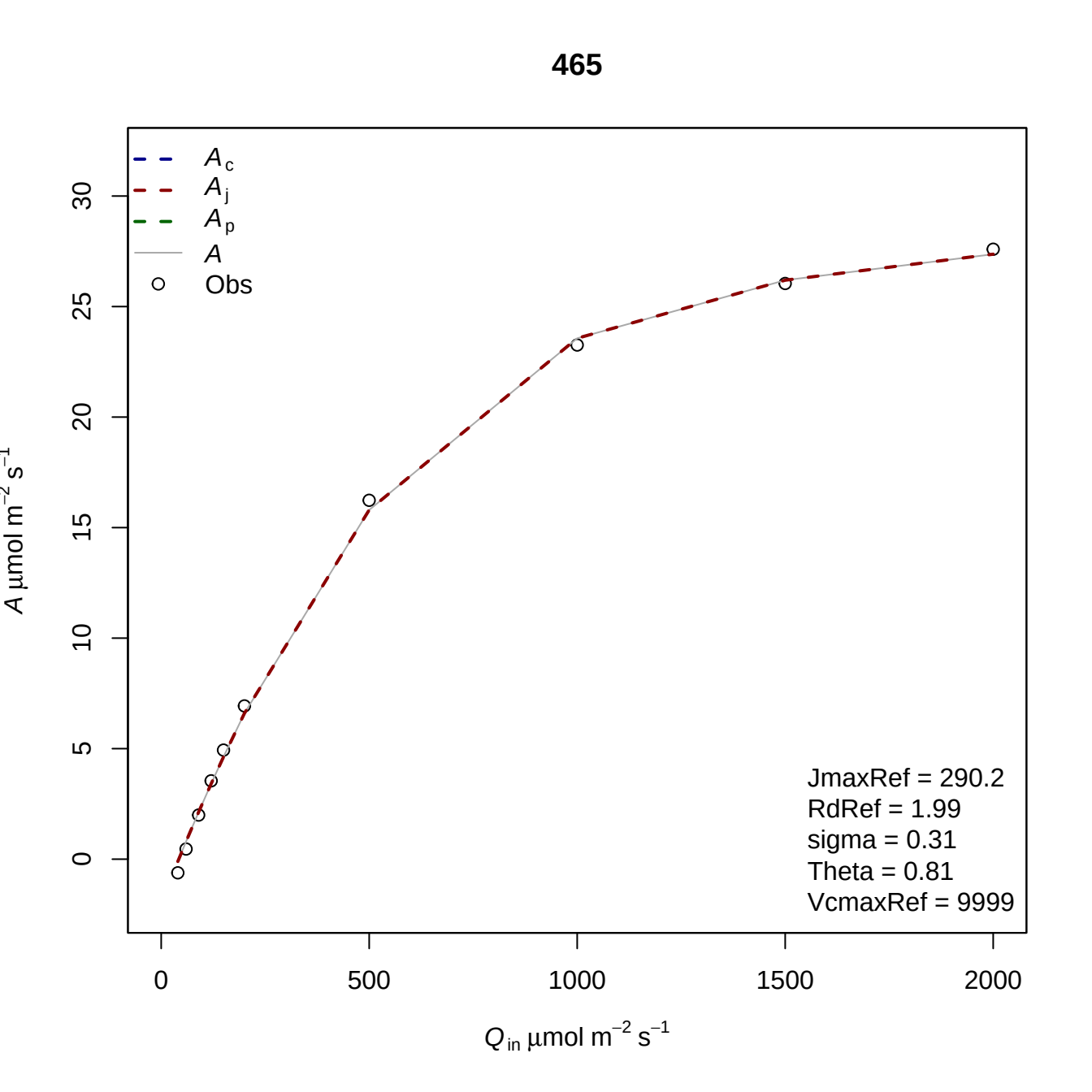

460
A_c
A_j
A_p
A
Obs
JmaxRef = 288.58
RdRef = 4.17
sigma = 0.29
Theta = 0.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

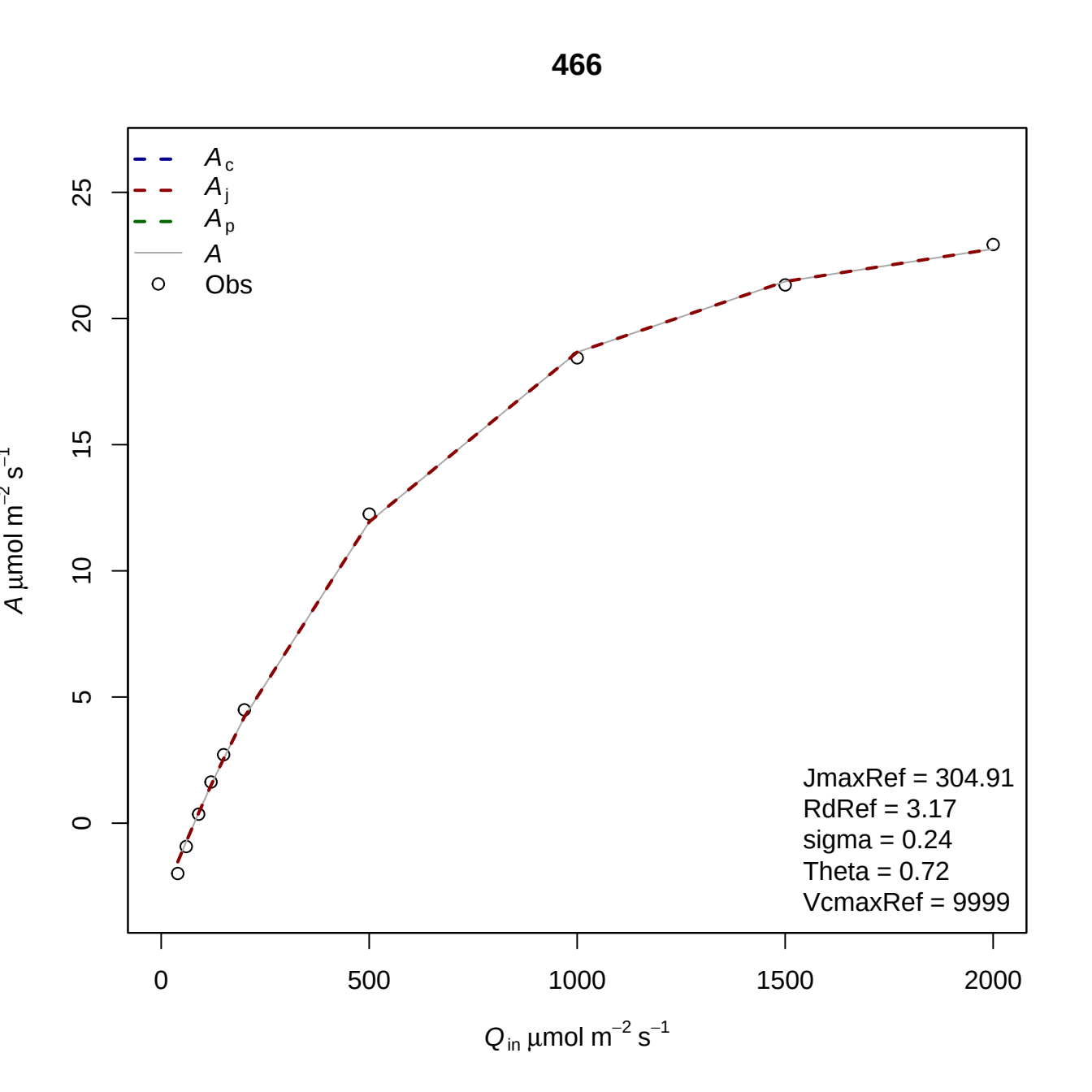

461
A_c
A_j
A_p
A
Obs
JmaxRef = 160.48
RdRef = 0.68
sigma = 0.12
Theta = 0.21
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

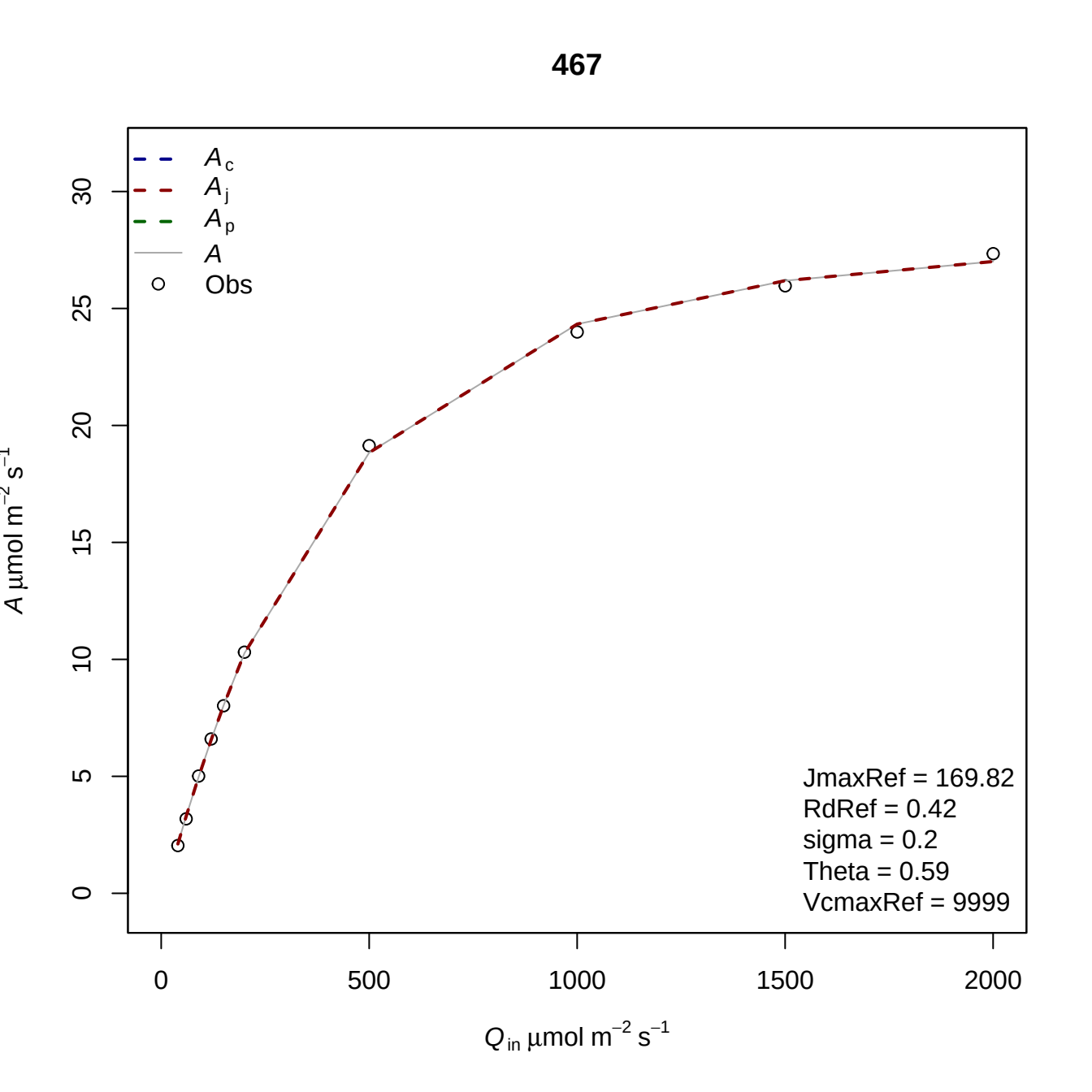

462
A_c
A_j
A_p
A
Obs
JmaxRef = 192.16
RdRef = 0.69
sigma = 0.22
Theta = 0.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

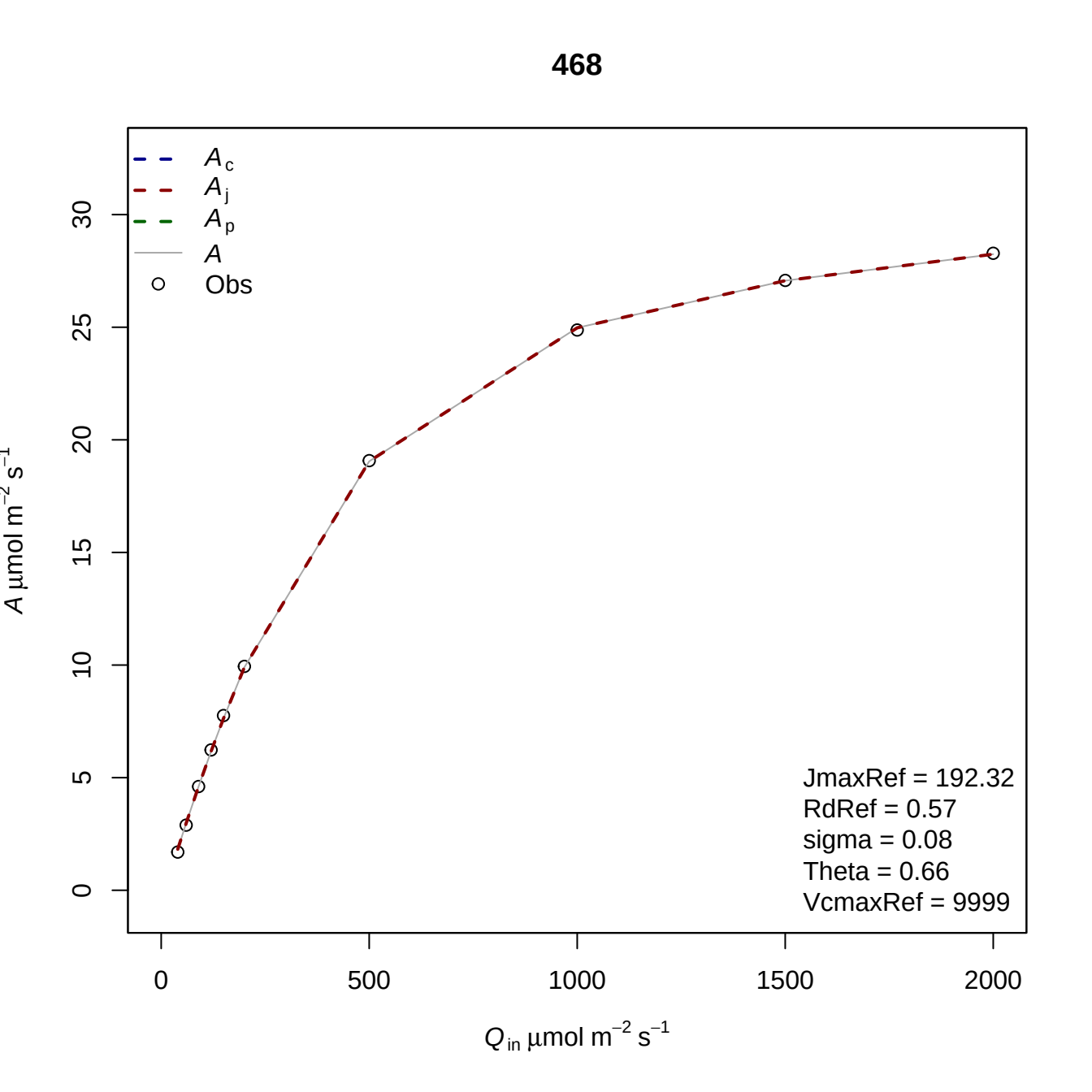

463
A_c
A_j
A_p
A
Obs
JmaxRef = 234.07
RdRef = 0.96
sigma = 0.14
Theta = 0.74
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

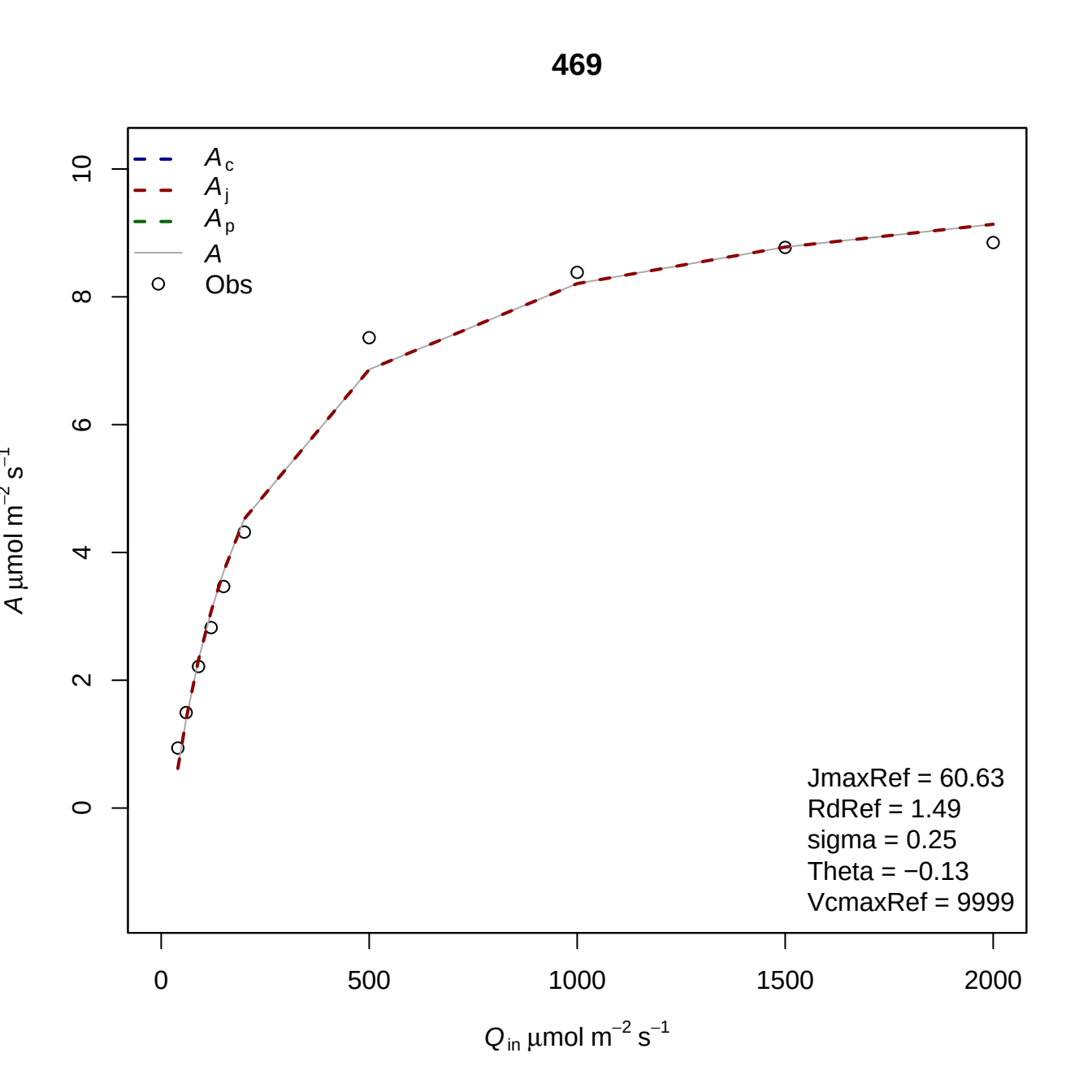

464
A_c
A_j
A_p
A
Obs
JmaxRef = 251.6
RdRef = 1.2
sigma = 0.31
Theta = 0.82
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

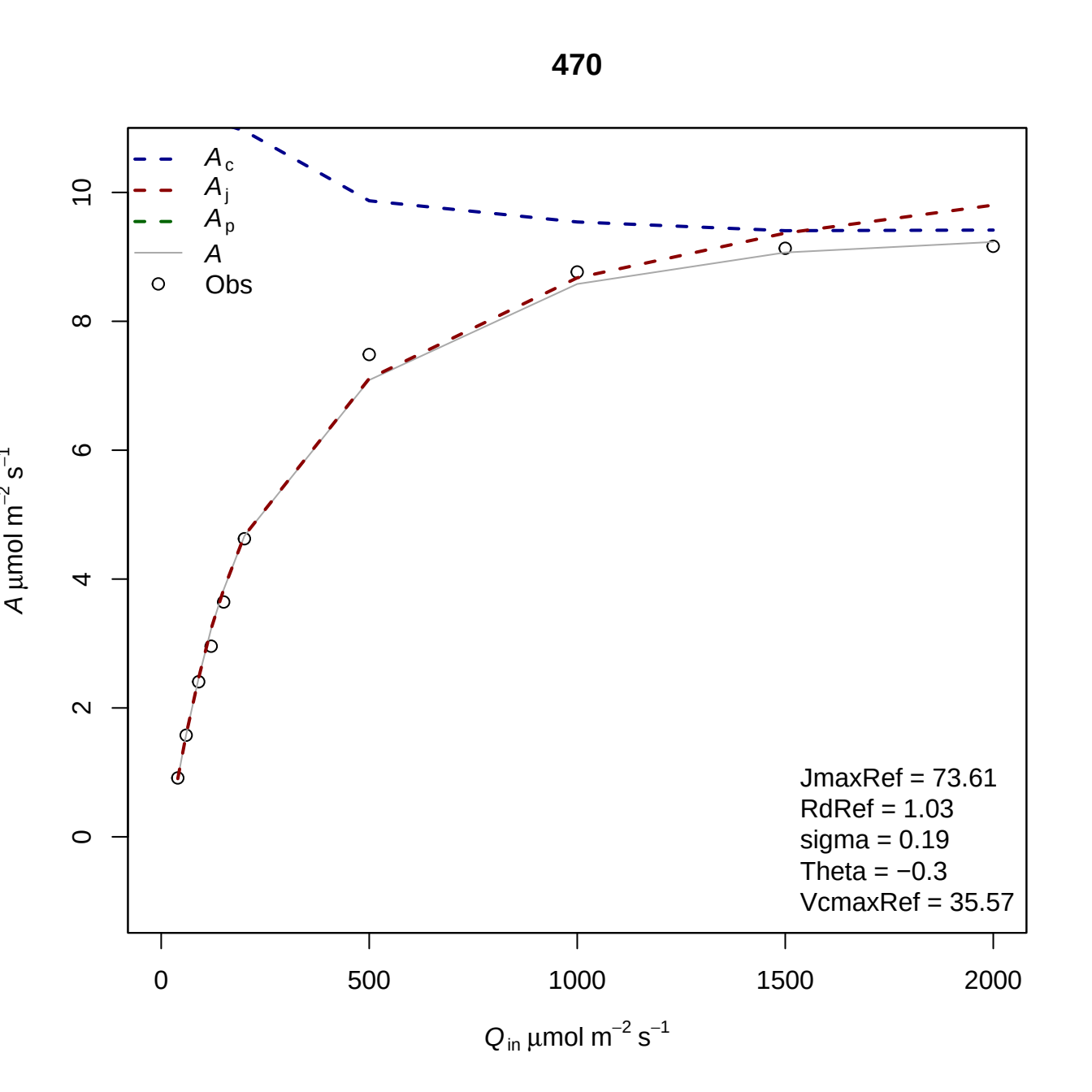

465
A_c
A_j
A_p
A
Obs
JmaxRef = 290.2
RdRef = 1.99
sigma = 0.31
Theta = 0.81
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

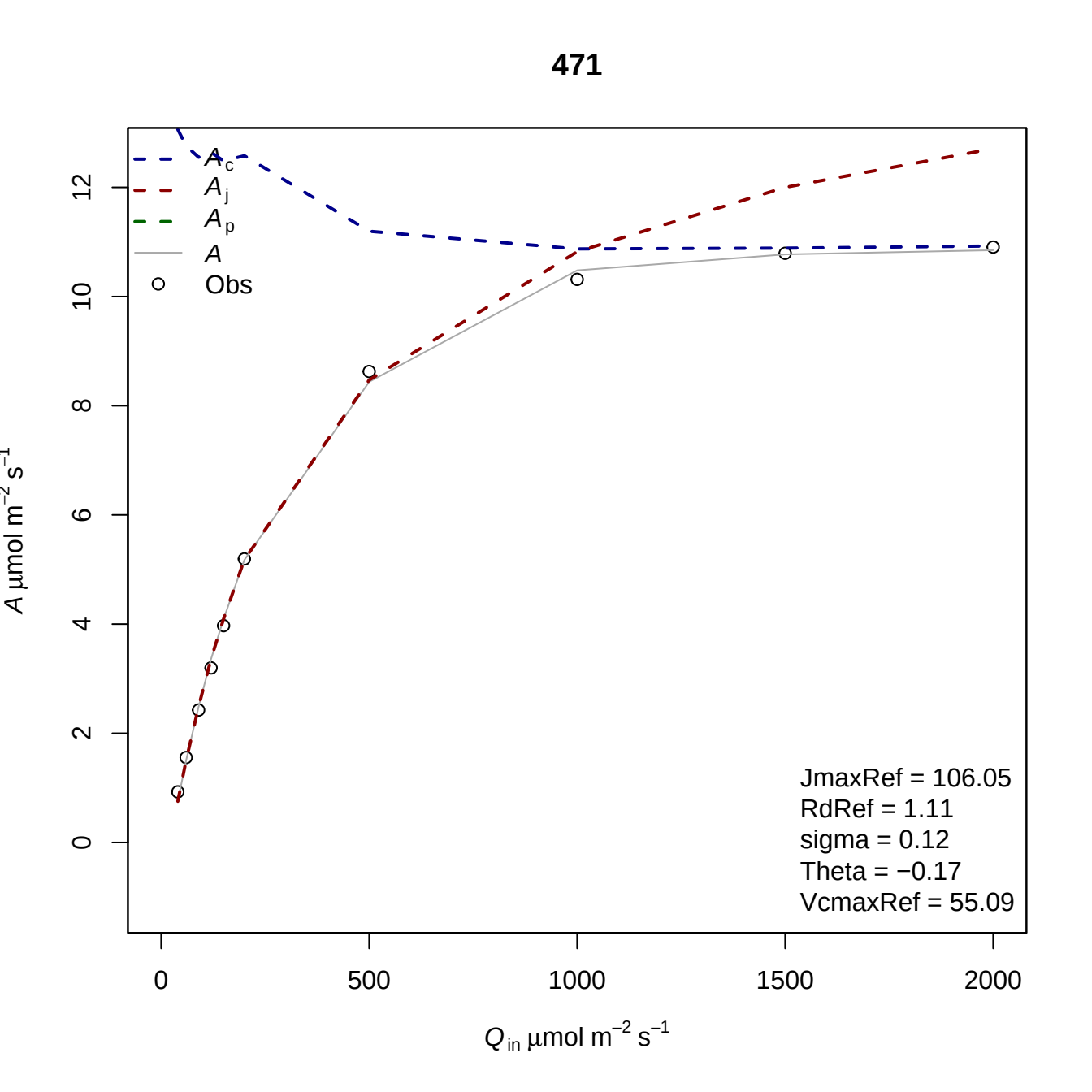

466
A_c
A_j
A_p
A
Obs
JmaxRef = 304.91
RdRef = 3.17
sigma = 0.24
Theta = 0.72
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

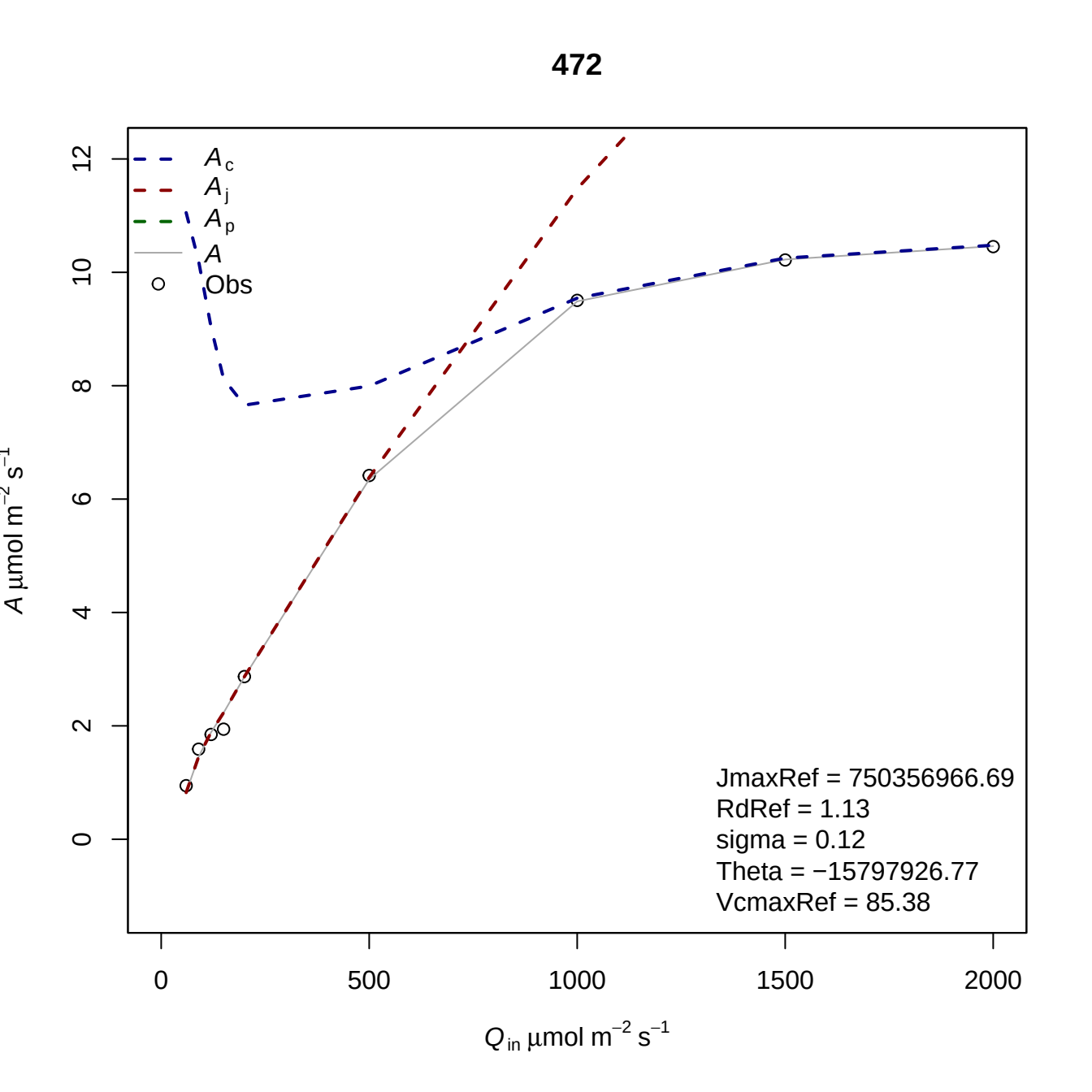

467
A_c
A_j
A_p
A
Obs
JmaxRef = 169.82
RdRef = 0.42
sigma = 0.2
Theta = 0.59
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

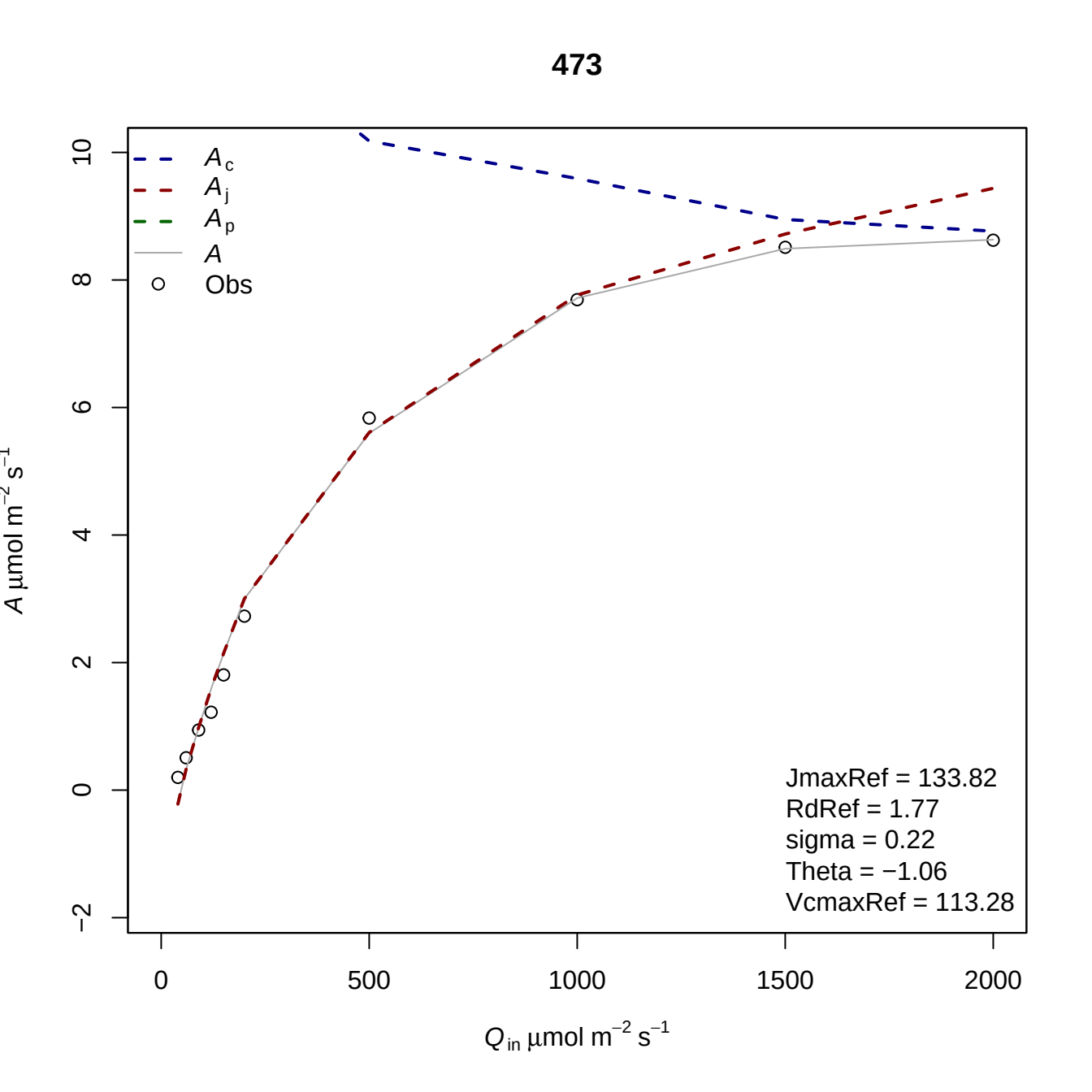

468
A_c
A_j
A_p
A
Obs
JmaxRef = 192.32
RdRef = 0.57
sigma = 0.08
Theta = 0.66
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

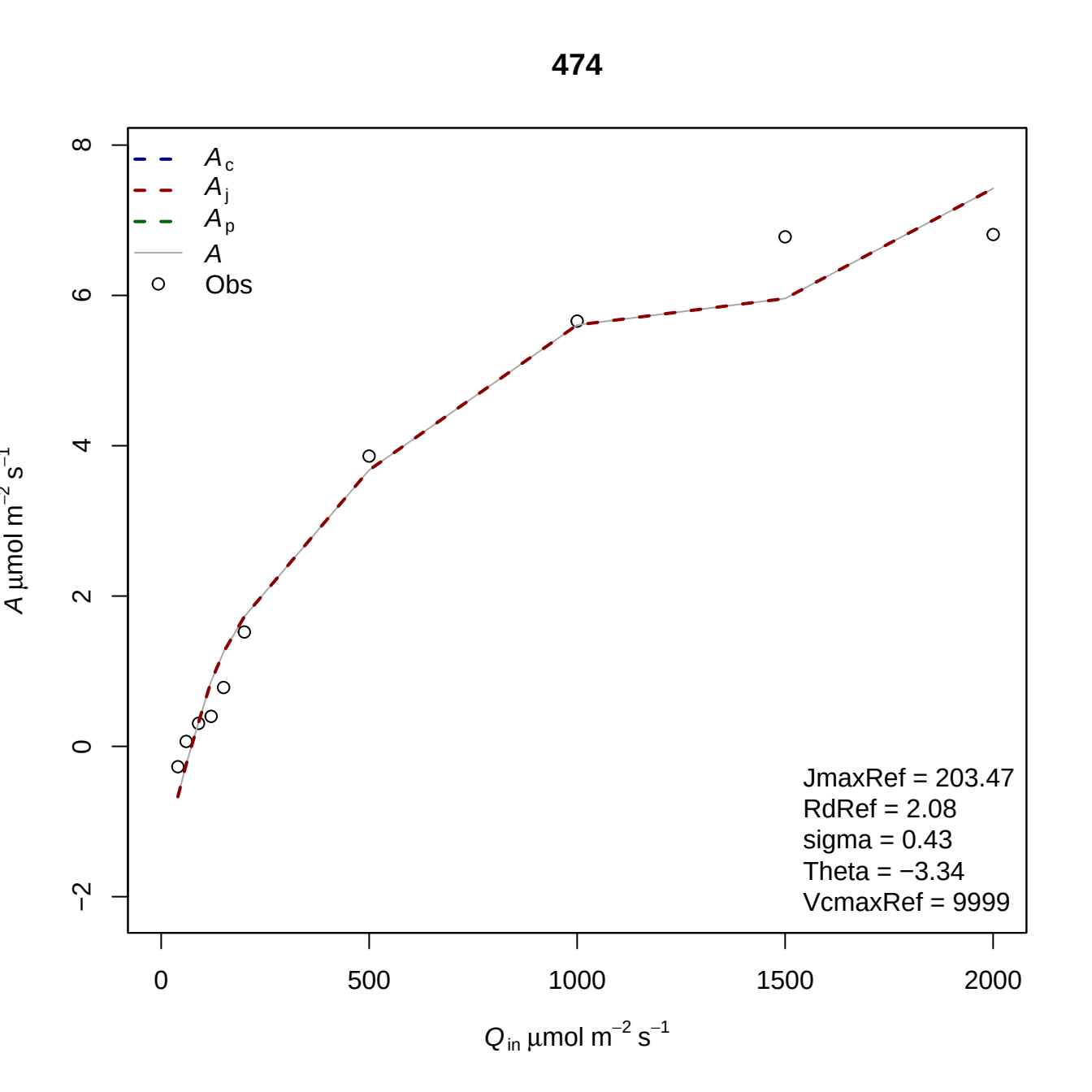

469
A_c
A_j
A_p
A
Obs
JmaxRef = 60.63
RdRef = 1.49
sigma = 0.25
Theta = −0.13
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

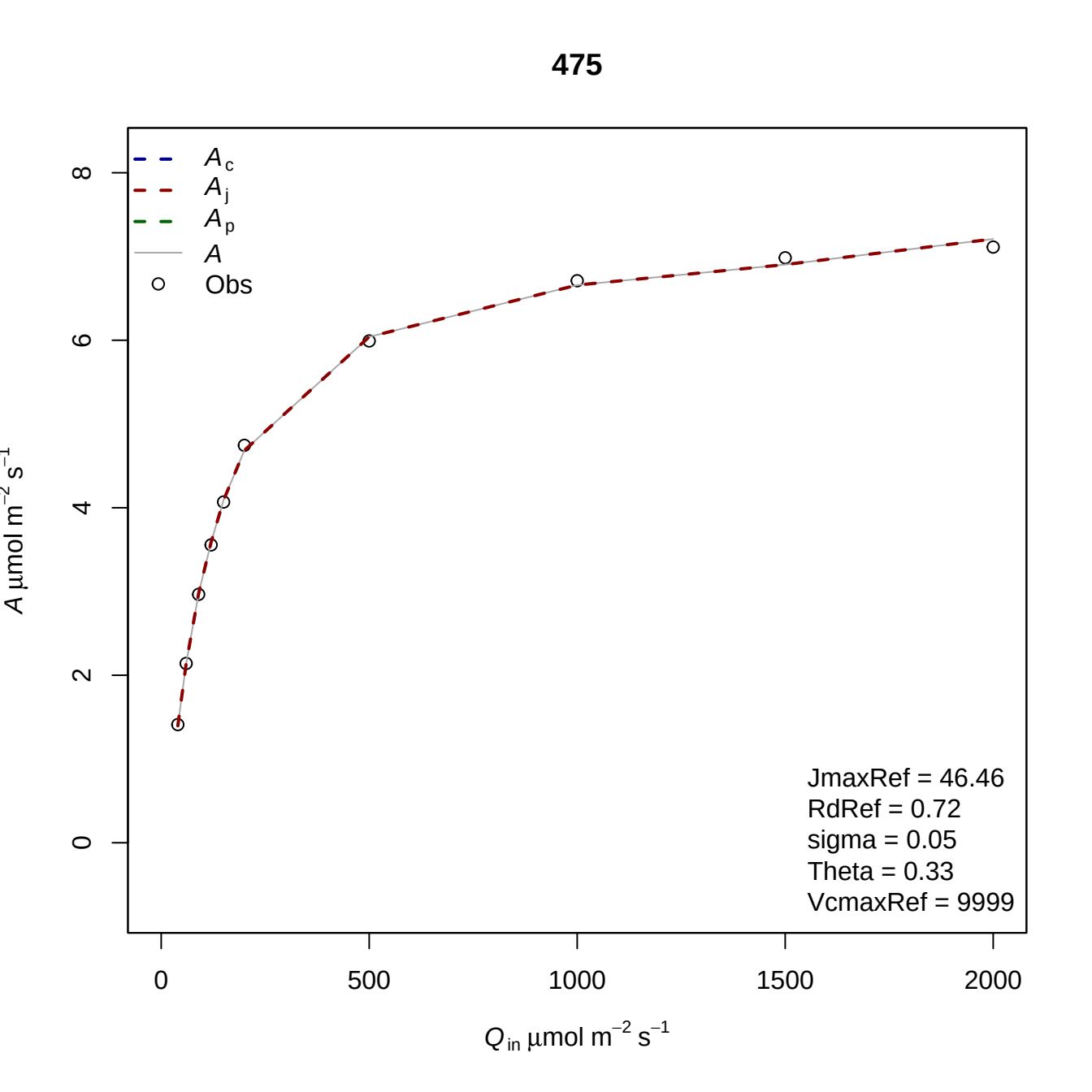

470
A_c
A_j
A_p
A
Obs
JmaxRef = 73.61
RdRef = 1.03
sigma = 0.19
Theta = −0.3
VcmaxRef = 35.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

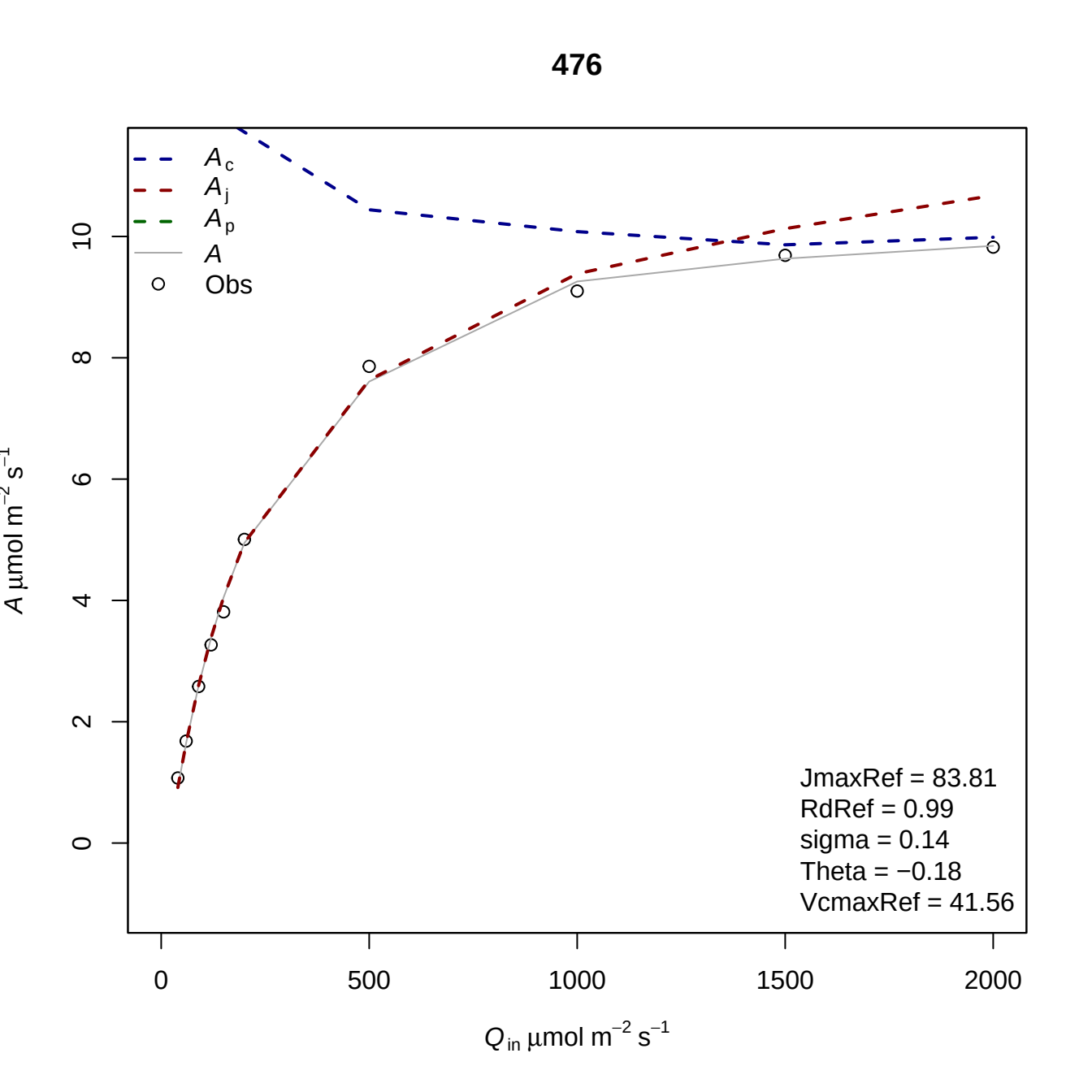

471
A_c
A_j
A_p
A
Obs
JmaxRef = 106.05
RdRef = 1.11
sigma = 0.12
Theta = −0.17
VcmaxRef = 55.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

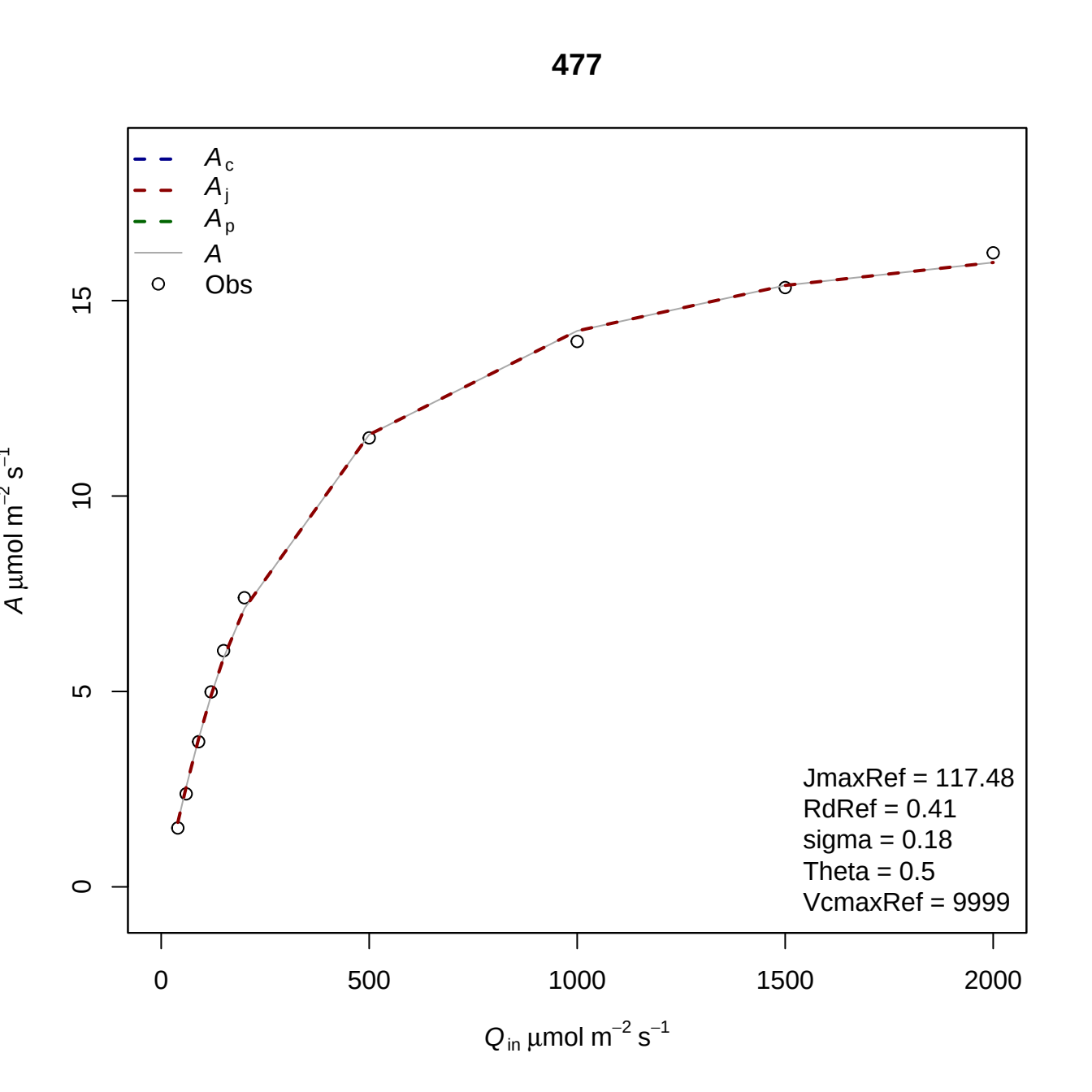

472
A_c
A_j
A_p
A
Obs
JmaxRef = 750356966.69
RdRef = 1.13
sigma = 0.12
Theta = −15797926.77
VcmaxRef = 85.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

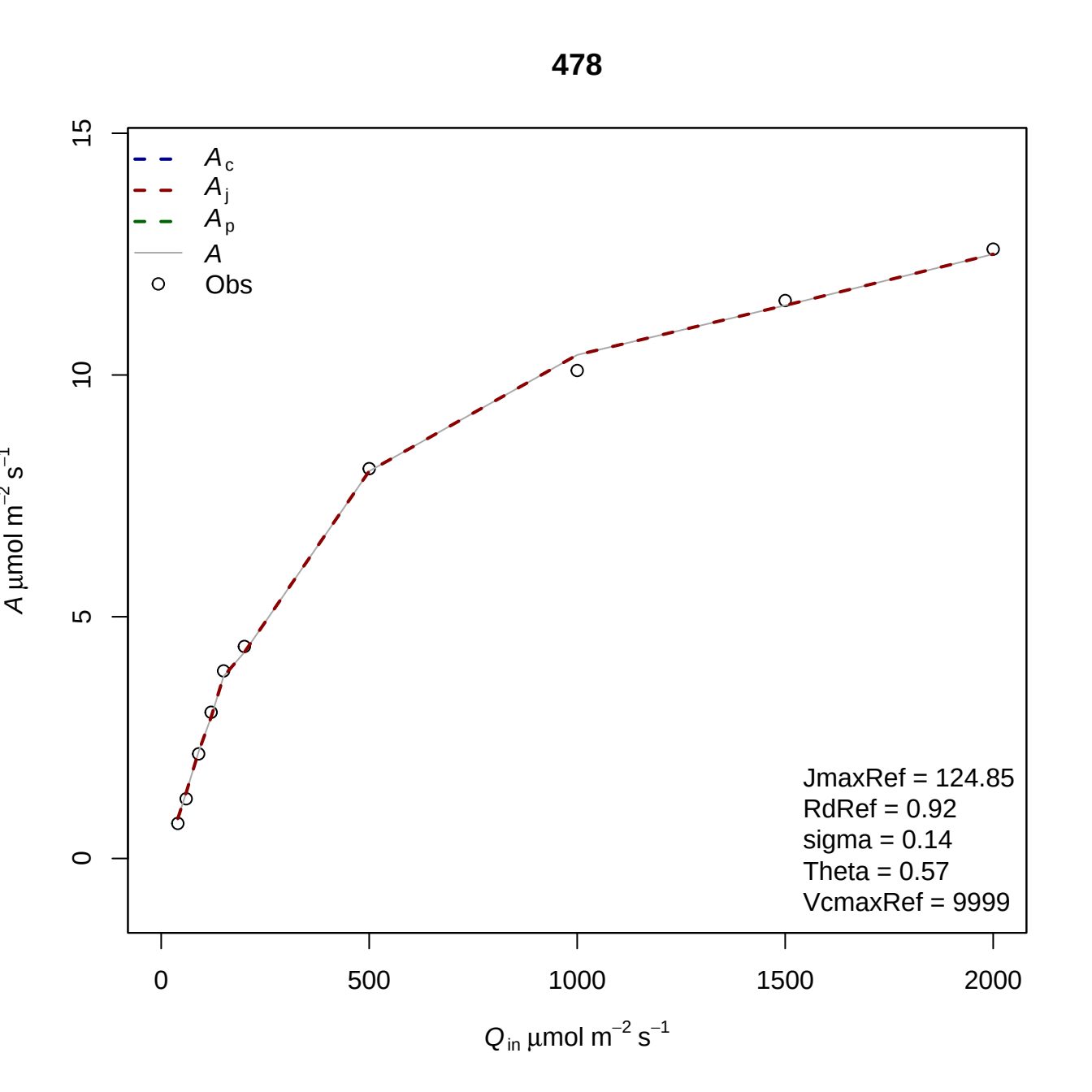

473
A_c
A_j
A_p
A
Obs
JmaxRef = 133.82
RdRef = 1.77
sigma = 0.22
Theta = −1.06
VcmaxRef = 113.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

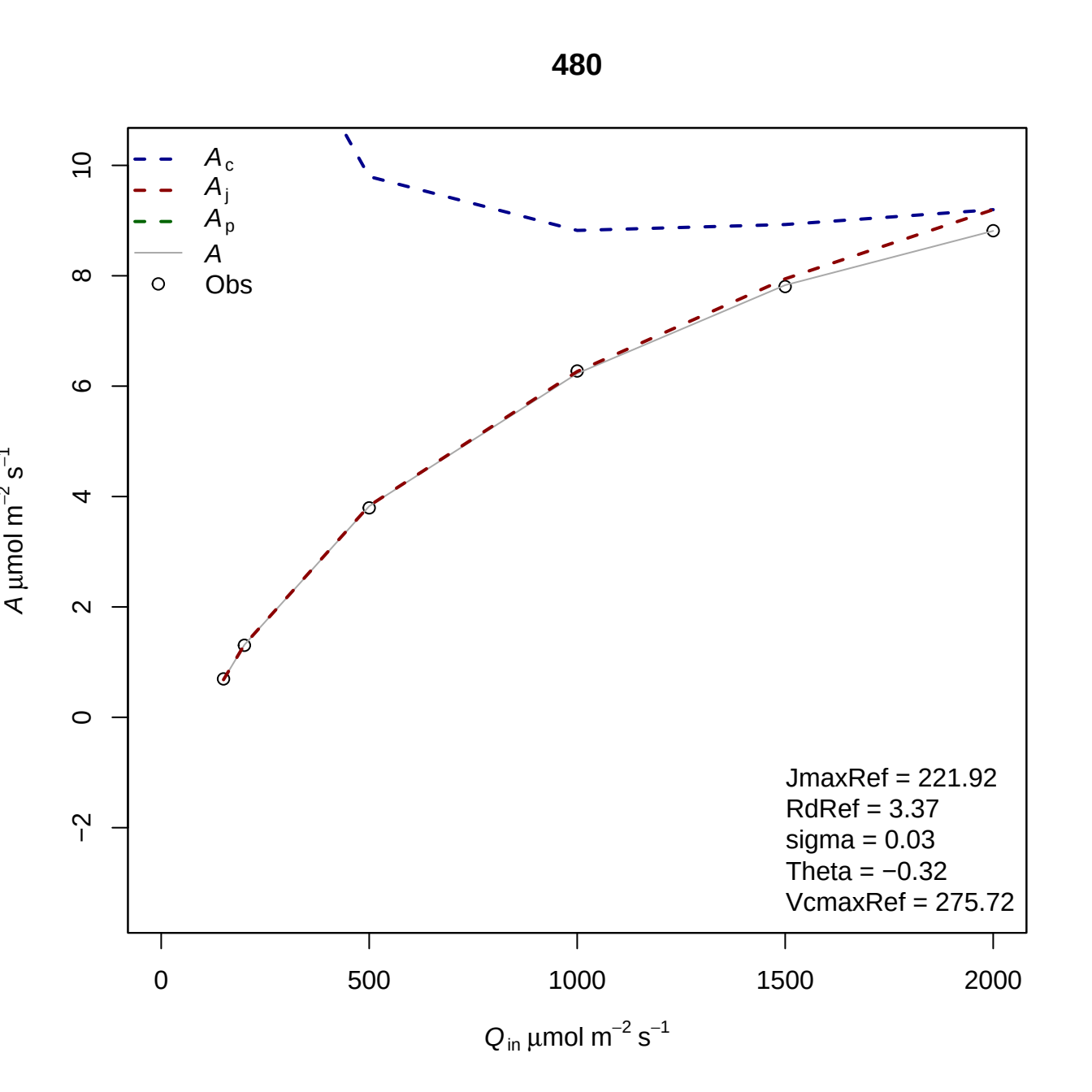

474
A_c
A_j
A_p
A
Obs
JmaxRef = 203.47
RdRef = 2.08
sigma = 0.43
Theta = −3.34
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

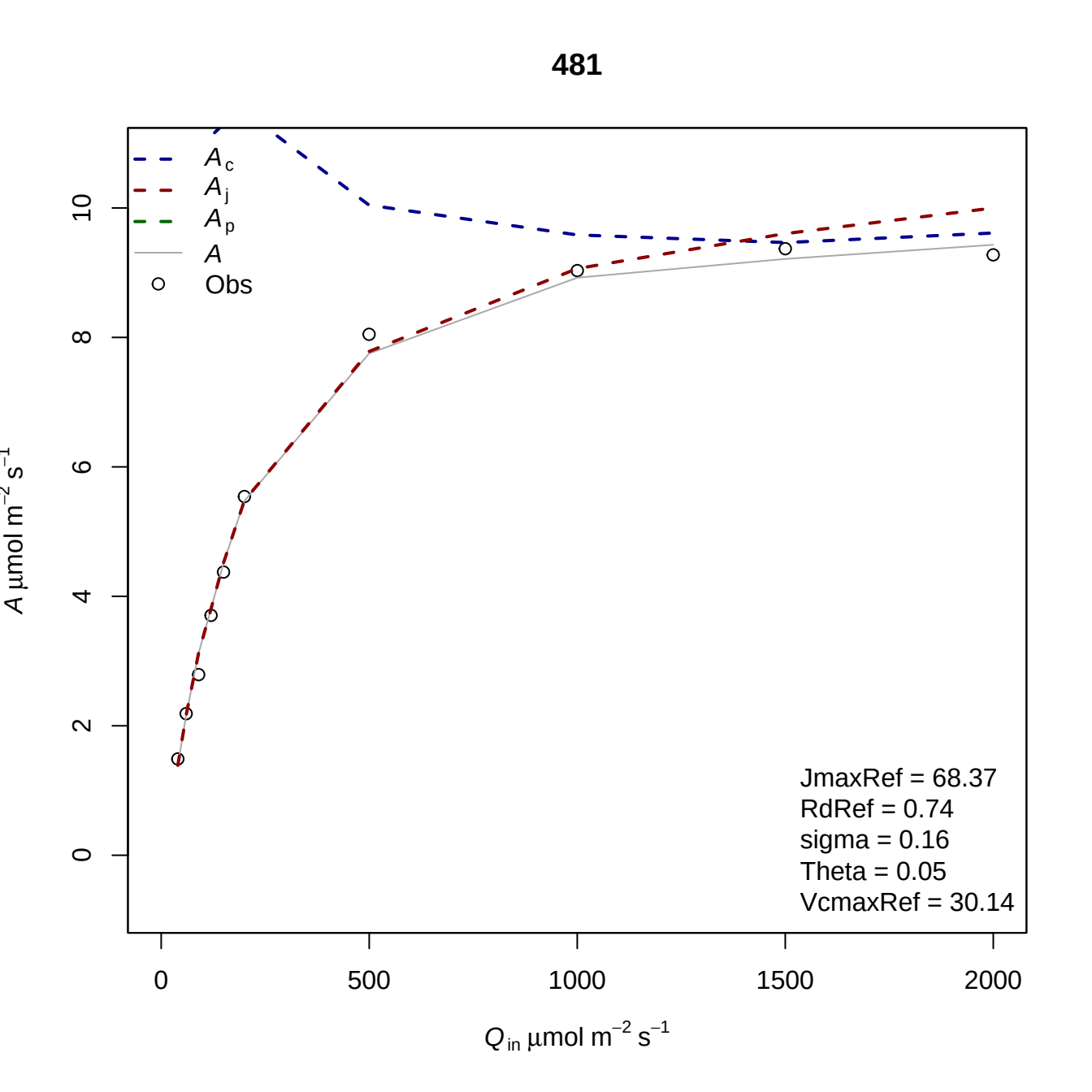

475
A_c
A_j
A_p
A
Obs
JmaxRef = 46.46
RdRef = 0.72
sigma = 0.05
Theta = 0.33
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

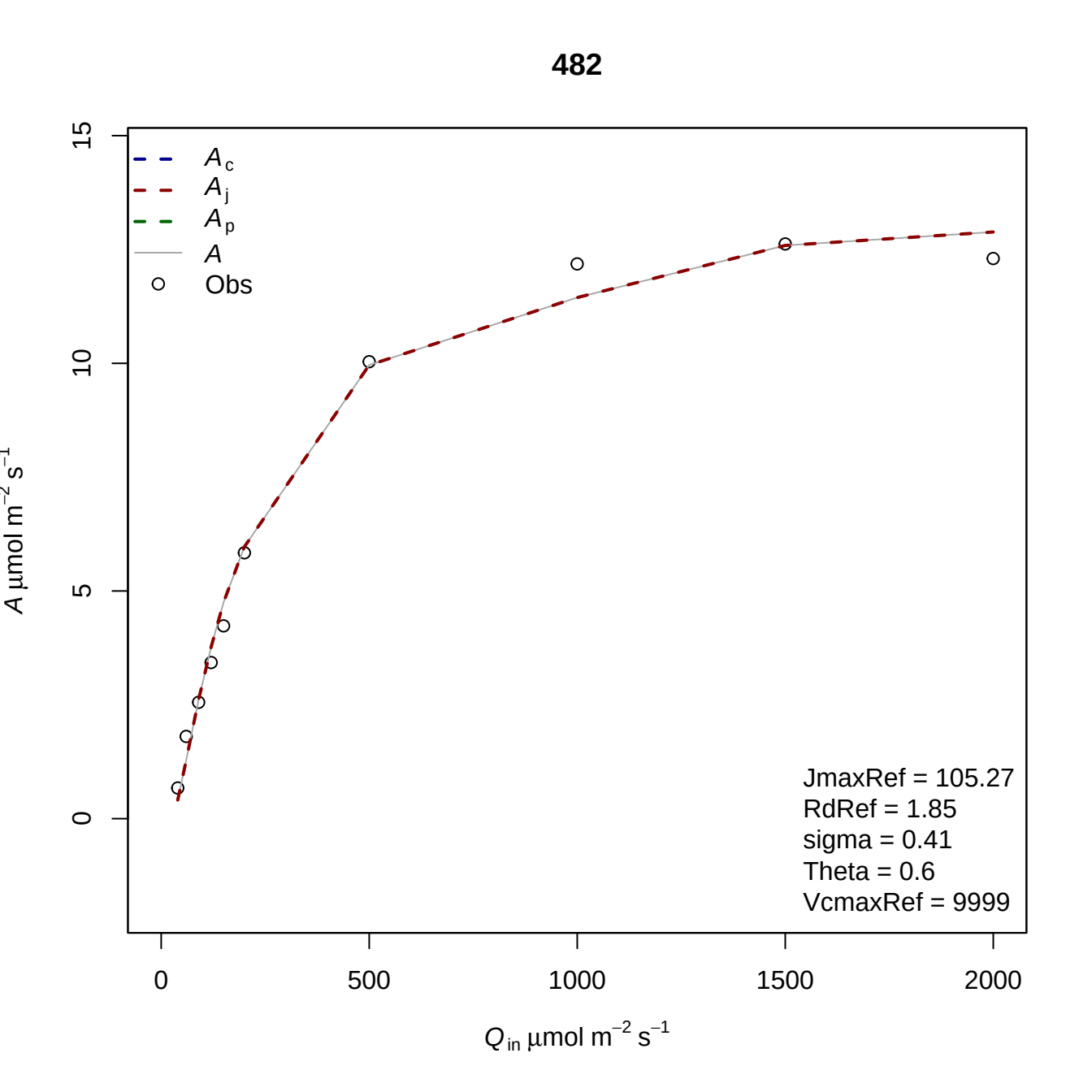

476
A_c
A_j
A_p
A
Obs
JmaxRef = 83.81
RdRef = 0.99
sigma = 0.14
Theta = −0.18
VcmaxRef = 41.56
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

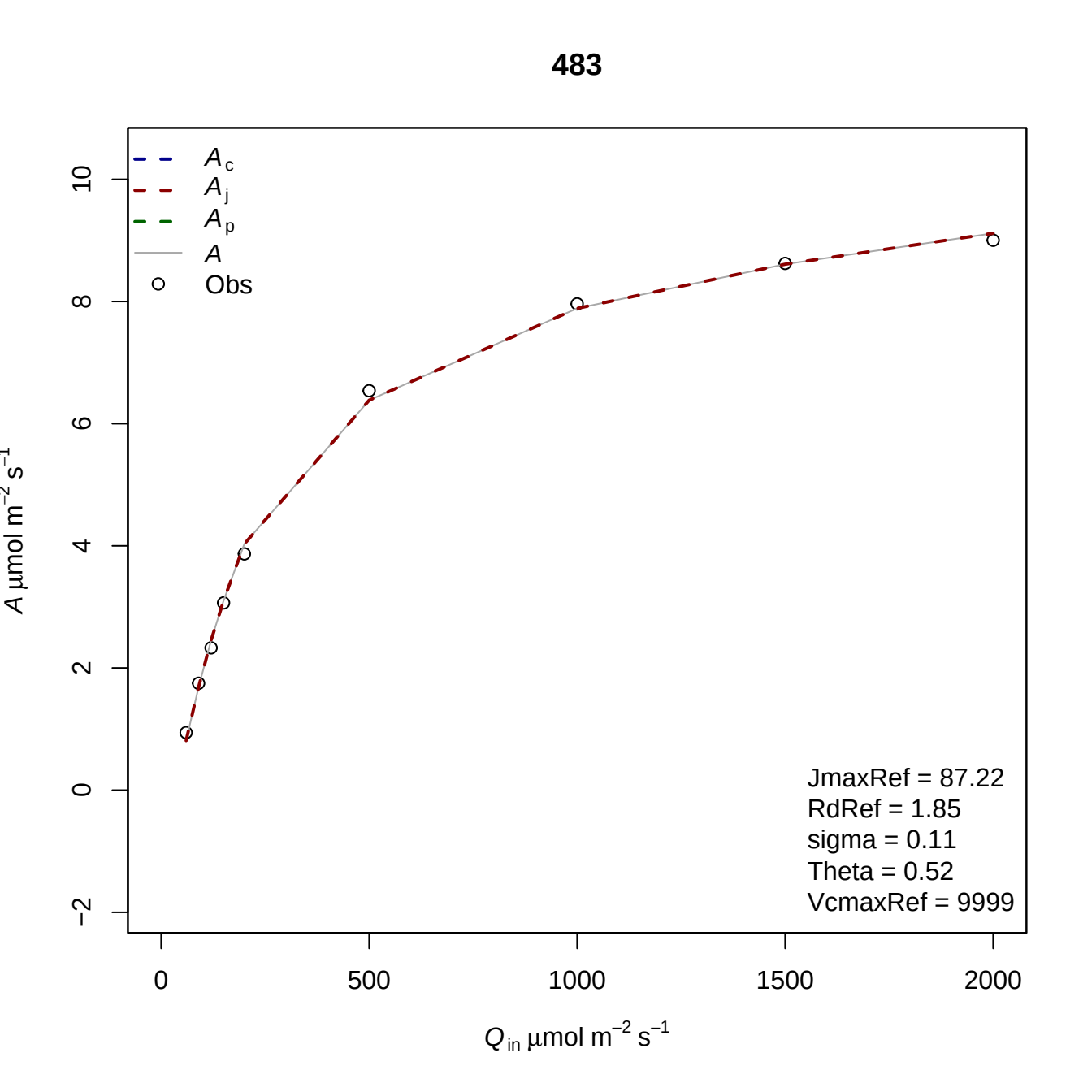

477
A_c
A_j
A_p
A
Obs
JmaxRef = 117.48
RdRef = 0.41
sigma = 0.18
Theta = 0.5
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

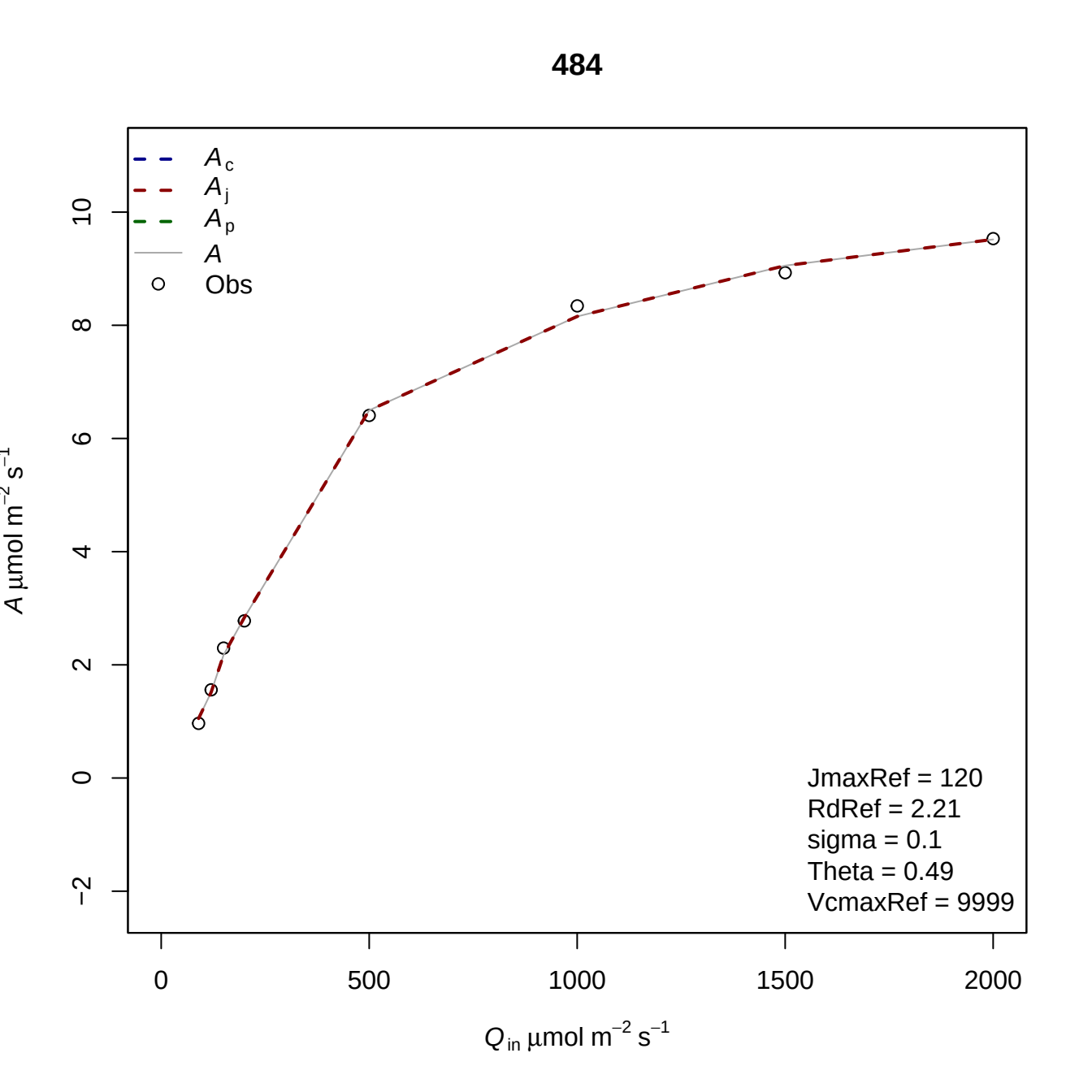

478
A_c
A_j
A_p
A
Obs
JmaxRef = 124.85
RdRef = 0.92
sigma = 0.14
Theta = 0.57
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

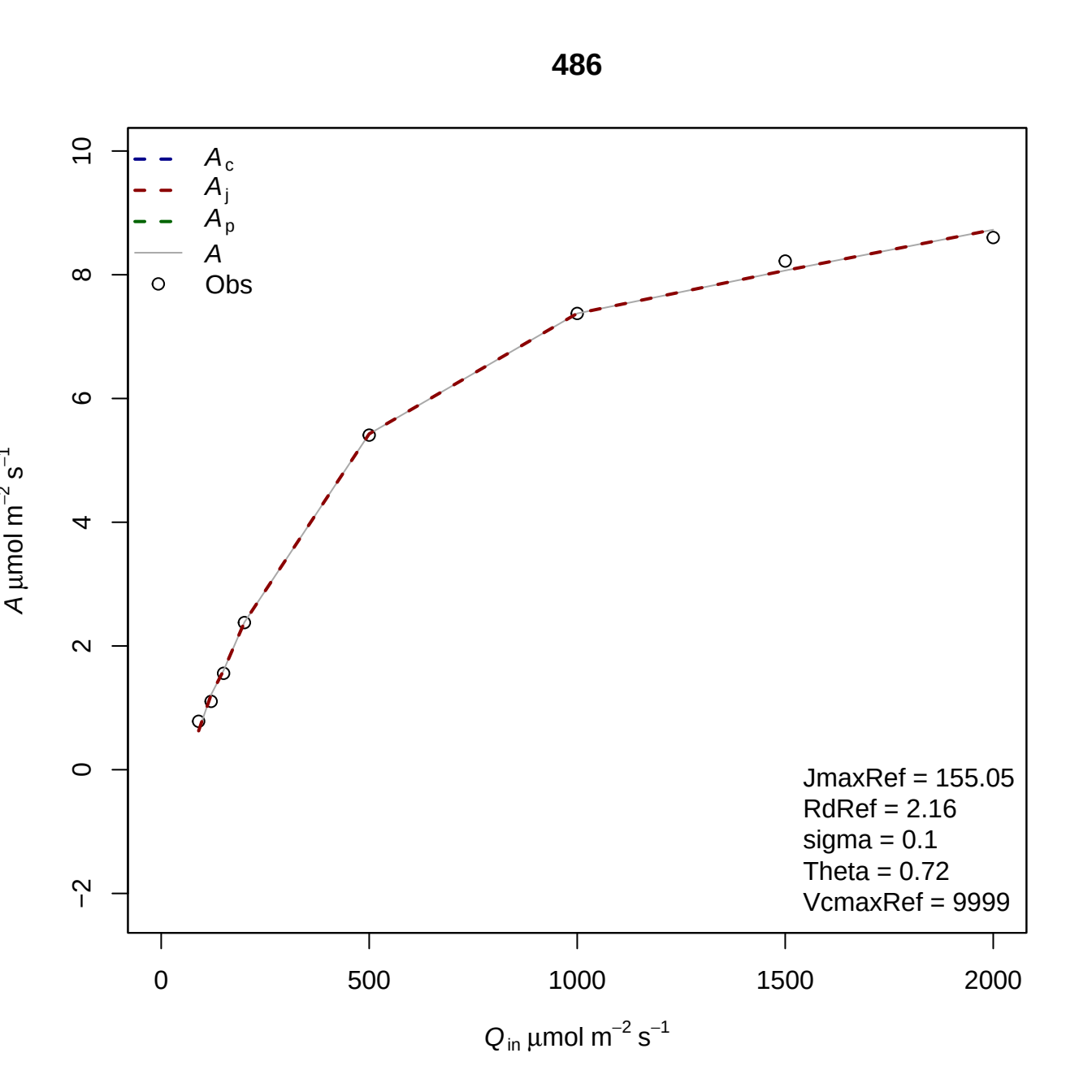

480
A_c
A_j
A_p
A
Obs
JmaxRef = 221.92
RdRef = 3.37
sigma = 0.03
Theta = −0.32
VcmaxRef = 275.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

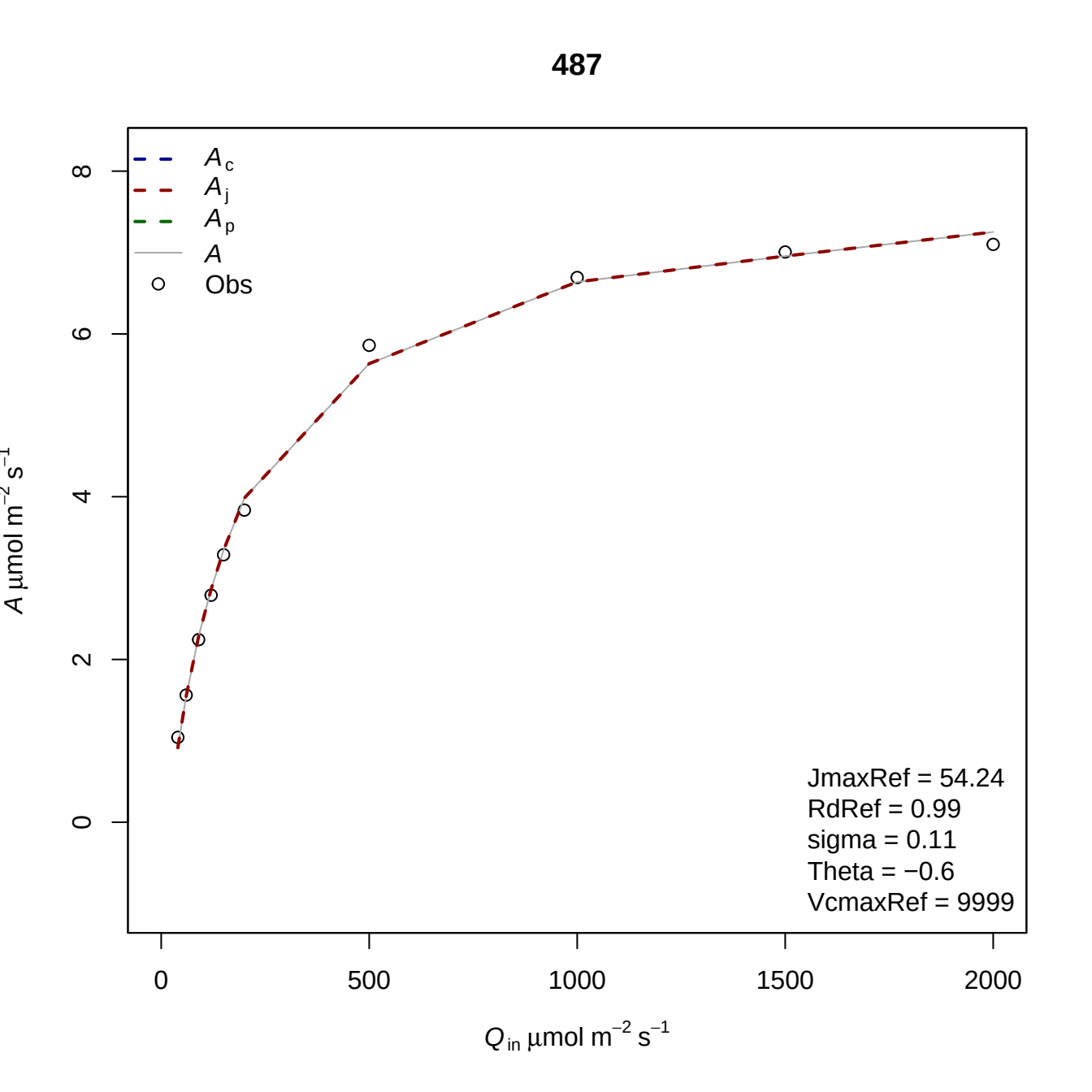

481
A_c
A_j
A_p
A
Obs
JmaxRef = 68.37
RdRef = 0.74
sigma = 0.16
Theta = 0.05
VcmaxRef = 30.14
0
2
4
6
8
10
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

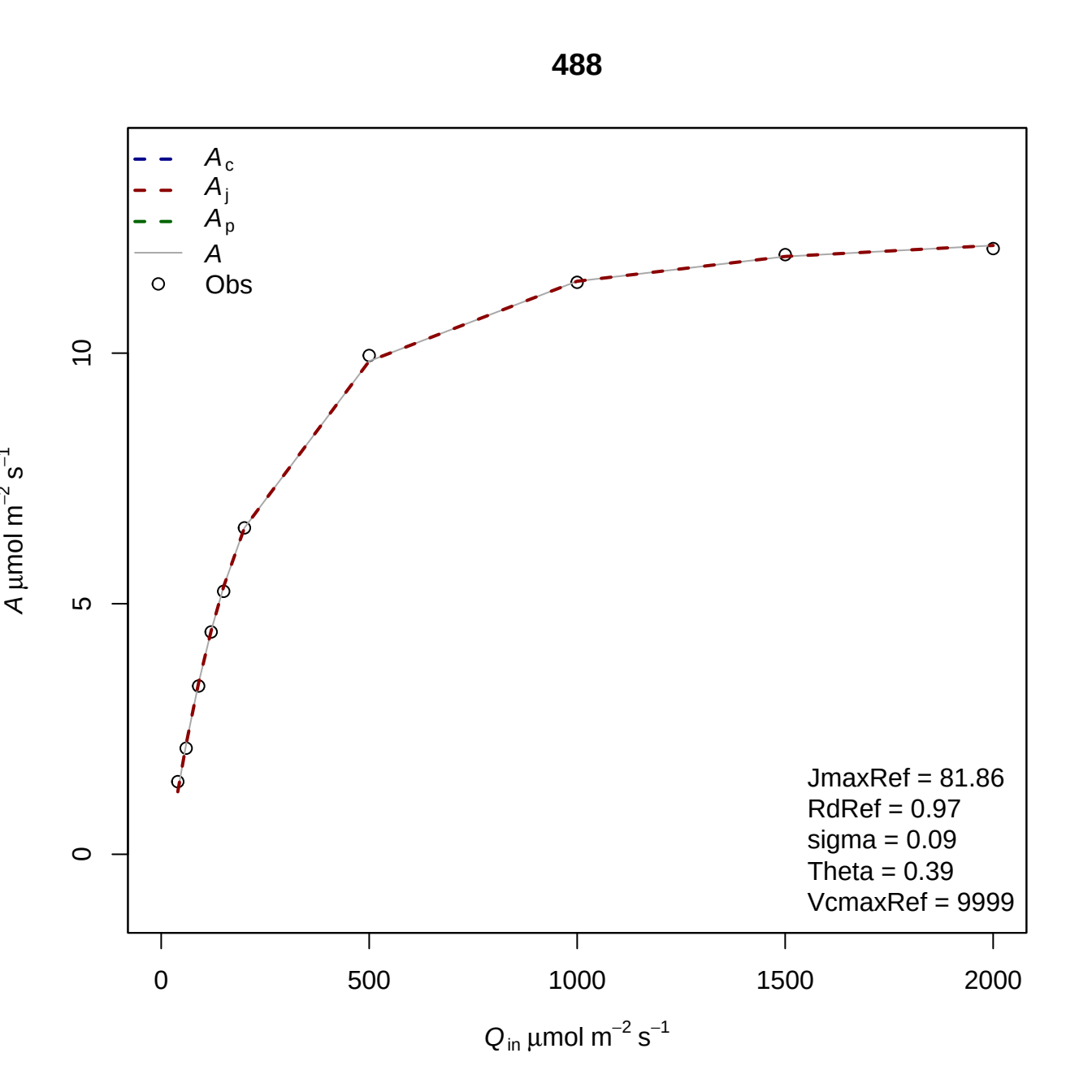

482
A_c
A_j
A_p
A
Obs
JmaxRef = 105.27
RdRef = 1.85
sigma = 0.41
Theta = 0.6
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

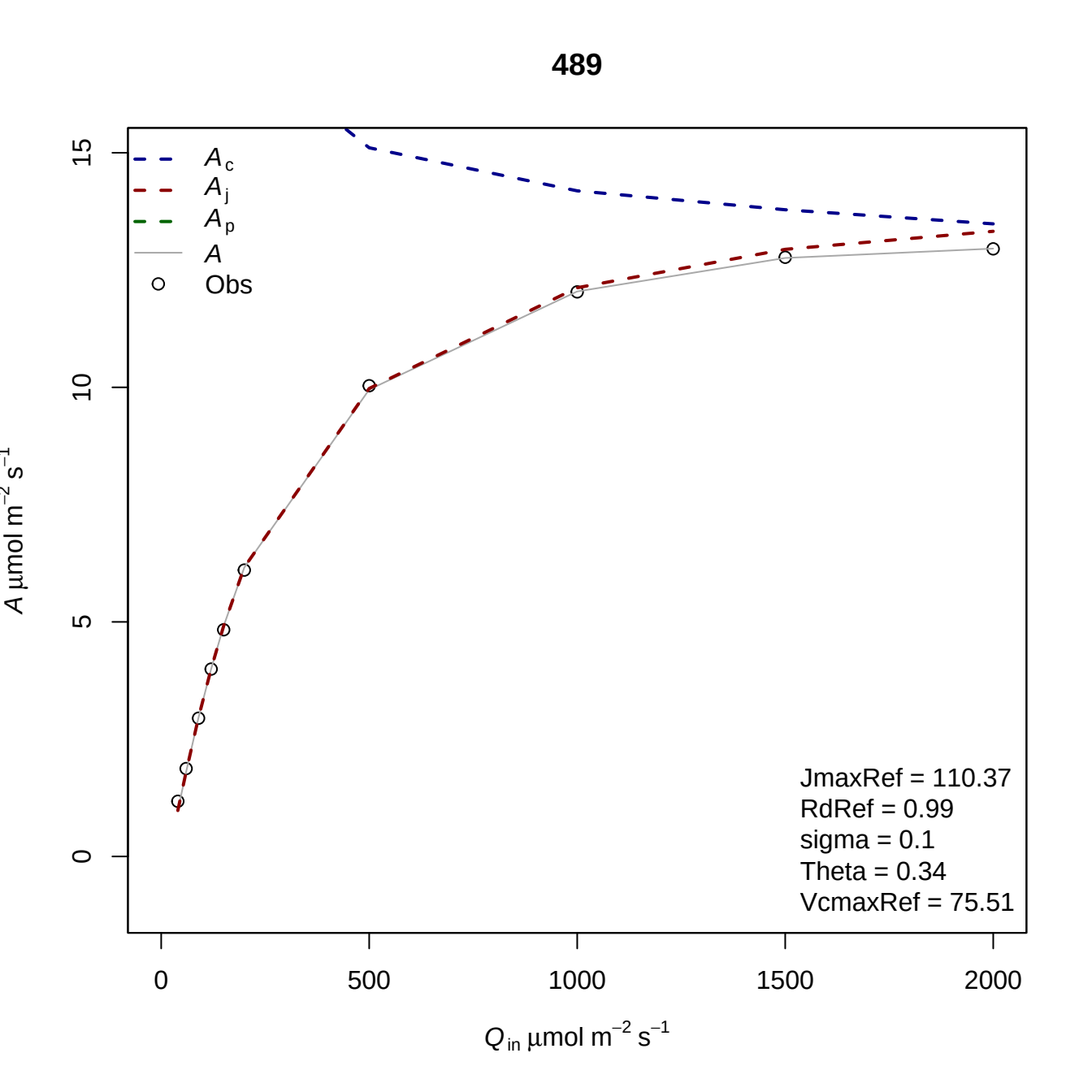

483
A_c
A_j
A_p
A
Obs
JmaxRef = 87.22
RdRef = 1.85
sigma = 0.11
Theta = 0.52
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

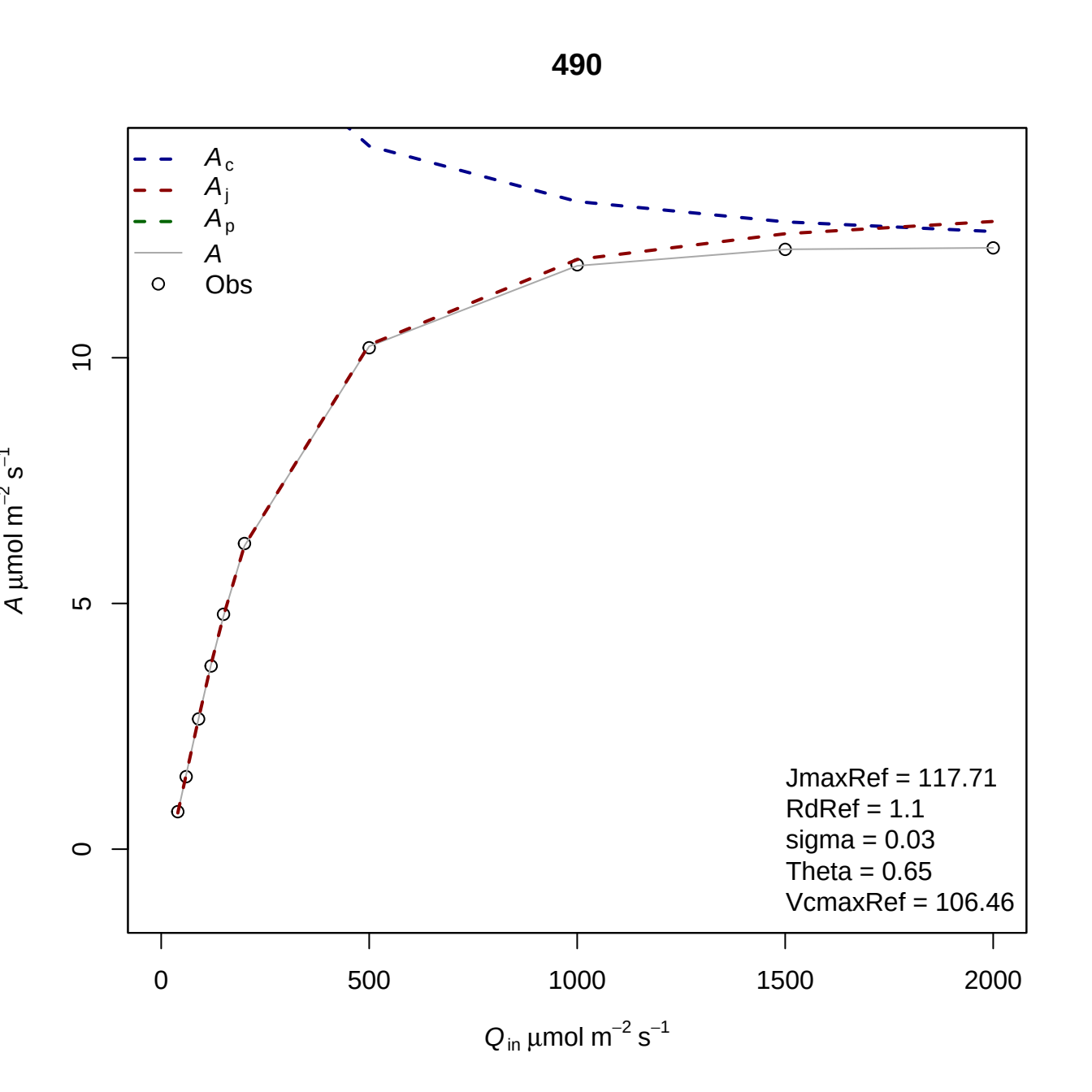

484
A_c
A_j
A_p
A
Obs
JmaxRef = 120
RdRef = 2.21
sigma = 0.1
Theta = 0.49
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

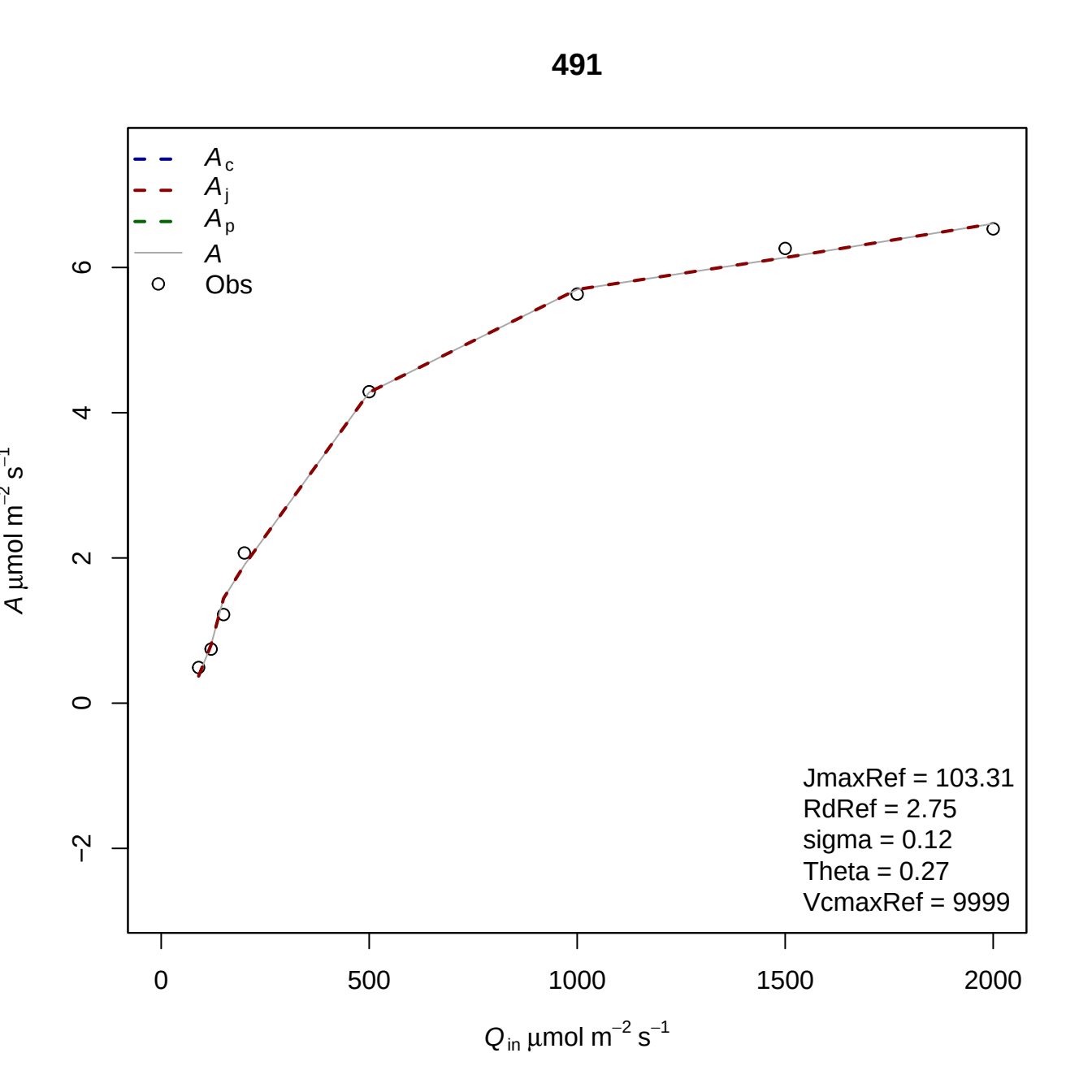

486
A_c
A_j
A_p
A
Obs
JmaxRef = 155.05
RdRef = 2.16
sigma = 0.1
Theta = 0.72
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

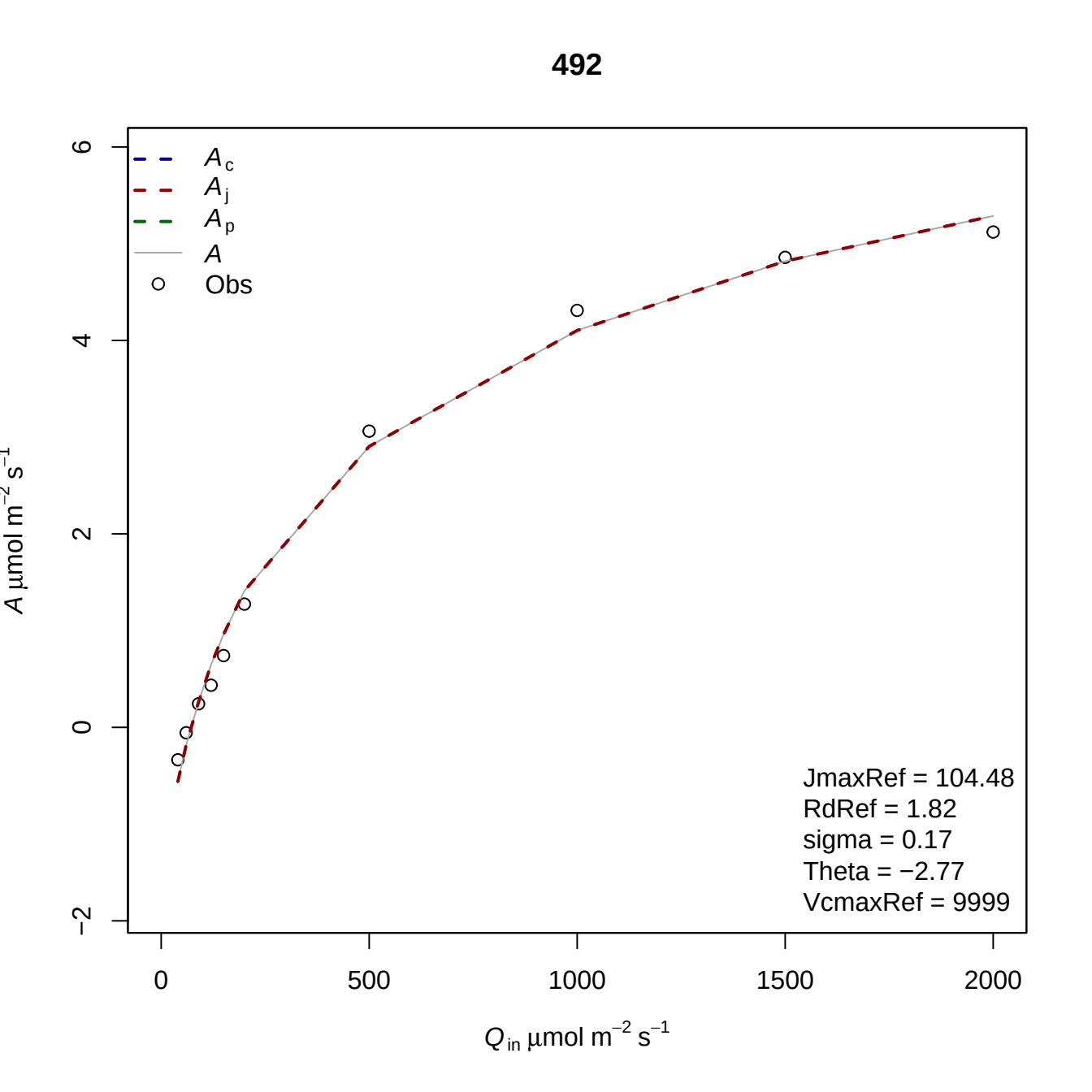

487
A_c
A_j
A_p
A
Obs
JmaxRef = 54.24
RdRef = 0.99
sigma = 0.11
Theta = −0.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

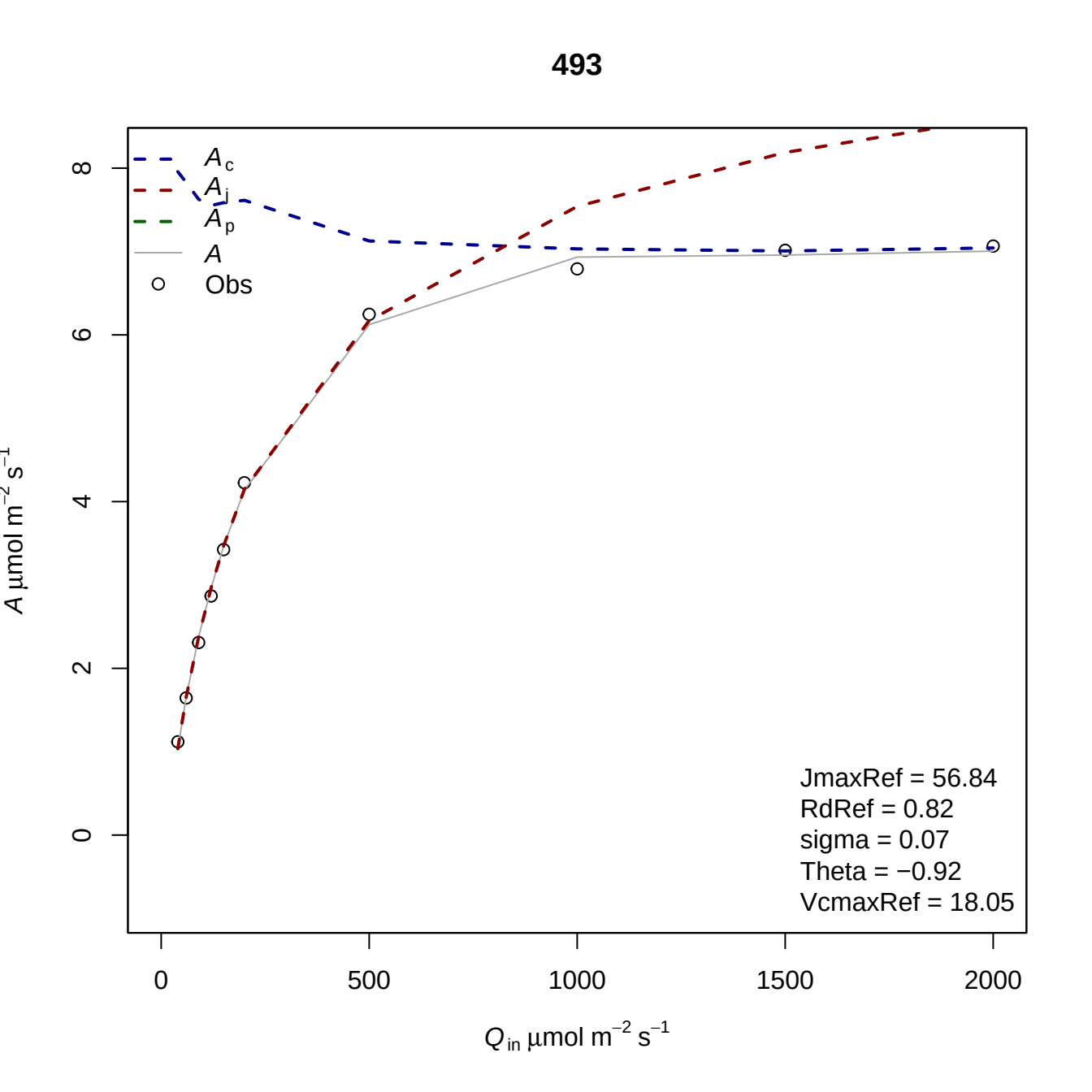

488
A_c
A_j
A_p
A
Obs
JmaxRef = 81.86
RdRef = 0.97
sigma = 0.09
Theta = 0.39
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

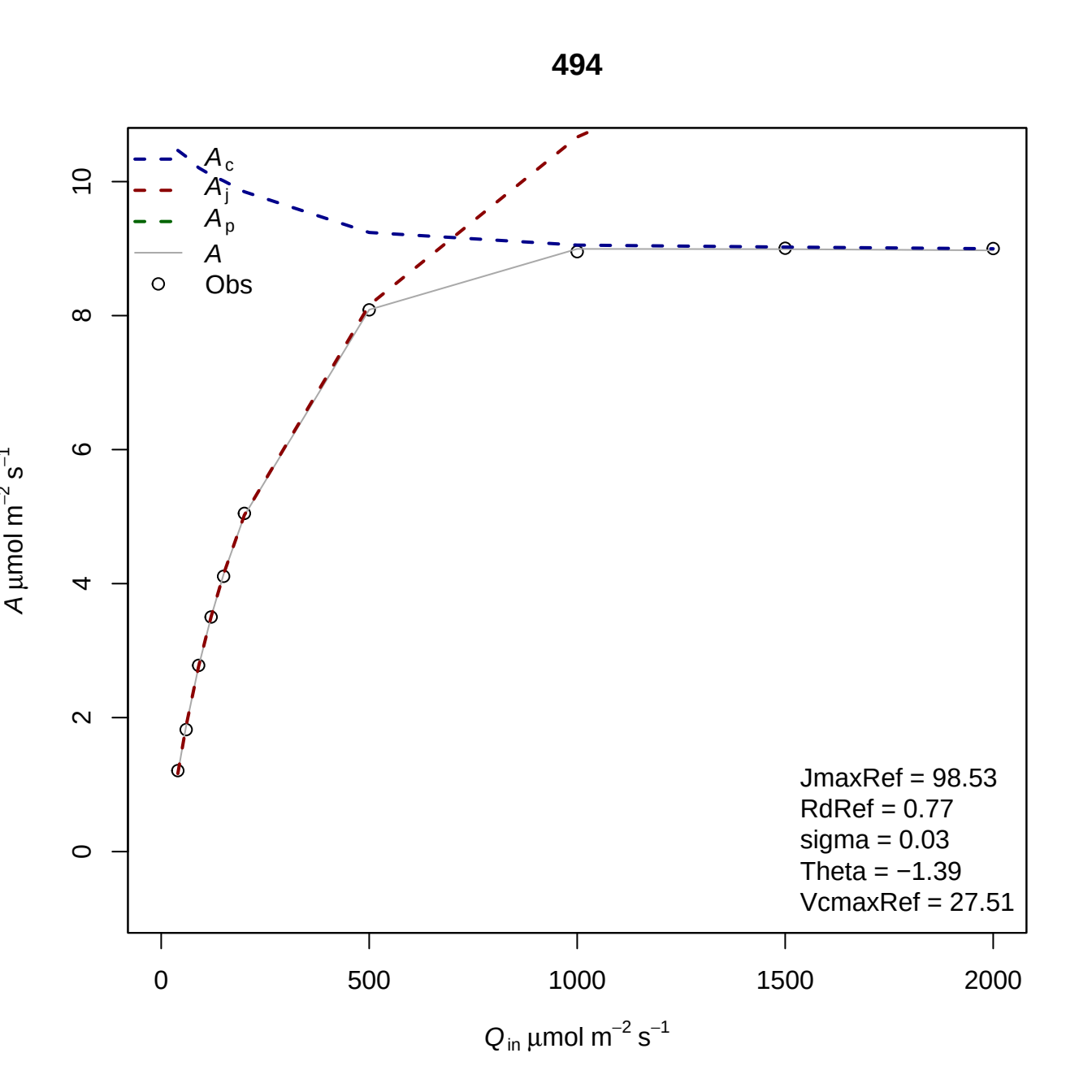

489
A_c
A_j
A_p
A
Obs
JmaxRef = 110.37
RdRef = 0.99
sigma = 0.1
Theta = 0.34
VcmaxRef = 75.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

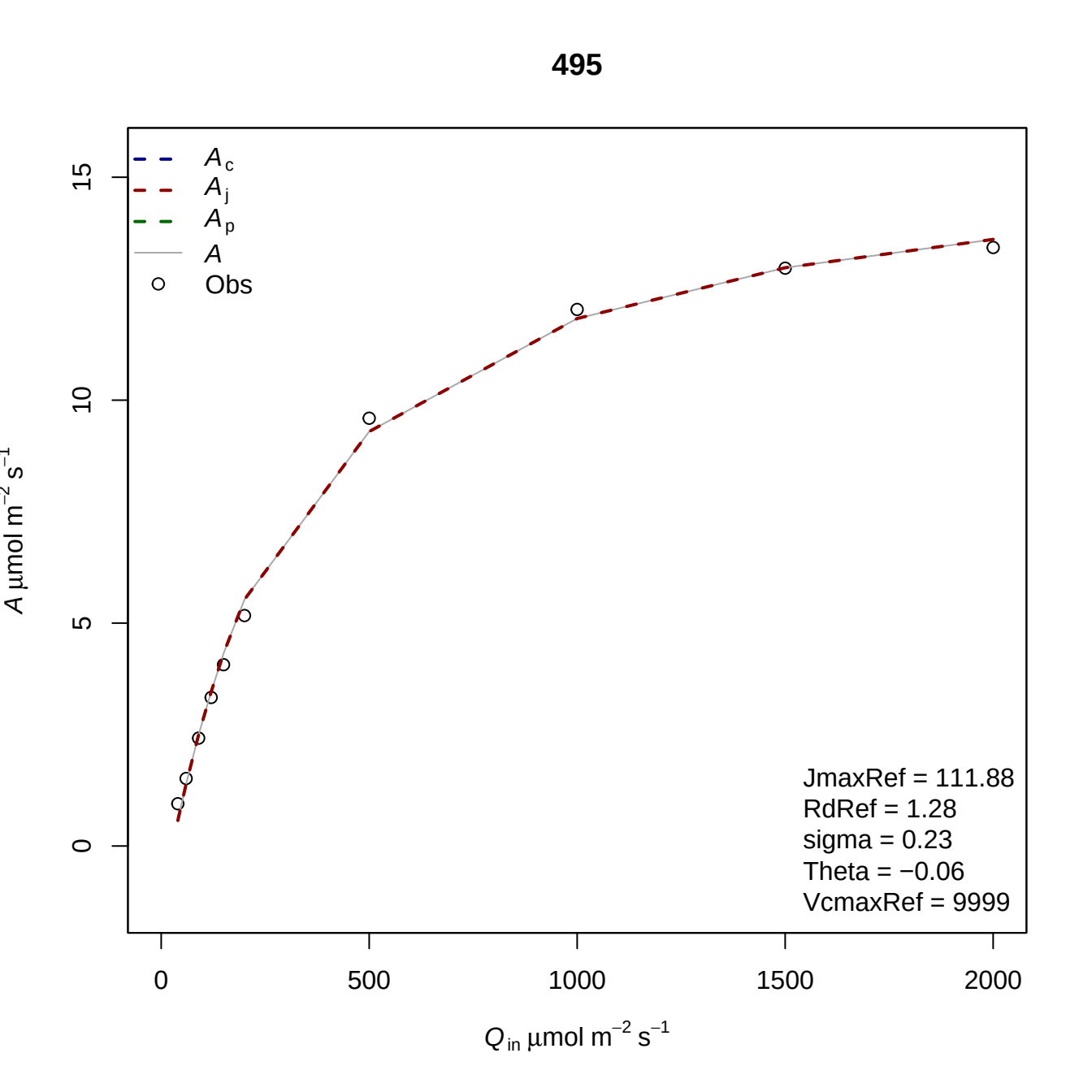

490
A_c
A_j
A_p
A
Obs
JmaxRef = 117.71
RdRef = 1.1
sigma = 0.03
Theta = 0.65
VcmaxRef = 106.46
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

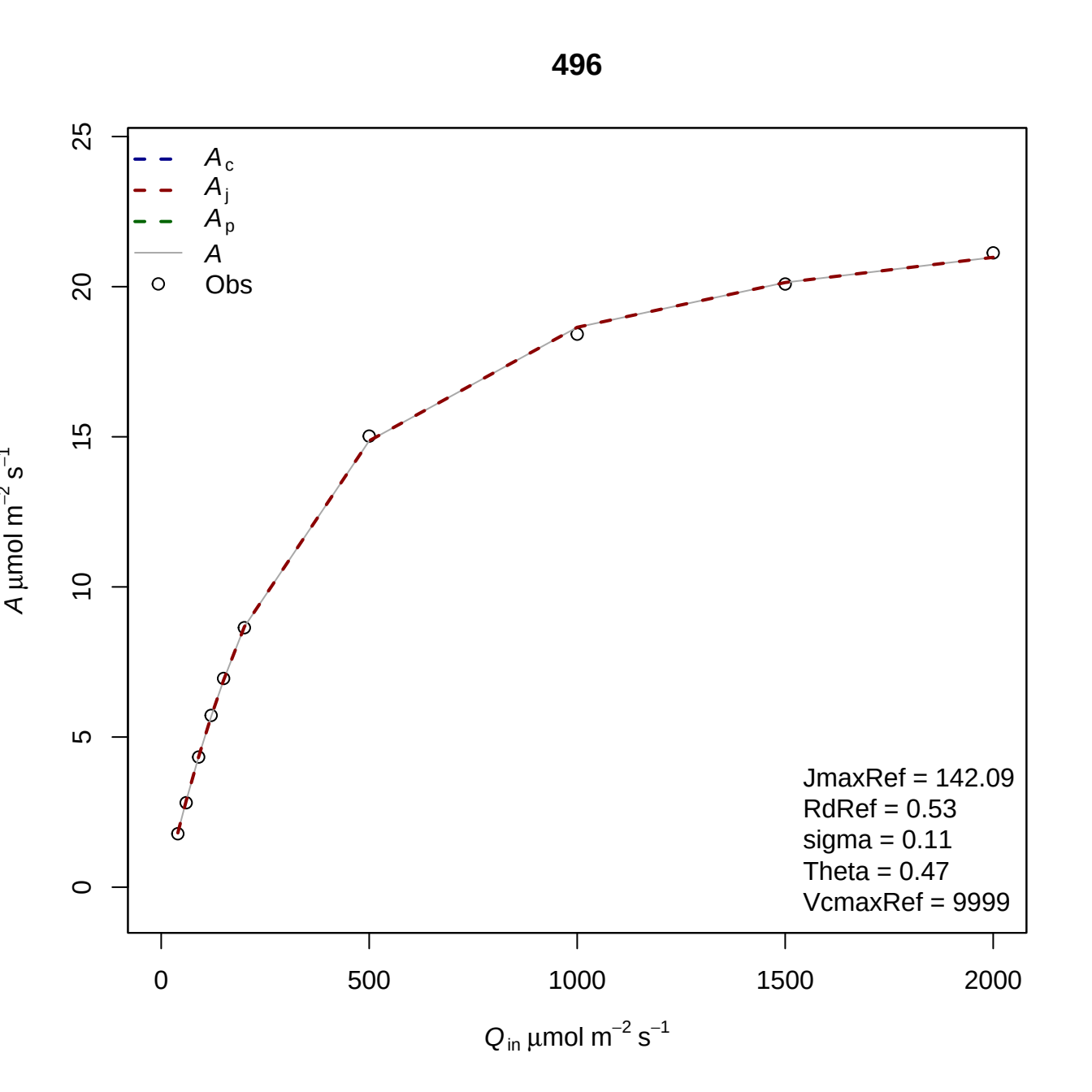

491
A_c
A_j
A_p
A
Obs
JmaxRef = 103.31
RdRef = 2.75
sigma = 0.12
Theta = 0.27
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

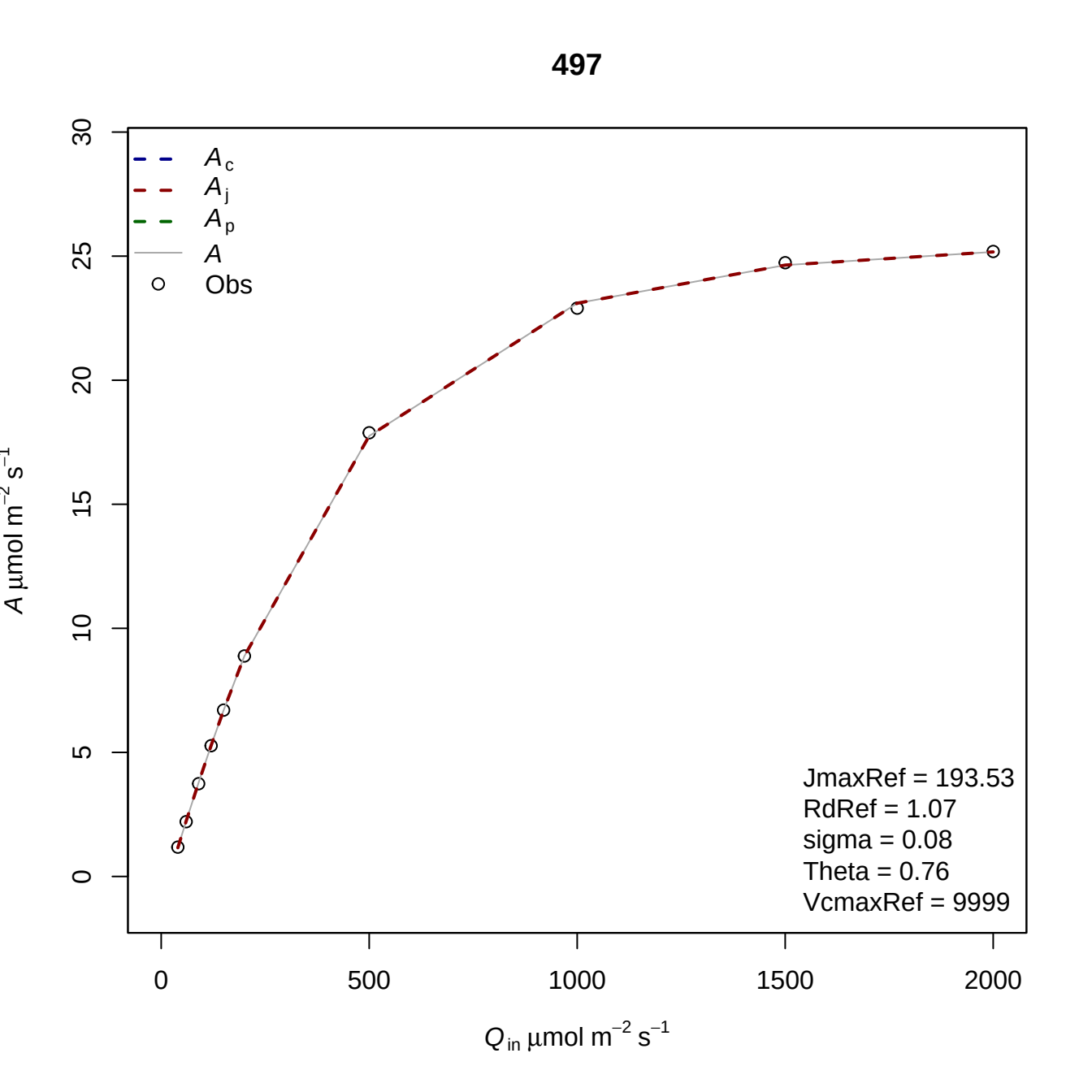

492
A_c
A_j
A_p
A
Obs
JmaxRef = 104.48
RdRef = 1.82
sigma = 0.17
Theta = −2.77
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

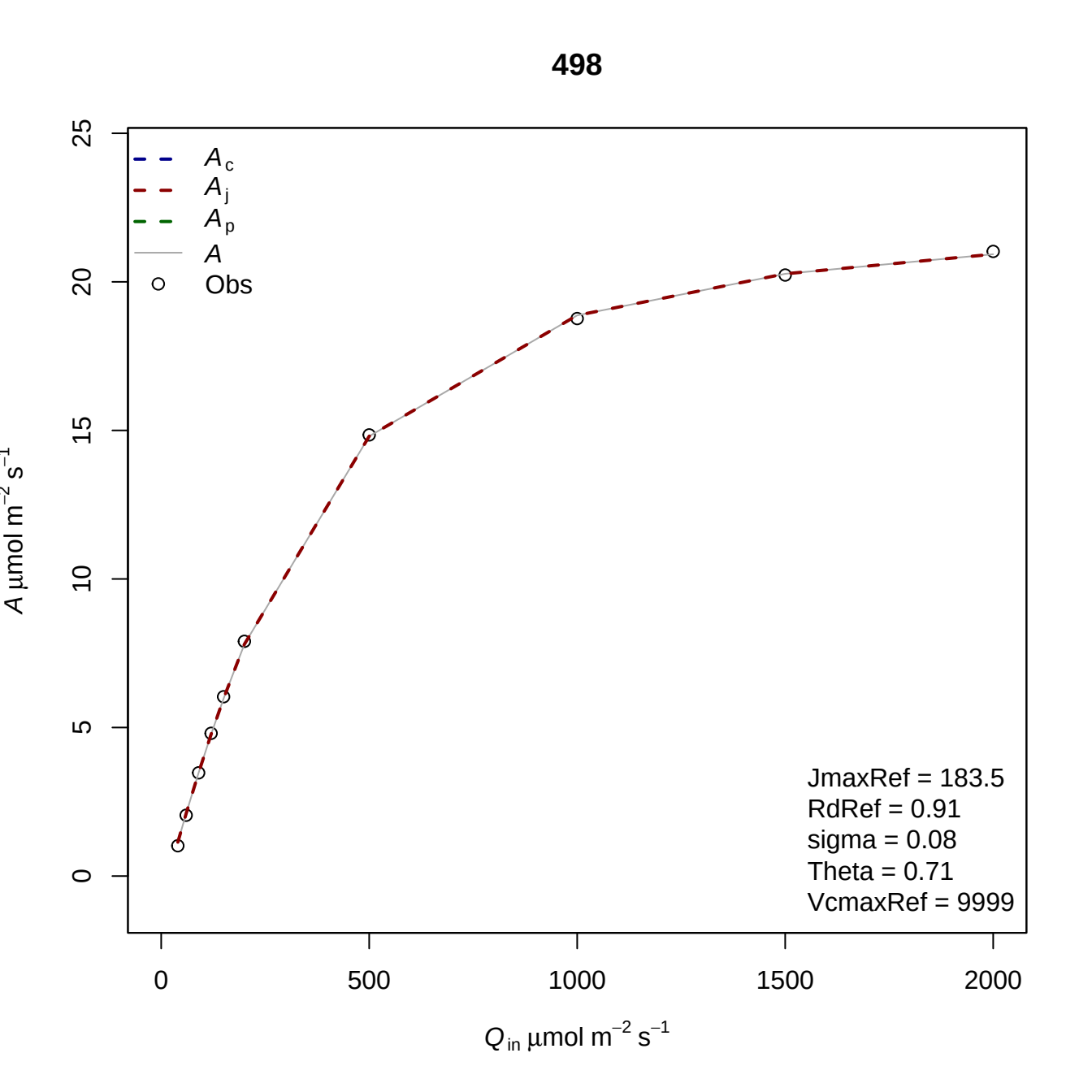

493
A_c
A_j
A_p
A
Obs
JmaxRef = 56.84
RdRef = 0.82
sigma = 0.07
Theta = −0.92
VcmaxRef = 18.05
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

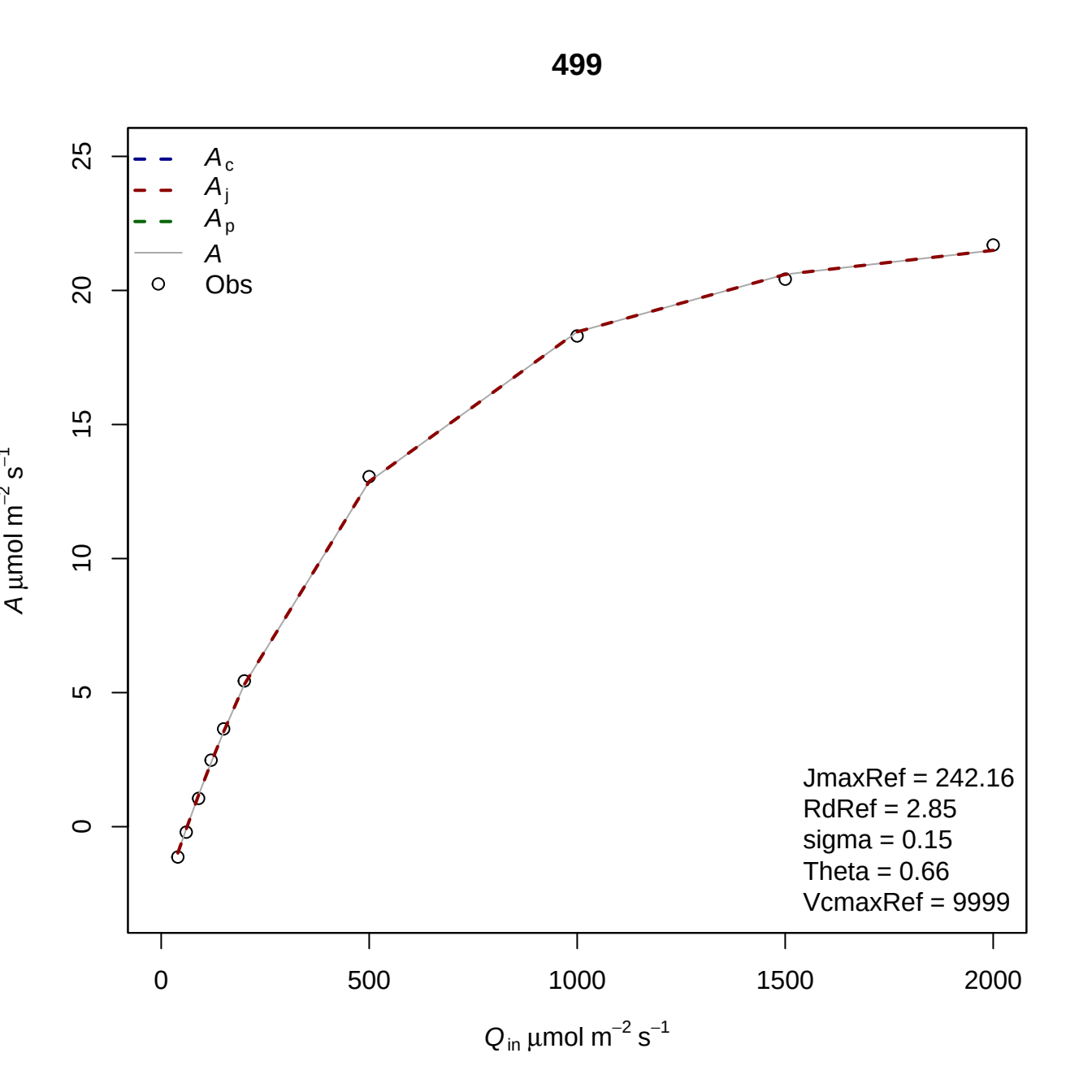

494
A_c
A_j
A_p
A
Obs
JmaxRef = 98.53
RdRef = 0.77
sigma = 0.03
Theta = −1.39
VcmaxRef = 27.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

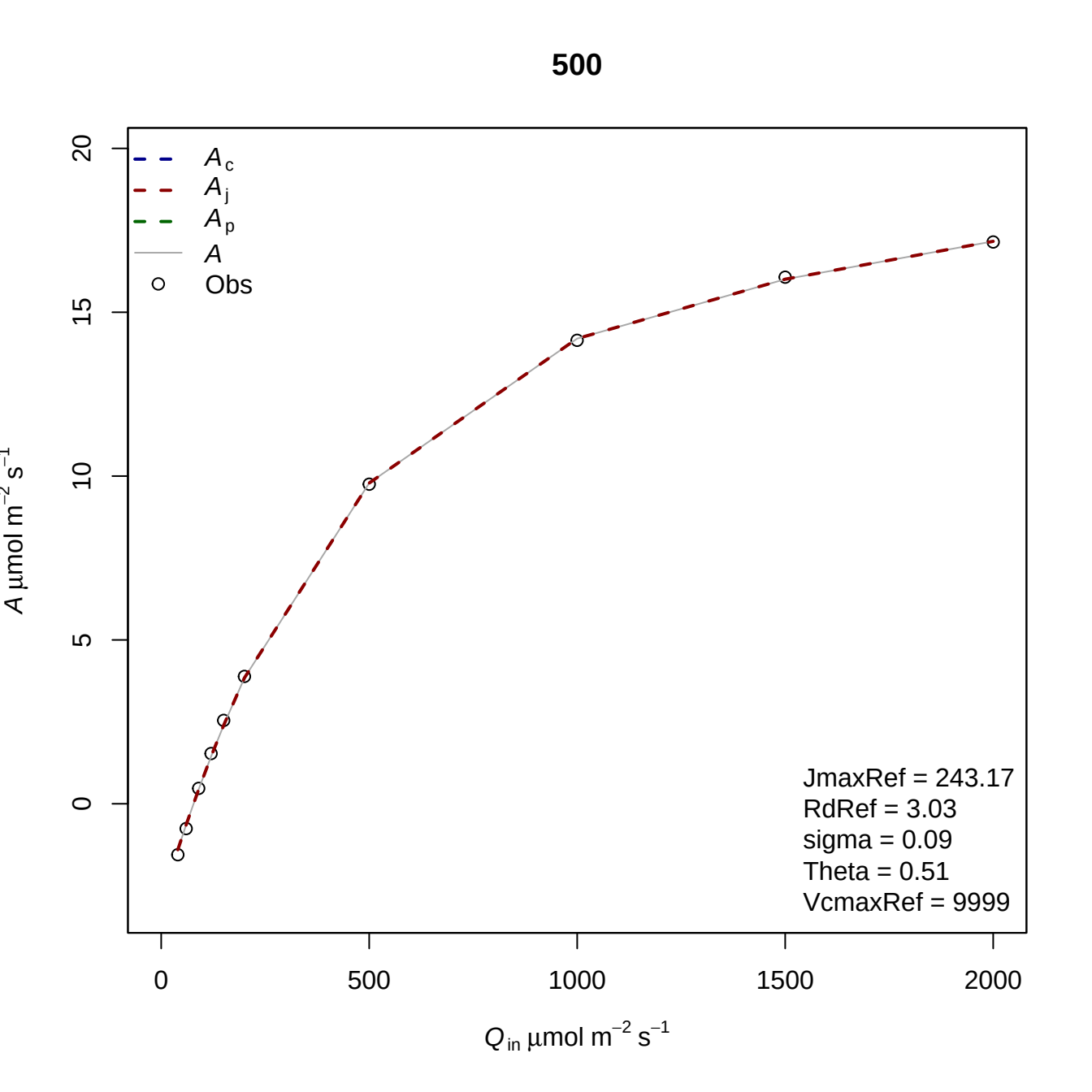

495
A_c
A_j
A_p
A
Obs
JmaxRef = 111.88
RdRef = 1.28
sigma = 0.23
Theta = −0.06
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

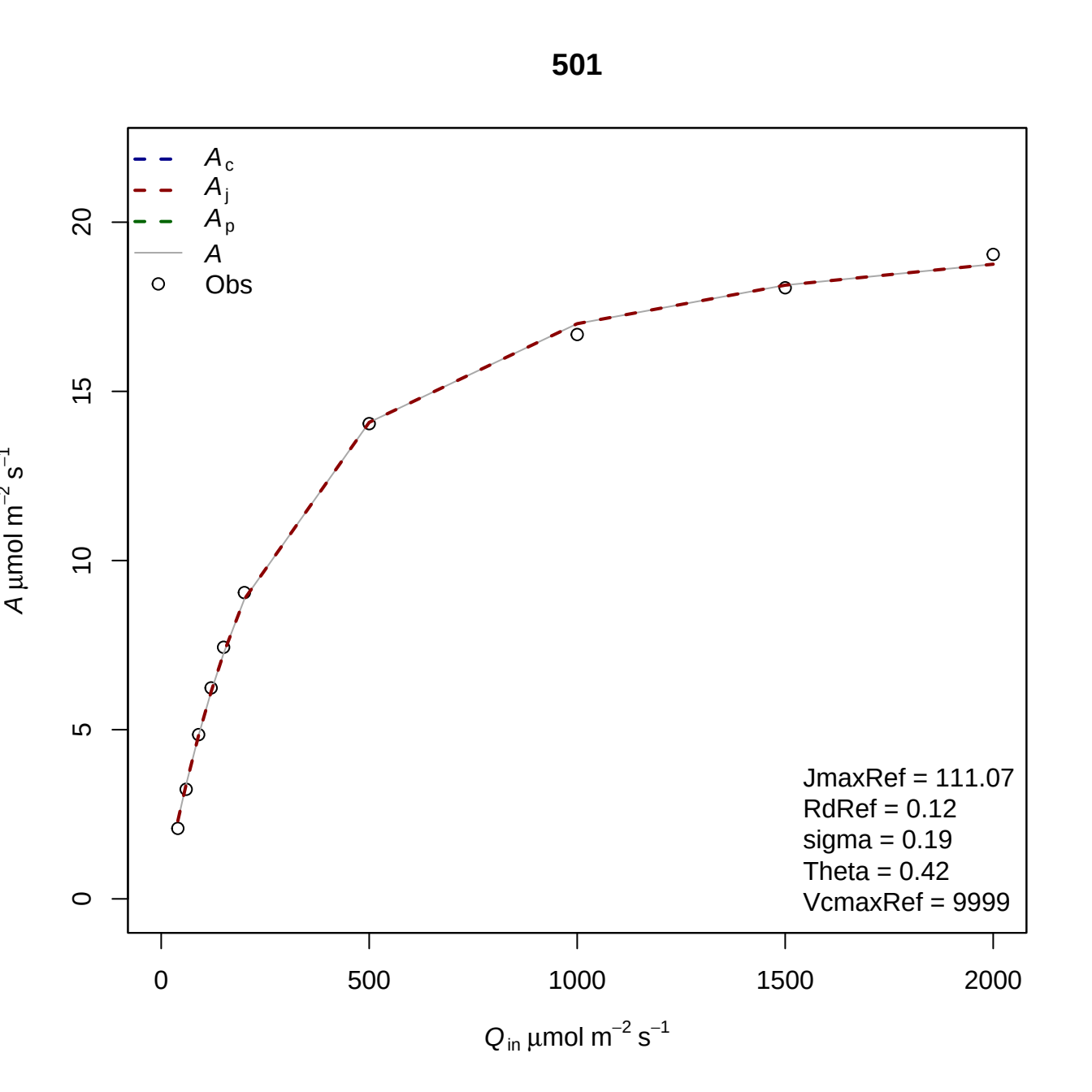

496
A_c
A_j
A_p
A
Obs
JmaxRef = 142.09
RdRef = 0.53
sigma = 0.11
Theta = 0.47
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

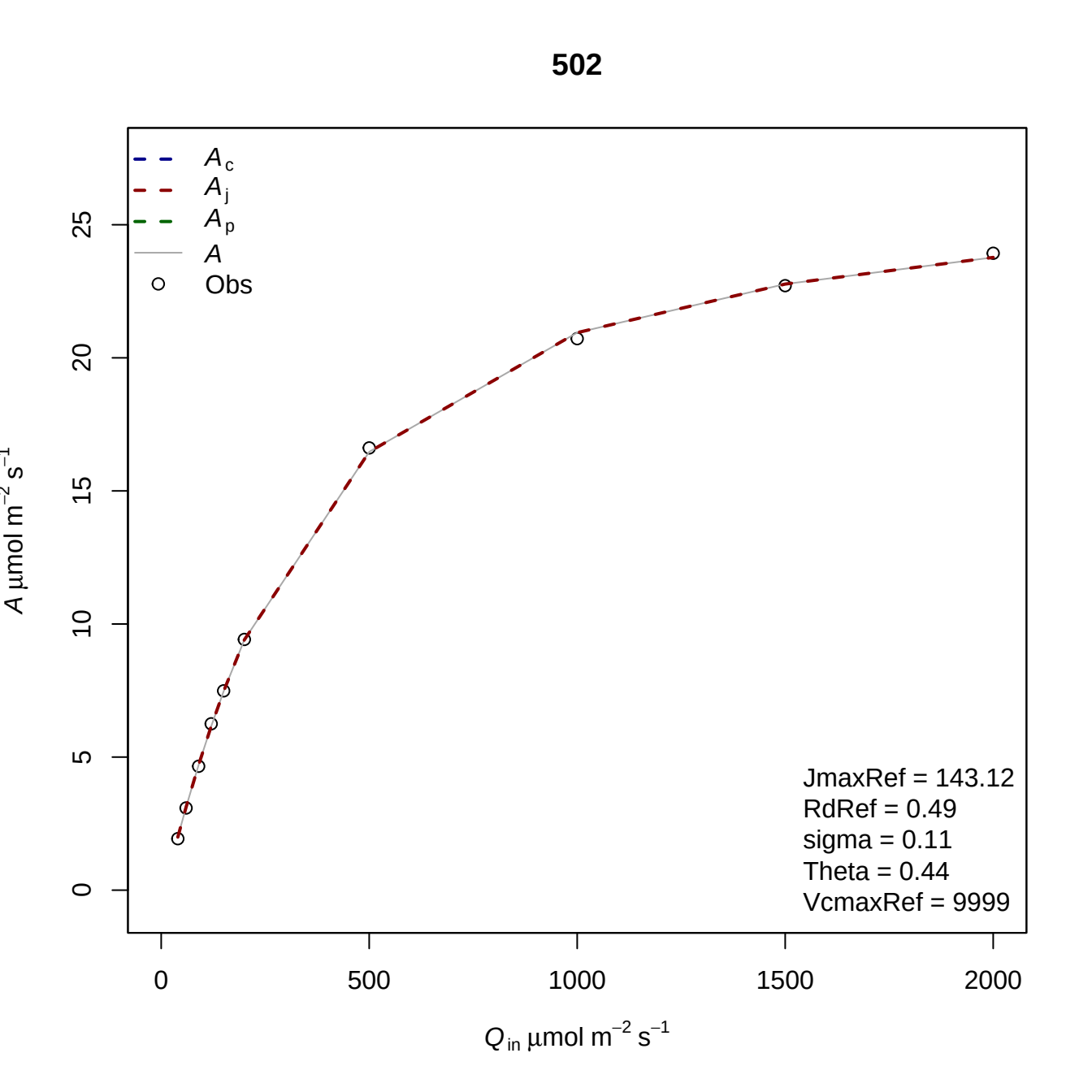

497
A_c
A_j
A_p
A
Obs
JmaxRef = 193.53
RdRef = 1.07
sigma = 0.08
Theta = 0.76
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

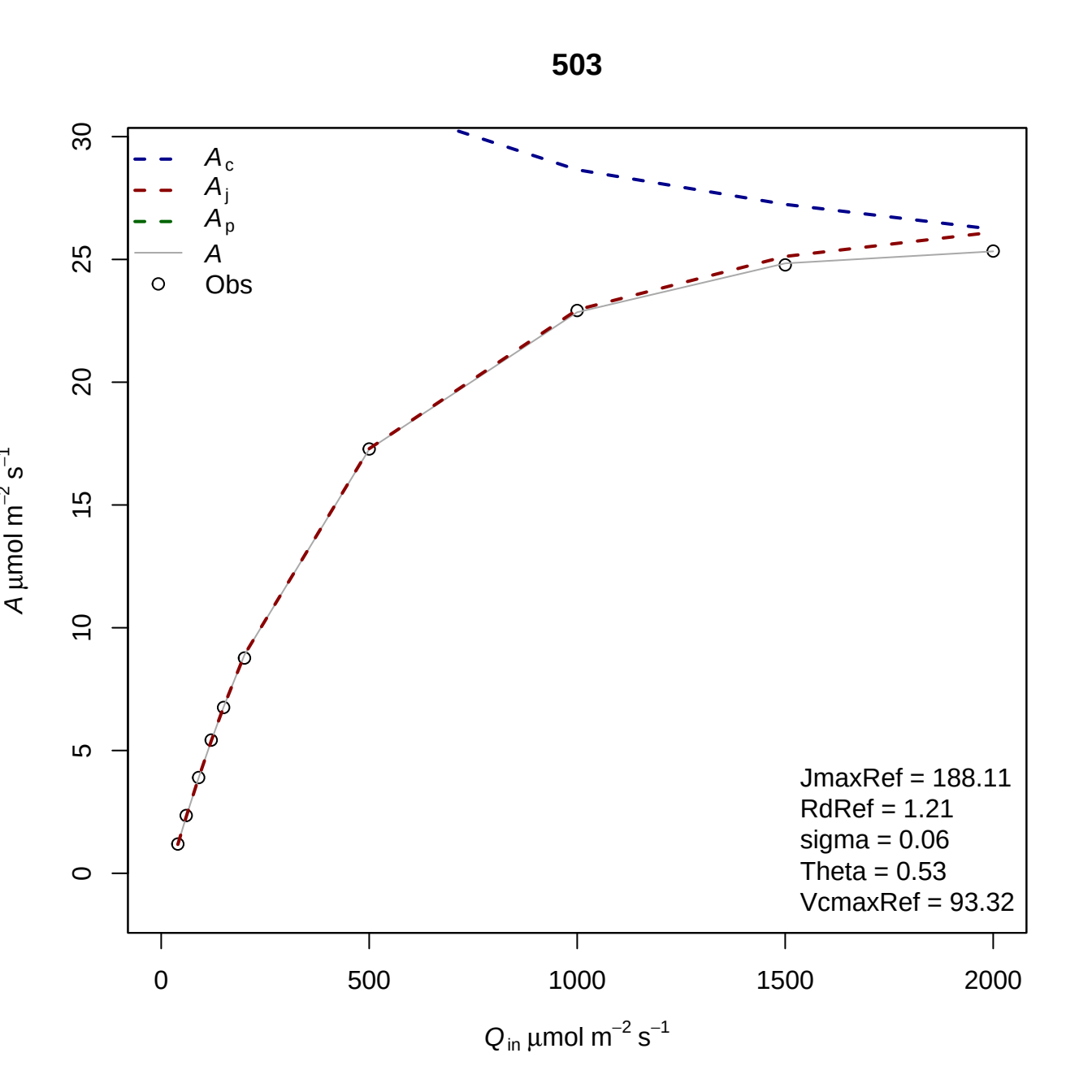

498
A_c
A_j
A_p
A
Obs
JmaxRef = 183.5
RdRef = 0.91
sigma = 0.08
Theta = 0.71
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

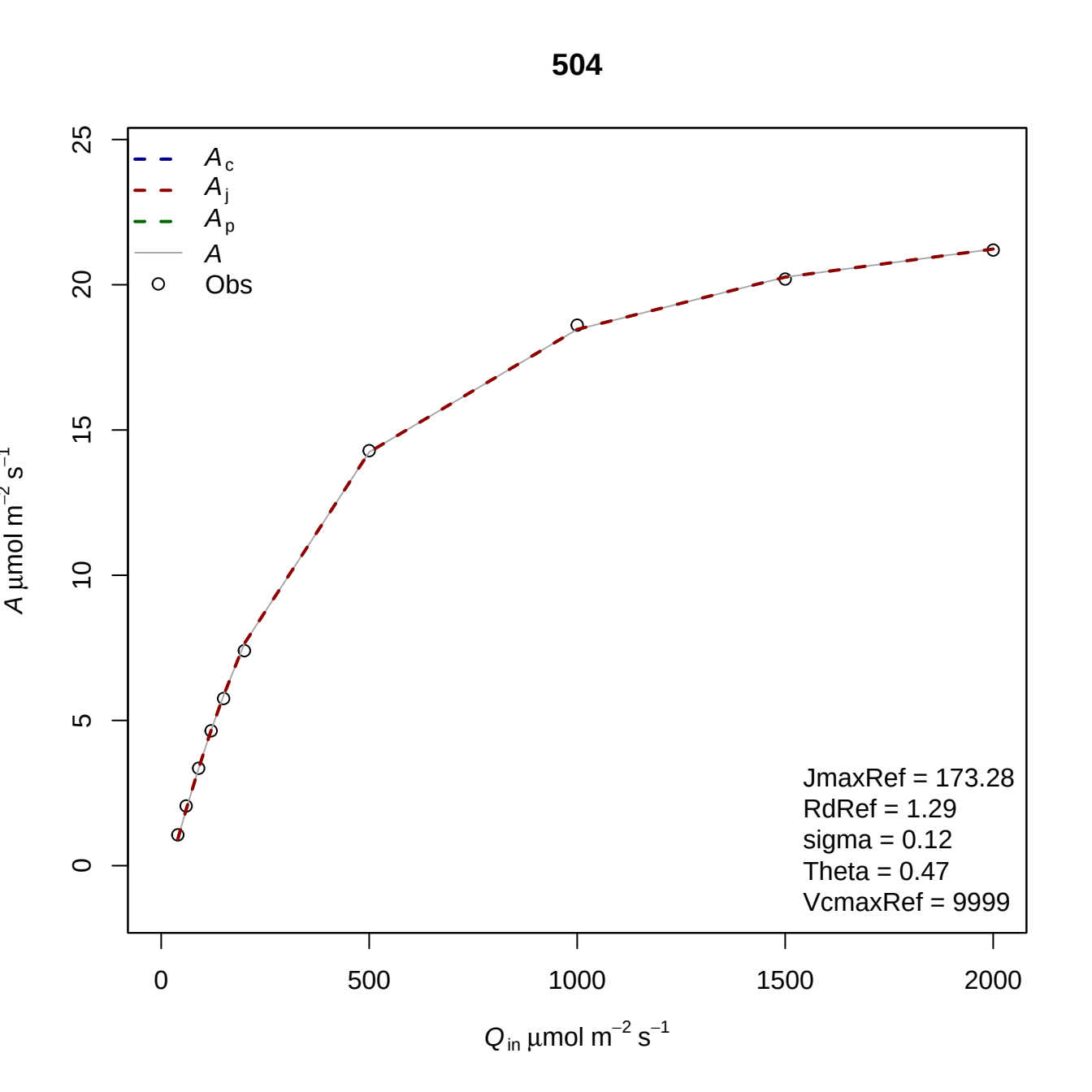

499
A_c
A_j
A_p
A
Obs
JmaxRef = 242.16
RdRef = 2.85
sigma = 0.15
Theta = 0.66
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

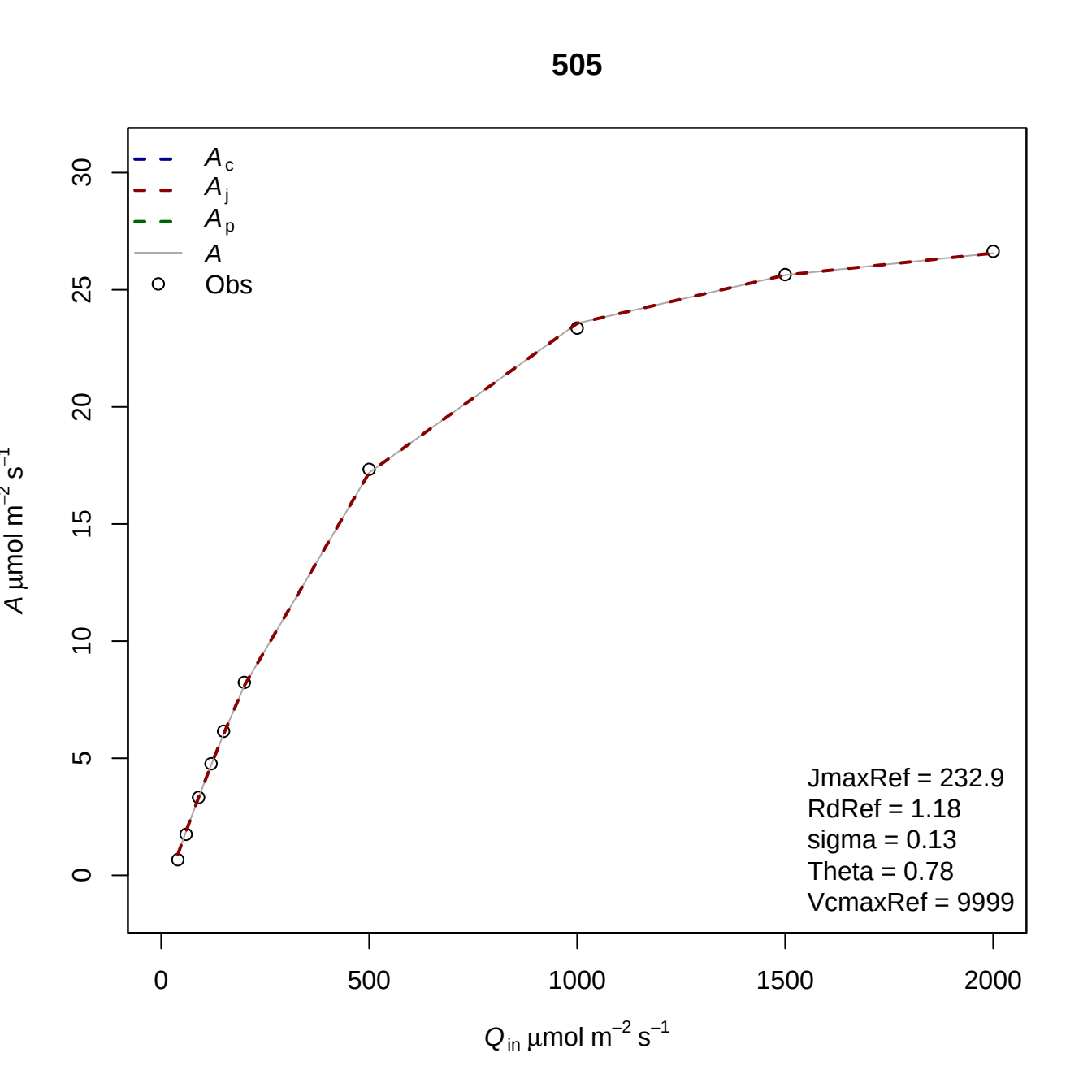

500
A_c
A_j
A_p
A
Obs
JmaxRef = 243.17
RdRef = 3.03
sigma = 0.09
Theta = 0.51
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

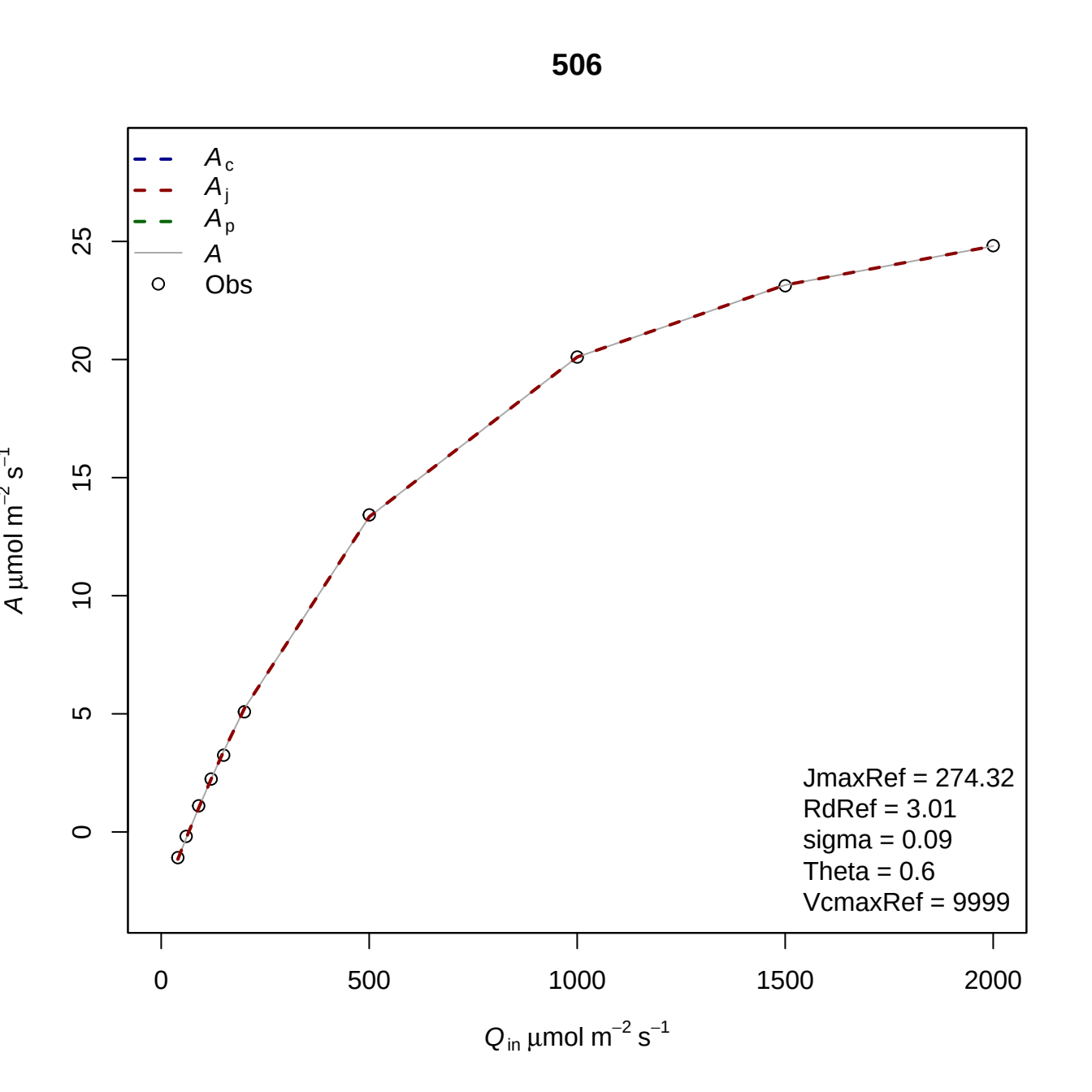

501
A_c
A_j
A_p
A
Obs
JmaxRef = 111.07
RdRef = 0.12
sigma = 0.19
Theta = 0.42
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

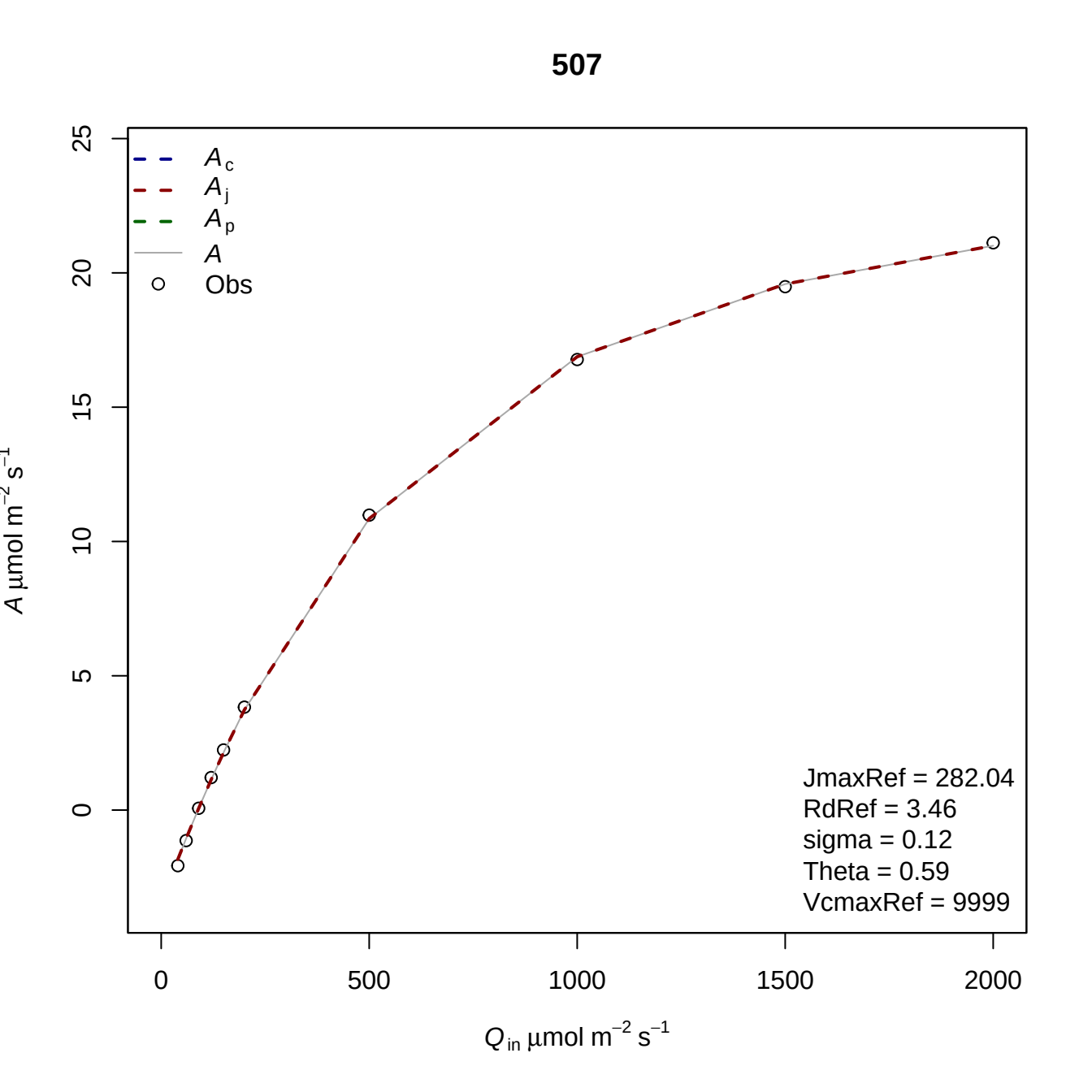

502
A_c
A_j
A_p
A
Obs
JmaxRef = 143.12
RdRef = 0.49
sigma = 0.11
Theta = 0.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

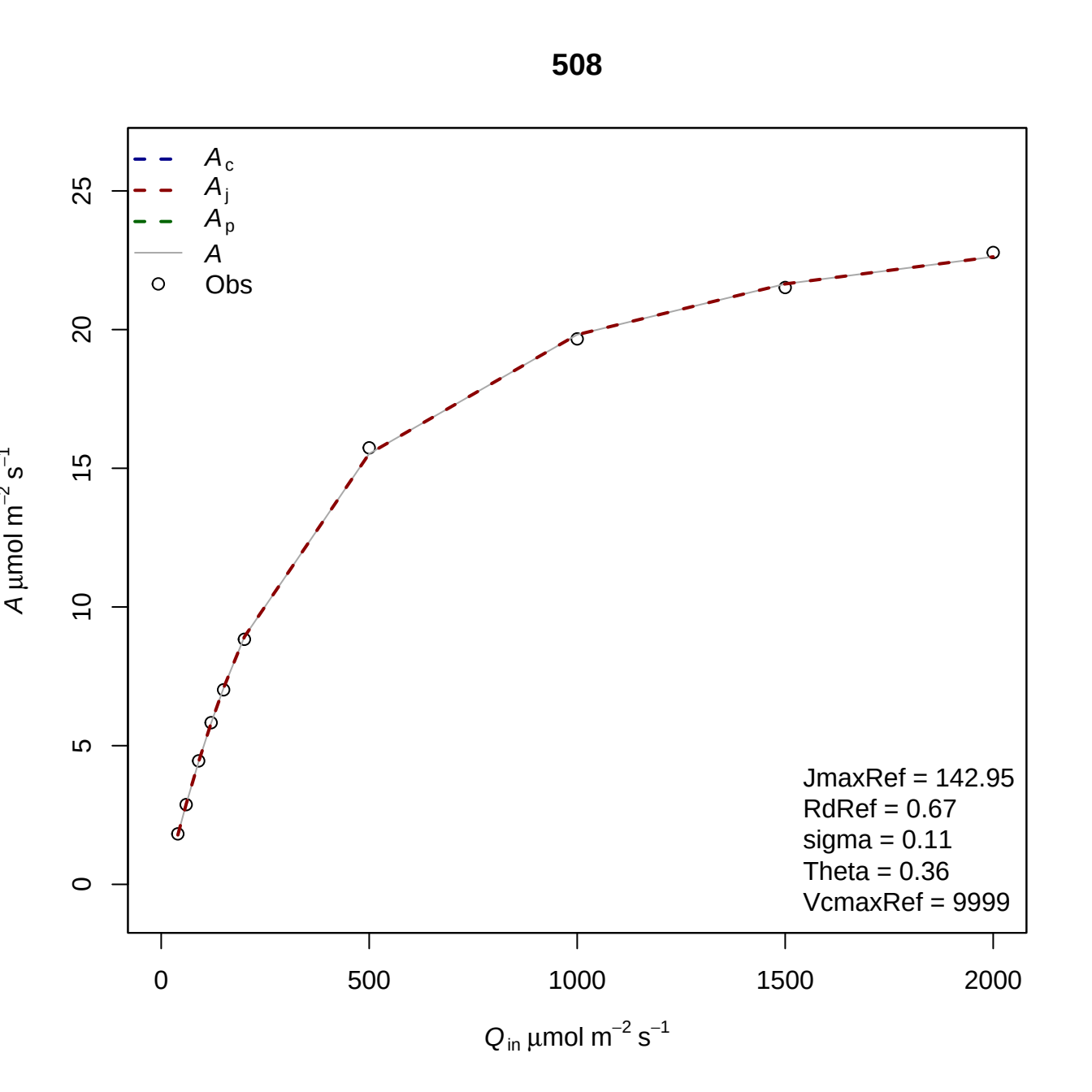

503
A_c
A_j
A_p
A
Obs
JmaxRef = 188.11
RdRef = 1.21
sigma = 0.06
Theta = 0.53
VcmaxRef = 93.32
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

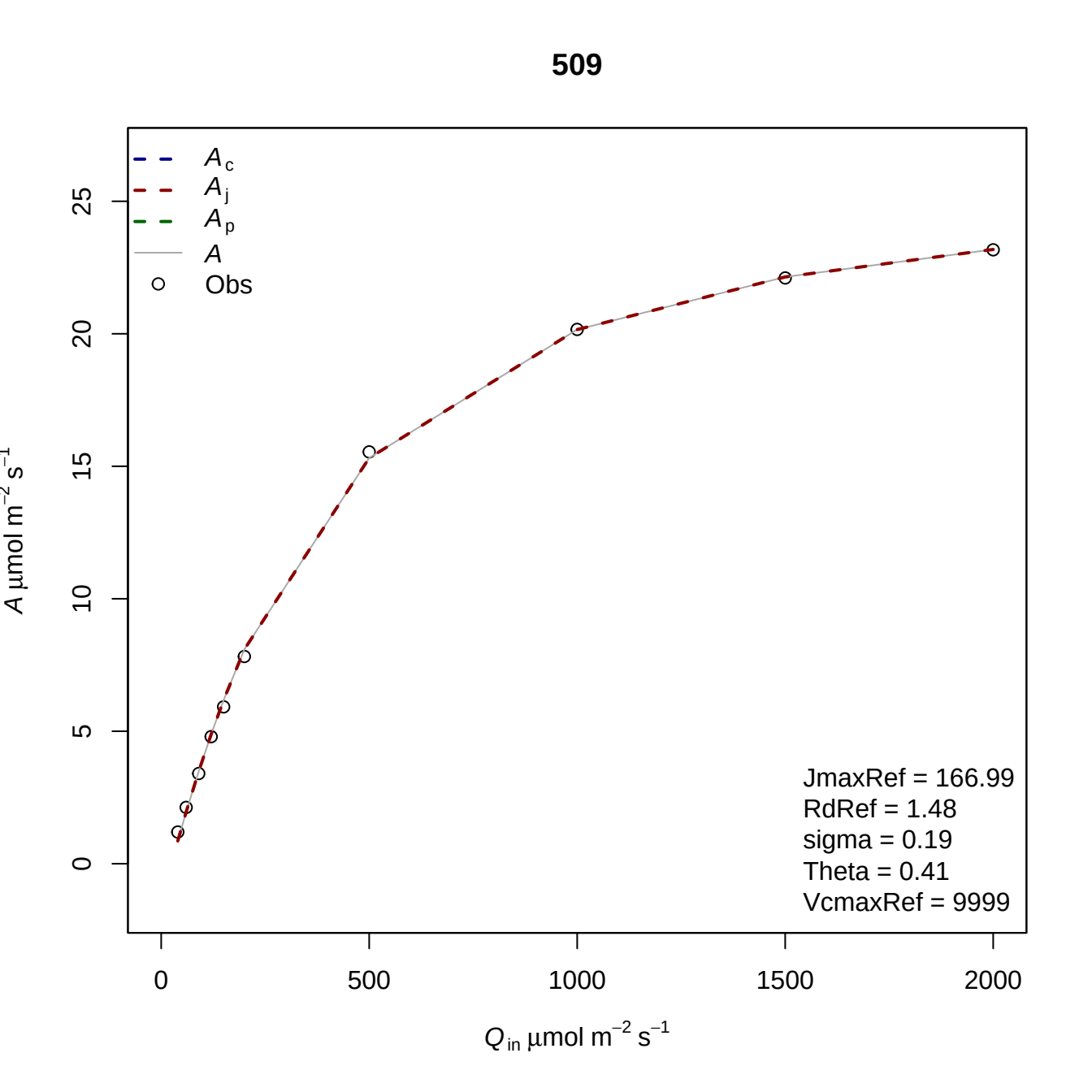

504
A_c
A_j
A_p
A
Obs
JmaxRef = 173.28
RdRef = 1.29
sigma = 0.12
Theta = 0.47
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

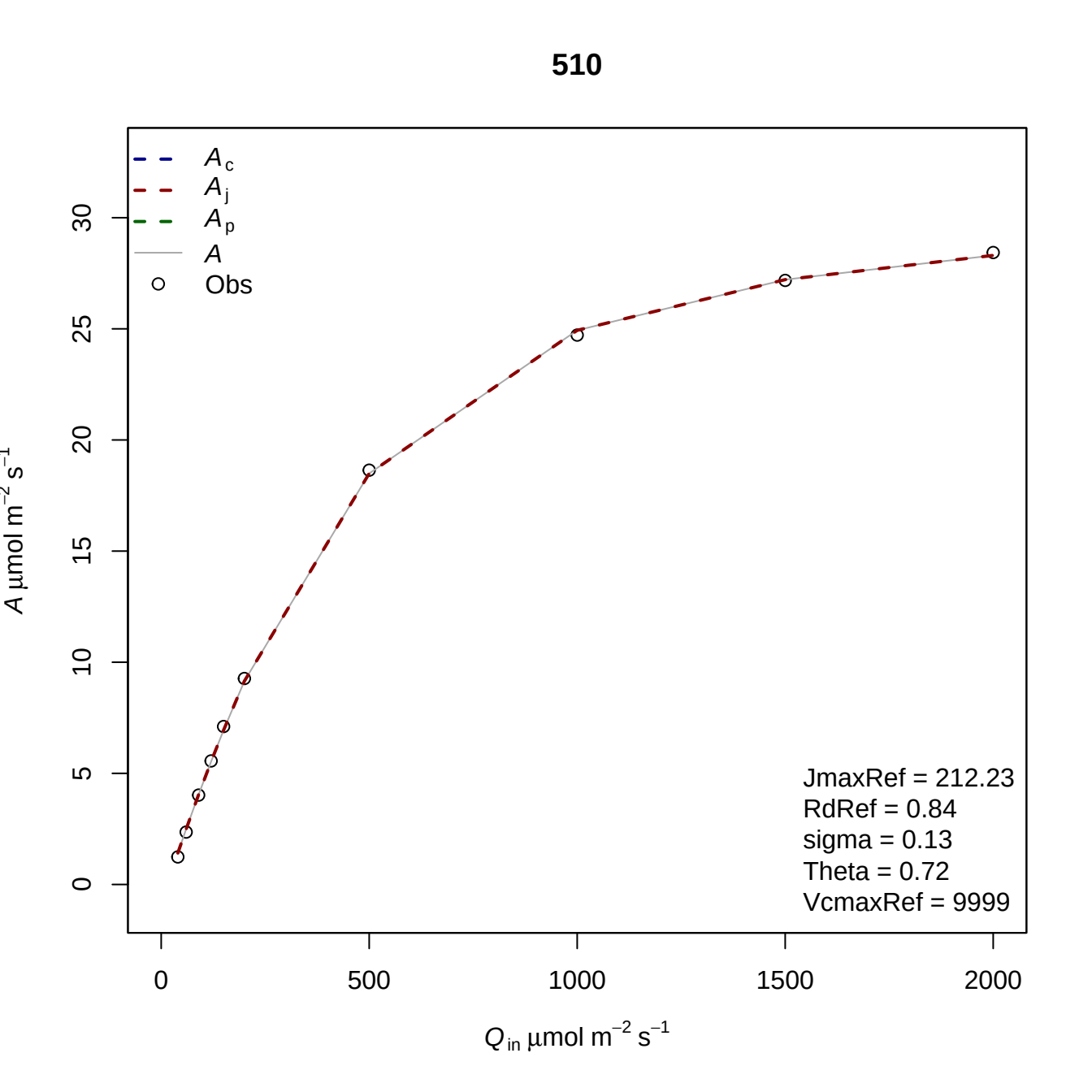

505
A_c
A_j
A_p
A
Obs
JmaxRef = 232.9
RdRef = 1.18
sigma = 0.13
Theta = 0.78
VcmaxRef = 9999
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

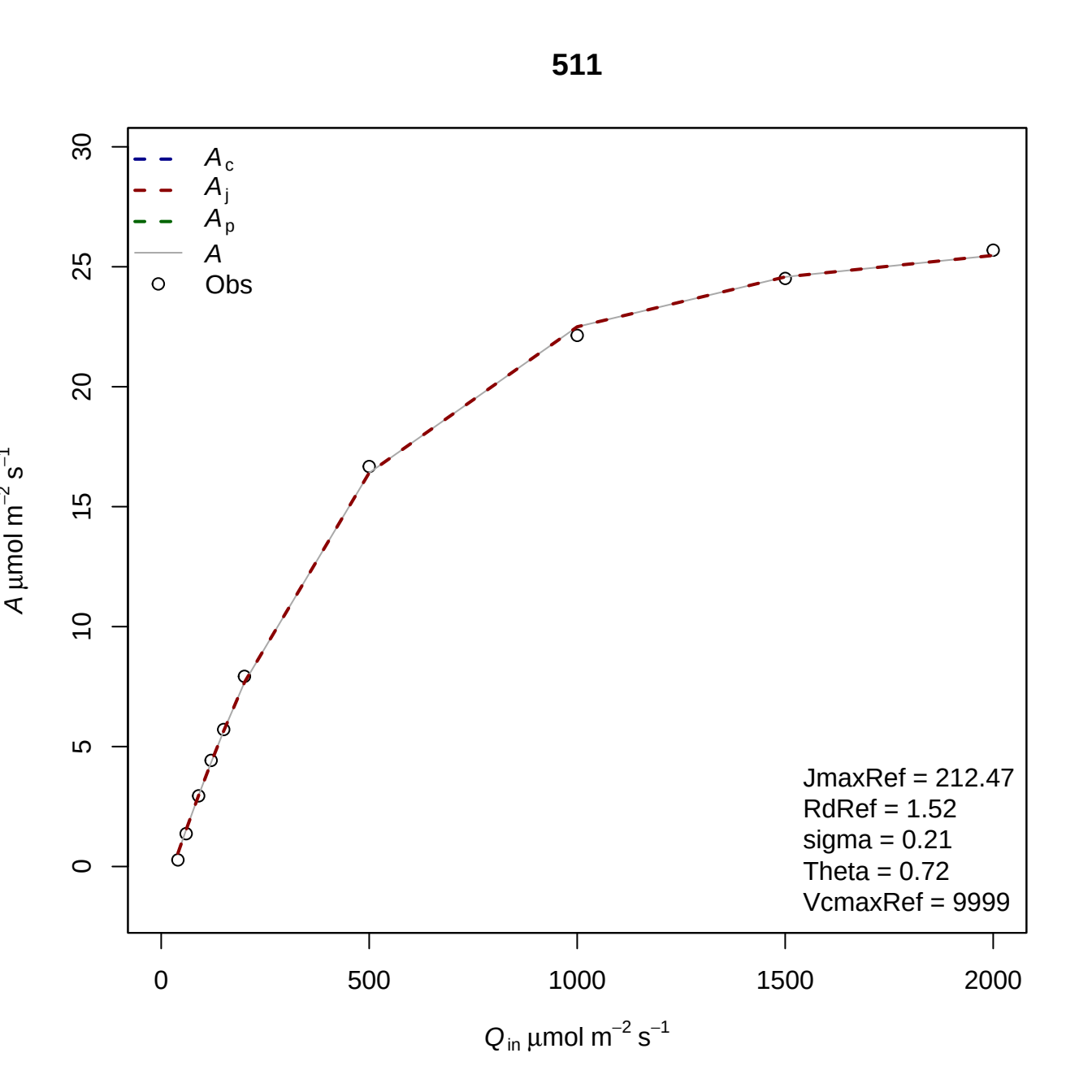

506
A_c
A_j
A_p
A
Obs
JmaxRef = 274.32
RdRef = 3.01
sigma = 0.09
Theta = 0.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

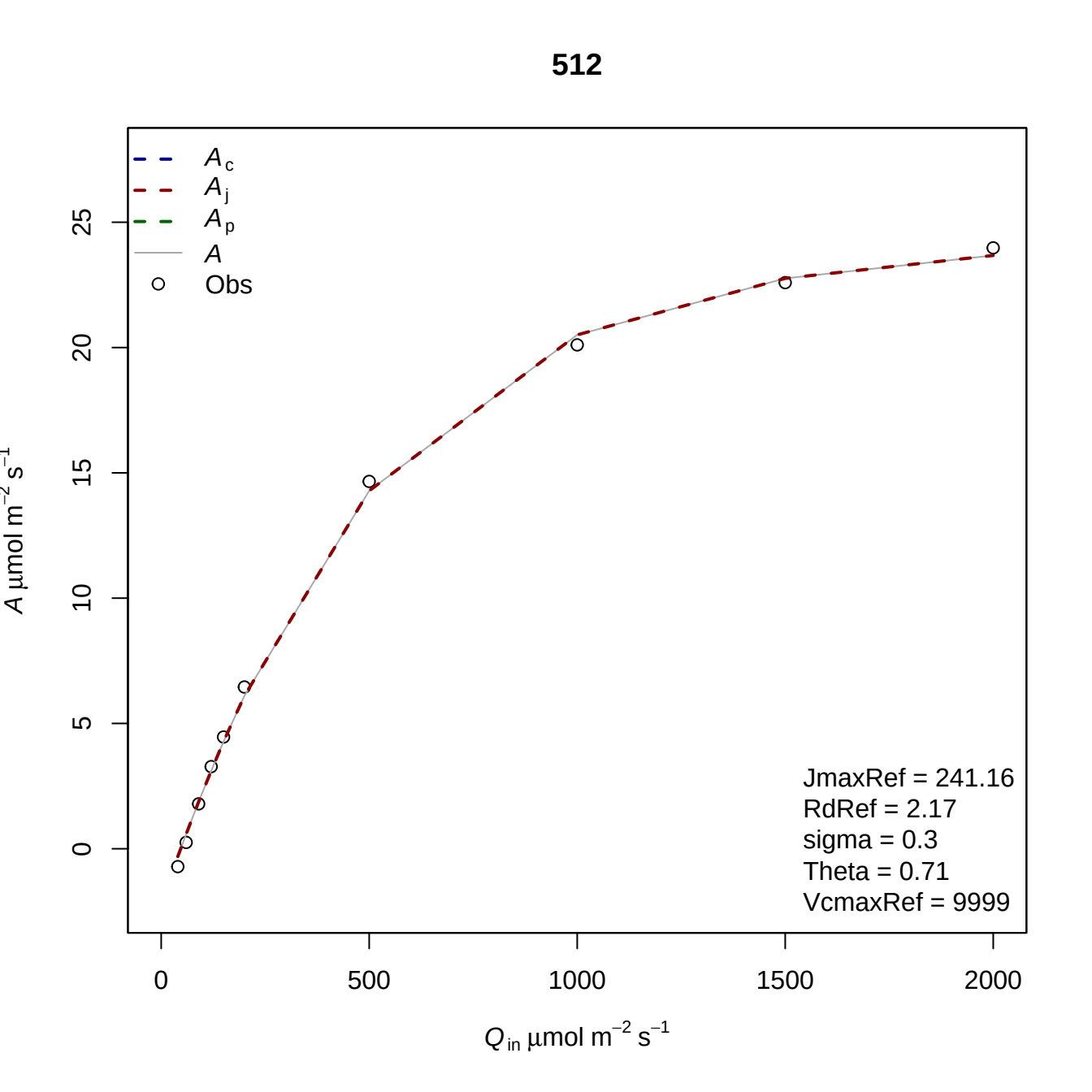

507
A_c
A_j
A_p
A
Obs
JmaxRef = 282.04
RdRef = 3.46
sigma = 0.12
Theta = 0.59
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

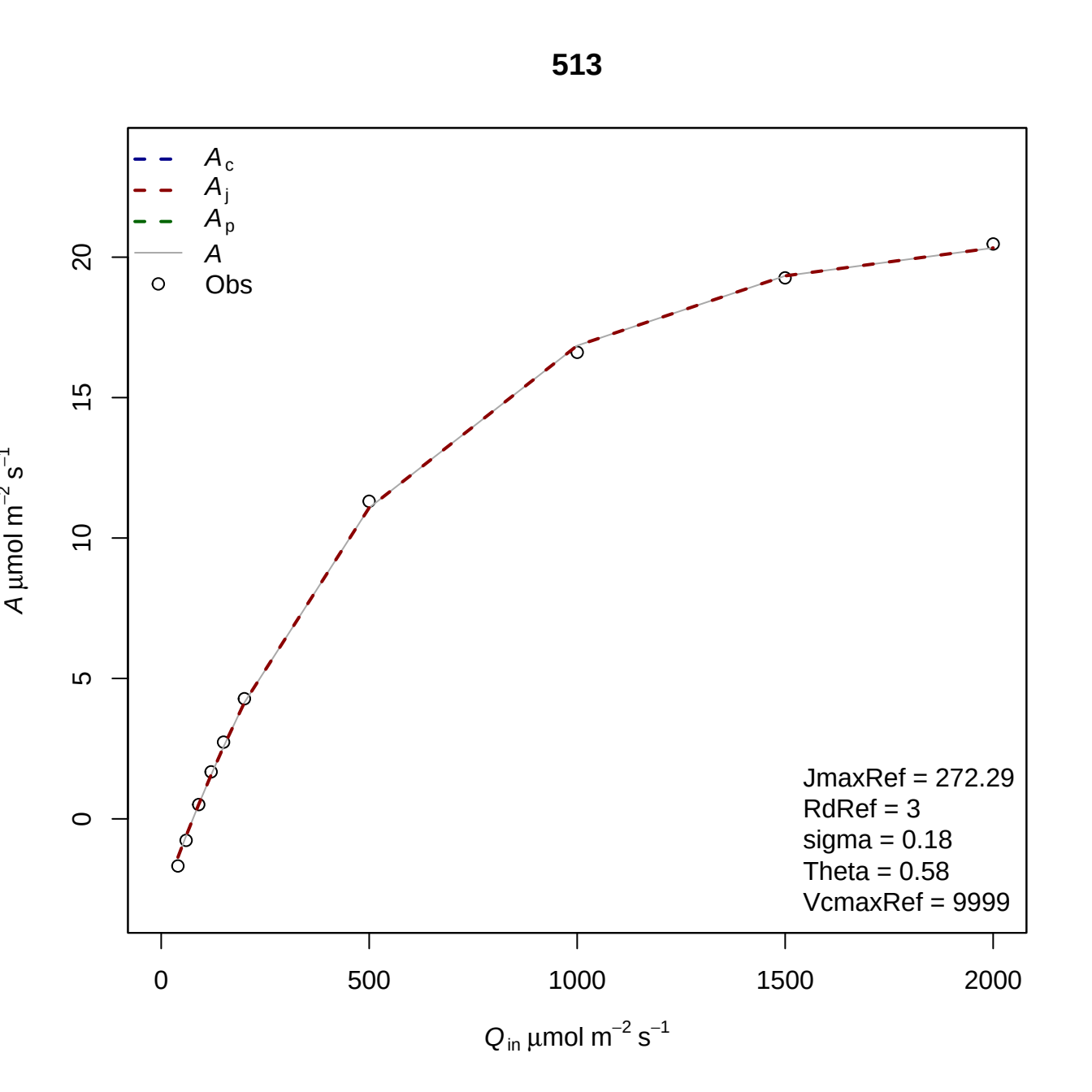

508
A_c
A_j
A_p
A
Obs
JmaxRef = 142.95
RdRef = 0.67
sigma = 0.11
Theta = 0.36
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

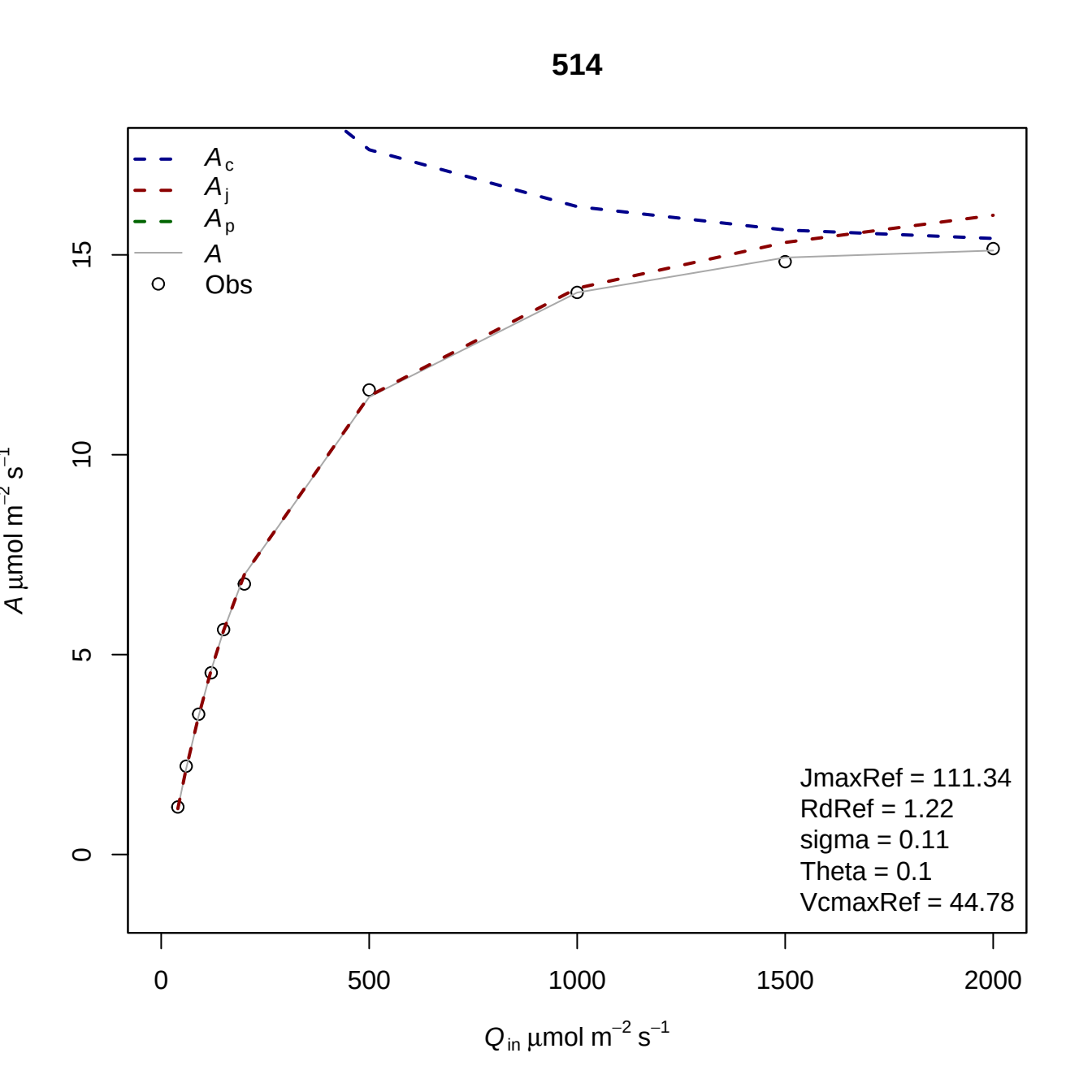

509
A_c
A_j
A_p
A
Obs
JmaxRef = 166.99
RdRef = 1.48
sigma = 0.19
Theta = 0.41
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

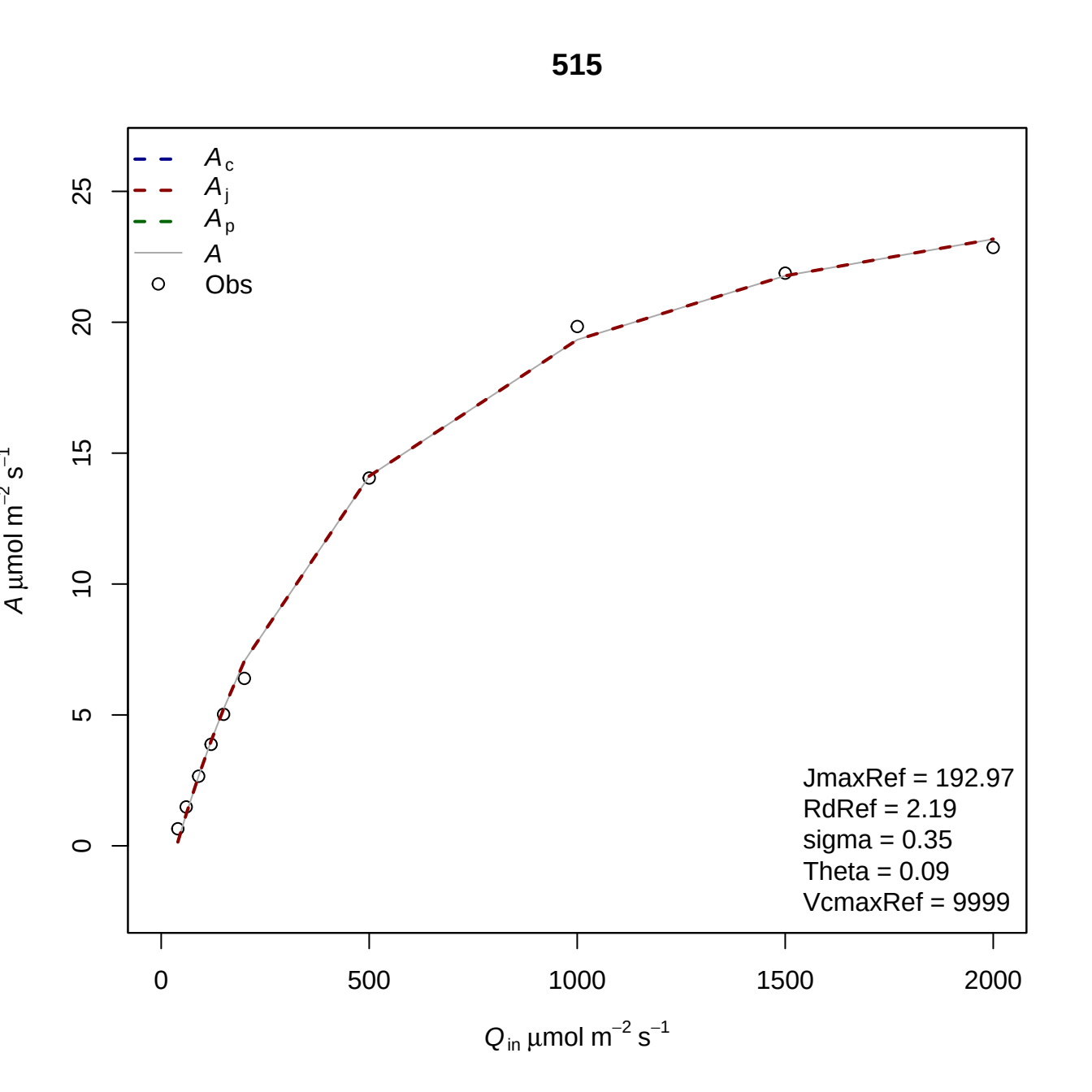

510
A_c
A_j
A_p
A
Obs
JmaxRef = 212.23
RdRef = 0.84
sigma = 0.13
Theta = 0.72
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

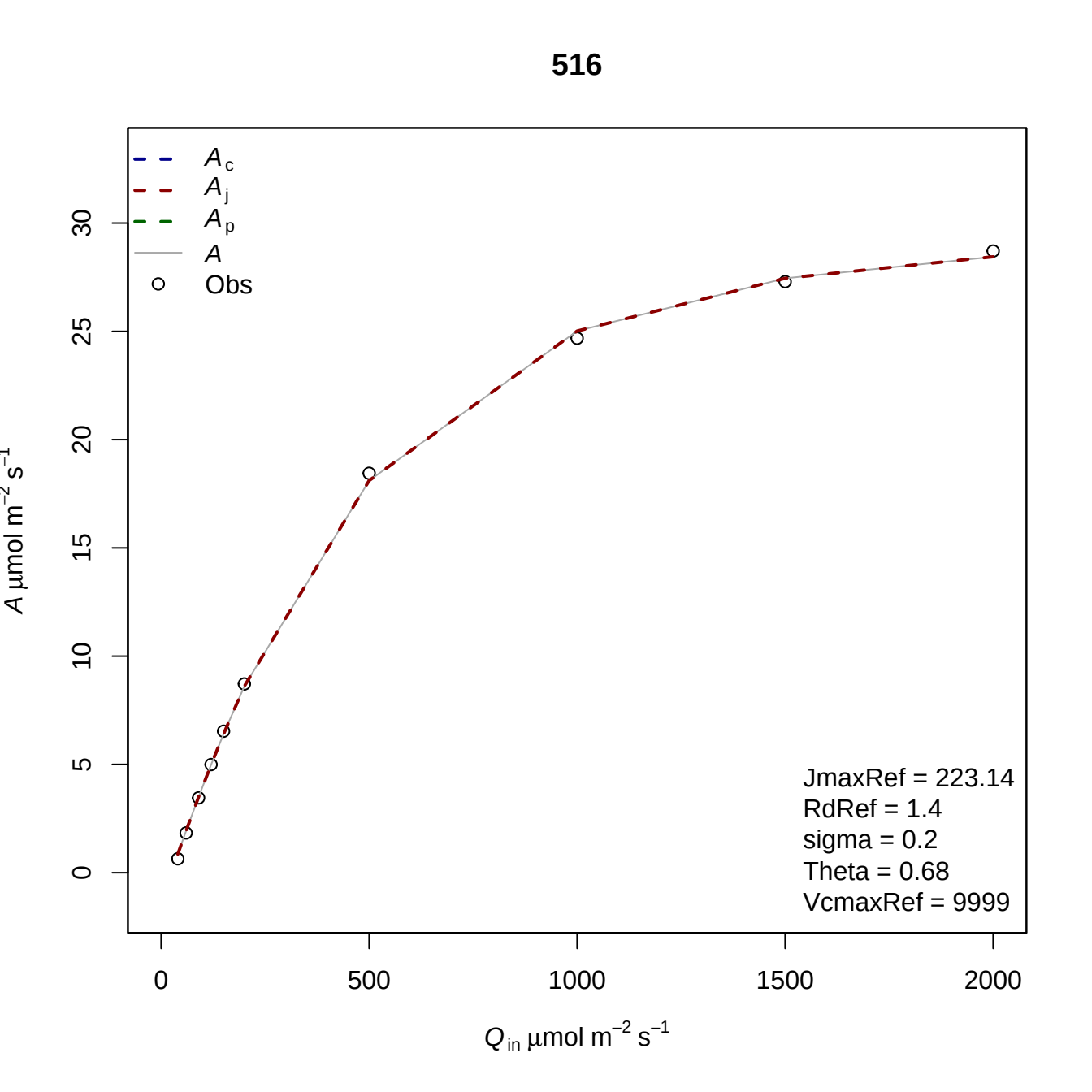

511
A_c
A_j
A_p
A
Obs
JmaxRef = 212.47
RdRef = 1.52
sigma = 0.21
Theta = 0.72
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

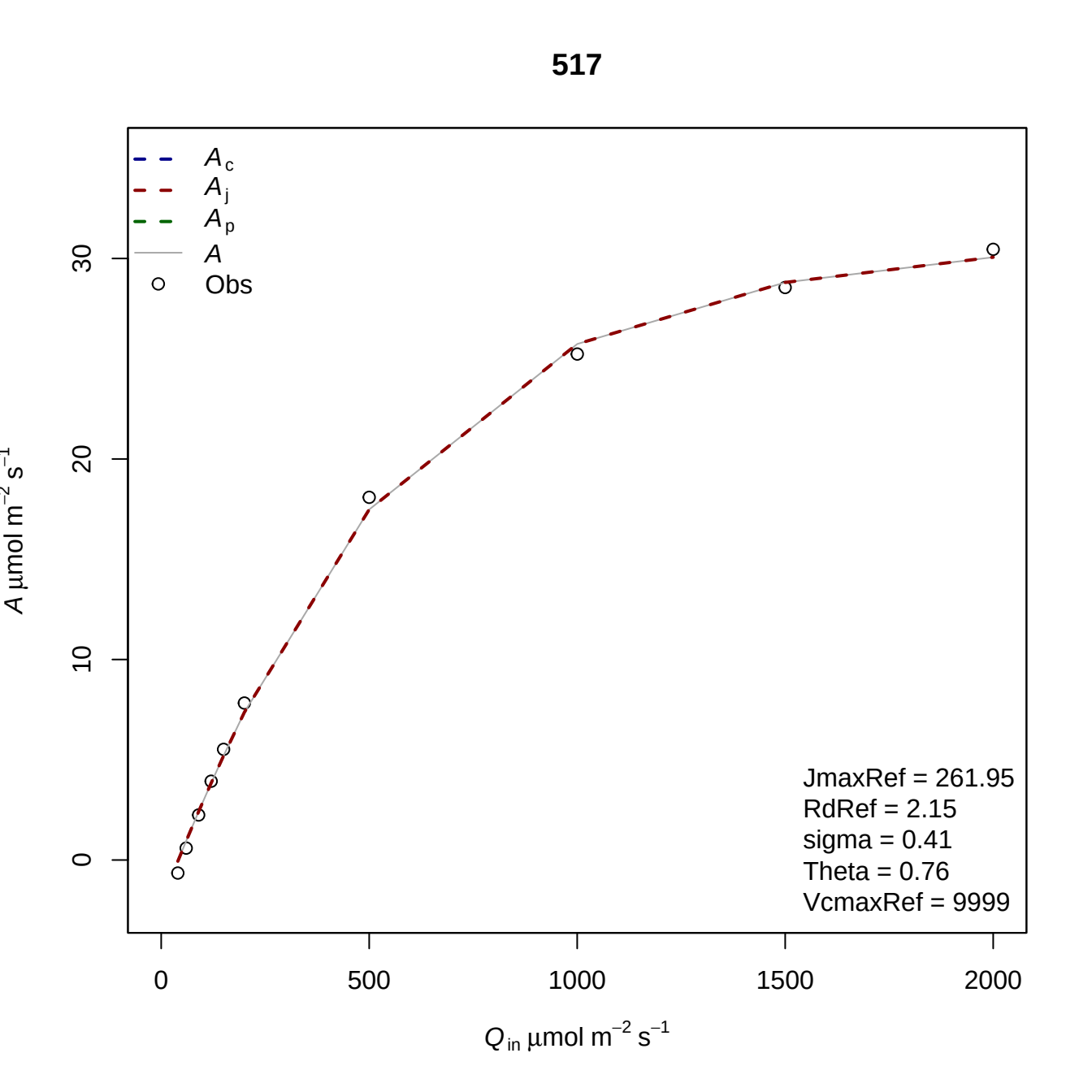

512
A_c
A_j
A_p
A
Obs
JmaxRef = 241.16
RdRef = 2.17
sigma = 0.3
Theta = 0.71
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

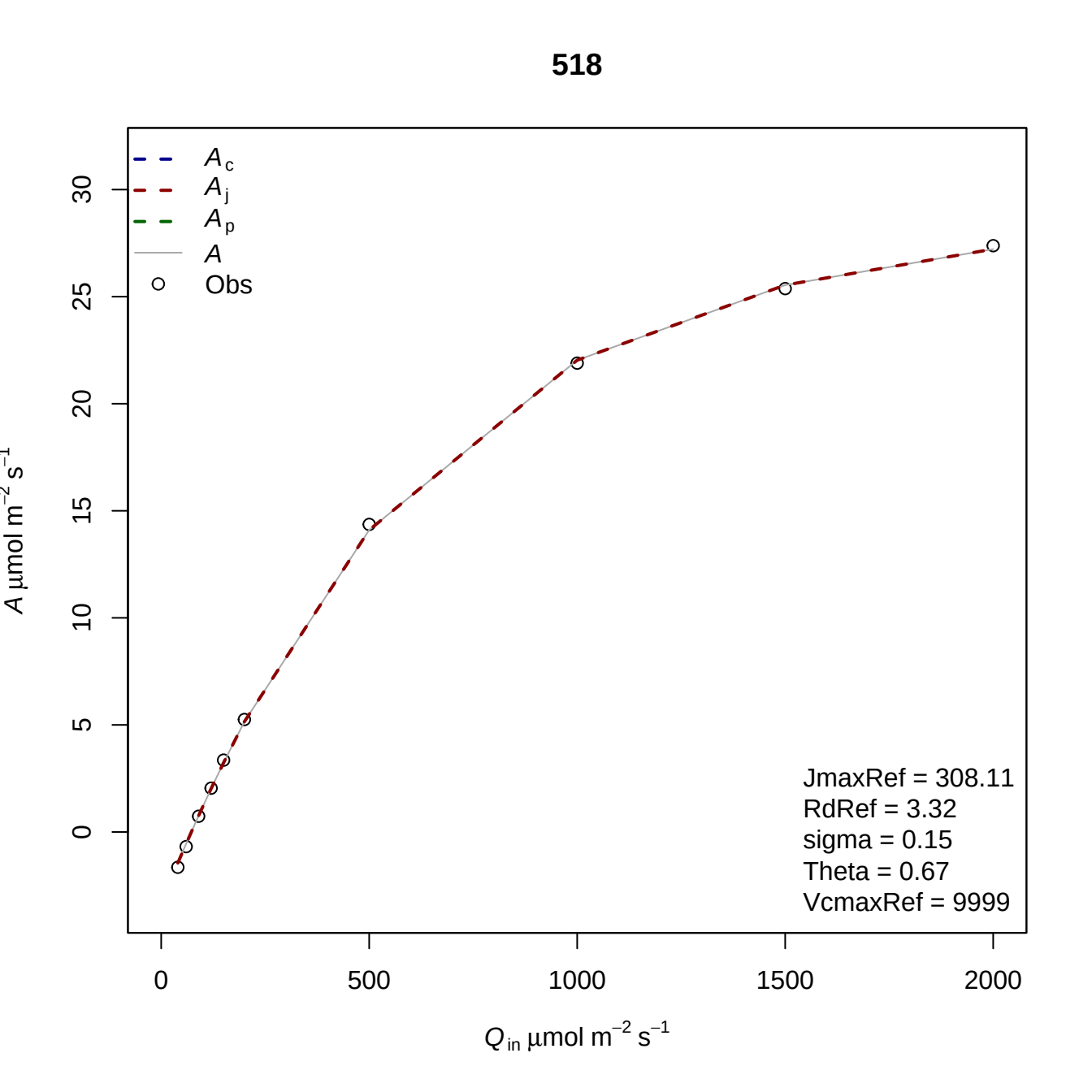

513
A_c
A_j
A_p
A
Obs
JmaxRef = 272.29
RdRef = 3
sigma = 0.18
Theta = 0.58
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

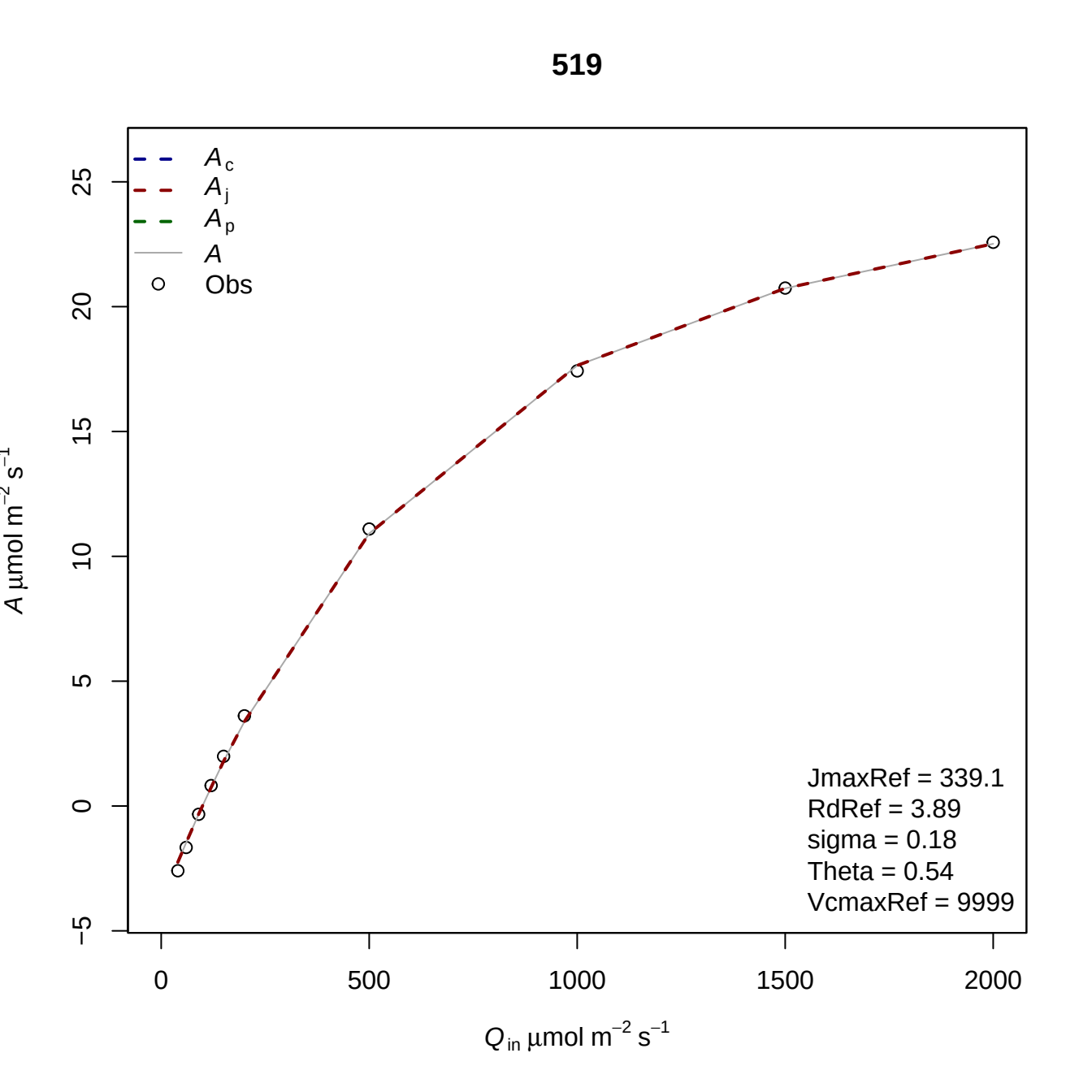

514
A_c
A_j
A_p
A
Obs
JmaxRef = 111.34
RdRef = 1.22
sigma = 0.11
Theta = 0.1
VcmaxRef = 44.78
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

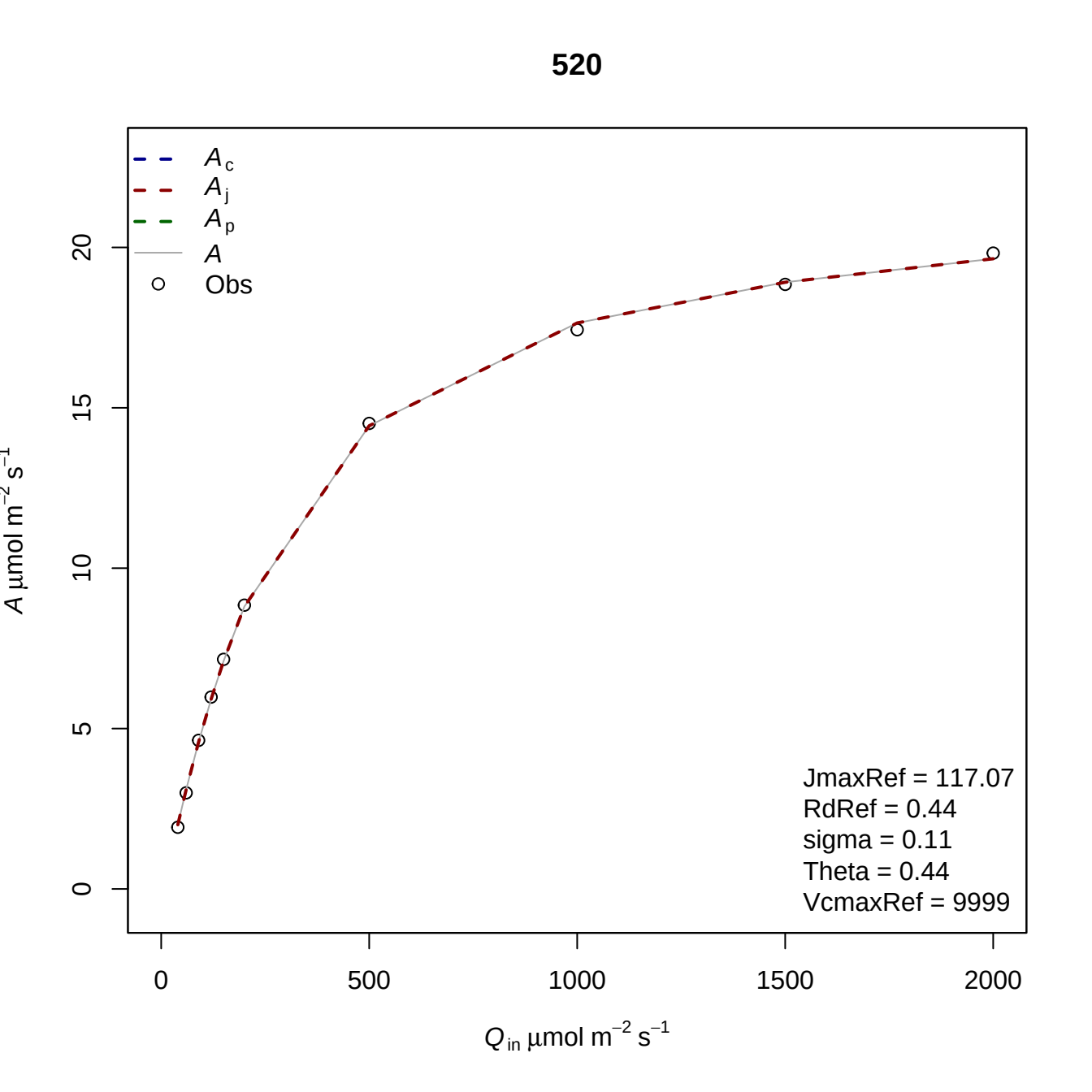

515
A_c
A_j
A_p
A
Obs
JmaxRef = 192.97
RdRef = 2.19
sigma = 0.35
Theta = 0.09
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

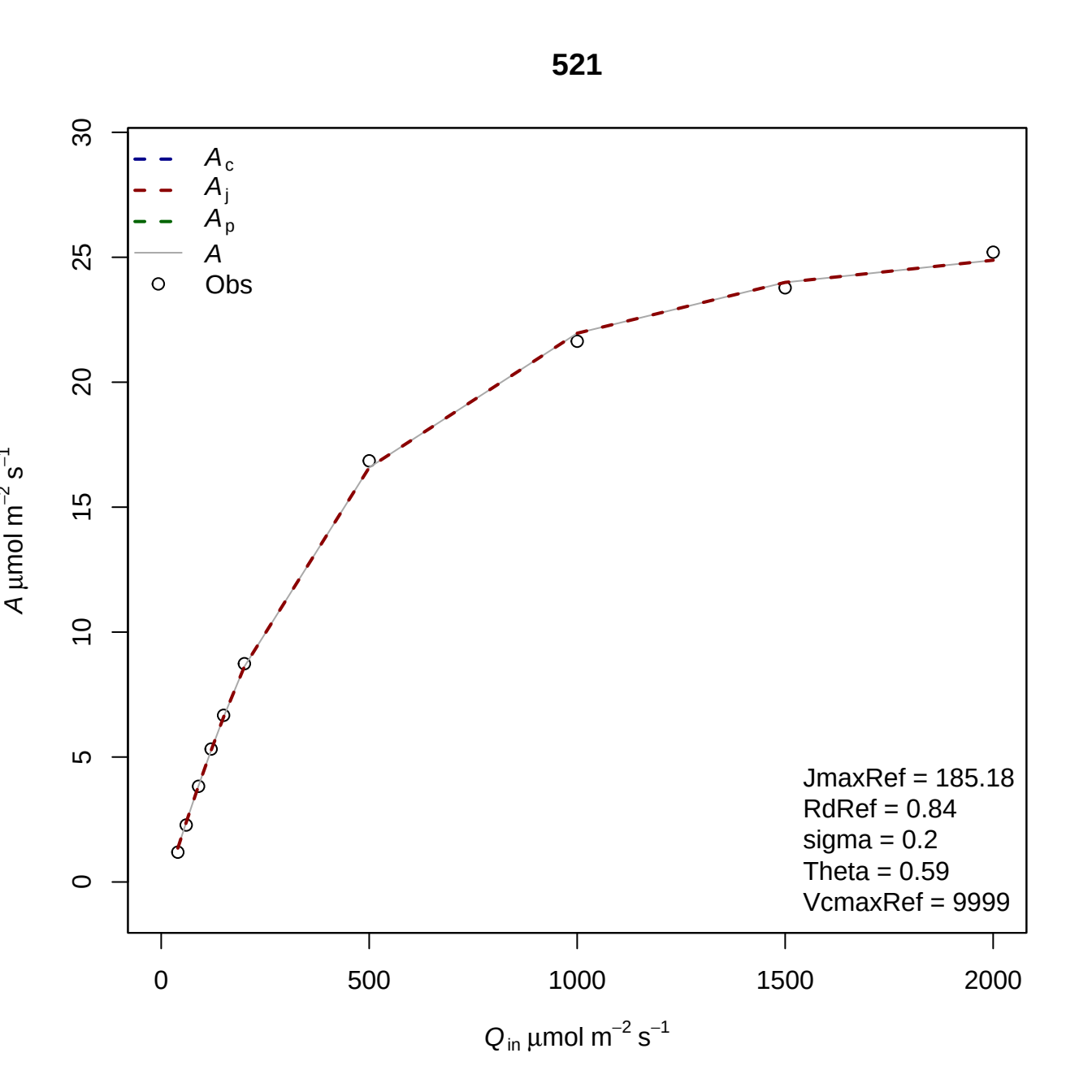

516
A_c
A_j
A_p
A
Obs
JmaxRef = 223.14
RdRef = 1.4
sigma = 0.2
Theta = 0.68
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

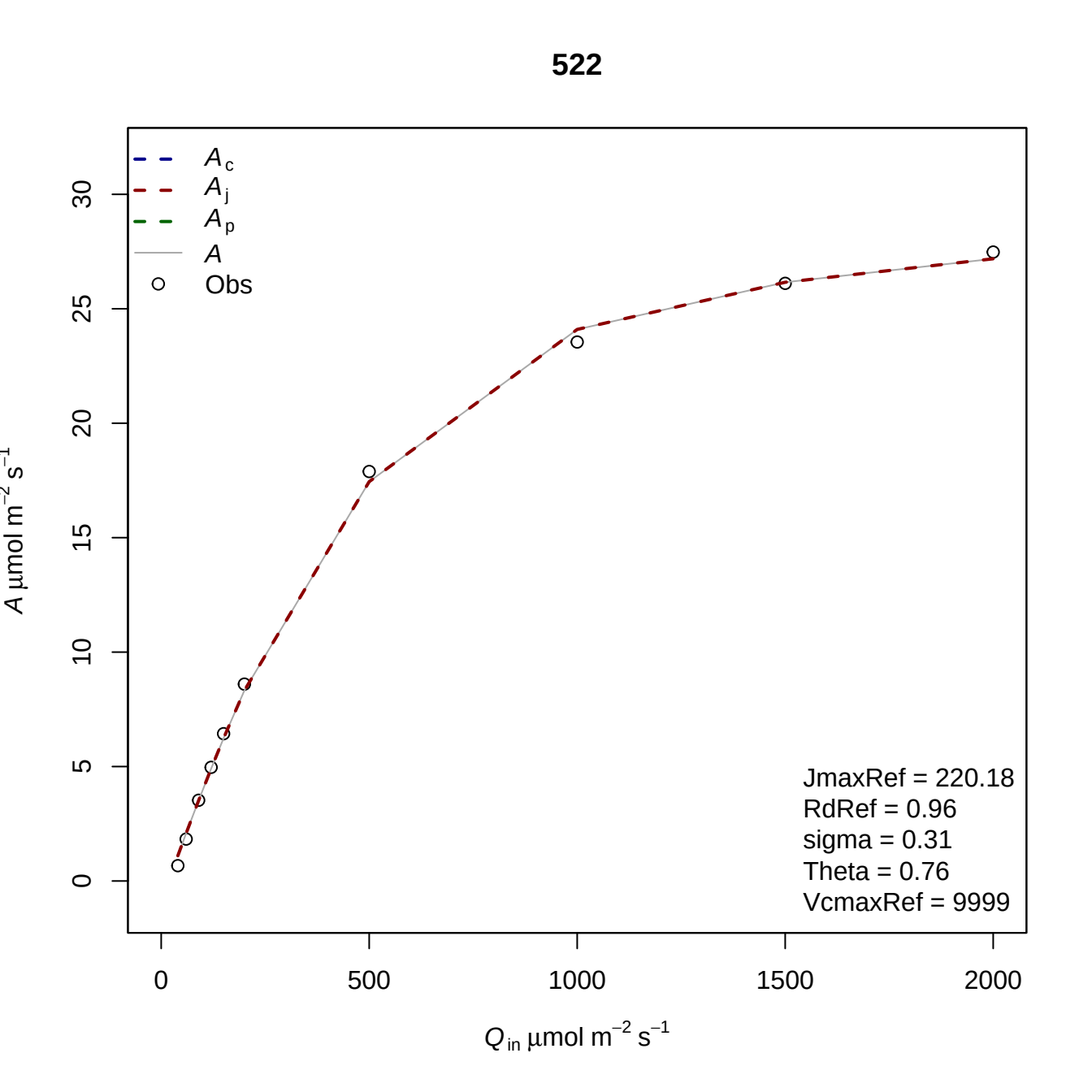

517
A_c
A_j
A_p
A
Obs
JmaxRef = 261.95
RdRef = 2.15
sigma = 0.41
Theta = 0.76
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

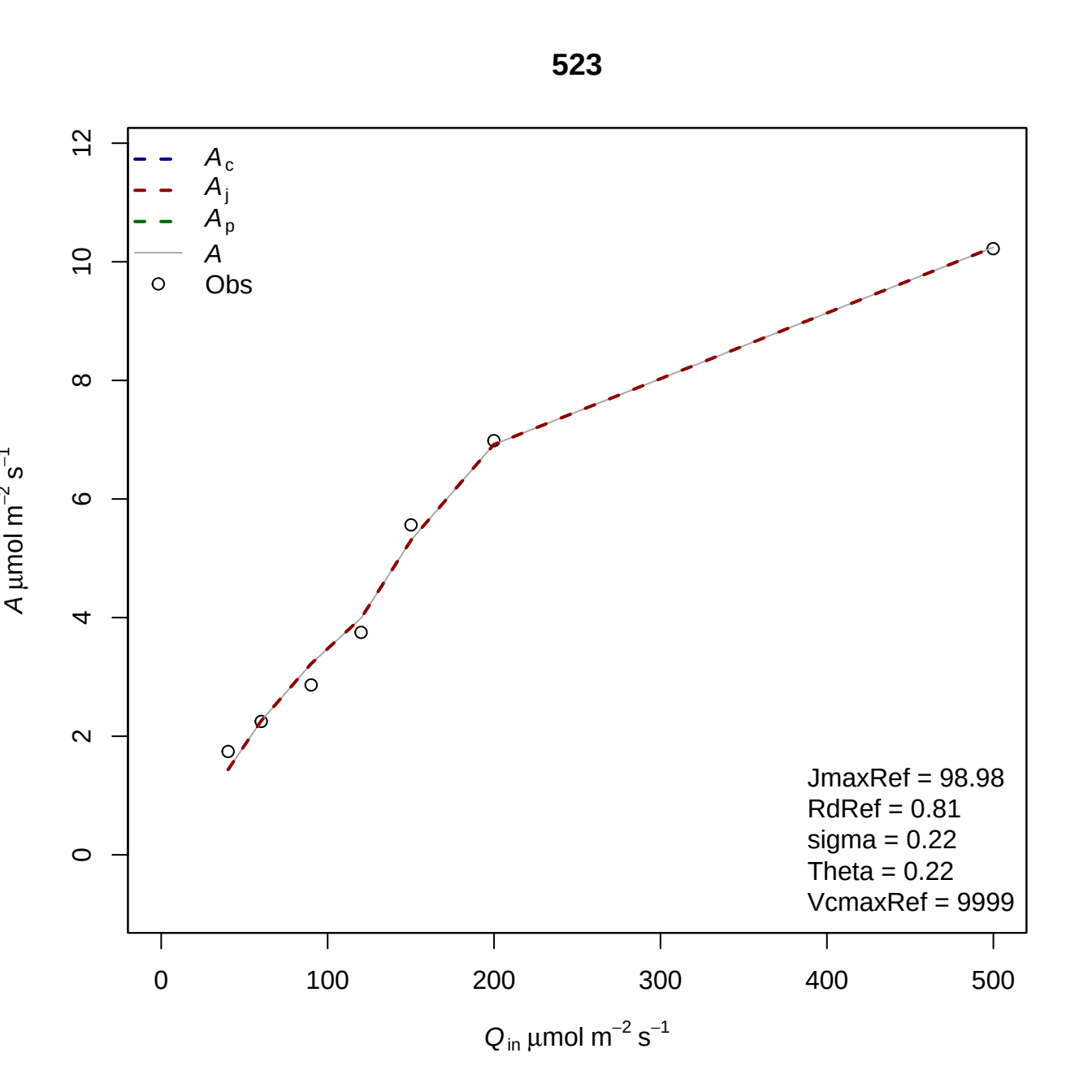

518
A_c
A_j
A_p
A
Obs
JmaxRef = 308.11
RdRef = 3.32
sigma = 0.15
Theta = 0.67
VcmaxRef = 9999
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

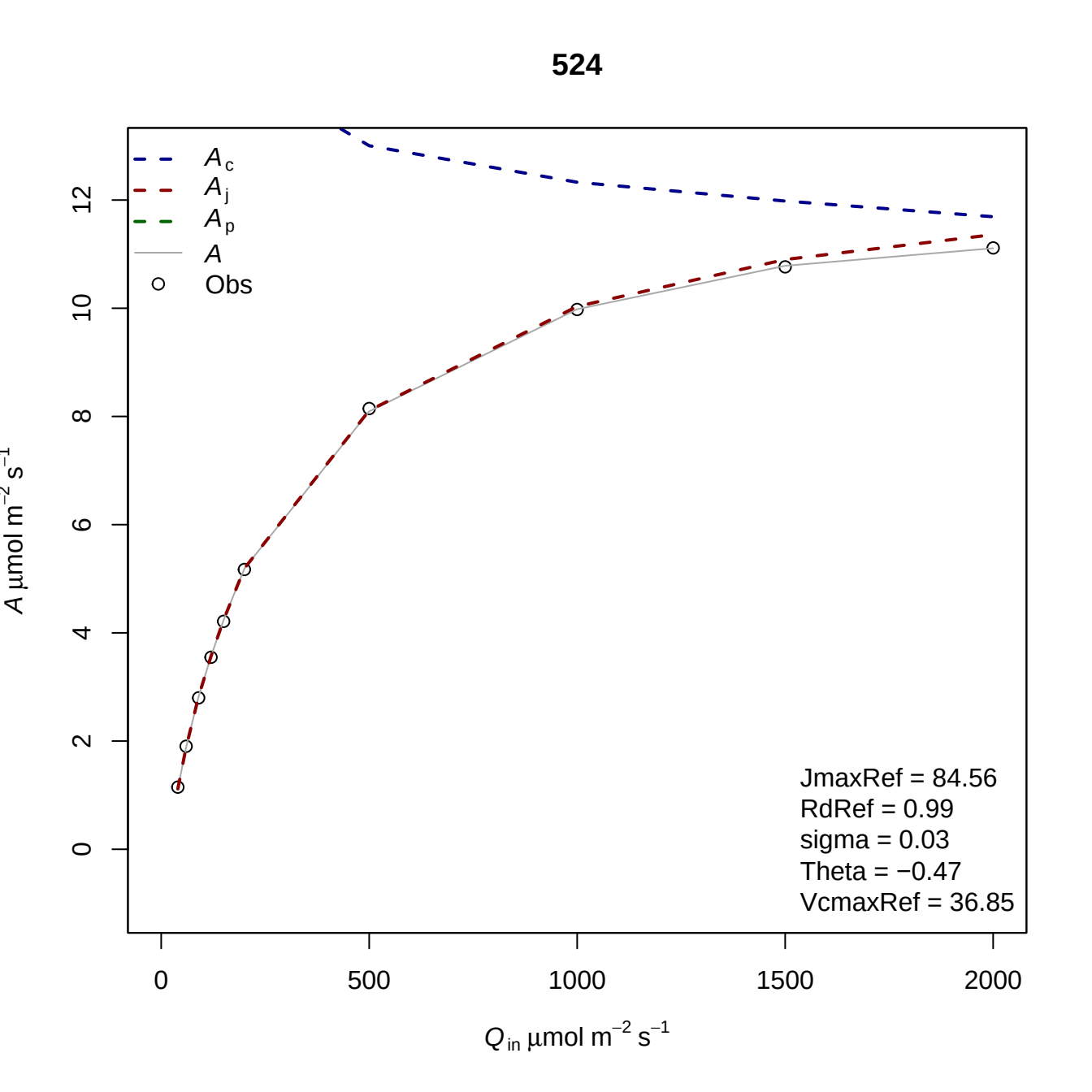

519
A_c
A_j
A_p
A
Obs
JmaxRef = 339.1
RdRef = 3.89
sigma = 0.18
Theta = 0.54
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500
2000

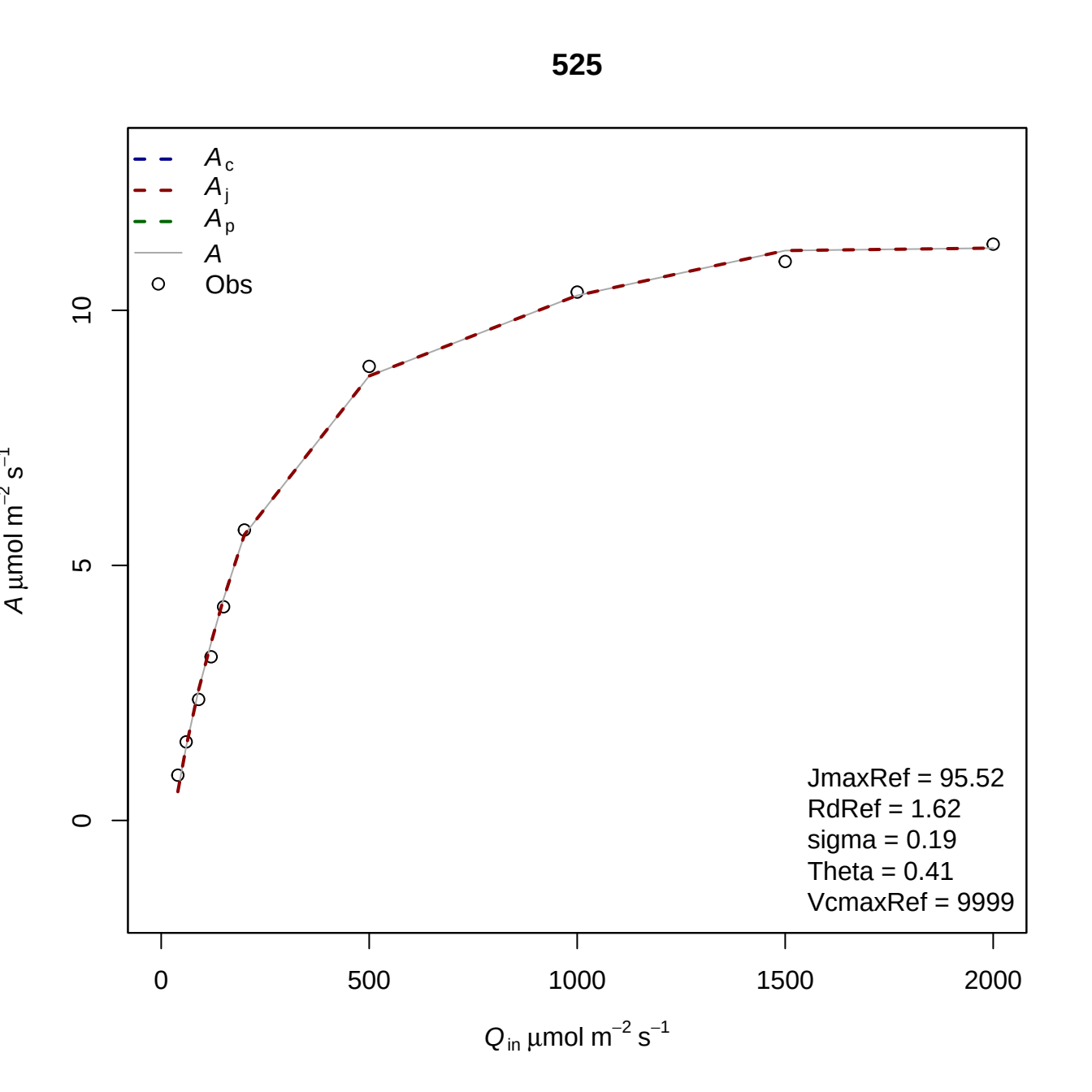

520
A_c
A_j
A_p
A
Obs
JmaxRef = 117.07
RdRef = 0.44
sigma = 0.11
Theta = 0.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

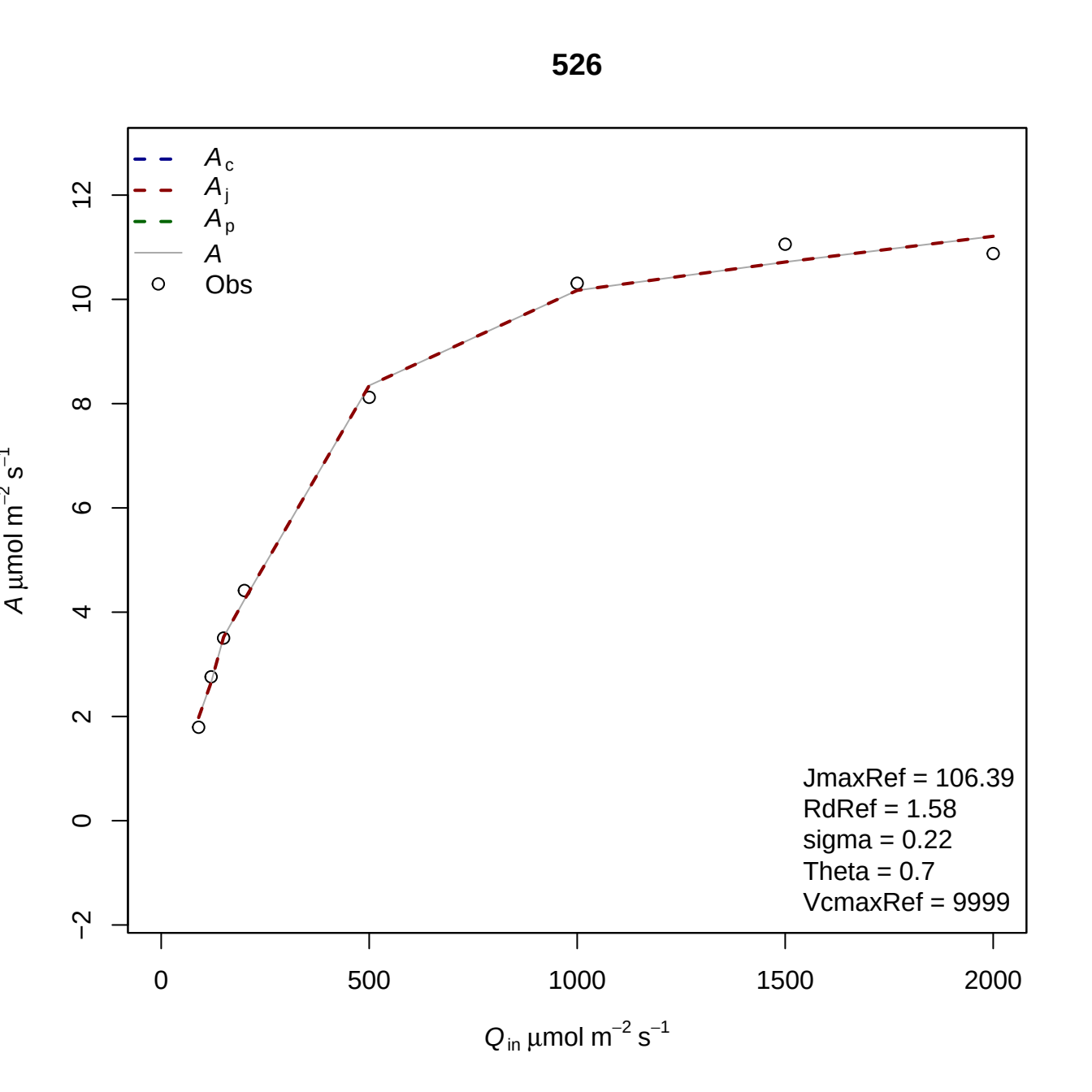

521
A_c
A_j
A_p
A
Obs
JmaxRef = 185.18
RdRef = 0.84
sigma = 0.2
Theta = 0.59
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

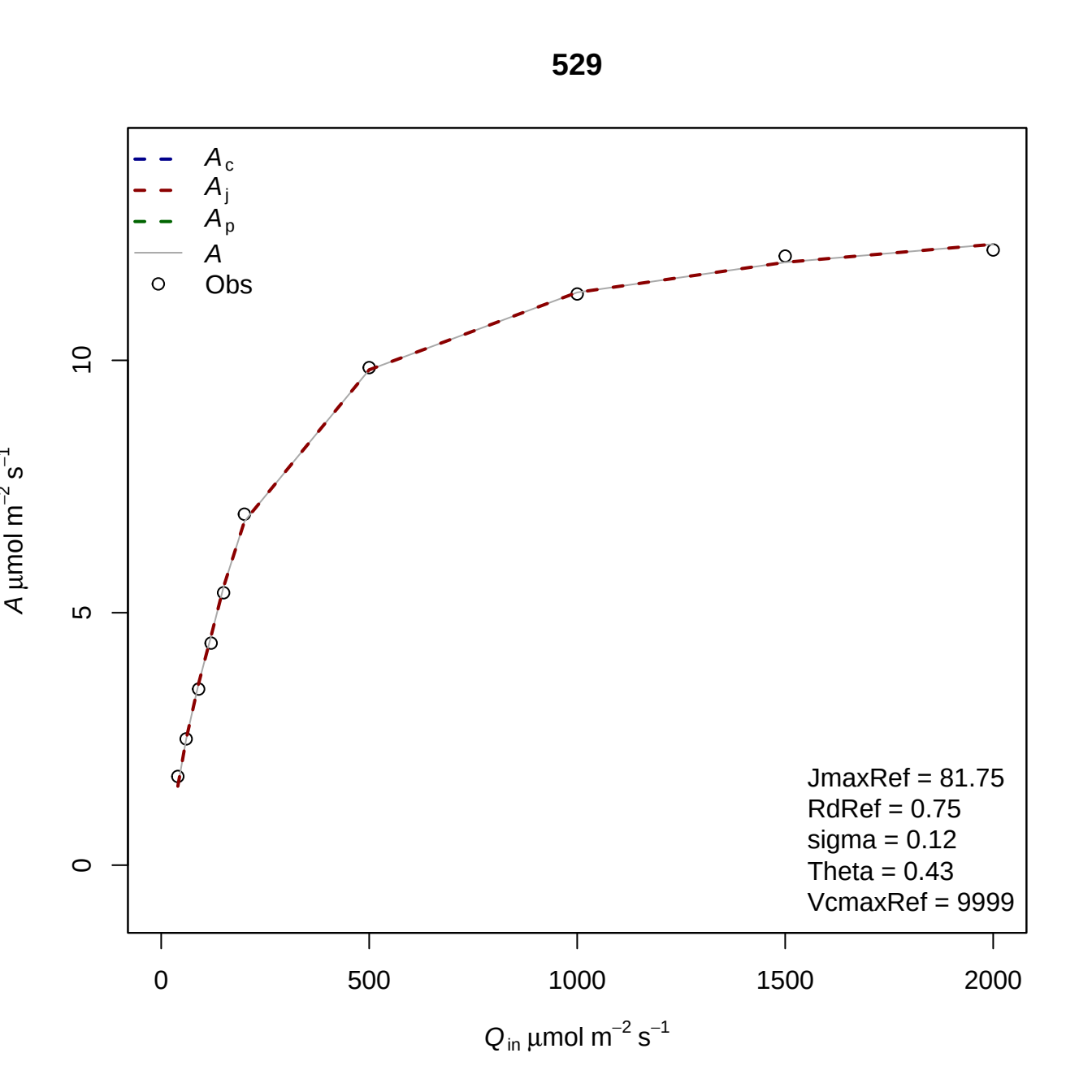

522
A_c
A_j
A_p
A
Obs
JmaxRef = 220.18
RdRef = 0.96
sigma = 0.31
Theta = 0.76
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

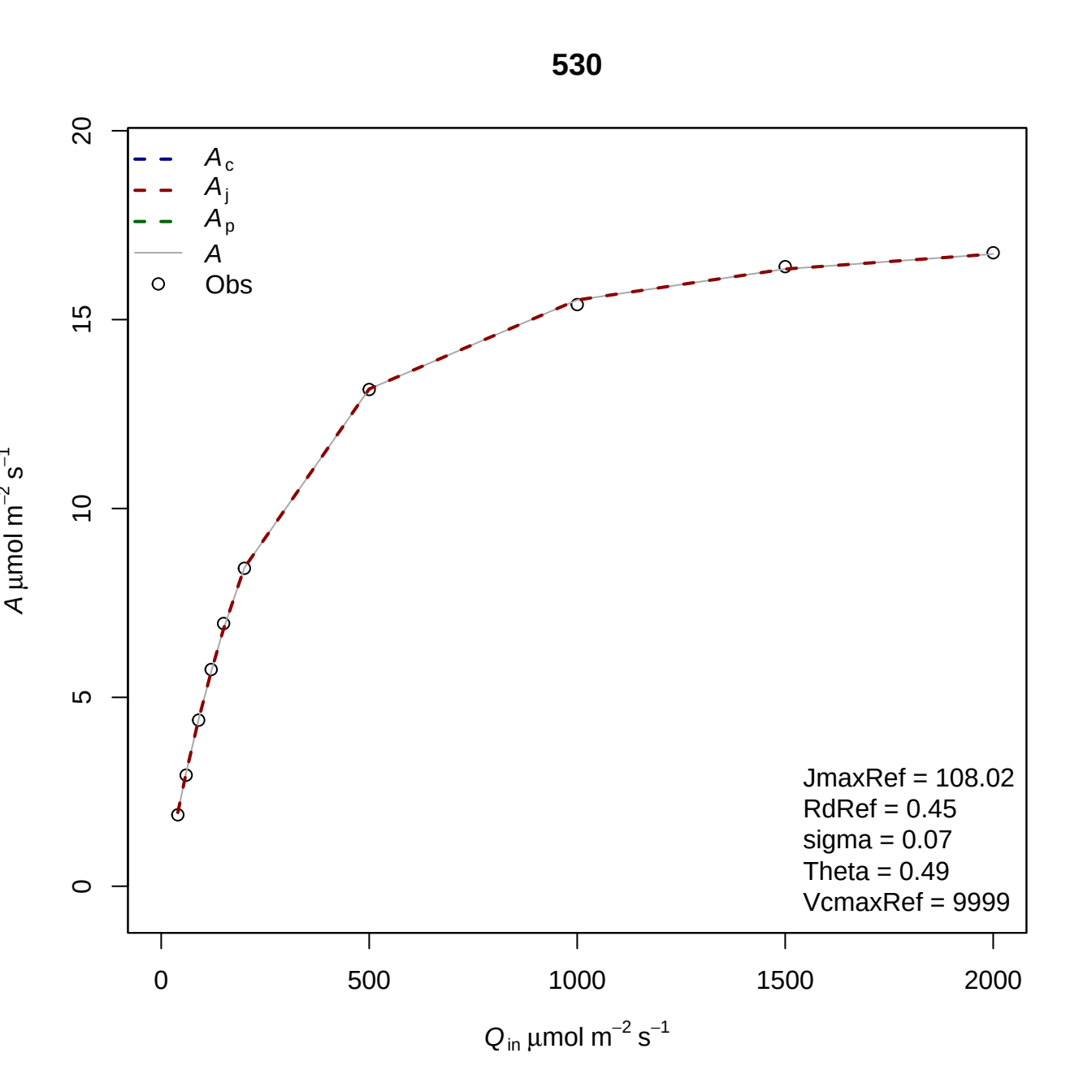

523
A_c
A_j
A_p
A
Obs
JmaxRef = 98.98
RdRef = 0.81
sigma = 0.22
Theta = 0.22
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
100
200
300
400
500

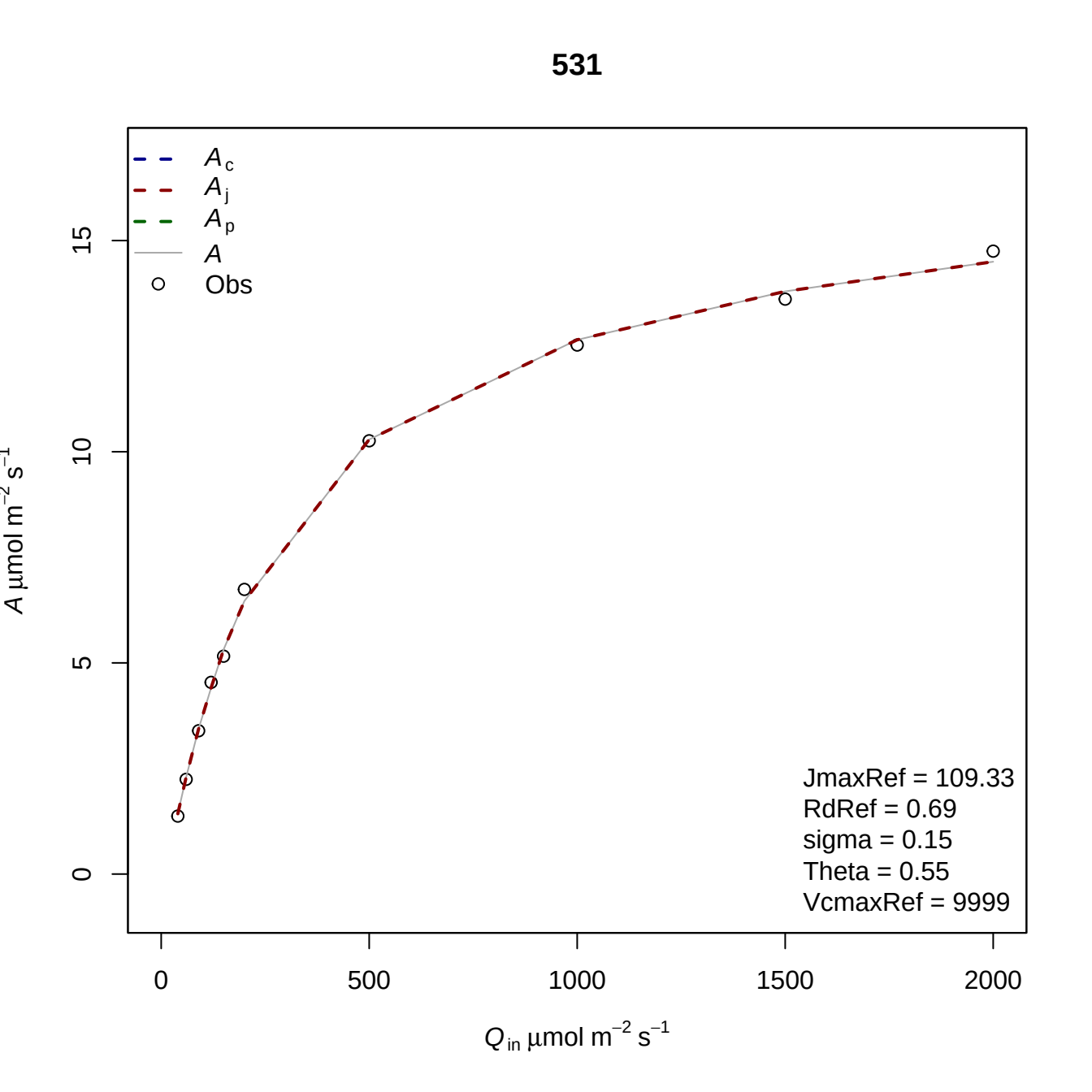

524
A_c
A_j
A_p
A
Obs
JmaxRef = 84.56
RdRef = 0.99
sigma = 0.03
Theta = −0.47
VcmaxRef = 36.85
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

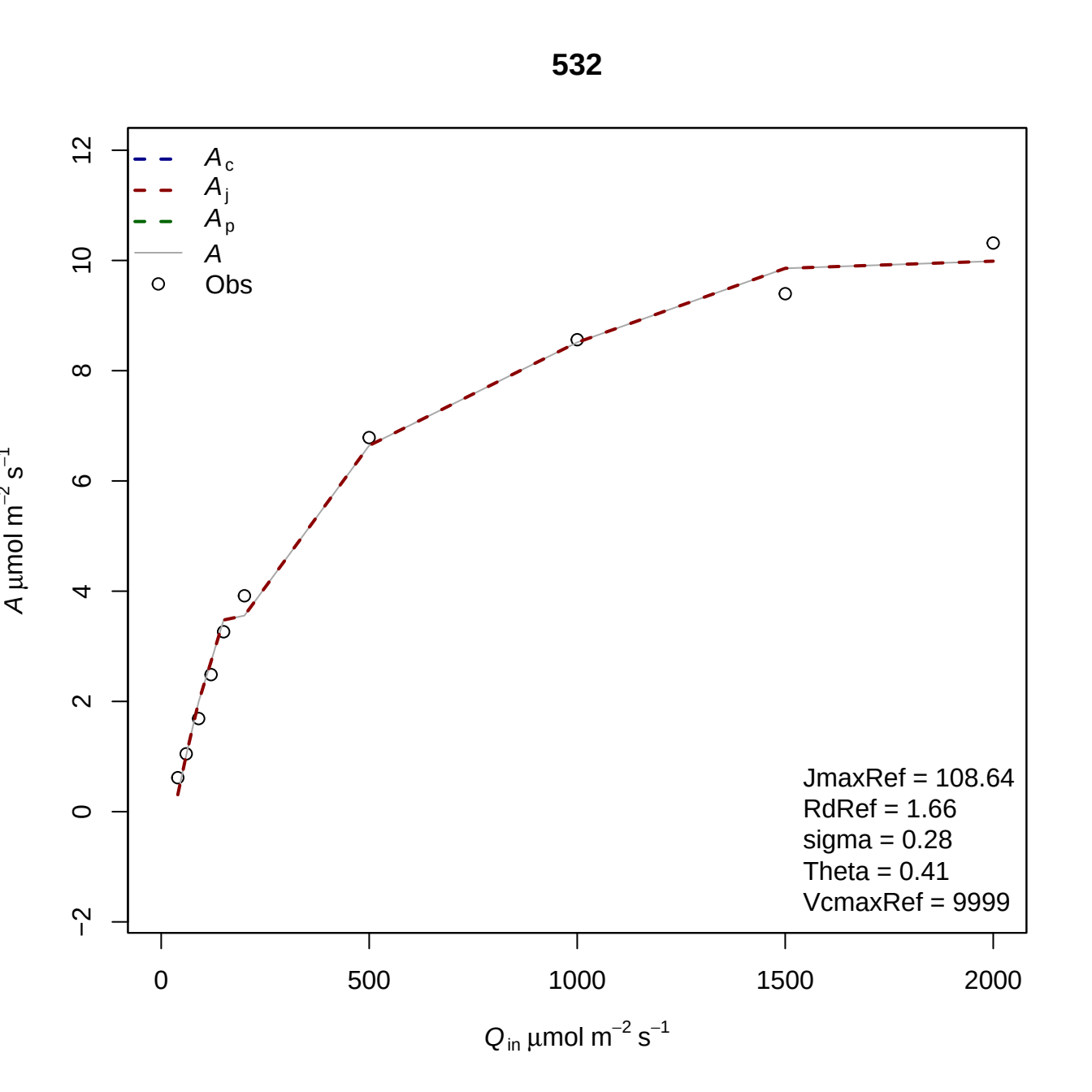

525
A_c
A_j
A_p
A
Obs
JmaxRef = 95.52
RdRef = 1.62
sigma = 0.19
Theta = 0.41
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

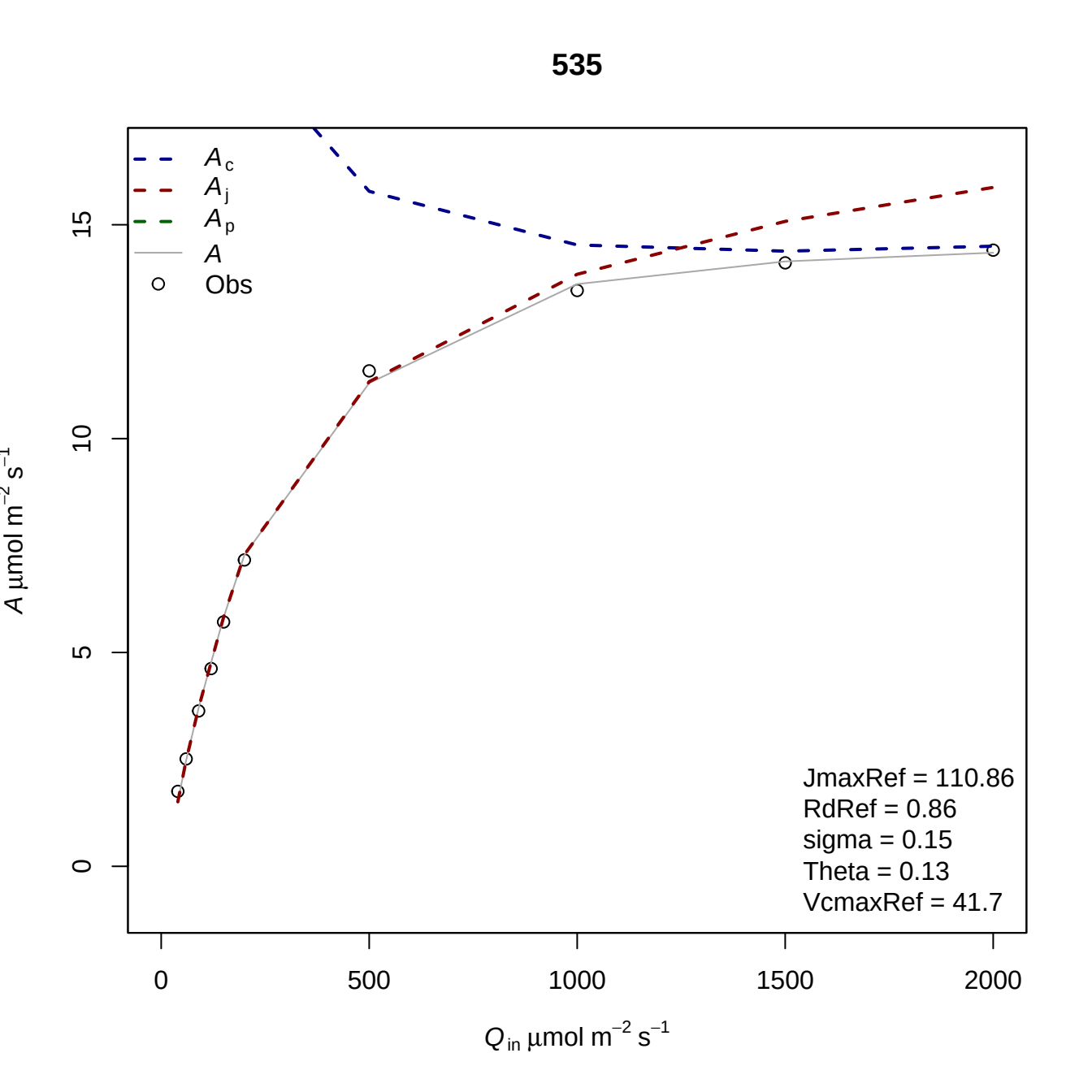

526
A_c
A_j
A_p
A
Obs
JmaxRef = 106.39
RdRef = 1.58
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

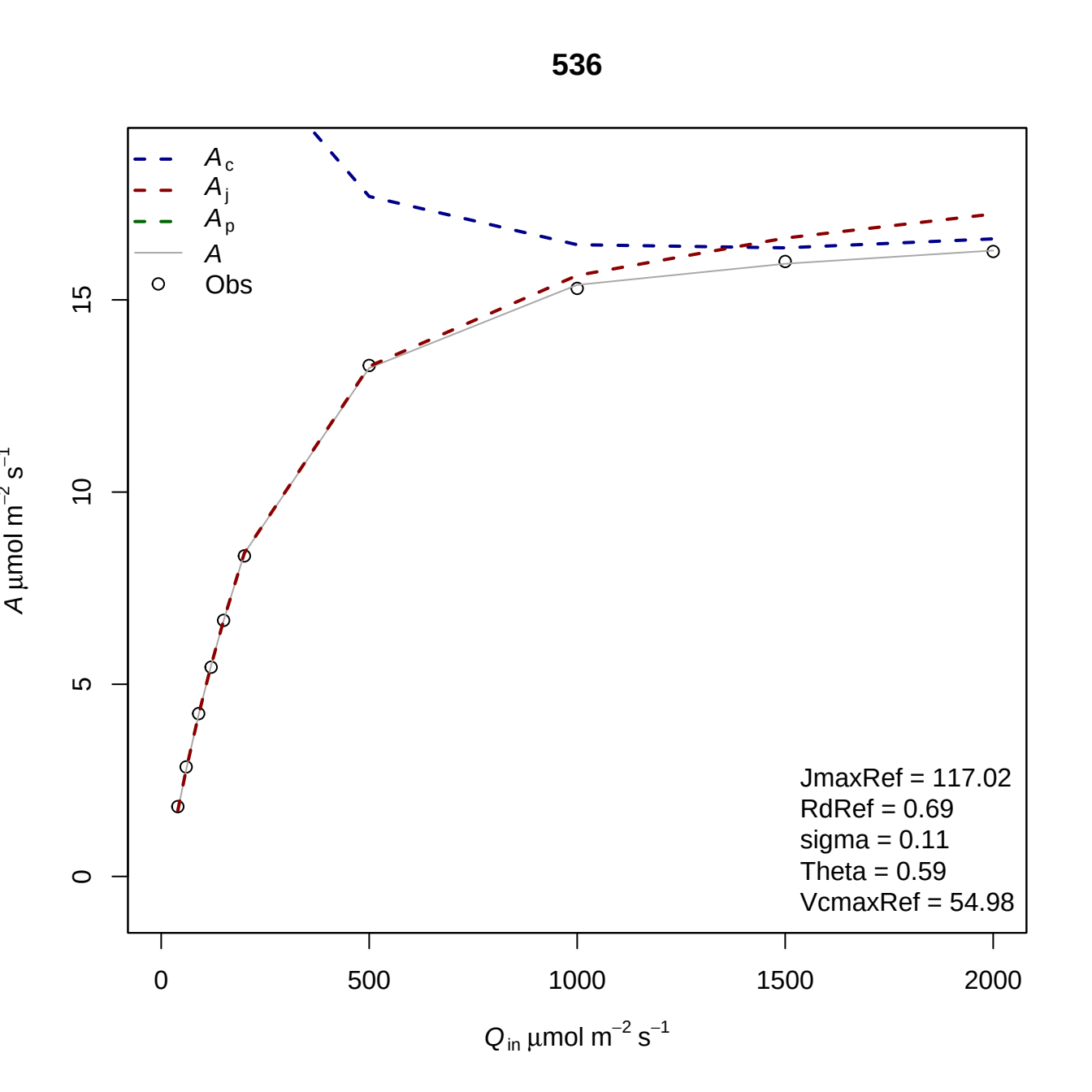

529
A_c
A_j
A_p
A
Obs
JmaxRef = 81.75
RdRef = 0.75
sigma = 0.12
Theta = 0.43
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

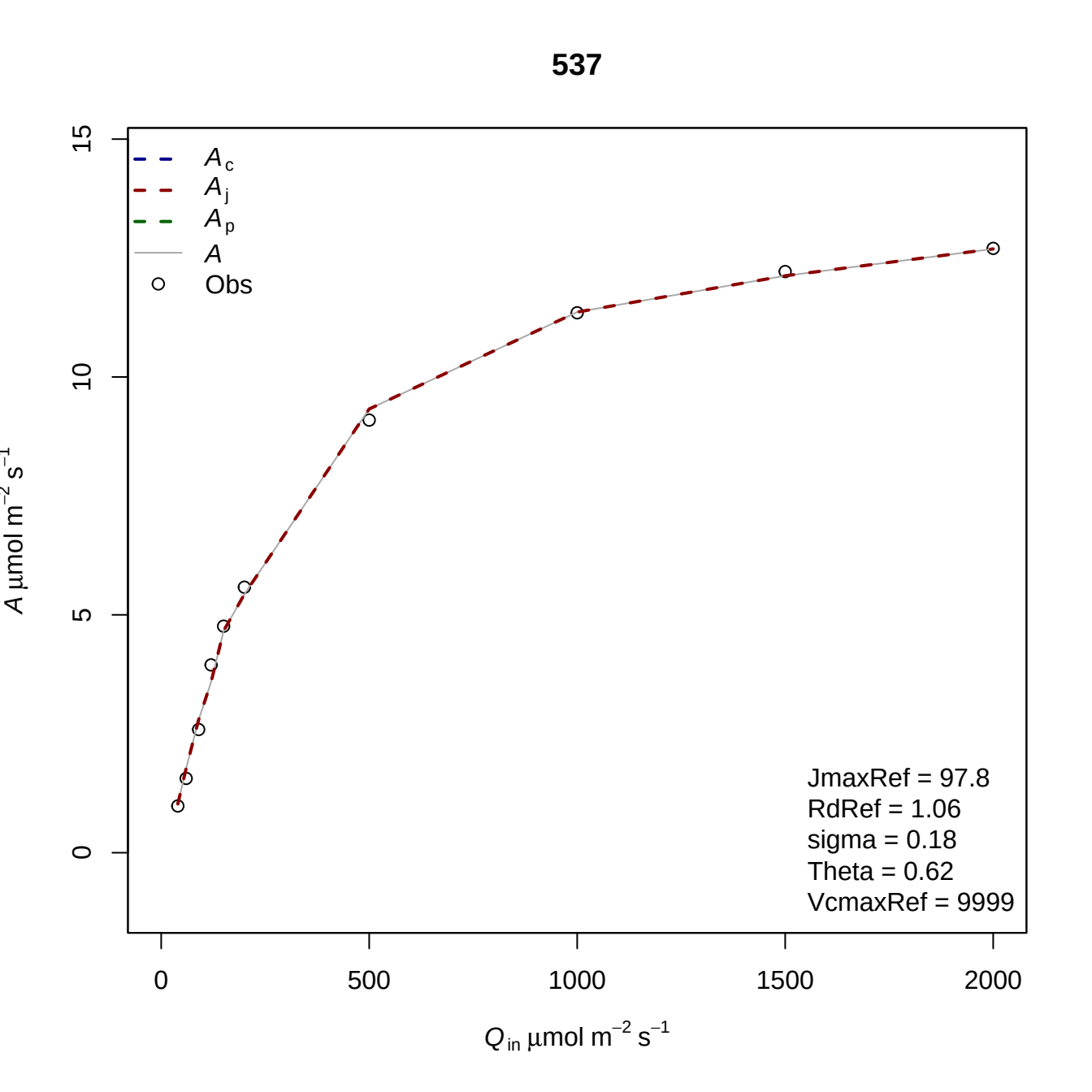

530
A_c
A_j
A_p
A
Obs
JmaxRef = 108.02
RdRef = 0.45
sigma = 0.07
Theta = 0.49
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

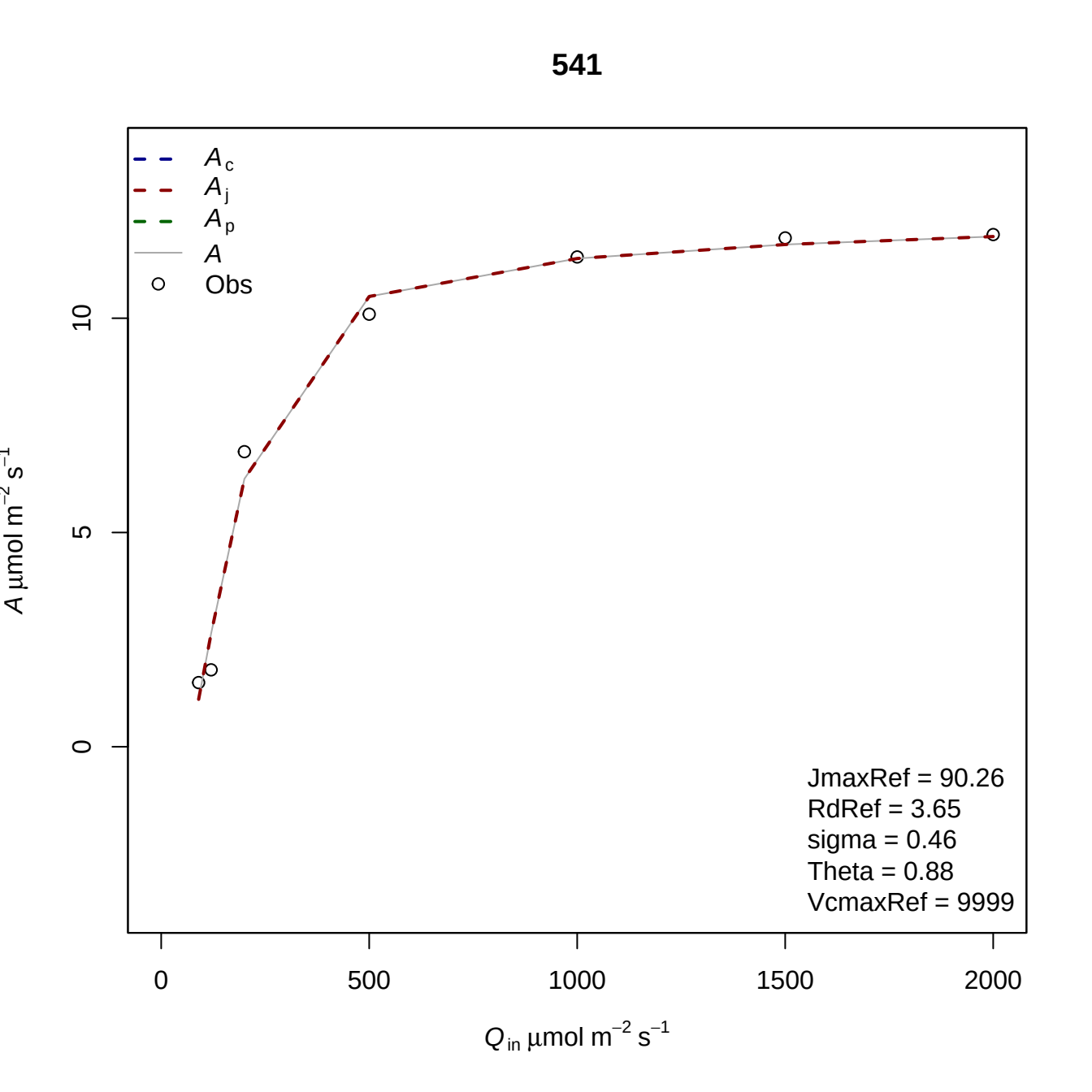

531
A_c
A_j
A_p
A
Obs
JmaxRef = 109.33
RdRef = 0.69
sigma = 0.15
Theta = 0.55
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

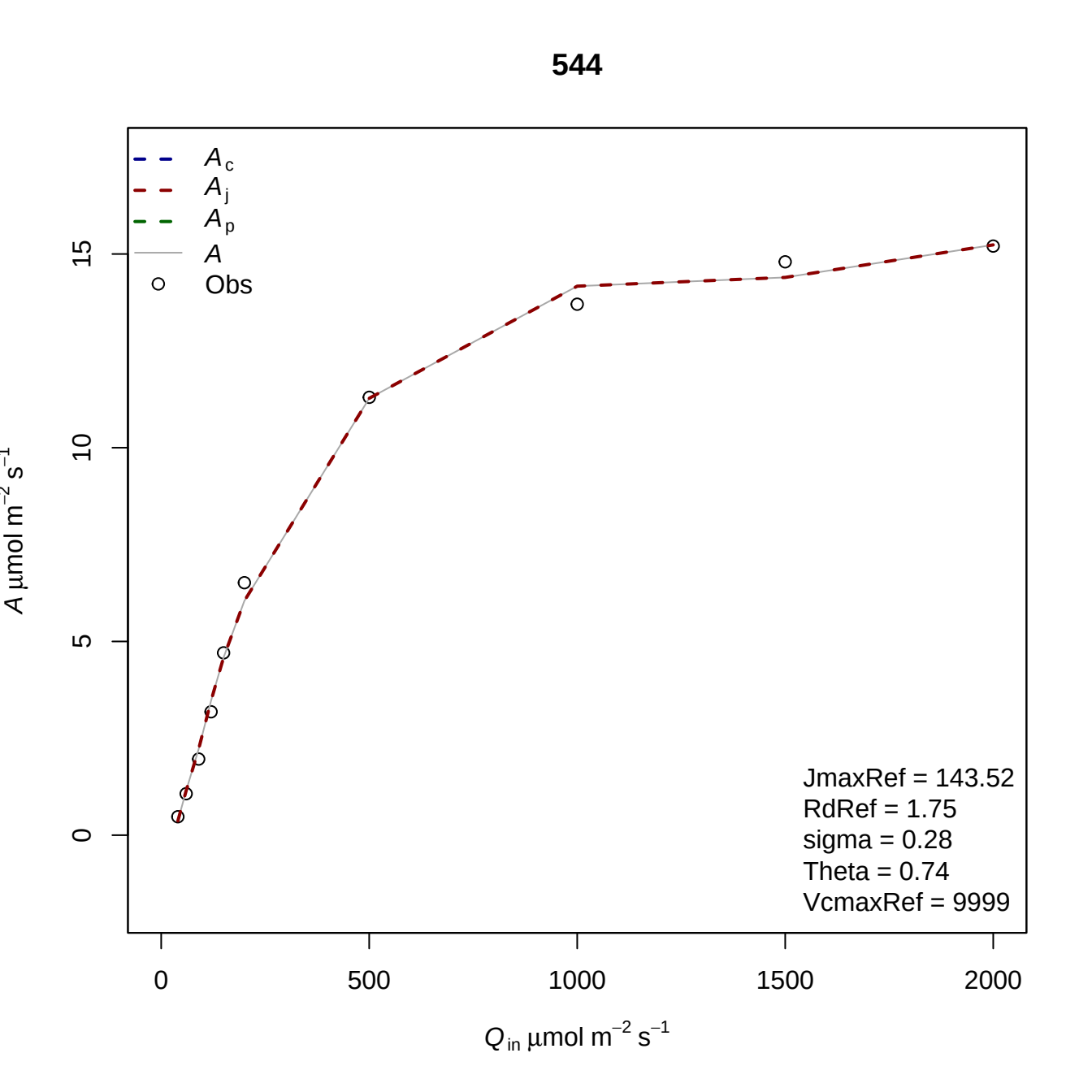

532
A_c
A_j
A_p
A
Obs
JmaxRef = 108.64
RdRef = 1.66
sigma = 0.28
Theta = 0.41
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

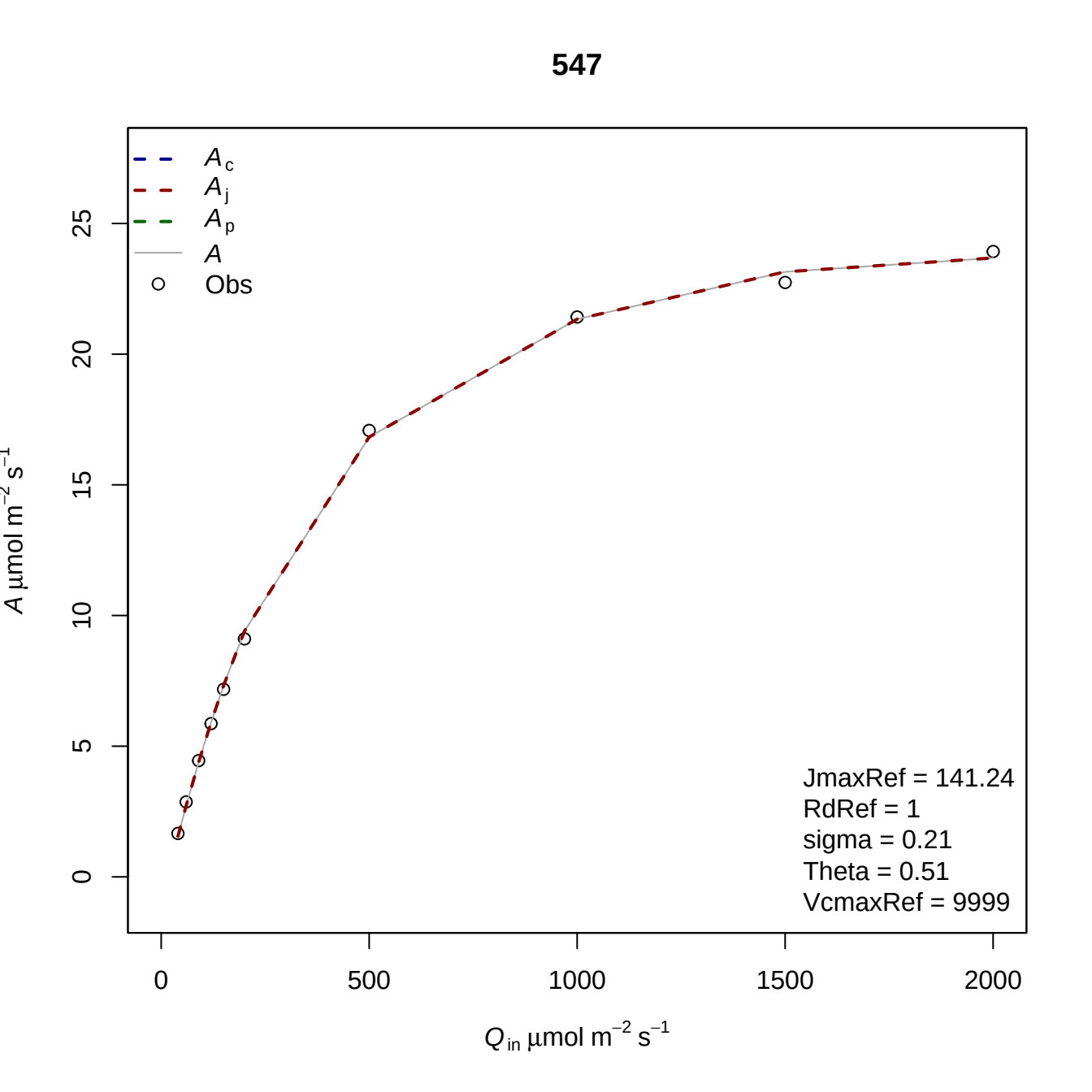

535
A_c
A_j
A_p
A
Obs
JmaxRef = 110.86
RdRef = 0.86
sigma = 0.15
Theta = 0.13
VcmaxRef = 41.7
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

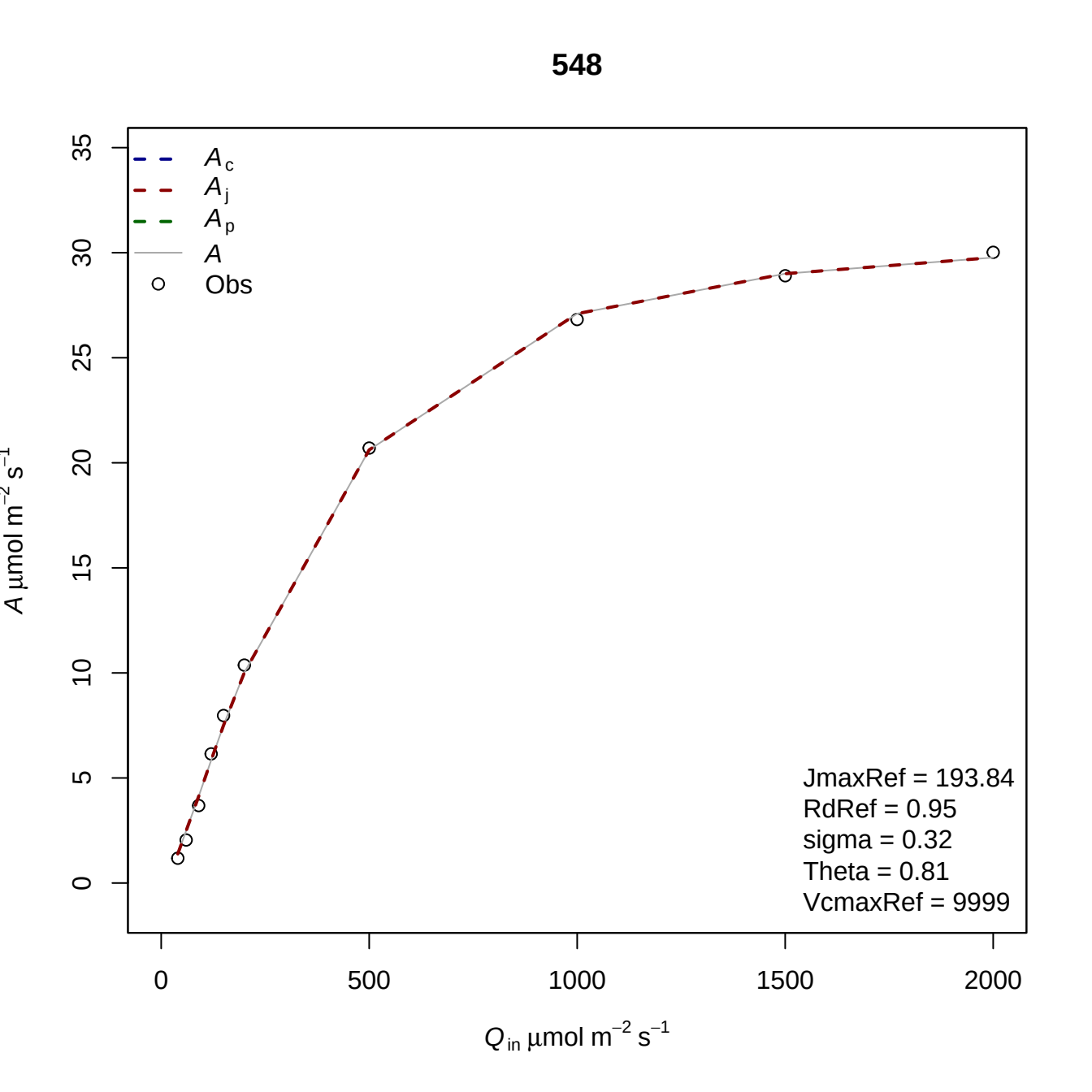

536
A_c
A_j
A_p
A
Obs
JmaxRef = 117.02
RdRef = 0.69
sigma = 0.11
Theta = 0.59
VcmaxRef = 54.98
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

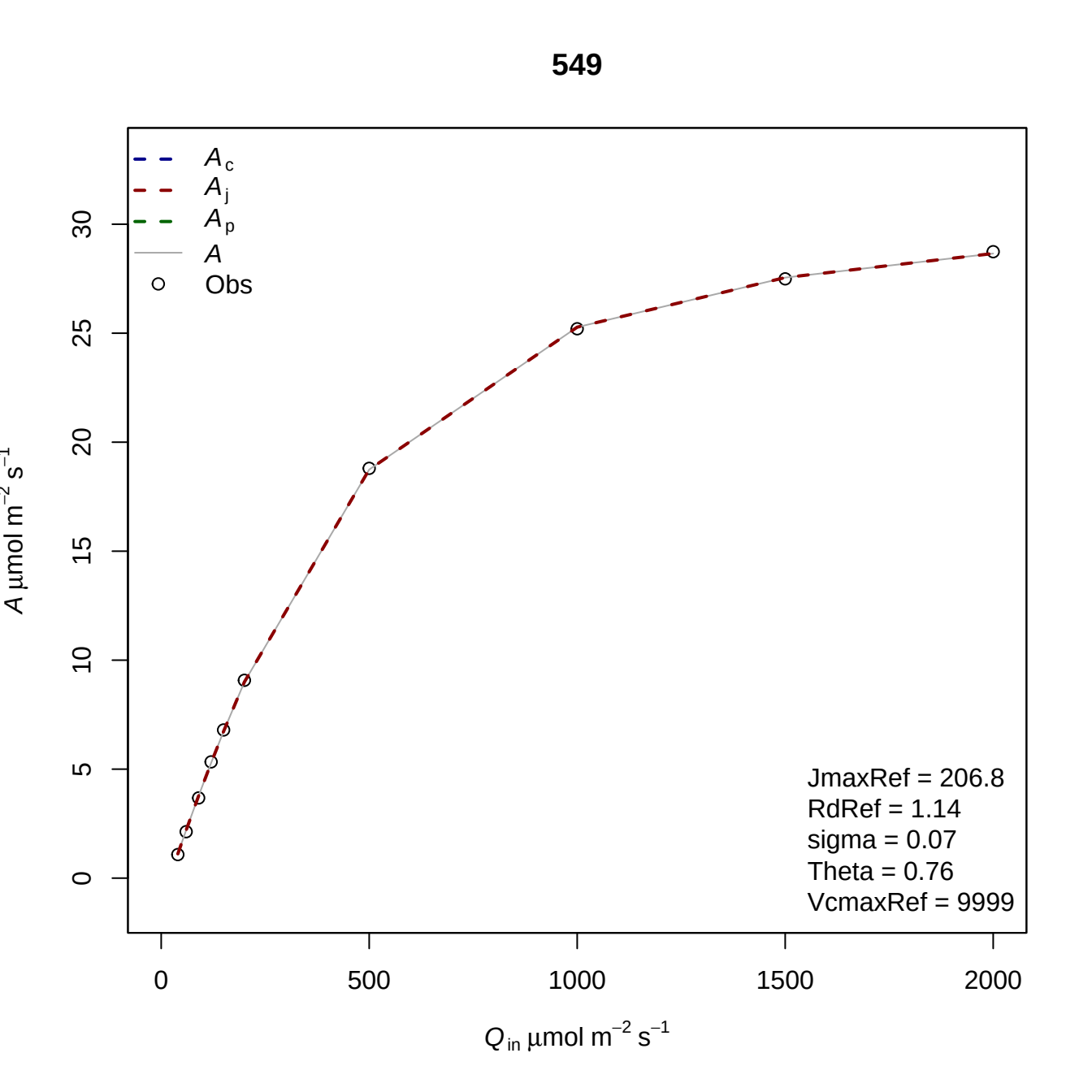

537
A_c
A_j
A_p
A
Obs
JmaxRef = 97.8
RdRef = 1.06
sigma = 0.18
Theta = 0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

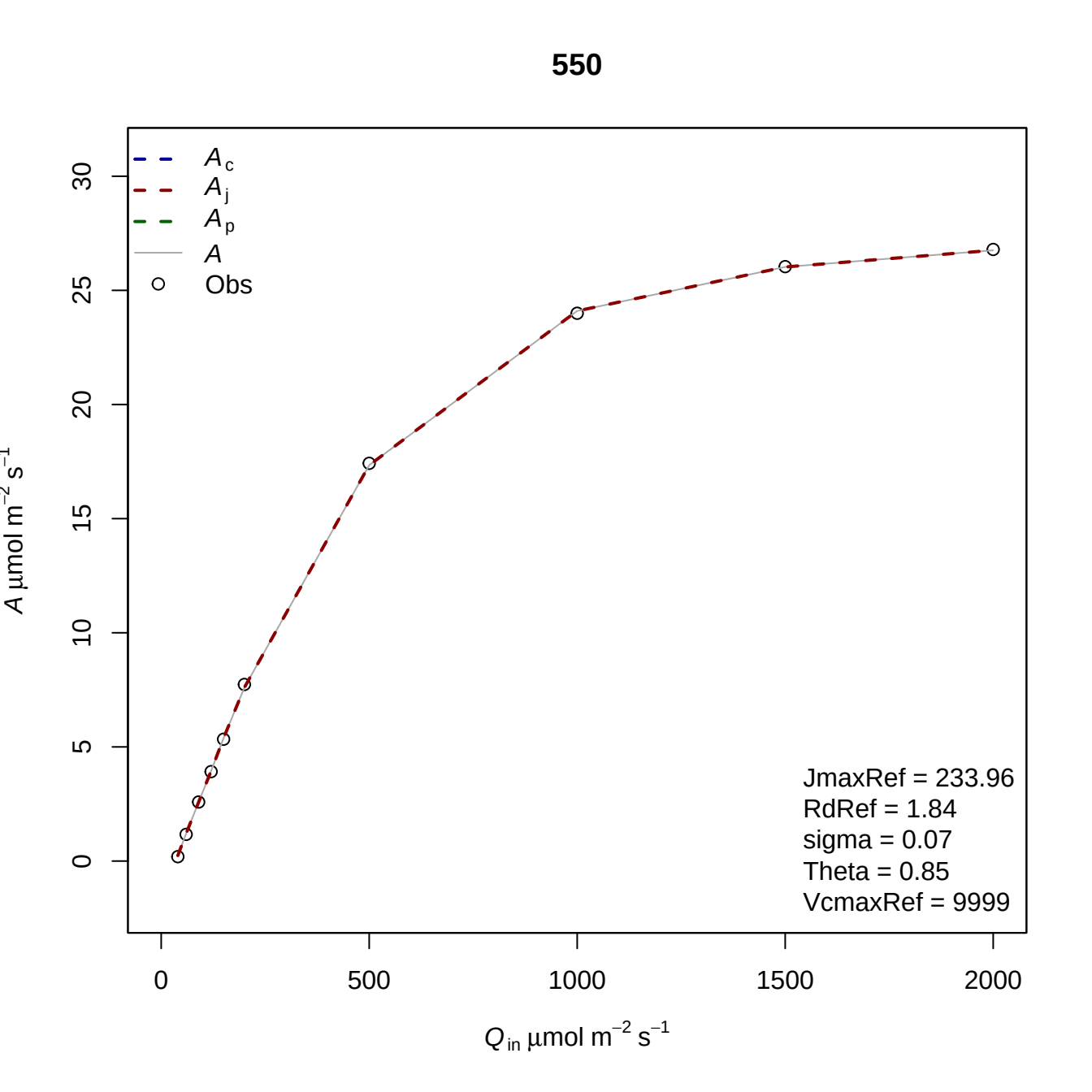

541
A_c
A_j
A_p
A
Obs
JmaxRef = 90.26
RdRef = 3.65
sigma = 0.46
Theta = 0.88
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

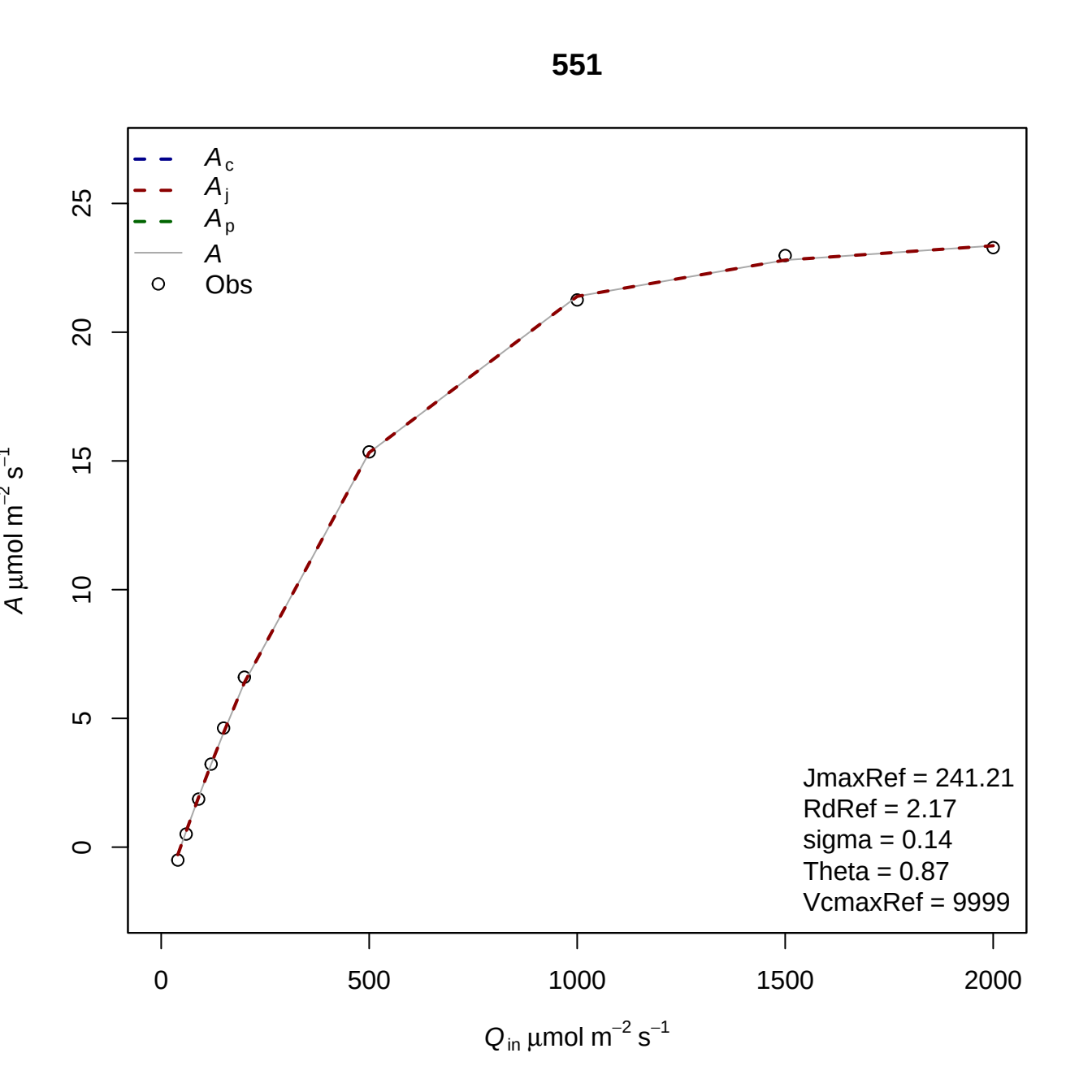

544
A_c
A_j
A_p
A
Obs
JmaxRef = 143.52
RdRef = 1.75
sigma = 0.28
Theta = 0.74
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

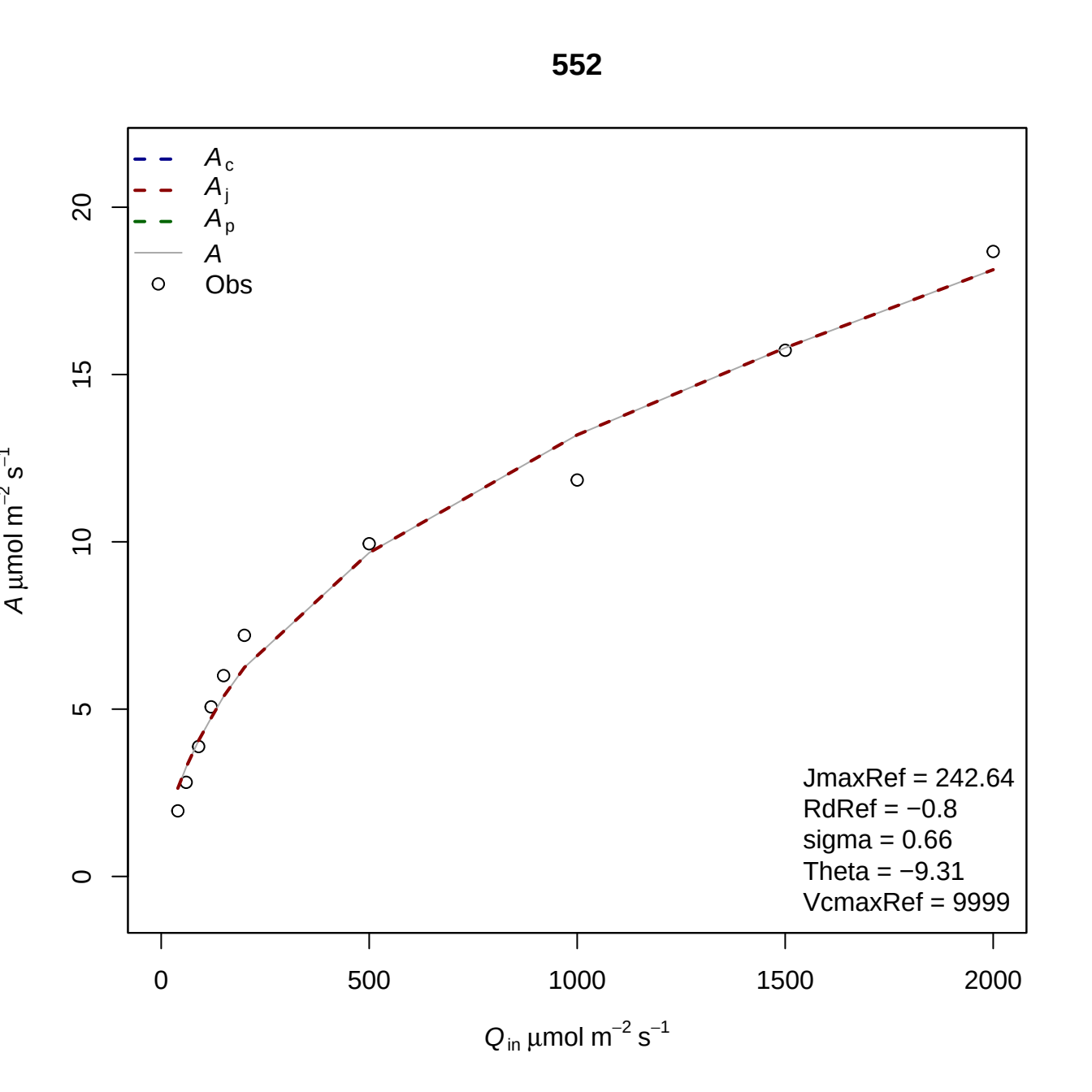

547
A_c
A_j
A_p
A
Obs
JmaxRef = 141.24
RdRef = 1
sigma = 0.21
Theta = 0.51
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

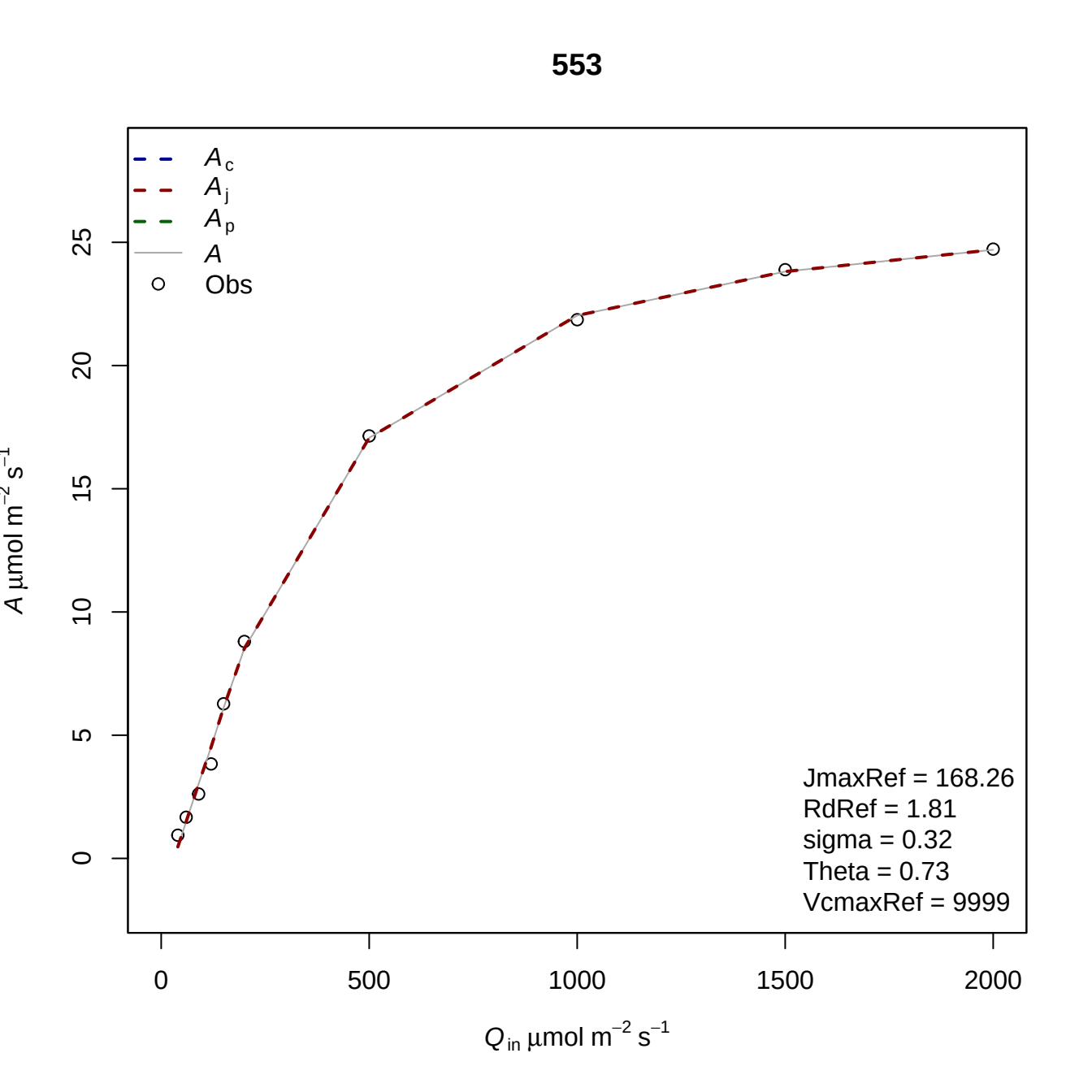

548
A_c
A_j
A_p
A
Obs
JmaxRef = 193.84
RdRef = 0.95
sigma = 0.32
Theta = 0.81
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

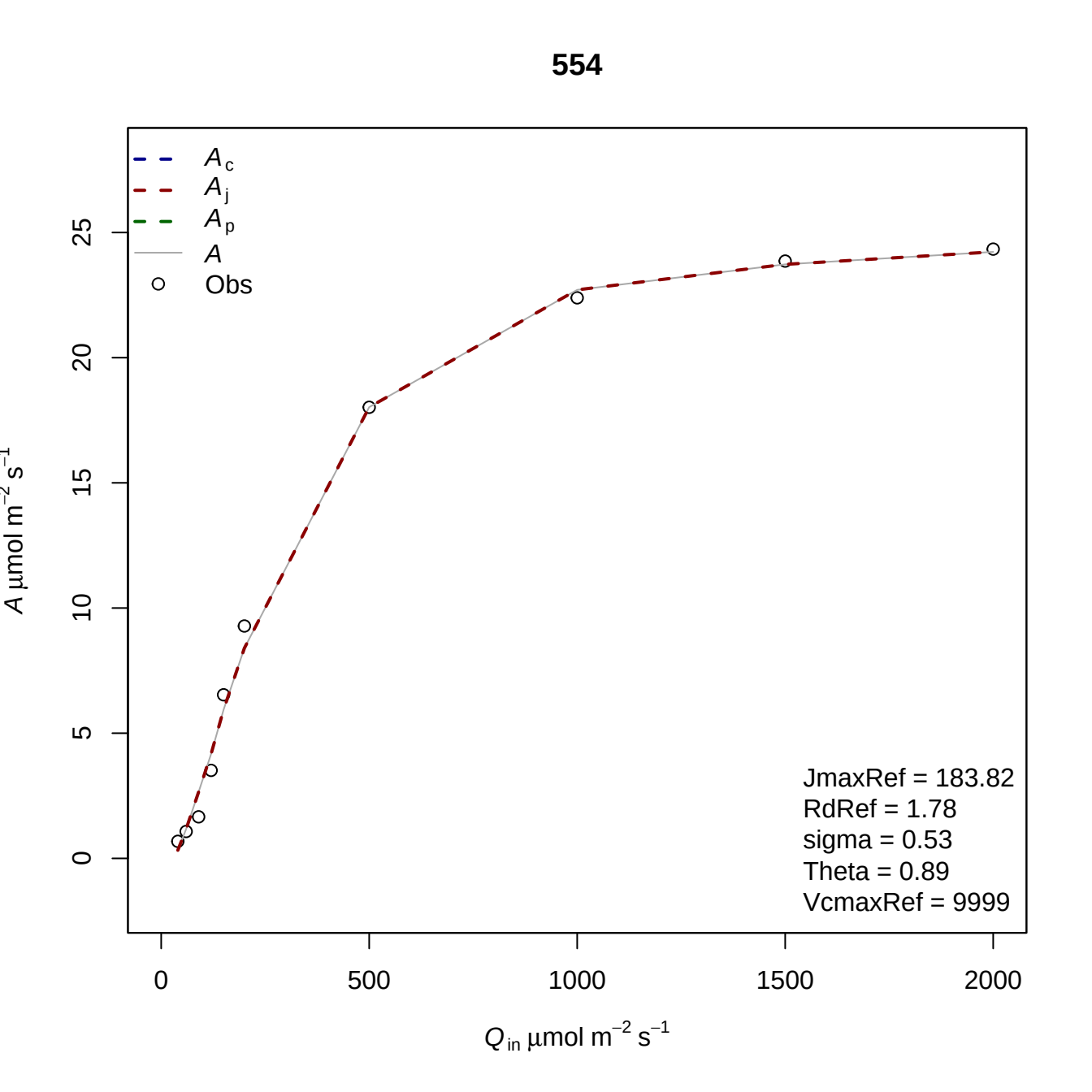

549
A_c
A_j
A_p
A
Obs
JmaxRef = 206.8
RdRef = 1.14
sigma = 0.07
Theta = 0.76
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

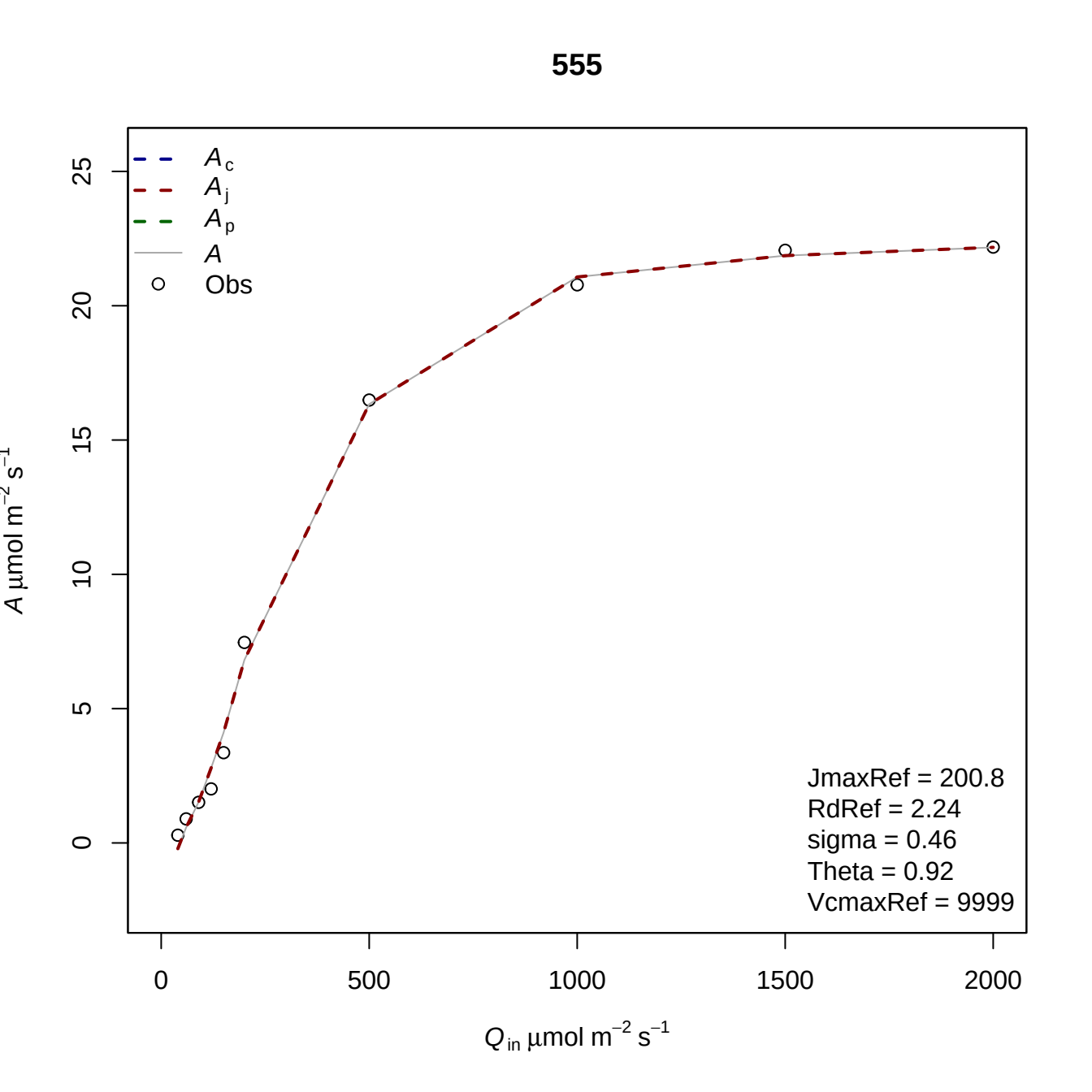

550
A_c
A_j
A_p
A
Obs
JmaxRef = 233.96
RdRef = 1.84
sigma = 0.07
Theta = 0.85
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

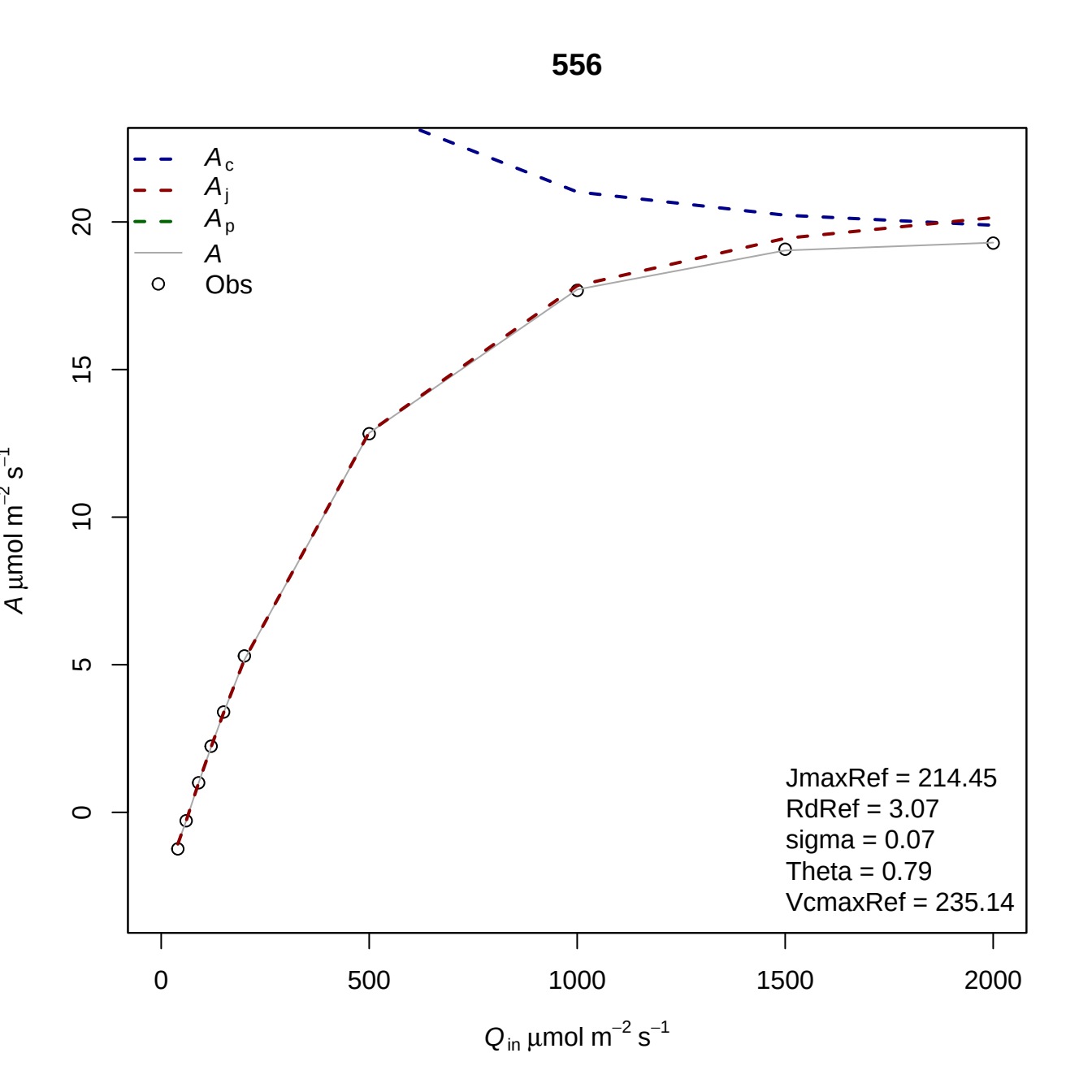

551
A_c
A_j
A_p
A
Obs
JmaxRef = 241.21
RdRef = 2.17
sigma = 0.14
Theta = 0.87
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

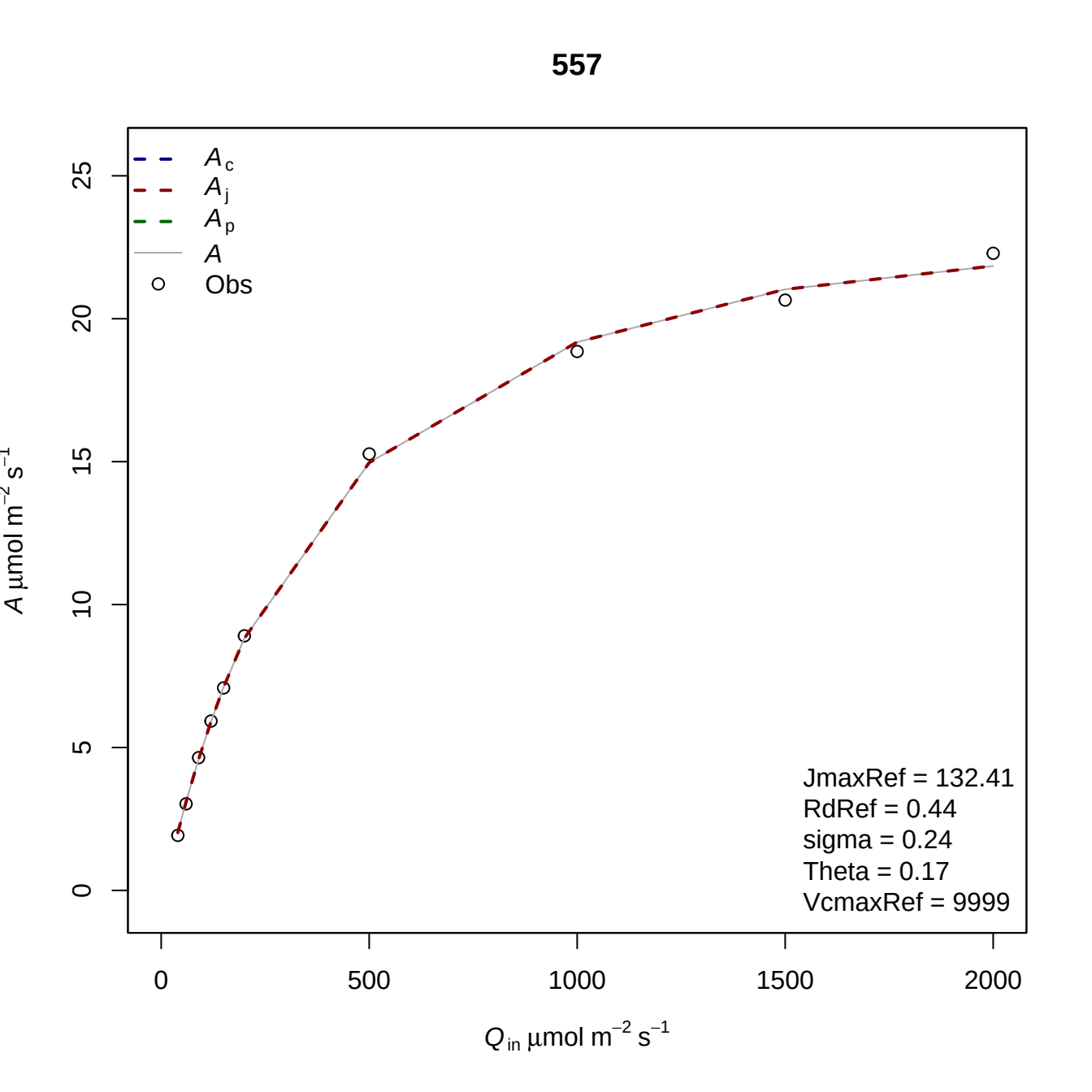

552
A_c
A_j
A_p
A
Obs
JmaxRef = 242.64
RdRef = −0.8
sigma = 0.66
Theta = −9.31
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

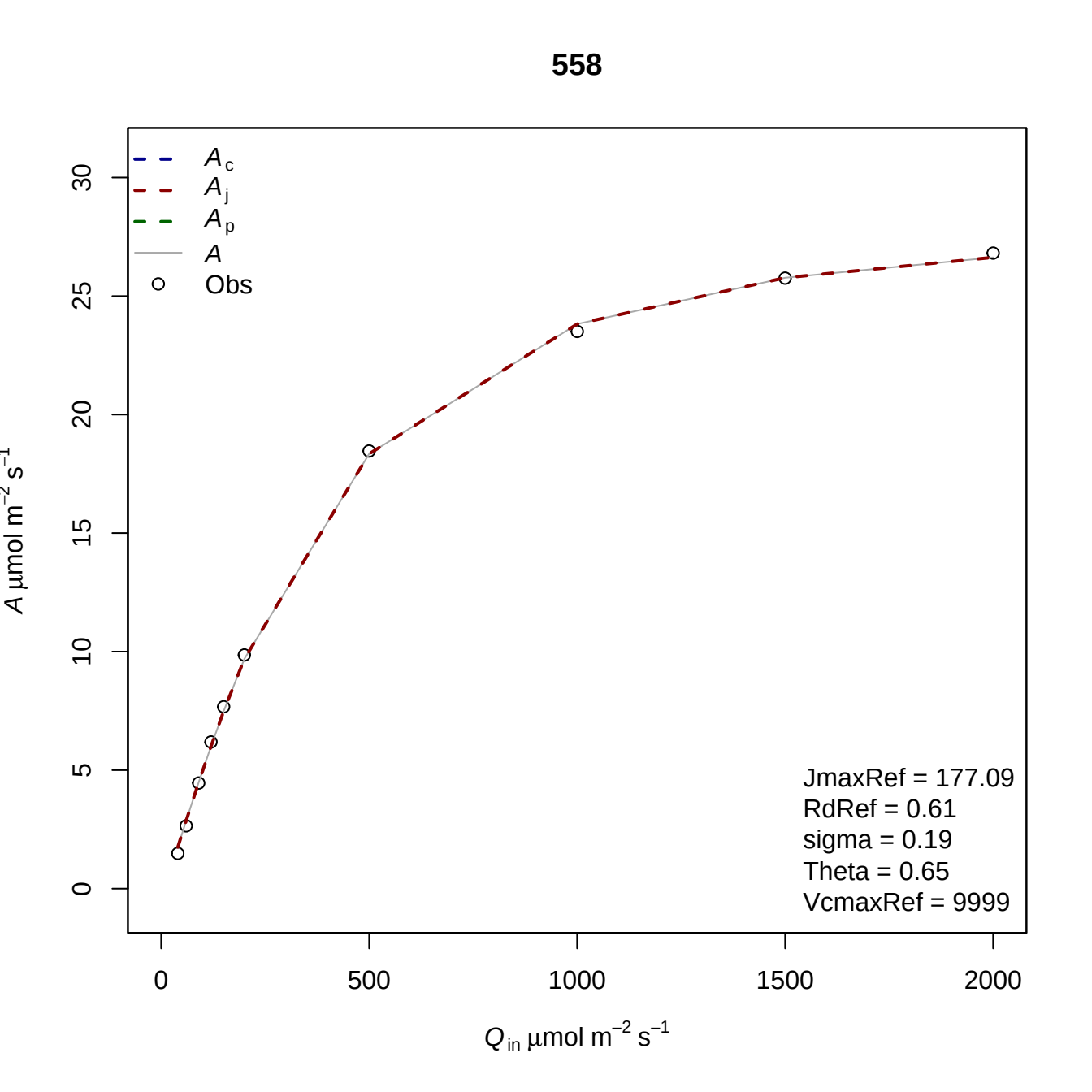

553
A_c
A_j
A_p
A
Obs
JmaxRef = 168.26
RdRef = 1.81
sigma = 0.32
Theta = 0.73
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

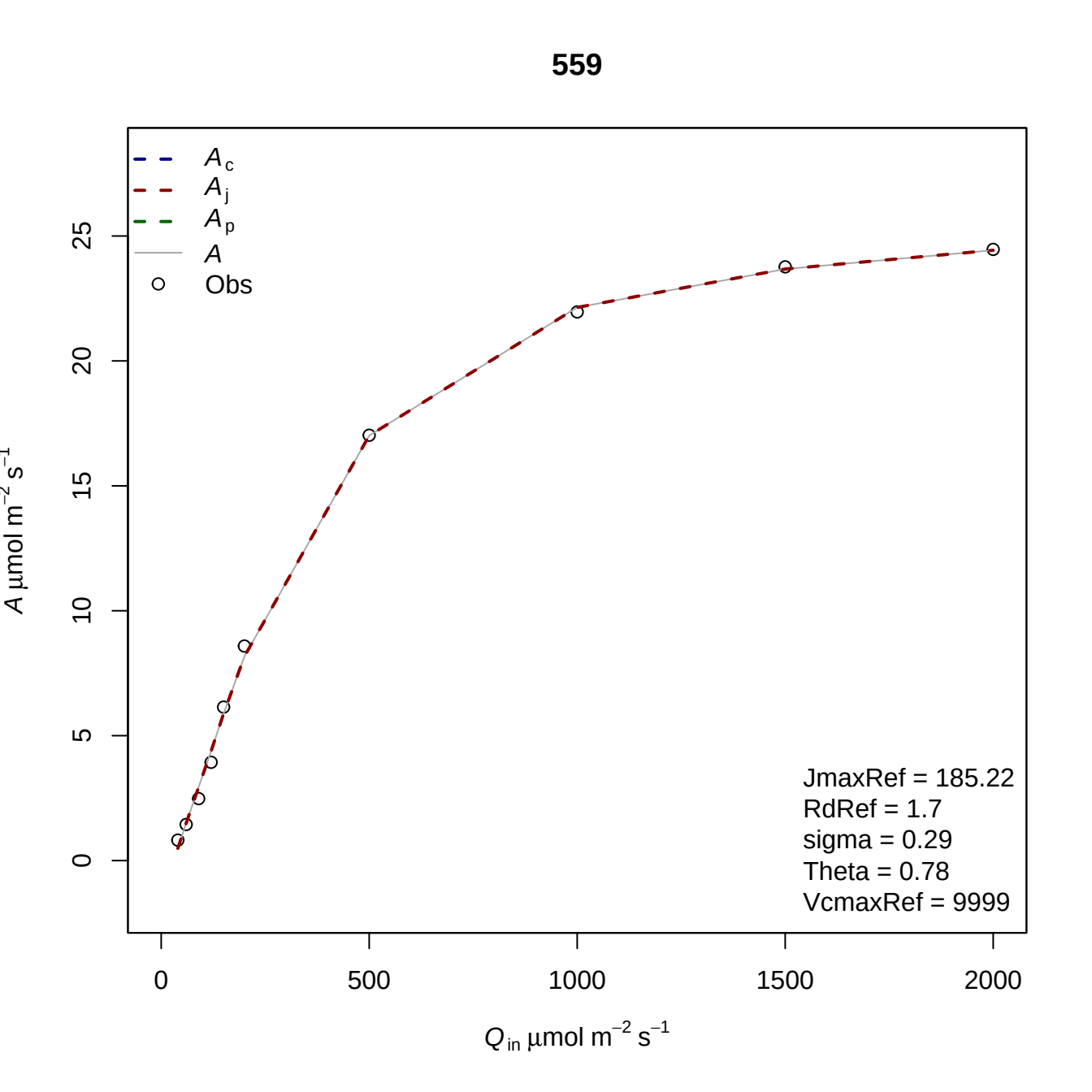

554
A_c
A_j
A_p
A
Obs
JmaxRef = 183.82
RdRef = 1.78
sigma = 0.53
Theta = 0.89
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

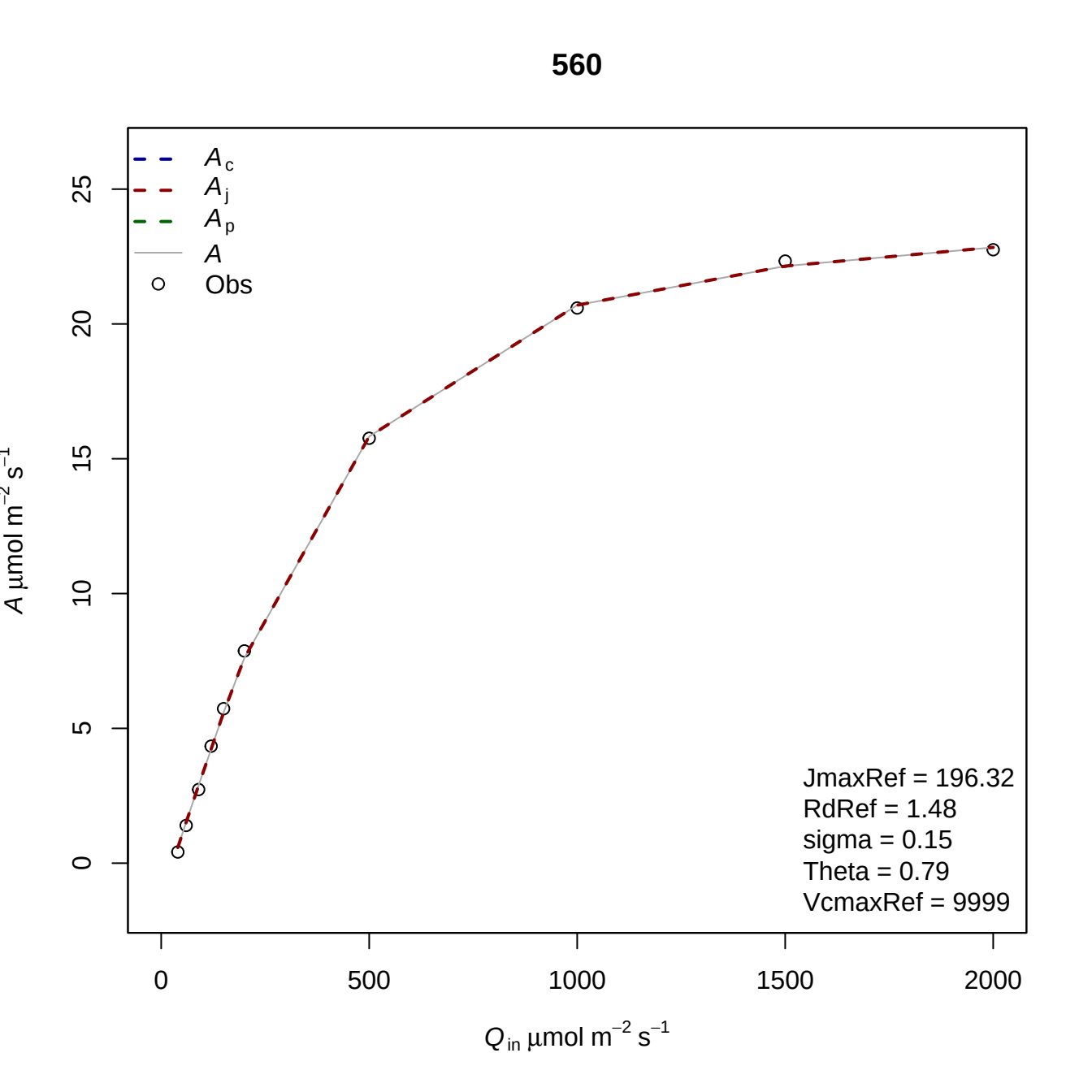

555
A_c
A_j
A_p
A
Obs
JmaxRef = 200.8
RdRef = 2.24
sigma = 0.46
Theta = 0.92
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

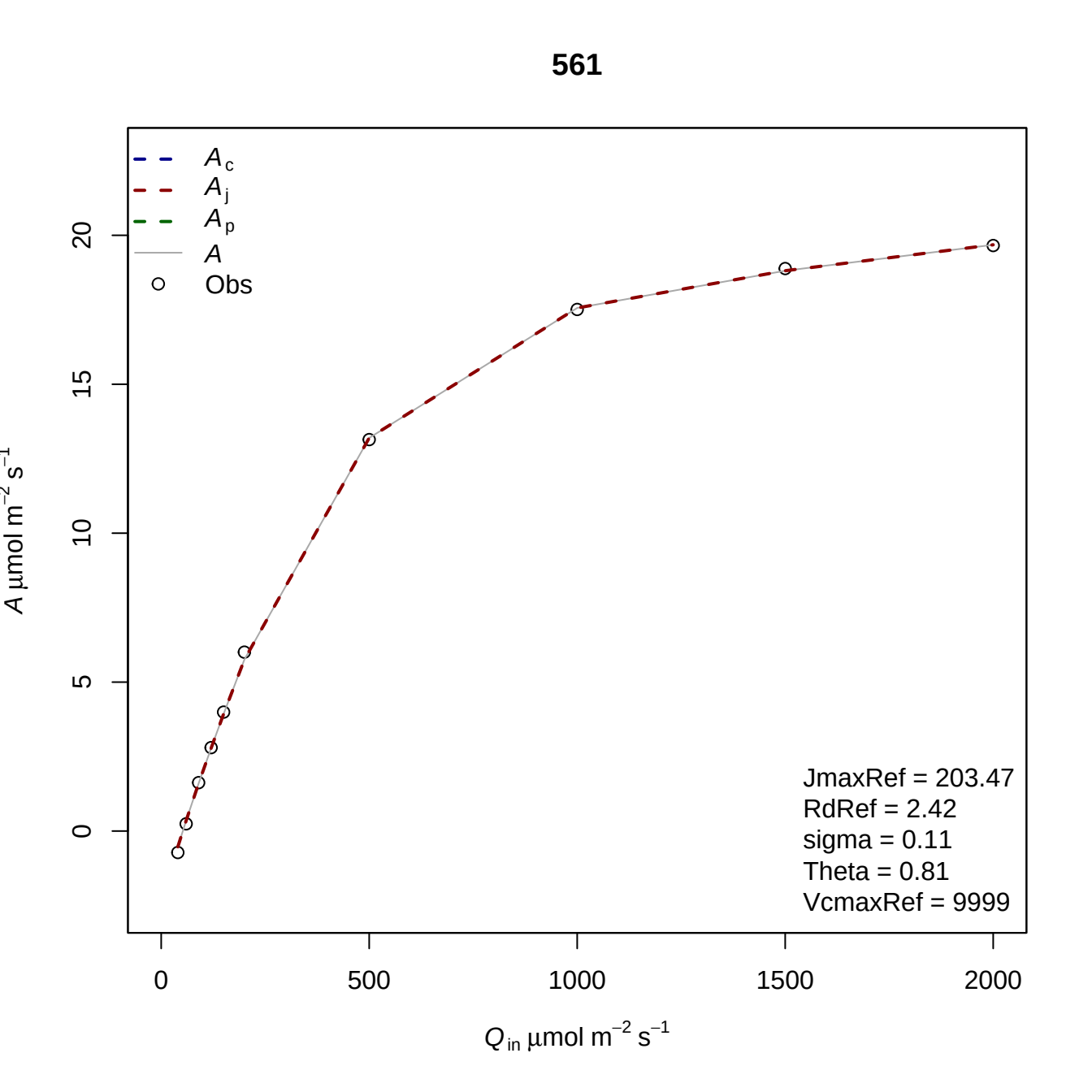

556
A_c
A_j
A_p
A
Obs
JmaxRef = 214.45
RdRef = 3.07
sigma = 0.07
Theta = 0.79
VcmaxRef = 235.14
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

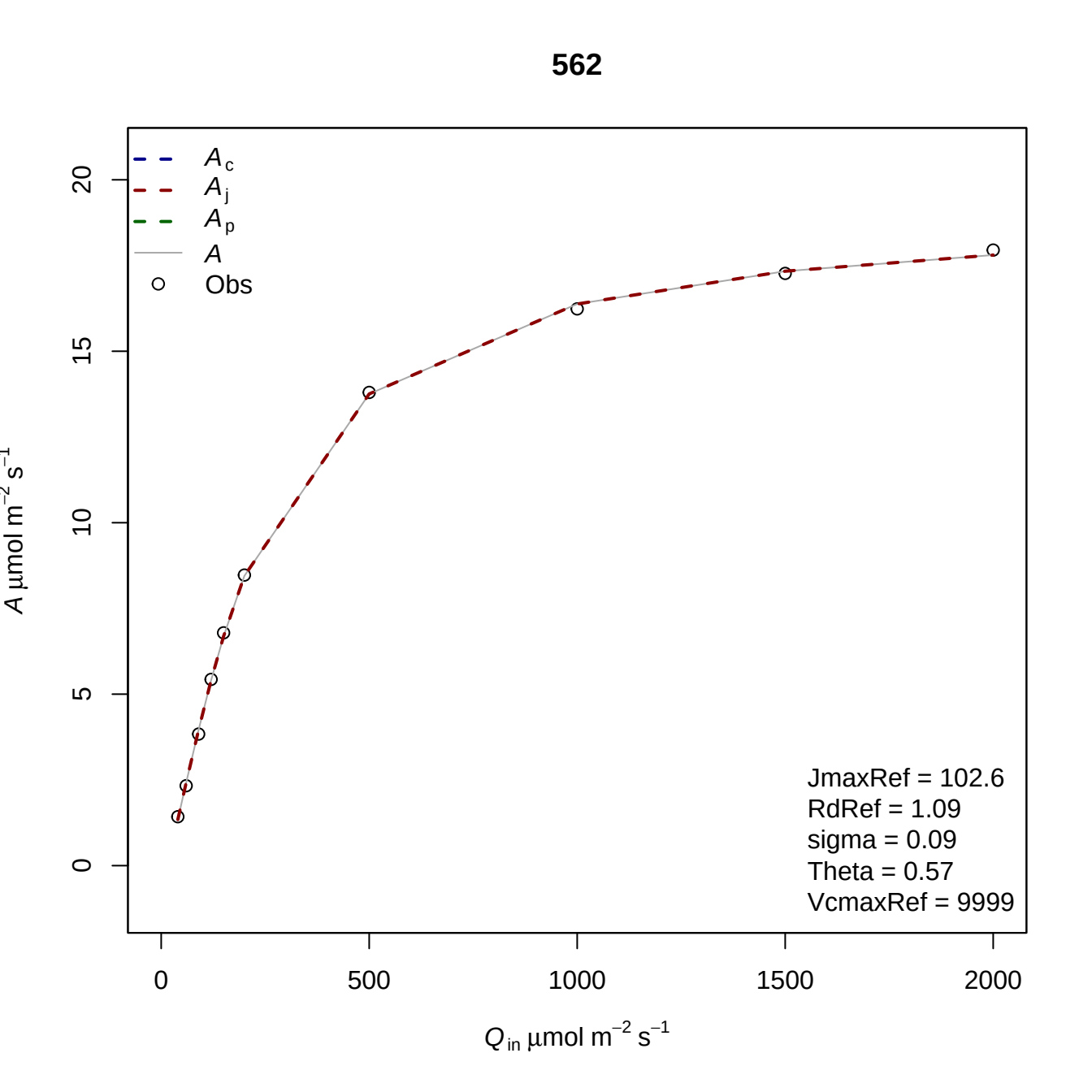

557
A_c
A_j
A_p
A
Obs
JmaxRef = 132.41
RdRef = 0.44
sigma = 0.24
Theta = 0.17
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

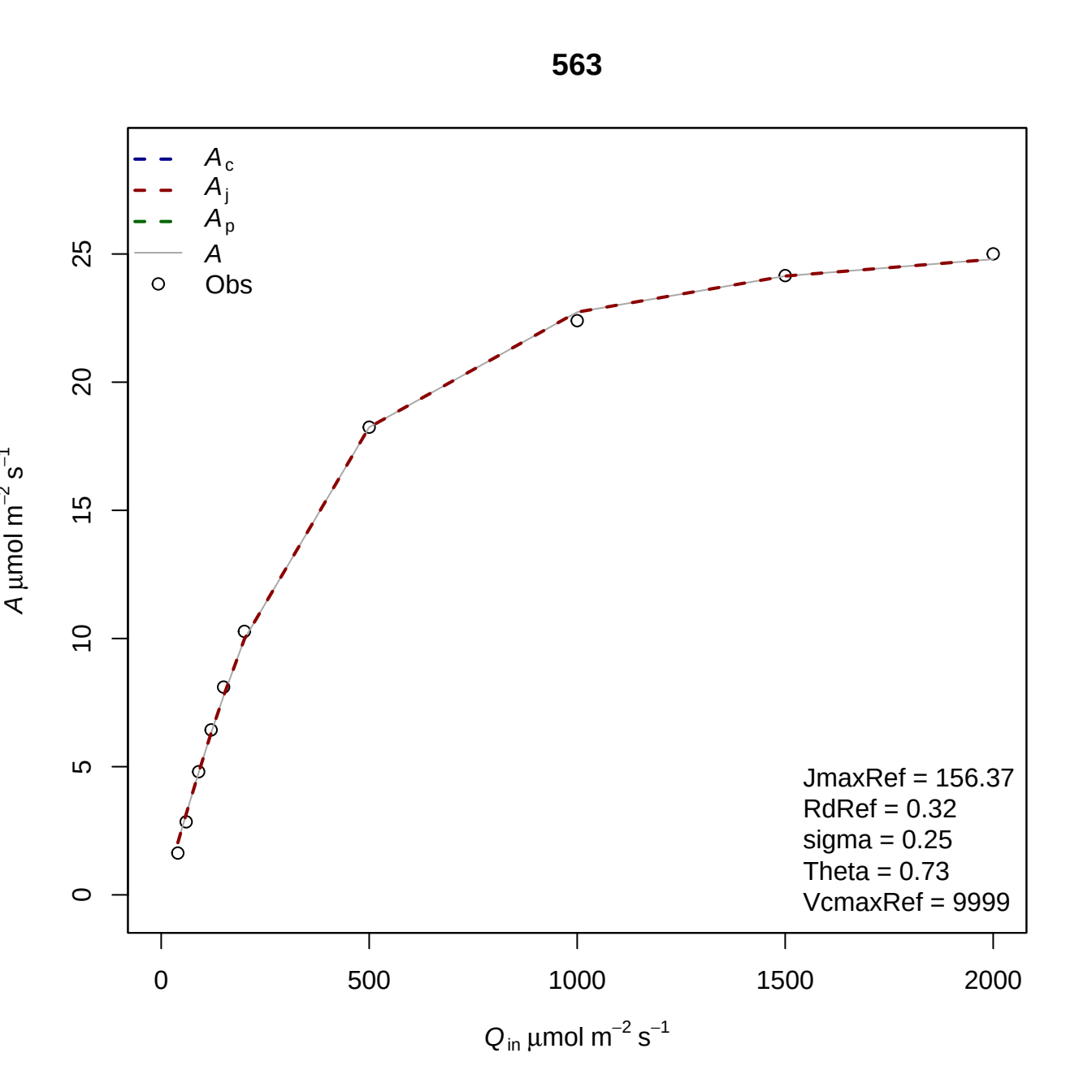

558
A_c
A_j
A_p
A
Obs
JmaxRef = 177.09
RdRef = 0.61
sigma = 0.19
Theta = 0.65
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

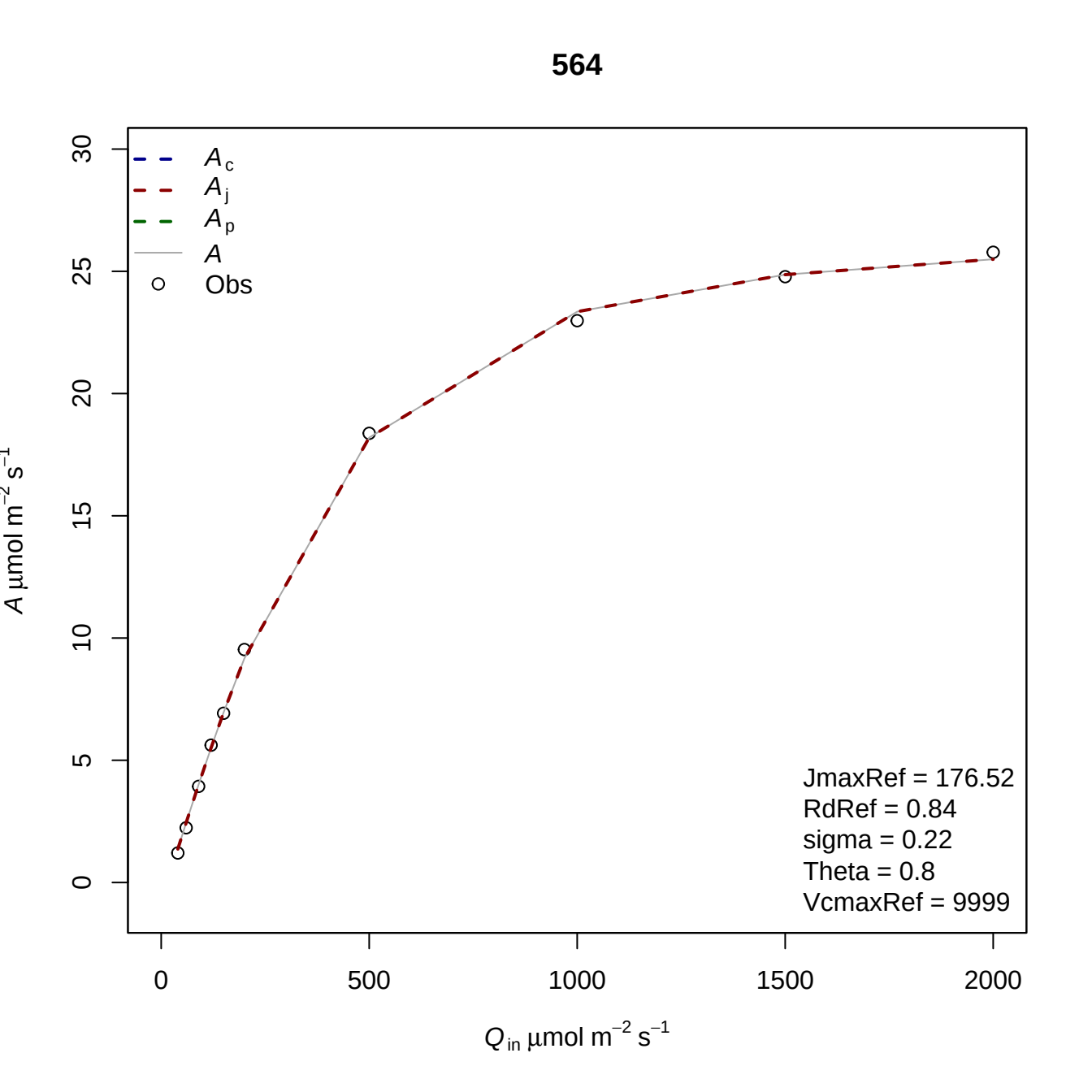

559
A_c
A_j
A_p
A
Obs
JmaxRef = 185.22
RdRef = 1.7
sigma = 0.29
Theta = 0.78
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

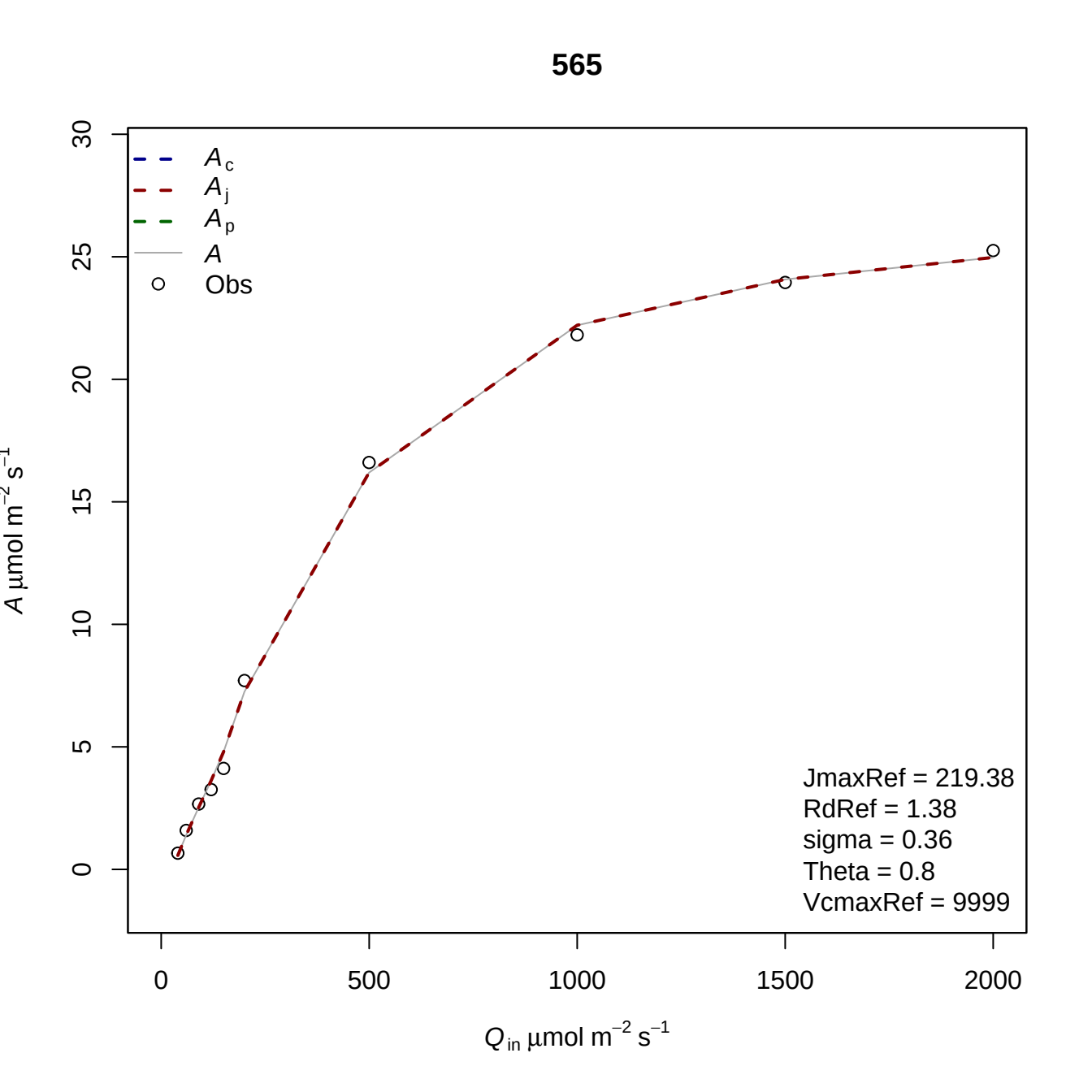

560
A_c
A_j
A_p
A
Obs
JmaxRef = 196.32
RdRef = 1.48
sigma = 0.15
Theta = 0.79
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

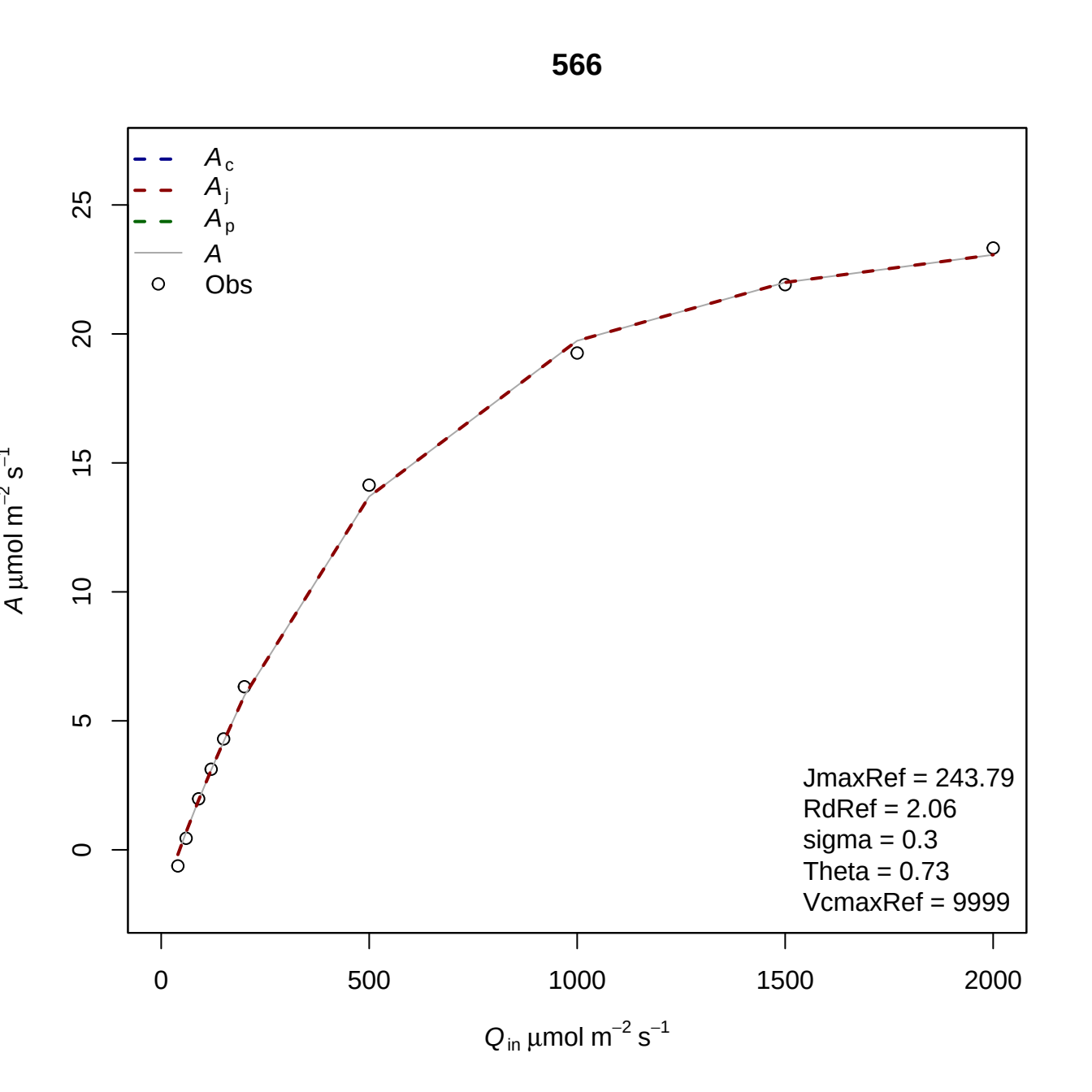

561
A_c
A_j
A_p
A
Obs
JmaxRef = 203.47
RdRef = 2.42
sigma = 0.11
Theta = 0.81
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

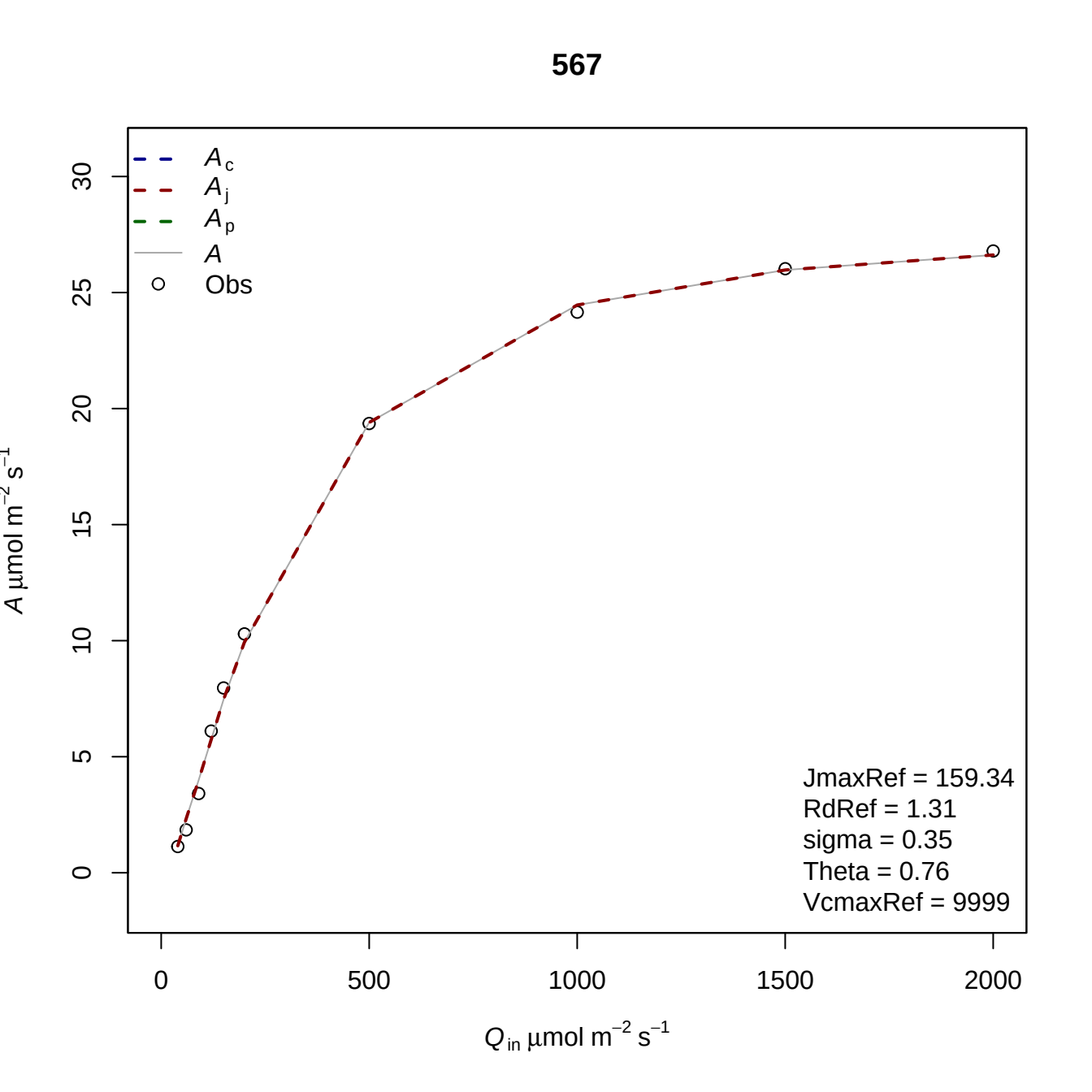

562
A_c
A_j
A_p
A
Obs
JmaxRef = 102.6
RdRef = 1.09
sigma = 0.09
Theta = 0.57
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

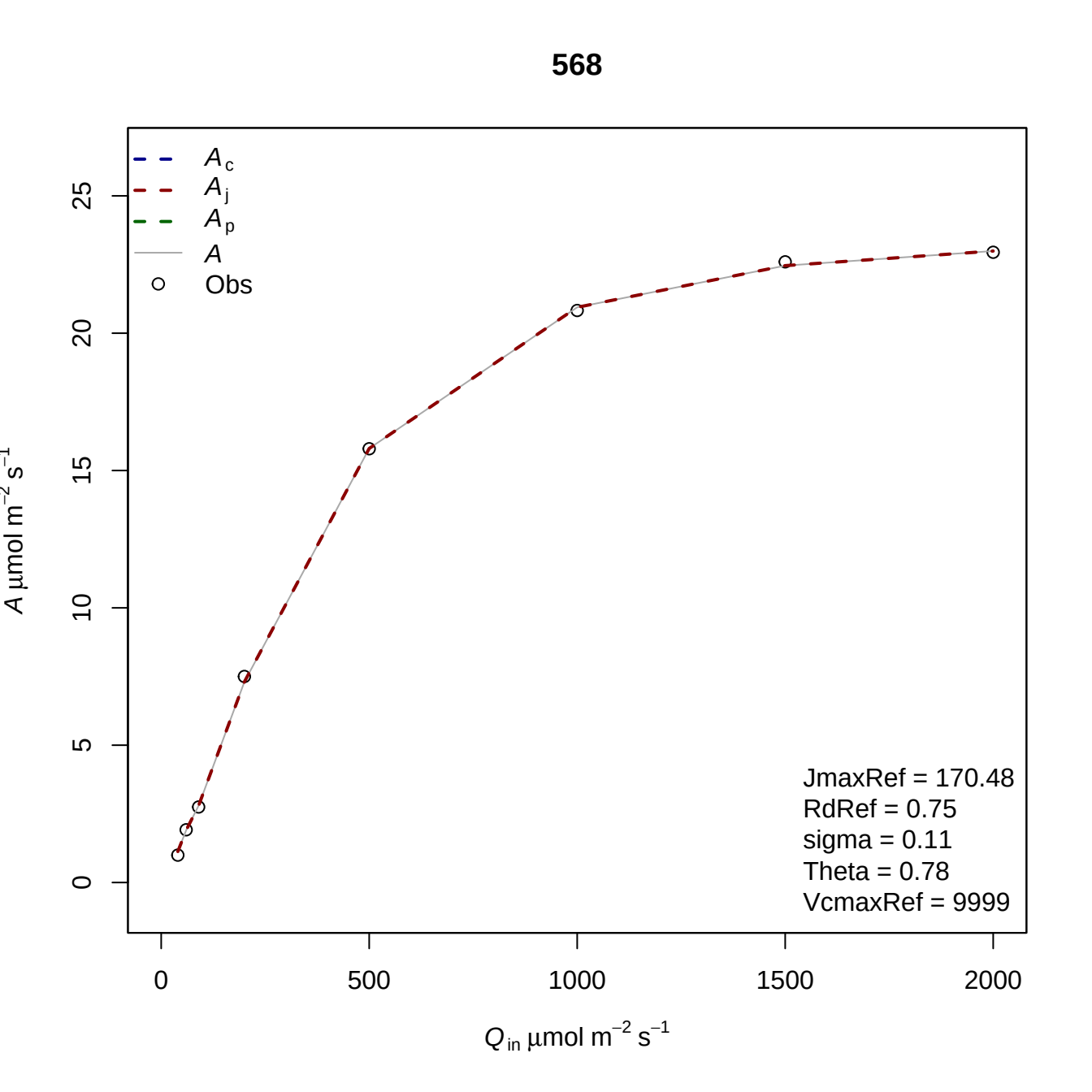

563
A_c
A_j
A_p
A
Obs
JmaxRef = 156.37
RdRef = 0.32
sigma = 0.25
Theta = 0.73
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

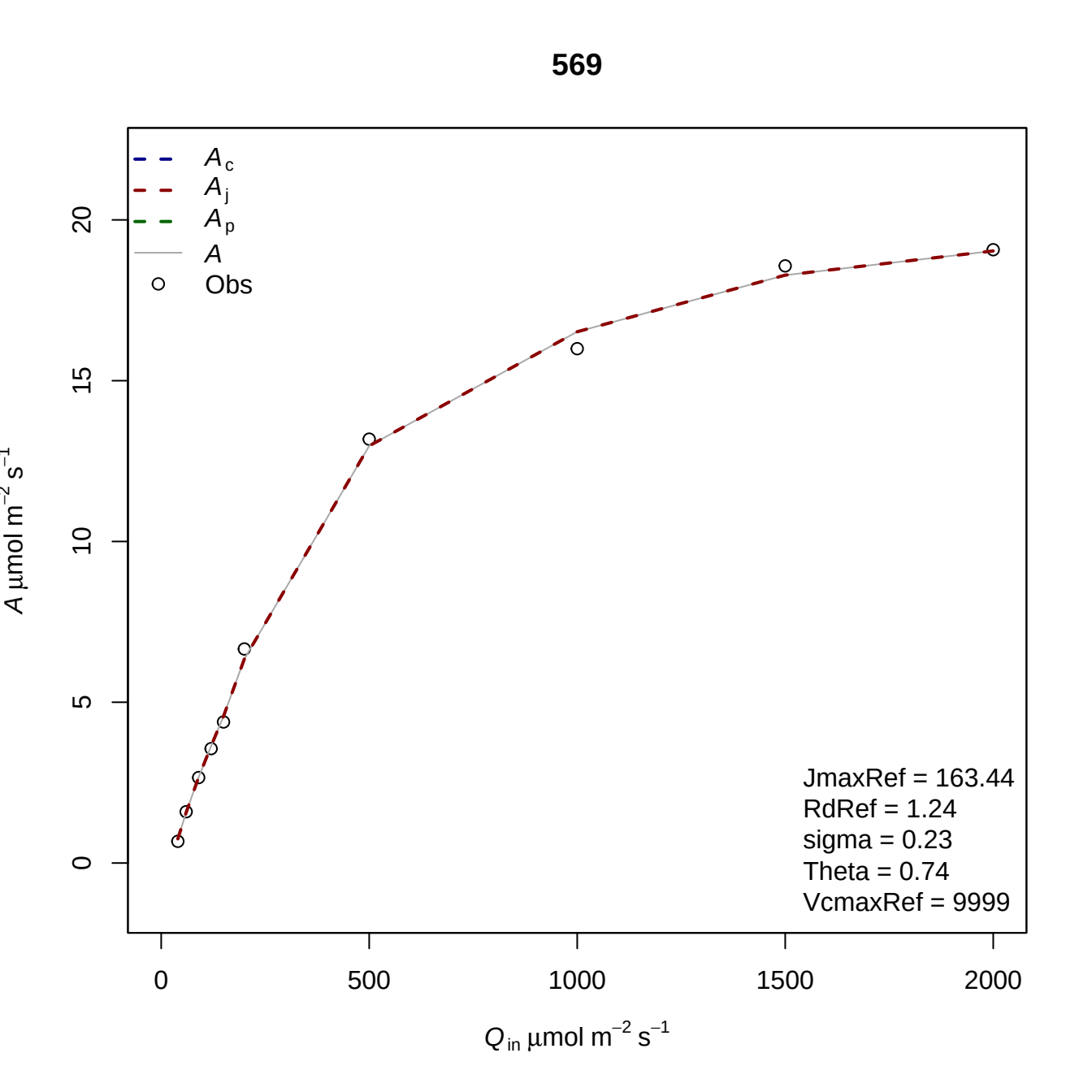

564
A_c
A_j
A_p
A
Obs
JmaxRef = 176.52
RdRef = 0.84
sigma = 0.22
Theta = 0.8
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

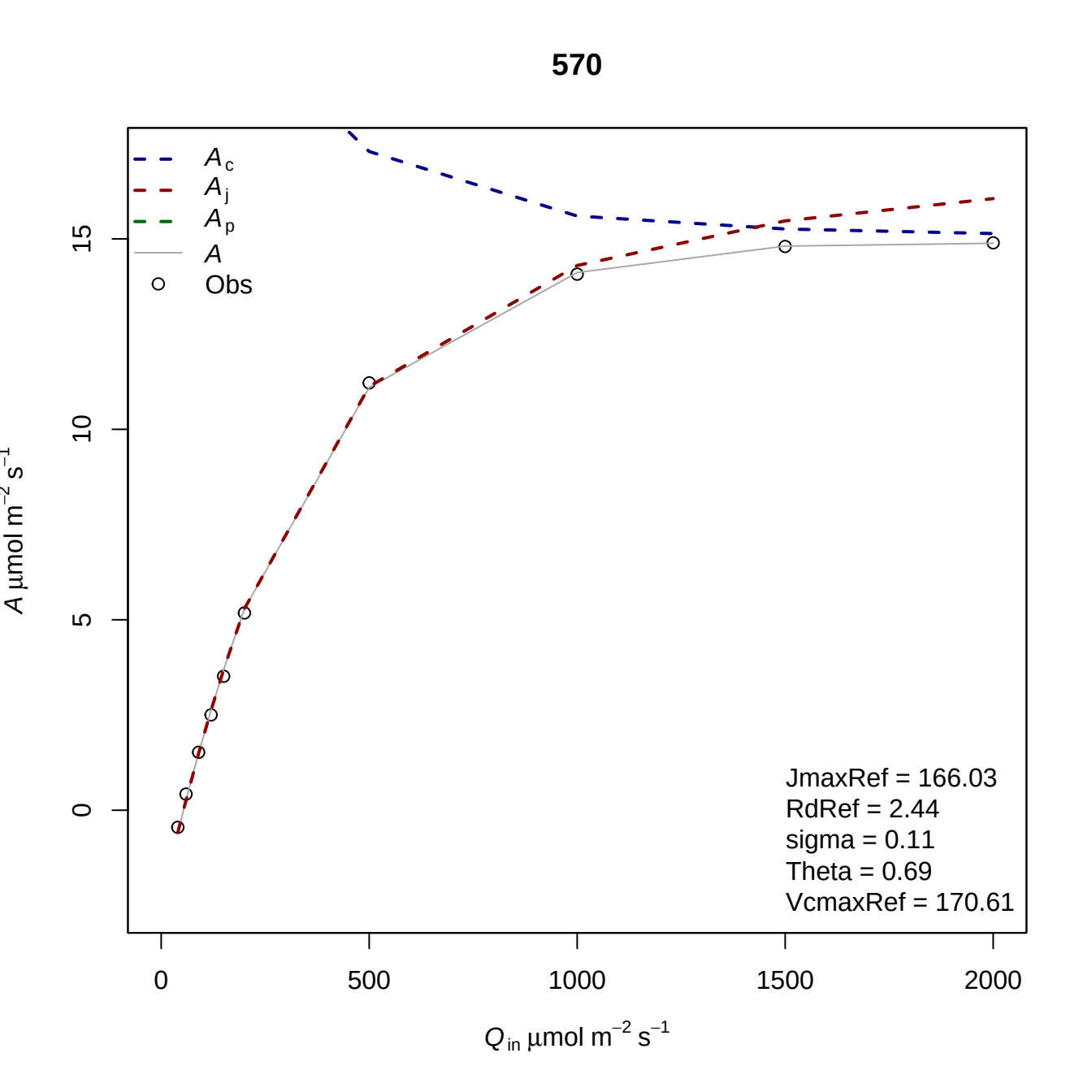

565
A_c
A_j
A_p
A
Obs
JmaxRef = 219.38
RdRef = 1.38
sigma = 0.36
Theta = 0.8
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

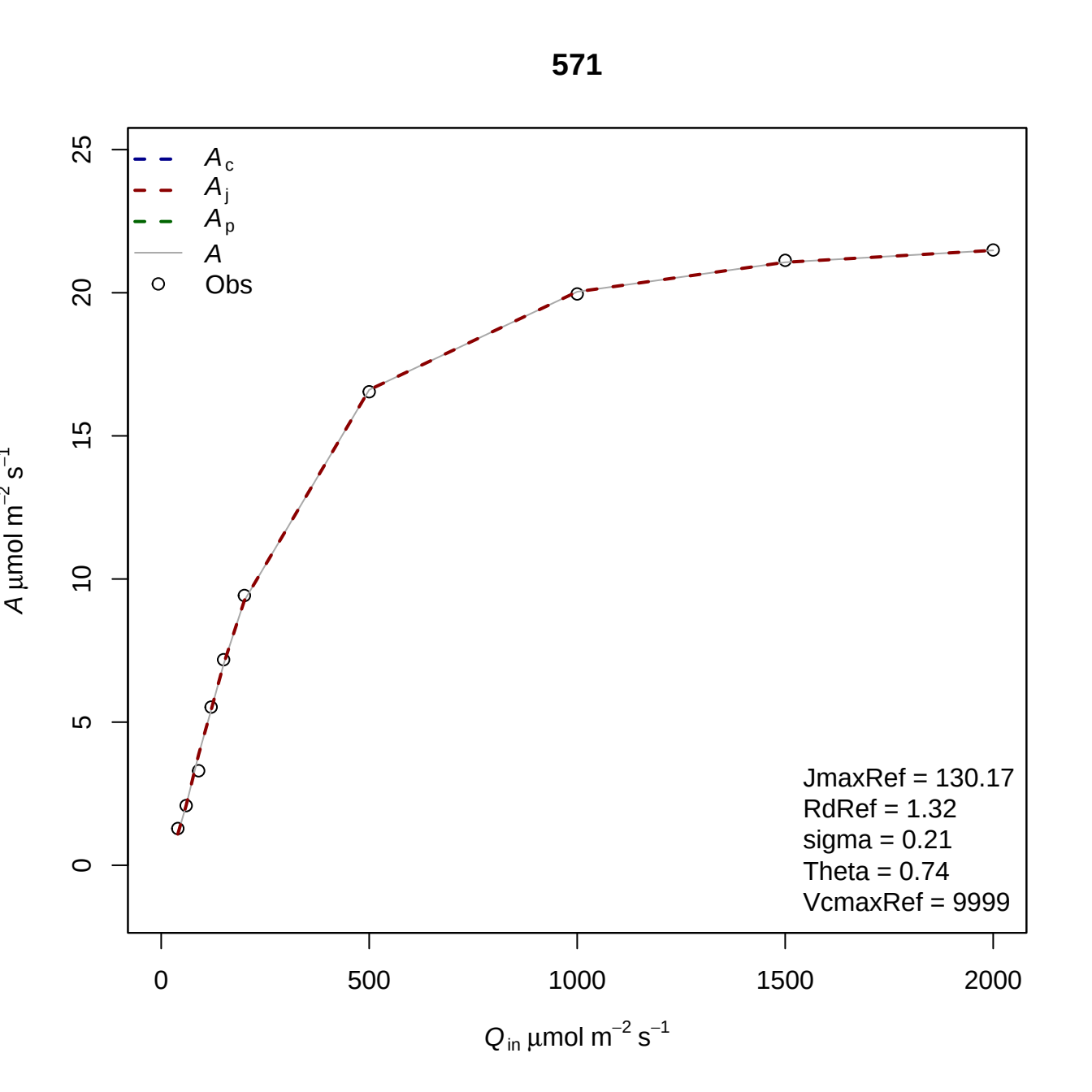

566
A_c
A_j
A_p
A
Obs
JmaxRef = 243.79
RdRef = 2.06
sigma = 0.3
Theta = 0.73
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

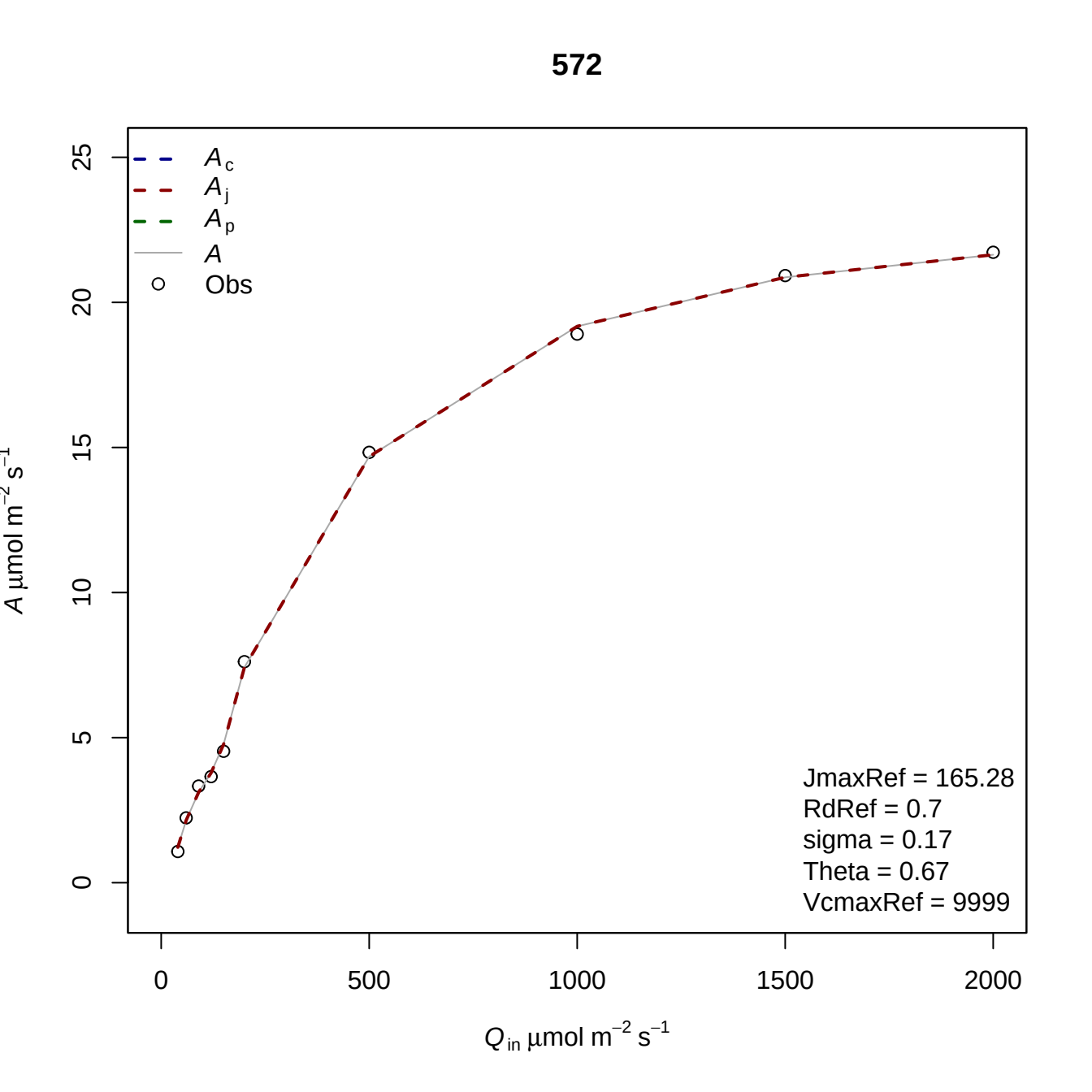

567
A_c
A_j
A_p
A
Obs
JmaxRef = 159.34
RdRef = 1.31
sigma = 0.35
Theta = 0.76
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

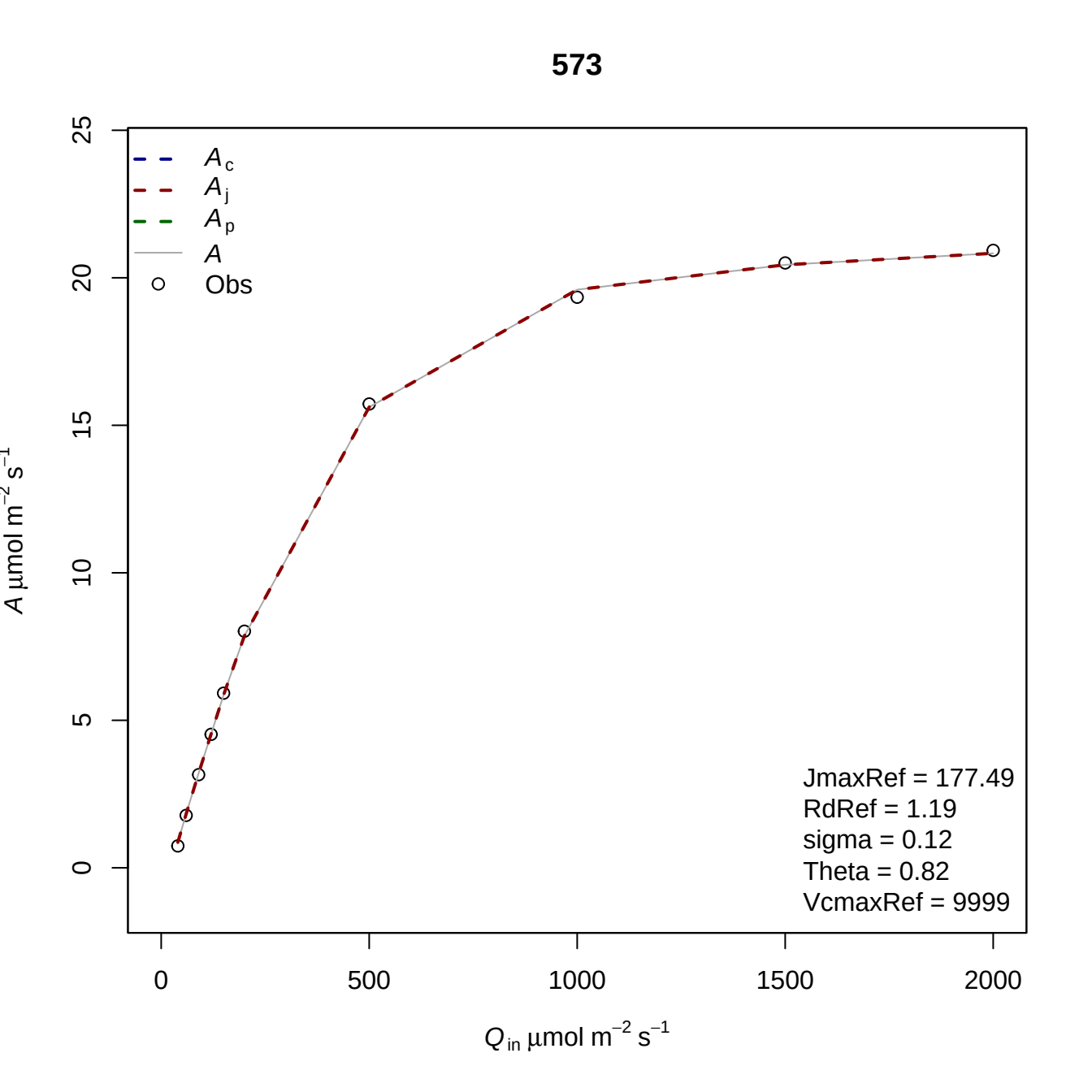

568
A_c
A_j
A_p
A
Obs
JmaxRef = 170.48
RdRef = 0.75
sigma = 0.11
Theta = 0.78
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

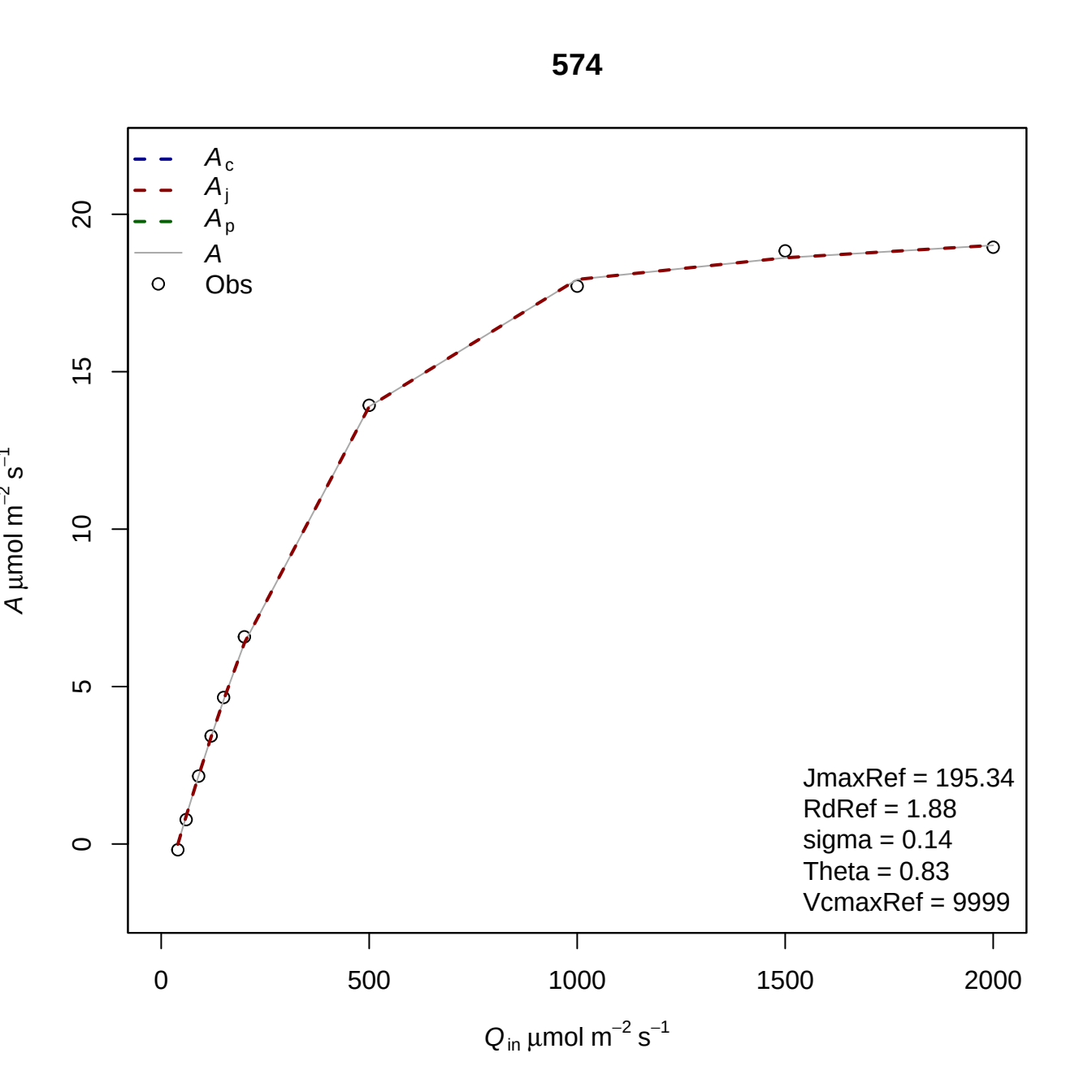

569
A_c
A_j
A_p
A
Obs
JmaxRef = 163.44
RdRef = 1.24
sigma = 0.23
Theta = 0.74
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

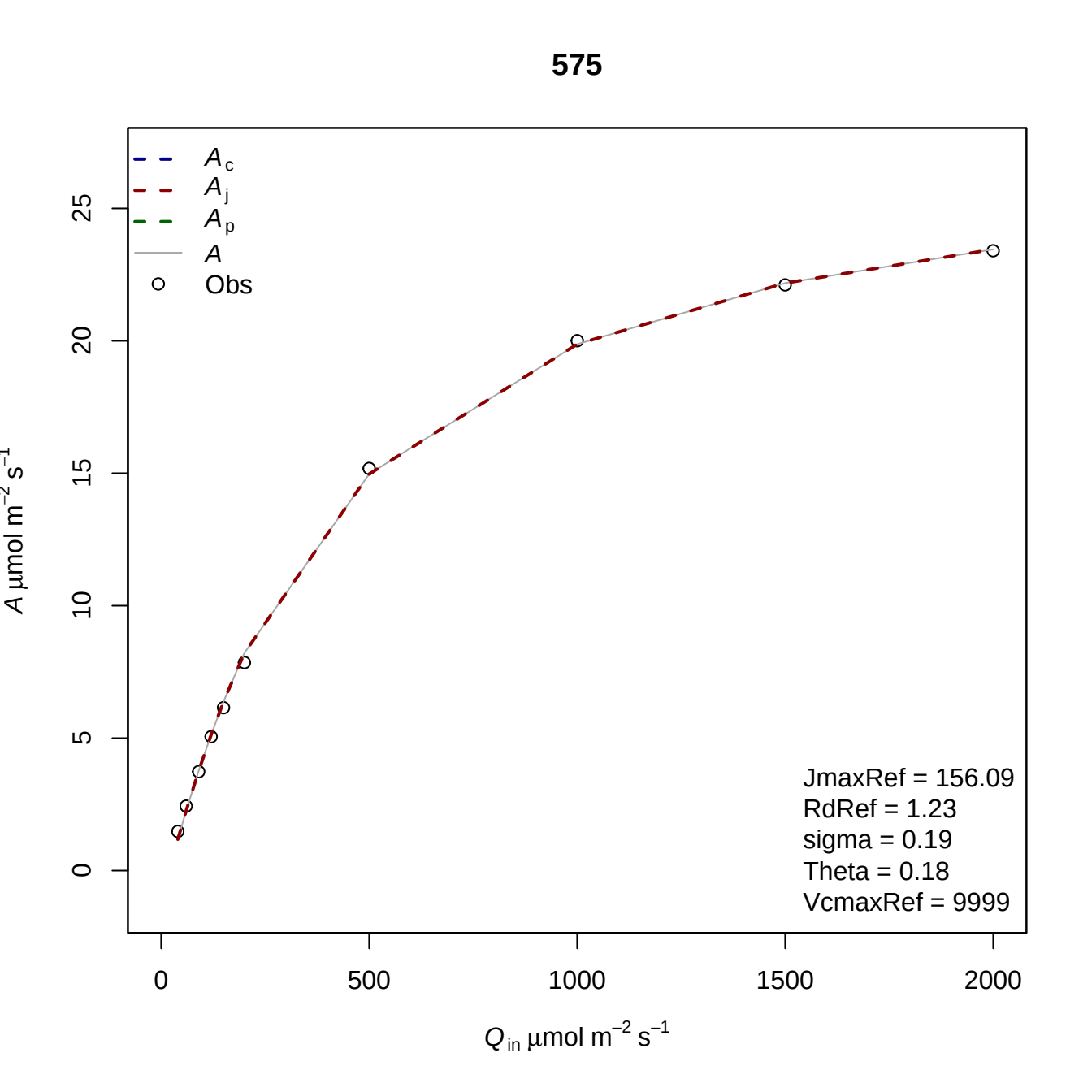

570
A_c
A_j
A_p
A
Obs
JmaxRef = 166.03
RdRef = 2.44
sigma = 0.11
Theta = 0.69
VcmaxRef = 170.61
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

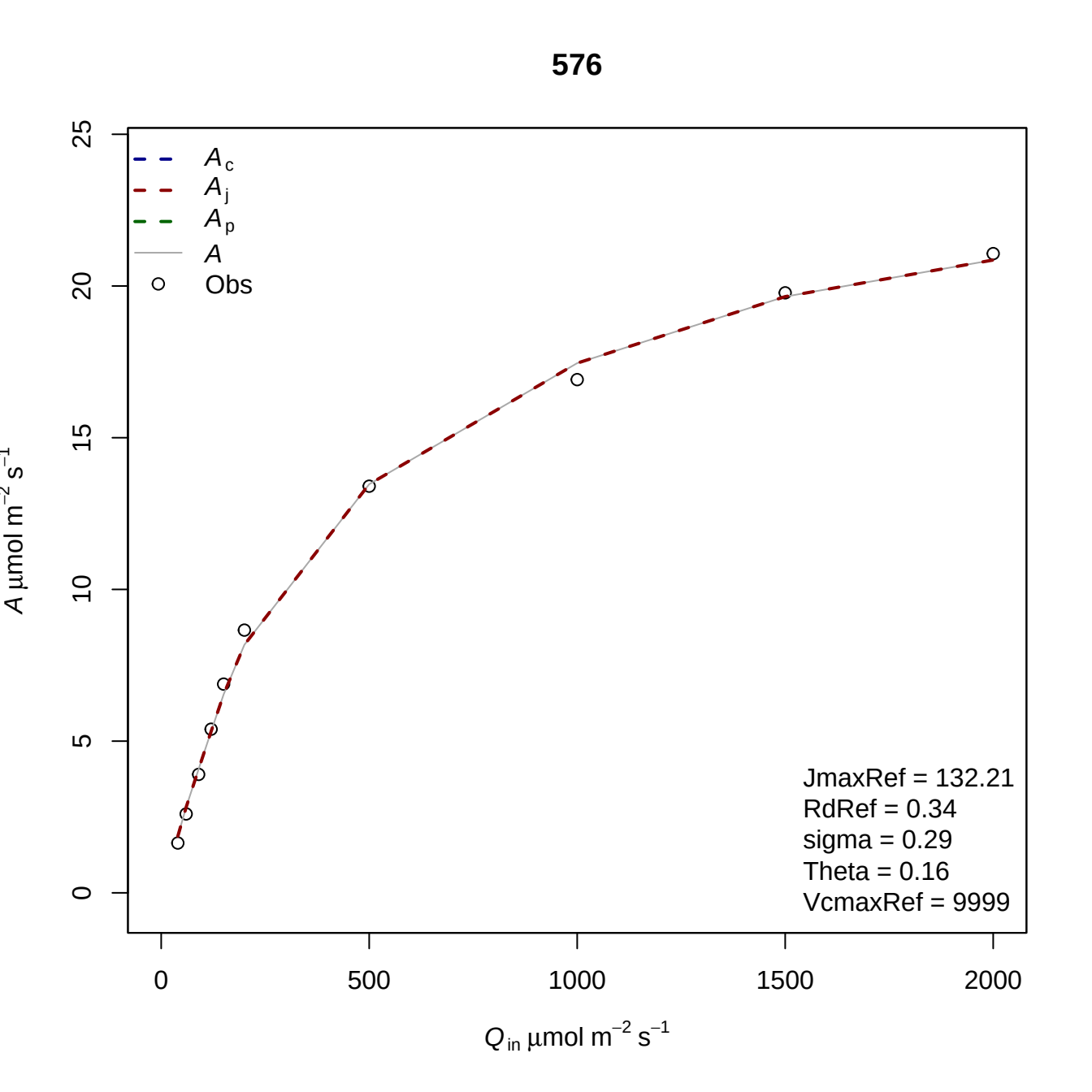

571
A_c
A_j
A_p
A
Obs
JmaxRef = 130.17
RdRef = 1.32
sigma = 0.21
Theta = 0.74
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

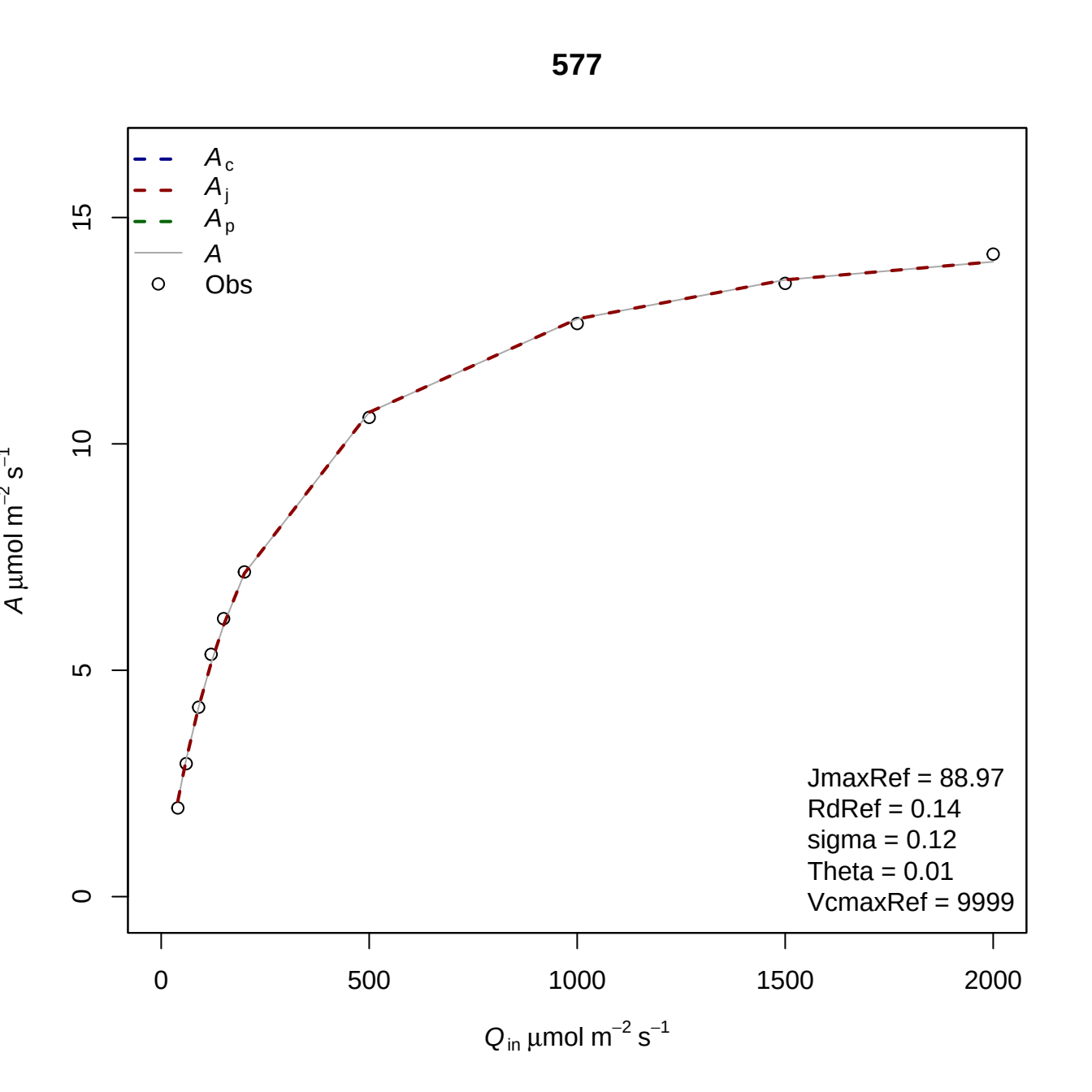

572
A_c
A_j
A_p
A
Obs
JmaxRef = 165.28
RdRef = 0.7
sigma = 0.17
Theta = 0.67
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

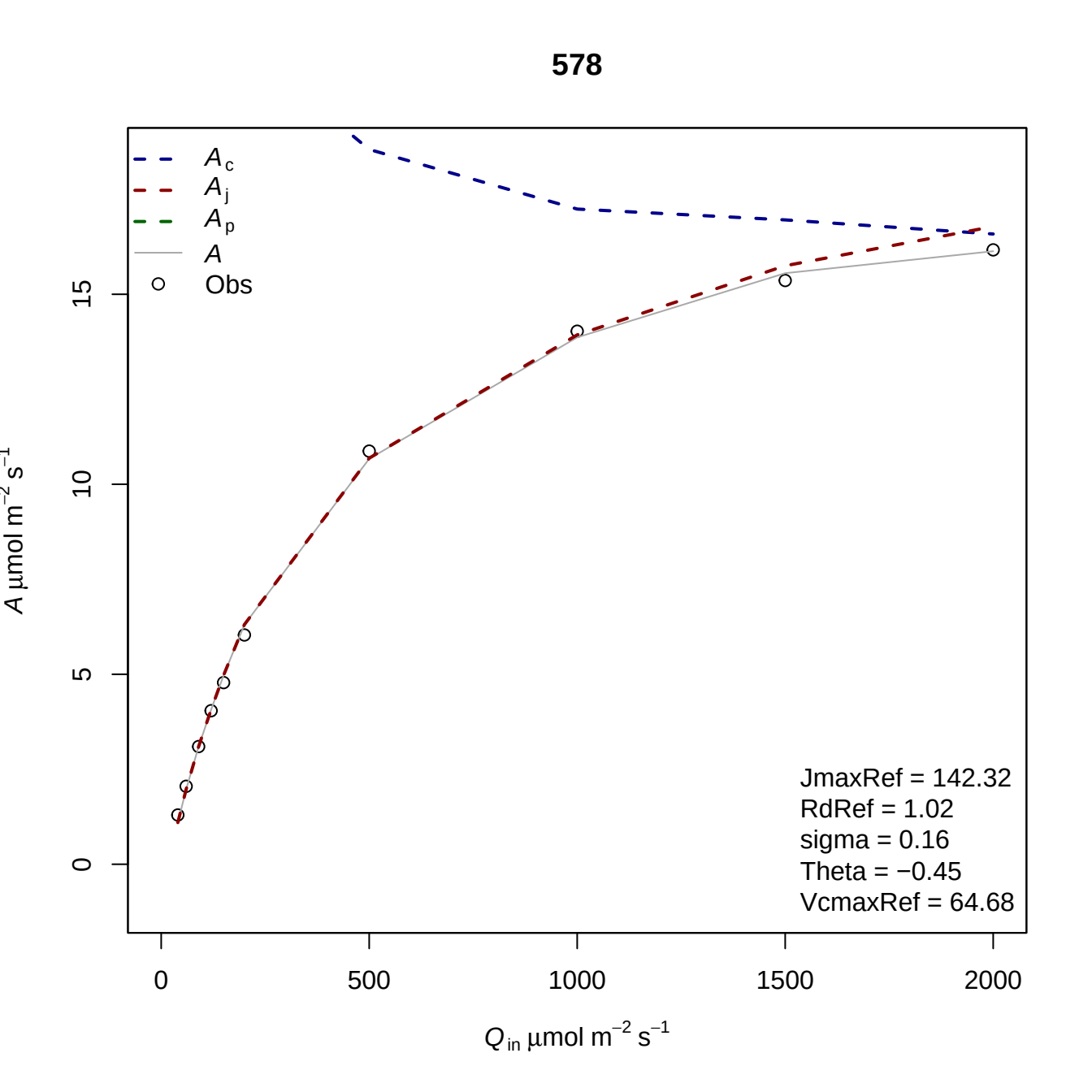

573
A_c
A_j
A_p
A
Obs
JmaxRef = 177.49
RdRef = 1.19
sigma = 0.12
Theta = 0.82
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

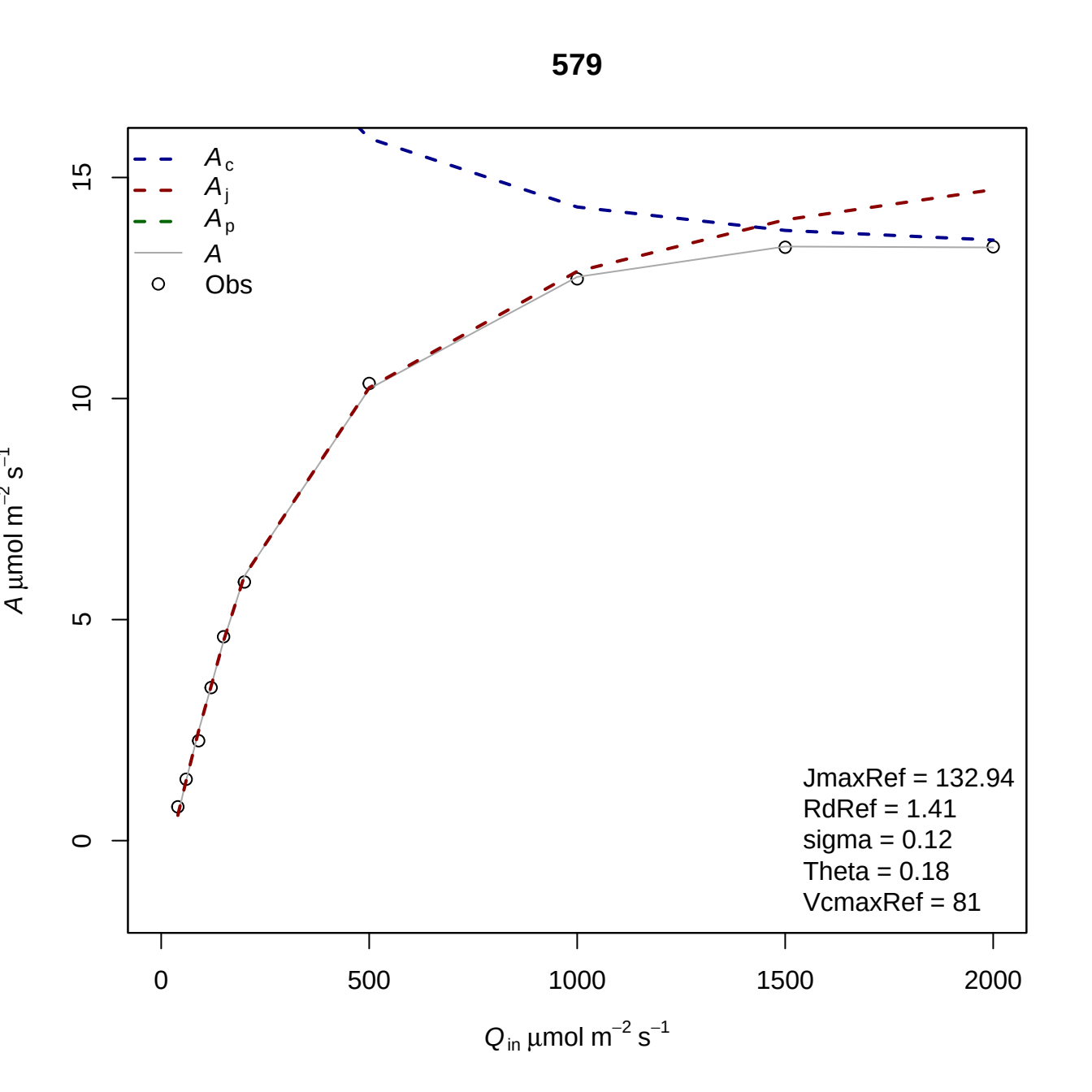

574
A_c
A_j
A_p
A
Obs
JmaxRef = 195.34
RdRef = 1.88
sigma = 0.14
Theta = 0.83
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

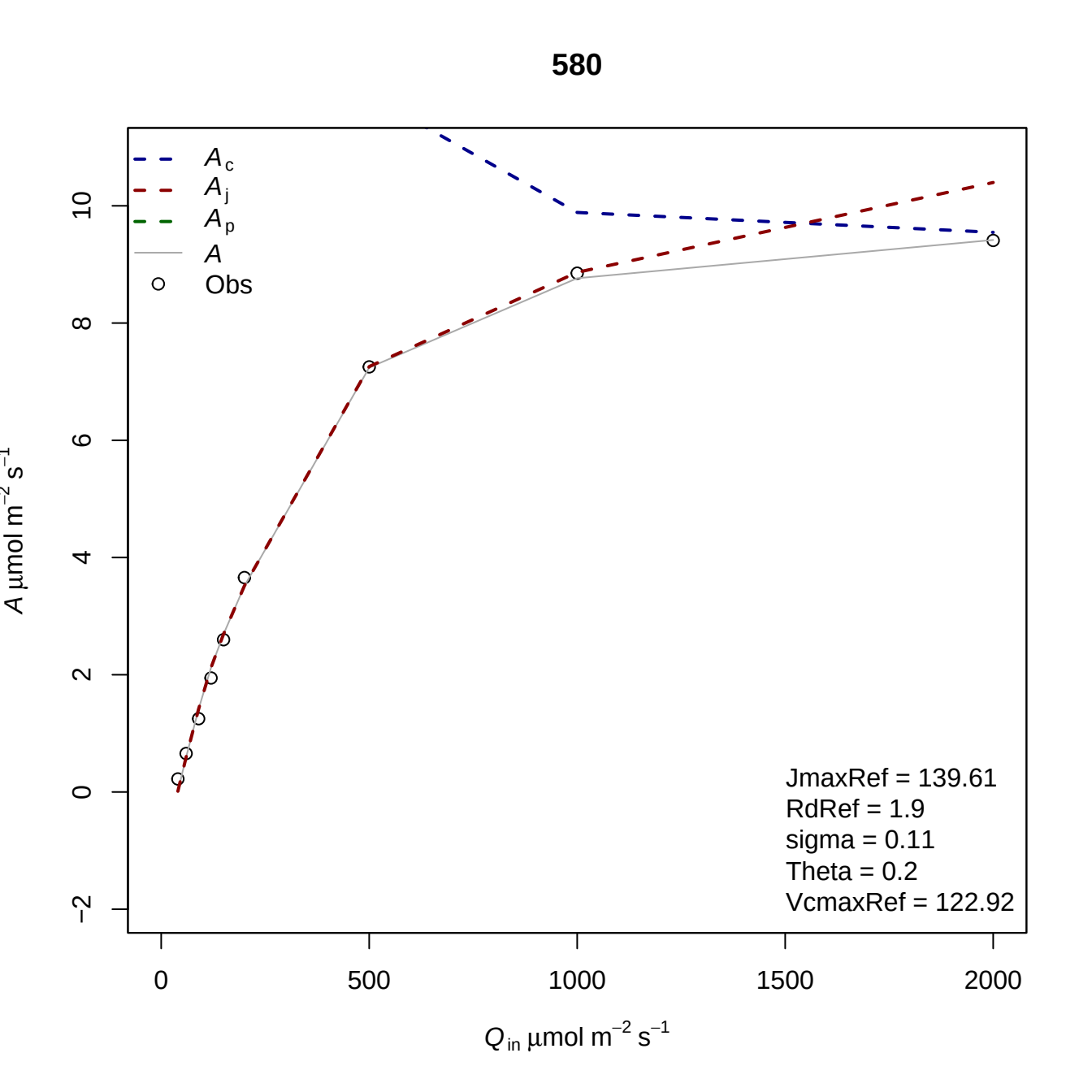

575
A_c
A_j
A_p
A
Obs
JmaxRef = 156.09
RdRef = 1.23
sigma = 0.19
Theta = 0.18
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

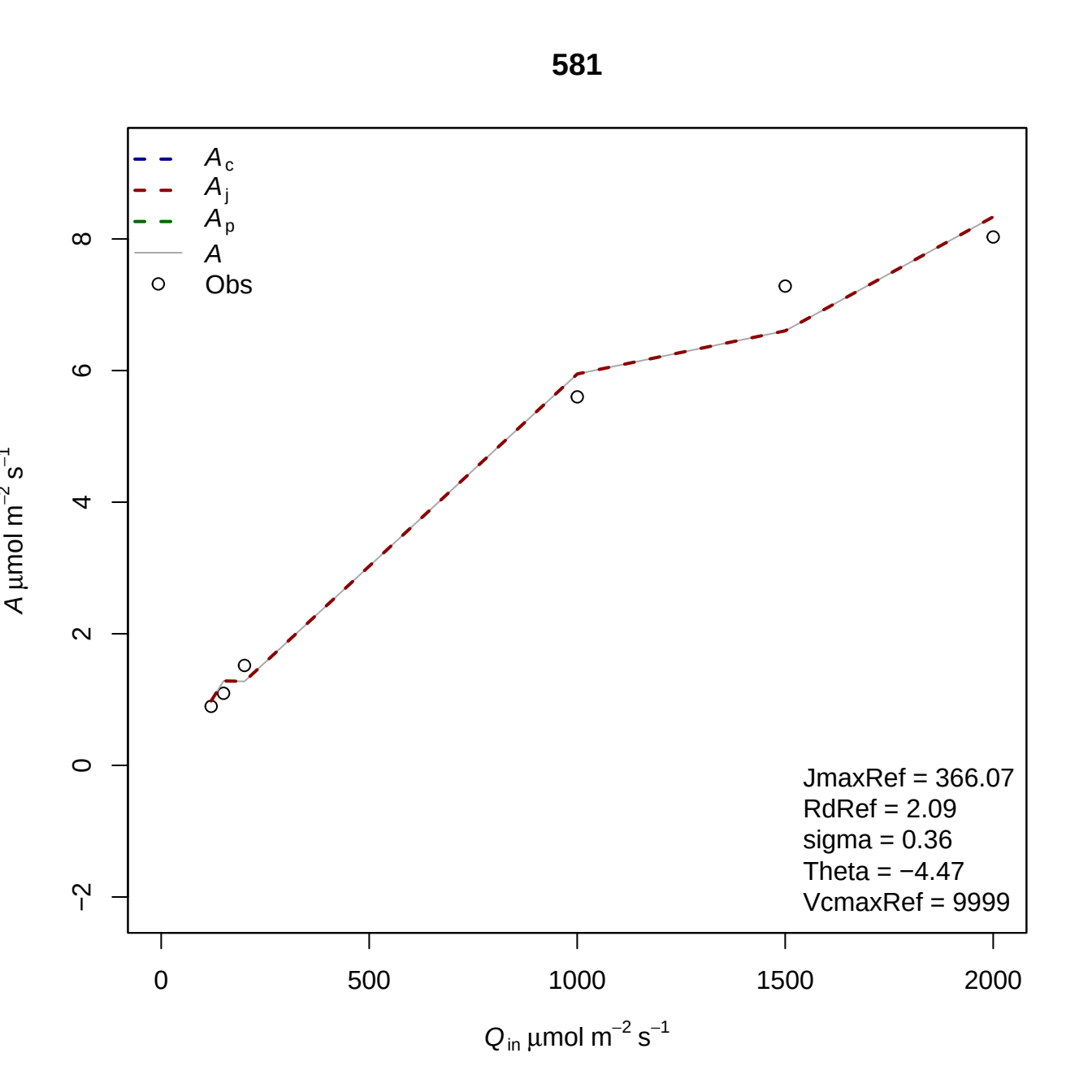

576
A_c
A_j
A_p
A
Obs
JmaxRef = 132.21
RdRef = 0.34
sigma = 0.29
Theta = 0.16
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

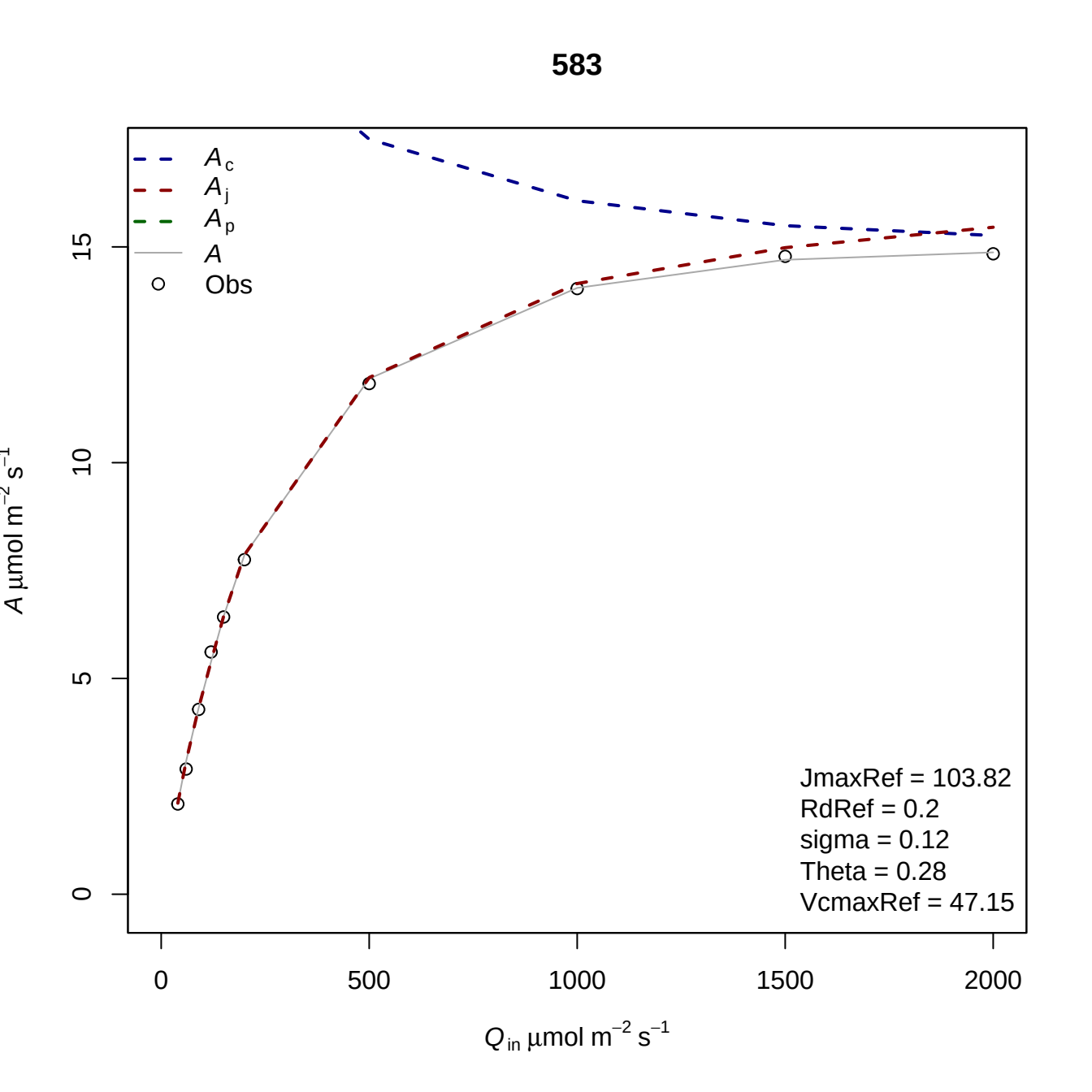

577
A_c
A_j
A_p
A
Obs
JmaxRef = 88.97
RdRef = 0.14
sigma = 0.12
Theta = 0.01
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

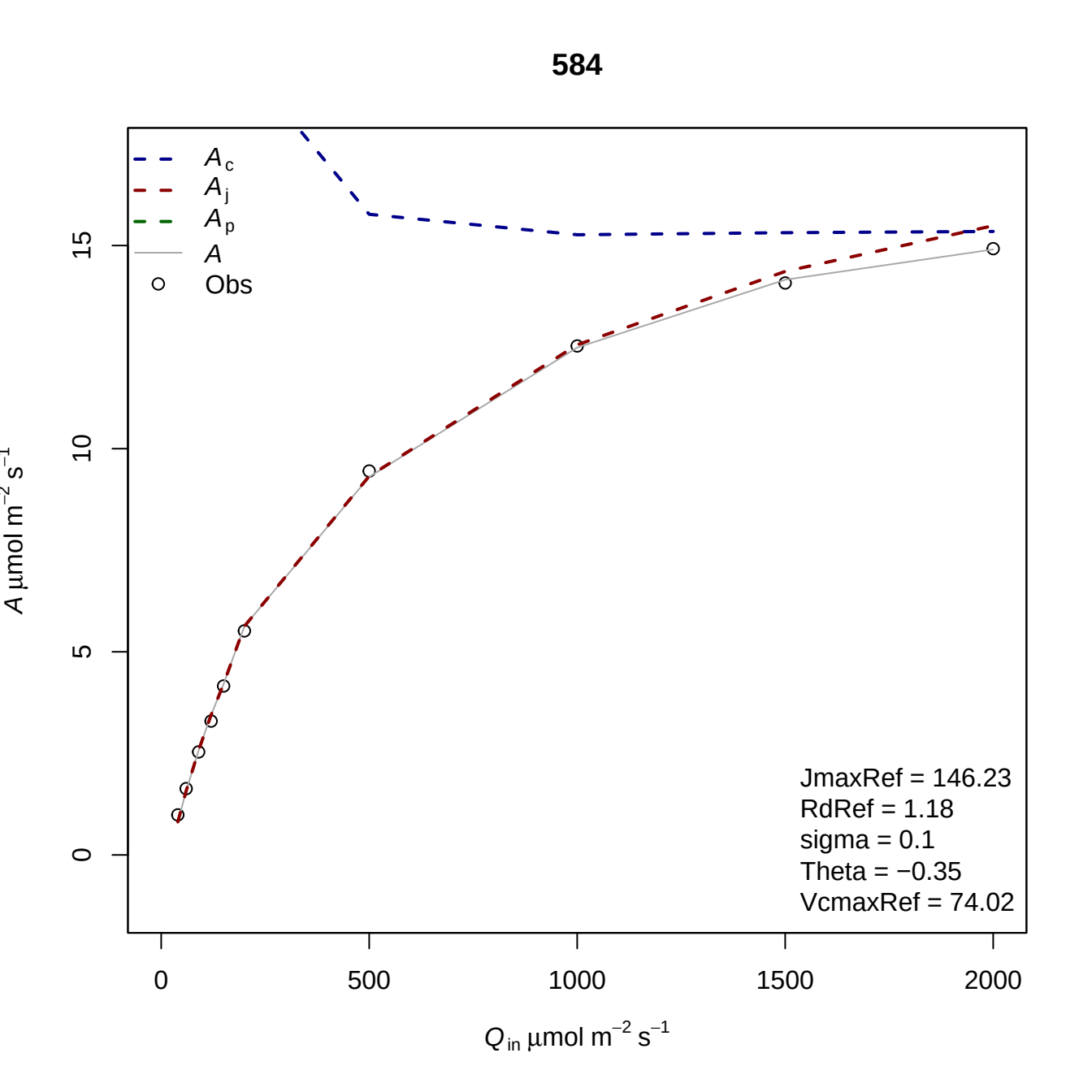

578
A_c
A_j
A_p
A
Obs
JmaxRef = 142.32
RdRef = 1.02
sigma = 0.16
Theta = −0.45
VcmaxRef = 64.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

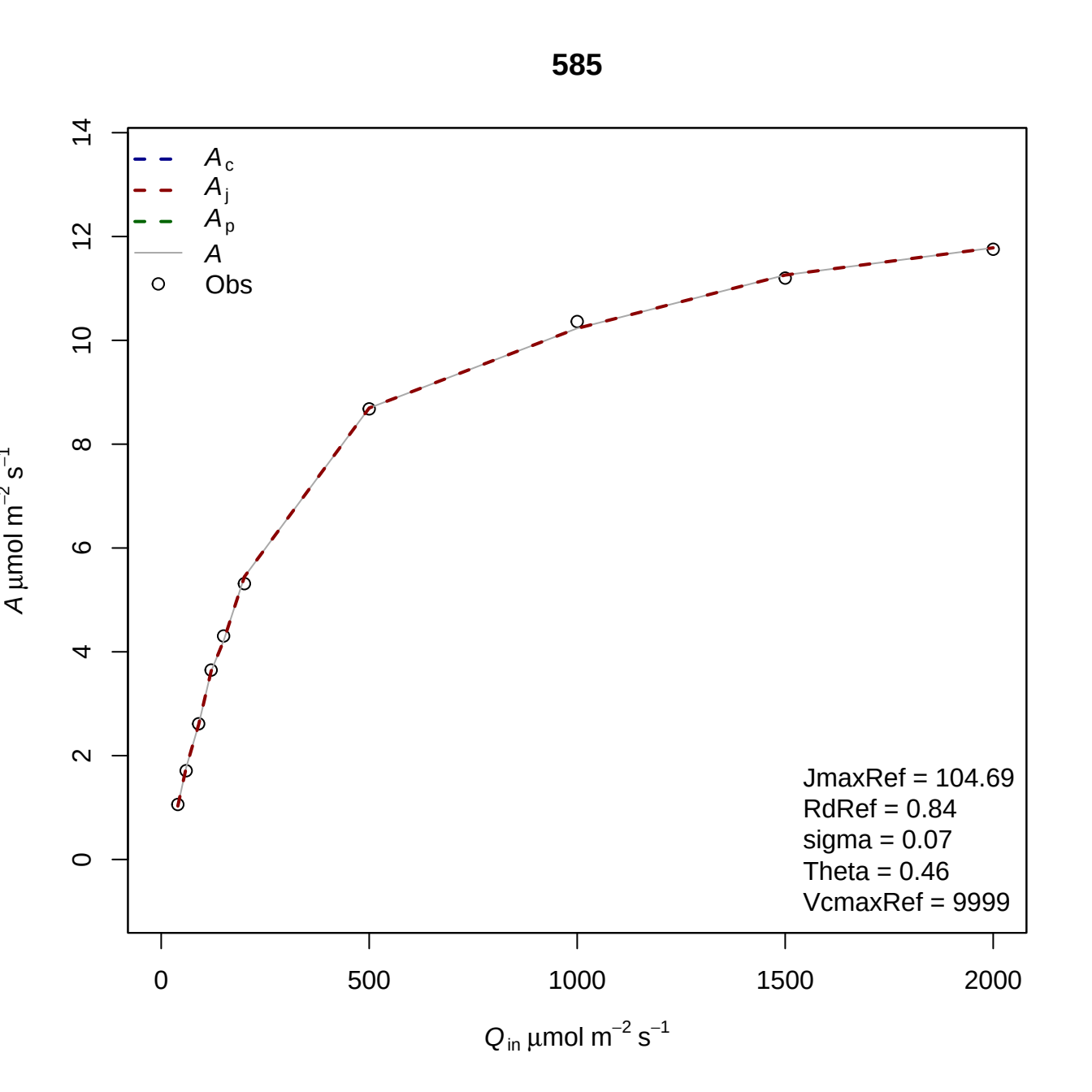

579
A_c
A_j
A_p
A
Obs
JmaxRef = 132.94
RdRef = 1.41
sigma = 0.12
Theta = 0.18
VcmaxRef = 81
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

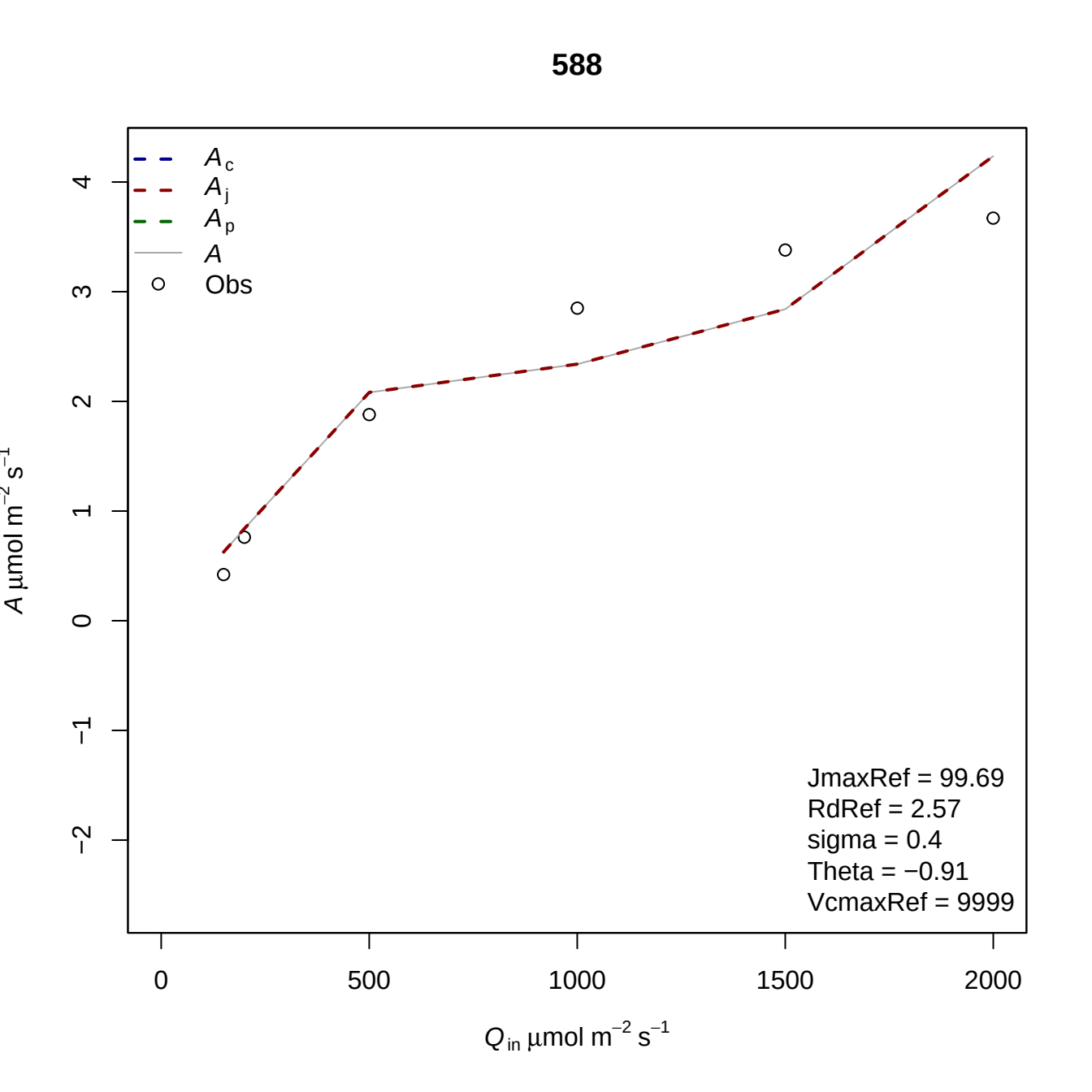

580
A_c
A_j
A_p
A
Obs
JmaxRef = 139.61
RdRef = 1.9
sigma = 0.11
Theta = 0.2
VcmaxRef = 122.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

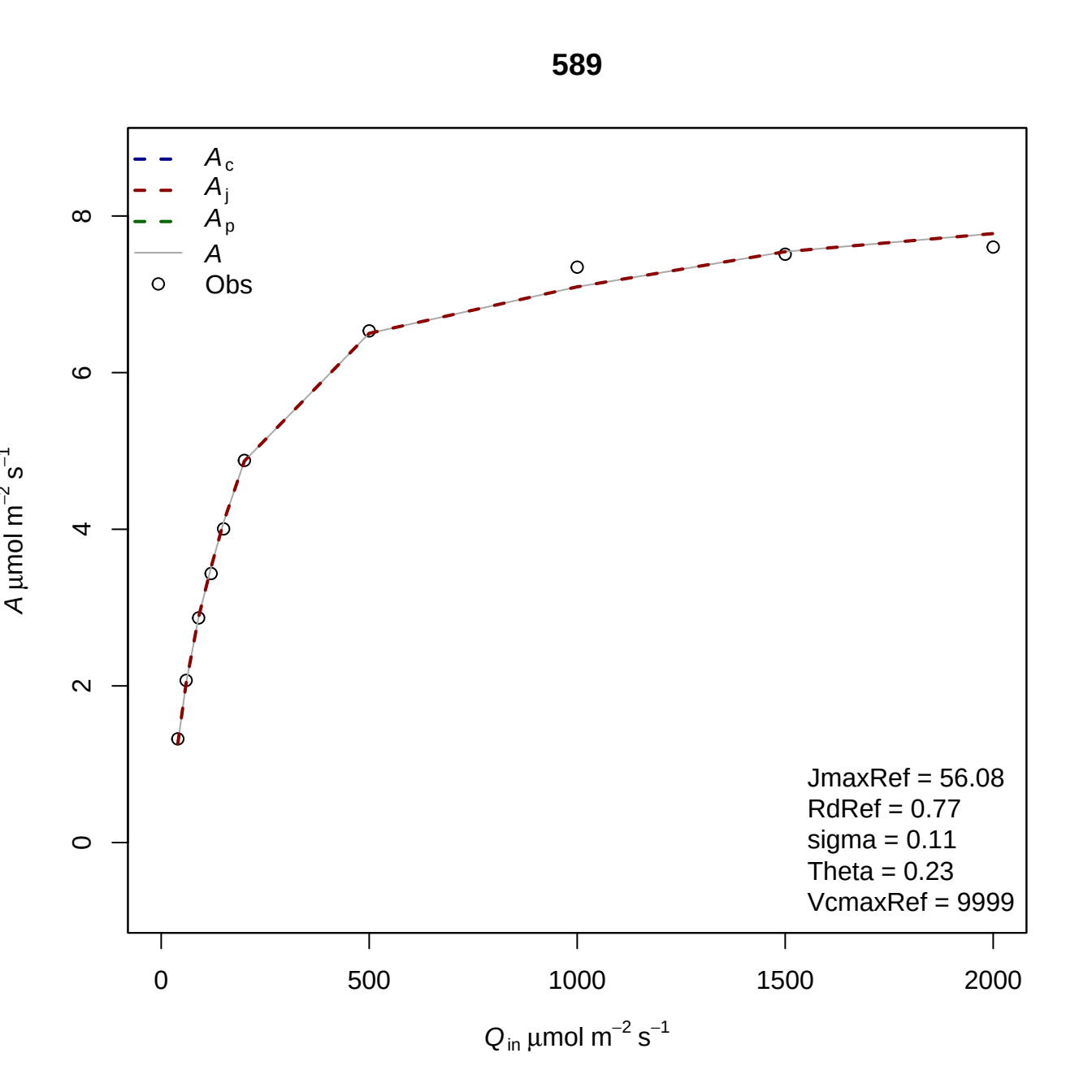

581
A_c
A_j
A_p
A
Obs
JmaxRef = 366.07
RdRef = 2.09
sigma = 0.36
Theta = −4.47
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

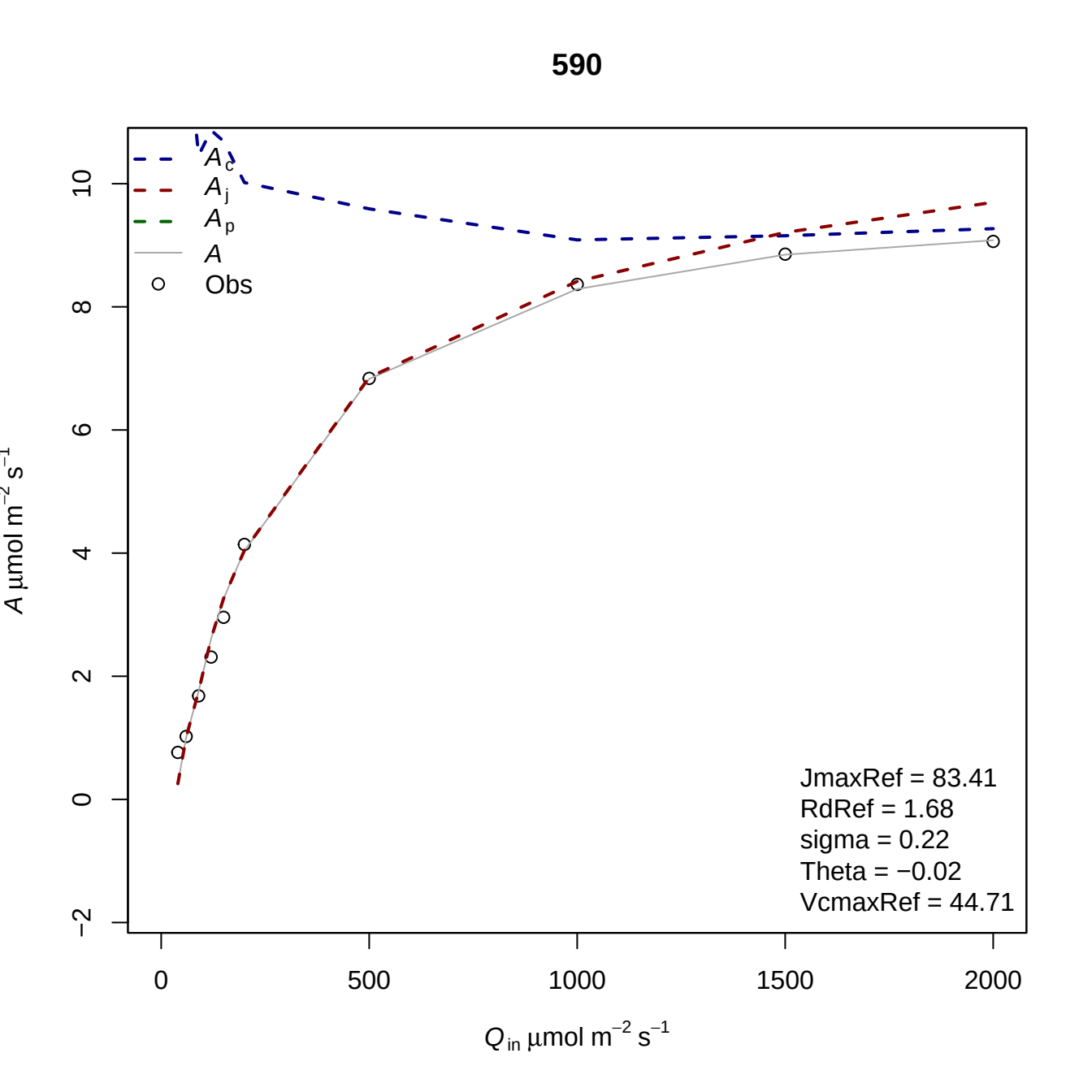

583
A_c
A_j
A_p
A
Obs
JmaxRef = 103.82
RdRef = 0.2
sigma = 0.12
Theta = 0.28
VcmaxRef = 47.15
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

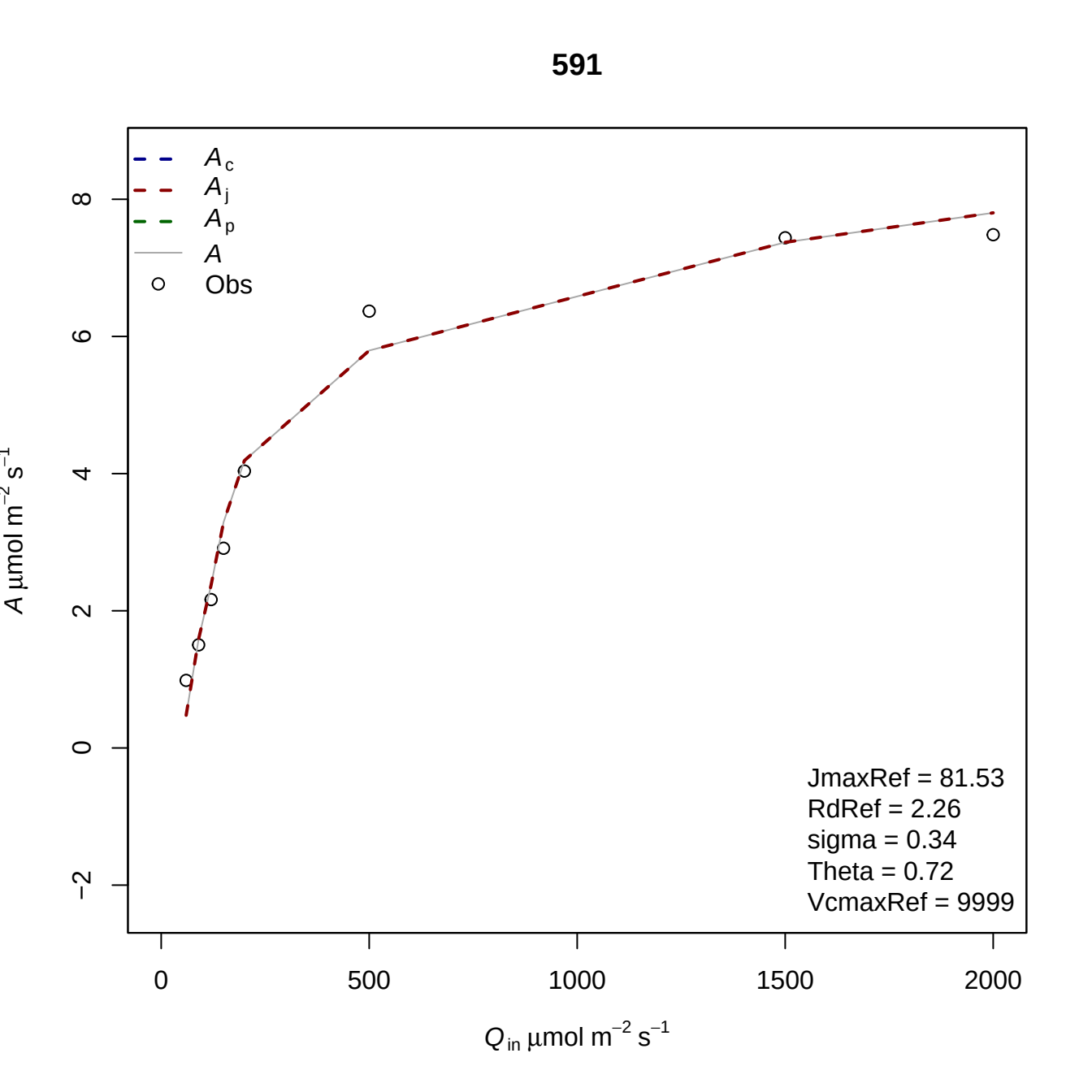

584
A_c
A_j
A_p
A
Obs
JmaxRef = 146.23
RdRef = 1.18
sigma = 0.1
Theta = −0.35
VcmaxRef = 74.02
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

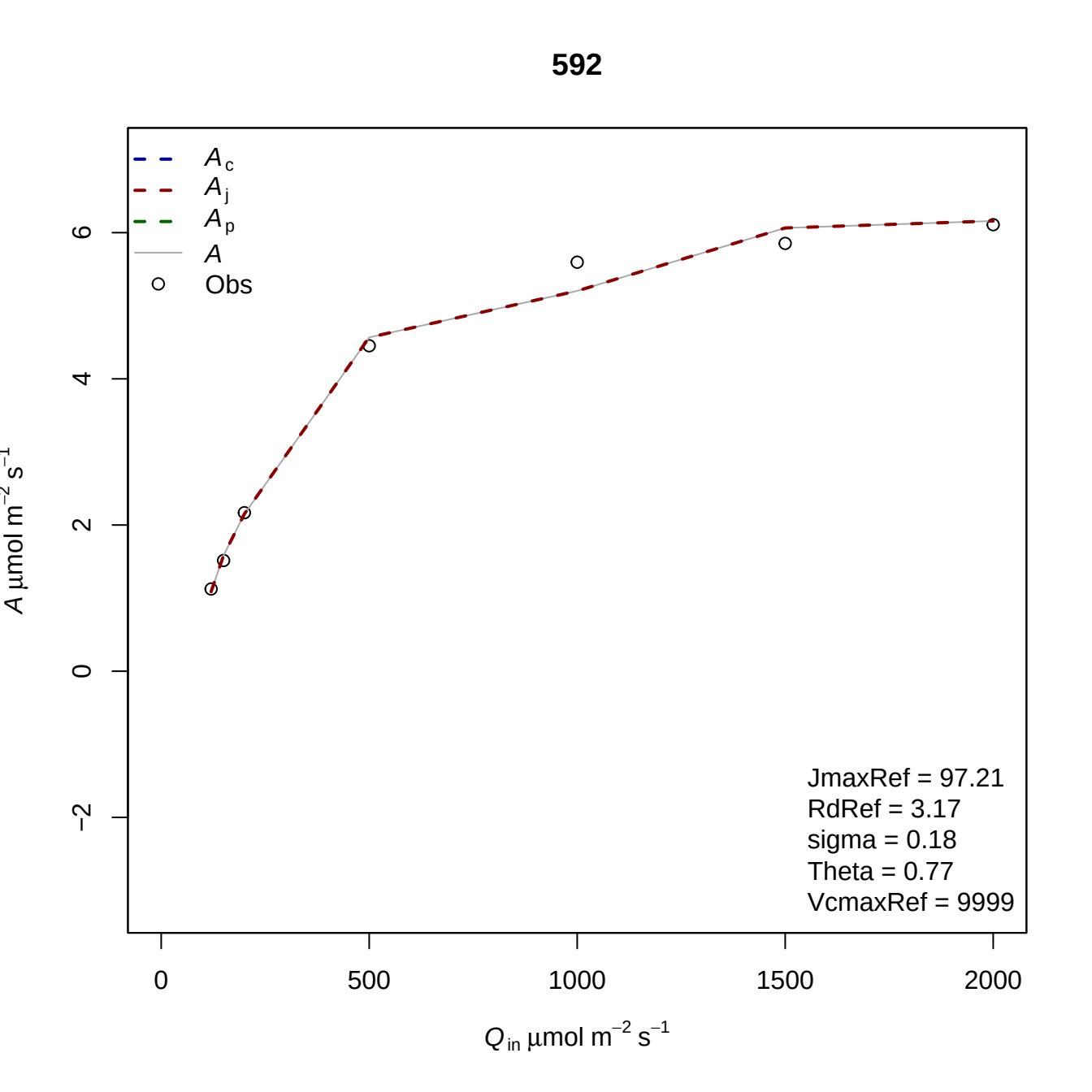

585
A_c
A_j
A_p
A
Obs
JmaxRef = 104.69
RdRef = 0.84
sigma = 0.07
Theta = 0.46
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500
2000

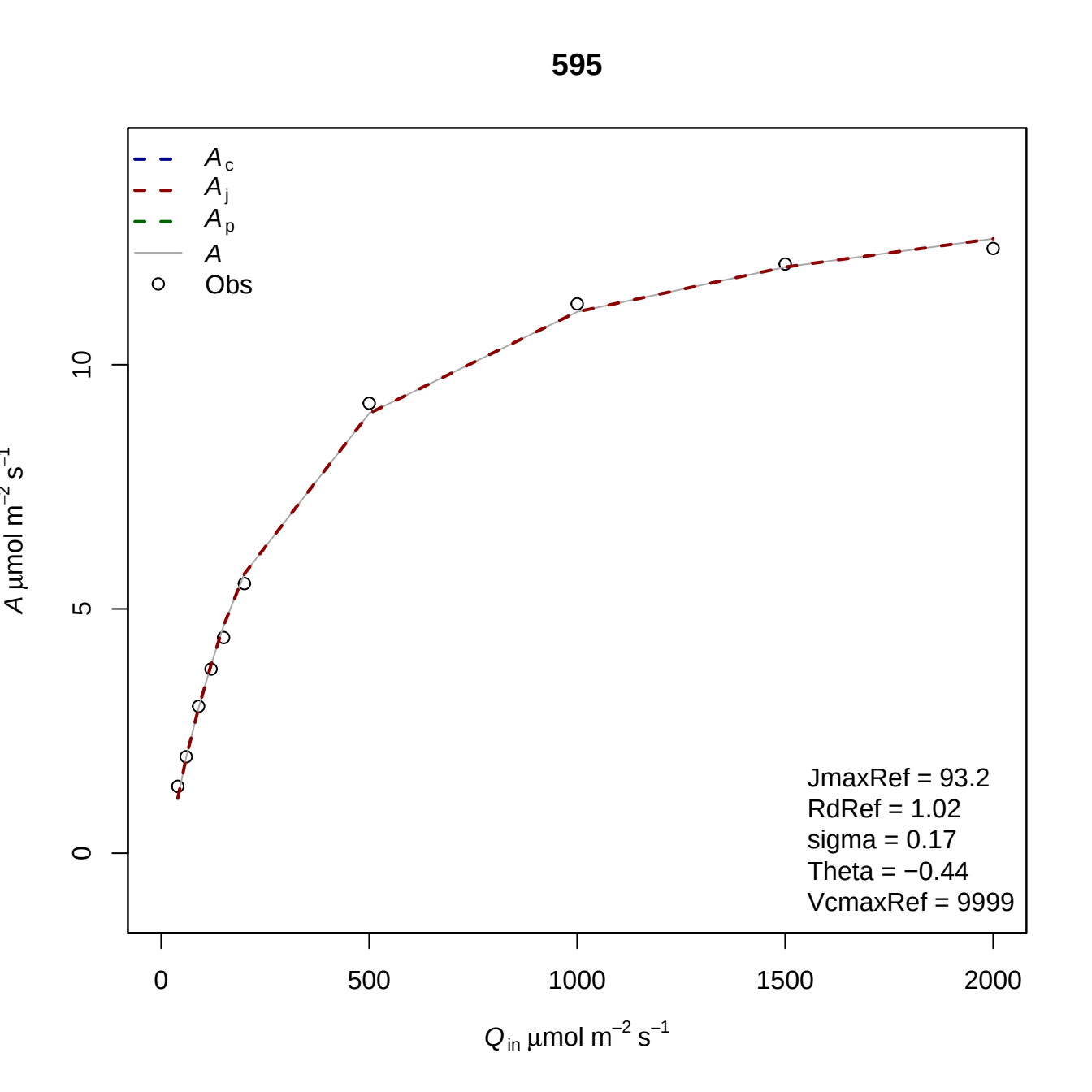

588
A_c
A_j
A_p
A
Obs
JmaxRef = 99.69
RdRef = 2.57
sigma = 0.4
Theta = −0.91
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

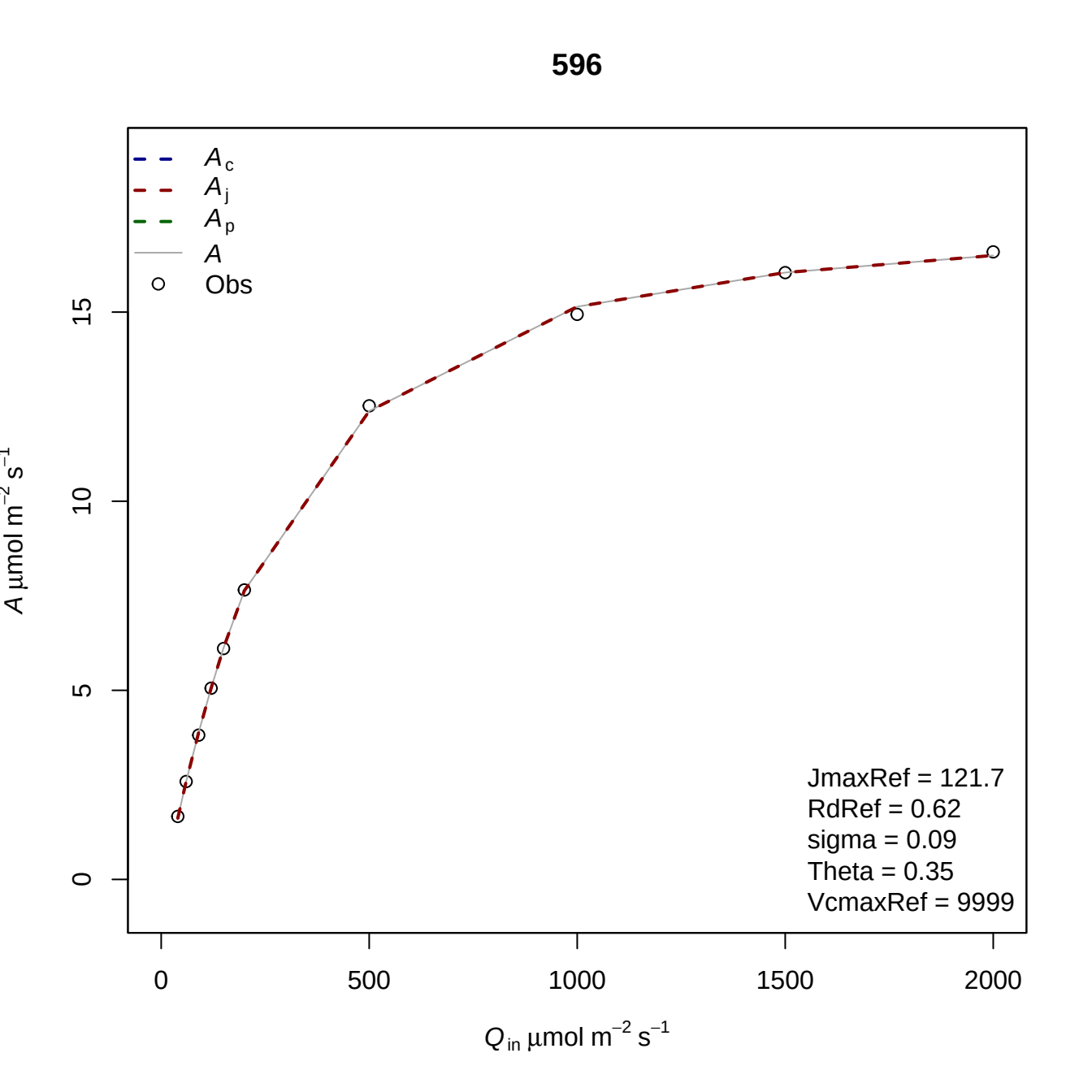

589
A_c
A_j
A_p
A
Obs
JmaxRef = 56.08
RdRef = 0.77
sigma = 0.11
Theta = 0.23
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

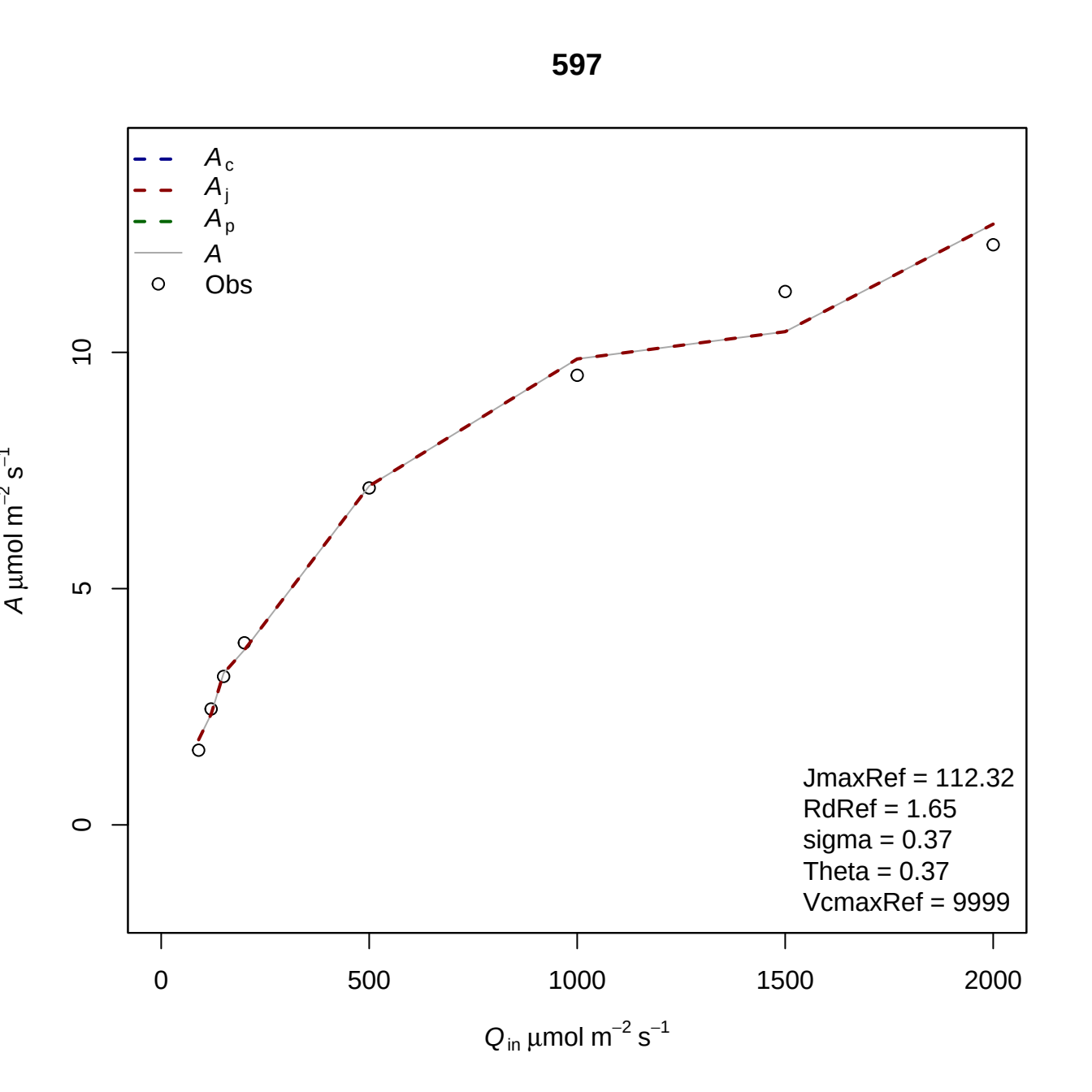

590
A_c
A_j
A_p
A
Obs
JmaxRef = 83.41
RdRef = 1.68
sigma = 0.22
Theta = −0.02
VcmaxRef = 44.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

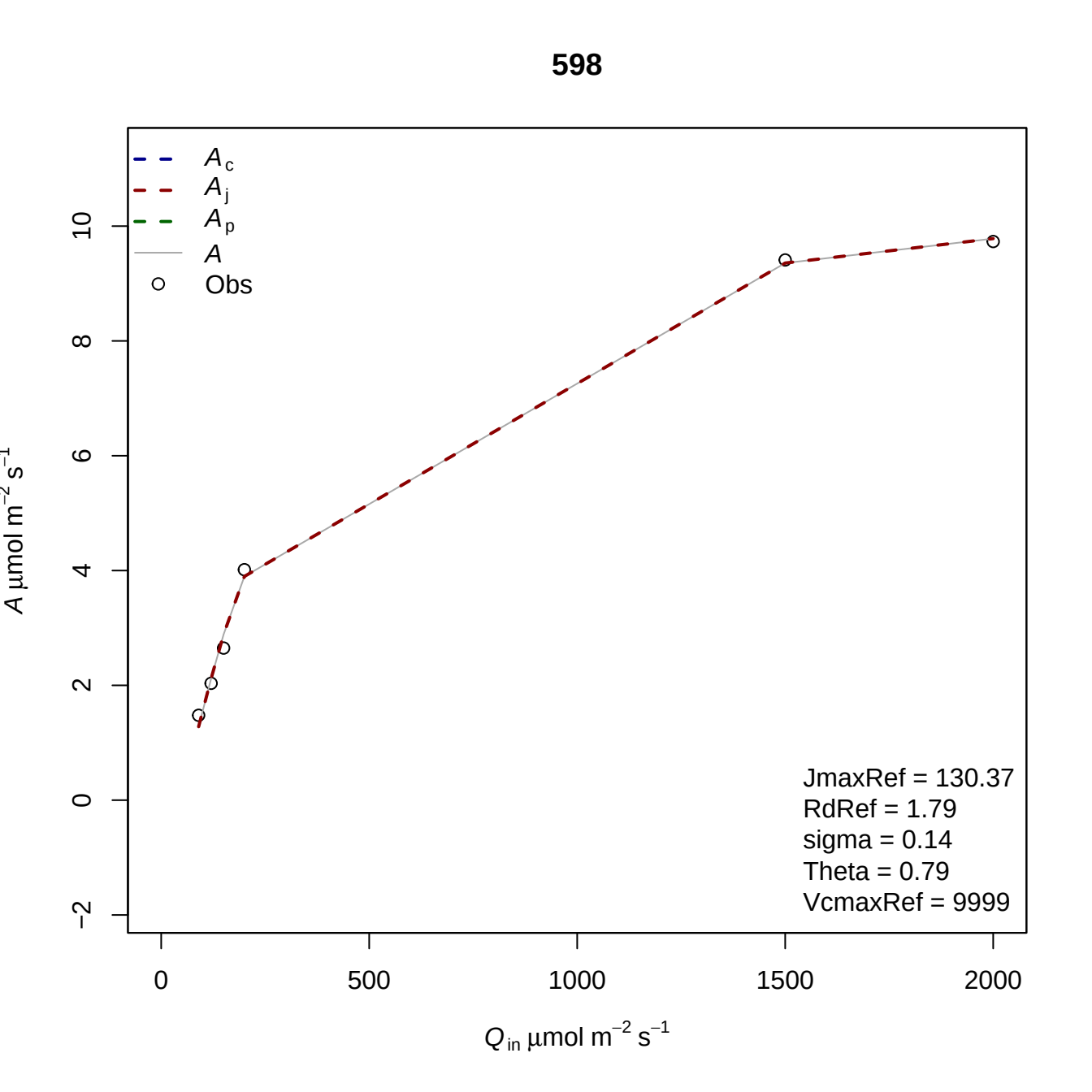

591
A_c
A_j
A_p
A
Obs
JmaxRef = 81.53
RdRef = 2.26
sigma = 0.34
Theta = 0.72
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

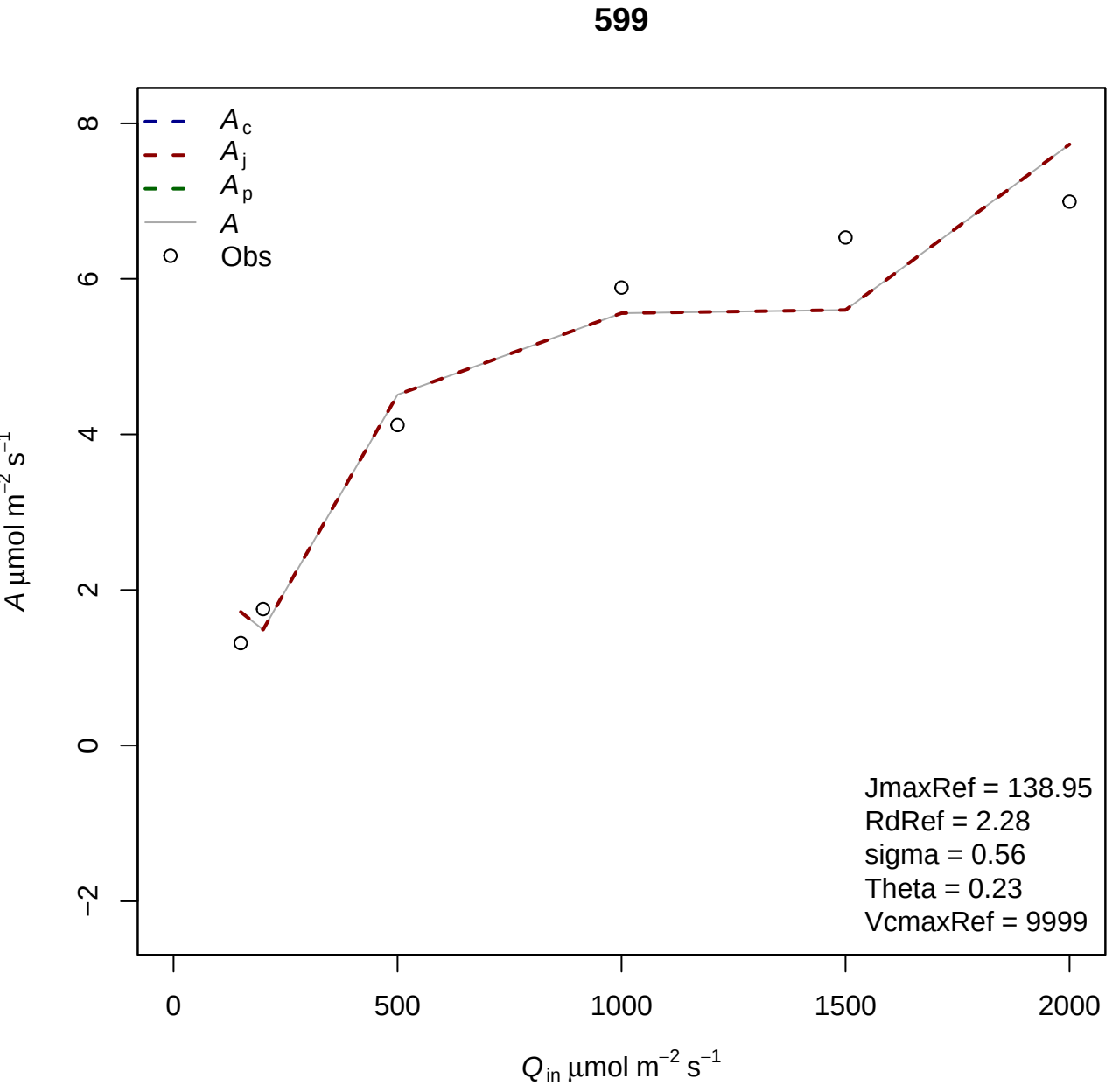

592
A_c
A_j
A_p
A
Obs
JmaxRef = 97.21
RdRef = 3.17
sigma = 0.18
Theta = 0.77
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

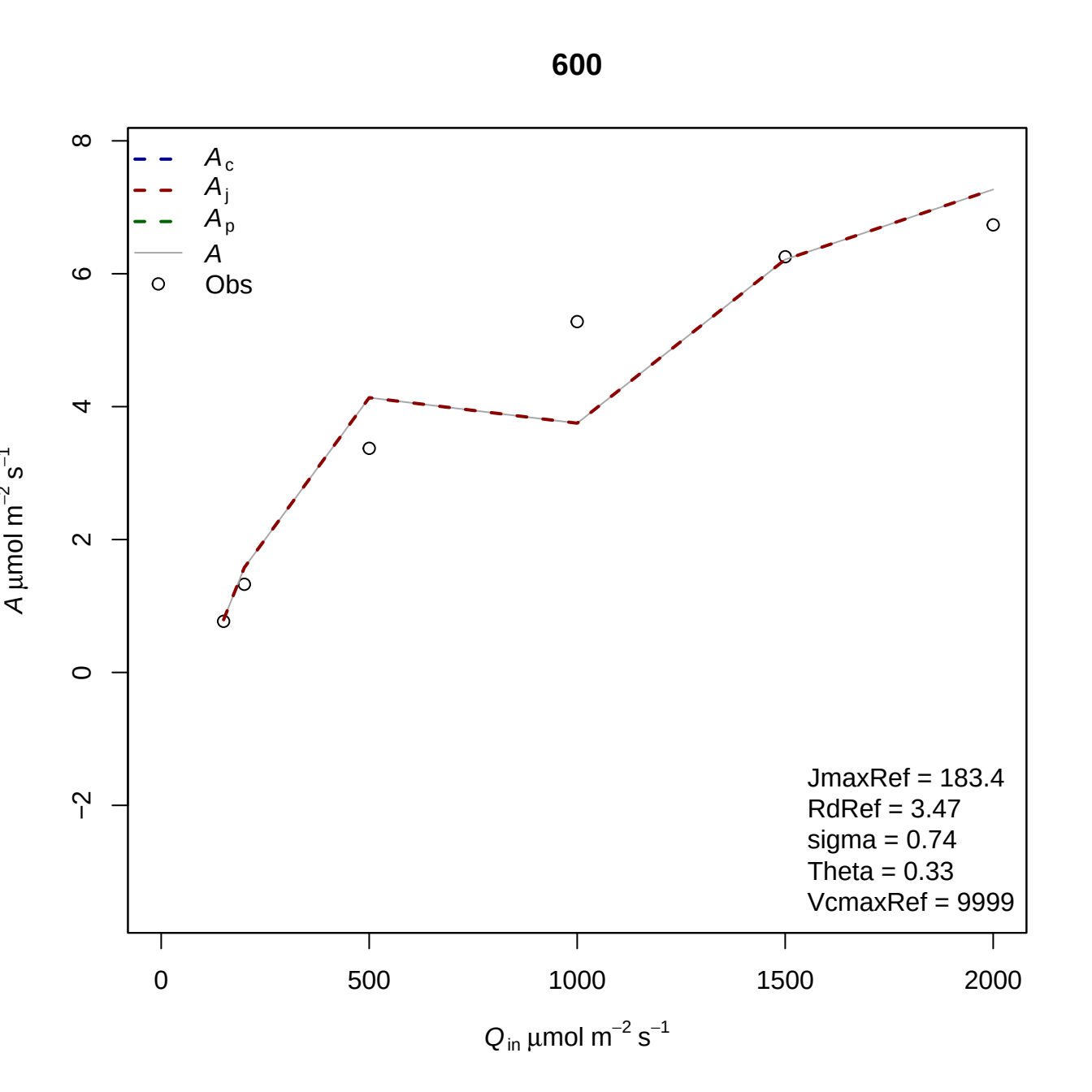

595
A_c
A_j
A_p
A
Obs
JmaxRef = 93.2
RdRef = 1.02
sigma = 0.17
Theta = −0.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

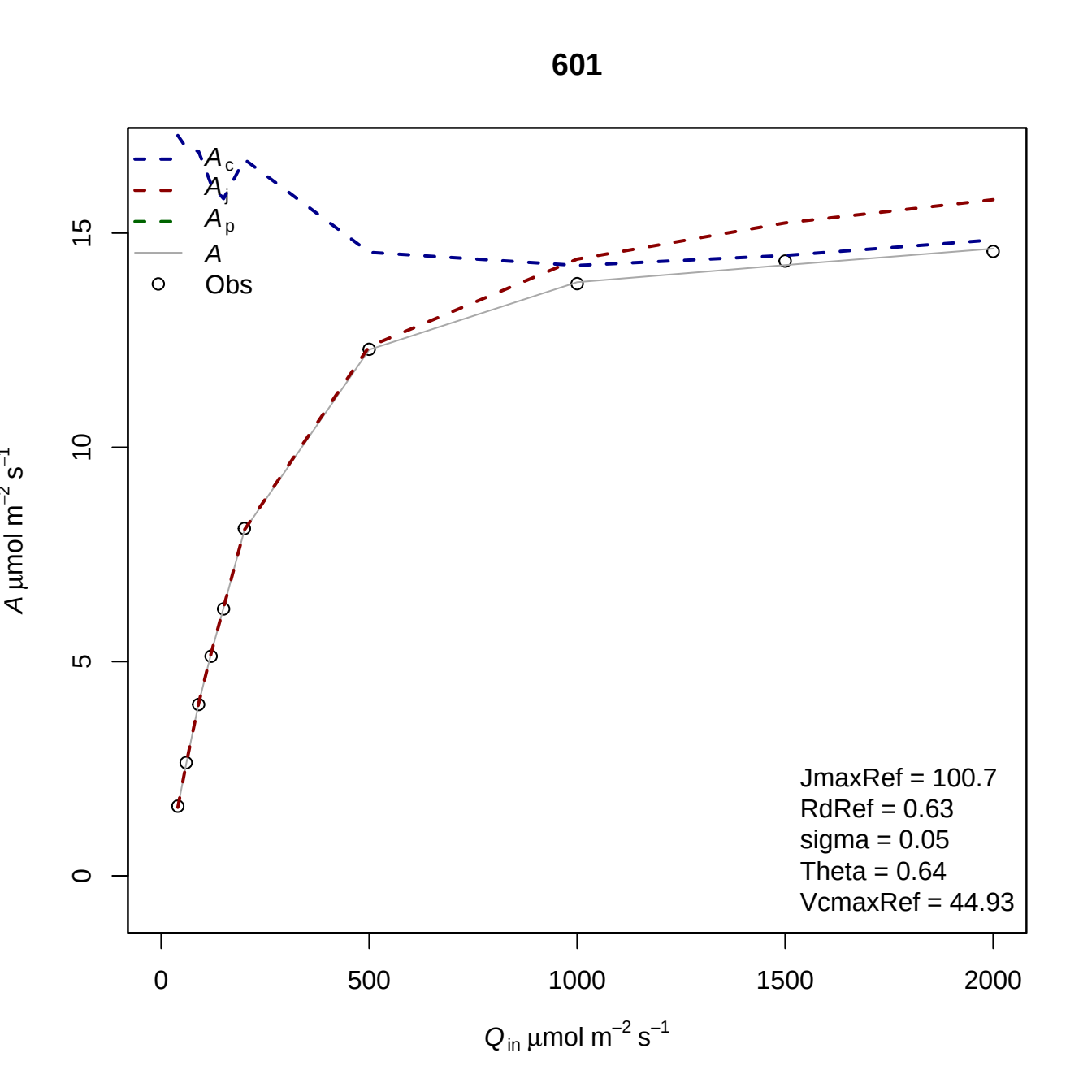

596
A_c
A_j
A_p
A
Obs
JmaxRef = 121.7
RdRef = 0.62
sigma = 0.09
Theta = 0.35
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

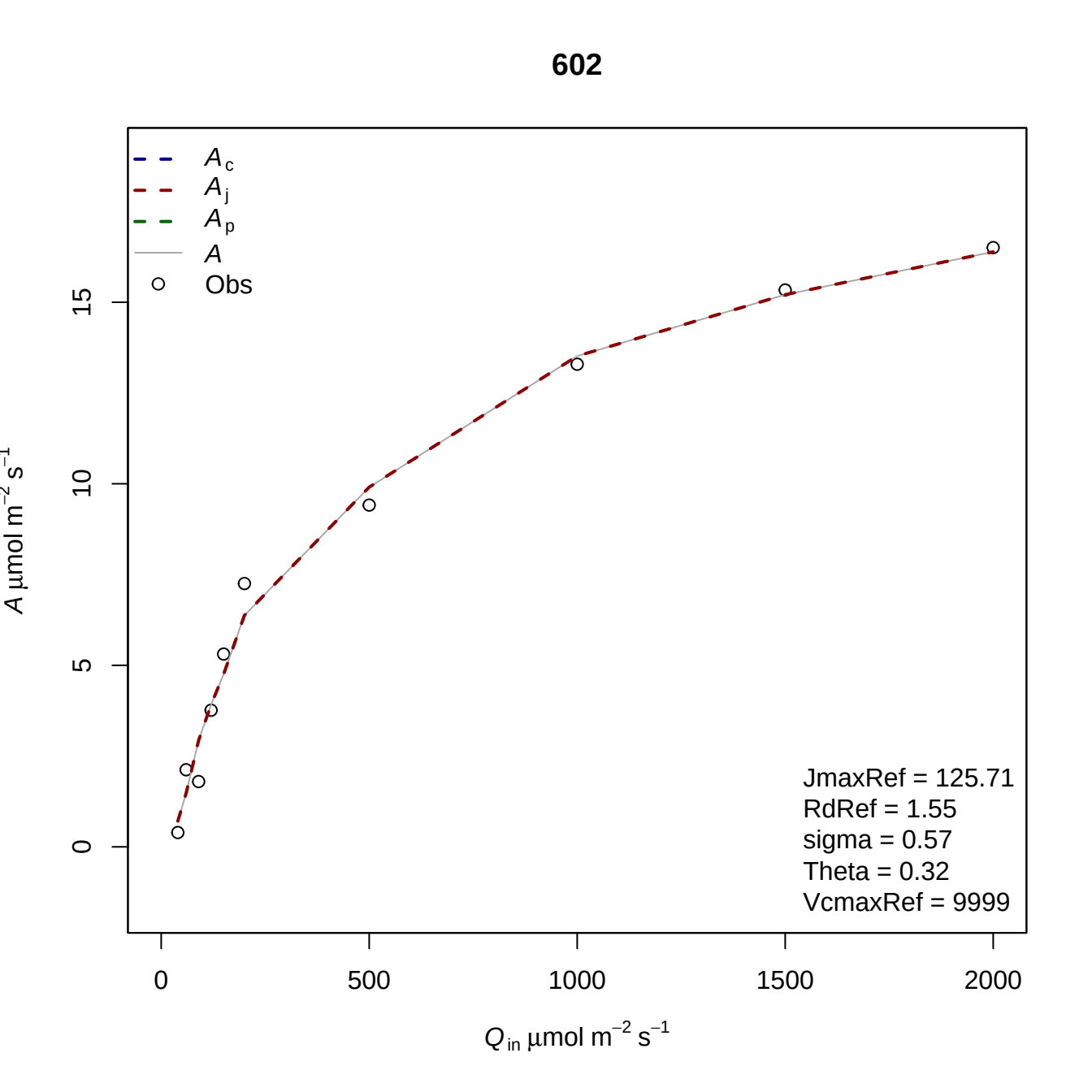

597
A_c
A_j
A_p
A
Obs
JmaxRef = 112.32
RdRef = 1.65
sigma = 0.37
Theta = 0.37
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

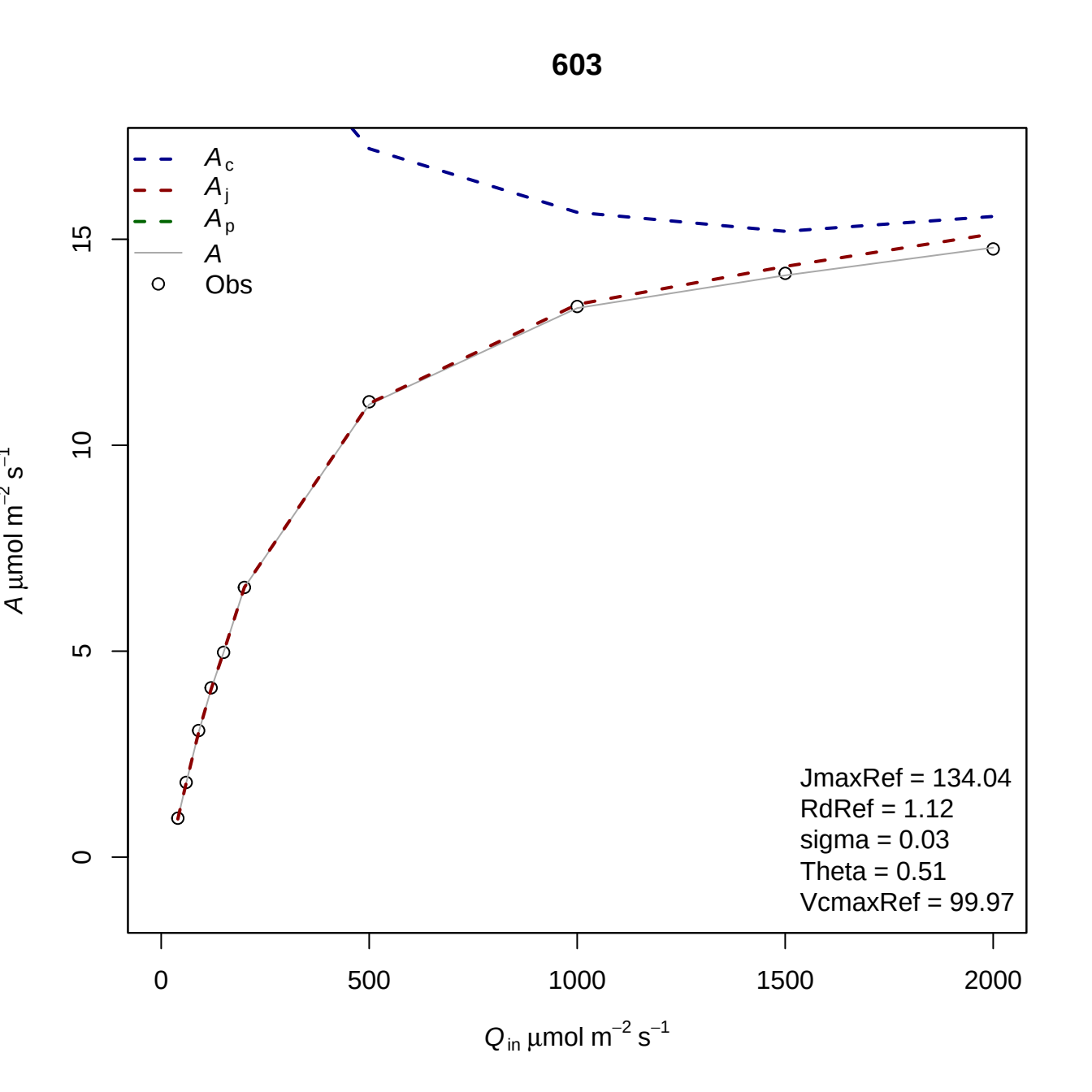

598
A_c
A_j
A_p
A
Obs
JmaxRef = 130.37
RdRef = 1.79
sigma = 0.14
Theta = 0.79
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

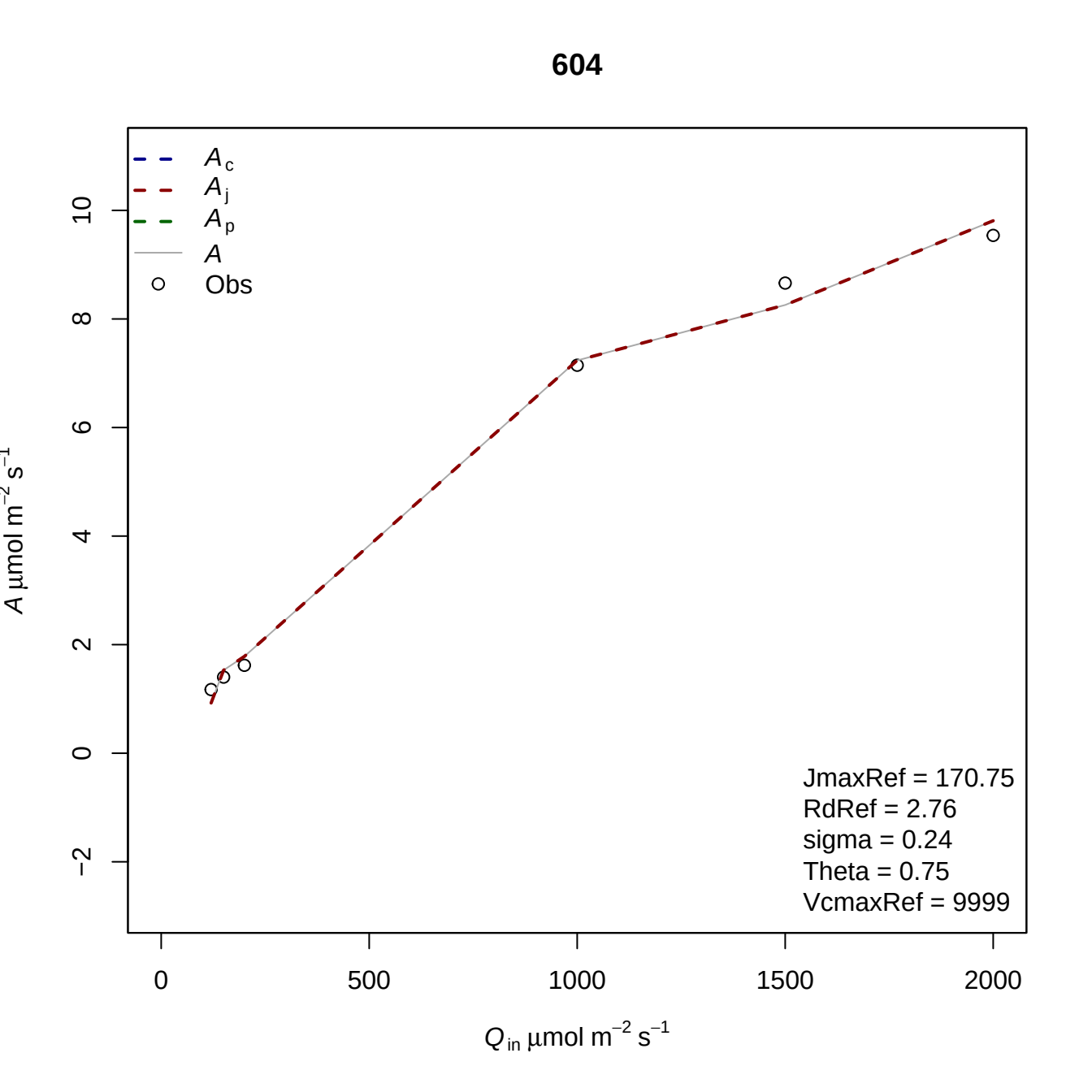

599
A_c
A_j
A_p
A
Obs
JmaxRef = 138.95
RdRef = 2.28
sigma = 0.56
Theta = 0.23
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

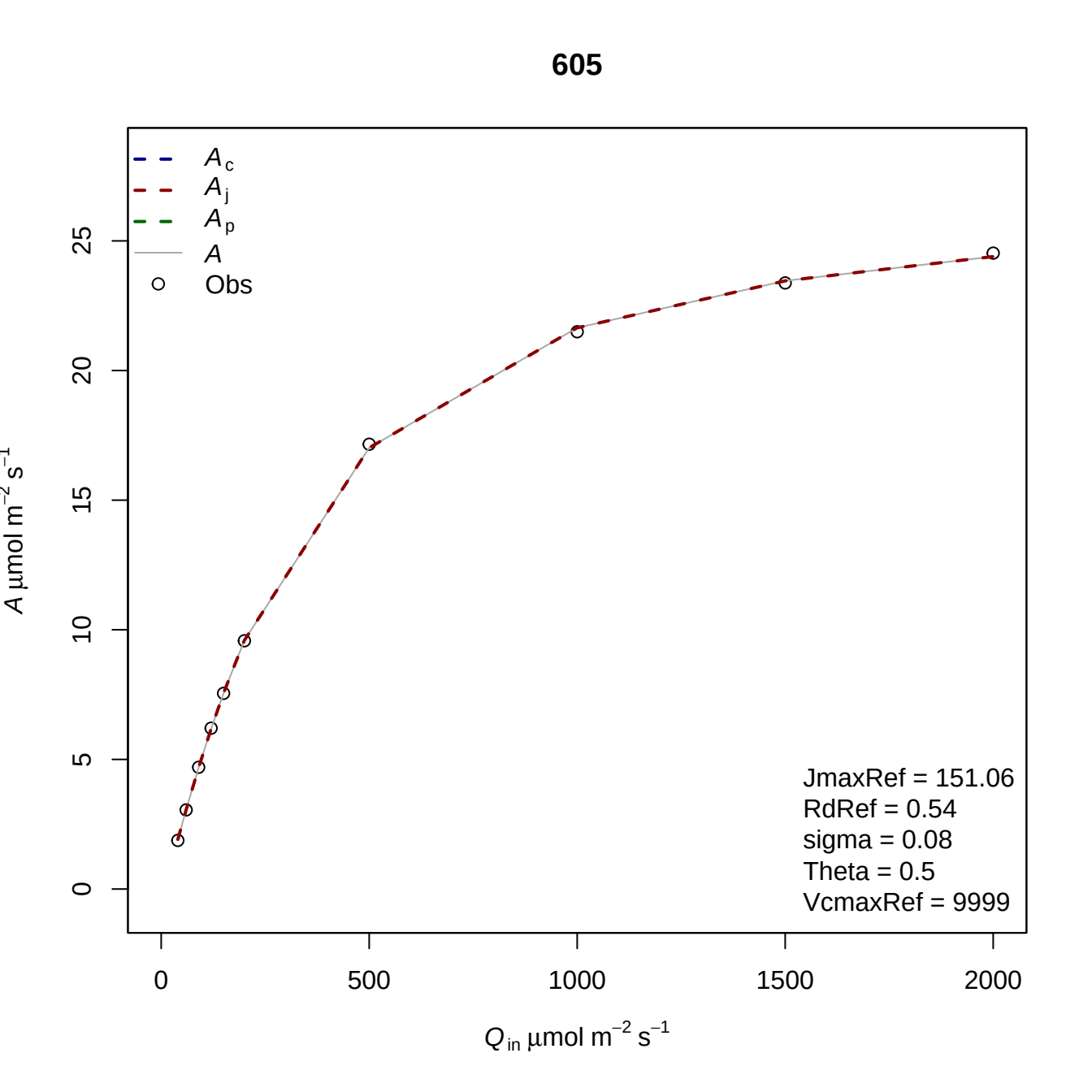

600
A_c
A_j
A_p
A
Obs
JmaxRef = 183.4
RdRef = 3.47
sigma = 0.74
Theta = 0.33
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

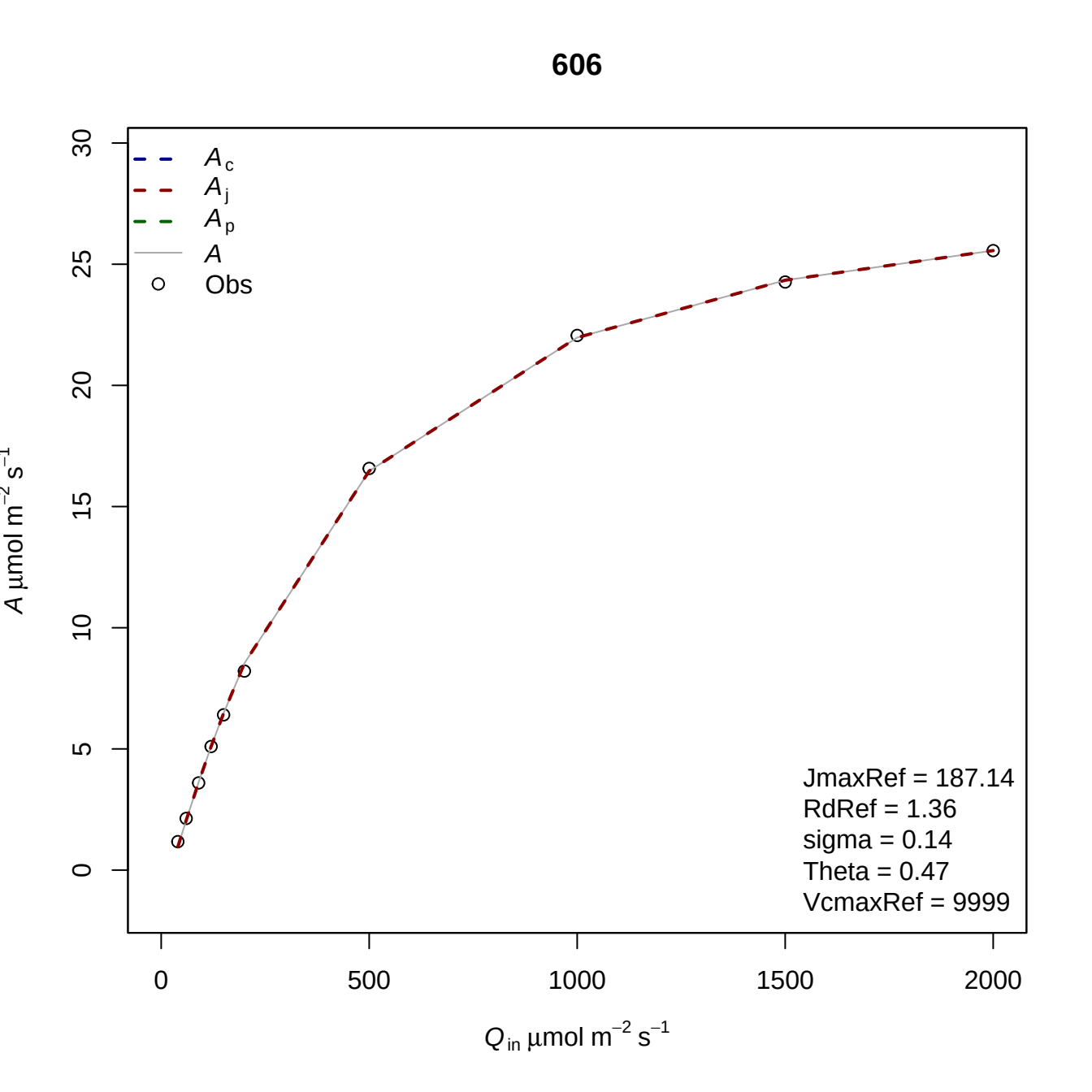

601
A_c
A_j
A_p
A
Obs
JmaxRef = 100.7
RdRef = 0.63
sigma = 0.05
Theta = 0.64
VcmaxRef = 44.93
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

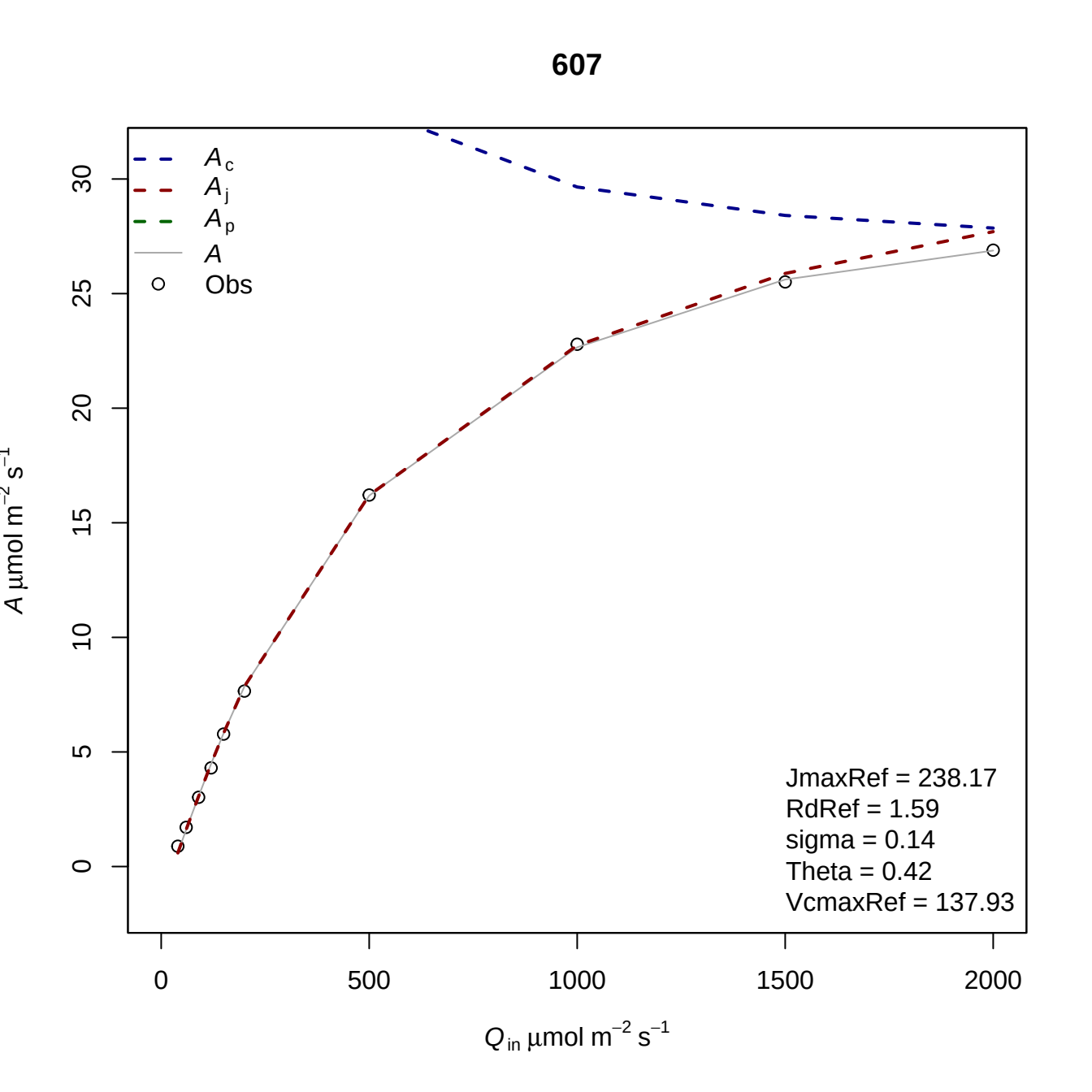

602
A_c
A_j
A_p
A
Obs
JmaxRef = 125.71
RdRef = 1.55
sigma = 0.57
Theta = 0.32
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

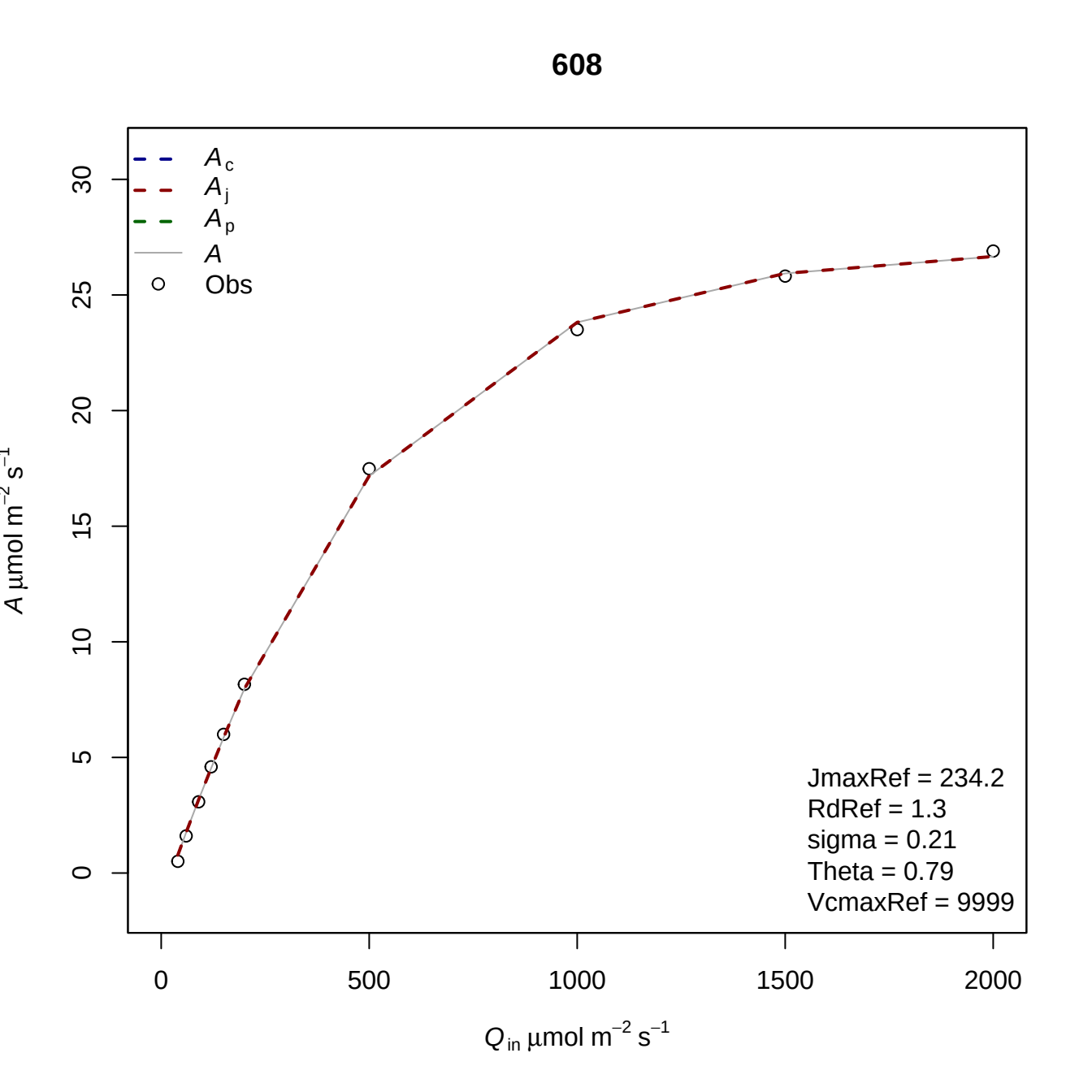

603
A_c
A_j
A_p
A
Obs
JmaxRef = 134.04
RdRef = 1.12
sigma = 0.03
Theta = 0.51
VcmaxRef = 99.97
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

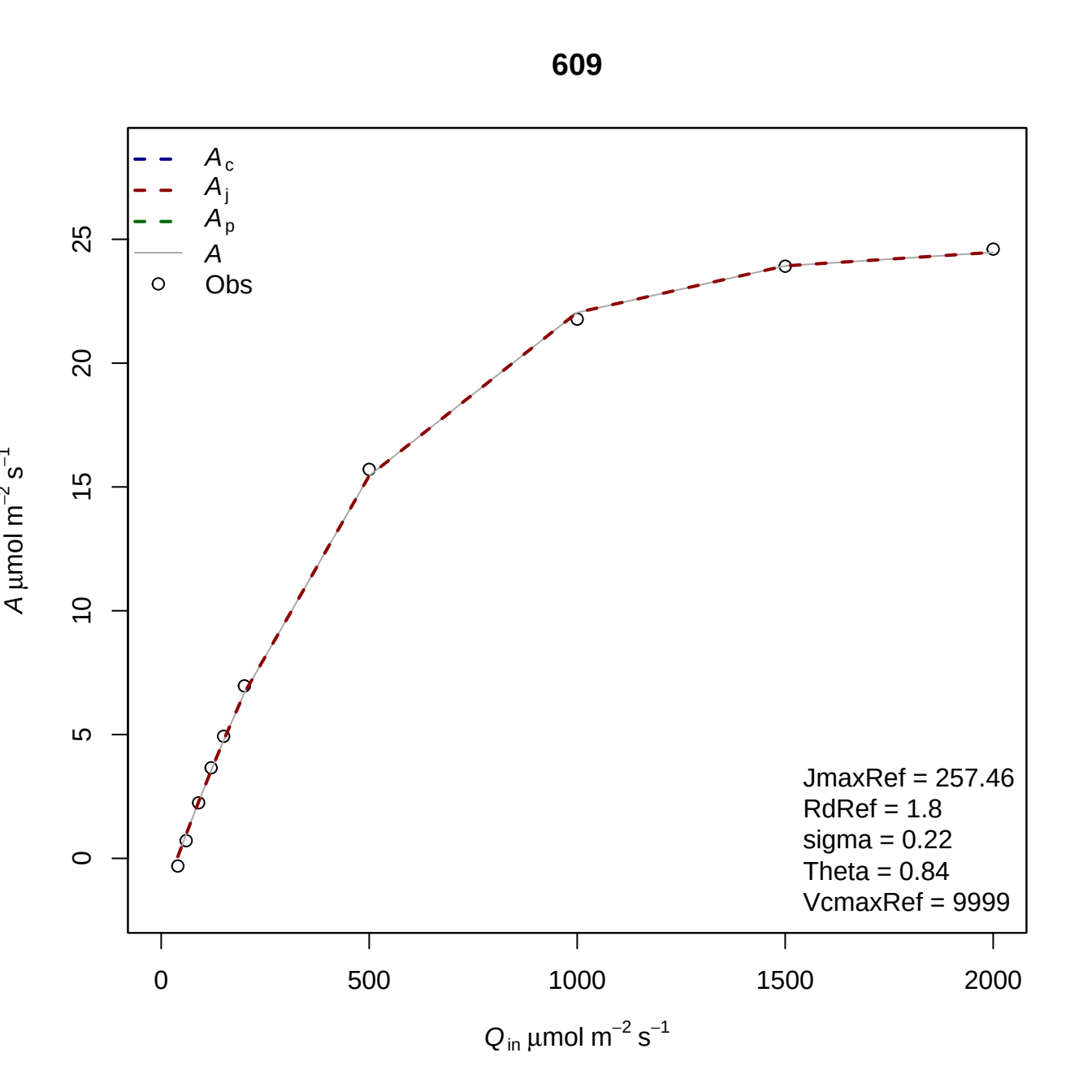

604
A_c
A_j
A_p
A
Obs
JmaxRef = 170.75
RdRef = 2.76
sigma = 0.24
Theta = 0.75
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

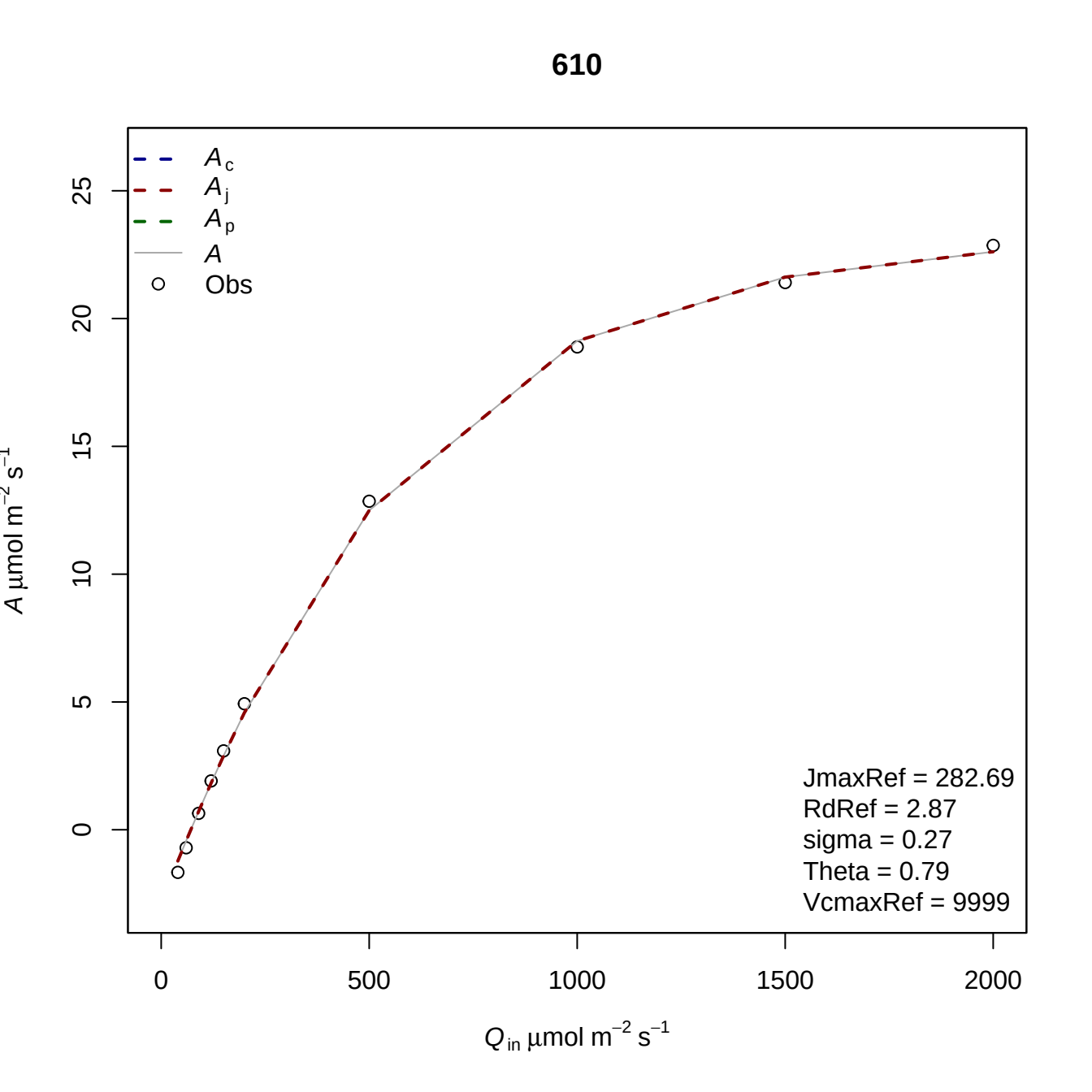

605
A_c
A_j
A_p
A
Obs
JmaxRef = 151.06
RdRef = 0.54
sigma = 0.08
Theta = 0.5
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

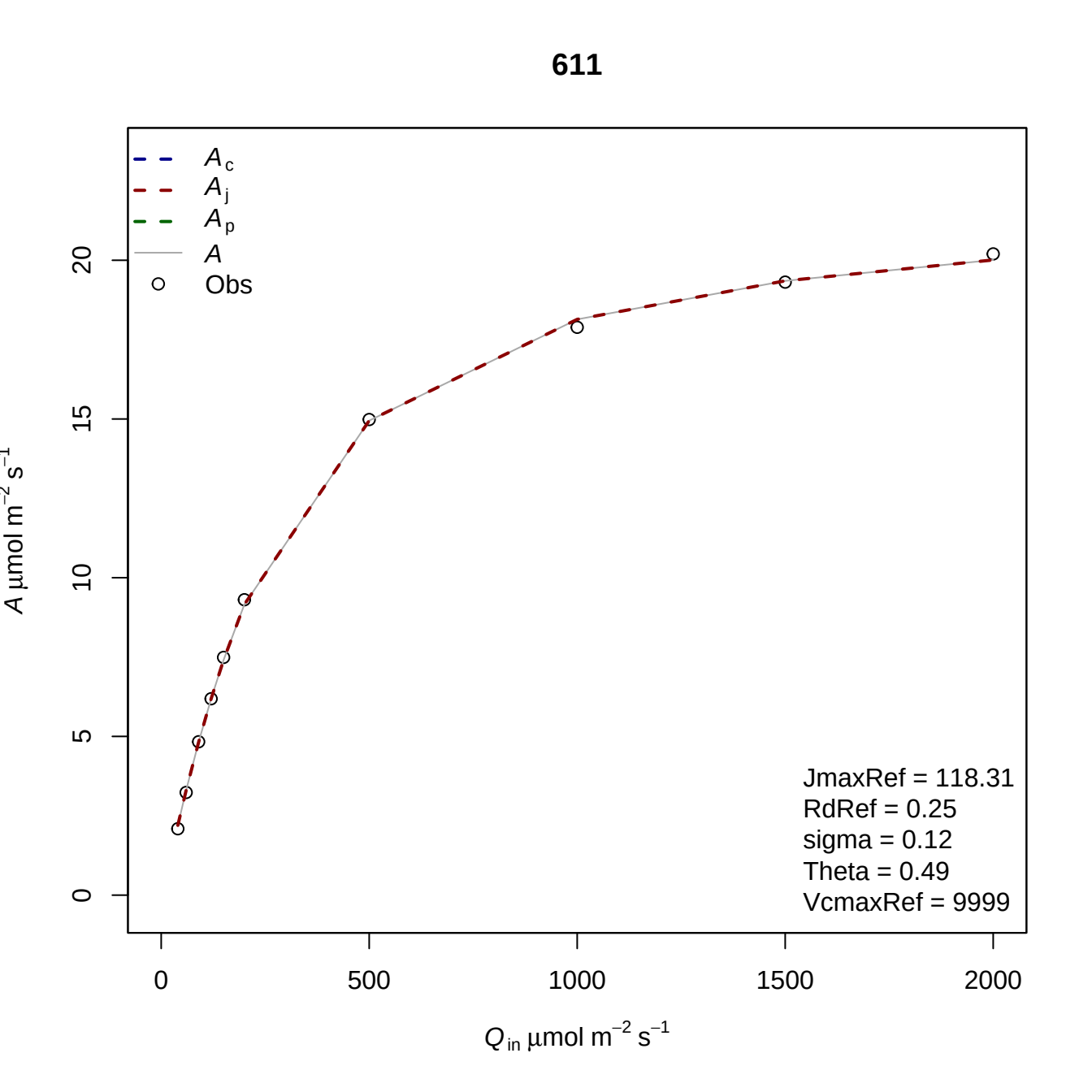

606
A_c
A_j
A_p
A
Obs
JmaxRef = 187.14
RdRef = 1.36
sigma = 0.14
Theta = 0.47
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

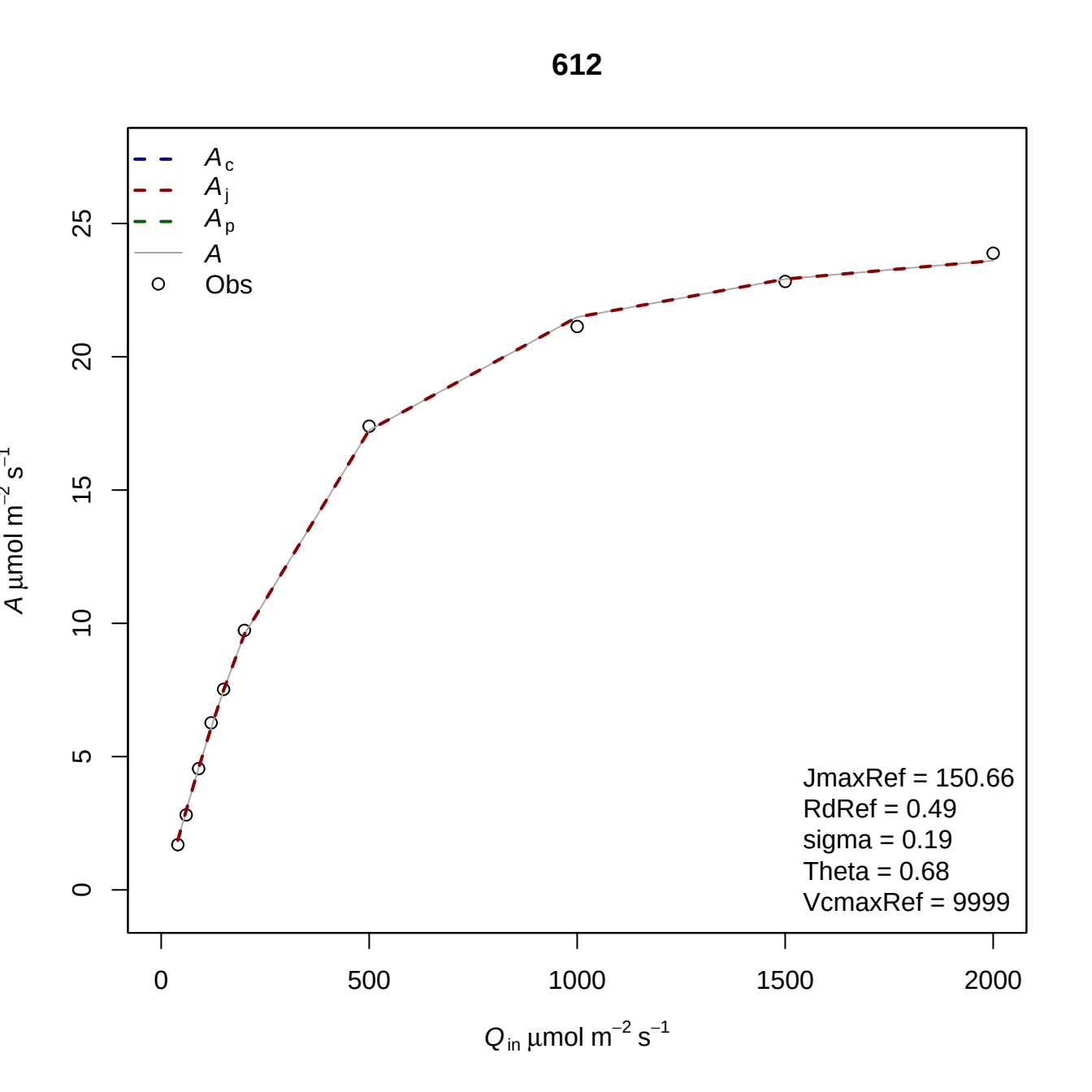

607
A_c
A_j
A_p
A
Obs
JmaxRef = 238.17
RdRef = 1.59
sigma = 0.14
Theta = 0.42
VcmaxRef = 137.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

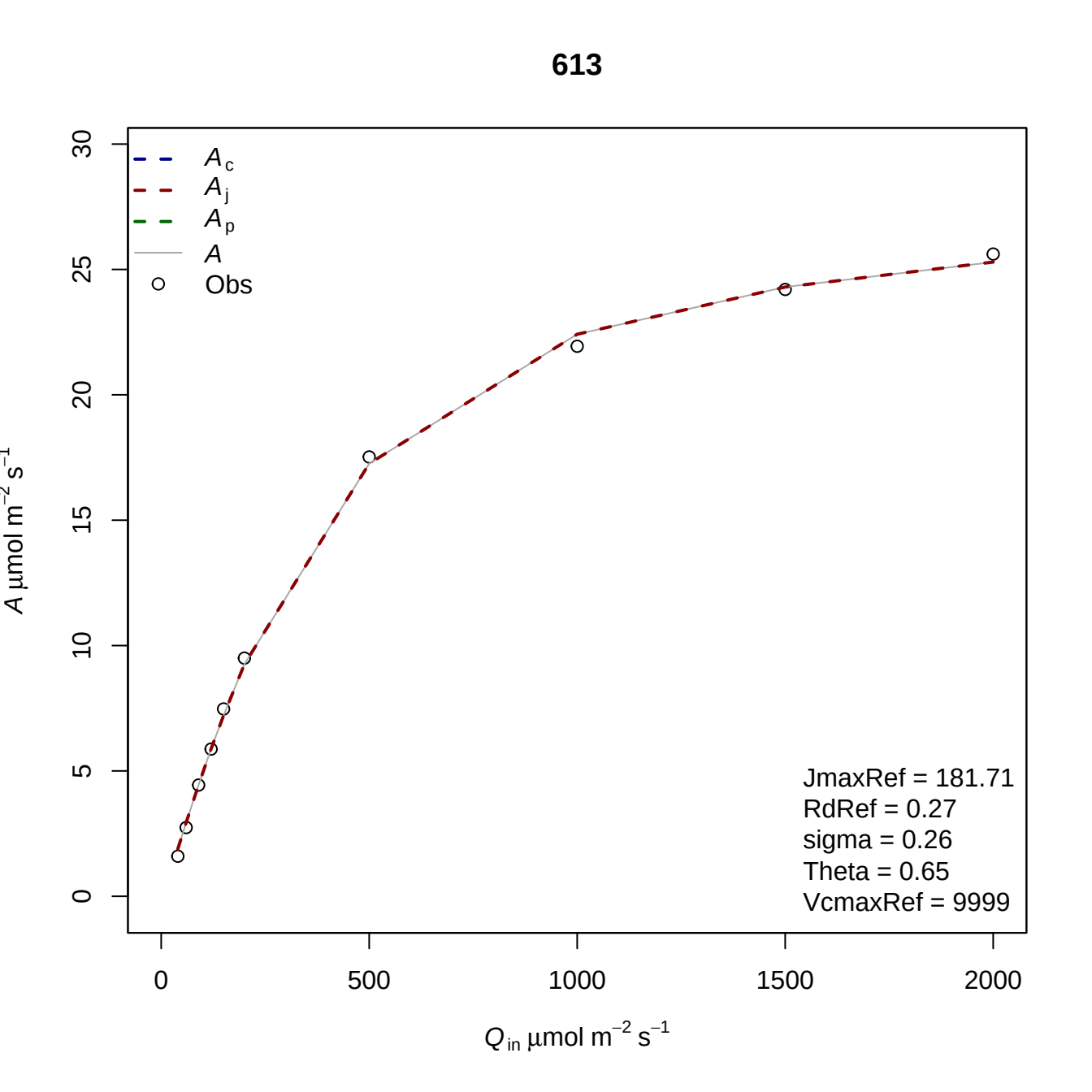

608
A_c
A_j
A_p
A
Obs
JmaxRef = 234.2
RdRef = 1.3
sigma = 0.21
Theta = 0.79
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

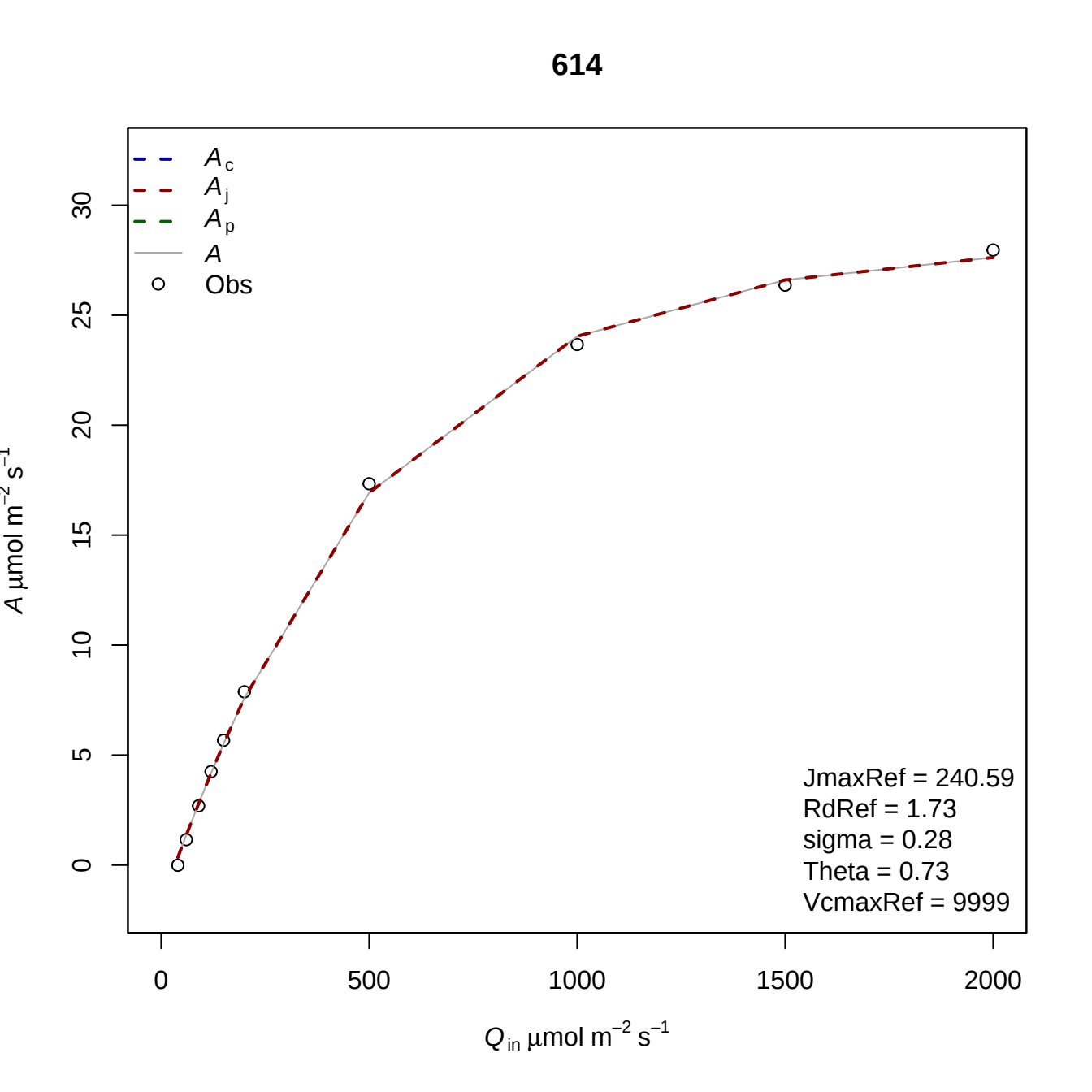

609
A_c
A_j
A_p
A
Obs
JmaxRef = 257.46
RdRef = 1.8
sigma = 0.22
Theta = 0.84
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

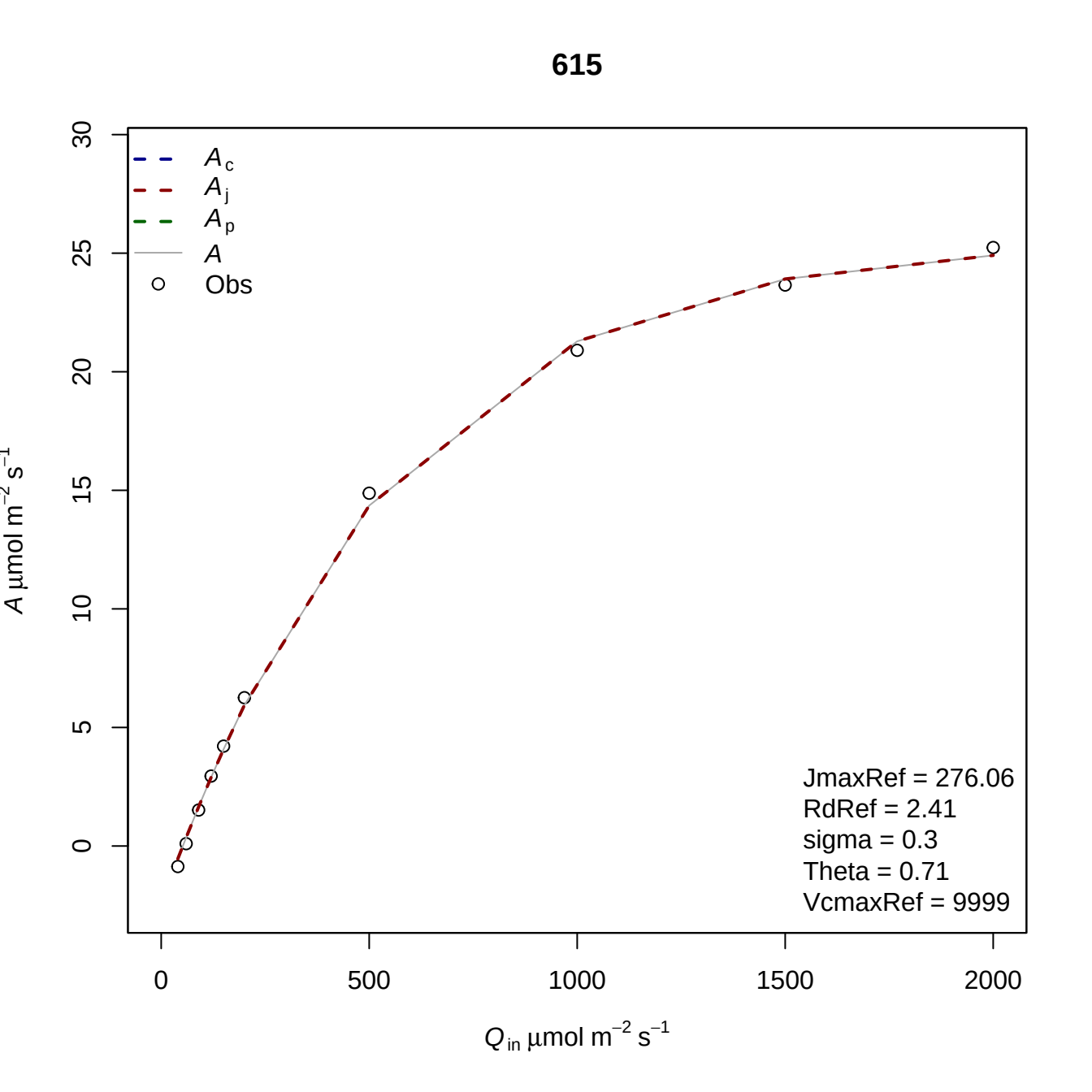

610
A_c
A_j
A_p
A
Obs
JmaxRef = 282.69
RdRef = 2.87
sigma = 0.27
Theta = 0.79
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

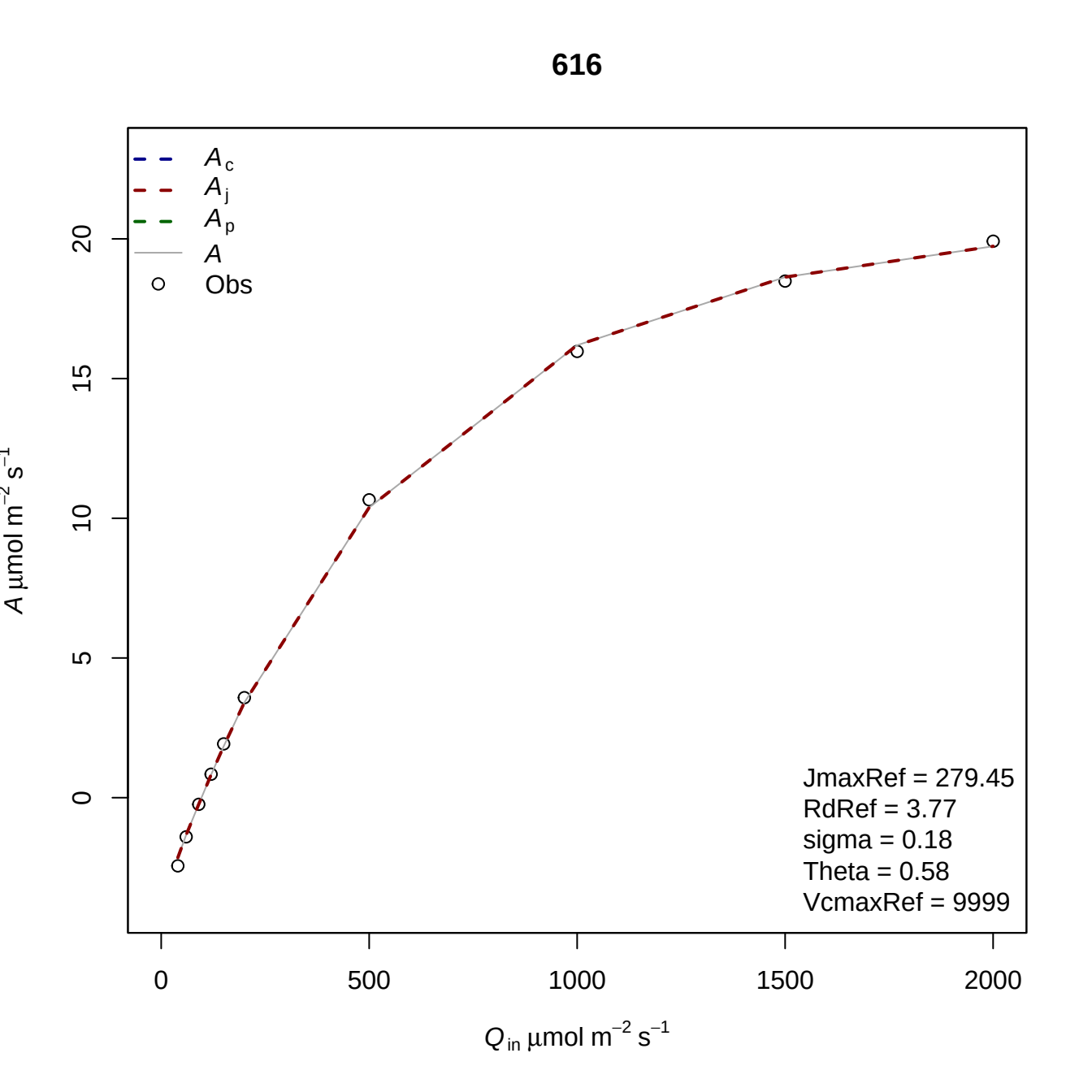

611
A_c
A_j
A_p
A
Obs
JmaxRef = 118.31
RdRef = 0.25
sigma = 0.12
Theta = 0.49
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

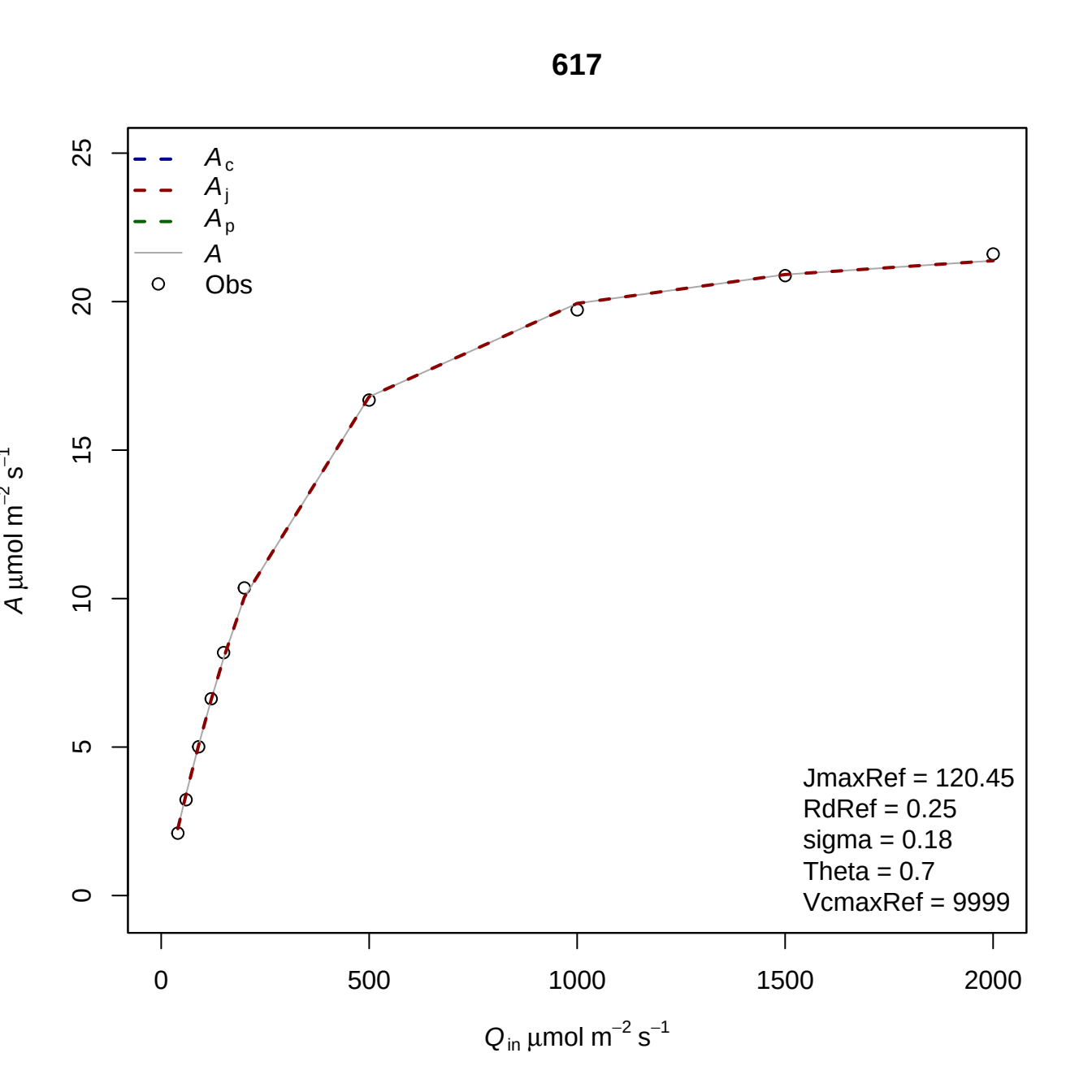

612
A_c
A_j
A_p
A
Obs
JmaxRef = 150.66
RdRef = 0.49
sigma = 0.19
Theta = 0.68
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

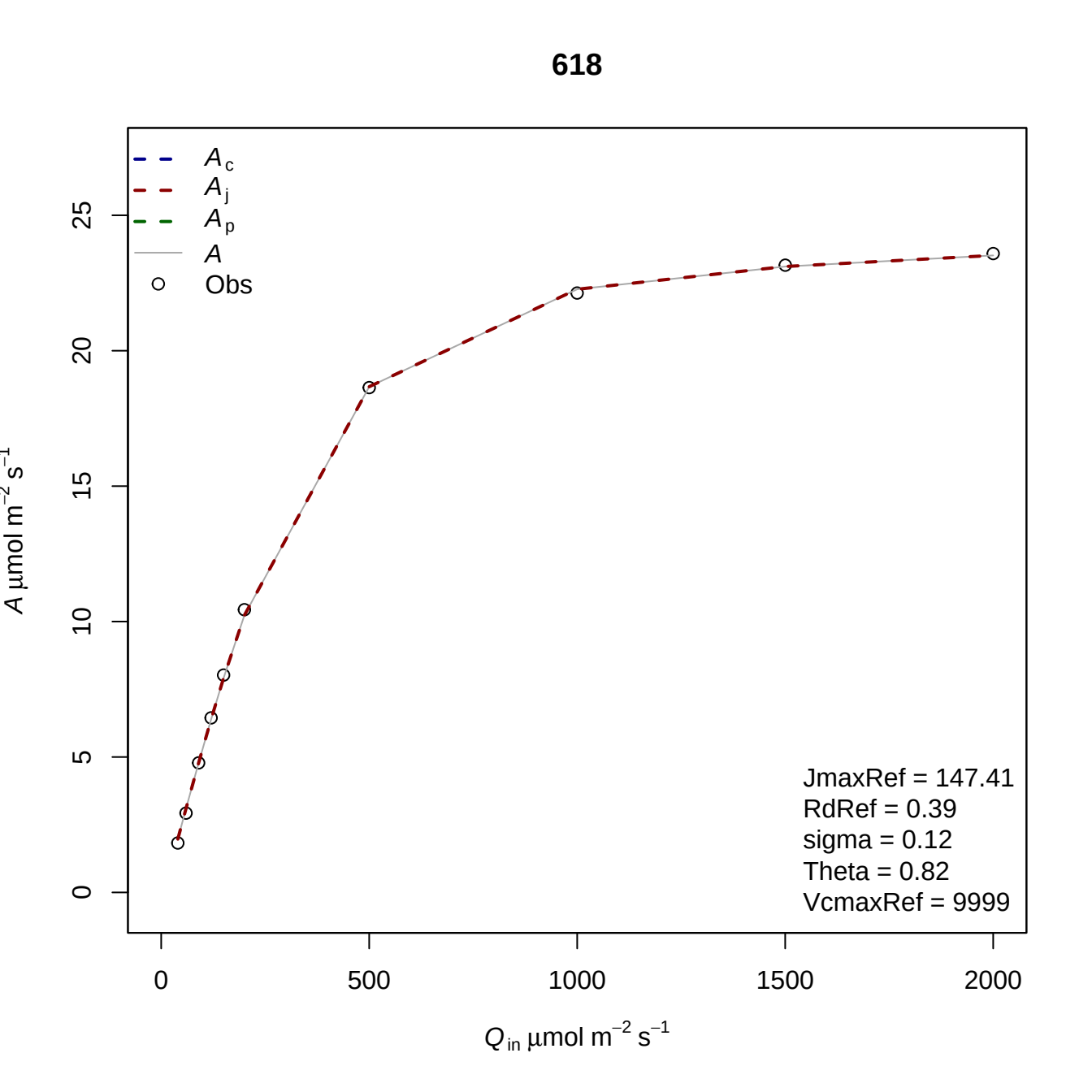

613
A_c
A_j
A_p
A
Obs
JmaxRef = 181.71
RdRef = 0.27
sigma = 0.26
Theta = 0.65
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

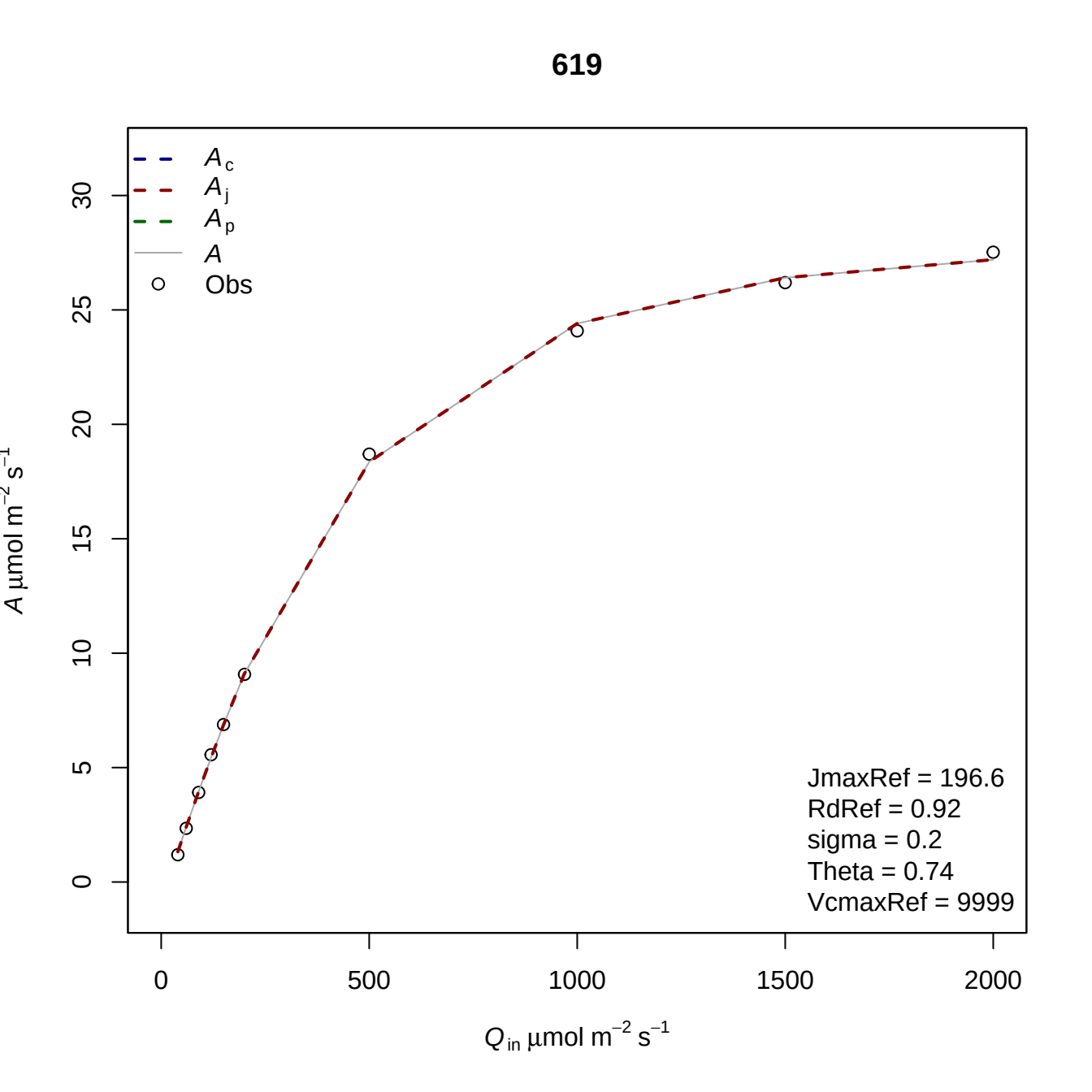

614
A_c
A_j
A_p
A
Obs
JmaxRef = 240.59
RdRef = 1.73
sigma = 0.28
Theta = 0.73
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

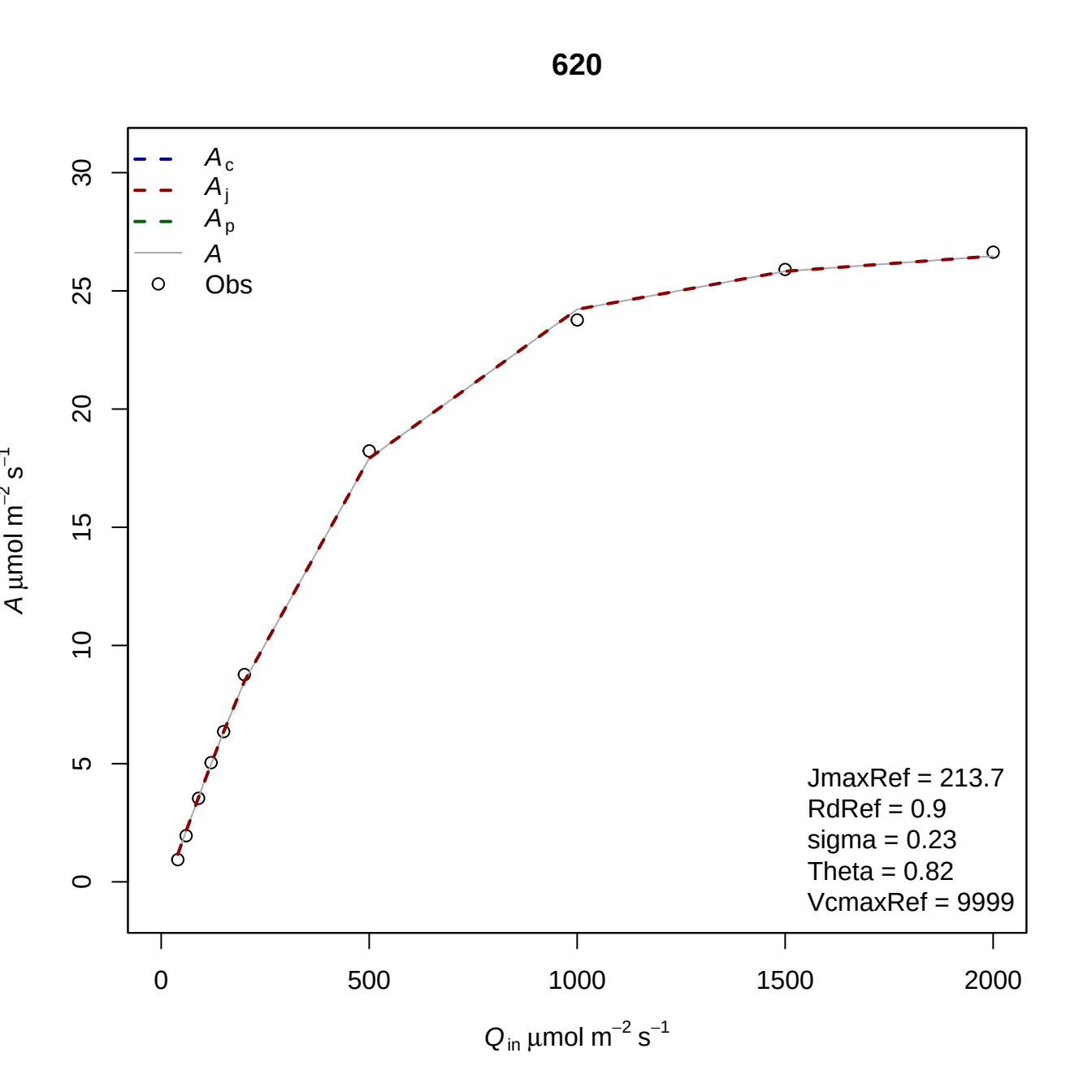

615
A_c
A_j
A_p
A
Obs
JmaxRef = 276.06
RdRef = 2.41
sigma = 0.3
Theta = 0.71
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

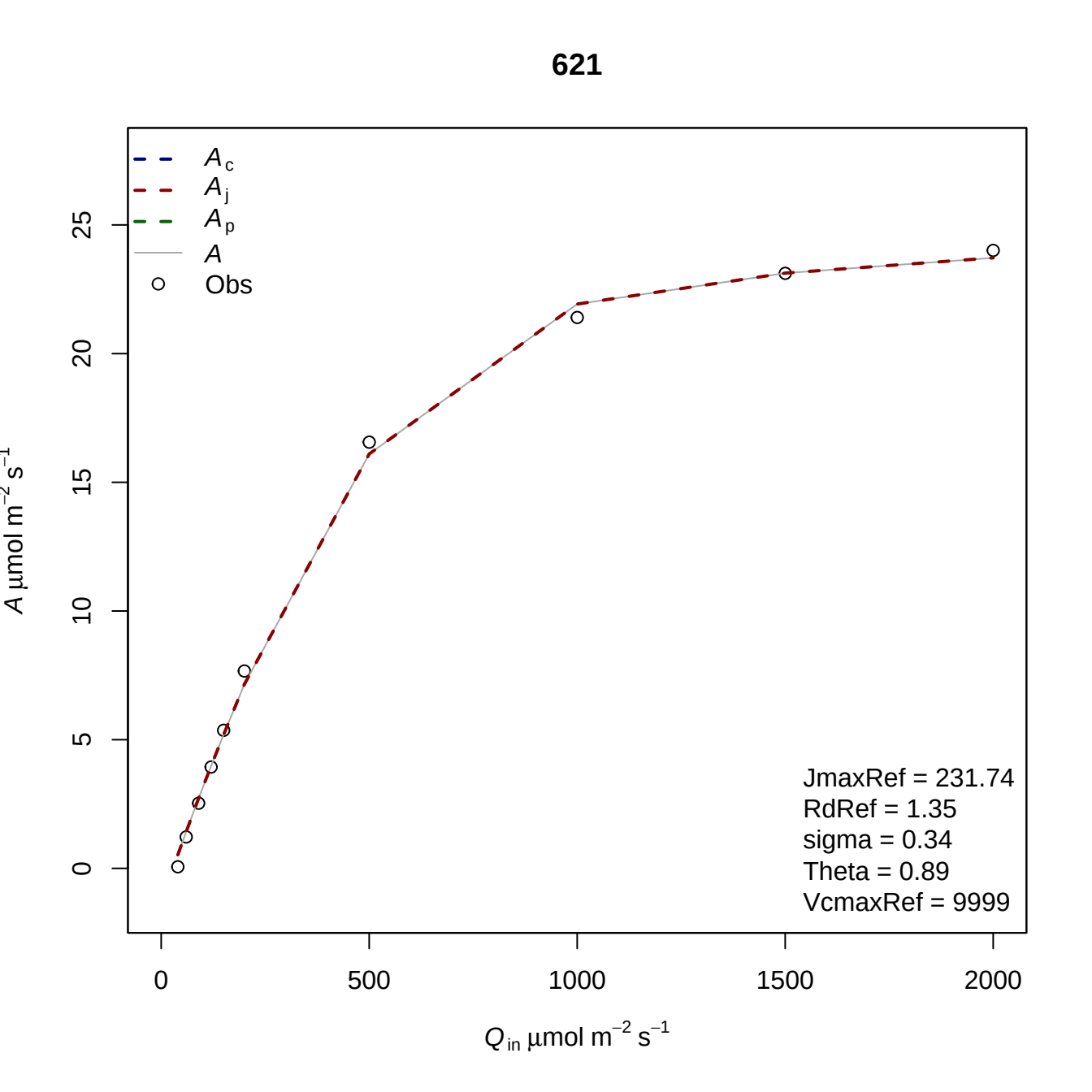

616
A_c
A_j
A_p
A
Obs
JmaxRef = 279.45
RdRef = 3.77
sigma = 0.18
Theta = 0.58
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

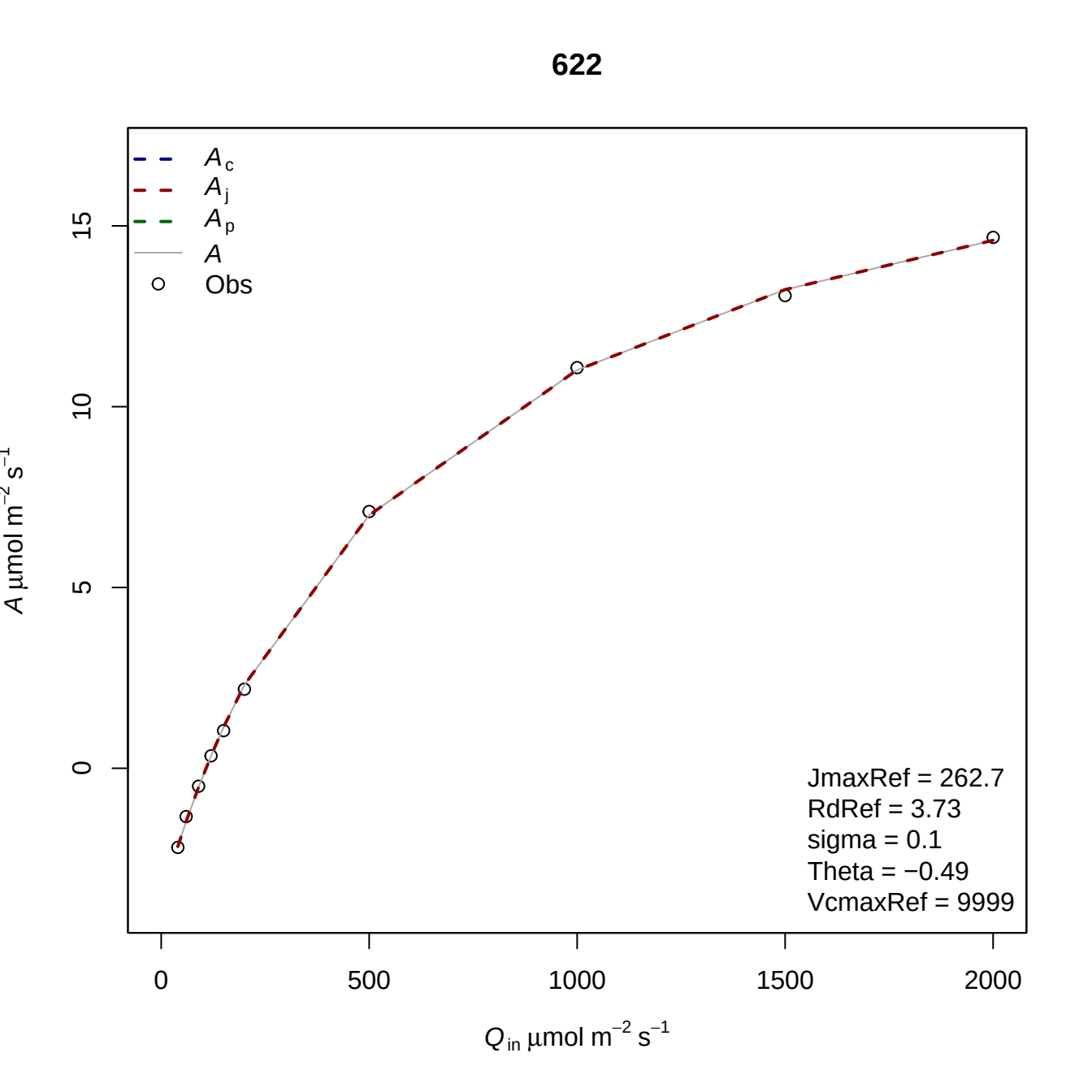

617
A_c
A_j
A_p
A
Obs
JmaxRef = 120.45
RdRef = 0.25
sigma = 0.18
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

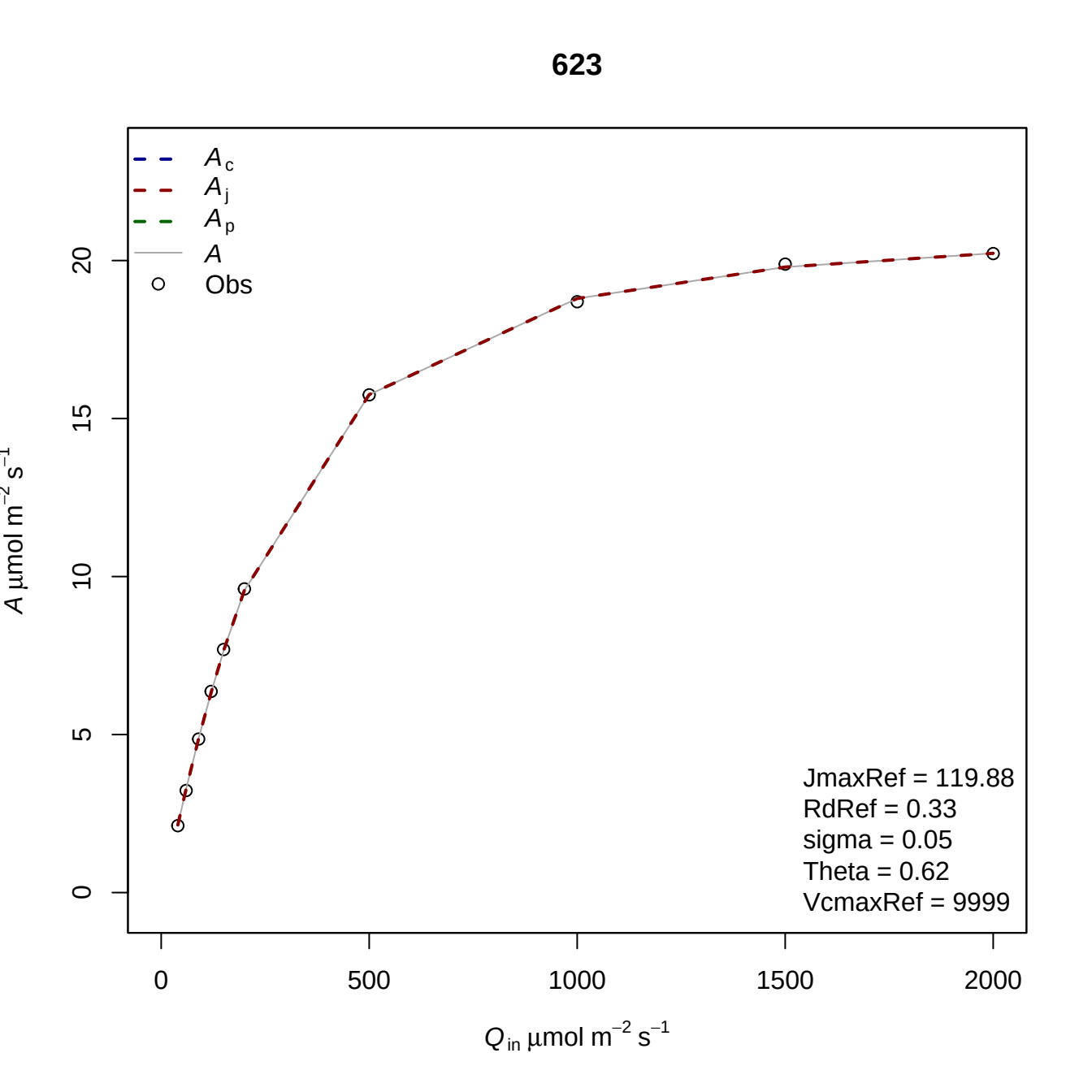

618
A_c
A_j
A_p
A
Obs
JmaxRef = 147.41
RdRef = 0.39
sigma = 0.12
Theta = 0.82
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

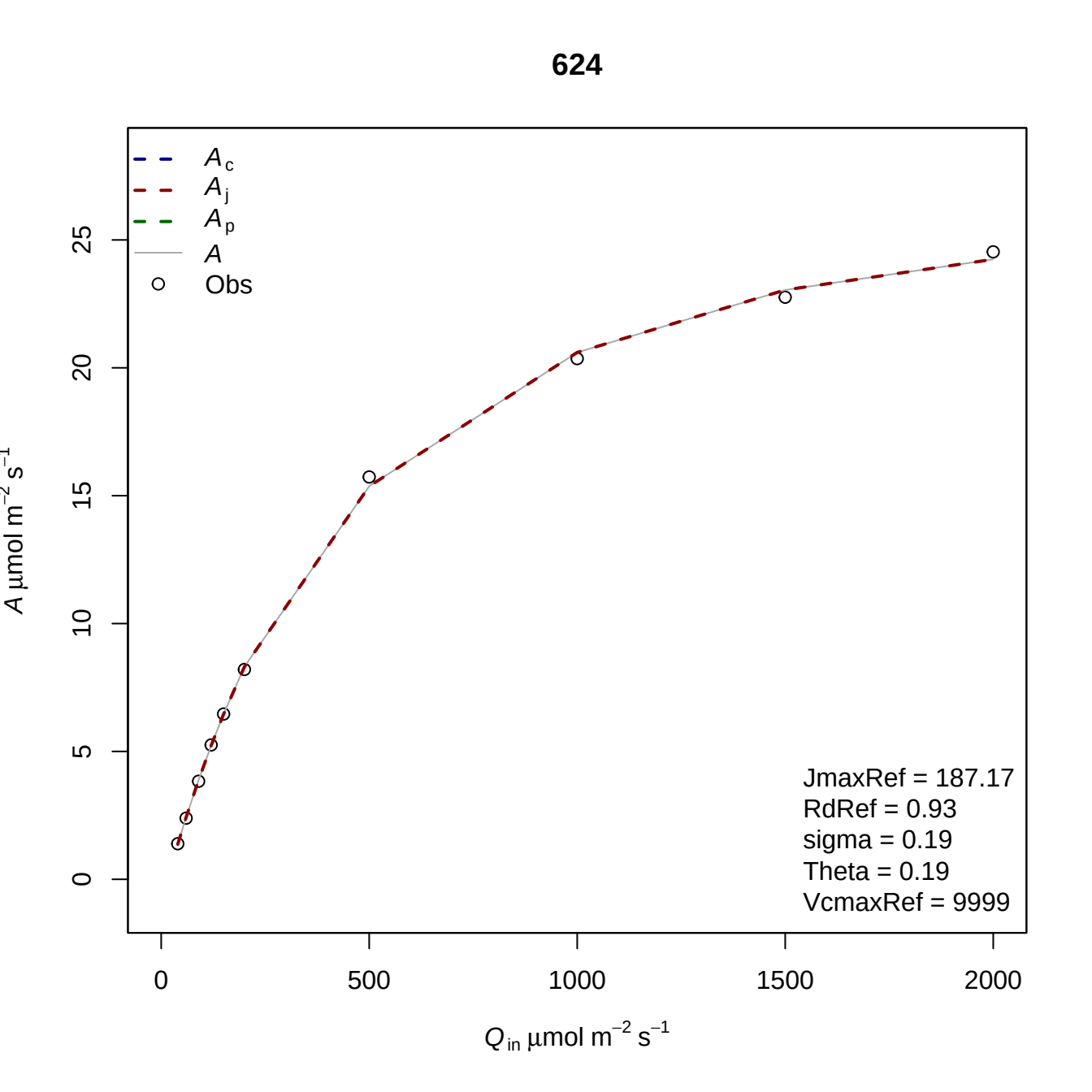

619
A_c
A_j
A_p
A
Obs
JmaxRef = 196.6
RdRef = 0.92
sigma = 0.2
Theta = 0.74
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

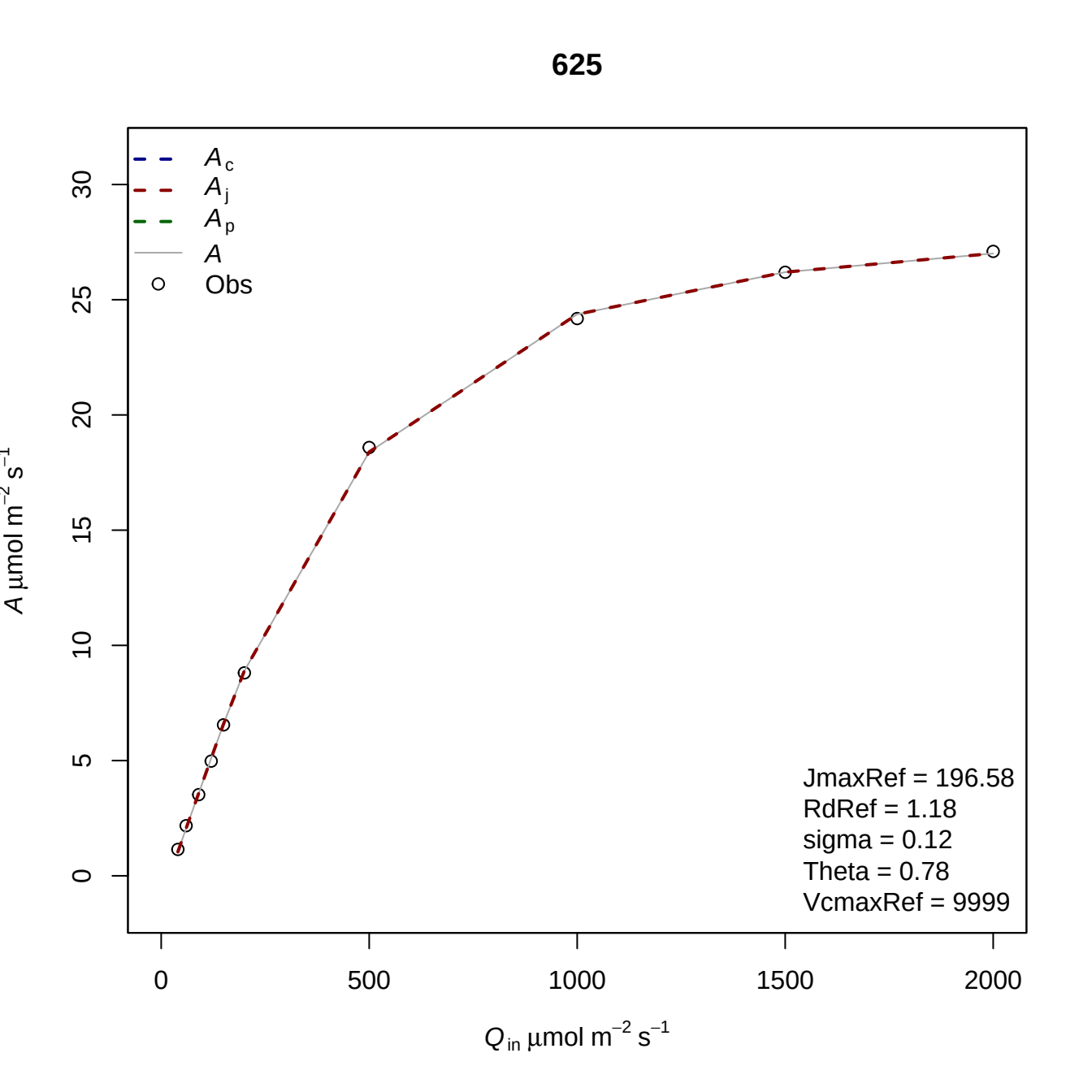

620
A_c
A_j
A_p
A
Obs
JmaxRef = 213.7
RdRef = 0.9
sigma = 0.23
Theta = 0.82
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

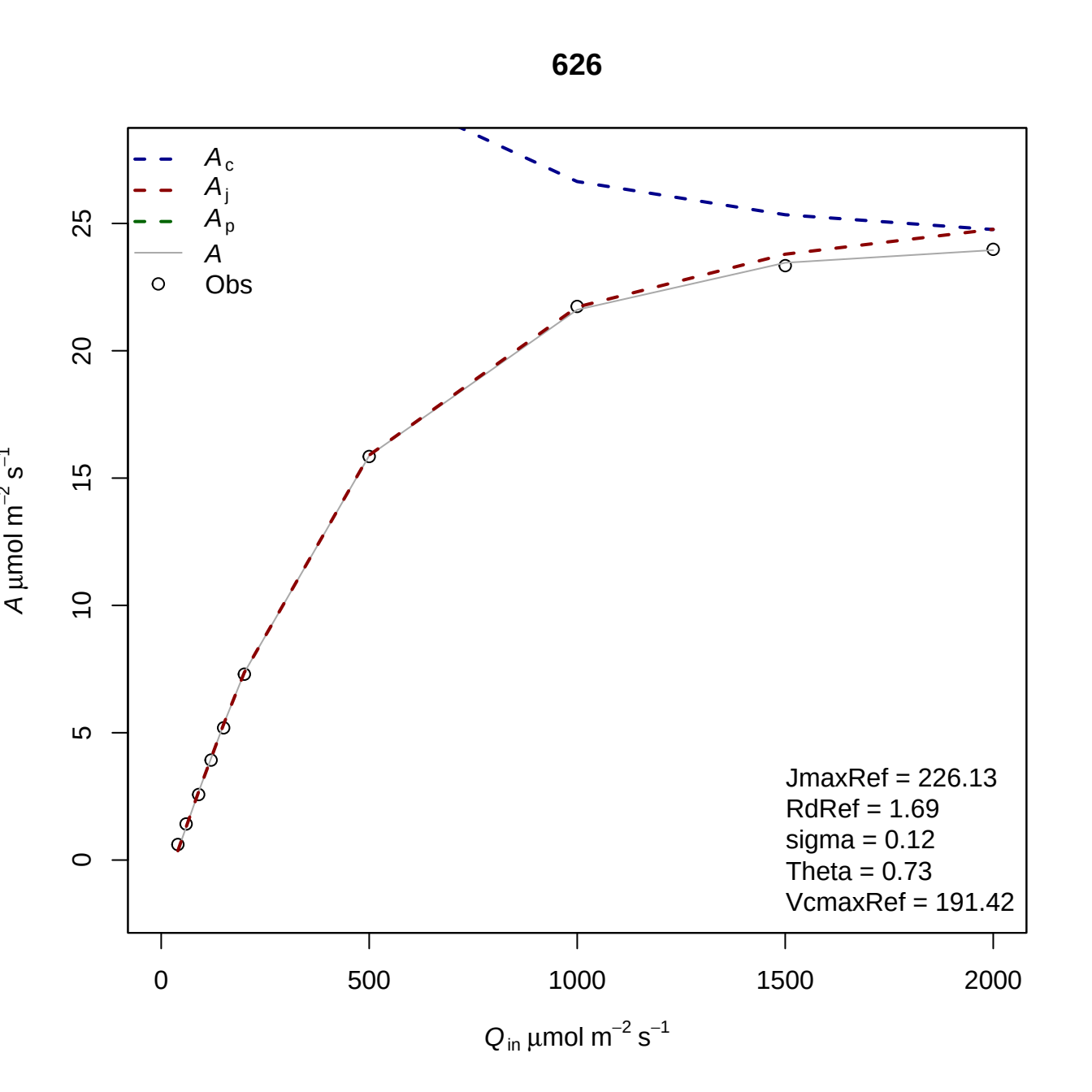

621
A_c
A_j
A_p
A
Obs
JmaxRef = 231.74
RdRef = 1.35
sigma = 0.34
Theta = 0.89
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

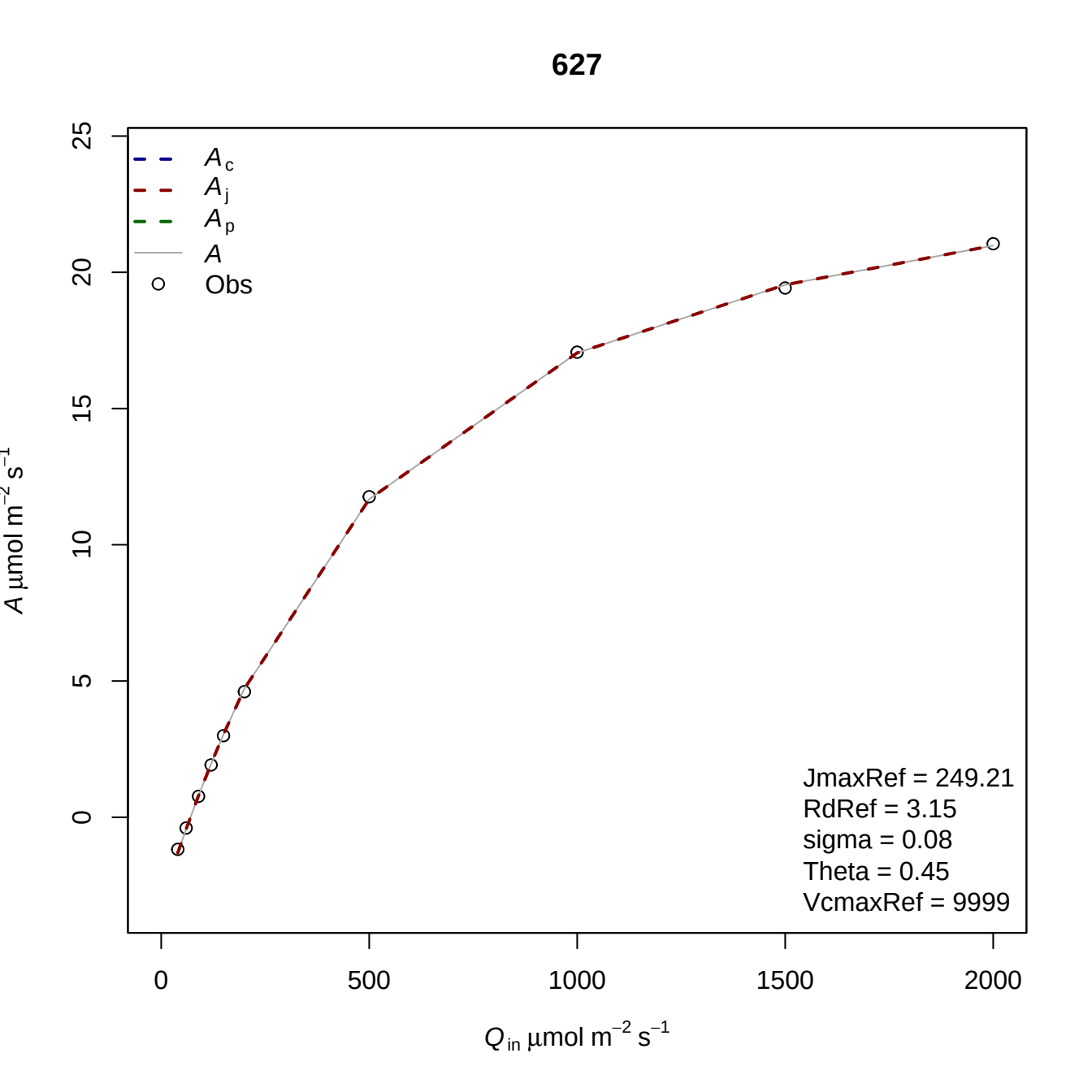

622
A_c
A_j
A_p
A
Obs
JmaxRef = 262.7
RdRef = 3.73
sigma = 0.1
Theta = −0.49
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

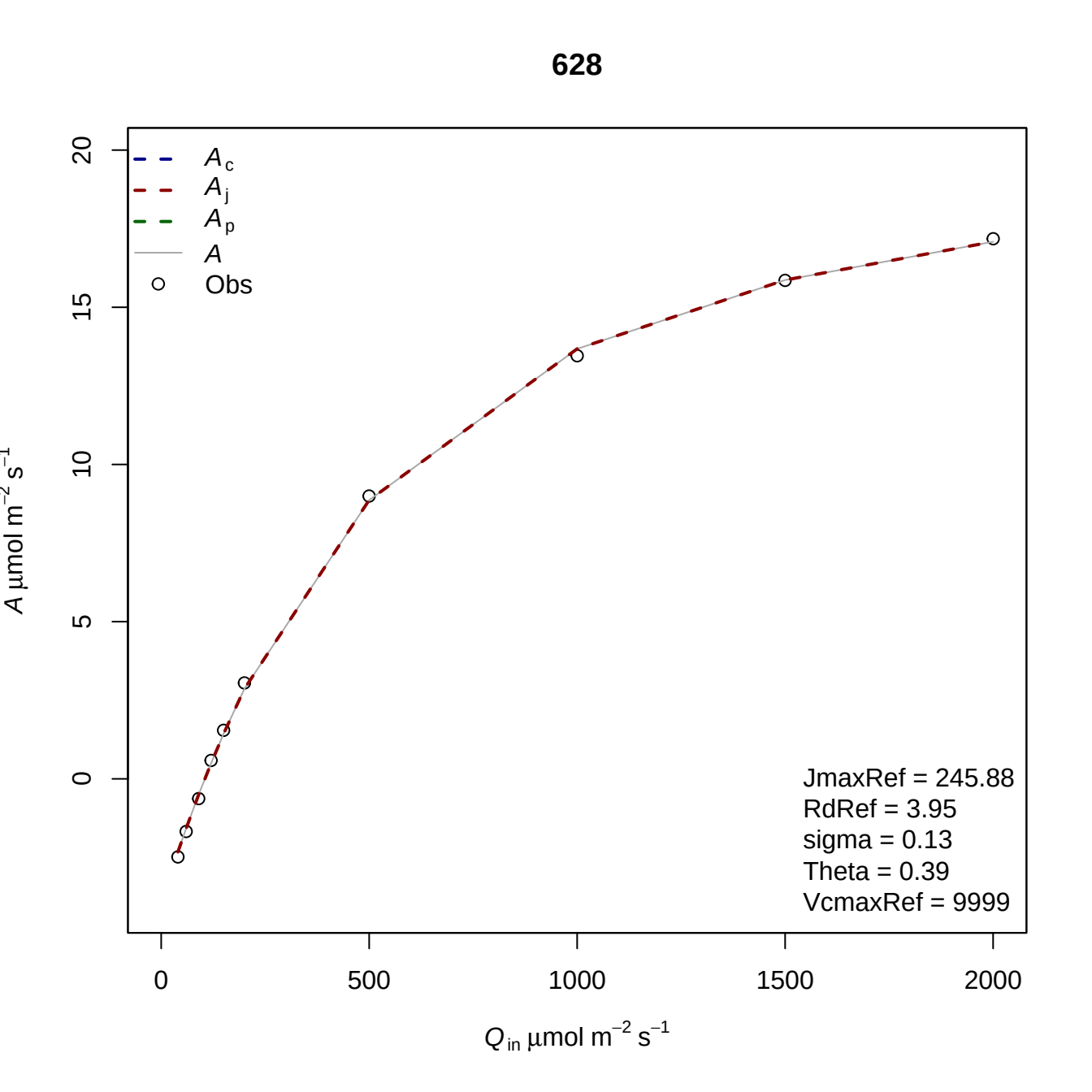

623
A_c
A_j
A_p
A
Obs
JmaxRef = 119.88
RdRef = 0.33
sigma = 0.05
Theta = 0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

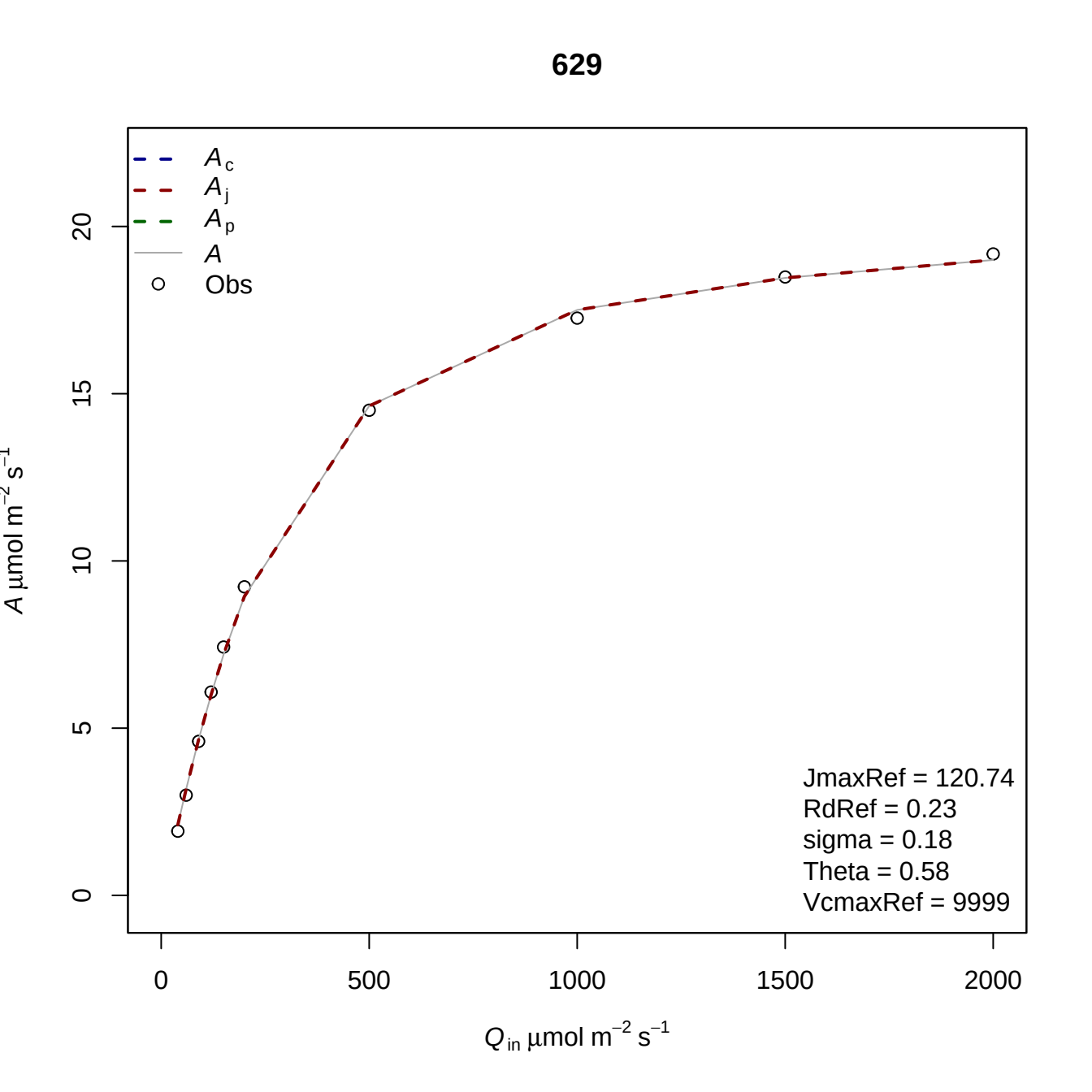

624
A_c
A_j
A_p
A
Obs
JmaxRef = 187.17
RdRef = 0.93
sigma = 0.19
Theta = 0.19
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

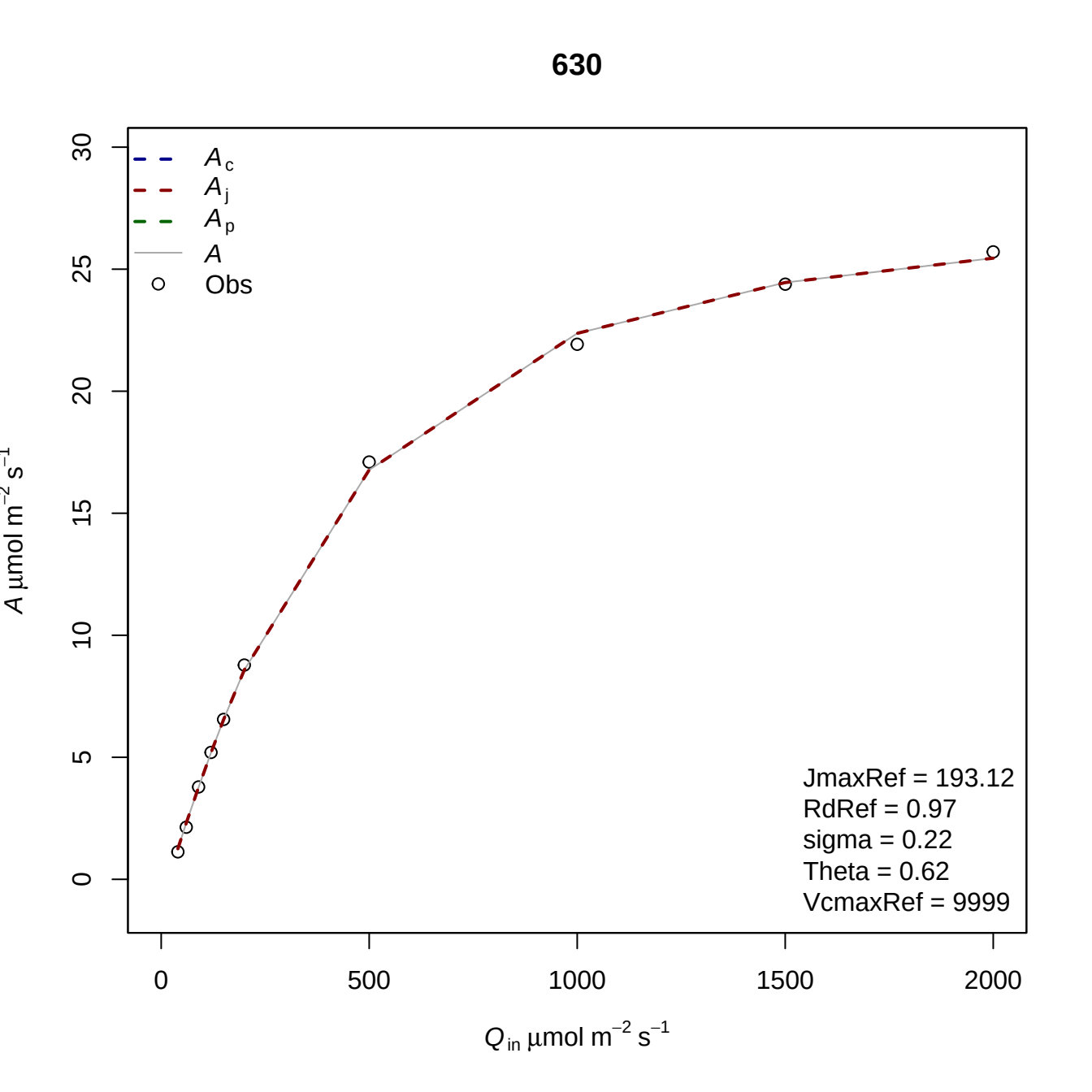

625
A_c
A_j
A_p
A
Obs
JmaxRef = 196.58
RdRef = 1.18
sigma = 0.12
Theta = 0.78
VcmaxRef = 9999
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

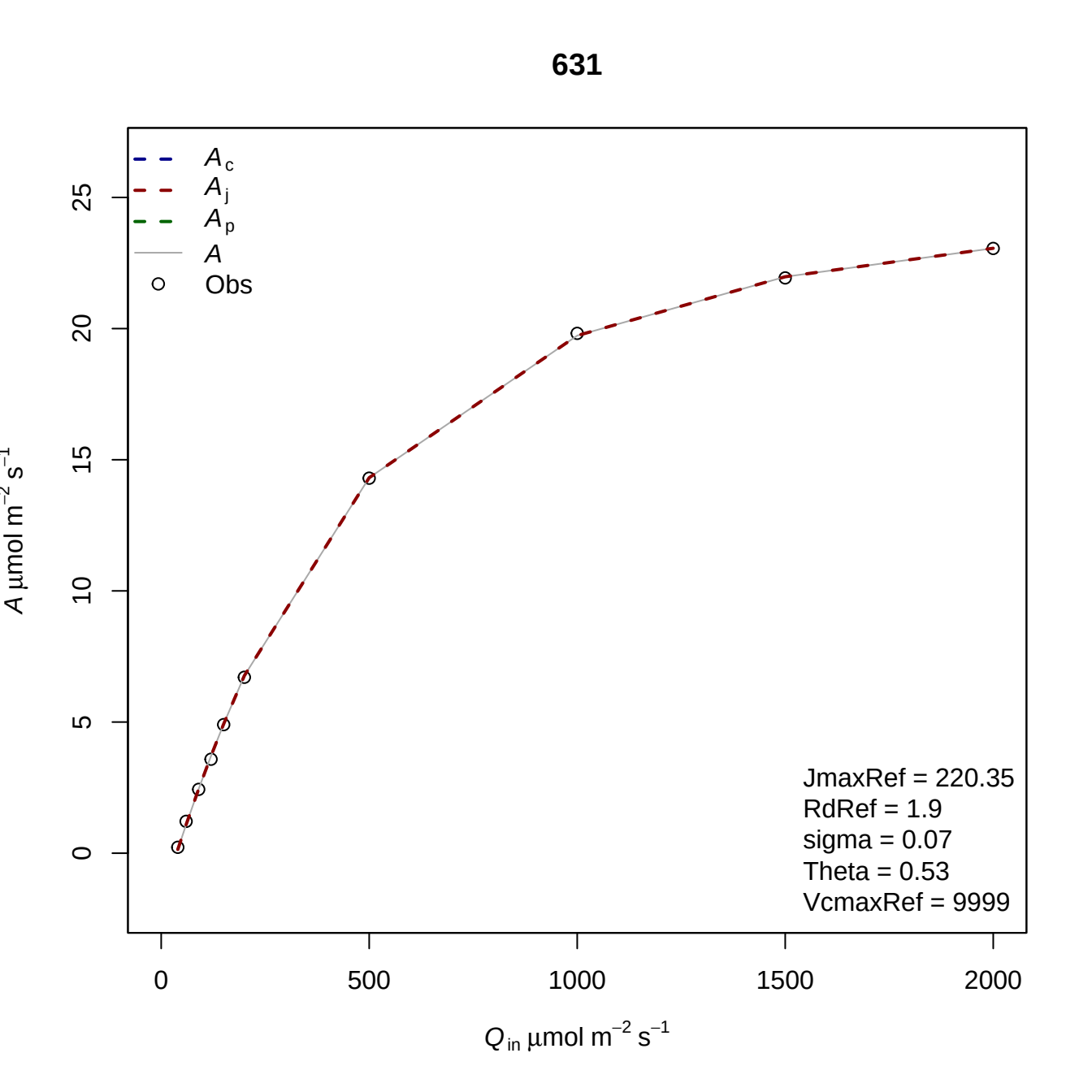

626
A_c
A_j
A_p
A
Obs
JmaxRef = 226.13
RdRef = 1.69
sigma = 0.12
Theta = 0.73
VcmaxRef = 191.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

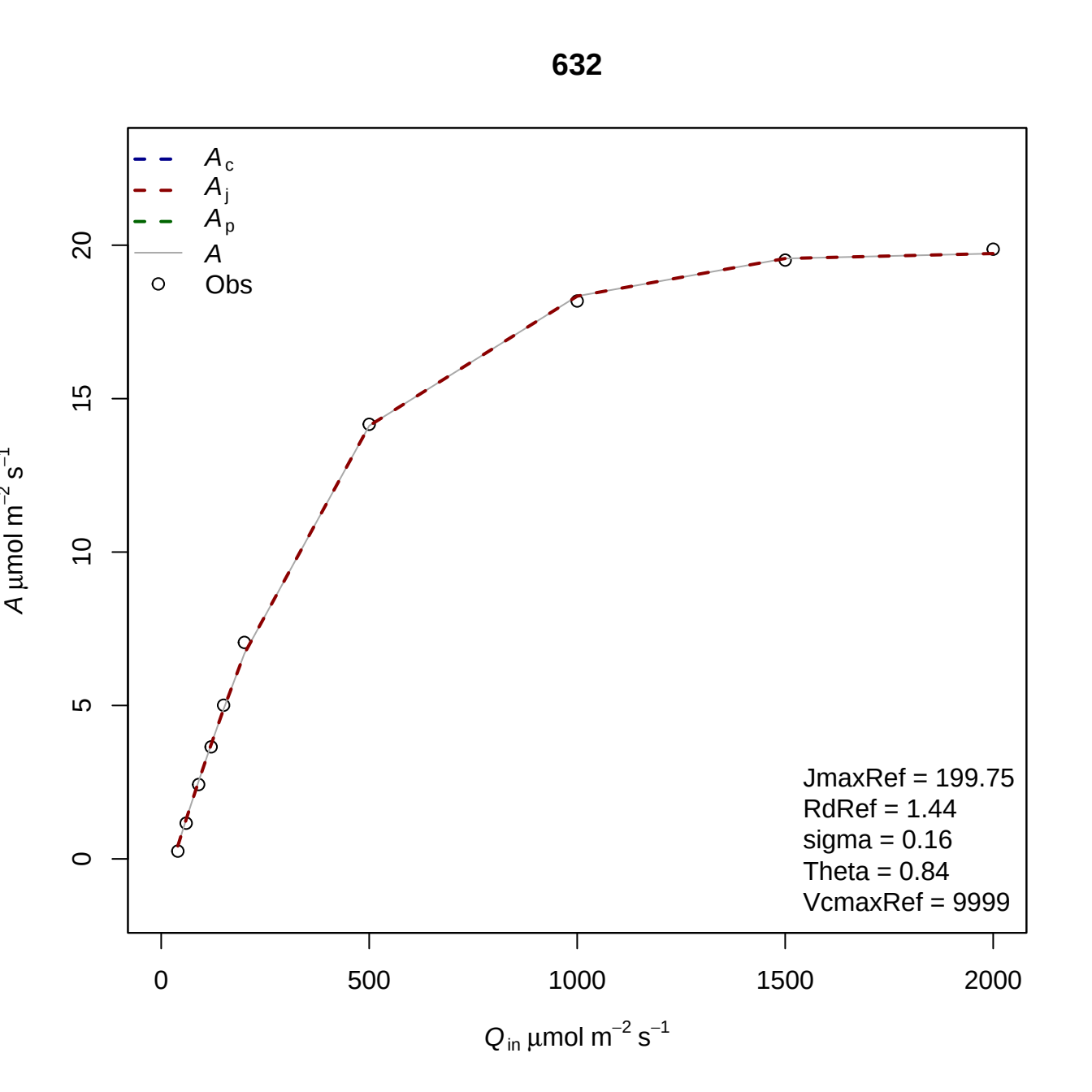

627
A_c
A_j
A_p
A
Obs
JmaxRef = 249.21
RdRef = 3.15
sigma = 0.08
Theta = 0.45
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

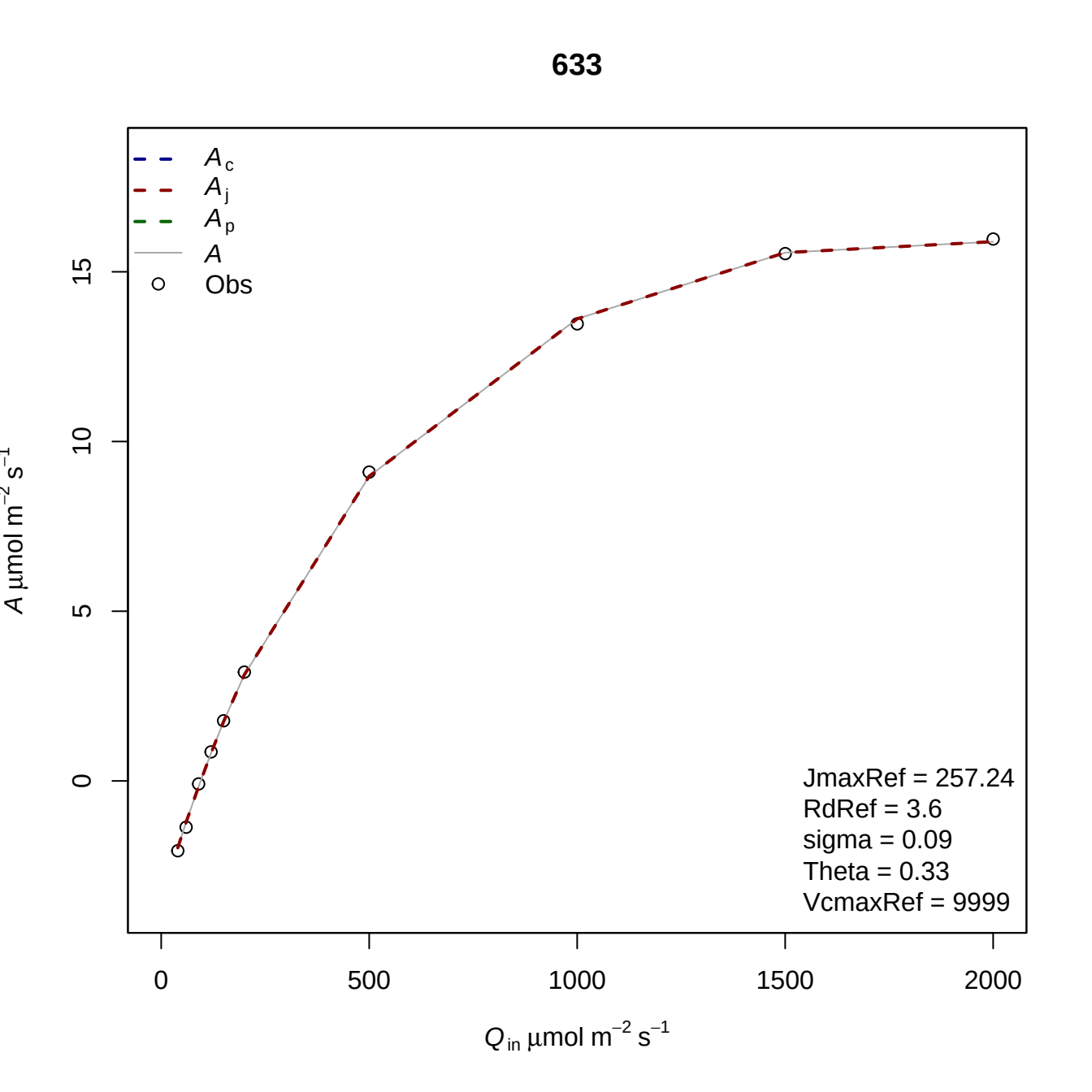

628
A_c
A_j
A_p
A
Obs
JmaxRef = 245.88
RdRef = 3.95
sigma = 0.13
Theta = 0.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

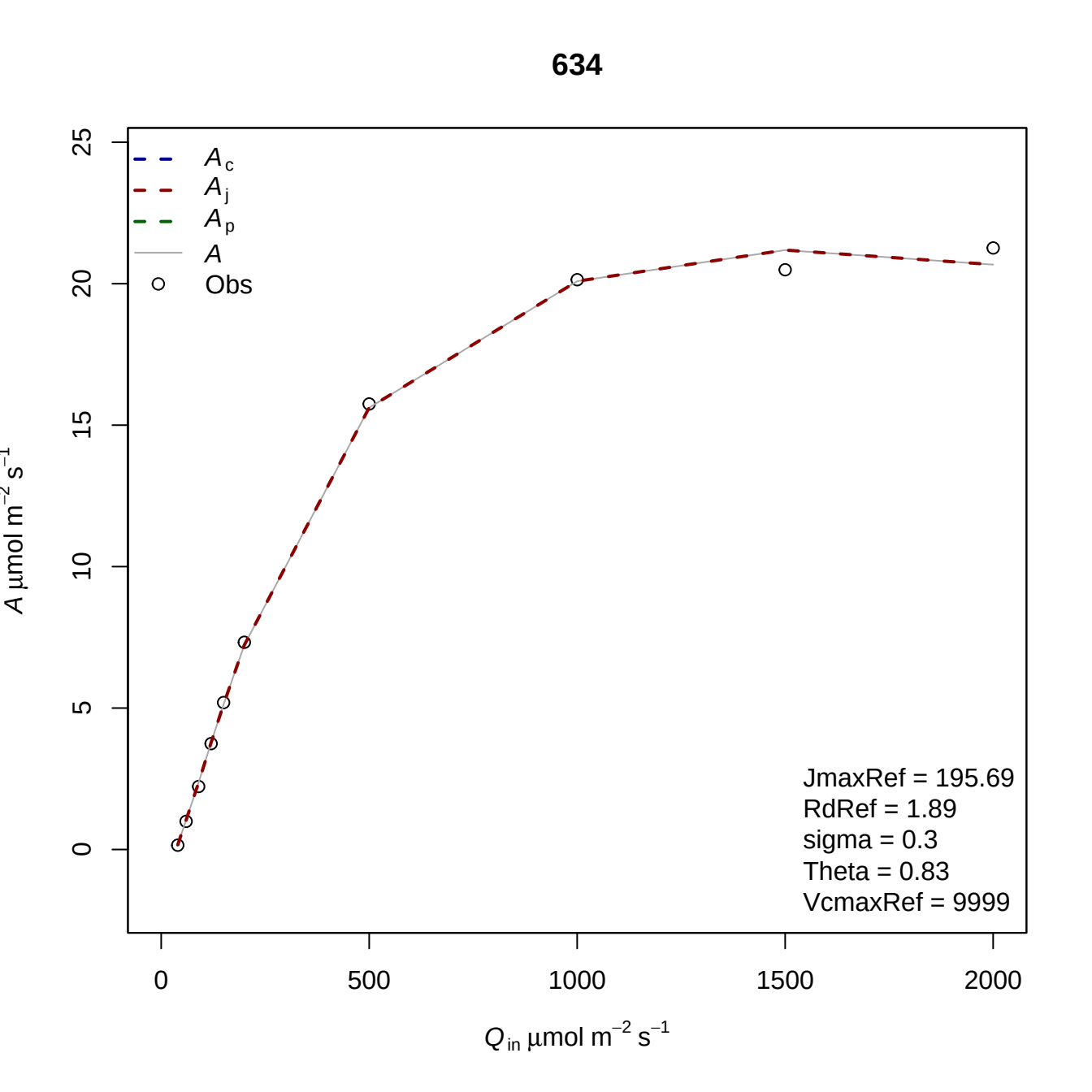

629
A_c
A_j
A_p
A
Obs
JmaxRef = 120.74
RdRef = 0.23
sigma = 0.18
Theta = 0.58
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

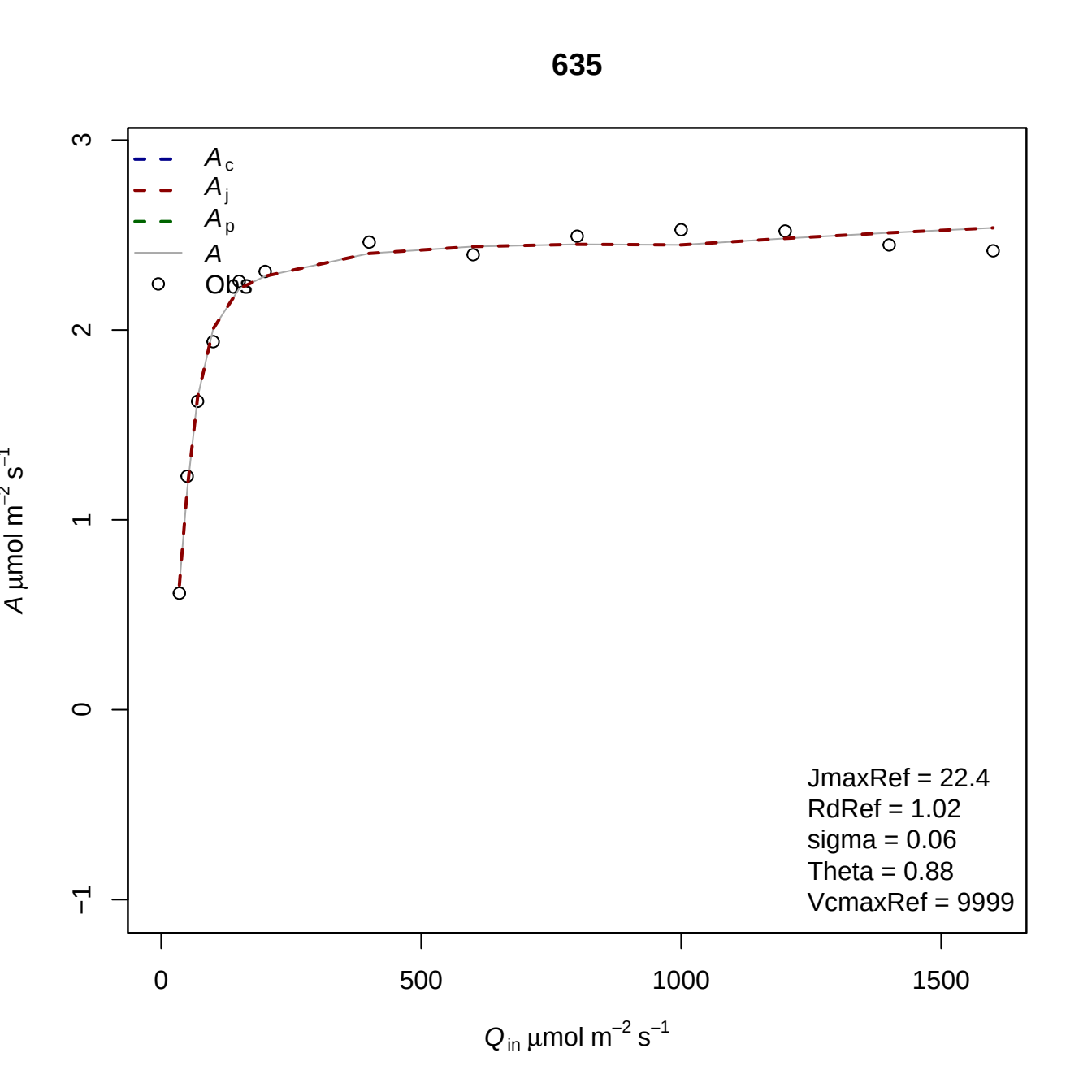

630
A_c
A_j
A_p
A
Obs
JmaxRef = 193.12
RdRef = 0.97
sigma = 0.22
Theta = 0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

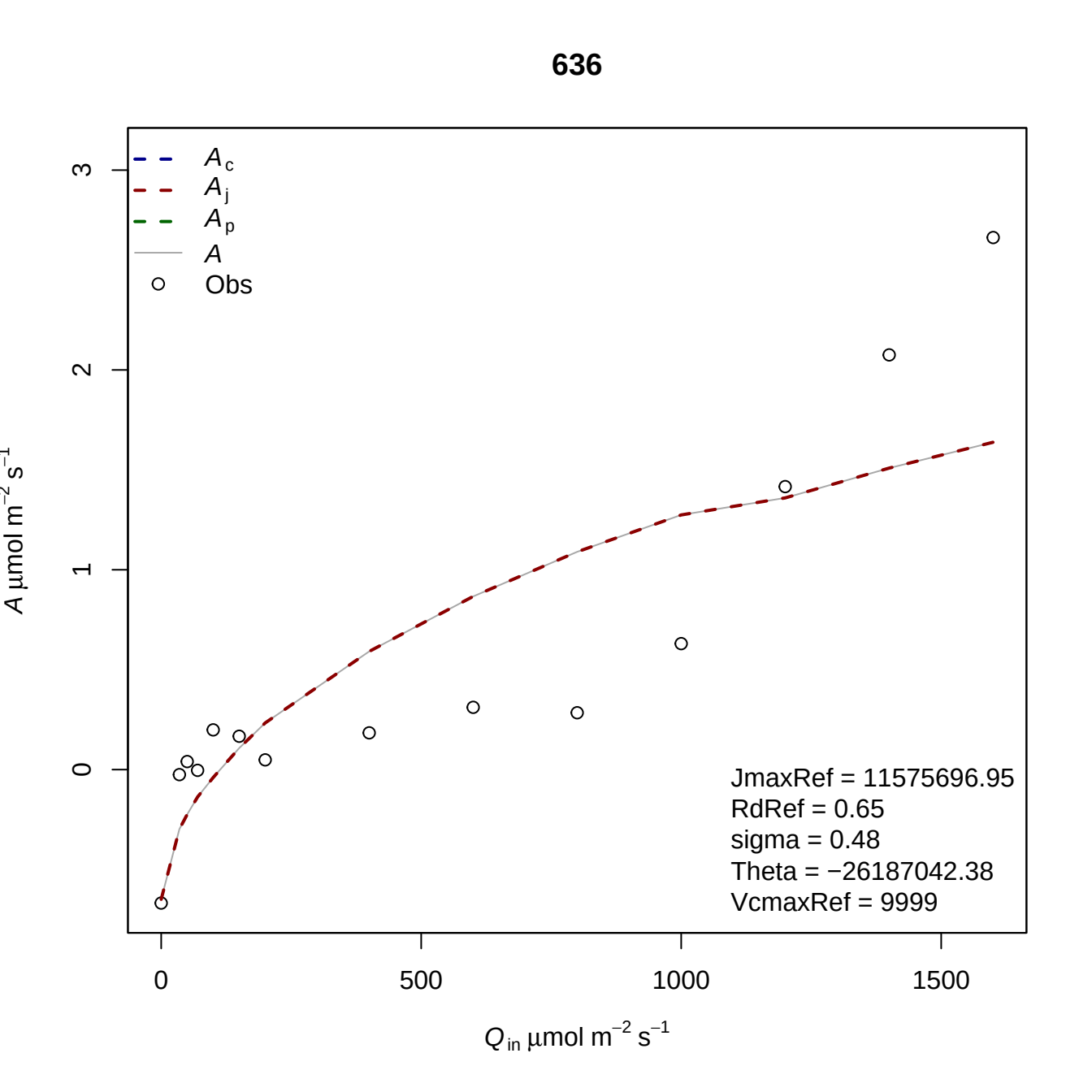

631
A_c
A_j
A_p
A
Obs
JmaxRef = 220.35
RdRef = 1.9
sigma = 0.07
Theta = 0.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

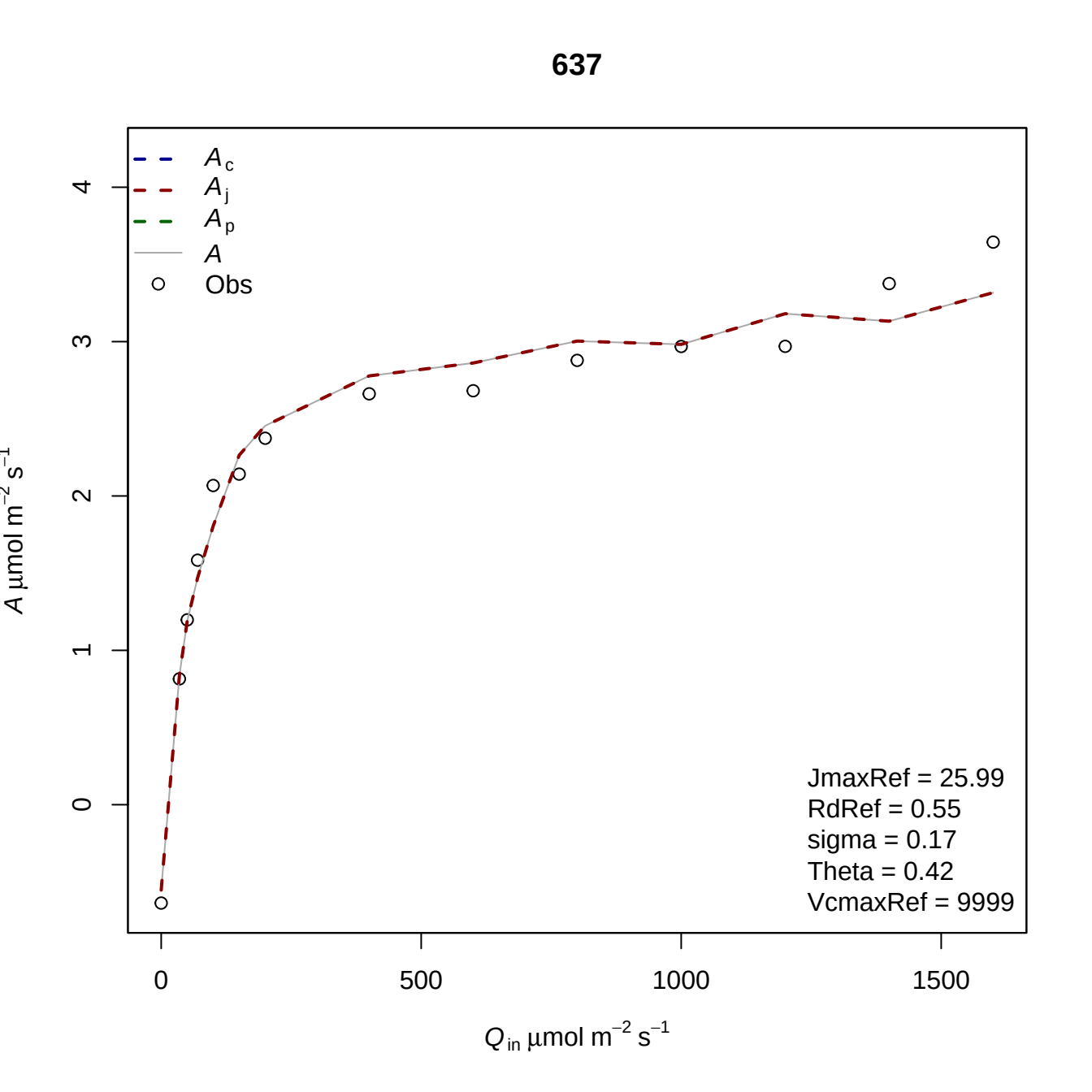

632
A_c
A_j
A_p
A
Obs
JmaxRef = 199.75
RdRef = 1.44
sigma = 0.16
Theta = 0.84
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

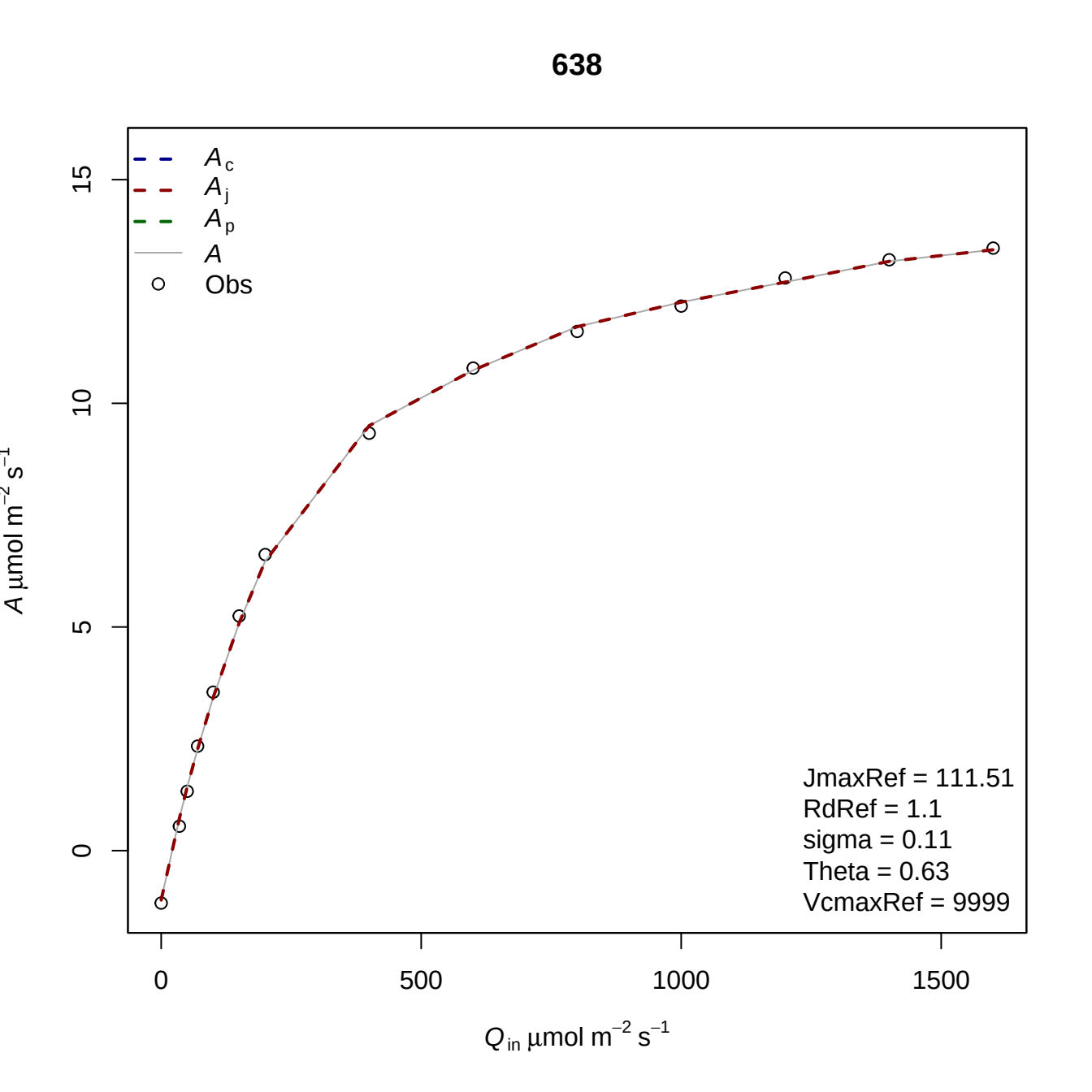

633
A_c
A_j
A_p
A
Obs
JmaxRef = 257.24
RdRef = 3.6
sigma = 0.09
Theta = 0.33
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

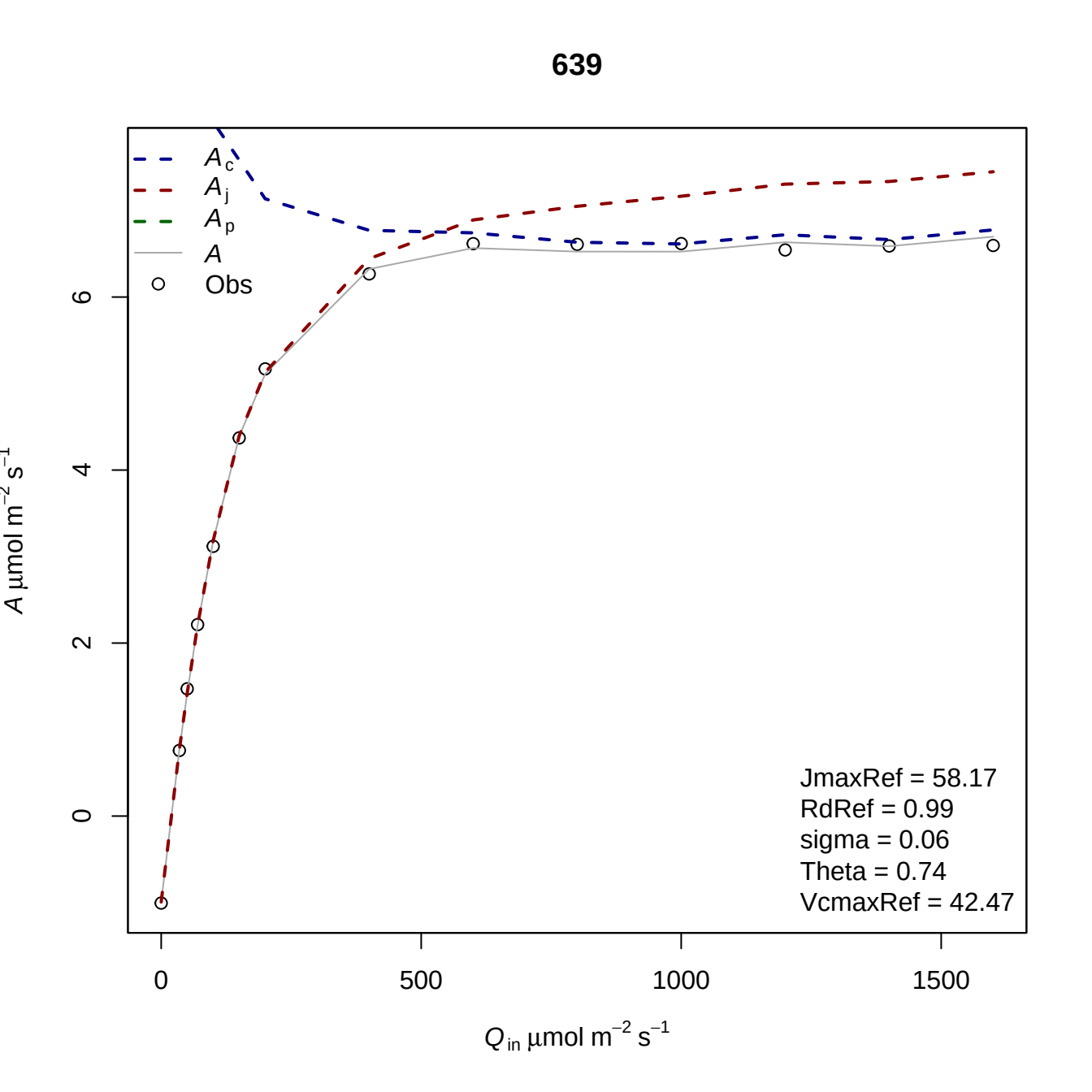

634
A_c
A_j
A_p
A
Obs
JmaxRef = 195.69
RdRef = 1.89
sigma = 0.3
Theta = 0.83
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

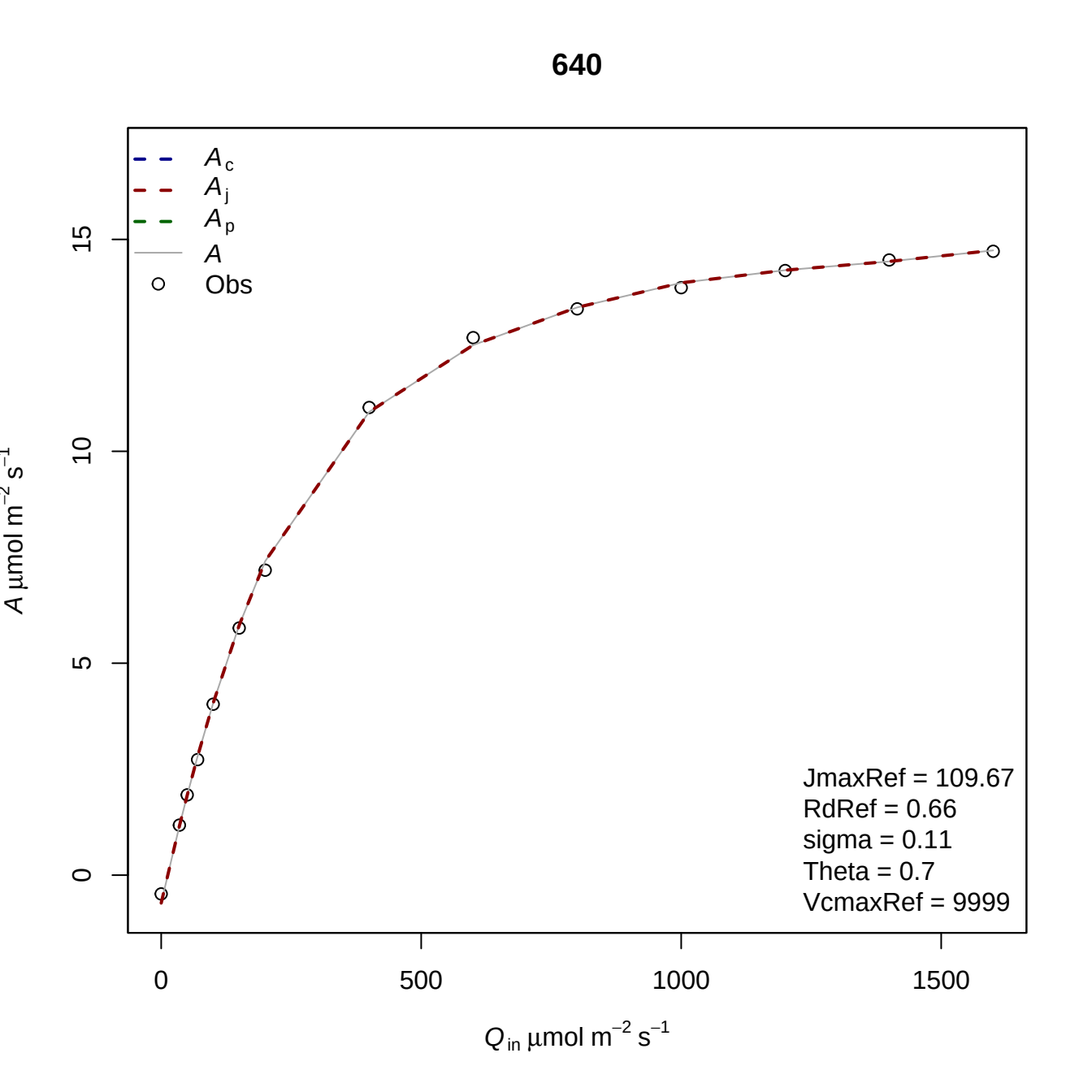

635
A_c
A_j
A_p
A
Obs
JmaxRef = 22.4
RdRef = 1.02
sigma = 0.06
Theta = 0.88
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

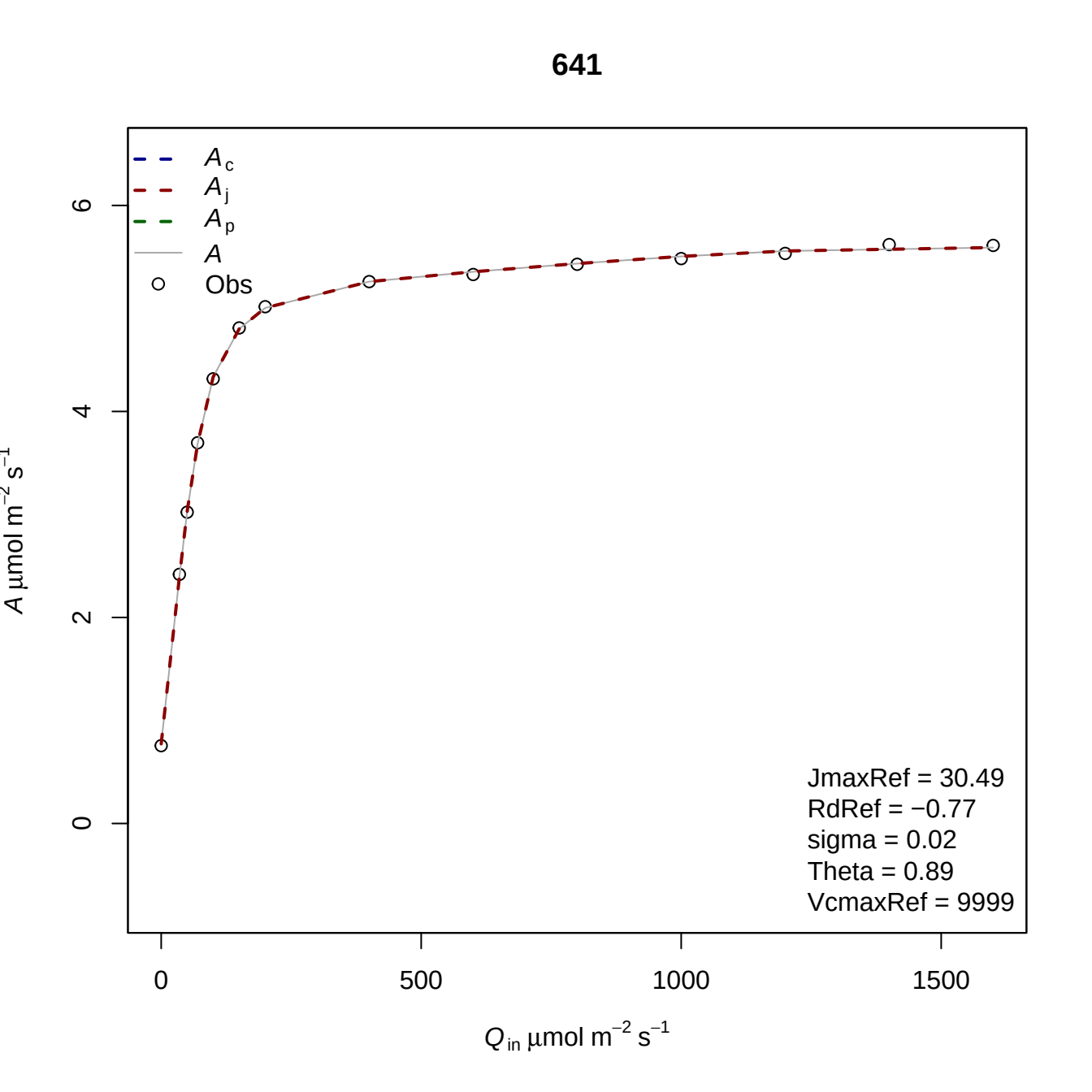

636
A_c
A_j
A_p
A
Obs
JmaxRef = 11575696.95
RdRef = 0.65
sigma = 0.48
Theta = −26187042.38
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

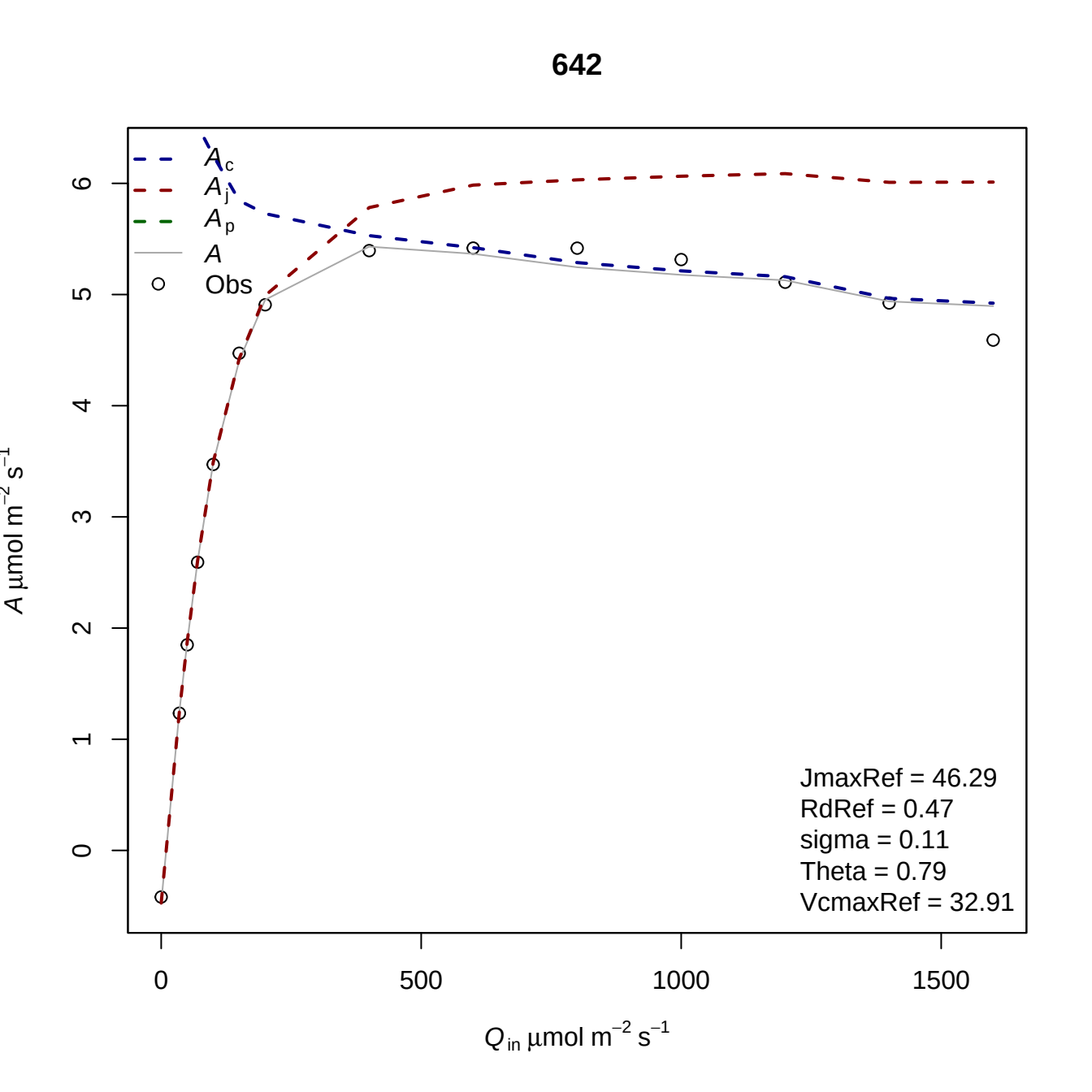

637
A_c
A_j
A_p
A
Obs
JmaxRef = 25.99
RdRef = 0.55
sigma = 0.17
Theta = 0.42
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

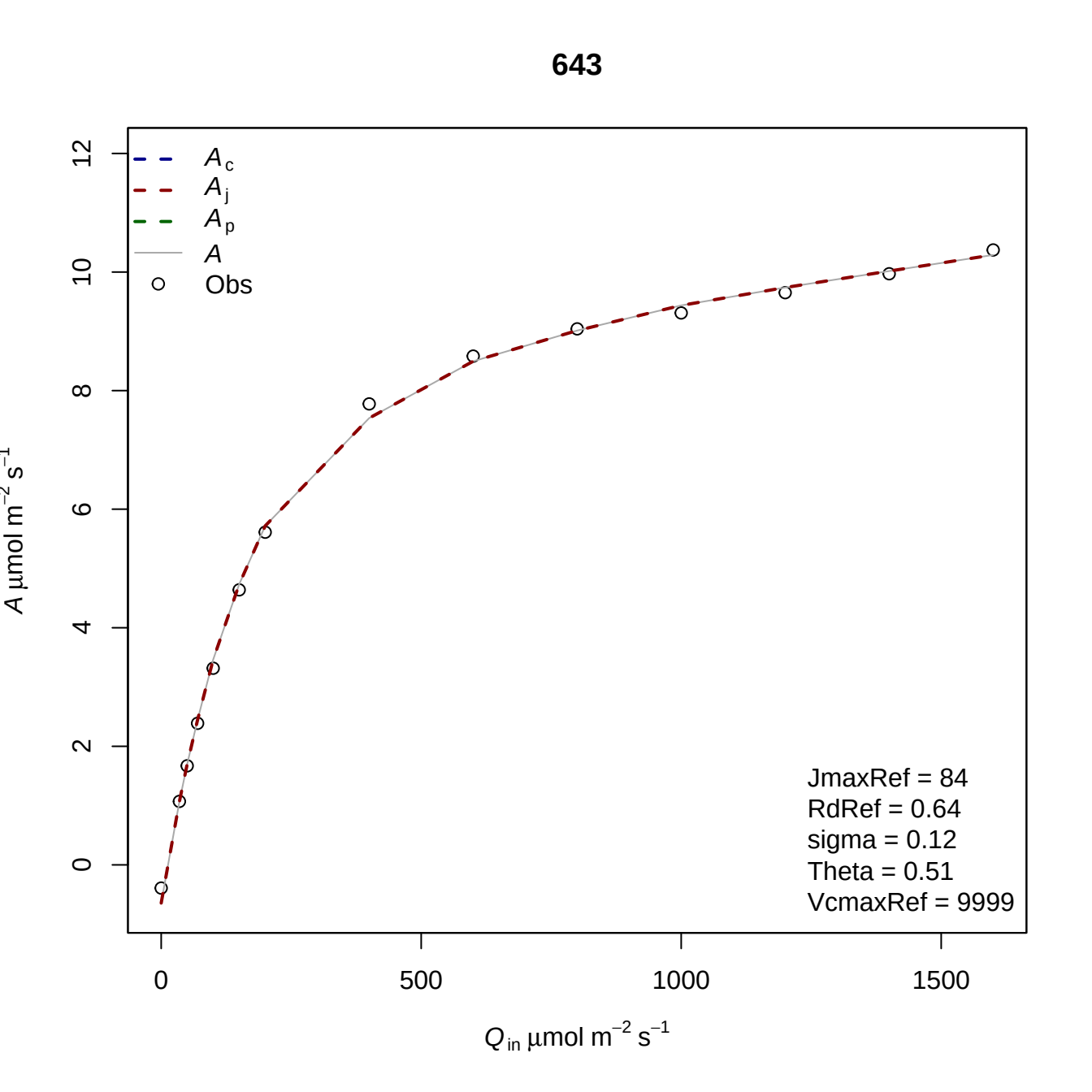

638
A_c
A_j
A_p
A
Obs
JmaxRef = 111.51
RdRef = 1.1
sigma = 0.11
Theta = 0.63
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

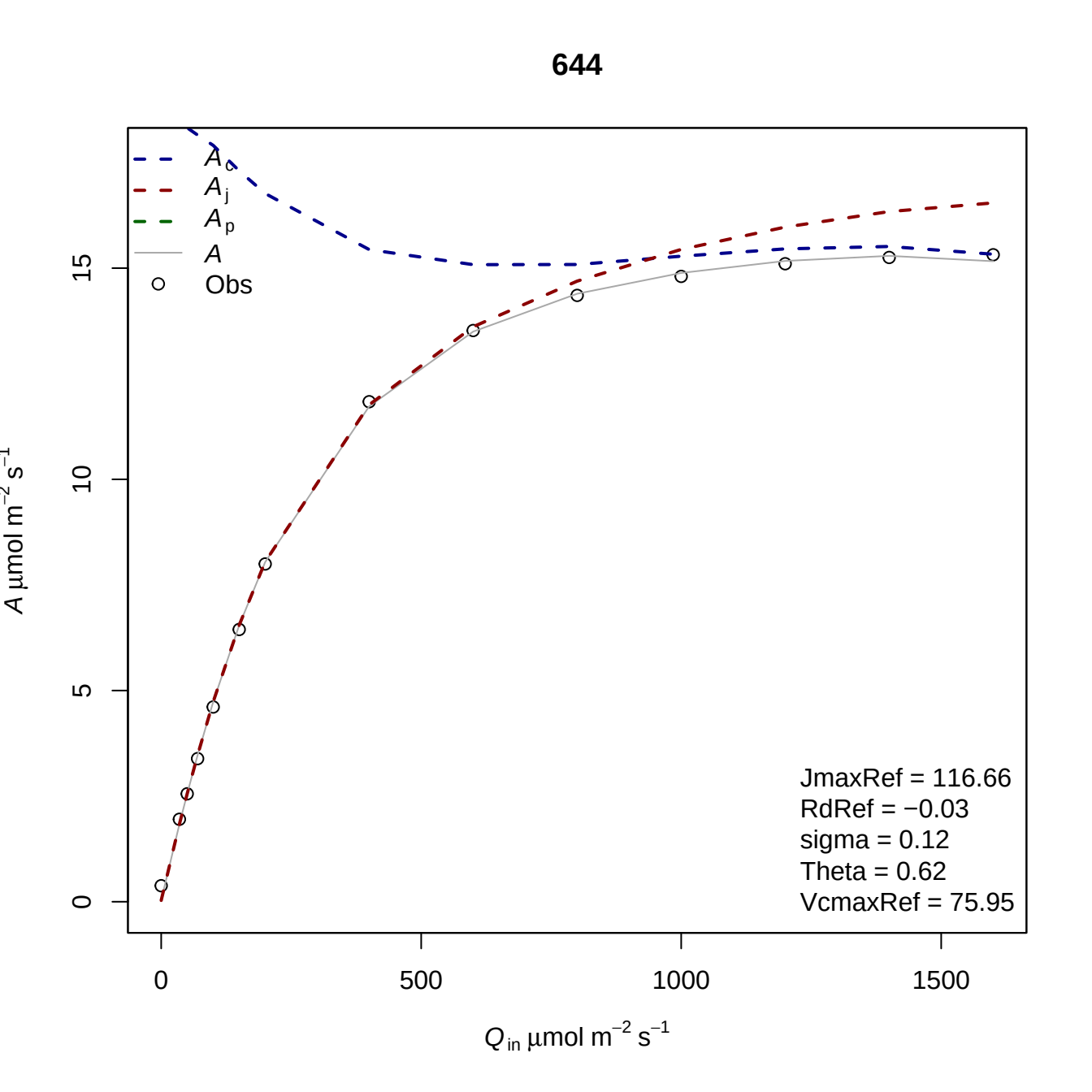

639
A_c
A_j
A_p
A
Obs
JmaxRef = 58.17
RdRef = 0.99
sigma = 0.06
Theta = 0.74
VcmaxRef = 42.47
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

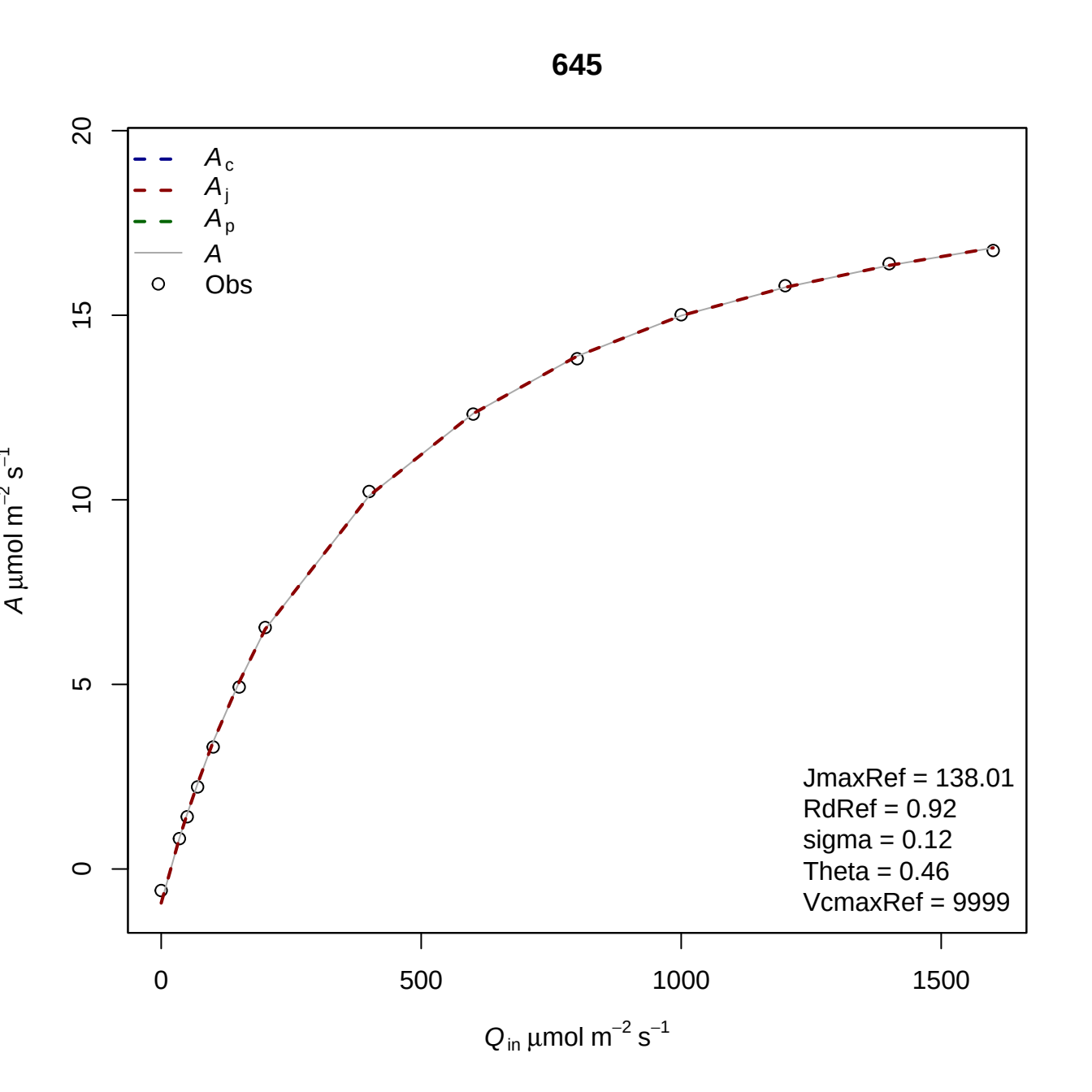

640
A_c
A_j
A_p
A
Obs
JmaxRef = 109.67
RdRef = 0.66
sigma = 0.11
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

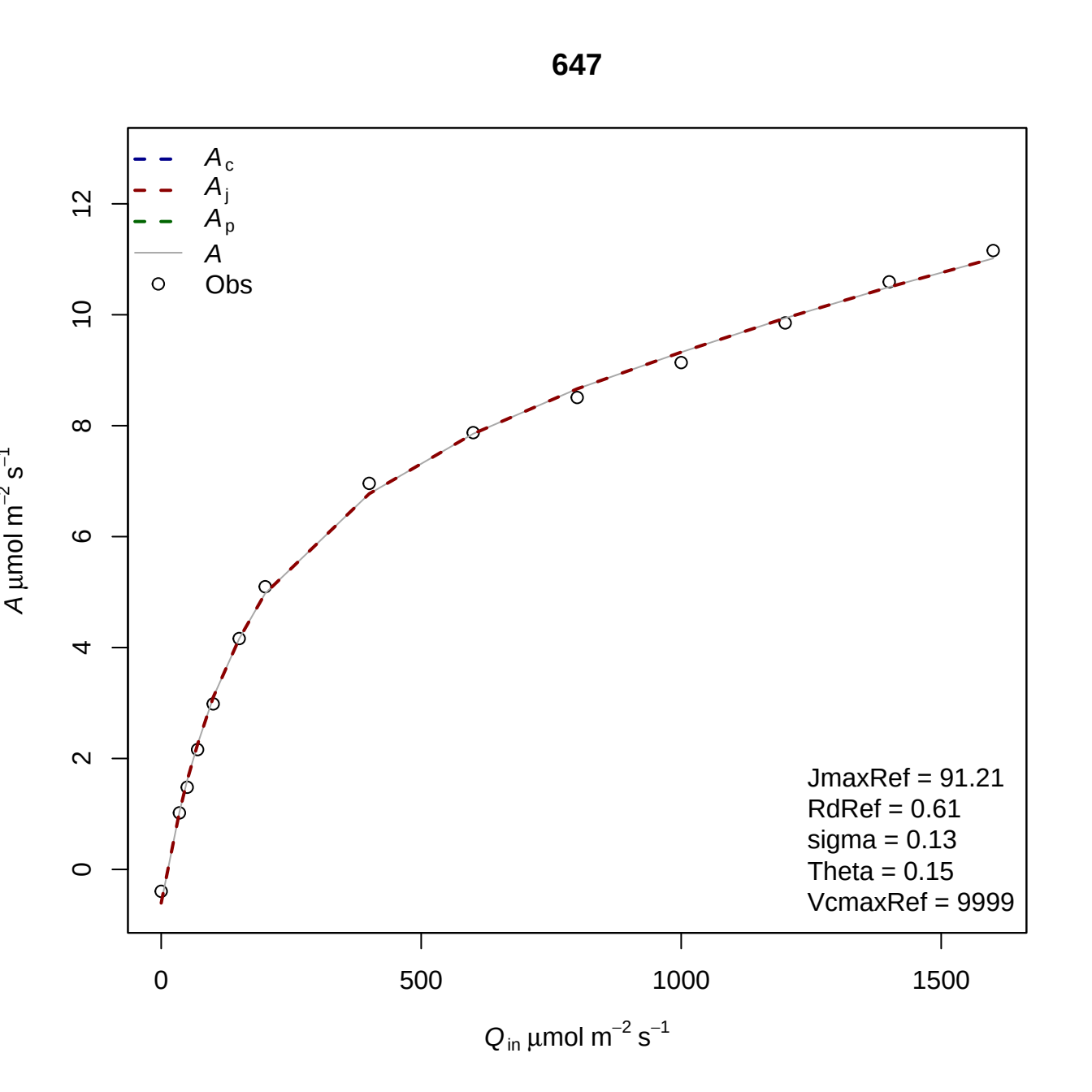

641
A_c
A_j
A_p
A
Obs
JmaxRef = 30.49
RdRef = −0.77
sigma = 0.02
Theta = 0.89
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

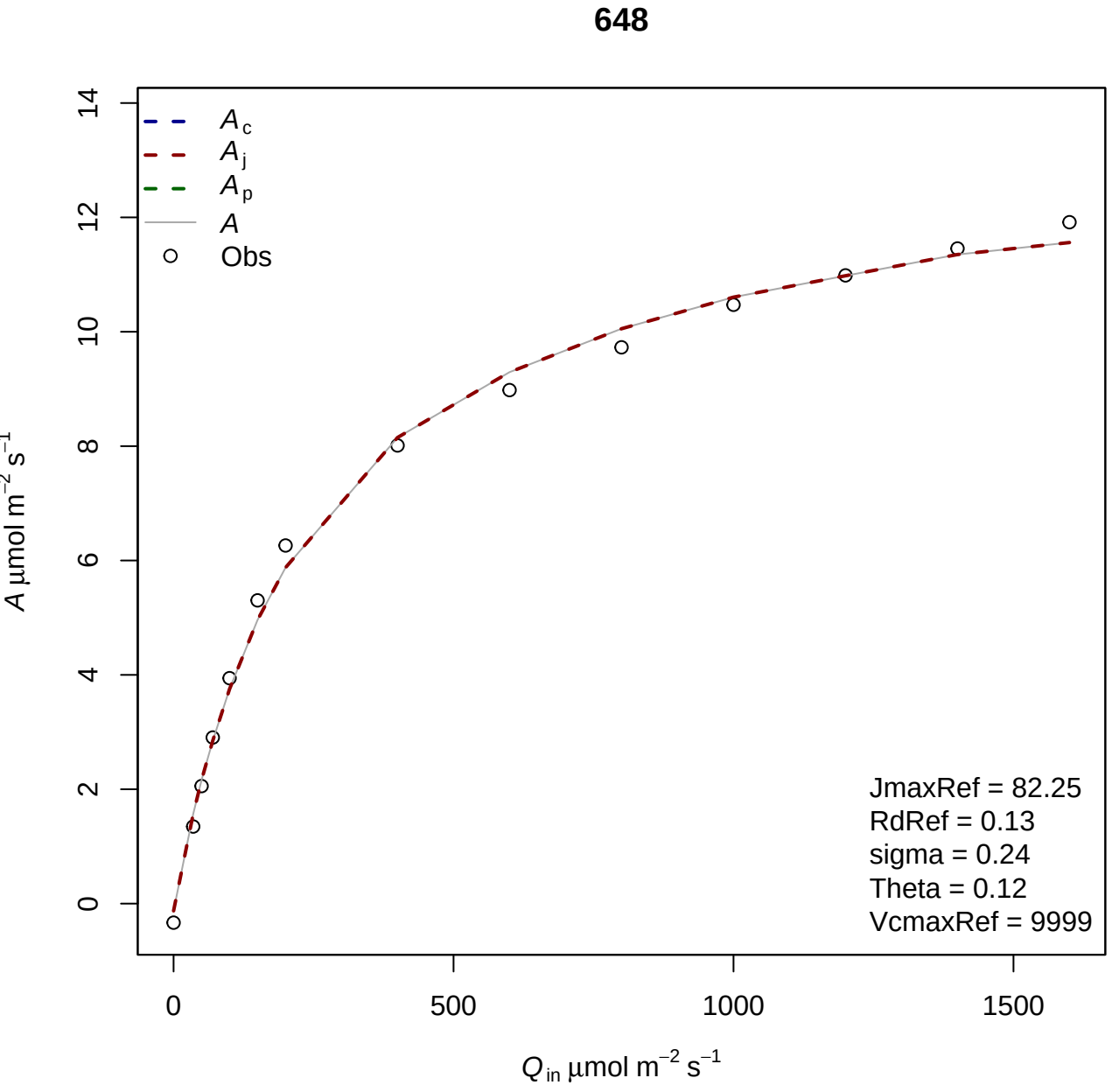

642
A_c
A_j
A_p
A
Obs
JmaxRef = 46.29
RdRef = 0.47
sigma = 0.11
Theta = 0.79
VcmaxRef = 32.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

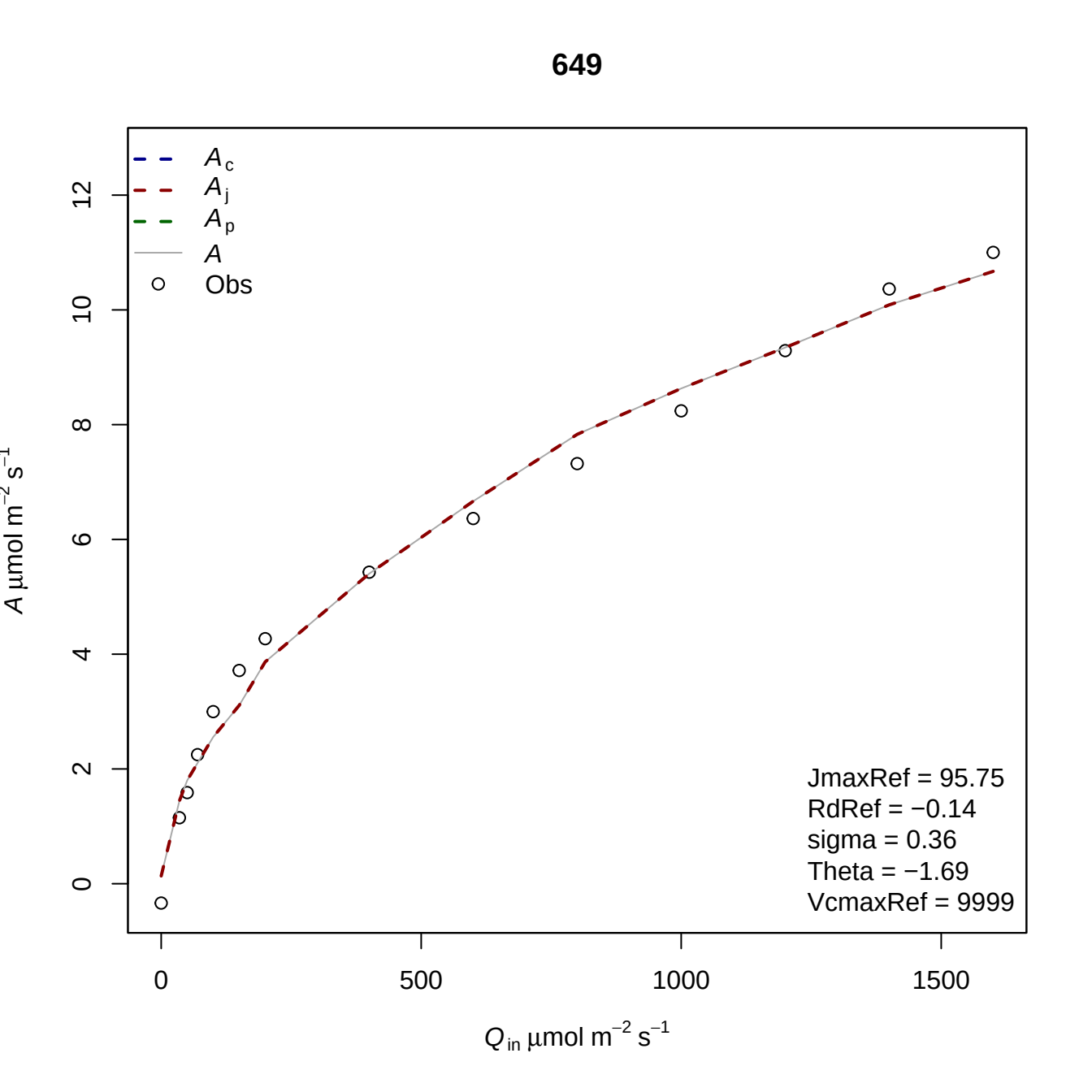

643
A_c
A_j
A_p
A
Obs
JmaxRef = 84
RdRef = 0.64
sigma = 0.12
Theta = 0.51
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

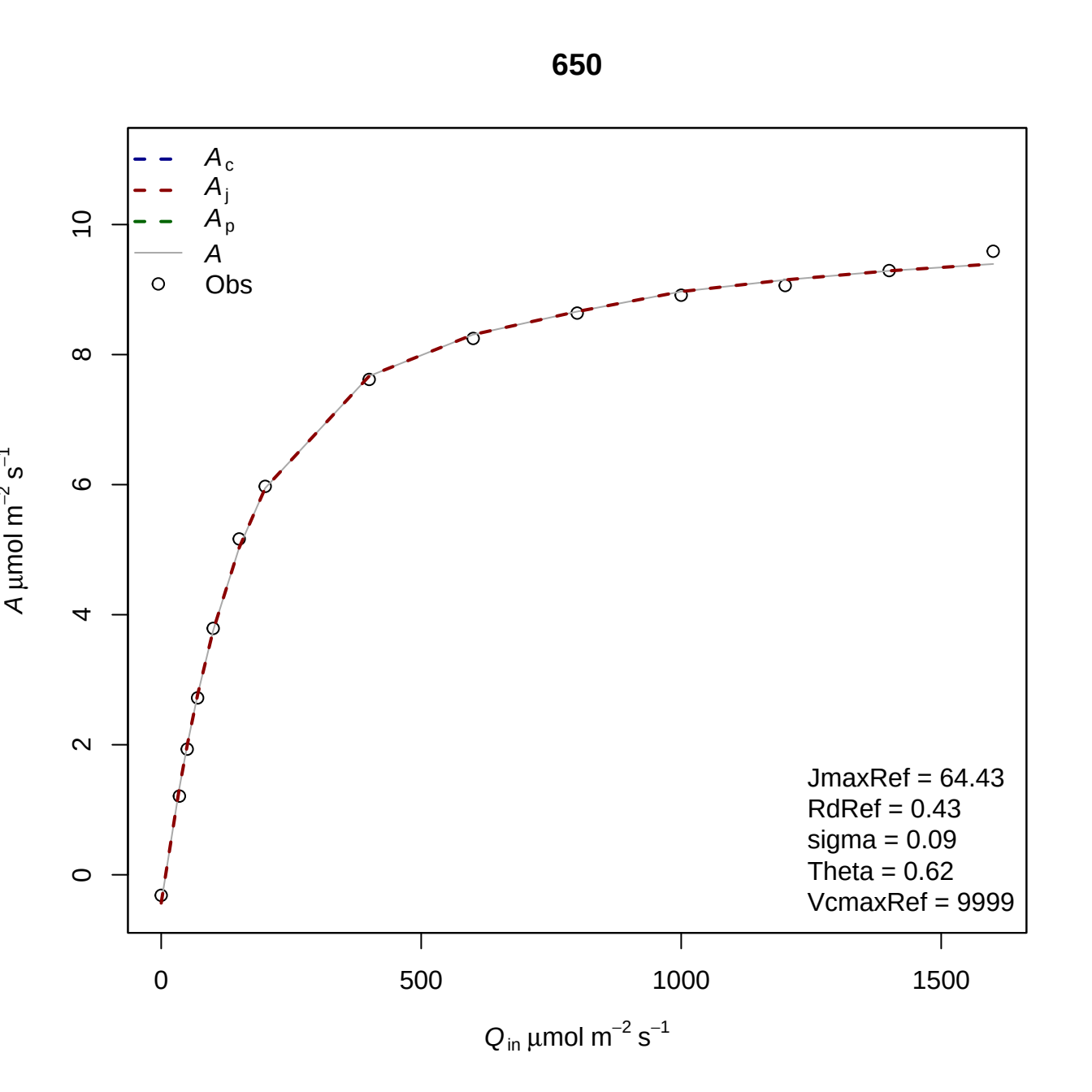

644
A_c
A_j
A_p
A
Obs
JmaxRef = 116.66
RdRef = −0.03
sigma = 0.12
Theta = 0.62
VcmaxRef = 75.95
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

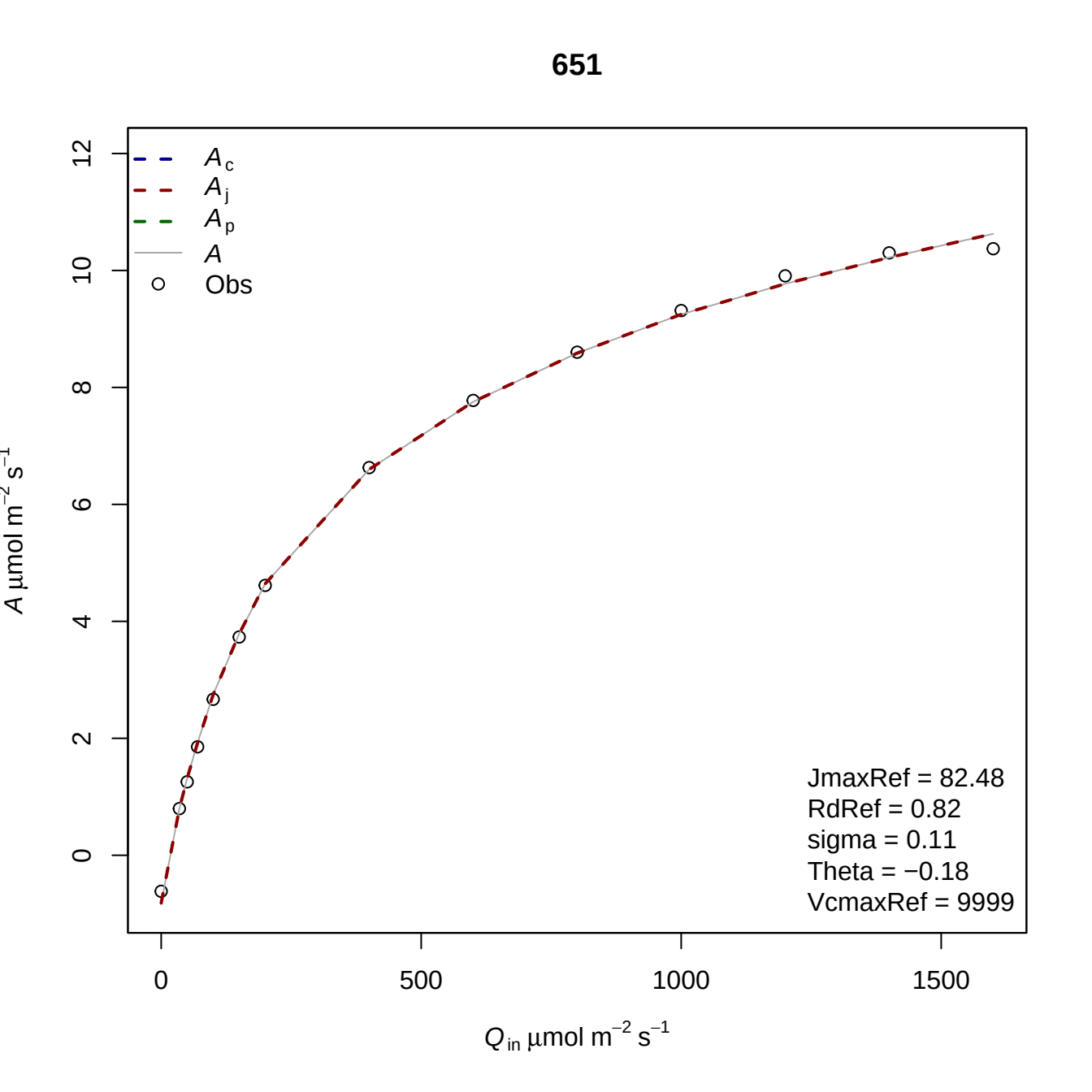

645
A_c
A_j
A_p
A
Obs
JmaxRef = 138.01
RdRef = 0.92
sigma = 0.12
Theta = 0.46
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

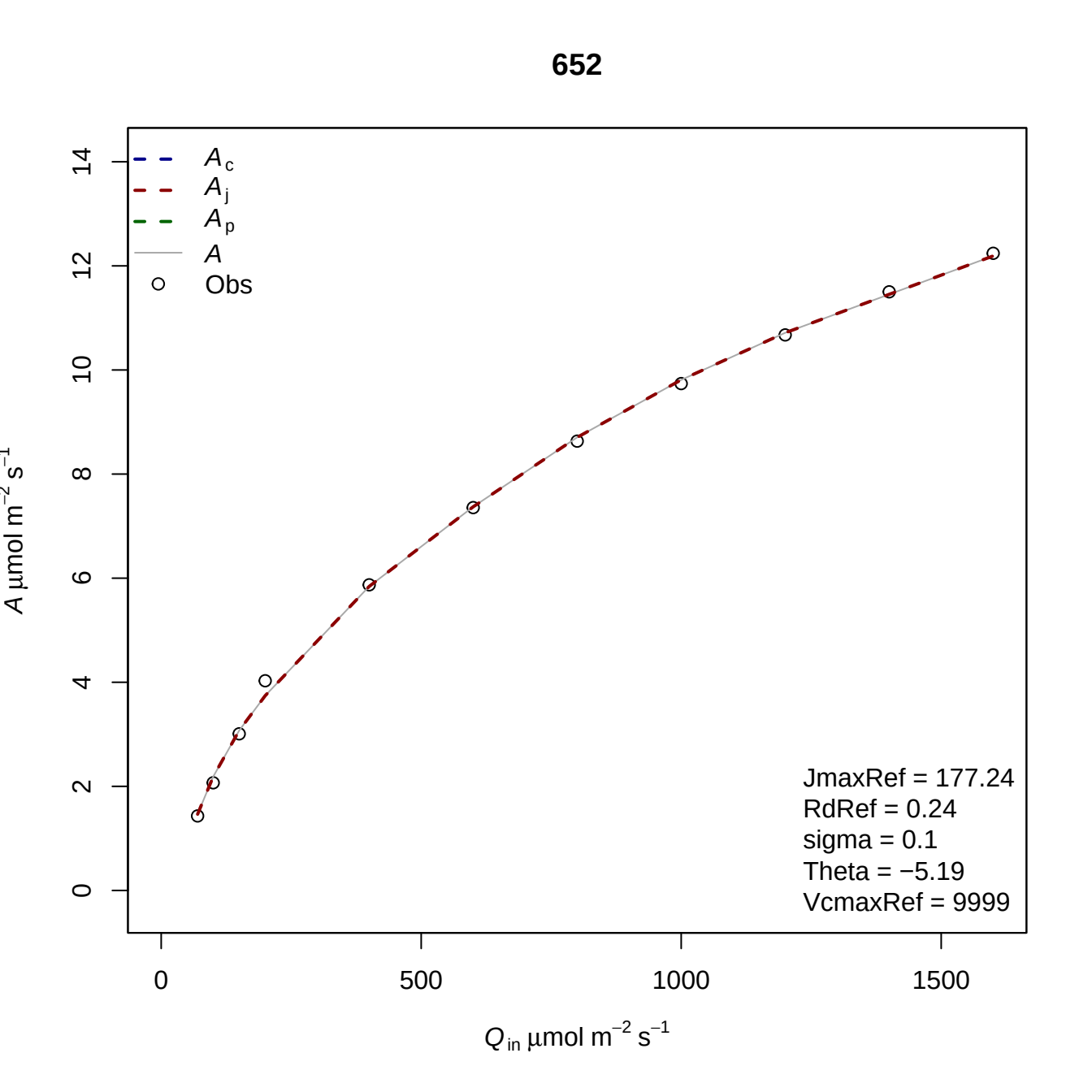

647
A_c
A_j
A_p
A
Obs
JmaxRef = 91.21
RdRef = 0.61
sigma = 0.13
Theta = 0.15
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

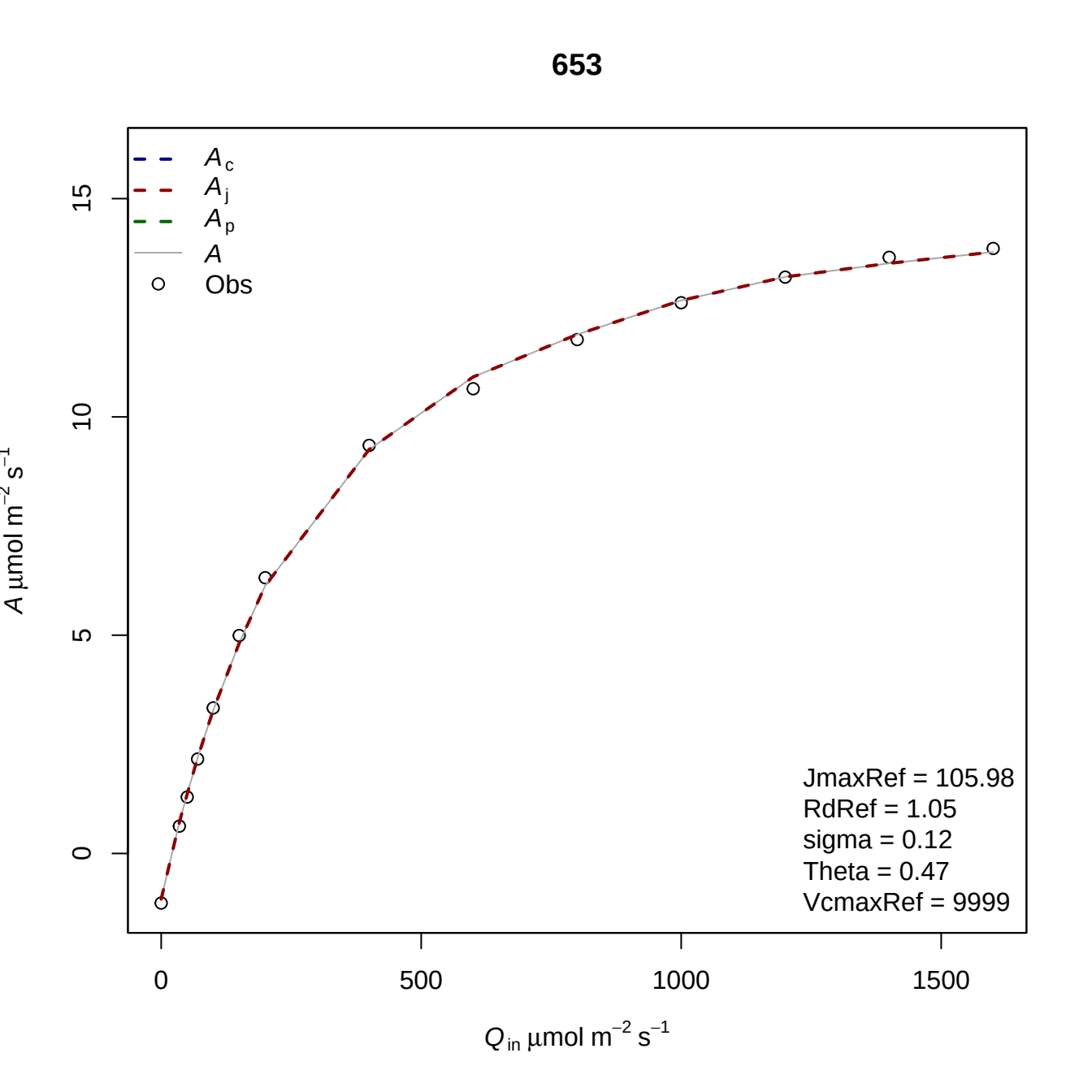

648
A_c
A_j
A_p
A
Obs
JmaxRef = 82.25
RdRef = 0.13
sigma = 0.24
Theta = 0.12
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

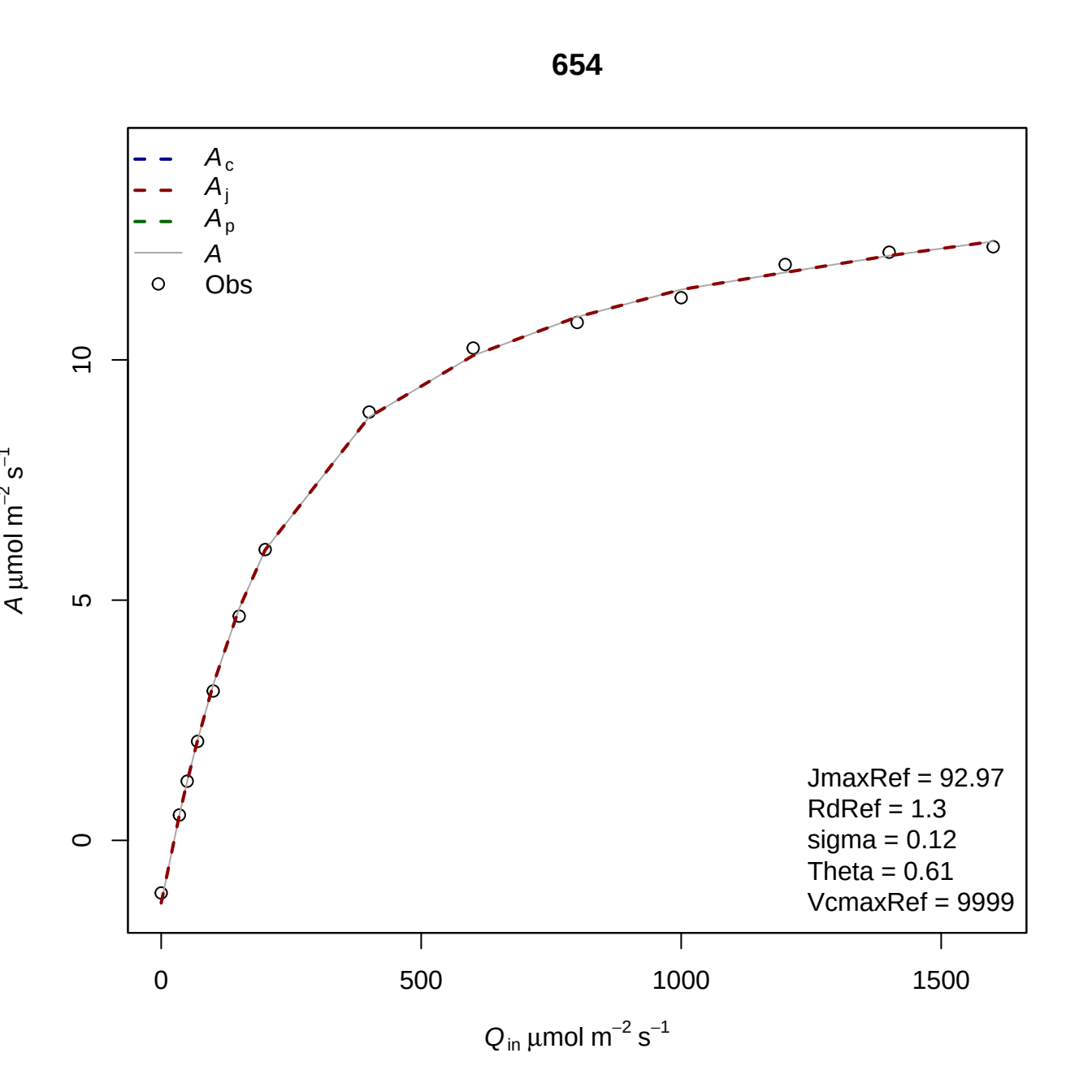

649
A_c
A_j
A_p
A
Obs
JmaxRef = 95.75
RdRef = −0.14
sigma = 0.36
Theta = −1.69
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

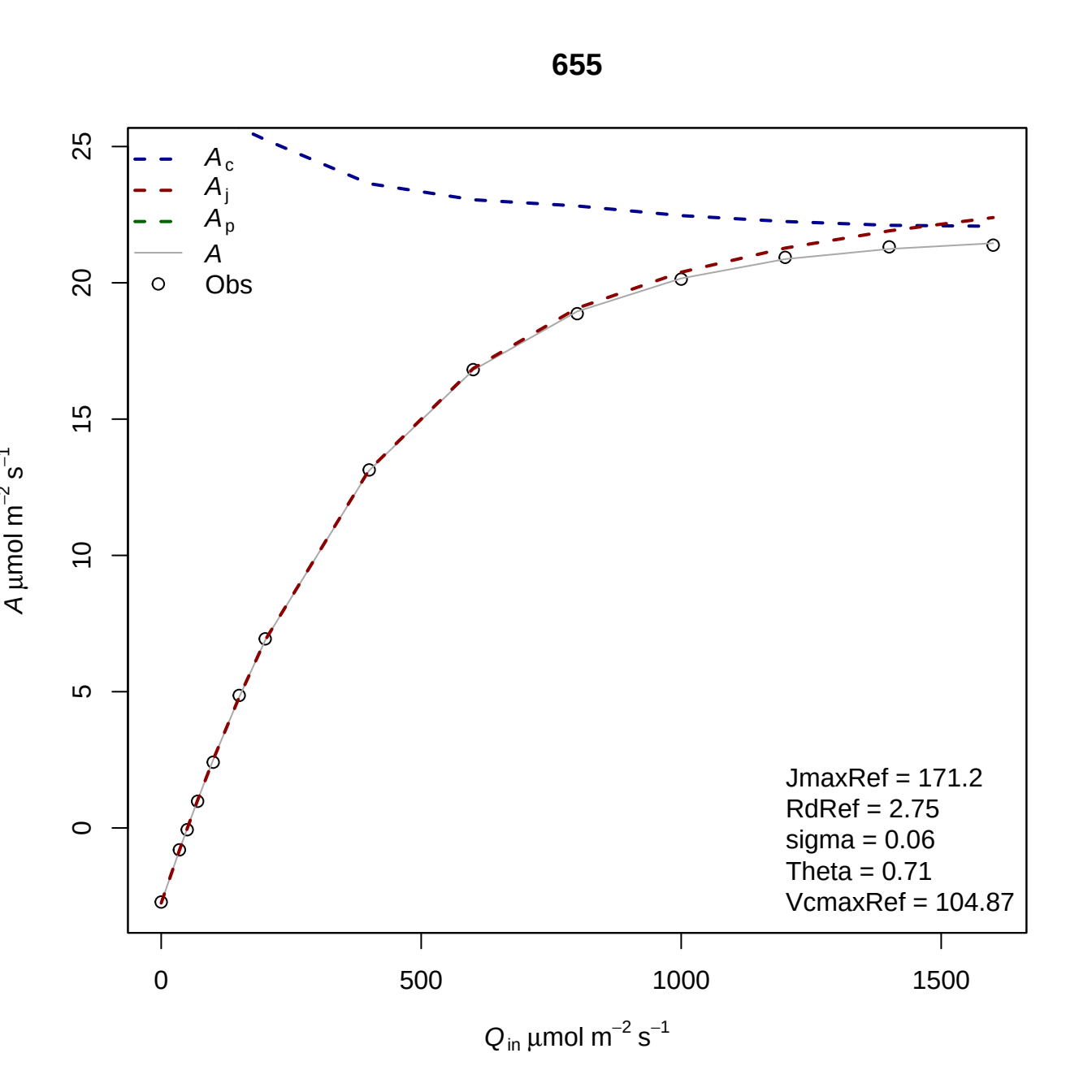

650
A_c
A_j
A_p
A
Obs
JmaxRef = 64.43
RdRef = 0.43
sigma = 0.09
Theta = 0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

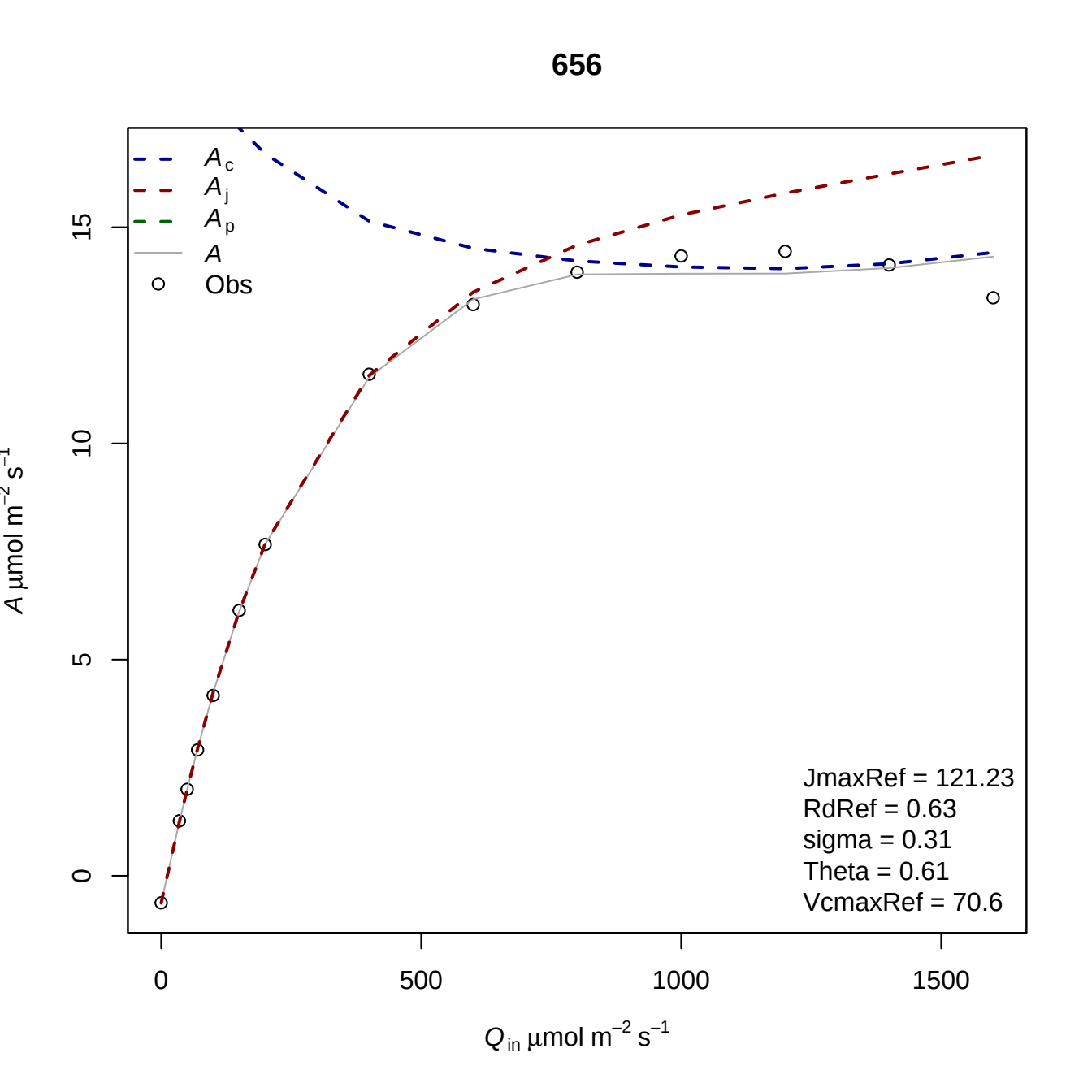

651
A_c
A_j
A_p
A
Obs
JmaxRef = 82.48
RdRef = 0.82
sigma = 0.11
Theta = −0.18
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

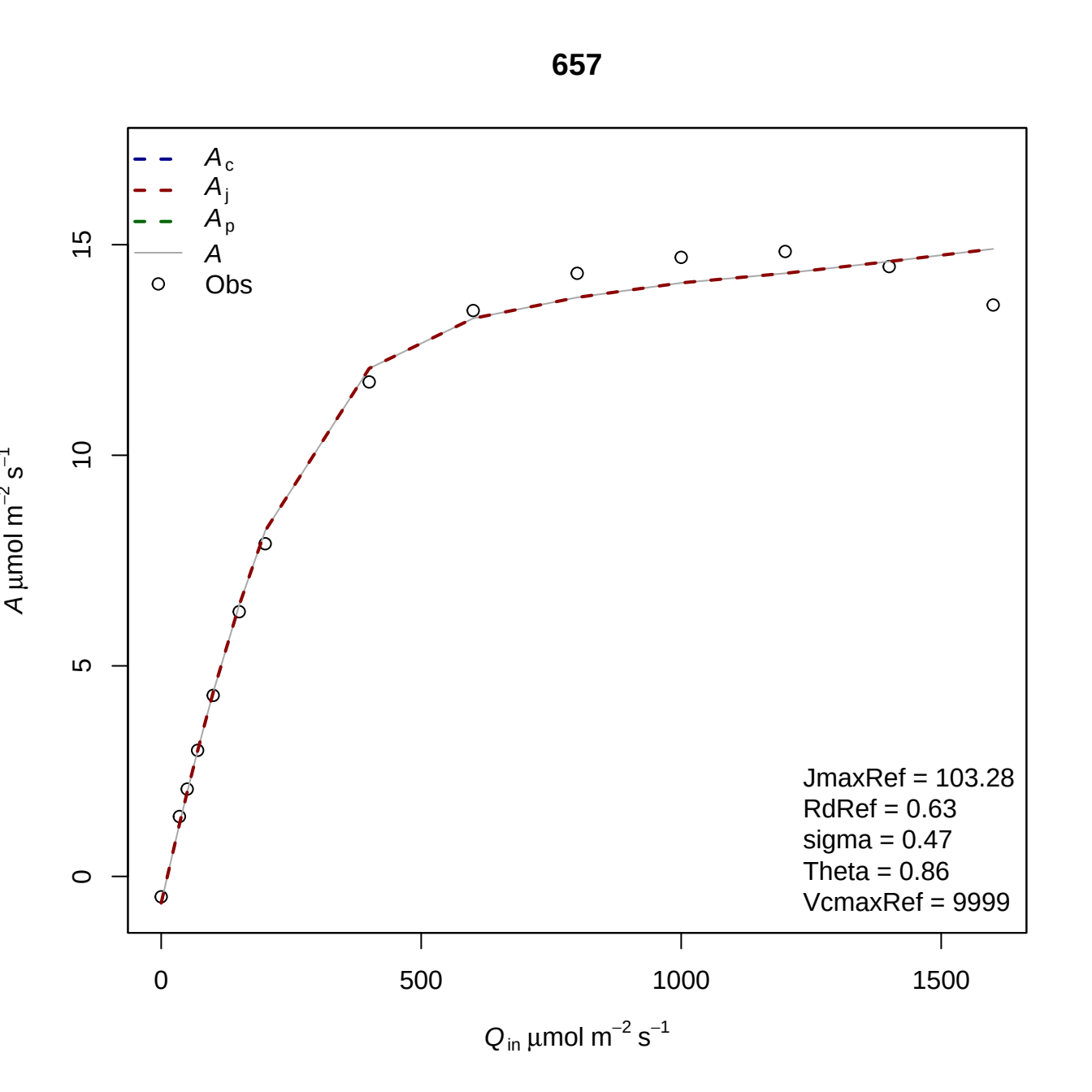

652
A_c
A_j
A_p
A
Obs
JmaxRef = 177.24
RdRef = 0.24
sigma = 0.1
Theta = −5.19
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

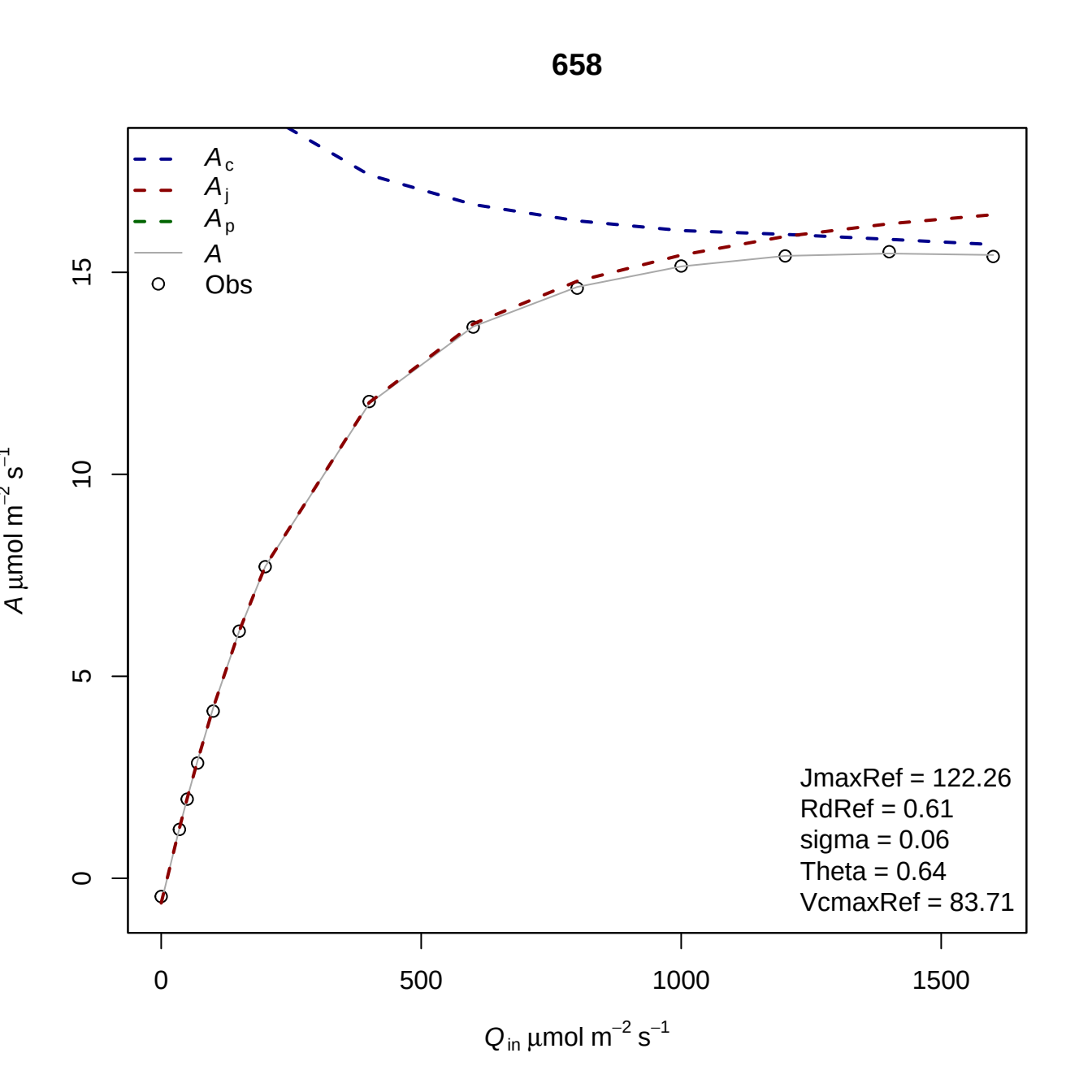

653
A_c
A_j
A_p
A
Obs
JmaxRef = 105.98
RdRef = 1.05
sigma = 0.12
Theta = 0.47
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

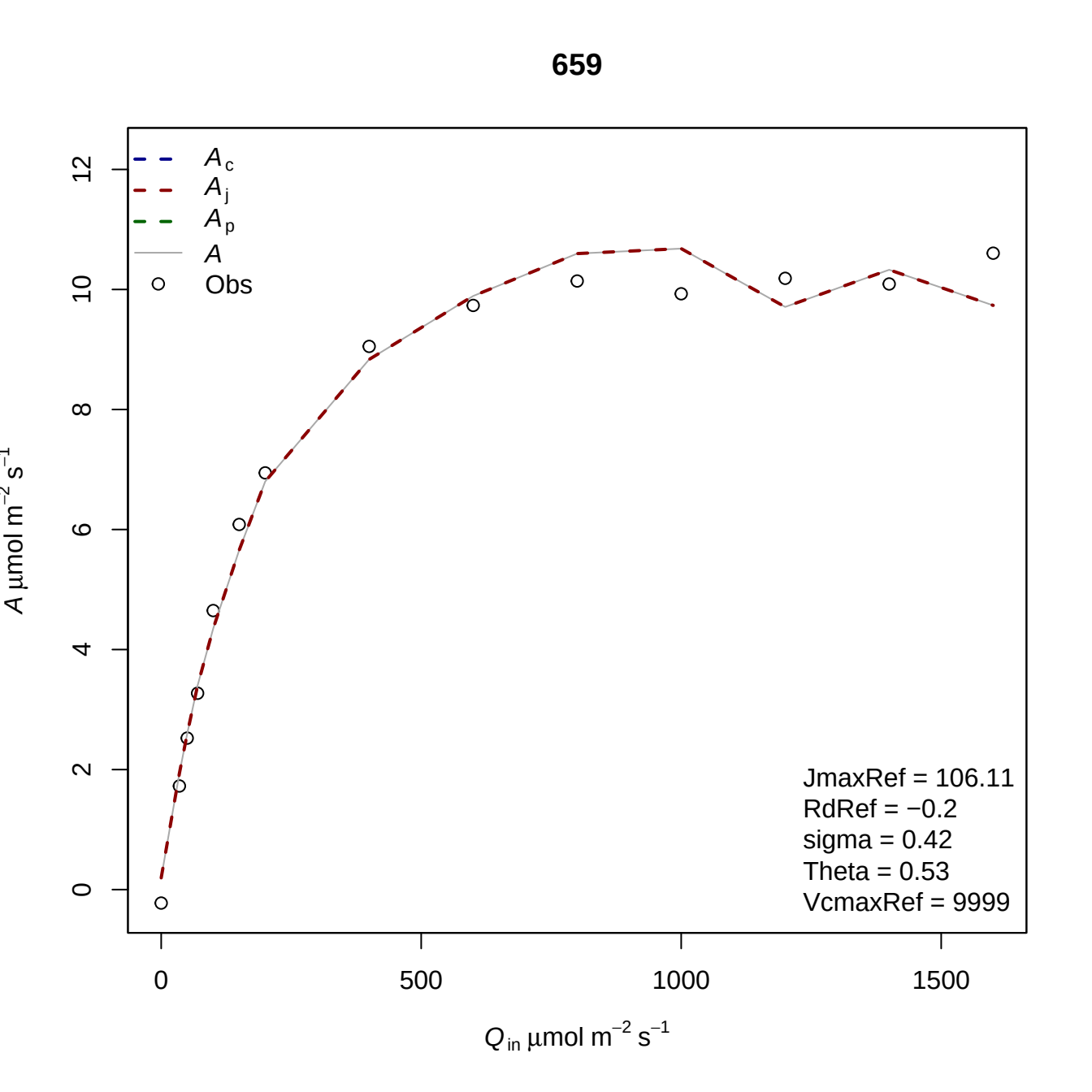

654
A_c
A_j
A_p
A
Obs
JmaxRef = 92.97
RdRef = 1.3
sigma = 0.12
Theta = 0.61
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

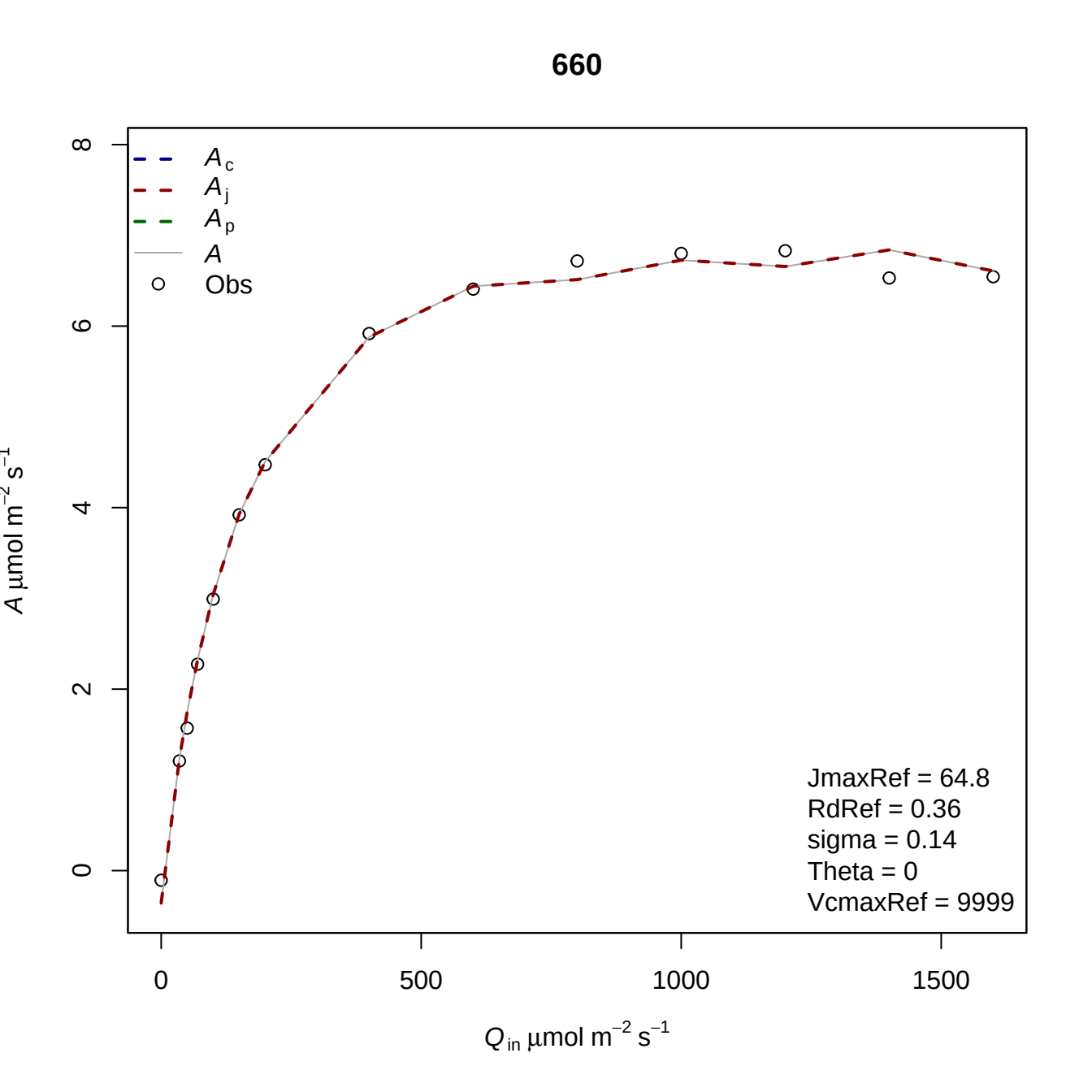

655
A_c
A_j
A_p
A
Obs
JmaxRef = 171.2
RdRef = 2.75
sigma = 0.06
Theta = 0.71
VcmaxRef = 104.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

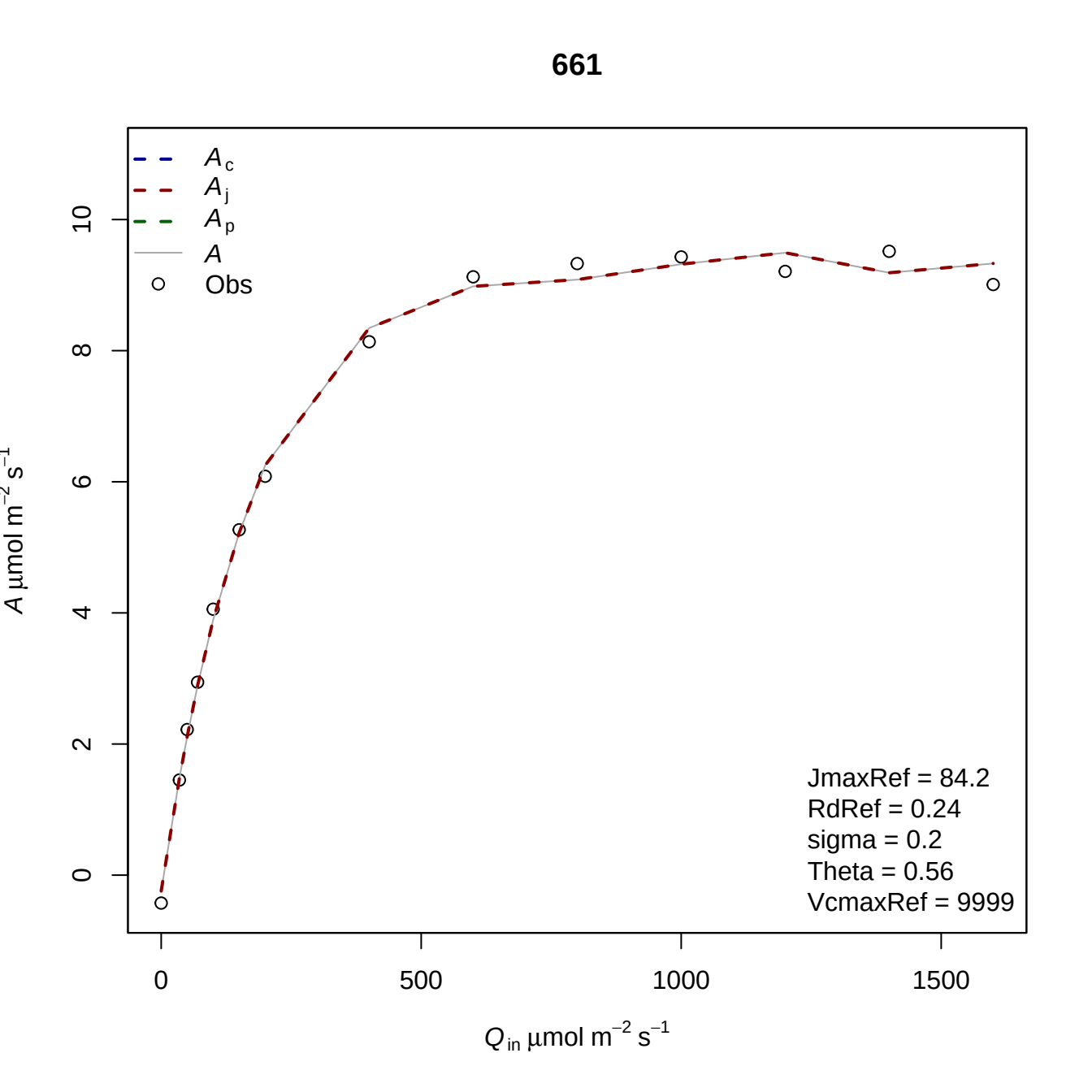

656
A_c
A_j
A_p
A
Obs
JmaxRef = 121.23
RdRef = 0.63
sigma = 0.31
Theta = 0.61
VcmaxRef = 70.6
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

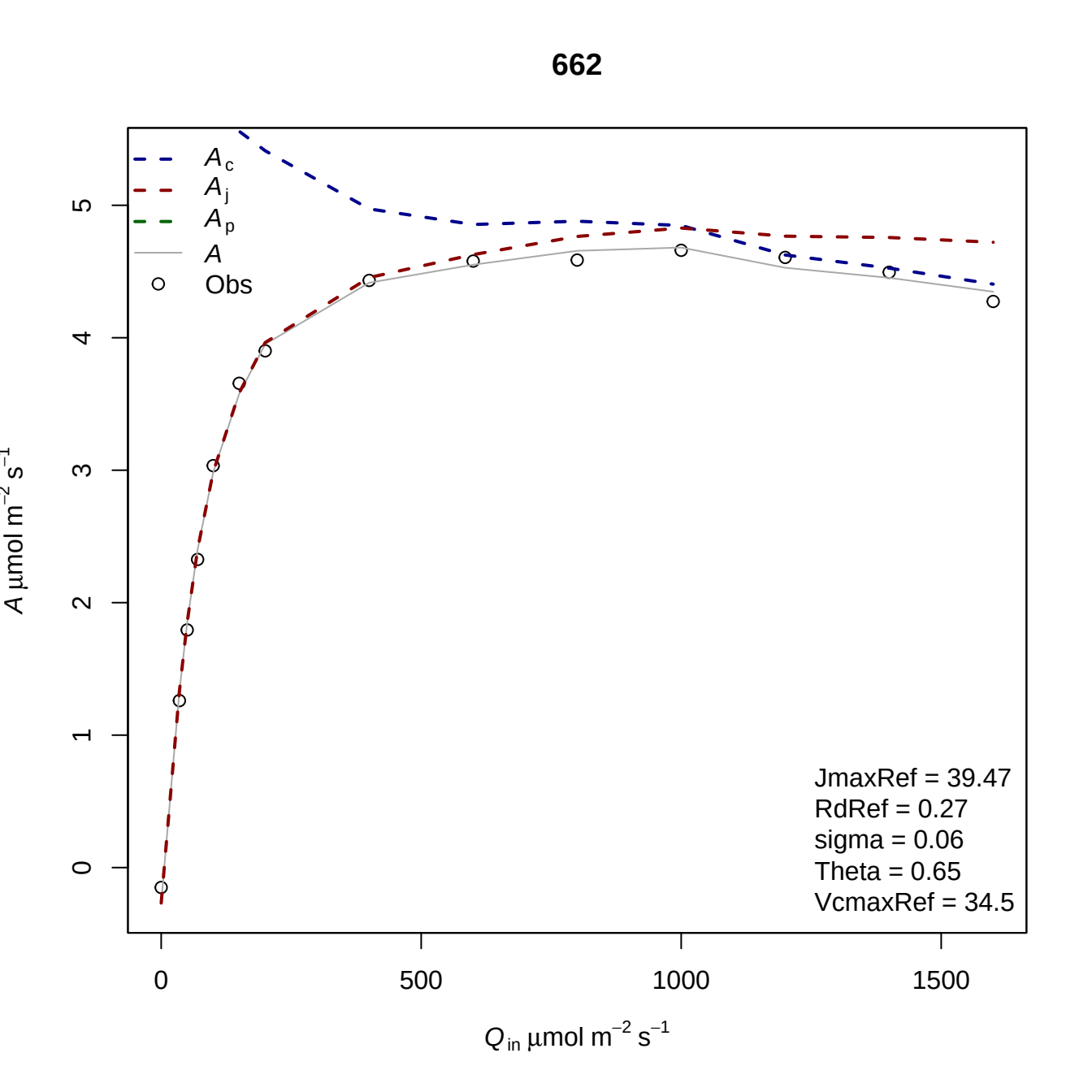

657
A_c
A_j
A_p
A
Obs
JmaxRef = 103.28
RdRef = 0.63
sigma = 0.47
Theta = 0.86
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

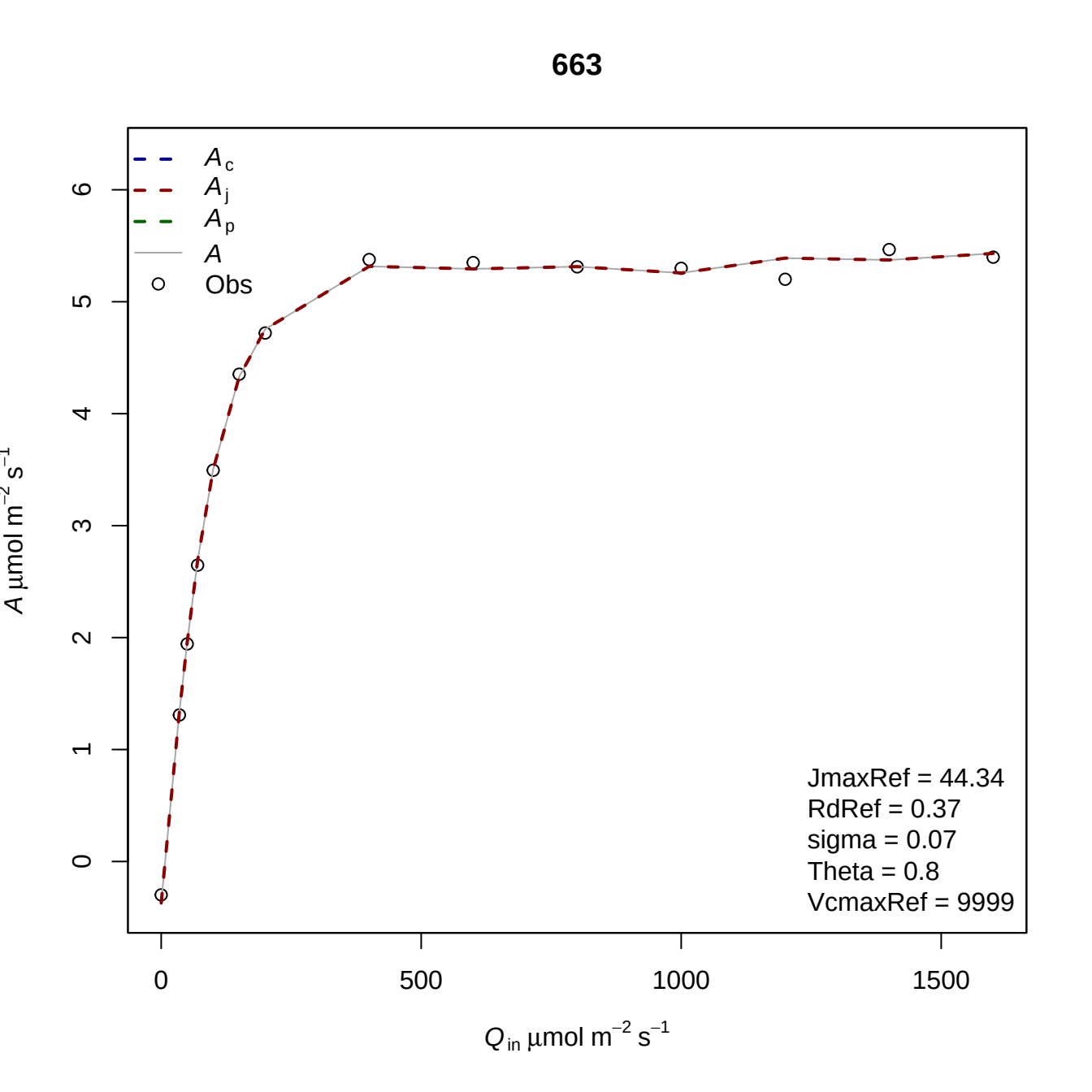

658
A_c
A_j
A_p
A
Obs
JmaxRef = 122.26
RdRef = 0.61
sigma = 0.06
Theta = 0.64
VcmaxRef = 83.71
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

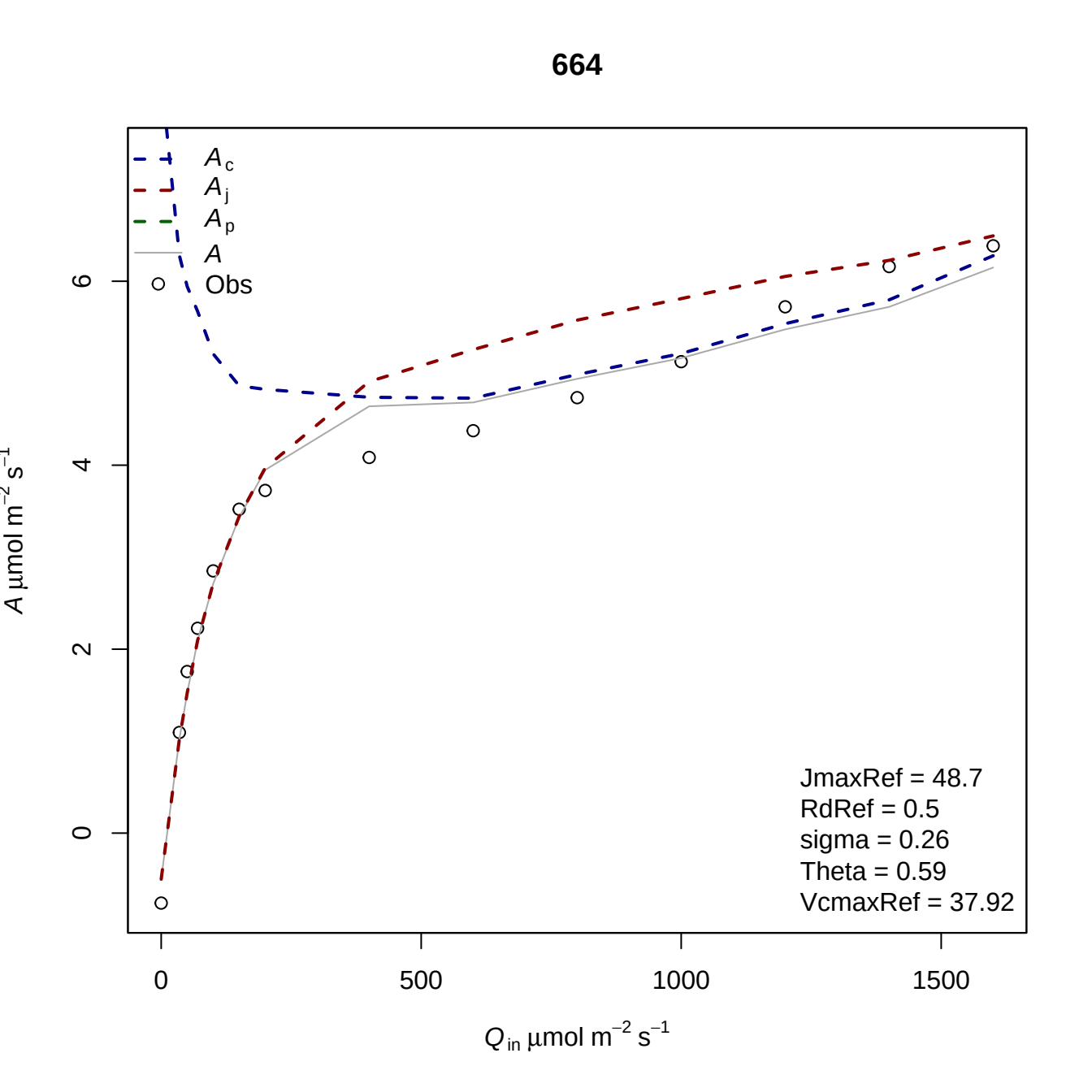

659
A_c
A_j
A_p
A
Obs
JmaxRef = 106.11
RdRef = −0.2
sigma = 0.42
Theta = 0.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

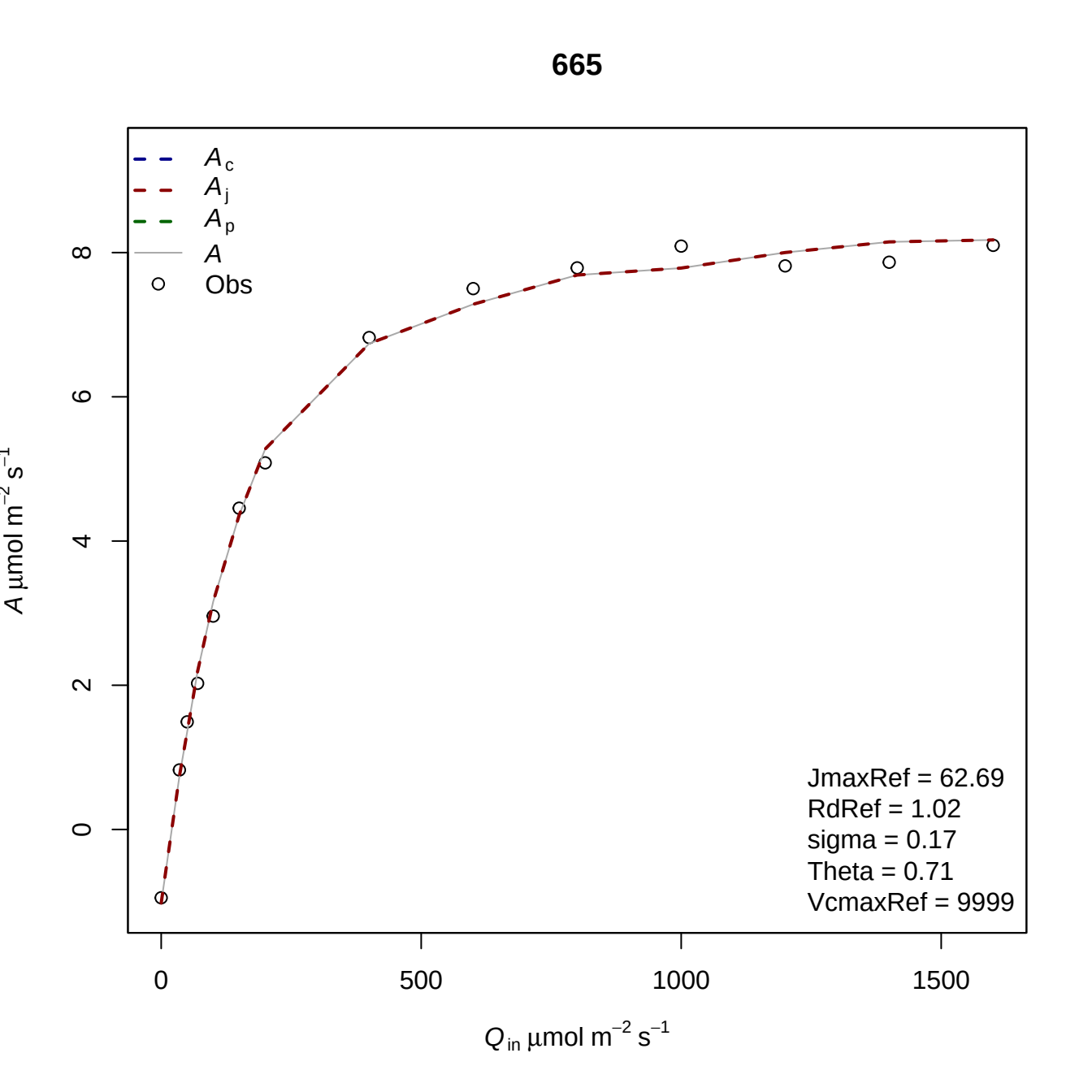

660
A_c
A_j
A_p
A
Obs
JmaxRef = 64.8
RdRef = 0.36
sigma = 0.14
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

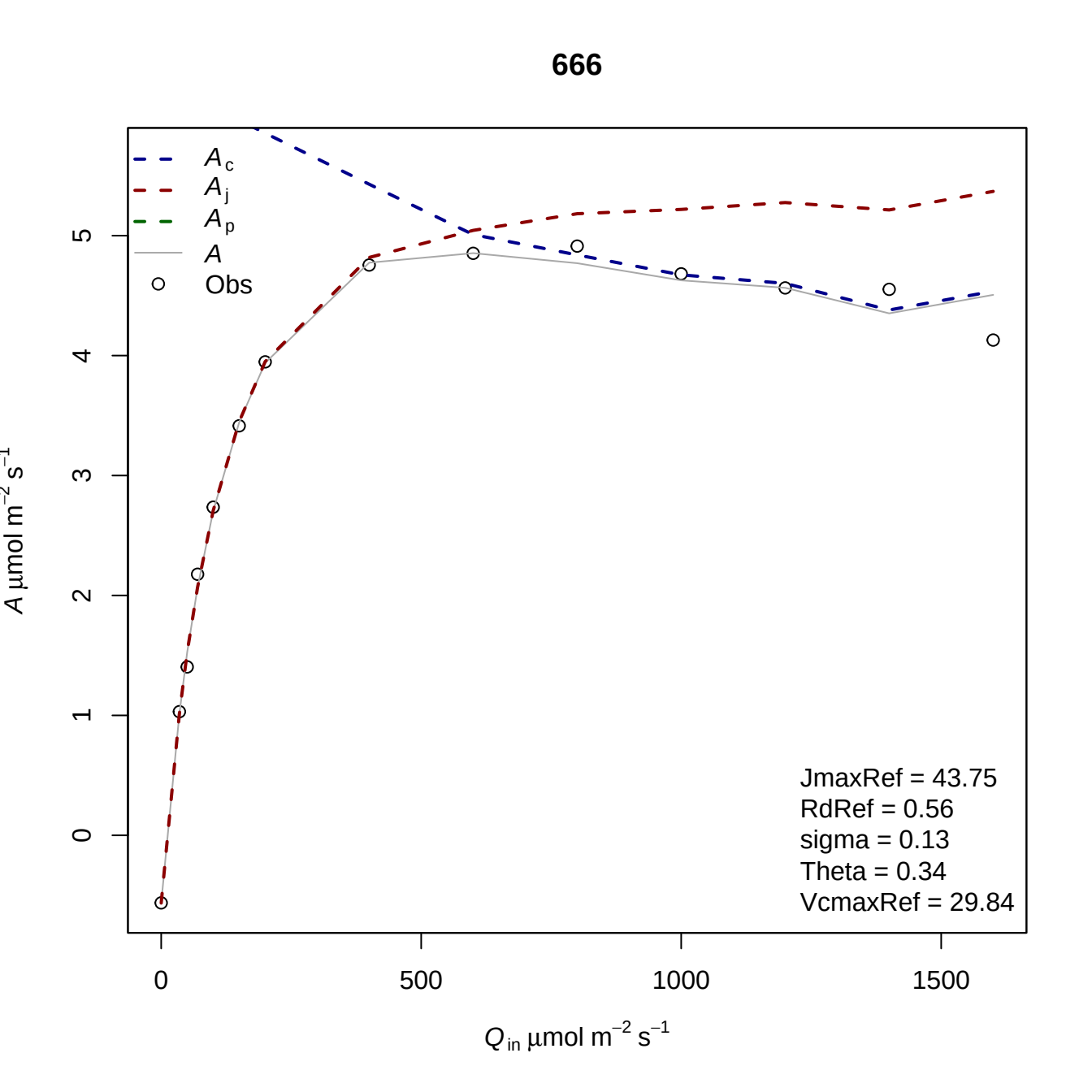

661
A_c
A_j
A_p
A
Obs
JmaxRef = 84.2
RdRef = 0.24
sigma = 0.2
Theta = 0.56
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

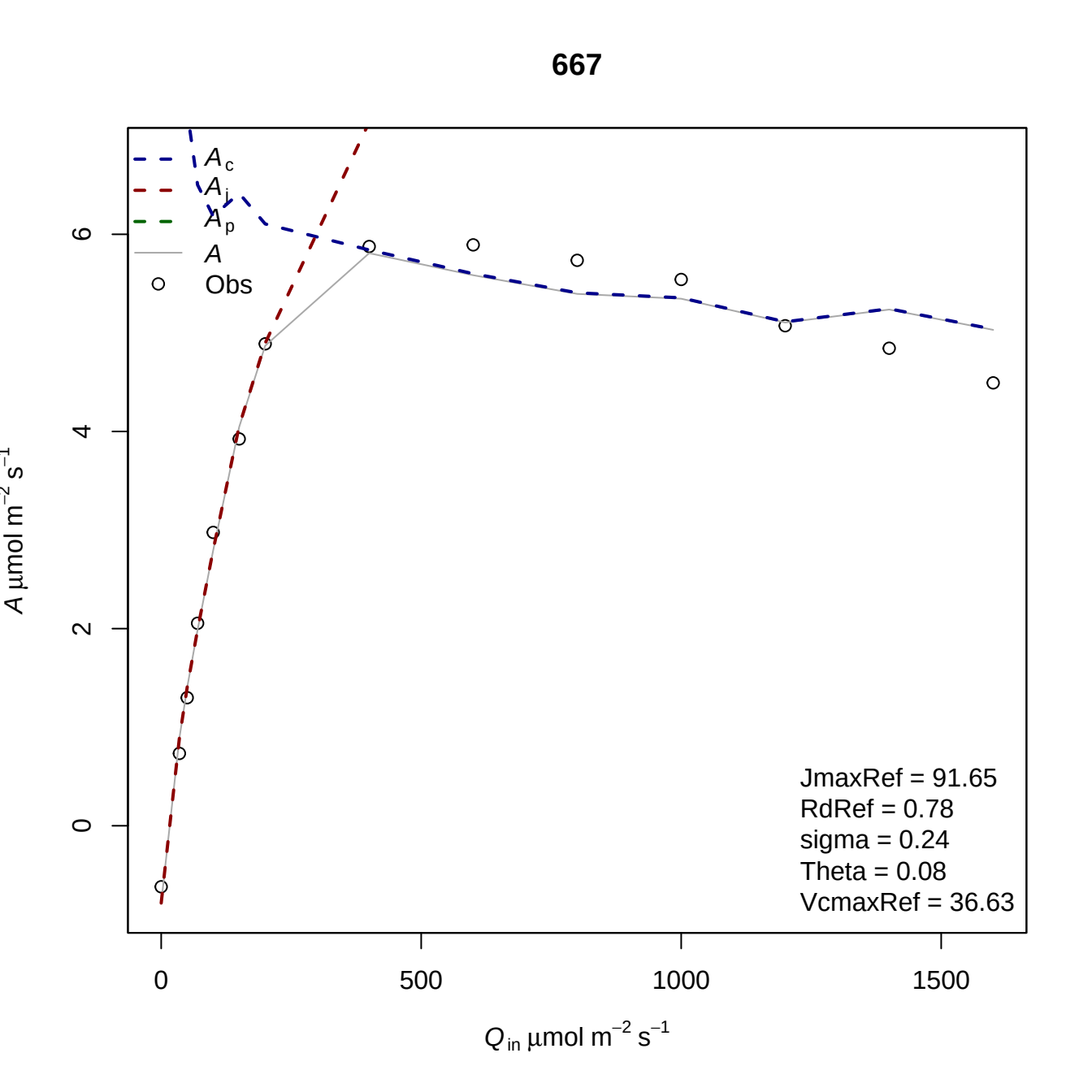

662
A_c
A_j
A_p
A
Obs
JmaxRef = 39.47
RdRef = 0.27
sigma = 0.06
Theta = 0.65
VcmaxRef = 34.5
0
1
2
3
4
5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

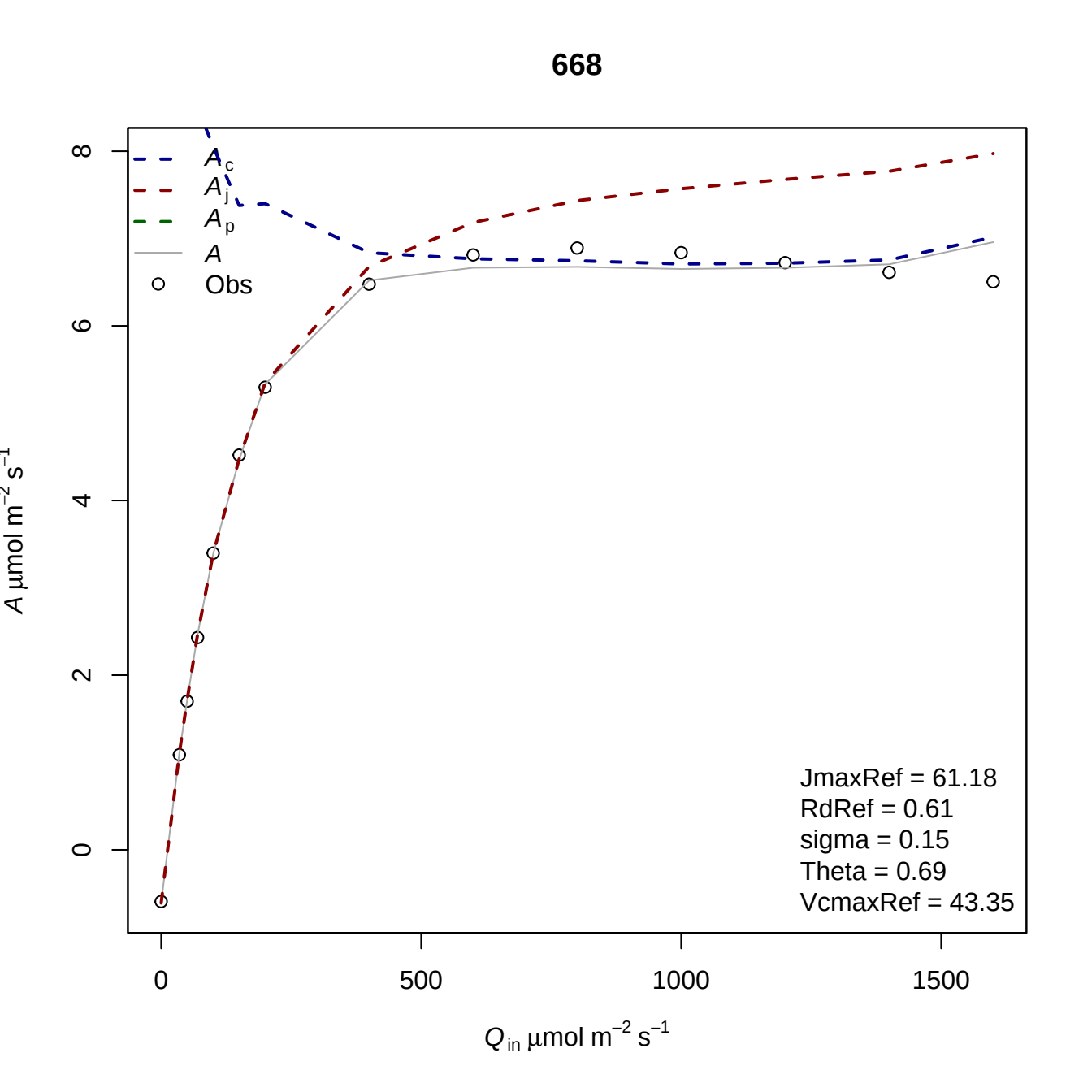

663
A_c
A_j
A_p
A
Obs
JmaxRef = 44.34
RdRef = 0.37
sigma = 0.07
Theta = 0.8
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

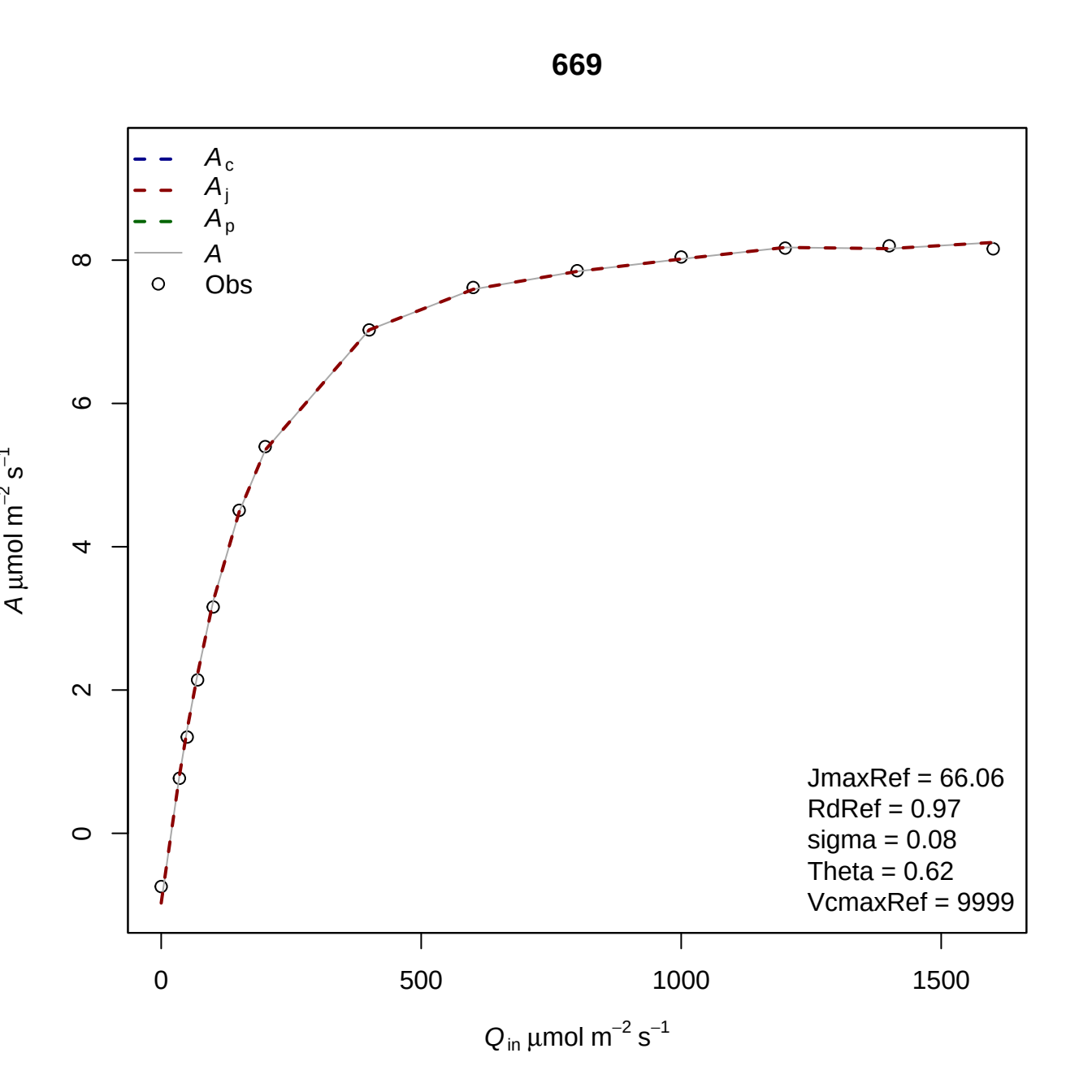

664
A_c
A_j
A_p
A
Obs
JmaxRef = 48.7
RdRef = 0.5
sigma = 0.26
Theta = 0.59
VcmaxRef = 37.92
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

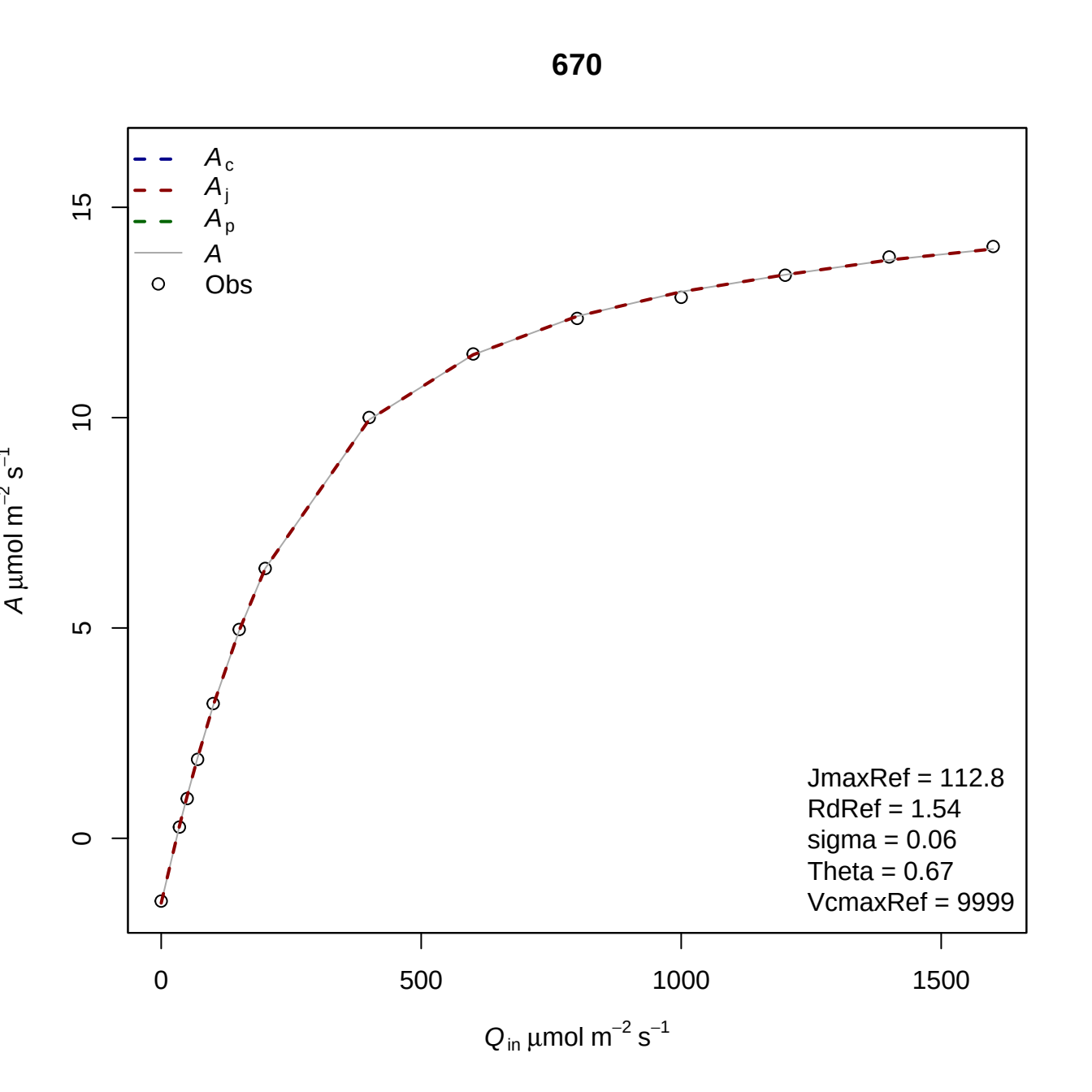

665
A_c
A_j
A_p
A
Obs
JmaxRef = 62.69
RdRef = 1.02
sigma = 0.17
Theta = 0.71
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

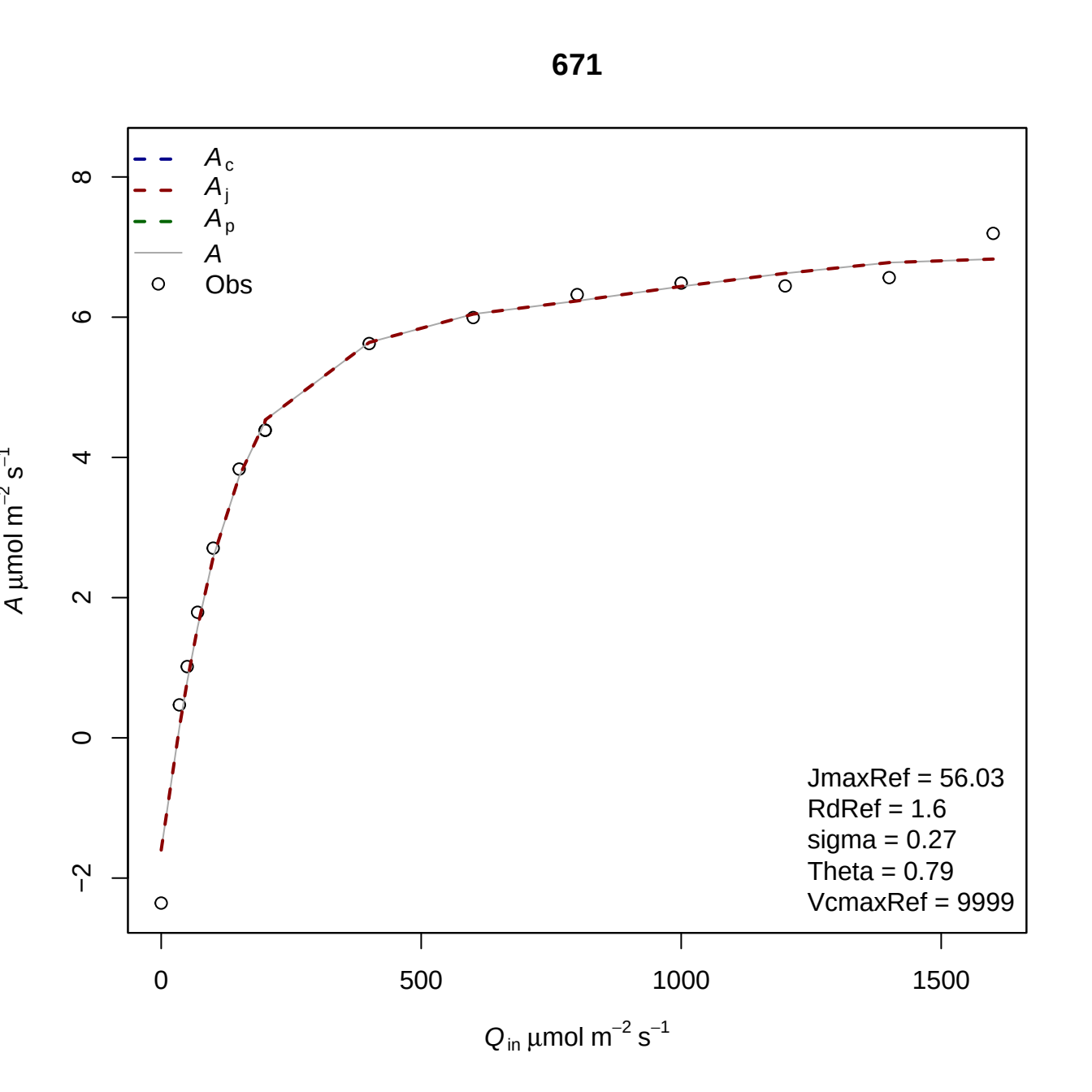

666
A_c
A_j
A_p
A
Obs
JmaxRef = 43.75
RdRef = 0.56
sigma = 0.13
Theta = 0.34
VcmaxRef = 29.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

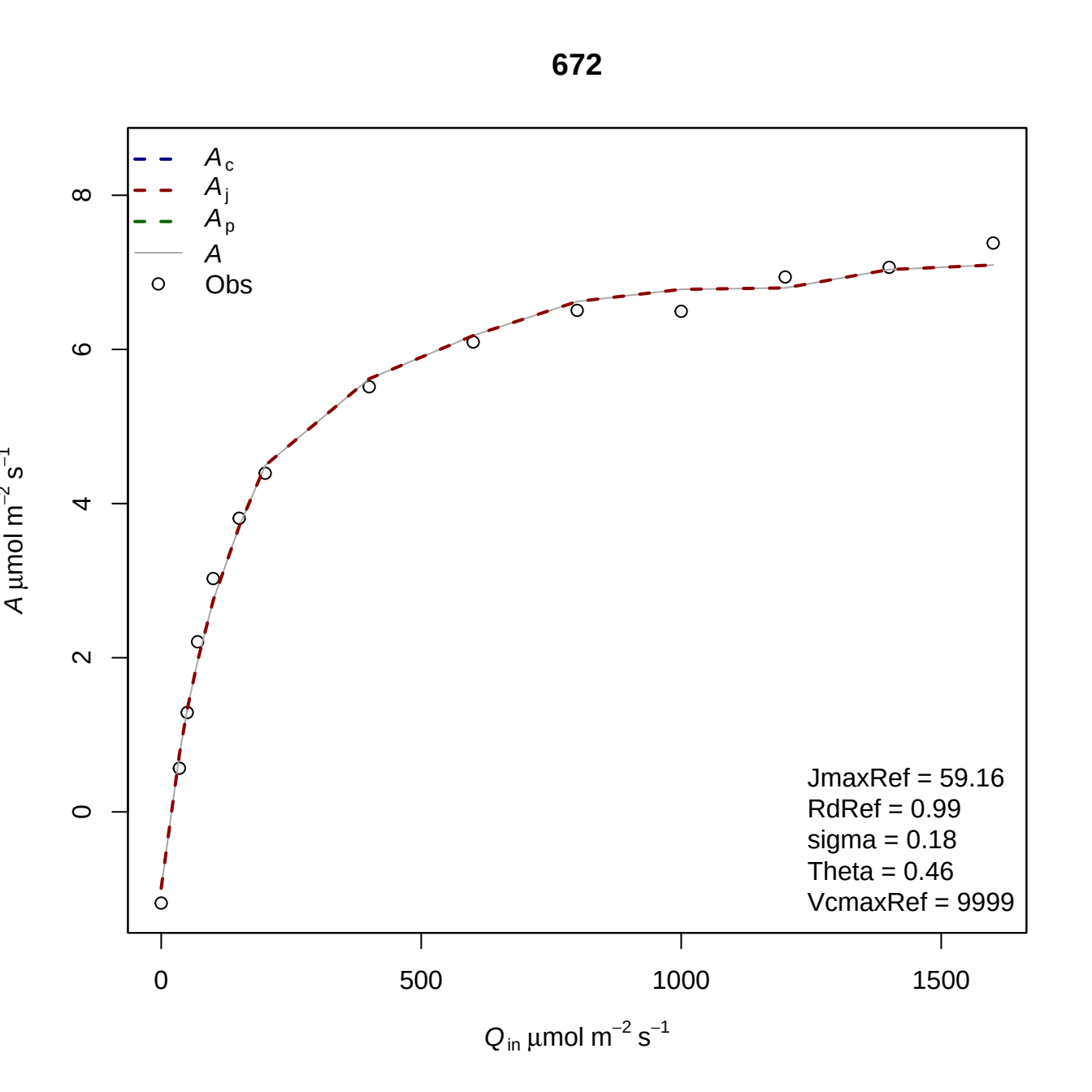

667
A_c
A_j
A_p
A
Obs
JmaxRef = 91.65
RdRef = 0.78
sigma = 0.24
Theta = 0.08
VcmaxRef = 36.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

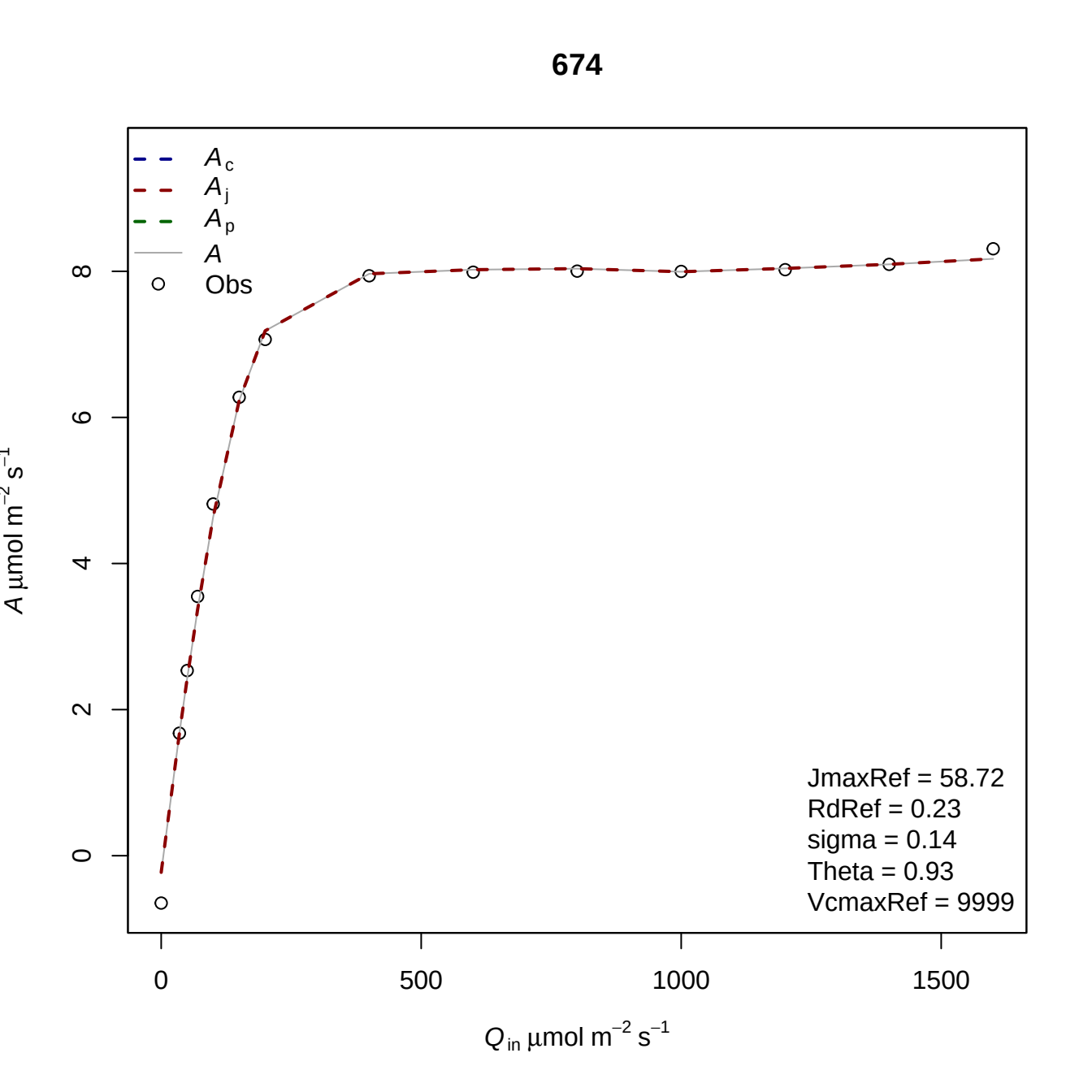

668
A_c
A_j
A_p
A
Obs
JmaxRef = 61.18
RdRef = 0.61
sigma = 0.15
Theta = 0.69
VcmaxRef = 43.35
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

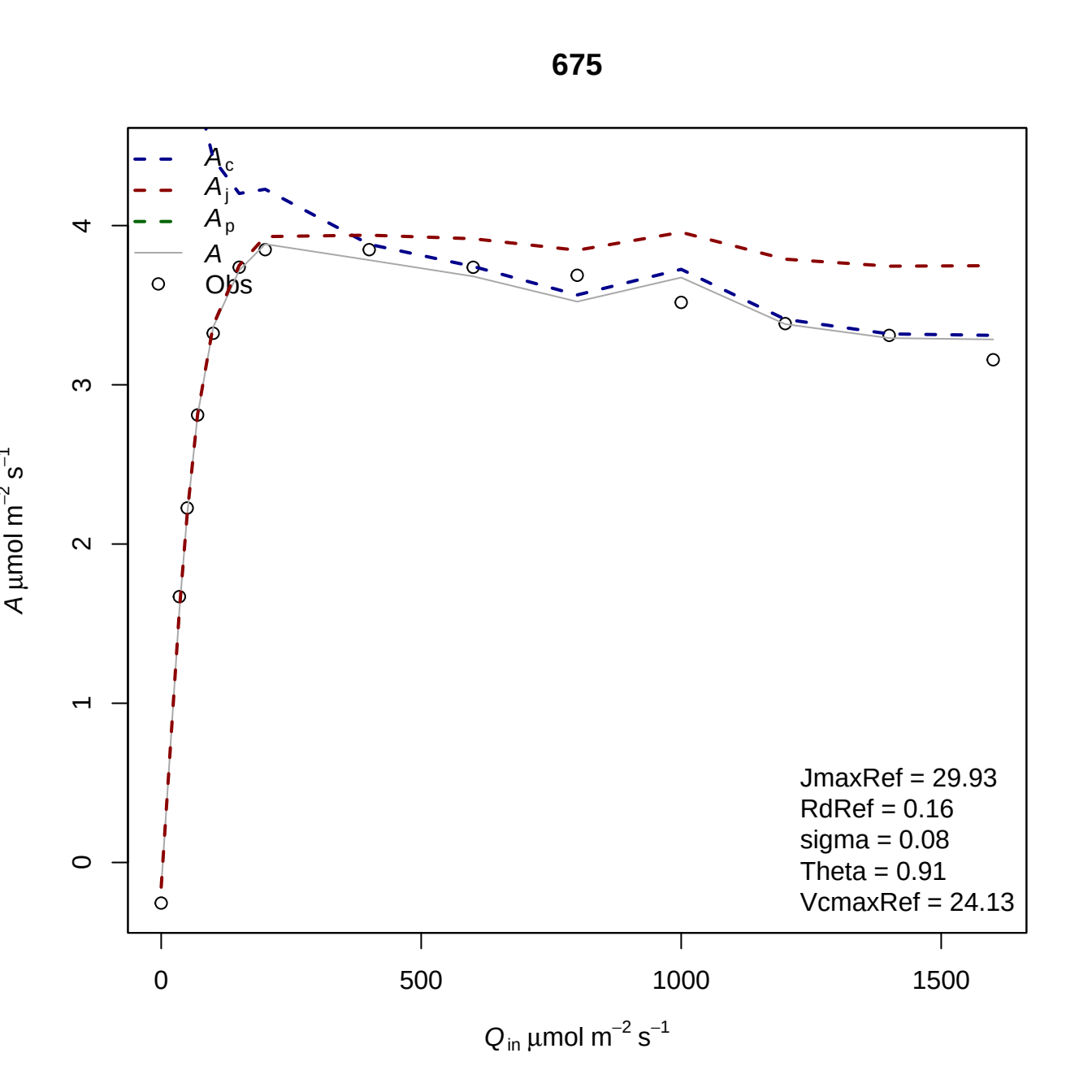

669
A_c
A_j
A_p
A
Obs
JmaxRef = 66.06
RdRef = 0.97
sigma = 0.08
Theta = 0.62
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

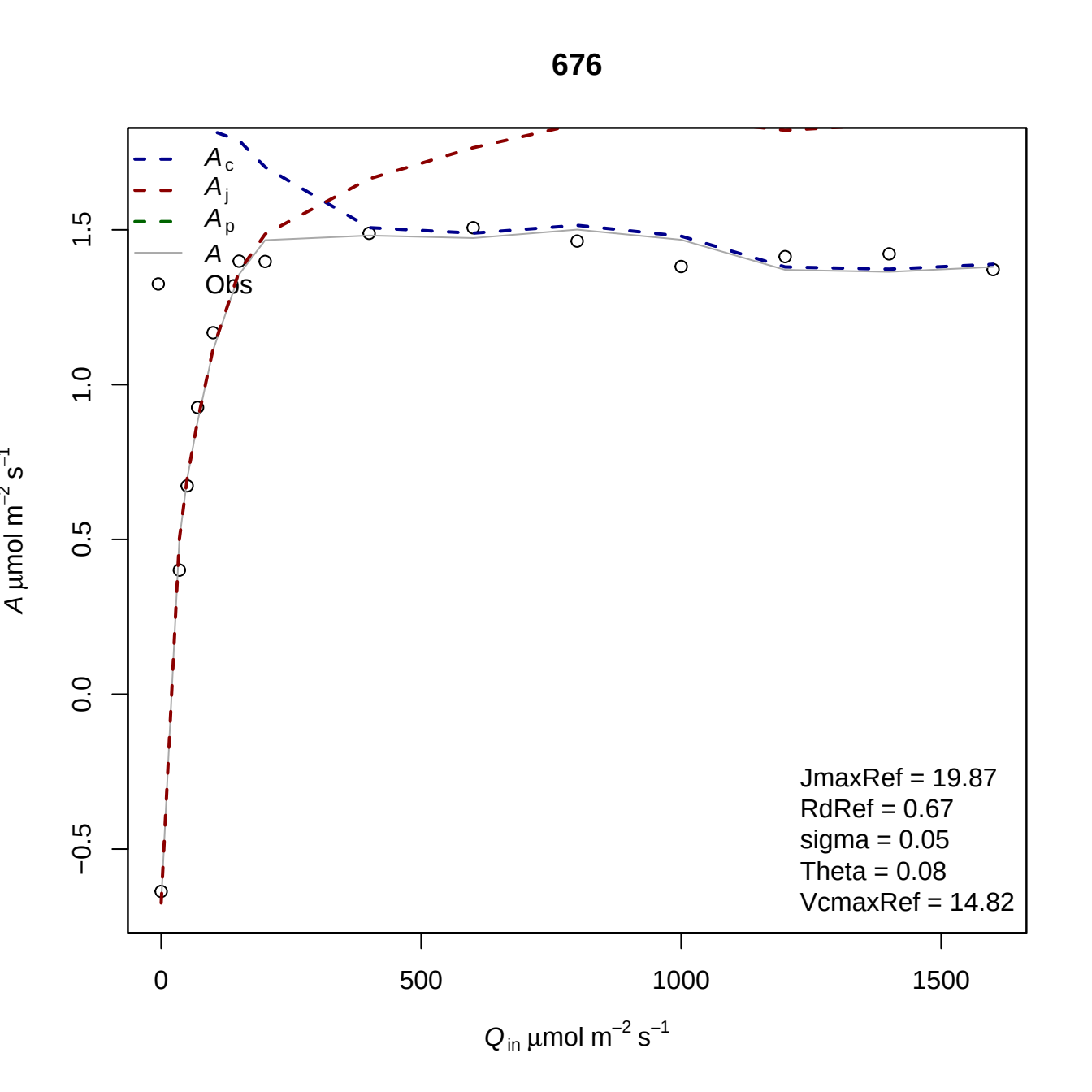

670
A_c
A_j
A_p
A
Obs
JmaxRef = 112.8
RdRef = 1.54
sigma = 0.06
Theta = 0.67
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

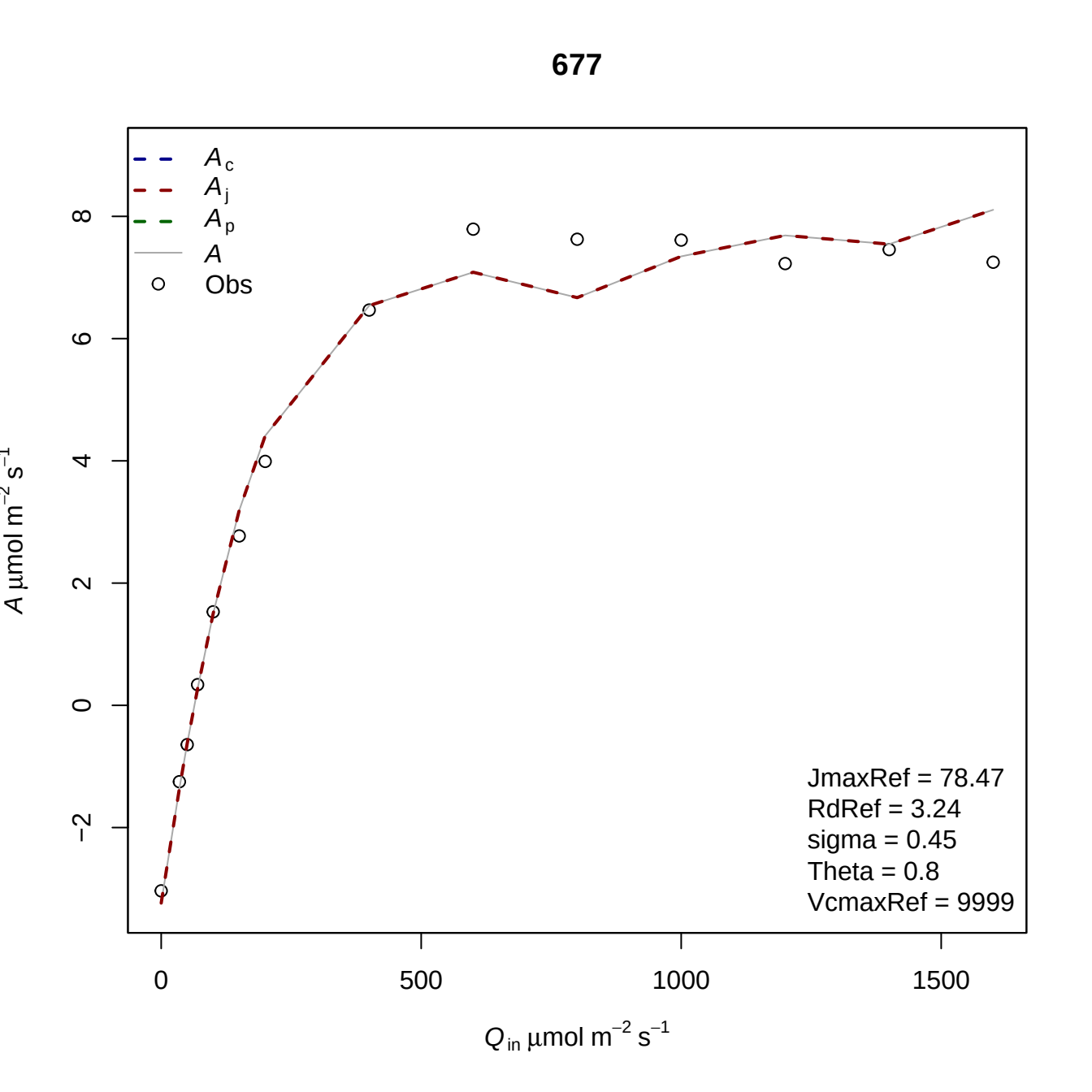

671
A_c
A_j
A_p
A
Obs
JmaxRef = 56.03
RdRef = 1.6
sigma = 0.27
Theta = 0.79
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

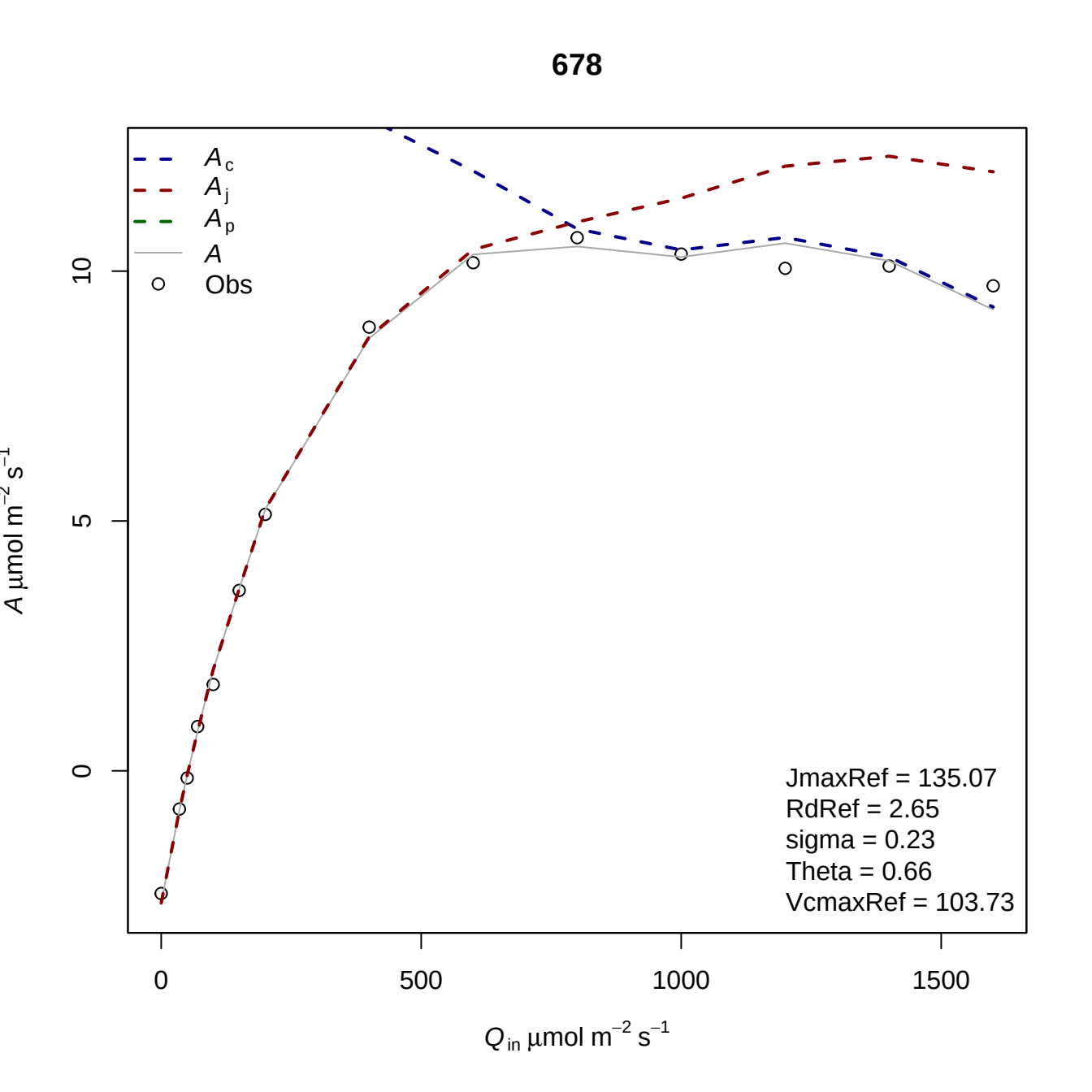

672
A_c
A_j
A_p
A
Obs
JmaxRef = 59.16
RdRef = 0.99
sigma = 0.18
Theta = 0.46
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

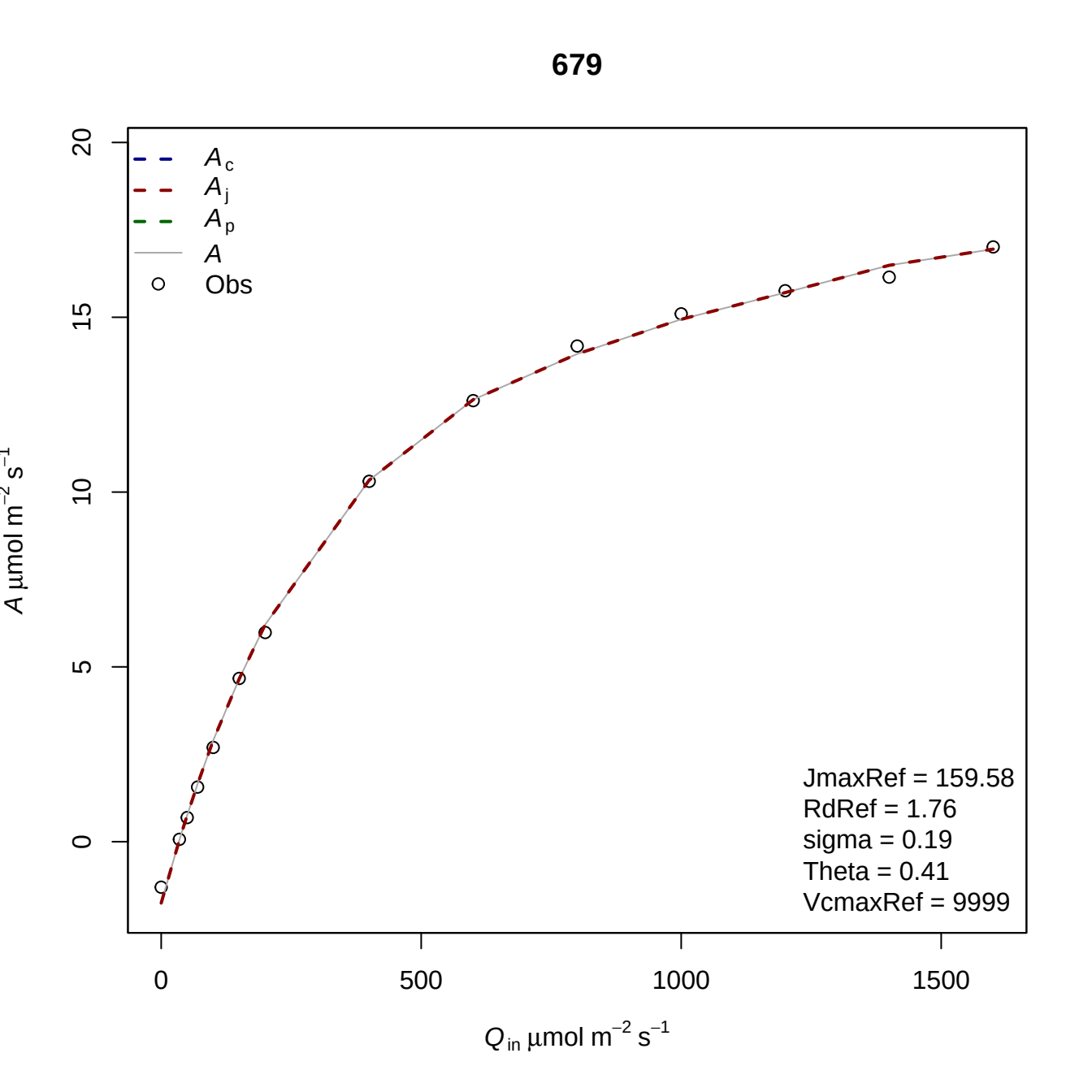

674
A_c
A_j
A_p
A
Obs
JmaxRef = 58.72
RdRef = 0.23
sigma = 0.14
Theta = 0.93
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

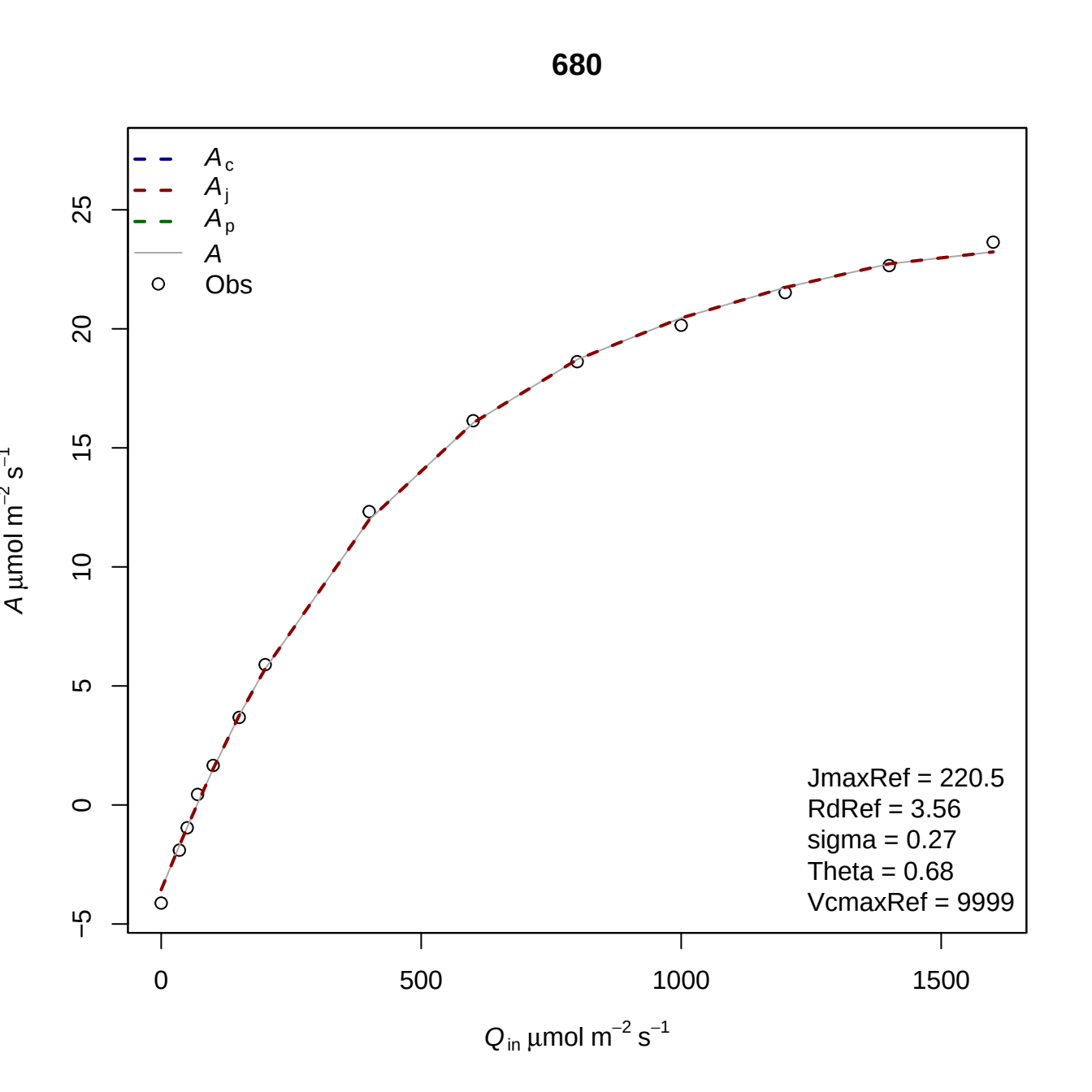

675
A_c
A_j
A_p
A
Obs
JmaxRef = 29.93
RdRef = 0.16
sigma = 0.08
Theta = 0.91
VcmaxRef = 24.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

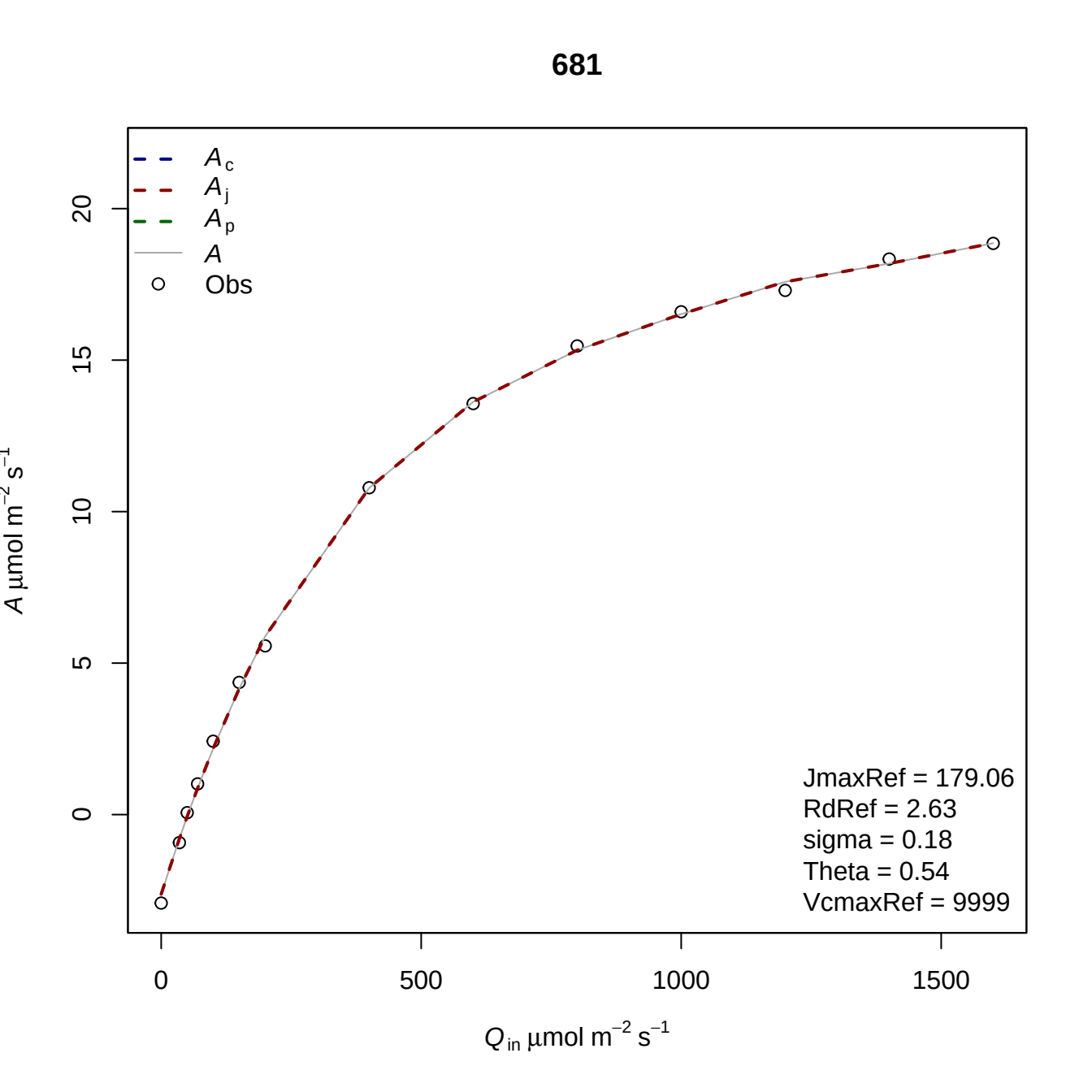

676
A_c
A_j
A_p
A
Obs
JmaxRef = 19.87
RdRef = 0.67
sigma = 0.05
Theta = 0.08
VcmaxRef = 14.82
−0.5
0.0
0.5
1.0
1.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

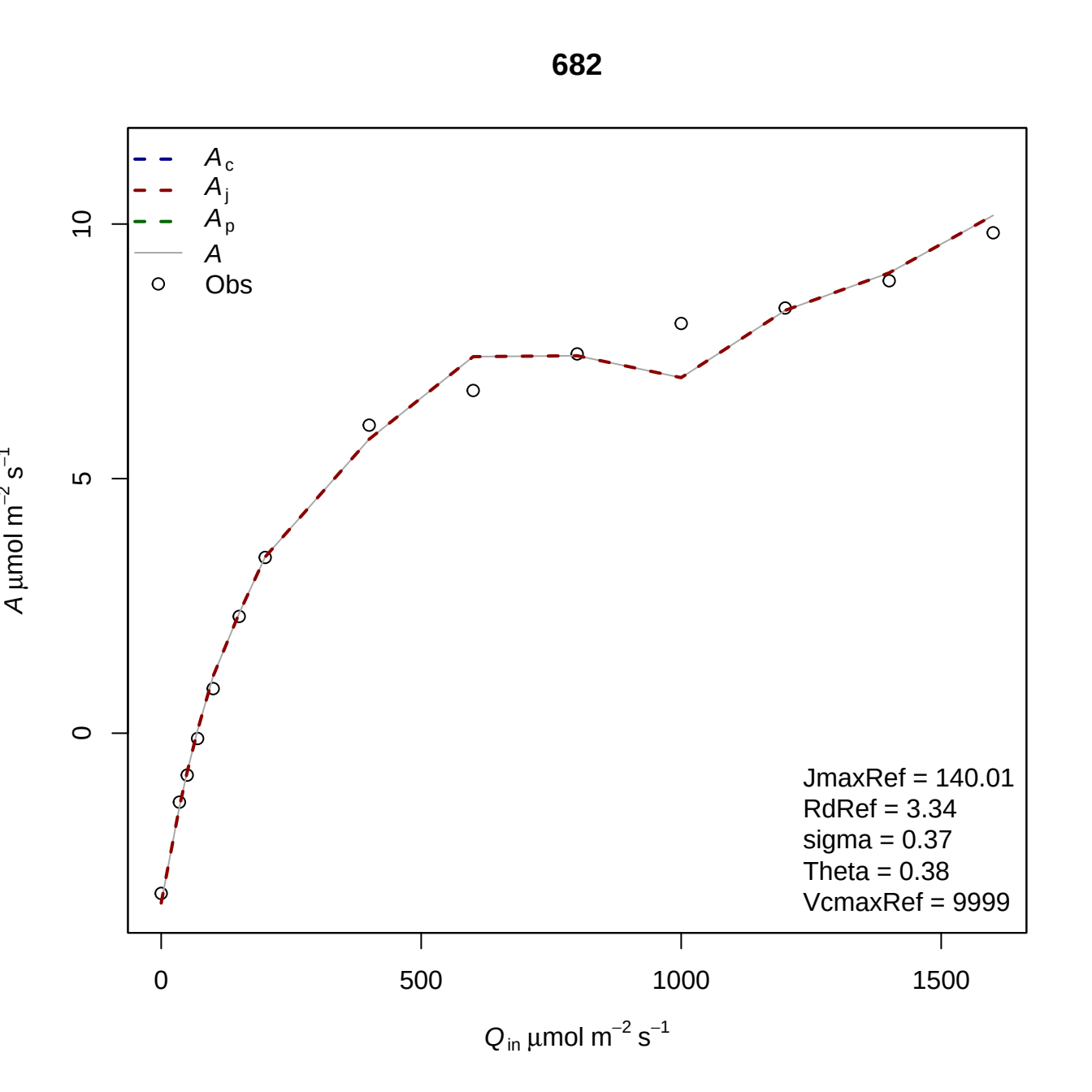

677
A_c
A_j
A_p
A
Obs
JmaxRef = 78.47
RdRef = 3.24
sigma = 0.45
Theta = 0.8
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

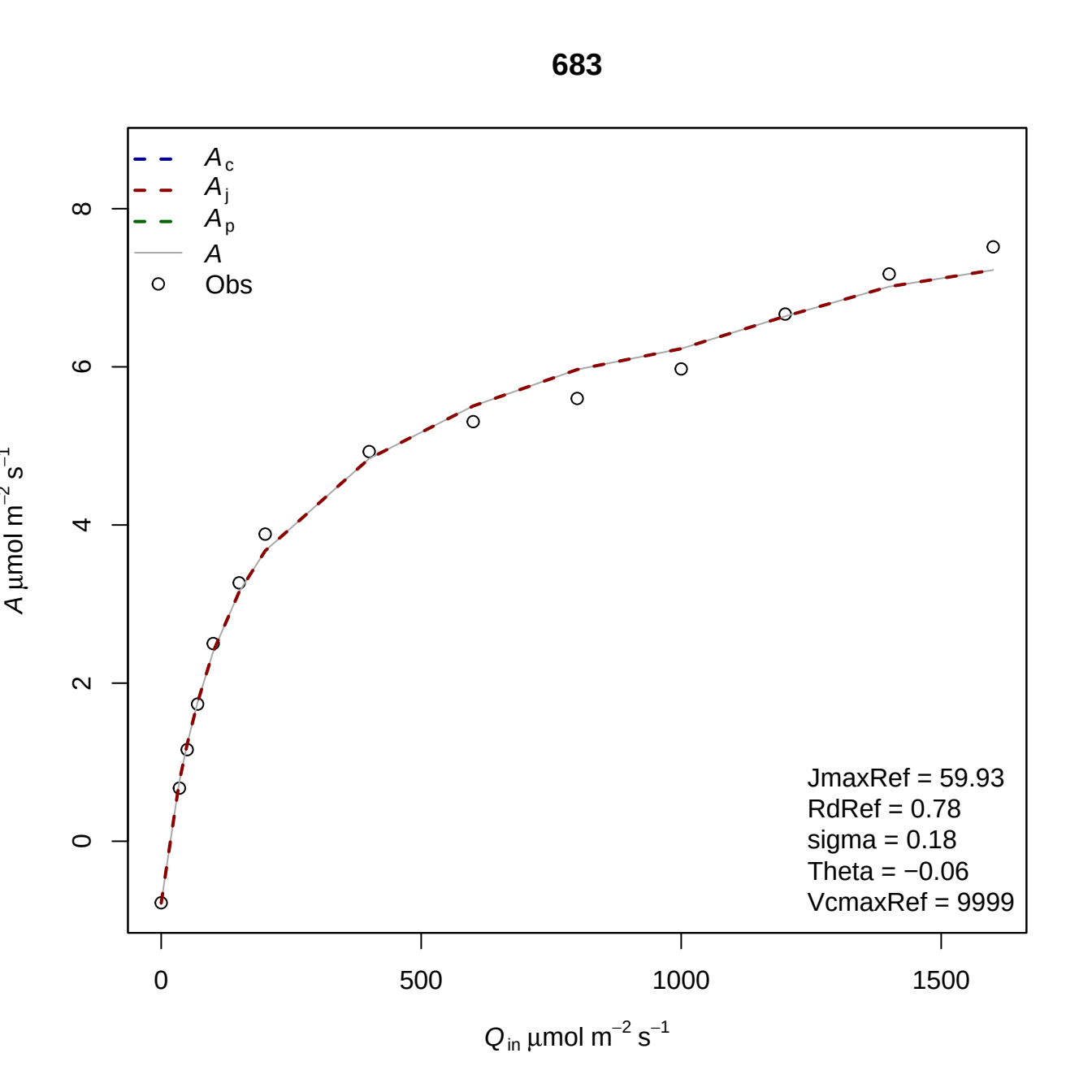

678
A_c
A_j
A_p
A
Obs
JmaxRef = 135.07
RdRef = 2.65
sigma = 0.23
Theta = 0.66
VcmaxRef = 103.73
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

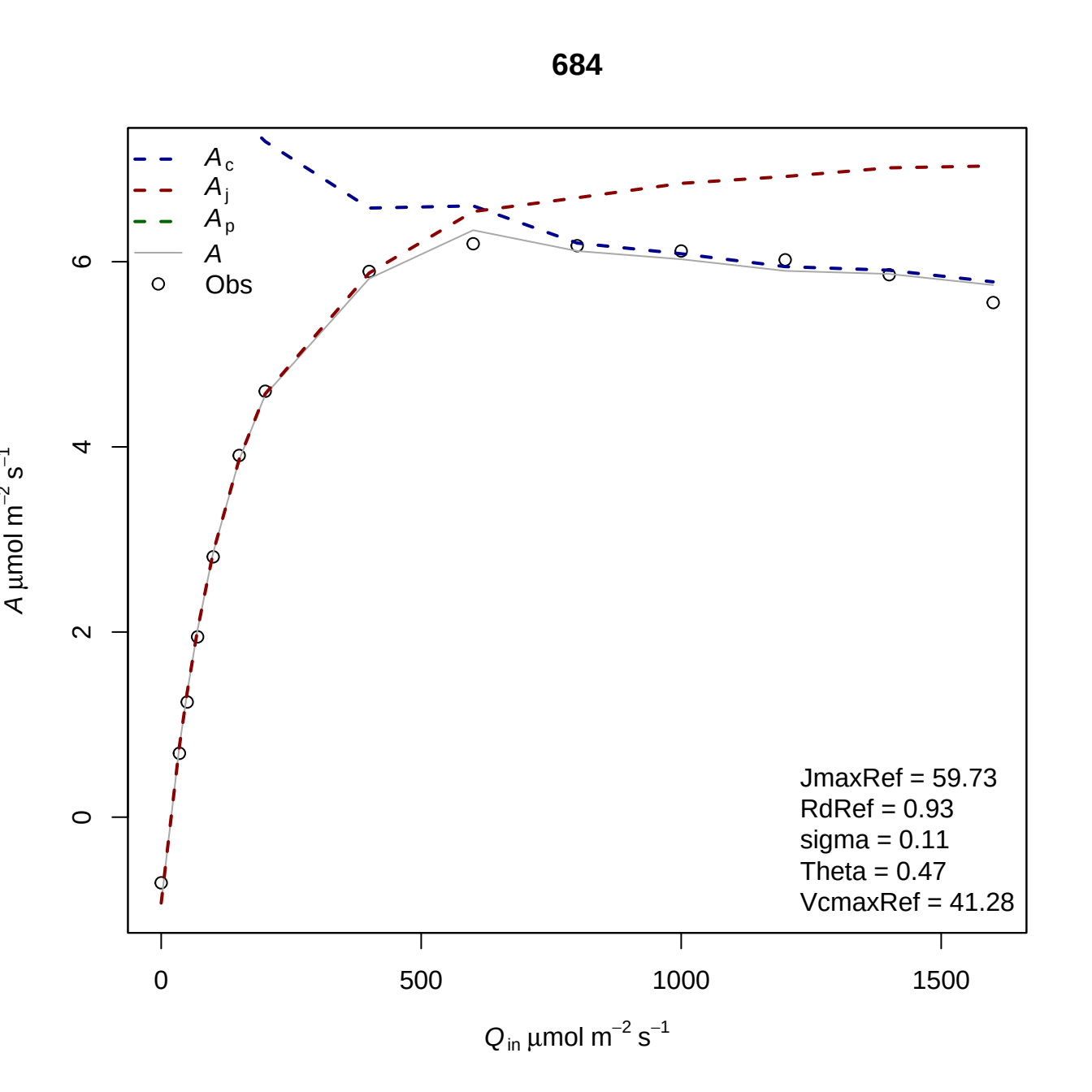

679
A_c
A_j
A_p
A
Obs
JmaxRef = 159.58
RdRef = 1.76
sigma = 0.19
Theta = 0.41
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

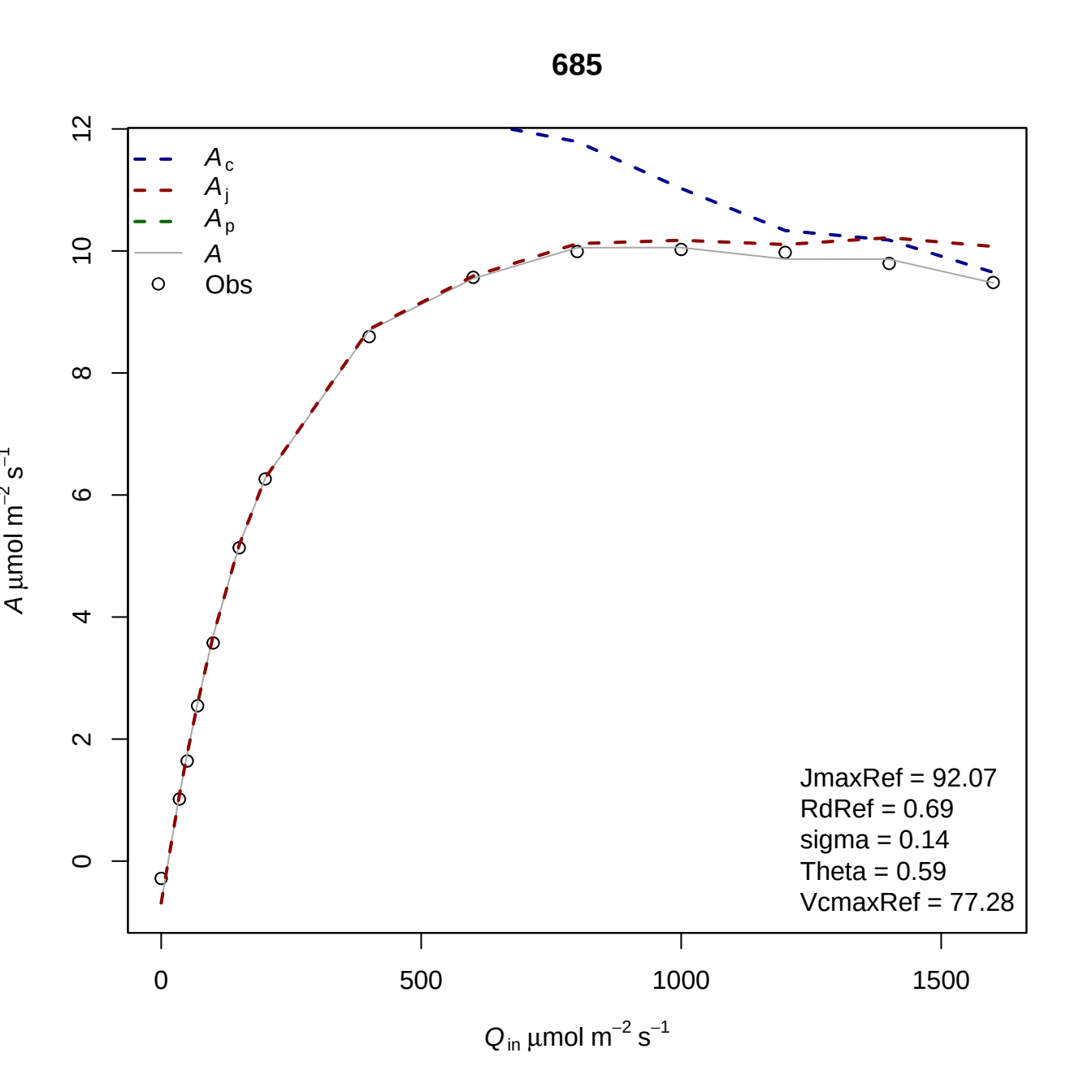

680
A_c
A_j
A_p
A
Obs
JmaxRef = 220.5
RdRef = 3.56
sigma = 0.27
Theta = 0.68
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500

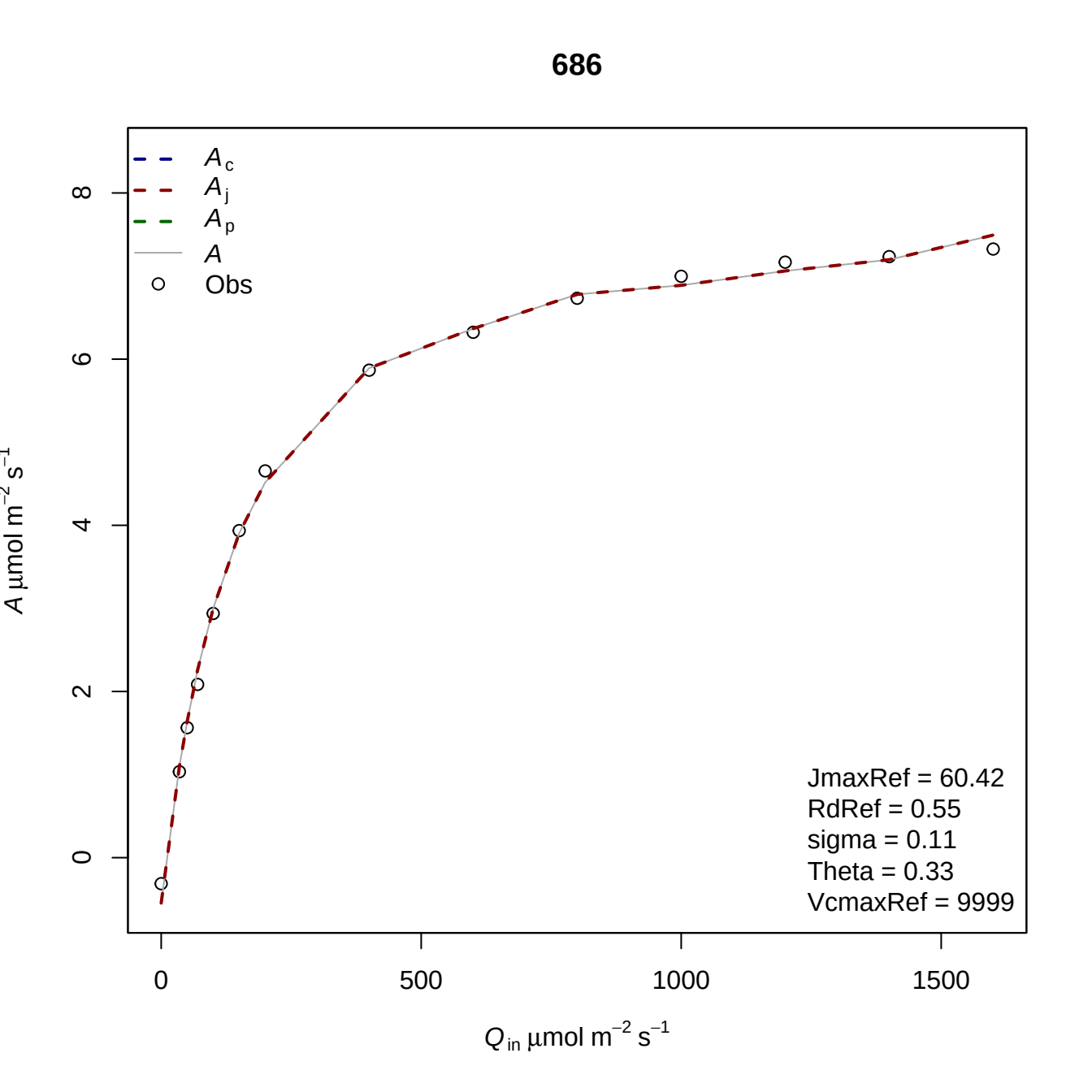

681
A_c
A_j
A_p
A
Obs
JmaxRef = 179.06
RdRef = 2.63
sigma = 0.18
Theta = 0.54
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

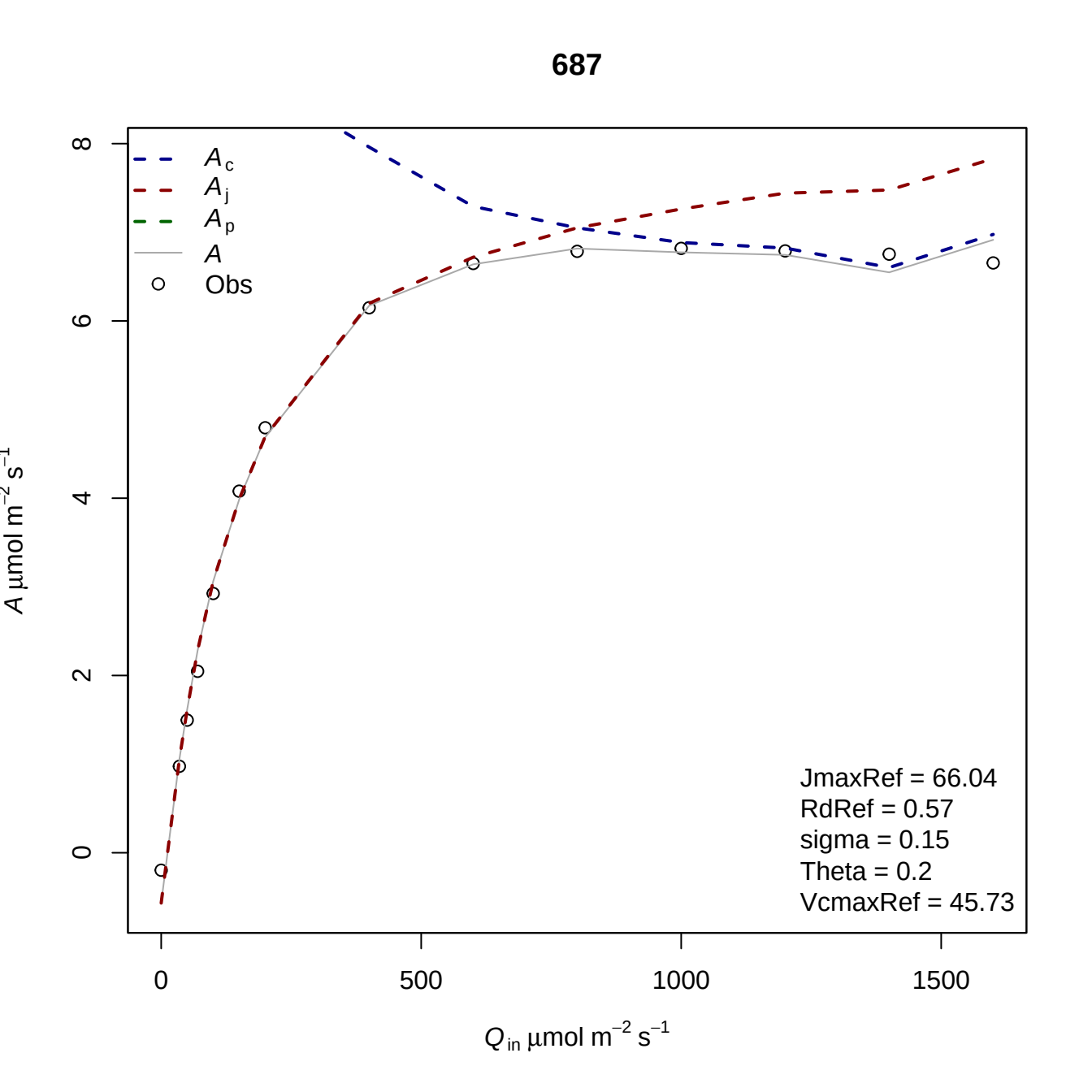

682
A_c
A_j
A_p
A
Obs
JmaxRef = 140.01
RdRef = 3.34
sigma = 0.37
Theta = 0.38
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

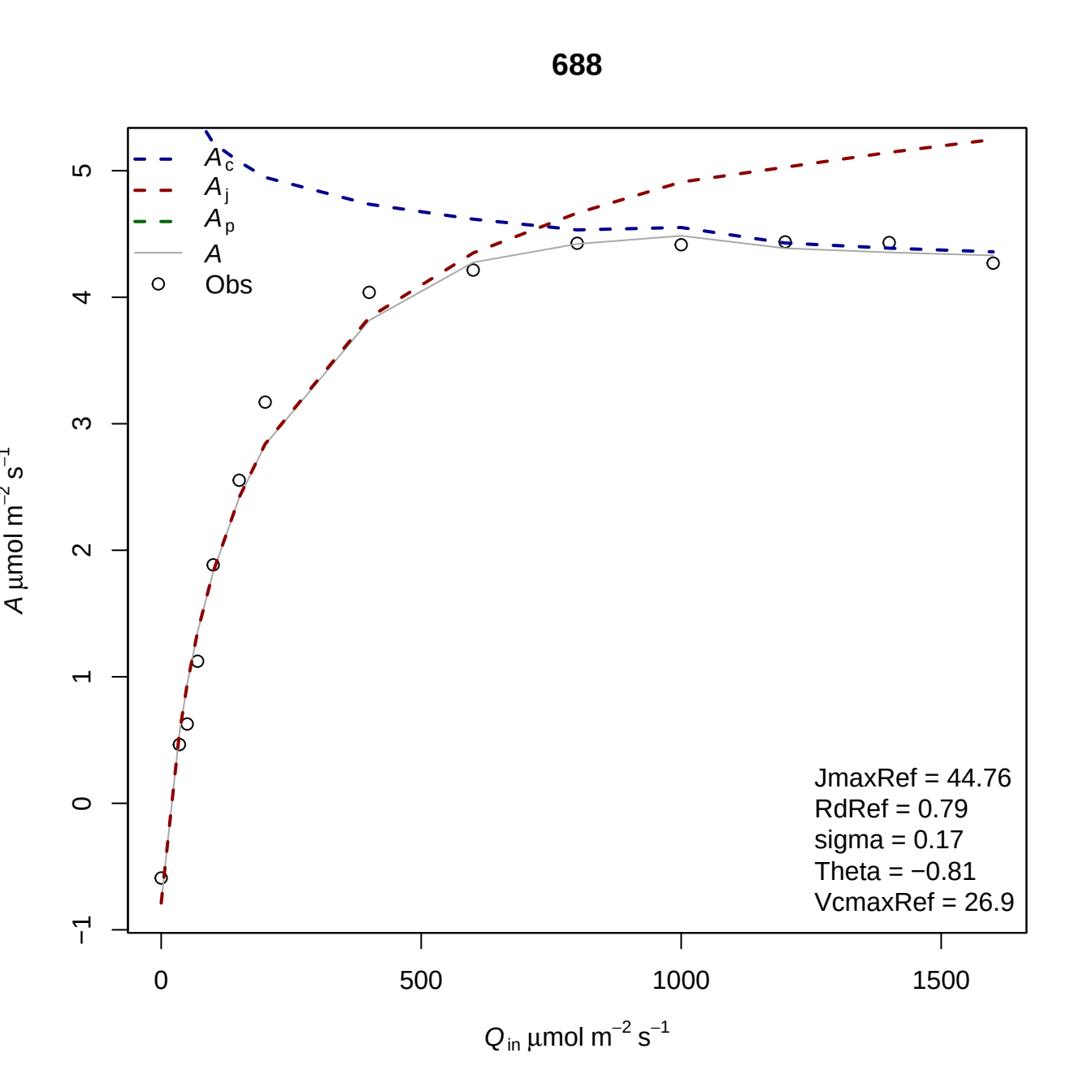

683
A_c
A_j
A_p
A
Obs
JmaxRef = 59.93
RdRef = 0.78
sigma = 0.18
Theta = −0.06
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

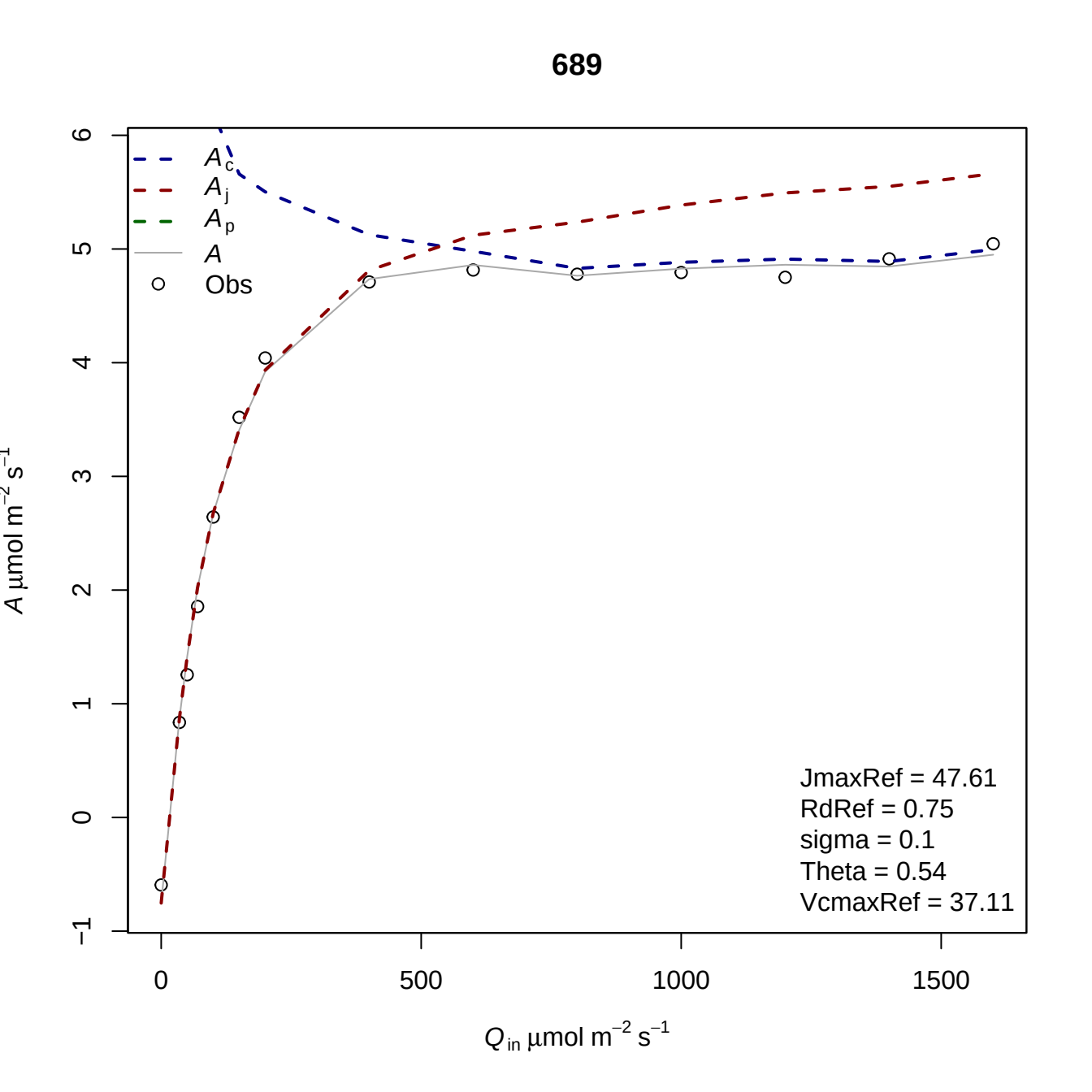

684
A_c
A_j
A_p
A
Obs
JmaxRef = 59.73
RdRef = 0.93
sigma = 0.11
Theta = 0.47
VcmaxRef = 41.28
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

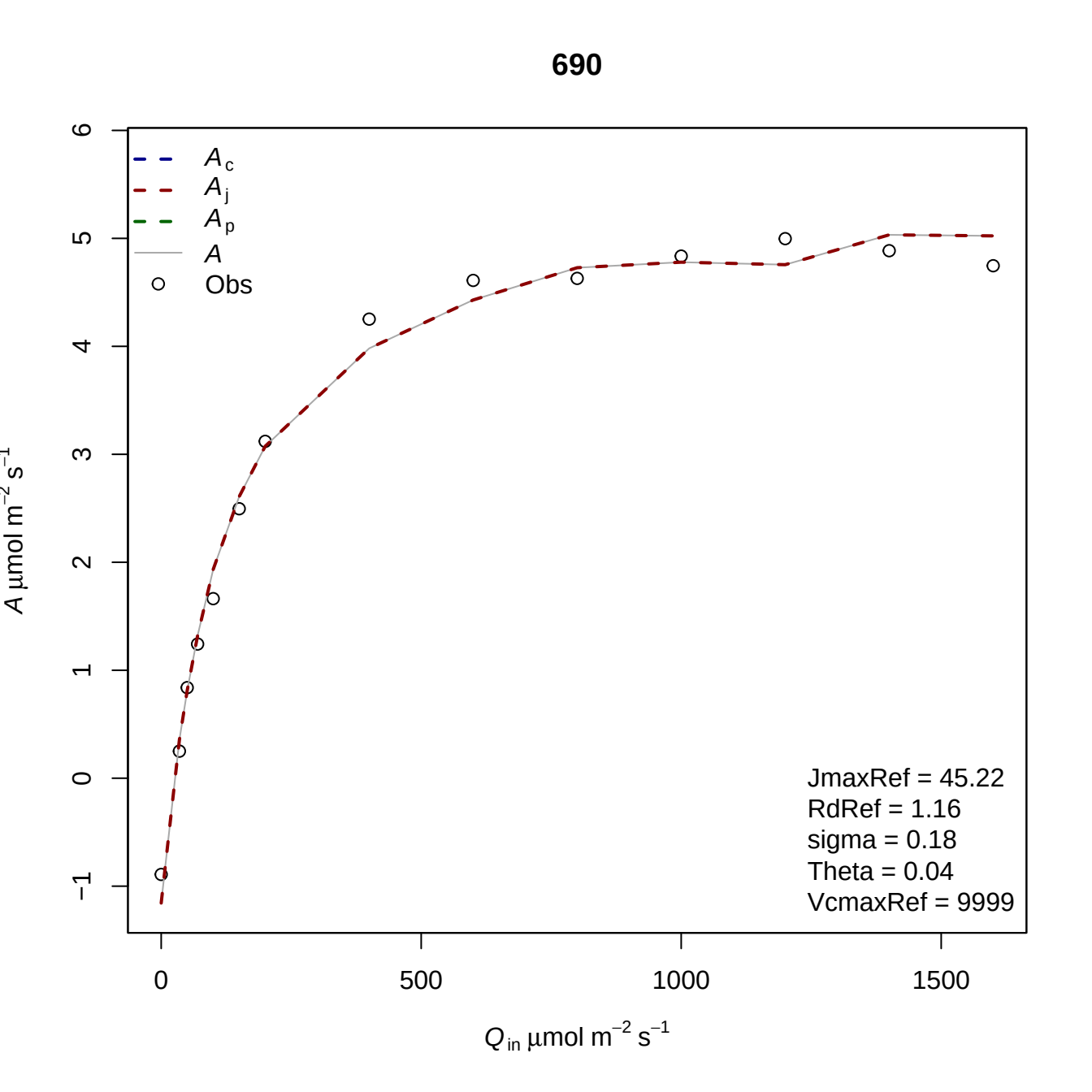

685
A_c
A_j
A_p
A
Obs
JmaxRef = 92.07
RdRef = 0.69
sigma = 0.14
Theta = 0.59
VcmaxRef = 77.28
0
2
4
6
8
10
12
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

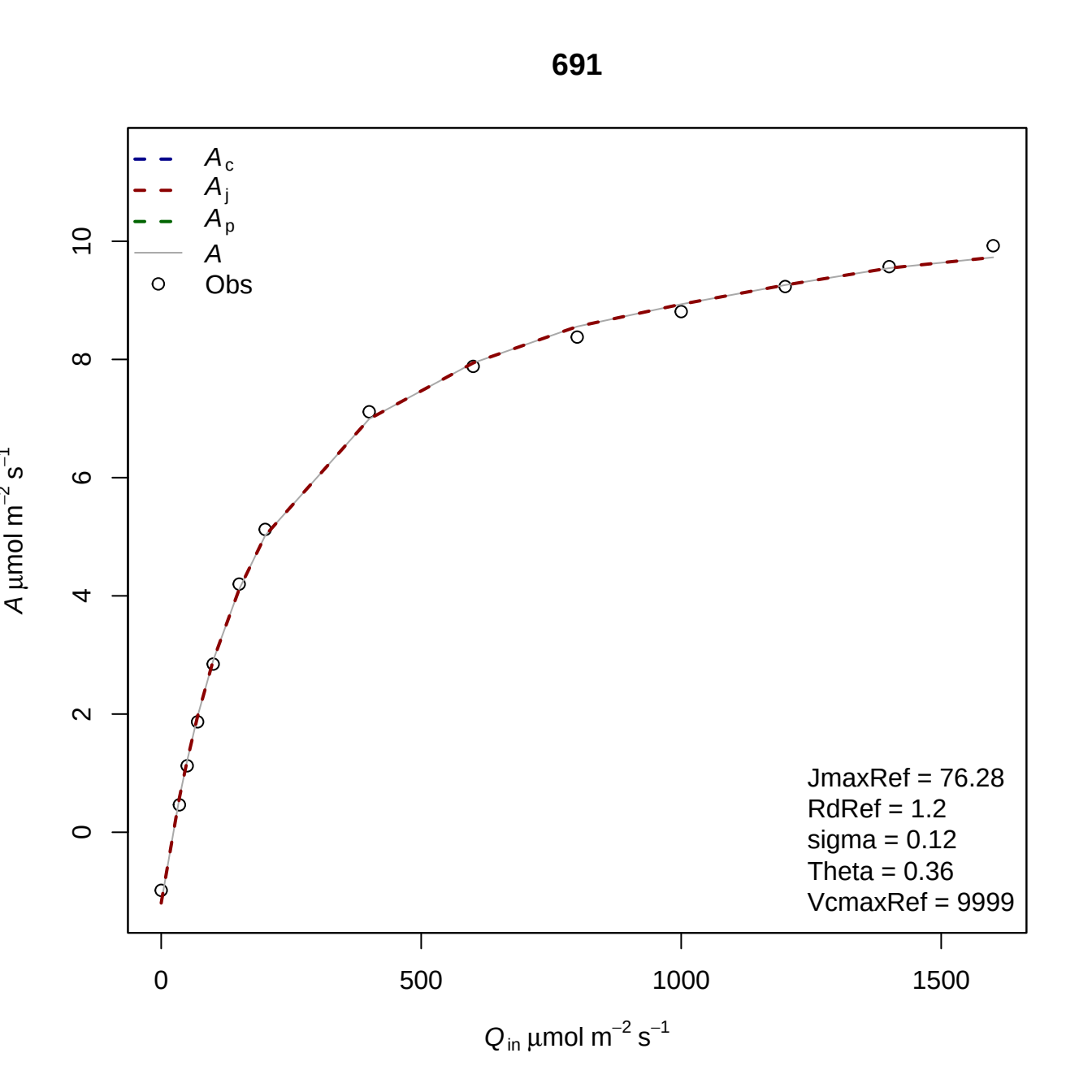

686
A_c
A_j
A_p
A
Obs
JmaxRef = 60.42
RdRef = 0.55
sigma = 0.11
Theta = 0.33
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

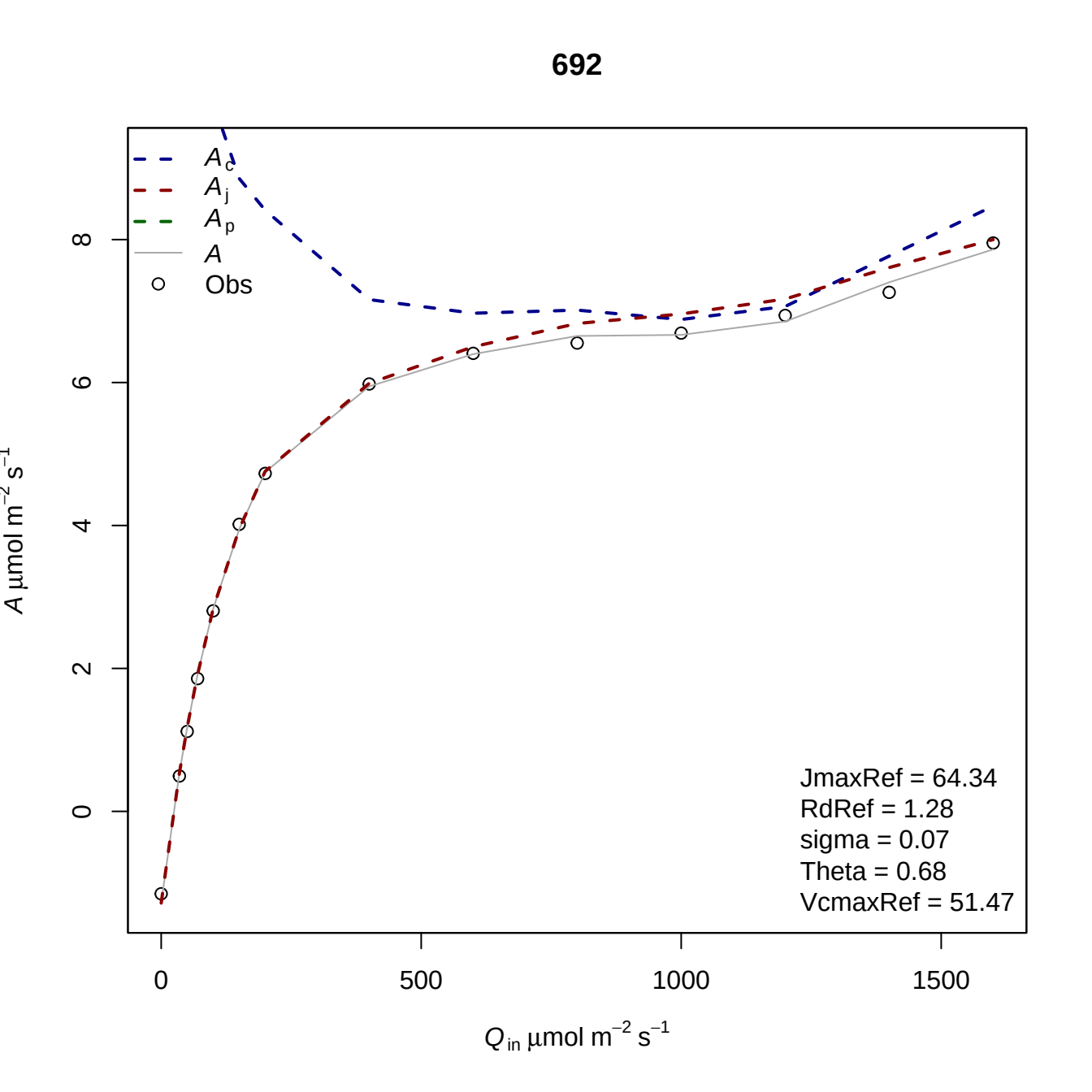

687
A_c
A_j
A_p
A
Obs
JmaxRef = 66.04
RdRef = 0.57
sigma = 0.15
Theta = 0.2
VcmaxRef = 45.73
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

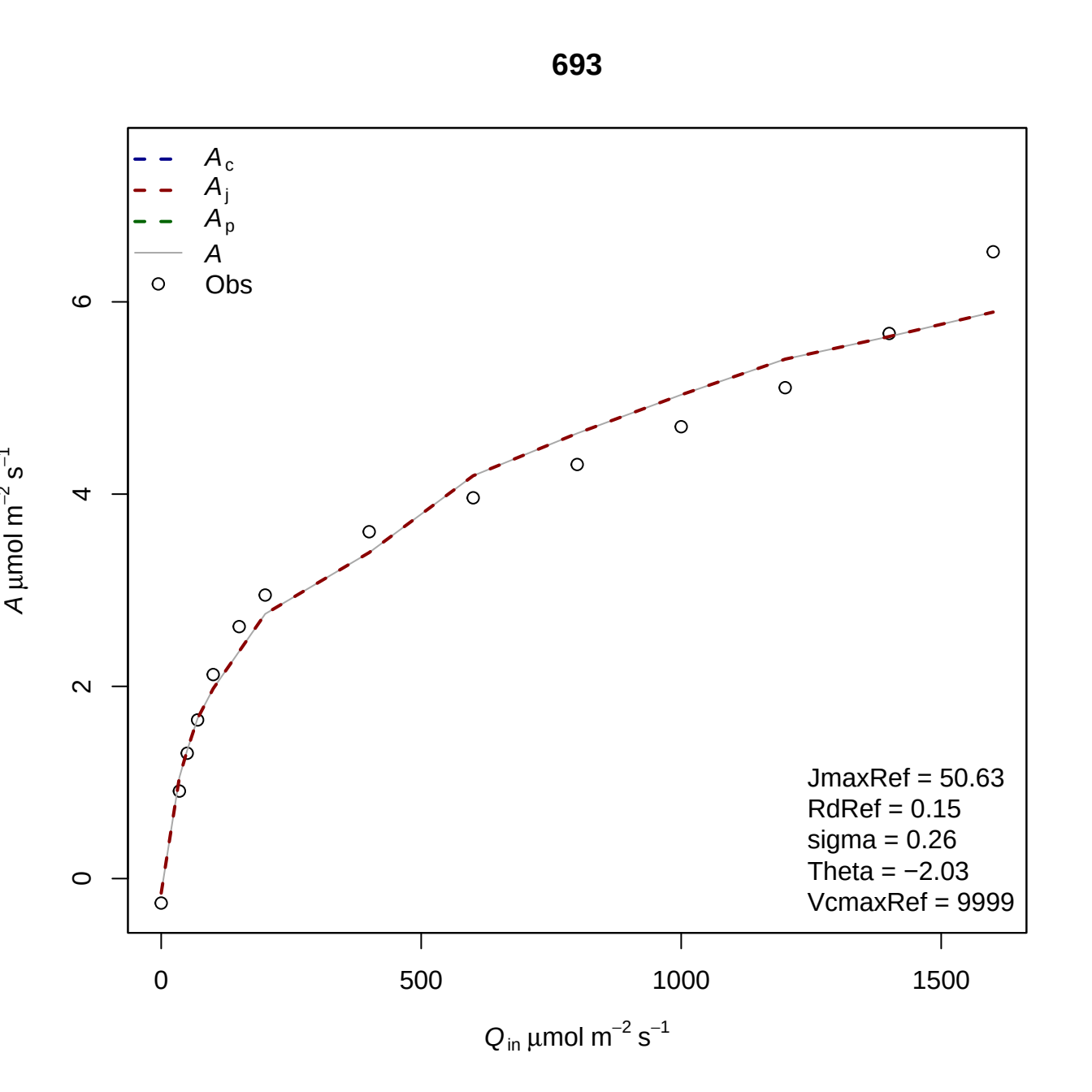

688
A_c
A_j
A_p
A
Obs
JmaxRef = 44.76
RdRef = 0.79
sigma = 0.17
Theta = −0.81
VcmaxRef = 26.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

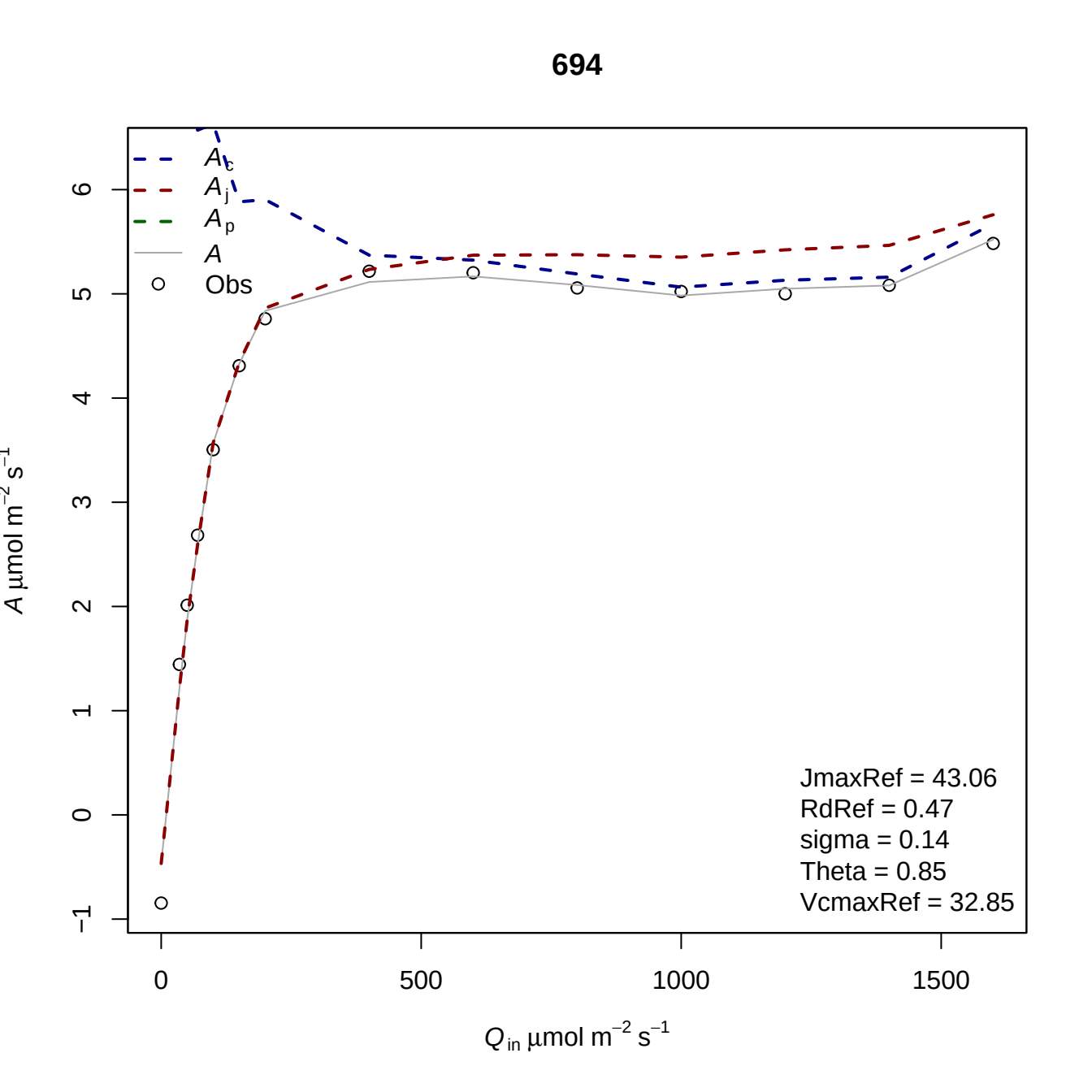

689
A_c
A_j
A_p
A
Obs
JmaxRef = 47.61
RdRef = 0.75
sigma = 0.1
Theta = 0.54
VcmaxRef = 37.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

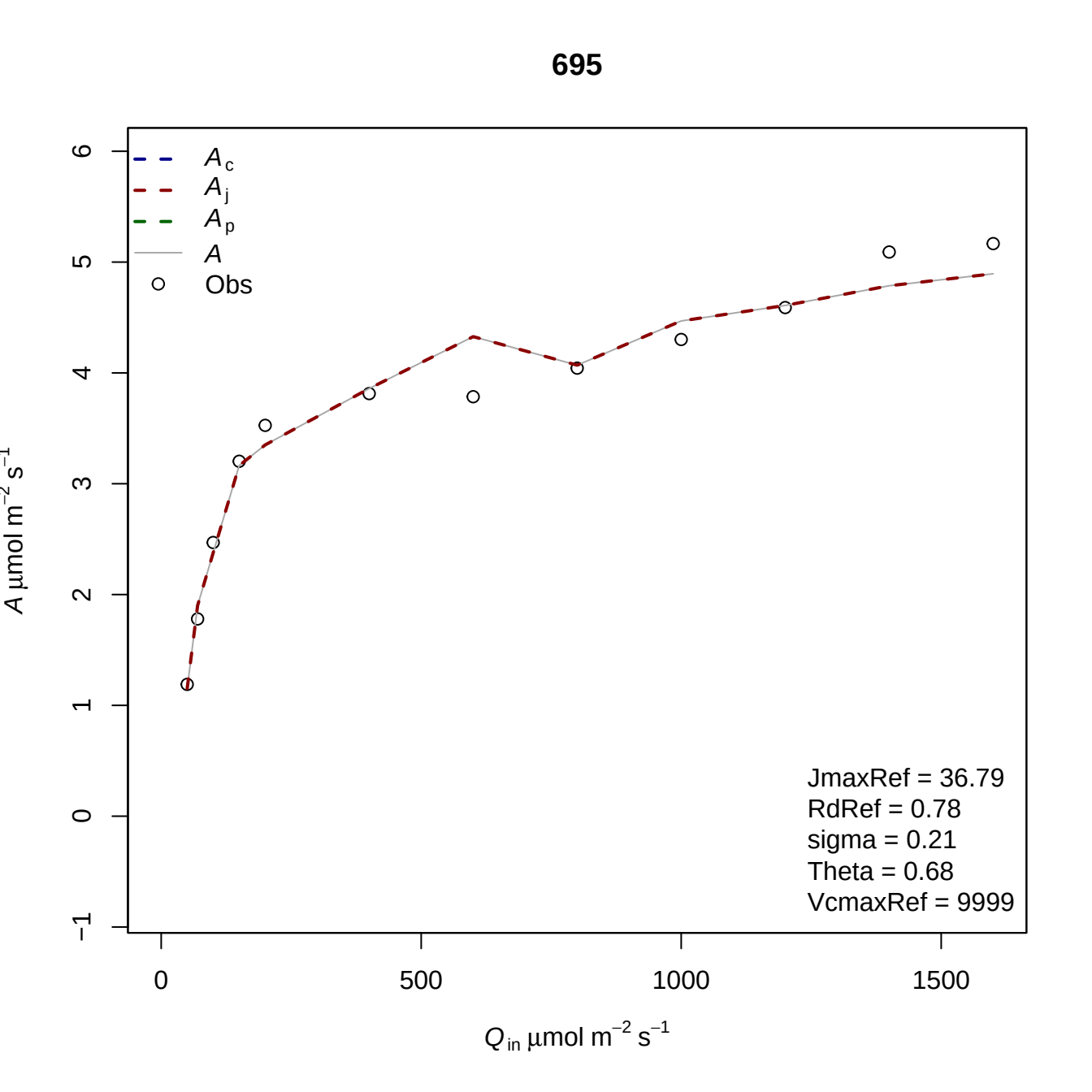

690
A_c
A_j
A_p
A
Obs
JmaxRef = 45.22
RdRef = 1.16
sigma = 0.18
Theta = 0.04
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

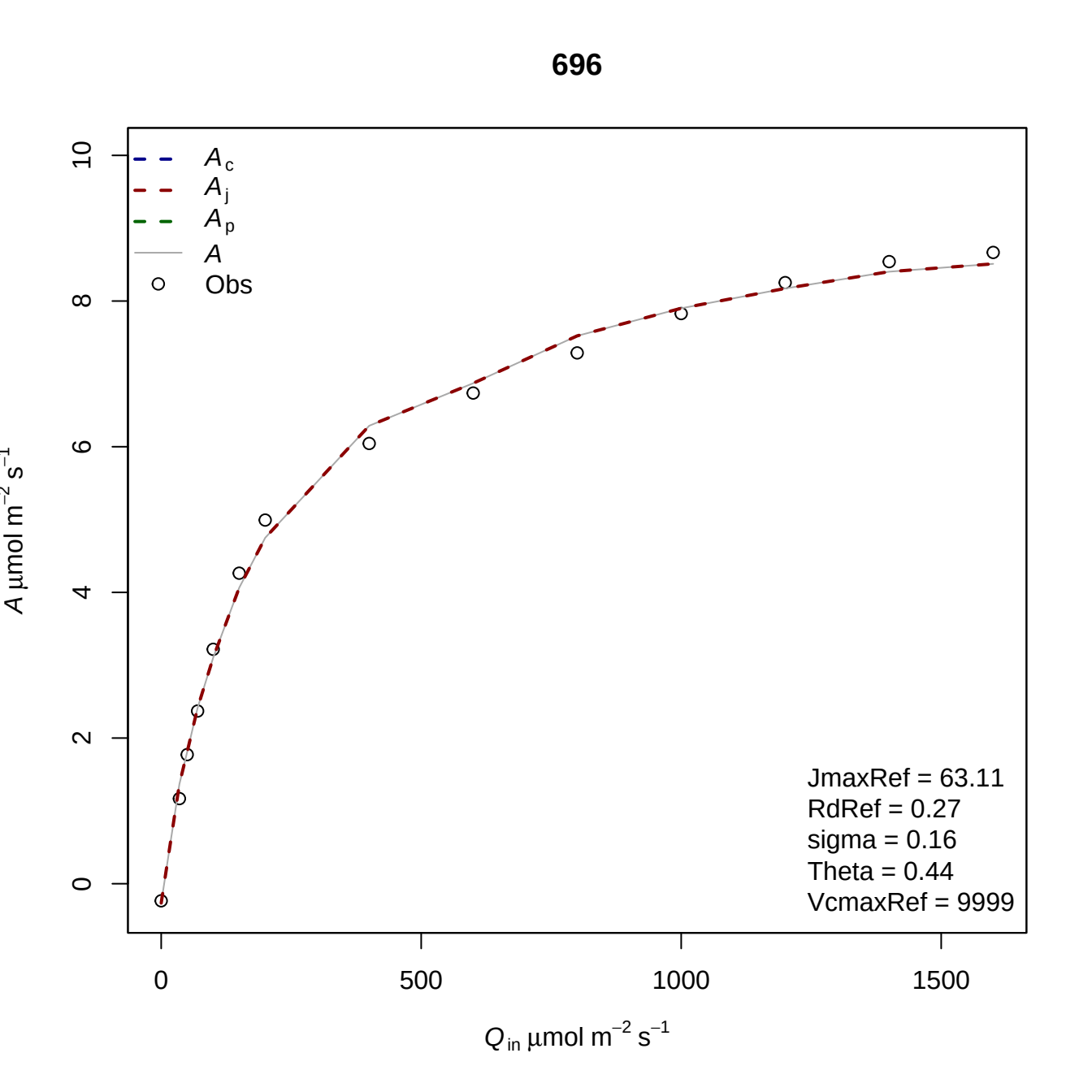

691
A_c
A_j
A_p
A
Obs
JmaxRef = 76.28
RdRef = 1.2
sigma = 0.12
Theta = 0.36
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

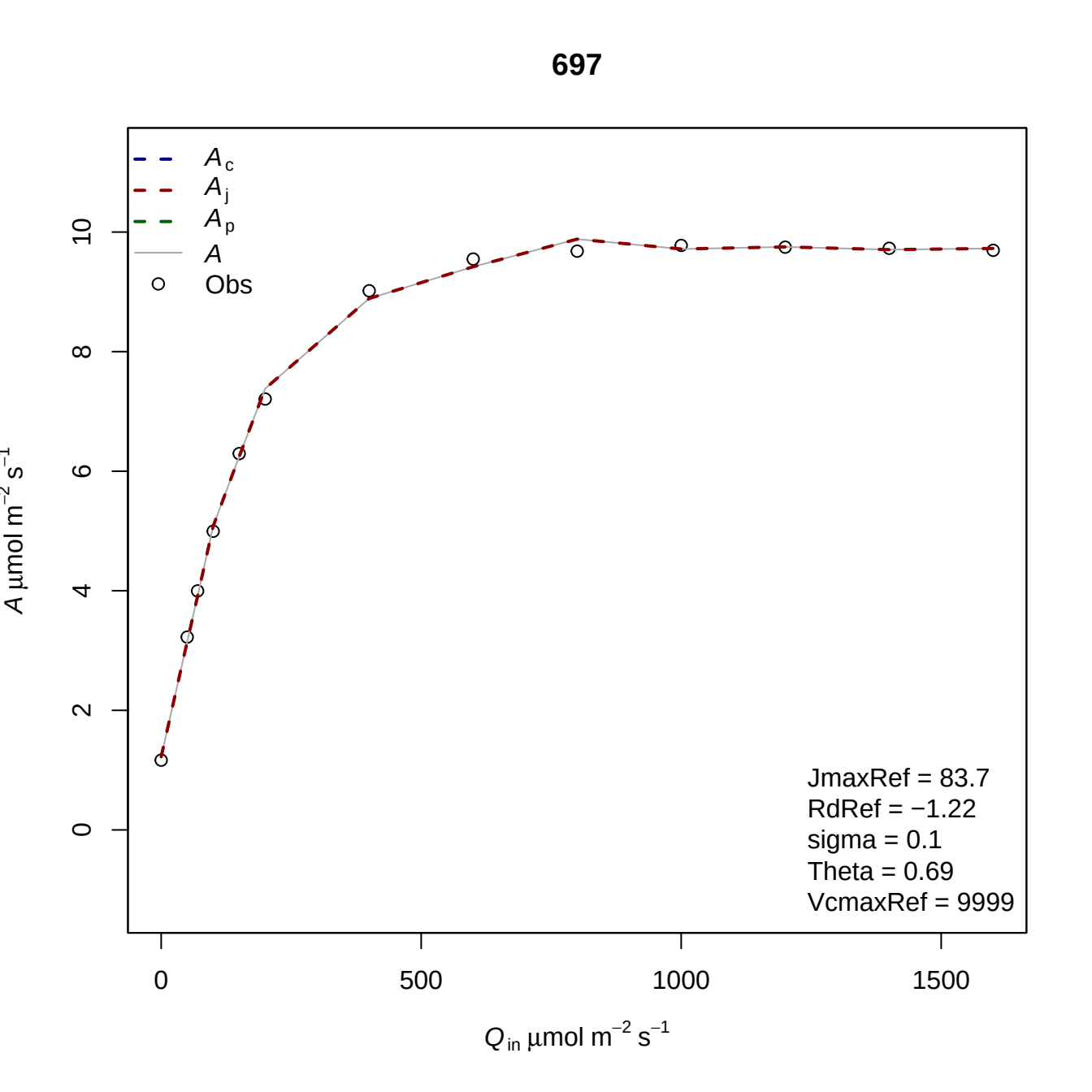

692
A_c
A_j
A_p
A
Obs
JmaxRef = 64.34
RdRef = 1.28
sigma = 0.07
Theta = 0.68
VcmaxRef = 51.47
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

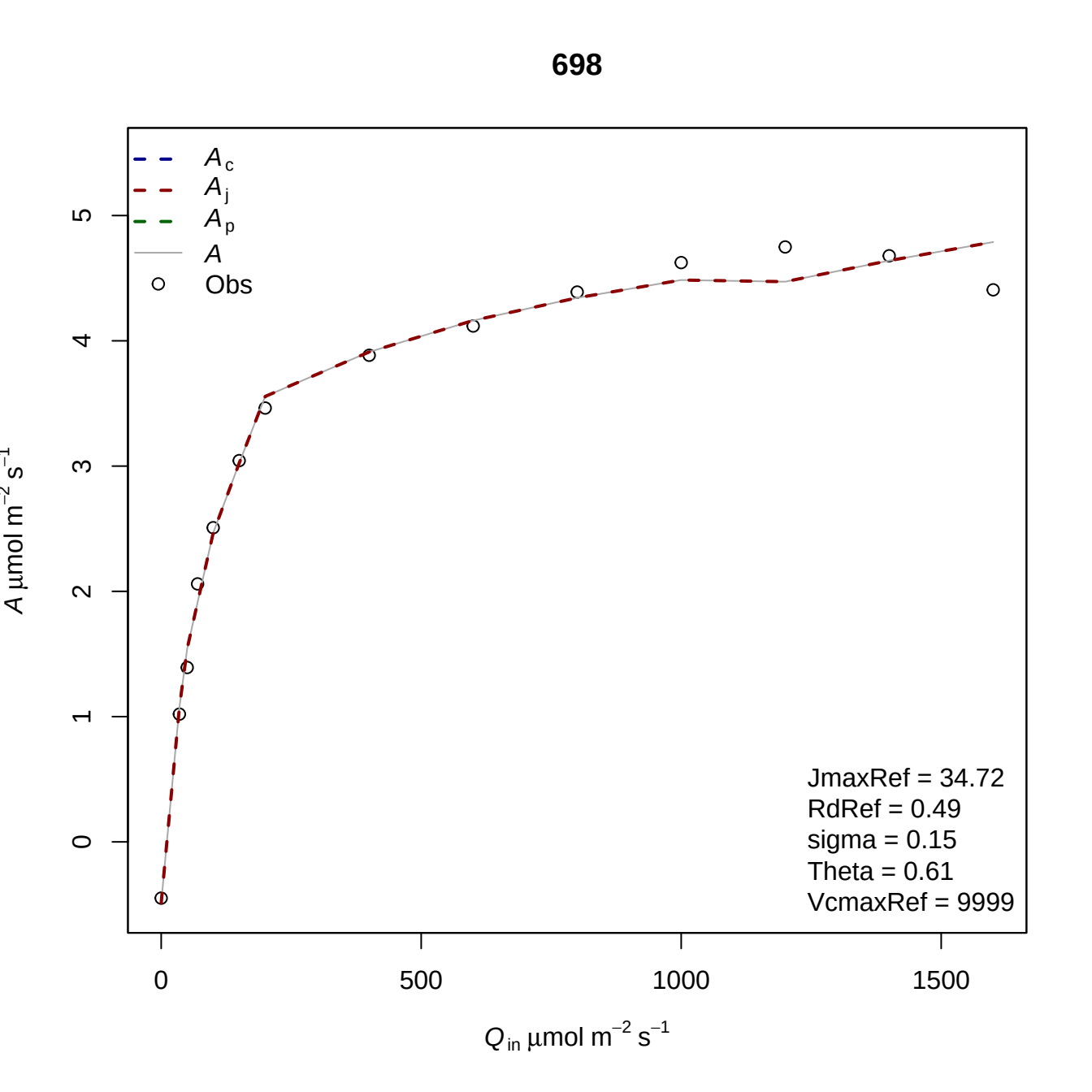

693
A_c
A_j
A_p
A
Obs
JmaxRef = 50.63
RdRef = 0.15
sigma = 0.26
Theta = −2.03
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

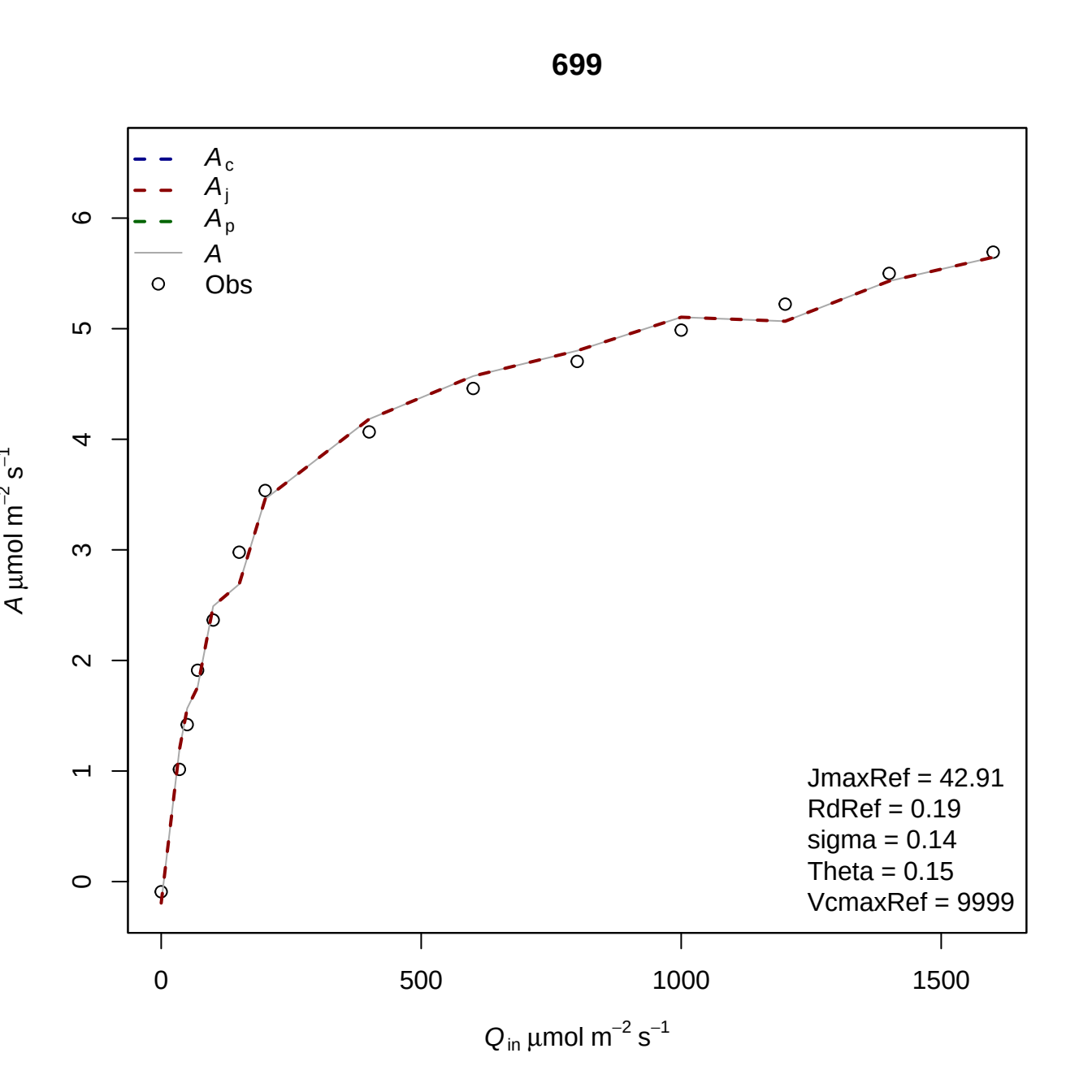

694
A_c
A_j
A_p
A
Obs
JmaxRef = 43.06
RdRef = 0.47
sigma = 0.14
Theta = 0.85
VcmaxRef = 32.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

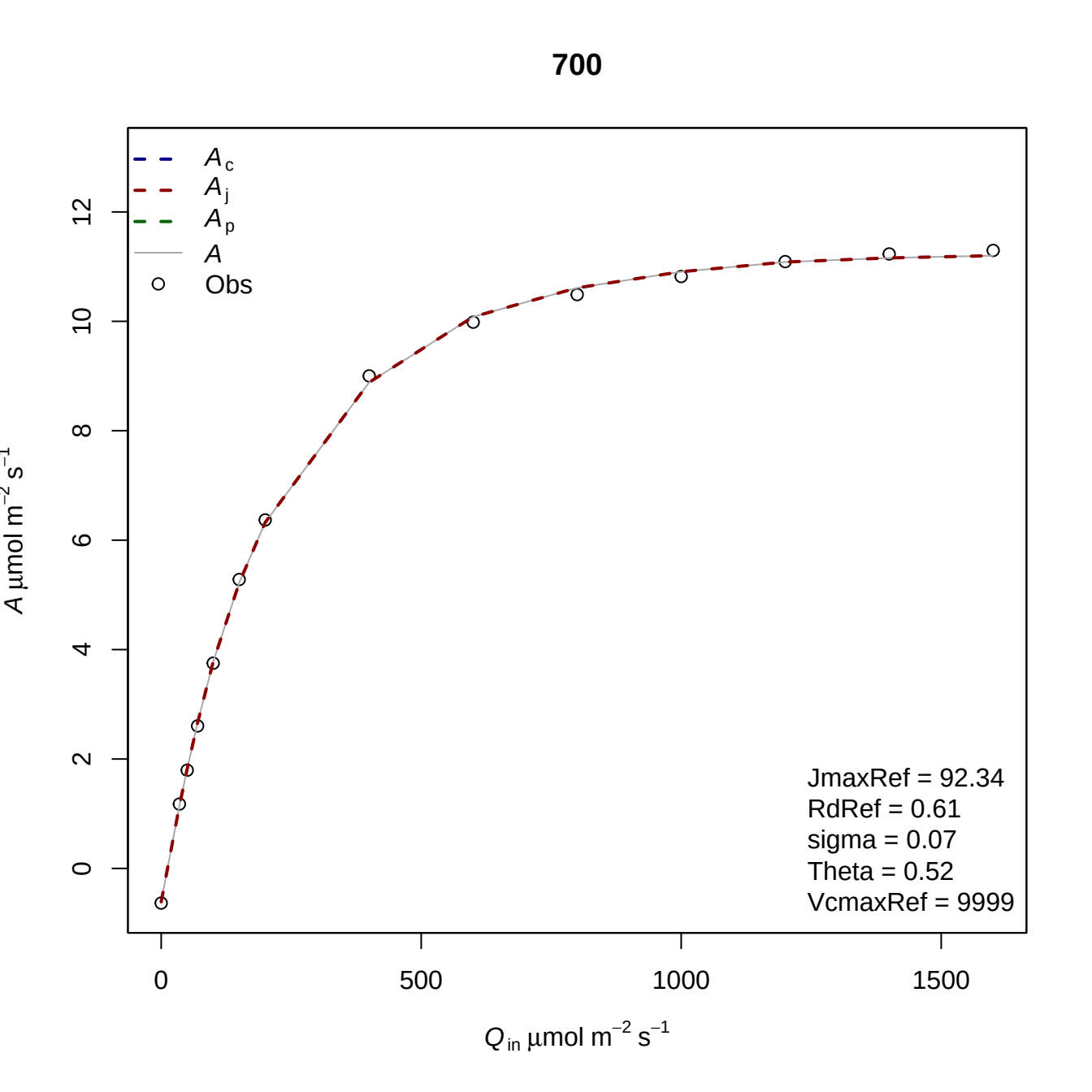

695
A_c
A_j
A_p
A
Obs
JmaxRef = 36.79
RdRef = 0.78
sigma = 0.21
Theta = 0.68
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

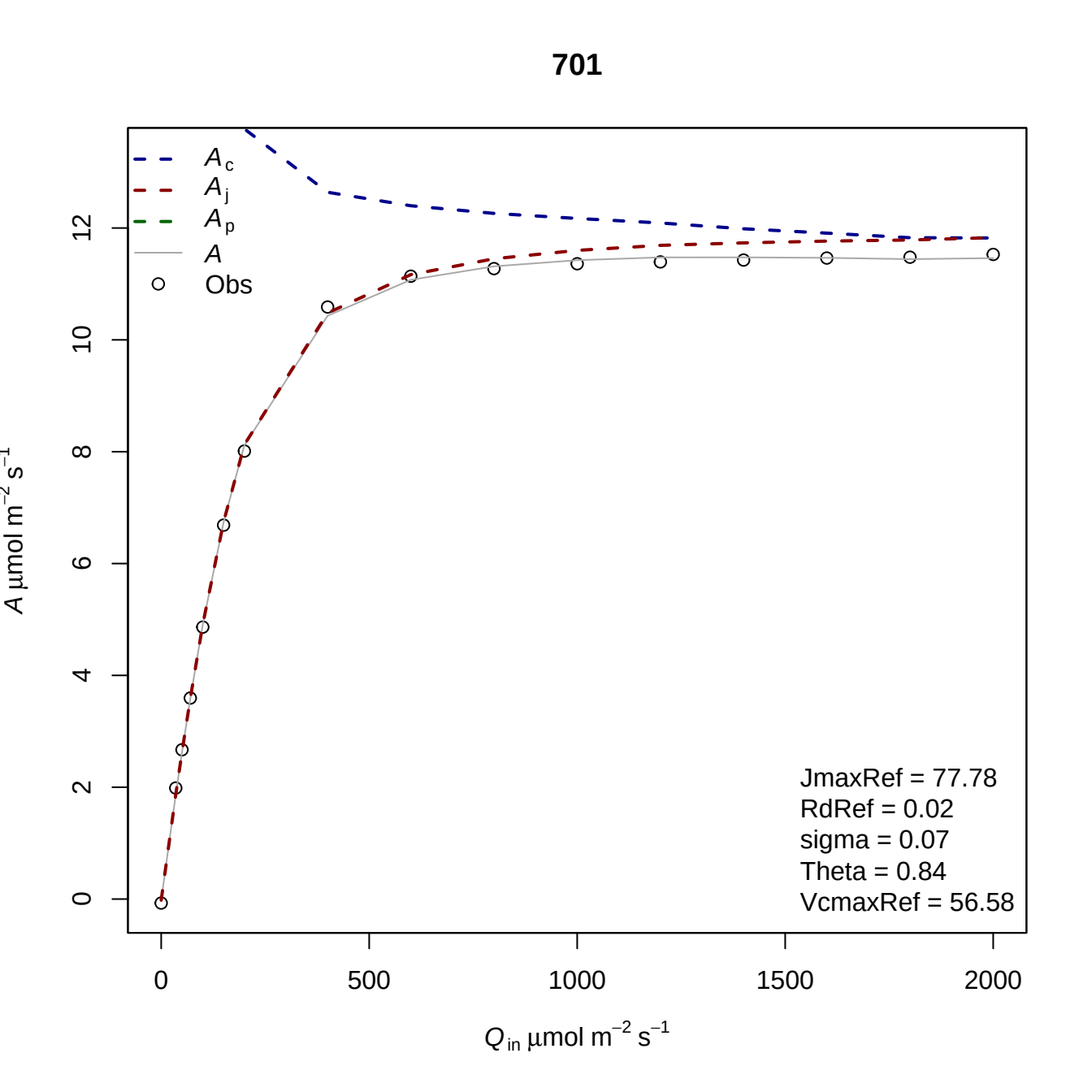

696
A_c
A_j
A_p
A
Obs
JmaxRef = 63.11
RdRef = 0.27
sigma = 0.16
Theta = 0.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

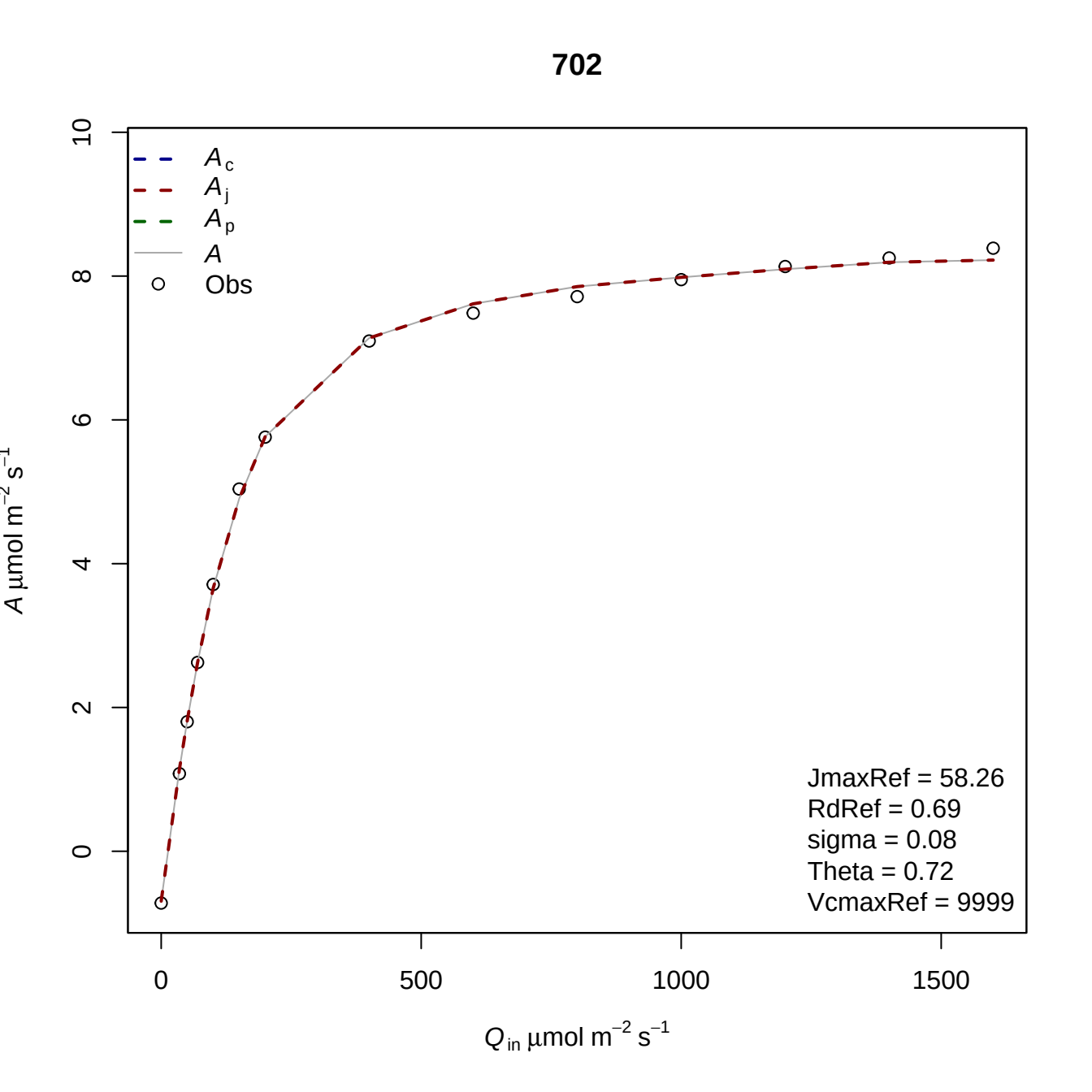

697
A_c
A_j
A_p
A
Obs
JmaxRef = 83.7
RdRef = −1.22
sigma = 0.1
Theta = 0.69
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

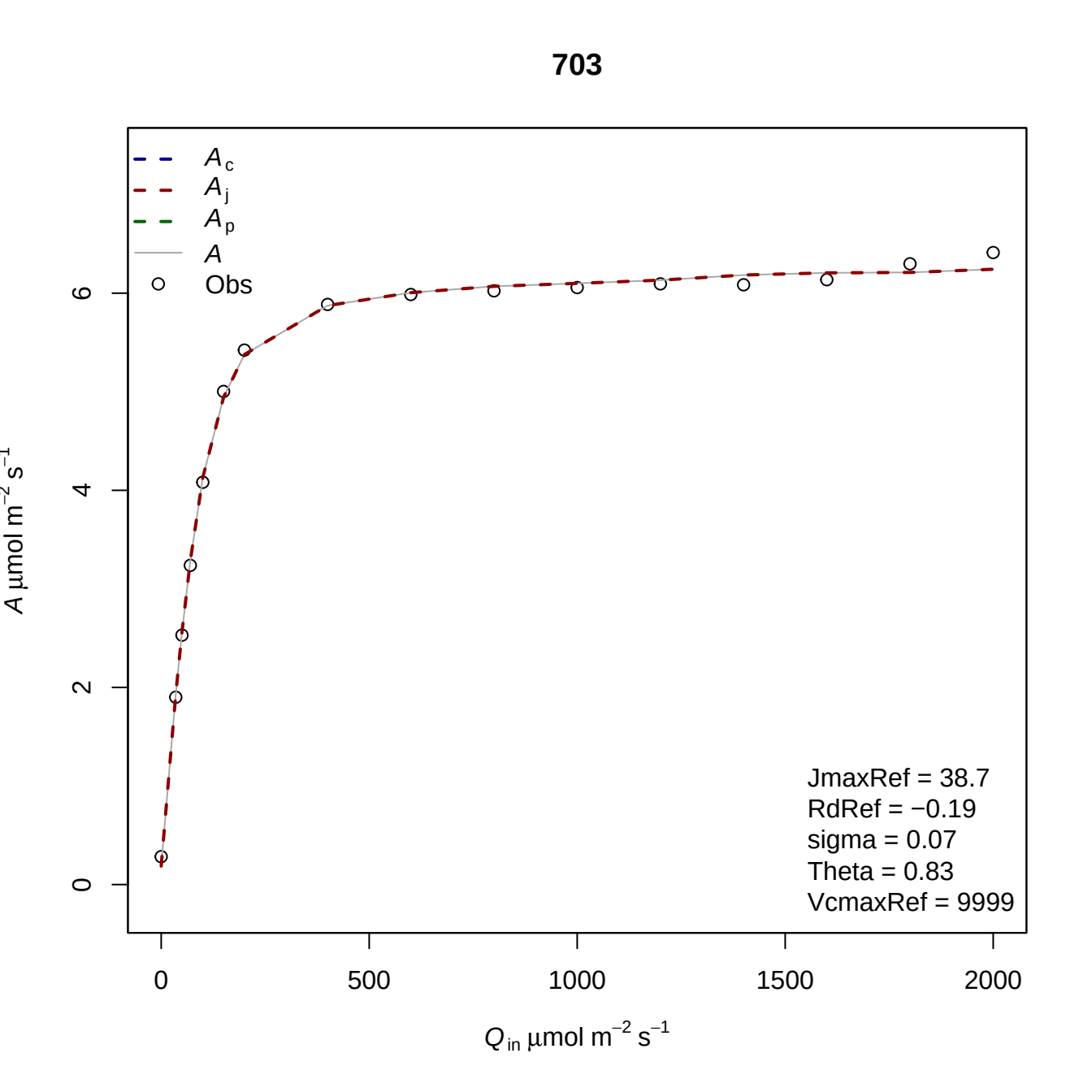

698
A_c
A_j
A_p
A
Obs
JmaxRef = 34.72
RdRef = 0.49
sigma = 0.15
Theta = 0.61
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

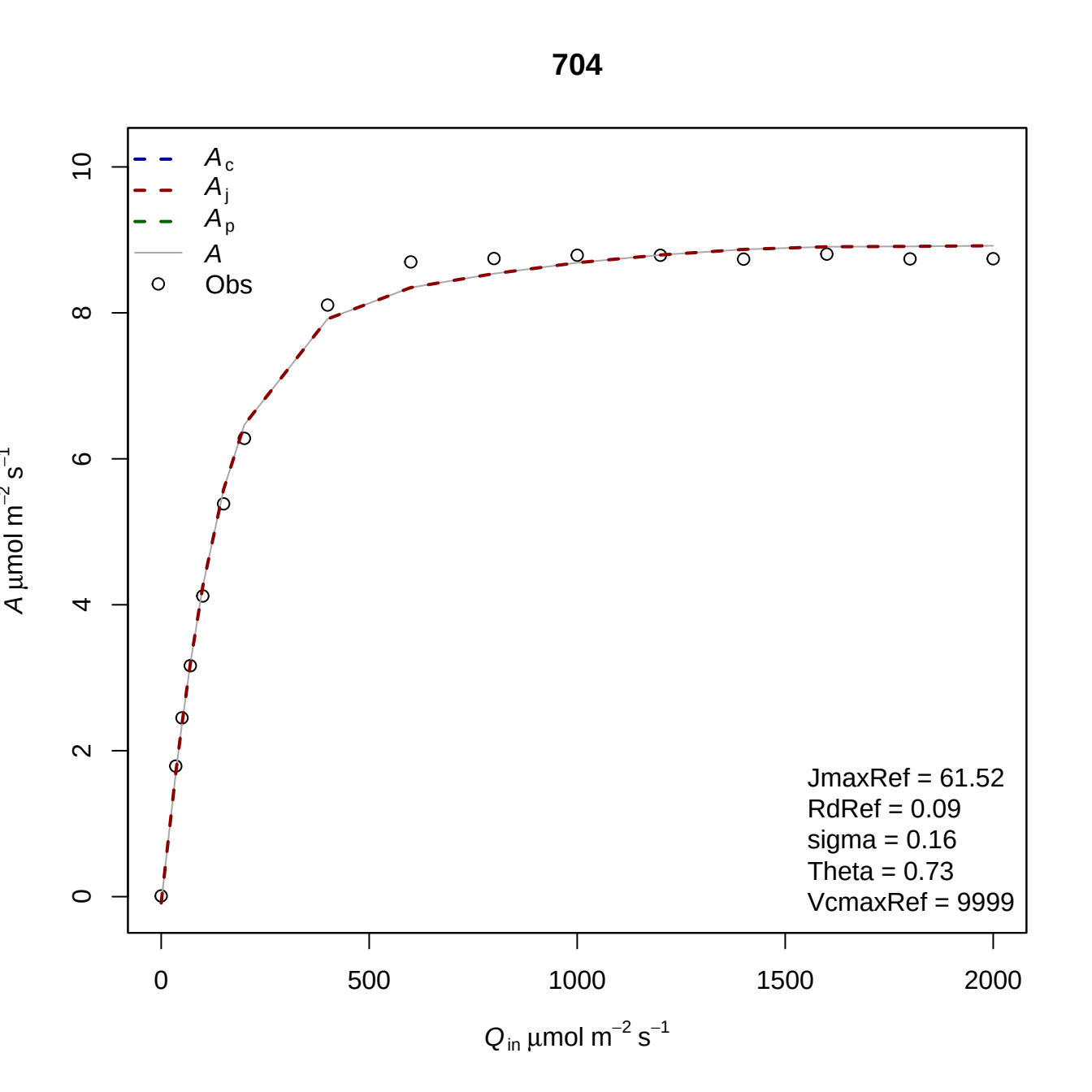

699
A_c
A_j
A_p
A
Obs
JmaxRef = 42.91
RdRef = 0.19
sigma = 0.14
Theta = 0.15
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

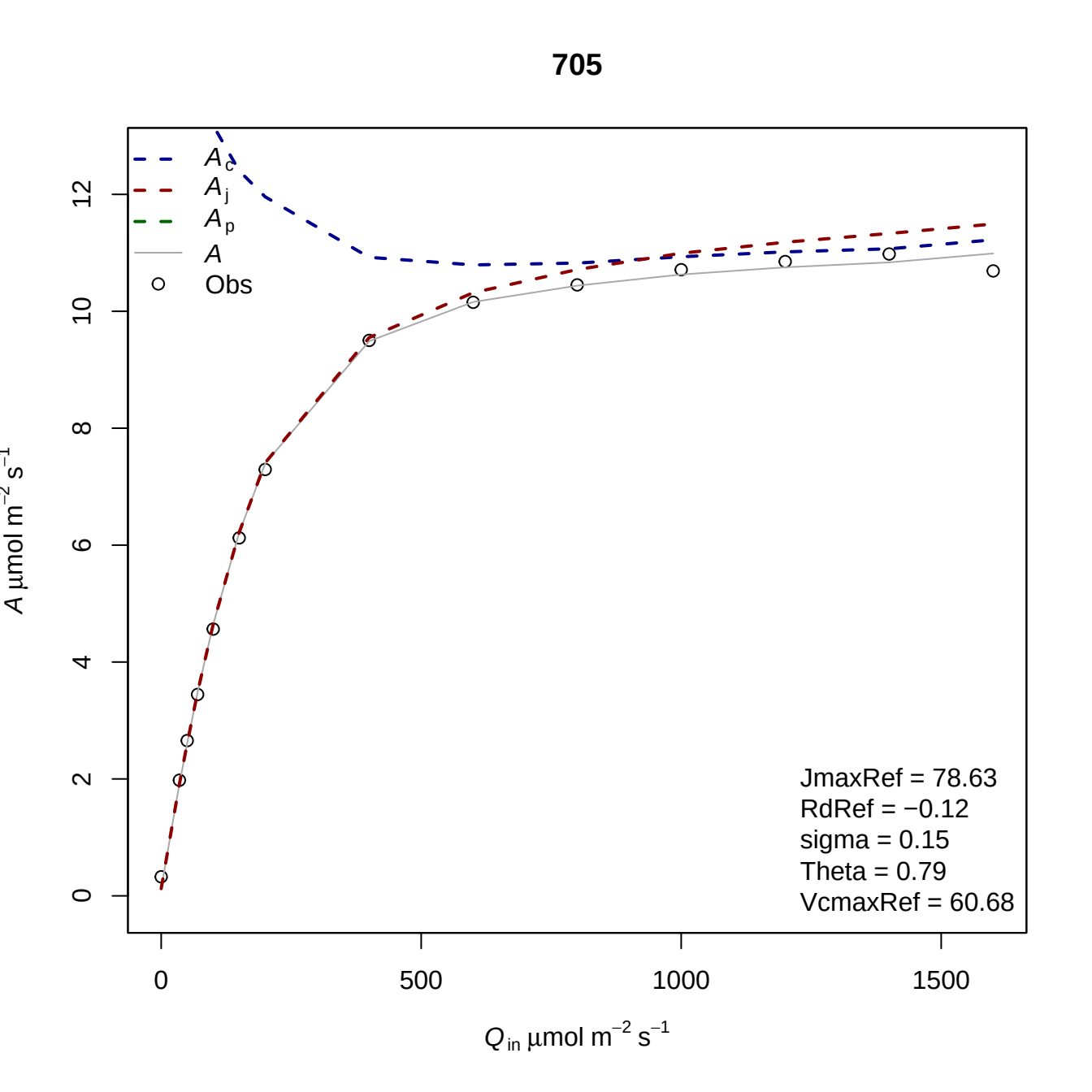

700
A_c
A_j
A_p
A
Obs
JmaxRef = 92.34
RdRef = 0.61
sigma = 0.07
Theta = 0.52
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

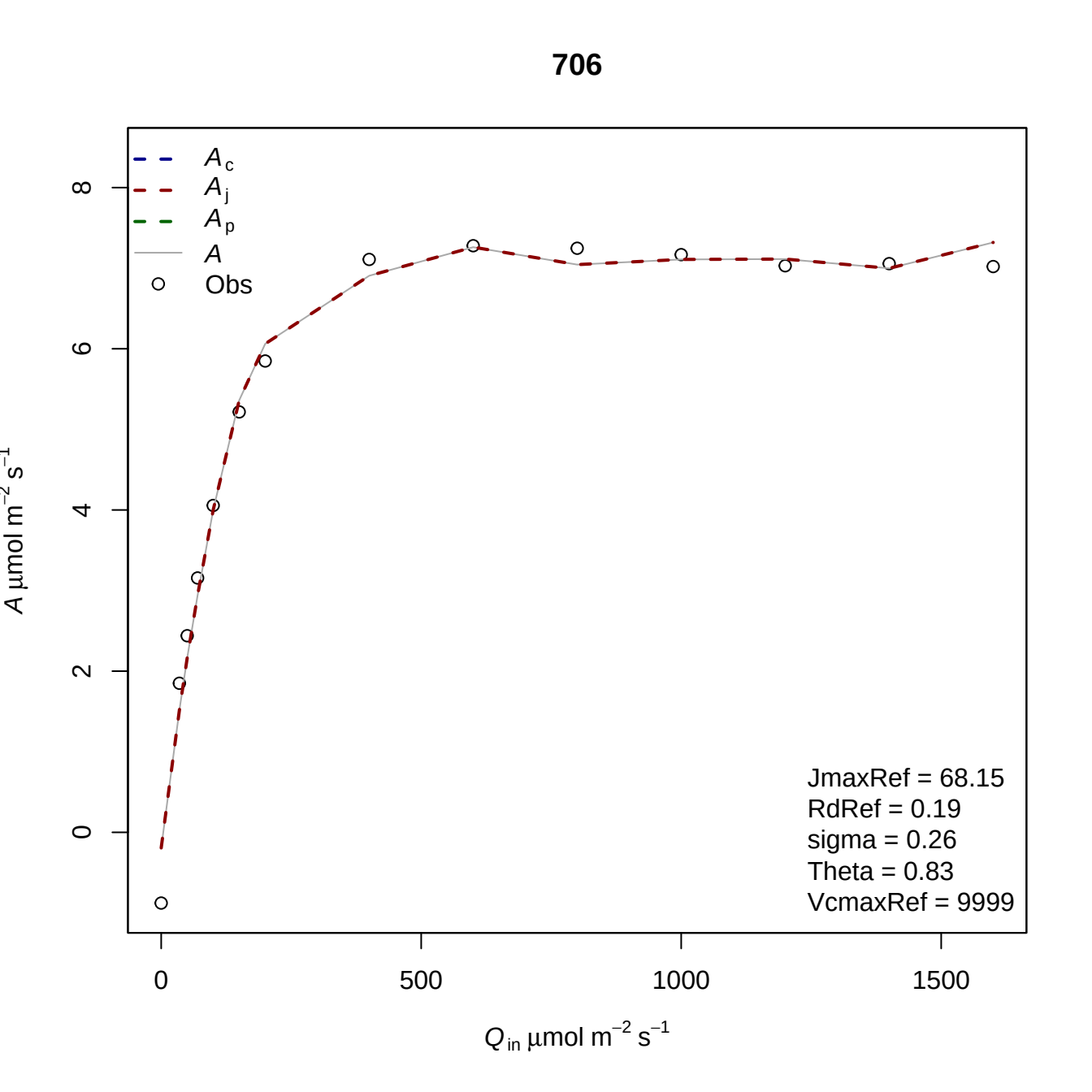

701
A_c
A_j
A_p
A
Obs
JmaxRef = 77.78
RdRef = 0.02
sigma = 0.07
Theta = 0.84
VcmaxRef = 56.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

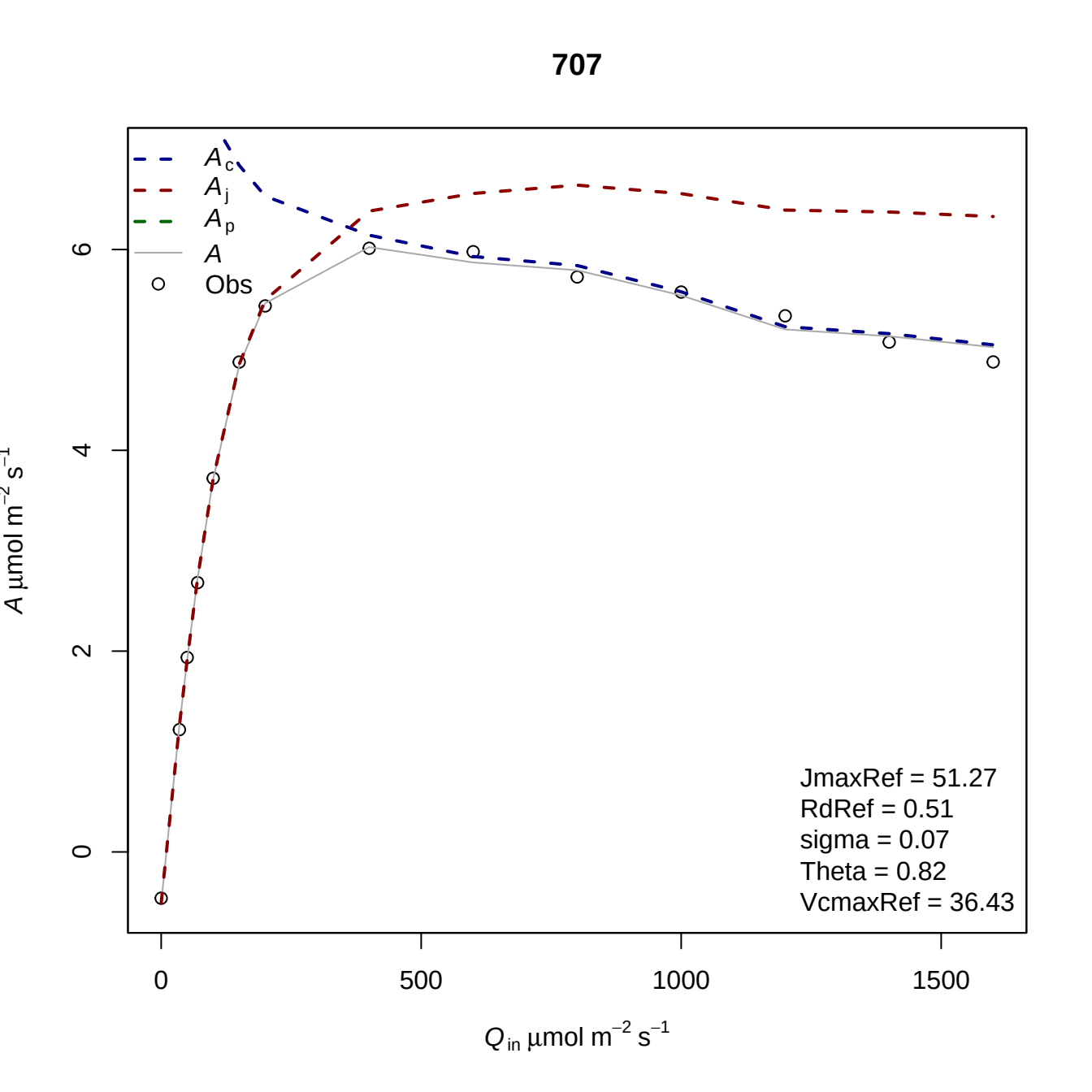

702
A_c
A_j
A_p
A
Obs
JmaxRef = 58.26
RdRef = 0.69
sigma = 0.08
Theta = 0.72
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

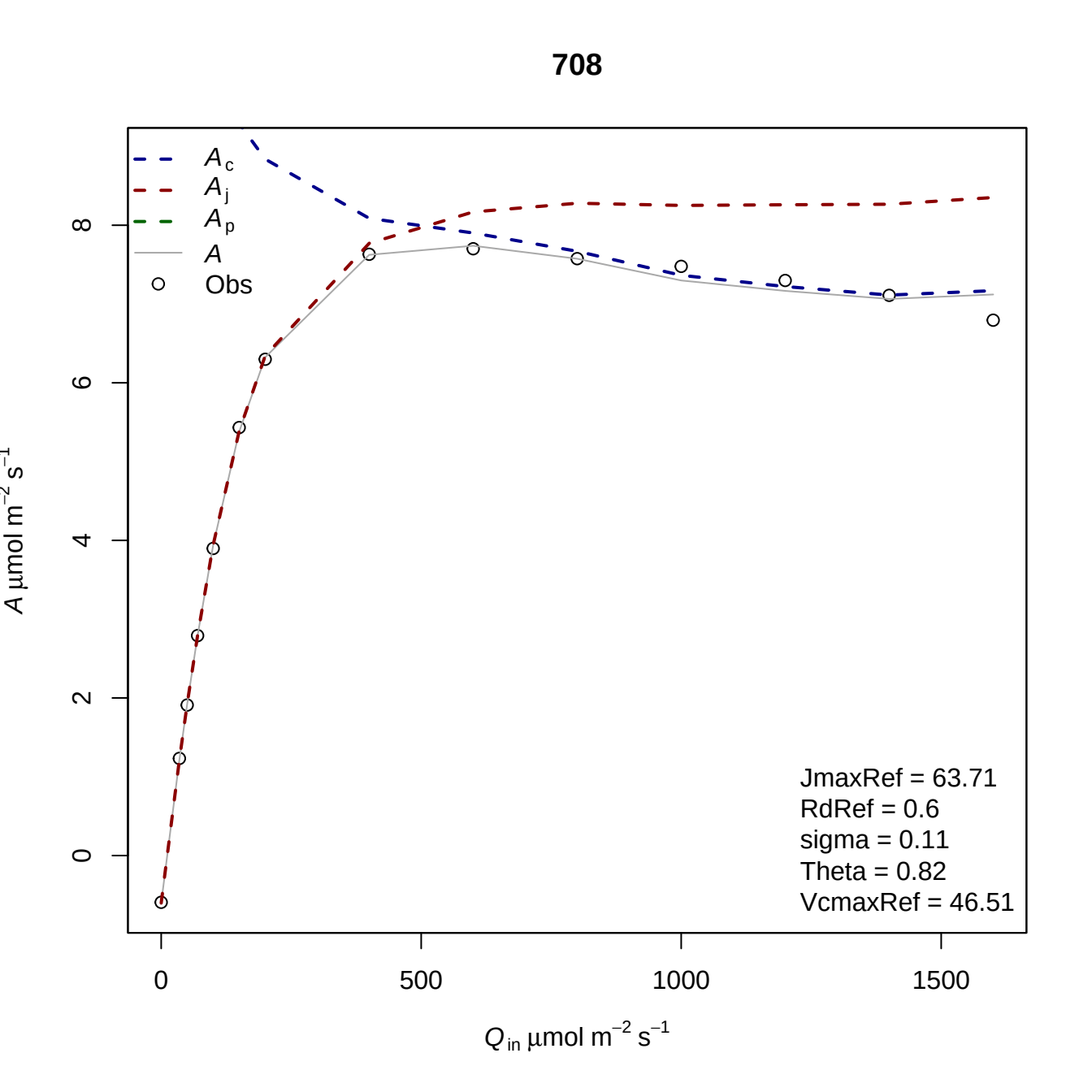

703
A_c
A_j
A_p
A
Obs
JmaxRef = 38.7
RdRef = −0.19
sigma = 0.07
Theta = 0.83
VcmaxRef = 9999
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

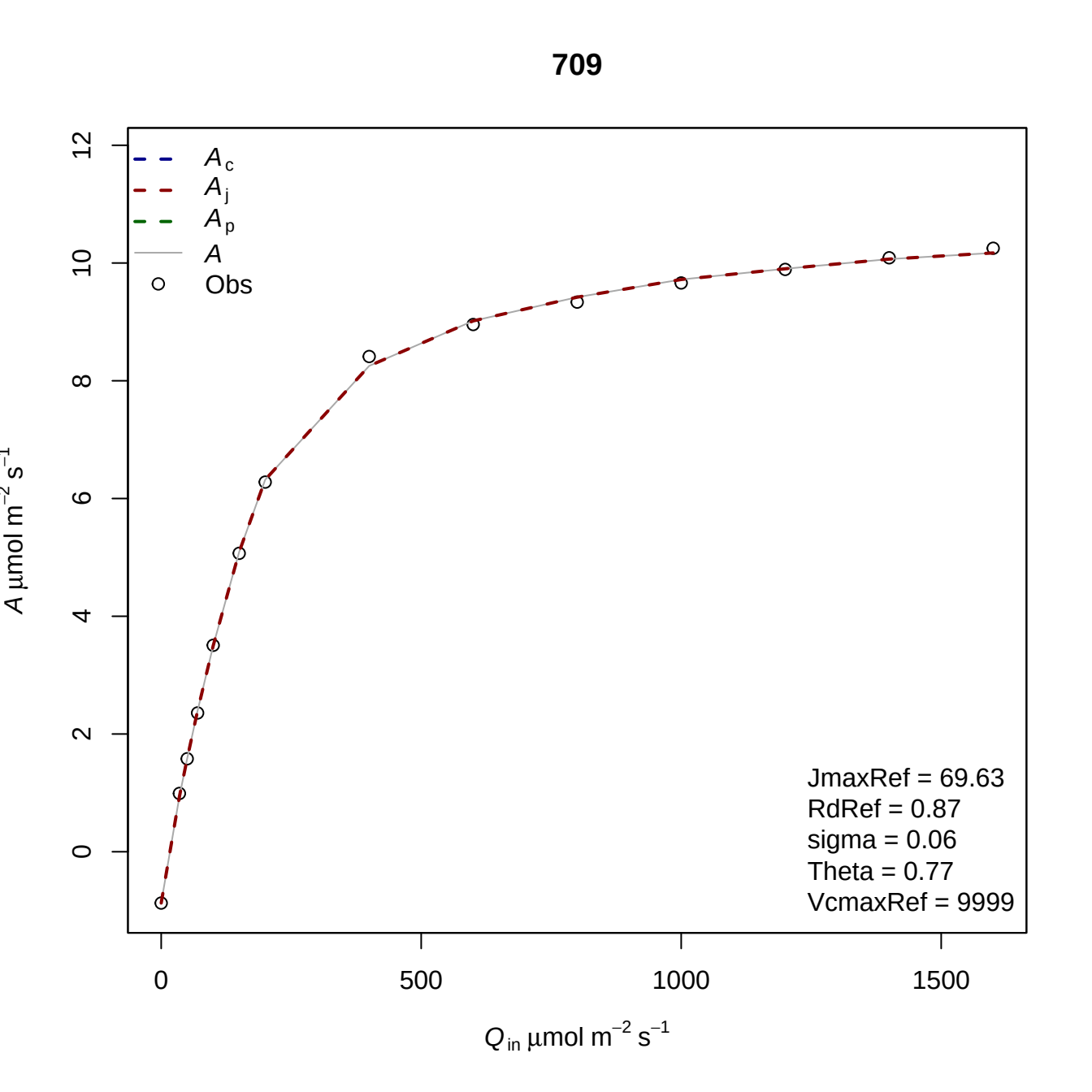

704
A_c
A_j
A_p
A
Obs
JmaxRef = 61.52
RdRef = 0.09
sigma = 0.16
Theta = 0.73
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

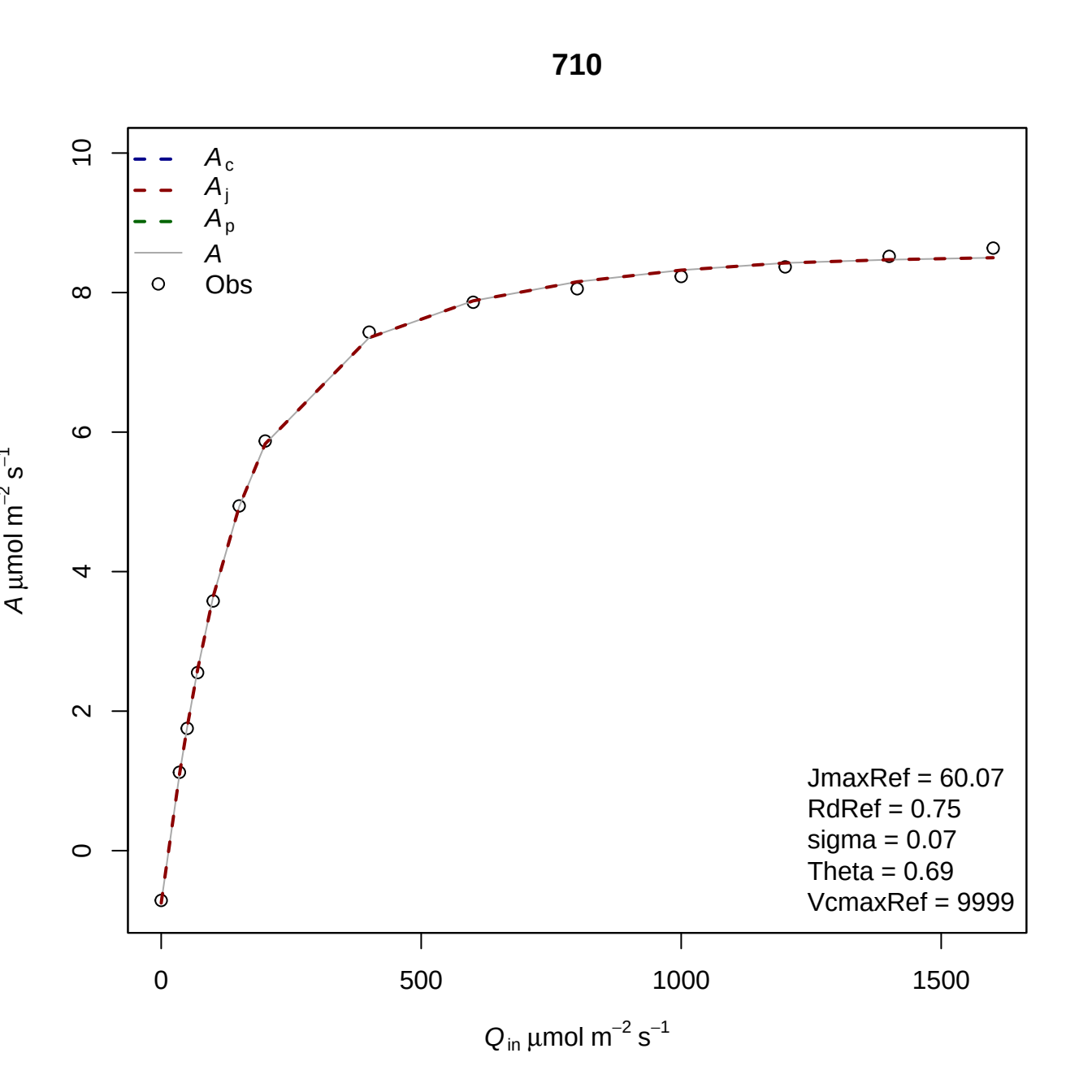

705
A_c
A_j
A_p
A
Obs
JmaxRef = 78.63
RdRef = −0.12
sigma = 0.15
Theta = 0.79
VcmaxRef = 60.68
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

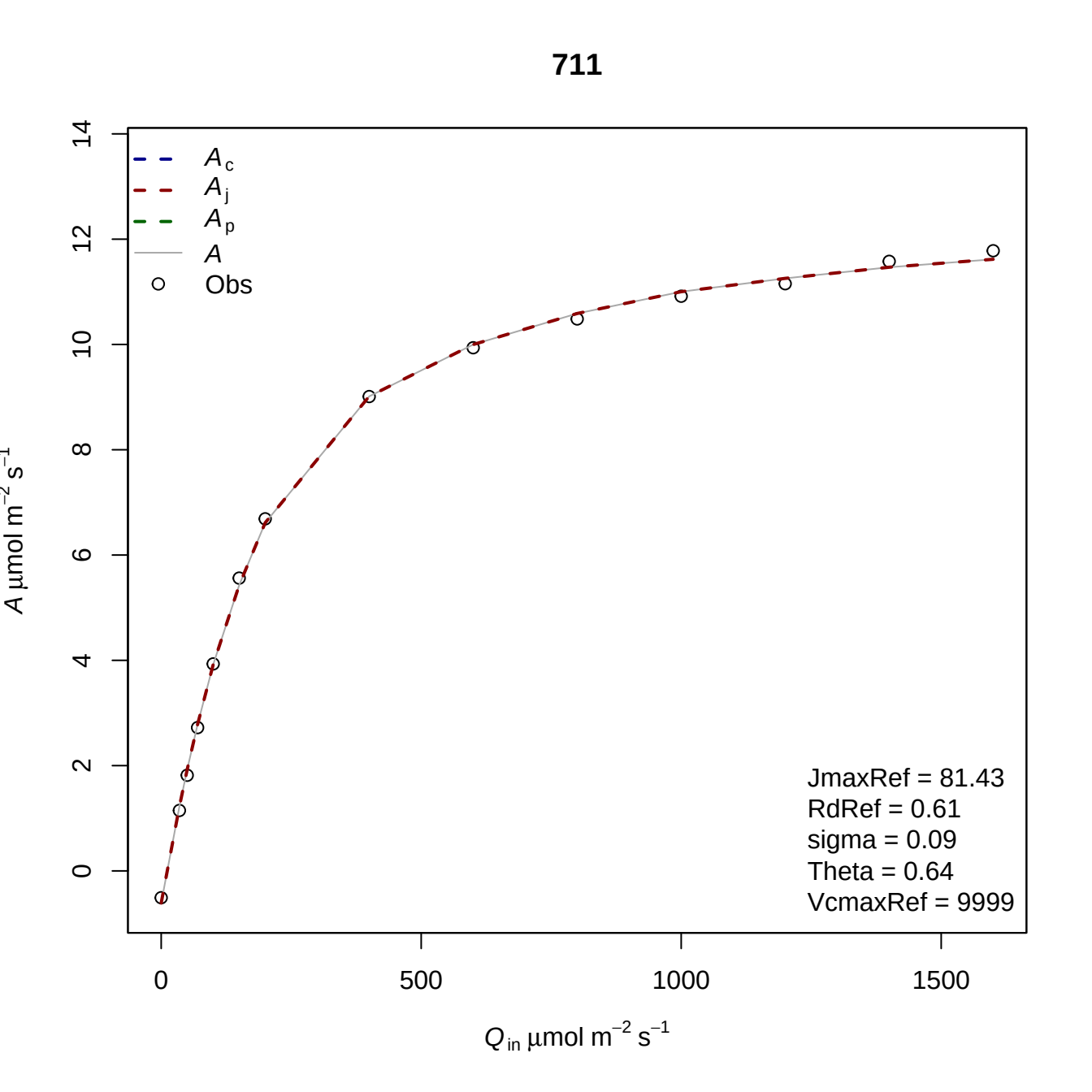

706
A_c
A_j
A_p
A
Obs
JmaxRef = 68.15
RdRef = 0.19
sigma = 0.26
Theta = 0.83
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

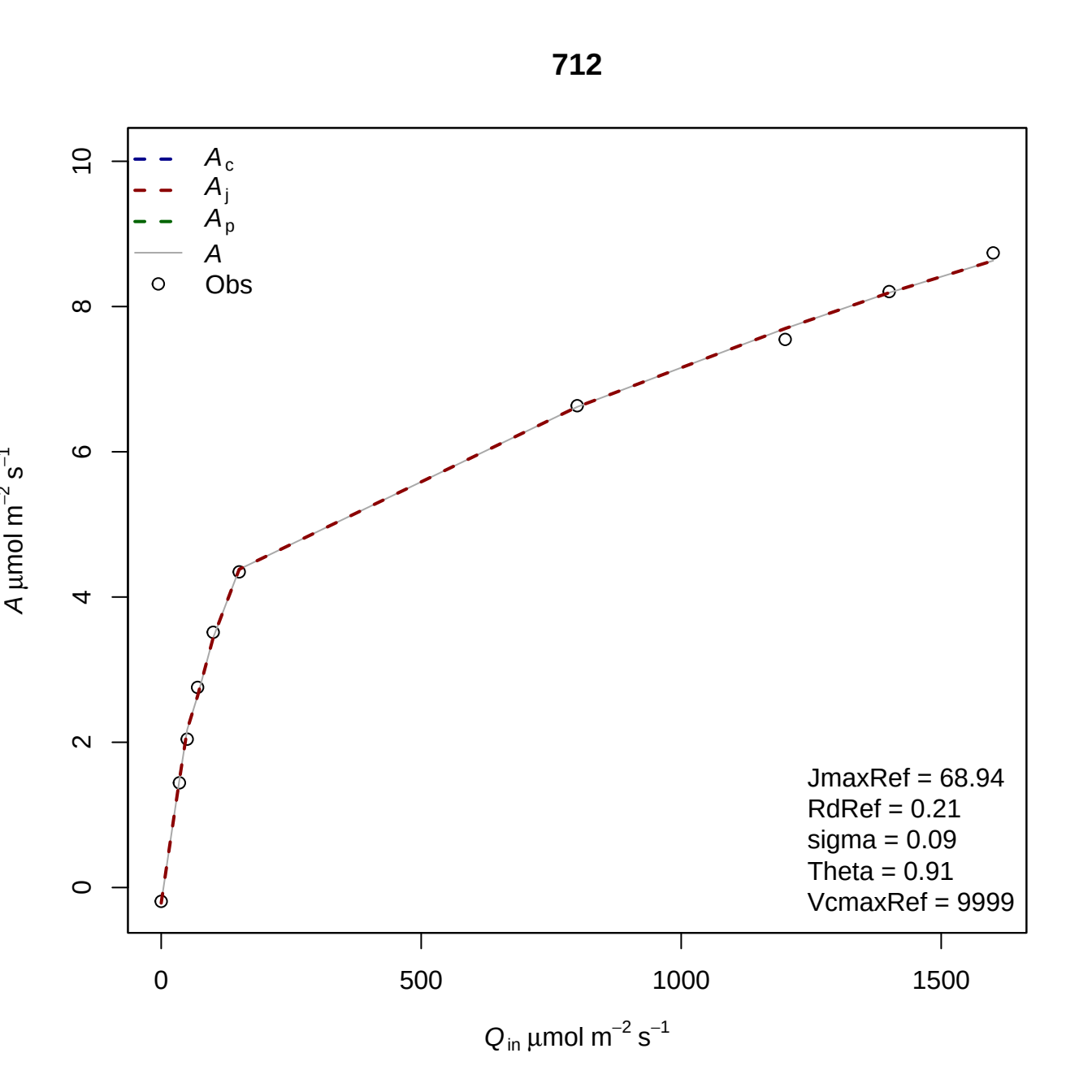

707
A_c
A_j
A_p
A
Obs
JmaxRef = 51.27
RdRef = 0.51
sigma = 0.07
Theta = 0.82
VcmaxRef = 36.43
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

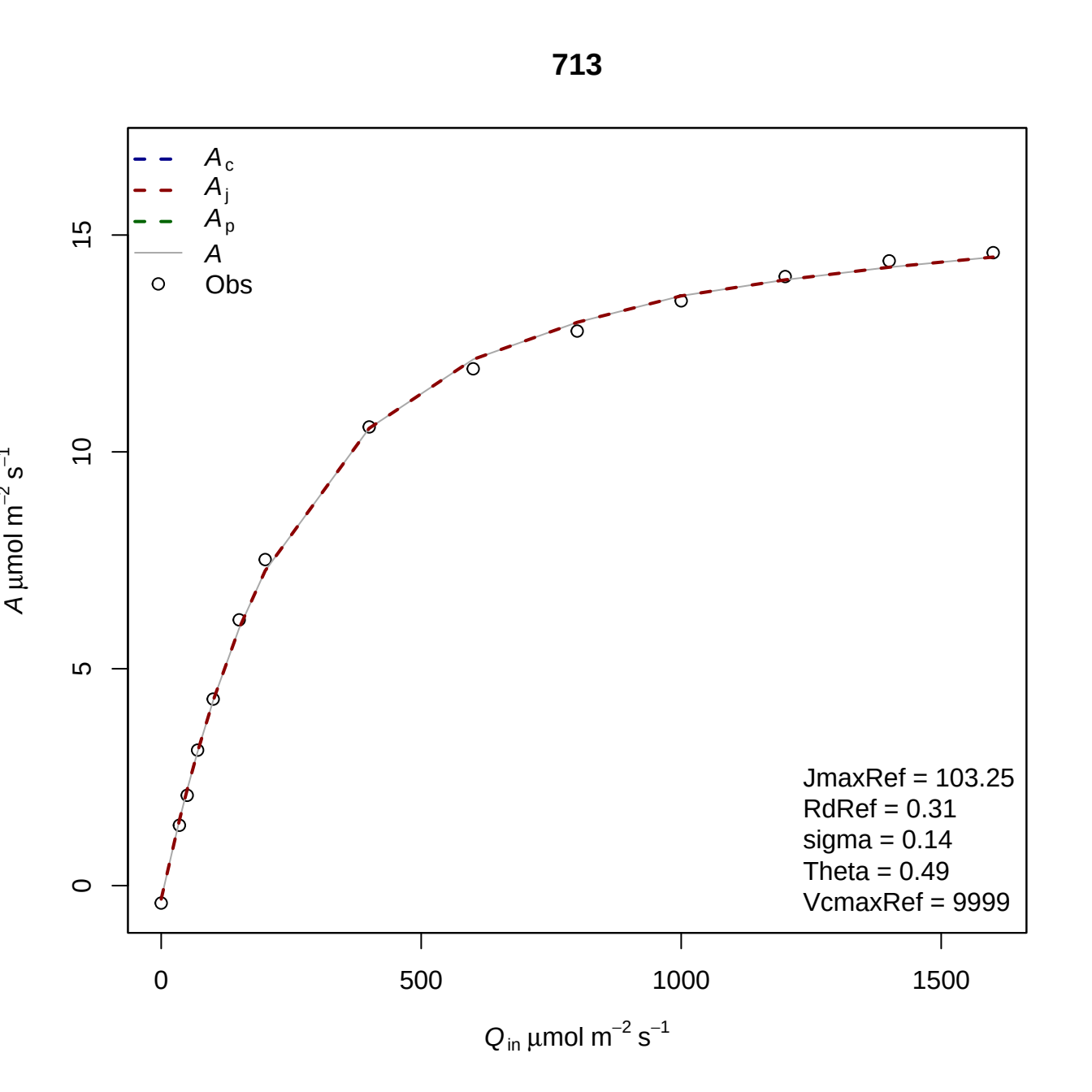

708
A_c
A_j
A_p
A
Obs
JmaxRef = 63.71
RdRef = 0.6
sigma = 0.11
Theta = 0.82
VcmaxRef = 46.51
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

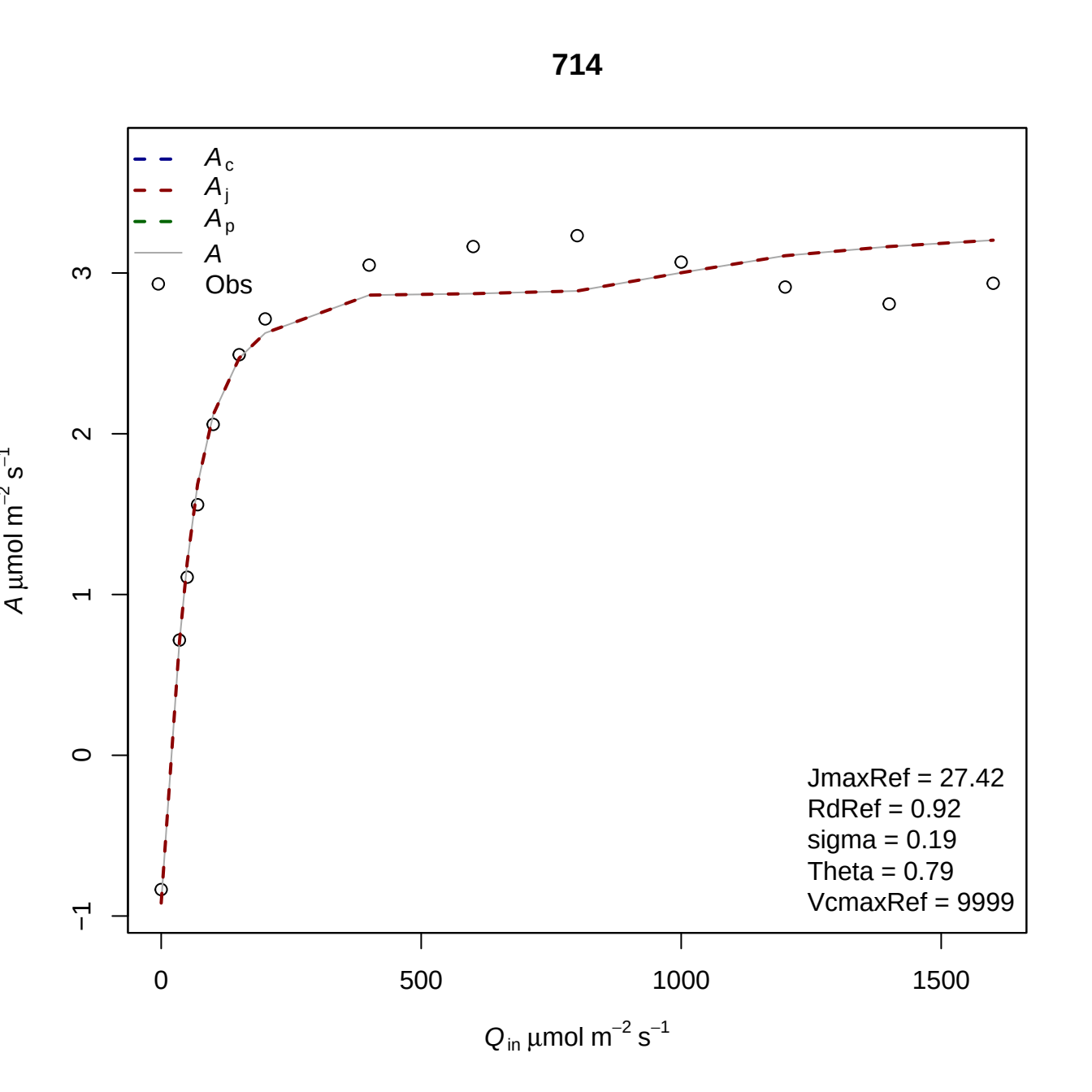

709
A_c
A_j
A_p
A
Obs
JmaxRef = 69.63
RdRef = 0.87
sigma = 0.06
Theta = 0.77
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

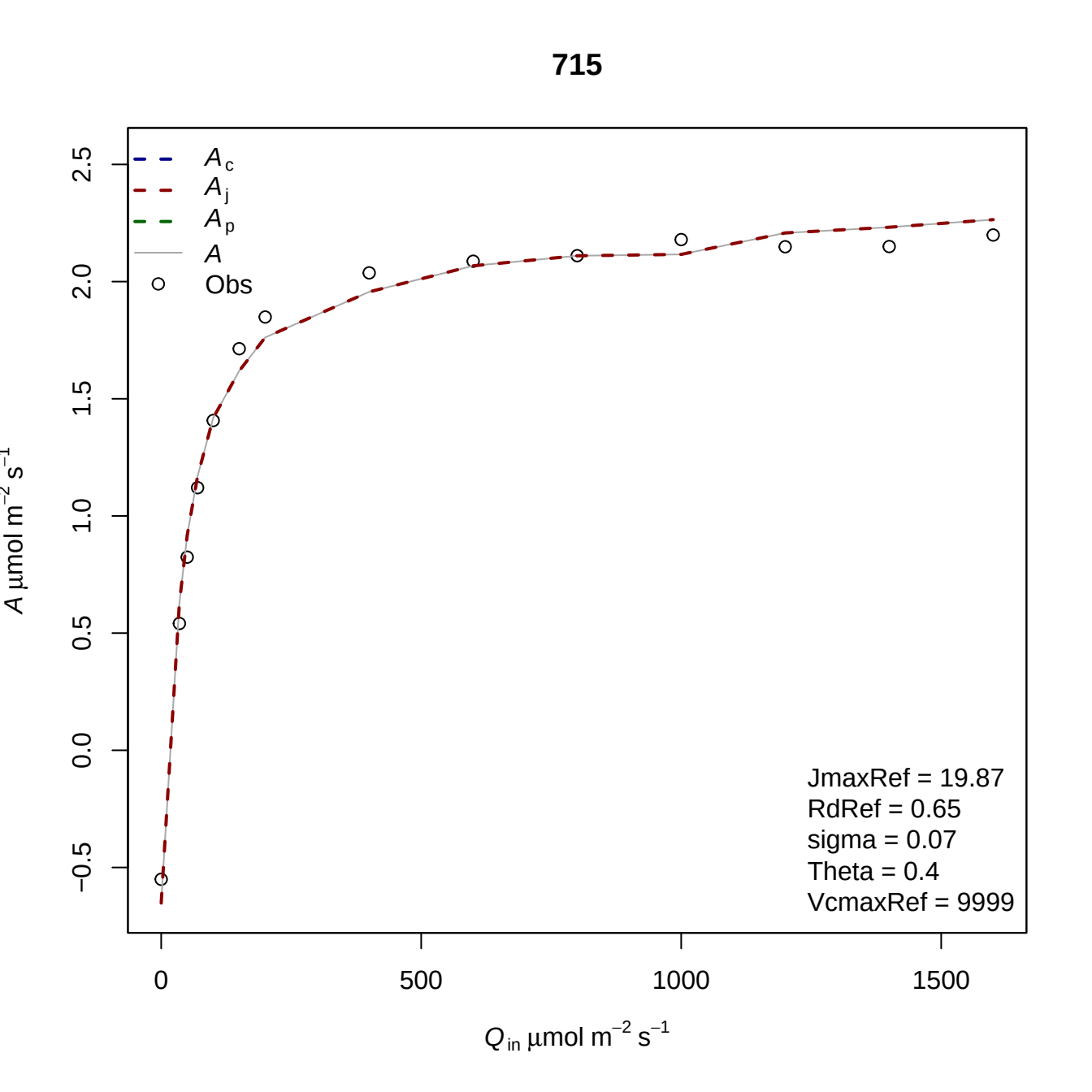

710
A_c
A_j
A_p
A
Obs
JmaxRef = 60.07
RdRef = 0.75
sigma = 0.07
Theta = 0.69
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

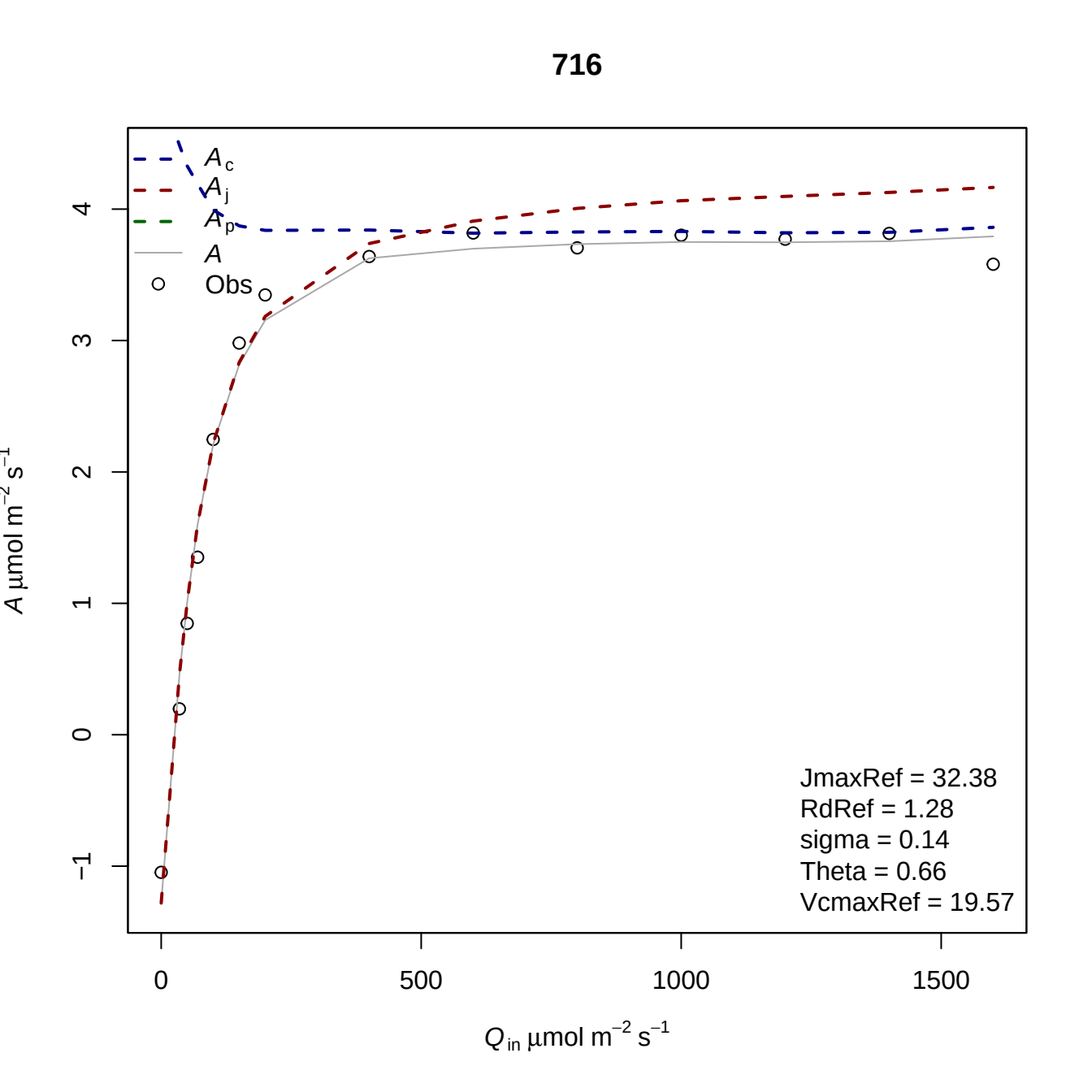

711
A_c
A_j
A_p
A
Obs
JmaxRef = 81.43
RdRef = 0.61
sigma = 0.09
Theta = 0.64
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

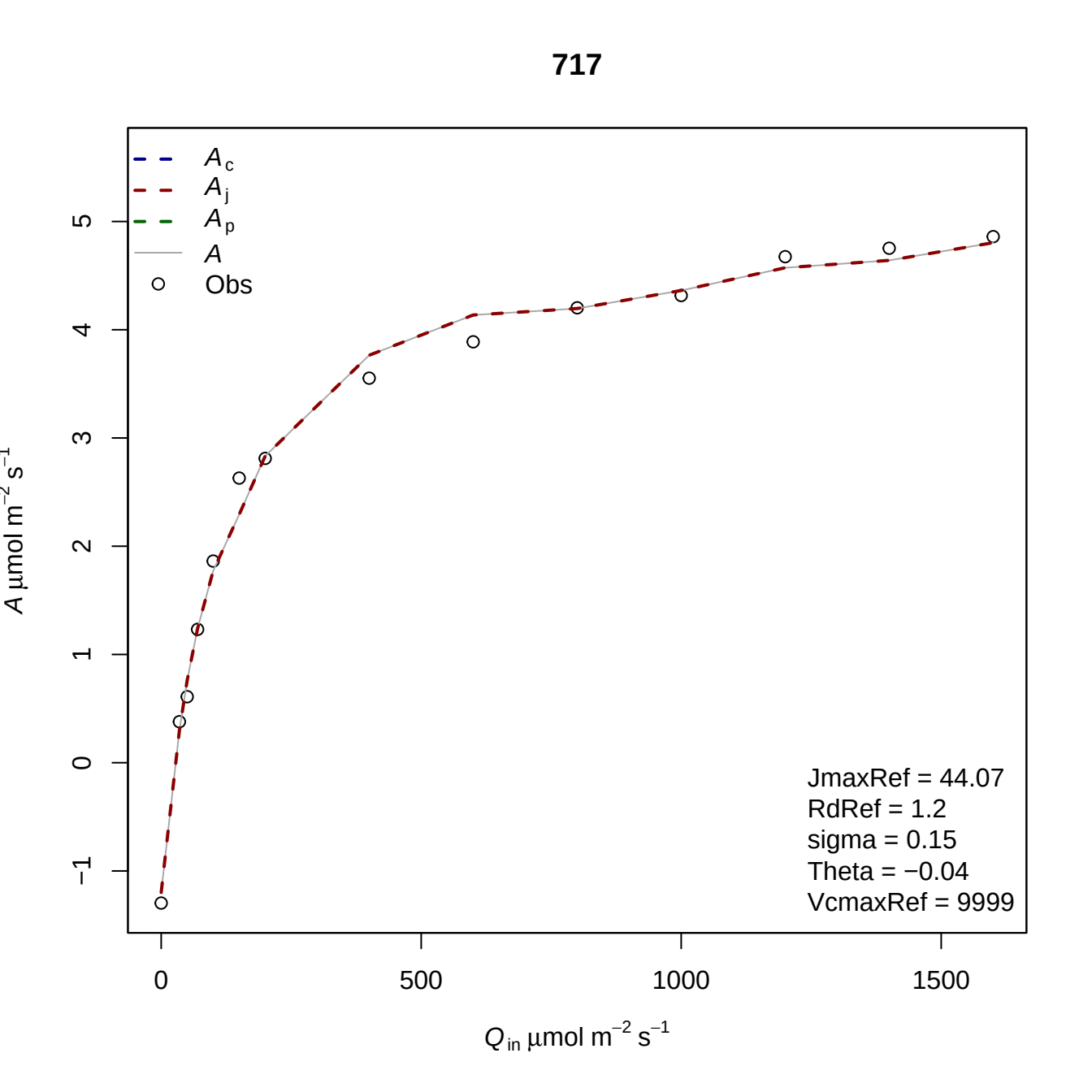

712
A_c
A_j
A_p
A
Obs
JmaxRef = 68.94
RdRef = 0.21
sigma = 0.09
Theta = 0.91
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

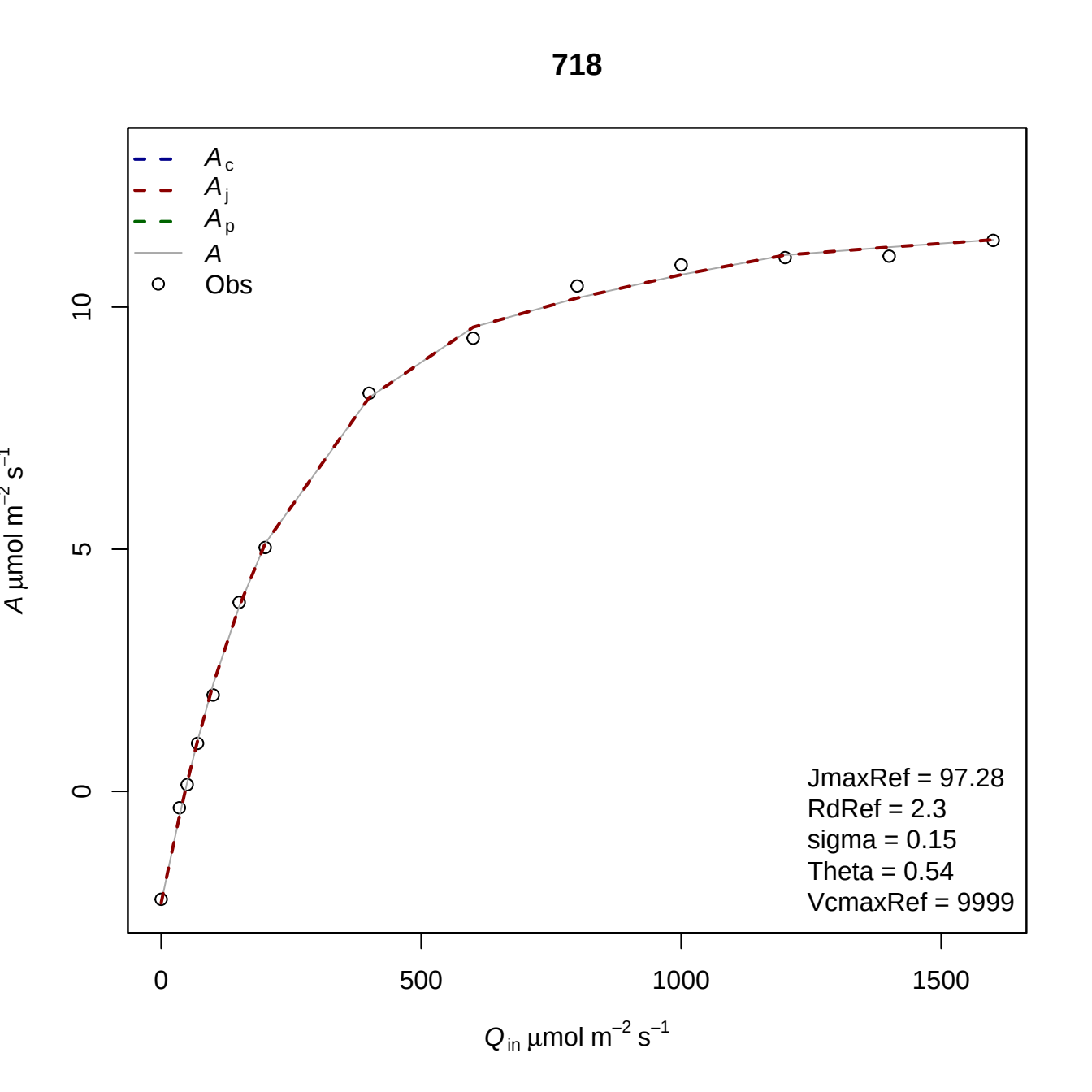

713
A_c
A_j
A_p
A
Obs
JmaxRef = 103.25
RdRef = 0.31
sigma = 0.14
Theta = 0.49
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

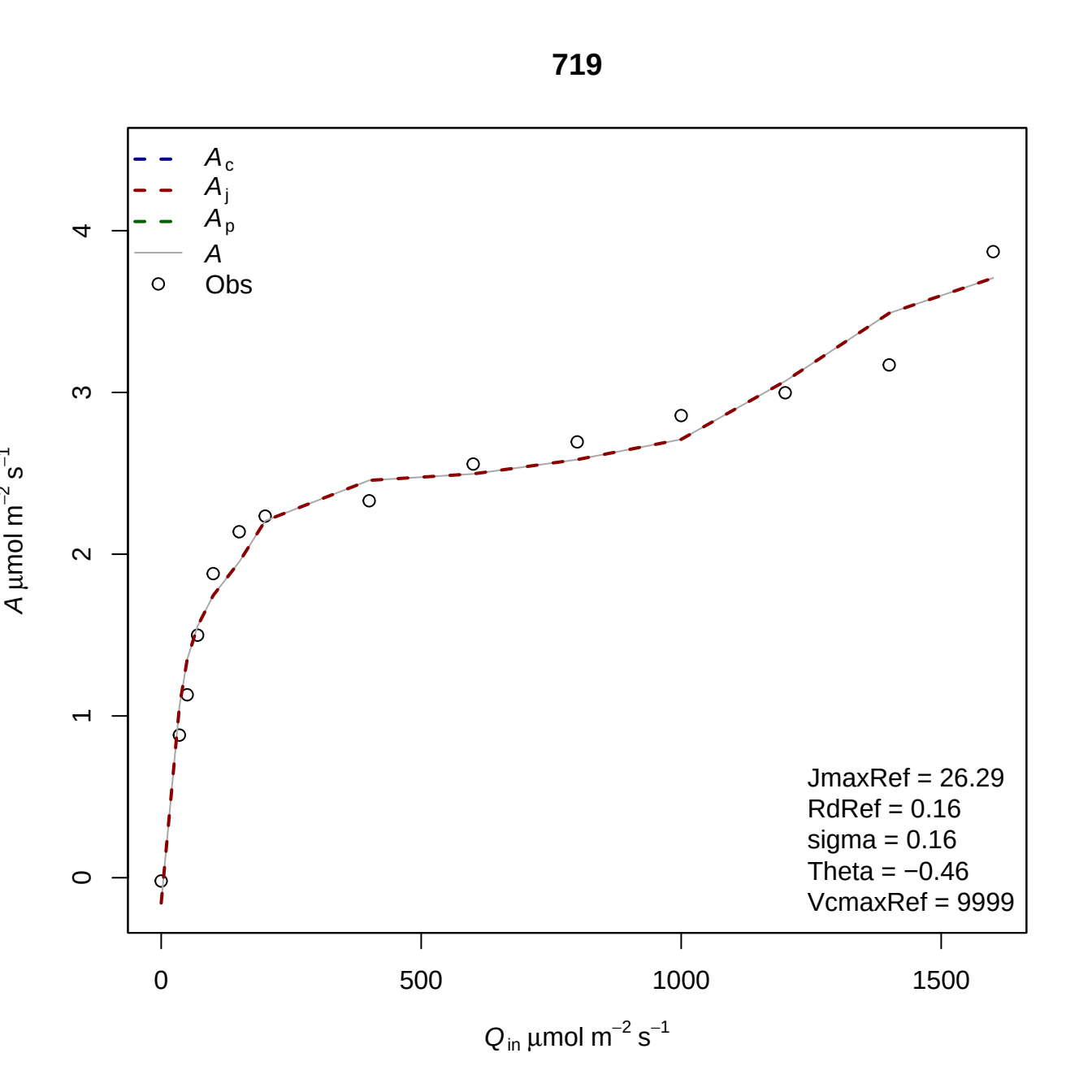

714
A_c
A_j
A_p
A
Obs
JmaxRef = 27.42
RdRef = 0.92
sigma = 0.19
Theta = 0.79
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

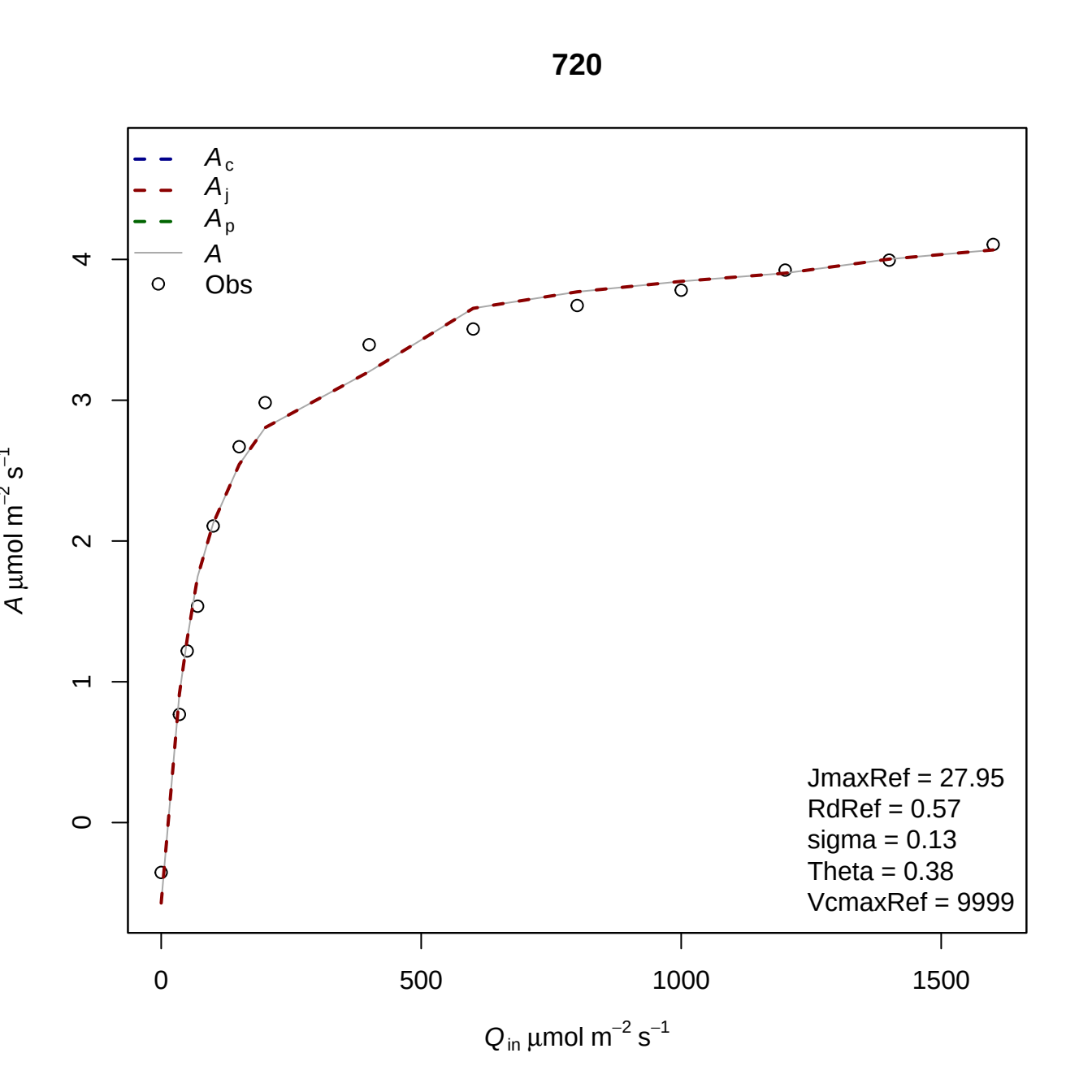

715
A_c
A_j
A_p
A
Obs
JmaxRef = 19.87
RdRef = 0.65
sigma = 0.07
Theta = 0.4
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500

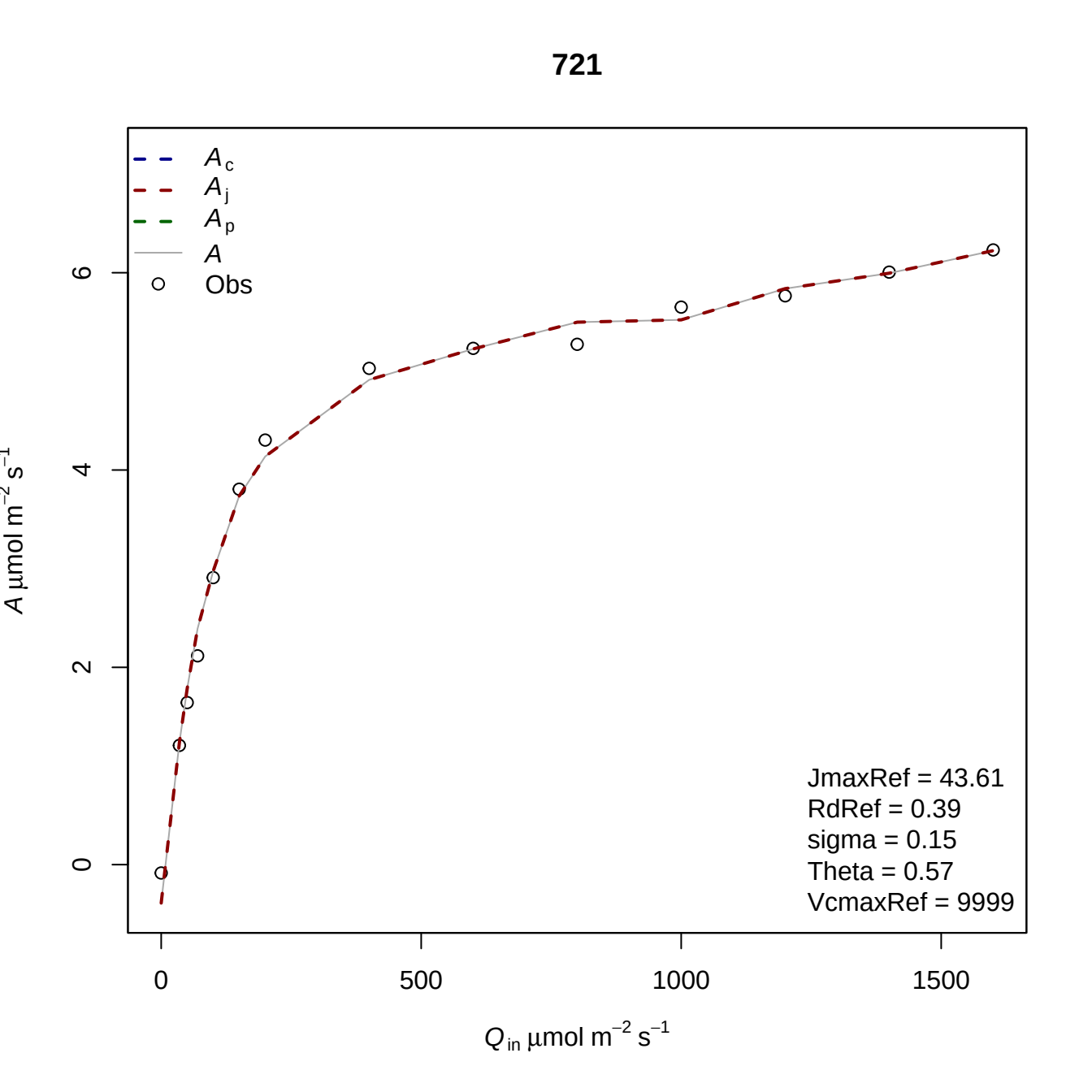

716
A_c
A_j
A_p
A
Obs
JmaxRef = 32.38
RdRef = 1.28
sigma = 0.14
Theta = 0.66
VcmaxRef = 19.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

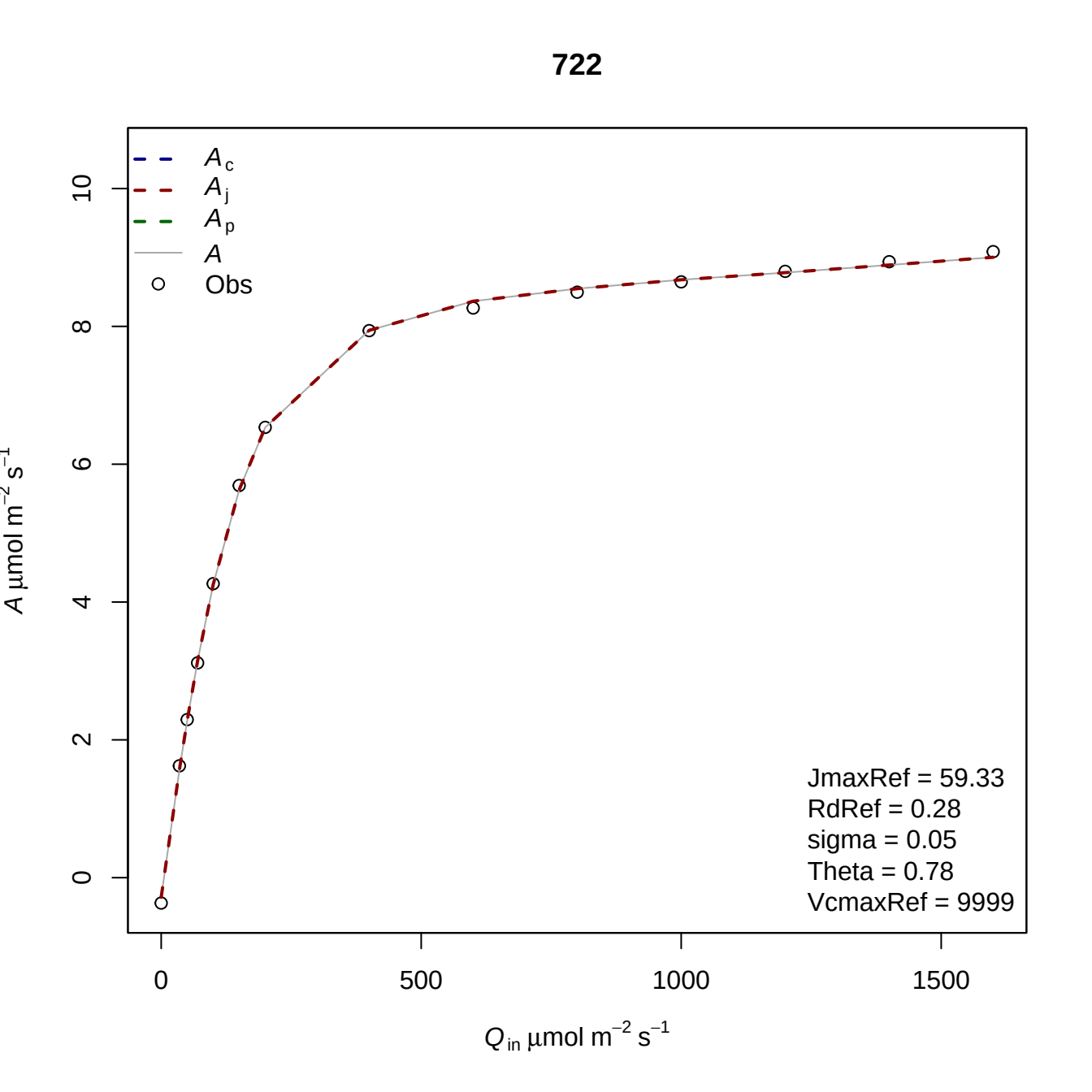

717
A_c
A_j
A_p
A
Obs
JmaxRef = 44.07
RdRef = 1.2
sigma = 0.15
Theta = −0.04
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

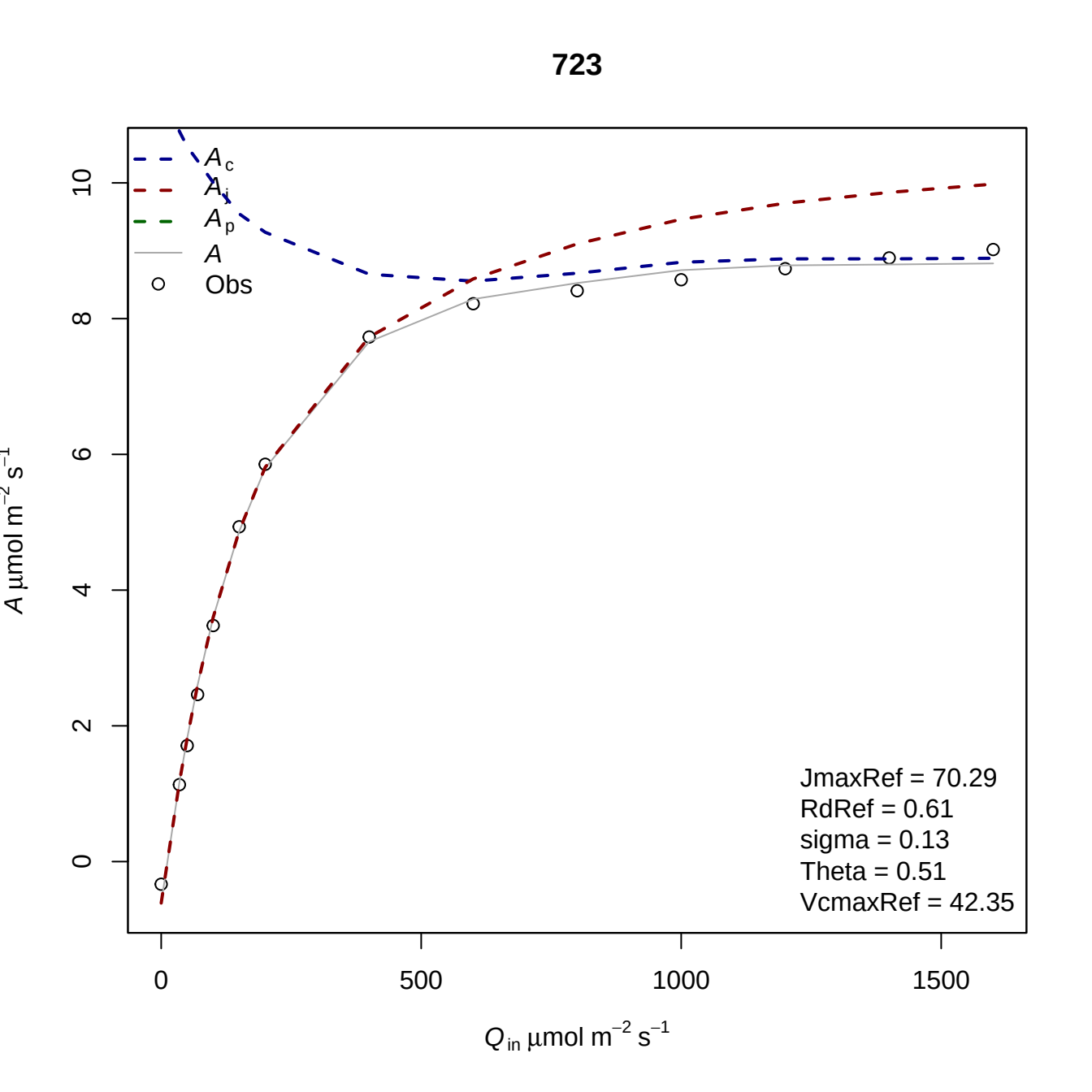

718
A_c
A_j
A_p
A
Obs
JmaxRef = 97.28
RdRef = 2.3
sigma = 0.15
Theta = 0.54
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

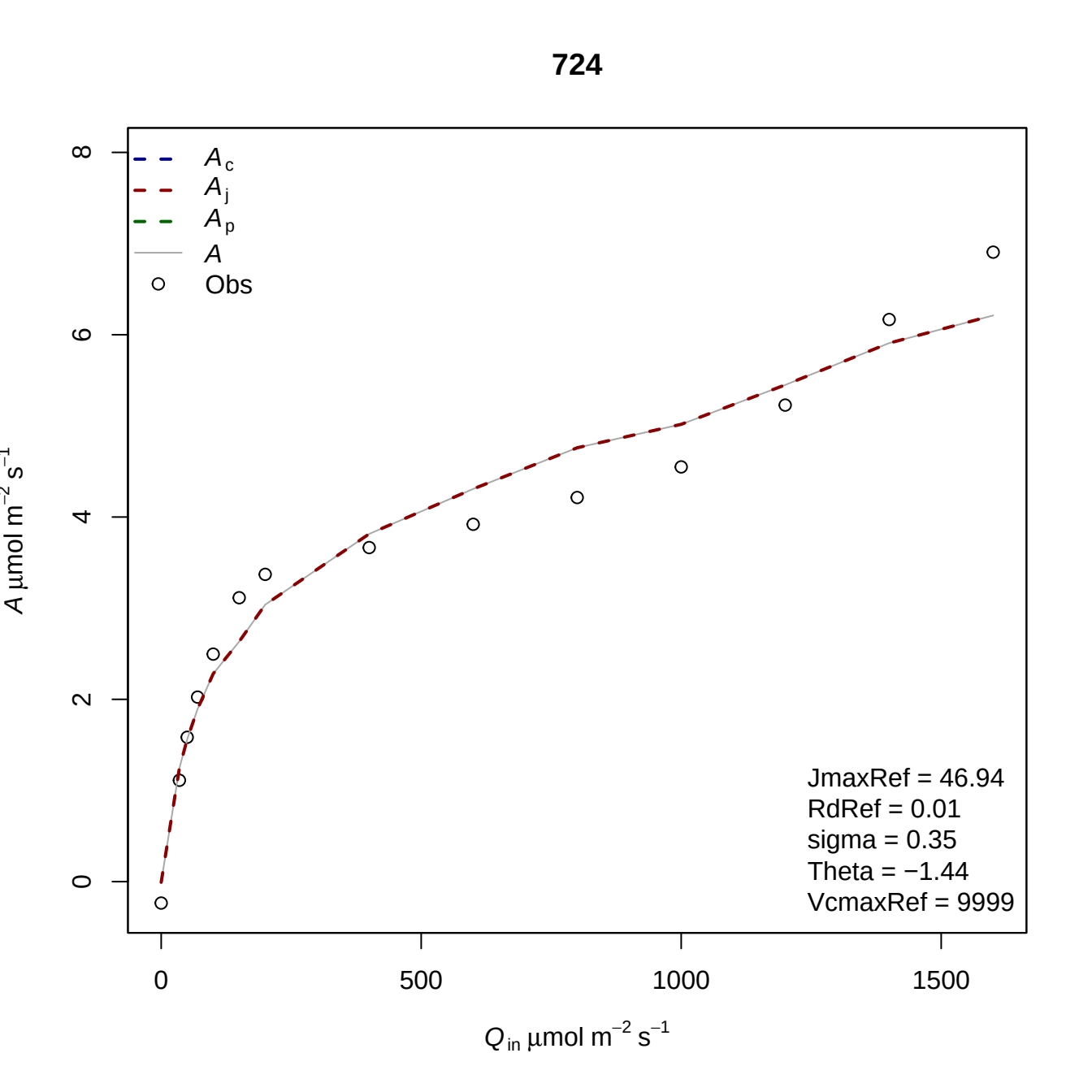

719
A_c
A_j
A_p
A
Obs
JmaxRef = 26.29
RdRef = 0.16
sigma = 0.16
Theta = −0.46
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

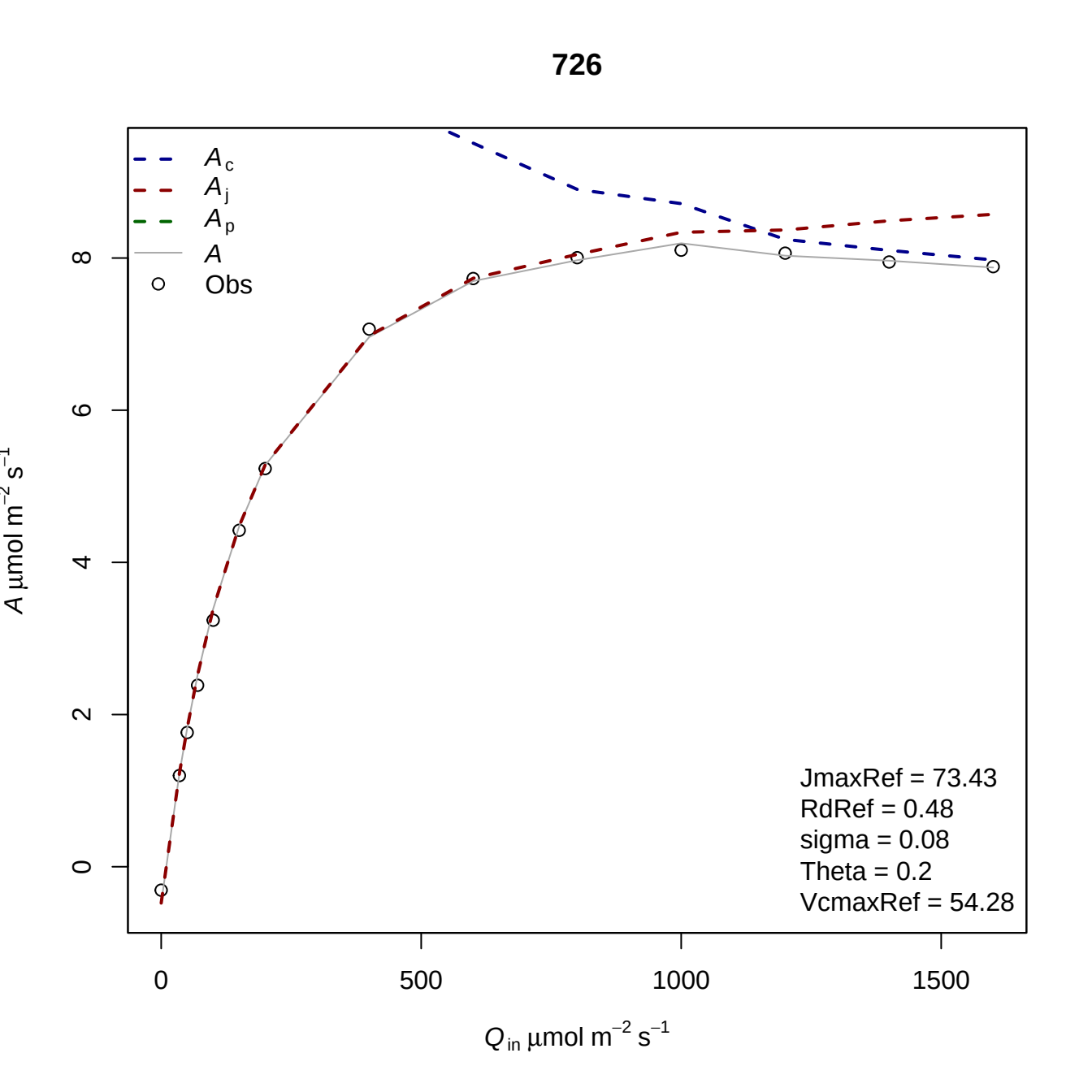

720
A_c
A_j
A_p
A
Obs
JmaxRef = 27.95
RdRef = 0.57
sigma = 0.13
Theta = 0.38
VcmaxRef = 9999
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

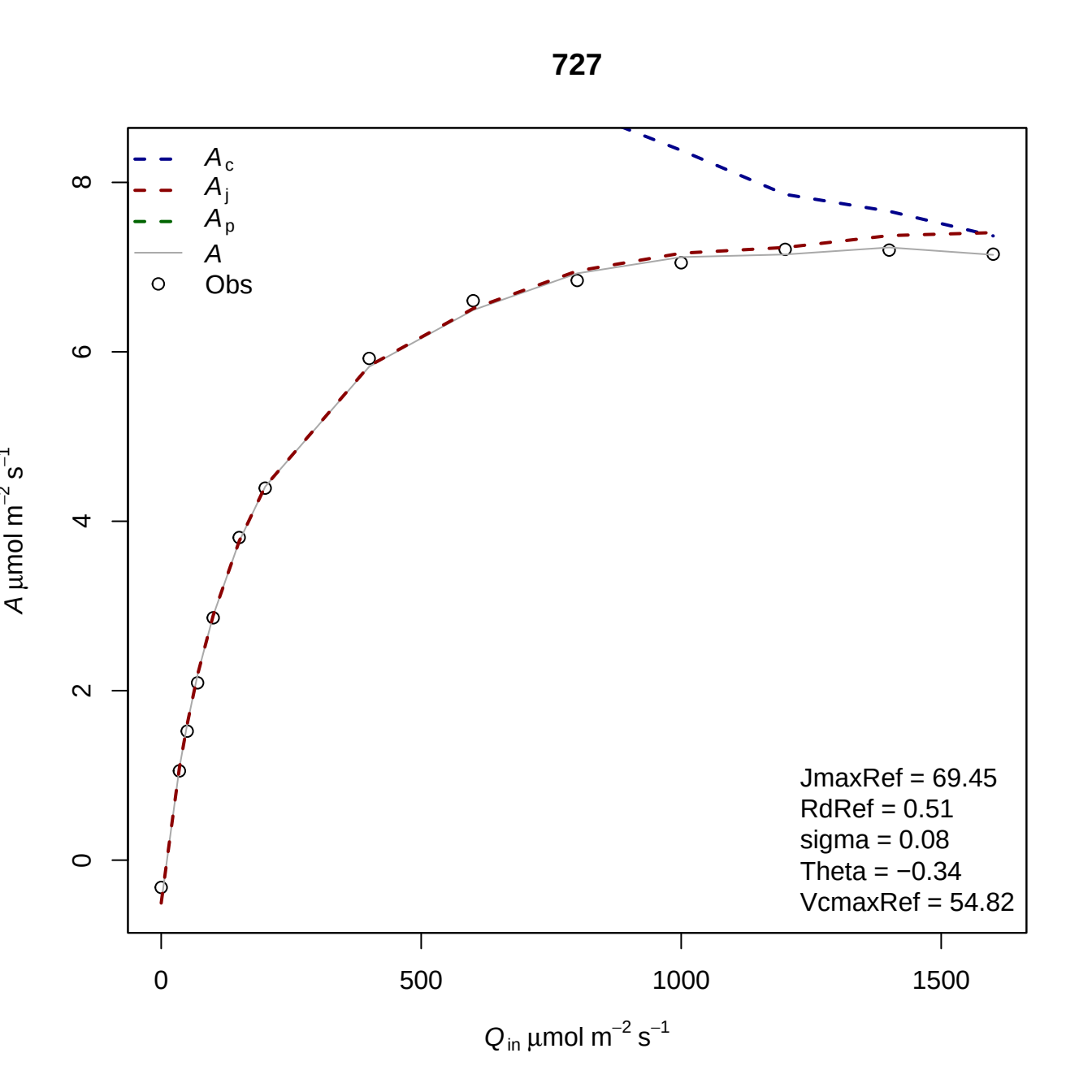

721
A_c
A_j
A_p
A
Obs
JmaxRef = 43.61
RdRef = 0.39
sigma = 0.15
Theta = 0.57
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

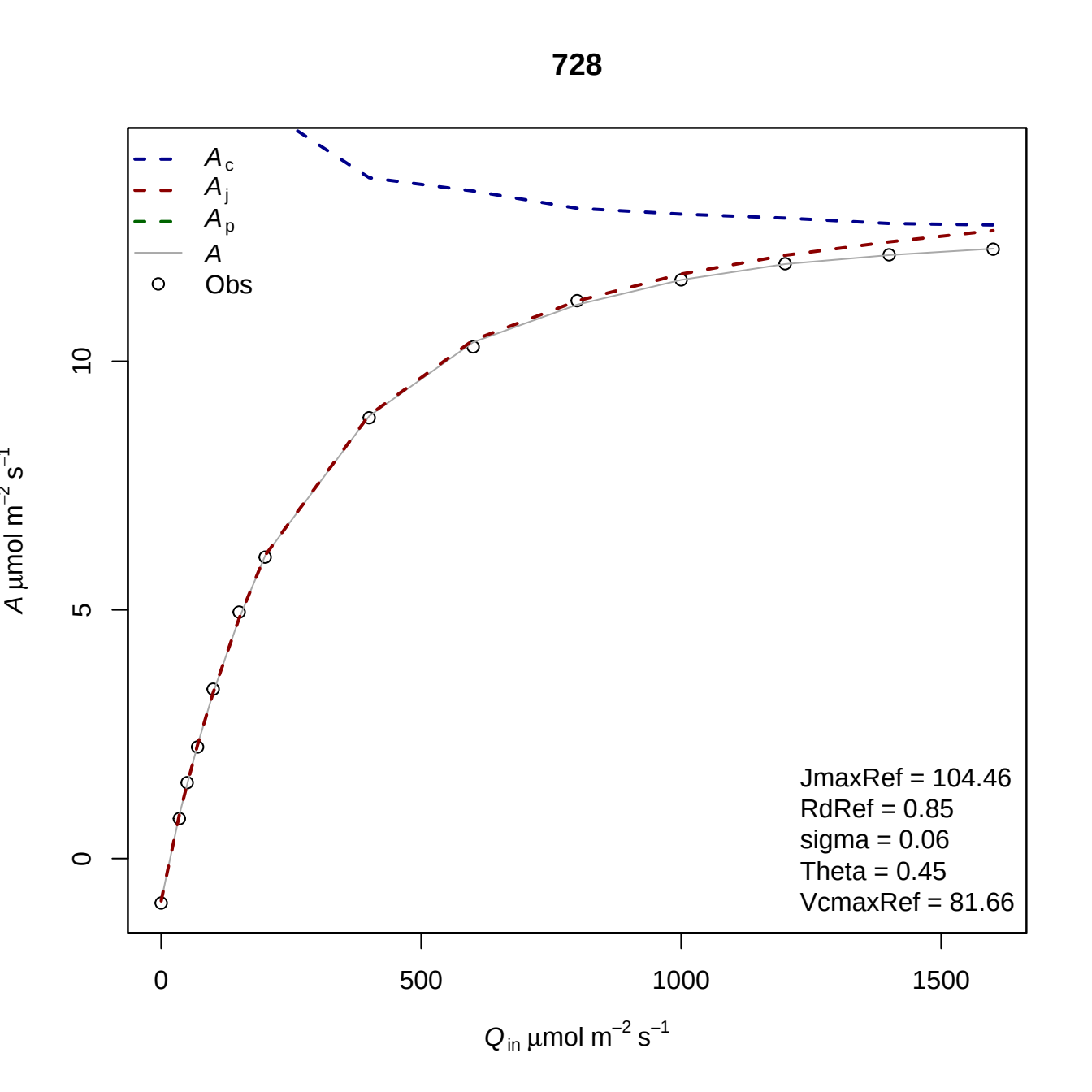

722
A_c
A_j
A_p
A
Obs
JmaxRef = 59.33
RdRef = 0.28
sigma = 0.05
Theta = 0.78
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

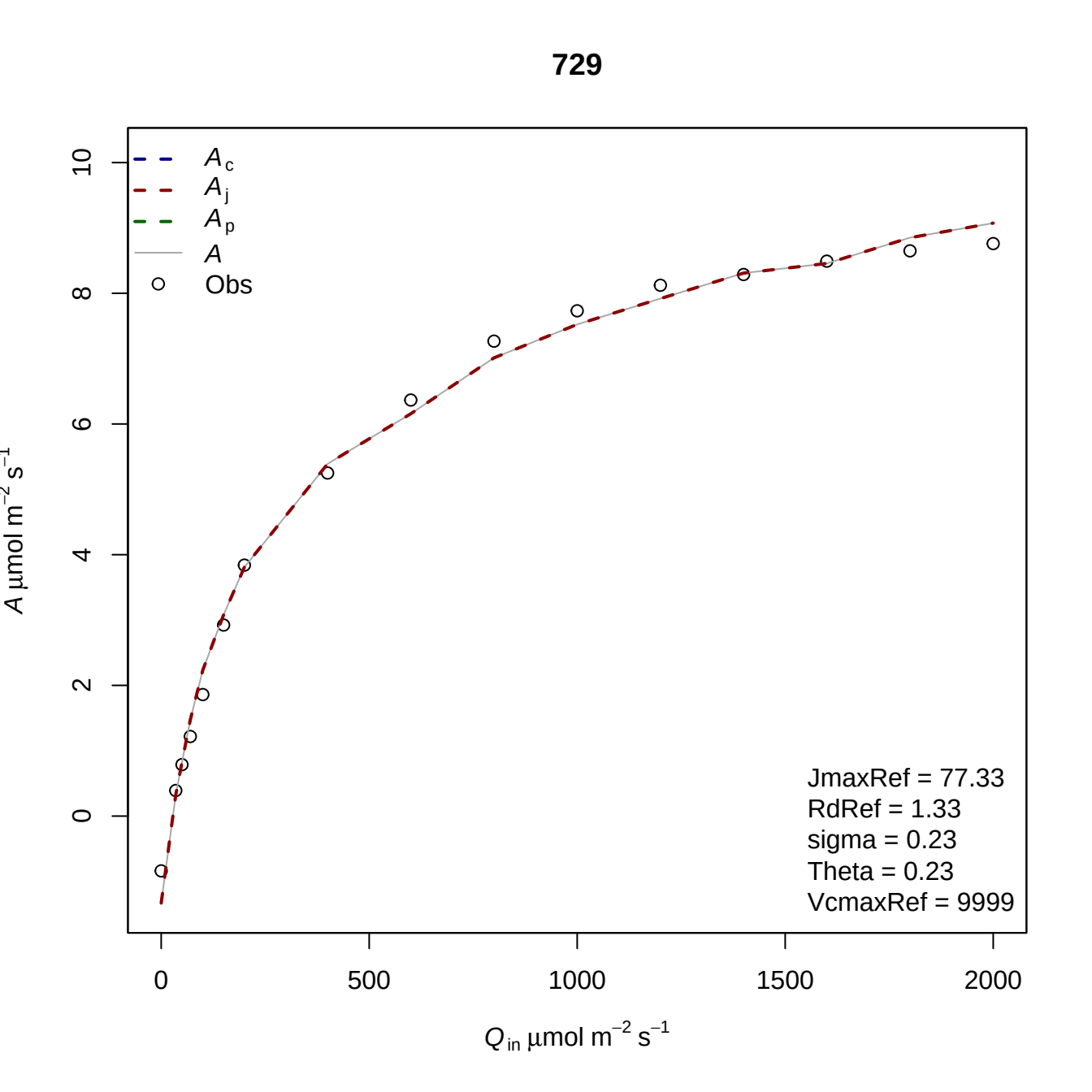

723
A_c
A_j
A_p
A
Obs
JmaxRef = 70.29
RdRef = 0.61
sigma = 0.13
Theta = 0.51
VcmaxRef = 42.35
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

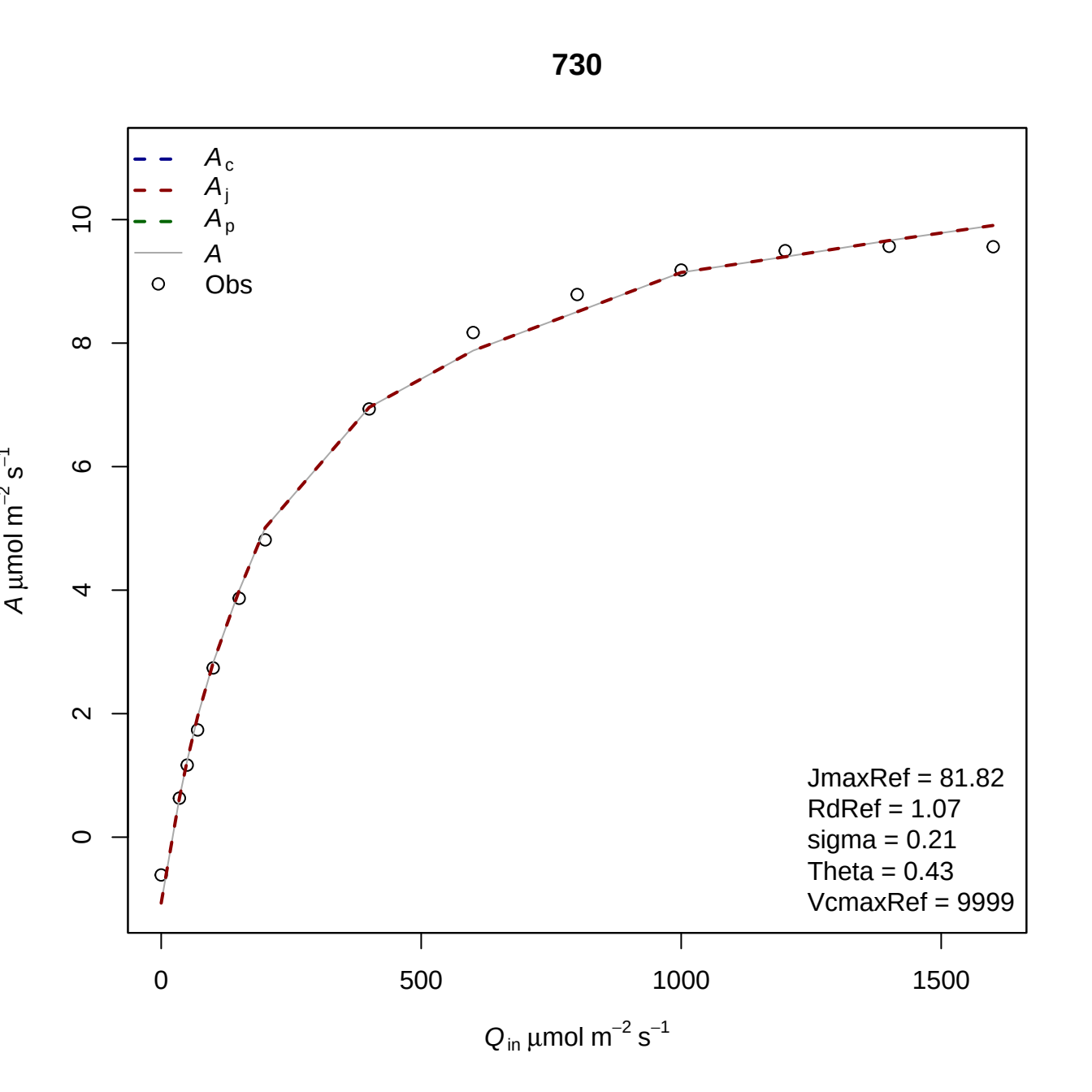

724
A_c
A_j
A_p
A
Obs
JmaxRef = 46.94
RdRef = 0.01
sigma = 0.35
Theta = −1.44
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

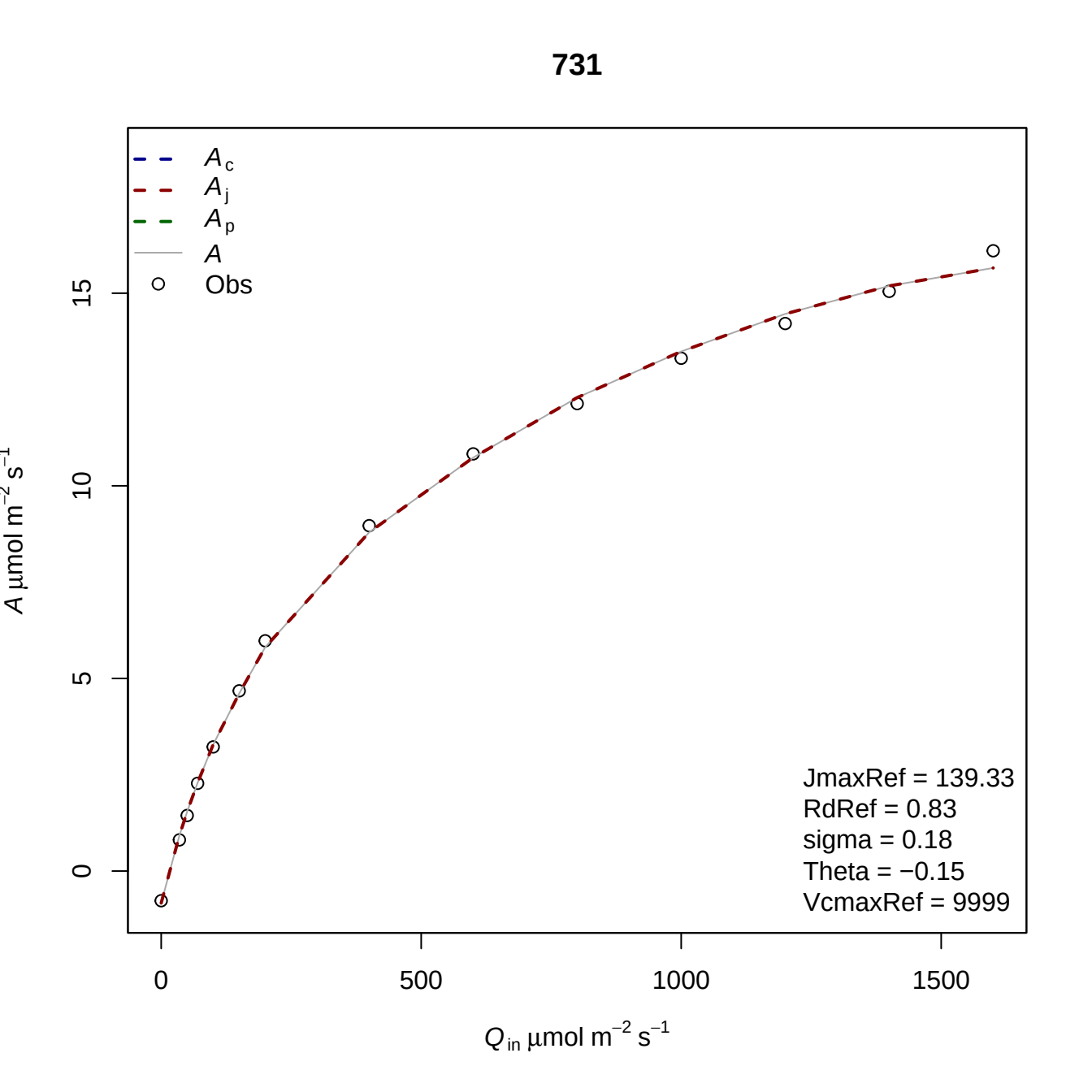

726
A_c
A_j
A_p
A
Obs
JmaxRef = 73.43
RdRef = 0.48
sigma = 0.08
Theta = 0.2
VcmaxRef = 54.28
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

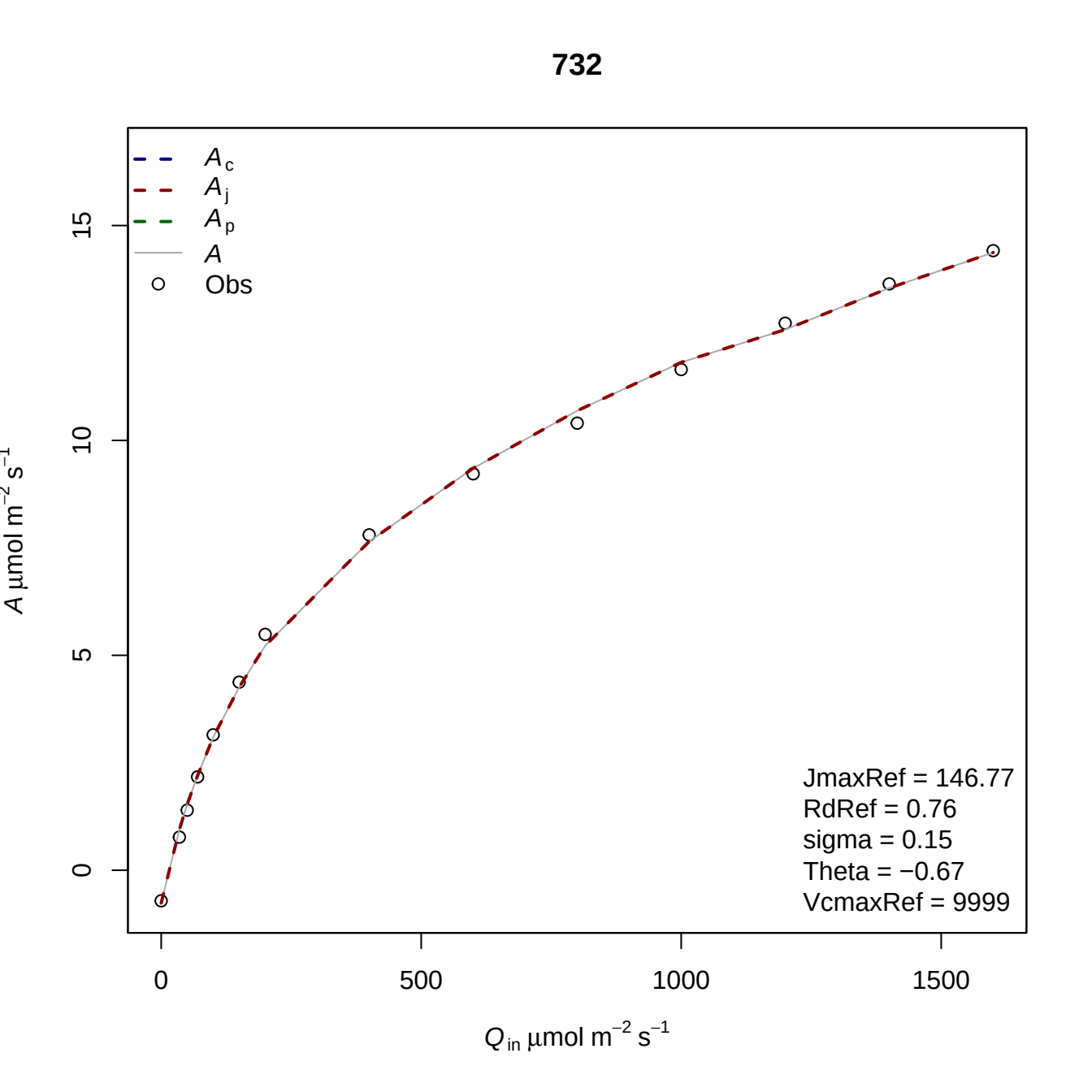

727
A_c
A_j
A_p
A
Obs
JmaxRef = 69.45
RdRef = 0.51
sigma = 0.08
Theta = −0.34
VcmaxRef = 54.82
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

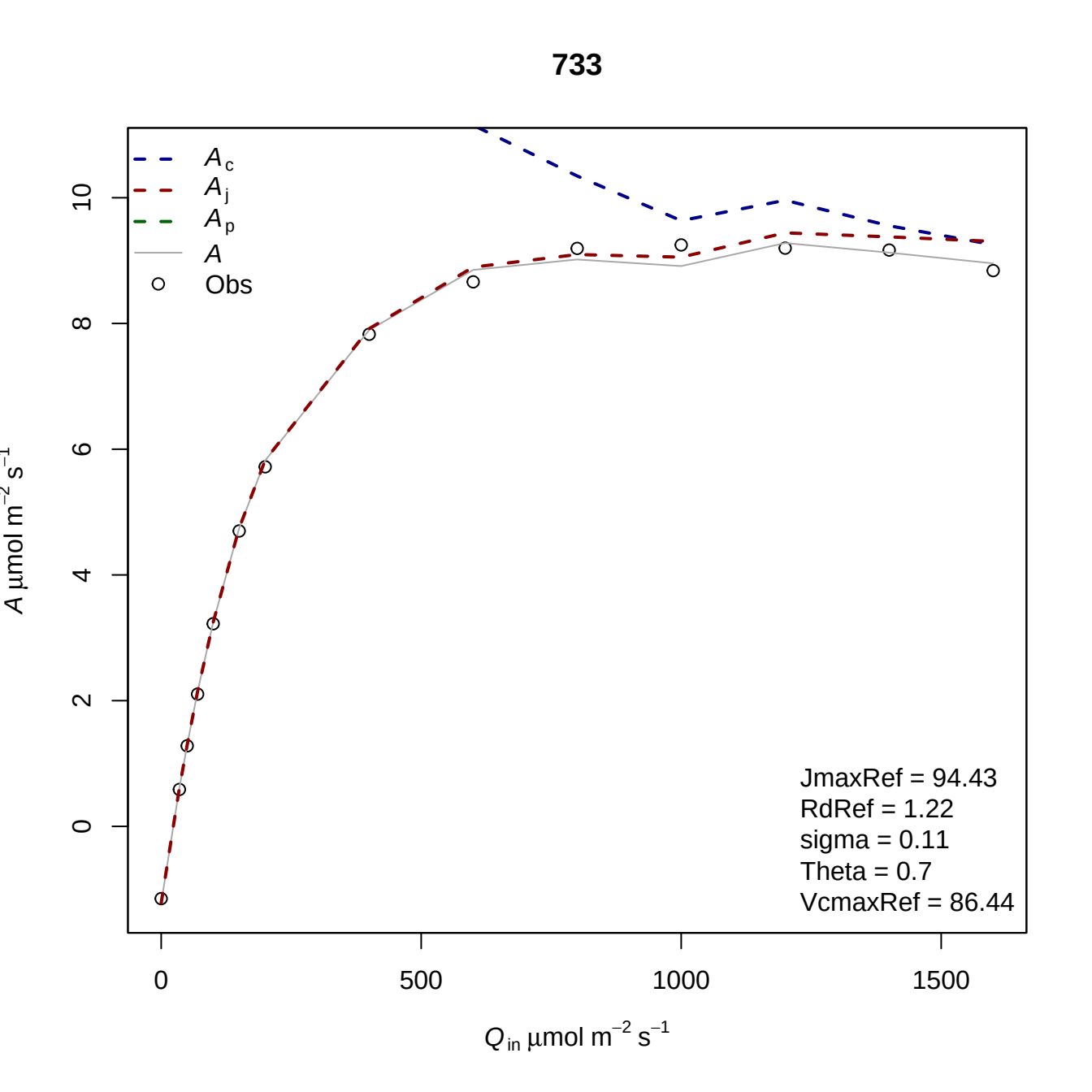

728
A_c
A_j
A_p
A
Obs
JmaxRef = 104.46
RdRef = 0.85
sigma = 0.06
Theta = 0.45
VcmaxRef = 81.66
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

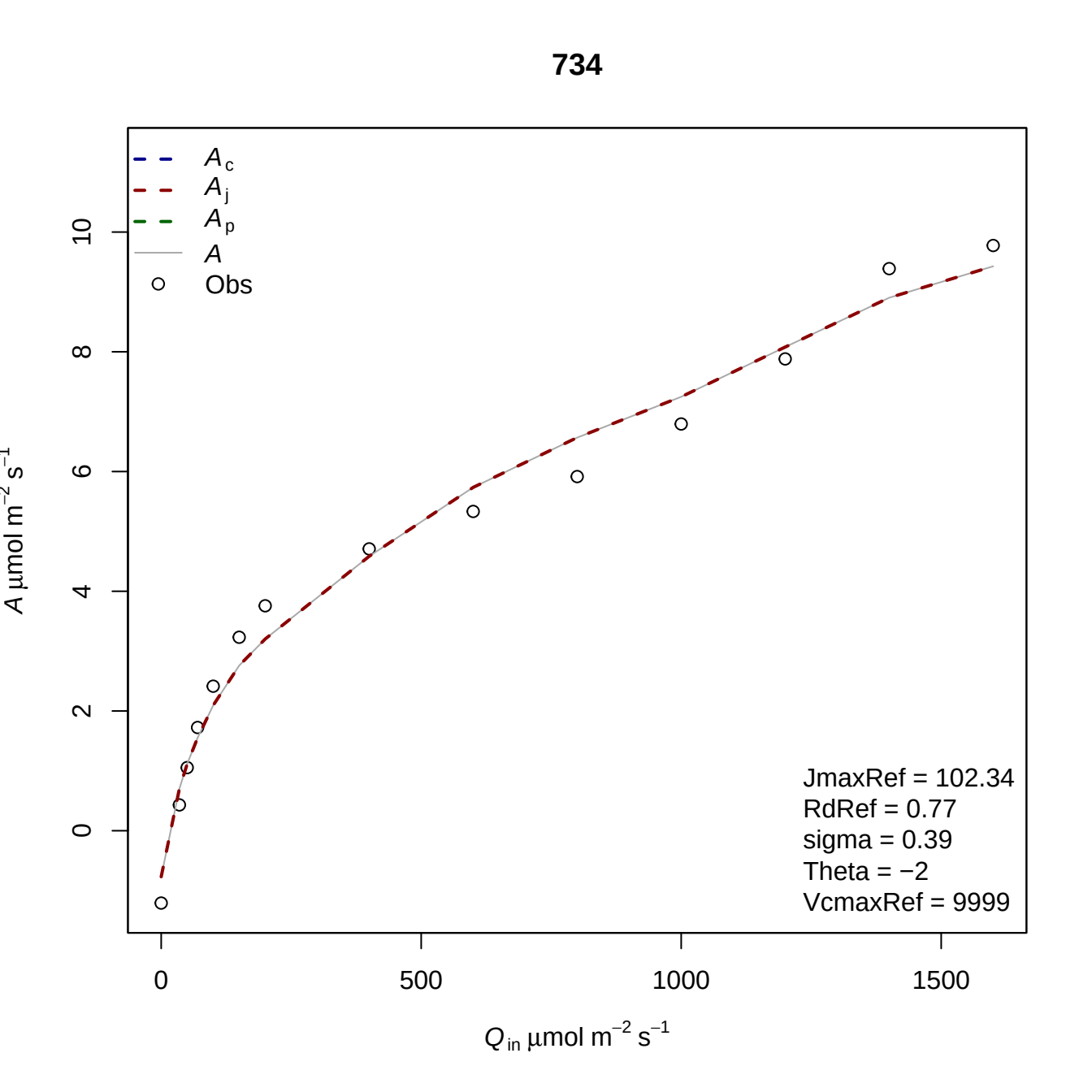

729
A_c
A_j
A_p
A
Obs
JmaxRef = 77.33
RdRef = 1.33
sigma = 0.23
Theta = 0.23
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

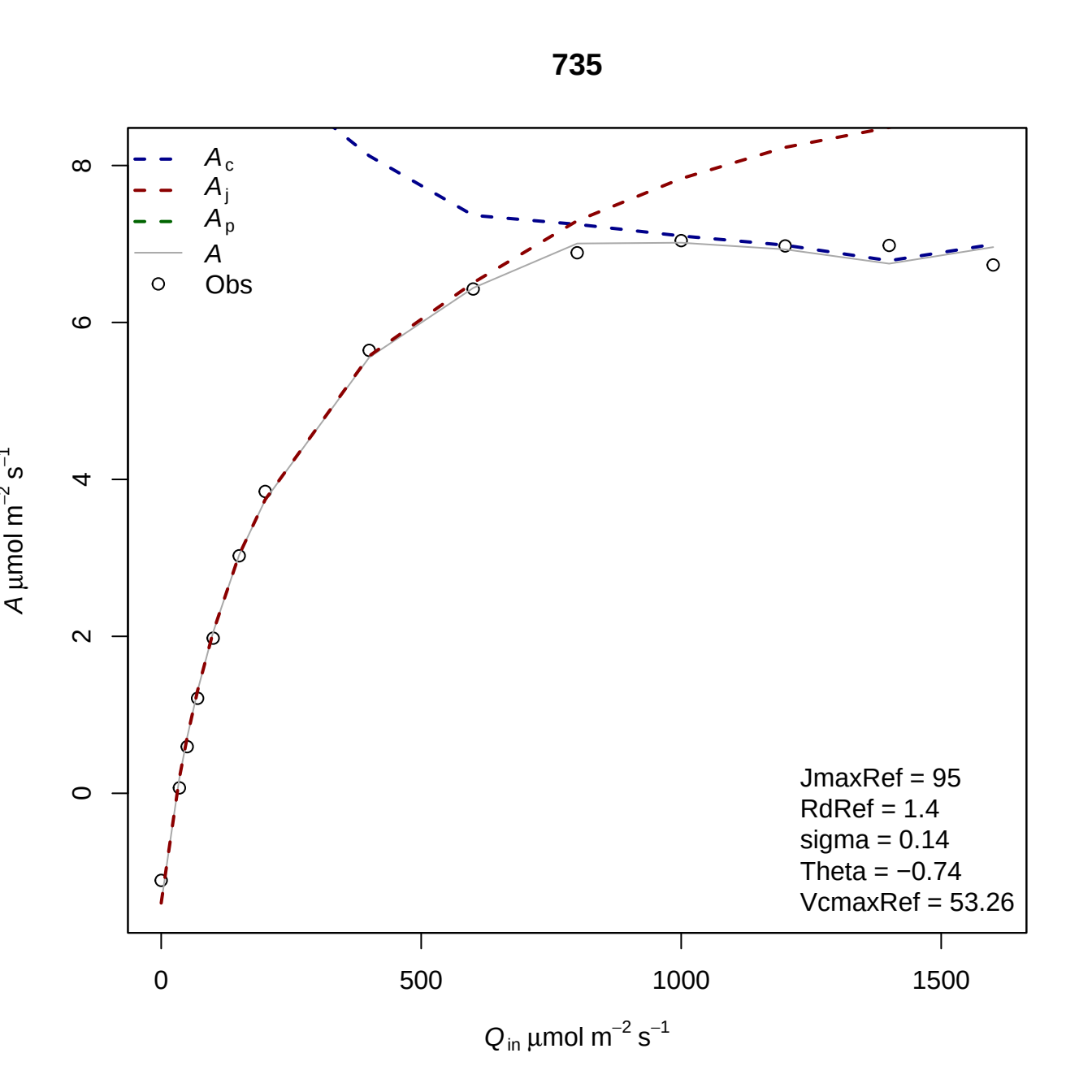

730
A_c
A_j
A_p
A
Obs
JmaxRef = 81.82
RdRef = 1.07
sigma = 0.21
Theta = 0.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

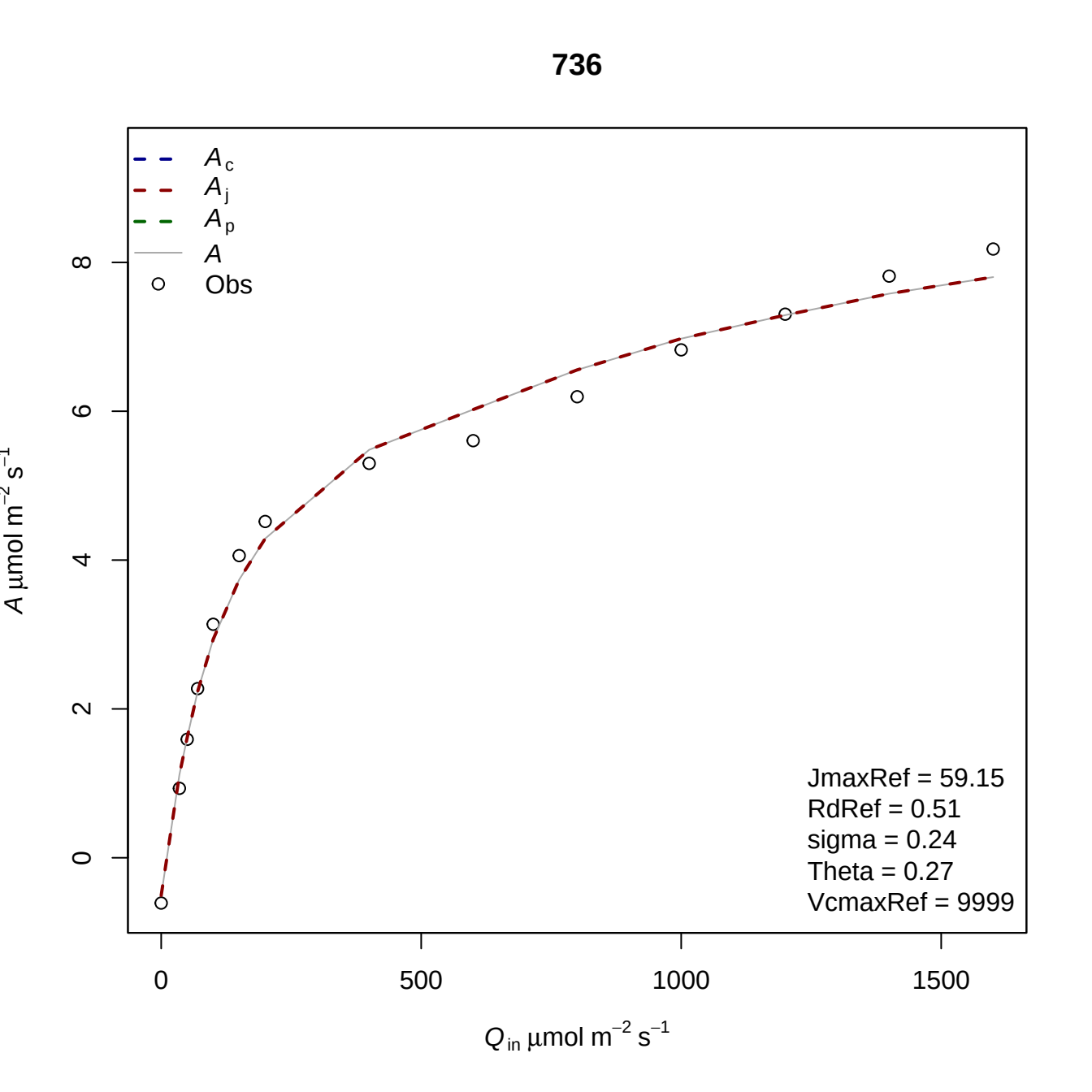

731
A_c
A_j
A_p
A
Obs
JmaxRef = 139.33
RdRef = 0.83
sigma = 0.18
Theta = −0.15
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

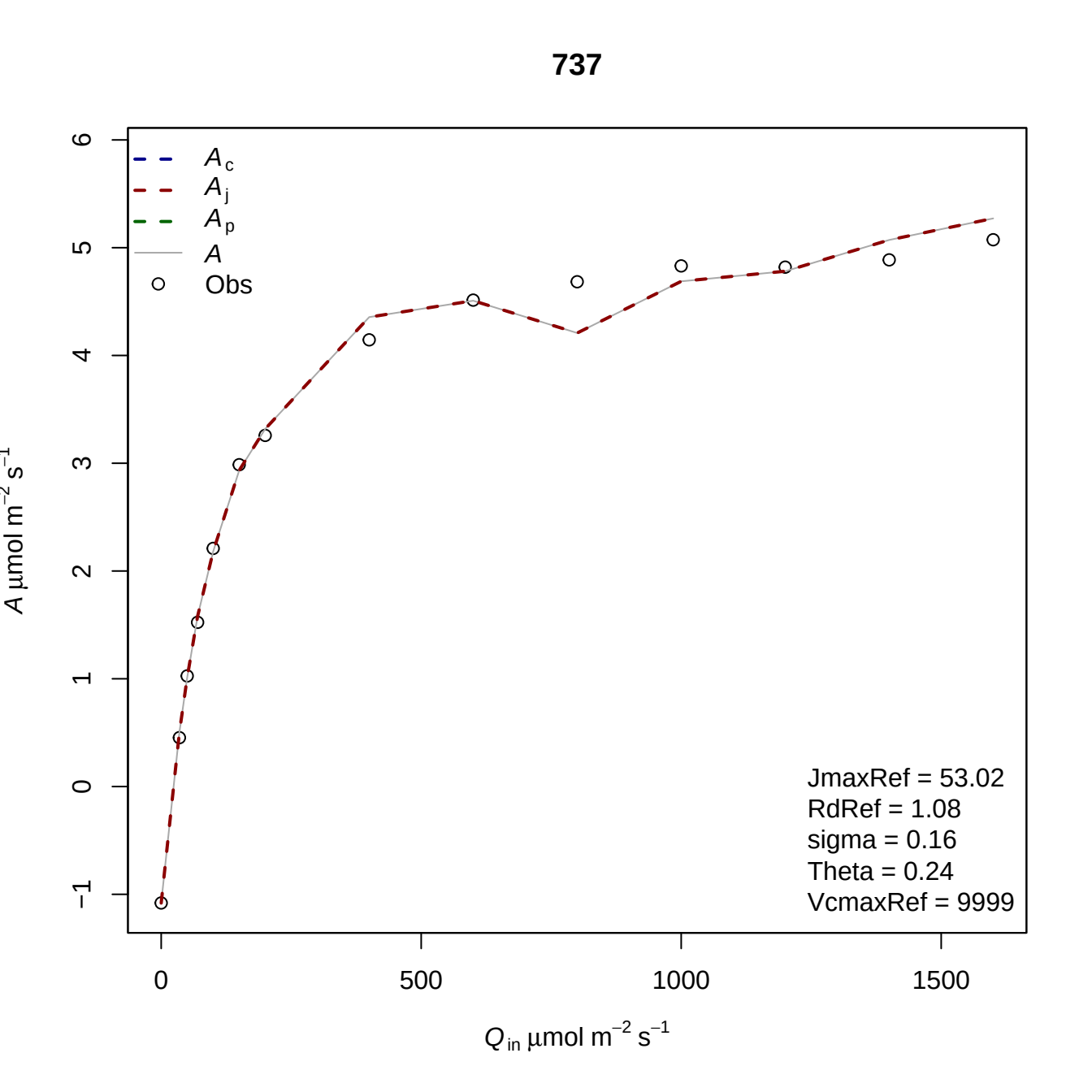

732
A_c
A_j
A_p
A
Obs
JmaxRef = 146.77
RdRef = 0.76
sigma = 0.15
Theta = −0.67
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

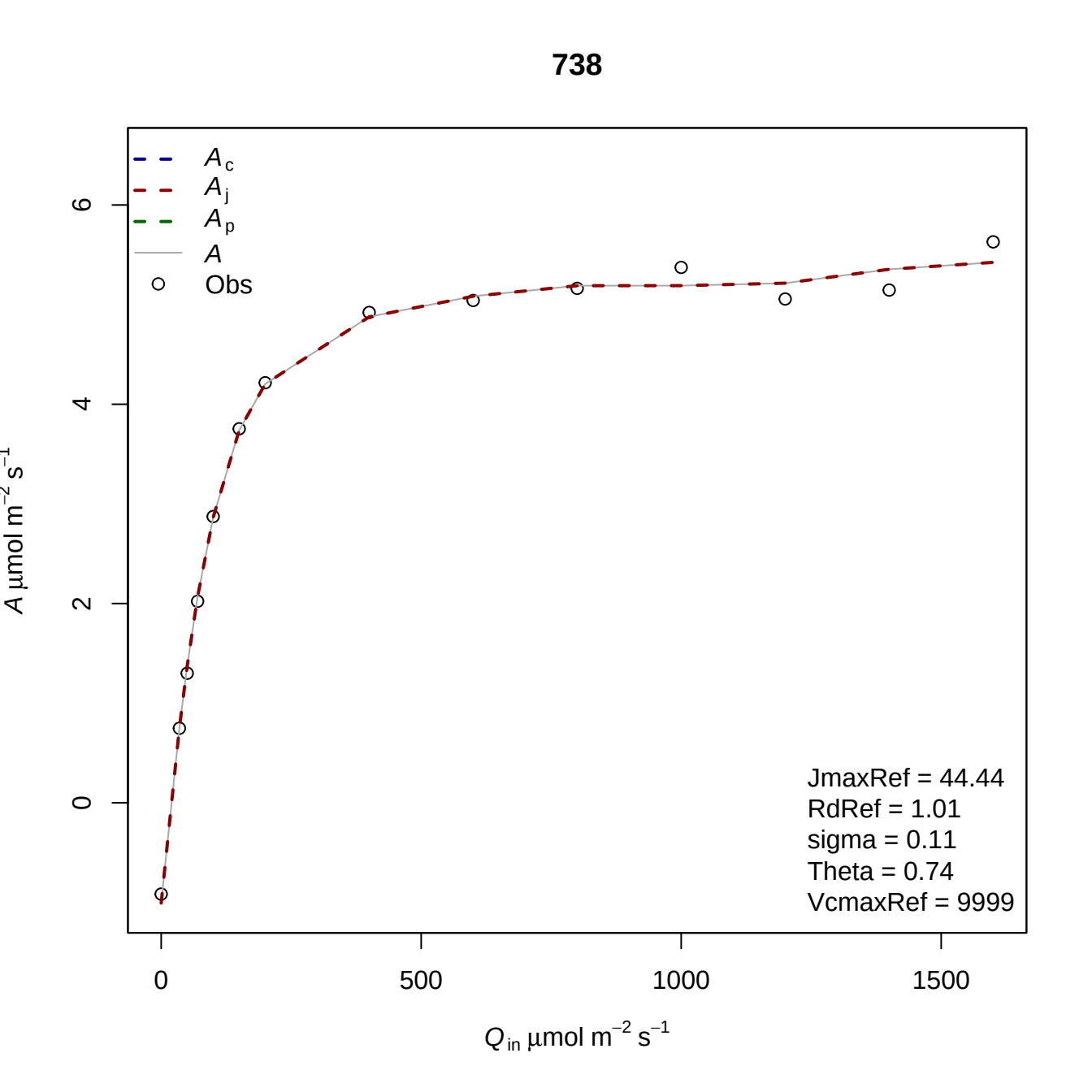

733
A_c
A_j
A_p
A
Obs
JmaxRef = 94.43
RdRef = 1.22
sigma = 0.11
Theta = 0.7
VcmaxRef = 86.44
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

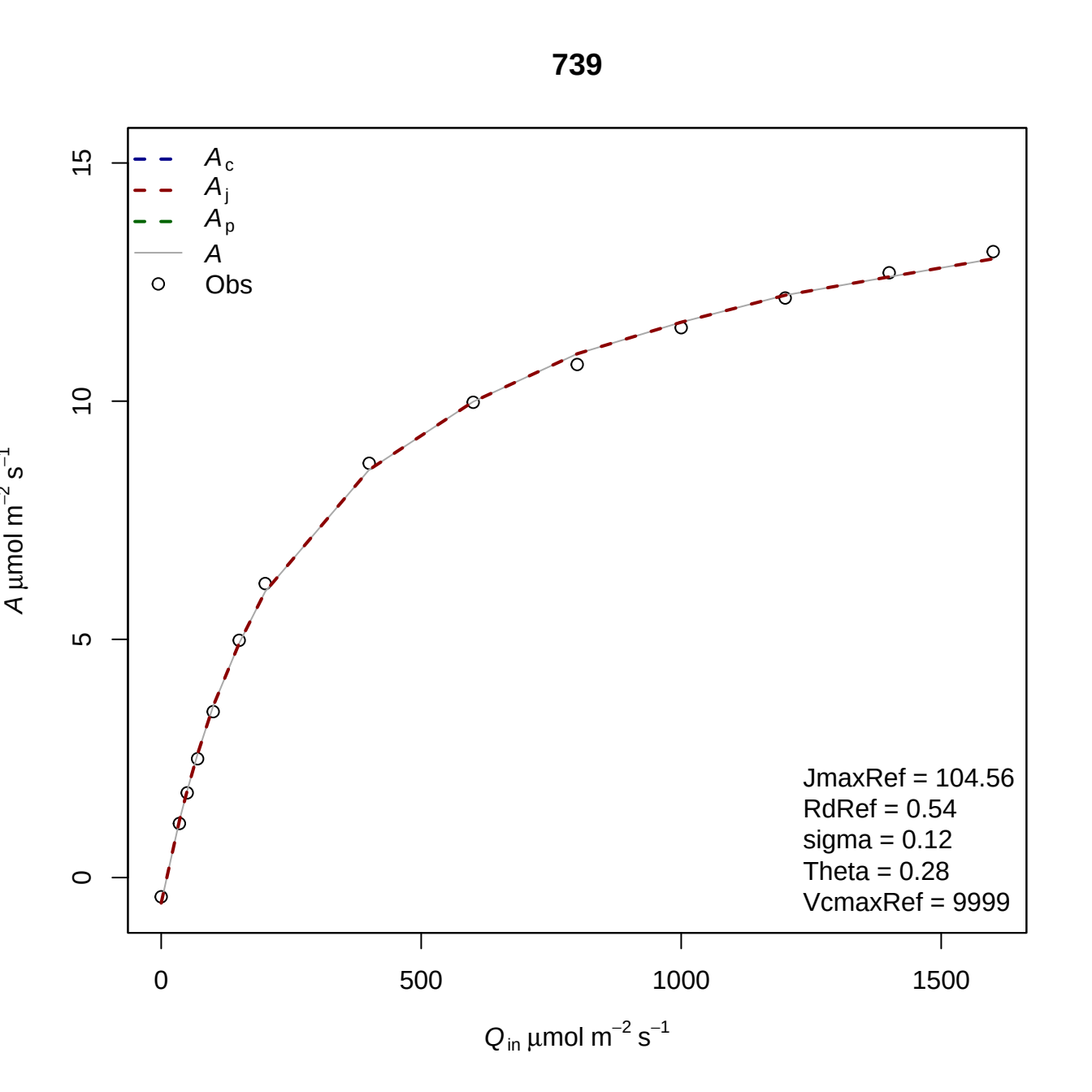

734
A_c
A_j
A_p
A
Obs
JmaxRef = 102.34
RdRef = 0.77
sigma = 0.39
Theta = −2
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

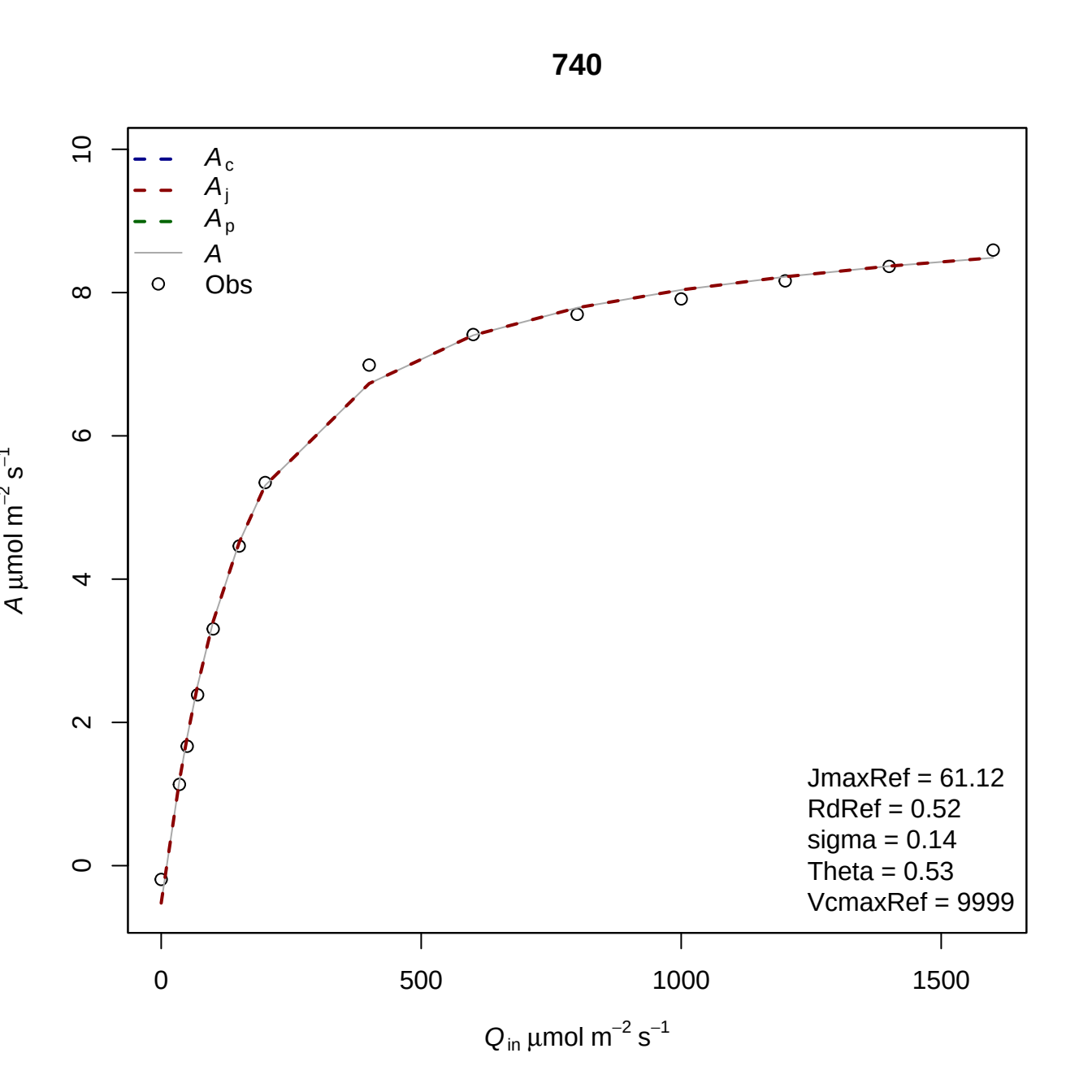

735
A_c
A_j
A_p
A
Obs
JmaxRef = 95
RdRef = 1.4
sigma = 0.14
Theta = −0.74
VcmaxRef = 53.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

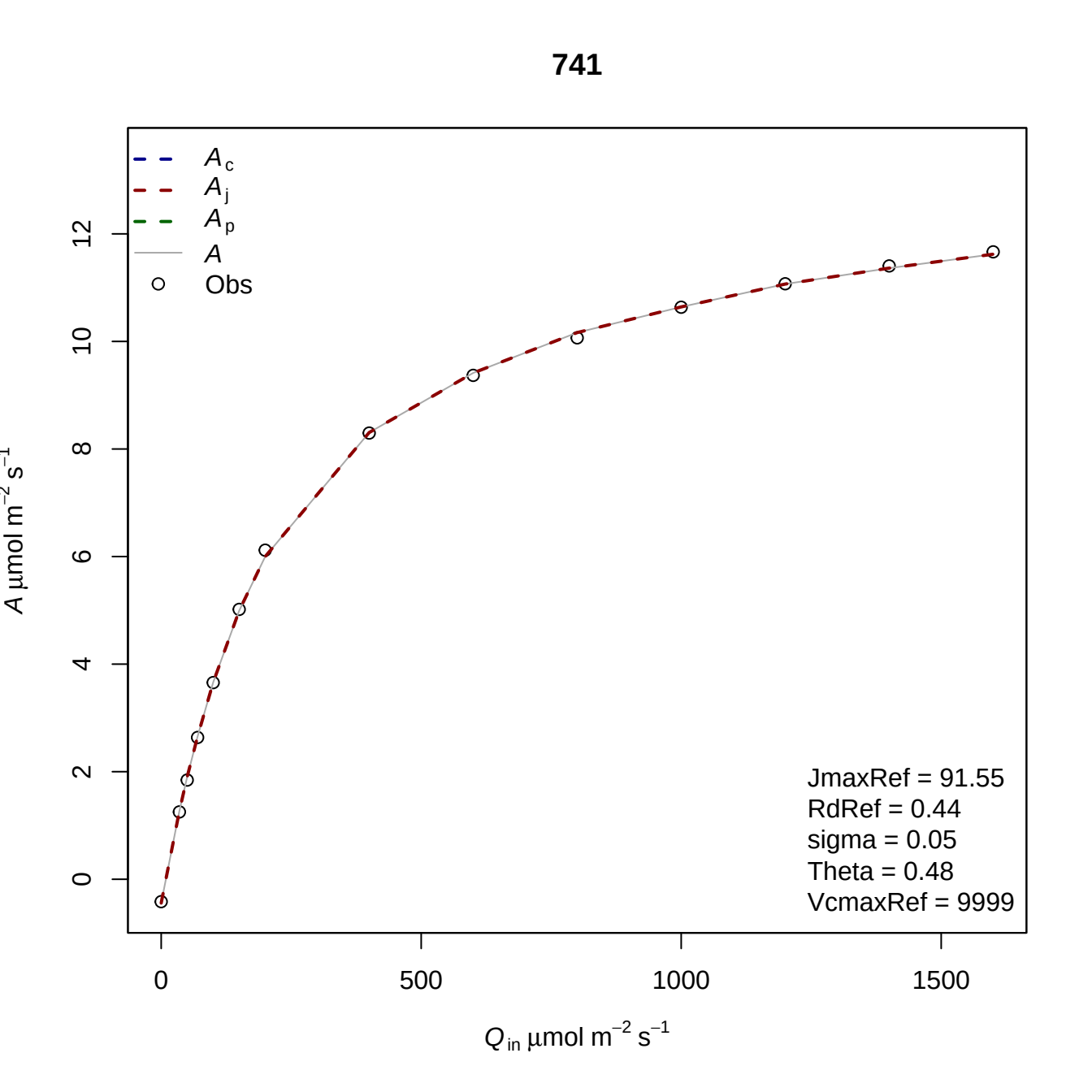

736
A_c
A_j
A_p
A
Obs
JmaxRef = 59.15
RdRef = 0.51
sigma = 0.24
Theta = 0.27
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

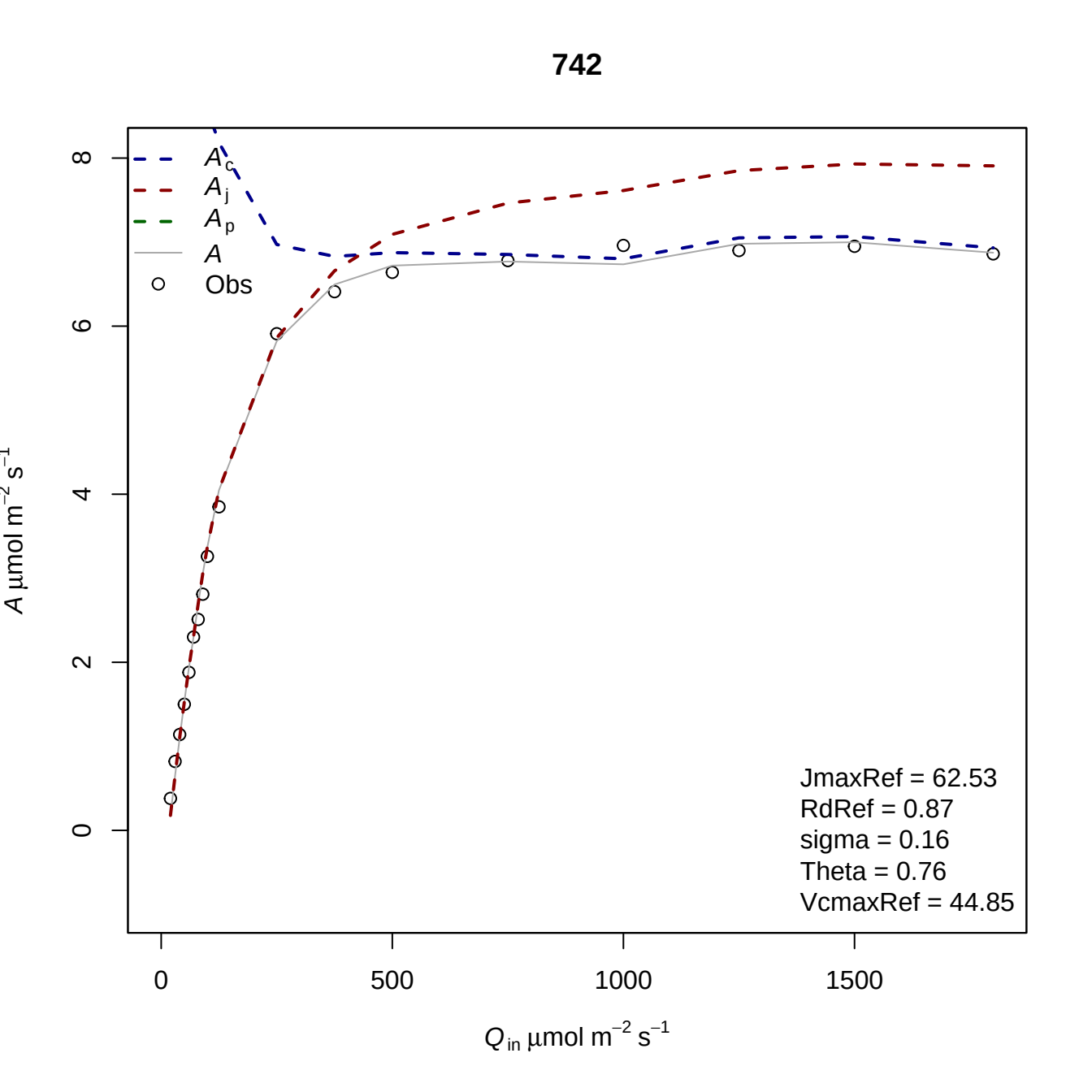

737
A_c
A_j
A_p
A
Obs
JmaxRef = 53.02
RdRef = 1.08
sigma = 0.16
Theta = 0.24
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

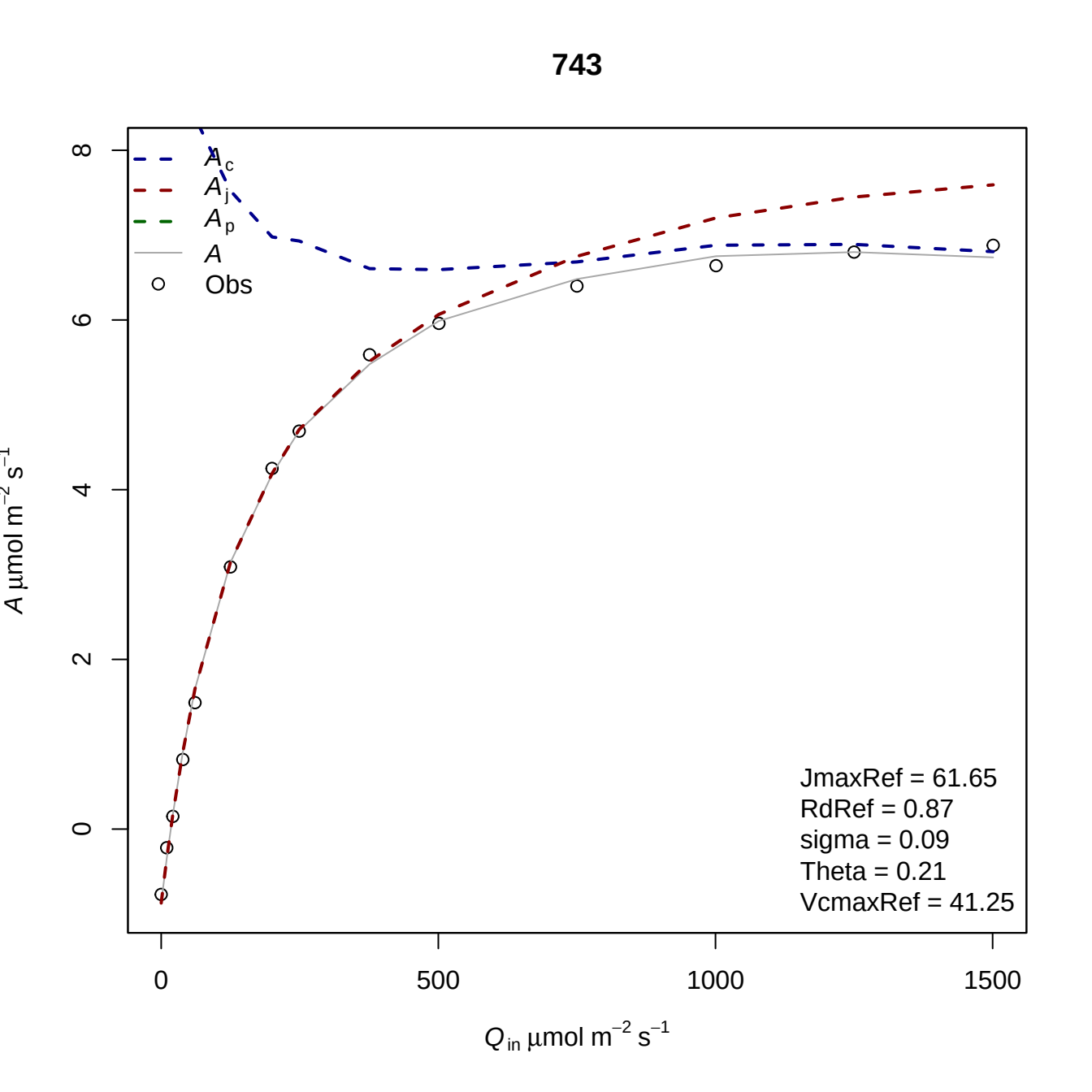

738
A_c
A_j
A_p
A
Obs
JmaxRef = 44.44
RdRef = 1.01
sigma = 0.11
Theta = 0.74
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

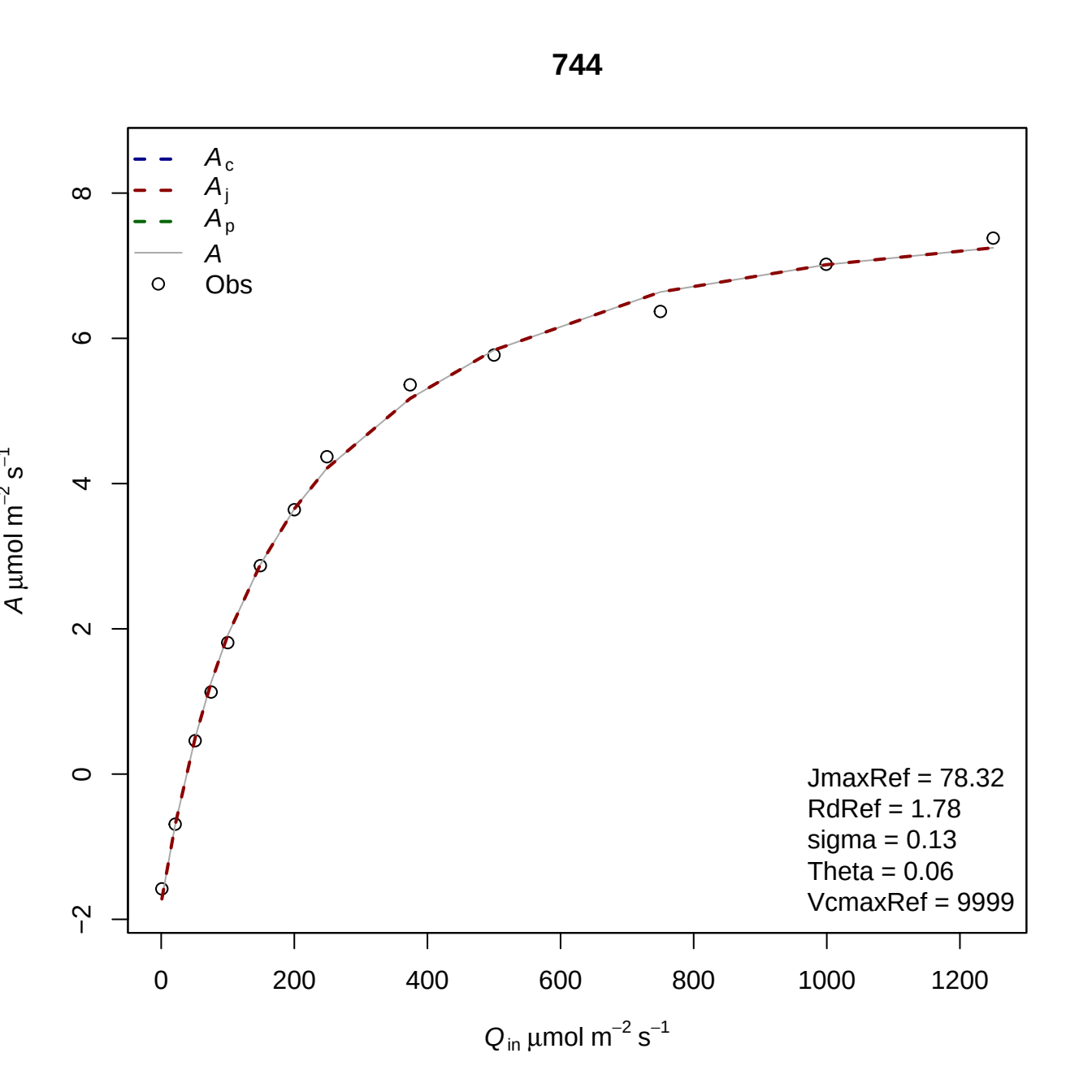

739
A_c
A_j
A_p
A
Obs
JmaxRef = 104.56
RdRef = 0.54
sigma = 0.12
Theta = 0.28
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

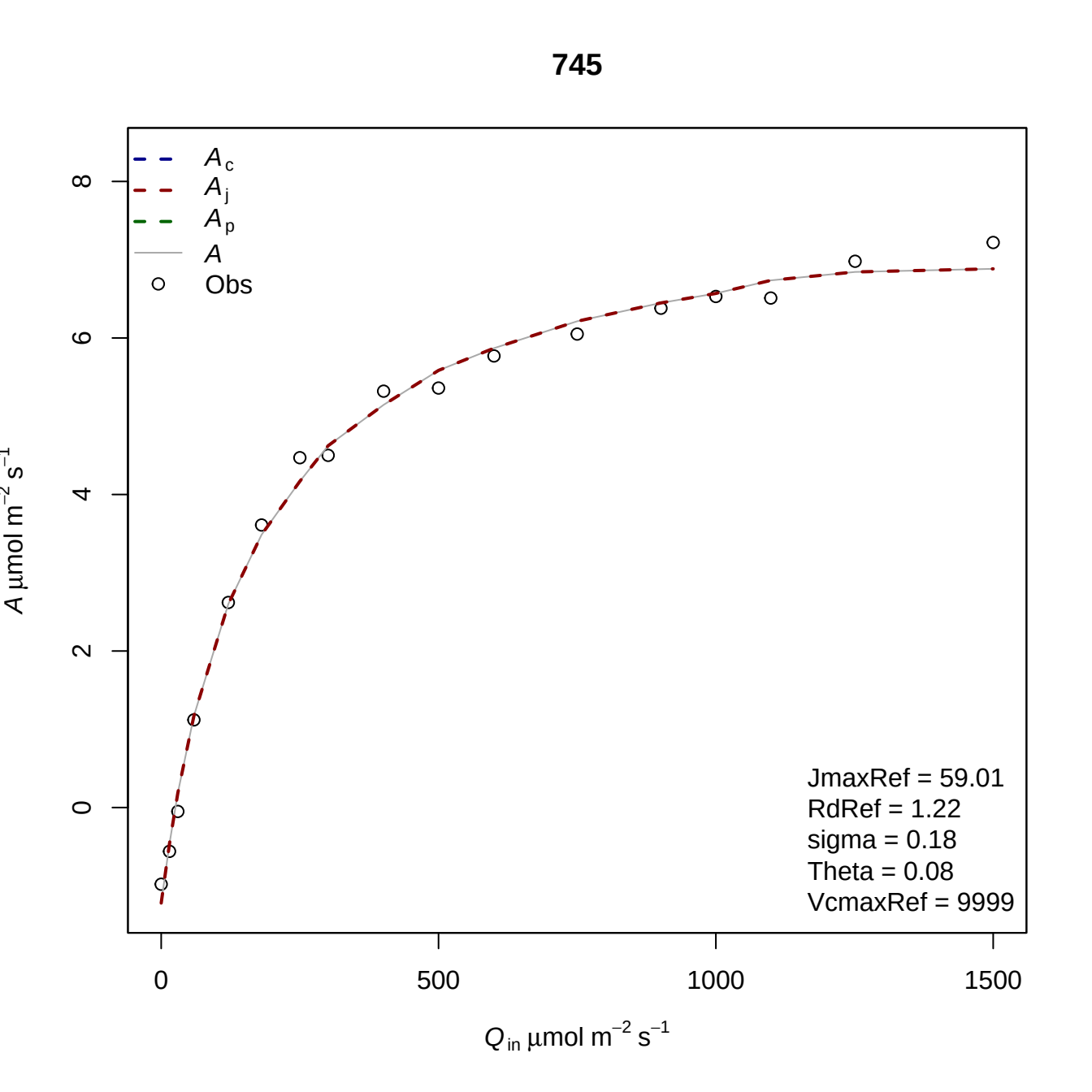

740
A_c
A_j
A_p
A
Obs
JmaxRef = 61.12
RdRef = 0.52
sigma = 0.14
Theta = 0.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

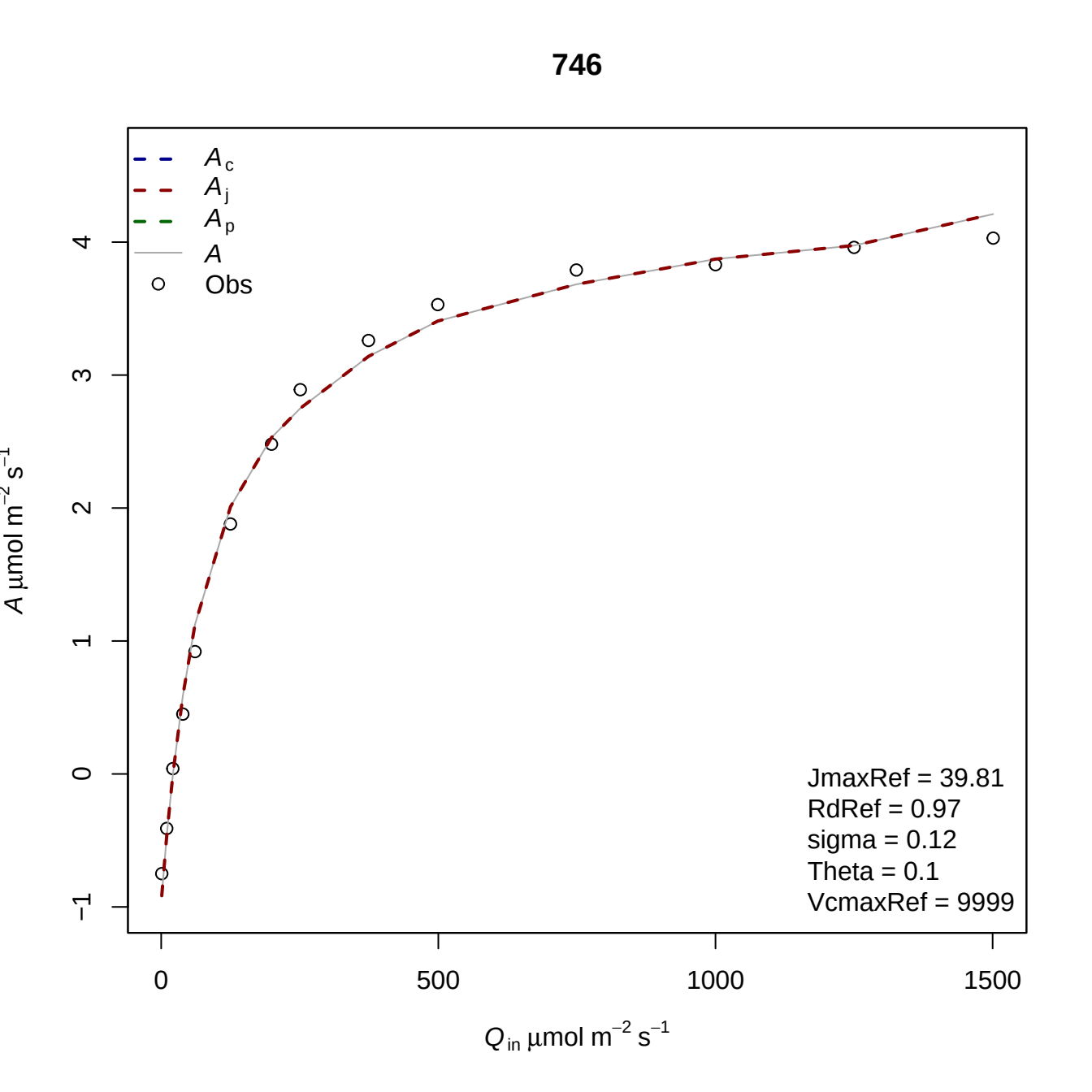

741
A_c
A_j
A_p
A
Obs
JmaxRef = 91.55
RdRef = 0.44
sigma = 0.05
Theta = 0.48
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

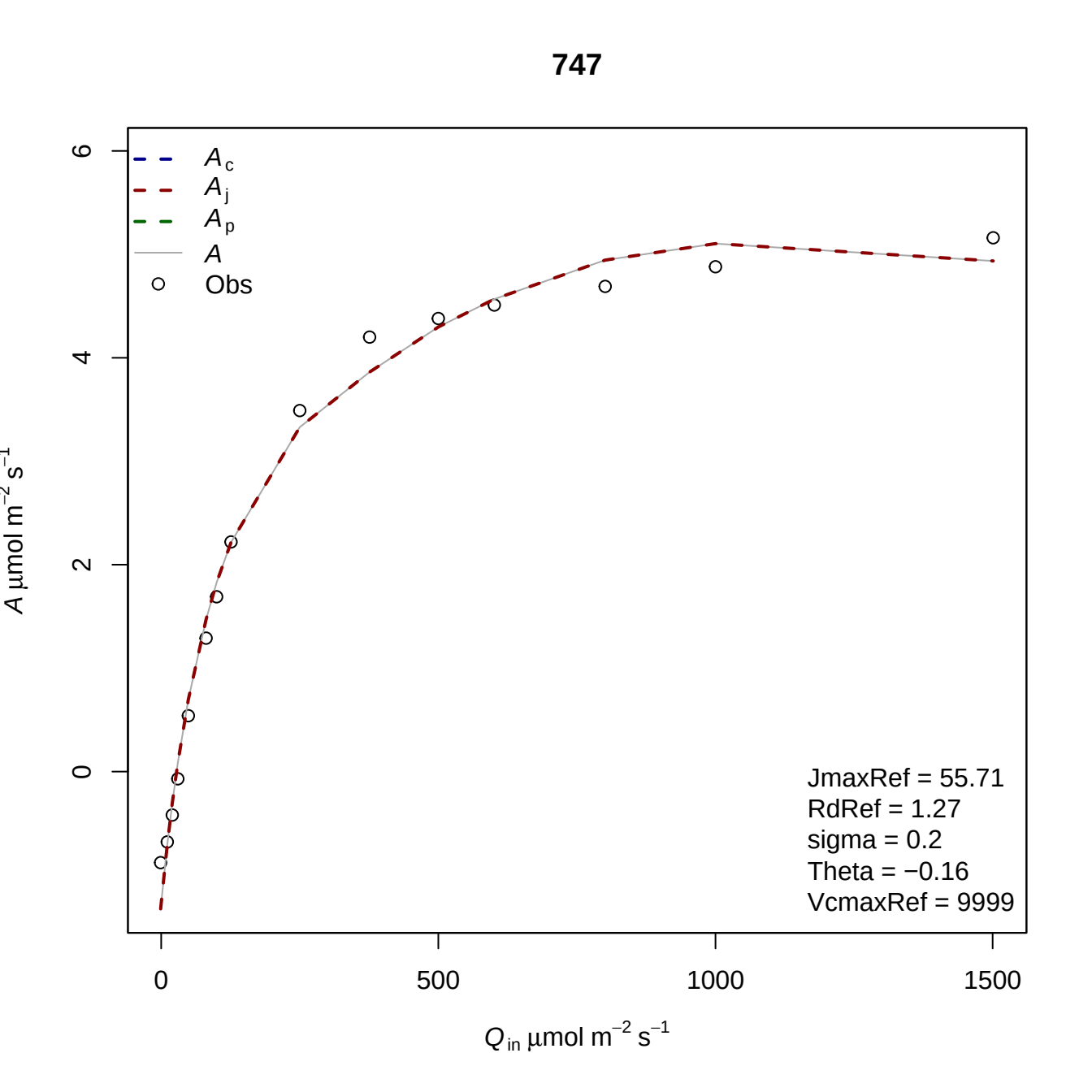

742
A_c
A_j
A_p
A
Obs
JmaxRef = 62.53
RdRef = 0.87
sigma = 0.16
Theta = 0.76
VcmaxRef = 44.85
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

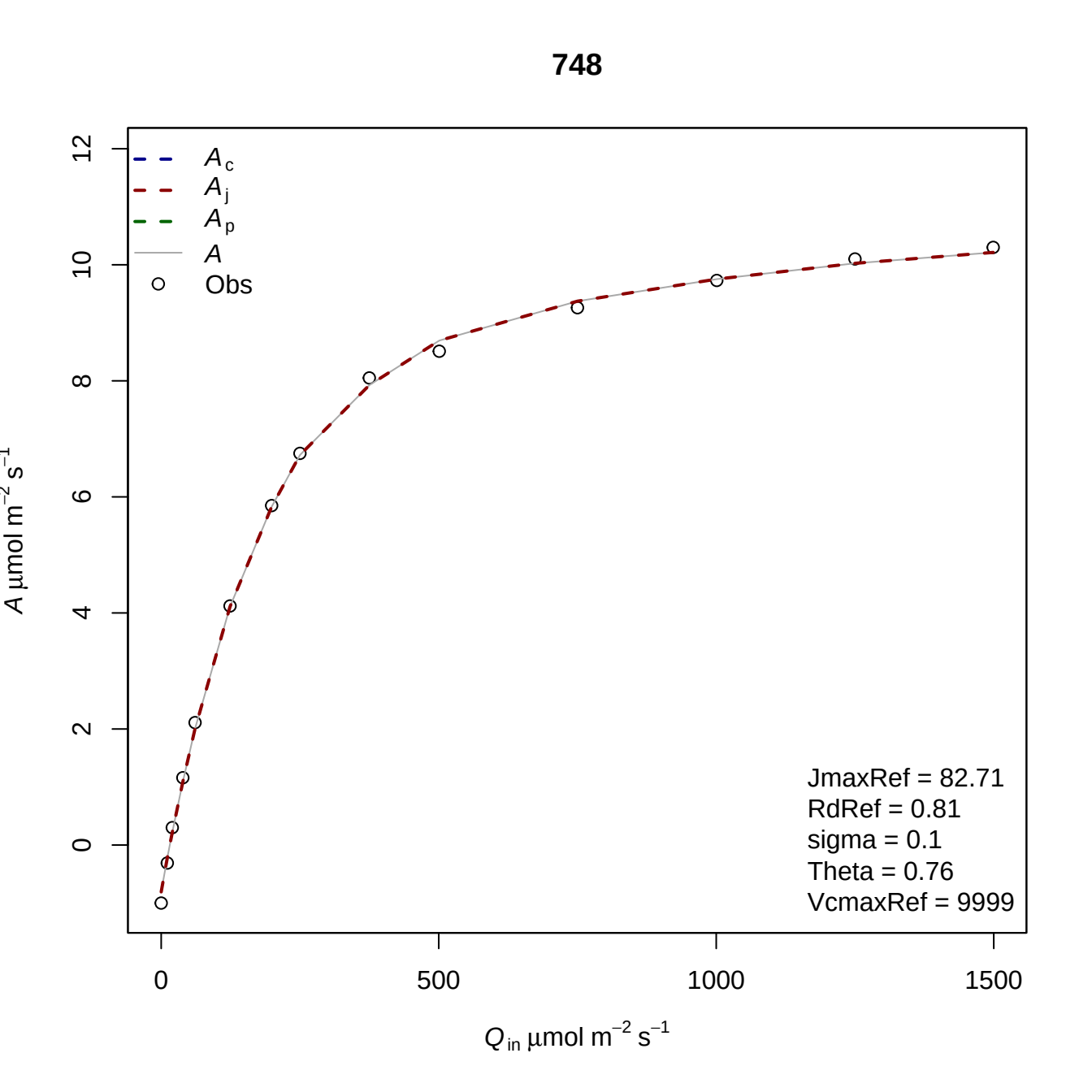

743
A_c
A_j
A_p
A
Obs
JmaxRef = 61.65
RdRef = 0.87
sigma = 0.09
Theta = 0.21
VcmaxRef = 41.25
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

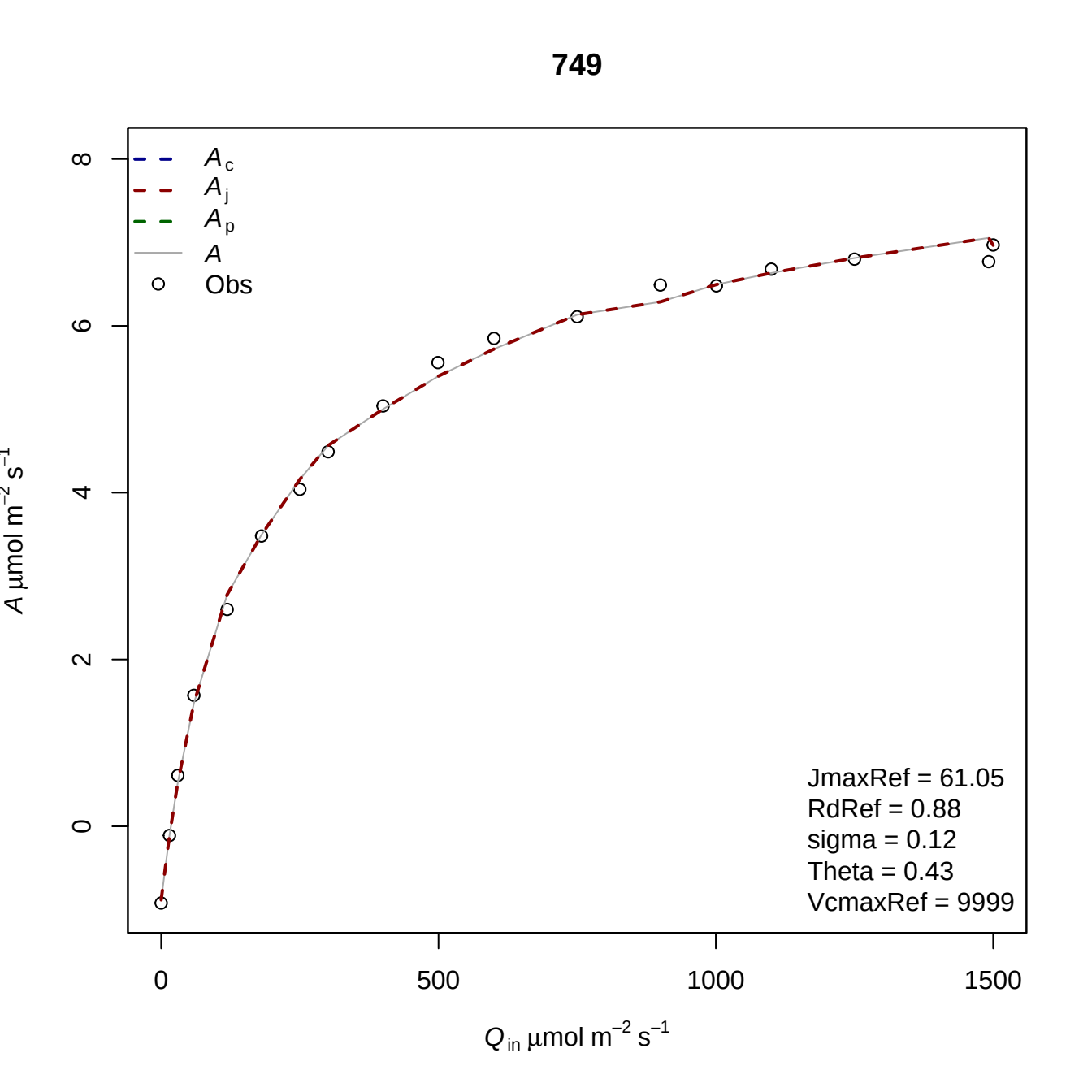

744
A_c
A_j
A_p
A
Obs
JmaxRef = 78.32
RdRef = 1.78
sigma = 0.13
Theta = 0.06
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200

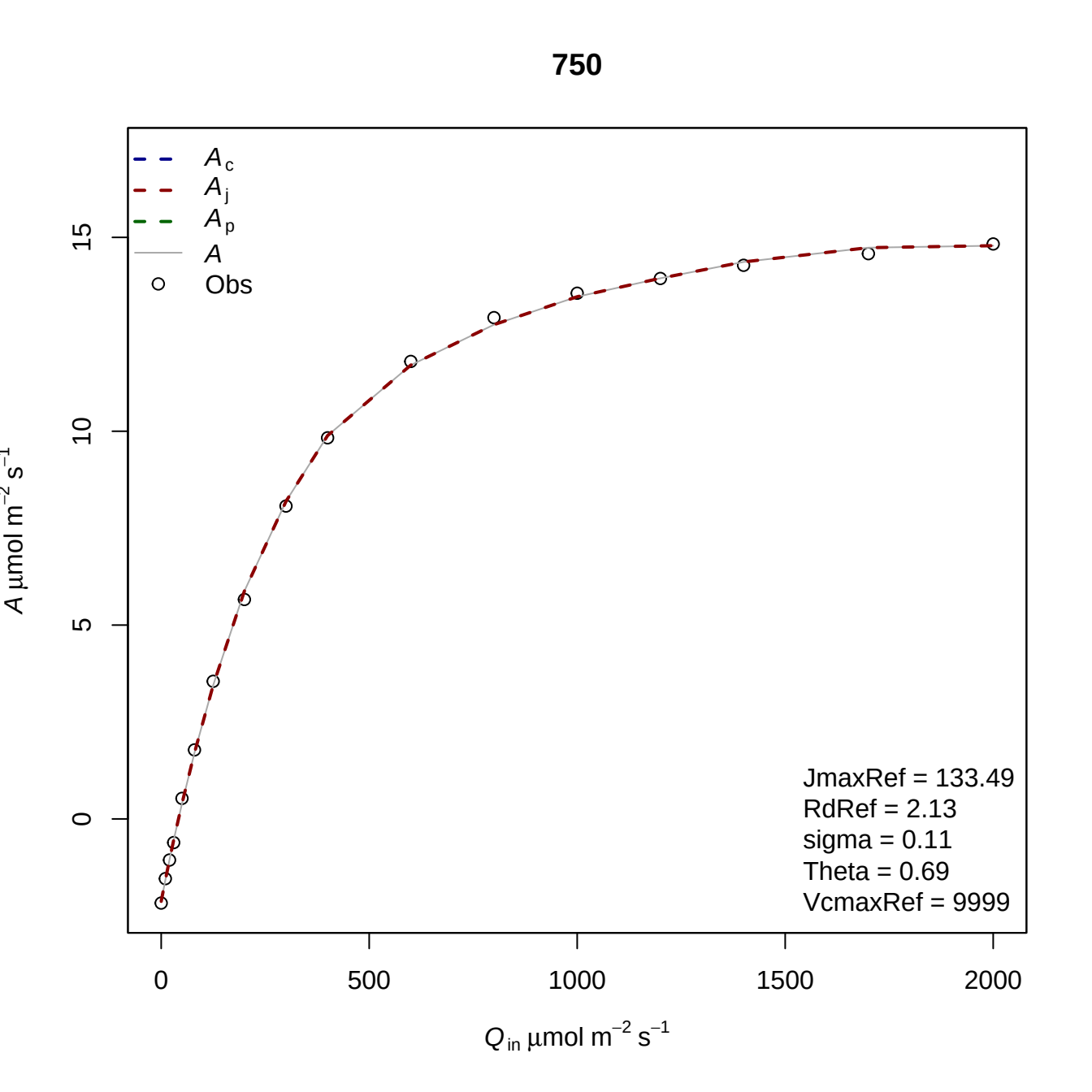

745
A_c
A_j
A_p
A
Obs
JmaxRef = 59.01
RdRef = 1.22
sigma = 0.18
Theta = 0.08
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

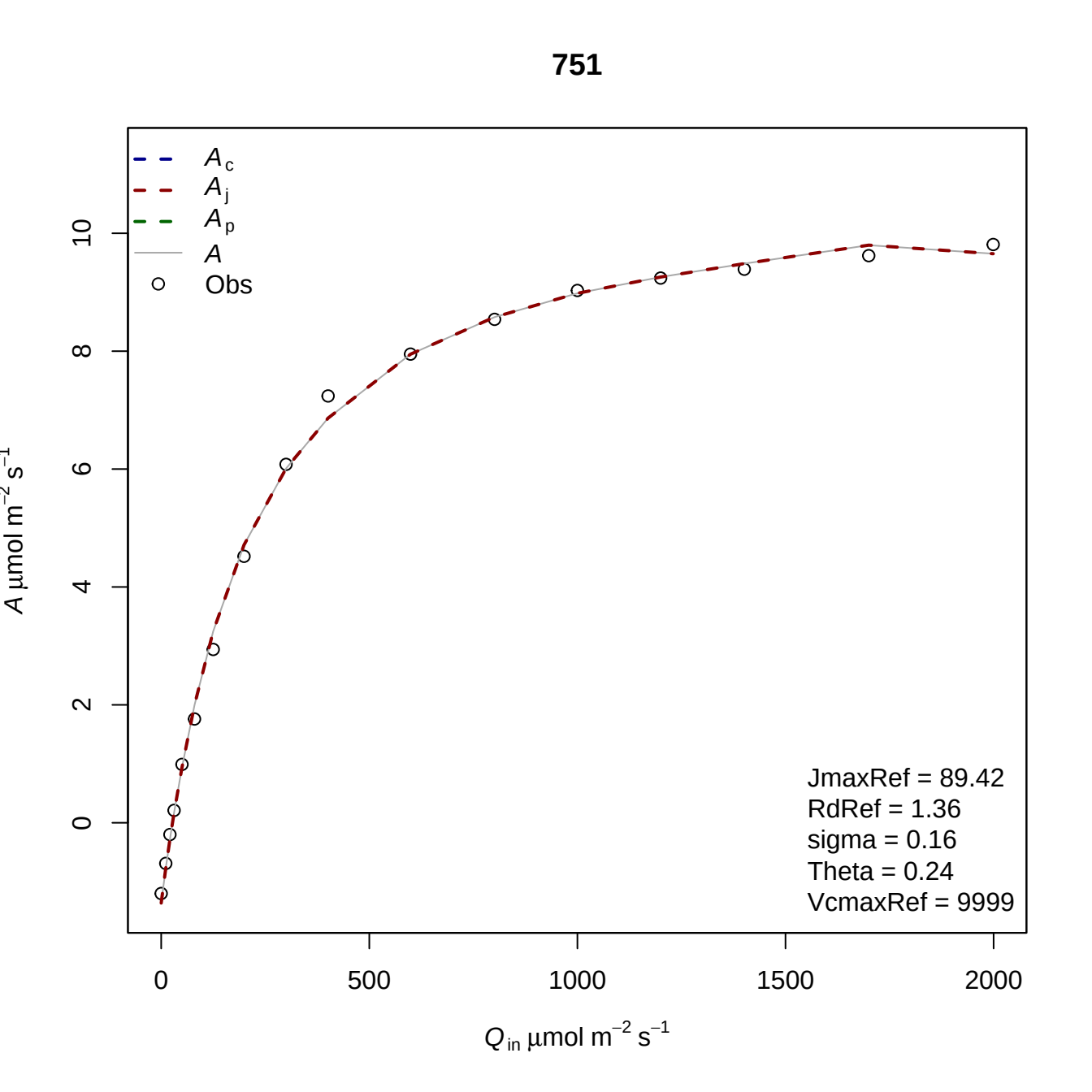

746
A_c
A_j
A_p
A
Obs
JmaxRef = 39.81
RdRef = 0.97
sigma = 0.12
Theta = 0.1
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

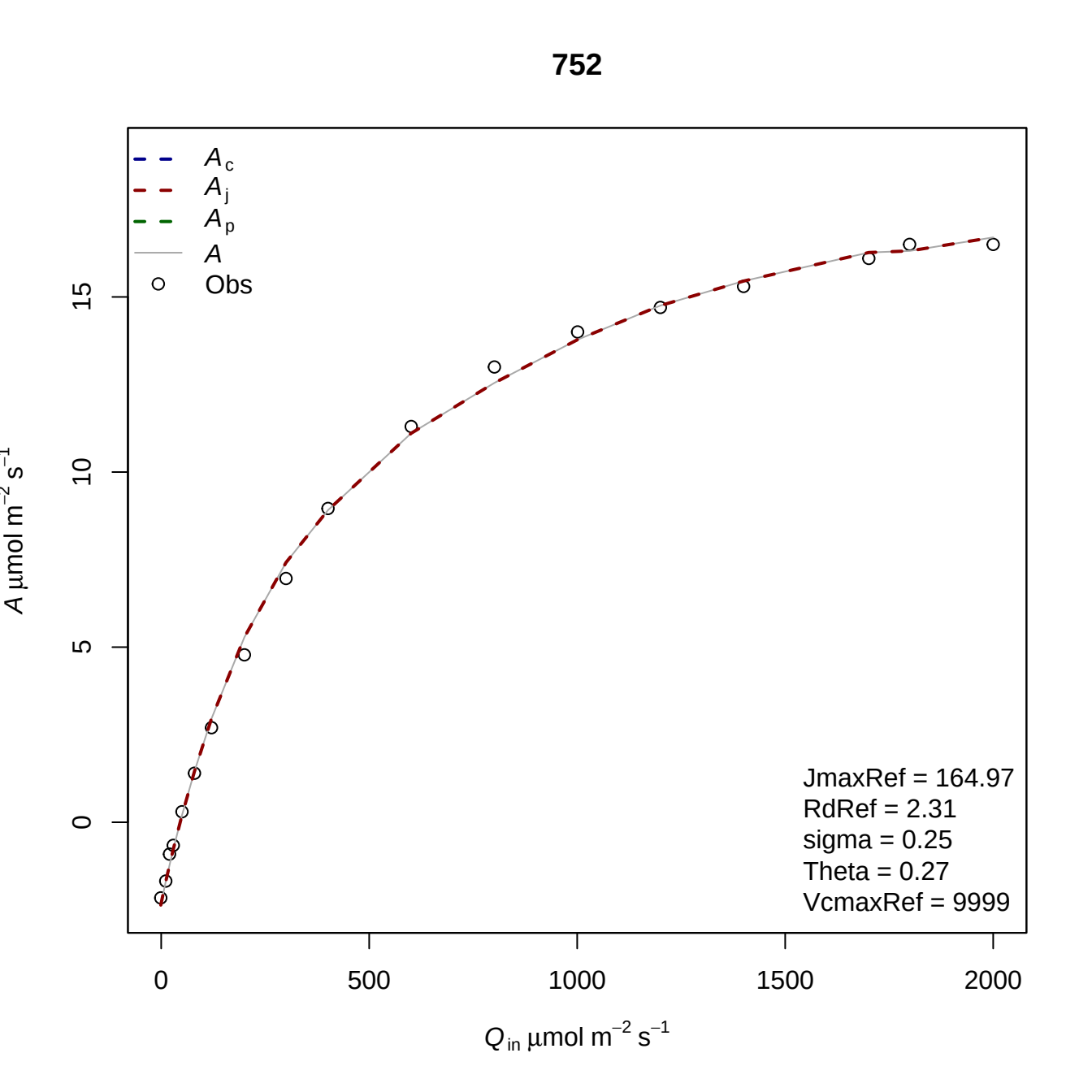

747
A_c
A_j
A_p
A
Obs
JmaxRef = 55.71
RdRef = 1.27
sigma = 0.2
Theta = −0.16
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

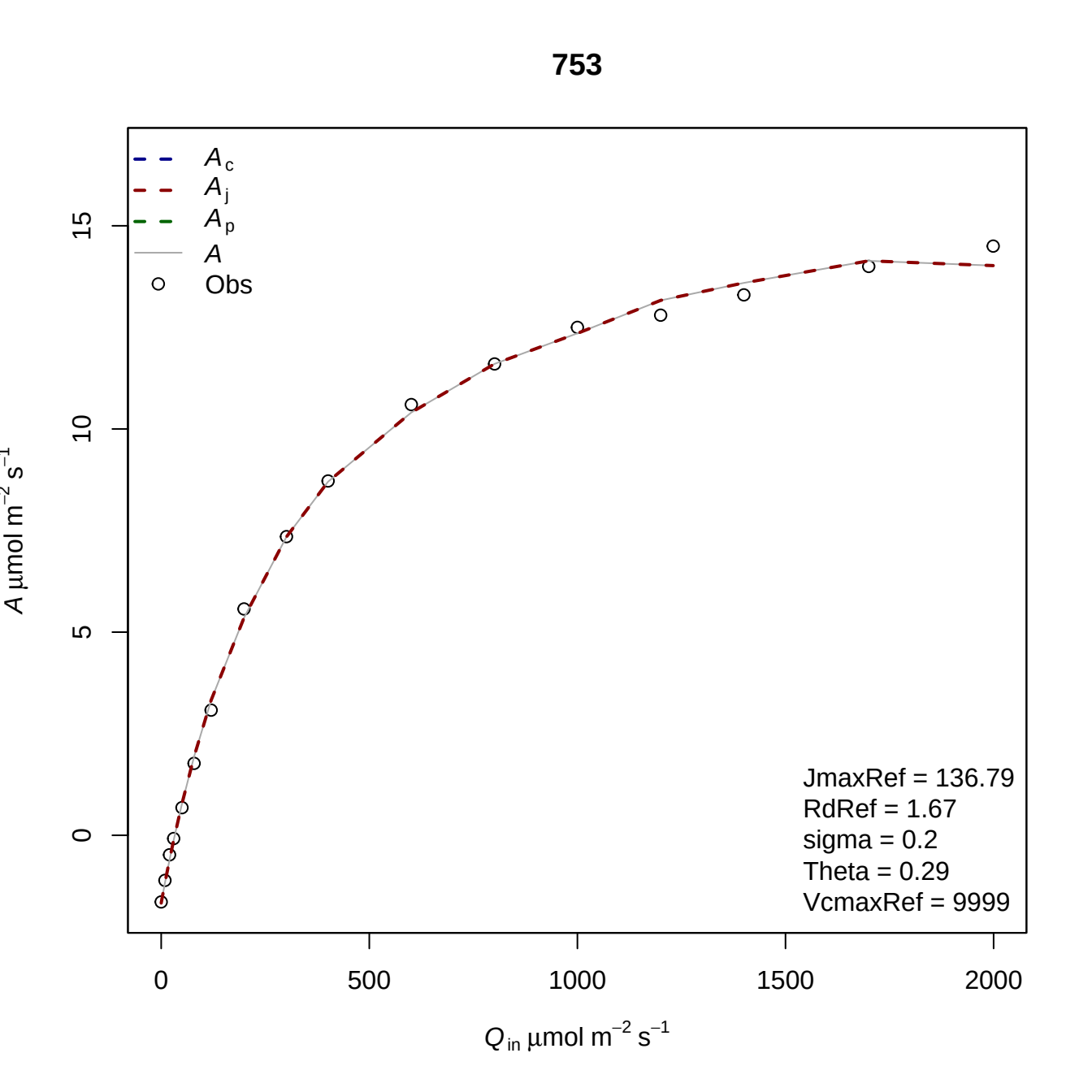

748
A_c
A_j
A_p
A
Obs
JmaxRef = 82.71
RdRef = 0.81
sigma = 0.1
Theta = 0.76
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

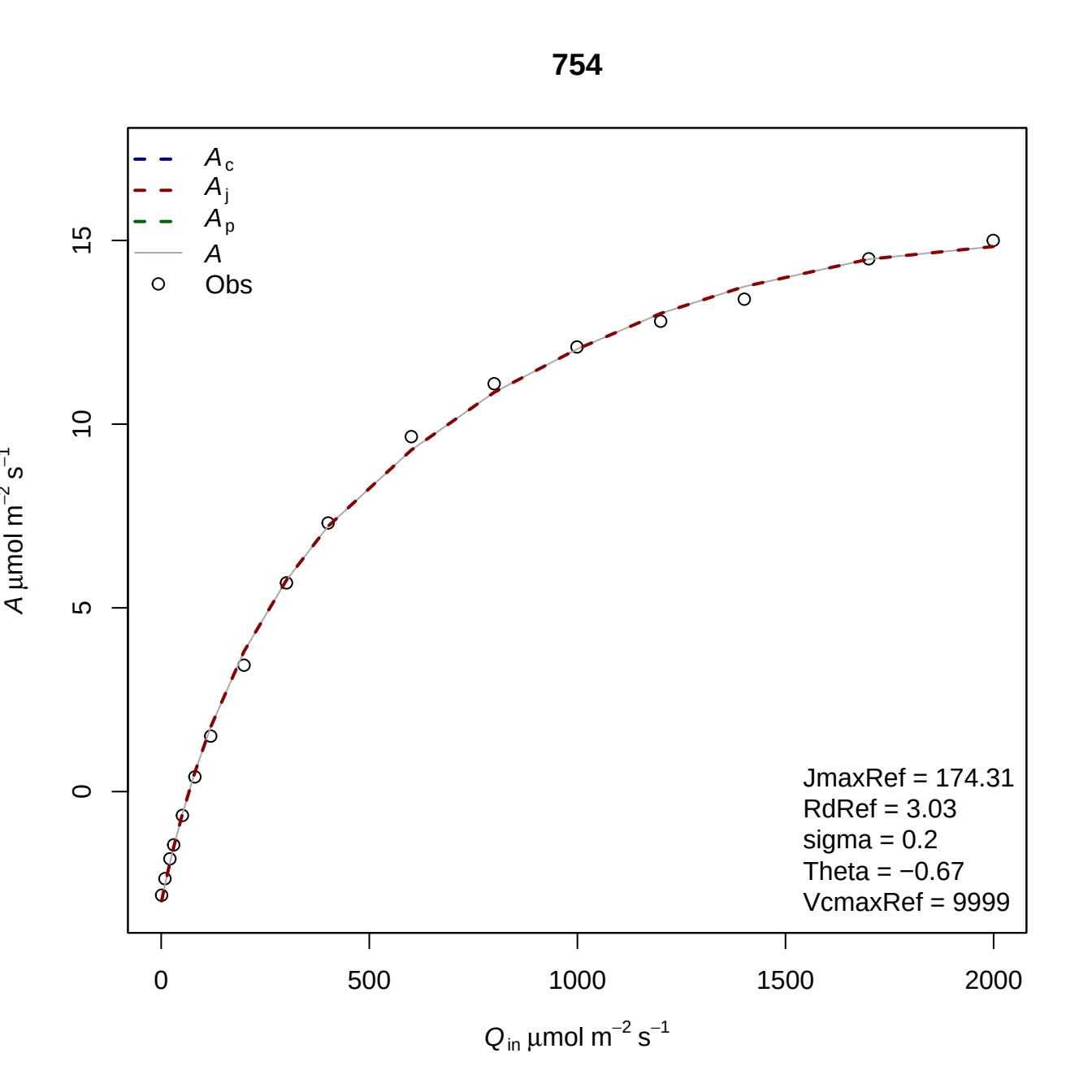

749
A_c
A_j
A_p
A
Obs
JmaxRef = 61.05
RdRef = 0.88
sigma = 0.12
Theta = 0.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

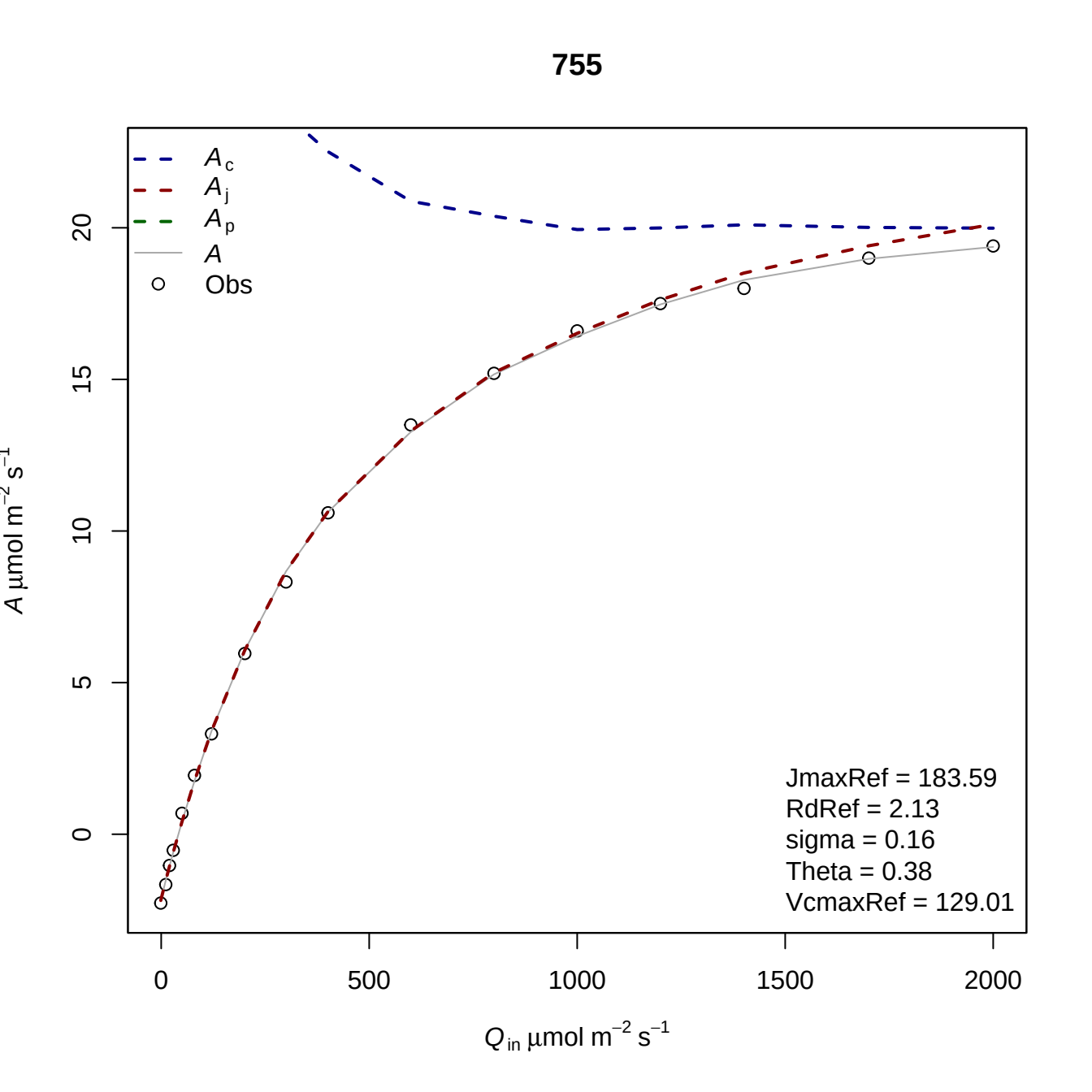

750
A_c
A_j
A_p
A
Obs
JmaxRef = 133.49
RdRef = 2.13
sigma = 0.11
Theta = 0.69
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

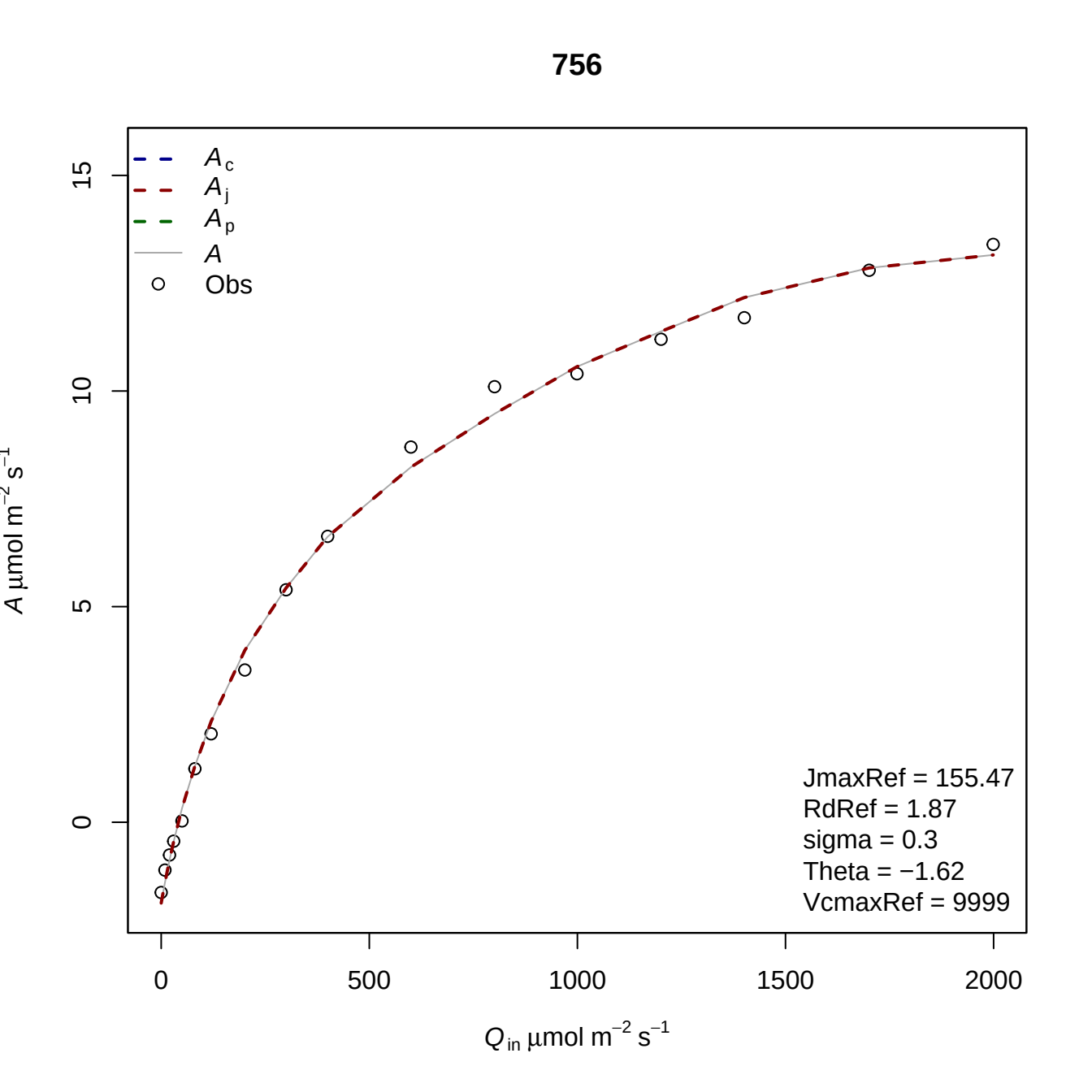

751
A_c
A_j
A_p
A
Obs
JmaxRef = 89.42
RdRef = 1.36
sigma = 0.16
Theta = 0.24
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

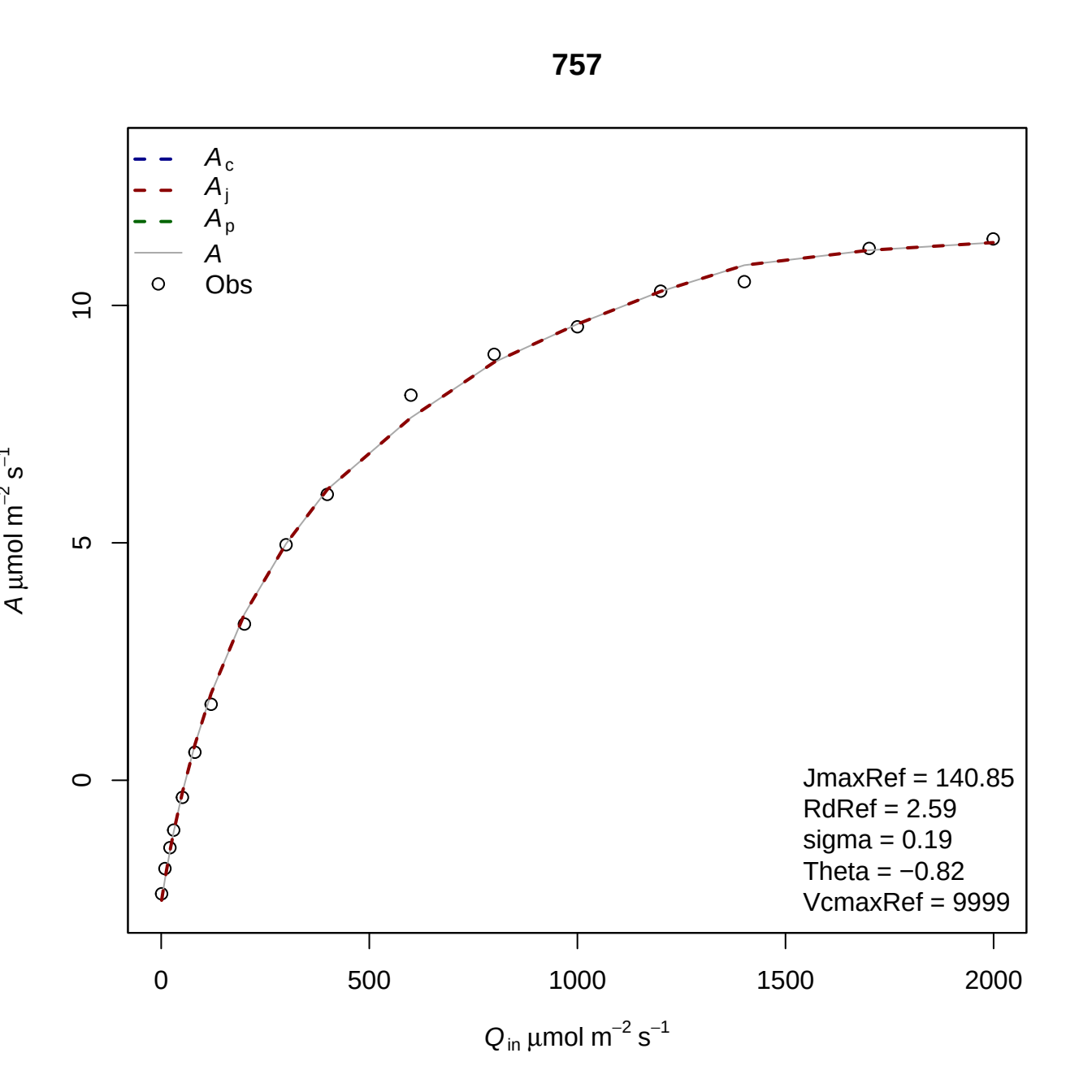

752
A_c
A_j
A_p
A
Obs
JmaxRef = 164.97
RdRef = 2.31
sigma = 0.25
Theta = 0.27
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

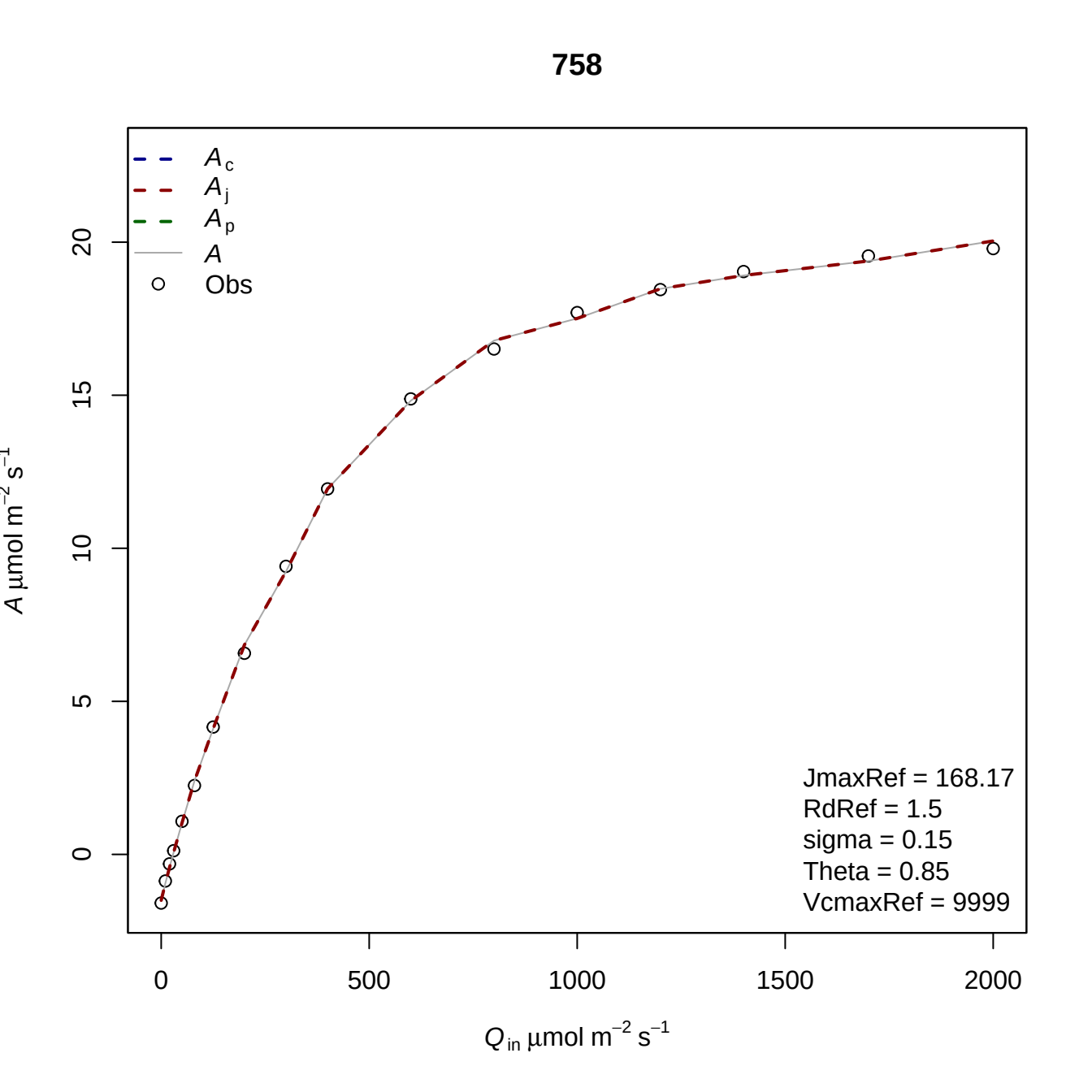

753
A_c
A_j
A_p
A
Obs
JmaxRef = 136.79
RdRef = 1.67
sigma = 0.2
Theta = 0.29
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

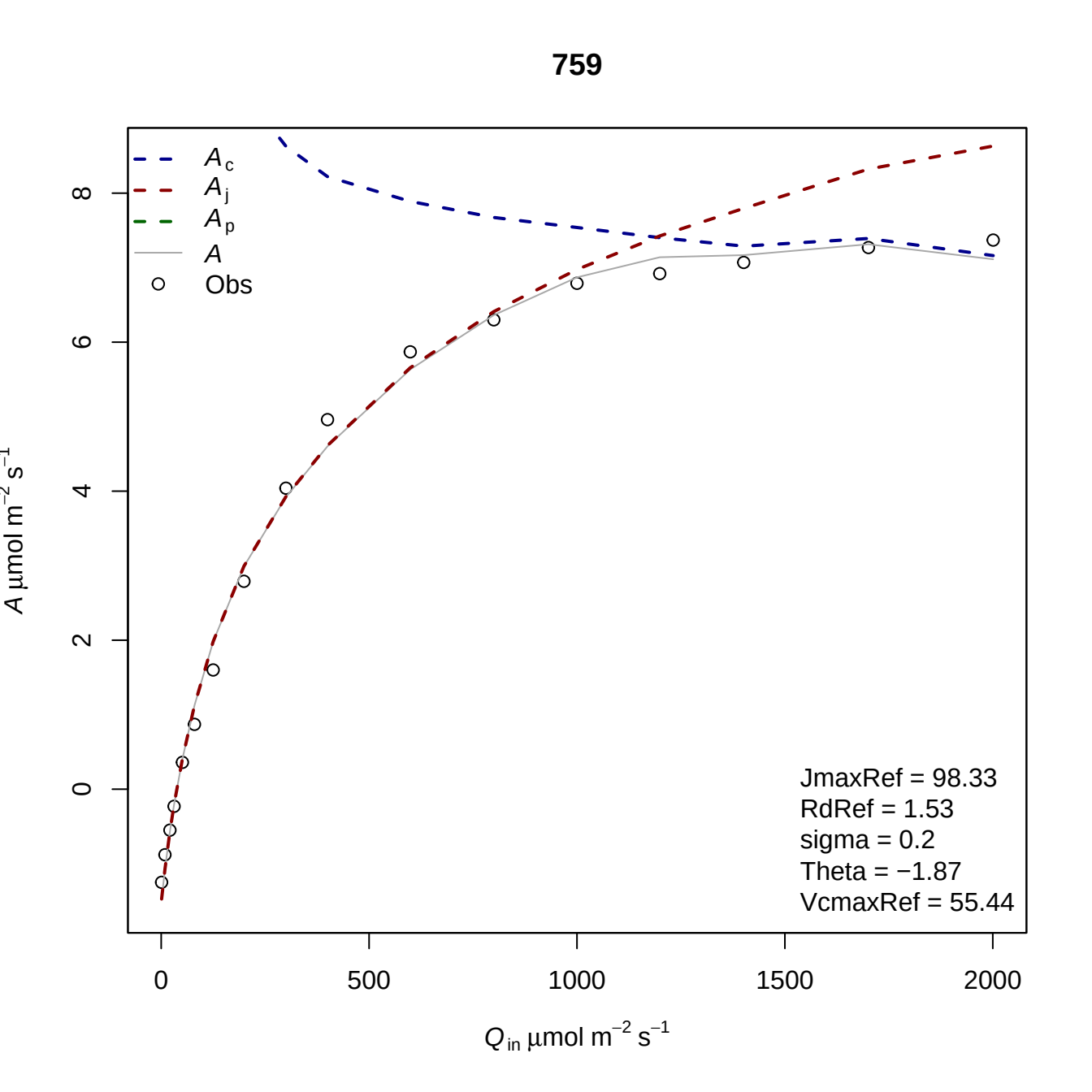

754
A_c
A_j
A_p
A
Obs
JmaxRef = 174.31
RdRef = 3.03
sigma = 0.2
Theta = −0.67
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

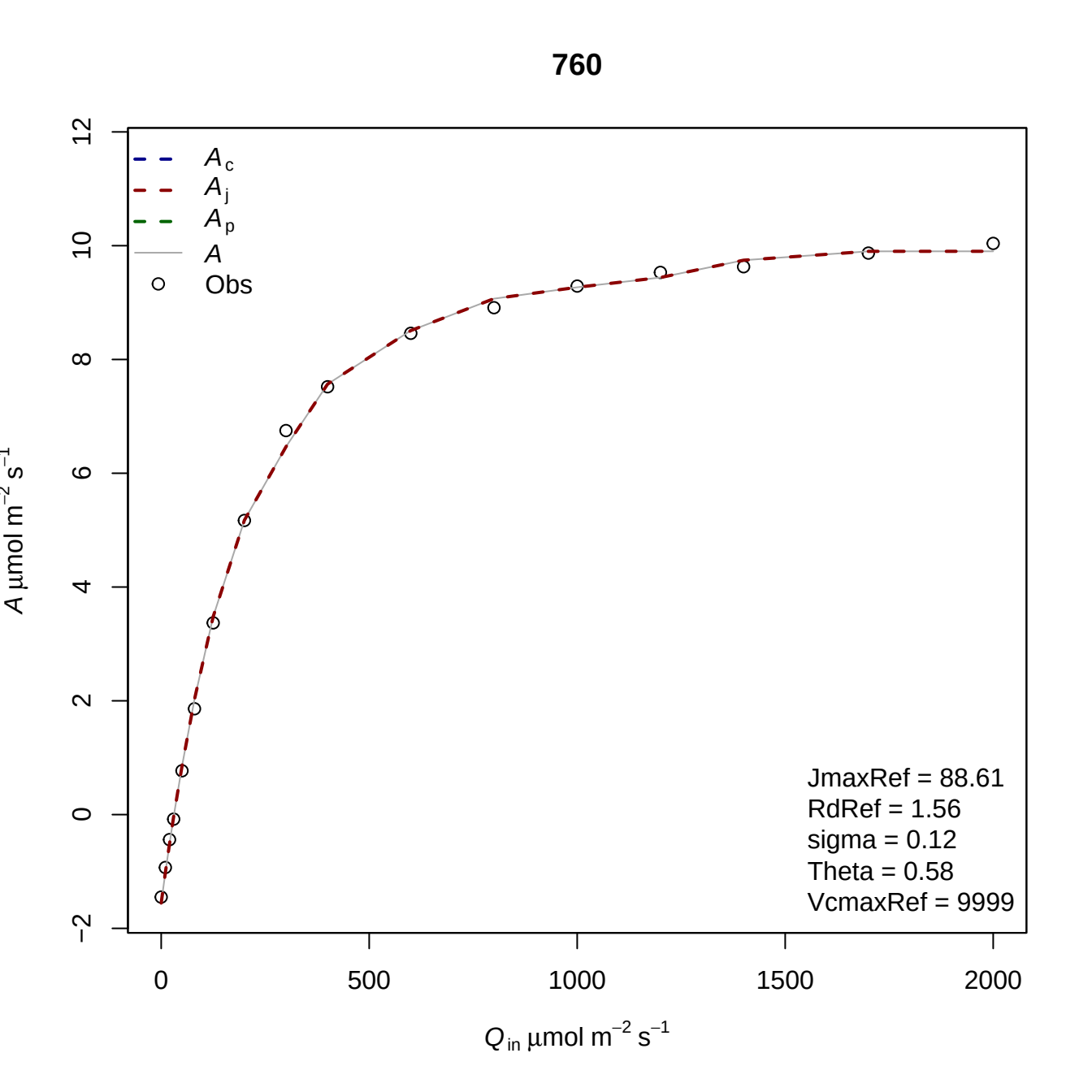

755
A_c
A_j
A_p
A
Obs
JmaxRef = 183.59
RdRef = 2.13
sigma = 0.16
Theta = 0.38
VcmaxRef = 129.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

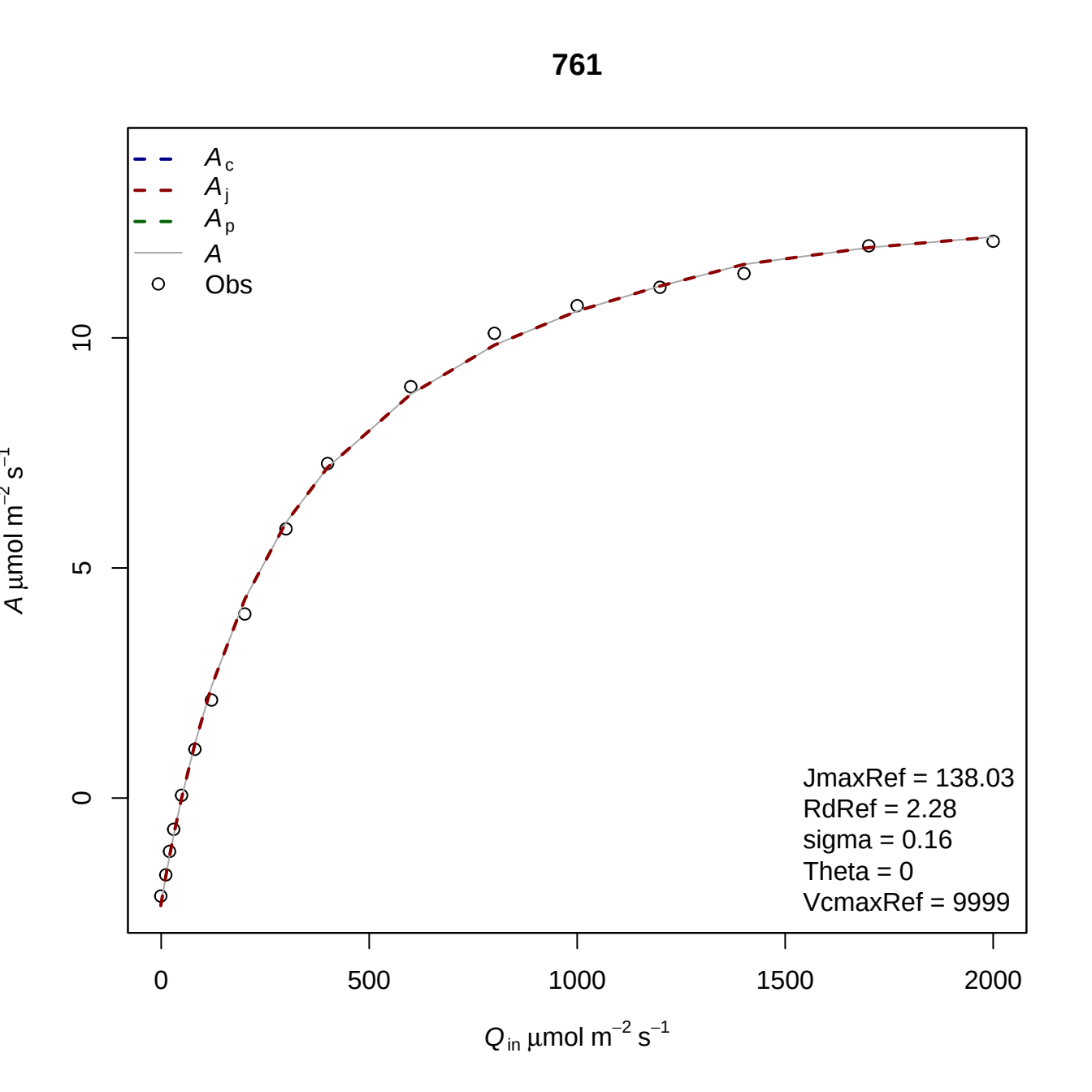

756
A_c
A_j
A_p
A
Obs
JmaxRef = 155.47
RdRef = 1.87
sigma = 0.3
Theta = −1.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

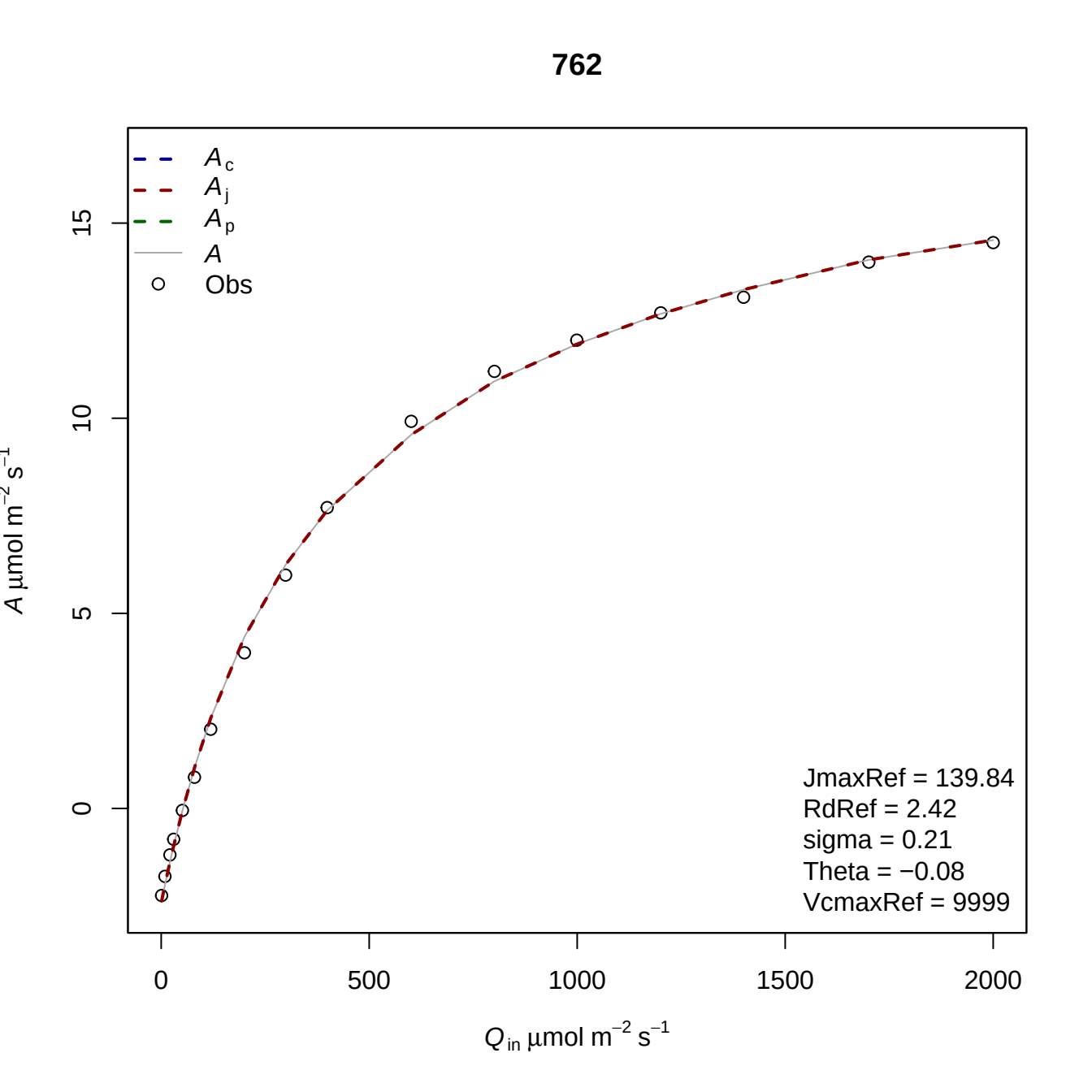

757
A_c
A_j
A_p
A
Obs
JmaxRef = 140.85
RdRef = 2.59
sigma = 0.19
Theta = −0.82
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

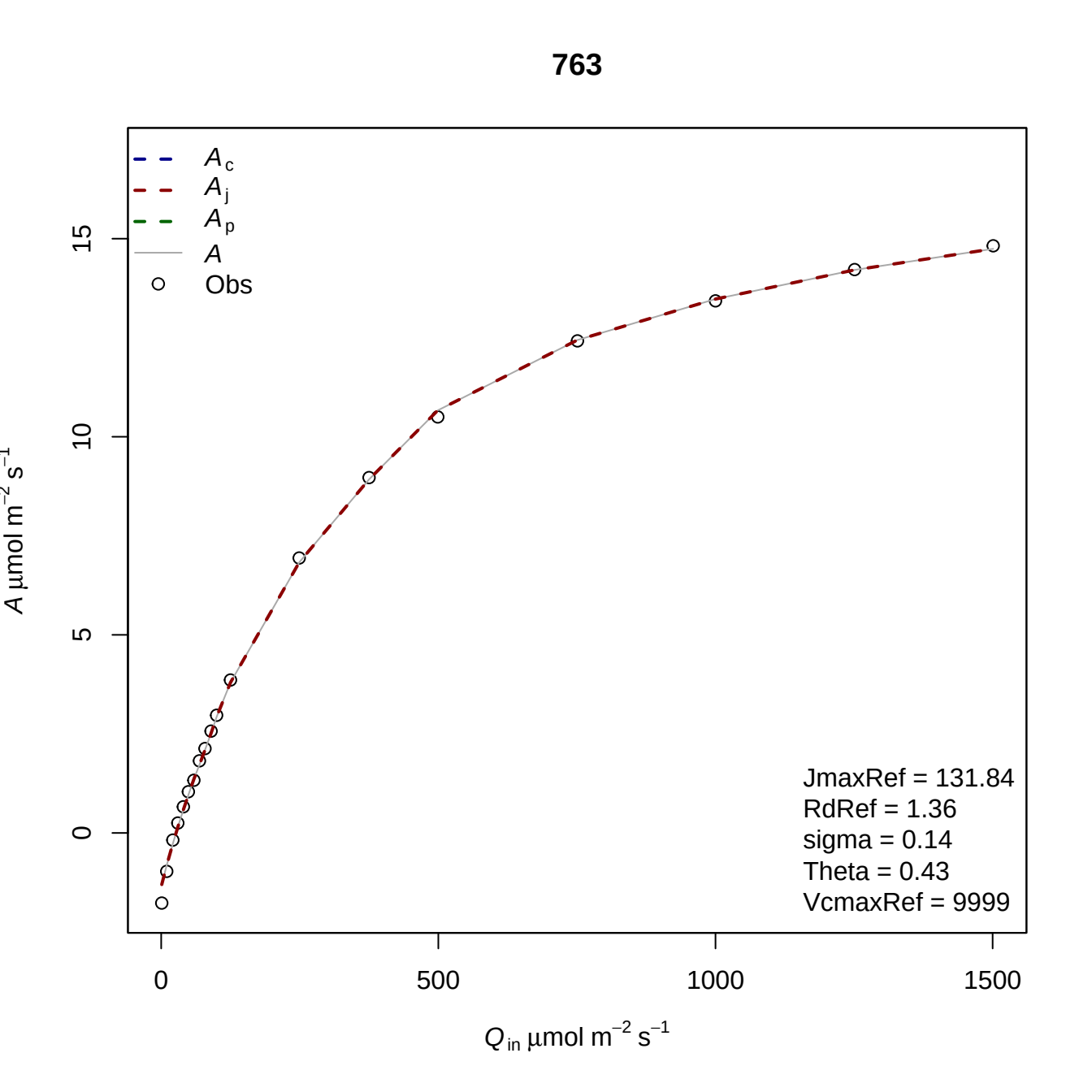

758
A_c
A_j
A_p
A
Obs
JmaxRef = 168.17
RdRef = 1.5
sigma = 0.15
Theta = 0.85
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

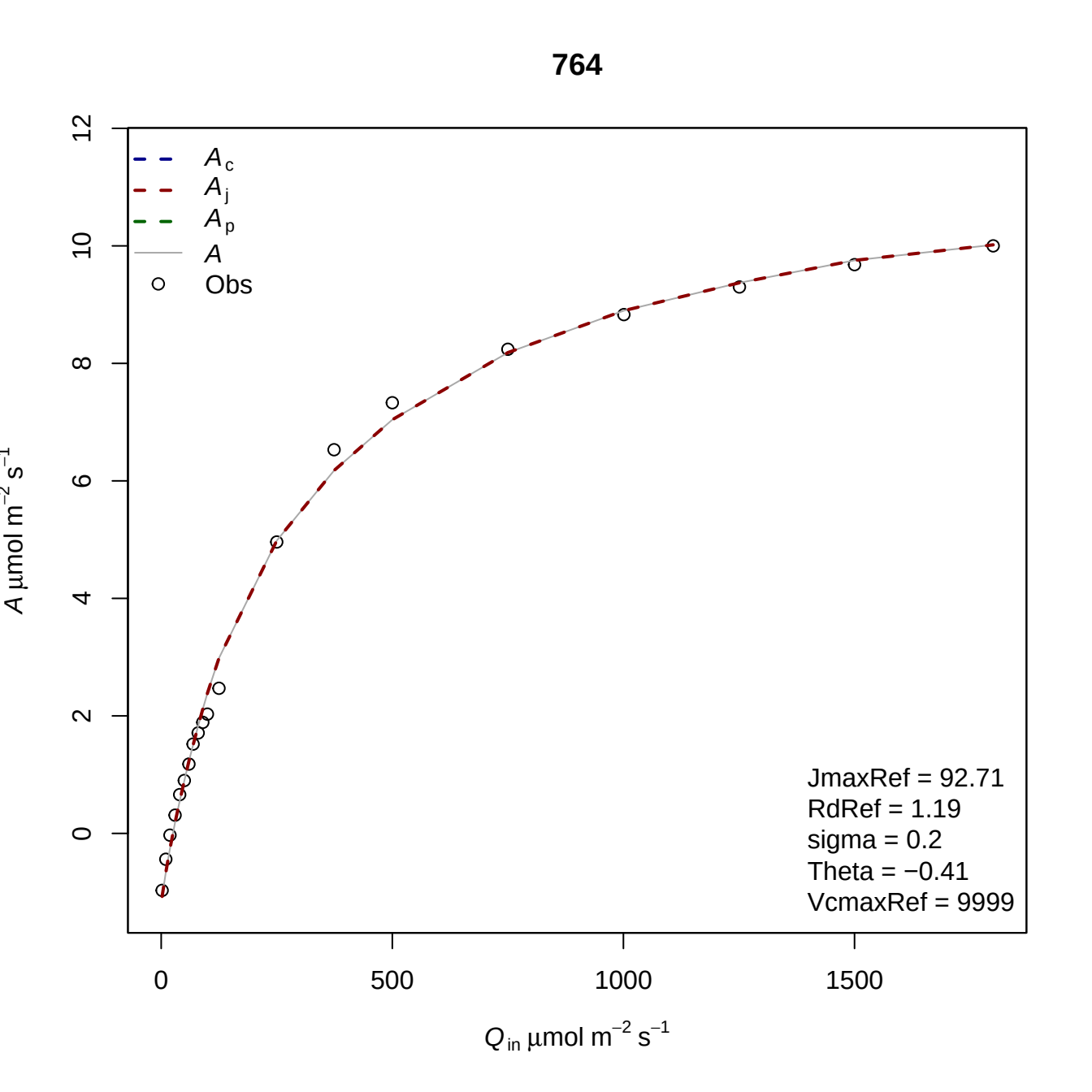

759
A_c
A_j
A_p
A
Obs
JmaxRef = 98.33
RdRef = 1.53
sigma = 0.2
Theta = −1.87
VcmaxRef = 55.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

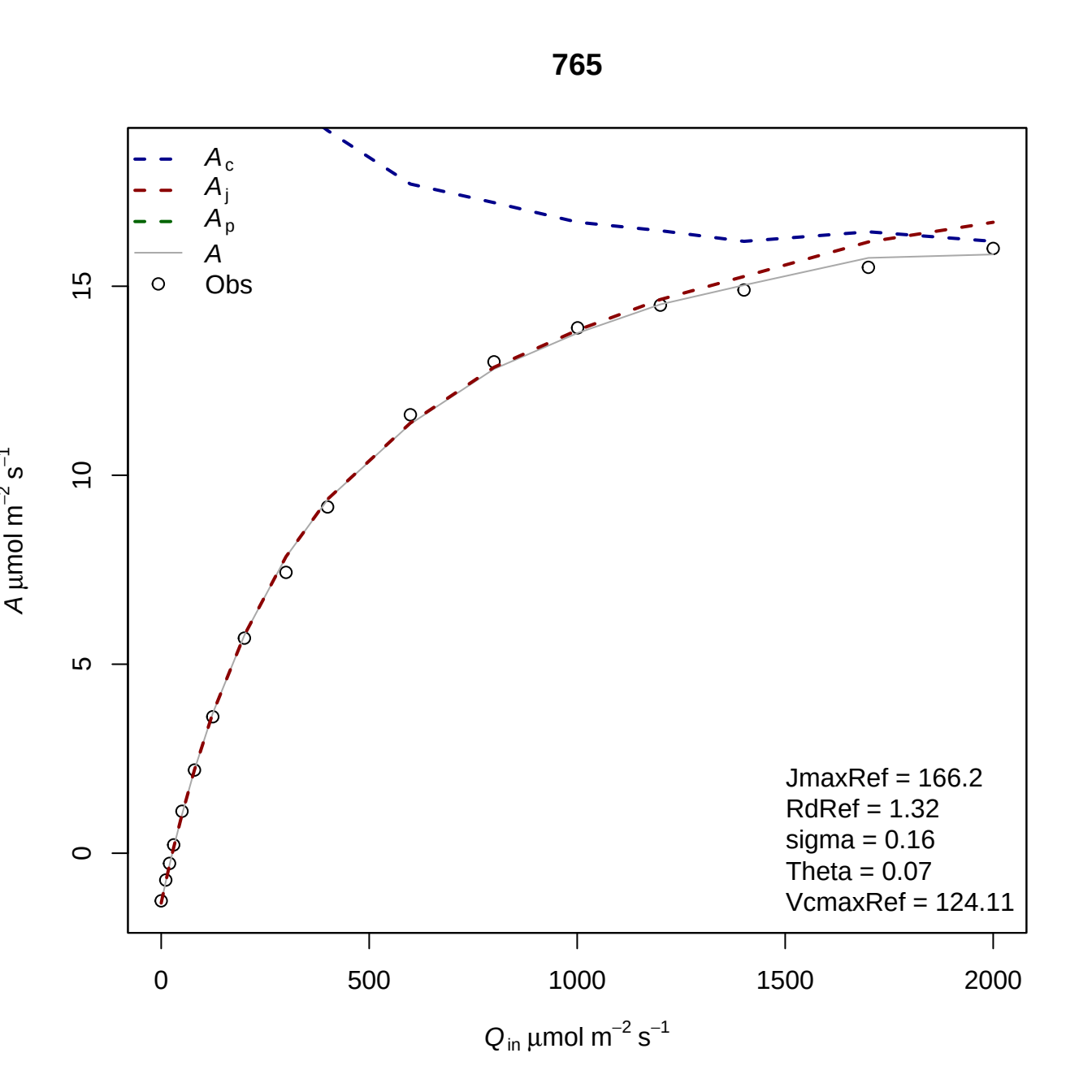

760
A_c
A_j
A_p
A
Obs
JmaxRef = 88.61
RdRef = 1.56
sigma = 0.12
Theta = 0.58
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

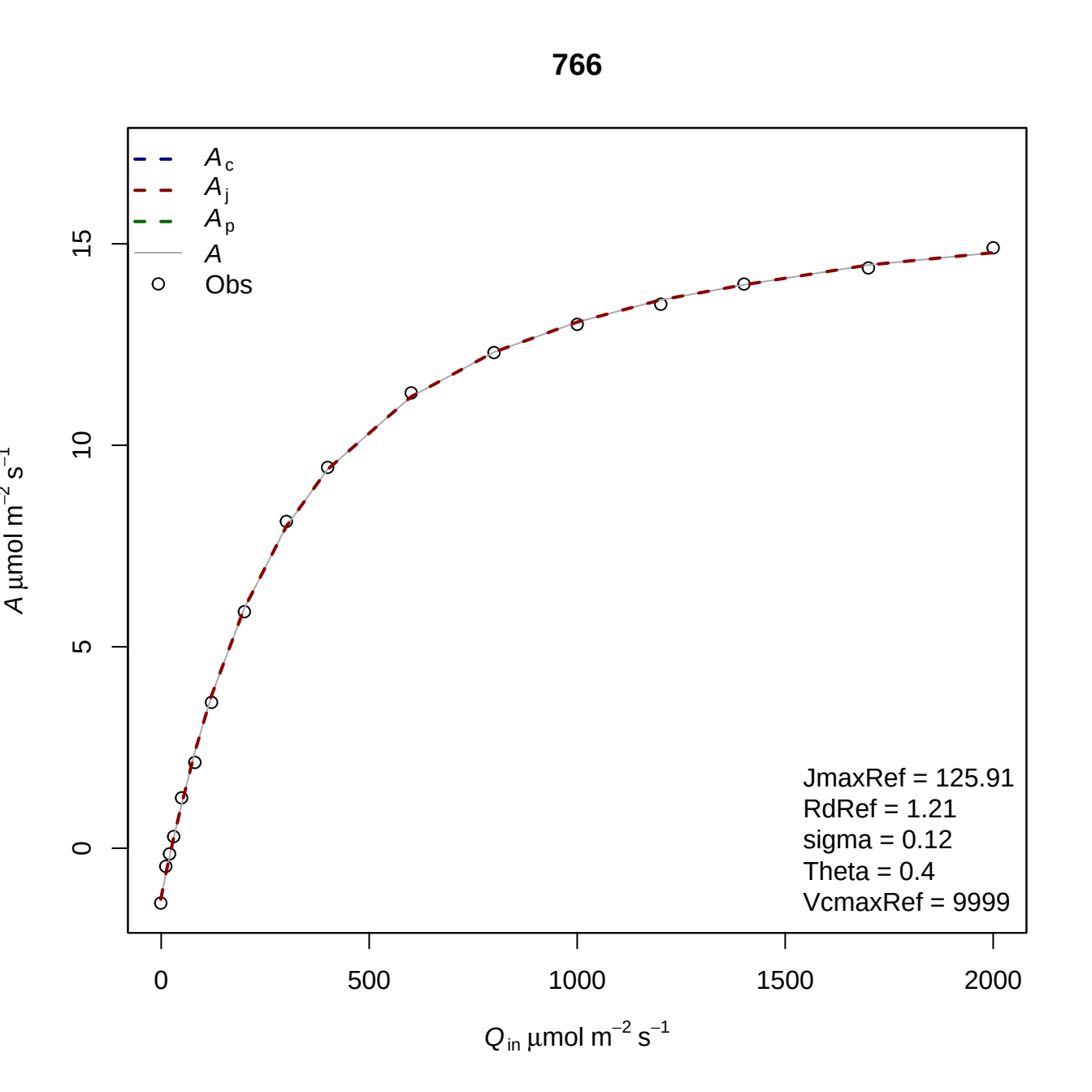

761
A_c
A_j
A_p
A
Obs
JmaxRef = 138.03
RdRef = 2.28
sigma = 0.16
Theta = 0
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

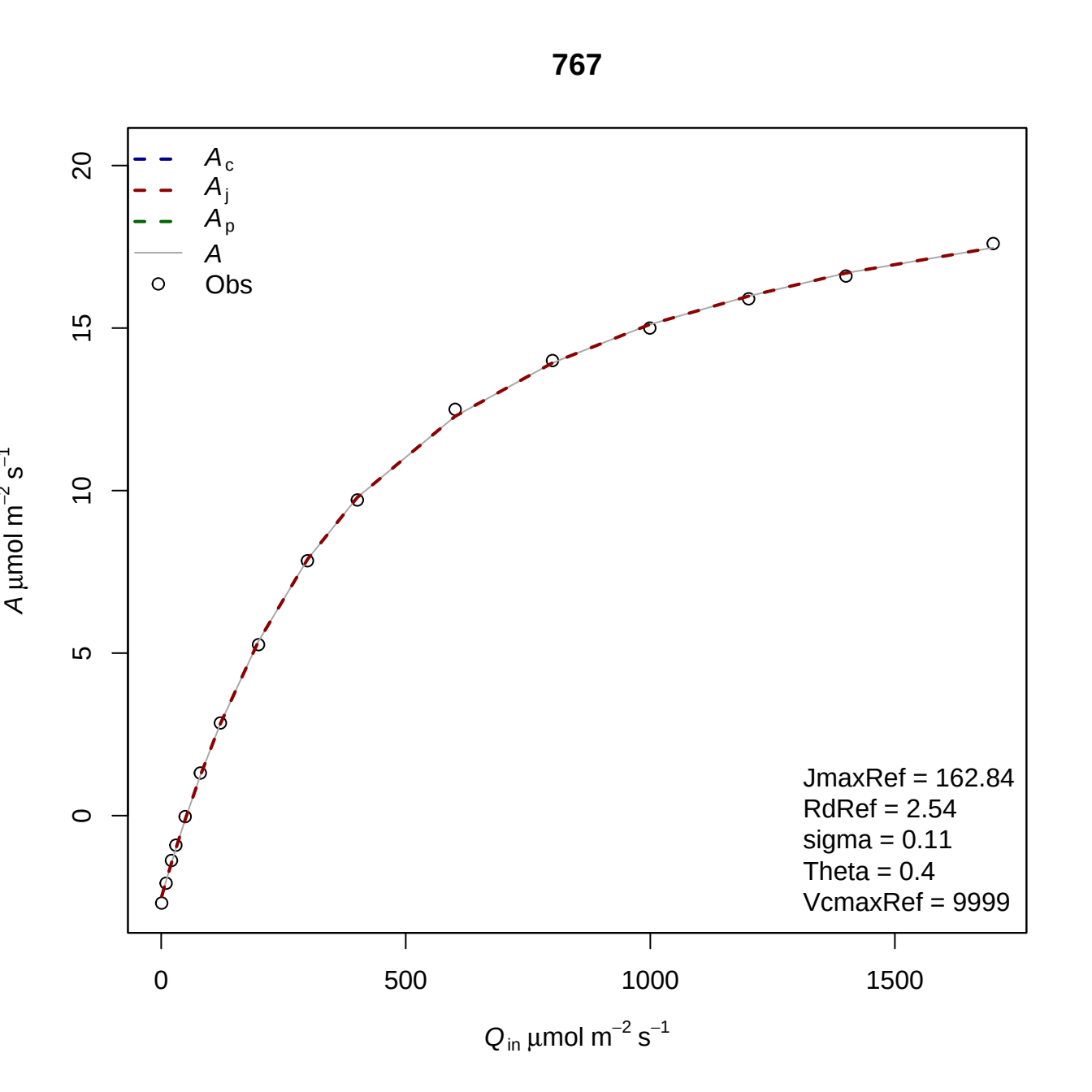

762
A_c
A_j
A_p
A
Obs
JmaxRef = 139.84
RdRef = 2.42
sigma = 0.21
Theta = −0.08
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

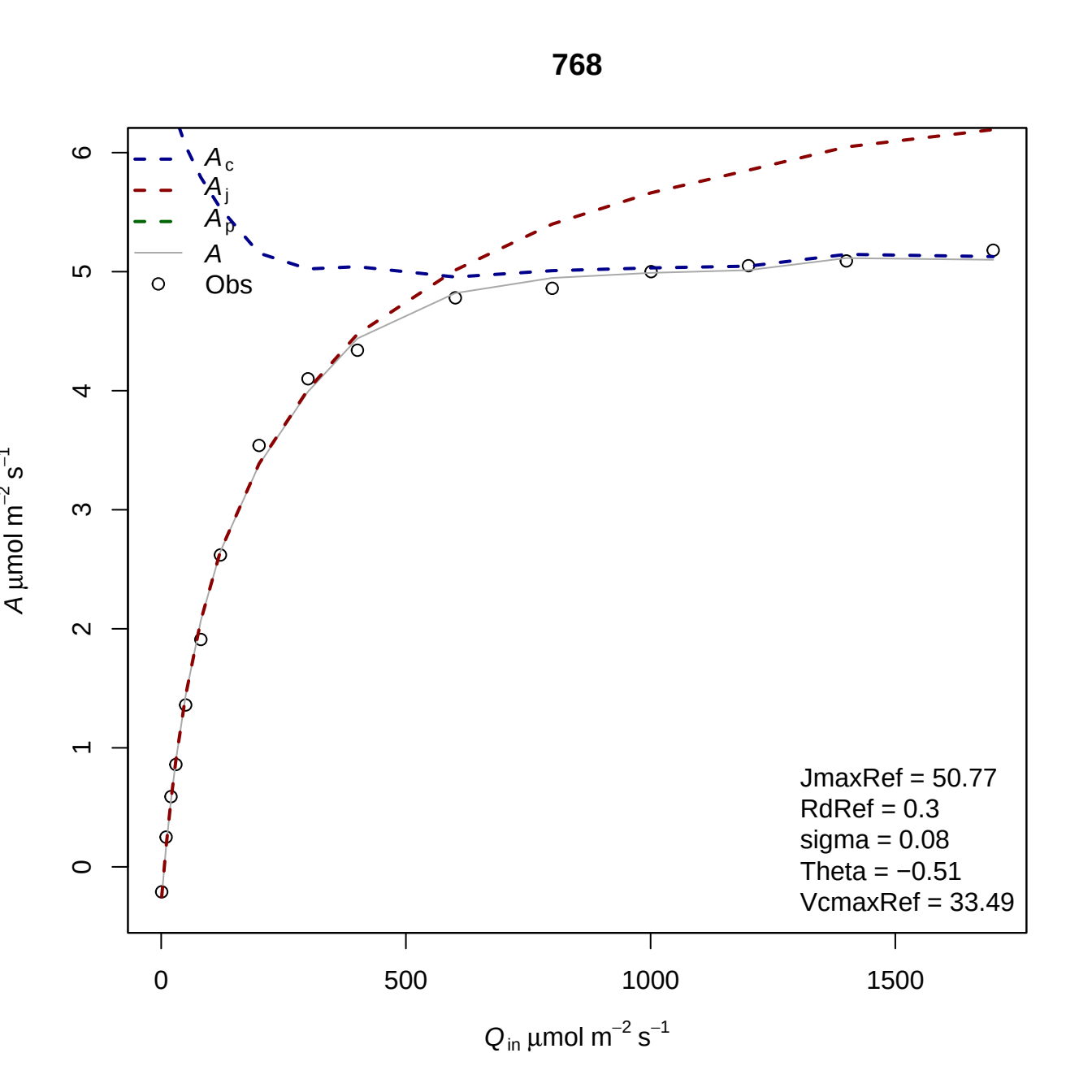

763
A_c
A_j
A_p
A
Obs
JmaxRef = 131.84
RdRef = 1.36
sigma = 0.14
Theta = 0.43
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

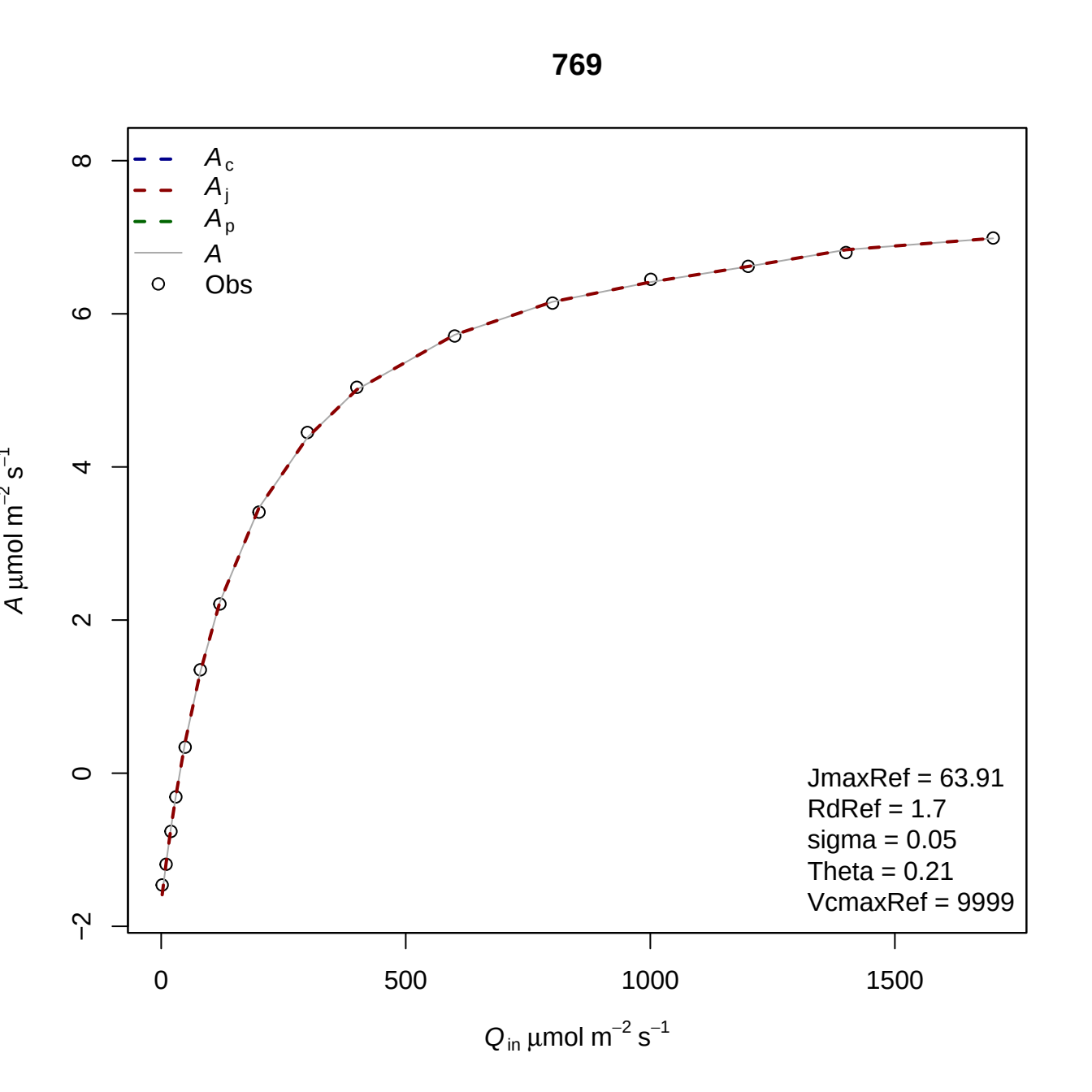

764
A_c
A_j
A_p
A
Obs
JmaxRef = 92.71
RdRef = 1.19
sigma = 0.2
Theta = −0.41
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

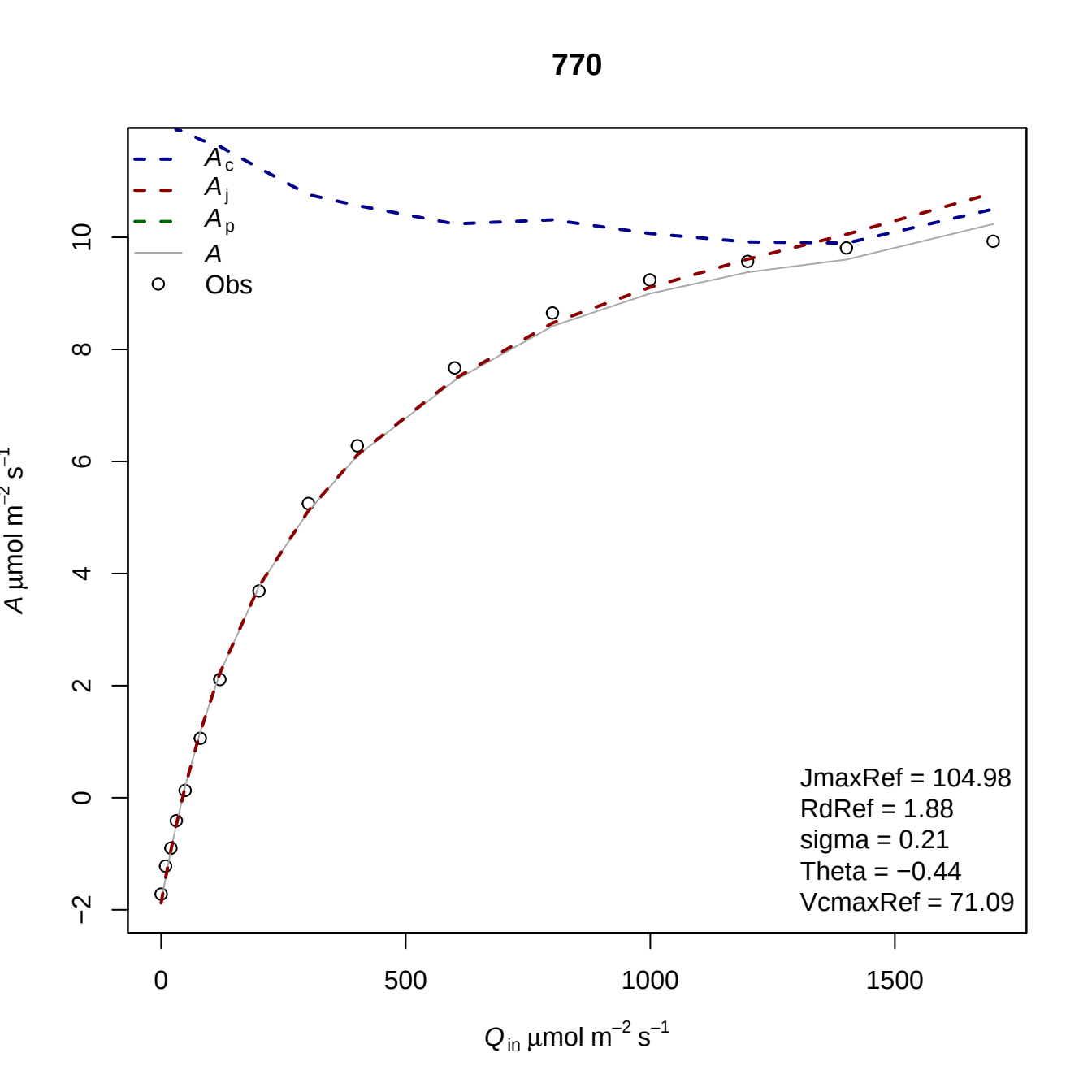

765
A_c
A_j
A_p
A
Obs
JmaxRef = 166.2
RdRef = 1.32
sigma = 0.16
Theta = 0.07
VcmaxRef = 124.11
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

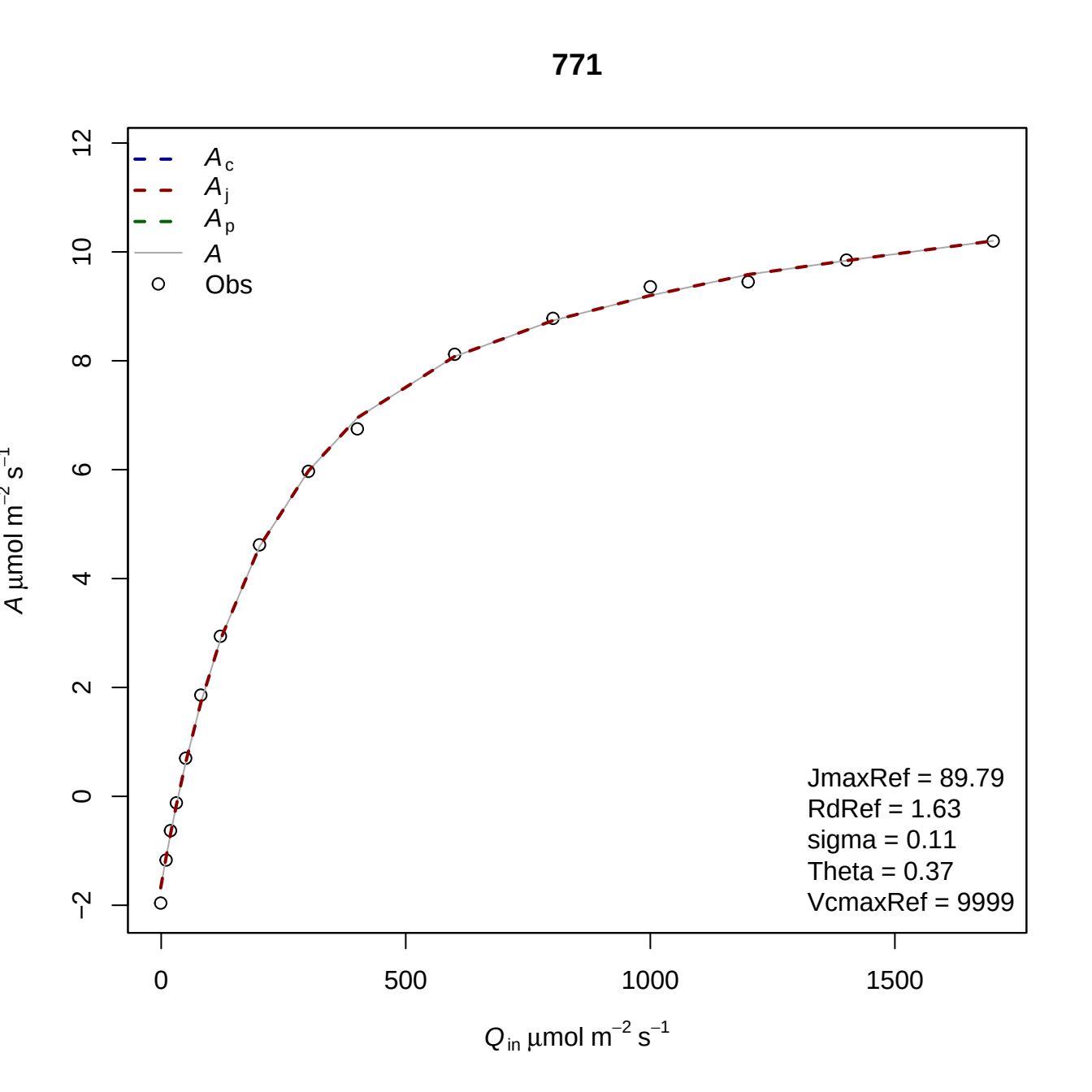

766
A_c
A_j
A_p
A
Obs
JmaxRef = 125.91
RdRef = 1.21
sigma = 0.12
Theta = 0.4
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

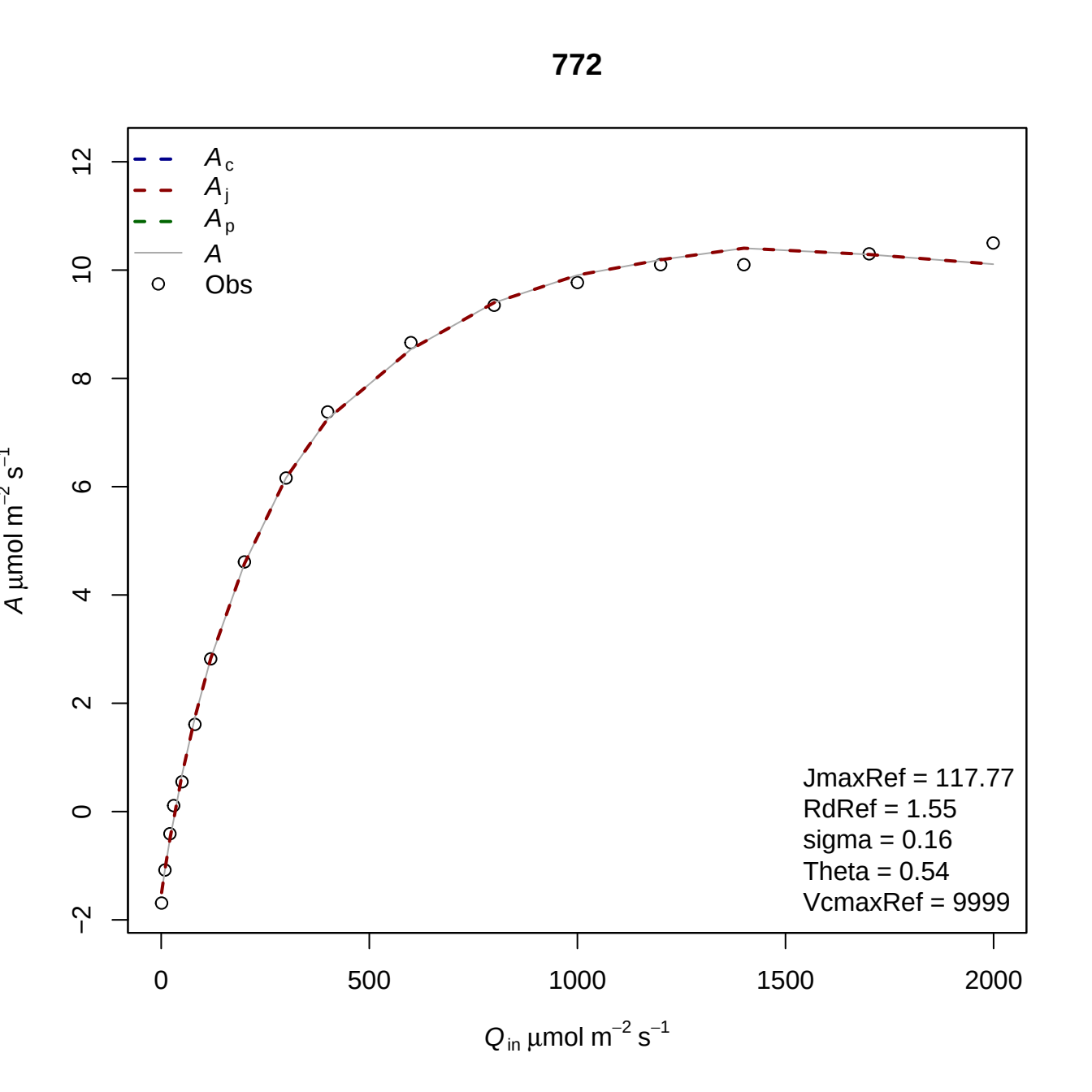

767
A_c
A_j
A_p
A
Obs
JmaxRef = 162.84
RdRef = 2.54
sigma = 0.11
Theta = 0.4
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

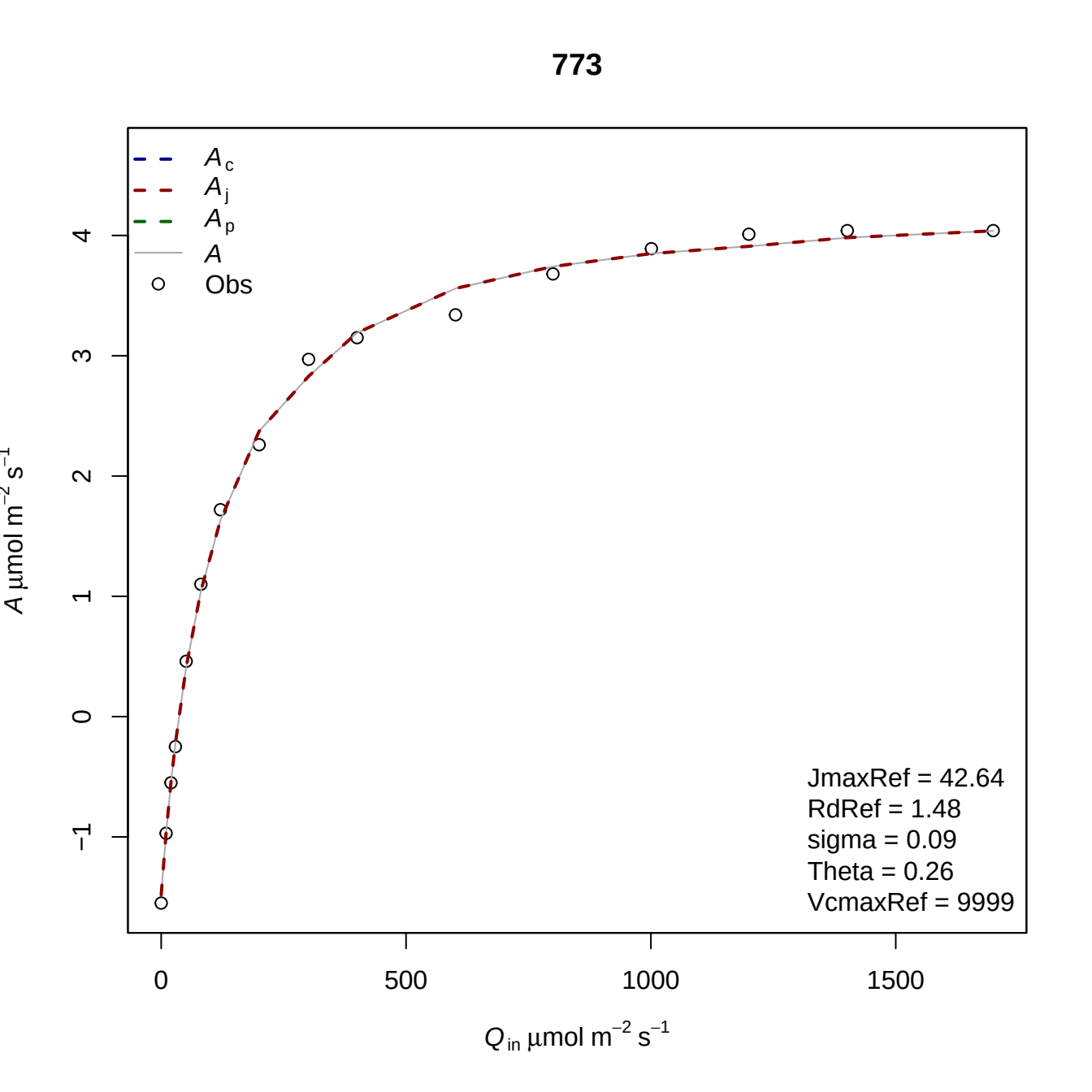

768
A_c
A_j
A_p
A
Obs
JmaxRef = 50.77
RdRef = 0.3
sigma = 0.08
Theta = −0.51
VcmaxRef = 33.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

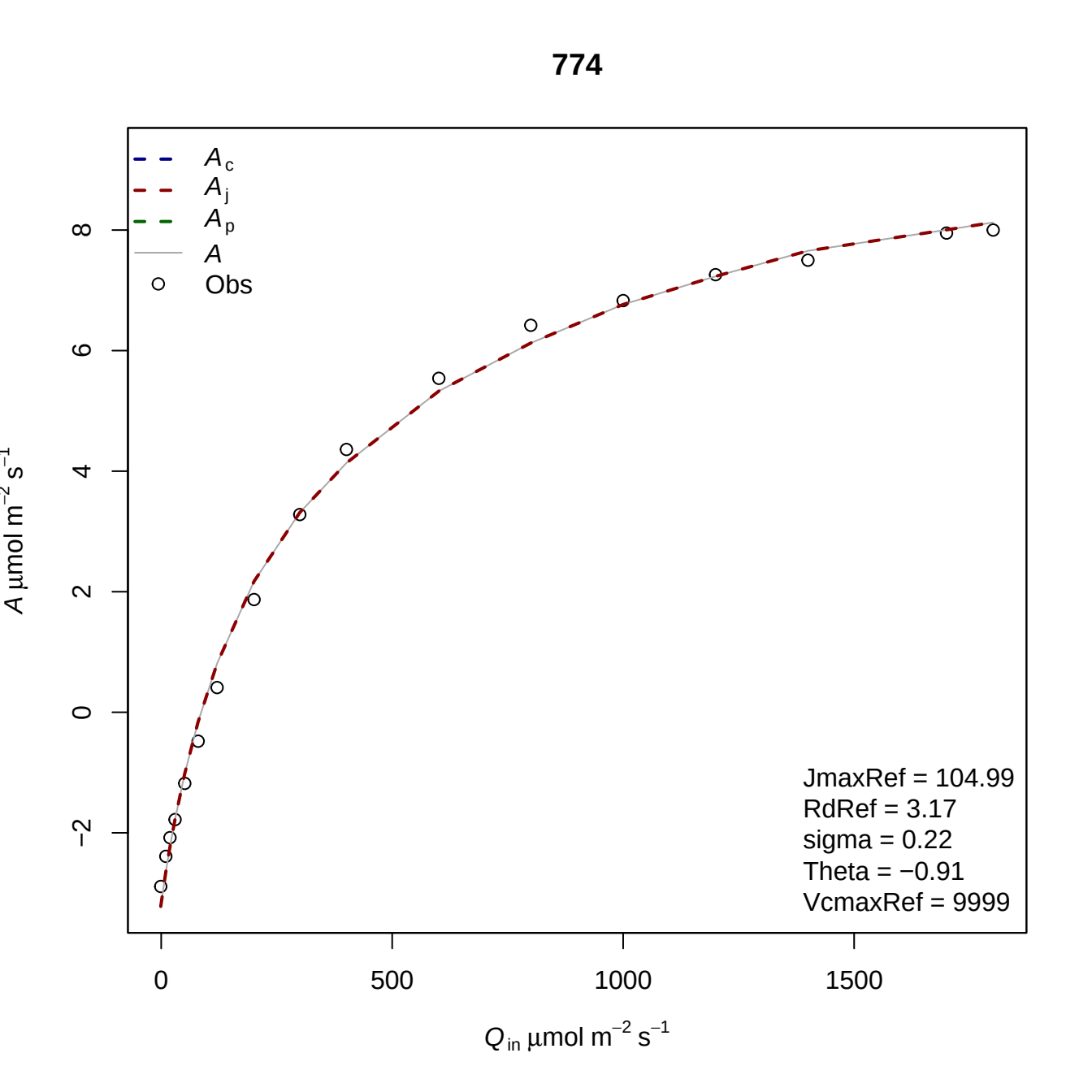

769
A_c
A_j
A_p
A
Obs
JmaxRef = 63.91
RdRef = 1.7
sigma = 0.05
Theta = 0.21
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

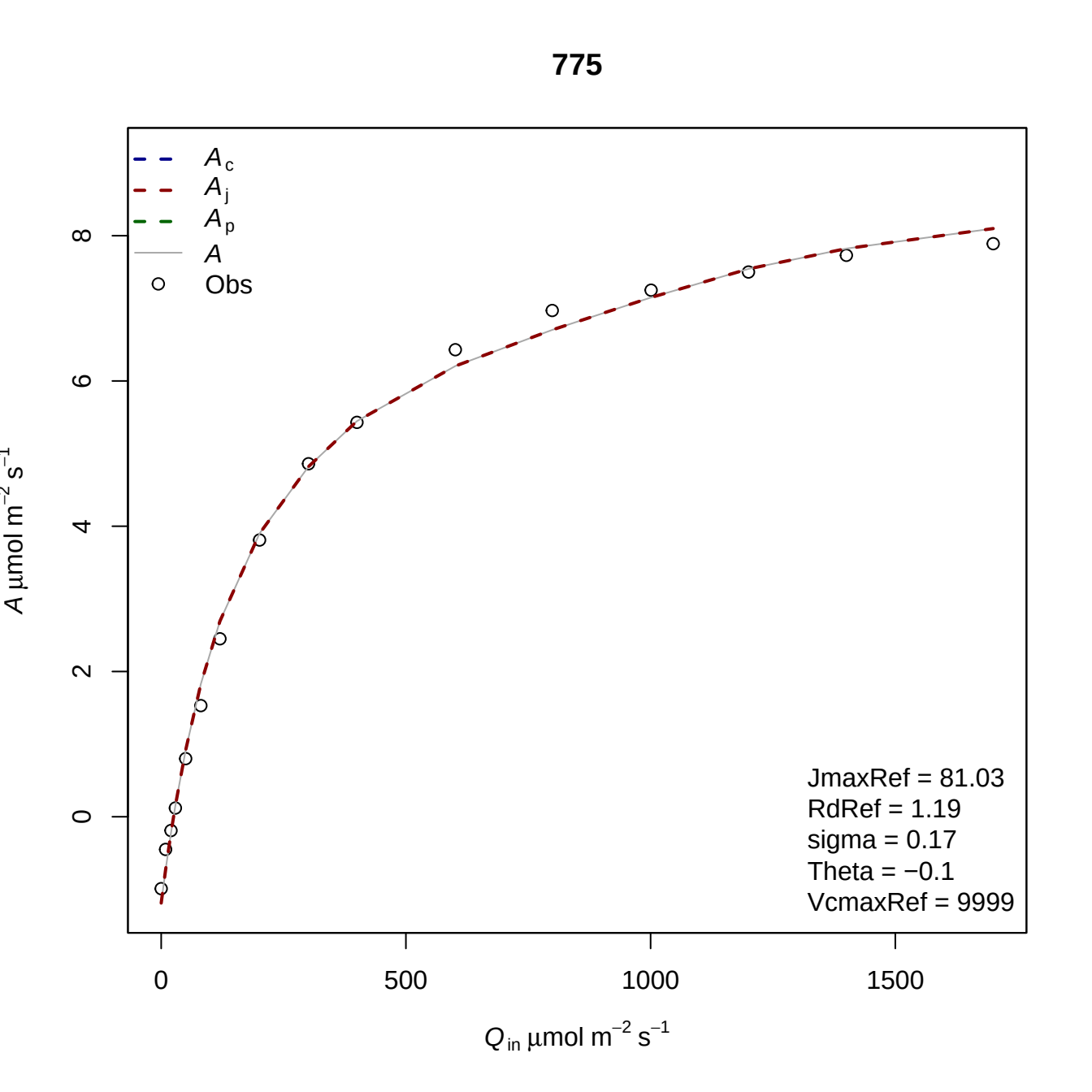

770
A_c
A_j
A_p
A
Obs
JmaxRef = 104.98
RdRef = 1.88
sigma = 0.21
Theta = −0.44
VcmaxRef = 71.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

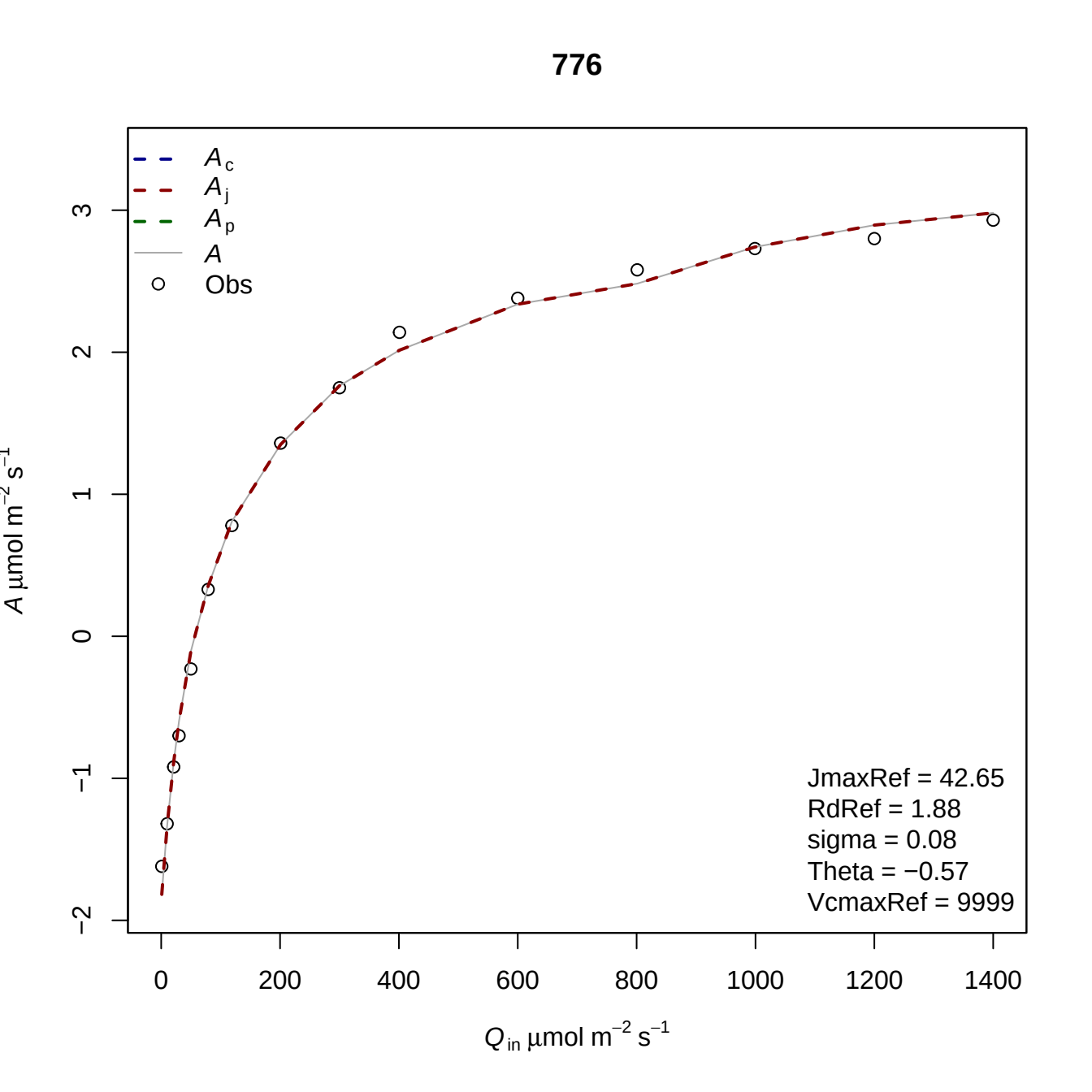

771
A_c
A_j
A_p
A
Obs
JmaxRef = 89.79
RdRef = 1.63
sigma = 0.11
Theta = 0.37
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

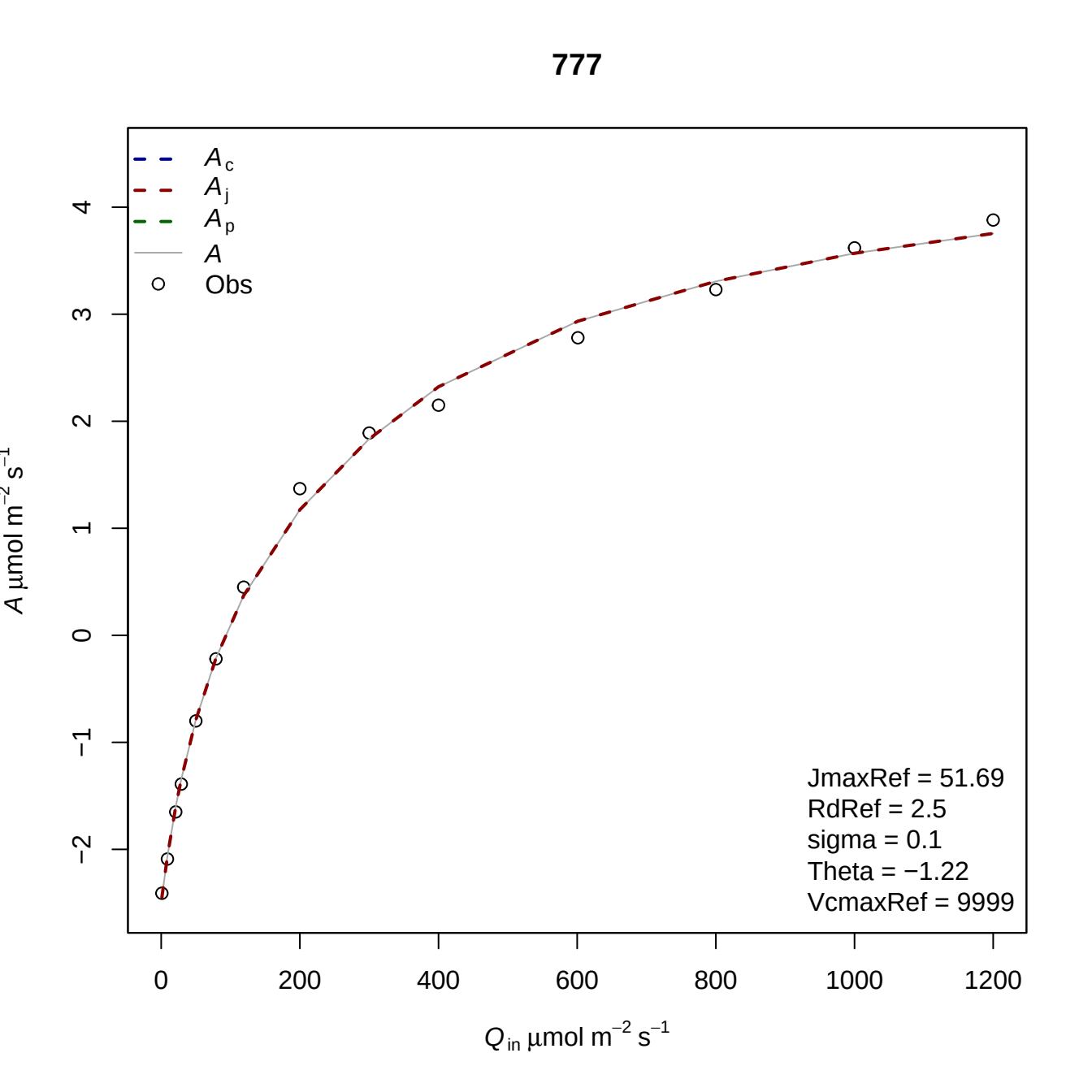

772
A_c
A_j
A_p
A
Obs
JmaxRef = 117.77
RdRef = 1.55
sigma = 0.16
Theta = 0.54
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

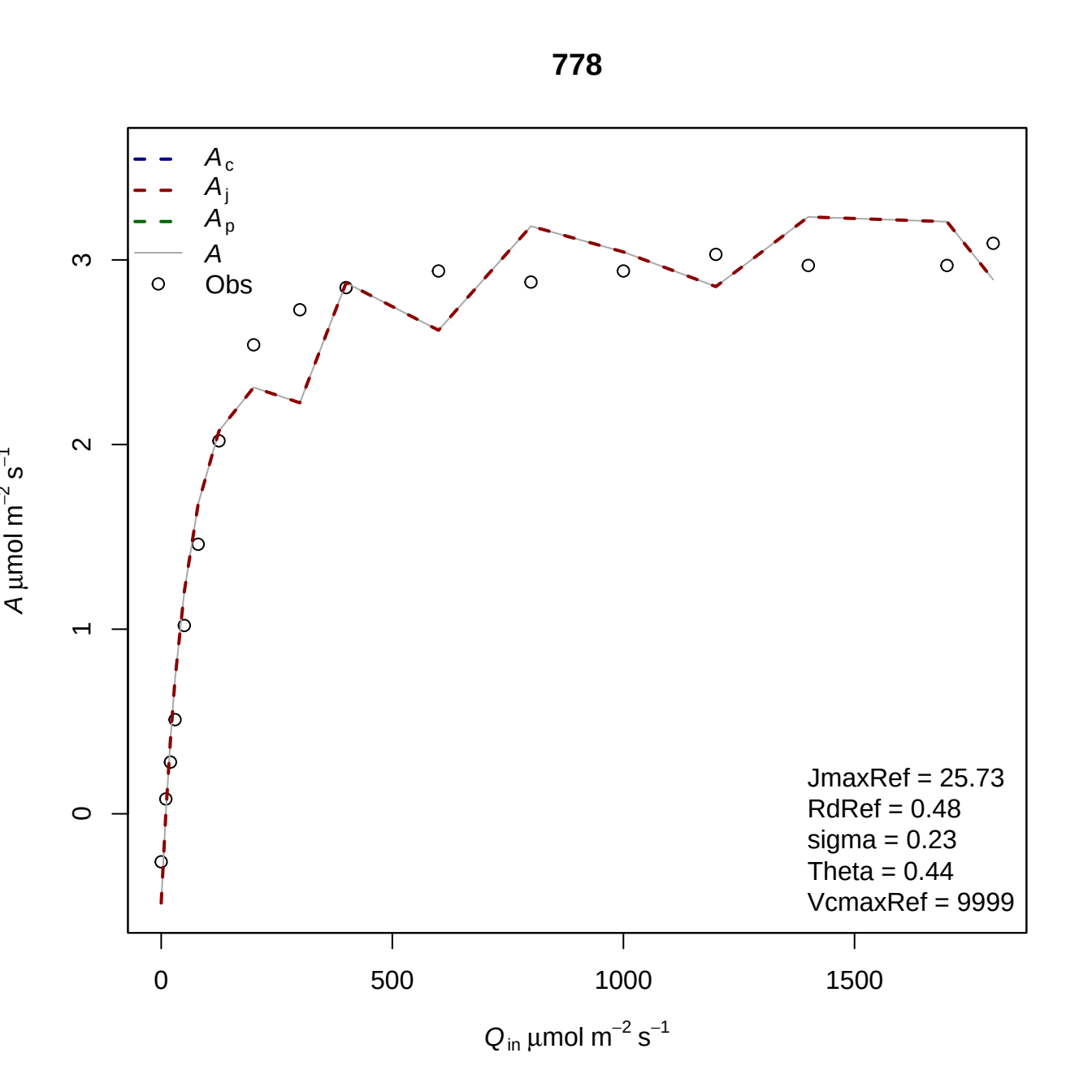

773
A_c
A_j
A_p
A
Obs
JmaxRef = 42.64
RdRef = 1.48
sigma = 0.09
Theta = 0.26
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

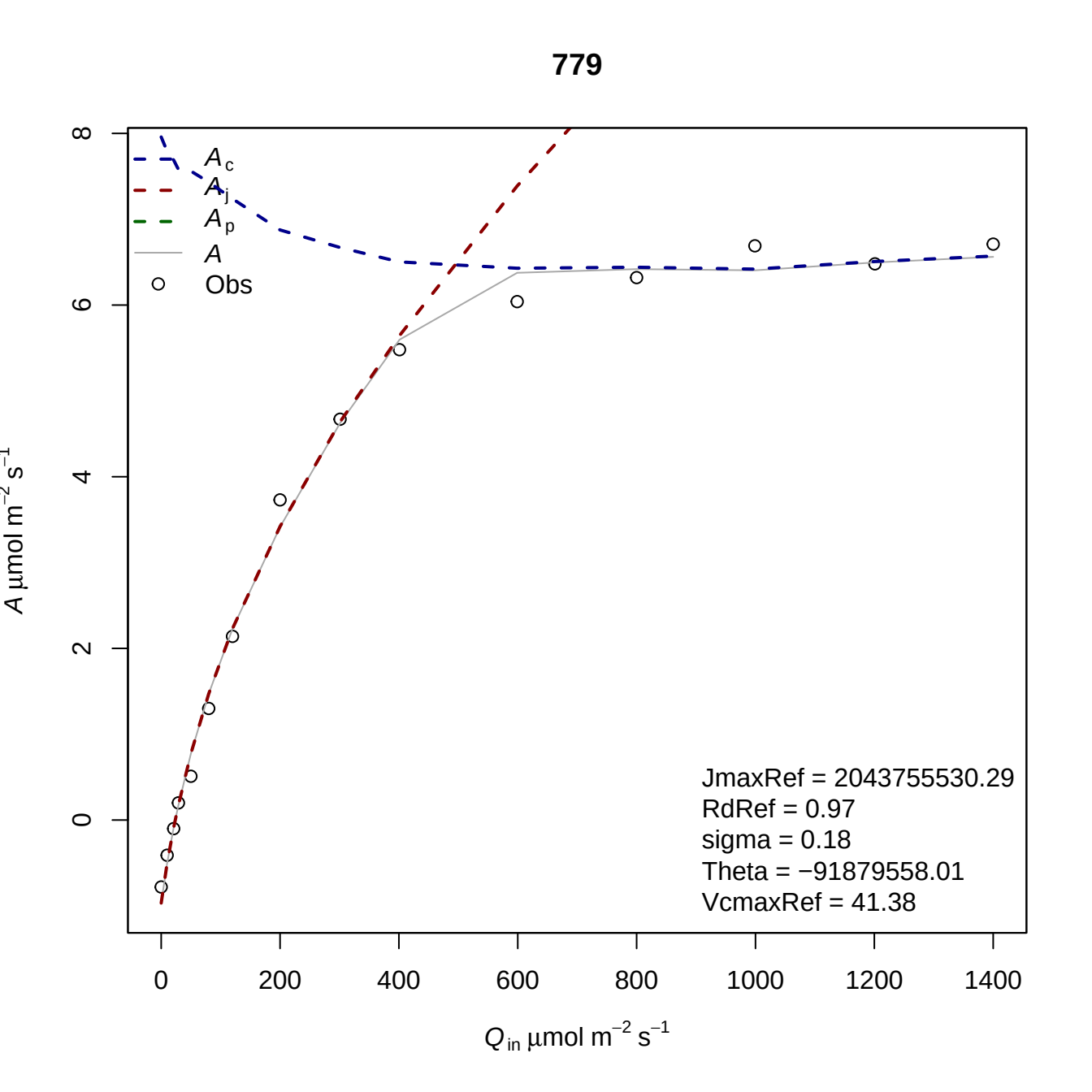

774
A_c
A_j
A_p
A
Obs
JmaxRef = 104.99
RdRef = 3.17
sigma = 0.22
Theta = −0.91
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

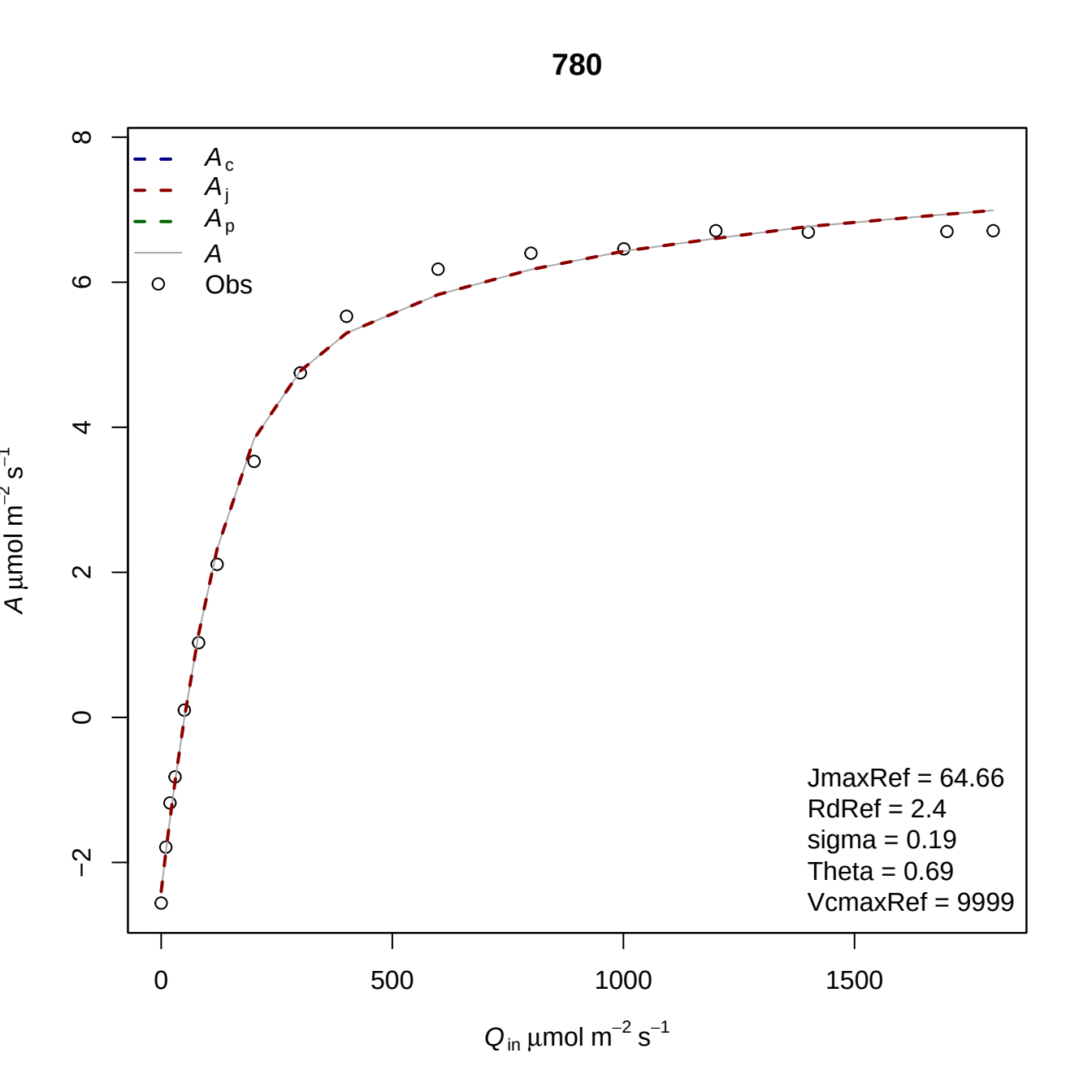

775
A_c
A_j
A_p
A
Obs
JmaxRef = 81.03
RdRef = 1.19
sigma = 0.17
Theta = −0.1
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

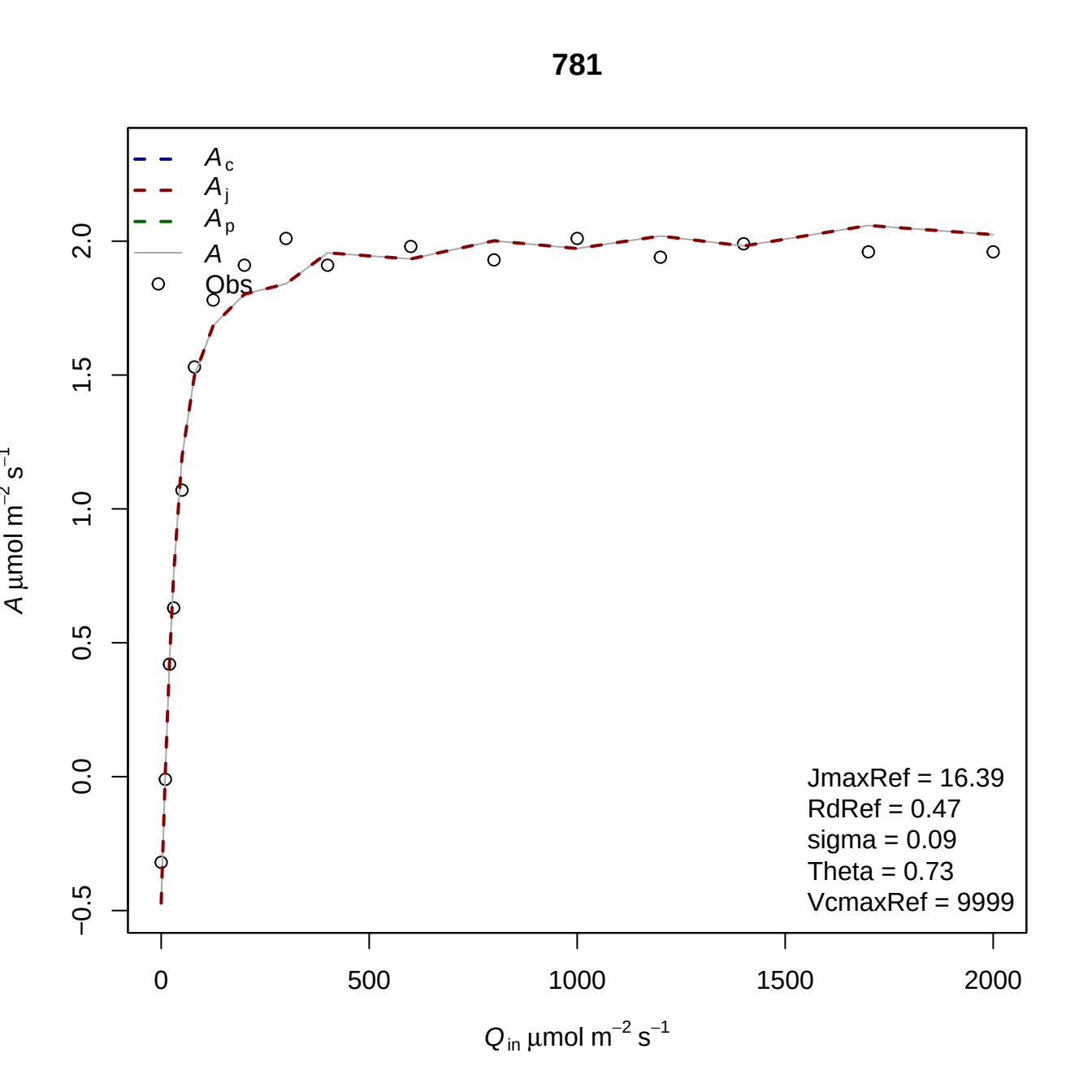

776
A_c
A_j
A_p
A
Obs
JmaxRef = 42.65
RdRef = 1.88
sigma = 0.08
Theta = −0.57
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

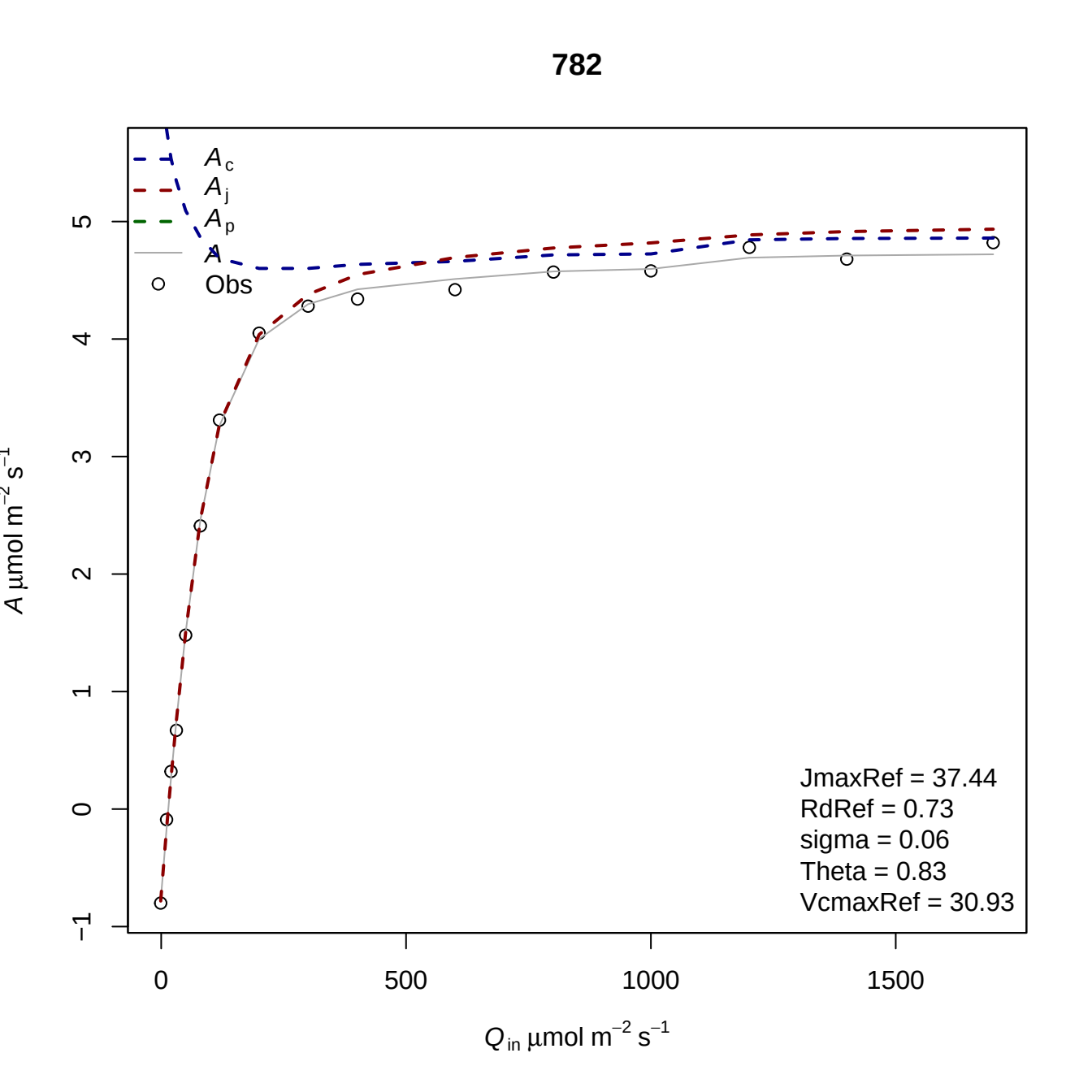

777
A_c
A_j
A_p
A
Obs
JmaxRef = 51.69
RdRef = 2.5
sigma = 0.1
Theta = −1.22
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
200
400
600
800
1000
1200

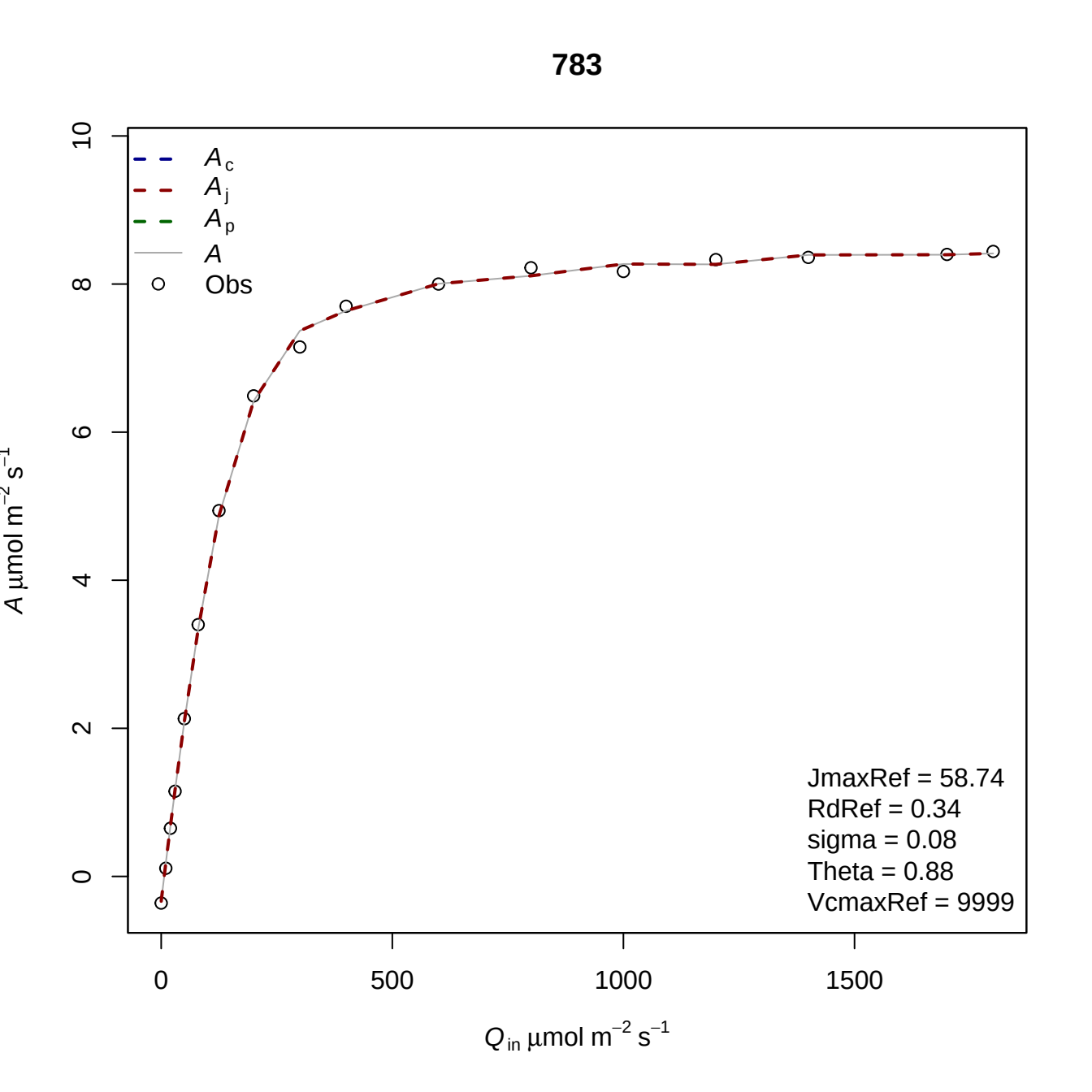

778
A_c
A_j
A_p
A
Obs
JmaxRef = 25.73
RdRef = 0.48
sigma = 0.23
Theta = 0.44
VcmaxRef = 9999
0
1
2
3
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

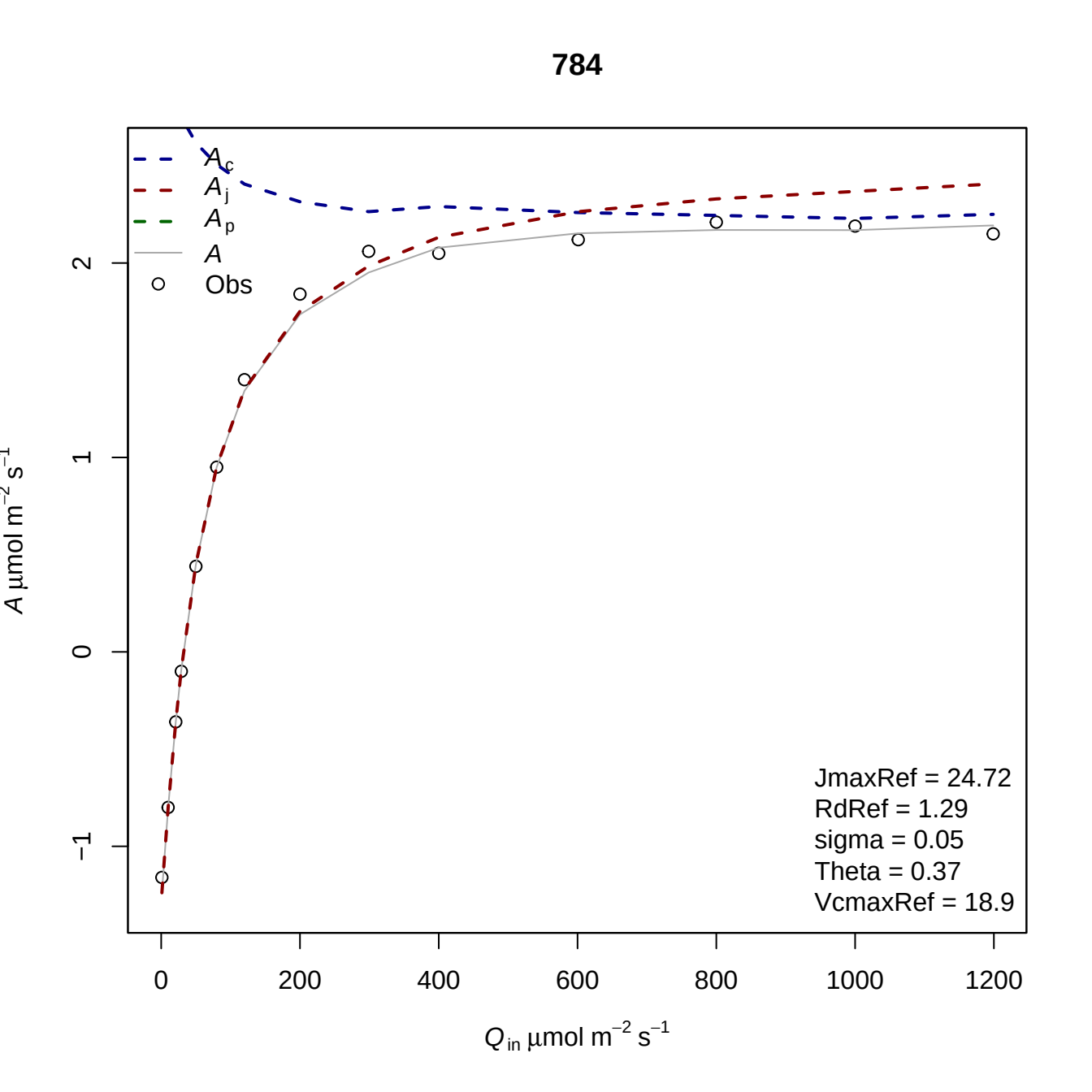

779
A_c
A_j
A_p
A
Obs
JmaxRef = 2043755530.29
RdRef = 0.97
sigma = 0.18
Theta = −91879558.01
VcmaxRef = 41.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
200
400
600
800
1000
1200
1400

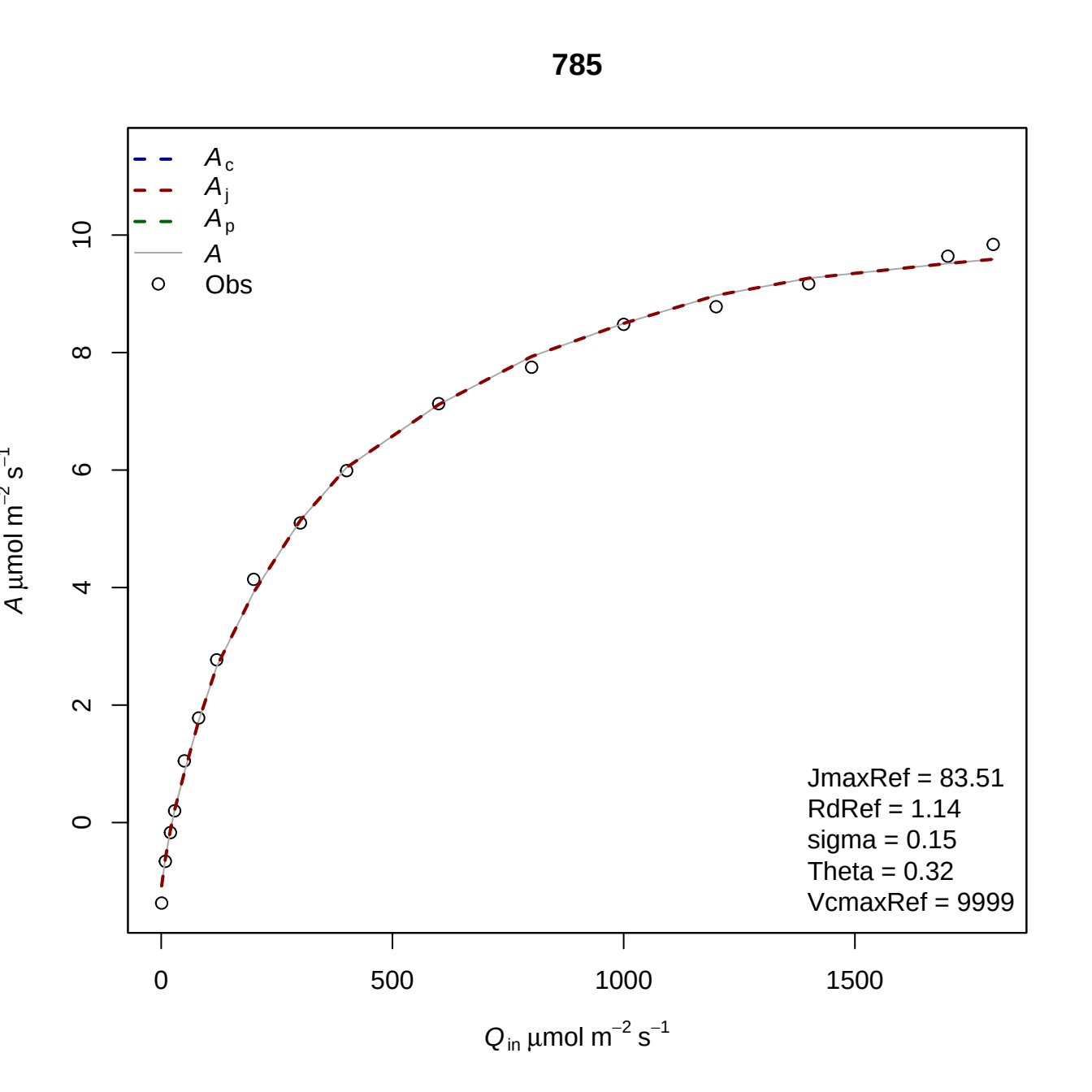

780
A_c
A_j
A_p
A
Obs
JmaxRef = 64.66
RdRef = 2.4
sigma = 0.19
Theta = 0.69
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

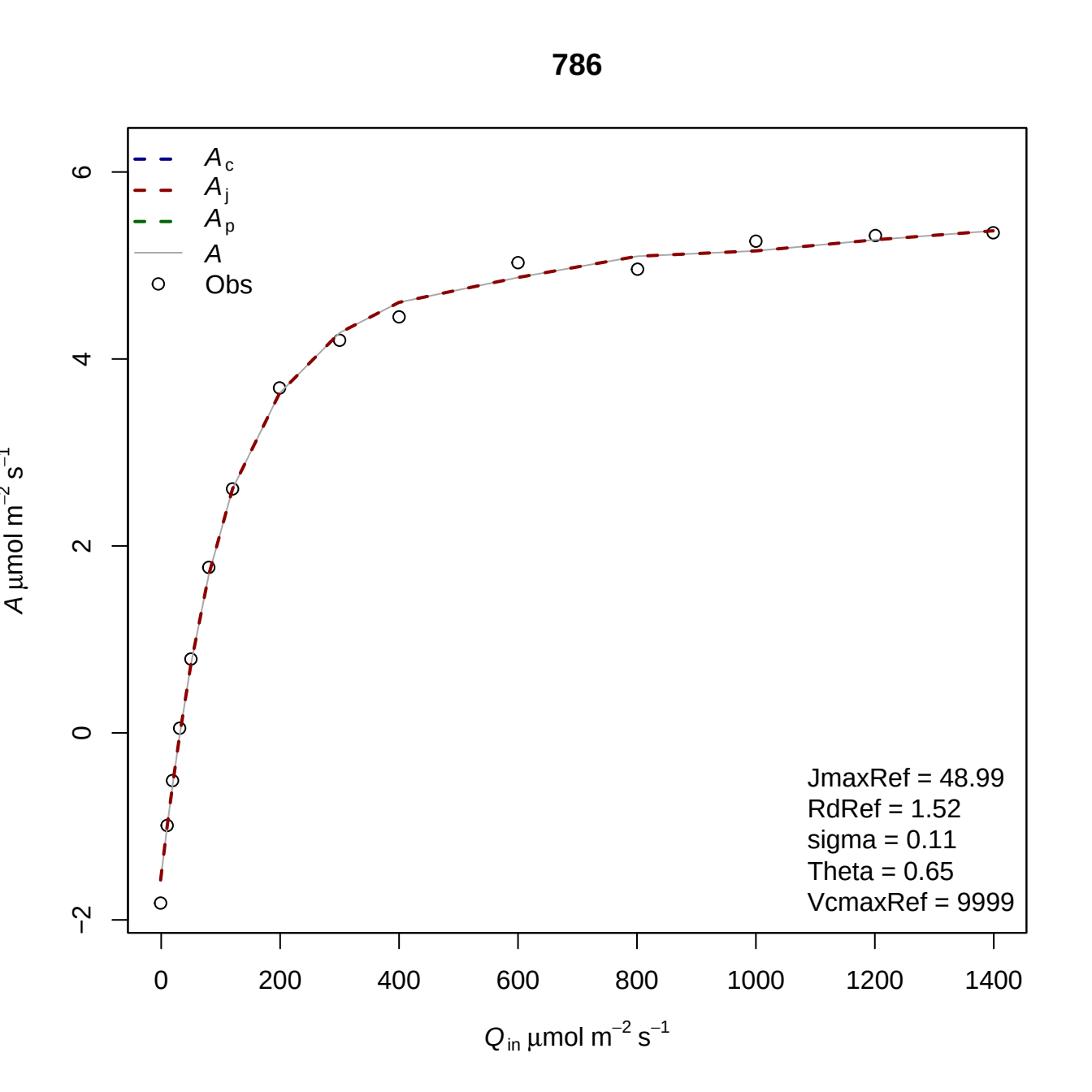

781
A_c
A_j
A_p
A
Obs
JmaxRef = 16.39
RdRef = 0.47
sigma = 0.09
Theta = 0.73
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
2000

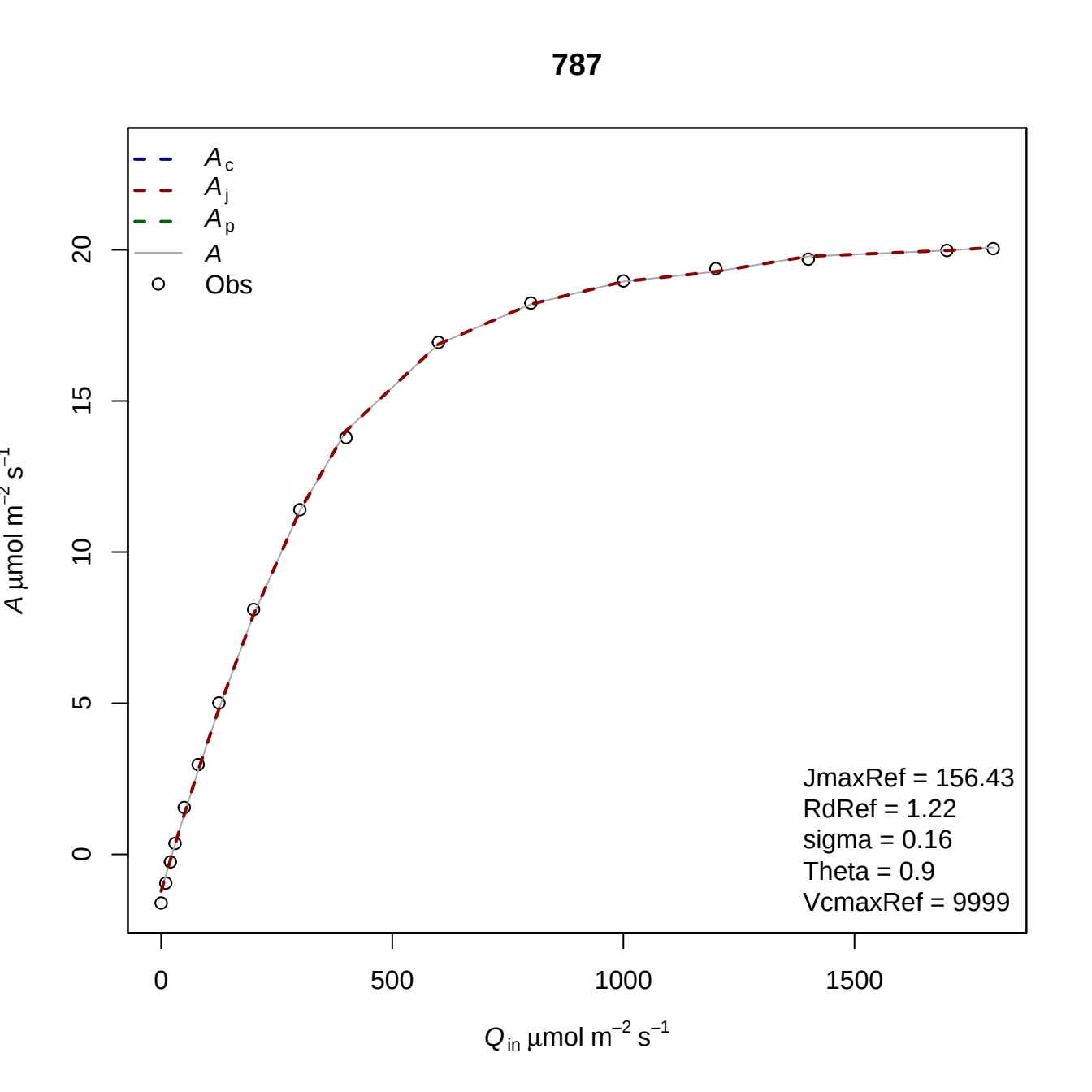

782
A_c
A_j
A_p
A
Obs
JmaxRef = 37.44
RdRef = 0.73
sigma = 0.06
Theta = 0.83
VcmaxRef = 30.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

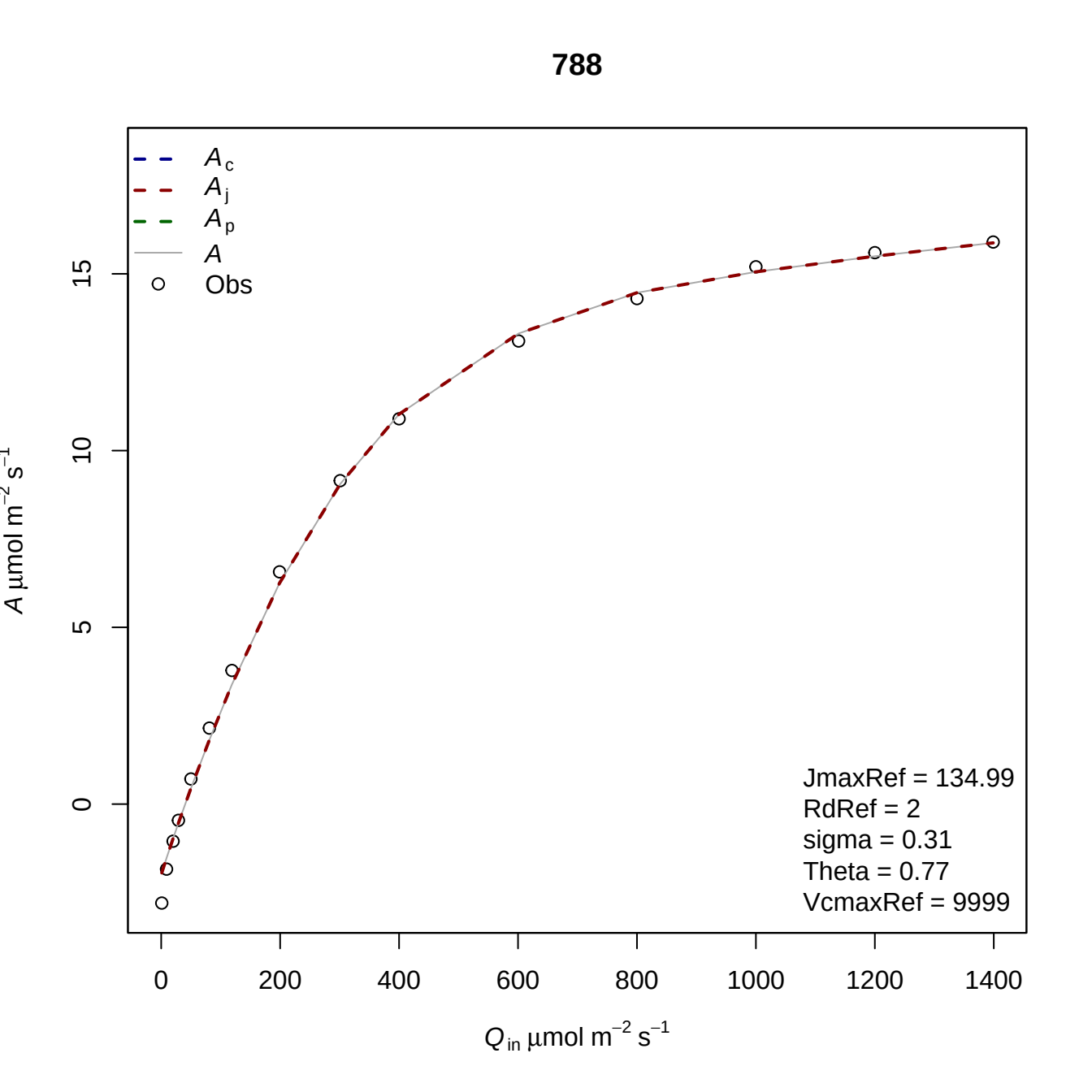

783
A_c
A_j
A_p
A
Obs
JmaxRef = 58.74
RdRef = 0.34
sigma = 0.08
Theta = 0.88
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

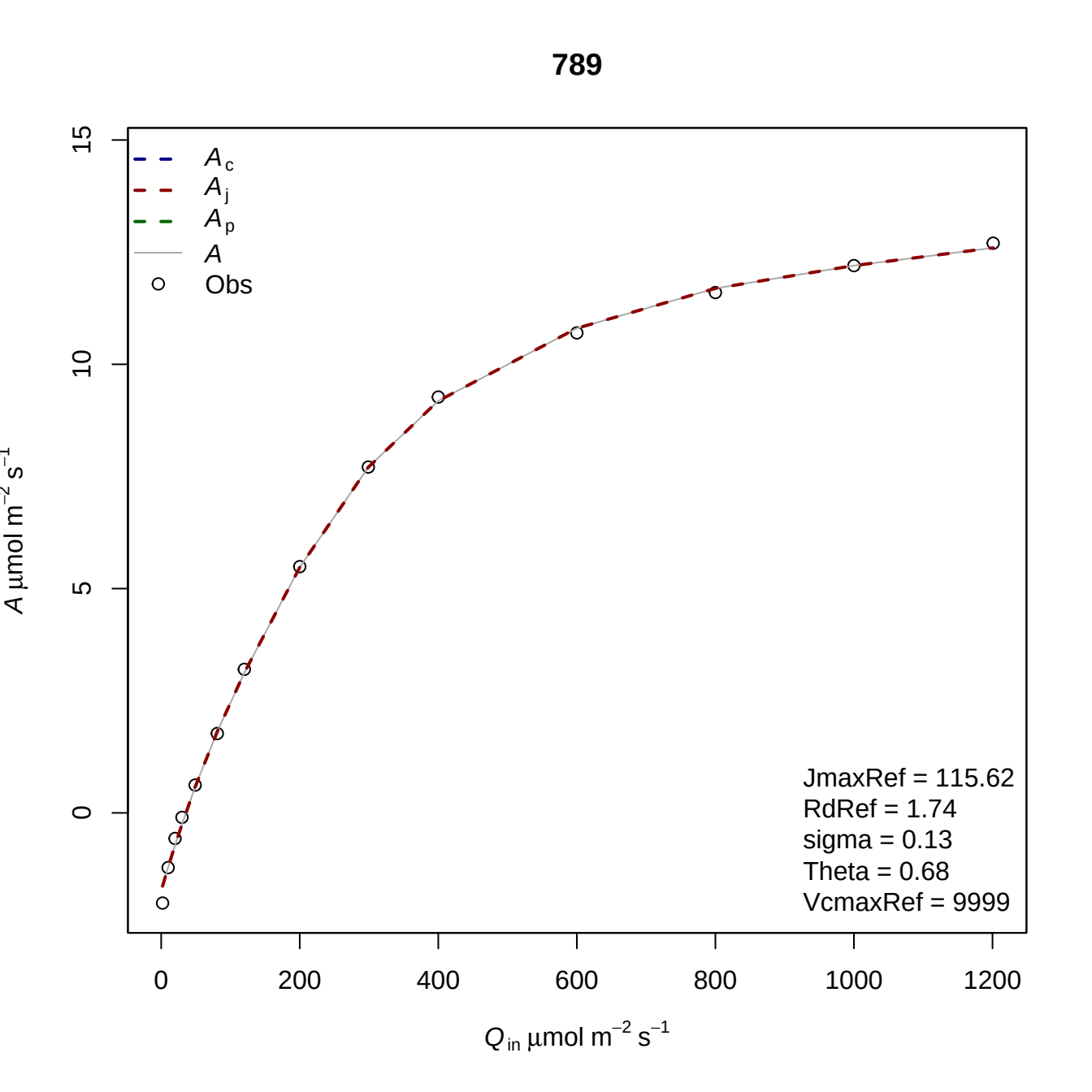

784
A_c
A_j
A_p
A
Obs
JmaxRef = 24.72
RdRef = 1.29
sigma = 0.05
Theta = 0.37
VcmaxRef = 18.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
200
400
600
800
1000
1200

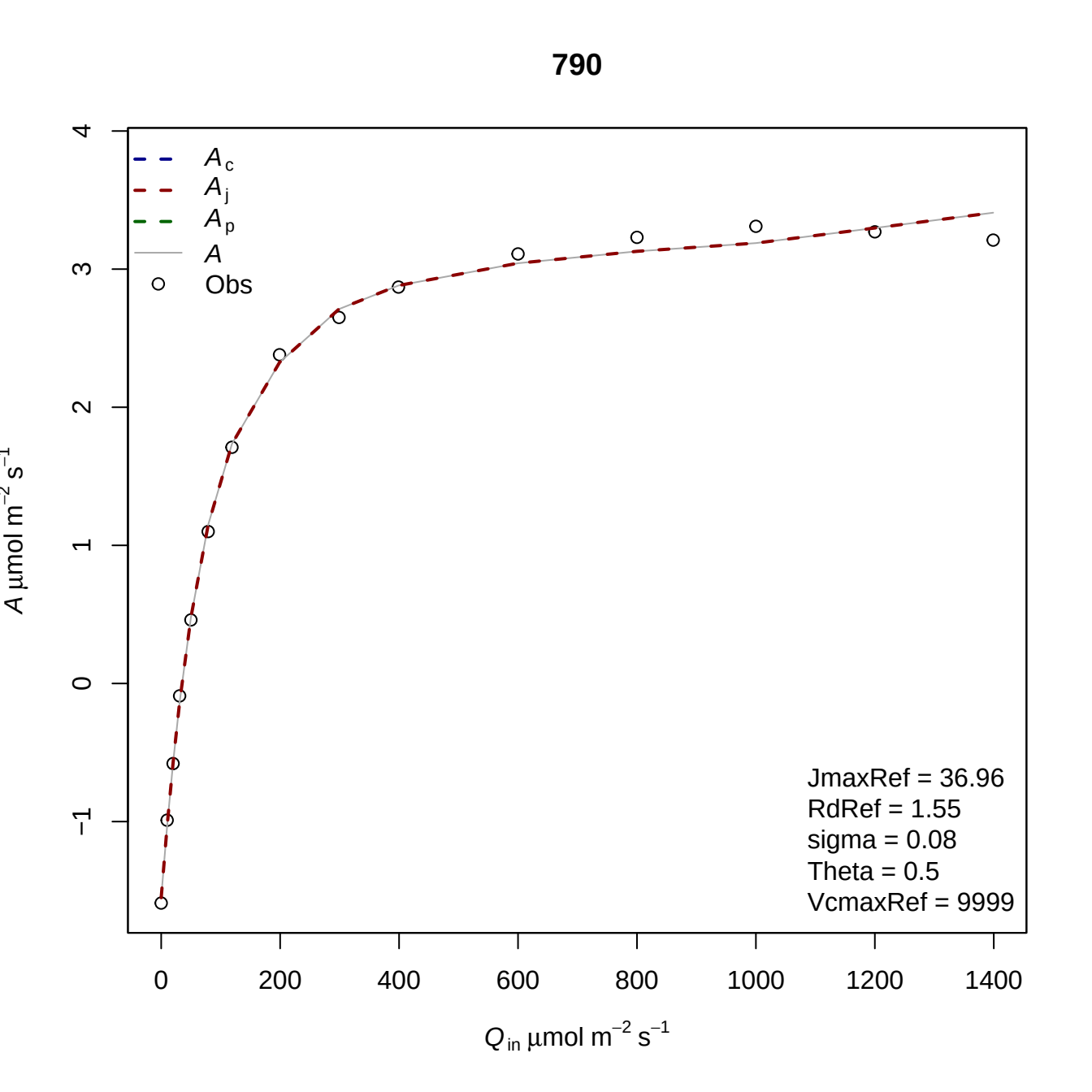

785
A_c
A_j
A_p
A
Obs
JmaxRef = 83.51
RdRef = 1.14
sigma = 0.15
Theta = 0.32
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

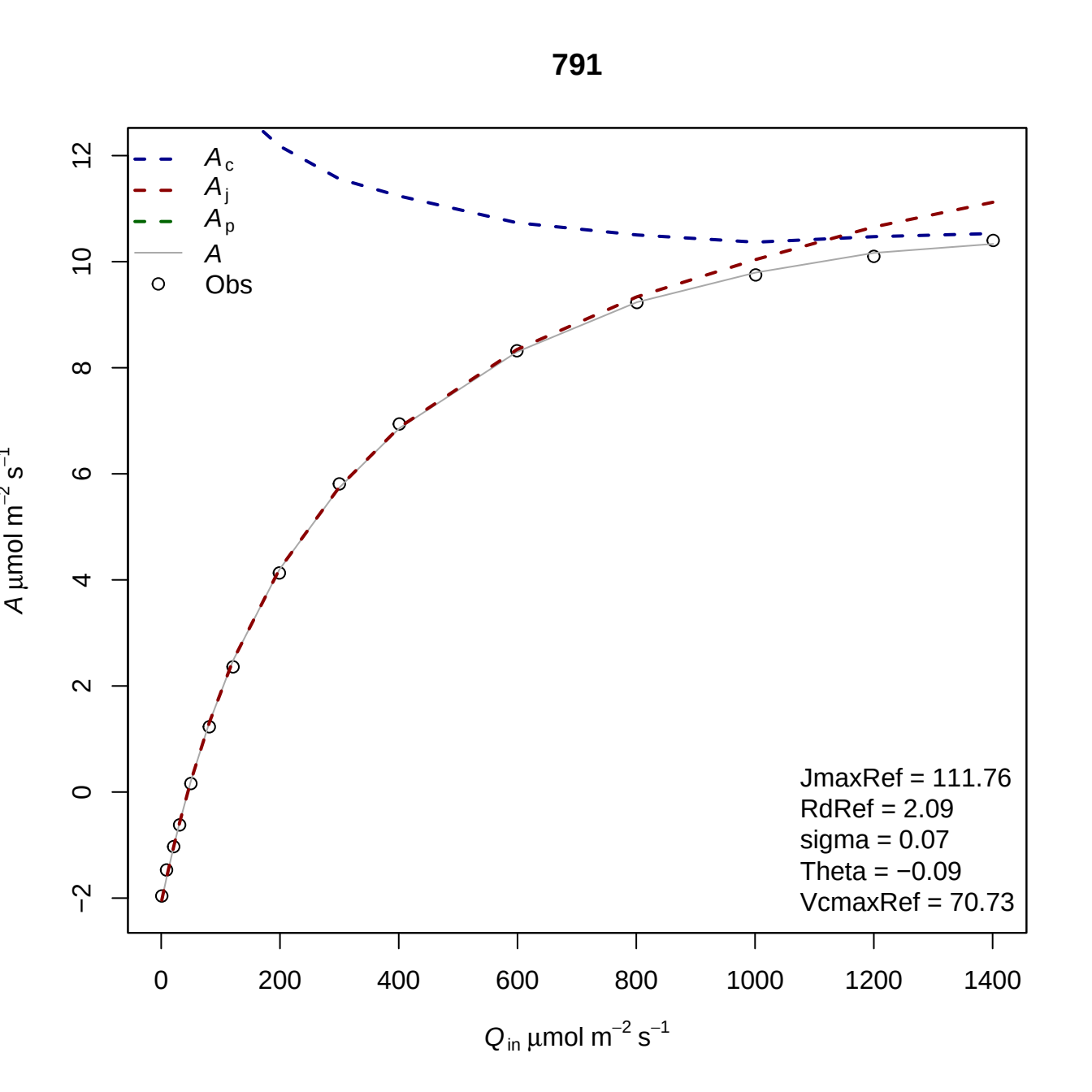

786
A_c
A_j
A_p
A
Obs
JmaxRef = 48.99
RdRef = 1.52
sigma = 0.11
Theta = 0.65
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

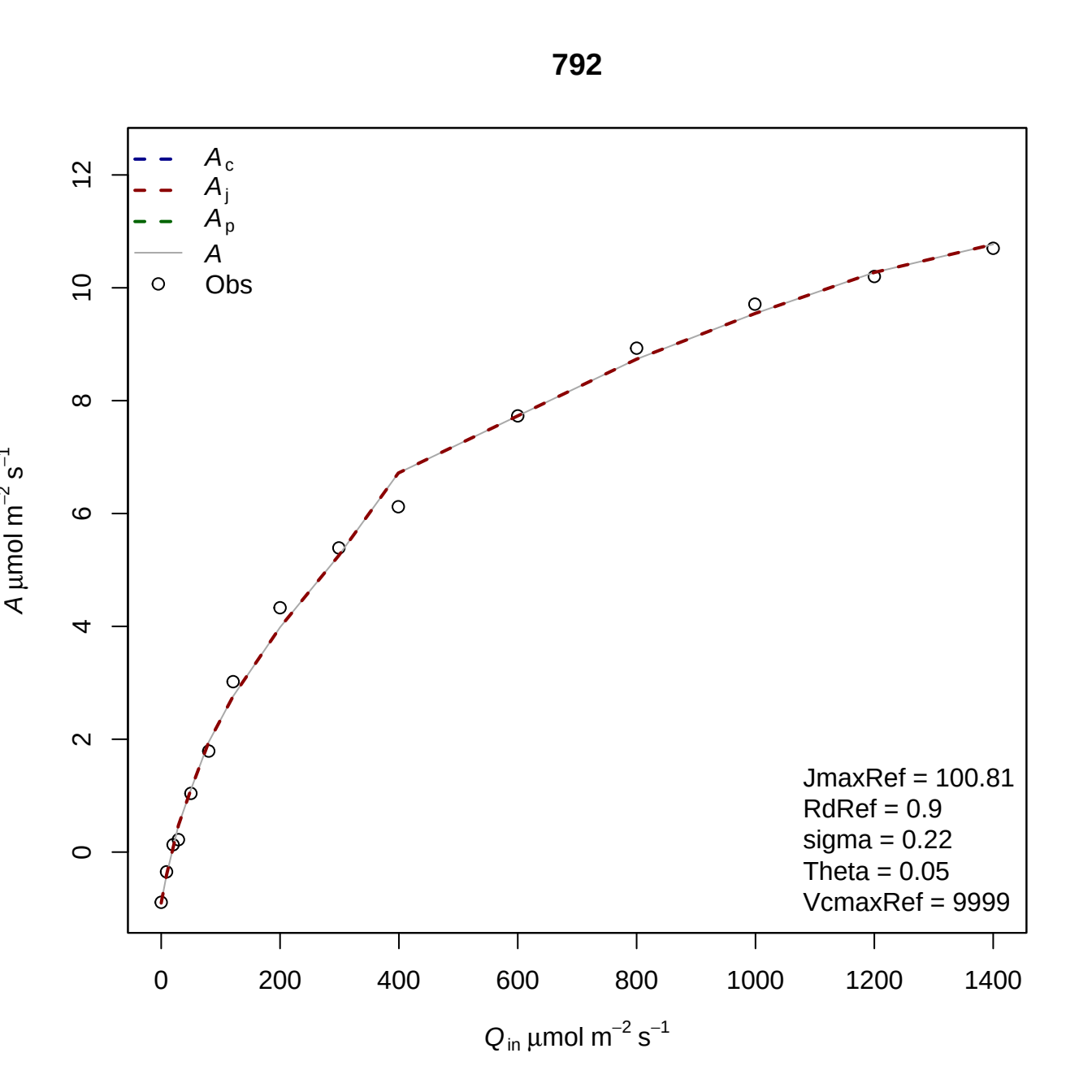

787
A_c
A_j
A_p
A
Obs
JmaxRef = 156.43
RdRef = 1.22
sigma = 0.16
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

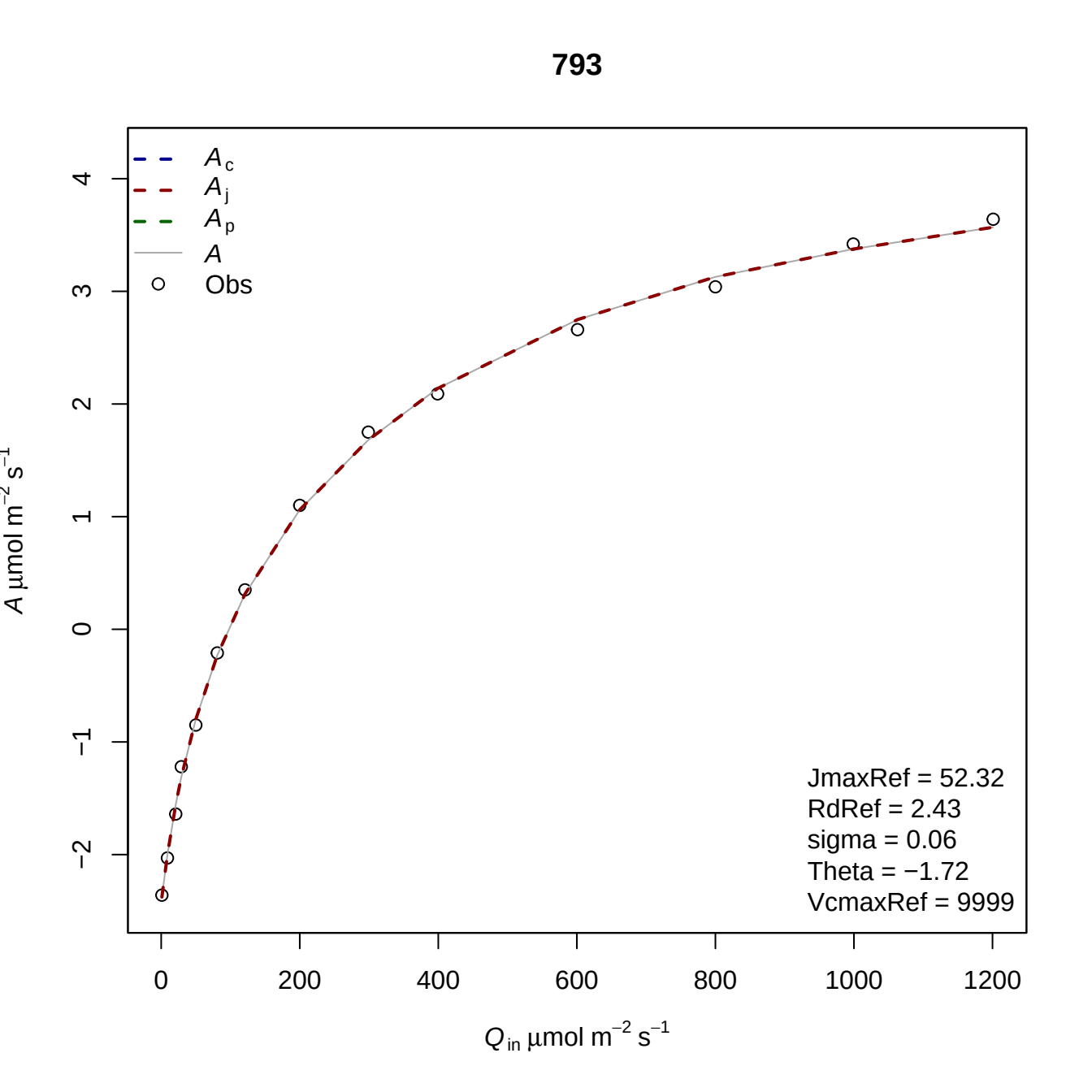

788
A_c
A_j
A_p
A
Obs
JmaxRef = 134.99
RdRef = 2
sigma = 0.31
Theta = 0.77
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200
1400

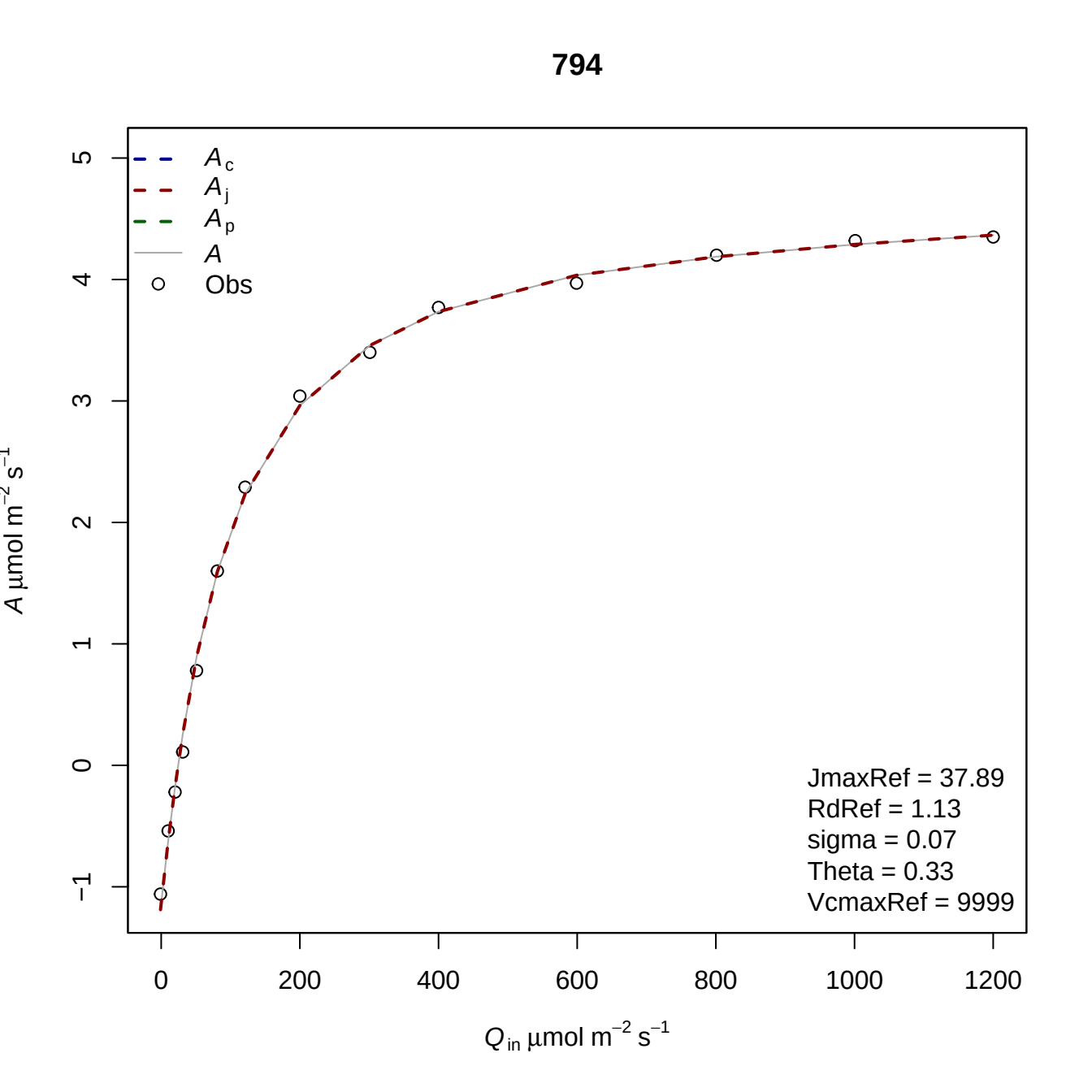

789
A_c
A_j
A_p
A
Obs
JmaxRef = 115.62
RdRef = 1.74
sigma = 0.13
Theta = 0.68
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200

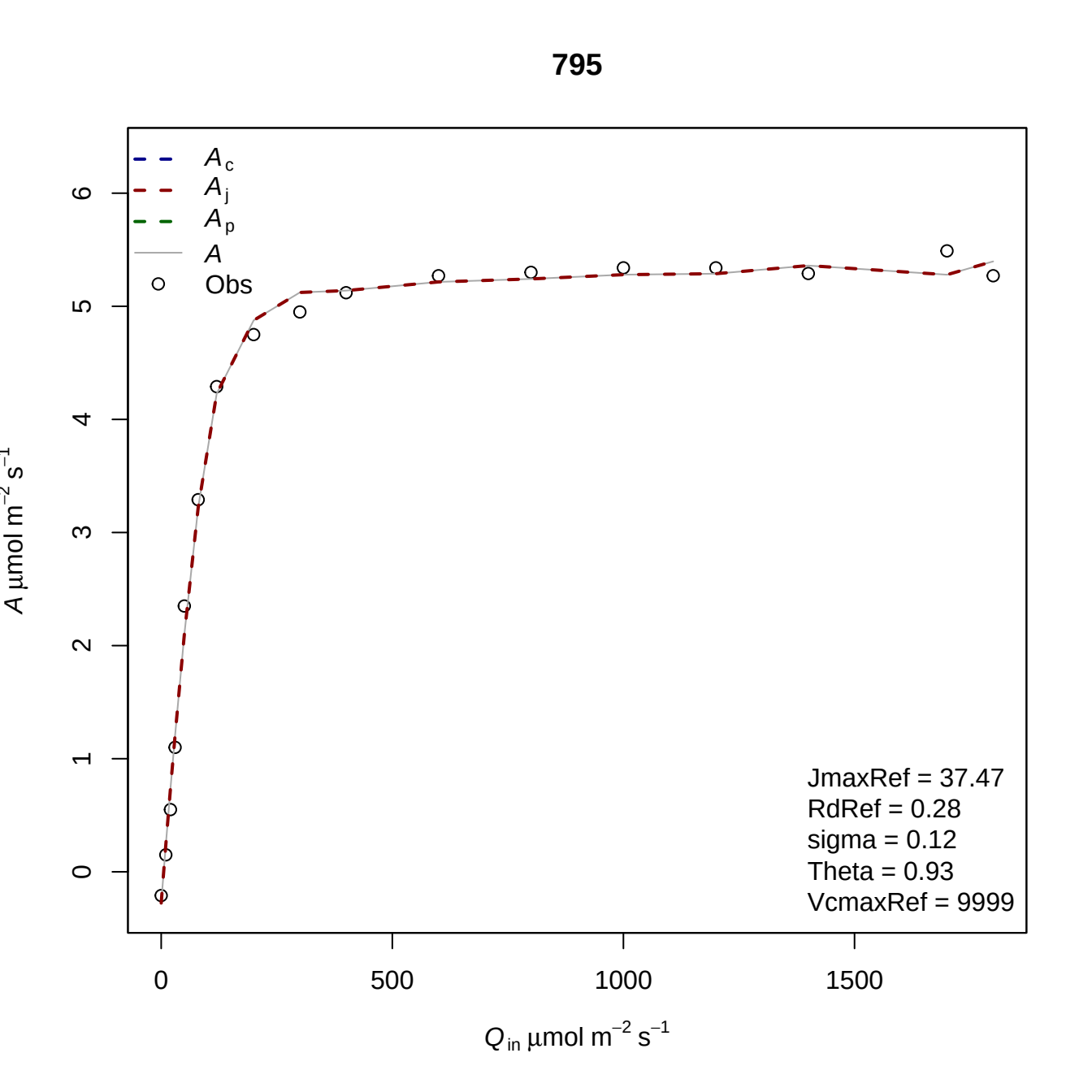

790
A_c
A_j
A_p
A
Obs
JmaxRef = 36.96
RdRef = 1.55
sigma = 0.08
Theta = 0.5
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
200
400
600
800
1000
1200
1400

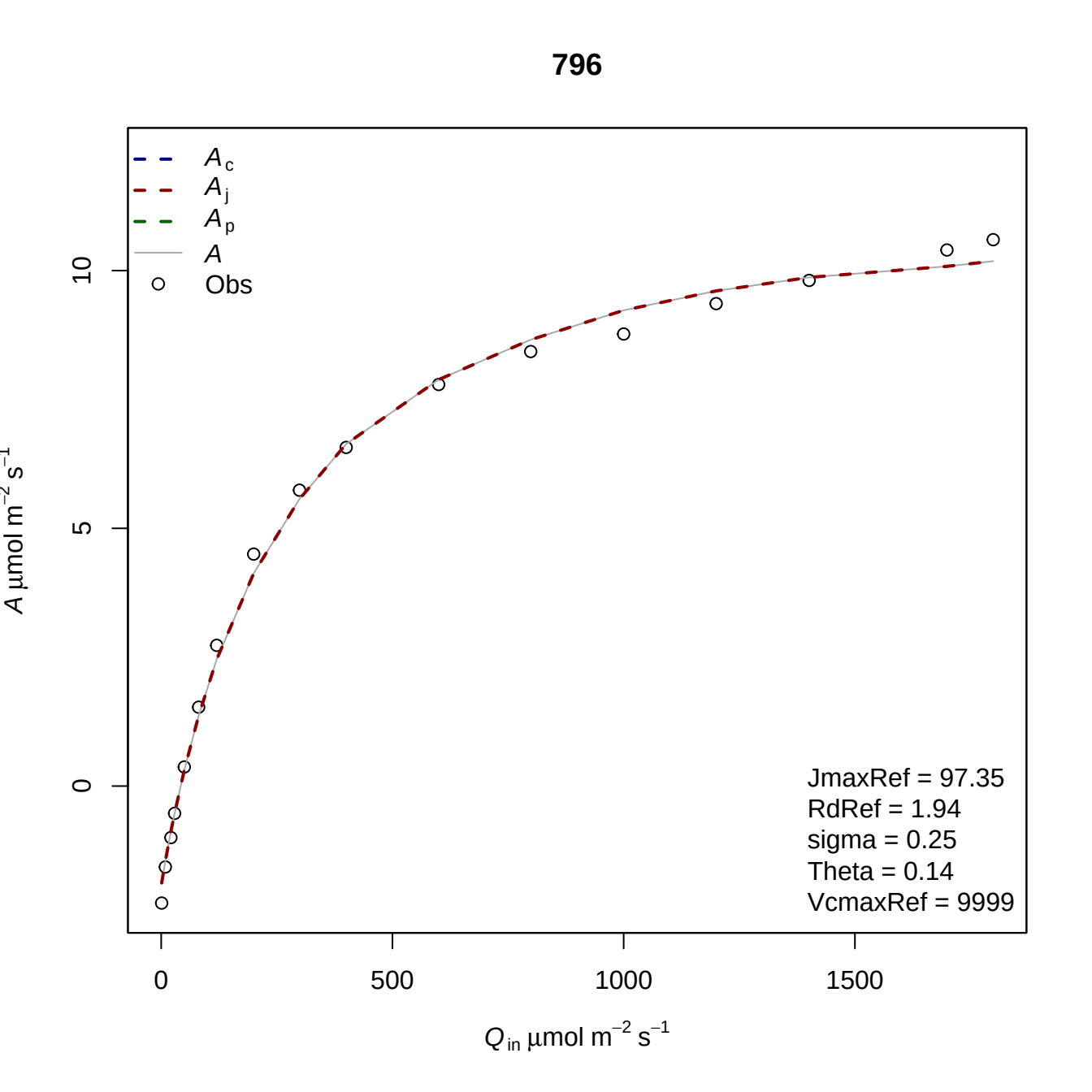

791
A_c
A_j
A_p
A
Obs
JmaxRef = 111.76
RdRef = 2.09
sigma = 0.07
Theta = −0.09
VcmaxRef = 70.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

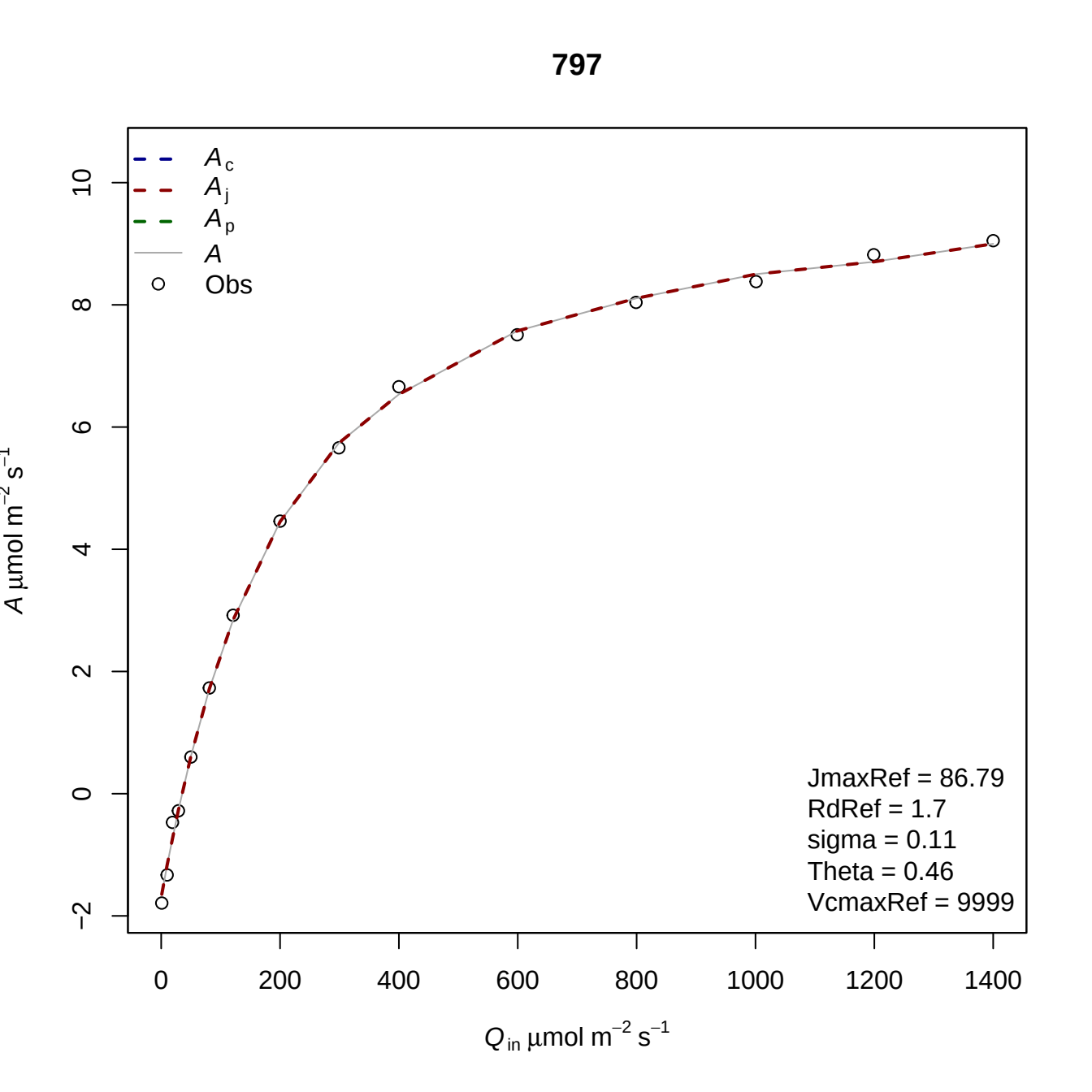

792
A_c
A_j
A_p
A
Obs
JmaxRef = 100.81
RdRef = 0.9
sigma = 0.22
Theta = 0.05
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

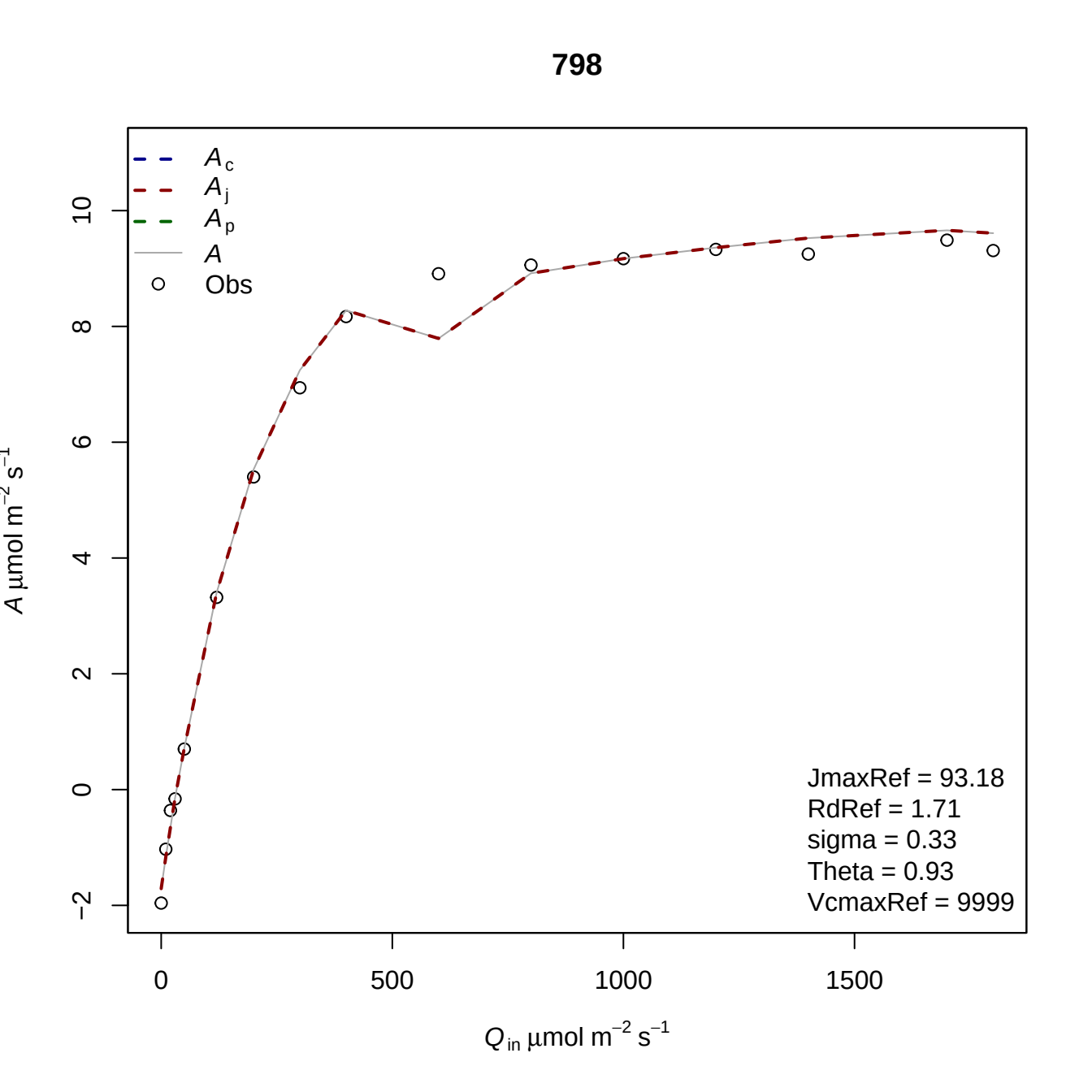

793
A_c
A_j
A_p
A
Obs
JmaxRef = 52.32
RdRef = 2.43
sigma = 0.06
Theta = −1.72
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200

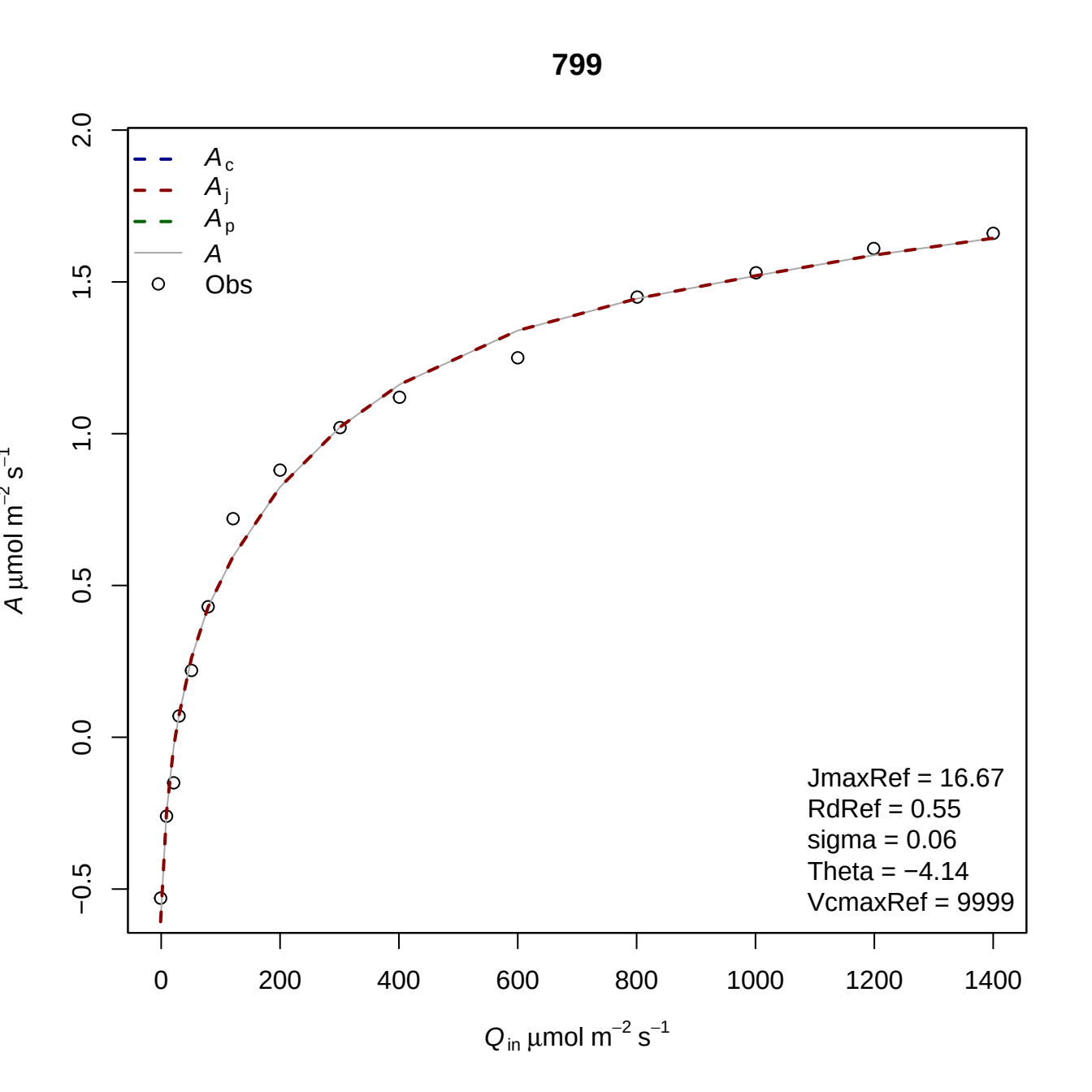

794
A_c
A_j
A_p
A
Obs
JmaxRef = 37.89
RdRef = 1.13
sigma = 0.07
Theta = 0.33
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000
1200

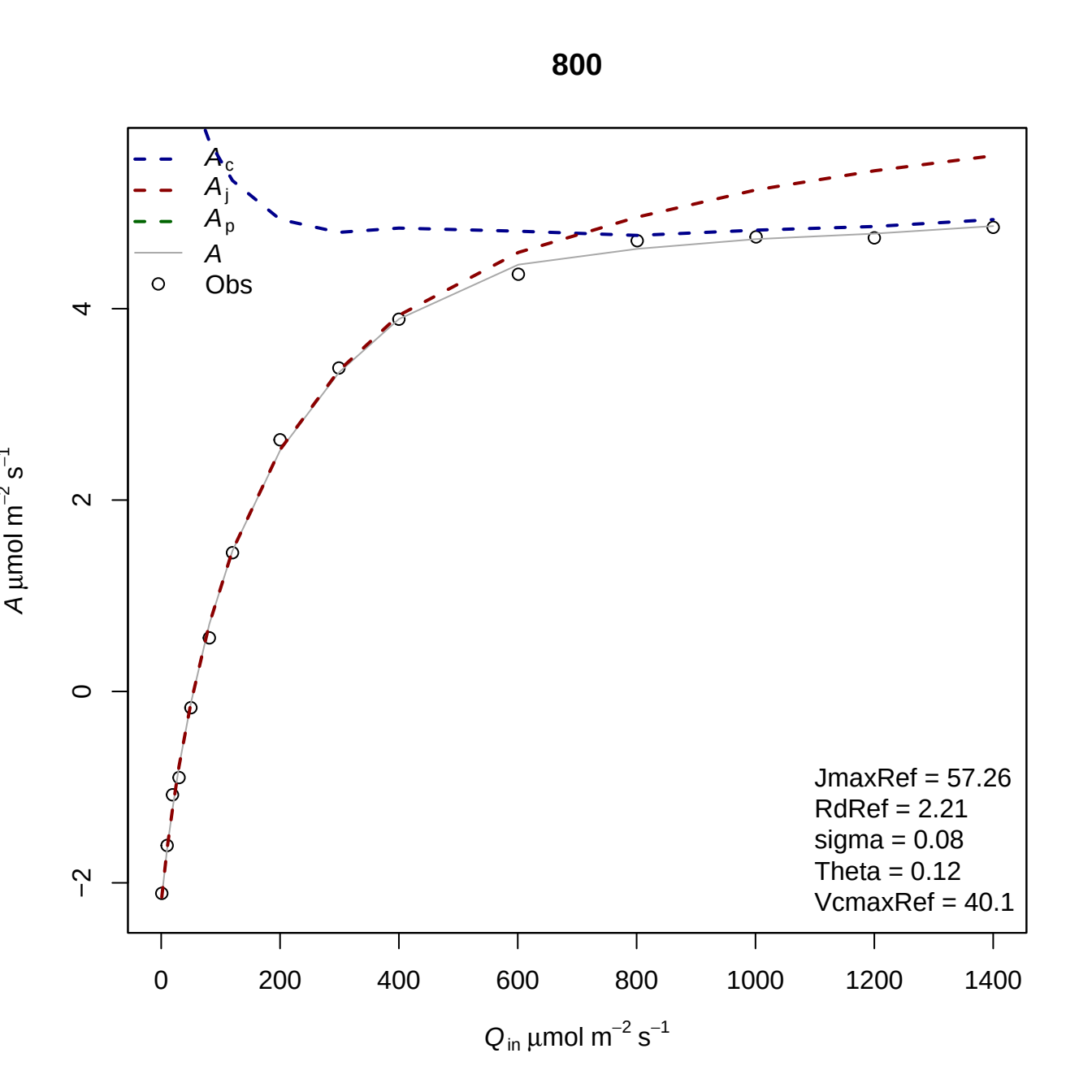

795
A_c
A_j
A_p
A
Obs
JmaxRef = 37.47
RdRef = 0.28
sigma = 0.12
Theta = 0.93
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

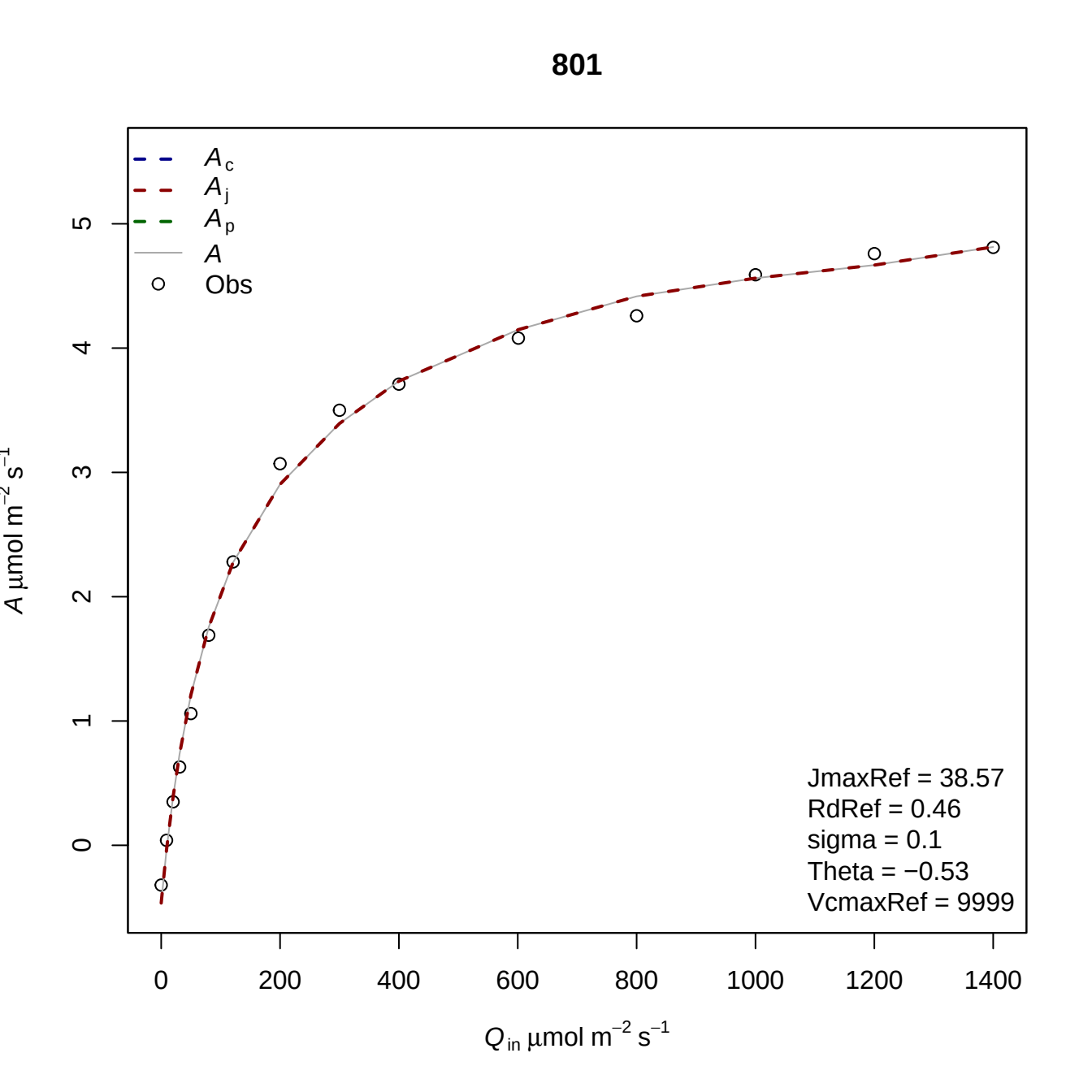

796
A_c
A_j
A_p
A
Obs
JmaxRef = 97.35
RdRef = 1.94
sigma = 0.25
Theta = 0.14
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

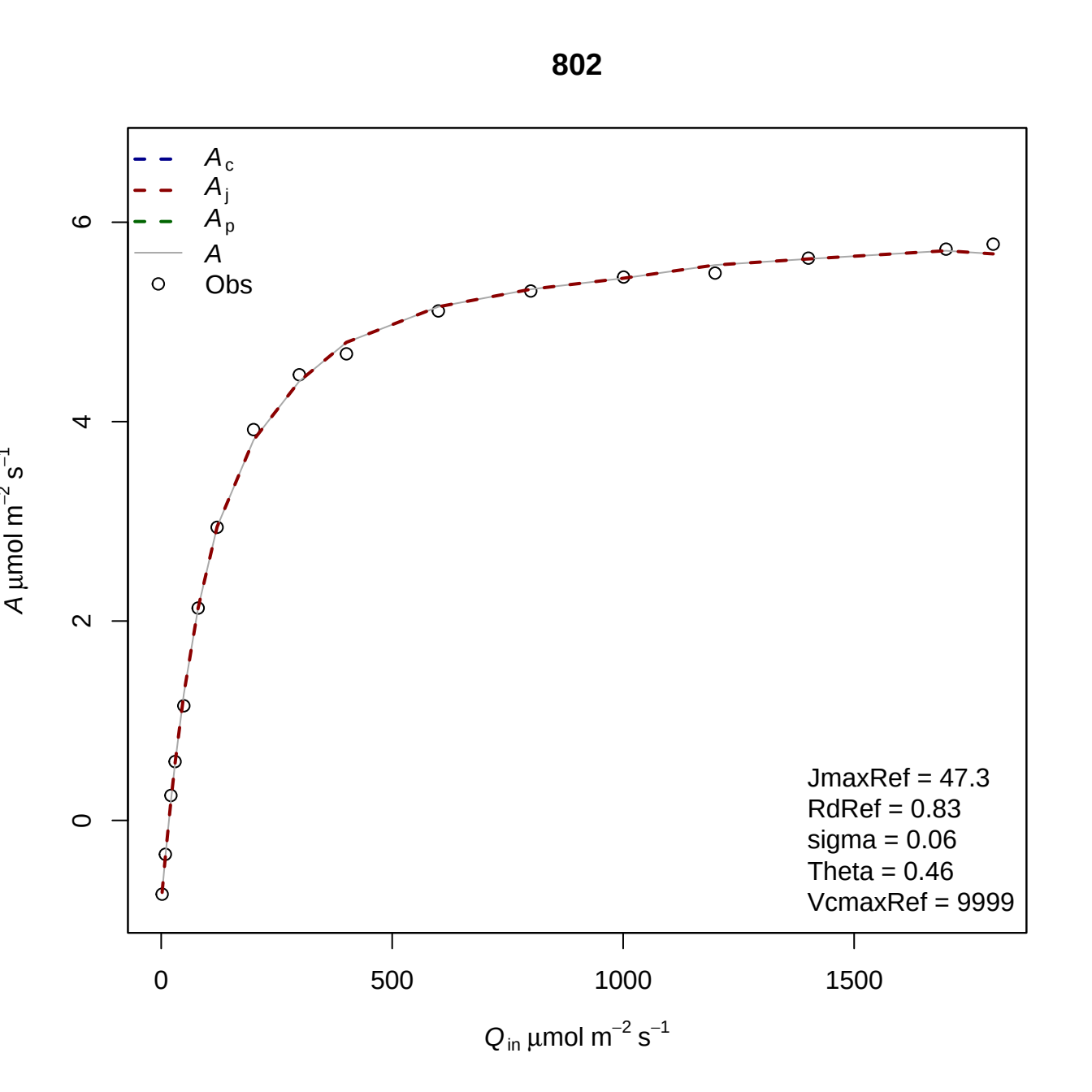

797
A_c
A_j
A_p
A
Obs
JmaxRef = 86.79
RdRef = 1.7
sigma = 0.11
Theta = 0.46
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

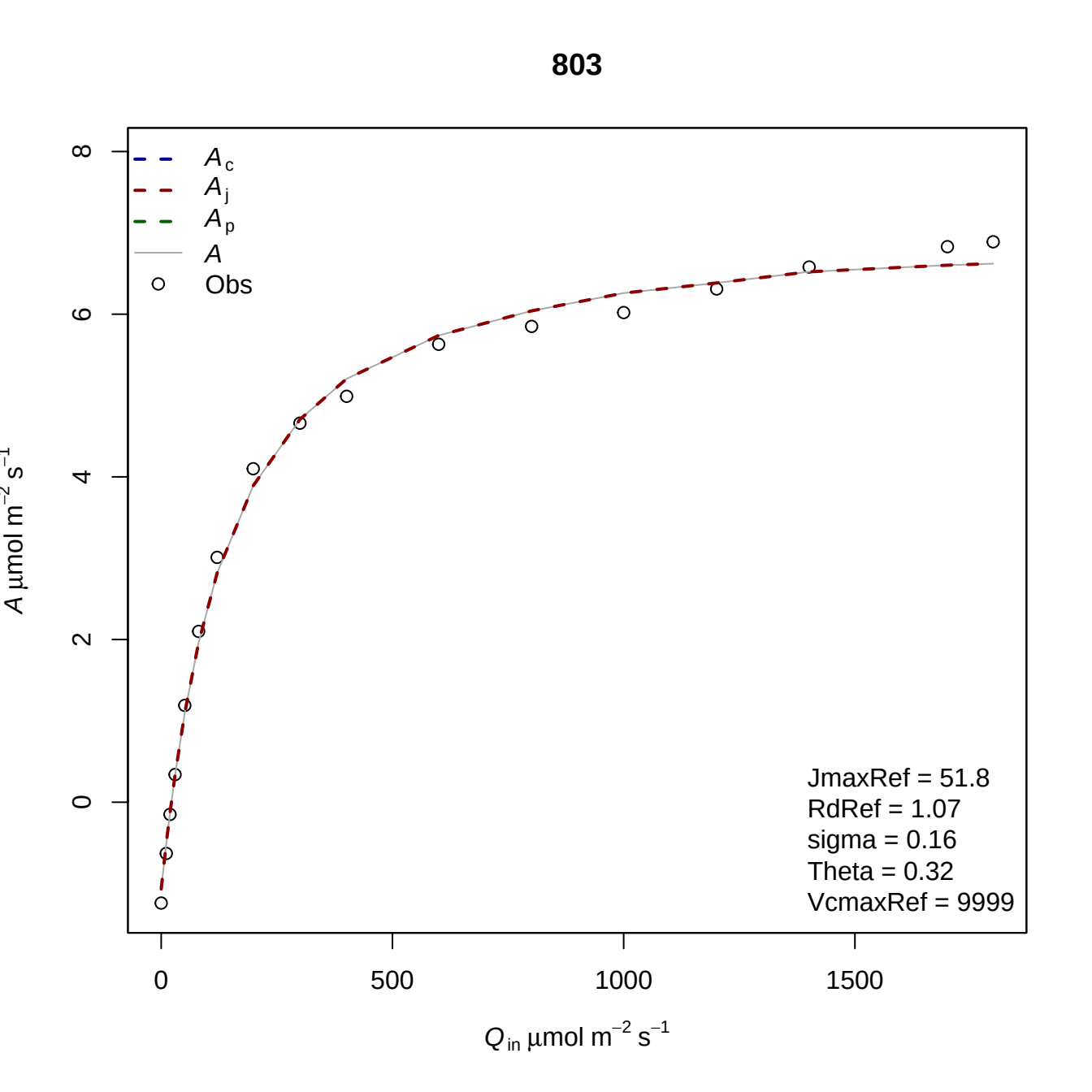

798
A_c
A_j
A_p
A
Obs
JmaxRef = 93.18
RdRef = 1.71
sigma = 0.33
Theta = 0.93
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

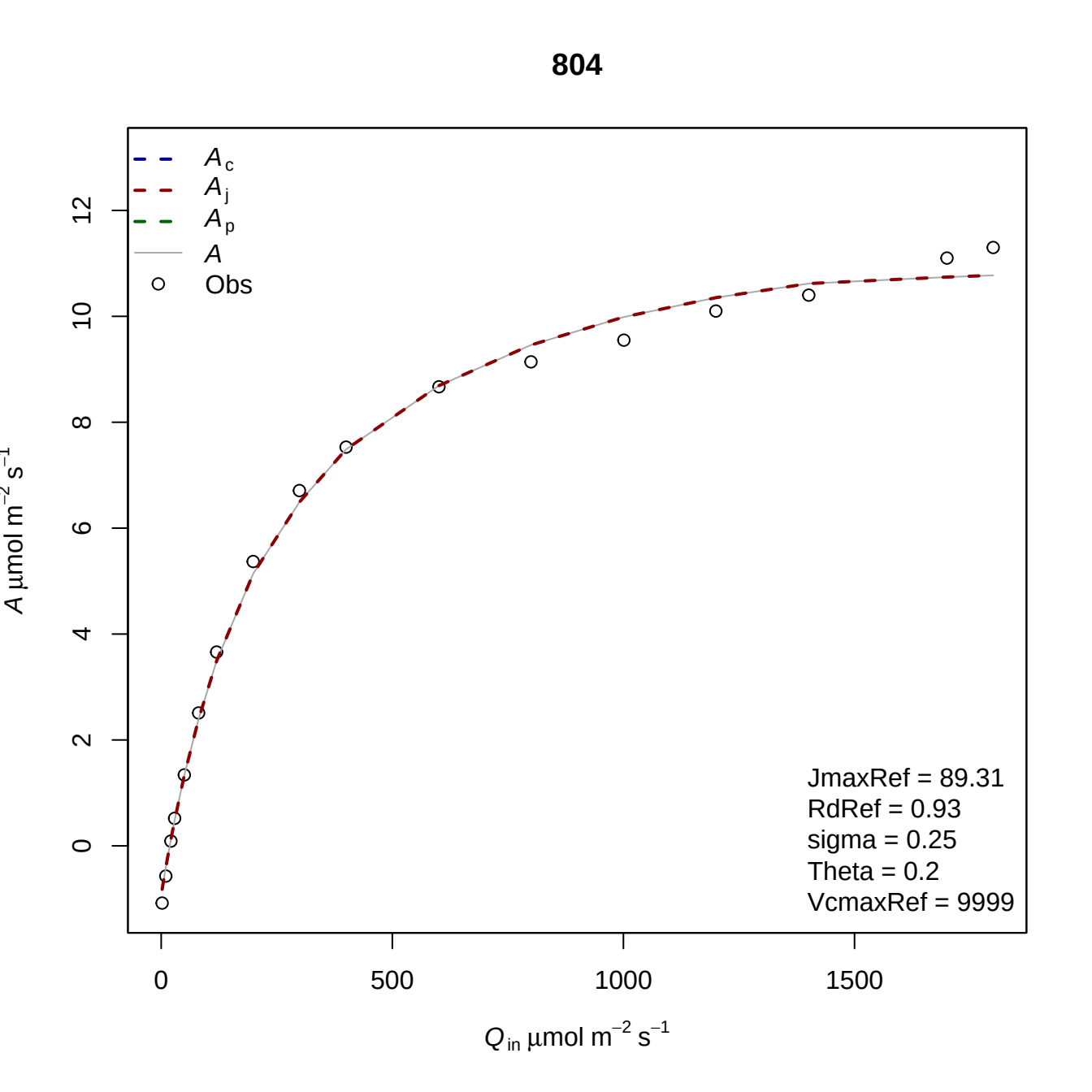

799
A_c
A_j
A_p
A
Obs
JmaxRef = 16.67
RdRef = 0.55
sigma = 0.06
Theta = −4.14
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.0
1.5
1.0
0.5
0.0
−0.5
0
200
400
600
800
1000
1200
1400

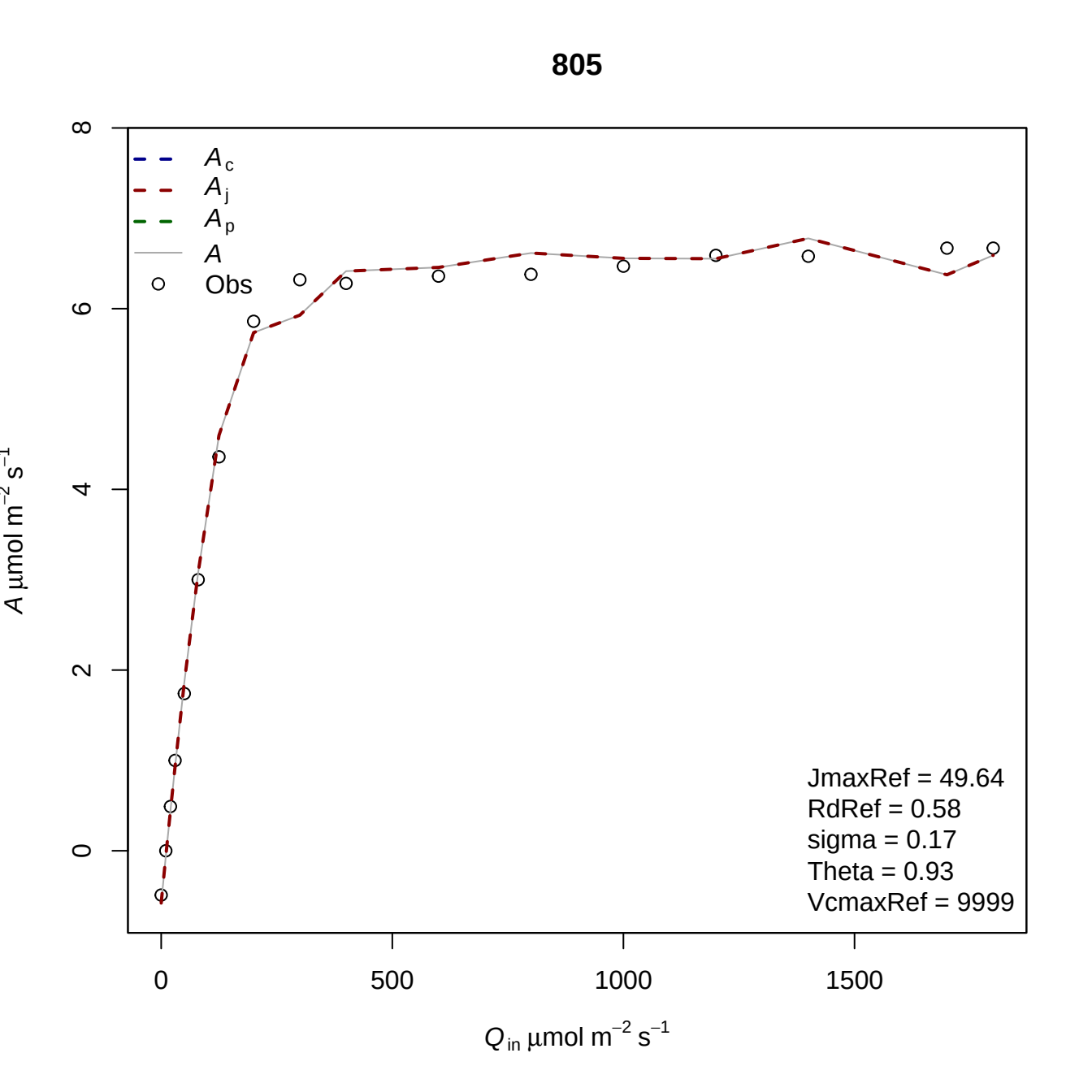

800
A_c
A_j
A_p
A
Obs
JmaxRef = 57.26
RdRef = 2.21
sigma = 0.08
Theta = 0.12
VcmaxRef = 40.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000
1200
1400

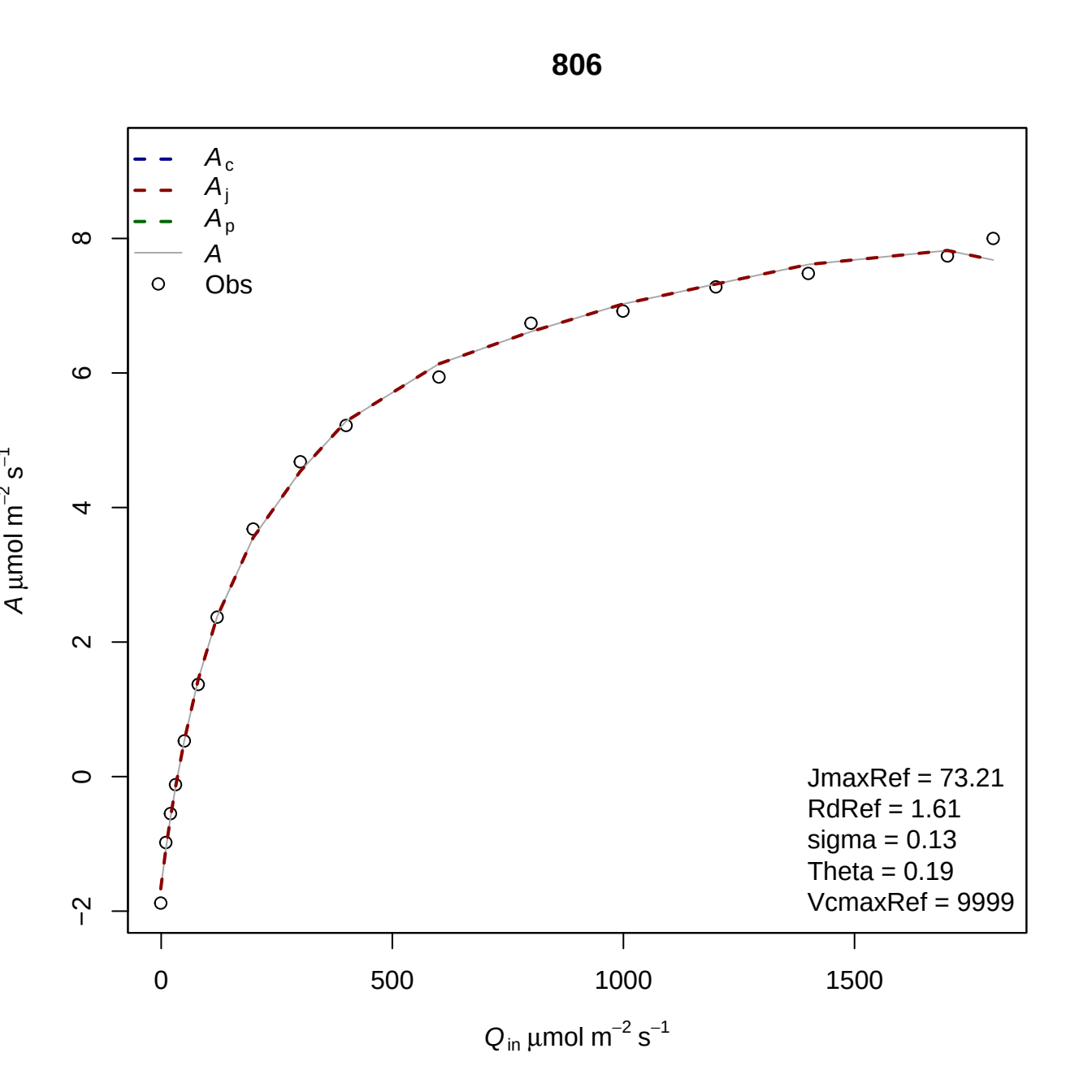

801
A_c
A_j
A_p
A
Obs
JmaxRef = 38.57
RdRef = 0.46
sigma = 0.1
Theta = −0.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
200
400
600
800
1000
1200
1400

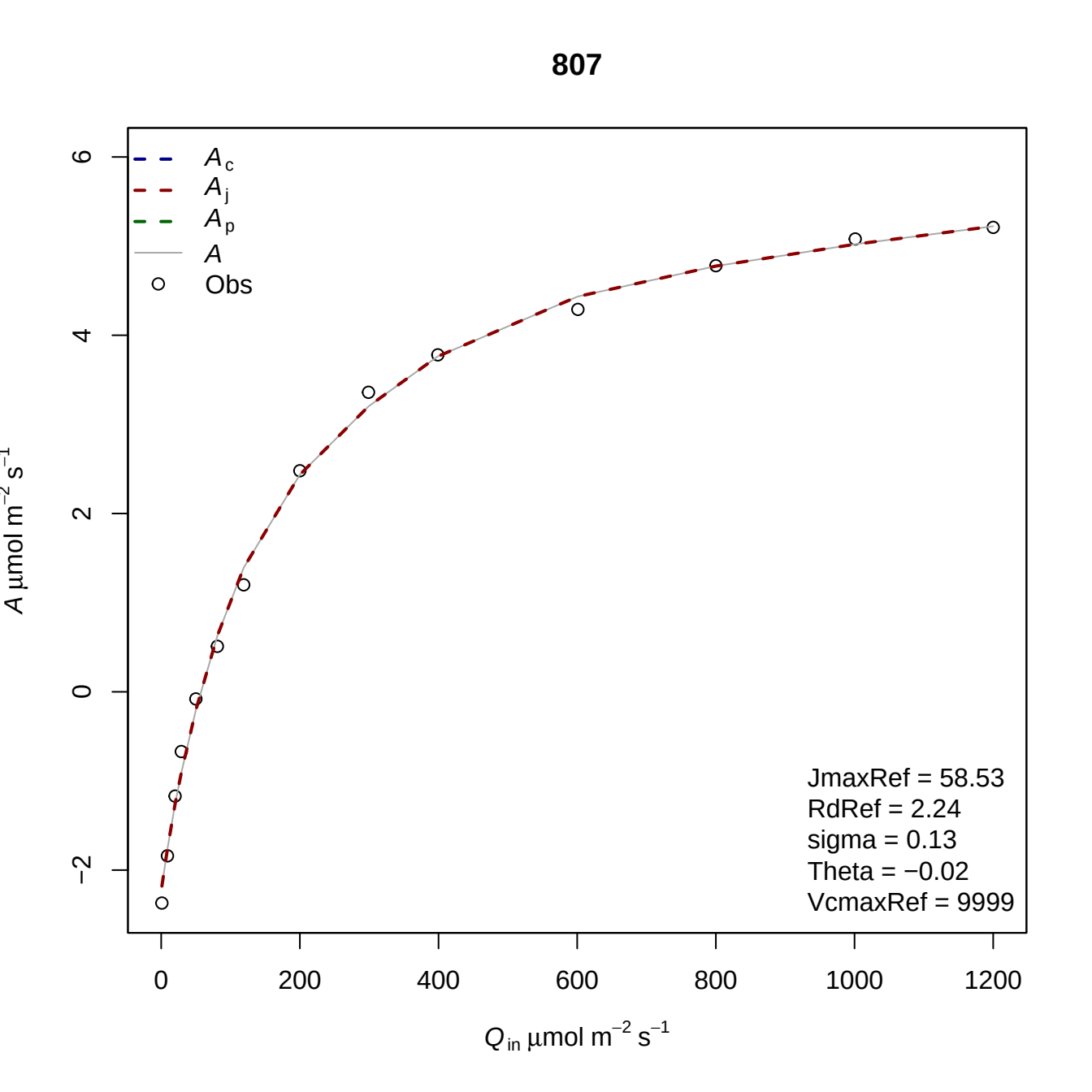

802
A_c
A_j
A_p
A
Obs
JmaxRef = 47.3
RdRef = 0.83
sigma = 0.06
Theta = 0.46
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

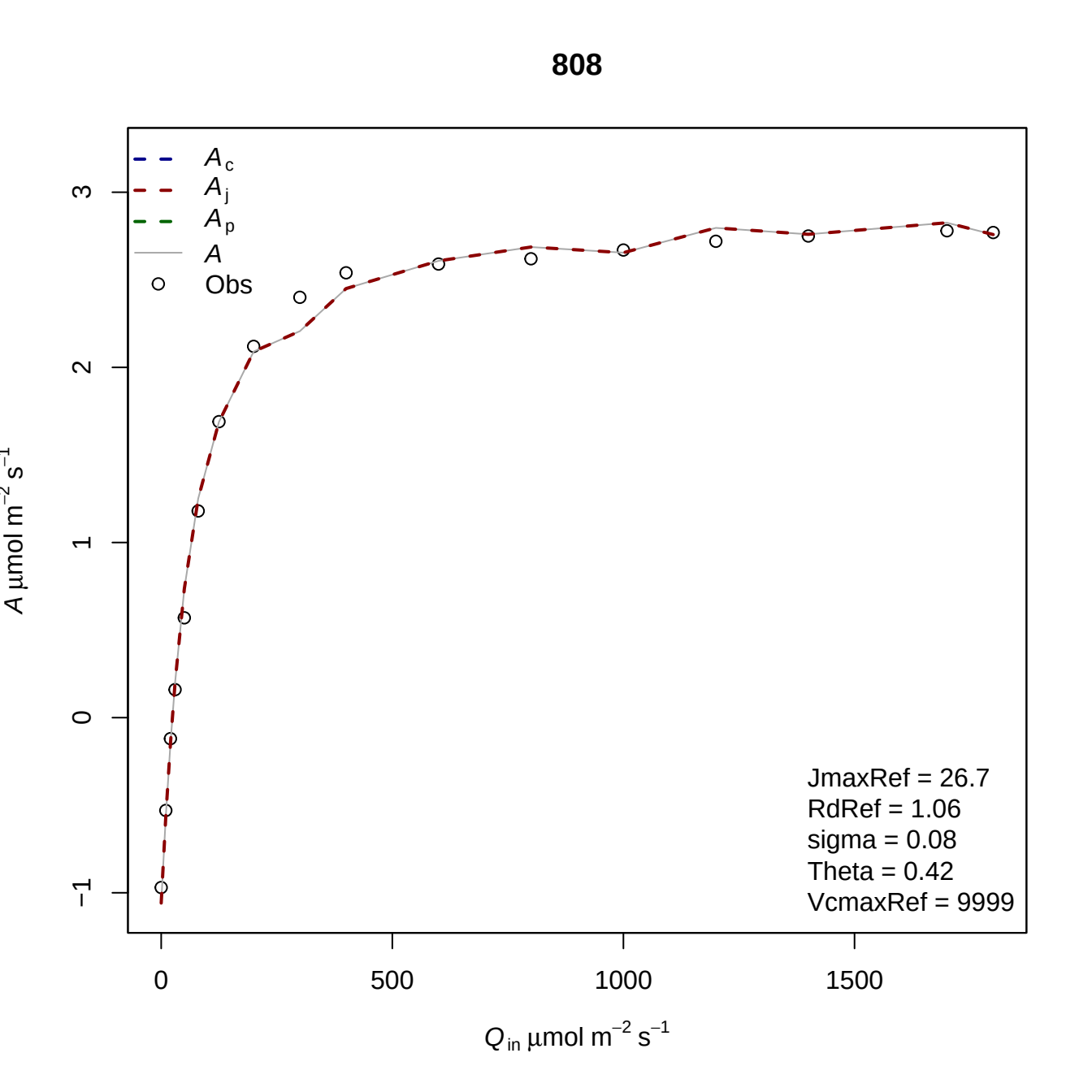

803
A_c
A_j
A_p
A
Obs
JmaxRef = 51.8
RdRef = 1.07
sigma = 0.16
Theta = 0.32
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

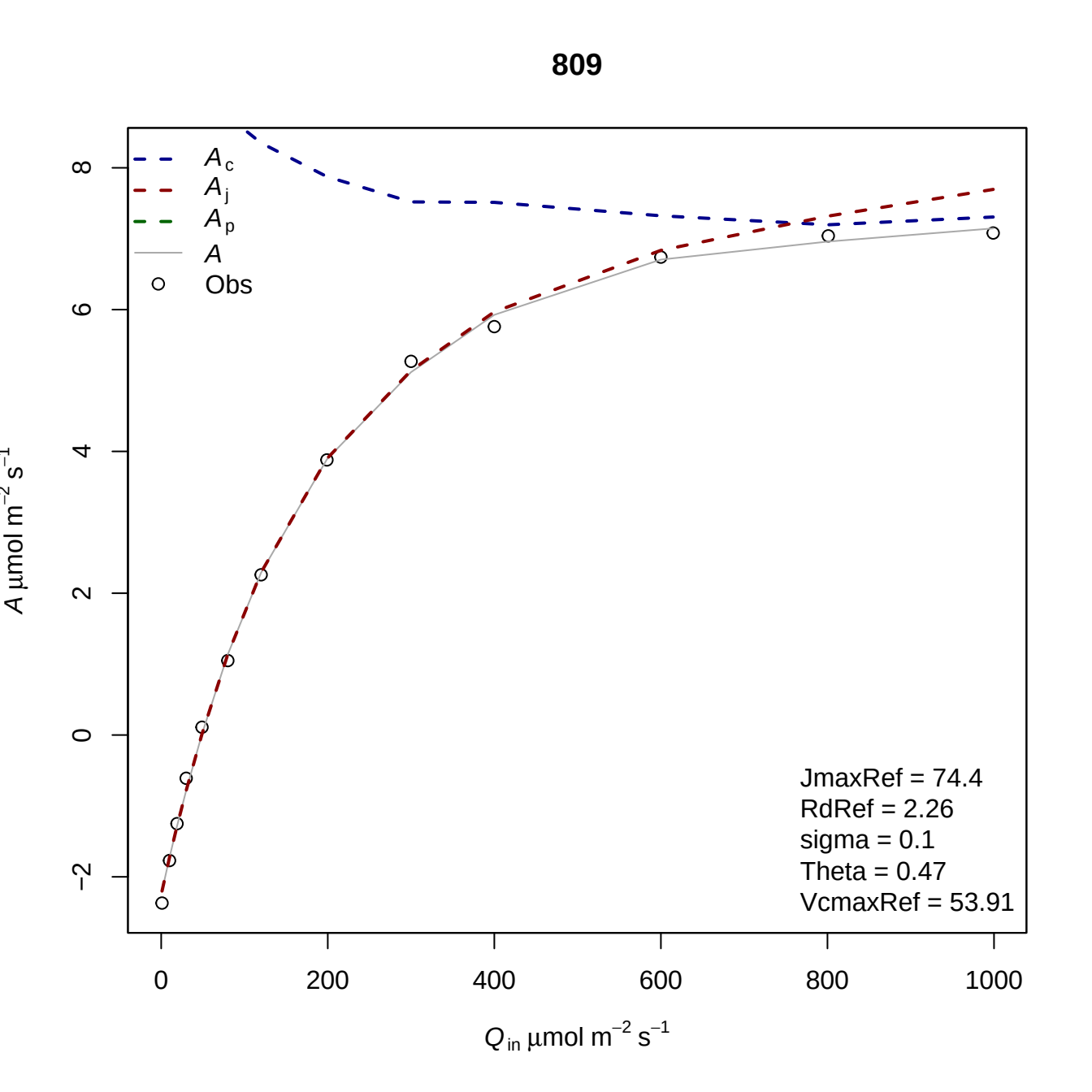

804
A_c
A_j
A_p
A
Obs
JmaxRef = 89.31
RdRef = 0.93
sigma = 0.25
Theta = 0.2
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

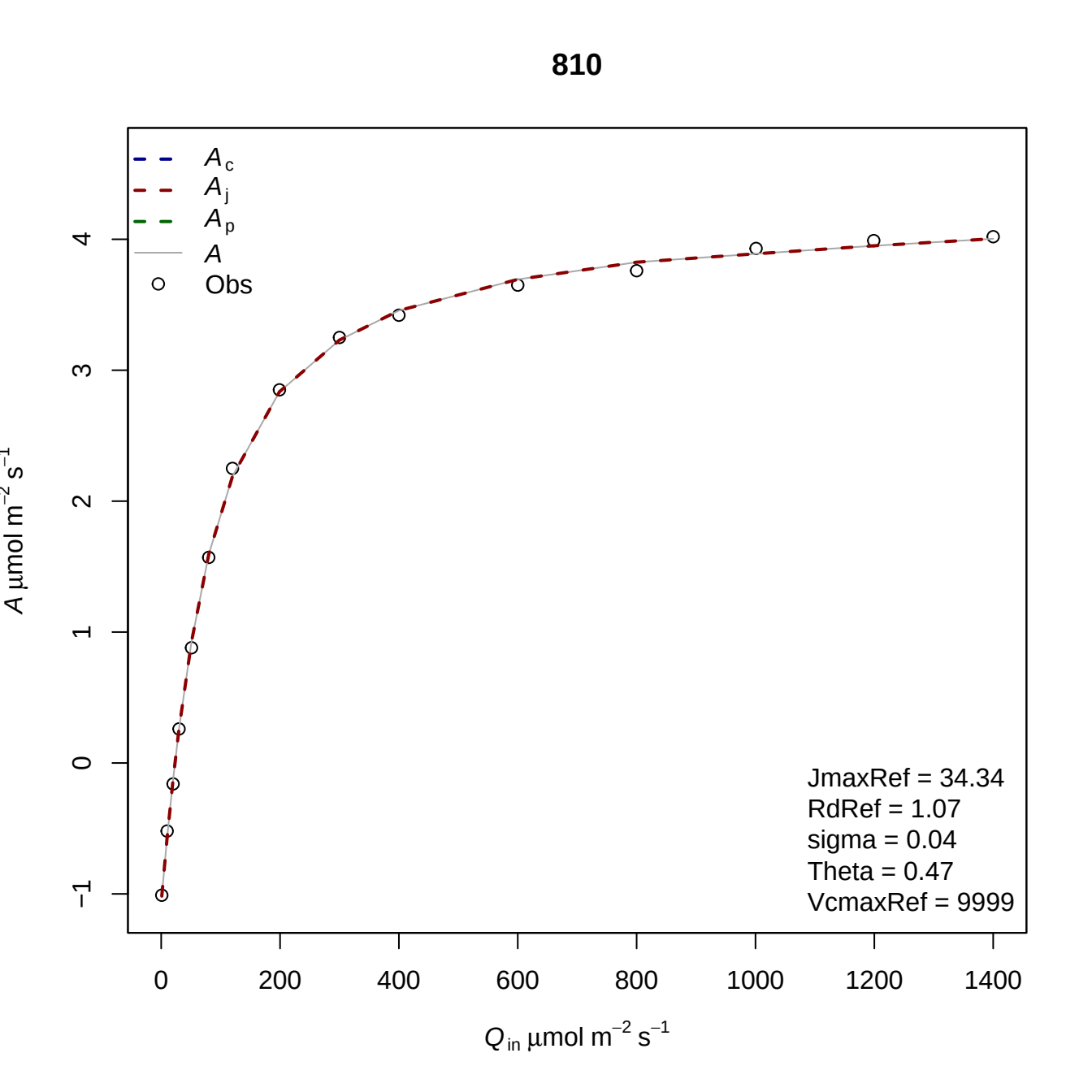

805
A_c
A_j
A_p
A
Obs
JmaxRef = 49.64
RdRef = 0.58
sigma = 0.17
Theta = 0.93
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

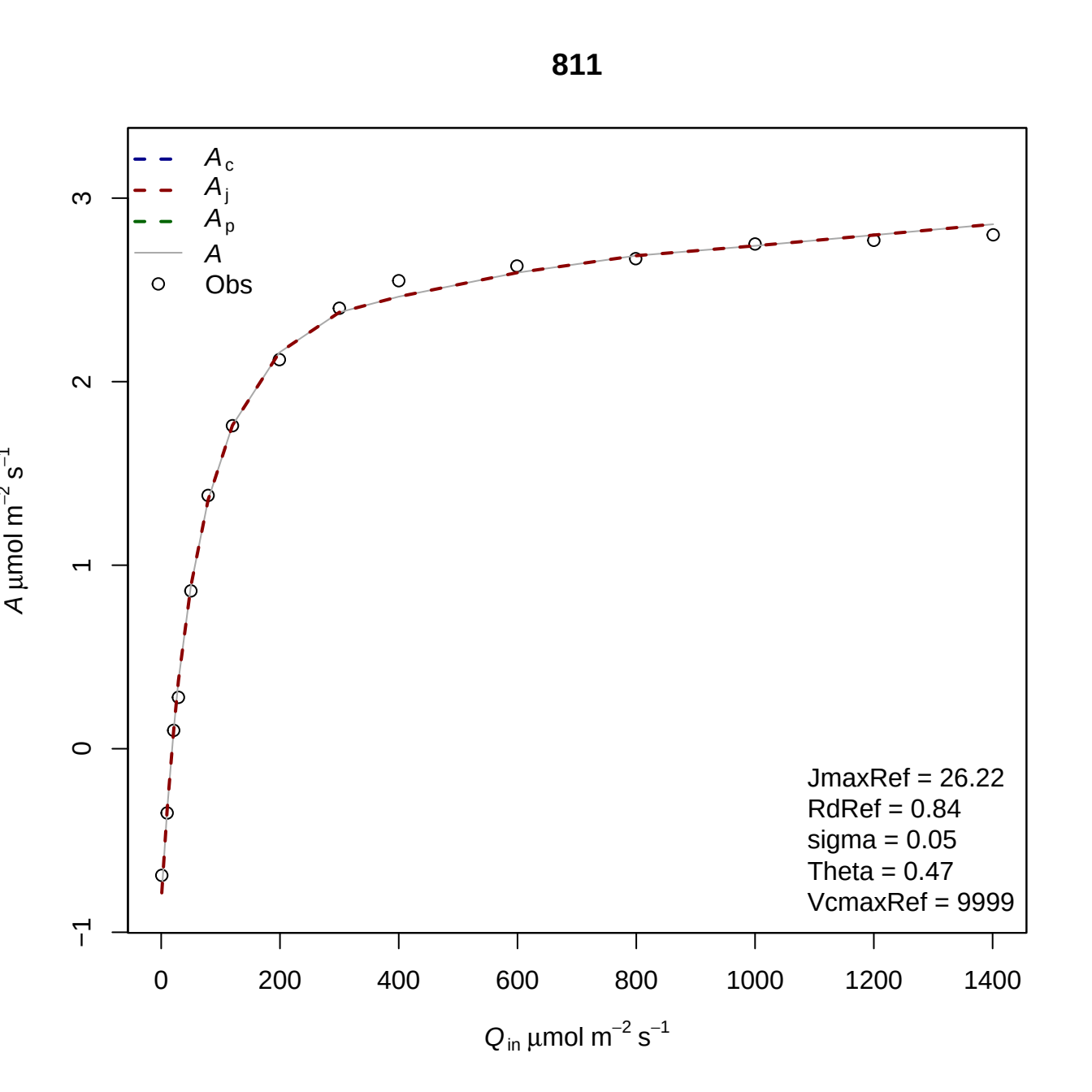

806
A_c
A_j
A_p
A
Obs
JmaxRef = 73.21
RdRef = 1.61
sigma = 0.13
Theta = 0.19
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

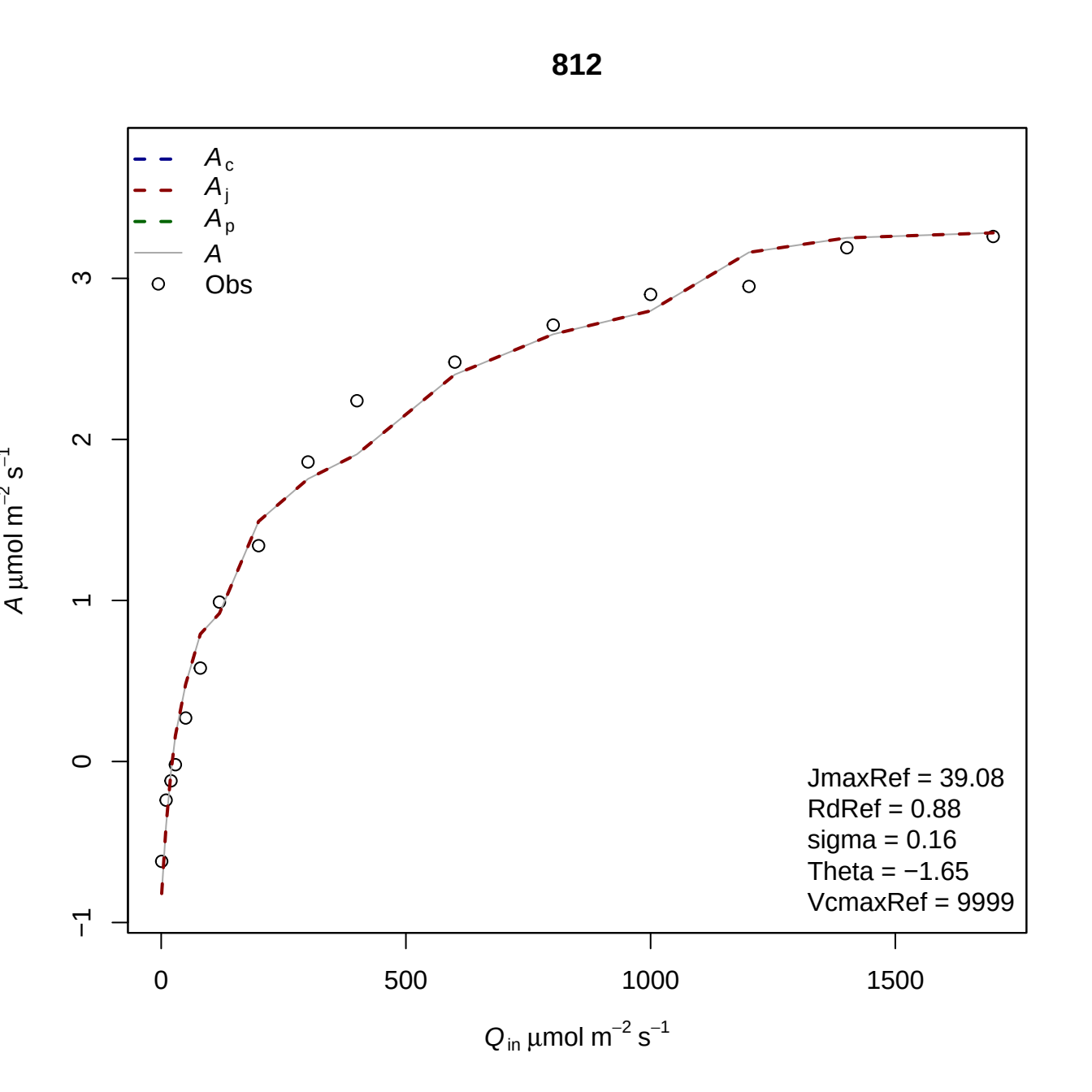

807
A_c
A_j
A_p
A
Obs
JmaxRef = 58.53
RdRef = 2.24
sigma = 0.13
Theta = −0.02
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200

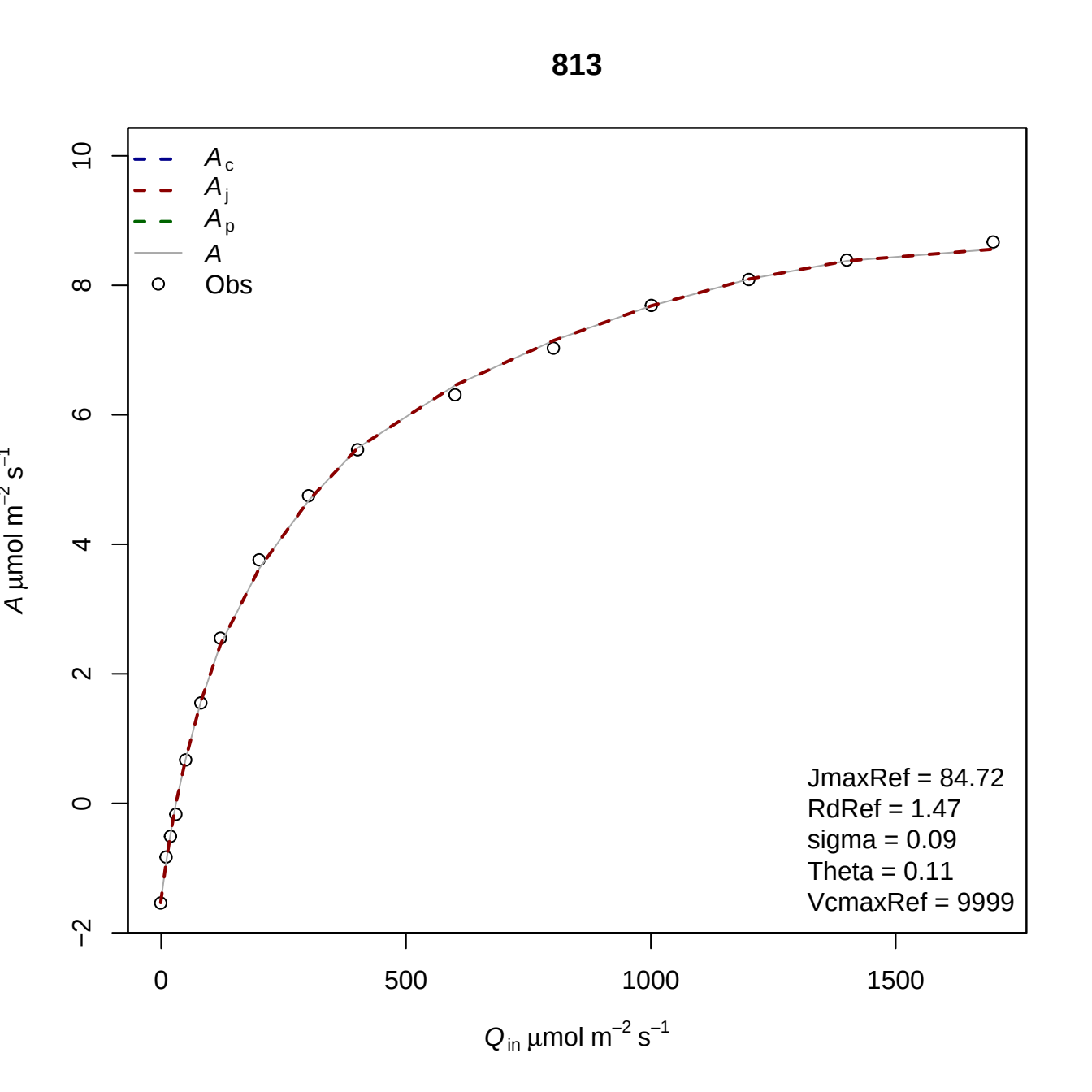

808
A_c
A_j
A_p
A
Obs
JmaxRef = 26.7
RdRef = 1.06
sigma = 0.08
Theta = 0.42
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

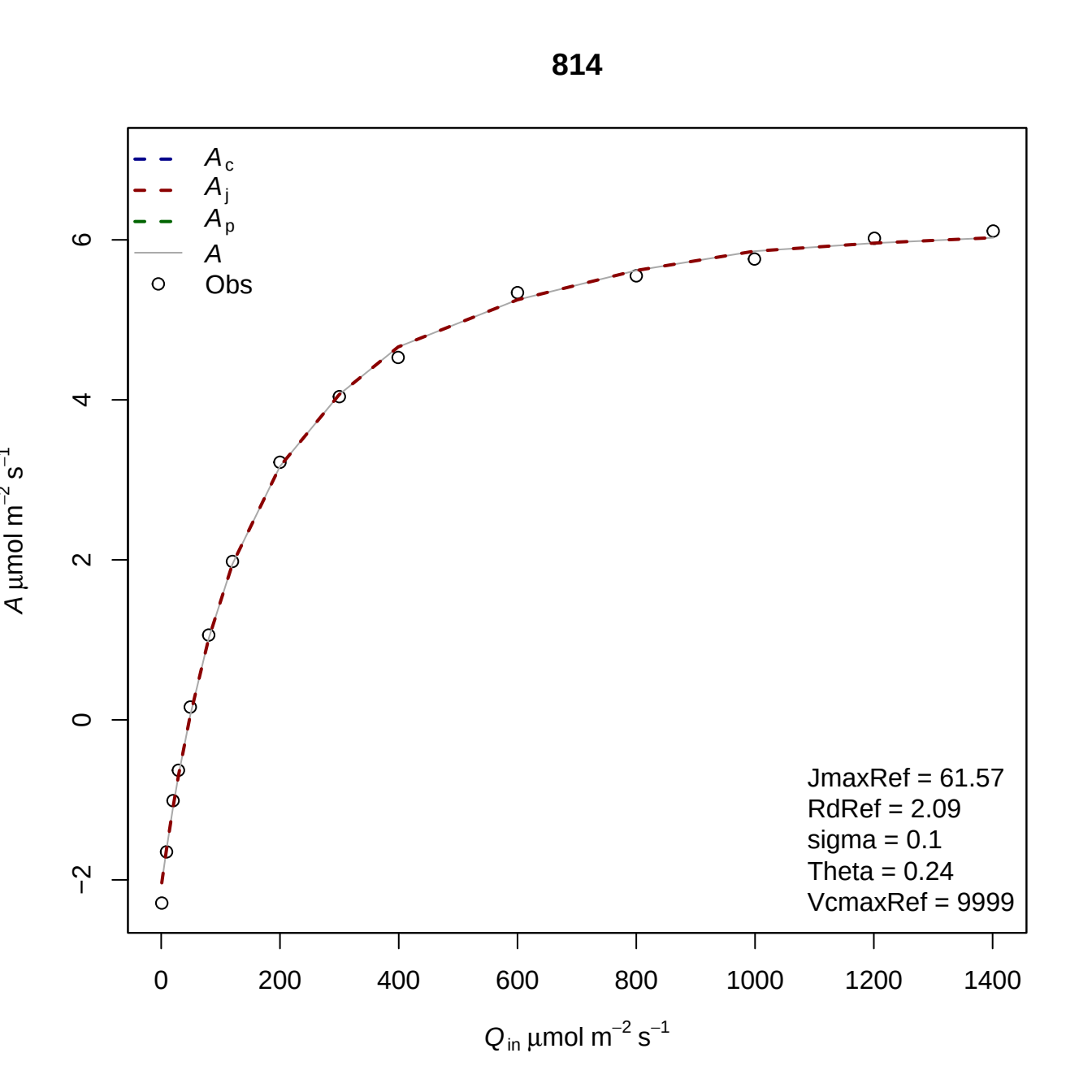

809
A_c
A_j
A_p
A
Obs
JmaxRef = 74.4
RdRef = 2.26
sigma = 0.1
Theta = 0.47
VcmaxRef = 53.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
200
400
600
800
1000

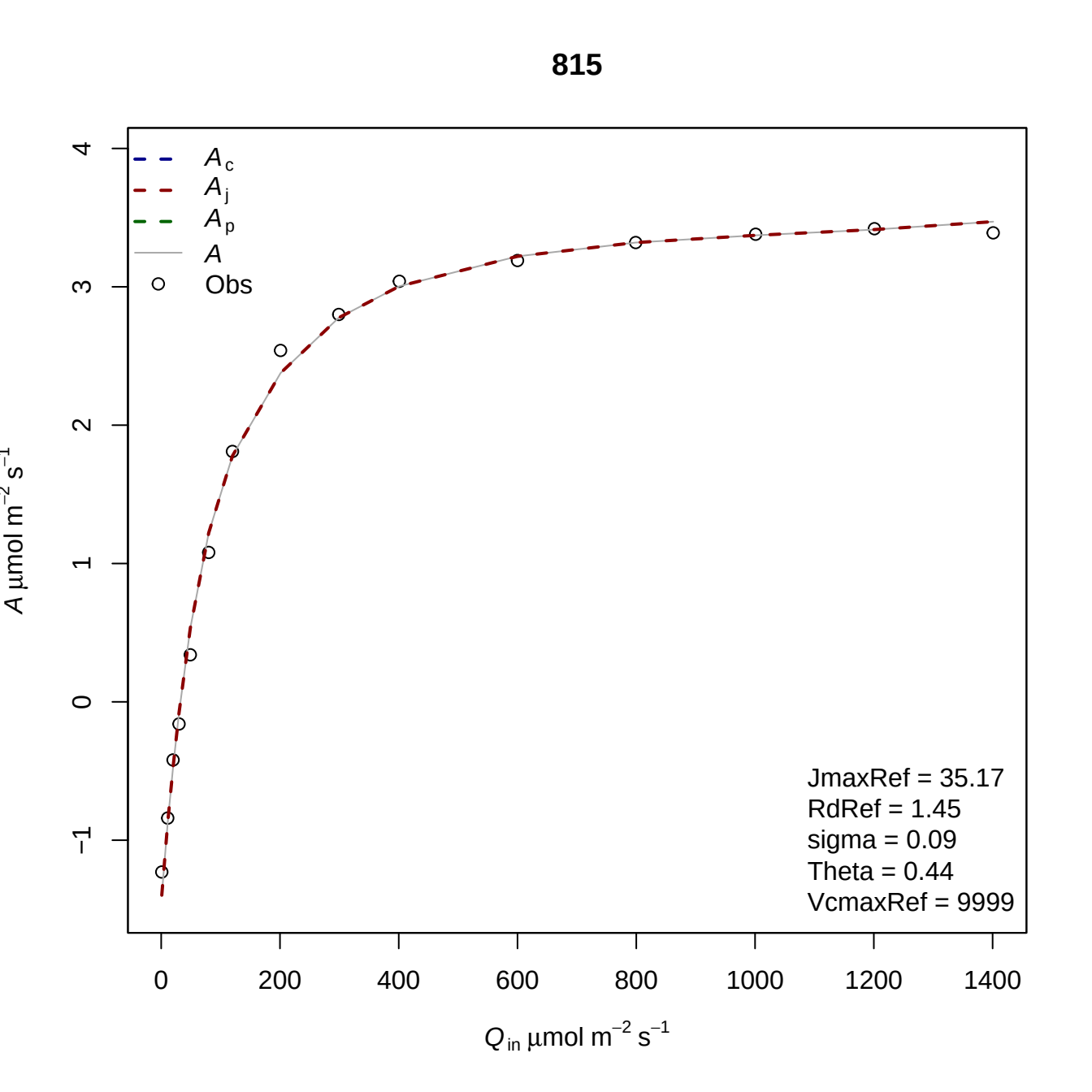

810
A_c
A_j
A_p
A
Obs
JmaxRef = 34.34
RdRef = 1.07
sigma = 0.04
Theta = 0.47
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

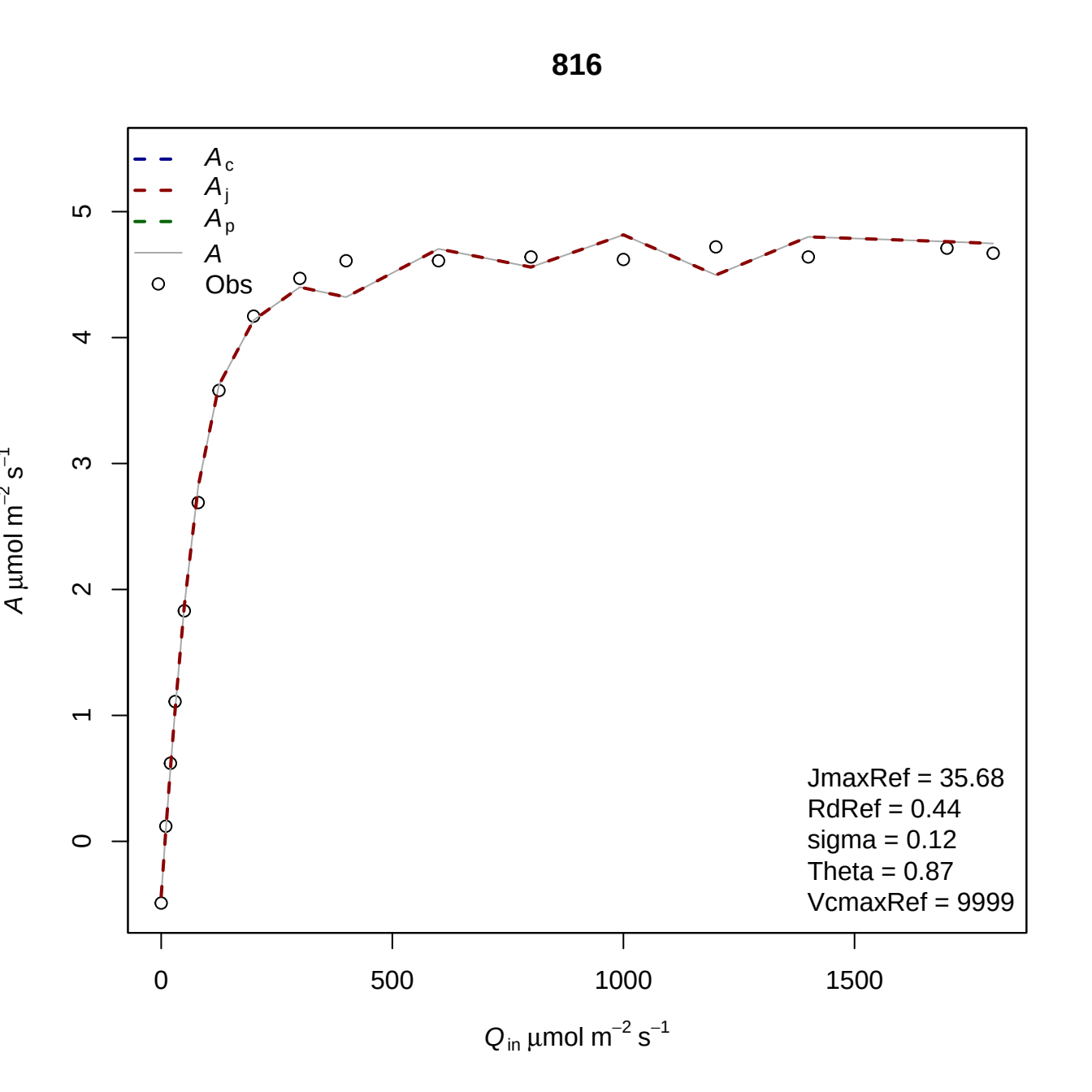

811
A_c
A_j
A_p
A
Obs
JmaxRef = 26.22
RdRef = 0.84
sigma = 0.05
Theta = 0.47
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

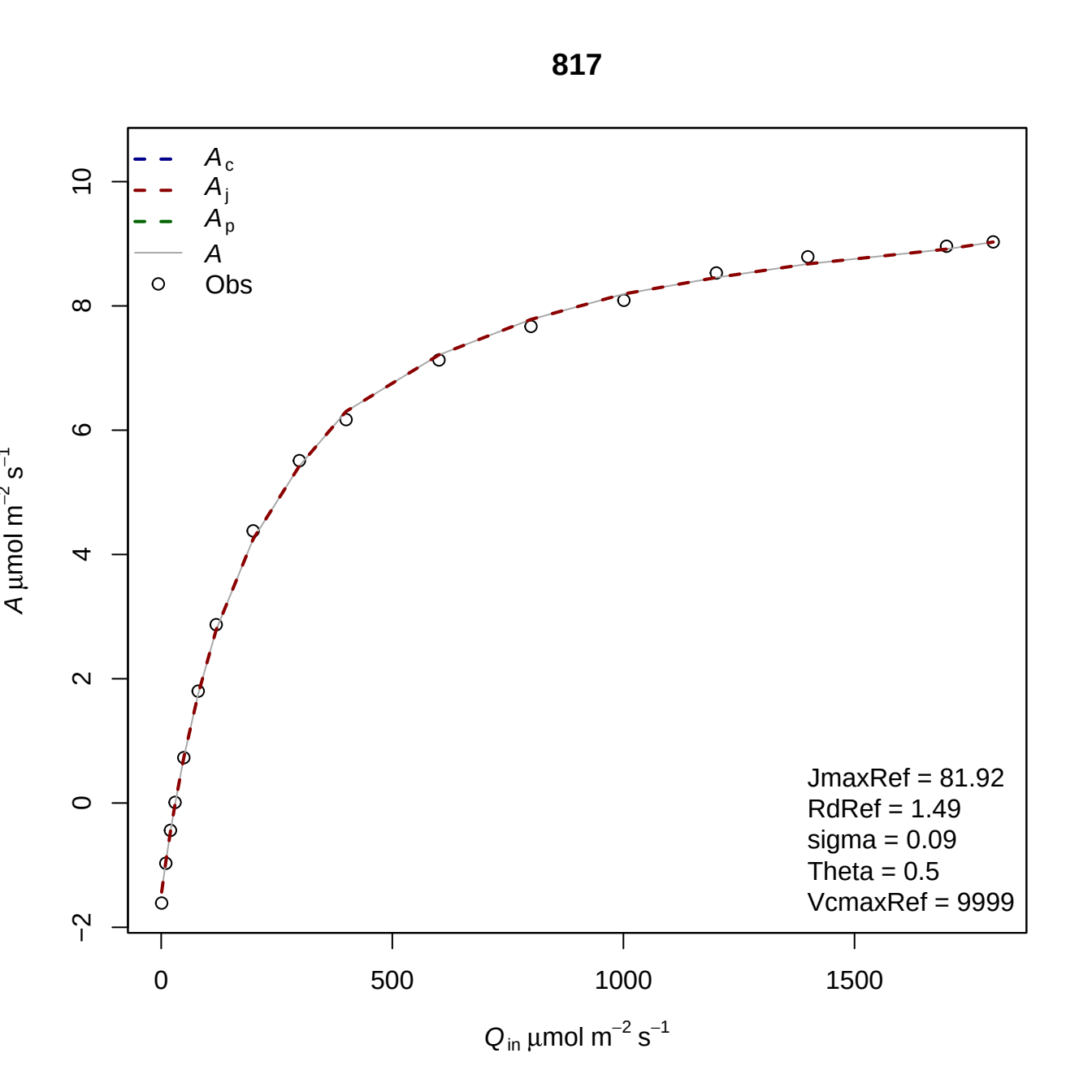

812
A_c
A_j
A_p
A
Obs
JmaxRef = 39.08
RdRef = 0.88
sigma = 0.16
Theta = −1.65
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

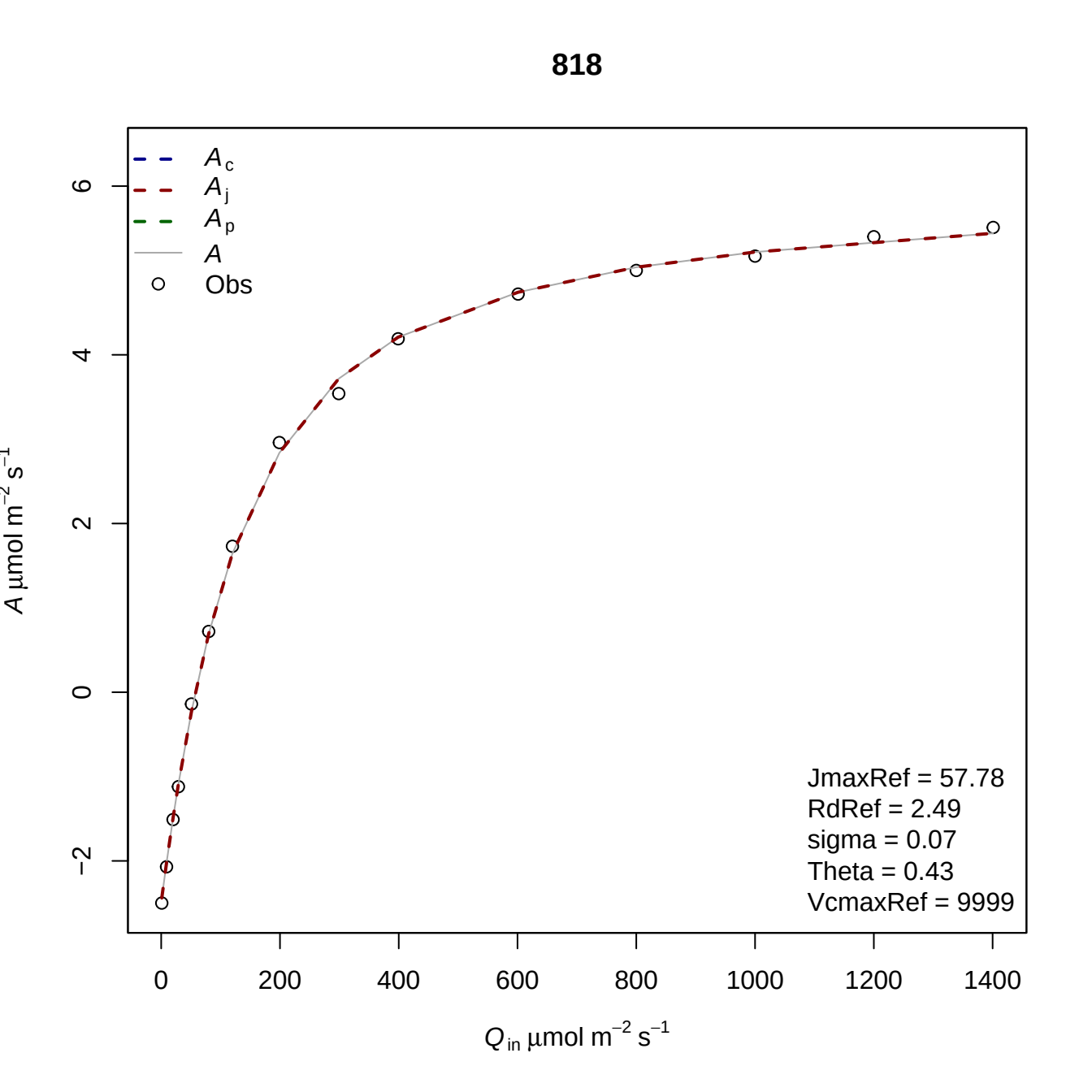

813
A_c
A_j
A_p
A
Obs
JmaxRef = 84.72
RdRef = 1.47
sigma = 0.09
Theta = 0.11
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

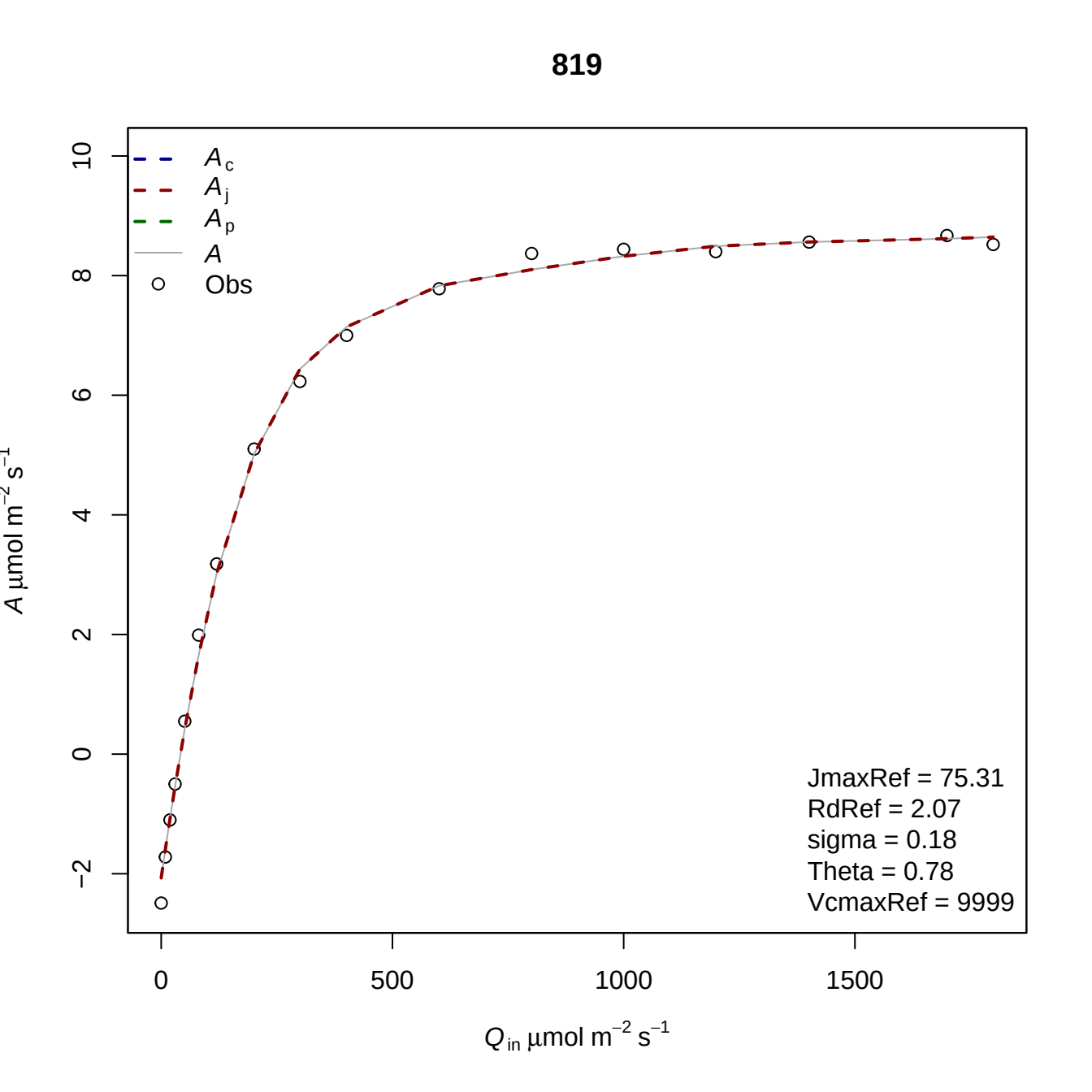

814
A_c
A_j
A_p
A
Obs
JmaxRef = 61.57
RdRef = 2.09
sigma = 0.1
Theta = 0.24
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

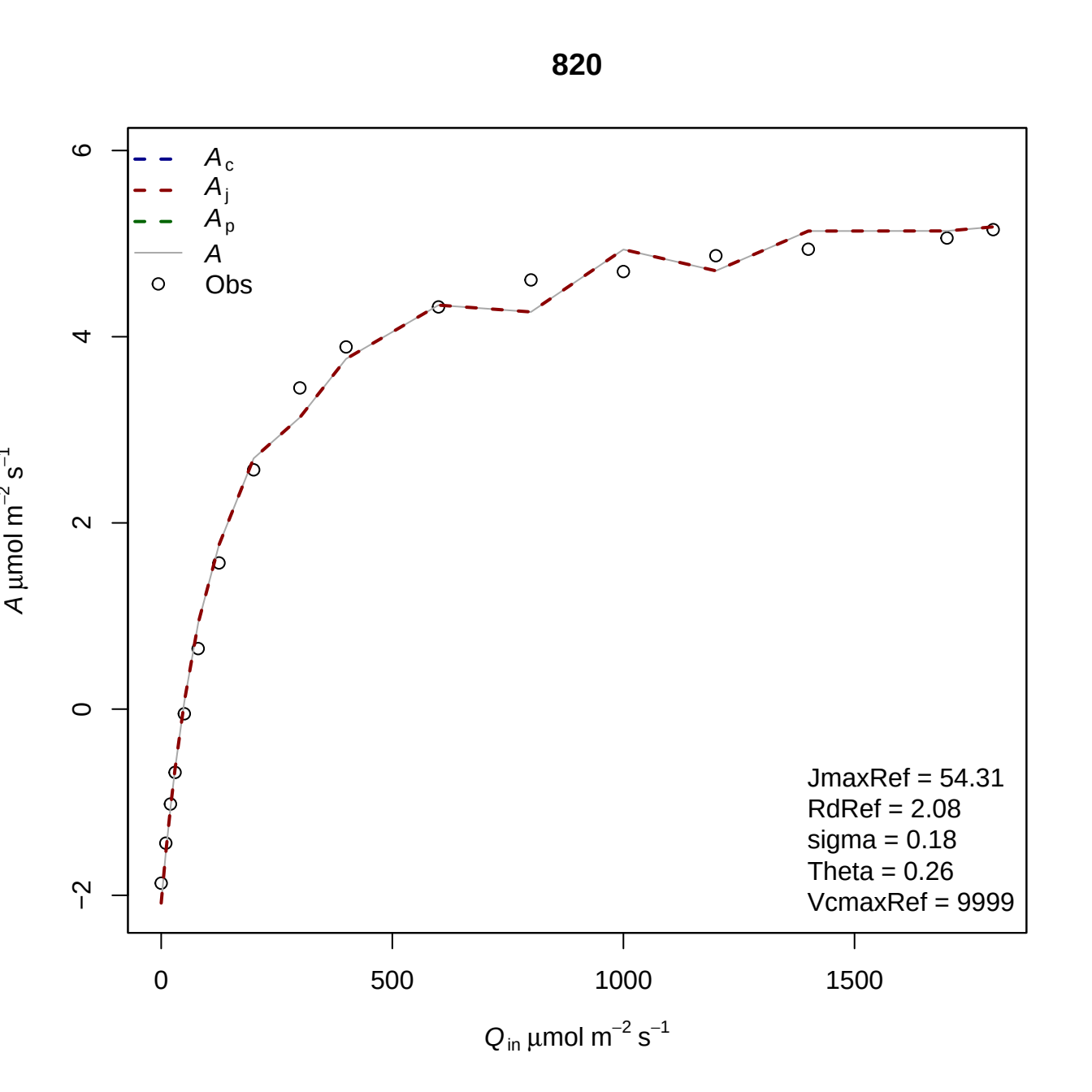

815
A_c
A_j
A_p
A
Obs
JmaxRef = 35.17
RdRef = 1.45
sigma = 0.09
Theta = 0.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
200
400
600
800
1000
1200
1400

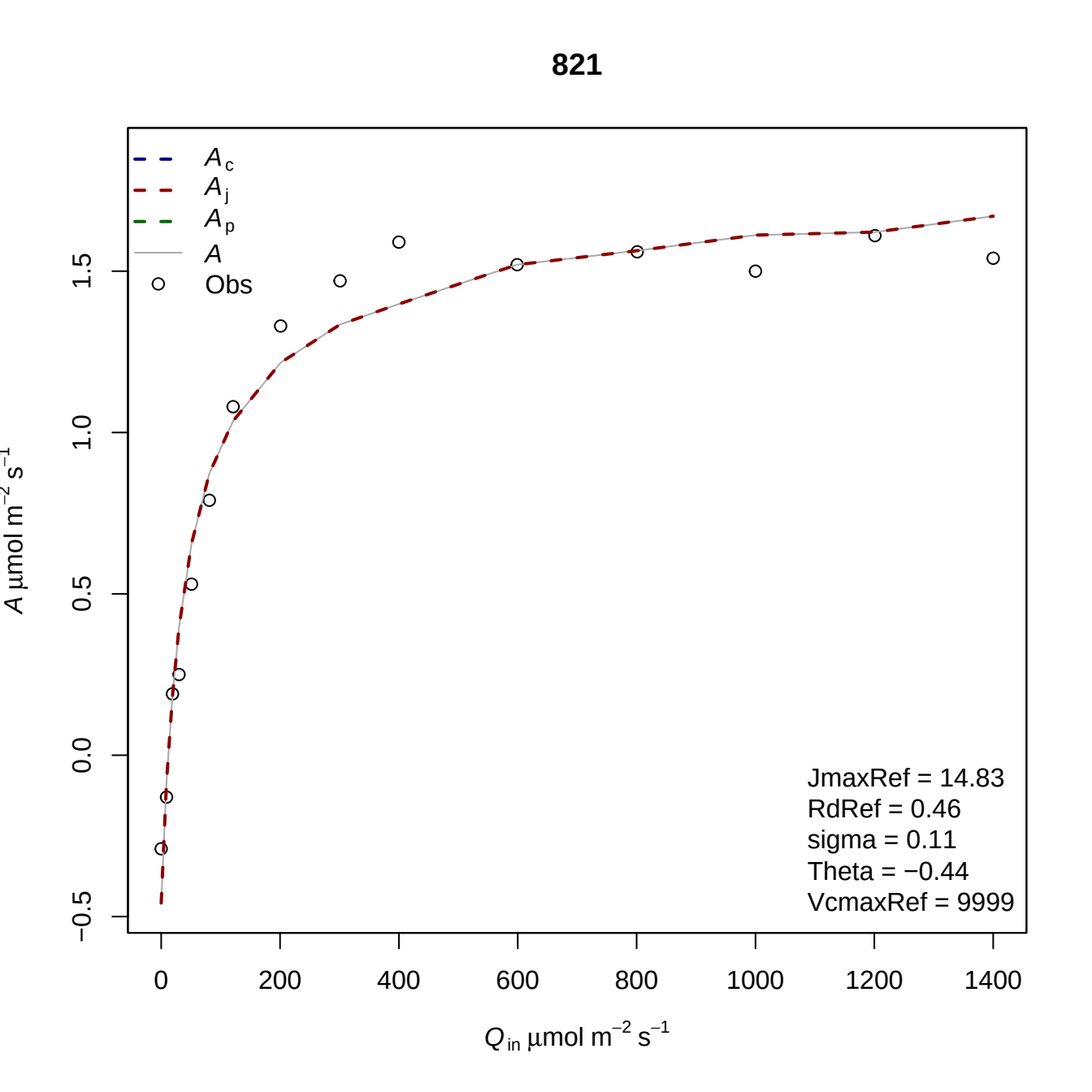

816
A_c
A_j
A_p
A
Obs
JmaxRef = 35.68
RdRef = 0.44
sigma = 0.12
Theta = 0.87
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

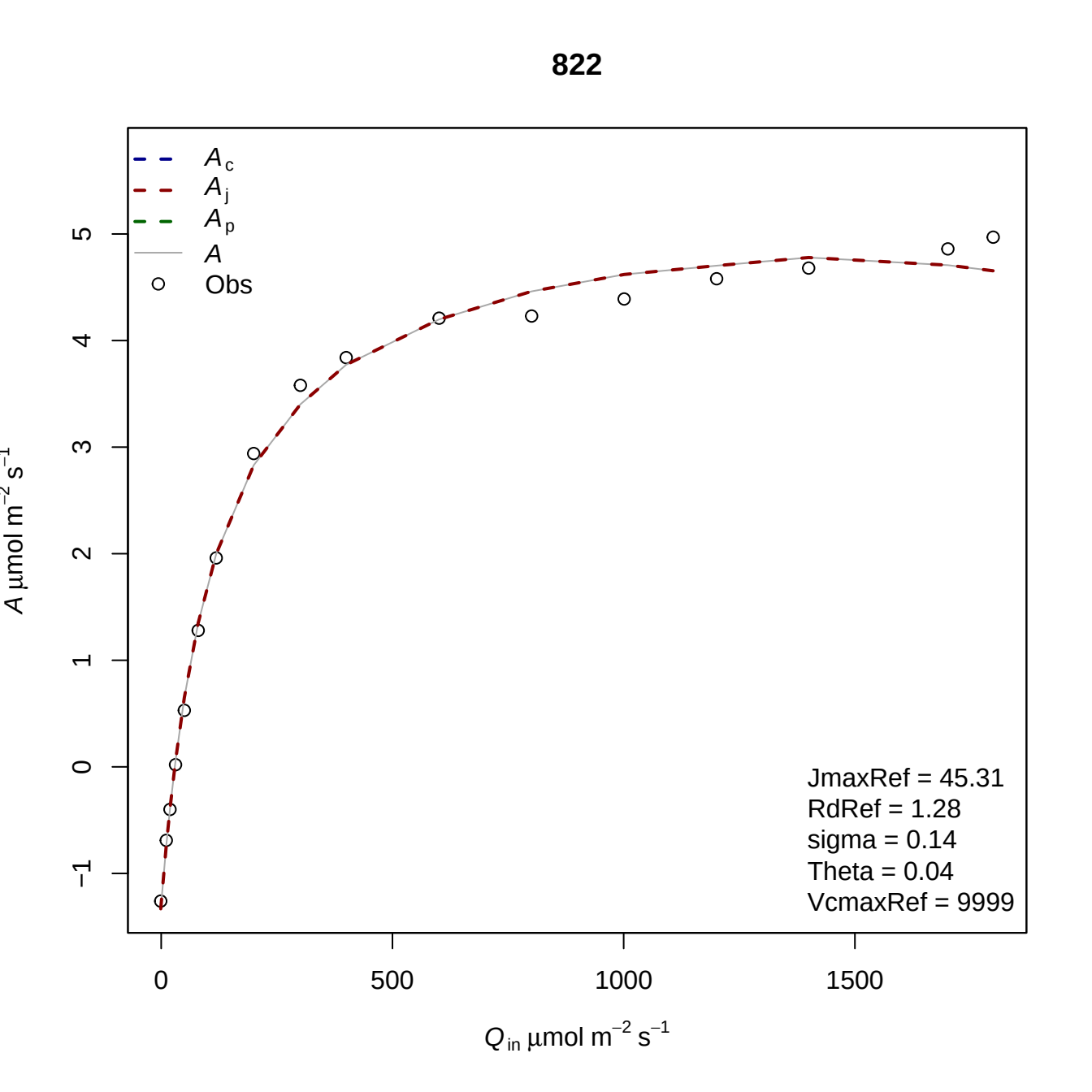

817
A_c
A_j
A_p
A
Obs
JmaxRef = 81.92
RdRef = 1.49
sigma = 0.09
Theta = 0.5
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

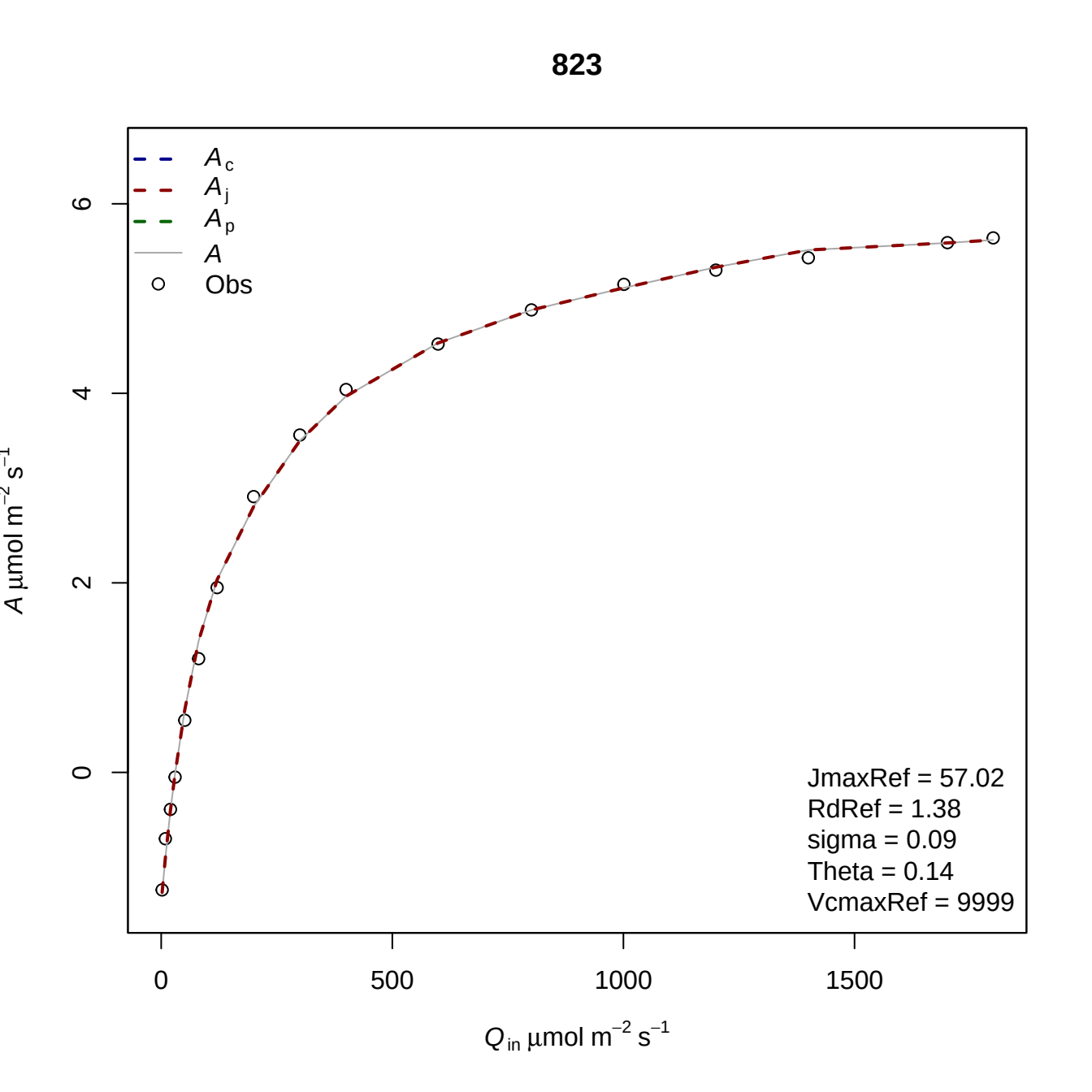

818
A_c
A_j
A_p
A
Obs
JmaxRef = 57.78
RdRef = 2.49
sigma = 0.07
Theta = 0.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200
1400

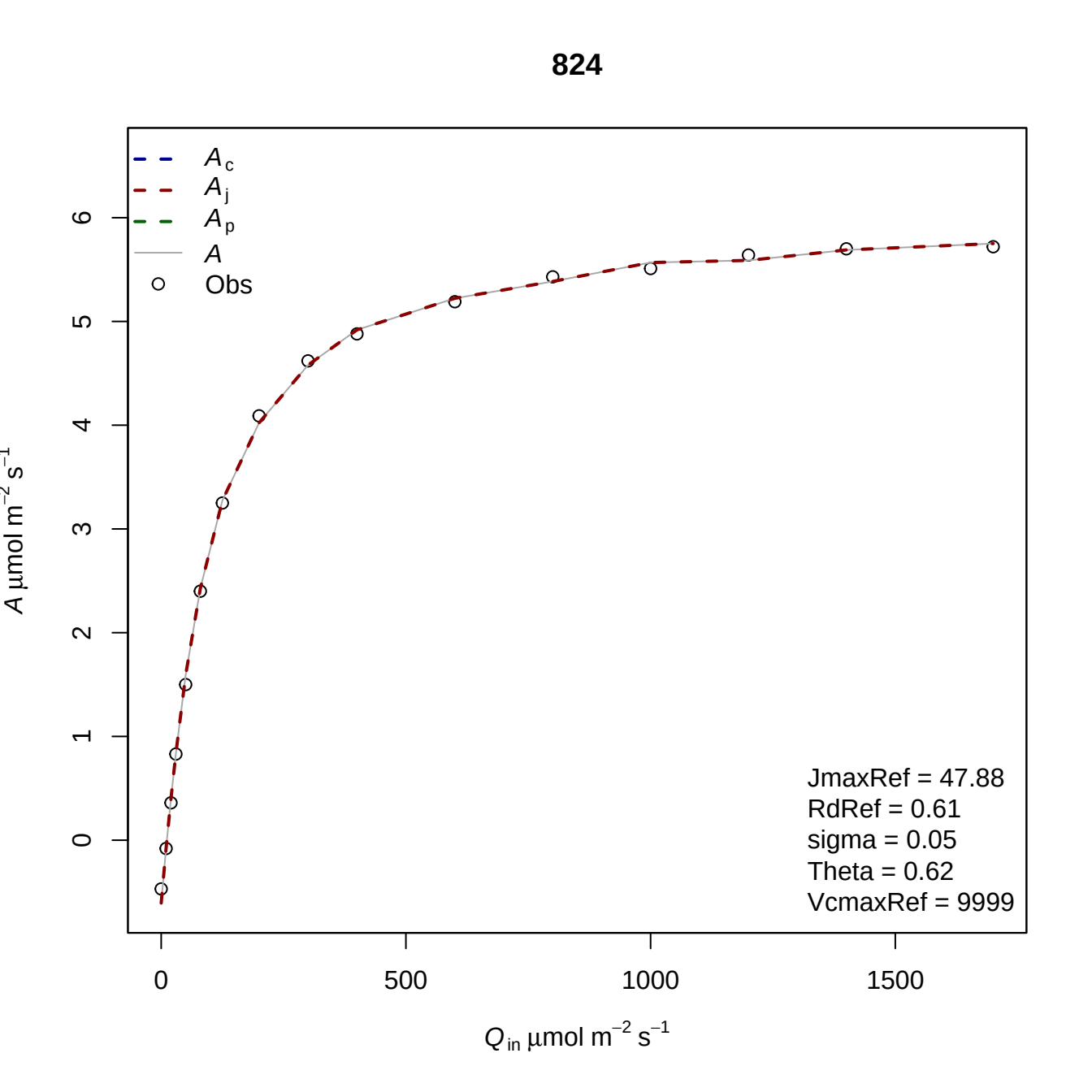

819
A_c
A_j
A_p
A
Obs
JmaxRef = 75.31
RdRef = 2.07
sigma = 0.18
Theta = 0.78
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

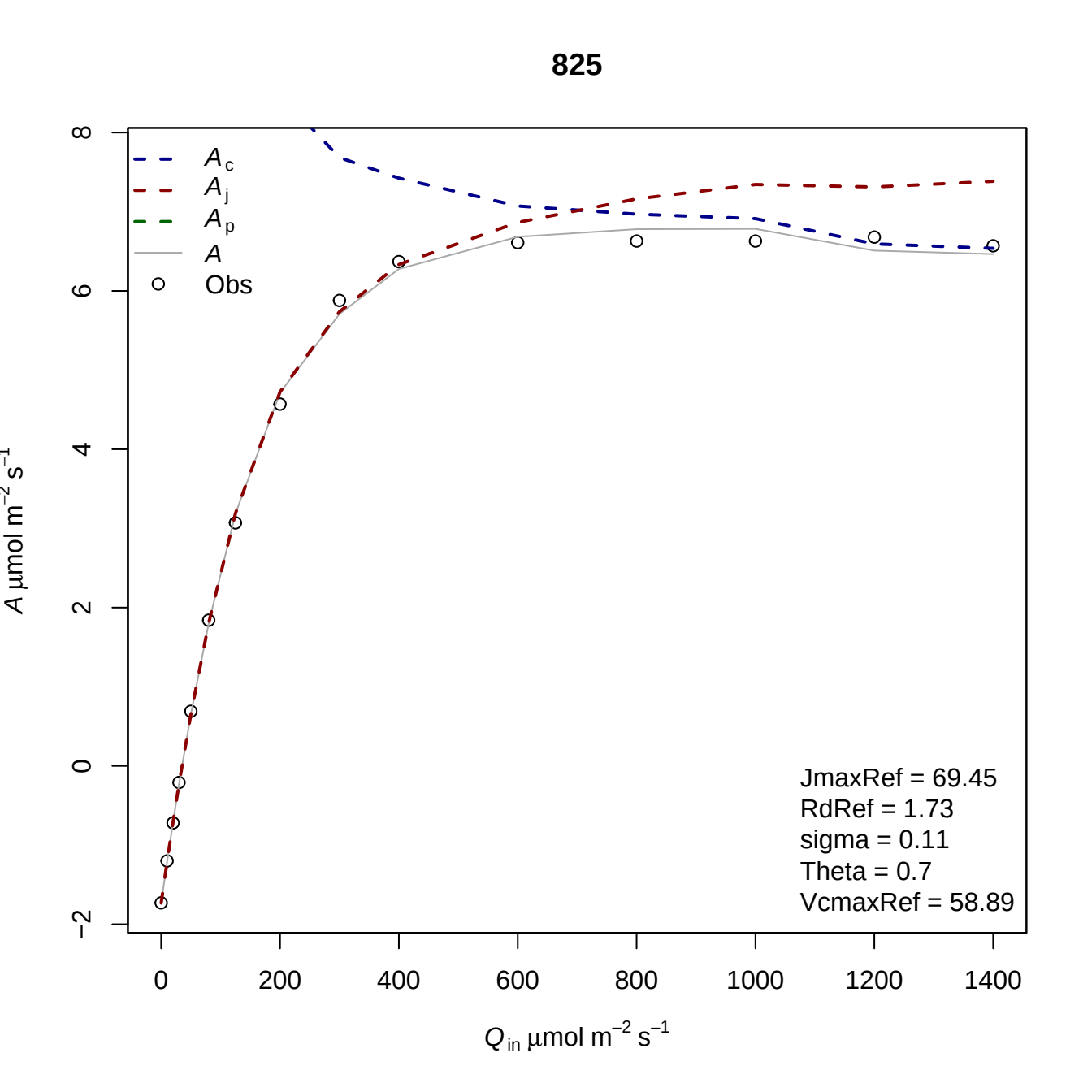

820
A_c
A_j
A_p
A
Obs
JmaxRef = 54.31
RdRef = 2.08
sigma = 0.18
Theta = 0.26
VcmaxRef = 9999
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

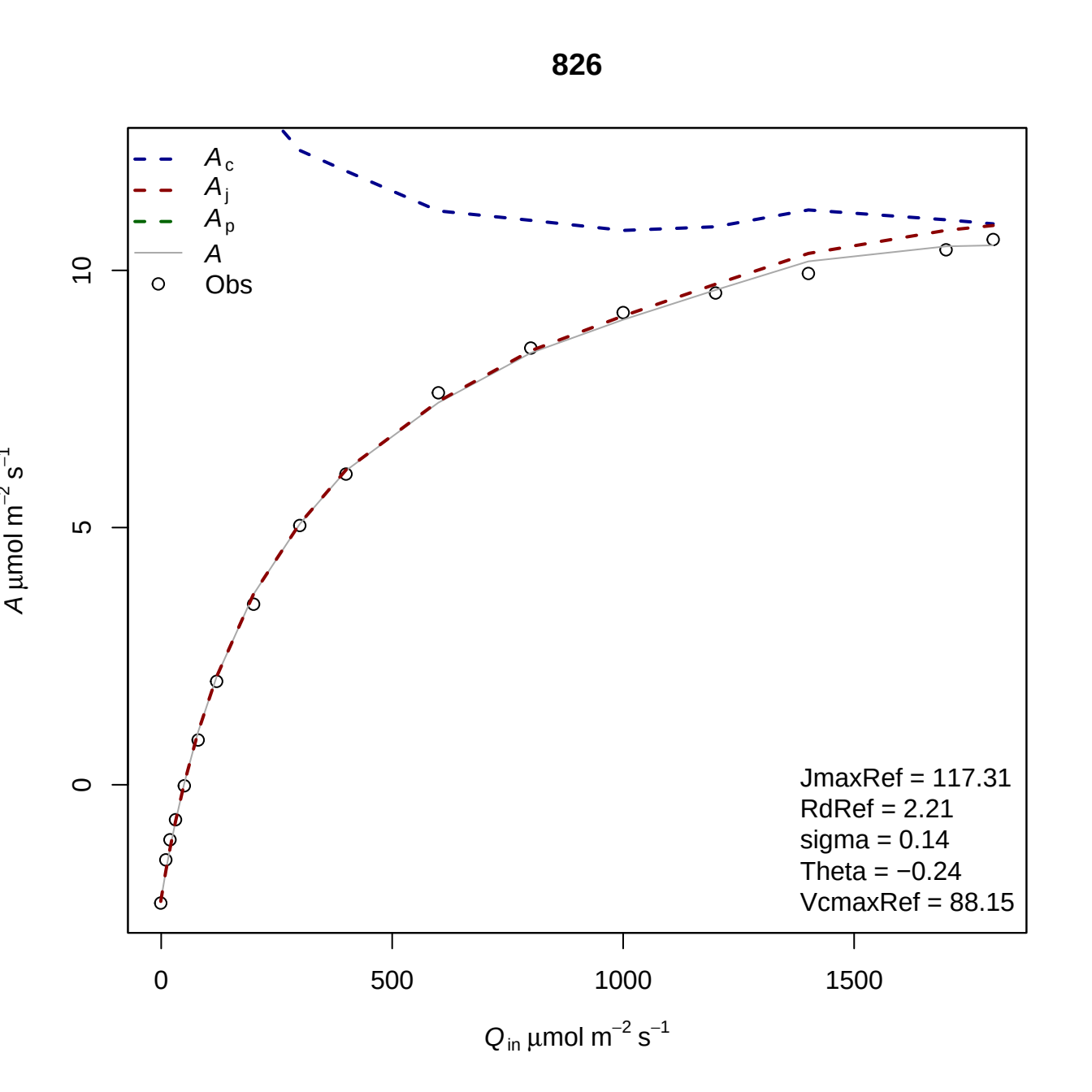

821
A_c
A_j
A_p
A
Obs
JmaxRef = 14.83
RdRef = 0.46
sigma = 0.11
Theta = −0.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
0
200
400
600
800
1000
1200
1400
Q_{in} µmol m^{-2} s^{-1}

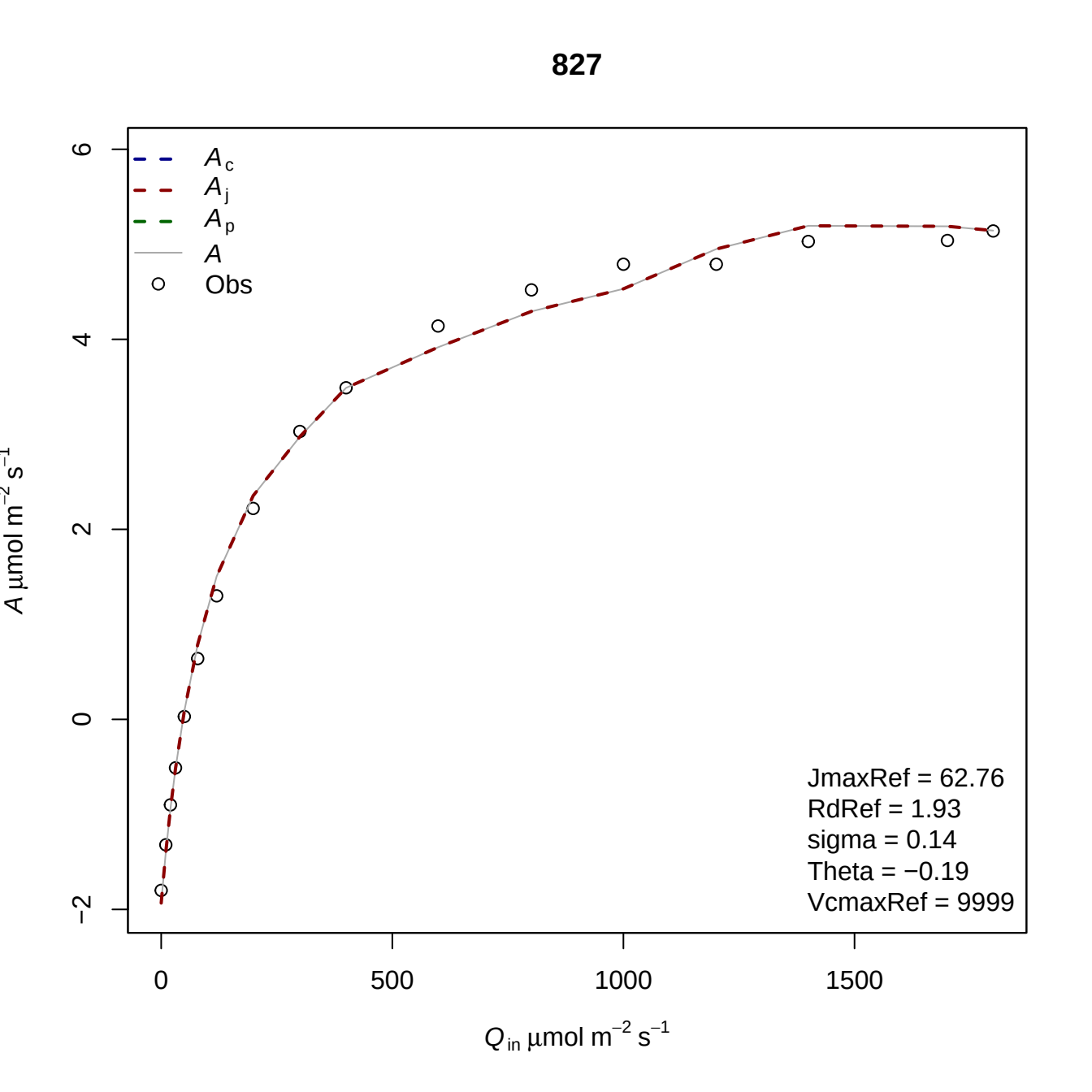

822
A_c
A_j
A_p
A
Obs
JmaxRef = 45.31
RdRef = 1.28
sigma = 0.14
Theta = 0.04
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

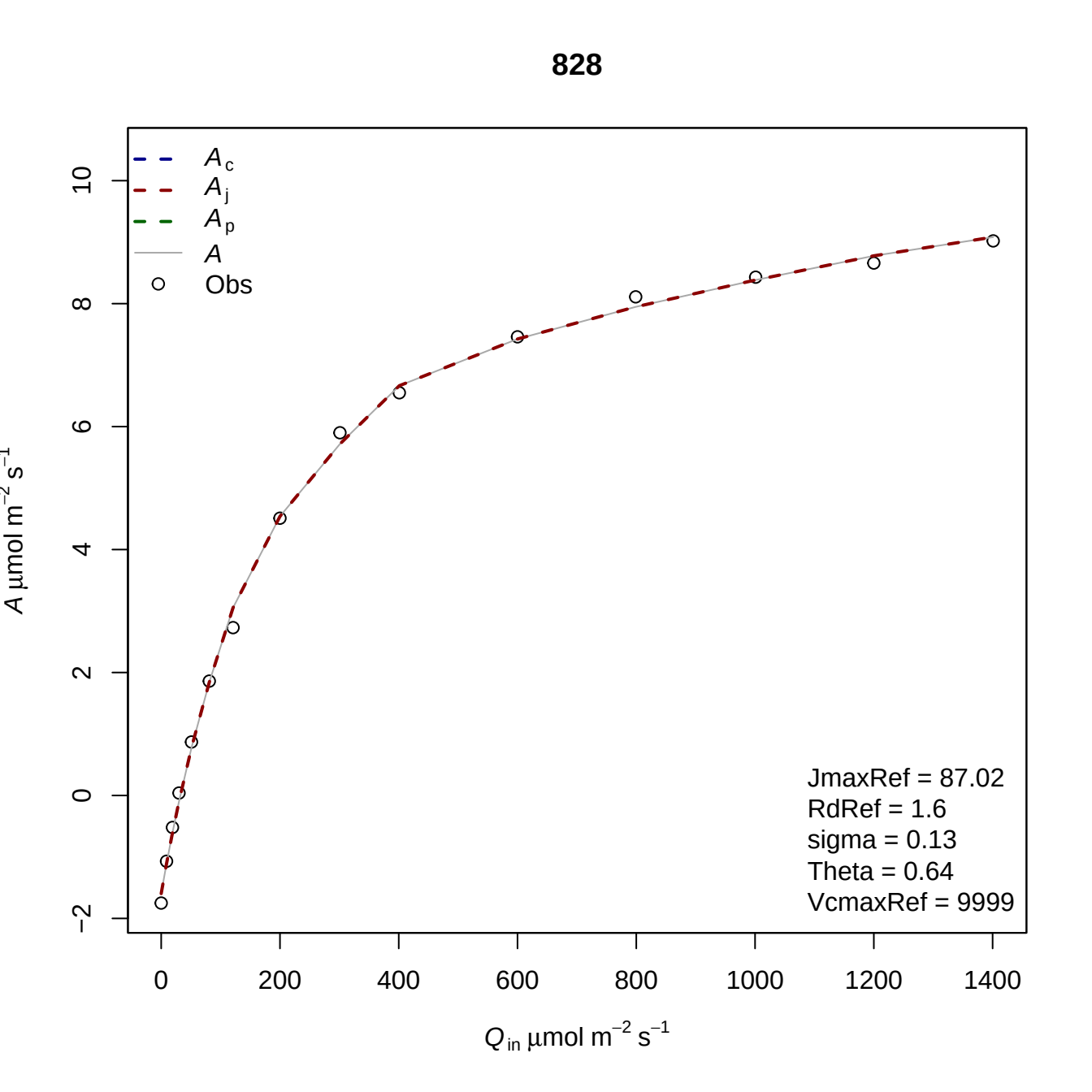

823
A_c
A_j
A_p
A
Obs
JmaxRef = 57.02
RdRef = 1.38
sigma = 0.09
Theta = 0.14
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

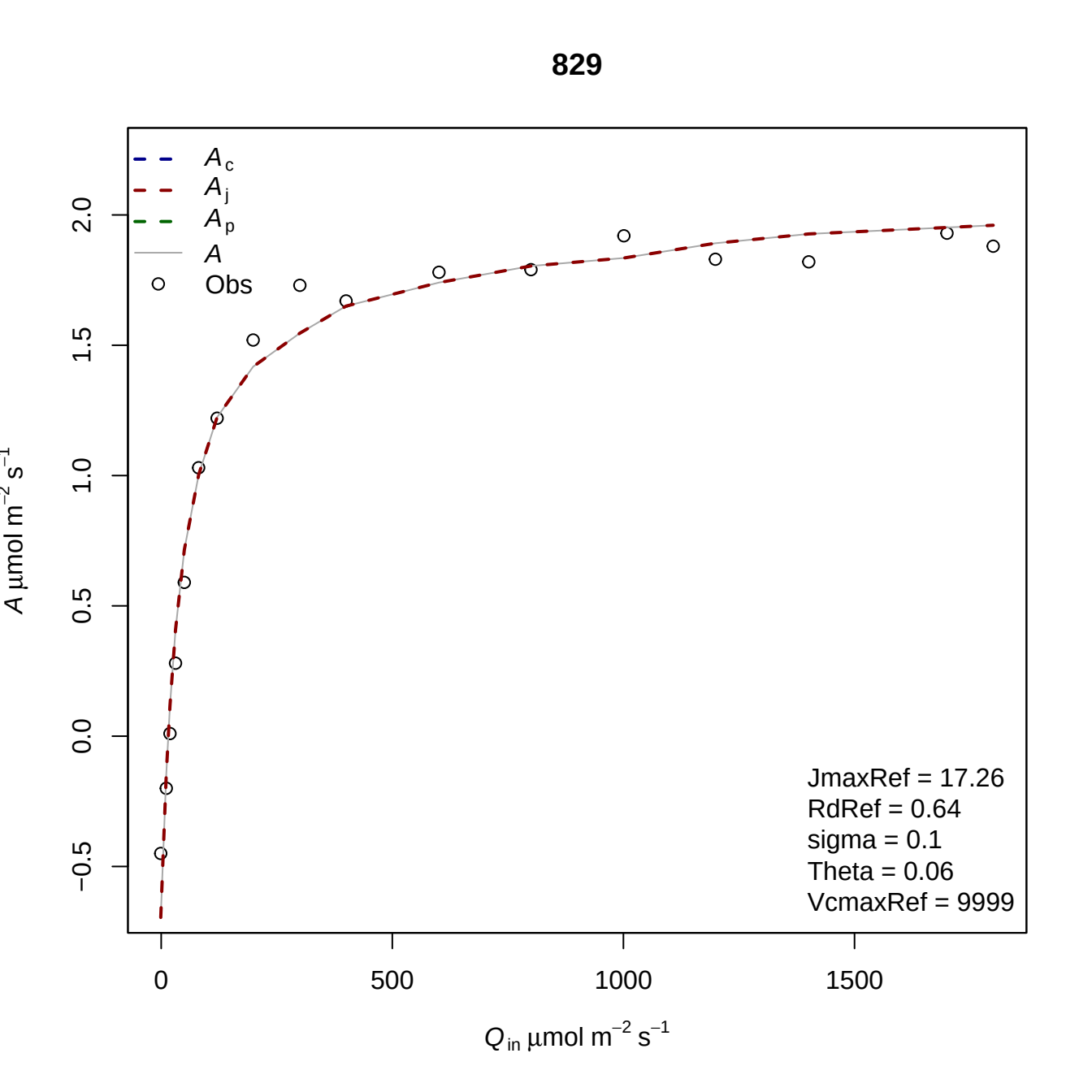

824
A_c
A_j
A_p
A
Obs
JmaxRef = 47.88
RdRef = 0.61
sigma = 0.05
Theta = 0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

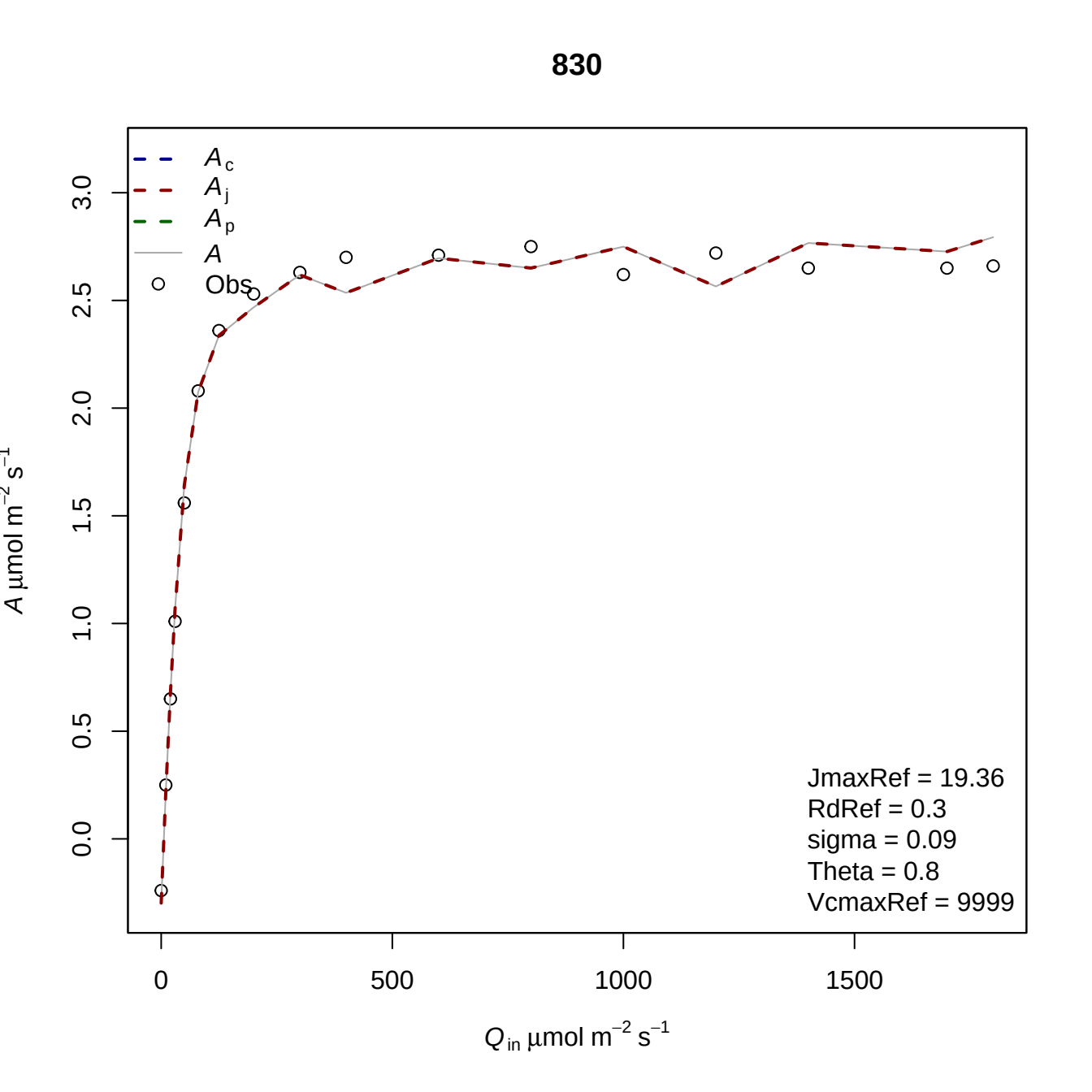

825
A_c
A_j
A_p
A
Obs
JmaxRef = 69.45
RdRef = 1.73
sigma = 0.11
Theta = 0.7
VcmaxRef = 58.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200
1400

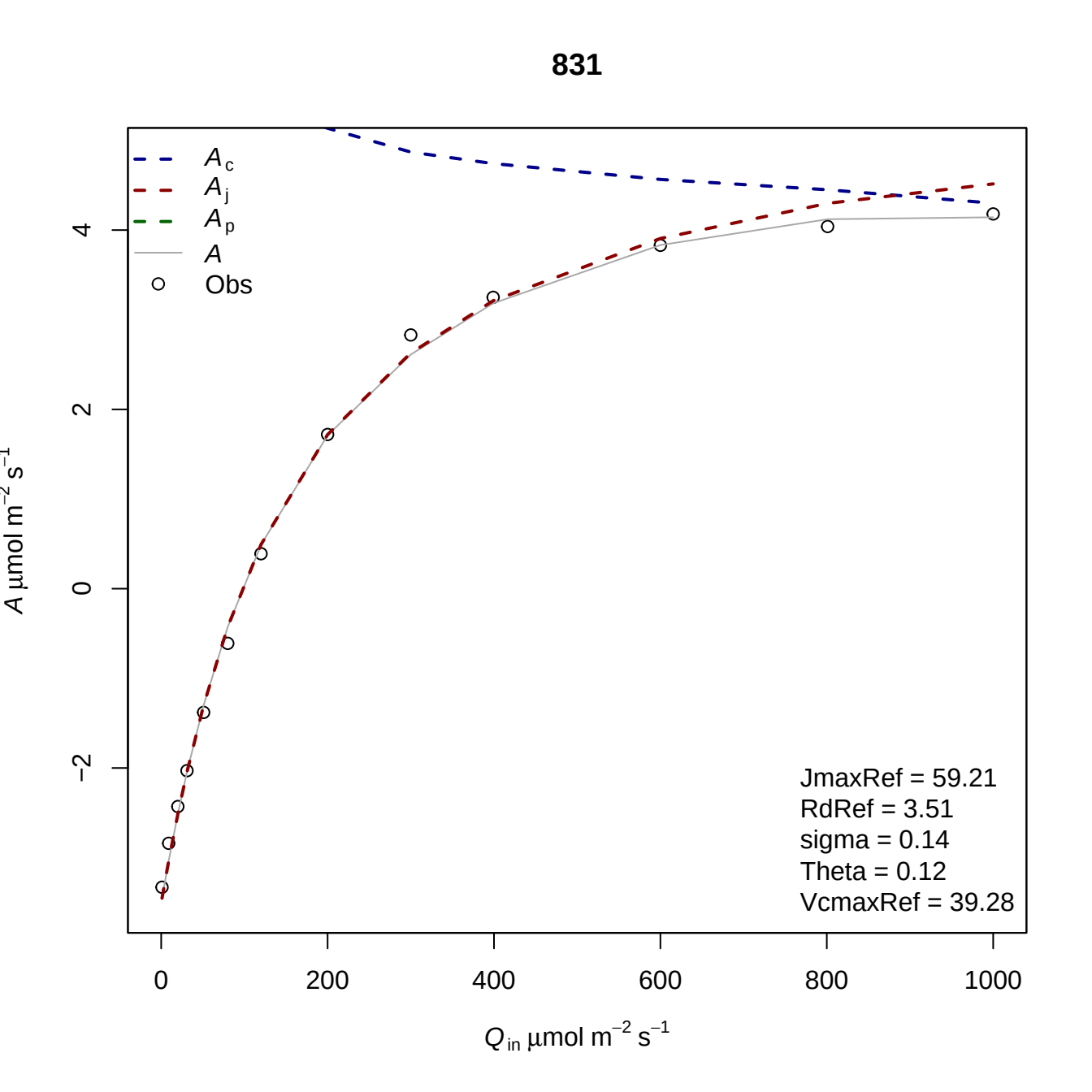

826
A_c
A_j
A_p
A
Obs
JmaxRef = 117.31
RdRef = 2.21
sigma = 0.14
Theta = −0.24
VcmaxRef = 88.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

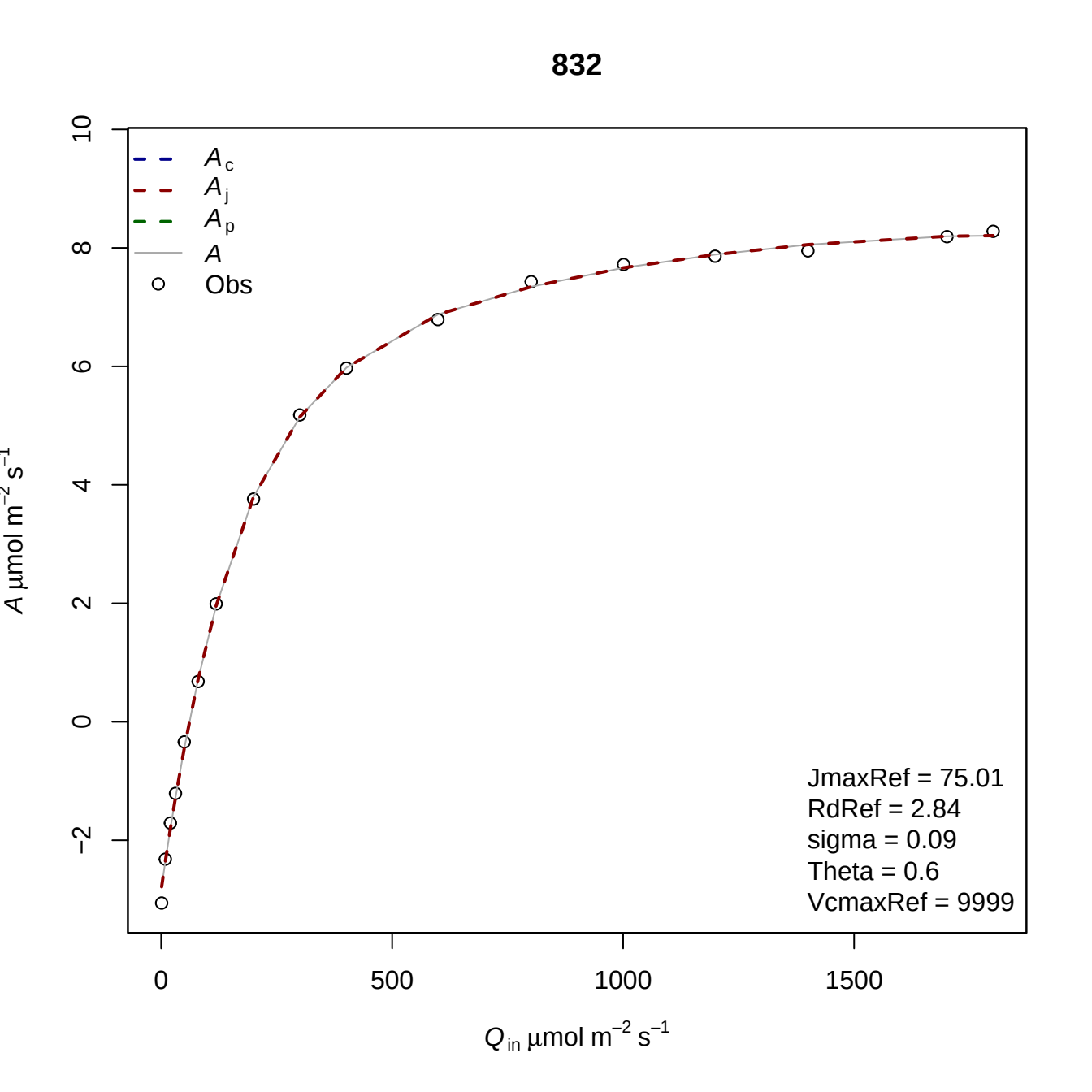

827
A_c
A_j
A_p
A
Obs
JmaxRef = 62.76
RdRef = 1.93
sigma = 0.14
Theta = −0.19
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

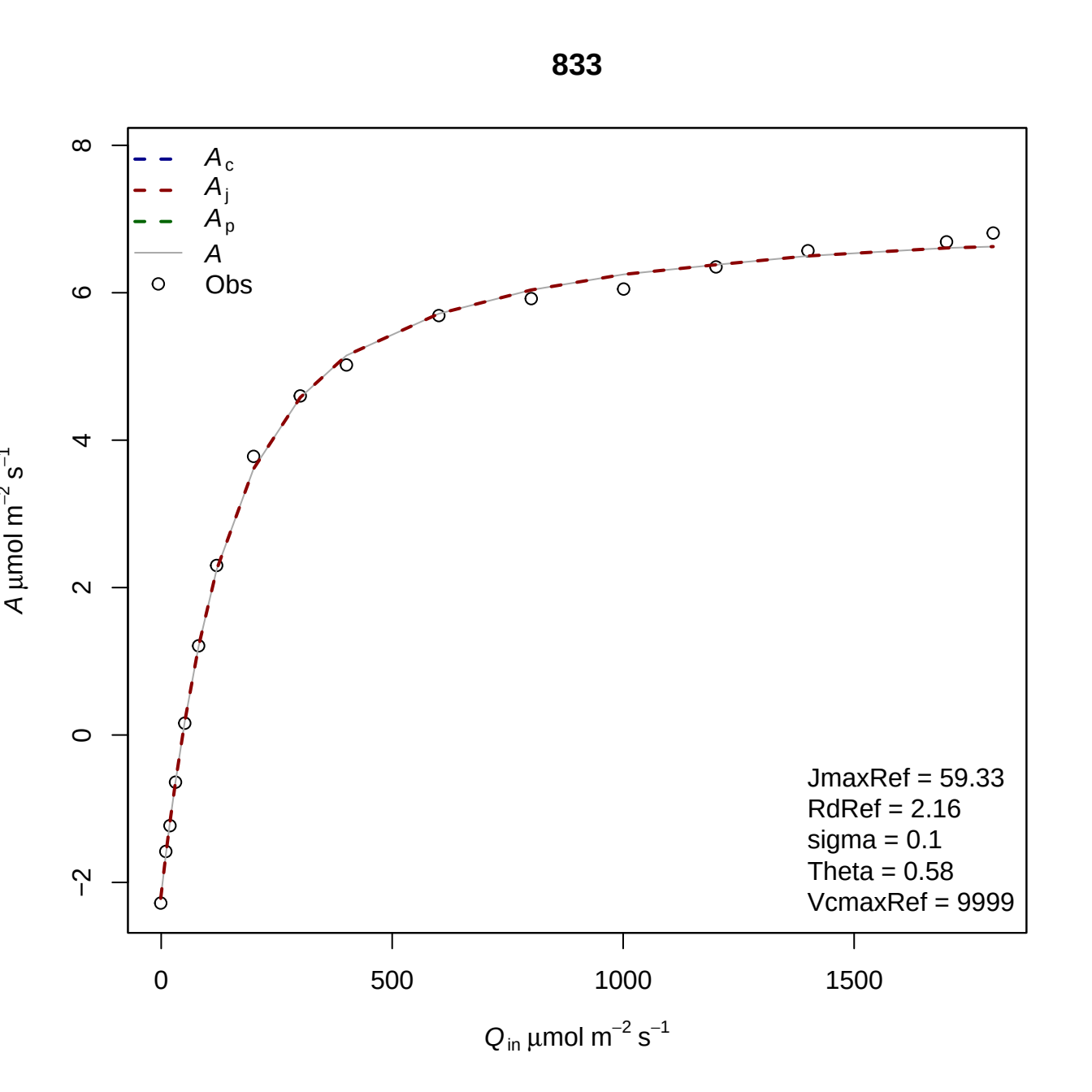

828
A_c
A_j
A_p
A
Obs
JmaxRef = 87.02
RdRef = 1.6
sigma = 0.13
Theta = 0.64
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

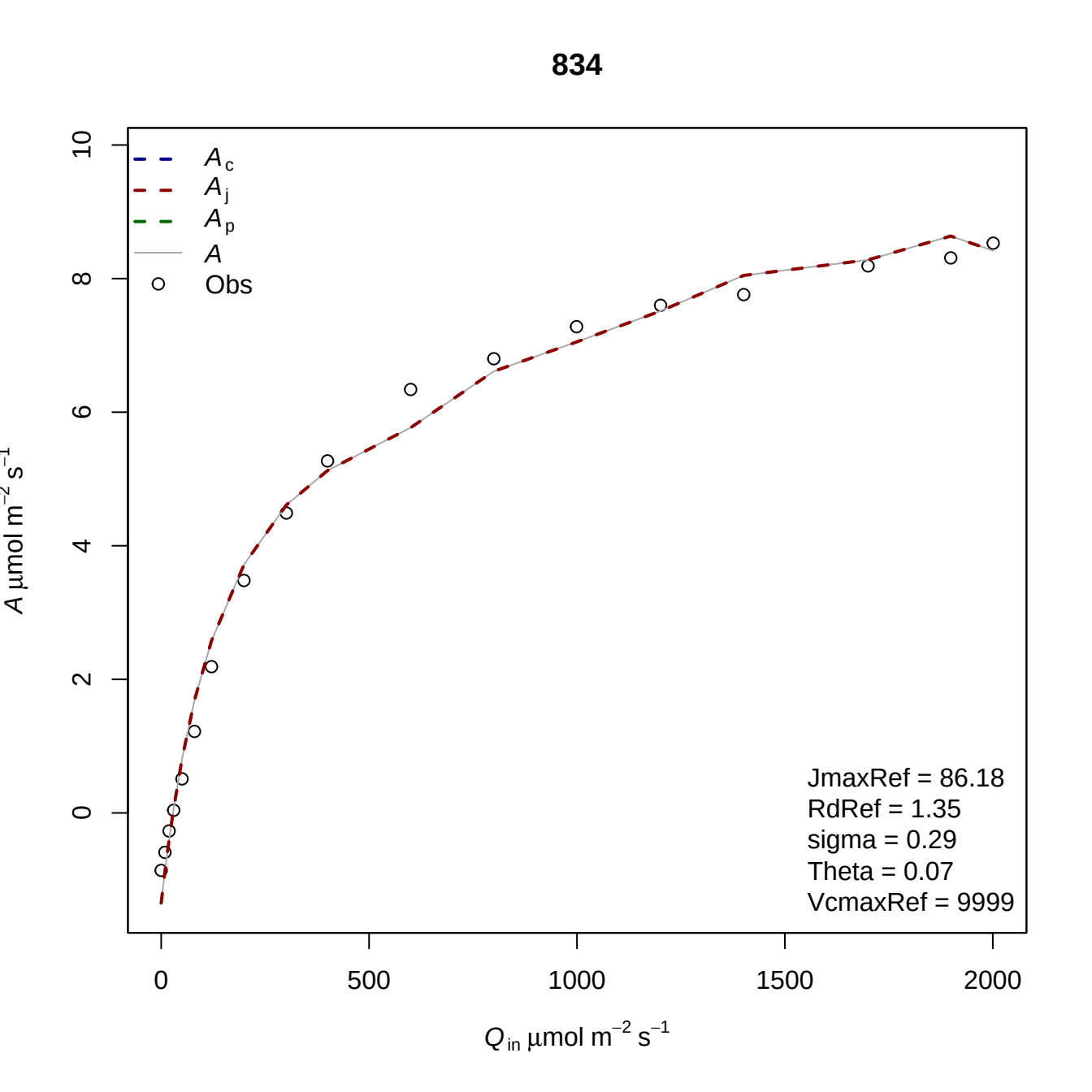

829
A_c
A_j
A_p
A
Obs
JmaxRef = 17.26
RdRef = 0.64
sigma = 0.1
Theta = 0.06
VcmaxRef = 9999
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

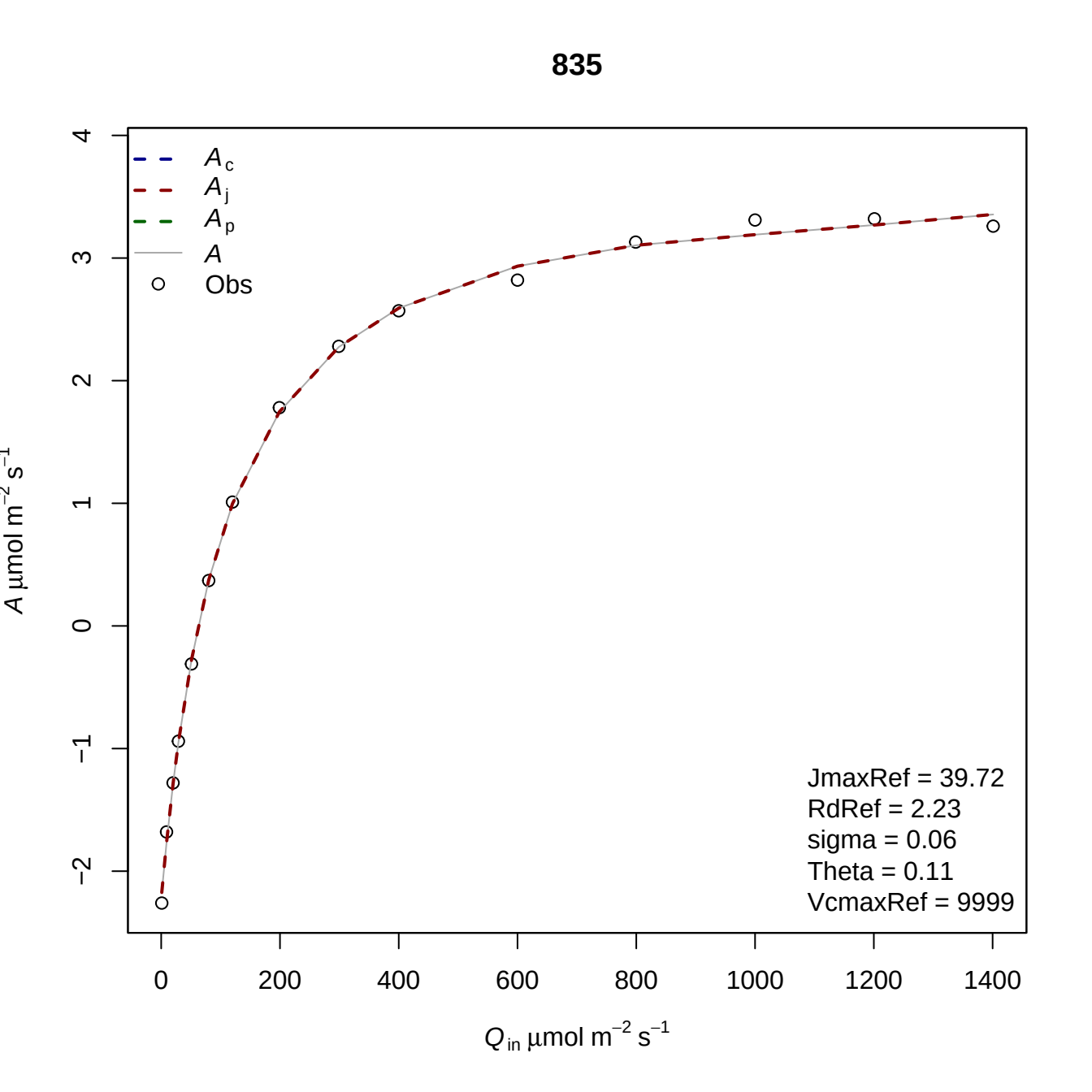

830
A_c
A_j
A_p
A
Obs
JmaxRef = 19.36
RdRef = 0.3
sigma = 0.09
Theta = 0.8
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
500
1000
1500

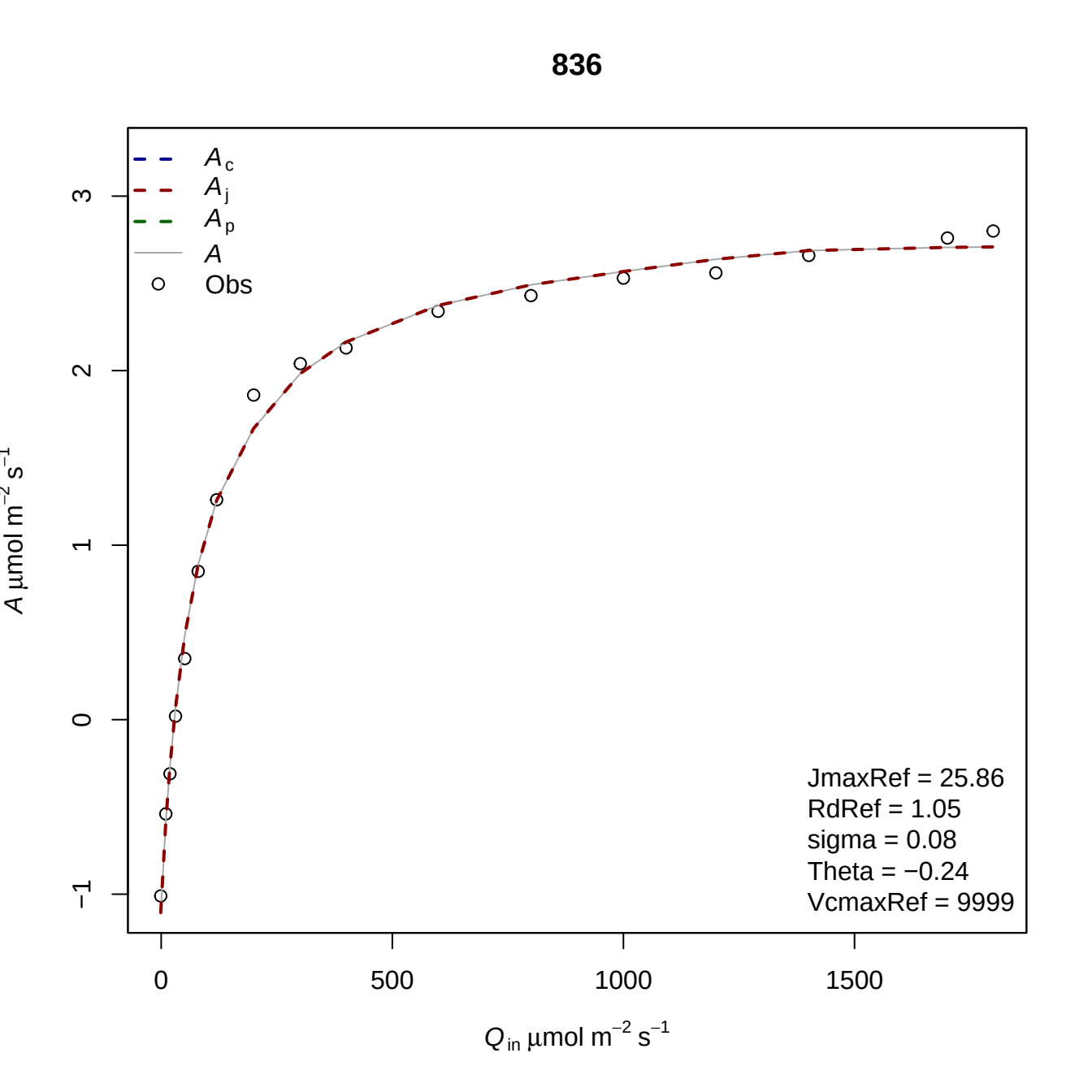

831
A_c
A_j
A_p
A
Obs
JmaxRef = 59.21
RdRef = 3.51
sigma = 0.14
Theta = 0.12
VcmaxRef = 39.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

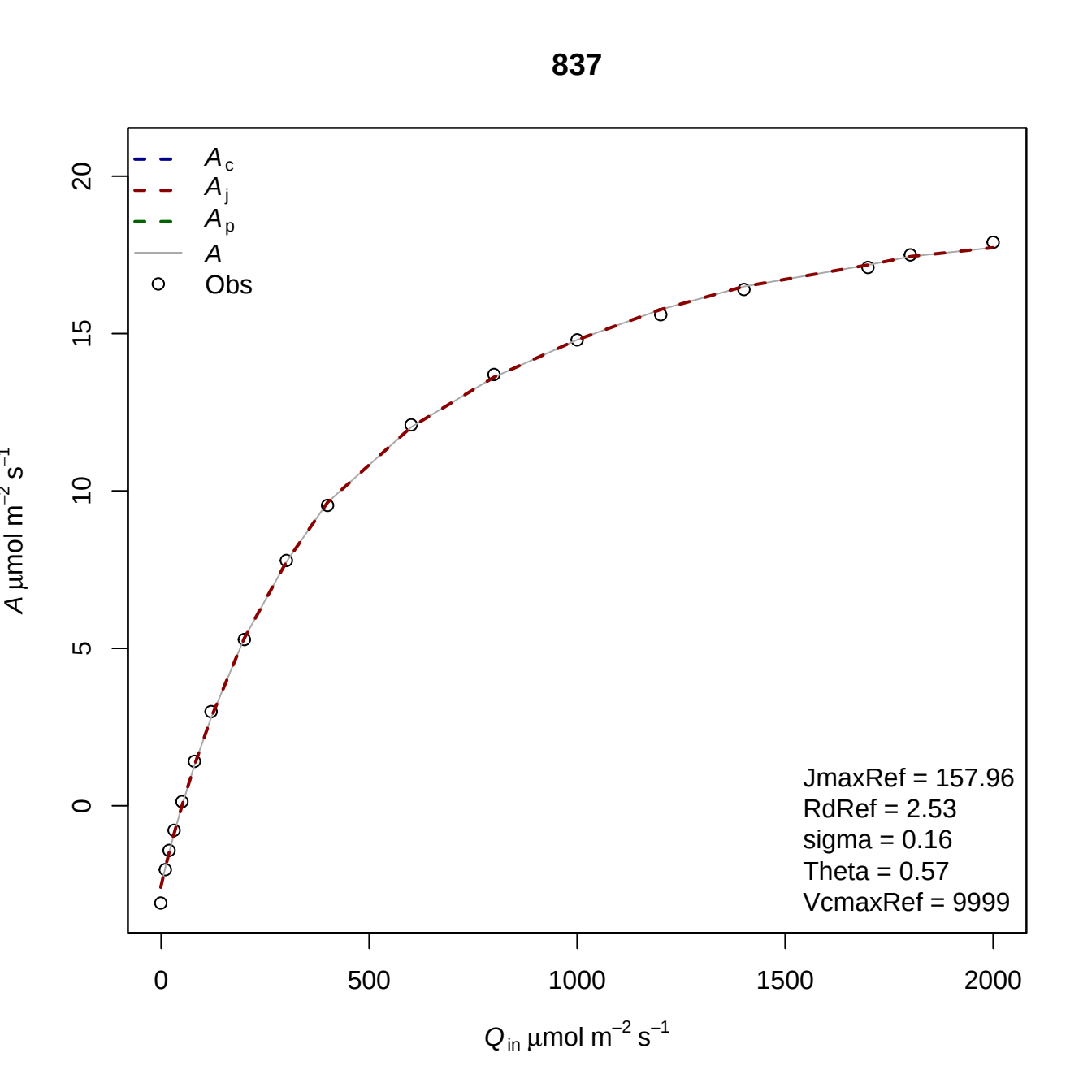

832
A_c
A_j
A_p
A
Obs
JmaxRef = 75.01
RdRef = 2.84
sigma = 0.09
Theta = 0.6
VcmaxRef = 9999
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

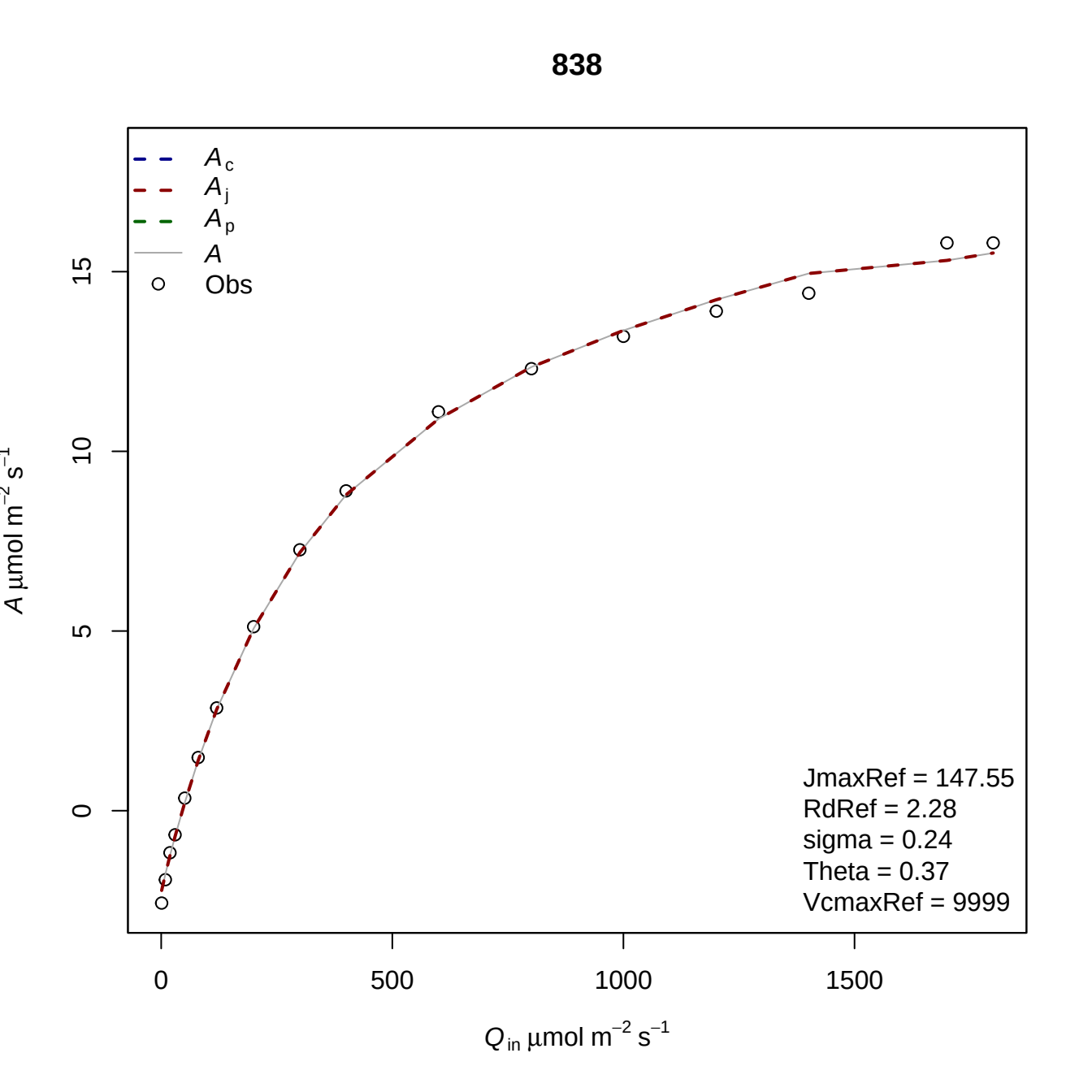

833
A_c
A_j
A_p
A
Obs
JmaxRef = 59.33
RdRef = 2.16
sigma = 0.1
Theta = 0.58
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

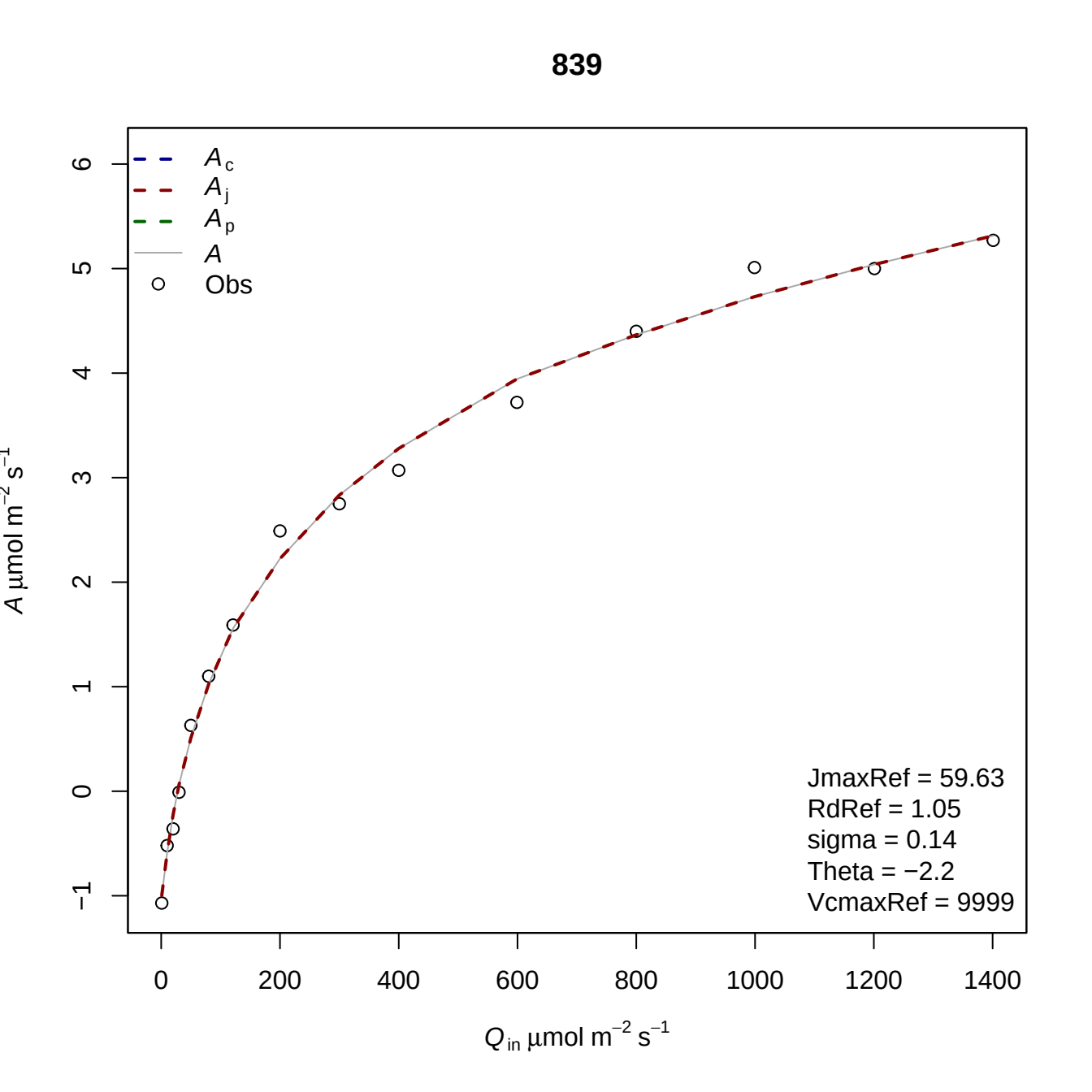

834
A_c
A_j
A_p
A
Obs
JmaxRef = 86.18
RdRef = 1.35
sigma = 0.29
Theta = 0.07
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

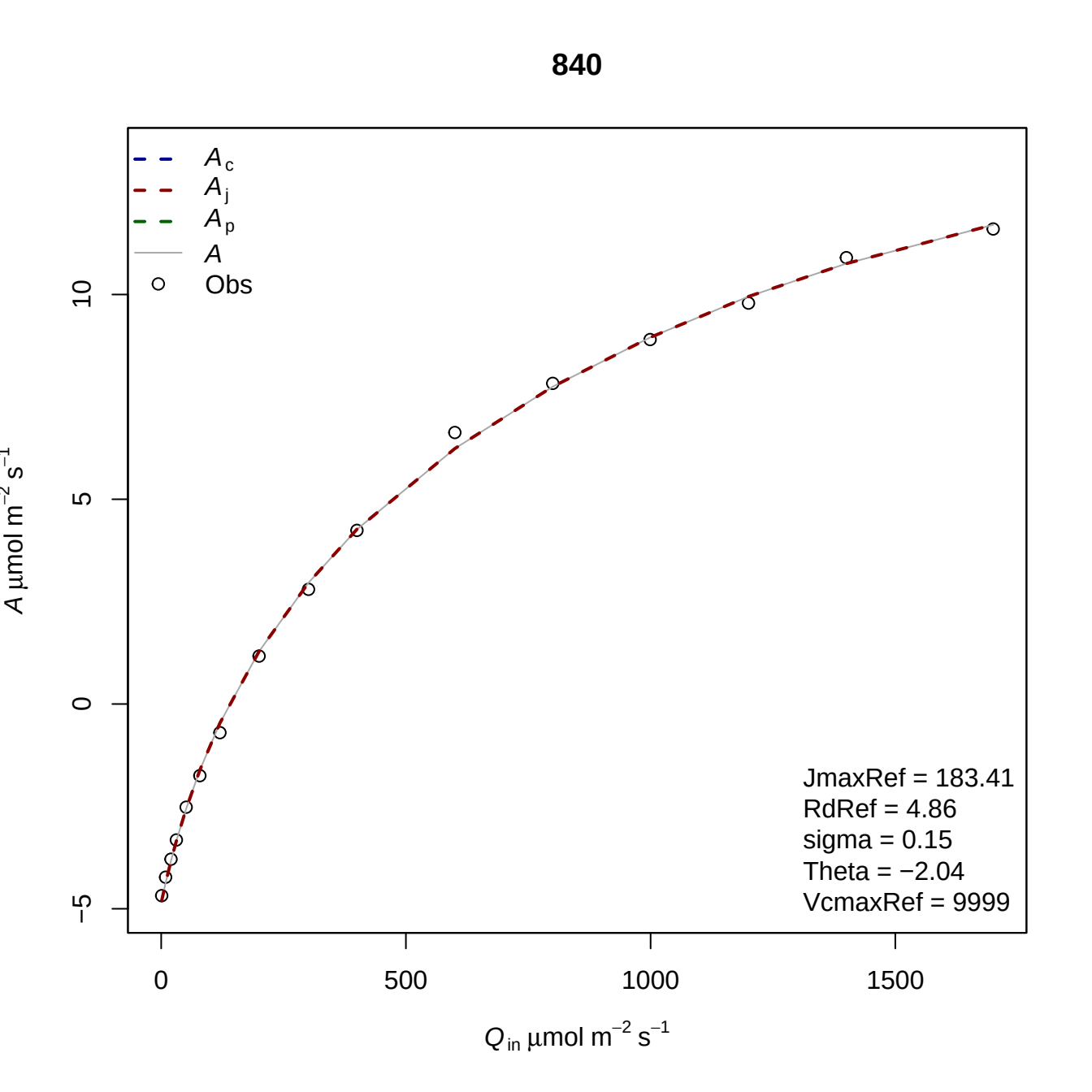

835
A_c
A_j
A_p
A
Obs
JmaxRef = 39.72
RdRef = 2.23
sigma = 0.06
Theta = 0.11
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200
1400

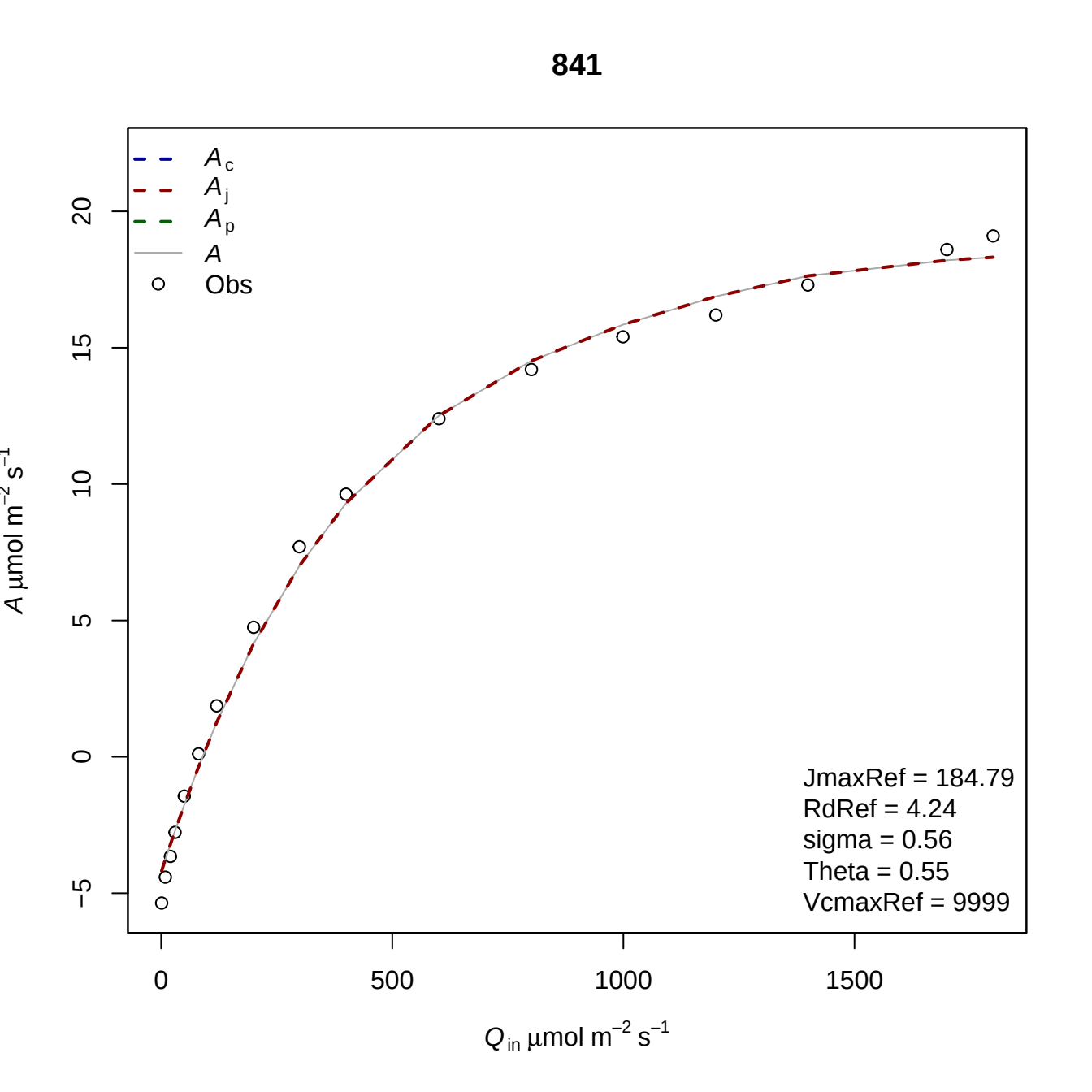

836
A_c
A_j
A_p
A
Obs
JmaxRef = 25.86
RdRef = 1.05
sigma = 0.08
Theta = −0.24
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

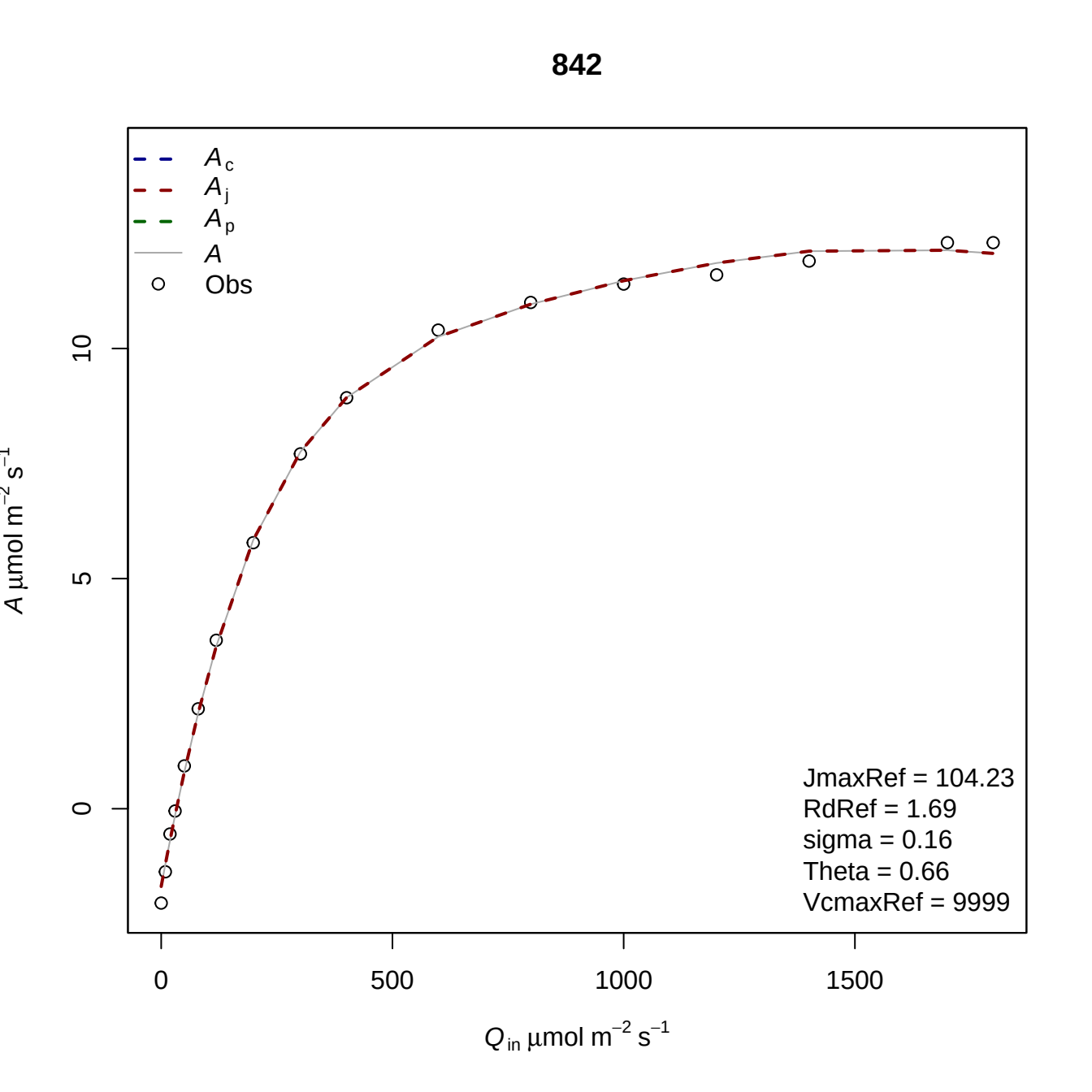

837
A_c
A_j
A_p
A
Obs
JmaxRef = 157.96
RdRef = 2.53
sigma = 0.16
Theta = 0.57
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

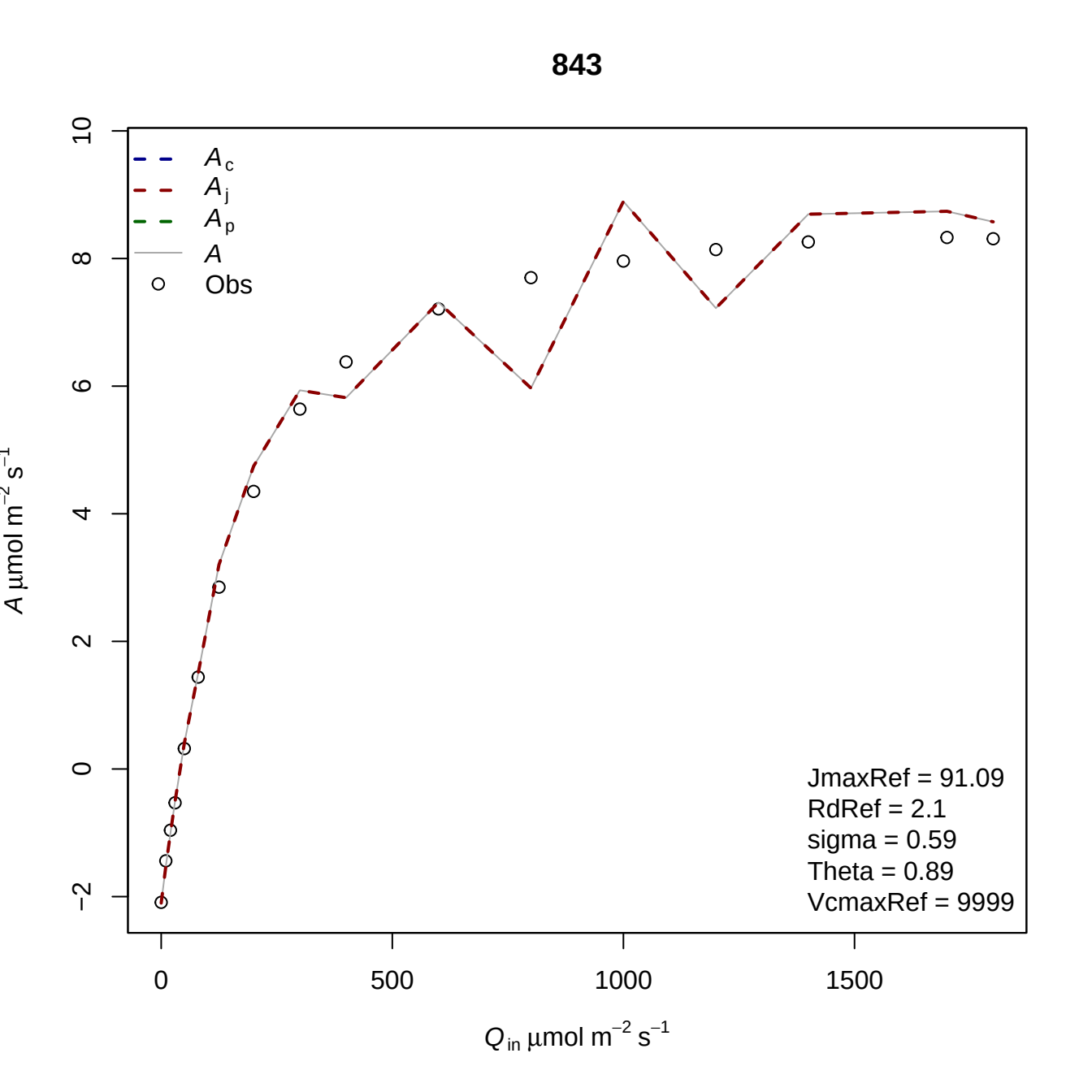

838
A_c
A_j
A_p
A
Obs
JmaxRef = 147.55
RdRef = 2.28
sigma = 0.24
Theta = 0.37
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

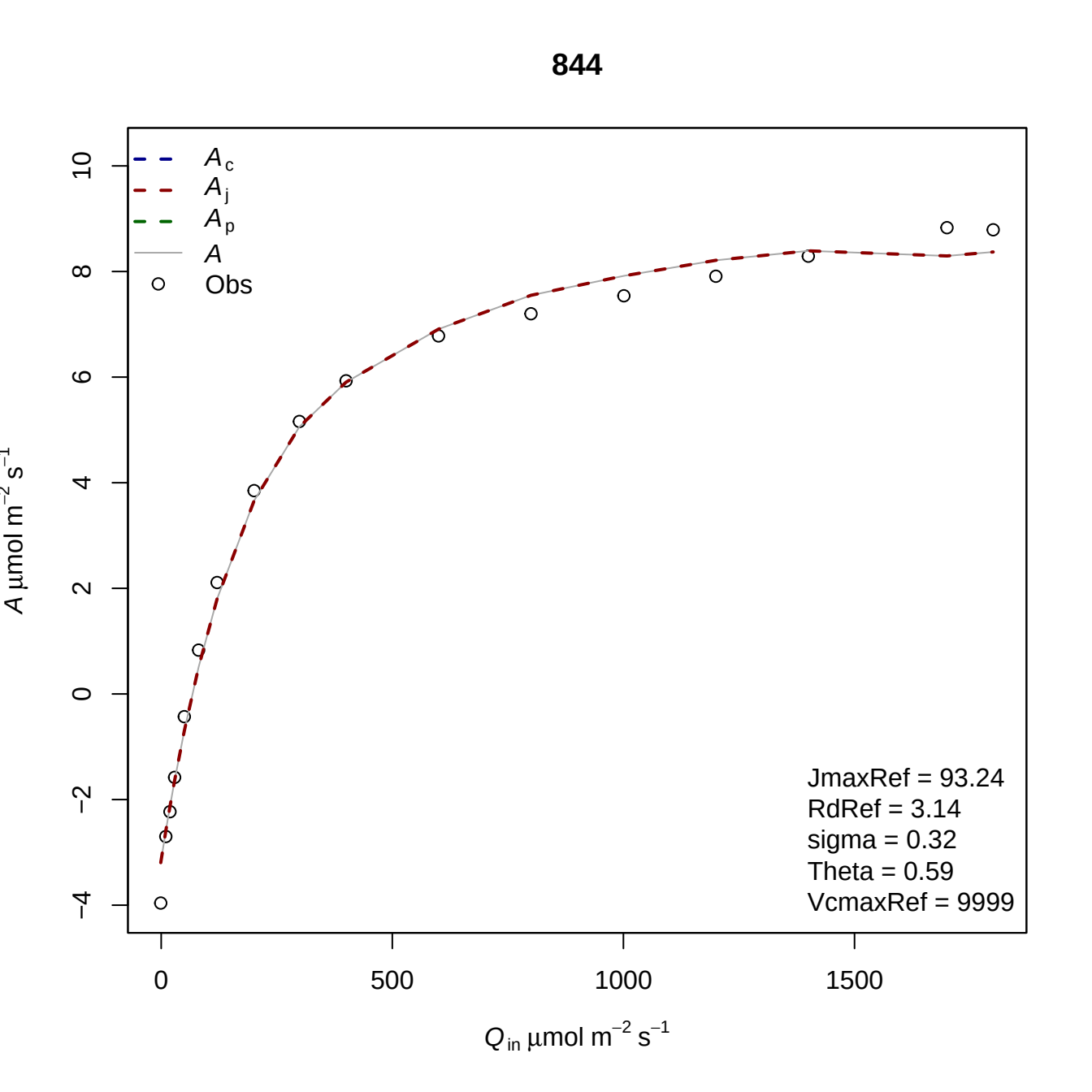

839
A_c
A_j
A_p
A
Obs
JmaxRef = 59.63
RdRef = 1.05
sigma = 0.14
Theta = −2.2
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
1400

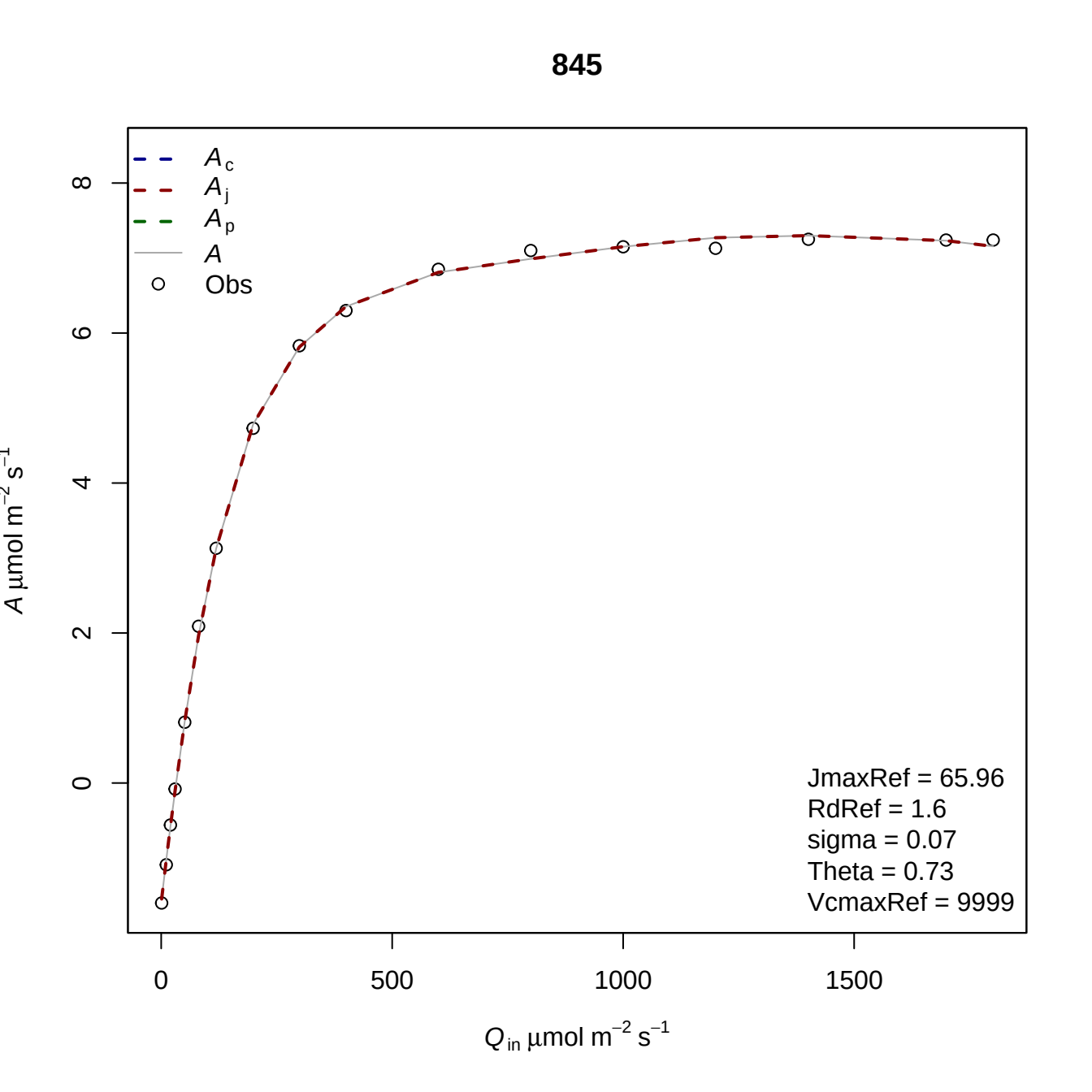

840
A_c
A_j
A_p
A
Obs
JmaxRef = 183.41
RdRef = 4.86
sigma = 0.15
Theta = −2.04
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

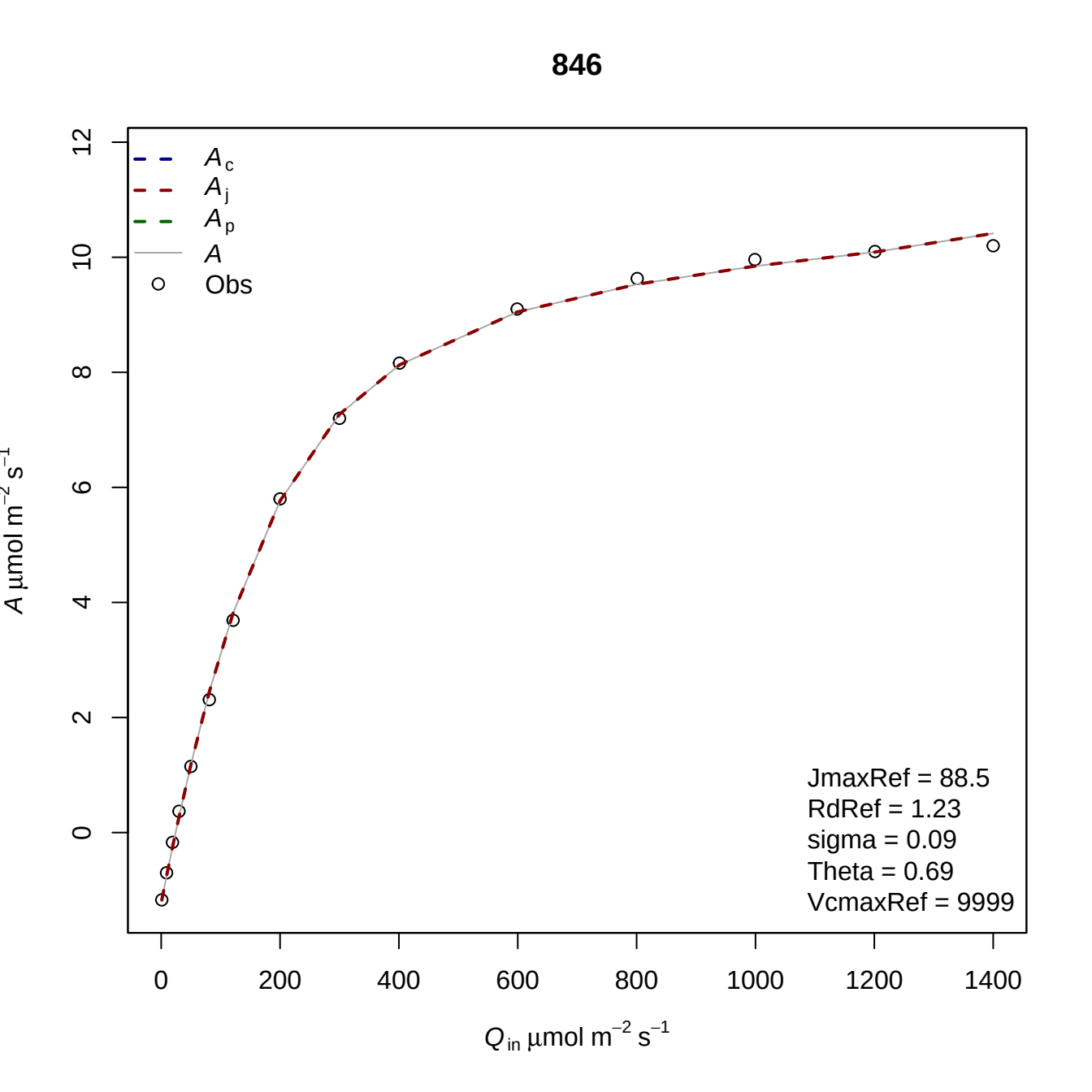

841
A_c
A_j
A_p
A
Obs
JmaxRef = 184.79
RdRef = 4.24
sigma = 0.56
Theta = 0.55
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500

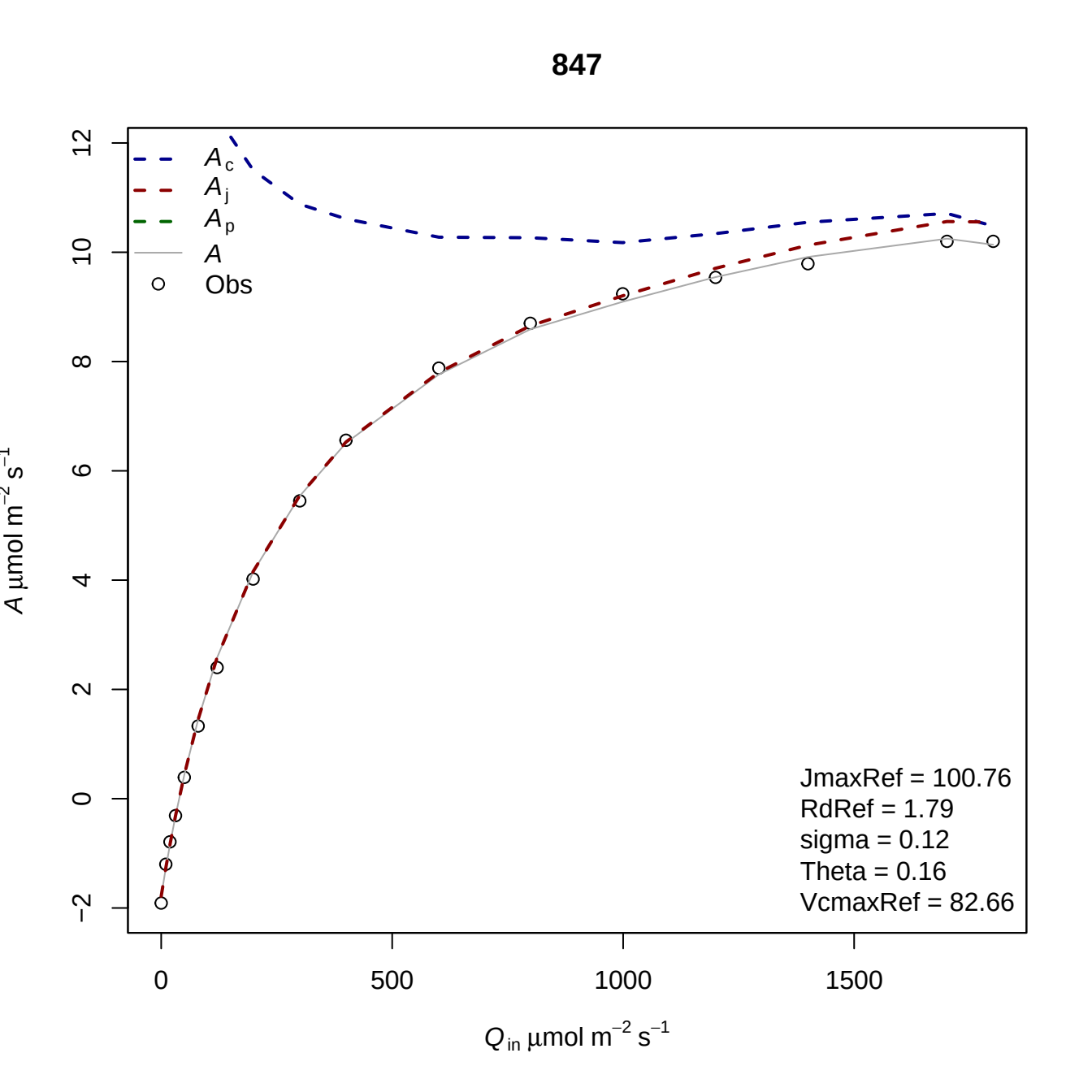

842
A_c
A_j
A_p
A
Obs
JmaxRef = 104.23
RdRef = 1.69
sigma = 0.16
Theta = 0.66
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

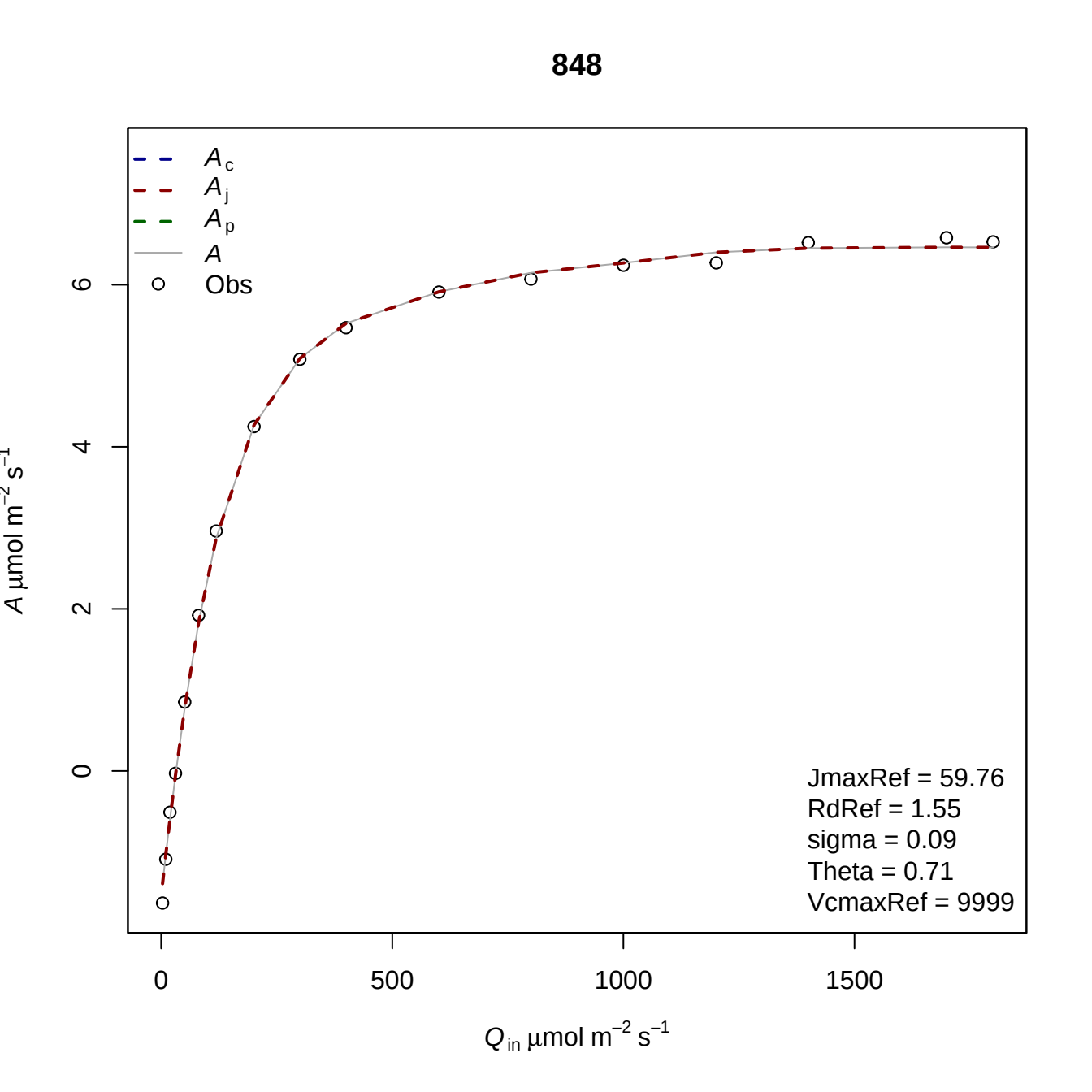

843
A_c
A_j
A_p
A
Obs
JmaxRef = 91.09
RdRef = 2.1
sigma = 0.59
Theta = 0.89
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

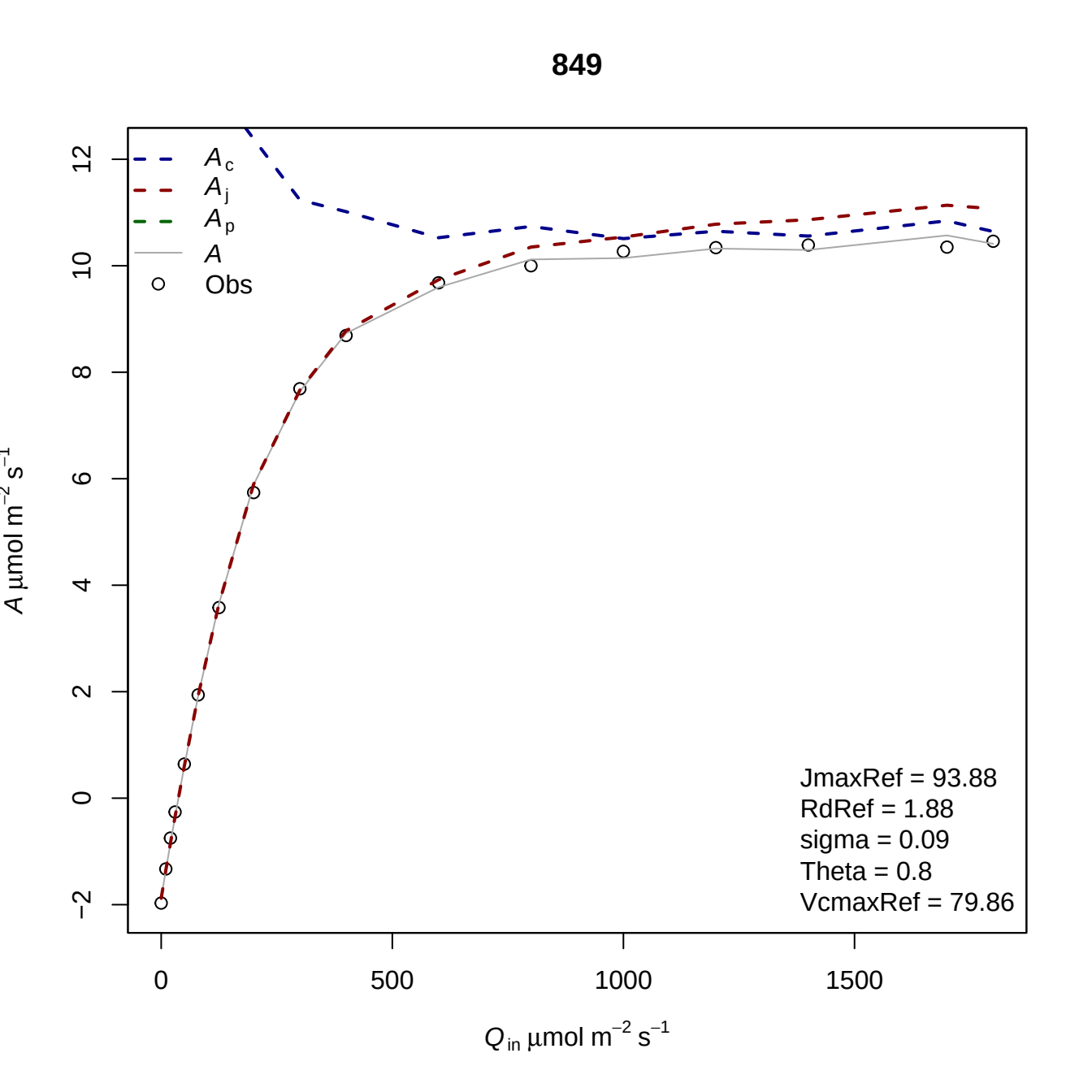

844
A_c
A_j
A_p
A
Obs
JmaxRef = 93.24
RdRef = 3.14
sigma = 0.32
Theta = 0.59
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

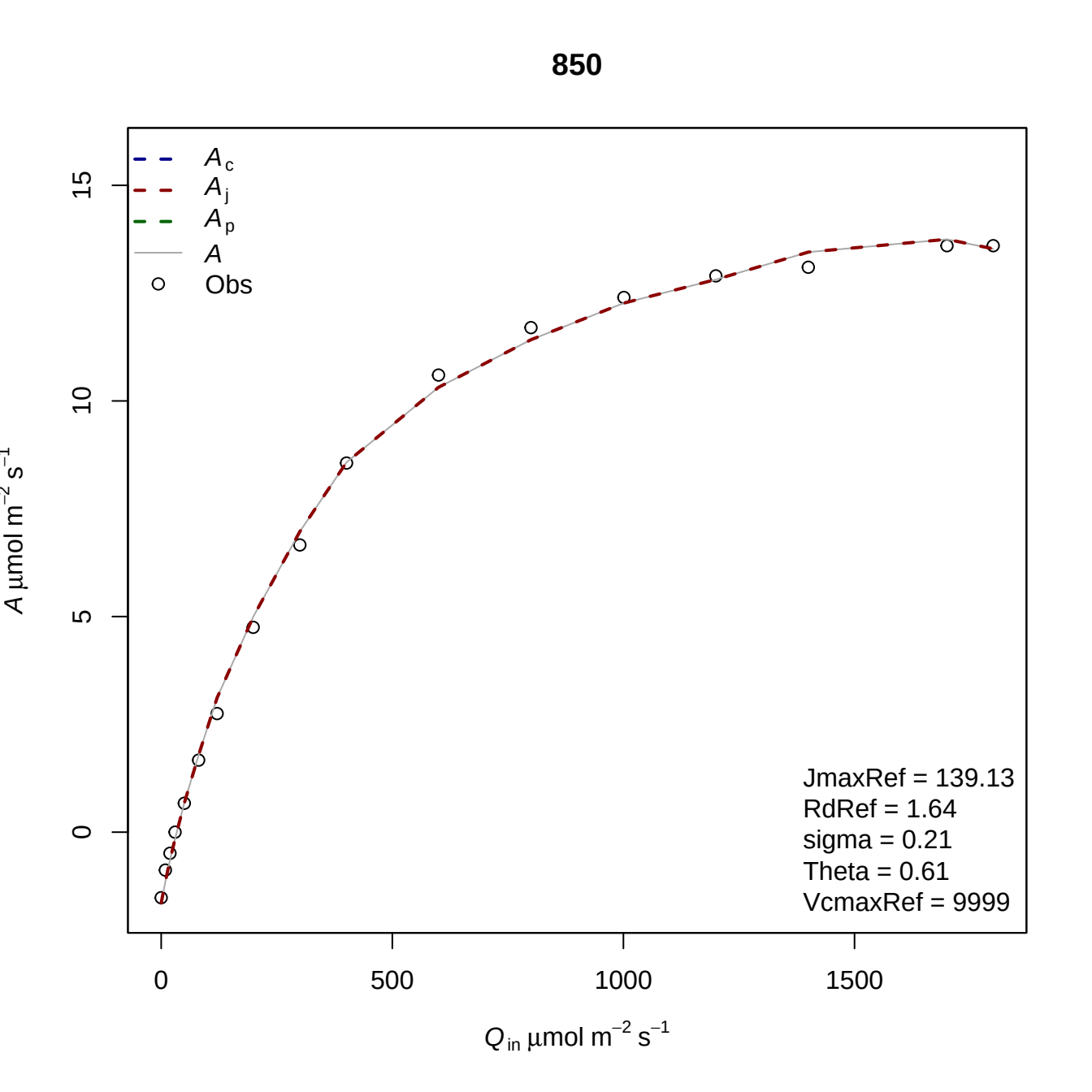

845
A_c
A_j
A_p
A
Obs
JmaxRef = 65.96
RdRef = 1.6
sigma = 0.07
Theta = 0.73
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

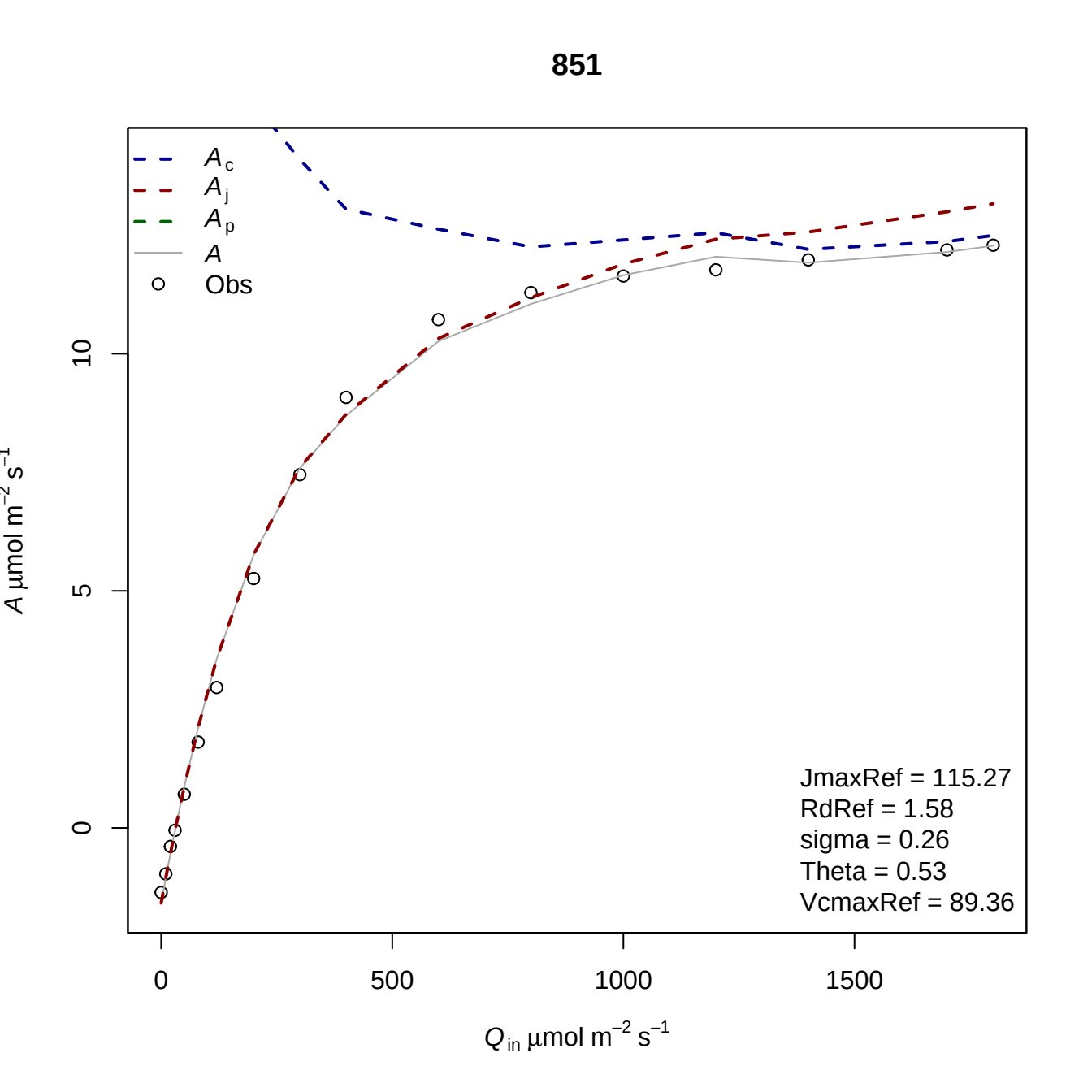

846
A_c
A_j
A_p
A
Obs
JmaxRef = 88.5
RdRef = 1.23
sigma = 0.09
Theta = 0.69
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

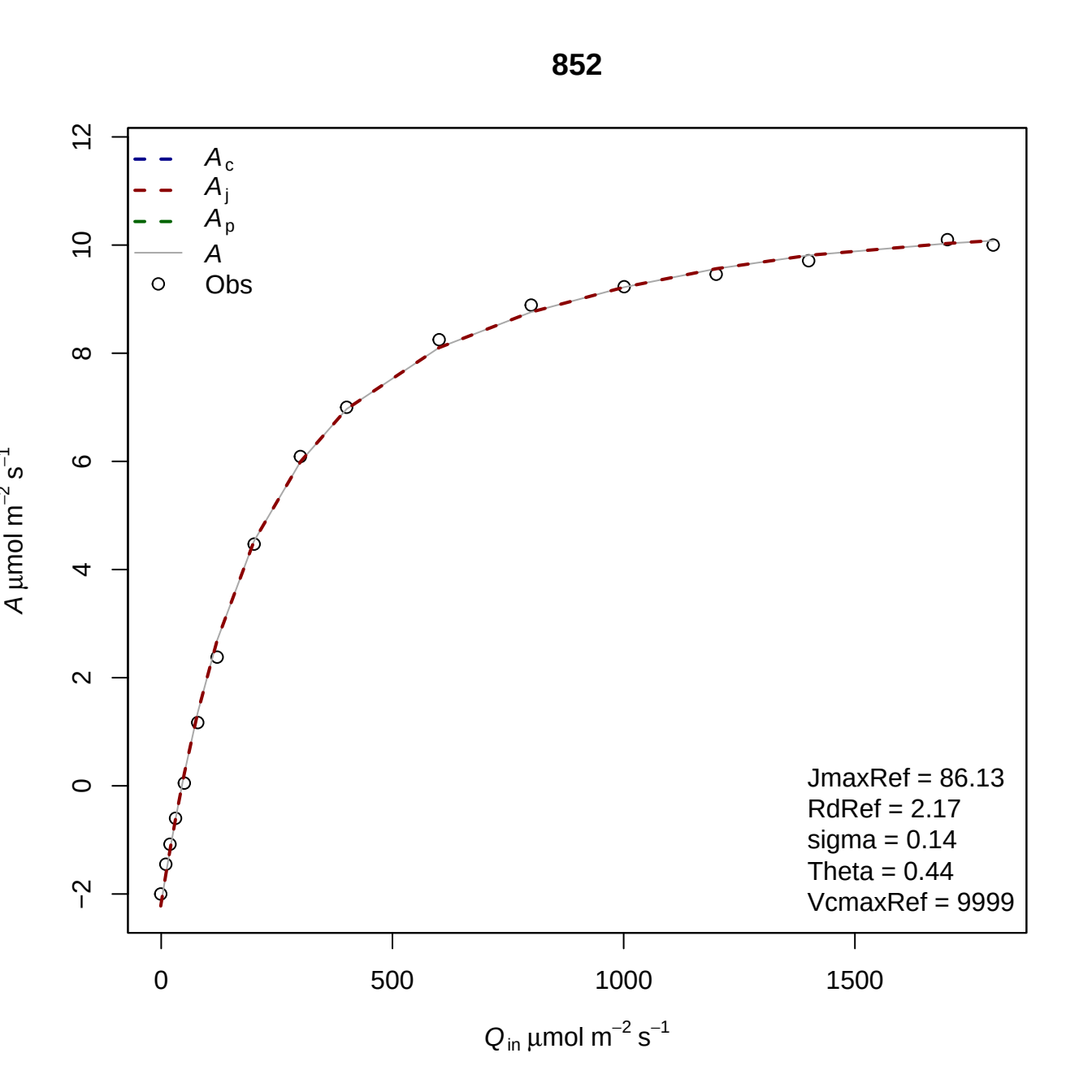

847
A_c
A_j
A_p
A
Obs
JmaxRef = 100.76
RdRef = 1.79
sigma = 0.12
Theta = 0.16
VcmaxRef = 82.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

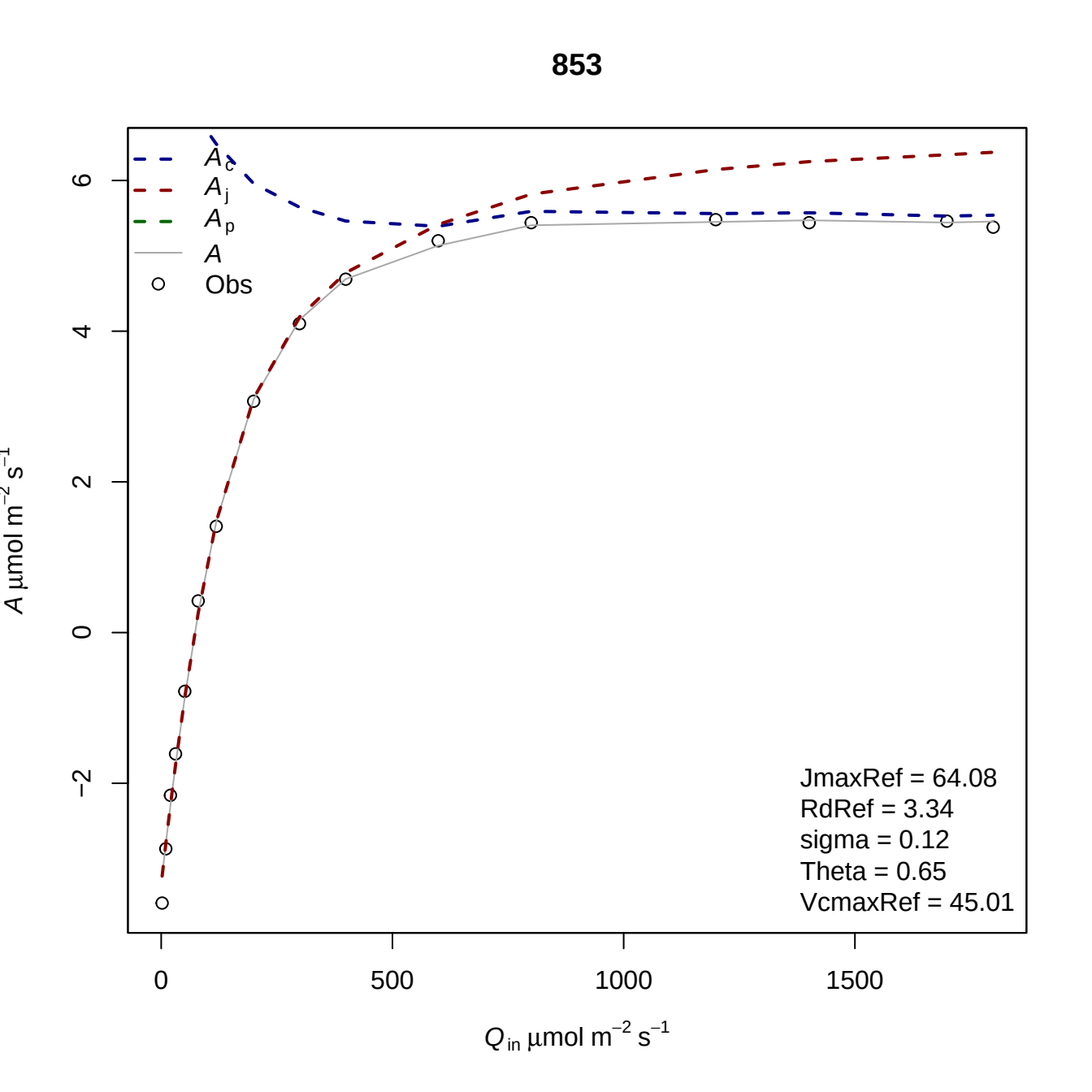

848
A_c
A_j
A_p
A
Obs
JmaxRef = 59.76
RdRef = 1.55
sigma = 0.09
Theta = 0.71
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

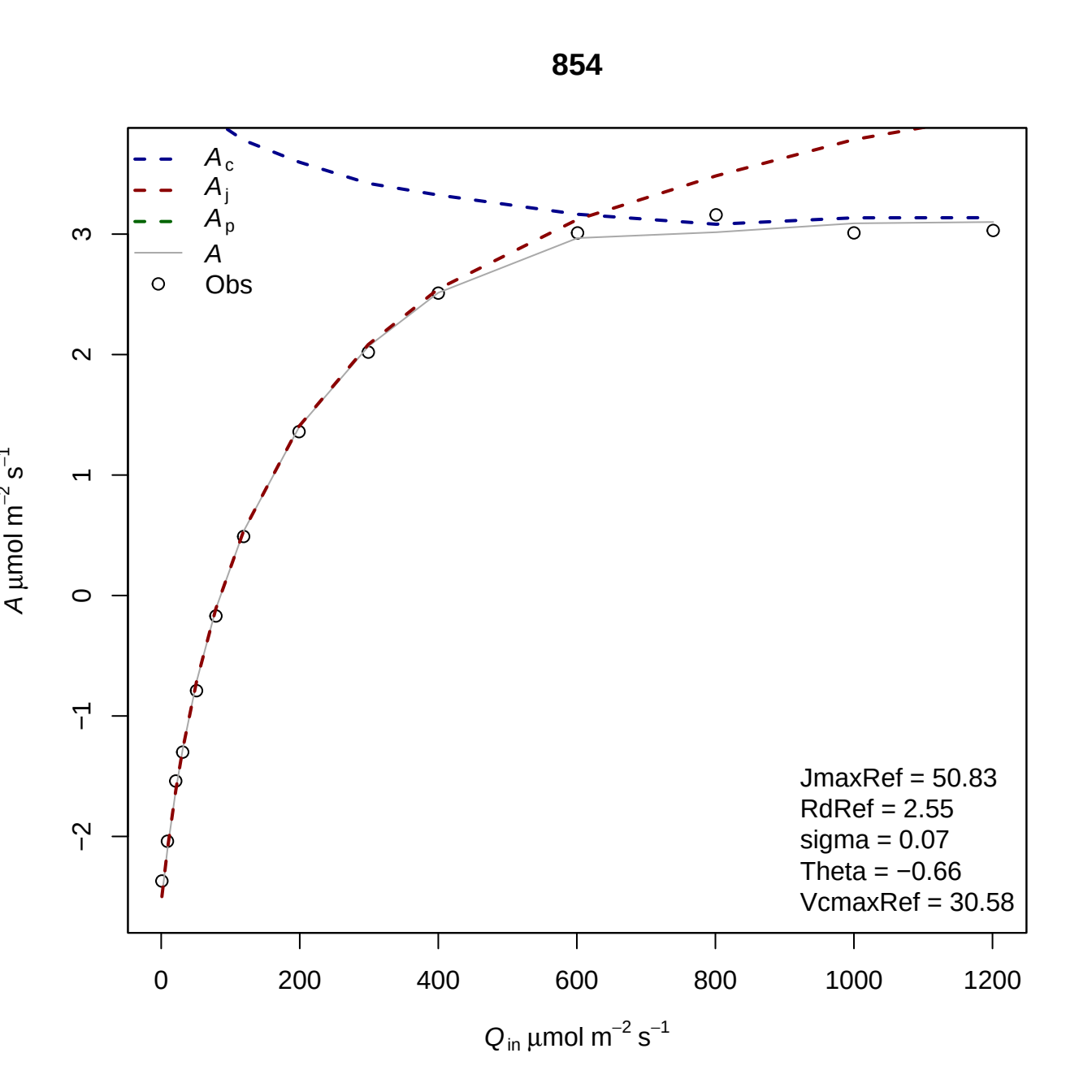

849
A_c
A_j
A_p
A
Obs
JmaxRef = 93.88
RdRef = 1.88
sigma = 0.09
Theta = 0.8
VcmaxRef = 79.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

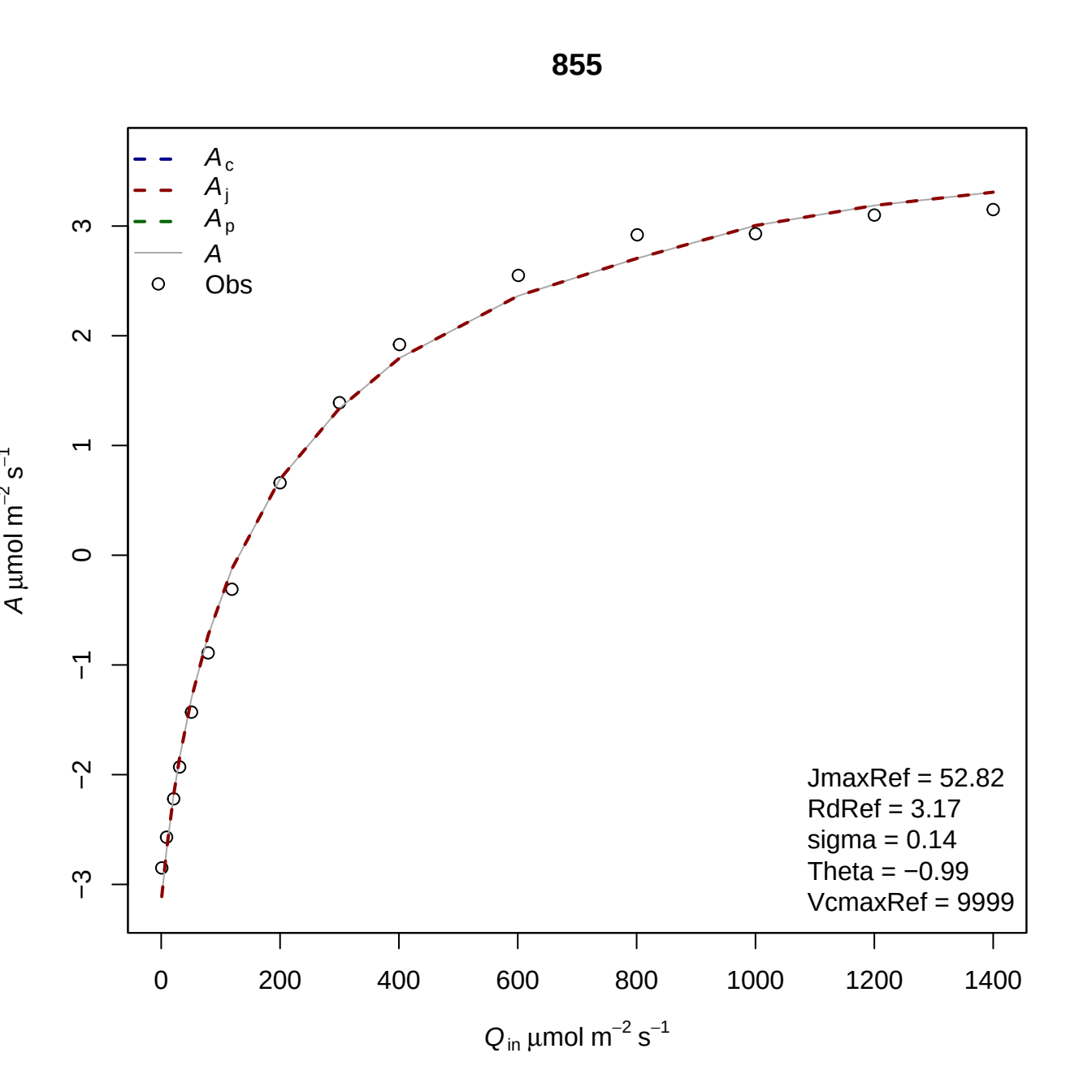

850
A_c
A_j
A_p
A
Obs
JmaxRef = 139.13
RdRef = 1.64
sigma = 0.21
Theta = 0.61
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

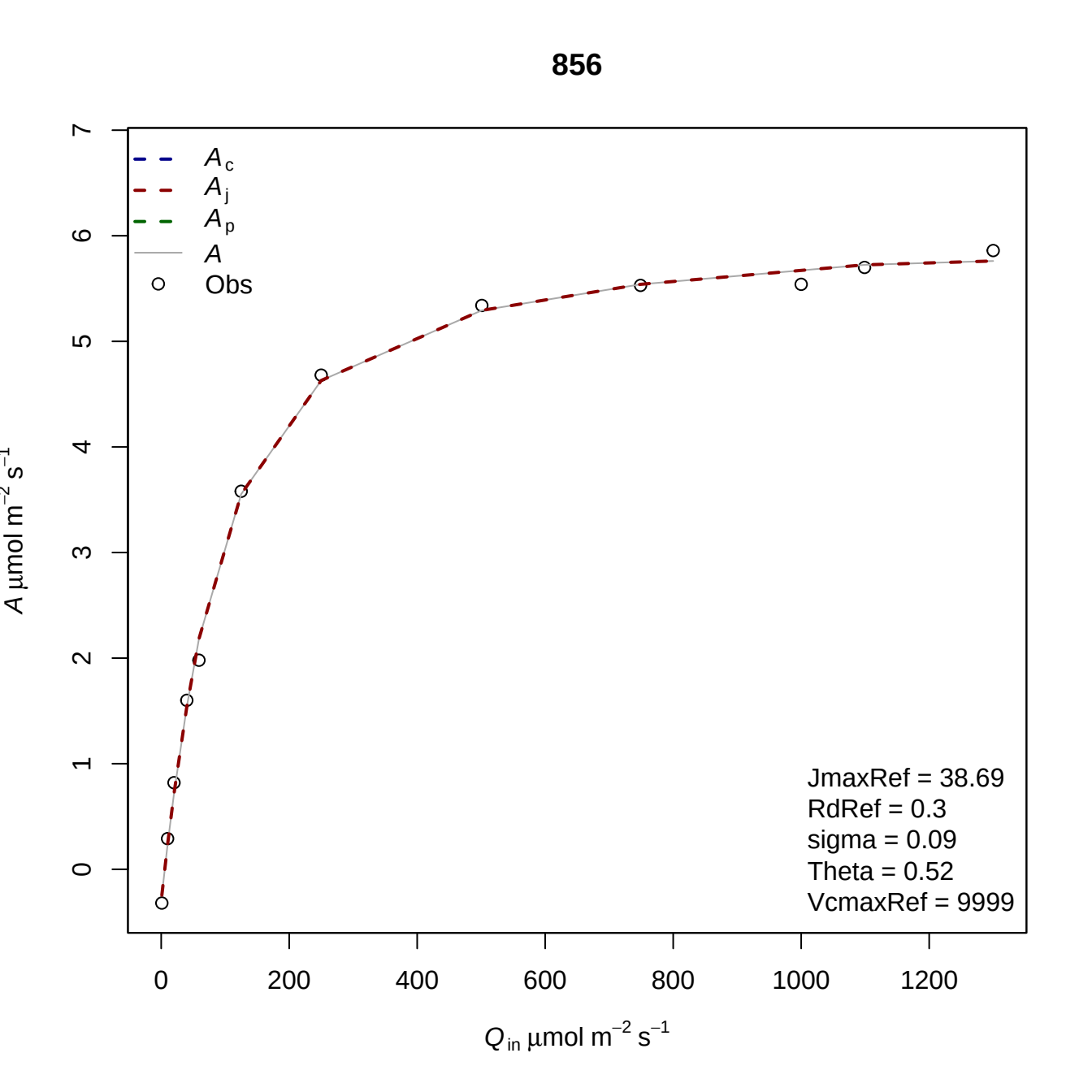

851
A_c
A_j
A_p
A
Obs
JmaxRef = 115.27
RdRef = 1.58
sigma = 0.26
Theta = 0.53
VcmaxRef = 89.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

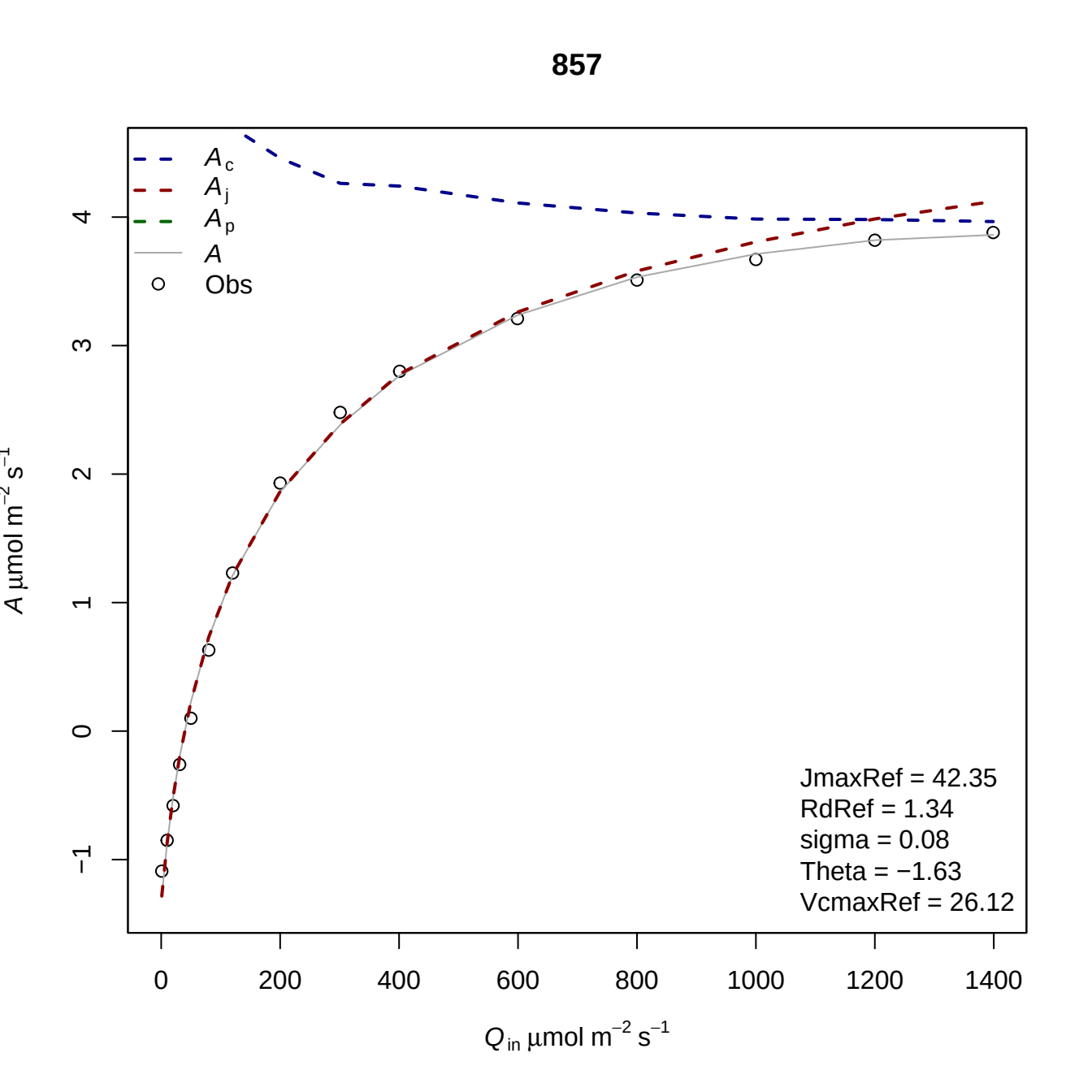

852
A_c
A_j
A_p
A
Obs
JmaxRef = 86.13
RdRef = 2.17
sigma = 0.14
Theta = 0.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

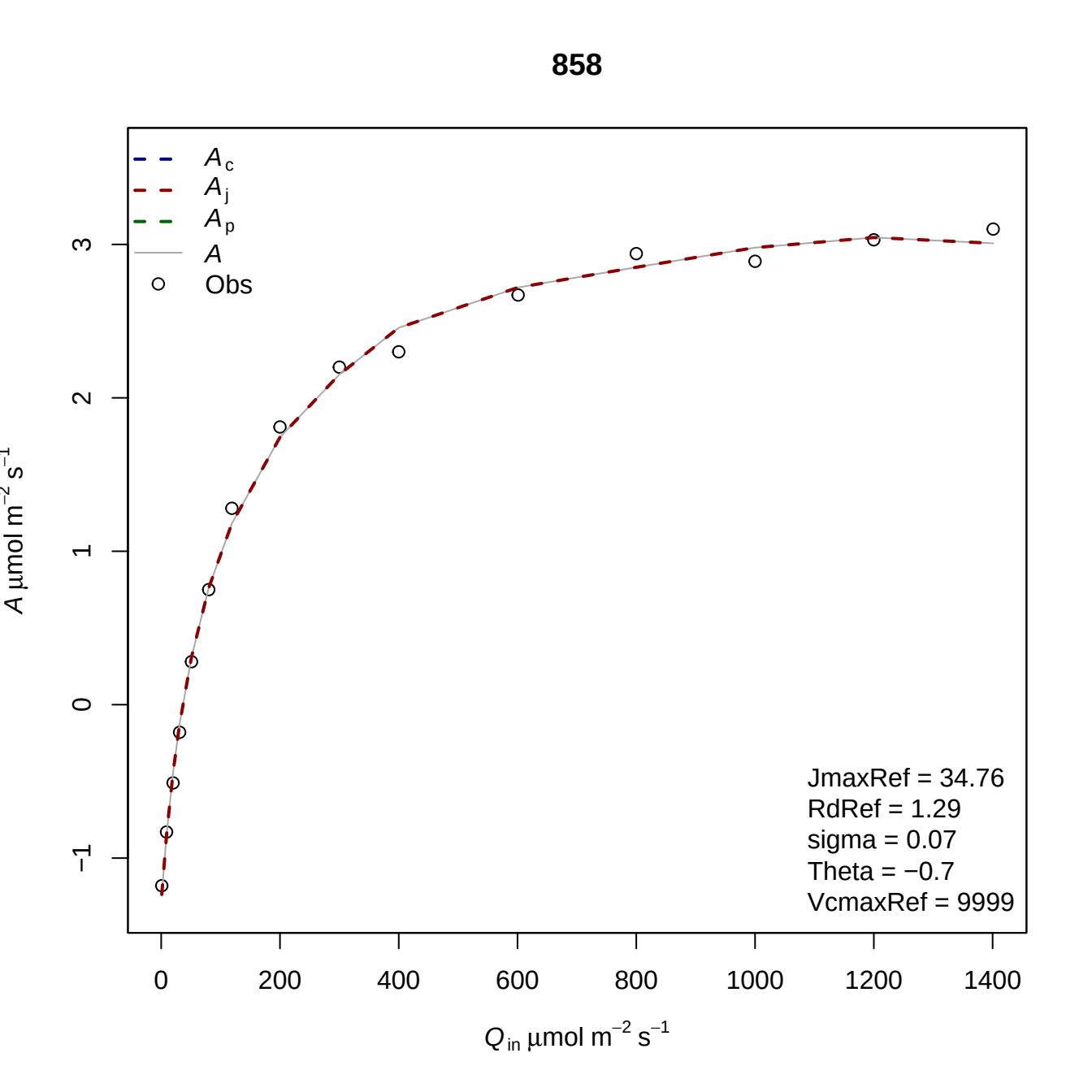

853
A_c
A_j
A_p
A
Obs
JmaxRef = 64.08
RdRef = 3.34
sigma = 0.12
Theta = 0.65
VcmaxRef = 45.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

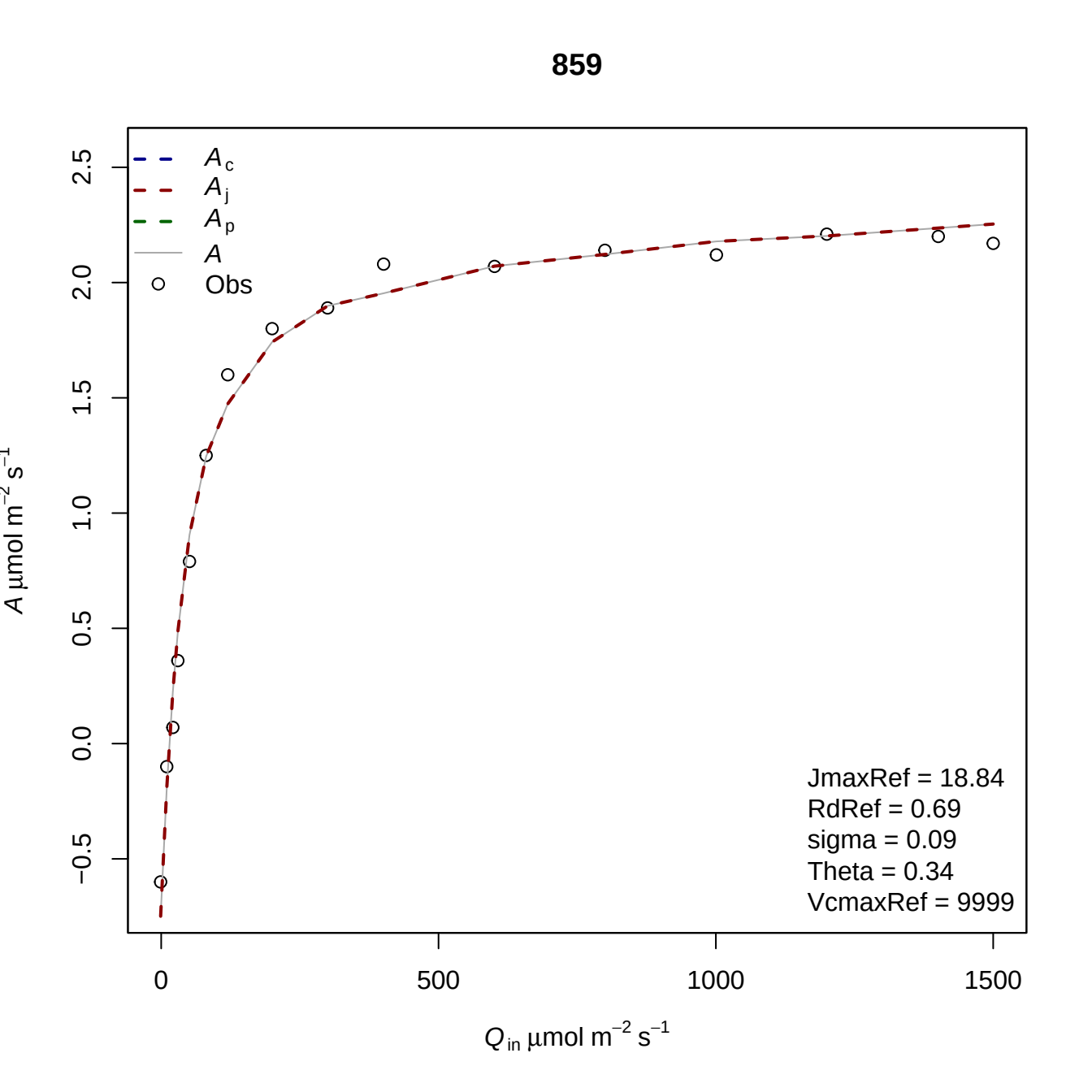

854
A_c
A_j
A_p
A
Obs
JmaxRef = 50.83
RdRef = 2.55
sigma = 0.07
Theta = −0.66
VcmaxRef = 30.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200

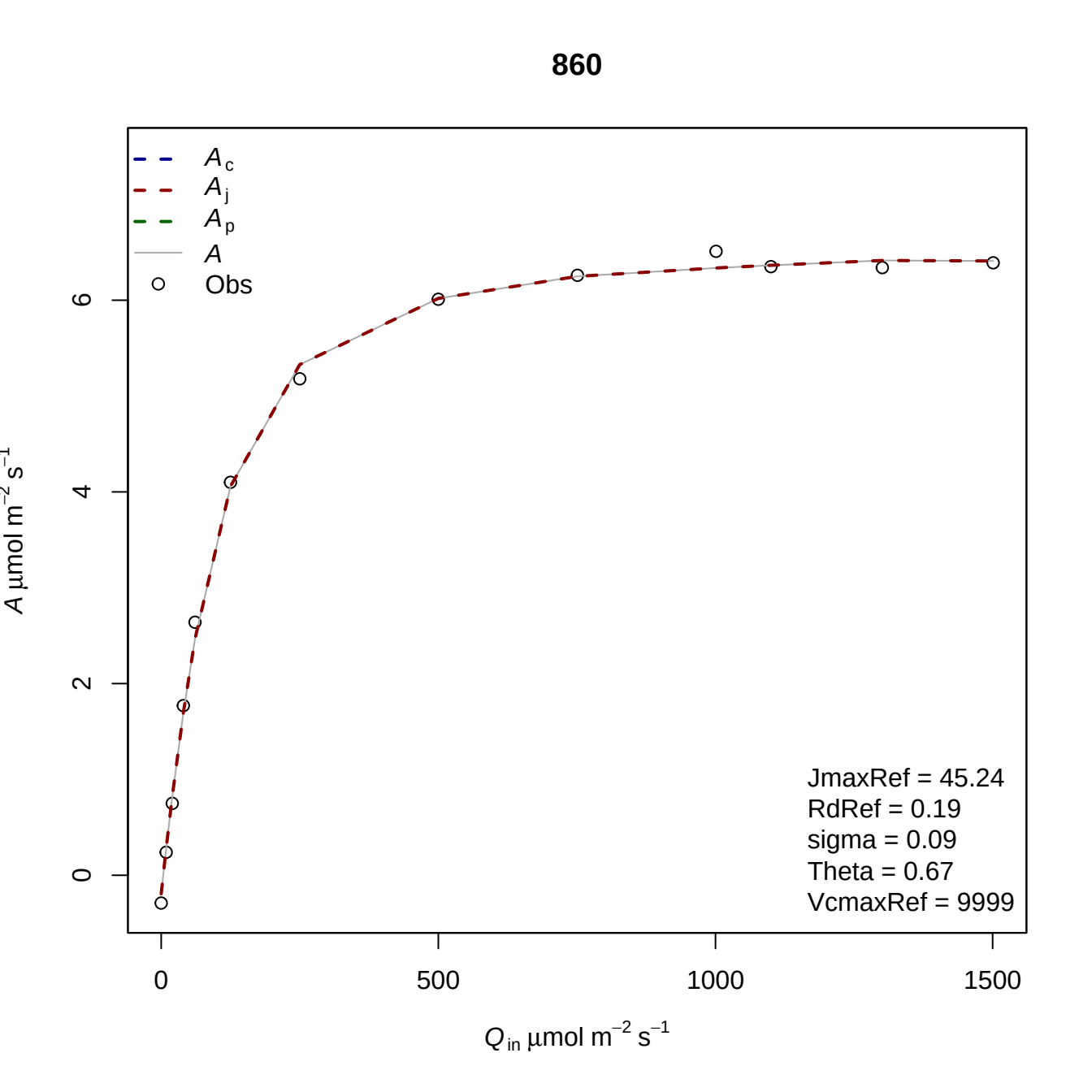

855
A_c
A_j
A_p
A
Obs
JmaxRef = 52.82
RdRef = 3.17
sigma = 0.14
Theta = −0.99
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−3
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

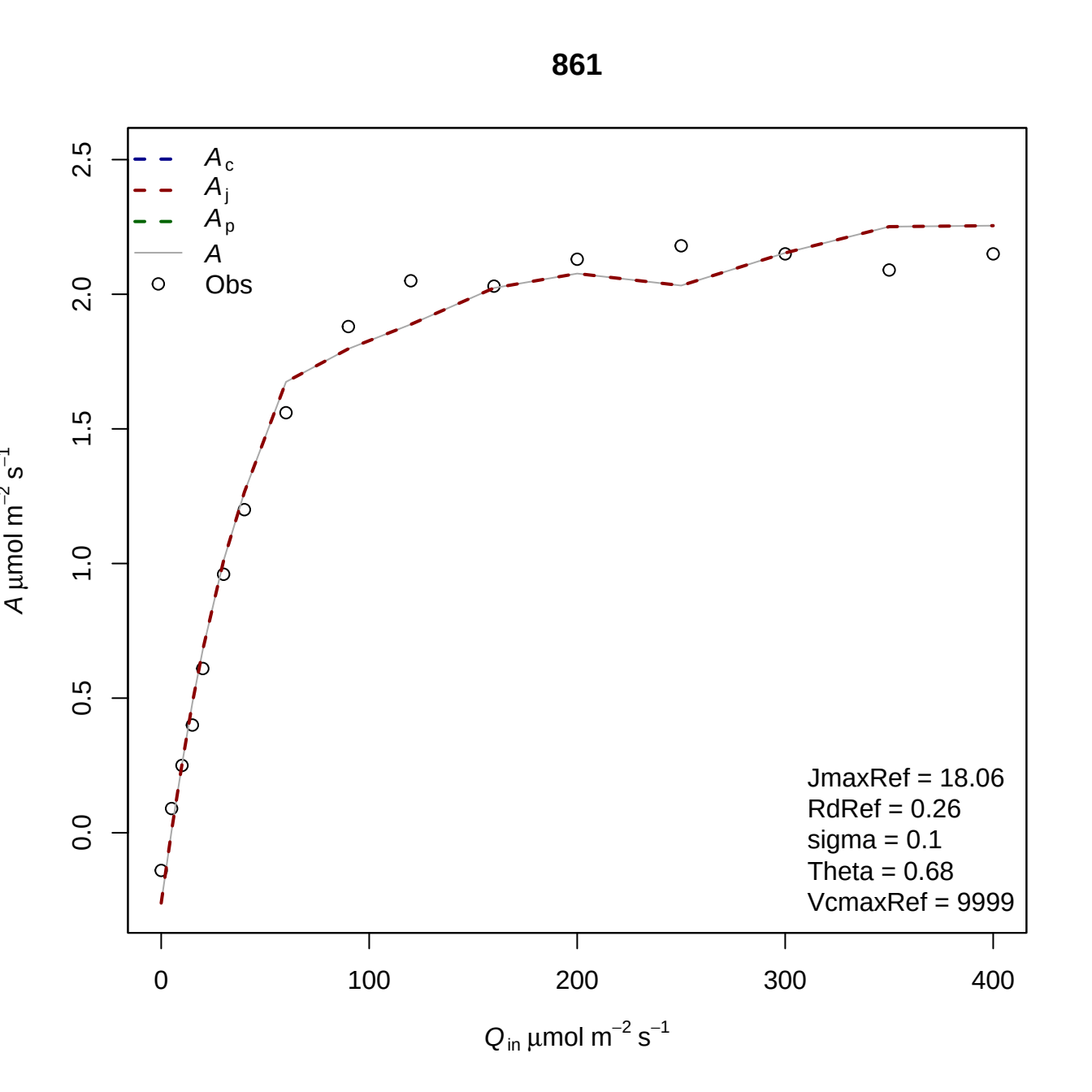

856
A_c
A_j
A_p
A
Obs
JmaxRef = 38.69
RdRef = 0.3
sigma = 0.09
Theta = 0.52
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200

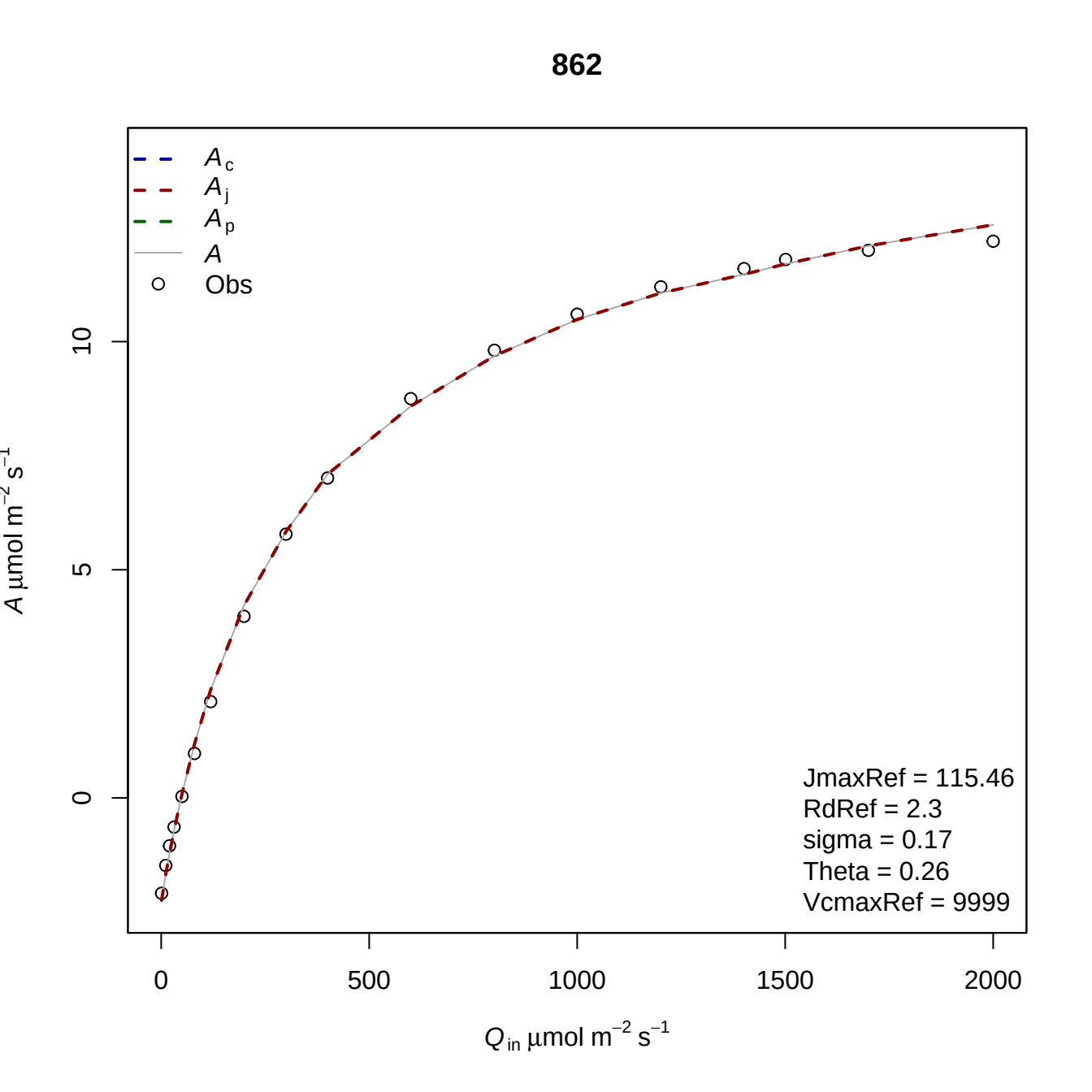

857
A_c
A_j
A_p
A
Obs
JmaxRef = 42.35
RdRef = 1.34
sigma = 0.08
Theta = −1.63
VcmaxRef = 26.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

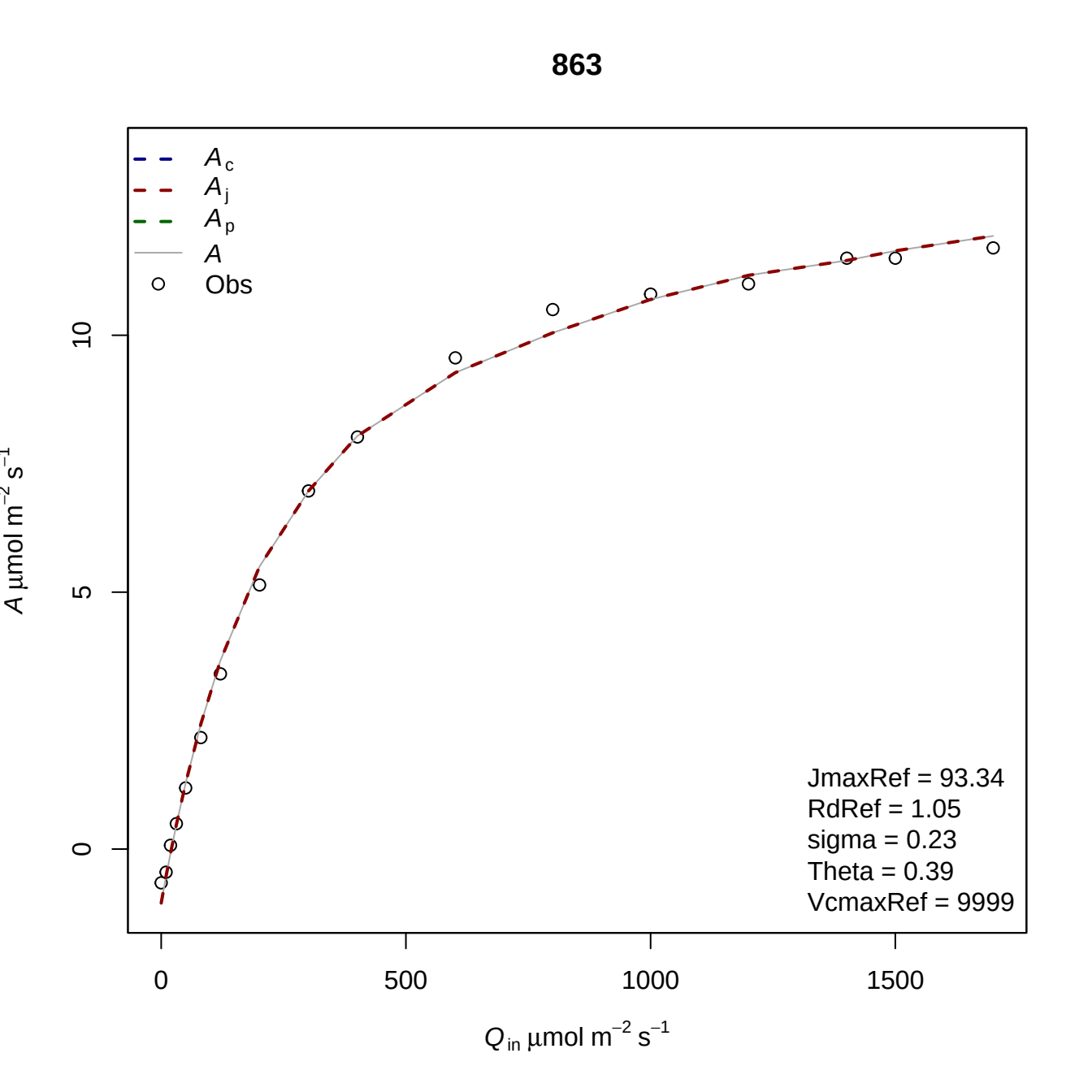

858
A_c
A_j
A_p
A
Obs
JmaxRef = 34.76
RdRef = 1.29
sigma = 0.07
Theta = −0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

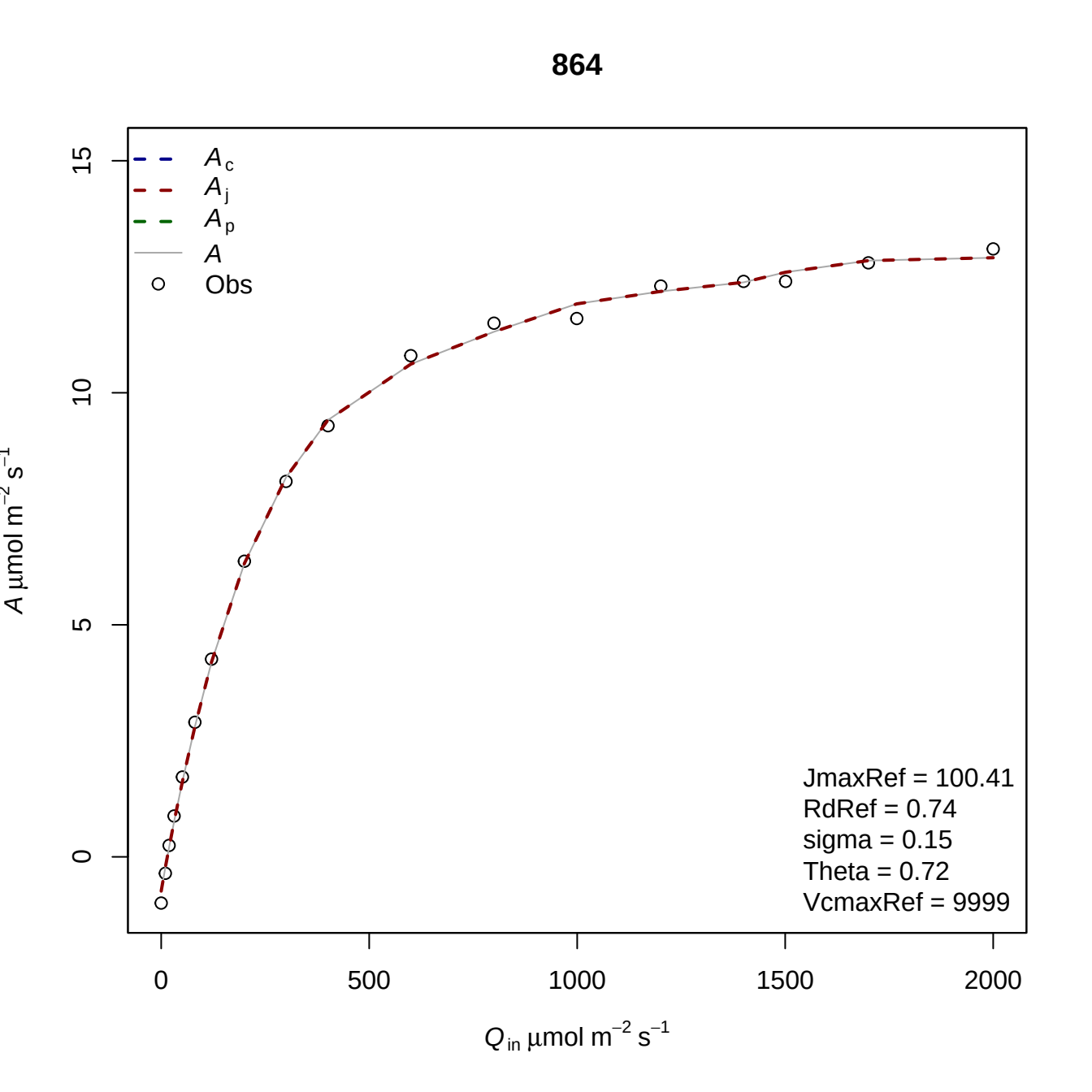

859
A_c
A_j
A_p
A
Obs
JmaxRef = 18.84
RdRef = 0.69
sigma = 0.09
Theta = 0.34
VcmaxRef = 9999
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

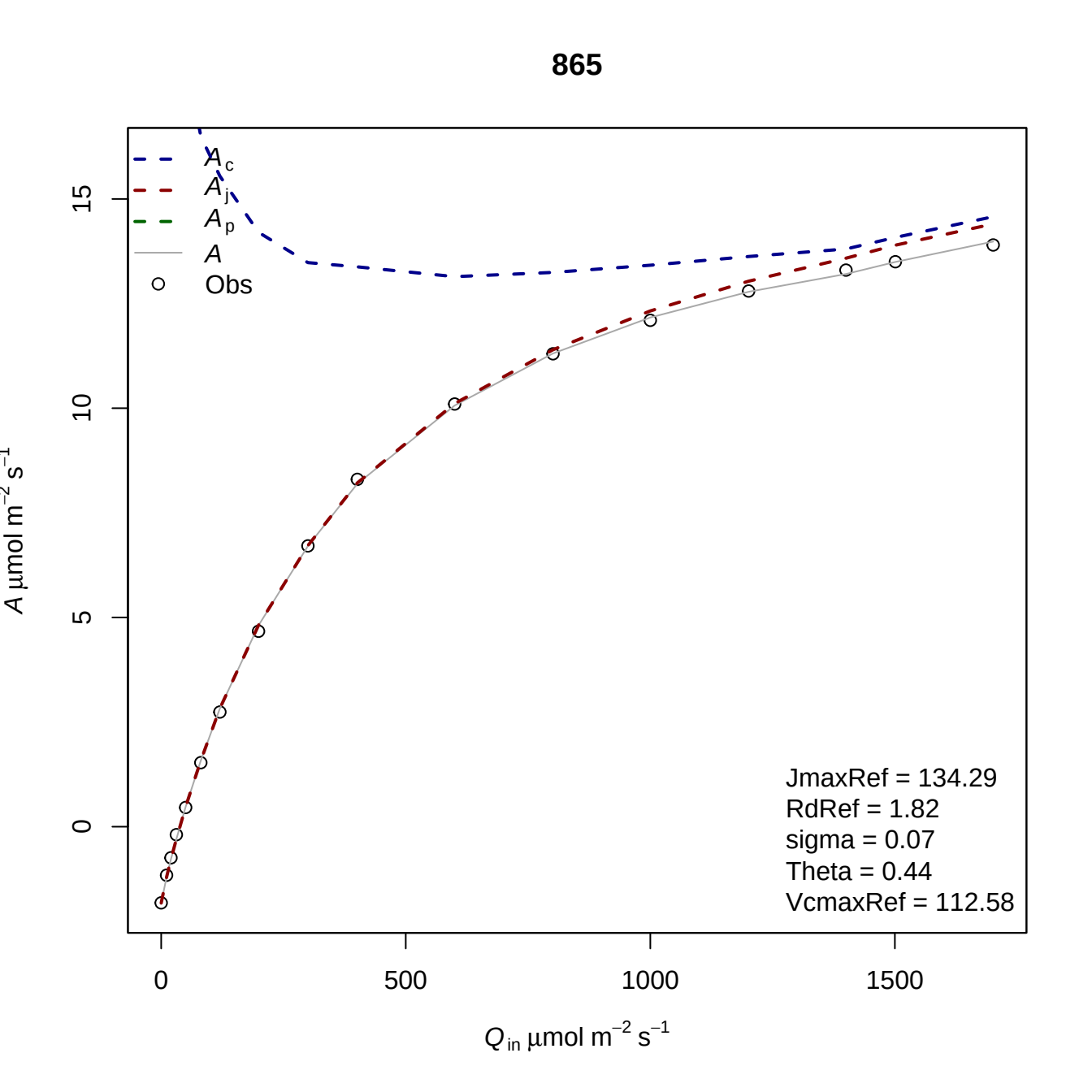

860
A_c
A_j
A_p
A
Obs
JmaxRef = 45.24
RdRef = 0.19
sigma = 0.09
Theta = 0.67
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

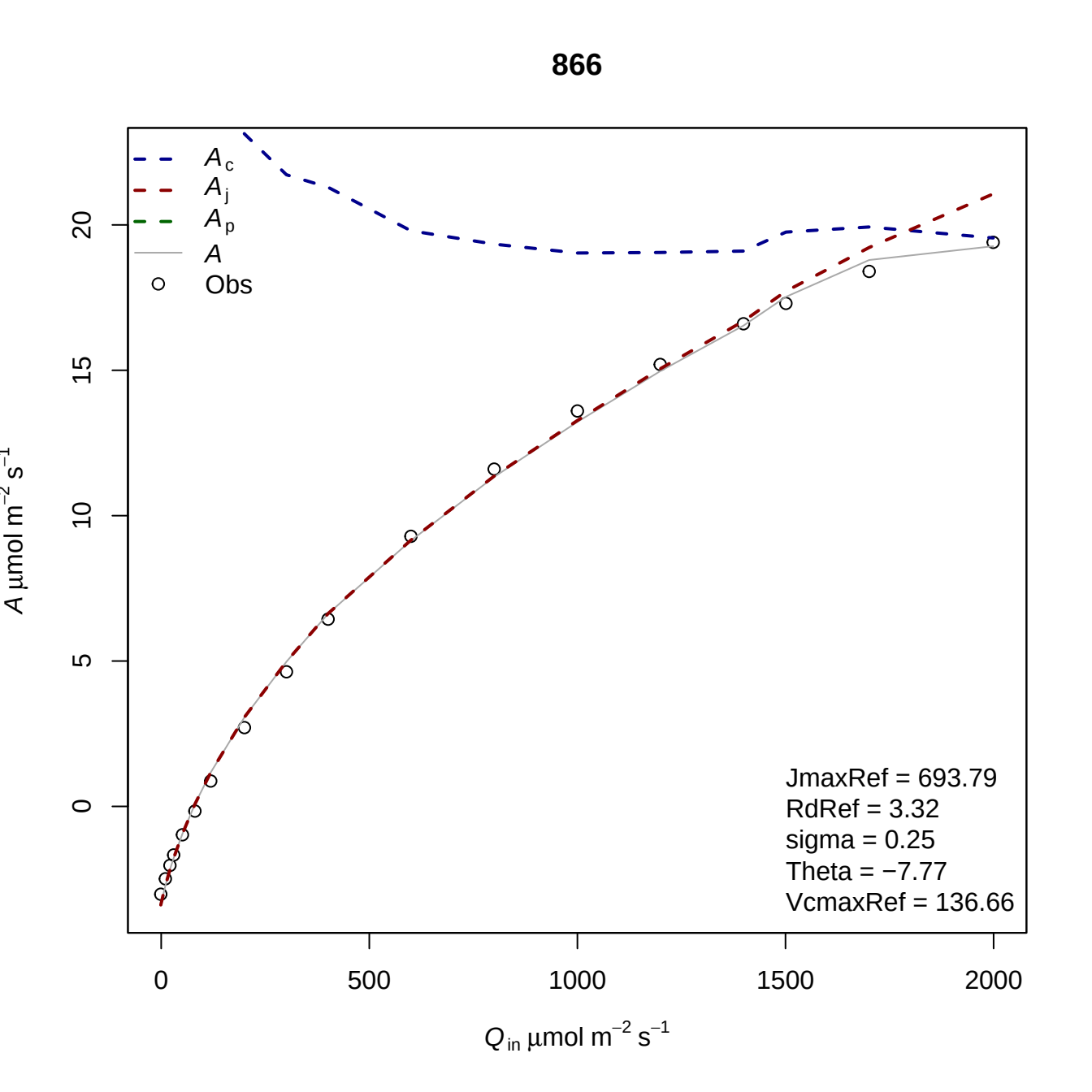

861
A_c
A_j
A_p
A
Obs
JmaxRef = 18.06
RdRef = 0.26
sigma = 0.1
Theta = 0.68
VcmaxRef = 9999
0.0
0.5
1.0
1.5
2.0
2.5
0
100
200
300
400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

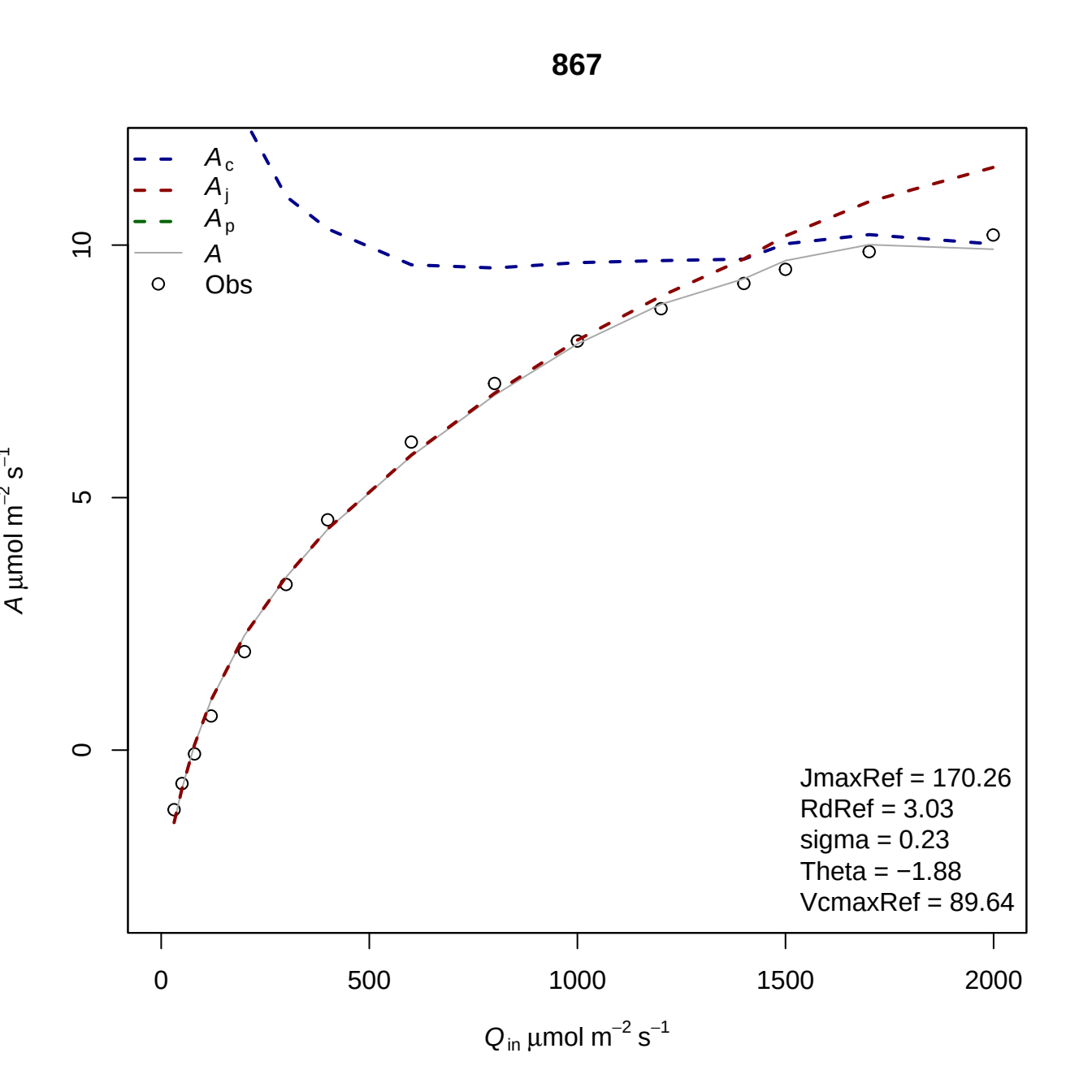

862
A_c
A_j
A_p
A
Obs
JmaxRef = 115.46
RdRef = 2.3
sigma = 0.17
Theta = 0.26
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

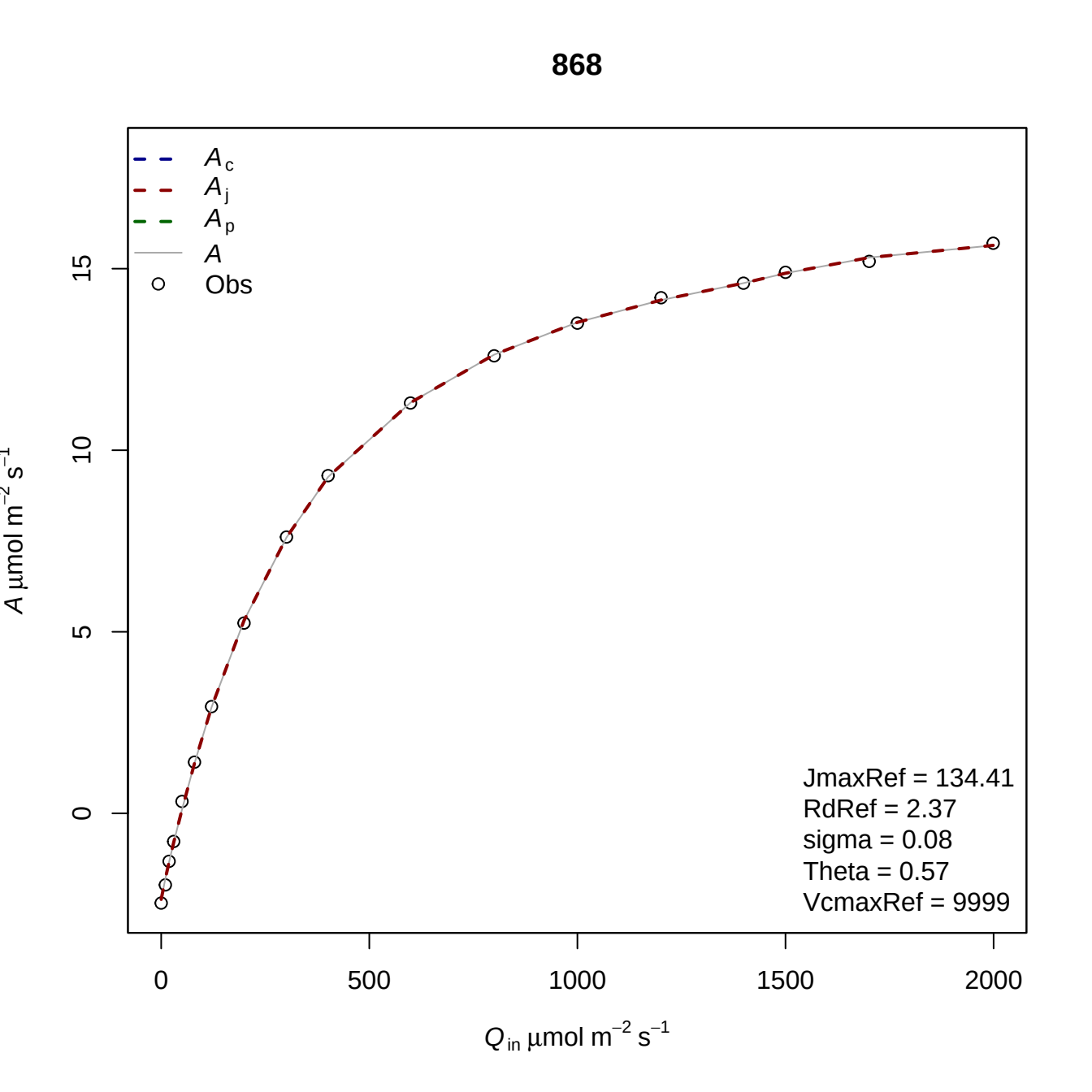

863
A_c
A_j
A_p
A
Obs
JmaxRef = 93.34
RdRef = 1.05
sigma = 0.23
Theta = 0.39
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

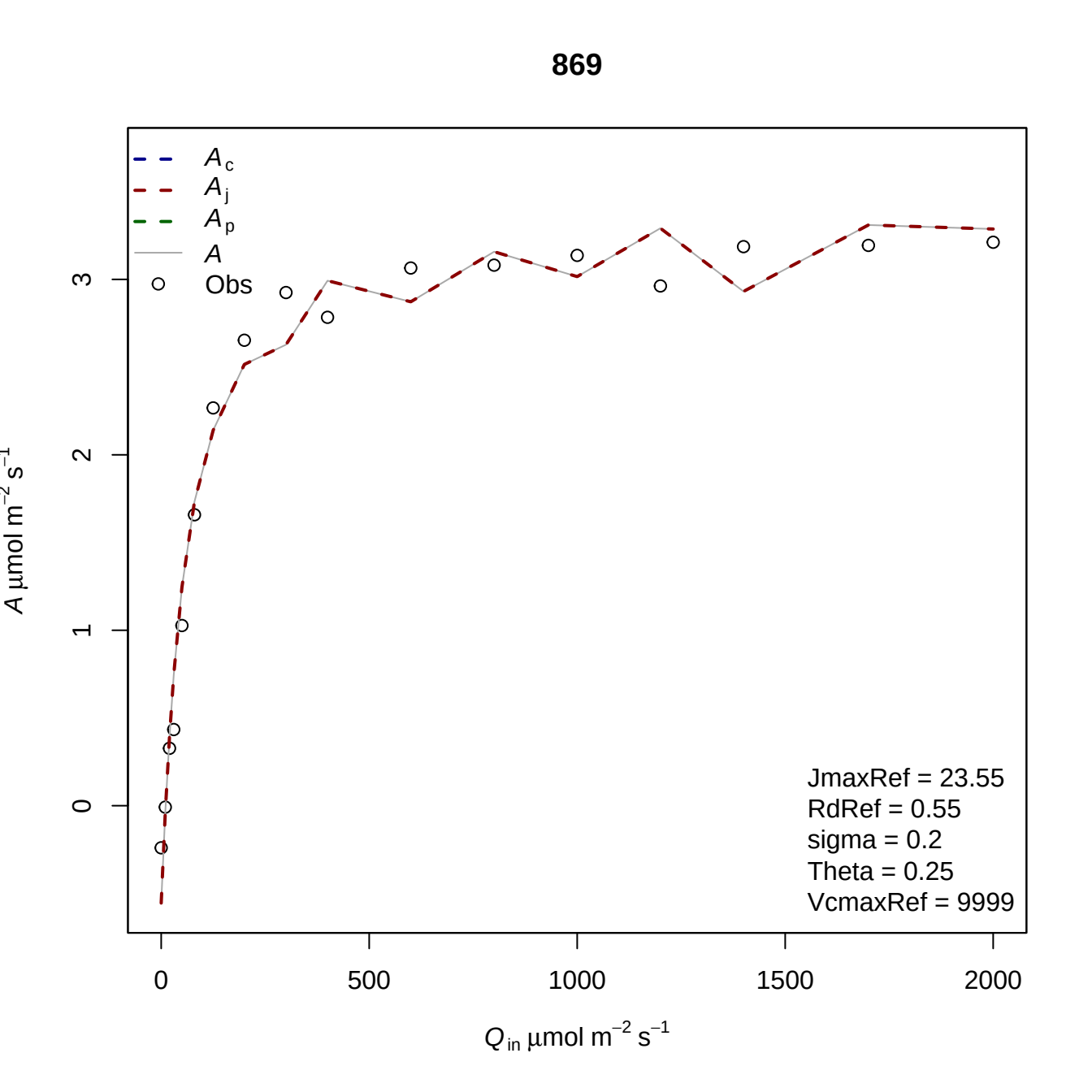

864
A_c
A_j
A_p
A
Obs
JmaxRef = 100.41
RdRef = 0.74
sigma = 0.15
Theta = 0.72
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

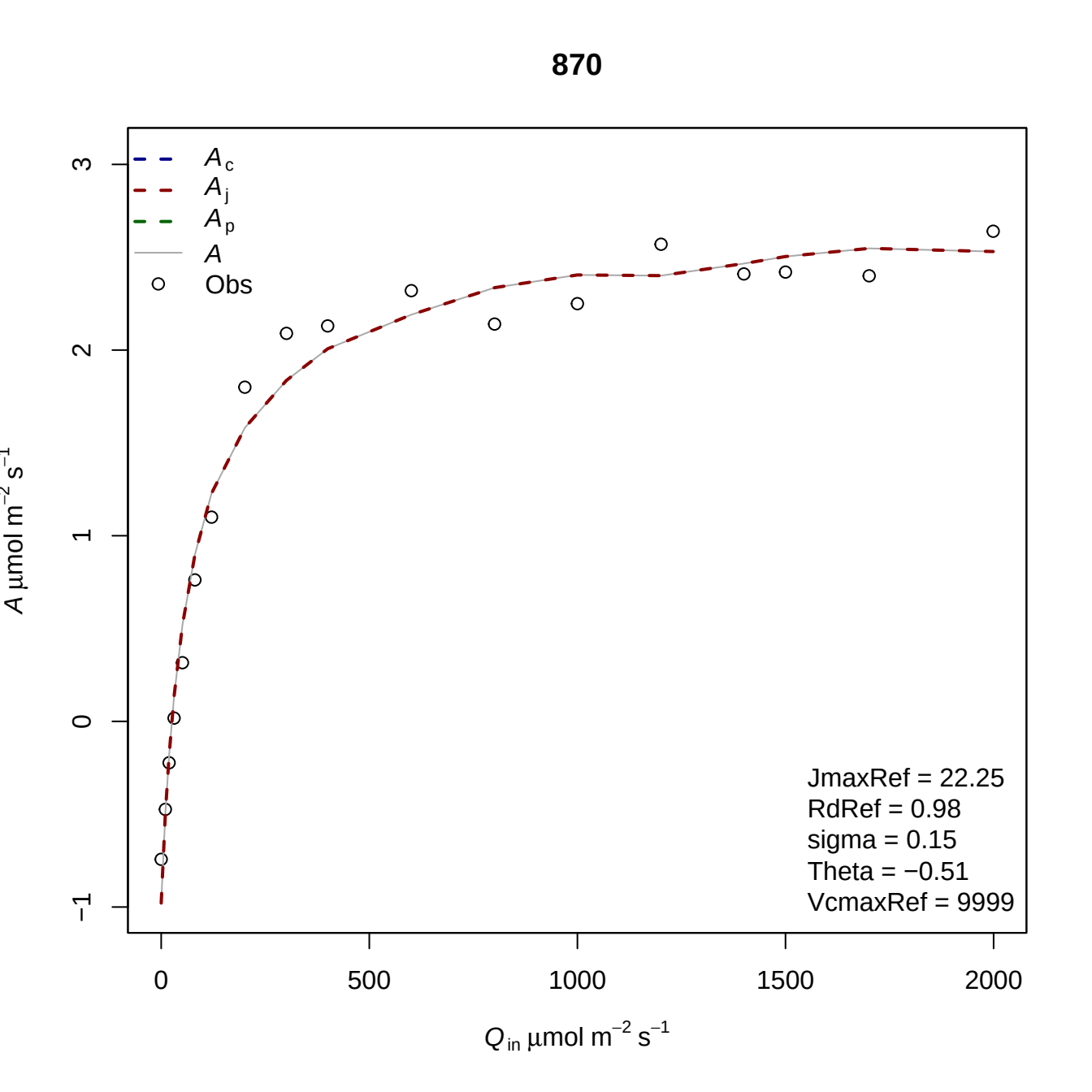

865
A_c
A_j
A_p
A
Obs
JmaxRef = 134.29
RdRef = 1.82
sigma = 0.07
Theta = 0.44
VcmaxRef = 112.58
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

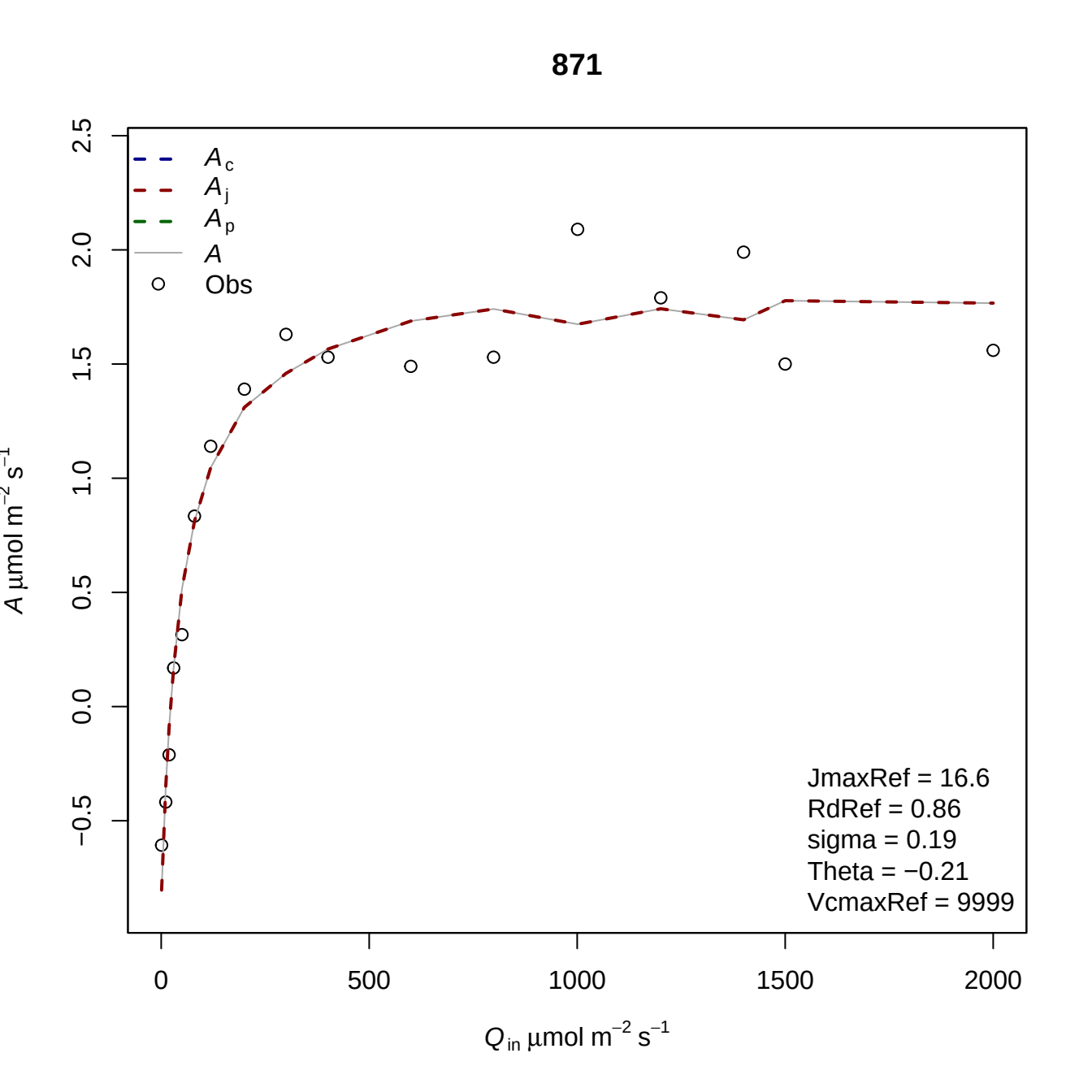

866
A_c
A_j
A_p
A
Obs
JmaxRef = 693.79
RdRef = 3.32
sigma = 0.25
Theta = −7.77
VcmaxRef = 136.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

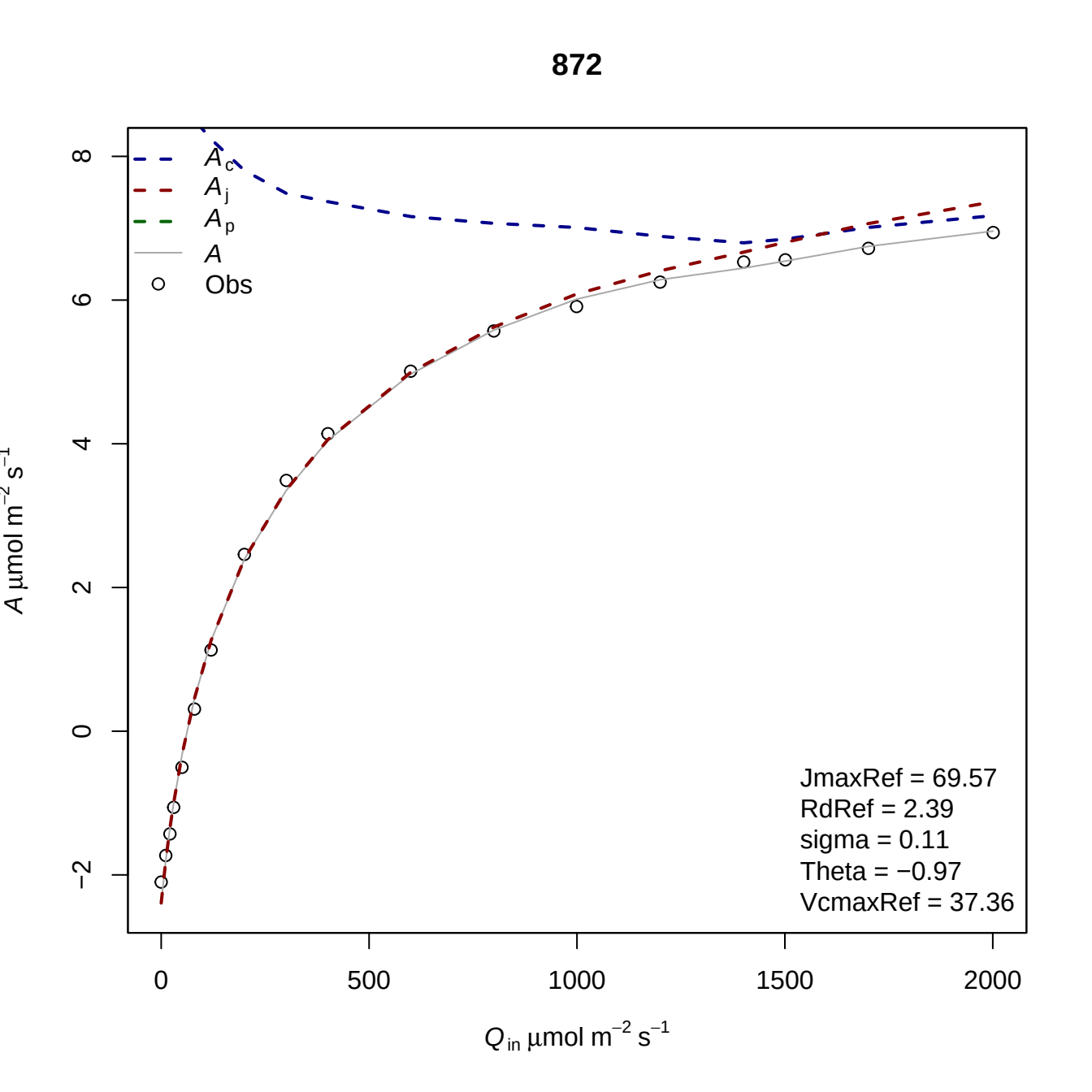

867
A_c
A_j
A_p
A
Obs
JmaxRef = 170.26
RdRef = 3.03
sigma = 0.23
Theta = −1.88
VcmaxRef = 89.64
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

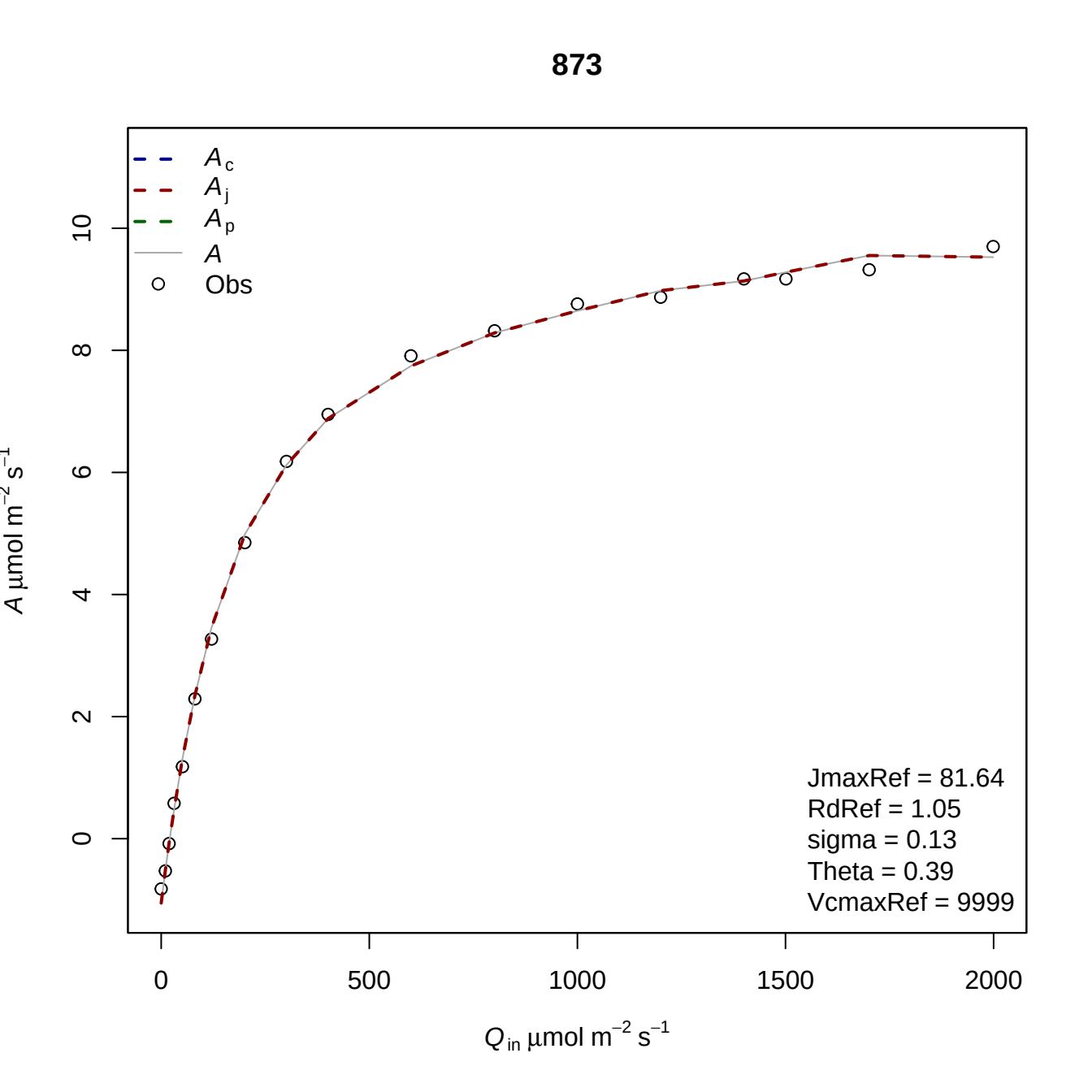

868
A_c
A_j
A_p
A
Obs
JmaxRef = 134.41
RdRef = 2.37
sigma = 0.08
Theta = 0.57
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

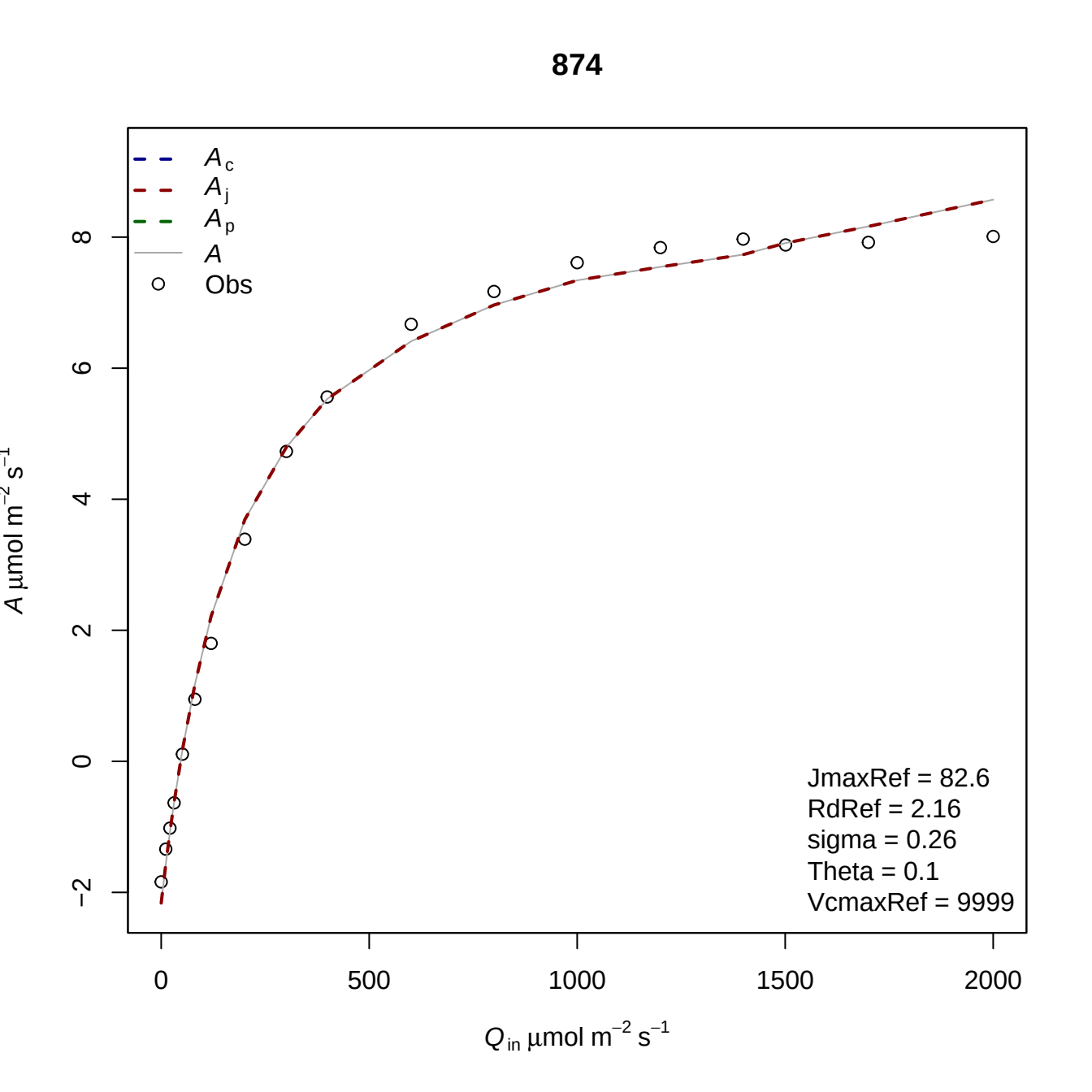

869
A_c
A_j
A_p
A
Obs
JmaxRef = 23.55
RdRef = 0.55
sigma = 0.2
Theta = 0.25
VcmaxRef = 9999
0
1
2
3
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

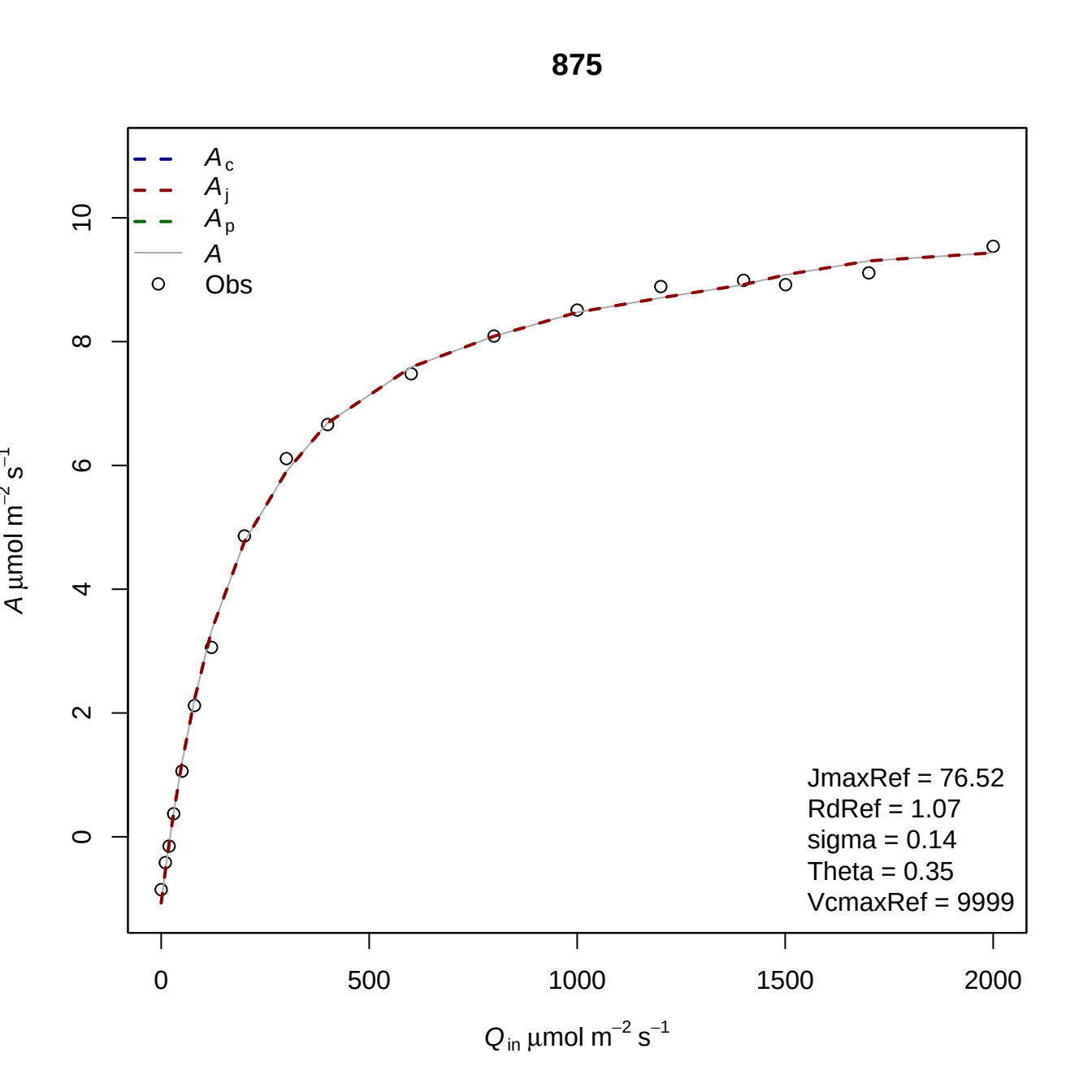

870
A_c
A_j
A_p
A
Obs
JmaxRef = 22.25
RdRef = 0.98
sigma = 0.15
Theta = −0.51
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

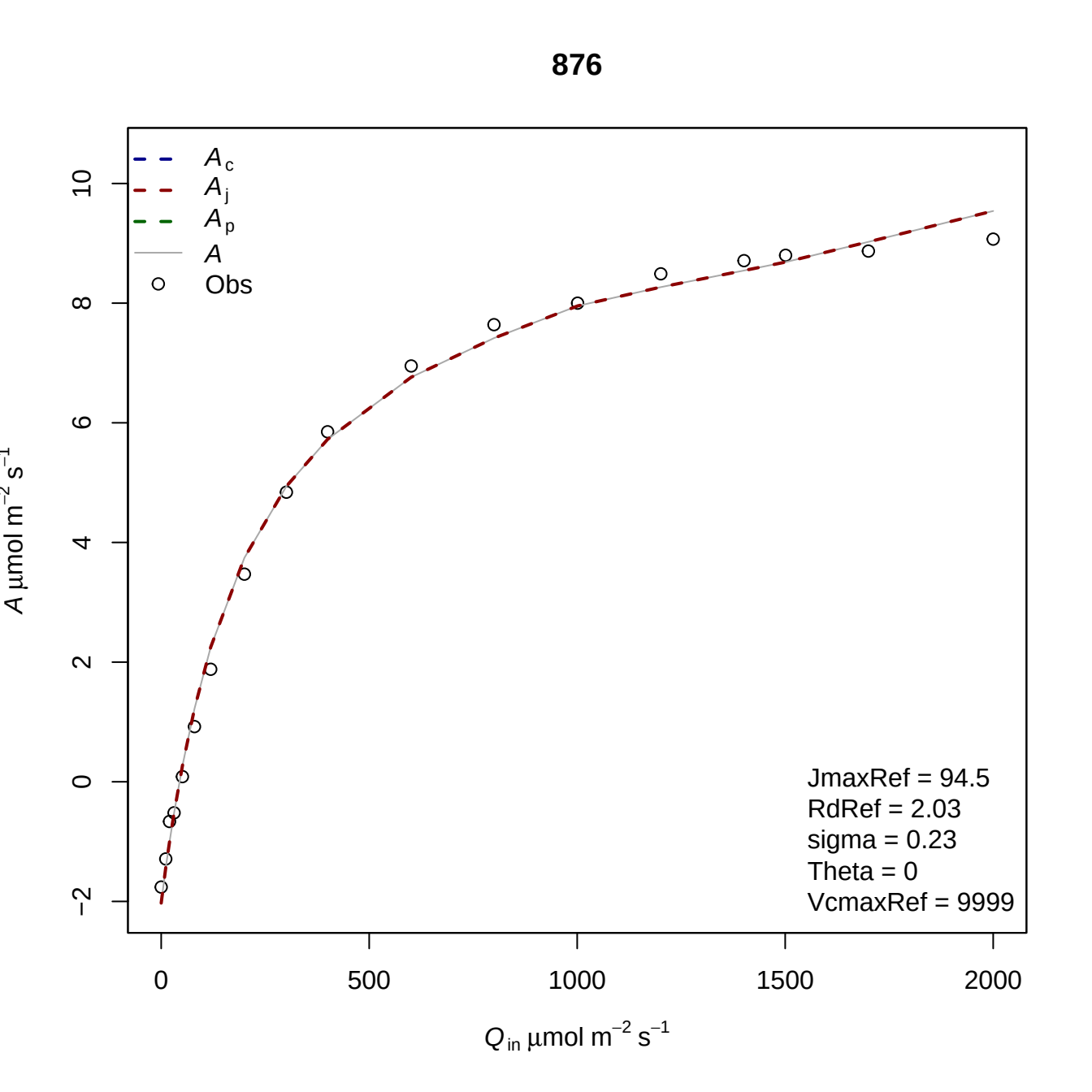

871
A_c
A_j
A_p
A
Obs
JmaxRef = 16.6
RdRef = 0.86
sigma = 0.19
Theta = −0.21
VcmaxRef = 9999
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

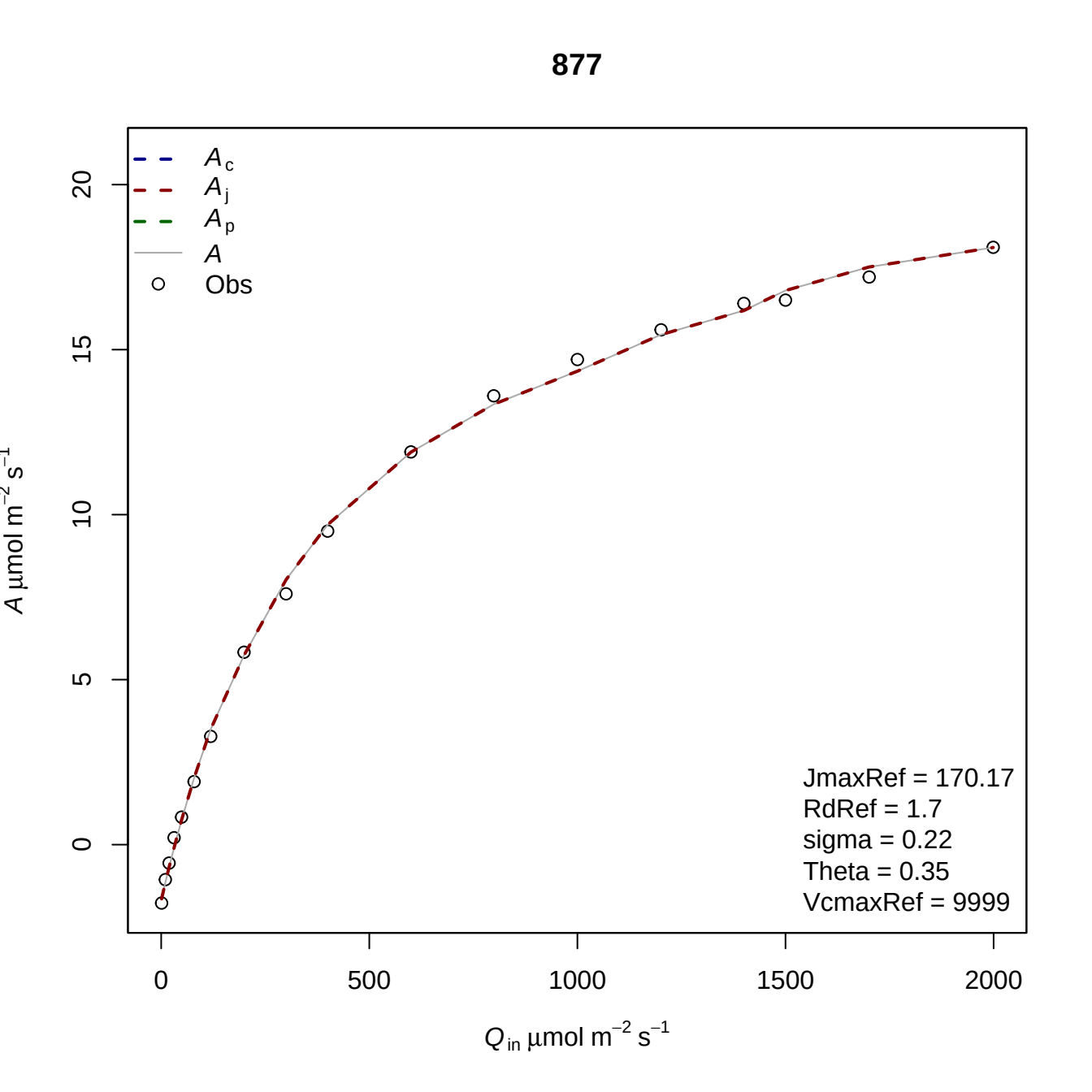

872
A_c
A_j
A_p
A
Obs
JmaxRef = 69.57
RdRef = 2.39
sigma = 0.11
Theta = −0.97
VcmaxRef = 37.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

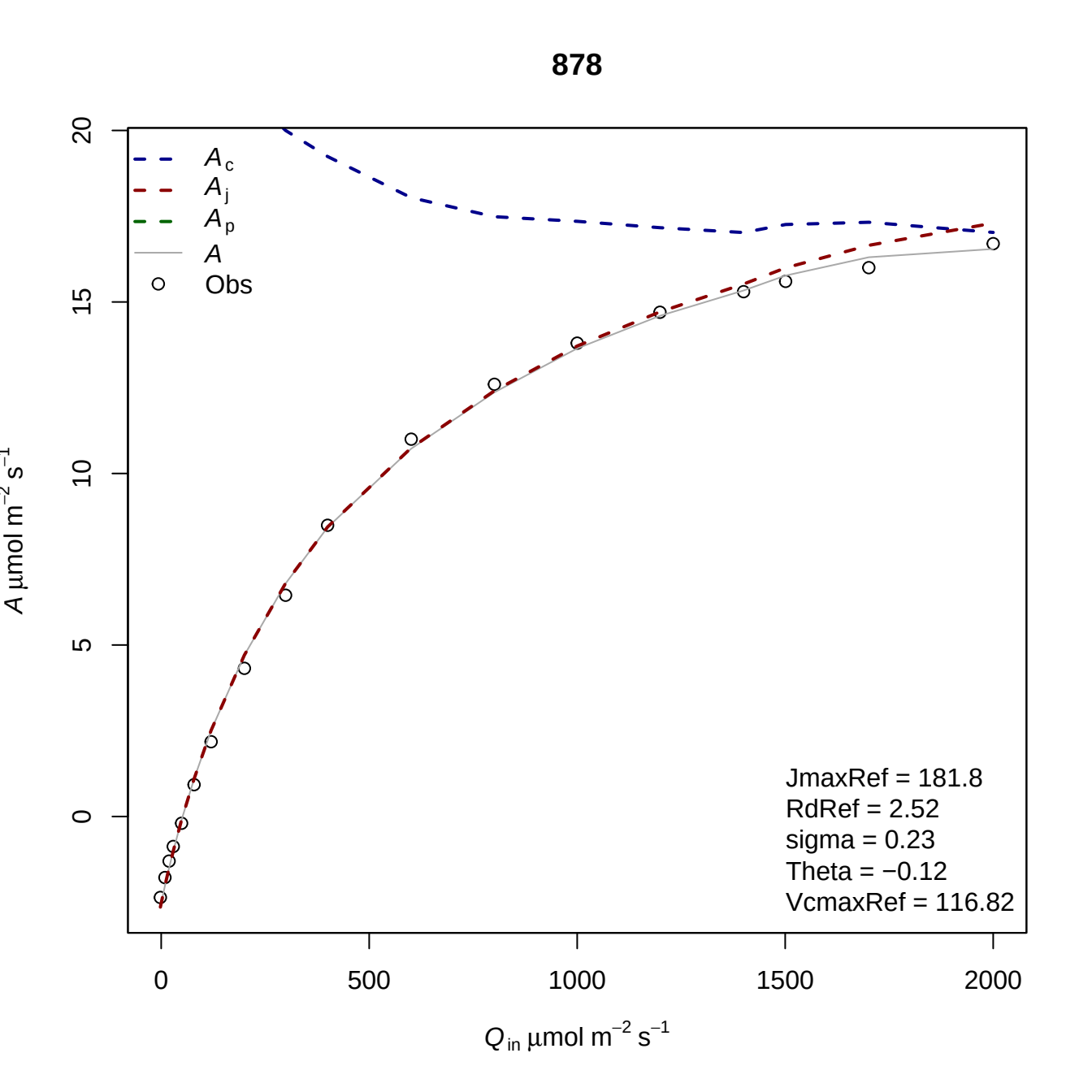

873
A_c
A_j
A_p
A
Obs
JmaxRef = 81.64
RdRef = 1.05
sigma = 0.13
Theta = 0.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

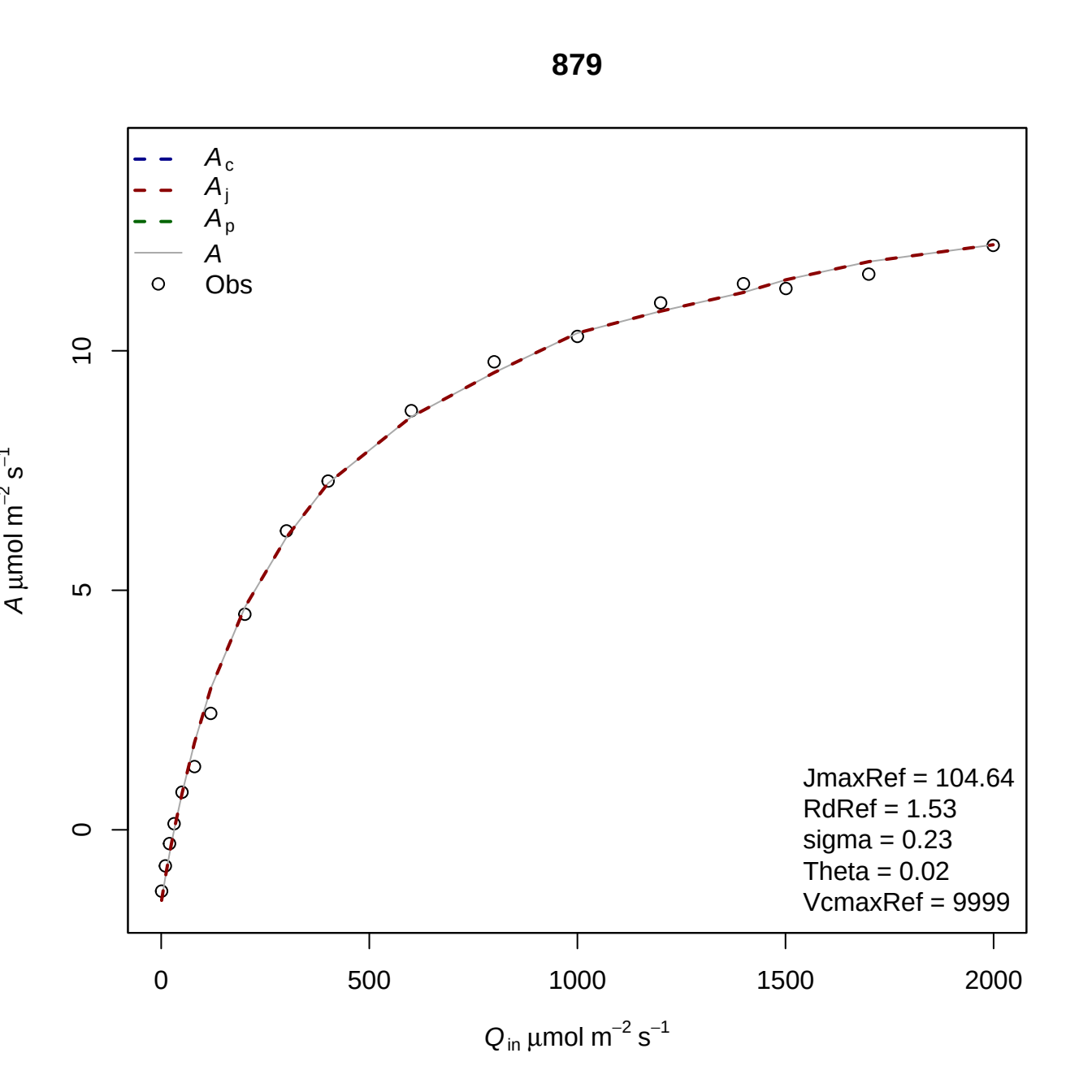

874
A_c
A_j
A_p
A
Obs
JmaxRef = 82.6
RdRef = 2.16
sigma = 0.26
Theta = 0.1
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

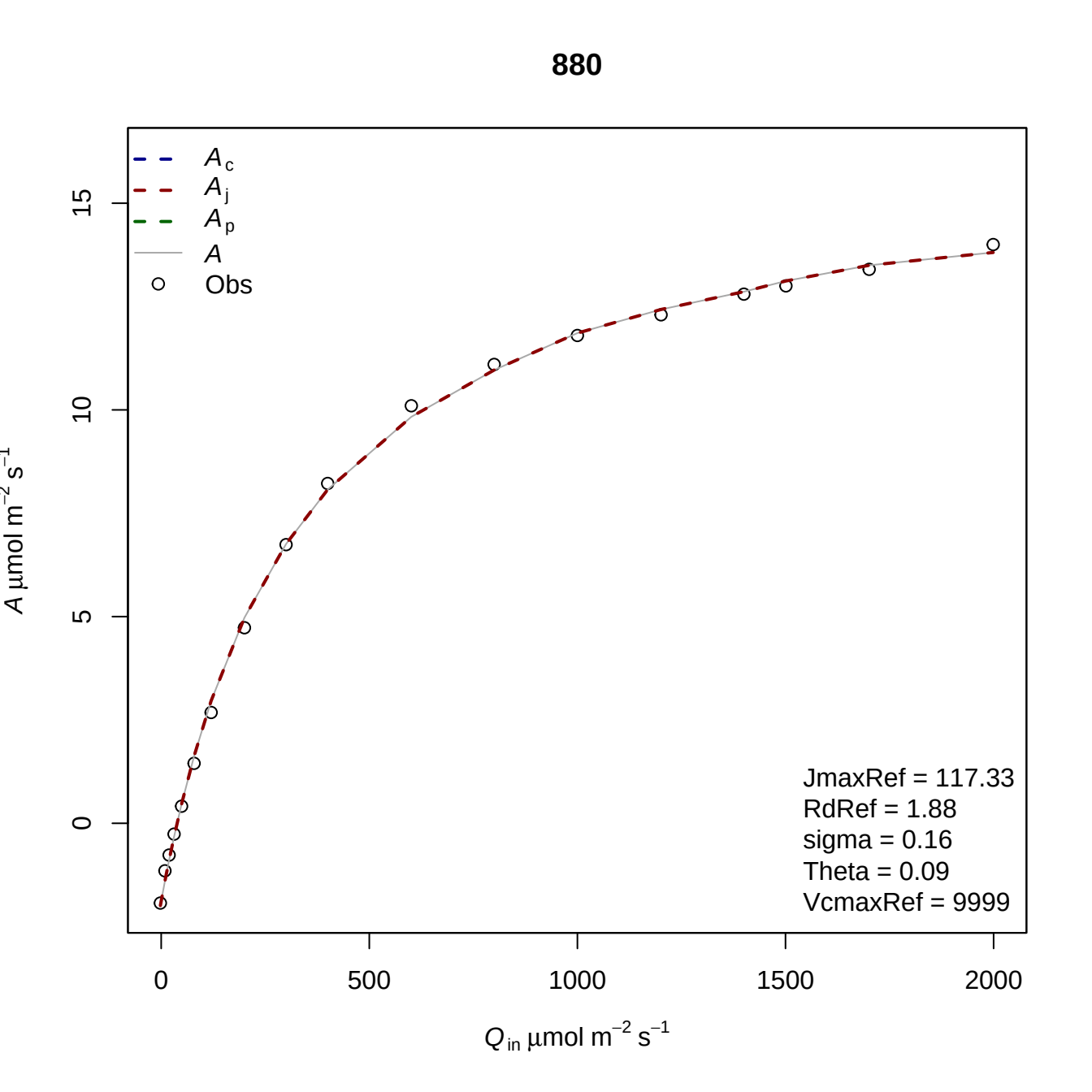

875
A_c
A_j
A_p
A
Obs
JmaxRef = 76.52
RdRef = 1.07
sigma = 0.14
Theta = 0.35
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

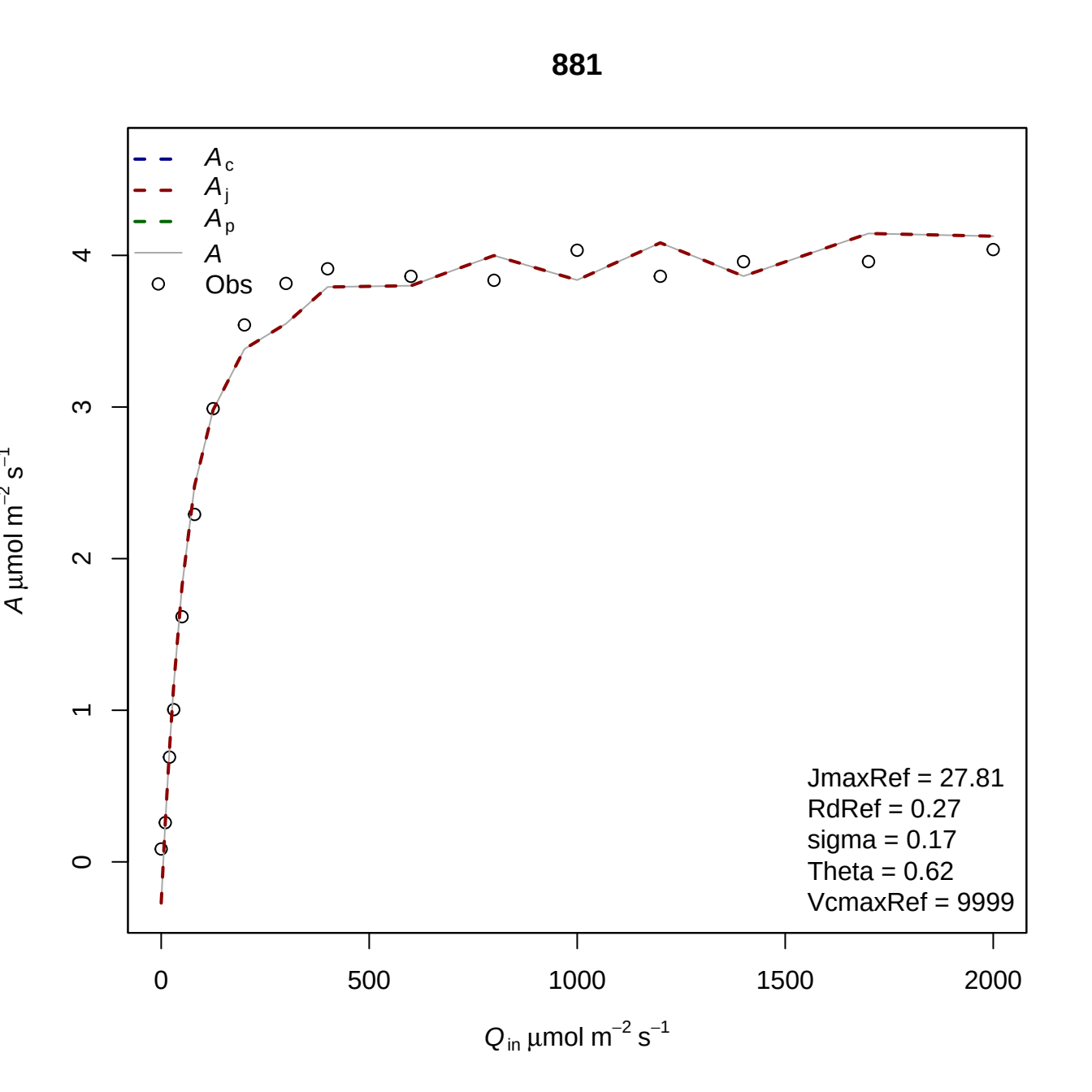

876
A_c
A_j
A_p
A
Obs
JmaxRef = 94.5
RdRef = 2.03
sigma = 0.23
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

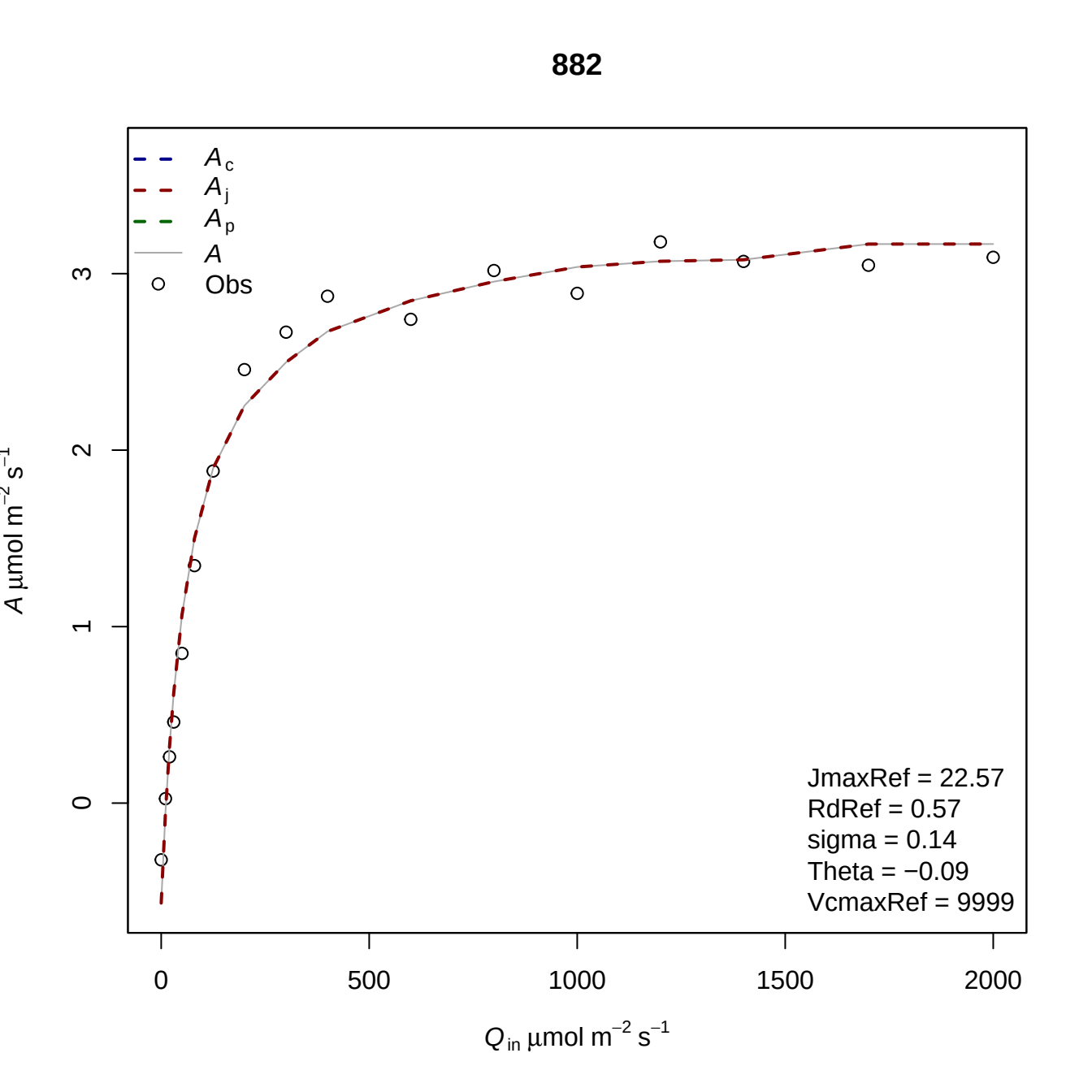

877
A_c
A_j
A_p
A
Obs
JmaxRef = 170.17
RdRef = 1.7
sigma = 0.22
Theta = 0.35
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

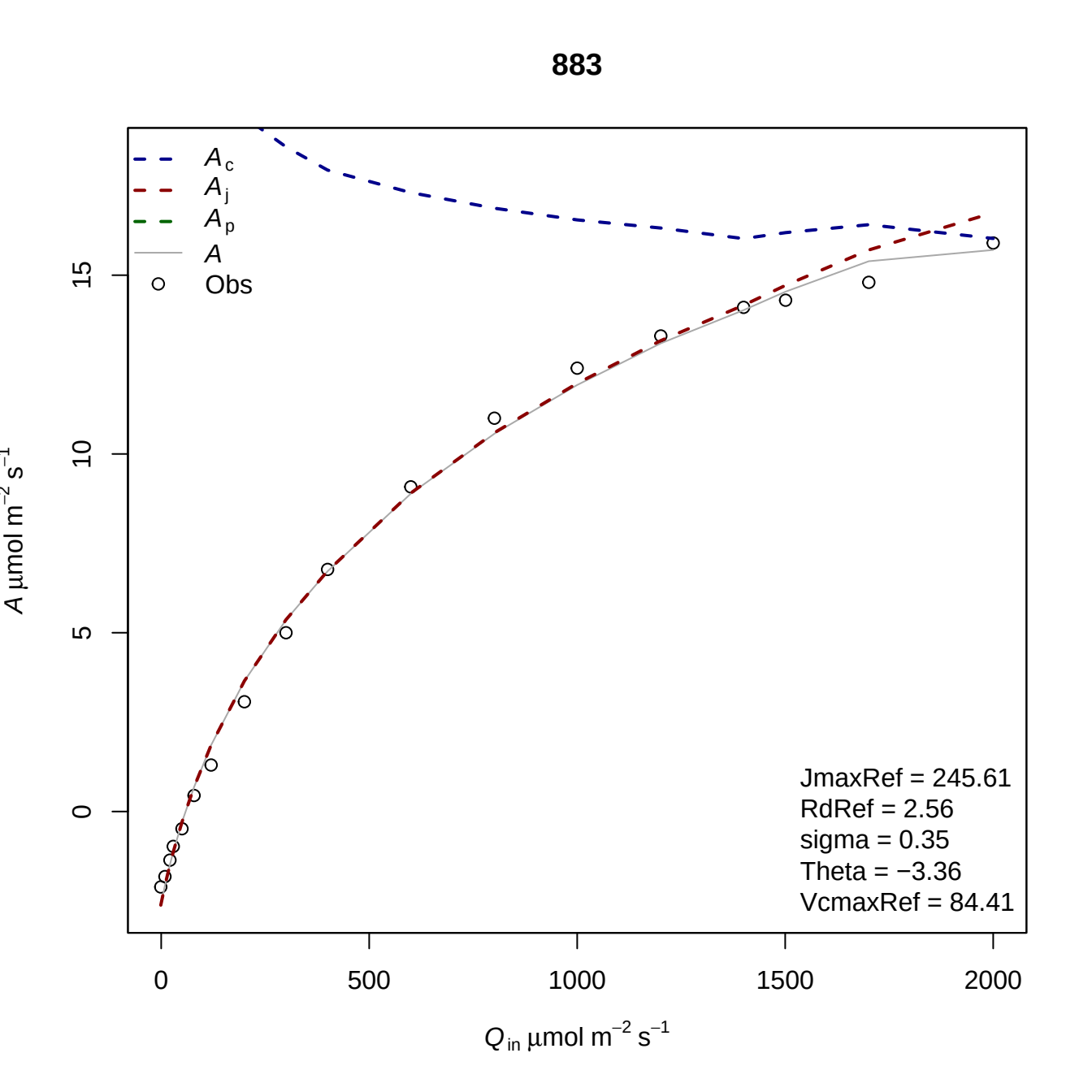

878
A_c
A_j
A_p
A
Obs
JmaxRef = 181.8
RdRef = 2.52
sigma = 0.23
Theta = −0.12
VcmaxRef = 116.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

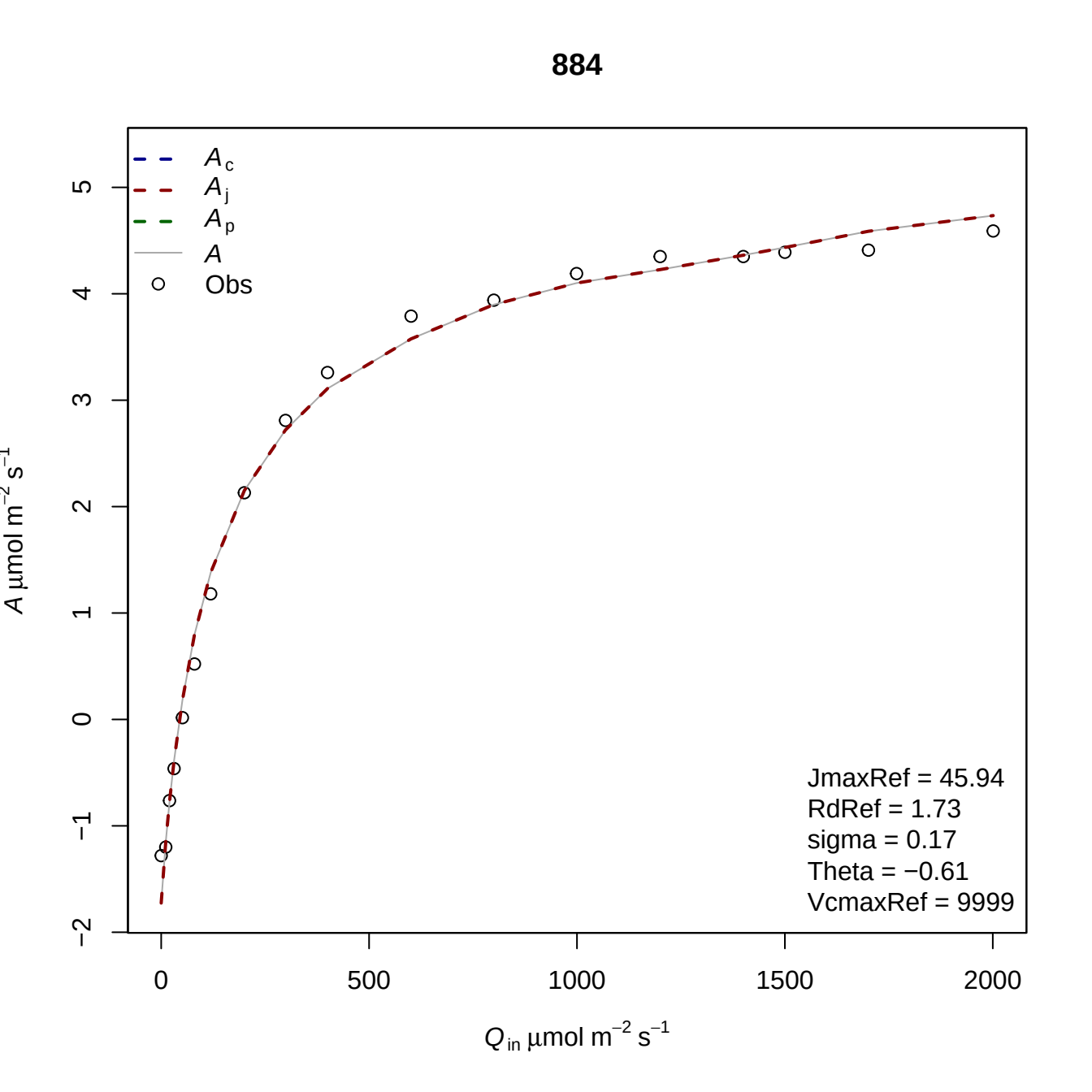

879
A_c
A_j
A_p
A
Obs
JmaxRef = 104.64
RdRef = 1.53
sigma = 0.23
Theta = 0.02
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

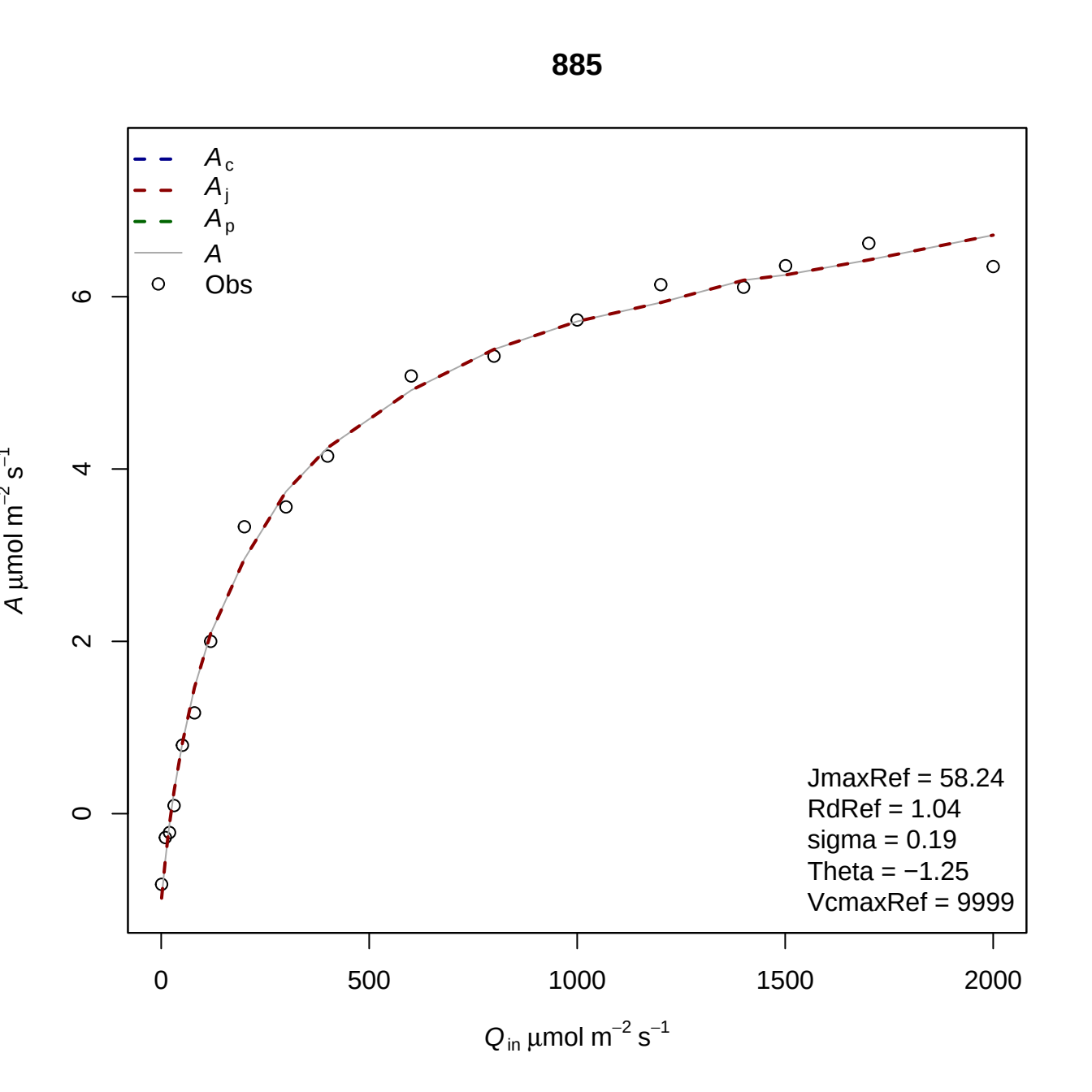

880
A_c
A_j
A_p
A
Obs
JmaxRef = 117.33
RdRef = 1.88
sigma = 0.16
Theta = 0.09
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

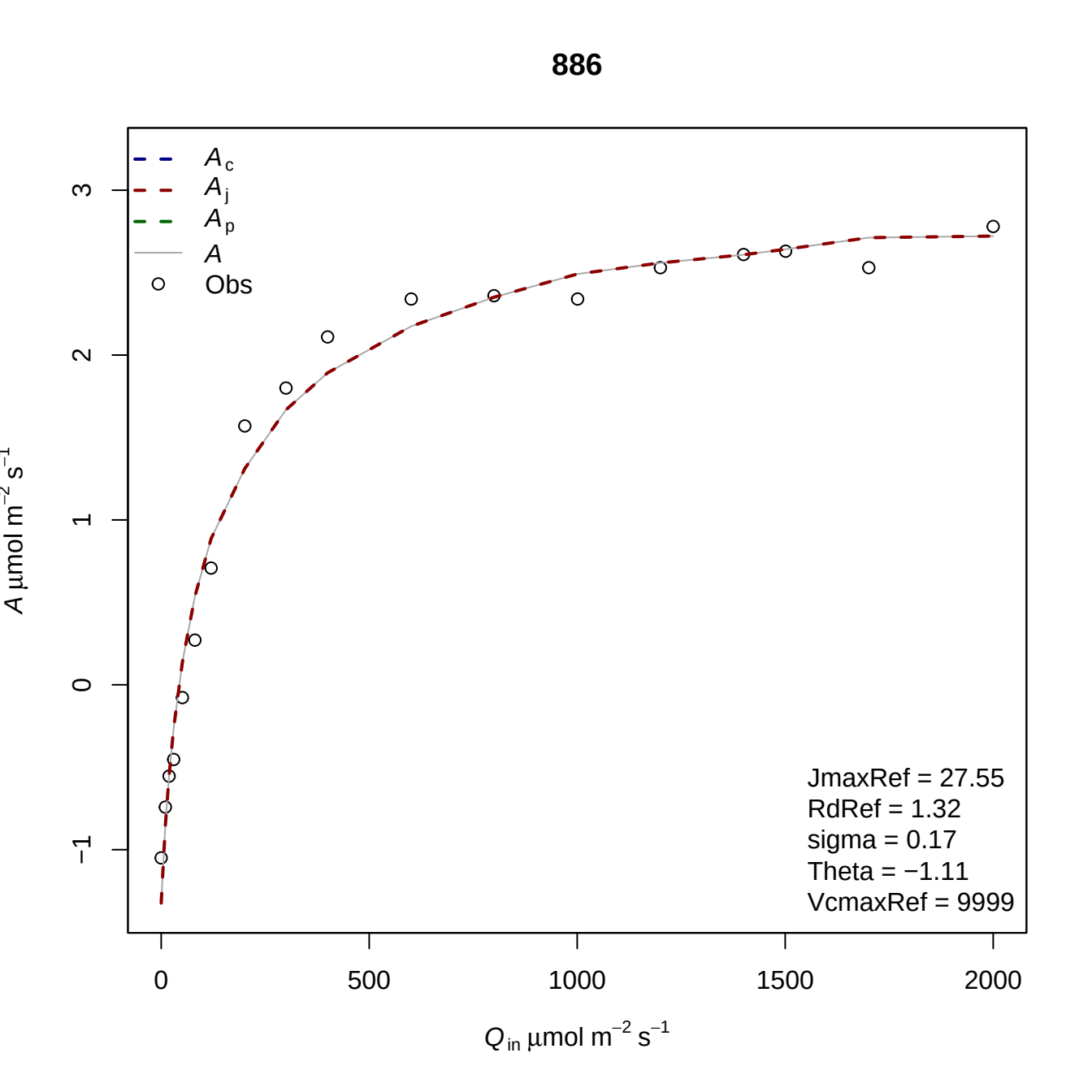

881
A_c
A_j
A_p
A
Obs
JmaxRef = 27.81
RdRef = 0.27
sigma = 0.17
Theta = 0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

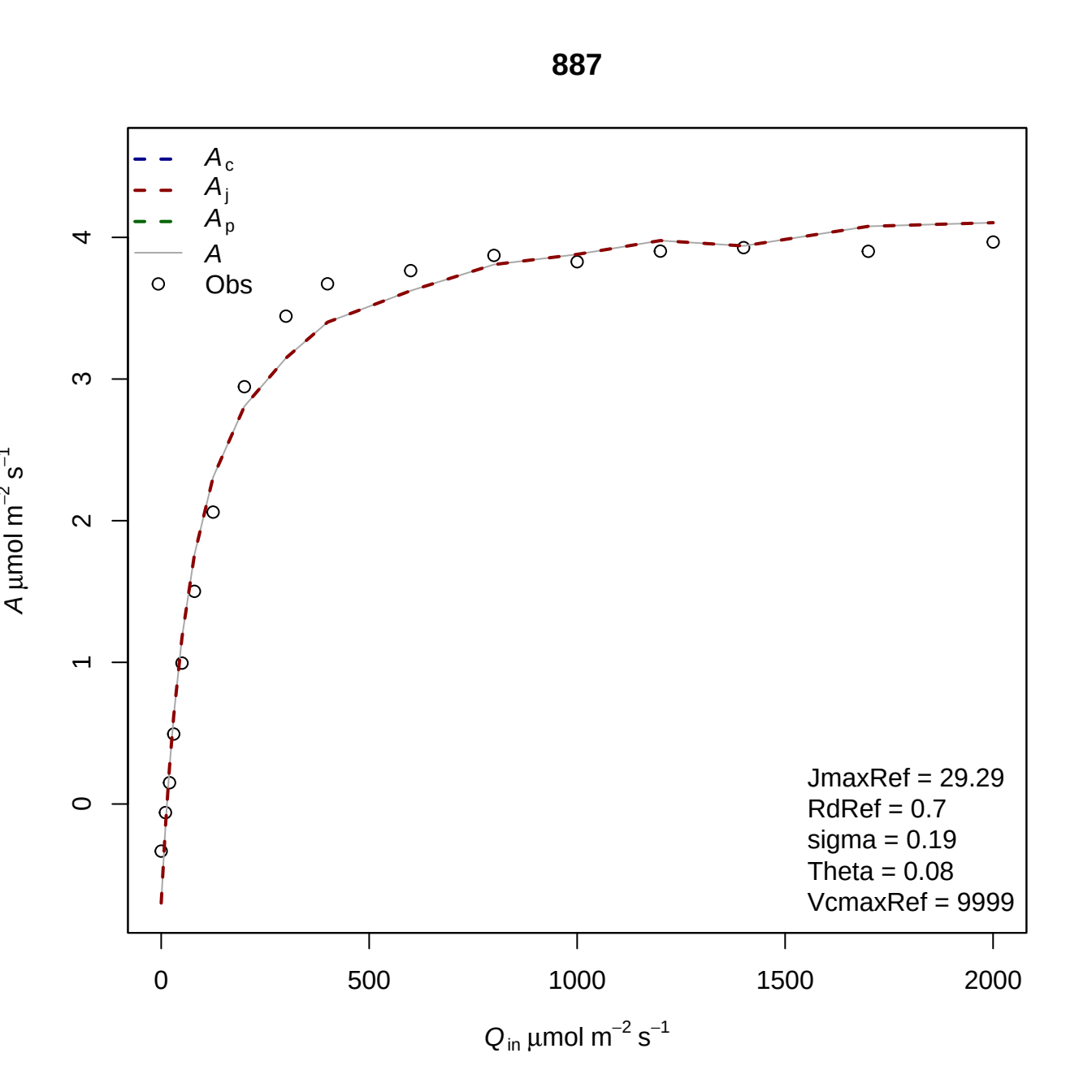

882
A_c
A_j
A_p
A
Obs
JmaxRef = 22.57
RdRef = 0.57
sigma = 0.14
Theta = −0.09
VcmaxRef = 9999
0
1
2
3
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

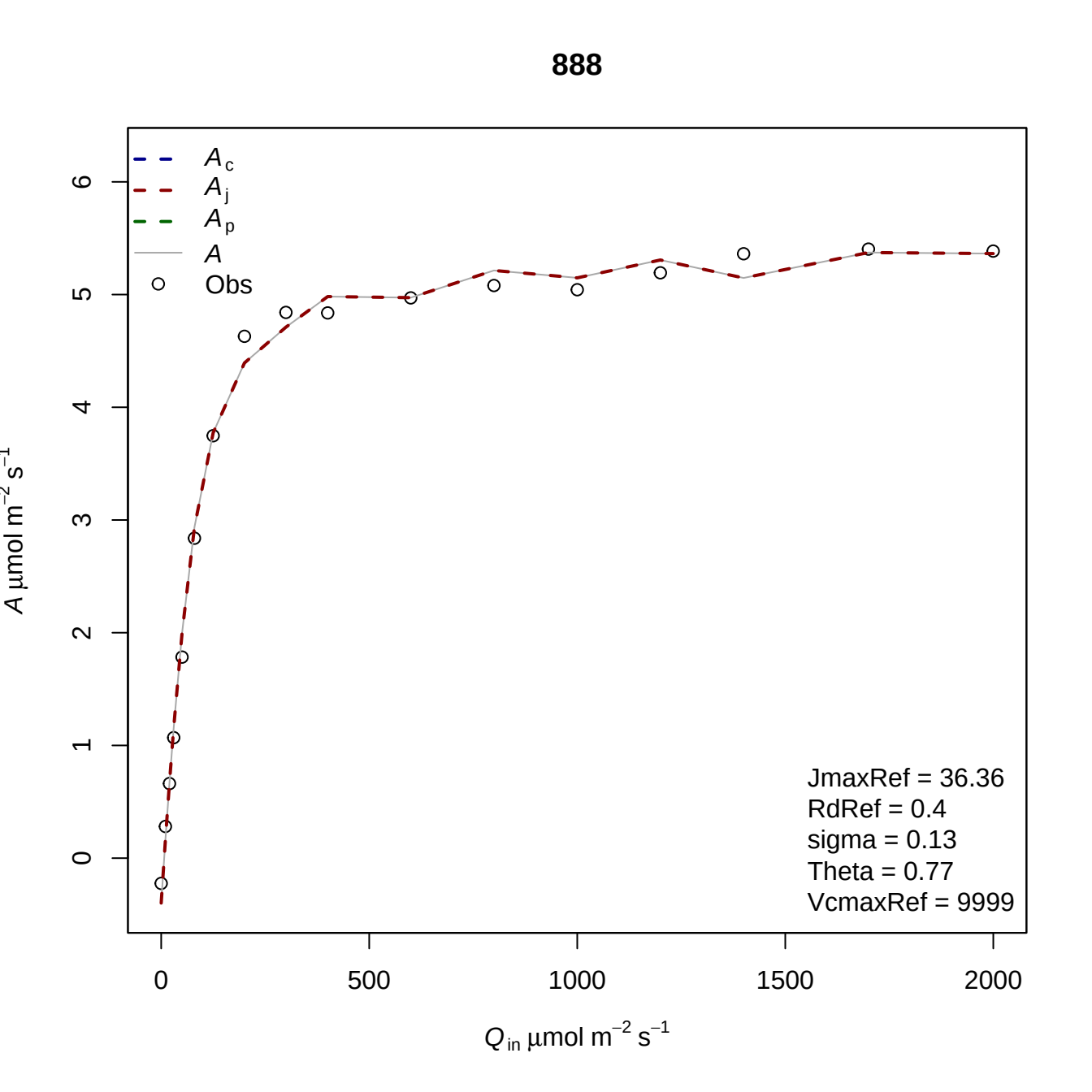

883
A_c
A_j
A_p
A
Obs
JmaxRef = 245.61
RdRef = 2.56
sigma = 0.35
Theta = −3.36
VcmaxRef = 84.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

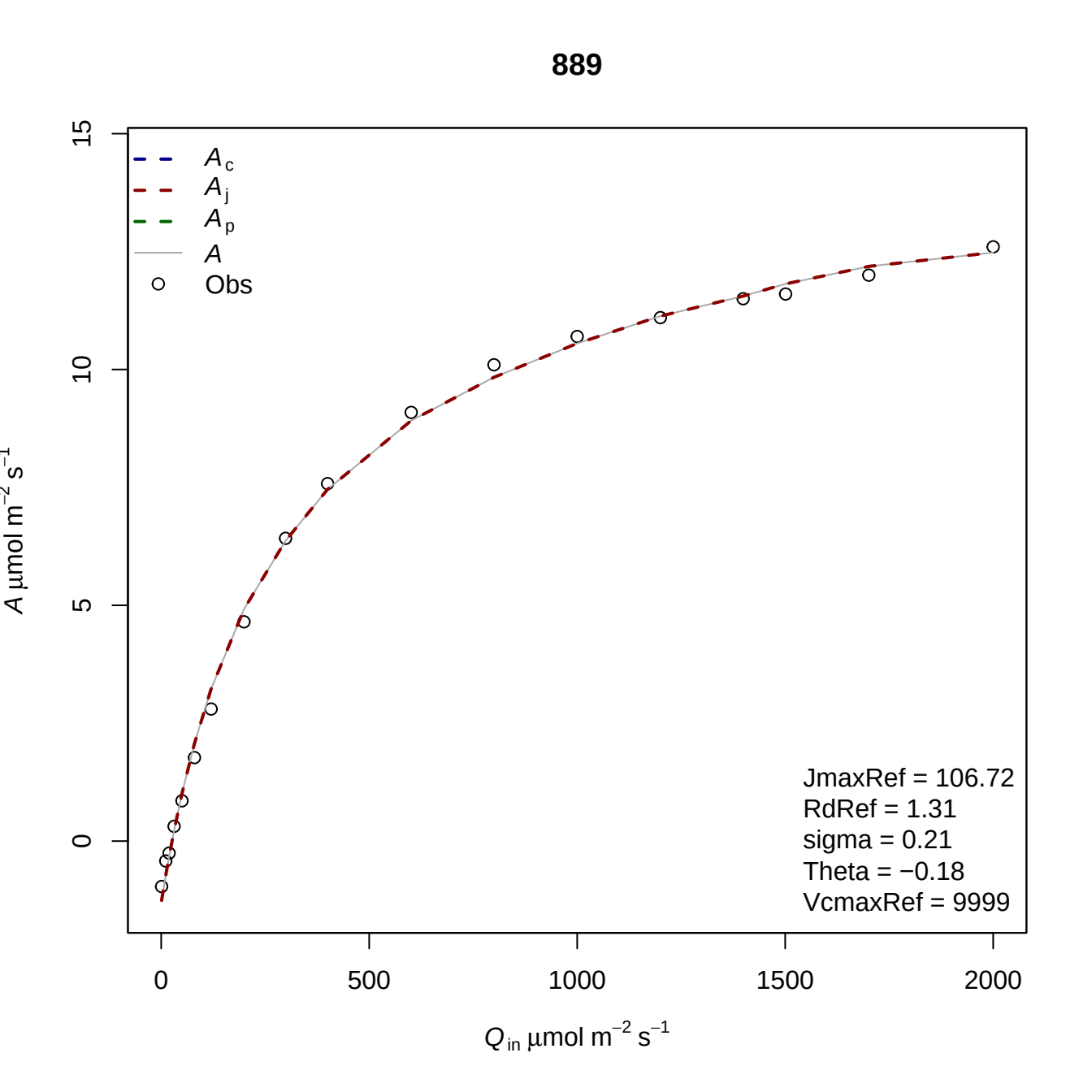

884
A_c
A_j
A_p
A
Obs
JmaxRef = 45.94
RdRef = 1.73
sigma = 0.17
Theta = −0.61
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
5
0
500
1000
1500
2000

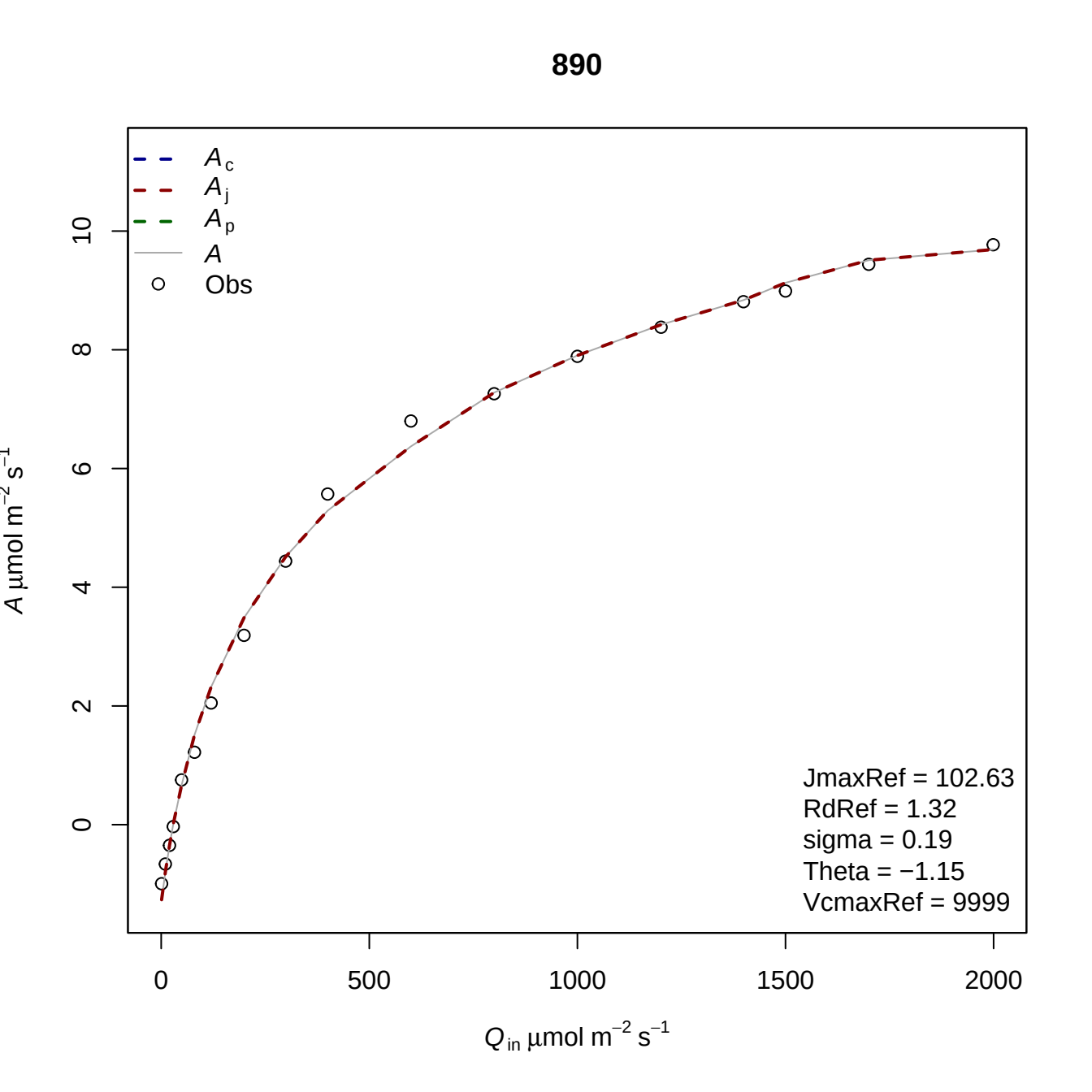

885
A_c
A_j
A_p
A
Obs
JmaxRef = 58.24
RdRef = 1.04
sigma = 0.19
Theta = −1.25
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

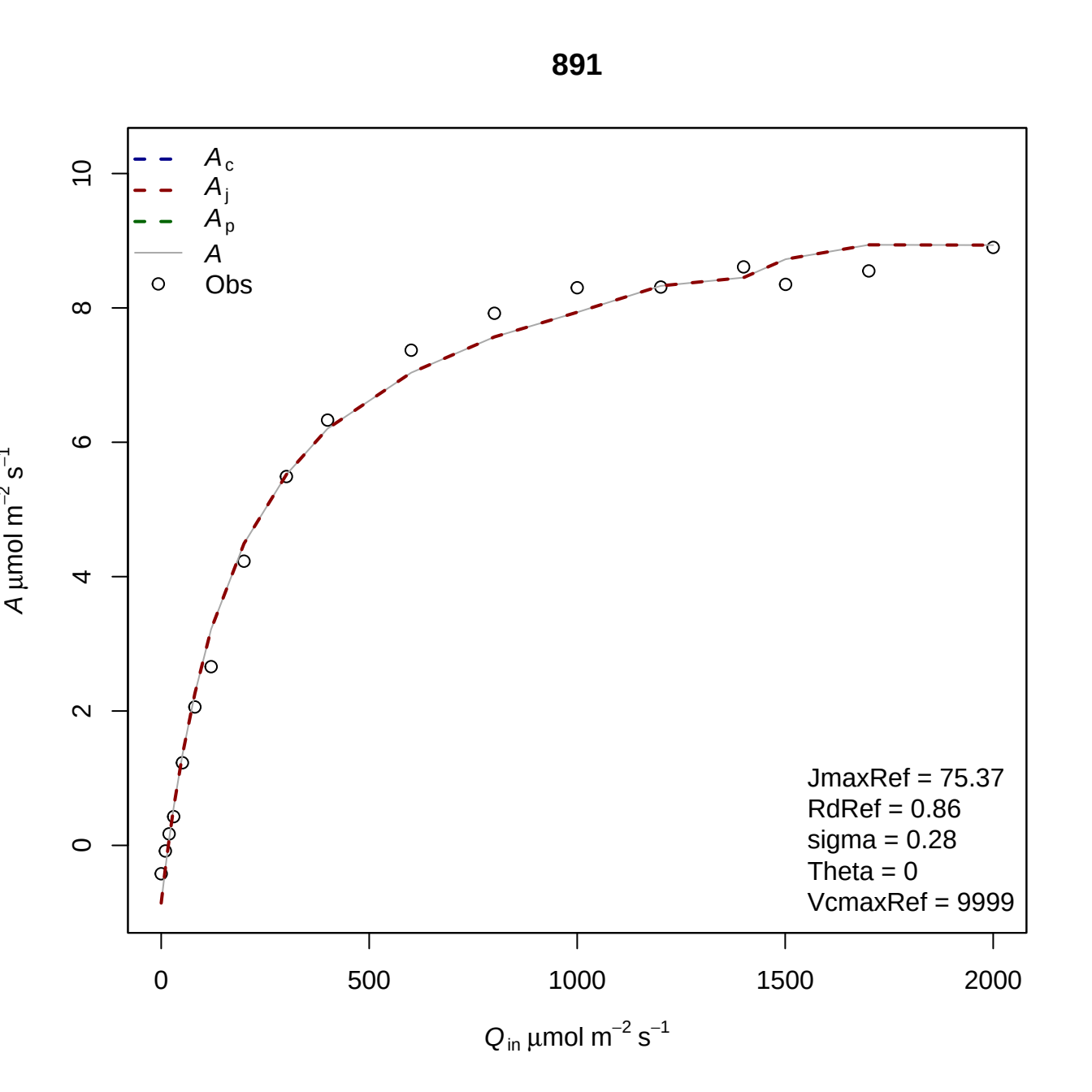

886
A_c
A_j
A_p
A
Obs
JmaxRef = 27.55
RdRef = 1.32
sigma = 0.17
Theta = −1.11
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500
2000

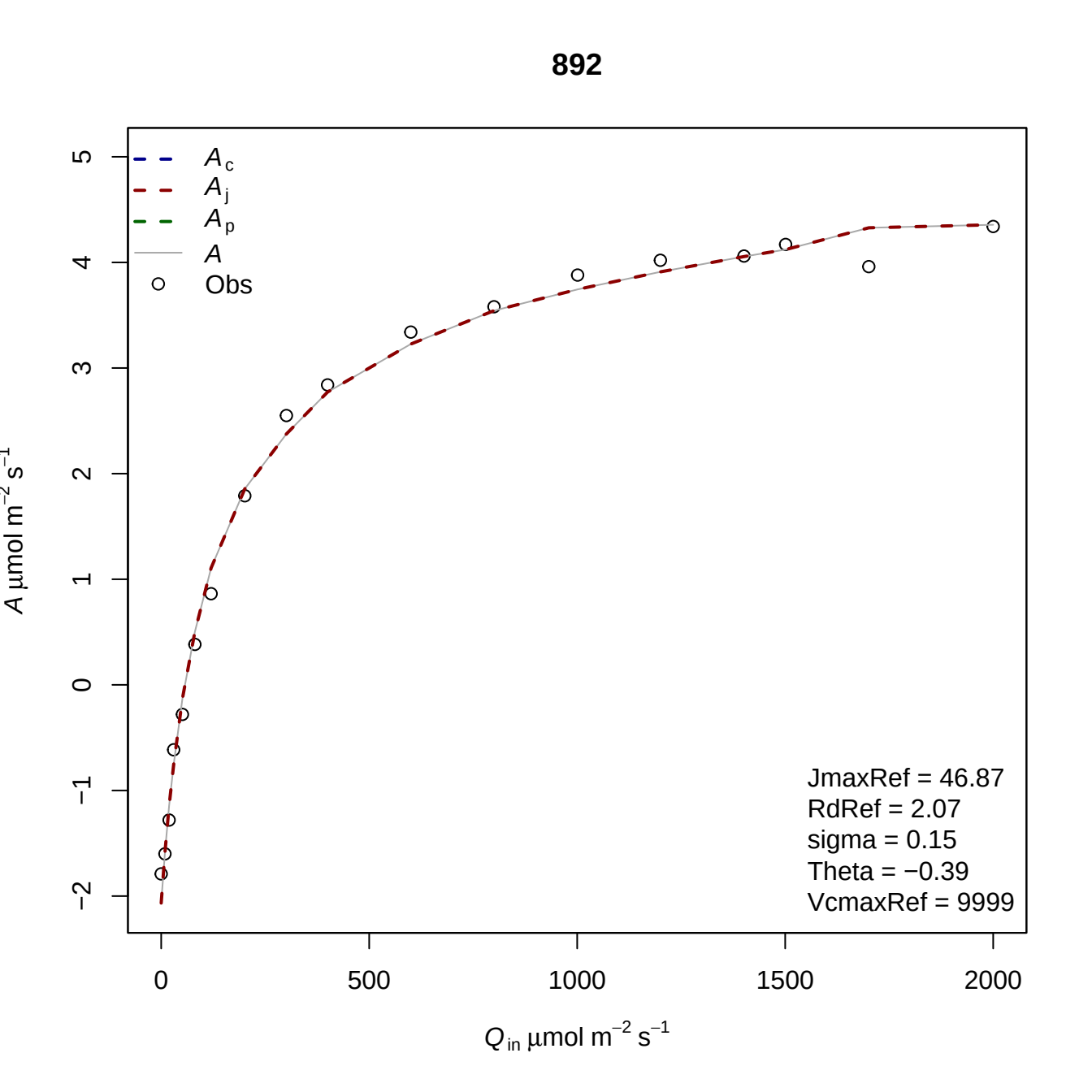

887
A_c
A_j
A_p
A
Obs
JmaxRef = 29.29
RdRef = 0.7
sigma = 0.19
Theta = 0.08
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

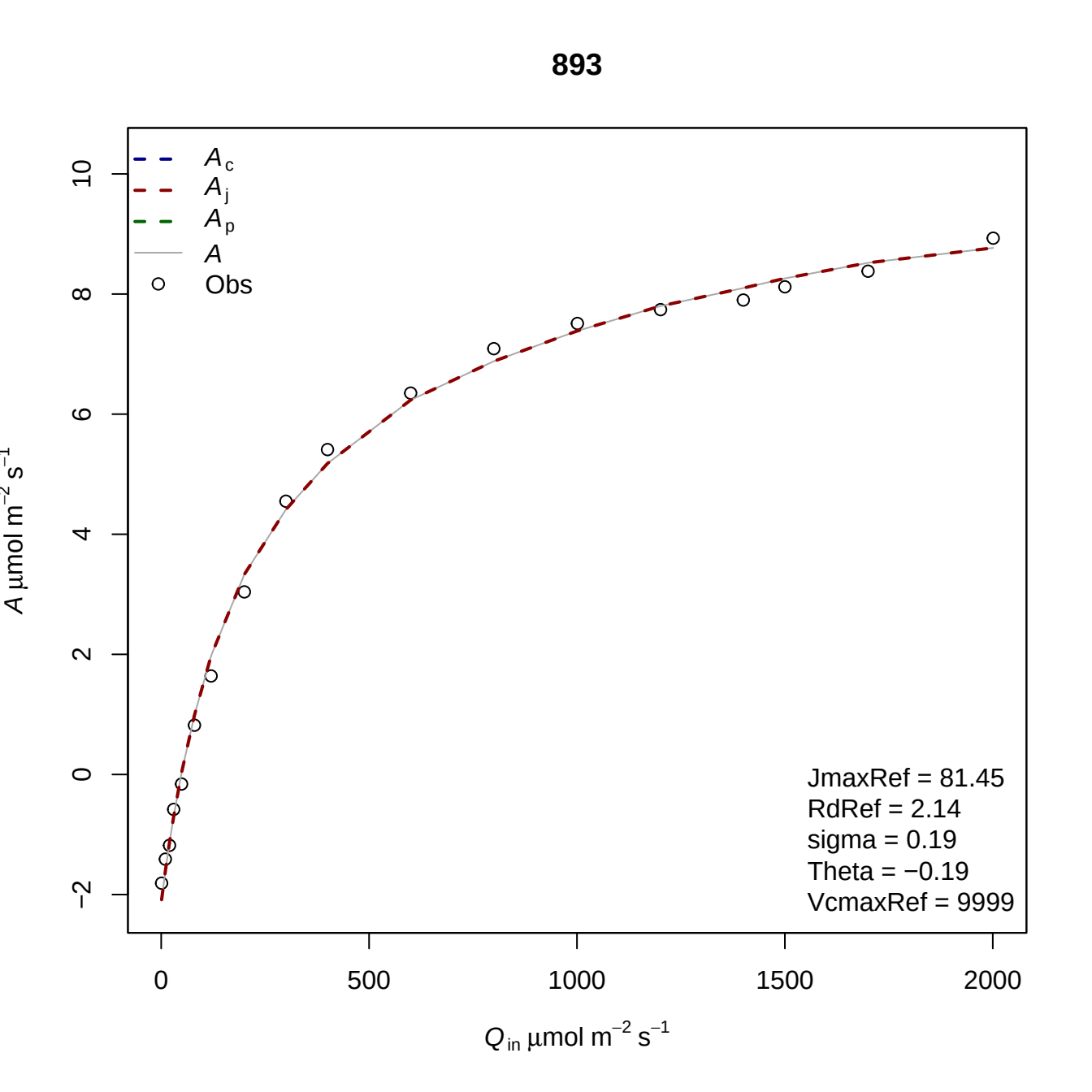

888
A_c
A_j
A_p
A
Obs
JmaxRef = 36.36
RdRef = 0.4
sigma = 0.13
Theta = 0.77
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500
2000

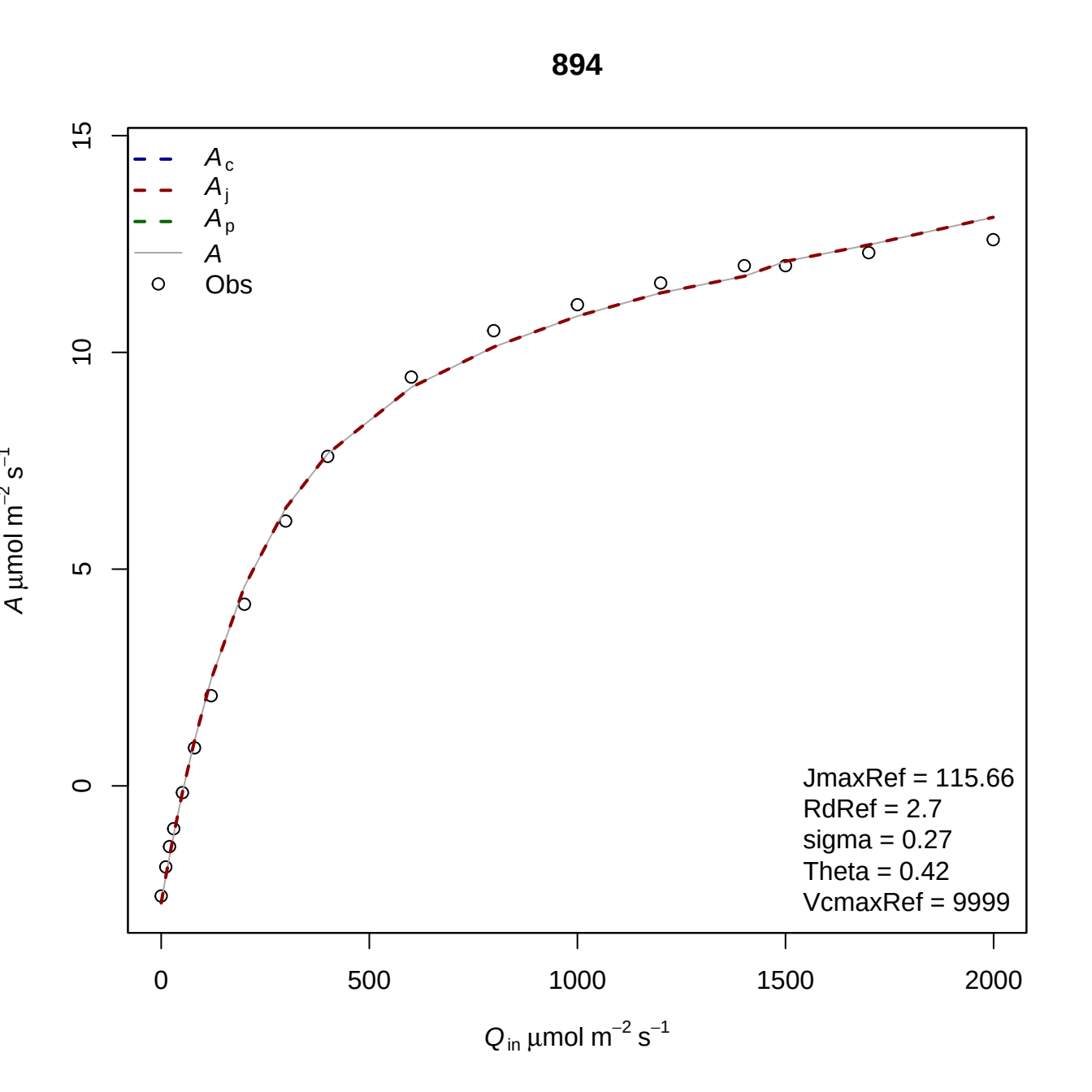

889
A_c
A_j
A_p
A
Obs
JmaxRef = 106.72
RdRef = 1.31
sigma = 0.21
Theta = −0.18
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

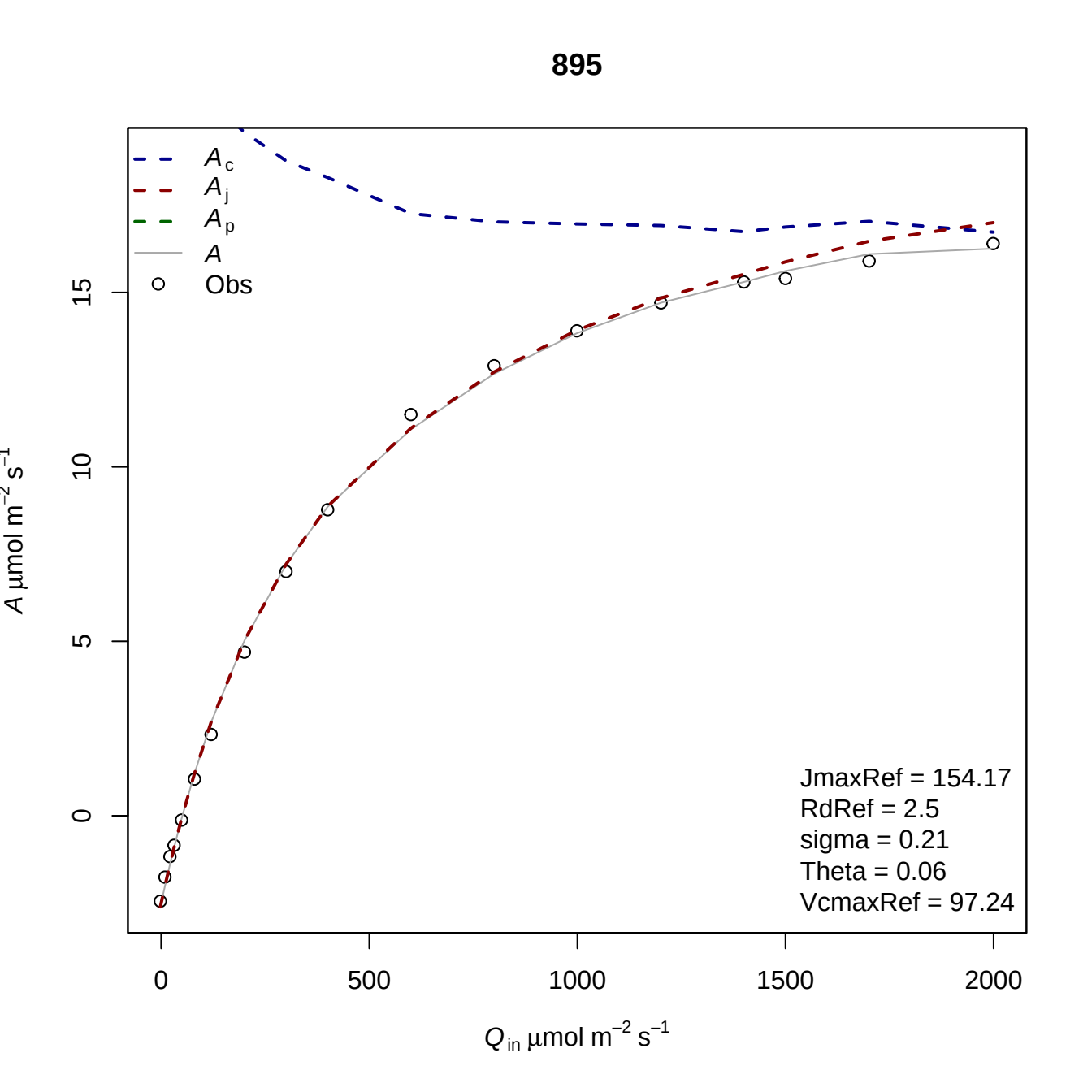

890
A_c
A_j
A_p
A
Obs
JmaxRef = 102.63
RdRef = 1.32
sigma = 0.19
Theta = −1.15
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

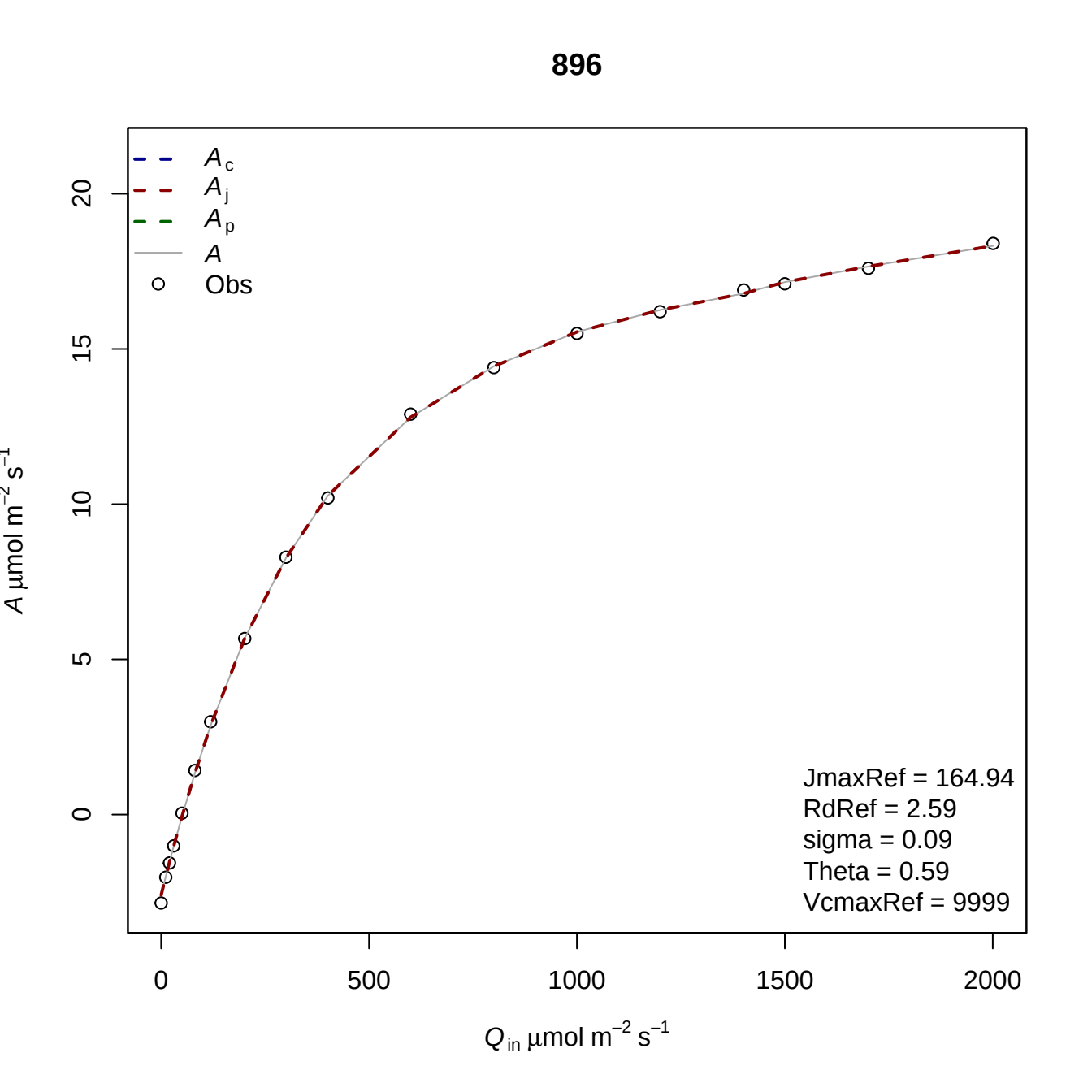

891
A_c
A_j
A_p
A
Obs
JmaxRef = 75.37
RdRef = 0.86
sigma = 0.28
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

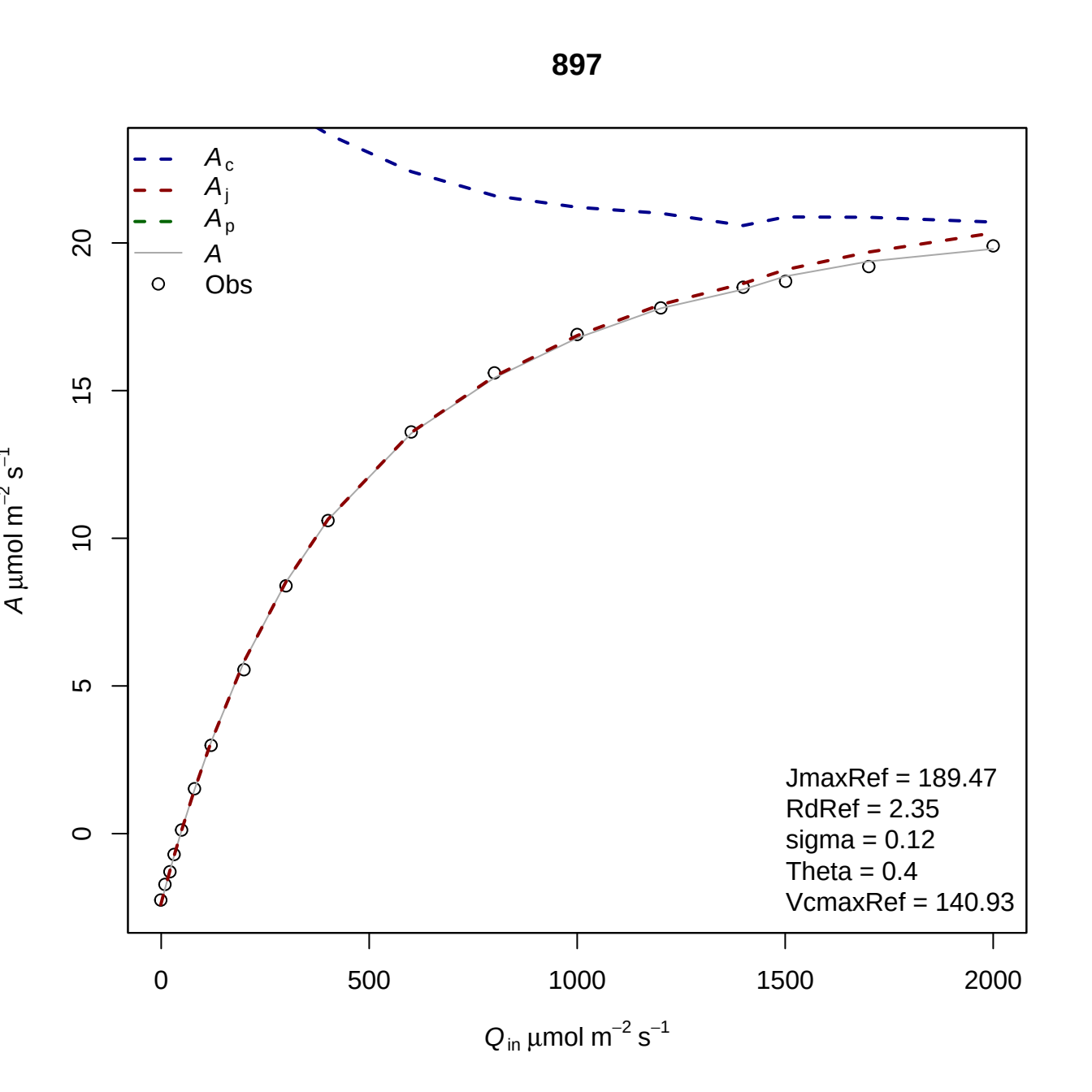

892
A_c
A_j
A_p
A
Obs
JmaxRef = 46.87
RdRef = 2.07
sigma = 0.15
Theta = −0.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

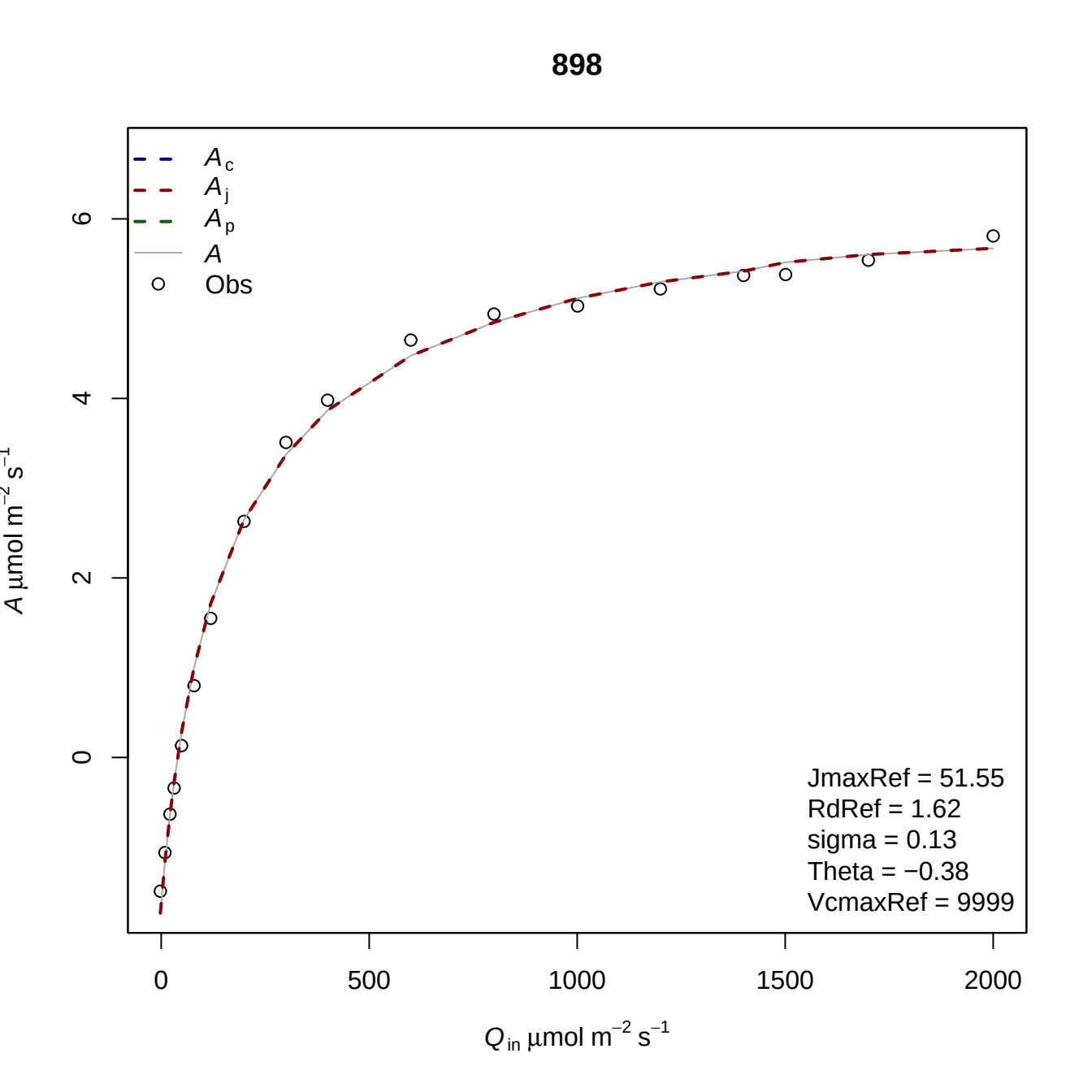

893
A_c
A_j
A_p
A
Obs
JmaxRef = 81.45
RdRef = 2.14
sigma = 0.19
Theta = −0.19
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

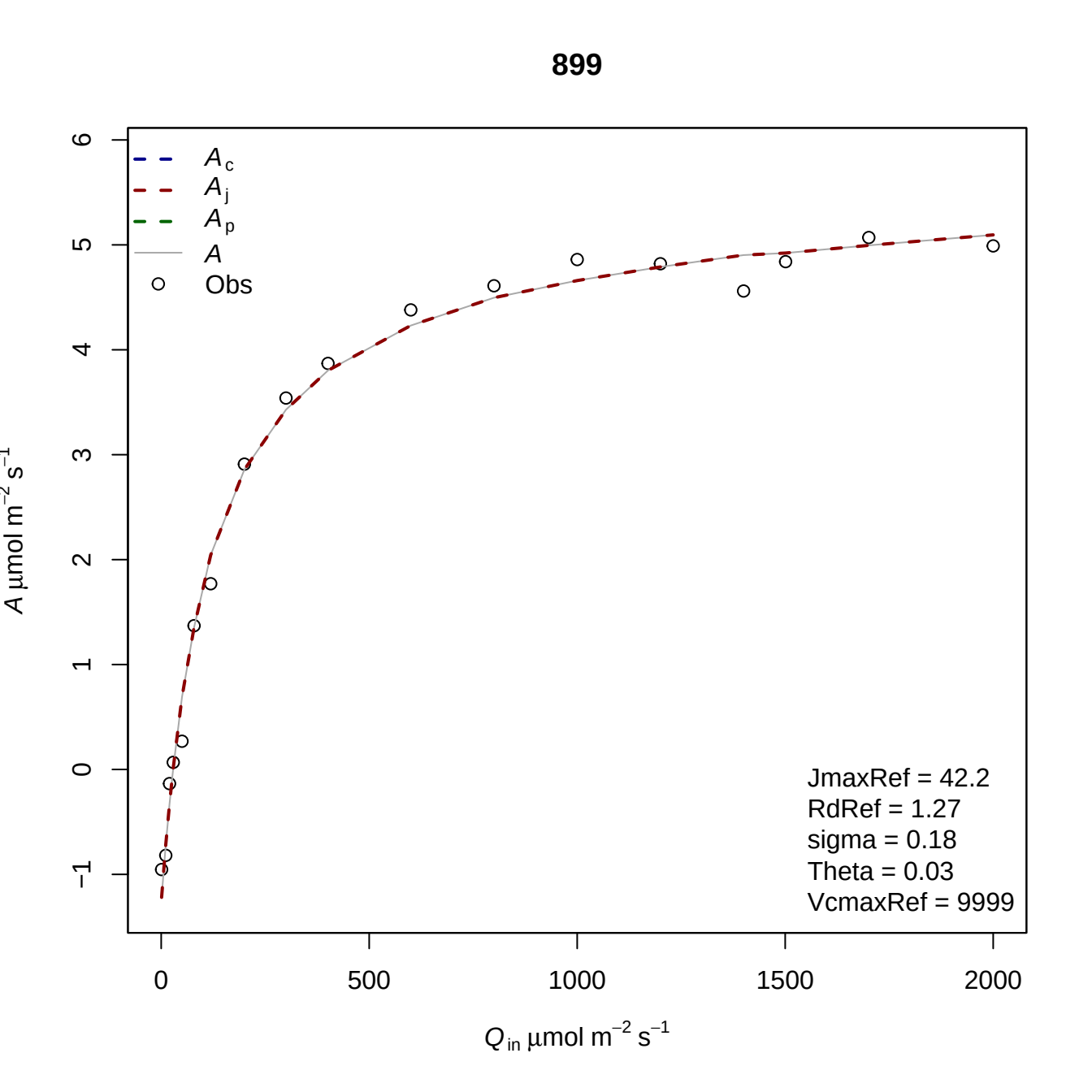

894
A_c
A_j
A_p
A
Obs
JmaxRef = 115.66
RdRef = 2.7
sigma = 0.27
Theta = 0.42
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

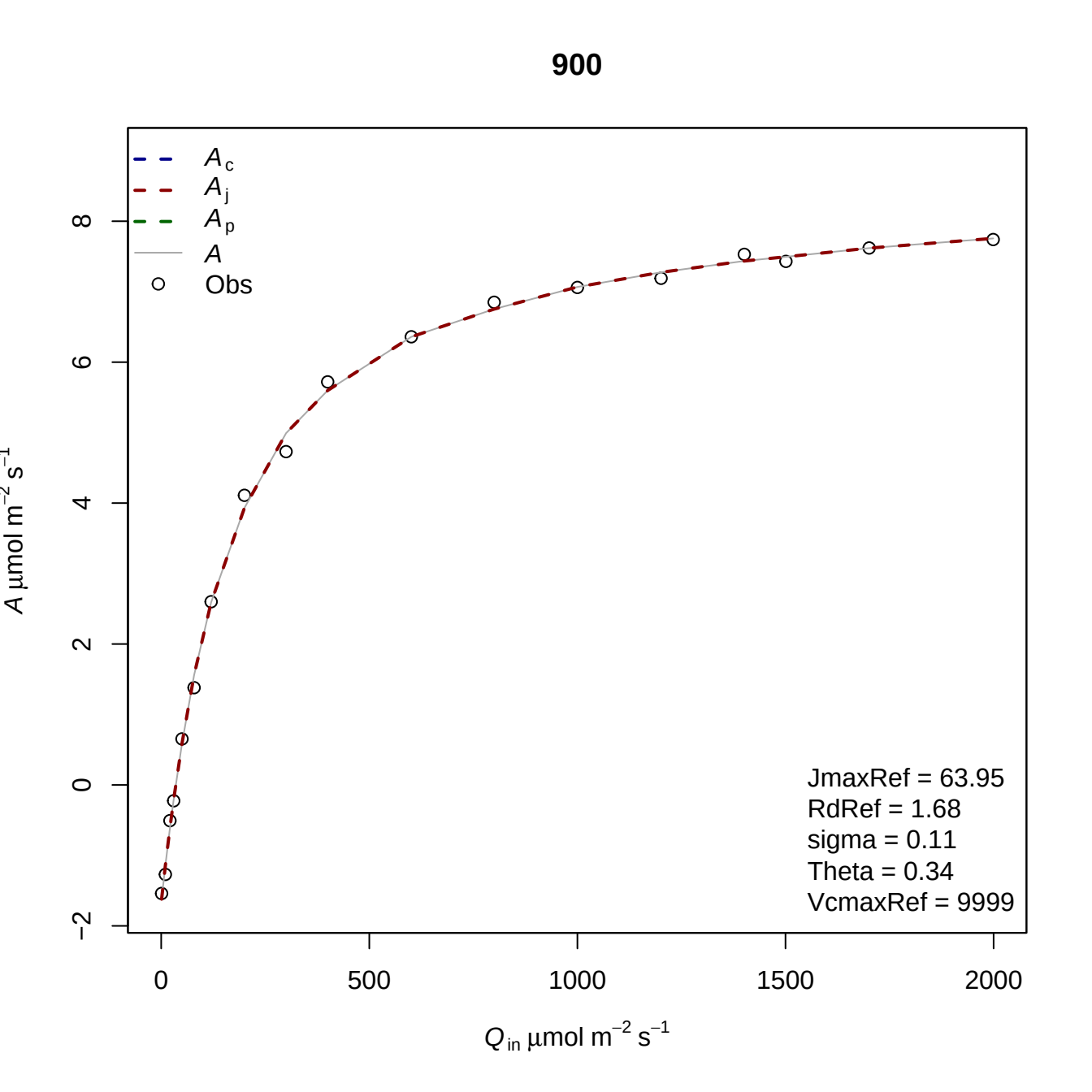

895
A_c
A_j
A_p
A
Obs
JmaxRef = 154.17
RdRef = 2.5
sigma = 0.21
Theta = 0.06
VcmaxRef = 97.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

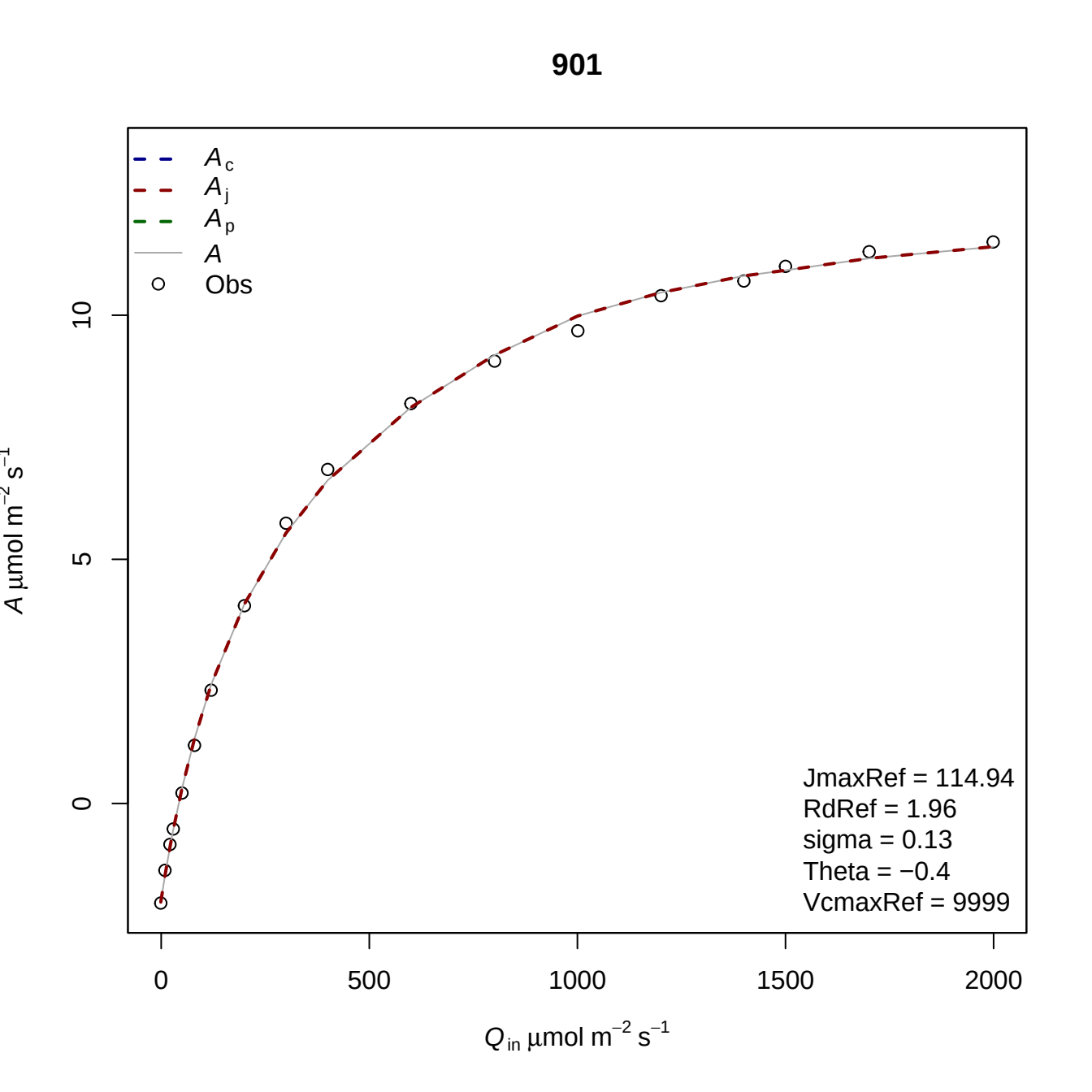

896
A_c
A_j
A_p
A
Obs
JmaxRef = 164.94
RdRef = 2.59
sigma = 0.09
Theta = 0.59
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

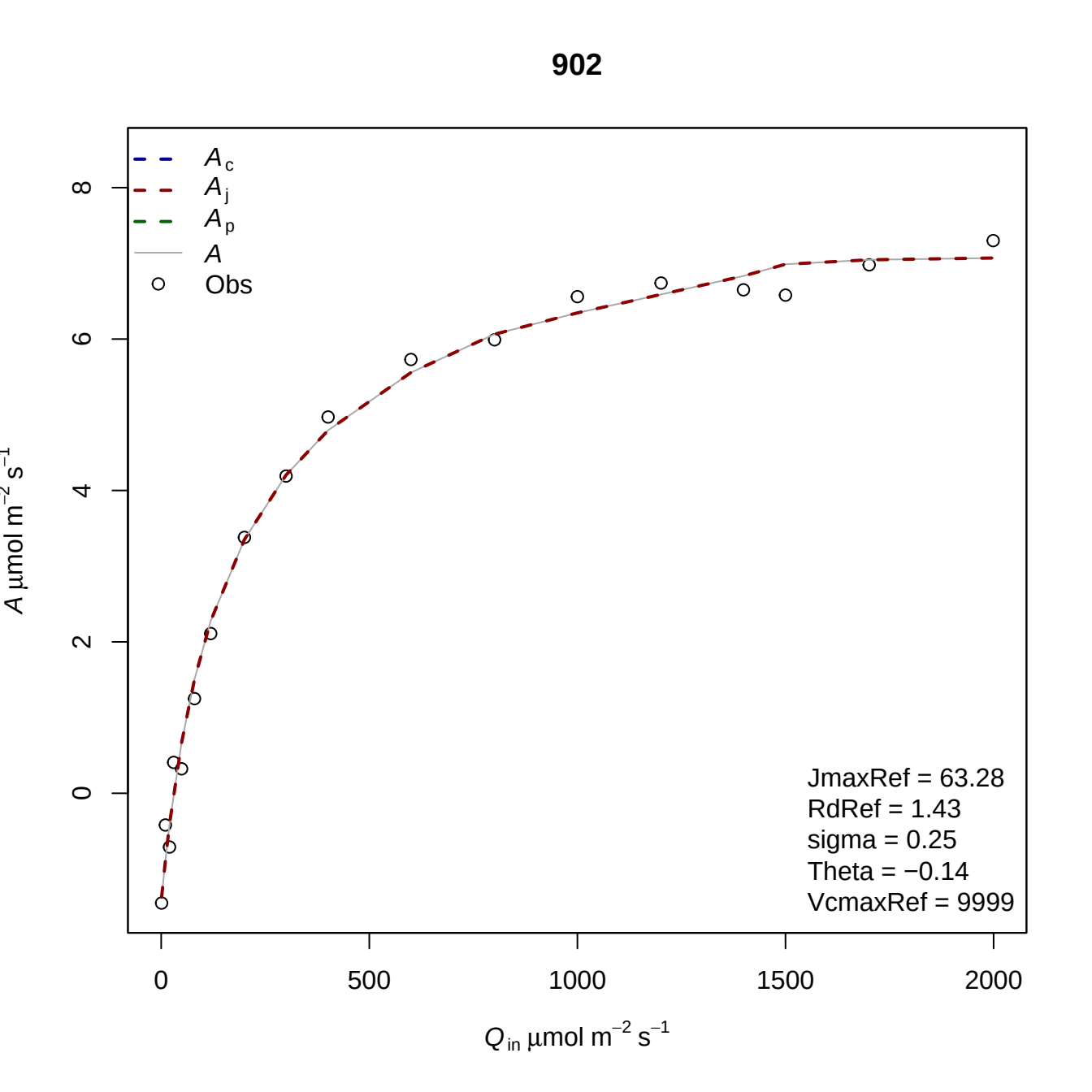

897
A_c
A_j
A_p
A
Obs
JmaxRef = 189.47
RdRef = 2.35
sigma = 0.12
Theta = 0.4
VcmaxRef = 140.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

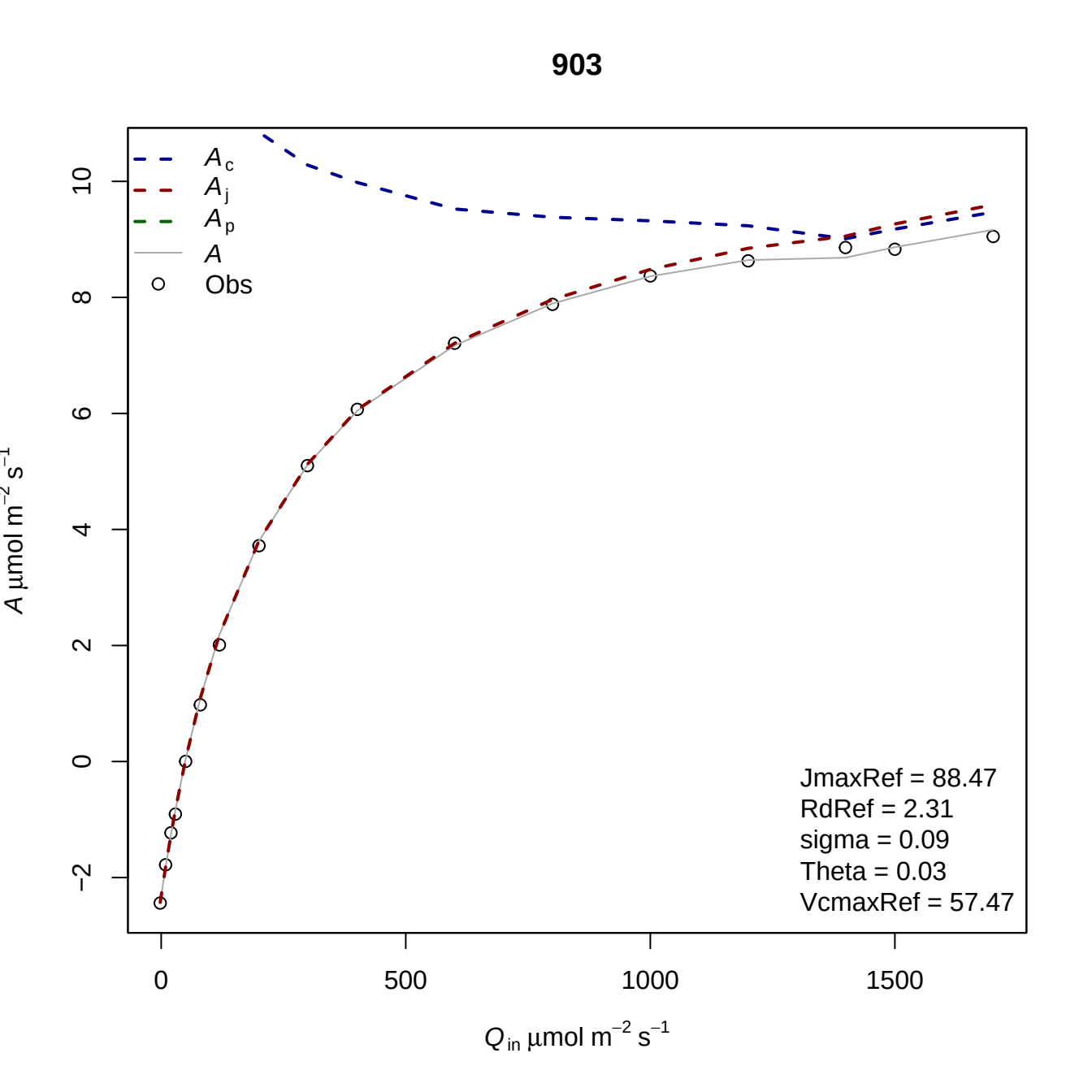

898
A_c
A_j
A_p
A
Obs
JmaxRef = 51.55
RdRef = 1.62
sigma = 0.13
Theta = −0.38
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

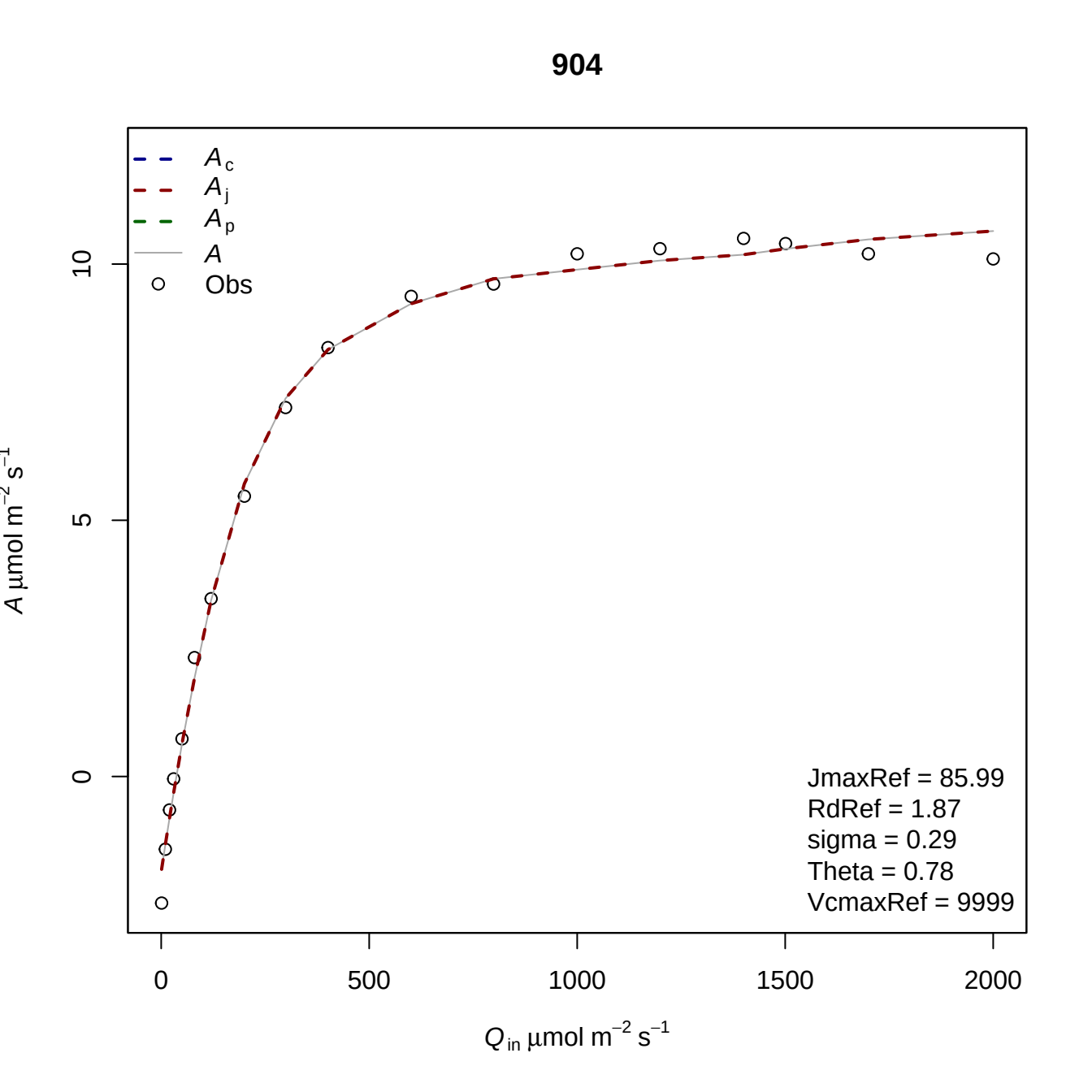

899
A_c
A_j
A_p
A
Obs
JmaxRef = 42.2
RdRef = 1.27
sigma = 0.18
Theta = 0.03
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

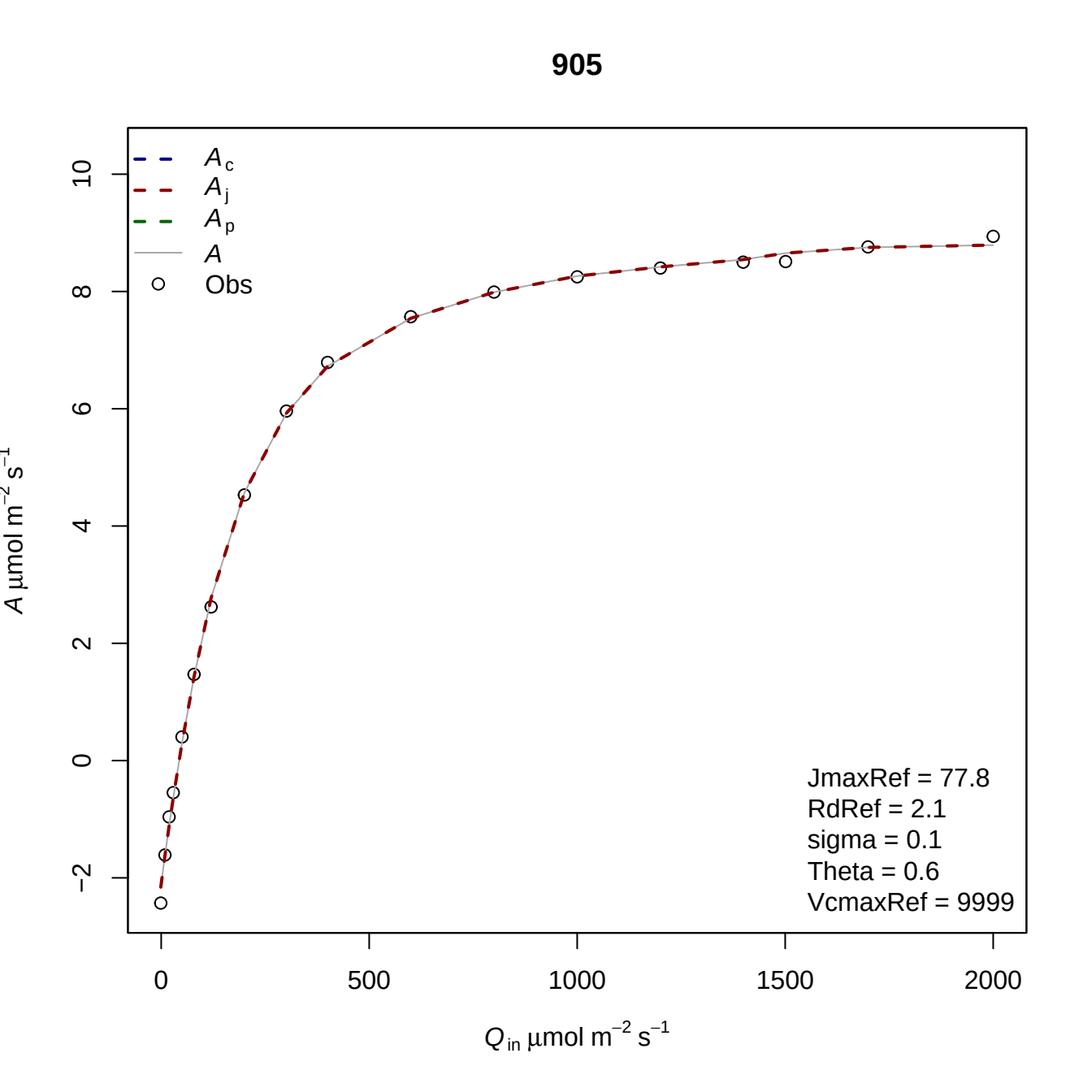

900
A_c
A_j
A_p
A
Obs
JmaxRef = 63.95
RdRef = 1.68
sigma = 0.11
Theta = 0.34
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

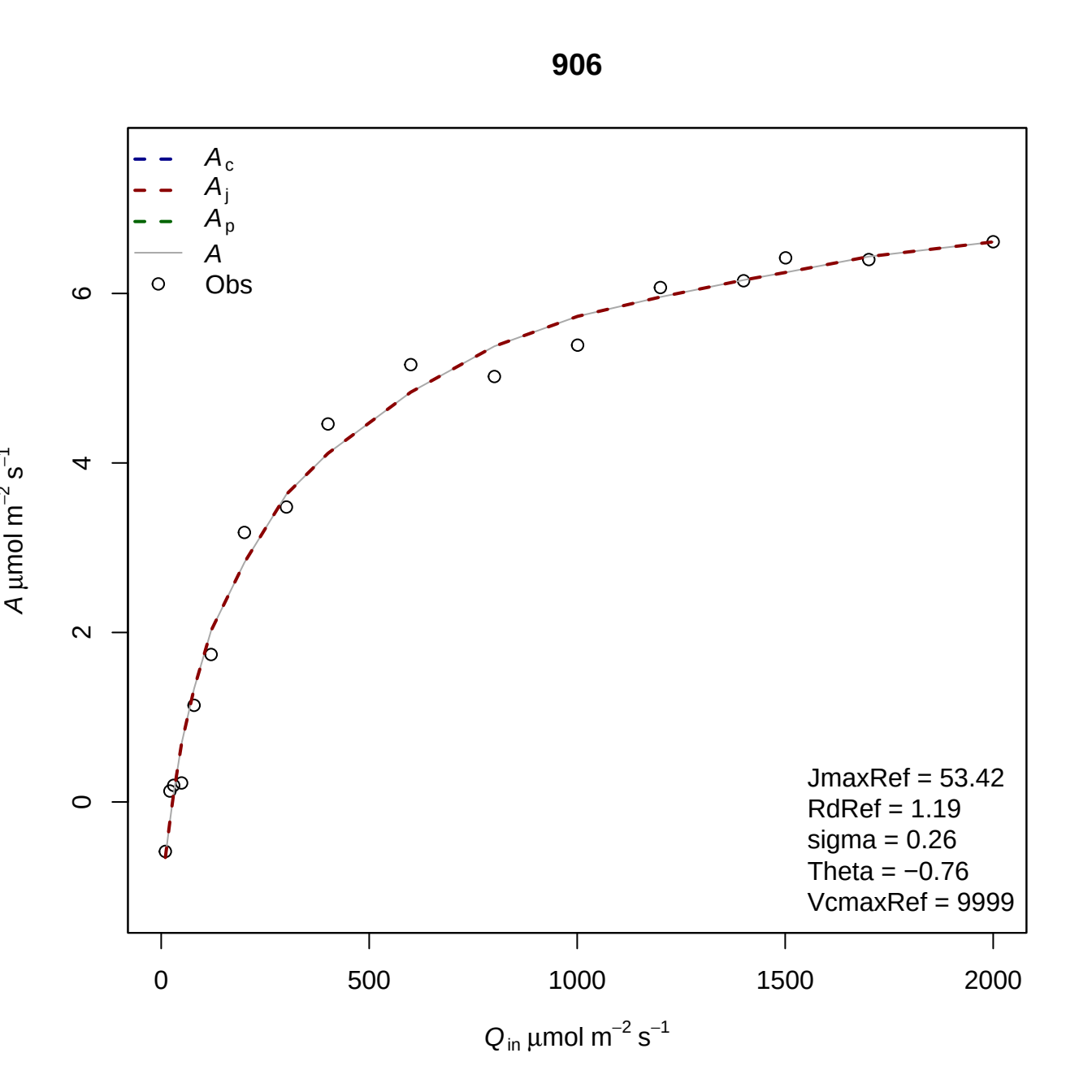

901
A_c
A_j
A_p
A
Obs
JmaxRef = 114.94
RdRef = 1.96
sigma = 0.13
Theta = −0.4
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

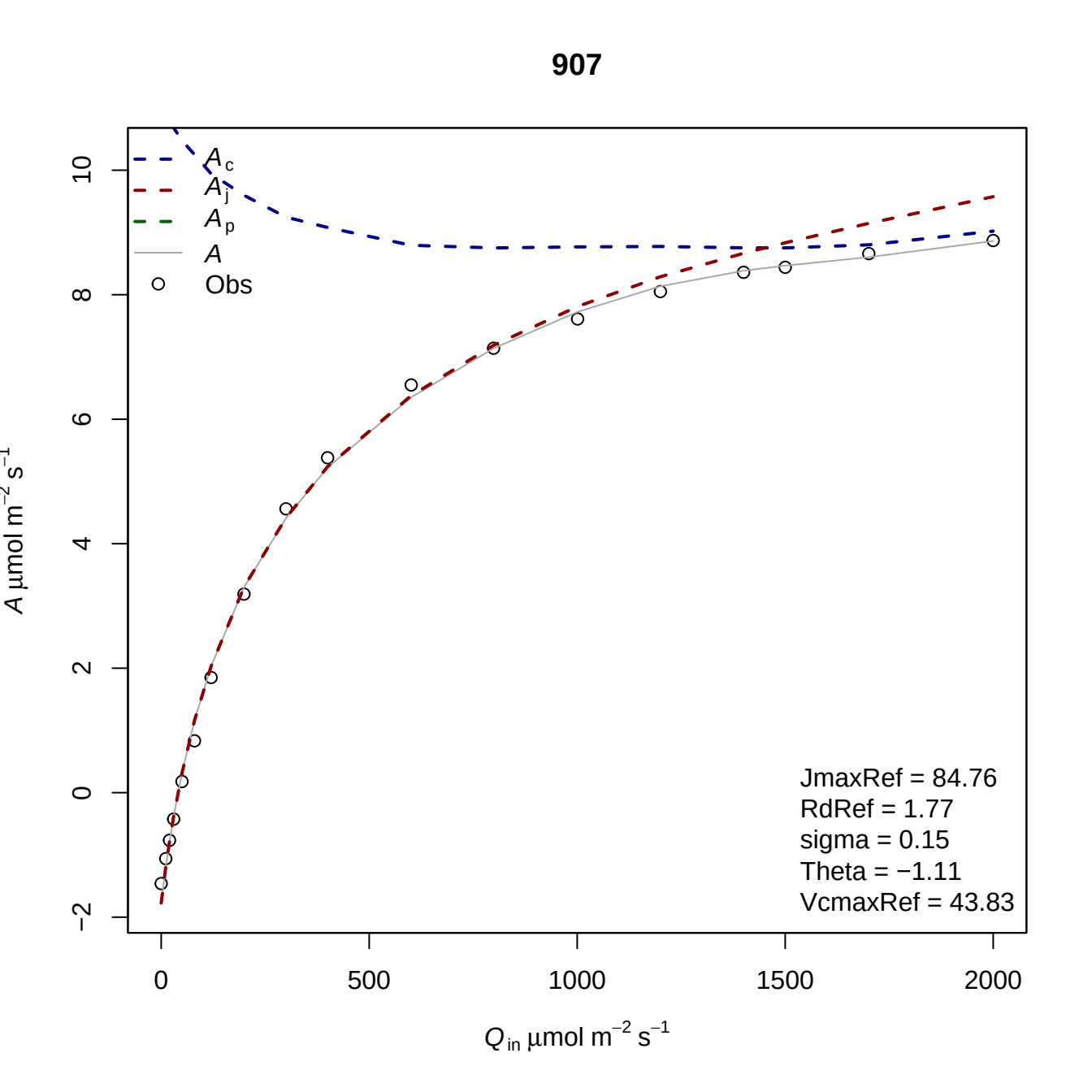

902
A_c
A_j
A_p
A
Obs
JmaxRef = 63.28
RdRef = 1.43
sigma = 0.25
Theta = −0.14
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

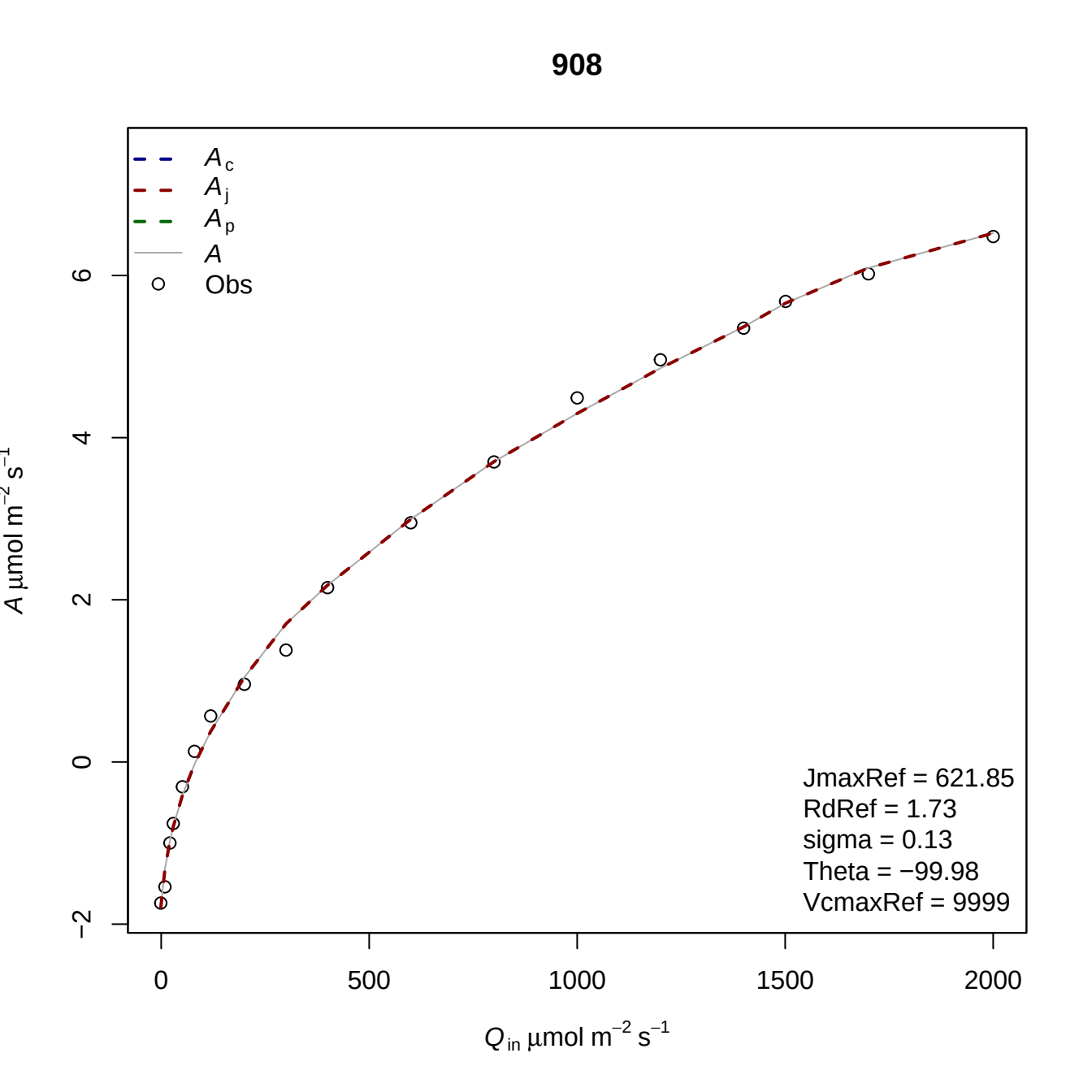

903
A_c
A_j
A_p
A
Obs
JmaxRef = 88.47
RdRef = 2.31
sigma = 0.09
Theta = 0.03
VcmaxRef = 57.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

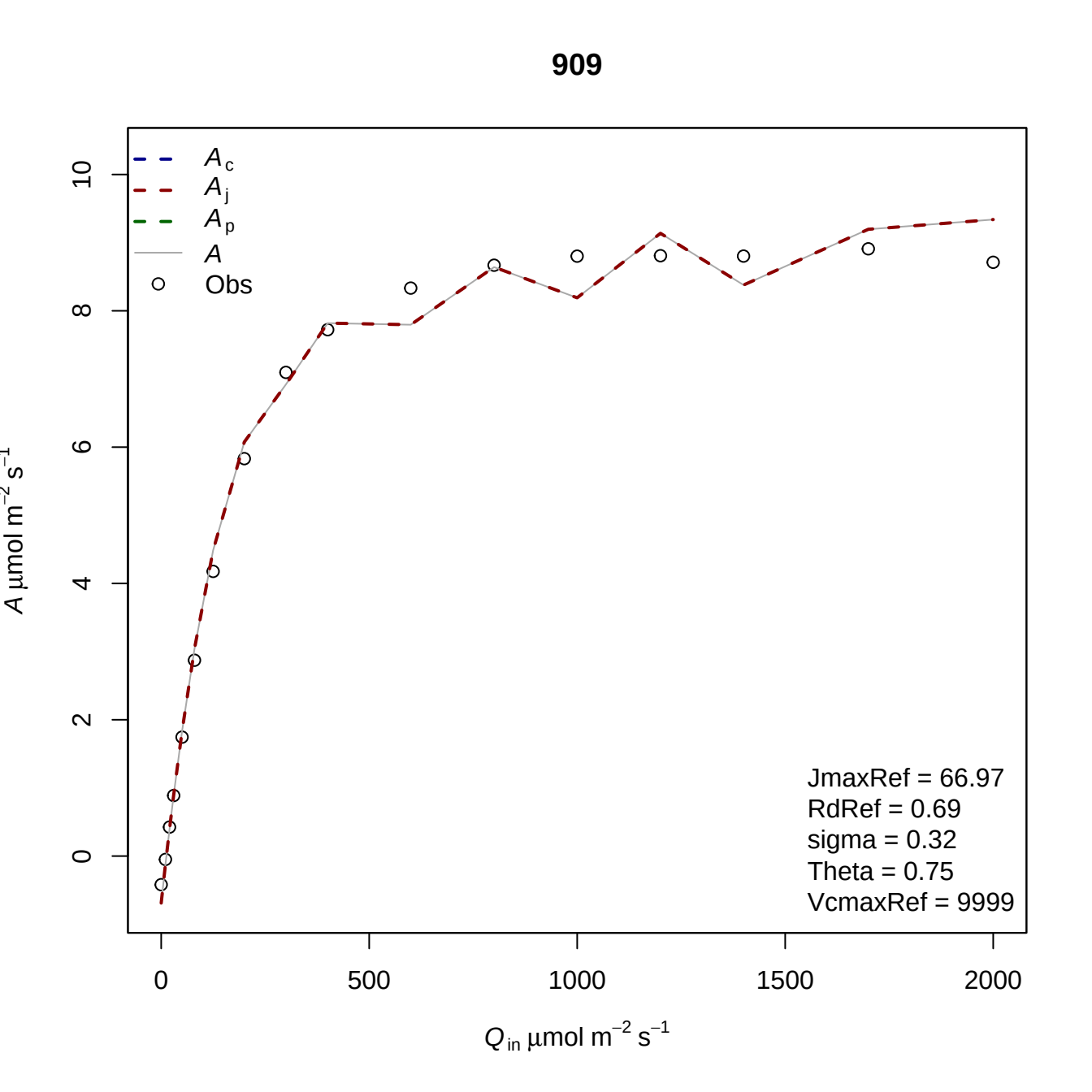

904
A_c
A_j
A_p
A
Obs
JmaxRef = 85.99
RdRef = 1.87
sigma = 0.29
Theta = 0.78
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

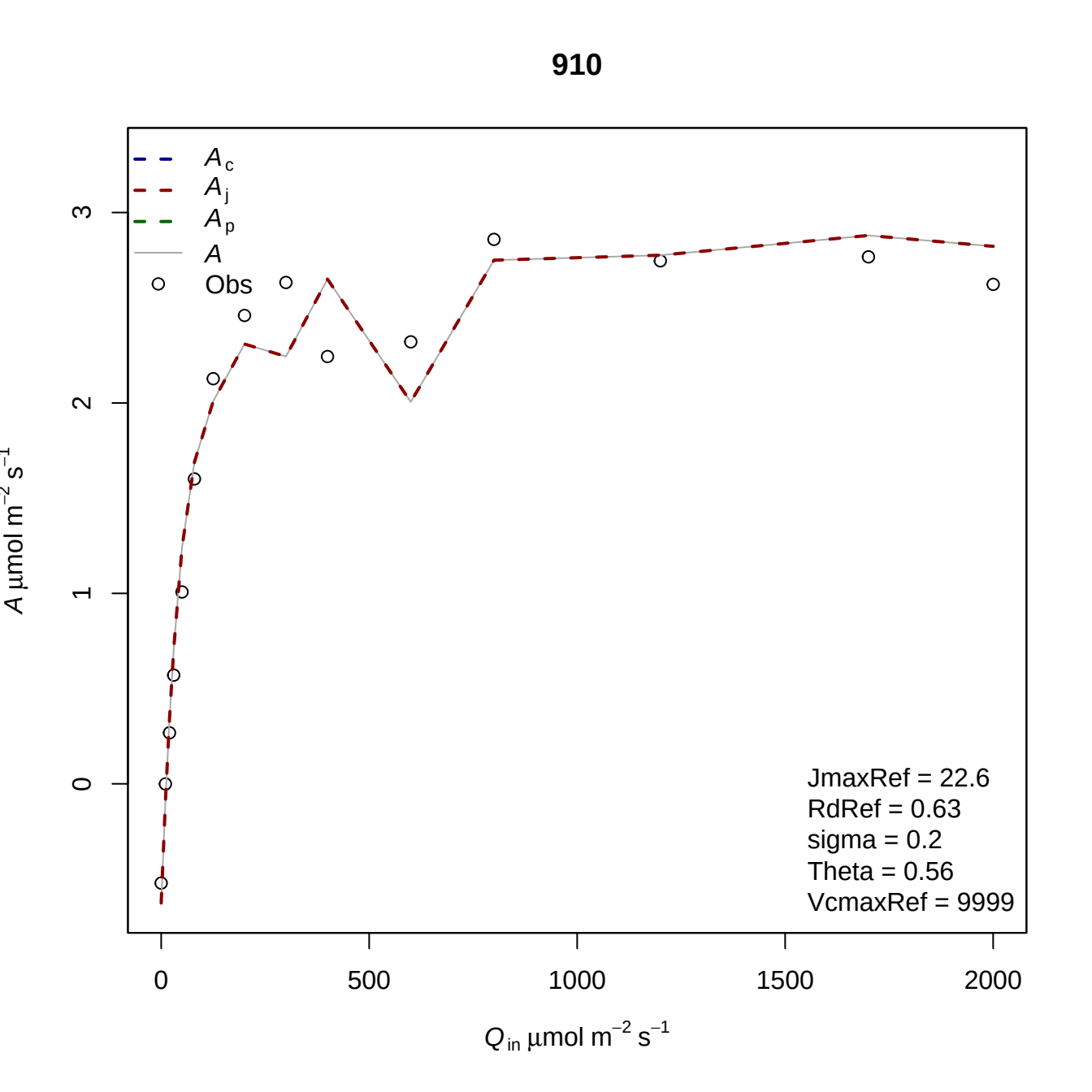

905
A_c
A_j
A_p
A
Obs
JmaxRef = 77.8
RdRef = 2.1
sigma = 0.1
Theta = 0.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

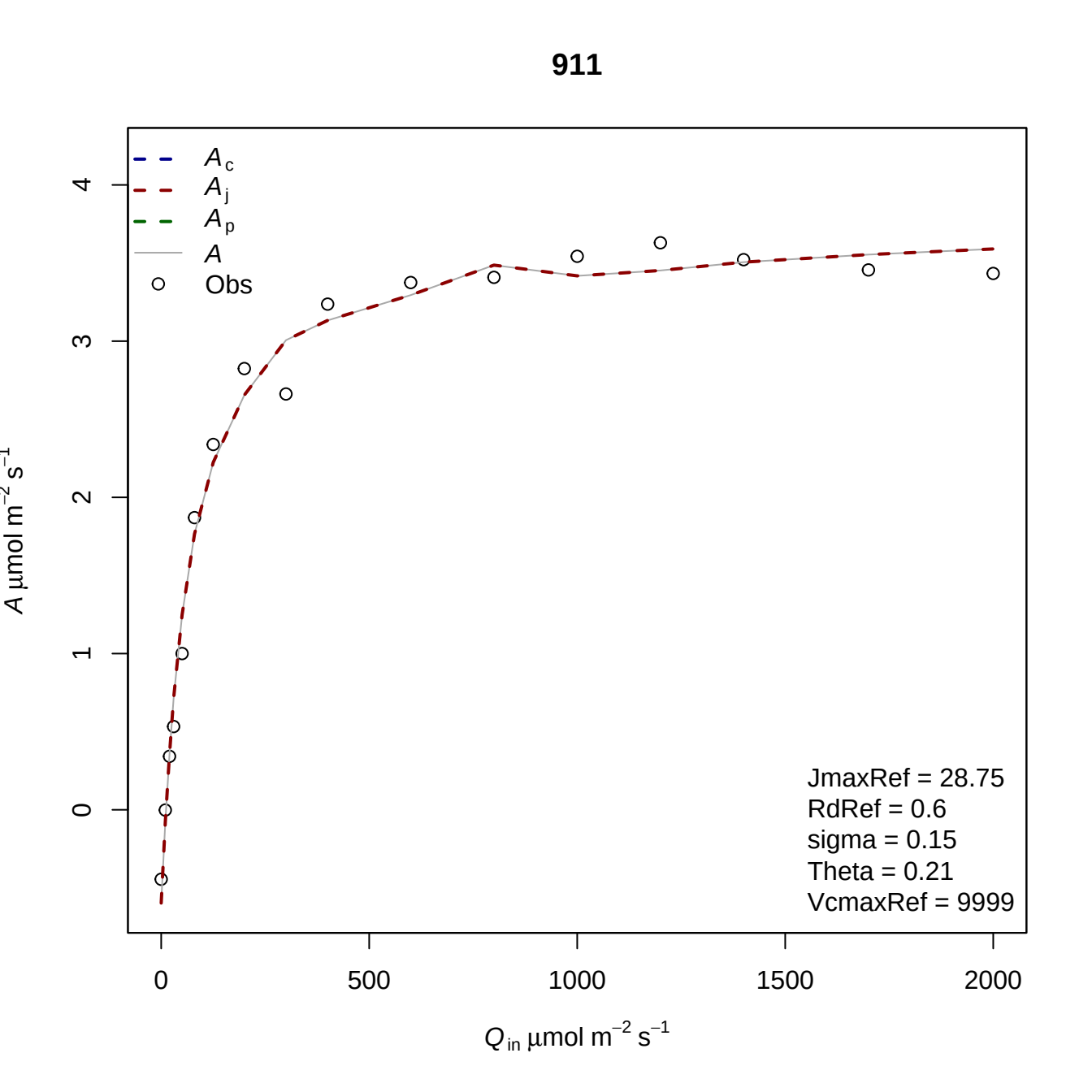

906
A_c
A_j
A_p
A
Obs
JmaxRef = 53.42
RdRef = 1.19
sigma = 0.26
Theta = −0.76
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

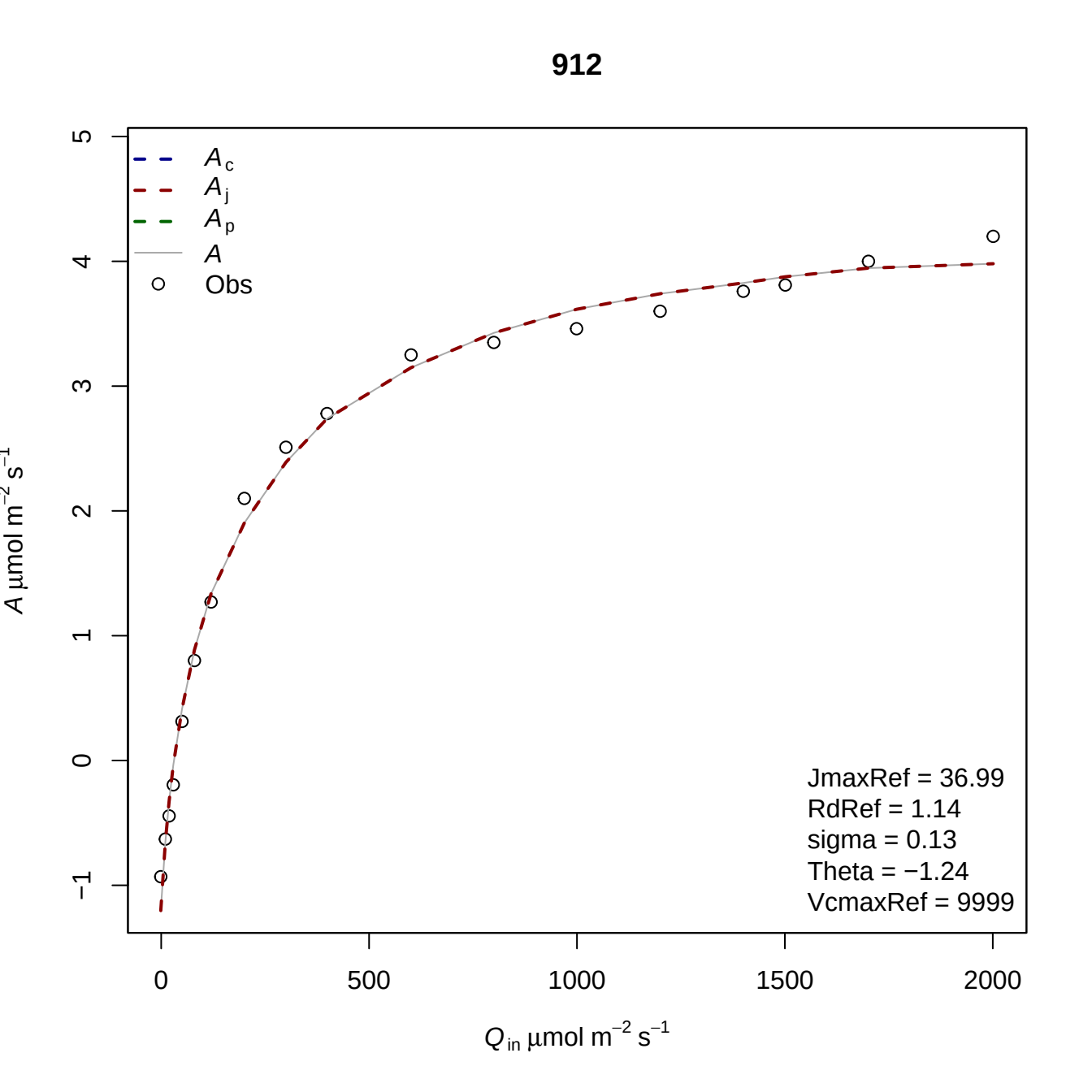

907
A_c
A_j
A_p
A
Obs
JmaxRef = 84.76
RdRef = 1.77
sigma = 0.15
Theta = −1.11
VcmaxRef = 43.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

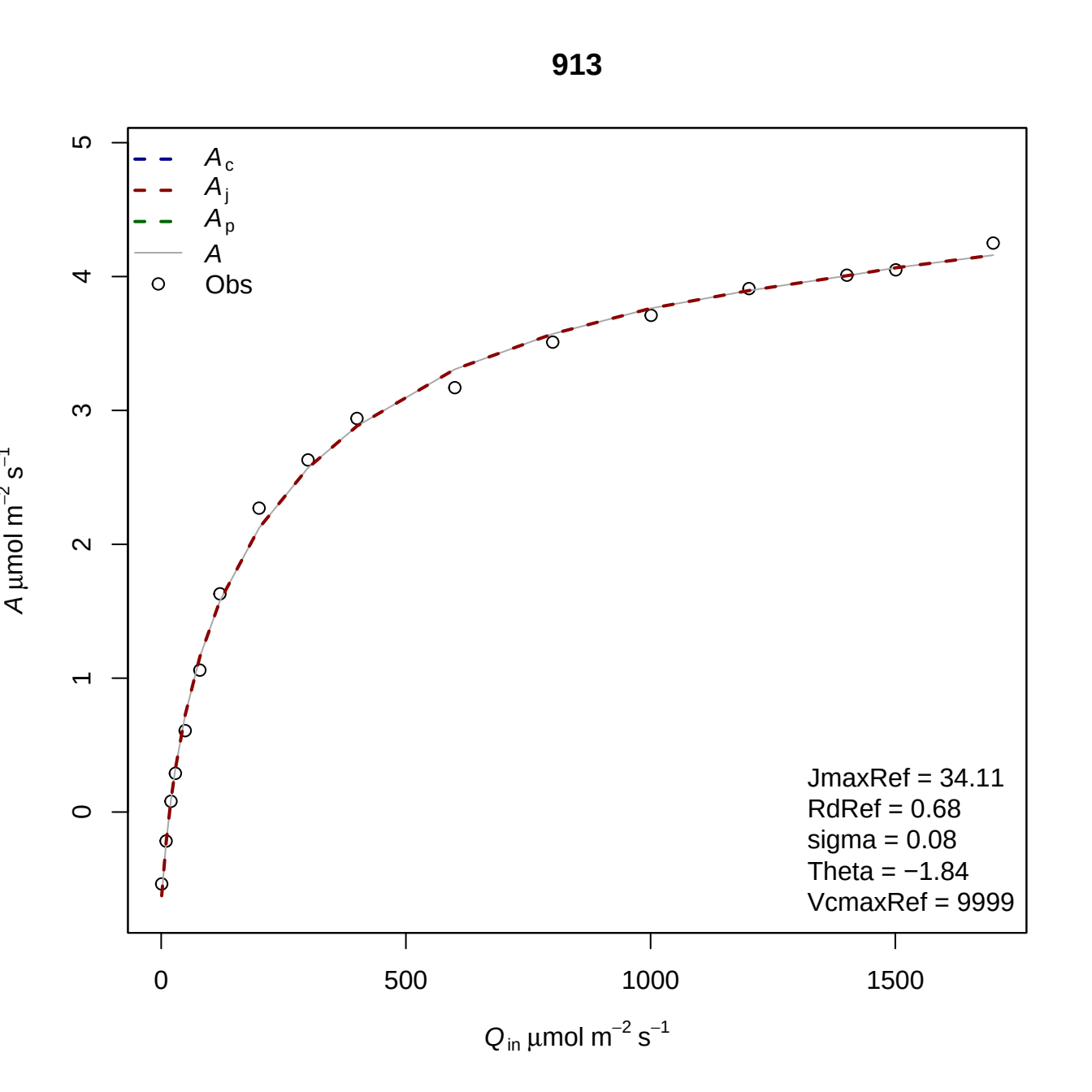

908
A_c
A_j
A_p
A
Obs
JmaxRef = 621.85
RdRef = 1.73
sigma = 0.13
Theta = −99.98
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

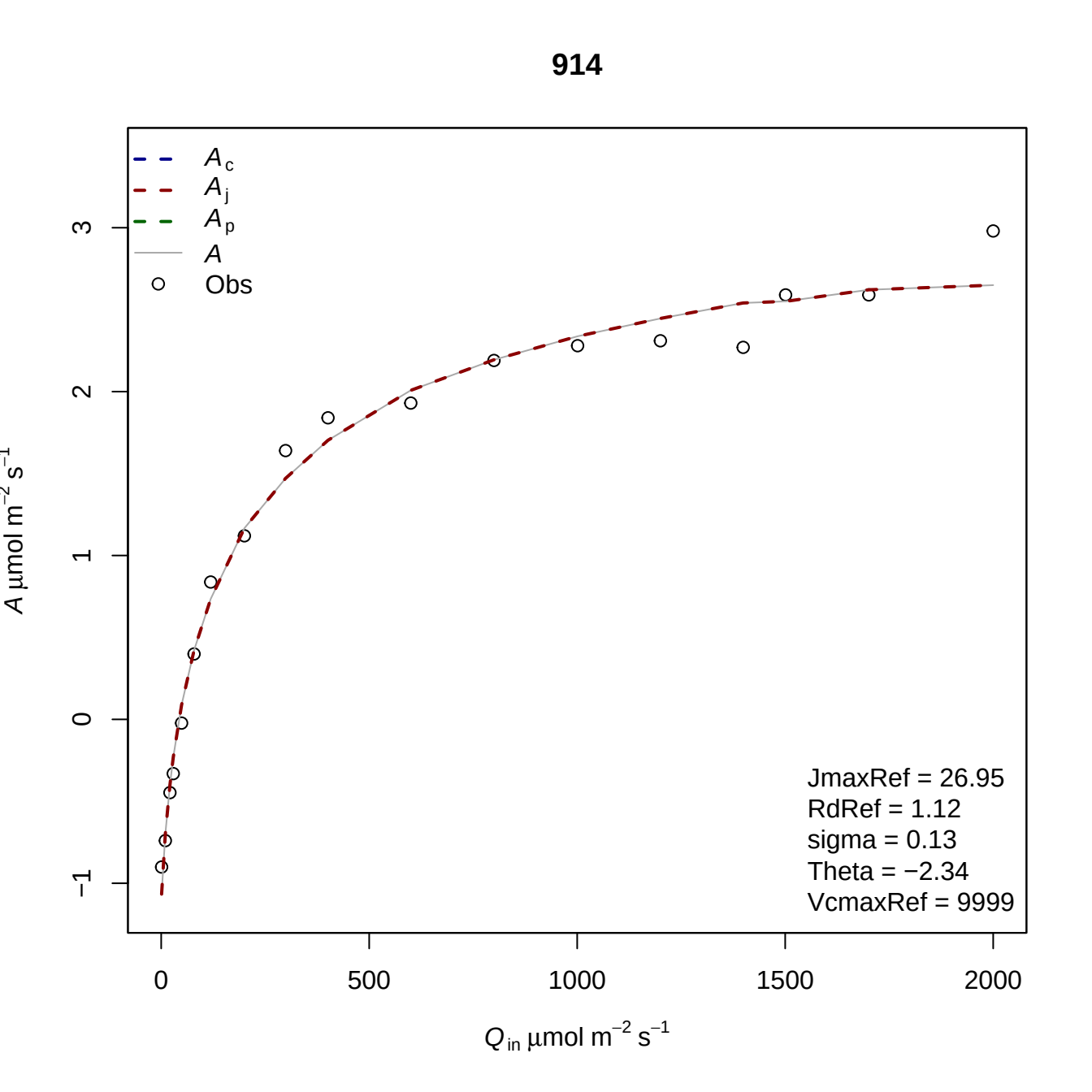

909
A_c
A_j
A_p
A
Obs
JmaxRef = 66.97
RdRef = 0.69
sigma = 0.32
Theta = 0.75
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

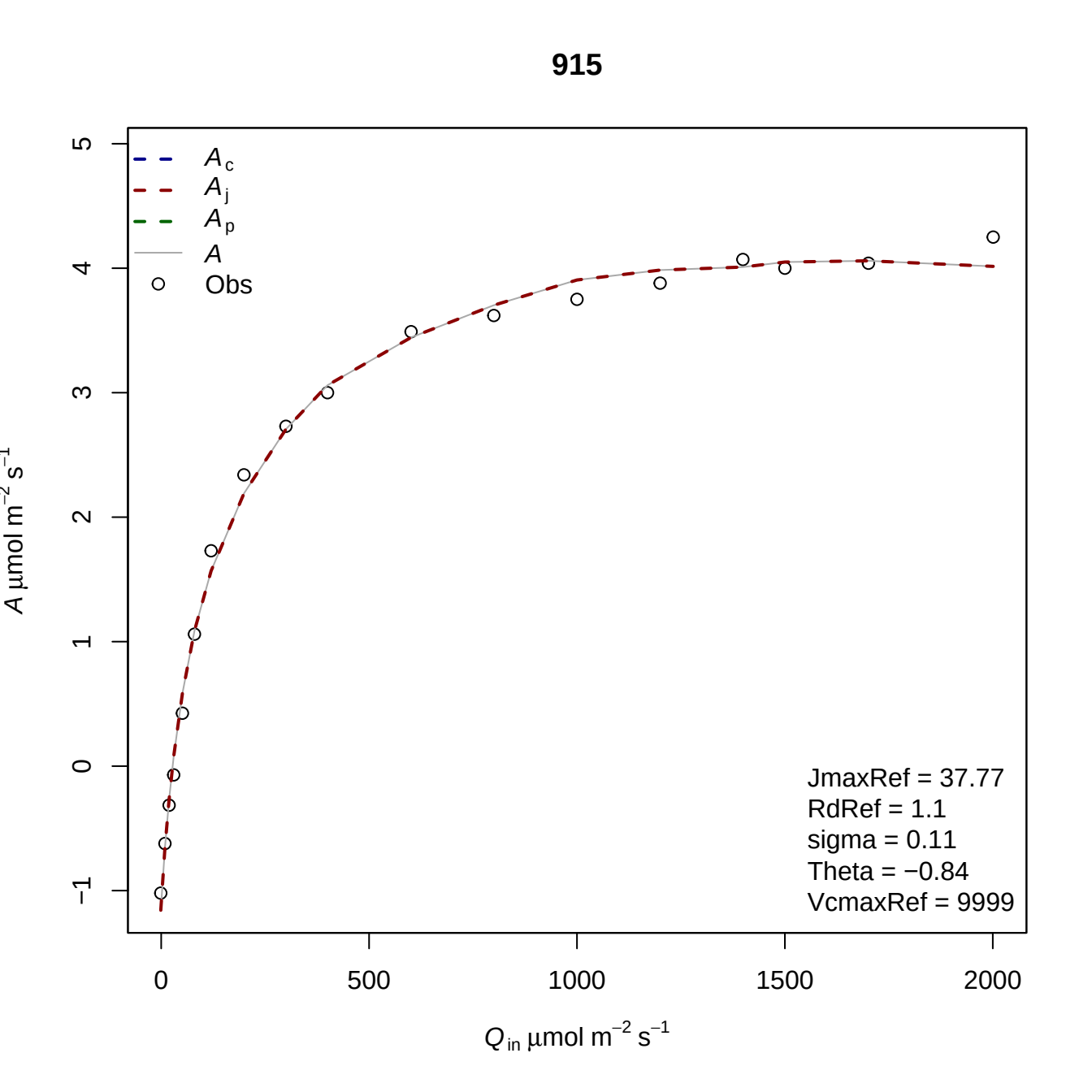

910
A_c
A_j
A_p
A
Obs
JmaxRef = 22.6
RdRef = 0.63
sigma = 0.2
Theta = 0.56
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

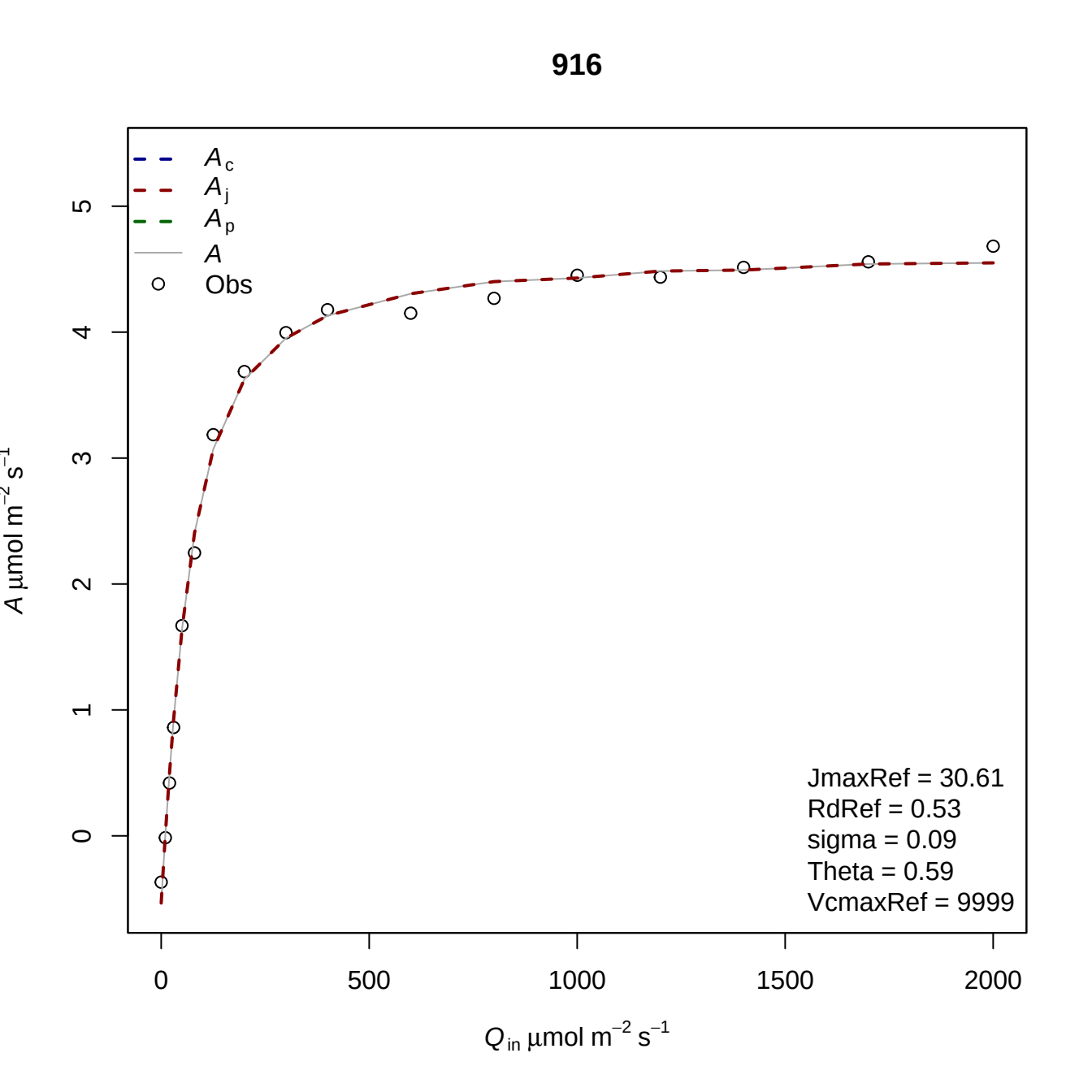

911
A_c
A_j
A_p
A
Obs
JmaxRef = 28.75
RdRef = 0.6
sigma = 0.15
Theta = 0.21
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

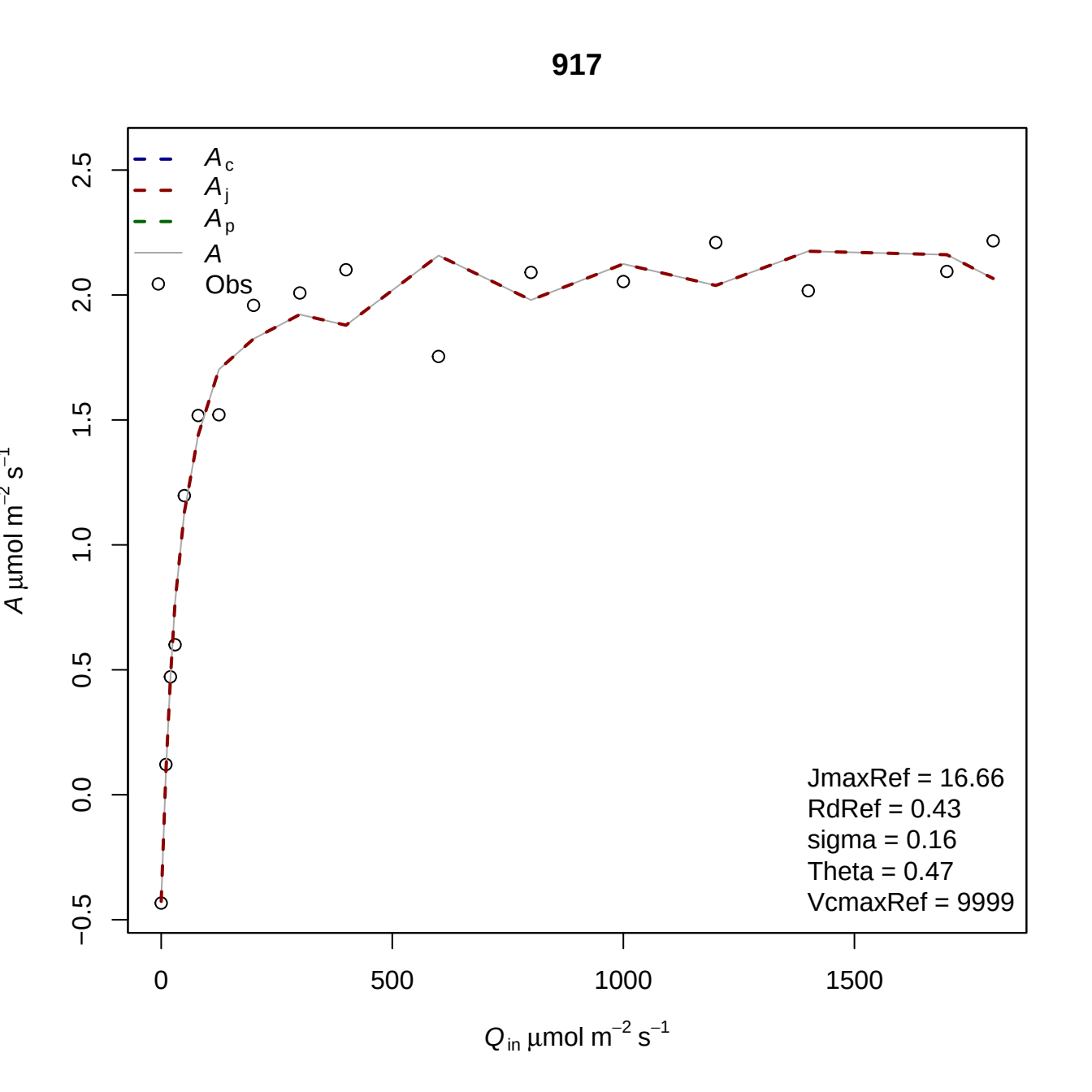

912
A_c
A_j
A_p
A
Obs
JmaxRef = 36.99
RdRef = 1.14
sigma = 0.13
Theta = −1.24
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

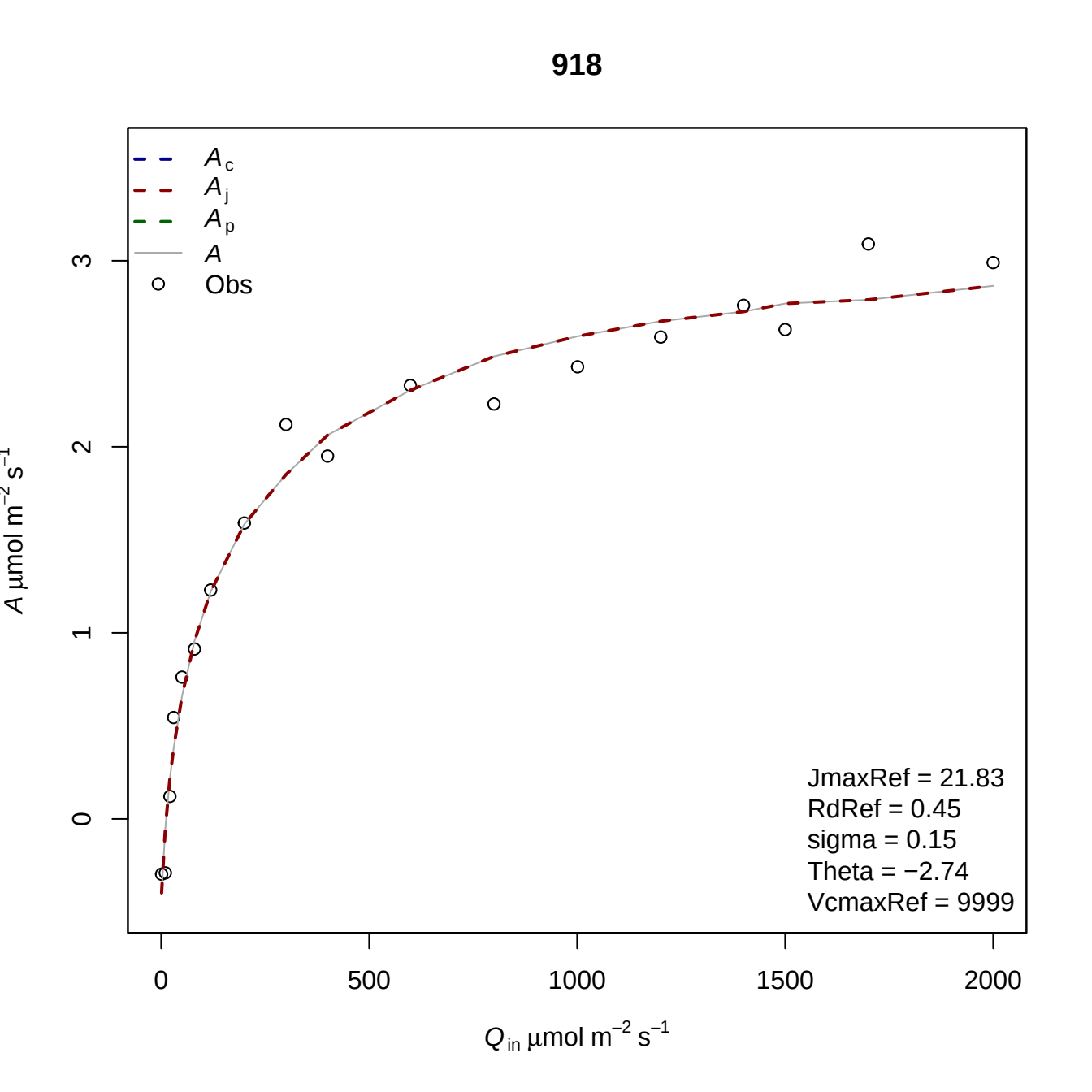

913
A_c
A_j
A_p
A
Obs
JmaxRef = 34.11
RdRef = 0.68
sigma = 0.08
Theta = −1.84
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

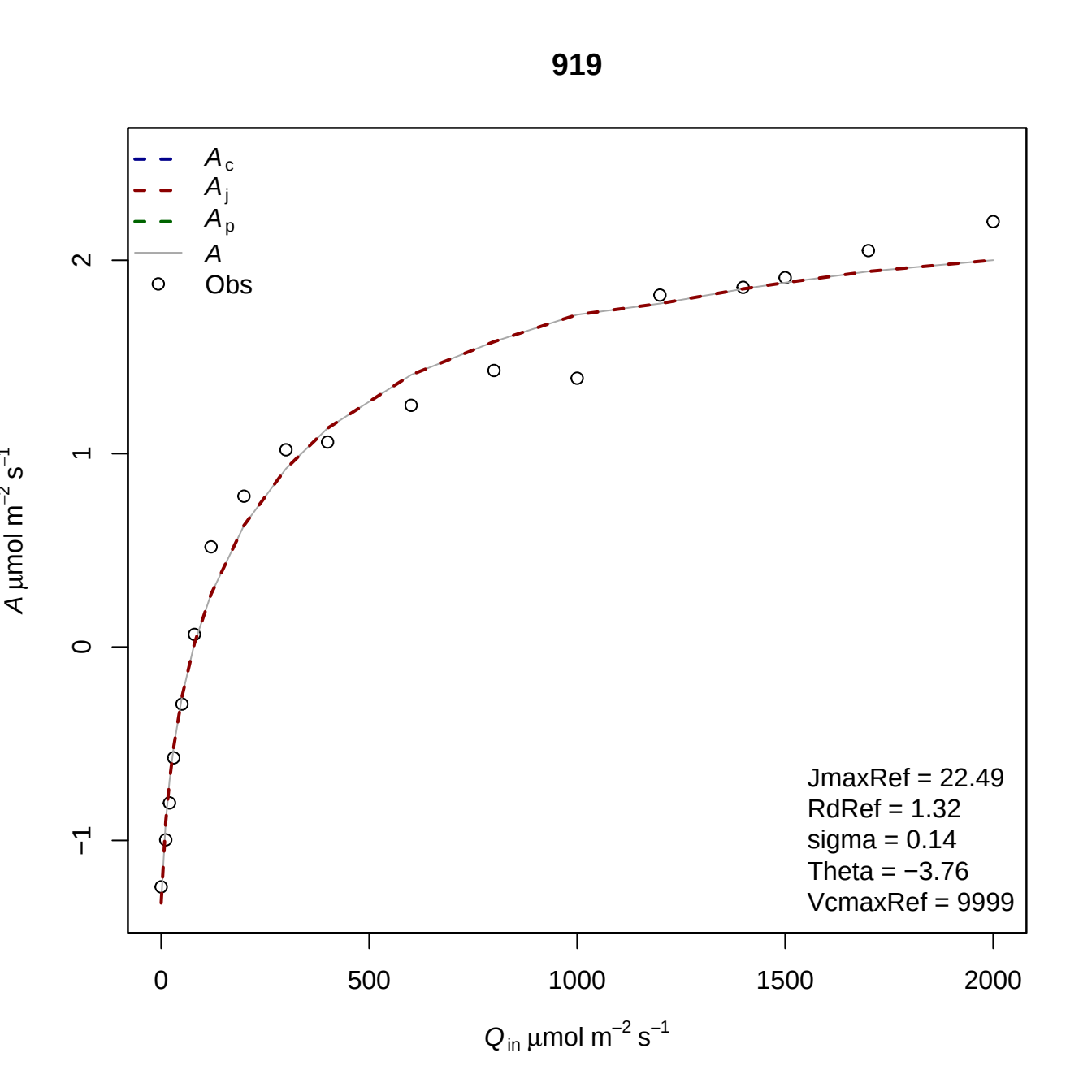

914
A_c
A_j
A_p
A
Obs
JmaxRef = 26.95
RdRef = 1.12
sigma = 0.13
Theta = −2.34
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

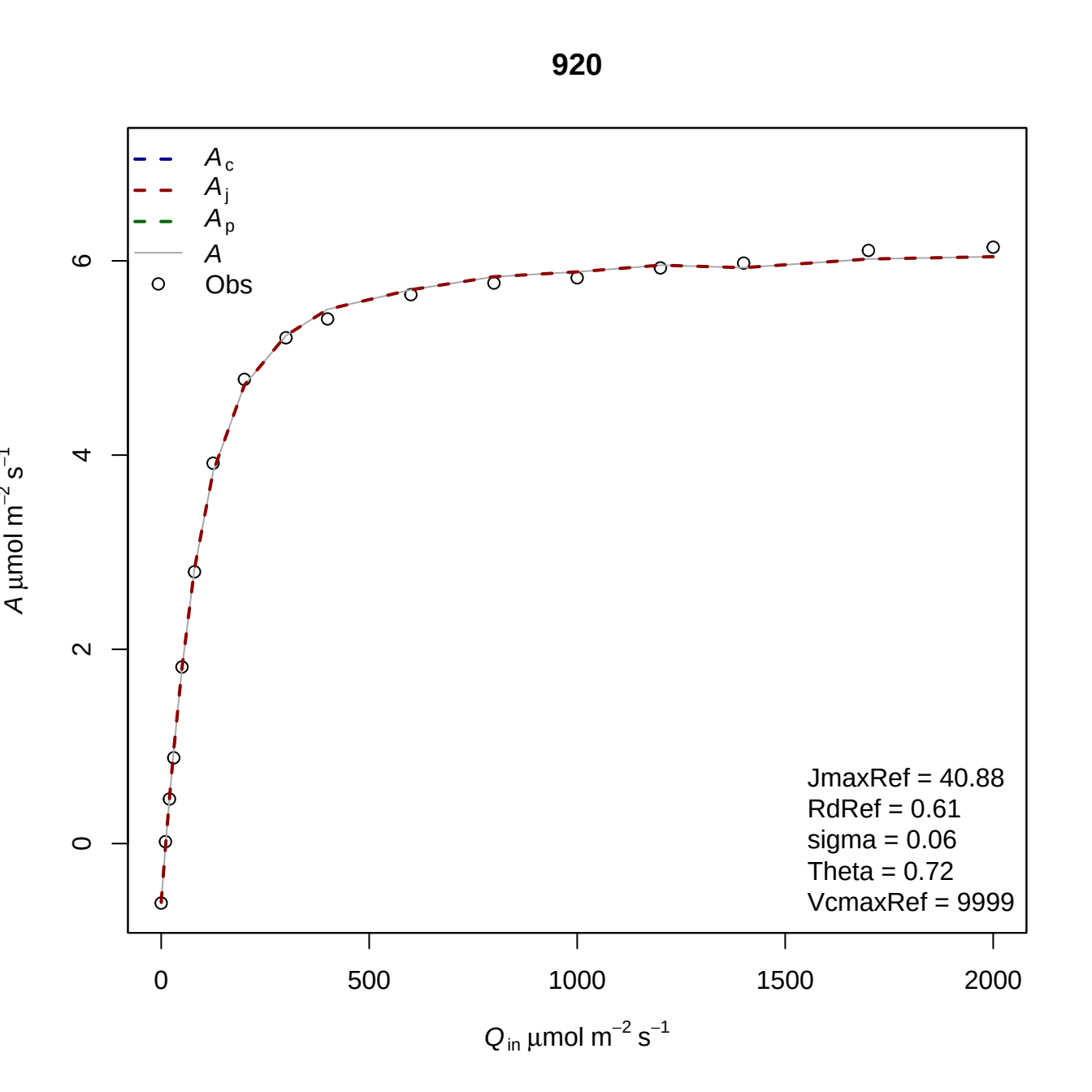

915
A_c
A_j
A_p
A
Obs
JmaxRef = 37.77
RdRef = 1.1
sigma = 0.11
Theta = −0.84
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

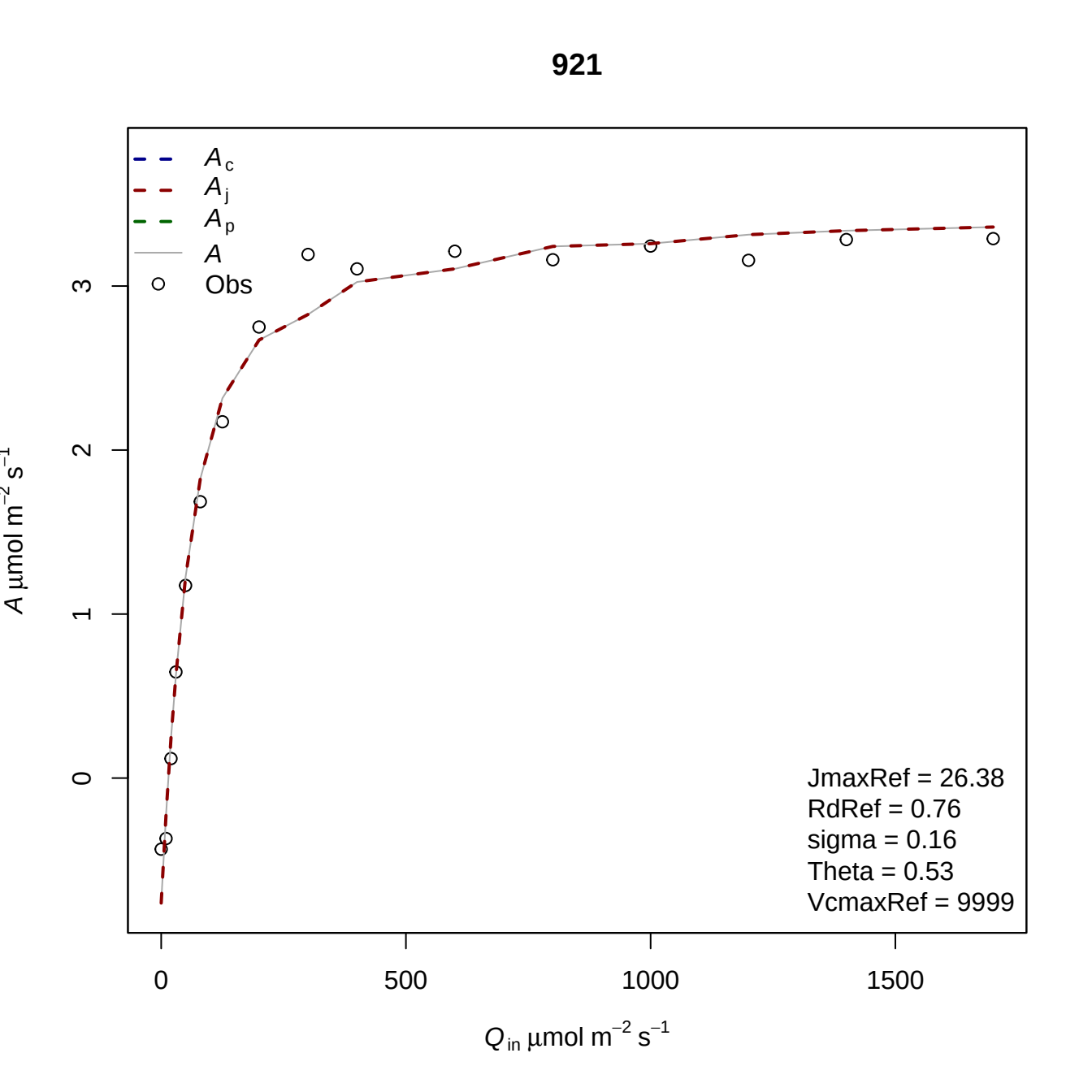

916
A_c
A_j
A_p
A
Obs
JmaxRef = 30.61
RdRef = 0.53
sigma = 0.09
Theta = 0.59
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500
2000

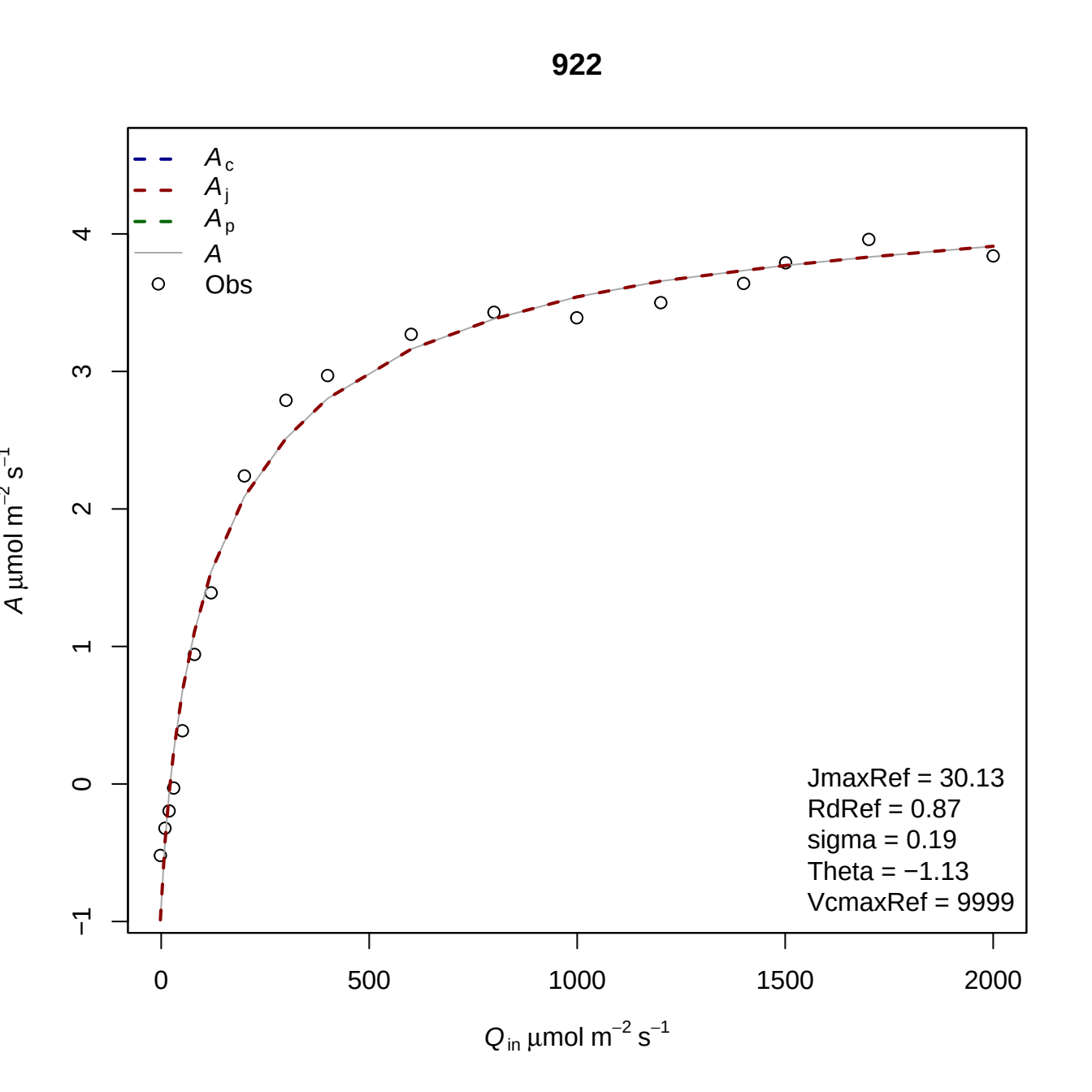

917
A_c
A_j
A_p
A
Obs
JmaxRef = 16.66
RdRef = 0.43
sigma = 0.16
Theta = 0.47
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500

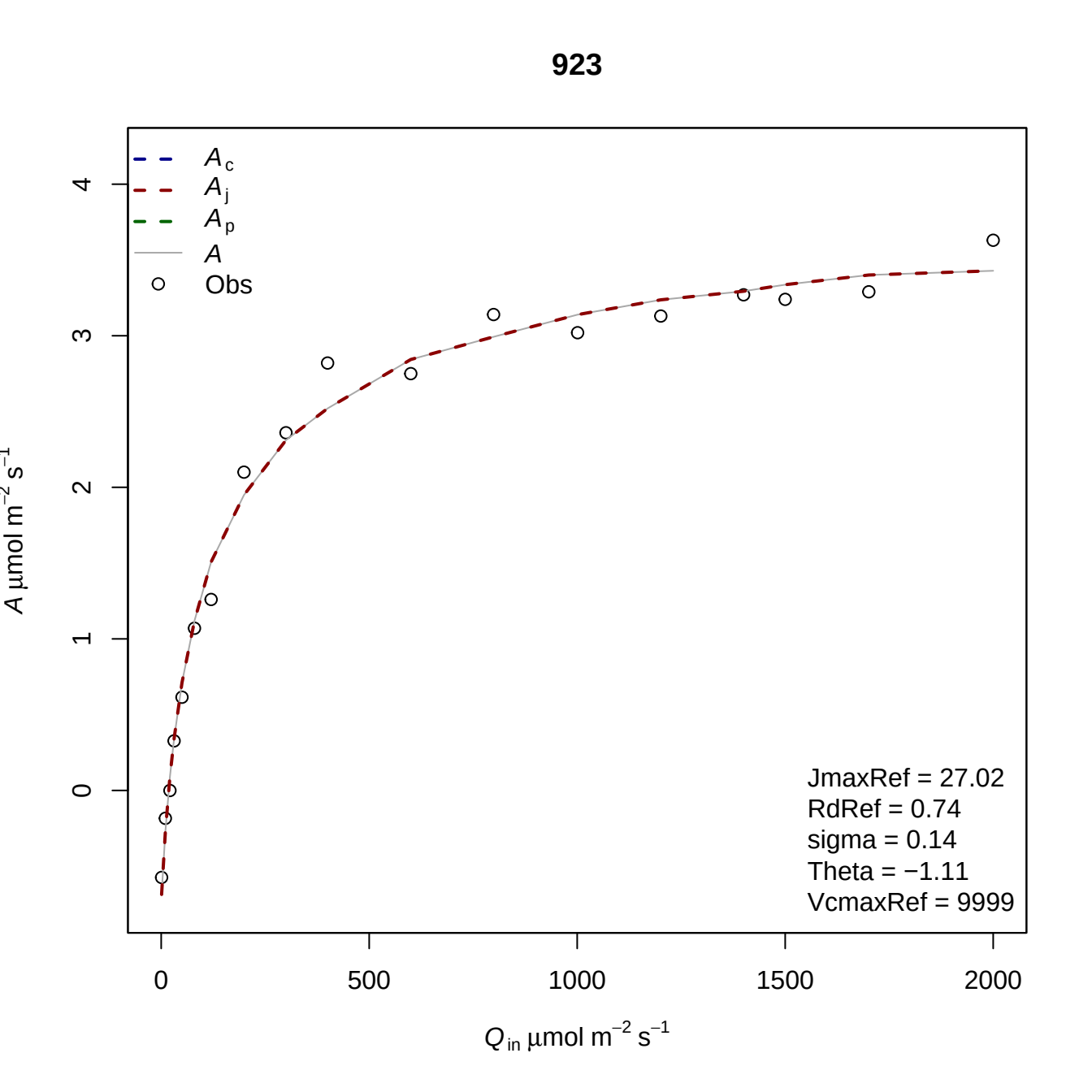

918
A_c
A_j
A_p
A
Obs
JmaxRef = 21.83
RdRef = 0.45
sigma = 0.15
Theta = −2.74
VcmaxRef = 9999
0
1
2
3
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

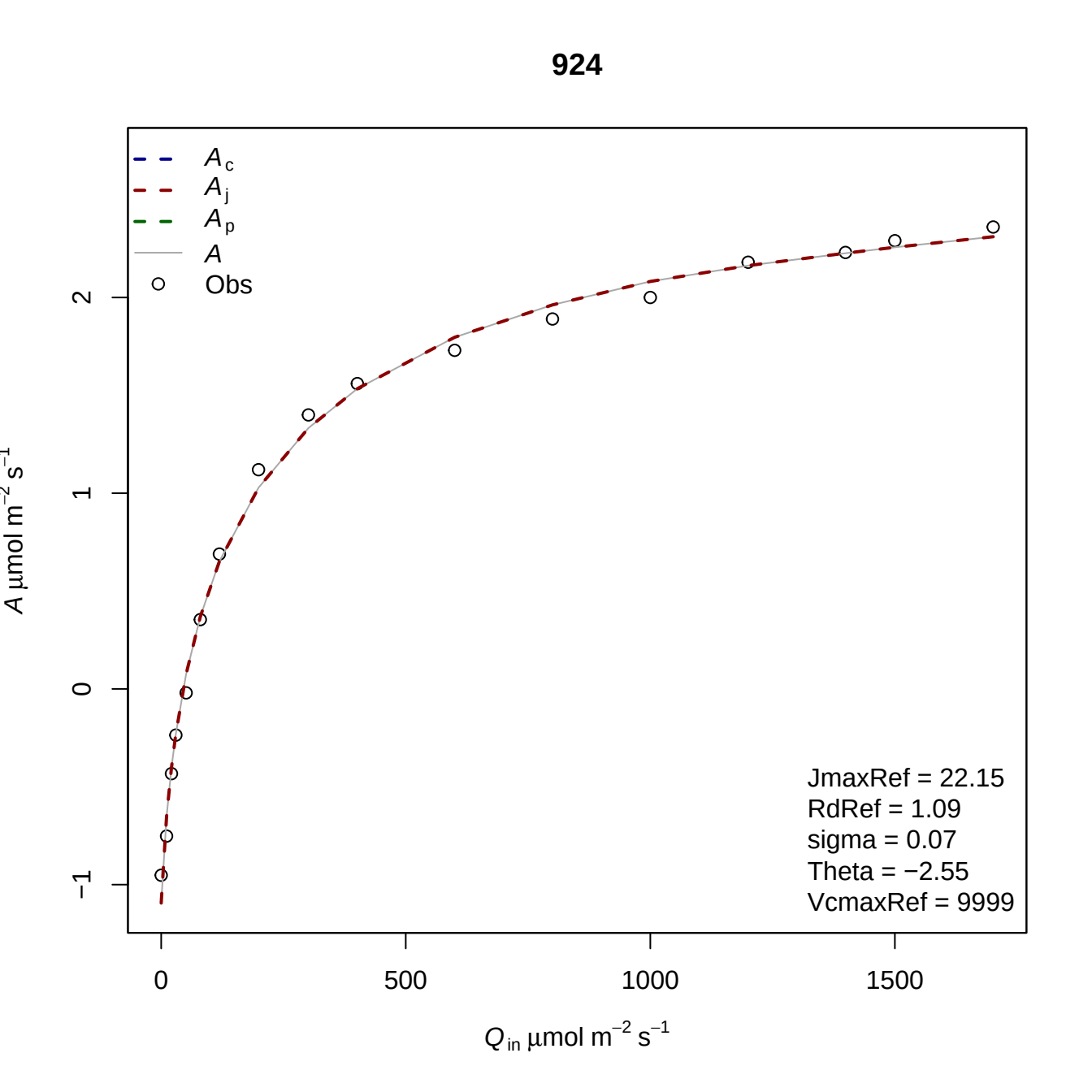

919
A_c
A_j
A_p
A
Obs
JmaxRef = 22.49
RdRef = 1.32
sigma = 0.14
Theta = −3.76
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500
2000

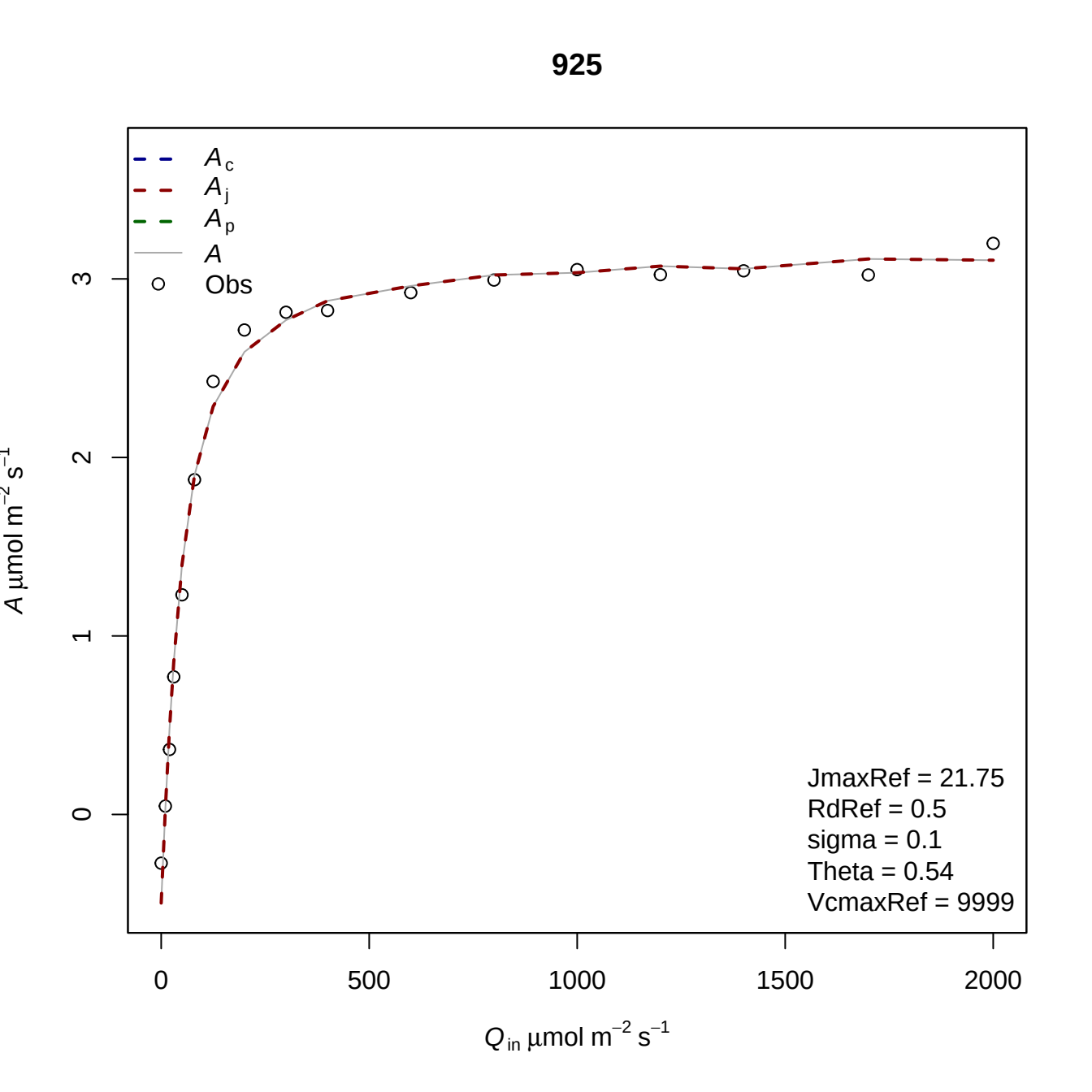

920
A_c
A_j
A_p
A
Obs
JmaxRef = 40.88
RdRef = 0.61
sigma = 0.06
Theta = 0.72
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

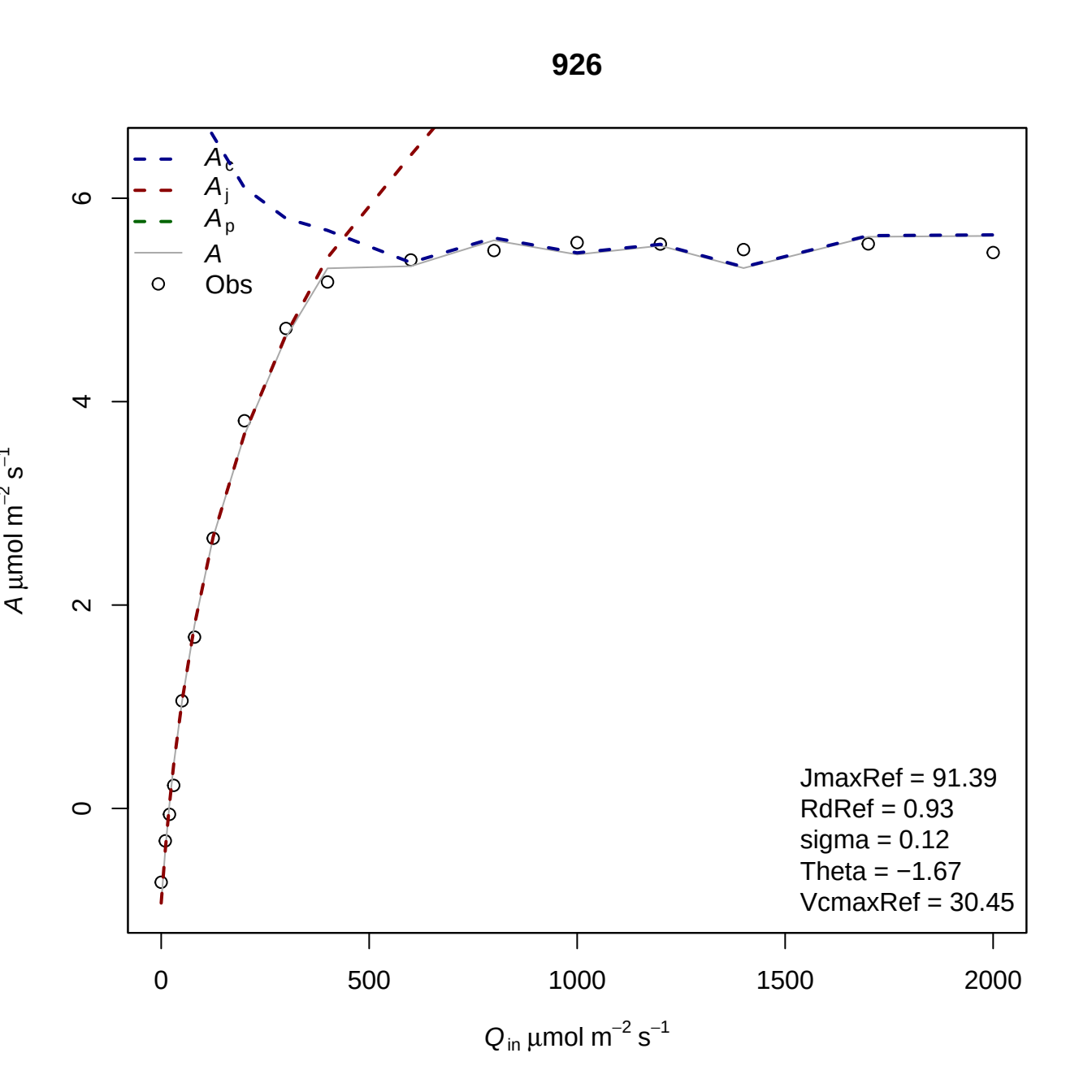

921
A_c
A_j
A_p
A
Obs
JmaxRef = 26.38
RdRef = 0.76
sigma = 0.16
Theta = 0.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

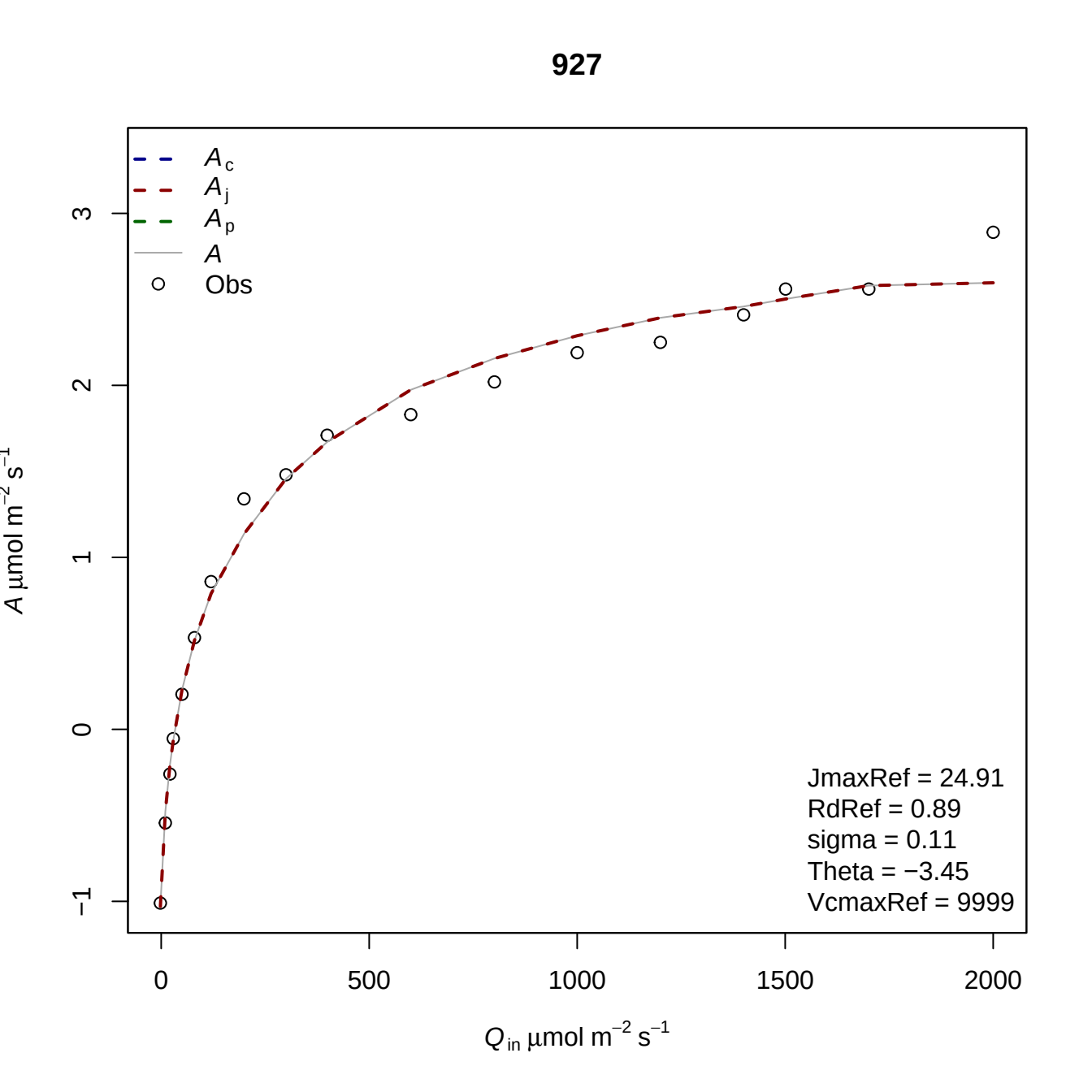

922
A_c
A_j
A_p
A
Obs
JmaxRef = 30.13
RdRef = 0.87
sigma = 0.19
Theta = −1.13
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

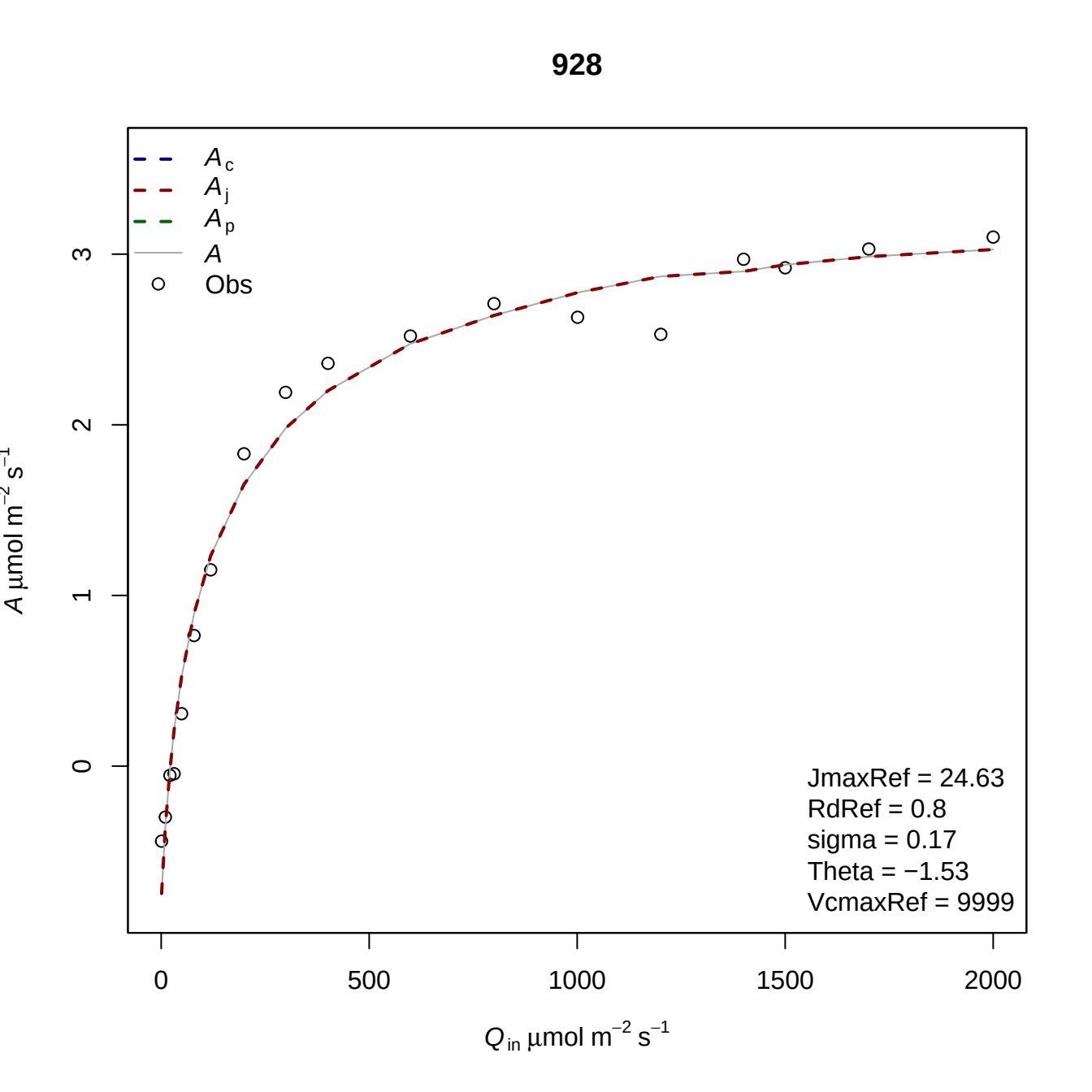

923
A_c
A_j
A_p
A
Obs
JmaxRef = 27.02
RdRef = 0.74
sigma = 0.14
Theta = −1.11
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

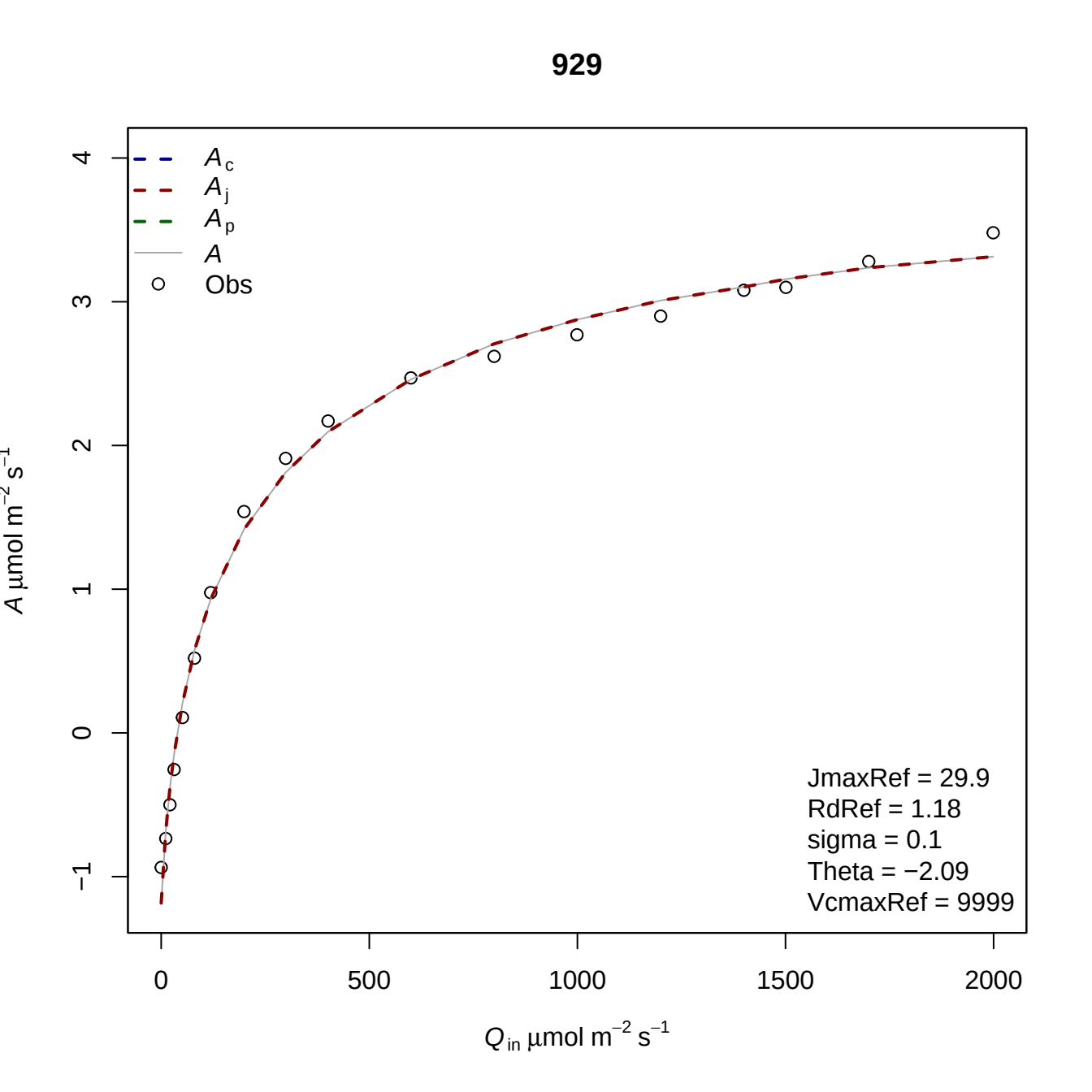

924
A_c
A_j
A_p
A
Obs
JmaxRef = 22.15
RdRef = 1.09
sigma = 0.07
Theta = −2.55
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

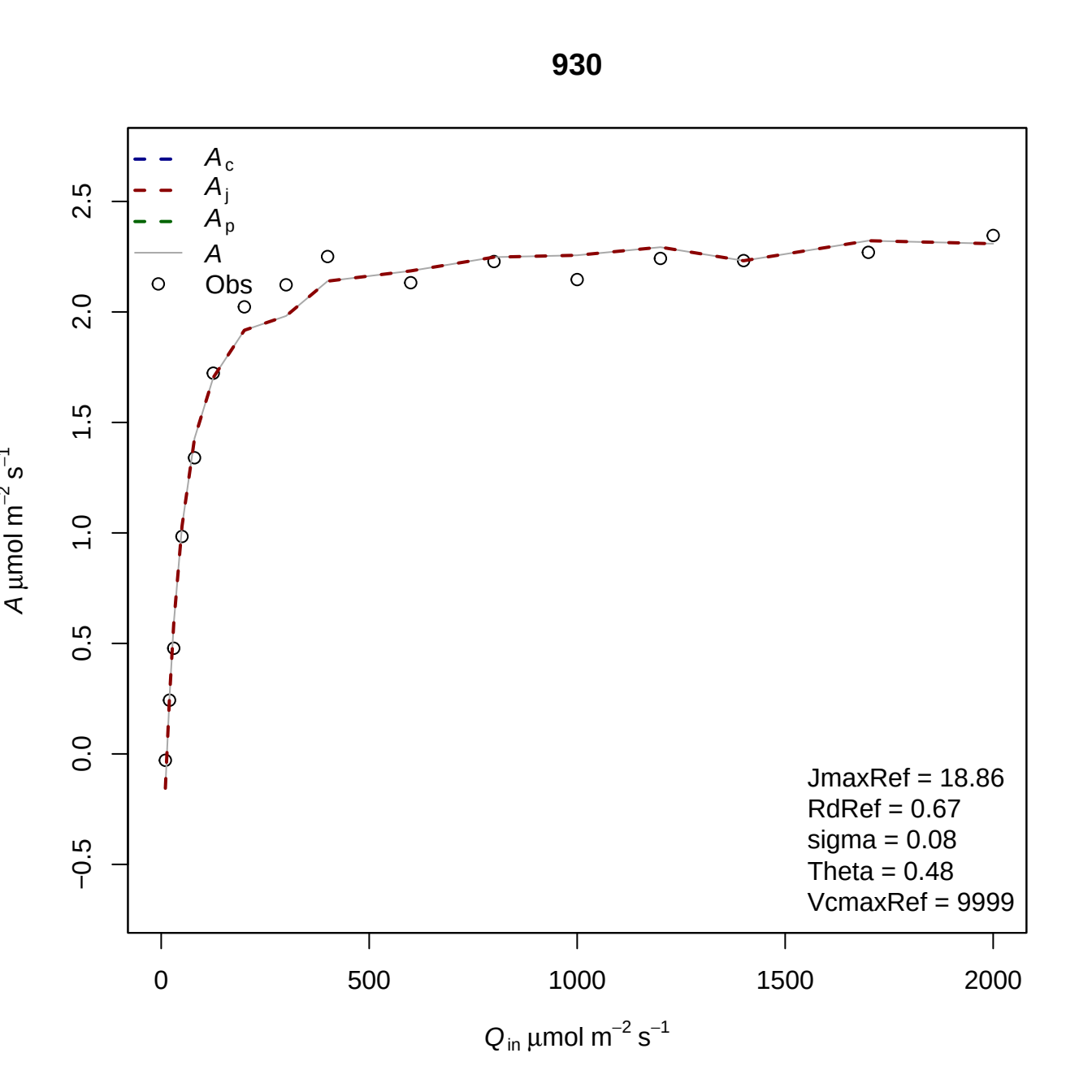

925
A_c
A_j
A_p
A
Obs
JmaxRef = 21.75
RdRef = 0.5
sigma = 0.1
Theta = 0.54
VcmaxRef = 9999
0
1
2
3
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

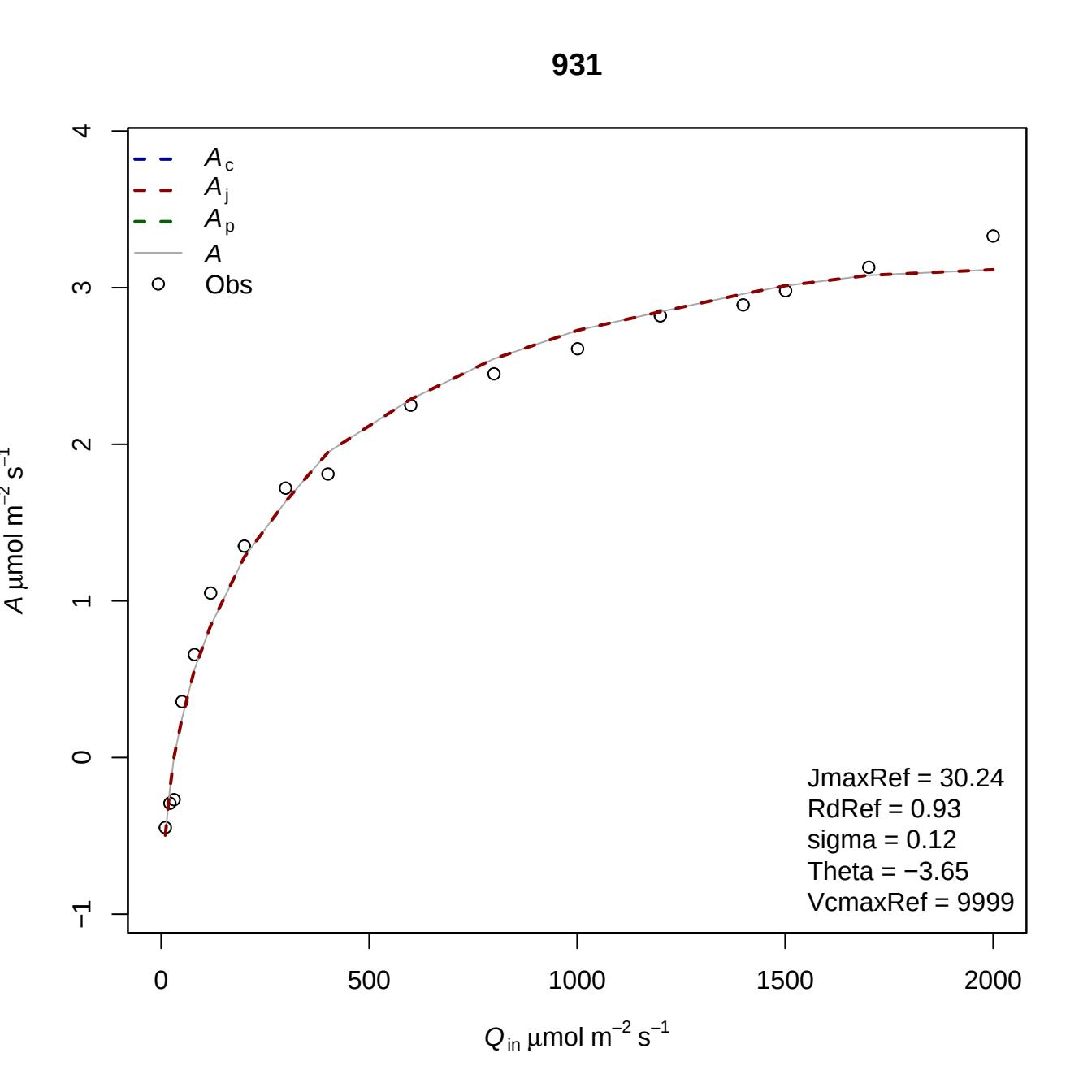

926
A_c
A_j
A_p
A
Obs
JmaxRef = 91.39
RdRef = 0.93
sigma = 0.12
Theta = −1.67
VcmaxRef = 30.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

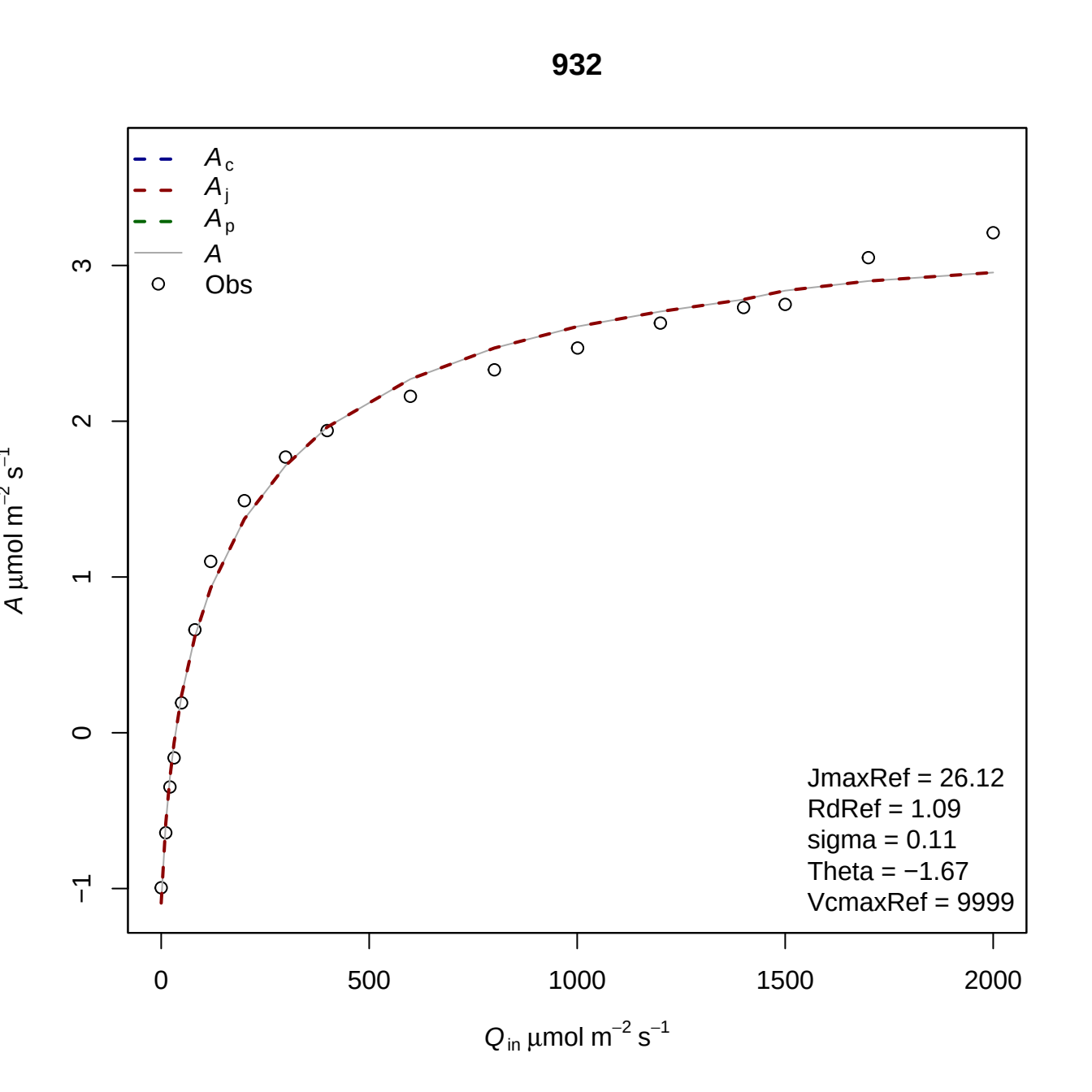

927
A_c
A_j
A_p
A
Obs
JmaxRef = 24.91
RdRef = 0.89
sigma = 0.11
Theta = −3.45
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

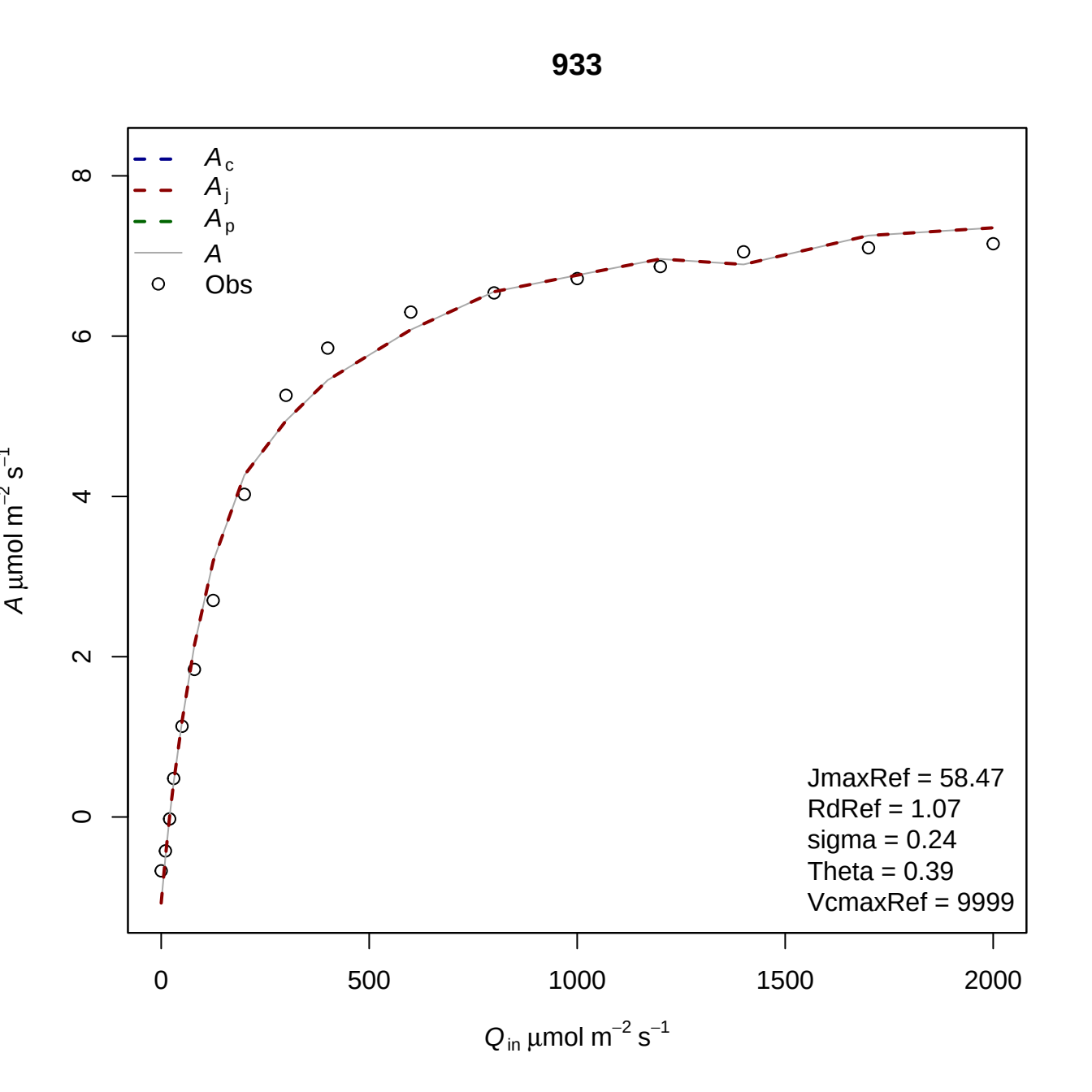

928
A_c
A_j
A_p
A
Obs
JmaxRef = 24.63
RdRef = 0.8
sigma = 0.17
Theta = −1.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

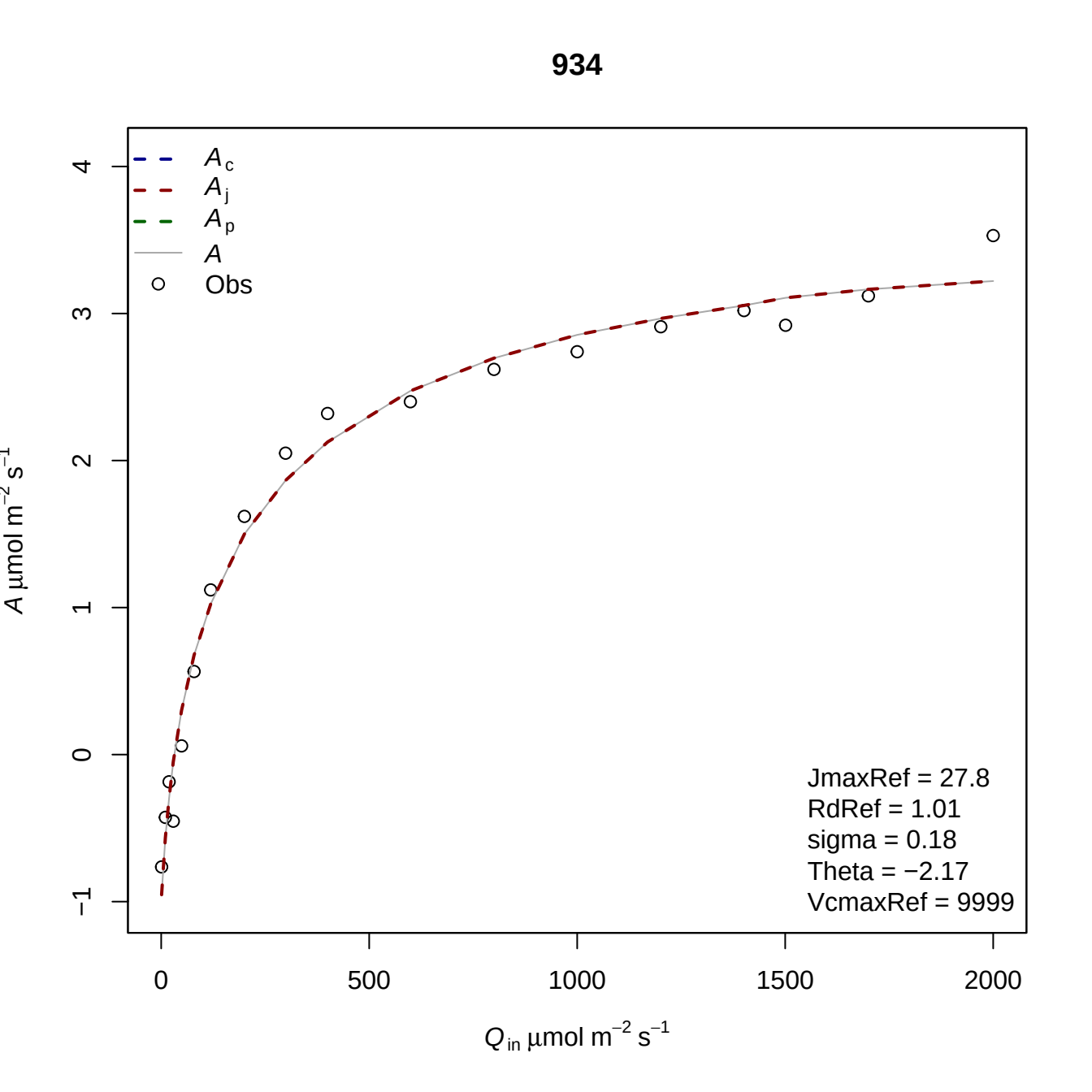

929
A_c
A_j
A_p
A
Obs
JmaxRef = 29.9
RdRef = 1.18
sigma = 0.1
Theta = −2.09
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

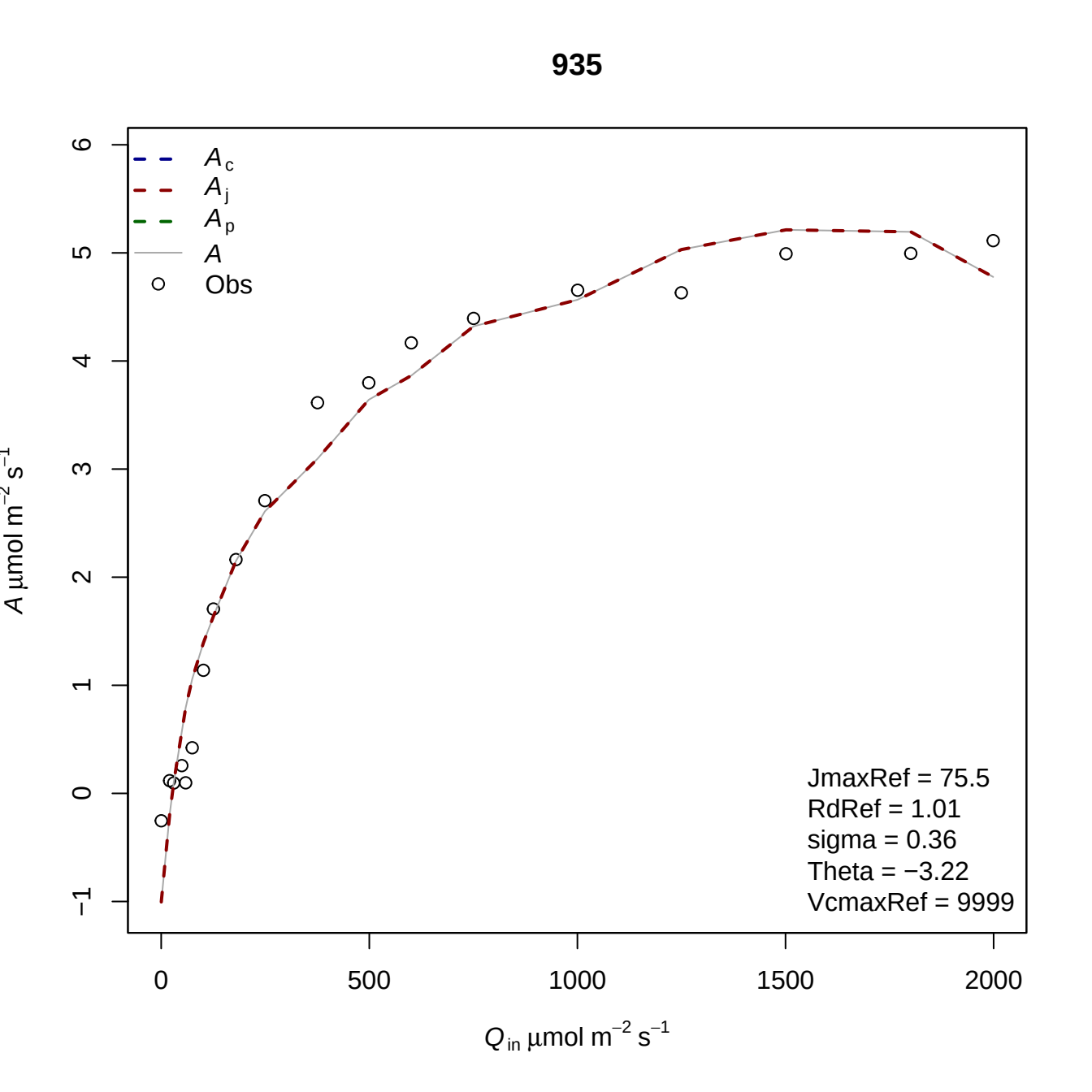

930
A_c
A_j
A_p
A
Obs
JmaxRef = 18.86
RdRef = 0.67
sigma = 0.08
Theta = 0.48
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500
2000

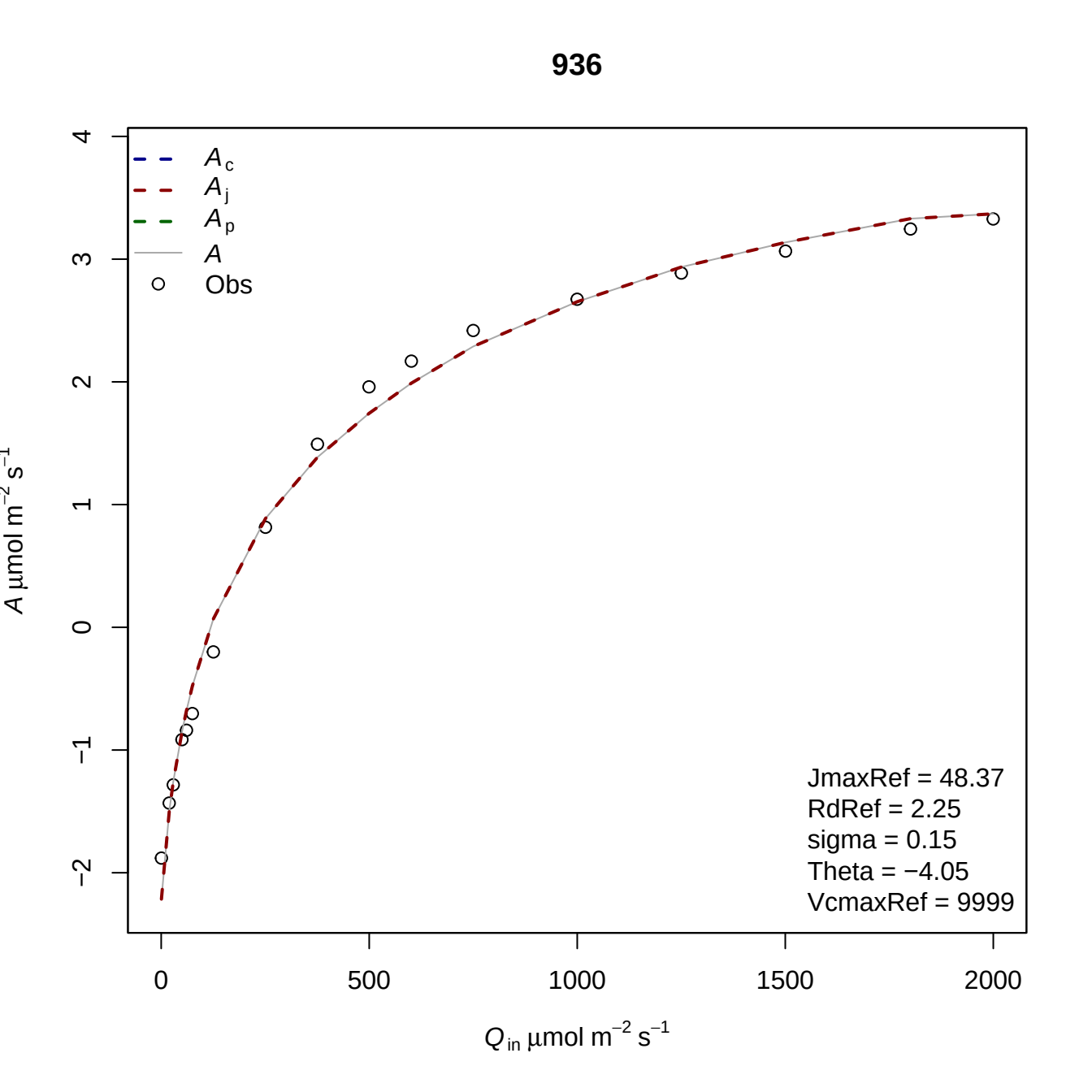

931
A_c
A_j
A_p
A
Obs
JmaxRef = 30.24
RdRef = 0.93
sigma = 0.12
Theta = −3.65
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

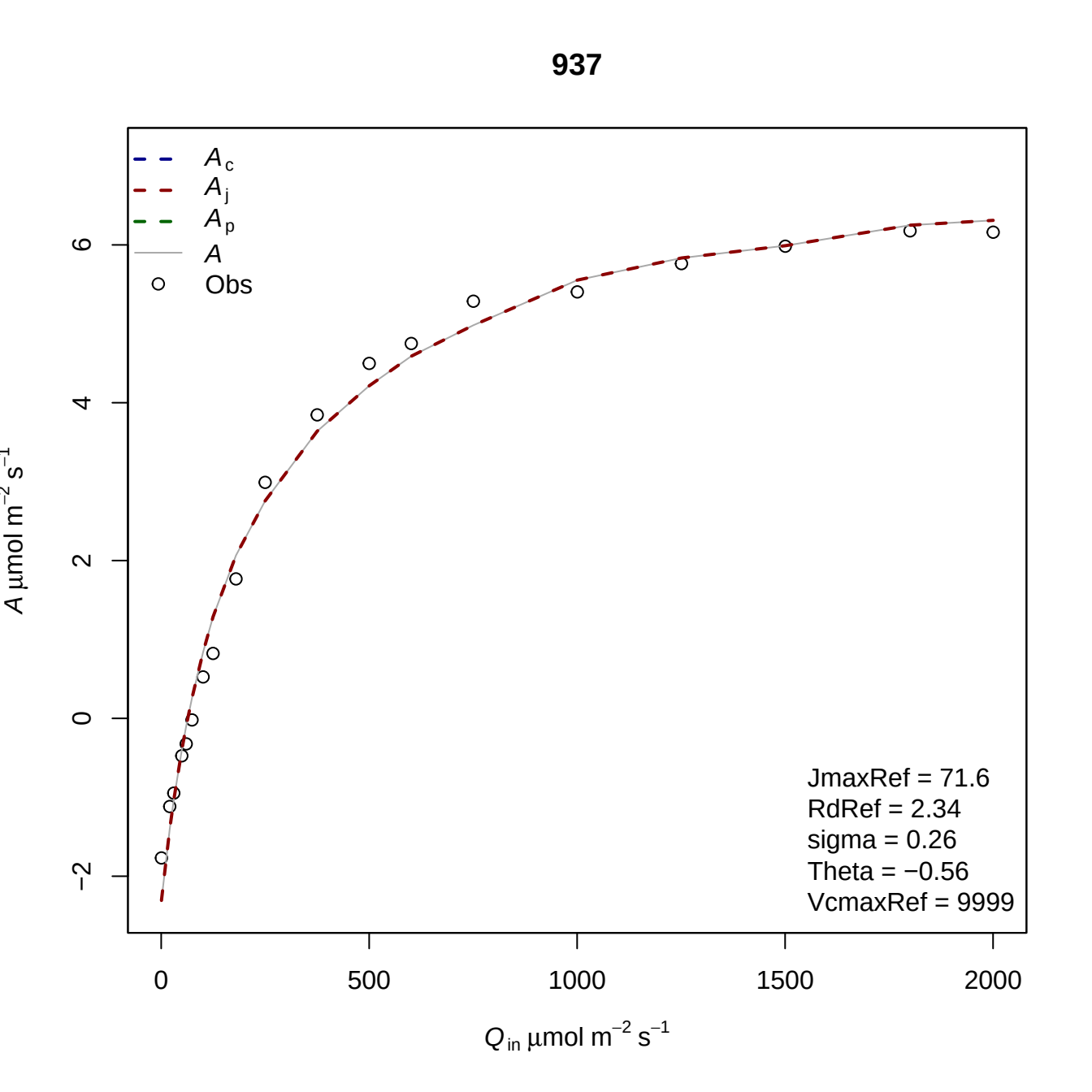

932
A_c
A_j
A_p
A
Obs
JmaxRef = 26.12
RdRef = 1.09
sigma = 0.11
Theta = −1.67
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

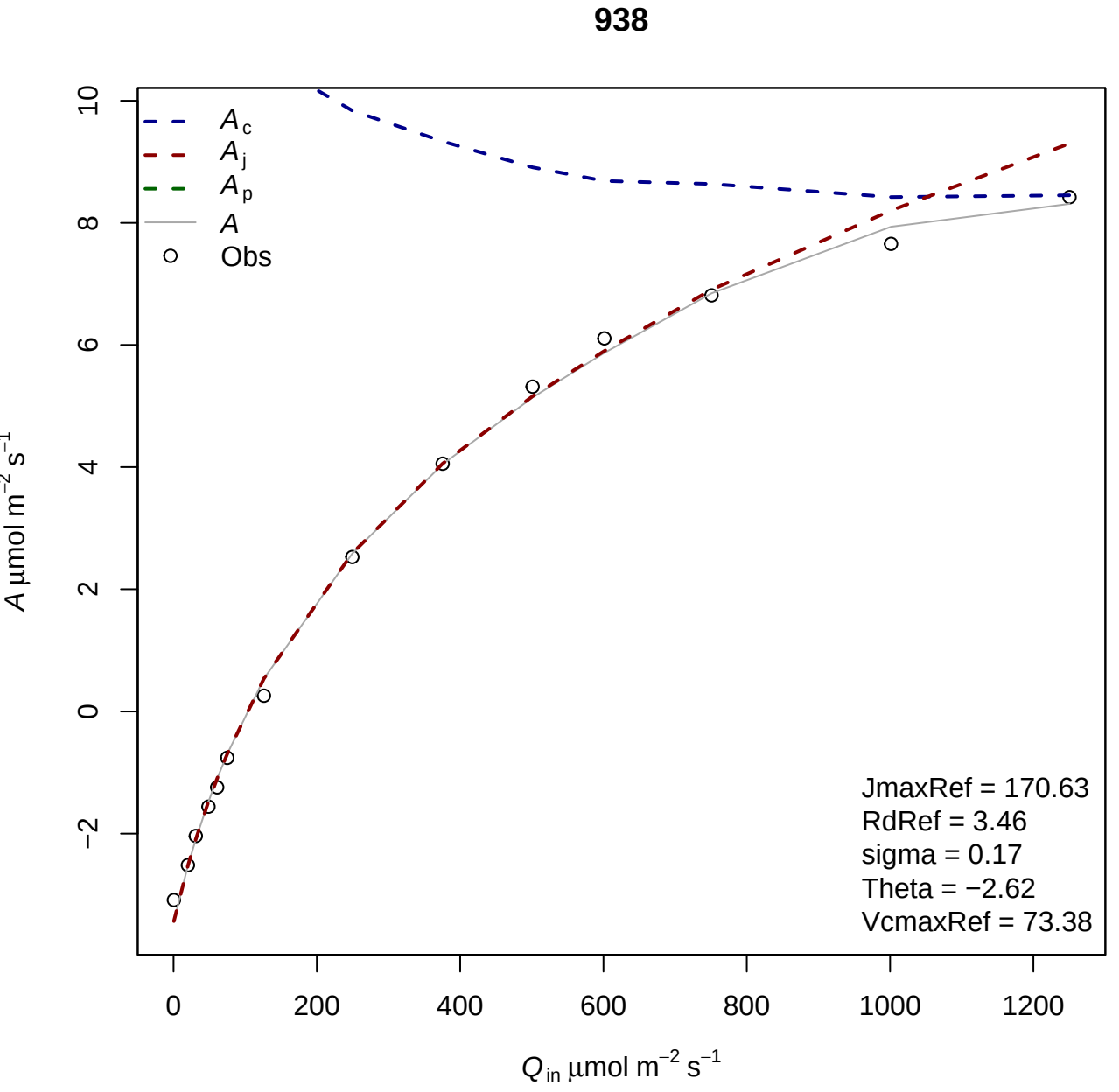

933
A_c
A_j
A_p
A
Obs
JmaxRef = 58.47
RdRef = 1.07
sigma = 0.24
Theta = 0.39
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

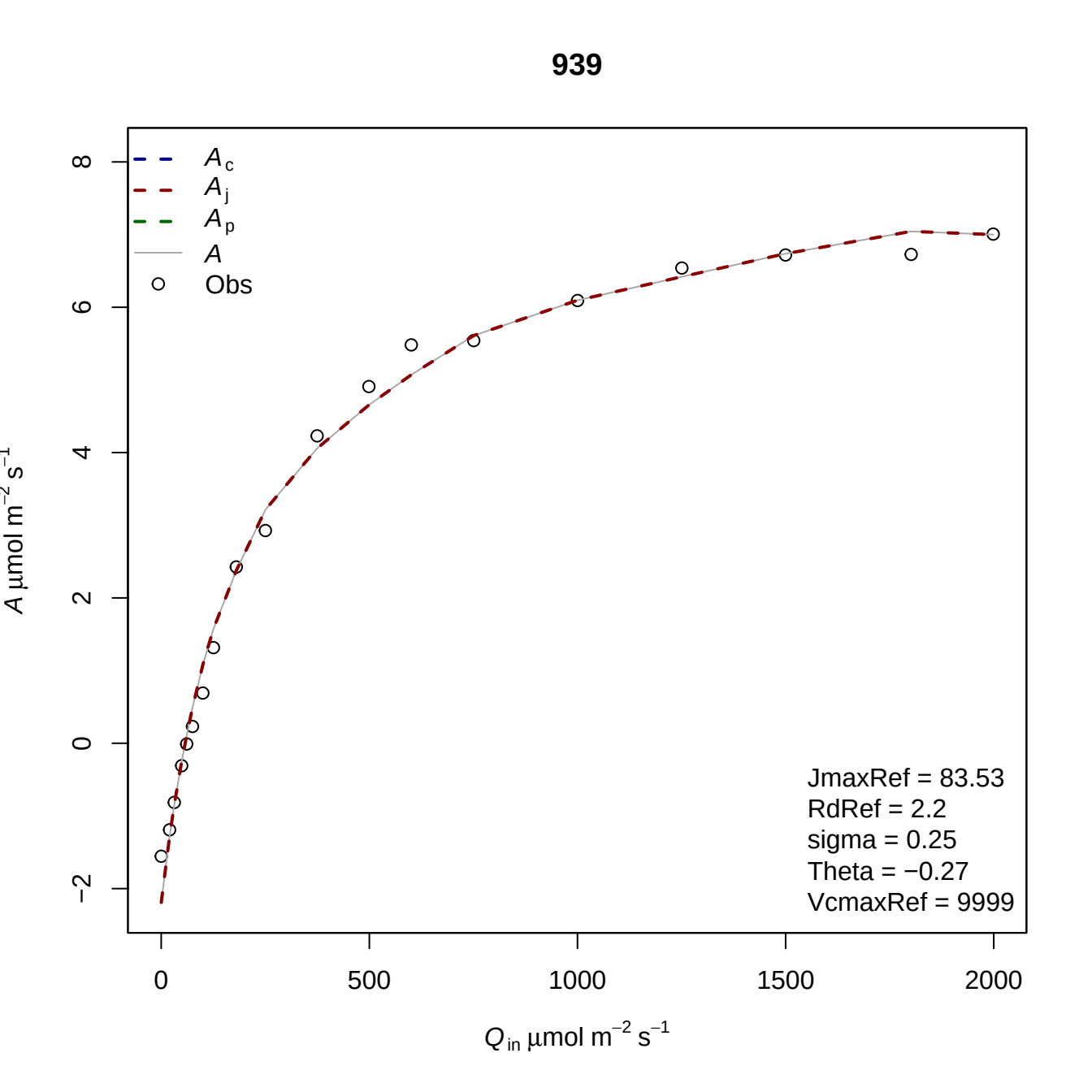

934
A_c
A_j
A_p
A
Obs
JmaxRef = 27.8
RdRef = 1.01
sigma = 0.18
Theta = −2.17
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

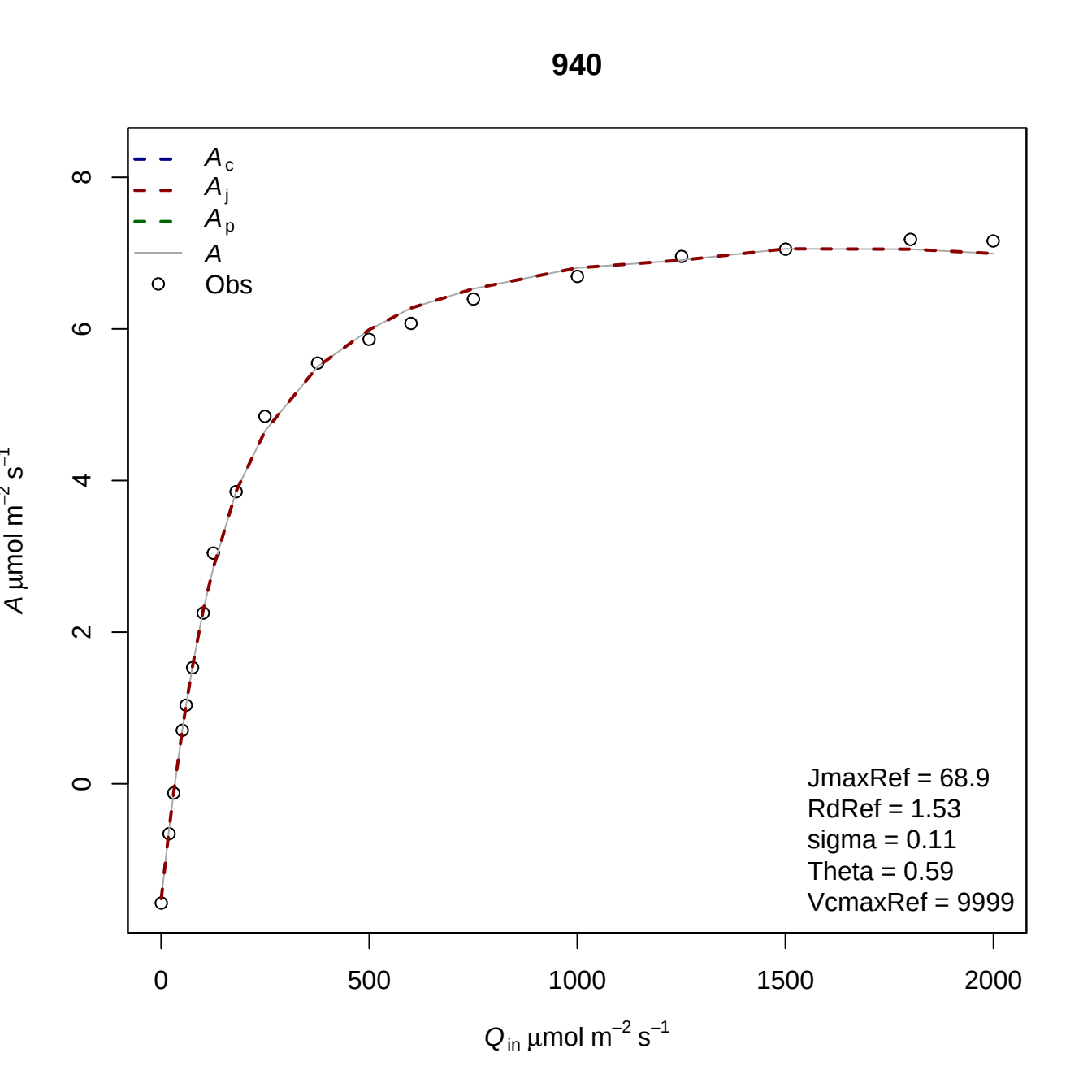

935
A_c
A_j
A_p
A
Obs
JmaxRef = 75.5
RdRef = 1.01
sigma = 0.36
Theta = −3.22
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

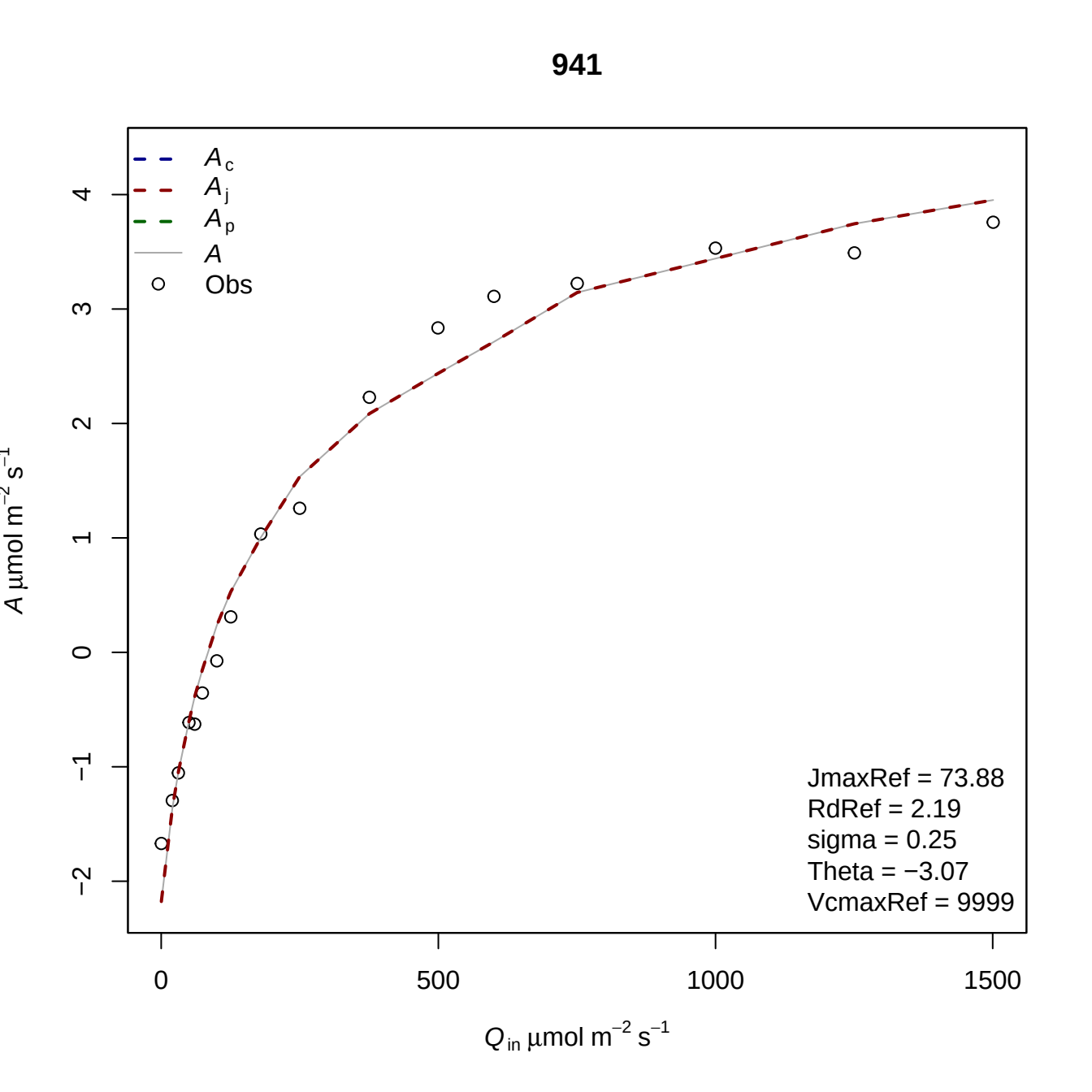

936
A_c
A_j
A_p
A
Obs
JmaxRef = 48.37
RdRef = 2.25
sigma = 0.15
Theta = −4.05
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

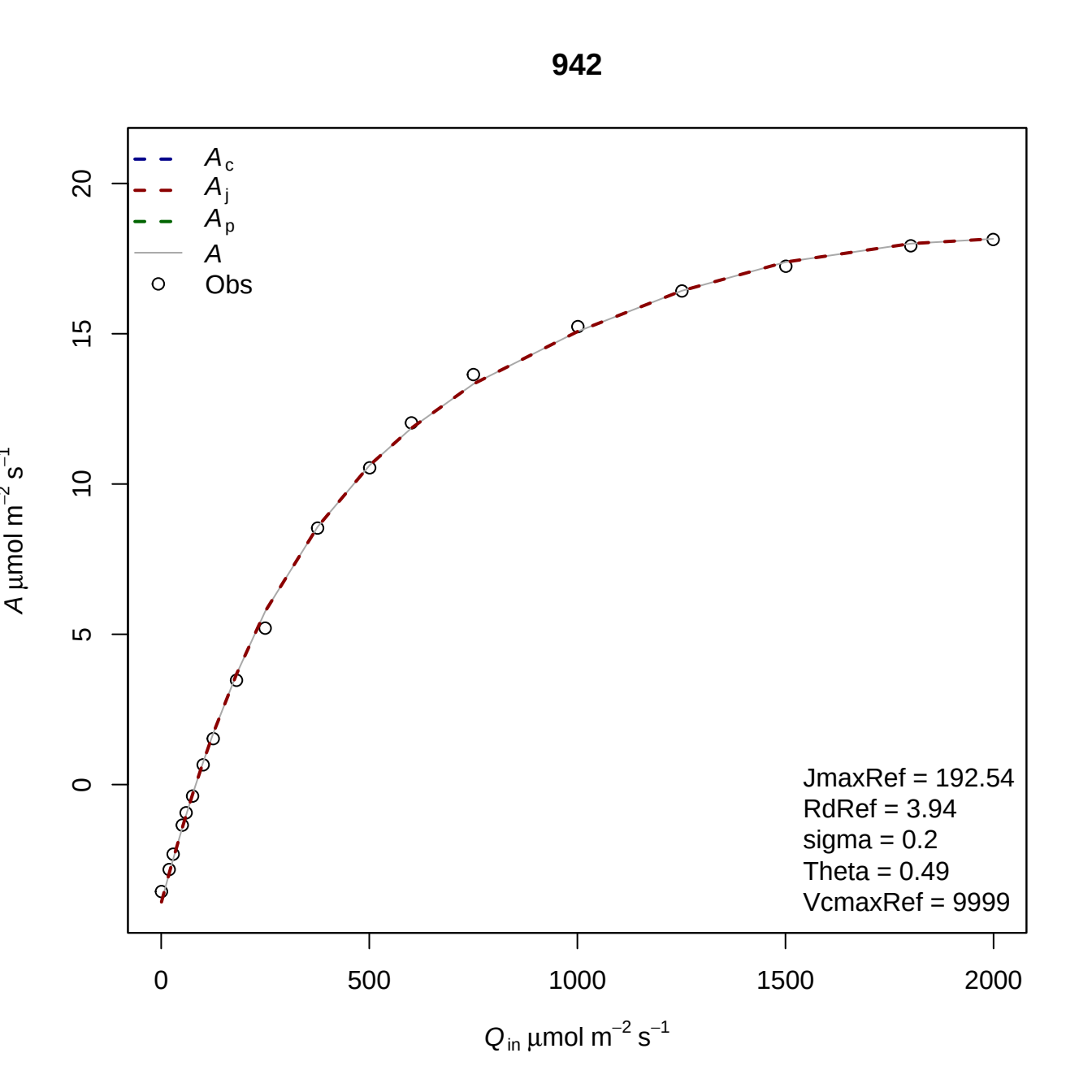

937
A_c
A_j
A_p
A
Obs
JmaxRef = 71.6
RdRef = 2.34
sigma = 0.26
Theta = −0.56
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

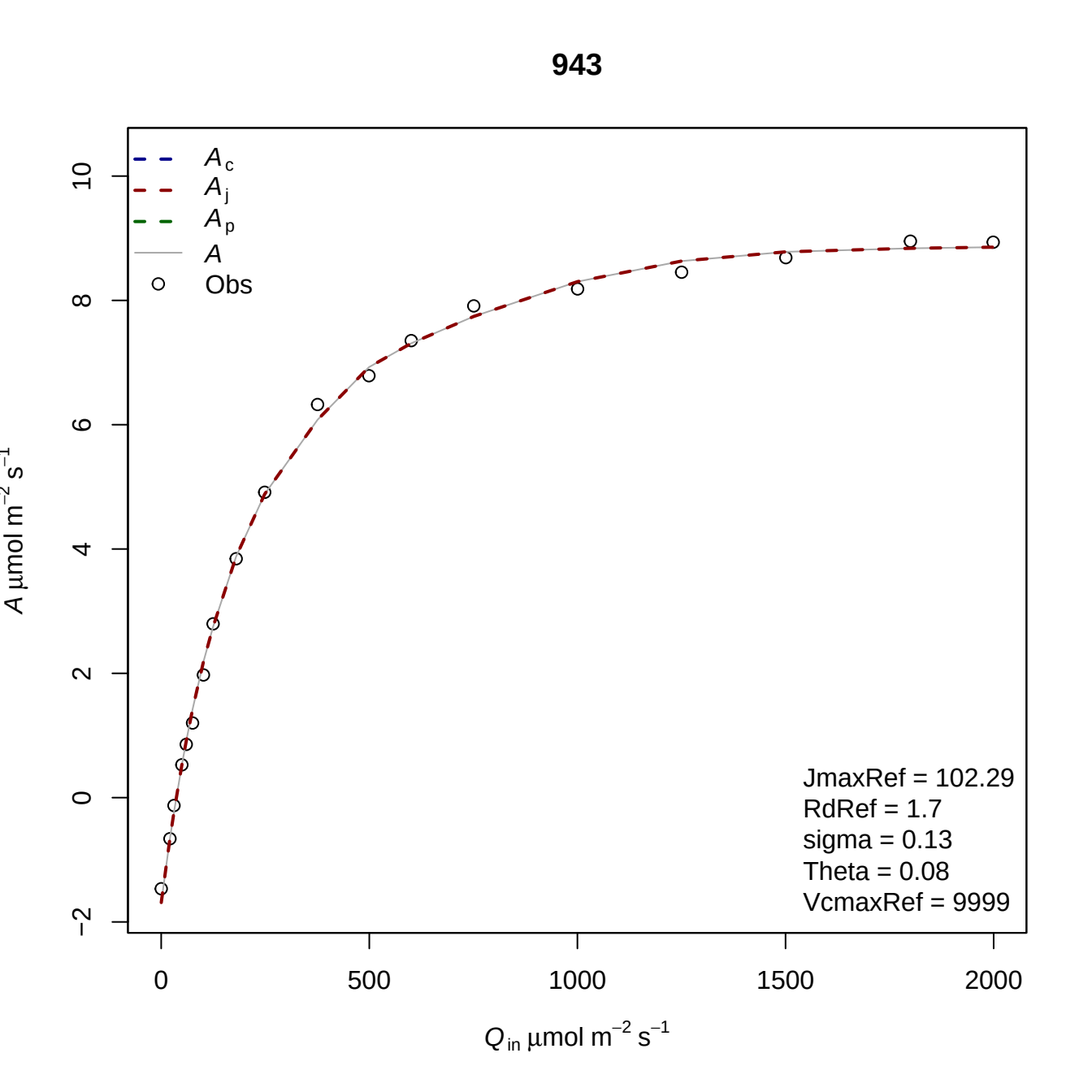

938
A_c
A_j
A_p
A
Obs
JmaxRef = 170.63
RdRef = 3.46
sigma = 0.17
Theta = −2.62
VcmaxRef = 73.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
200
400
600
800
1000
1200

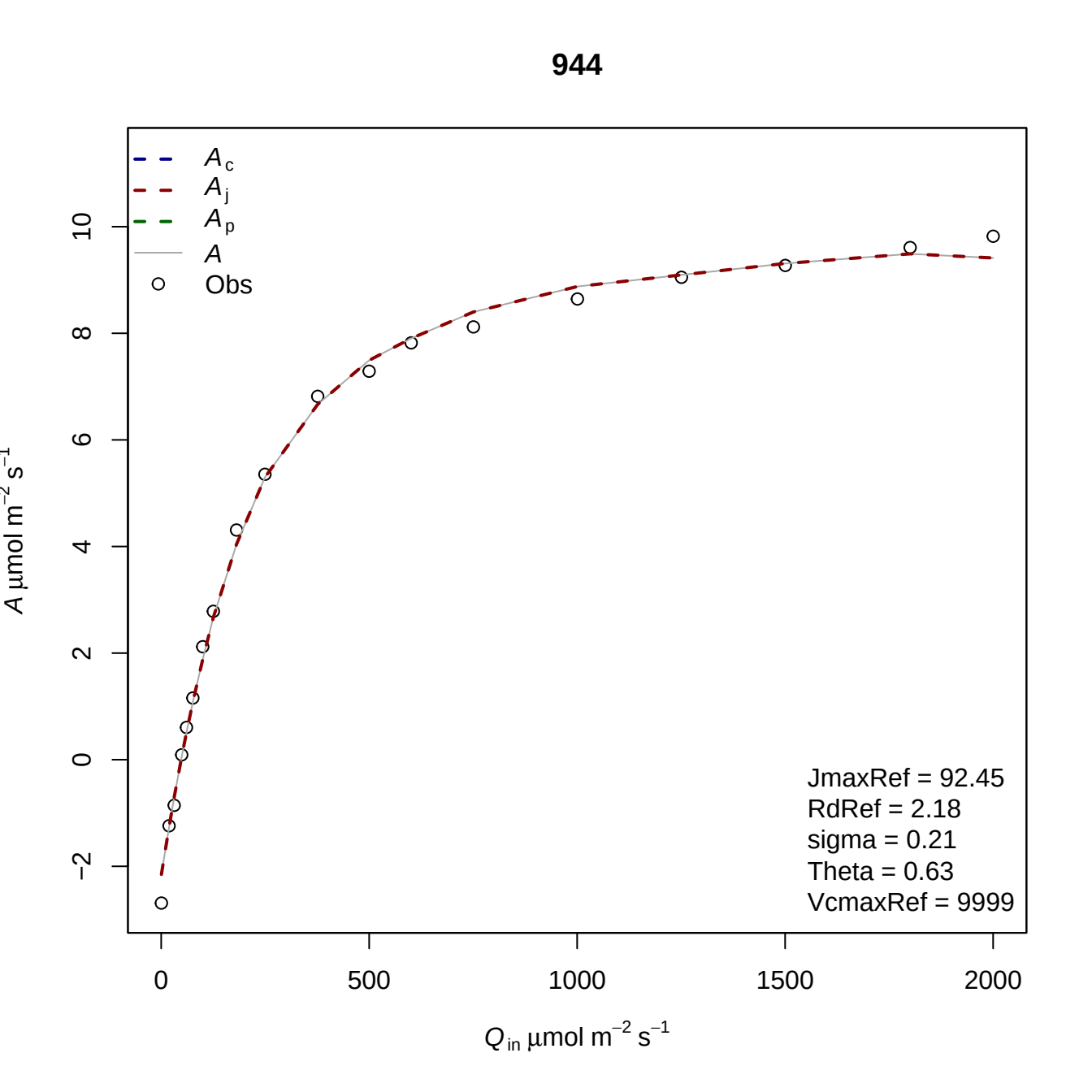

939
A_c
A_j
A_p
A
Obs
JmaxRef = 83.53
RdRef = 2.2
sigma = 0.25
Theta = −0.27
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

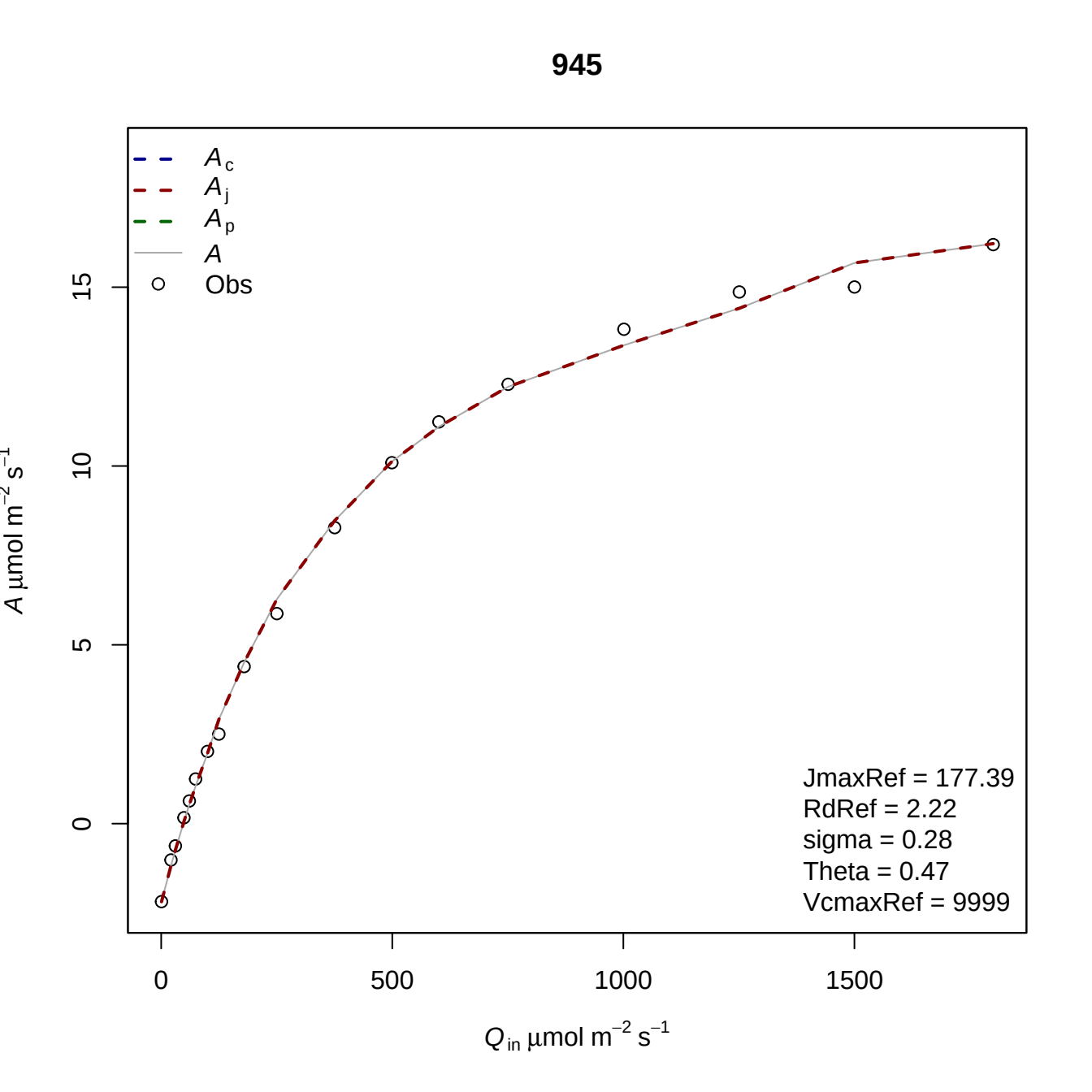

940
A_c
A_j
A_p
A
Obs
JmaxRef = 68.9
RdRef = 1.53
sigma = 0.11
Theta = 0.59
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

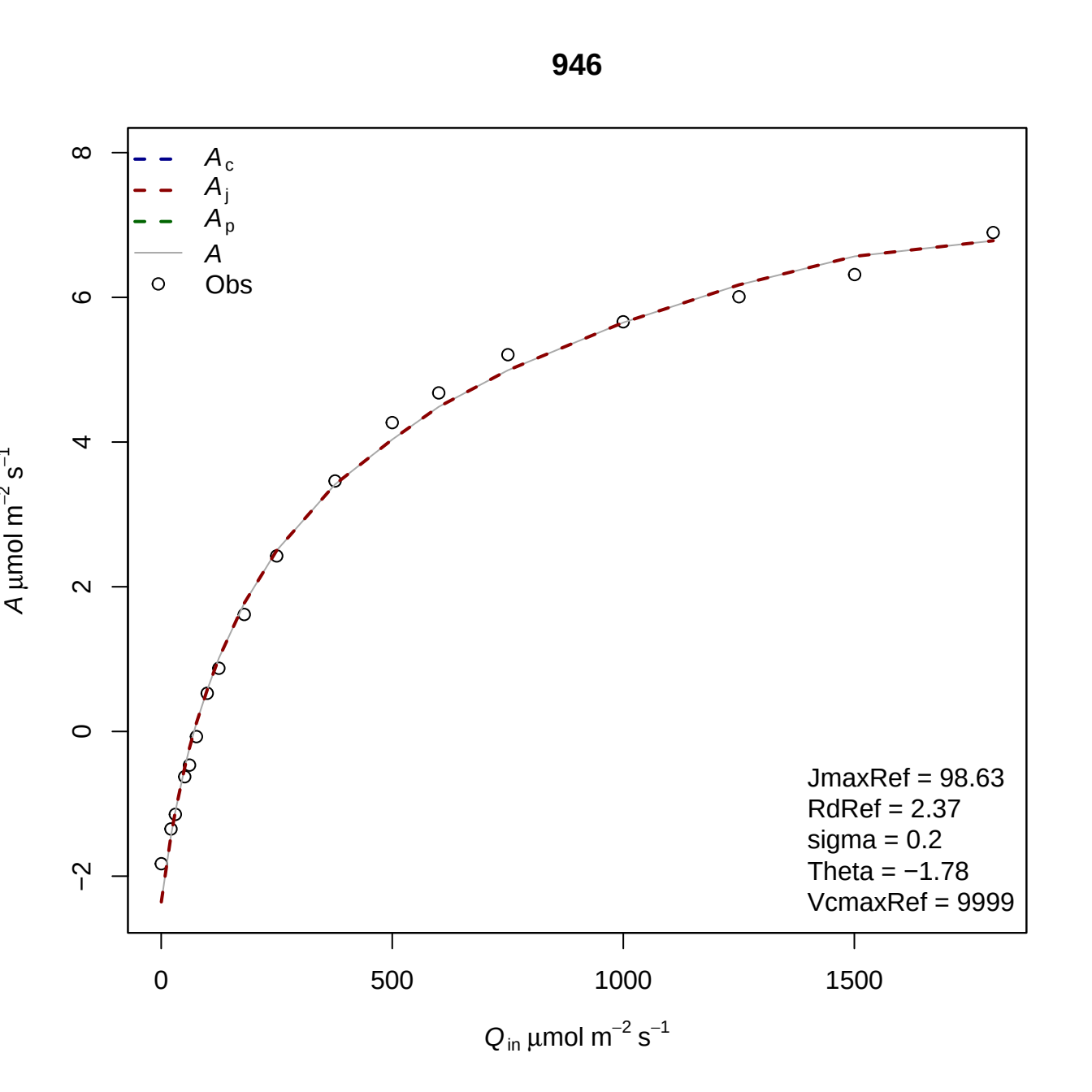

941
A_c
A_j
A_p
A
Obs
JmaxRef = 73.88
RdRef = 2.19
sigma = 0.25
Theta = −3.07
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
500
1000
1500

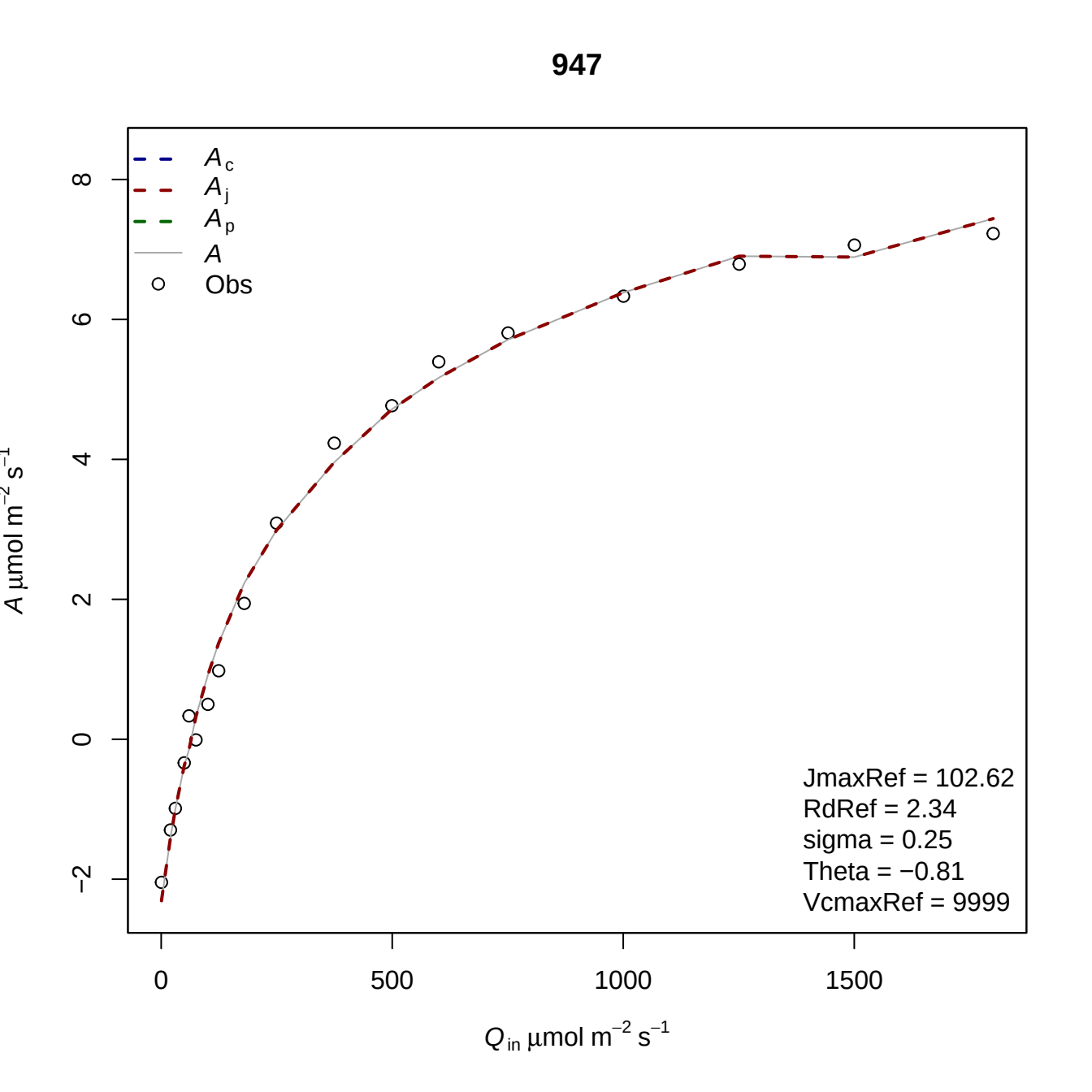

942
A_c
A_j
A_p
A
Obs
JmaxRef = 192.54
RdRef = 3.94
sigma = 0.2
Theta = 0.49
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

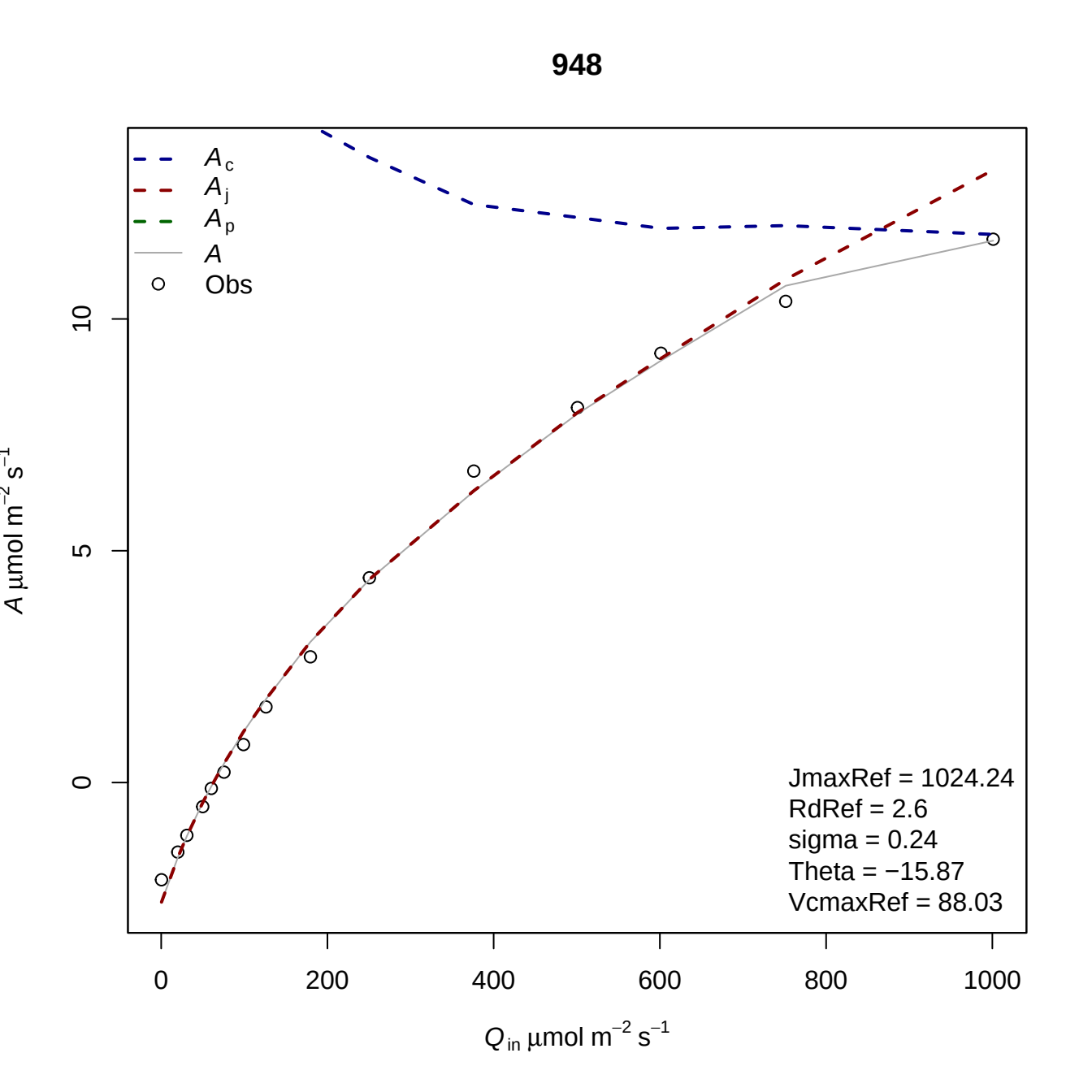

943
A_c
A_j
A_p
A
Obs
JmaxRef = 102.29
RdRef = 1.7
sigma = 0.13
Theta = 0.08
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

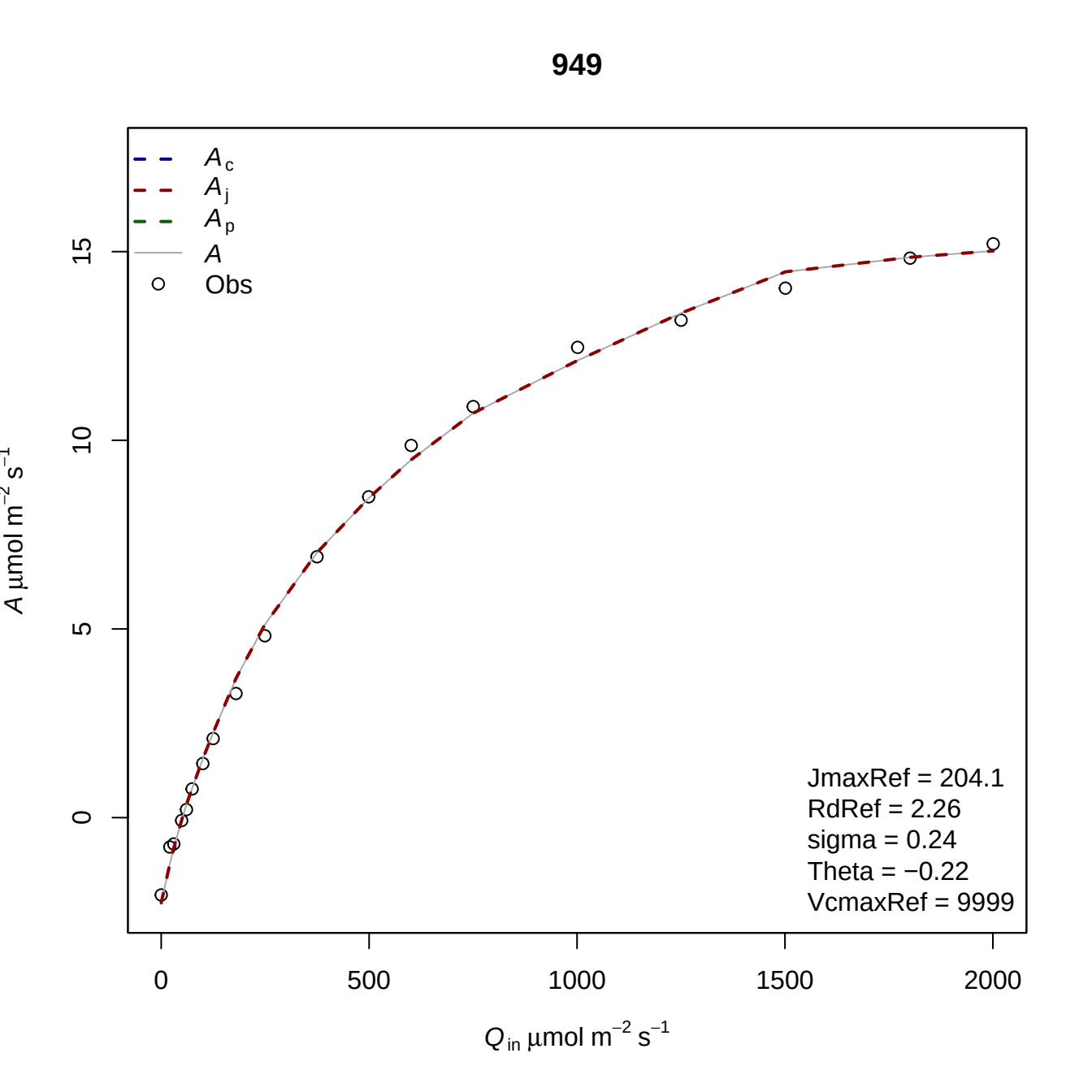

944
A_c
A_j
A_p
A
Obs
JmaxRef = 92.45
RdRef = 2.18
sigma = 0.21
Theta = 0.63
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

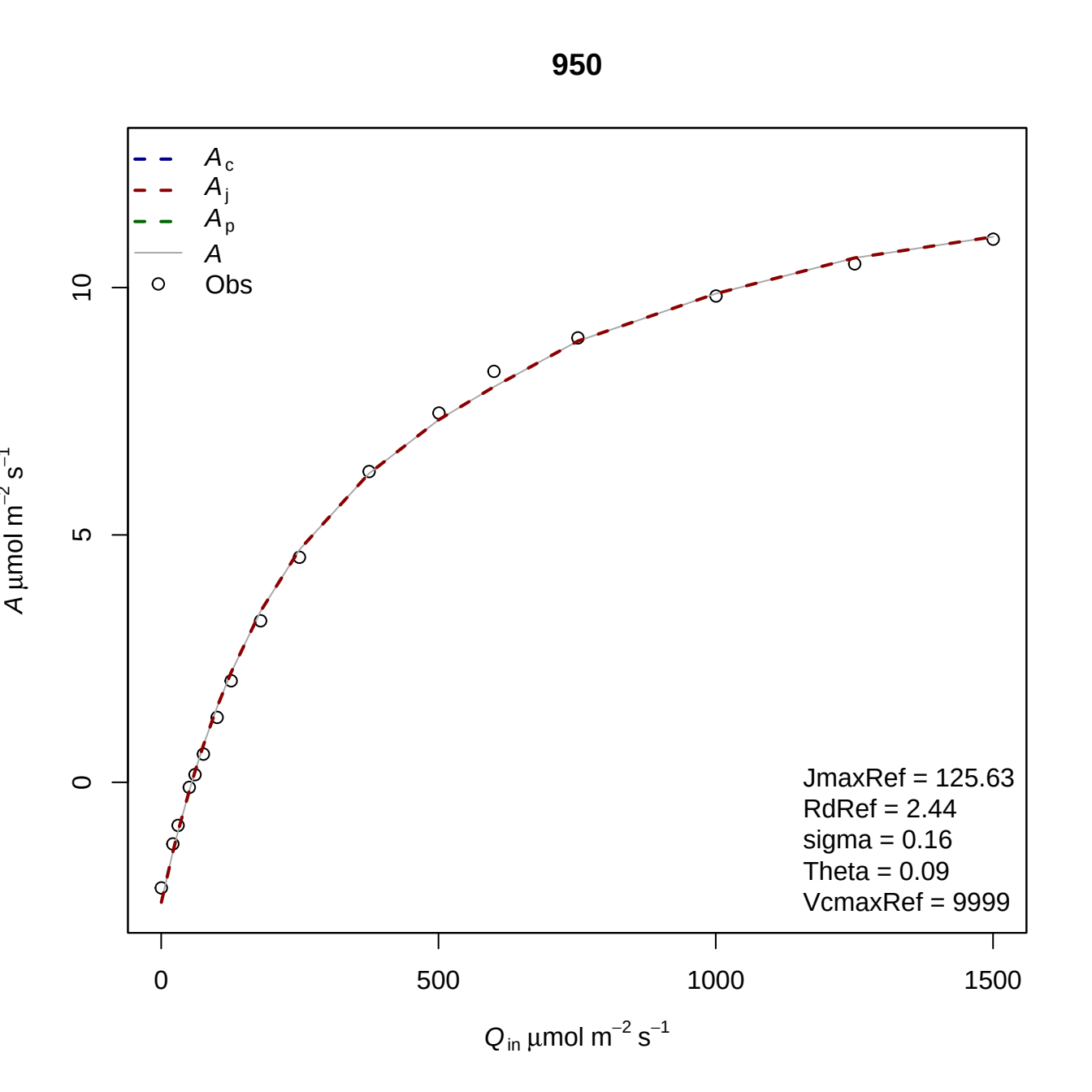

945
A_c
A_j
A_p
A
Obs
JmaxRef = 177.39
RdRef = 2.22
sigma = 0.28
Theta = 0.47
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

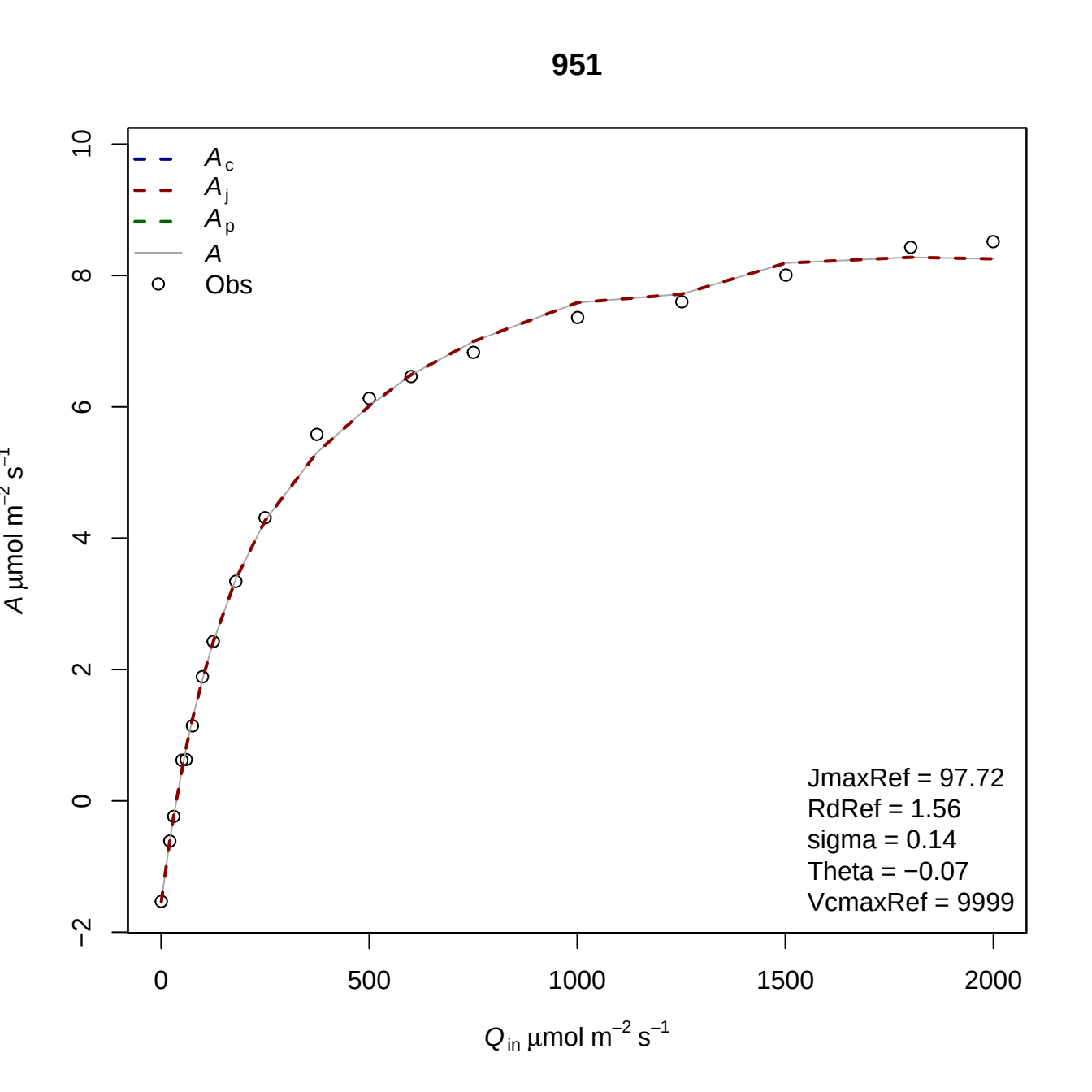

946
A_c
A_j
A_p
A
Obs
JmaxRef = 98.63
RdRef = 2.37
sigma = 0.2
Theta = −1.78
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

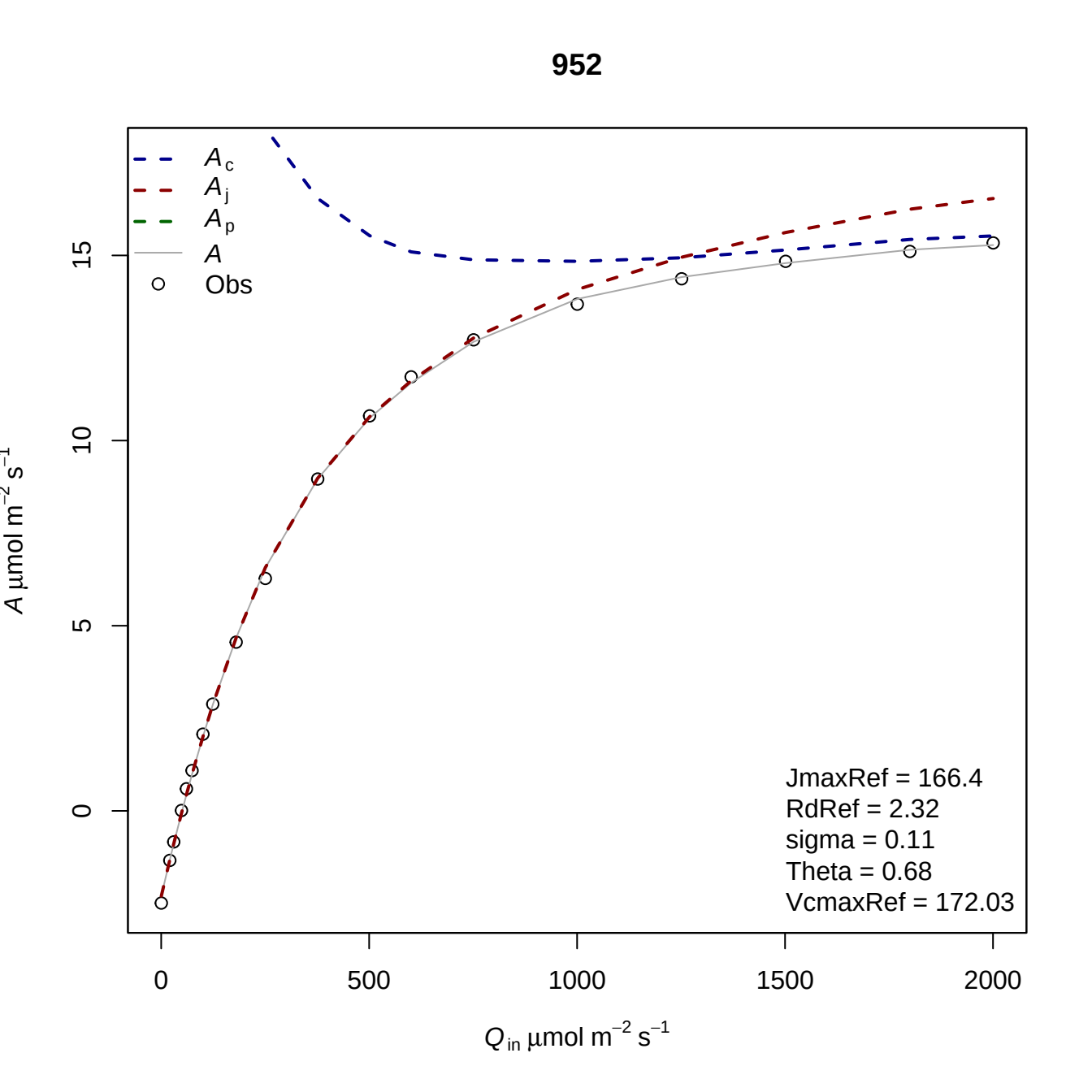

947
A_c
A_j
A_p
A
Obs
JmaxRef = 102.62
RdRef = 2.34
sigma = 0.25
Theta = −0.81
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

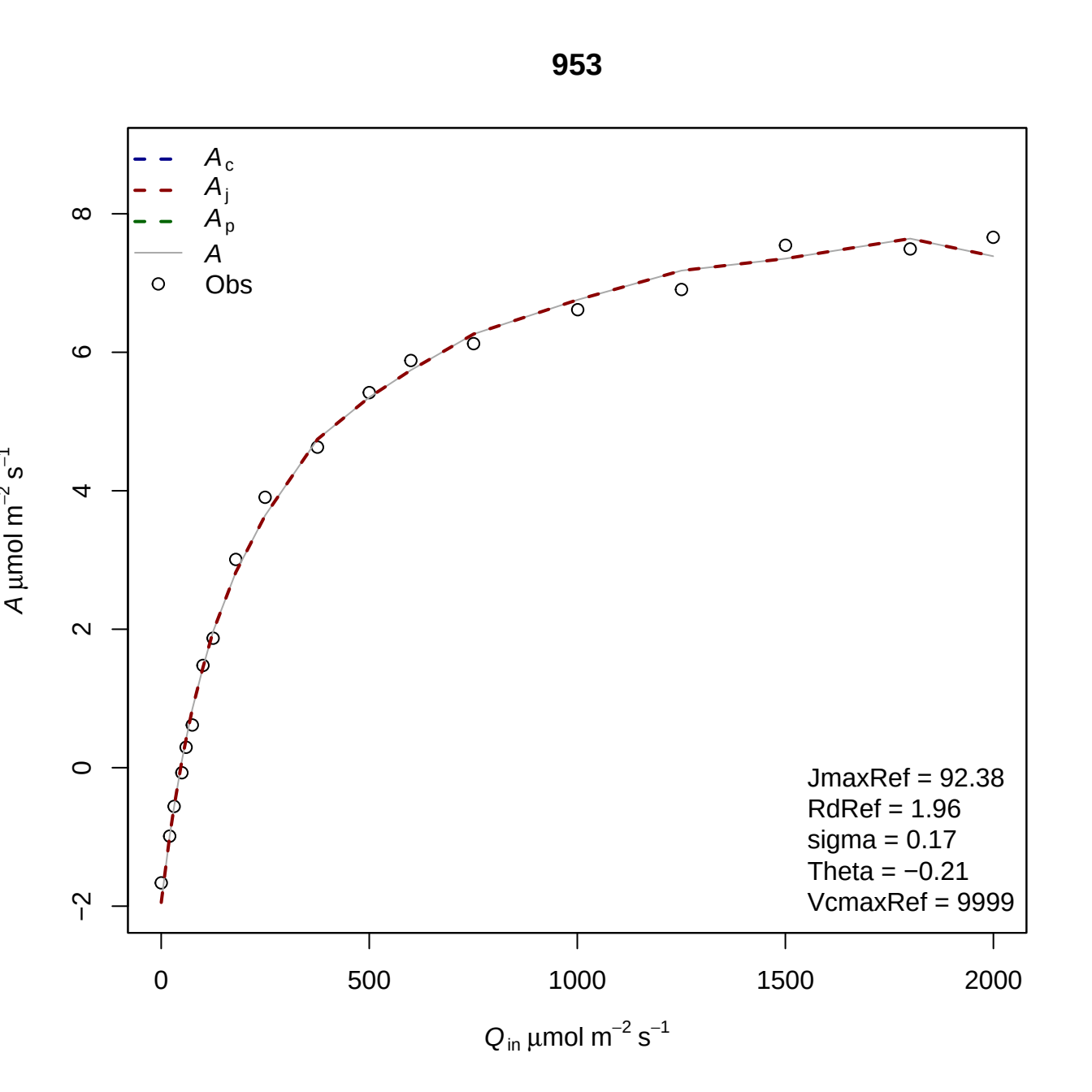

948
A_c
A_j
A_p
A
Obs
JmaxRef = 1024.24
RdRef = 2.6
sigma = 0.24
Theta = −15.87
VcmaxRef = 88.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000

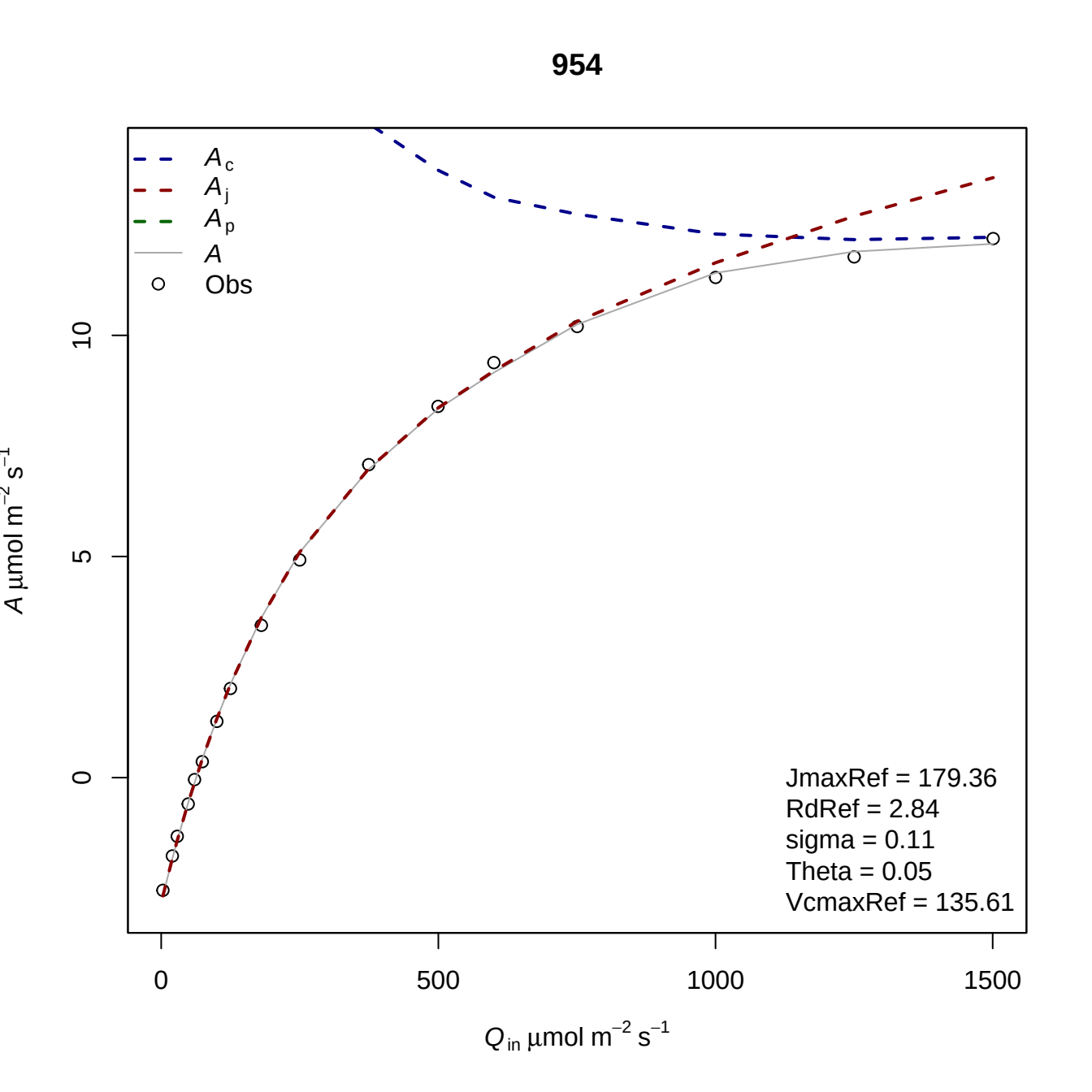

949
A_c
A_j
A_p
A
Obs
JmaxRef = 204.1
RdRef = 2.26
sigma = 0.24
Theta = −0.22
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

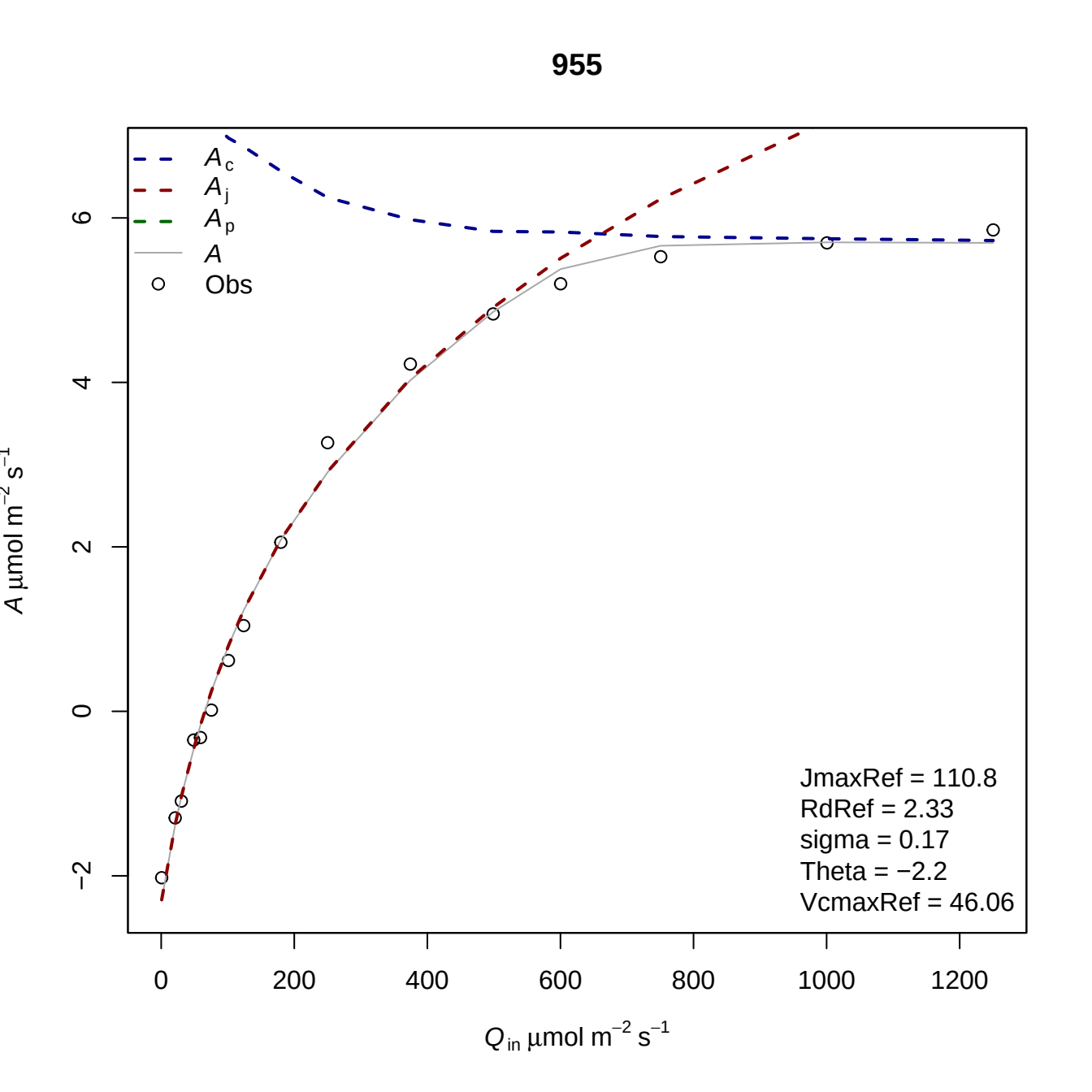

950
A_c
A_j
A_p
A
Obs
JmaxRef = 125.63
RdRef = 2.44
sigma = 0.16
Theta = 0.09
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

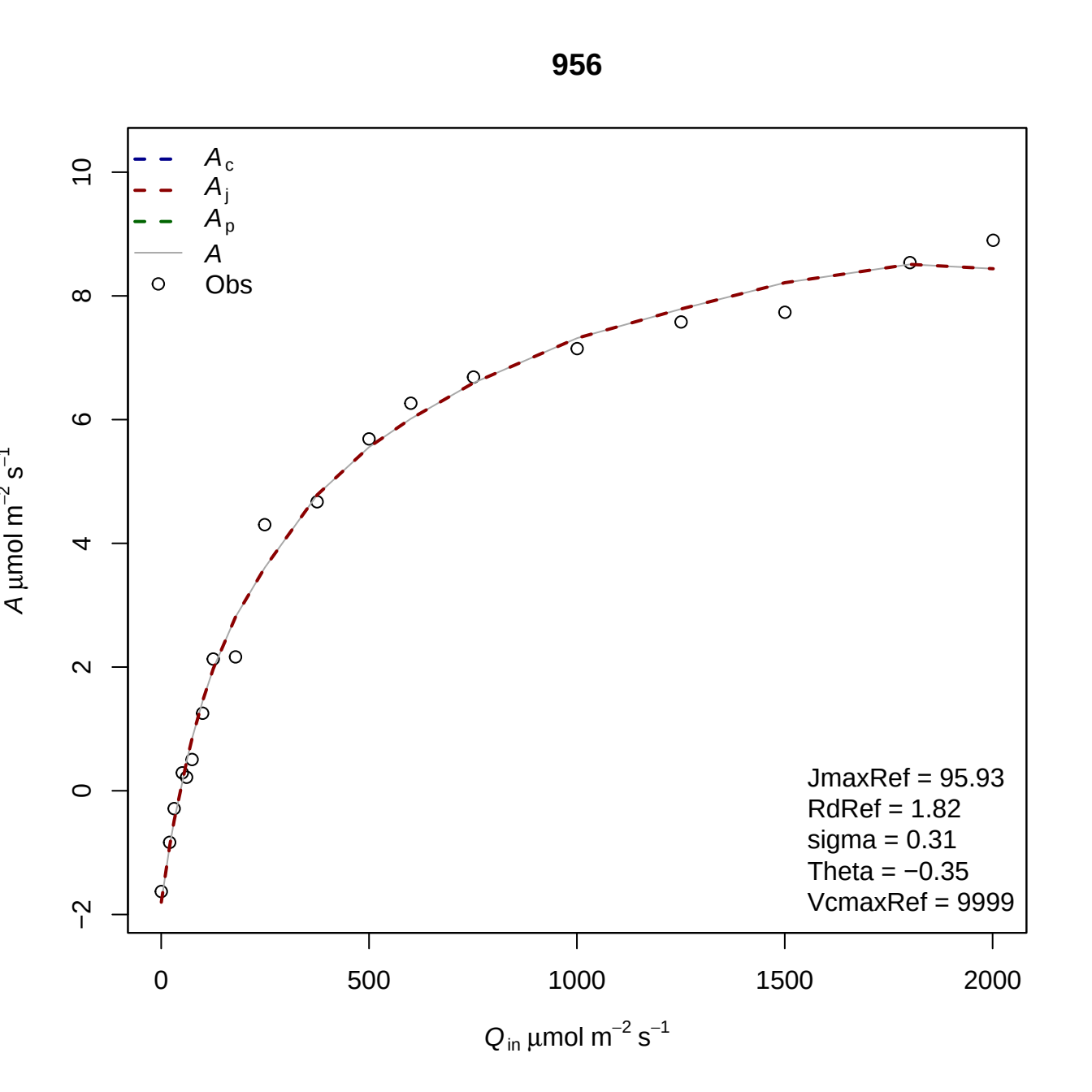

951
A_c
A_j
A_p
A
Obs
JmaxRef = 97.72
RdRef = 1.56
sigma = 0.14
Theta = −0.07
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

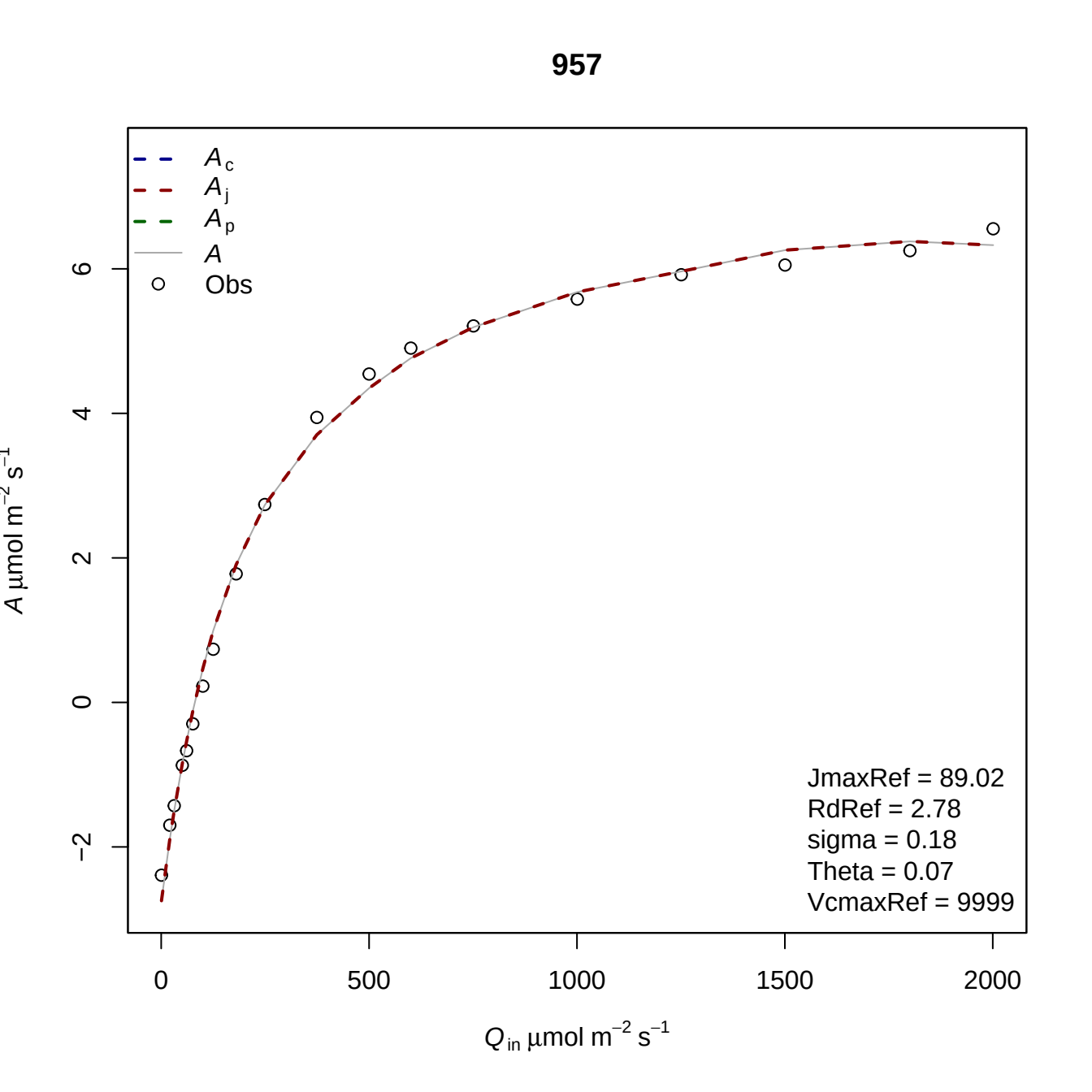

952
A_c
A_j
A_p
A
Obs
JmaxRef = 166.4
RdRef = 2.32
sigma = 0.11
Theta = 0.68
VcmaxRef = 172.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

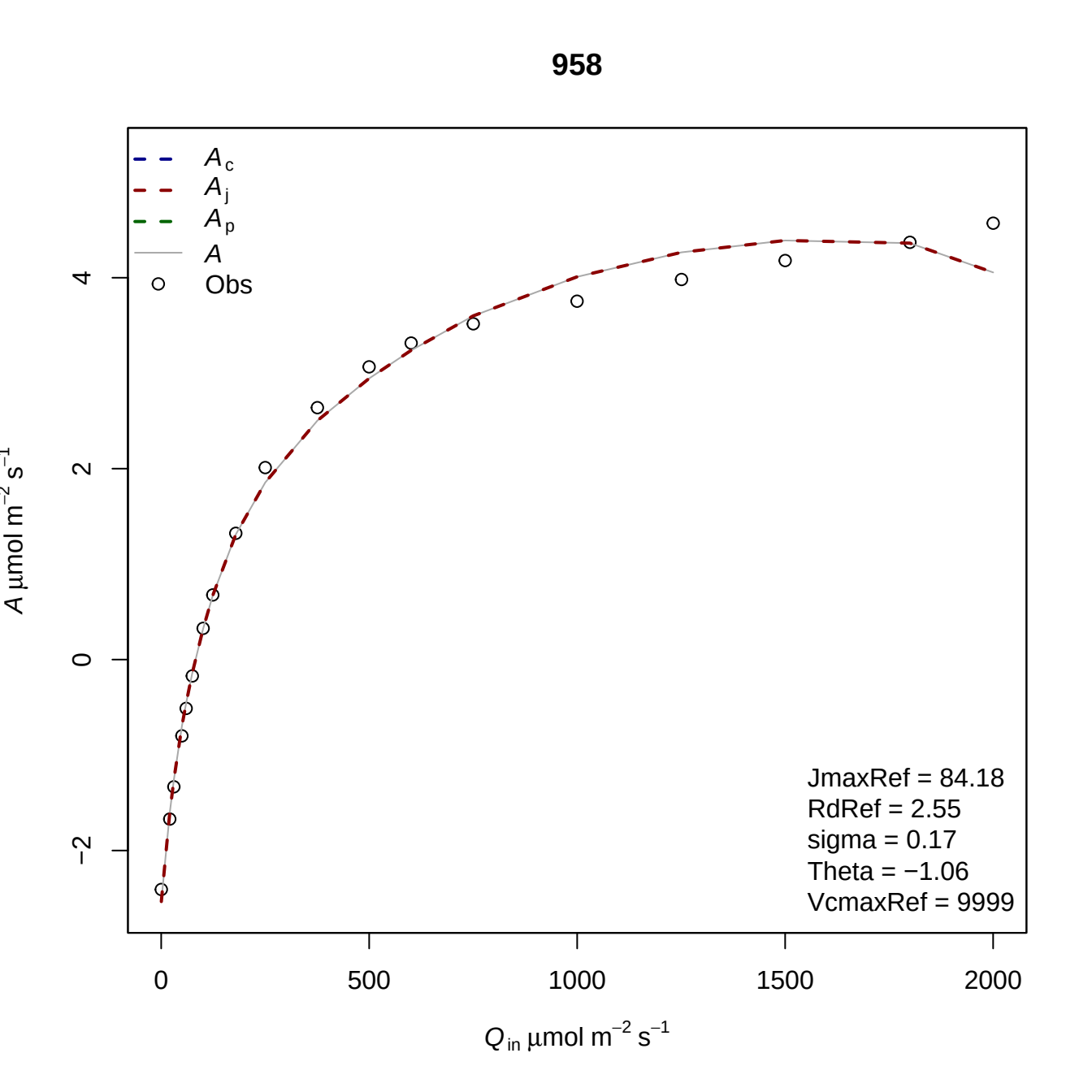

953
A_c
A_j
A_p
A
Obs
JmaxRef = 92.38
RdRef = 1.96
sigma = 0.17
Theta = −0.21
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

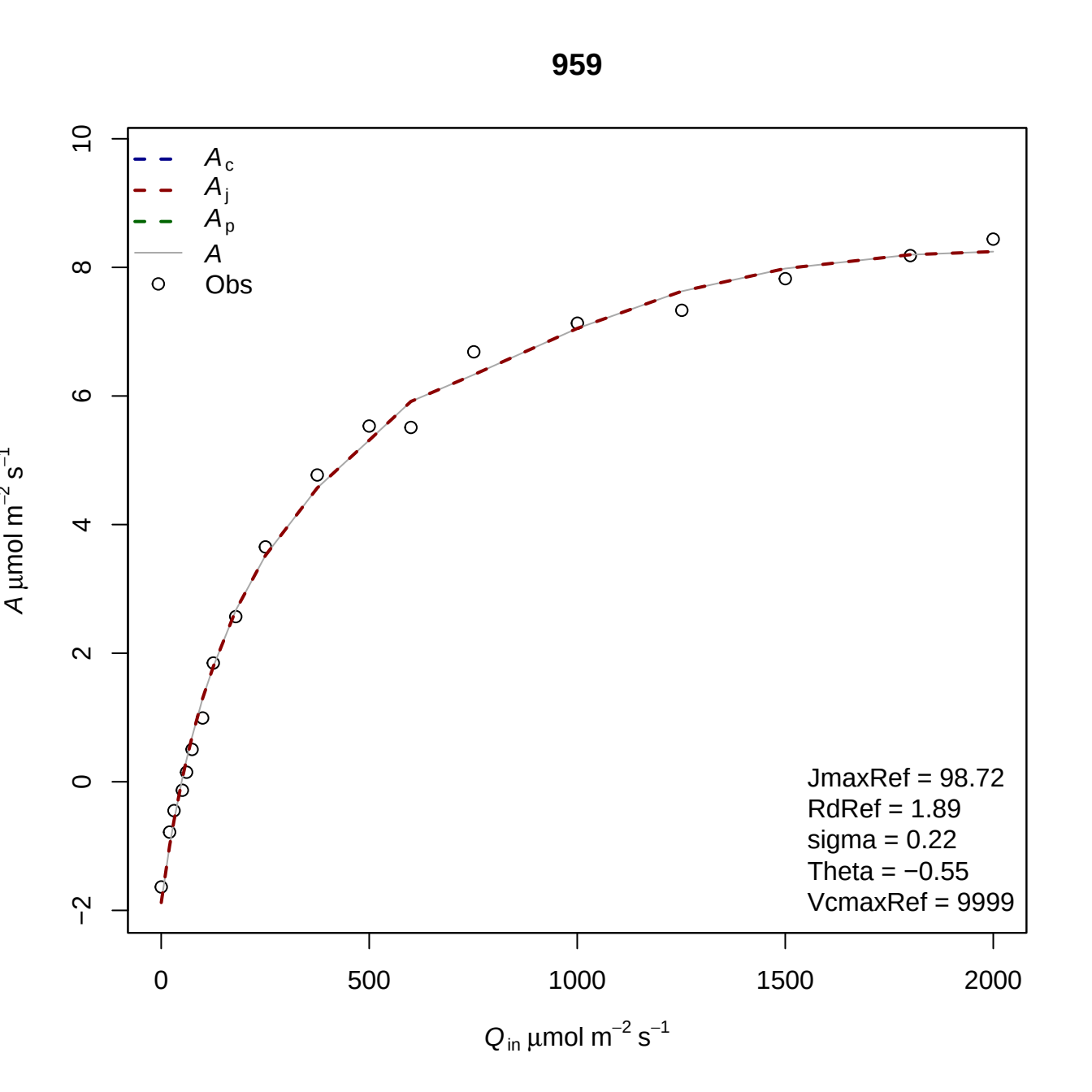

954
A_c
A_j
A_p
A
Obs
JmaxRef = 179.36
RdRef = 2.84
sigma = 0.11
Theta = 0.05
VcmaxRef = 135.61
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

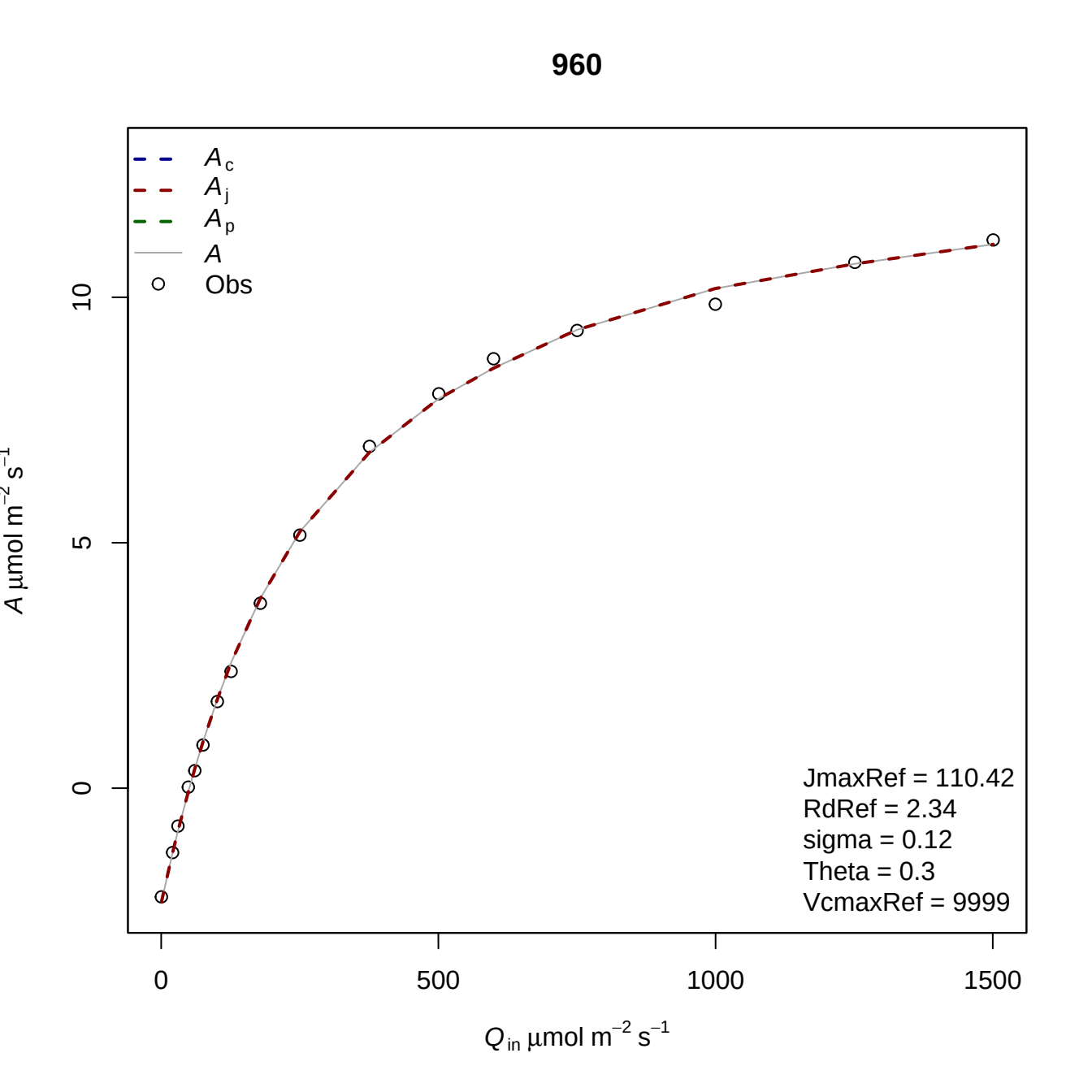

955
A_c
A_j
A_p
A
Obs
JmaxRef = 110.8
RdRef = 2.33
sigma = 0.17
Theta = −2.2
VcmaxRef = 46.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

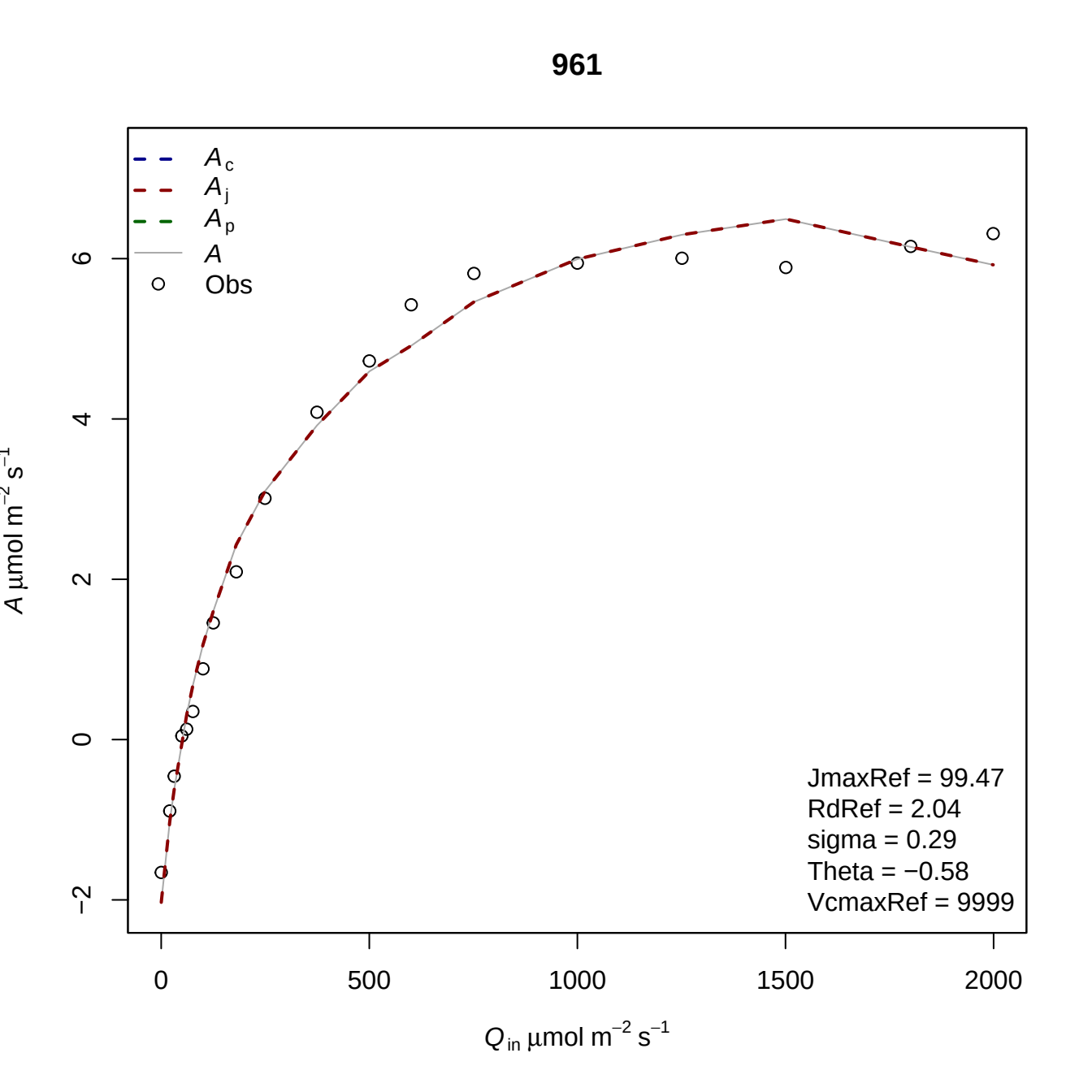

956
A_c
A_j
A_p
A
Obs
JmaxRef = 95.93
RdRef = 1.82
sigma = 0.31
Theta = −0.35
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

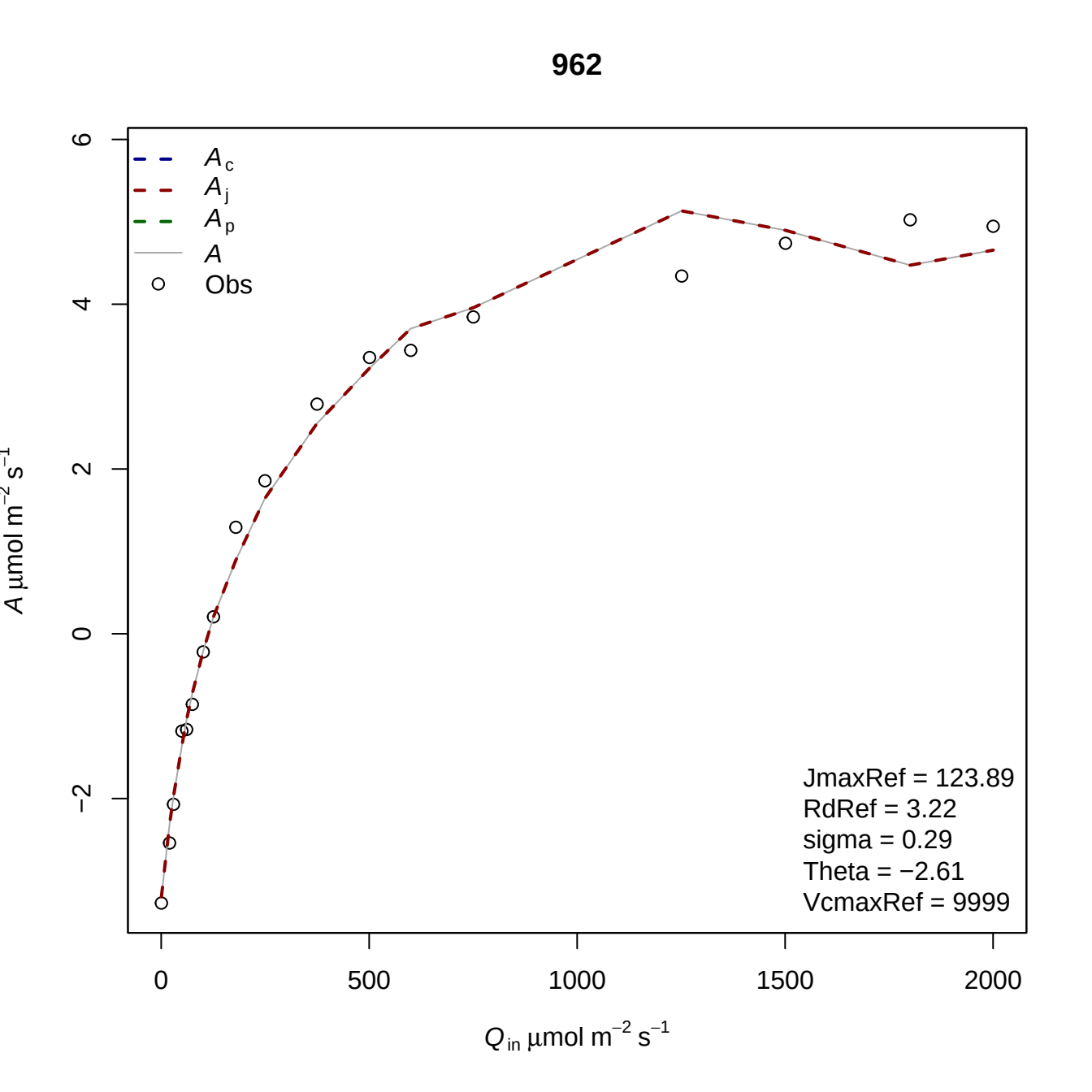

957
A_c
A_j
A_p
A
Obs
JmaxRef = 89.02
RdRef = 2.78
sigma = 0.18
Theta = 0.07
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

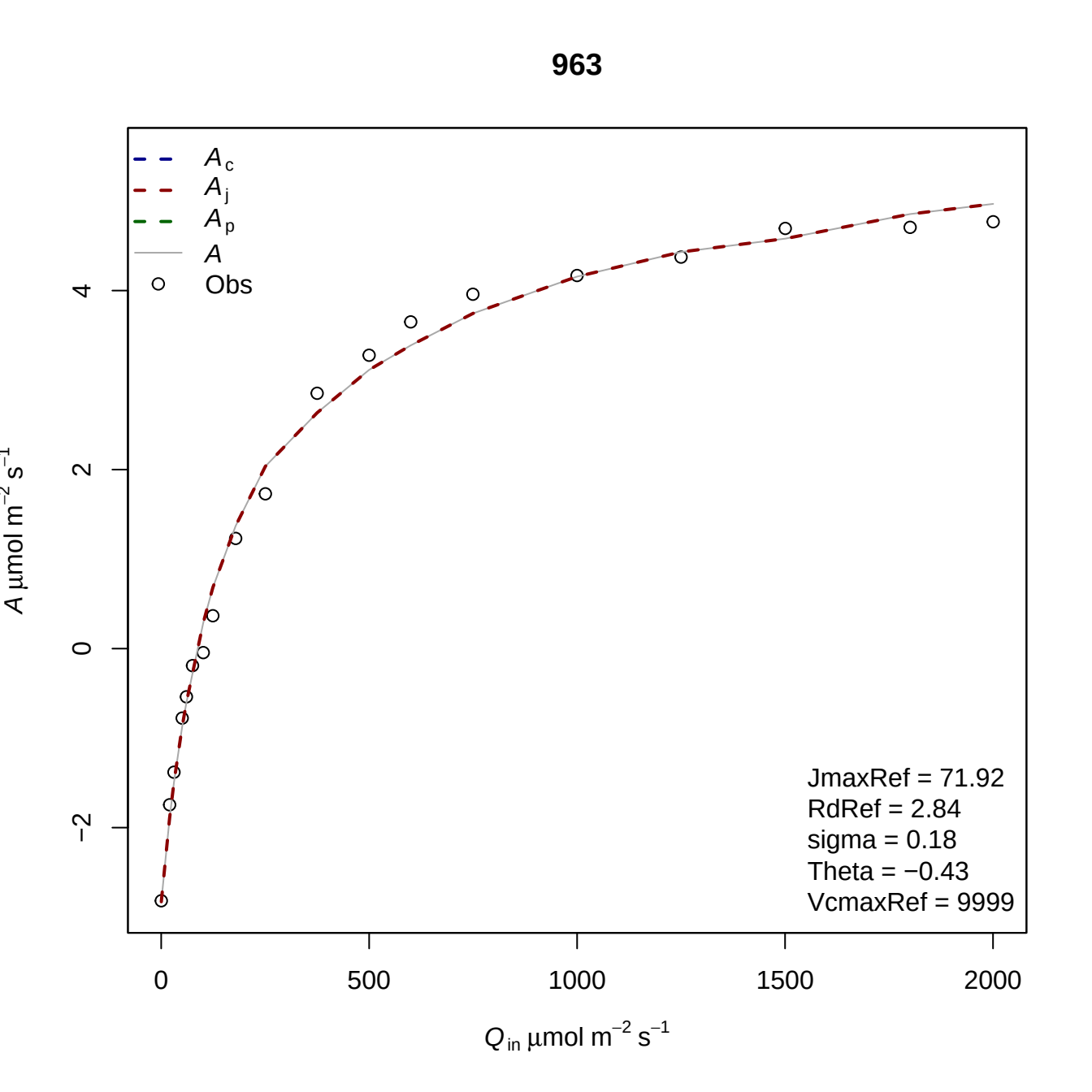

958
A_c
A_j
A_p
A
Obs
JmaxRef = 84.18
RdRef = 2.55
sigma = 0.17
Theta = −1.06
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

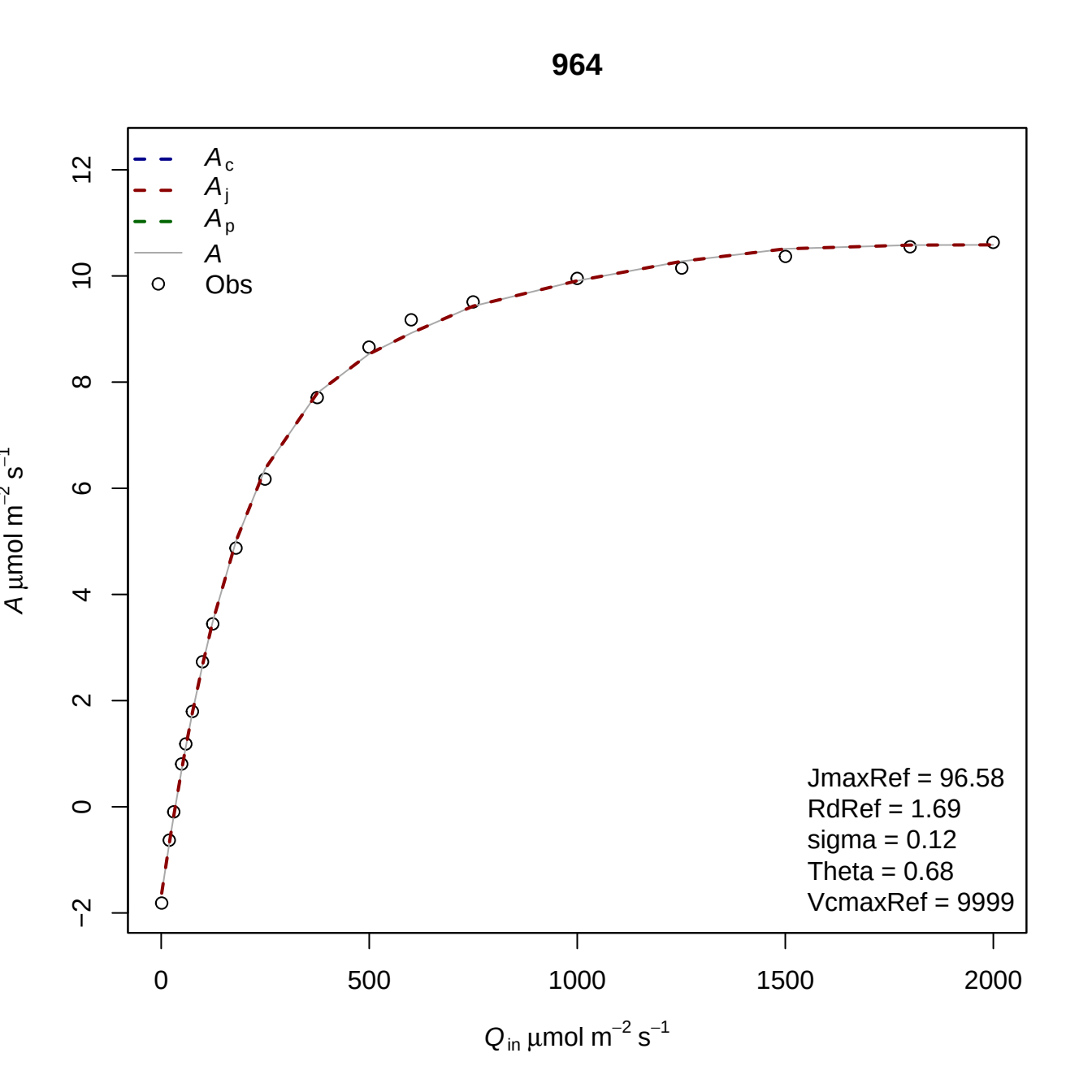

959
A_c
A_j
A_p
A
Obs
JmaxRef = 98.72
RdRef = 1.89
sigma = 0.22
Theta = −0.55
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

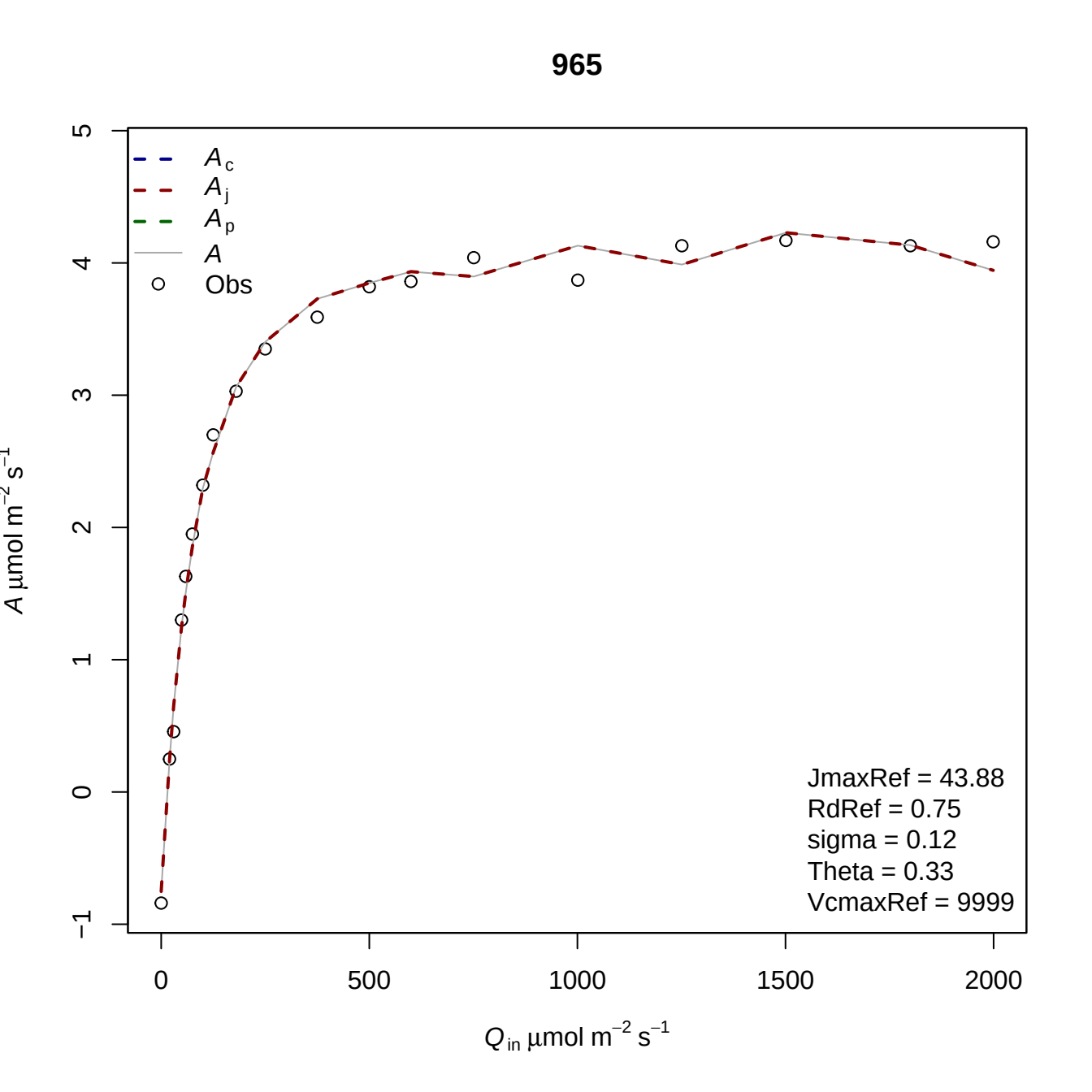

960
A_c
A_j
A_p
A
Obs
JmaxRef = 110.42
RdRef = 2.34
sigma = 0.12
Theta = 0.3
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

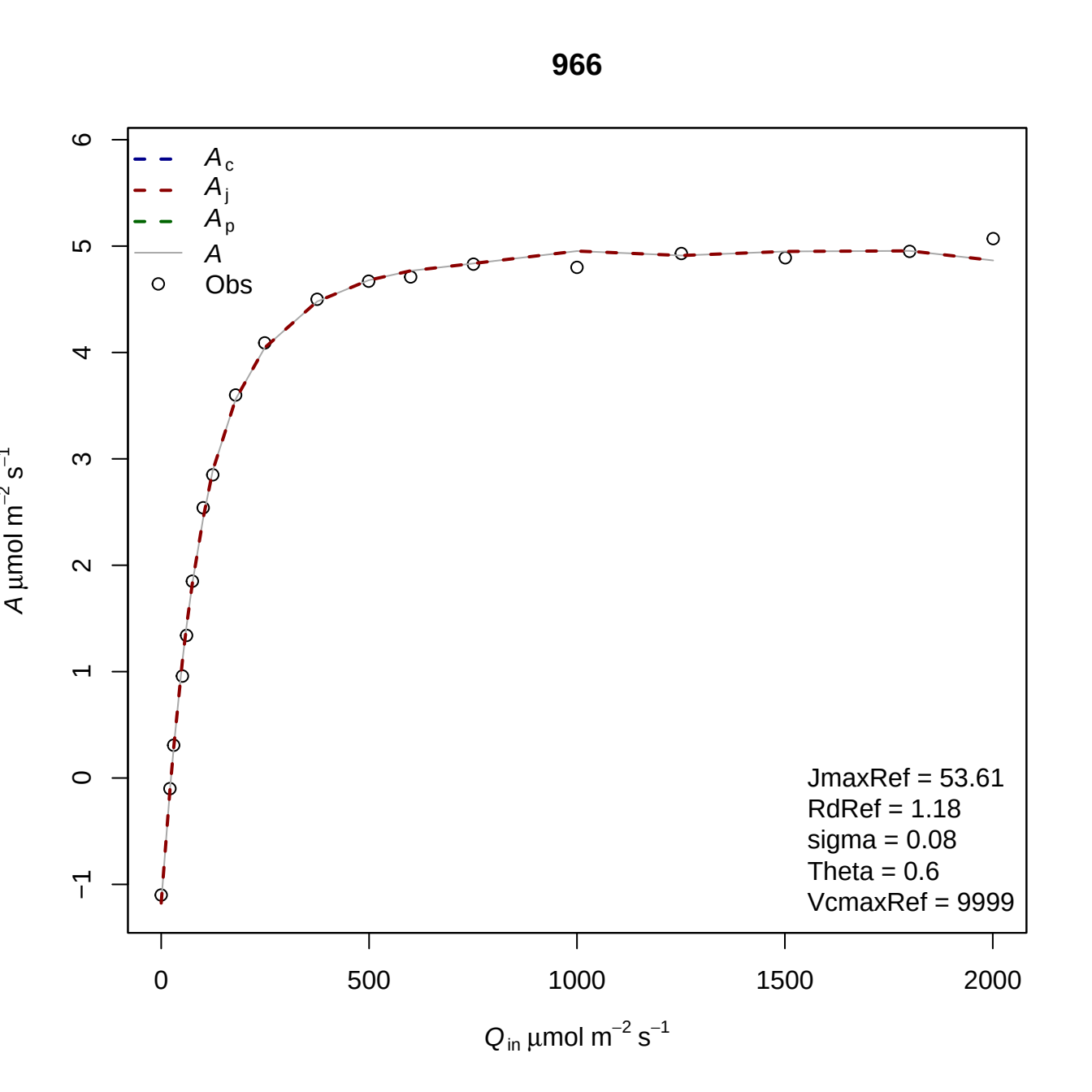

961
A_c
A_j
A_p
A
Obs
JmaxRef = 99.47
RdRef = 2.04
sigma = 0.29
Theta = −0.58
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

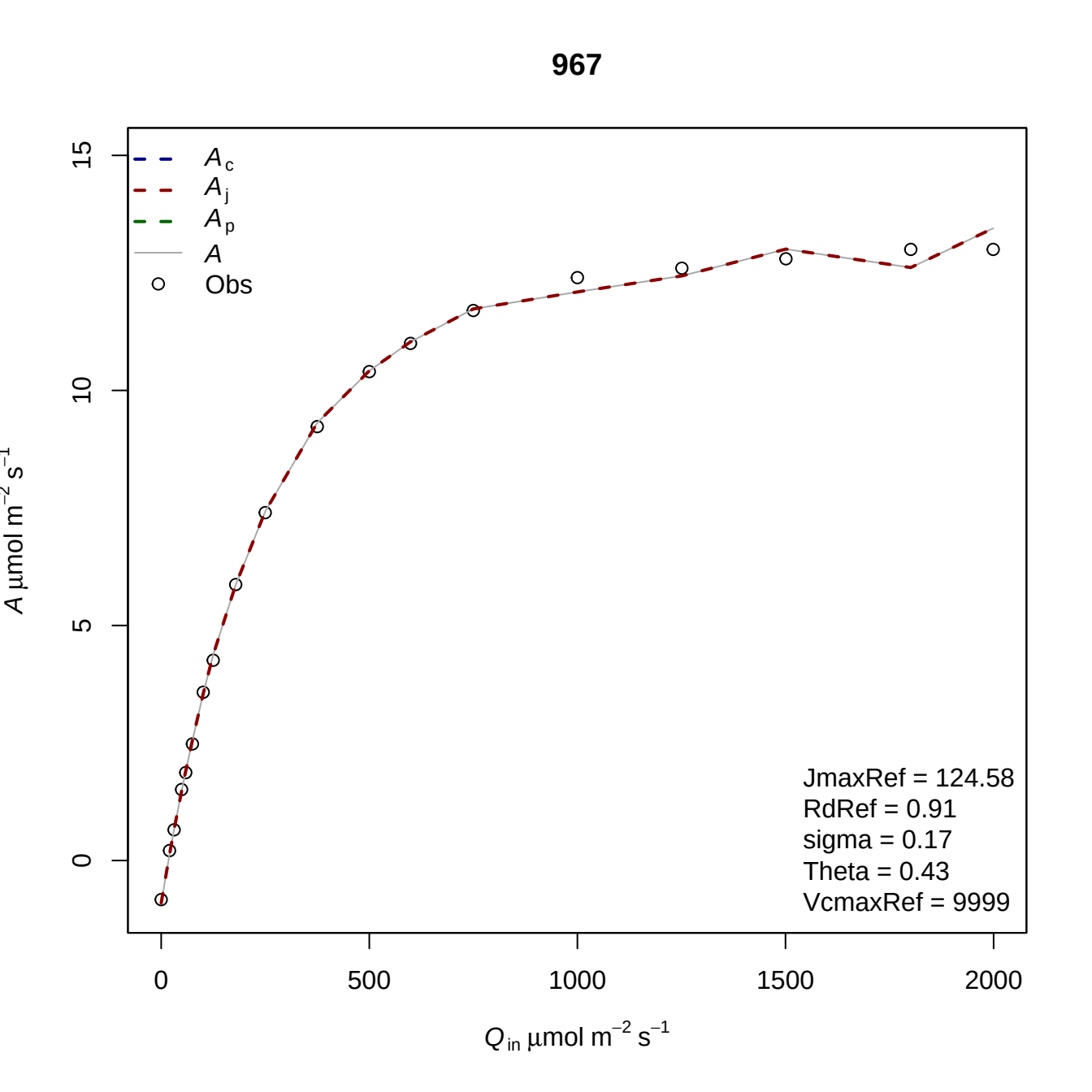

962
A_c
A_j
A_p
A
Obs
JmaxRef = 123.89
RdRef = 3.22
sigma = 0.29
Theta = −2.61
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

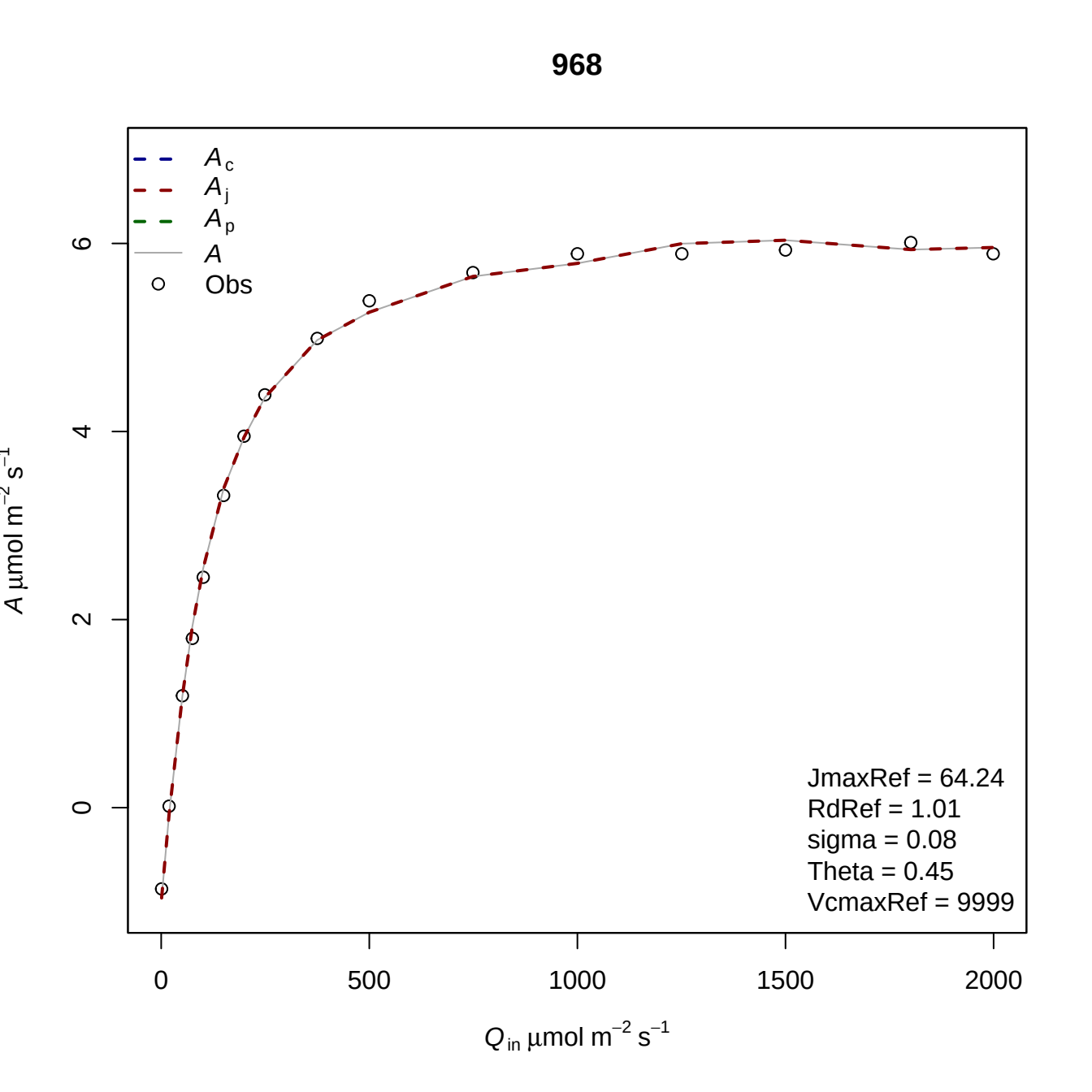

963
A_c
A_j
A_p
A
Obs
JmaxRef = 71.92
RdRef = 2.84
sigma = 0.18
Theta = −0.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

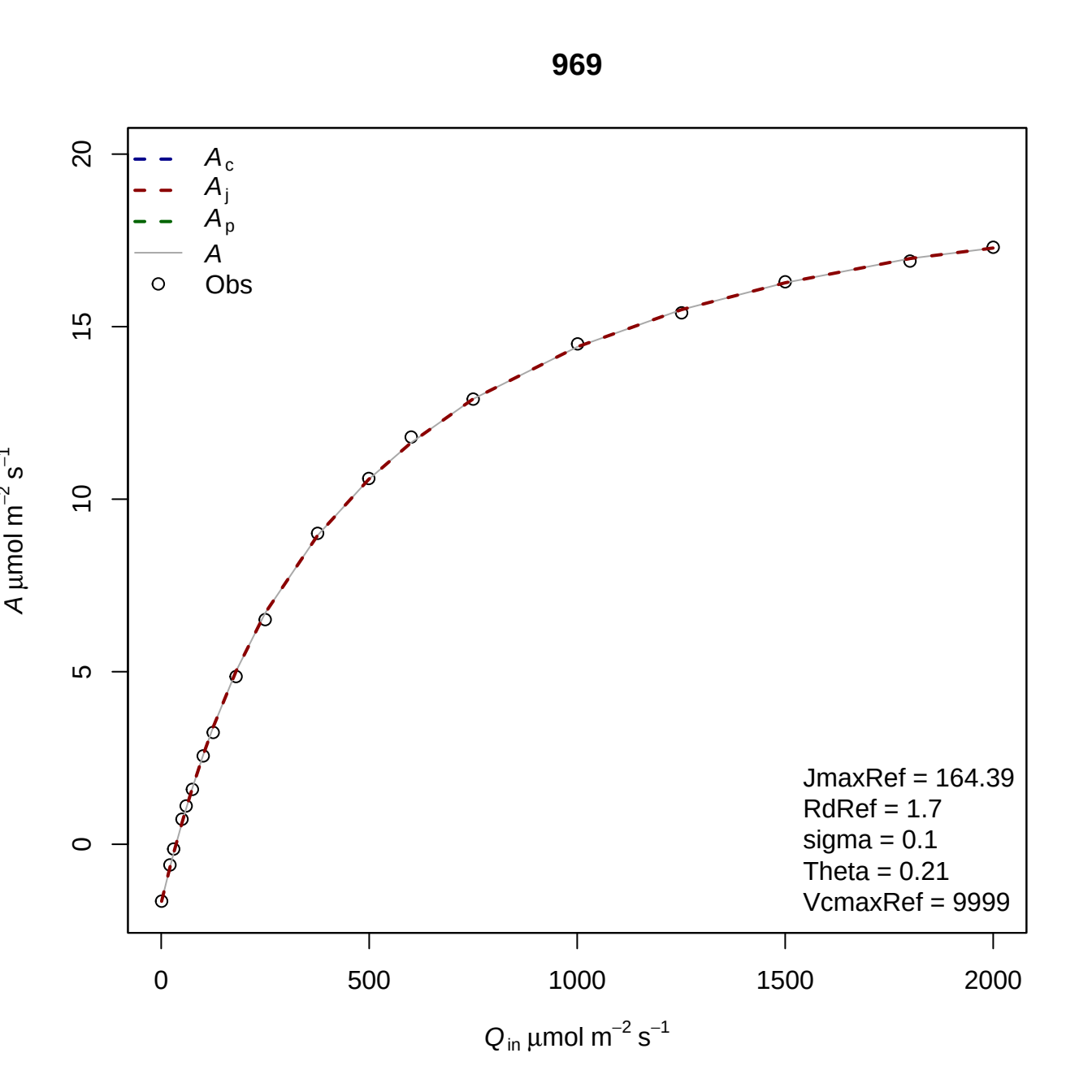

964
A_c
A_j
A_p
A
Obs
JmaxRef = 96.58
RdRef = 1.69
sigma = 0.12
Theta = 0.68
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

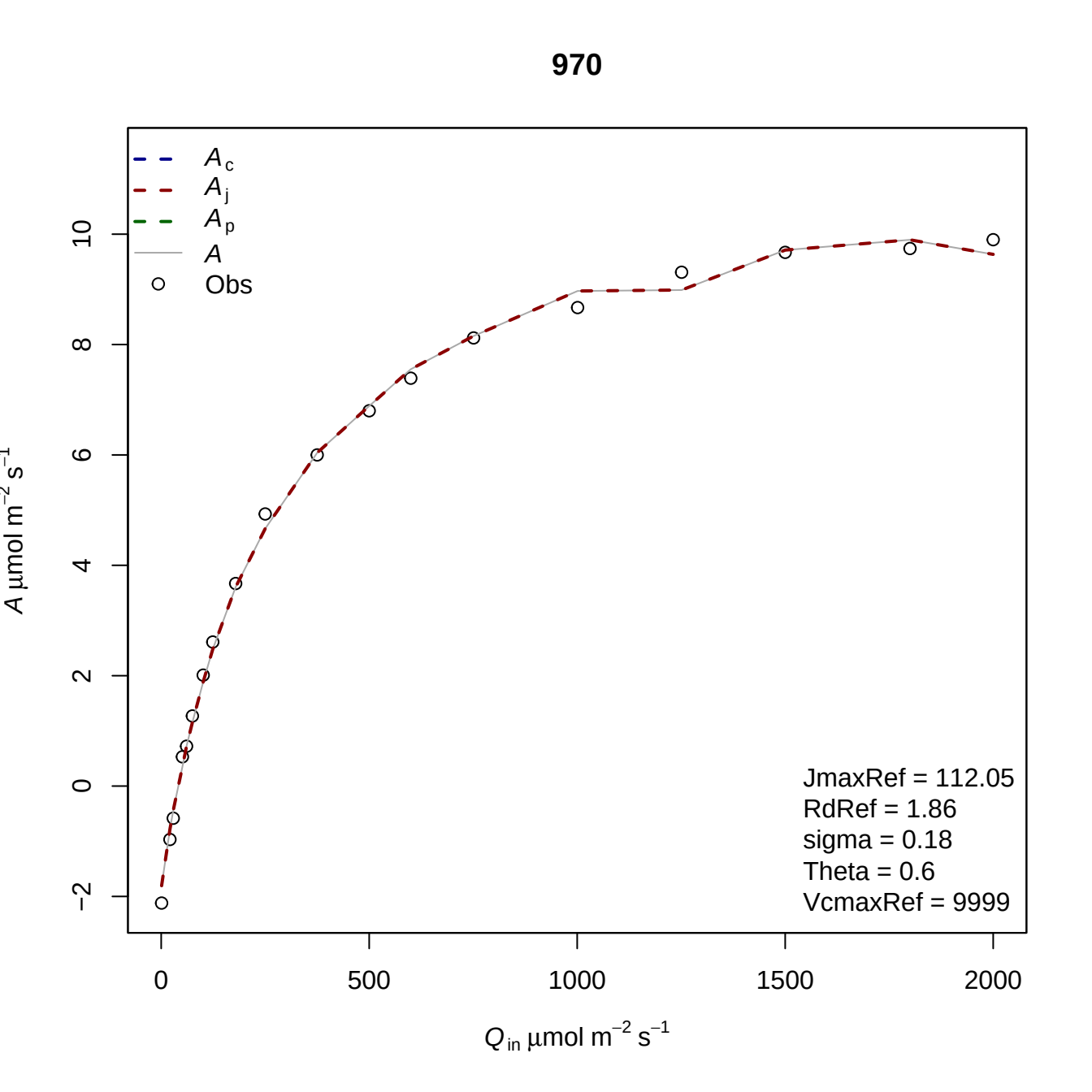

965
A_c
A_j
A_p
A
Obs
JmaxRef = 43.88
RdRef = 0.75
sigma = 0.12
Theta = 0.33
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

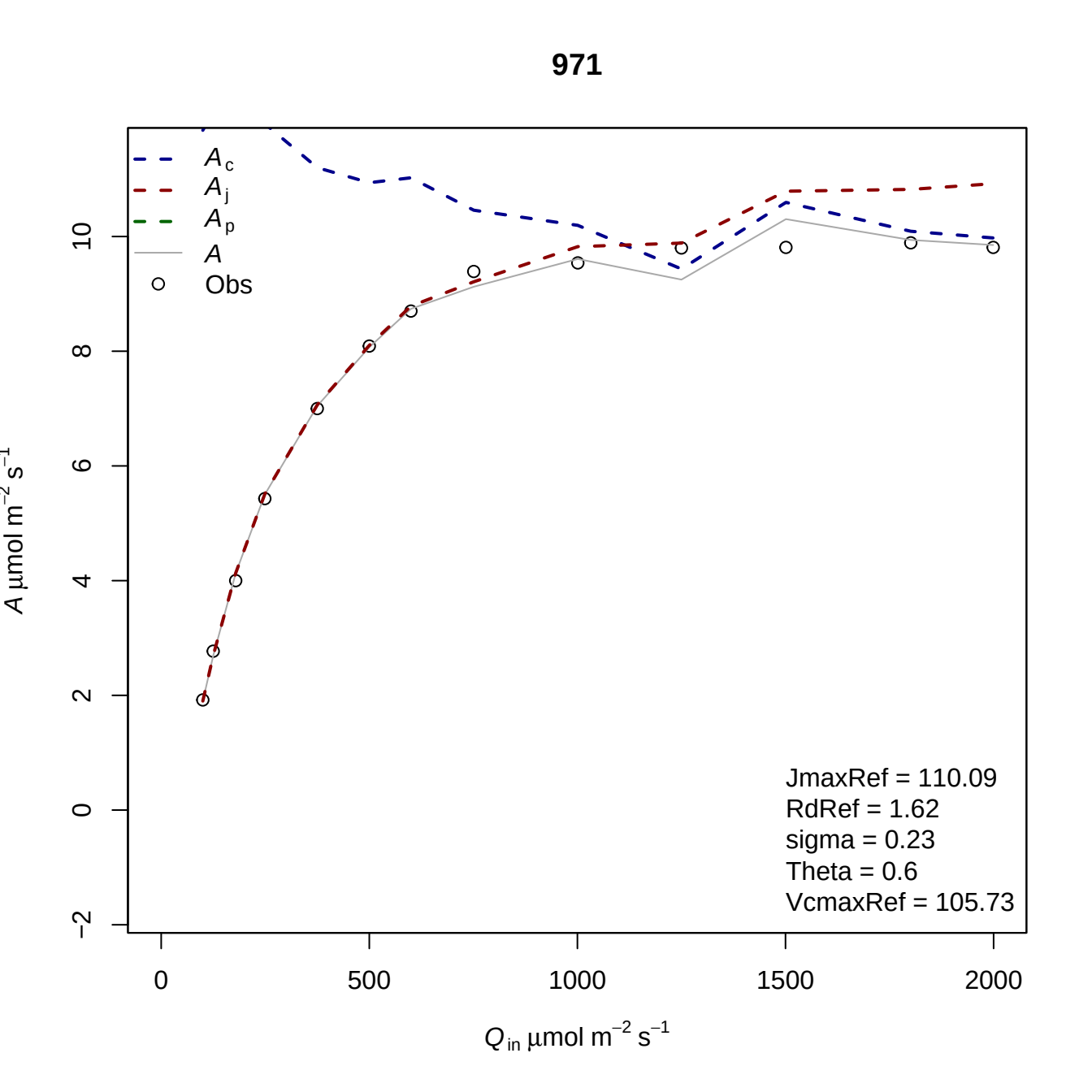

966
A_c
A_j
A_p
A
Obs
JmaxRef = 53.61
RdRef = 1.18
sigma = 0.08
Theta = 0.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

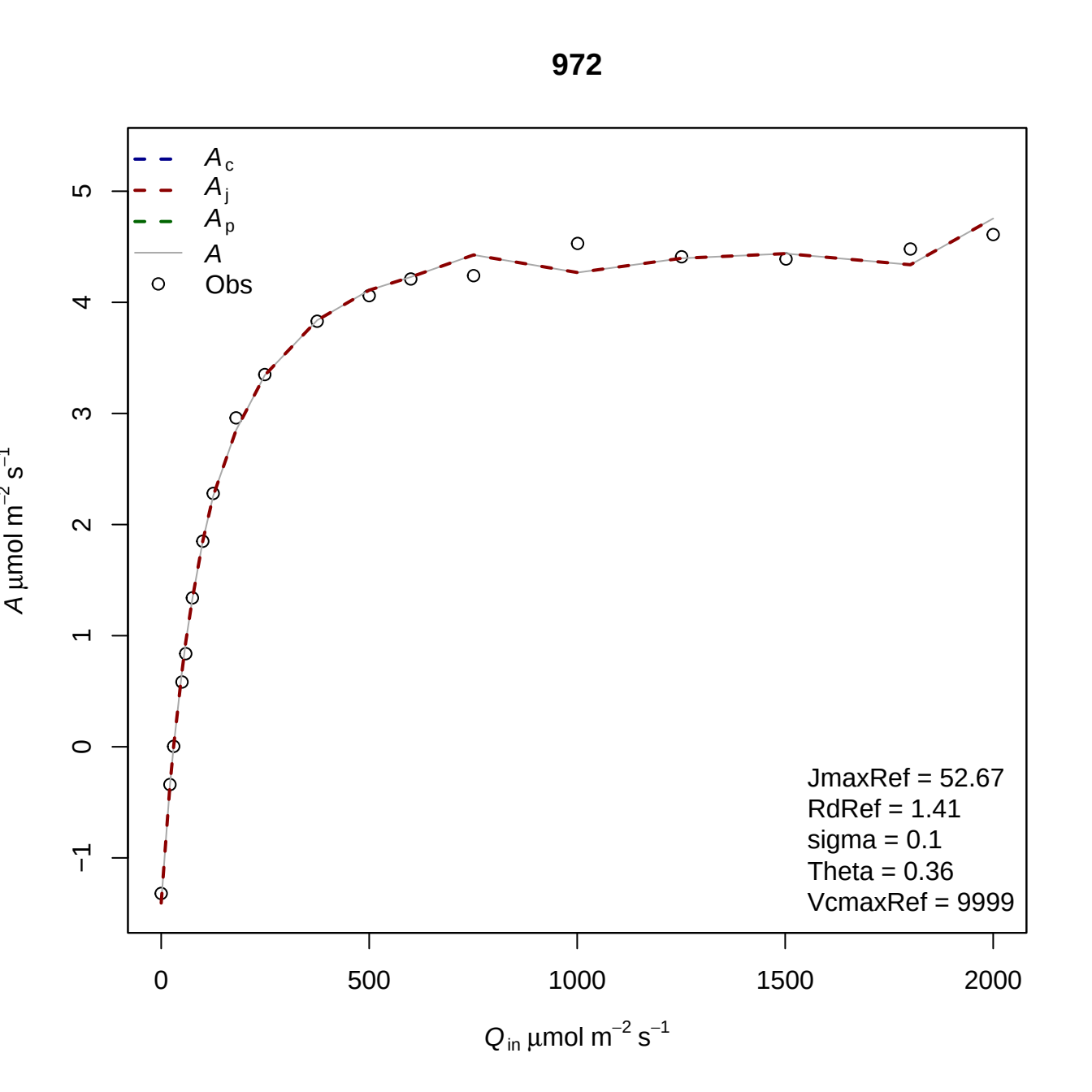

967
A_c
A_j
A_p
A
Obs
JmaxRef = 124.58
RdRef = 0.91
sigma = 0.17
Theta = 0.43
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

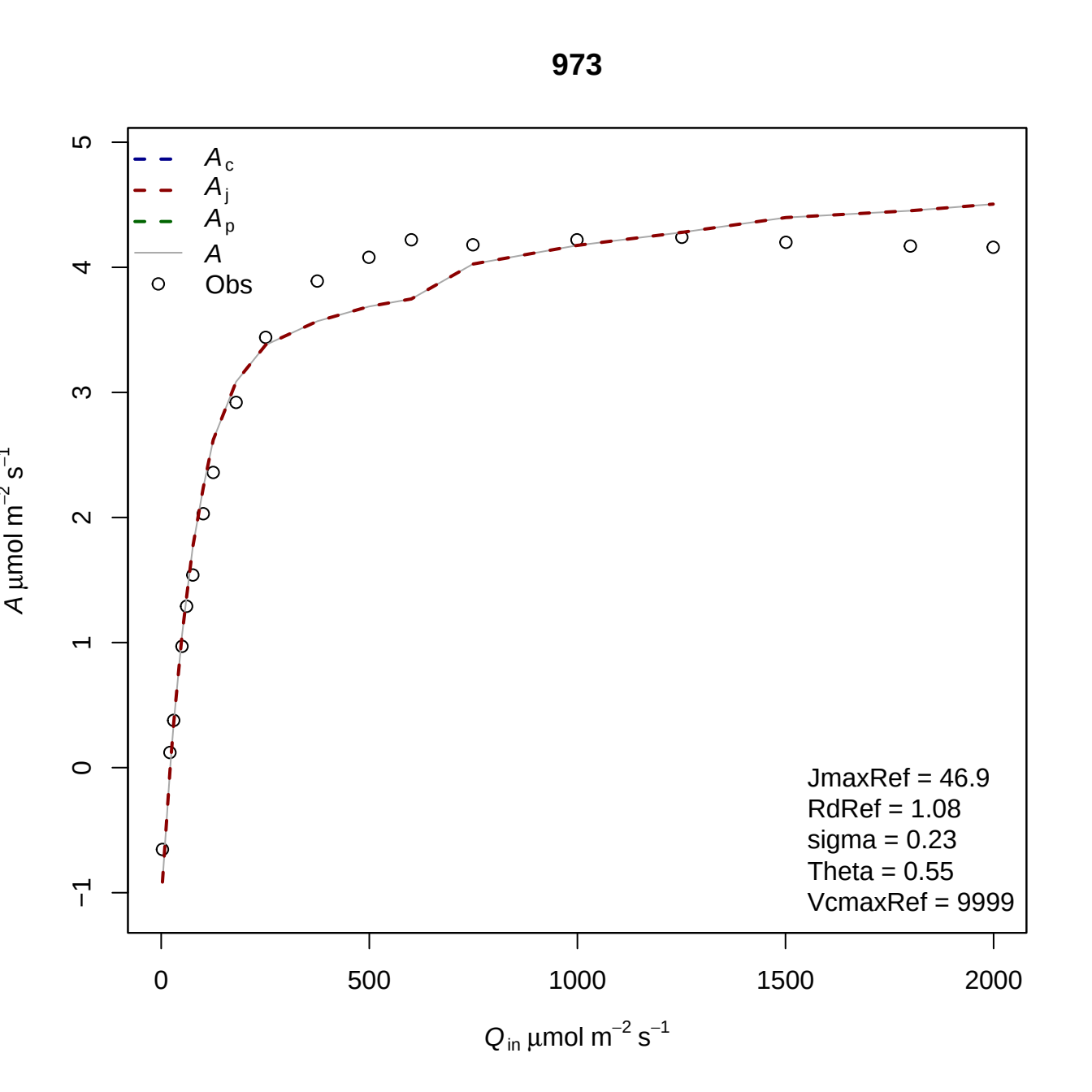

968
A_c
A_j
A_p
A
Obs
JmaxRef = 64.24
RdRef = 1.01
sigma = 0.08
Theta = 0.45
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

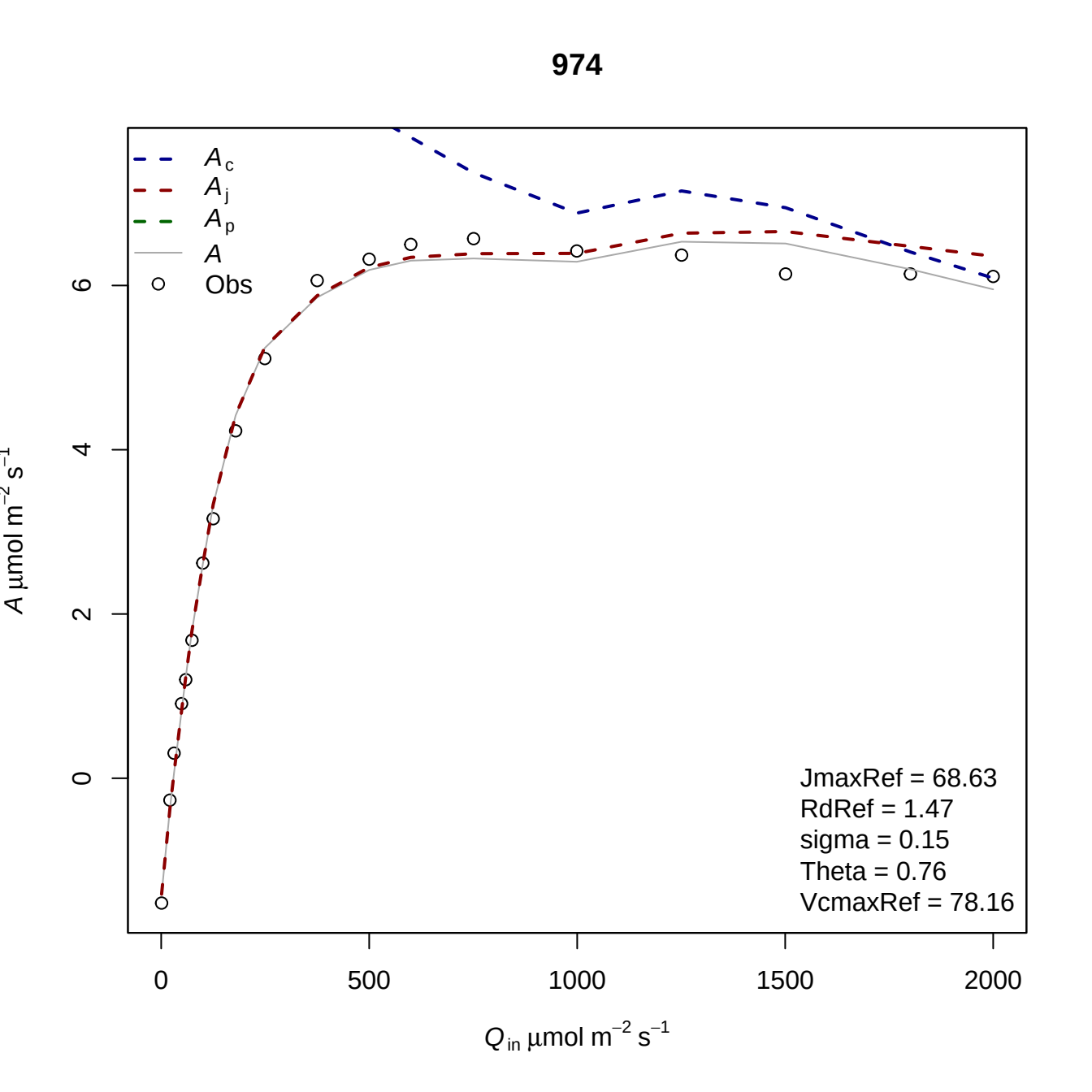

969
A_c
A_j
A_p
A
Obs
JmaxRef = 164.39
RdRef = 1.7
sigma = 0.1
Theta = 0.21
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

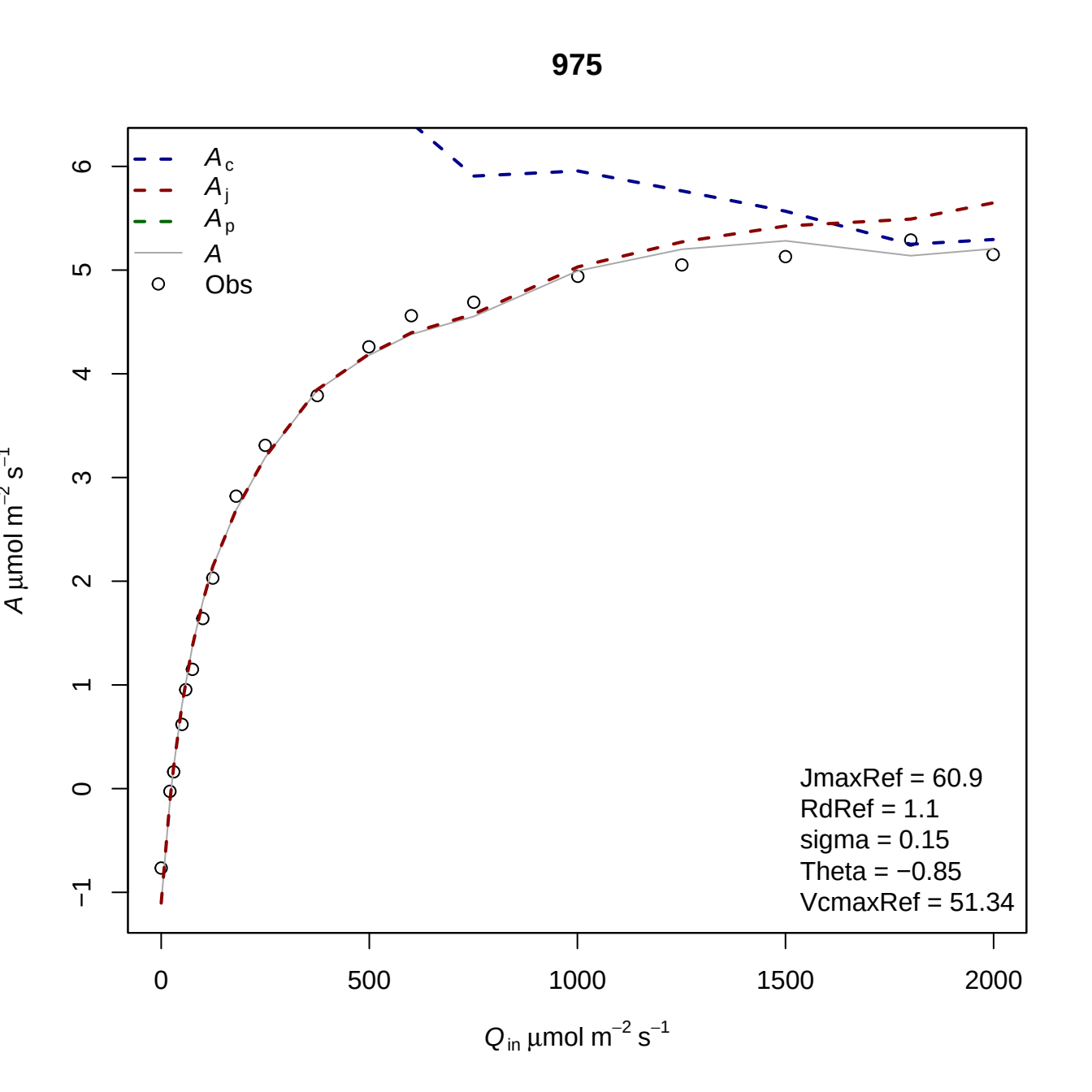

970
A_c
A_j
A_p
A
Obs
JmaxRef = 112.05
RdRef = 1.86
sigma = 0.18
Theta = 0.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

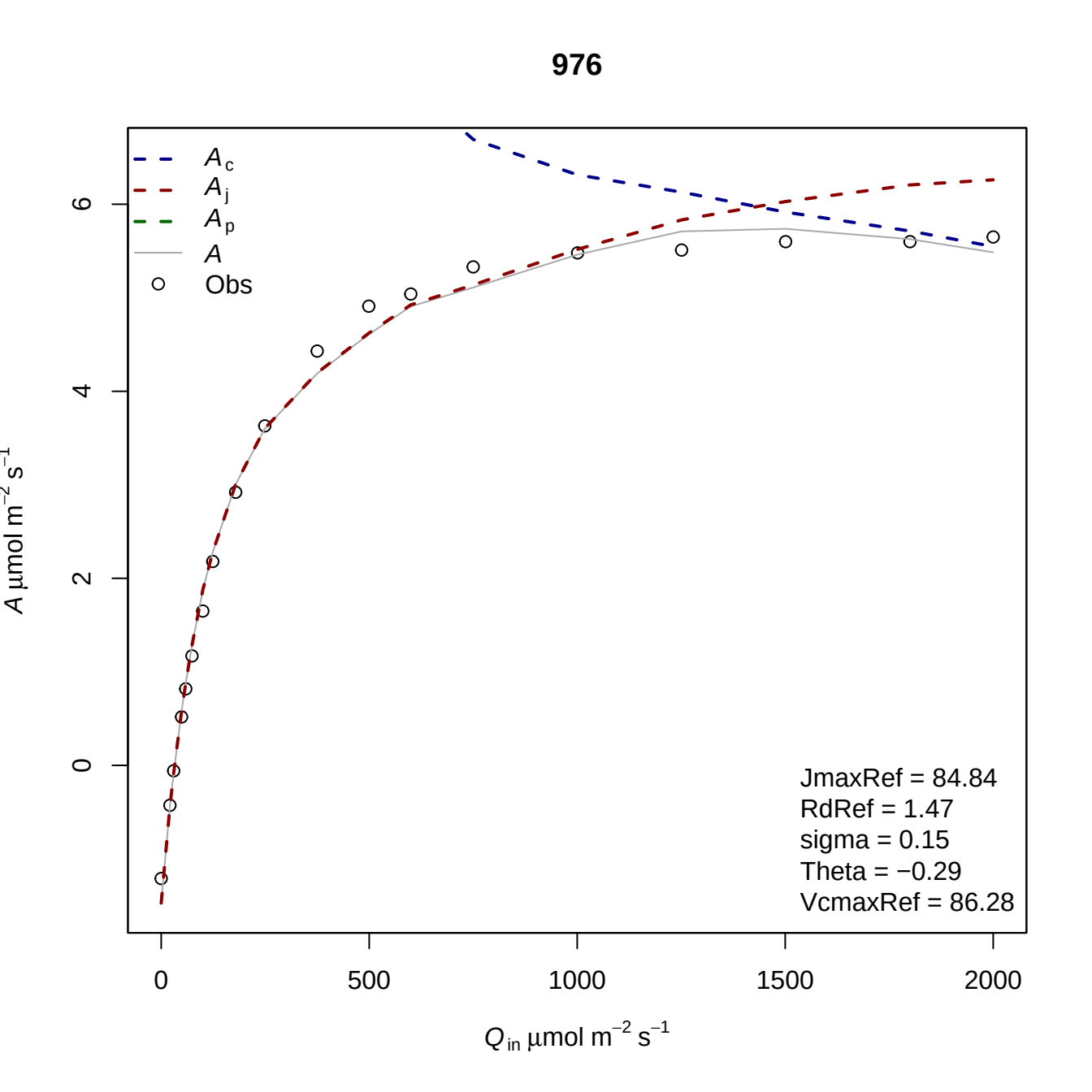

971
A_c
A_j
A_p
A
Obs
JmaxRef = 110.09
RdRef = 1.62
sigma = 0.23
Theta = 0.6
VcmaxRef = 105.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

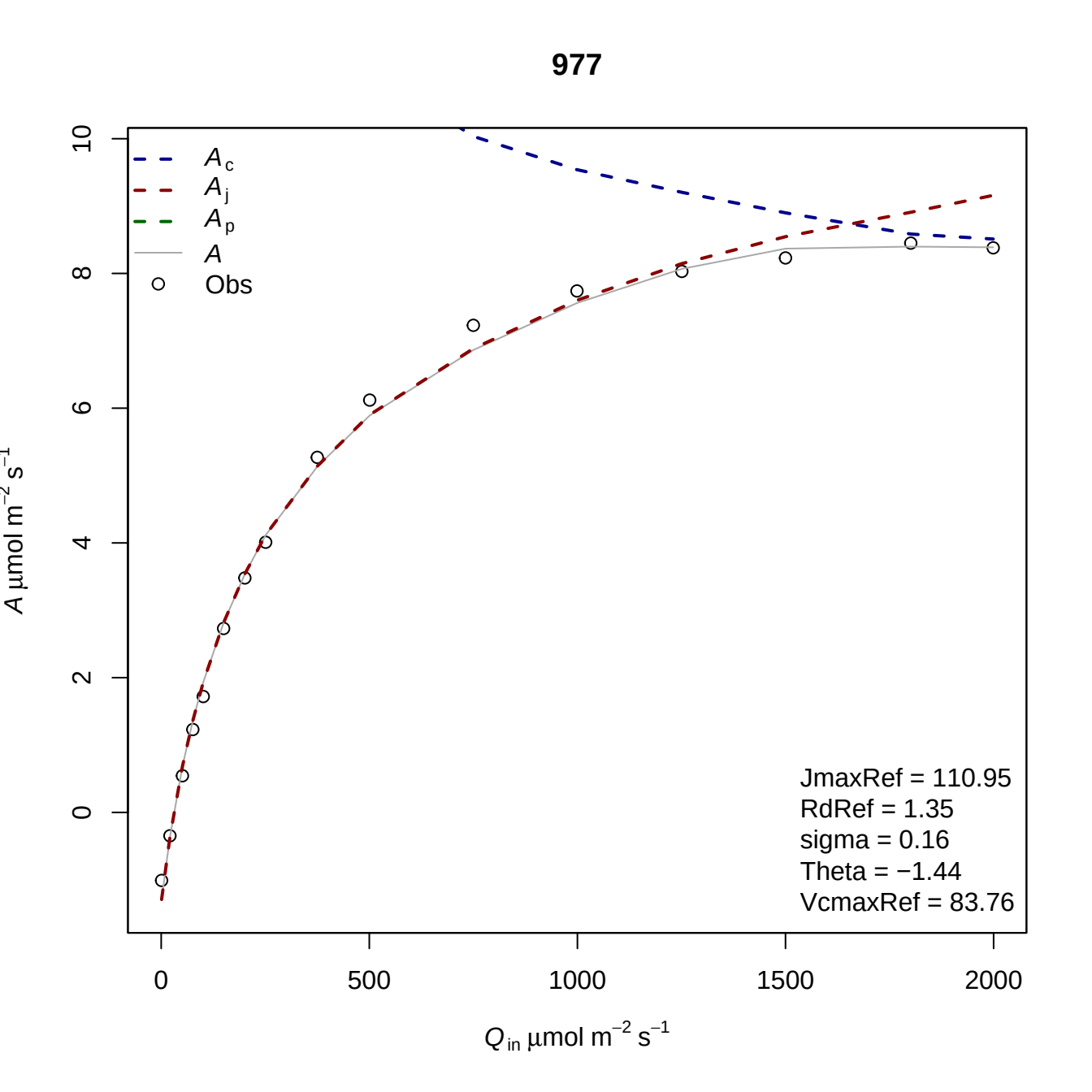

972
A_c
A_j
A_p
A
Obs
JmaxRef = 52.67
RdRef = 1.41
sigma = 0.1
Theta = 0.36
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

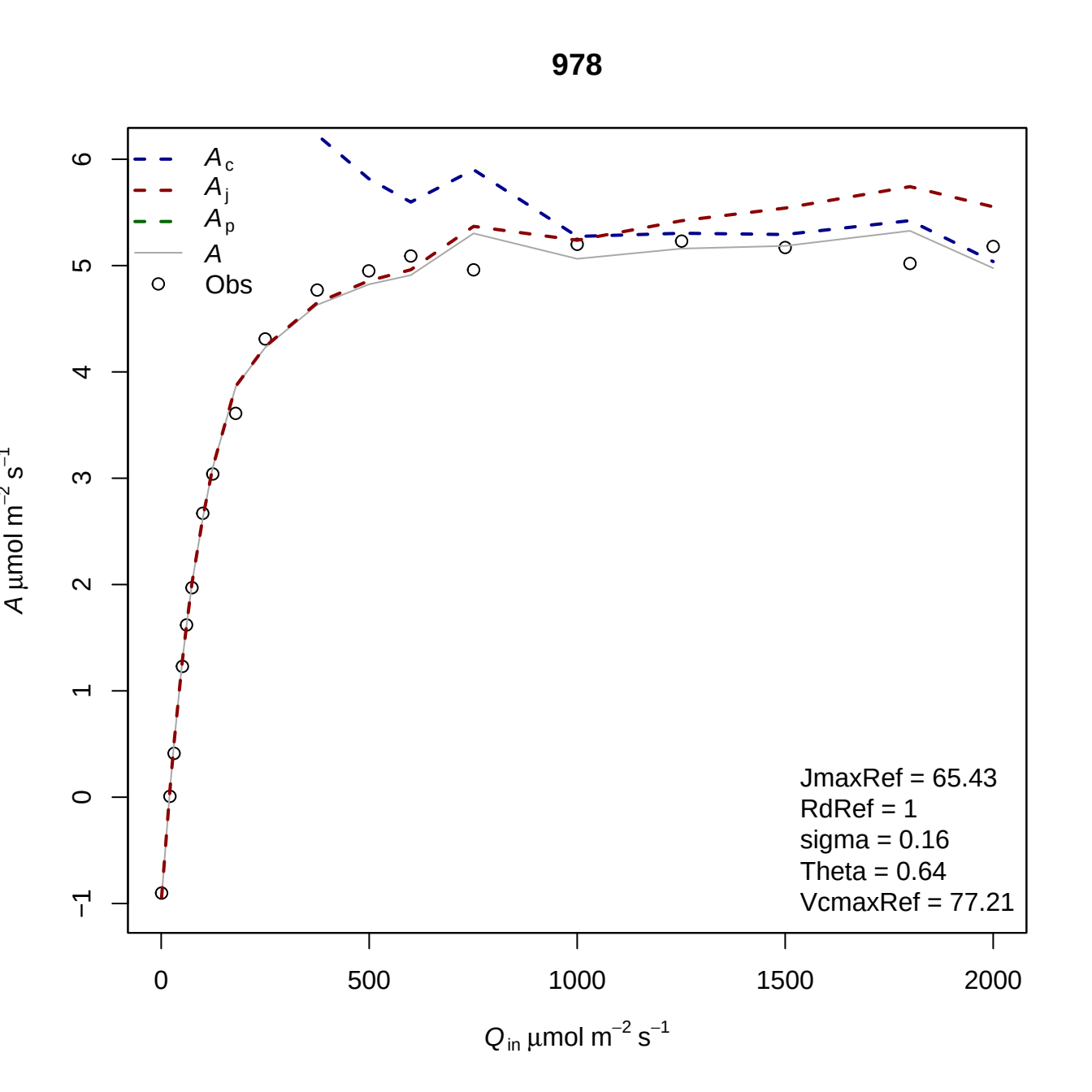

973
A_c
A_j
A_p
A
Obs
JmaxRef = 46.9
RdRef = 1.08
sigma = 0.23
Theta = 0.55
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

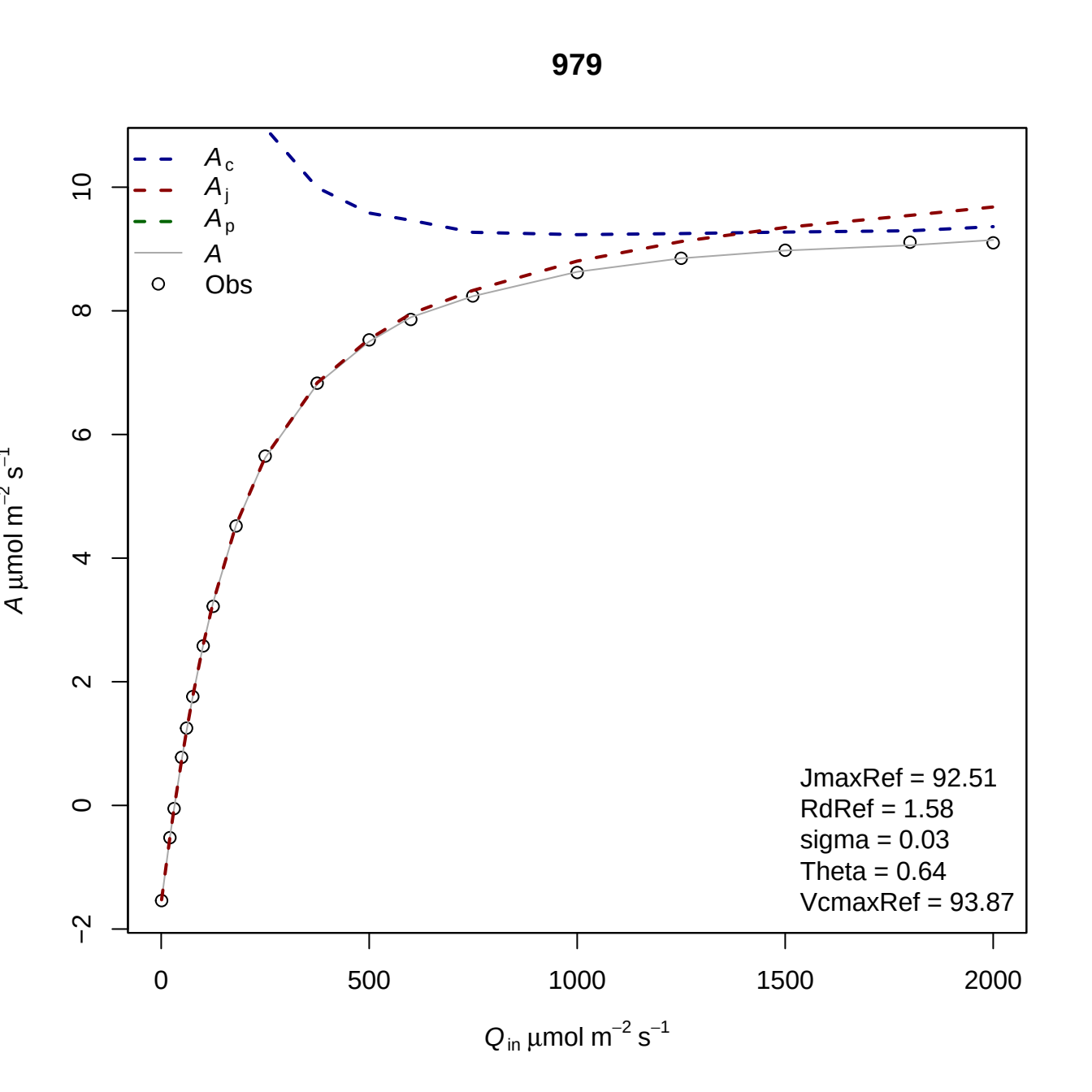

974
A_c
A_j
A_p
A
Obs
JmaxRef = 68.63
RdRef = 1.47
sigma = 0.15
Theta = 0.76
VcmaxRef = 78.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

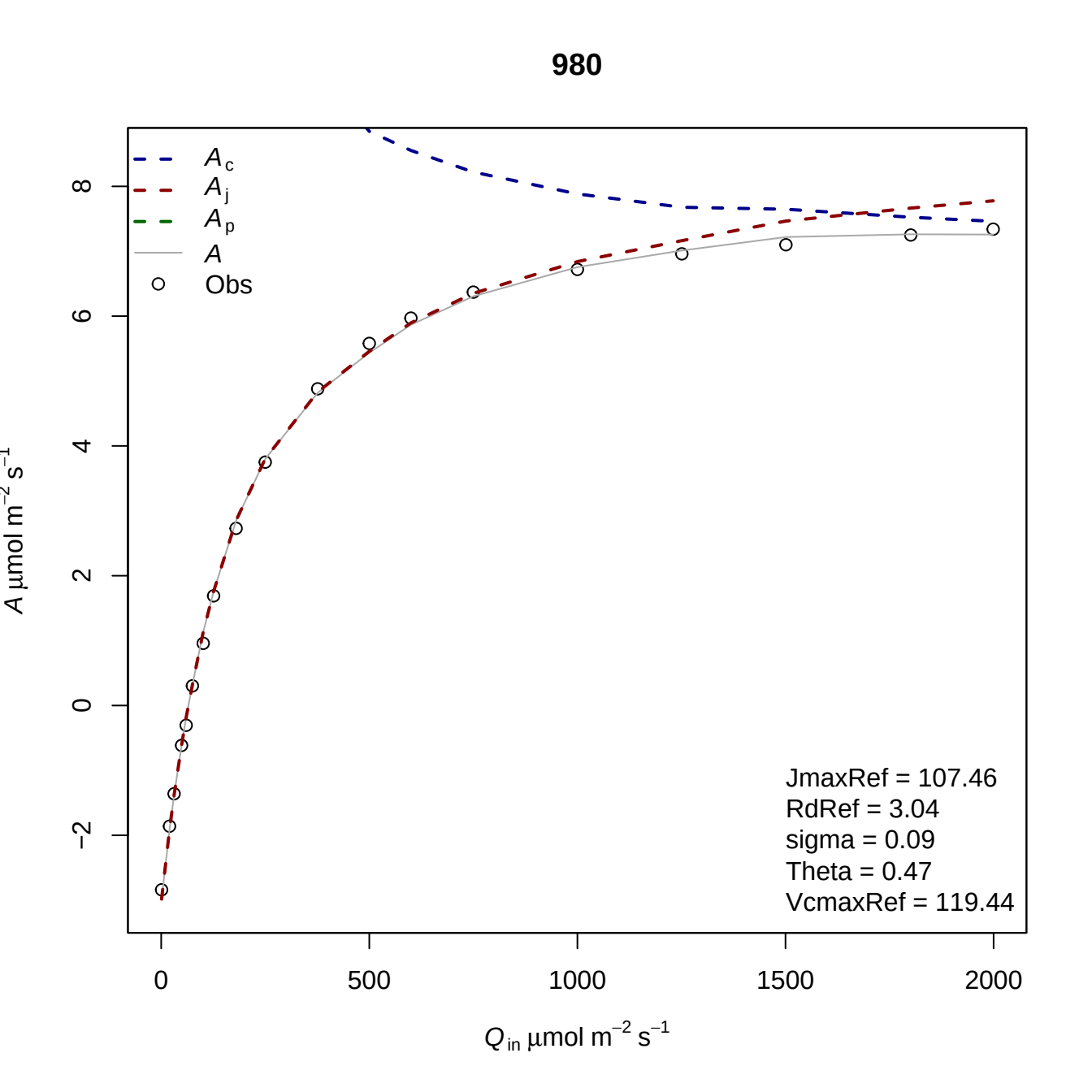

975
A_c
A_j
A_p
A
Obs
JmaxRef = 60.9
RdRef = 1.1
sigma = 0.15
Theta = −0.85
VcmaxRef = 51.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

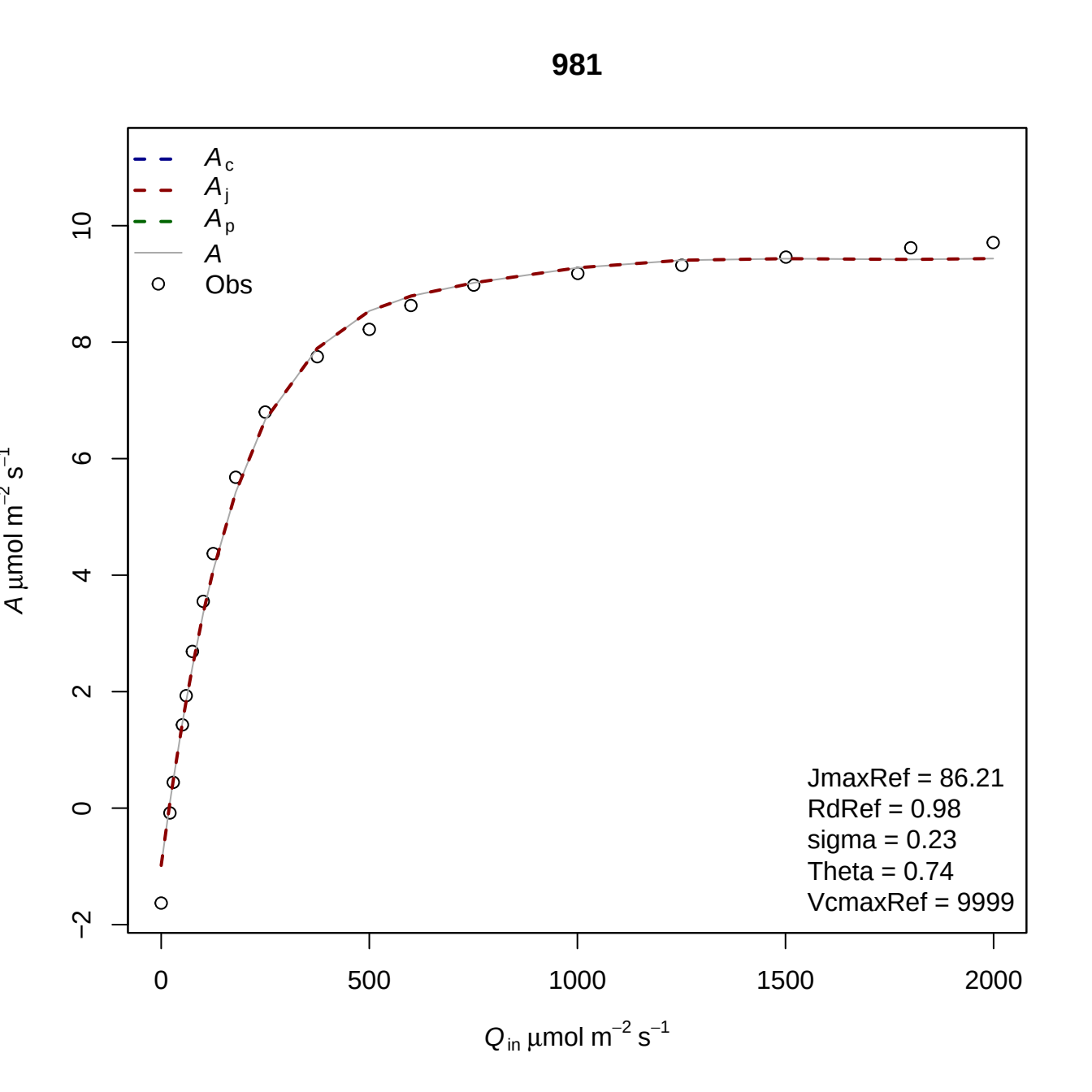

976
A_c
A_j
A_p
A
Obs
JmaxRef = 84.84
RdRef = 1.47
sigma = 0.15
Theta = −0.29
VcmaxRef = 86.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

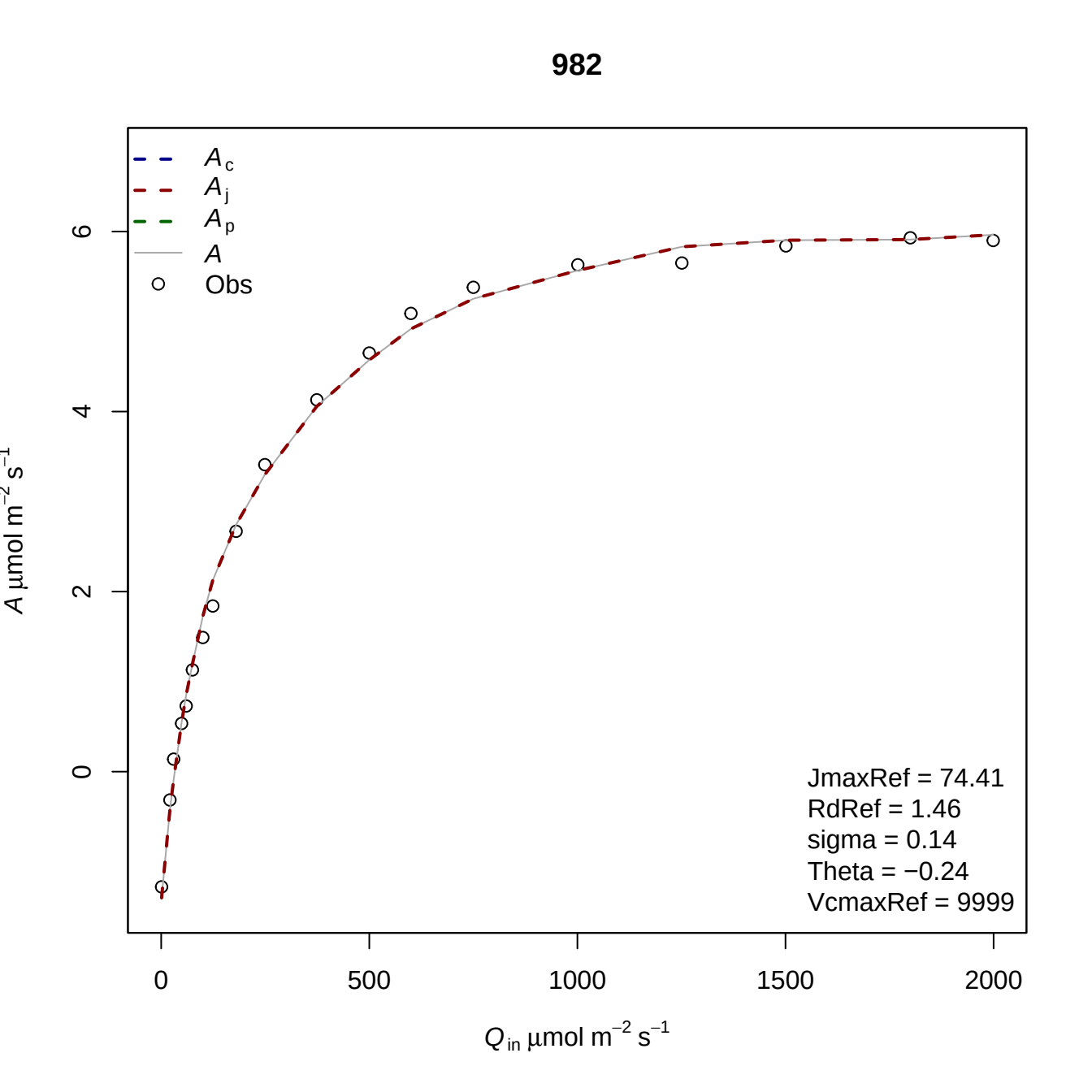

977
A_c
A_j
A_p
A
Obs
JmaxRef = 110.95
RdRef = 1.35
sigma = 0.16
Theta = −1.44
VcmaxRef = 83.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

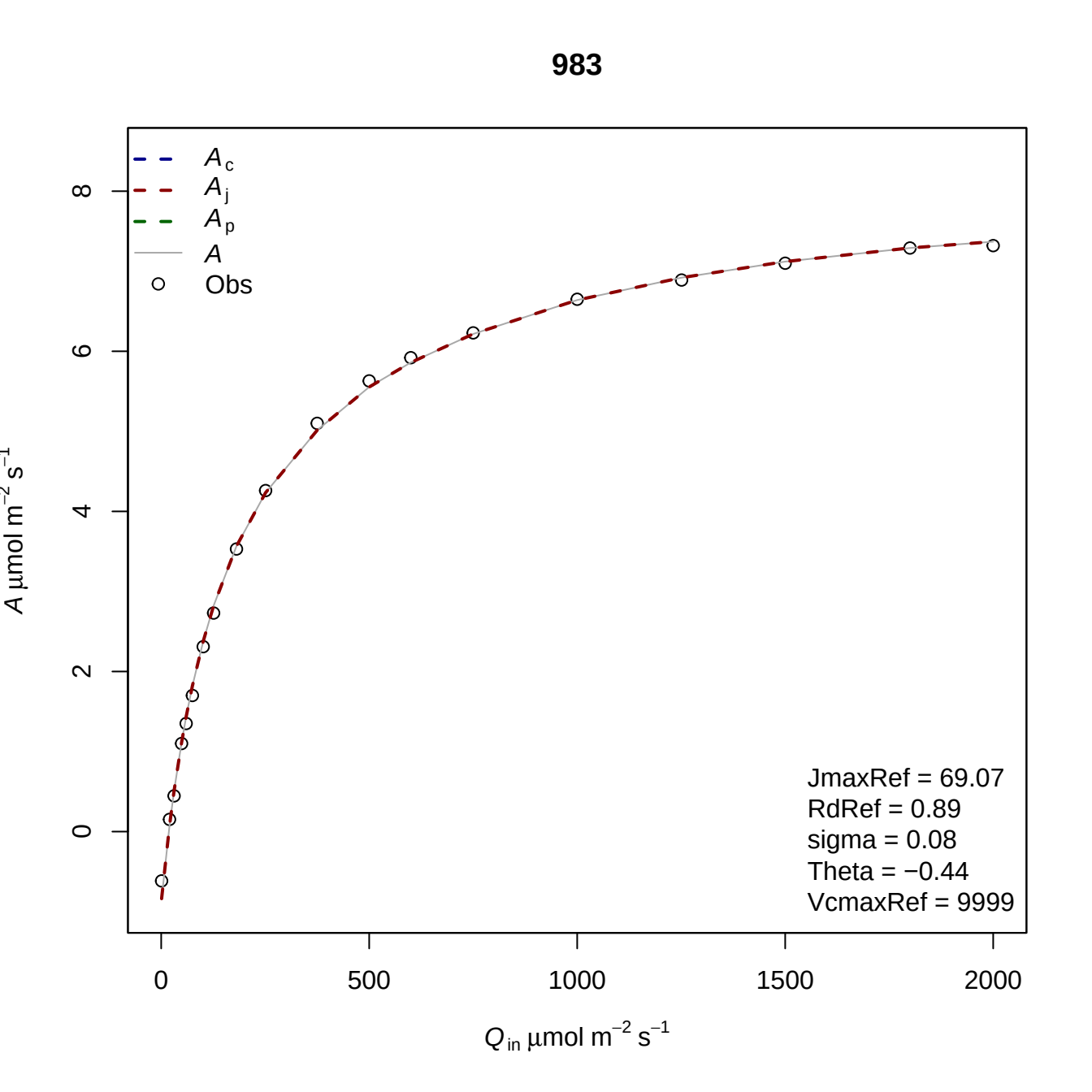

978
A_c
A_j
A_p
A
Obs
JmaxRef = 65.43
RdRef = 1
sigma = 0.16
Theta = 0.64
VcmaxRef = 77.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

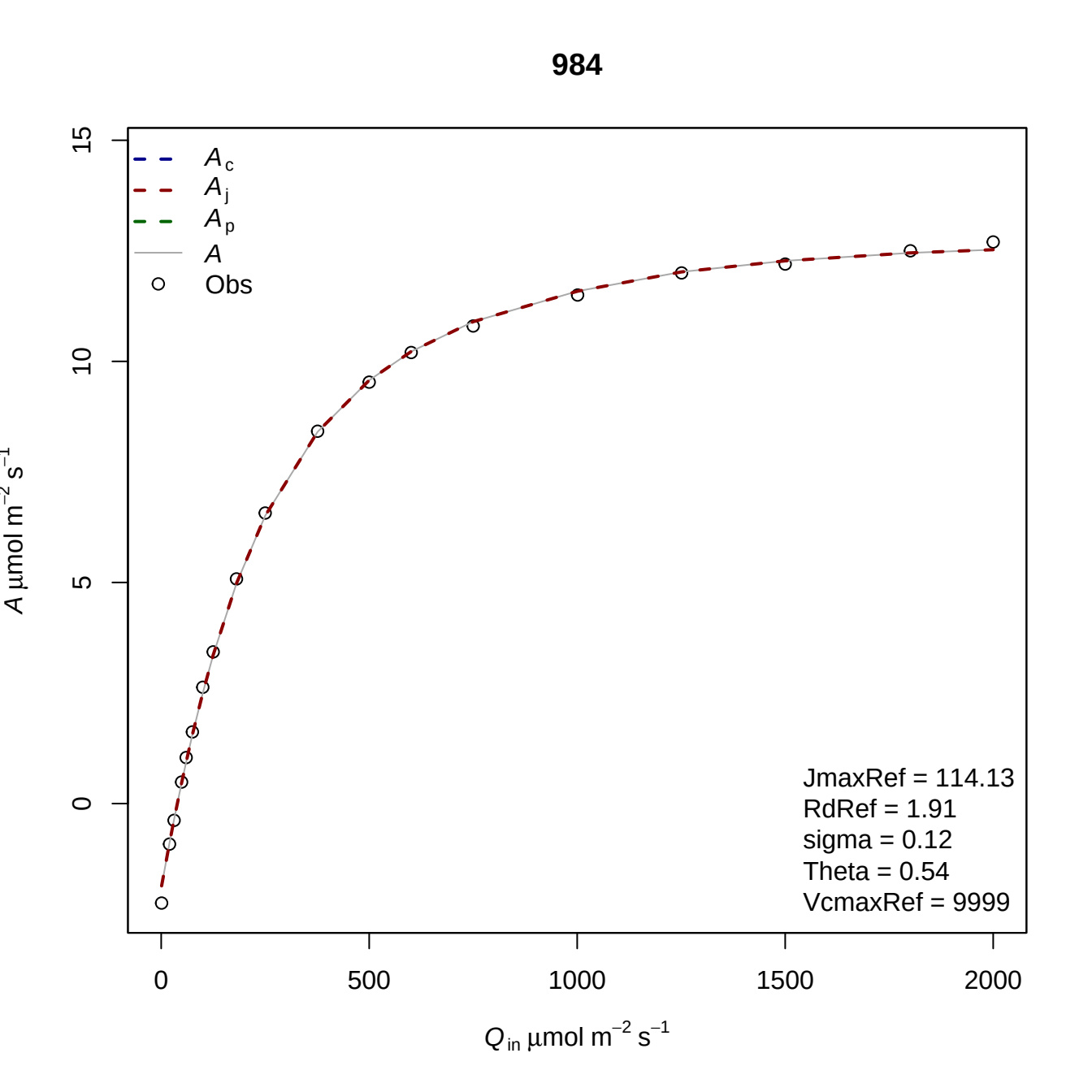

979
A_c
A_j
A_p
A
Obs
JmaxRef = 92.51
RdRef = 1.58
sigma = 0.03
Theta = 0.64
VcmaxRef = 93.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

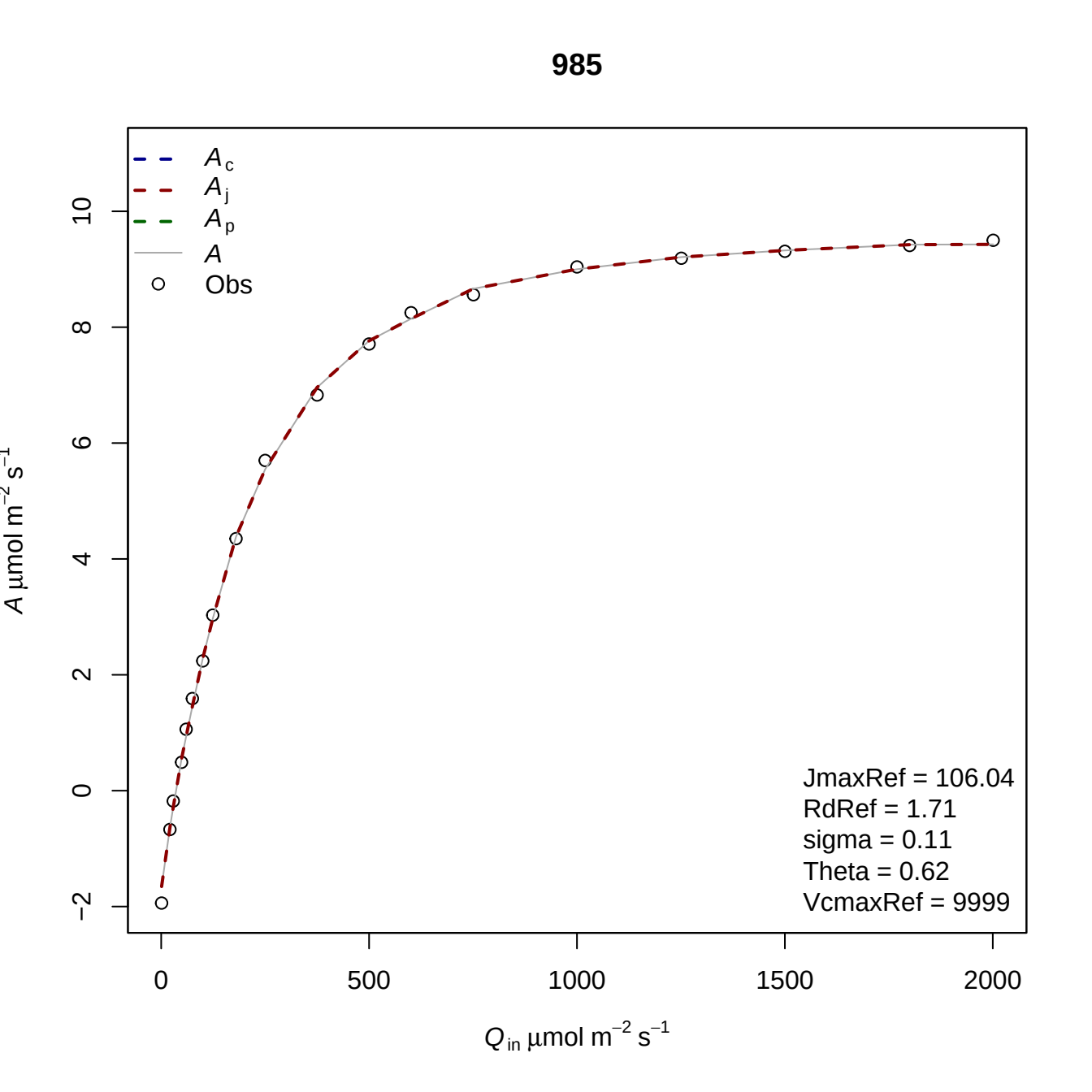

980
A_c
A_j
A_p
A
Obs
JmaxRef = 107.46
RdRef = 3.04
sigma = 0.09
Theta = 0.47
VcmaxRef = 119.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

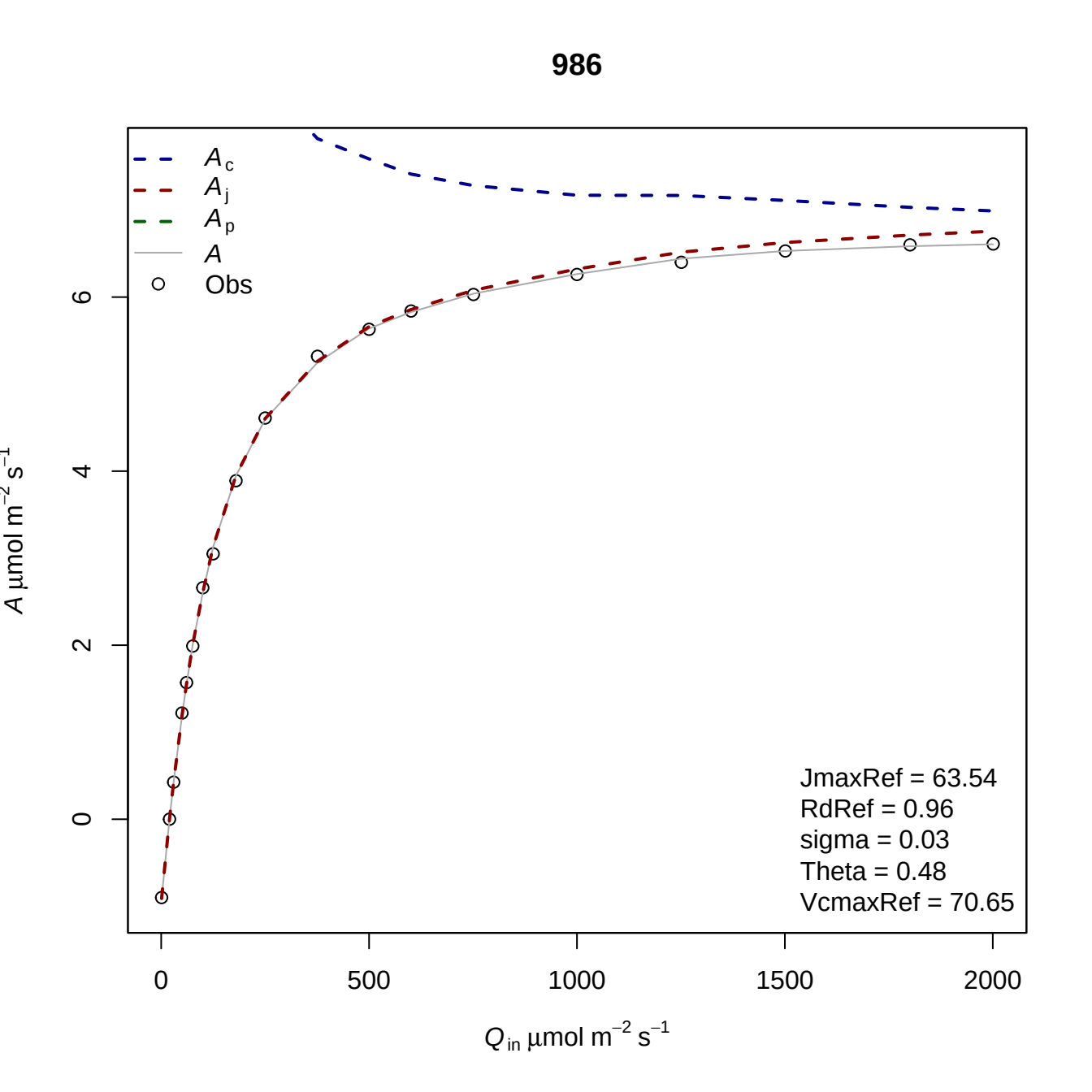

981
A_c
A_j
A_p
A
Obs
JmaxRef = 86.21
RdRef = 0.98
sigma = 0.23
Theta = 0.74
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

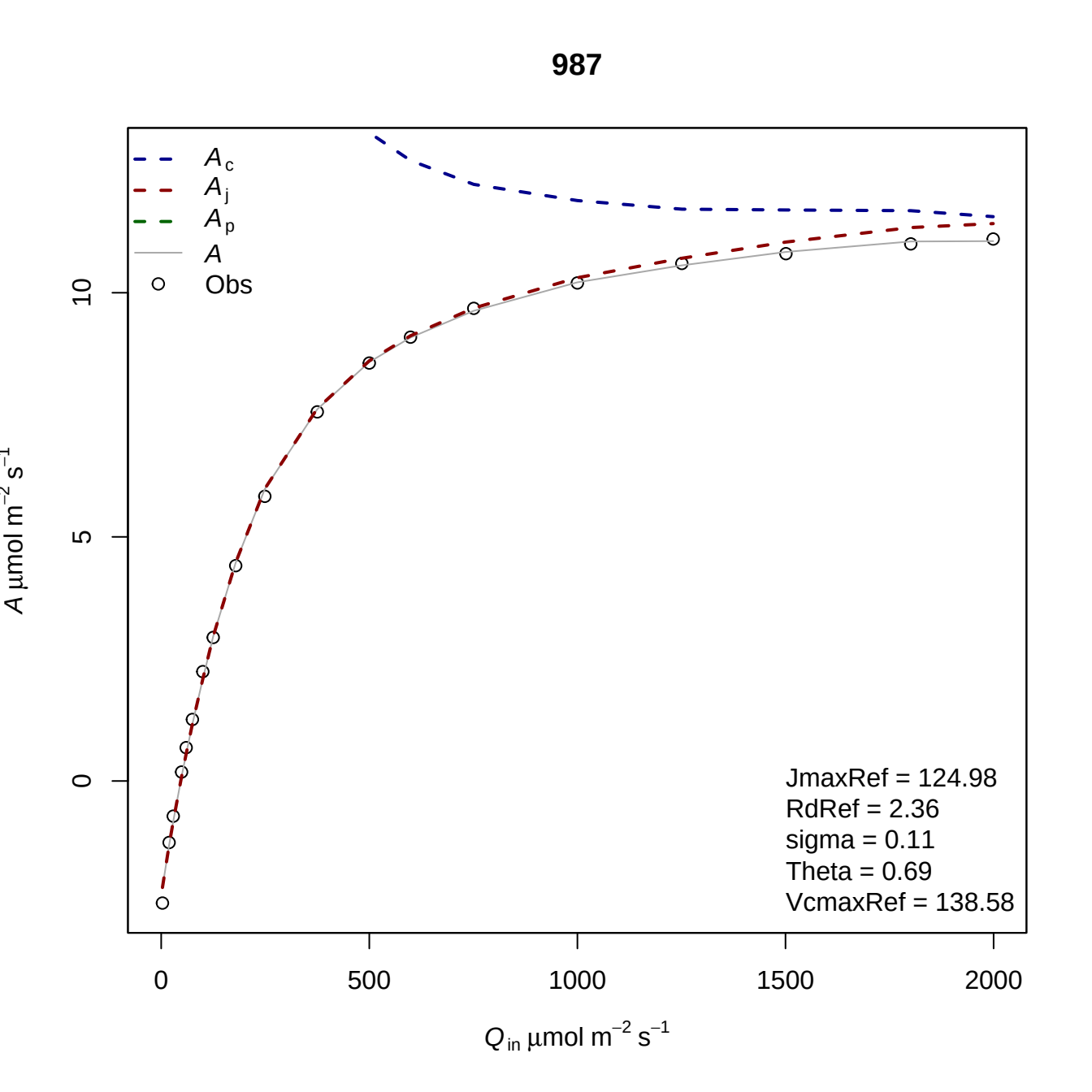

982
A_c
A_j
A_p
A
Obs
JmaxRef = 74.41
RdRef = 1.46
sigma = 0.14
Theta = −0.24
VcmaxRef = 9999
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

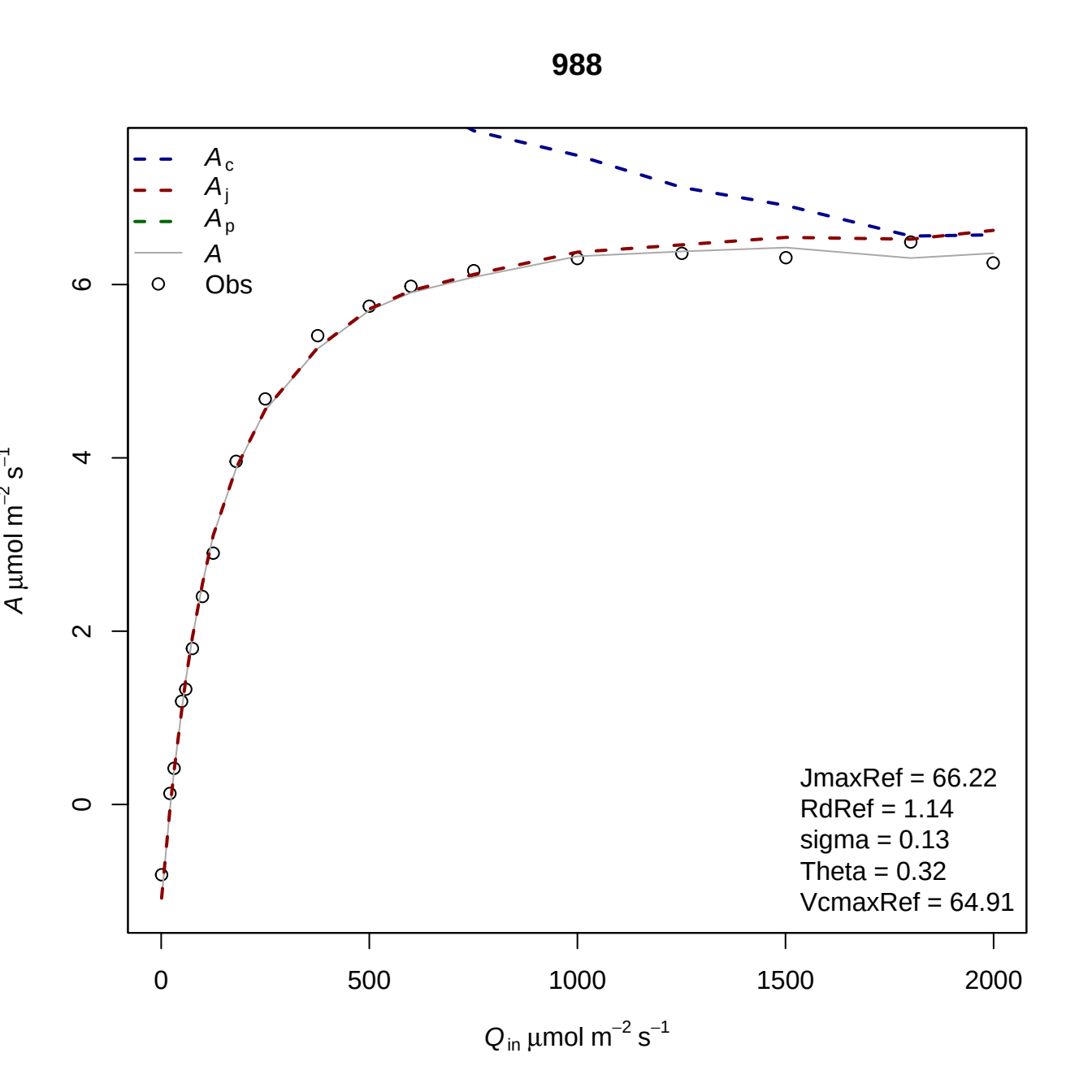

983
A_c
A_j
A_p
A
Obs
JmaxRef = 69.07
RdRef = 0.89
sigma = 0.08
Theta = −0.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

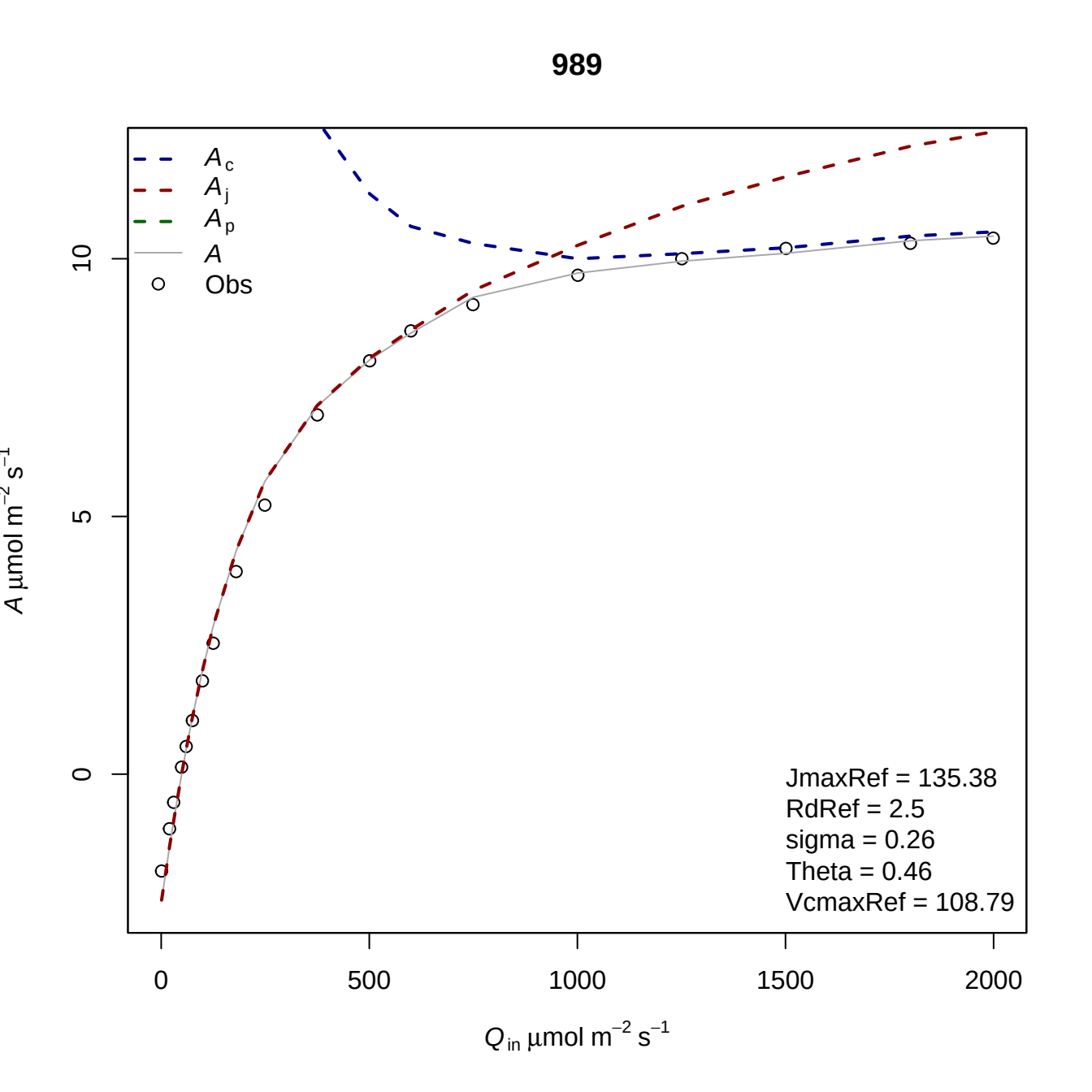

984
A_c
A_j
A_p
A
Obs
JmaxRef = 114.13
RdRef = 1.91
sigma = 0.12
Theta = 0.54
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

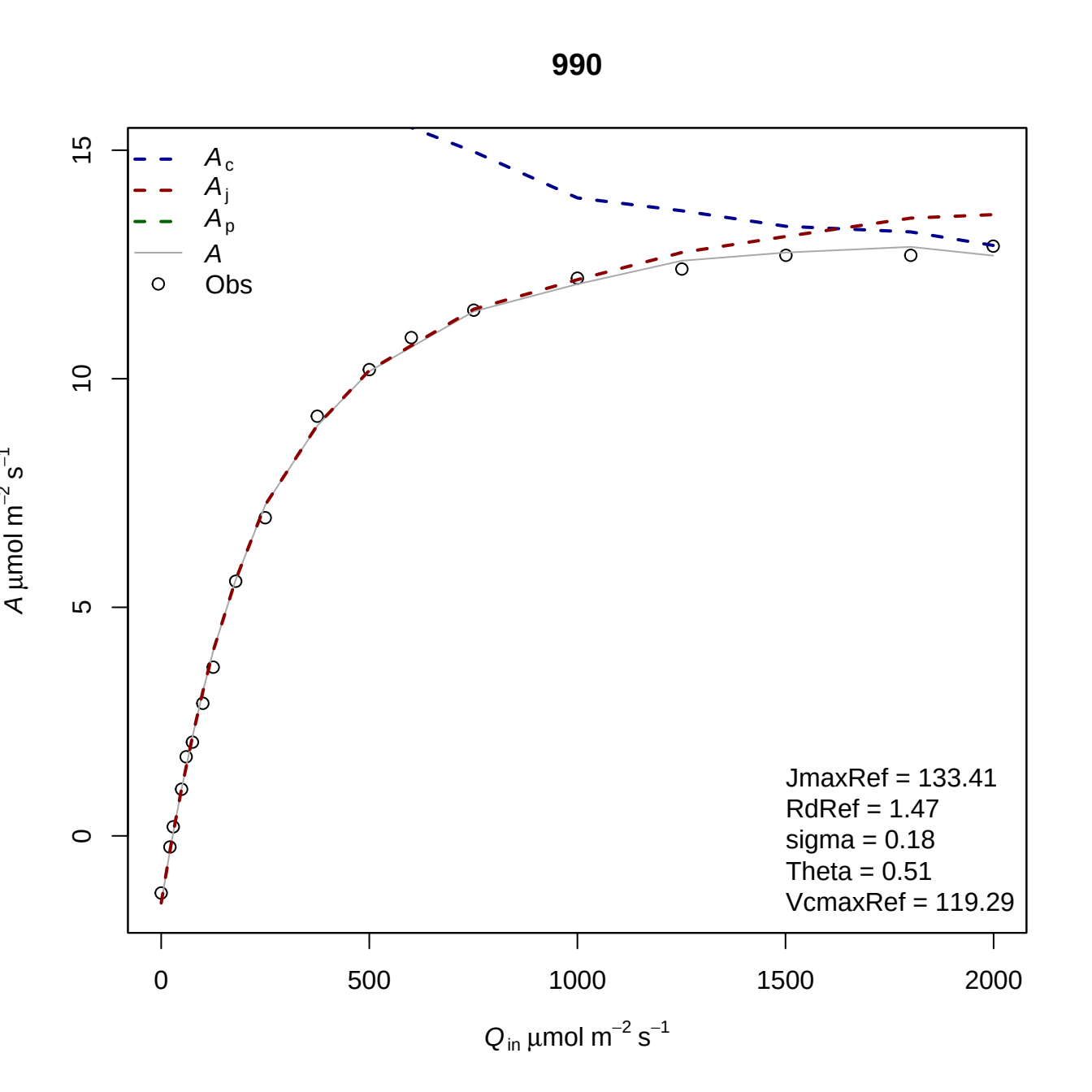

985
A_c
A_j
A_p
A
Obs
JmaxRef = 106.04
RdRef = 1.71
sigma = 0.11
Theta = 0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

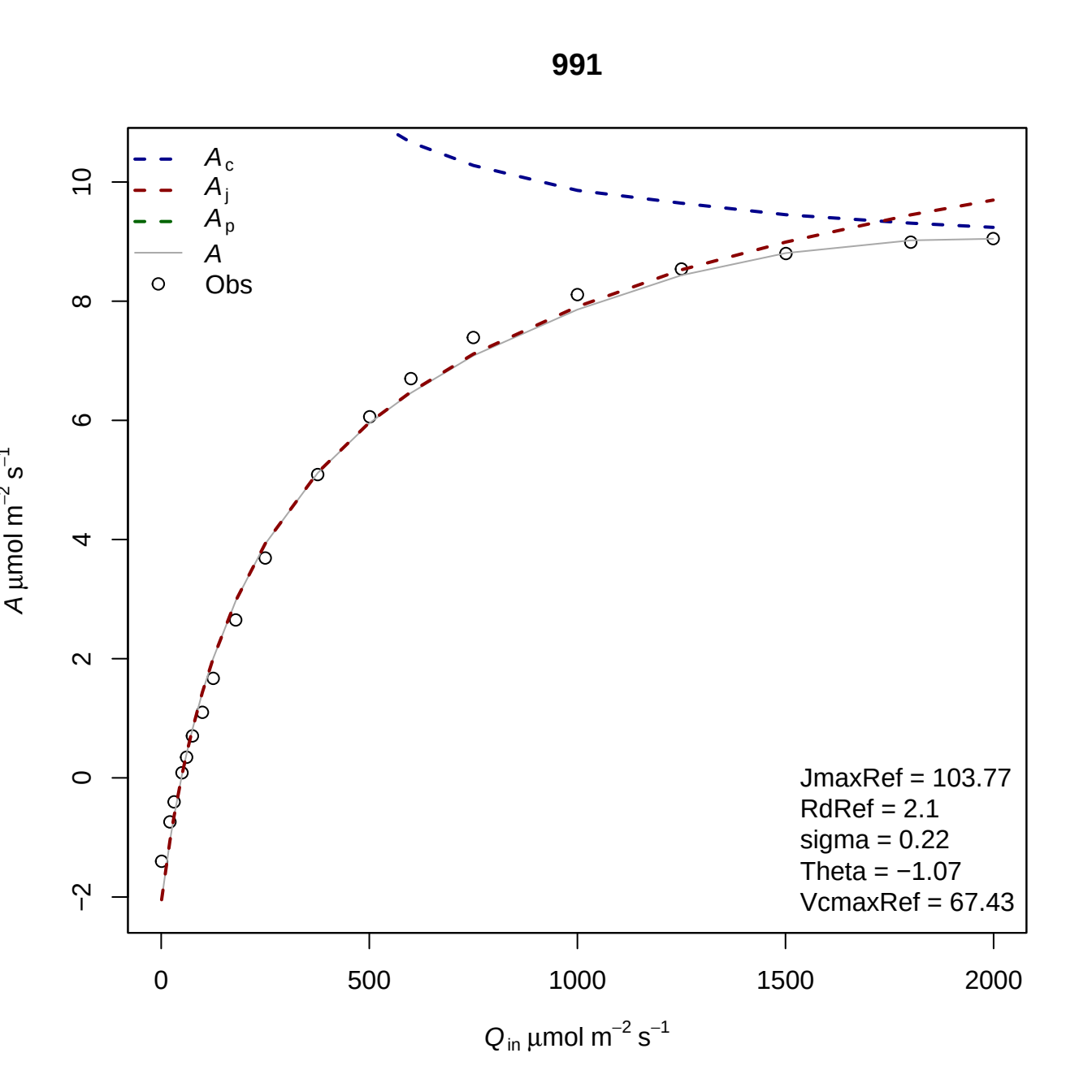

986
A_c
A_j
A_p
A
Obs
JmaxRef = 63.54
RdRef = 0.96
sigma = 0.03
Theta = 0.48
VcmaxRef = 70.65
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

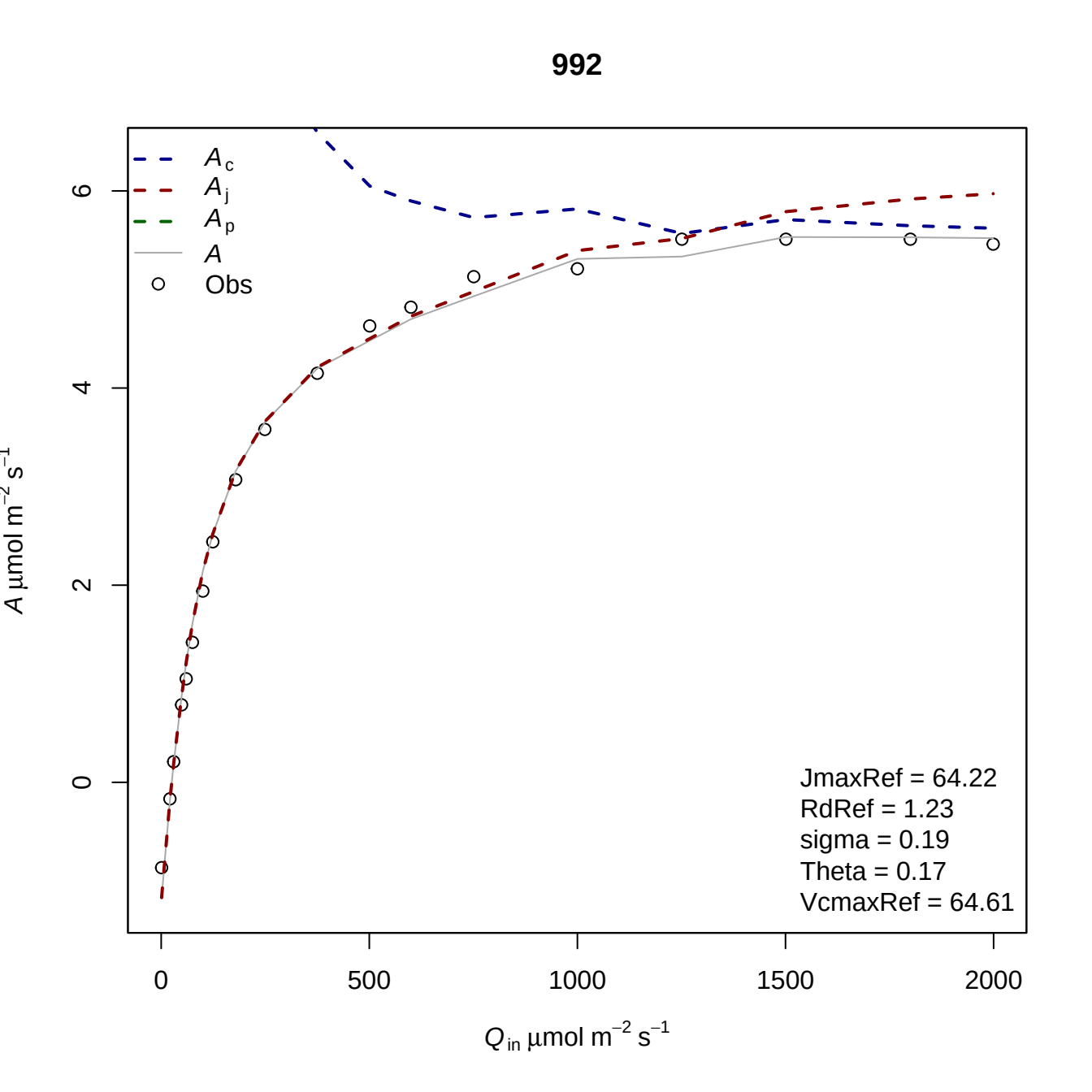

987
A_c
A_j
A_p
A
Obs
JmaxRef = 124.98
RdRef = 2.36
sigma = 0.11
Theta = 0.69
VcmaxRef = 138.58
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

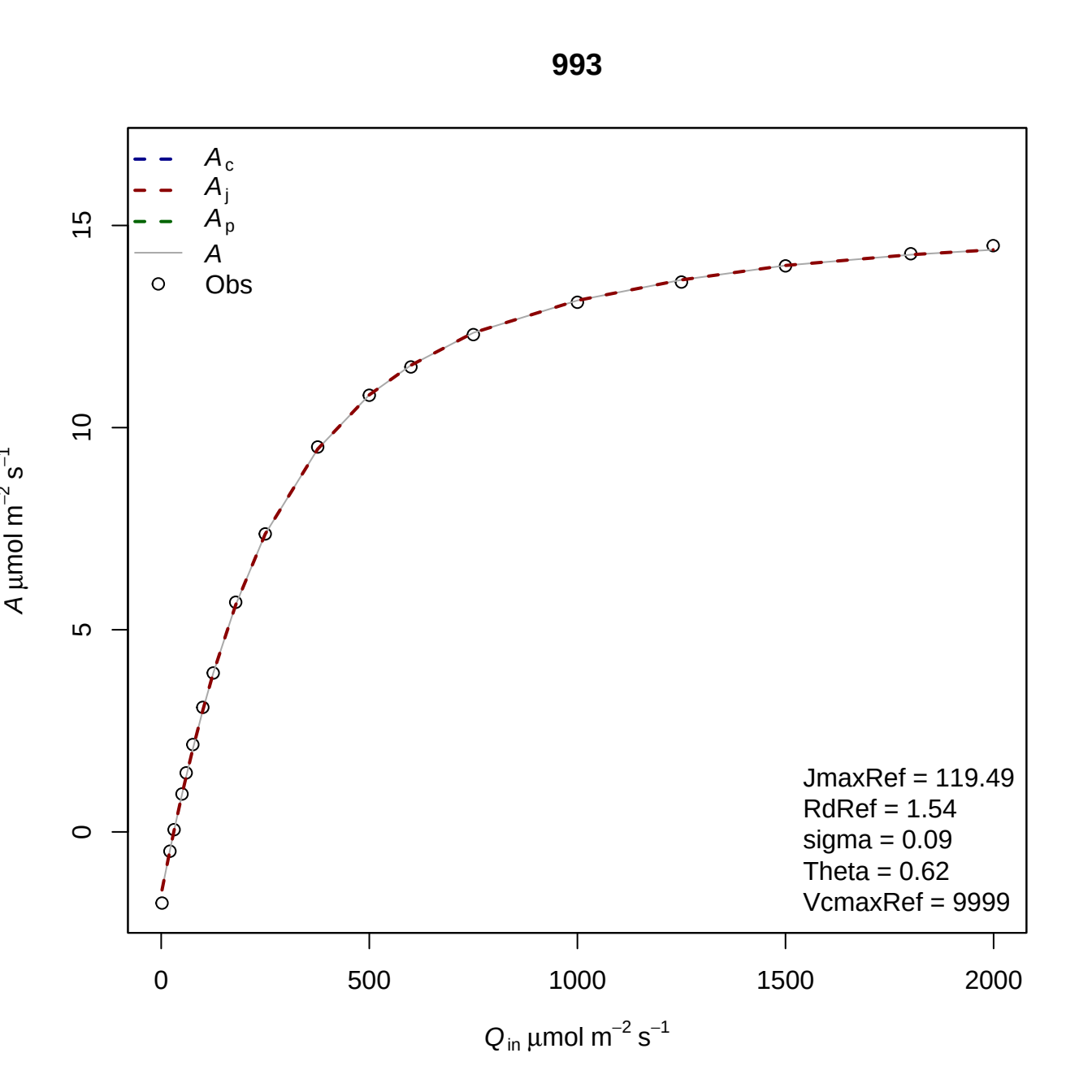

988
A_c
A_j
A_p
A
Obs
JmaxRef = 66.22
RdRef = 1.14
sigma = 0.13
Theta = 0.32
VcmaxRef = 64.91
A µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

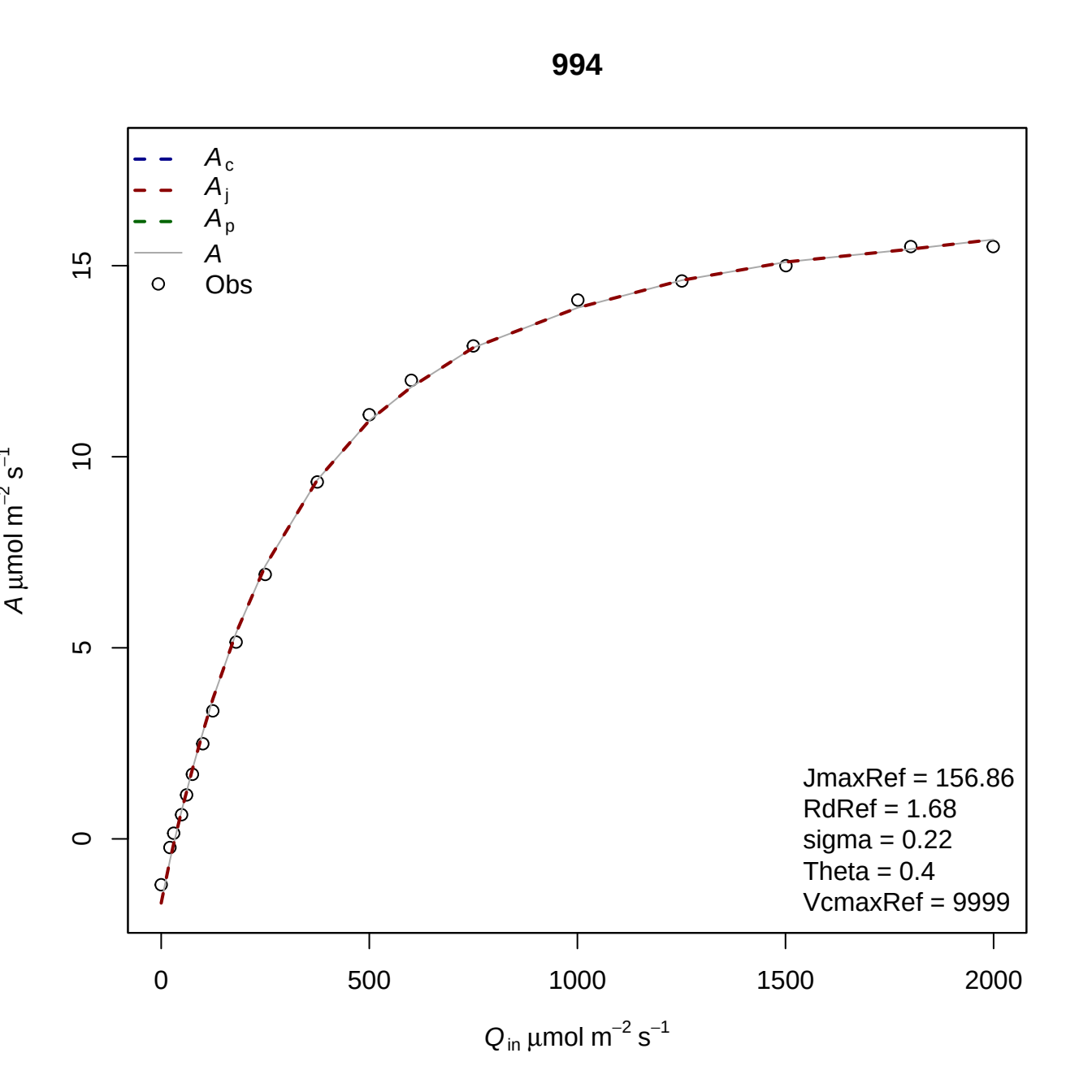

989
A_c
A_j
A_p
A
Obs
JmaxRef = 135.38
RdRef = 2.5
sigma = 0.26
Theta = 0.46
VcmaxRef = 108.79
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

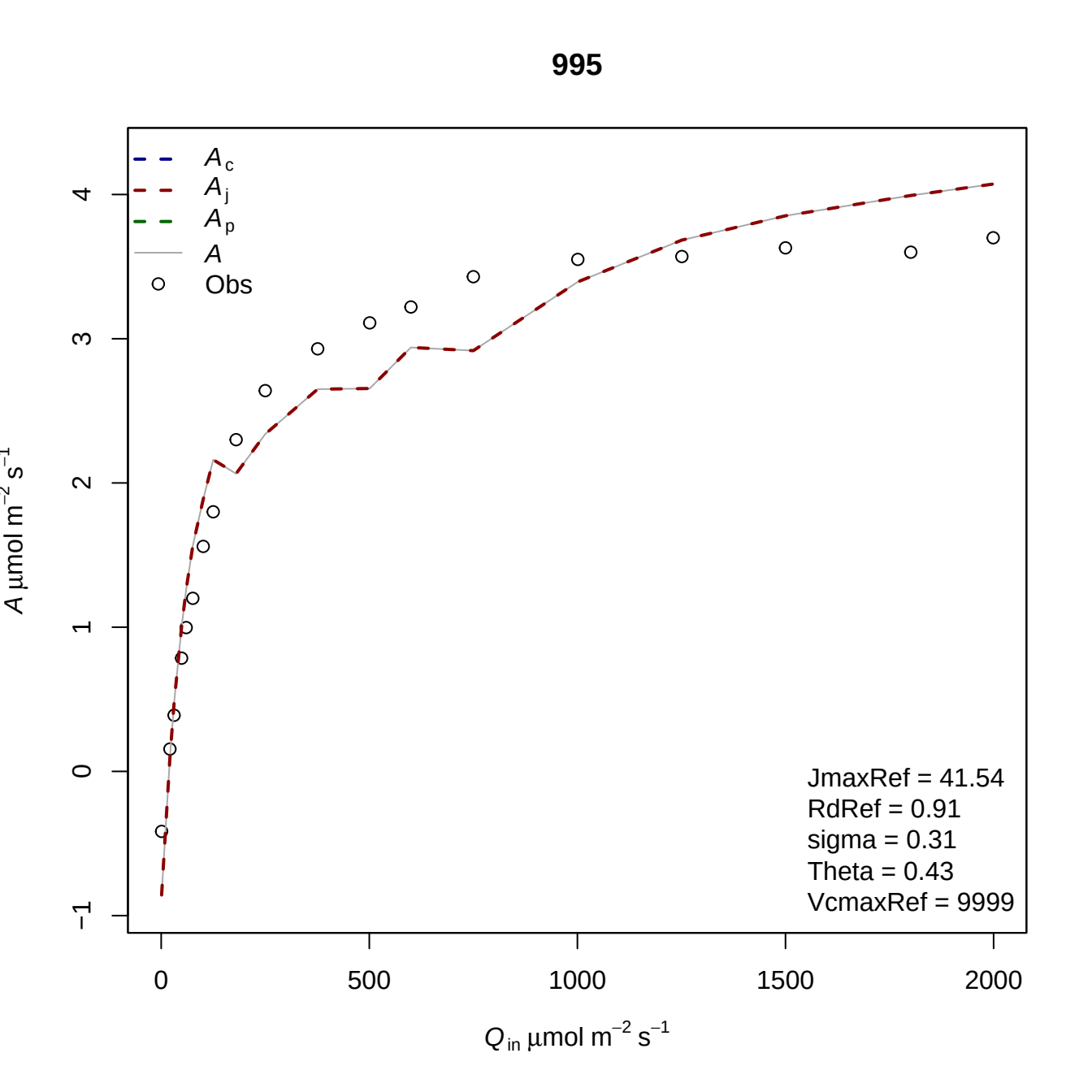

990
A_c
A_j
A_p
A
Obs
JmaxRef = 133.41
RdRef = 1.47
sigma = 0.18
Theta = 0.51
VcmaxRef = 119.29
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

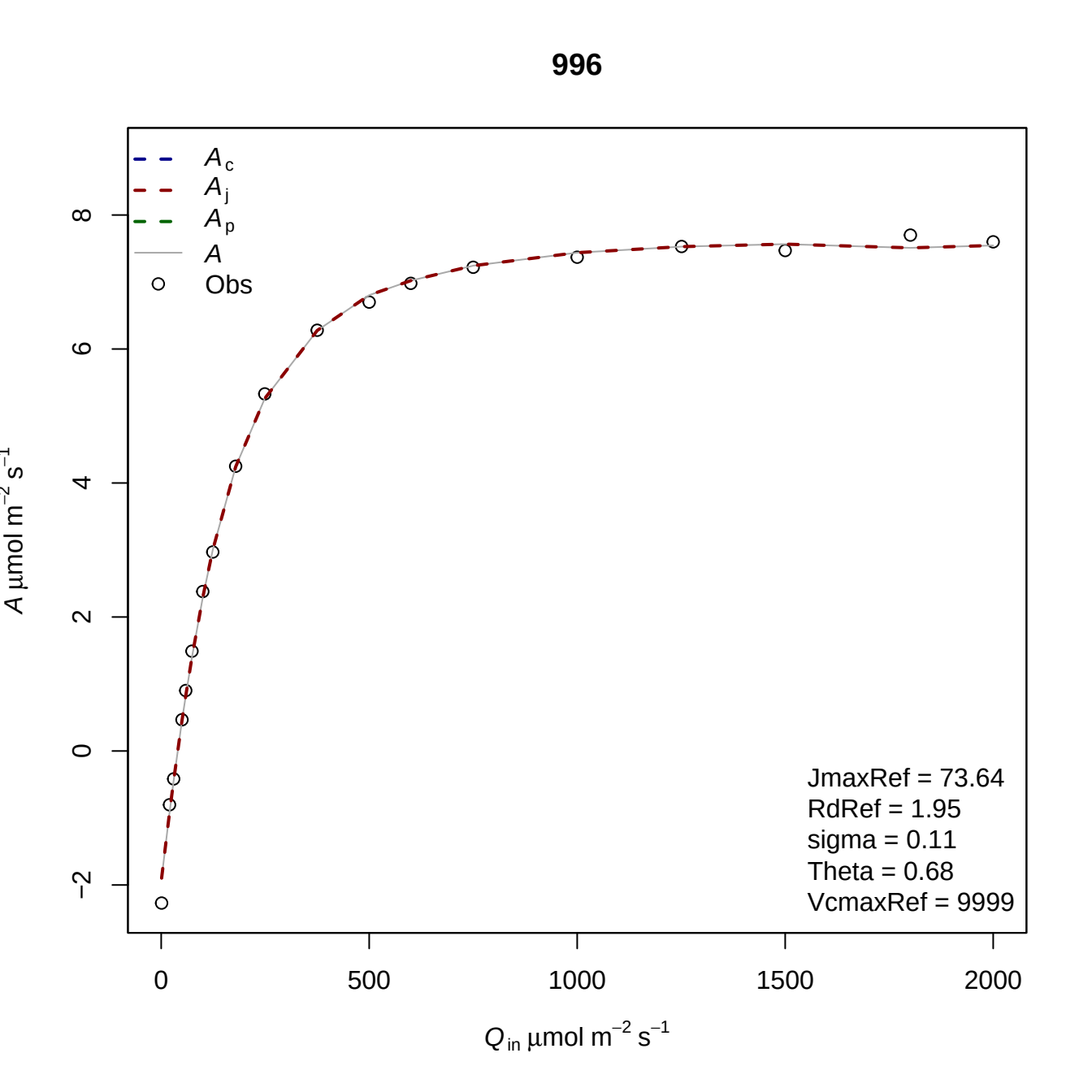

991
A_c
A_j
A_p
A
Obs
JmaxRef = 103.77
RdRef = 2.1
sigma = 0.22
Theta = −1.07
VcmaxRef = 67.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

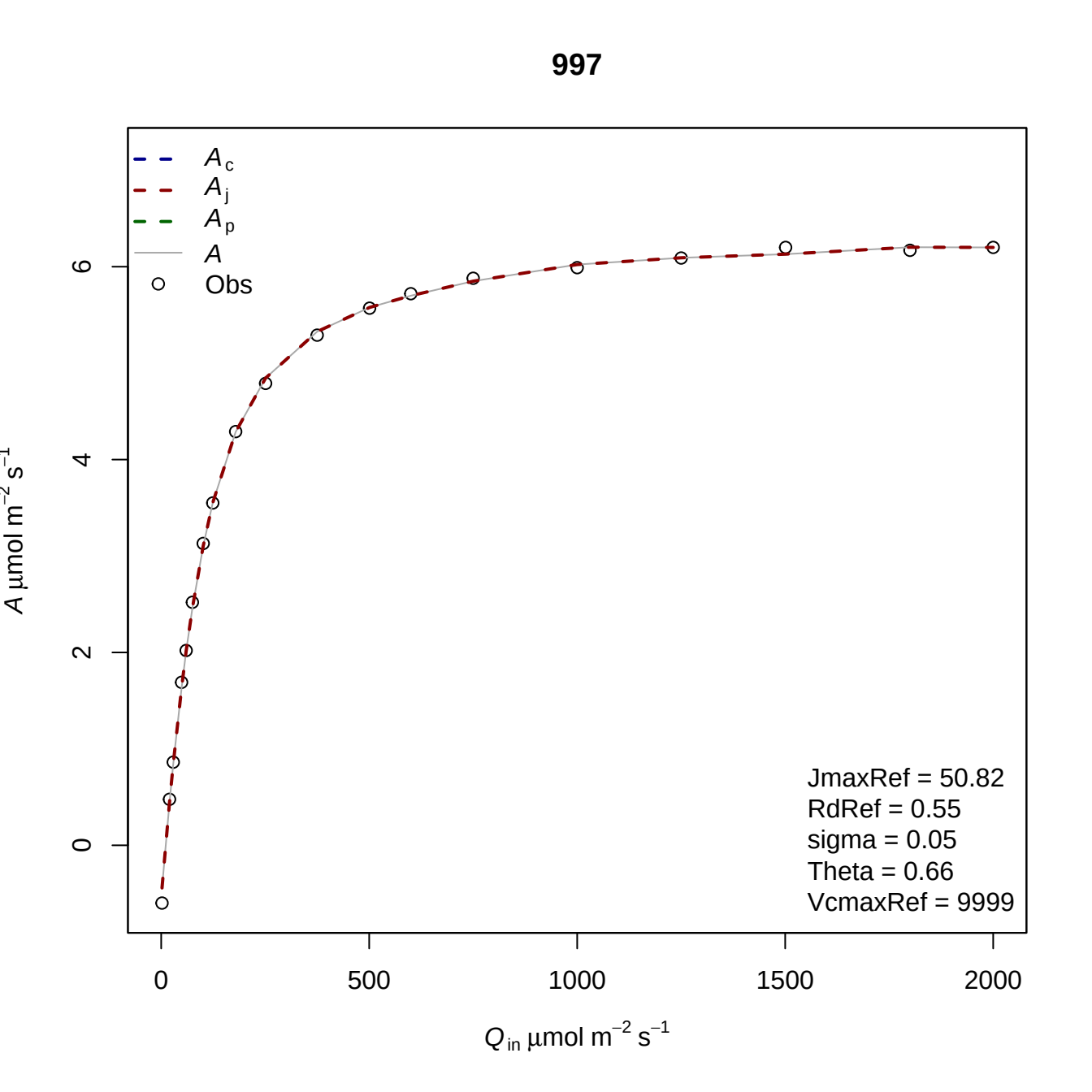

992
A_c
A_j
A_p
A
Obs
JmaxRef = 64.22
RdRef = 1.23
sigma = 0.19
Theta = 0.17
VcmaxRef = 64.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

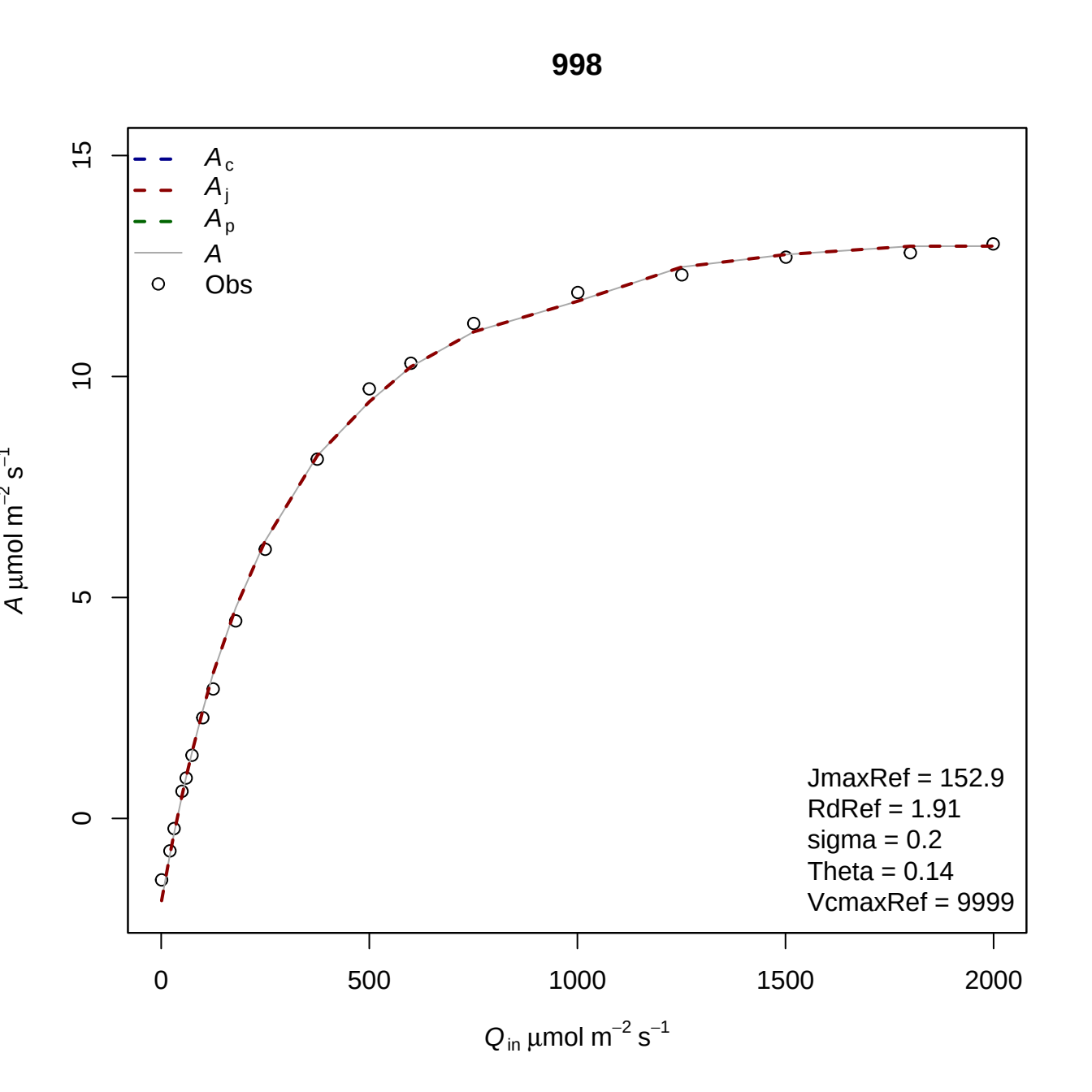

993
A_c
A_j
A_p
A
Obs
JmaxRef = 119.49
RdRef = 1.54
sigma = 0.09
Theta = 0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

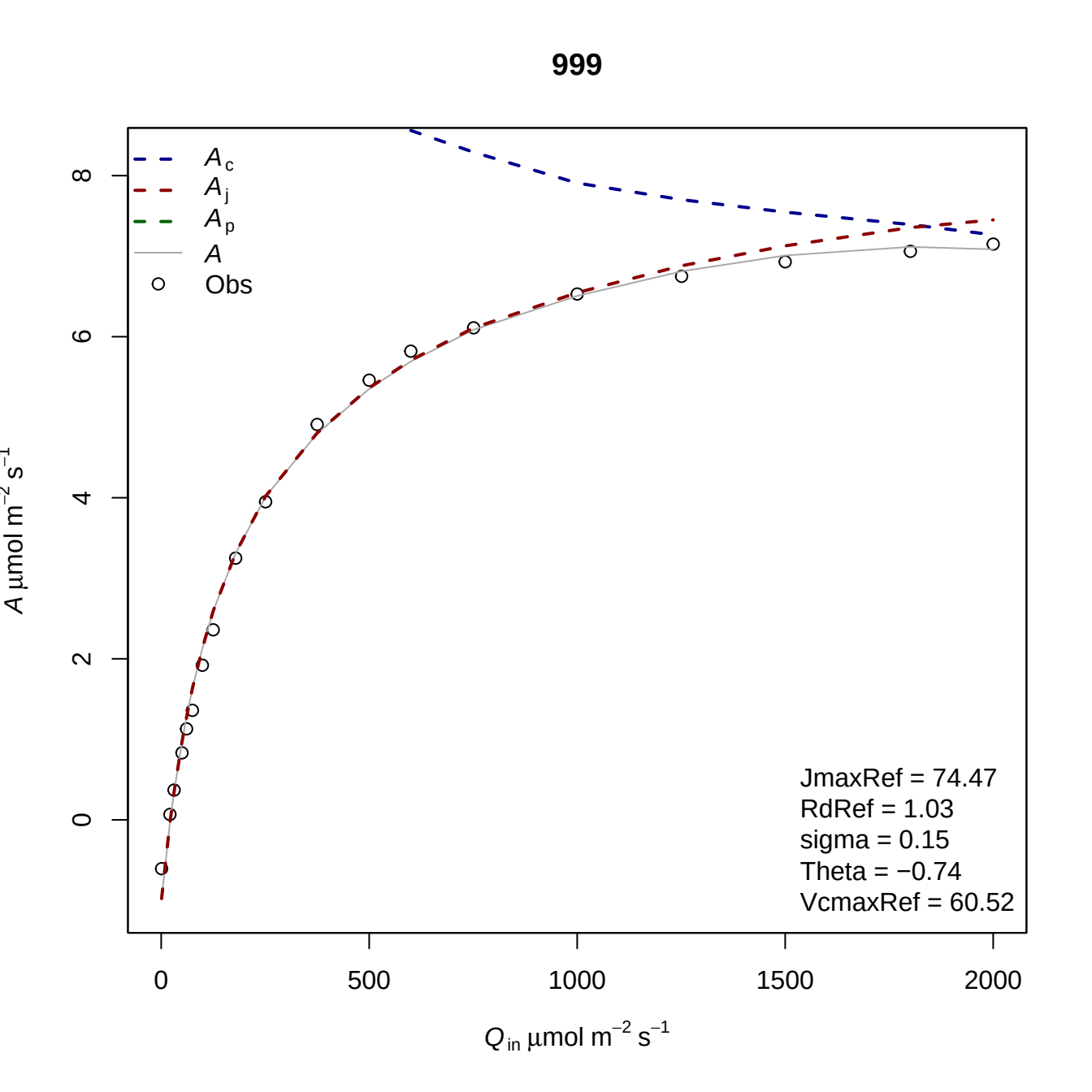

994
A_c
A_j
A_p
A
Obs
JmaxRef = 156.86
RdRef = 1.68
sigma = 0.22
Theta = 0.4
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

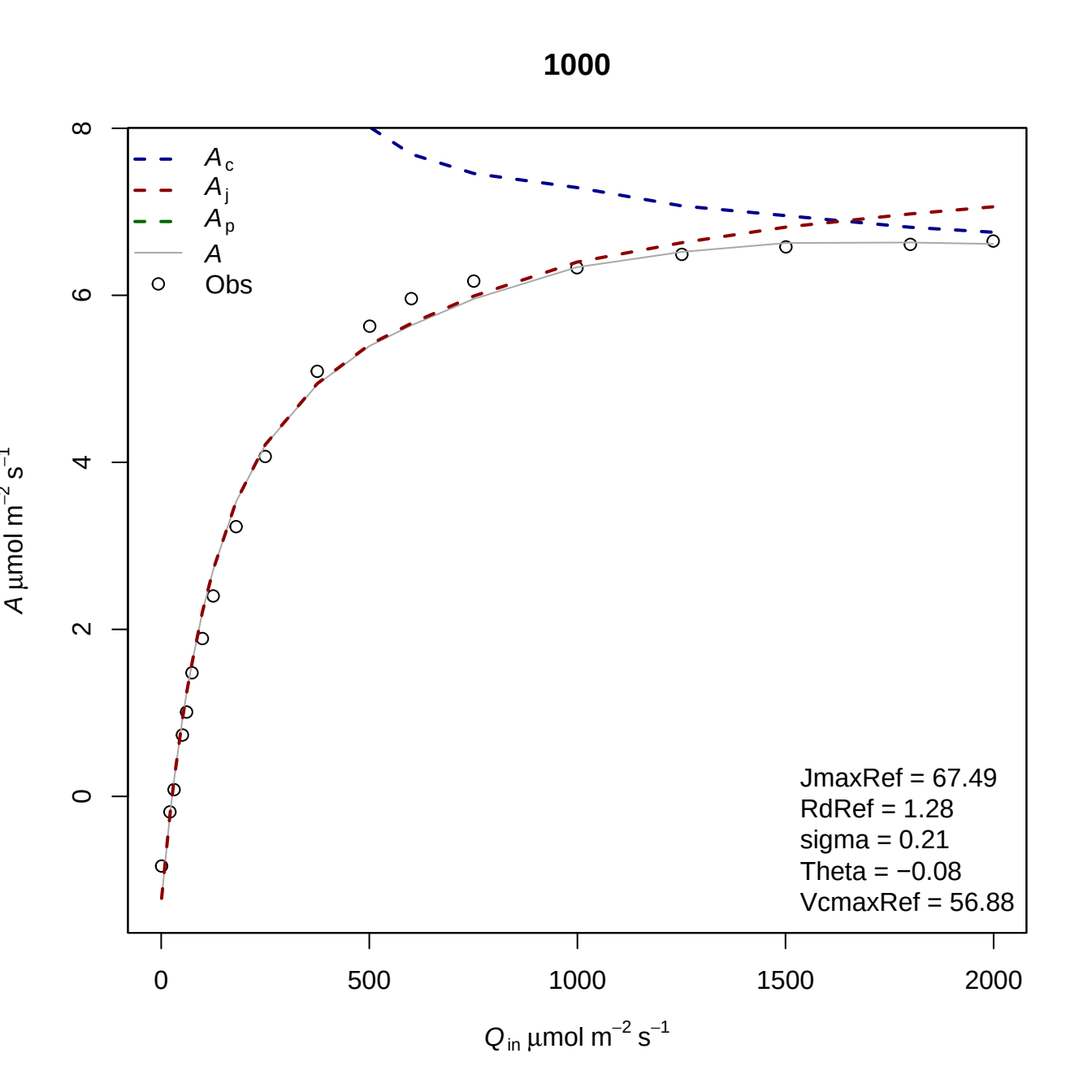

995
A_c
A_j
A_p
A
Obs
JmaxRef = 41.54
RdRef = 0.91
sigma = 0.31
Theta = 0.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

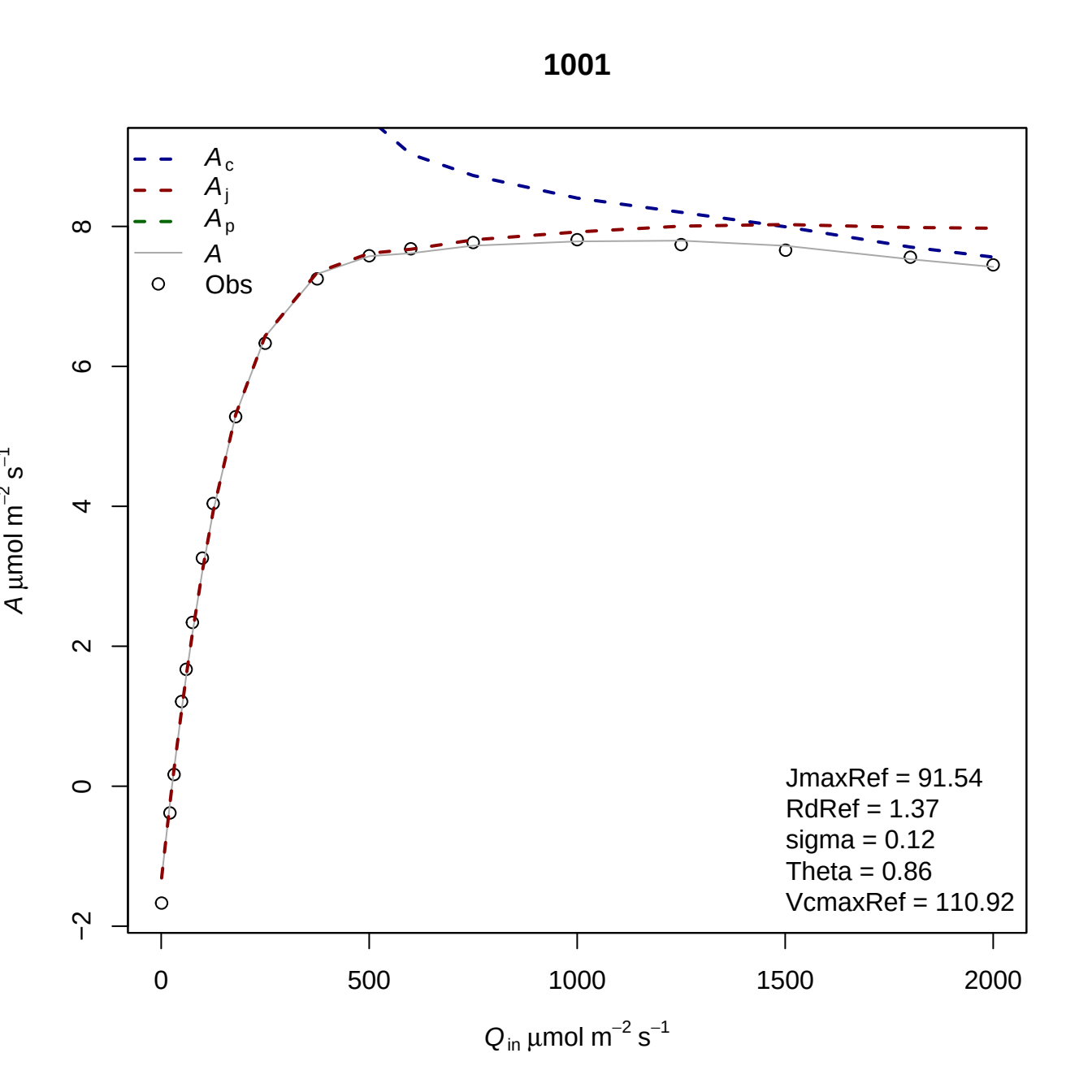

996
A_c
A_j
A_p
A
Obs
JmaxRef = 73.64
RdRef = 1.95
sigma = 0.11
Theta = 0.68
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

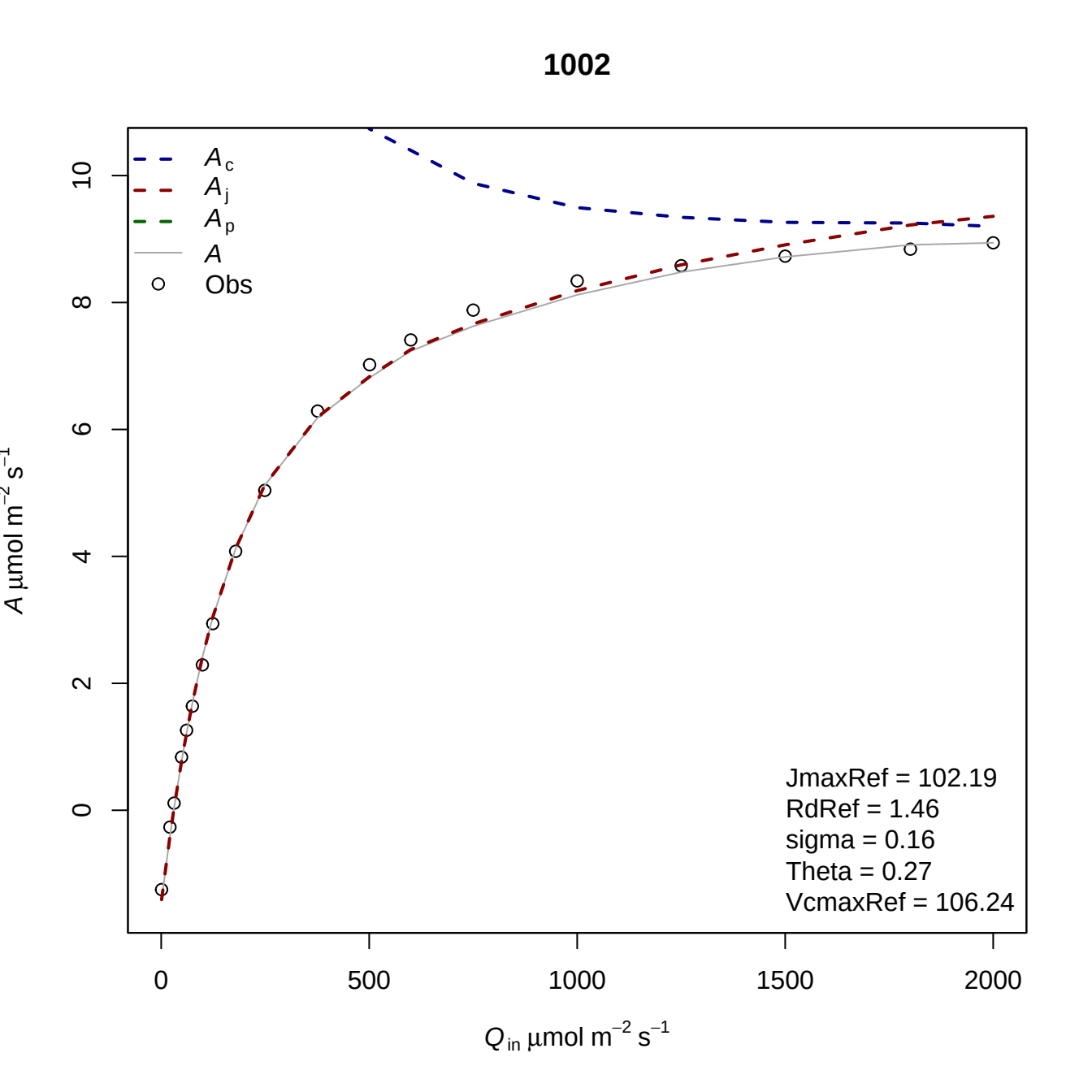

997
A_c
A_j
A_p
A
Obs
JmaxRef = 50.82
RdRef = 0.55
sigma = 0.05
Theta = 0.66
VcmaxRef = 9999
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

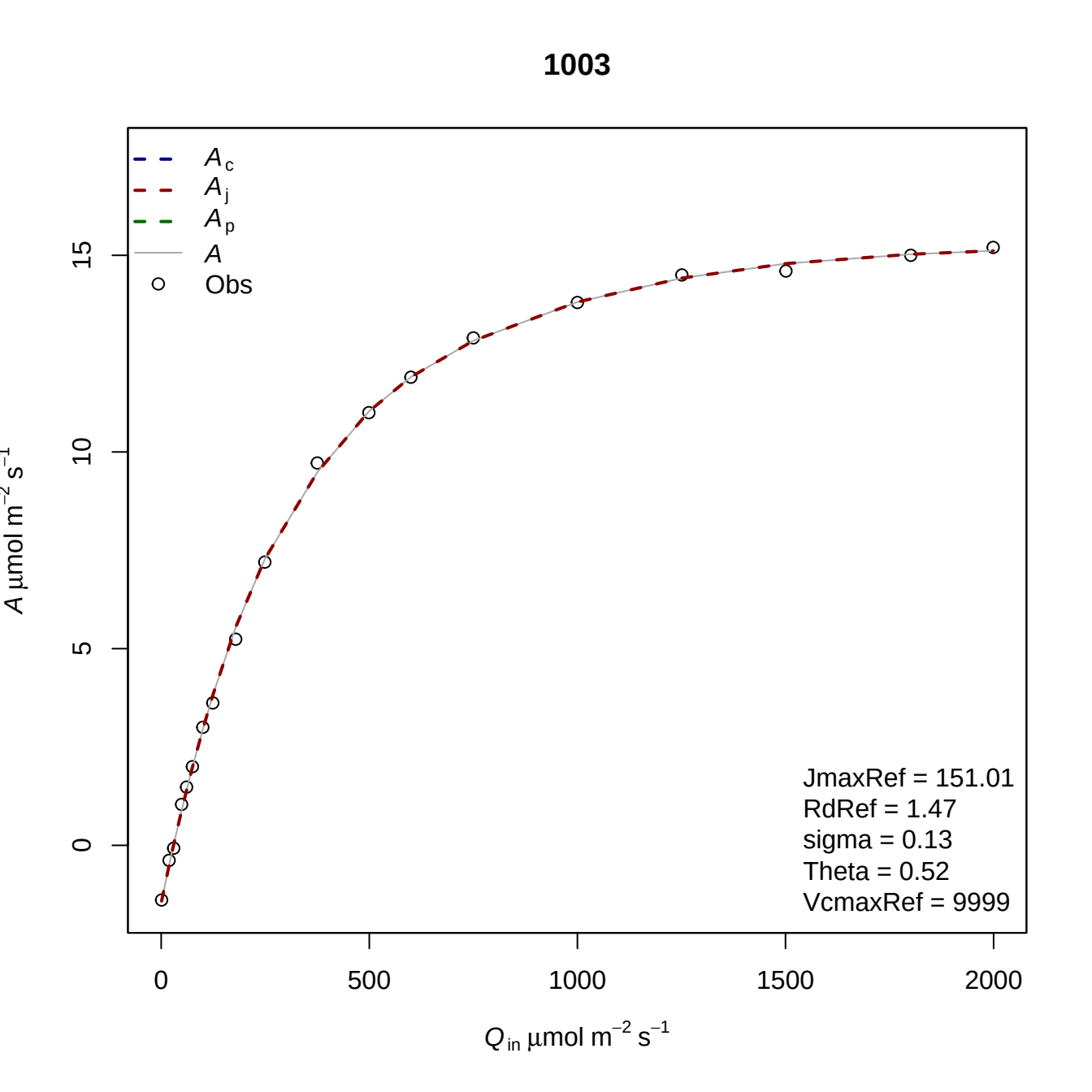

998
A_c
A_j
A_p
A
Obs
JmaxRef = 152.9
RdRef = 1.91
sigma = 0.2
Theta = 0.14
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

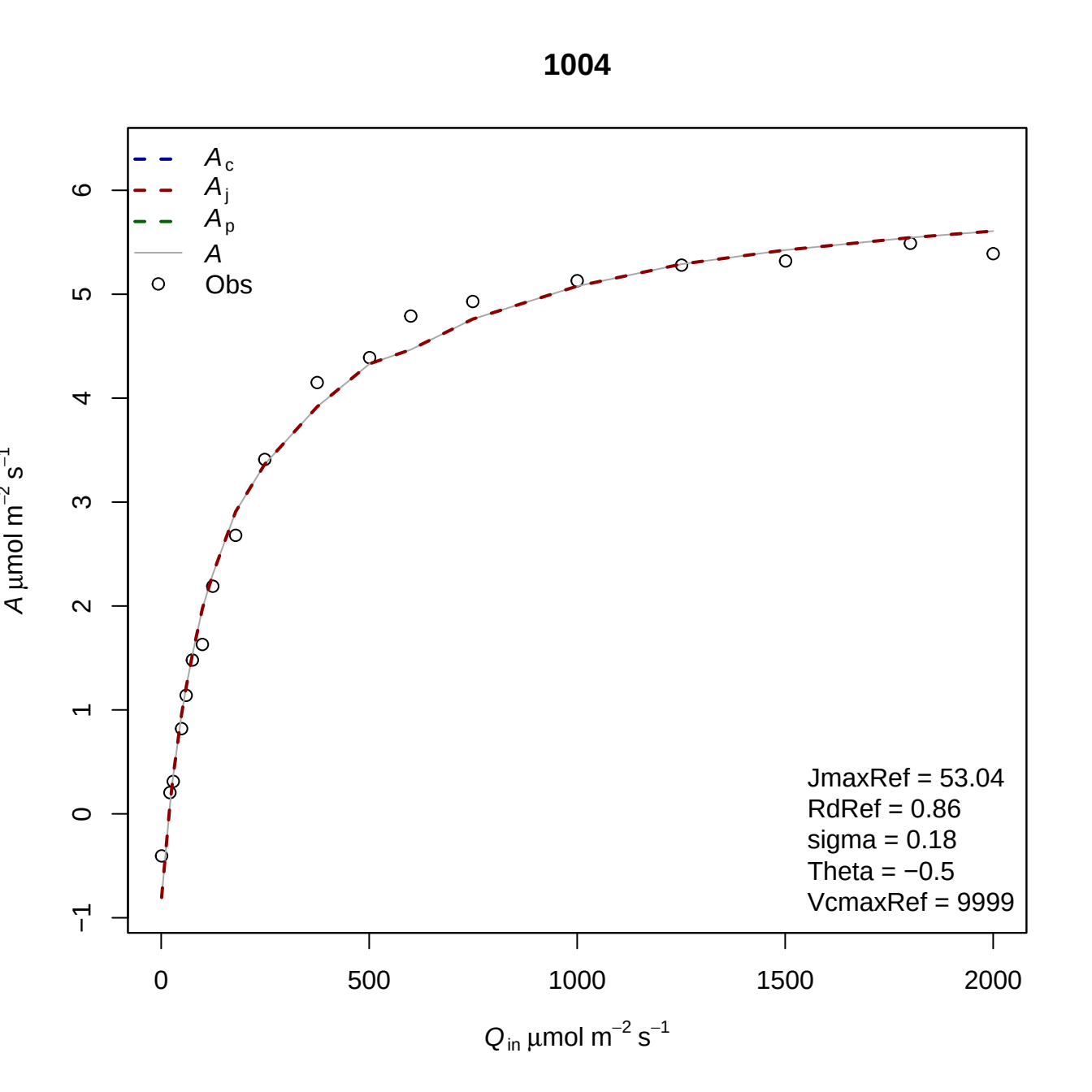

999
A_c
A_j
A_p
A
Obs
JmaxRef = 74.47
RdRef = 1.03
sigma = 0.15
Theta = −0.74
VcmaxRef = 60.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

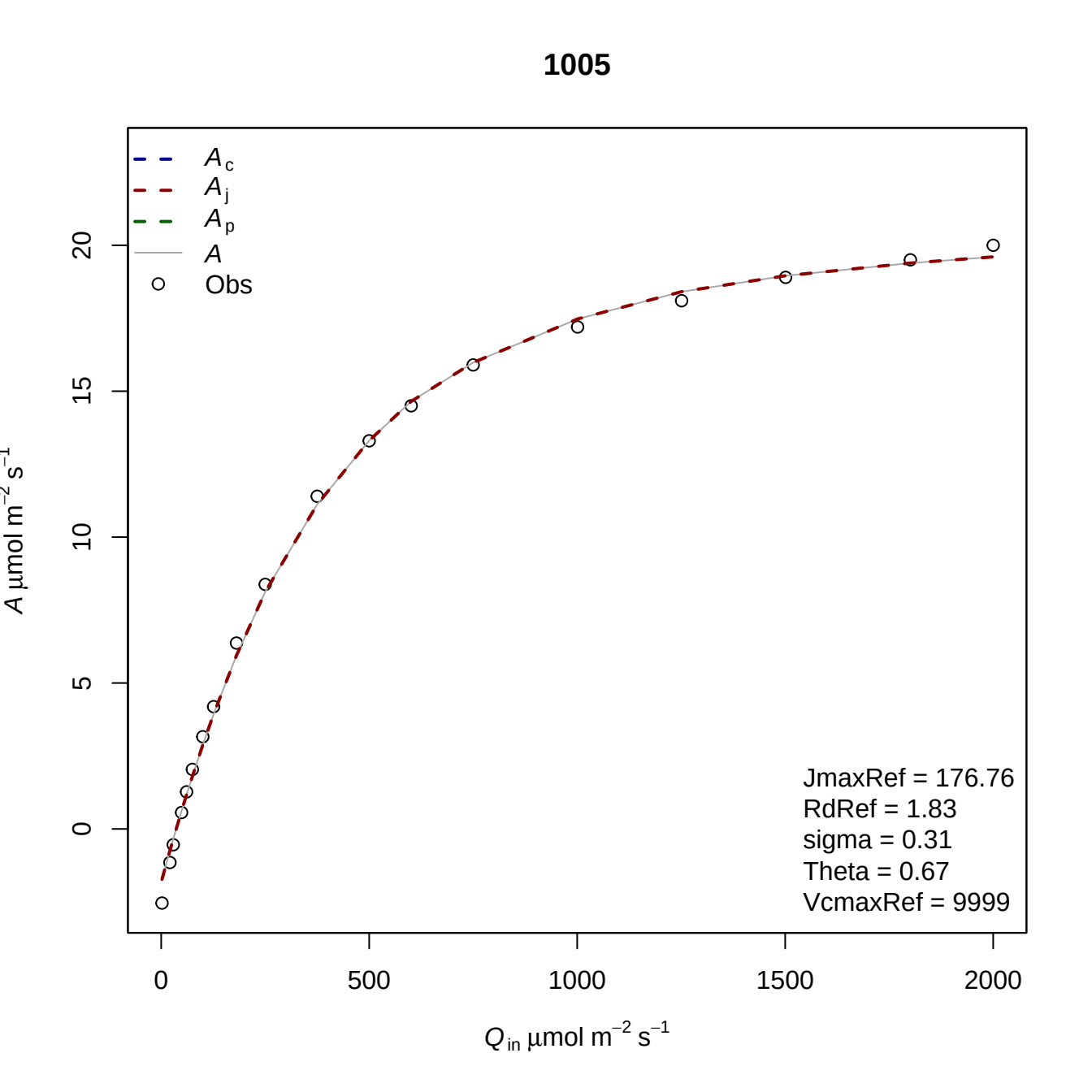

1000
A_c
A_j
A_p
A
Obs
JmaxRef = 67.49
RdRef = 1.28
sigma = 0.21
Theta = −0.08
VcmaxRef = 56.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

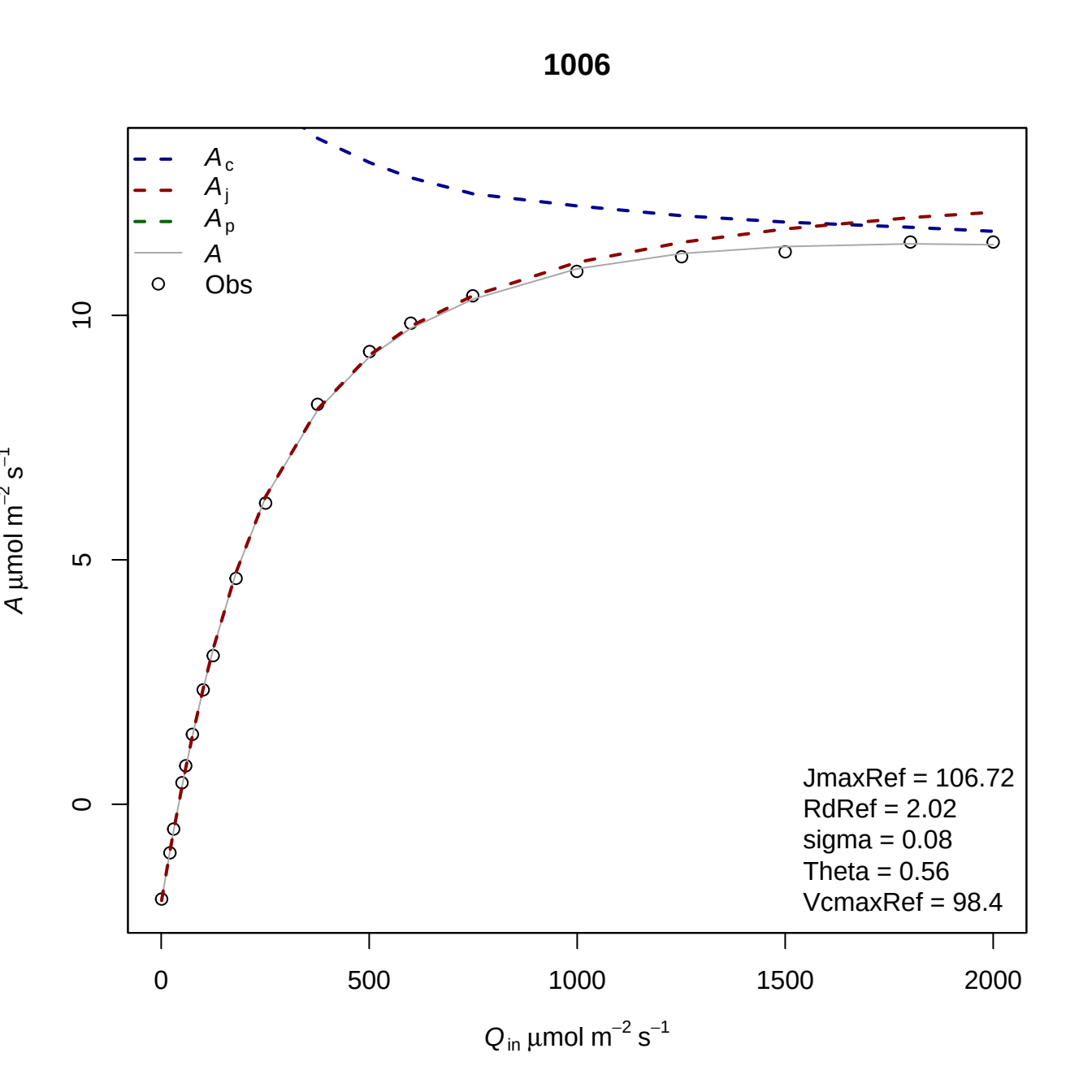

1001
A_c
A_j
A_p
A
Obs
JmaxRef = 91.54
RdRef = 1.37
sigma = 0.12
Theta = 0.86
VcmaxRef = 110.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

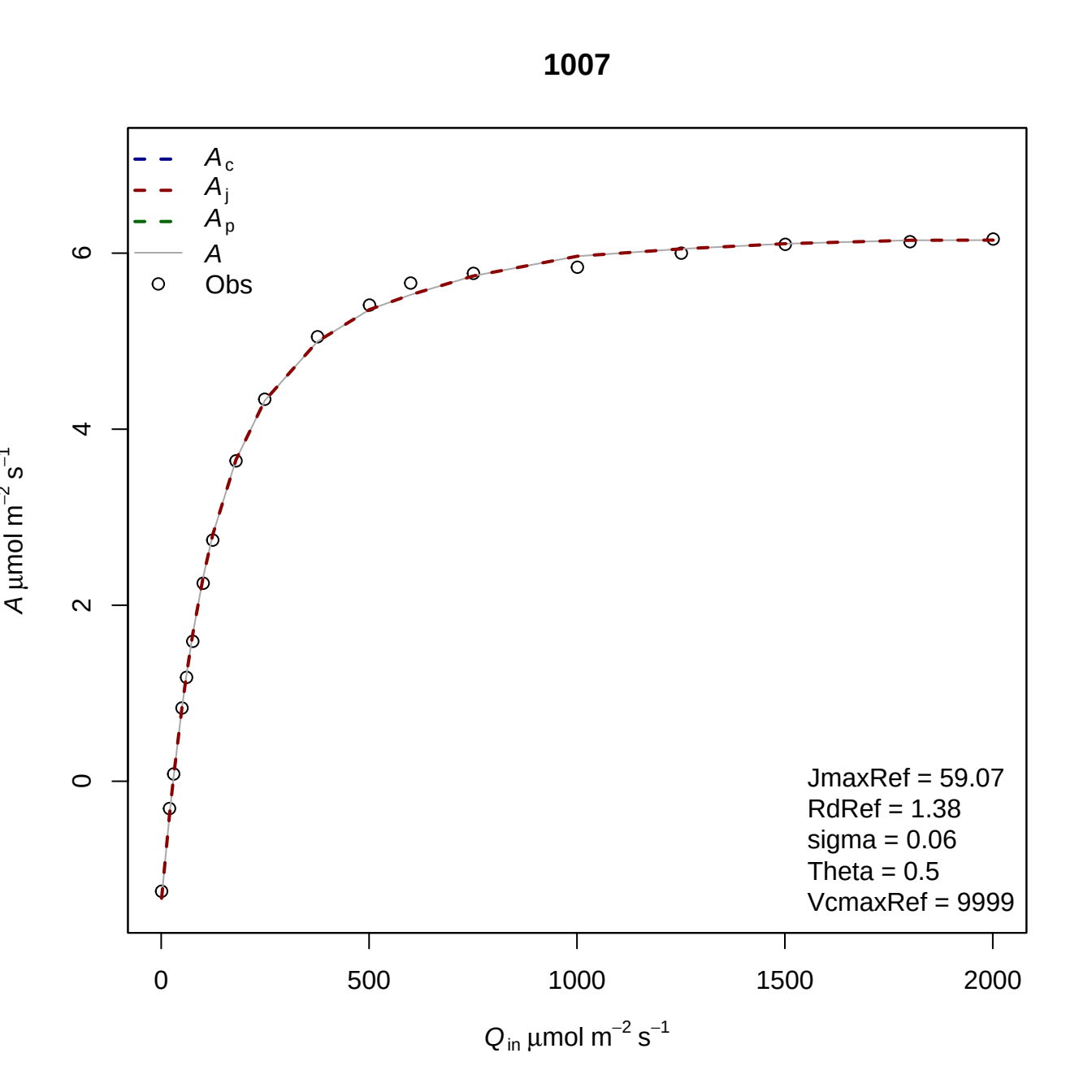

1002
A_c
A_j
A_p
A
Obs
JmaxRef = 102.19
RdRef = 1.46
sigma = 0.16
Theta = 0.27
VcmaxRef = 106.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

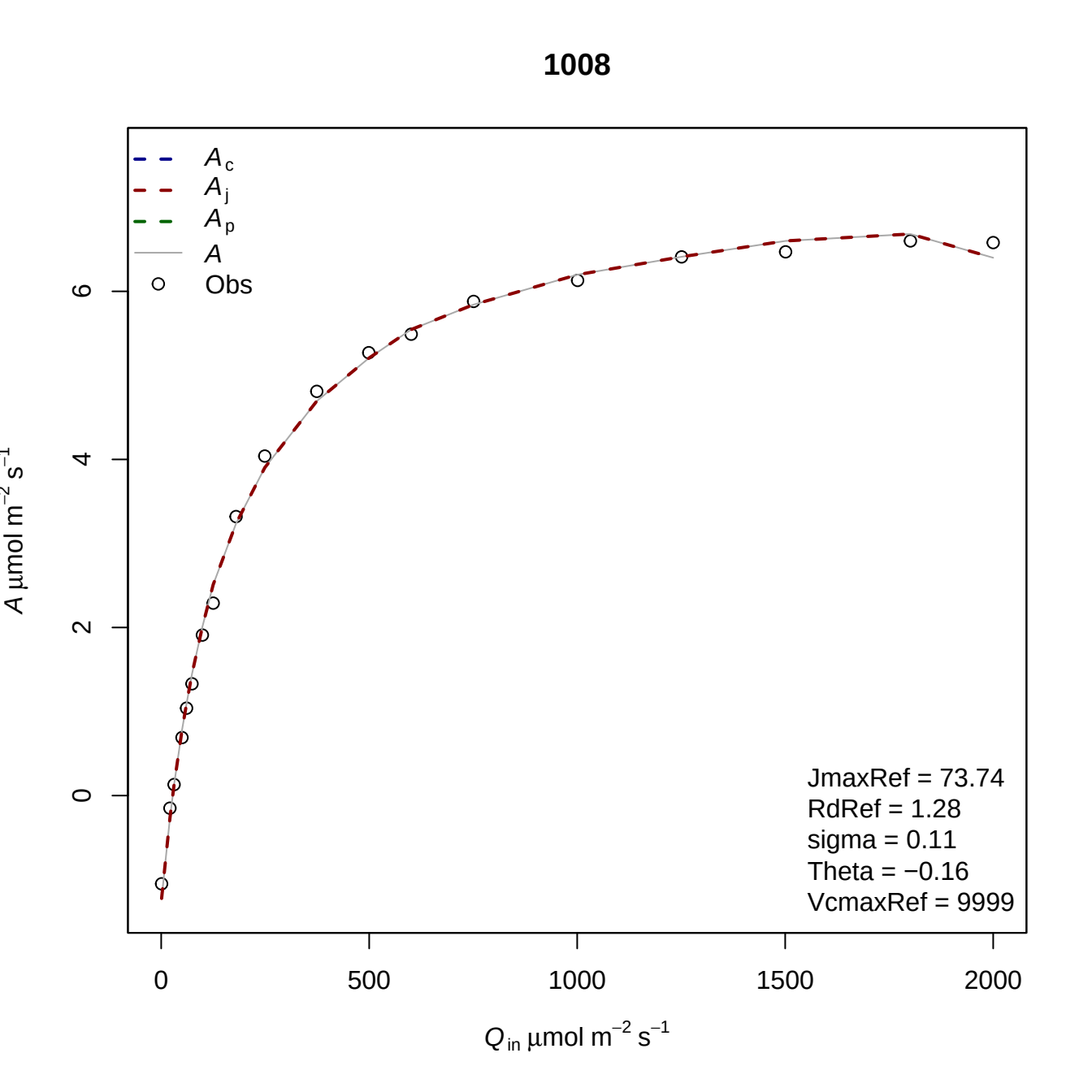

1003
A_c
A_j
A_p
A
Obs
JmaxRef = 151.01
RdRef = 1.47
sigma = 0.13
Theta = 0.52
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

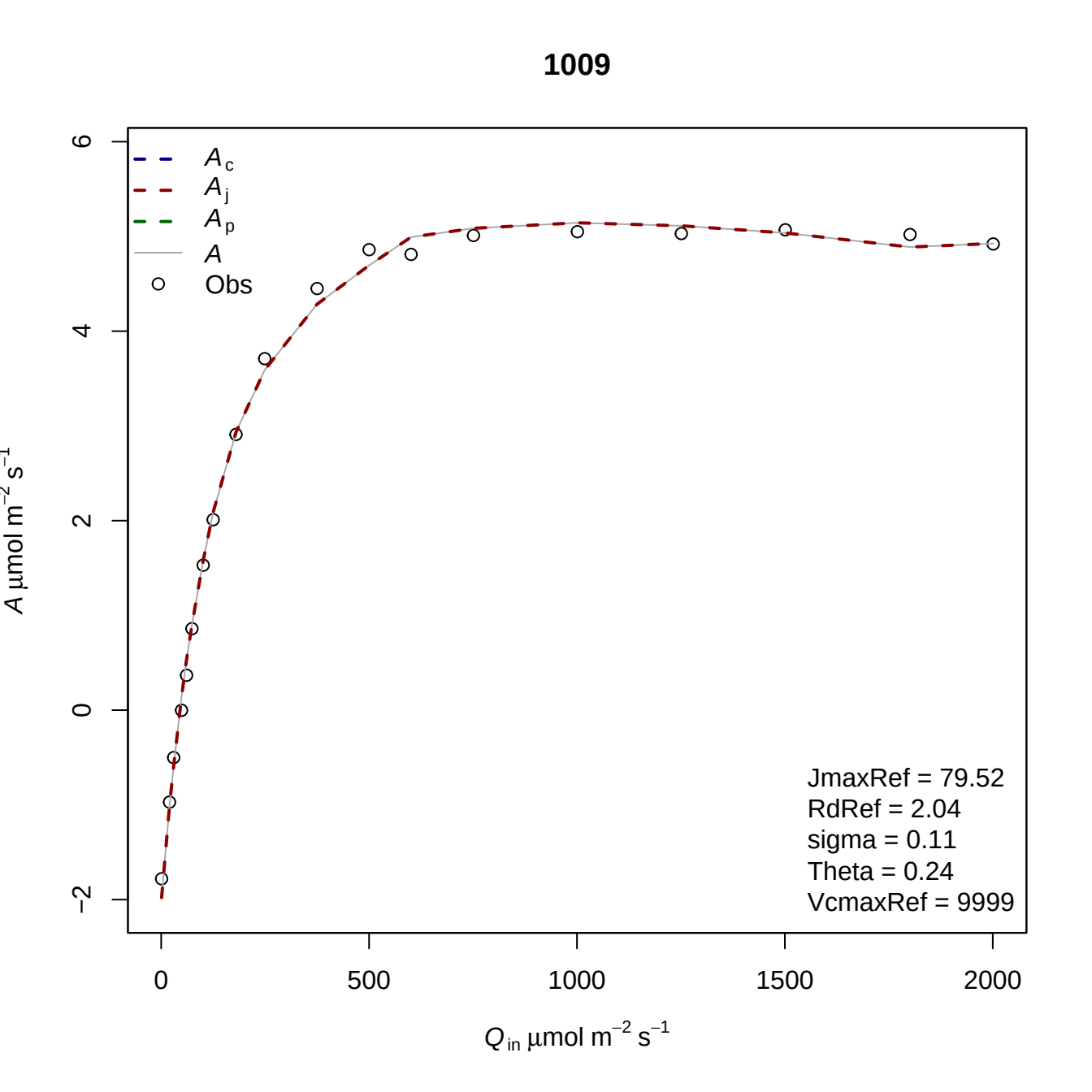

1004
A_c
A_j
A_p
A
Obs
JmaxRef = 53.04
RdRef = 0.86
sigma = 0.18
Theta = −0.5
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

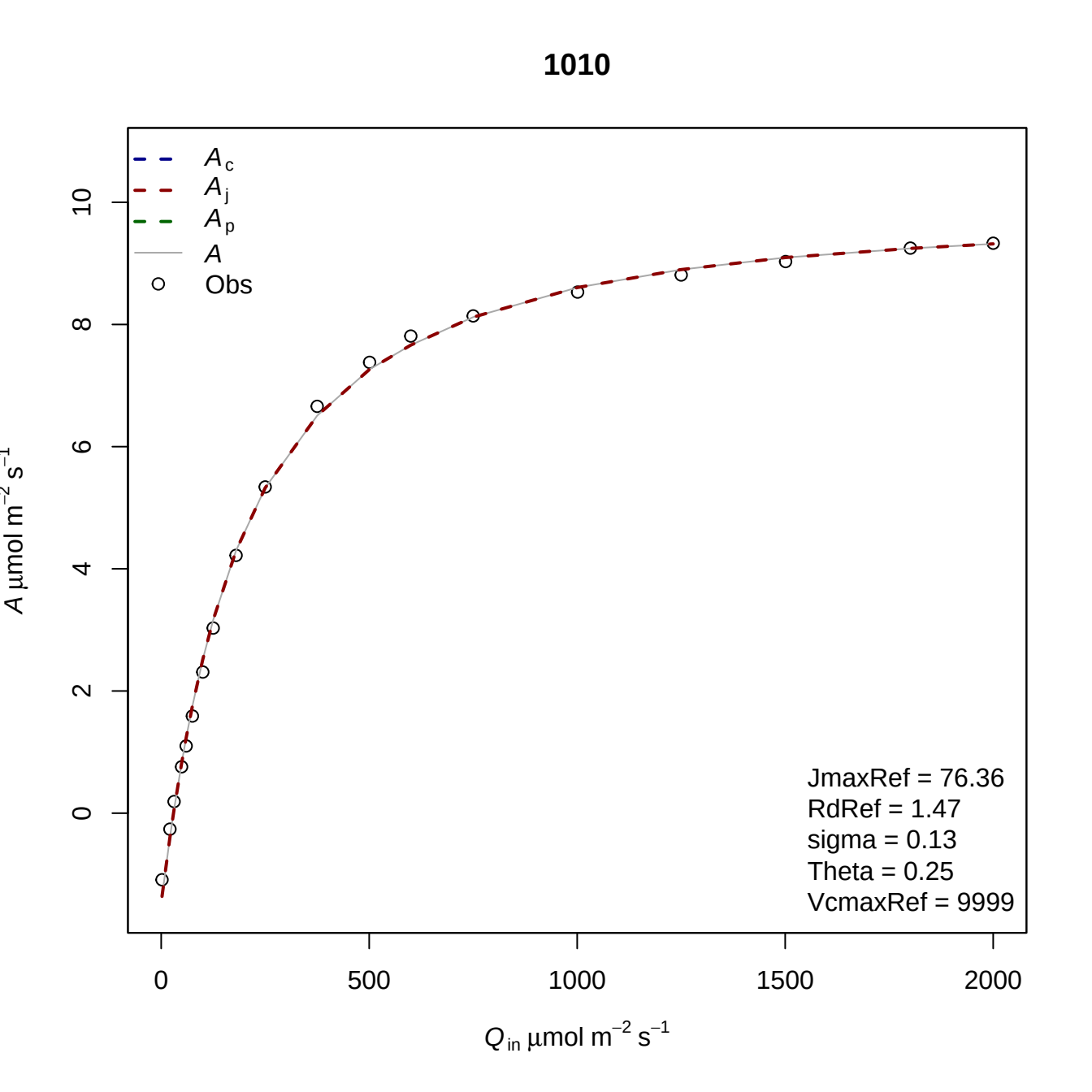

1005
A_c
A_j
A_p
A
Obs
JmaxRef = 176.76
RdRef = 1.83
sigma = 0.31
Theta = 0.67
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

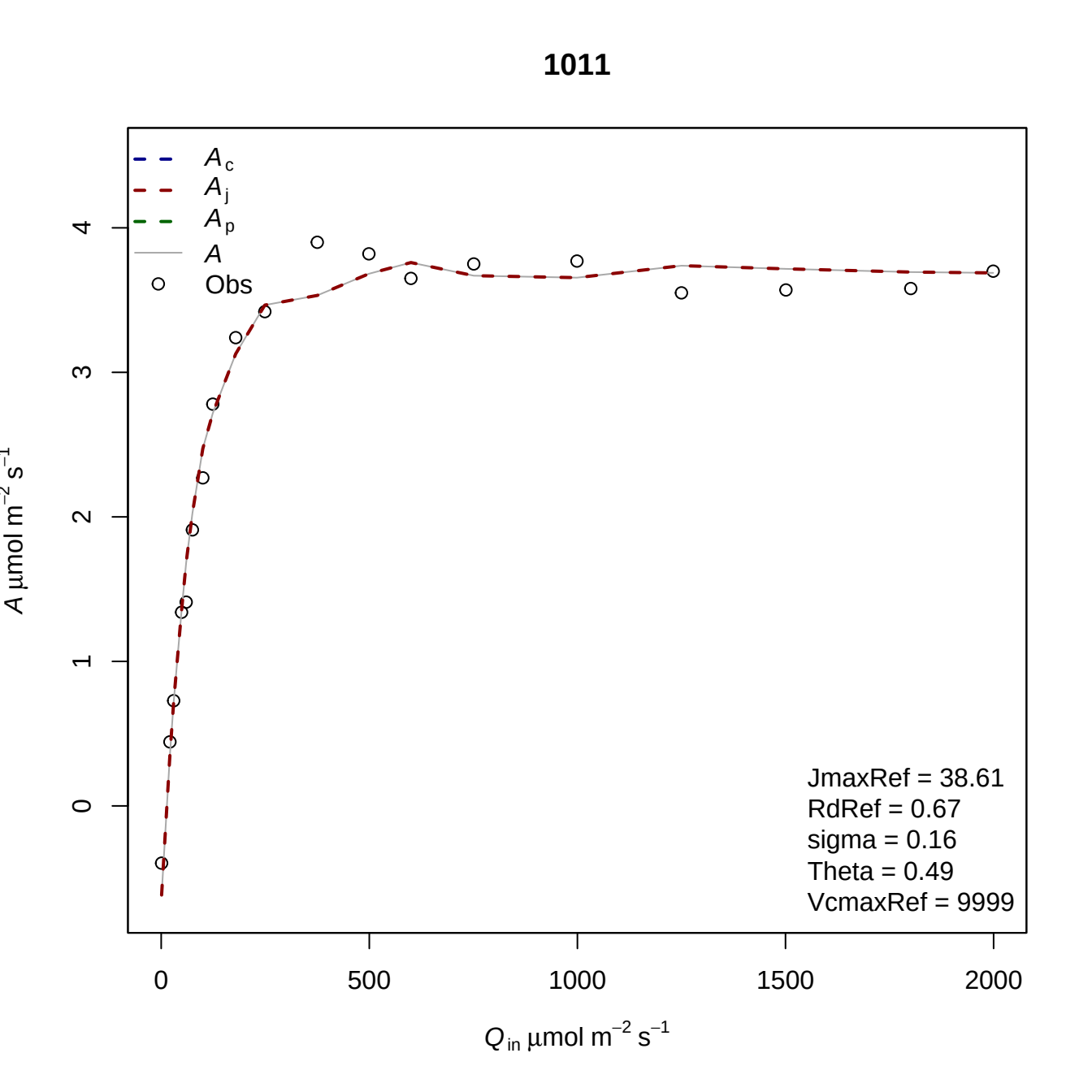

1006
A_c
A_j
A_p
A
Obs
JmaxRef = 106.72
RdRef = 2.02
sigma = 0.08
Theta = 0.56
VcmaxRef = 98.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

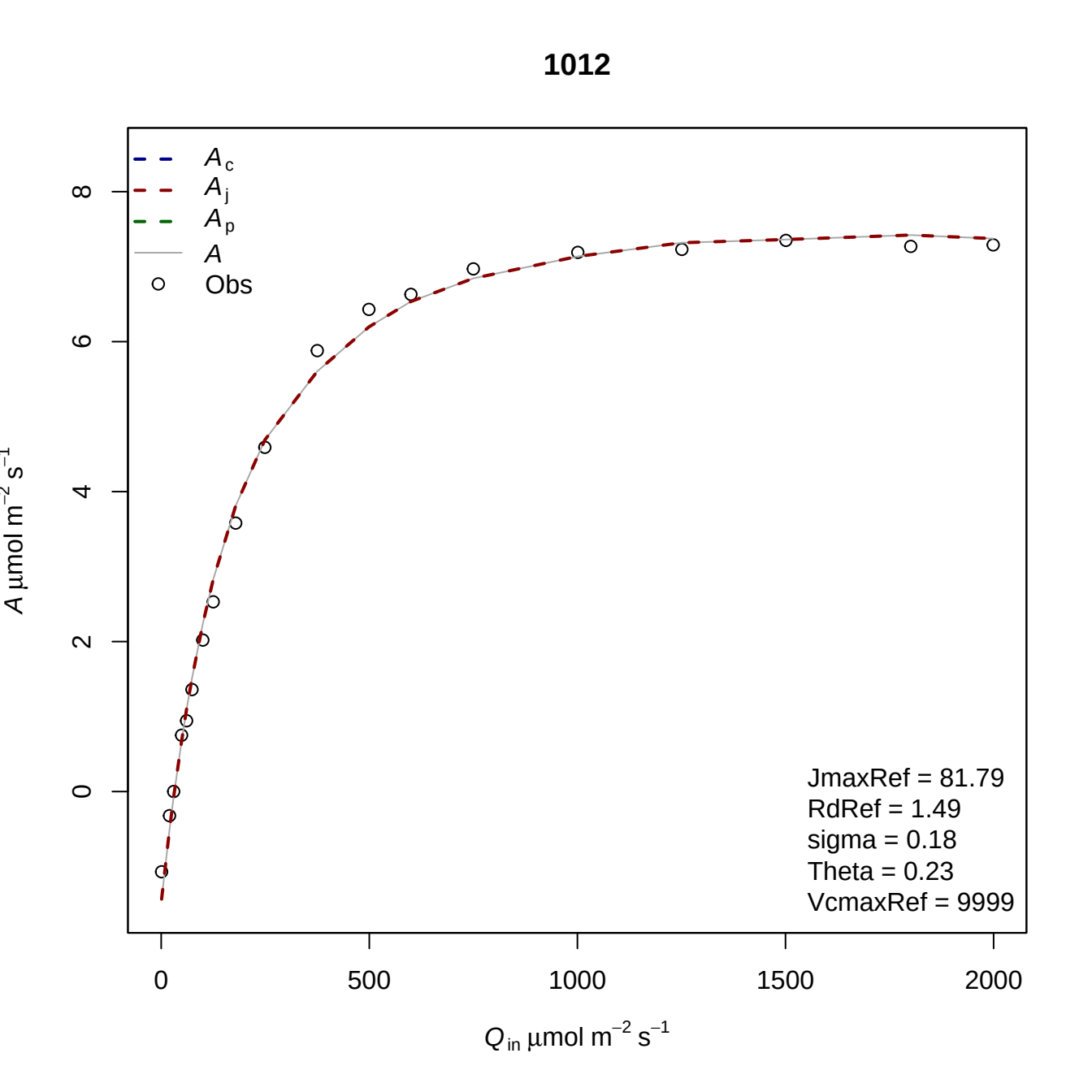

1007
A_c
A_j
A_p
A
Obs
JmaxRef = 59.07
RdRef = 1.38
sigma = 0.06
Theta = 0.5
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

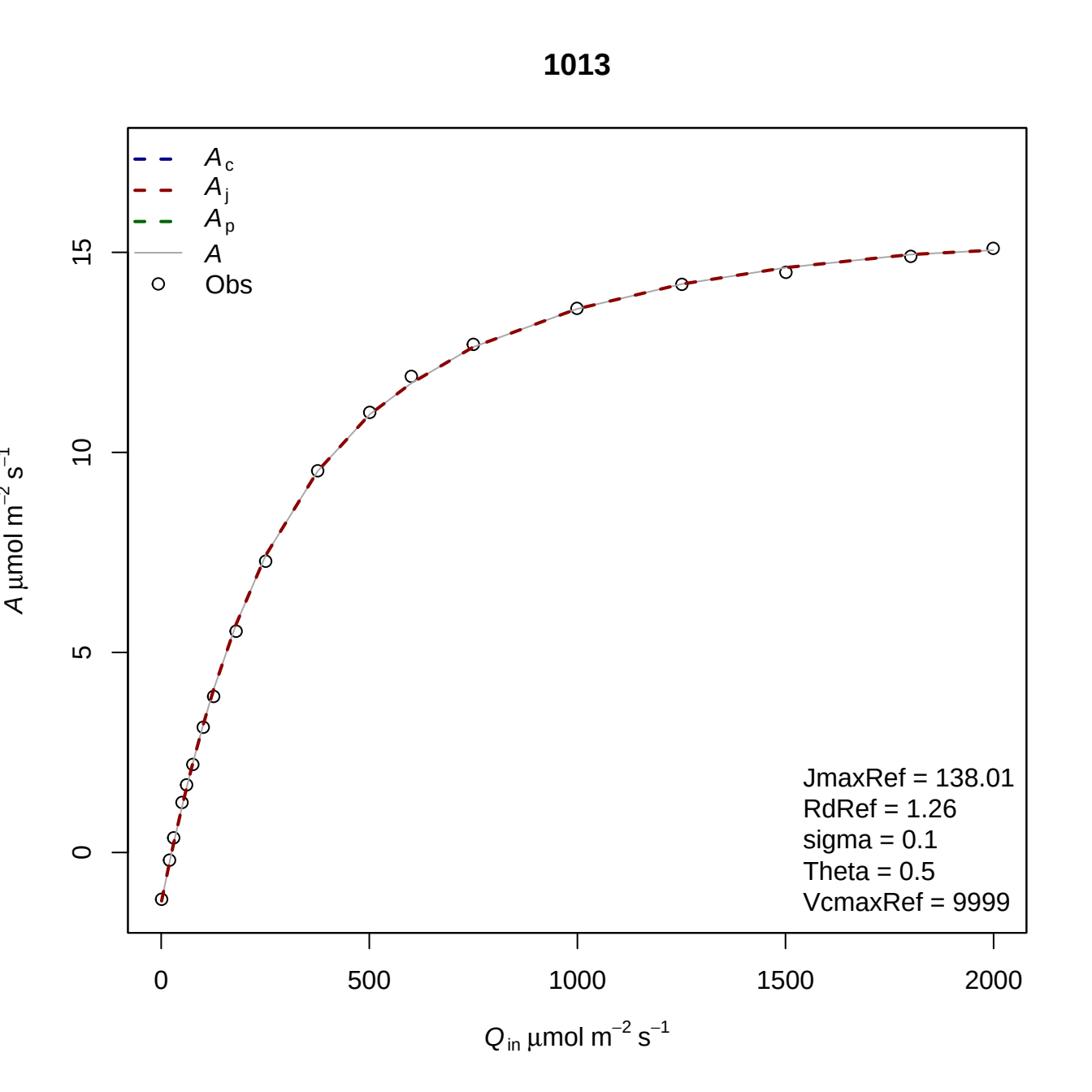

1008
A_c
A_j
A_p
A
Obs
JmaxRef = 73.74
RdRef = 1.28
sigma = 0.11
Theta = −0.16
VcmaxRef = 9999
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

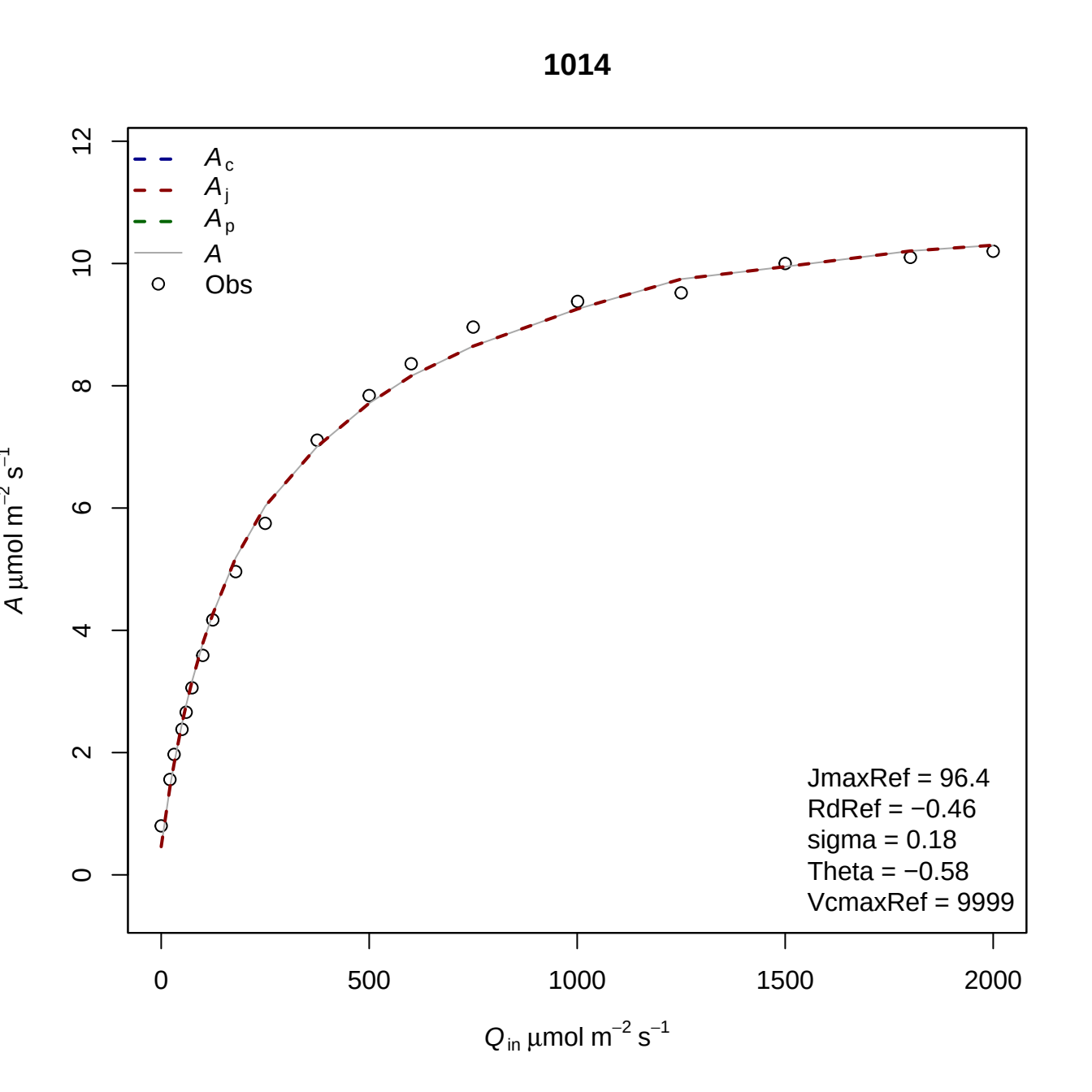

1009
A_c
A_j
A_p
A
Obs
JmaxRef = 79.52
RdRef = 2.04
sigma = 0.11
Theta = 0.24
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

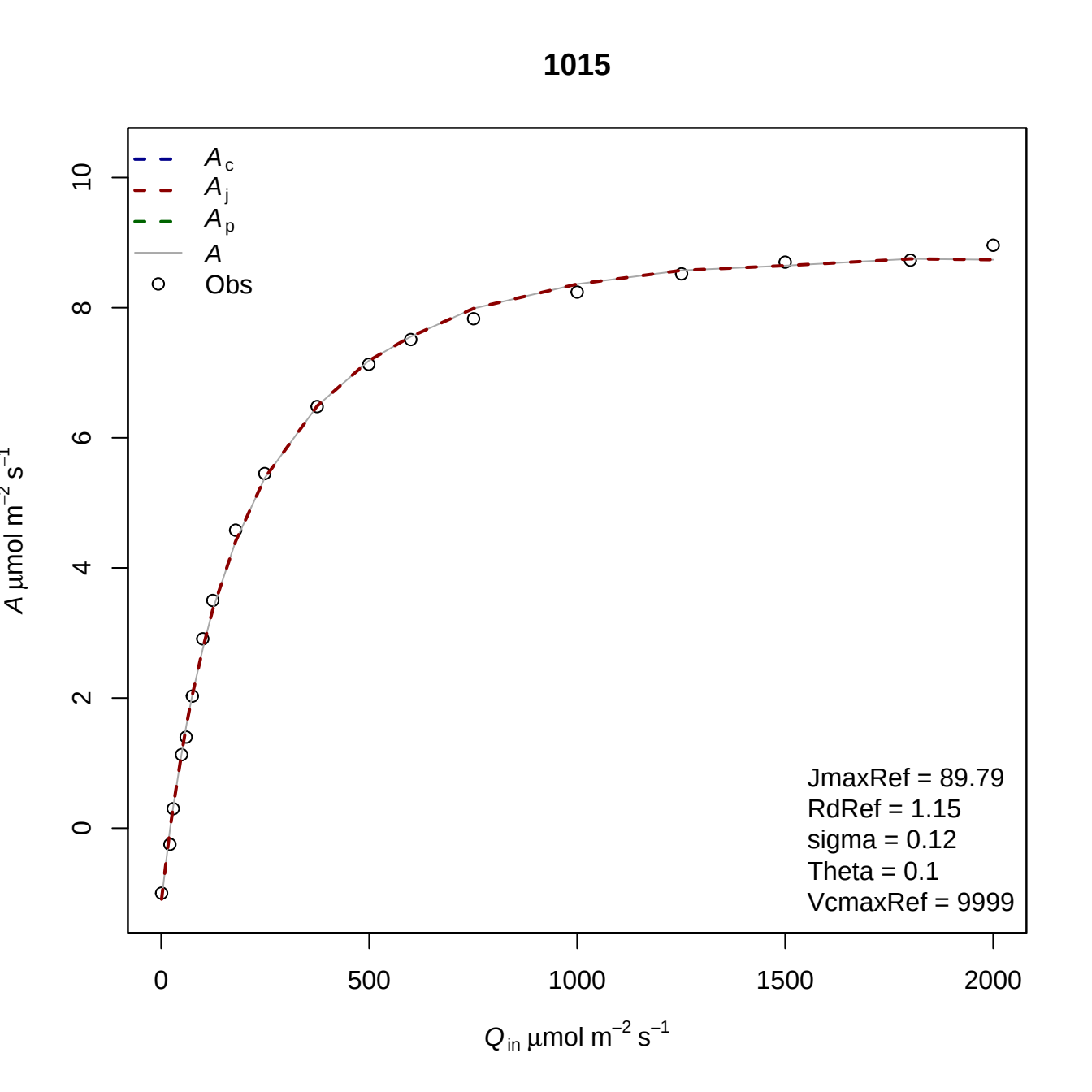

1010
A_c
A_j
A_p
A
Obs
JmaxRef = 76.36
RdRef = 1.47
sigma = 0.13
Theta = 0.25
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

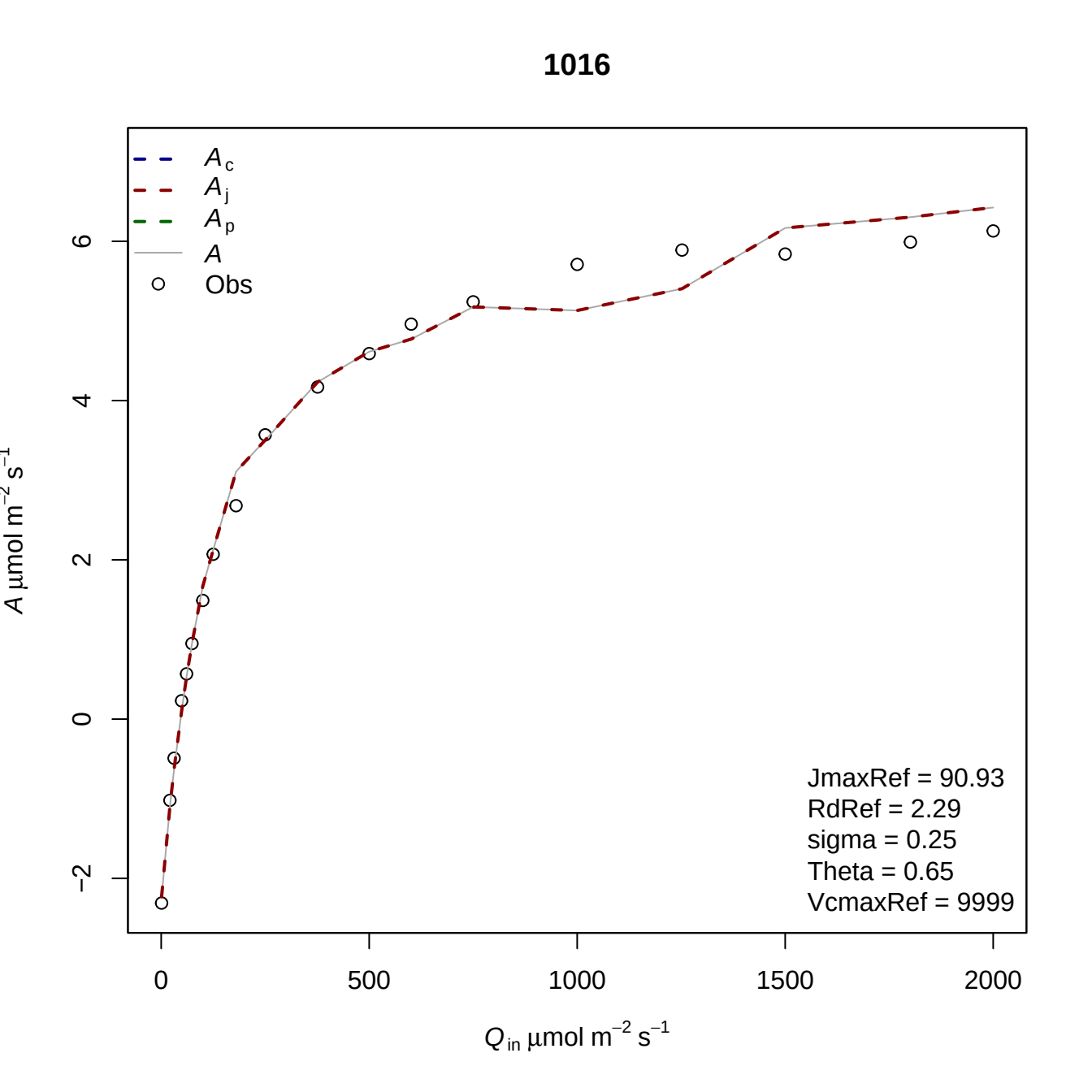

1011
A_c
A_j
A_p
A
Obs
JmaxRef = 38.61
RdRef = 0.67
sigma = 0.16
Theta = 0.49
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

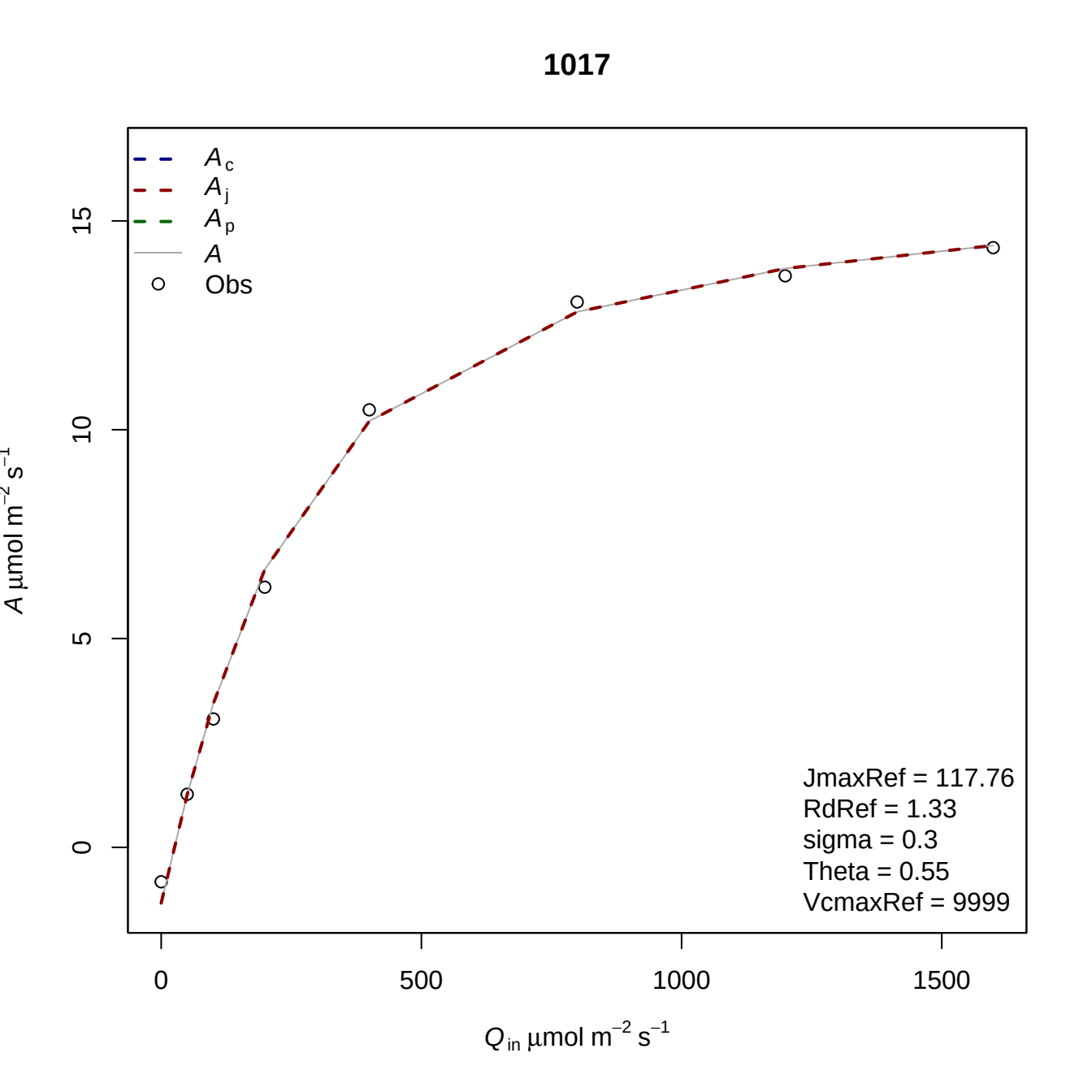

1012
A_c
A_j
A_p
A
Obs
JmaxRef = 81.79
RdRef = 1.49
sigma = 0.18
Theta = 0.23
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

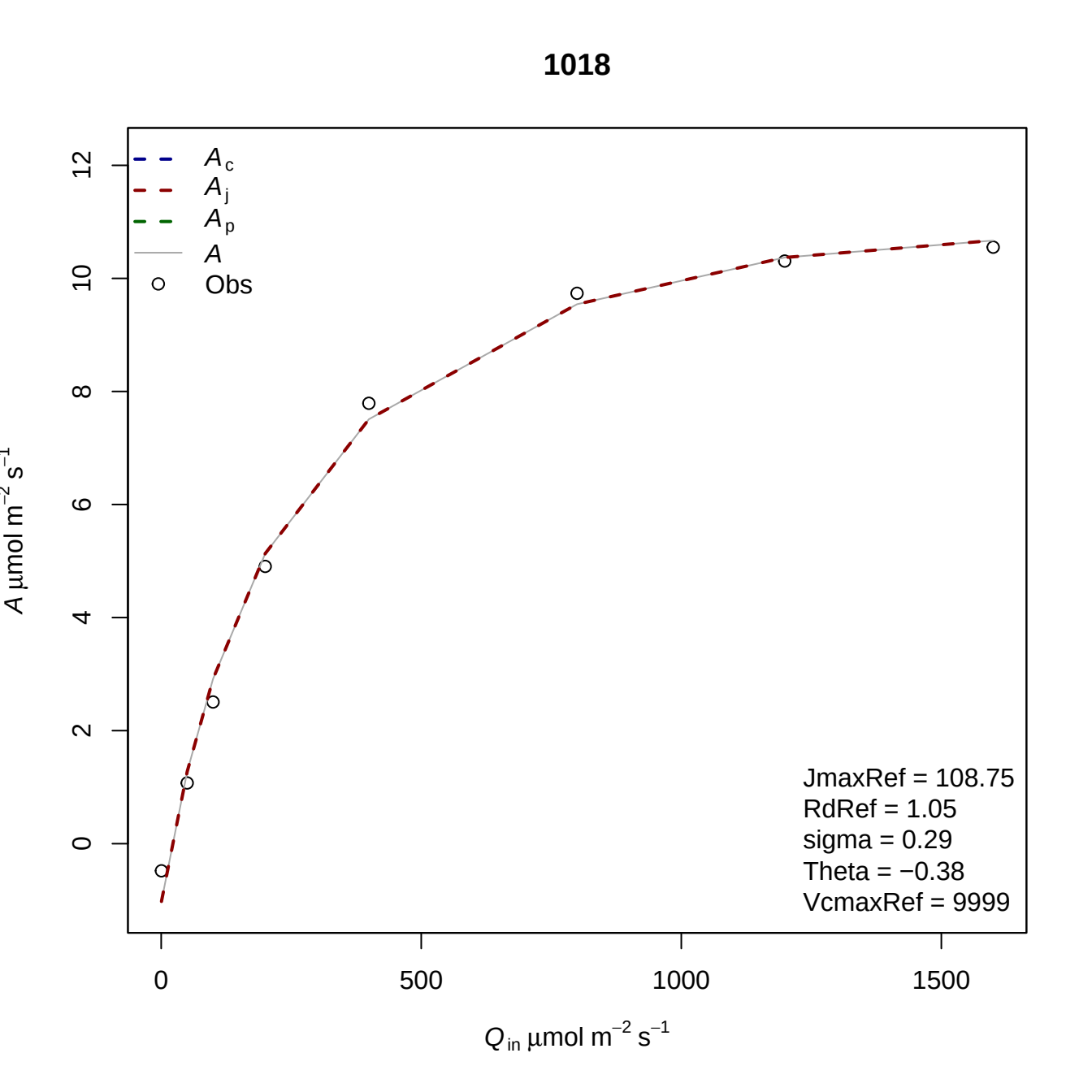

1013
A_c
A_j
A_p
A
Obs
JmaxRef = 138.01
RdRef = 1.26
sigma = 0.1
Theta = 0.5
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

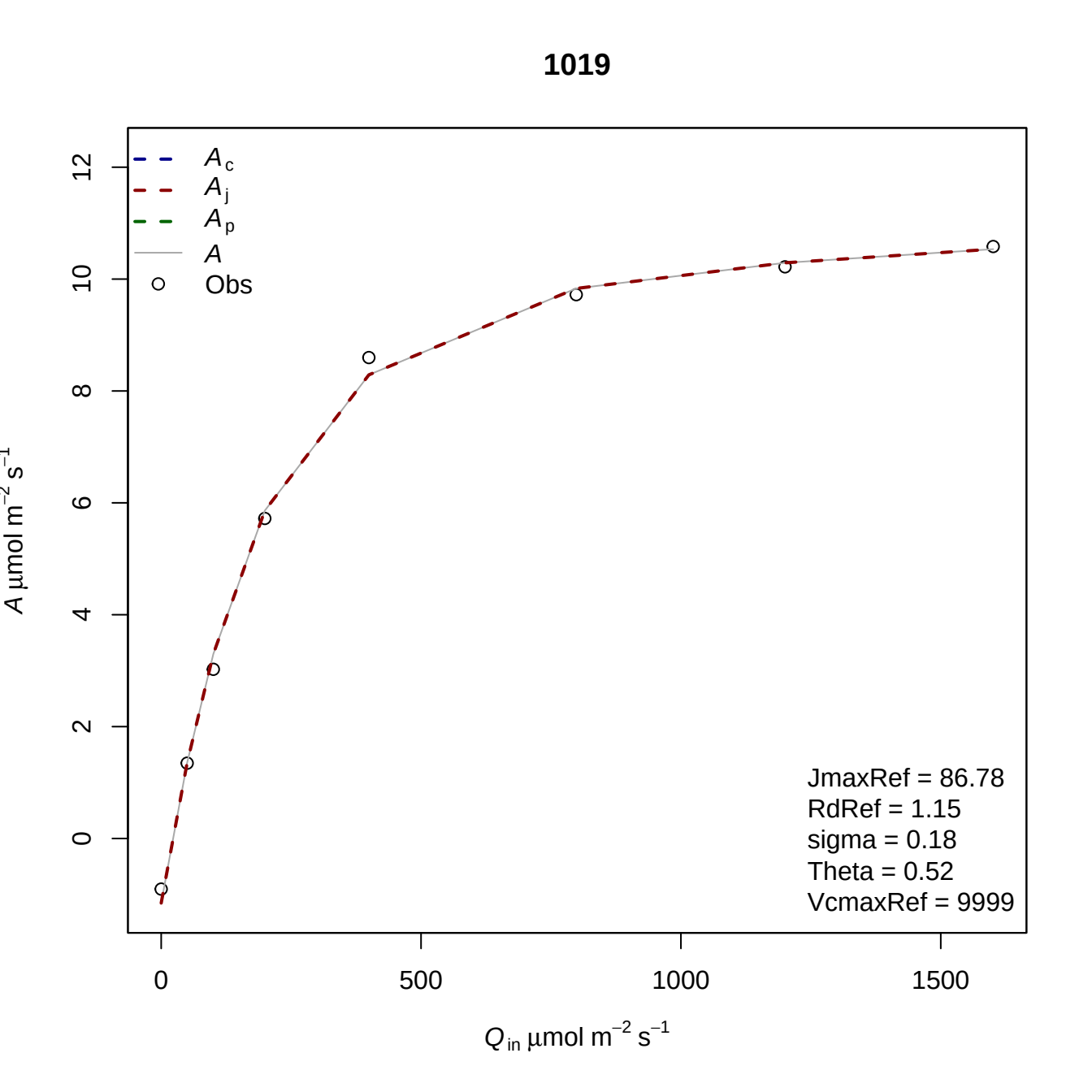

1014
A_c
A_j
A_p
A
Obs
JmaxRef = 96.4
RdRef = −0.46
sigma = 0.18
Theta = −0.58
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

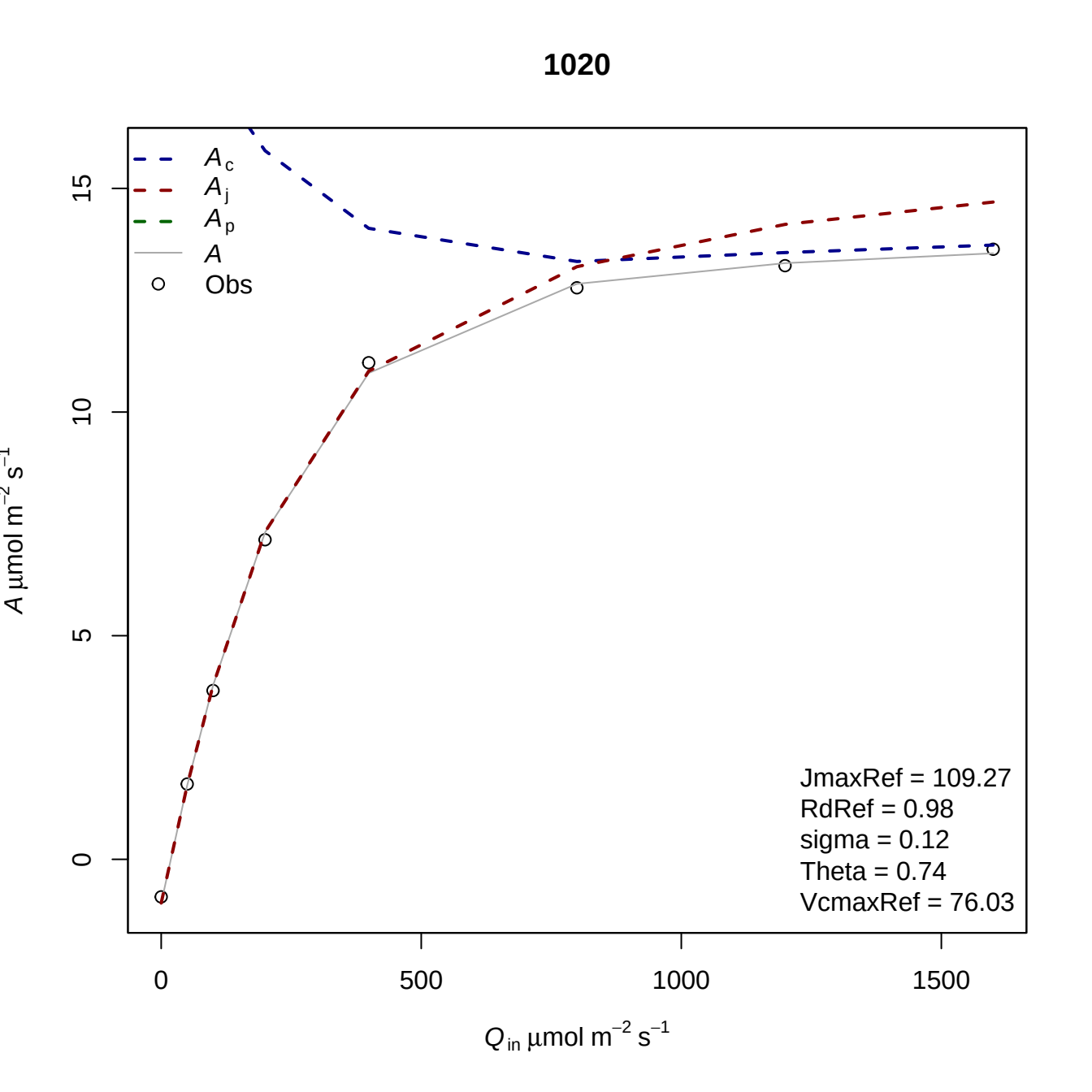

1015
A_c
A_j
A_p
A
Obs
JmaxRef = 89.79
RdRef = 1.15
sigma = 0.12
Theta = 0.1
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

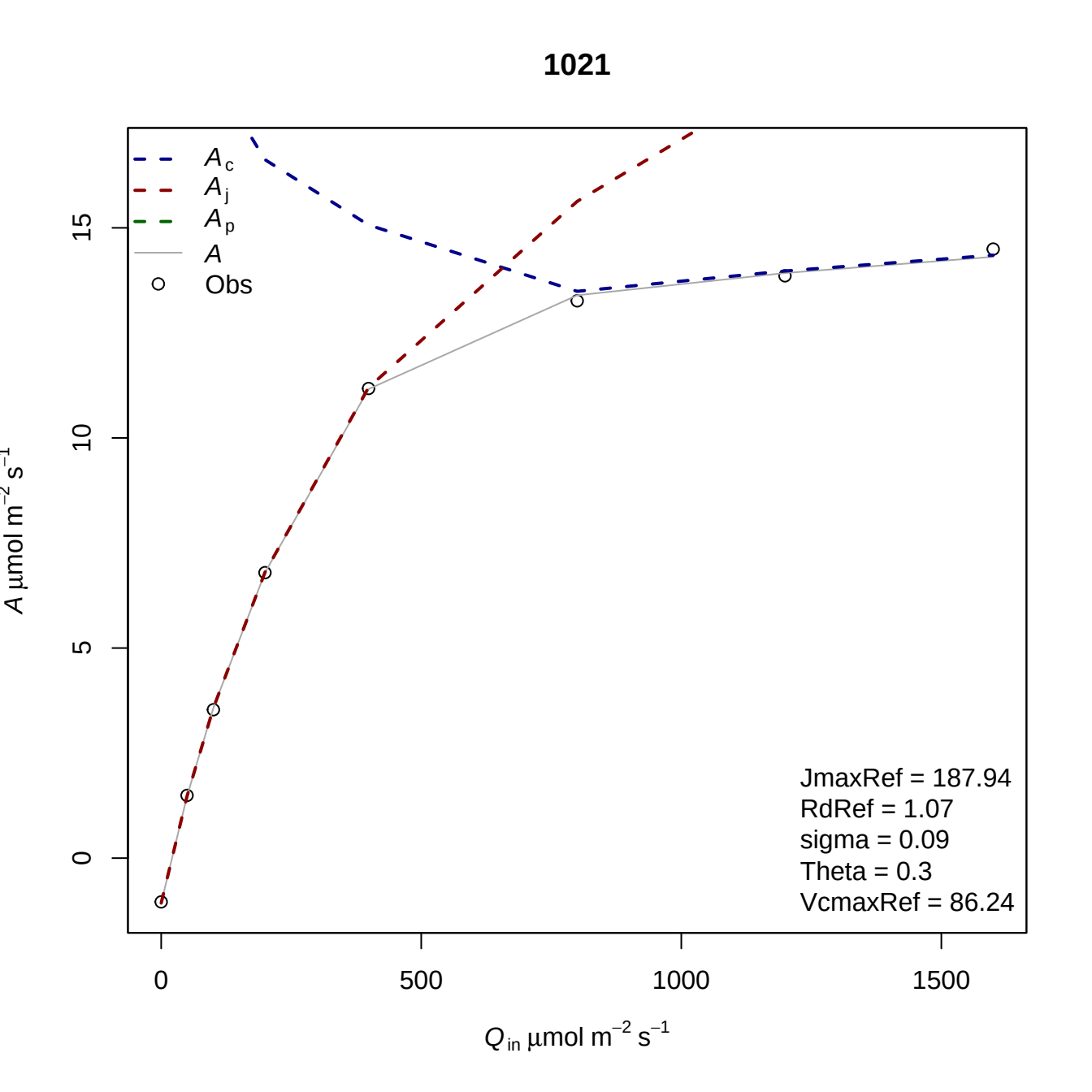

1016
A_c
A_j
A_p
A
Obs
JmaxRef = 90.93
RdRef = 2.29
sigma = 0.25
Theta = 0.65
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

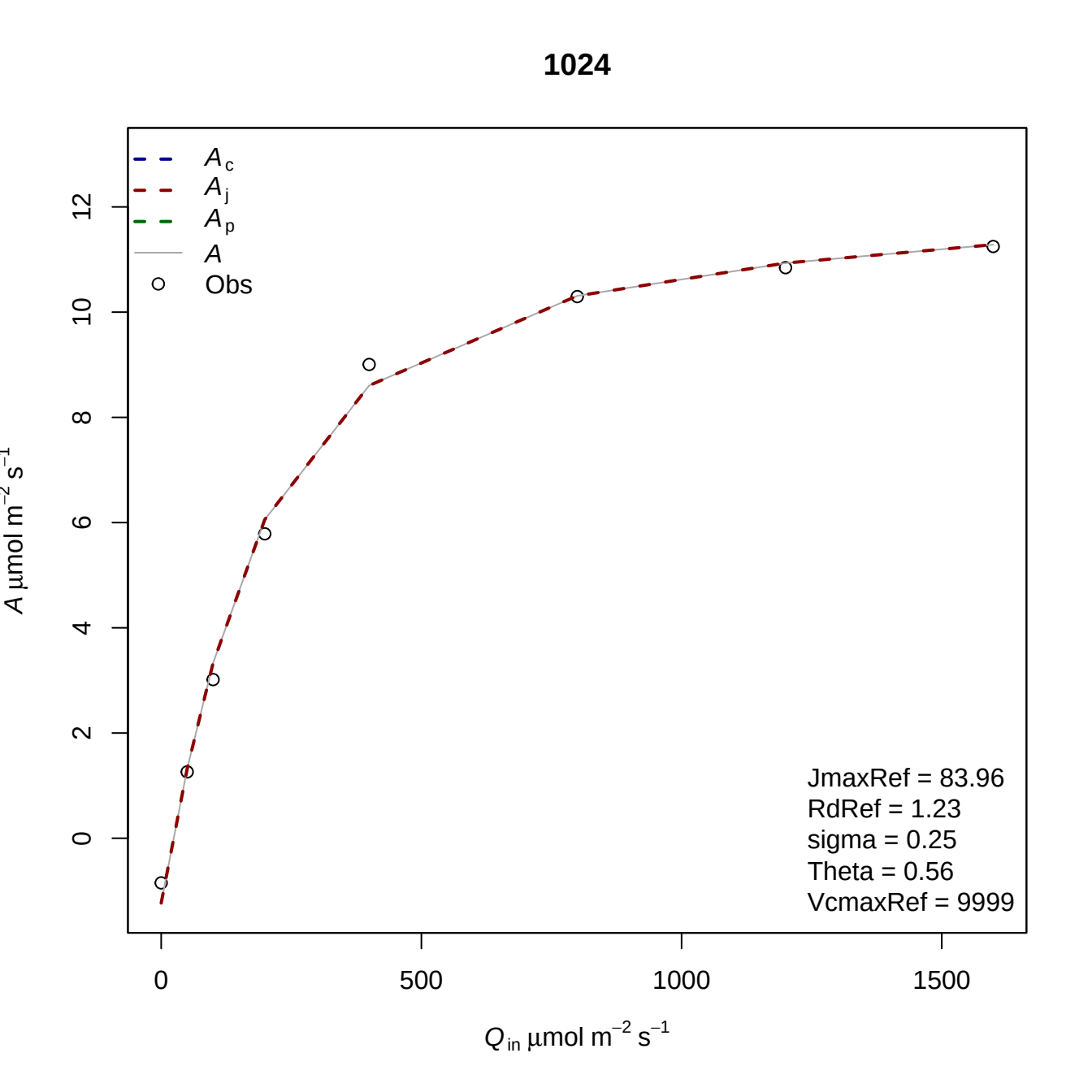

1017
A_c
A_j
A_p
A
Obs
JmaxRef = 117.76
RdRef = 1.33
sigma = 0.3
Theta = 0.55
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

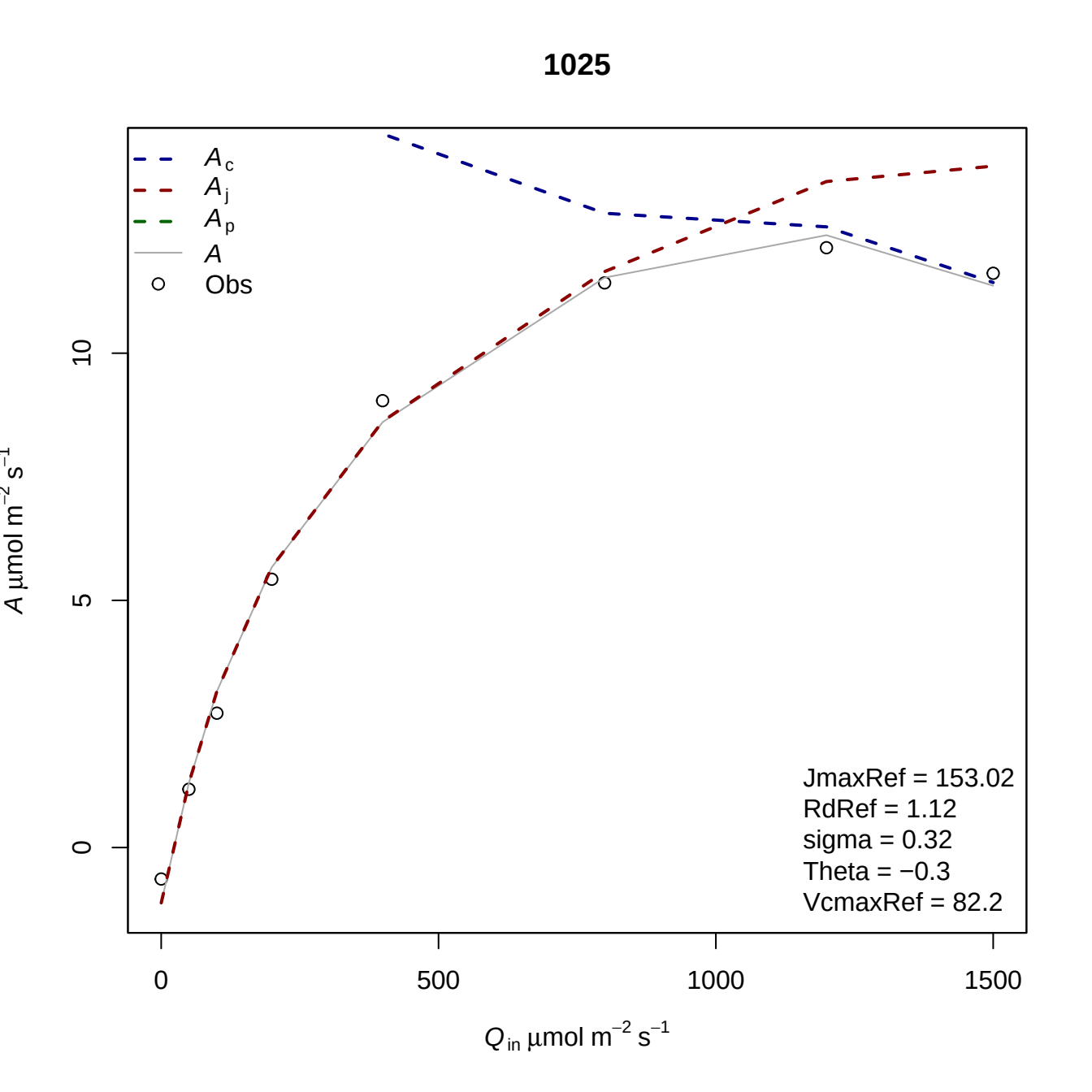

1018
A_c
A_j
A_p
A
Obs
JmaxRef = 108.75
RdRef = 1.05
sigma = 0.29
Theta = −0.38
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

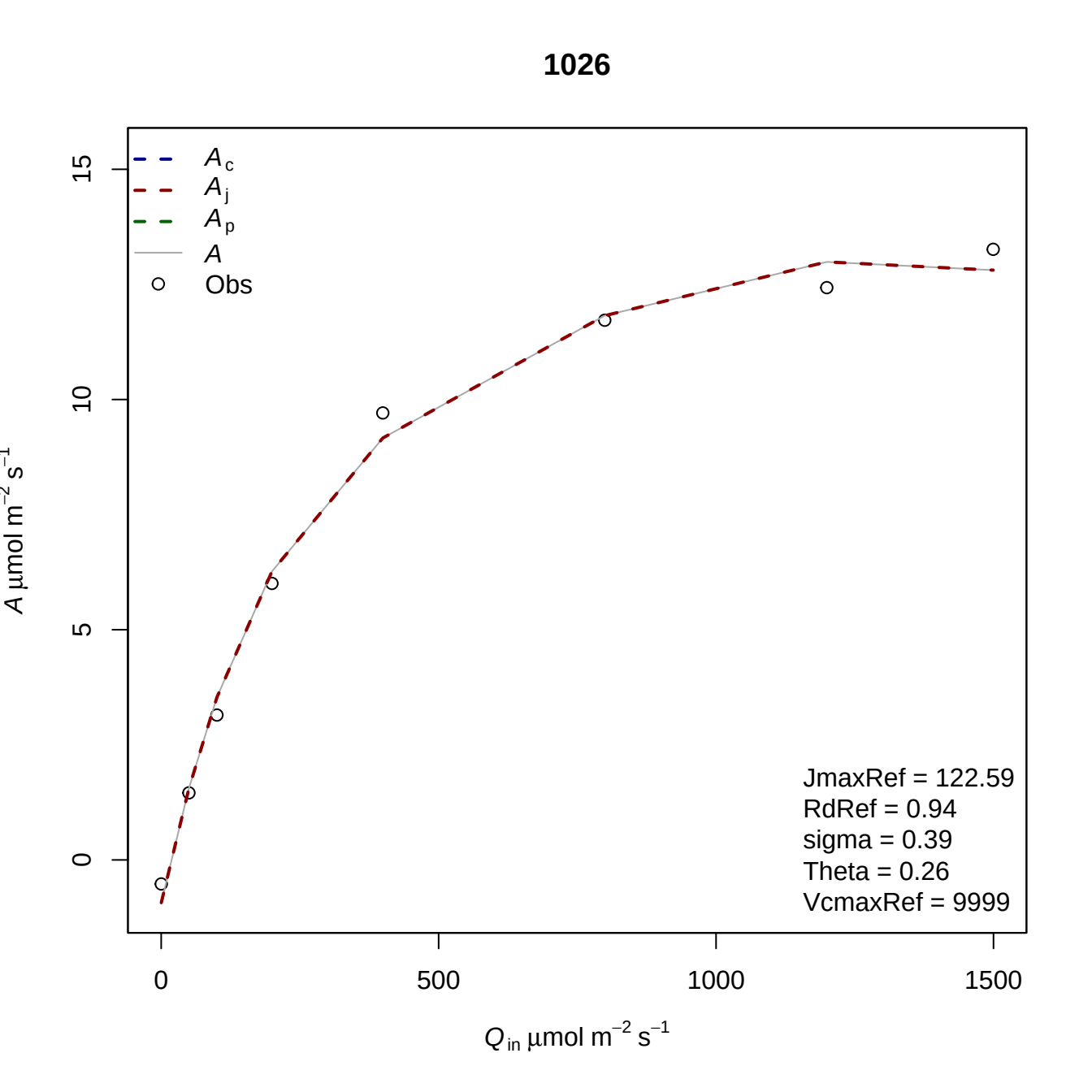

1019
A_c
A_j
A_p
A
Obs
JmaxRef = 86.78
RdRef = 1.15
sigma = 0.18
Theta = 0.52
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

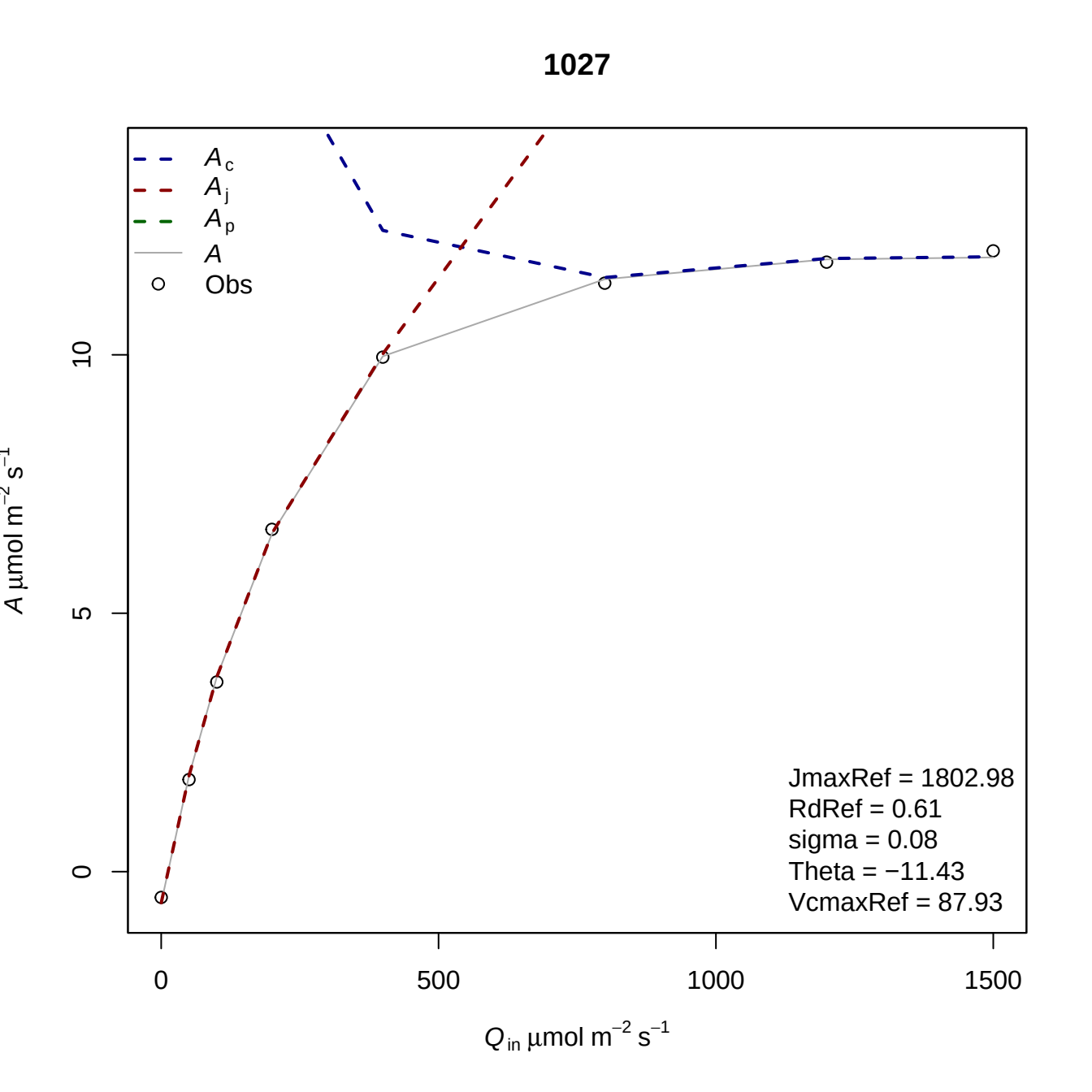

1020
A_c
A_j
A_p
A
Obs
JmaxRef = 109.27
RdRef = 0.98
sigma = 0.12
Theta = 0.74
VcmaxRef = 76.03
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

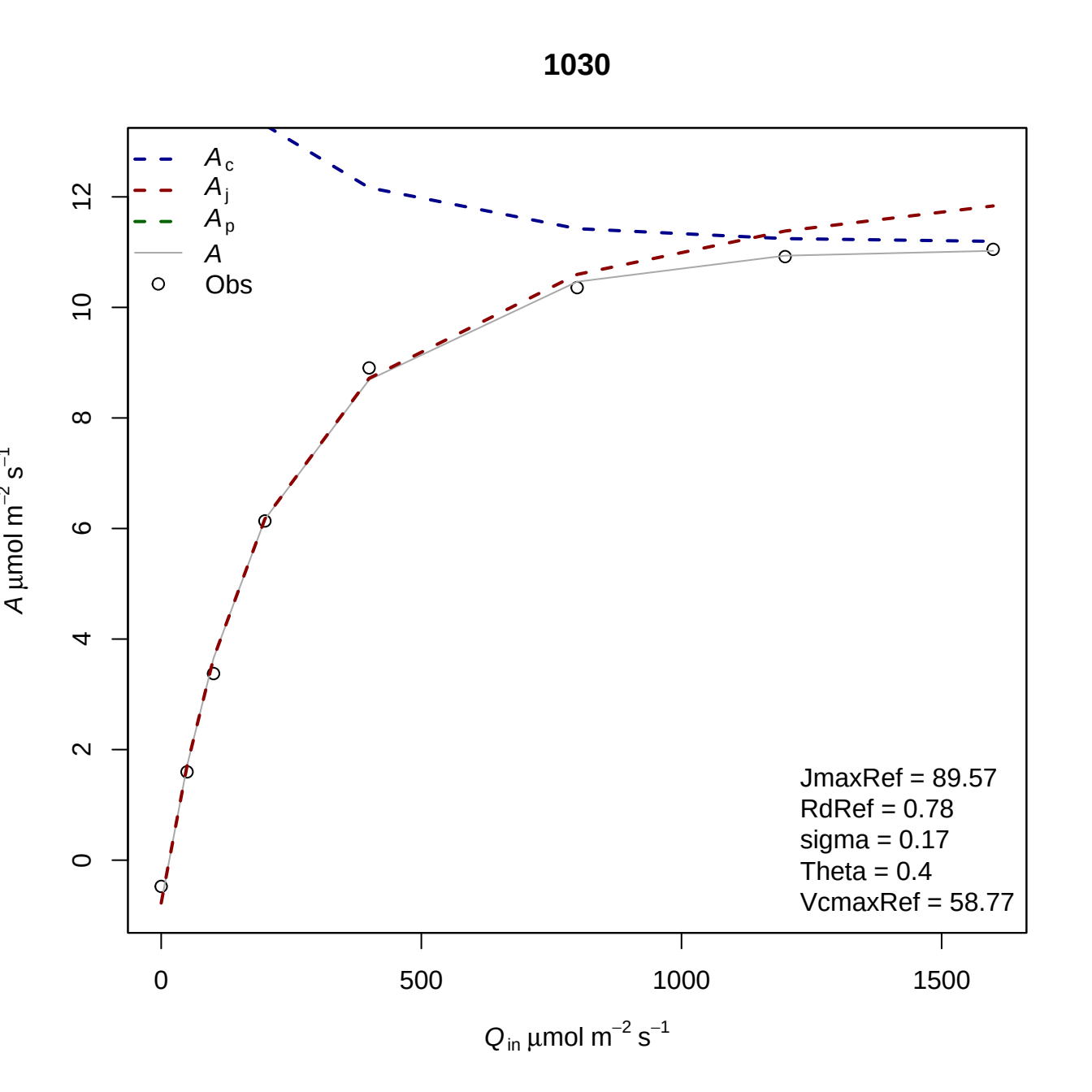

1021
A_c
A_j
A_p
A
Obs
JmaxRef = 187.94
RdRef = 1.07
sigma = 0.09
Theta = 0.3
VcmaxRef = 86.24
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

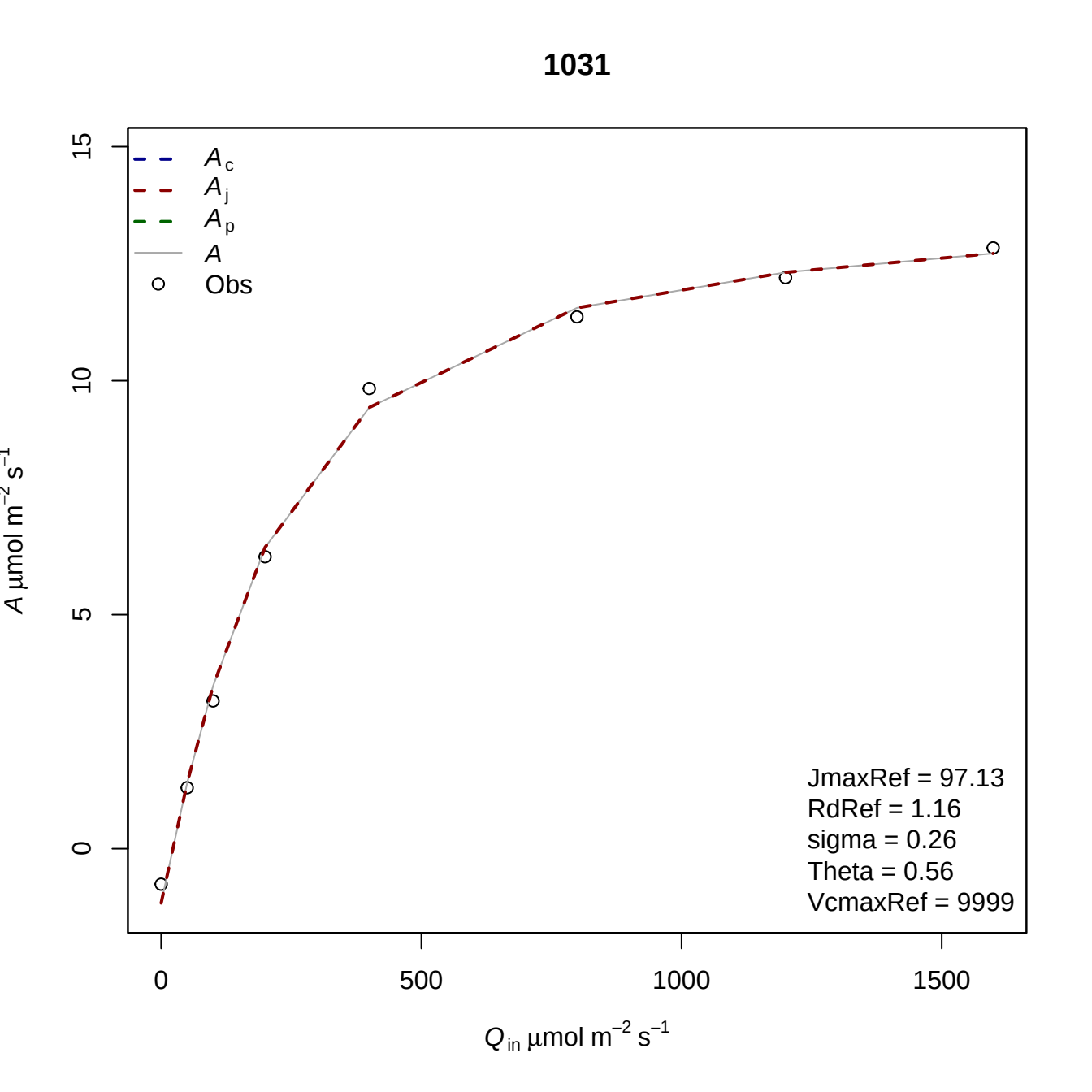

1024
A_c
A_j
A_p
A
Obs
JmaxRef = 83.96
RdRef = 1.23
sigma = 0.25
Theta = 0.56
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

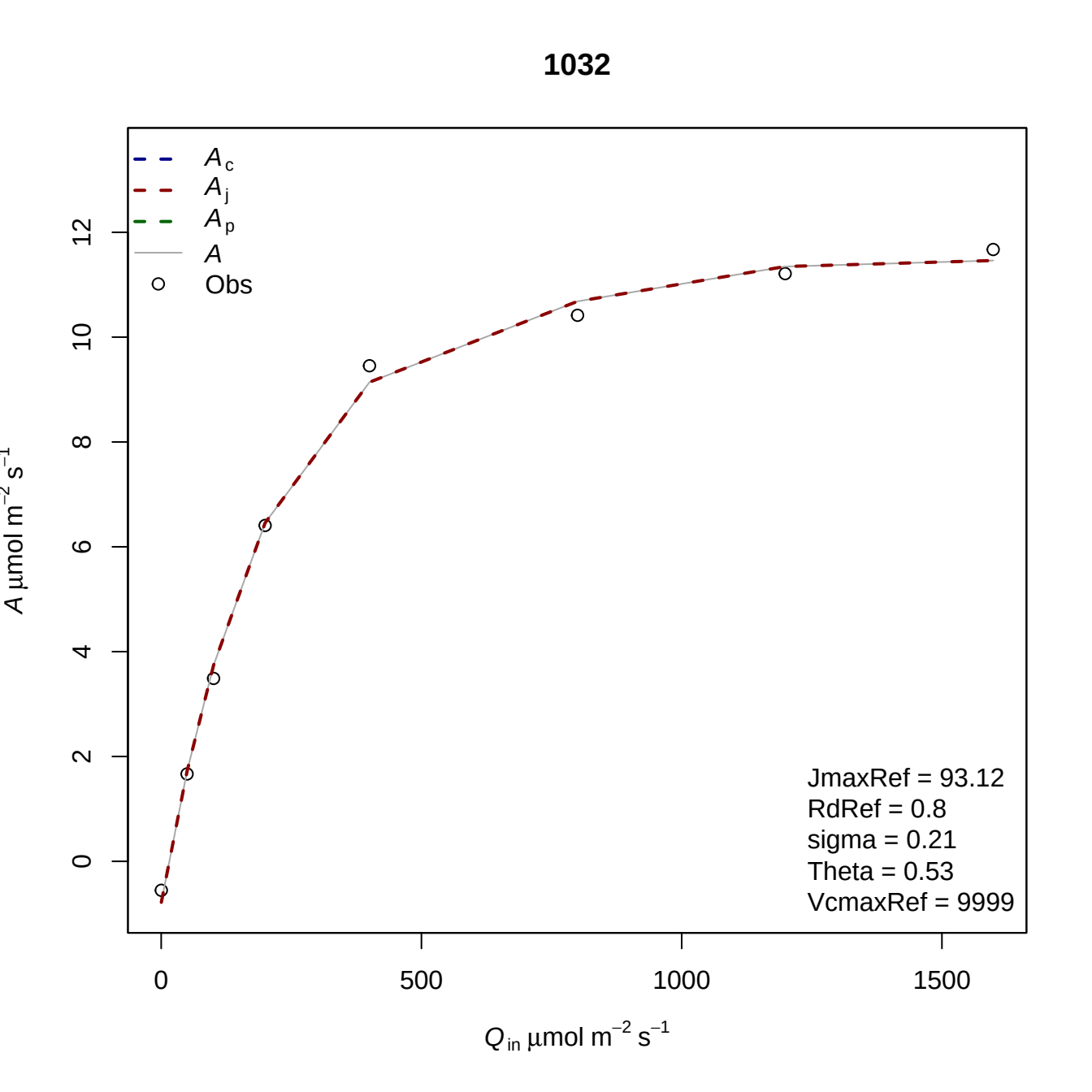

1025
A_c
A_j
A_p
A
Obs
JmaxRef = 153.02
RdRef = 1.12
sigma = 0.32
Theta = −0.3
VcmaxRef = 82.2
0
5
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

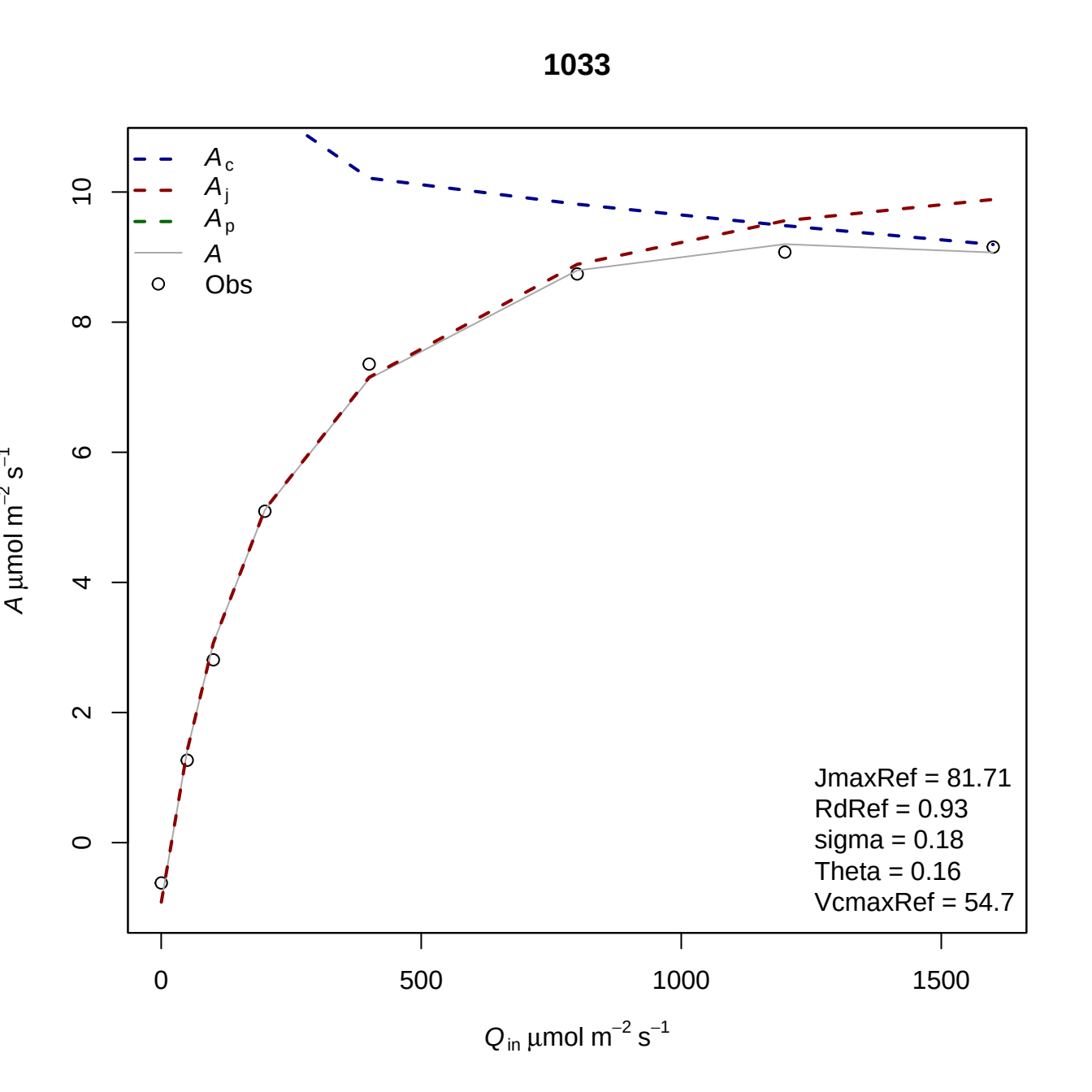

1026
A_c
A_j
A_p
A
Obs
JmaxRef = 122.59
RdRef = 0.94
sigma = 0.39
Theta = 0.26
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

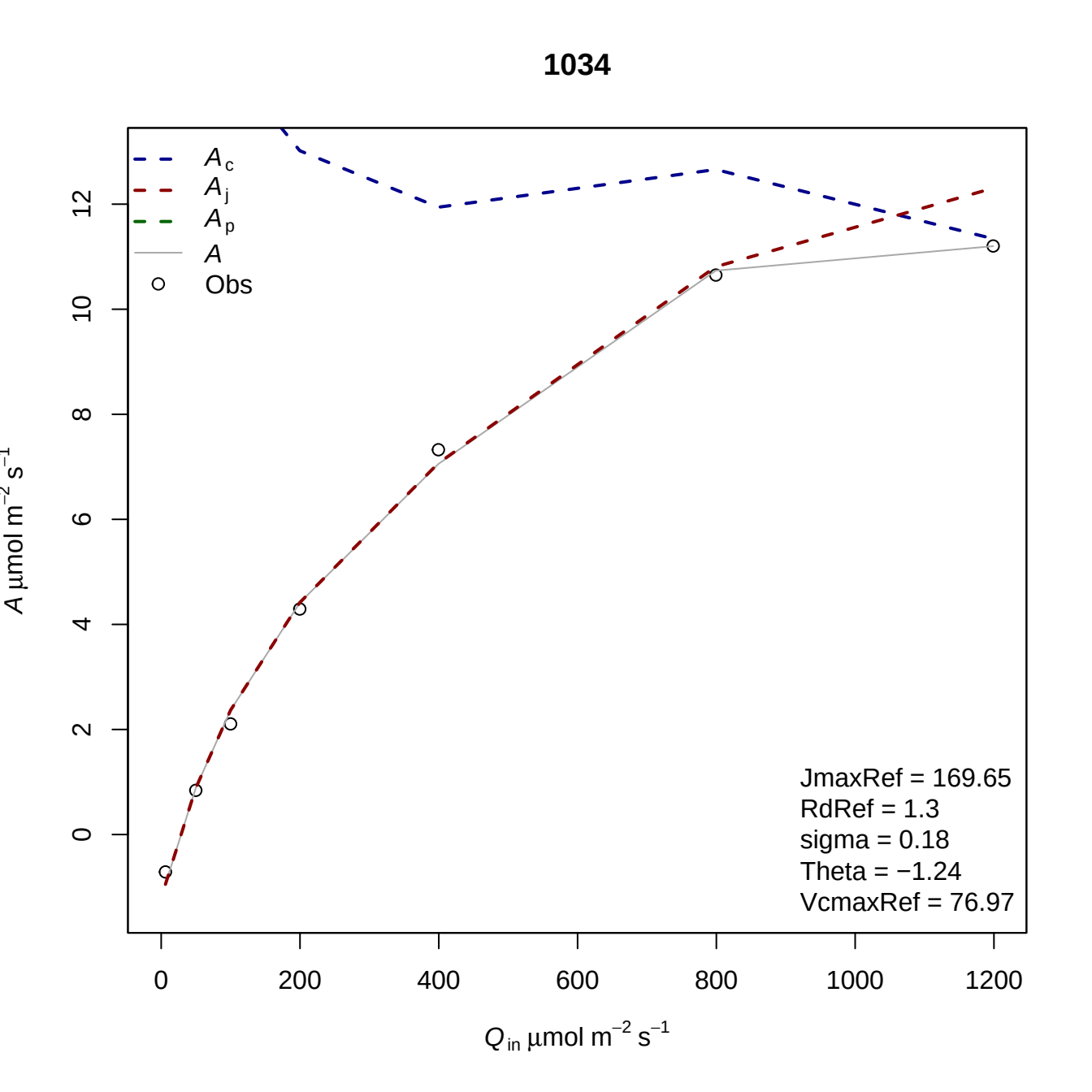

1027
A_c
A_j
A_p
A
Obs
JmaxRef = 1802.98
RdRef = 0.61
sigma = 0.08
Theta = −11.43
VcmaxRef = 87.93
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

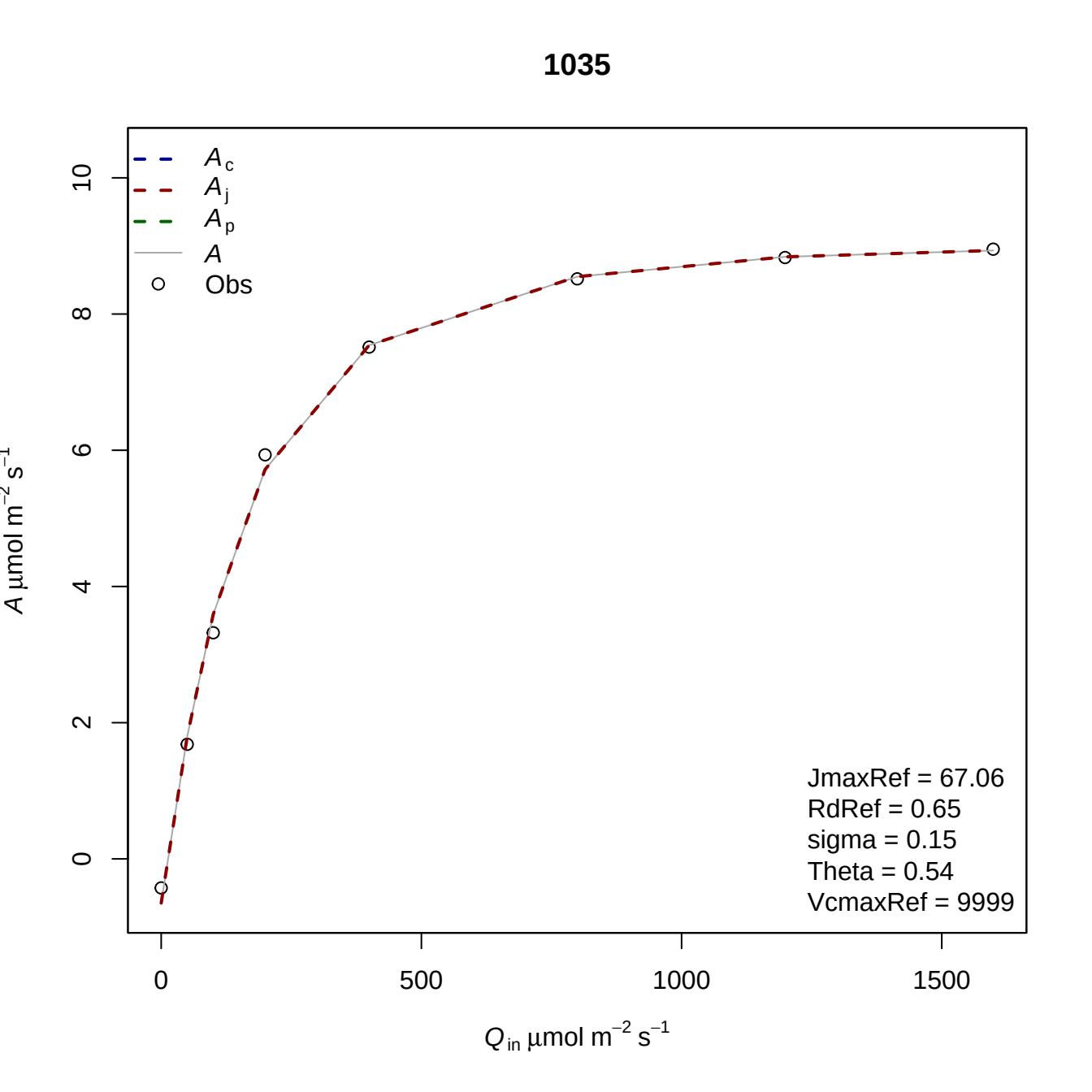

1030
A_c
A_j
A_p
A
Obs
JmaxRef = 89.57
RdRef = 0.78
sigma = 0.17
Theta = 0.4
VcmaxRef = 58.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

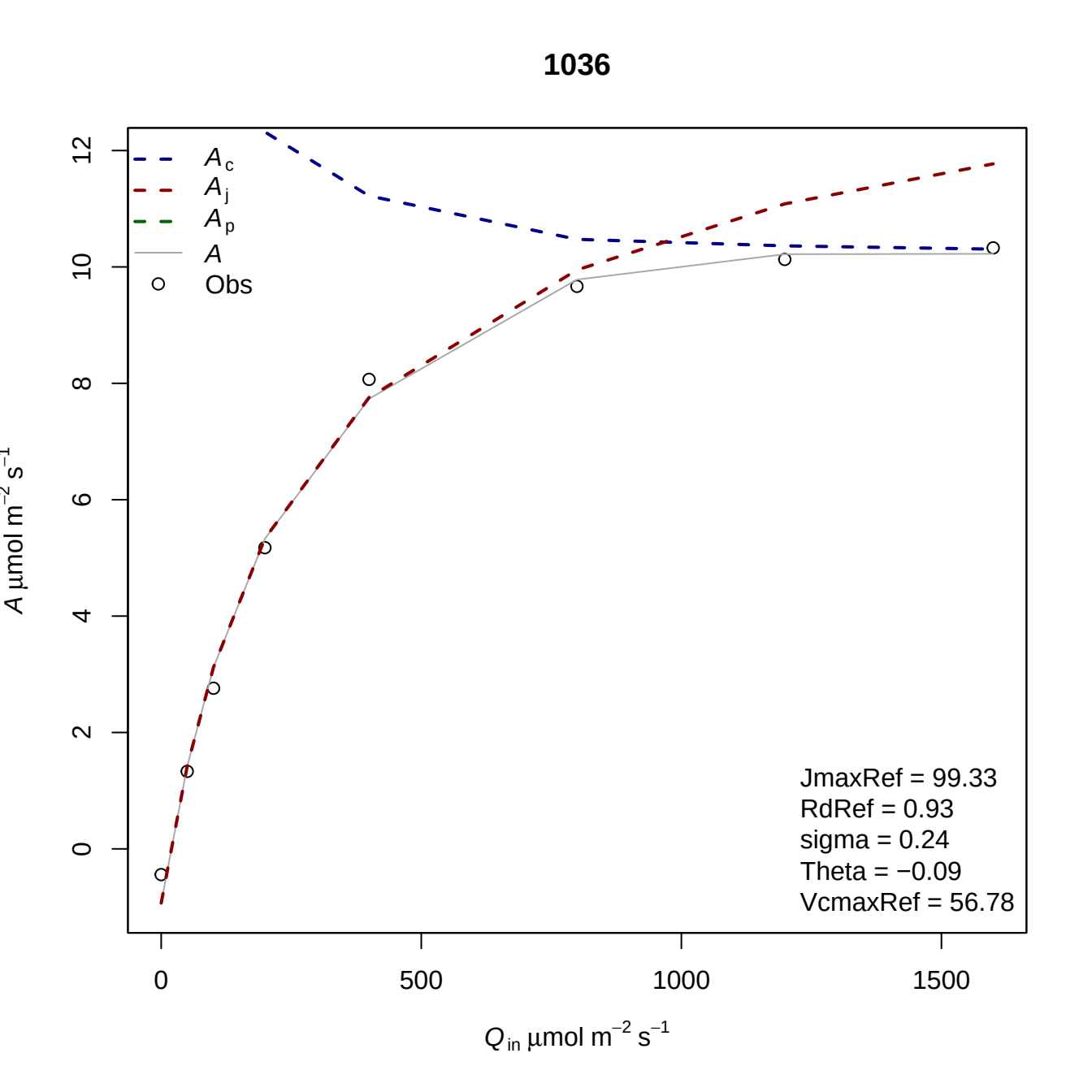

1031
A_c
A_j
A_p
A
Obs
JmaxRef = 97.13
RdRef = 1.16
sigma = 0.26
Theta = 0.56
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

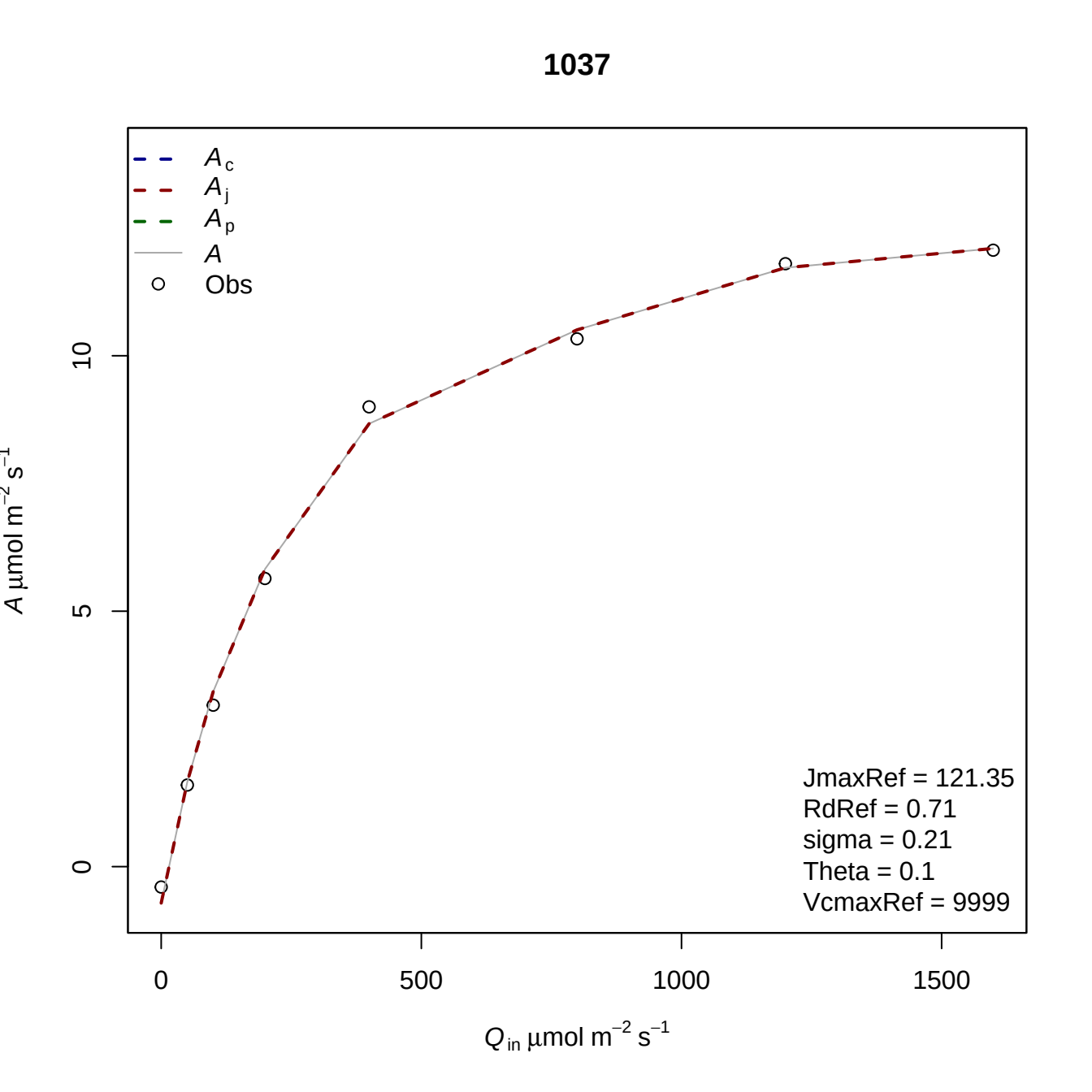

1032
A_c
A_j
A_p
A
Obs
JmaxRef = 93.12
RdRef = 0.8
sigma = 0.21
Theta = 0.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

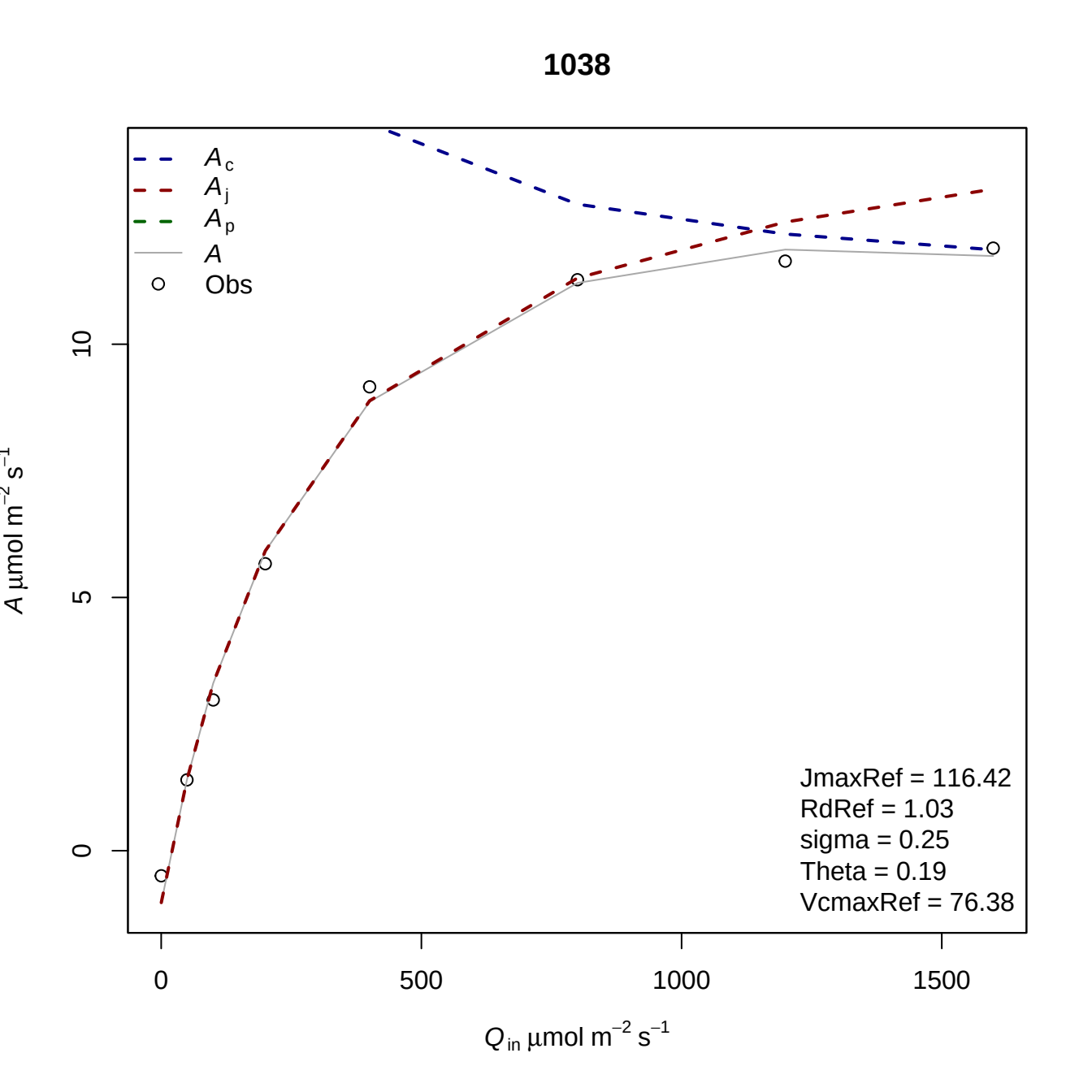

1033
A_c
A_j
A_p
A
Obs
JmaxRef = 81.71
RdRef = 0.93
sigma = 0.18
Theta = 0.16
VcmaxRef = 54.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

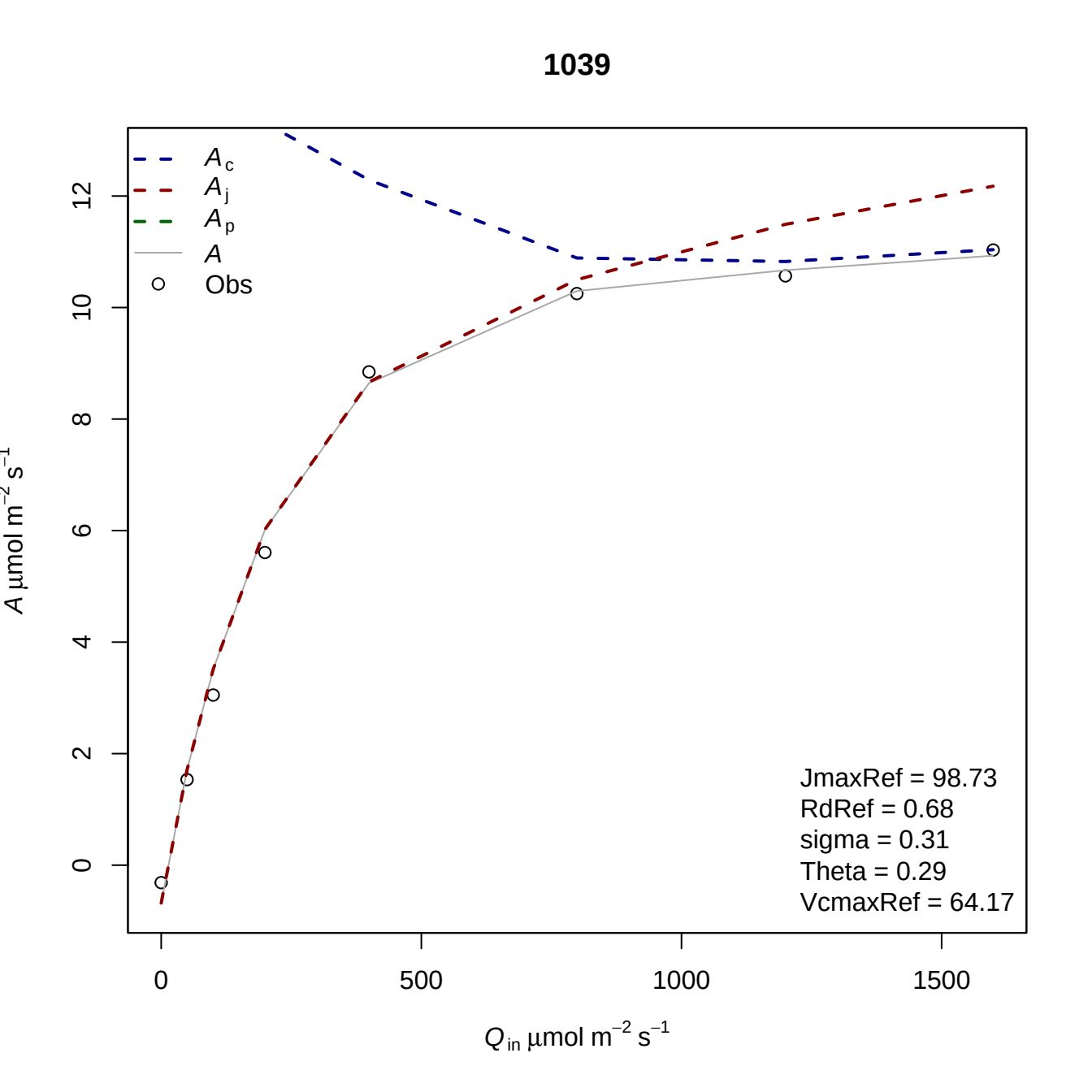

1034
A_c
A_j
A_p
A
Obs
JmaxRef = 169.65
RdRef = 1.3
sigma = 0.18
Theta = −1.24
VcmaxRef = 76.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

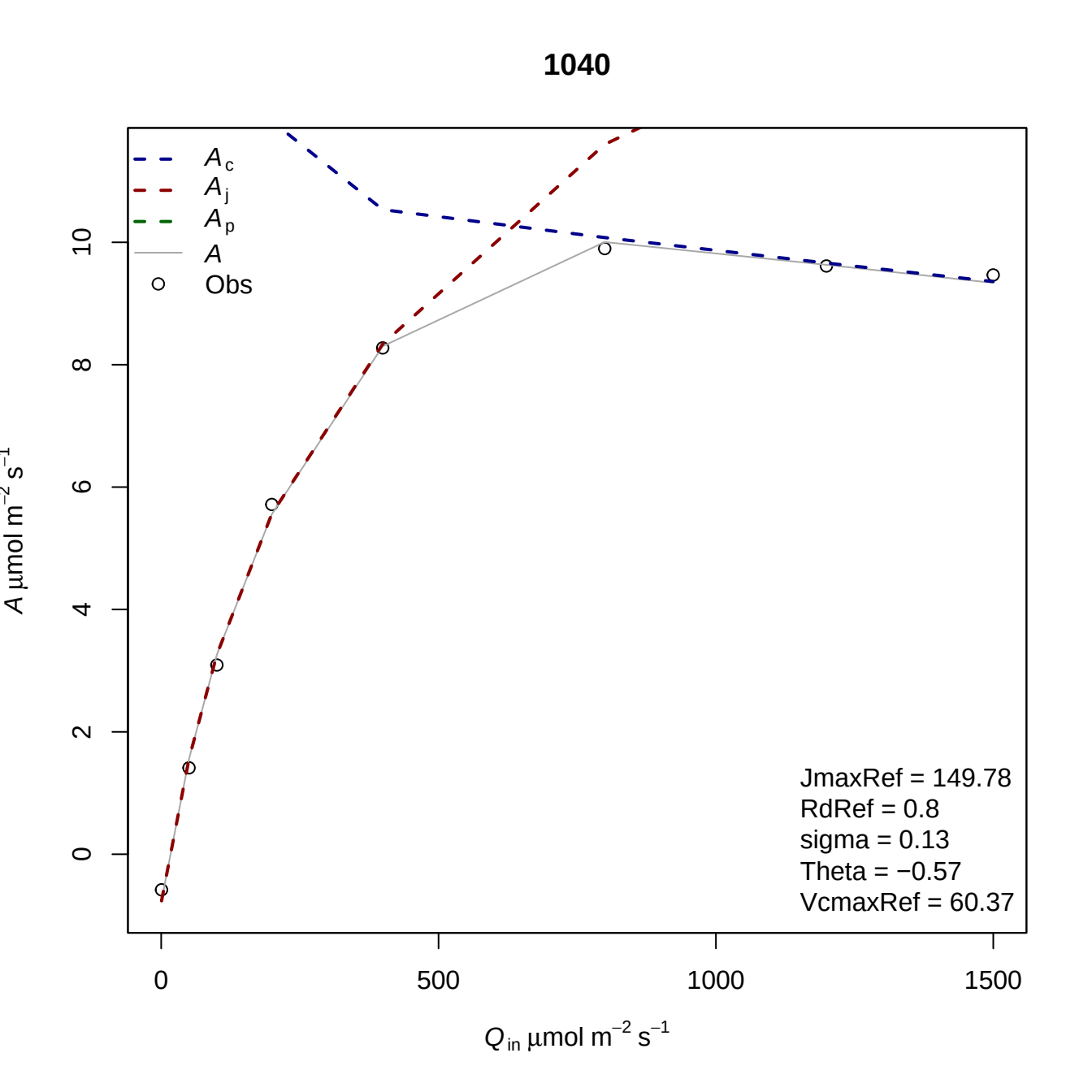

1035
A_c
A_j
A_p
A
Obs
JmaxRef = 67.06
RdRef = 0.65
sigma = 0.15
Theta = 0.54
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

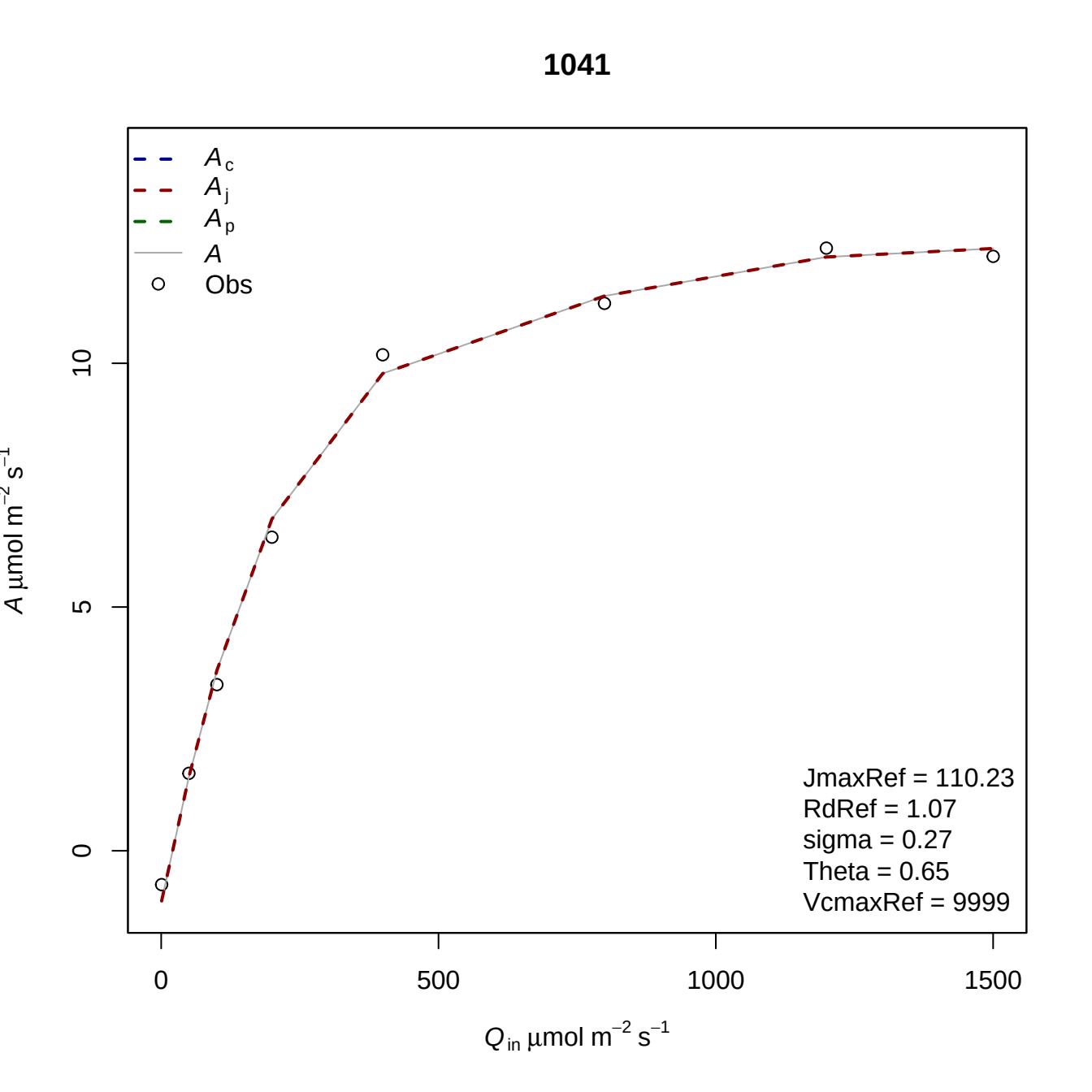

1036
A_c
A_j
A_p
A
Obs
JmaxRef = 99.33
RdRef = 0.93
sigma = 0.24
Theta = −0.09
VcmaxRef = 56.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

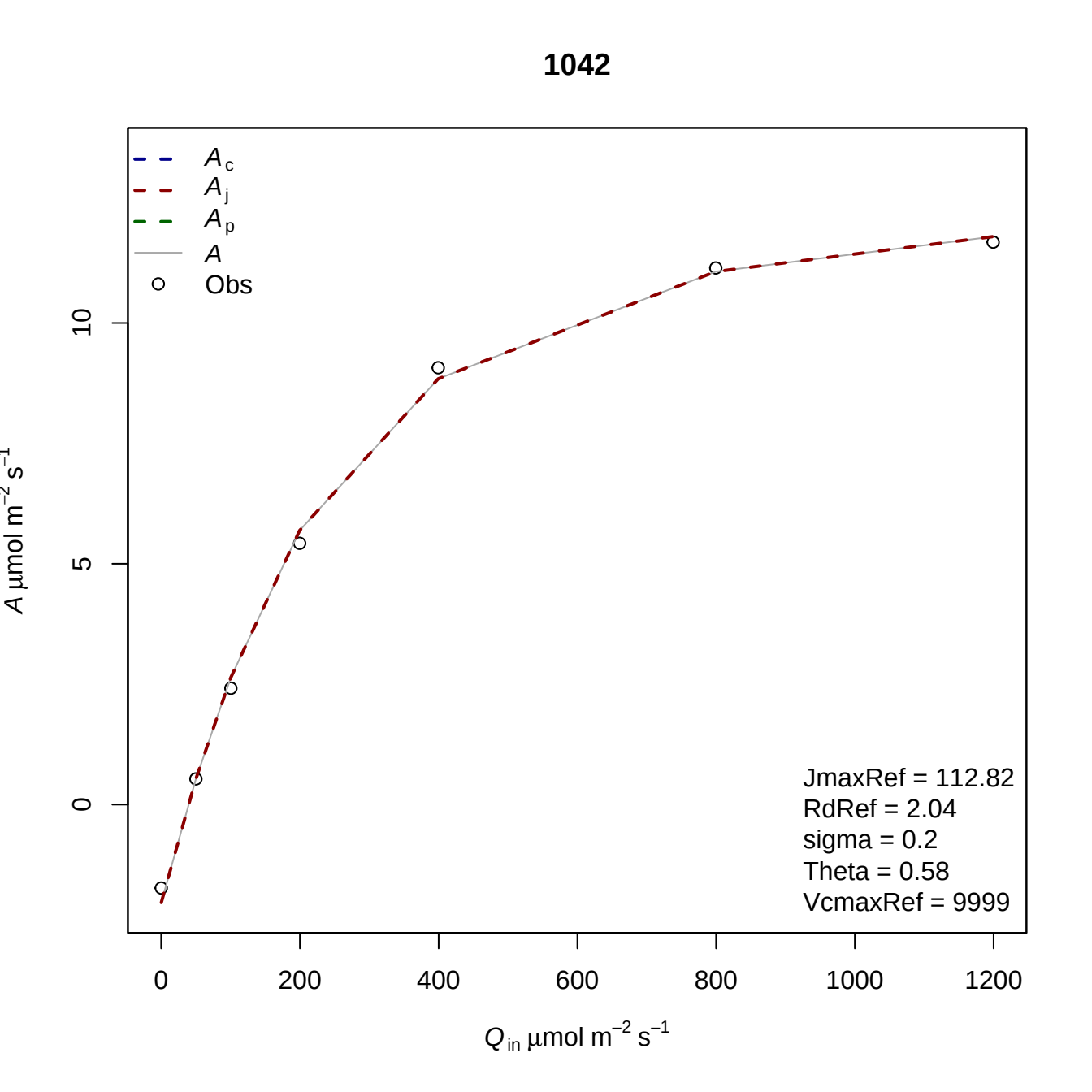

1037
A_c
A_j
A_p
A
Obs
JmaxRef = 121.35
RdRef = 0.71
sigma = 0.21
Theta = 0.1
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

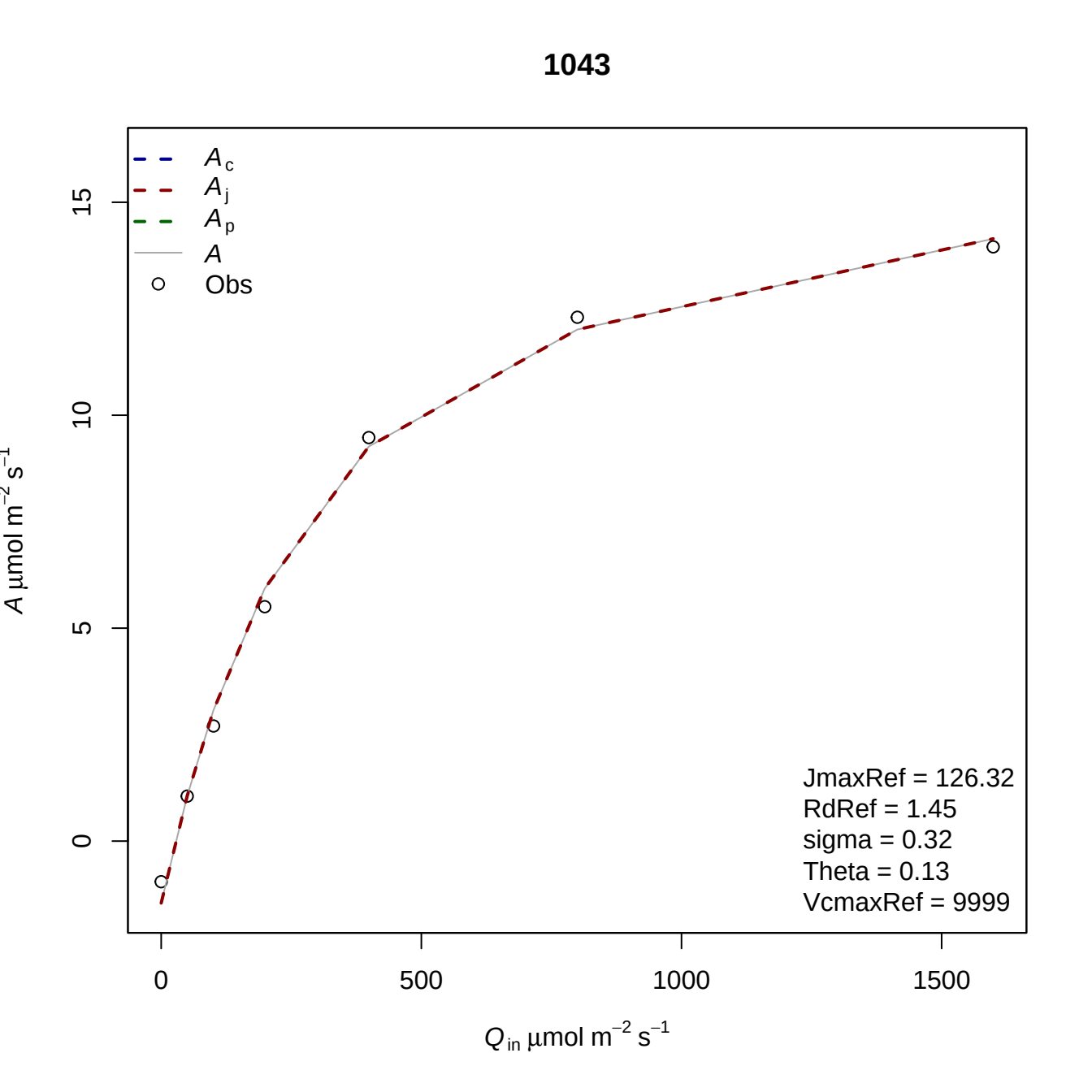

1038
A_c
A_j
A_p
A
Obs
JmaxRef = 116.42
RdRef = 1.03
sigma = 0.25
Theta = 0.19
VcmaxRef = 76.38
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

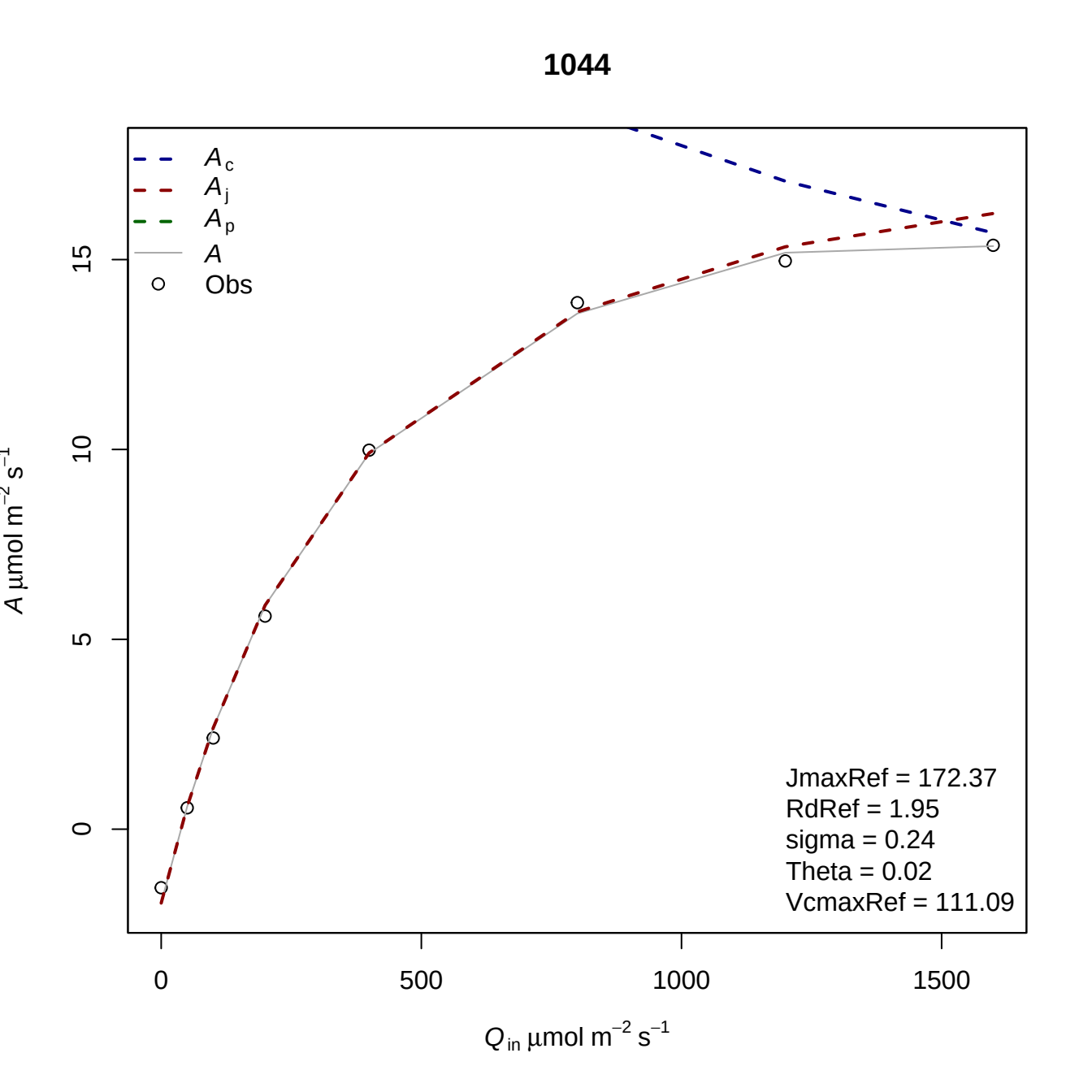

1039
A_c
A_j
A_p
A
Obs
JmaxRef = 98.73
RdRef = 0.68
sigma = 0.31
Theta = 0.29
VcmaxRef = 64.17
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

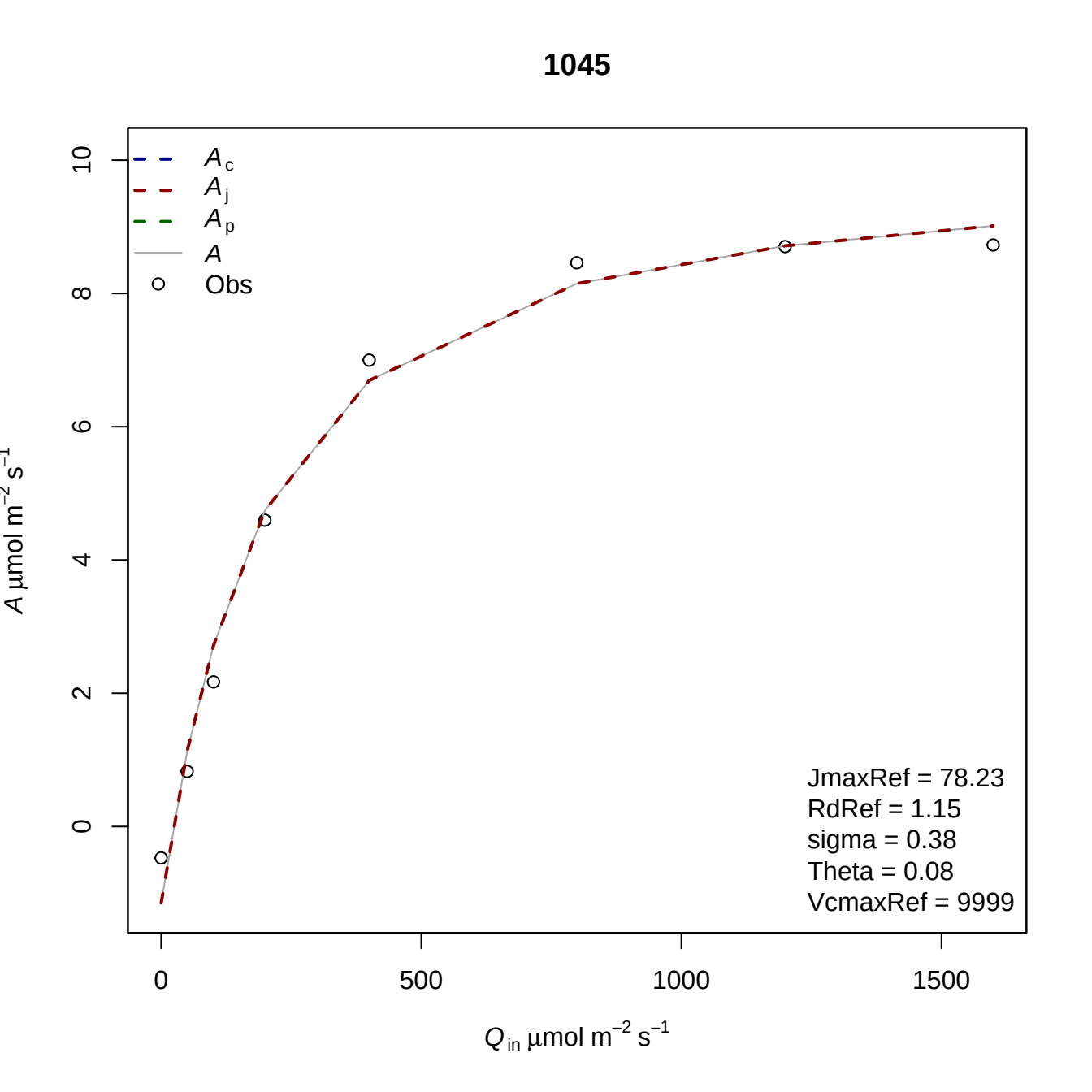

1040
A_c
A_j
A_p
A
Obs
JmaxRef = 149.78
RdRef = 0.8
sigma = 0.13
Theta = −0.57
VcmaxRef = 60.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

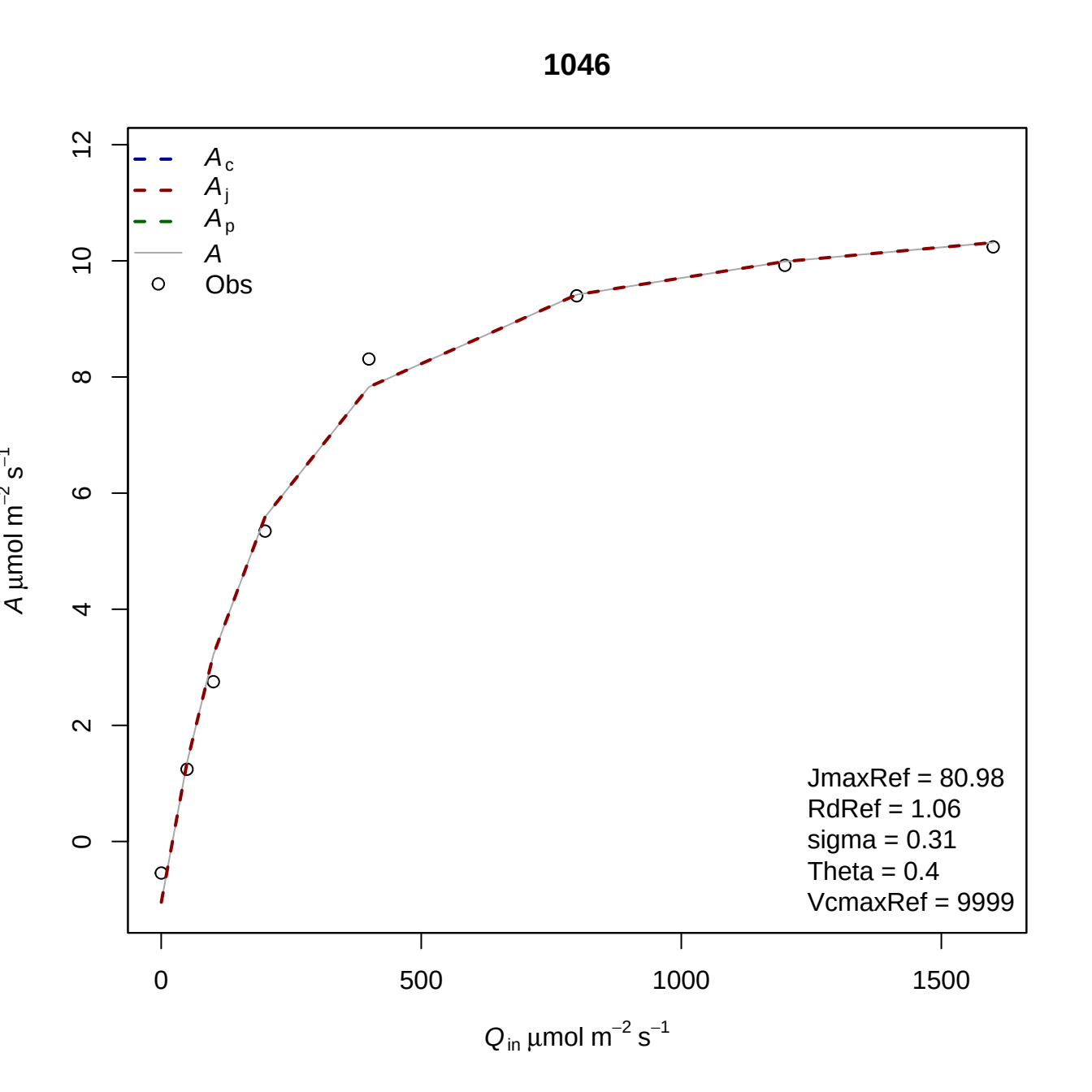

1041
A_c
A_j
A_p
A
Obs
JmaxRef = 110.23
RdRef = 1.07
sigma = 0.27
Theta = 0.65
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

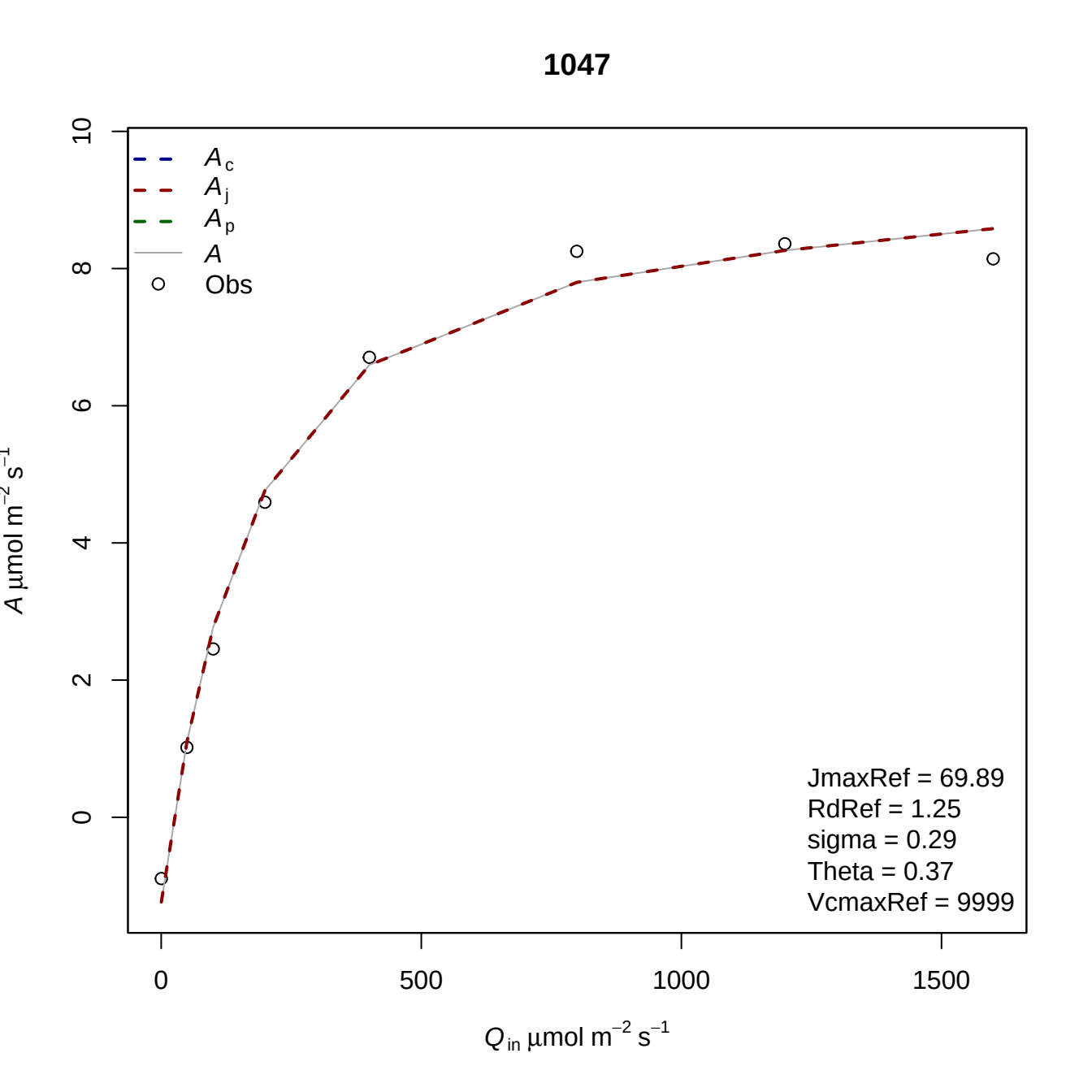

1042
A_c
A_j
A_p
A
Obs
JmaxRef = 112.82
RdRef = 2.04
sigma = 0.2
Theta = 0.58
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200

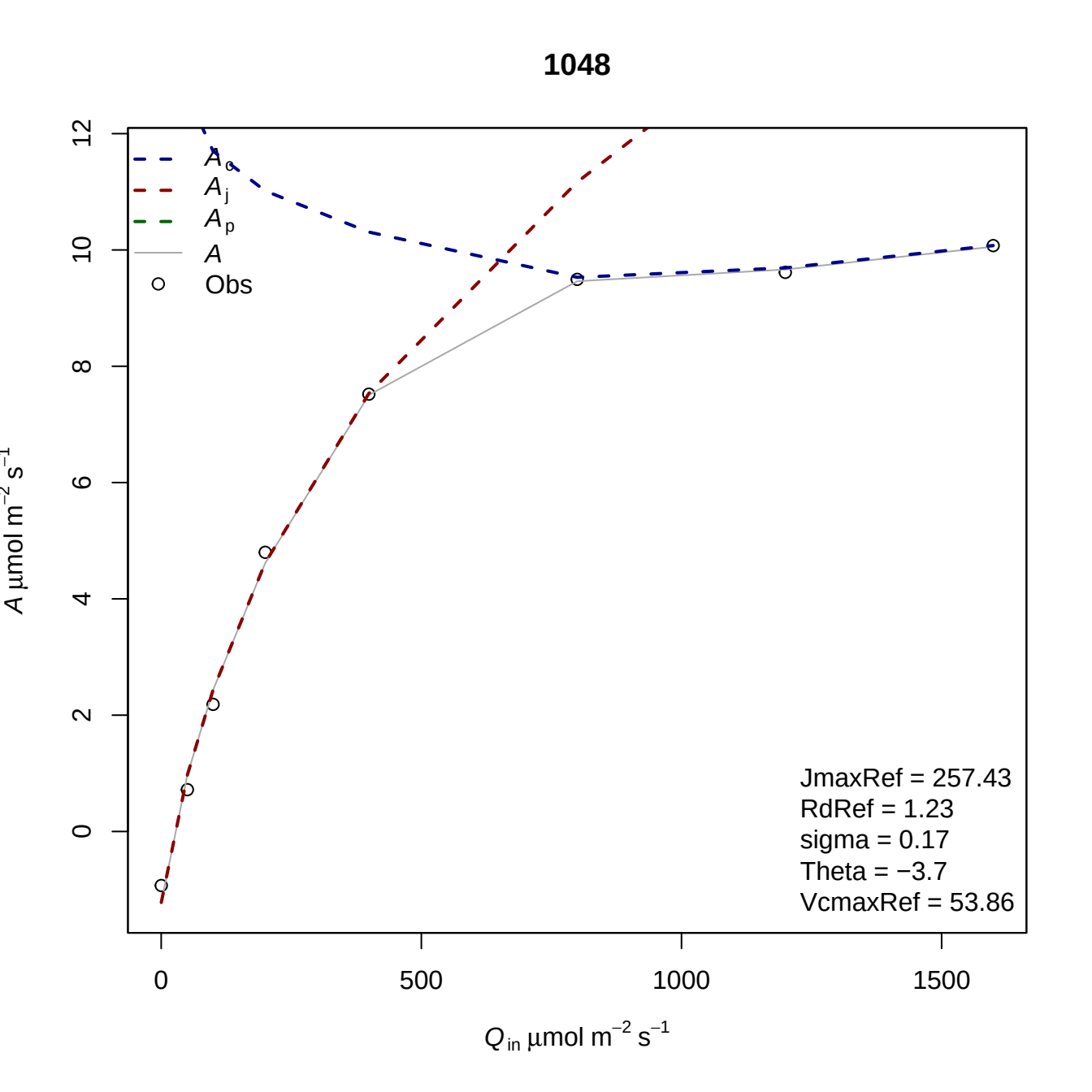

1043
A_c
A_j
A_p
A
Obs
JmaxRef = 126.32
RdRef = 1.45
sigma = 0.32
Theta = 0.13
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

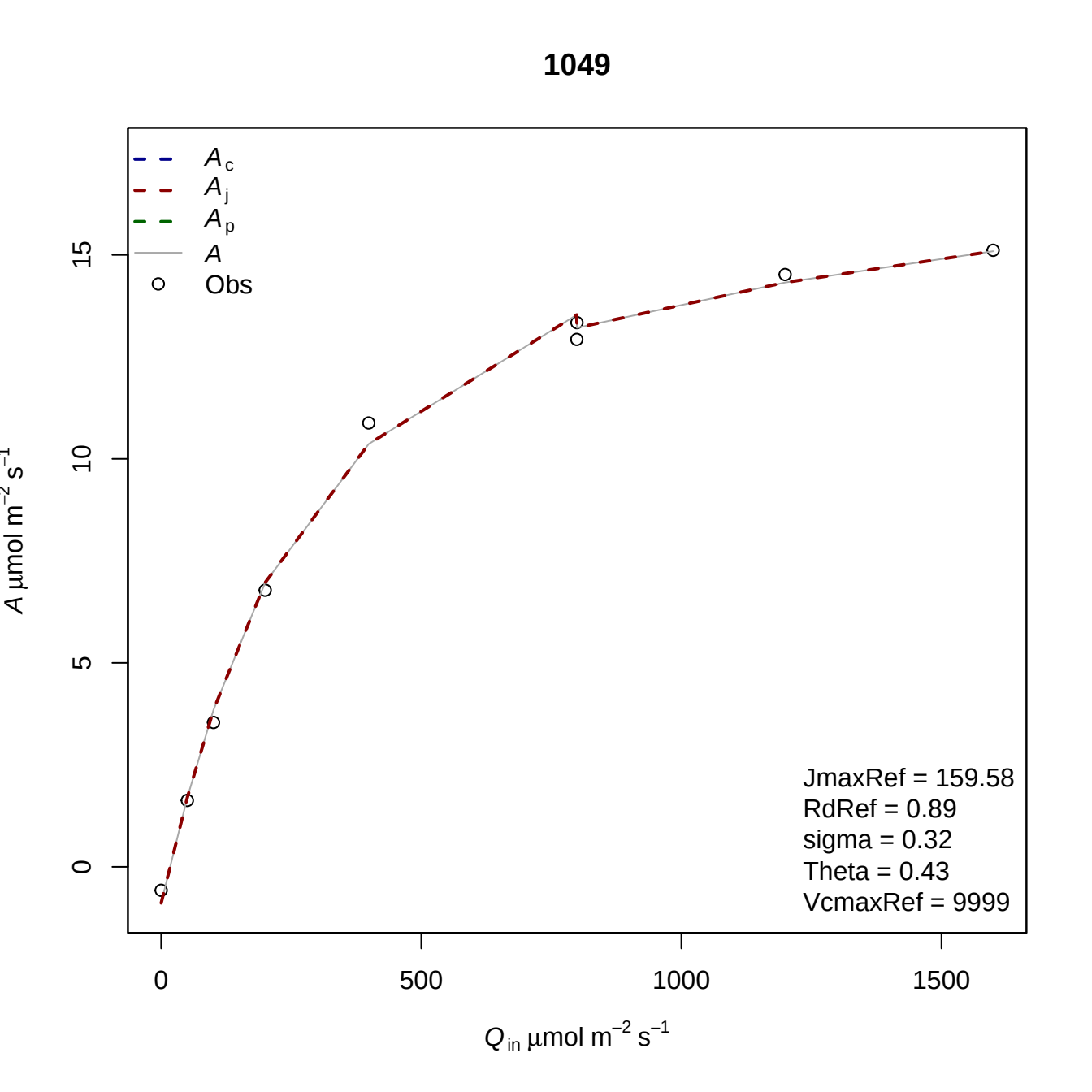

1044
A_c
A_j
A_p
A
Obs
JmaxRef = 172.37
RdRef = 1.95
sigma = 0.24
Theta = 0.02
VcmaxRef = 111.09
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

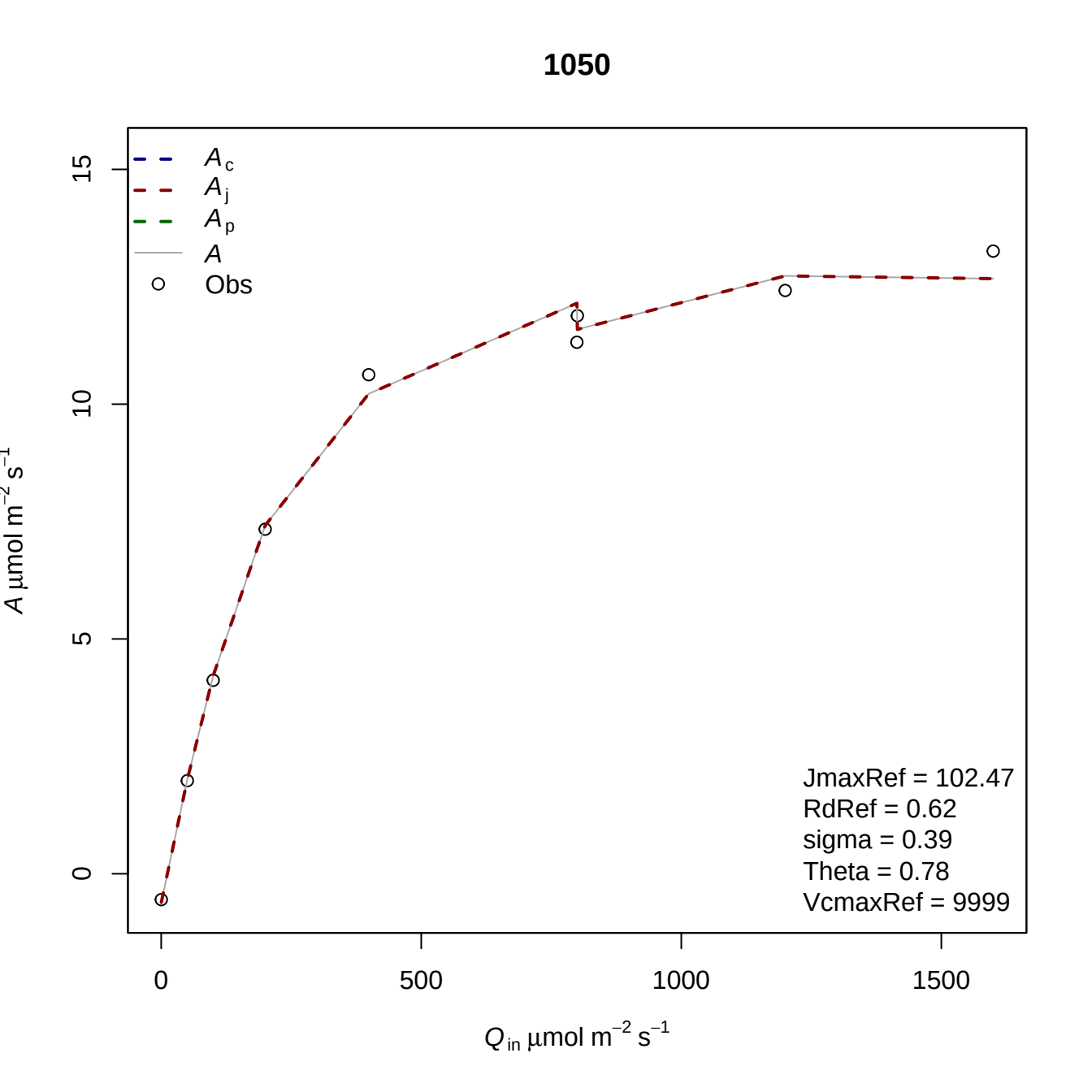

1045
A_c
A_j
A_p
A
Obs
JmaxRef = 78.23
RdRef = 1.15
sigma = 0.38
Theta = 0.08
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

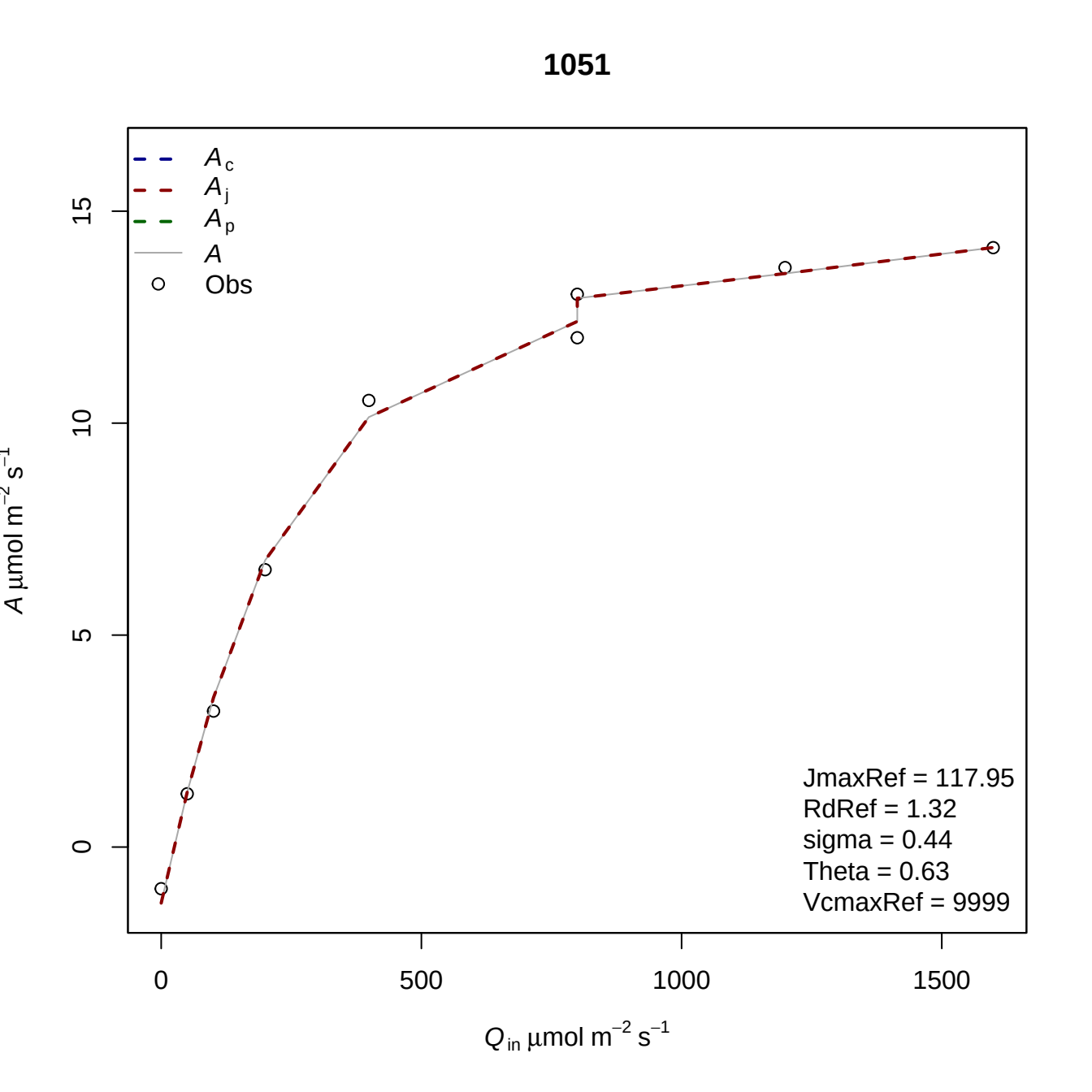

1046
A_c
A_j
A_p
A
Obs
JmaxRef = 80.98
RdRef = 1.06
sigma = 0.31
Theta = 0.4
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

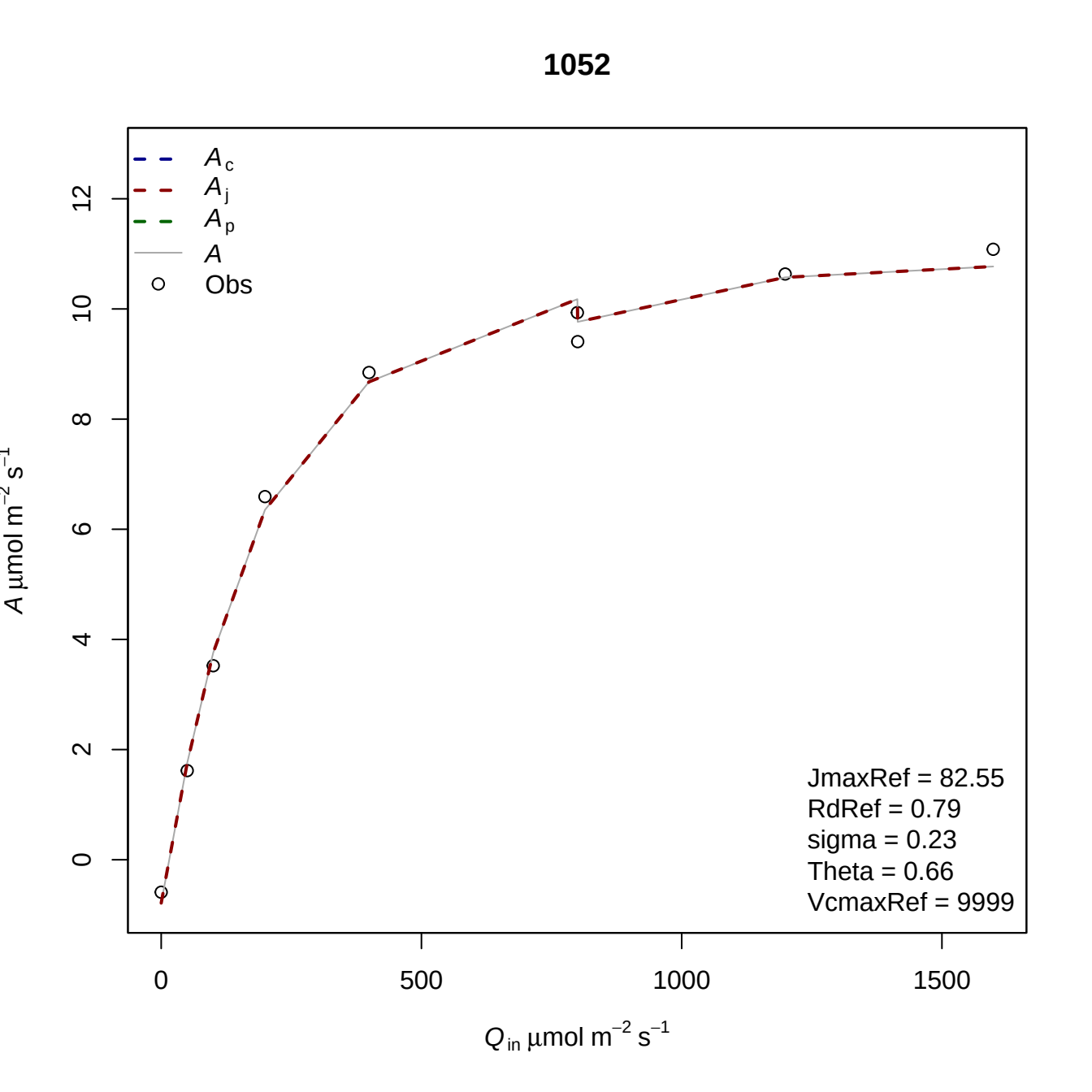

1047
A_c
A_j
A_p
A
Obs
JmaxRef = 69.89
RdRef = 1.25
sigma = 0.29
Theta = 0.37
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

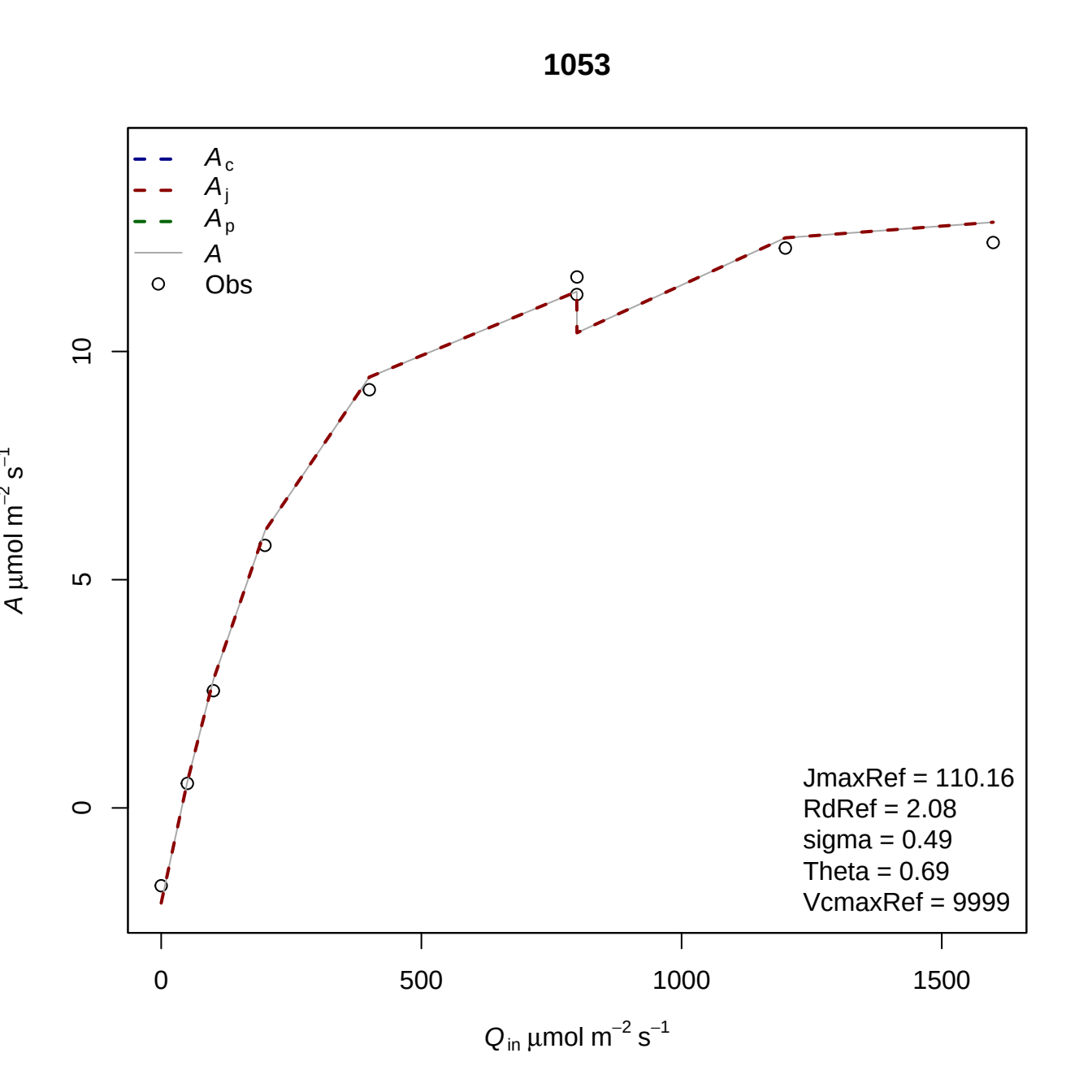

1048
A_c
A_j
A_p
A
Obs
JmaxRef = 257.43
RdRef = 1.23
sigma = 0.17
Theta = −3.7
VcmaxRef = 53.86
0
2
4
6
8
10
12
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

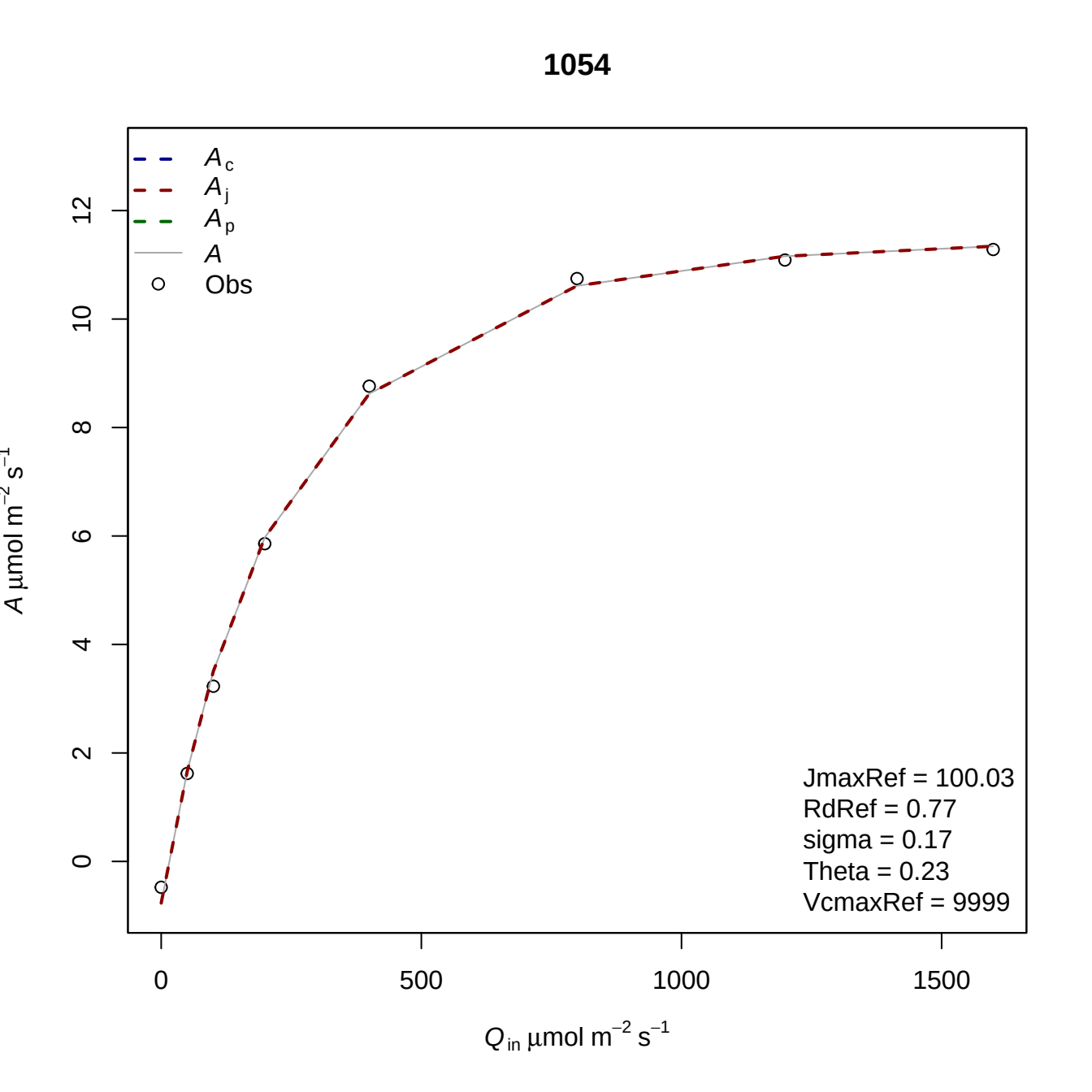

1049
A_c
A_j
A_p
A
Obs
JmaxRef = 159.58
RdRef = 0.89
sigma = 0.32
Theta = 0.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

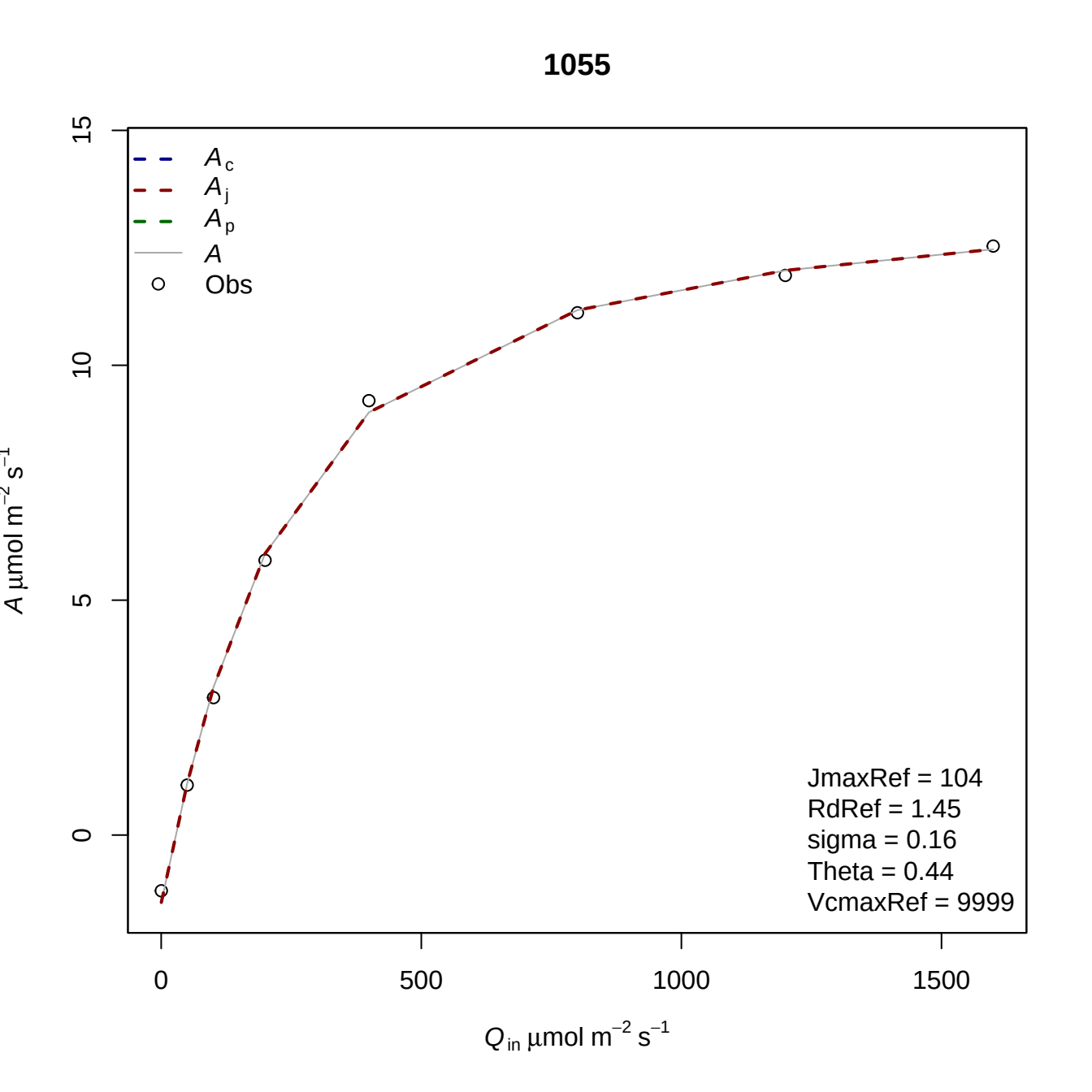

1050
A_c
A_j
A_p
A
Obs
JmaxRef = 102.47
RdRef = 0.62
sigma = 0.39
Theta = 0.78
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

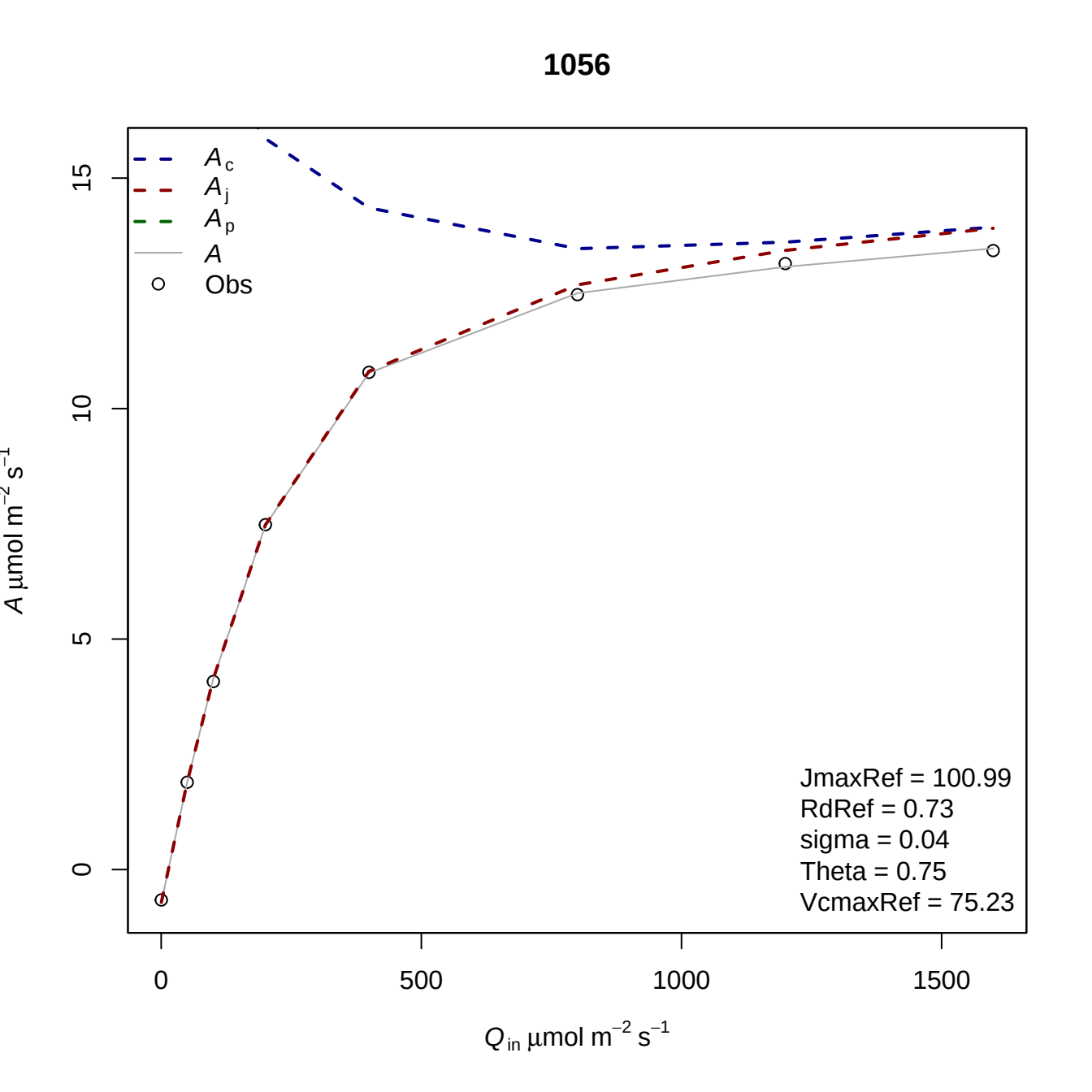

1051
A_c
A_j
A_p
A
Obs
JmaxRef = 117.95
RdRef = 1.32
sigma = 0.44
Theta = 0.63
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

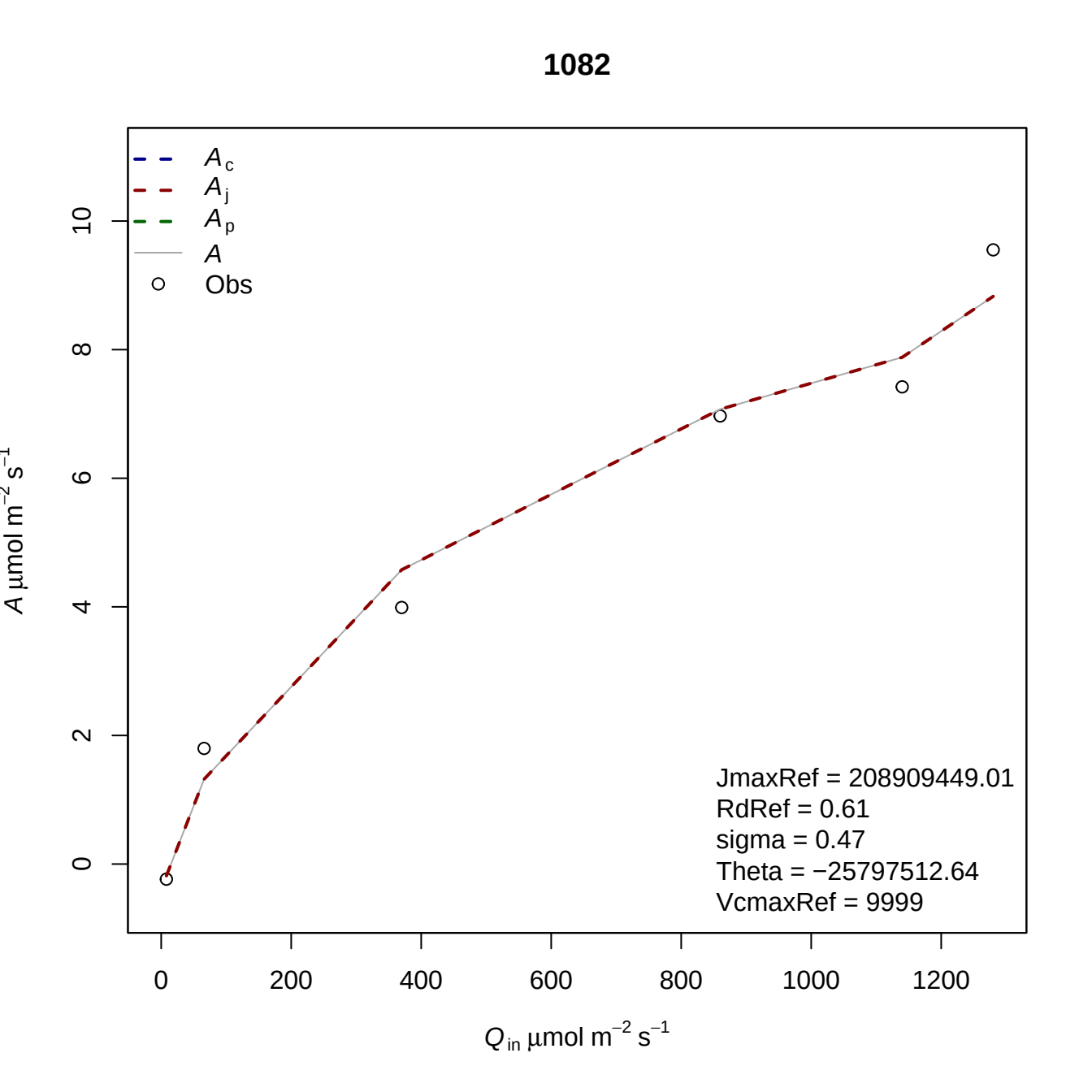

1052
A_c
A_j
A_p
A
Obs
JmaxRef = 82.55
RdRef = 0.79
sigma = 0.23
Theta = 0.66
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

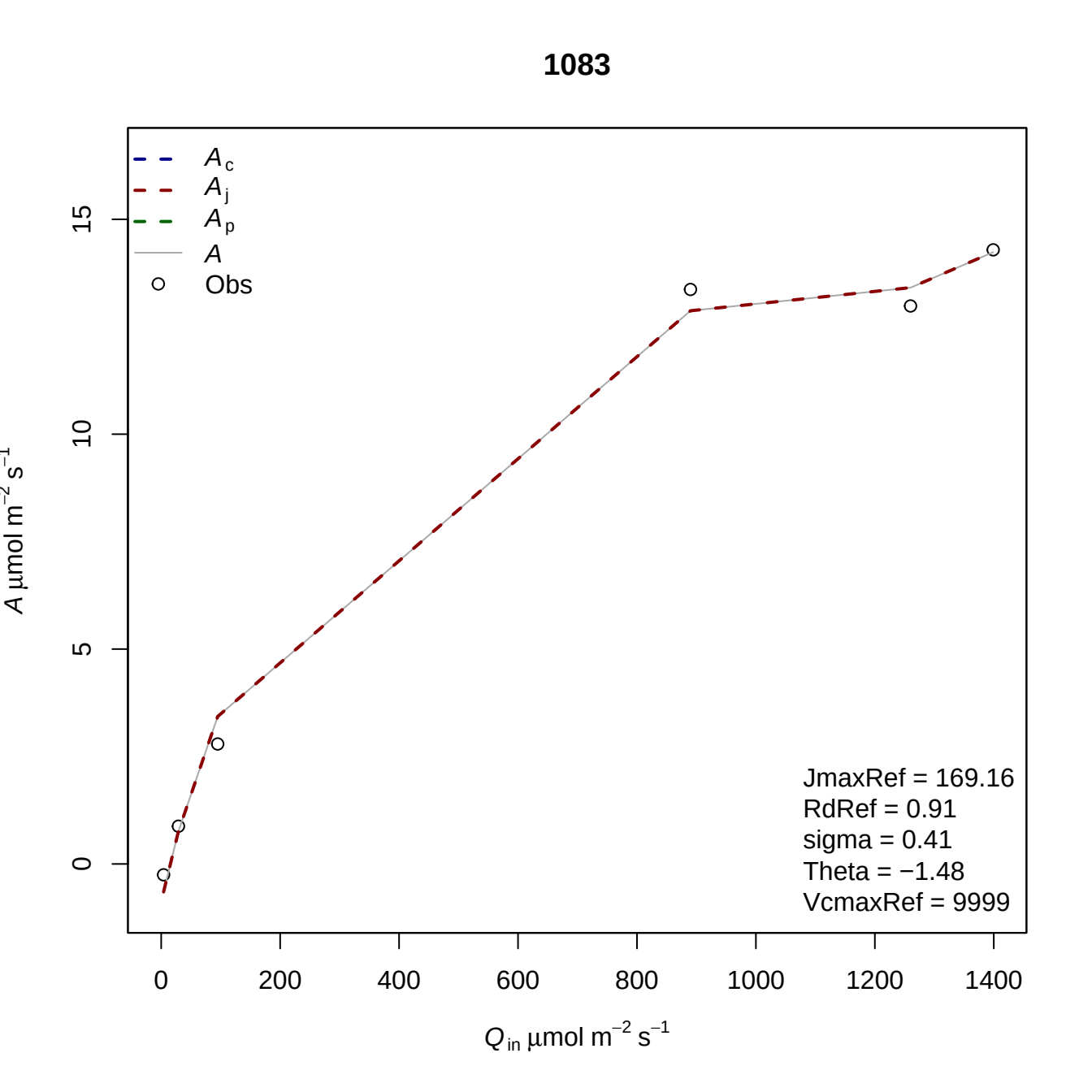

1053
A_c
A_j
A_p
A
Obs
JmaxRef = 110.16
RdRef = 2.08
sigma = 0.49
Theta = 0.69
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

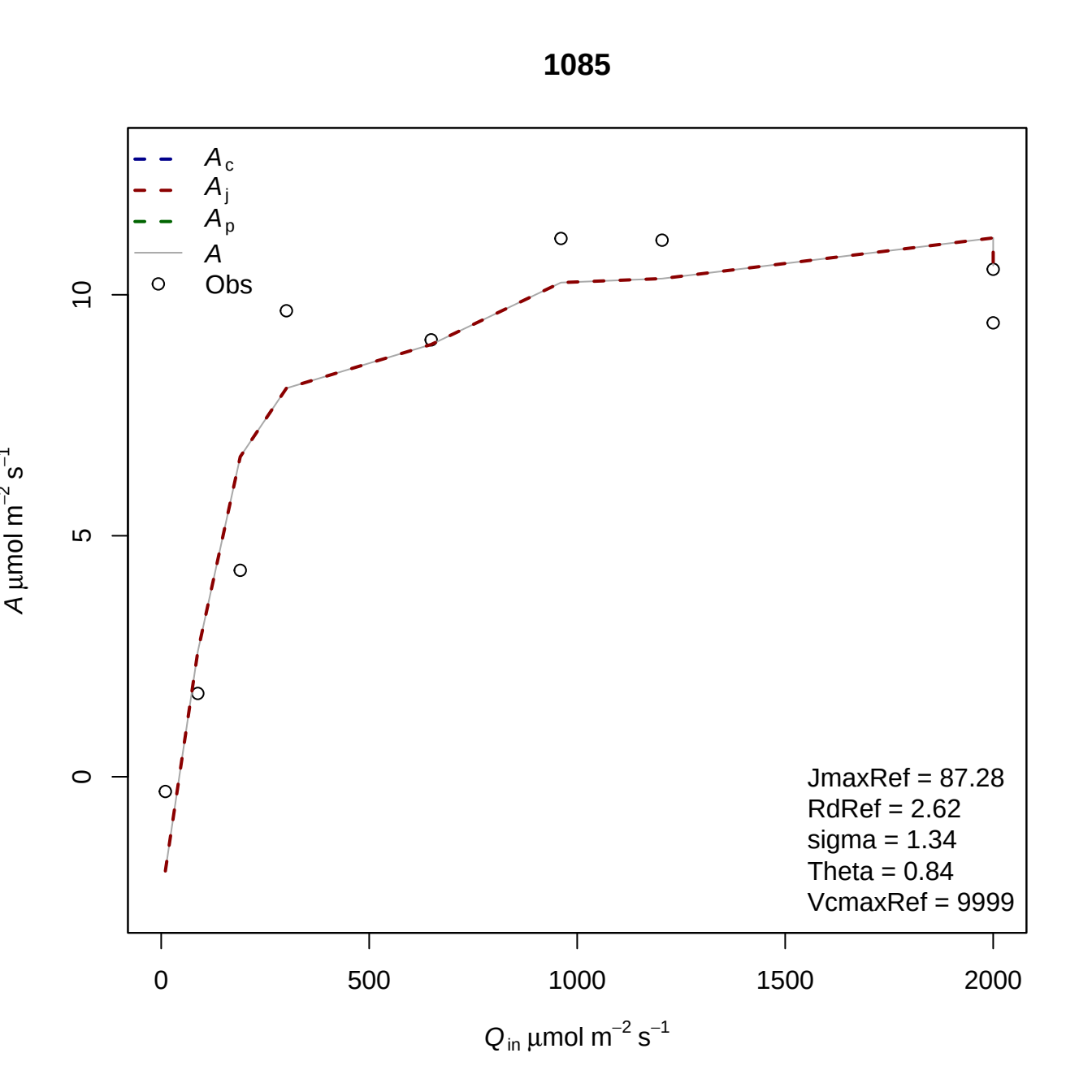

1054
A_c
A_j
A_p
A
Obs
JmaxRef = 100.03
RdRef = 0.77
sigma = 0.17
Theta = 0.23
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

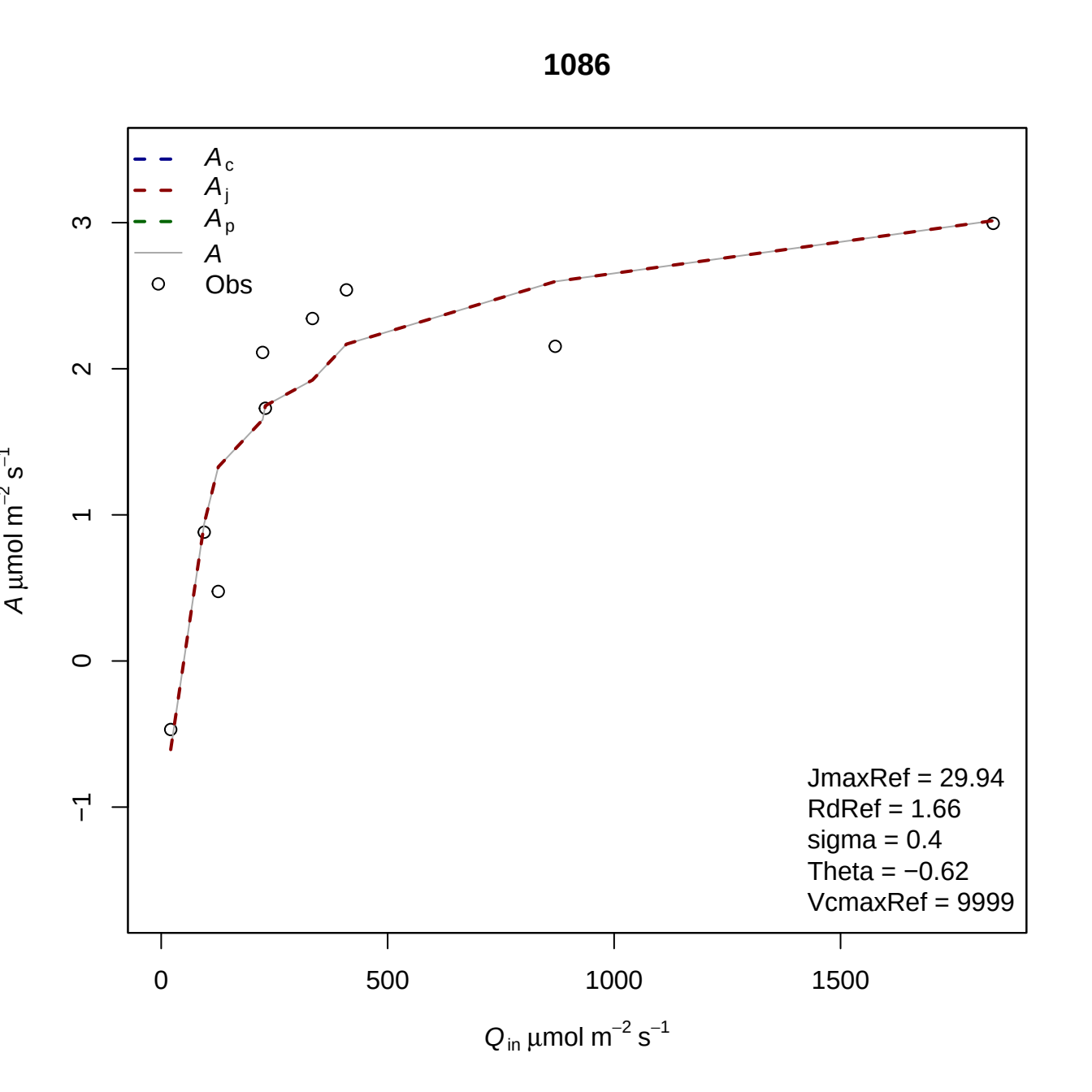

1055
A_c
A_j
A_p
A
Obs
JmaxRef = 104
RdRef = 1.45
sigma = 0.16
Theta = 0.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

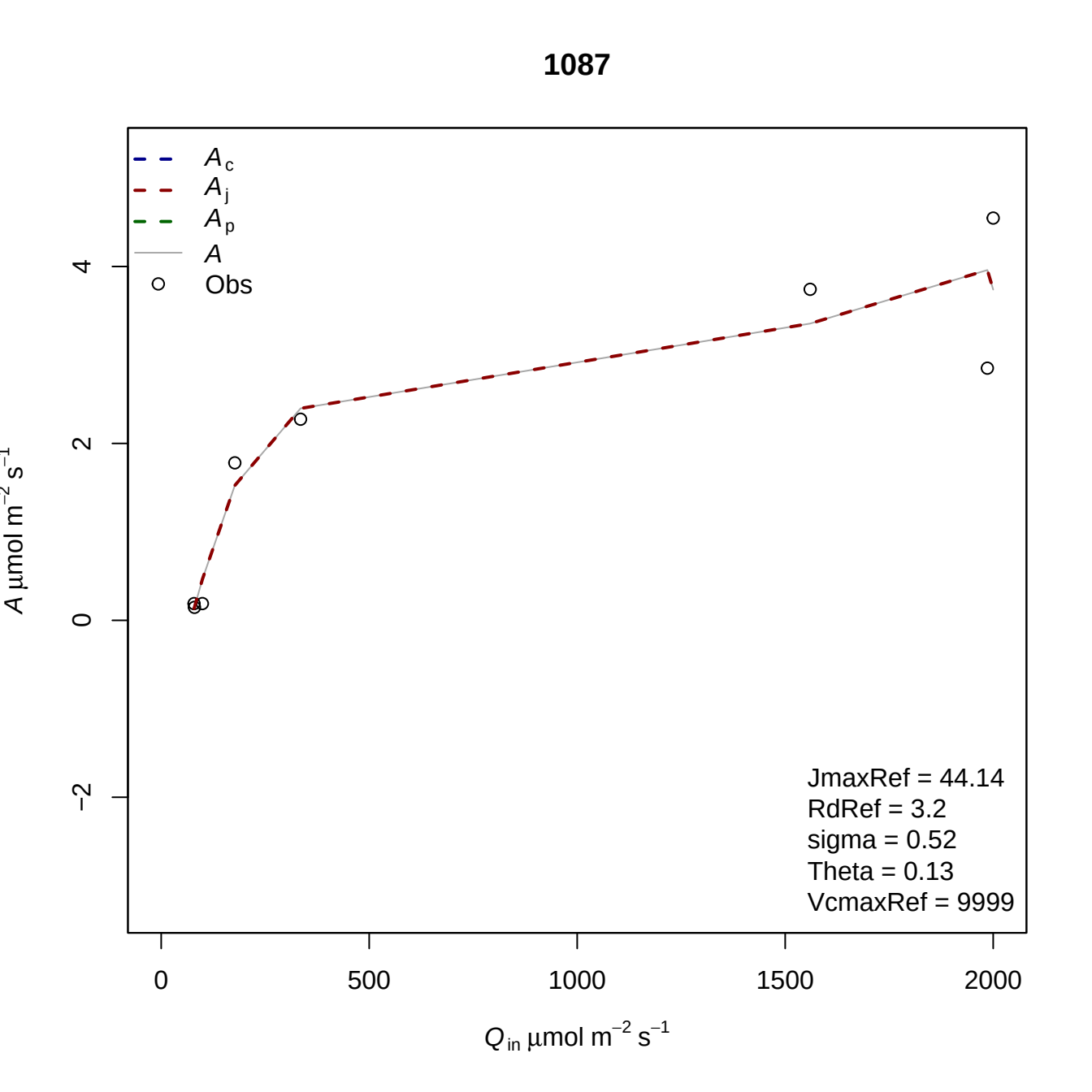

1056
A_c
A_j
A_p
A
Obs
JmaxRef = 100.99
RdRef = 0.73
sigma = 0.04
Theta = 0.75
VcmaxRef = 75.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

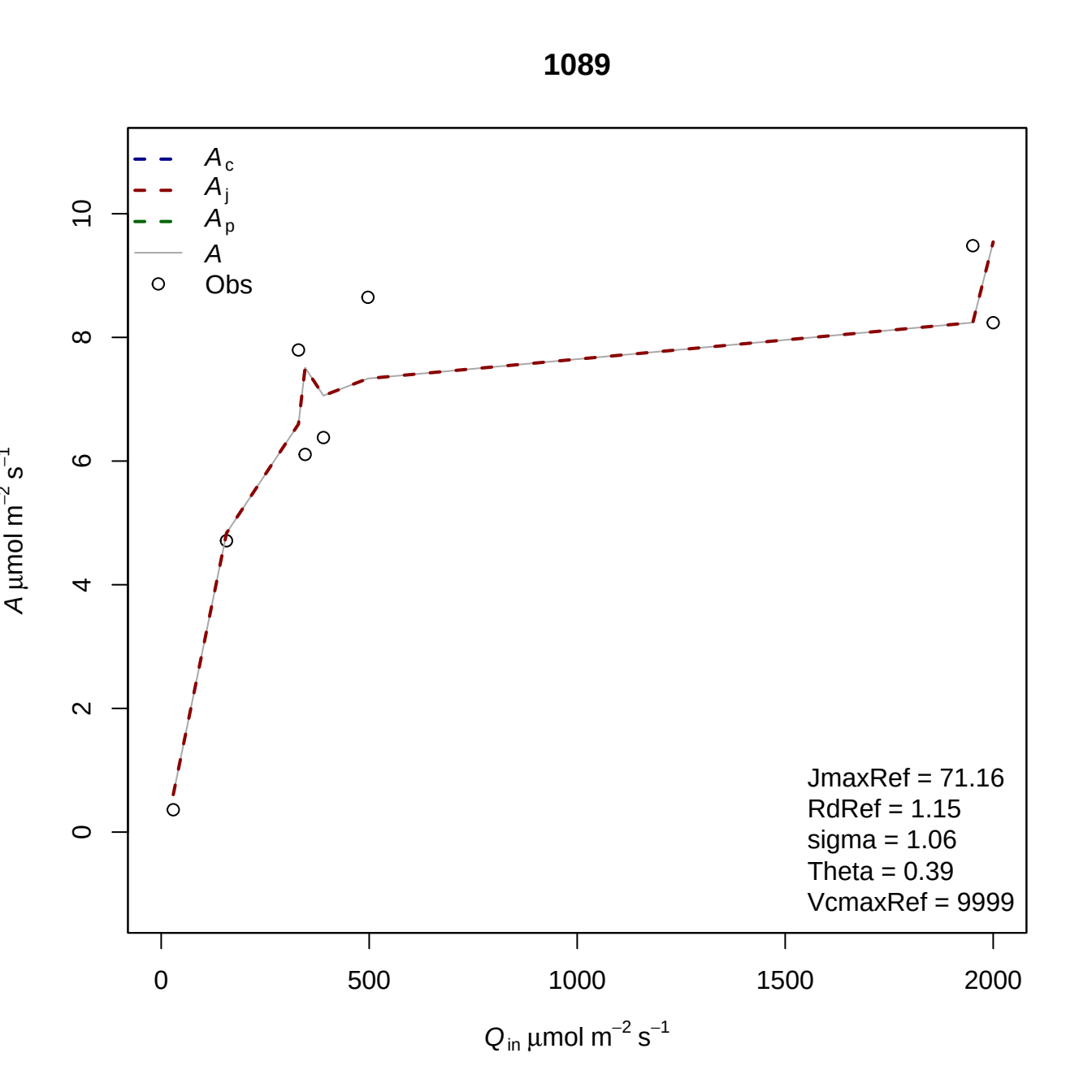

1082
A_c
A_j
A_p
A
Obs
JmaxRef = 208909449.01
RdRef = 0.61
sigma = 0.47
Theta = −25797512.64
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200

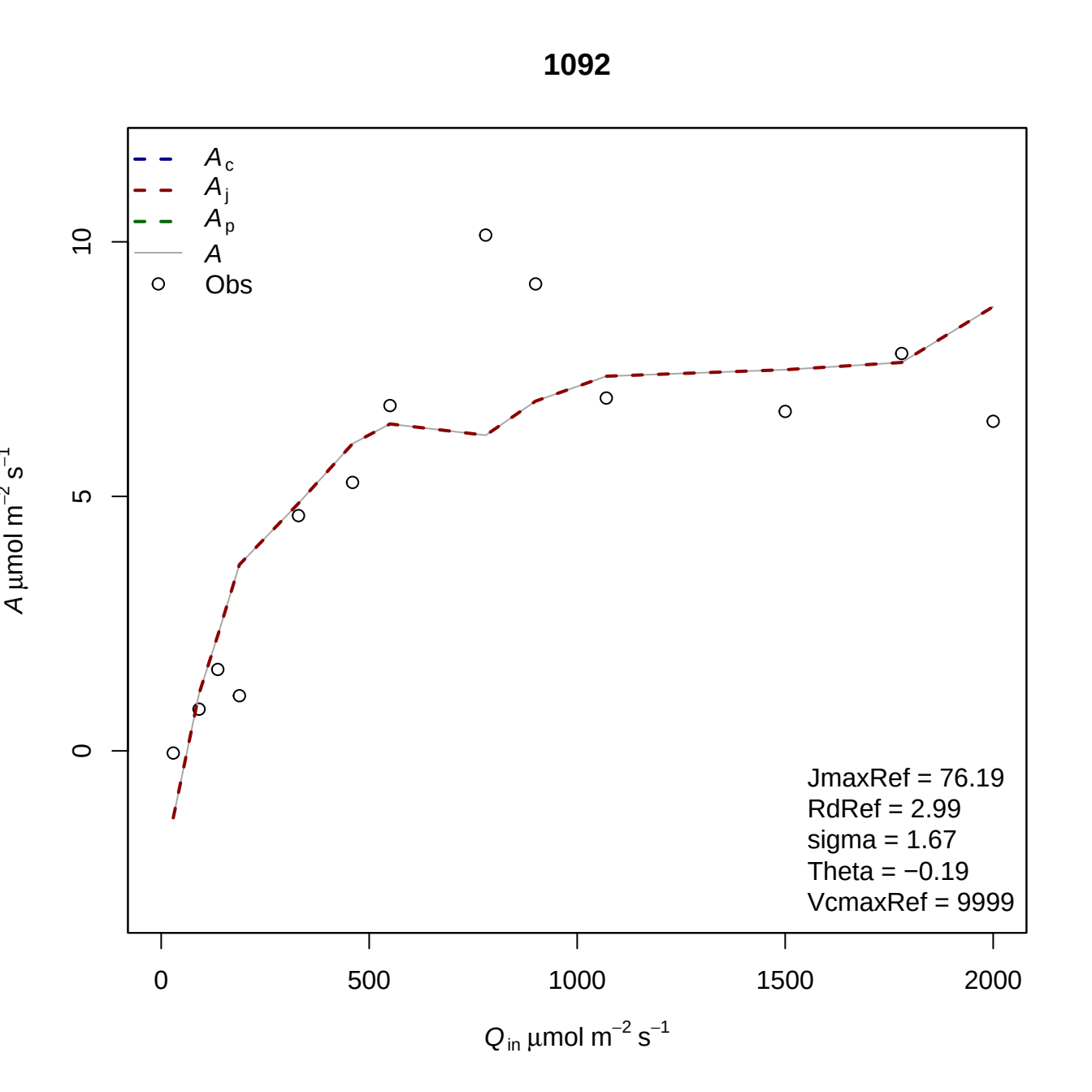

1083
A_c
A_j
A_p
A
Obs
JmaxRef = 169.16
RdRef = 0.91
sigma = 0.41
Theta = −1.48
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200
1400

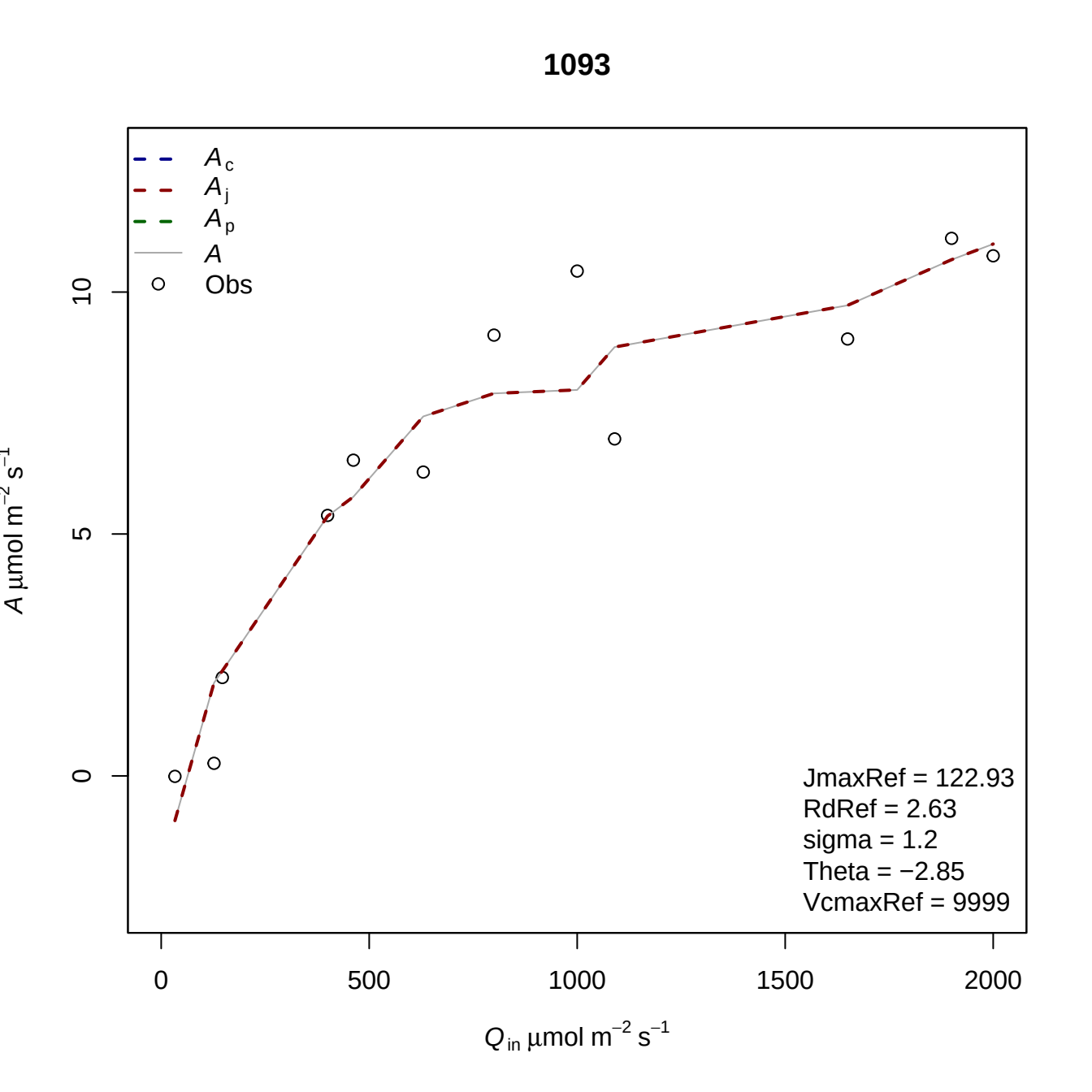

1085
A_c
A_j
A_p
A
Obs
JmaxRef = 87.28
RdRef = 2.62
sigma = 1.34
Theta = 0.84
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

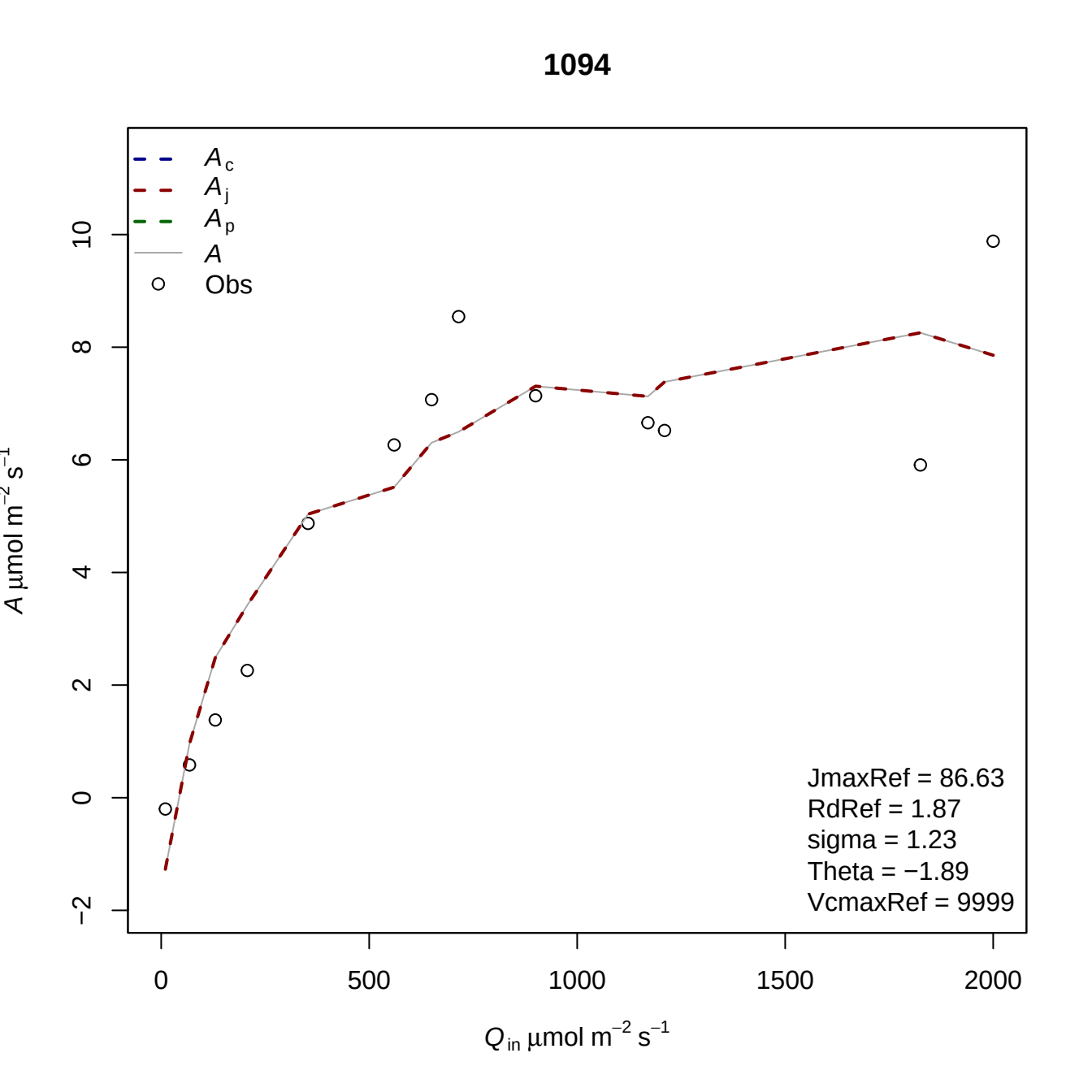

1086
A_c
A_j
A_p
A
Obs
JmaxRef = 29.94
RdRef = 1.66
sigma = 0.4
Theta = −0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

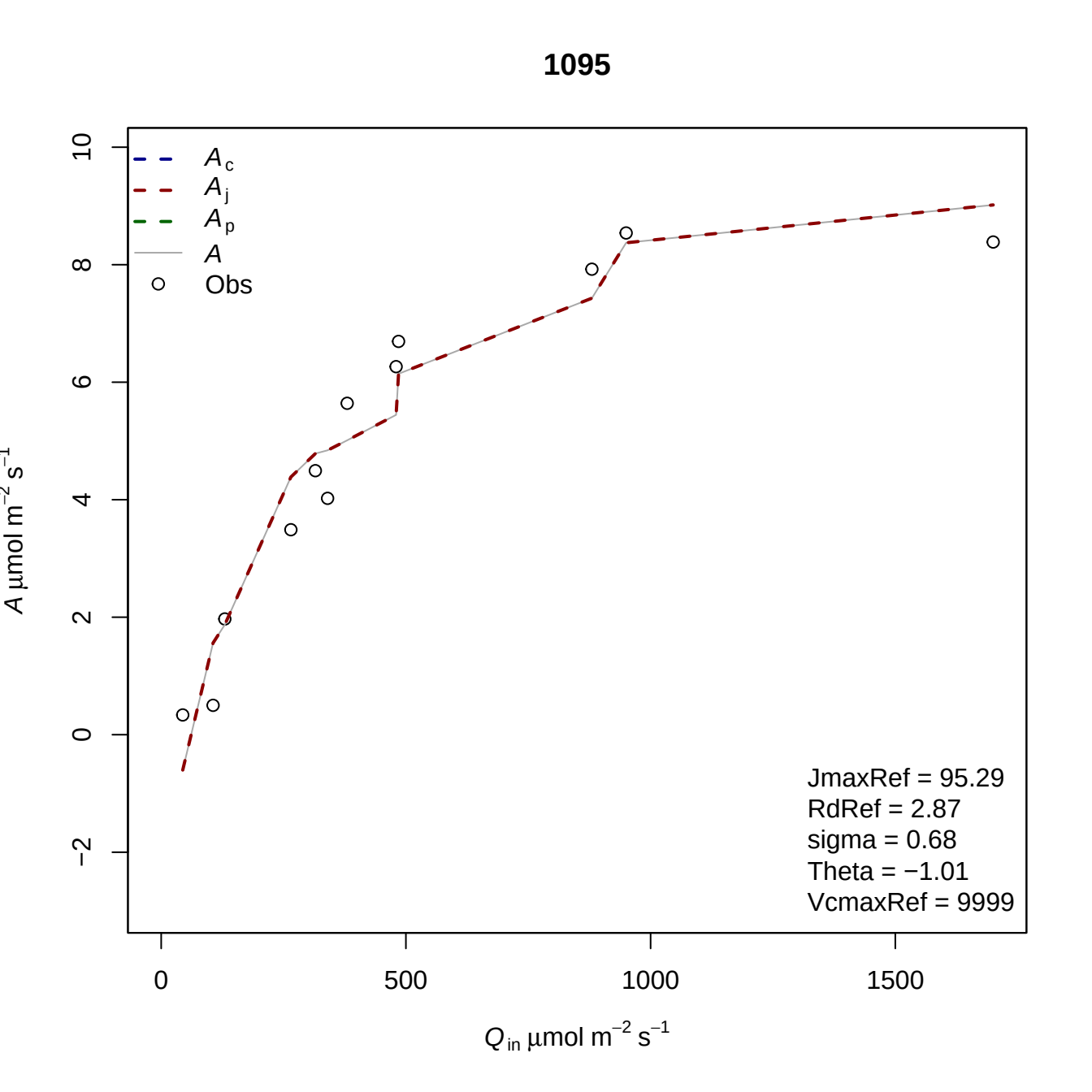

1087
A_c
A_j
A_p
A
Obs
JmaxRef = 44.14
RdRef = 3.2
sigma = 0.52
Theta = 0.13
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

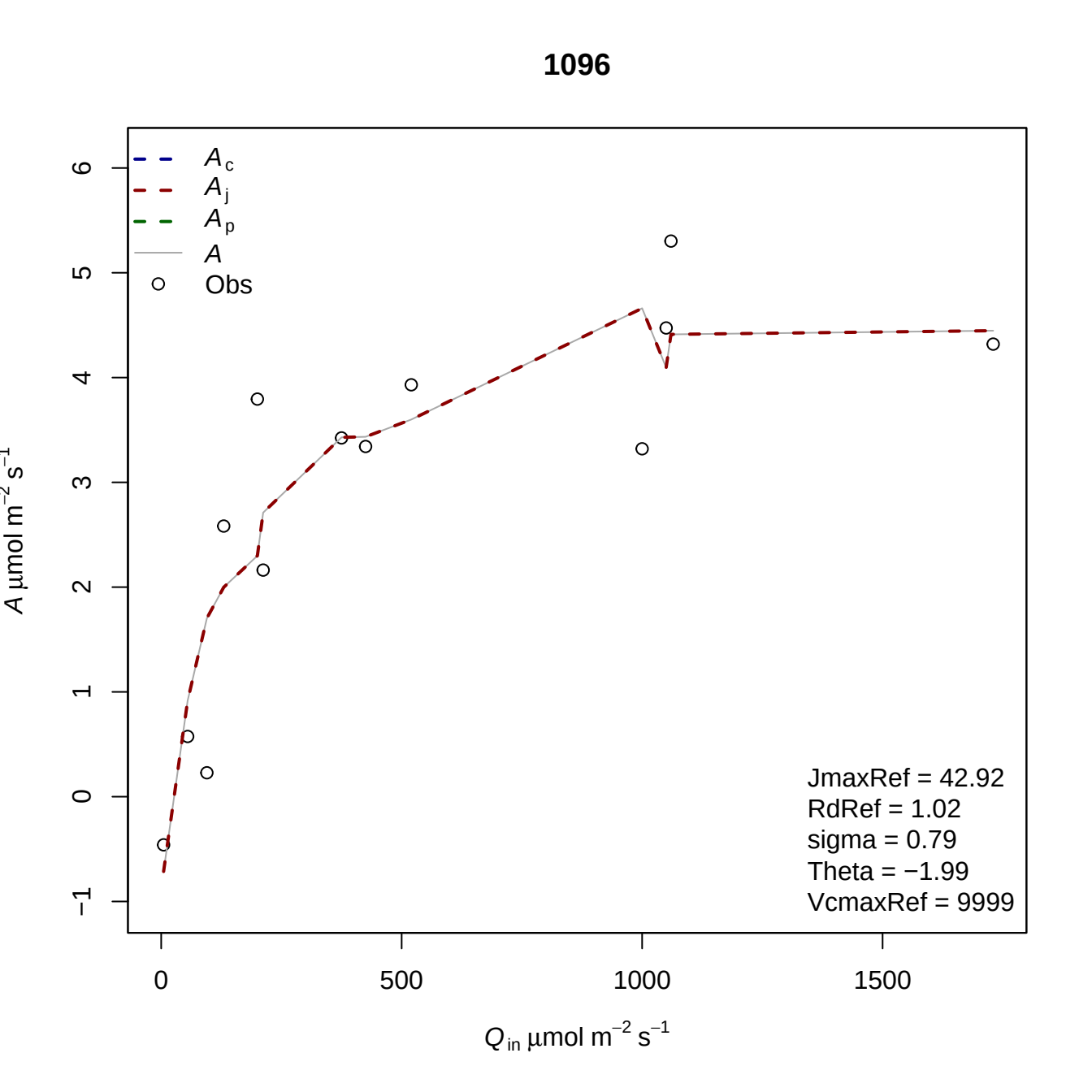

1089
A_c
A_j
A_p
A
Obs
JmaxRef = 71.16
RdRef = 1.15
sigma = 1.06
Theta = 0.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

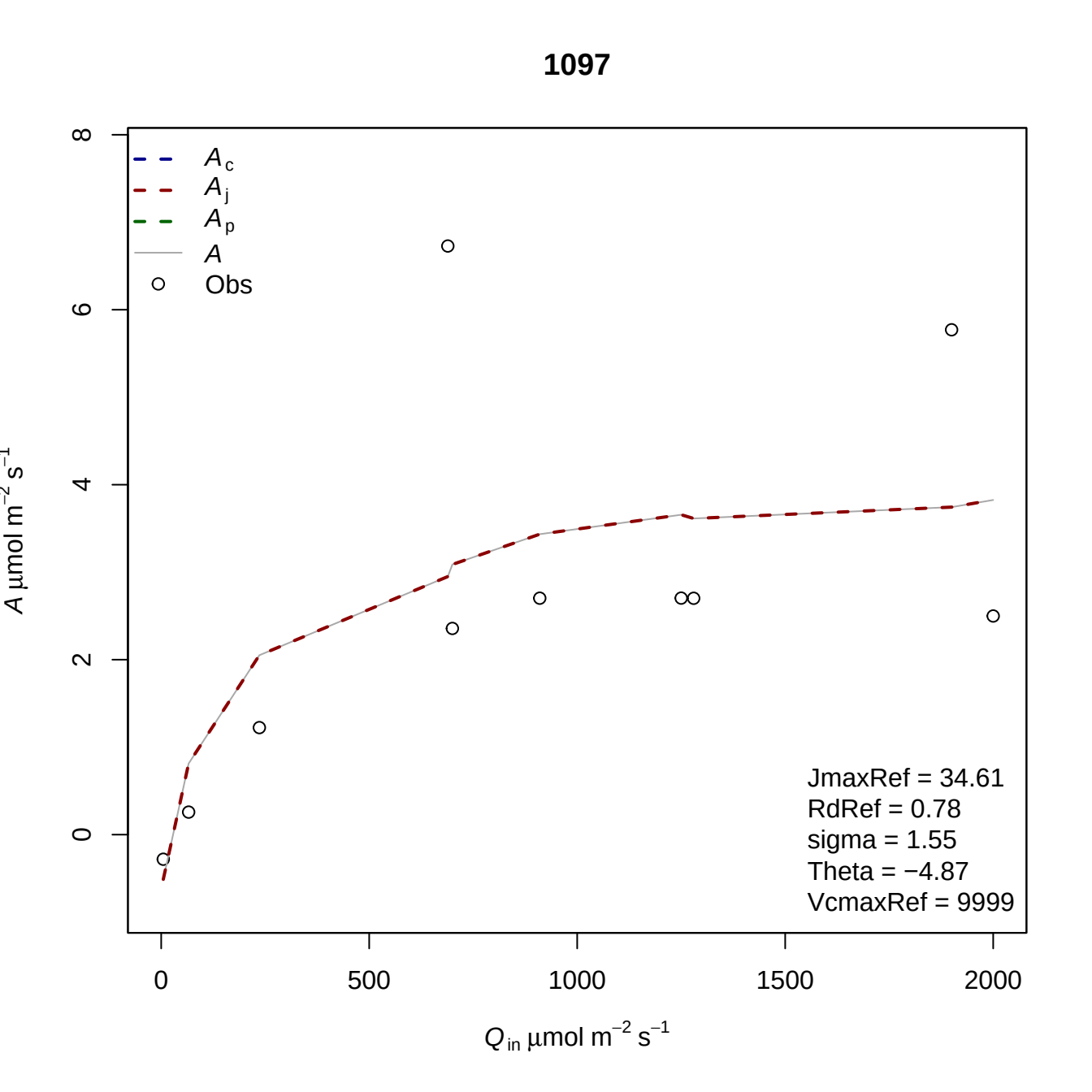

1092
A_c
A_j
A_p
A
Obs
JmaxRef = 76.19
RdRef = 2.99
sigma = 1.67
Theta = −0.19
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

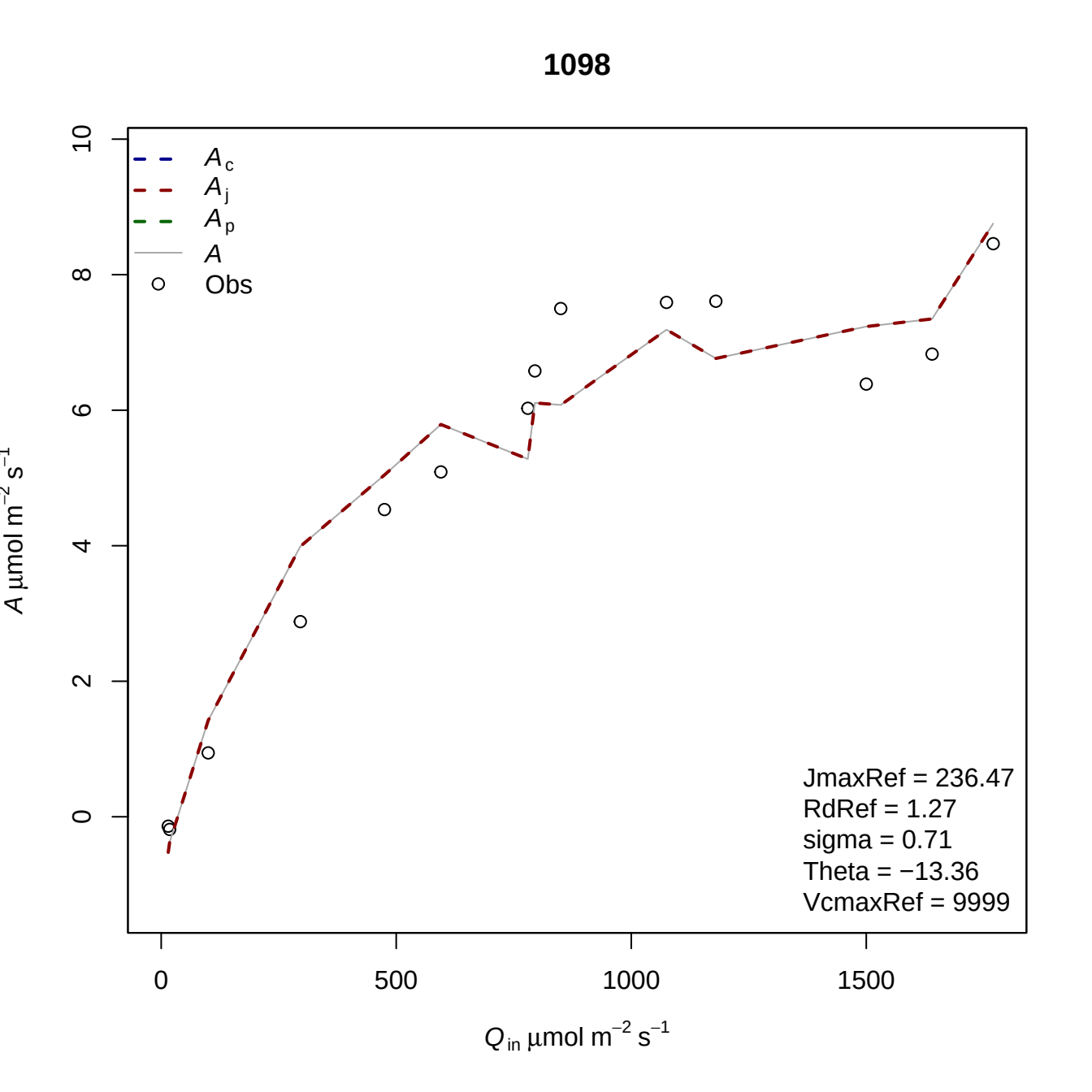

1093
A_c
A_j
A_p
A
Obs
JmaxRef = 122.93
RdRef = 2.63
sigma = 1.2
Theta = −2.85
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

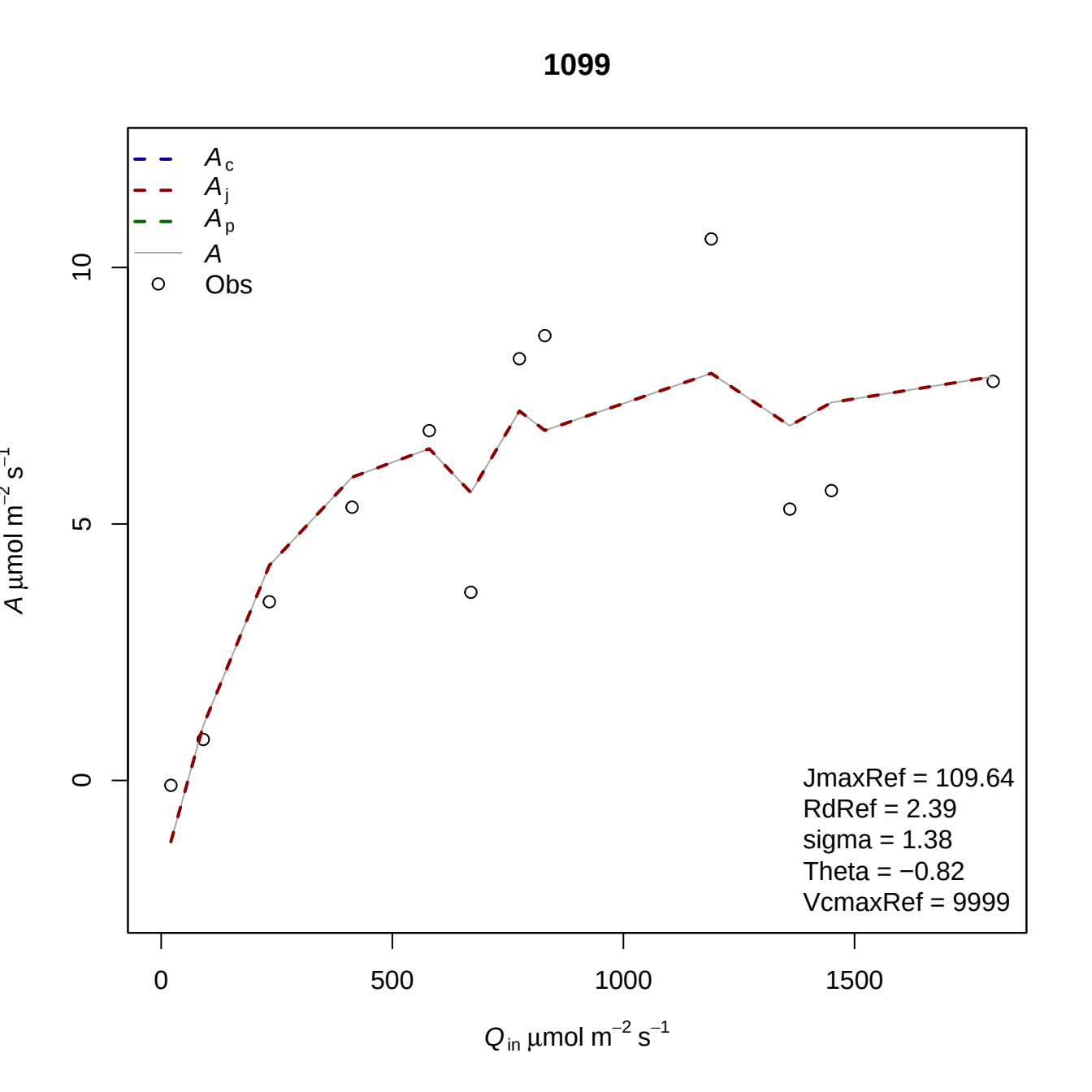

1094
A_c
A_j
A_p
A
Obs
JmaxRef = 86.63
RdRef = 1.87
sigma = 1.23
Theta = −1.89
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

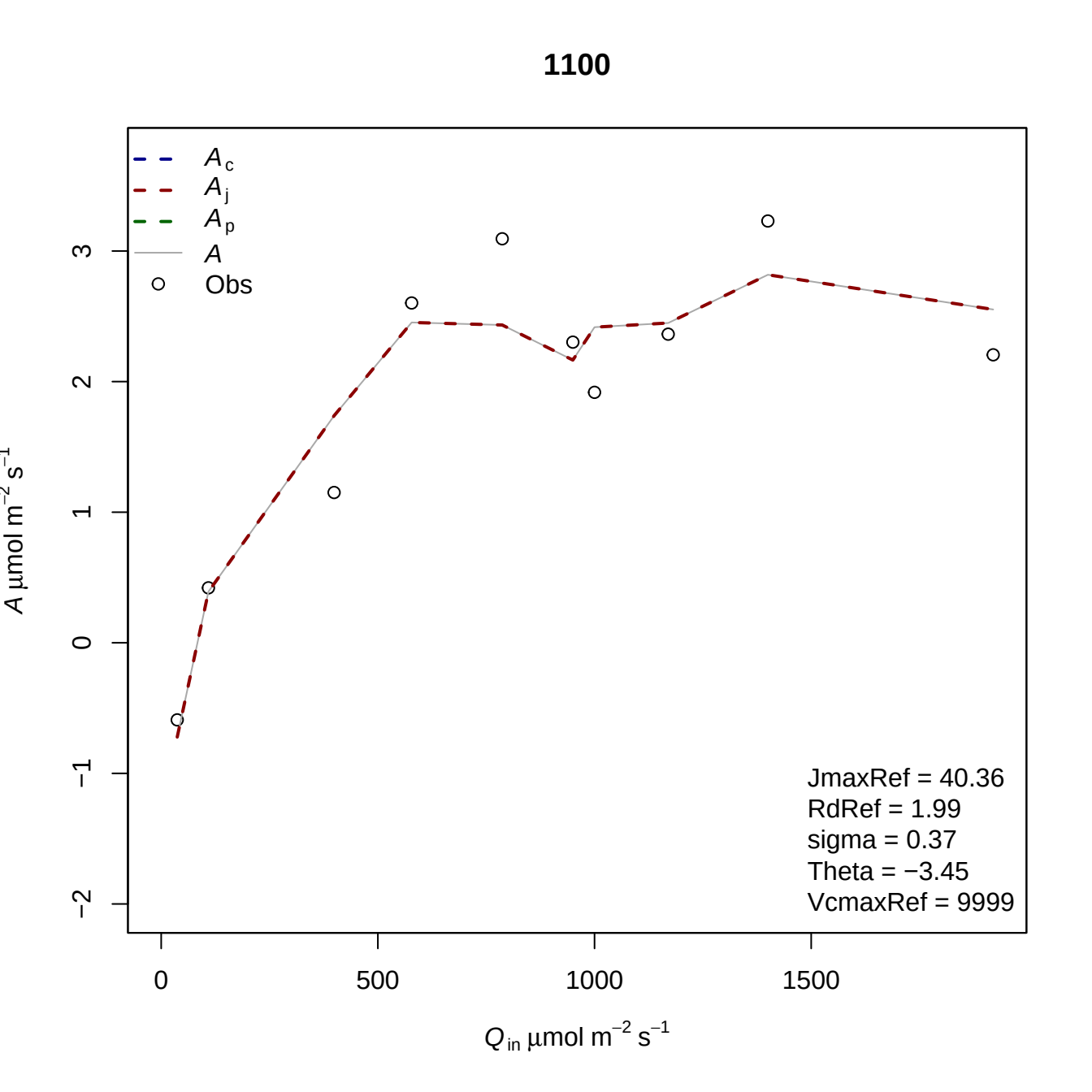

1095
A_c
A_j
A_p
A
Obs
JmaxRef = 95.29
RdRef = 2.87
sigma = 0.68
Theta = −1.01
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

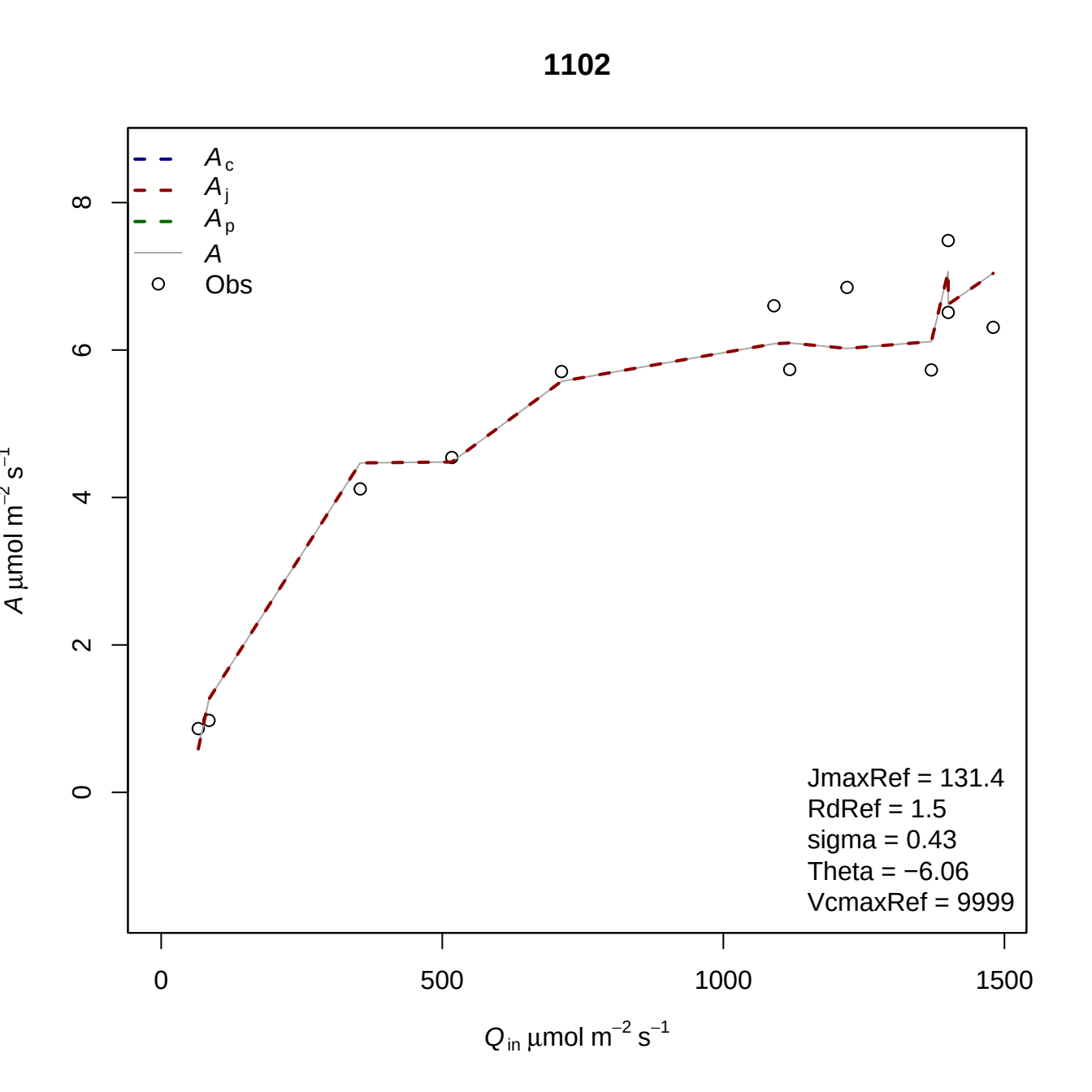

1096
A_c
A_j
A_p
A
Obs
JmaxRef = 42.92
RdRef = 1.02
sigma = 0.79
Theta = −1.99
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

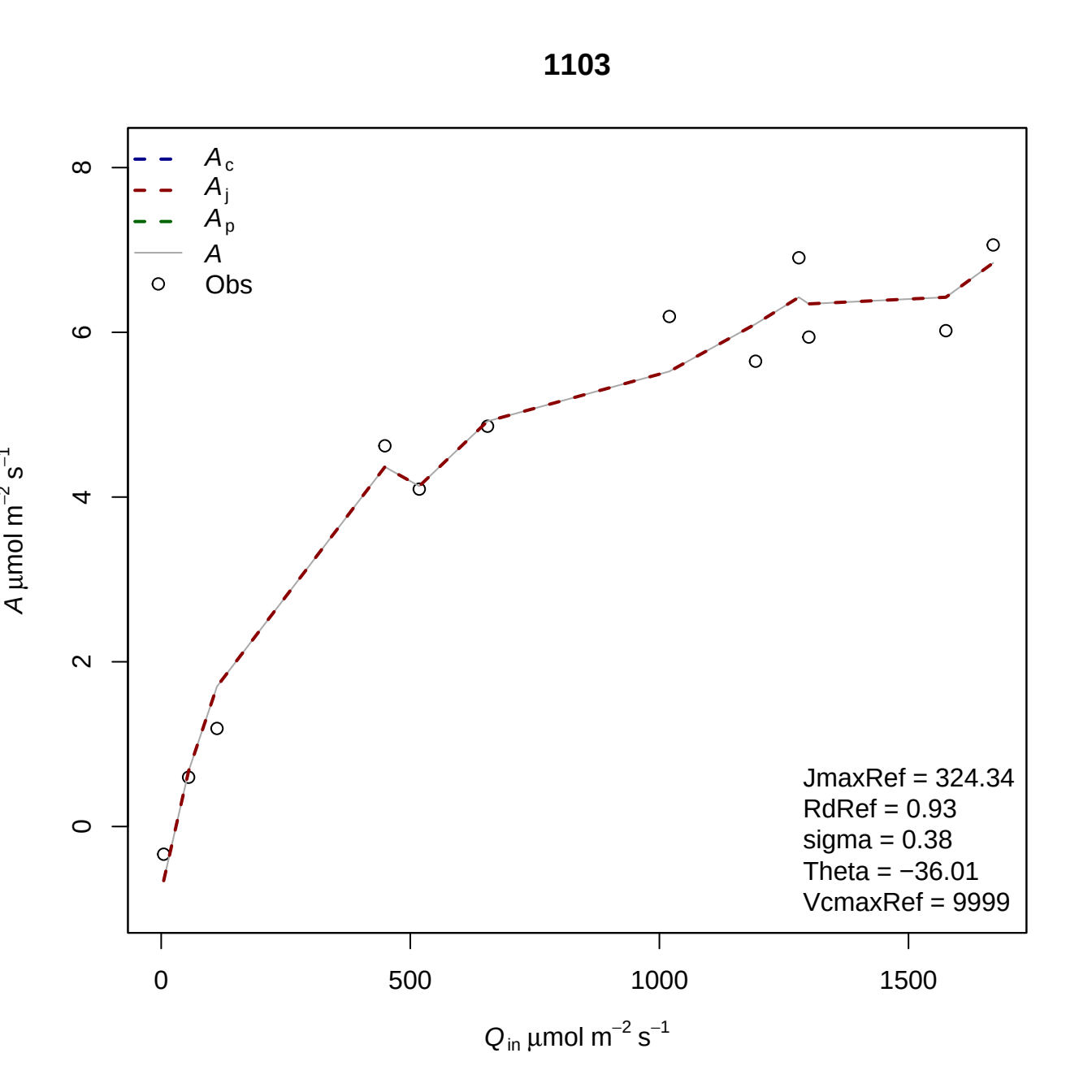

1097
A_c
A_j
A_p
A
Obs
JmaxRef = 34.61
RdRef = 0.78
sigma = 1.55
Theta = −4.87
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

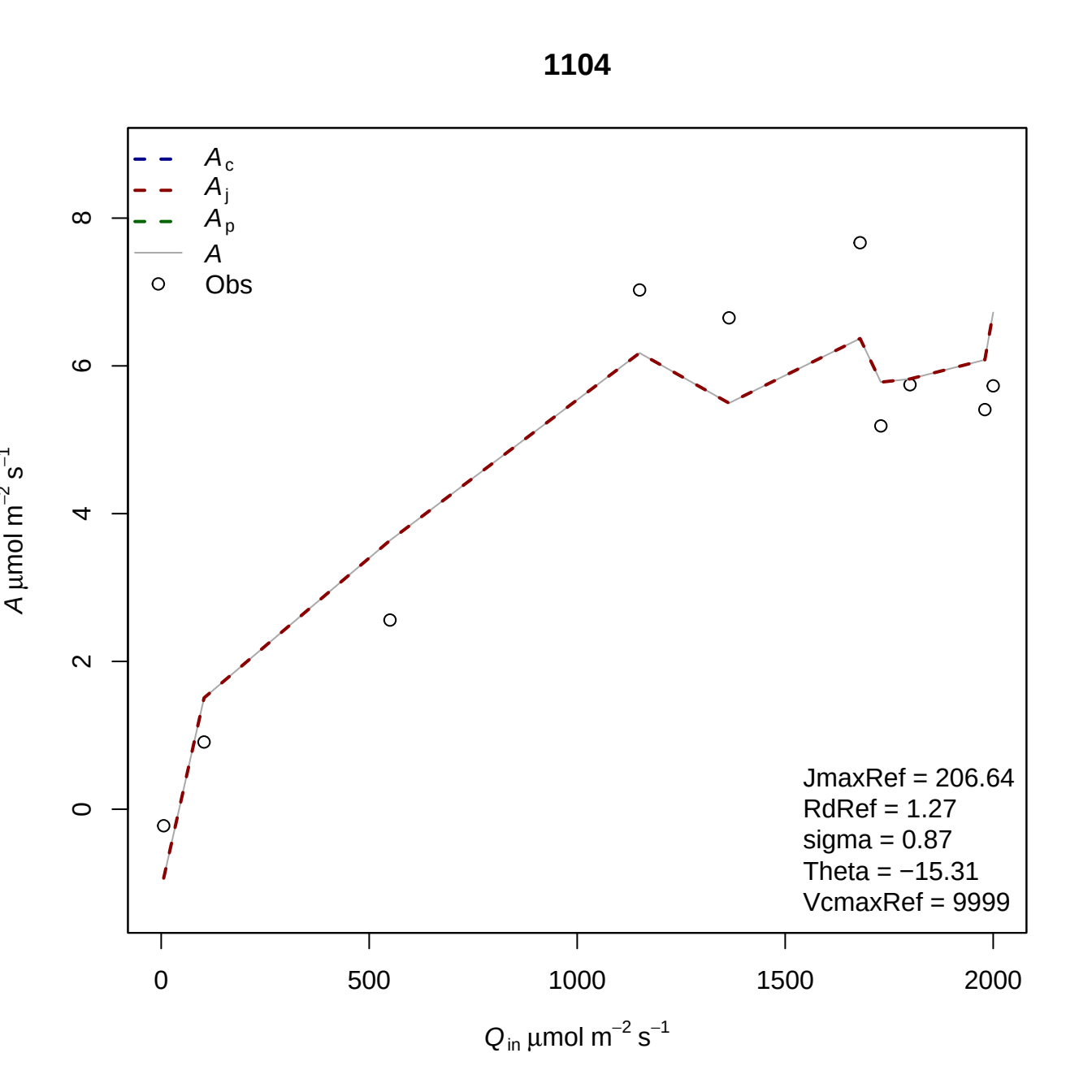

1098
A_c
A_j
A_p
A
Obs
JmaxRef = 236.47
RdRef = 1.27
sigma = 0.71
Theta = −13.36
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

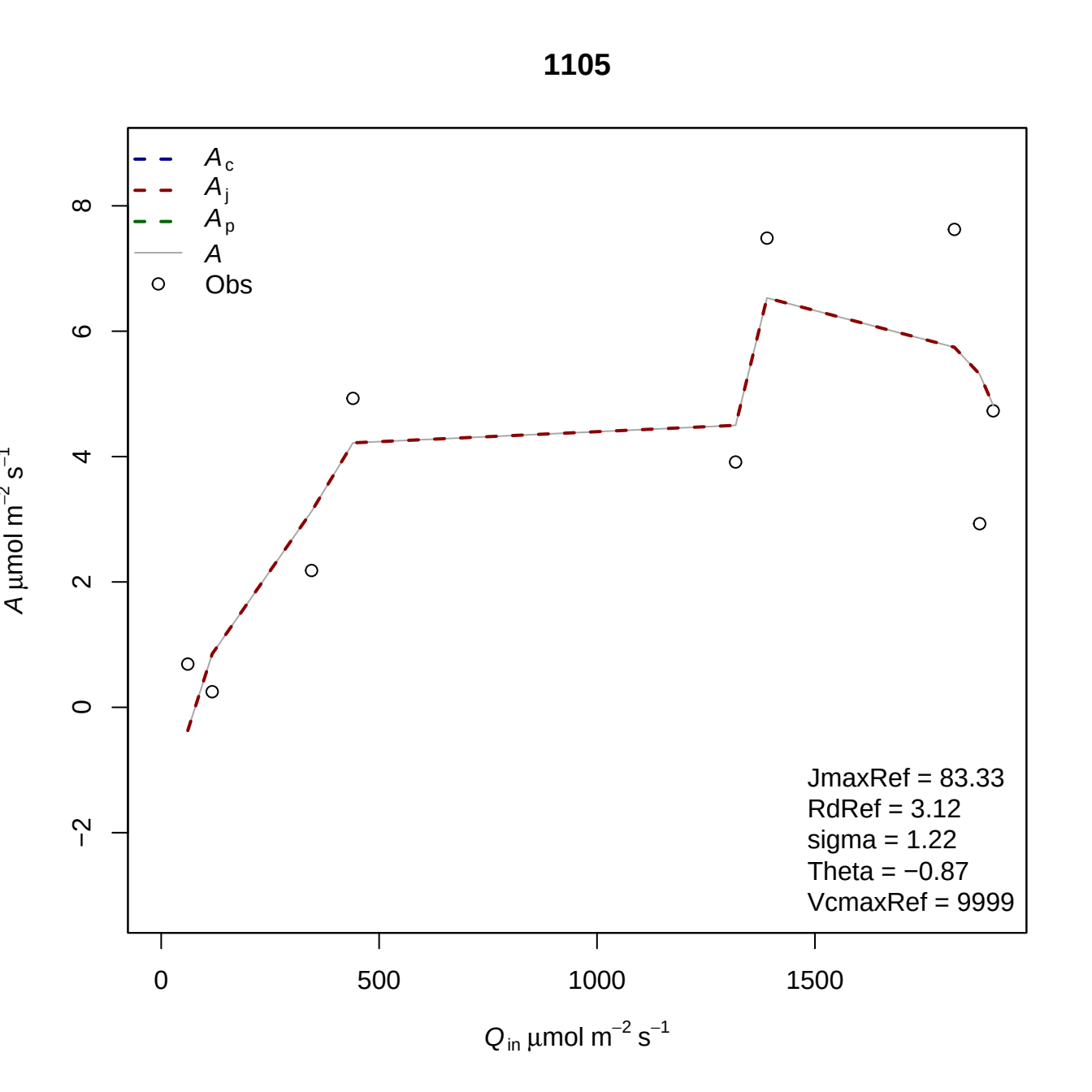

1099
A_c
A_j
A_p
A
Obs
JmaxRef = 109.64
RdRef = 2.39
sigma = 1.38
Theta = −0.82
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

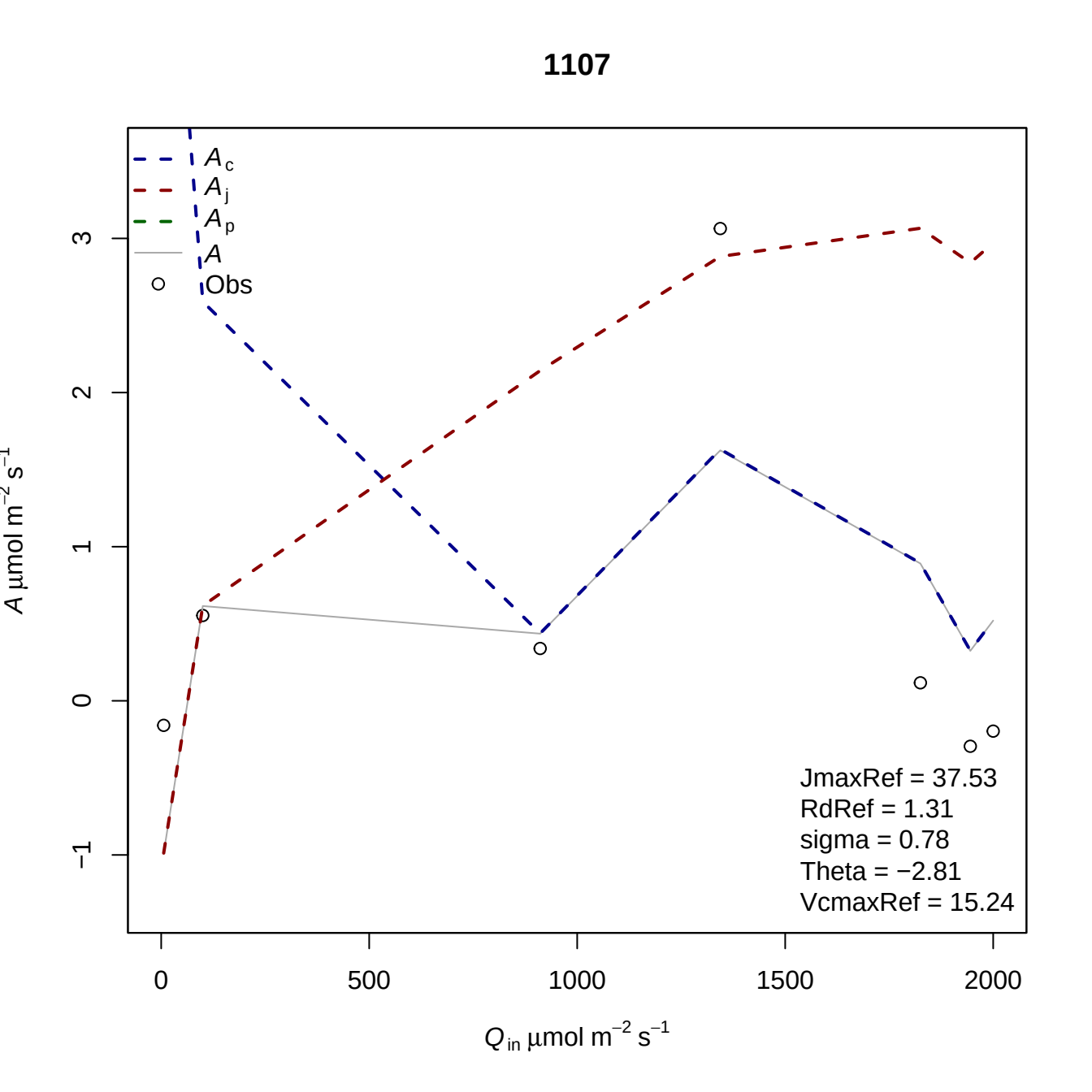

1100
A_c
A_j
A_p
A
Obs
JmaxRef = 40.36
RdRef = 1.99
sigma = 0.37
Theta = −3.45
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
500
1000
1500

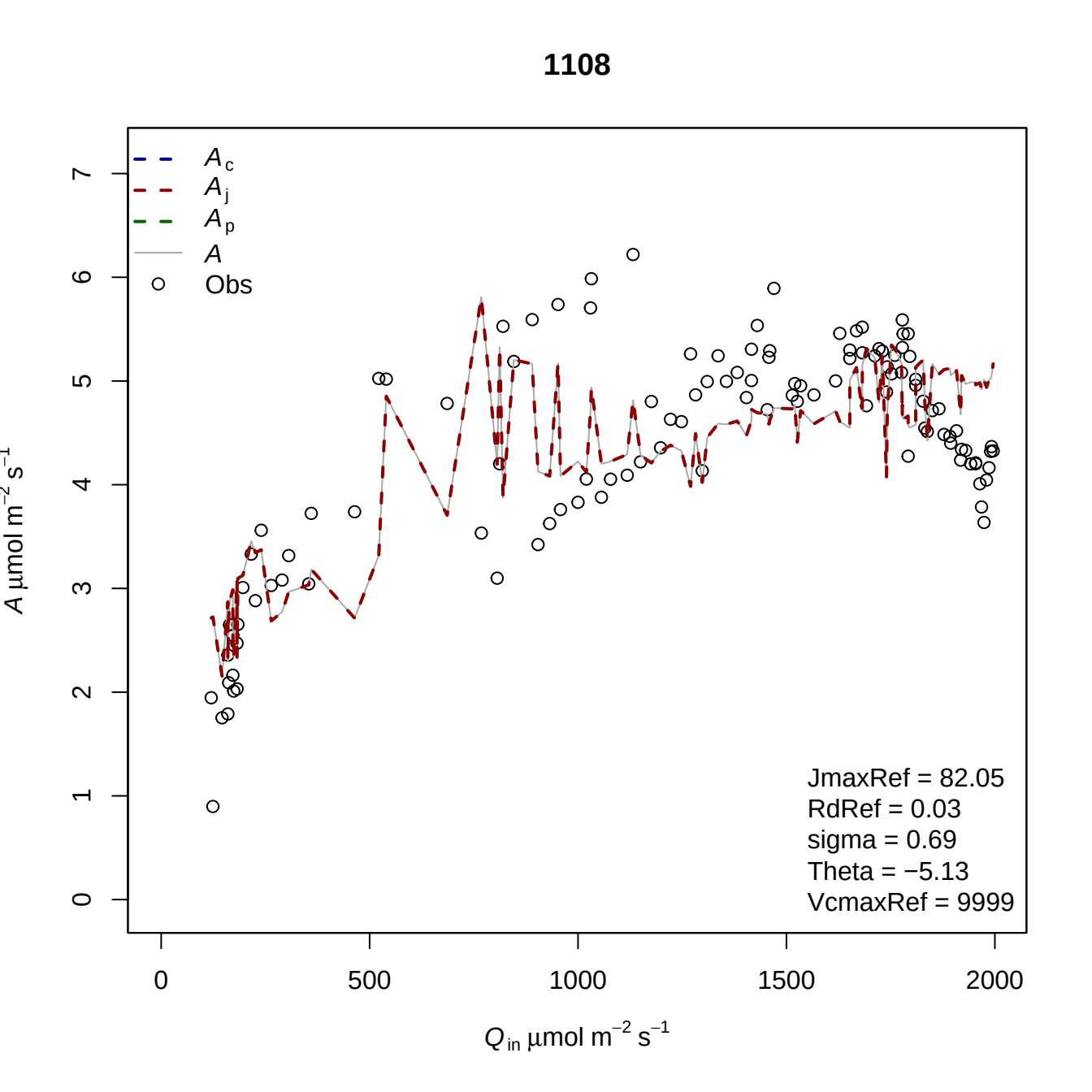

1102
A_c
A_j
A_p
A
Obs
JmaxRef = 131.4
RdRef = 1.5
sigma = 0.43
Theta = −6.06
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

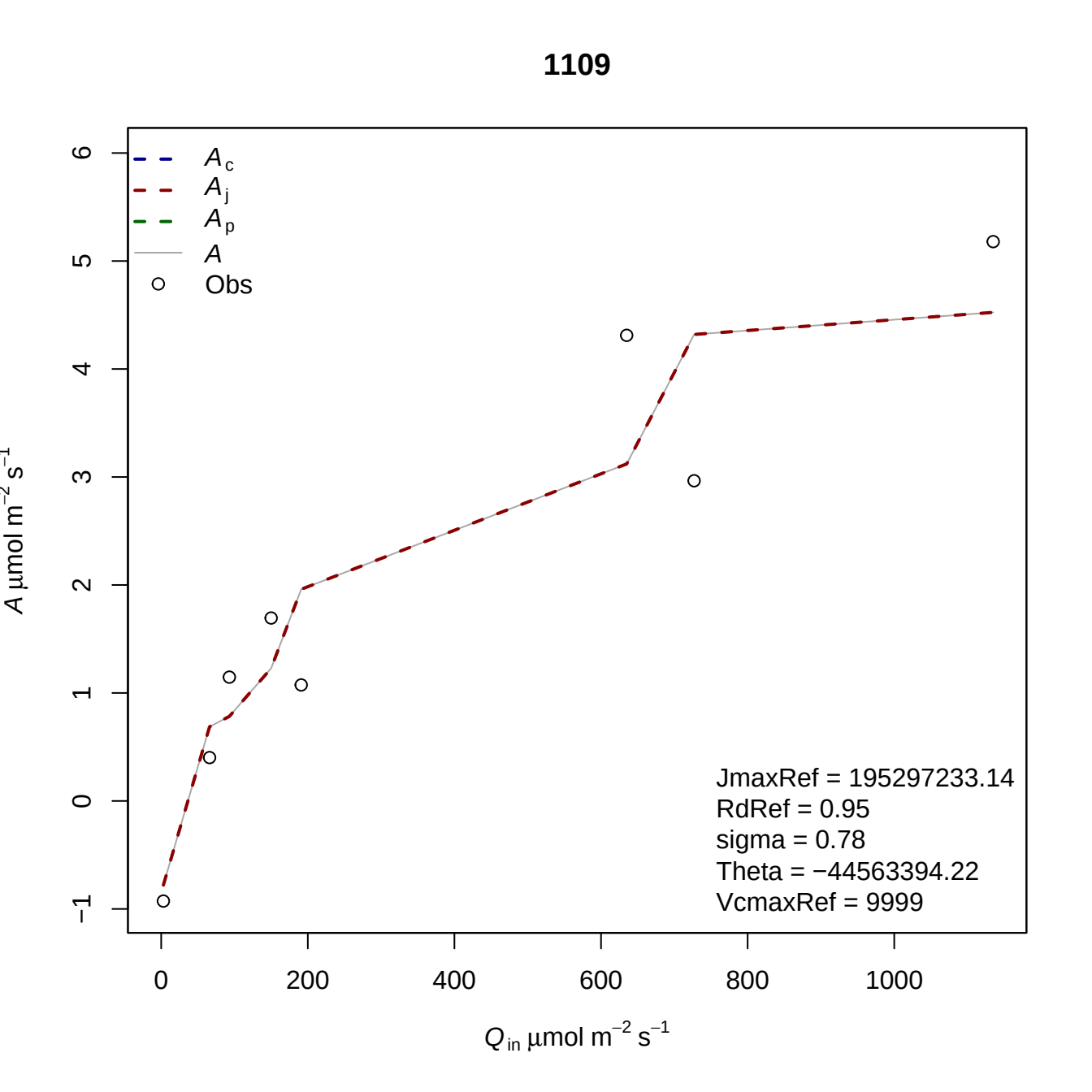

1103
A_c
A_j
A_p
A
Obs
JmaxRef = 324.34
RdRef = 0.93
sigma = 0.38
Theta = −36.01
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

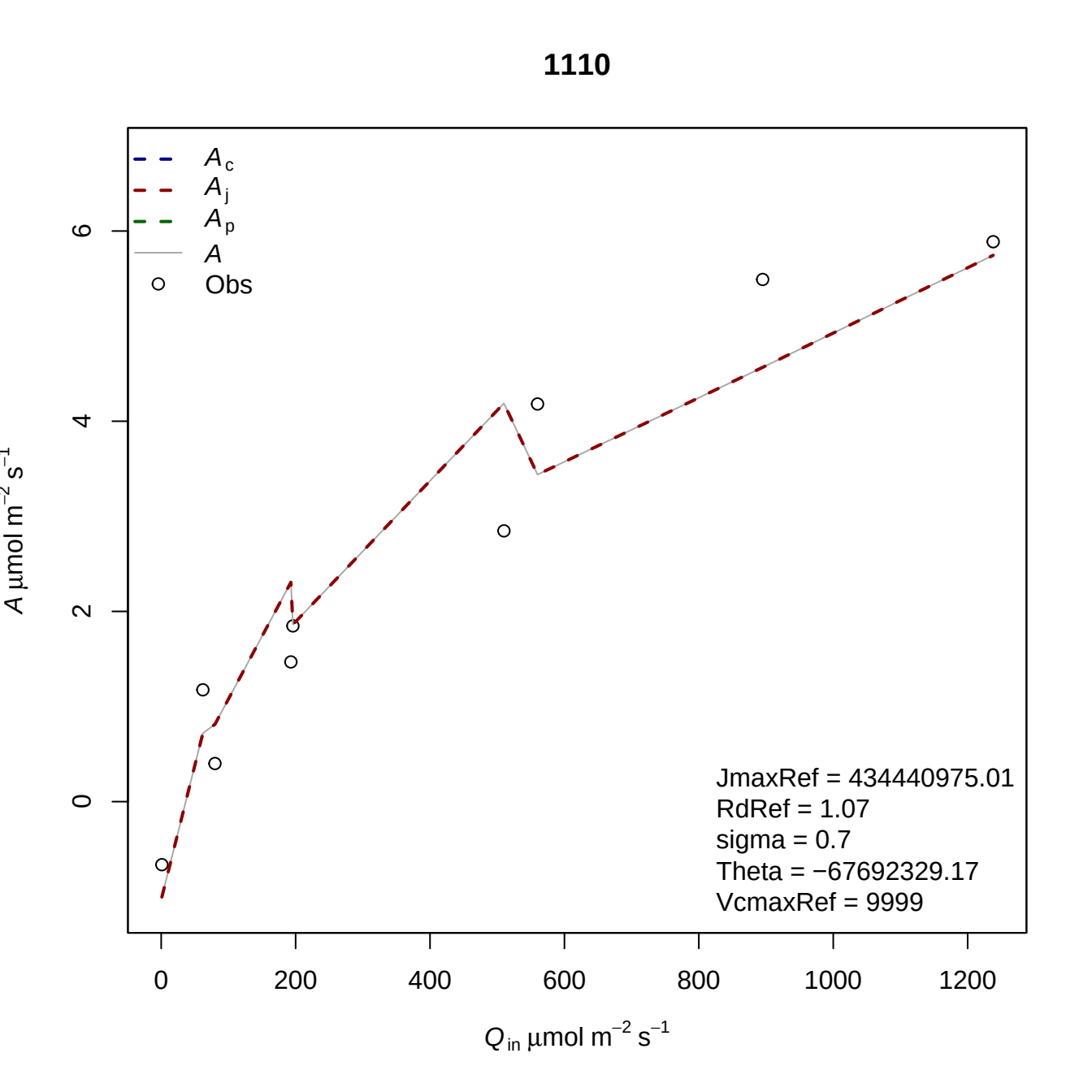

1104
A_c
A_j
A_p
A
Obs
JmaxRef = 206.64
RdRef = 1.27
sigma = 0.87
Theta = −15.31
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

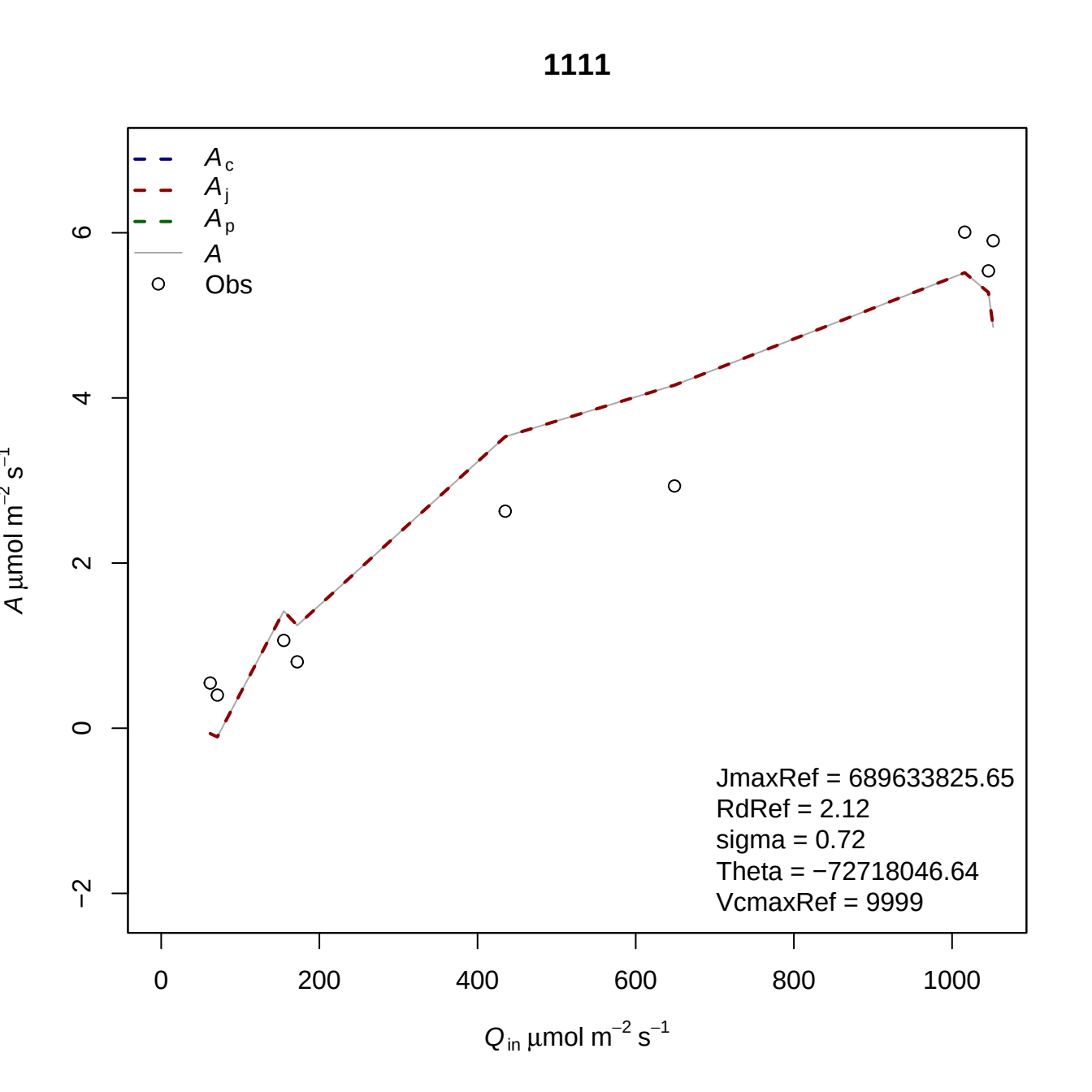

1105
A_c
A_j
A_p
A
Obs
JmaxRef = 83.33
RdRef = 3.12
sigma = 1.22
Theta = −0.87
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

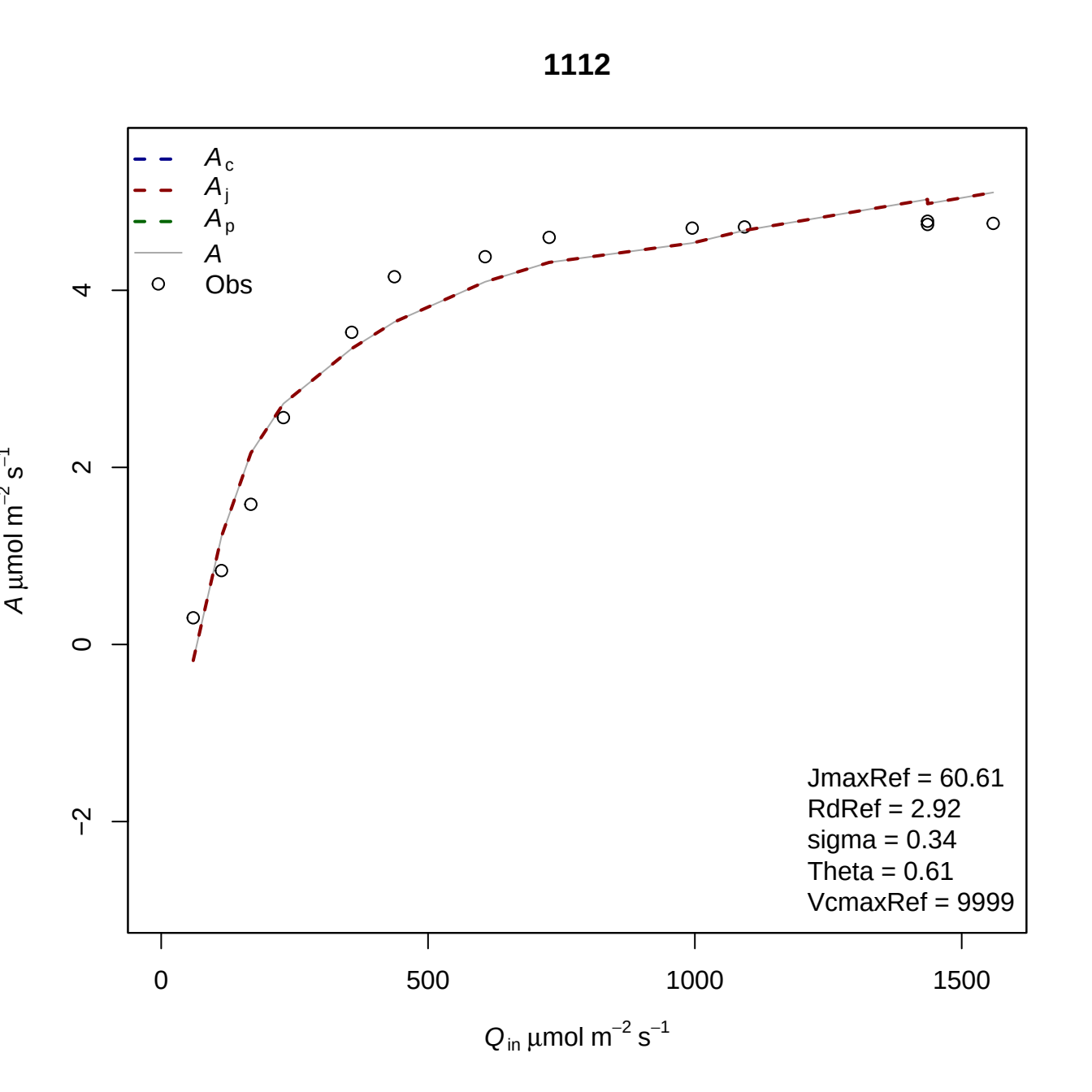

1107
A_c
A_j
A_p
A
Obs
JmaxRef = 37.53
RdRef = 1.31
sigma = 0.78
Theta = −2.81
VcmaxRef = 15.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

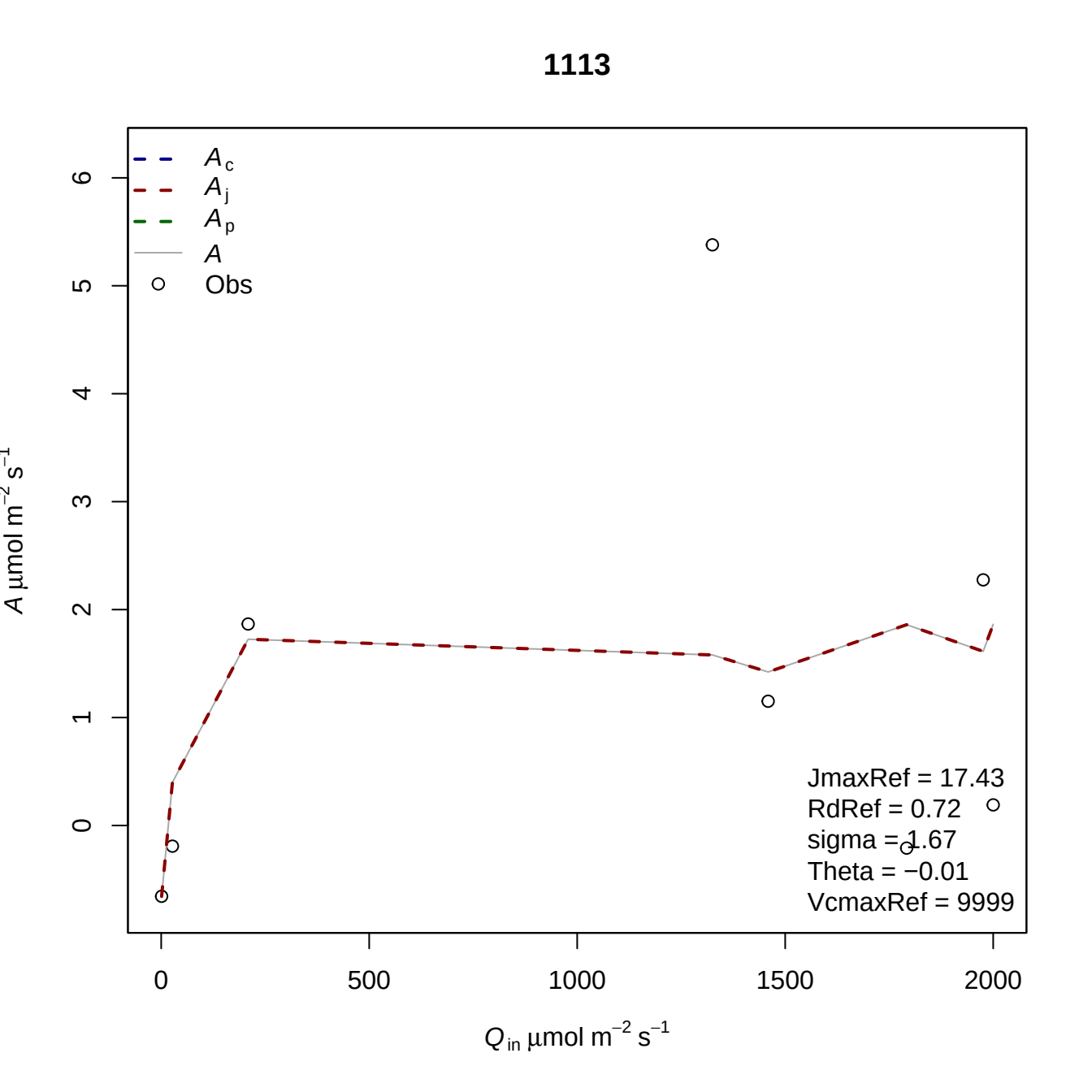

1108
A_c
A_j
A_p
A
Obs
JmaxRef = 82.05
RdRef = 0.03
sigma = 0.69
Theta = −5.13
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
7
0
500
1000
1500
2000

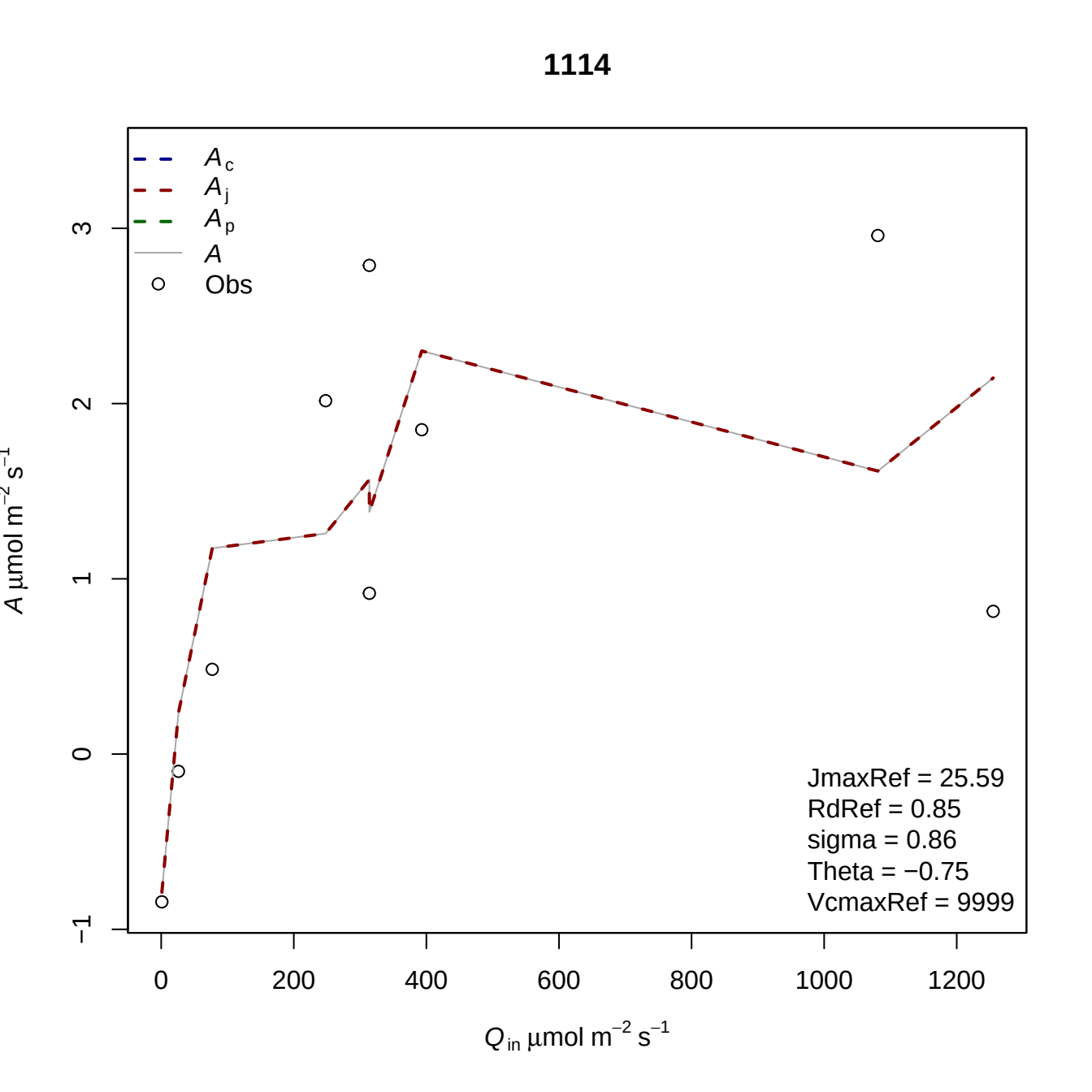

1109
A_c
A_j
A_p
A
Obs
JmaxRef = 195297233.14
RdRef = 0.95
sigma = 0.78
Theta = −44563394.22
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
200
400
600
800
1000

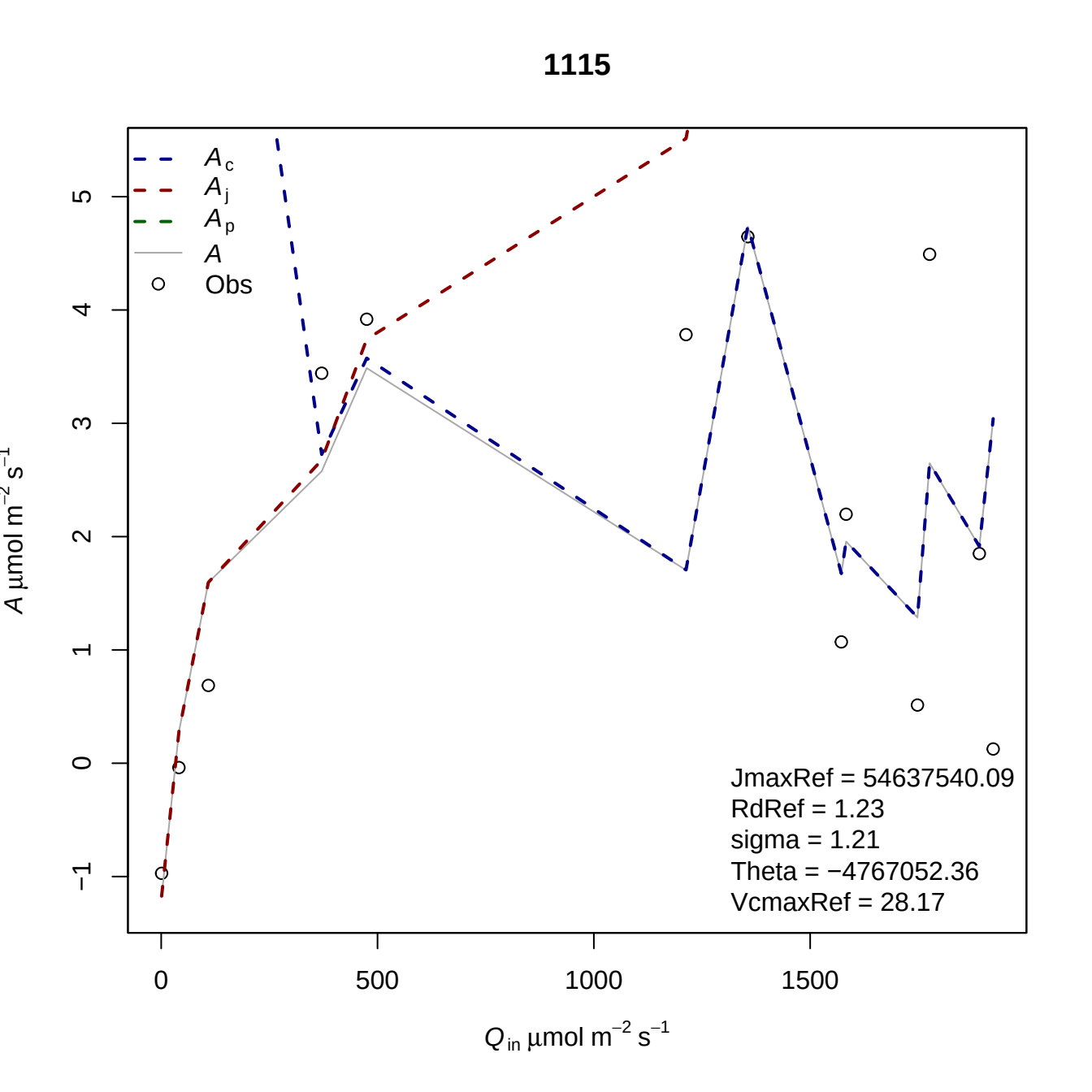

1110
A_c
A_j
A_p
A
Obs
JmaxRef = 434440975.01
RdRef = 1.07
sigma = 0.7
Theta = −67692329.17
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

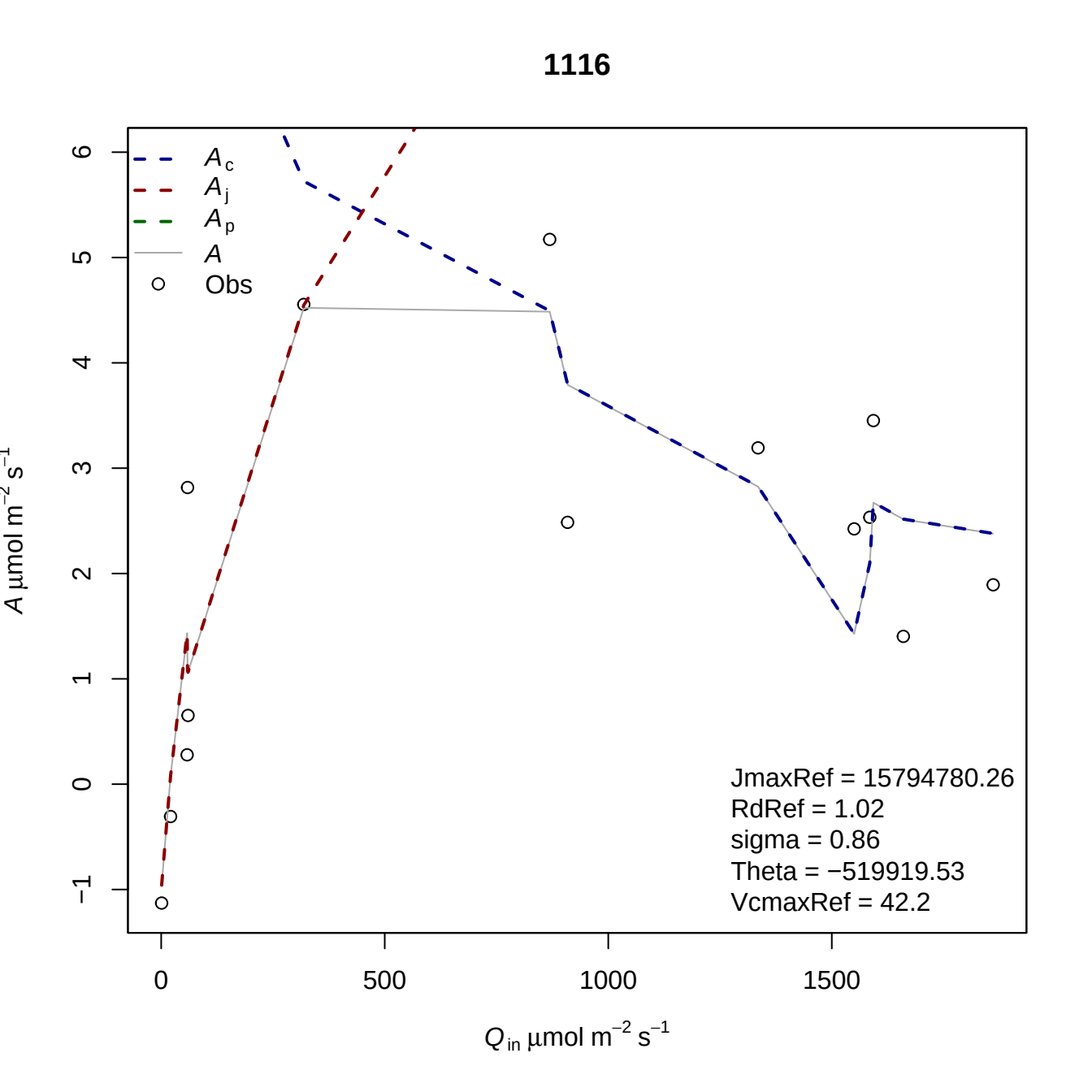

1111
A_c
A_j
A_p
A
Obs
JmaxRef = 689633825.65
RdRef = 2.12
sigma = 0.72
Theta = −72718046.64
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

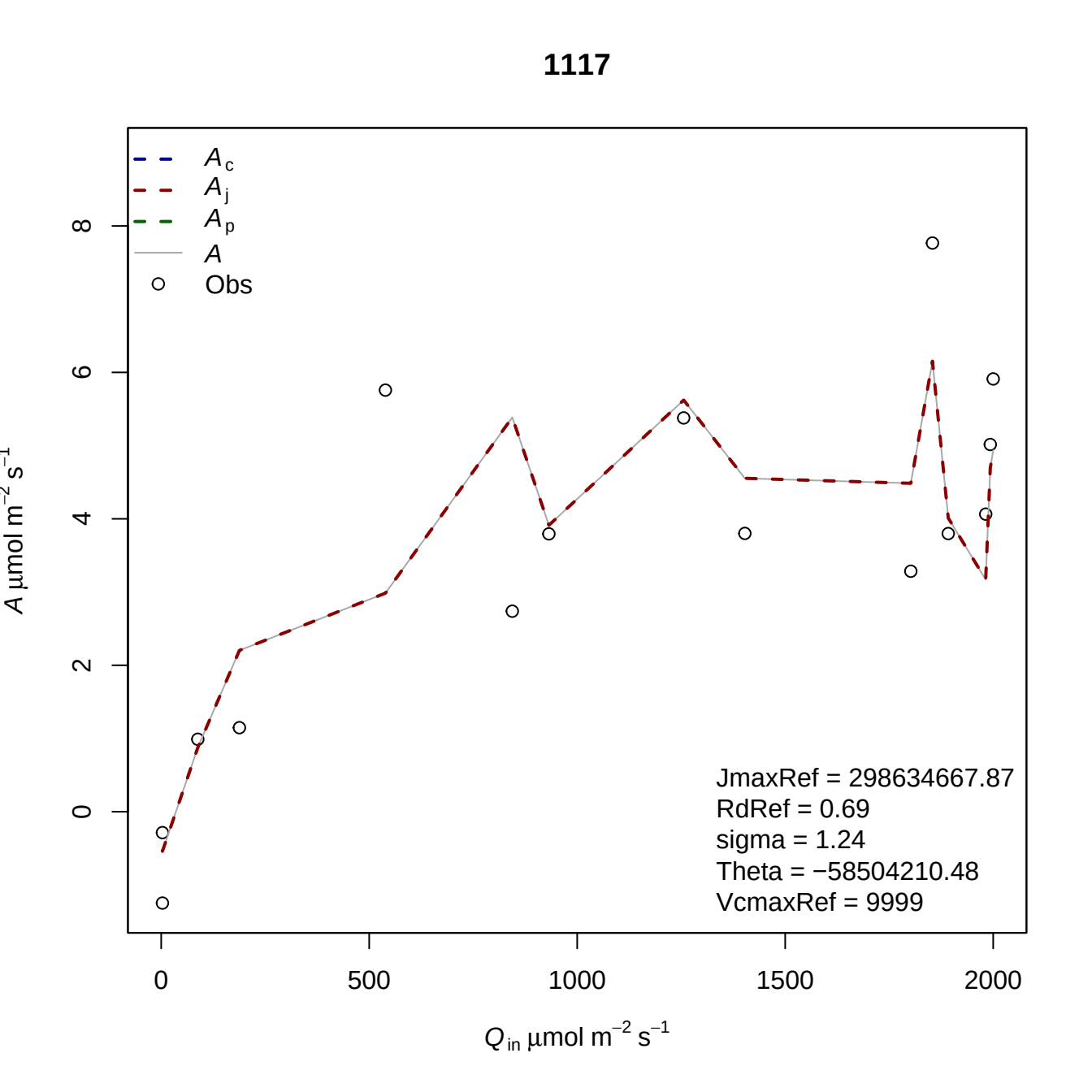

1112
A_c
A_j
A_p
A
Obs
JmaxRef = 60.61
RdRef = 2.92
sigma = 0.34
Theta = 0.61
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

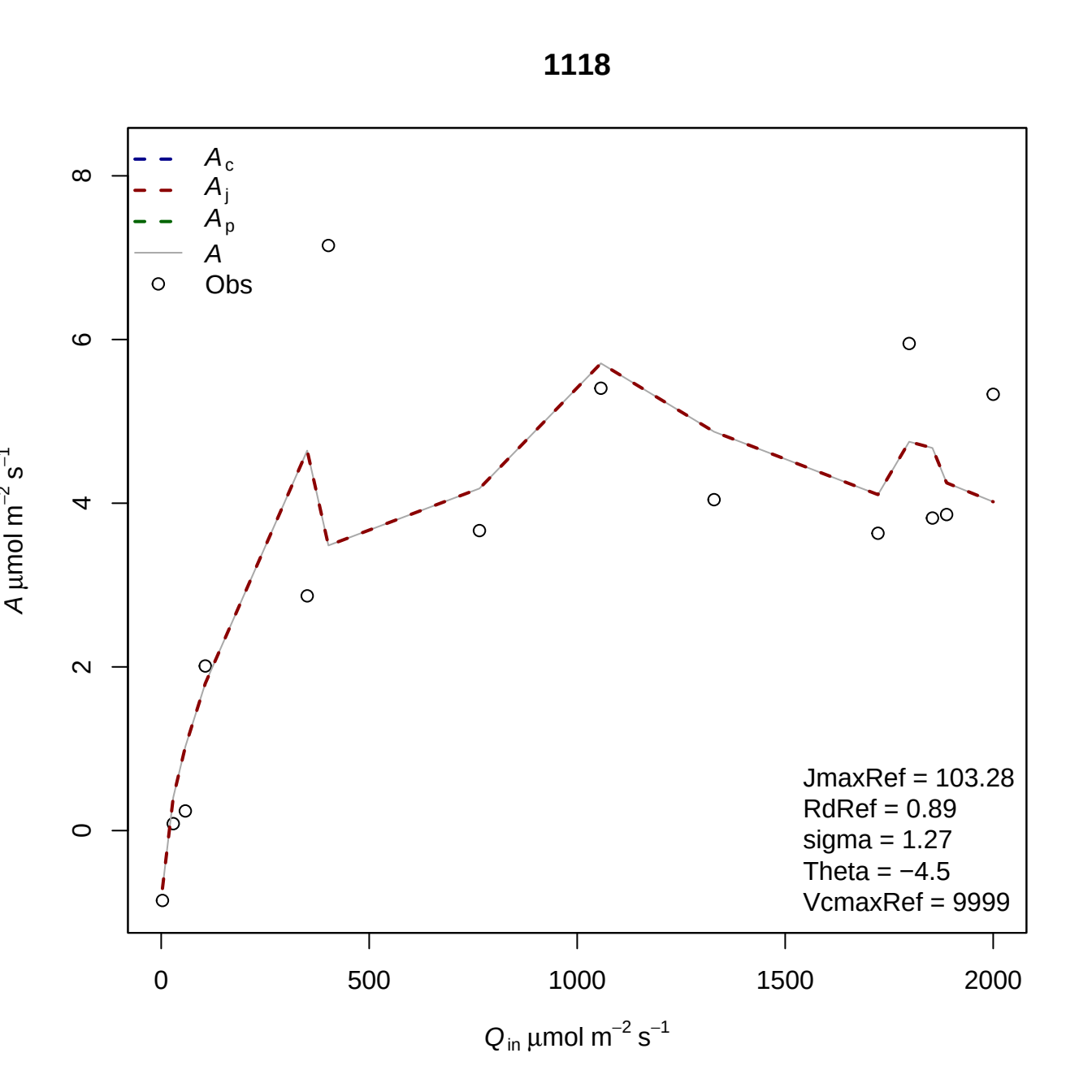

1113
A_c
A_j
A_p
A
Obs
JmaxRef = 17.43
RdRef = 0.72
sigma = 1.67
Theta = −0.01
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500
2000

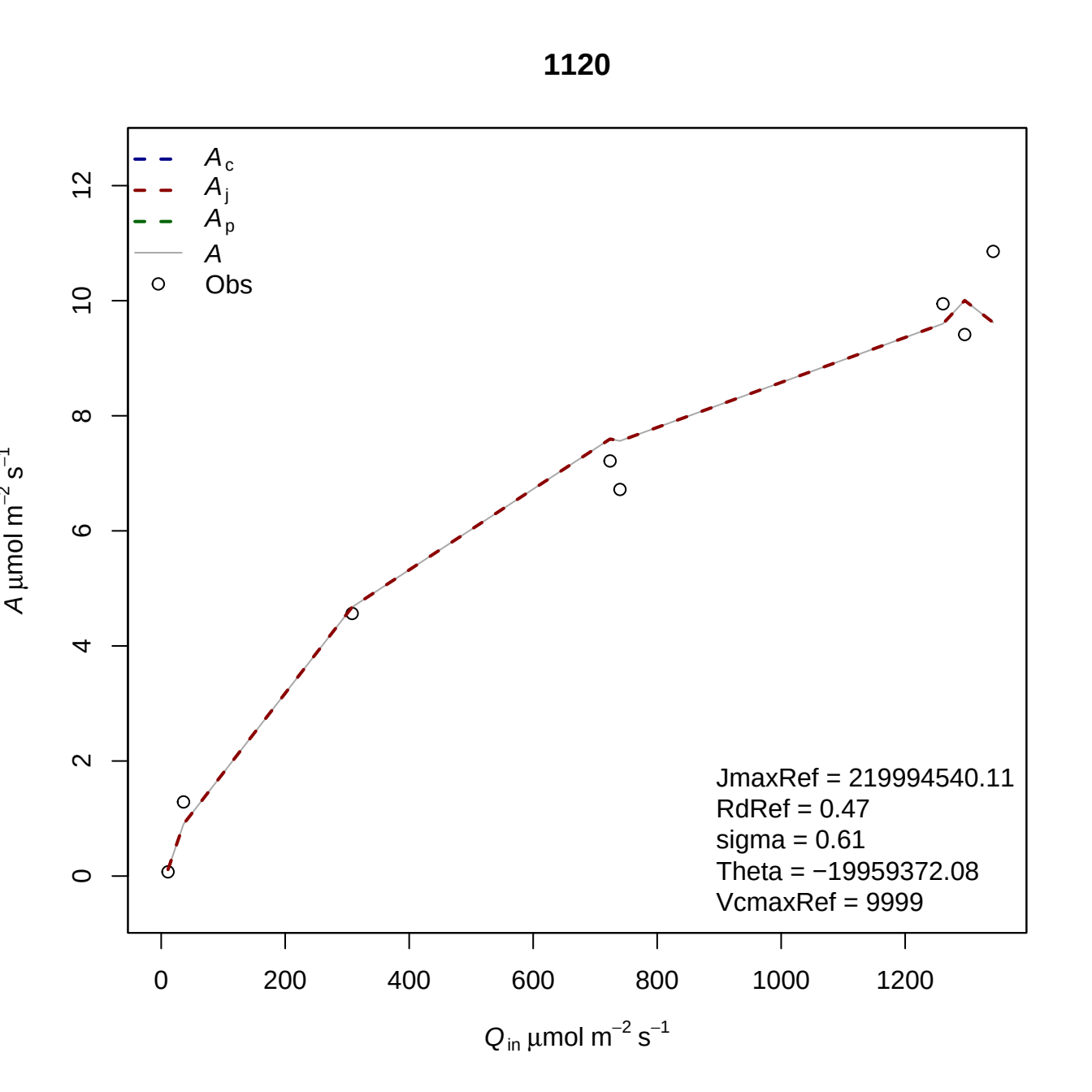

1114
A_c
A_j
A_p
A
Obs
JmaxRef = 25.59
RdRef = 0.85
sigma = 0.86
Theta = −0.75
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200

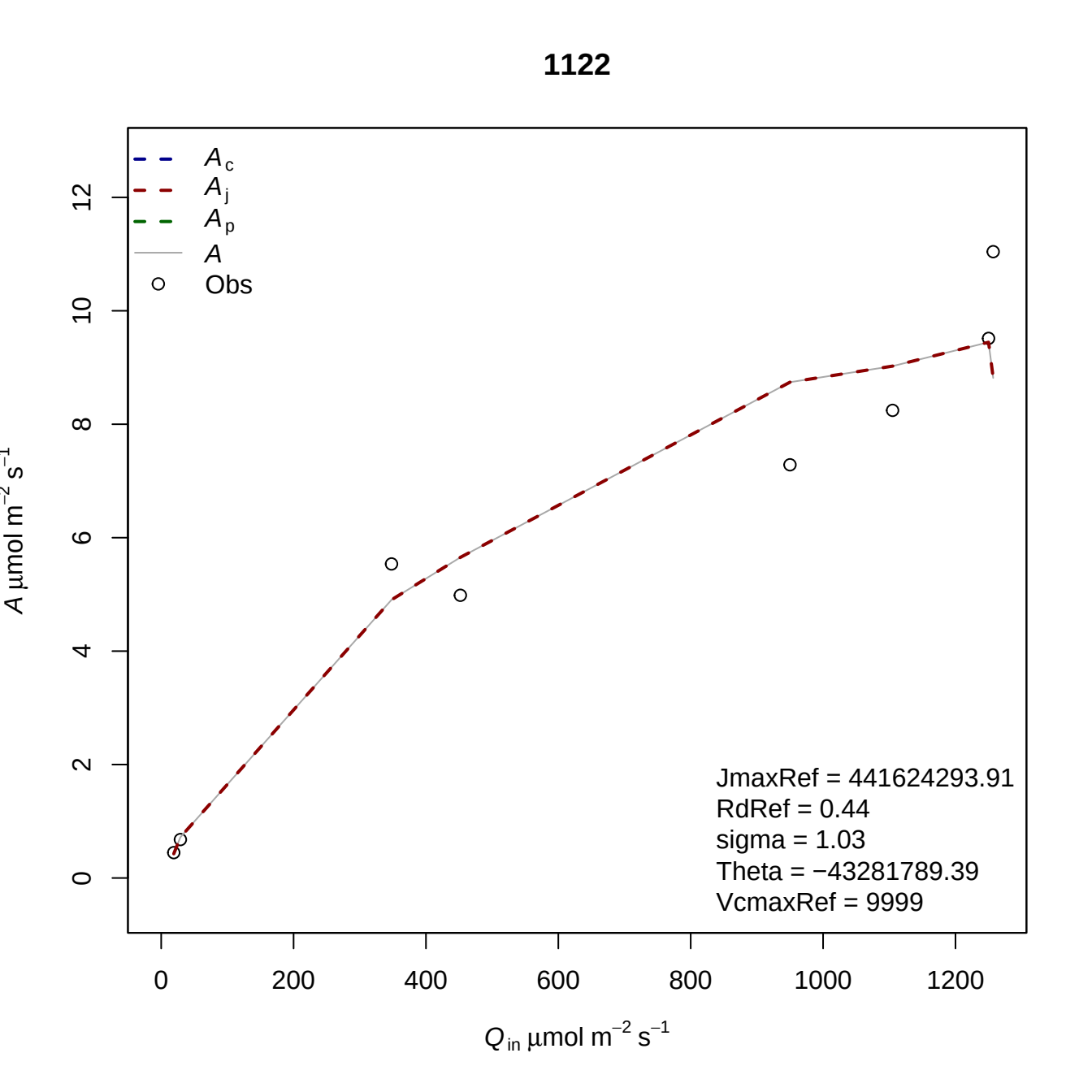

1115
A_c
A_j
A_p
A
Obs
JmaxRef = 54637540.09
RdRef = 1.23
sigma = 1.21
Theta = −4767052.36
VcmaxRef = 28.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

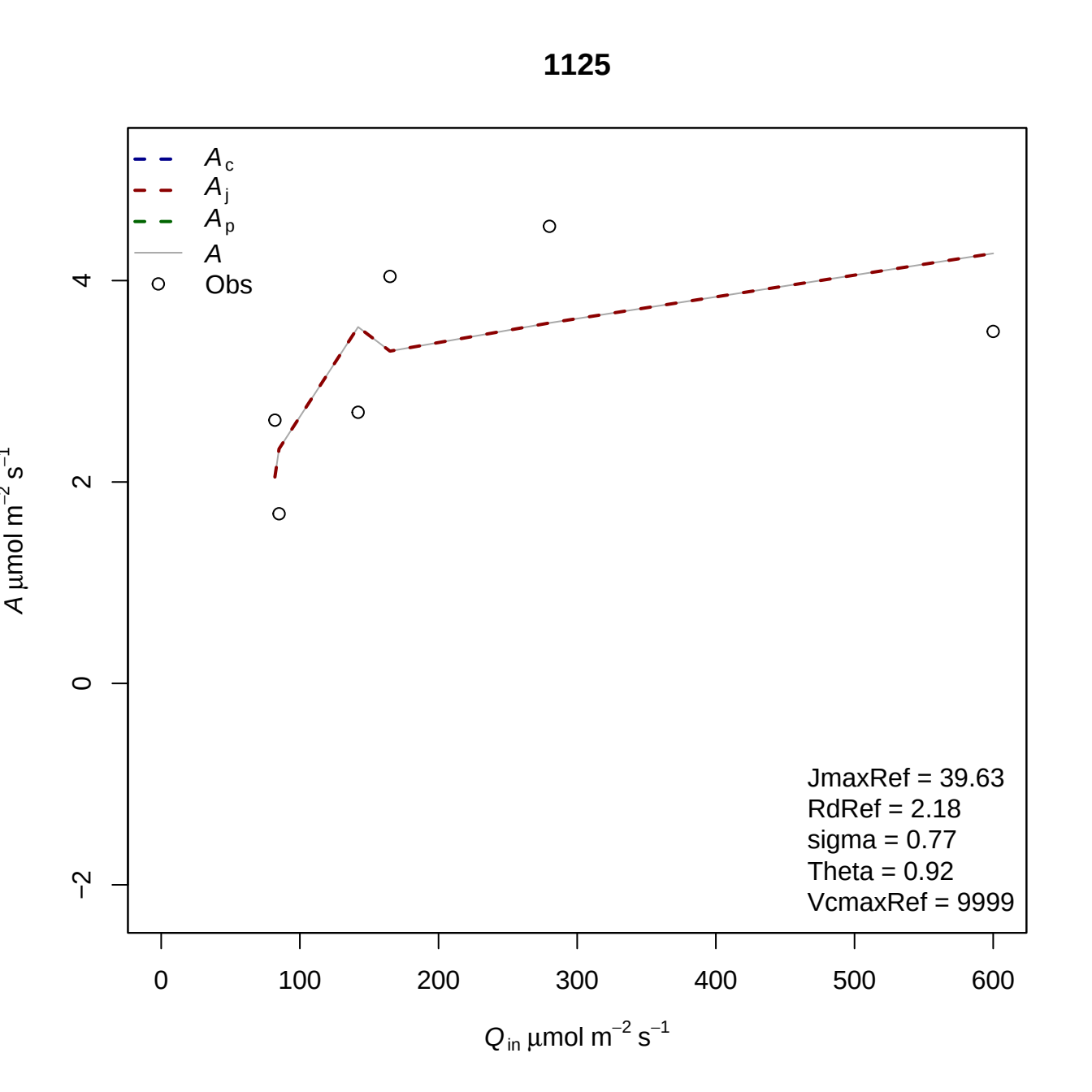

1116
A_c
A_j
A_p
A
Obs
JmaxRef = 15794780.26
RdRef = 1.02
sigma = 0.86
Theta = −519919.53
VcmaxRef = 42.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

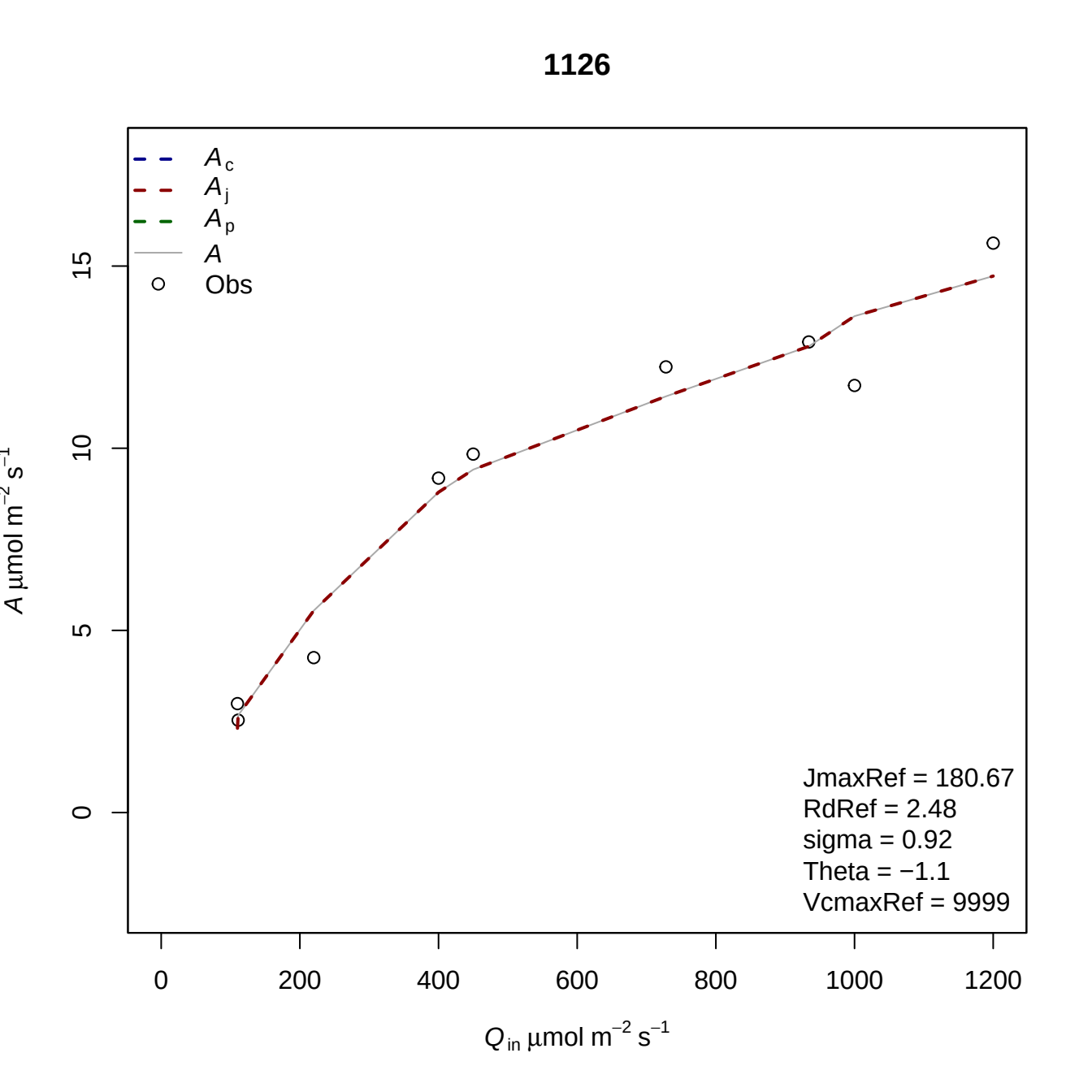

1117
A_c
A_j
A_p
A
Obs
JmaxRef = 298634667.87
RdRef = 0.69
sigma = 1.24
Theta = −58504210.48
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

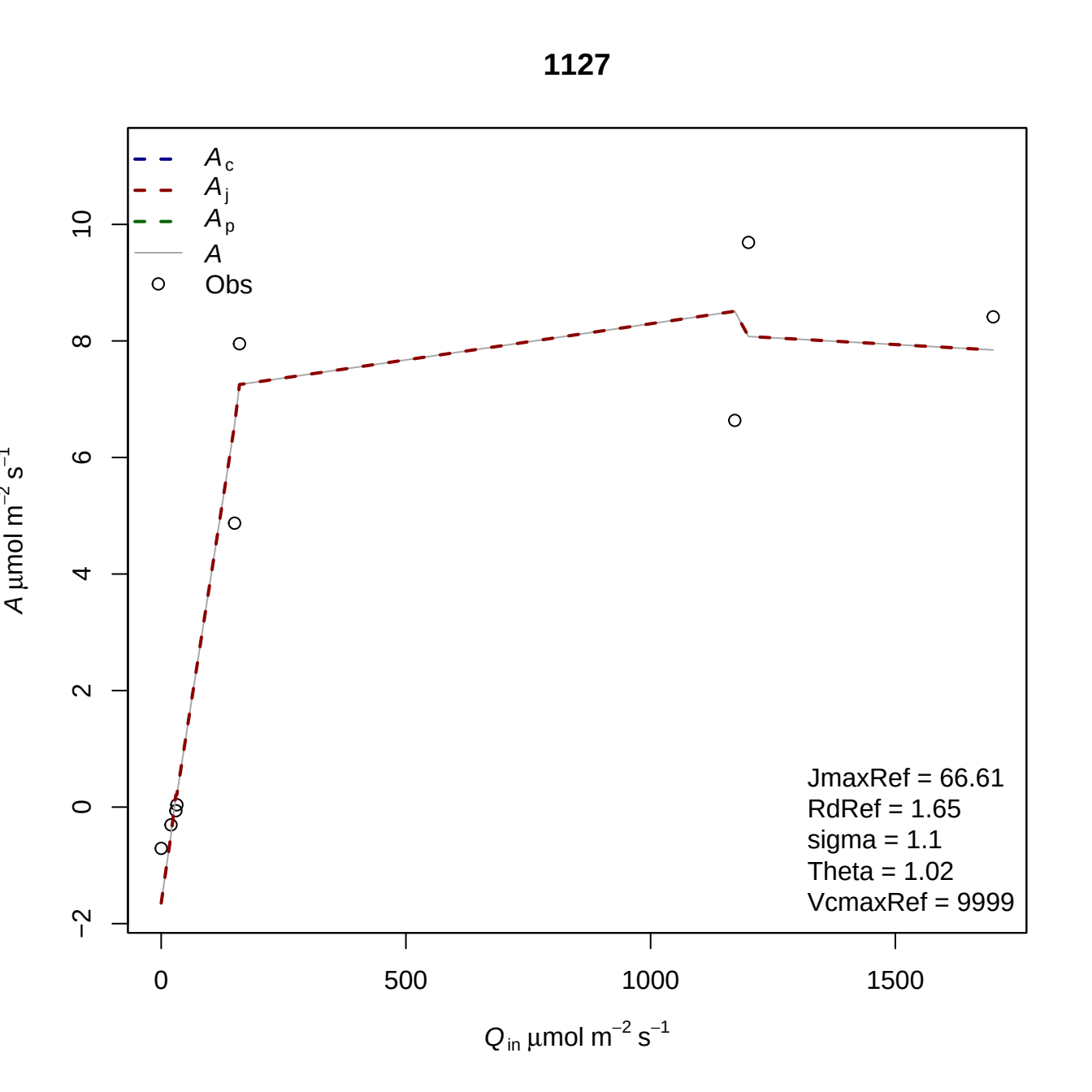

1118
A_c
A_j
A_p
A
Obs
JmaxRef = 103.28
RdRef = 0.89
sigma = 1.27
Theta = −4.5
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

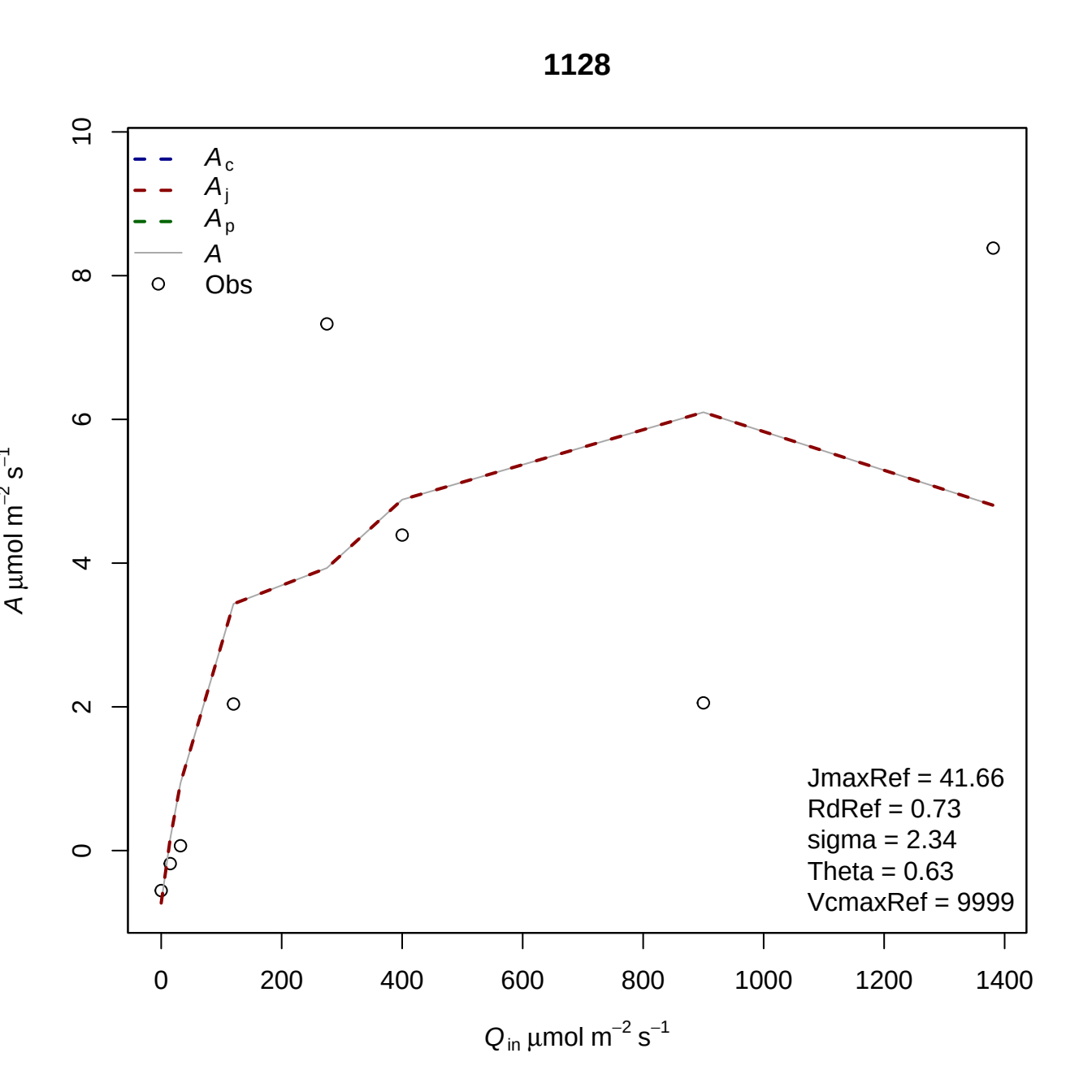

1120
A_c
A_j
A_p
A
Obs
JmaxRef = 219994540.11
RdRef = 0.47
sigma = 0.61
Theta = −19959372.08
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

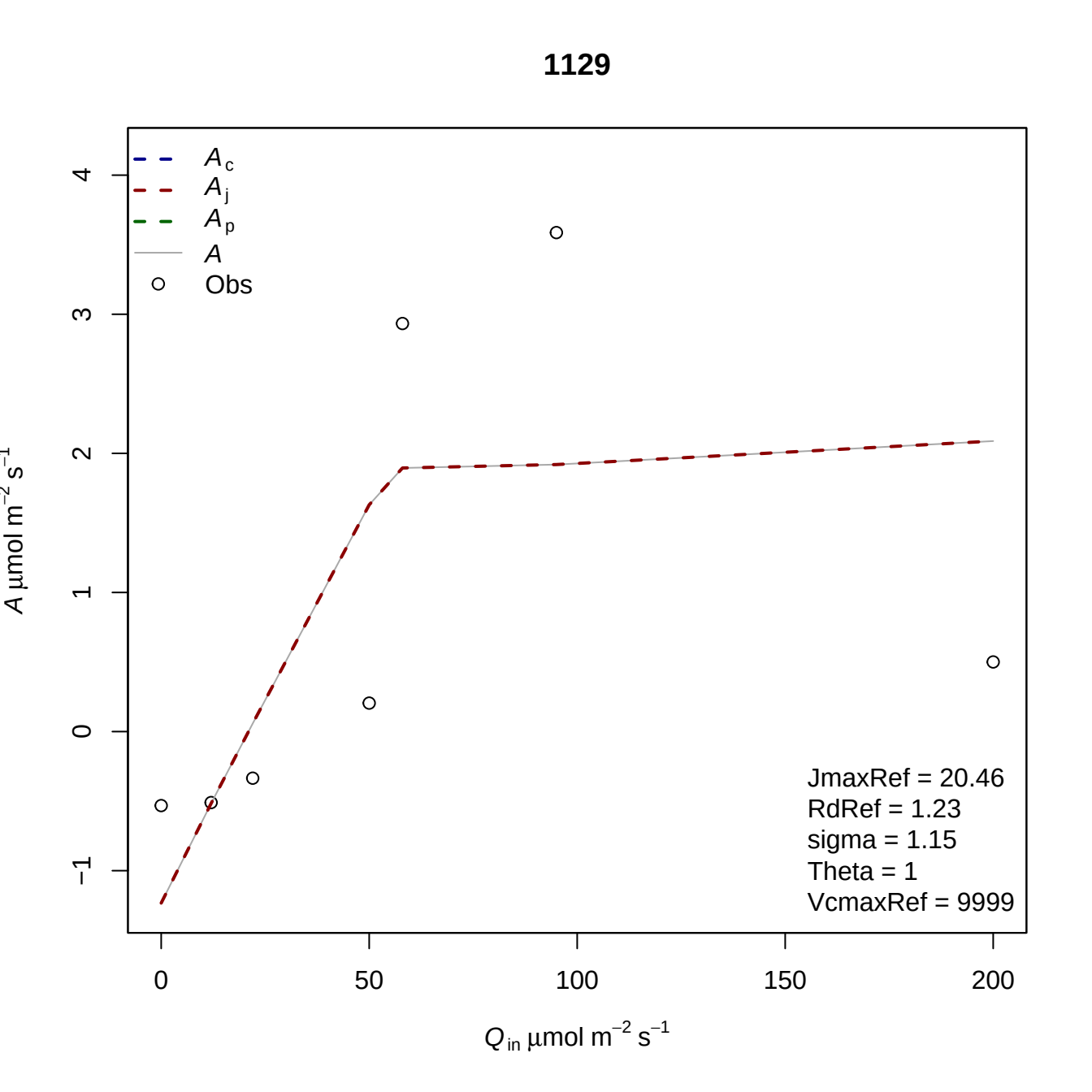

1122
A_c
A_j
A_p
A
Obs
JmaxRef = 441624293.91
RdRef = 0.44
sigma = 1.03
Theta = −43281789.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

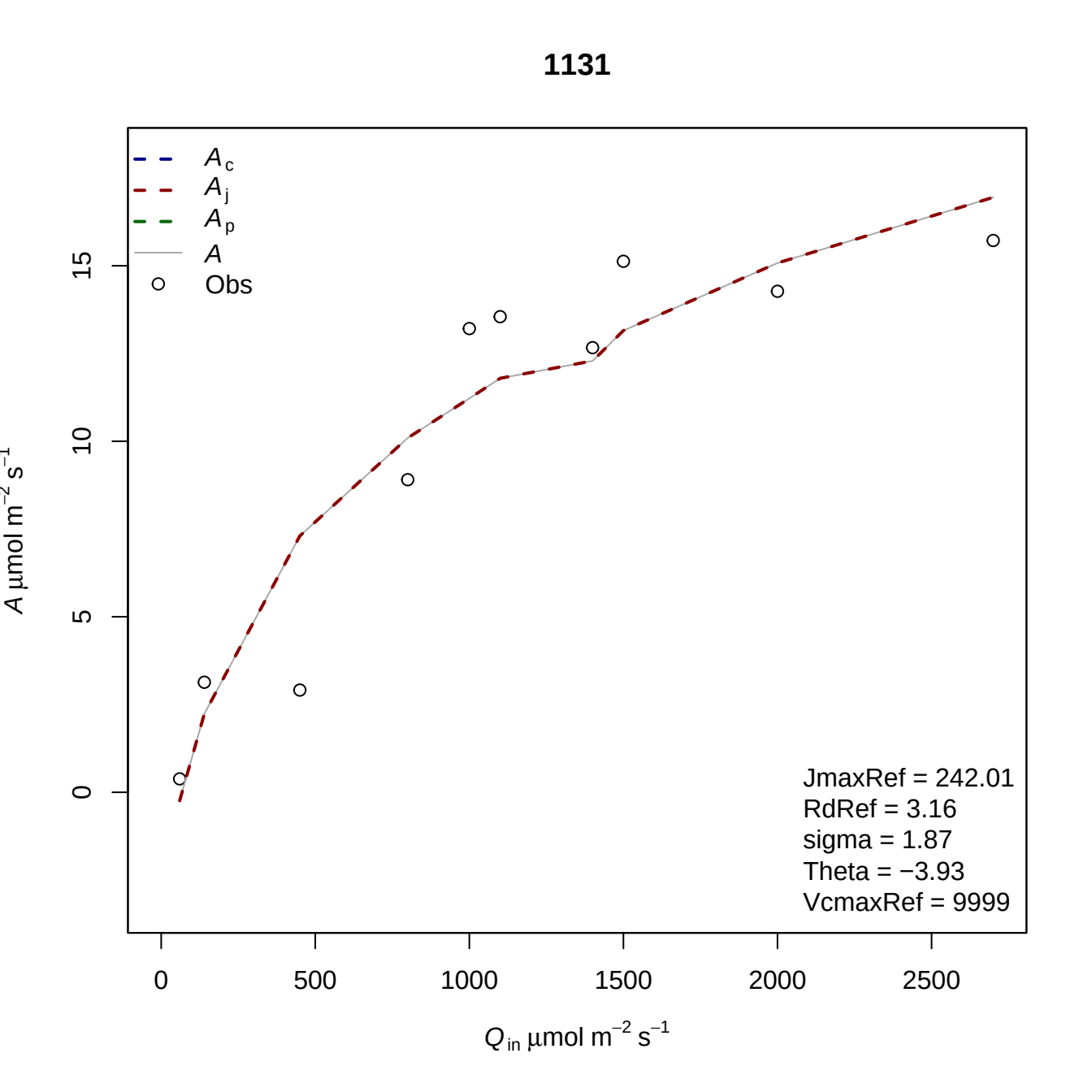

1125
A_c
A_j
A_p
A
Obs
JmaxRef = 39.63
RdRef = 2.18
sigma = 0.77
Theta = 0.92
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
100
200
300
400
500
600

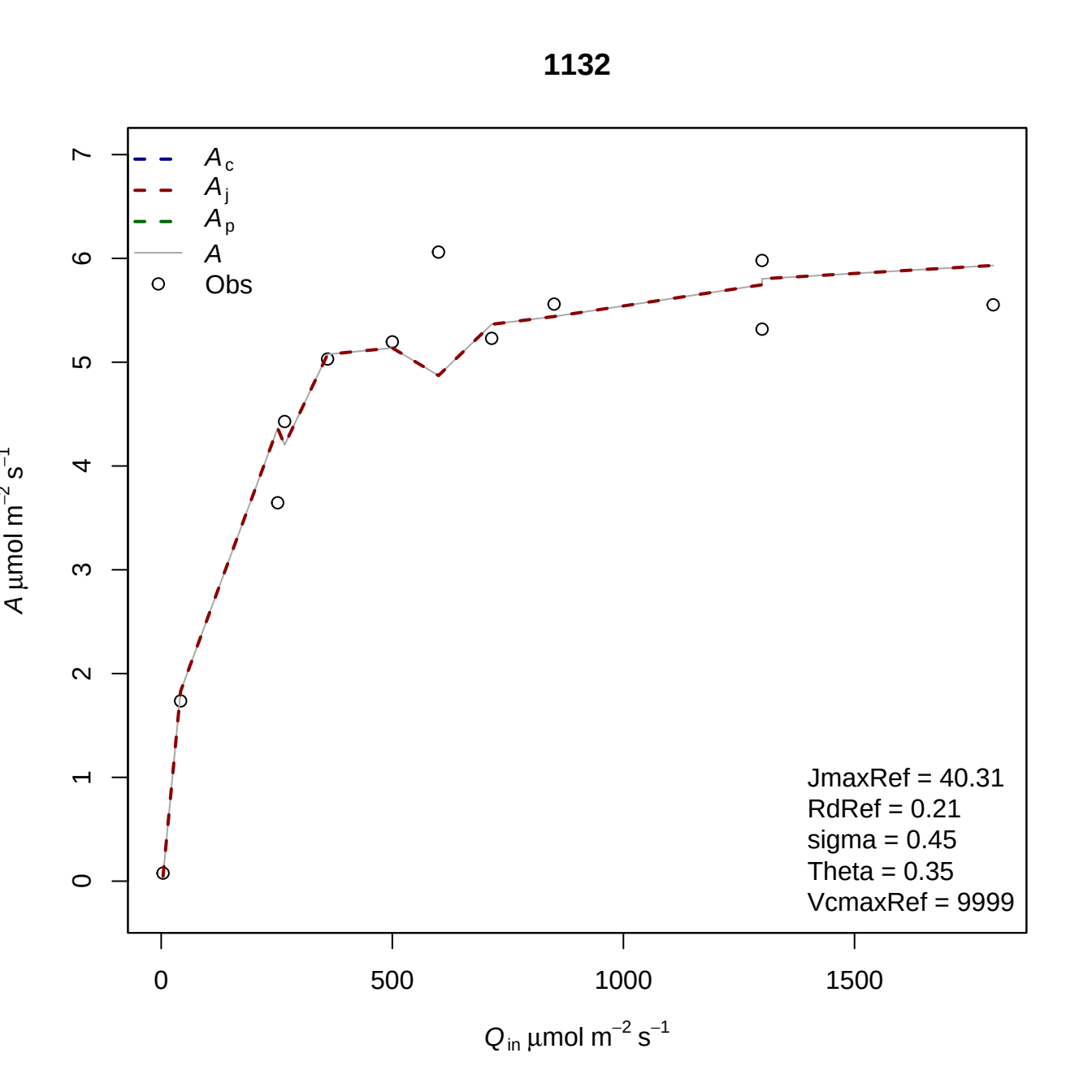

1126
A_c
A_j
A_p
A
Obs
JmaxRef = 180.67
RdRef = 2.48
sigma = 0.92
Theta = −1.1
VcmaxRef = 9999
0
5
10
15
0
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

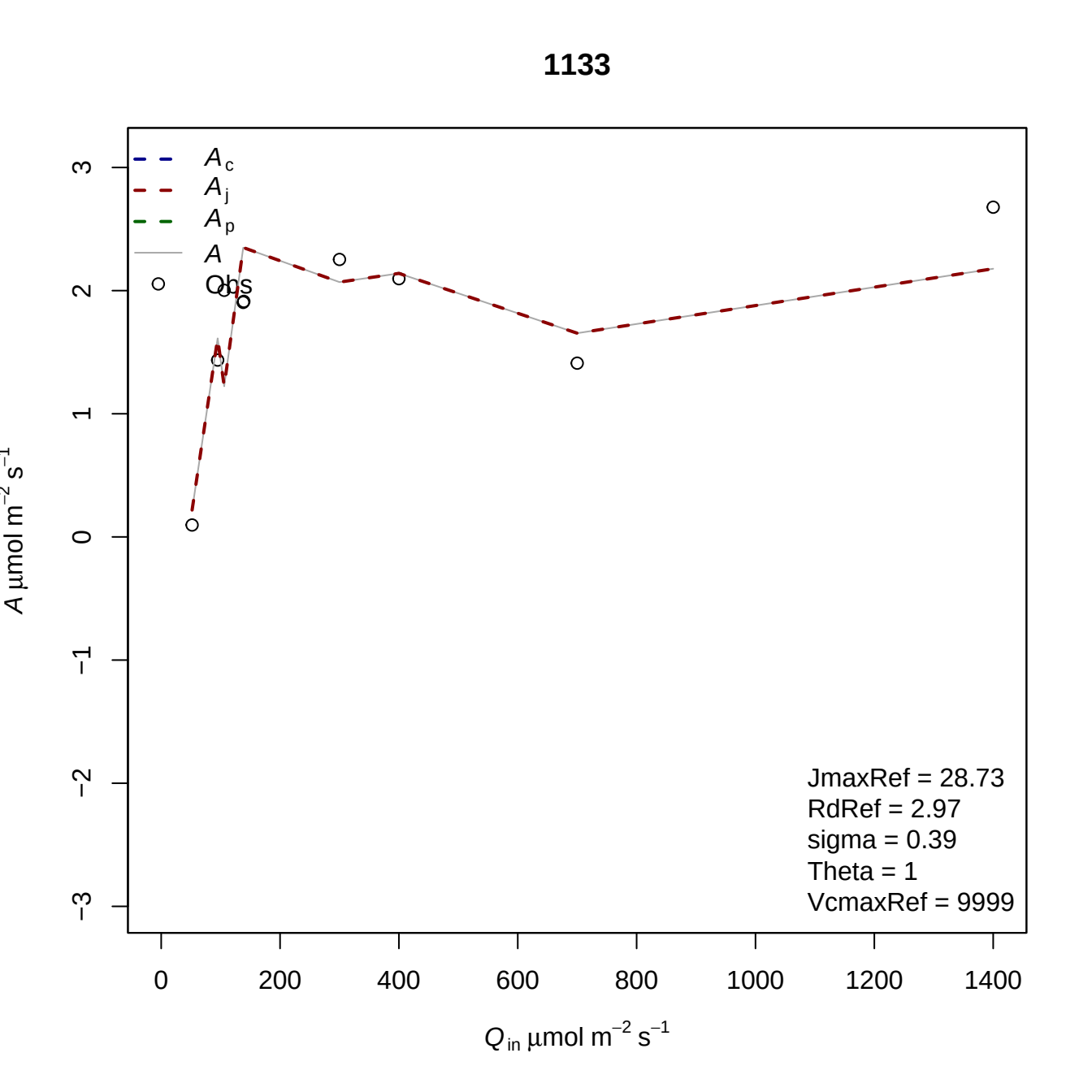

1127
A_c
A_j
A_p
A
Obs
JmaxRef = 66.61
RdRef = 1.65
sigma = 1.1
Theta = 1.02
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

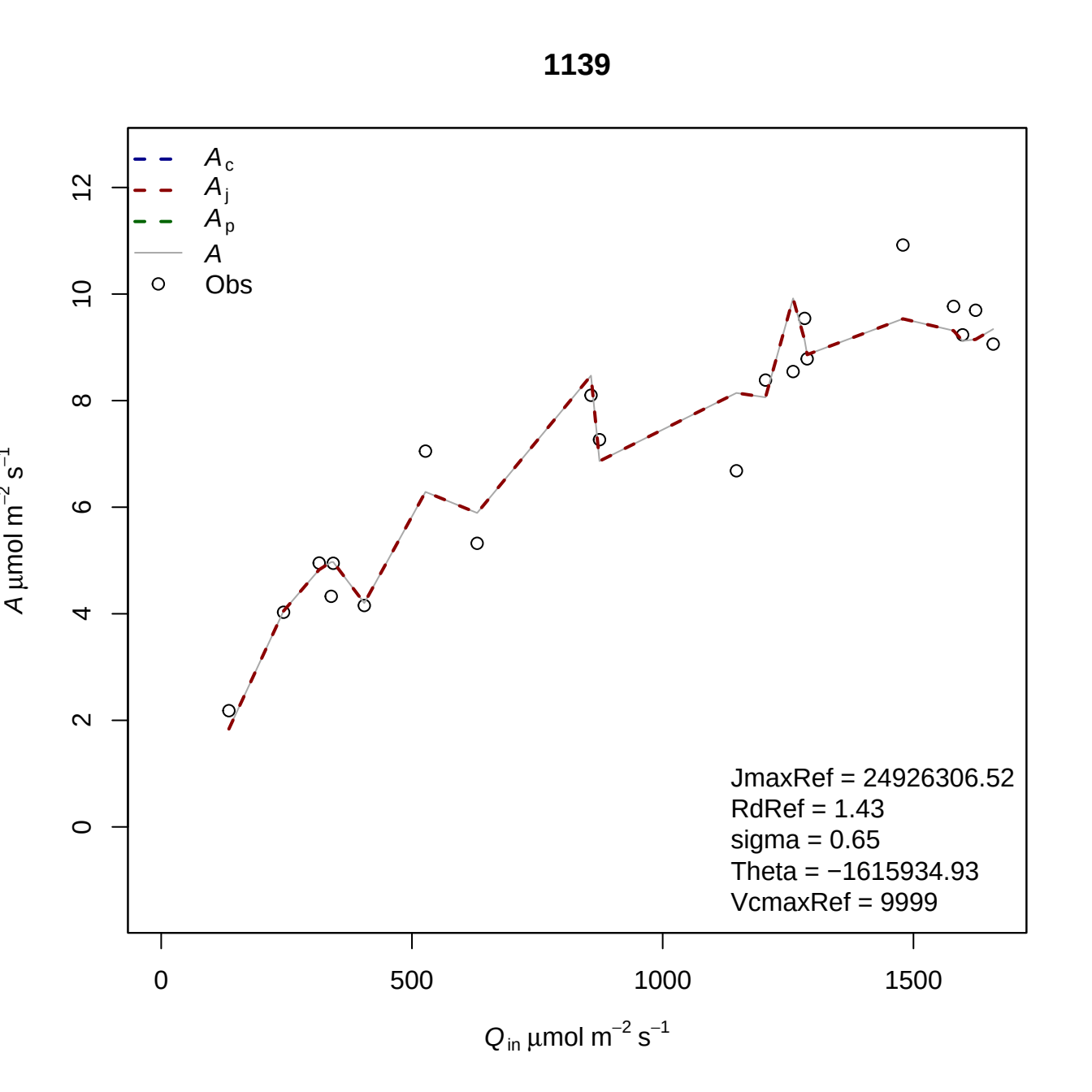

1128
A_c
A_j
A_p
A
Obs
JmaxRef = 41.66
RdRef = 0.73
sigma = 2.34
Theta = 0.63
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

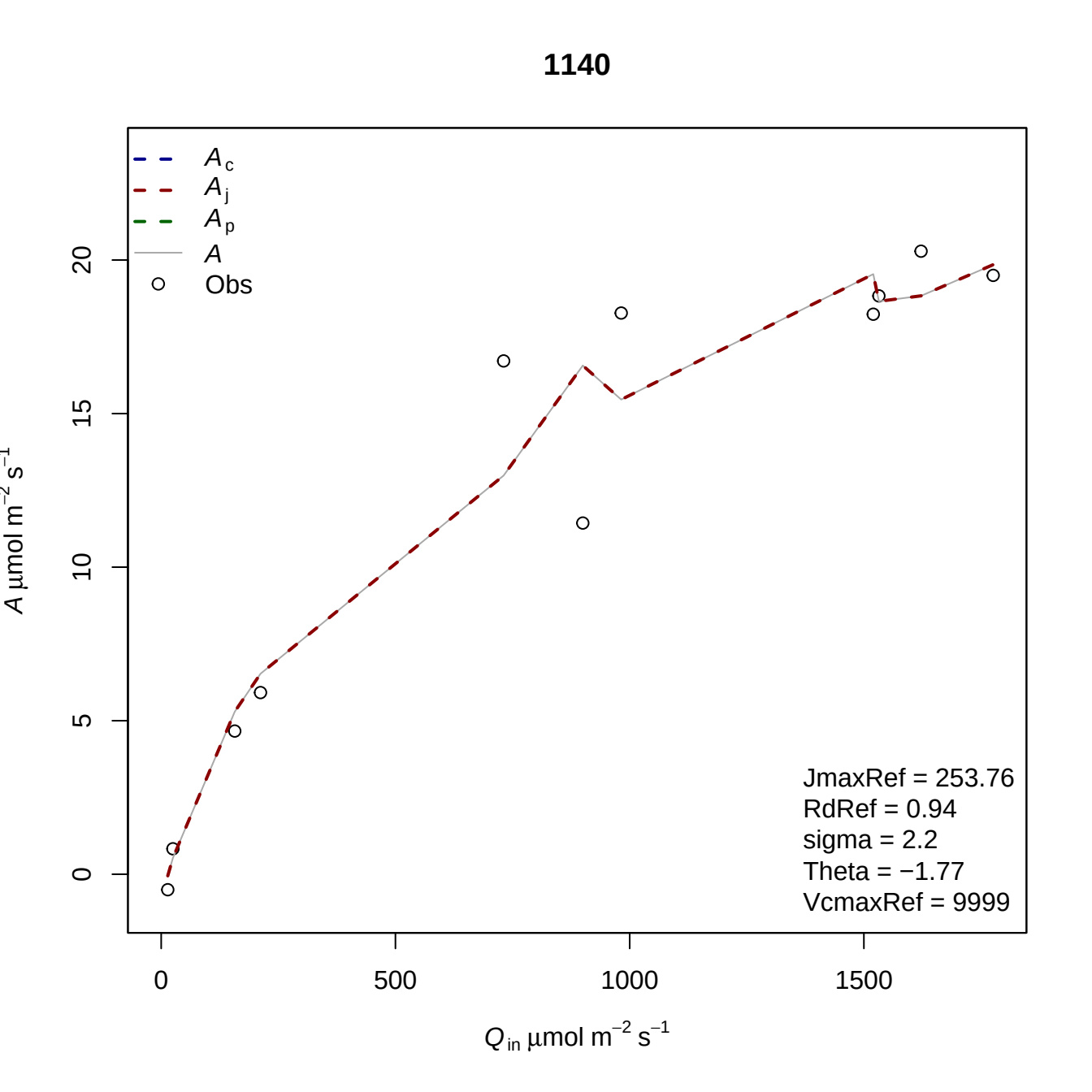

1129
A_c
A_j
A_p
A
Obs
JmaxRef = 20.46
RdRef = 1.23
sigma = 1.15
Theta = 1
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
50
100
150
200

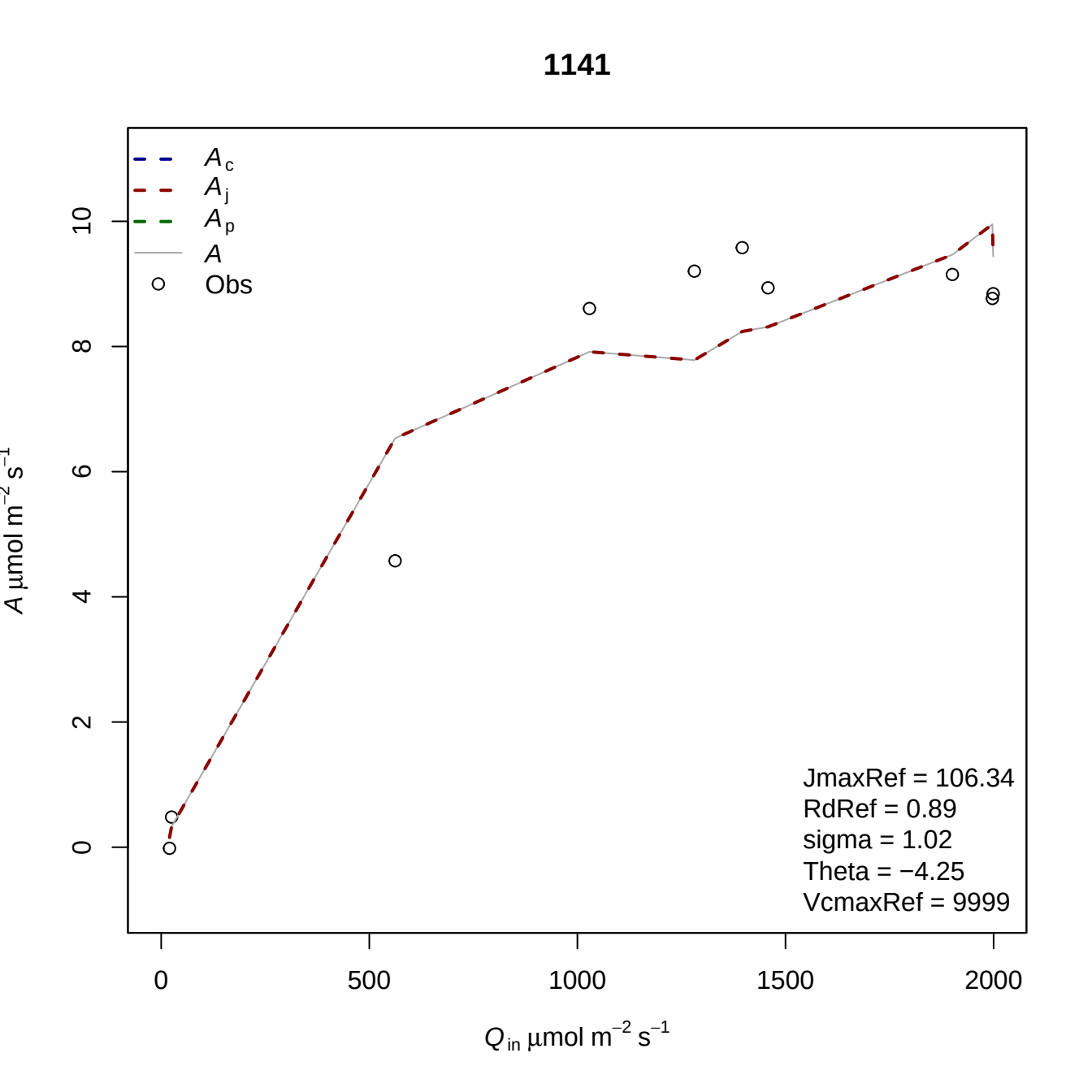

1131
A_c
A_j
A_p
A
Obs
JmaxRef = 242.01
RdRef = 3.16
sigma = 1.87
Theta = −3.93
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000
2500

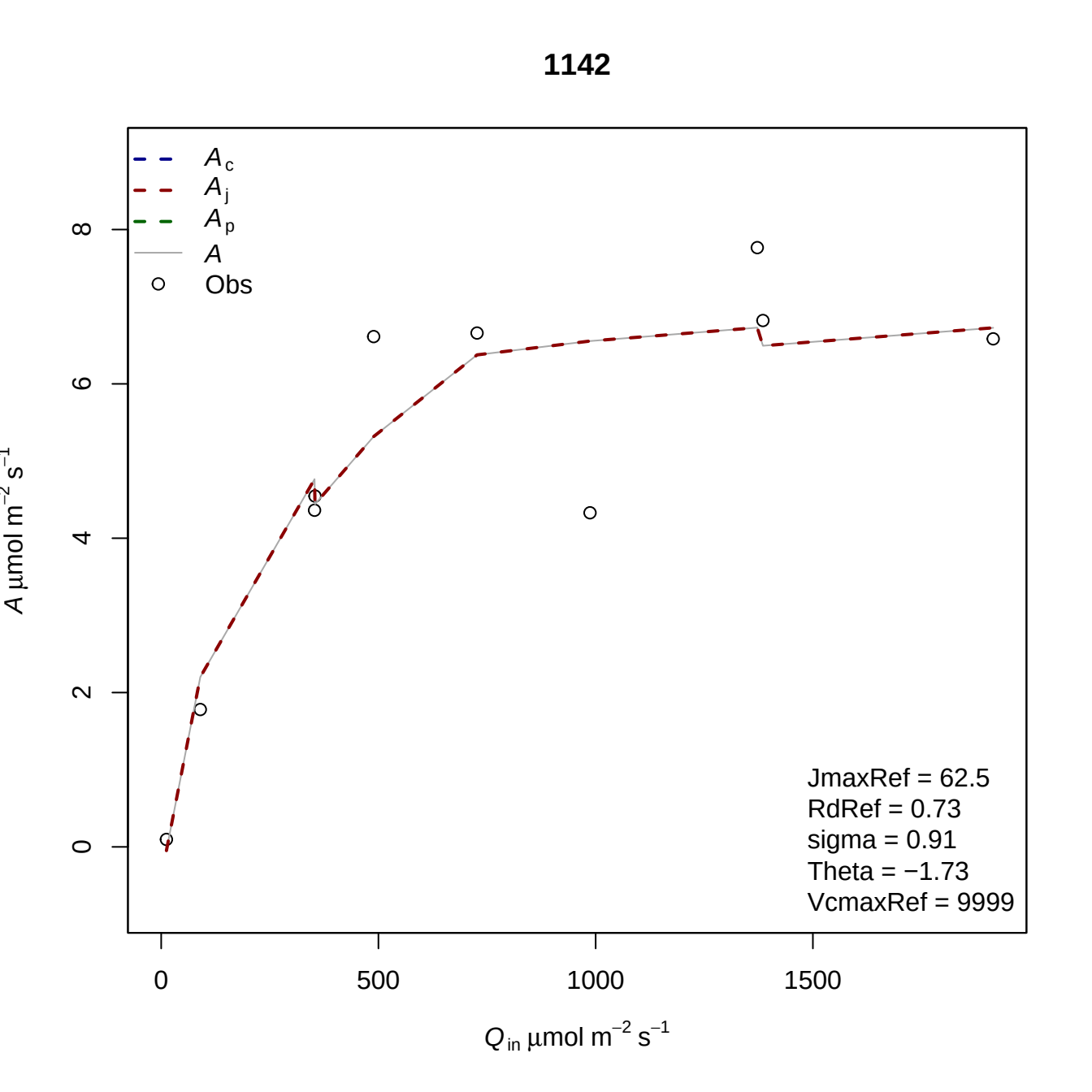

1132
A_c
A_j
A_p
A
Obs
JmaxRef = 40.31
RdRef = 0.21
sigma = 0.45
Theta = 0.35
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
7
0
500
1000
1500

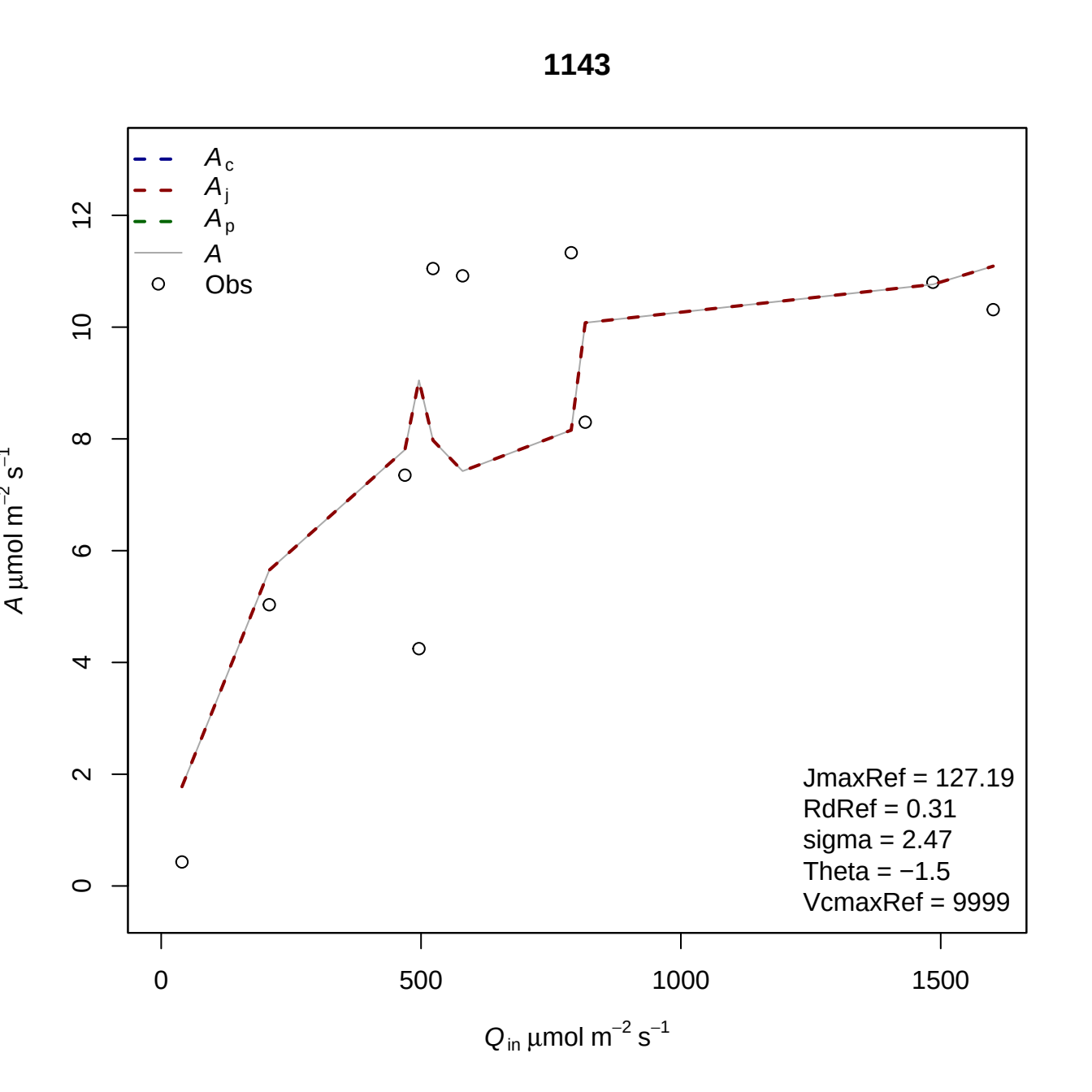

1133
A_c
A_j
A_p
A
Obs
JmaxRef = 28.73
RdRef = 2.97
sigma = 0.39
Theta = 1
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
−3
0
200
400
600
800
1000
1200
1400

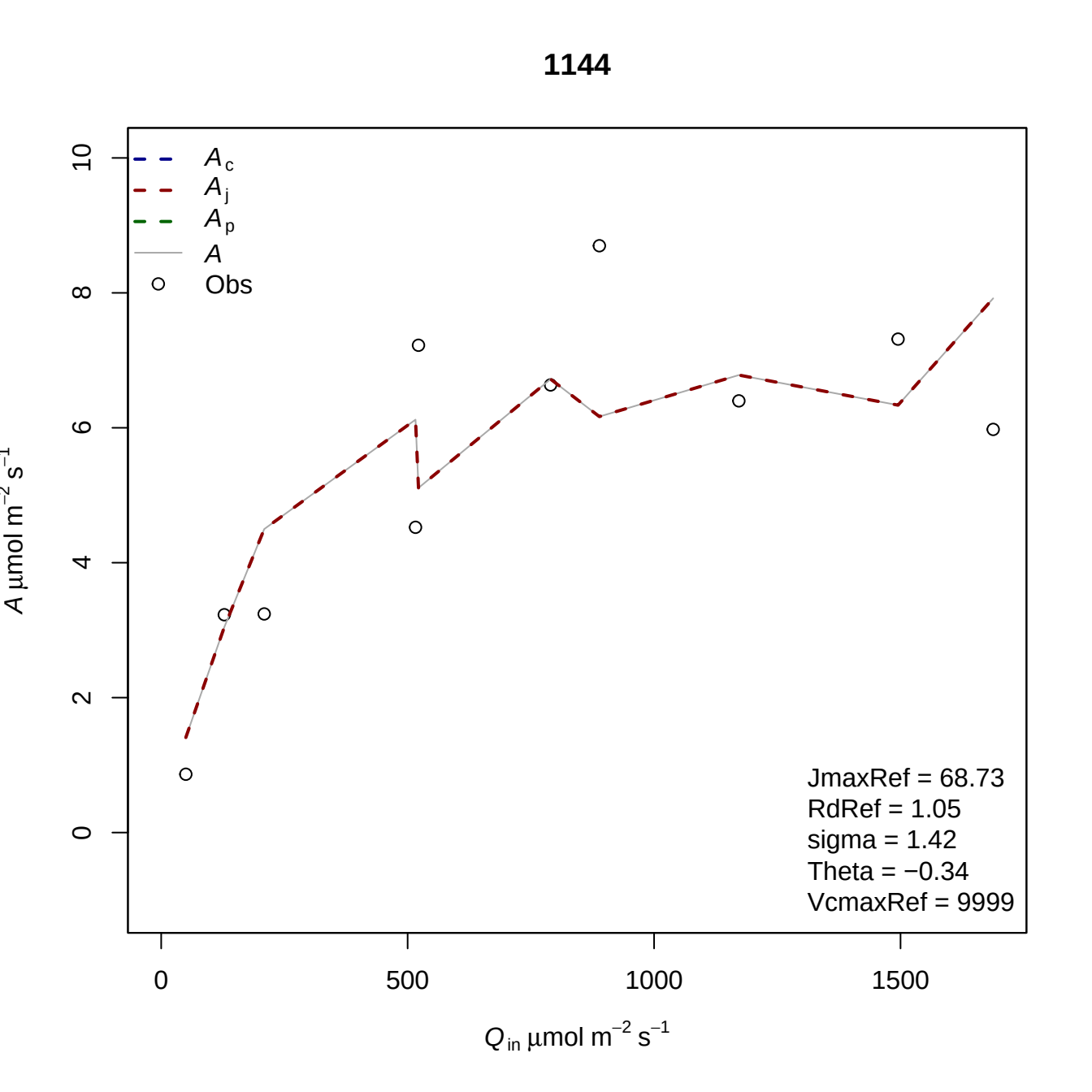

1139
A_c
A_j
A_p
A
Obs
JmaxRef = 24926306.52
RdRef = 1.43
sigma = 0.65
Theta = −1615934.93
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

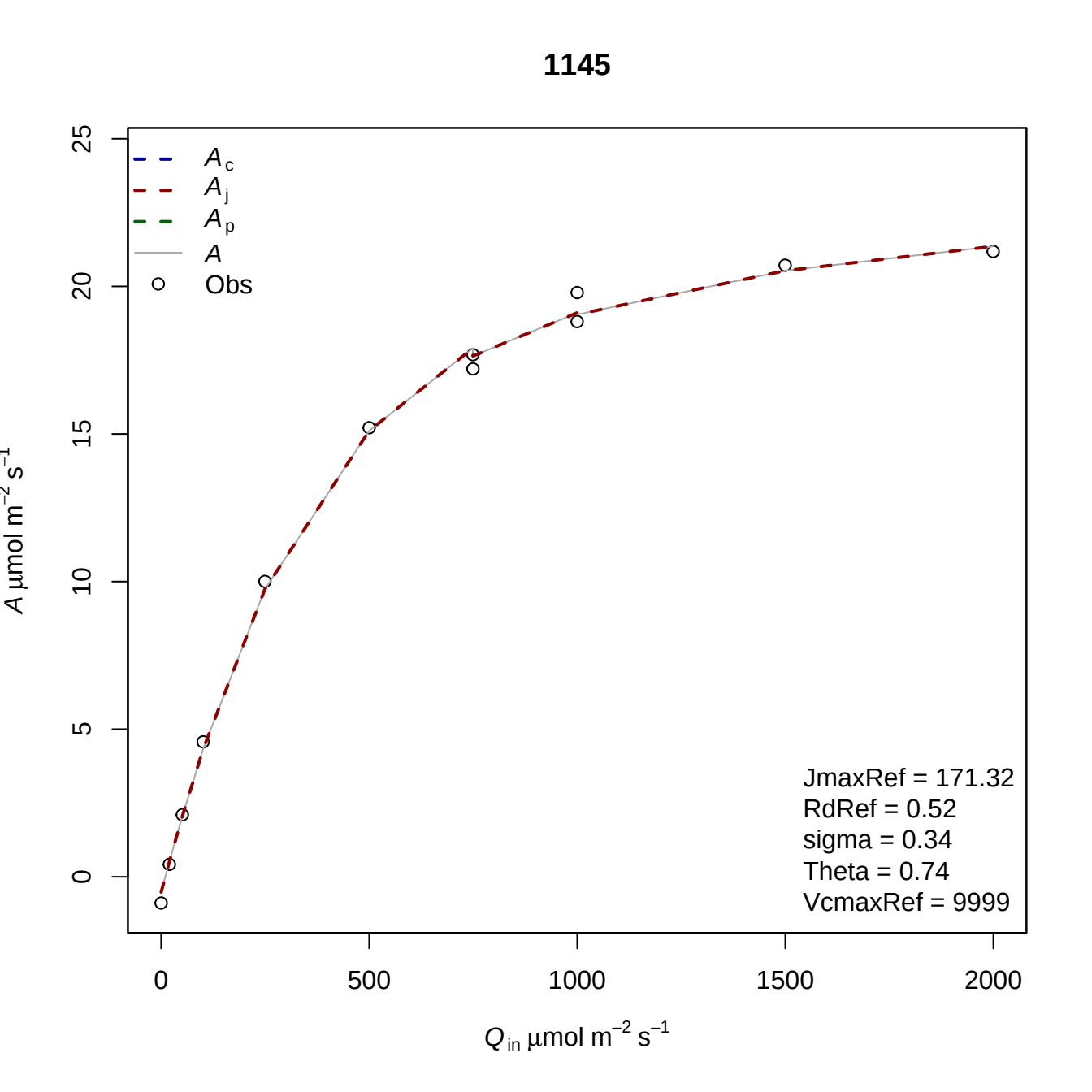

1140
A_c
A_j
A_p
A
Obs
JmaxRef = 253.76
RdRef = 0.94
sigma = 2.2
Theta = −1.77
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

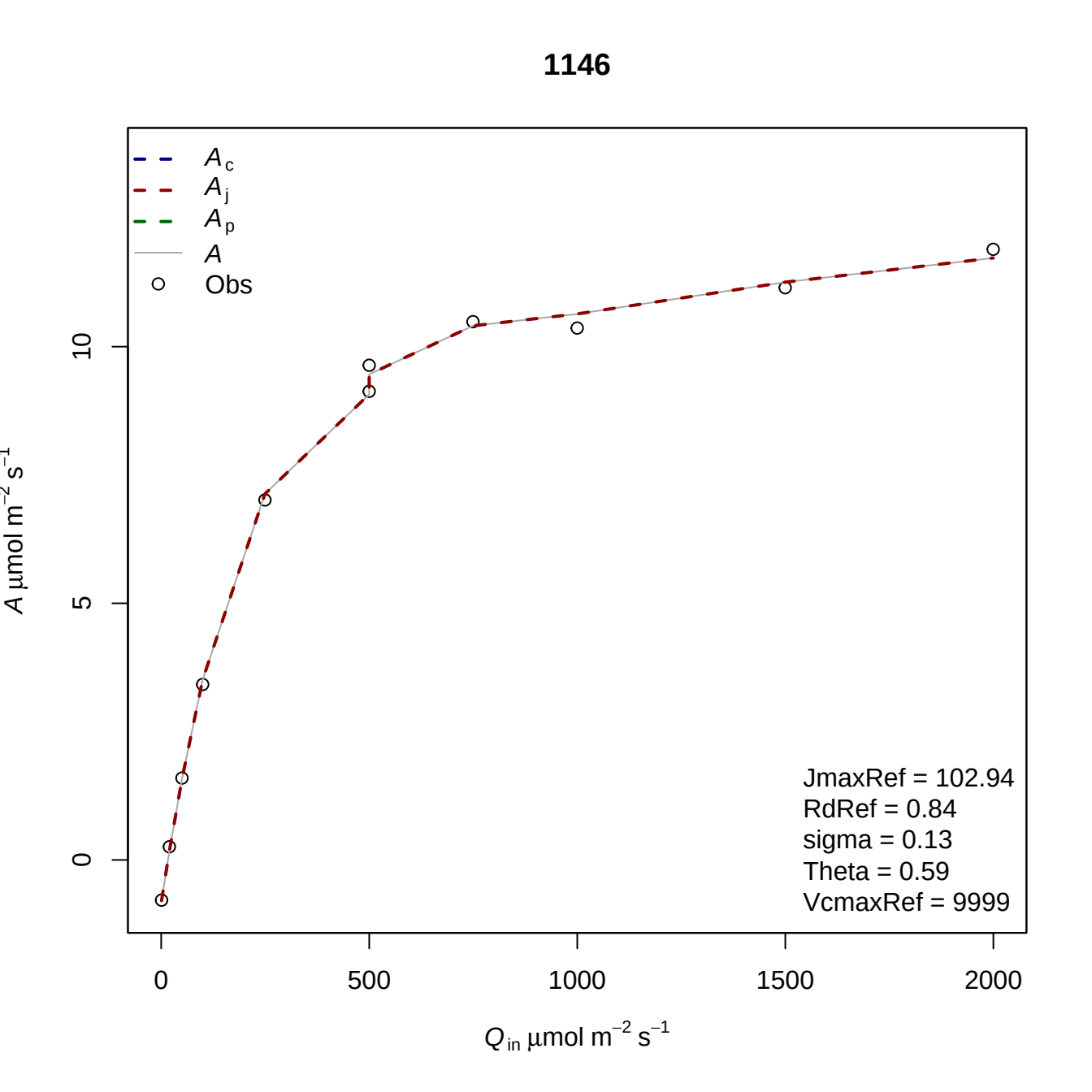

1141
A_c
A_j
A_p
A
Obs
JmaxRef = 106.34
RdRef = 0.89
sigma = 1.02
Theta = −4.25
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

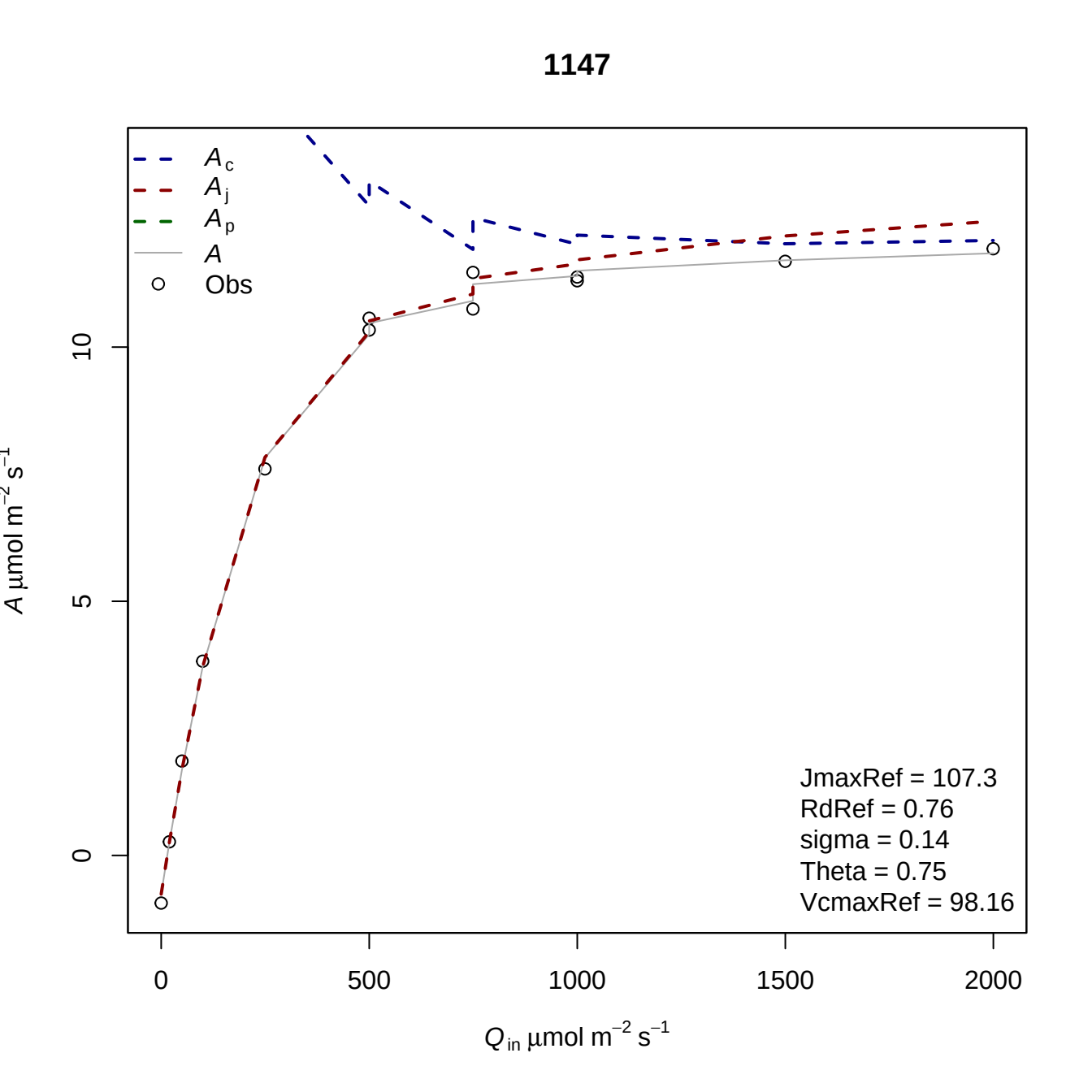

1142
A_c
A_j
A_p
A
Obs
JmaxRef = 62.5
RdRef = 0.73
sigma = 0.91
Theta = −1.73
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

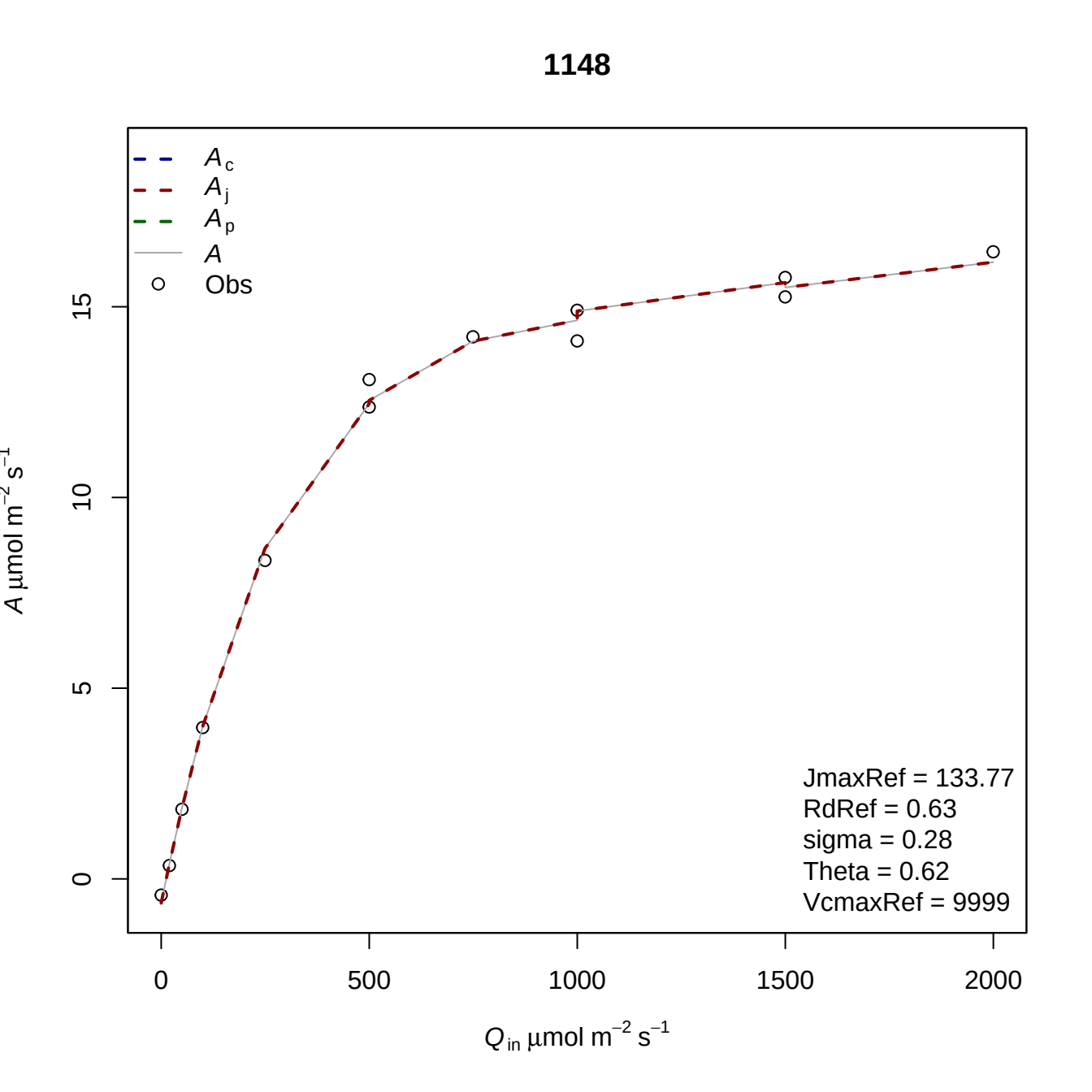

1143
A_c
A_j
A_p
A
Obs
JmaxRef = 127.19
RdRef = 0.31
sigma = 2.47
Theta = −1.5
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

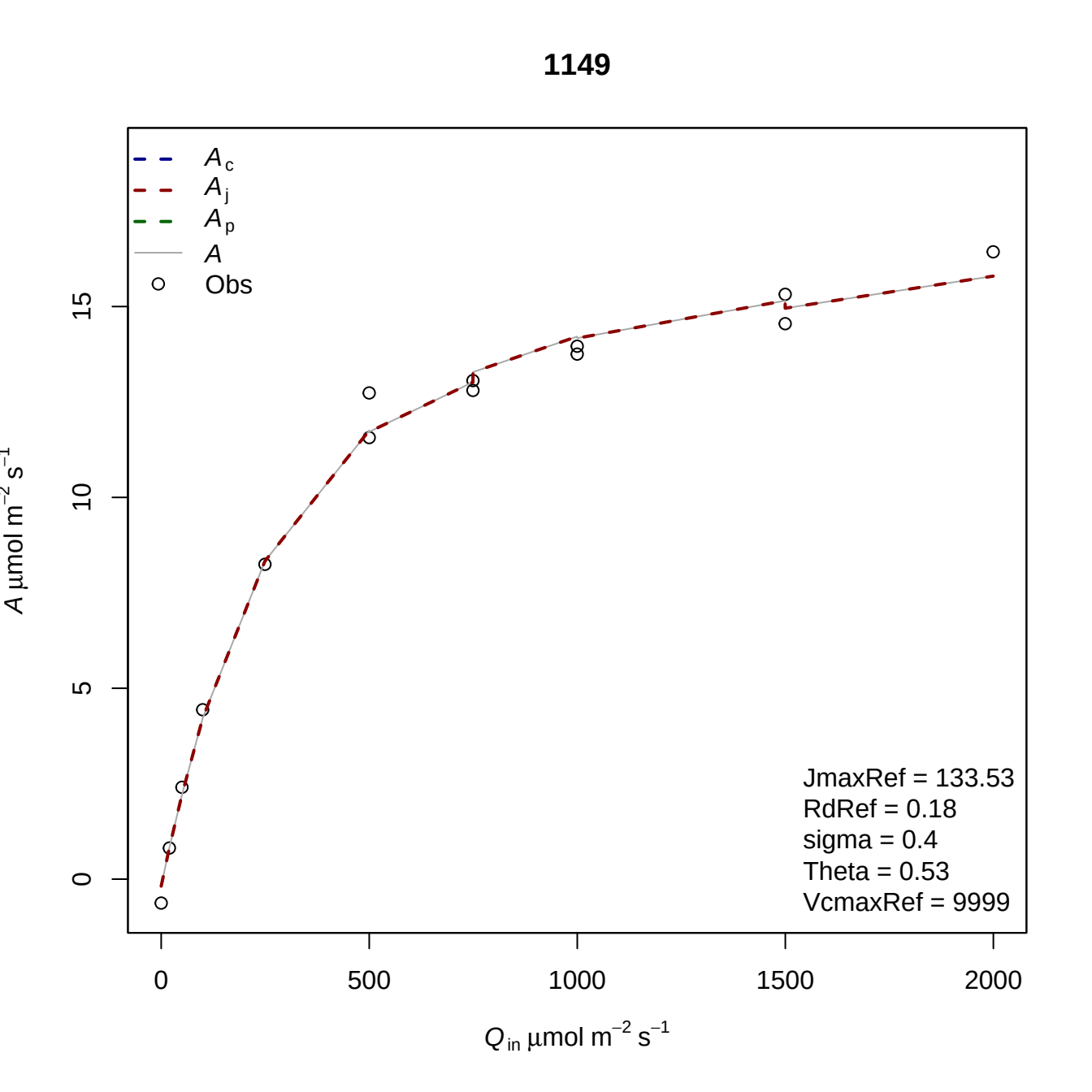

1144
A_c
A_j
A_p
A
Obs
JmaxRef = 68.73
RdRef = 1.05
sigma = 1.42
Theta = −0.34
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

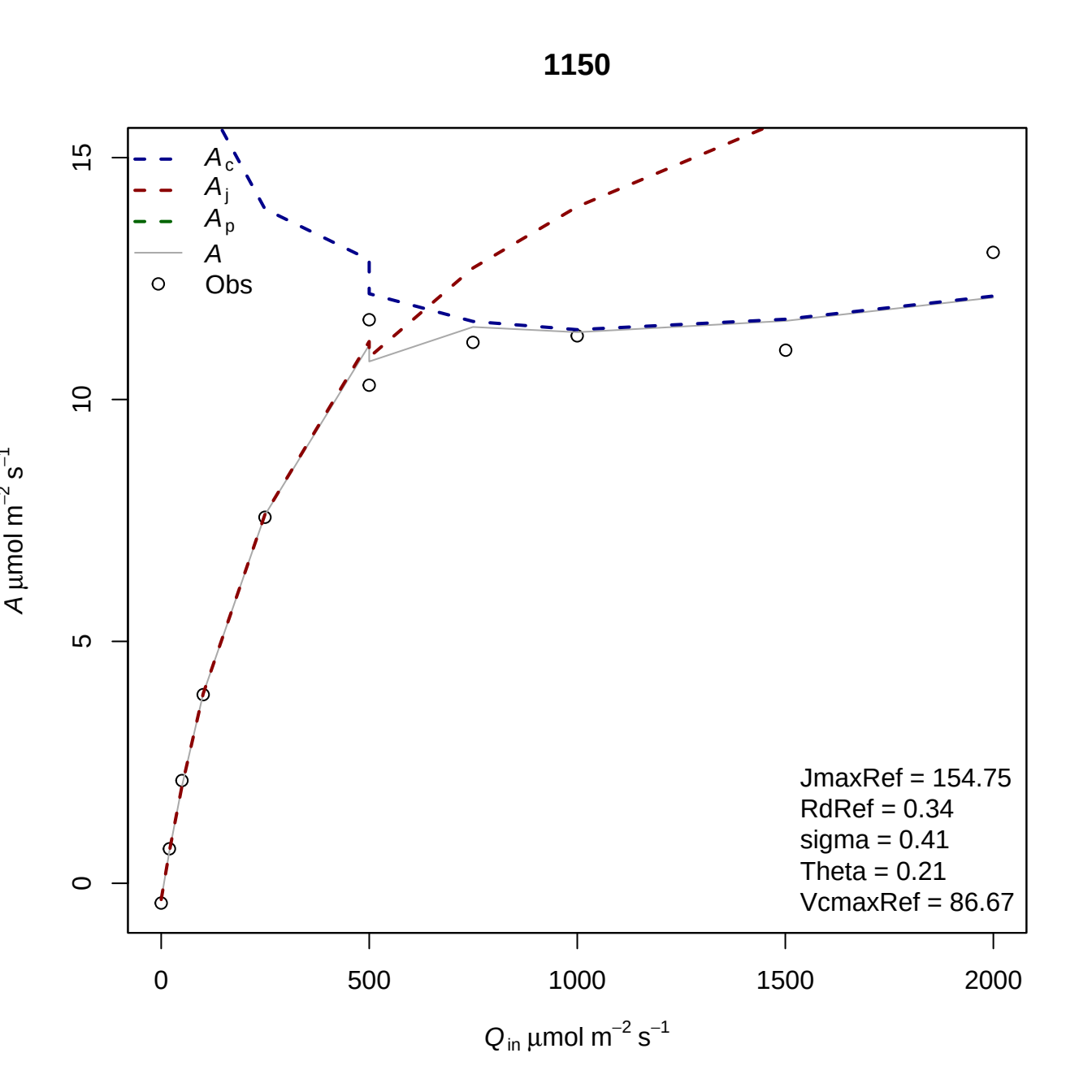

1145
A_c
A_j
A_p
A
Obs
JmaxRef = 171.32
RdRef = 0.52
sigma = 0.34
Theta = 0.74
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

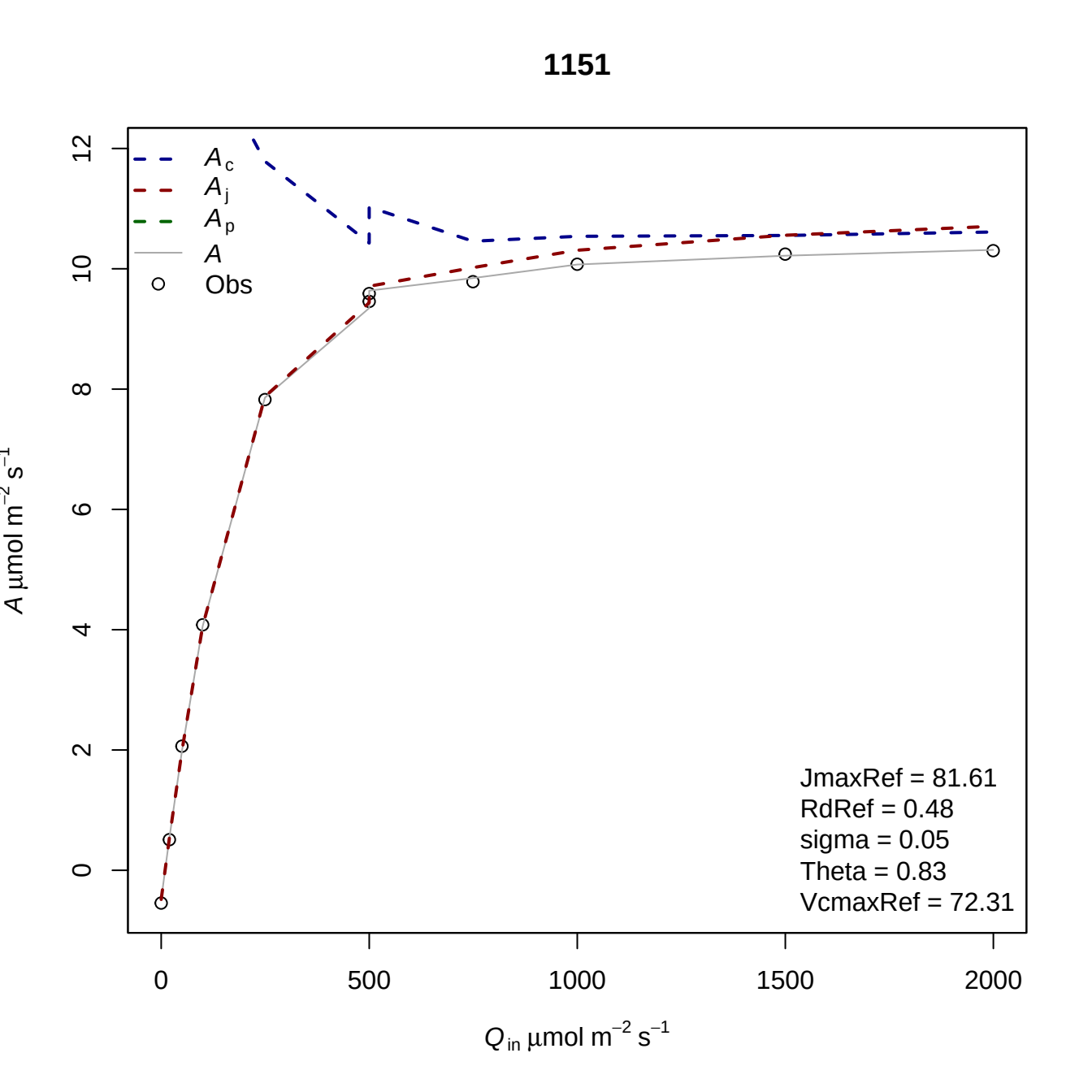

1146
A_c
A_j
A_p
A
Obs
JmaxRef = 102.94
RdRef = 0.84
sigma = 0.13
Theta = 0.59
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

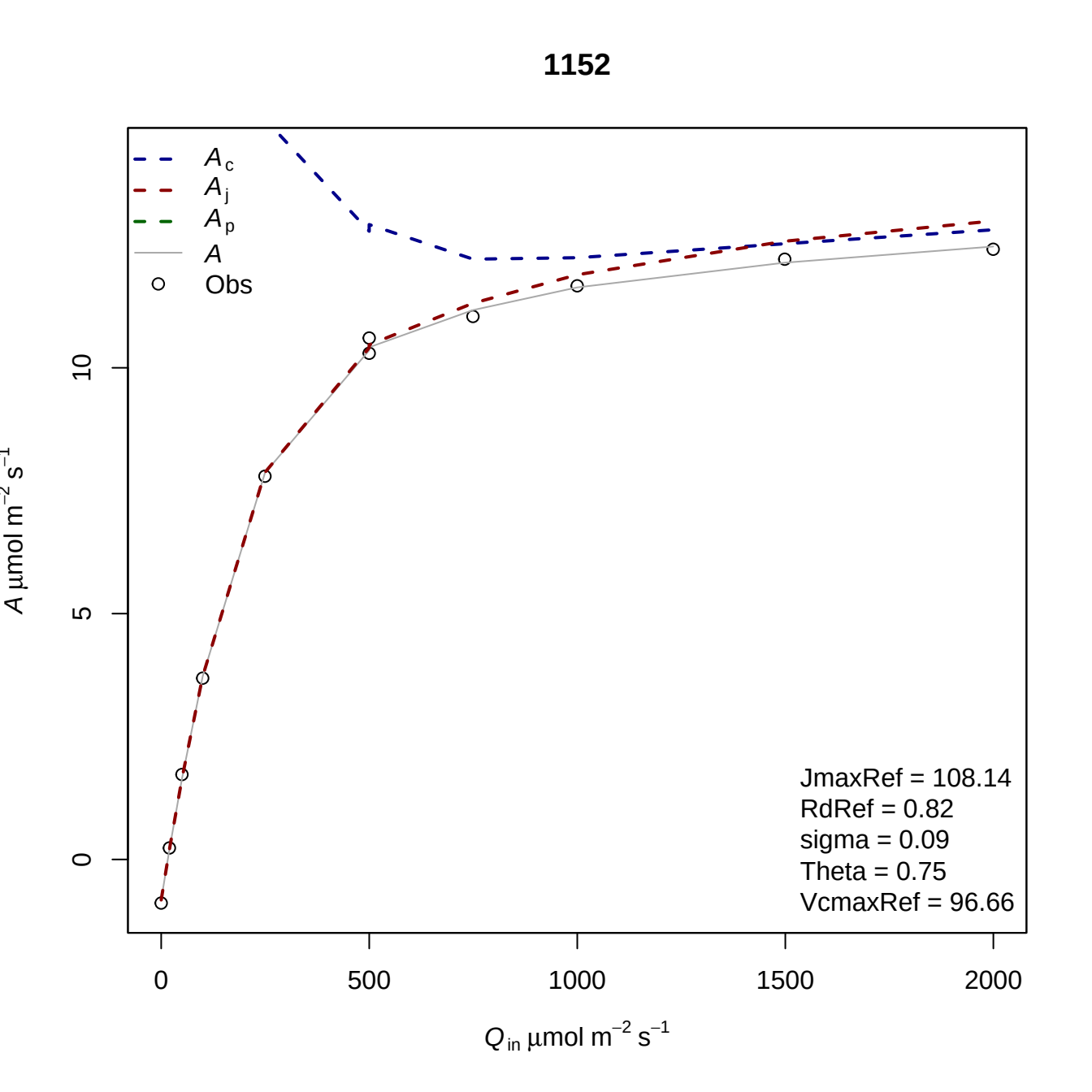

1147
A_c
A_j
A_p
A
Obs
JmaxRef = 107.3
RdRef = 0.76
sigma = 0.14
Theta = 0.75
VcmaxRef = 98.16
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

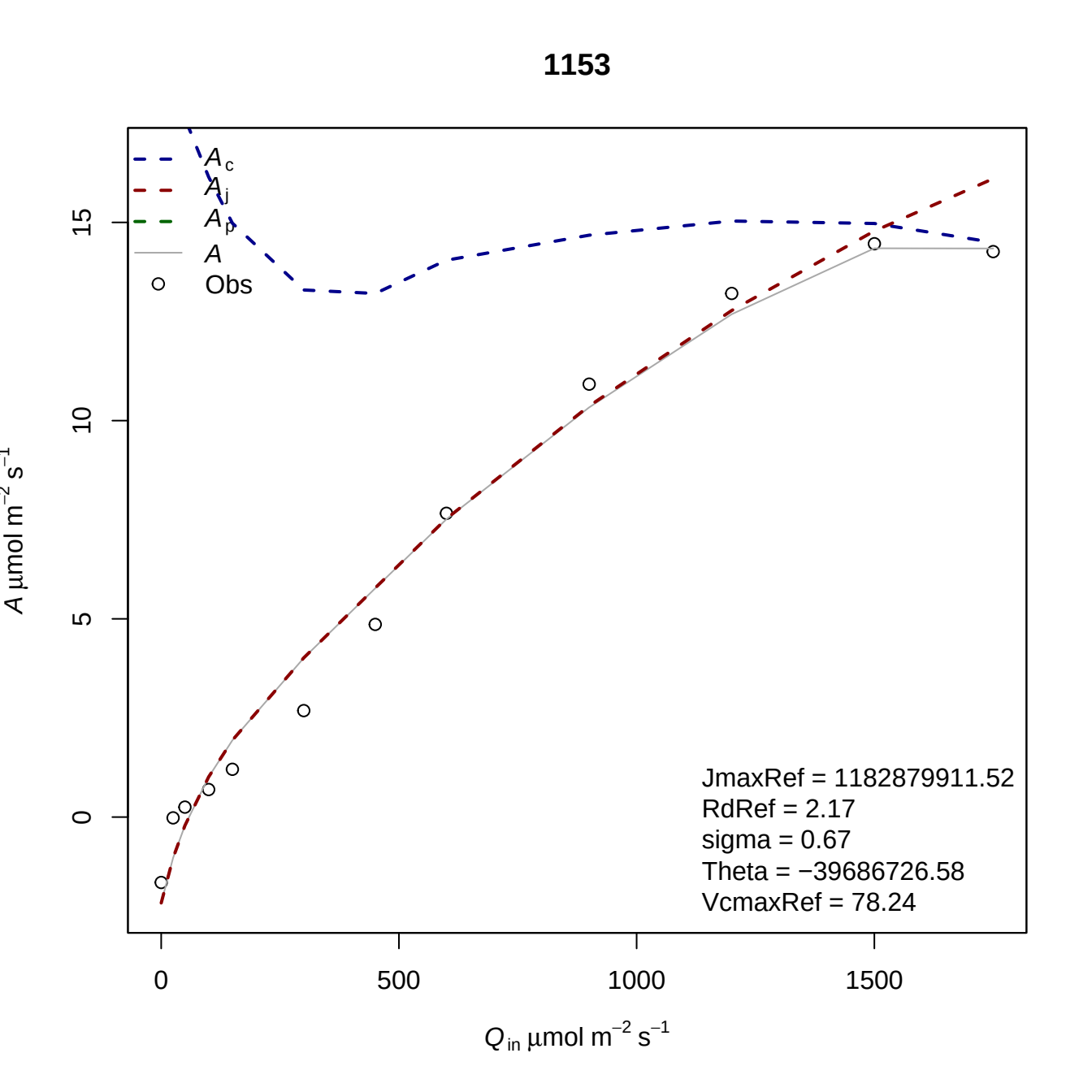

1148
A_c
A_j
A_p
A
Obs
JmaxRef = 133.77
RdRef = 0.63
sigma = 0.28
Theta = 0.62
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

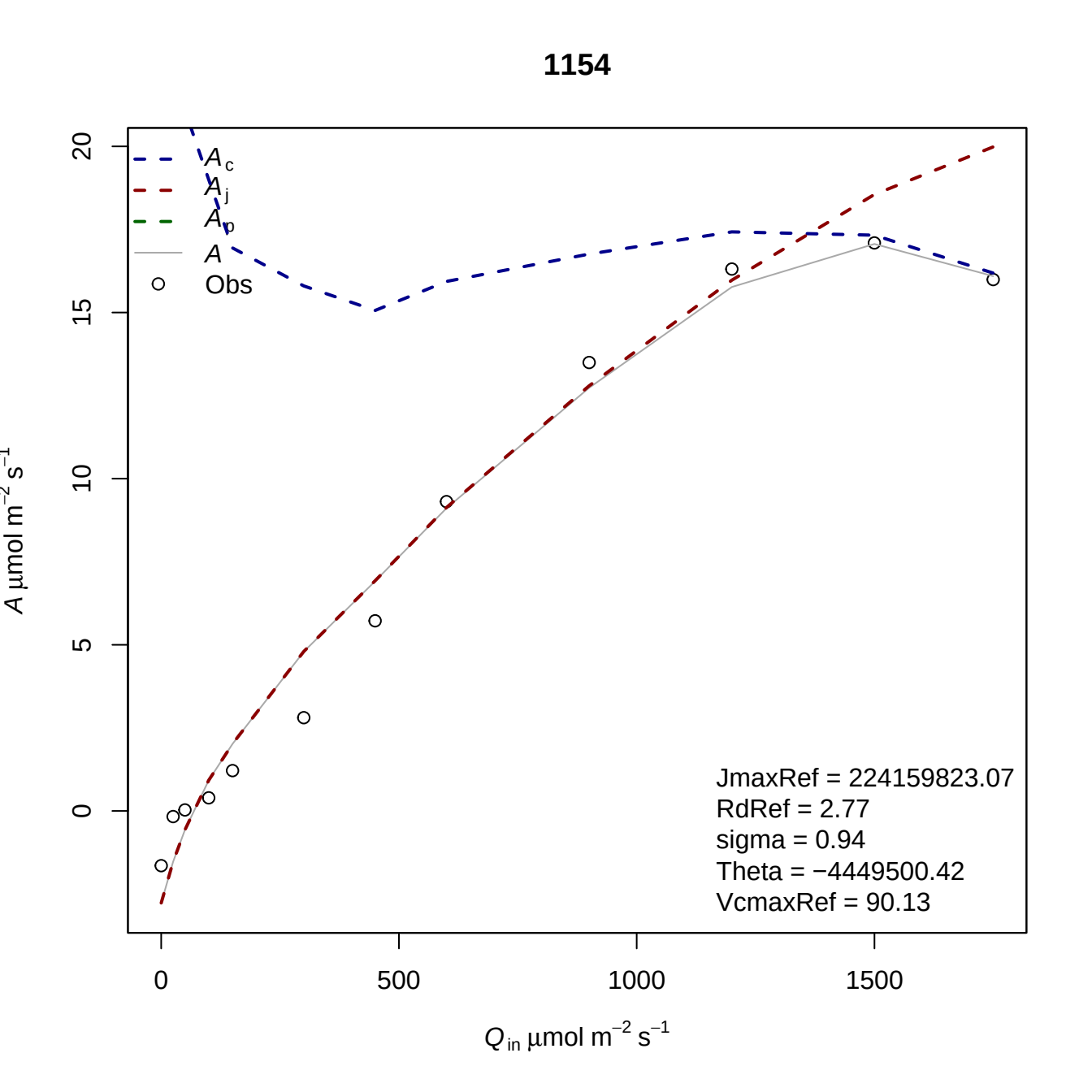

1149
A_c
A_j
A_p
A
Obs
JmaxRef = 133.53
RdRef = 0.18
sigma = 0.4
Theta = 0.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

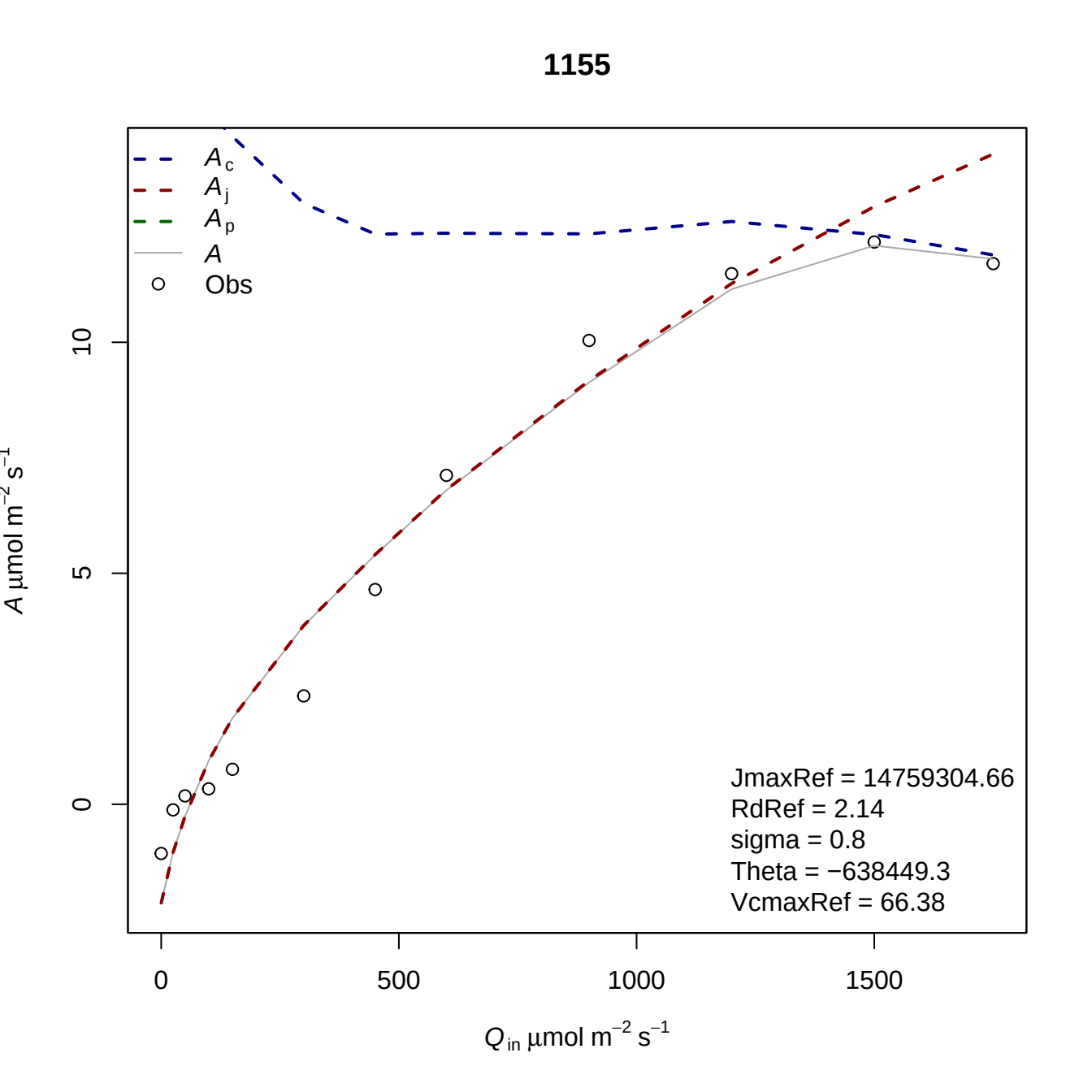

1150
A_c
A_j
A_p
A
Obs
JmaxRef = 154.75
RdRef = 0.34
sigma = 0.41
Theta = 0.21
VcmaxRef = 86.67
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

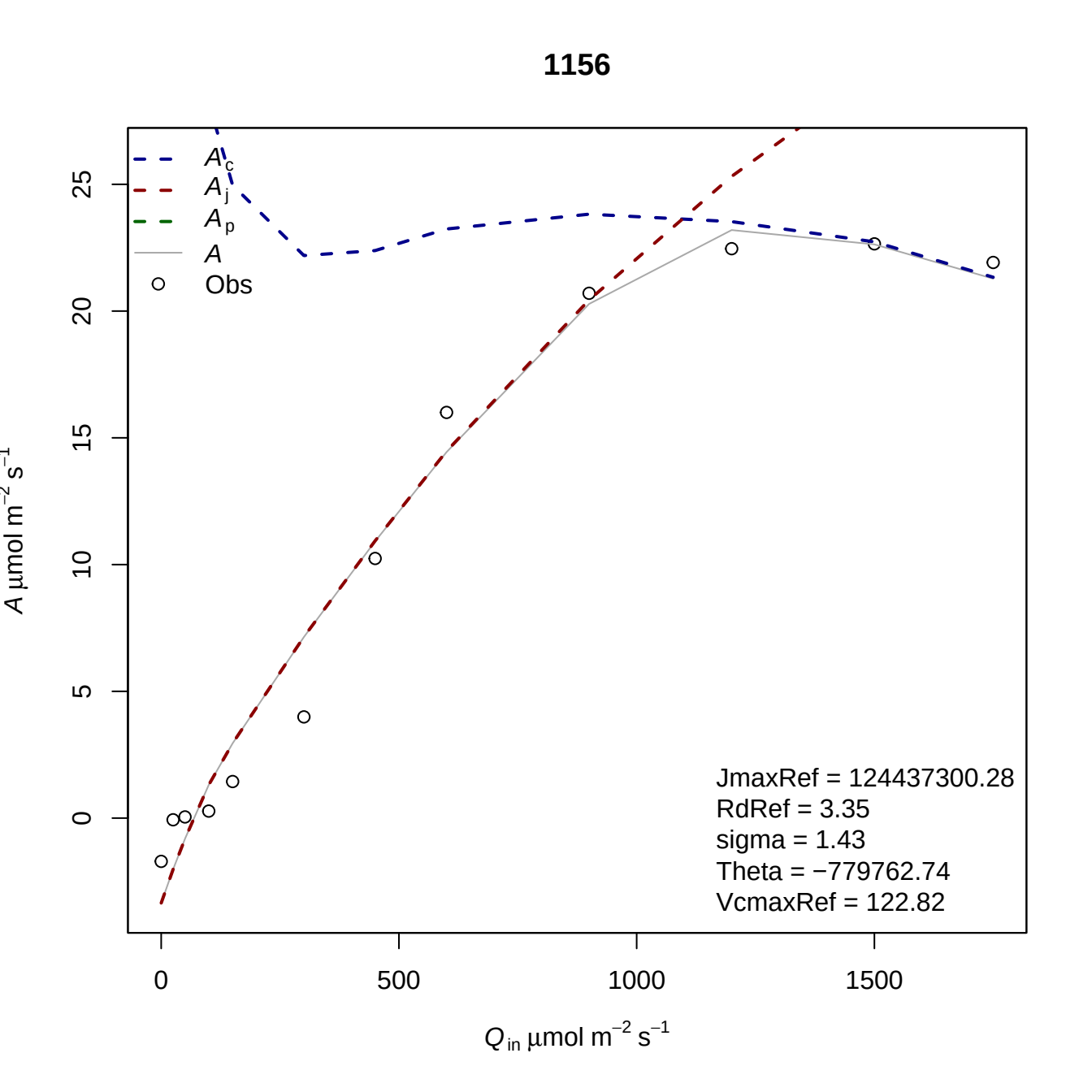

1151
A_c
A_j
A_p
A
Obs
JmaxRef = 81.61
RdRef = 0.48
sigma = 0.05
Theta = 0.83
VcmaxRef = 72.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

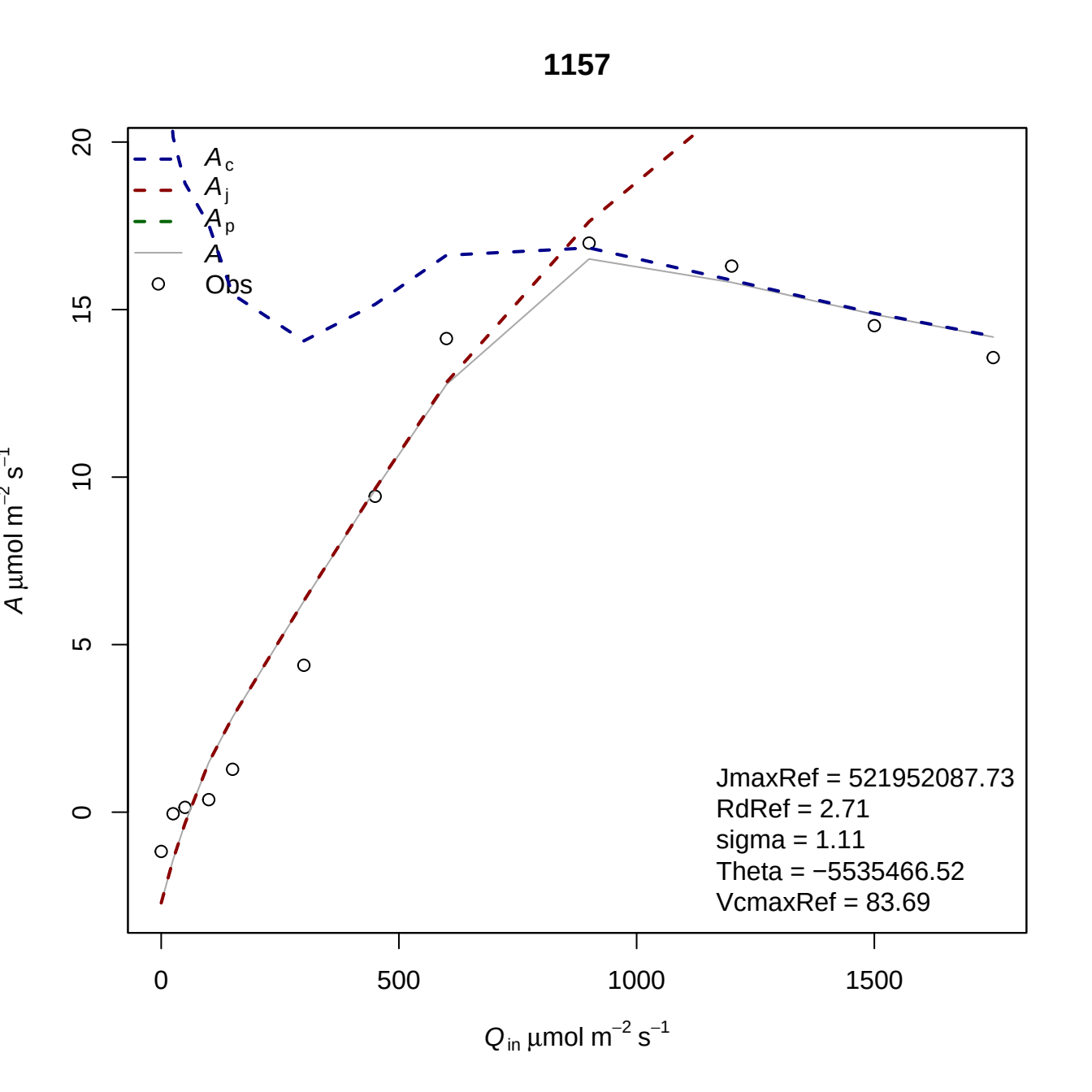

1152
A_c
A_j
A_p
A
Obs
JmaxRef = 108.14
RdRef = 0.82
sigma = 0.09
Theta = 0.75
VcmaxRef = 96.66
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

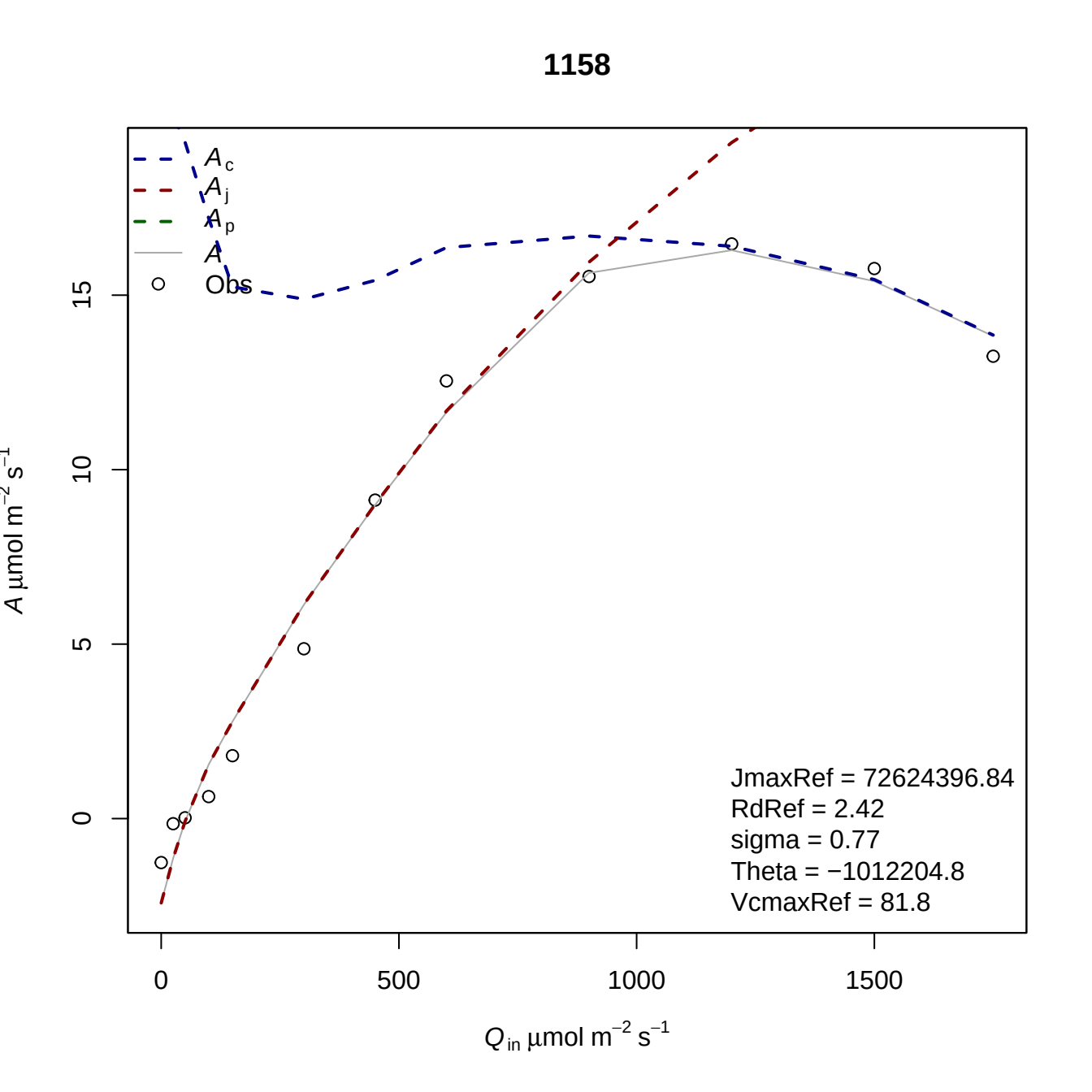

1153
A_c
A_j
A_p
A
Obs
JmaxRef = 1182879911.52
RdRef = 2.17
sigma = 0.67
Theta = −39686726.58
VcmaxRef = 78.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

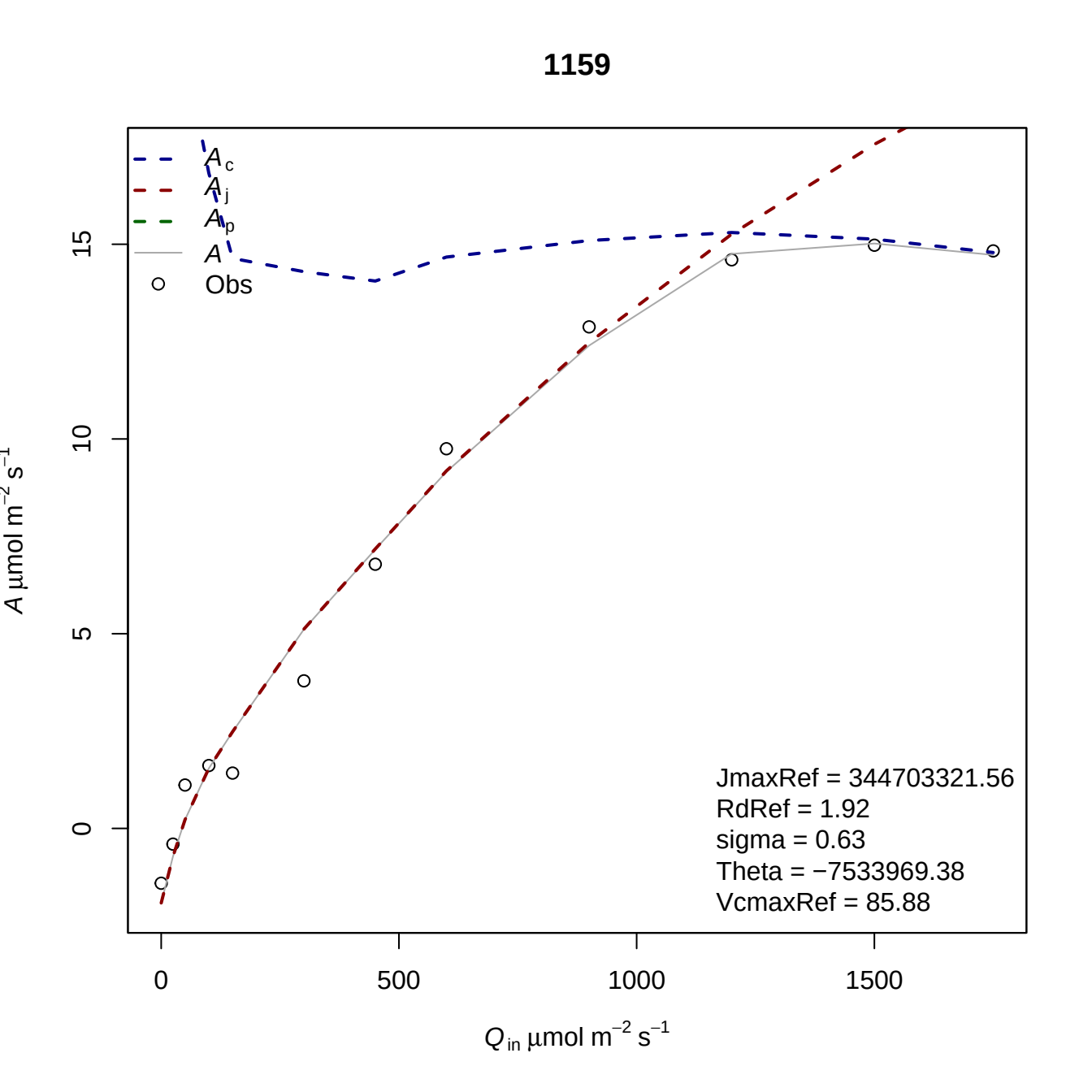

1154
A_c
A_j
A_p
A
Obs
JmaxRef = 224159823.07
RdRef = 2.77
sigma = 0.94
Theta = −4449500.42
VcmaxRef = 90.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

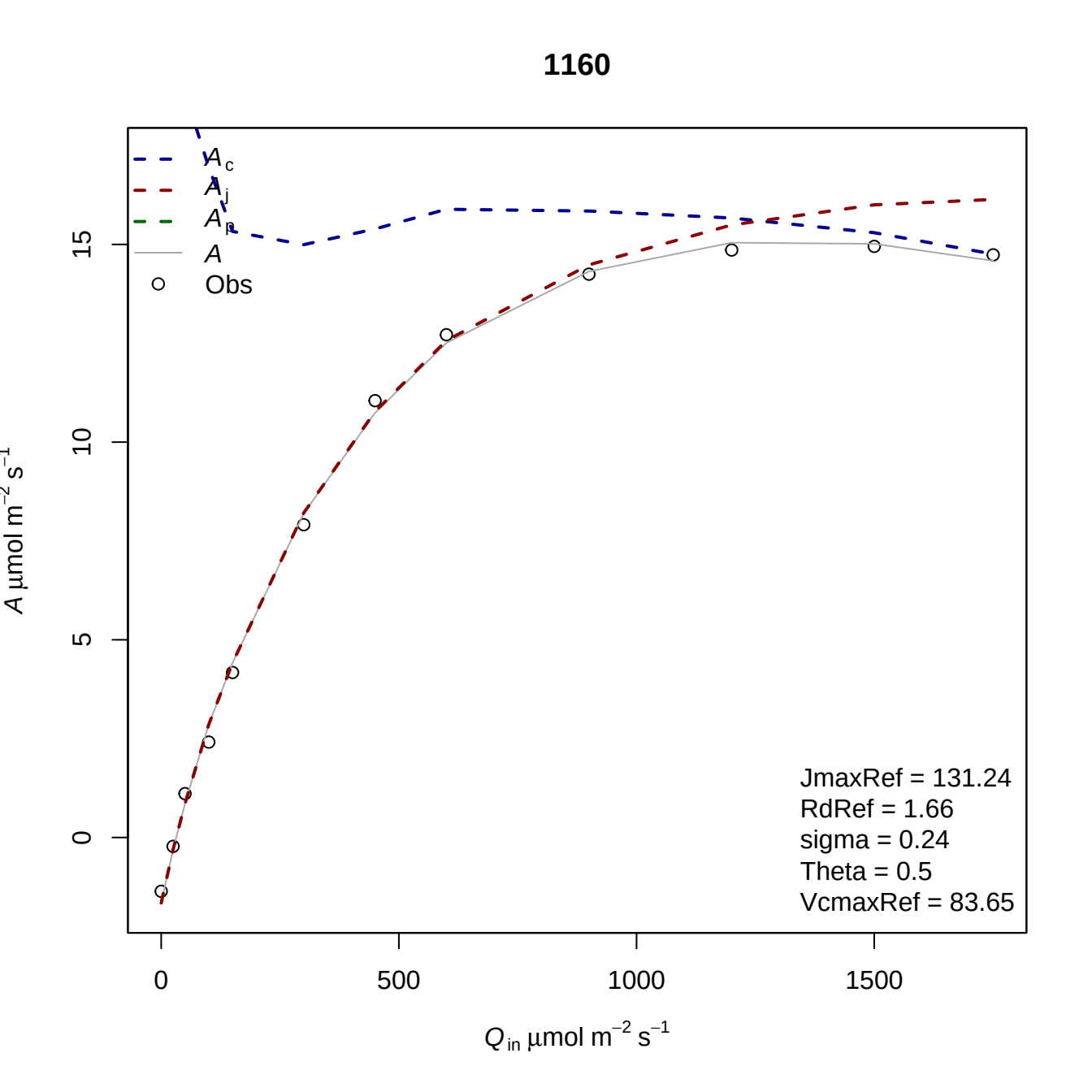

1155
A_c
A_j
A_p
A
Obs
JmaxRef = 14759304.66
RdRef = 2.14
sigma = 0.8
Theta = −638449.3
VcmaxRef = 66.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

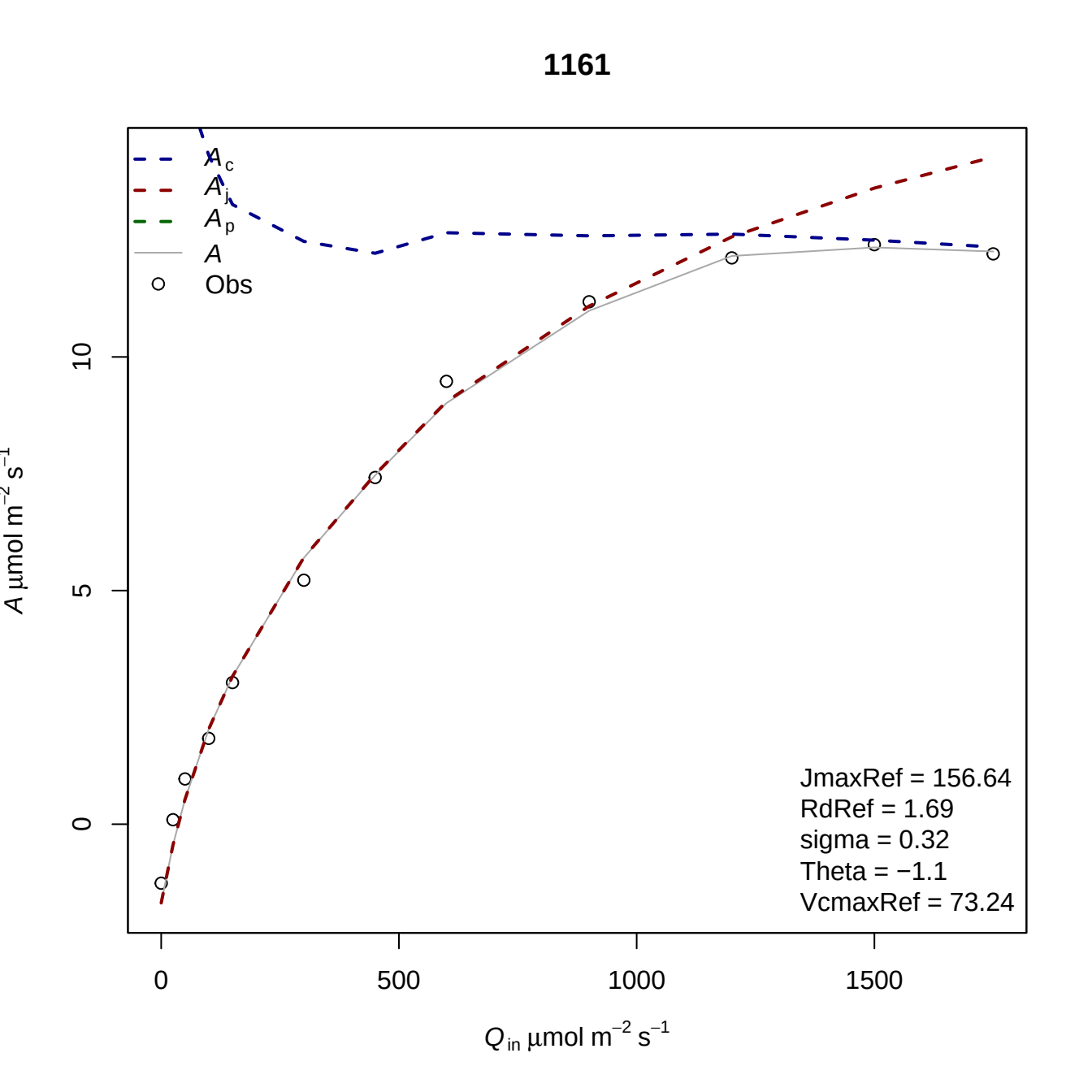

1156
A_c
A_j
A_p
A
Obs
JmaxRef = 124437300.28
RdRef = 3.35
sigma = 1.43
Theta = −779762.74
VcmaxRef = 122.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

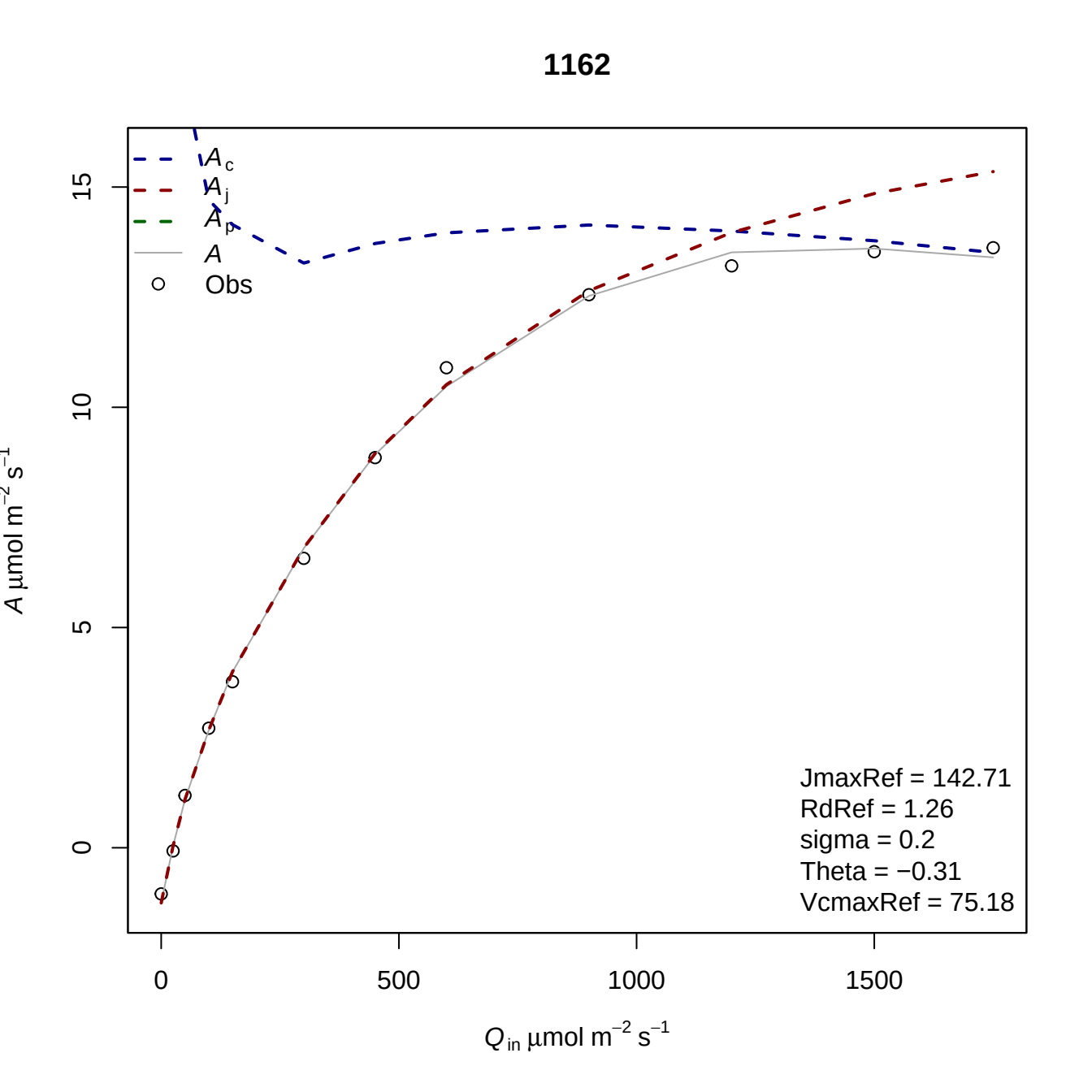

1157
A_c
A_j
A_p
A
Obs
JmaxRef = 521952087.73
RdRef = 2.71
sigma = 1.11
Theta = −5535466.52
VcmaxRef = 83.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

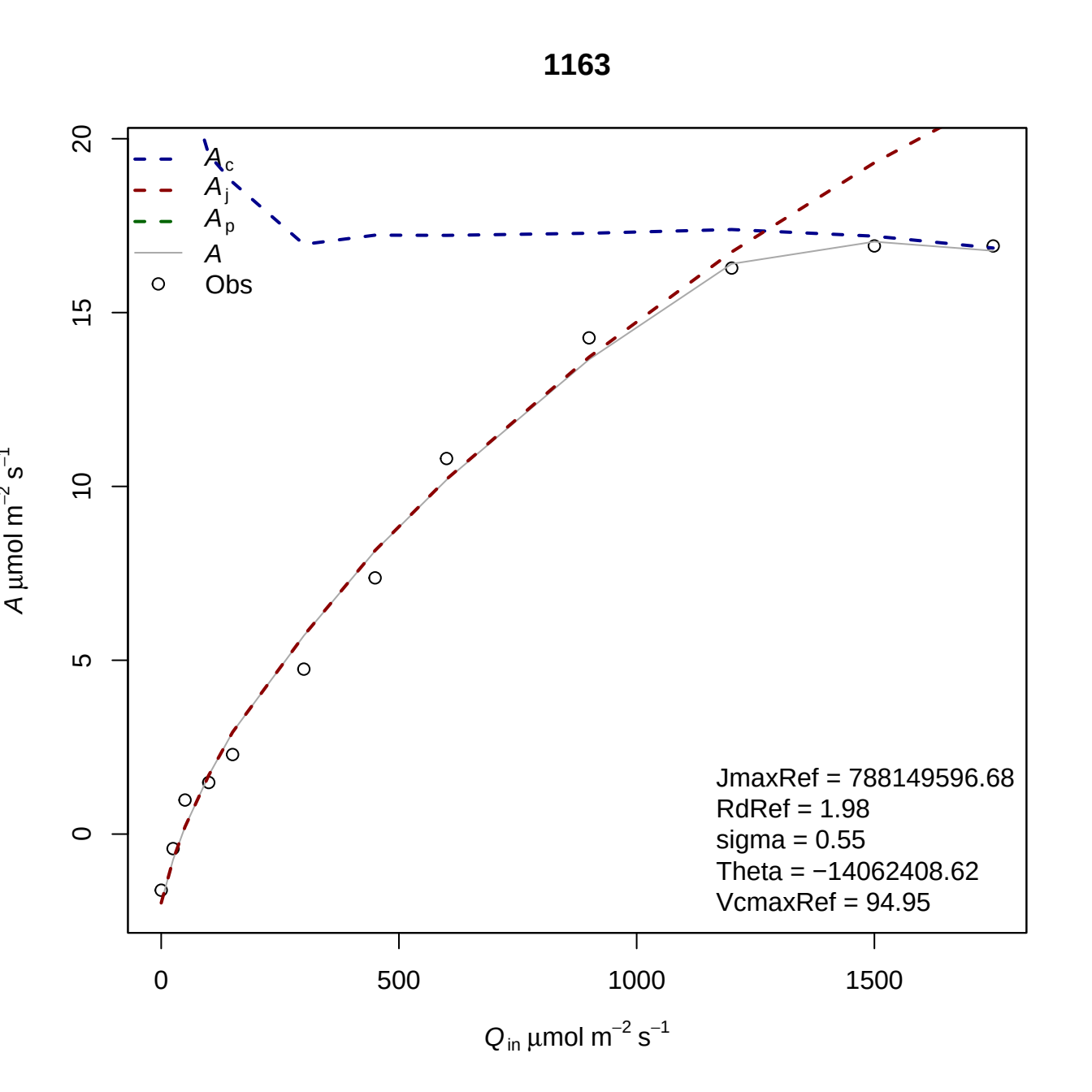

1158
A_c
A_j
A_p
A
Obs
JmaxRef = 72624396.84
RdRef = 2.42
sigma = 0.77
Theta = −1012204.8
VcmaxRef = 81.8
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

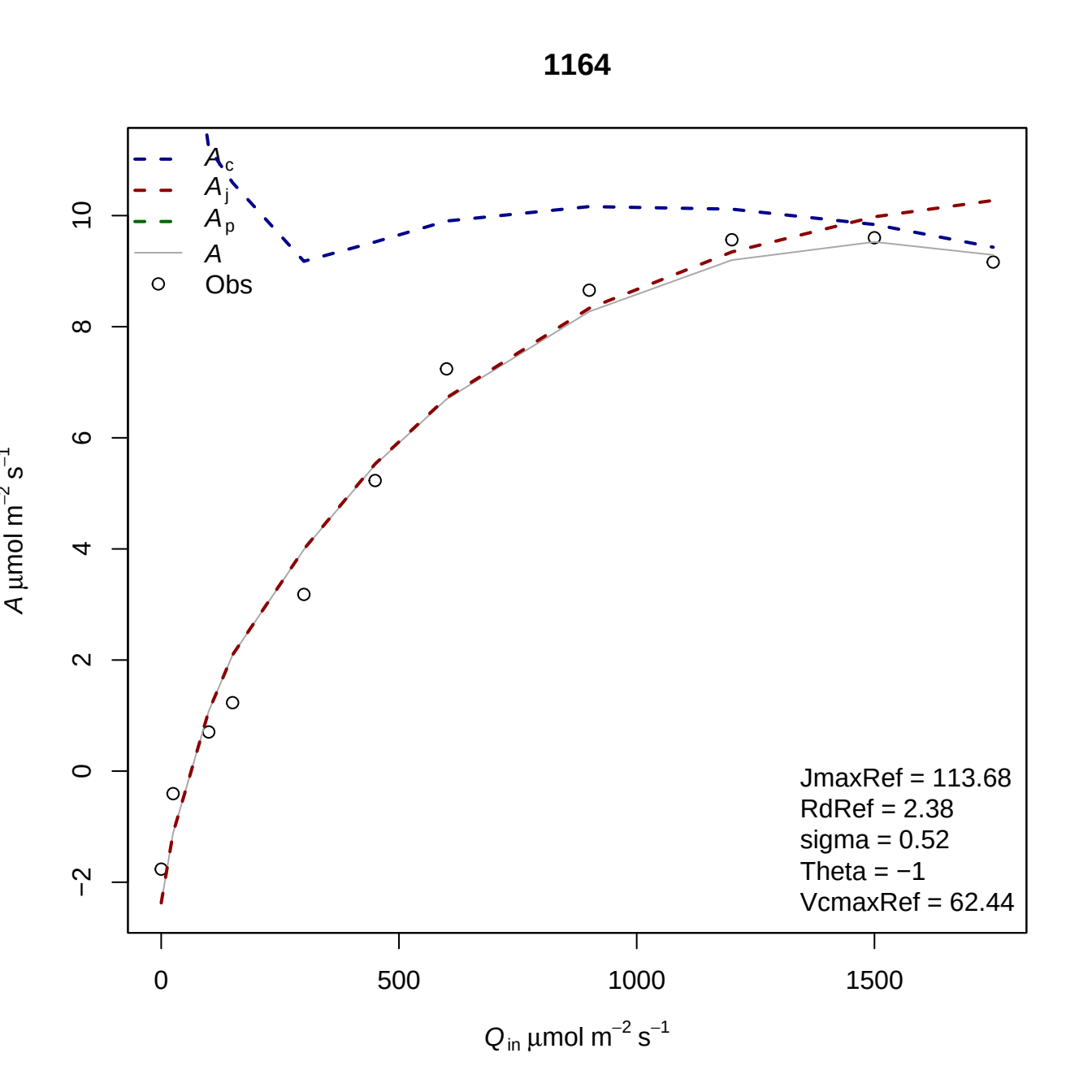

1159
A_c
A_j
A_p
A
Obs
JmaxRef = 344703321.56
RdRef = 1.92
sigma = 0.63
Theta = −7533969.38
VcmaxRef = 85.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

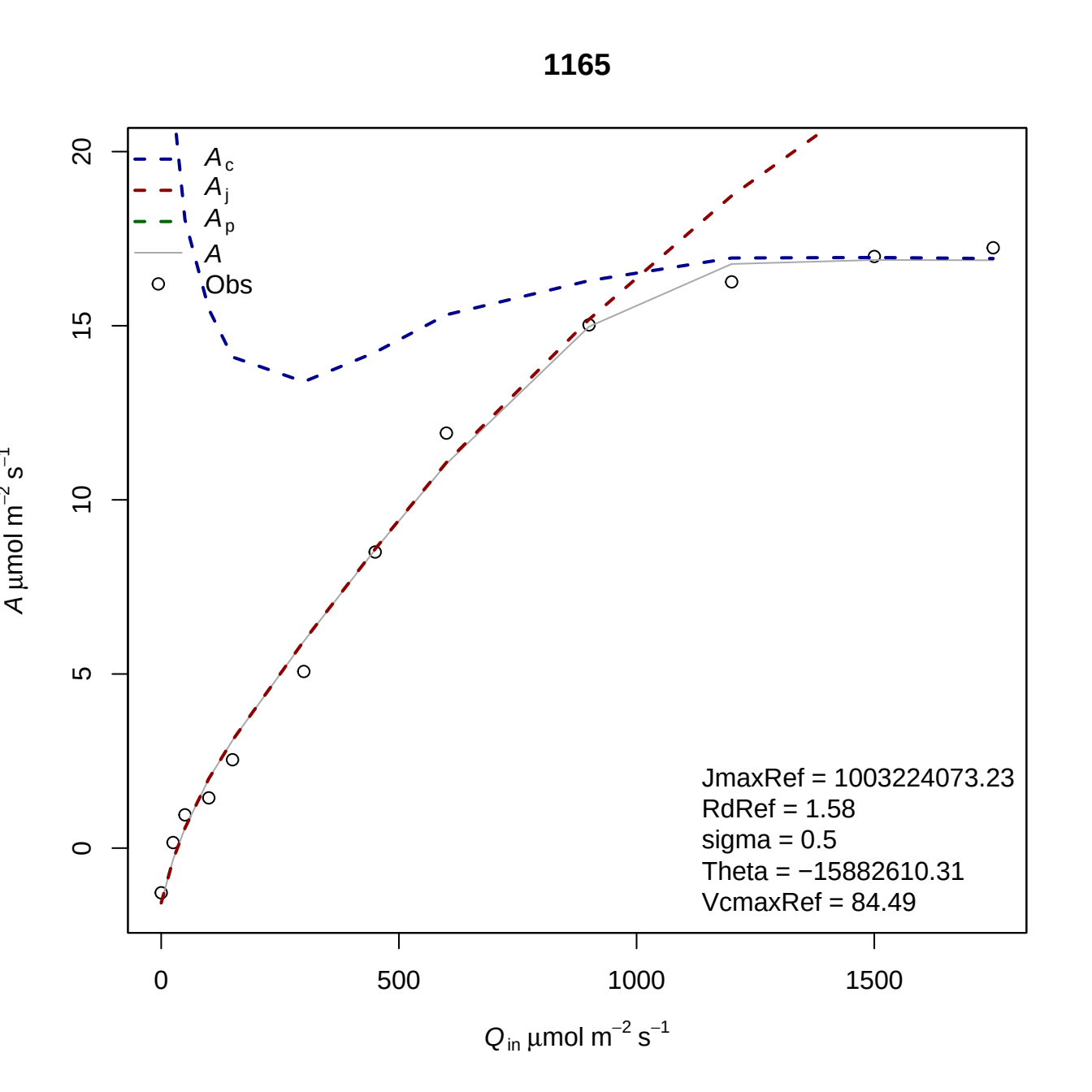

1160
A_c
A_j
A_p
A
Obs
JmaxRef = 131.24
RdRef = 1.66
sigma = 0.24
Theta = 0.5
VcmaxRef = 83.65
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

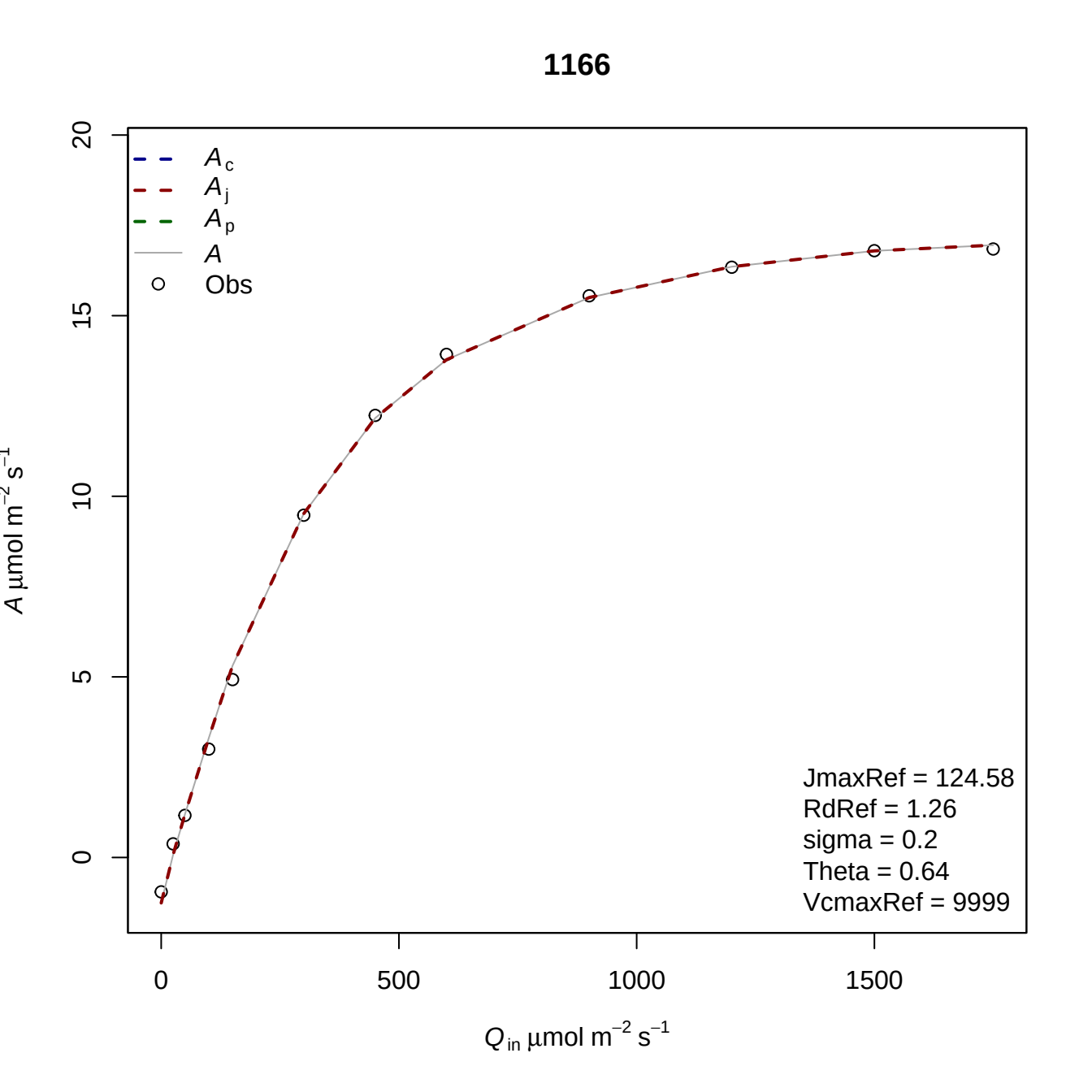

1161
A_c
A_j
A_p
A
Obs
JmaxRef = 156.64
RdRef = 1.69
sigma = 0.32
Theta = −1.1
VcmaxRef = 73.24
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

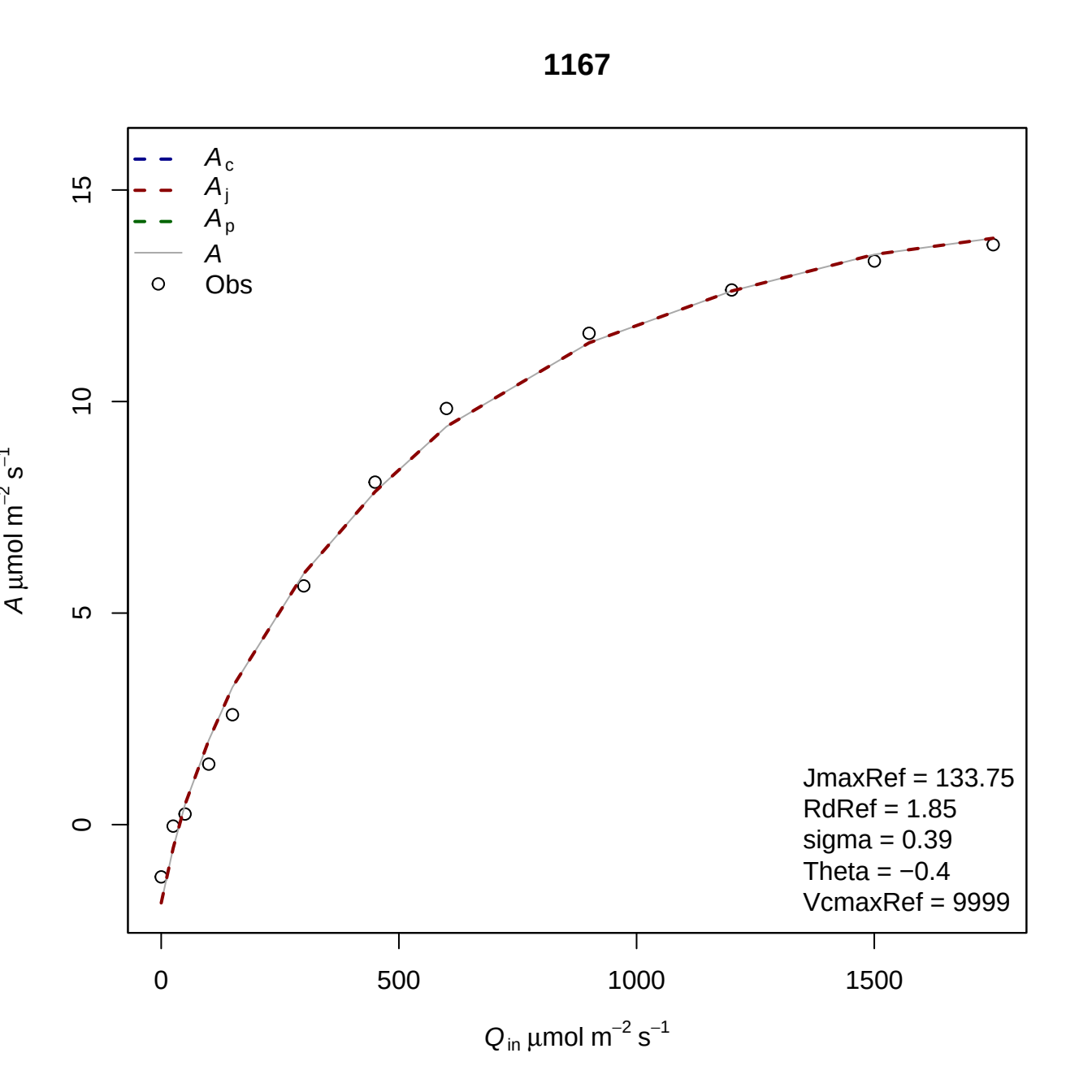

1162
A_c
A_j
A_p
A
Obs
JmaxRef = 142.71
RdRef = 1.26
sigma = 0.2
Theta = −0.31
VcmaxRef = 75.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

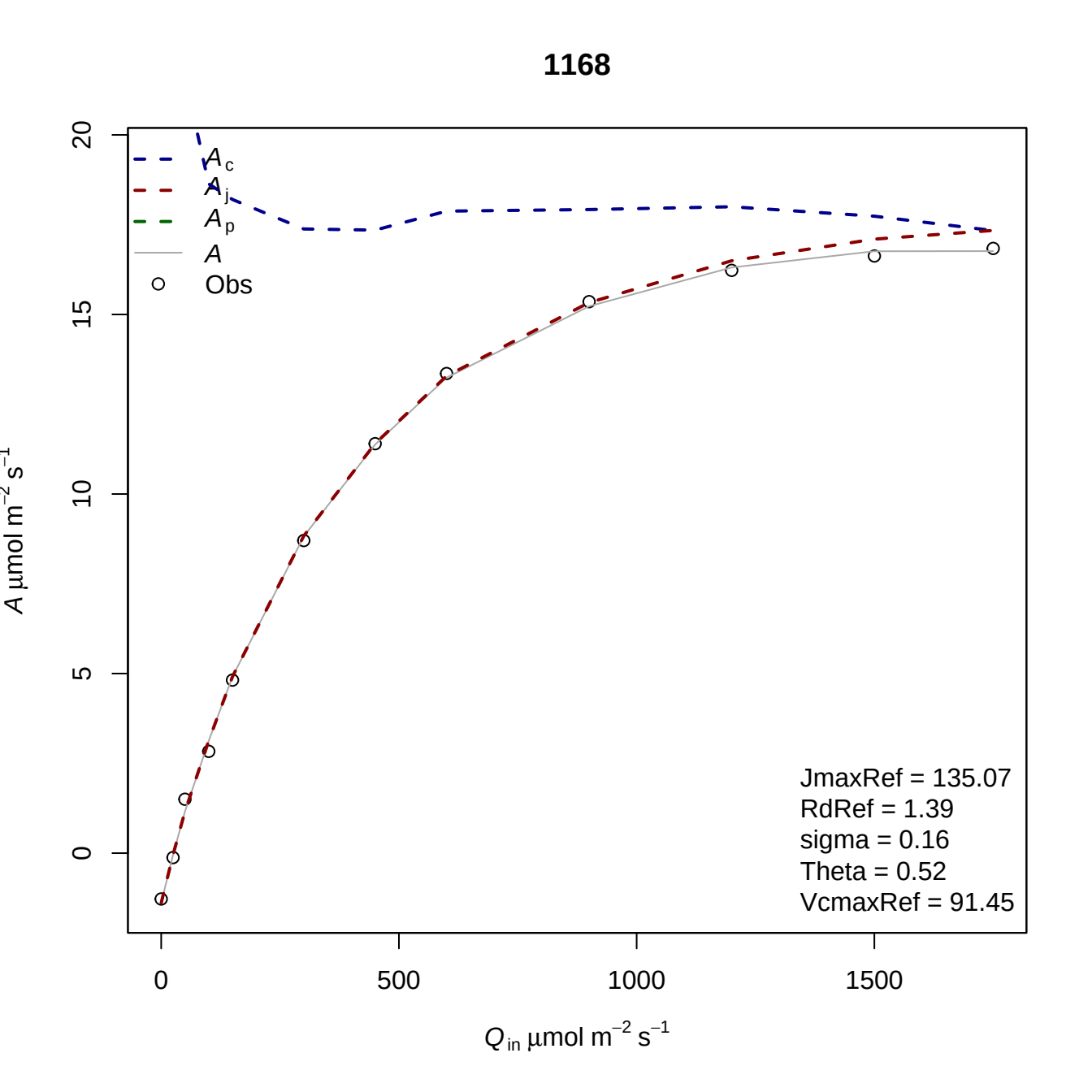

1163
A_c
A_j
A_p
A
Obs
JmaxRef = 788149596.68
RdRef = 1.98
sigma = 0.55
Theta = −14062408.62
VcmaxRef = 94.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

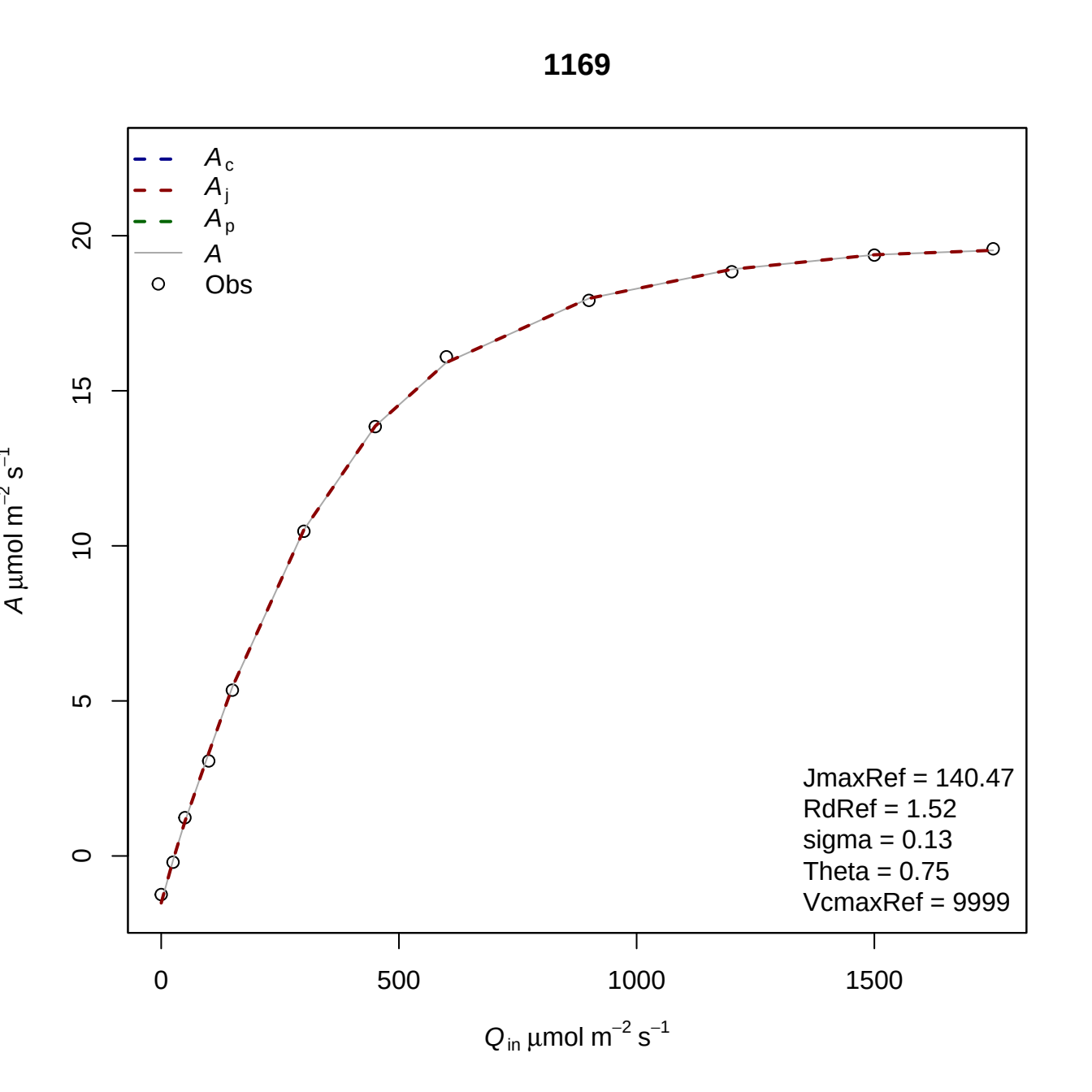

1164
A_c
A_j
A_p
A
Obs
JmaxRef = 113.68
RdRef = 2.38
sigma = 0.52
Theta = −1
VcmaxRef = 62.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

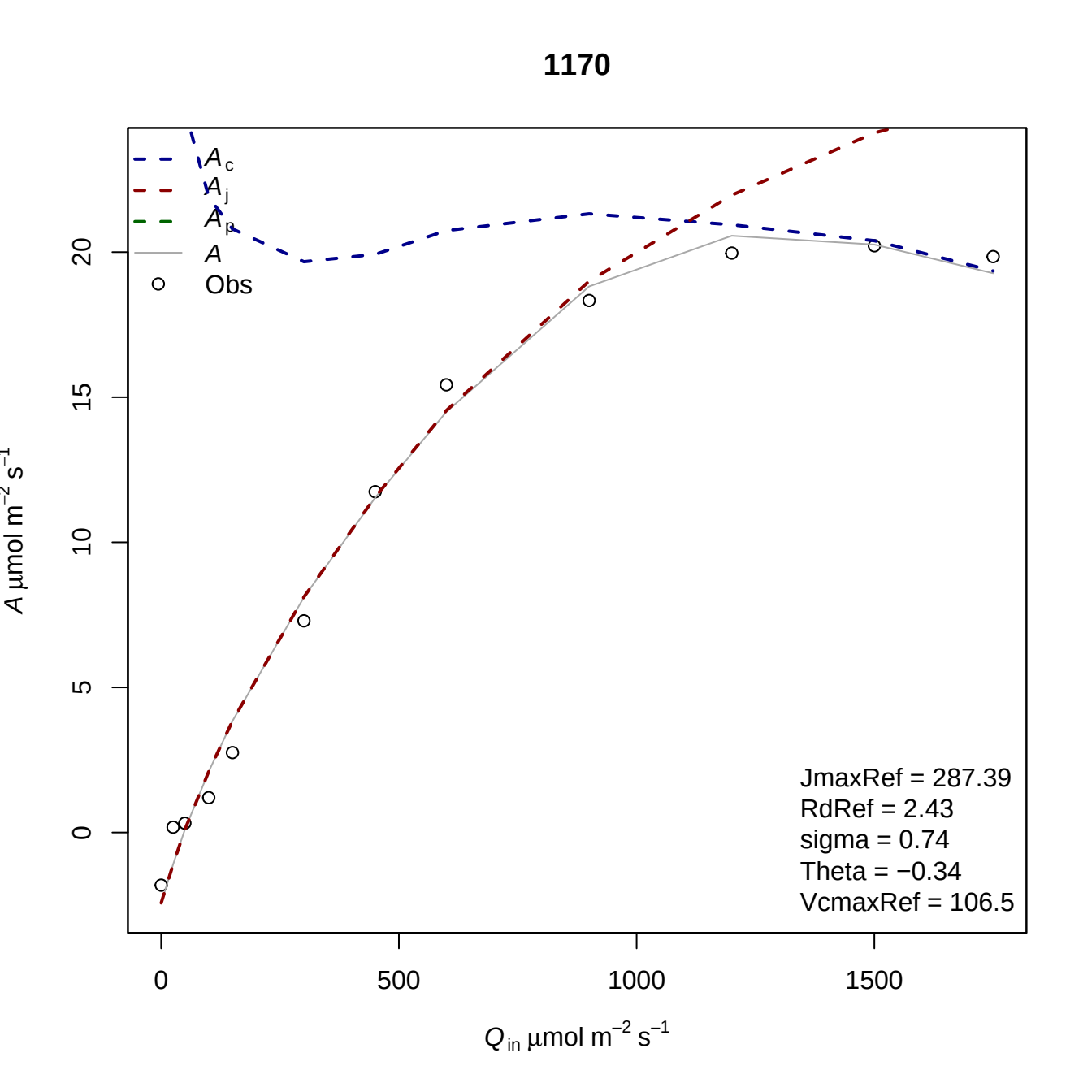

1165
A_c
A_j
A_p
A
Obs
JmaxRef = 1003224073.23
RdRef = 1.58
sigma = 0.5
Theta = −15882610.31
VcmaxRef = 84.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

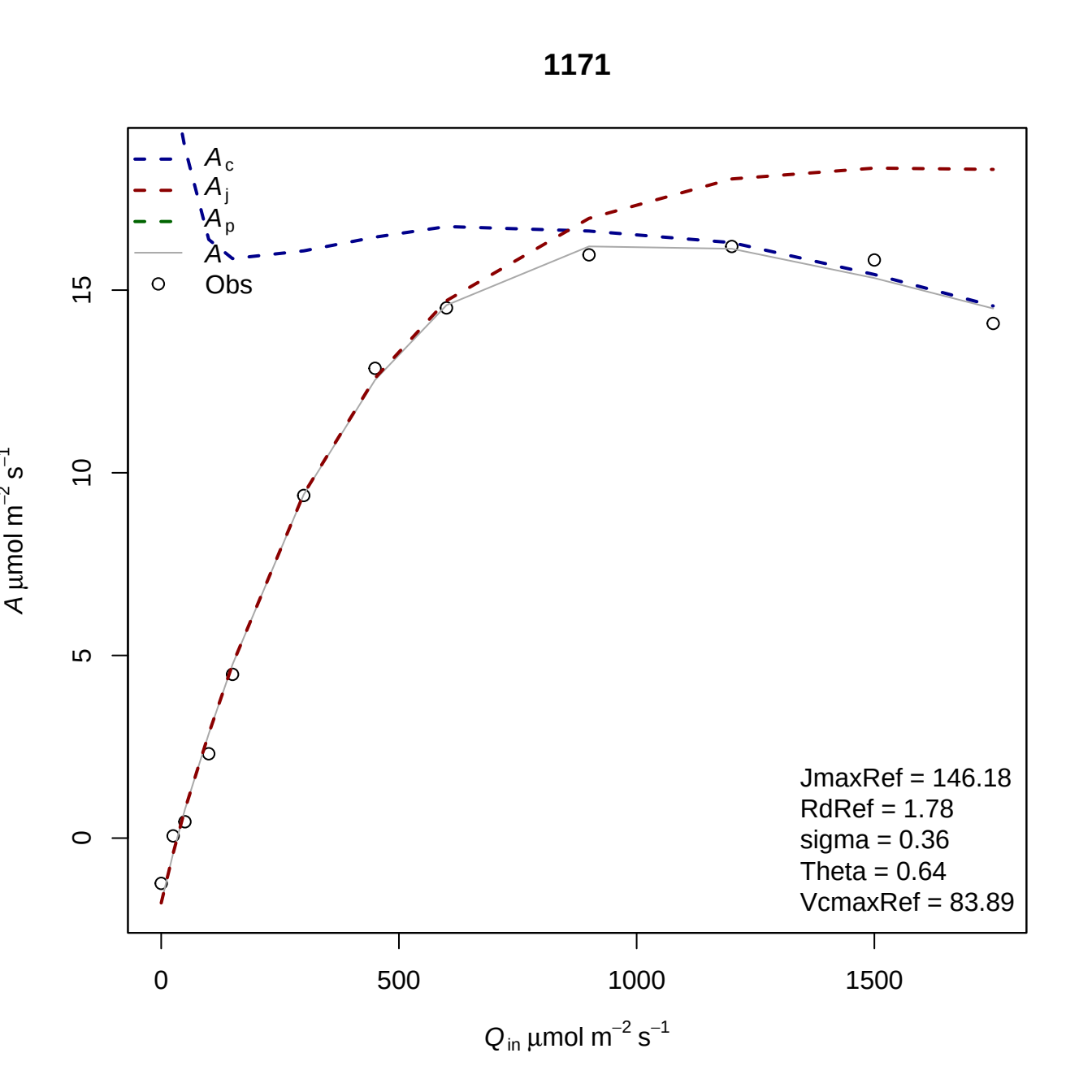

1166
A_c
A_j
A_p
A
Obs
JmaxRef = 124.58
RdRef = 1.26
sigma = 0.2
Theta = 0.64
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

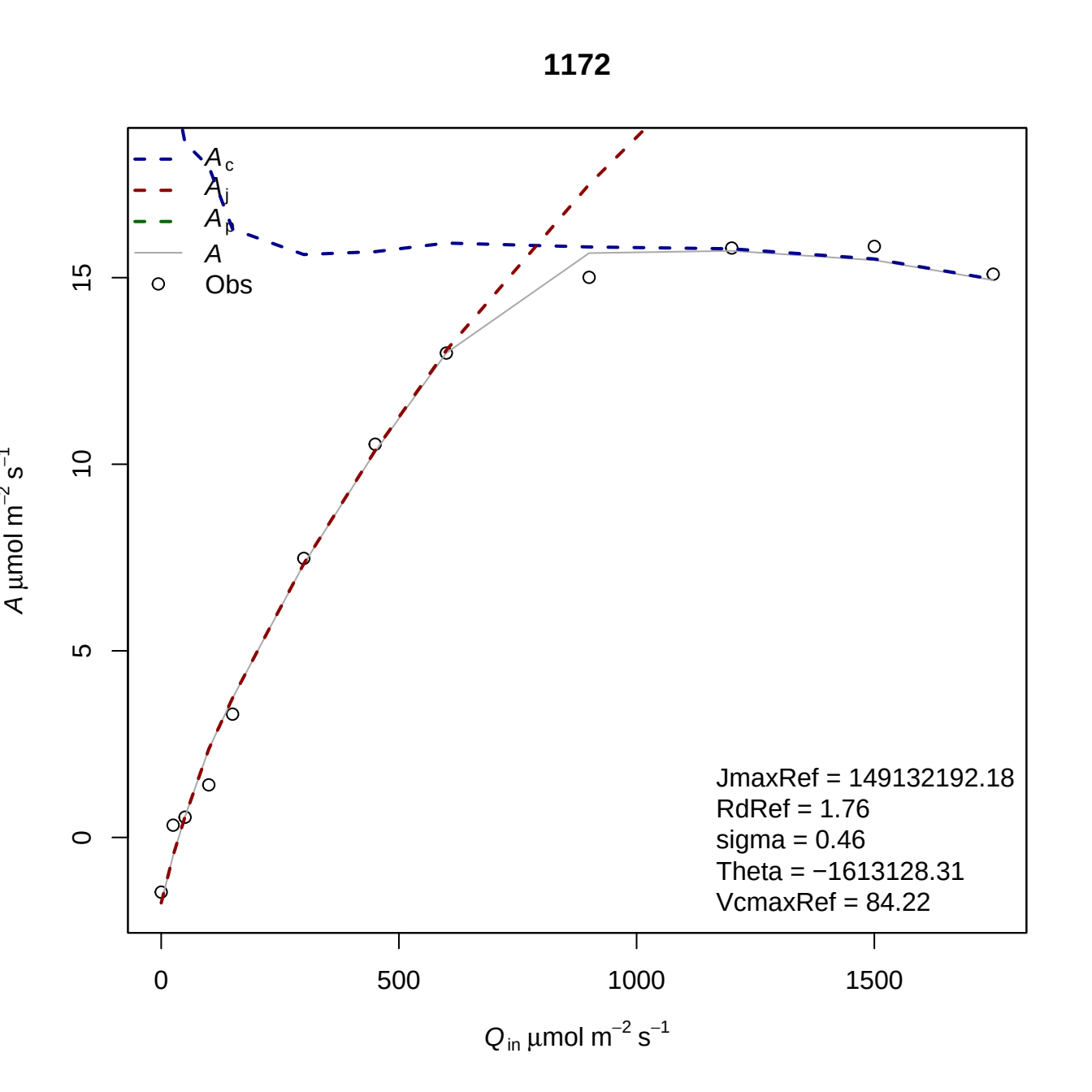

1167
A_c
A_j
A_p
A
Obs
JmaxRef = 133.75
RdRef = 1.85
sigma = 0.39
Theta = −0.4
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

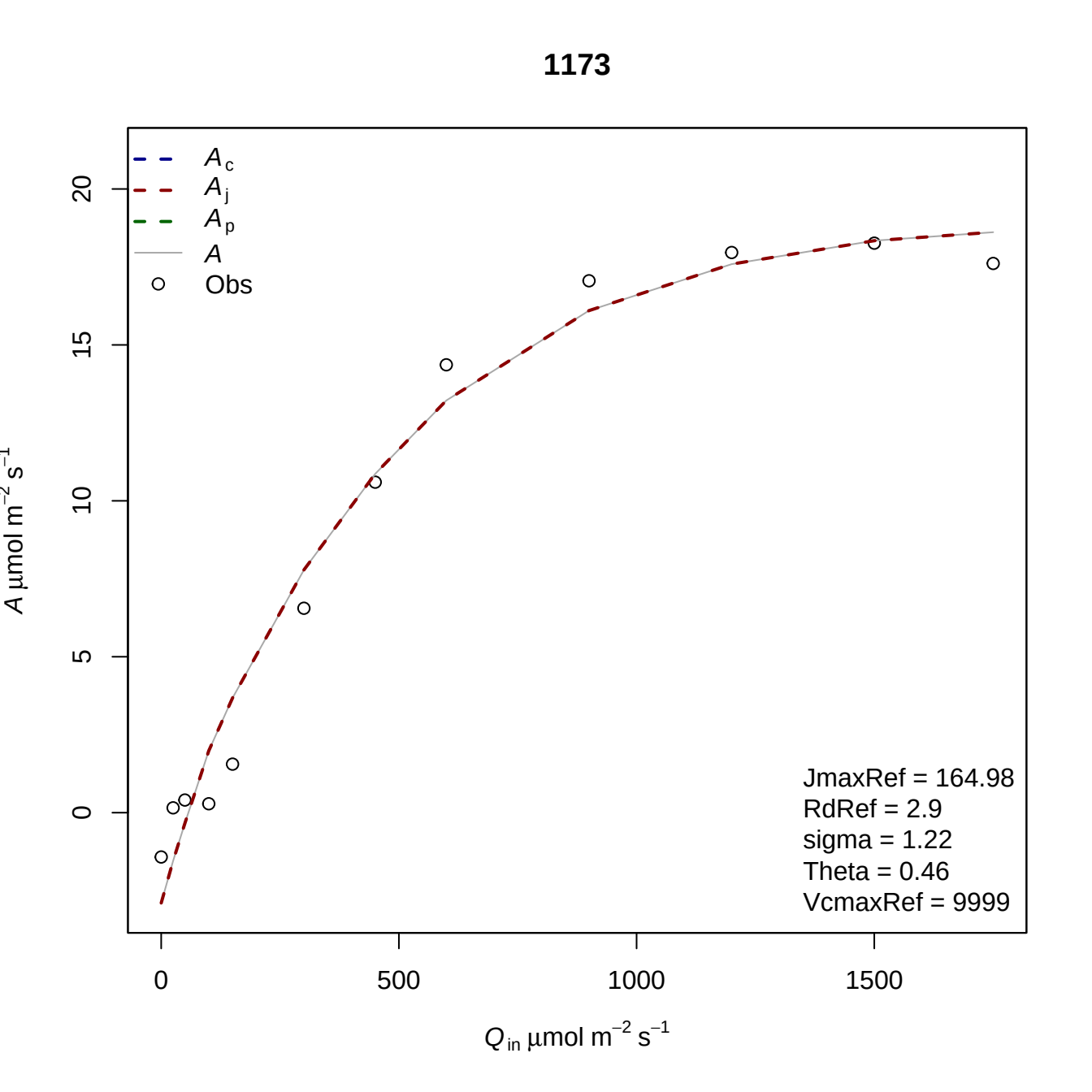

1168
A_c
A_j
A_p
A
Obs
JmaxRef = 135.07
RdRef = 1.39
sigma = 0.16
Theta = 0.52
VcmaxRef = 91.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

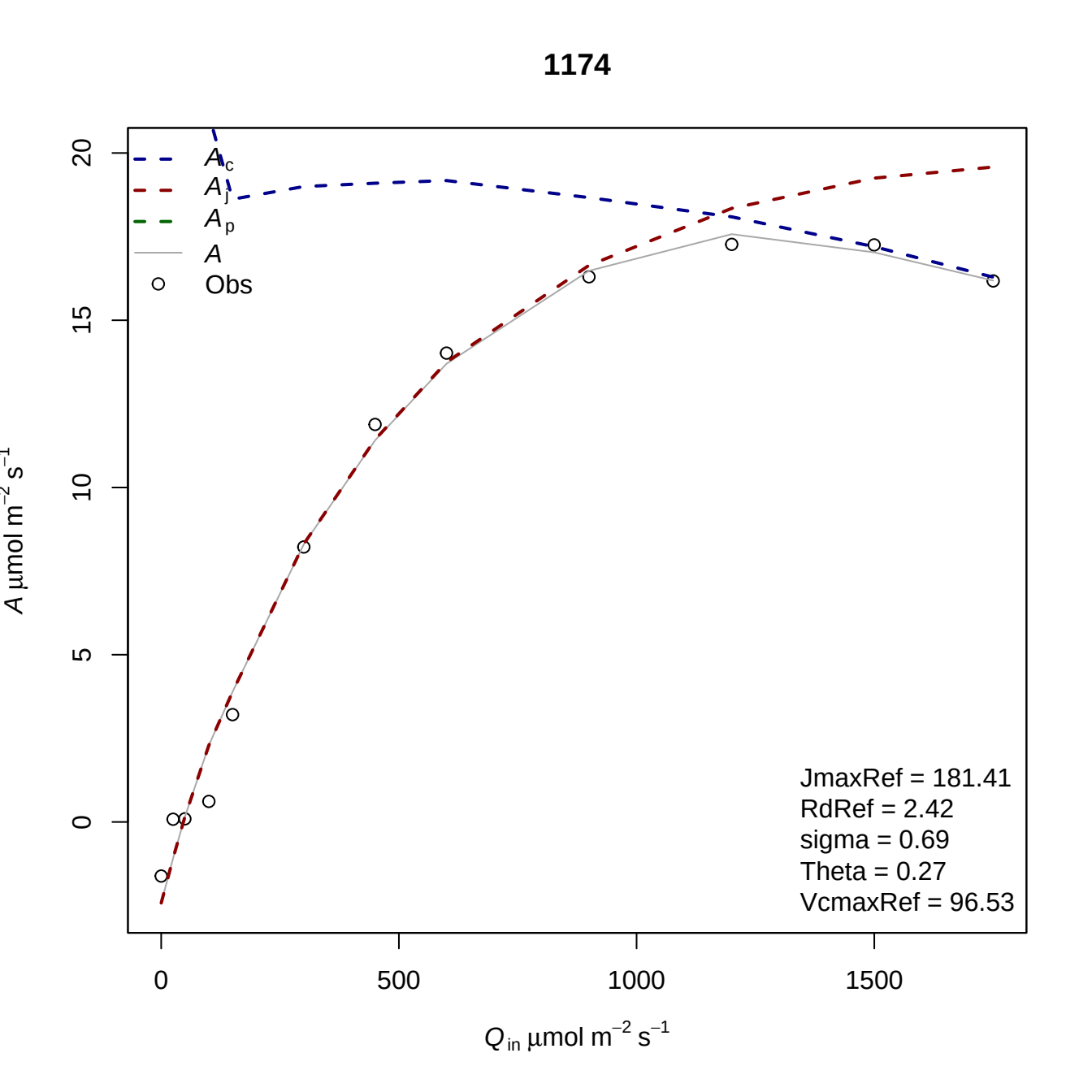

1169
A_c
A_j
A_p
A
Obs
JmaxRef = 140.47
RdRef = 1.52
sigma = 0.13
Theta = 0.75
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

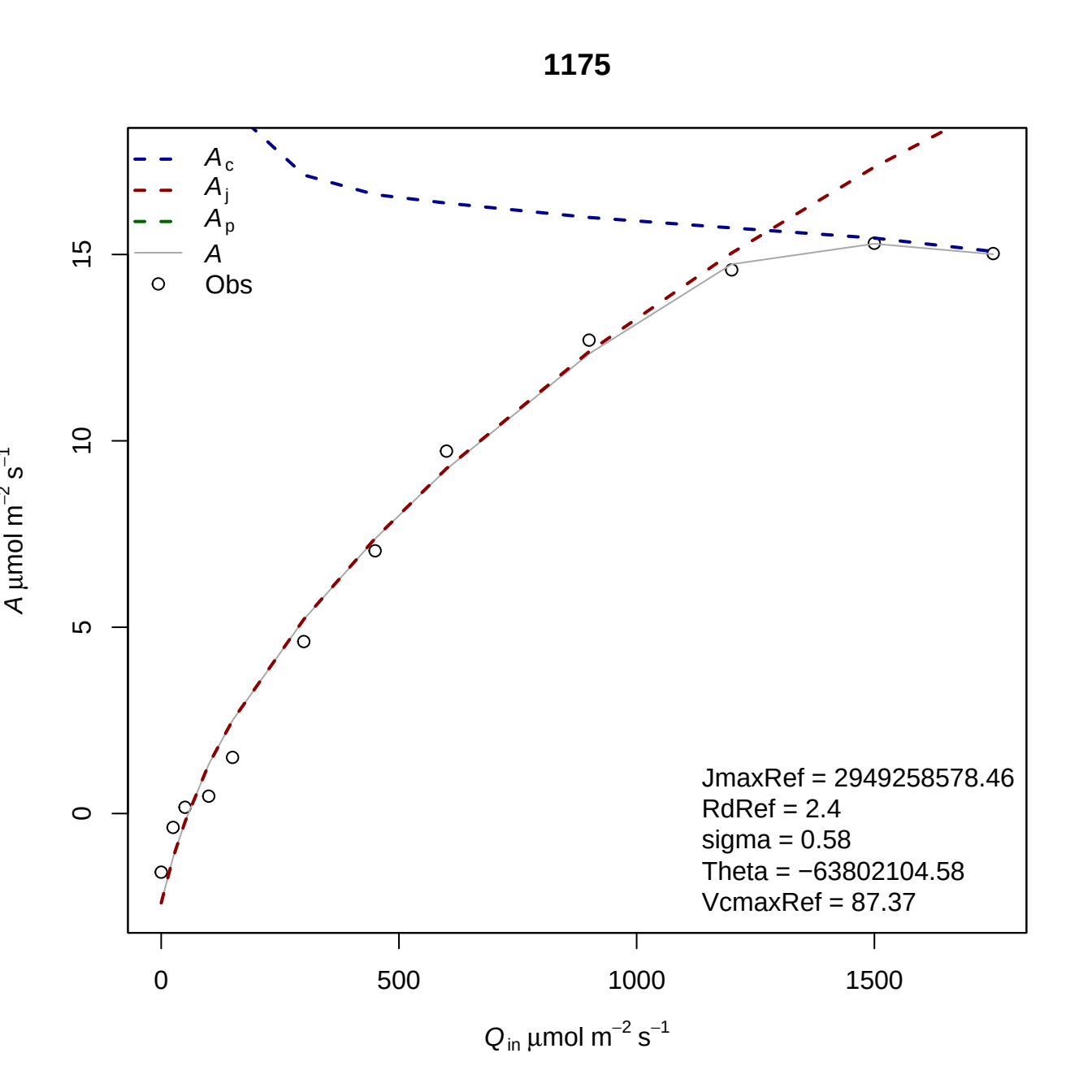

1170
A_c
A_j
A_p
A
Obs
JmaxRef = 287.39
RdRef = 2.43
sigma = 0.74
Theta = −0.34
VcmaxRef = 106.5
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

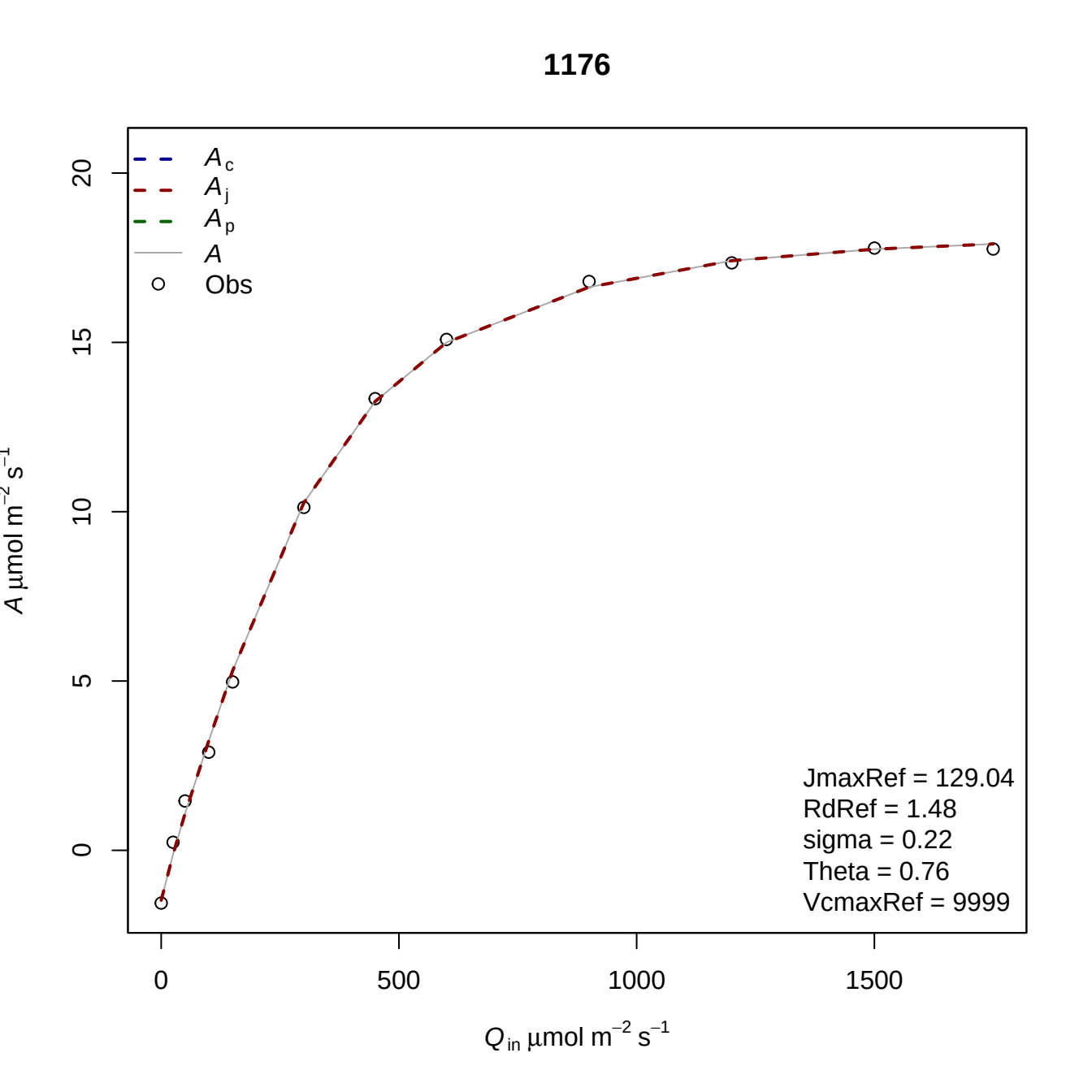

1171
A_c
A_j
A_p
A
Obs
JmaxRef = 146.18
RdRef = 1.78
sigma = 0.36
Theta = 0.64
VcmaxRef = 83.89
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

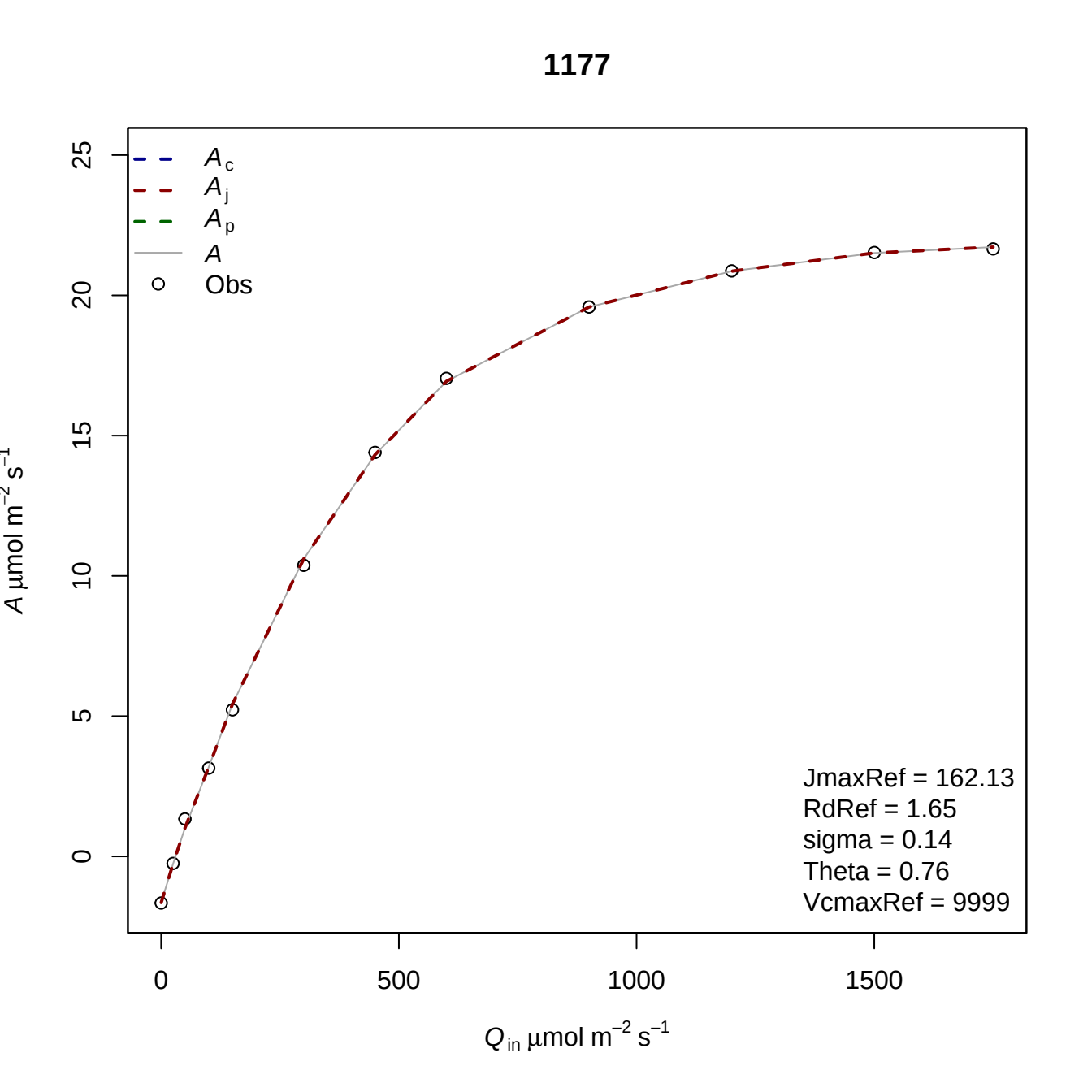

1172
A_c
A_j
A_p
A
Obs
JmaxRef = 149132192.18
RdRef = 1.76
sigma = 0.46
Theta = −1613128.31
VcmaxRef = 84.22
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

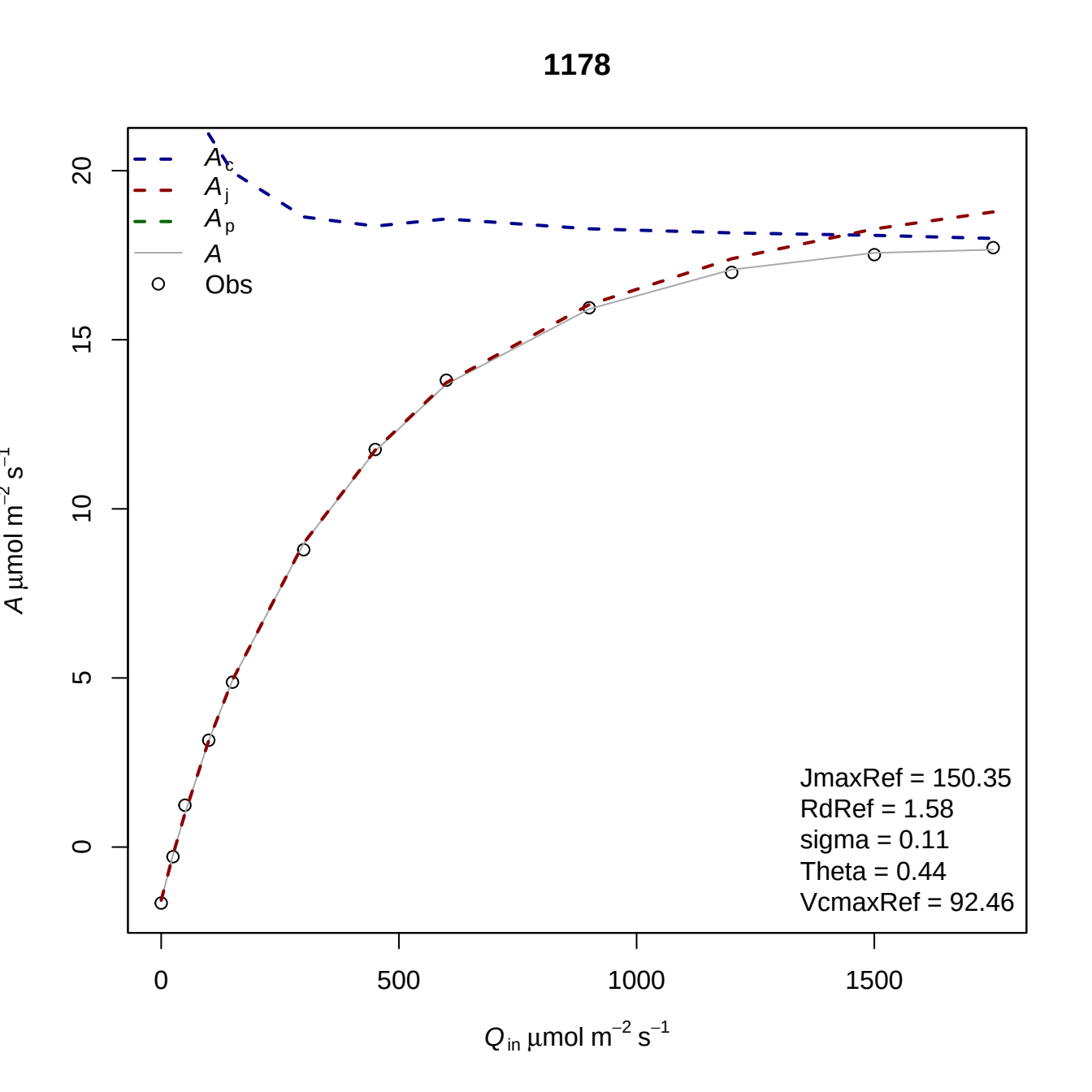

1173
A_c
A_j
A_p
A
Obs
JmaxRef = 164.98
RdRef = 2.9
sigma = 1.22
Theta = 0.46
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

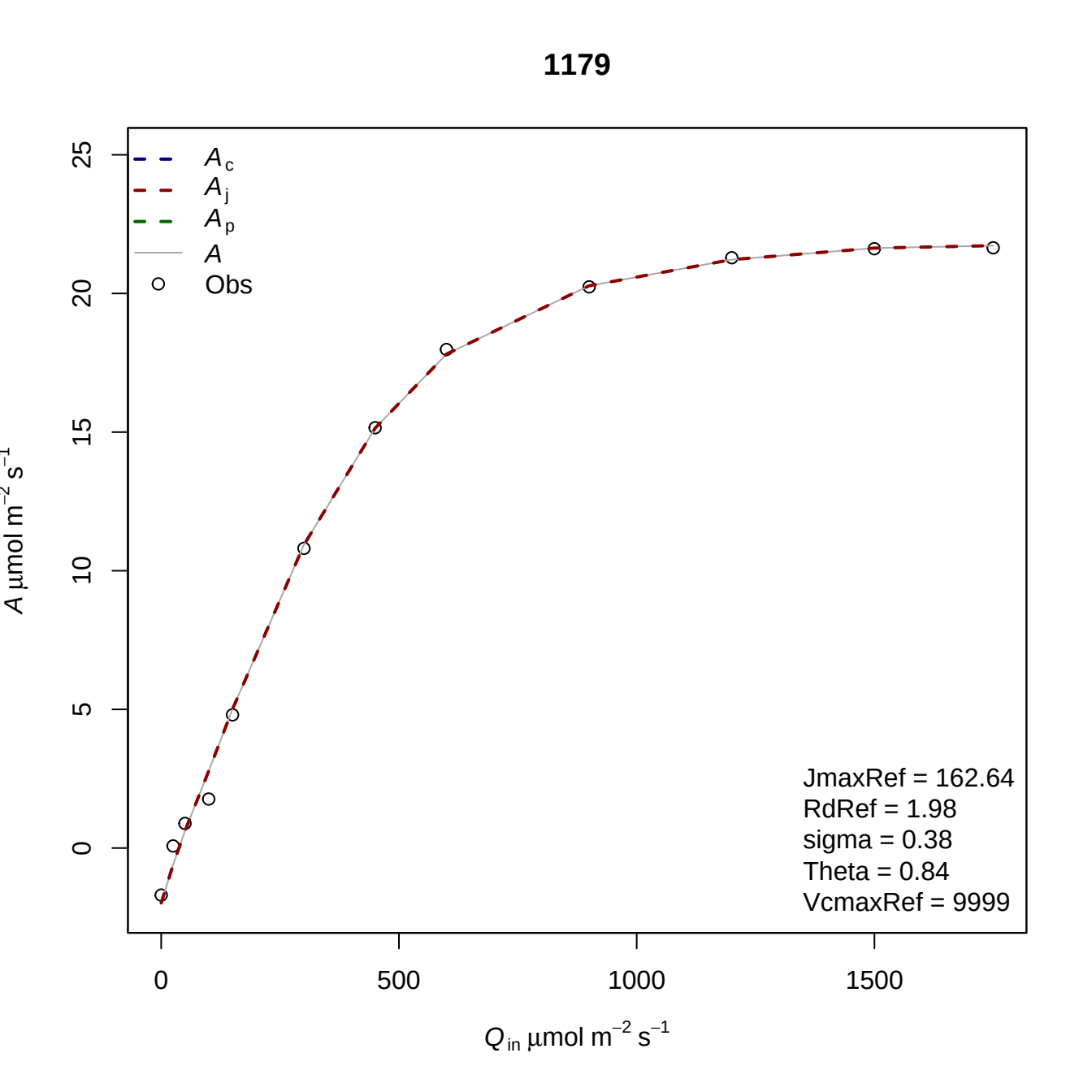

1174
A_c
A_j
A_p
A
Obs
JmaxRef = 181.41
RdRef = 2.42
sigma = 0.69
Theta = 0.27
VcmaxRef = 96.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

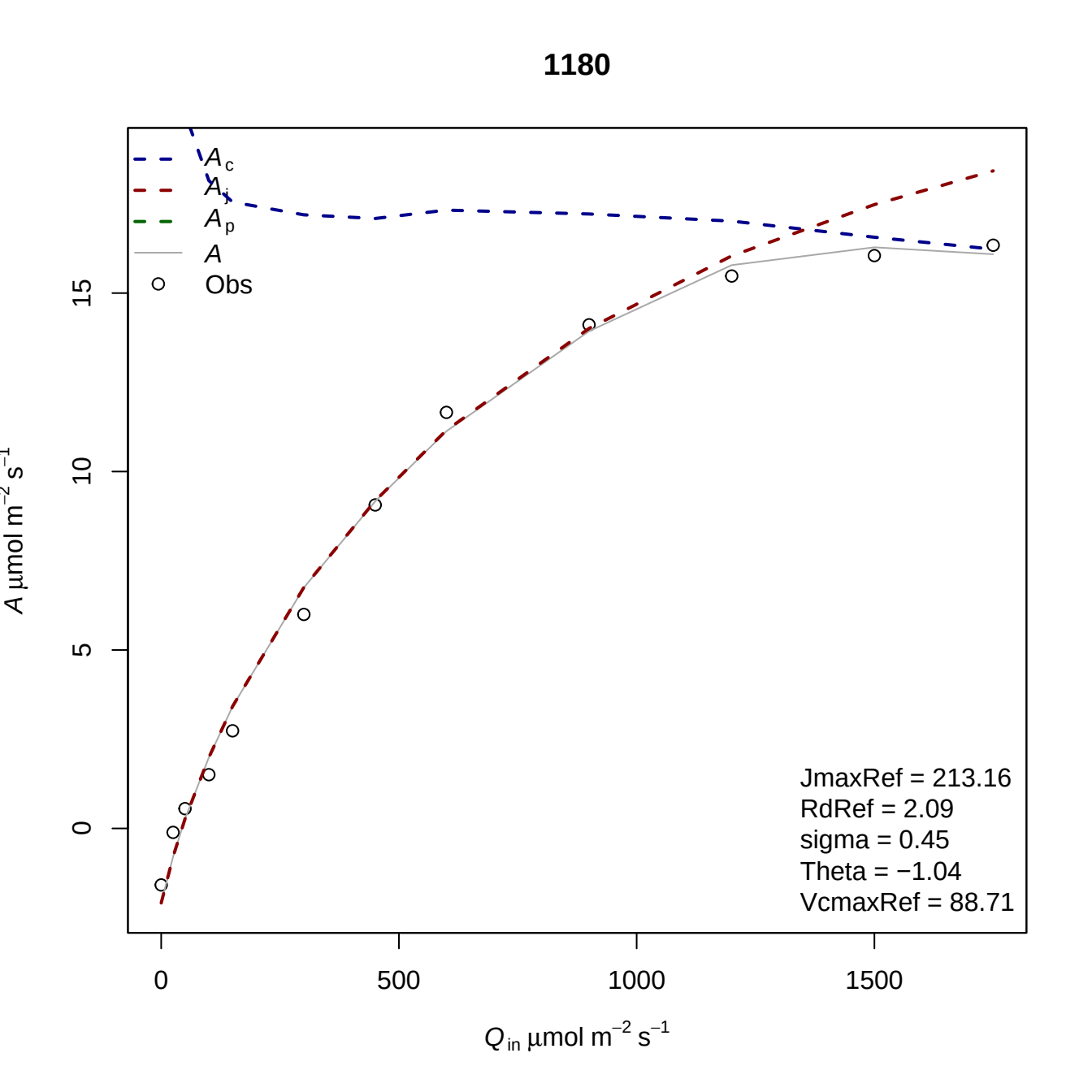

1175
A_c
A_j
A_p
A
Obs
JmaxRef = 2949258578.46
RdRef = 2.4
sigma = 0.58
Theta = −63802104.58
VcmaxRef = 87.37
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

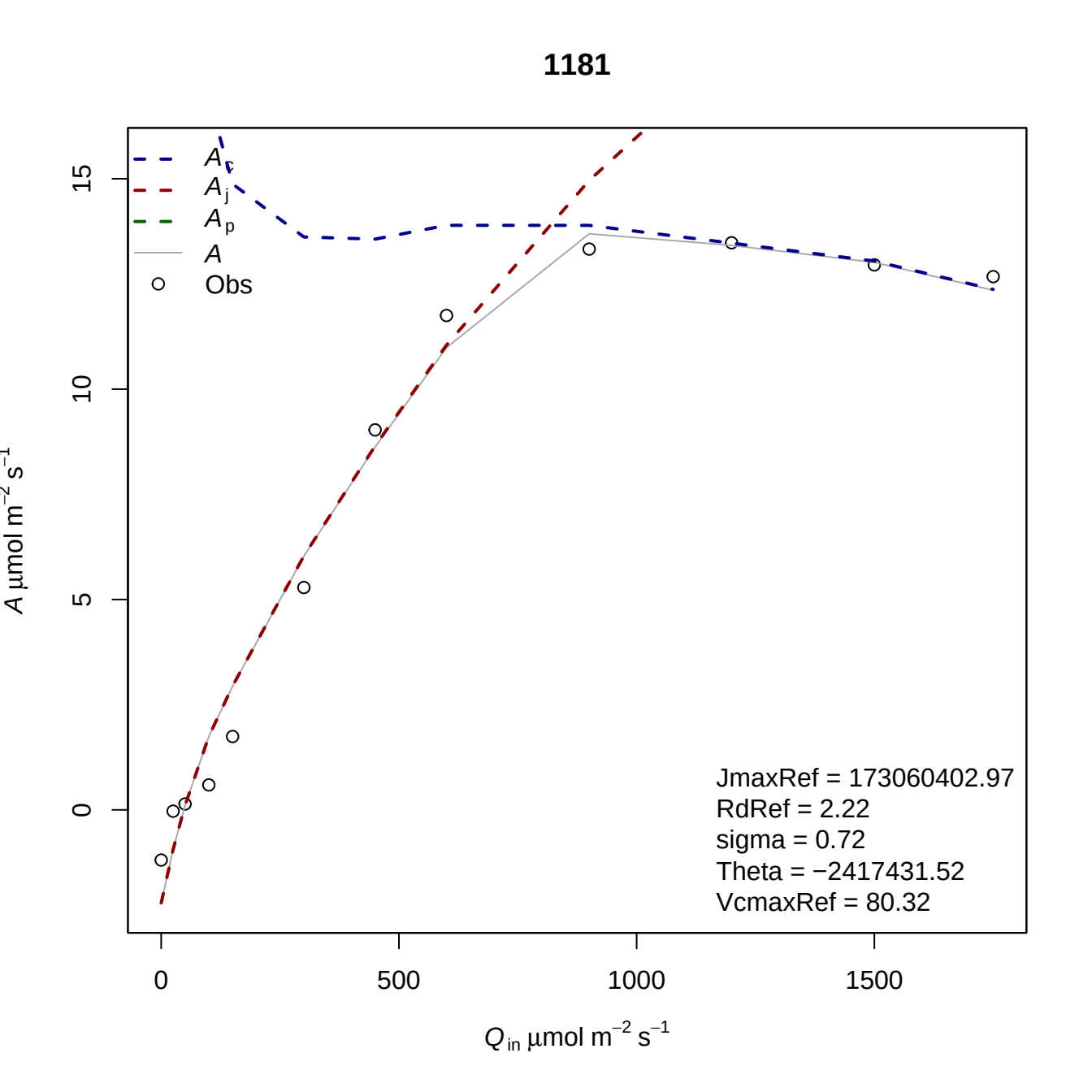

1176
A_c
A_j
A_p
A
Obs
JmaxRef = 129.04
RdRef = 1.48
sigma = 0.22
Theta = 0.76
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

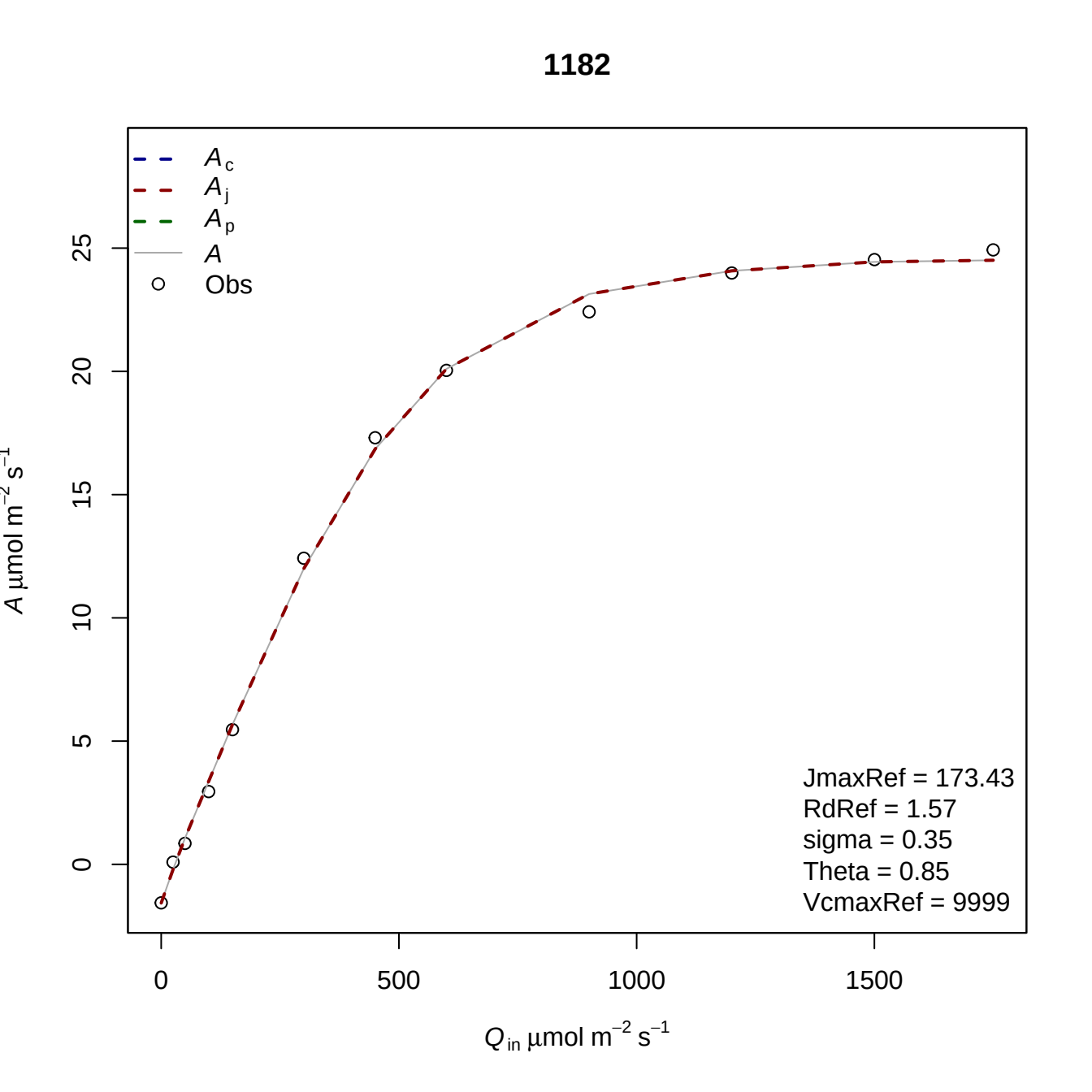

1177
A_c
A_j
A_p
A
Obs
JmaxRef = 162.13
RdRef = 1.65
sigma = 0.14
Theta = 0.76
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

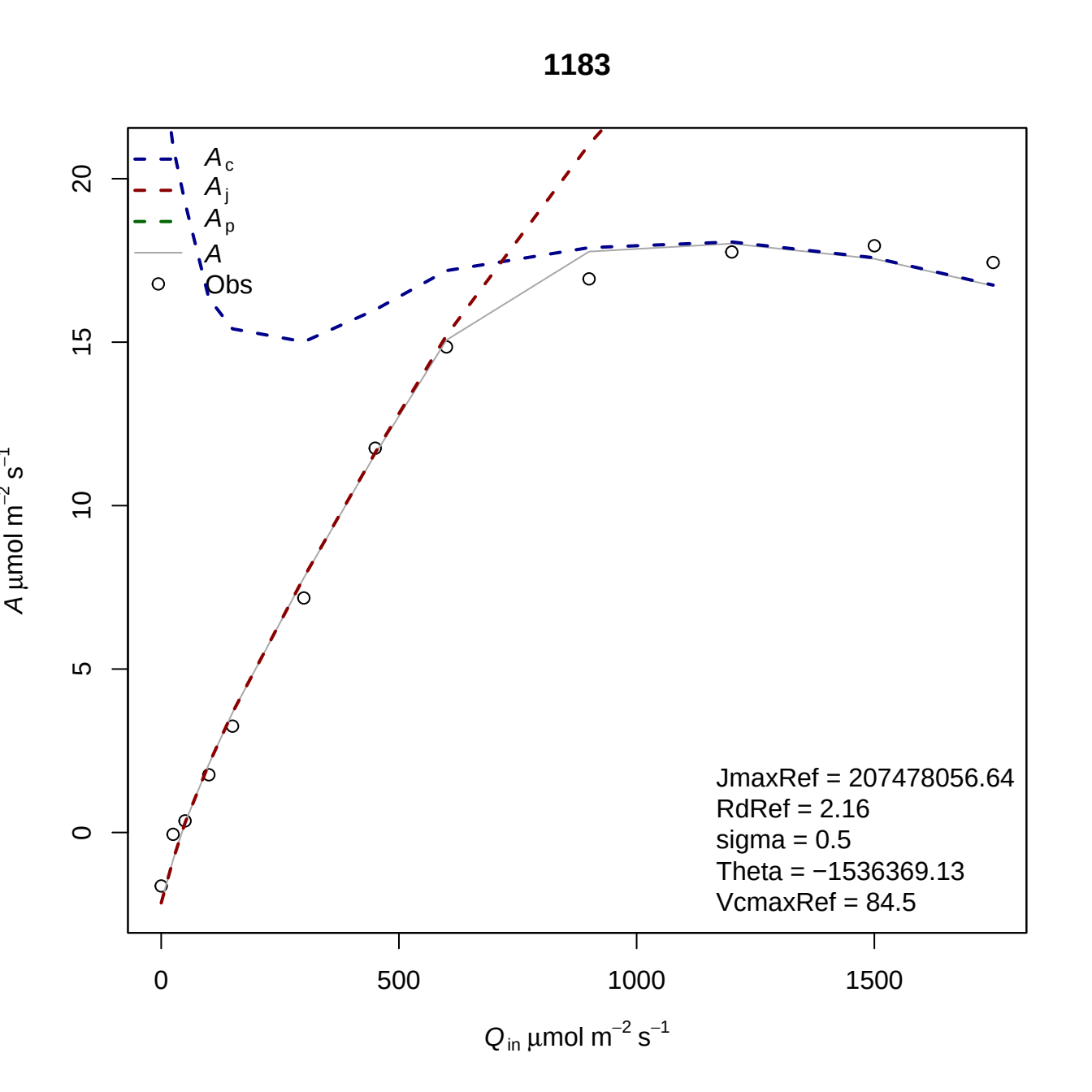

1178
A_c
A_j
A_p
A
Obs
JmaxRef = 150.35
RdRef = 1.58
sigma = 0.11
Theta = 0.44
VcmaxRef = 92.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

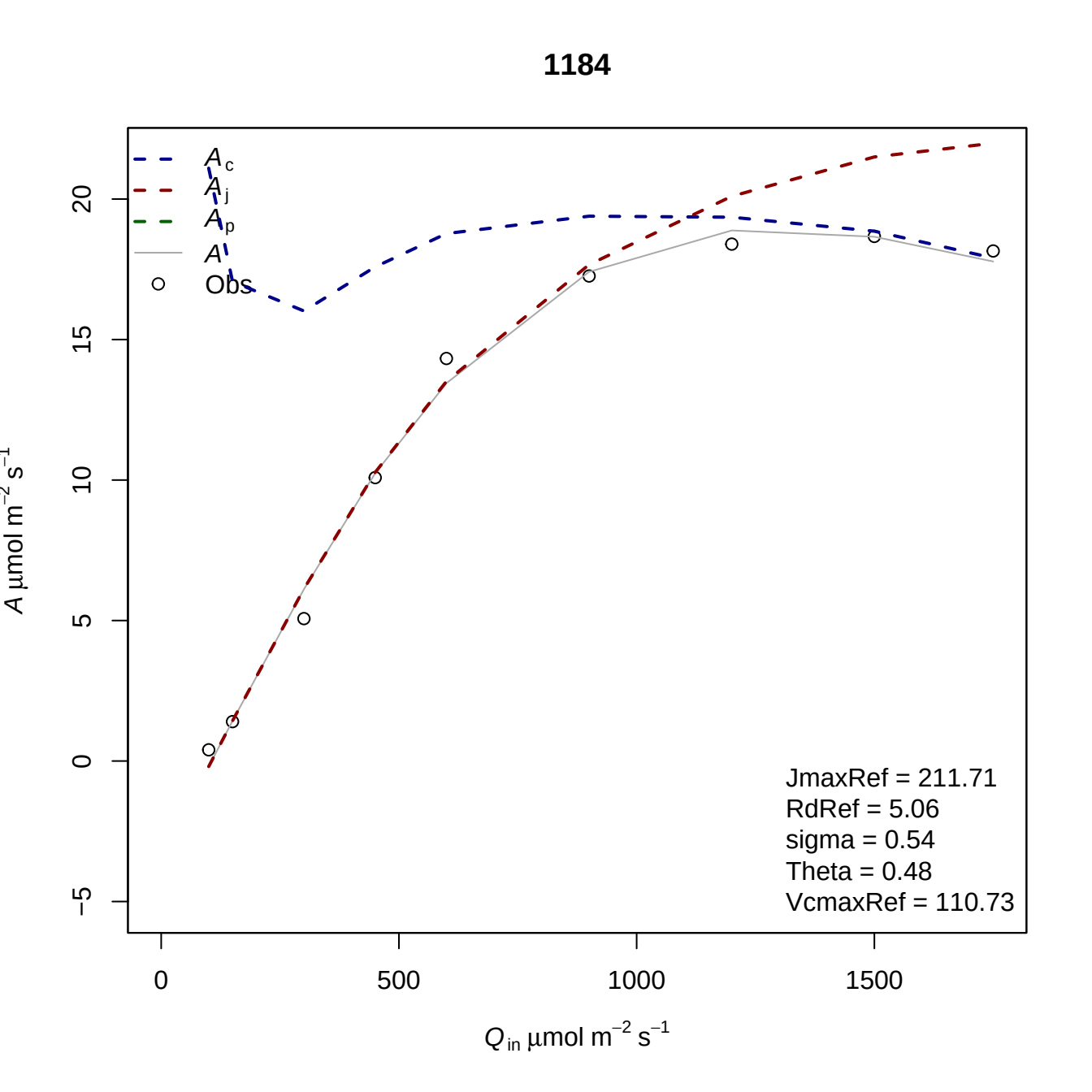

1179
A_c
A_j
A_p
A
Obs
JmaxRef = 162.64
RdRef = 1.98
sigma = 0.38
Theta = 0.84
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

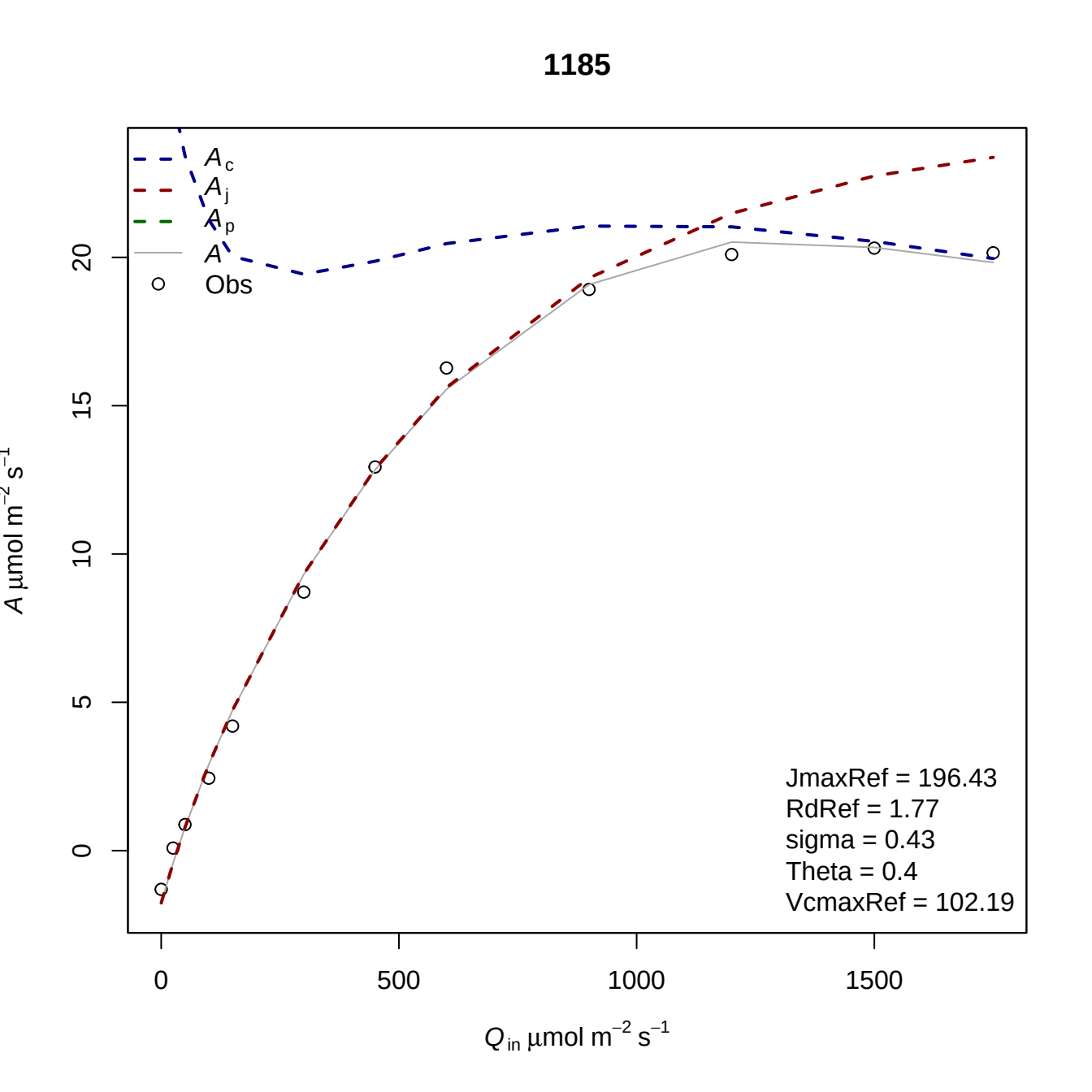

1180
A_c
A_j
A_p
A
Obs
JmaxRef = 213.16
RdRef = 2.09
sigma = 0.45
Theta = −1.04
VcmaxRef = 88.71
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

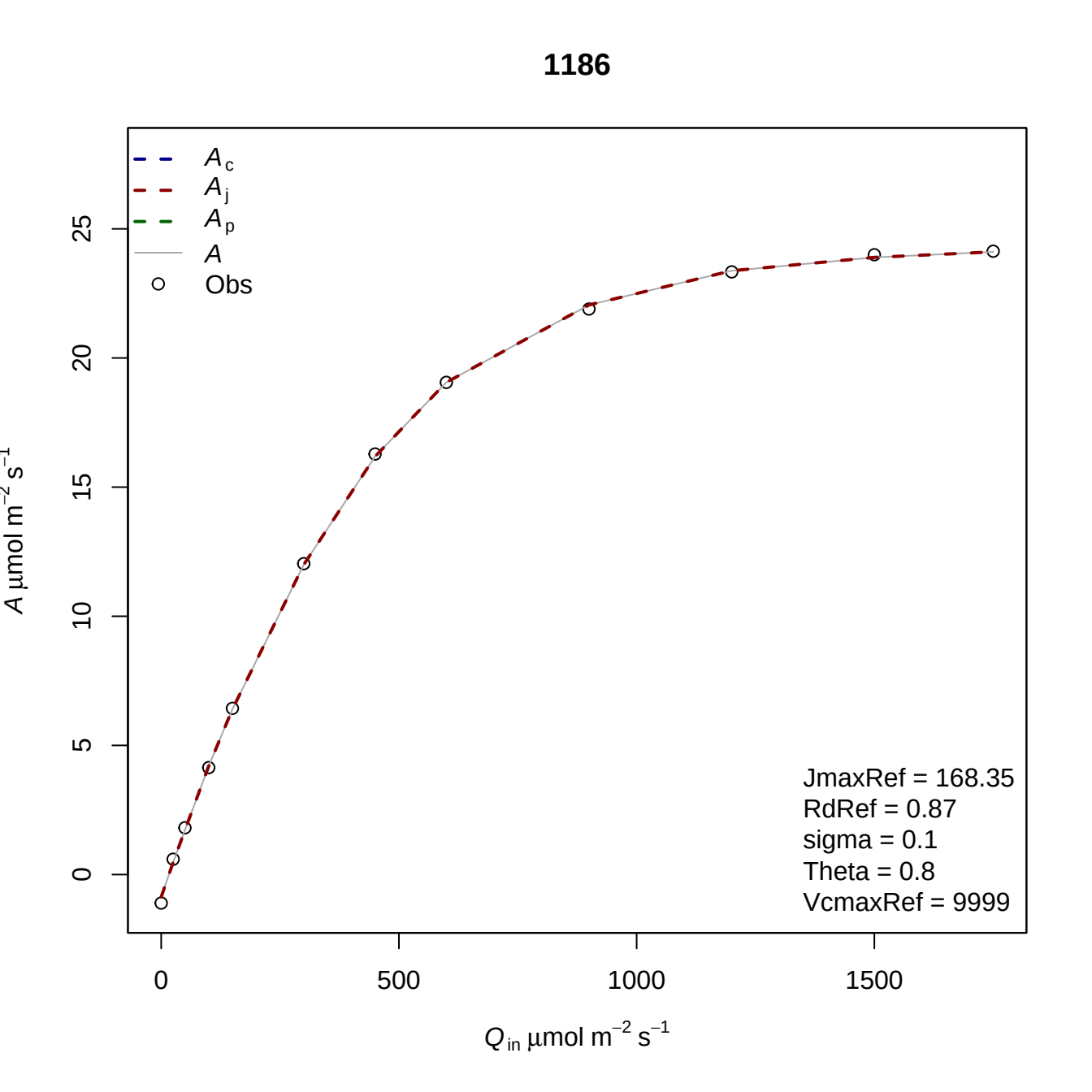

1181
A_c
A_j
A_p
A
Obs
JmaxRef = 173060402.97
RdRef = 2.22
sigma = 0.72
Theta = −2417431.52
VcmaxRef = 80.32
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

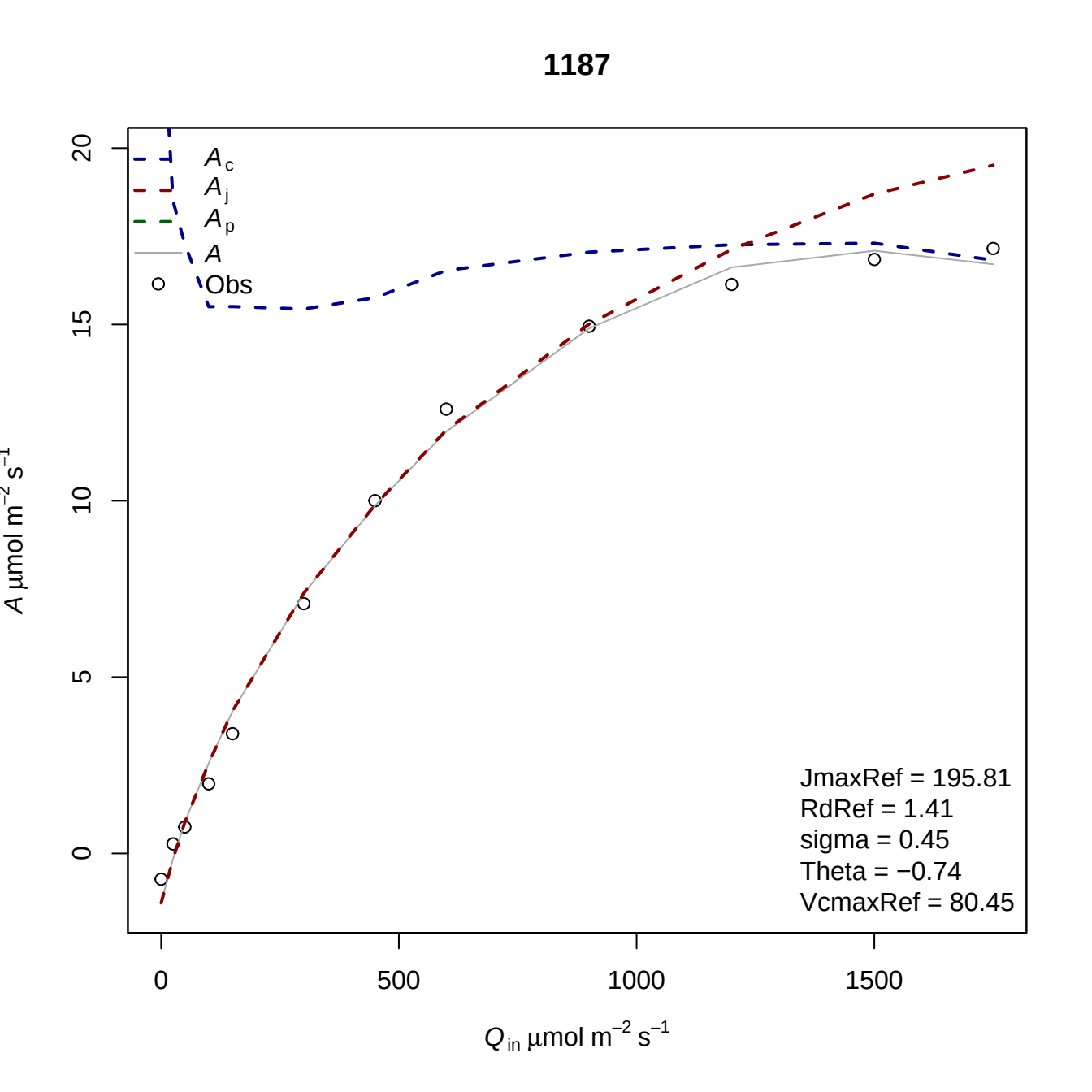

1182
A_c
A_j
A_p
A
Obs
JmaxRef = 173.43
RdRef = 1.57
sigma = 0.35
Theta = 0.85
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

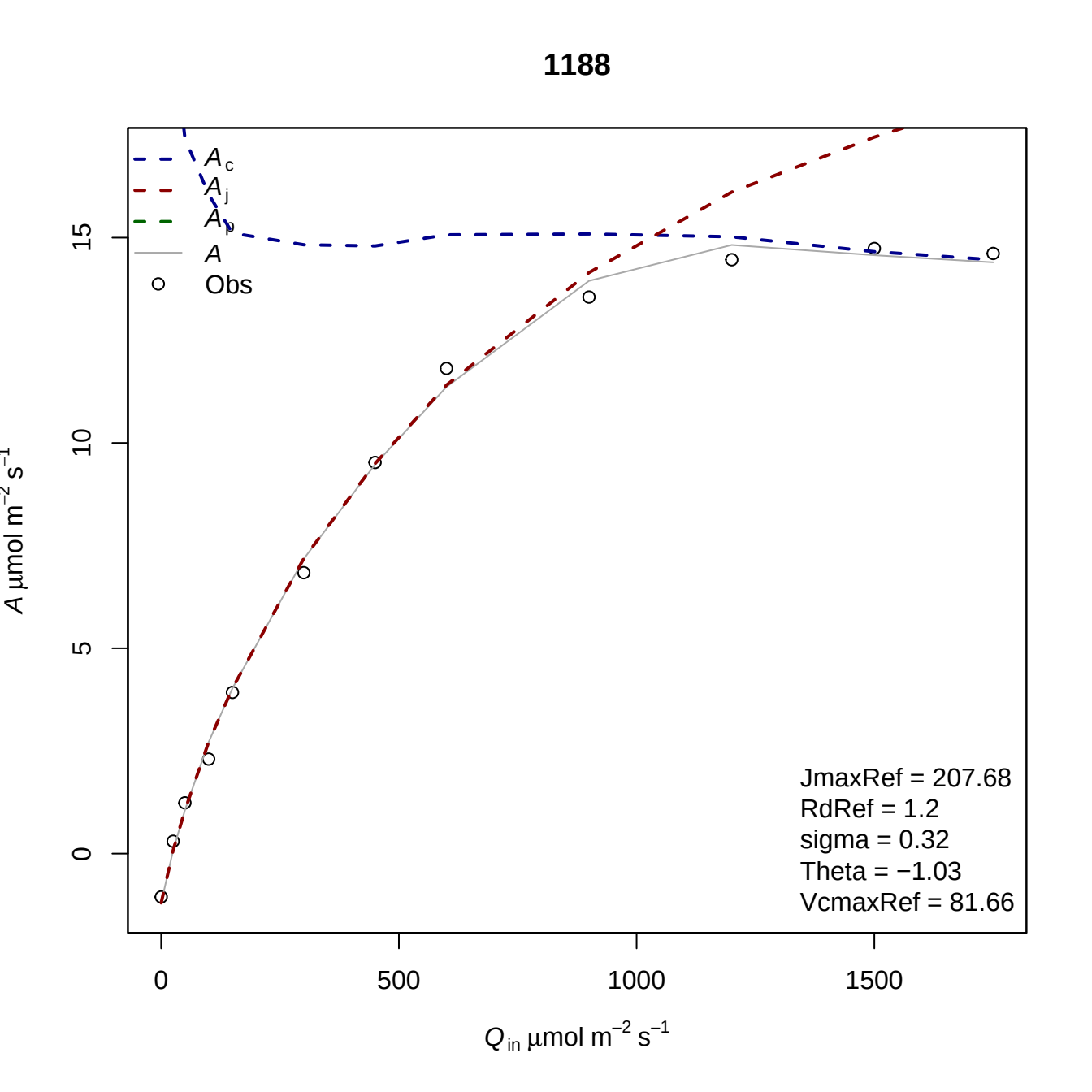

1183
A_c
A_j
A_p
A
Obs
JmaxRef = 207478056.64
RdRef = 2.16
sigma = 0.5
Theta = −1536369.13
VcmaxRef = 84.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

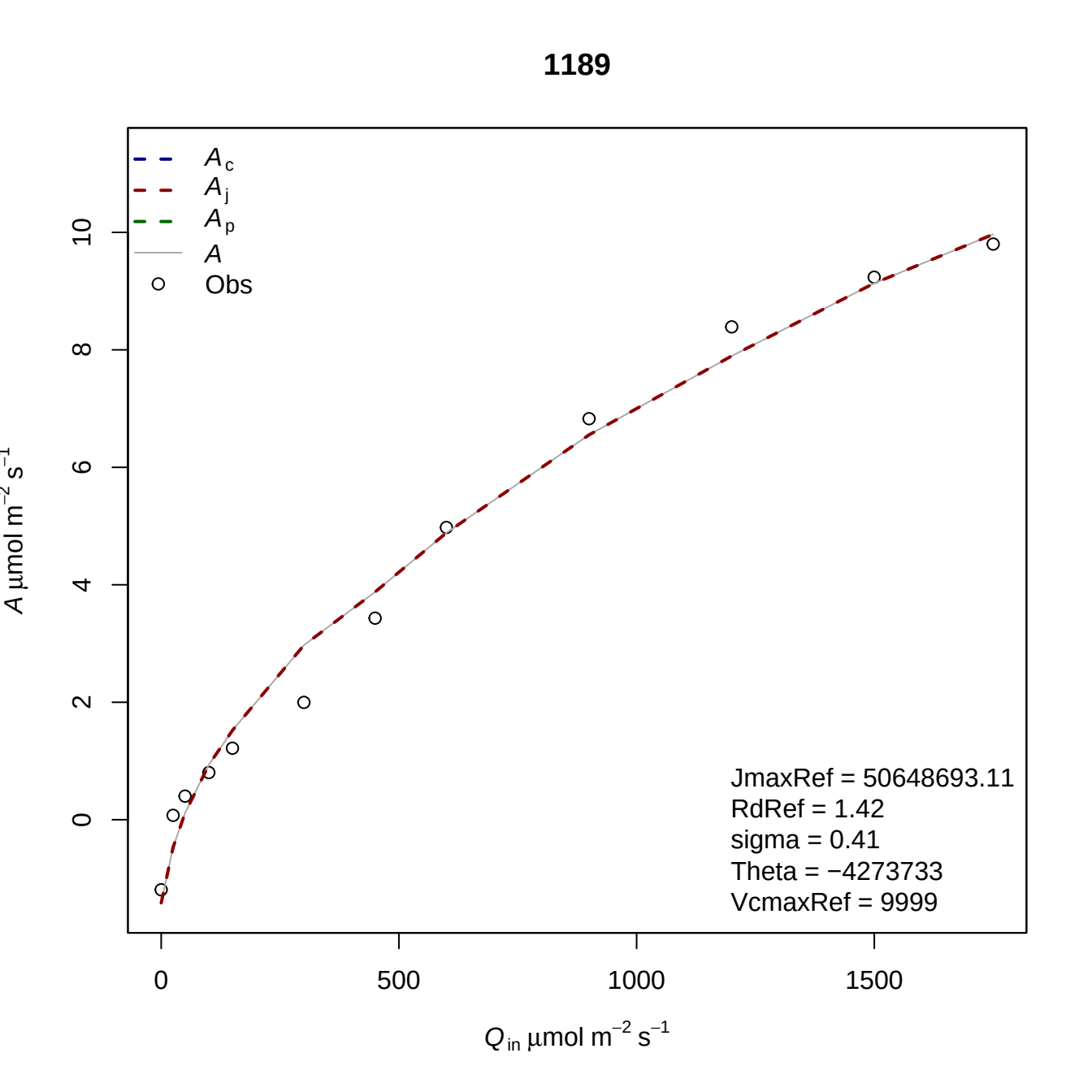

1184
A_c
A_j
A_p
A
Obs
JmaxRef = 211.71
RdRef = 5.06
sigma = 0.54
Theta = 0.48
VcmaxRef = 110.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500

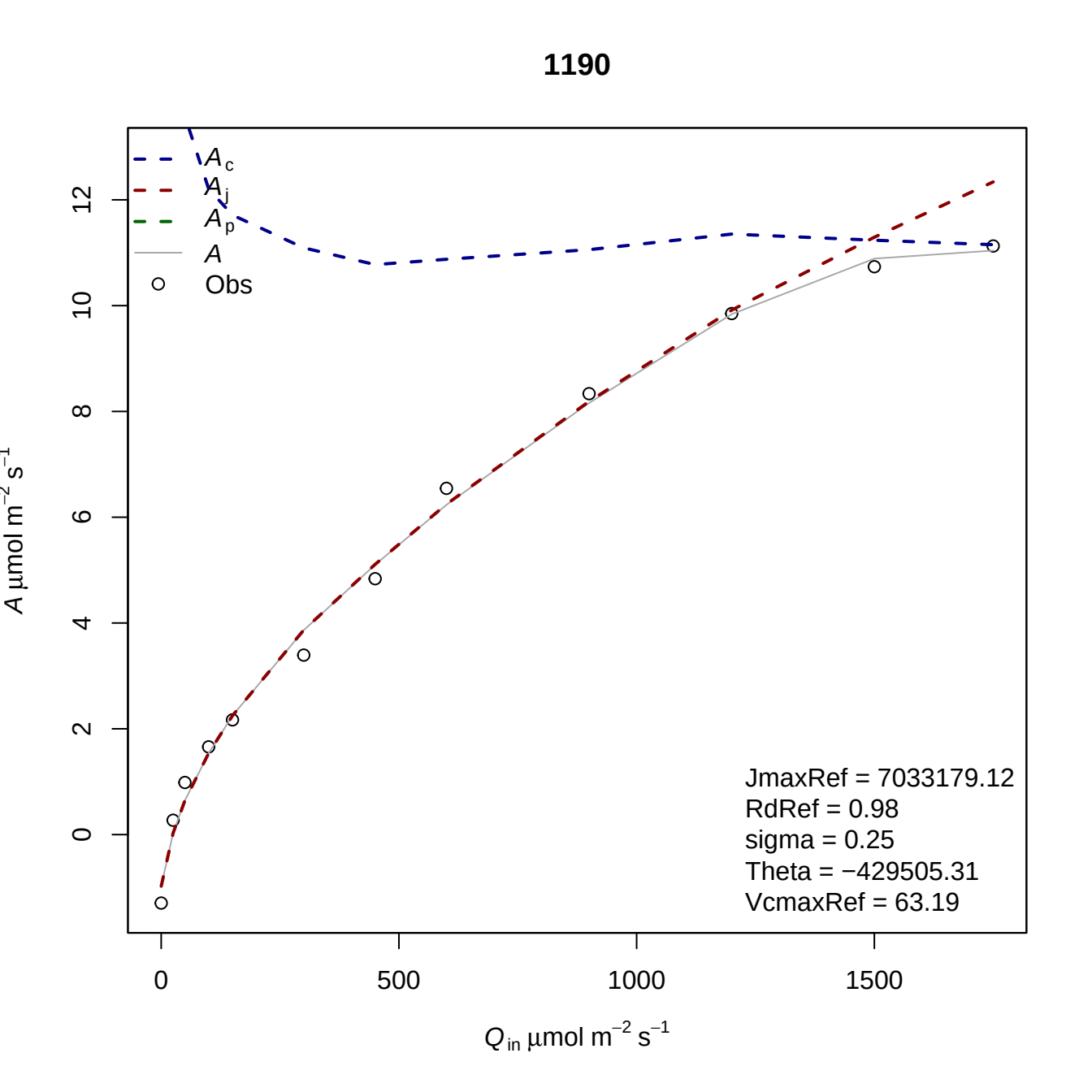

1185
A_c
A_j
A_p
A
Obs
JmaxRef = 196.43
RdRef = 1.77
sigma = 0.43
Theta = 0.4
VcmaxRef = 102.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

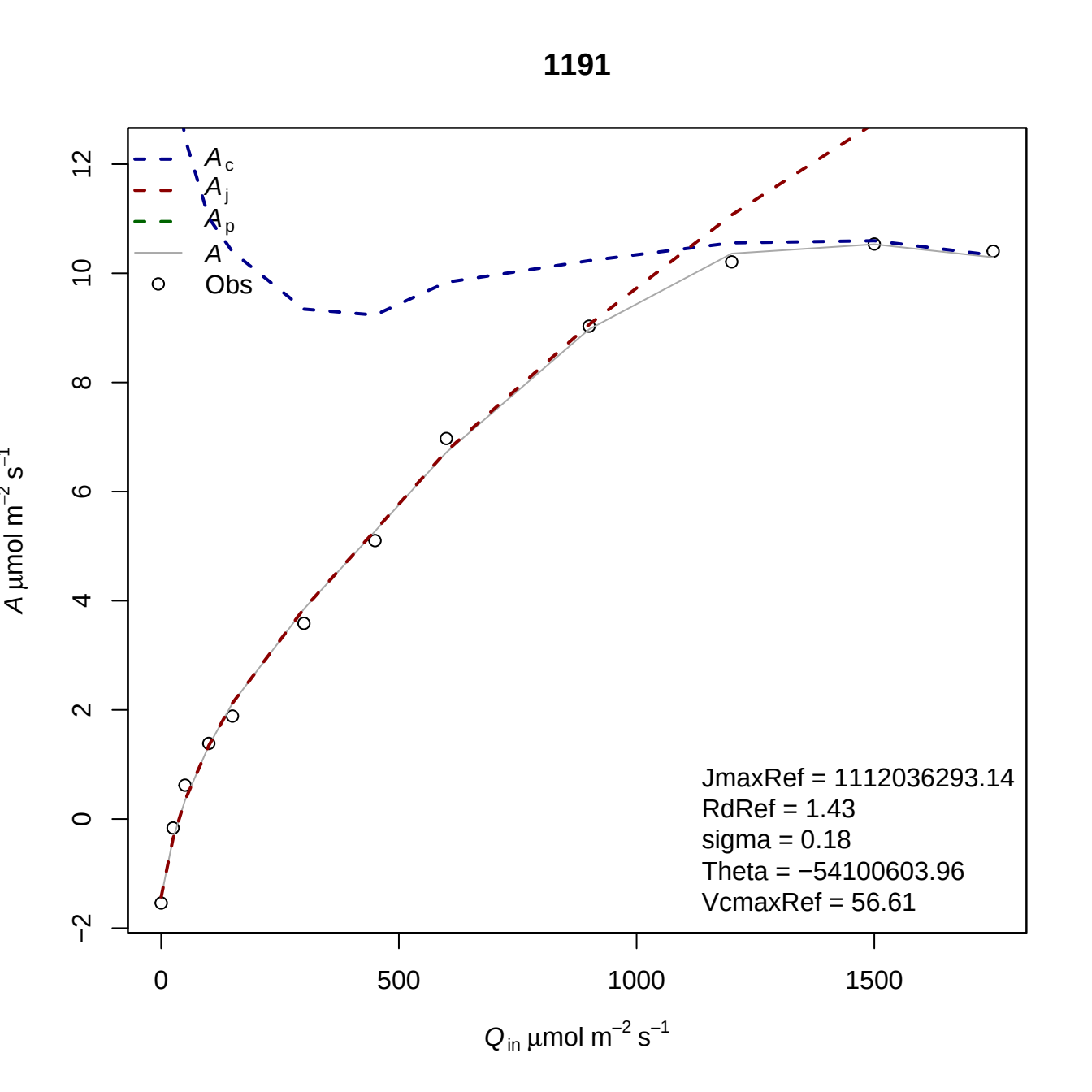

1186
A_c
A_j
A_p
A
Obs
JmaxRef = 168.35
RdRef = 0.87
sigma = 0.1
Theta = 0.8
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

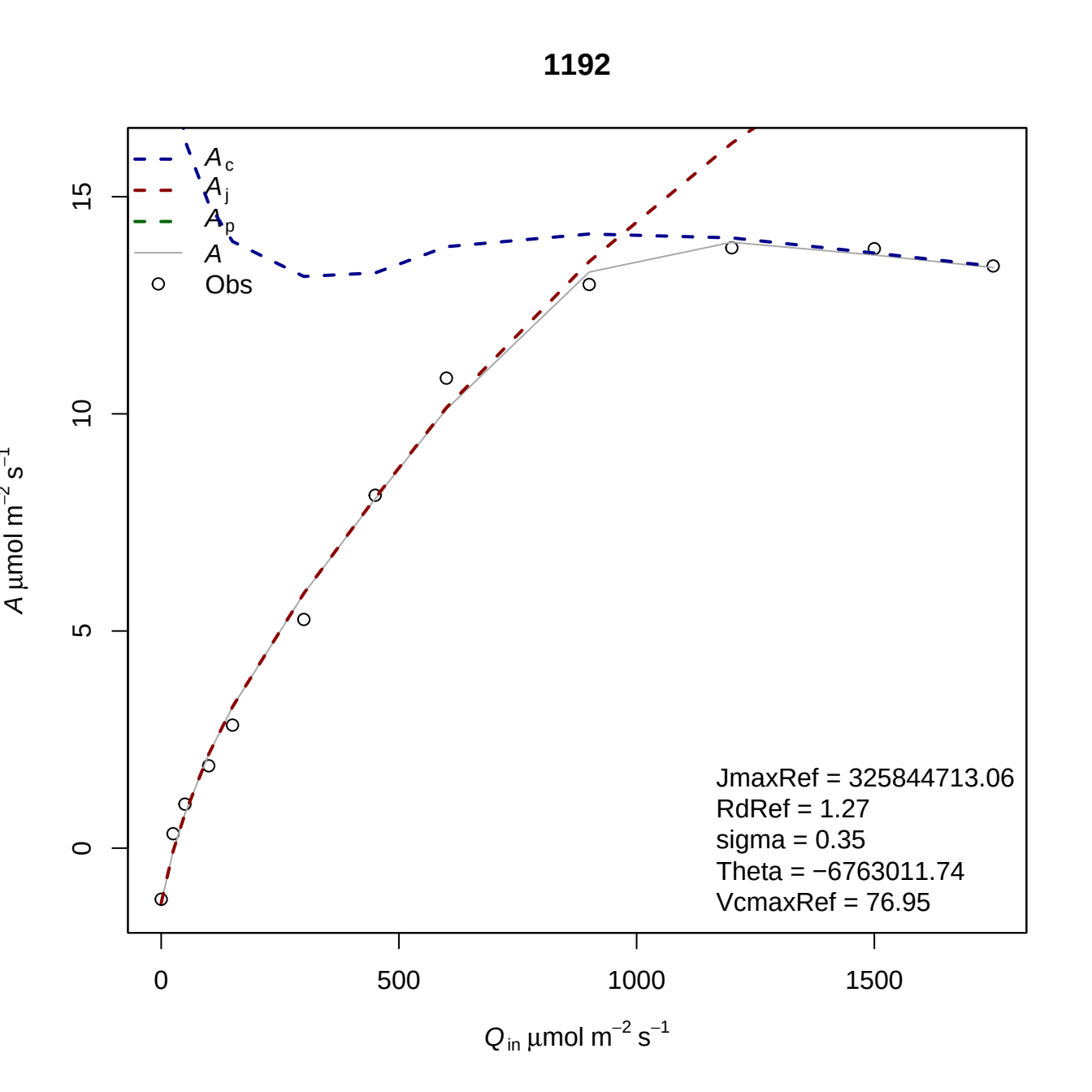

1187
A_c
A_j
A_p
A
Obs
JmaxRef = 195.81
RdRef = 1.41
sigma = 0.45
Theta = −0.74
VcmaxRef = 80.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

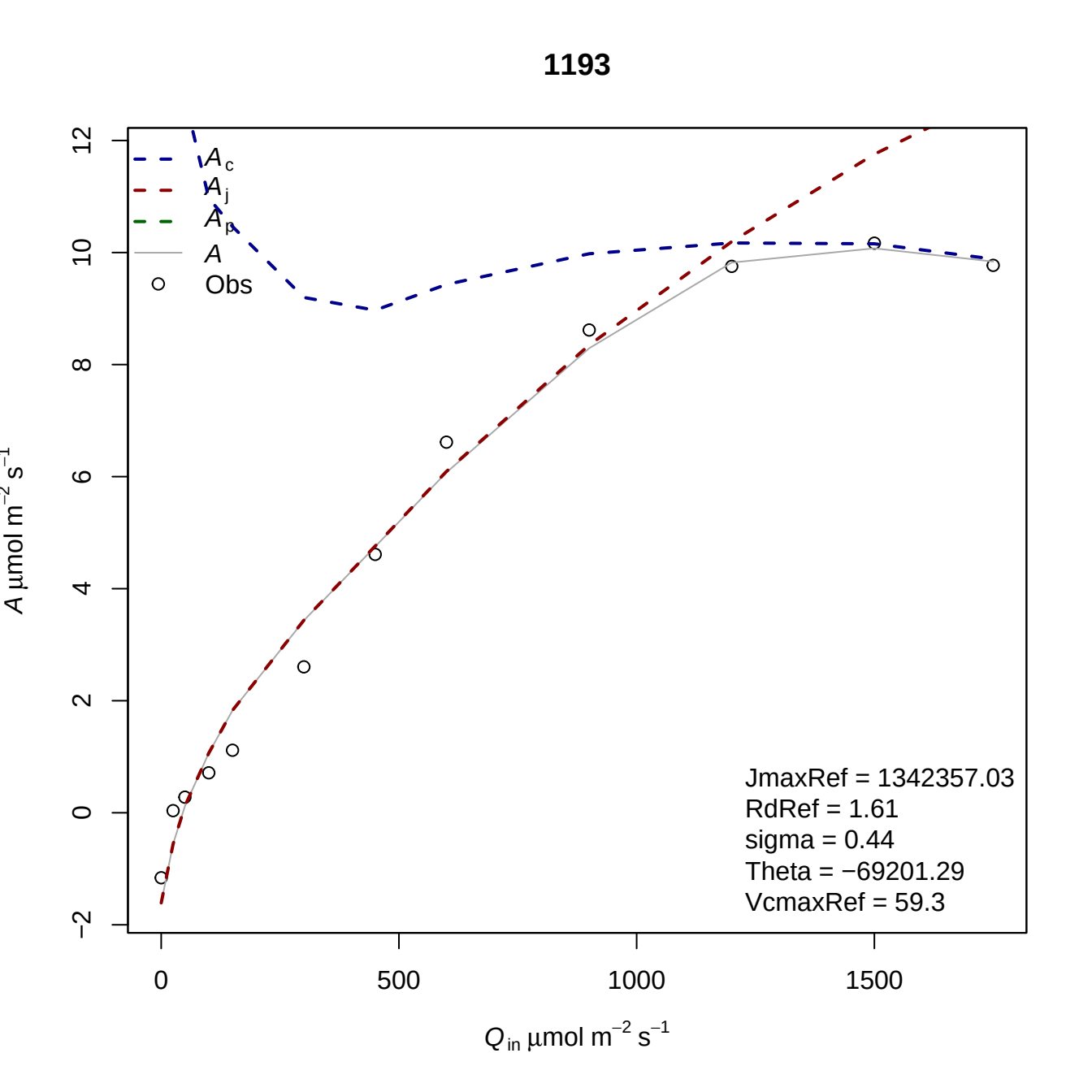

1188
A_c
A_j
A_p
A
Obs
JmaxRef = 207.68
RdRef = 1.2
sigma = 0.32
Theta = −1.03
VcmaxRef = 81.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

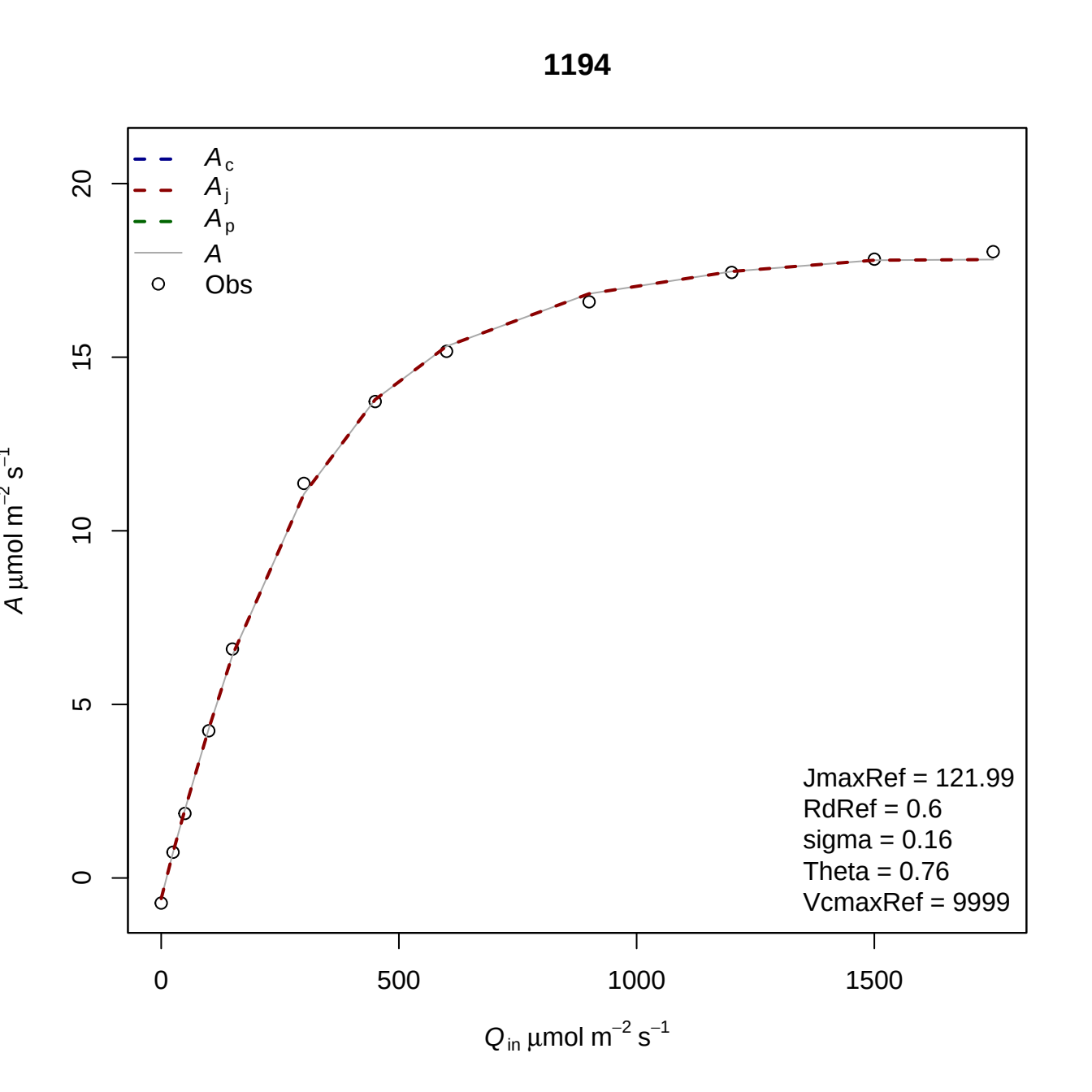

1189
A_c
A_j
A_p
A
Obs
JmaxRef = 50648693.11
RdRef = 1.42
sigma = 0.41
Theta = −4273733
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

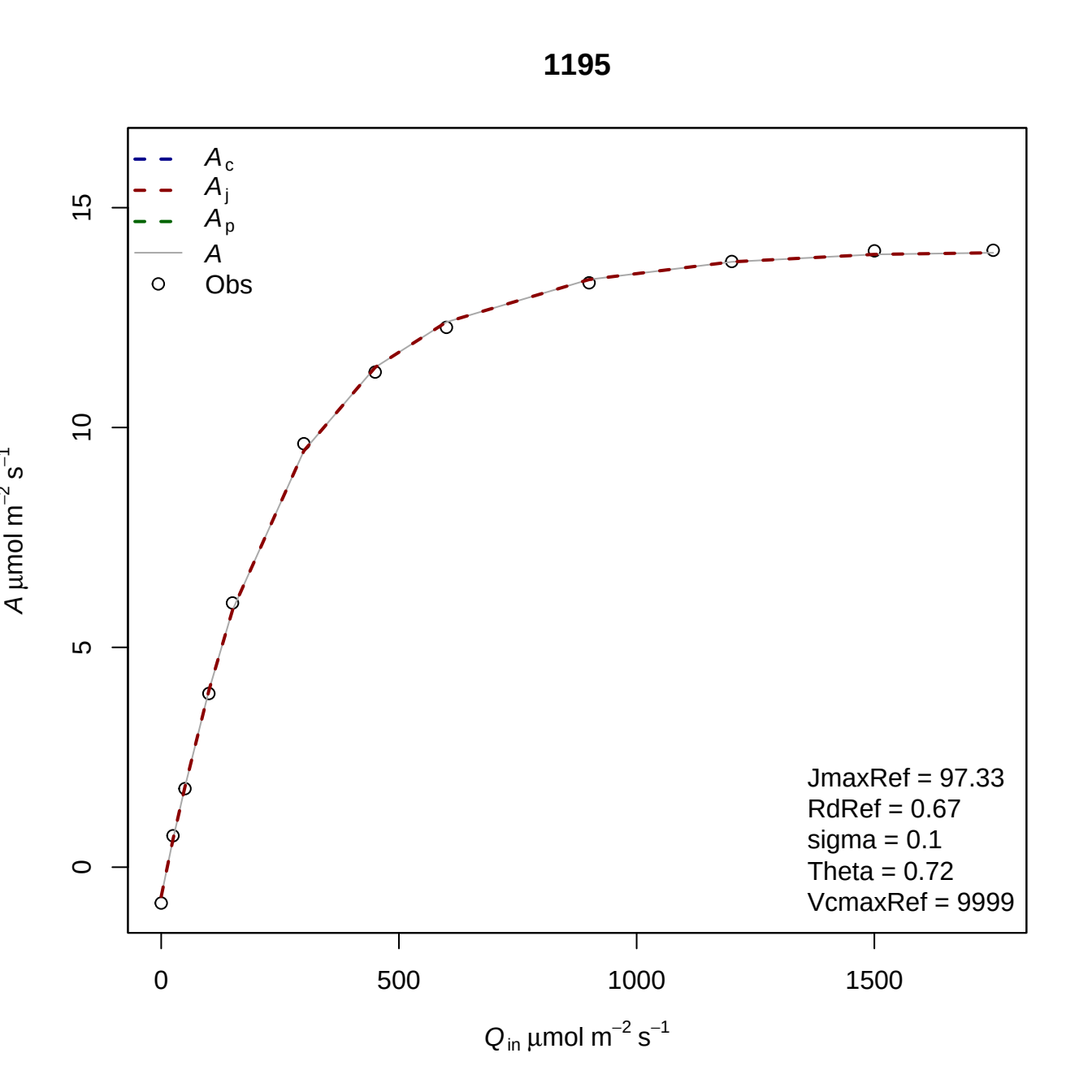

1190
A_c
A_j
A_p
A
Obs
JmaxRef = 7033179.12
RdRef = 0.98
sigma = 0.25
Theta = −429505.31
VcmaxRef = 63.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

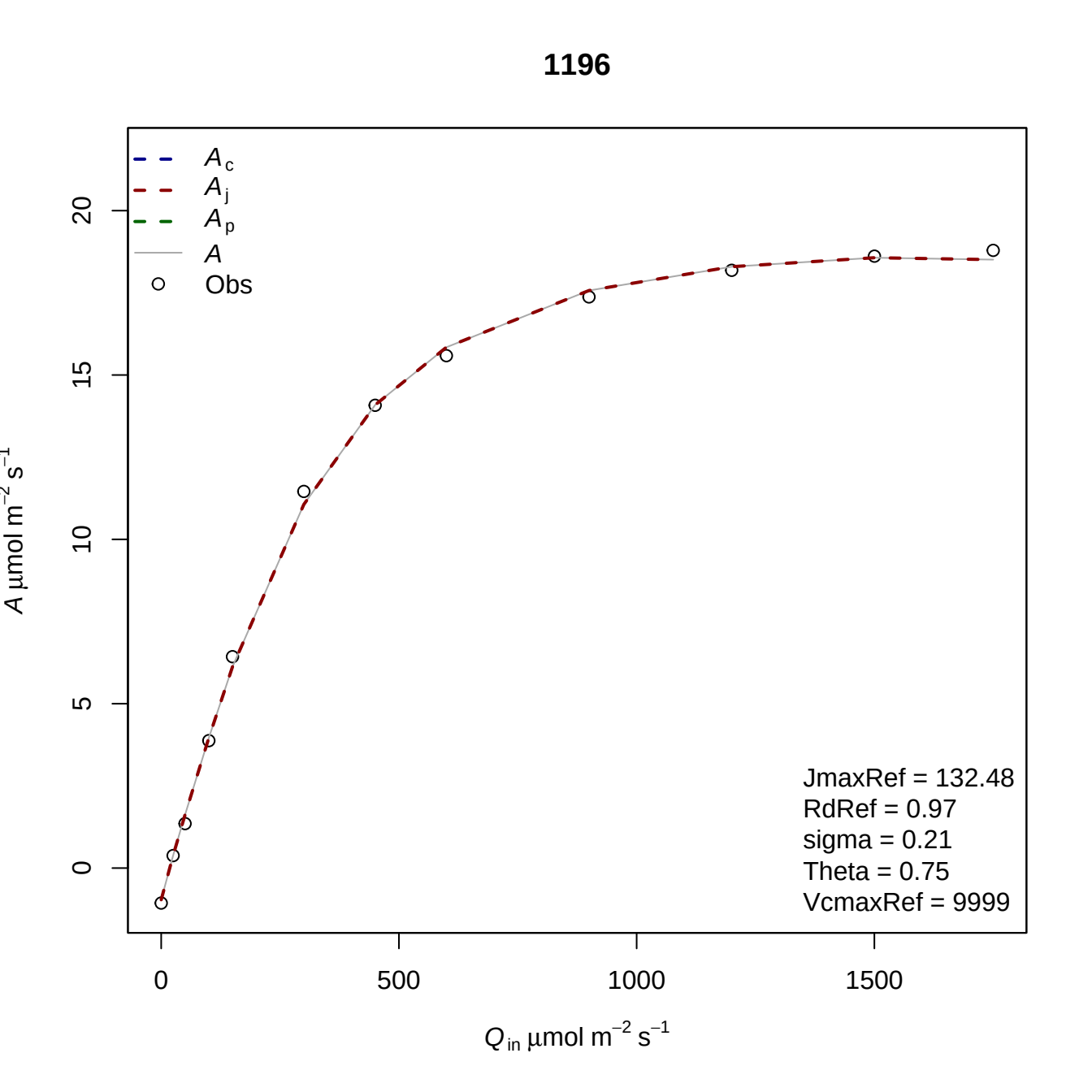

1191
A_c
A_j
A_p
A
Obs
JmaxRef = 1112036293.14
RdRef = 1.43
sigma = 0.18
Theta = −54100603.96
VcmaxRef = 56.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

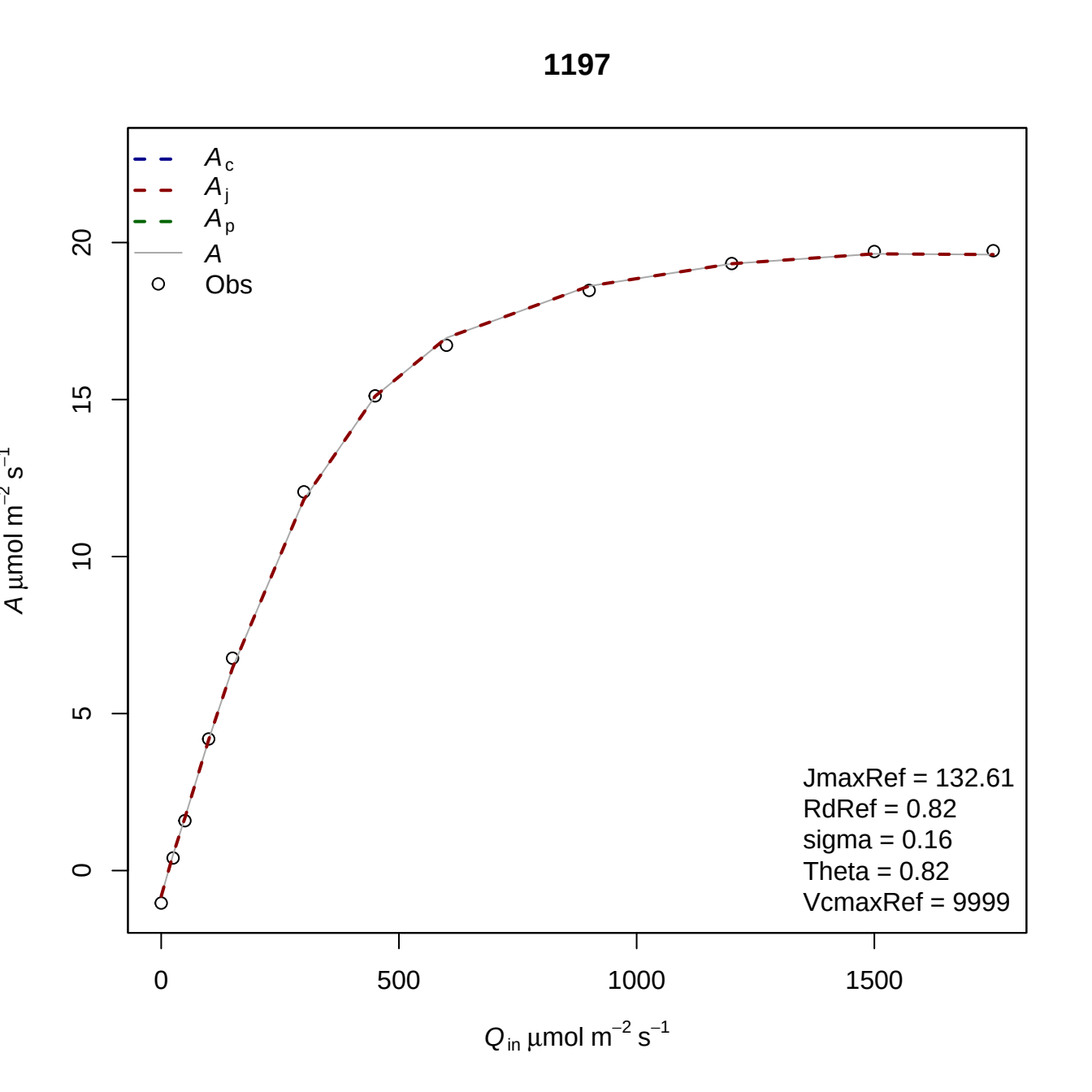

1192
A_c
A_j
A_p
A
Obs
JmaxRef = 325844713.06
RdRef = 1.27
sigma = 0.35
Theta = −6763011.74
VcmaxRef = 76.95
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

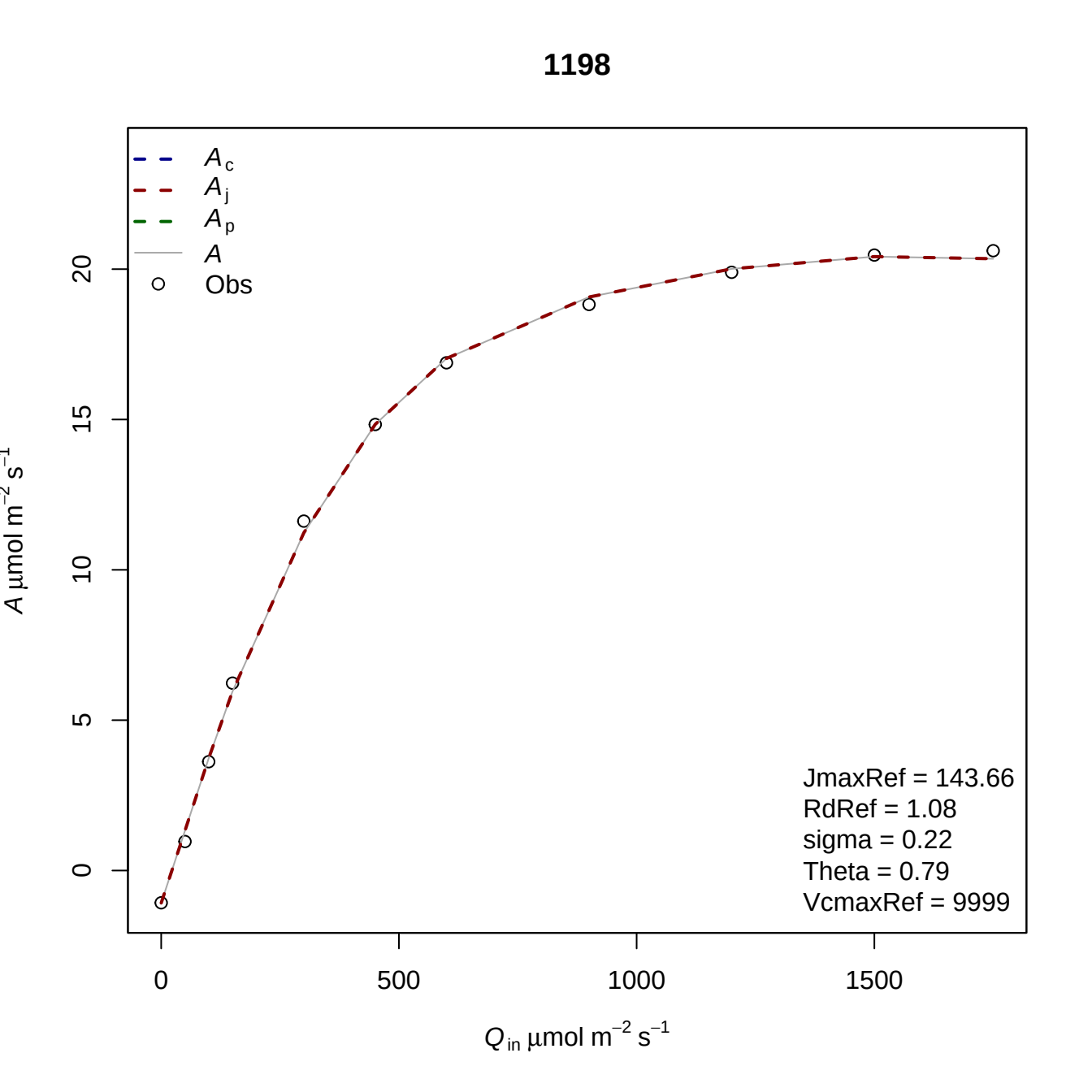

1193
A_c
A_j
A_p
A
Obs
JmaxRef = 1342357.03
RdRef = 1.61
sigma = 0.44
Theta = −69201.29
VcmaxRef = 59.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

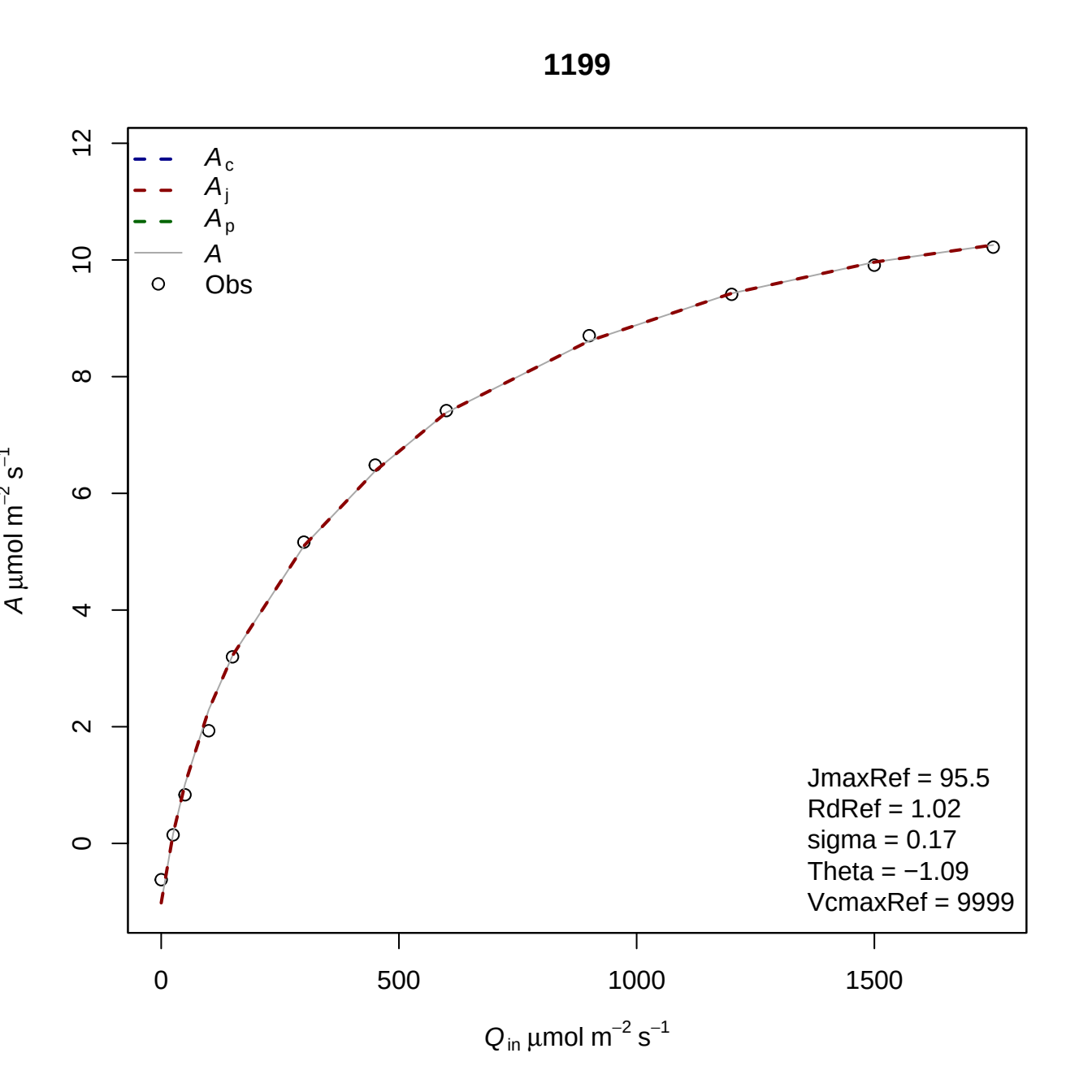

1194
A_c
A_j
A_p
A
Obs
JmaxRef = 121.99
RdRef = 0.6
sigma = 0.16
Theta = 0.76
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

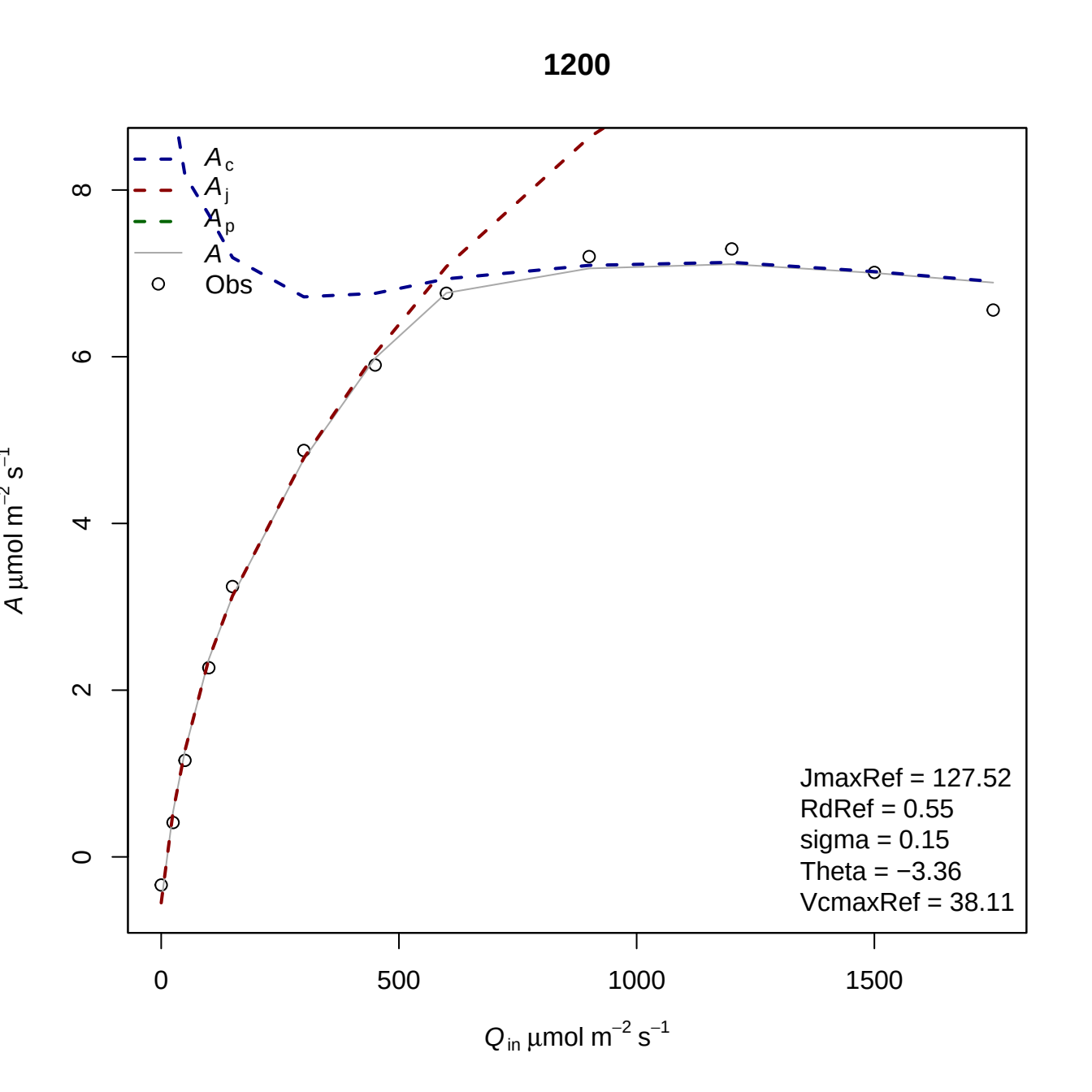

1195
A_c
A_j
A_p
A
Obs
JmaxRef = 97.33
RdRef = 0.67
sigma = 0.1
Theta = 0.72
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

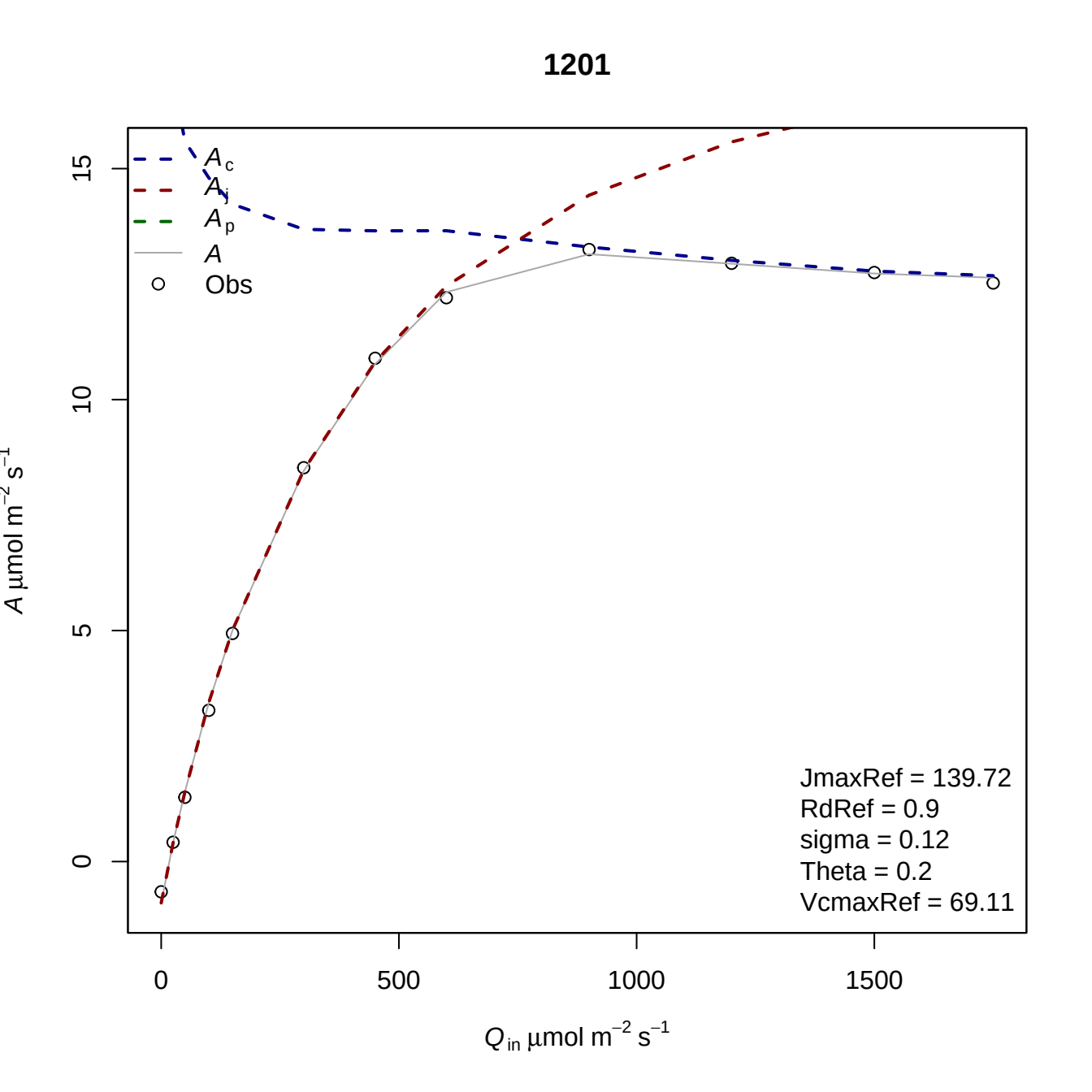

1196
A_c
A_j
A_p
A
Obs
JmaxRef = 132.48
RdRef = 0.97
sigma = 0.21
Theta = 0.75
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

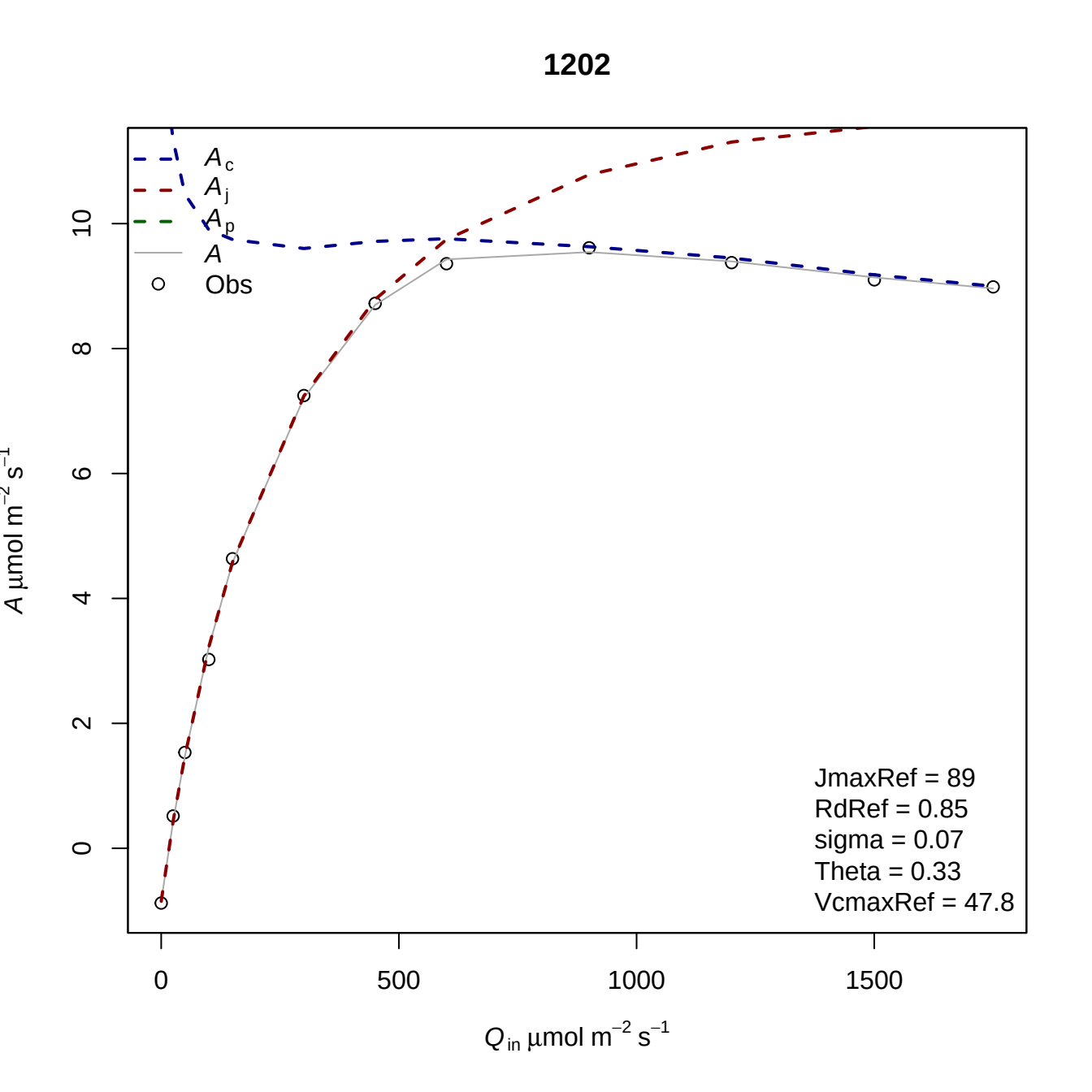

1197
A_c
A_j
A_p
A
Obs
JmaxRef = 132.61
RdRef = 0.82
sigma = 0.16
Theta = 0.82
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

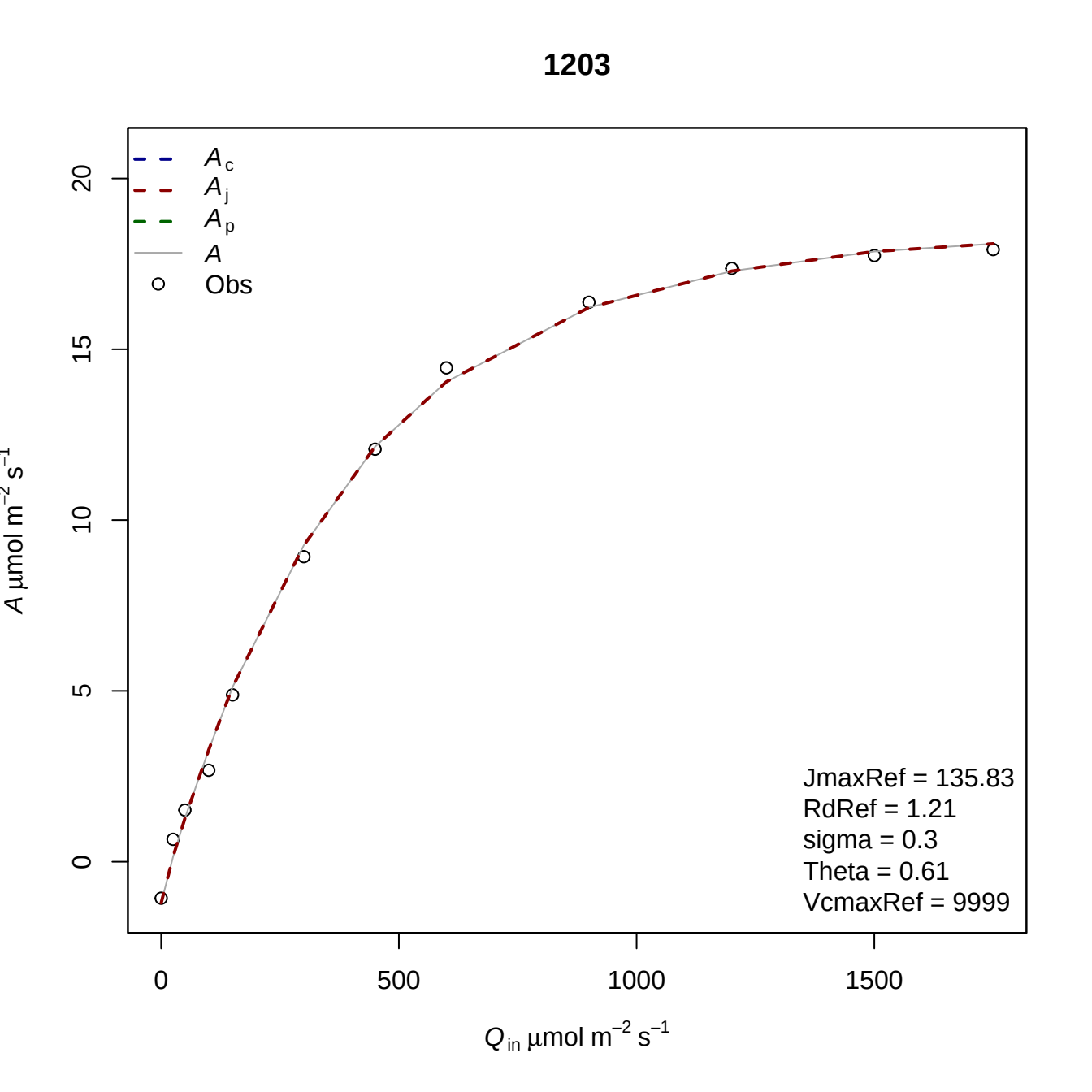

1198
A_c
A_j
A_p
A
Obs
JmaxRef = 143.66
RdRef = 1.08
sigma = 0.22
Theta = 0.79
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

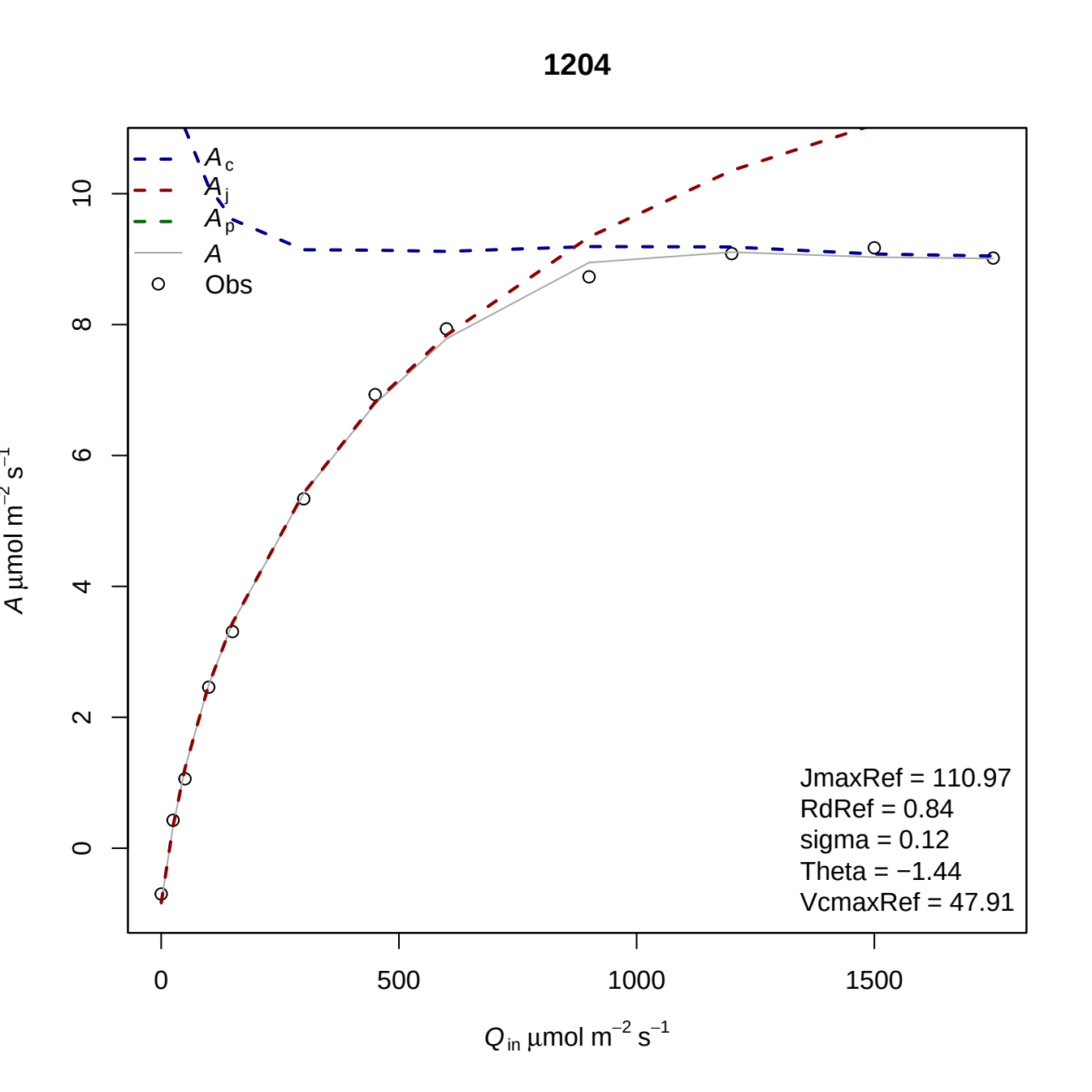

1199
A_c
A_j
A_p
A
Obs
JmaxRef = 95.5
RdRef = 1.02
sigma = 0.17
Theta = −1.09
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

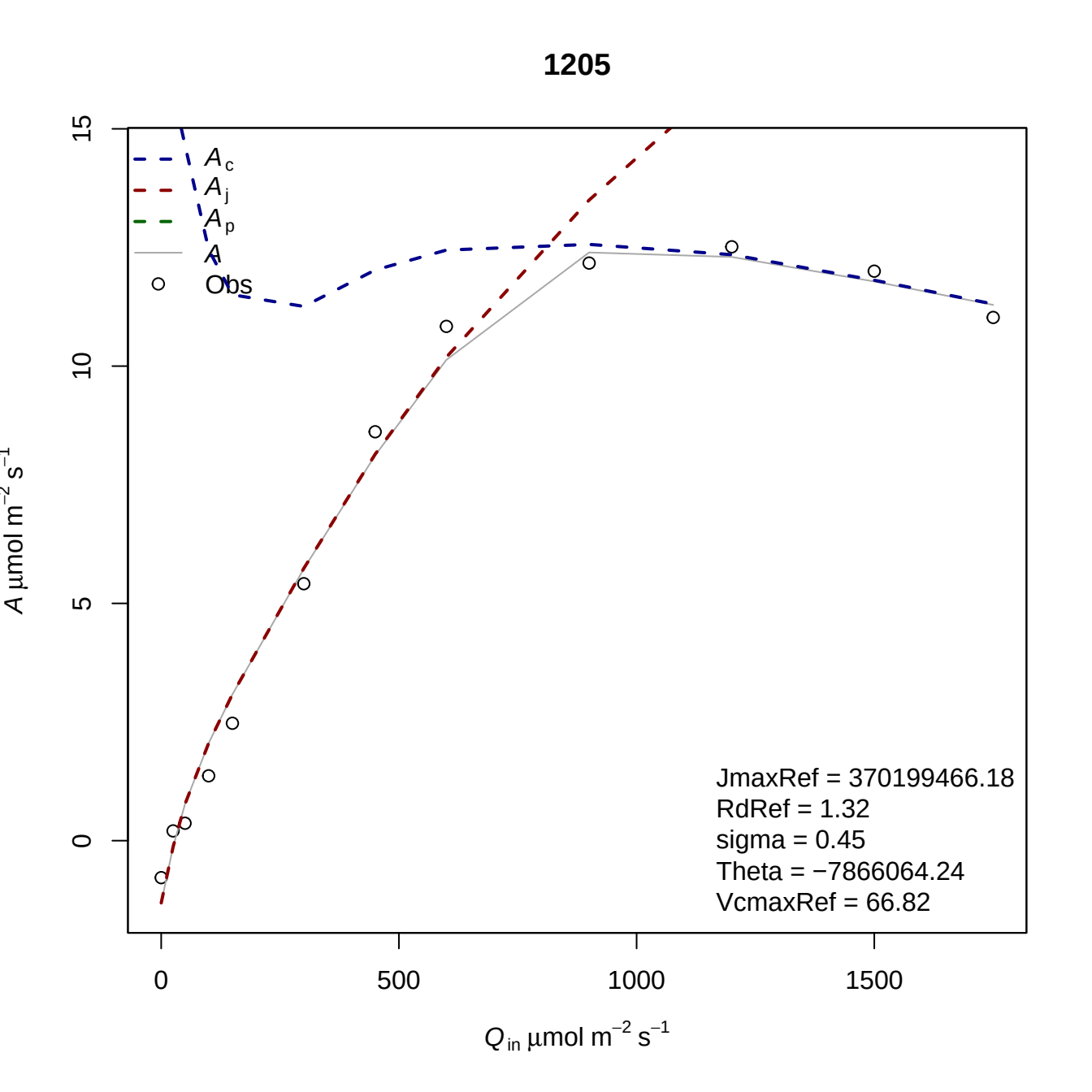

1200
A_c
A_j
A_p
A
Obs
JmaxRef = 127.52
RdRef = 0.55
sigma = 0.15
Theta = −3.36
VcmaxRef = 38.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

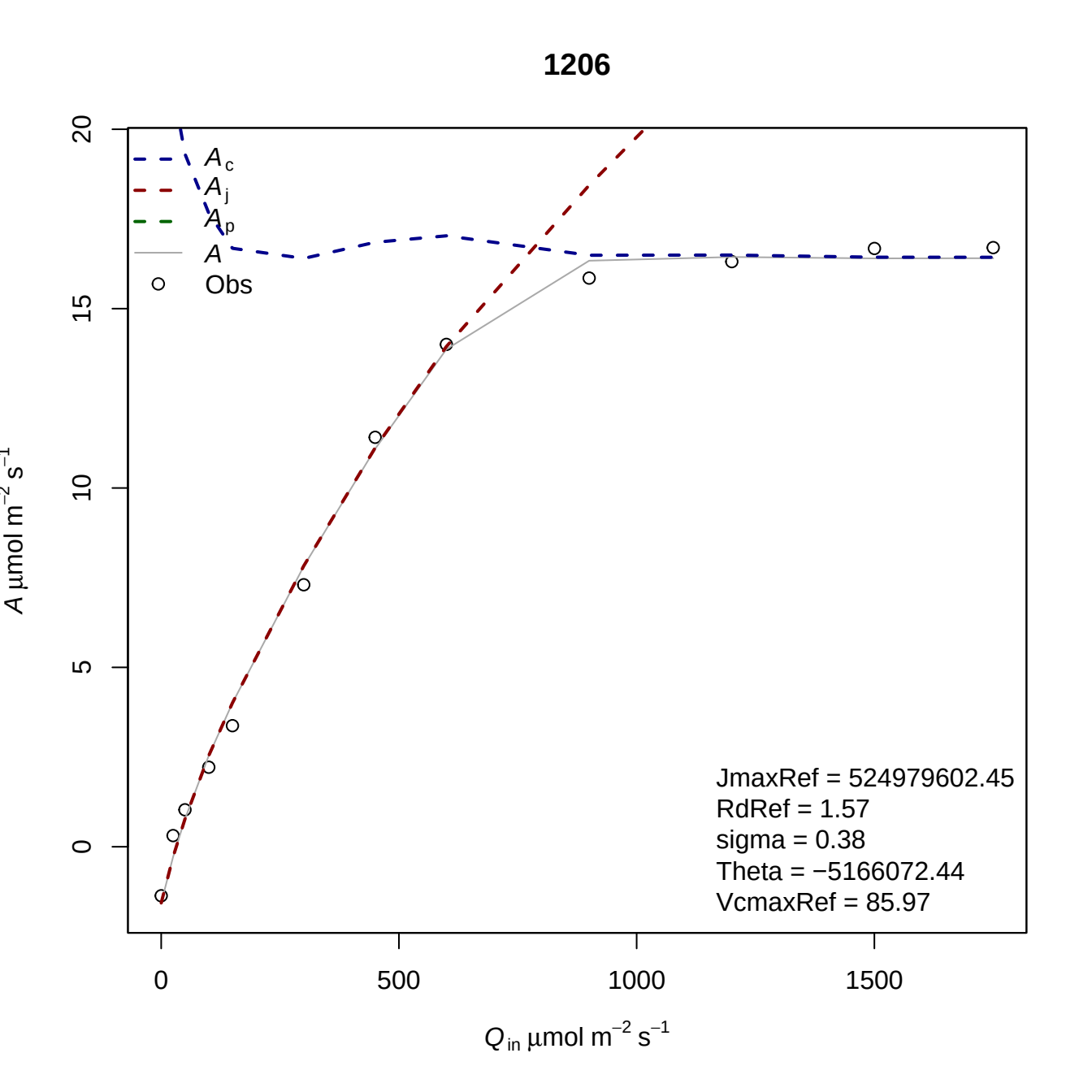

1201
A_c
A_j
A_p
A
Obs
JmaxRef = 139.72
RdRef = 0.9
sigma = 0.12
Theta = 0.2
VcmaxRef = 69.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

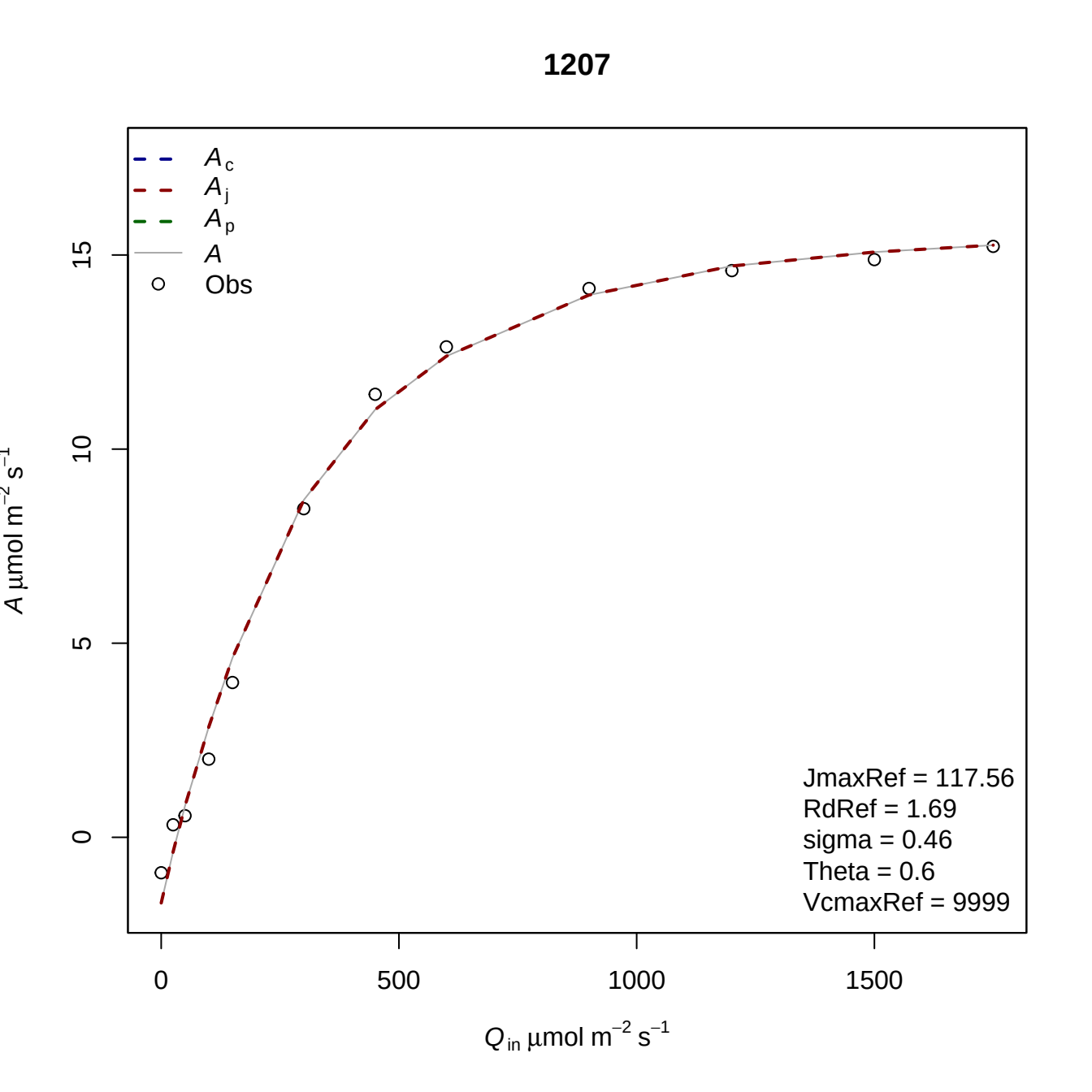

1202
A_c
A_j
A_p
A
Obs
JmaxRef = 89
RdRef = 0.85
sigma = 0.07
Theta = 0.33
VcmaxRef = 47.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

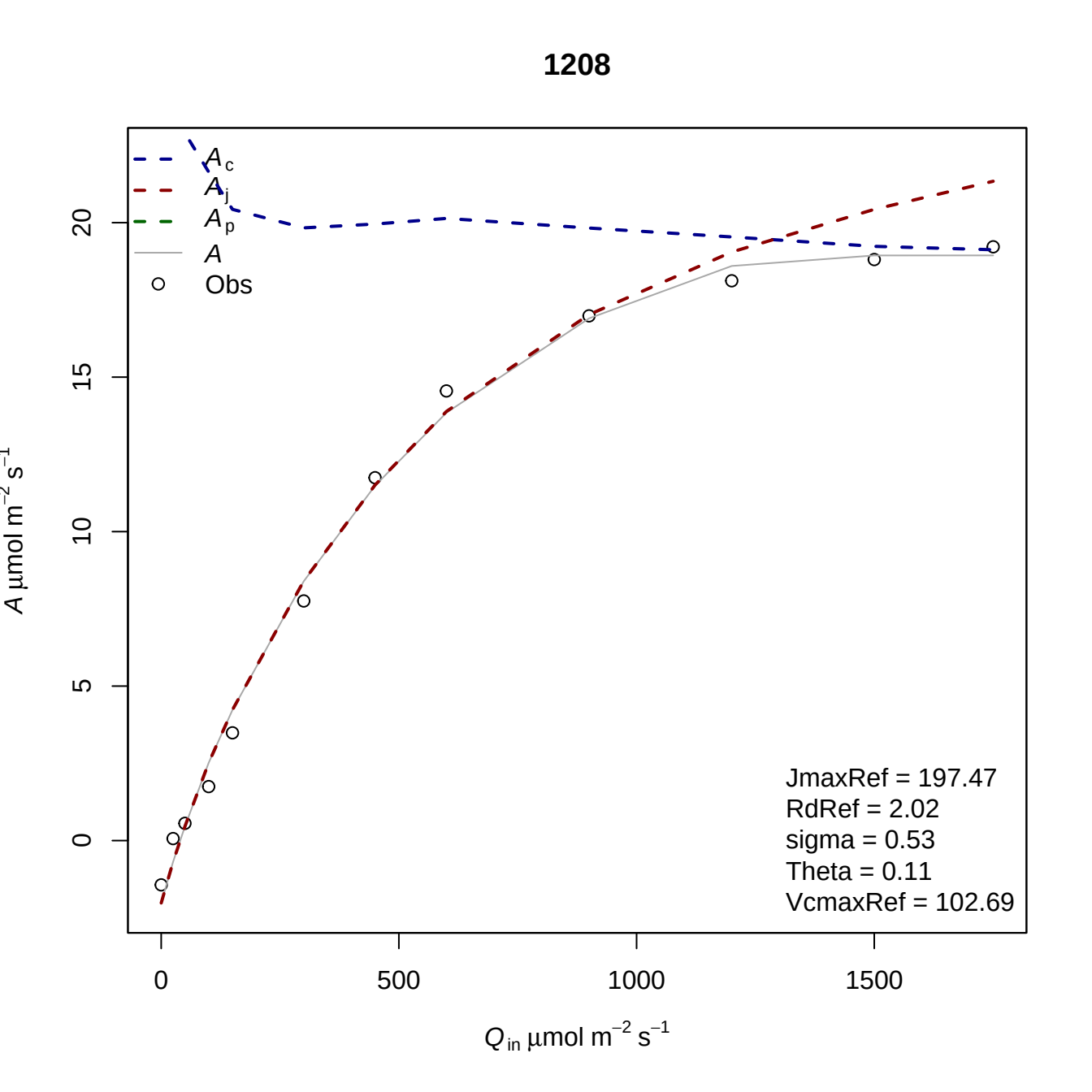

1203
A_c
A_j
A_p
A
Obs
JmaxRef = 135.83
RdRef = 1.21
sigma = 0.3
Theta = 0.61
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

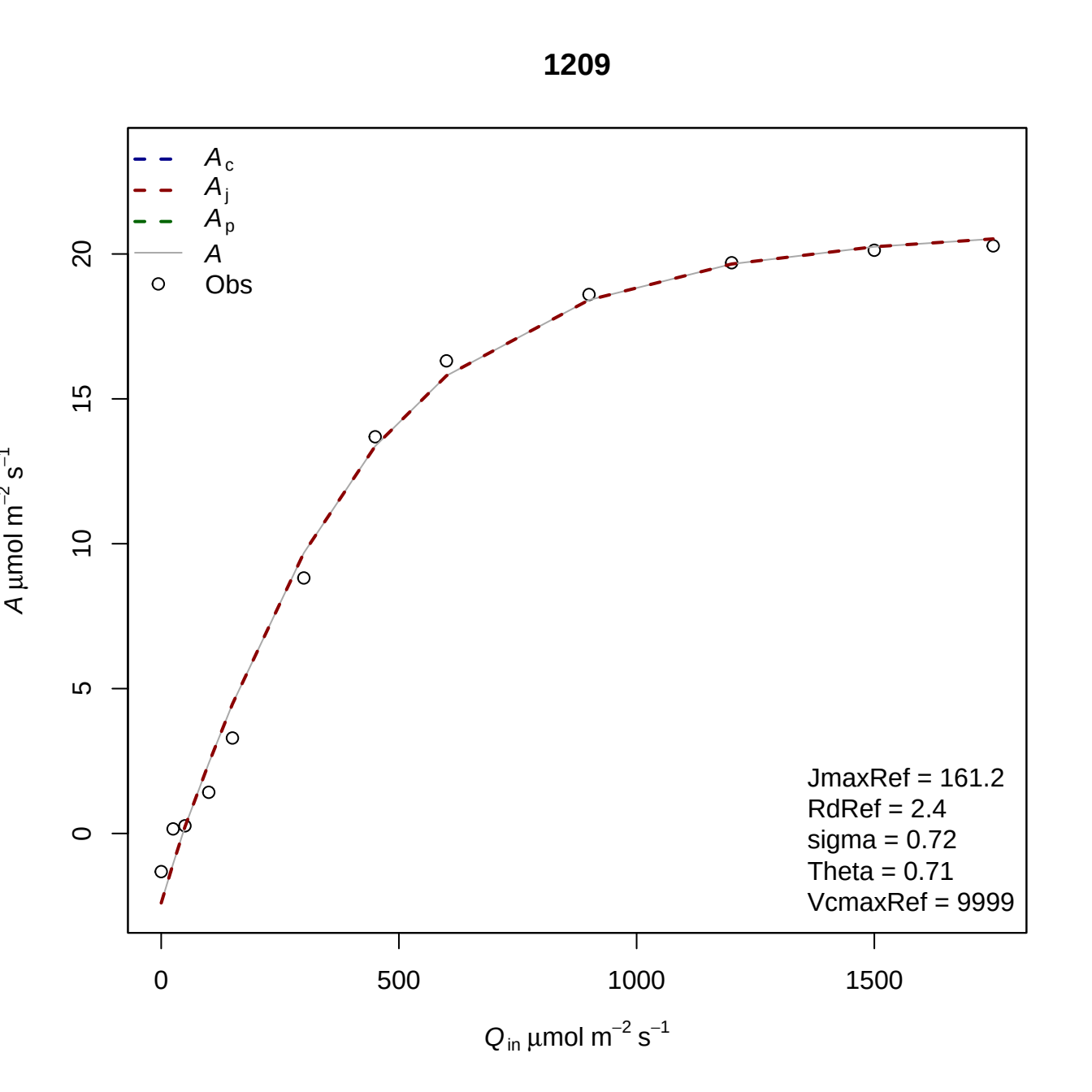

1204
A_c
A_j
A_p
A
Obs
JmaxRef = 110.97
RdRef = 0.84
sigma = 0.12
Theta = −1.44
VcmaxRef = 47.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

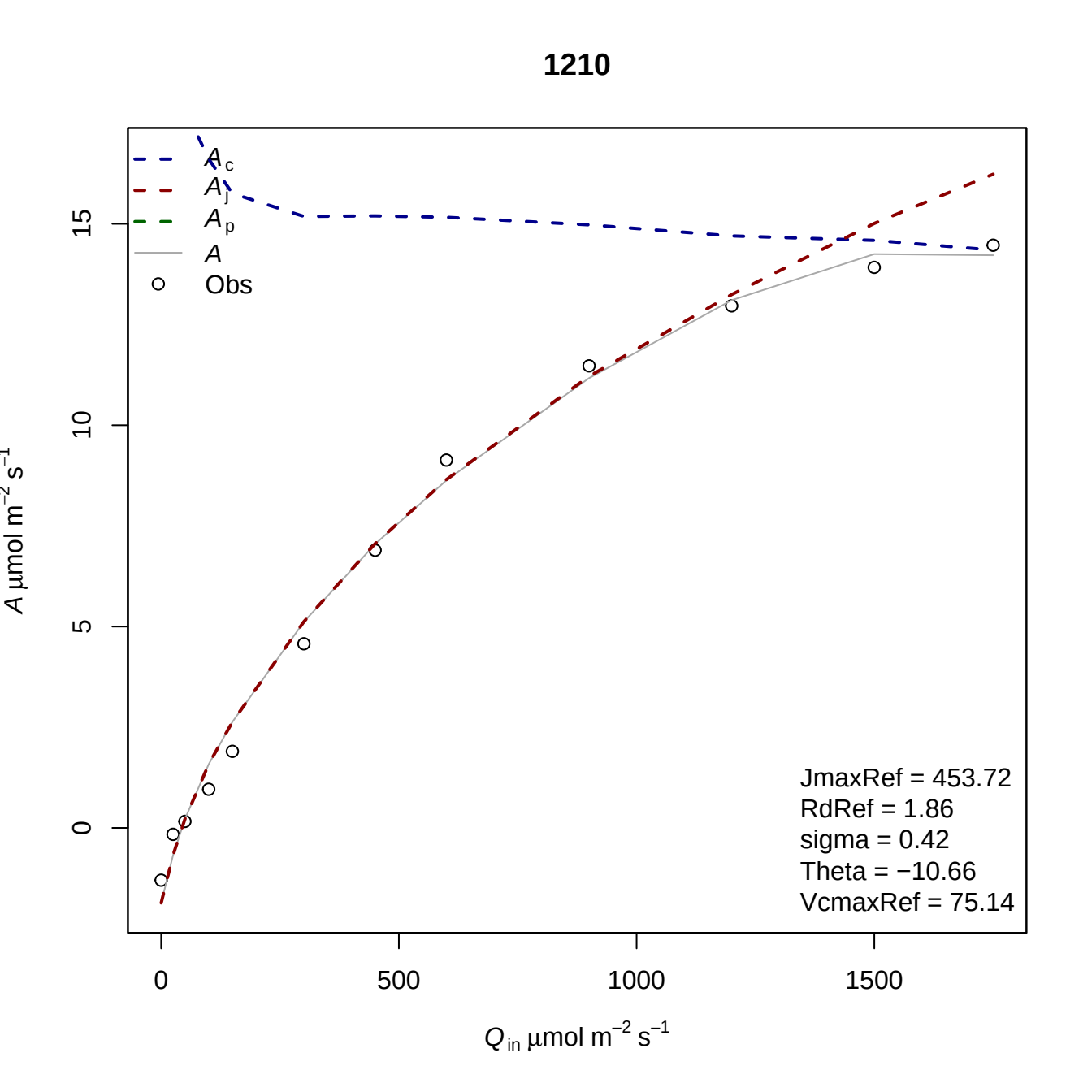

1205
A_c
A_j
A_p
A
Obs
JmaxRef = 370199466.18
RdRef = 1.32
sigma = 0.45
Theta = −7866064.24
VcmaxRef = 66.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

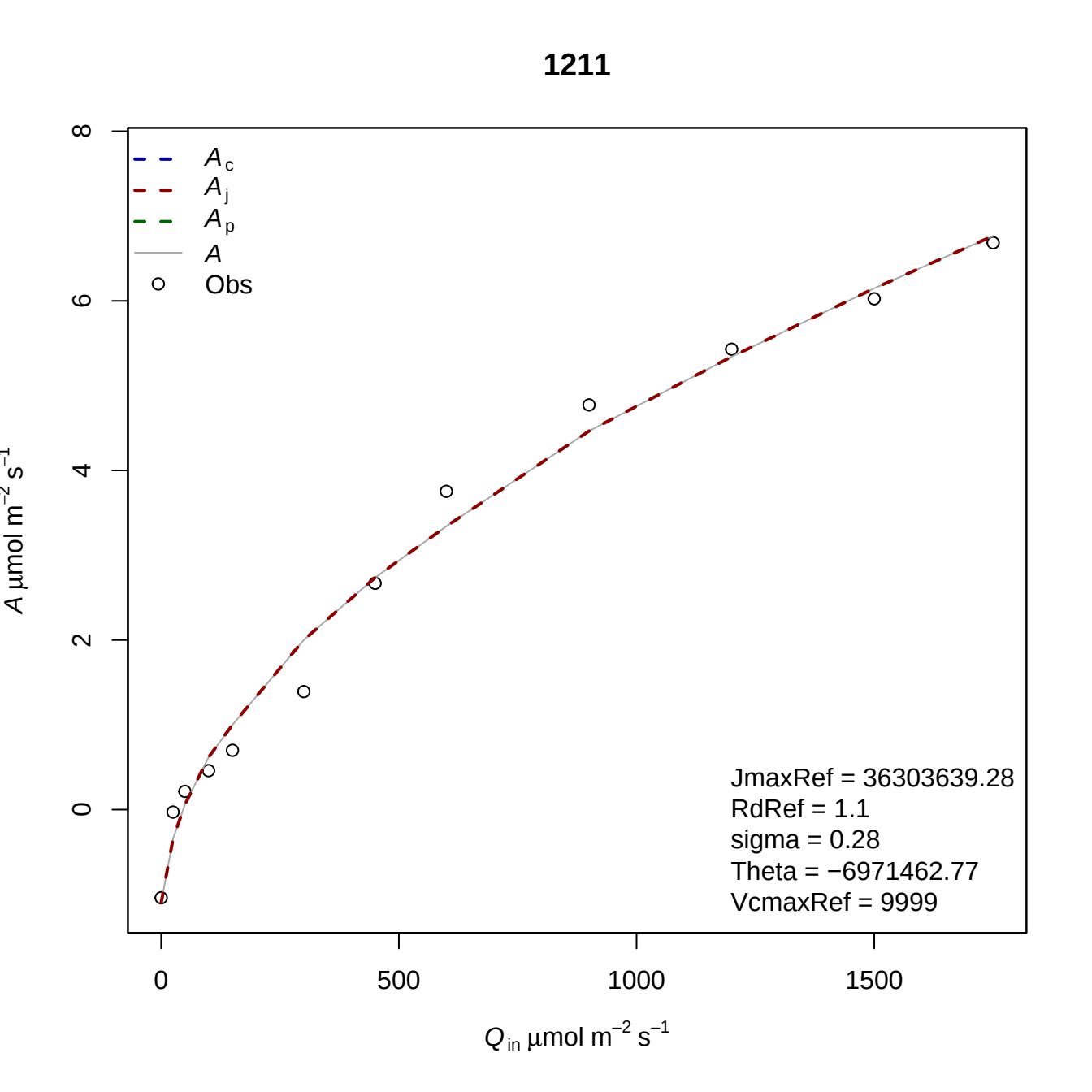

1206
A_c
A_j
A_p
A
Obs
JmaxRef = 524979602.45
RdRef = 1.57
sigma = 0.38
Theta = −5166072.44
VcmaxRef = 85.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

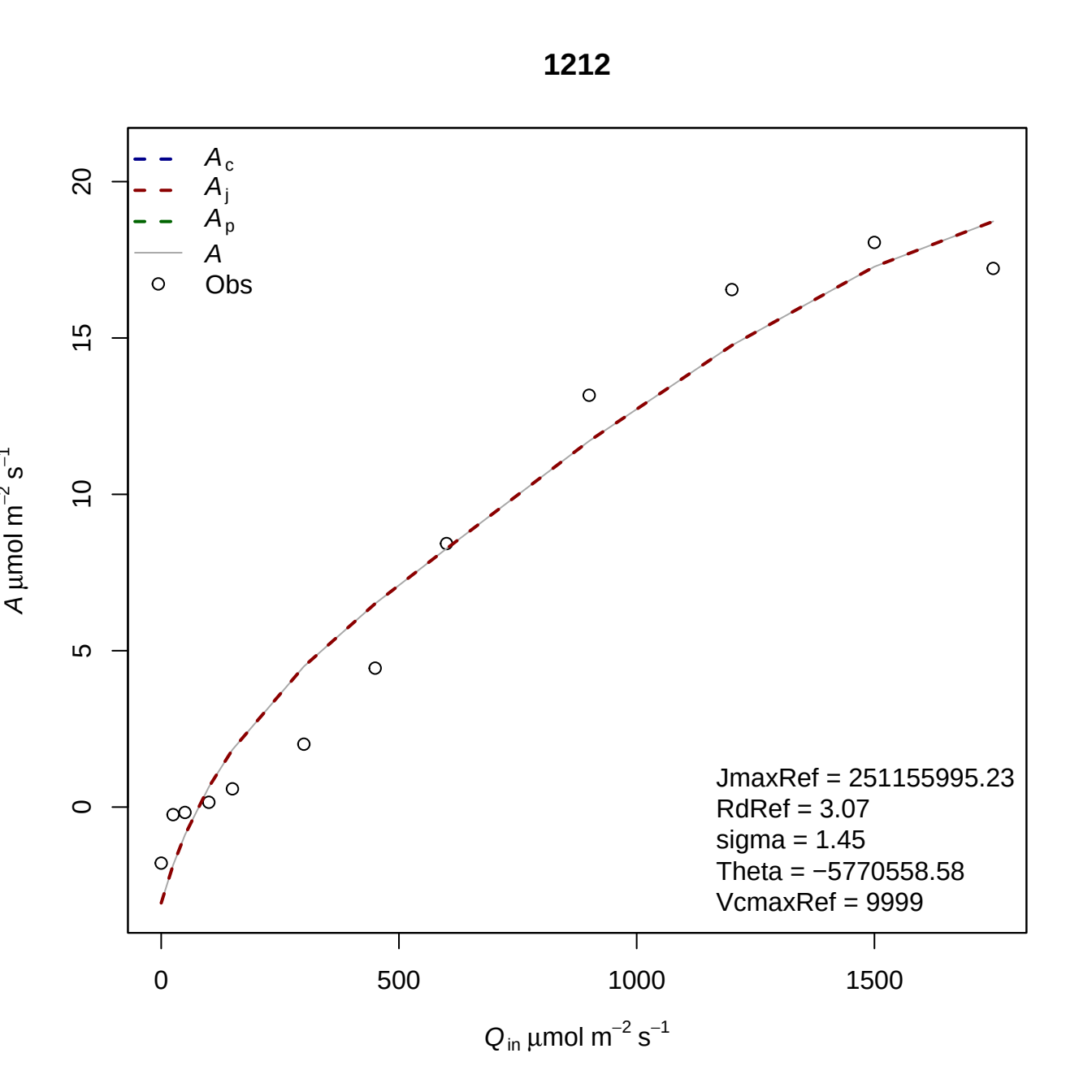

1207
A_c
A_j
A_p
A
Obs
JmaxRef = 117.56
RdRef = 1.69
sigma = 0.46
Theta = 0.6
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

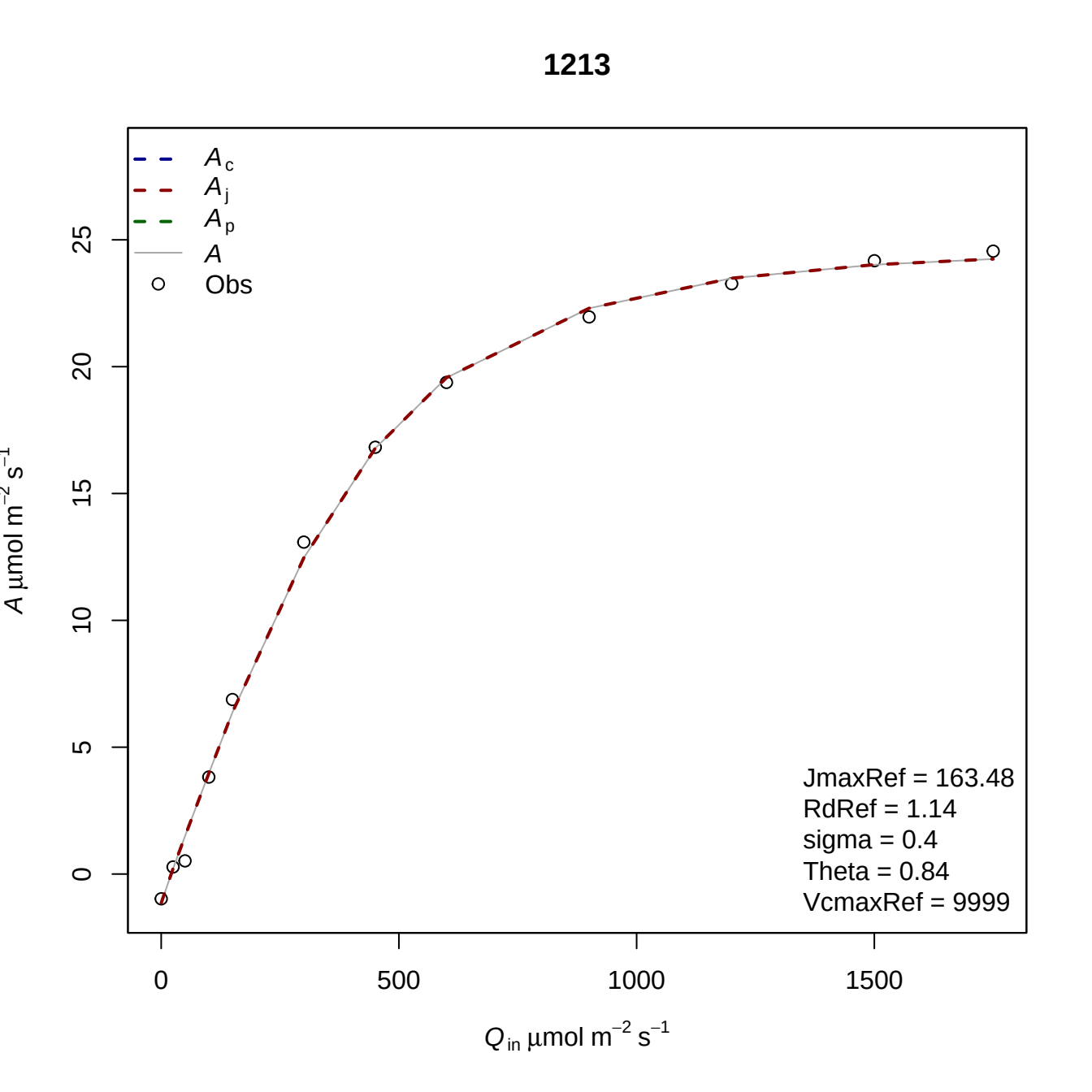

1208
A_c
A_j
A_p
A
Obs
JmaxRef = 197.47
RdRef = 2.02
sigma = 0.53
Theta = 0.11
VcmaxRef = 102.69
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

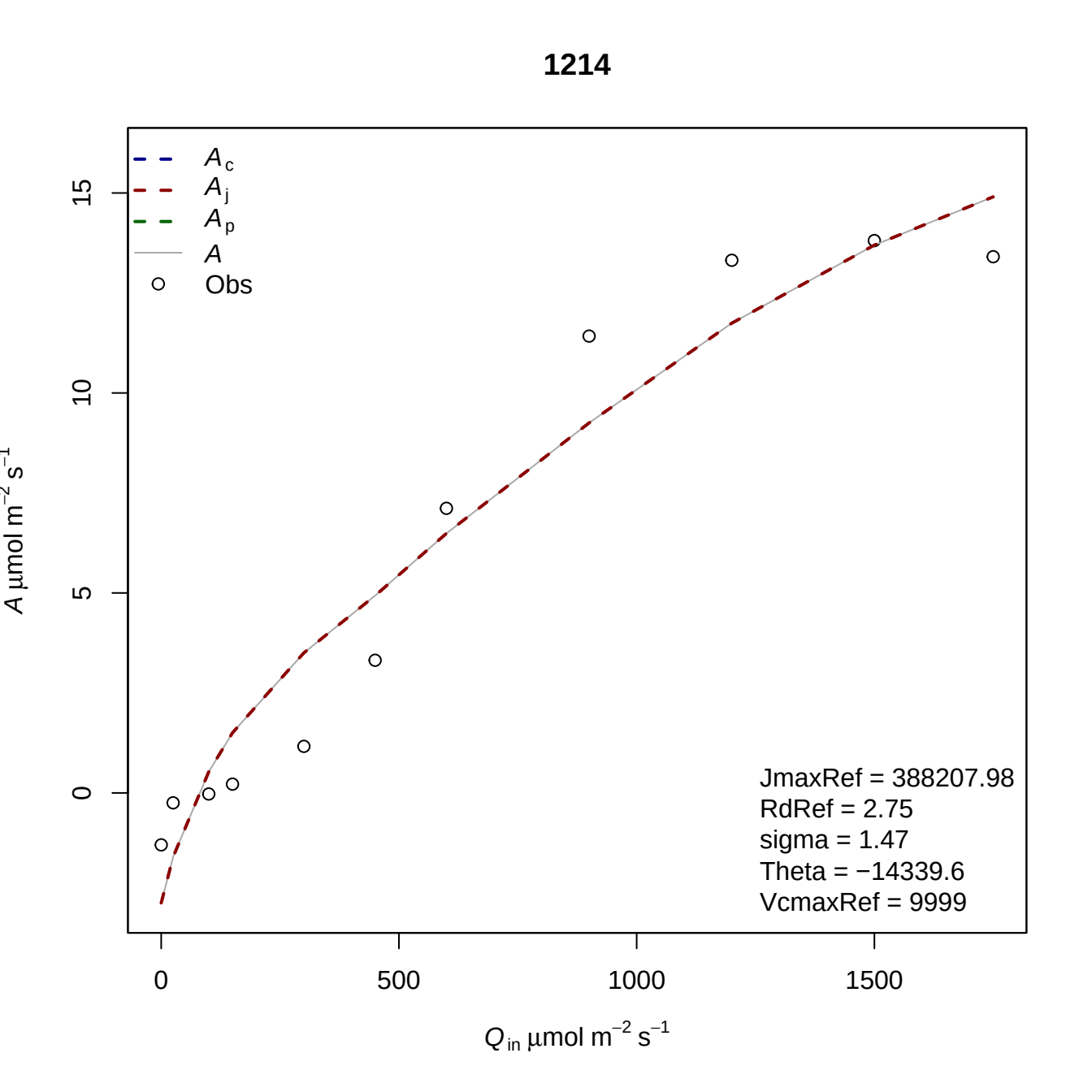

1209
A_c
A_j
A_p
A
Obs
JmaxRef = 161.2
RdRef = 2.4
sigma = 0.72
Theta = 0.71
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

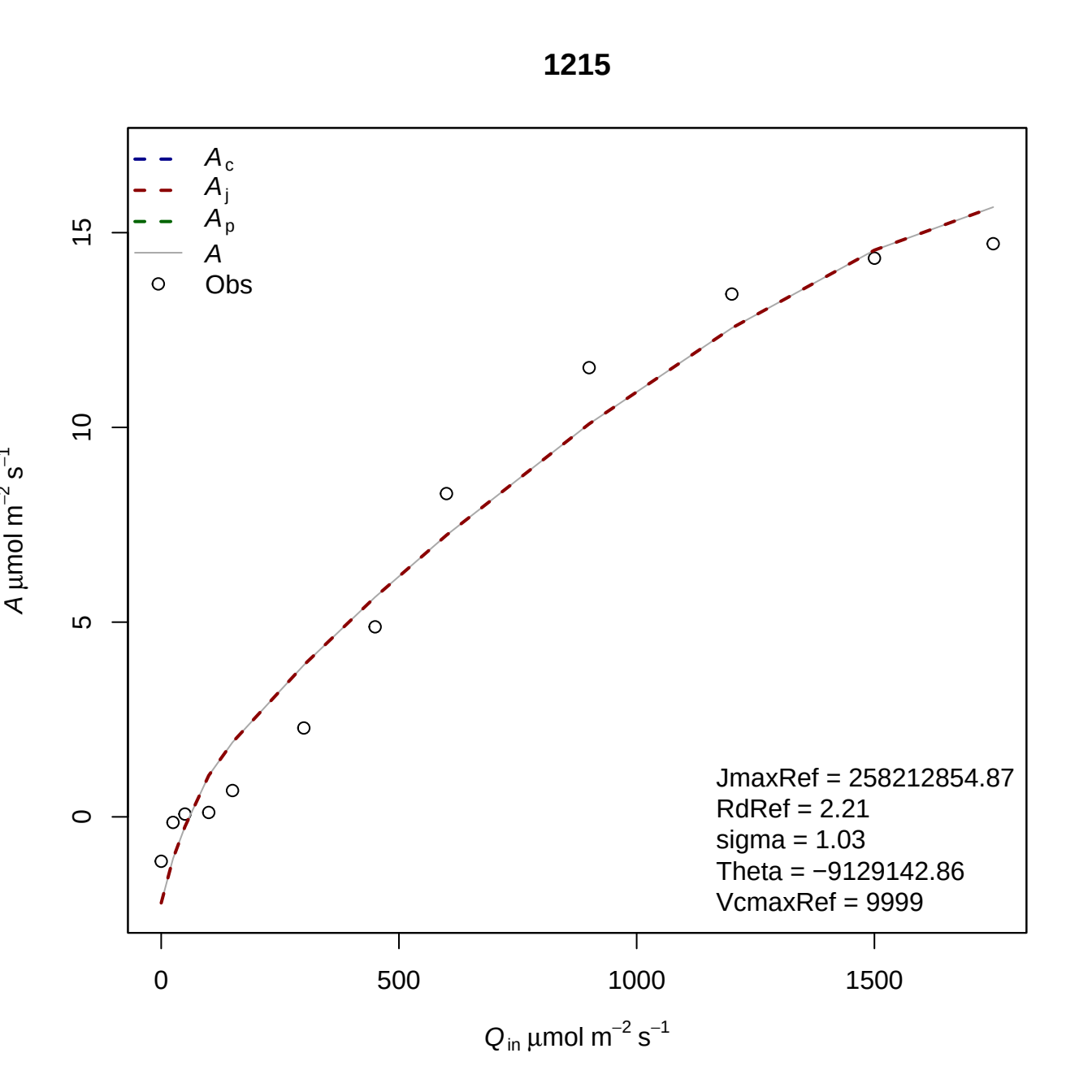

1210
A_c
A_j
A_p
A
Obs
JmaxRef = 453.72
RdRef = 1.86
sigma = 0.42
Theta = −10.66
VcmaxRef = 75.14
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

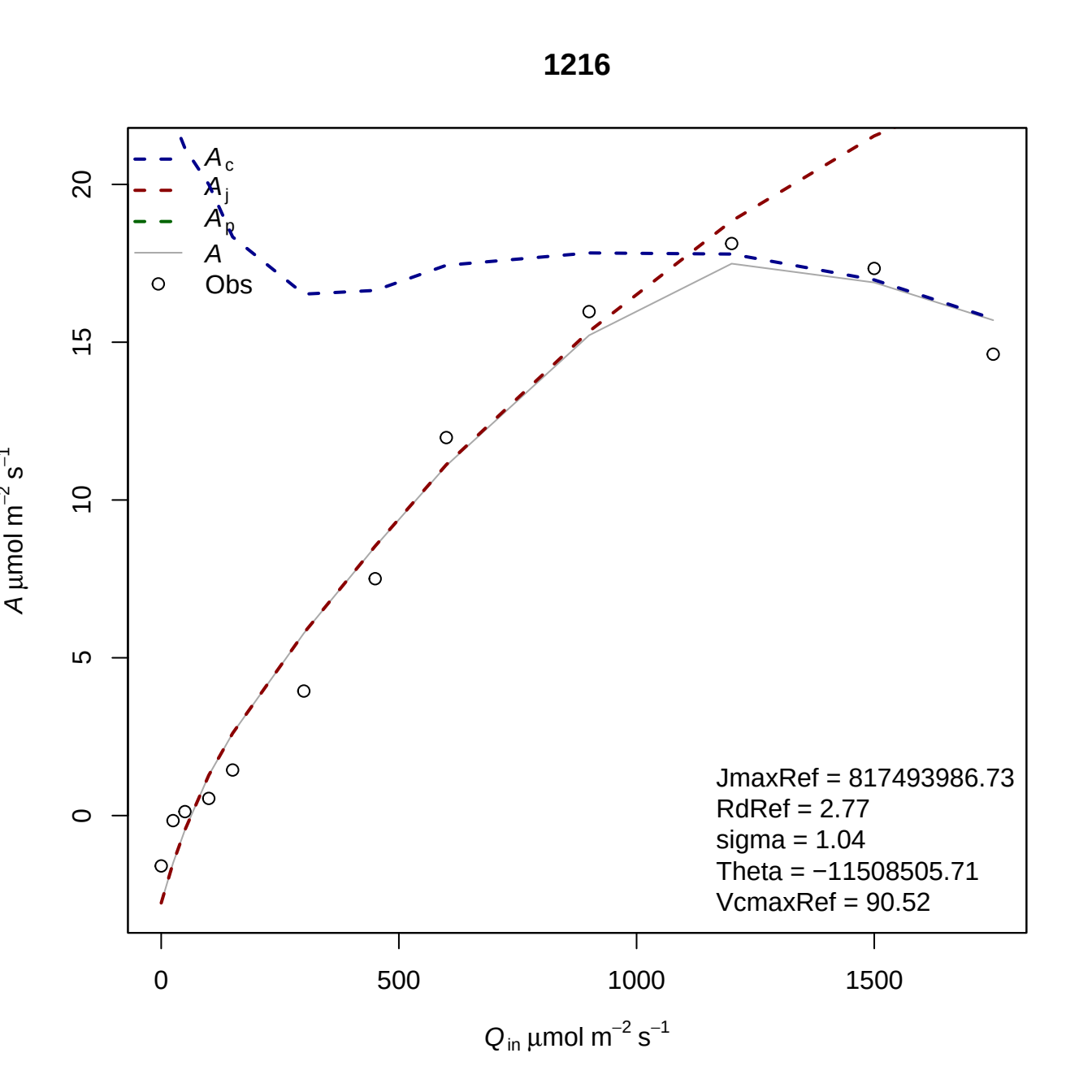

1211
A_c
A_j
A_p
A
Obs
JmaxRef = 36303639.28
RdRef = 1.1
sigma = 0.28
Theta = −6971462.77
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

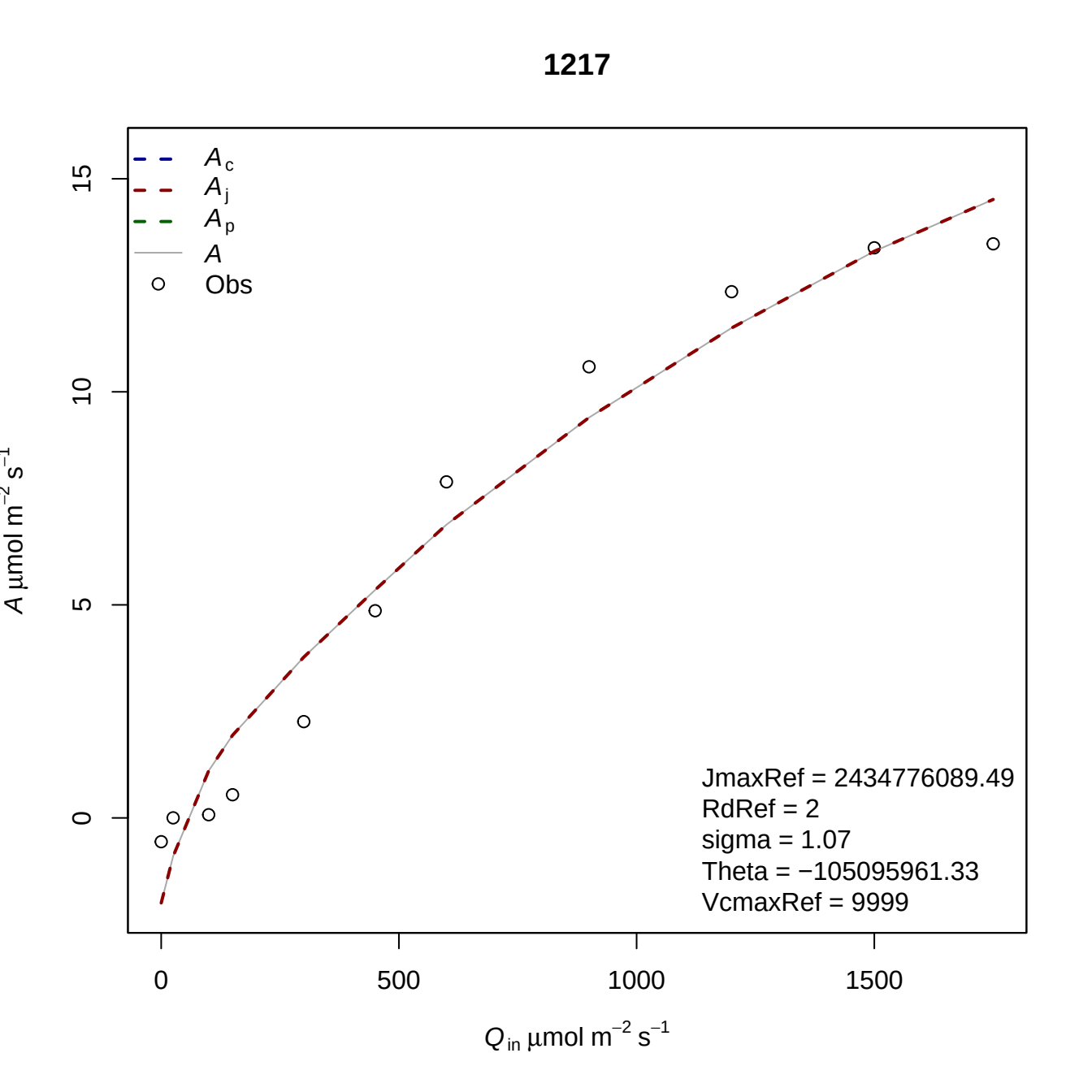

1212
A_c
A_j
A_p
A
Obs
JmaxRef = 251155995.23
RdRef = 3.07
sigma = 1.45
Theta = −5770558.58
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

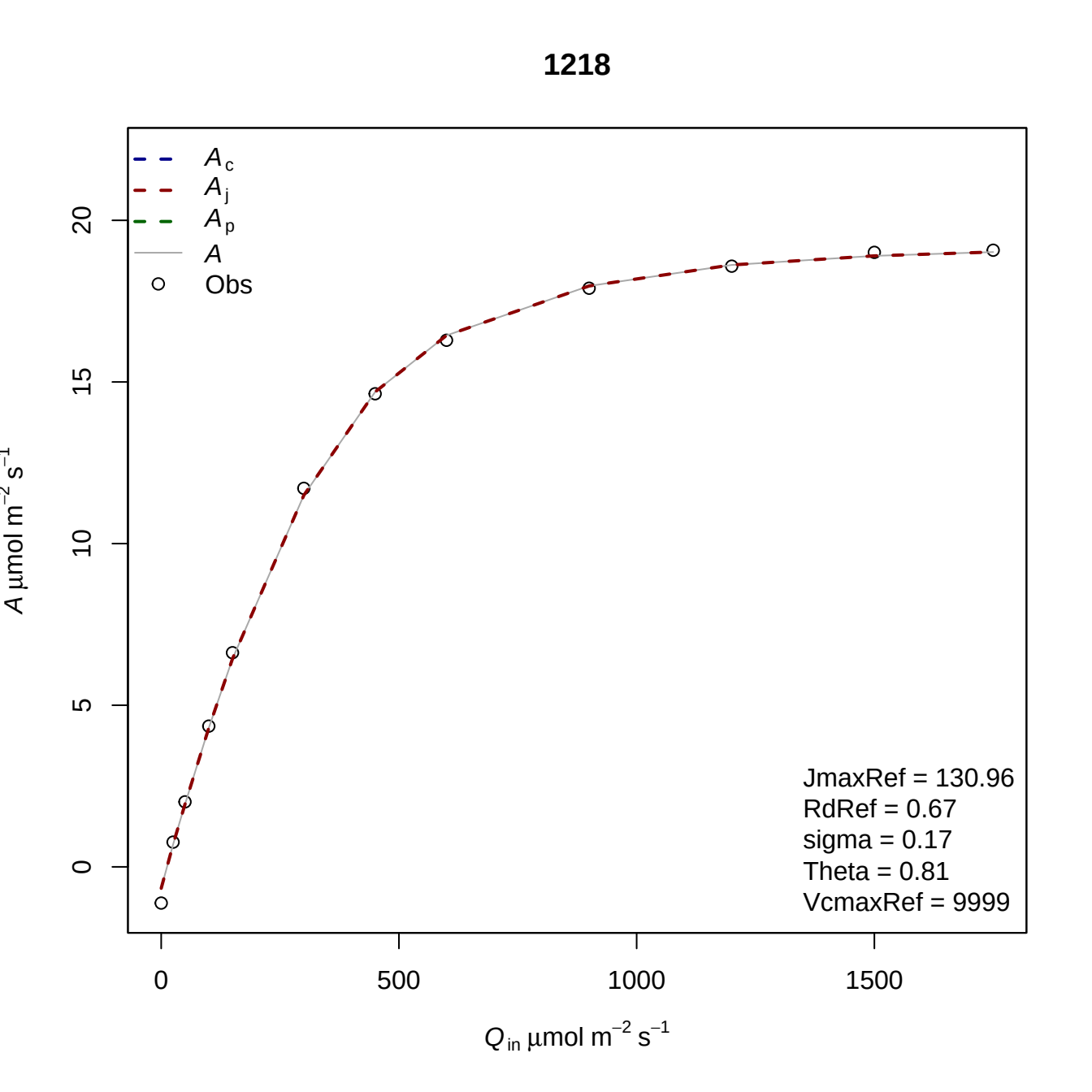

1213
A_c
A_j
A_p
A
Obs
JmaxRef = 163.48
RdRef = 1.14
sigma = 0.4
Theta = 0.84
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

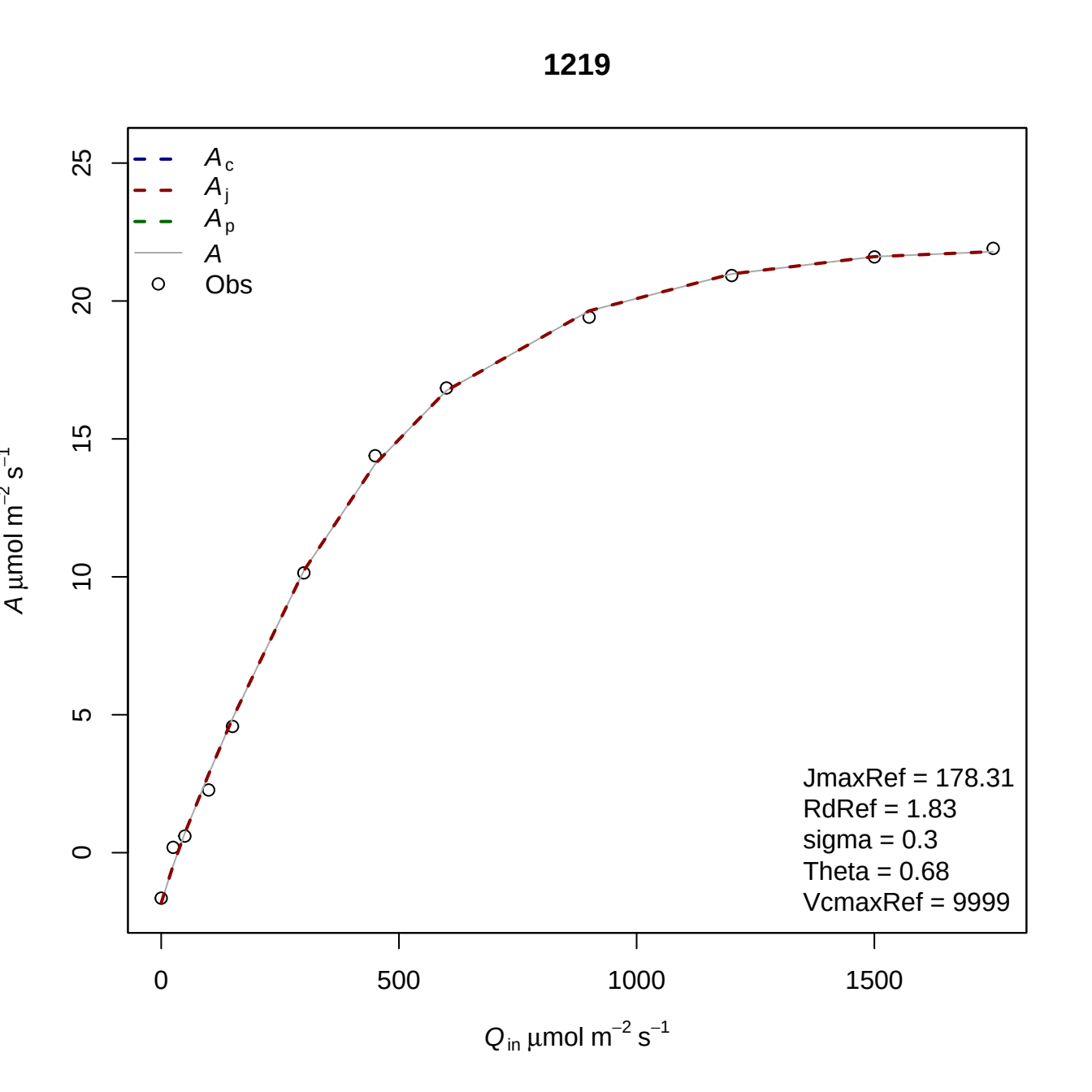

1214
A_c
A_j
A_p
A
Obs
JmaxRef = 388207.98
RdRef = 2.75
sigma = 1.47
Theta = −14339.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

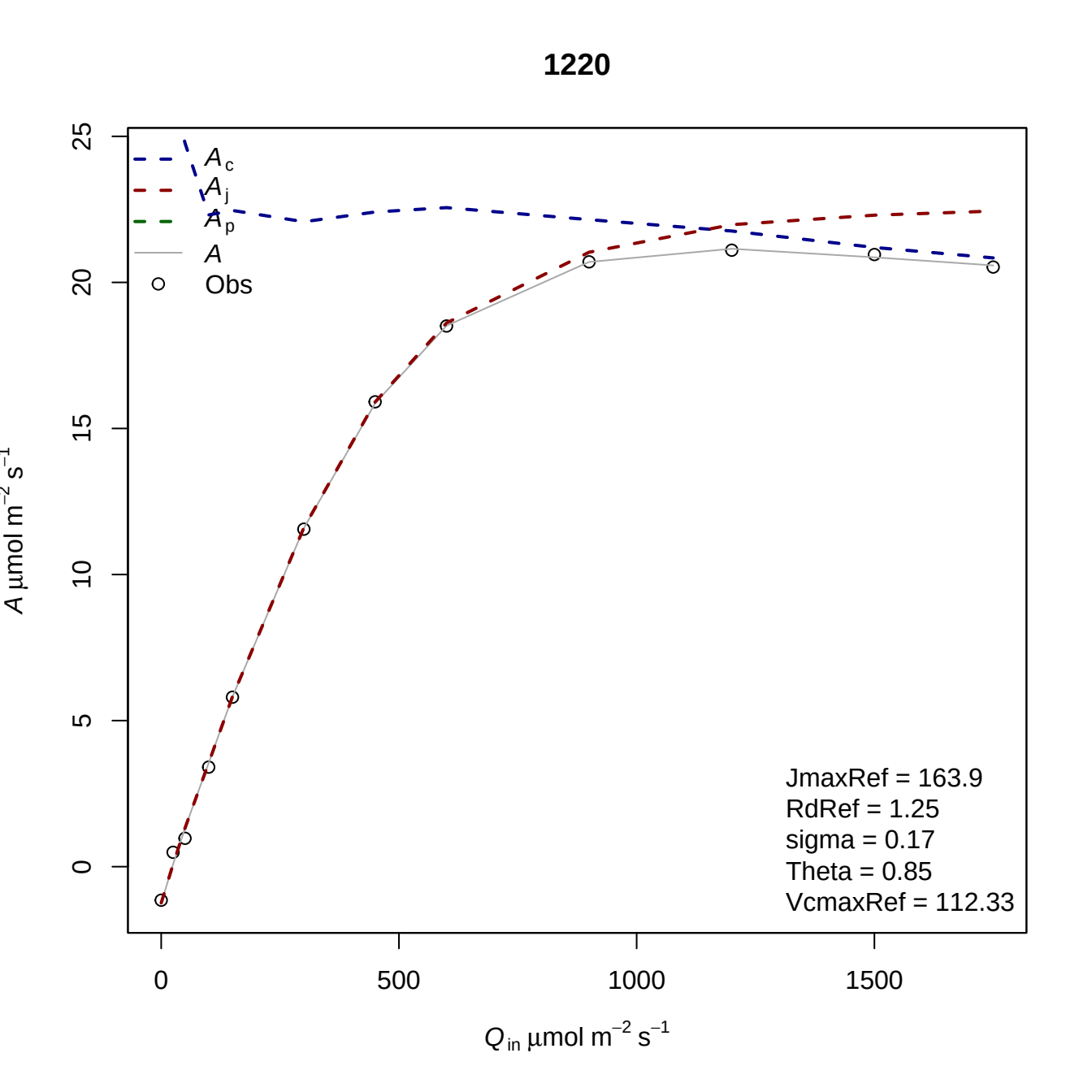

1215
A_c
A_j
A_p
A
Obs
JmaxRef = 258212854.87
RdRef = 2.21
sigma = 1.03
Theta = −9129142.86
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

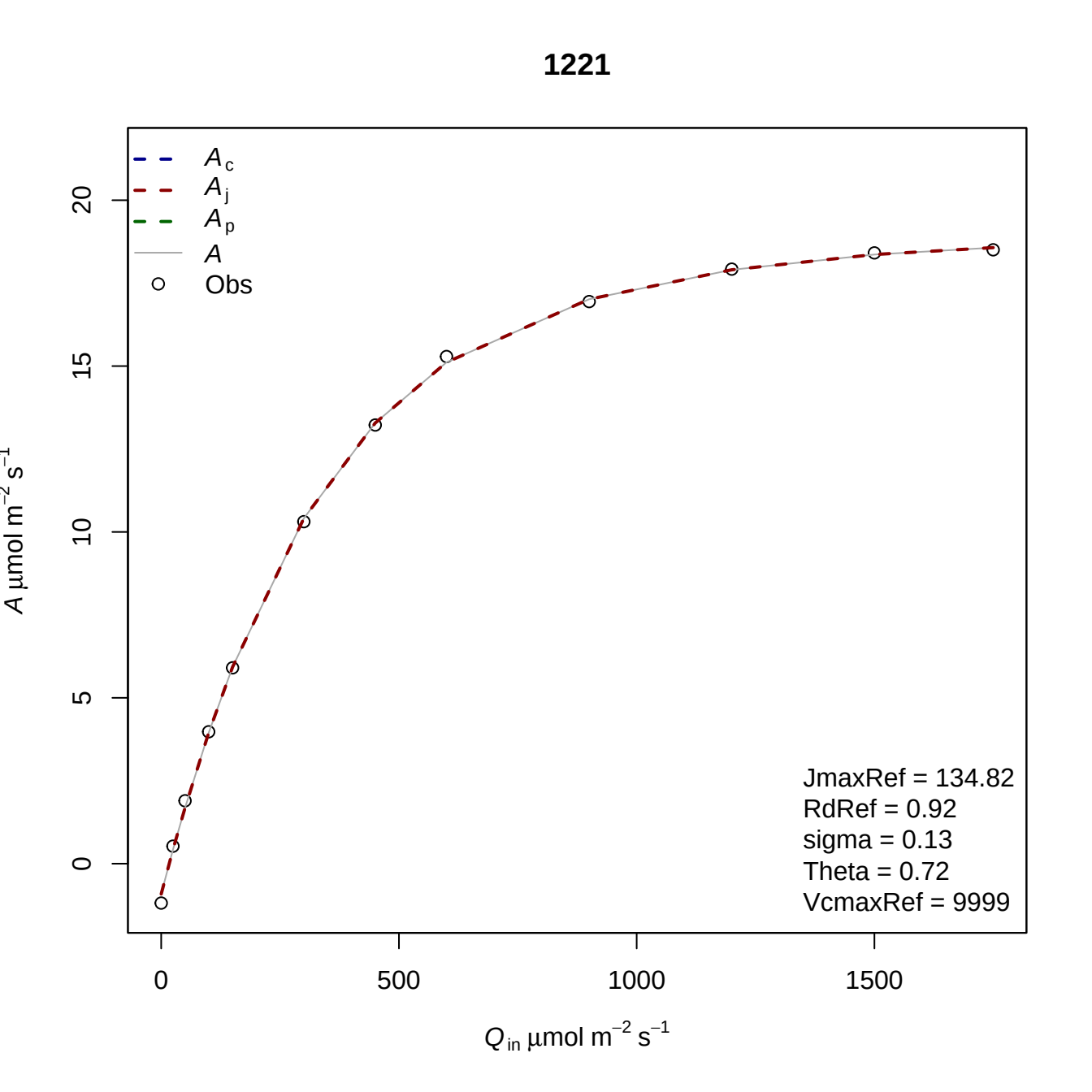

1216
A_c
A_j
A_p
A
Obs
JmaxRef = 817493986.73
RdRef = 2.77
sigma = 1.04
Theta = −11508505.71
VcmaxRef = 90.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

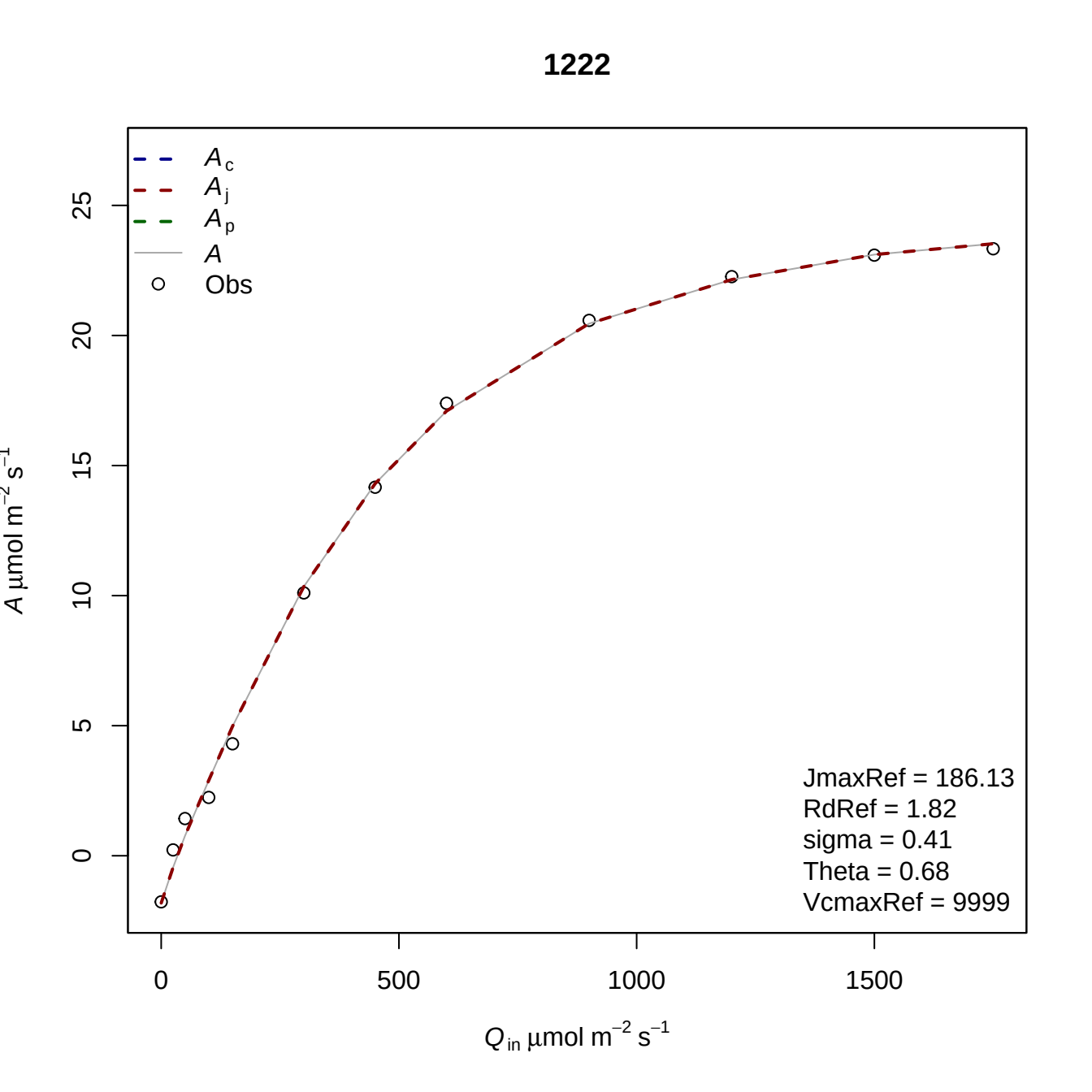

1217
A_c
A_j
A_p
A
Obs
JmaxRef = 2434776089.49
RdRef = 2
sigma = 1.07
Theta = −105095961.33
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

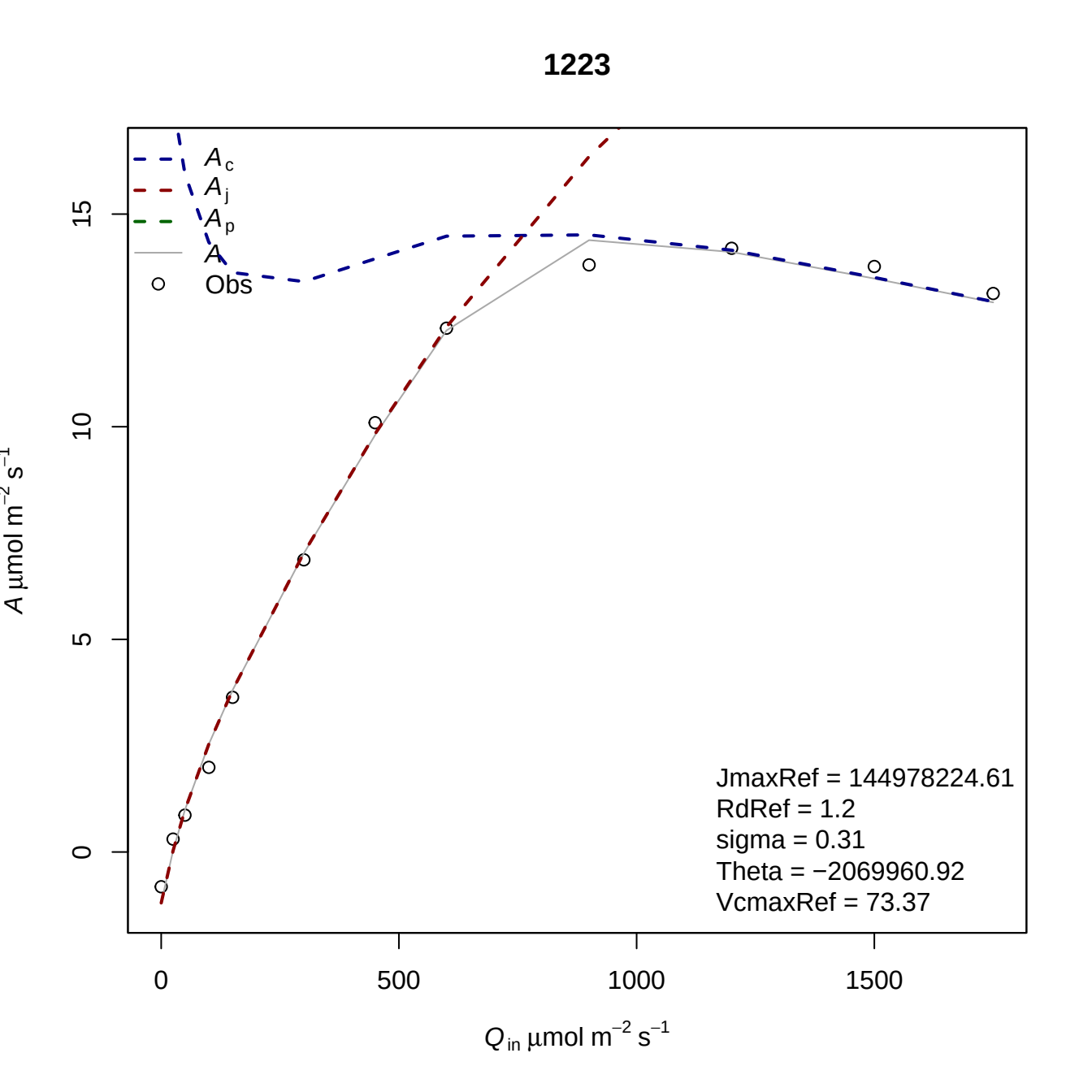

1218
A_c
A_j
A_p
A
Obs
JmaxRef = 130.96
RdRef = 0.67
sigma = 0.17
Theta = 0.81
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

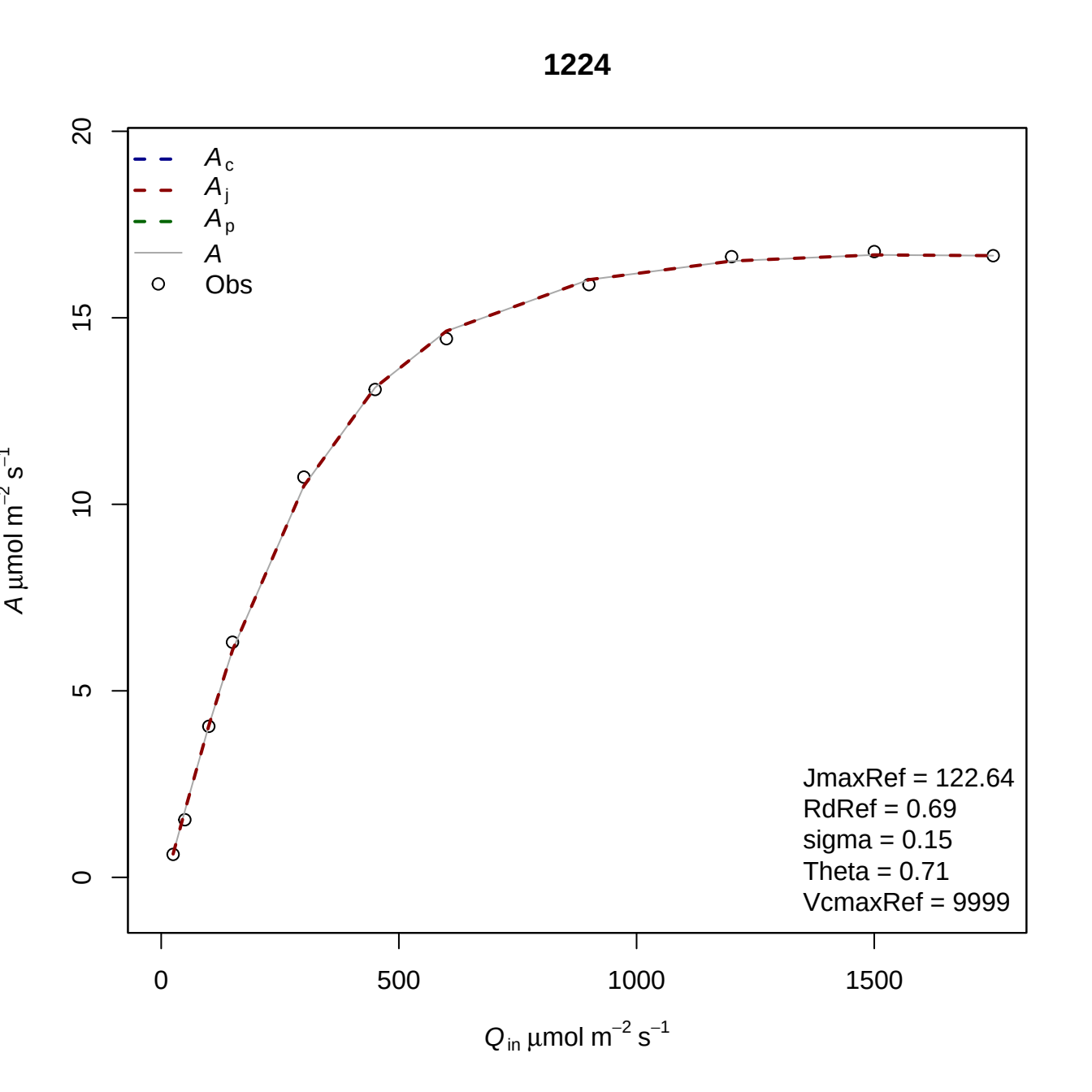

1219
A_c
A_j
A_p
A
Obs
JmaxRef = 178.31
RdRef = 1.83
sigma = 0.3
Theta = 0.68
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

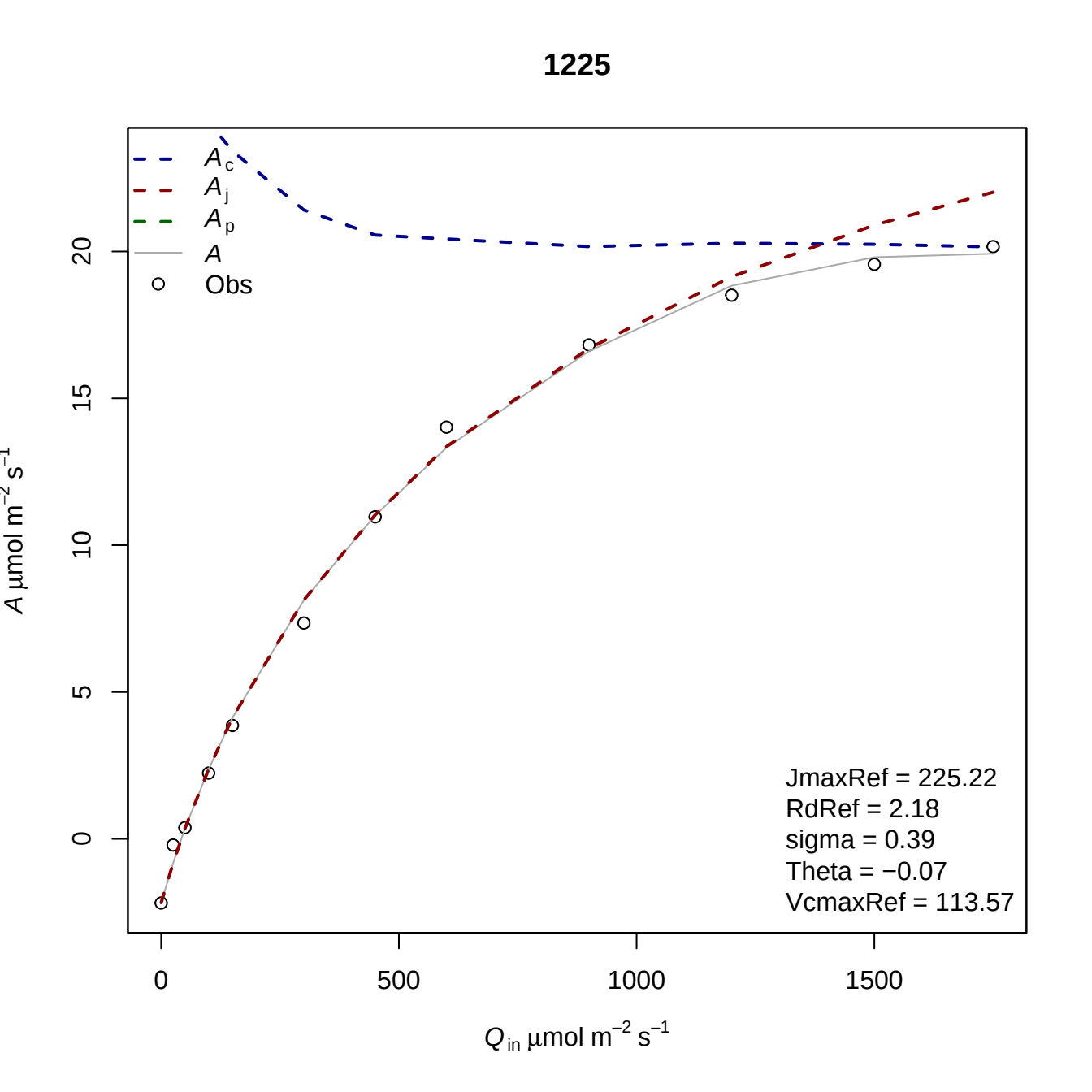

1220
A_c
A_j
A_p
A
Obs
JmaxRef = 163.9
RdRef = 1.25
sigma = 0.17
Theta = 0.85
VcmaxRef = 112.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

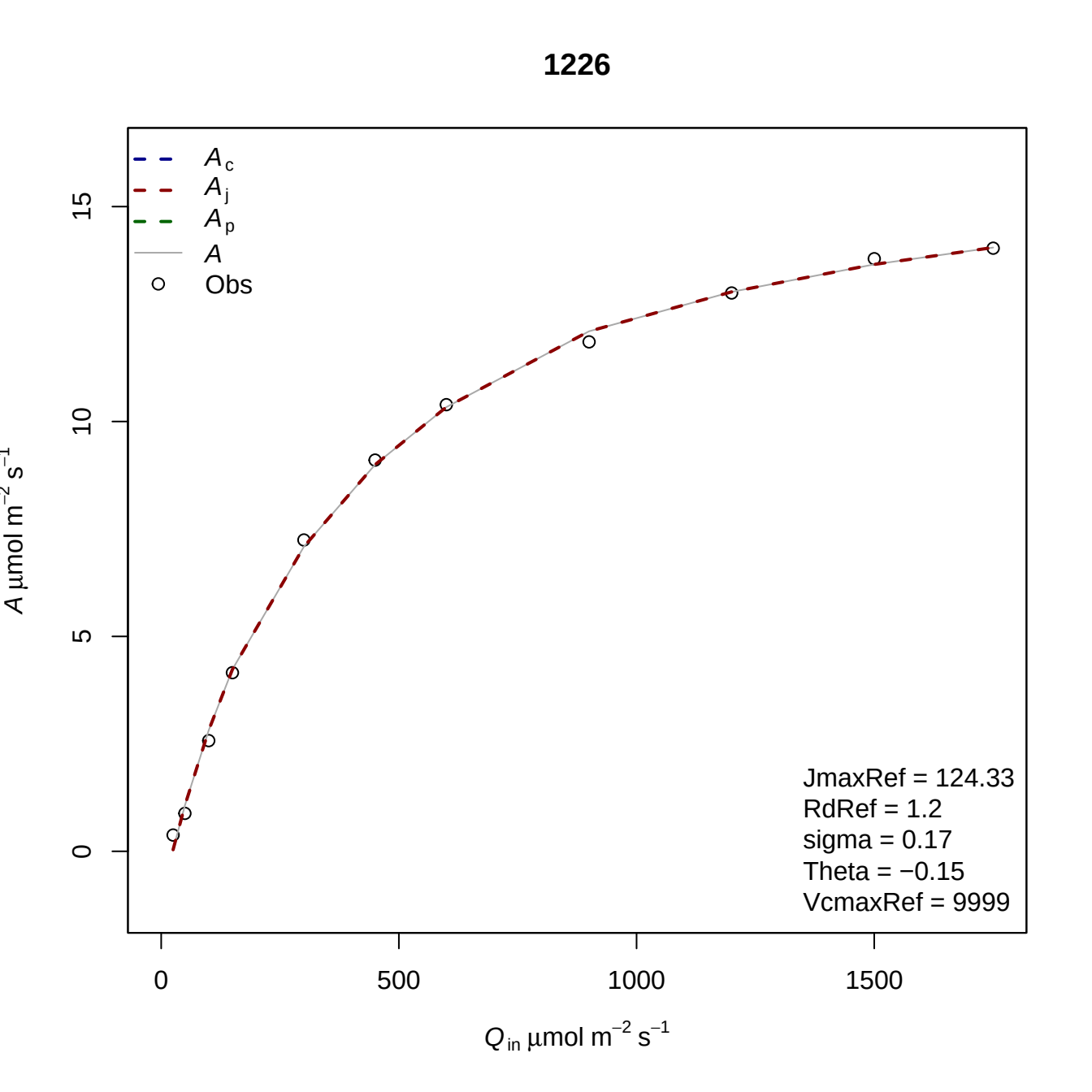

1221
A_c
A_j
A_p
A
Obs
JmaxRef = 134.82
RdRef = 0.92
sigma = 0.13
Theta = 0.72
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

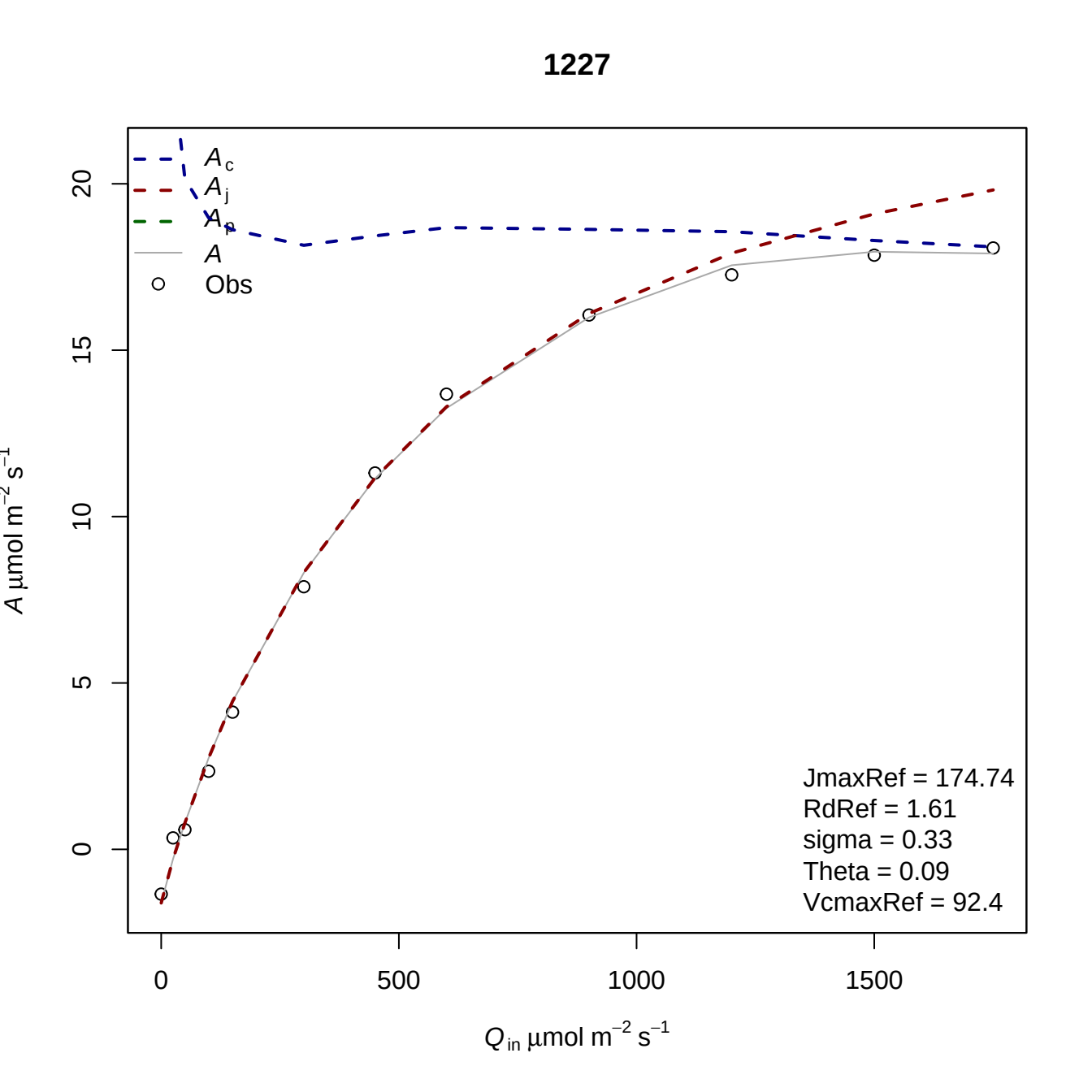

1222
A_c
A_j
A_p
A
Obs
JmaxRef = 186.13
RdRef = 1.82
sigma = 0.41
Theta = 0.68
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

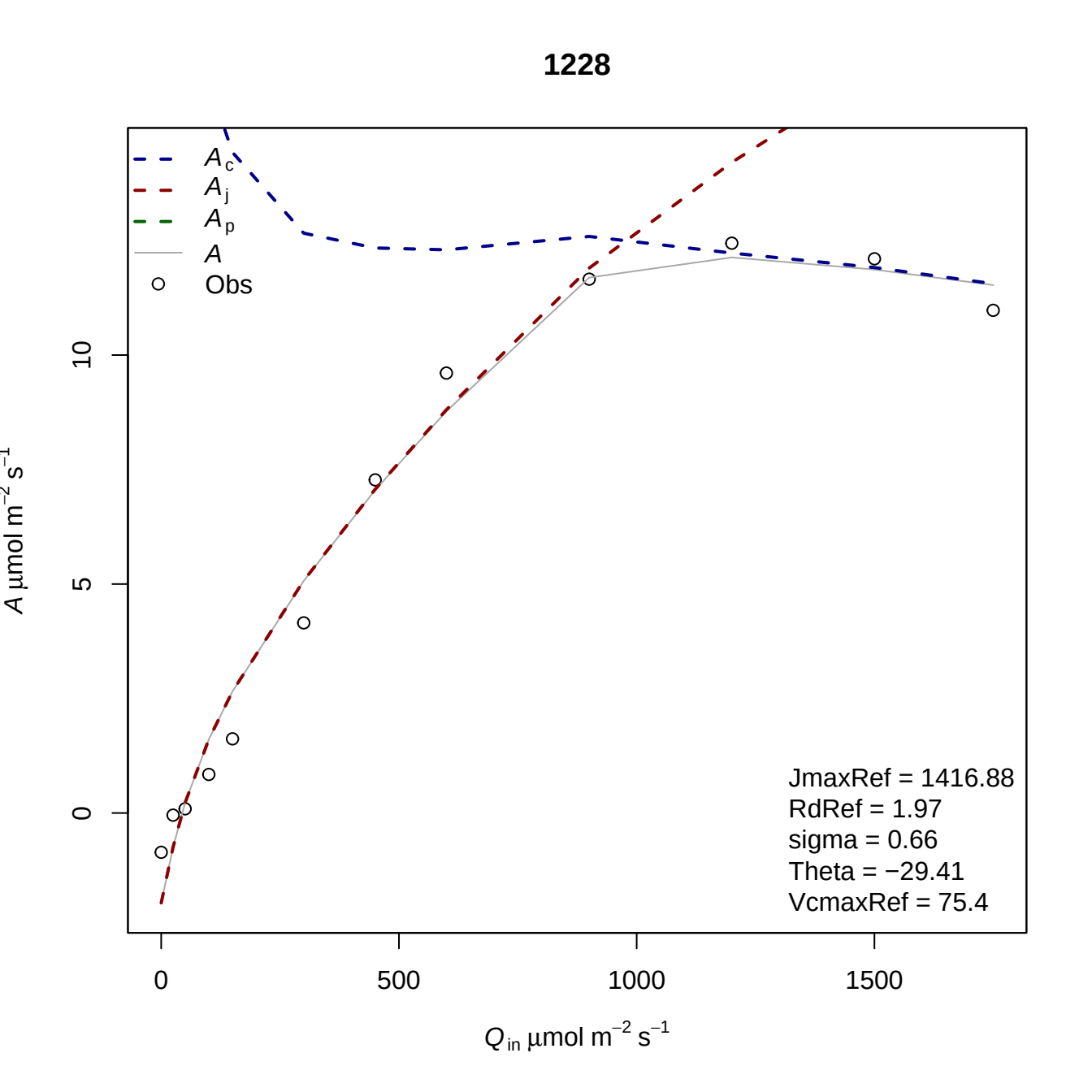

1223
A_c
A_j
A_p
A
Obs
JmaxRef = 144978224.61
RdRef = 1.2
sigma = 0.31
Theta = −2069960.92
VcmaxRef = 73.37
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

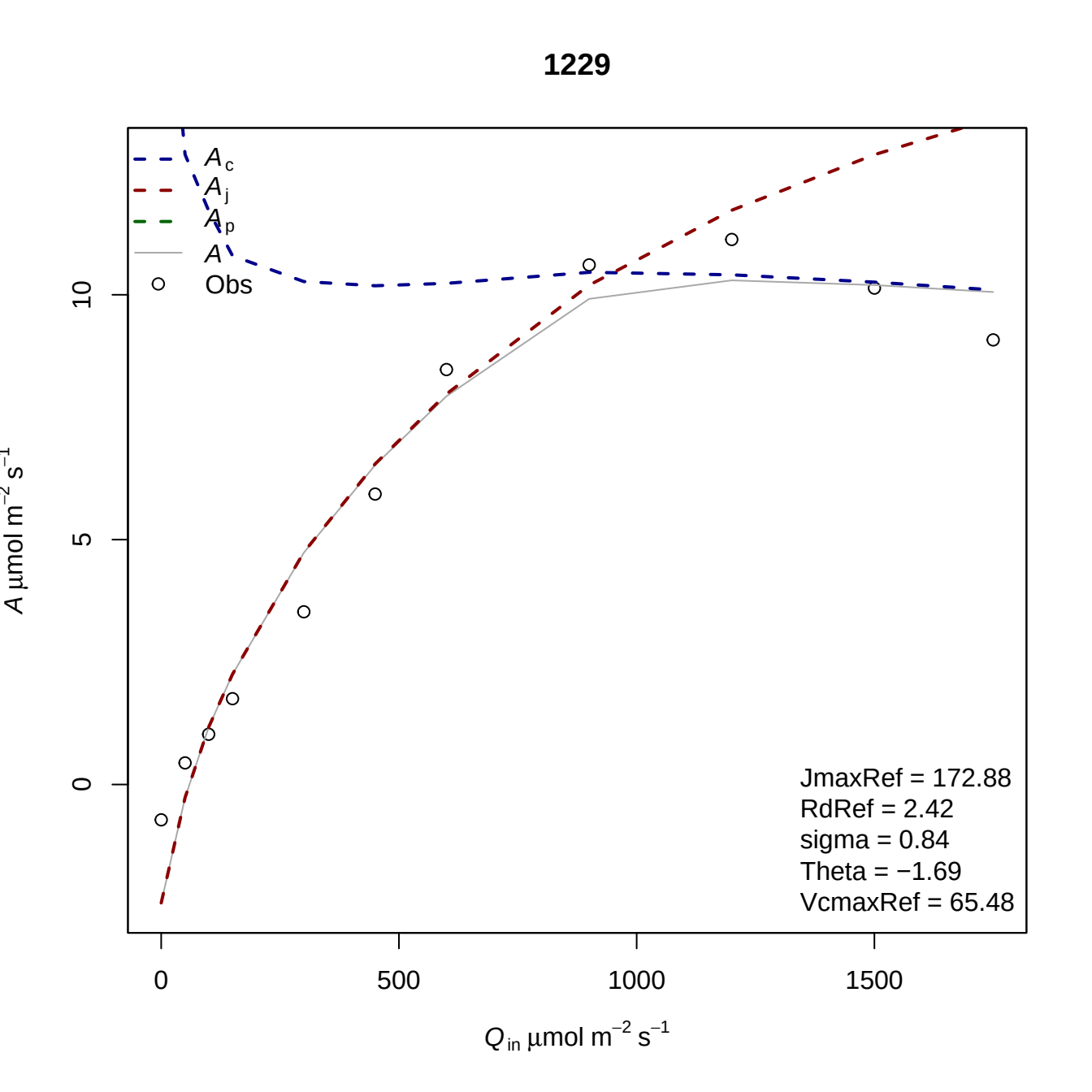

1224
A_c
A_j
A_p
A
Obs
JmaxRef = 122.64
RdRef = 0.69
sigma = 0.15
Theta = 0.71
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

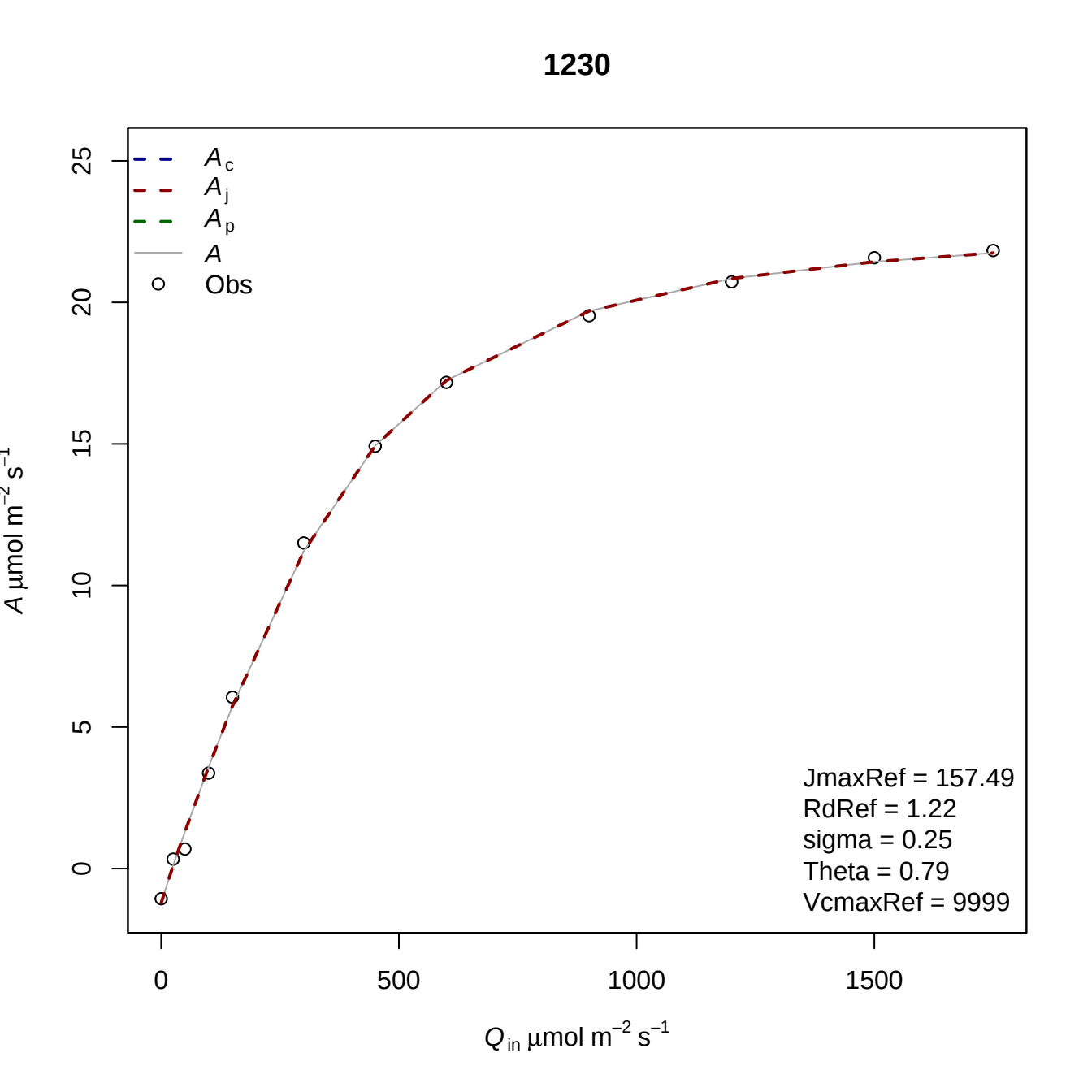

1225
A_c
A_j
A_p
A
Obs
JmaxRef = 225.22
RdRef = 2.18
sigma = 0.39
Theta = −0.07
VcmaxRef = 113.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

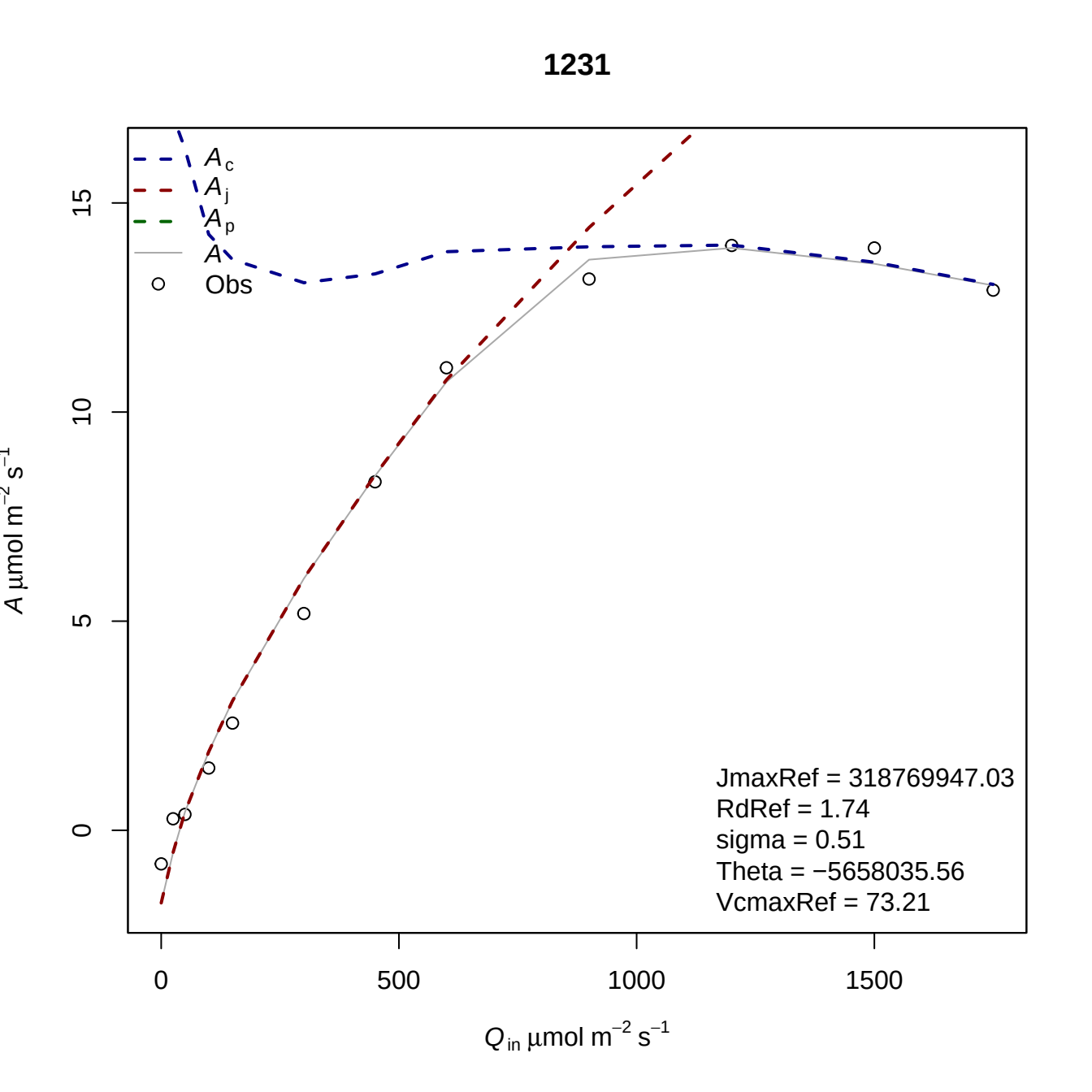

1226
A_c
A_j
A_p
A
Obs
JmaxRef = 124.33
RdRef = 1.2
sigma = 0.17
Theta = −0.15
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

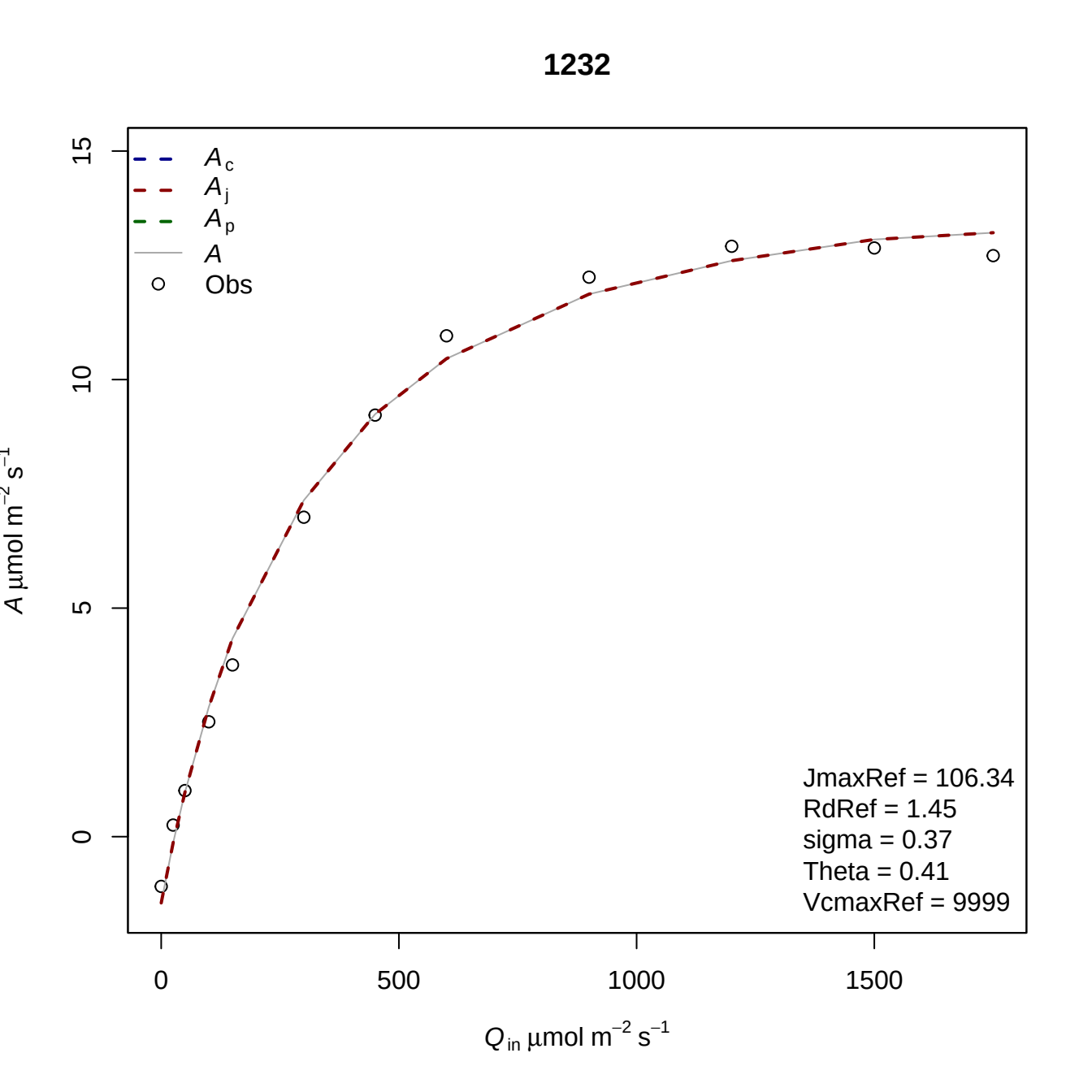

1227
A_c
A_j
A_p
A
Obs
JmaxRef = 174.74
RdRef = 1.61
sigma = 0.33
Theta = 0.09
VcmaxRef = 92.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

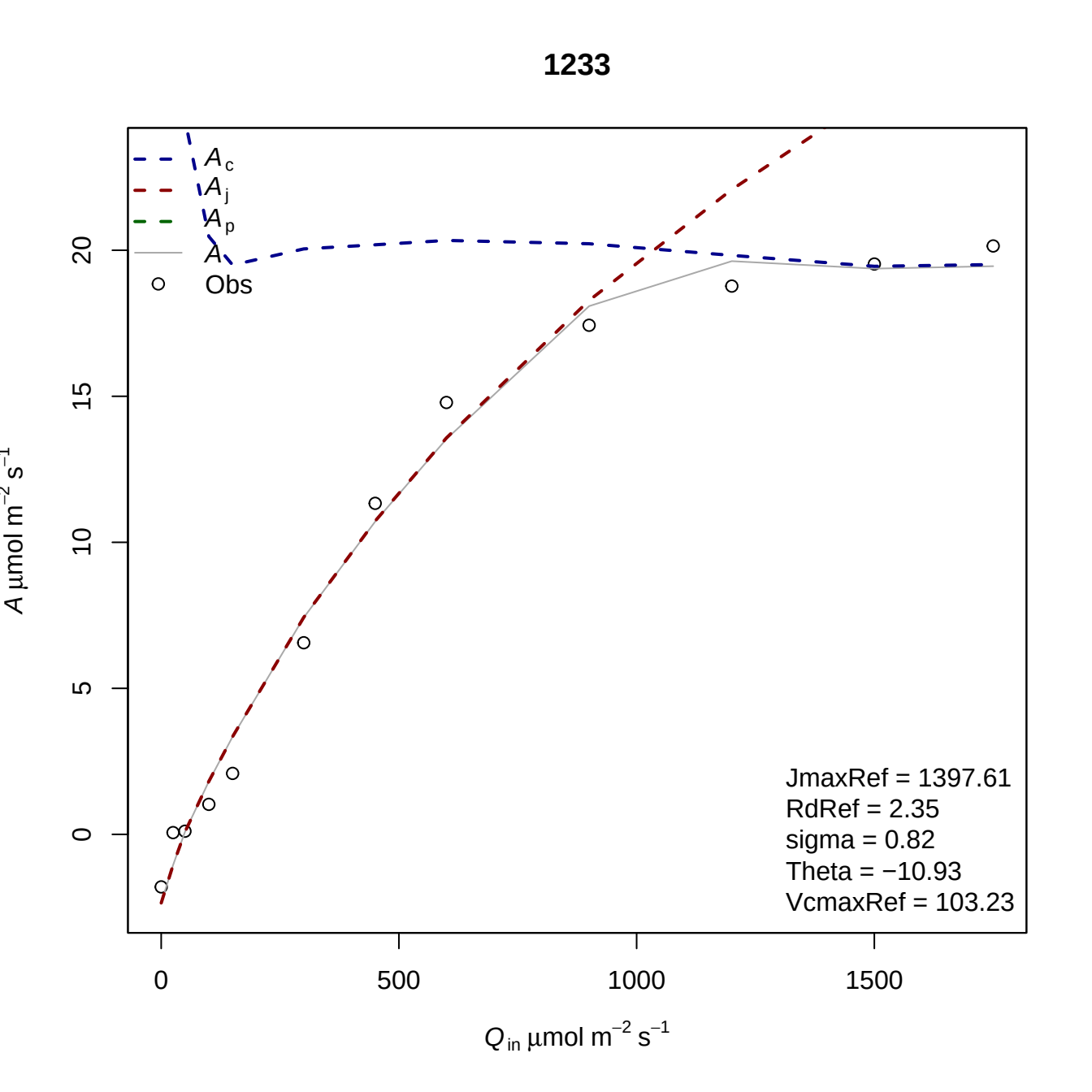

1228
A_c
A_j
A_p
A
Obs
JmaxRef = 1416.88
RdRef = 1.97
sigma = 0.66
Theta = −29.41
VcmaxRef = 75.4
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

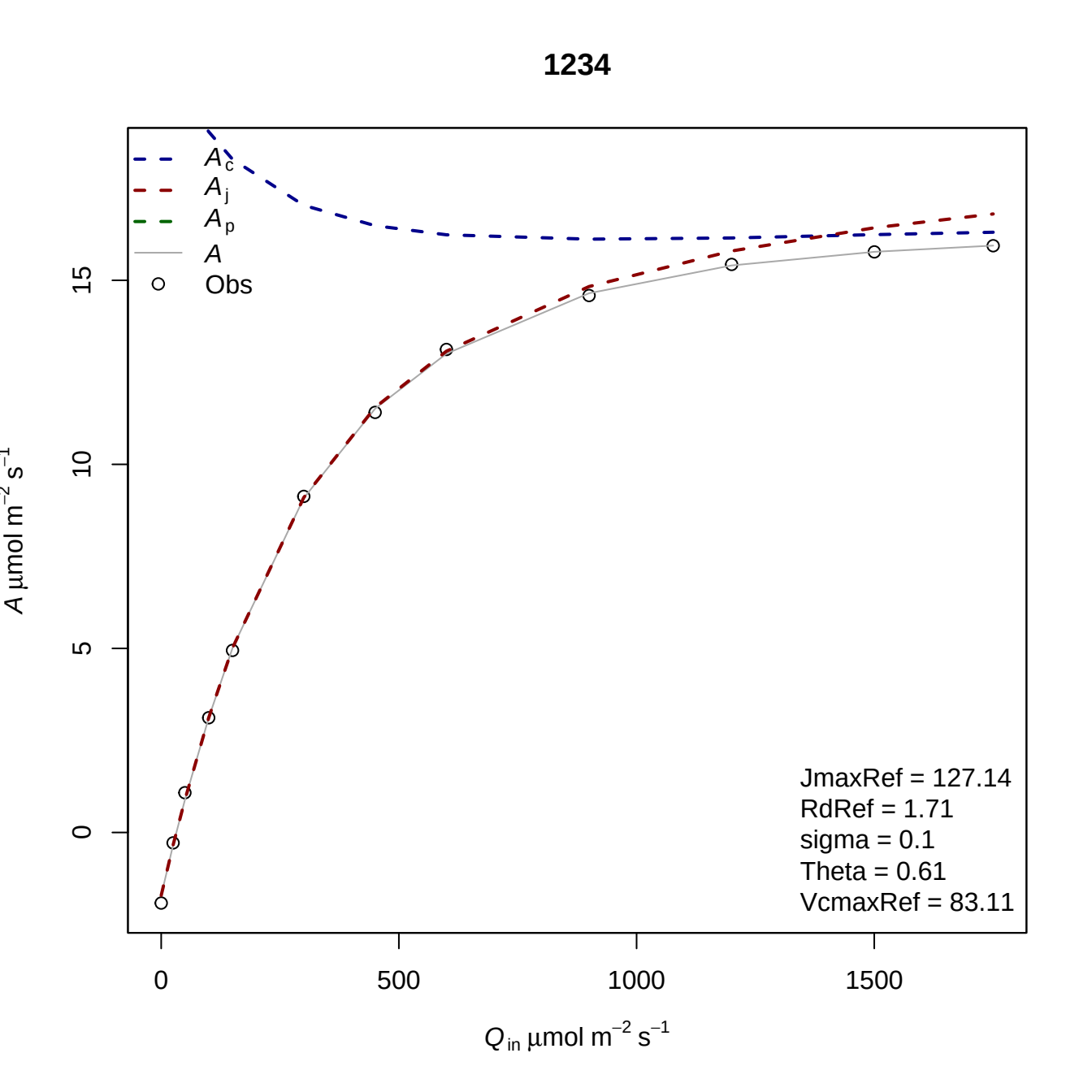

1229
A_c
A_j
A_p
A
Obs
JmaxRef = 172.88
RdRef = 2.42
sigma = 0.84
Theta = −1.69
VcmaxRef = 65.48
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

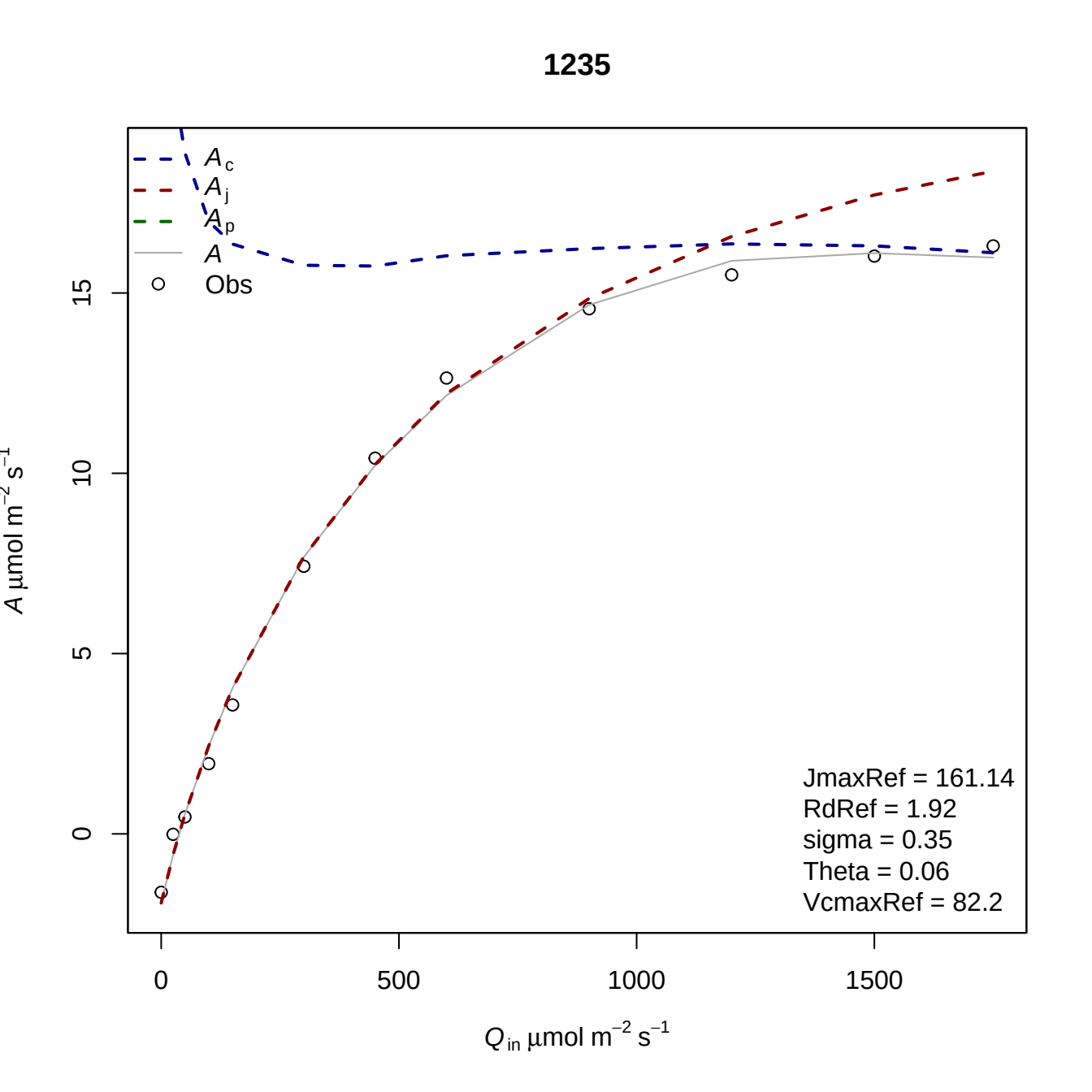

1230
A_c
A_j
A_p
A
Obs
JmaxRef = 157.49
RdRef = 1.22
sigma = 0.25
Theta = 0.79
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

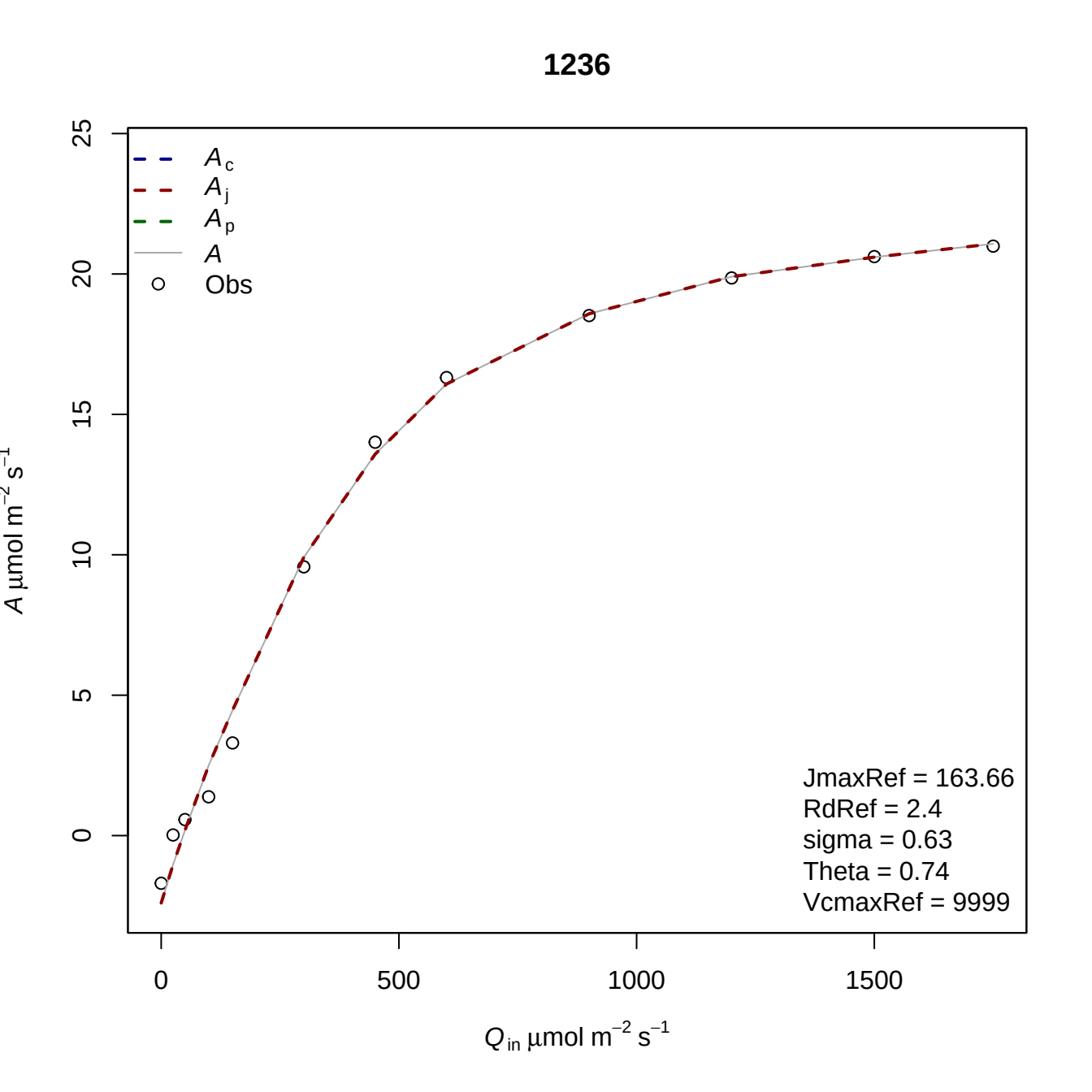

1231
A_c
A_j
A_p
A
Obs
JmaxRef = 318769947.03
RdRef = 1.74
sigma = 0.51
Theta = −5658035.56
VcmaxRef = 73.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

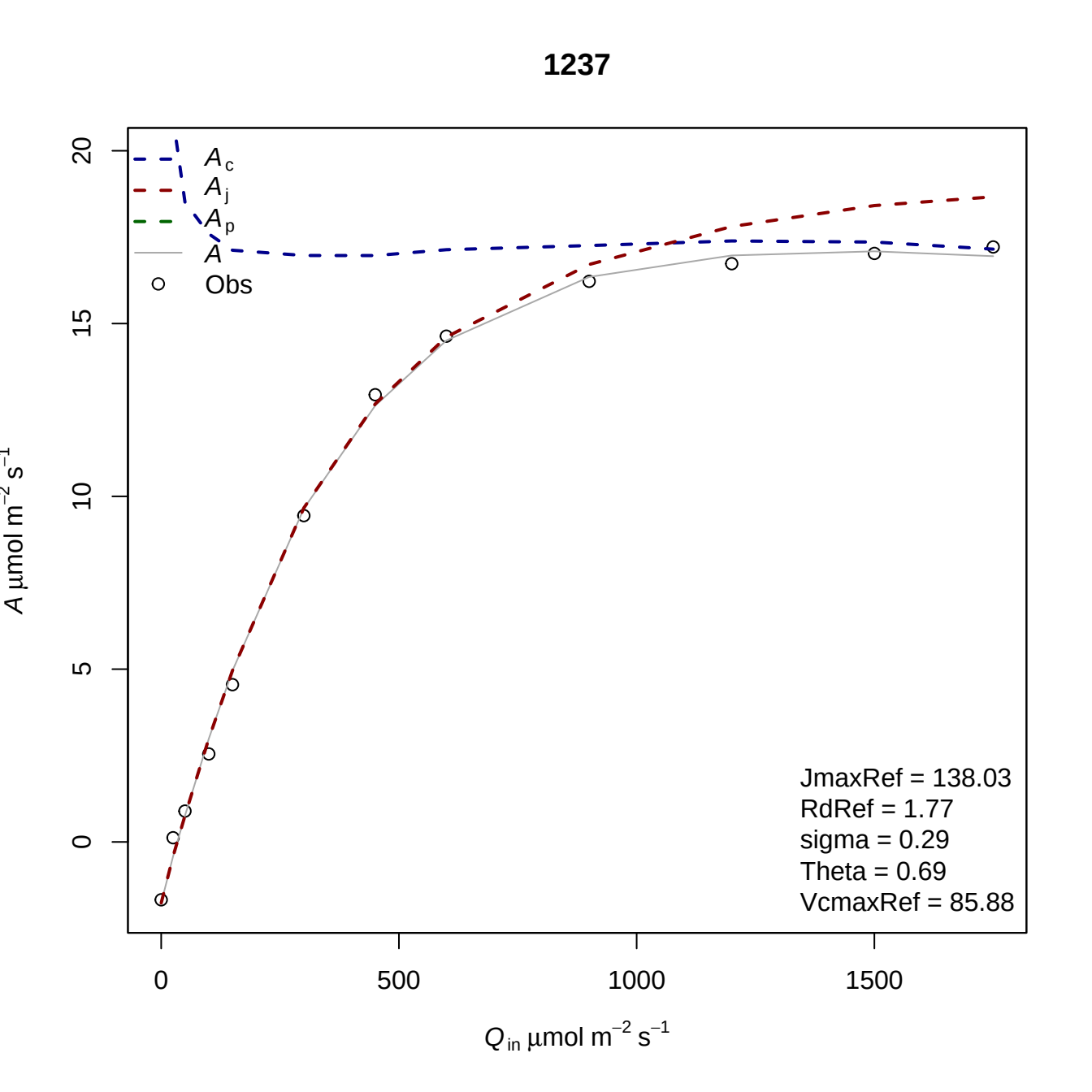

1232
A_c
A_j
A_p
A
Obs
JmaxRef = 106.34
RdRef = 1.45
sigma = 0.37
Theta = 0.41
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

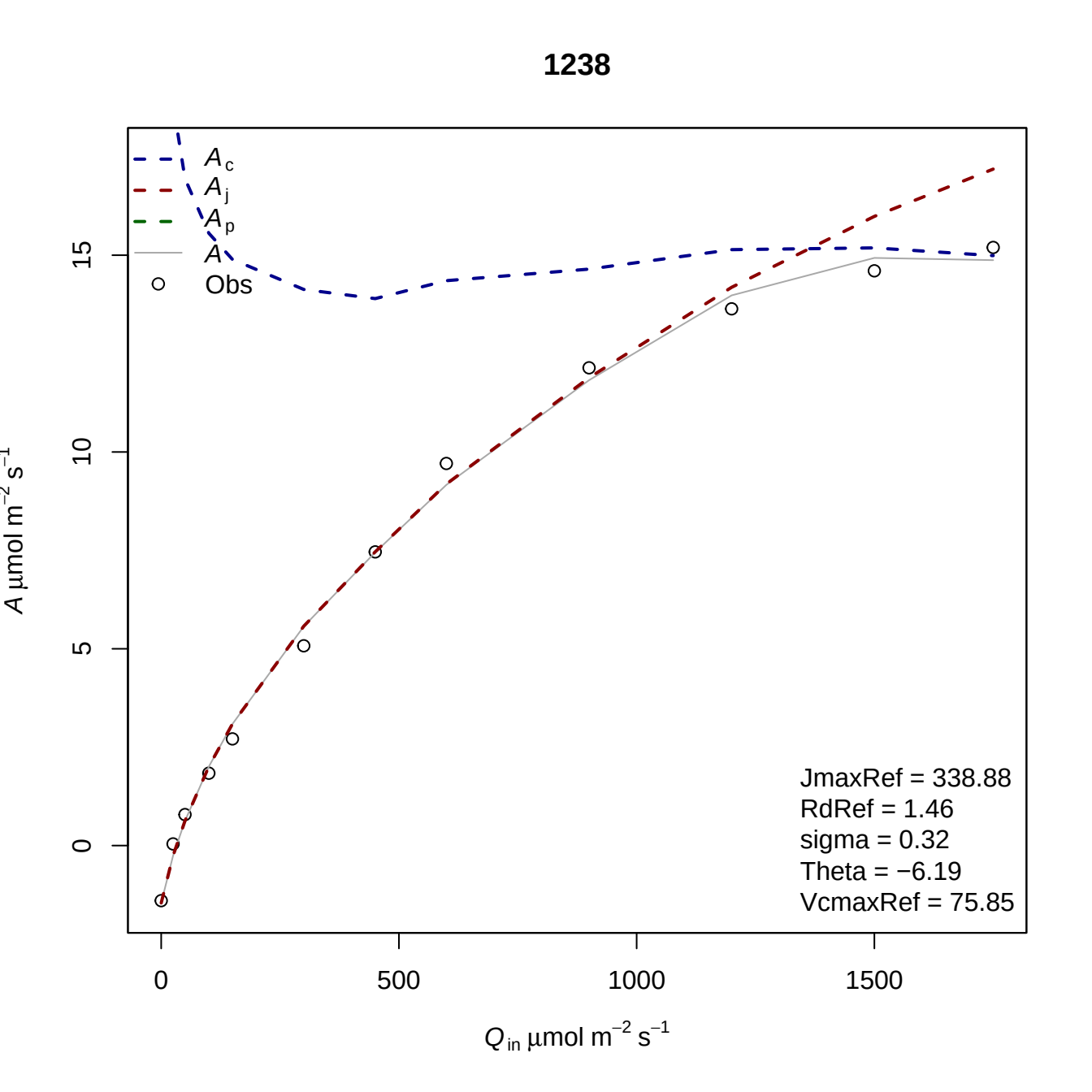

1233
A_c
A_j
A_p
A
Obs
JmaxRef = 1397.61
RdRef = 2.35
sigma = 0.82
Theta = −10.93
VcmaxRef = 103.23
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

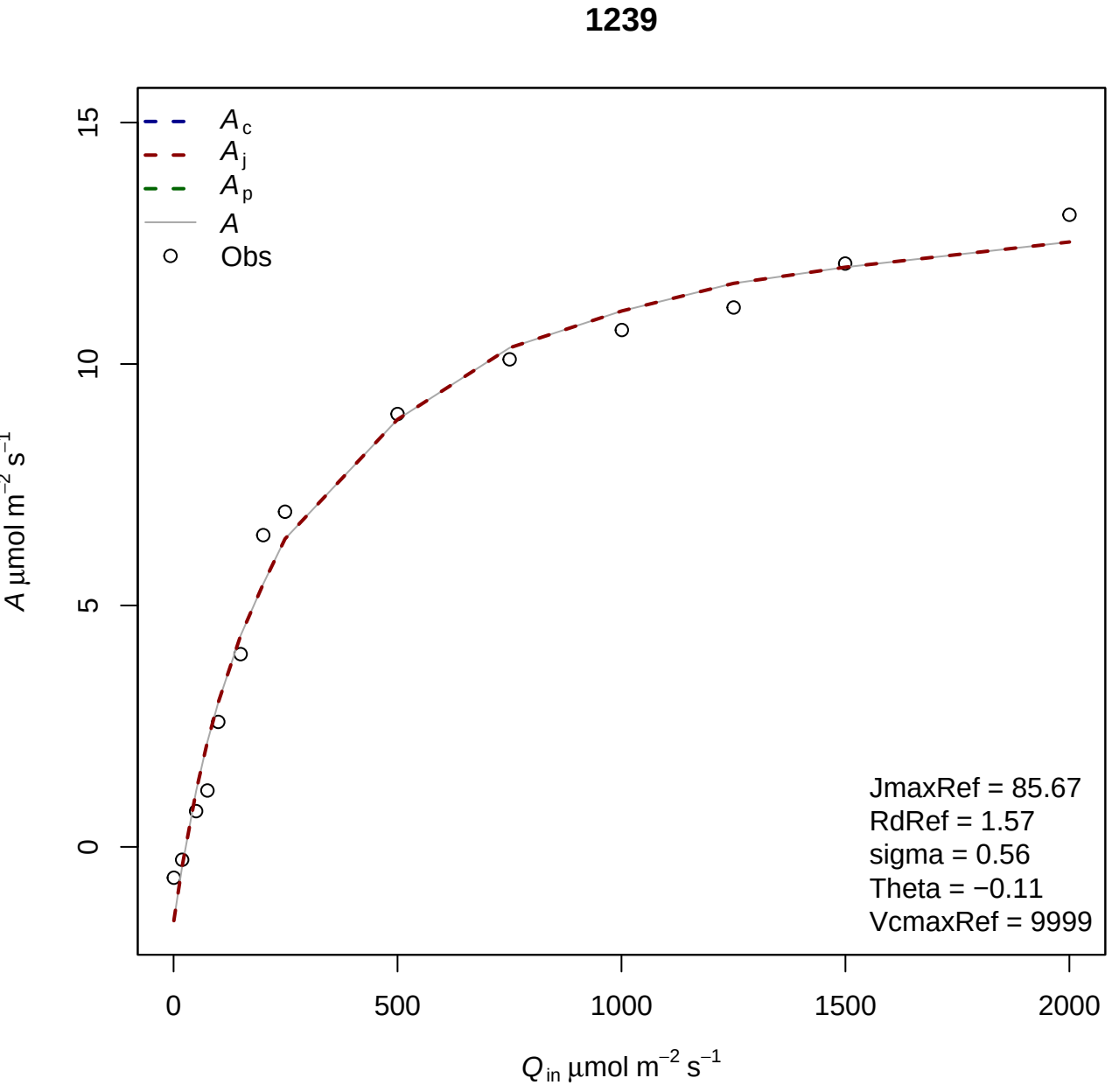

1234
A_c
A_j
A_p
A
Obs
JmaxRef = 127.14
RdRef = 1.71
sigma = 0.1
Theta = 0.61
VcmaxRef = 83.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

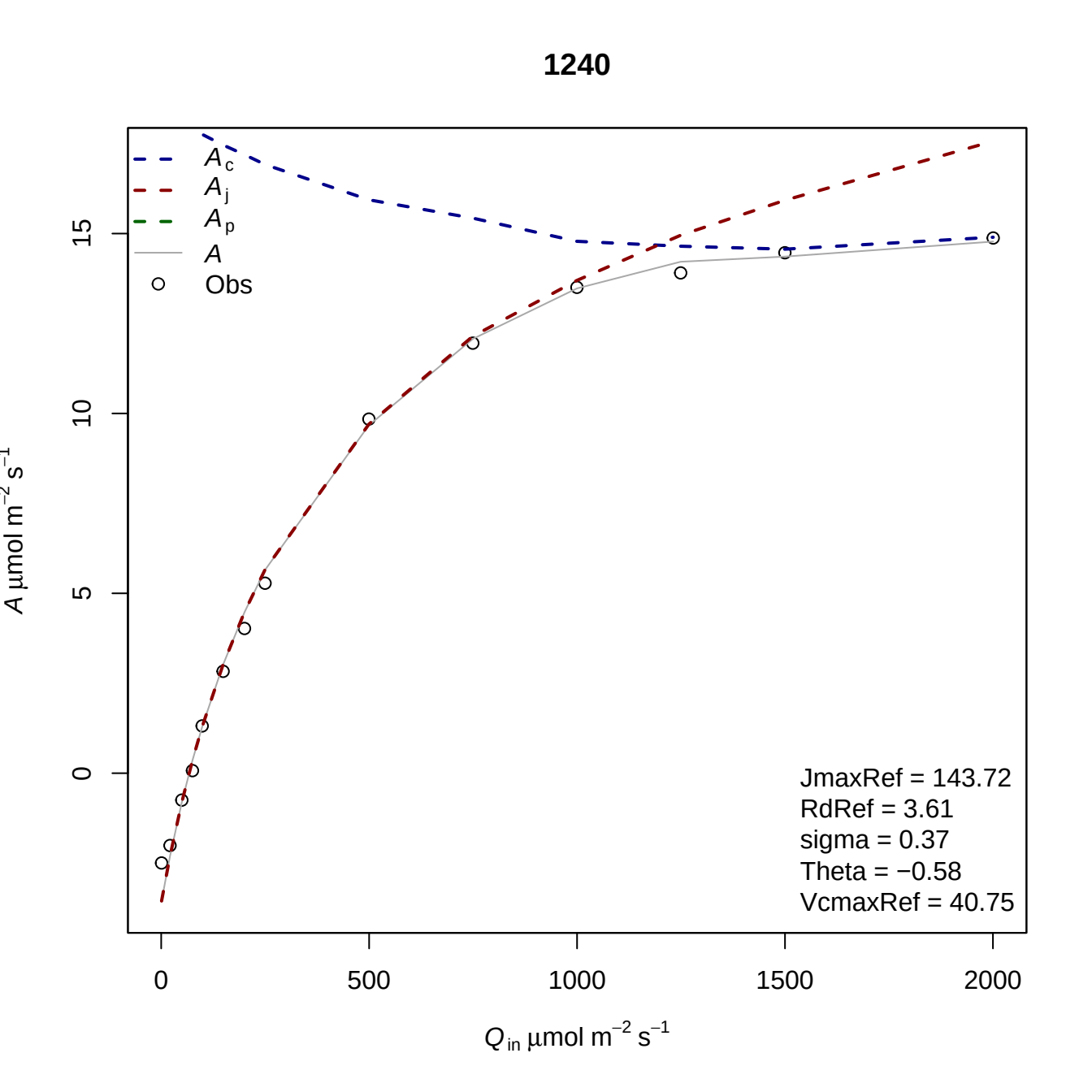

1235
A_c
A_j
A_p
A
Obs
JmaxRef = 161.14
RdRef = 1.92
sigma = 0.35
Theta = 0.06
VcmaxRef = 82.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

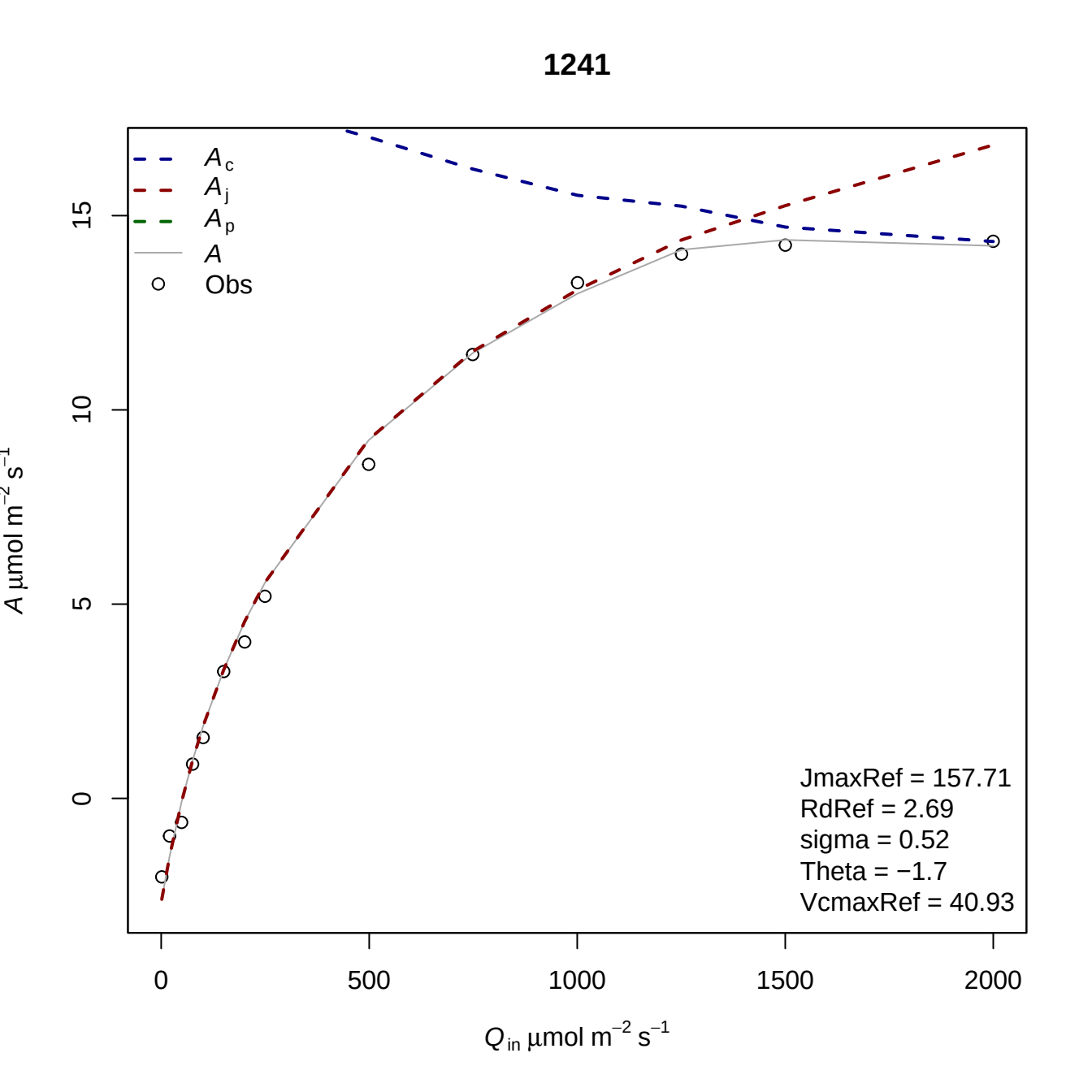

1236
A_c
A_j
A_p
A
Obs
JmaxRef = 163.66
RdRef = 2.4
sigma = 0.63
Theta = 0.74
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

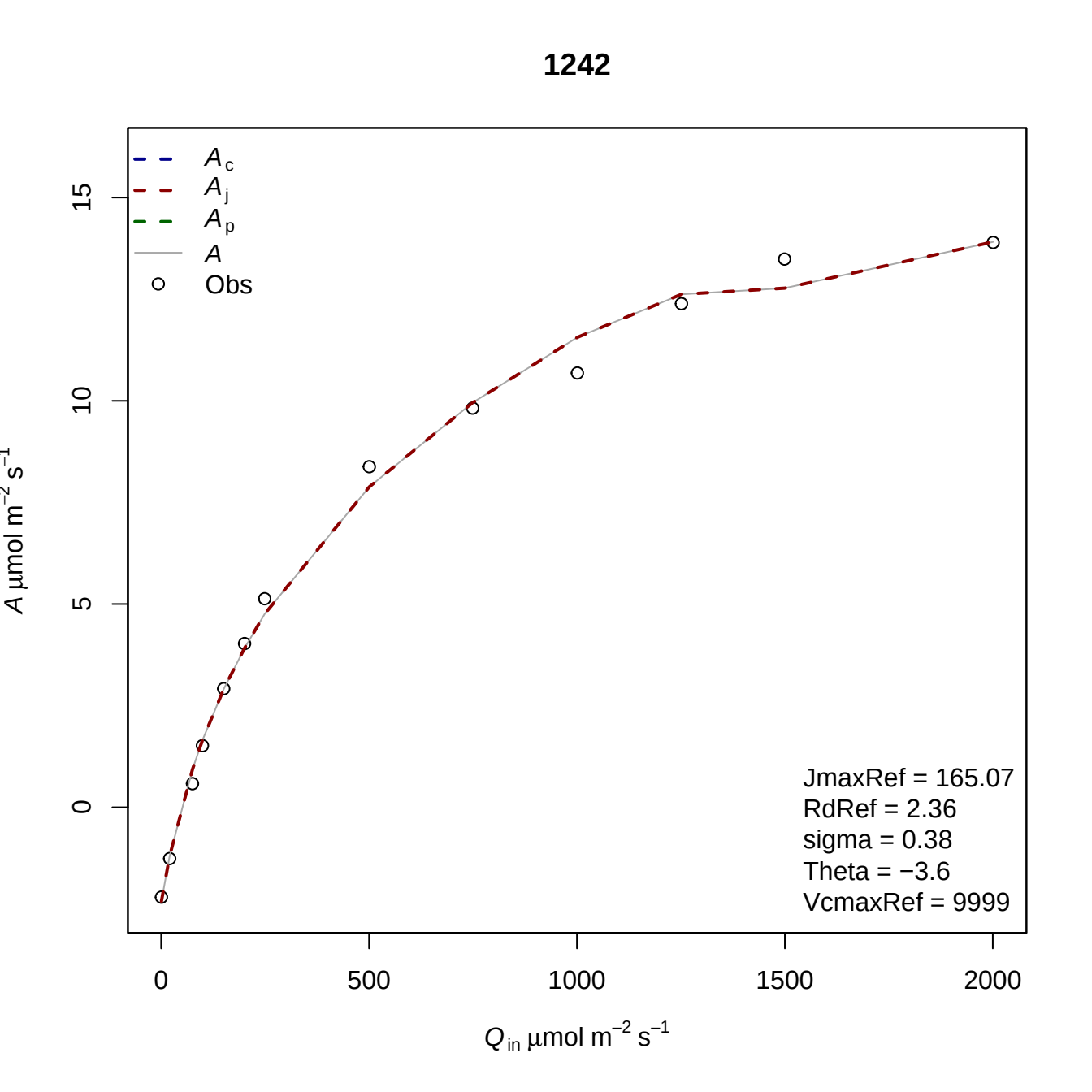

1237
A_c
A_j
A_p
A
Obs
JmaxRef = 138.03
RdRef = 1.77
sigma = 0.29
Theta = 0.69
VcmaxRef = 85.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

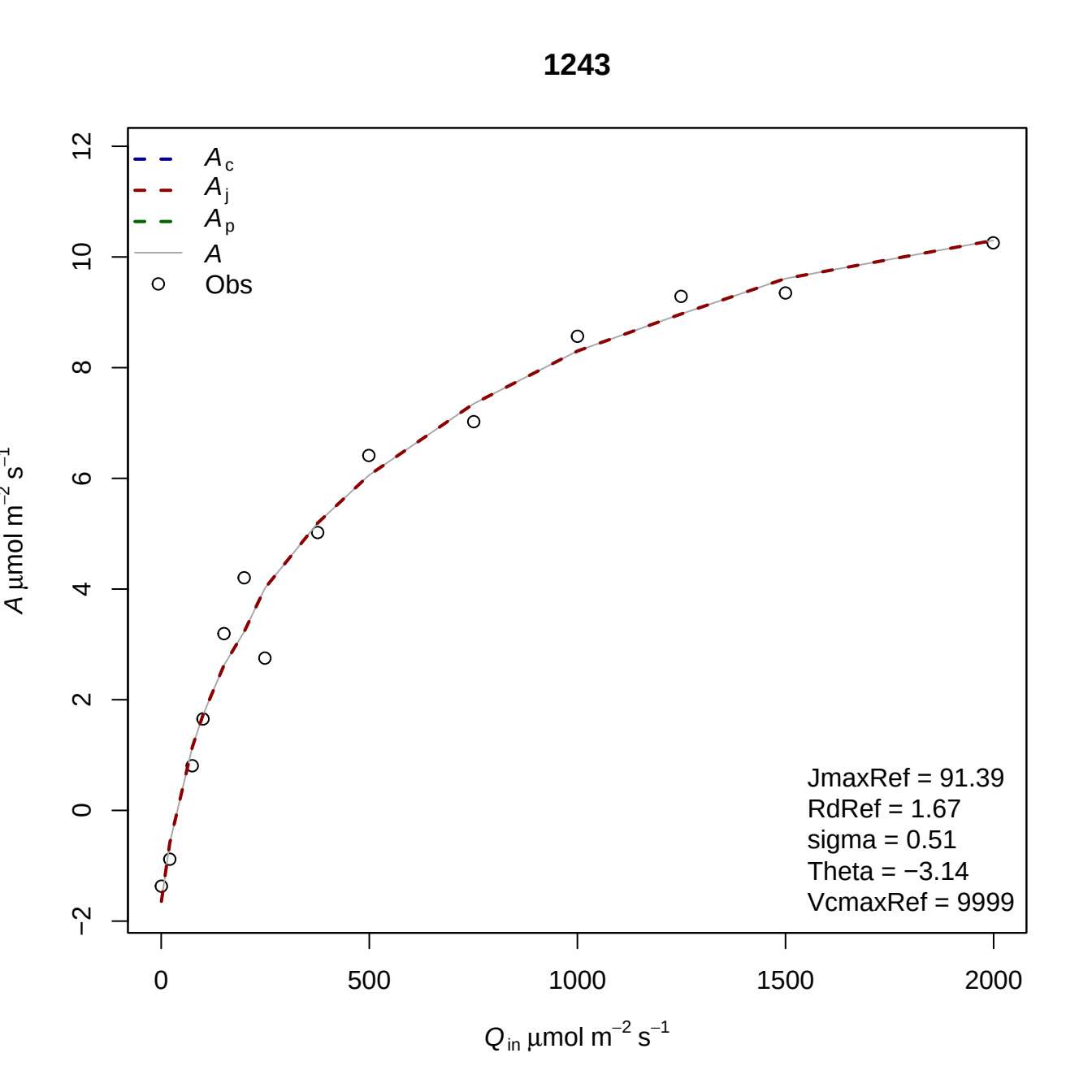

1238
A_c
A_j
A_p
A
Obs
JmaxRef = 338.88
RdRef = 1.46
sigma = 0.32
Theta = −6.19
VcmaxRef = 75.85
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

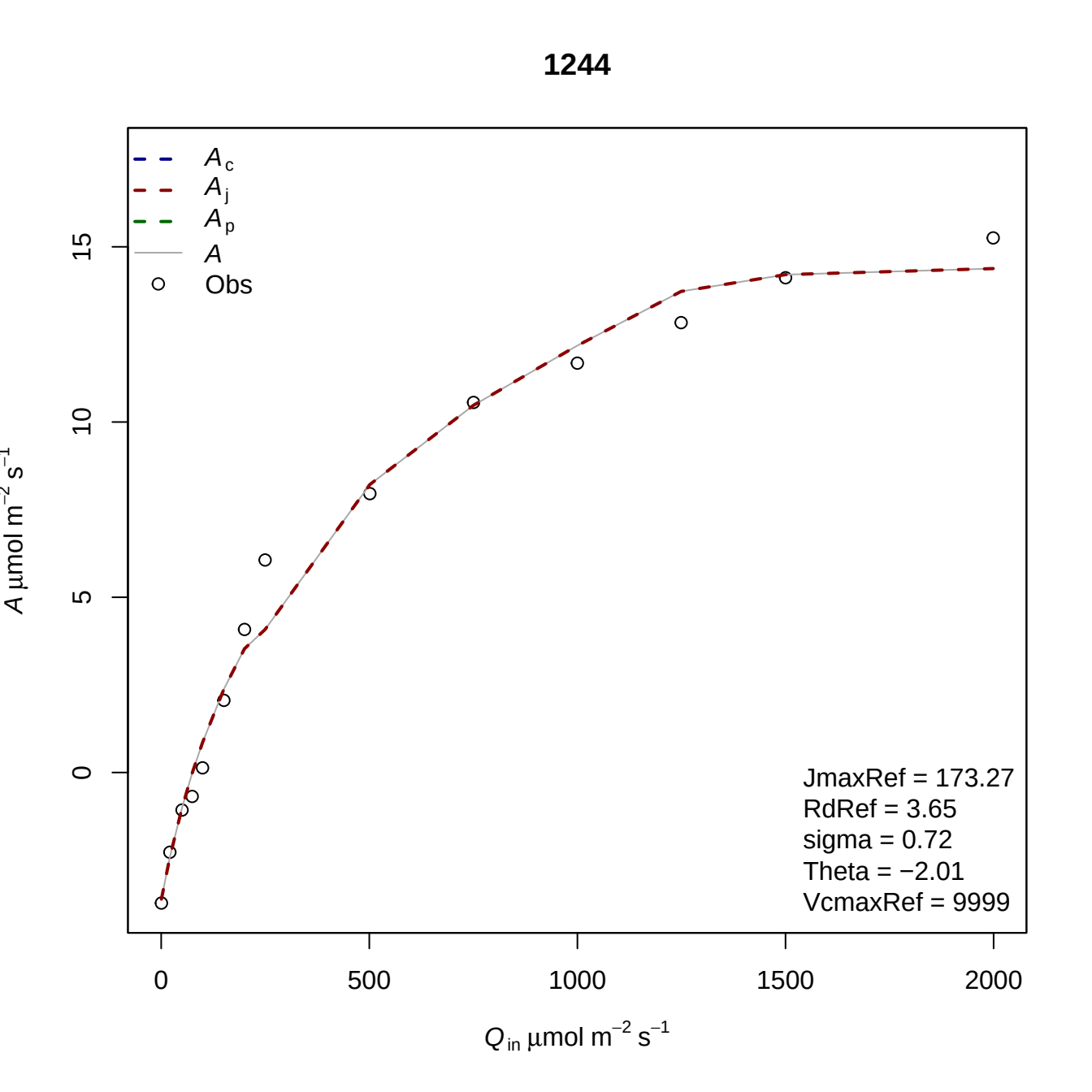

1239
A_c
A_j
A_p
A
Obs
JmaxRef = 85.67
RdRef = 1.57
sigma = 0.56
Theta = −0.11
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

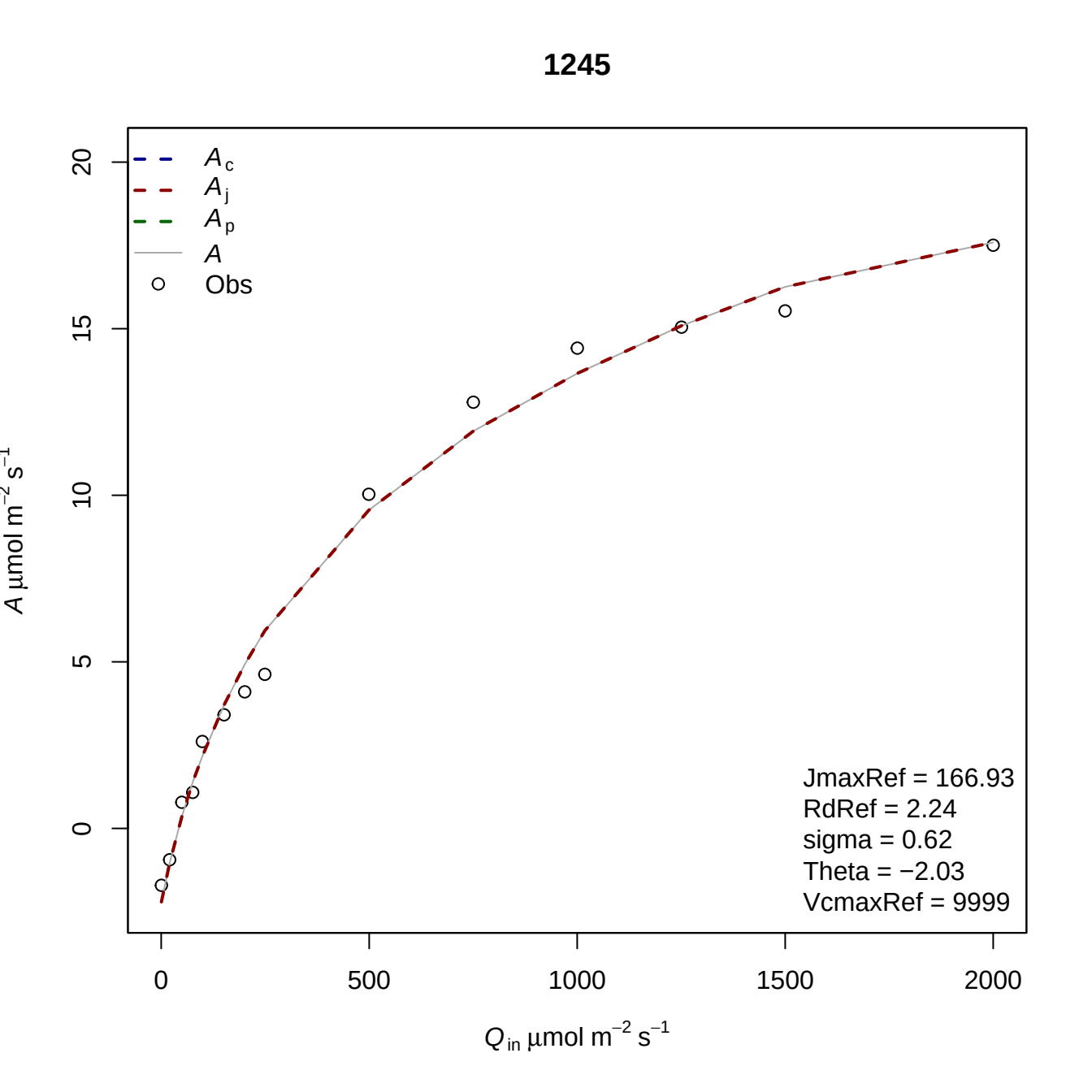

1240
A_c
A_j
A_p
A
Obs
JmaxRef = 143.72
RdRef = 3.61
sigma = 0.37
Theta = −0.58
VcmaxRef = 40.75
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

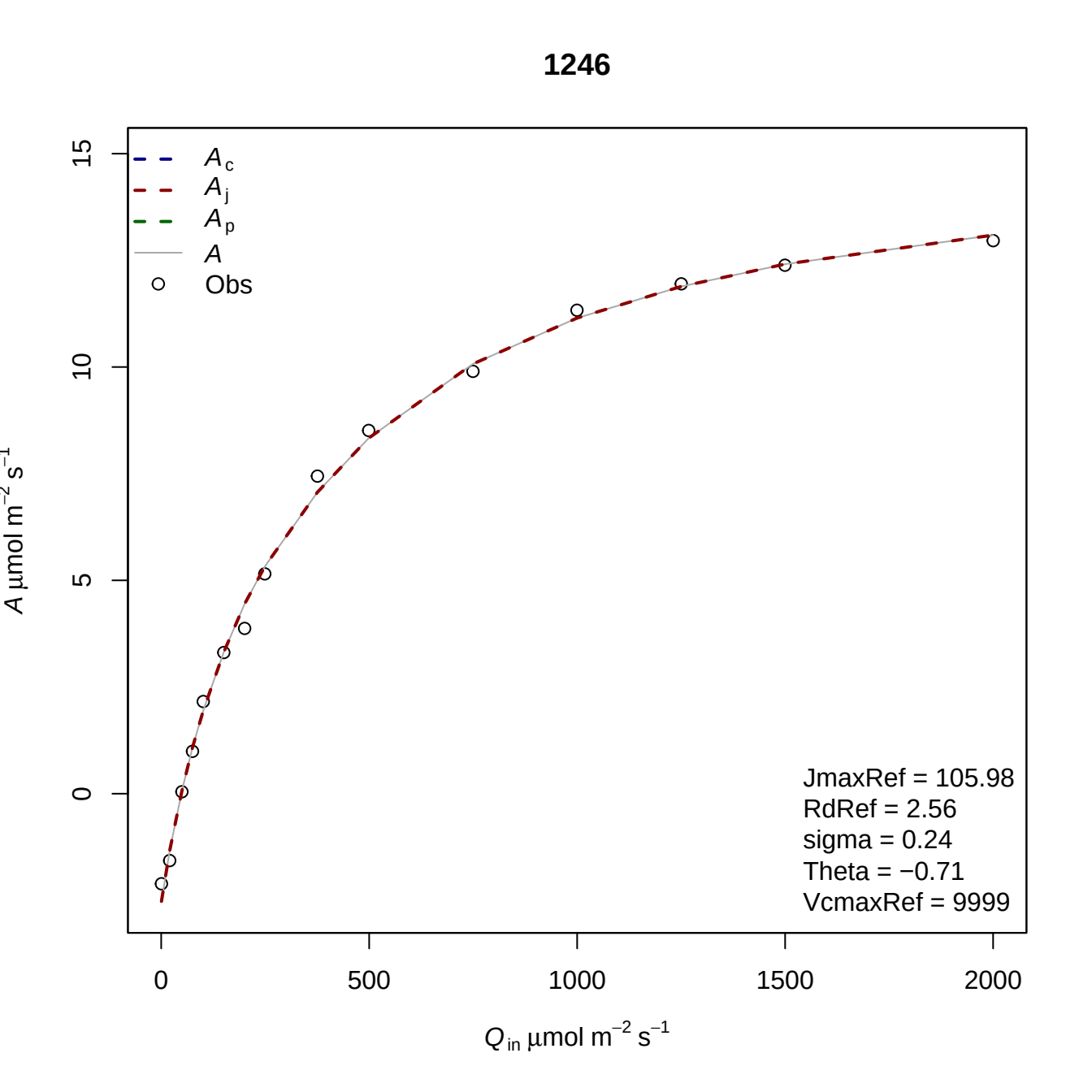

1241
A_c
A_j
A_p
A
Obs
JmaxRef = 157.71
RdRef = 2.69
sigma = 0.52
Theta = −1.7
VcmaxRef = 40.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

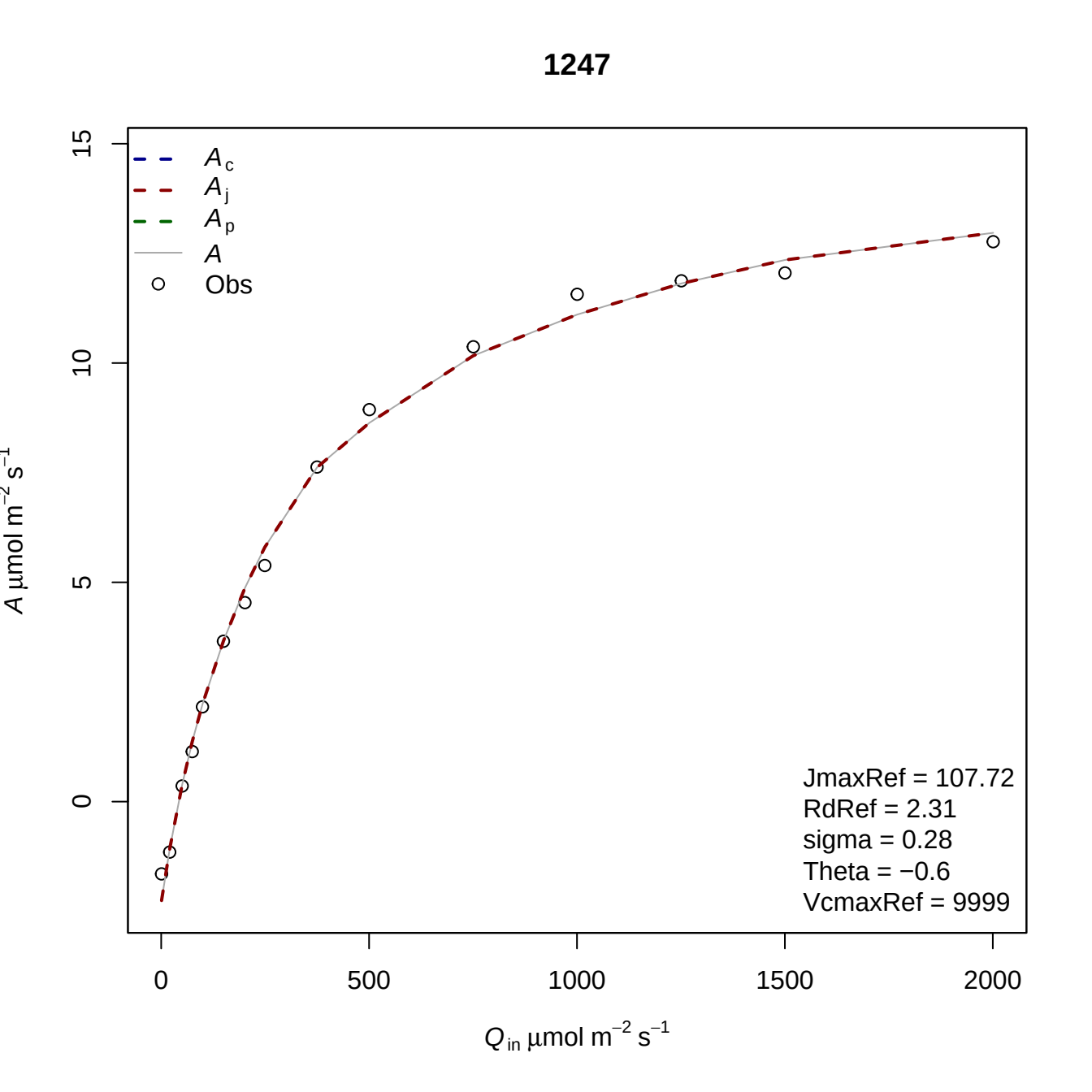

1242
A_c
A_j
A_p
A
Obs
JmaxRef = 165.07
RdRef = 2.36
sigma = 0.38
Theta = −3.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

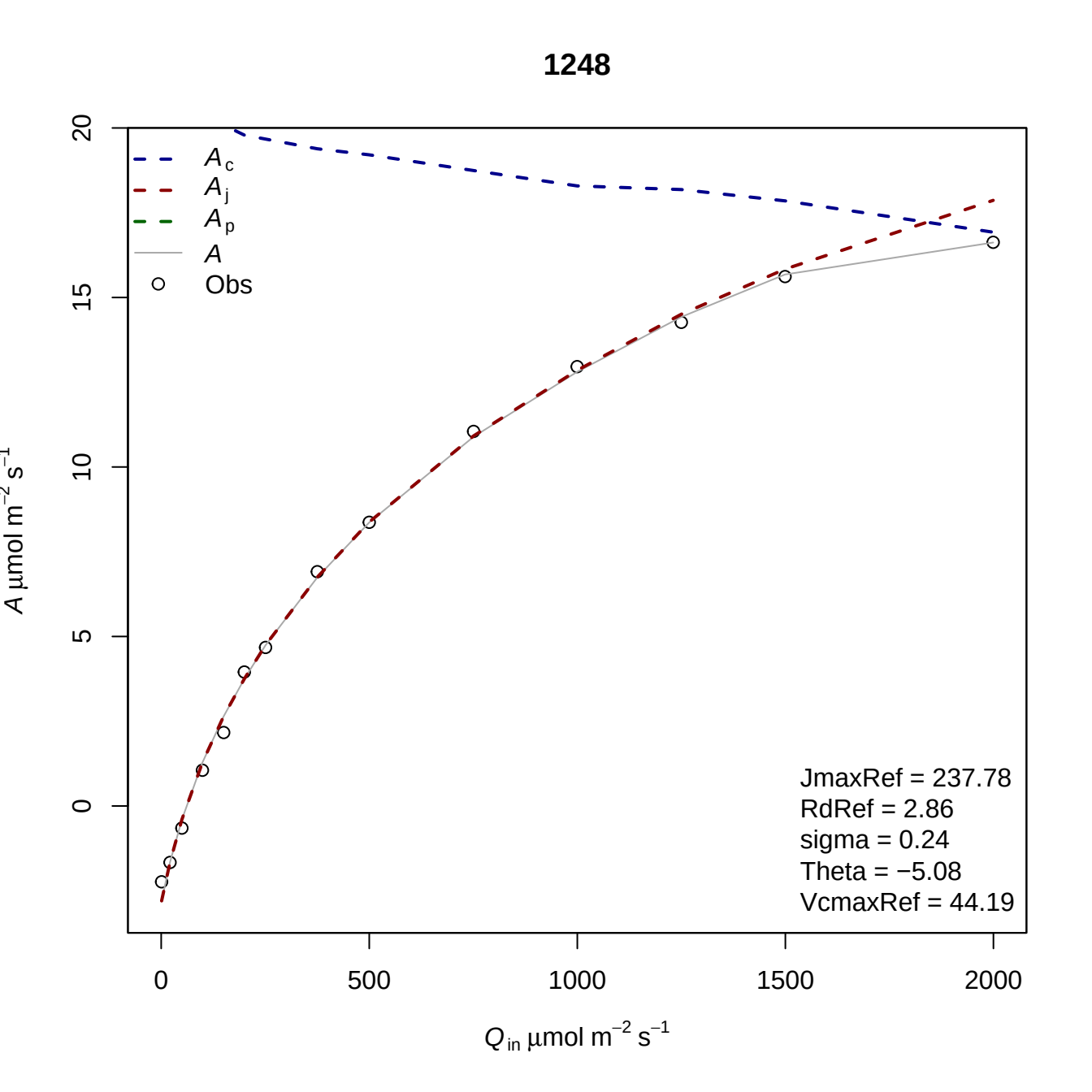

1243
A_c
A_j
A_p
A
Obs
JmaxRef = 91.39
RdRef = 1.67
sigma = 0.51
Theta = −3.14
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

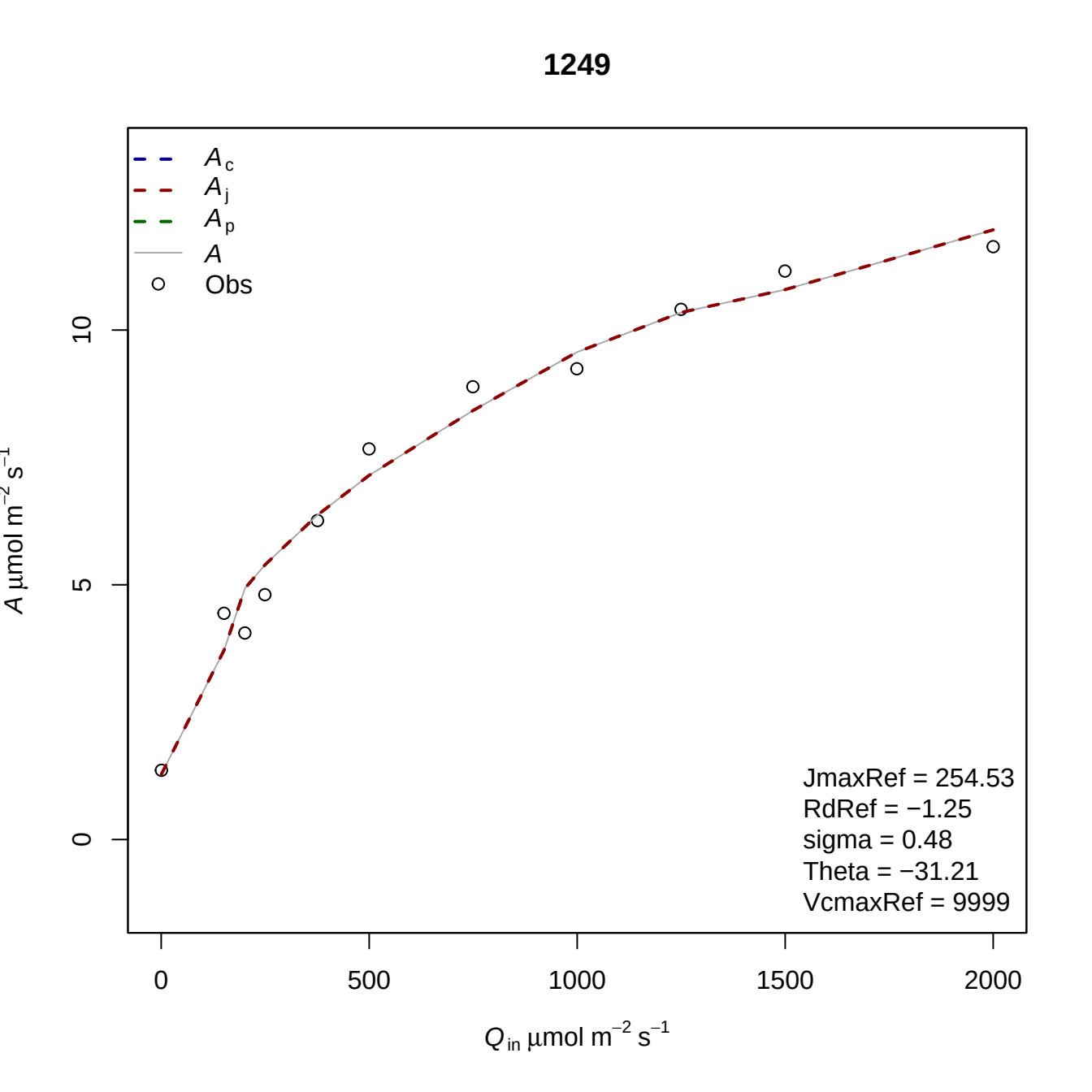

1244
A_c
A_j
A_p
A
Obs
JmaxRef = 173.27
RdRef = 3.65
sigma = 0.72
Theta = −2.01
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

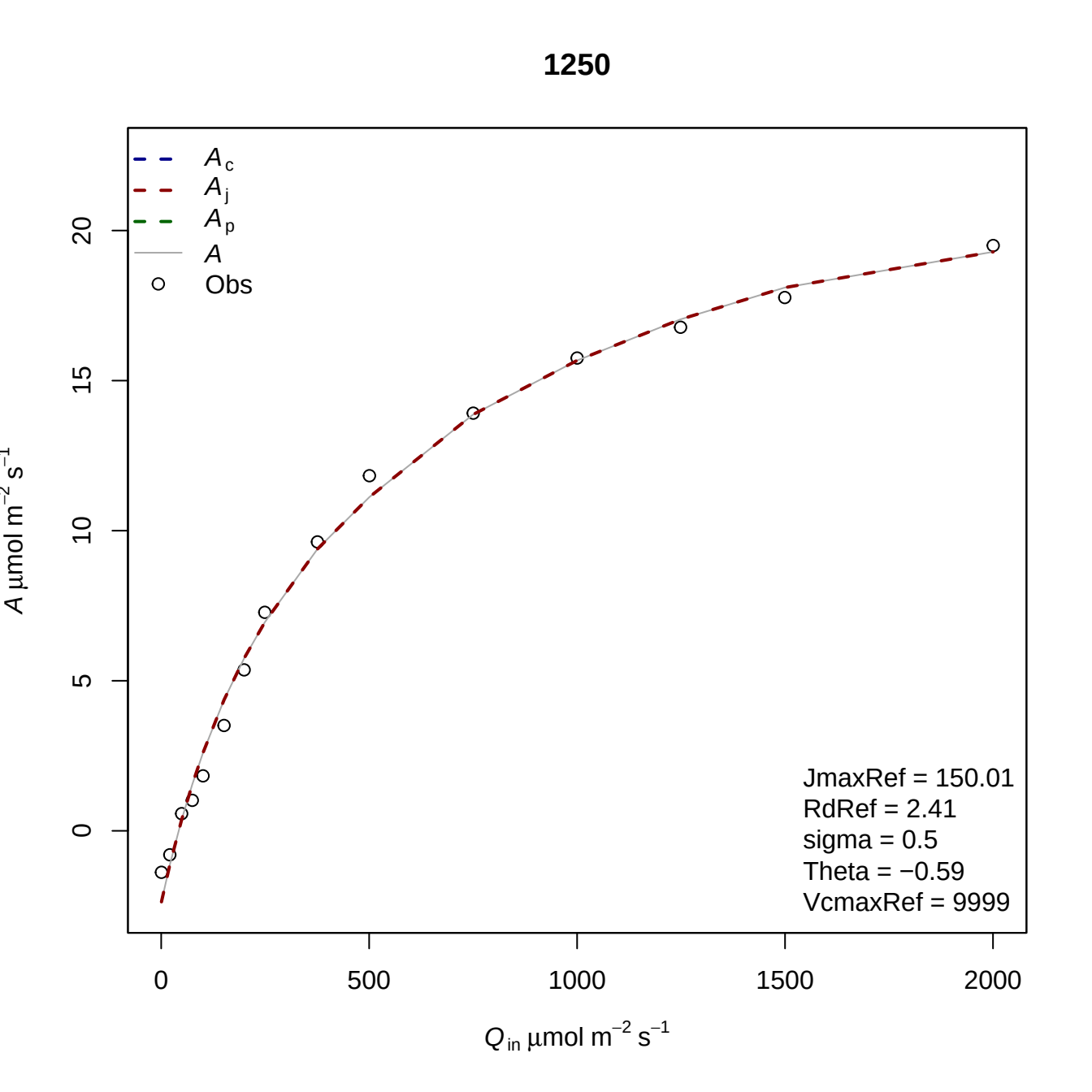

1245
A_c
A_j
A_p
A
Obs
JmaxRef = 166.93
RdRef = 2.24
sigma = 0.62
Theta = −2.03
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

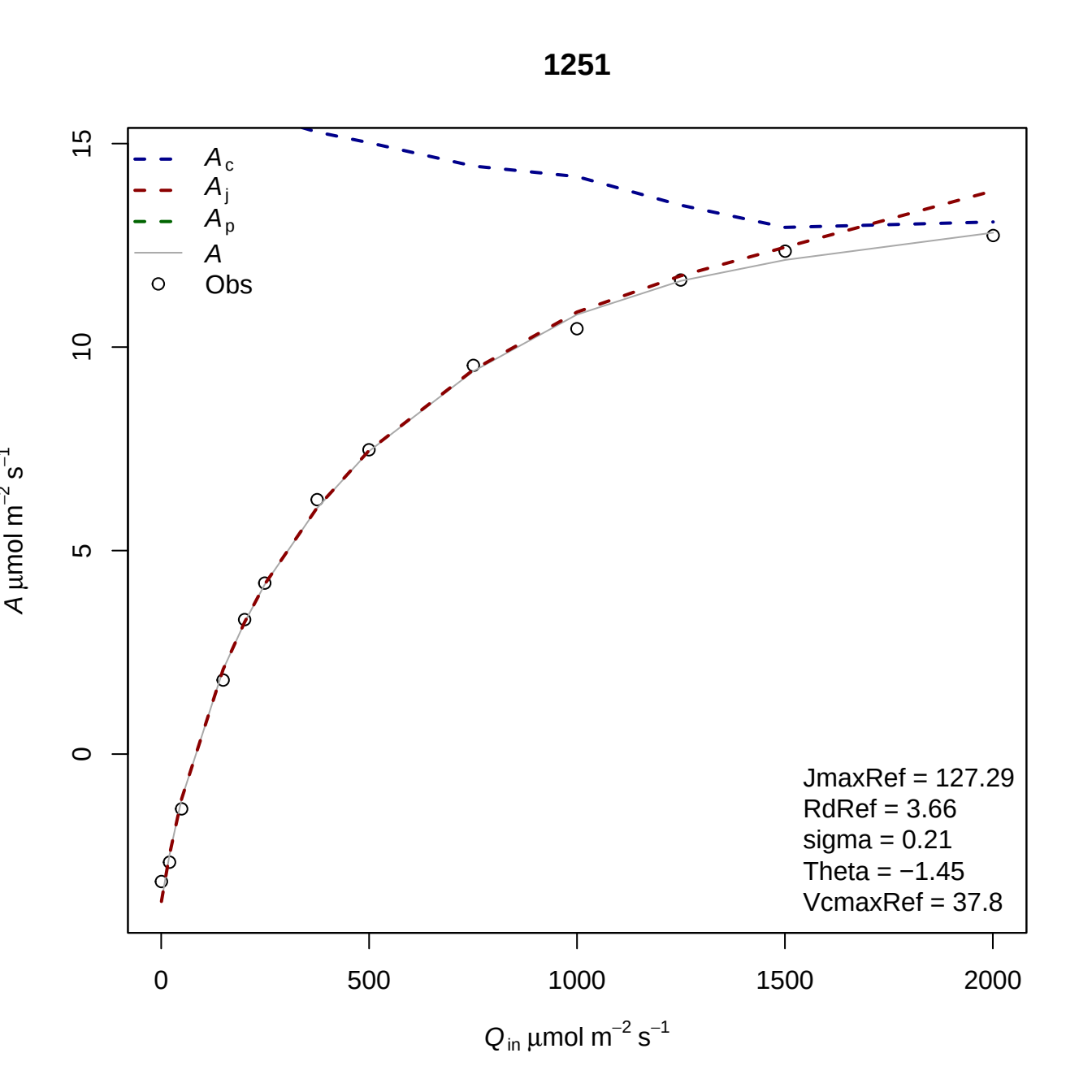

1246
A_c
A_j
A_p
A
Obs
JmaxRef = 105.98
RdRef = 2.56
sigma = 0.24
Theta = −0.71
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

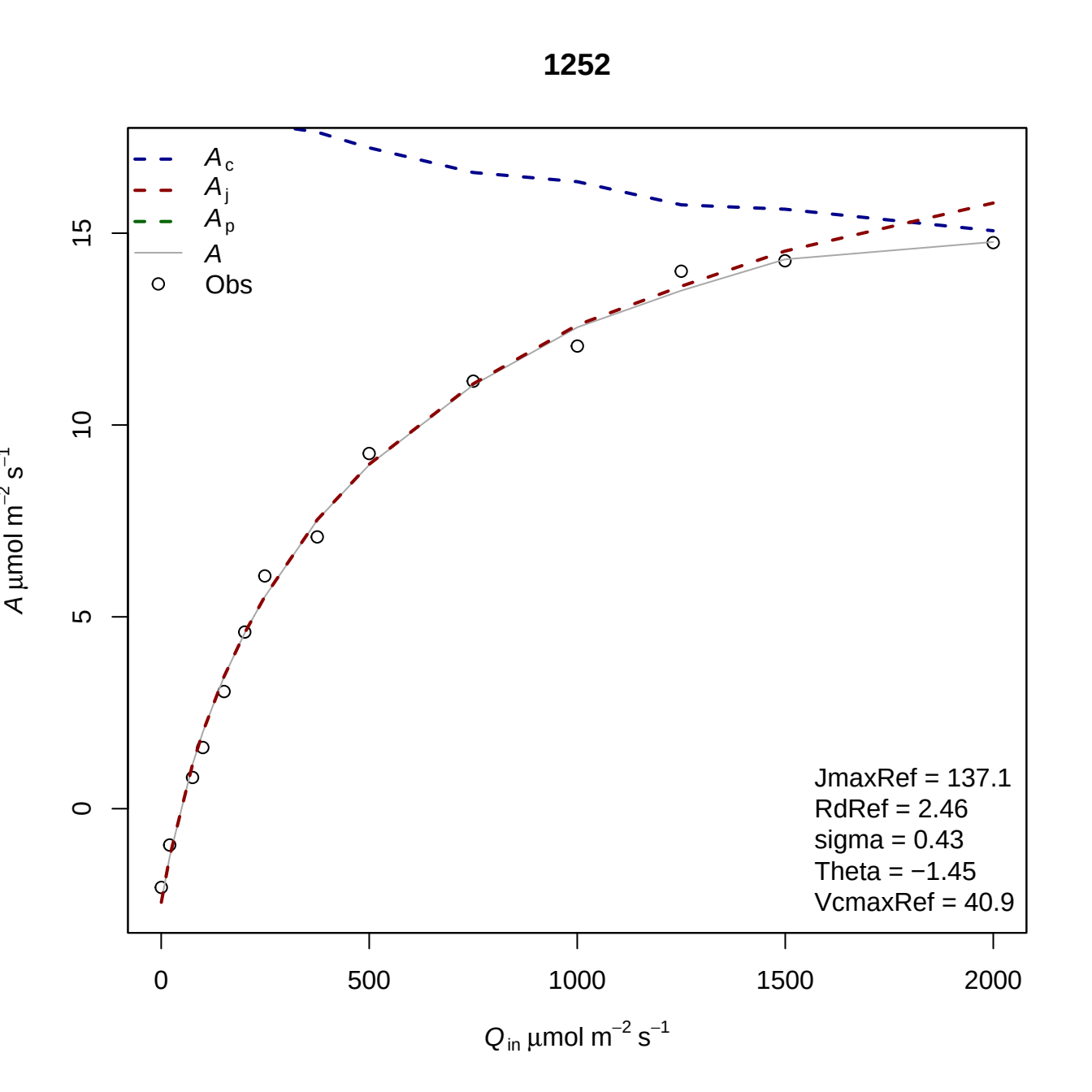

1247
A_c
A_j
A_p
A
Obs
JmaxRef = 107.72
RdRef = 2.31
sigma = 0.28
Theta = −0.6
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

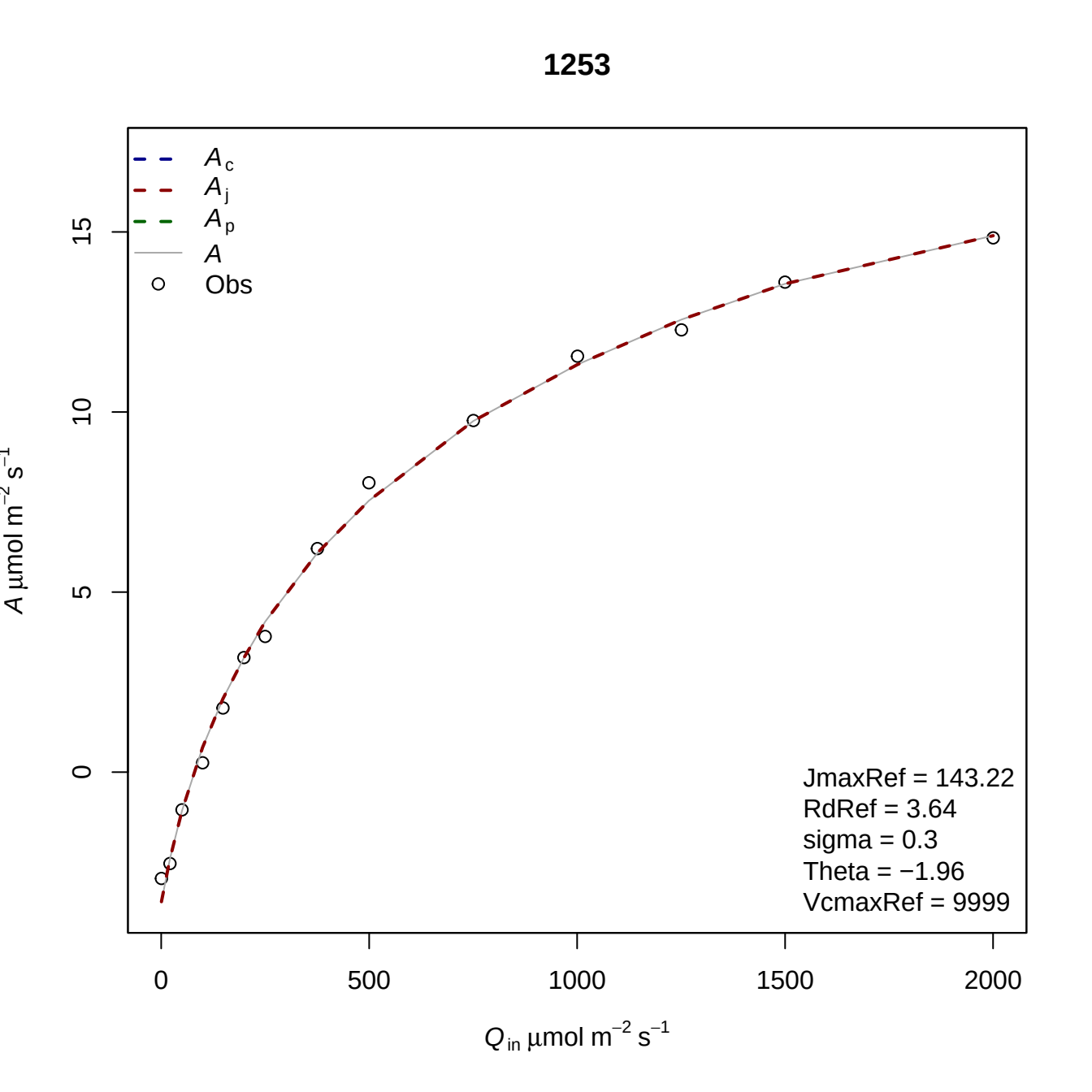

1248
A_c
A_j
A_p
A
Obs
JmaxRef = 237.78
RdRef = 2.86
sigma = 0.24
Theta = −5.08
VcmaxRef = 44.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

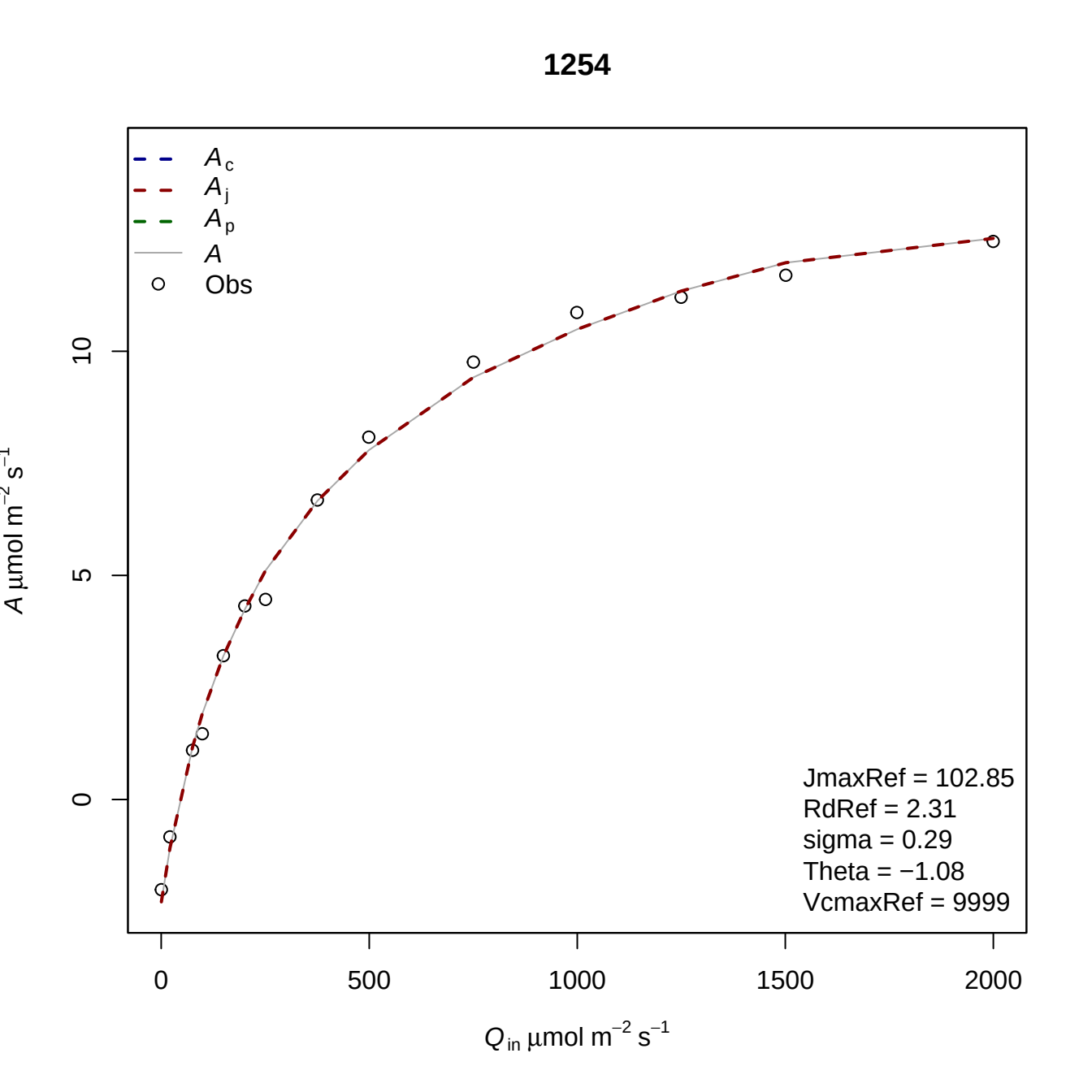

1249
A_c
A_j
A_p
A
Obs
JmaxRef = 254.53
RdRef = −1.25
sigma = 0.48
Theta = −31.21
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

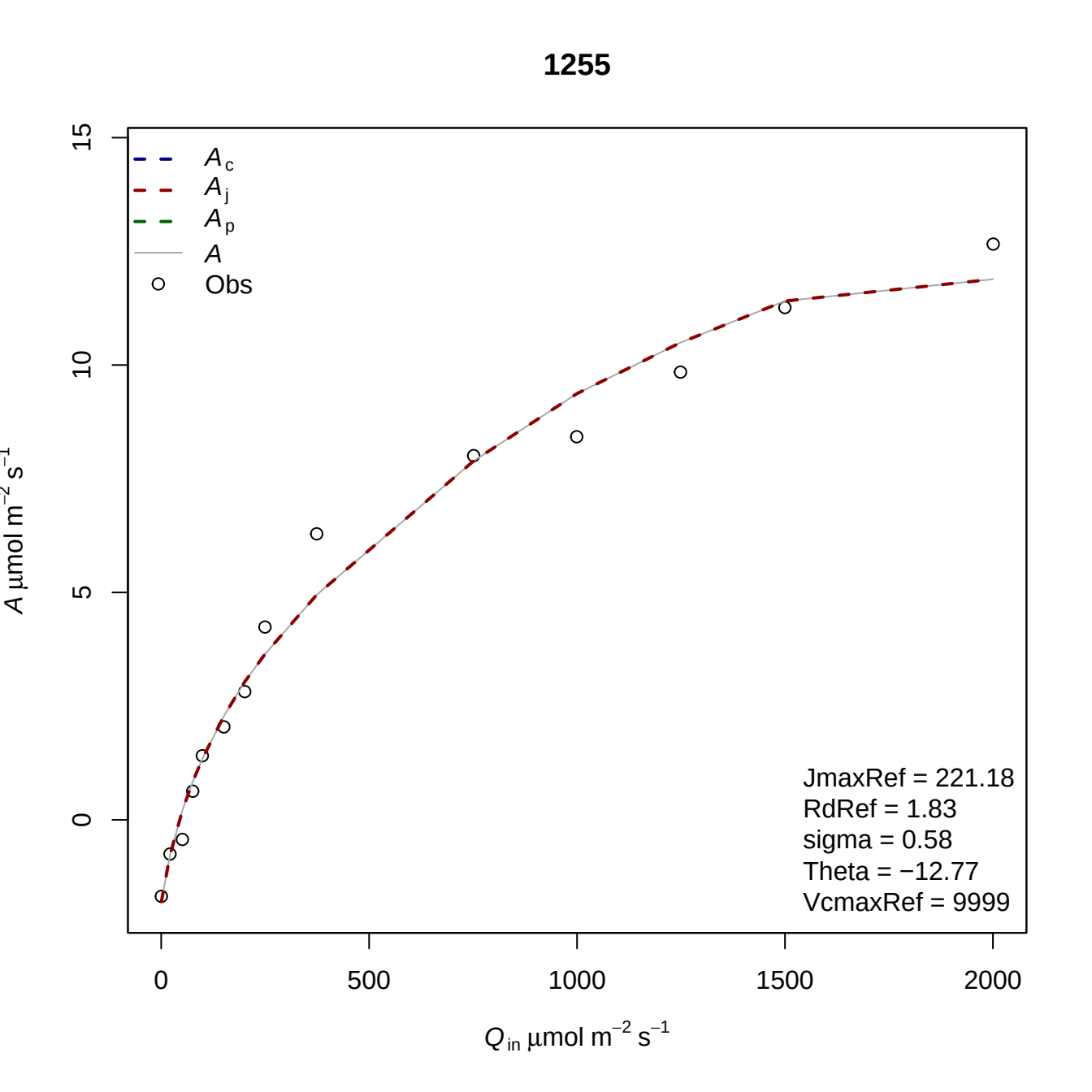

1250
A_c
A_j
A_p
A
Obs
JmaxRef = 150.01
RdRef = 2.41
sigma = 0.5
Theta = −0.59
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

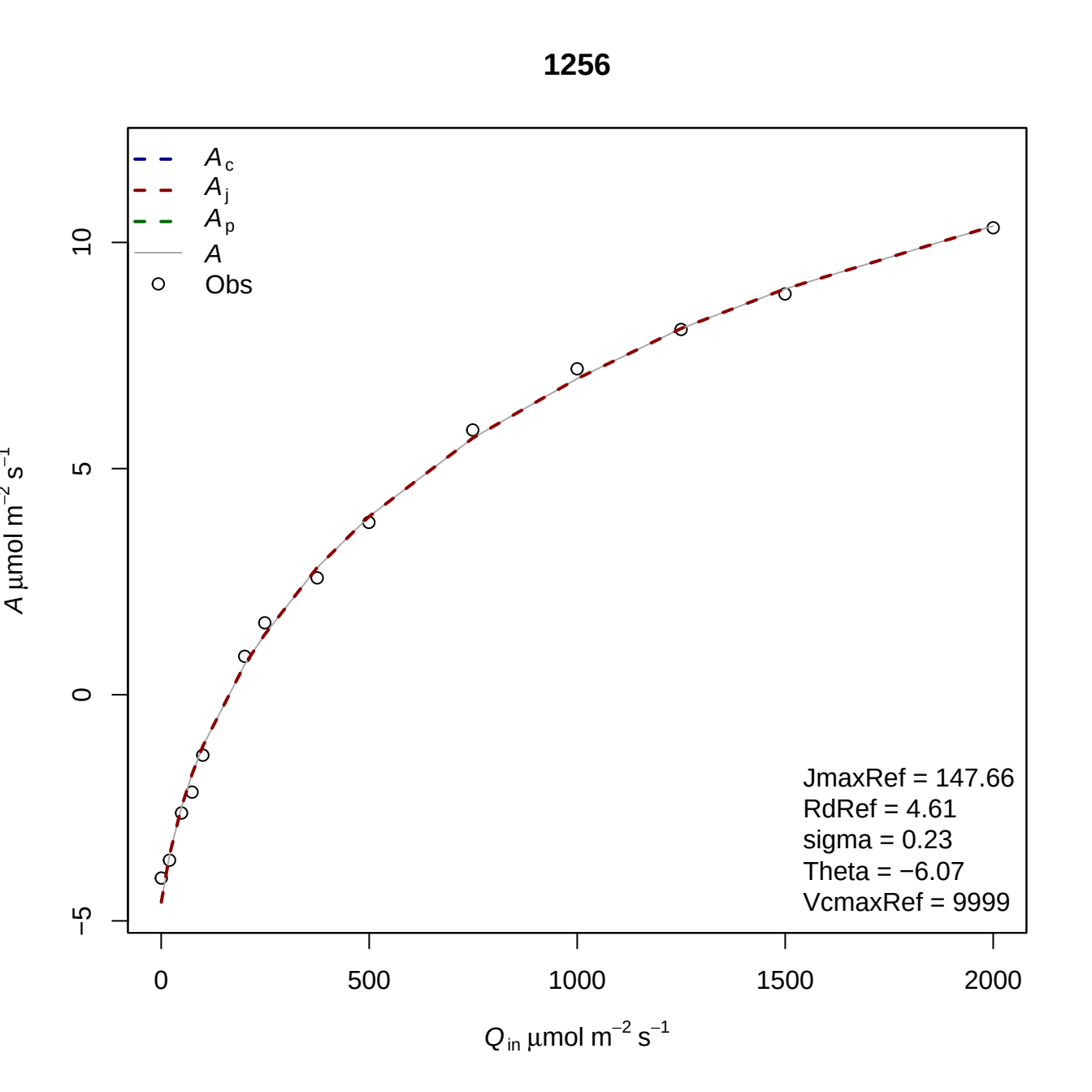

1251
A_c
A_j
A_p
A
Obs
JmaxRef = 127.29
RdRef = 3.66
sigma = 0.21
Theta = −1.45
VcmaxRef = 37.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

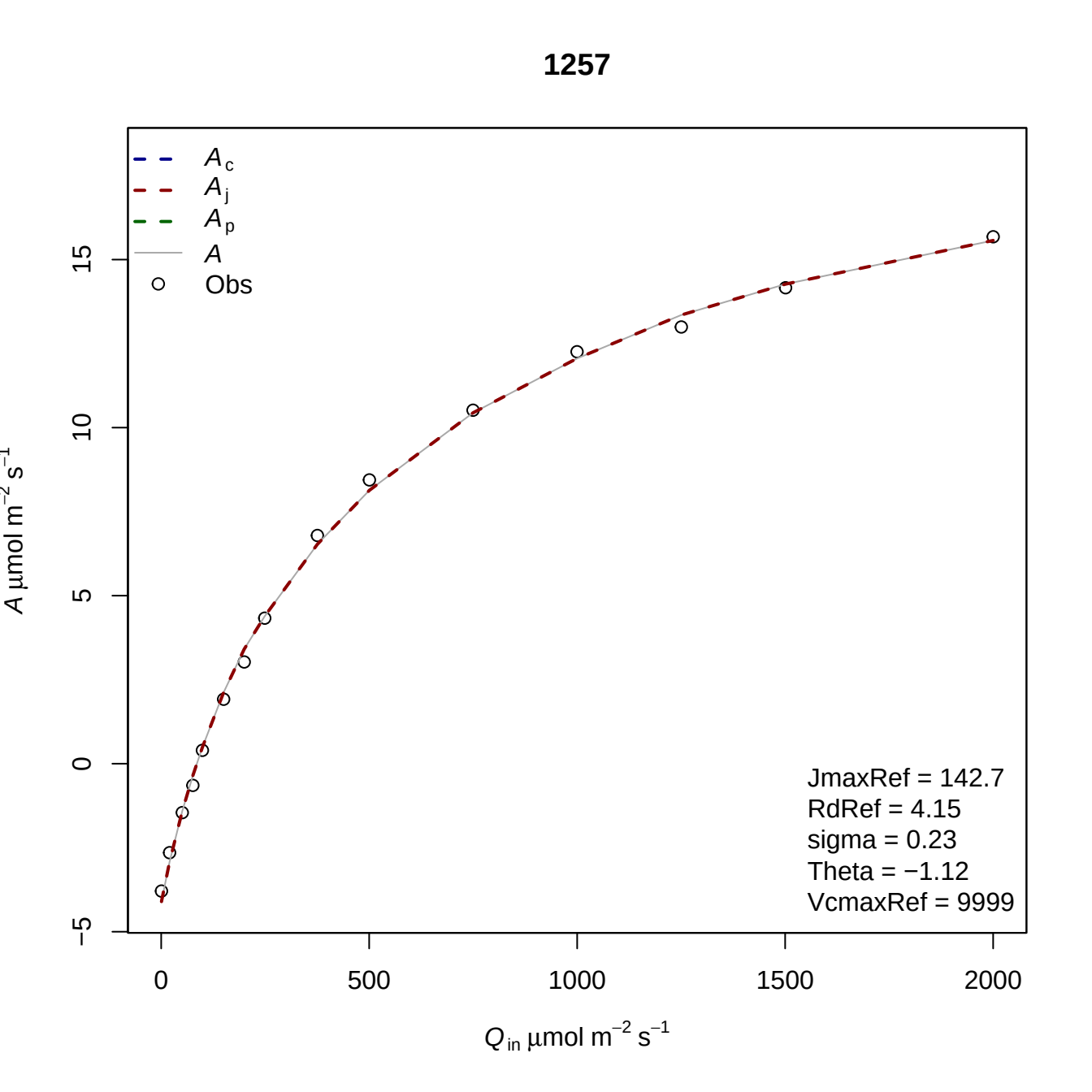

1252
A_c
A_j
A_p
A
Obs
JmaxRef = 137.1
RdRef = 2.46
sigma = 0.43
Theta = −1.45
VcmaxRef = 40.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

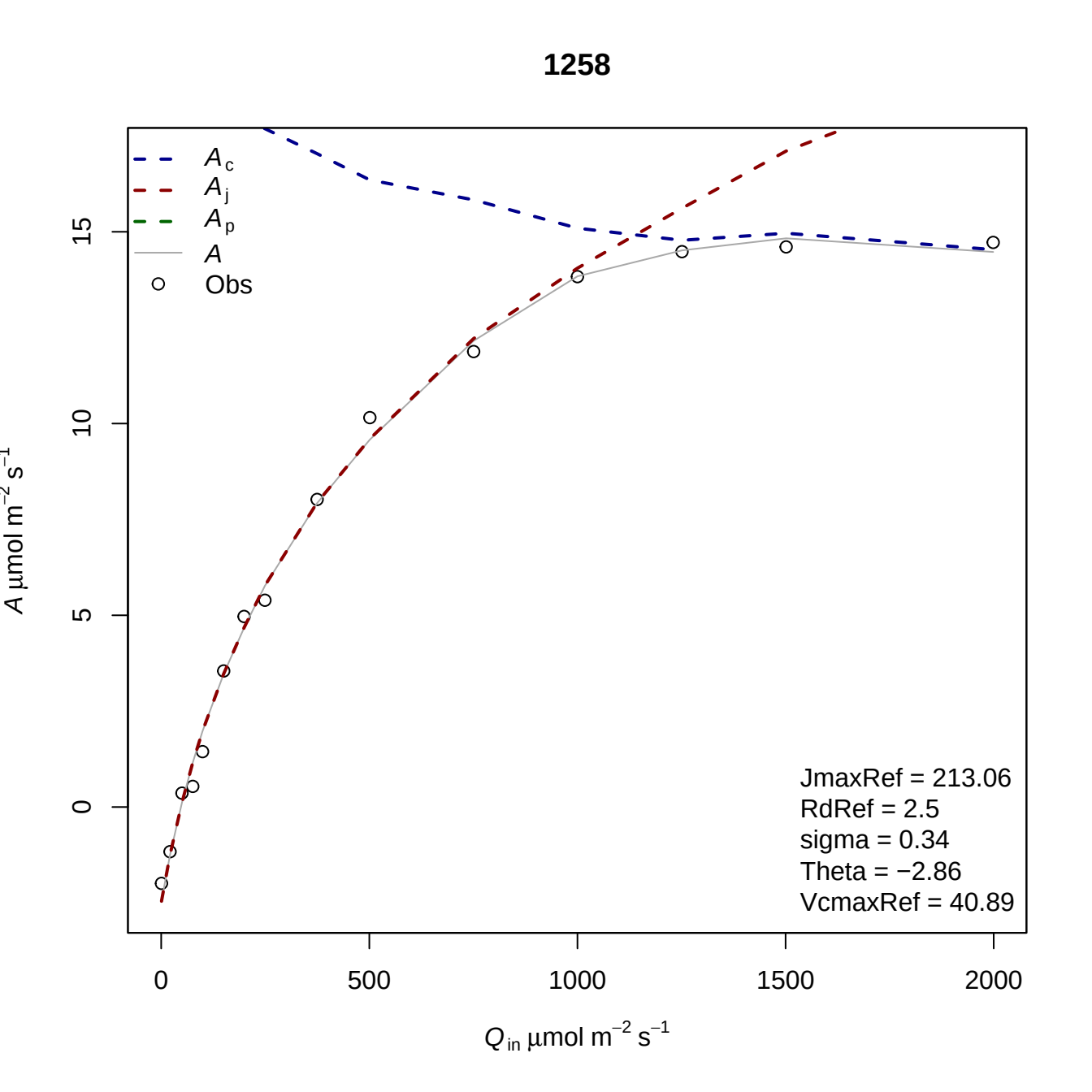

1253
A_c
A_j
A_p
A
Obs
JmaxRef = 143.22
RdRef = 3.64
sigma = 0.3
Theta = −1.96
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

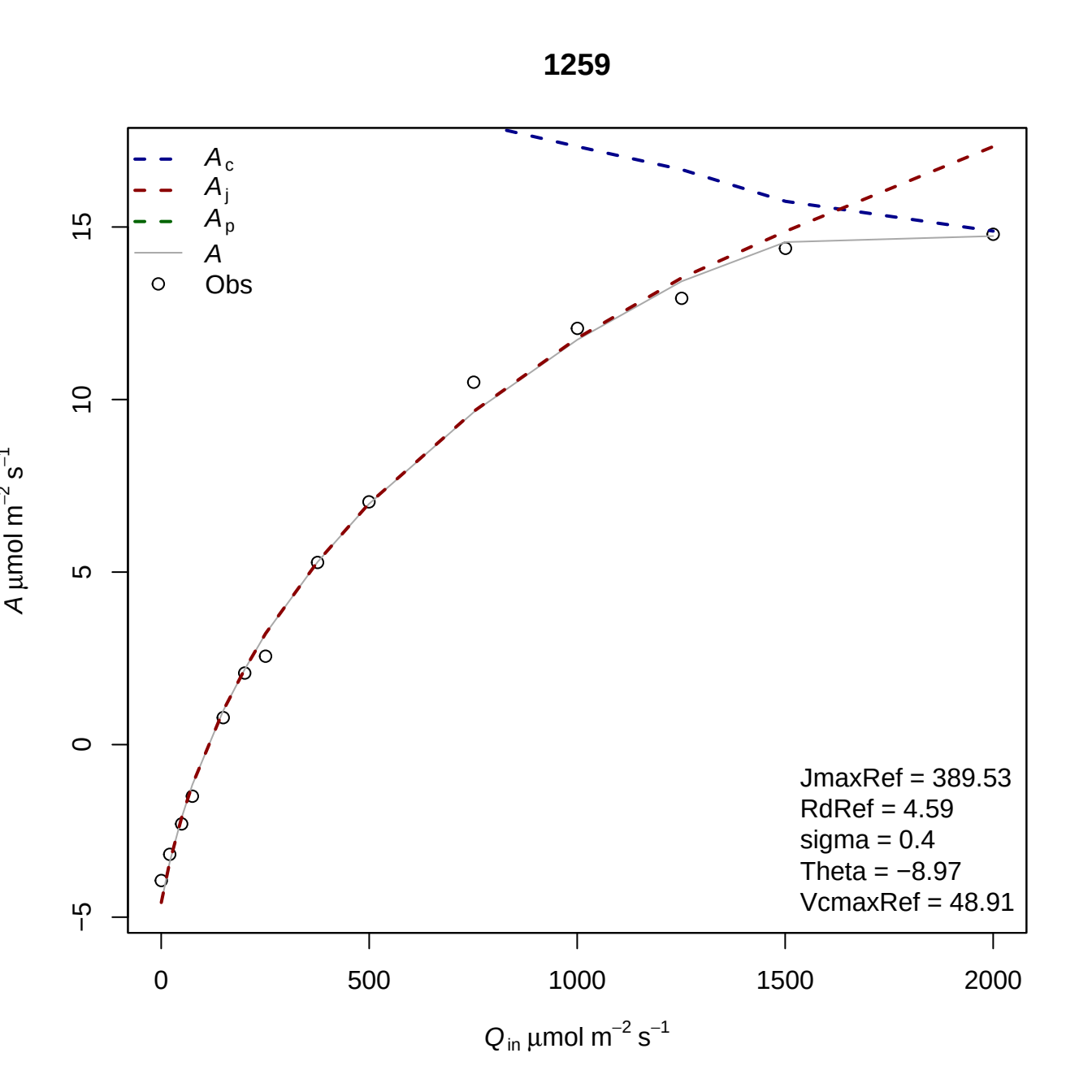

1254
A_c
A_j
A_p
A
Obs
JmaxRef = 102.85
RdRef = 2.31
sigma = 0.29
Theta = −1.08
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

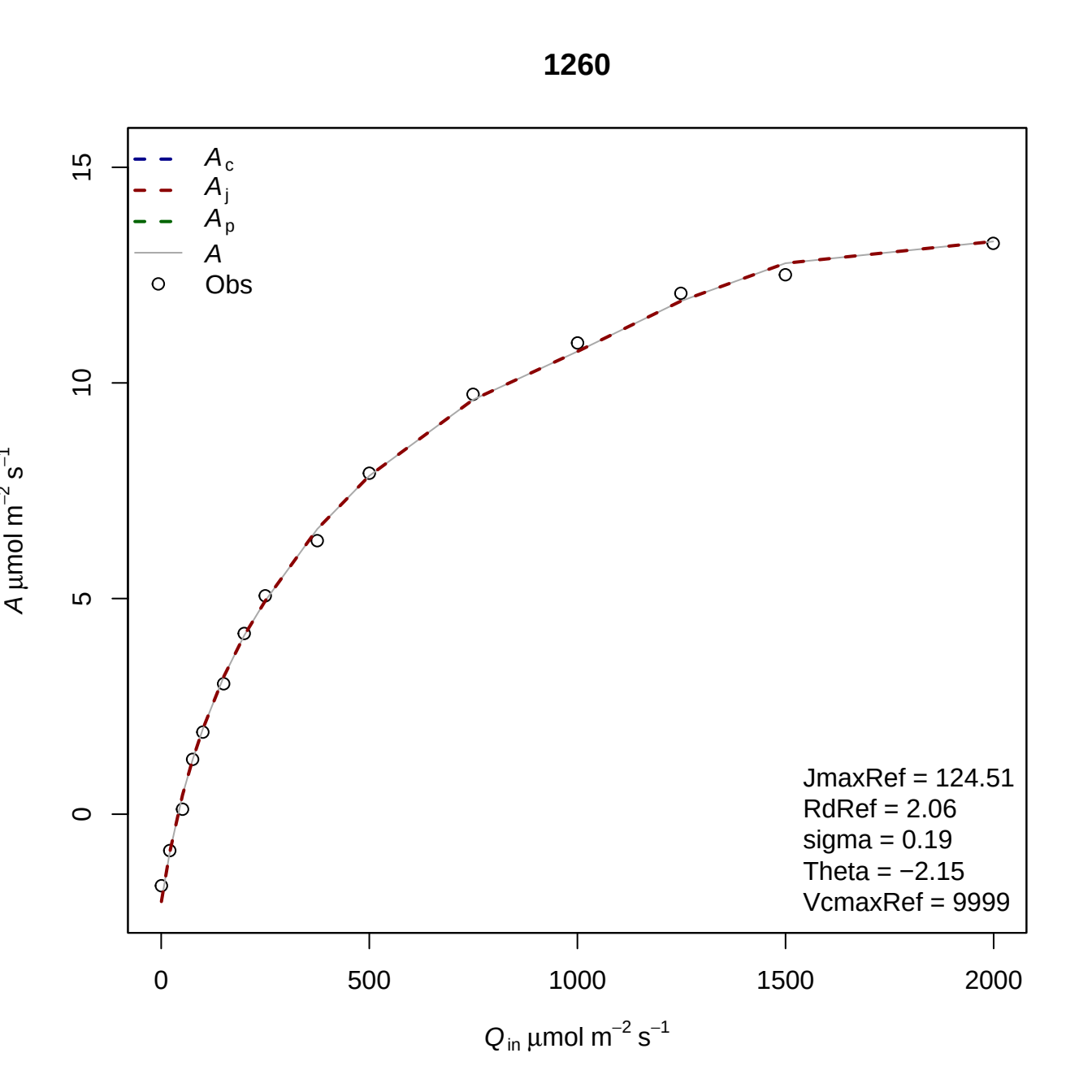

1255
A_c
A_j
A_p
A
Obs
JmaxRef = 221.18
RdRef = 1.83
sigma = 0.58
Theta = −12.77
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

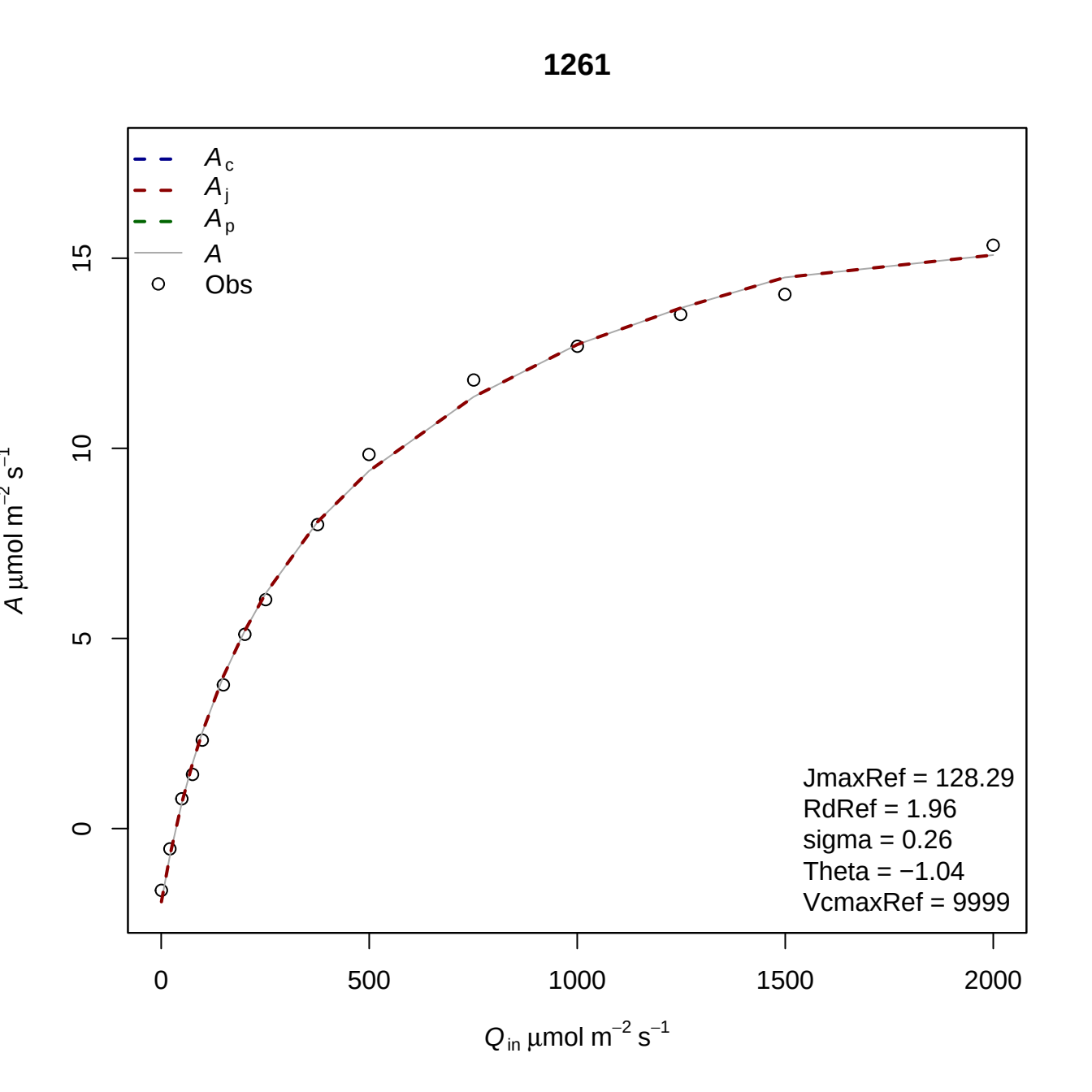

1256
A_c
A_j
A_p
A
Obs
JmaxRef = 147.66
RdRef = 4.61
sigma = 0.23
Theta = −6.07
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

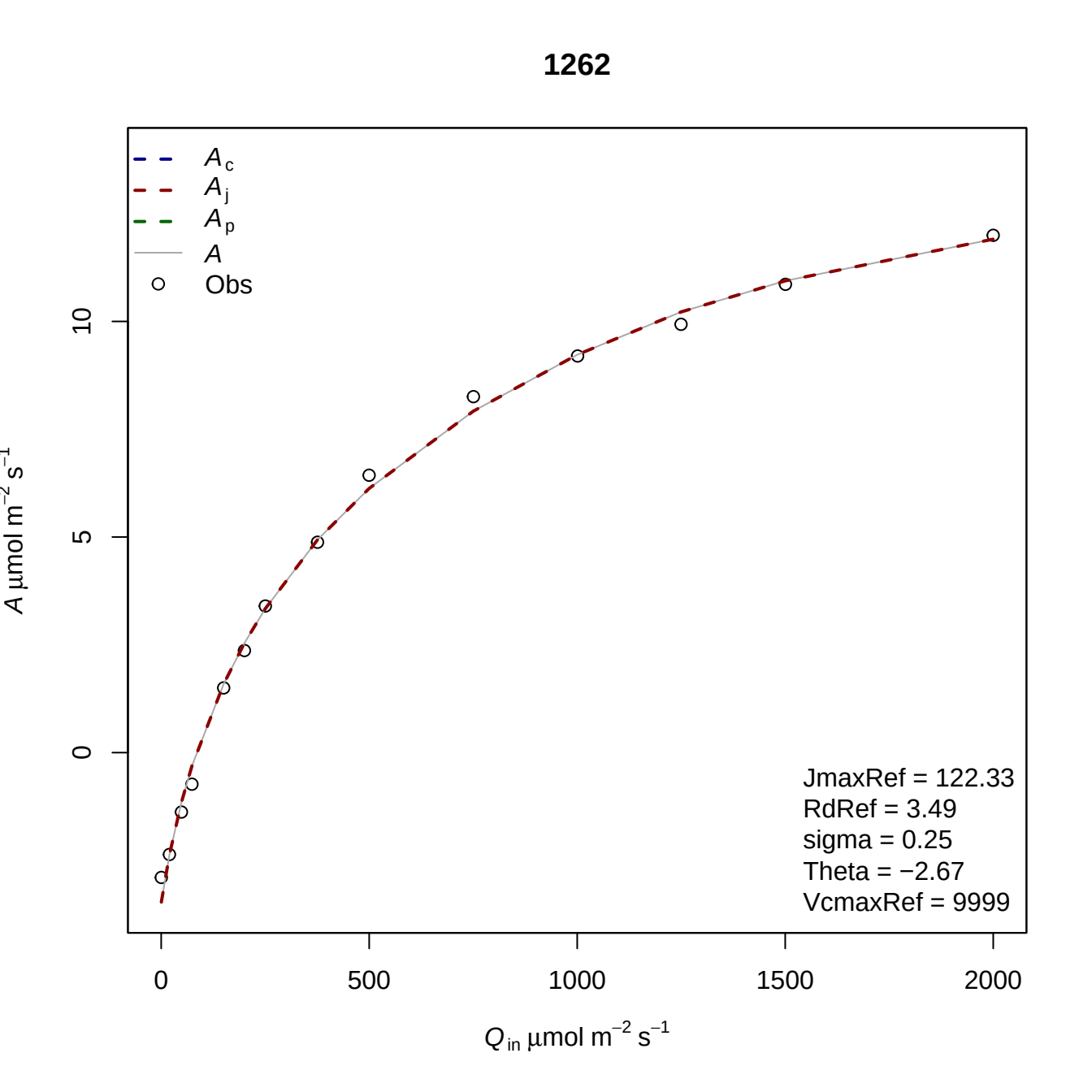

1257
A_c
A_j
A_p
A
Obs
JmaxRef = 142.7
RdRef = 4.15
sigma = 0.23
Theta = −1.12
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

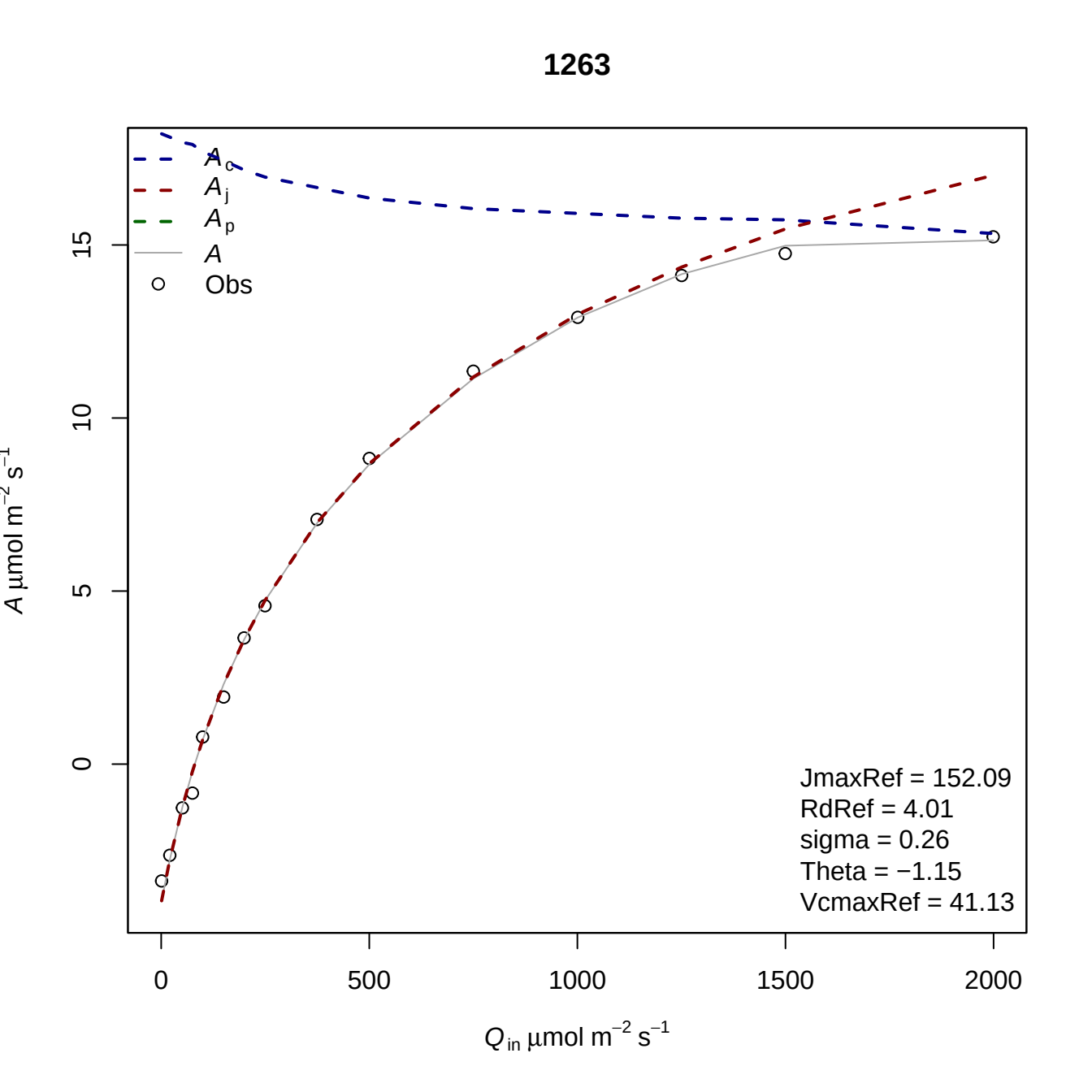

1258
A_c
A_j
A_p
A
Obs
JmaxRef = 213.06
RdRef = 2.5
sigma = 0.34
Theta = −2.86
VcmaxRef = 40.89
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

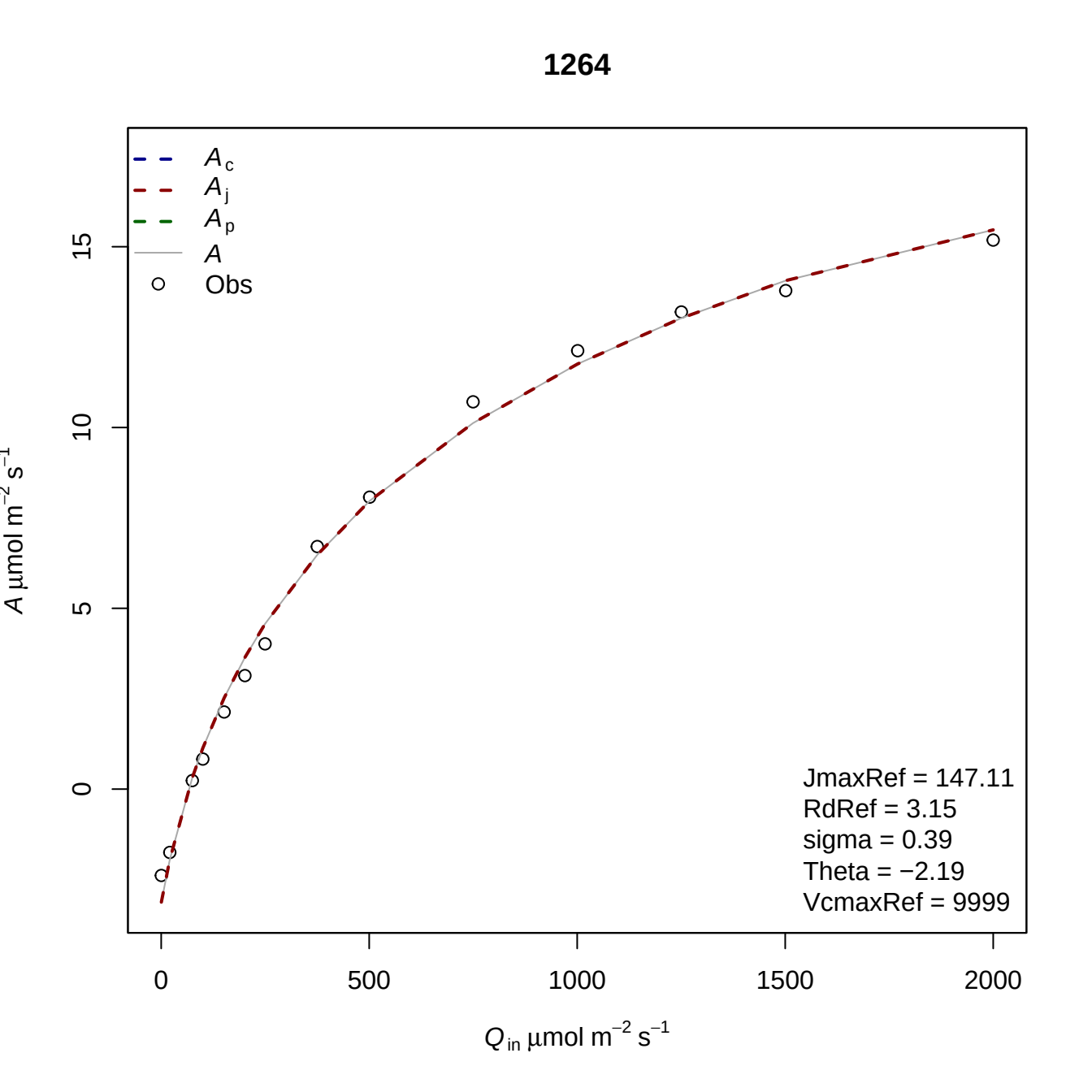

1259
A_c
A_j
A_p
A
Obs
JmaxRef = 389.53
RdRef = 4.59
sigma = 0.4
Theta = −8.97
VcmaxRef = 48.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

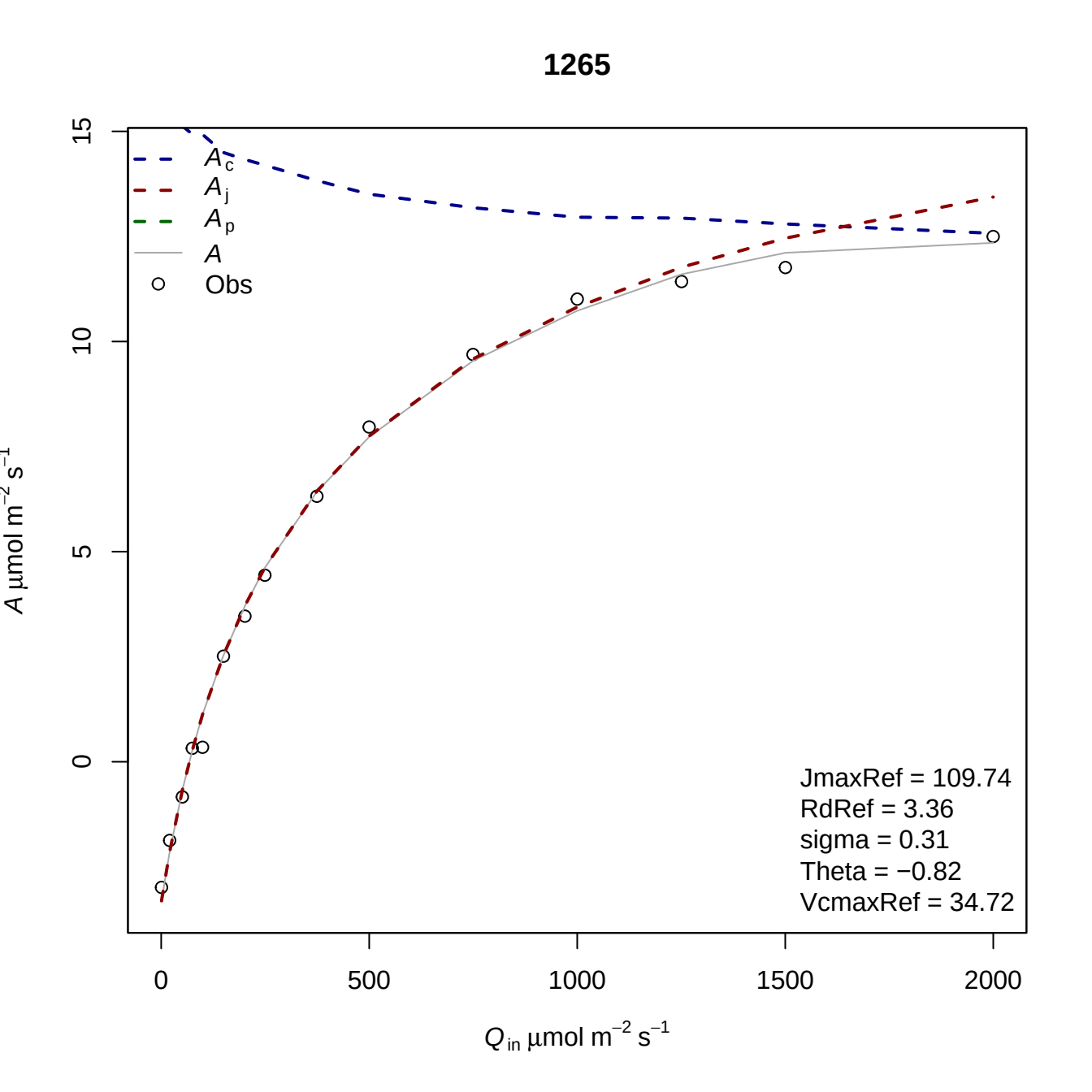

1260
A_c
A_j
A_p
A
Obs
JmaxRef = 124.51
RdRef = 2.06
sigma = 0.19
Theta = −2.15
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

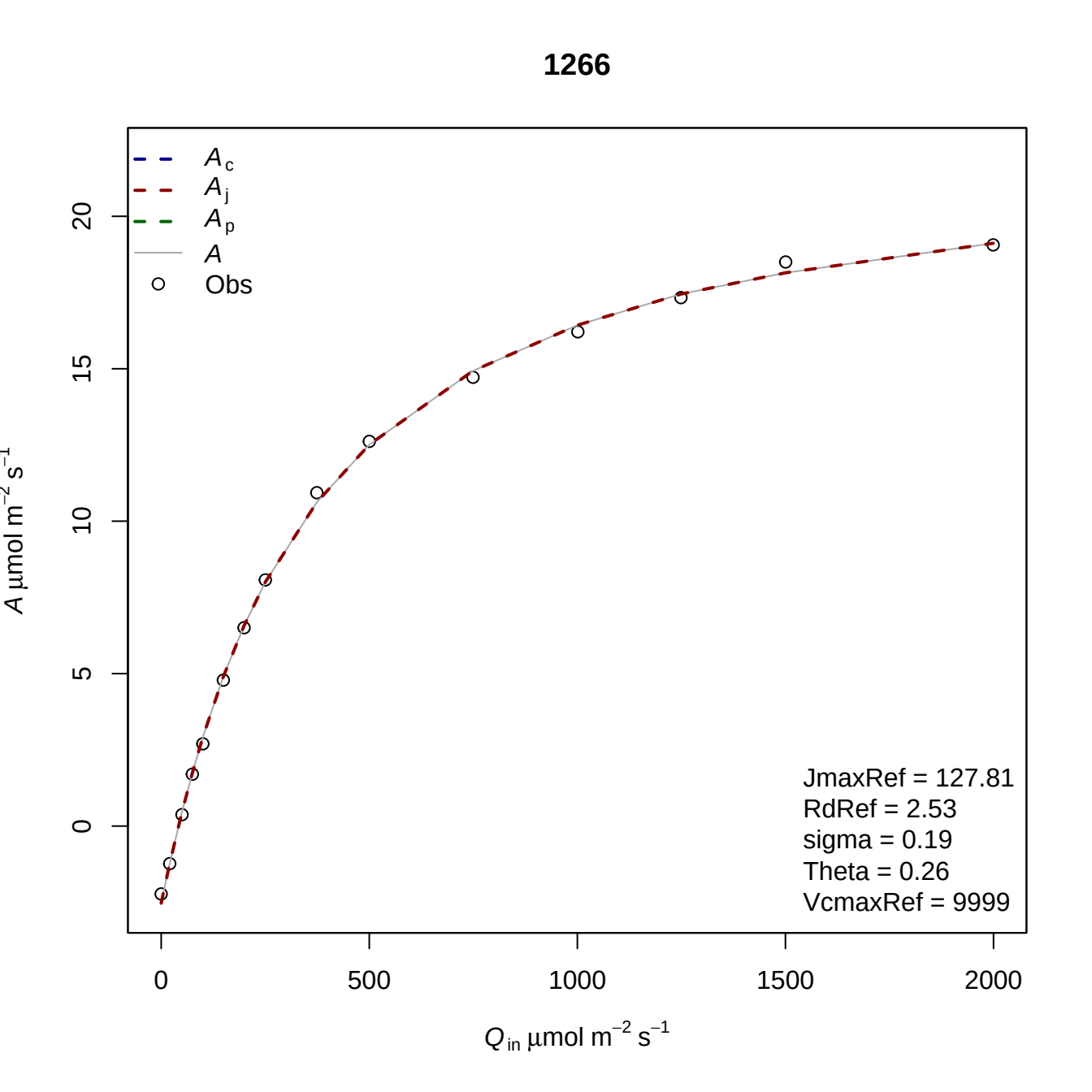

1261
A_c
A_j
A_p
A
Obs
JmaxRef = 128.29
RdRef = 1.96
sigma = 0.26
Theta = −1.04
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

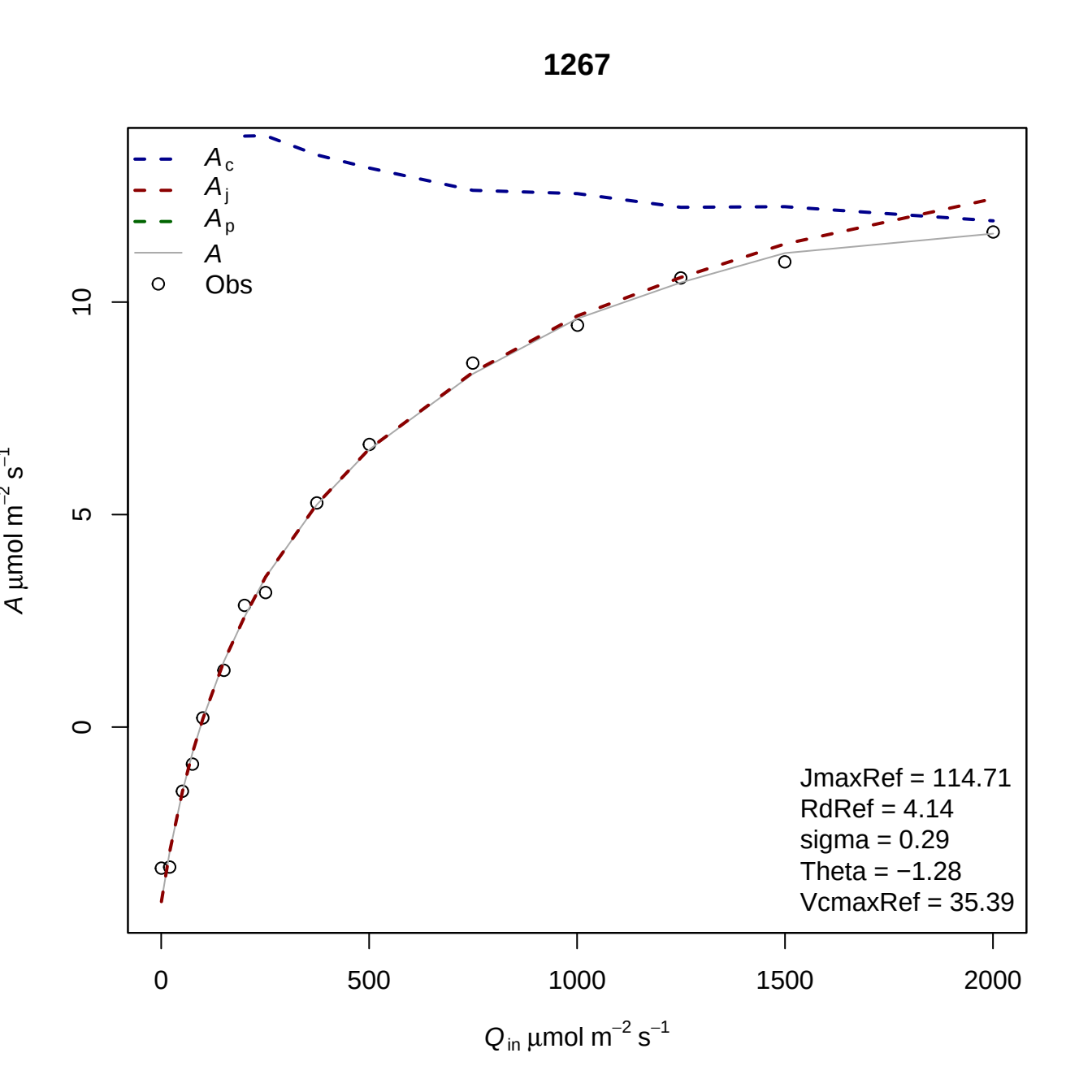

1262
A_c
A_j
A_p
A
Obs
JmaxRef = 122.33
RdRef = 3.49
sigma = 0.25
Theta = −2.67
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

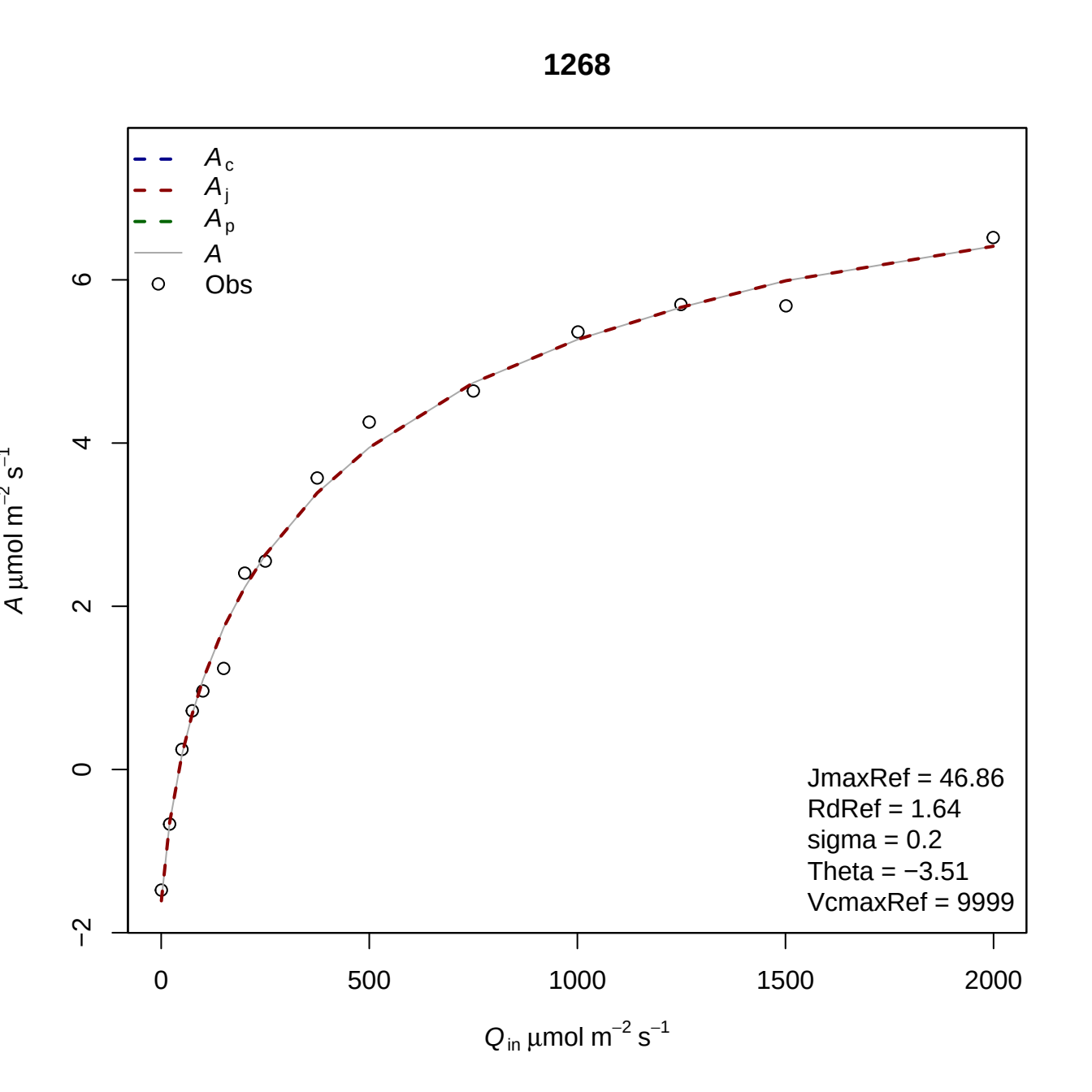

1263
A_c
A_j
A_p
A
Obs
JmaxRef = 152.09
RdRef = 4.01
sigma = 0.26
Theta = −1.15
VcmaxRef = 41.13
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

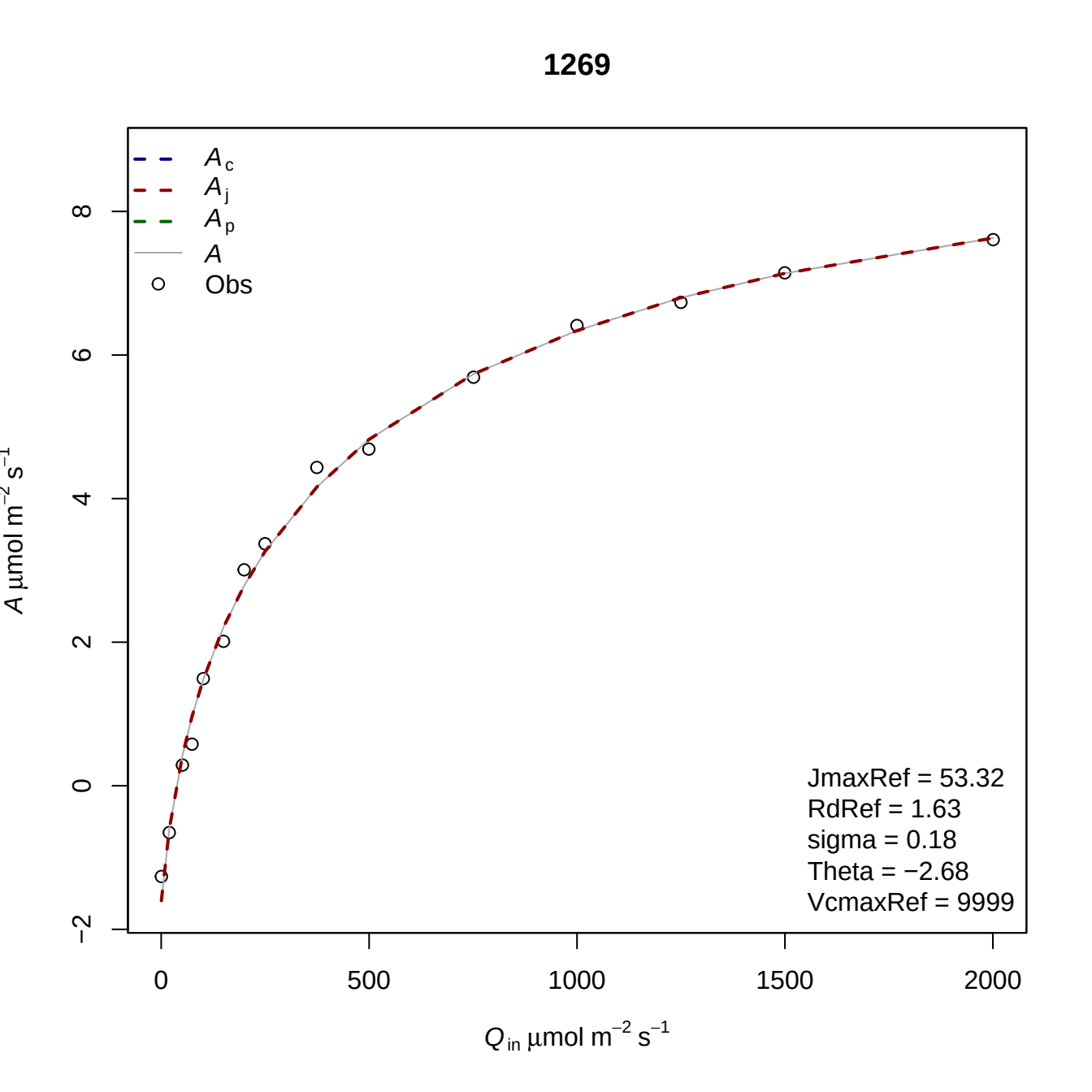

1264
A_c
A_j
A_p
A
Obs
JmaxRef = 147.11
RdRef = 3.15
sigma = 0.39
Theta = −2.19
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

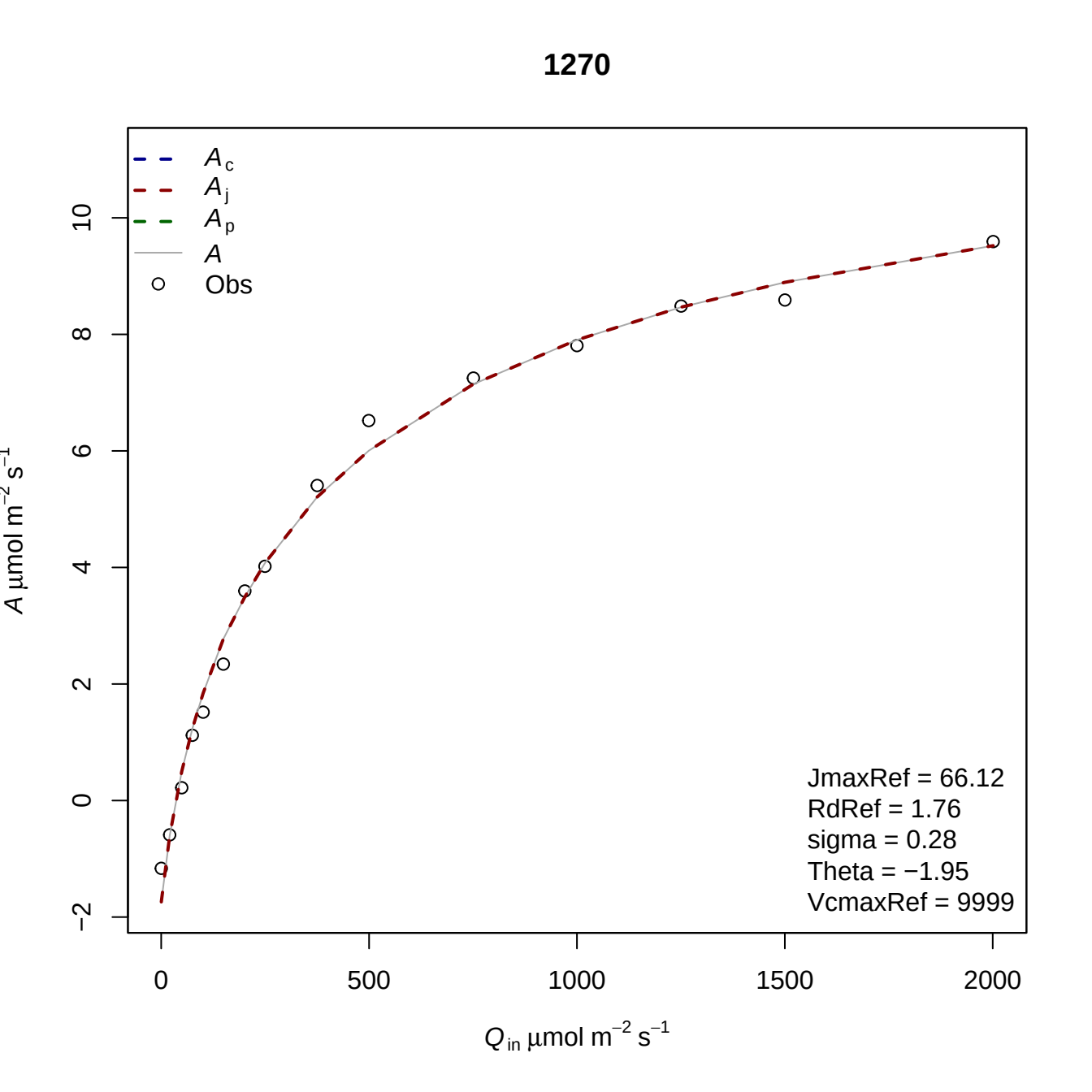

1265
A_c
A_j
A_p
A
Obs
JmaxRef = 109.74
RdRef = 3.36
sigma = 0.31
Theta = −0.82
VcmaxRef = 34.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

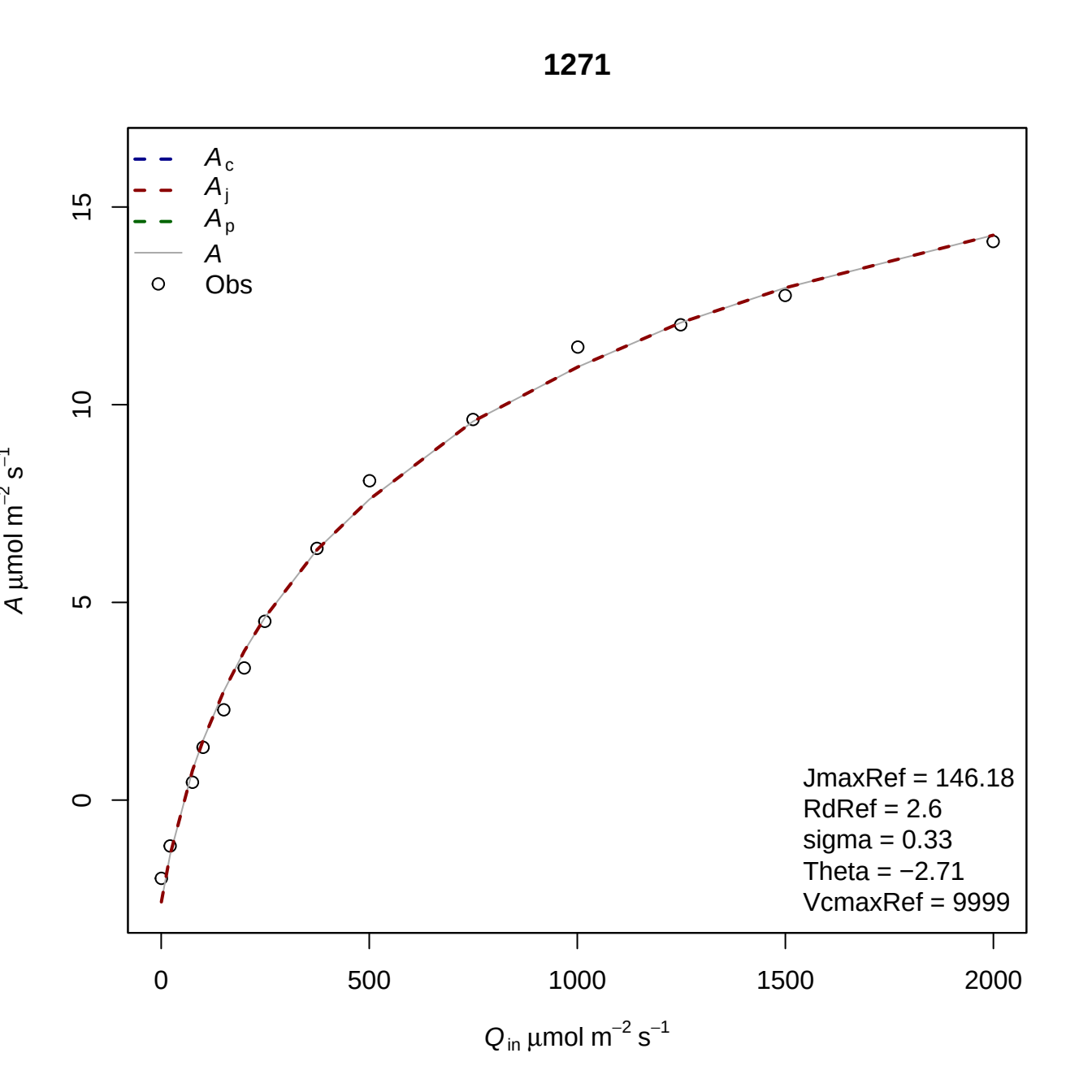

1266
A_c
A_j
A_p
A
Obs
JmaxRef = 127.81
RdRef = 2.53
sigma = 0.19
Theta = 0.26
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

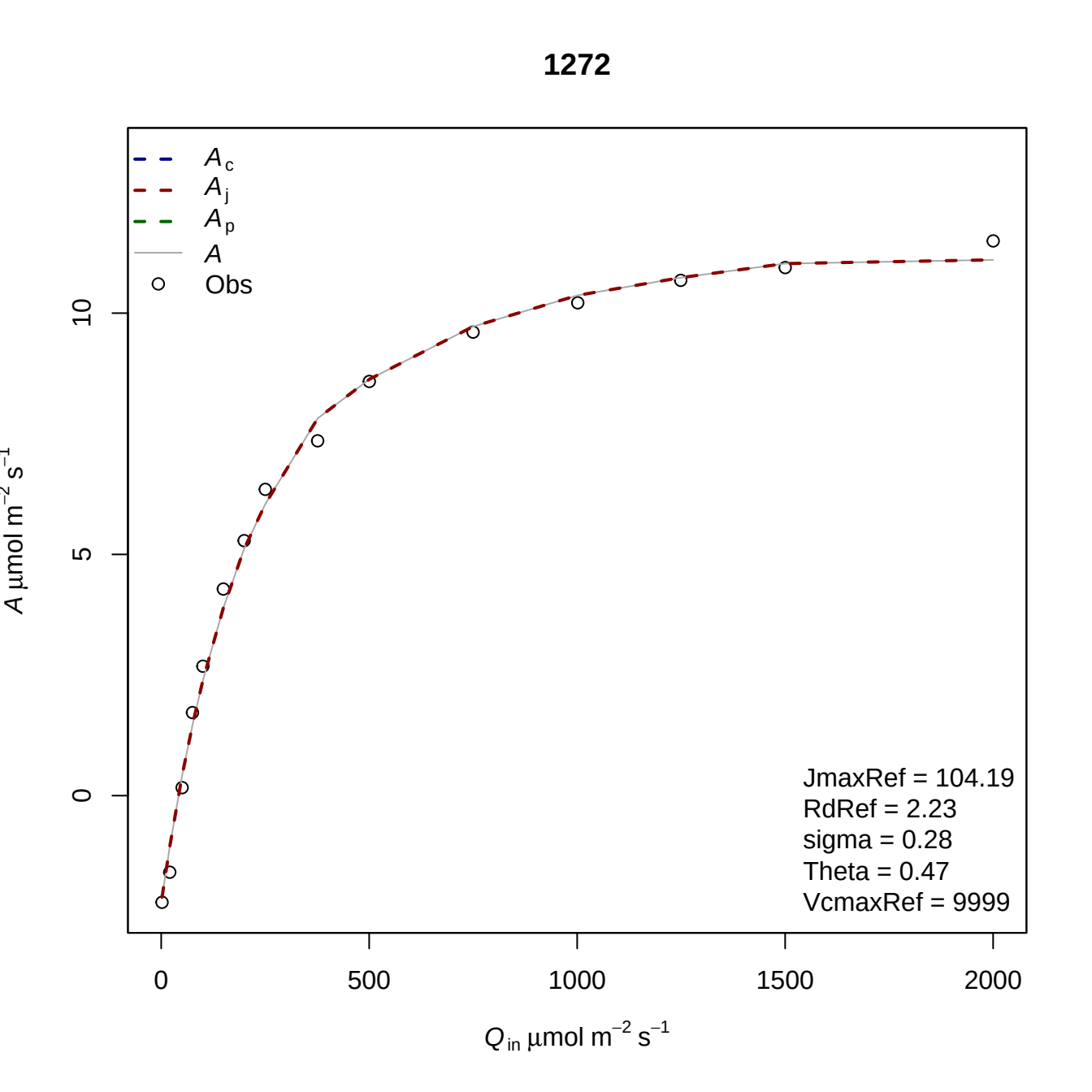

1267
A_c
A_j
A_p
A
Obs
JmaxRef = 114.71
RdRef = 4.14
sigma = 0.29
Theta = −1.28
VcmaxRef = 35.39
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

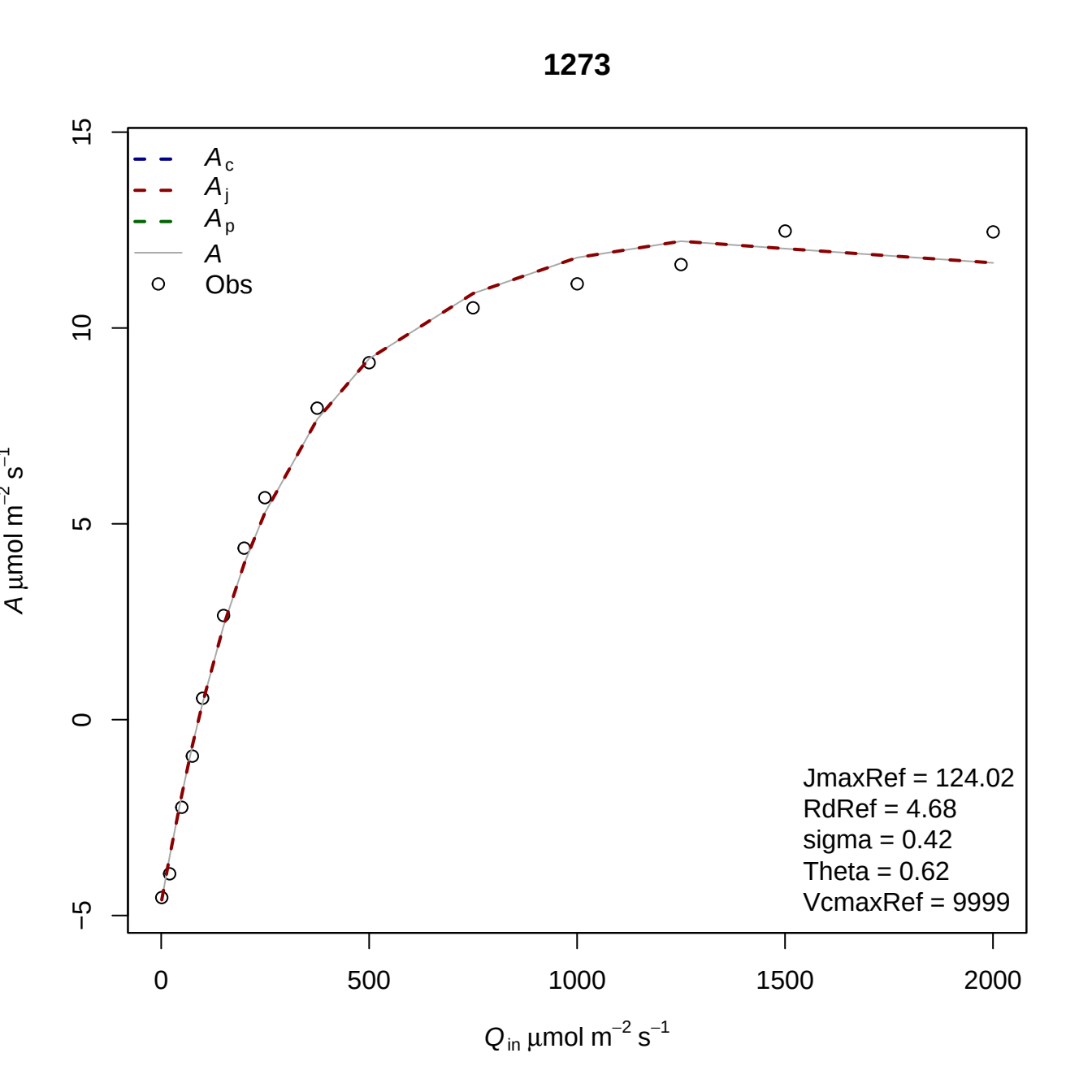

1268
A_c
A_j
A_p
A
Obs
JmaxRef = 46.86
RdRef = 1.64
sigma = 0.2
Theta = −3.51
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

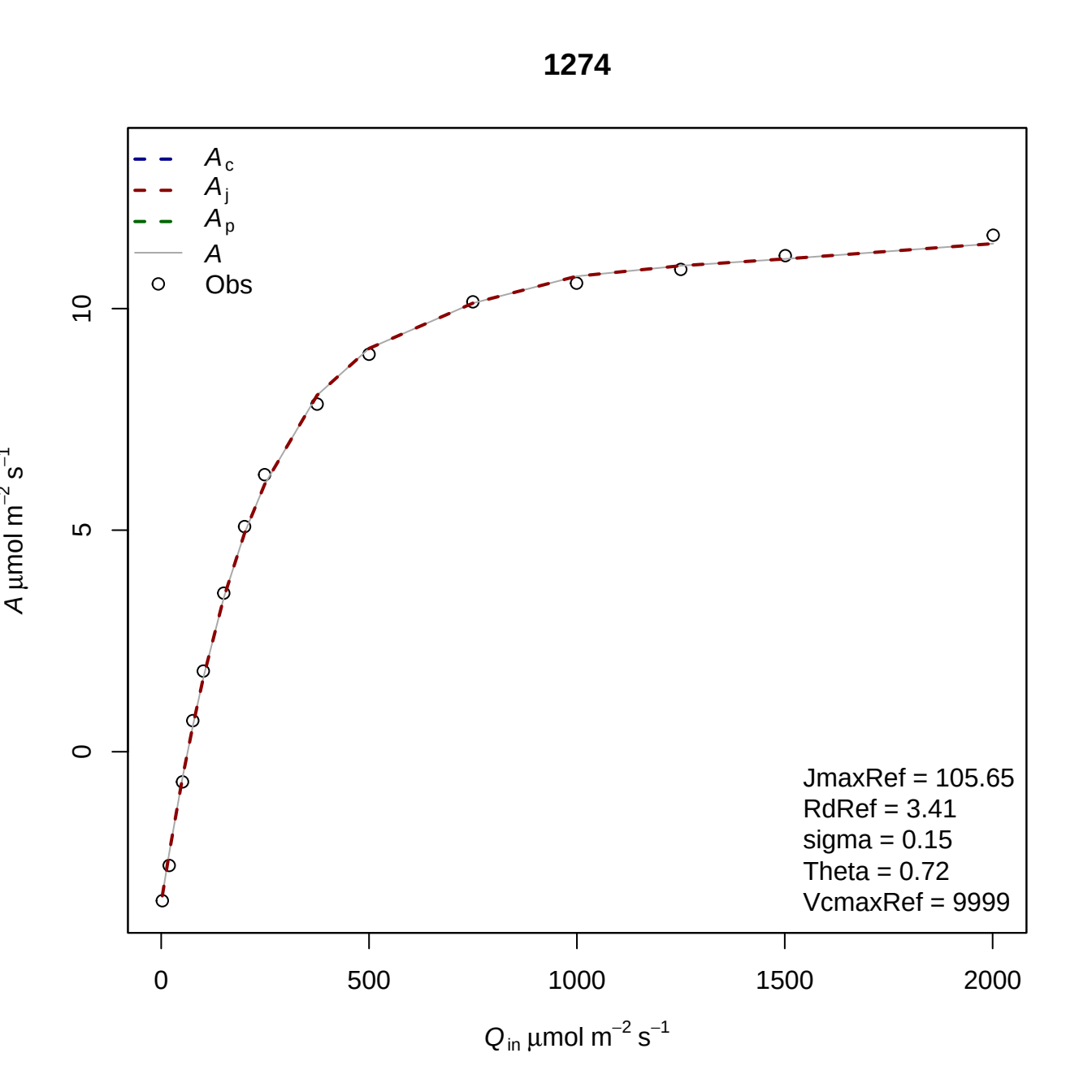

1269
A_c
A_j
A_p
A
Obs
JmaxRef = 53.32
RdRef = 1.63
sigma = 0.18
Theta = −2.68
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

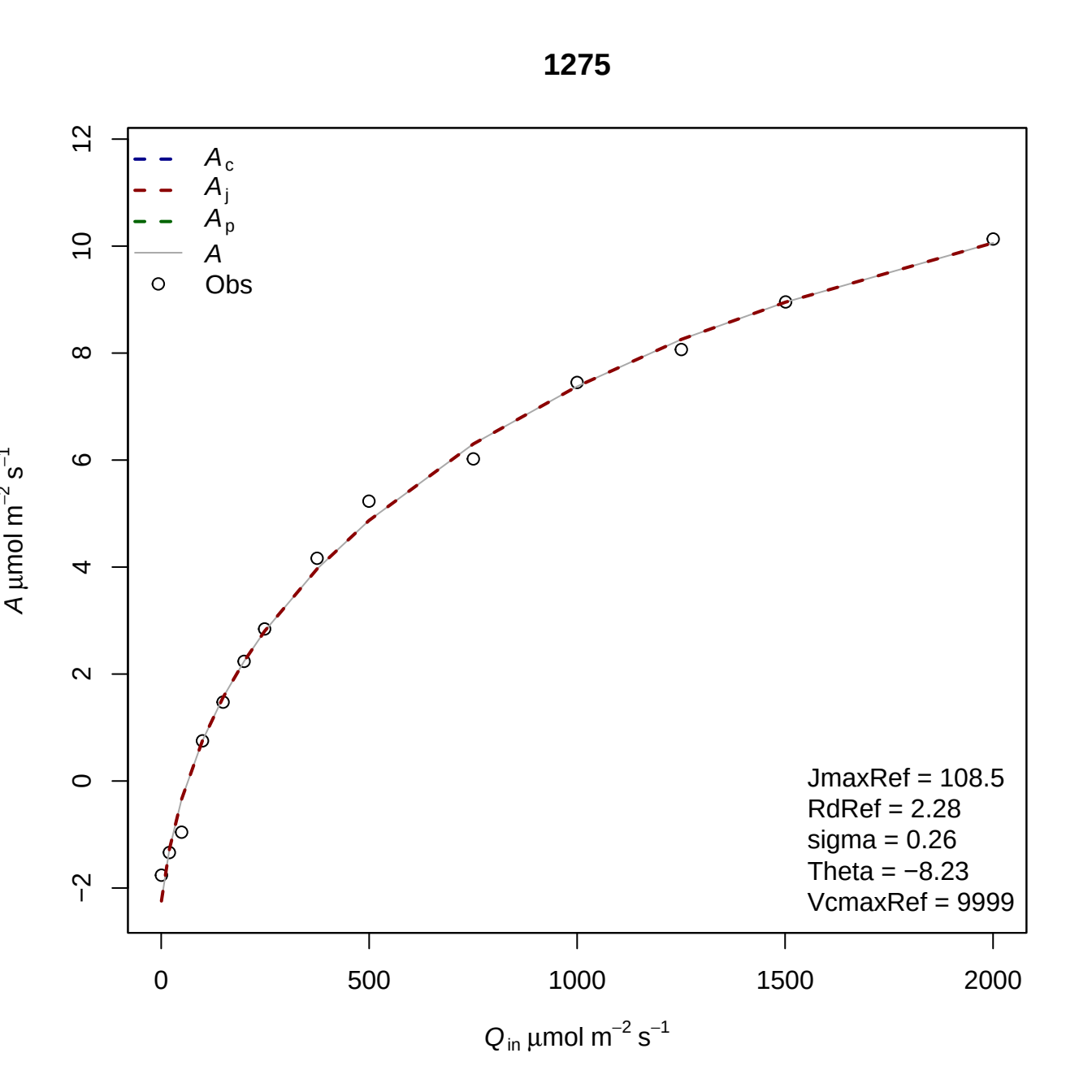

1270
A_c
A_j
A_p
A
Obs
JmaxRef = 66.12
RdRef = 1.76
sigma = 0.28
Theta = −1.95
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

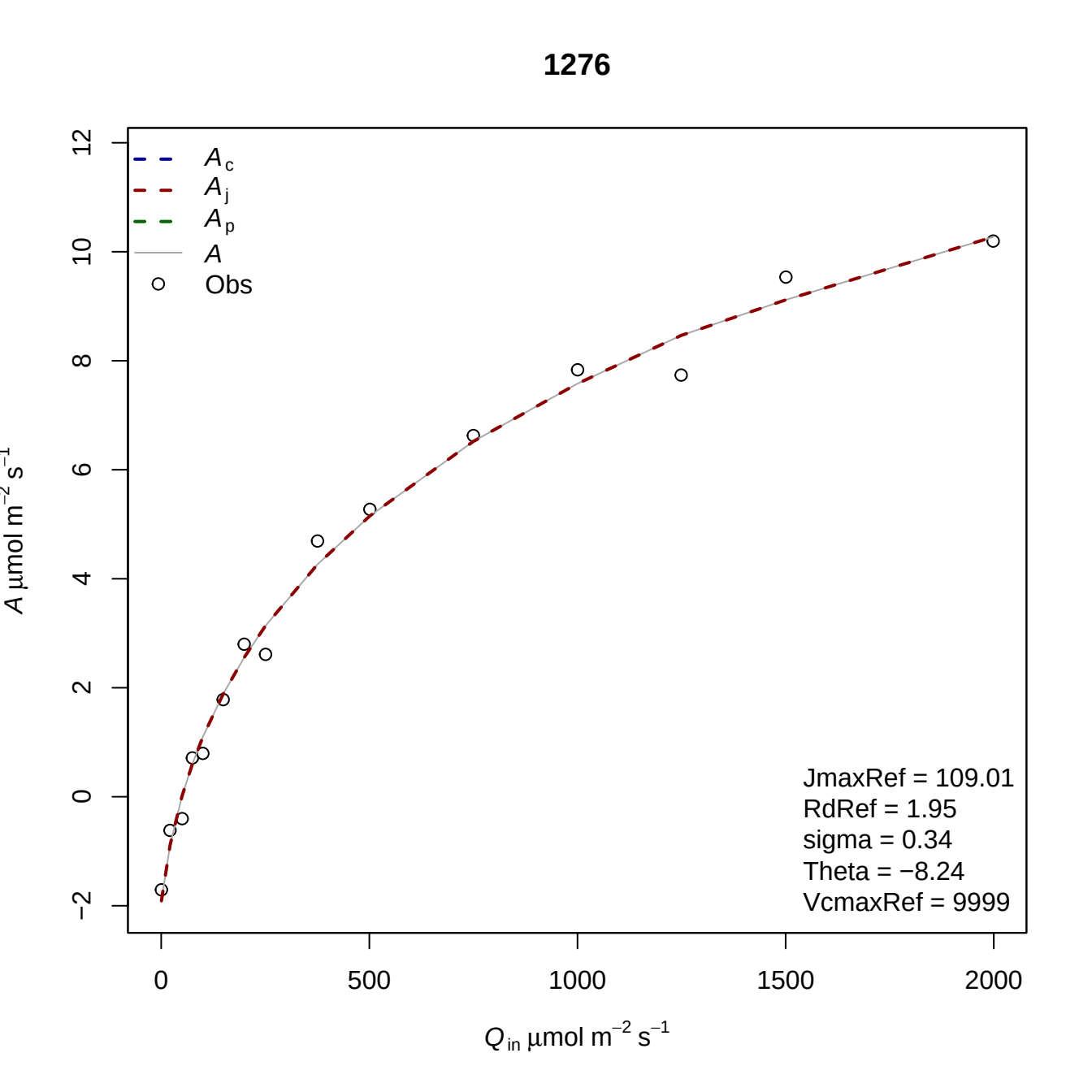

1271
A_c
A_j
A_p
A
Obs
JmaxRef = 146.18
RdRef = 2.6
sigma = 0.33
Theta = −2.71
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

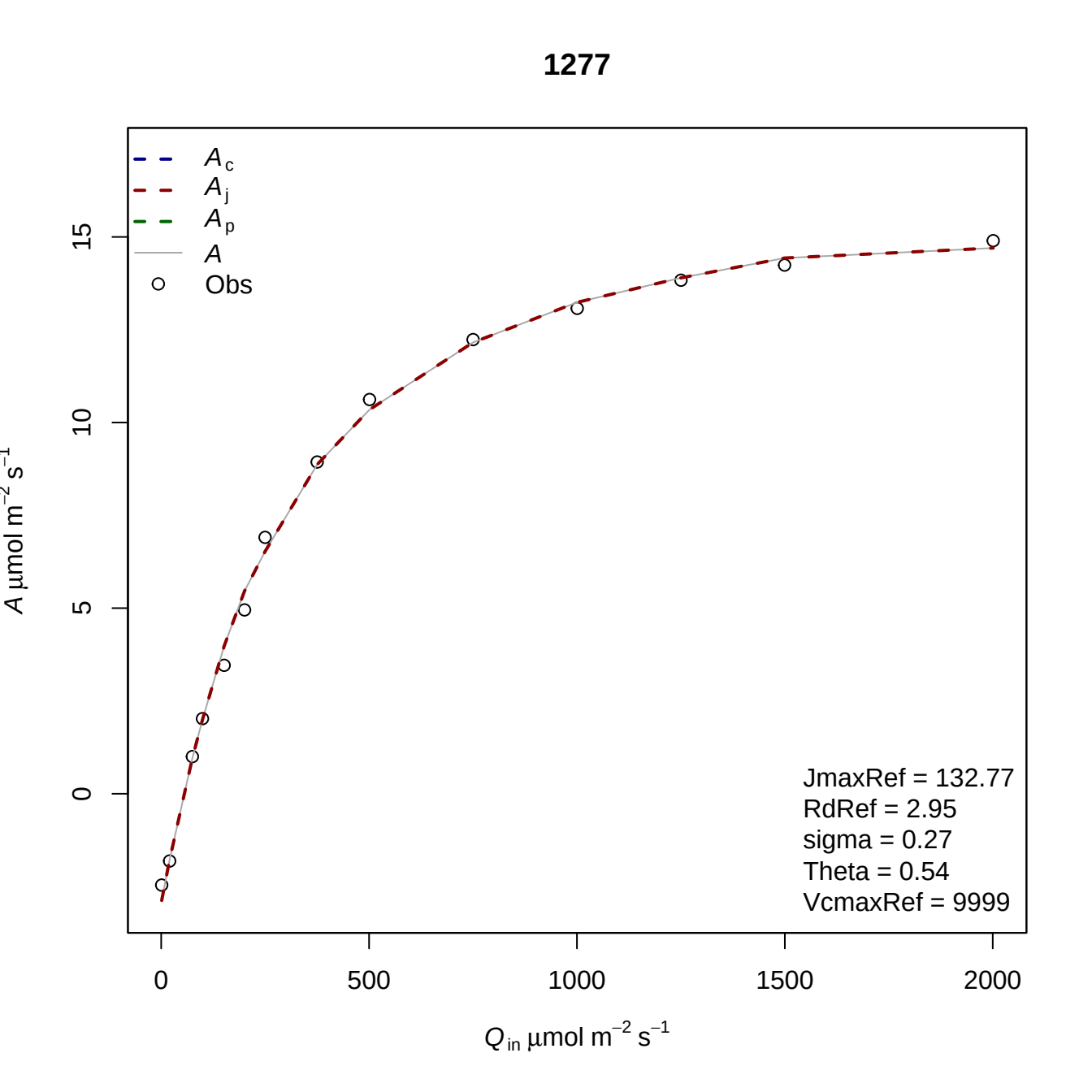

1272
A_c
A_j
A_p
A
Obs
JmaxRef = 104.19
RdRef = 2.23
sigma = 0.28
Theta = 0.47
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

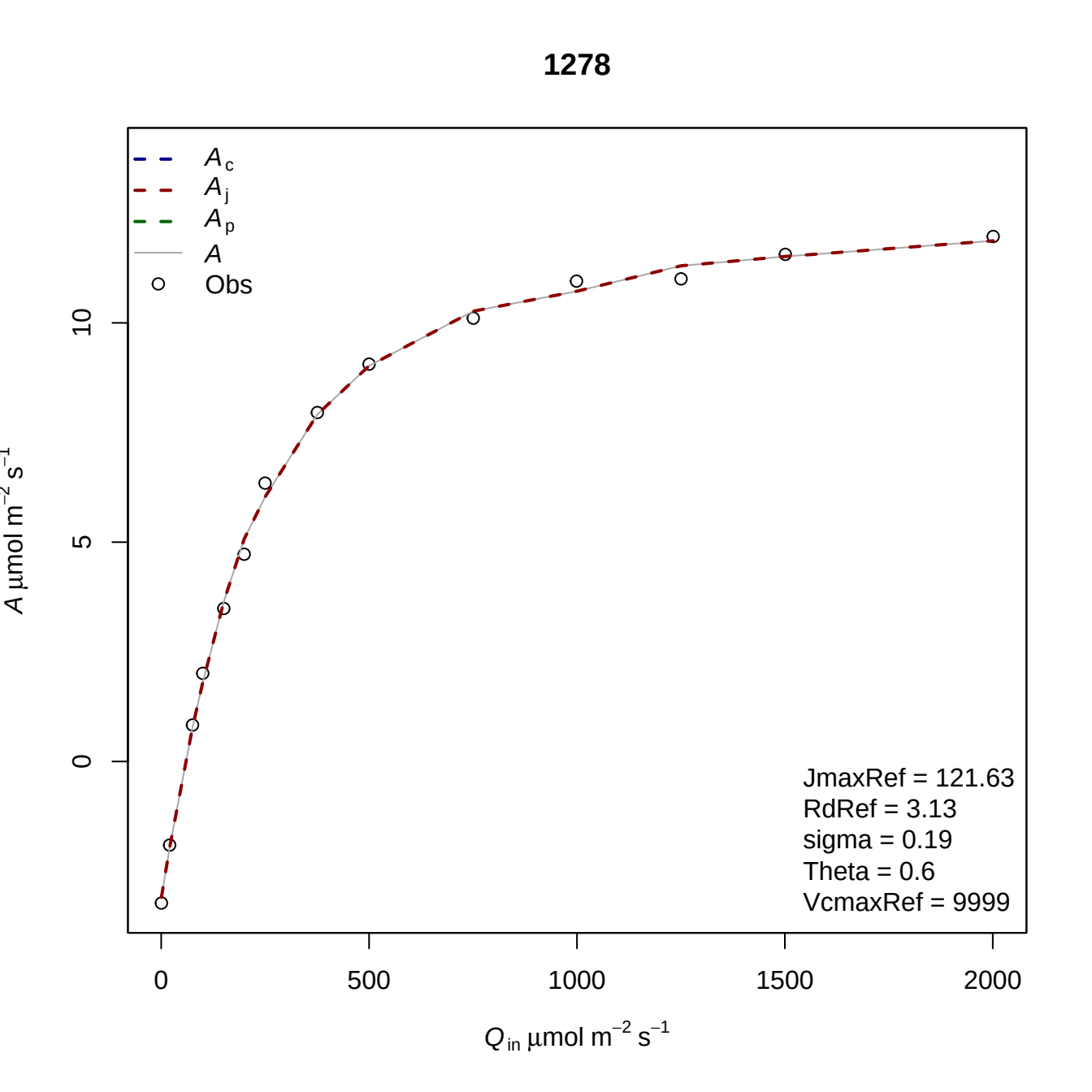

1273
A_c
A_j
A_p
A
Obs
JmaxRef = 124.02
RdRef = 4.68
sigma = 0.42
Theta = 0.62
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

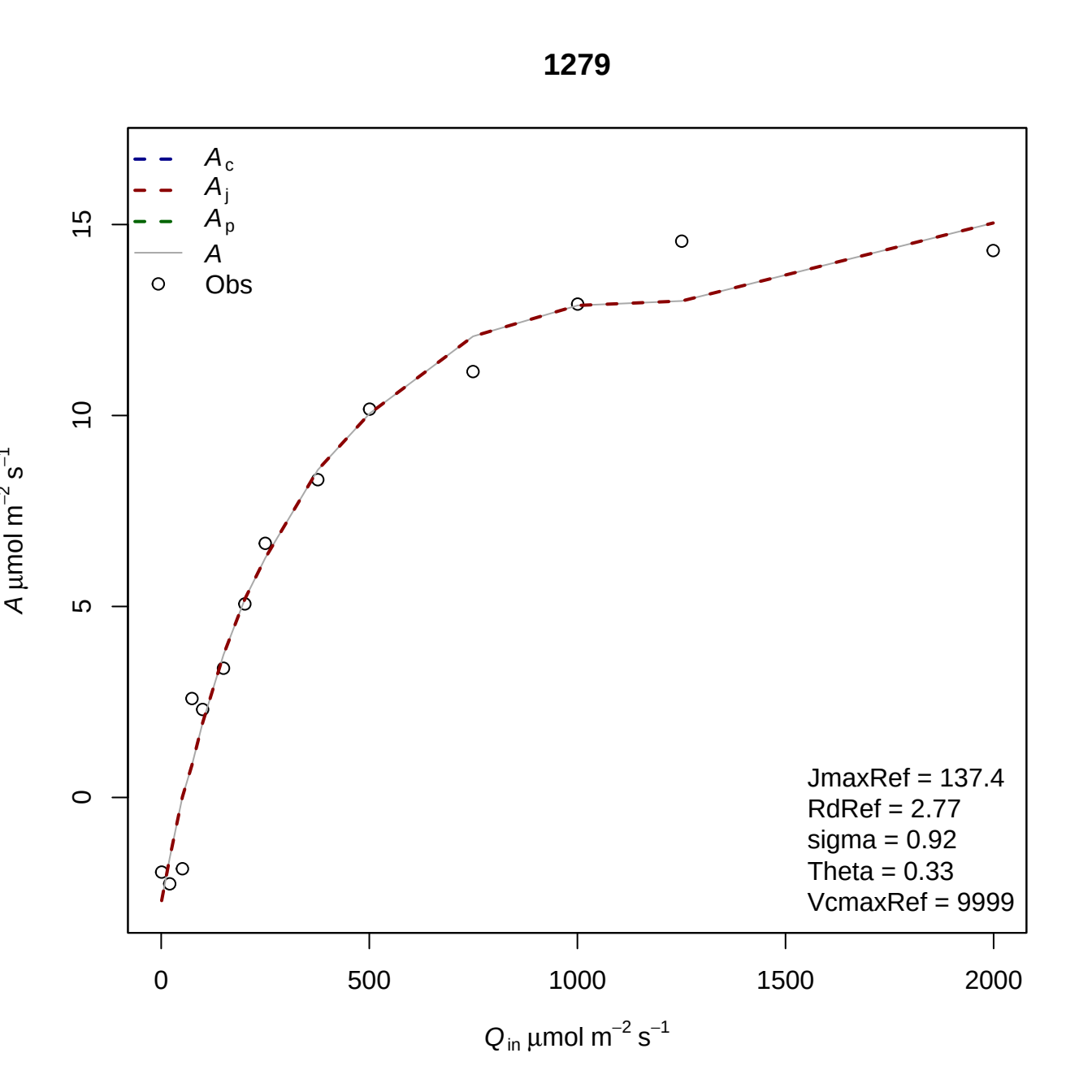

1274
A_c
A_j
A_p
A
Obs
JmaxRef = 105.65
RdRef = 3.41
sigma = 0.15
Theta = 0.72
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

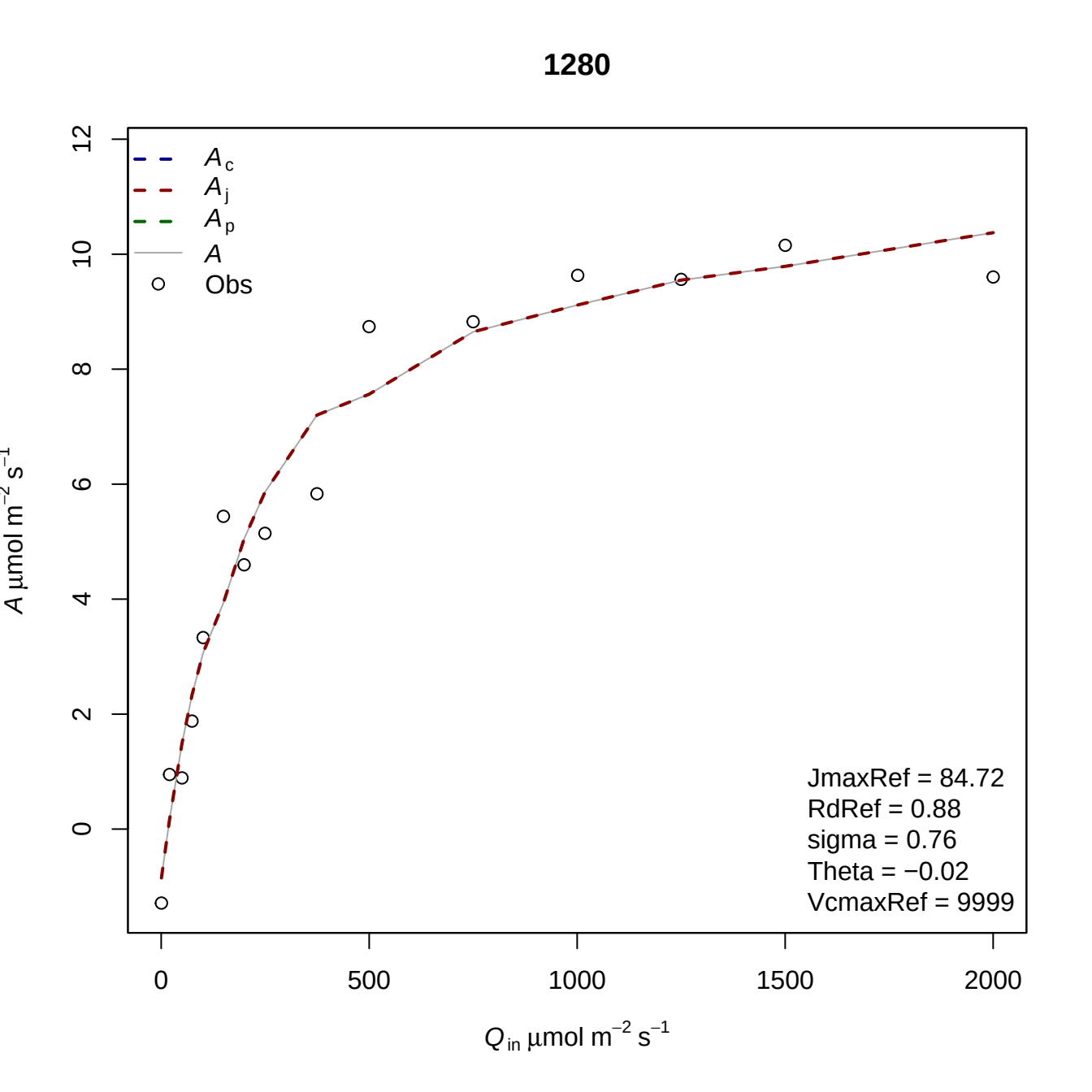

1275
A_c
A_j
A_p
A
Obs
JmaxRef = 108.5
RdRef = 2.28
sigma = 0.26
Theta = −8.23
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

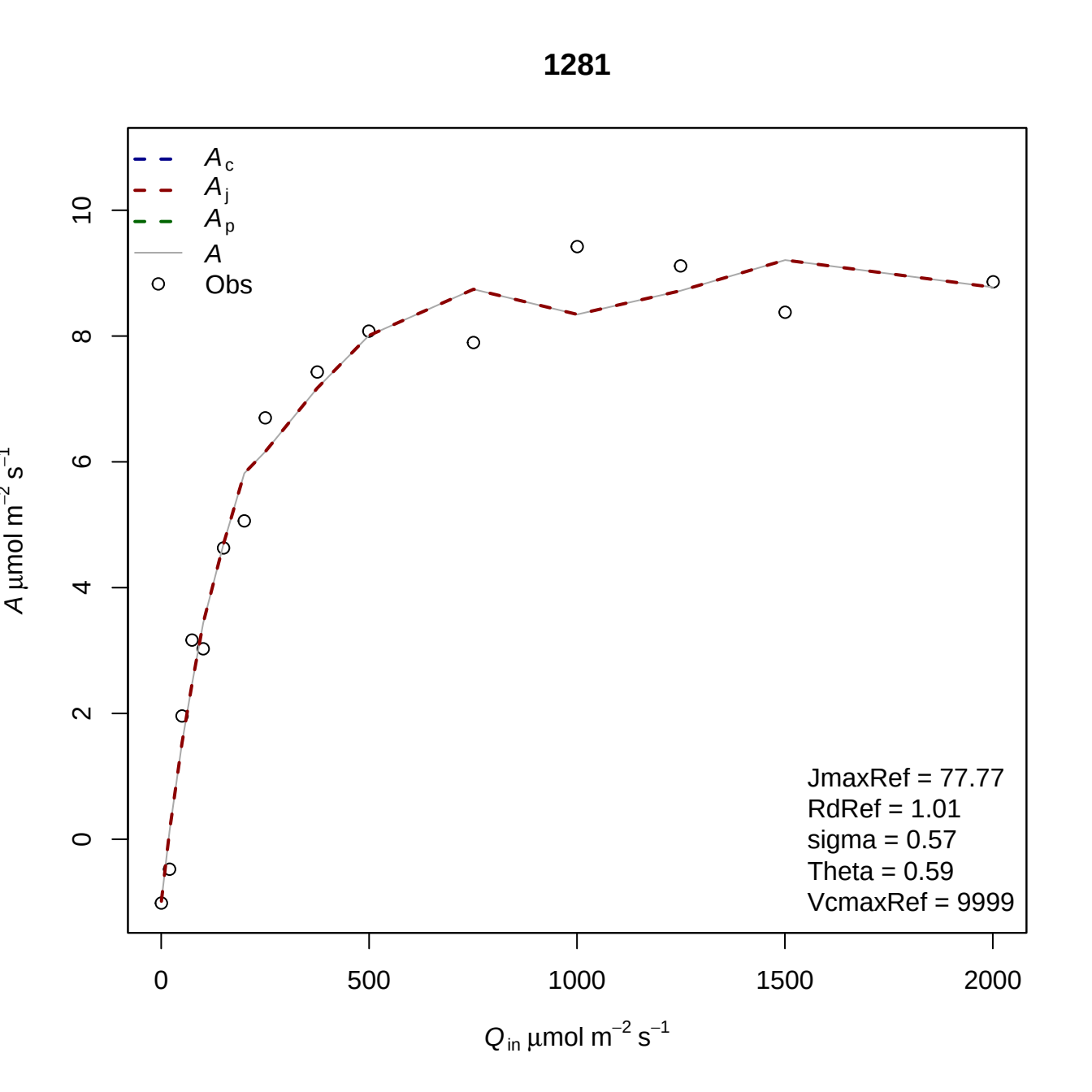

1276
A_c
A_j
A_p
A
Obs
JmaxRef = 109.01
RdRef = 1.95
sigma = 0.34
Theta = −8.24
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

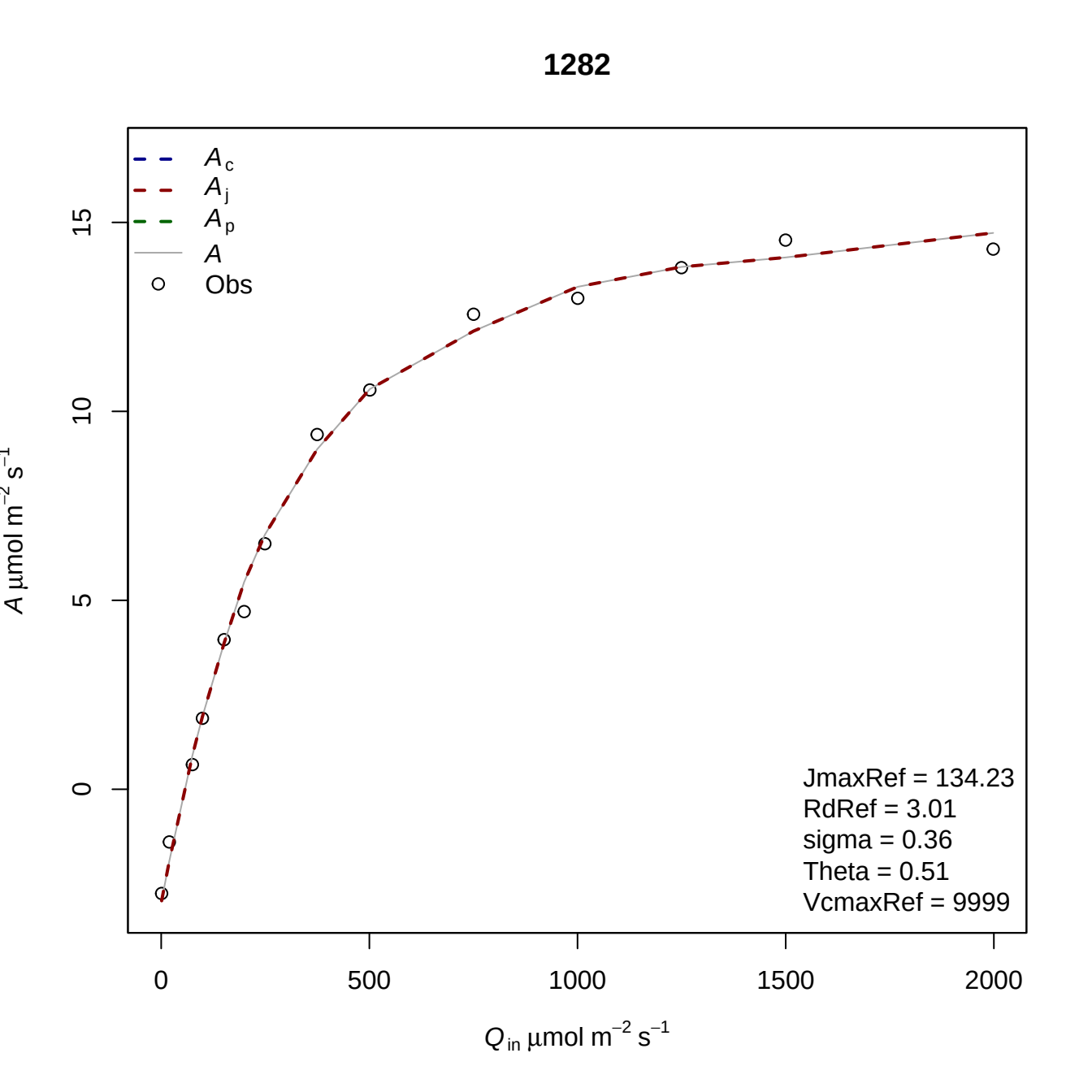

1277
A_c
A_j
A_p
A
Obs
JmaxRef = 132.77
RdRef = 2.95
sigma = 0.27
Theta = 0.54
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

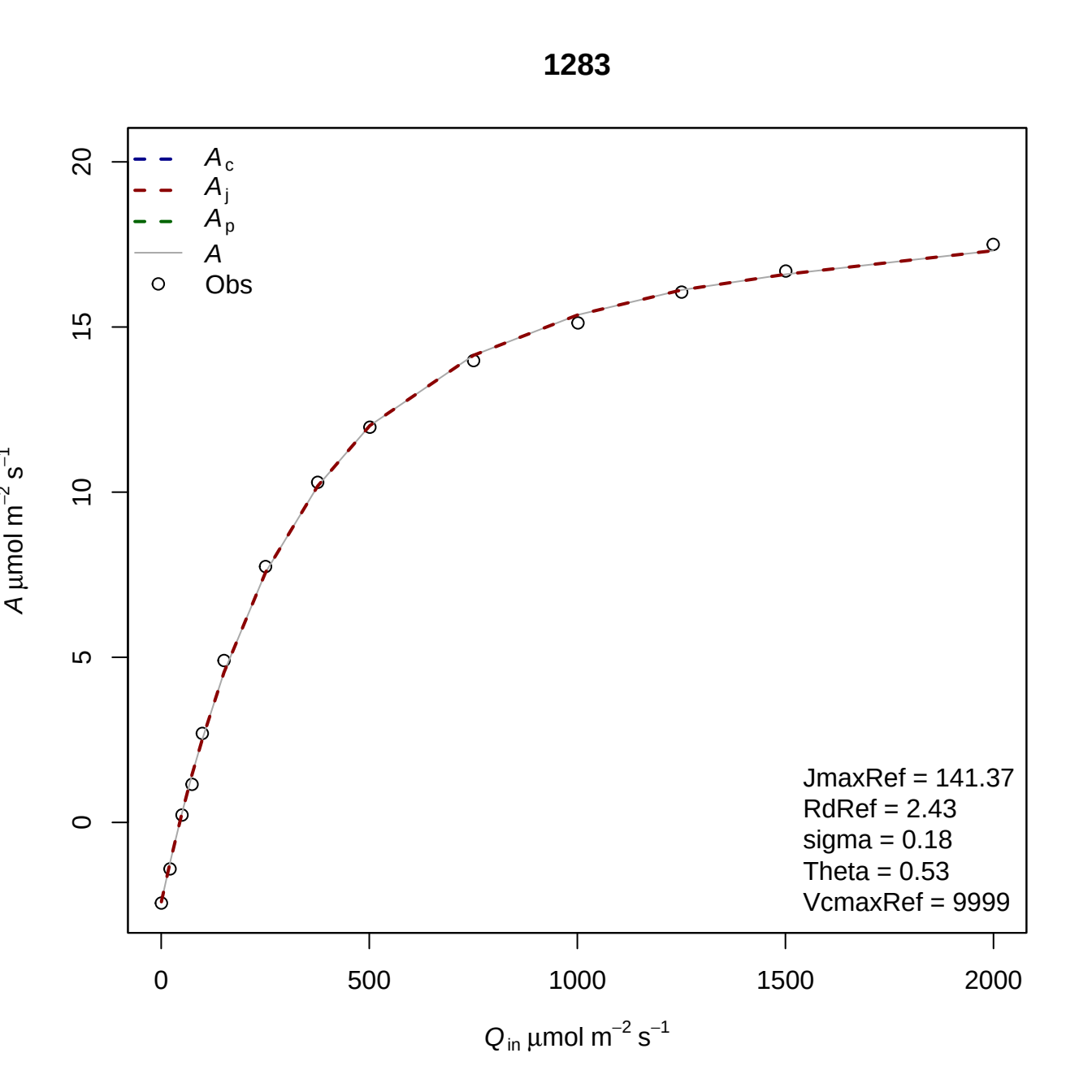

1278
A_c
A_j
A_p
A
Obs
JmaxRef = 121.63
RdRef = 3.13
sigma = 0.19
Theta = 0.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

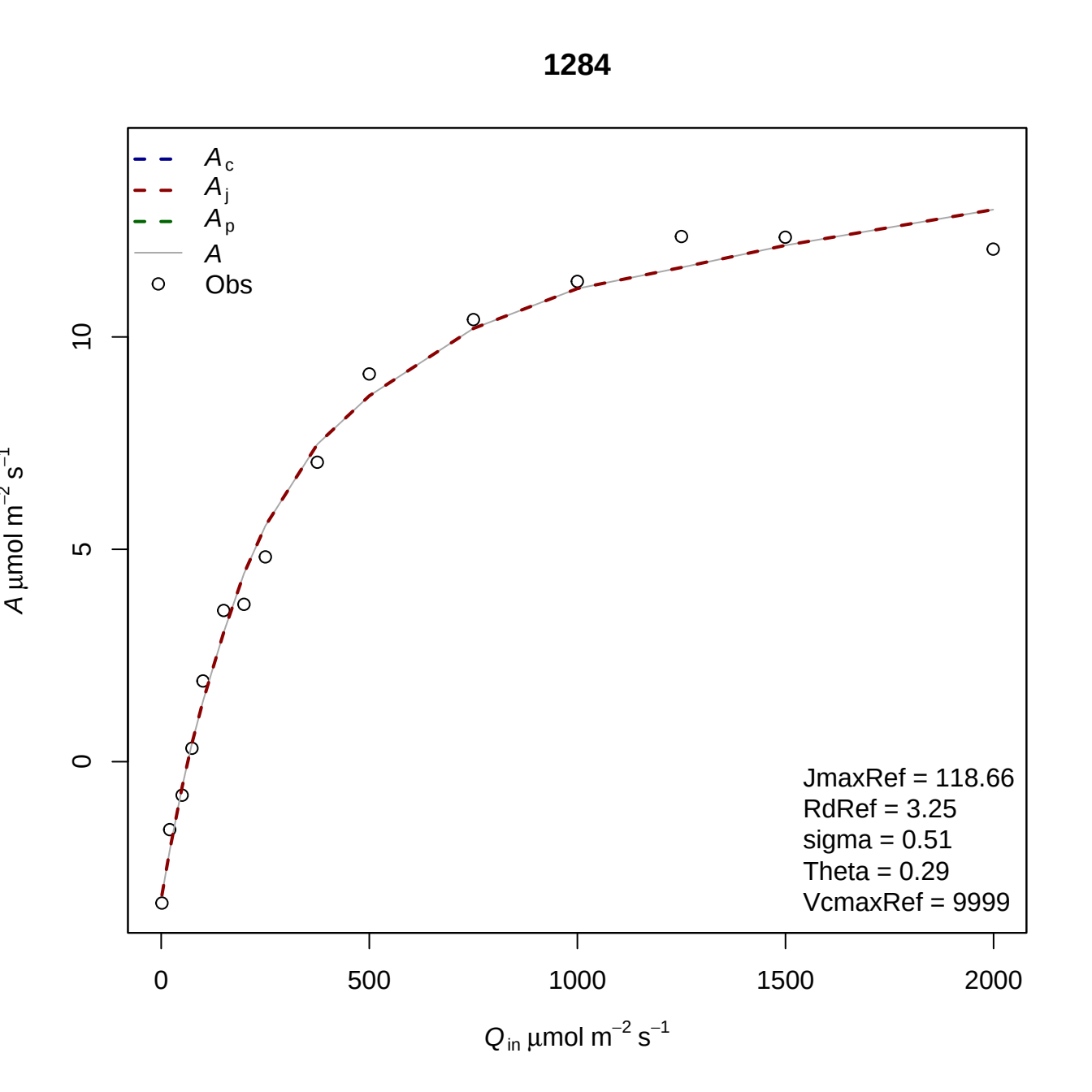

1279
A_c
A_j
A_p
A
Obs
JmaxRef = 137.4
RdRef = 2.77
sigma = 0.92
Theta = 0.33
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

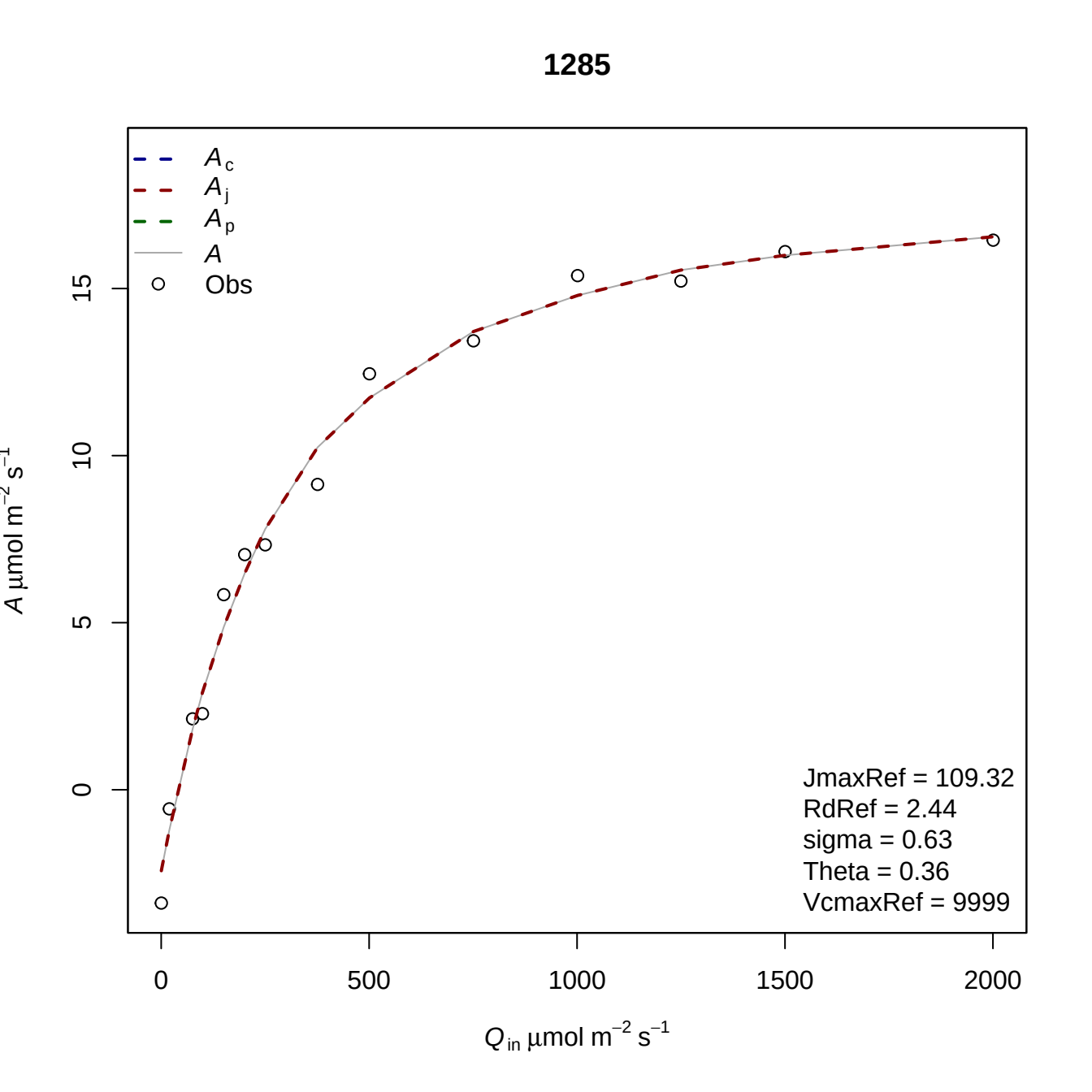

1280
A_c
A_j
A_p
A
Obs
JmaxRef = 84.72
RdRef = 0.88
sigma = 0.76
Theta = −0.02
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

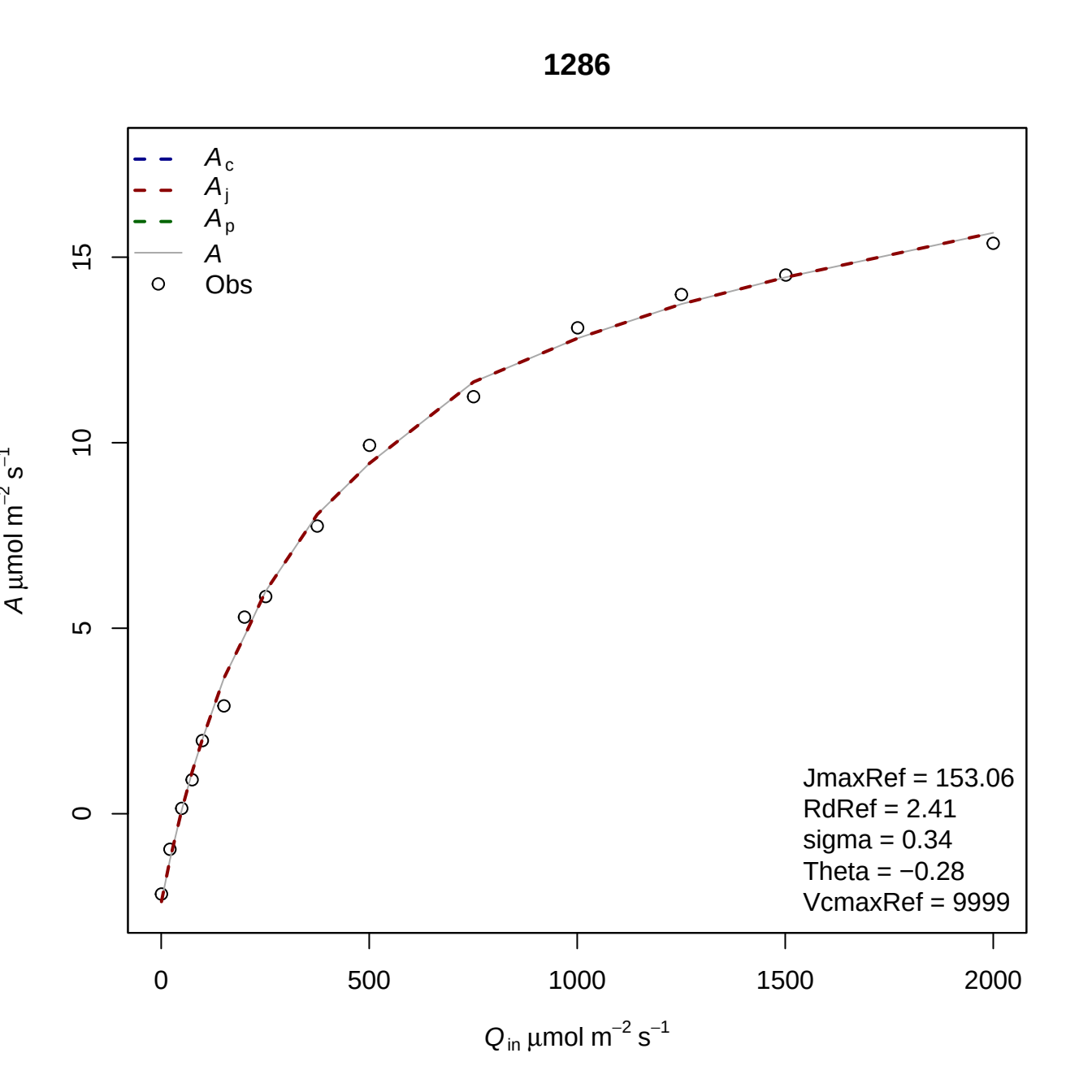

1281
A_c
A_j
A_p
A
Obs
JmaxRef = 77.77
RdRef = 1.01
sigma = 0.57
Theta = 0.59
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

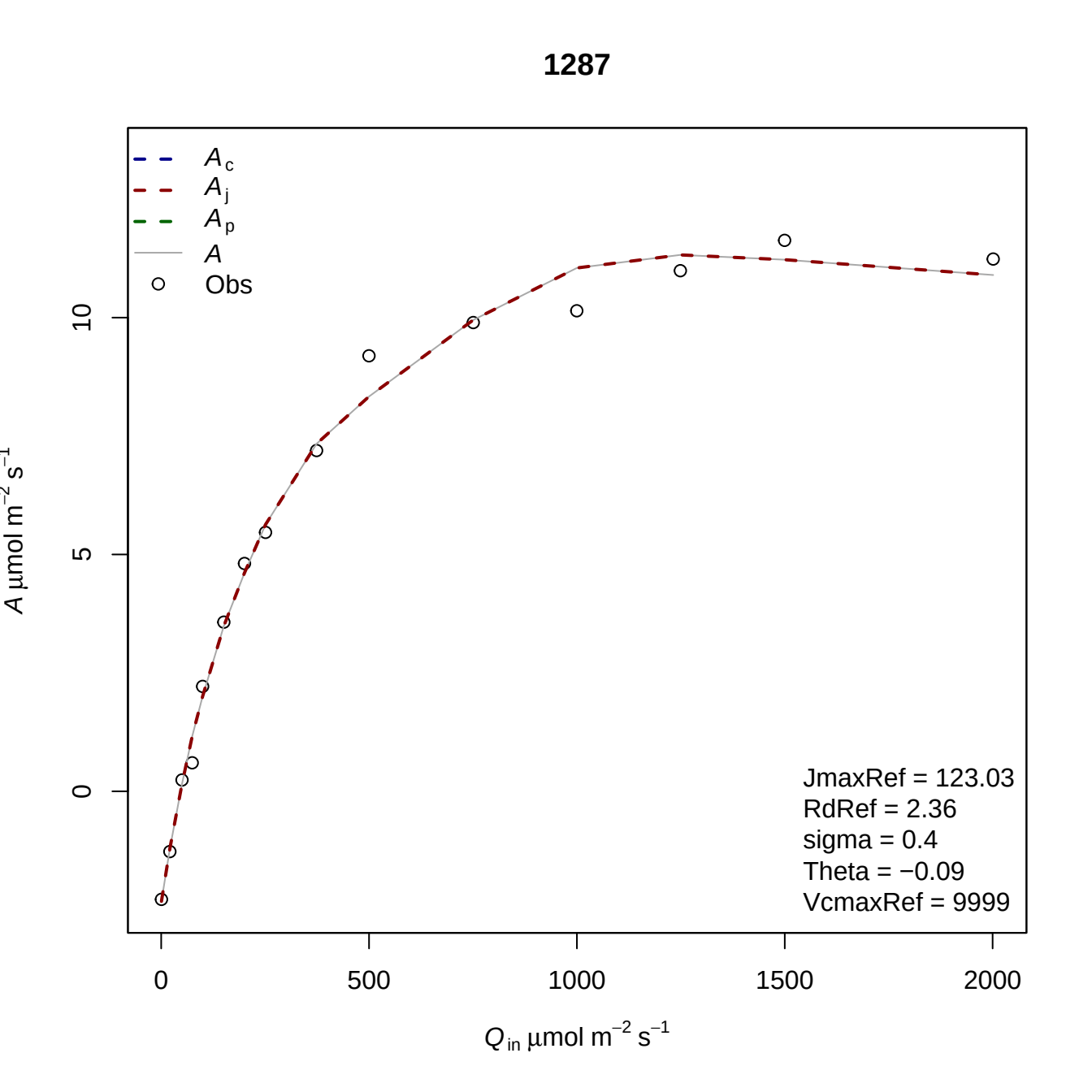

1282
A_c
A_j
A_p
A
Obs
JmaxRef = 134.23
RdRef = 3.01
sigma = 0.36
Theta = 0.51
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

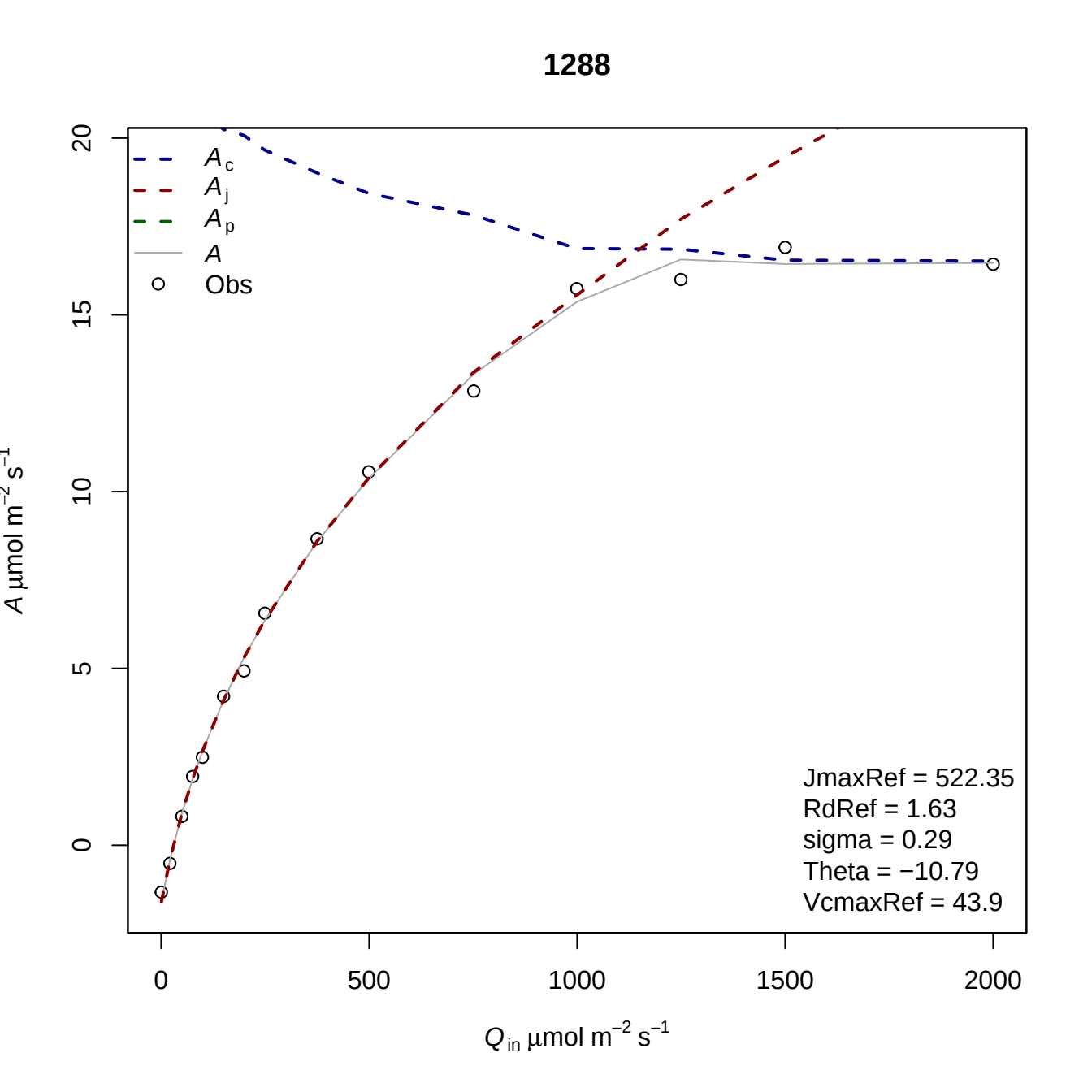

1283
A_c
A_j
A_p
A
Obs
JmaxRef = 141.37
RdRef = 2.43
sigma = 0.18
Theta = 0.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

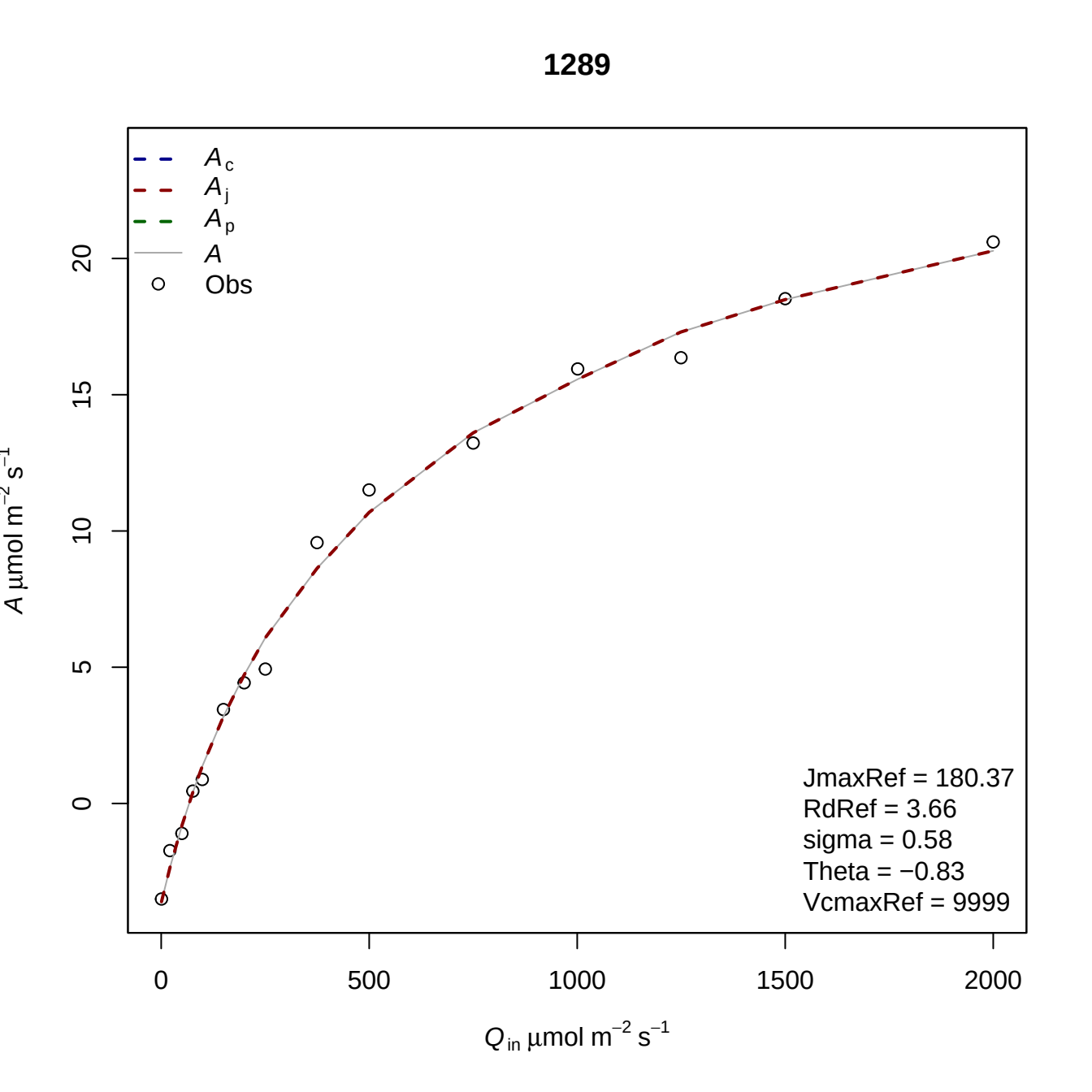

1284
A_c
A_j
A_p
A
Obs
JmaxRef = 118.66
RdRef = 3.25
sigma = 0.51
Theta = 0.29
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

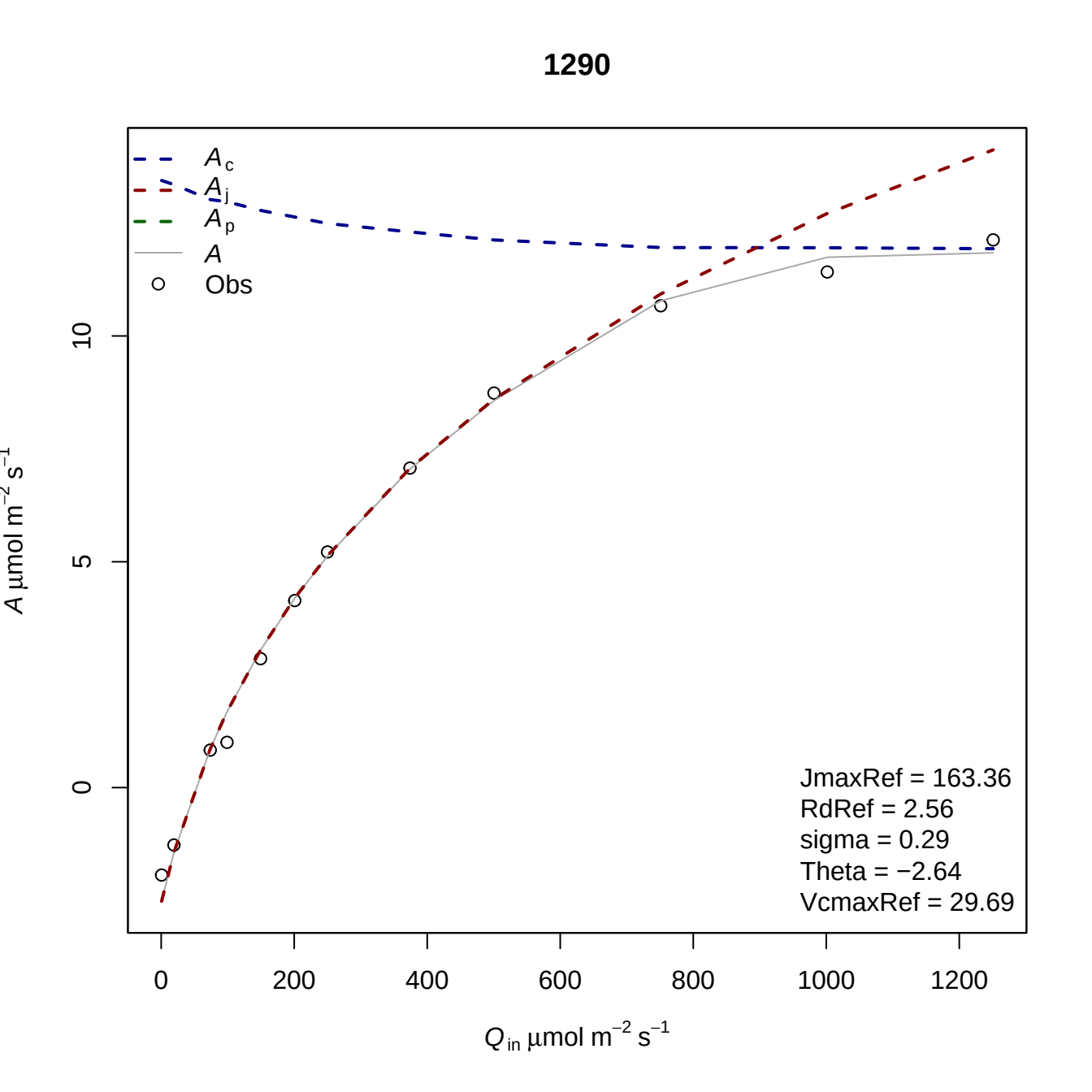

1285
A_c
A_j
A_p
A
Obs
JmaxRef = 109.32
RdRef = 2.44
sigma = 0.63
Theta = 0.36
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

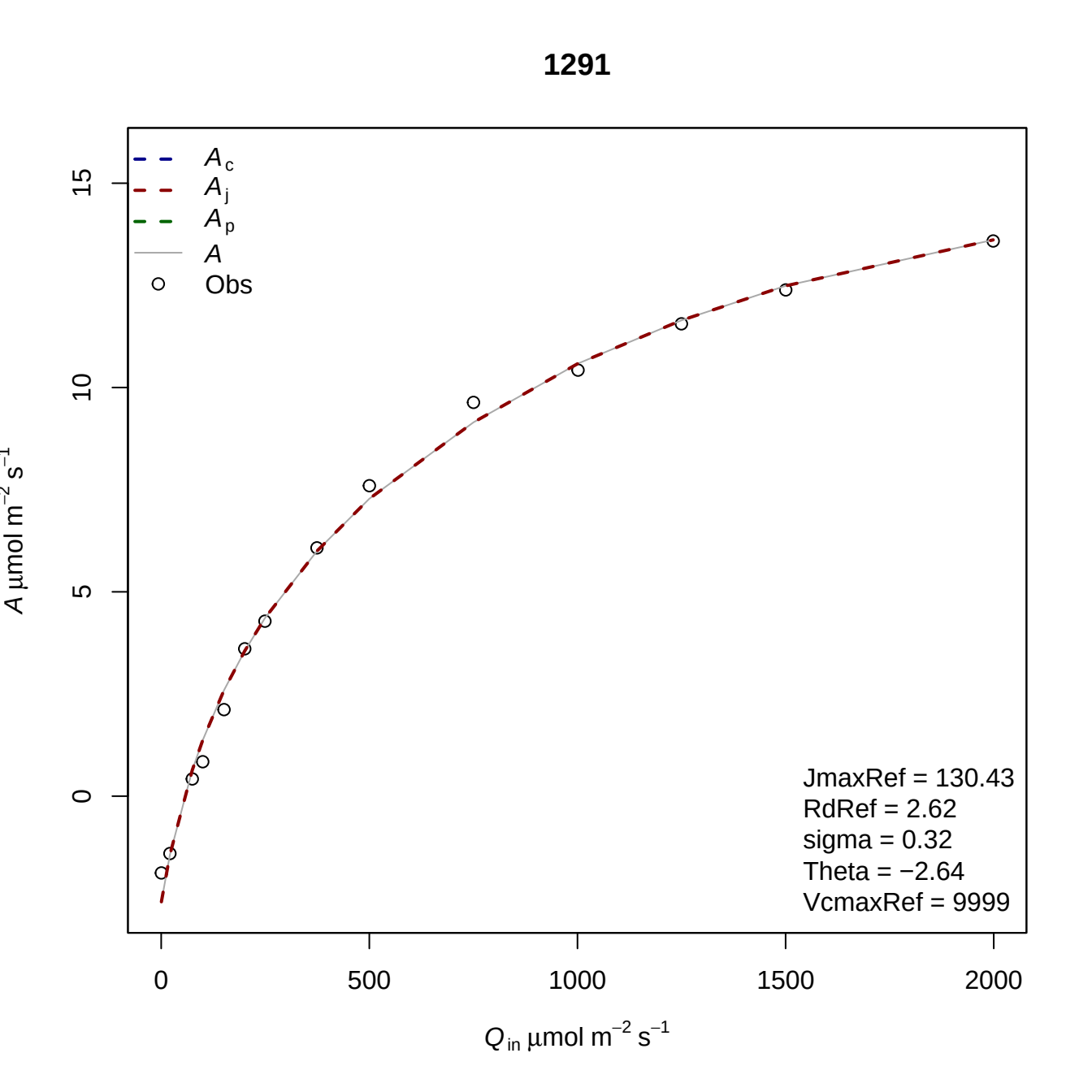

1286
A_c
A_j
A_p
A
Obs
JmaxRef = 153.06
RdRef = 2.41
sigma = 0.34
Theta = −0.28
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

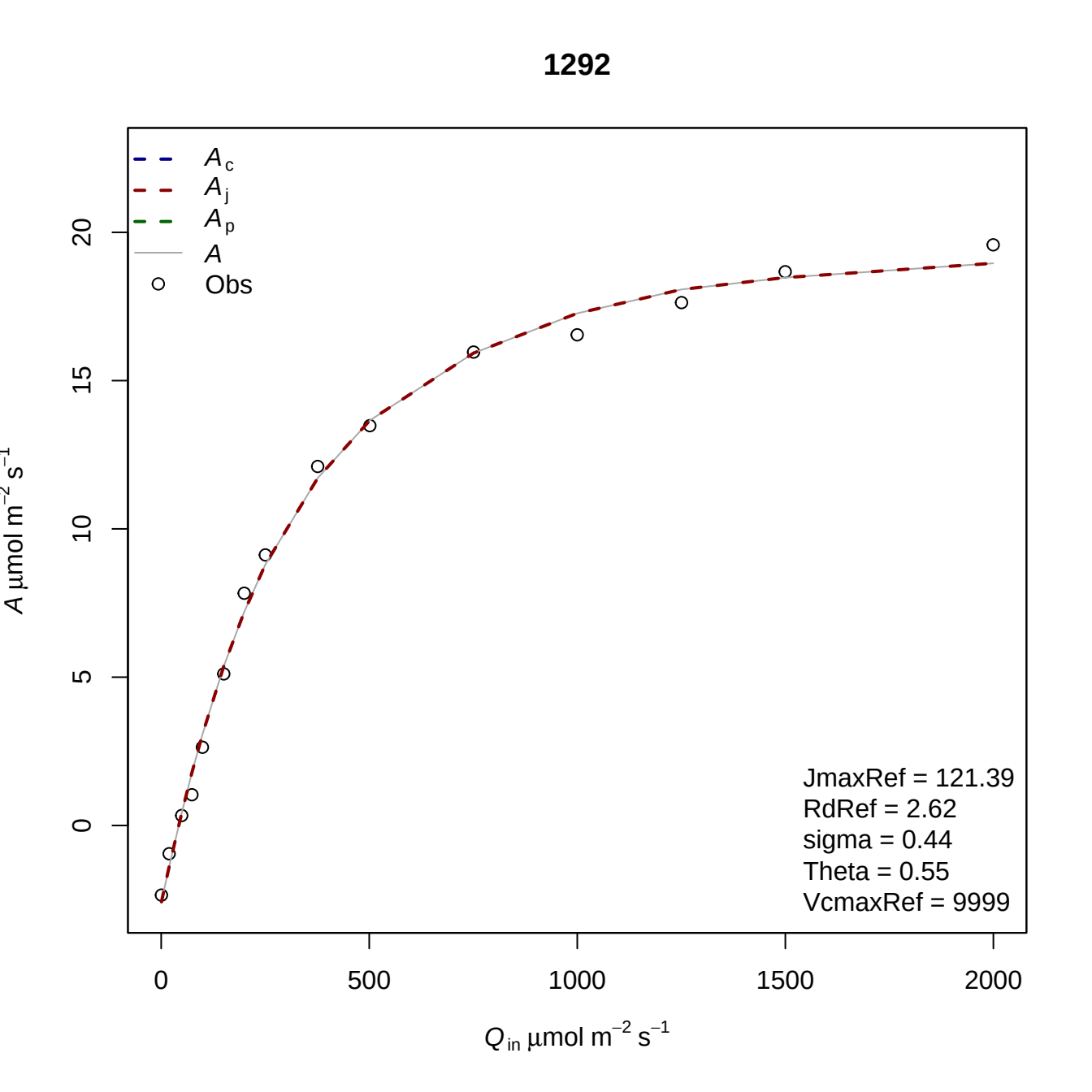

1287
A_c
A_j
A_p
A
Obs
JmaxRef = 123.03
RdRef = 2.36
sigma = 0.4
Theta = −0.09
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

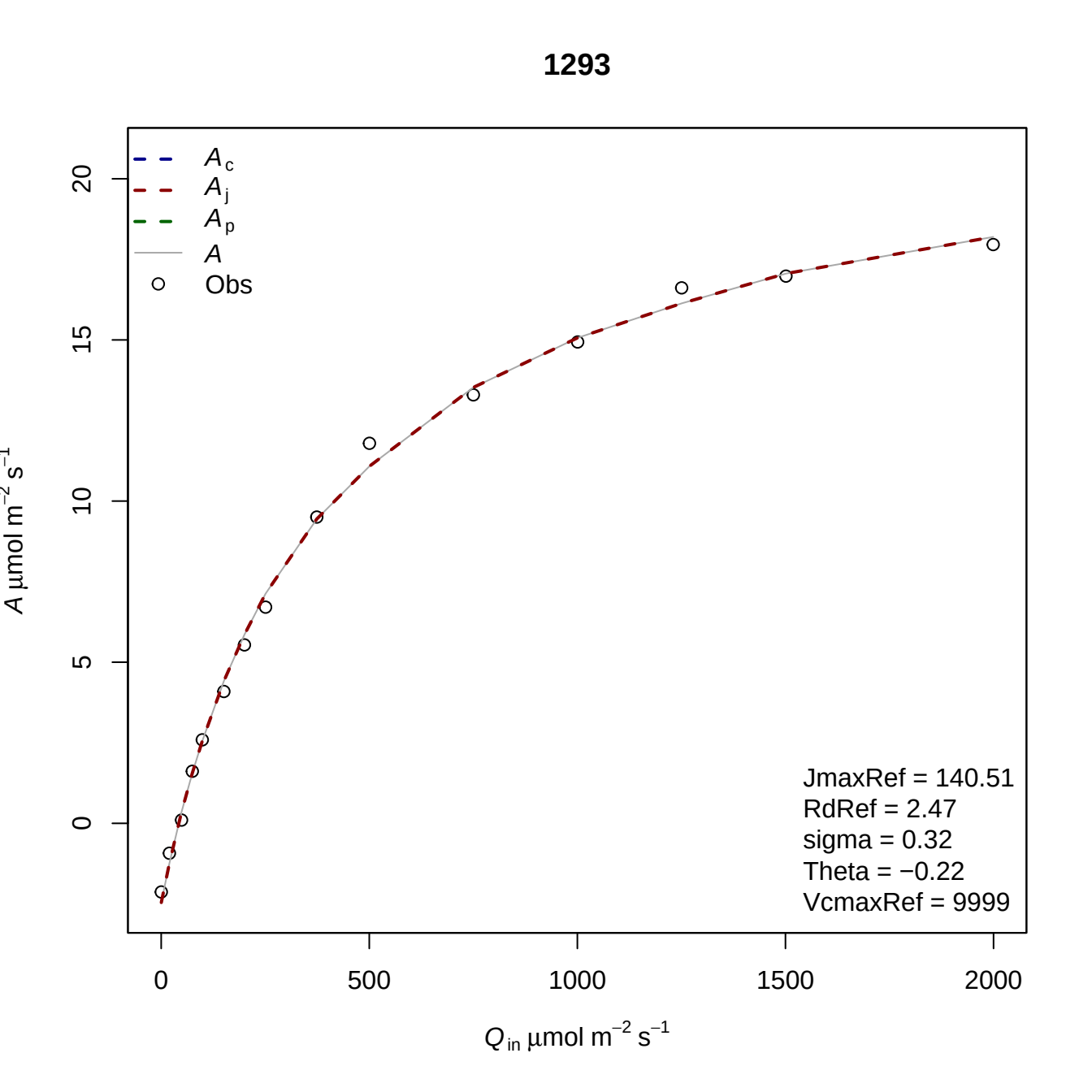

1288
A_c
A_j
A_p
A
Obs
JmaxRef = 522.35
RdRef = 1.63
sigma = 0.29
Theta = −10.79
VcmaxRef = 43.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

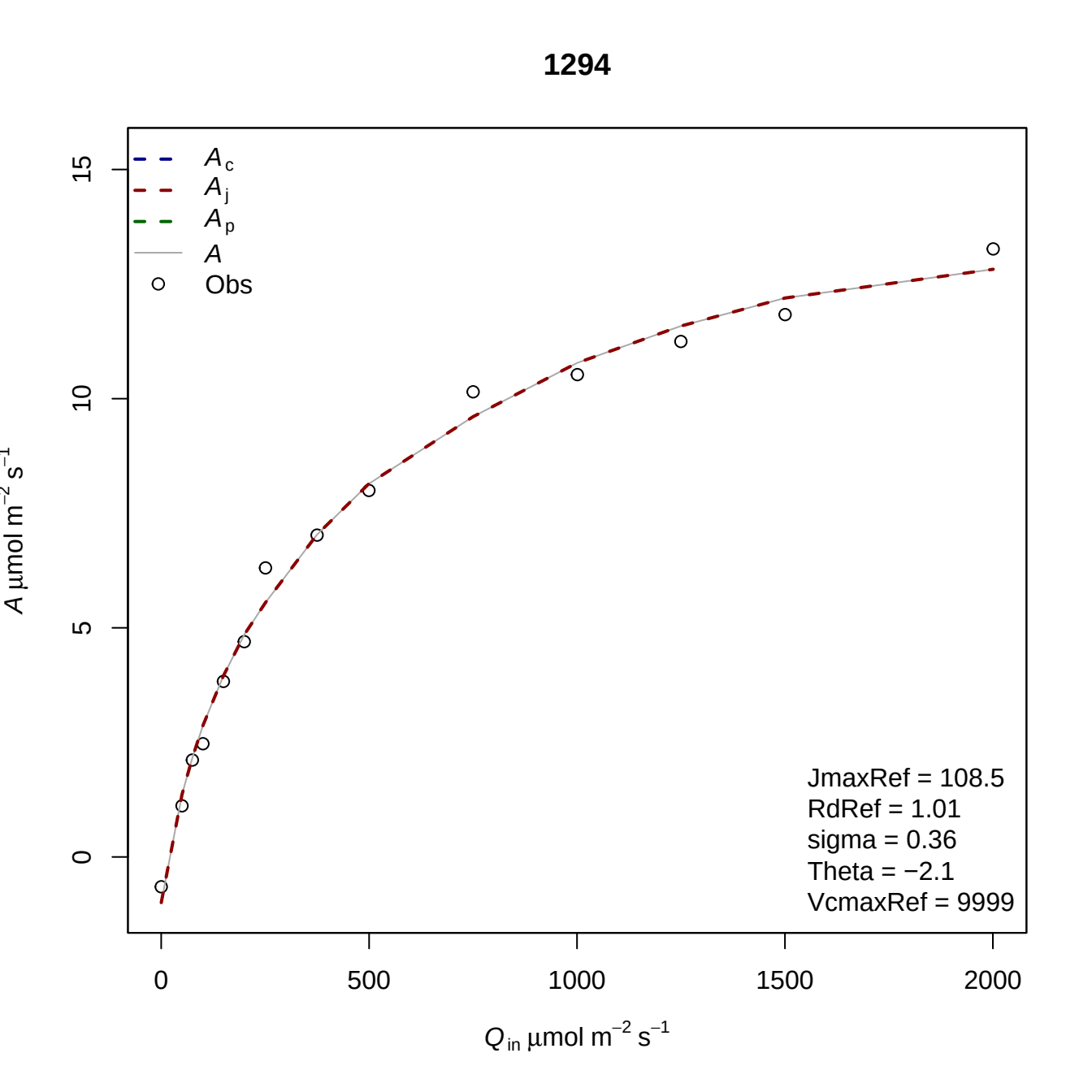

1289
A_c
A_j
A_p
A
Obs
JmaxRef = 180.37
RdRef = 3.66
sigma = 0.58
Theta = −0.83
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

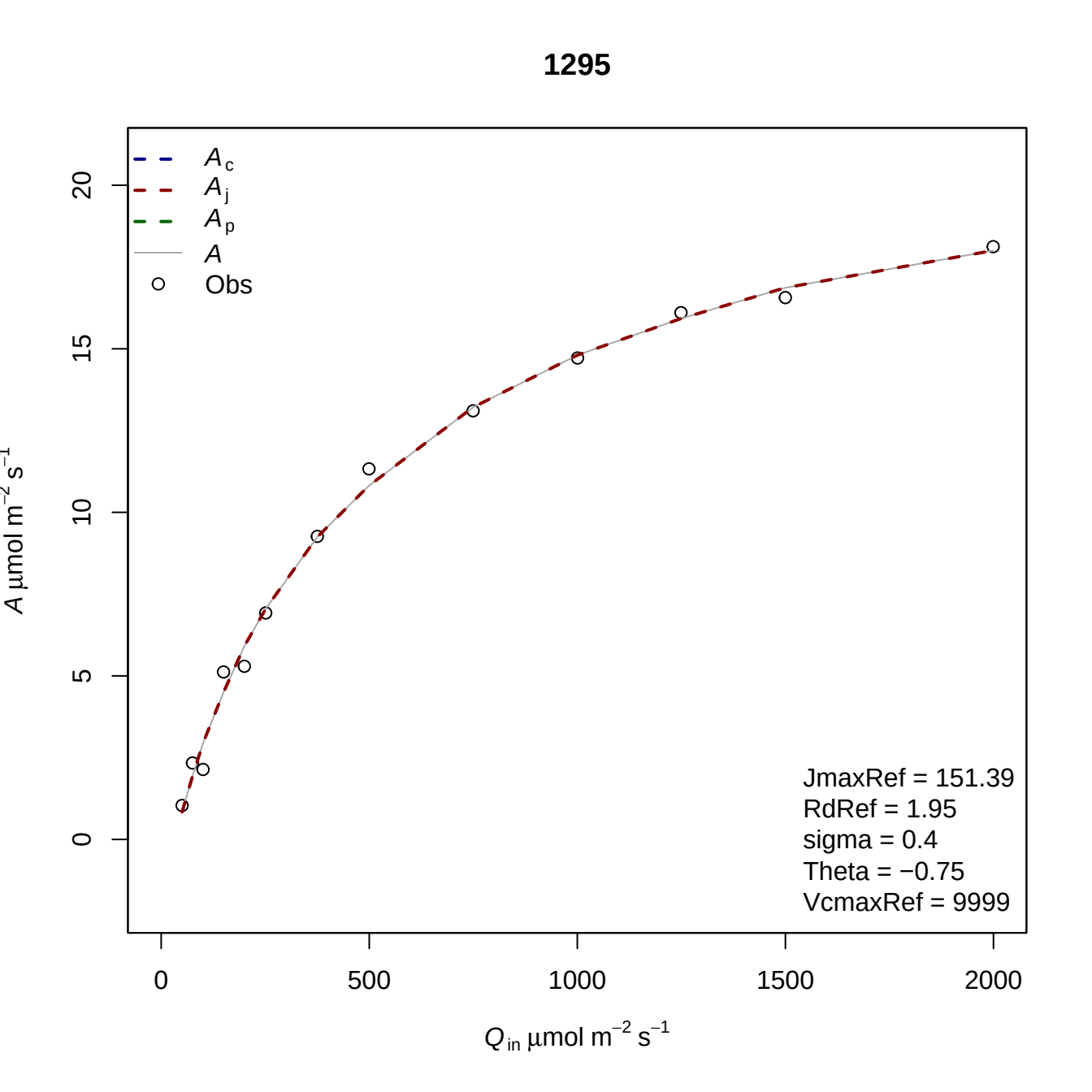

1290
A_c
A_j
A_p
A
Obs
JmaxRef = 163.36
RdRef = 2.56
sigma = 0.29
Theta = −2.64
VcmaxRef = 29.69
0
5
10
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

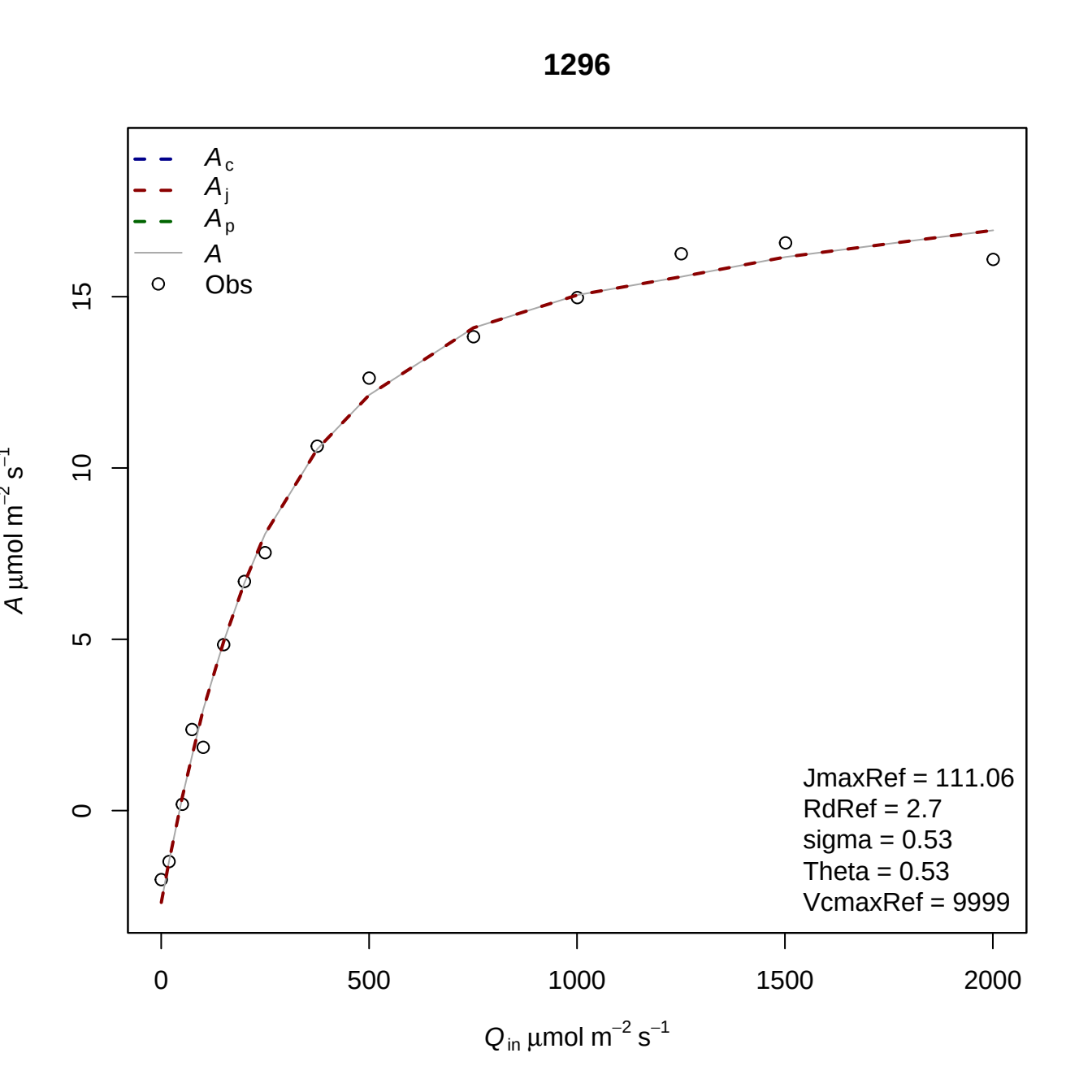

1291
A_c
A_j
A_p
A
Obs
JmaxRef = 130.43
RdRef = 2.62
sigma = 0.32
Theta = −2.64
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

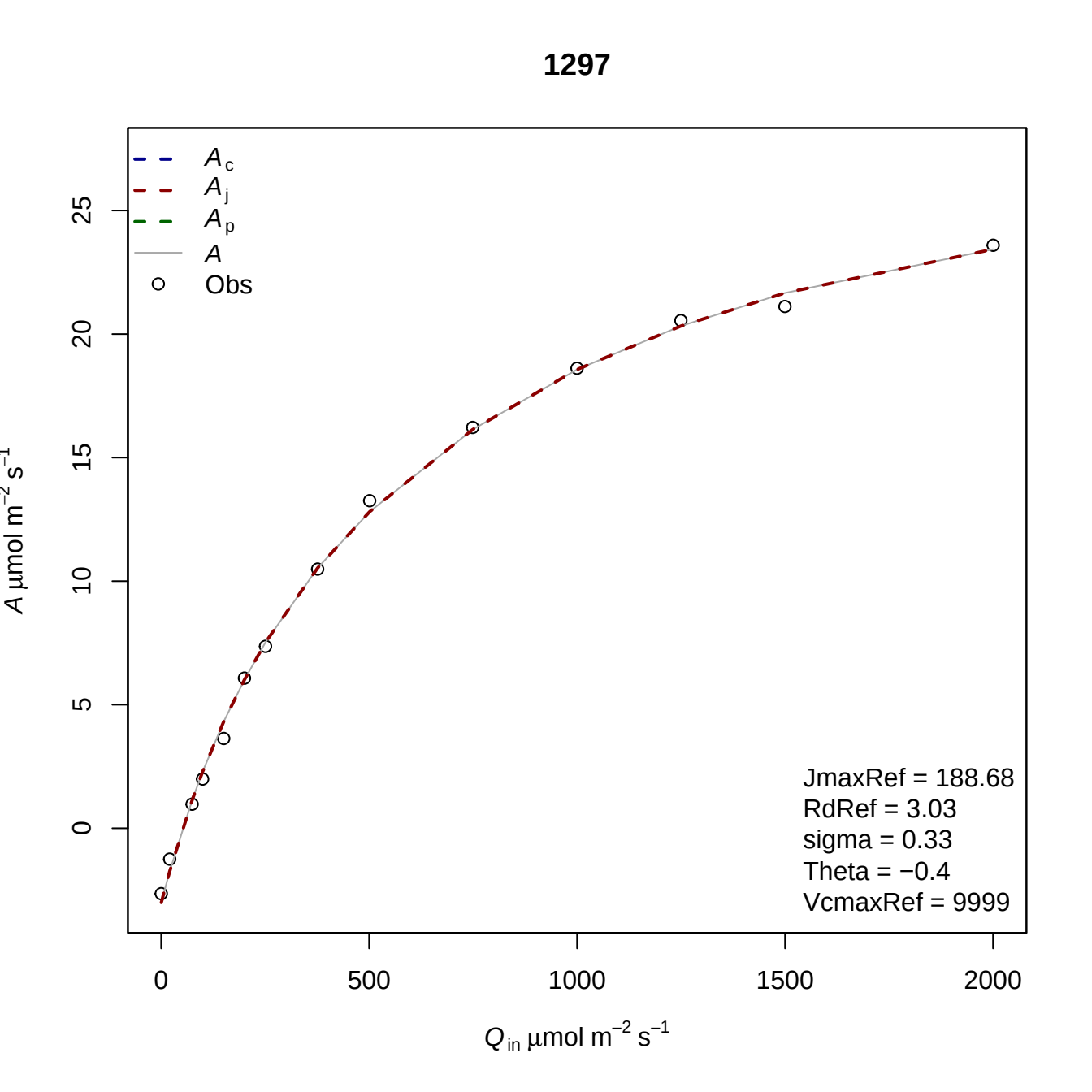

1292
A_c
A_j
A_p
A
Obs
JmaxRef = 121.39
RdRef = 2.62
sigma = 0.44
Theta = 0.55
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

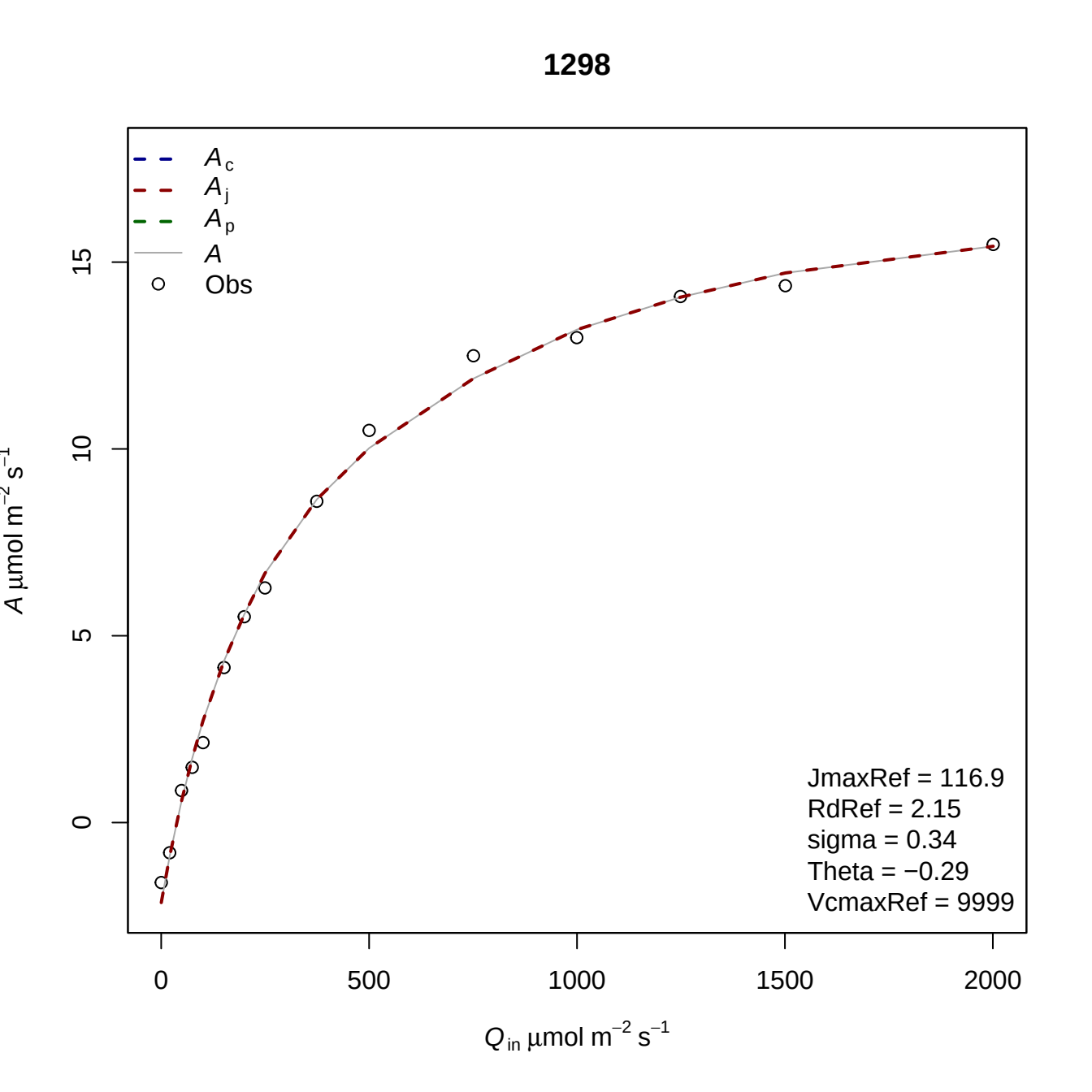

1293
A_c
A_j
A_p
A
Obs
JmaxRef = 140.51
RdRef = 2.47
sigma = 0.32
Theta = −0.22
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

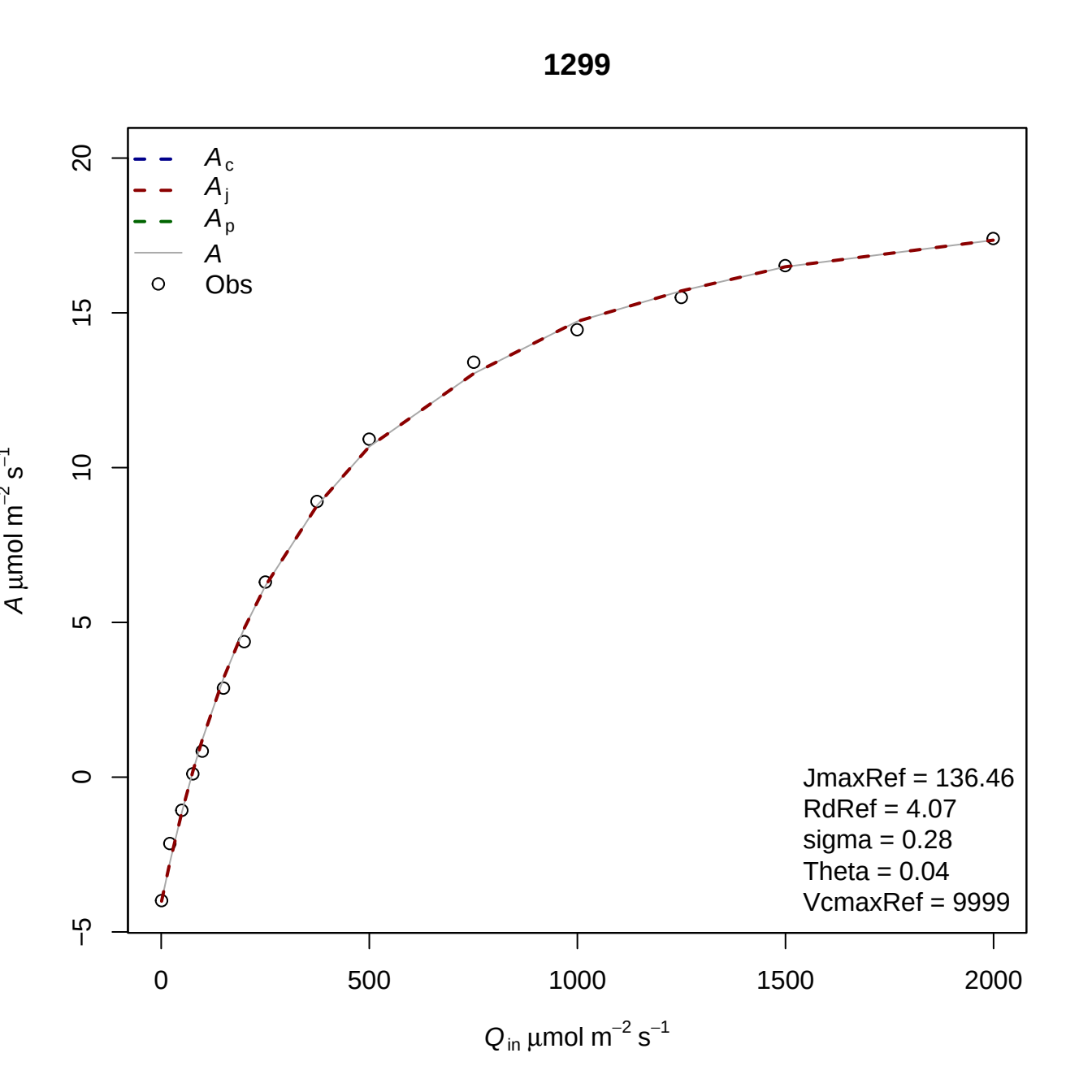

1294
A_c
A_j
A_p
A
Obs
JmaxRef = 108.5
RdRef = 1.01
sigma = 0.36
Theta = −2.1
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

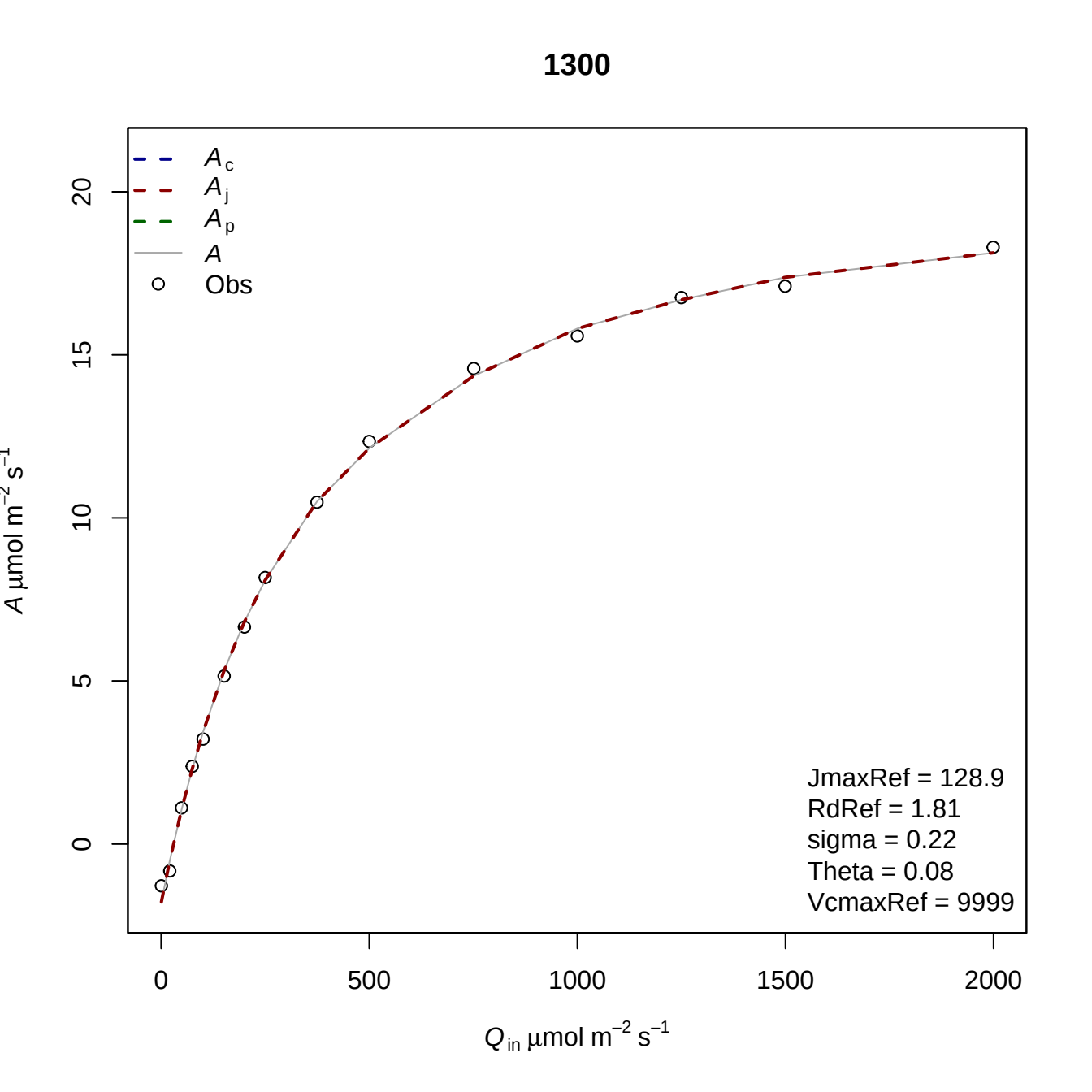

1295
A_c
A_j
A_p
A
Obs
JmaxRef = 151.39
RdRef = 1.95
sigma = 0.4
Theta = −0.75
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

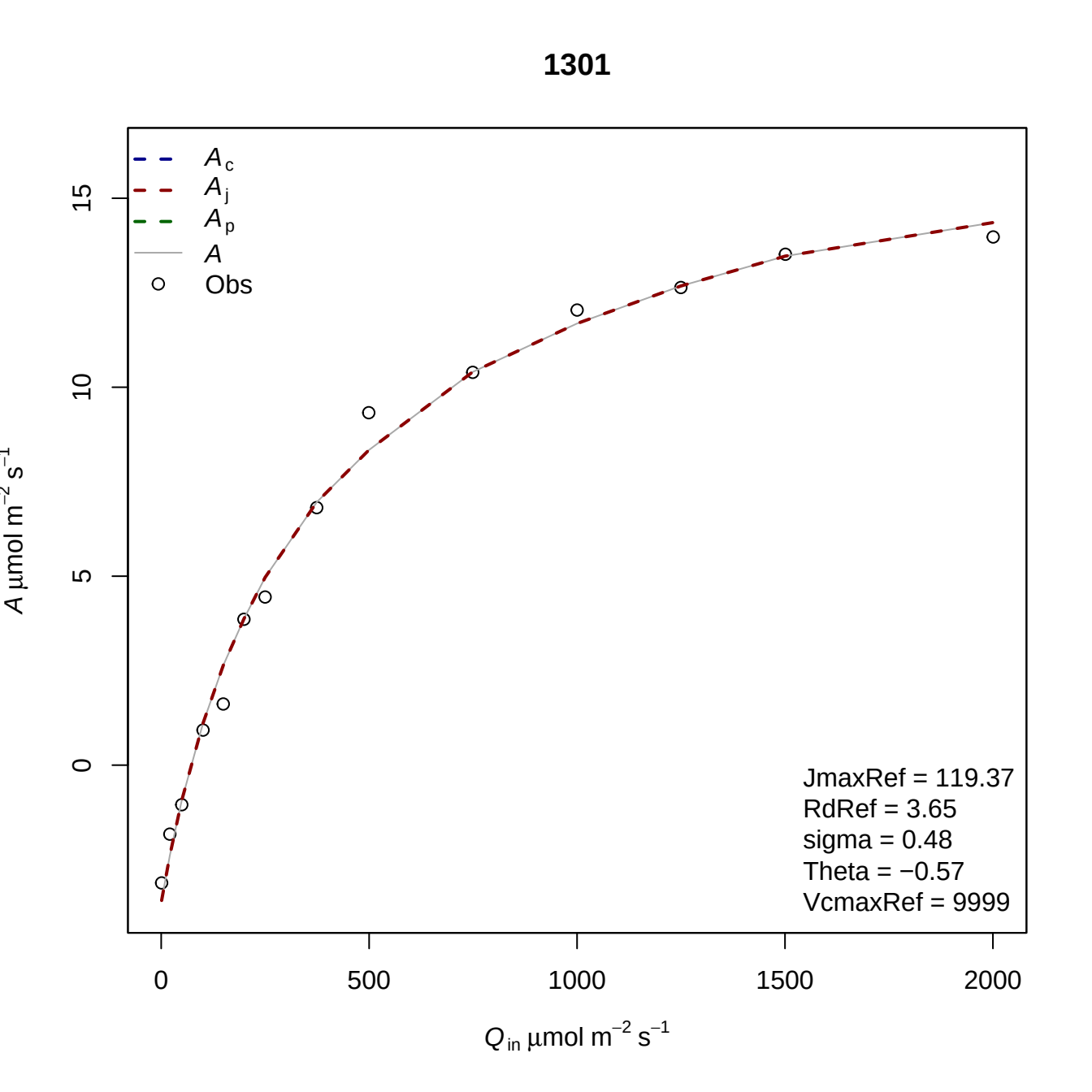

1296
A_c
A_j
A_p
A
Obs
JmaxRef = 111.06
RdRef = 2.7
sigma = 0.53
Theta = 0.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

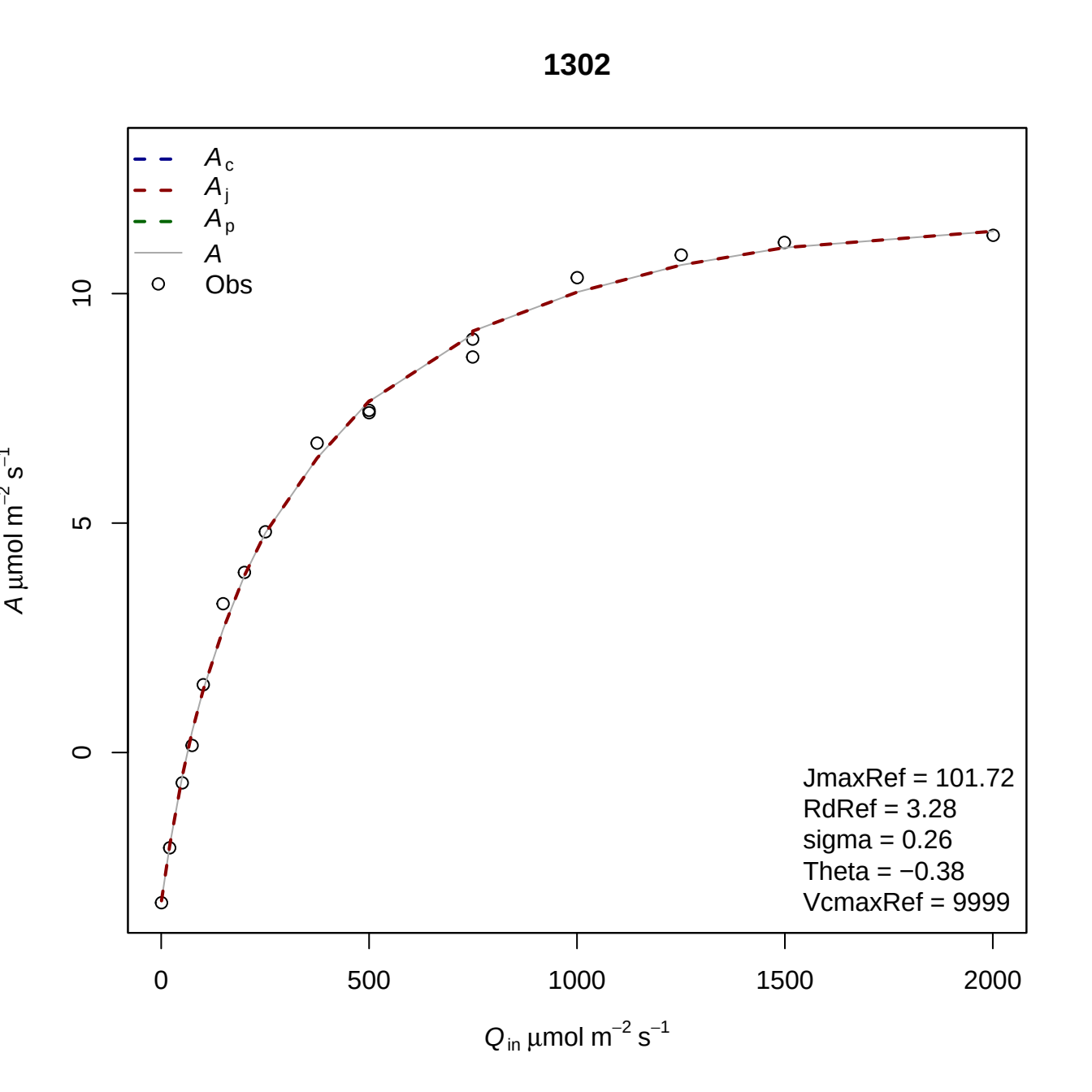

1297
A_c
A_j
A_p
A
Obs
JmaxRef = 188.68
RdRef = 3.03
sigma = 0.33
Theta = −0.4
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

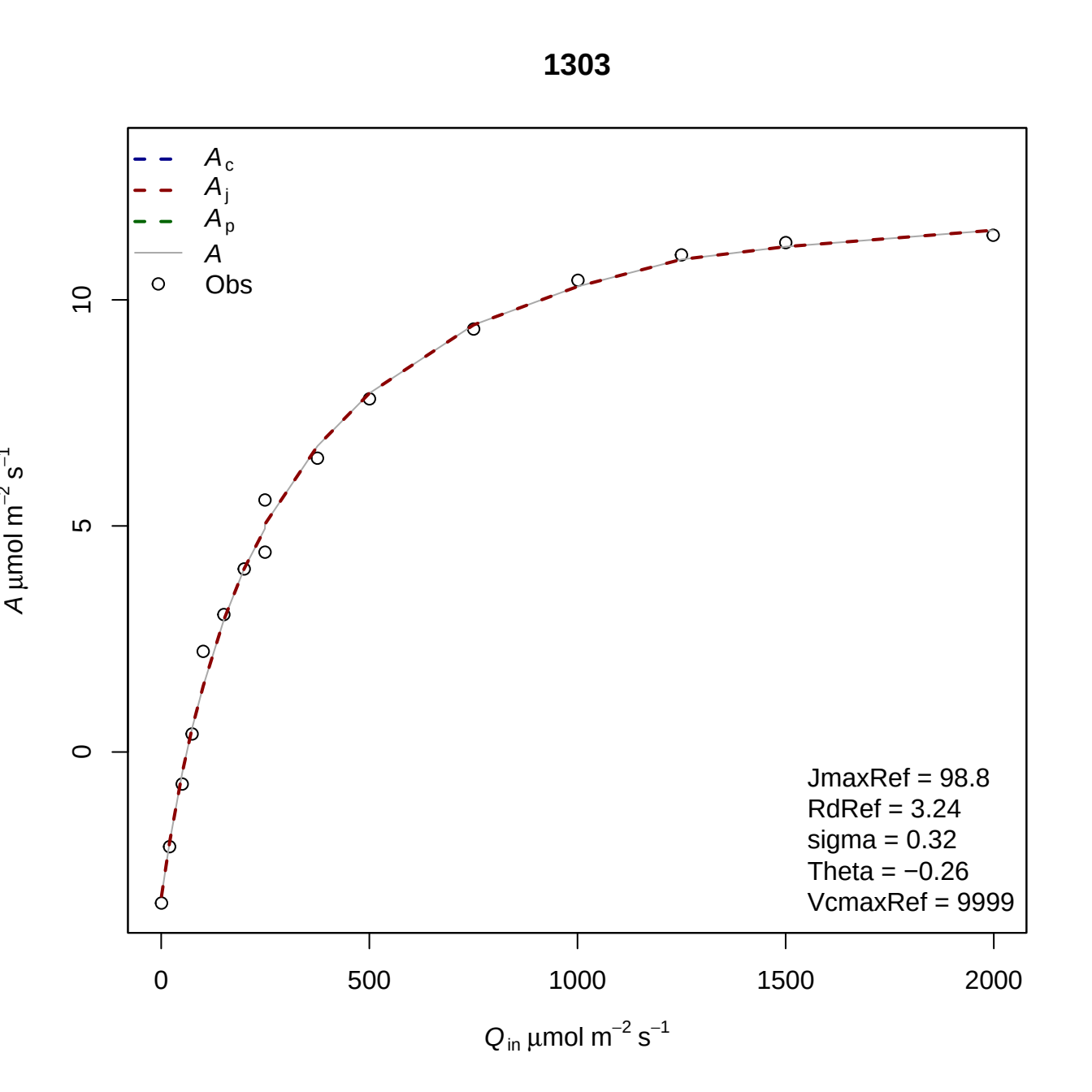

1298
A_c
A_j
A_p
A
Obs
JmaxRef = 116.9
RdRef = 2.15
sigma = 0.34
Theta = −0.29
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

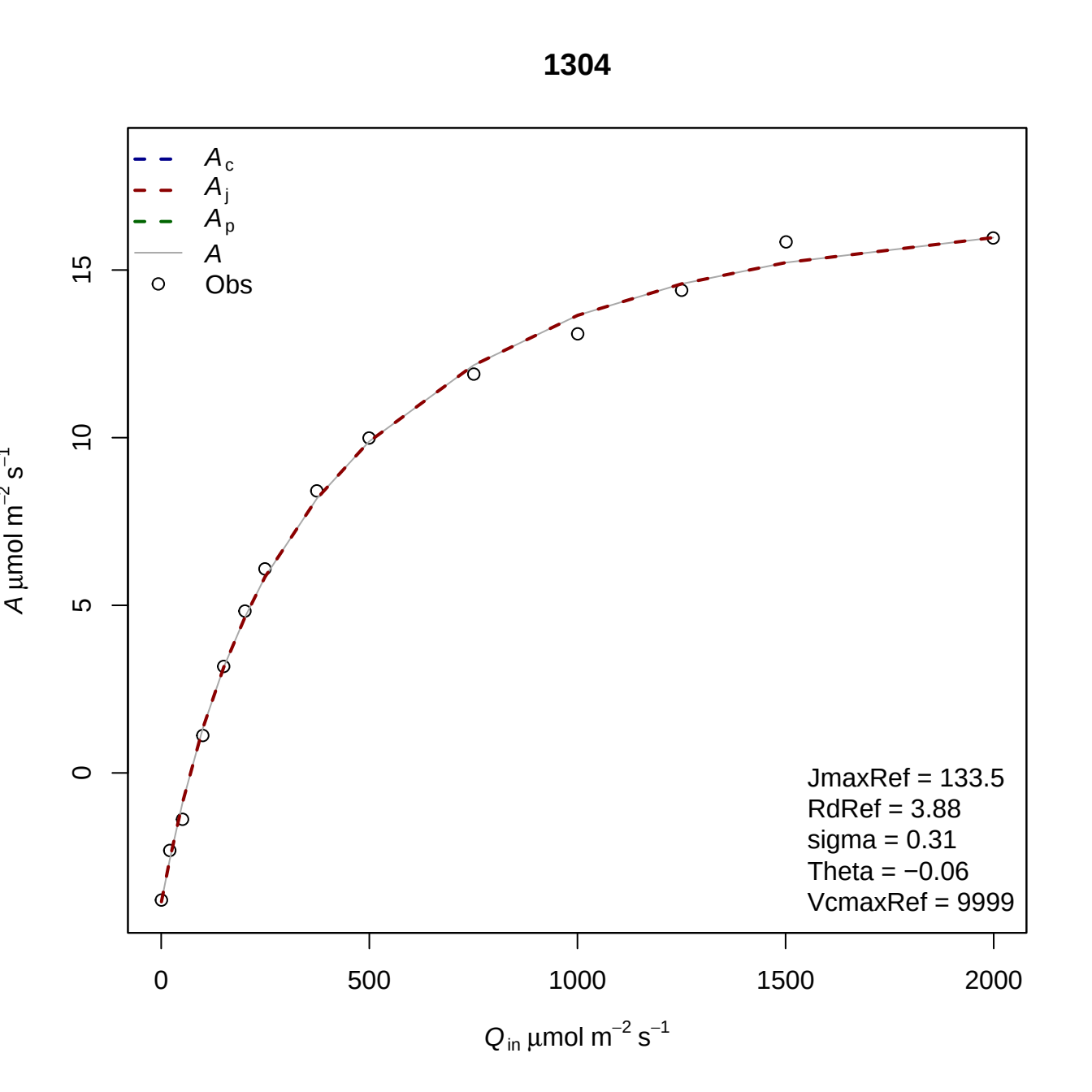

1299
A_c
A_j
A_p
A
Obs
JmaxRef = 136.46
RdRef = 4.07
sigma = 0.28
Theta = 0.04
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

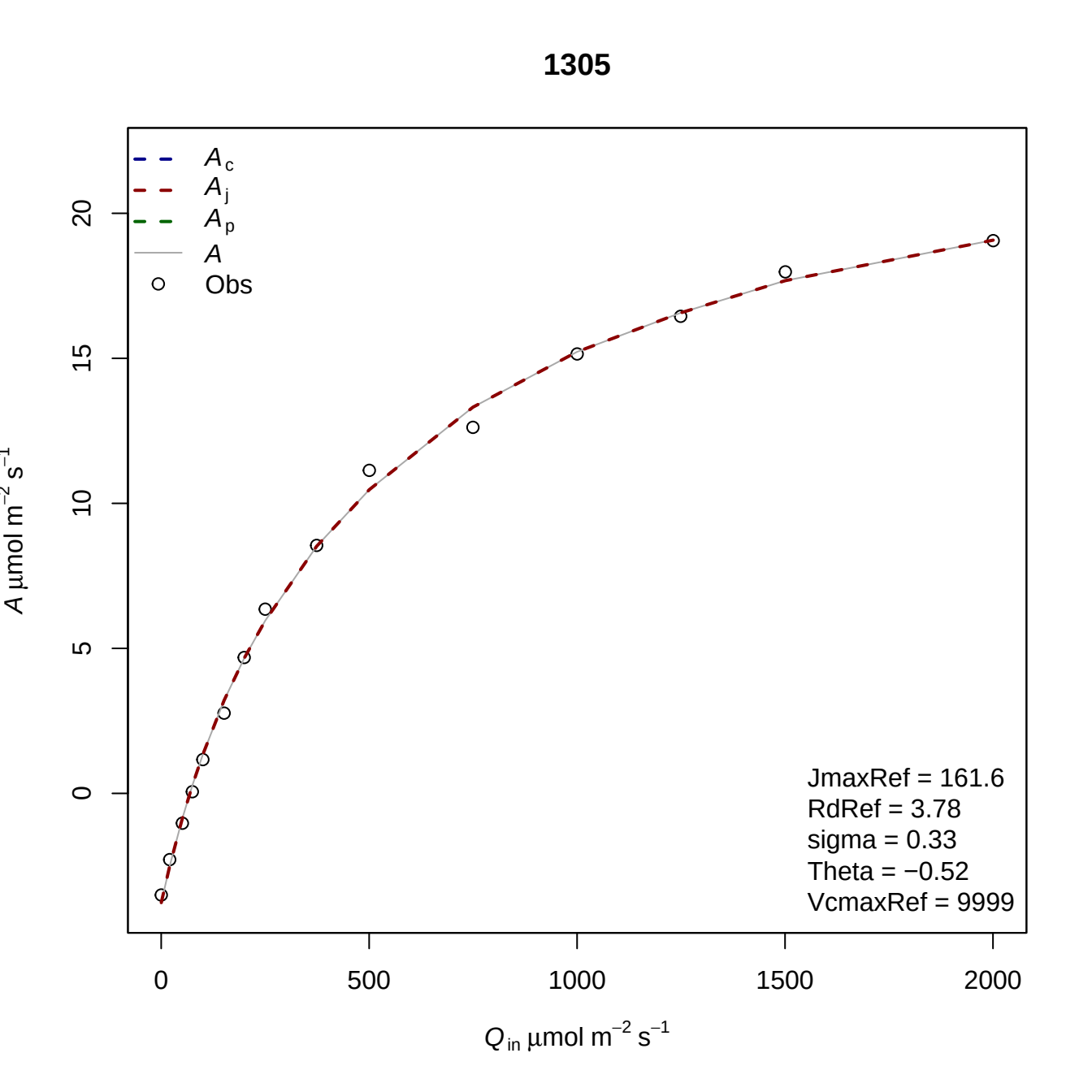

1300
A_c
A_j
A_p
A
Obs
JmaxRef = 128.9
RdRef = 1.81
sigma = 0.22
Theta = 0.08
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

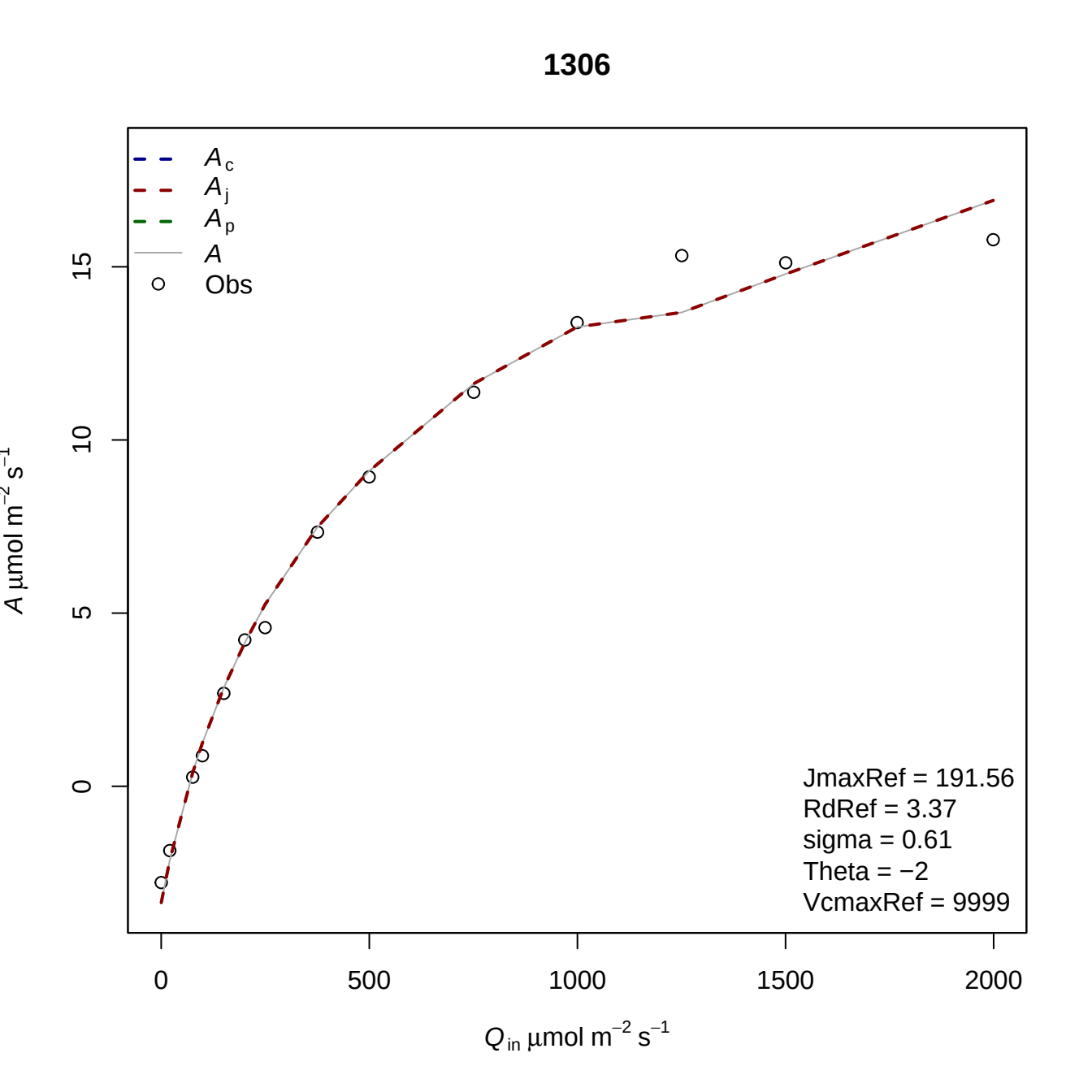

1301
A_c
A_j
A_p
A
Obs
JmaxRef = 119.37
RdRef = 3.65
sigma = 0.48
Theta = −0.57
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

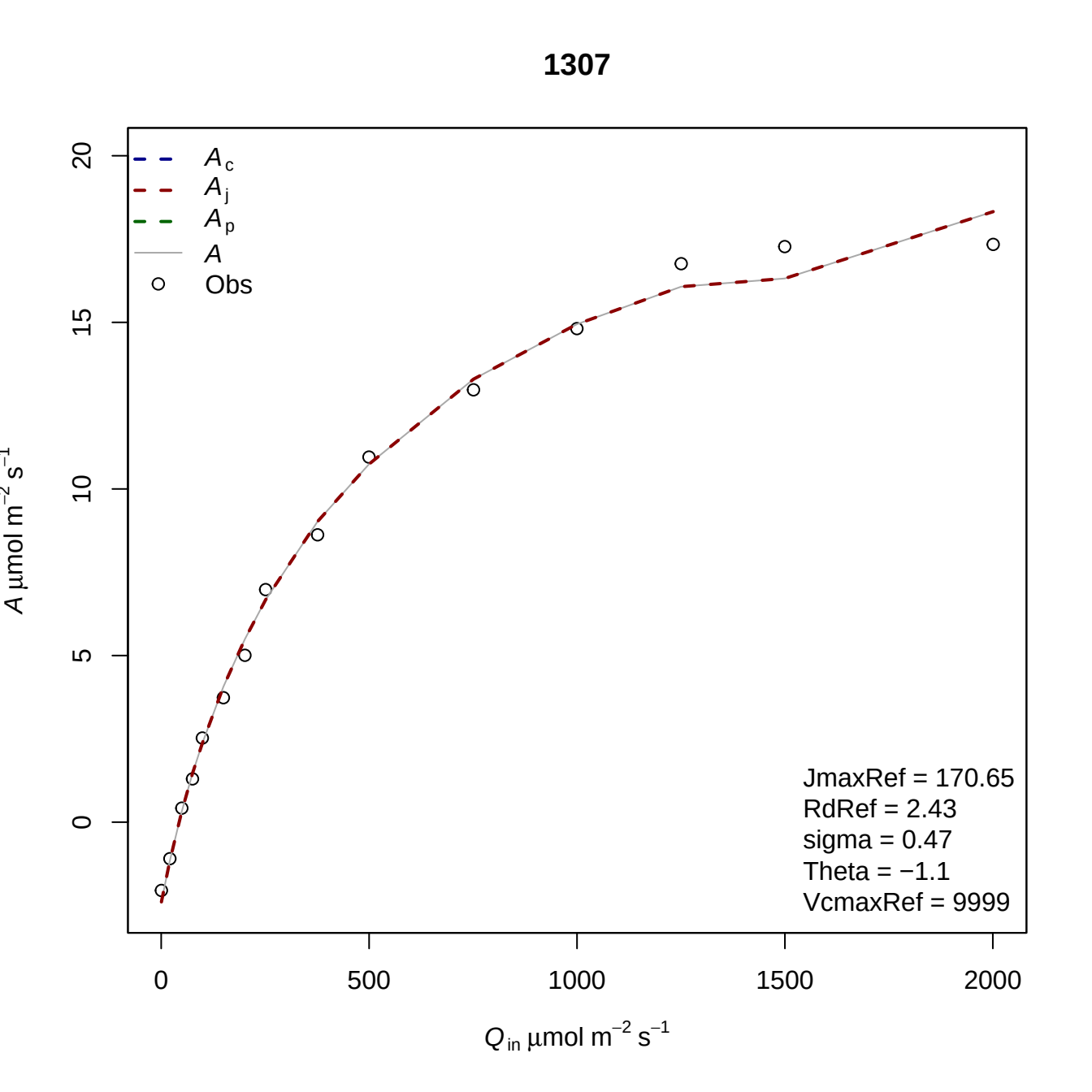

1302
A_c
A_j
A_p
A
Obs
JmaxRef = 101.72
RdRef = 3.28
sigma = 0.26
Theta = −0.38
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

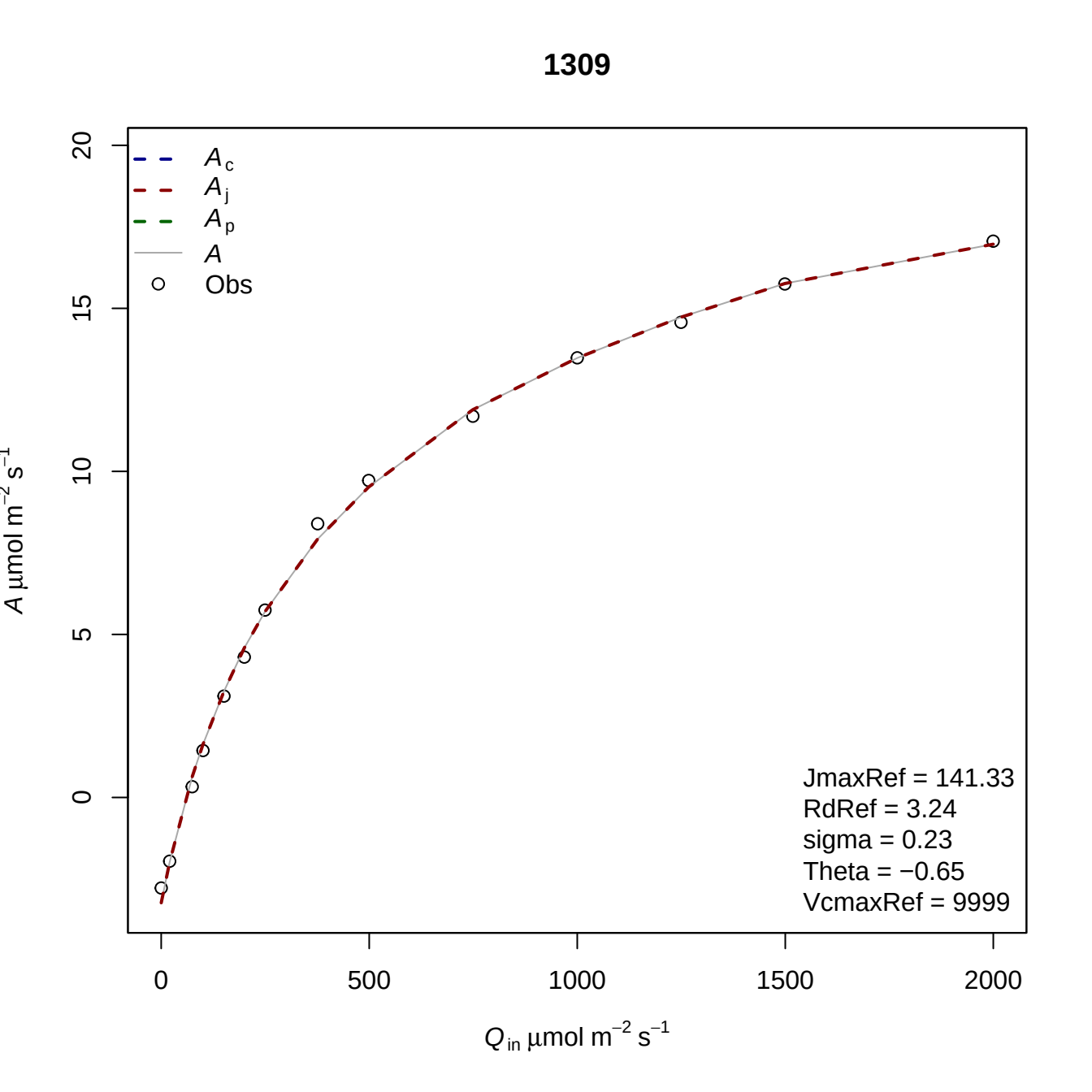

1303
A_c
A_j
A_p
A
Obs
JmaxRef = 98.8
RdRef = 3.24
sigma = 0.32
Theta = −0.26
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

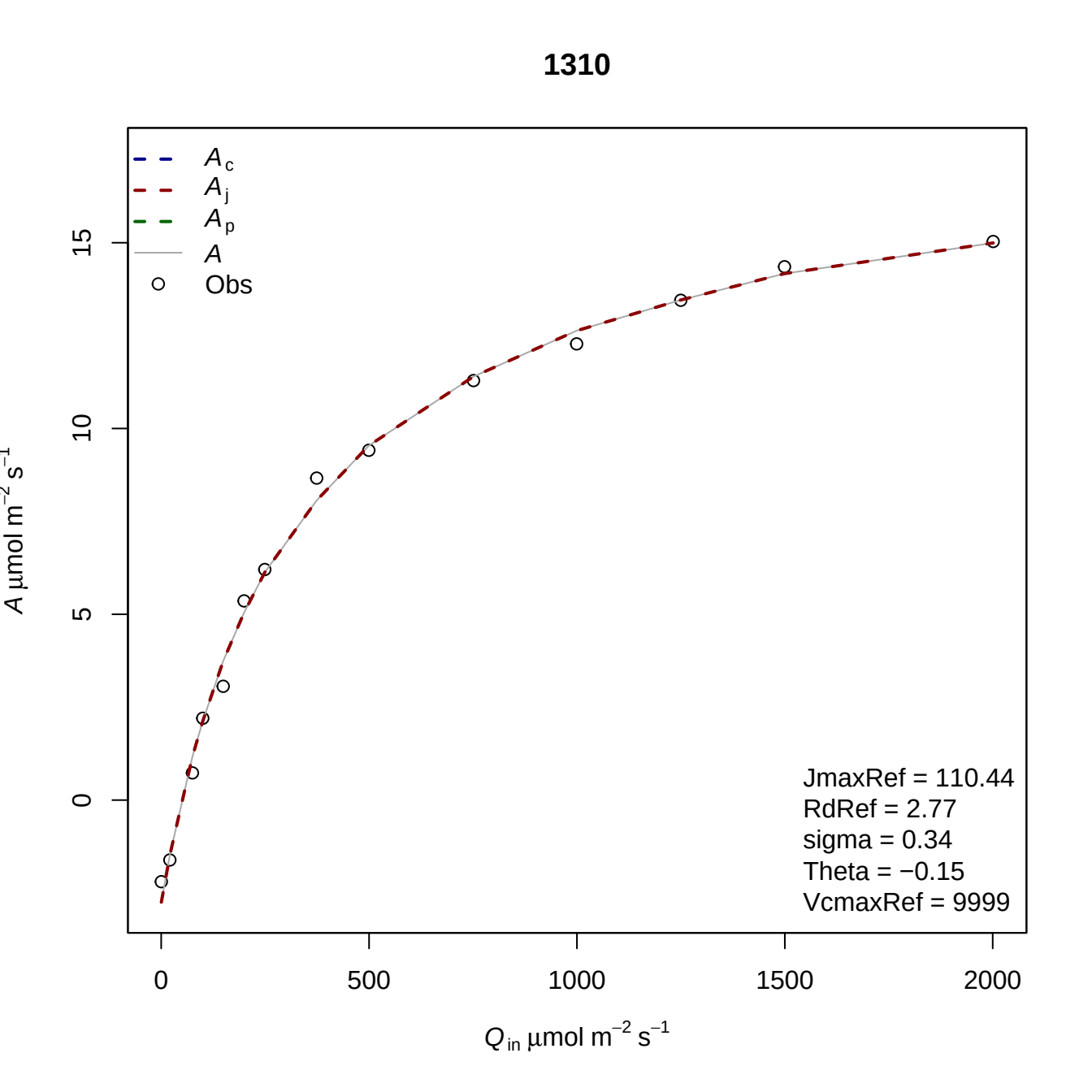

1304
A_c
A_j
A_p
A
Obs
JmaxRef = 133.5
RdRef = 3.88
sigma = 0.31
Theta = −0.06
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

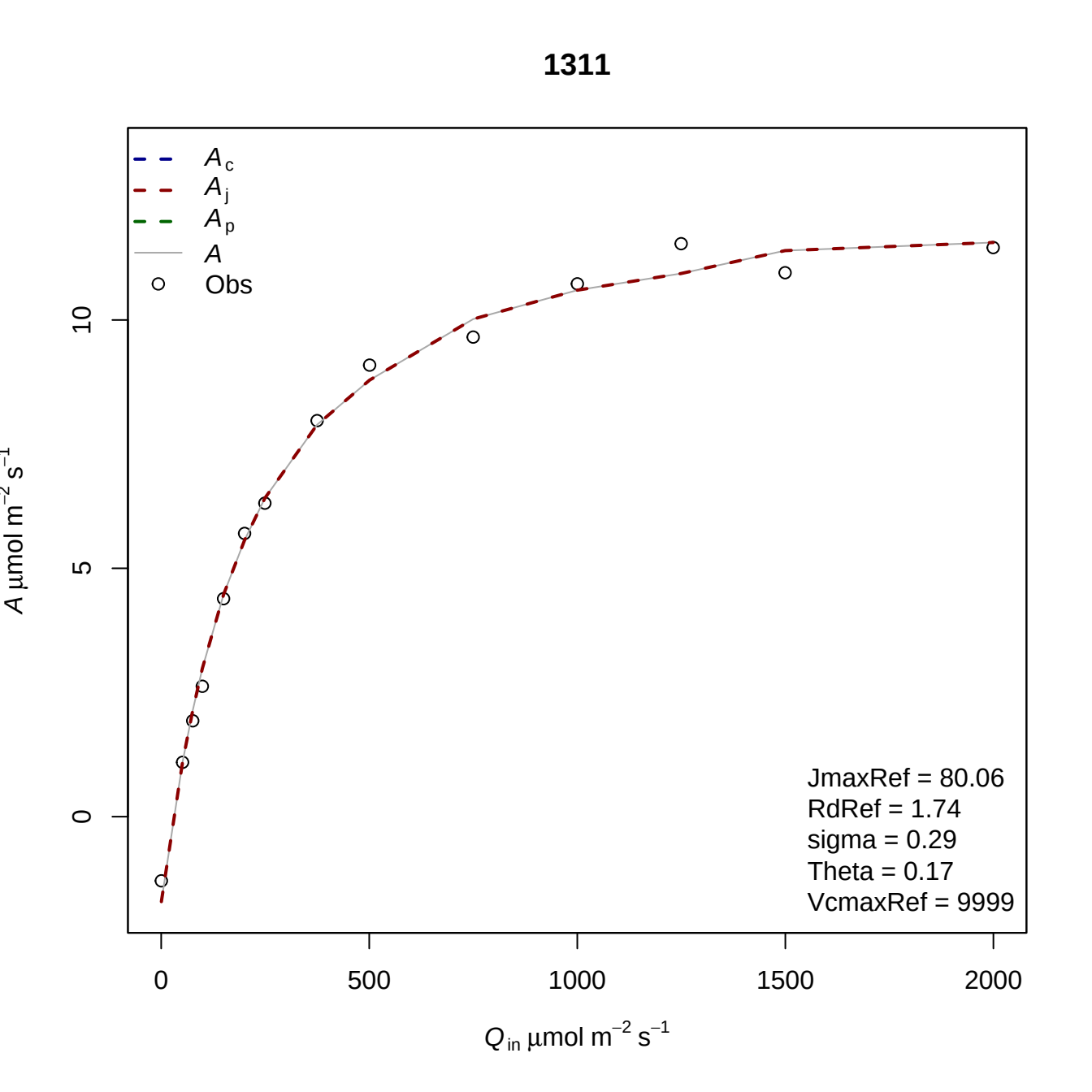

1305
A_c
A_j
A_p
A
Obs
JmaxRef = 161.6
RdRef = 3.78
sigma = 0.33
Theta = −0.52
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

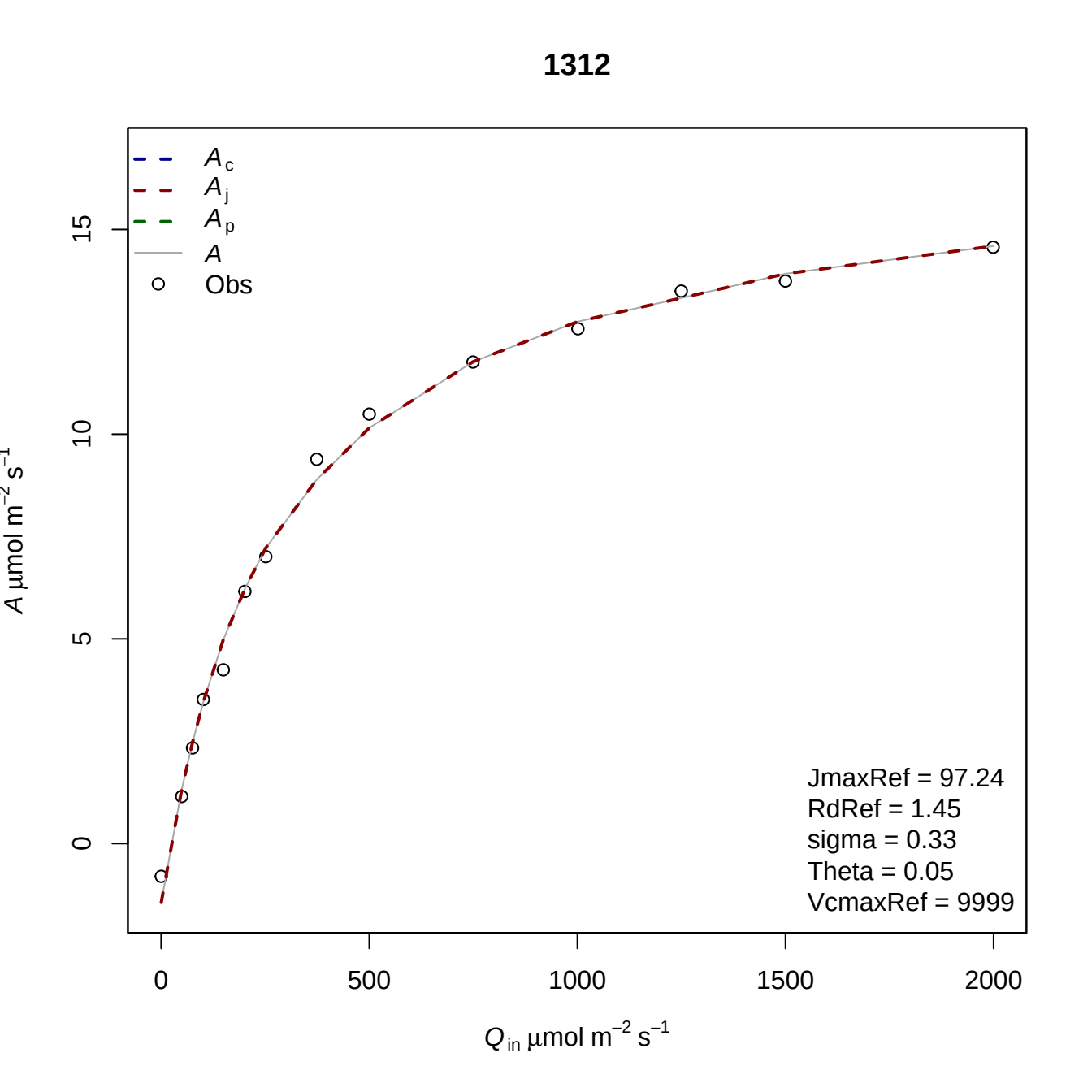

1306
A_c
A_j
A_p
A
Obs
JmaxRef = 191.56
RdRef = 3.37
sigma = 0.61
Theta = −2
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

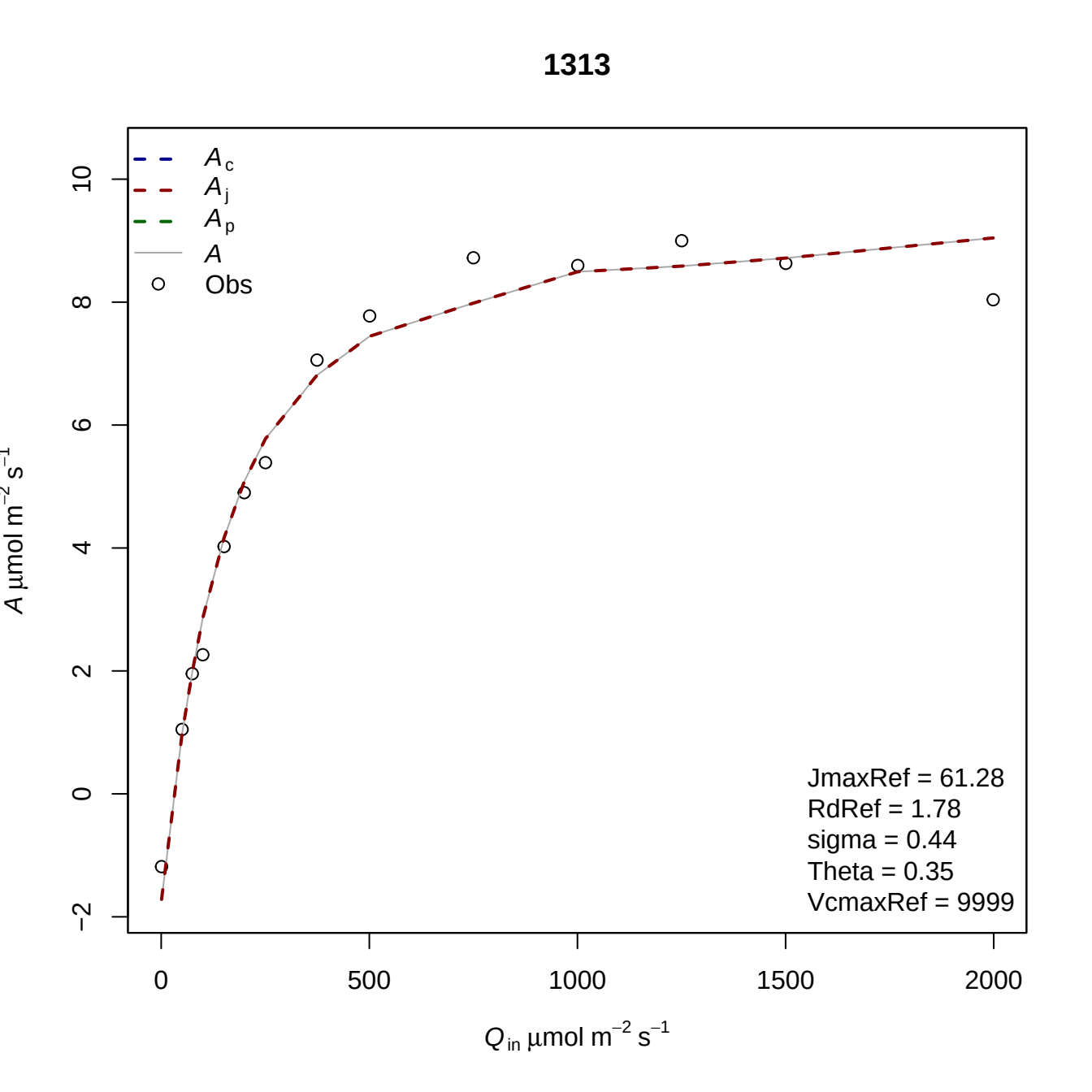

1307
A_c
A_j
A_p
A
Obs
JmaxRef = 170.65
RdRef = 2.43
sigma = 0.47
Theta = −1.1
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

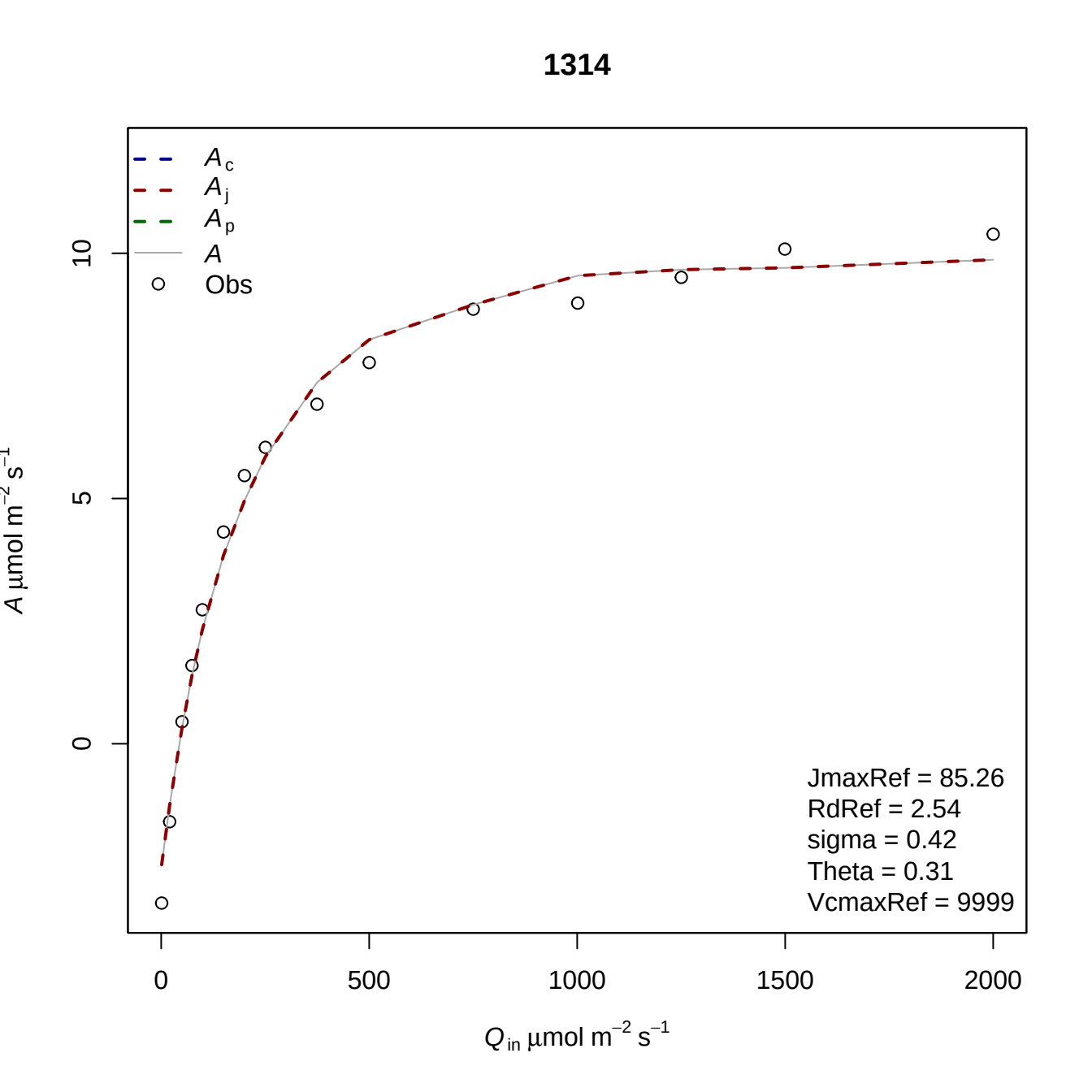

1309
A_c
A_j
A_p
A
Obs
JmaxRef = 141.33
RdRef = 3.24
sigma = 0.23
Theta = −0.65
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

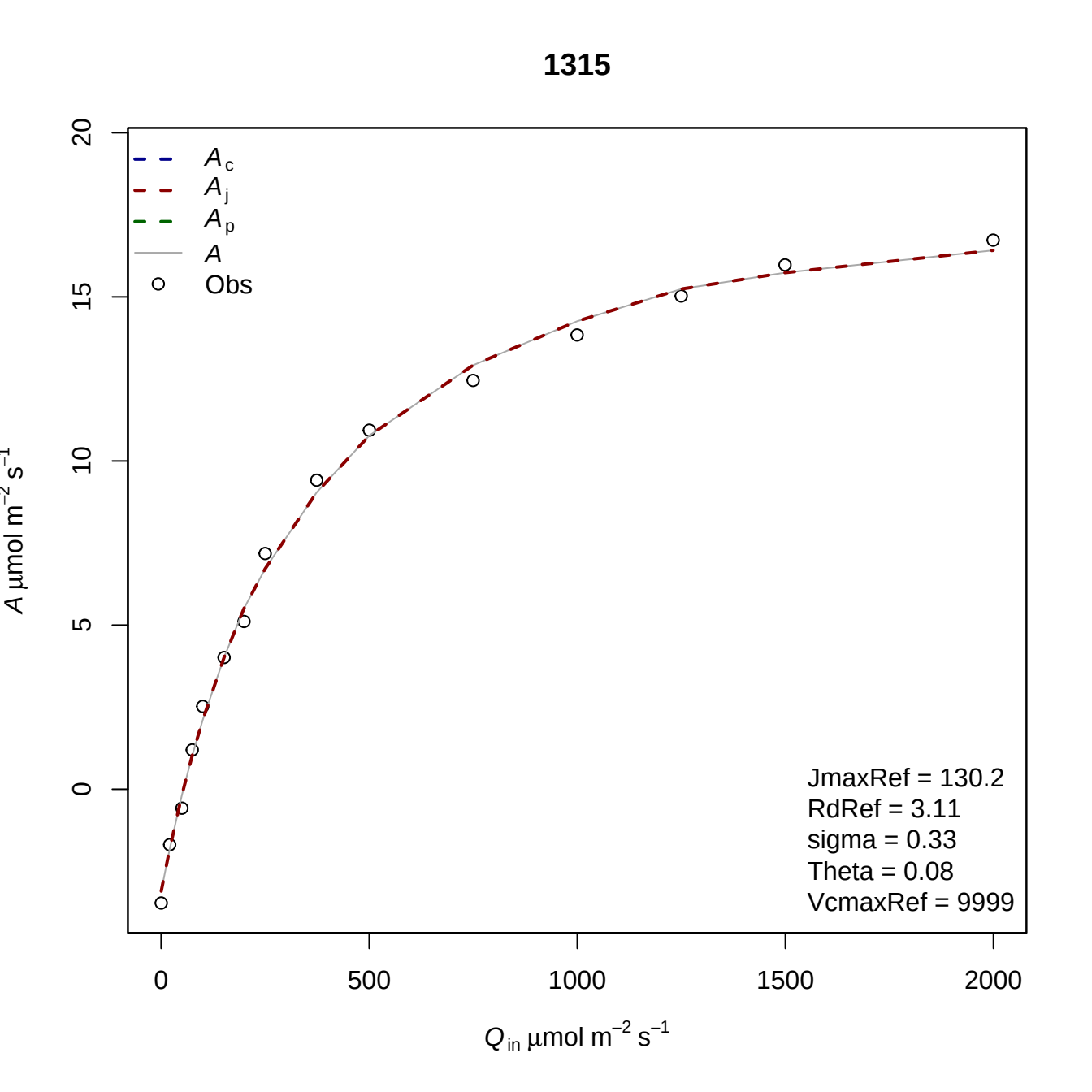

1310
A_c
A_j
A_p
A
Obs
JmaxRef = 110.44
RdRef = 2.77
sigma = 0.34
Theta = −0.15
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

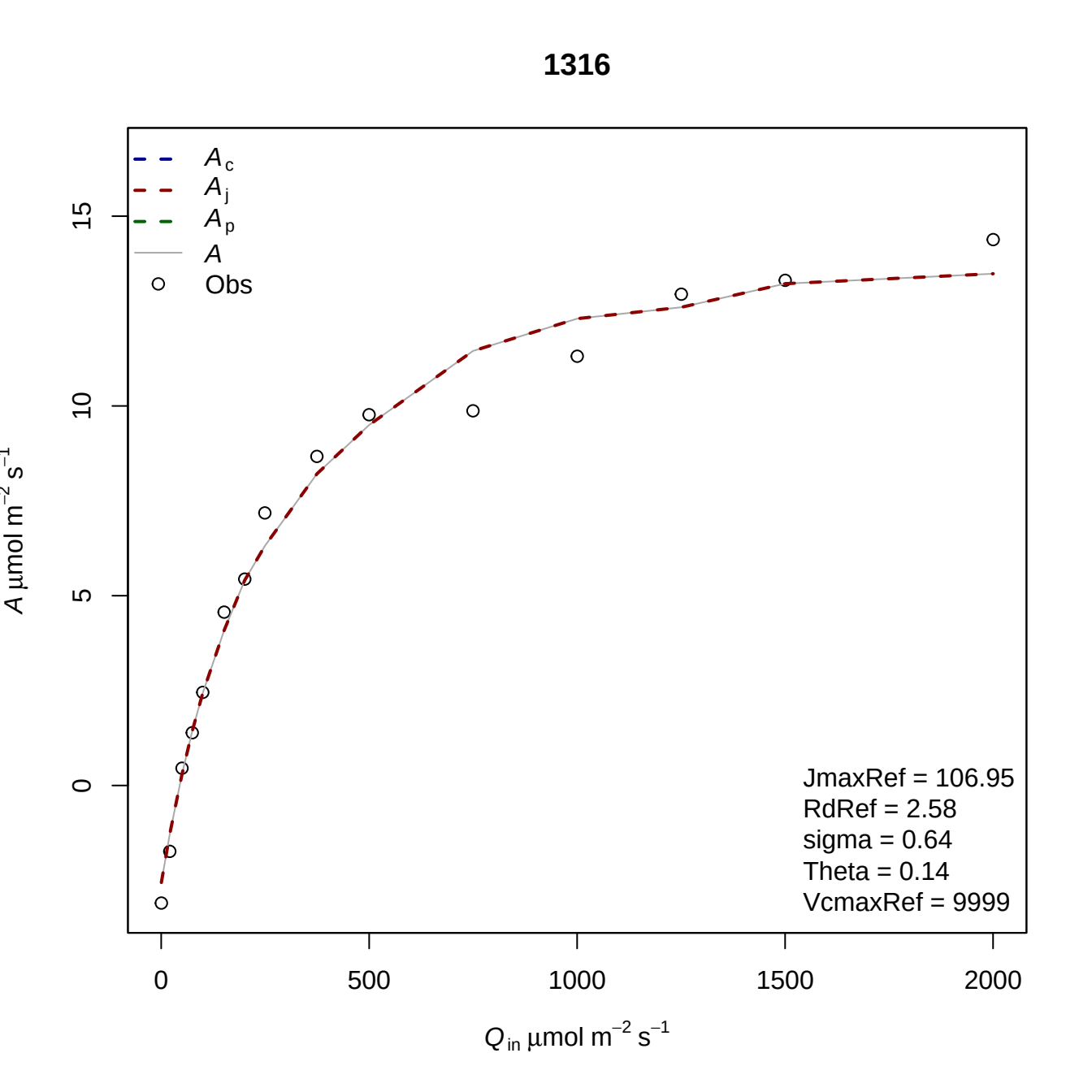

1311
A_c
A_j
A_p
A
Obs
JmaxRef = 80.06
RdRef = 1.74
sigma = 0.29
Theta = 0.17
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

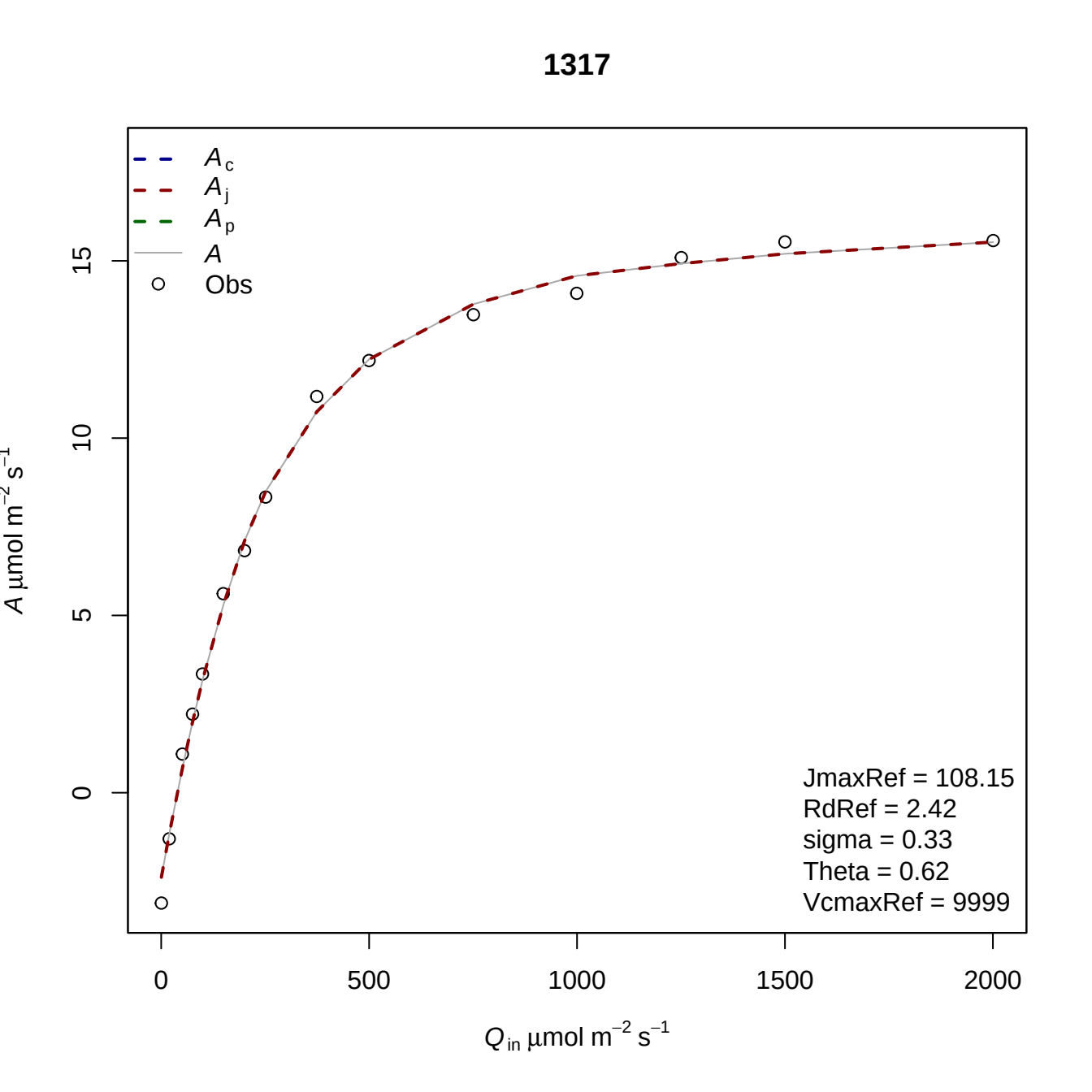

1312
A_c
A_j
A_p
A
Obs
JmaxRef = 97.24
RdRef = 1.45
sigma = 0.33
Theta = 0.05
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

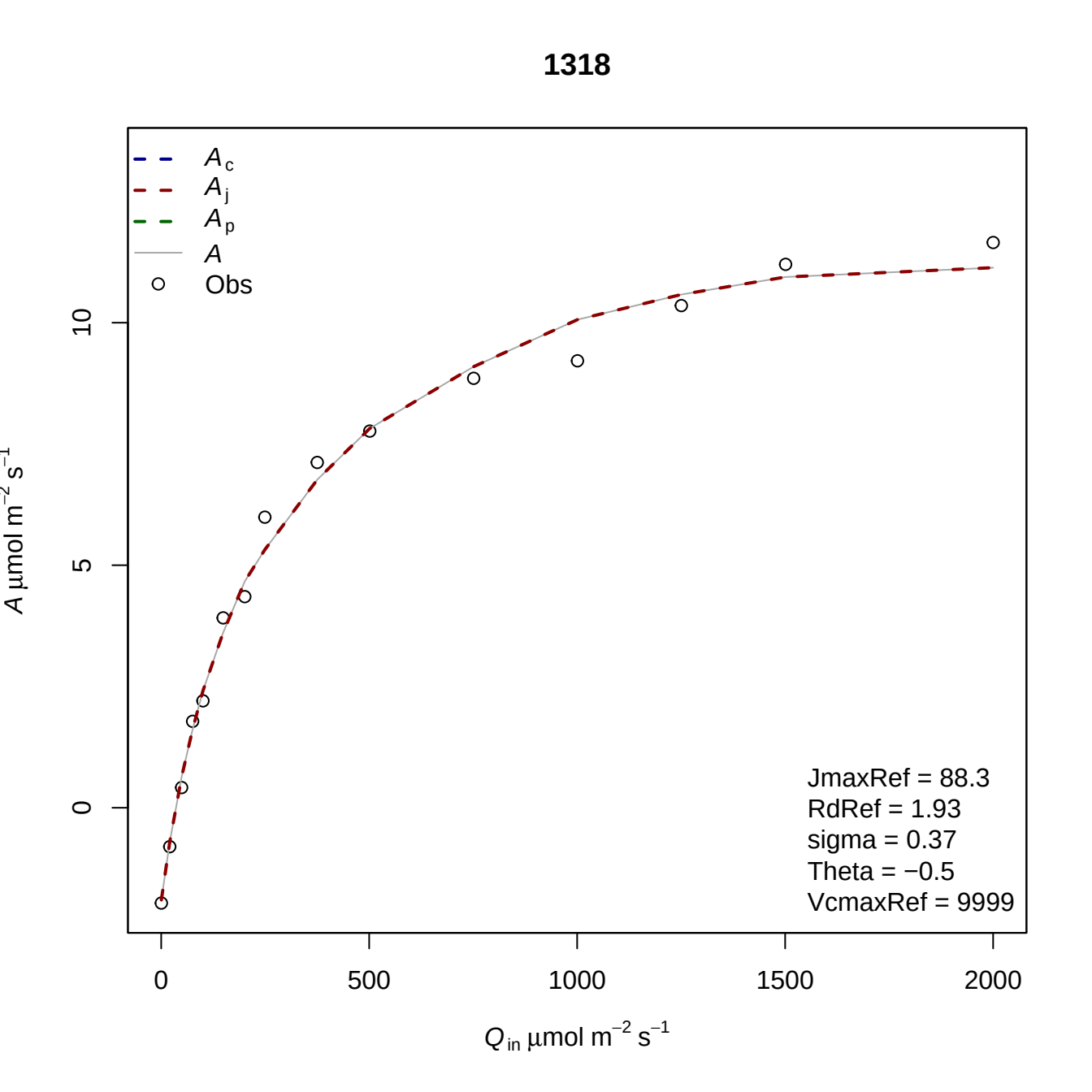

1313
A_c
A_j
A_p
A
Obs
JmaxRef = 61.28
RdRef = 1.78
sigma = 0.44
Theta = 0.35
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

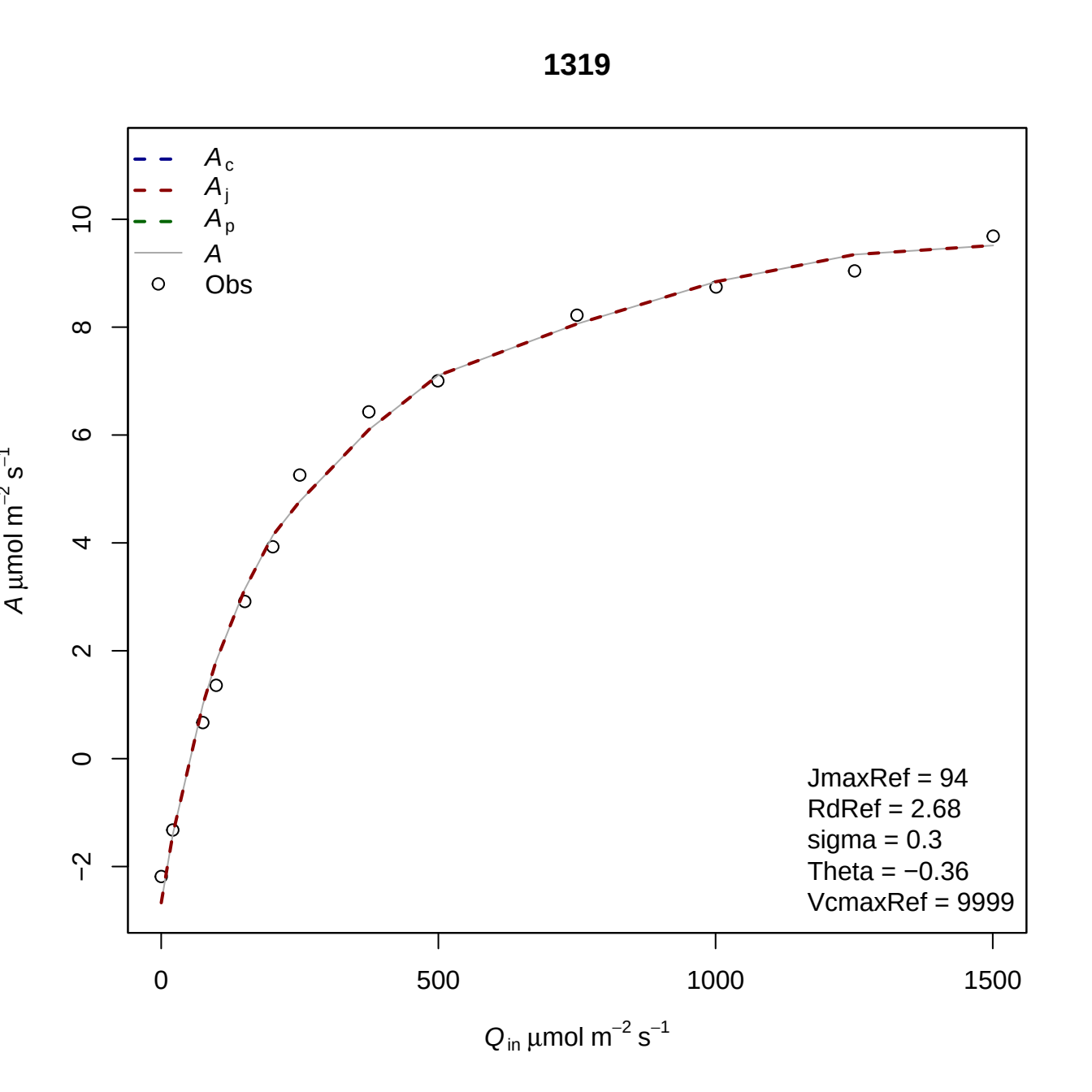

1314
A_c
A_j
A_p
A
Obs
JmaxRef = 85.26
RdRef = 2.54
sigma = 0.42
Theta = 0.31
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

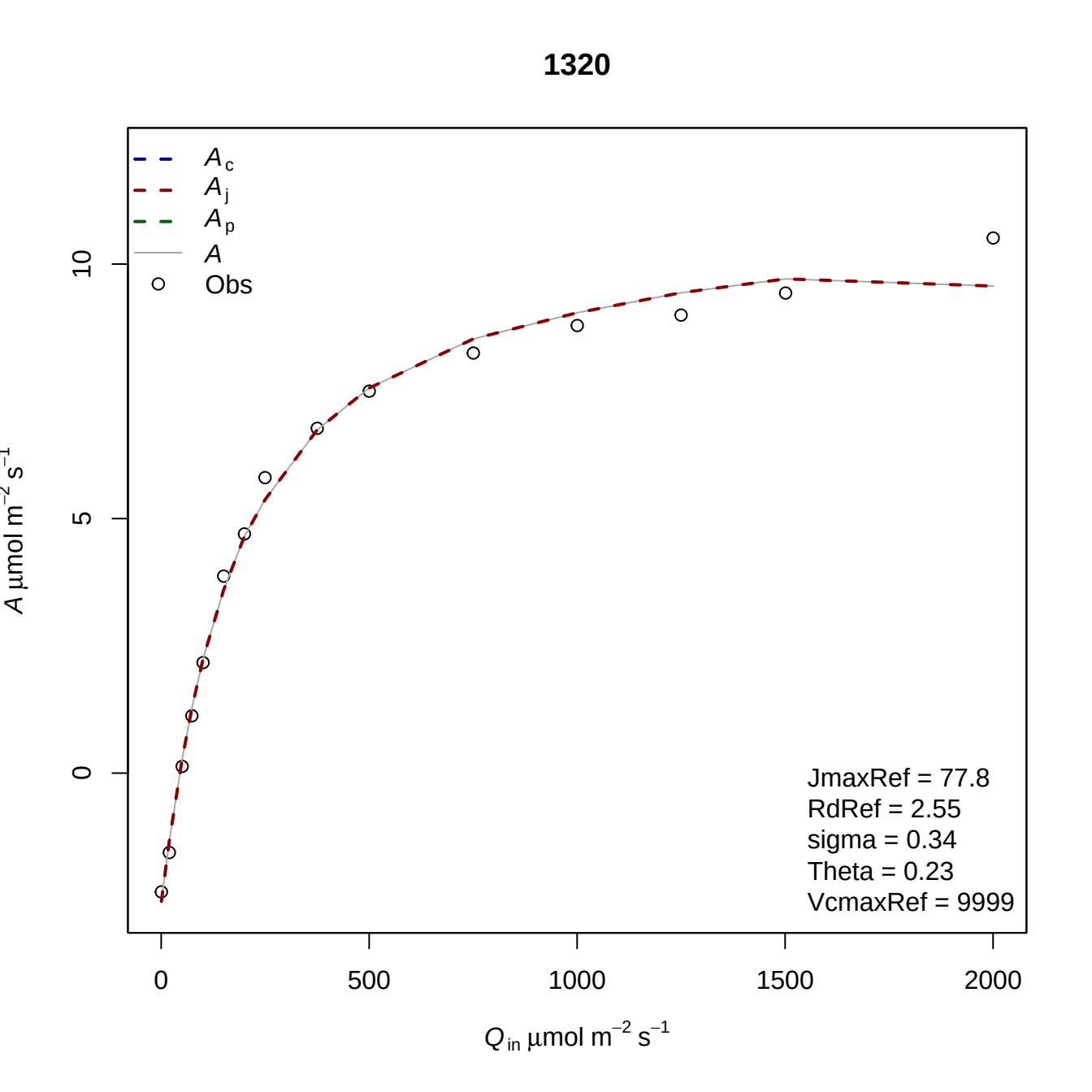

1315
A_c
A_j
A_p
A
Obs
JmaxRef = 130.2
RdRef = 3.11
sigma = 0.33
Theta = 0.08
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

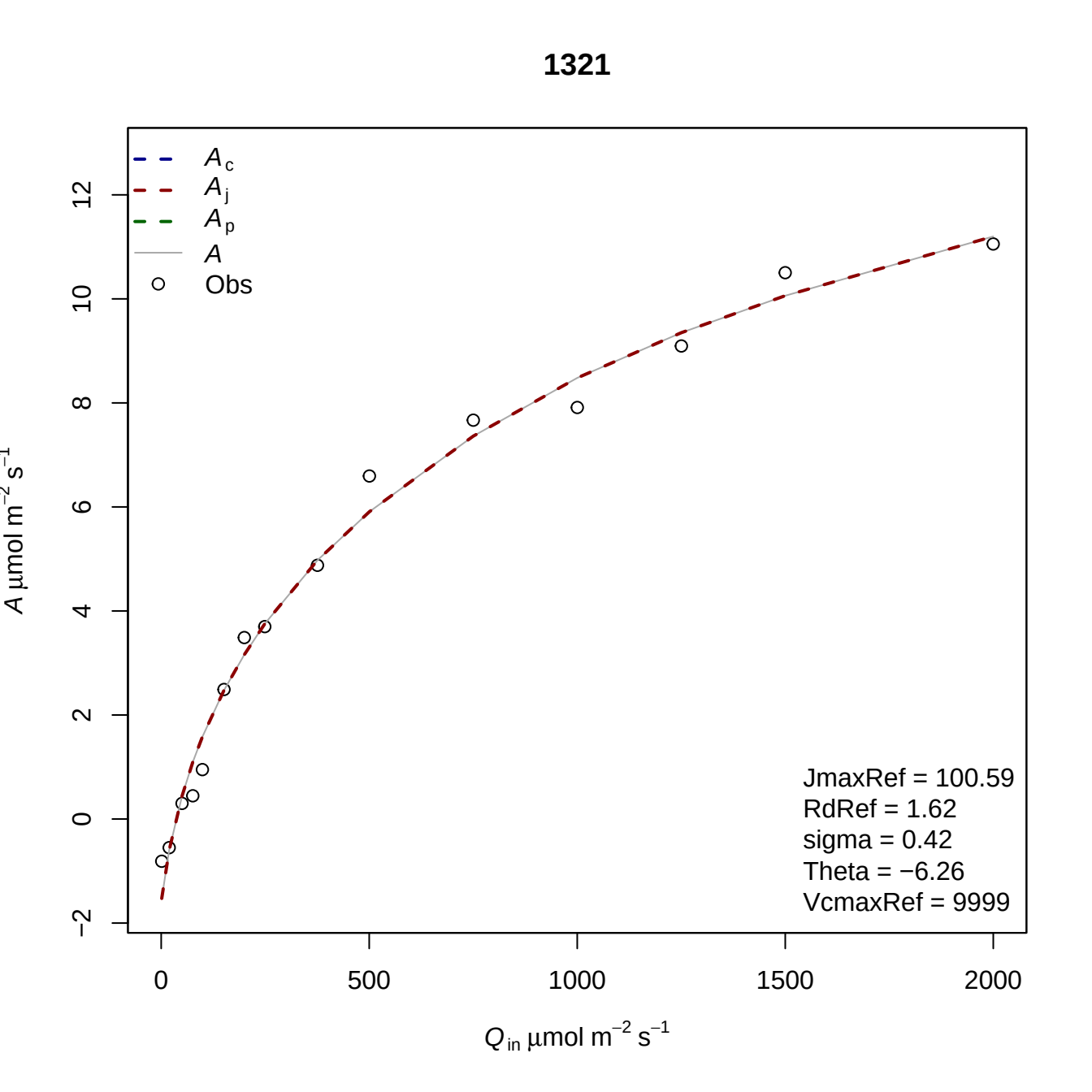

1316
A_c
A_j
A_p
A
Obs
JmaxRef = 106.95
RdRef = 2.58
sigma = 0.64
Theta = 0.14
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

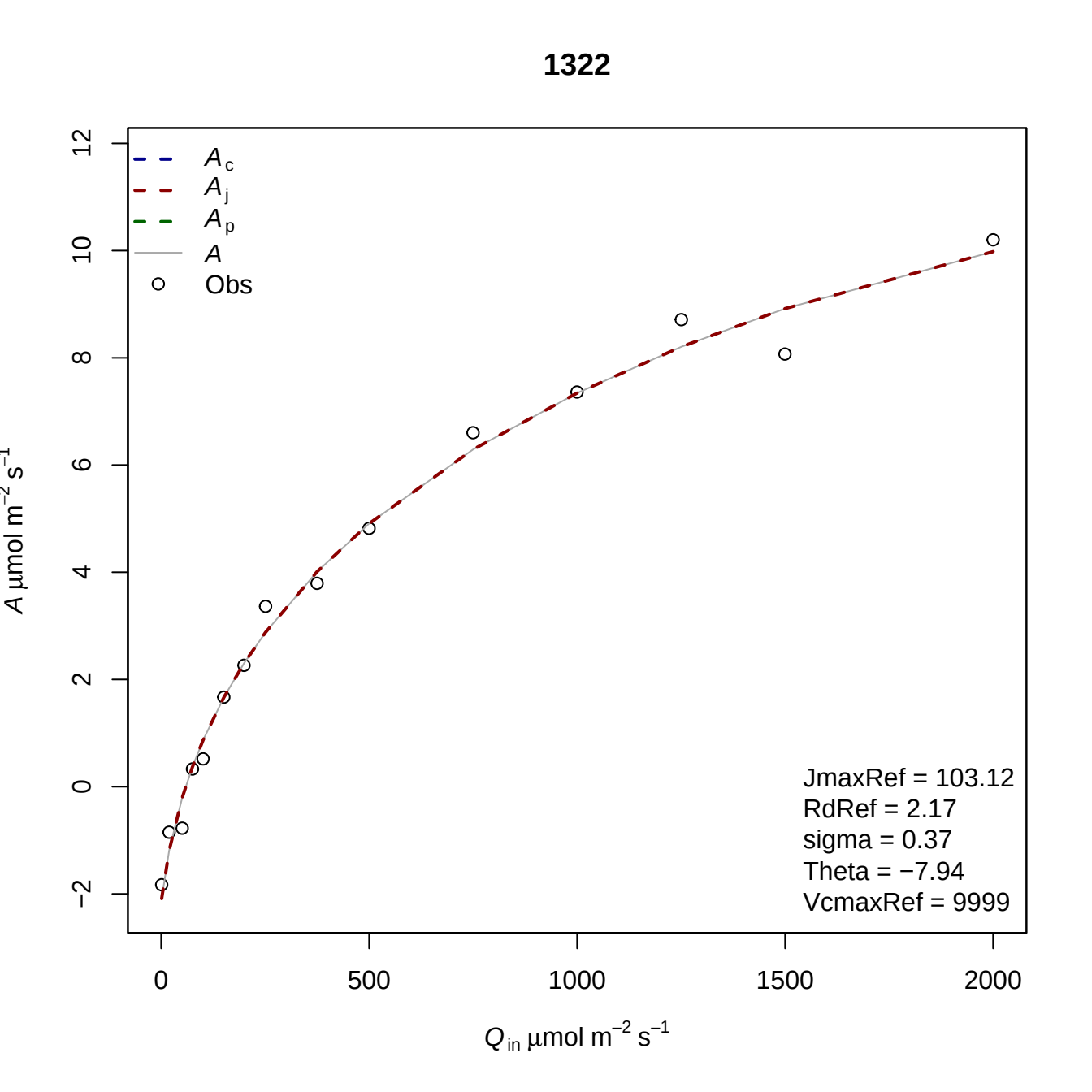

1317
A_c
A_j
A_p
A
Obs
JmaxRef = 108.15
RdRef = 2.42
sigma = 0.33
Theta = 0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

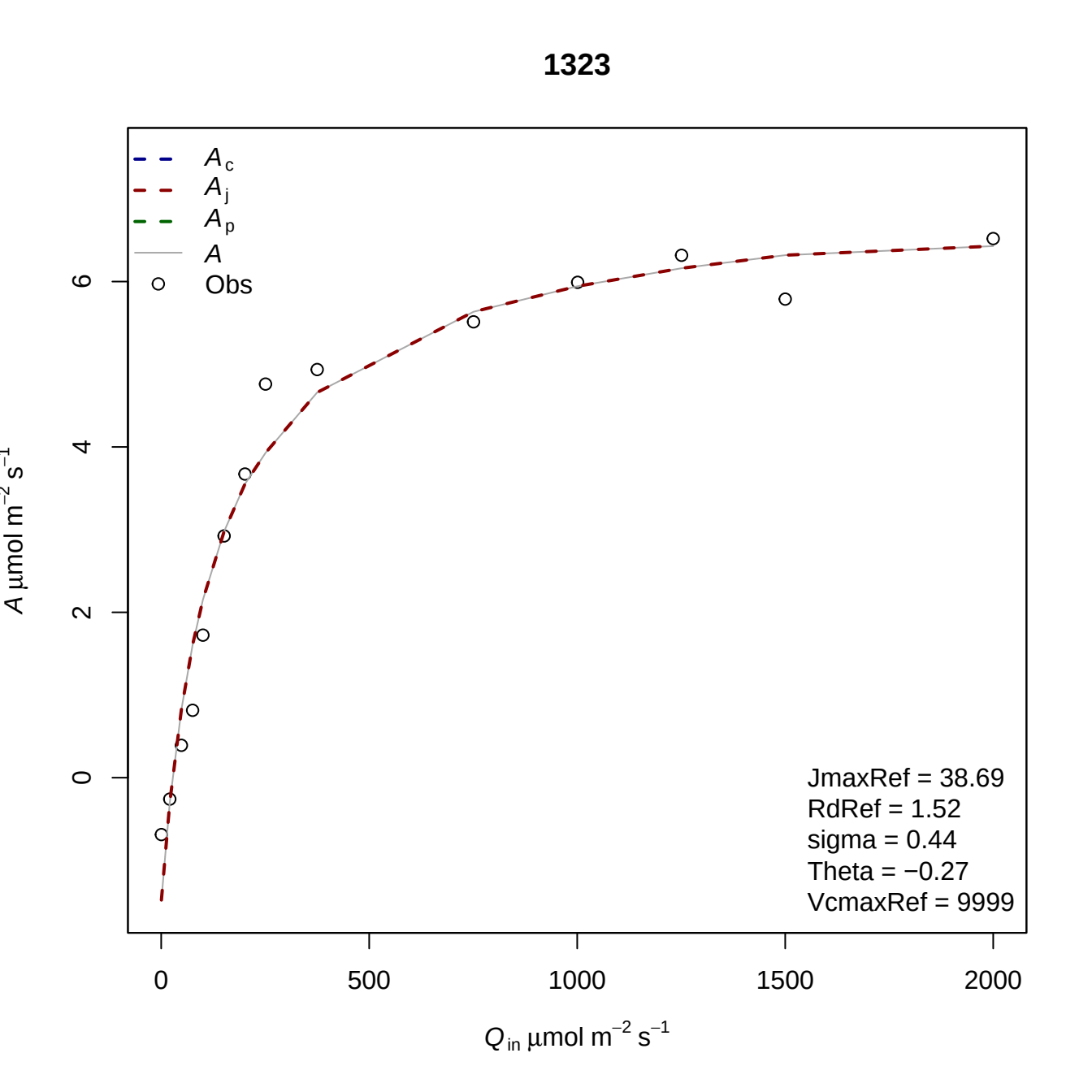

1318
A_c
A_j
A_p
A
Obs
JmaxRef = 88.3
RdRef = 1.93
sigma = 0.37
Theta = −0.5
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

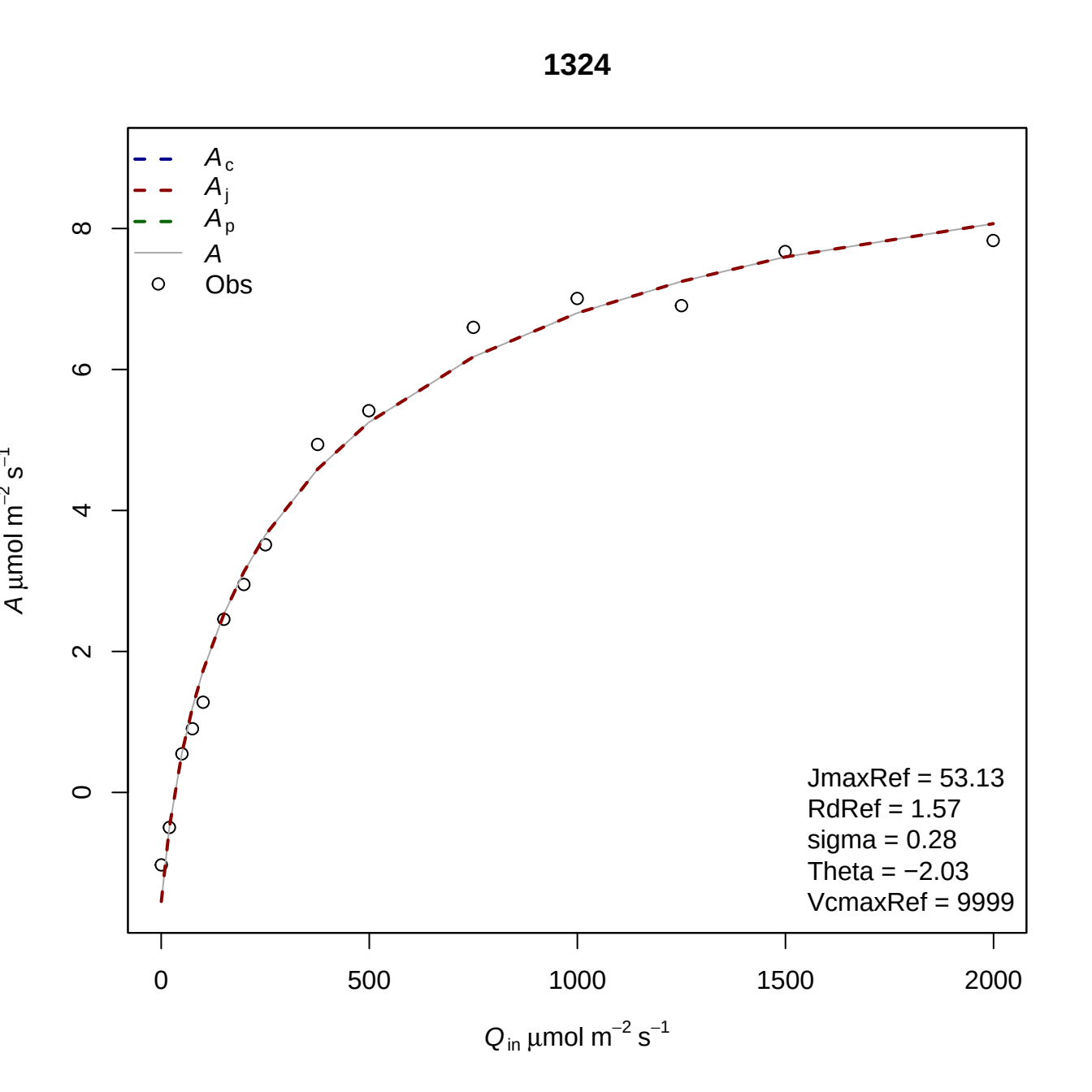

1319
A_c
A_j
A_p
A
Obs
JmaxRef = 94
RdRef = 2.68
sigma = 0.3
Theta = −0.36
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

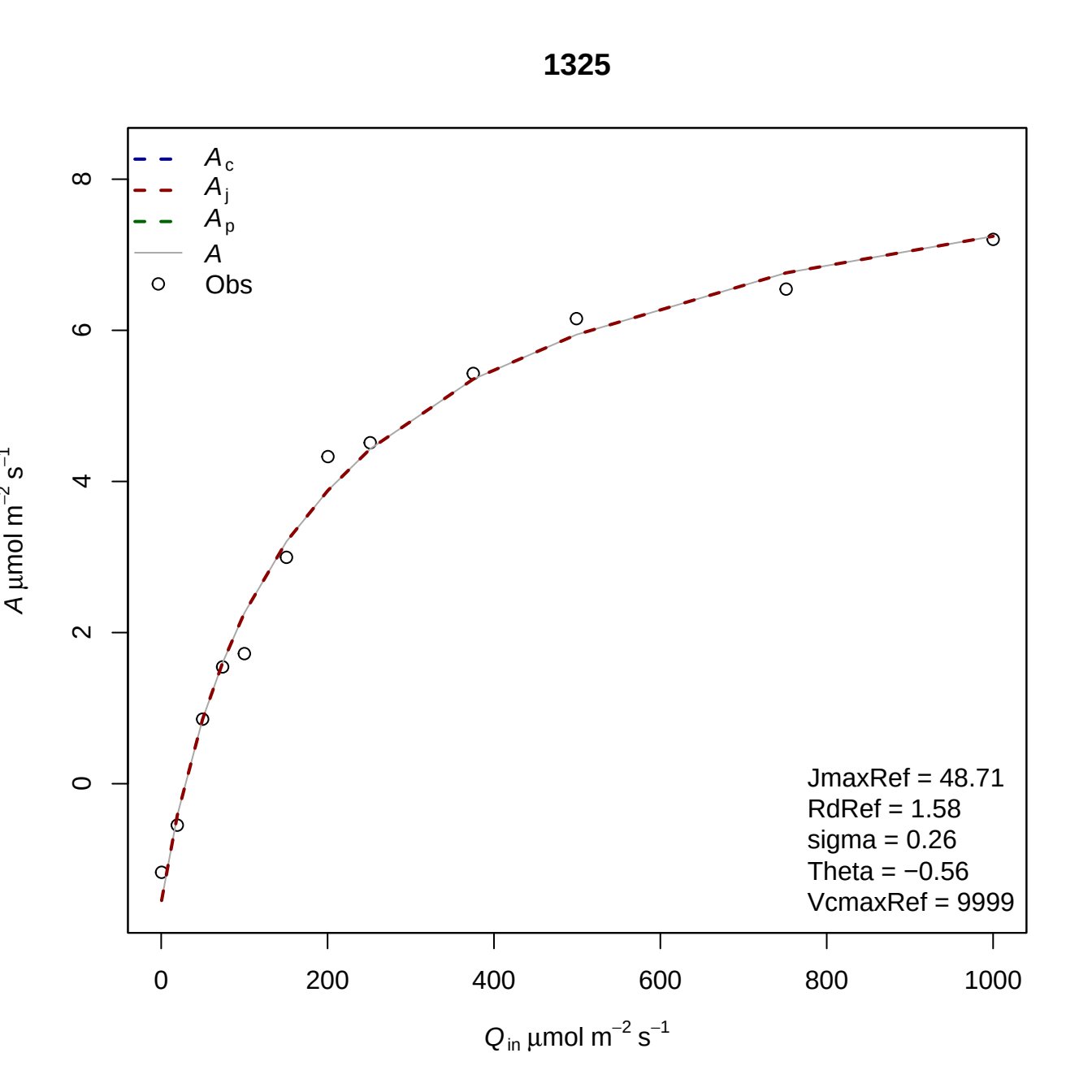

1320
A_c
A_j
A_p
A
Obs
JmaxRef = 77.8
RdRef = 2.55
sigma = 0.34
Theta = 0.23
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

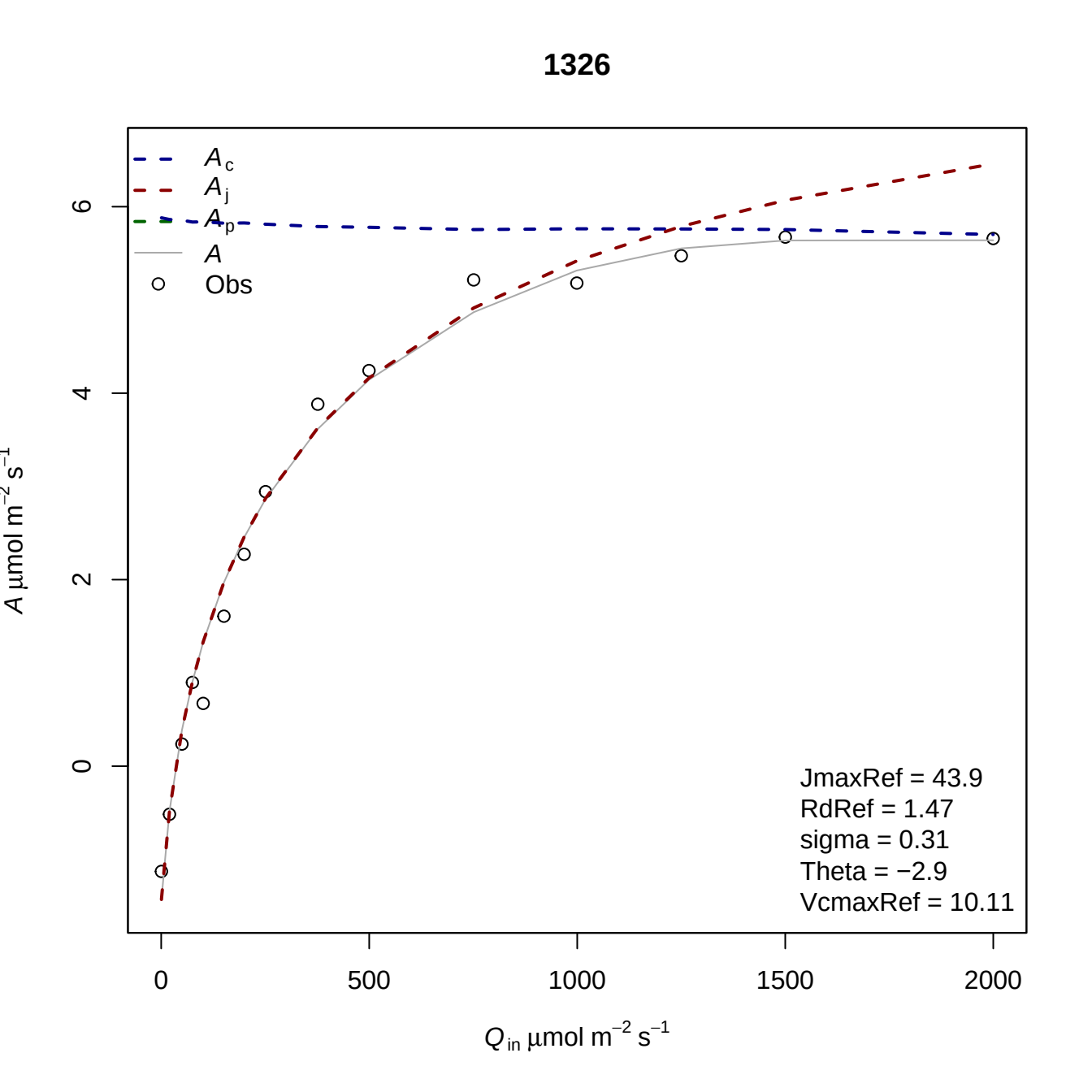

1321
A_c
A_j
A_p
A
Obs
JmaxRef = 100.59
RdRef = 1.62
sigma = 0.42
Theta = −6.26
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

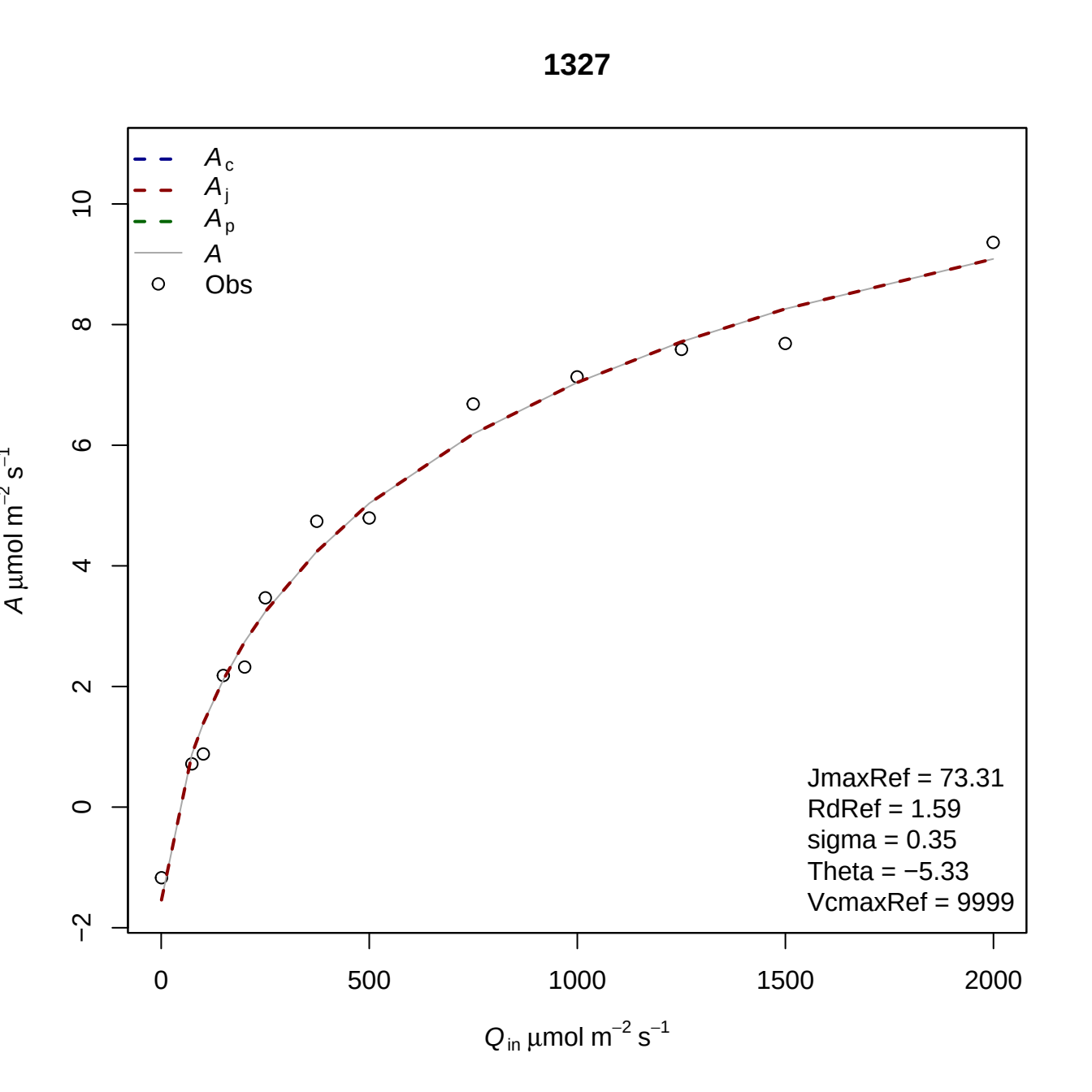

1322
A_c
A_j
A_p
A
Obs
JmaxRef = 103.12
RdRef = 2.17
sigma = 0.37
Theta = −7.94
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

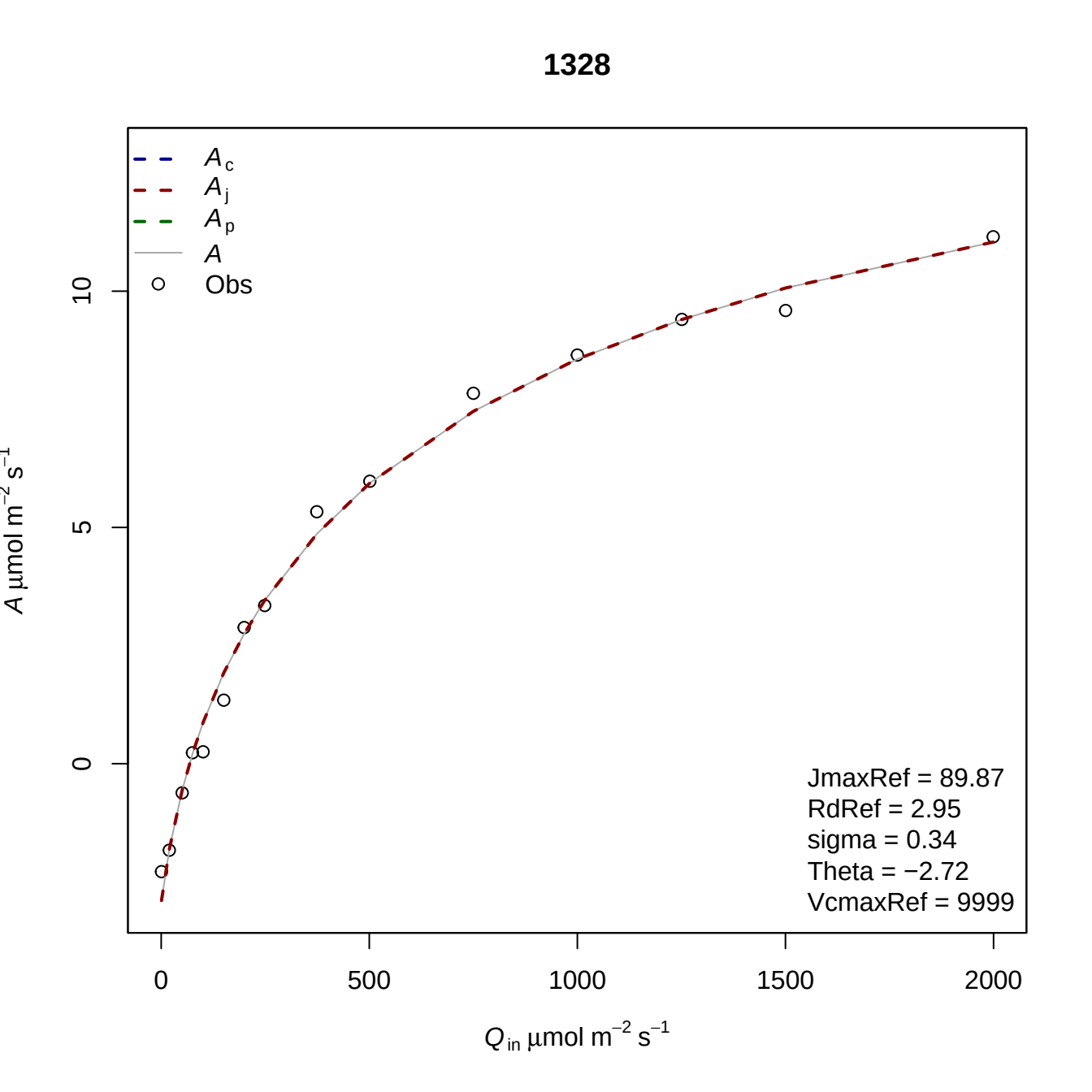

1323
A_c
A_j
A_p
A
Obs
JmaxRef = 38.69
RdRef = 1.52
sigma = 0.44
Theta = −0.27
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

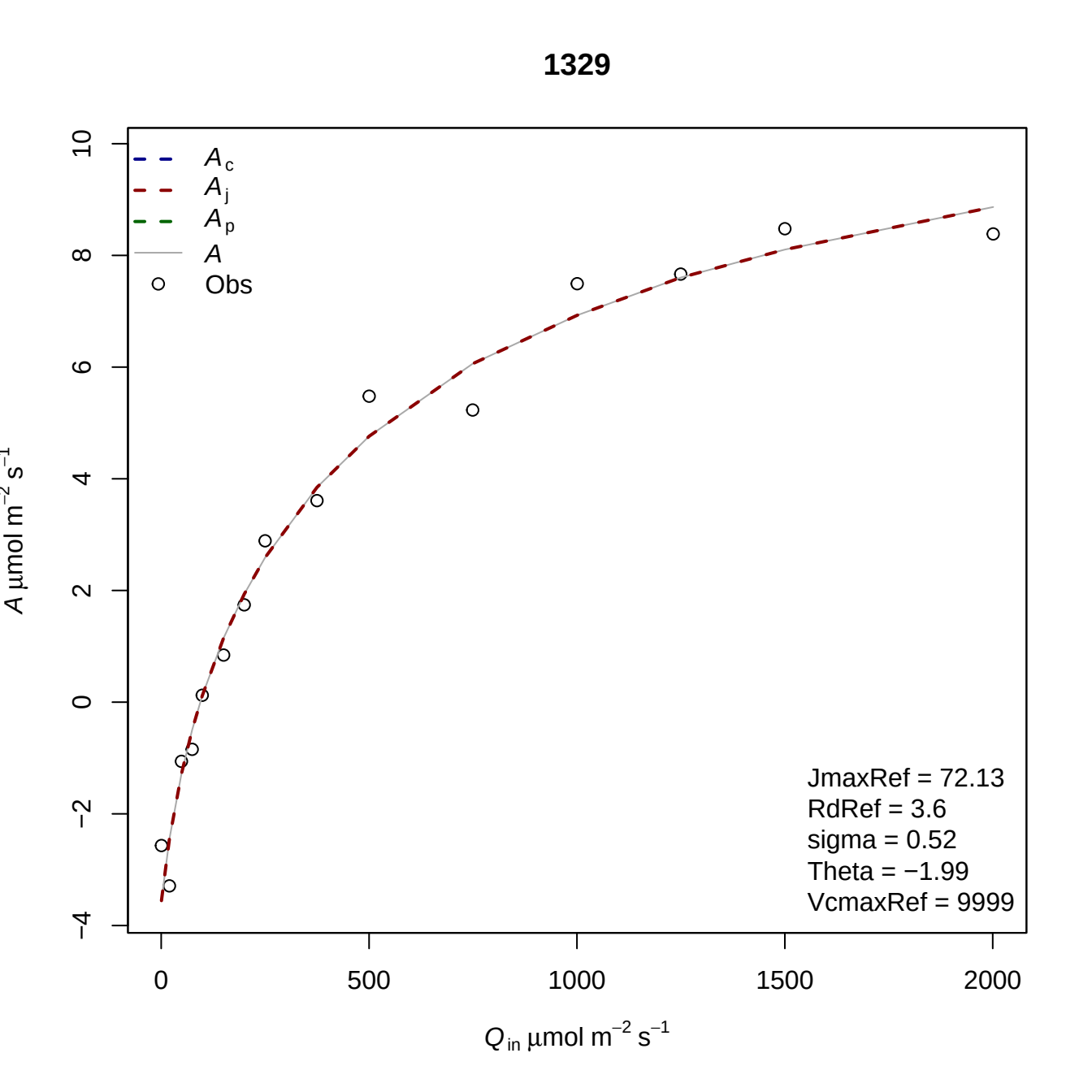

1324
A_c
A_j
A_p
A
Obs
JmaxRef = 53.13
RdRef = 1.57
sigma = 0.28
Theta = −2.03
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

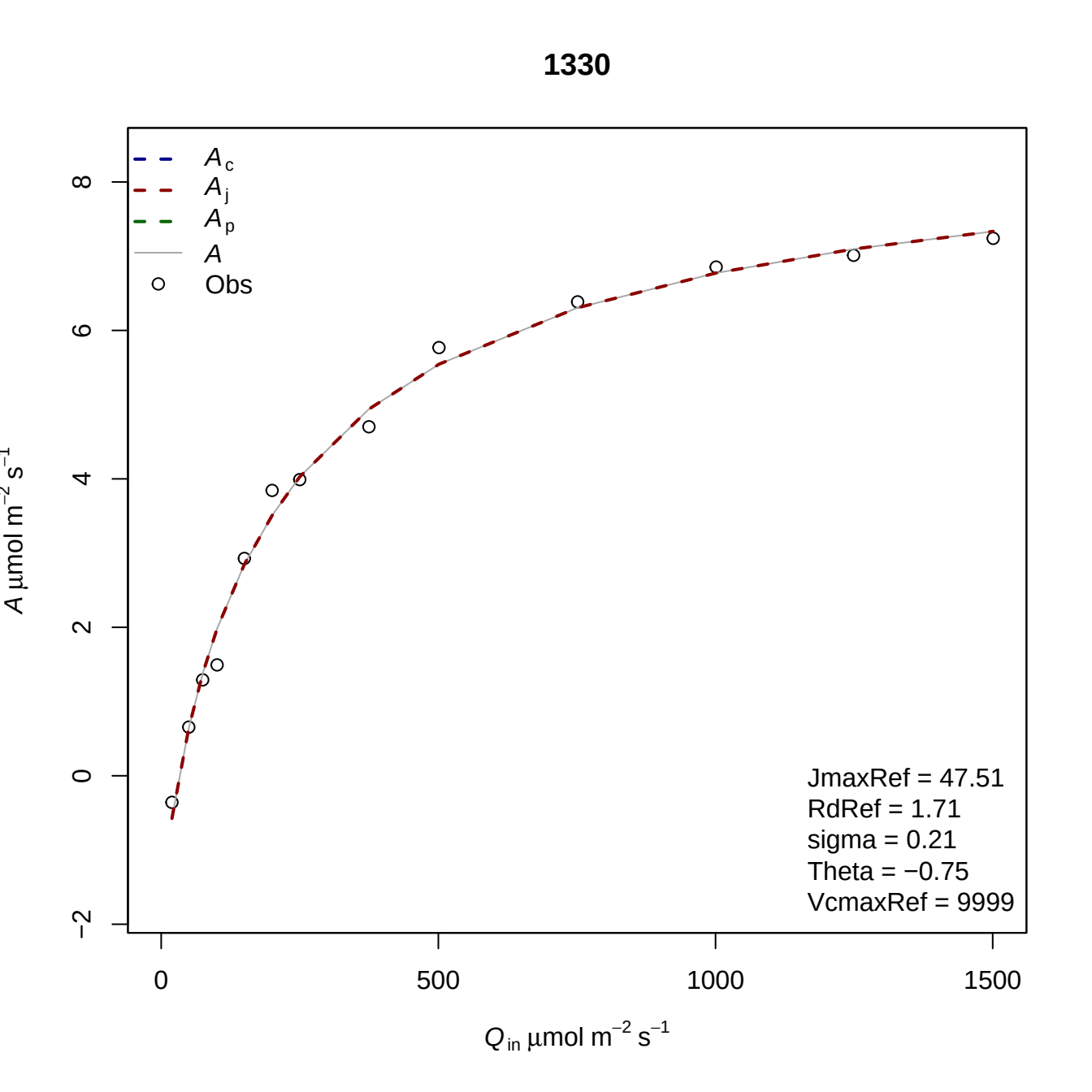

1325
A_c
A_j
A_p
A
Obs
JmaxRef = 48.71
RdRef = 1.58
sigma = 0.26
Theta = −0.56
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
200
400
600
800
1000

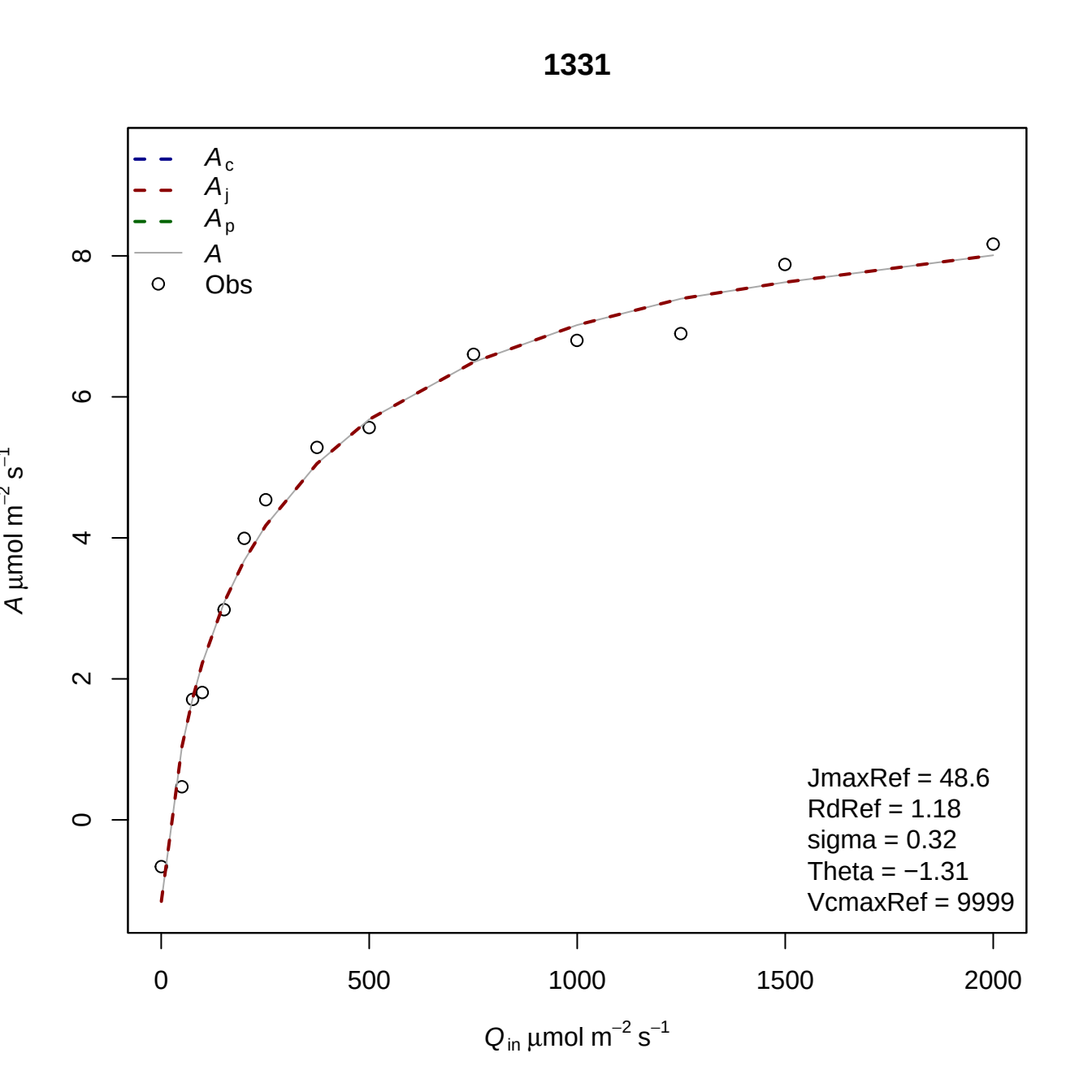

1326
A_c
A_j
A_p
A
Obs
JmaxRef = 43.9
RdRef = 1.47
sigma = 0.31
Theta = −2.9
VcmaxRef = 10.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

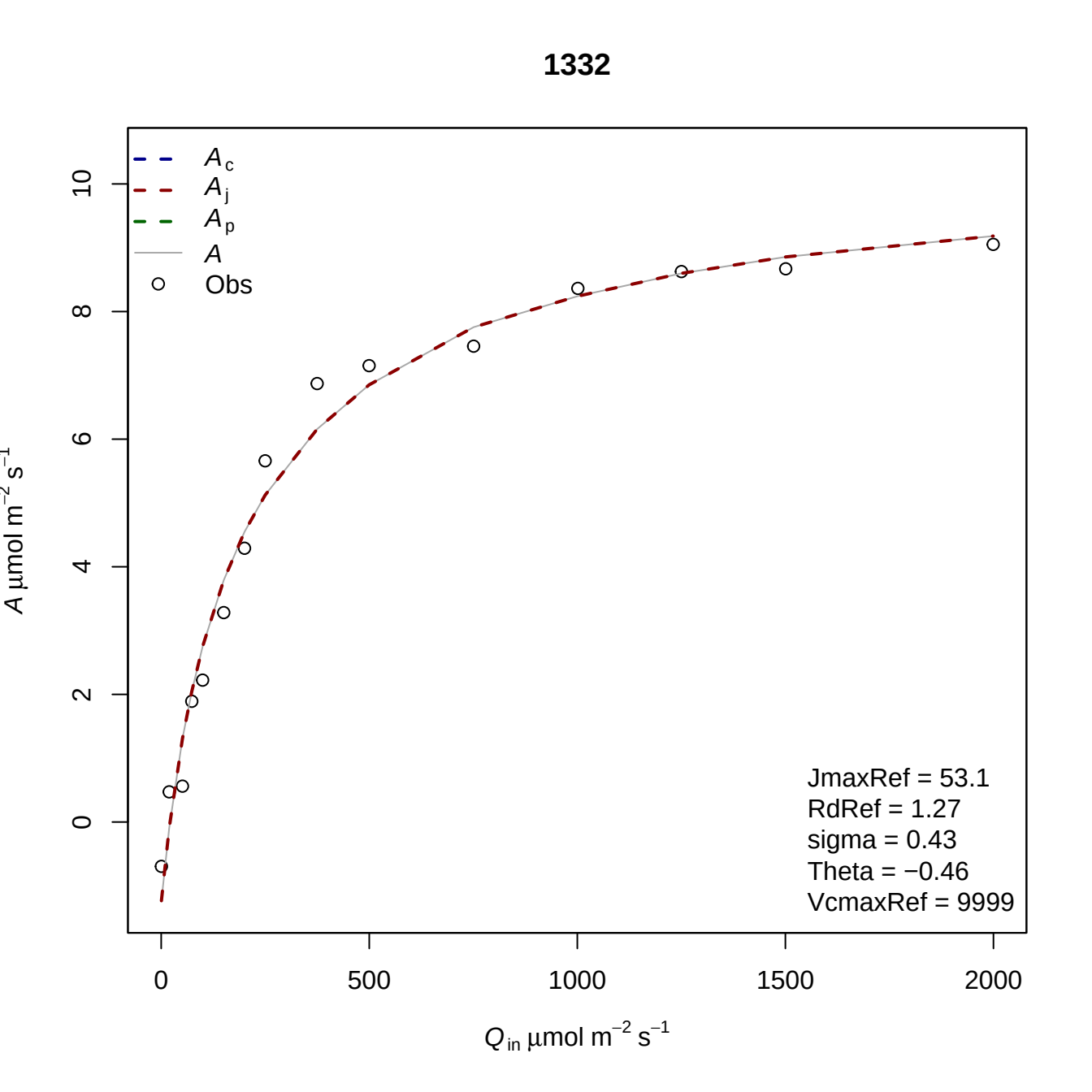

1327
A_c
A_j
A_p
A
Obs
JmaxRef = 73.31
RdRef = 1.59
sigma = 0.35
Theta = −5.33
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

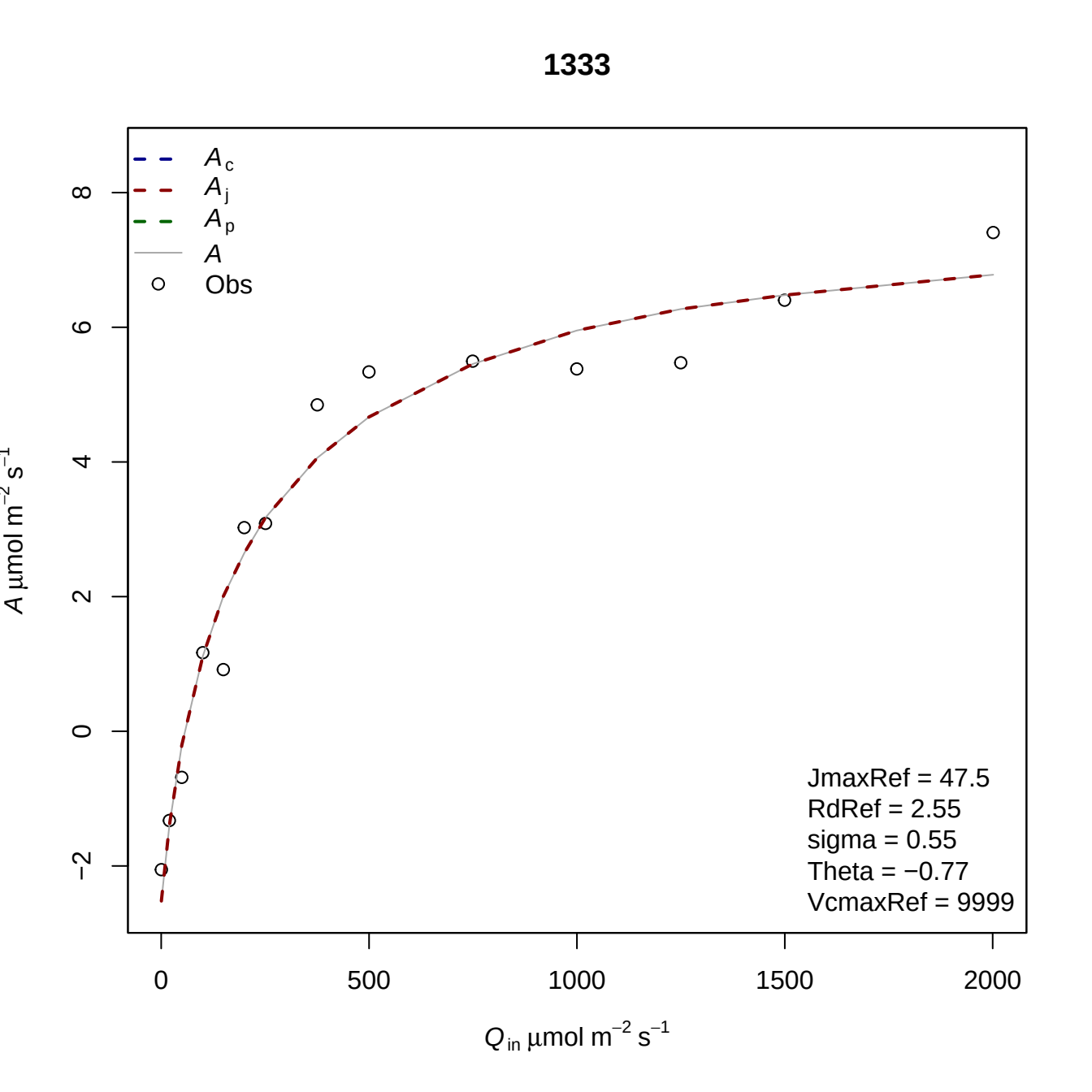

1328
A_c
A_j
A_p
A
Obs
JmaxRef = 89.87
RdRef = 2.95
sigma = 0.34
Theta = −2.72
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

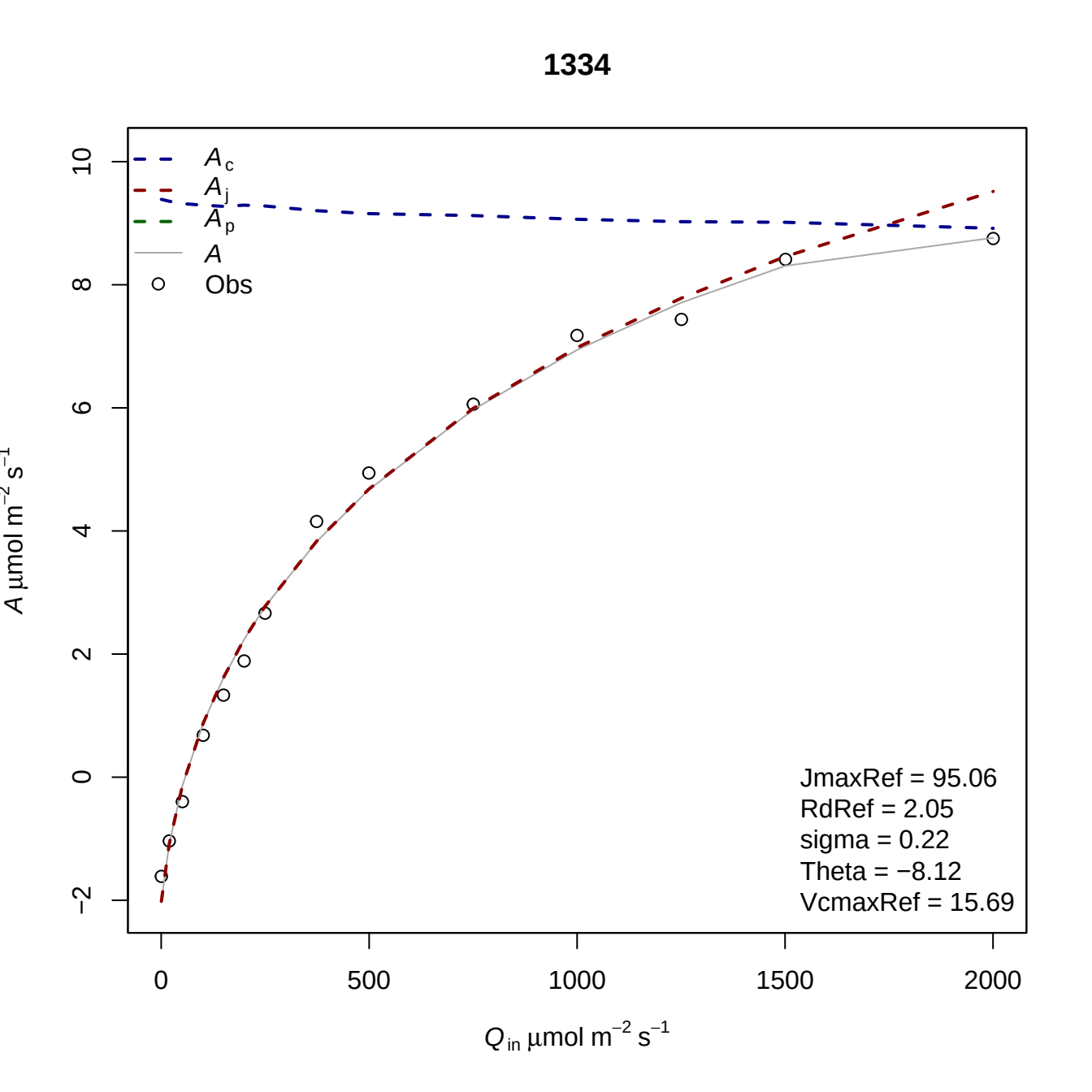

1329
A_c
A_j
A_p
A
Obs
JmaxRef = 72.13
RdRef = 3.6
sigma = 0.52
Theta = −1.99
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

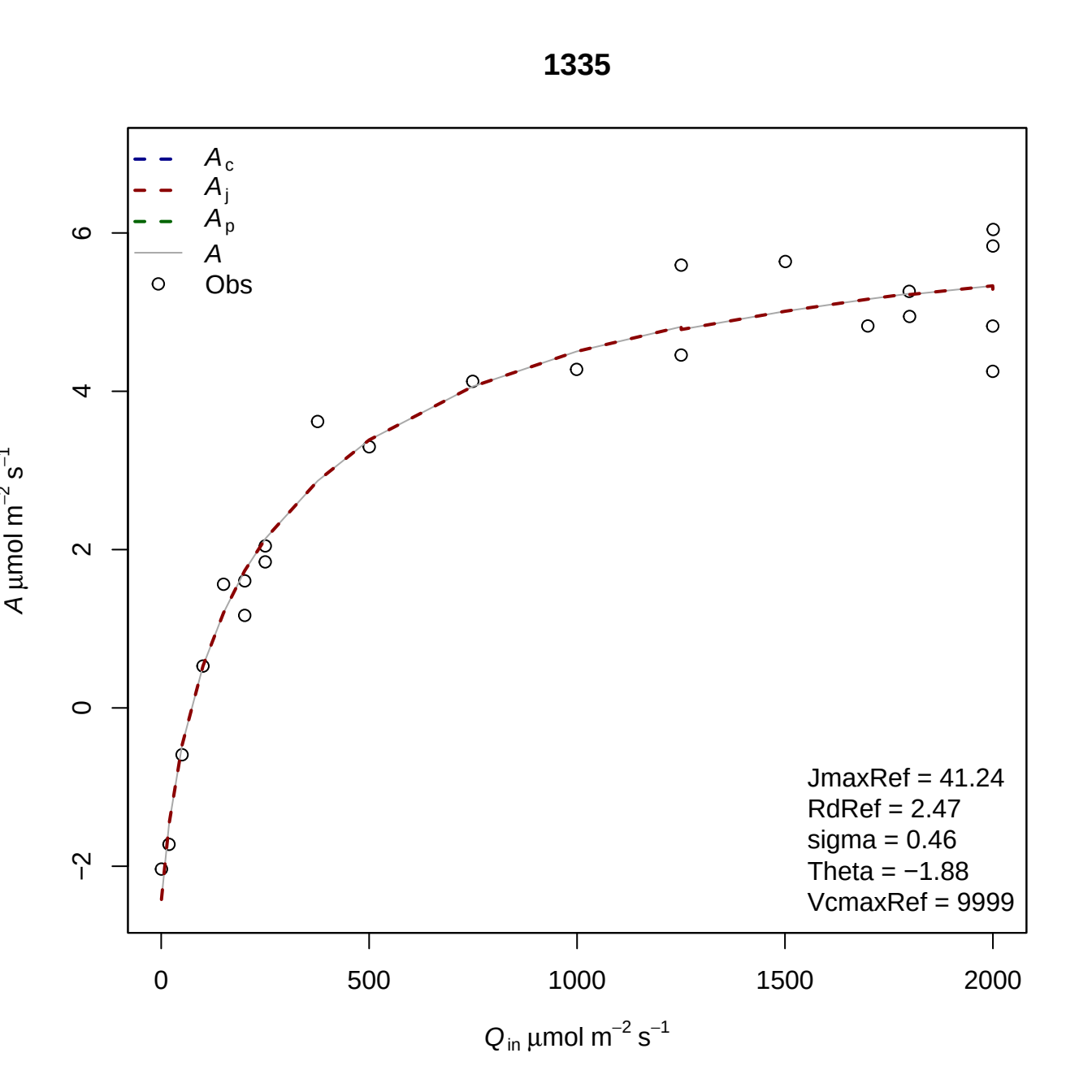

1330
A_c
A_j
A_p
A
Obs
JmaxRef = 47.51
RdRef = 1.71
sigma = 0.21
Theta = −0.75
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

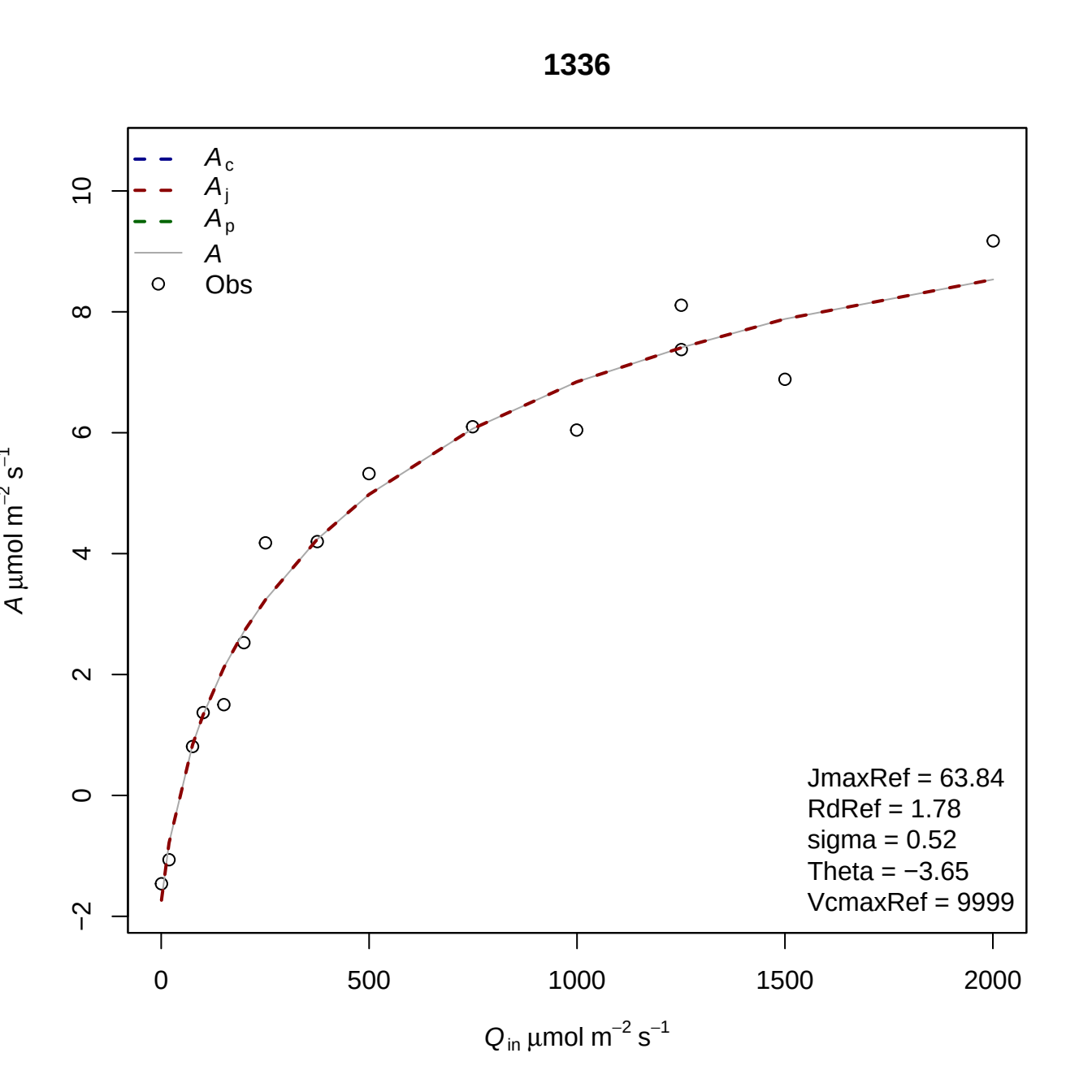

1331
A_c
A_j
A_p
A
Obs
JmaxRef = 48.6
RdRef = 1.18
sigma = 0.32
Theta = −1.31
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

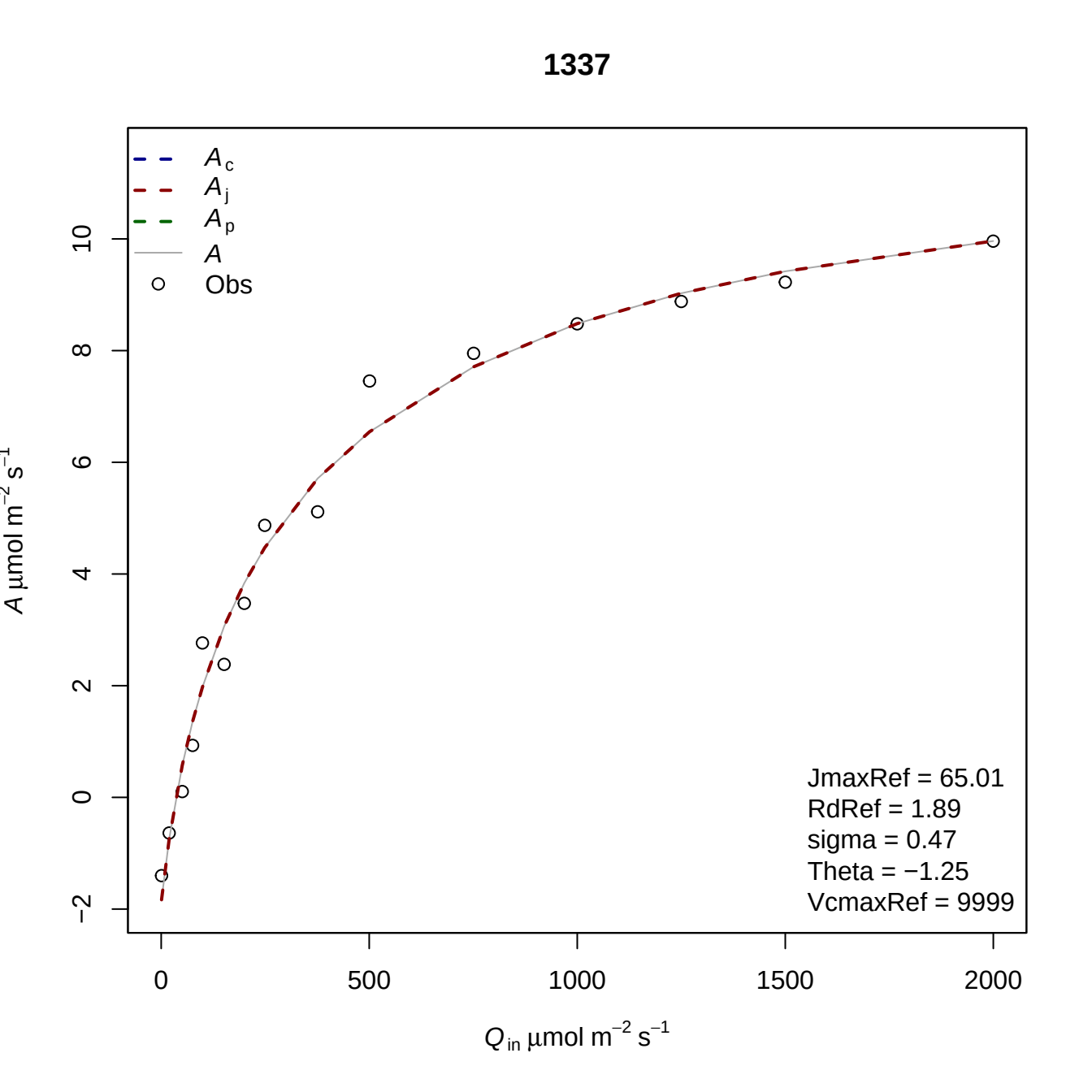

1332
A_c
A_j
A_p
A
Obs
JmaxRef = 53.1
RdRef = 1.27
sigma = 0.43
Theta = −0.46
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

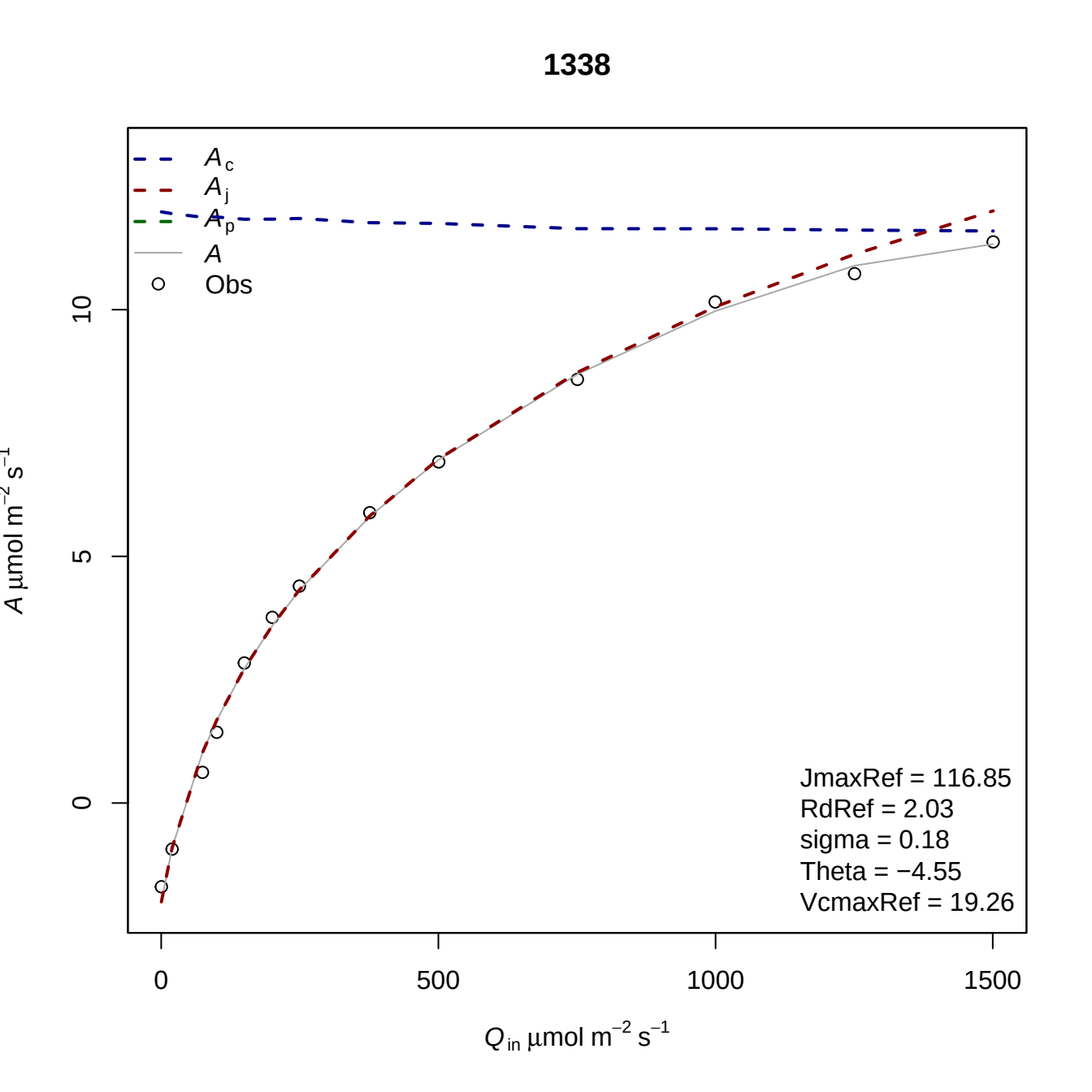

1333
A_c
A_j
A_p
A
Obs
JmaxRef = 47.5
RdRef = 2.55
sigma = 0.55
Theta = −0.77
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

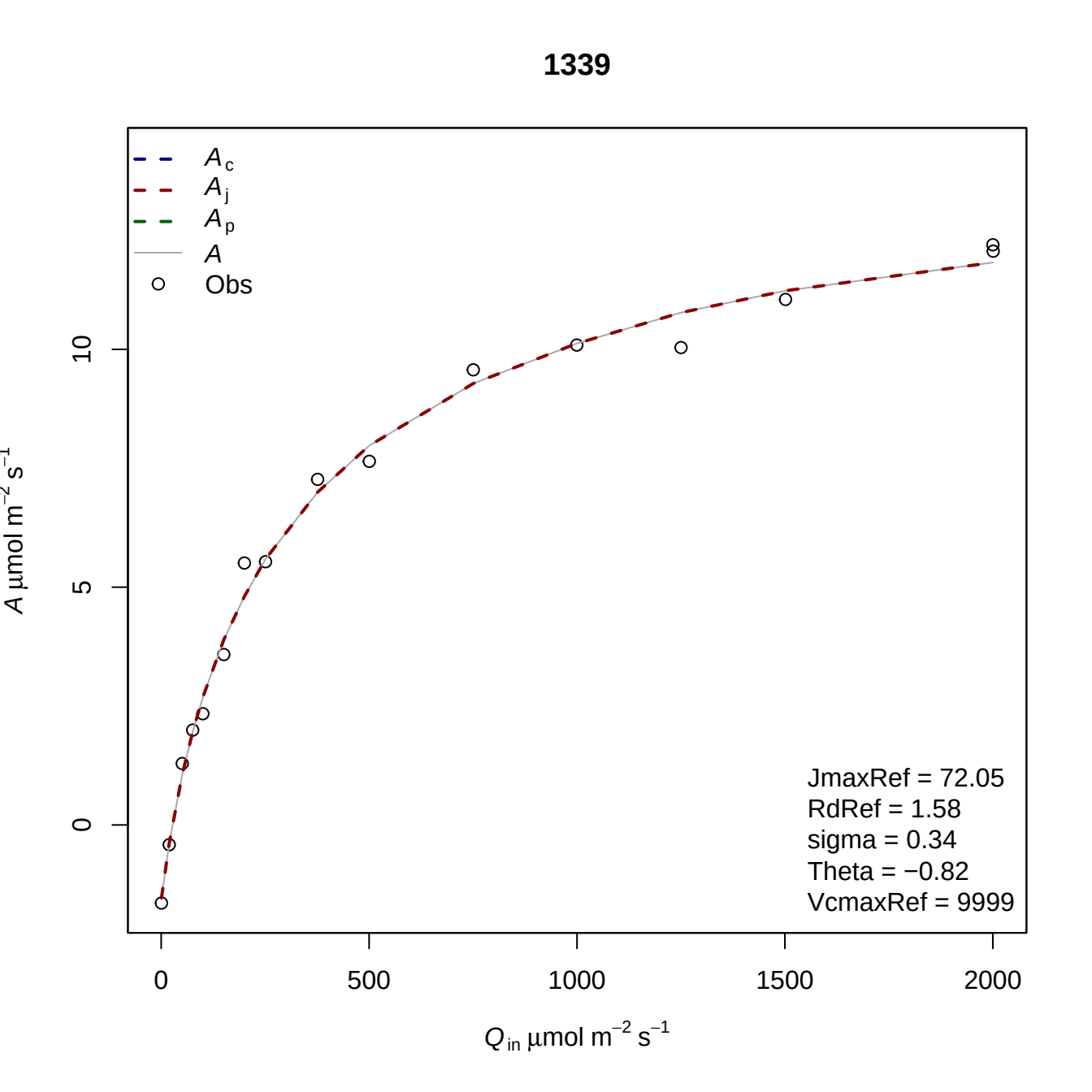

1334
A_c
A_j
A_p
A
Obs
JmaxRef = 95.06
RdRef = 2.05
sigma = 0.22
Theta = −8.12
VcmaxRef = 15.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

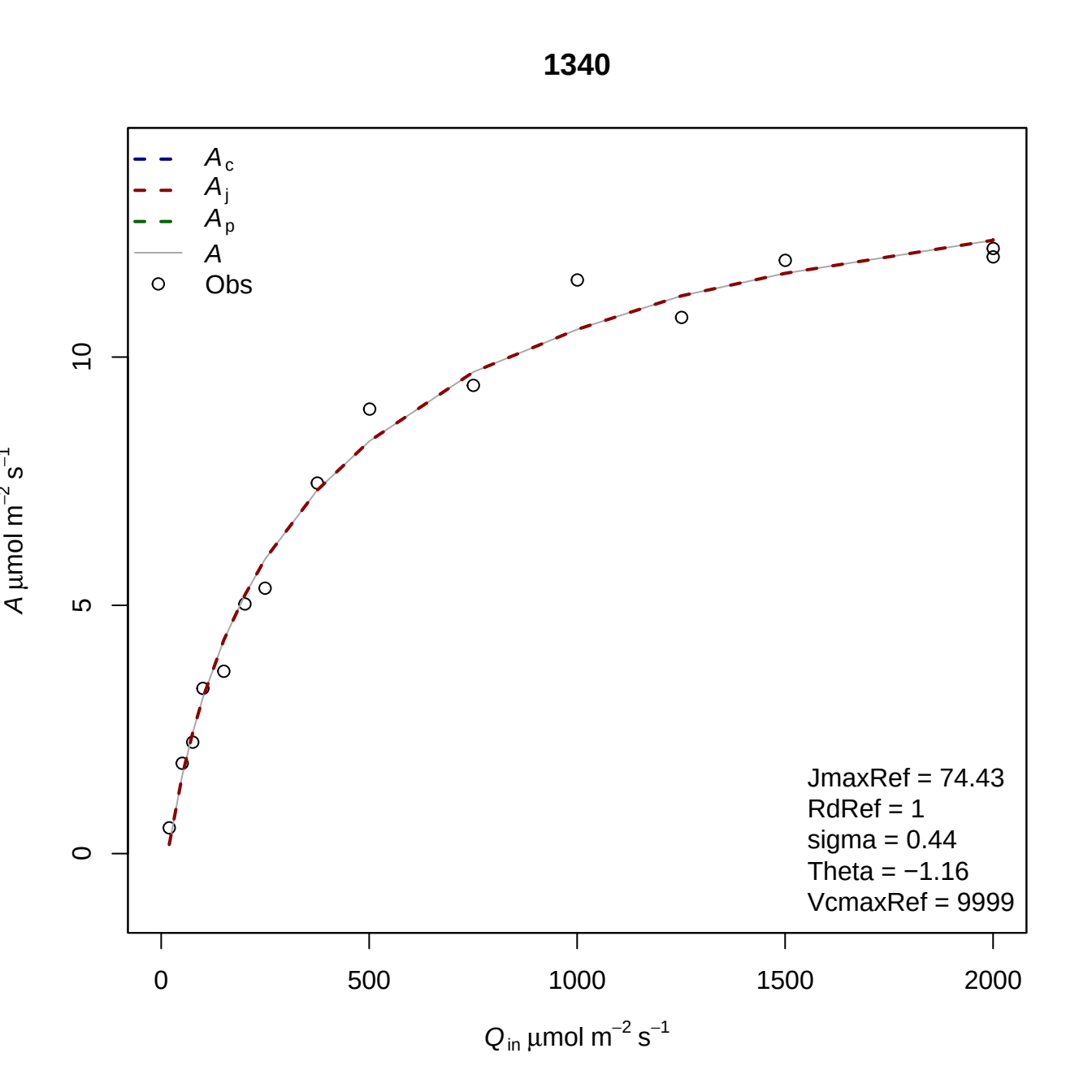

1335
A_c
A_j
A_p
A
Obs
JmaxRef = 41.24
RdRef = 2.47
sigma = 0.46
Theta = −1.88
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

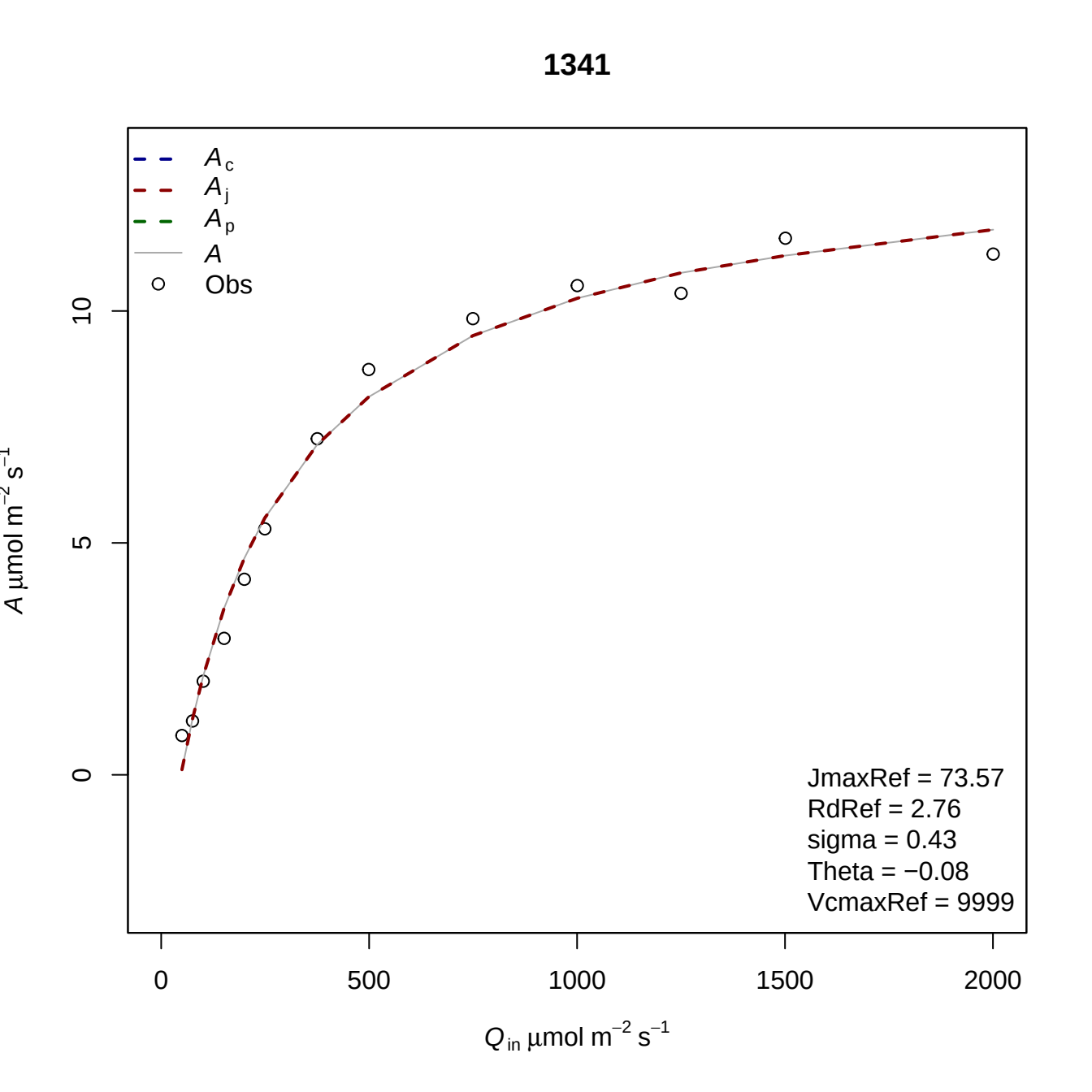

1336
A_c
A_j
A_p
A
Obs
JmaxRef = 63.84
RdRef = 1.78
sigma = 0.52
Theta = −3.65
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

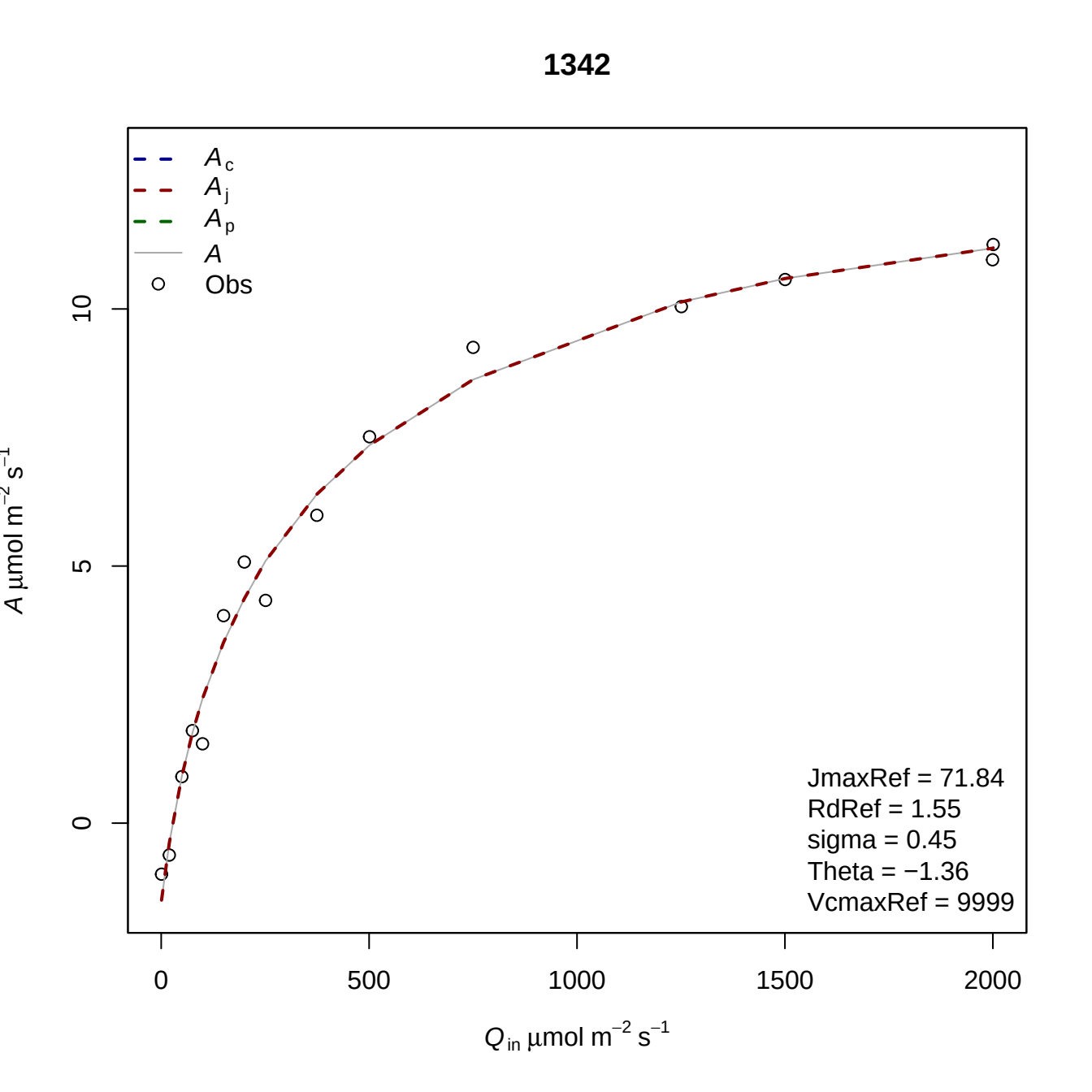

1337
A_c
A_j
A_p
A
Obs
JmaxRef = 65.01
RdRef = 1.89
sigma = 0.47
Theta = −1.25
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

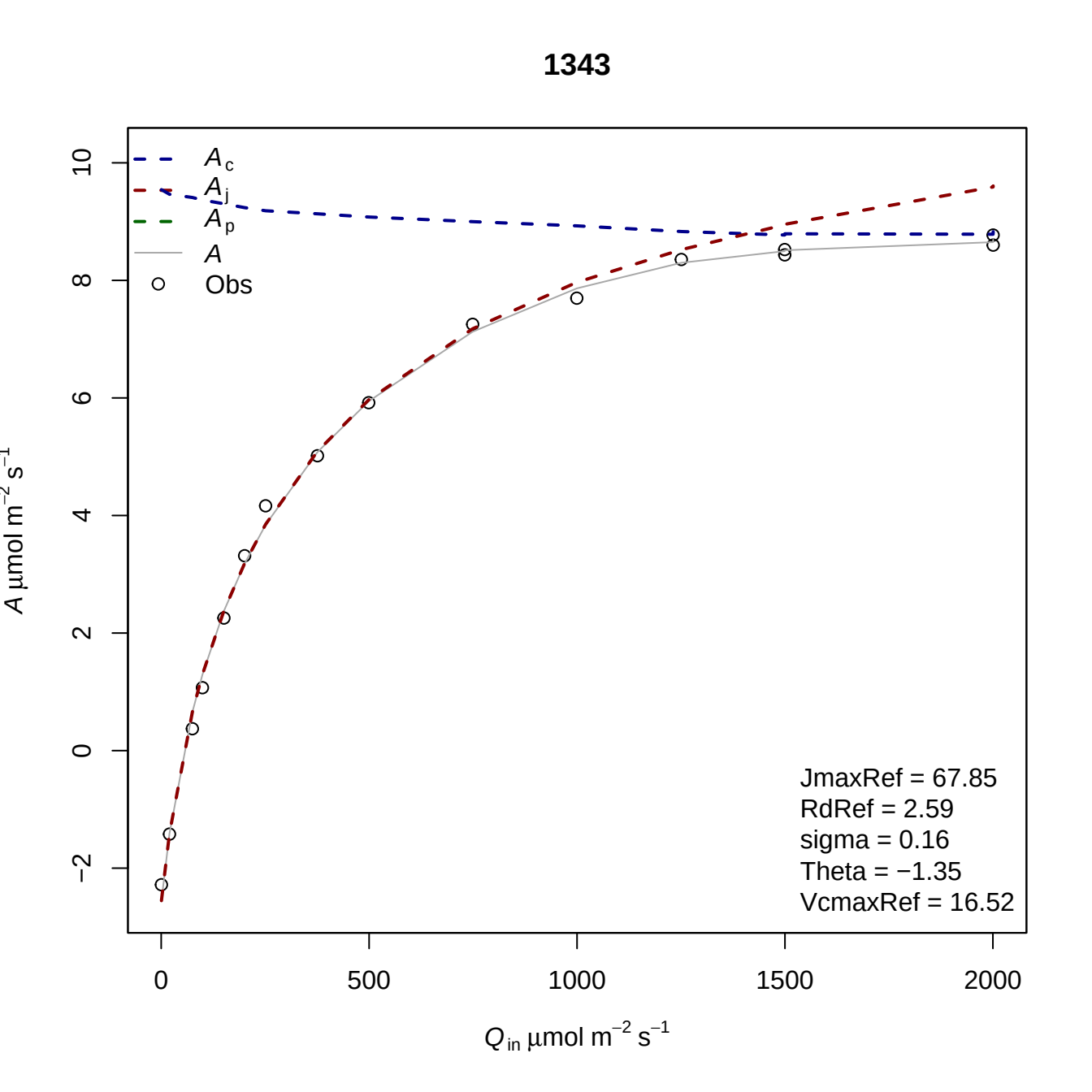

1338
A_c
A_j
A_p
A
Obs
JmaxRef = 116.85
RdRef = 2.03
sigma = 0.18
Theta = −4.55
VcmaxRef = 19.26
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

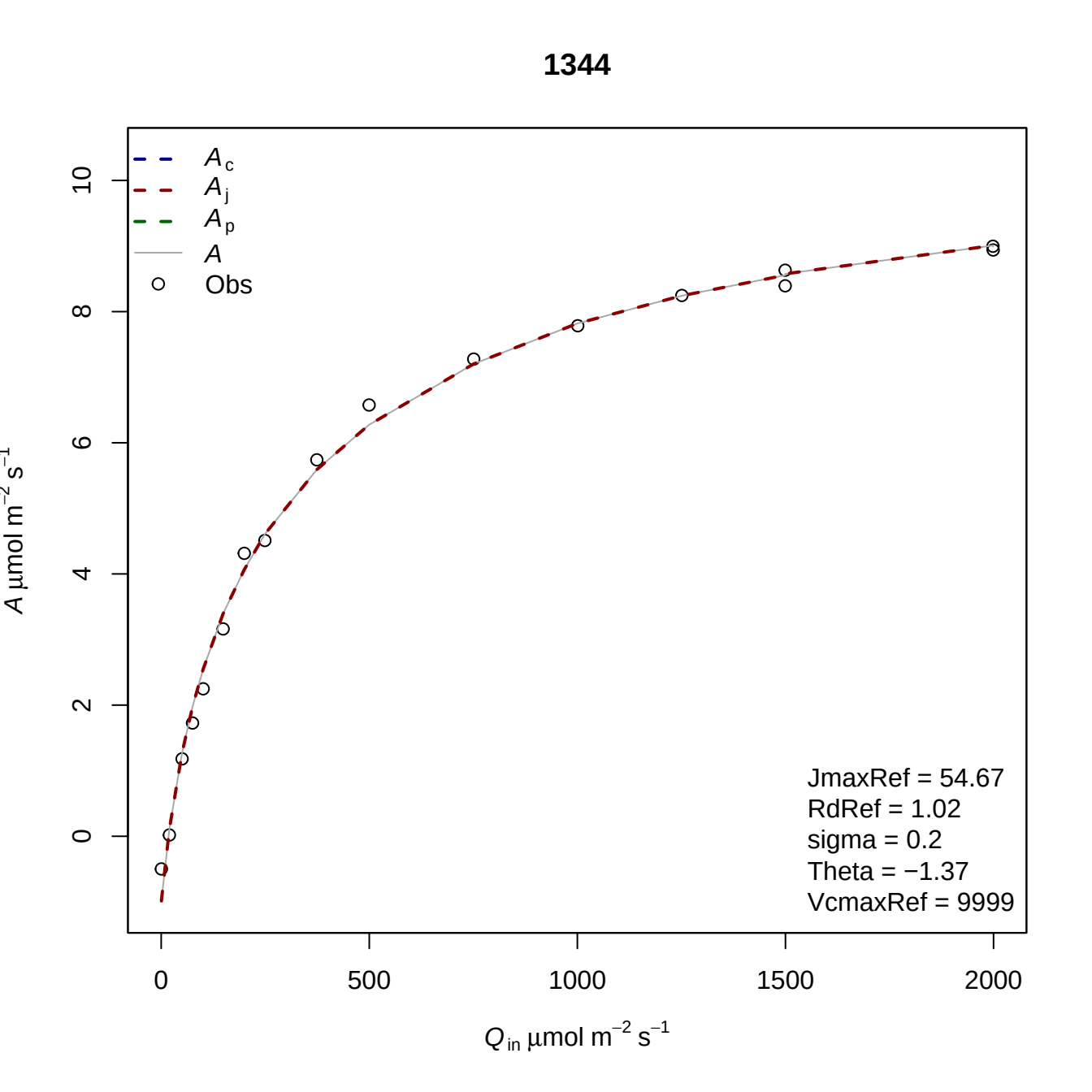

1339
A_c
A_j
A_p
A
Obs
JmaxRef = 72.05
RdRef = 1.58
sigma = 0.34
Theta = −0.82
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

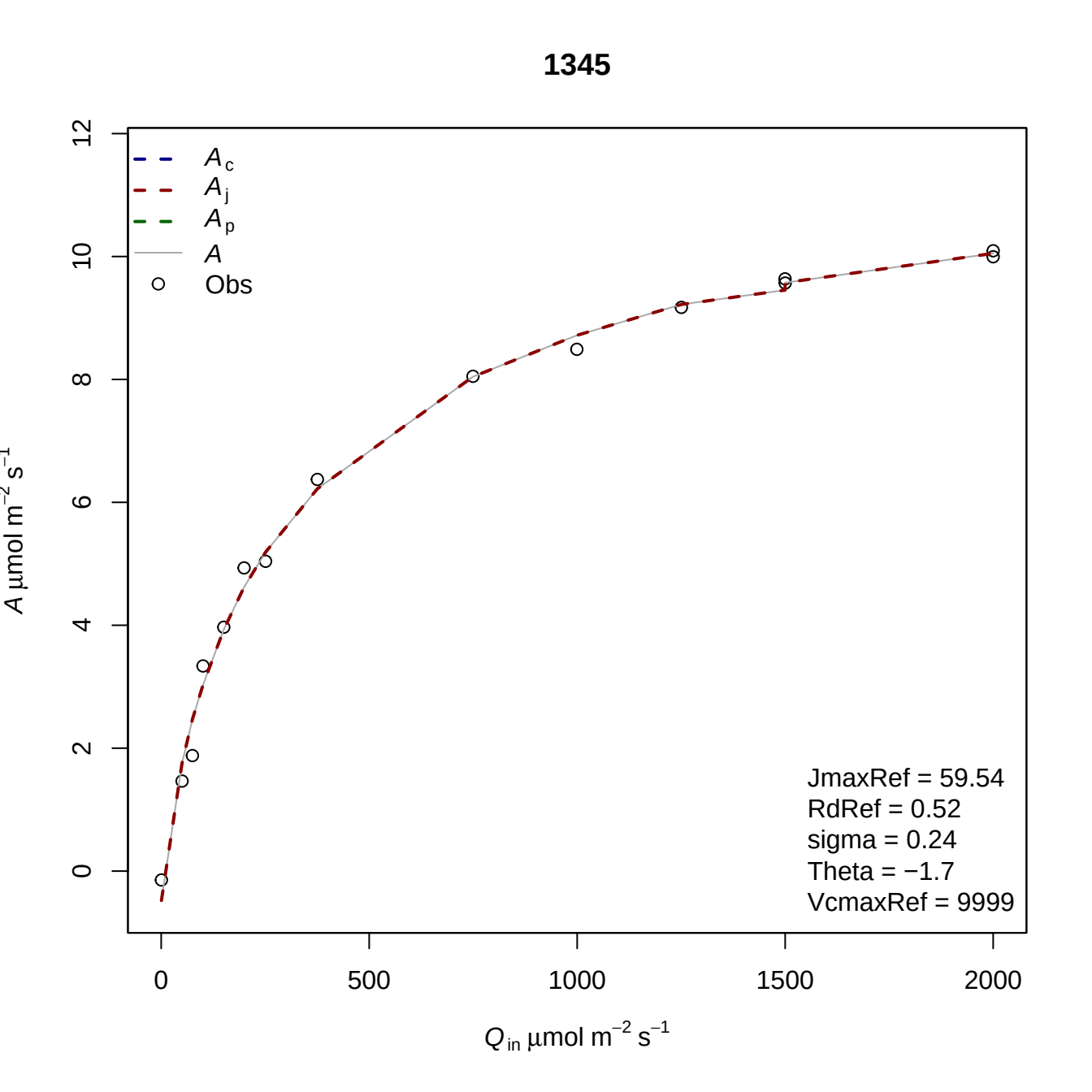

1340
A_c
A_j
A_p
A
Obs
JmaxRef = 74.43
RdRef = 1
sigma = 0.44
Theta = −1.16
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

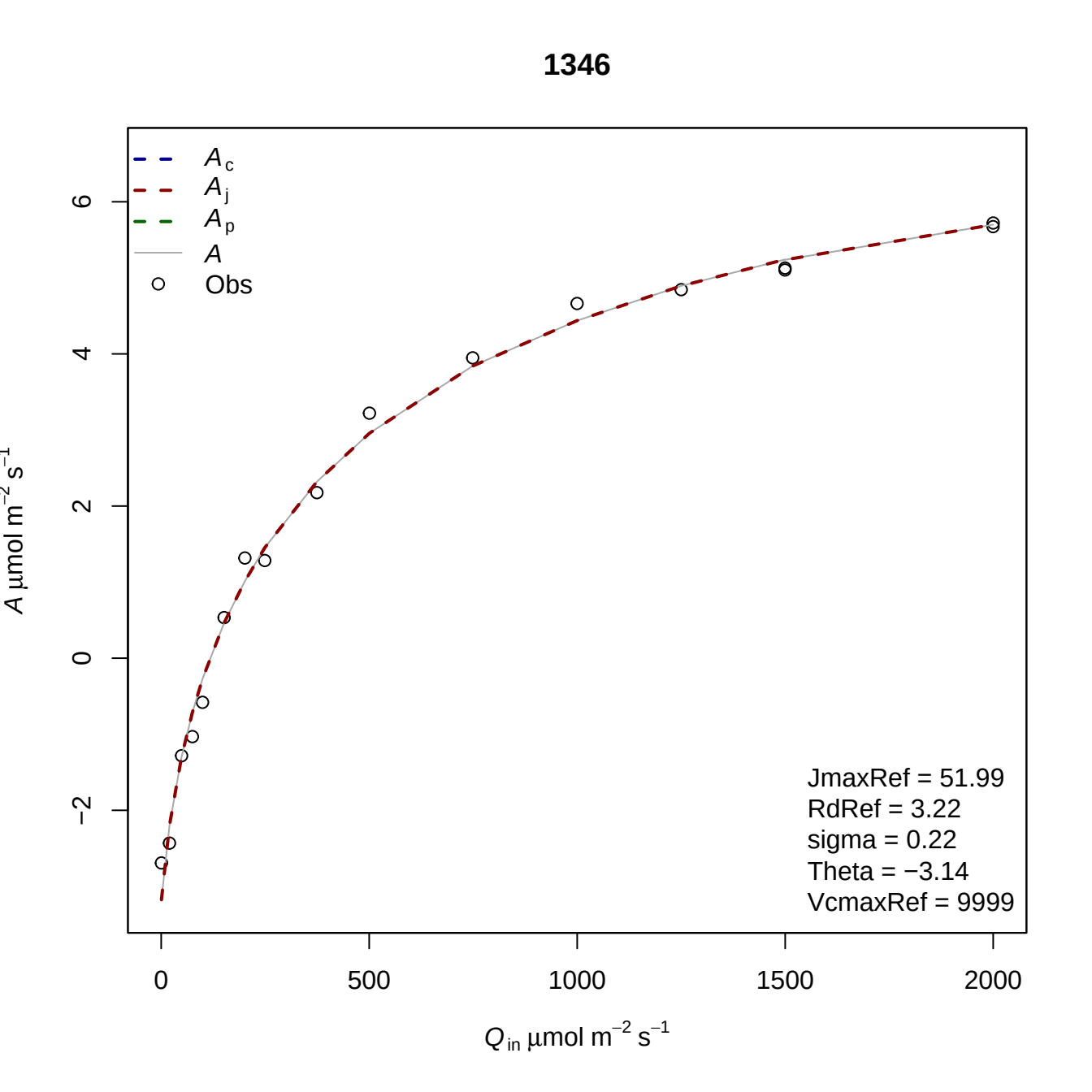

1341
A_c
A_j
A_p
A
Obs
JmaxRef = 73.57
RdRef = 2.76
sigma = 0.43
Theta = −0.08
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

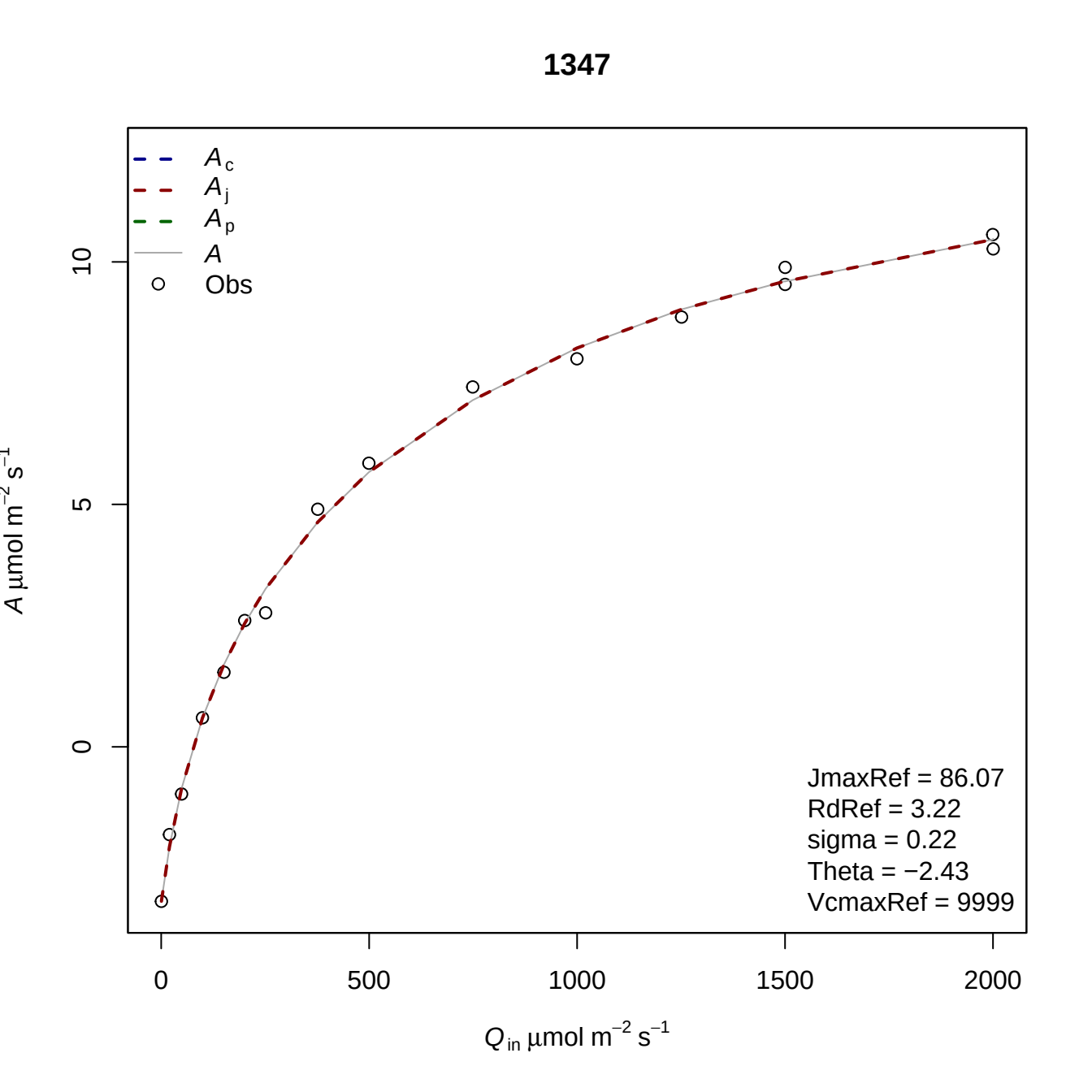

1342
A_c
A_j
A_p
A
Obs
JmaxRef = 71.84
RdRef = 1.55
sigma = 0.45
Theta = −1.36
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

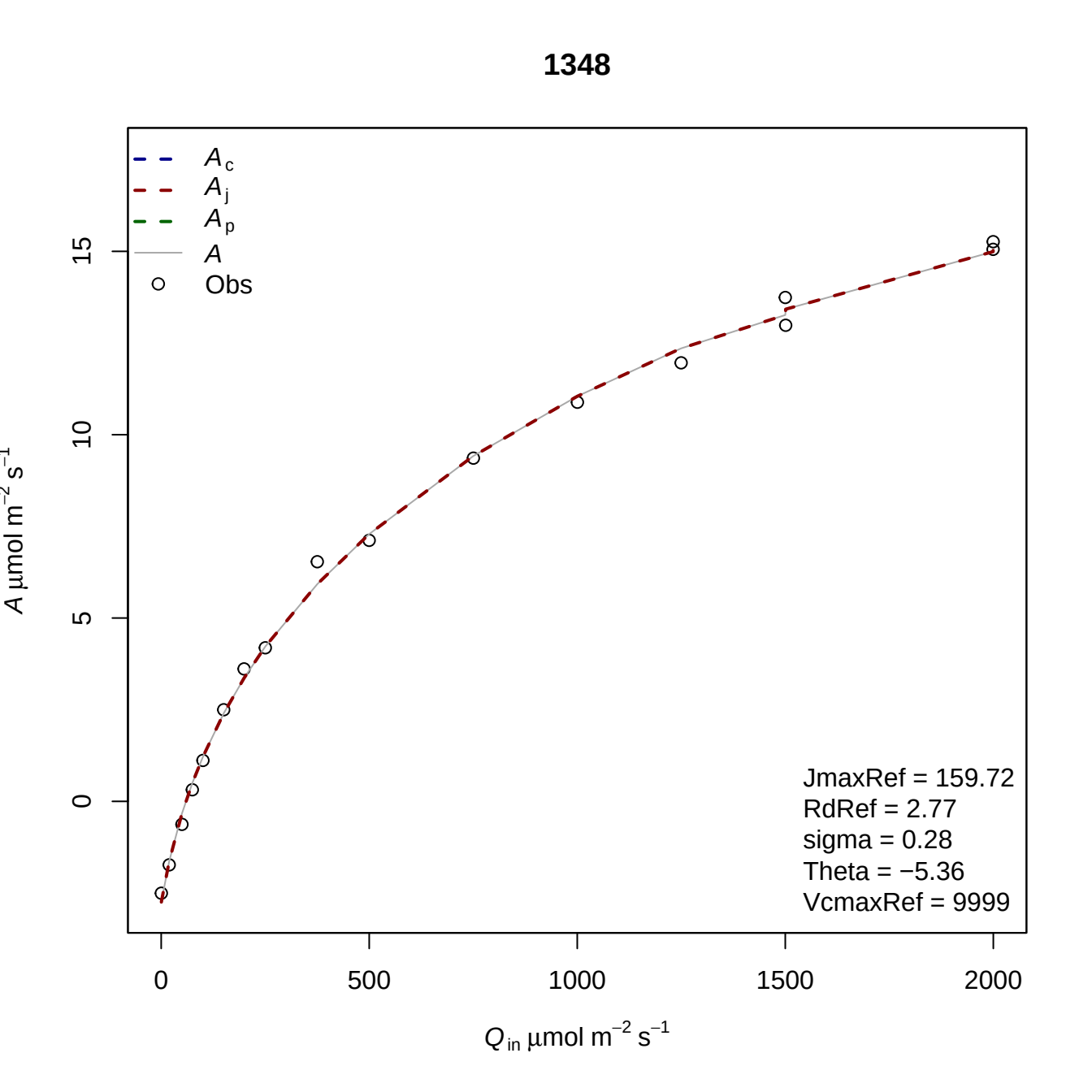

1343
A_c
A_j
A_p
A
Obs
JmaxRef = 67.85
RdRef = 2.59
sigma = 0.16
Theta = −1.35
VcmaxRef = 16.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

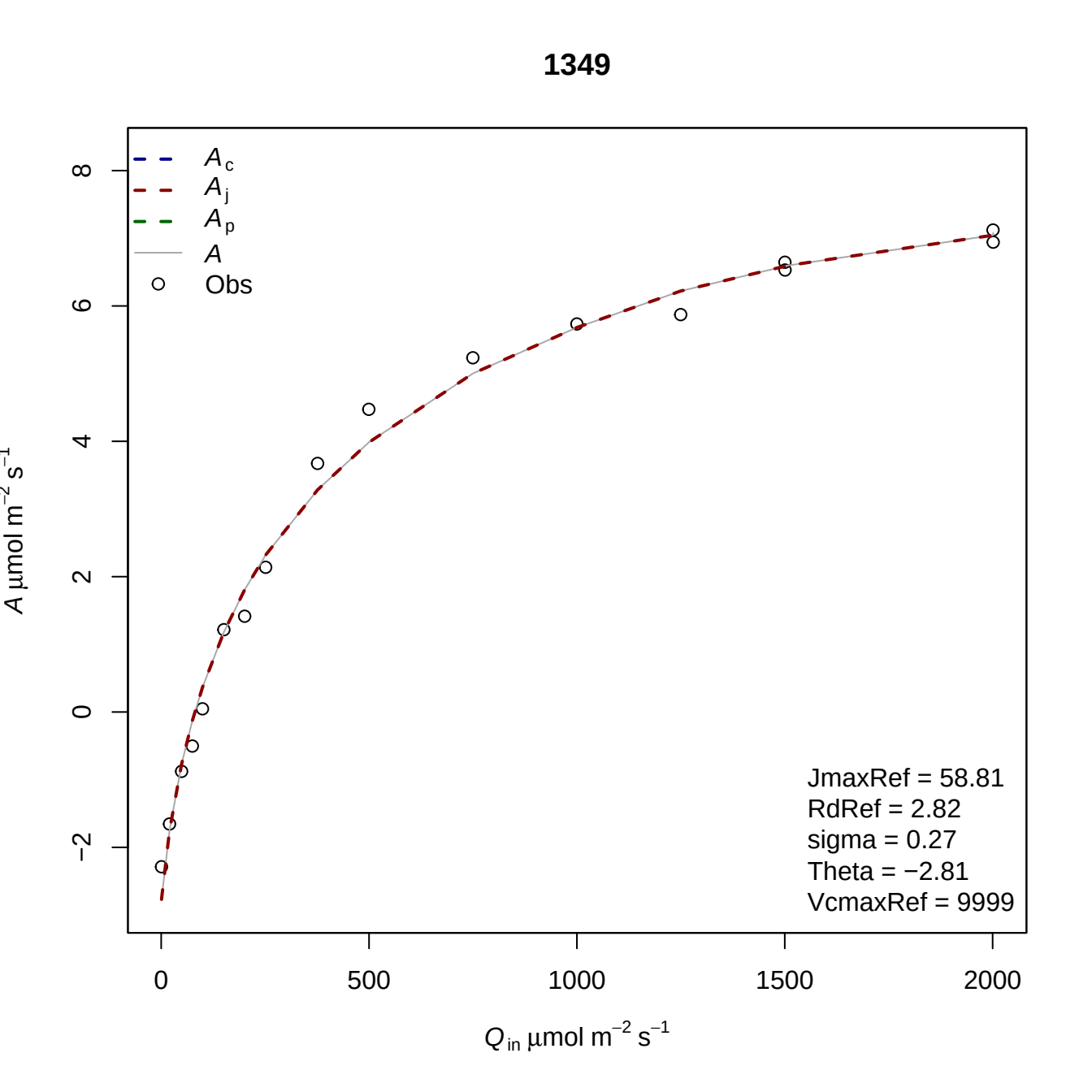

1344
A_c
A_j
A_p
A
Obs
JmaxRef = 54.67
RdRef = 1.02
sigma = 0.2
Theta = −1.37
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

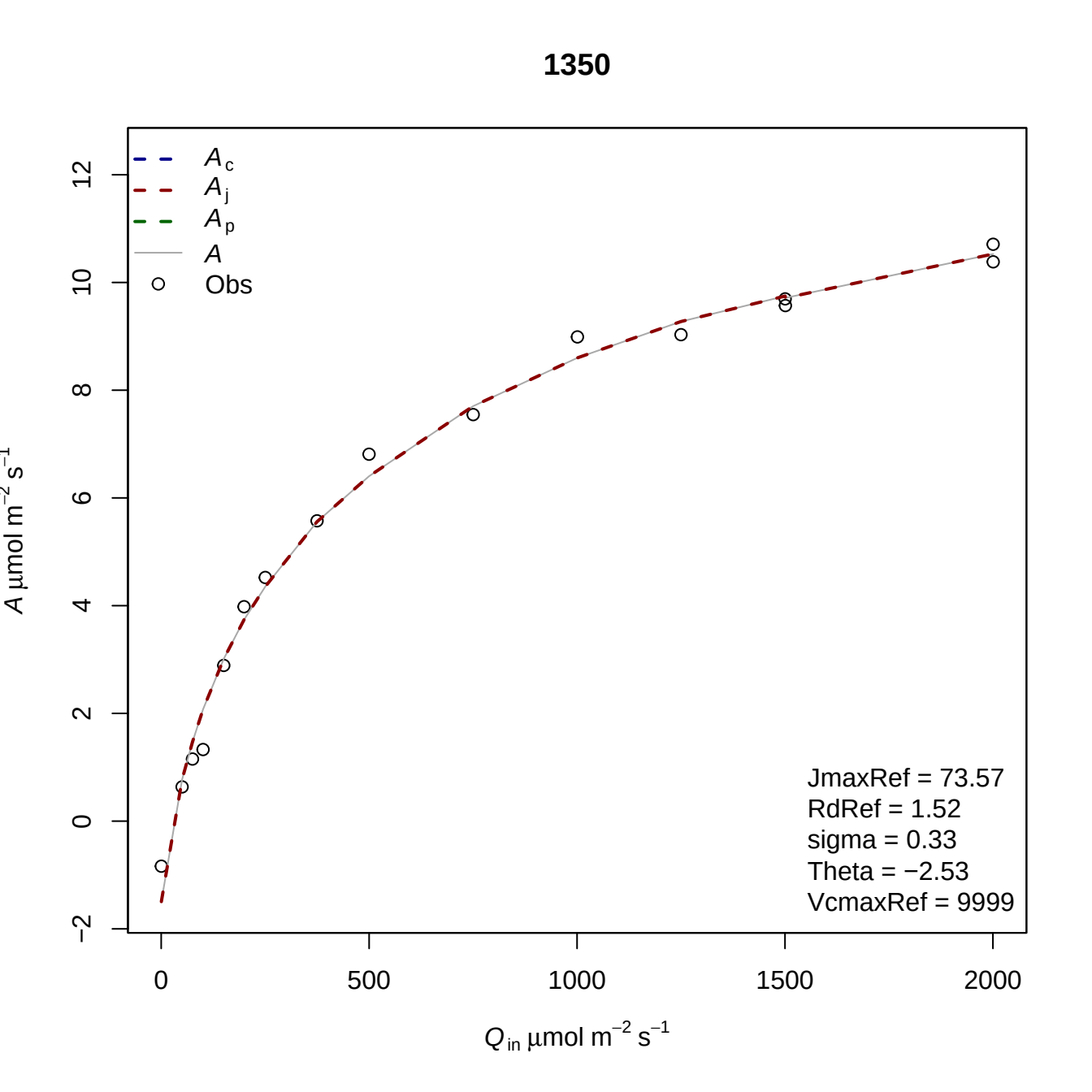

1345
A_c
A_j
A_p
A
Obs
JmaxRef = 59.54
RdRef = 0.52
sigma = 0.24
Theta = −1.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

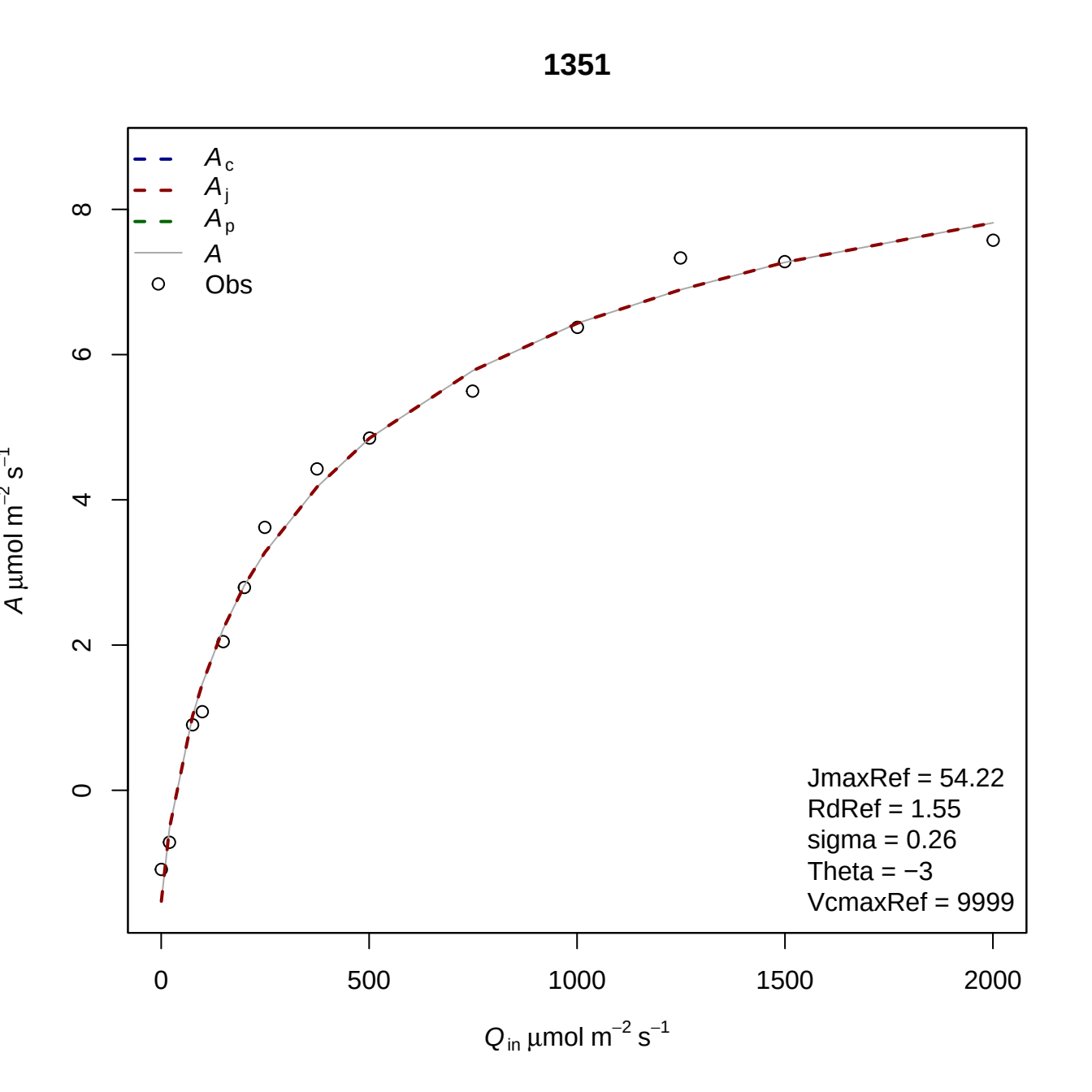

1346
A_c
A_j
A_p
A
Obs
JmaxRef = 51.99
RdRef = 3.22
sigma = 0.22
Theta = −3.14
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

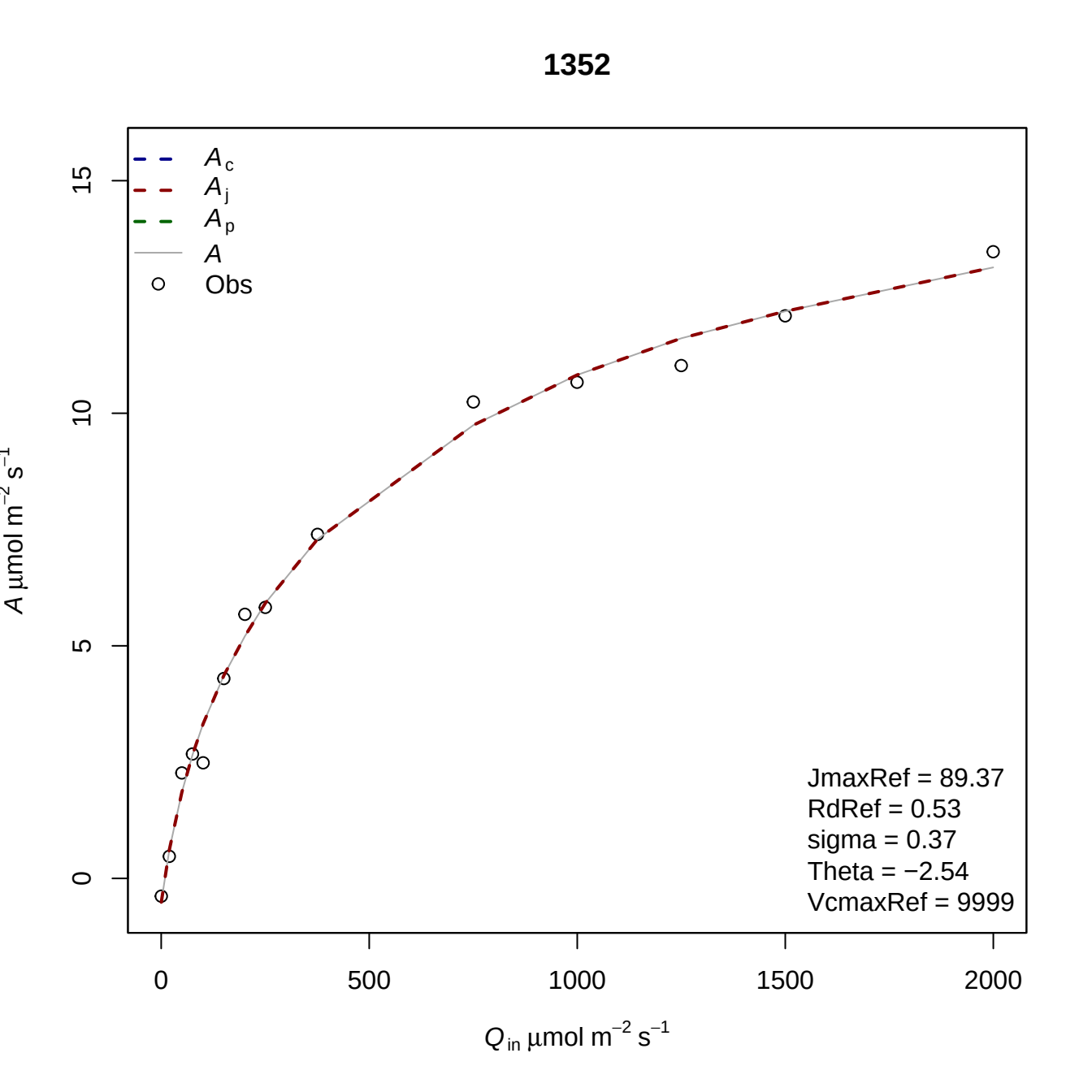

1347
A_c
A_j
A_p
A
Obs
JmaxRef = 86.07
RdRef = 3.22
sigma = 0.22
Theta = −2.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

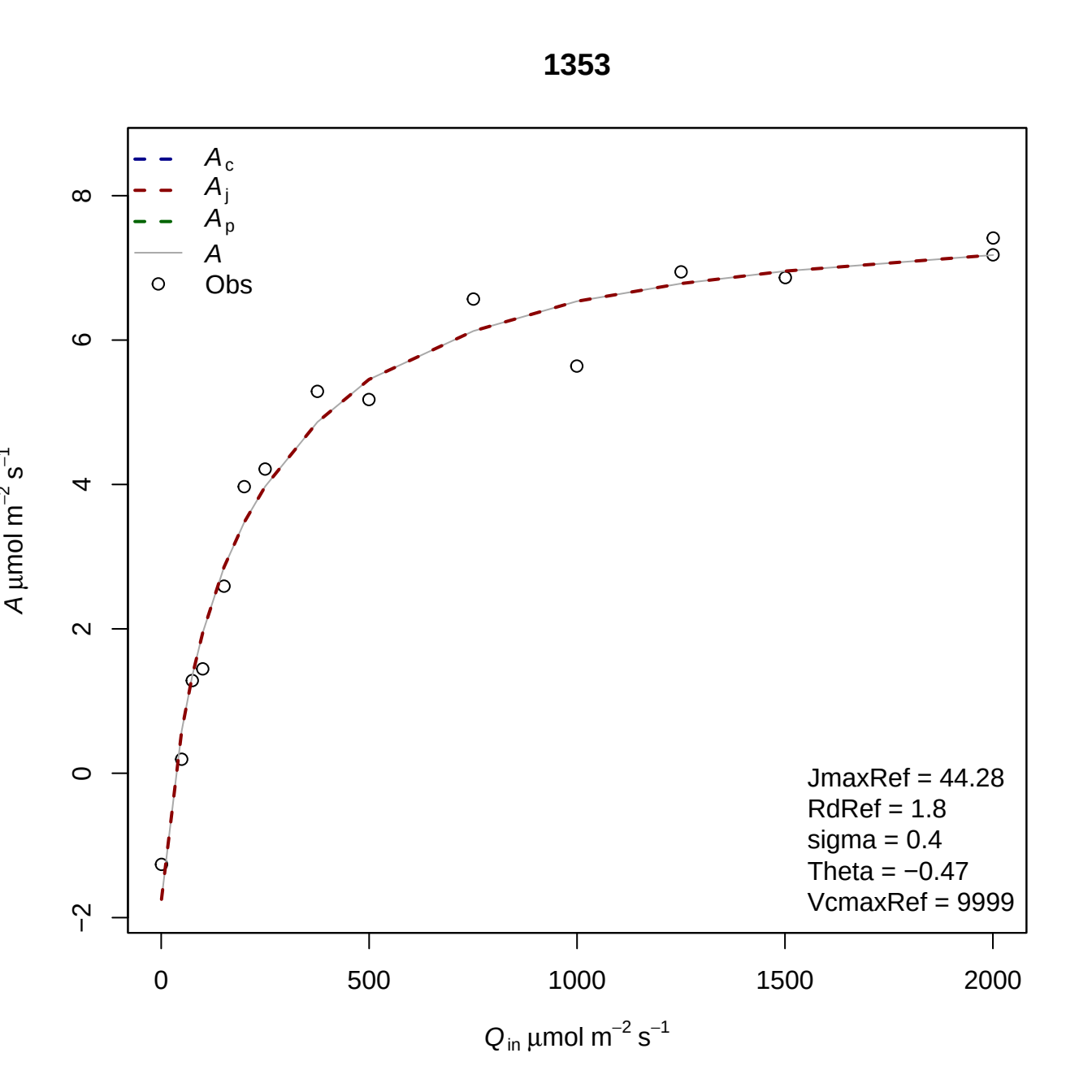

1348
A_c
A_j
A_p
A
Obs
JmaxRef = 159.72
RdRef = 2.77
sigma = 0.28
Theta = −5.36
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

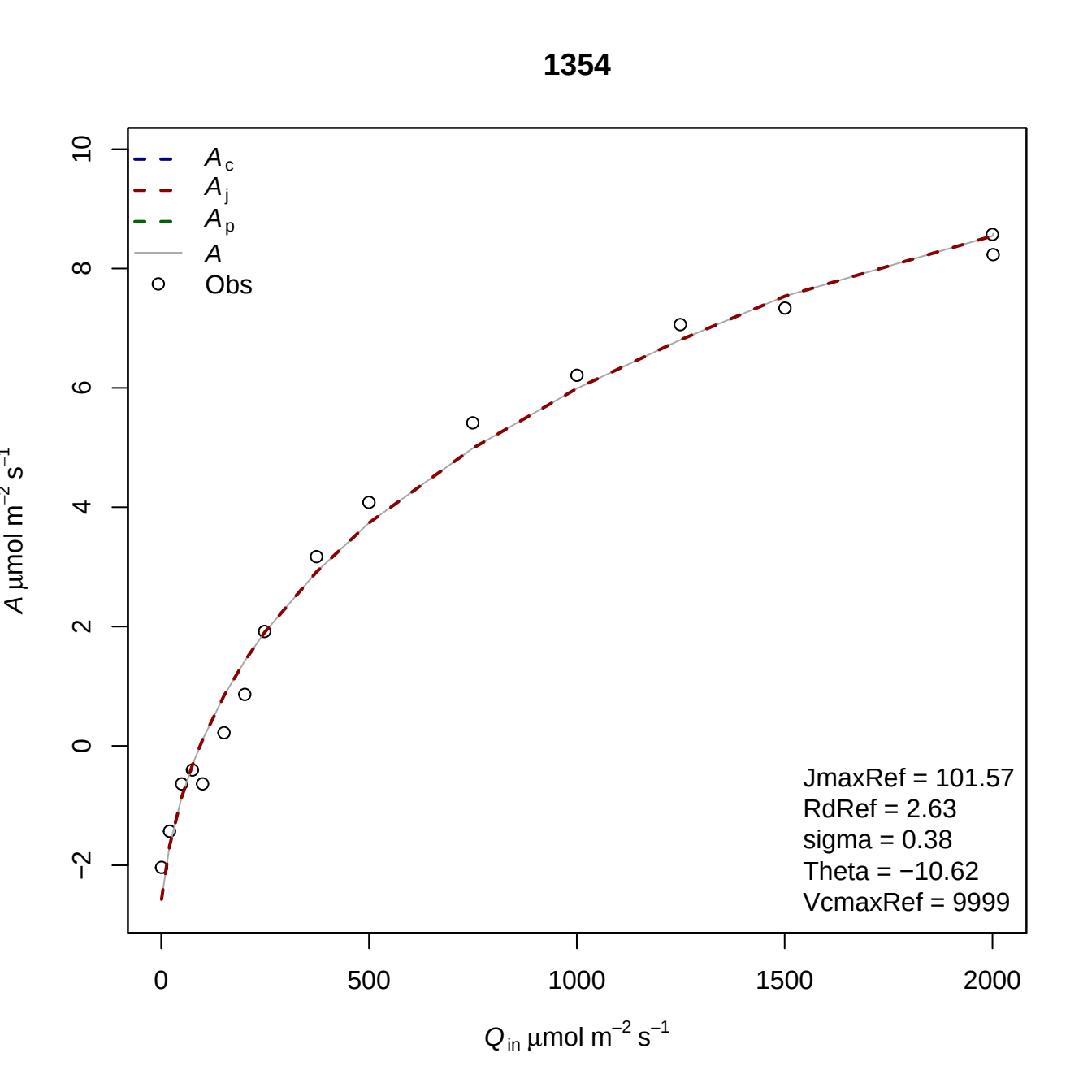

1349
A_c
A_j
A_p
A
Obs
JmaxRef = 58.81
RdRef = 2.82
sigma = 0.27
Theta = −2.81
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

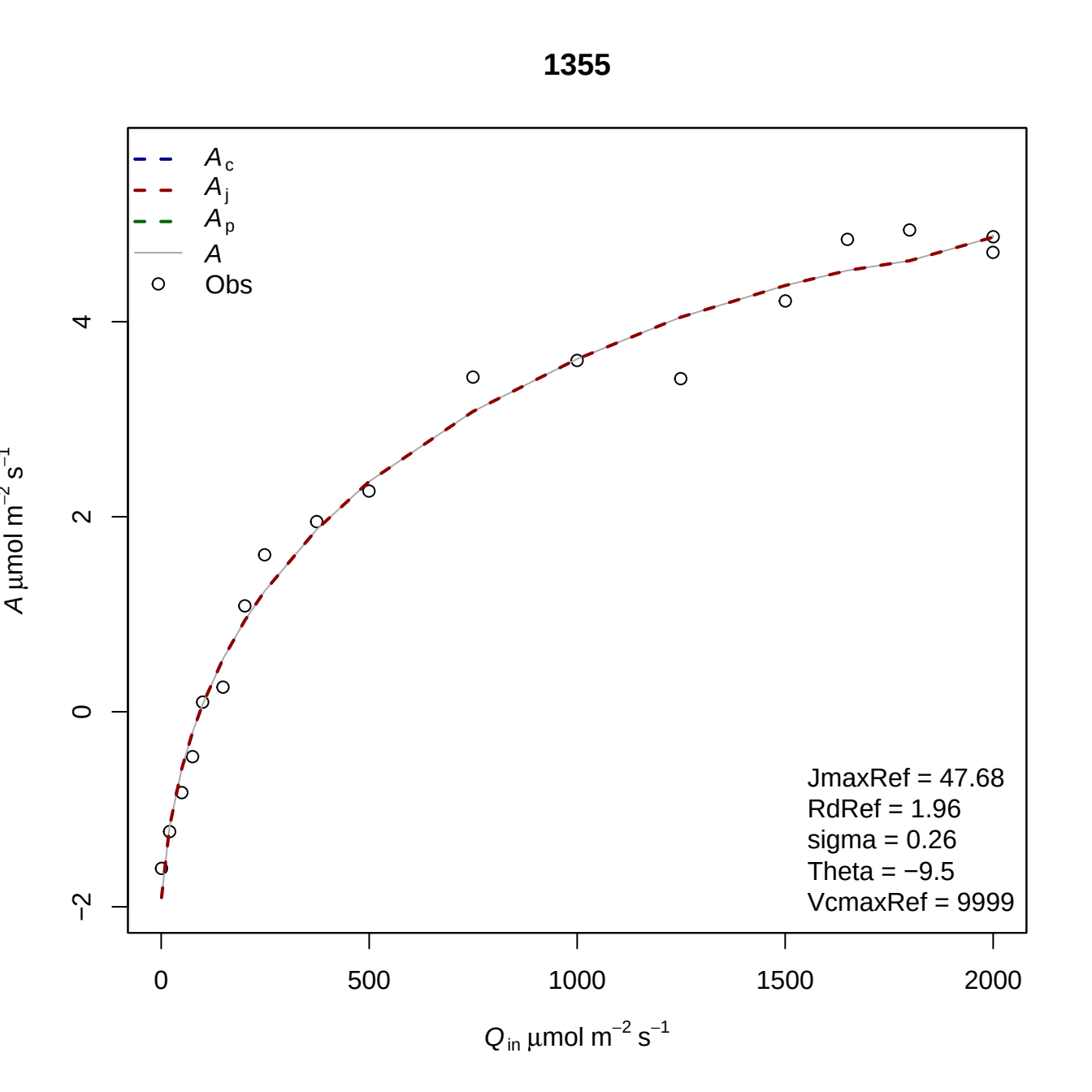

1350
A_c
A_j
A_p
A
Obs
JmaxRef = 73.57
RdRef = 1.52
sigma = 0.33
Theta = −2.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

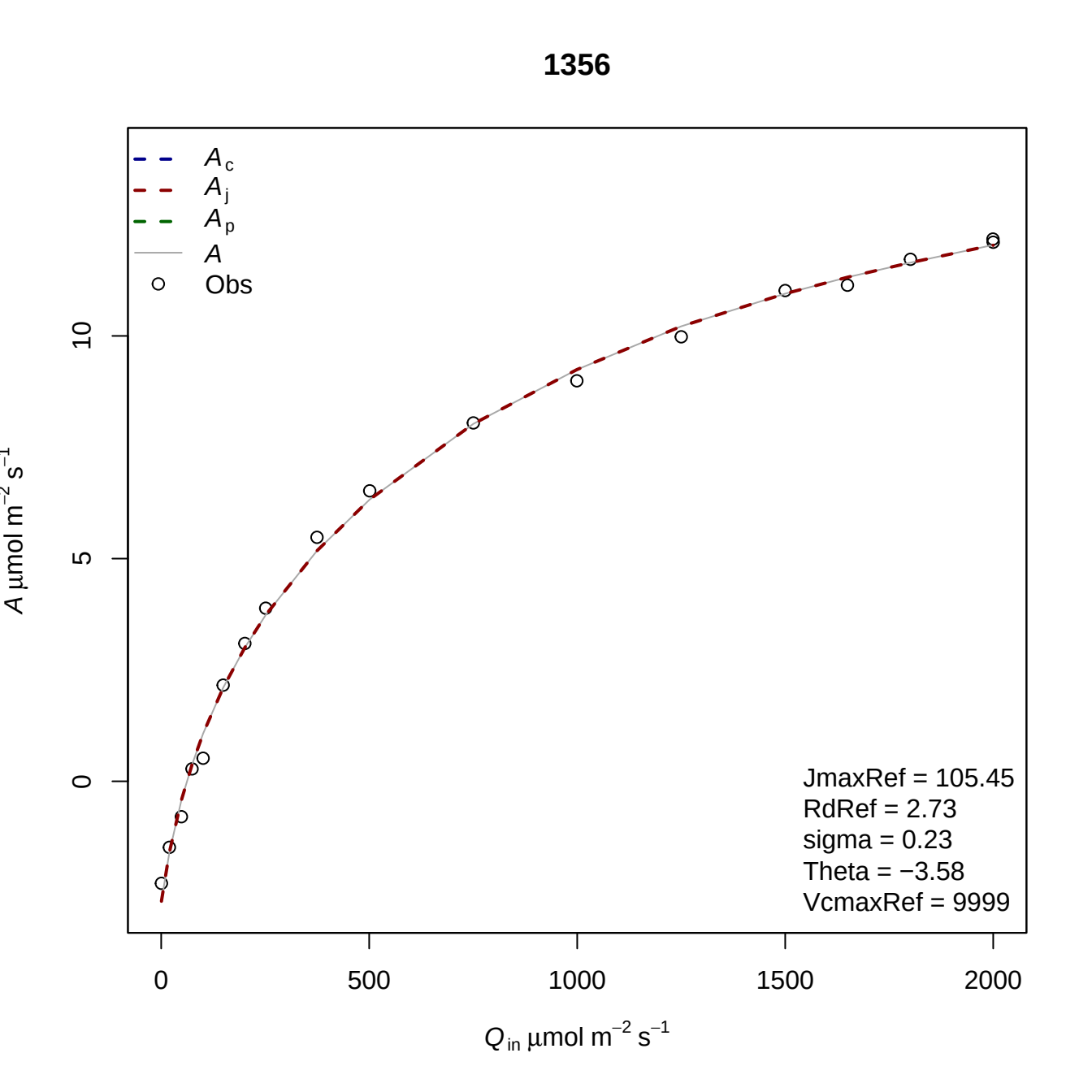

1351
A_c
A_j
A_p
A
Obs
JmaxRef = 54.22
RdRef = 1.55
sigma = 0.26
Theta = −3
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

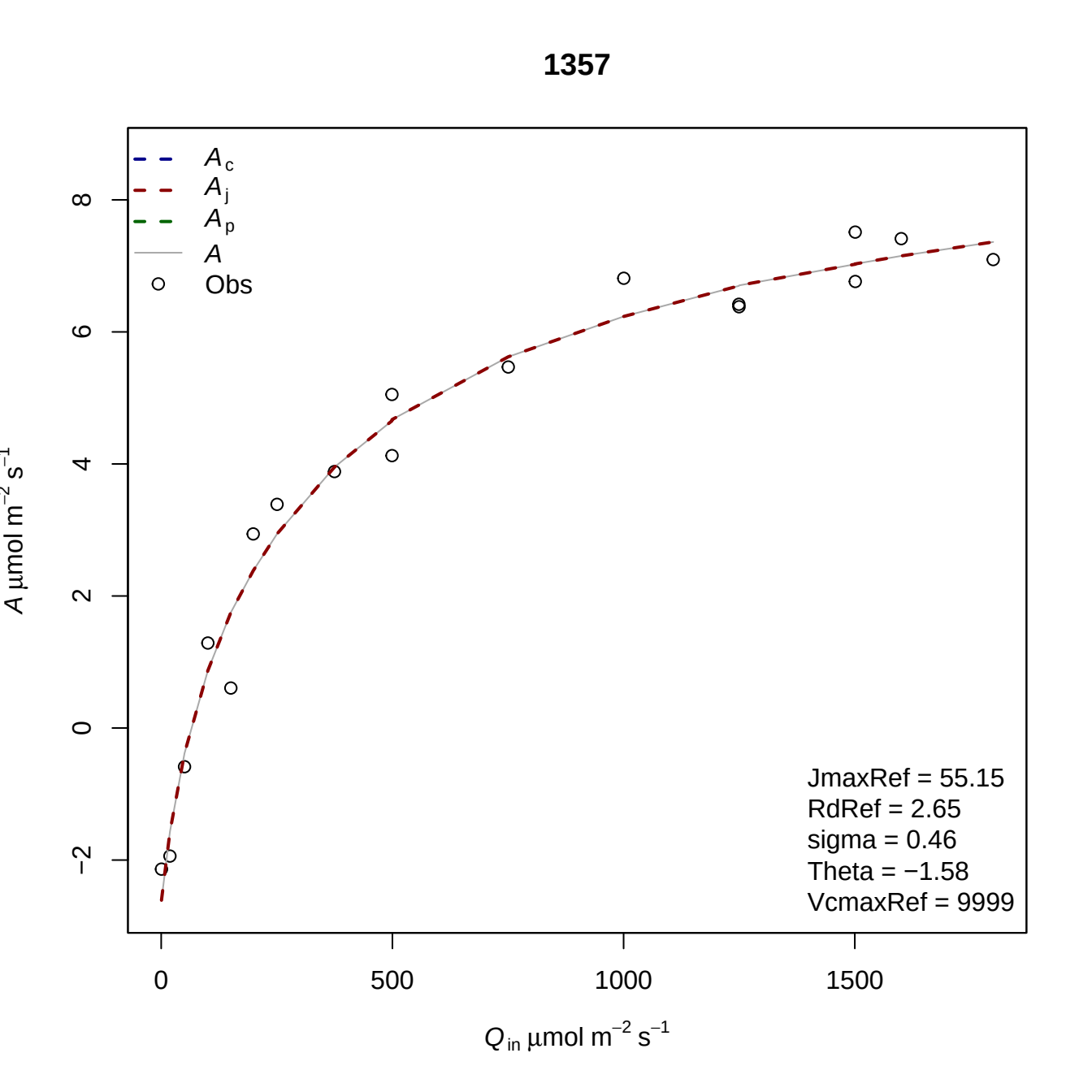

1352
A_c
A_j
A_p
A
Obs
JmaxRef = 89.37
RdRef = 0.53
sigma = 0.37
Theta = −2.54
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

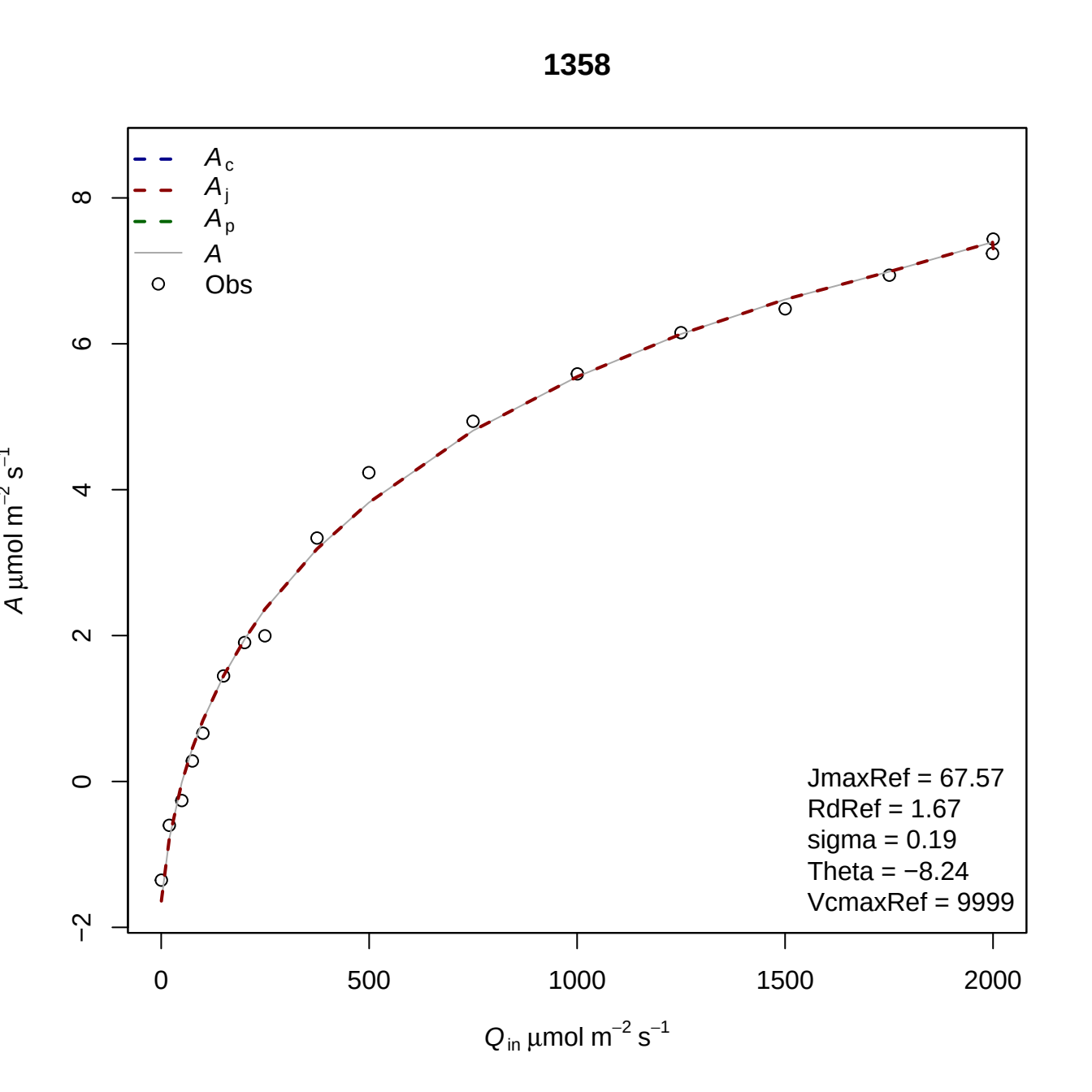

1353
A_c
A_j
A_p
A
Obs
JmaxRef = 44.28
RdRef = 1.8
sigma = 0.4
Theta = −0.47
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

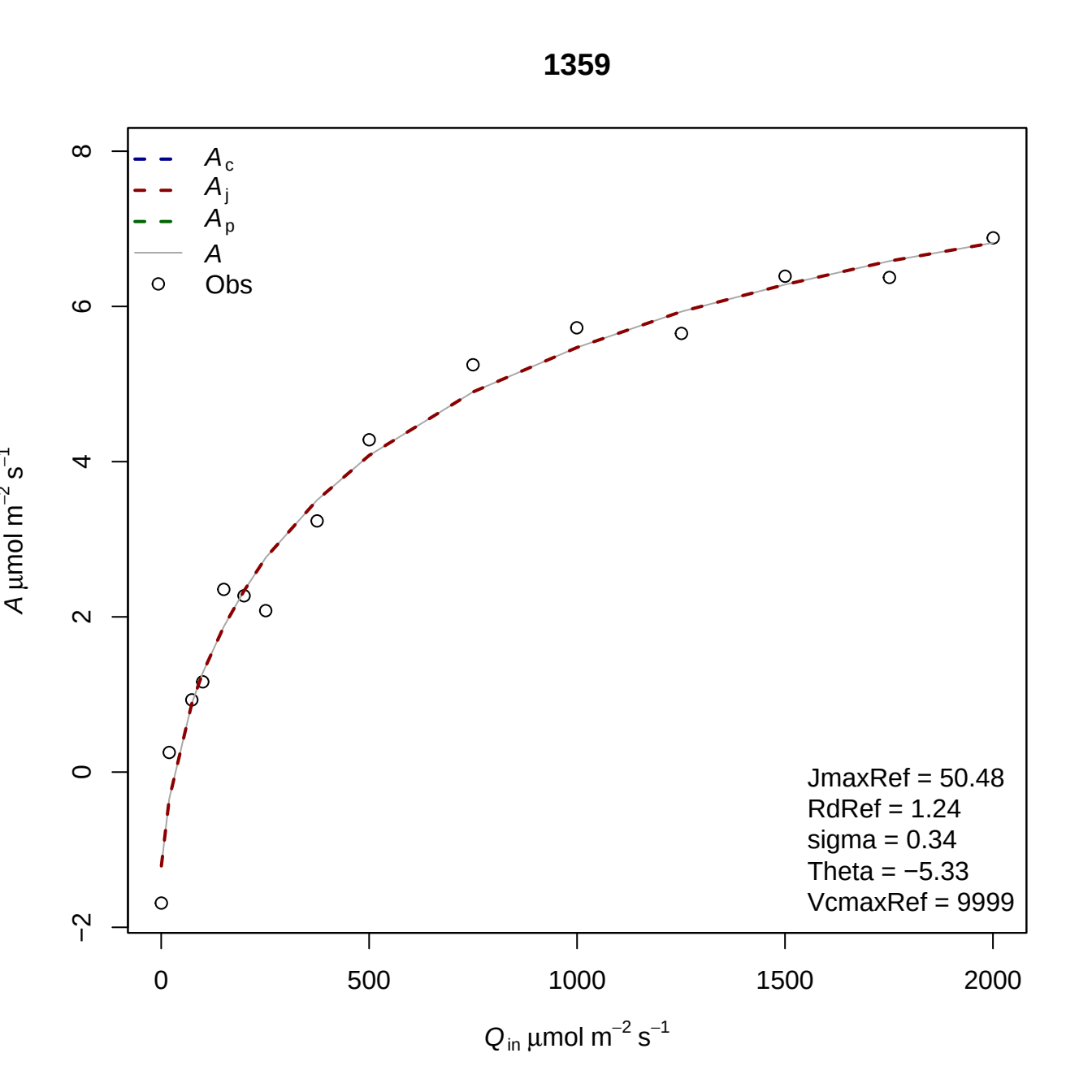

1354
A_c
A_j
A_p
A
Obs
JmaxRef = 101.57
RdRef = 2.63
sigma = 0.38
Theta = −10.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

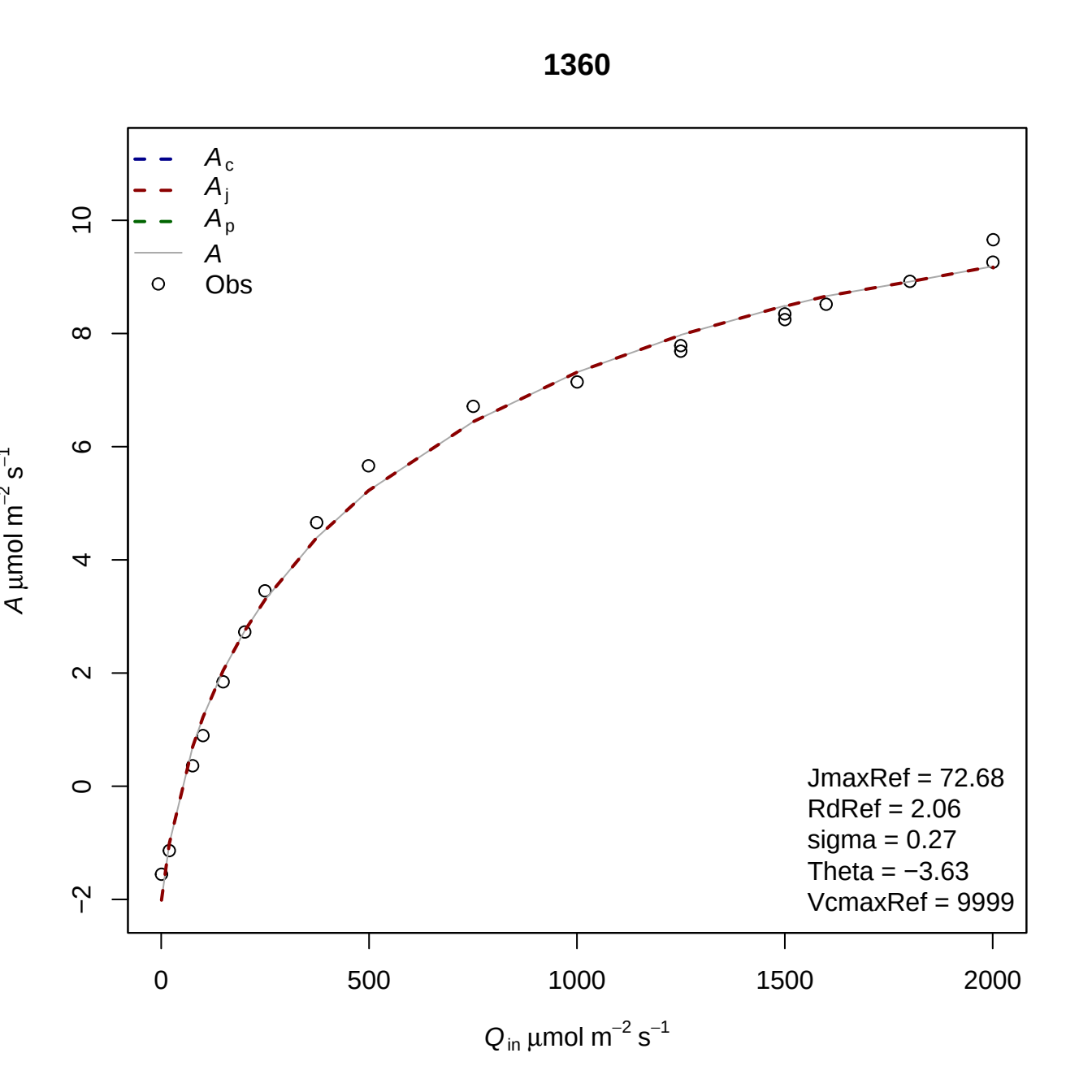

1355
A_c
A_j
A_p
A
Obs
JmaxRef = 47.68
RdRef = 1.96
sigma = 0.26
Theta = −9.5
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

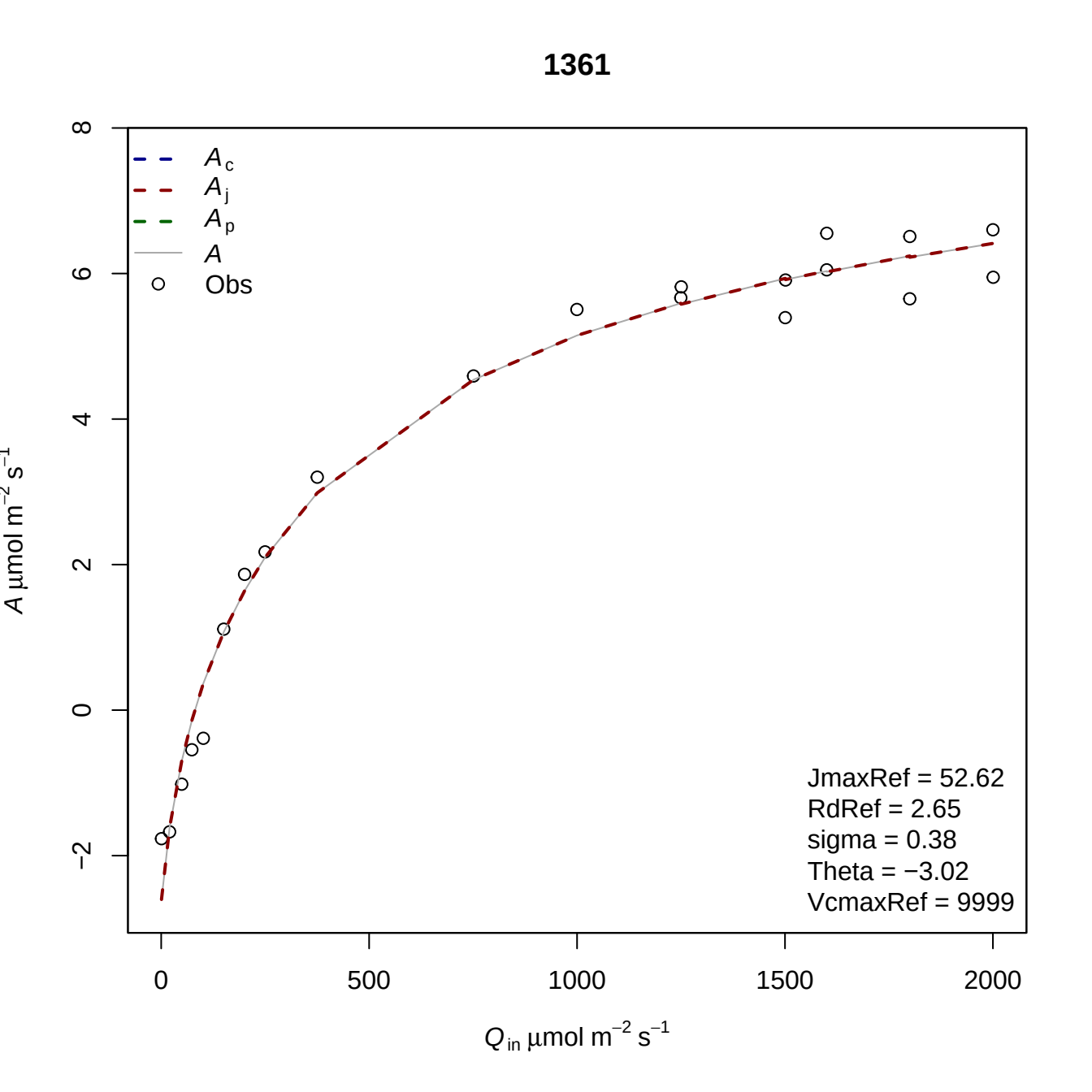

1356
A_c
A_j
A_p
A
Obs
JmaxRef = 105.45
RdRef = 2.73
sigma = 0.23
Theta = −3.58
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

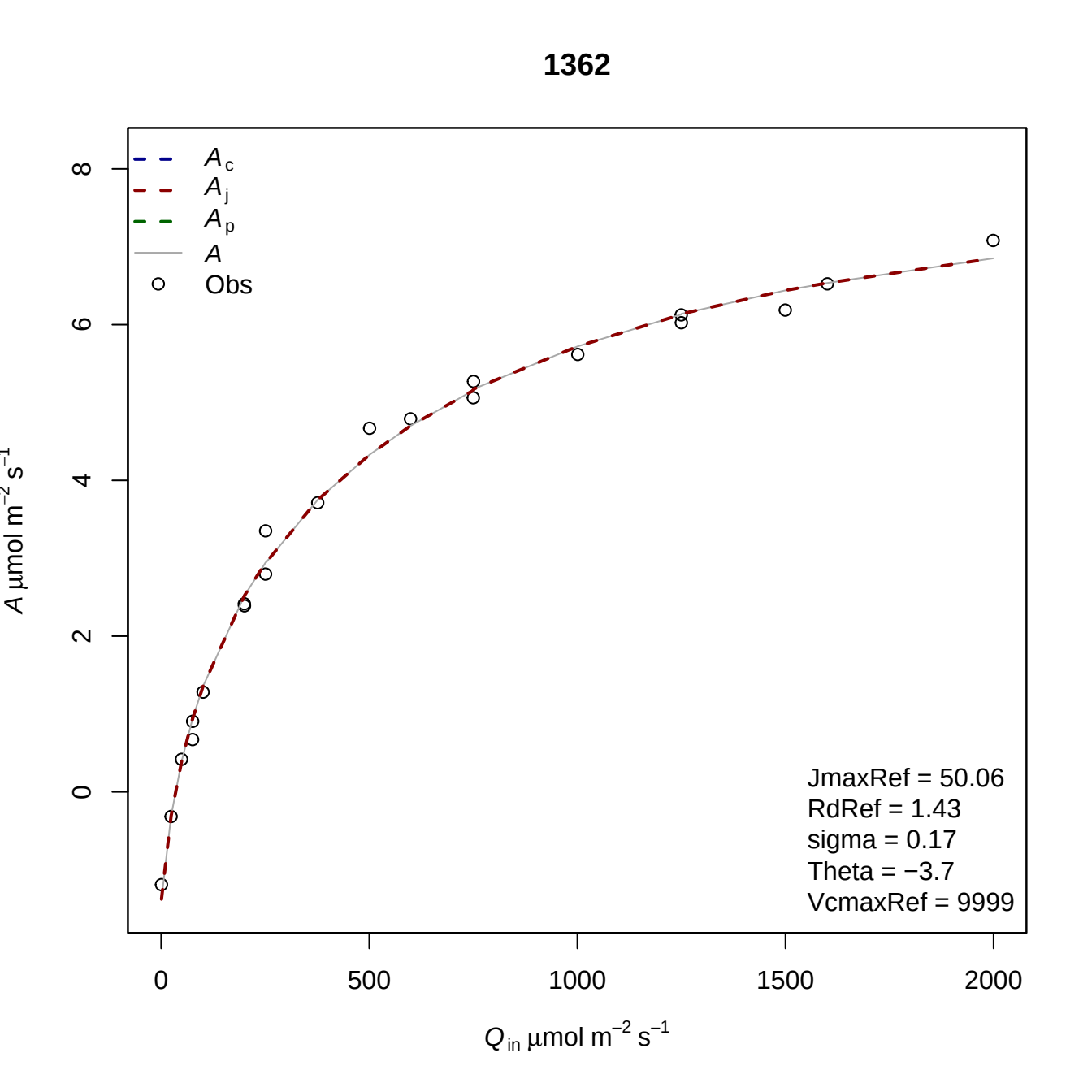

1357
A_c
A_j
A_p
A
Obs
JmaxRef = 55.15
RdRef = 2.65
sigma = 0.46
Theta = −1.58
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

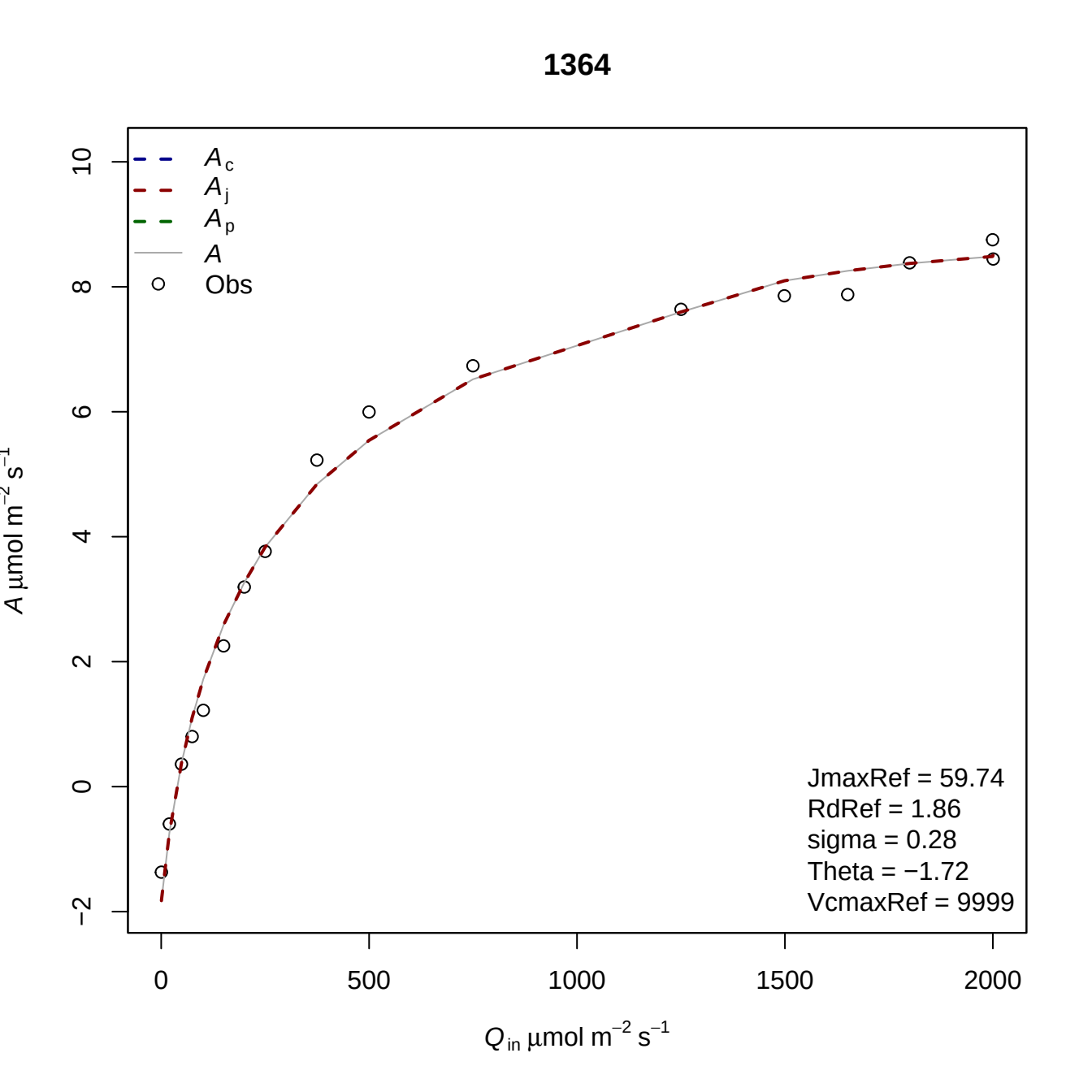

1358
A_c
A_j
A_p
A
Obs
JmaxRef = 67.57
RdRef = 1.67
sigma = 0.19
Theta = −8.24
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

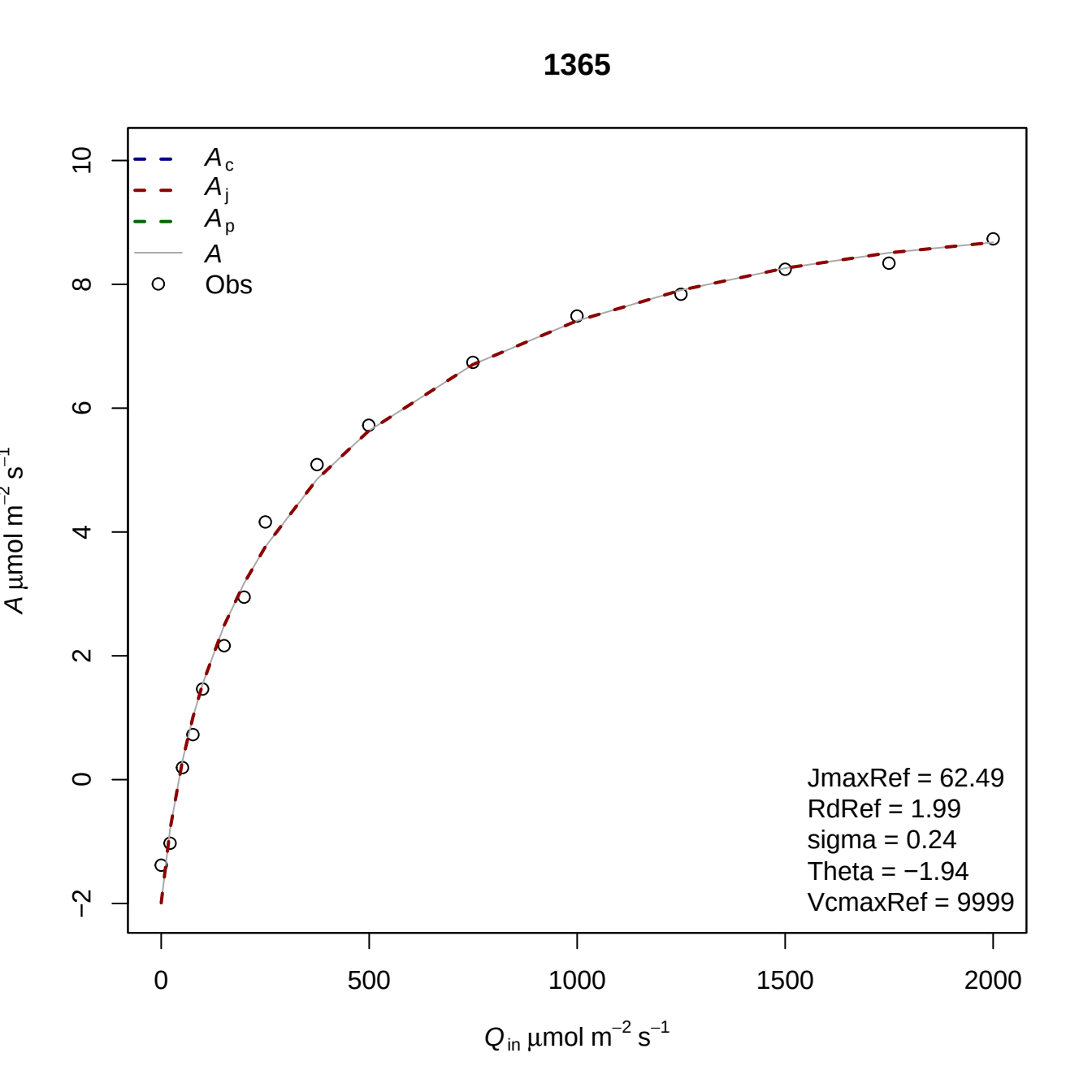

1359
A_c
A_j
A_p
A
Obs
JmaxRef = 50.48
RdRef = 1.24
sigma = 0.34
Theta = −5.33
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

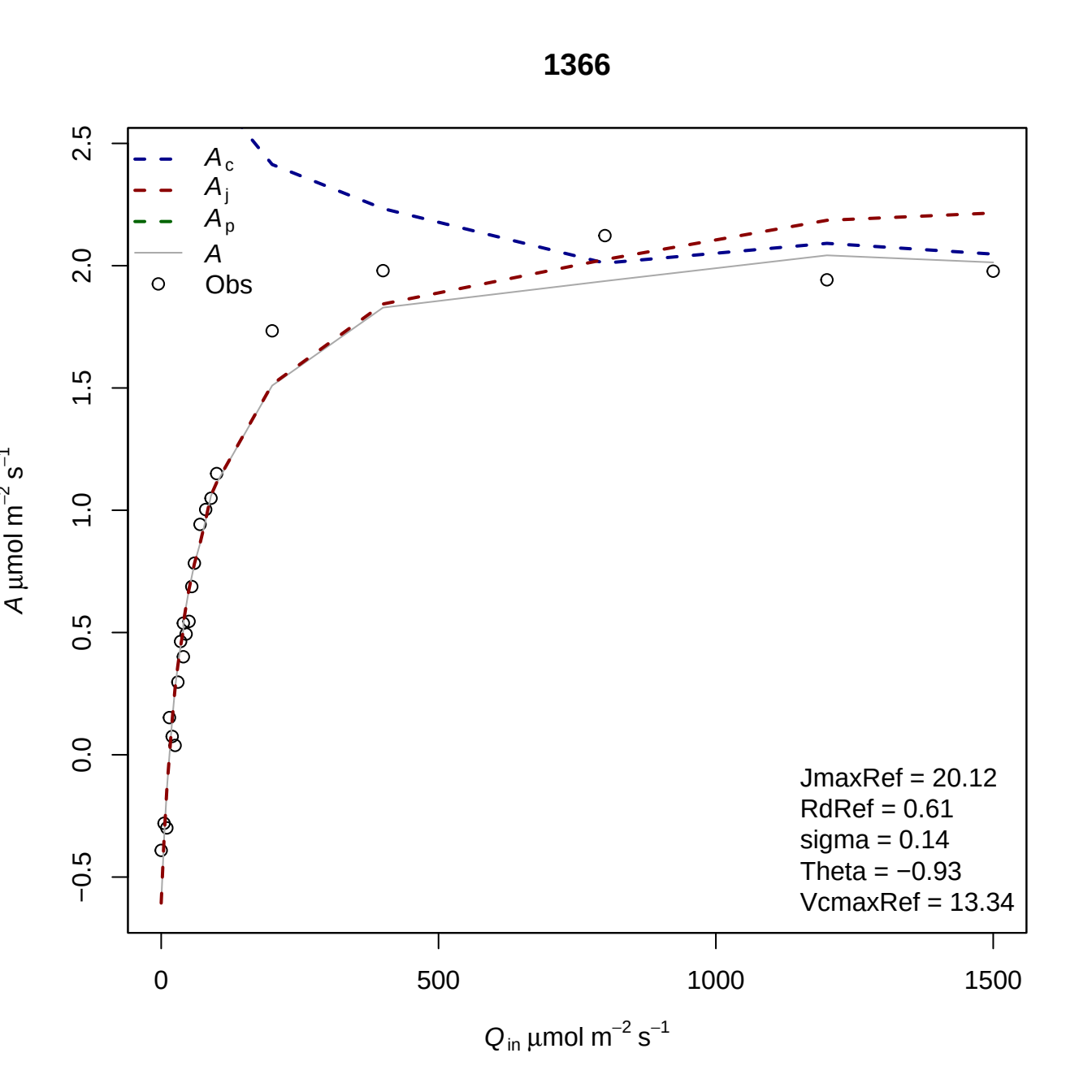

1360
A_c
A_j
A_p
A
Obs
JmaxRef = 72.68
RdRef = 2.06
sigma = 0.27
Theta = −3.63
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

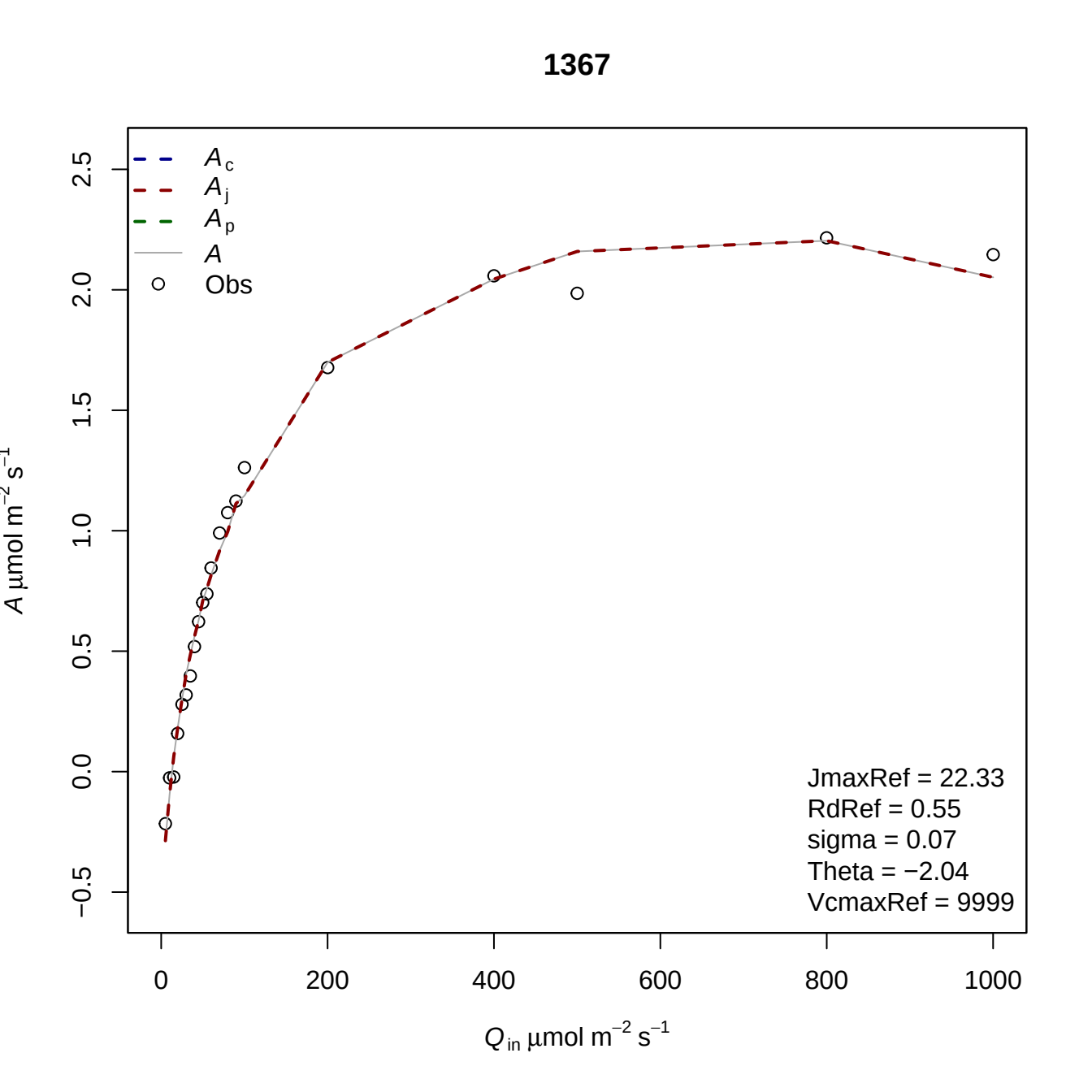

1361
A_c
A_j
A_p
A
Obs
JmaxRef = 52.62
RdRef = 2.65
sigma = 0.38
Theta = −3.02
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

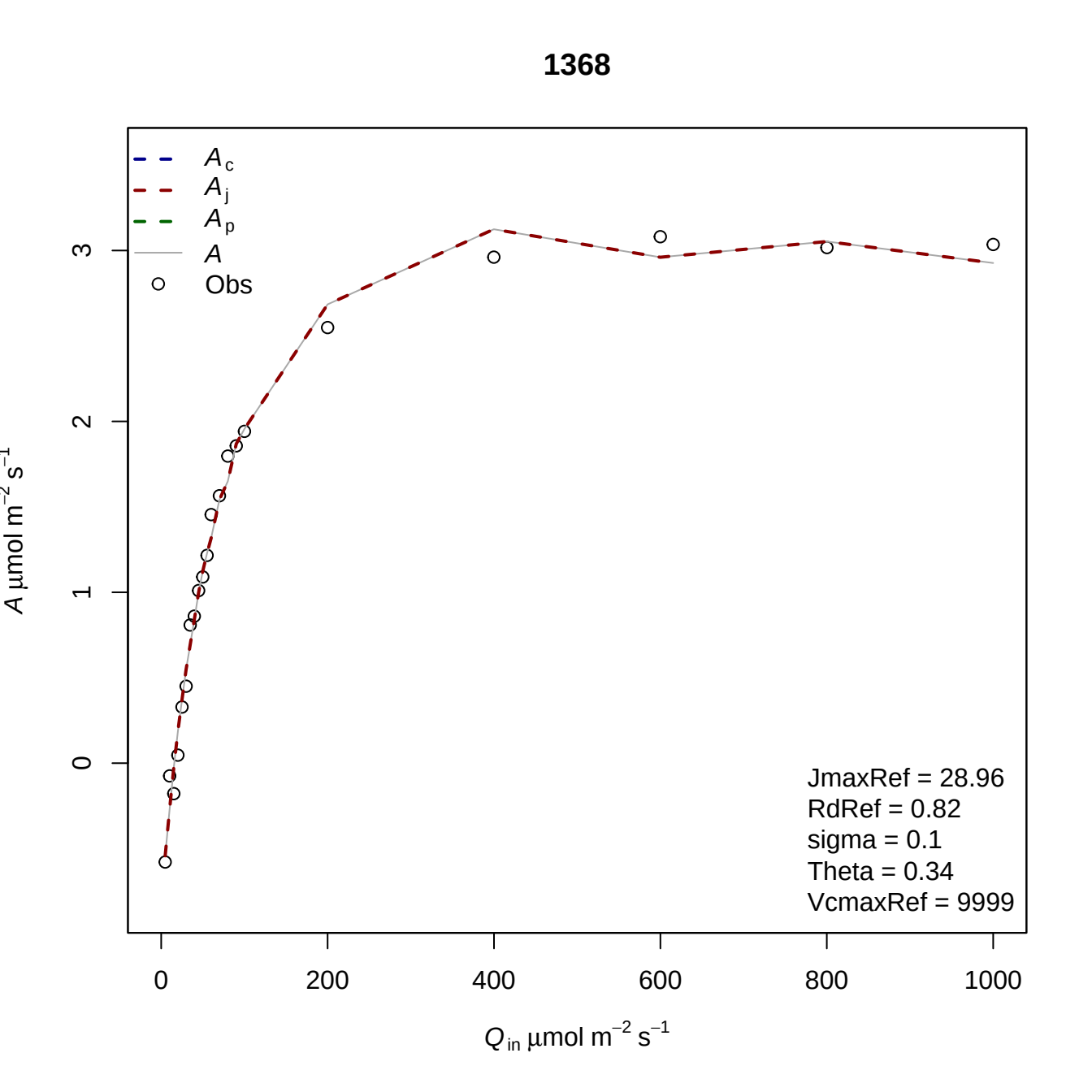

1362
A_c
A_j
A_p
A
Obs
JmaxRef = 50.06
RdRef = 1.43
sigma = 0.17
Theta = −3.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

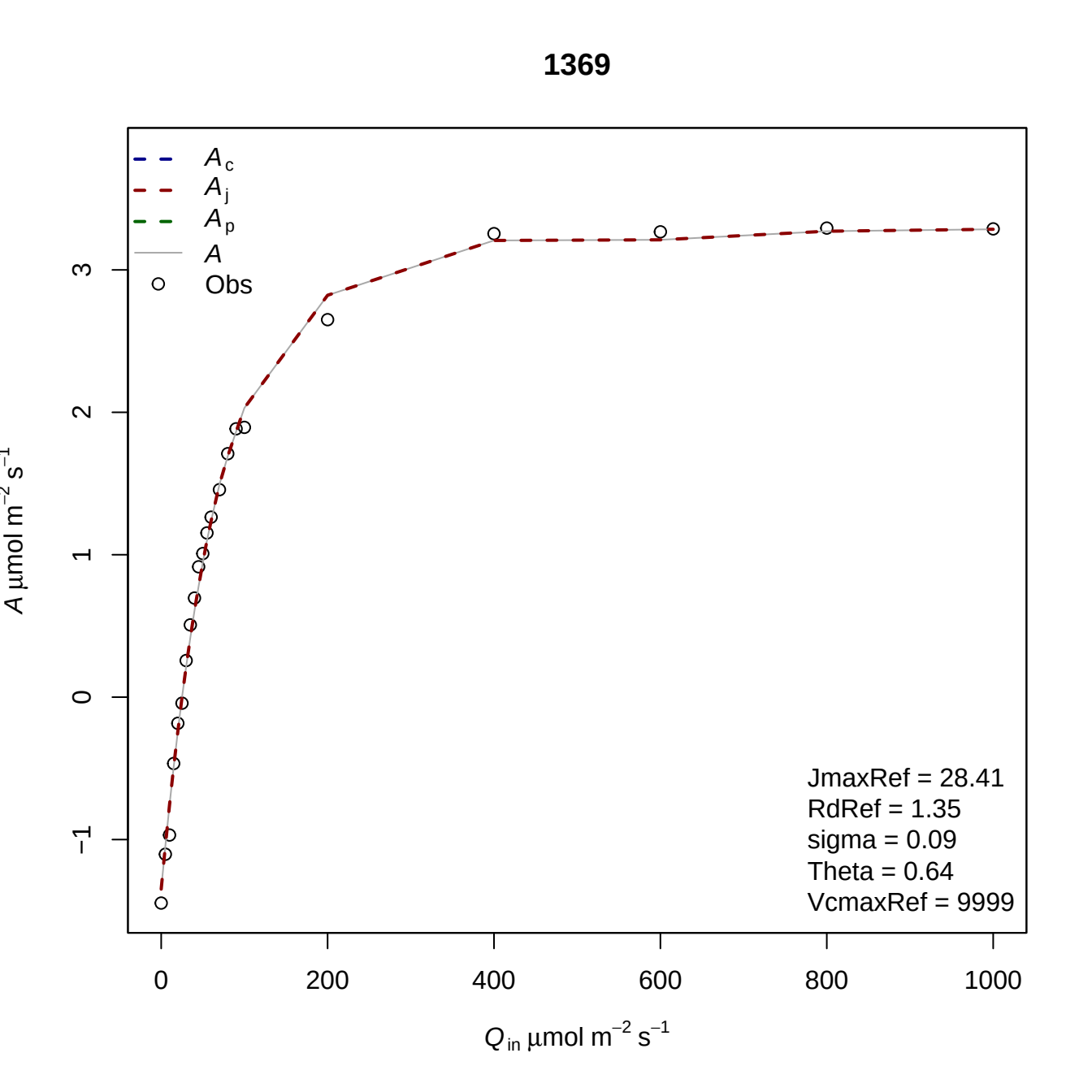

1364
A_c
A_j
A_p
A
Obs
JmaxRef = 59.74
RdRef = 1.86
sigma = 0.28
Theta = −1.72
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

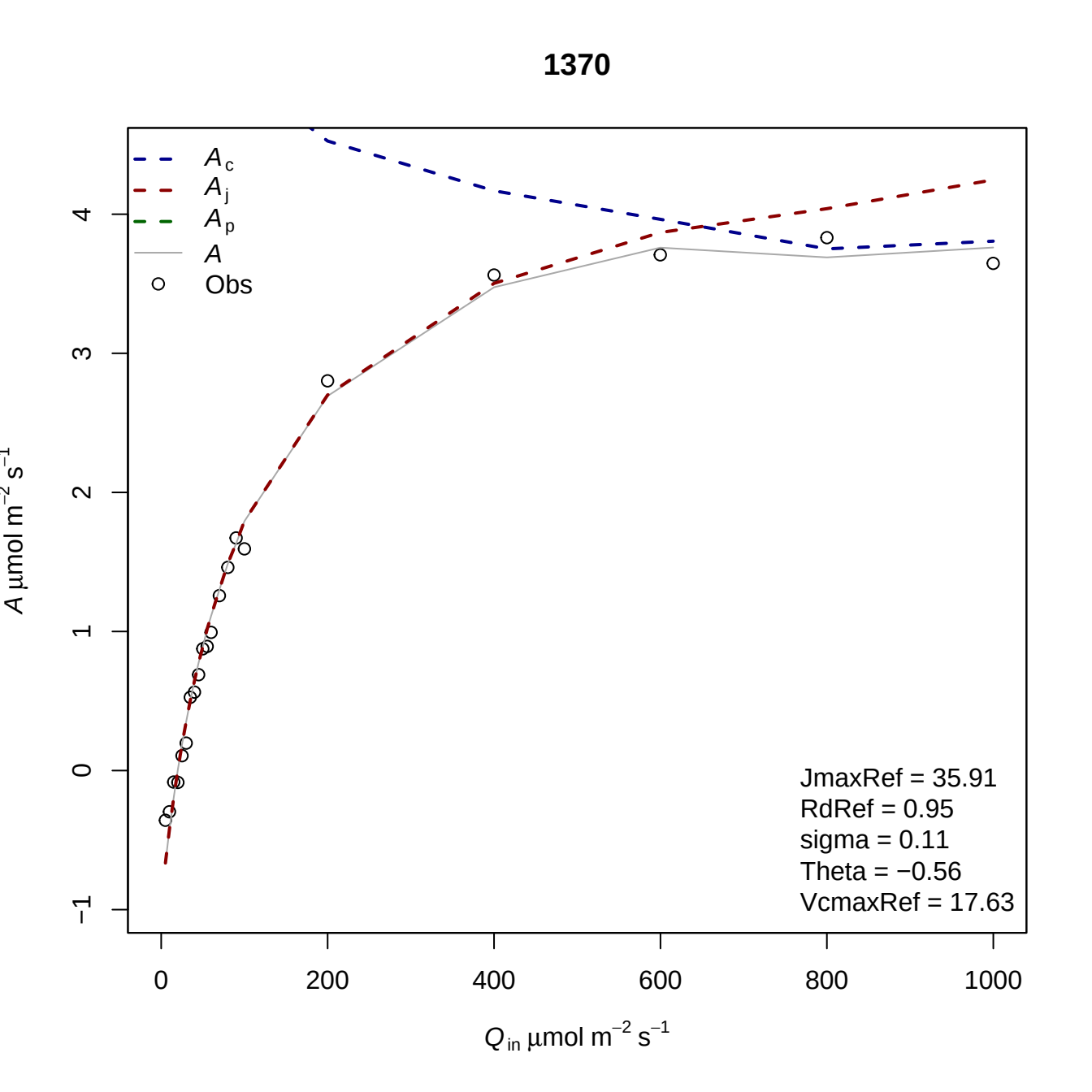

1365
A_c
A_j
A_p
A
Obs
JmaxRef = 62.49
RdRef = 1.99
sigma = 0.24
Theta = −1.94
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

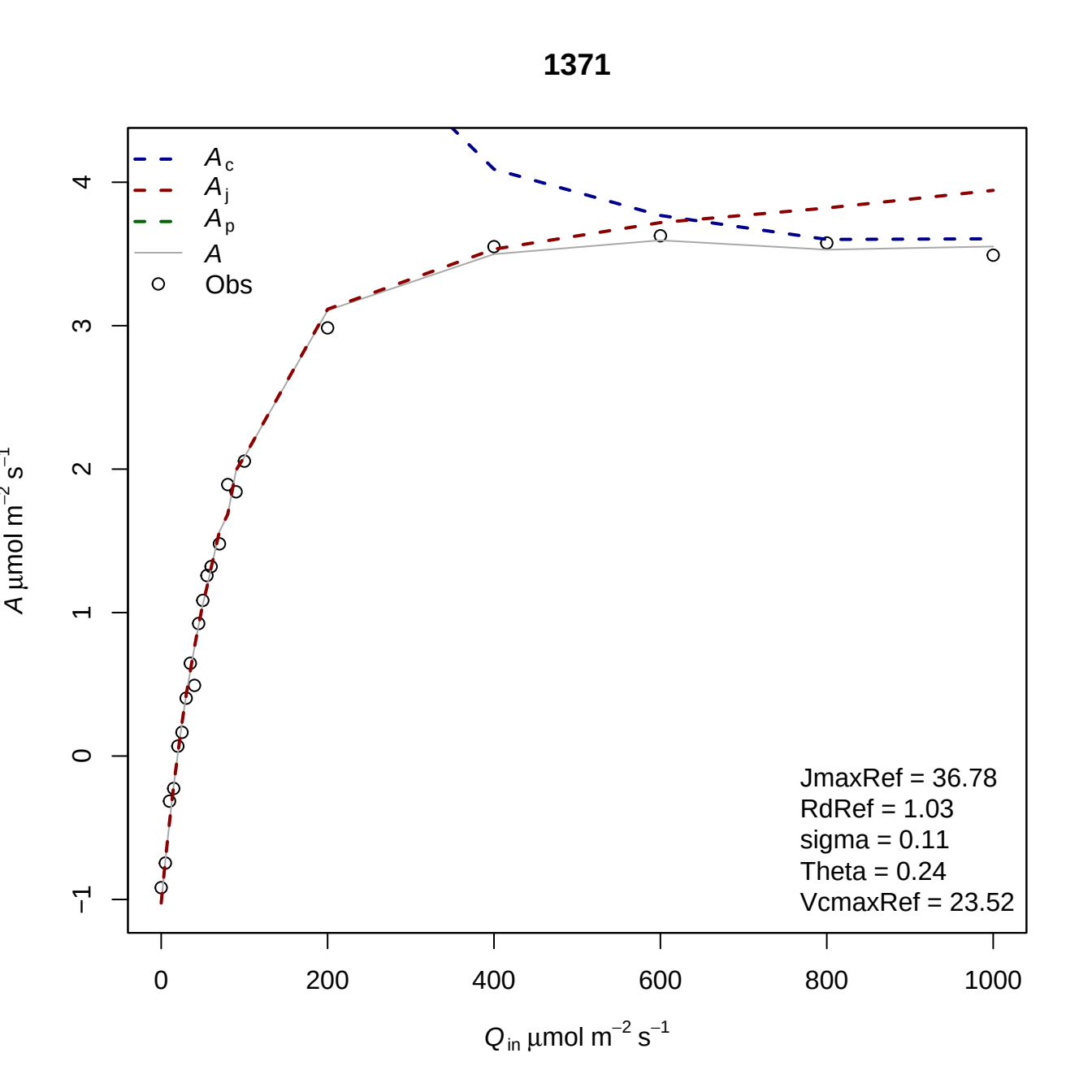

1366
A_c
A_j
A_p
A
Obs
JmaxRef = 20.12
RdRef = 0.61
sigma = 0.14
Theta = −0.93
VcmaxRef = 13.34
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

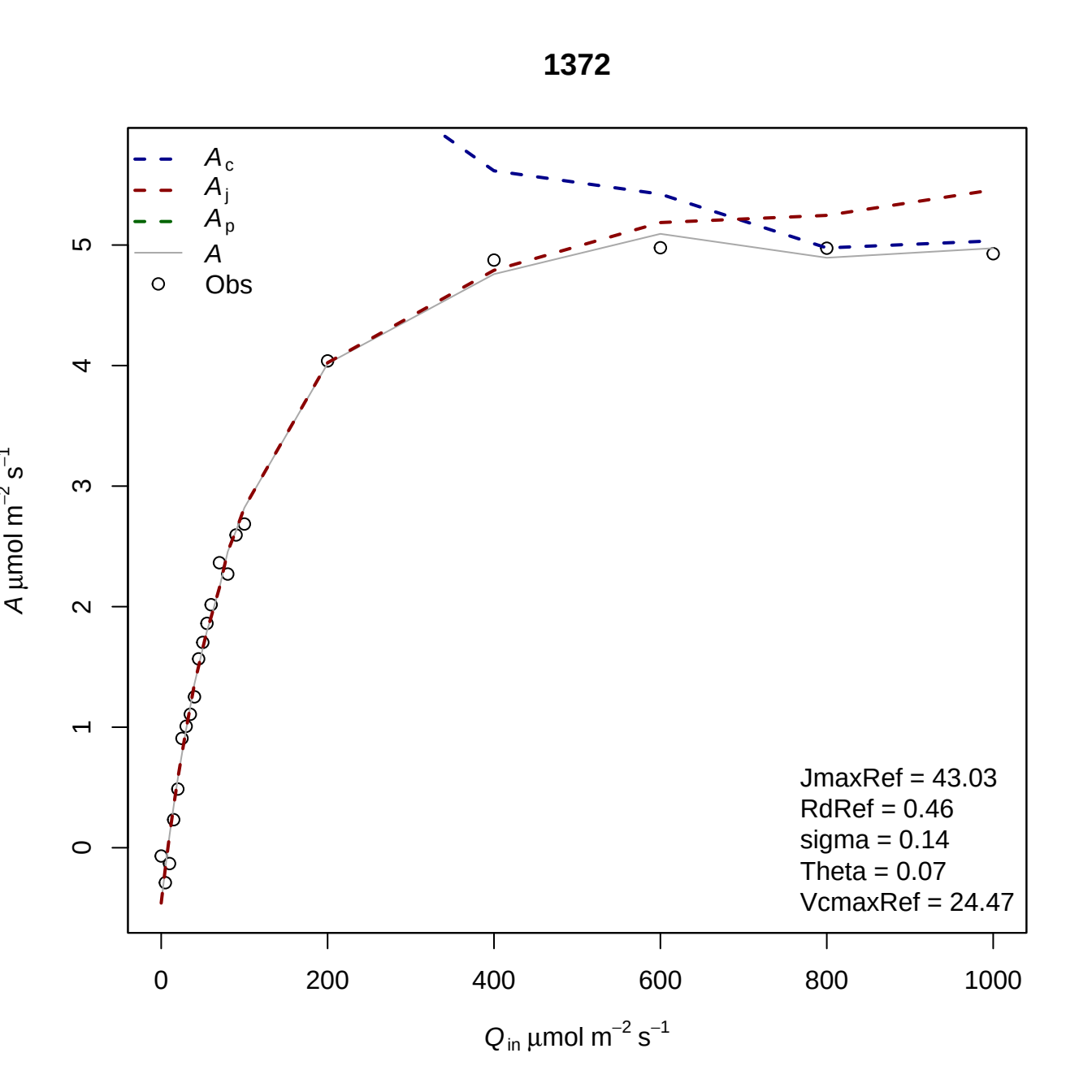

1367
A_c
A_j
A_p
A
Obs
JmaxRef = 22.33
RdRef = 0.55
sigma = 0.07
Theta = −2.04
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
200
400
600
800
1000

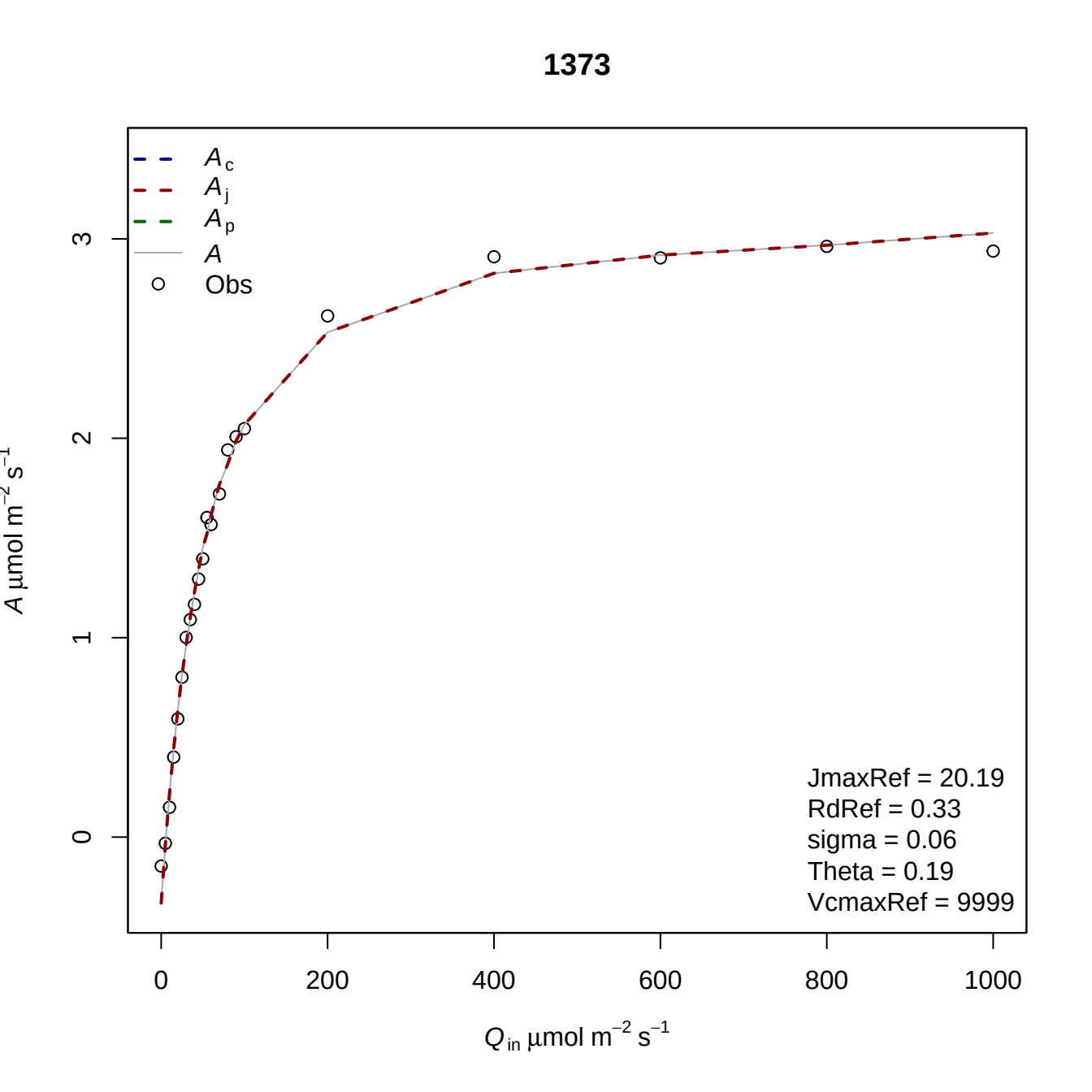

1368
A_c
A_j
A_p
A
Obs
JmaxRef = 28.96
RdRef = 0.82
sigma = 0.1
Theta = 0.34
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

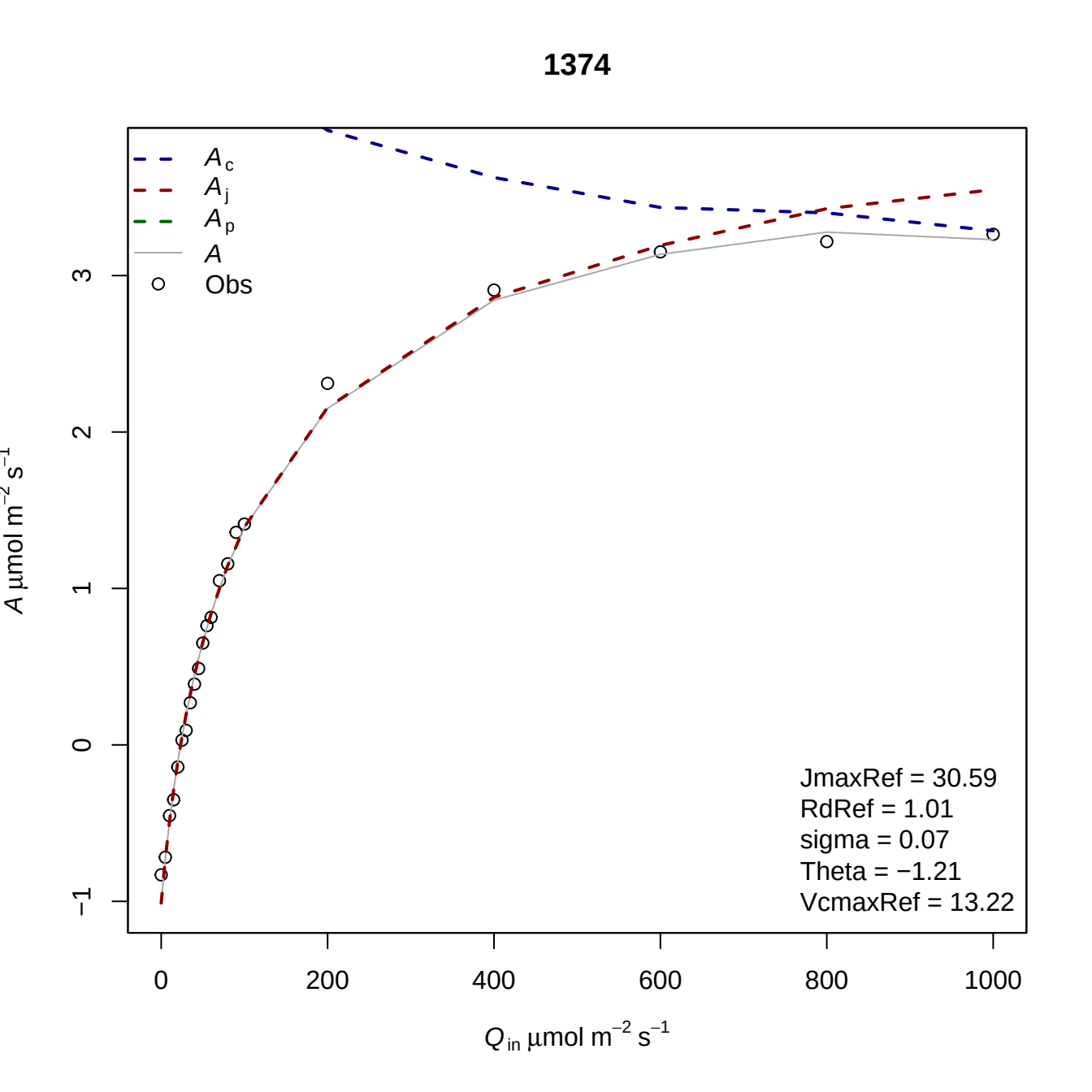

1369
A_c
A_j
A_p
A
Obs
JmaxRef = 28.41
RdRef = 1.35
sigma = 0.09
Theta = 0.64
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

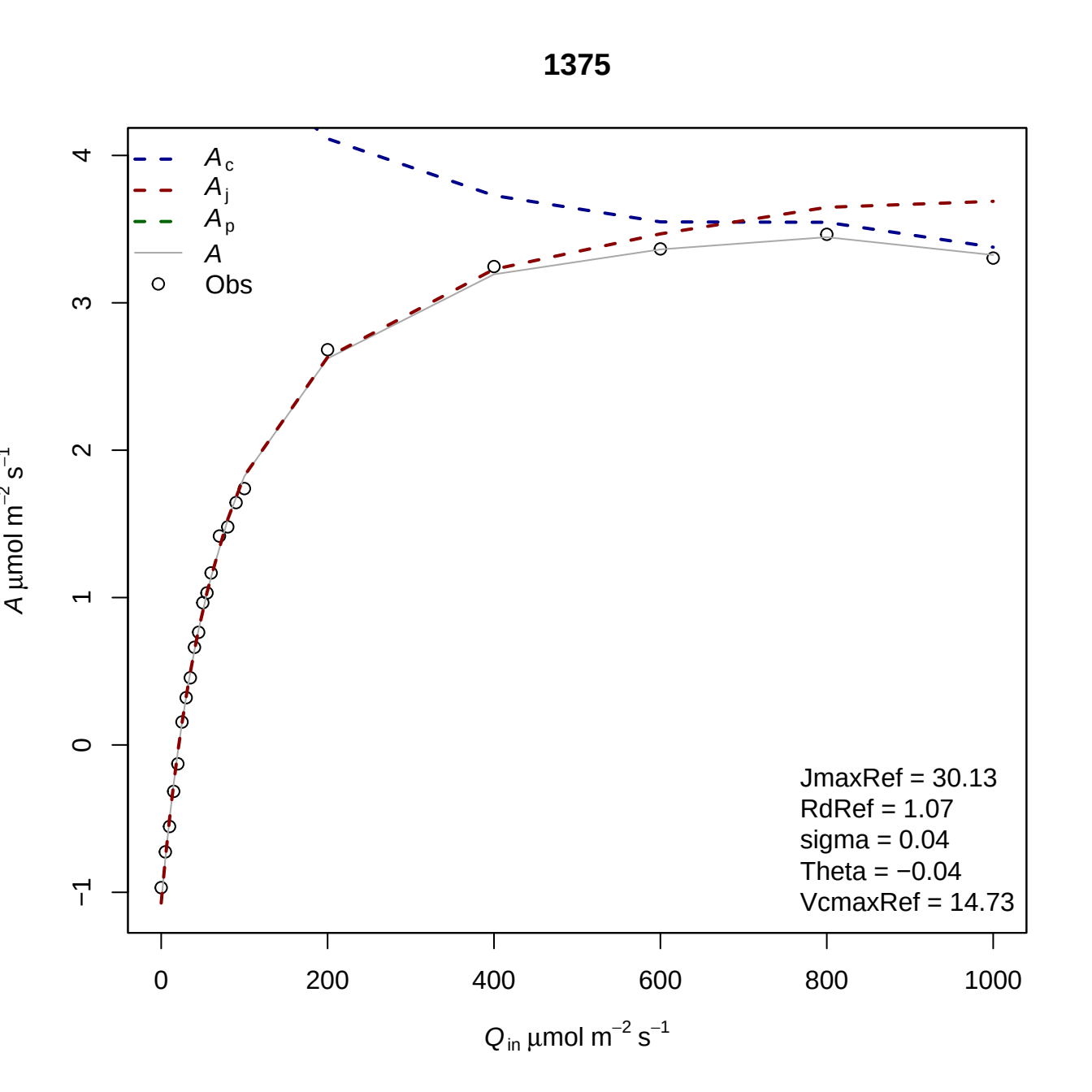

1370
A_c
A_j
A_p
A
Obs
JmaxRef = 35.91
RdRef = 0.95
sigma = 0.11
Theta = −0.56
VcmaxRef = 17.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

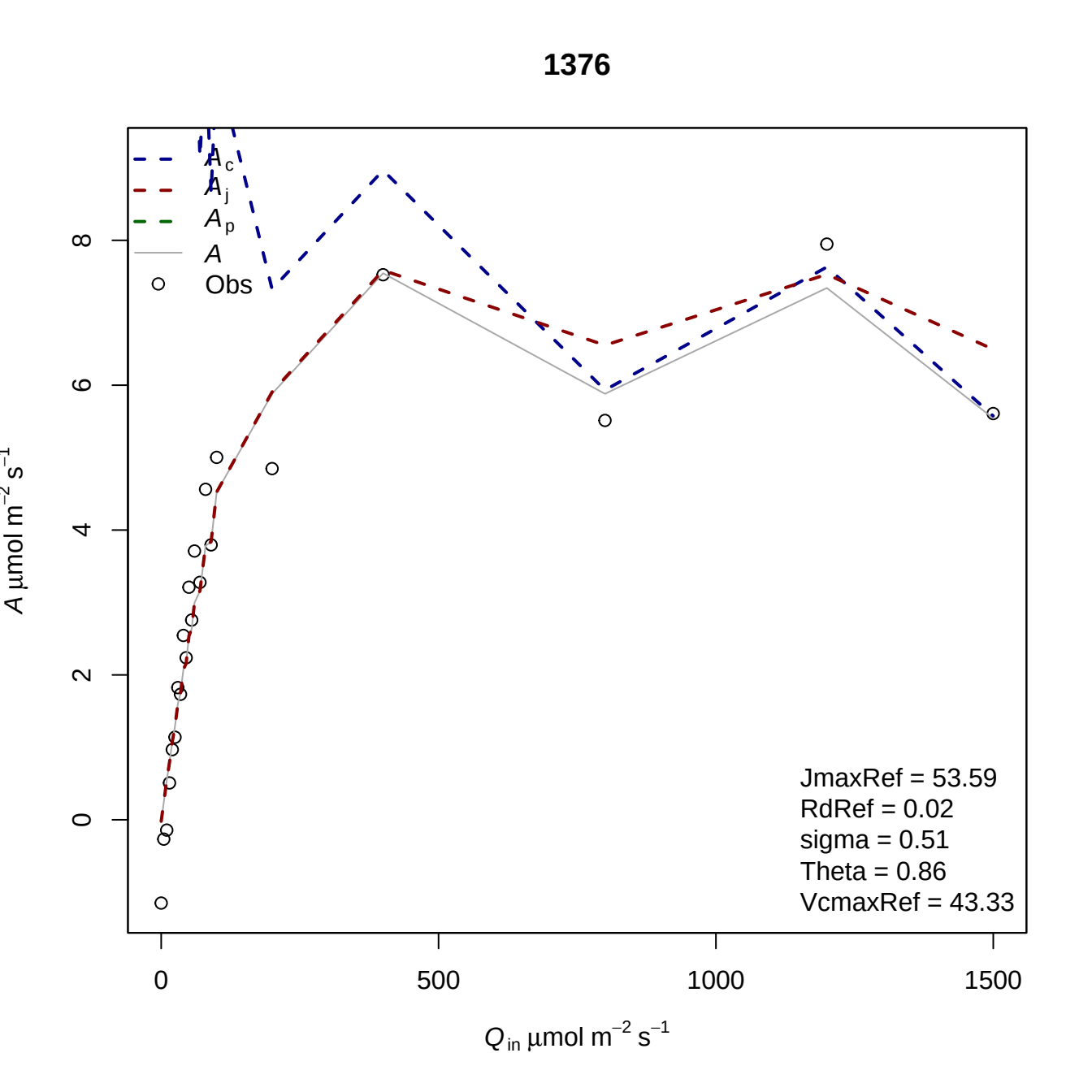

1371
A_c
A_j
A_p
A
Obs
JmaxRef = 36.78
RdRef = 1.03
sigma = 0.11
Theta = 0.24
VcmaxRef = 23.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

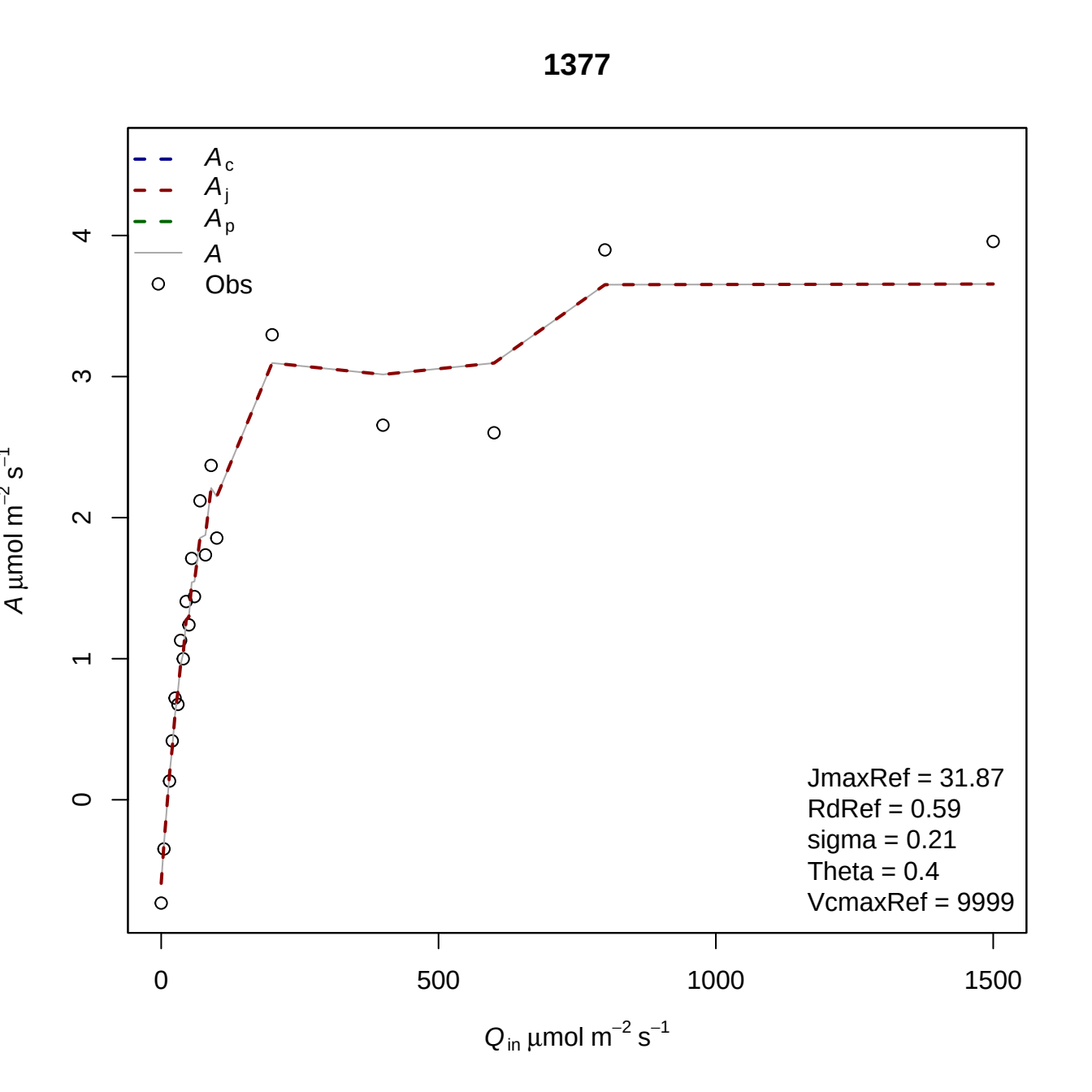

1372
A_c
A_j
A_p
A
Obs
JmaxRef = 43.03
RdRef = 0.46
sigma = 0.14
Theta = 0.07
VcmaxRef = 24.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
200
400
600
800
1000

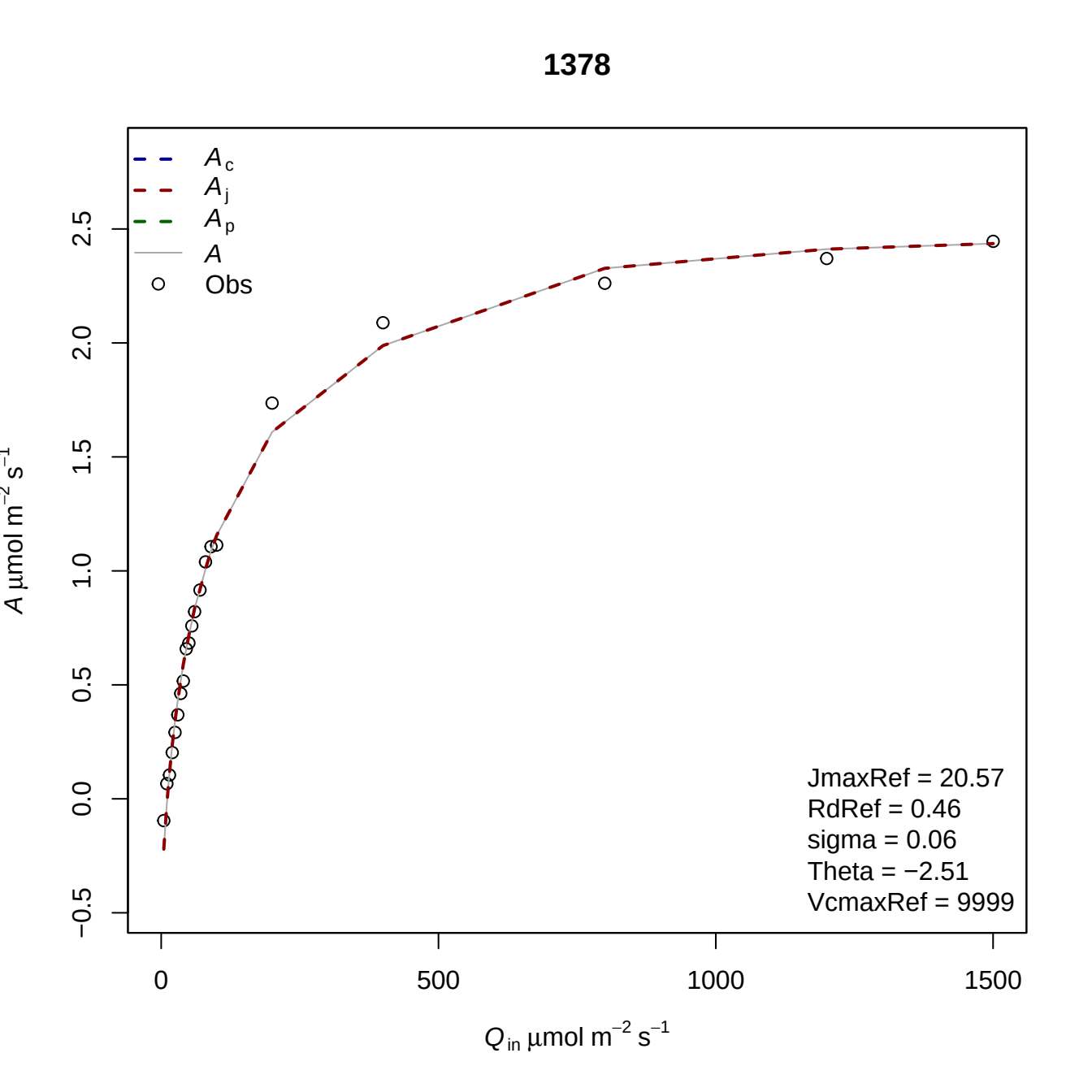

1373
A_c
A_j
A_p
A
Obs
JmaxRef = 20.19
RdRef = 0.33
sigma = 0.06
Theta = 0.19
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

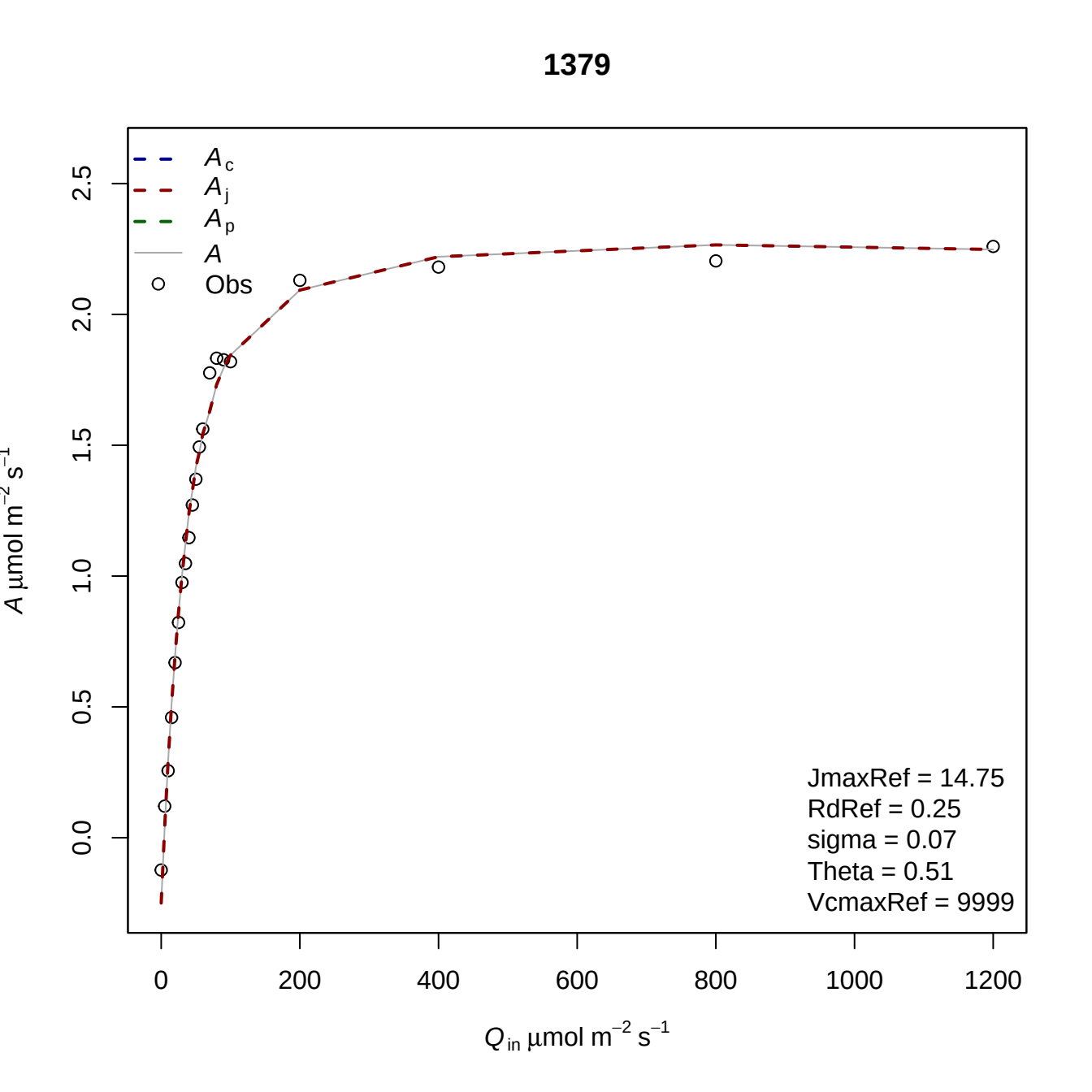

1374
A_c
A_j
A_p
A
Obs
JmaxRef = 30.59
RdRef = 1.01
sigma = 0.07
Theta = −1.21
VcmaxRef = 13.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

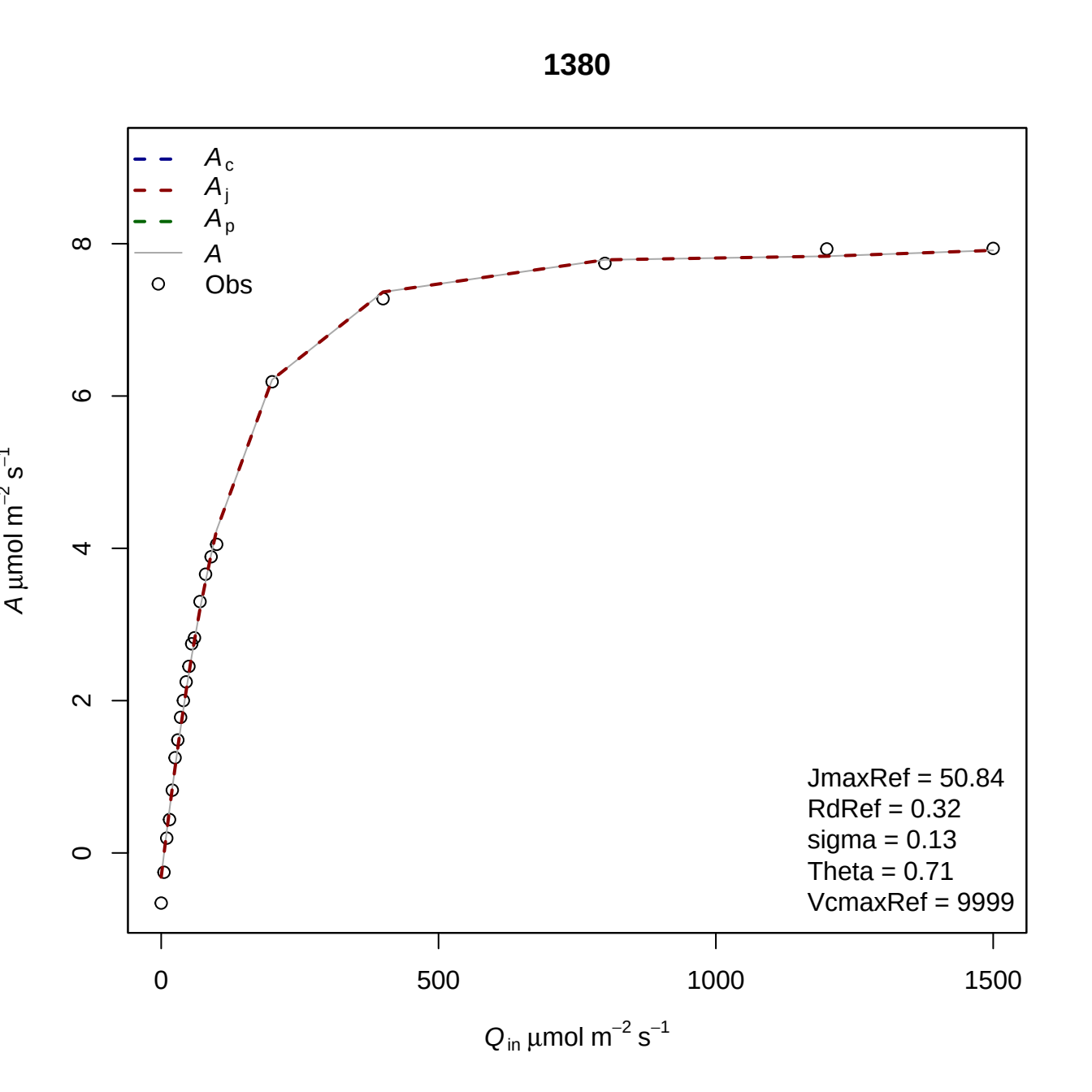

1375
A_c
A_j
A_p
A
Obs
JmaxRef = 30.13
RdRef = 1.07
sigma = 0.04
Theta = −0.04
VcmaxRef = 14.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

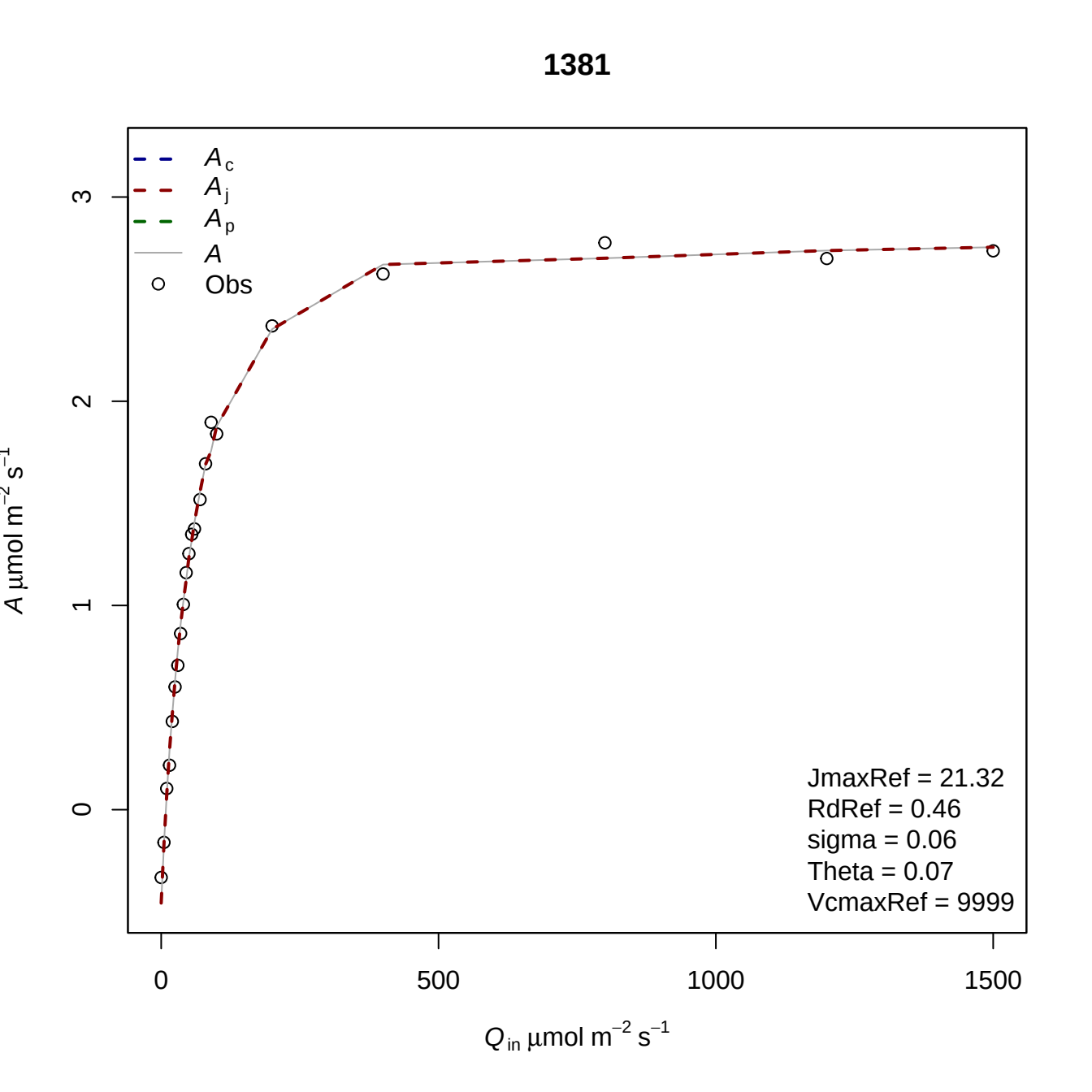

1376
A_c
A_j
A_p
A
Obs
JmaxRef = 53.59
RdRef = 0.02
sigma = 0.51
Theta = 0.86
VcmaxRef = 43.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

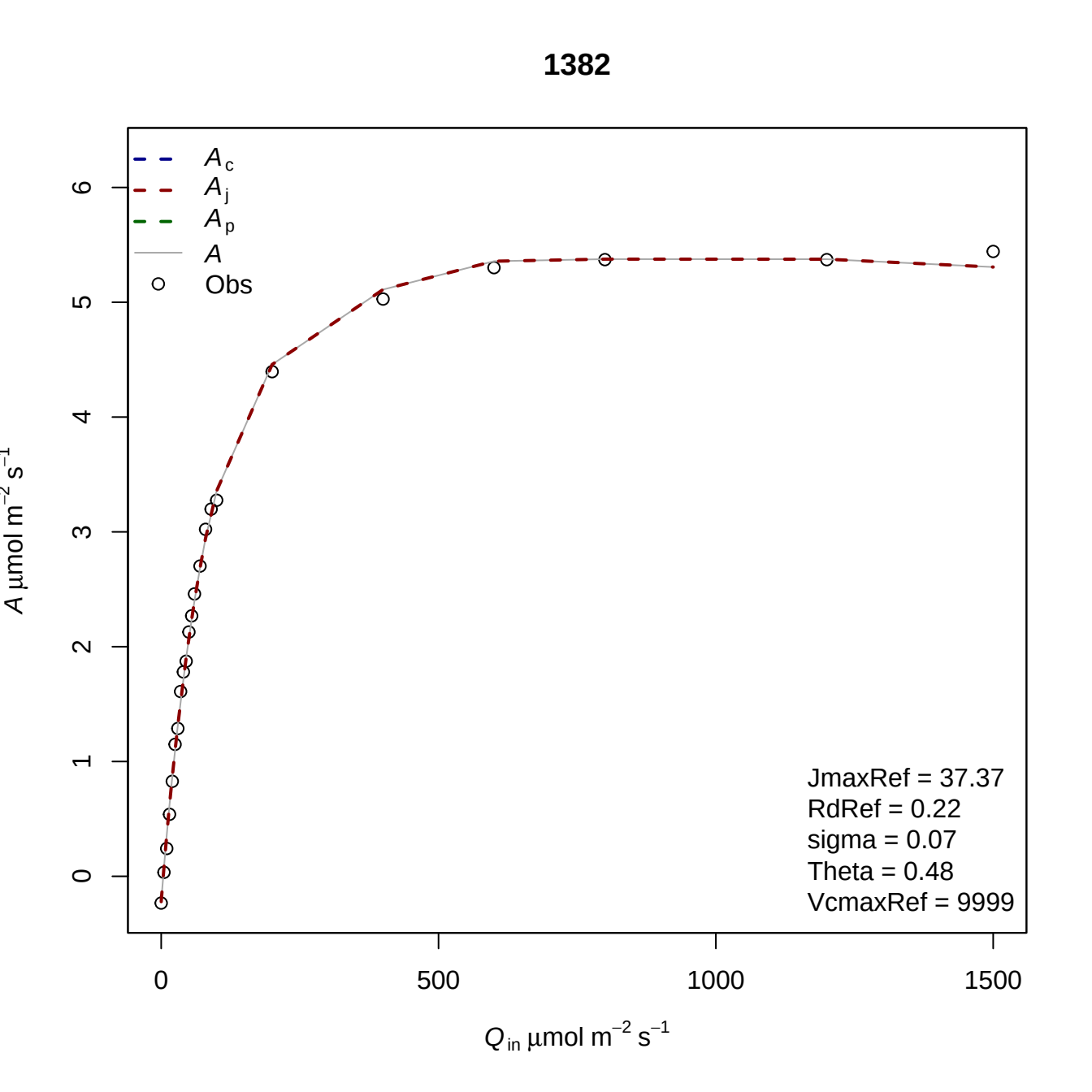

1377
A_c
A_j
A_p
A
Obs
JmaxRef = 31.87
RdRef = 0.59
sigma = 0.21
Theta = 0.4
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

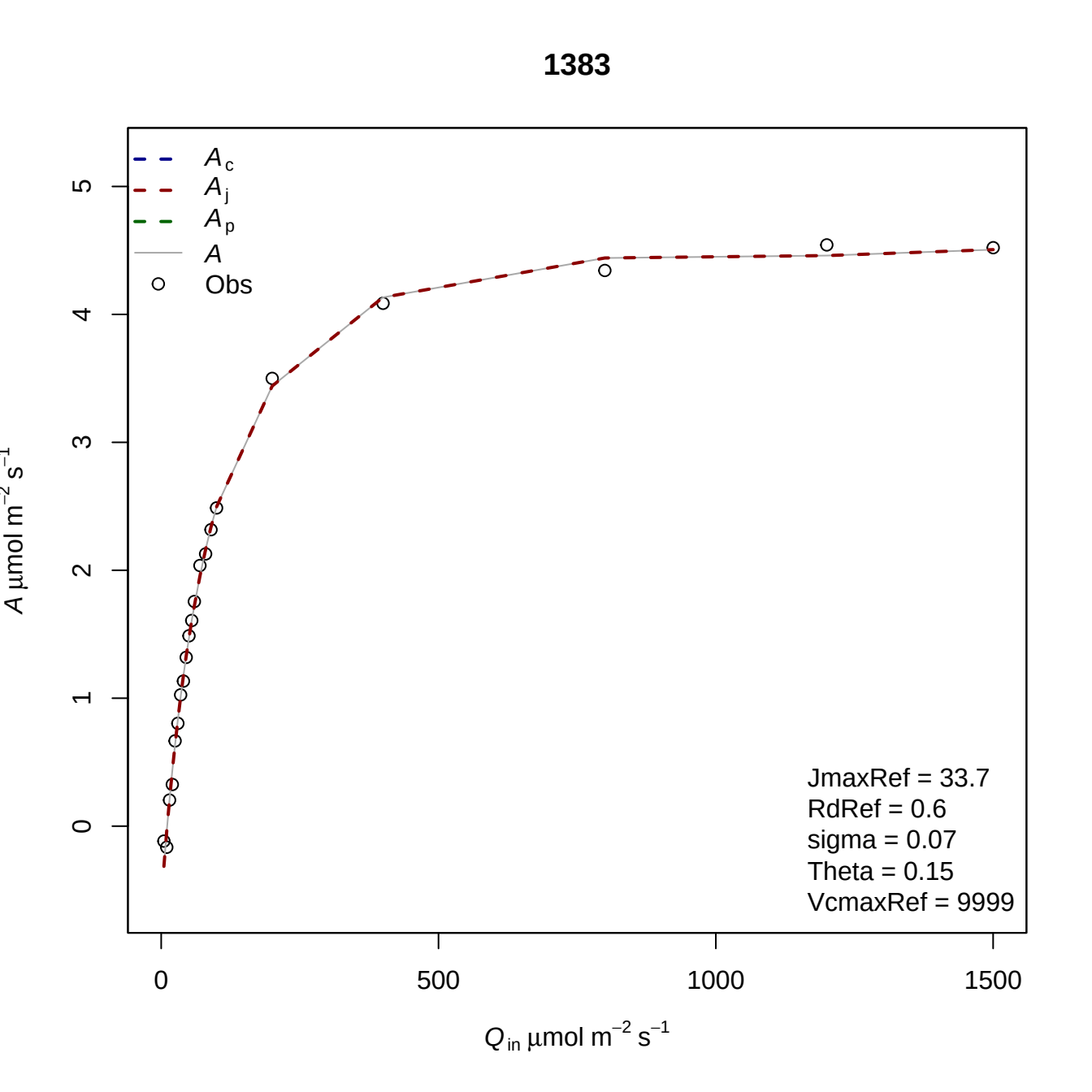

1378
A_c
A_j
A_p
A
Obs
JmaxRef = 20.57
RdRef = 0.46
sigma = 0.06
Theta = −2.51
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500

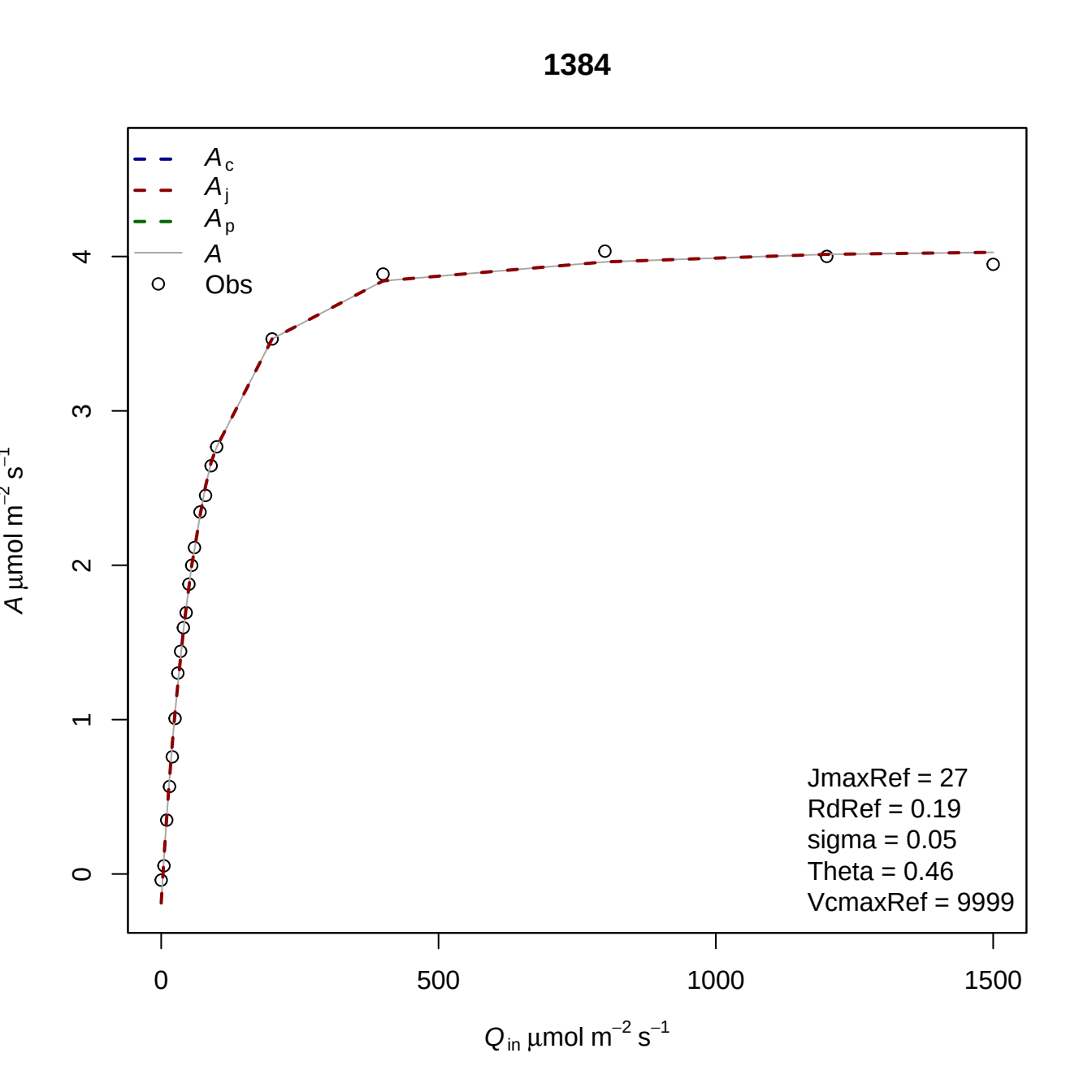

1379
A_c
A_j
A_p
A
Obs
JmaxRef = 14.75
RdRef = 0.25
sigma = 0.07
Theta = 0.51
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
0
200
400
600
800
1000
1200

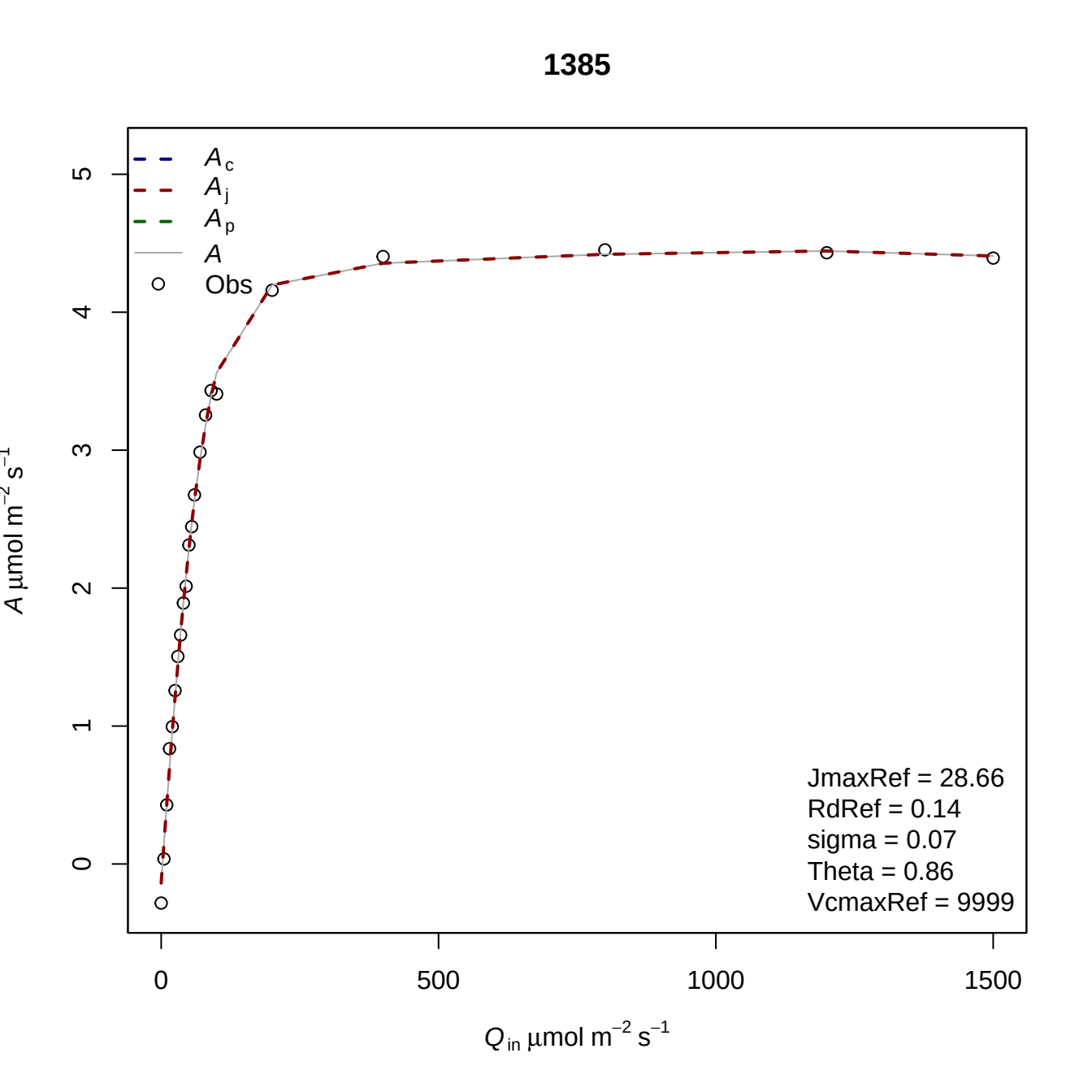

1380
A_c
A_j
A_p
A
Obs
JmaxRef = 50.84
RdRef = 0.32
sigma = 0.13
Theta = 0.71
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

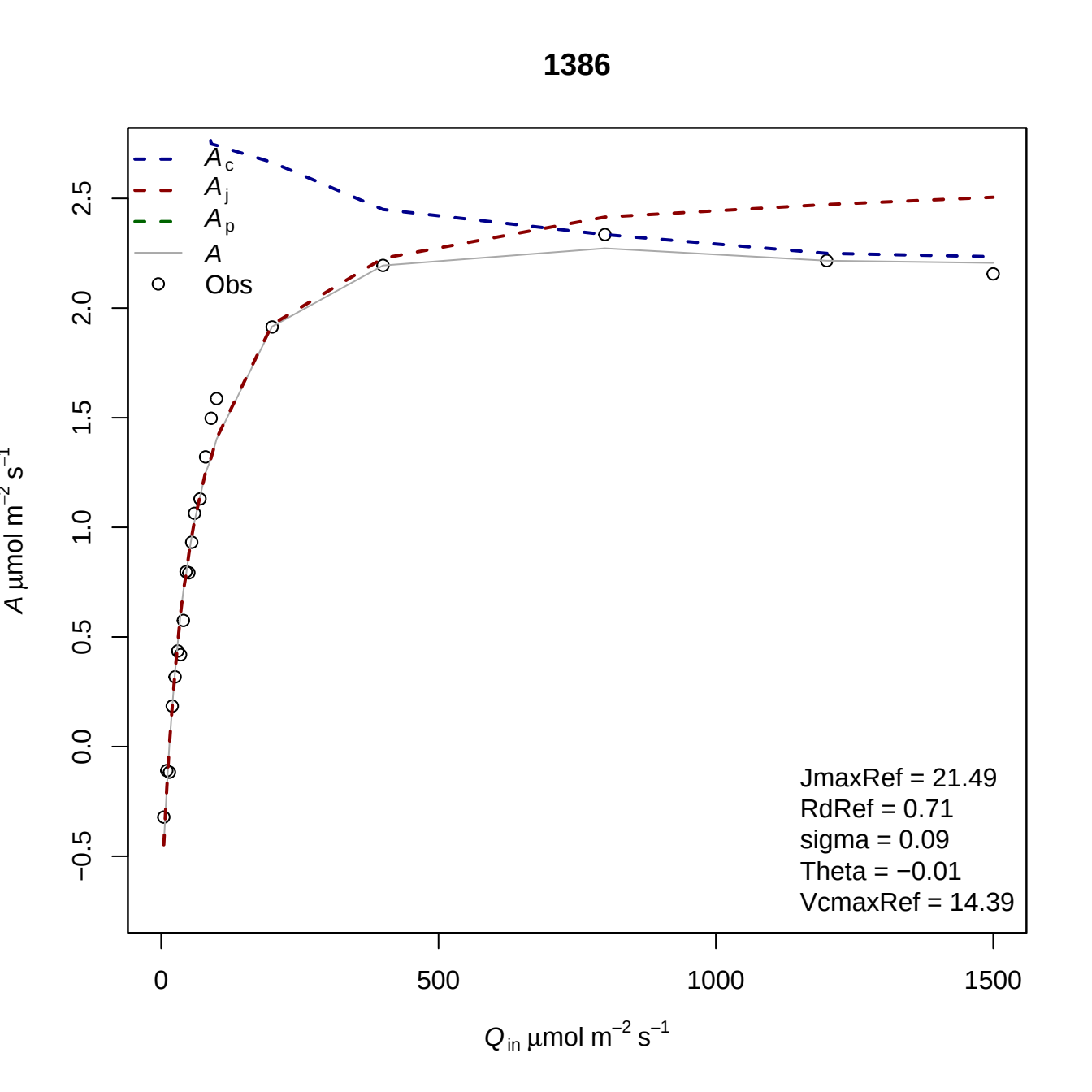

1381
A_c
A_j
A_p
A
Obs
JmaxRef = 21.32
RdRef = 0.46
sigma = 0.06
Theta = 0.07
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

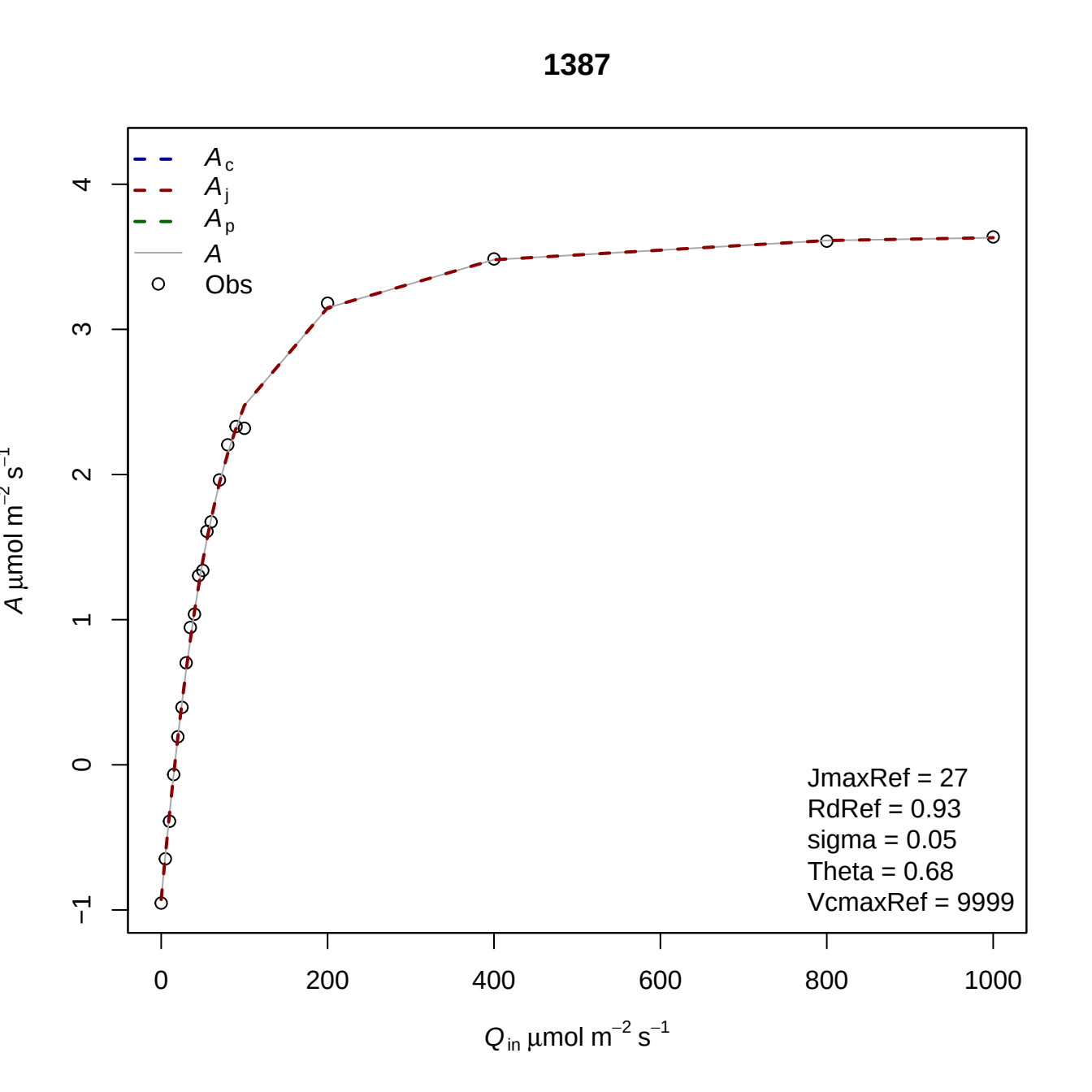

1382
A_c
A_j
A_p
A
Obs
JmaxRef = 37.37
RdRef = 0.22
sigma = 0.07
Theta = 0.48
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

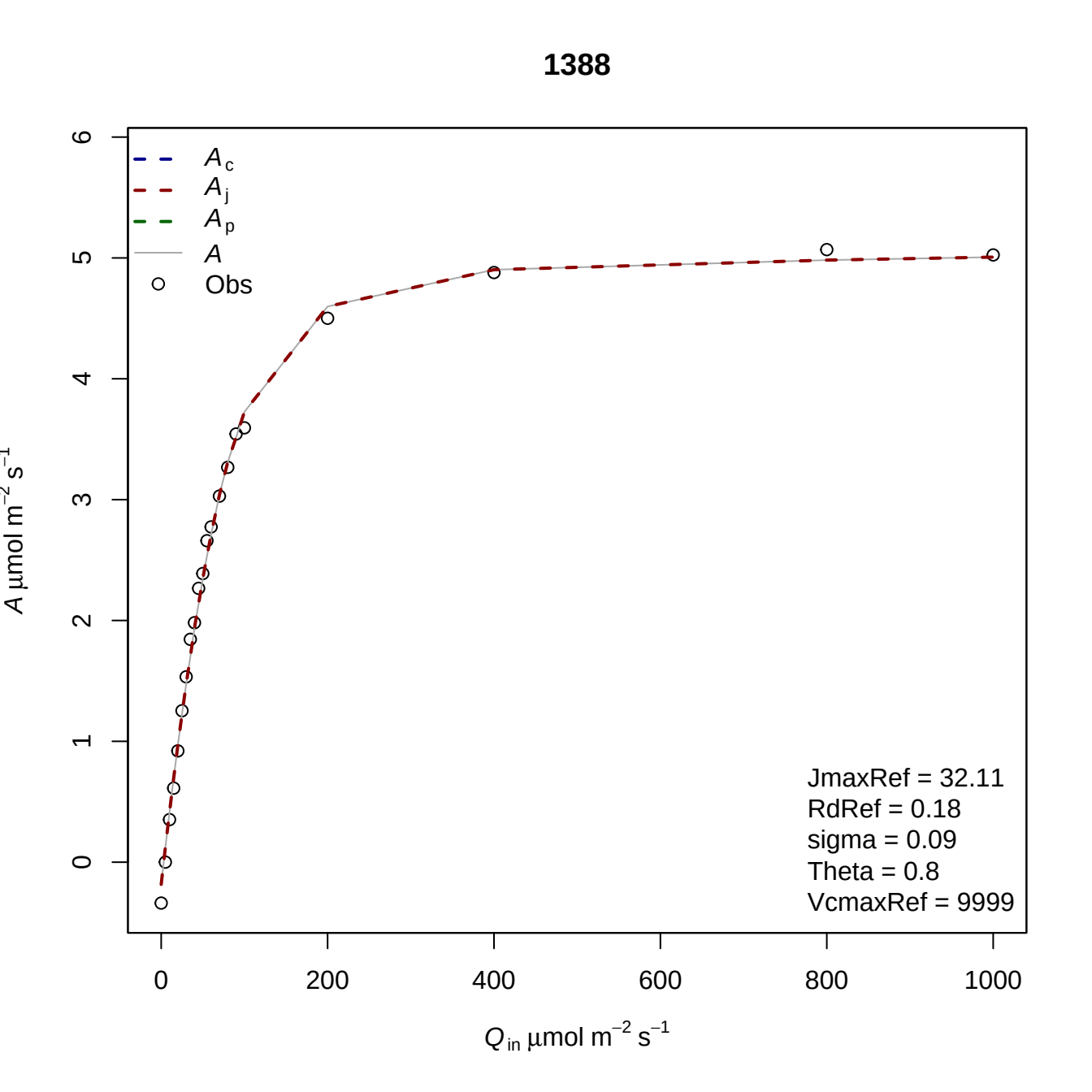

1383
A_c
A_j
A_p
A
Obs
JmaxRef = 33.7
RdRef = 0.6
sigma = 0.07
Theta = 0.15
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

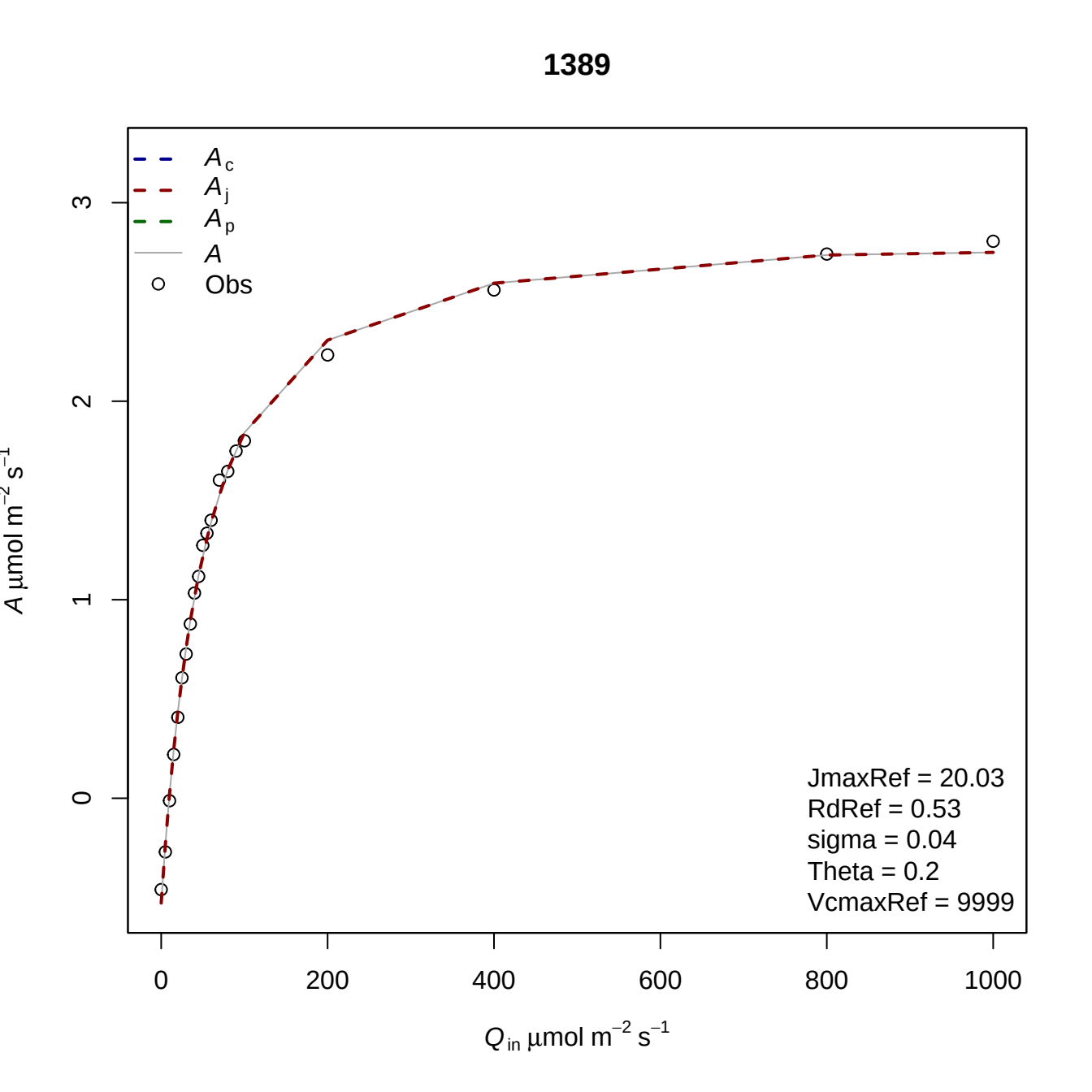

1384
A_c
A_j
A_p
A
Obs
JmaxRef = 27
RdRef = 0.19
sigma = 0.05
Theta = 0.46
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

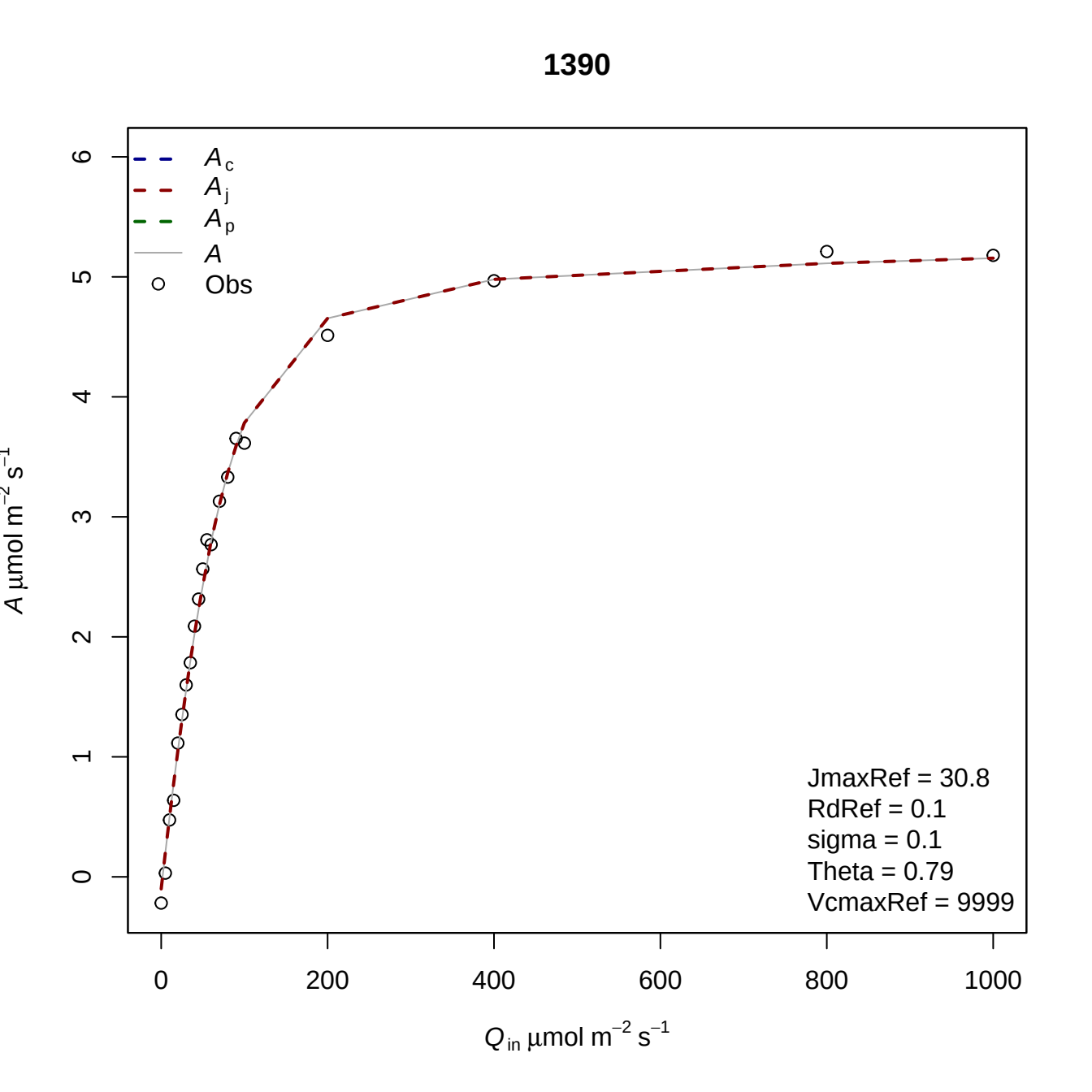

1385
A_c
A_j
A_p
A
Obs
JmaxRef = 28.66
RdRef = 0.14
sigma = 0.07
Theta = 0.86
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

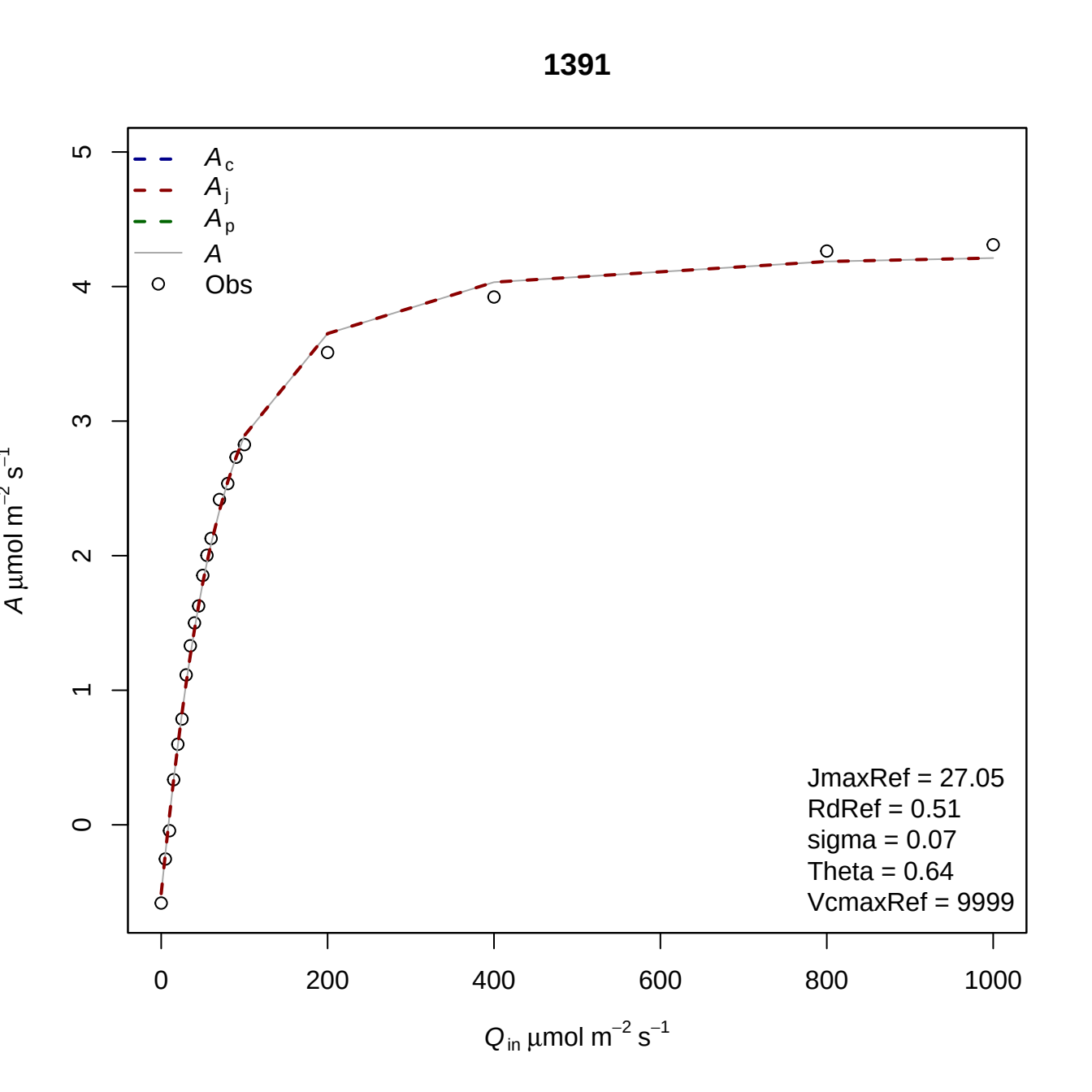

1386
A_c
A_j
A_p
A
Obs
JmaxRef = 21.49
RdRef = 0.71
sigma = 0.09
Theta = −0.01
VcmaxRef = 14.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500

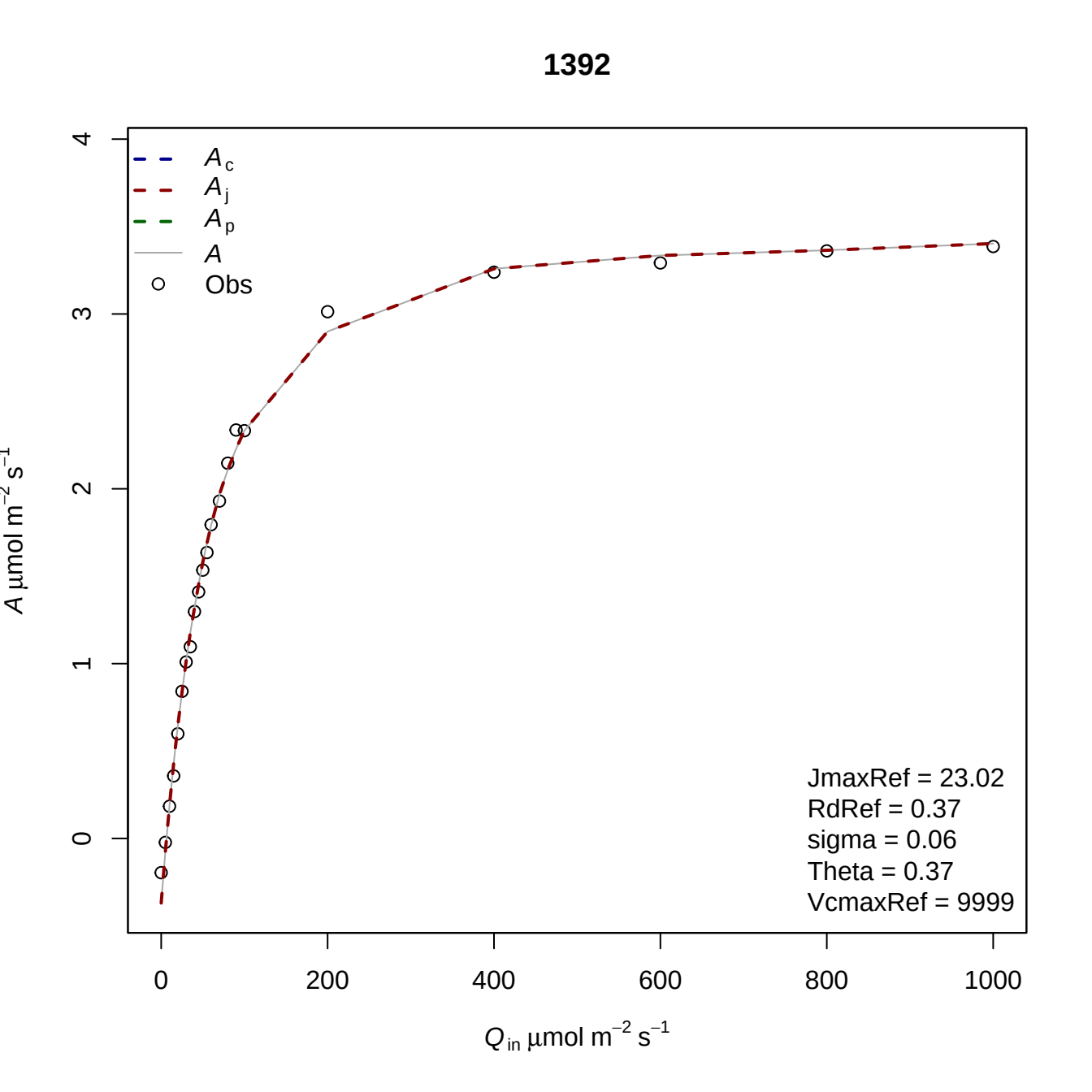

1387
A_c
A_j
A_p
A
Obs
JmaxRef = 27
RdRef = 0.93
sigma = 0.05
Theta = 0.68
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

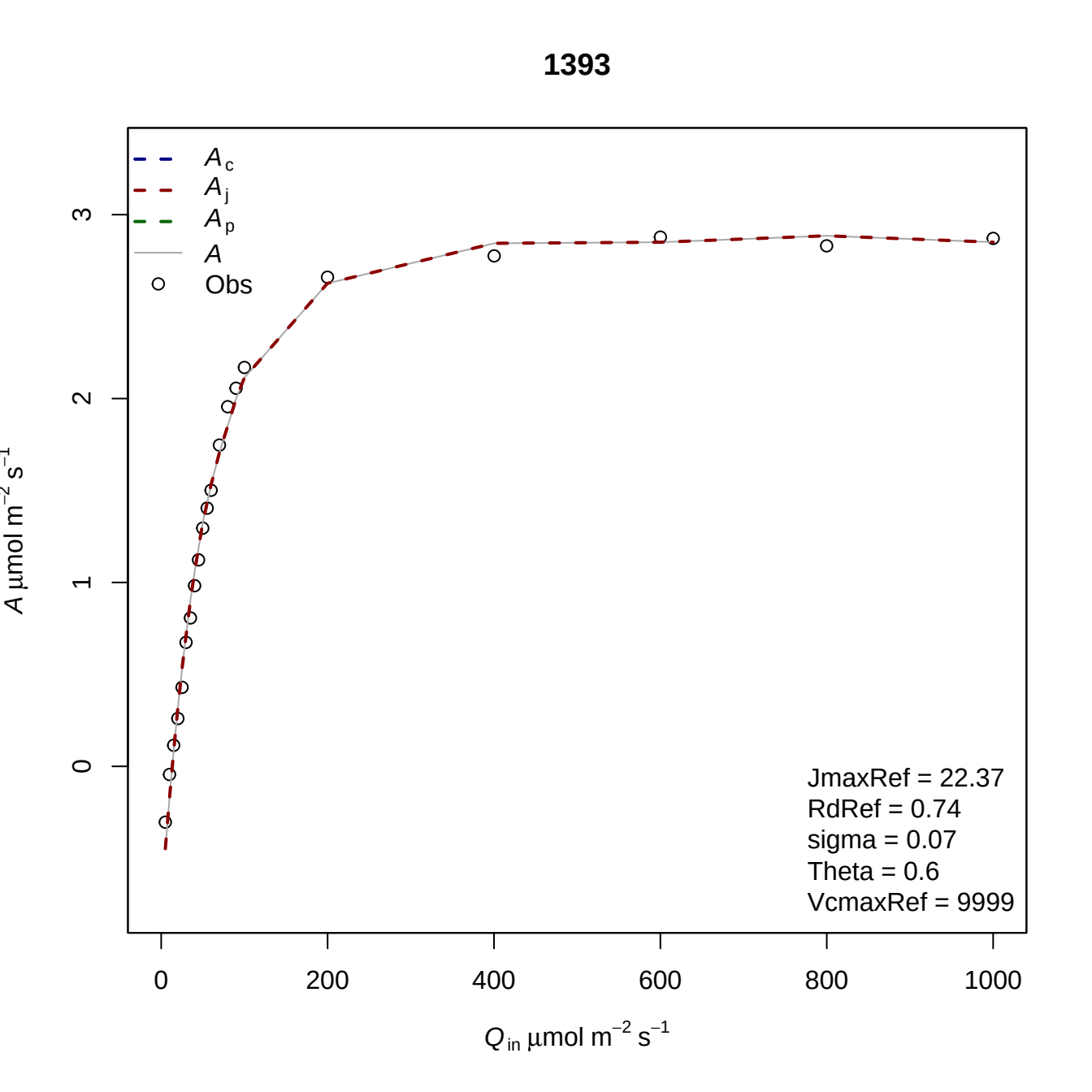

1388
A_c
A_j
A_p
A
Obs
JmaxRef = 32.11
RdRef = 0.18
sigma = 0.09
Theta = 0.8
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

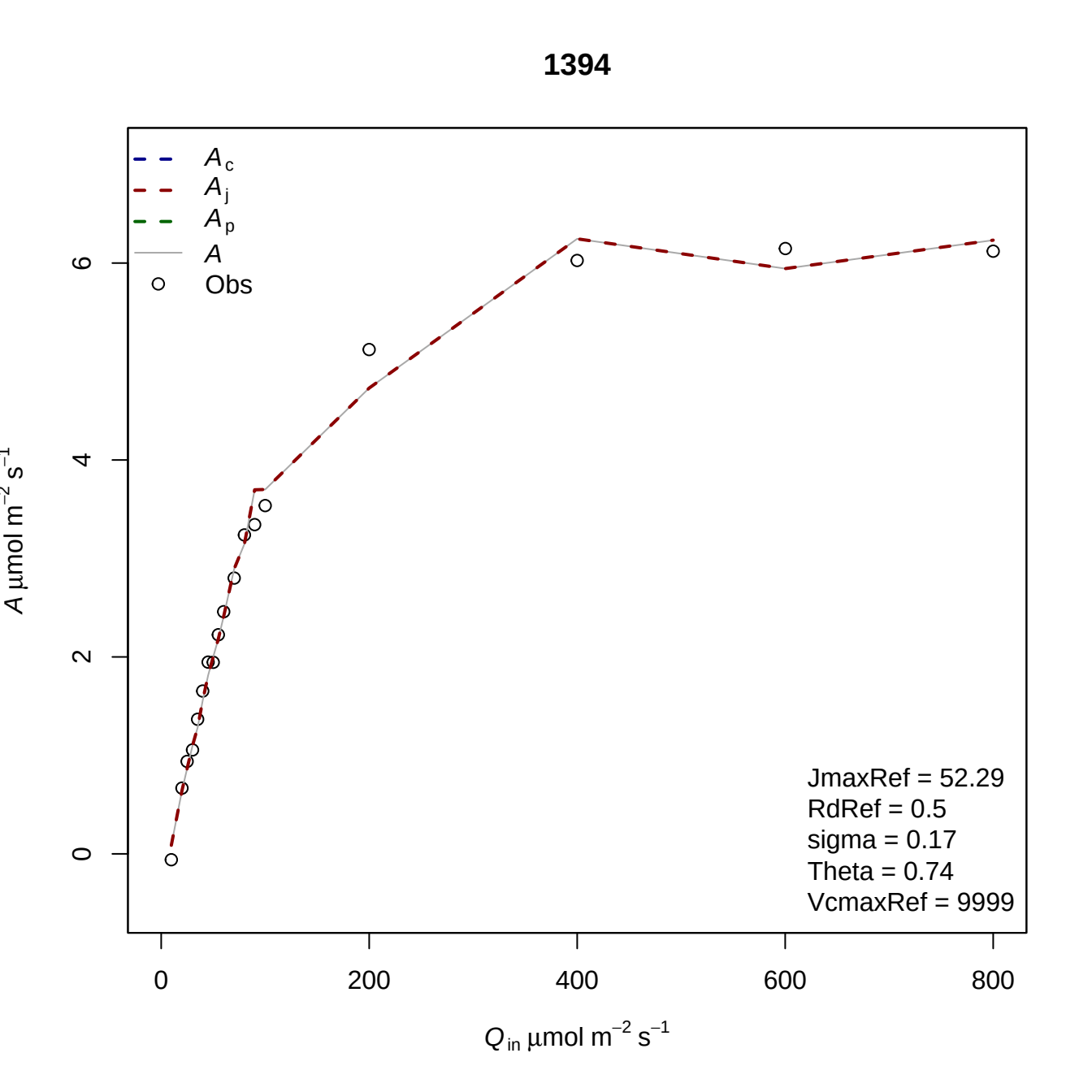

1389
A_c
A_j
A_p
A
Obs
JmaxRef = 20.03
RdRef = 0.53
sigma = 0.04
Theta = 0.2
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

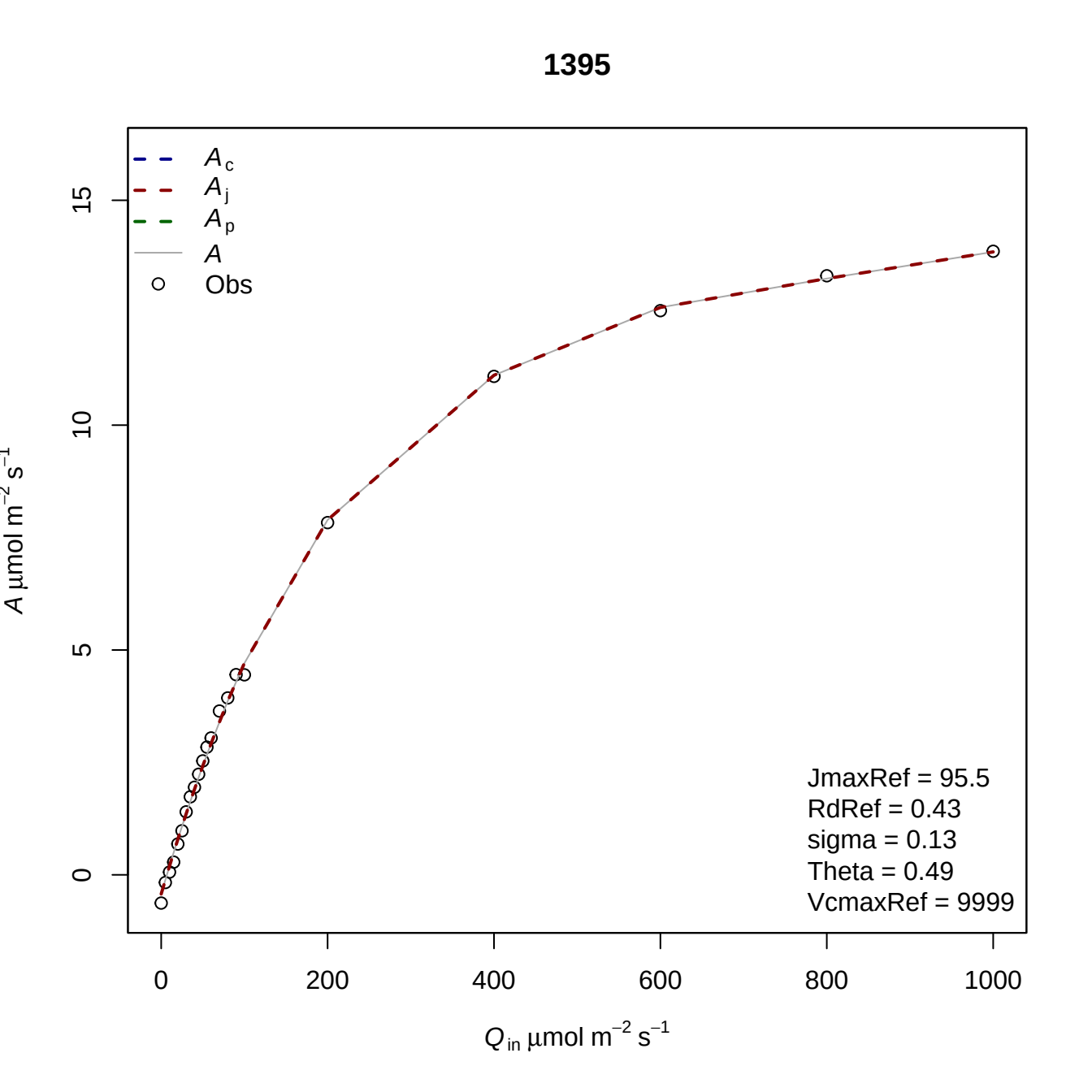

1390
A_c
A_j
A_p
A
Obs
JmaxRef = 30.8
RdRef = 0.1
sigma = 0.1
Theta = 0.79
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

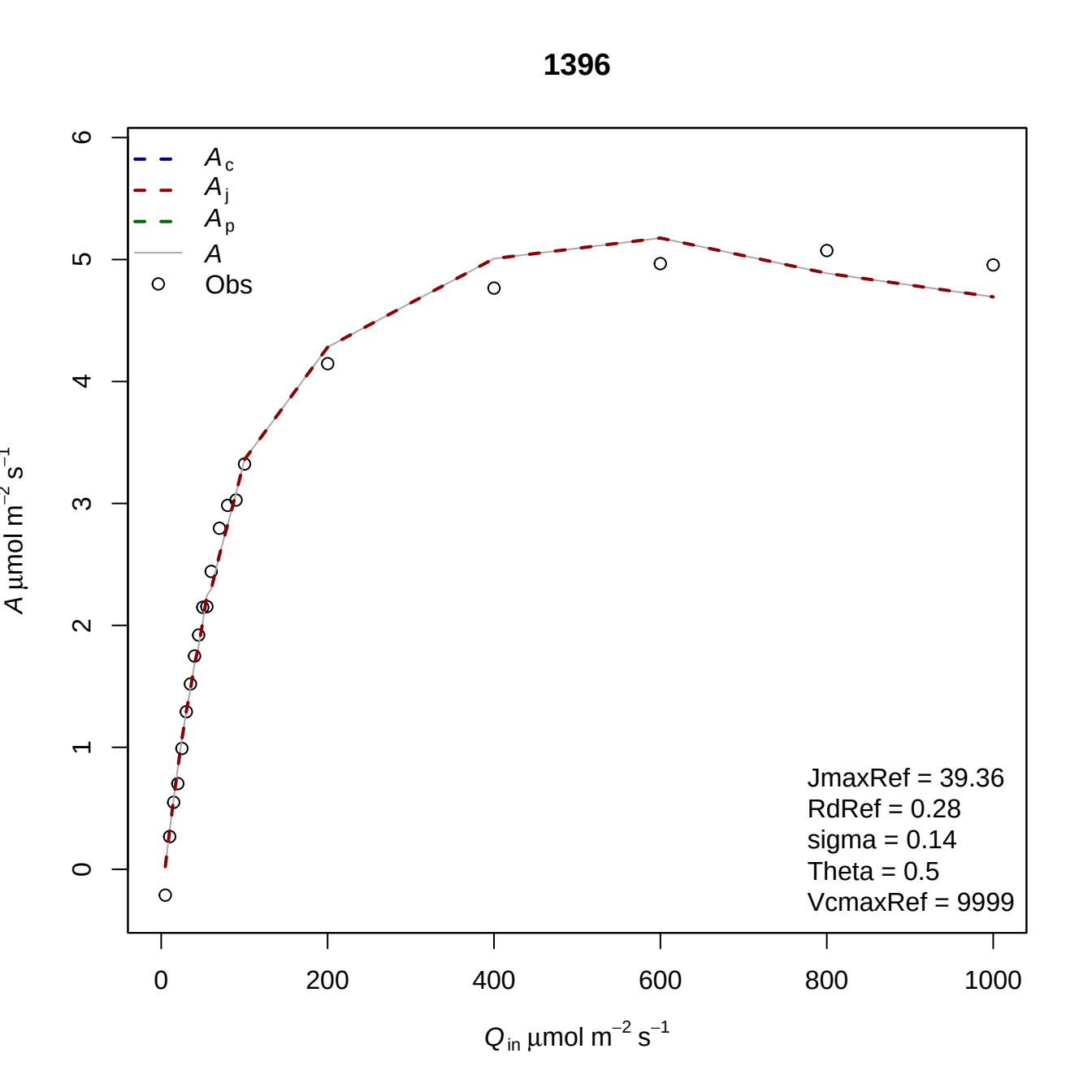

1391
A_c
A_j
A_p
A
Obs
JmaxRef = 27.05
RdRef = 0.51
sigma = 0.07
Theta = 0.64
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
200
400
600
800
1000

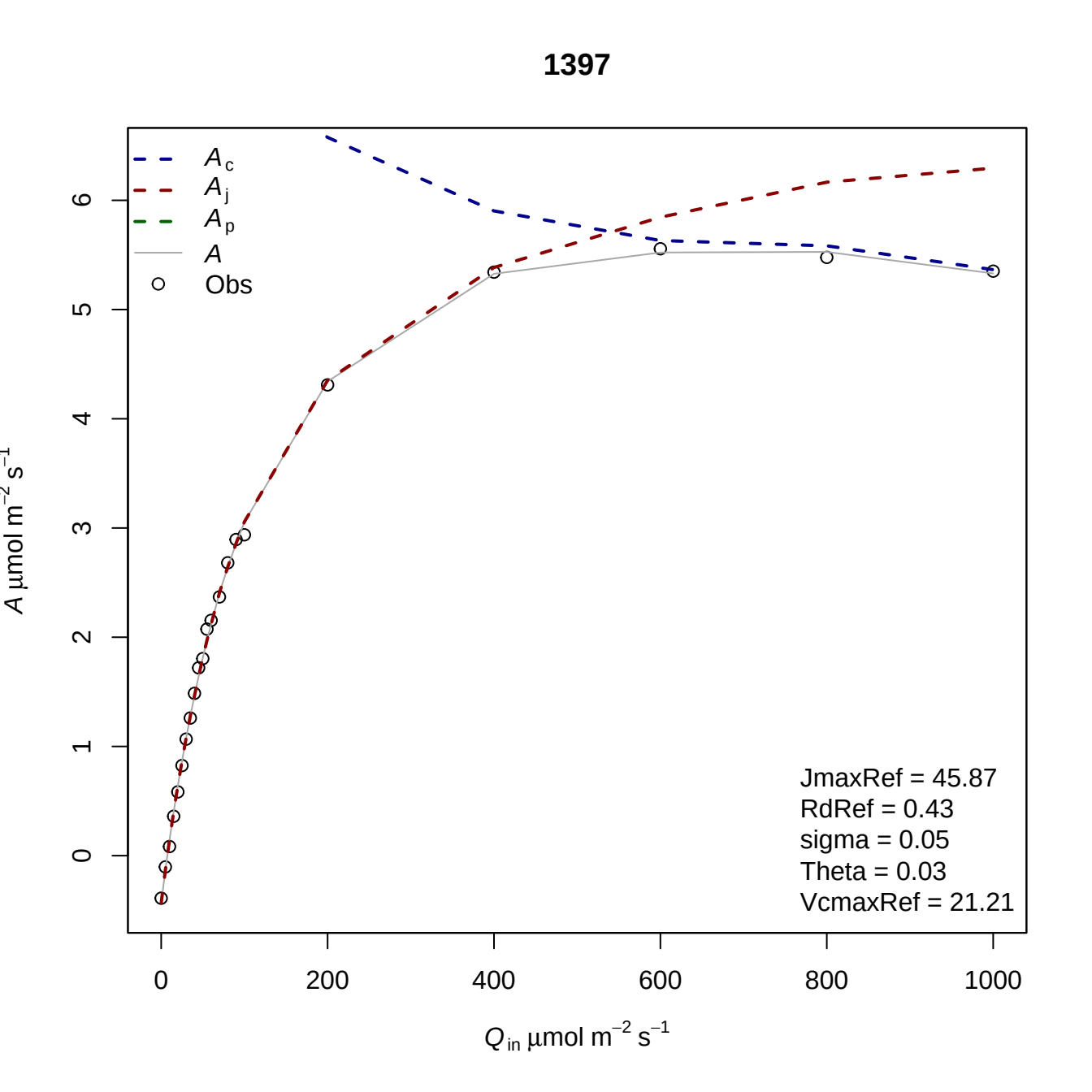

1392
A_c
A_j
A_p
A
Obs
JmaxRef = 23.02
RdRef = 0.37
sigma = 0.06
Theta = 0.37
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
200
400
600
800
1000

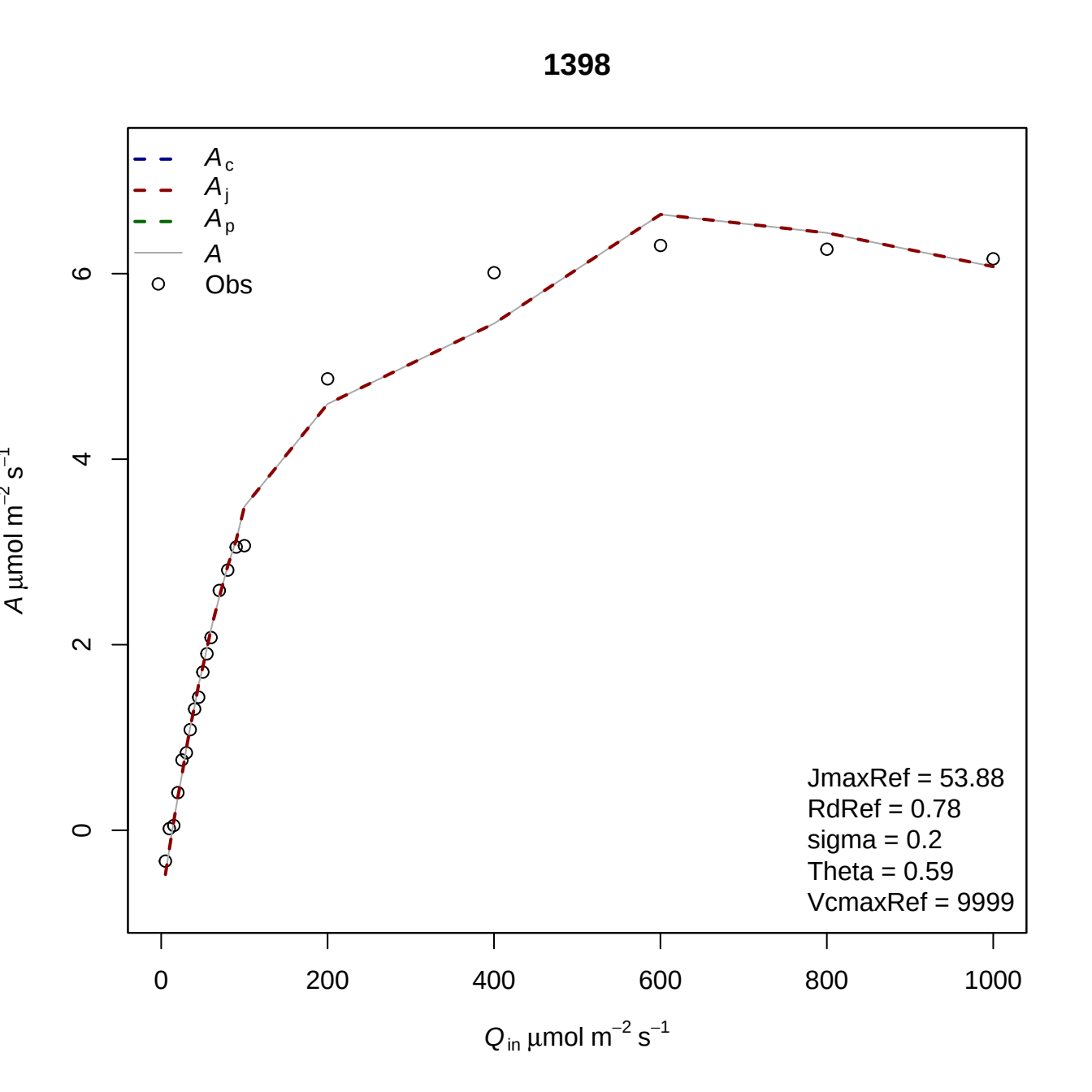

1393
A_c
A_j
A_p
A
Obs
JmaxRef = 22.37
RdRef = 0.74
sigma = 0.07
Theta = 0.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

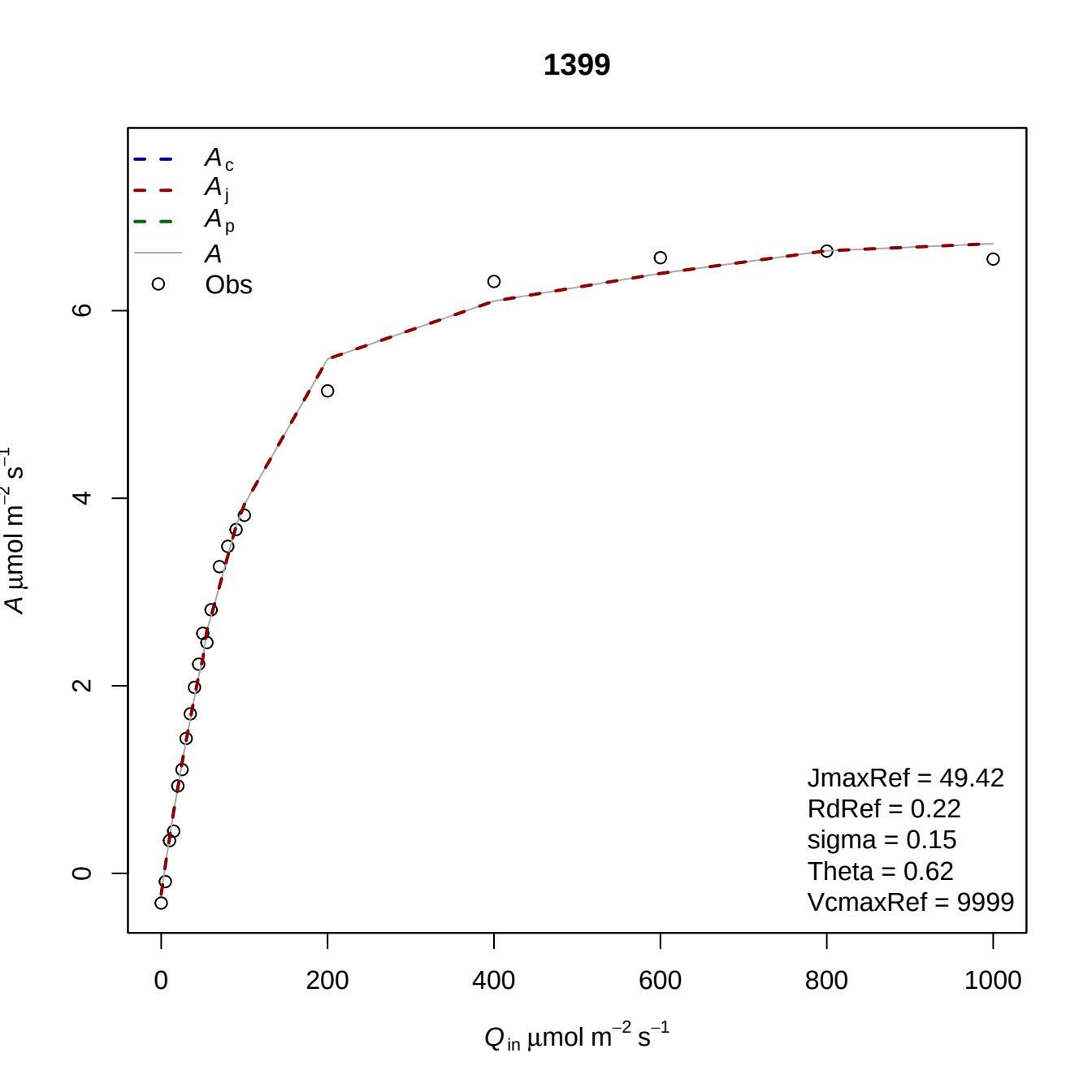

1394
A_c
A_j
A_p
A
Obs
JmaxRef = 52.29
RdRef = 0.5
sigma = 0.17
Theta = 0.74
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800

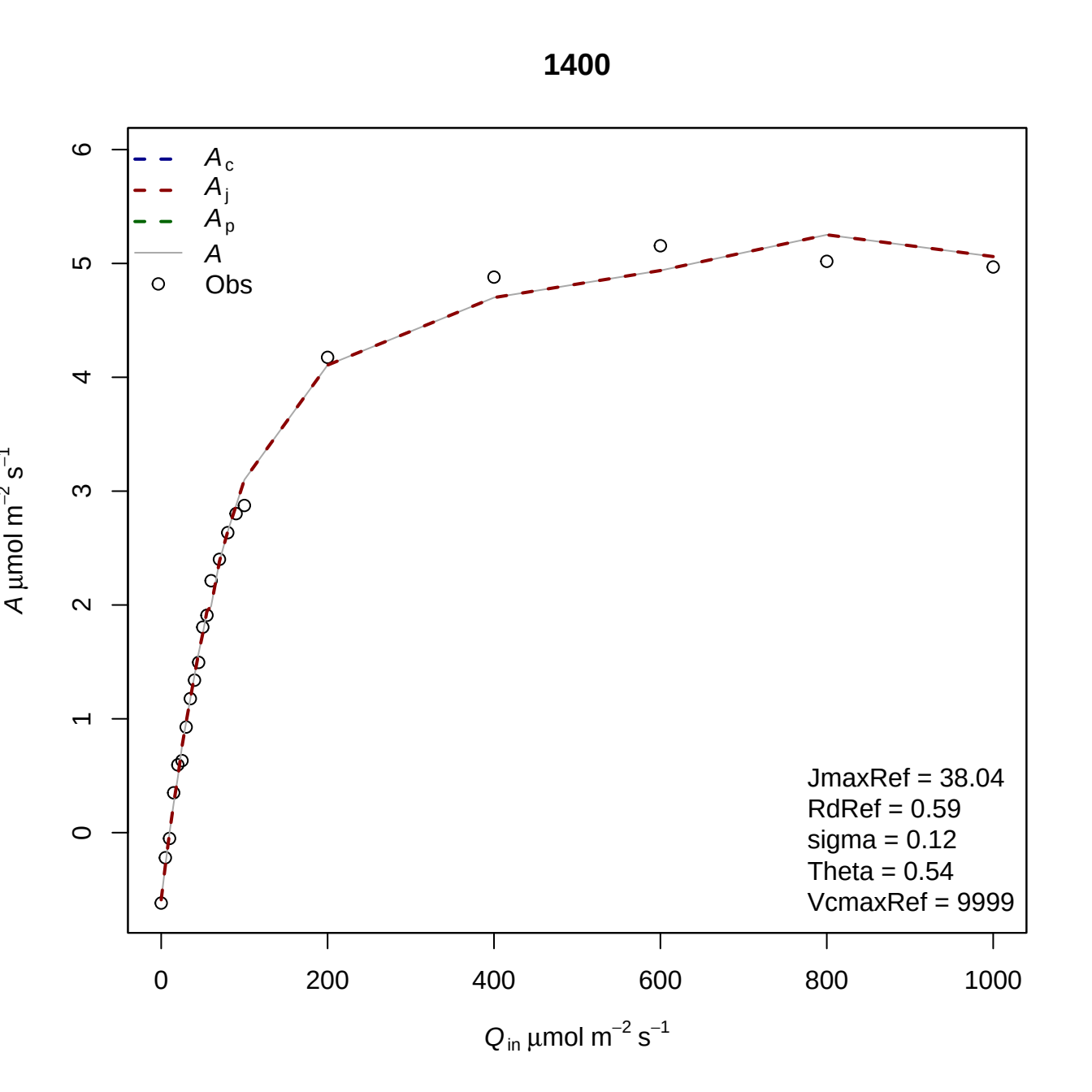

1395
A_c
A_j
A_p
A
Obs
JmaxRef = 95.5
RdRef = 0.43
sigma = 0.13
Theta = 0.49
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000

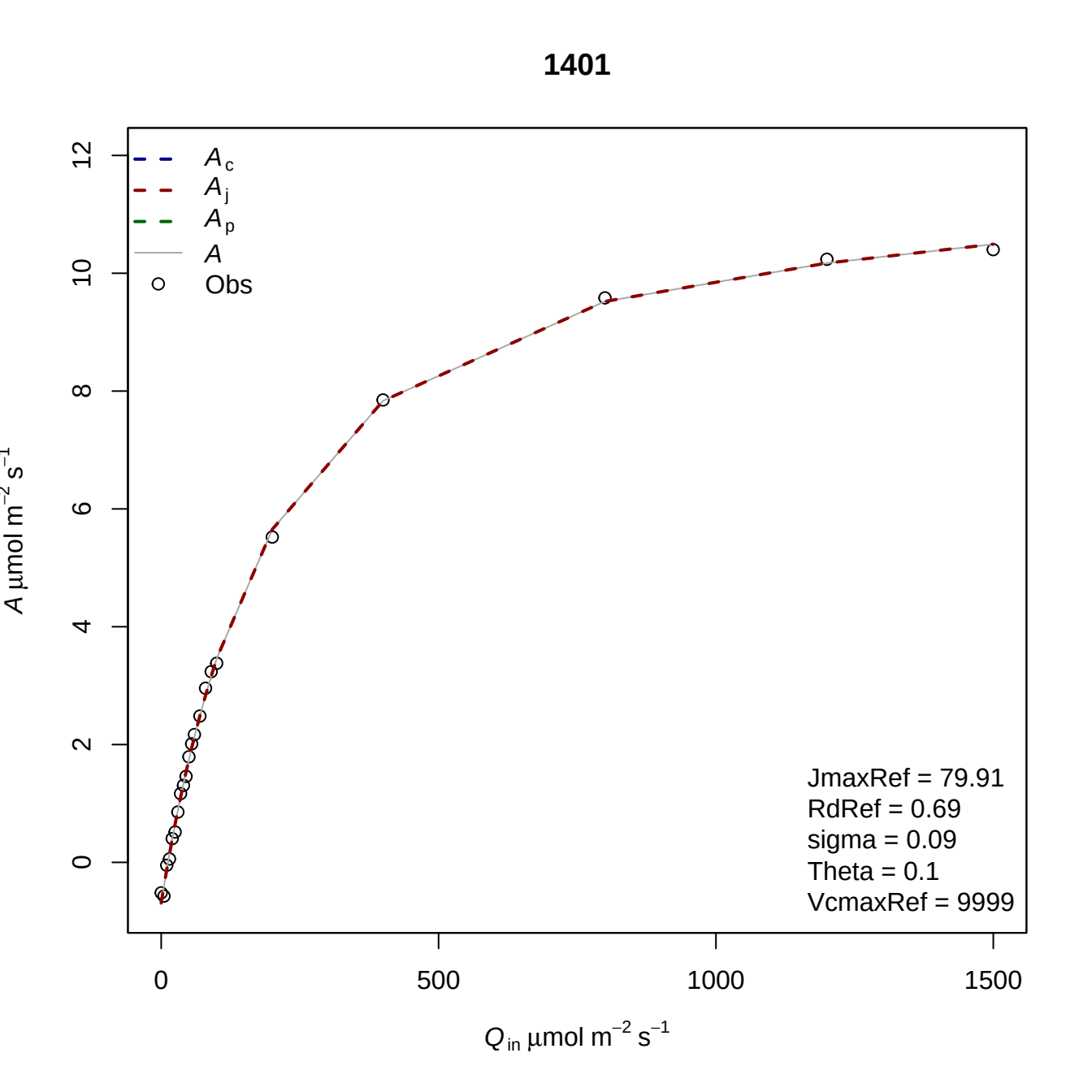

1396
A_c
A_j
A_p
A
Obs
JmaxRef = 39.36
RdRef = 0.28
sigma = 0.14
Theta = 0.5
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

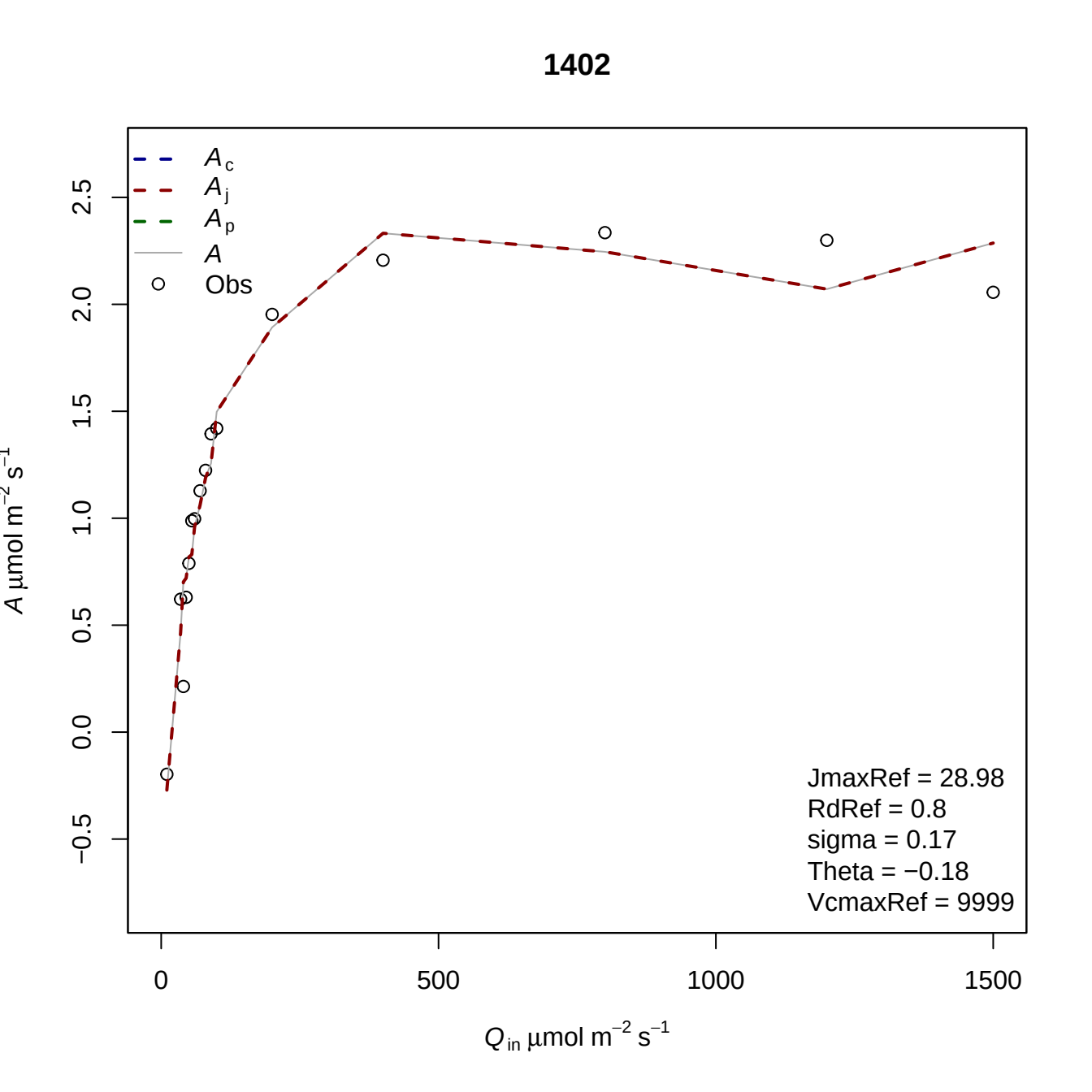

1397
A_c
A_j
A_p
A
Obs
JmaxRef = 45.87
RdRef = 0.43
sigma = 0.05
Theta = 0.03
VcmaxRef = 21.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

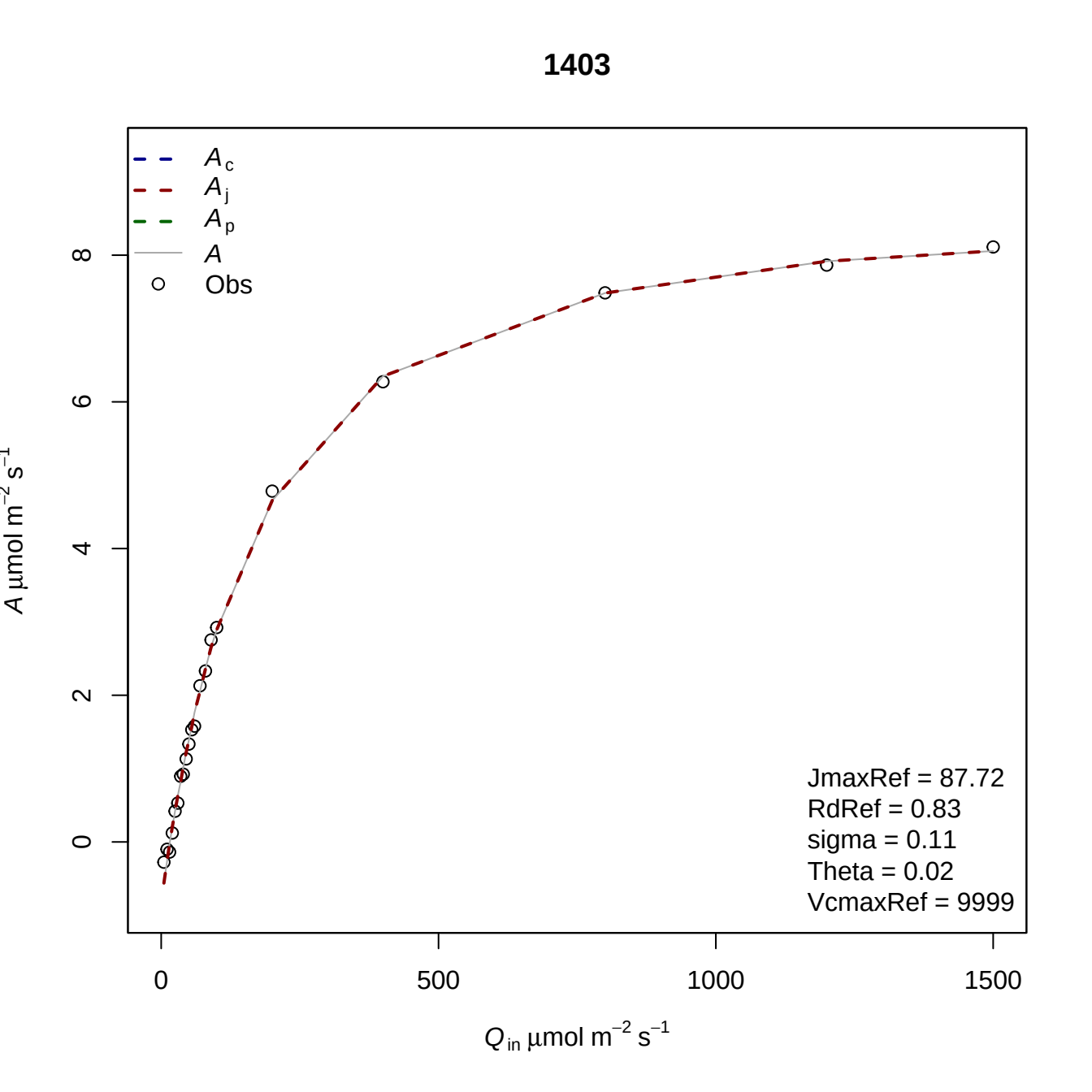

1398
A_c
A_j
A_p
A
Obs
JmaxRef = 53.88
RdRef = 0.78
sigma = 0.2
Theta = 0.59
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

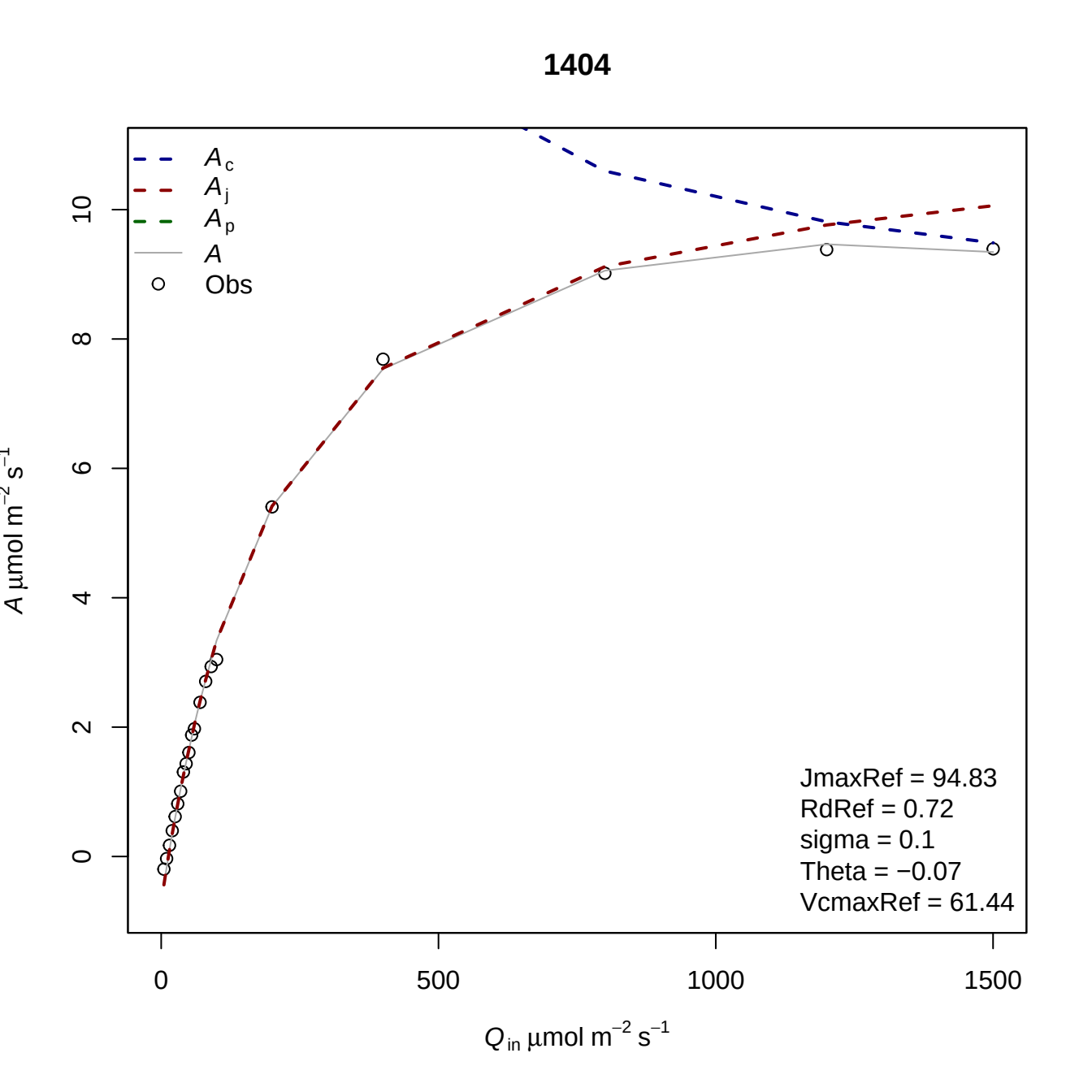

1399
A_c
A_j
A_p
A
Obs
JmaxRef = 49.42
RdRef = 0.22
sigma = 0.15
Theta = 0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

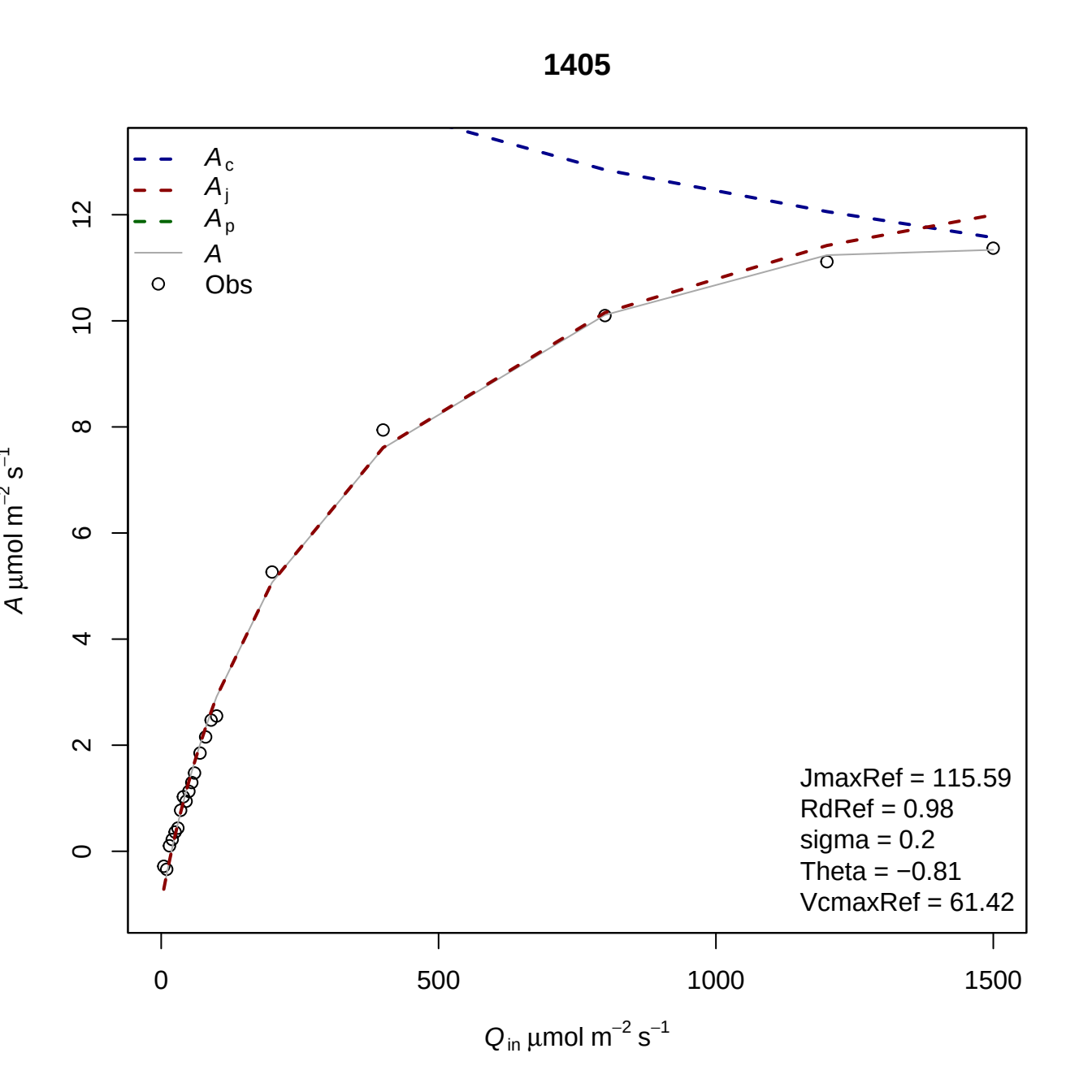

1400
A_c
A_j
A_p
A
Obs
JmaxRef = 38.04
RdRef = 0.59
sigma = 0.12
Theta = 0.54
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

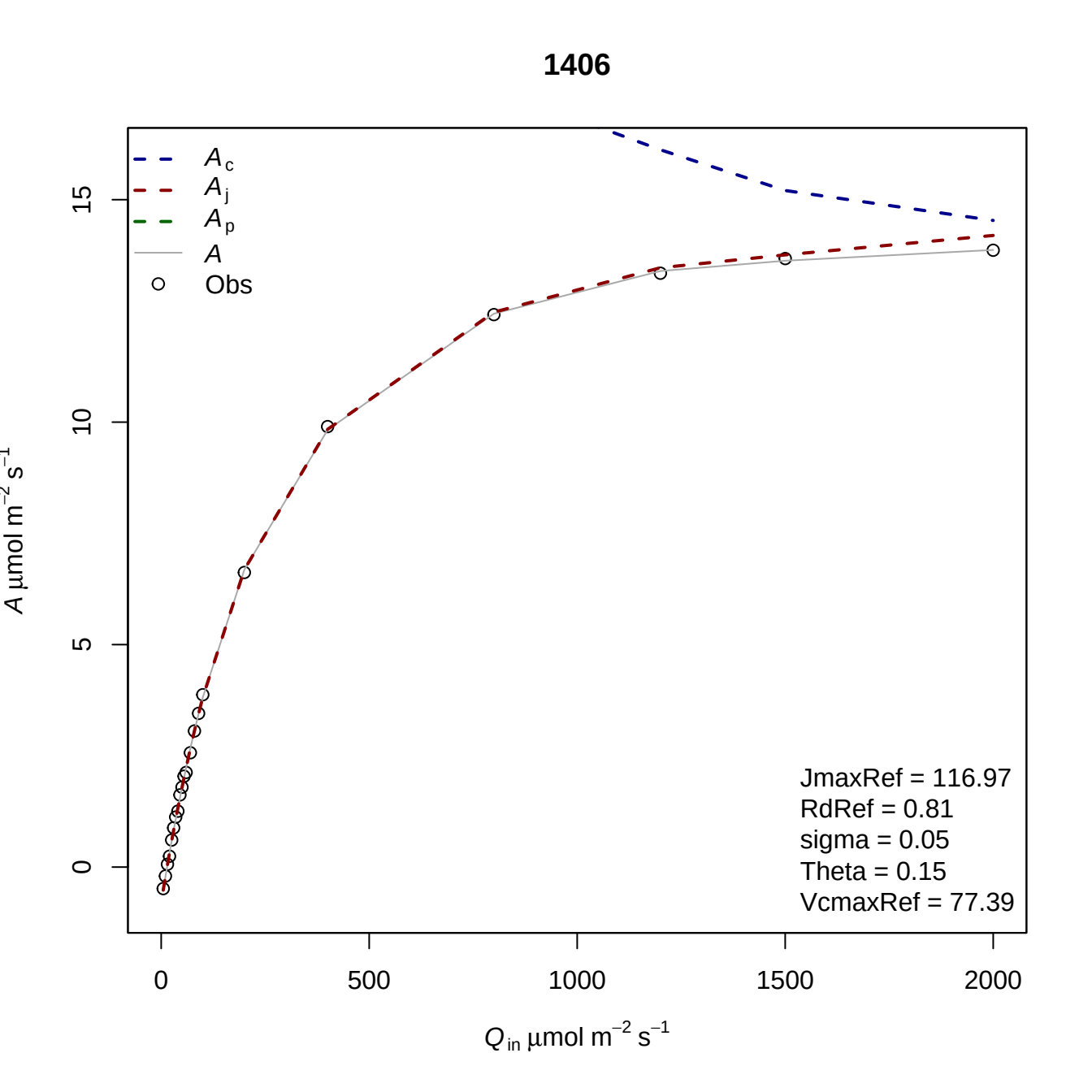

1401
A_c
A_j
A_p
A
Obs
JmaxRef = 79.91
RdRef = 0.69
sigma = 0.09
Theta = 0.1
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

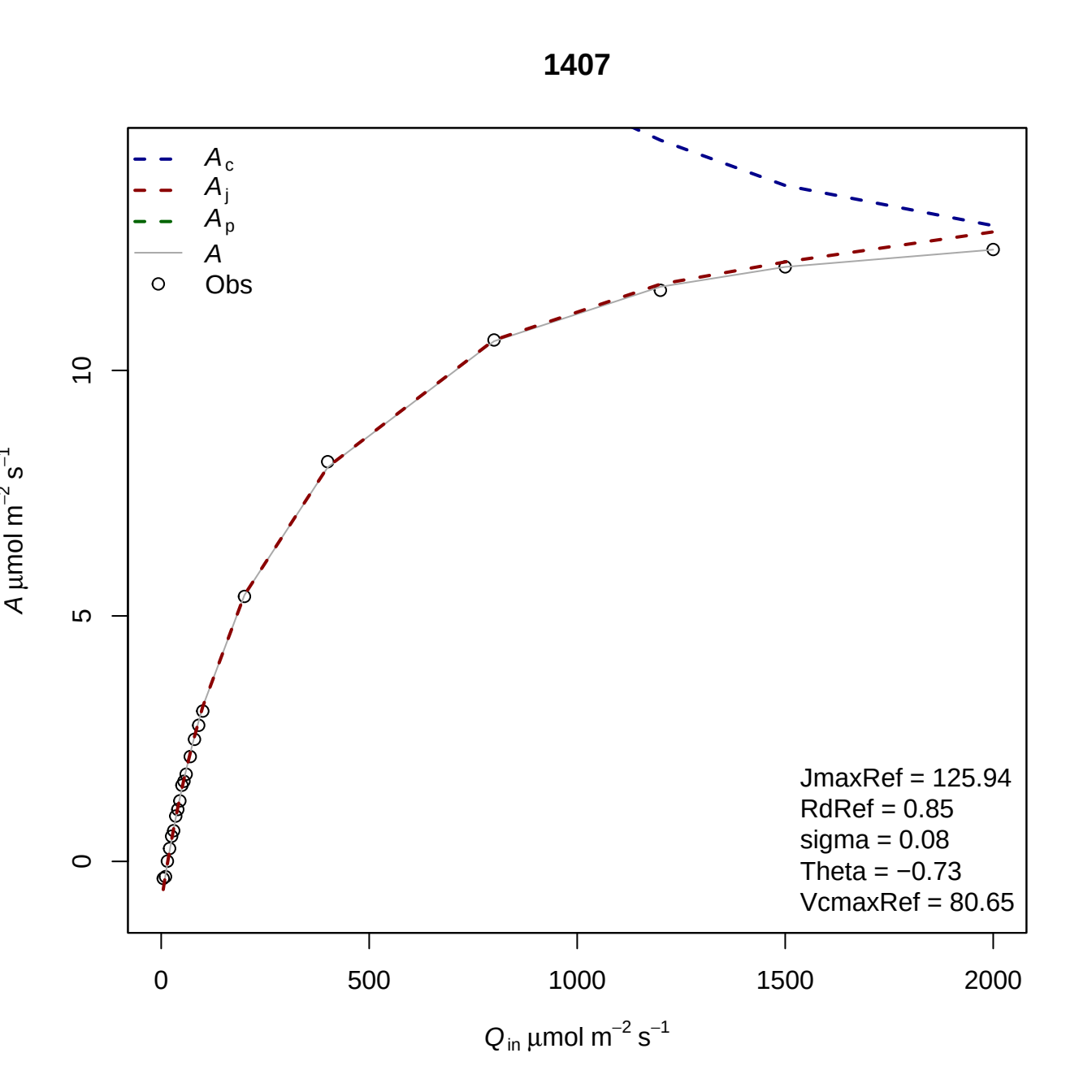

1402
A_c
A_j
A_p
A
Obs
JmaxRef = 28.98
RdRef = 0.8
sigma = 0.17
Theta = −0.18
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500

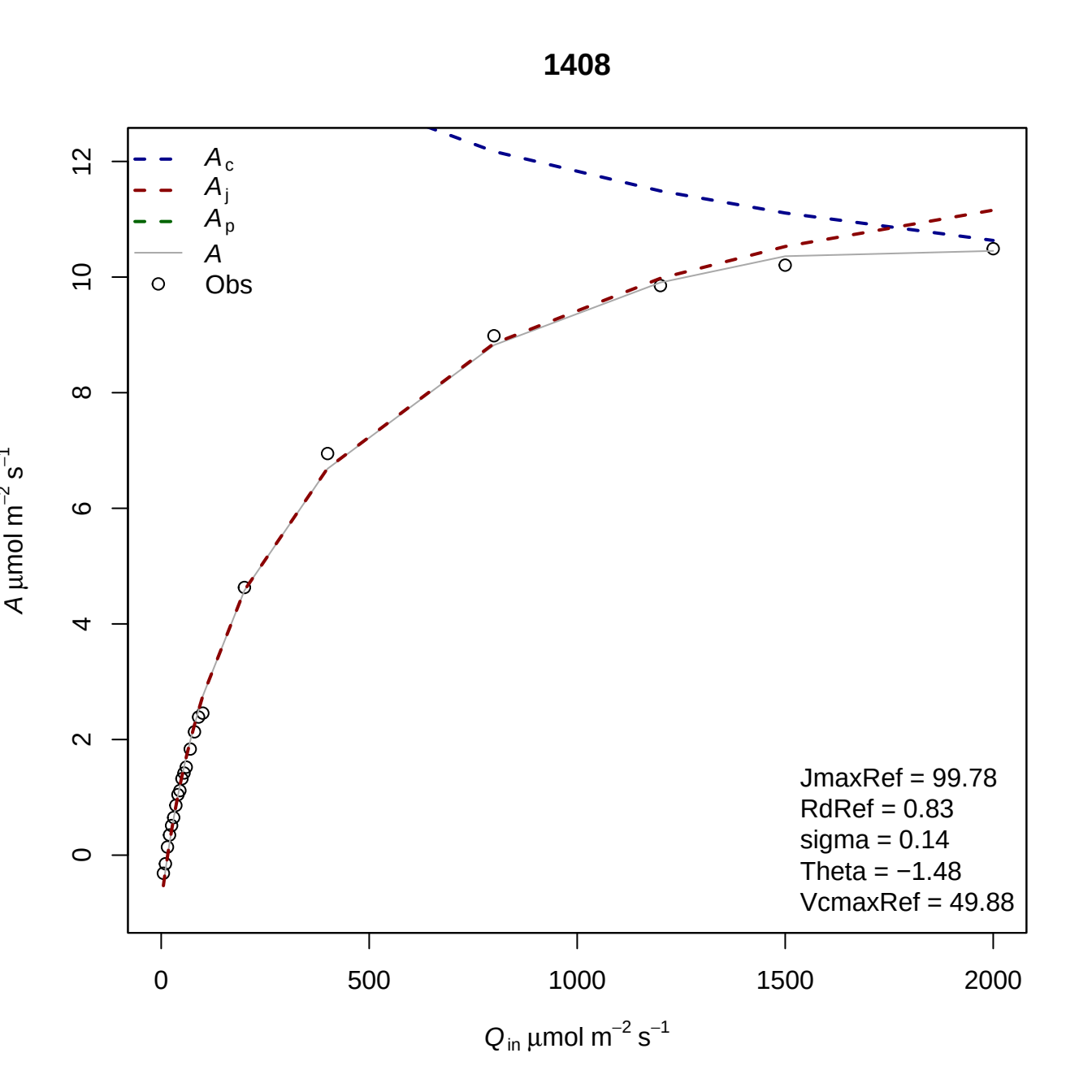

1403
A_c
A_j
A_p
A
Obs
JmaxRef = 87.72
RdRef = 0.83
sigma = 0.11
Theta = 0.02
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

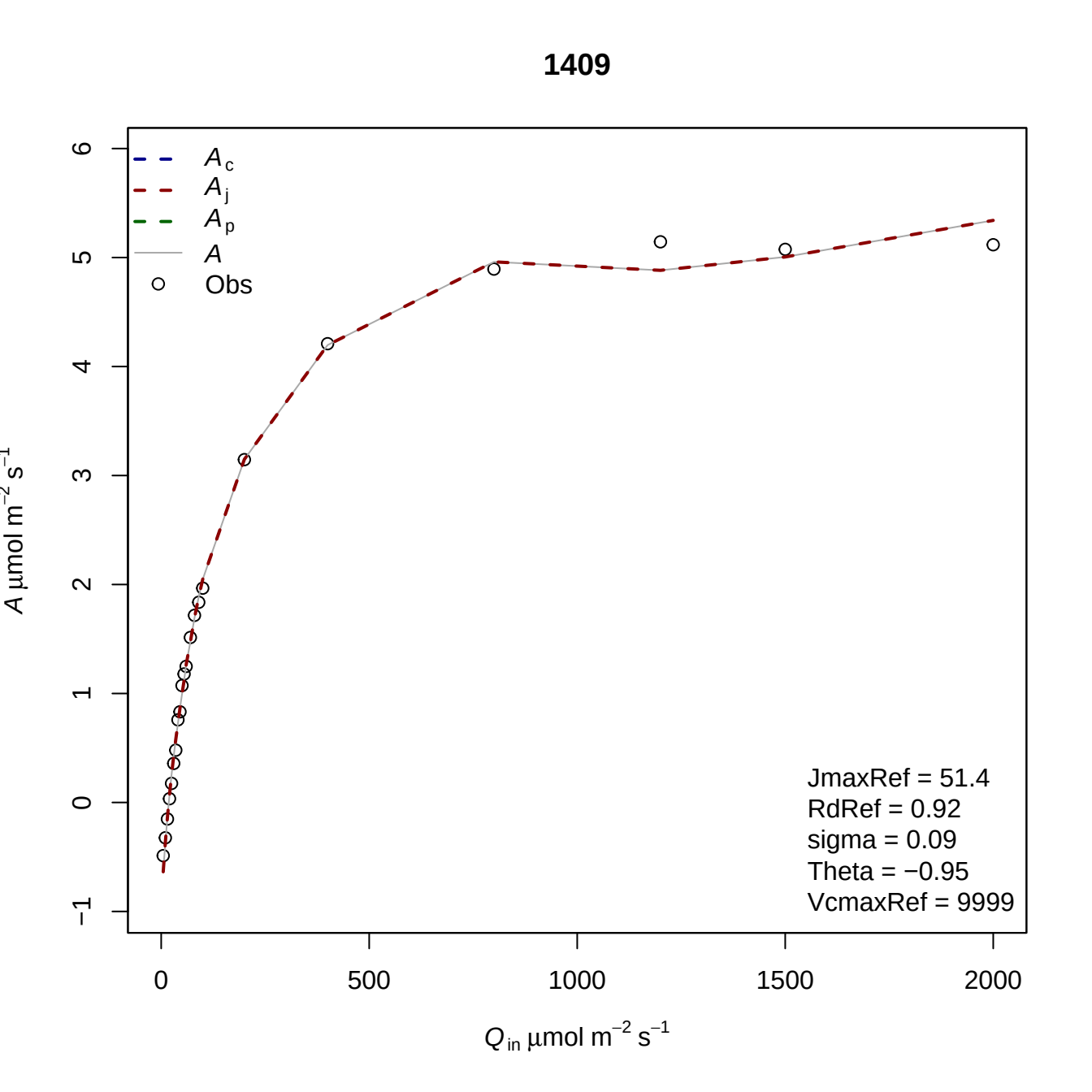

1404
A_c
A_j
A_p
A
Obs
JmaxRef = 94.83
RdRef = 0.72
sigma = 0.1
Theta = −0.07
VcmaxRef = 61.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

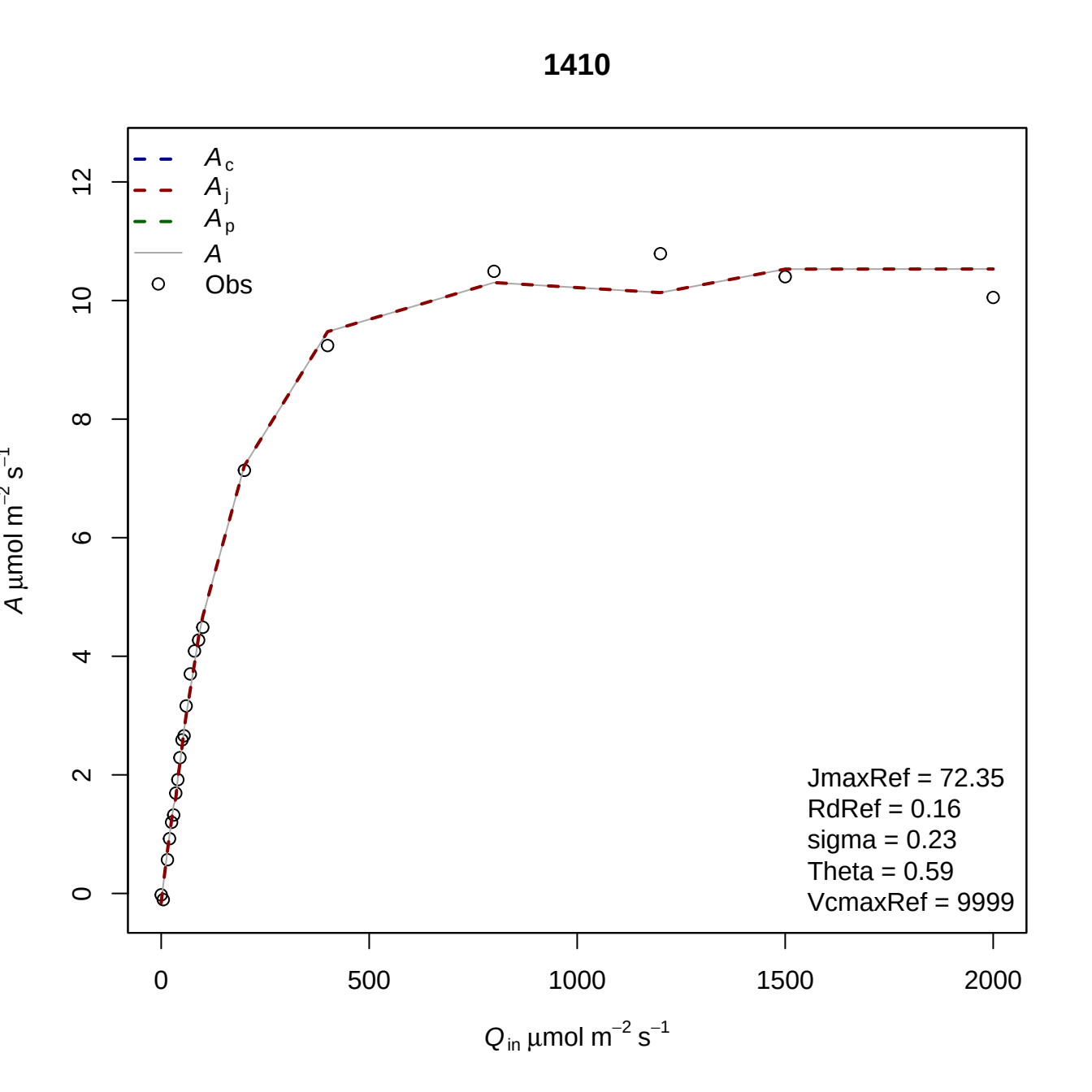

1405
A_c
A_j
A_p
A
Obs
JmaxRef = 115.59
RdRef = 0.98
sigma = 0.2
Theta = −0.81
VcmaxRef = 61.42
0
2
4
6
8
10
12
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

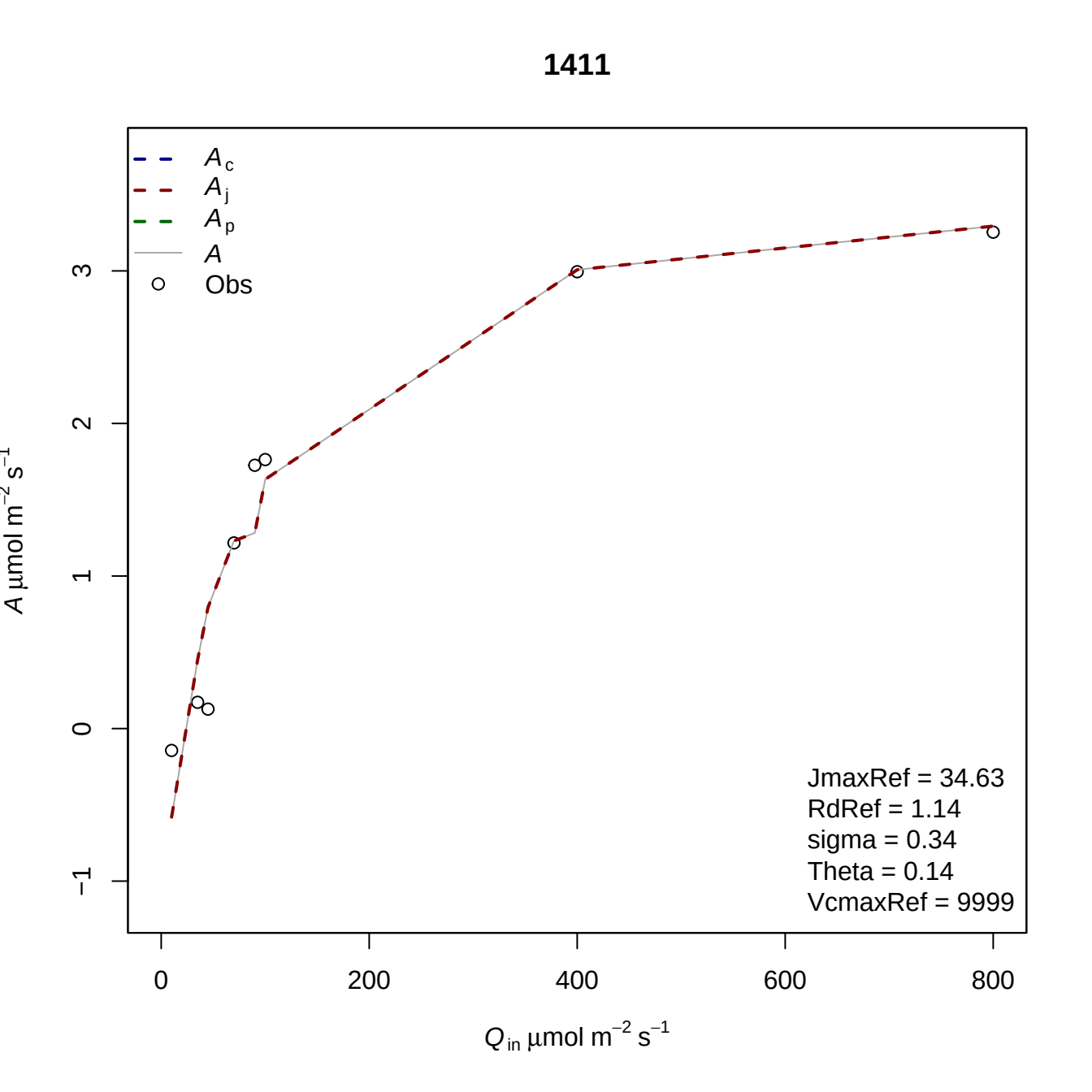

1406
A_c
A_j
A_p
A
Obs
JmaxRef = 116.97
RdRef = 0.81
sigma = 0.05
Theta = 0.15
VcmaxRef = 77.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

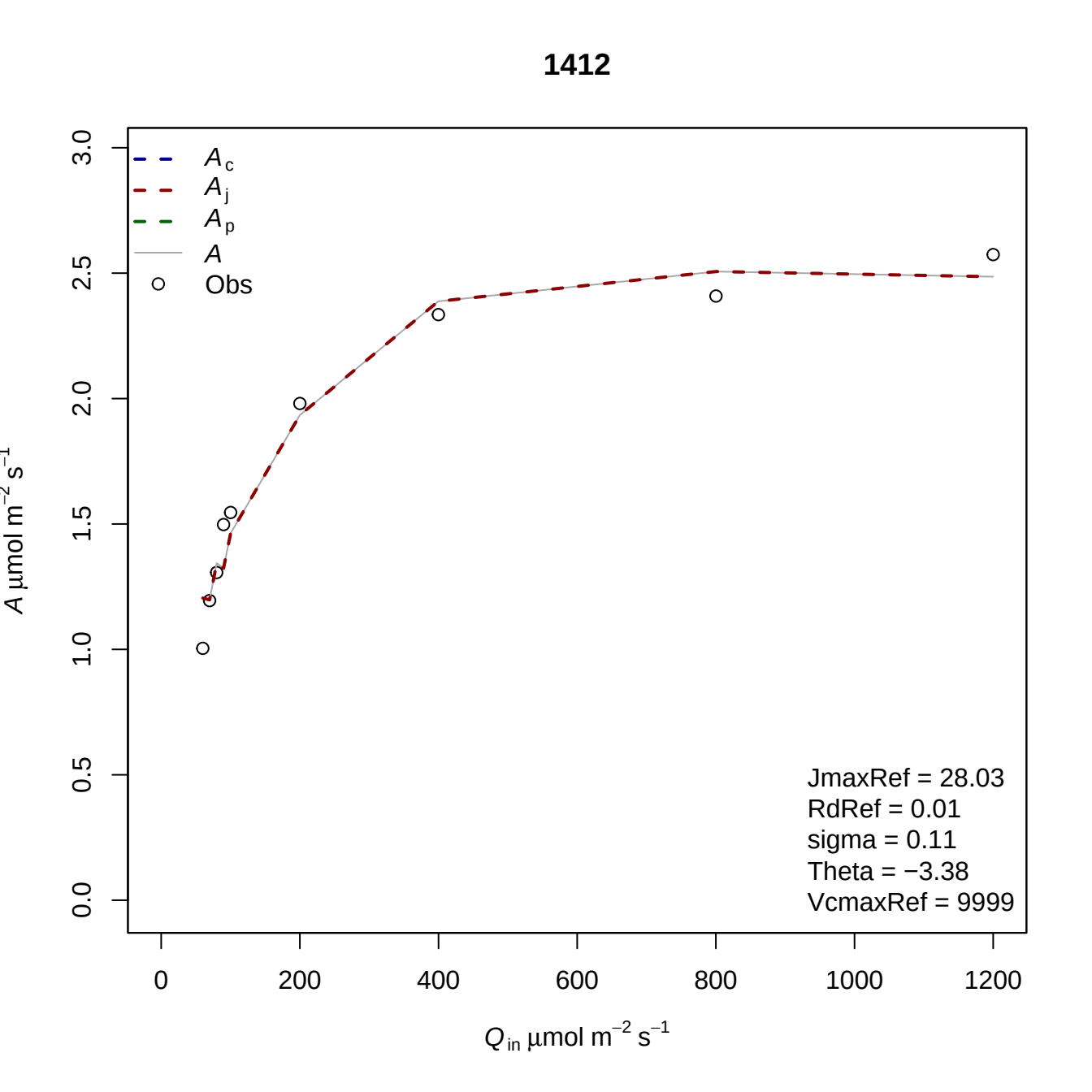

1407
A_c
A_j
A_p
A
Obs
JmaxRef = 125.94
RdRef = 0.85
sigma = 0.08
Theta = −0.73
VcmaxRef = 80.65
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

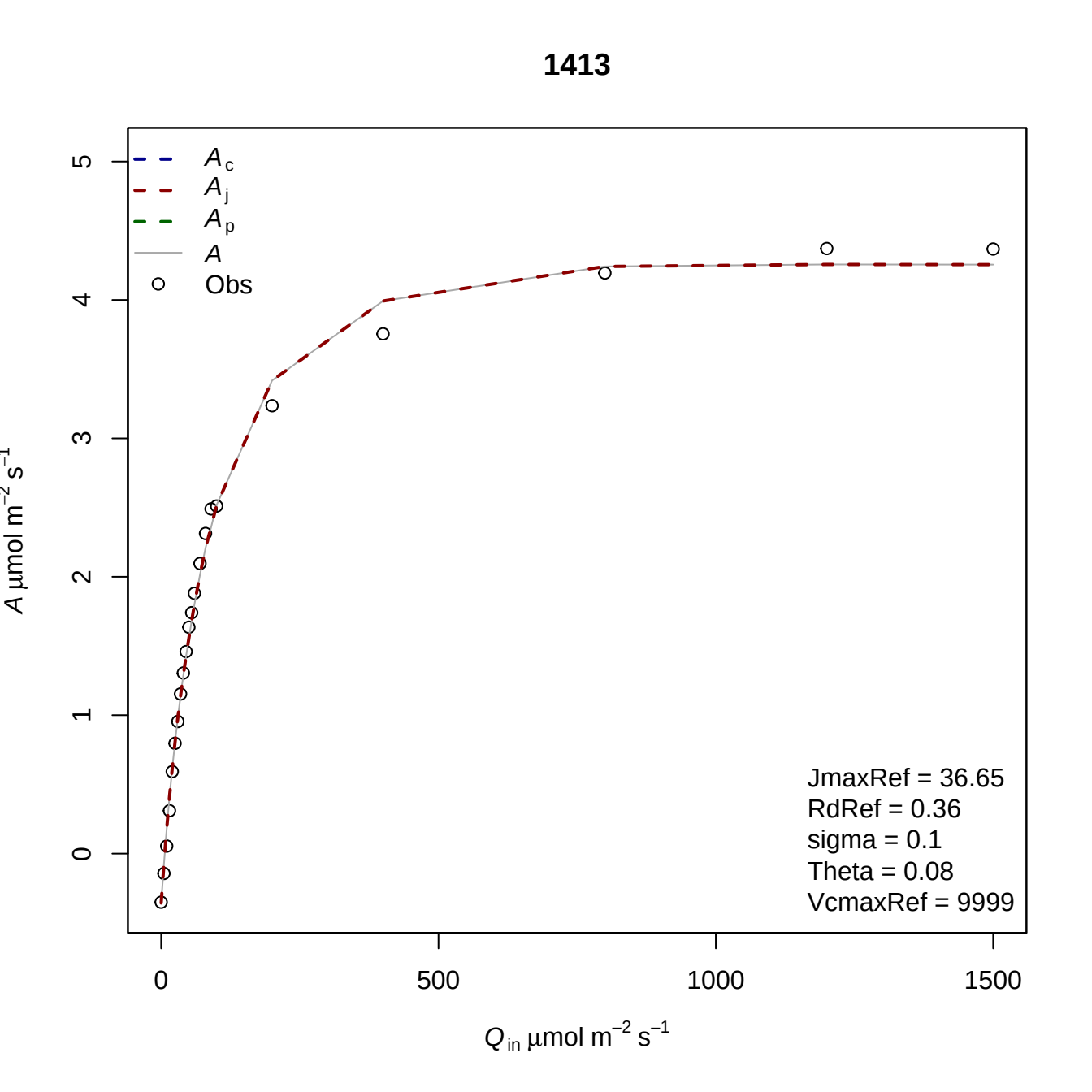

1408
A_c
A_j
A_p
A
Obs
JmaxRef = 99.78
RdRef = 0.83
sigma = 0.14
Theta = −1.48
VcmaxRef = 49.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

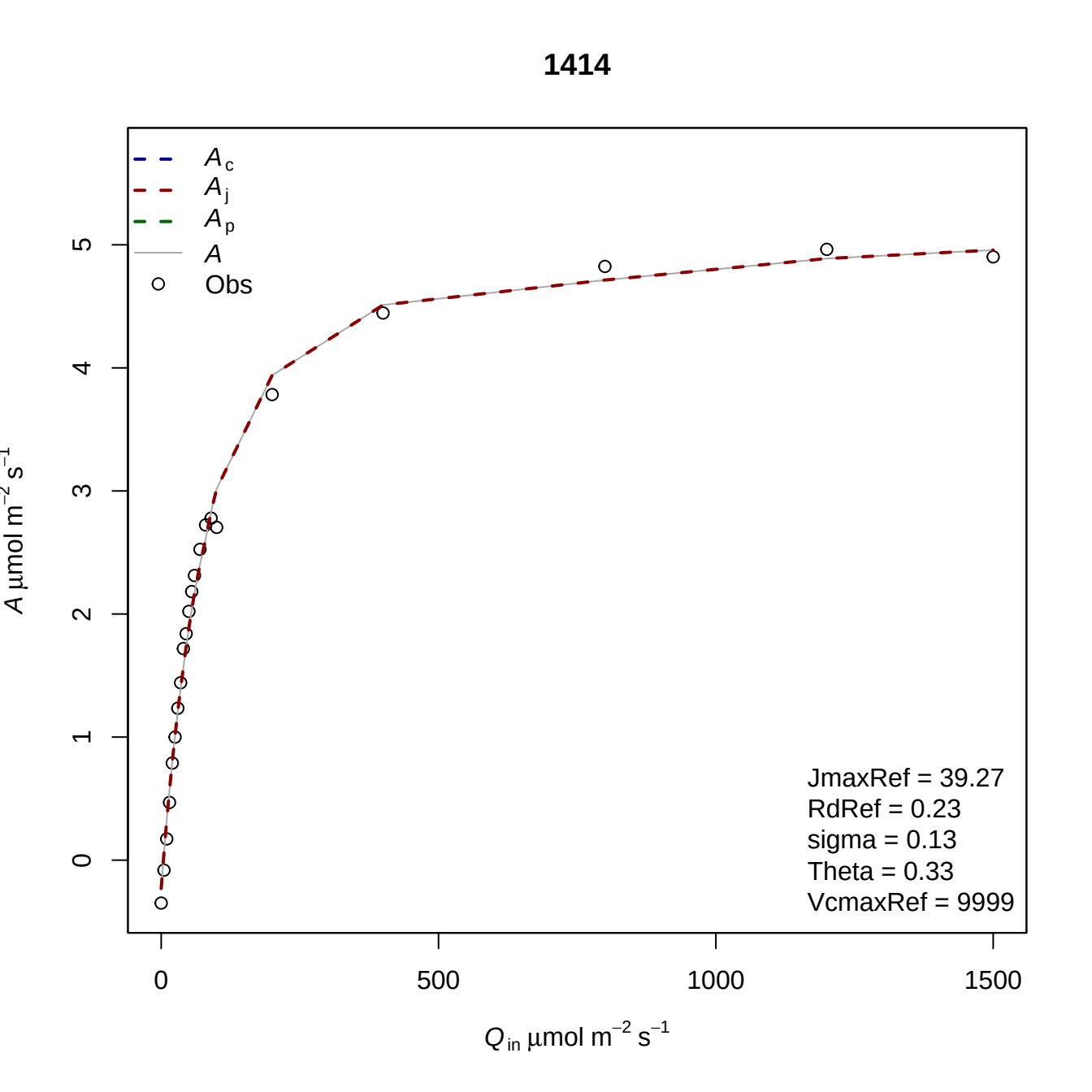

1409
A_c
A_j
A_p
A
Obs
JmaxRef = 51.4
RdRef = 0.92
sigma = 0.09
Theta = −0.95
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

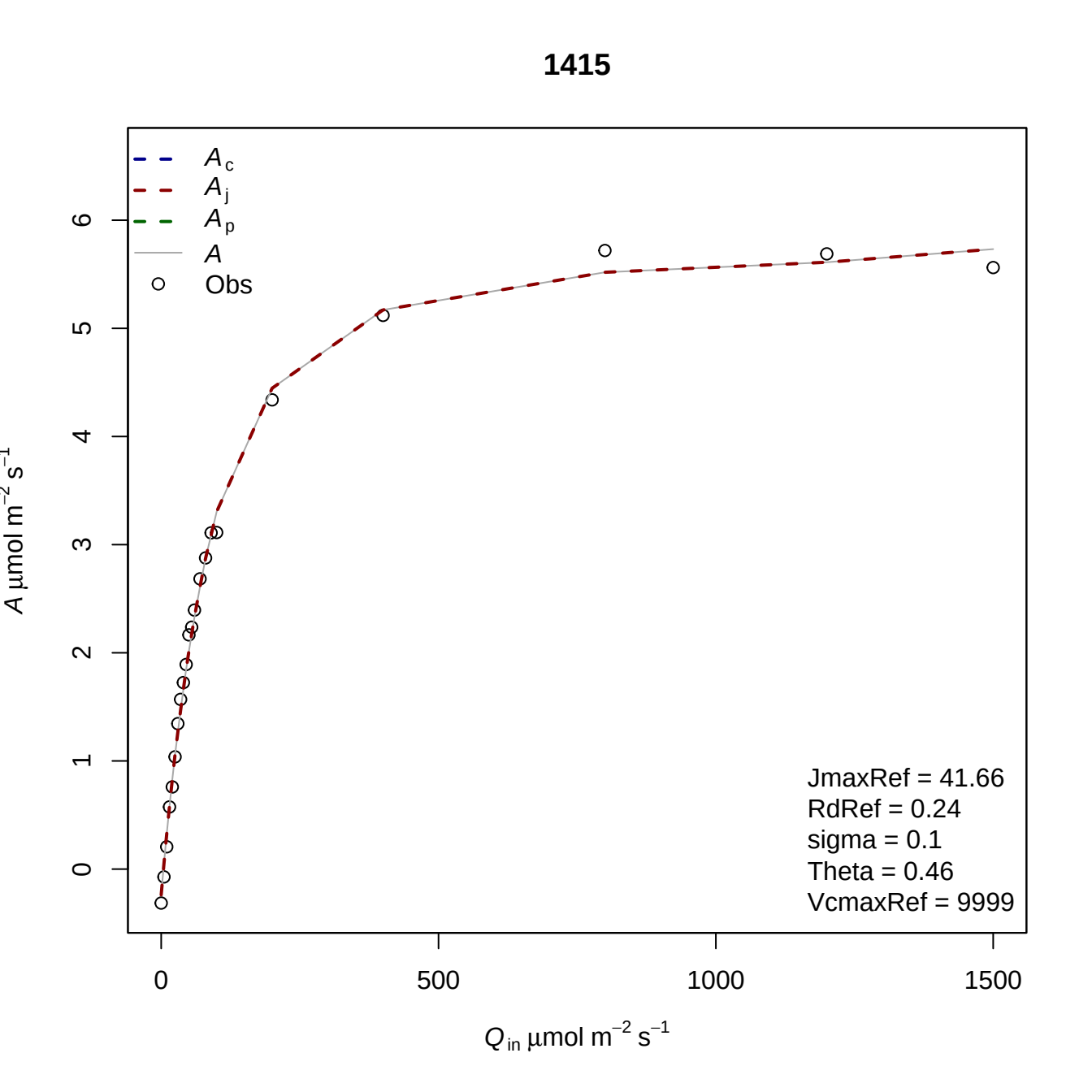

1410
A_c
A_j
A_p
A
Obs
JmaxRef = 72.35
RdRef = 0.16
sigma = 0.23
Theta = 0.59
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

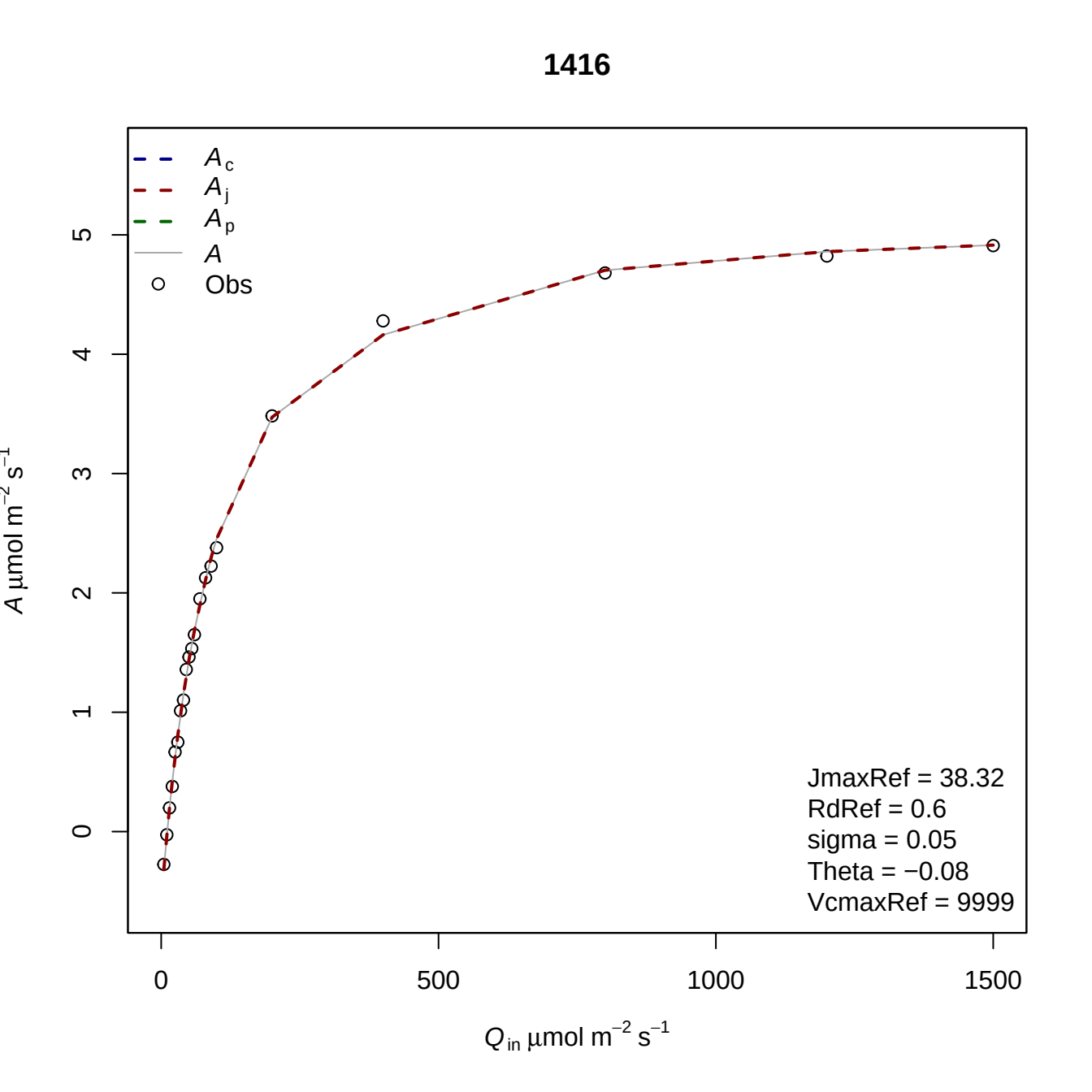

1411
A_c
A_j
A_p
A
Obs
JmaxRef = 34.63
RdRef = 1.14
sigma = 0.34
Theta = 0.14
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800

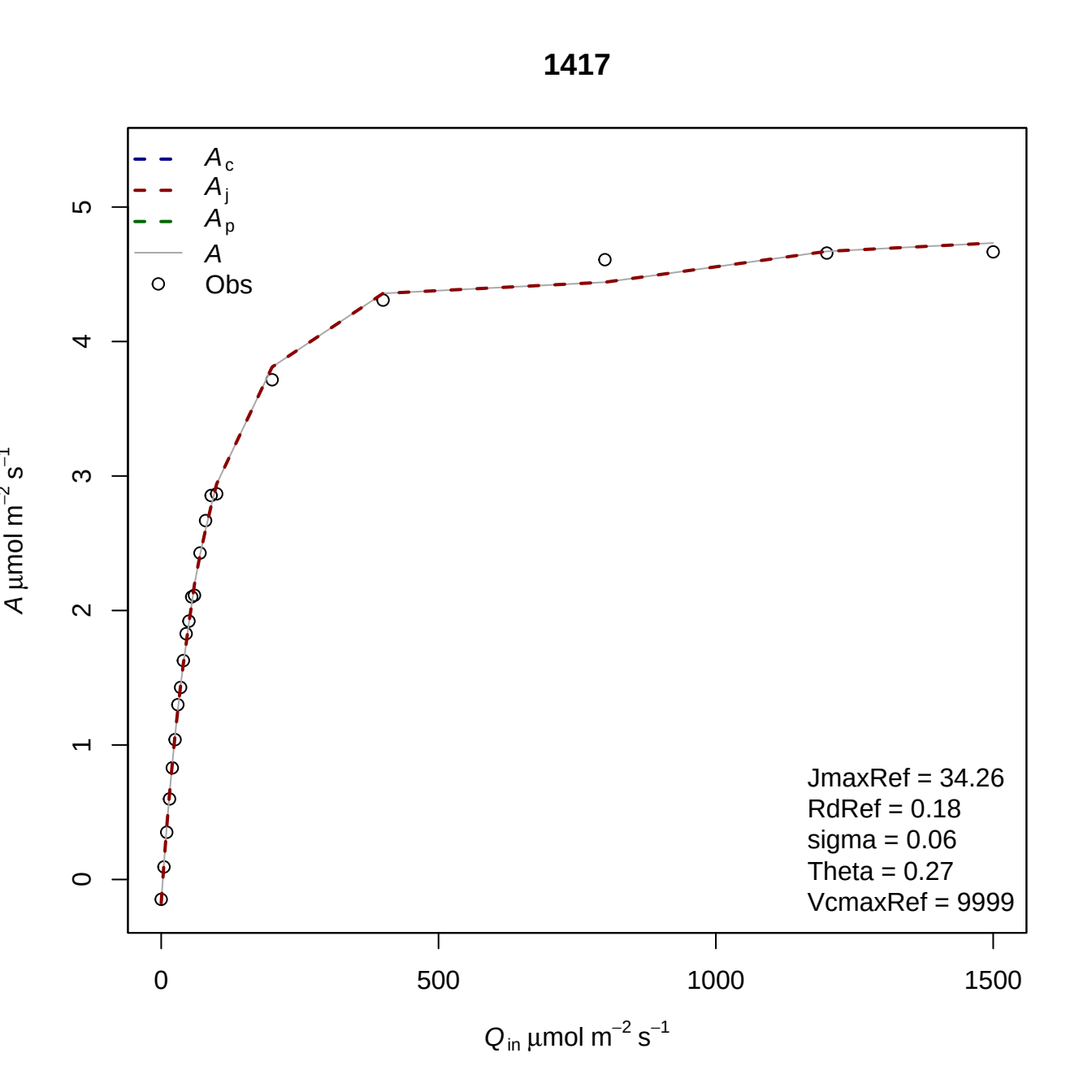

1412
A_c
A_j
A_p
A
Obs
JmaxRef = 28.03
RdRef = 0.01
sigma = 0.11
Theta = −3.38
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200

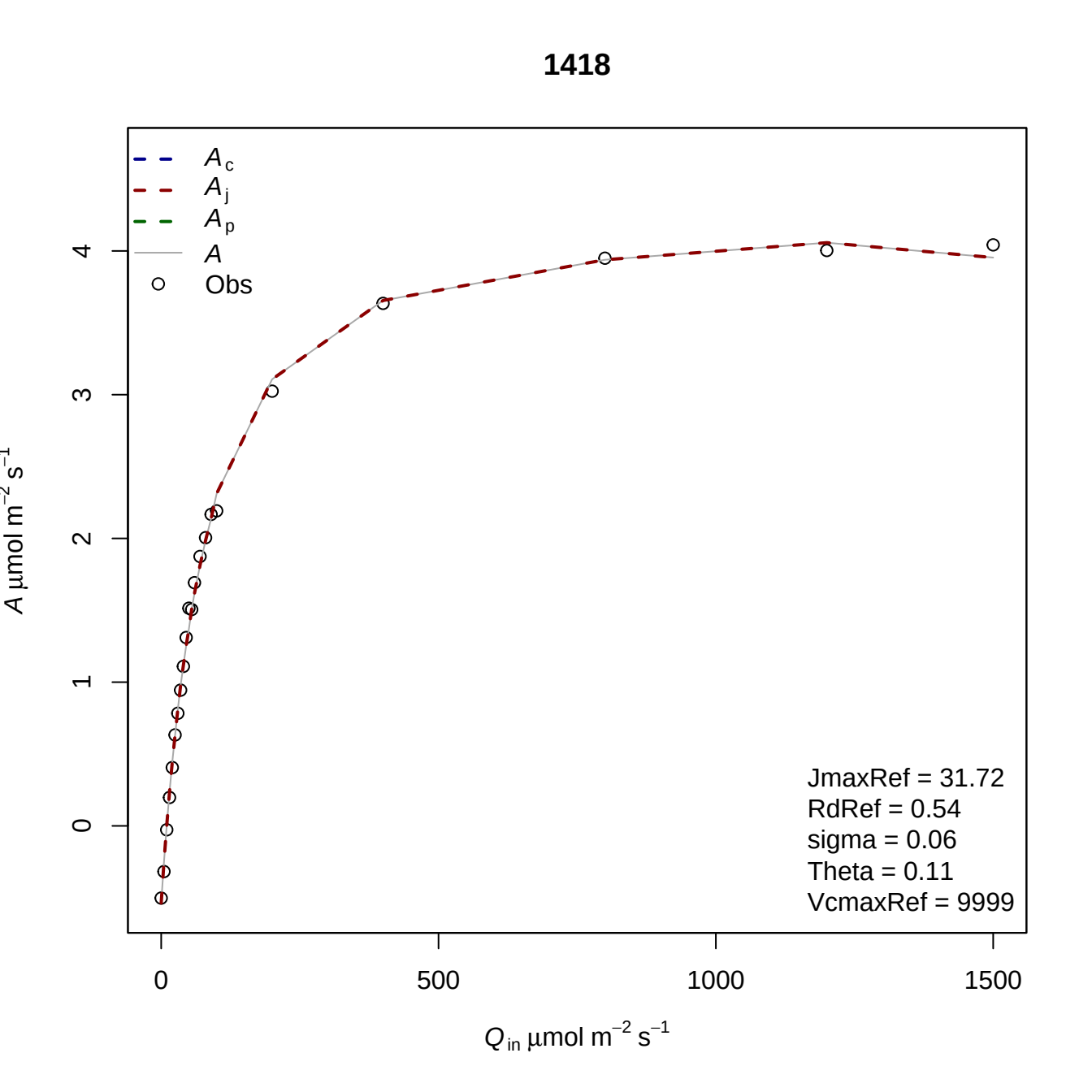

1413
A_c
A_j
A_p
A
Obs
JmaxRef = 36.65
RdRef = 0.36
sigma = 0.1
Theta = 0.08
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

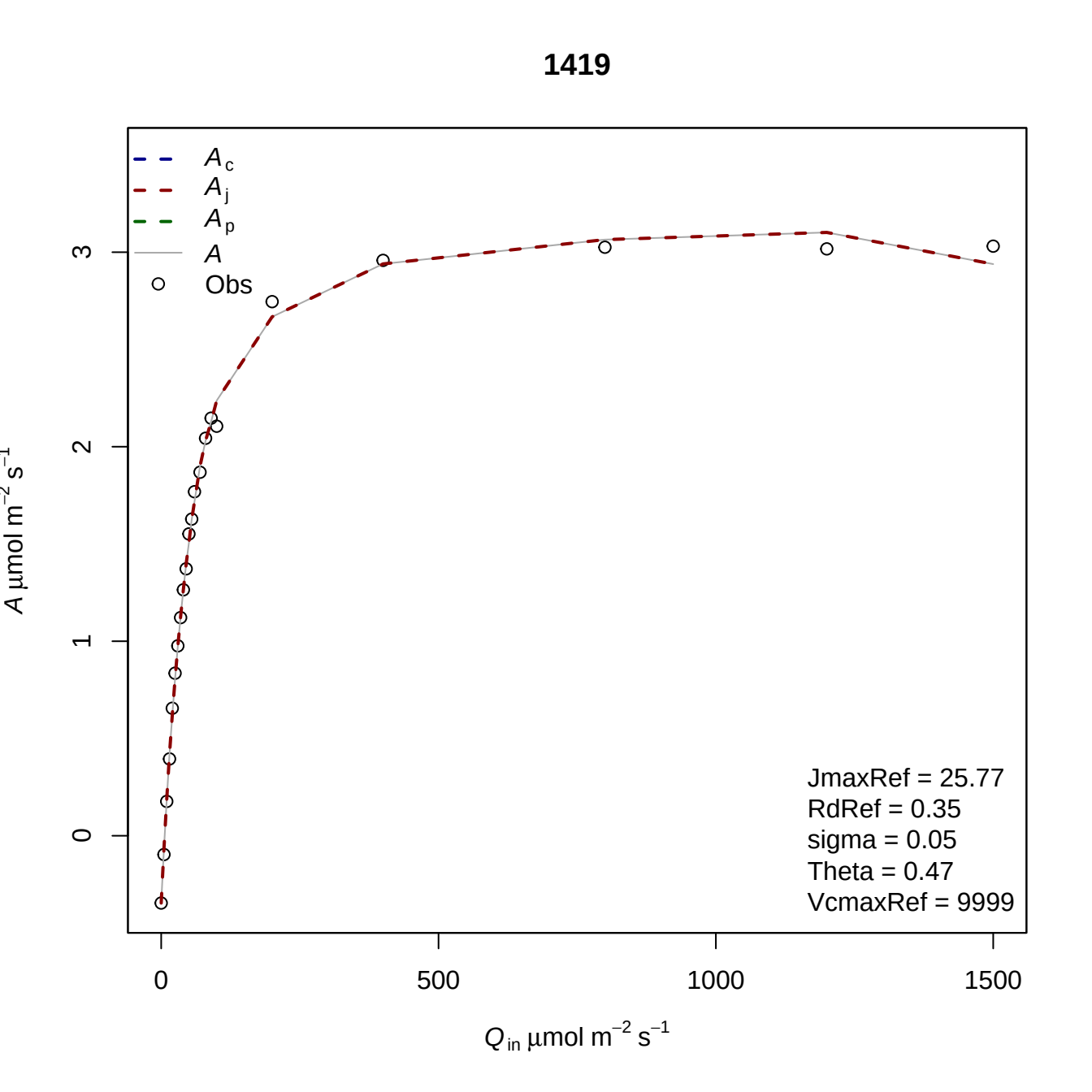

1414
A_c
A_j
A_p
A
Obs
JmaxRef = 39.27
RdRef = 0.23
sigma = 0.13
Theta = 0.33
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

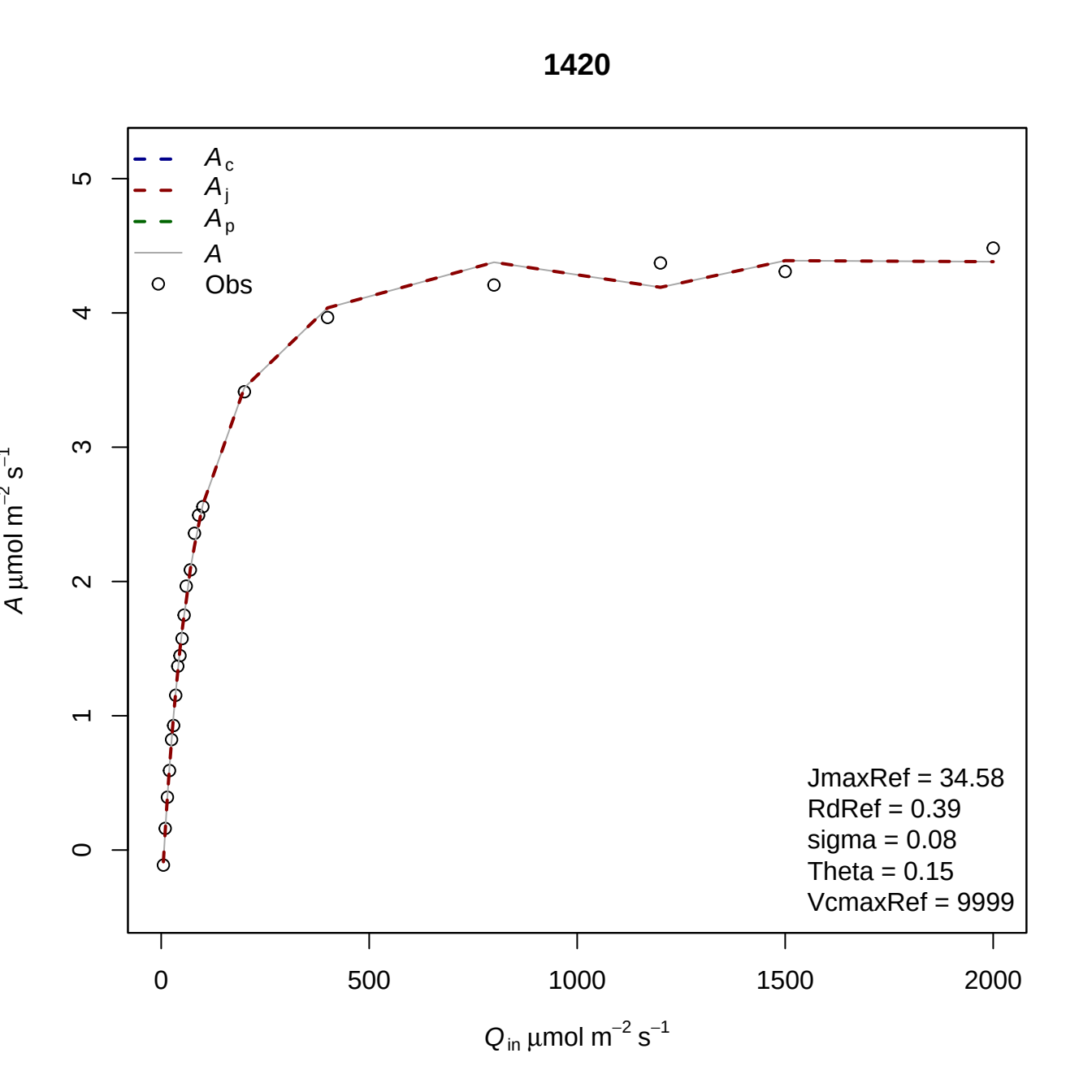

1415
A_c
A_j
A_p
A
Obs
JmaxRef = 41.66
RdRef = 0.24
sigma = 0.1
Theta = 0.46
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

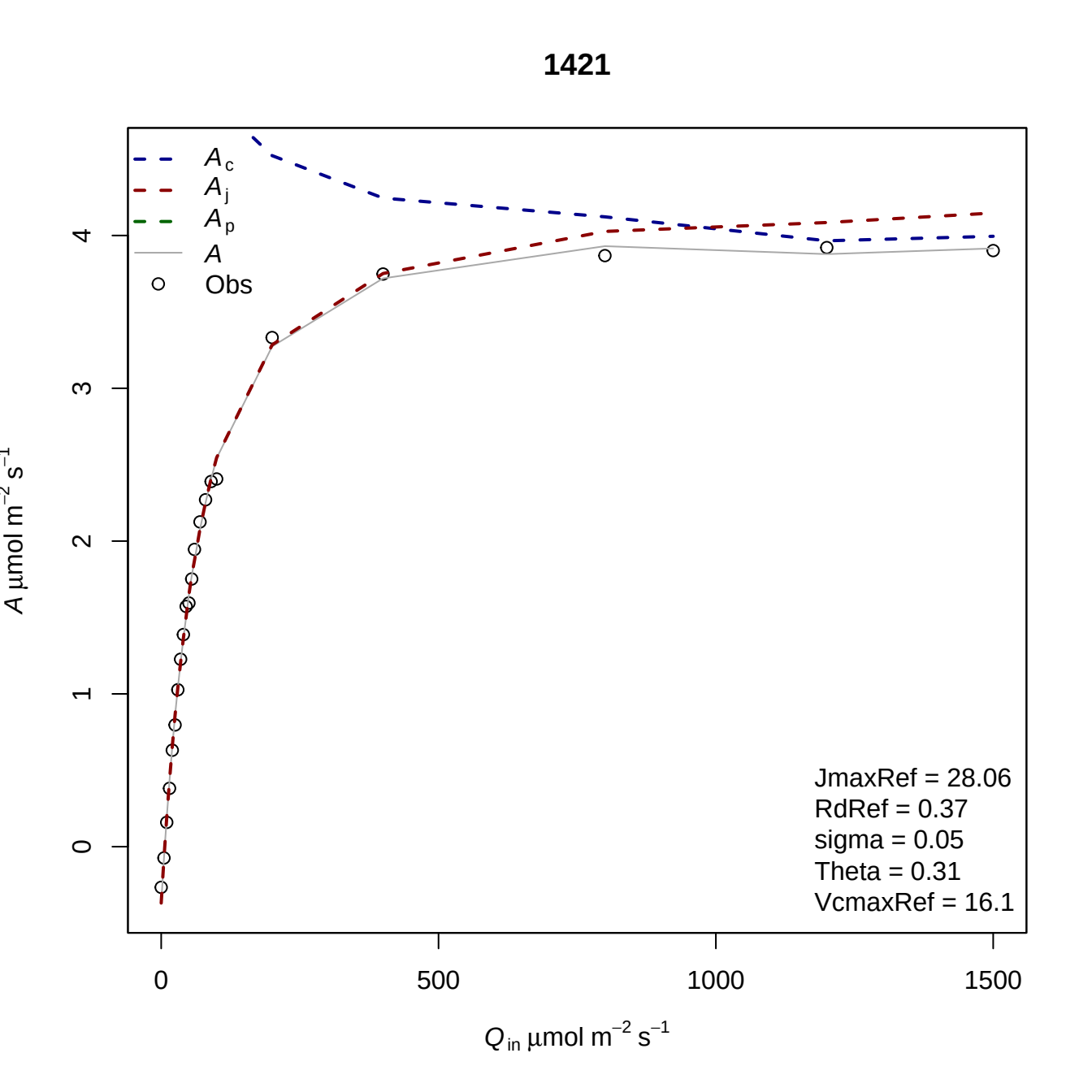

1416
A_c
A_j
A_p
A
Obs
JmaxRef = 38.32
RdRef = 0.6
sigma = 0.05
Theta = −0.08
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

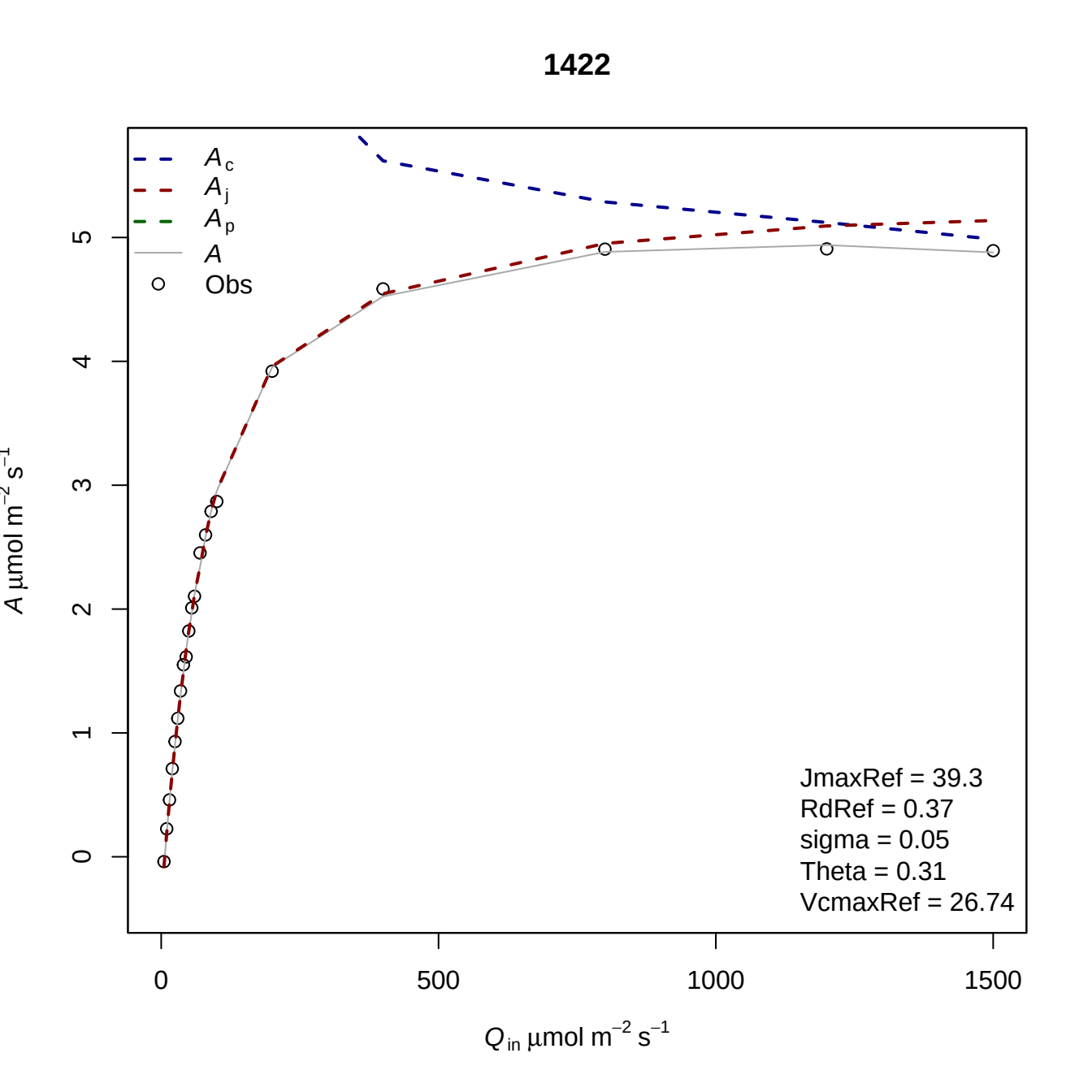

1417
A_c
A_j
A_p
A
Obs
JmaxRef = 34.26
RdRef = 0.18
sigma = 0.06
Theta = 0.27
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

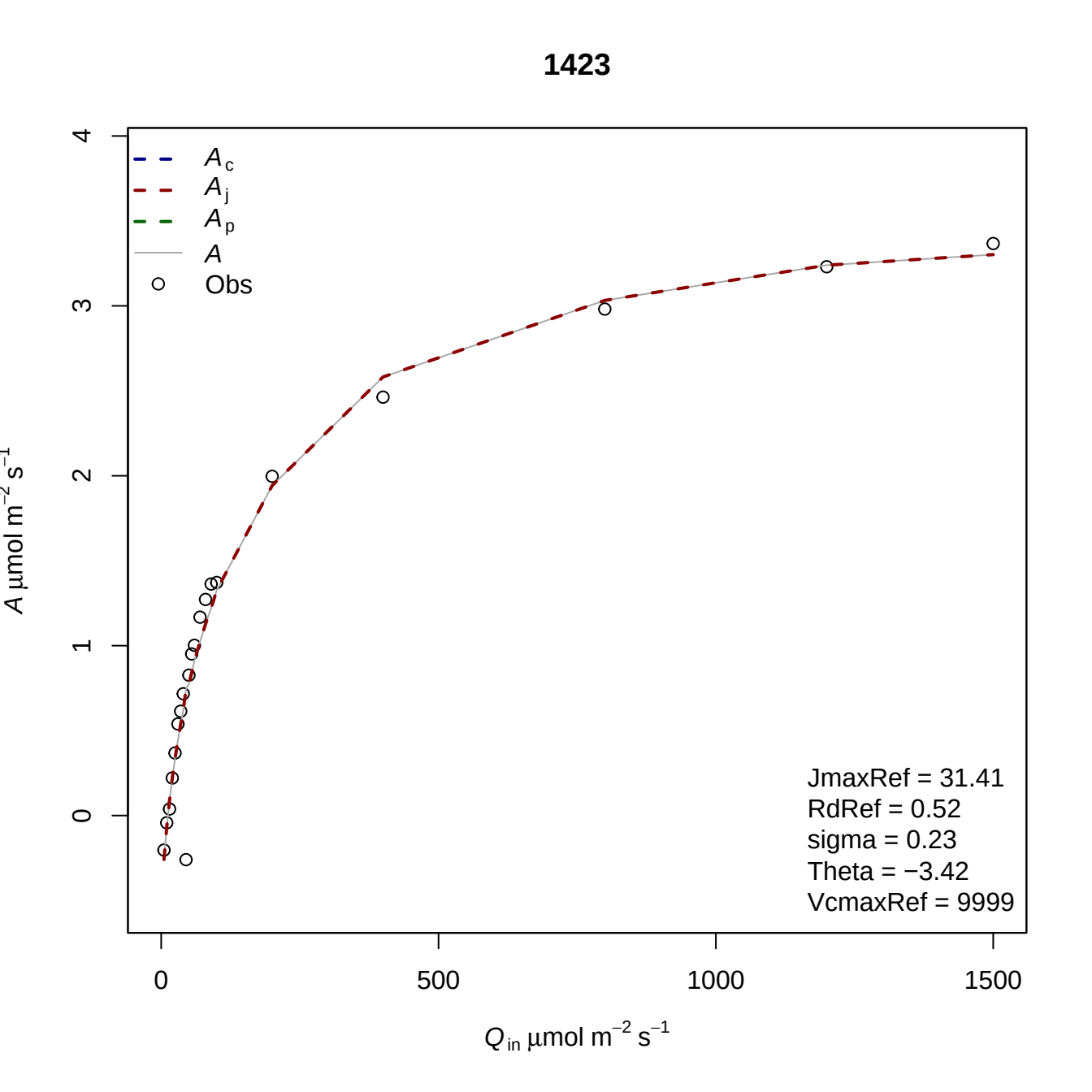

1418
A_c
A_j
A_p
A
Obs
JmaxRef = 31.72
RdRef = 0.54
sigma = 0.06
Theta = 0.11
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

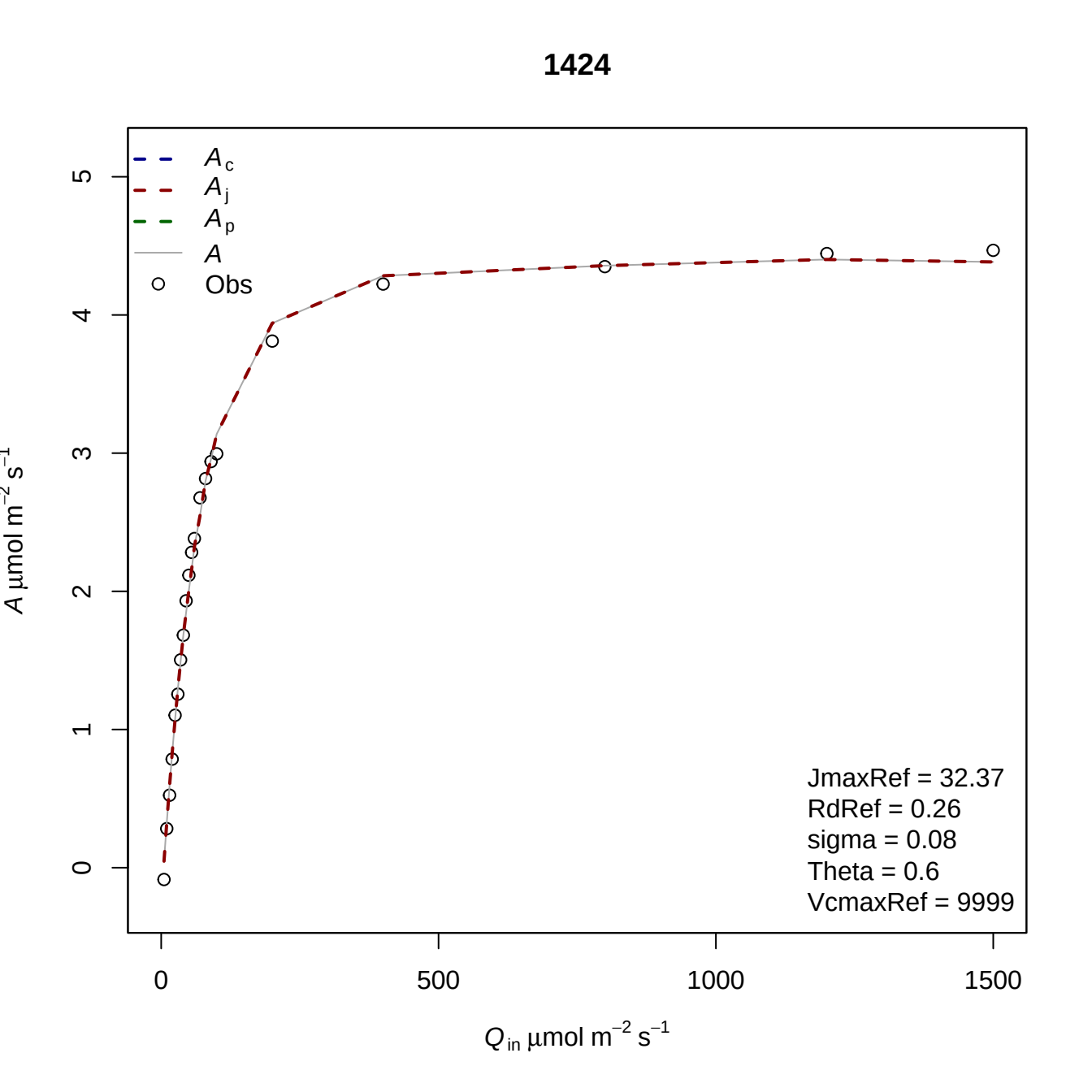

1419
A_c
A_j
A_p
A
Obs
JmaxRef = 25.77
RdRef = 0.35
sigma = 0.05
Theta = 0.47
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

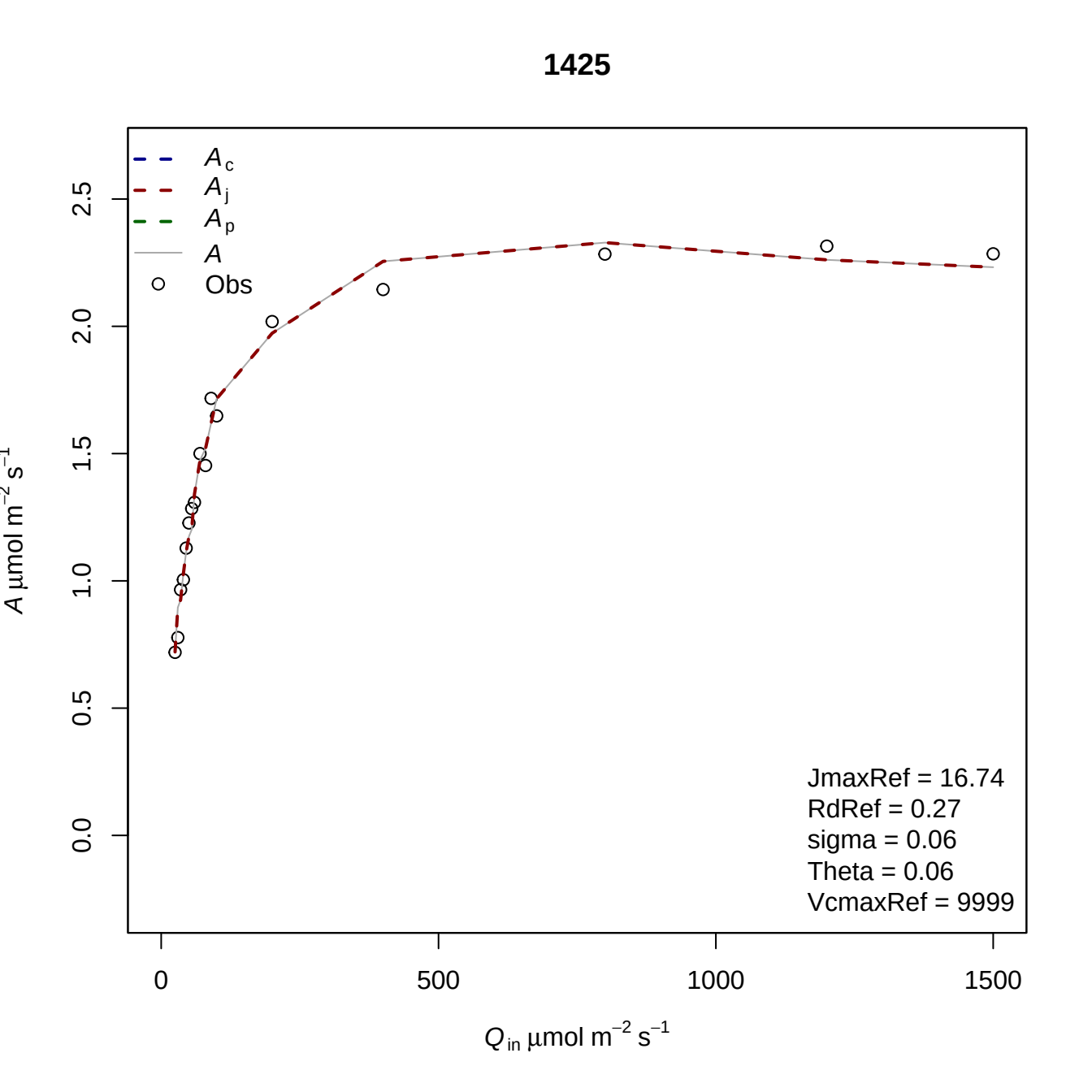

1420
A_c
A_j
A_p
A
Obs
JmaxRef = 34.58
RdRef = 0.39
sigma = 0.08
Theta = 0.15
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500
2000

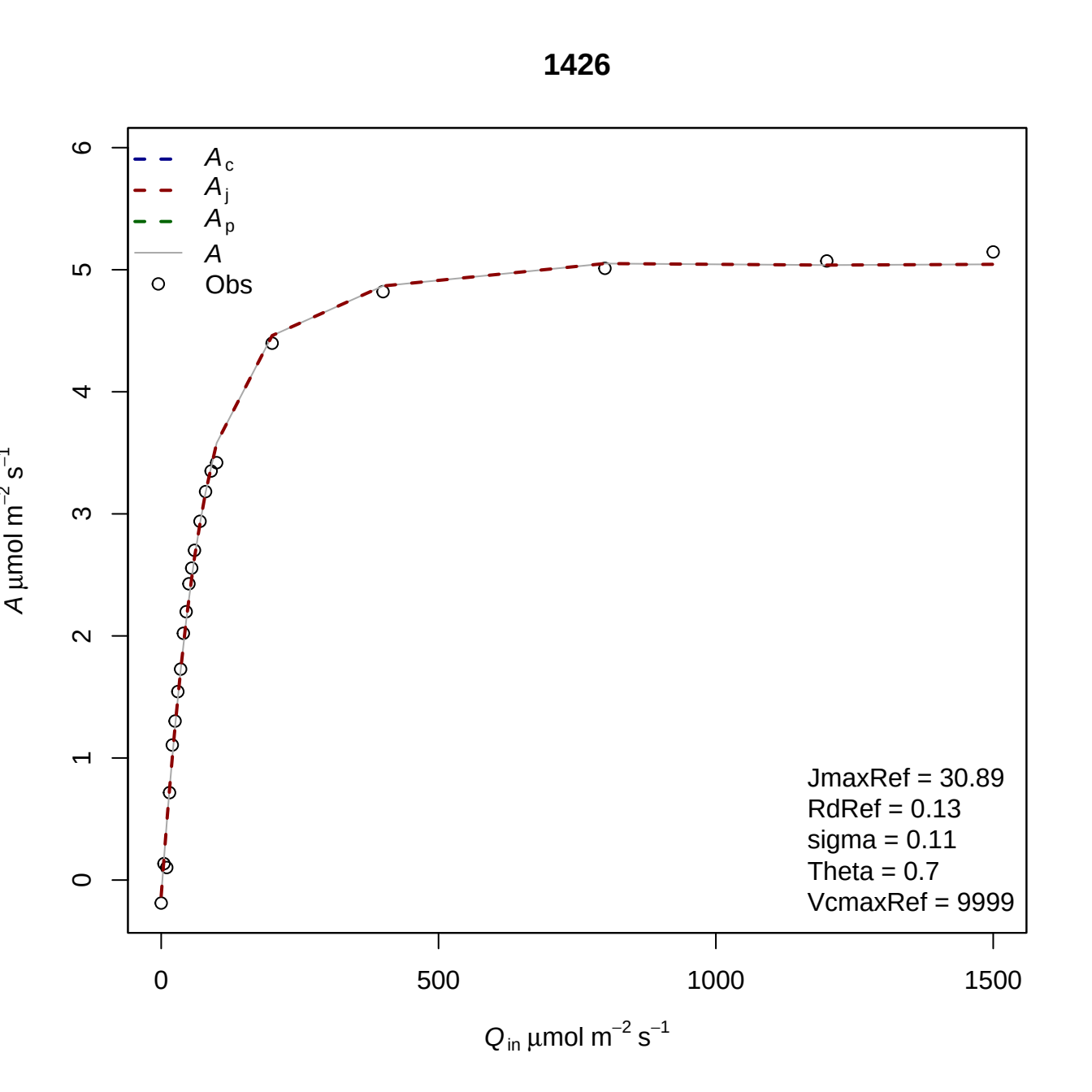

1421
A_c
A_j
A_p
A
Obs
JmaxRef = 28.06
RdRef = 0.37
sigma = 0.05
Theta = 0.31
VcmaxRef = 16.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

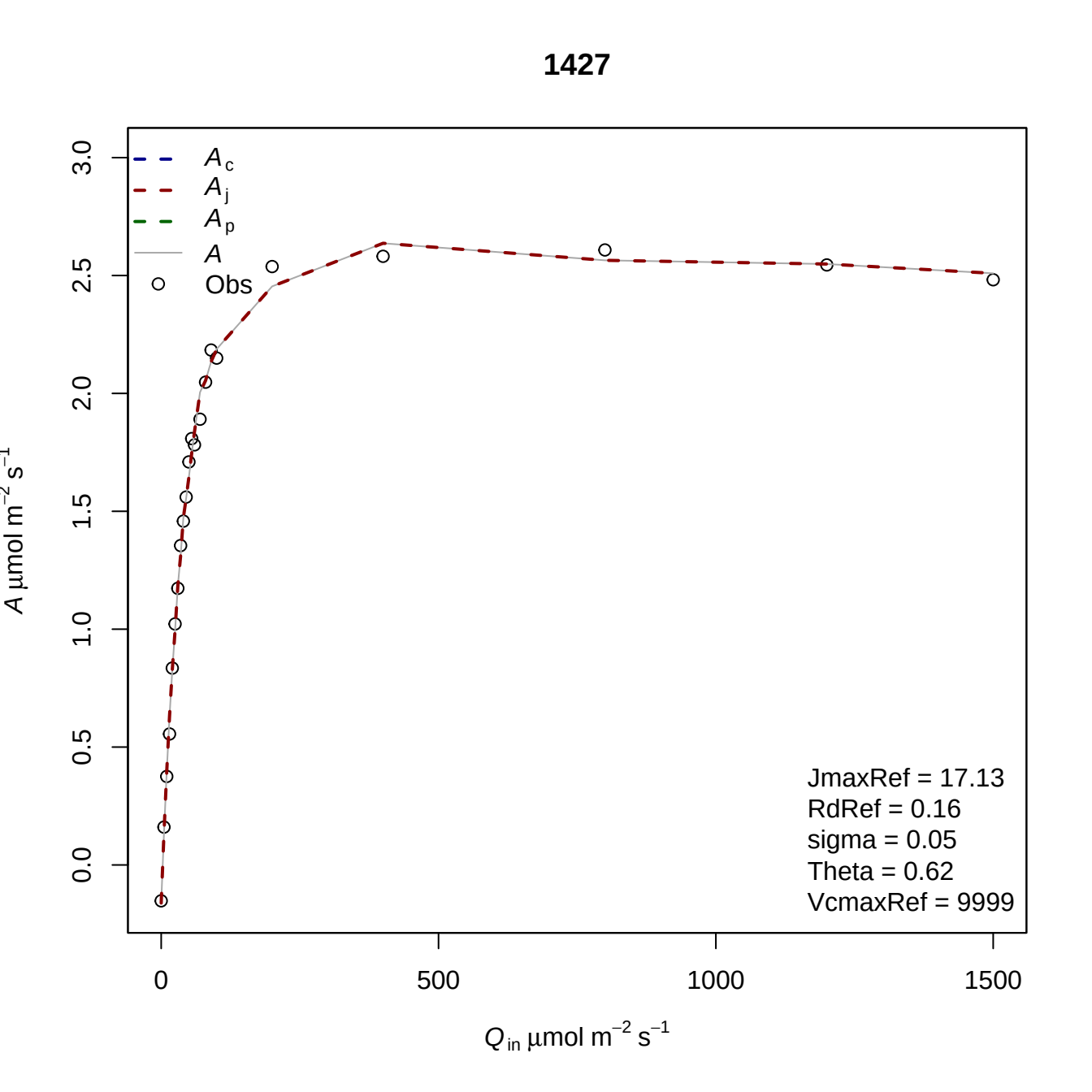

1422
A_c
A_j
A_p
A
Obs
JmaxRef = 39.3
RdRef = 0.37
sigma = 0.05
Theta = 0.31
VcmaxRef = 26.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

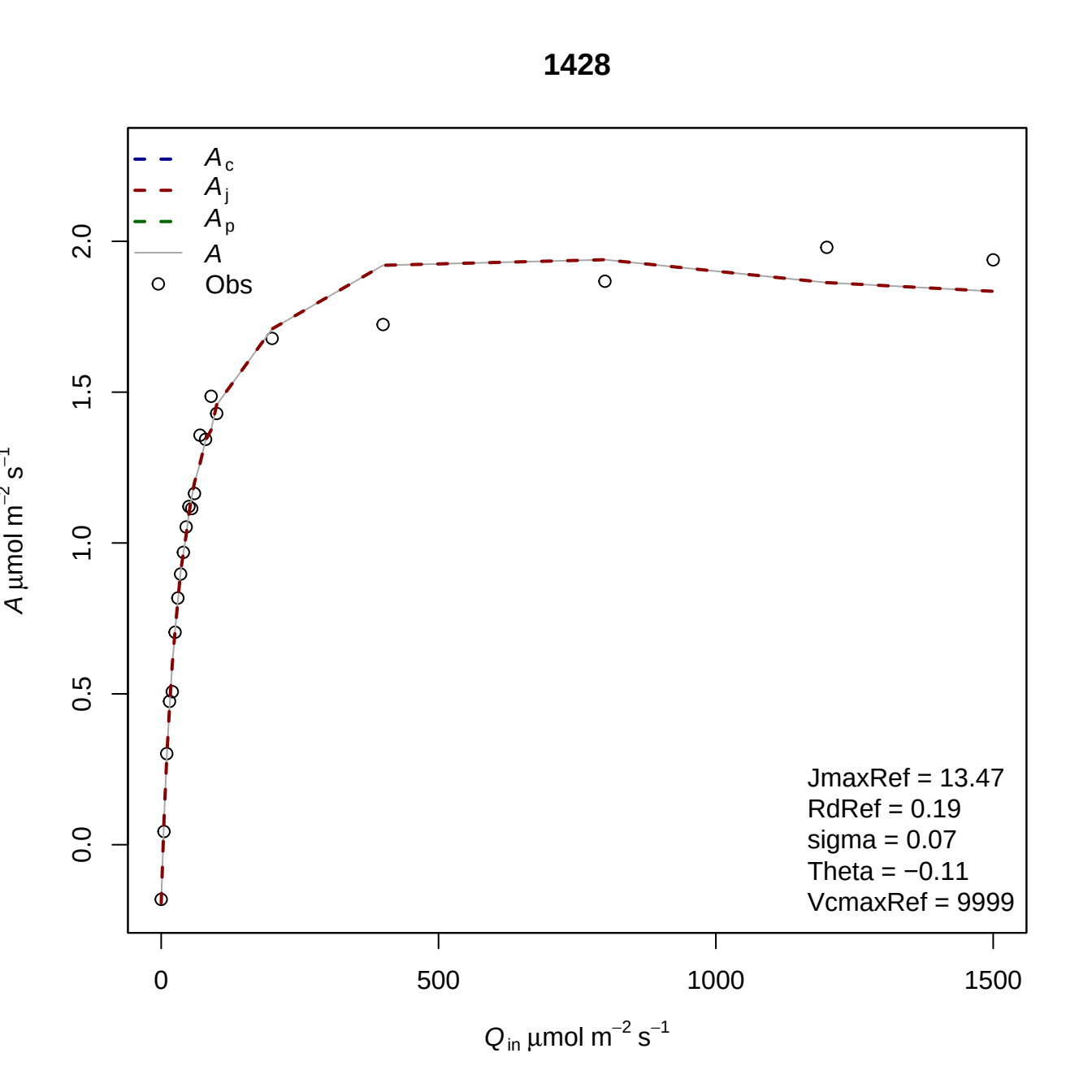

1423
A_c
A_j
A_p
A
Obs
JmaxRef = 31.41
RdRef = 0.52
sigma = 0.23
Theta = −3.42
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

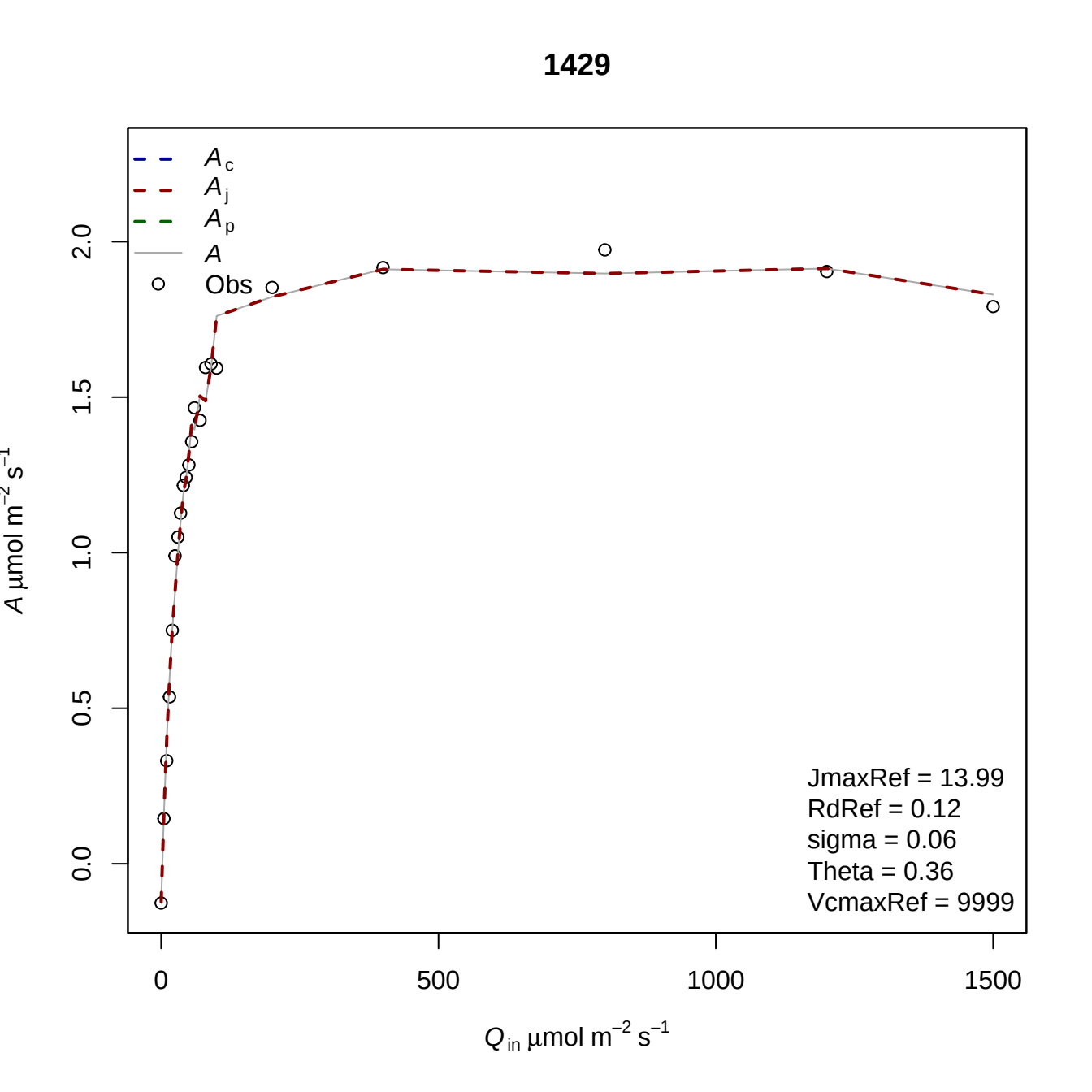

1424
A_c
A_j
A_p
A
Obs
JmaxRef = 32.37
RdRef = 0.26
sigma = 0.08
Theta = 0.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

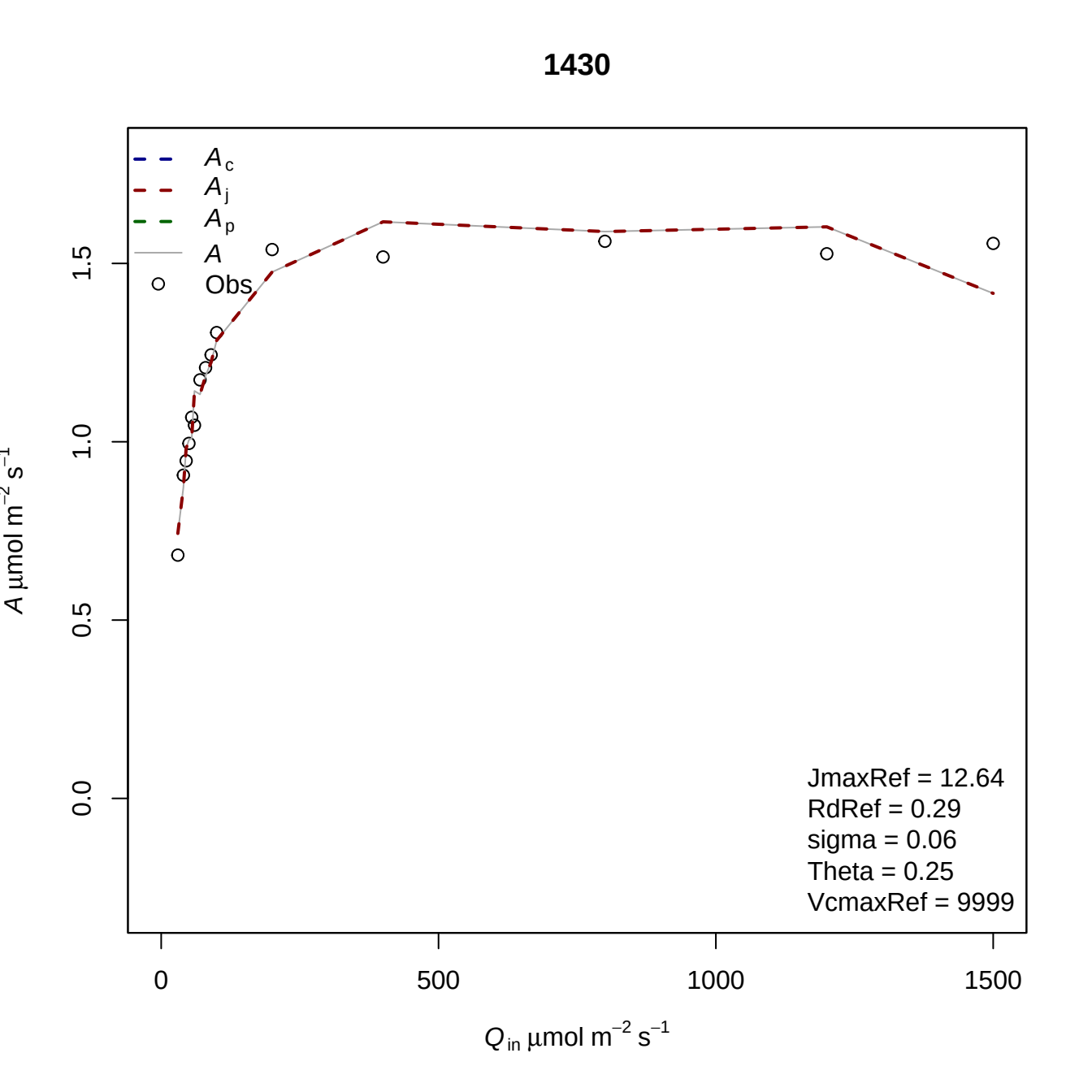

1425
A_c
A_j
A_p
A
Obs
JmaxRef = 16.74
RdRef = 0.27
sigma = 0.06
Theta = 0.06
VcmaxRef = 9999
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

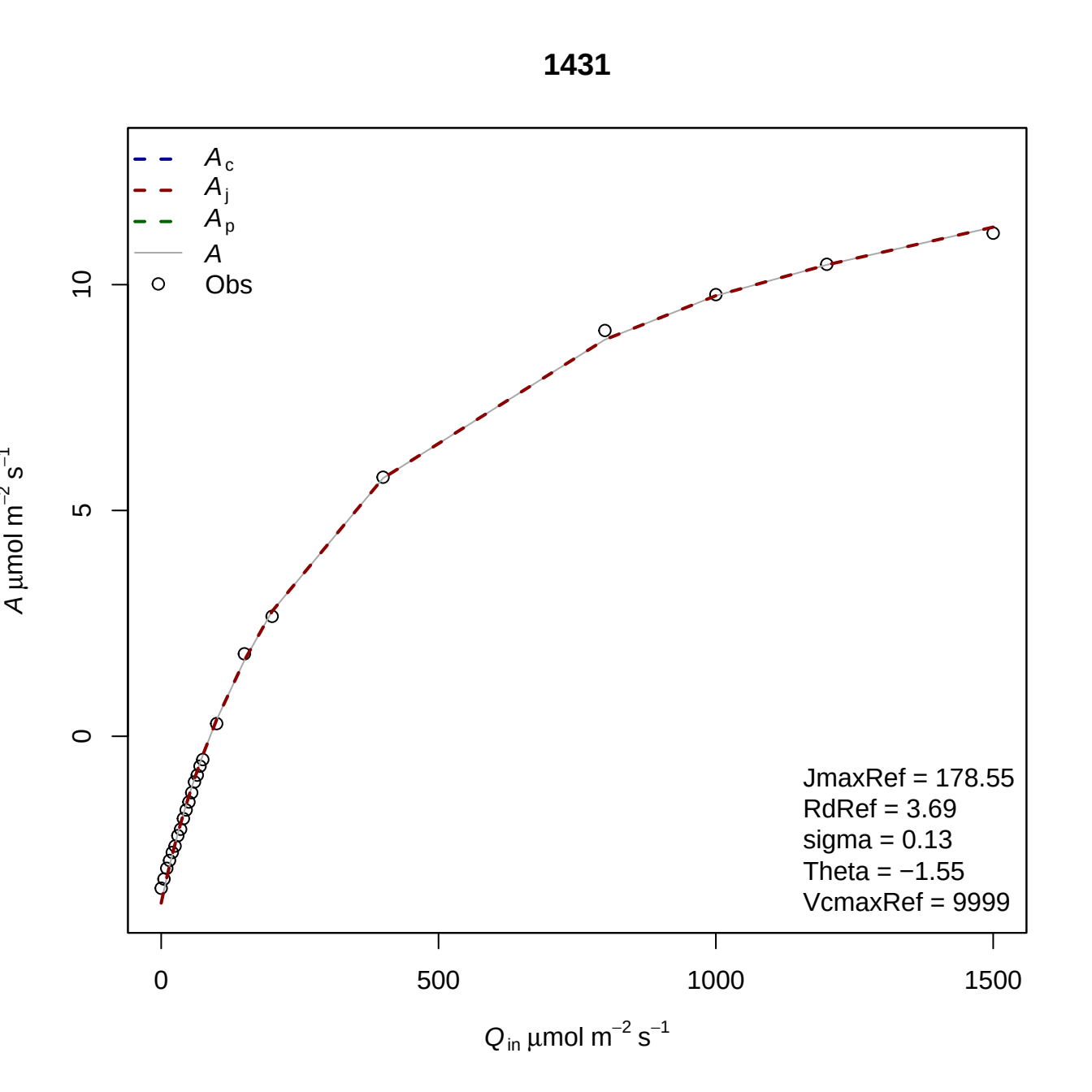

1426
A_c
A_j
A_p
A
Obs
JmaxRef = 30.89
RdRef = 0.13
sigma = 0.11
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

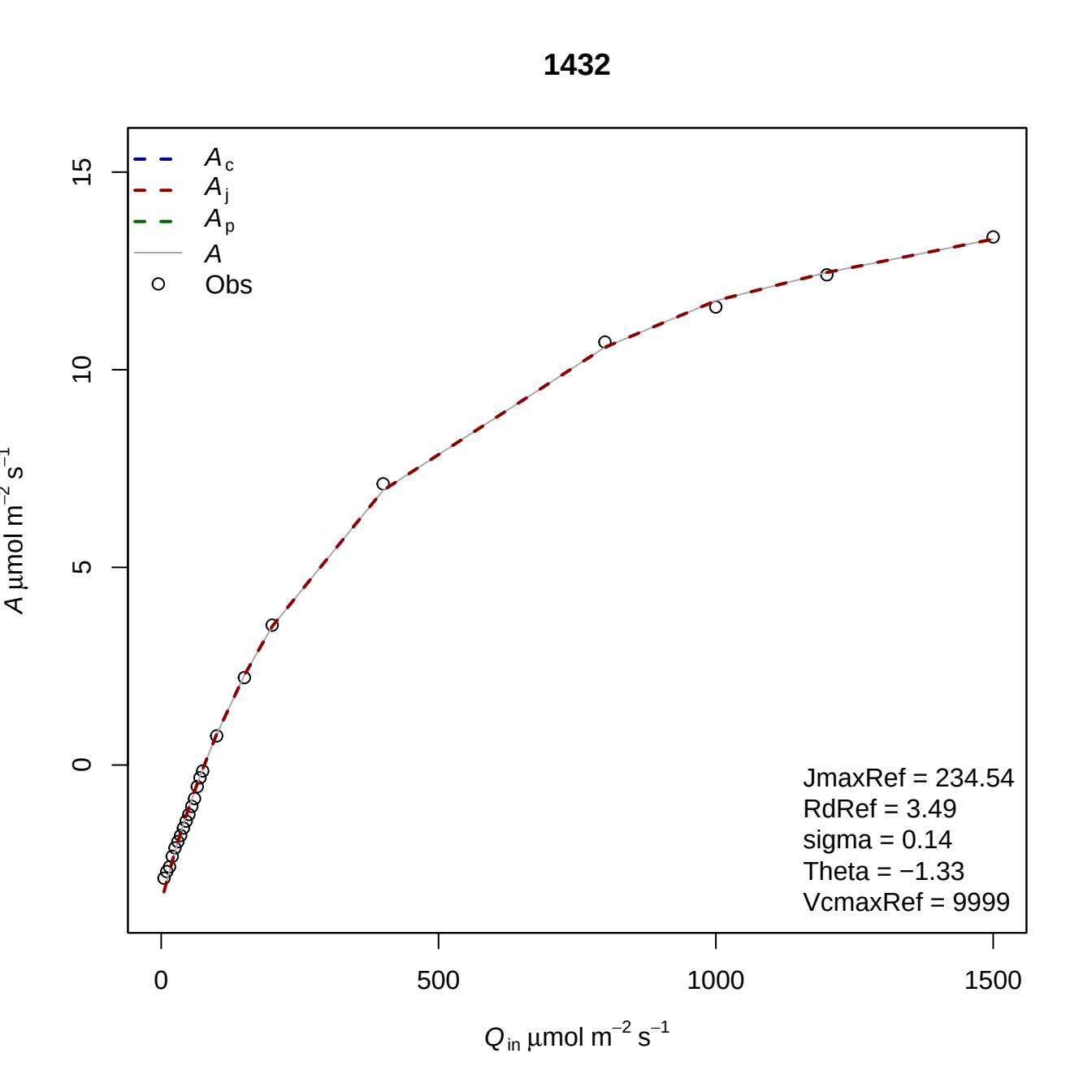

1427
A_c
A_j
A_p
A
Obs
JmaxRef = 17.13
RdRef = 0.16
sigma = 0.05
Theta = 0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
500
1000
1500

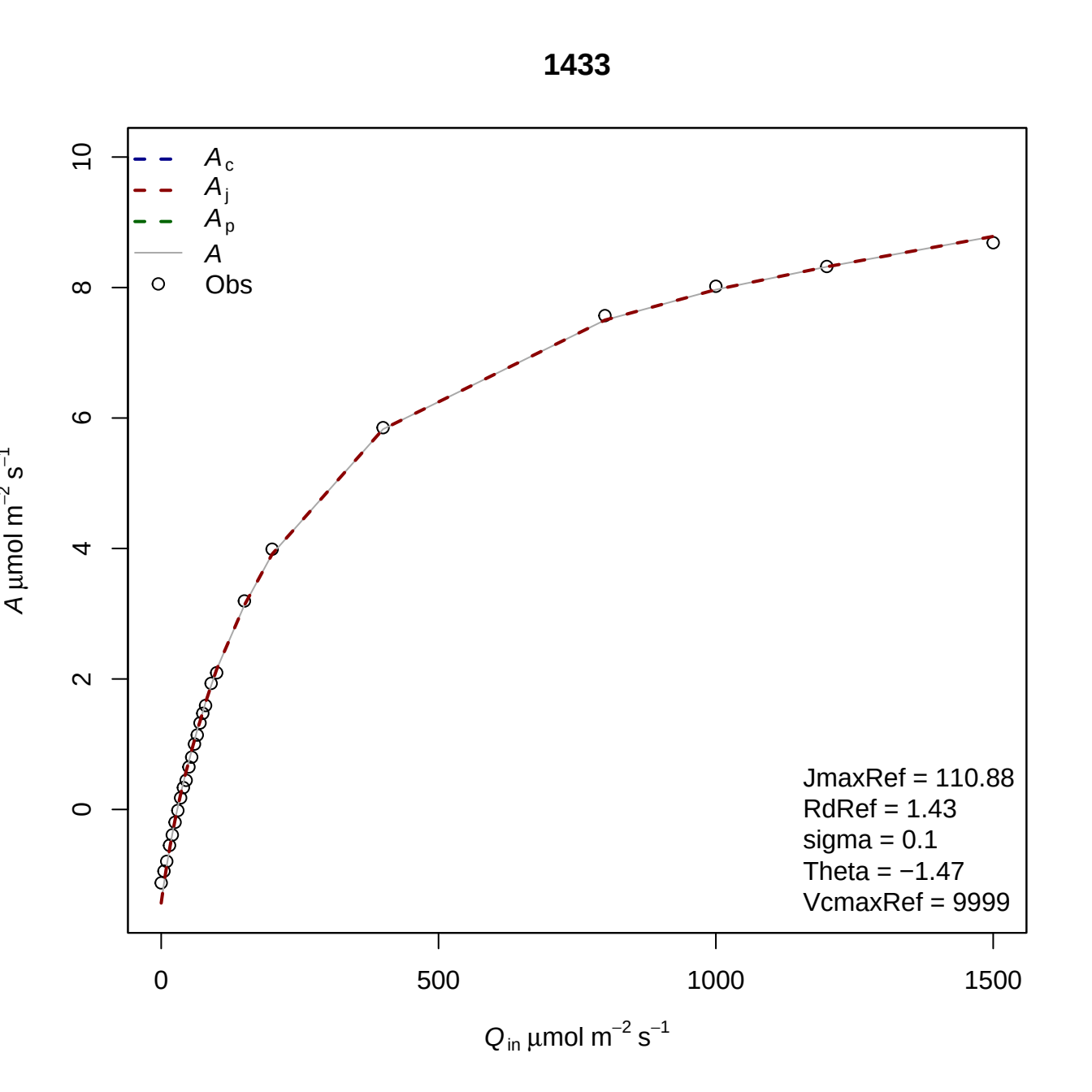

1428
A_c
A_j
A_p
A
Obs
JmaxRef = 13.47
RdRef = 0.19
sigma = 0.07
Theta = −0.11
VcmaxRef = 9999
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

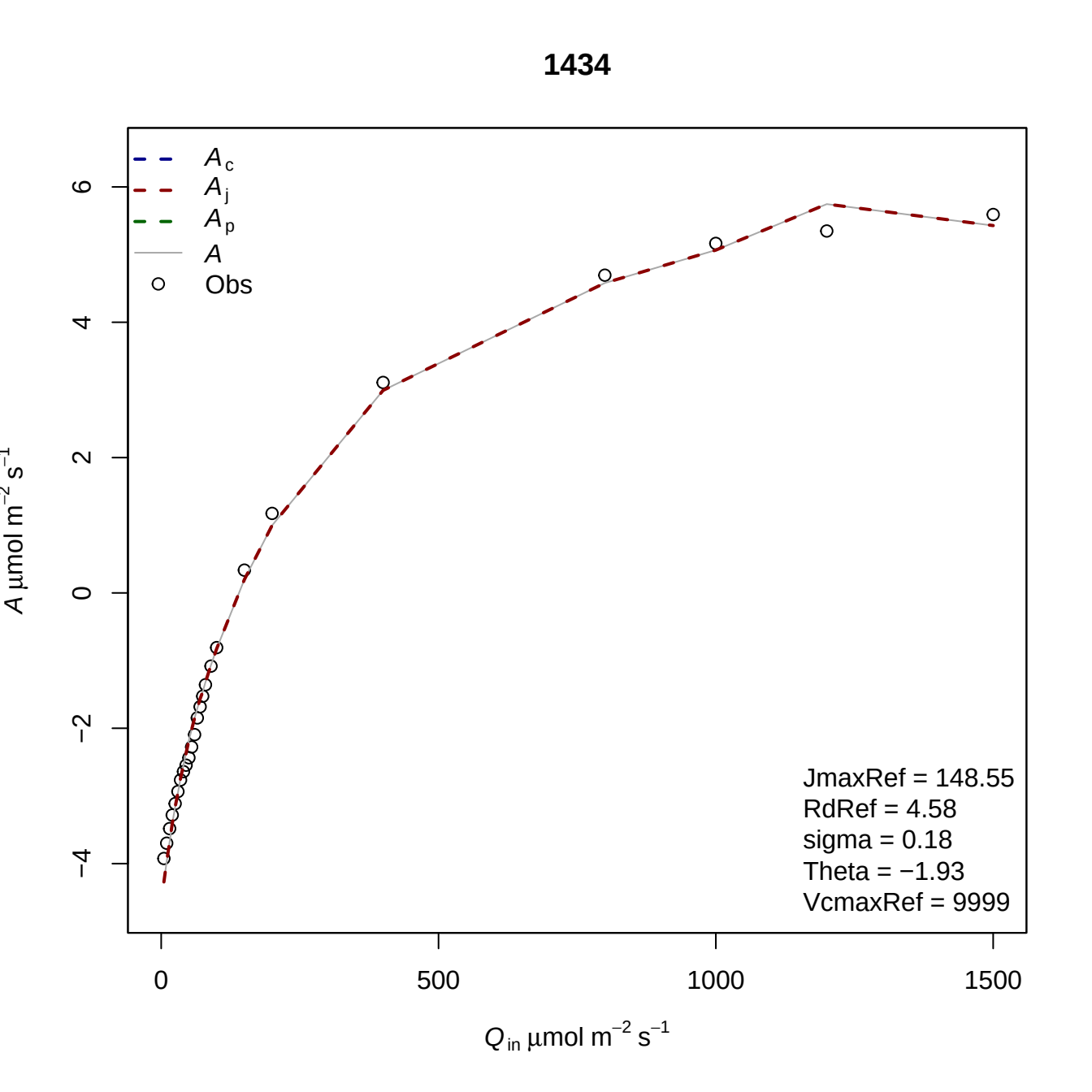

1429
A_c
A_j
A_p
A
Obs
JmaxRef = 13.99
RdRef = 0.12
sigma = 0.06
Theta = 0.36
VcmaxRef = 9999
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

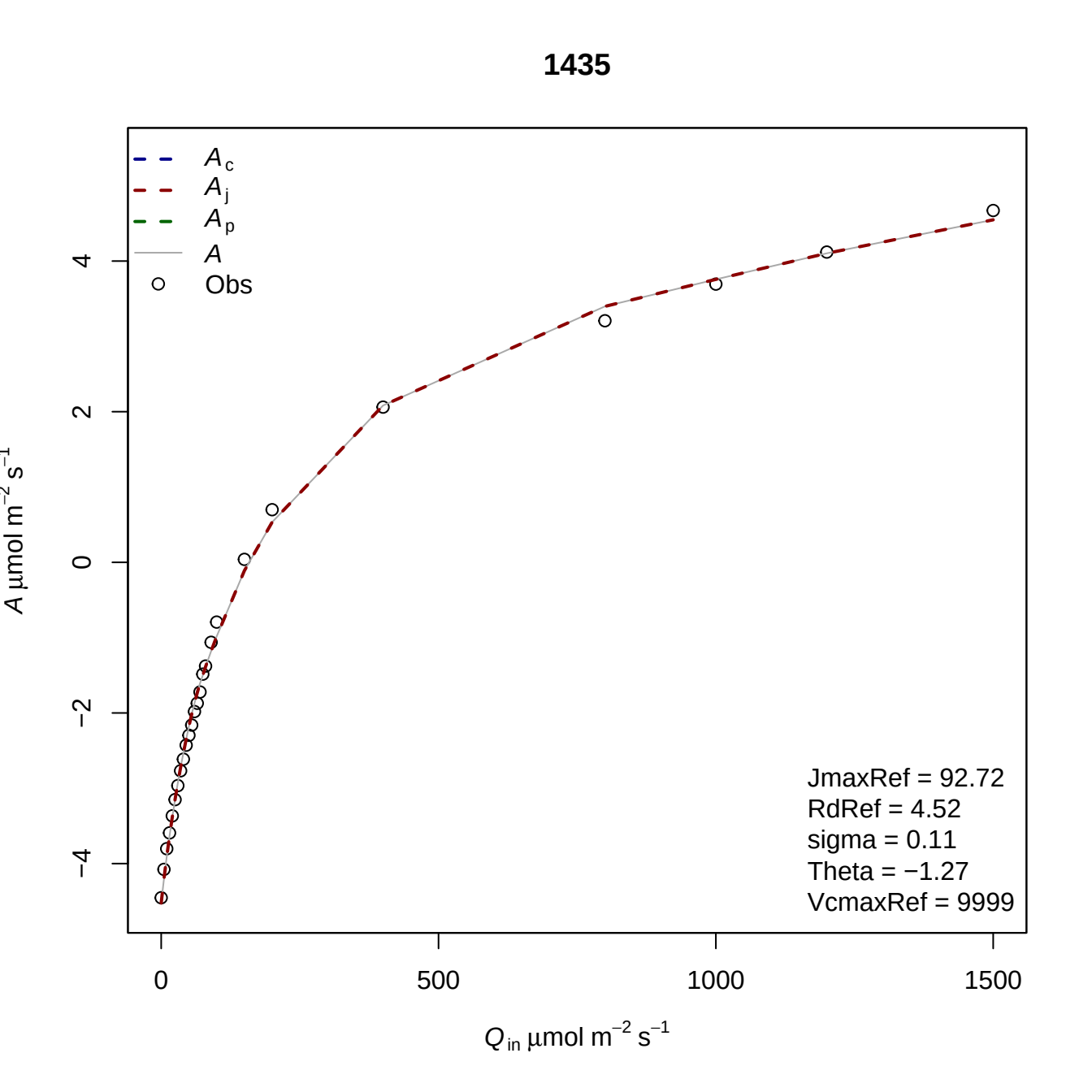

1430
A_c
A_j
A_p
A
Obs
JmaxRef = 12.64
RdRef = 0.29
sigma = 0.06
Theta = 0.25
VcmaxRef = 9999
0.0
0.5
1.0
1.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

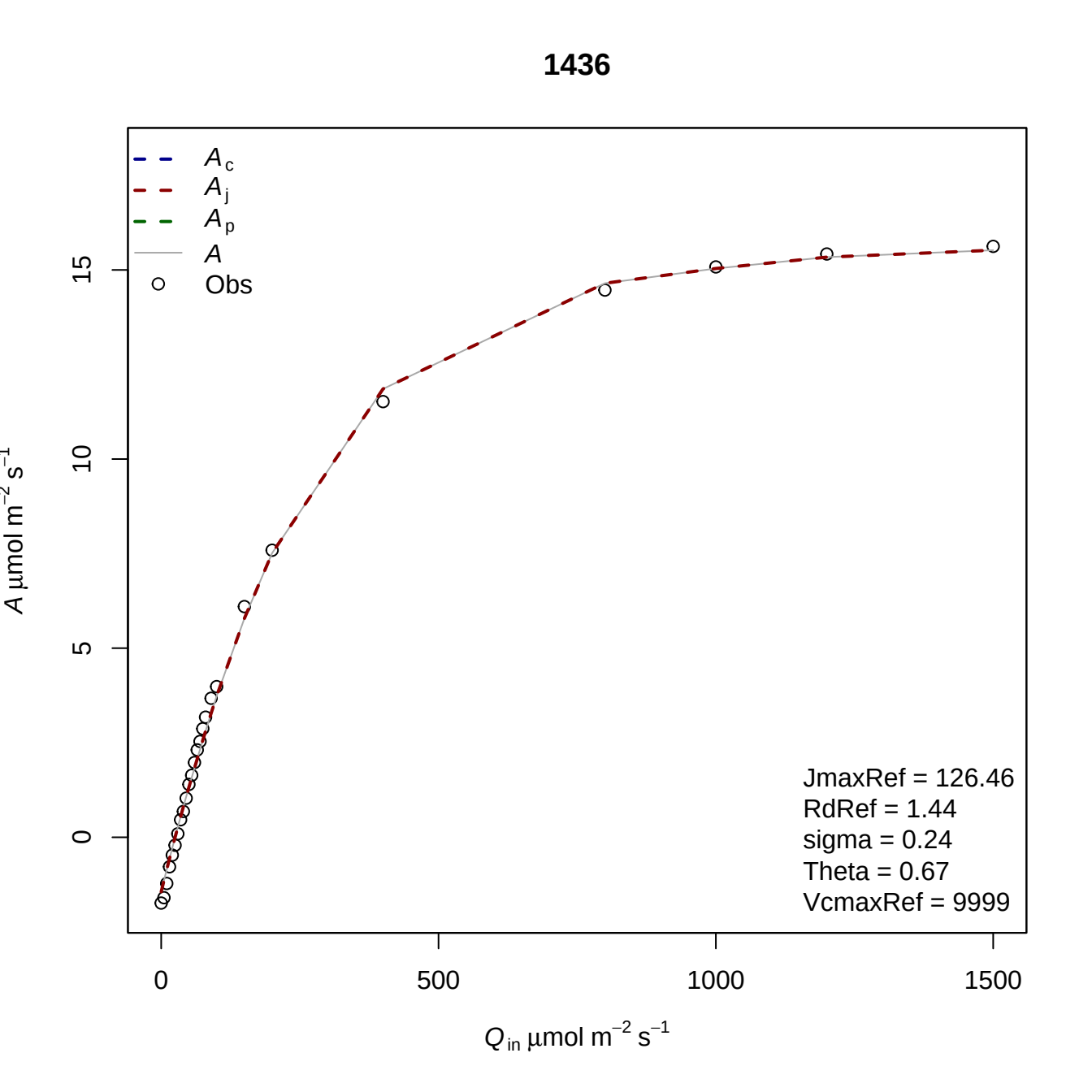

1431
A_c
A_j
A_p
A
Obs
JmaxRef = 178.55
RdRef = 3.69
sigma = 0.13
Theta = −1.55
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

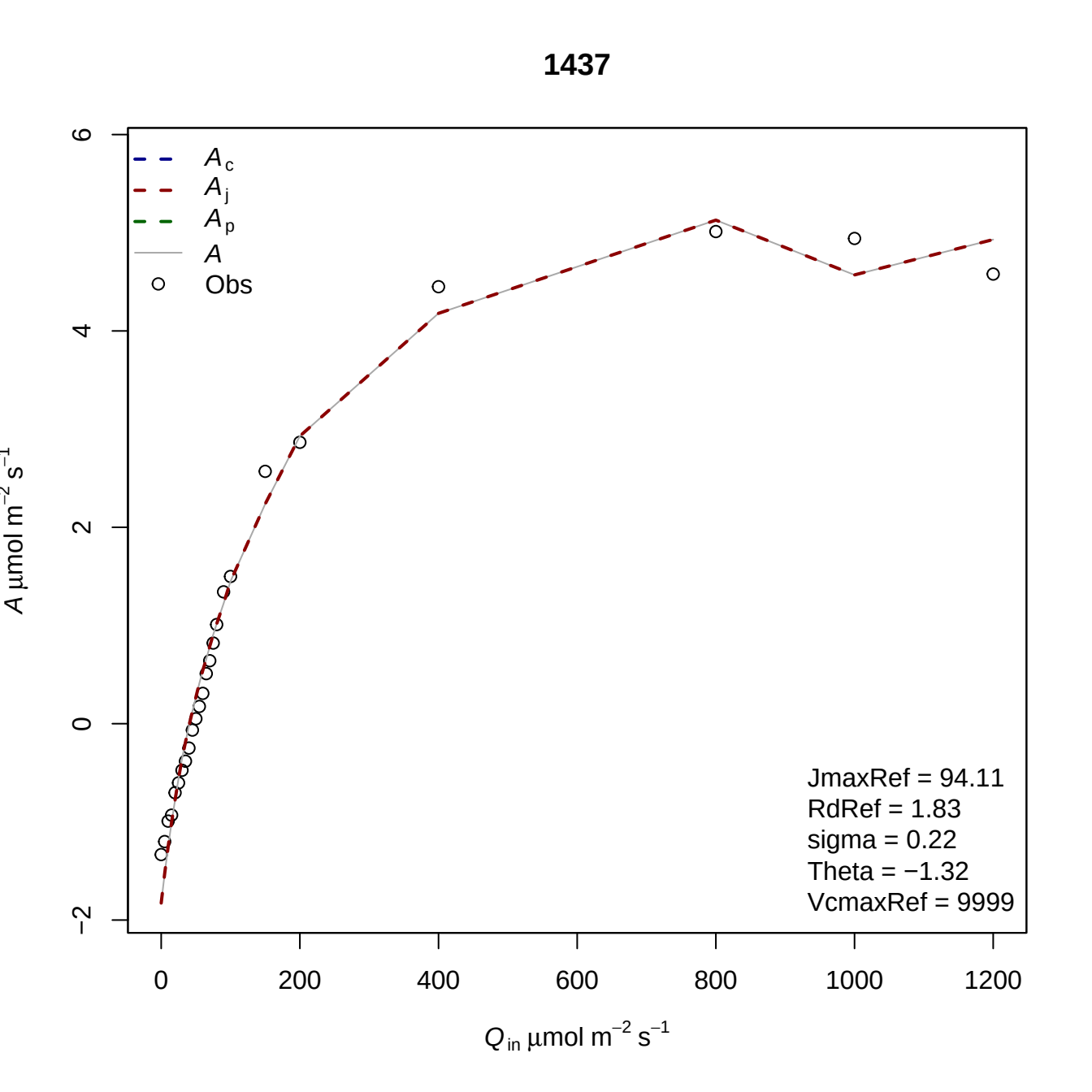

1432
A_c
A_j
A_p
A
Obs
JmaxRef = 234.54
RdRef = 3.49
sigma = 0.14
Theta = −1.33
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

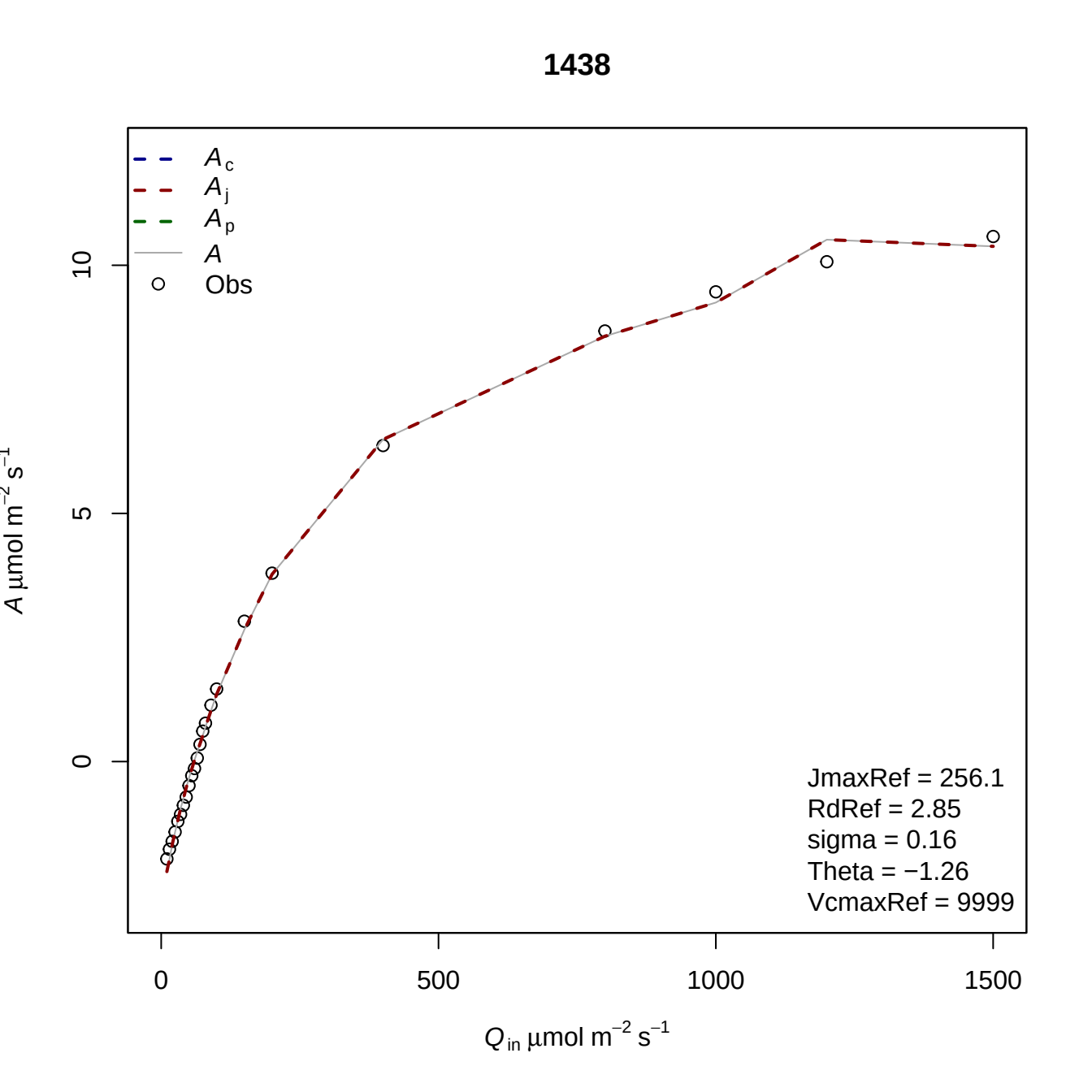

1433
A_c
A_j
A_p
A
Obs
JmaxRef = 110.88
RdRef = 1.43
sigma = 0.1
Theta = −1.47
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

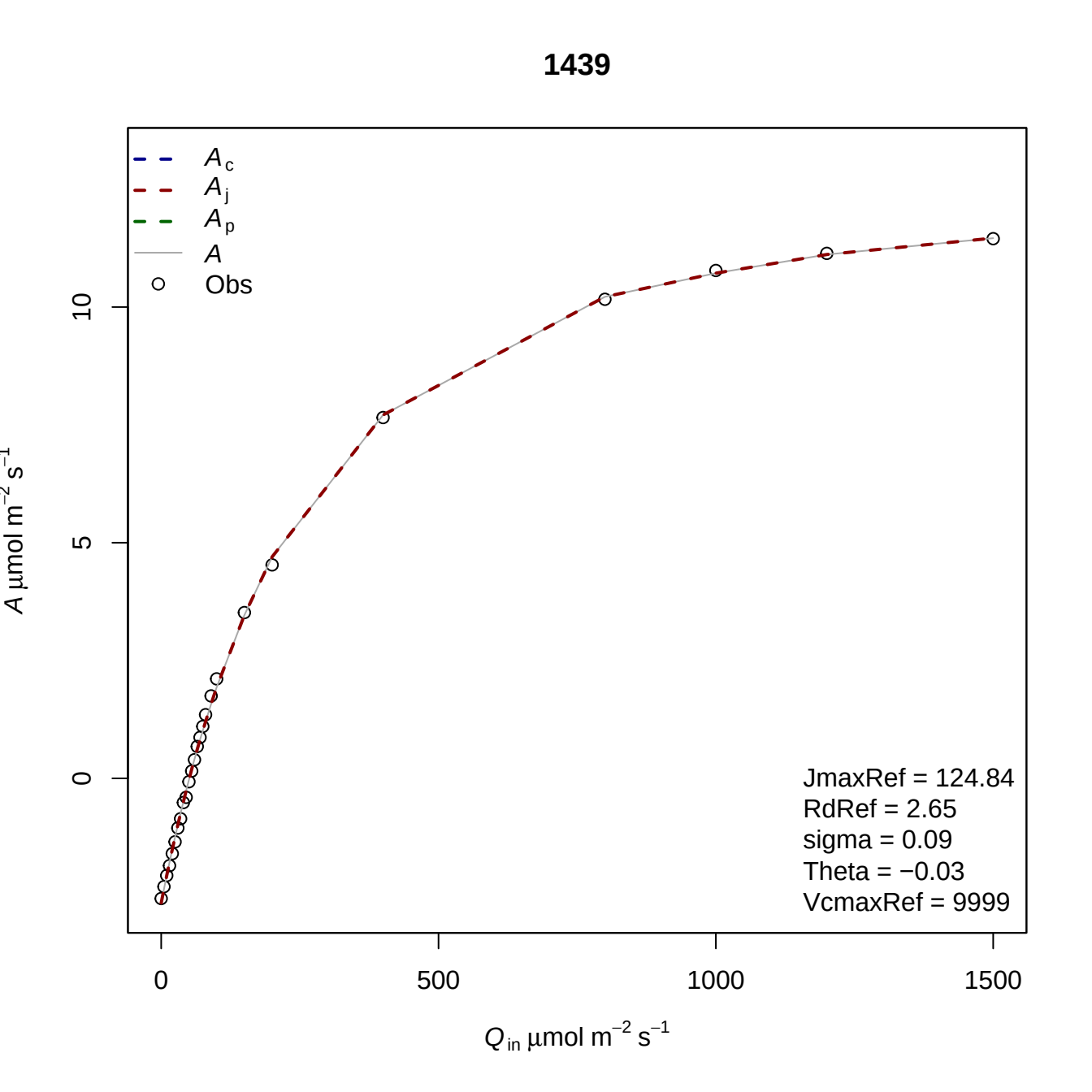

1434
A_c
A_j
A_p
A
Obs
JmaxRef = 148.55
RdRef = 4.58
sigma = 0.18
Theta = −1.93
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

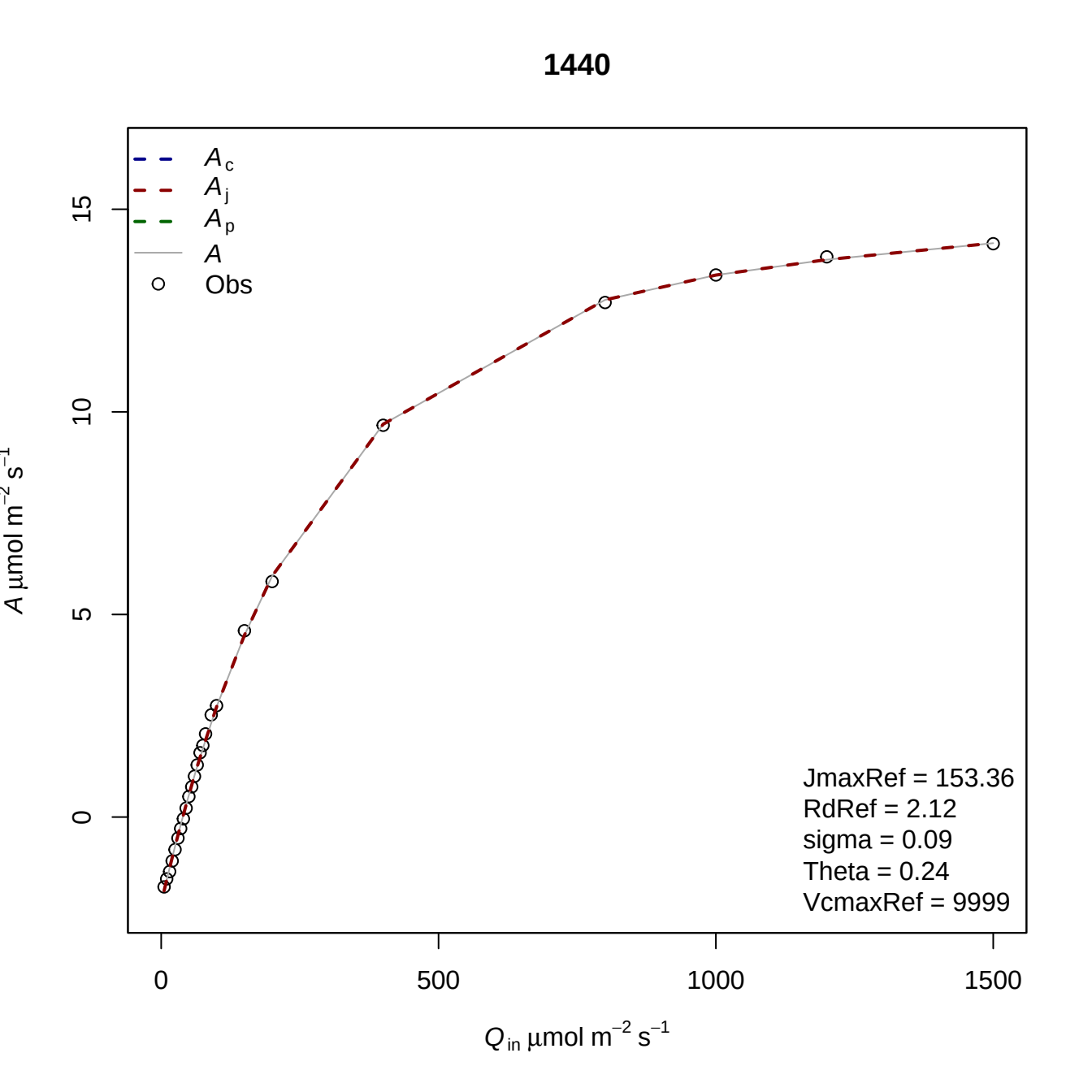

1435
A_c
A_j
A_p
A
Obs
JmaxRef = 92.72
RdRef = 4.52
sigma = 0.11
Theta = −1.27
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

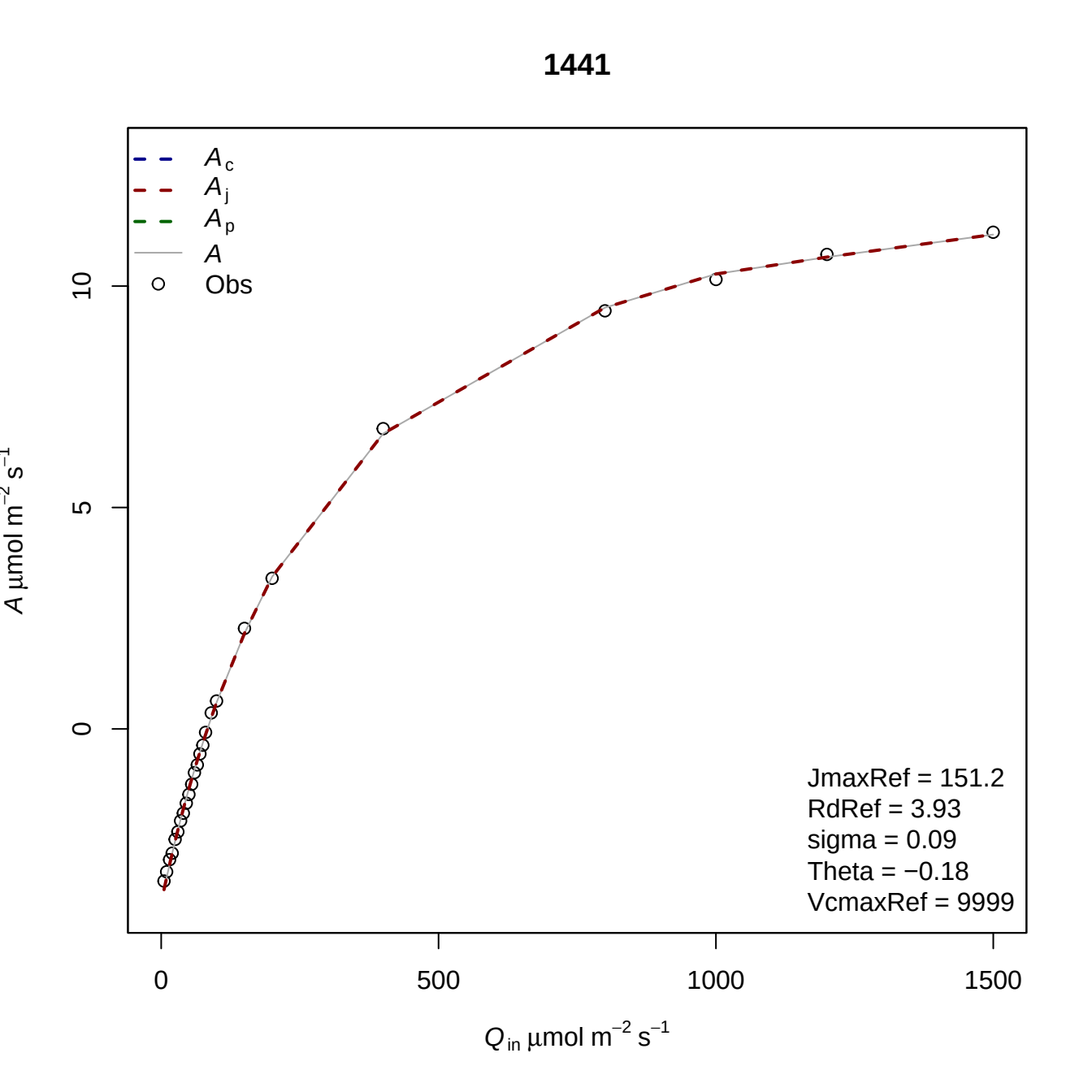

1436
A_c
A_j
A_p
A
Obs
JmaxRef = 126.46
RdRef = 1.44
sigma = 0.24
Theta = 0.67
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

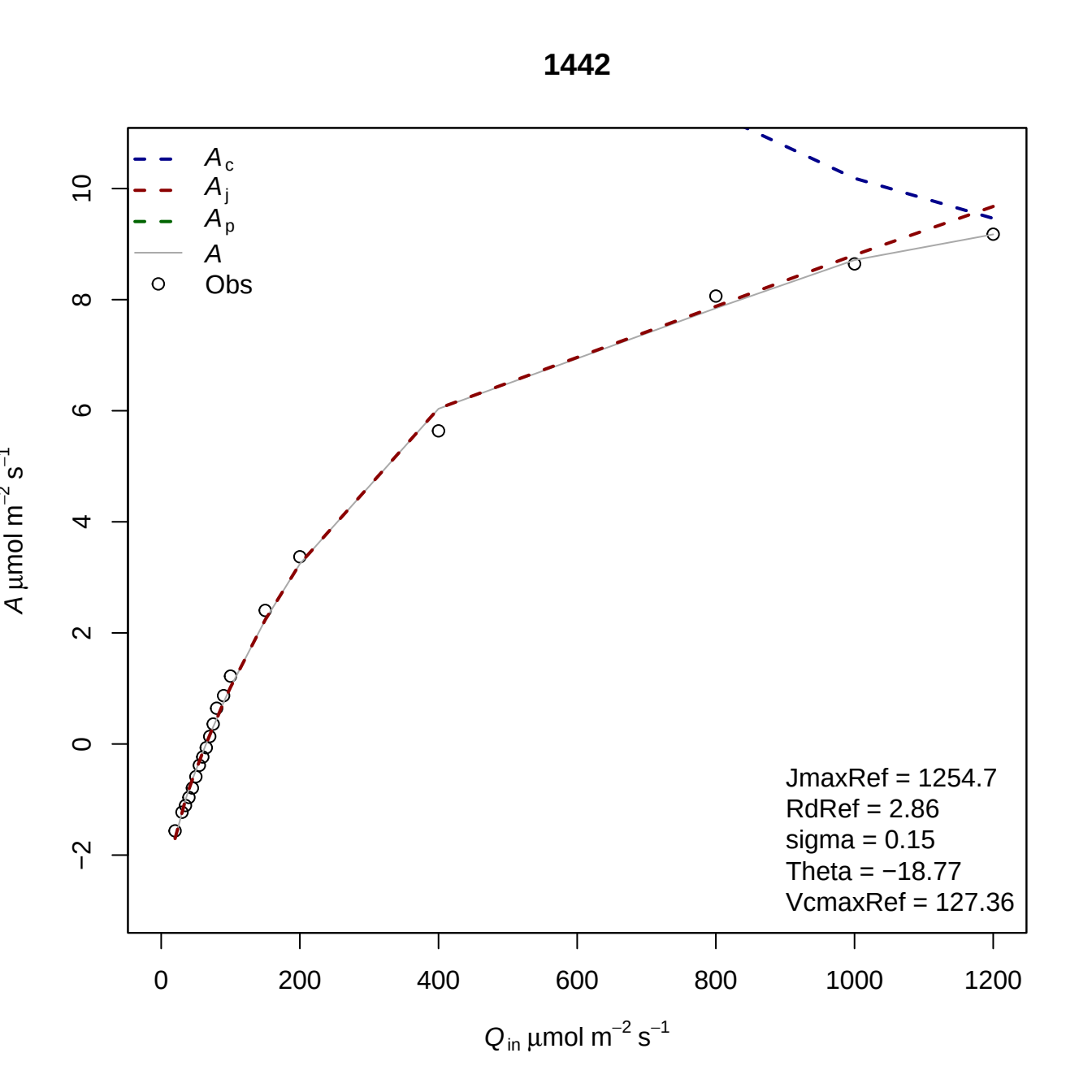

1437
A_c
A_j
A_p
A
Obs
JmaxRef = 94.11
RdRef = 1.83
sigma = 0.22
Theta = −1.32
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200

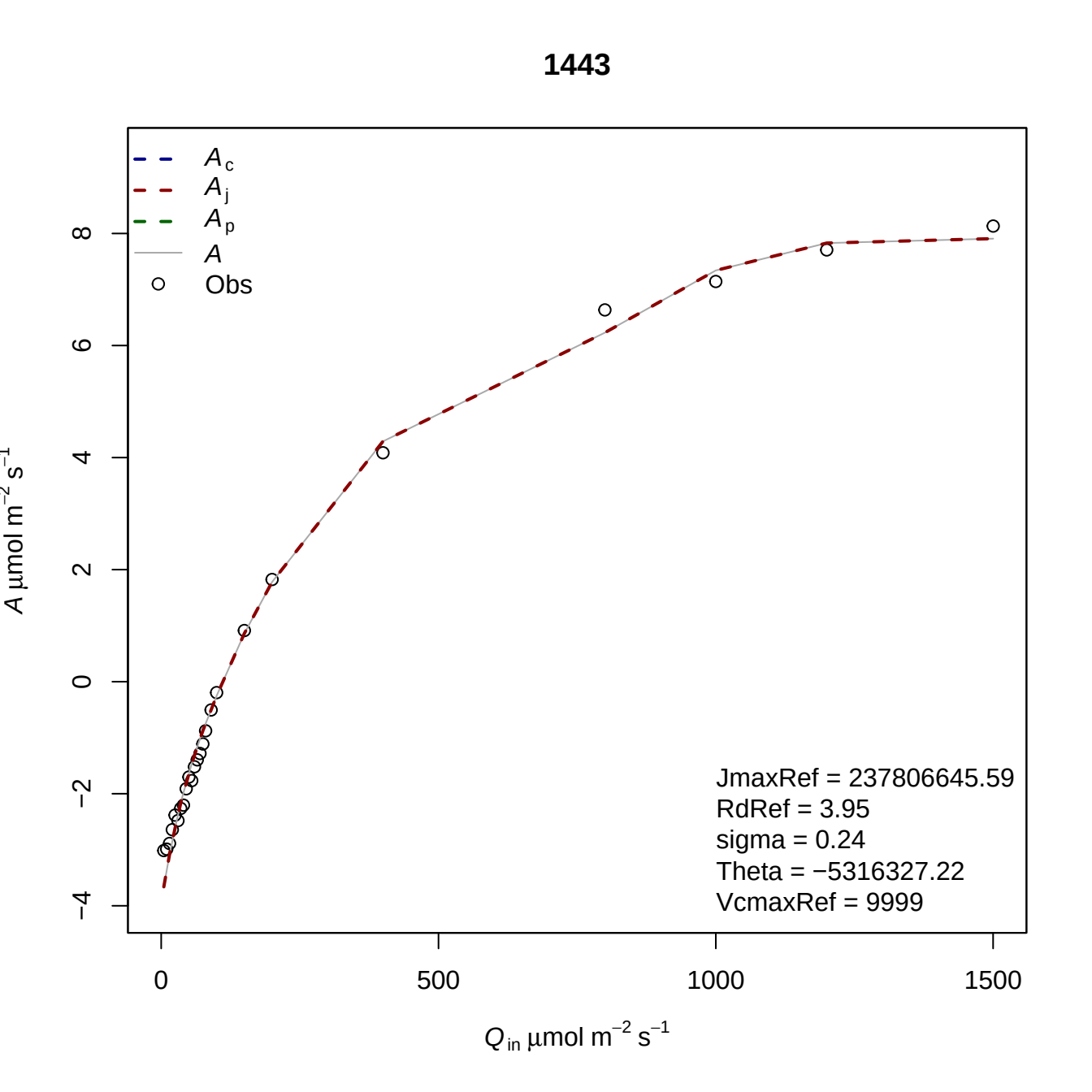

1438
A_c
A_j
A_p
A
Obs
JmaxRef = 256.1
RdRef = 2.85
sigma = 0.16
Theta = −1.26
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

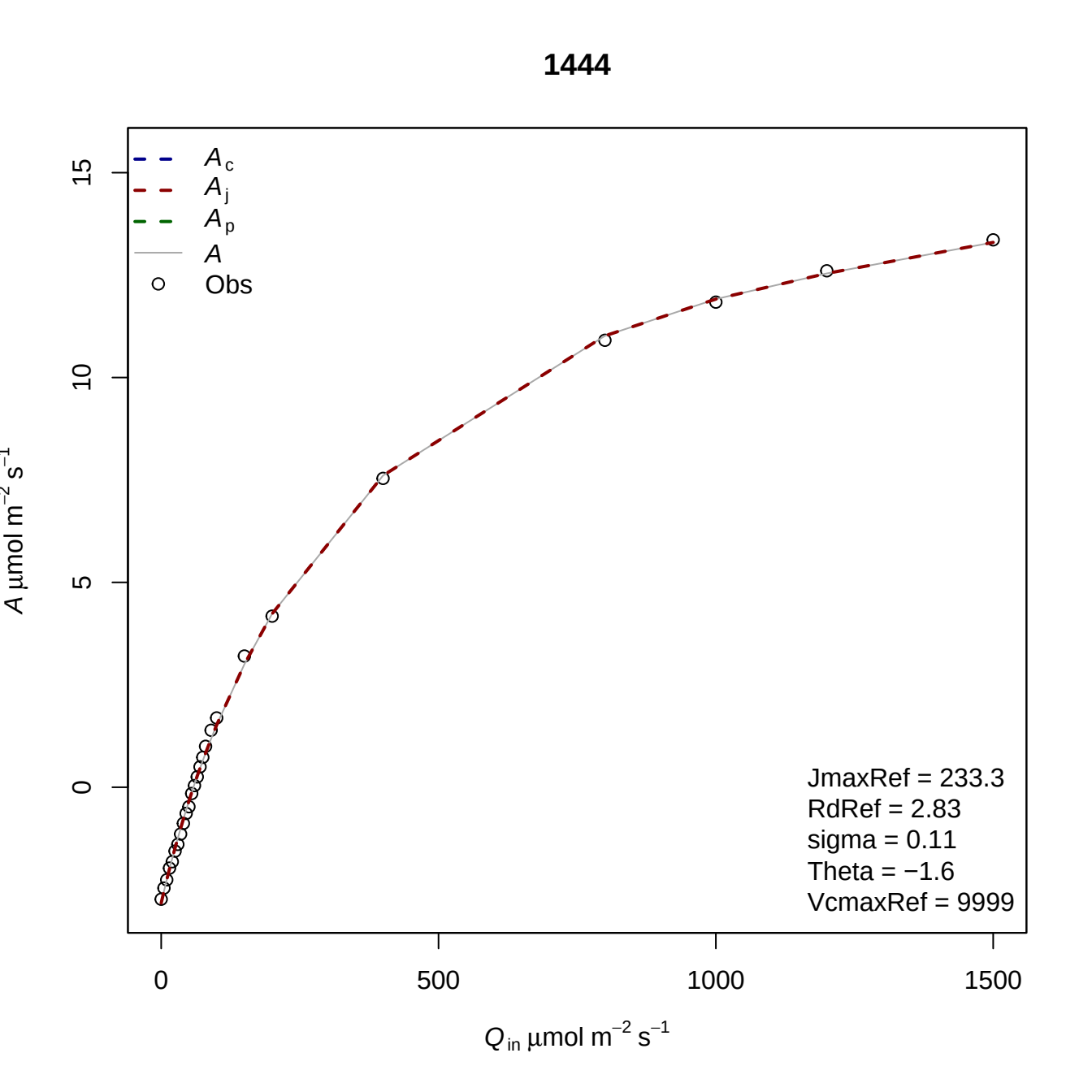

1439
A_c
A_j
A_p
A
Obs
JmaxRef = 124.84
RdRef = 2.65
sigma = 0.09
Theta = −0.03
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

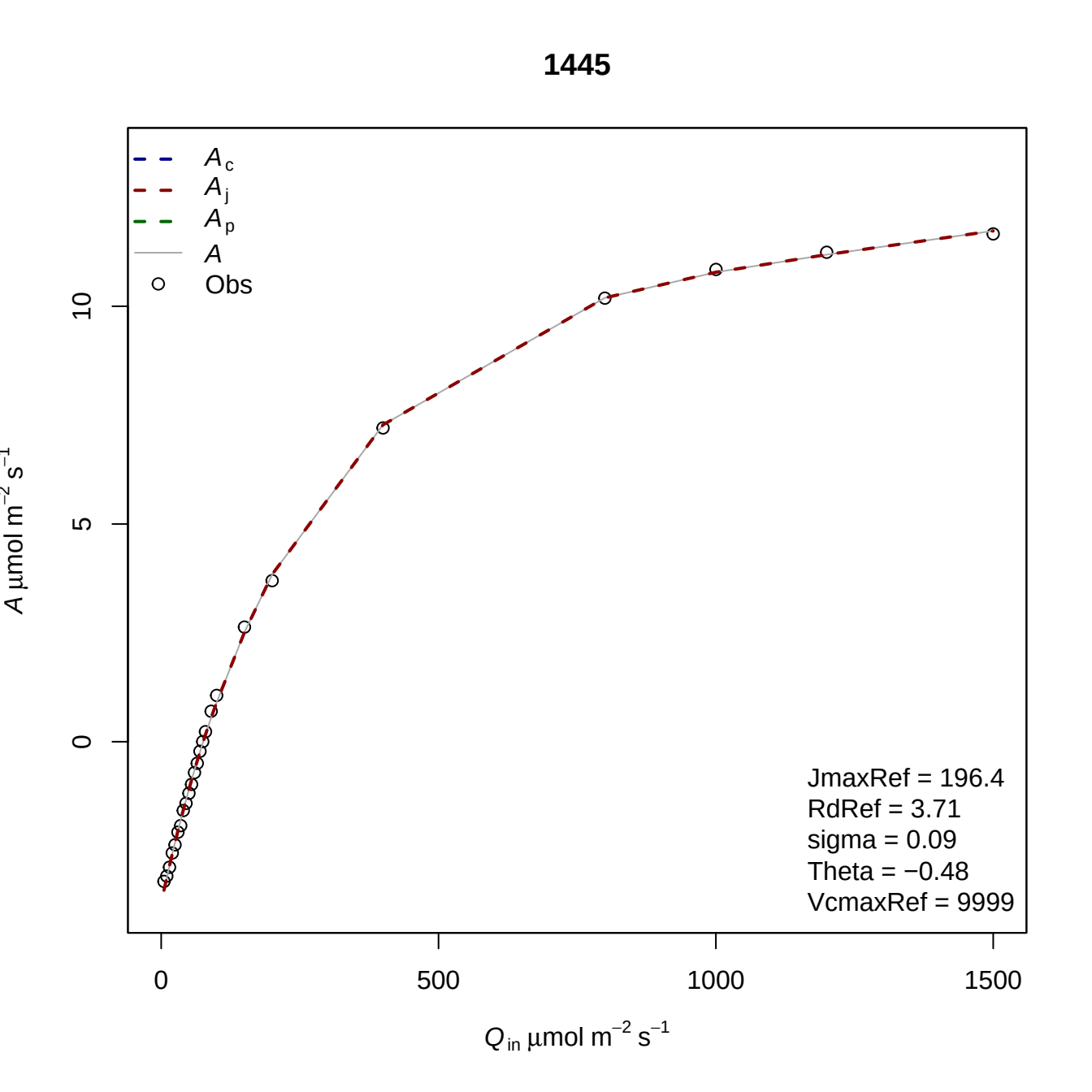

1440
A_c
A_j
A_p
A
Obs
JmaxRef = 153.36
RdRef = 2.12
sigma = 0.09
Theta = 0.24
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

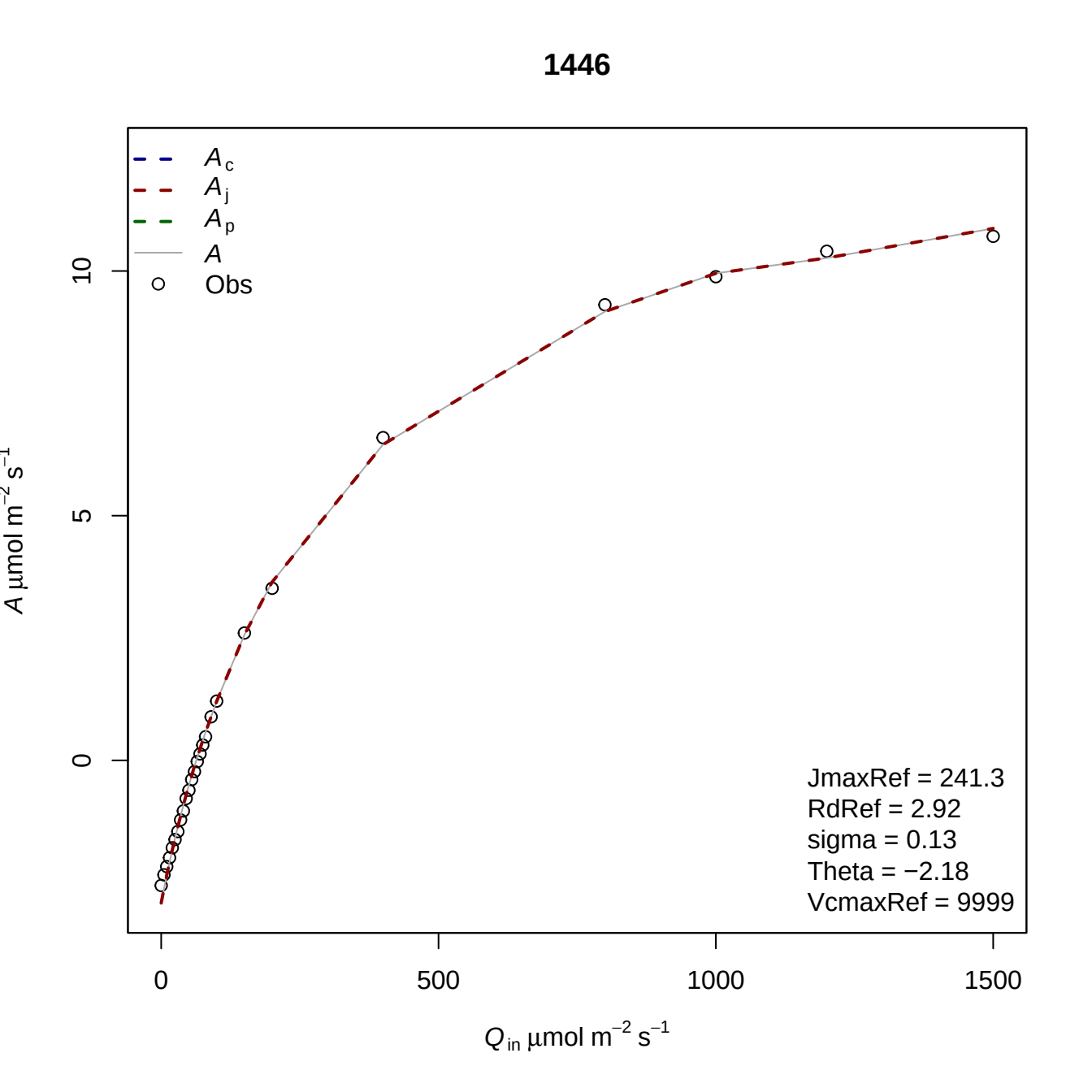

1441
A_c
A_j
A_p
A
Obs
JmaxRef = 151.2
RdRef = 3.93
sigma = 0.09
Theta = −0.18
VcmaxRef = 9999
0
5
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

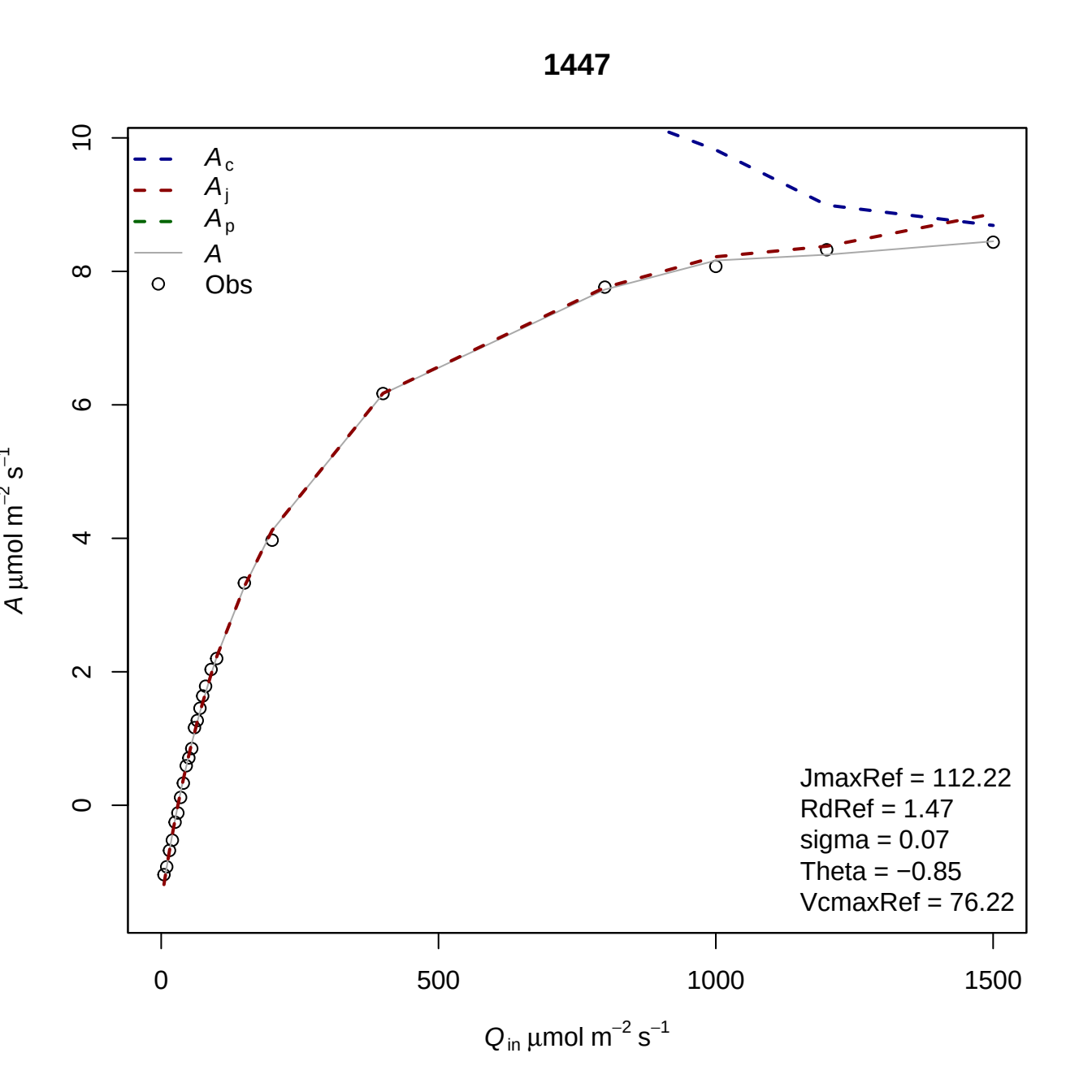

1442
A_c
A_j
A_p
A
Obs
JmaxRef = 1254.7
RdRef = 2.86
sigma = 0.15
Theta = −18.77
VcmaxRef = 127.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200

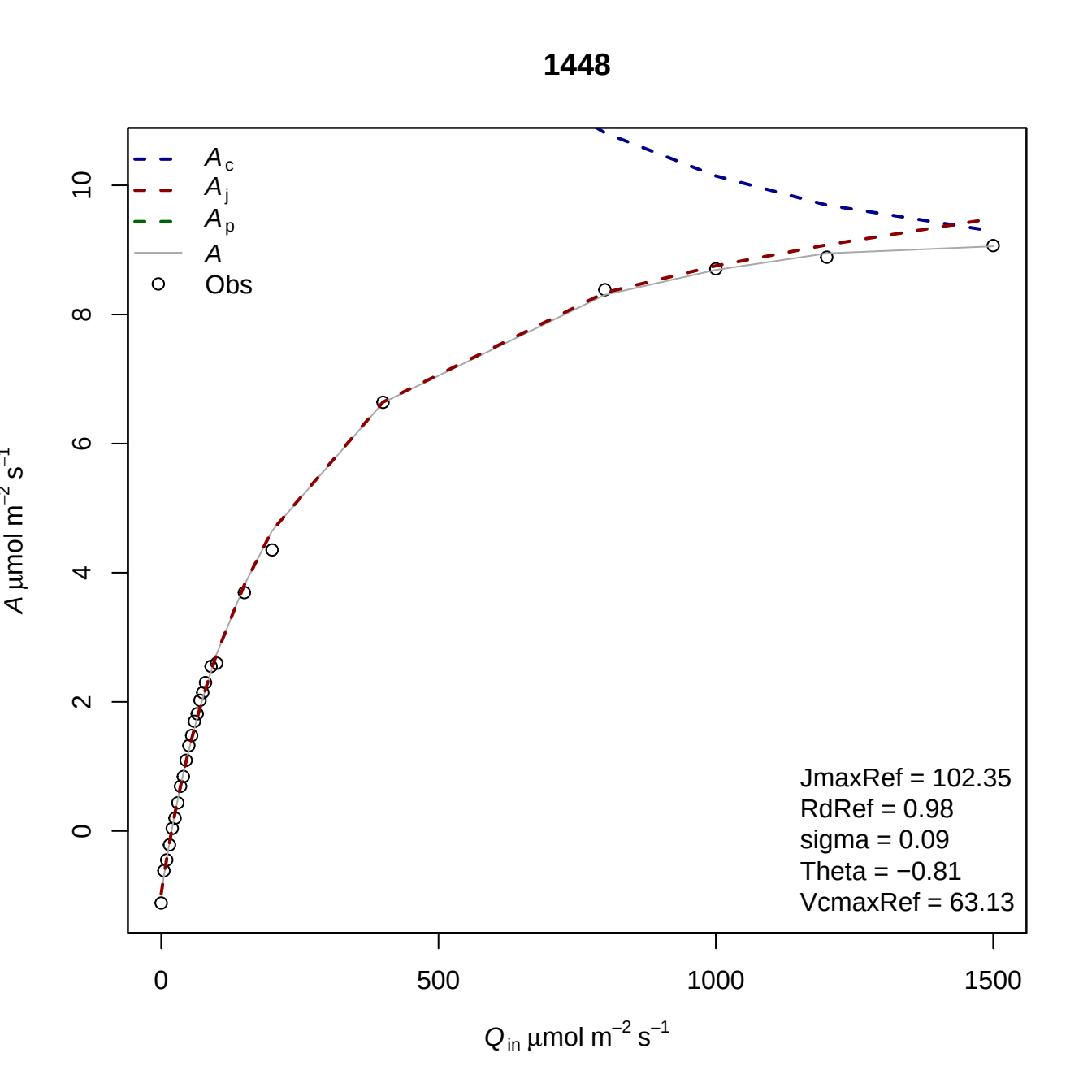

1443
A_c
A_j
A_p
A
Obs
JmaxRef = 237806645.59
RdRef = 3.95
sigma = 0.24
Theta = −5316327.22
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

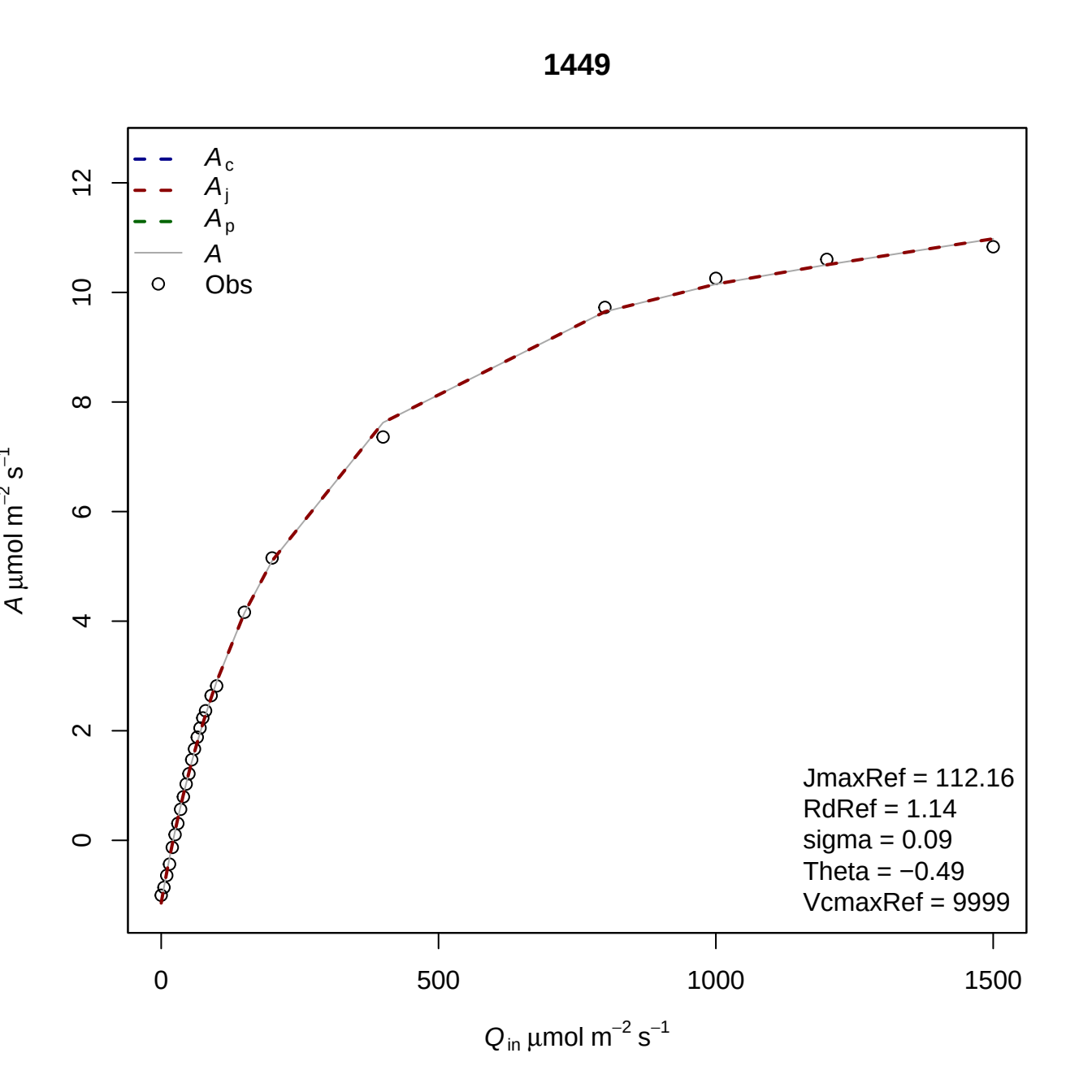

1444
A_c
A_j
A_p
A
Obs
JmaxRef = 233.3
RdRef = 2.83
sigma = 0.11
Theta = −1.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

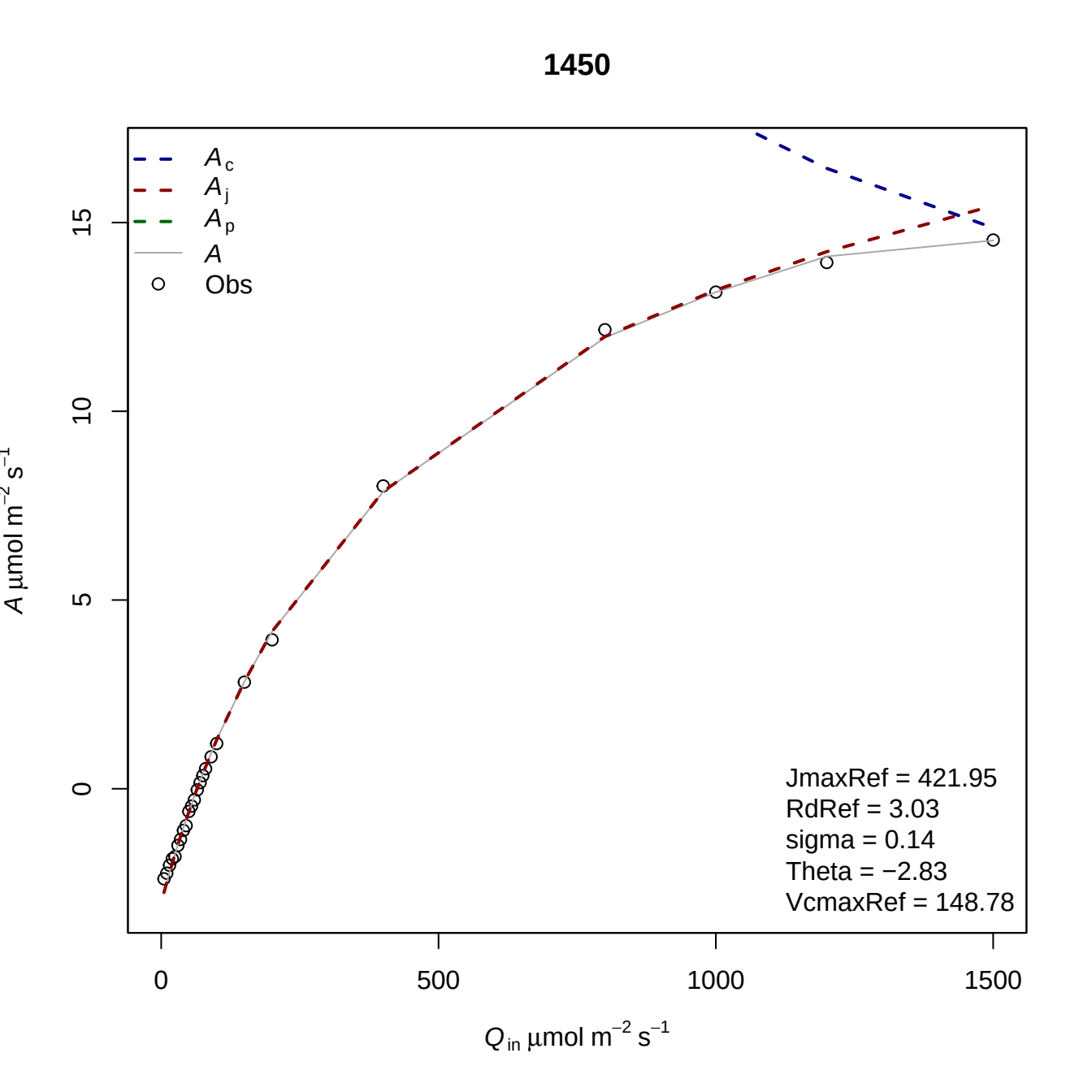

1445
A_c
A_j
A_p
A
Obs
JmaxRef = 196.4
RdRef = 3.71
sigma = 0.09
Theta = −0.48
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

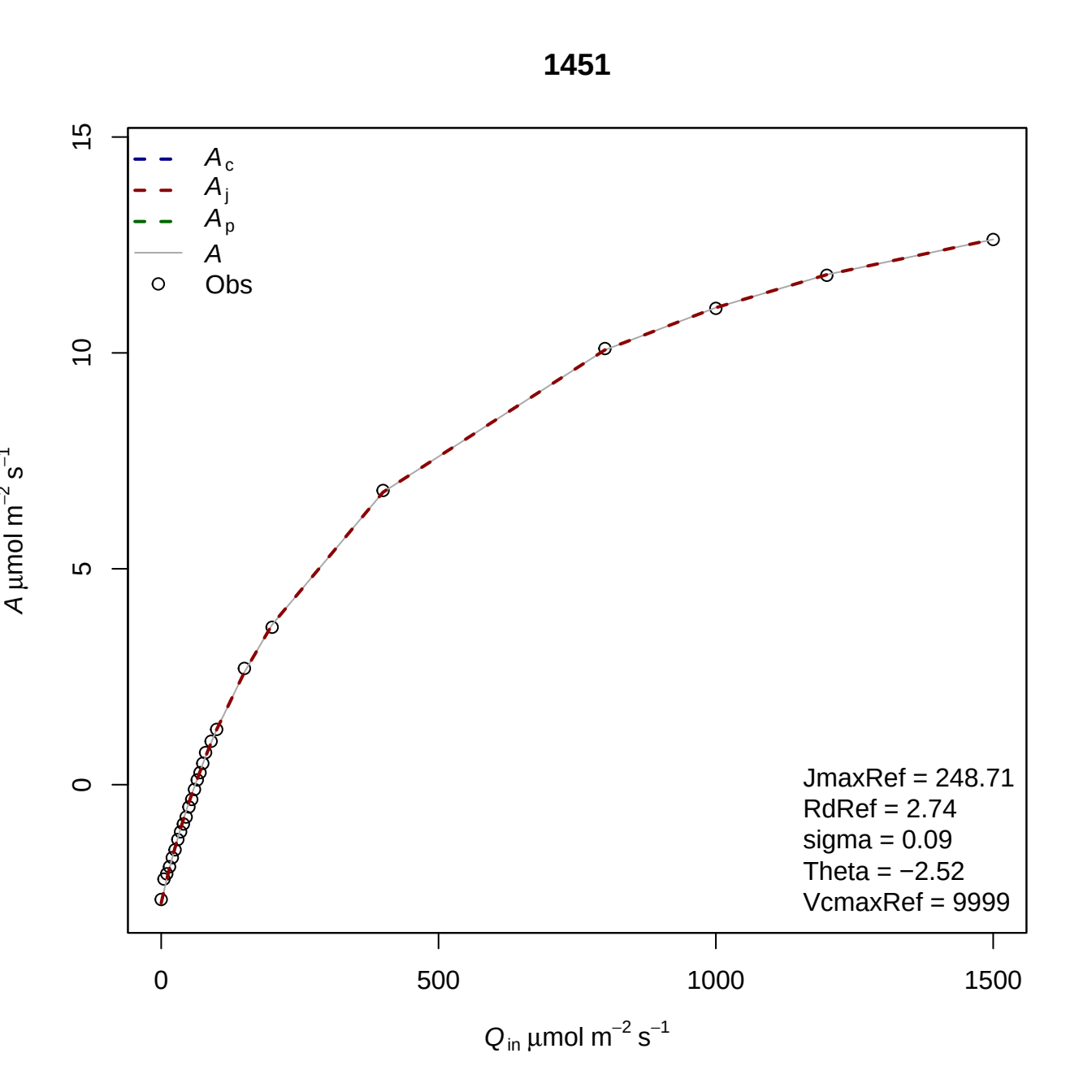

1446
A_c
A_j
A_p
A
Obs
JmaxRef = 241.3
RdRef = 2.92
sigma = 0.13
Theta = −2.18
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

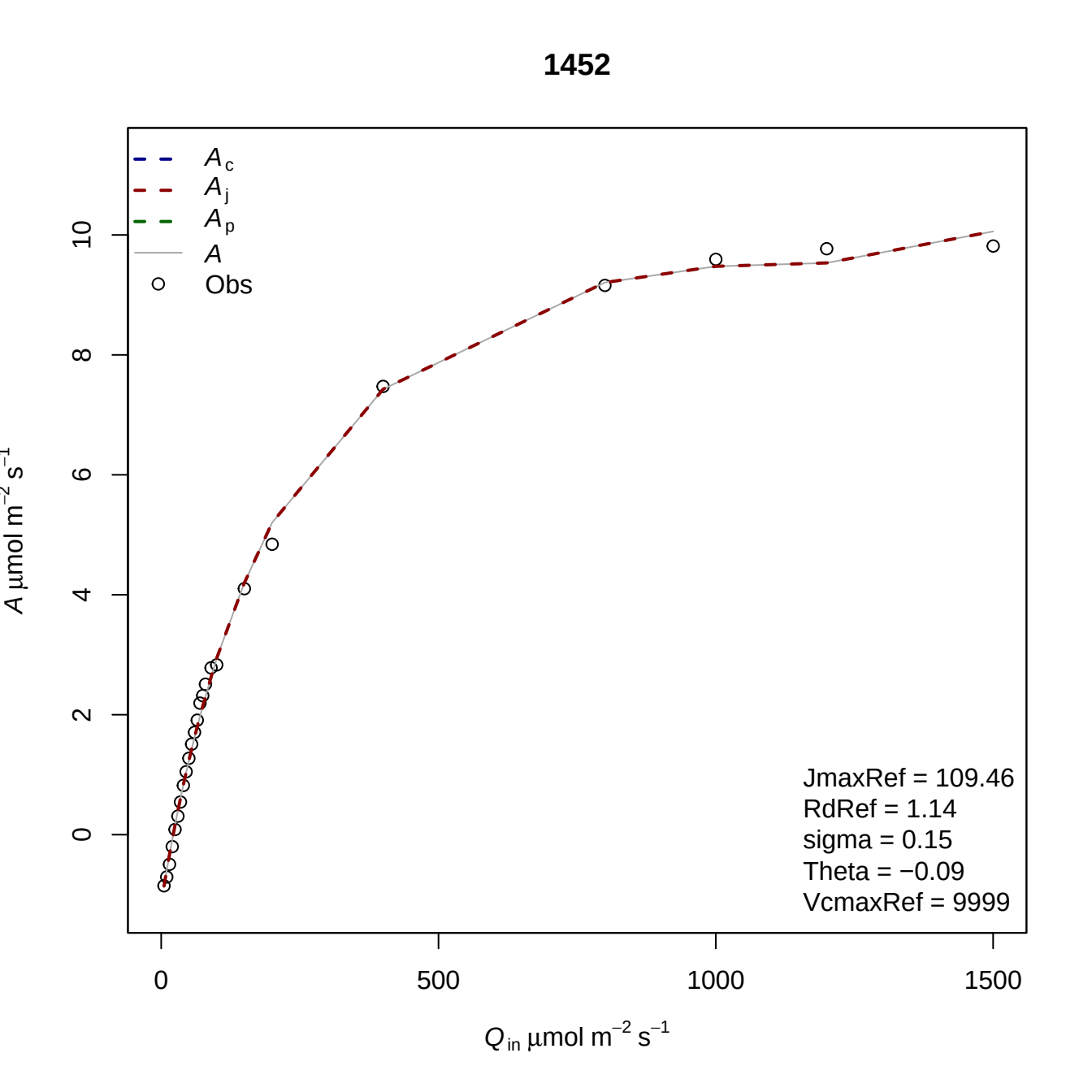

1447
A_c
A_j
A_p
A
Obs
JmaxRef = 112.22
RdRef = 1.47
sigma = 0.07
Theta = −0.85
VcmaxRef = 76.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

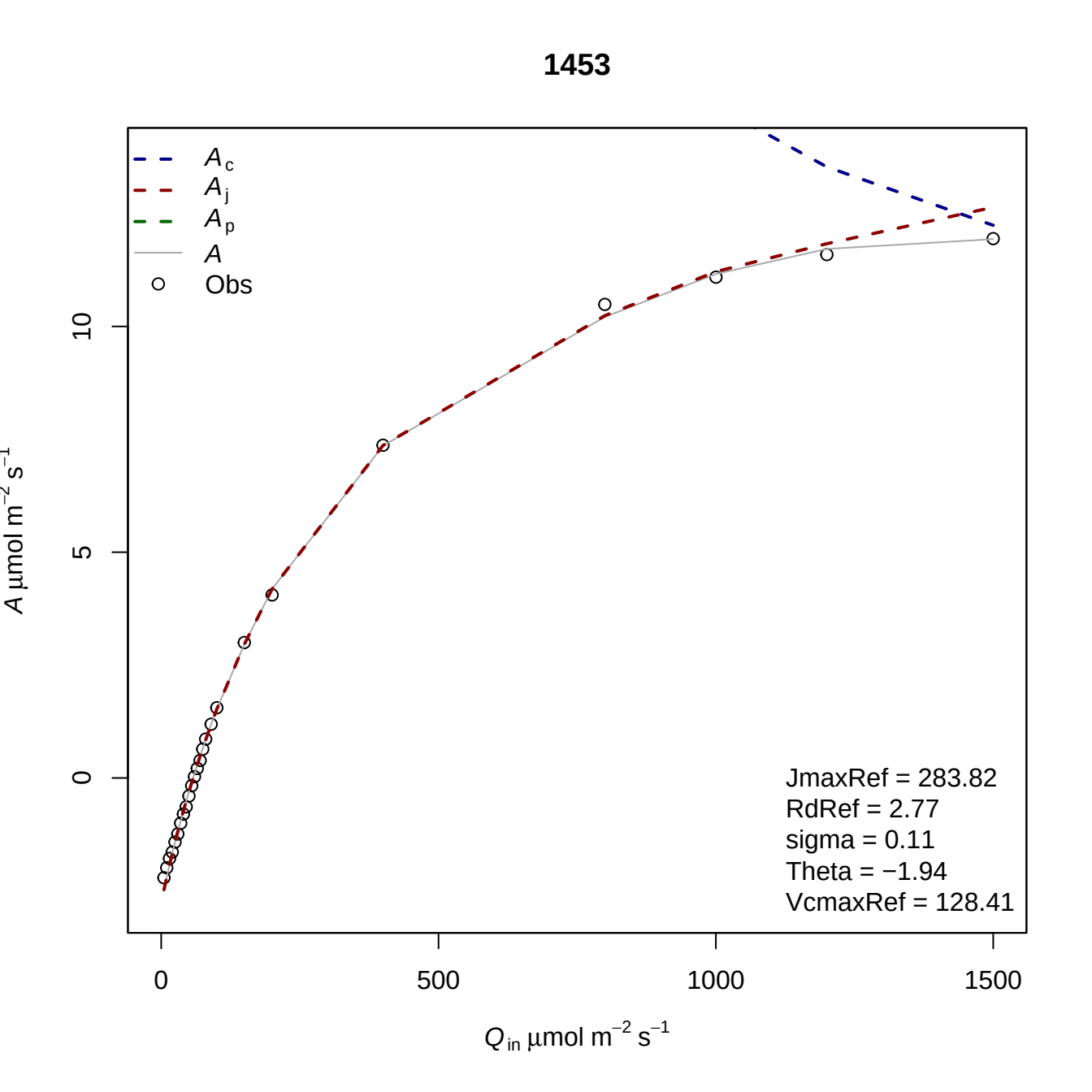

1448
A_c
A_j
A_p
A
Obs
JmaxRef = 102.35
RdRef = 0.98
sigma = 0.09
Theta = −0.81
VcmaxRef = 63.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

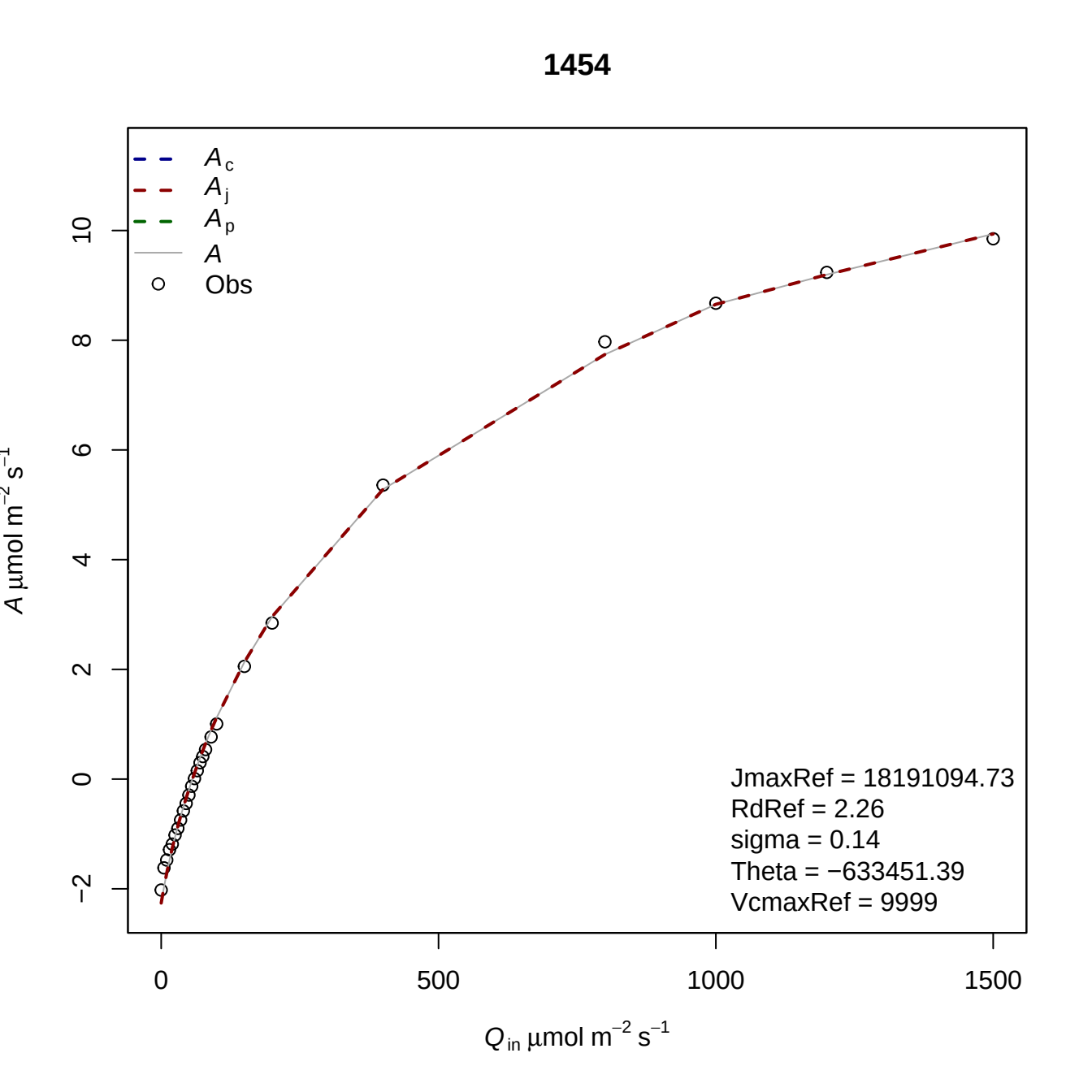

1449
A_c
A_j
A_p
A
Obs
JmaxRef = 112.16
RdRef = 1.14
sigma = 0.09
Theta = −0.49
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

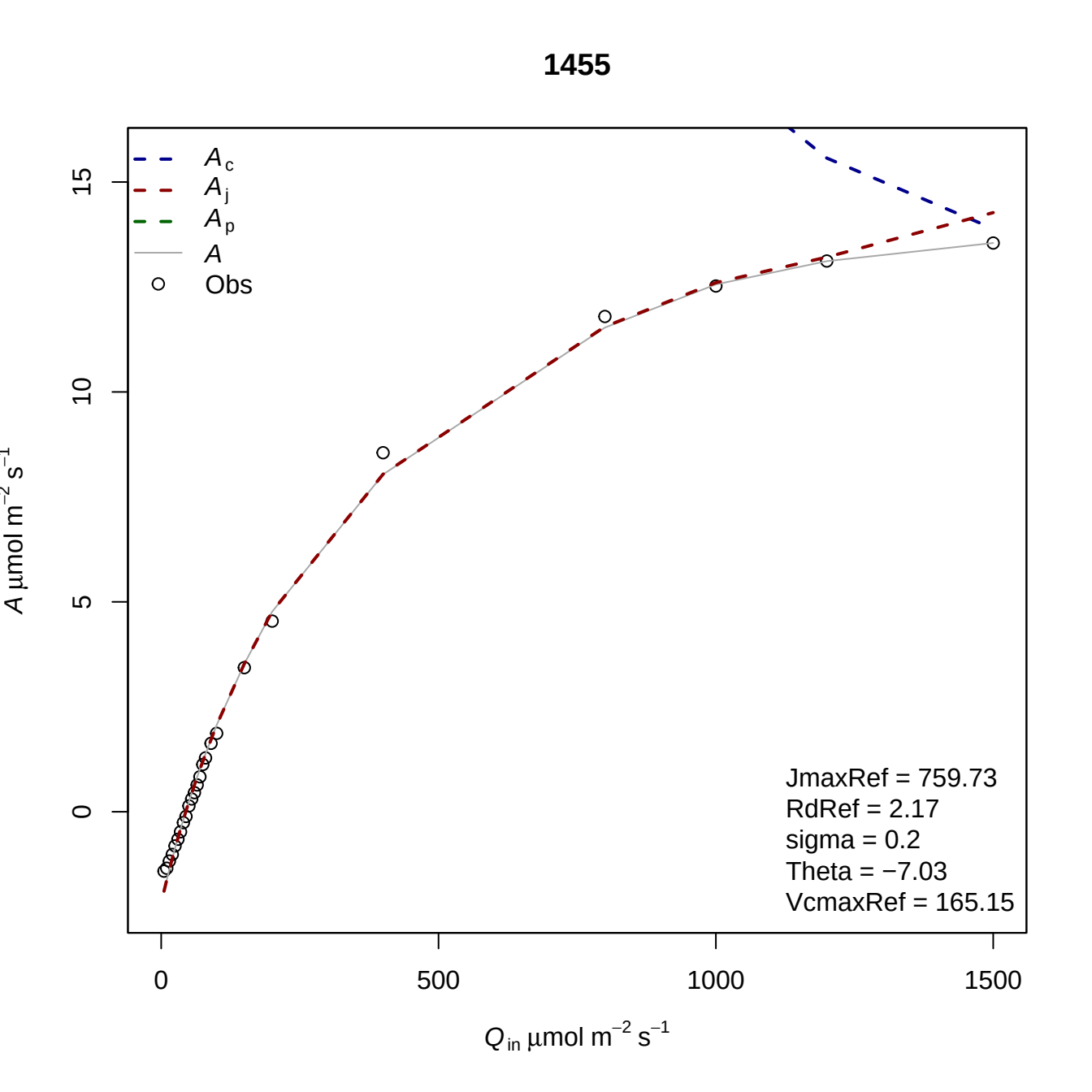

1450
A_c
A_j
A_p
A
Obs
JmaxRef = 421.95
RdRef = 3.03
sigma = 0.14
Theta = −2.83
VcmaxRef = 148.78
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

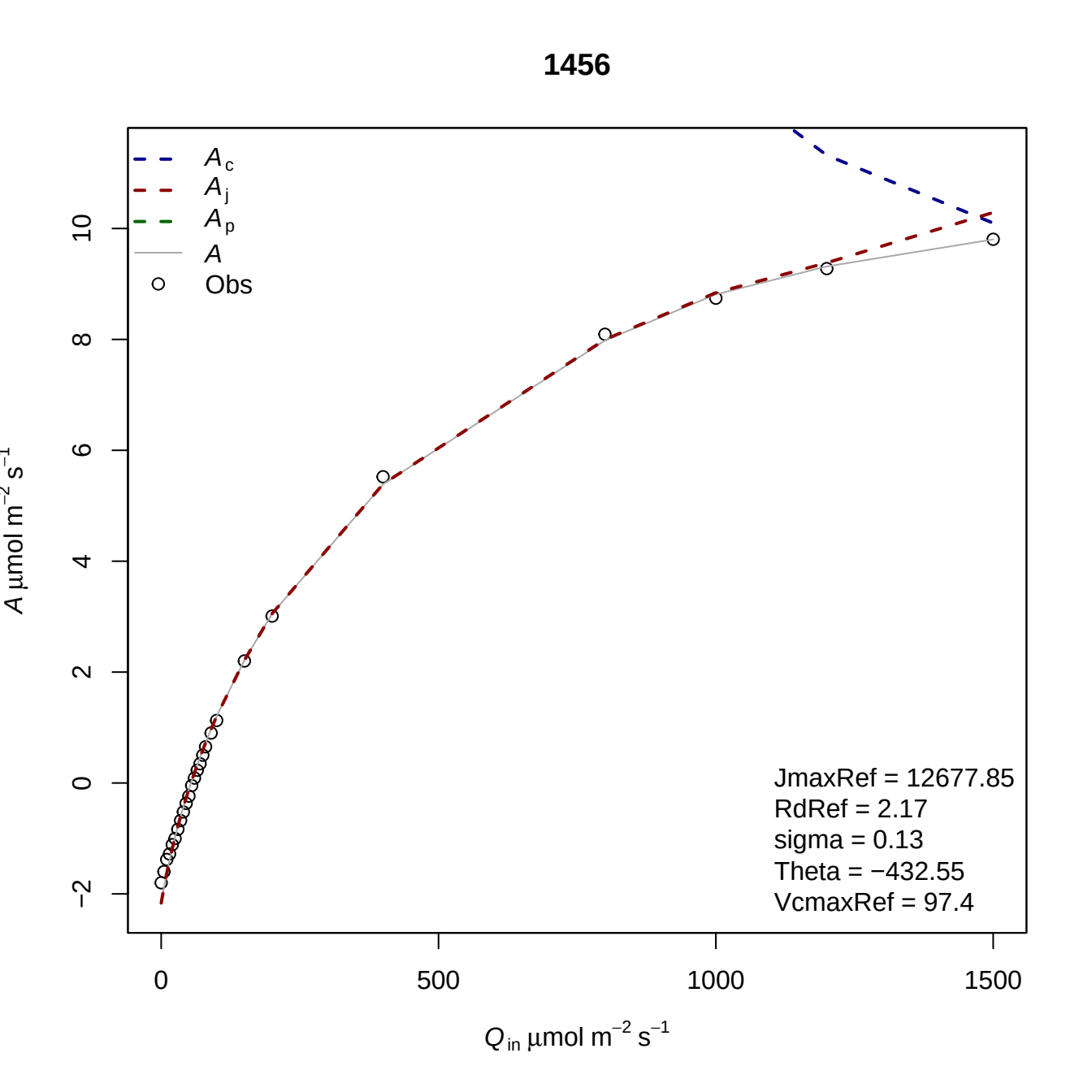

1451
A_c
A_j
A_p
A
Obs
JmaxRef = 248.71
RdRef = 2.74
sigma = 0.09
Theta = −2.52
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

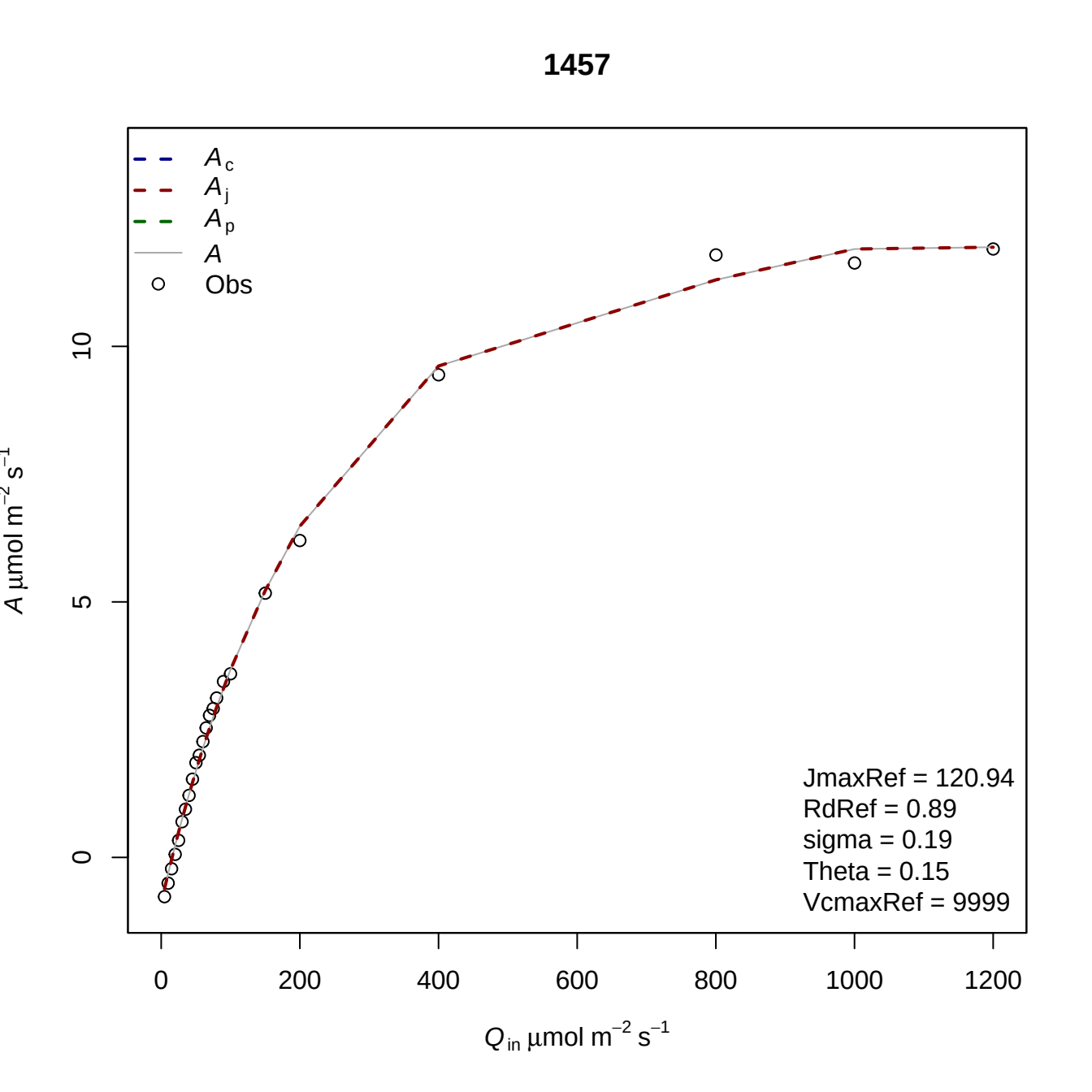

1452
A_c
A_j
A_p
A
Obs
JmaxRef = 109.46
RdRef = 1.14
sigma = 0.15
Theta = −0.09
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

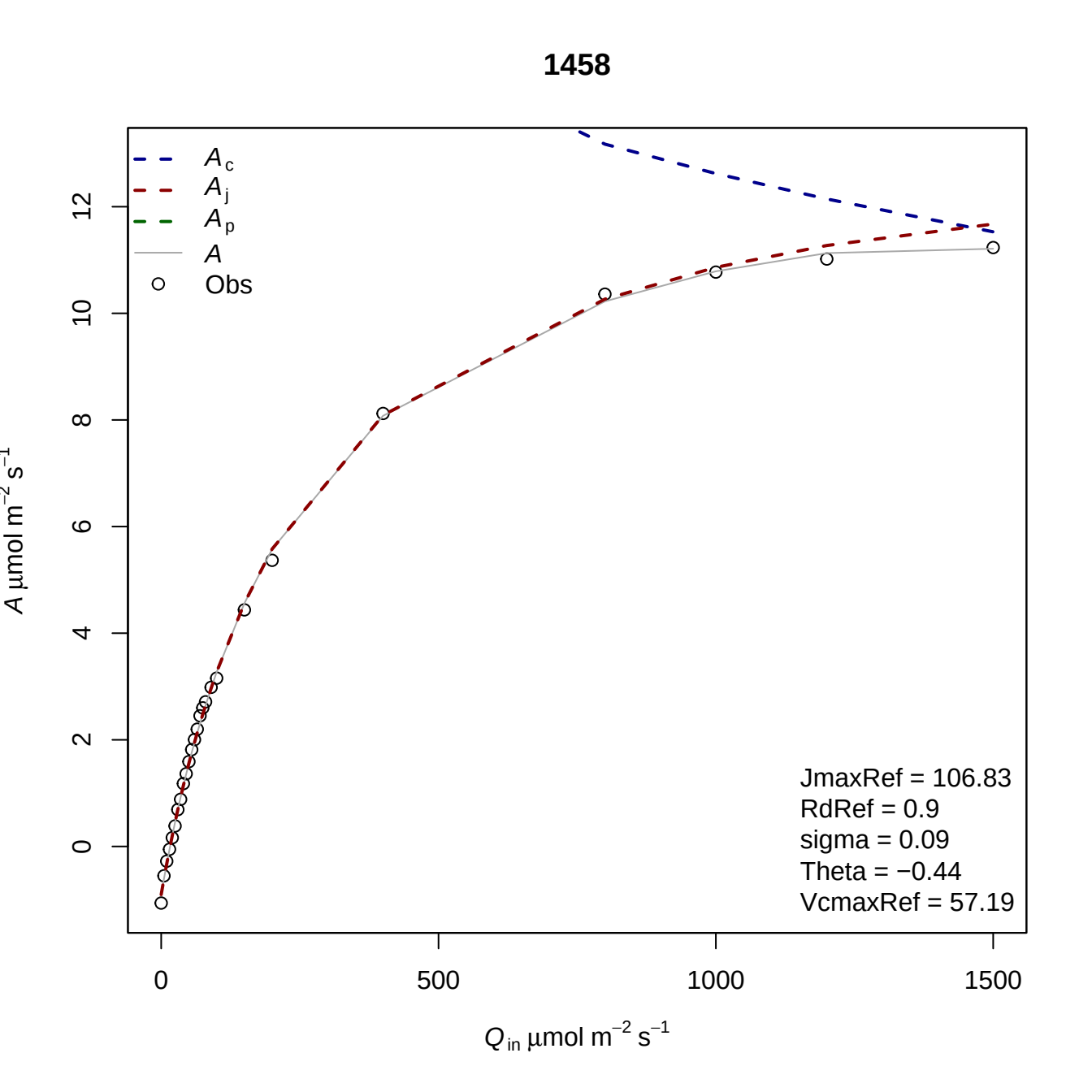

1453
A_c
A_j
A_p
A
Obs
JmaxRef = 283.82
RdRef = 2.77
sigma = 0.11
Theta = −1.94
VcmaxRef = 128.41
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

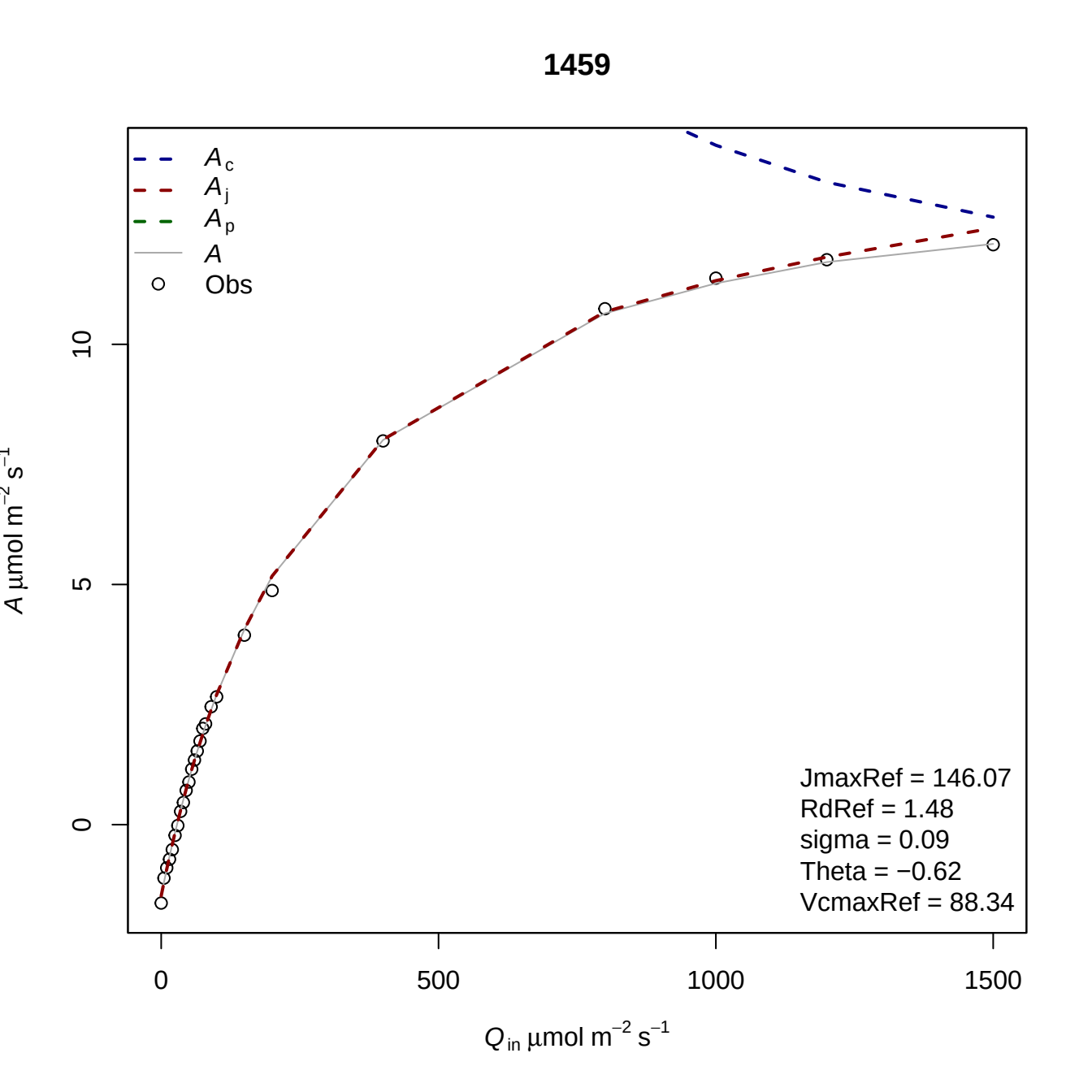

1454
A_c
A_j
A_p
A
Obs
JmaxRef = 18191094.73
RdRef = 2.26
sigma = 0.14
Theta = −633451.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

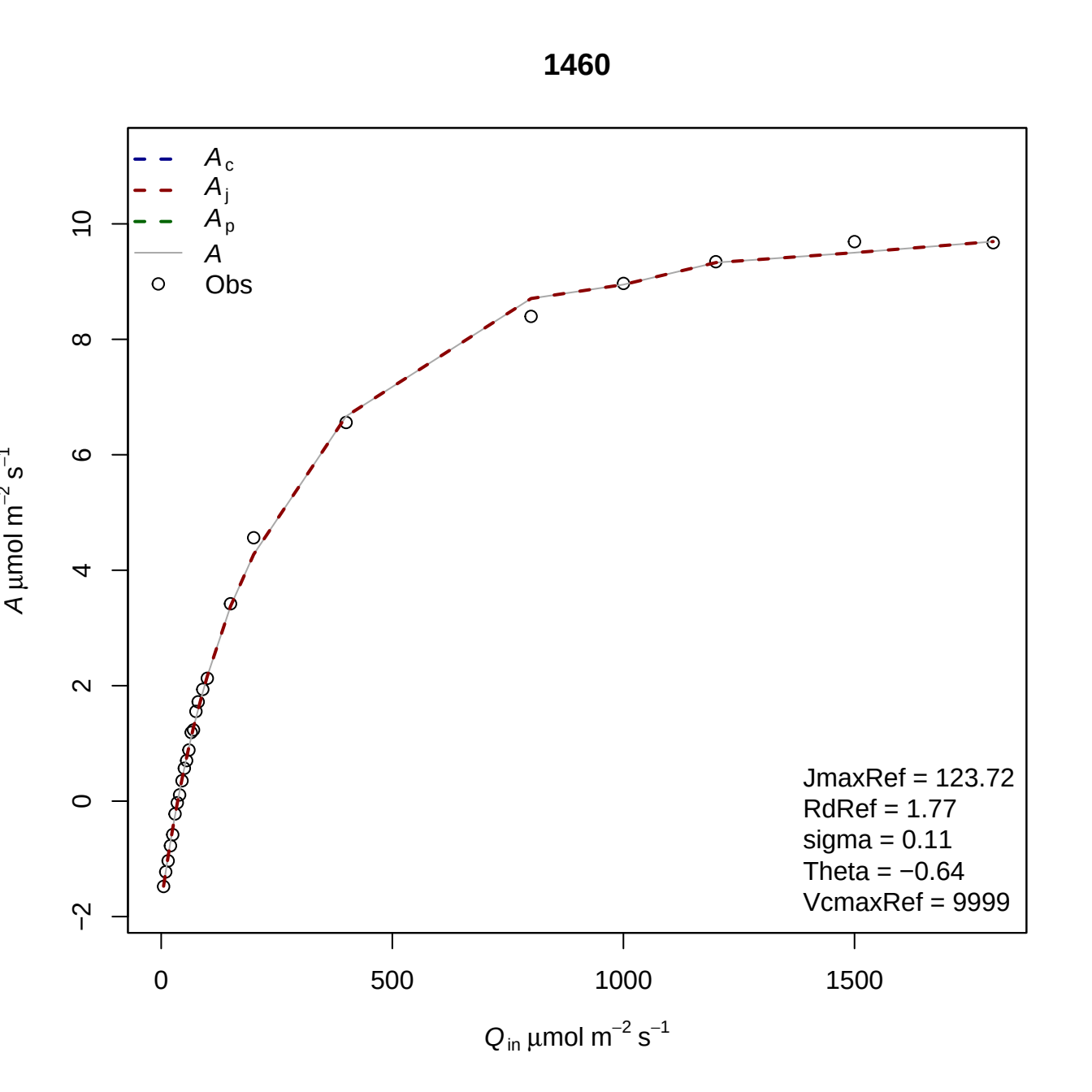

1455
A_c
A_j
A_p
A
Obs
JmaxRef = 759.73
RdRef = 2.17
sigma = 0.2
Theta = −7.03
VcmaxRef = 165.15
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

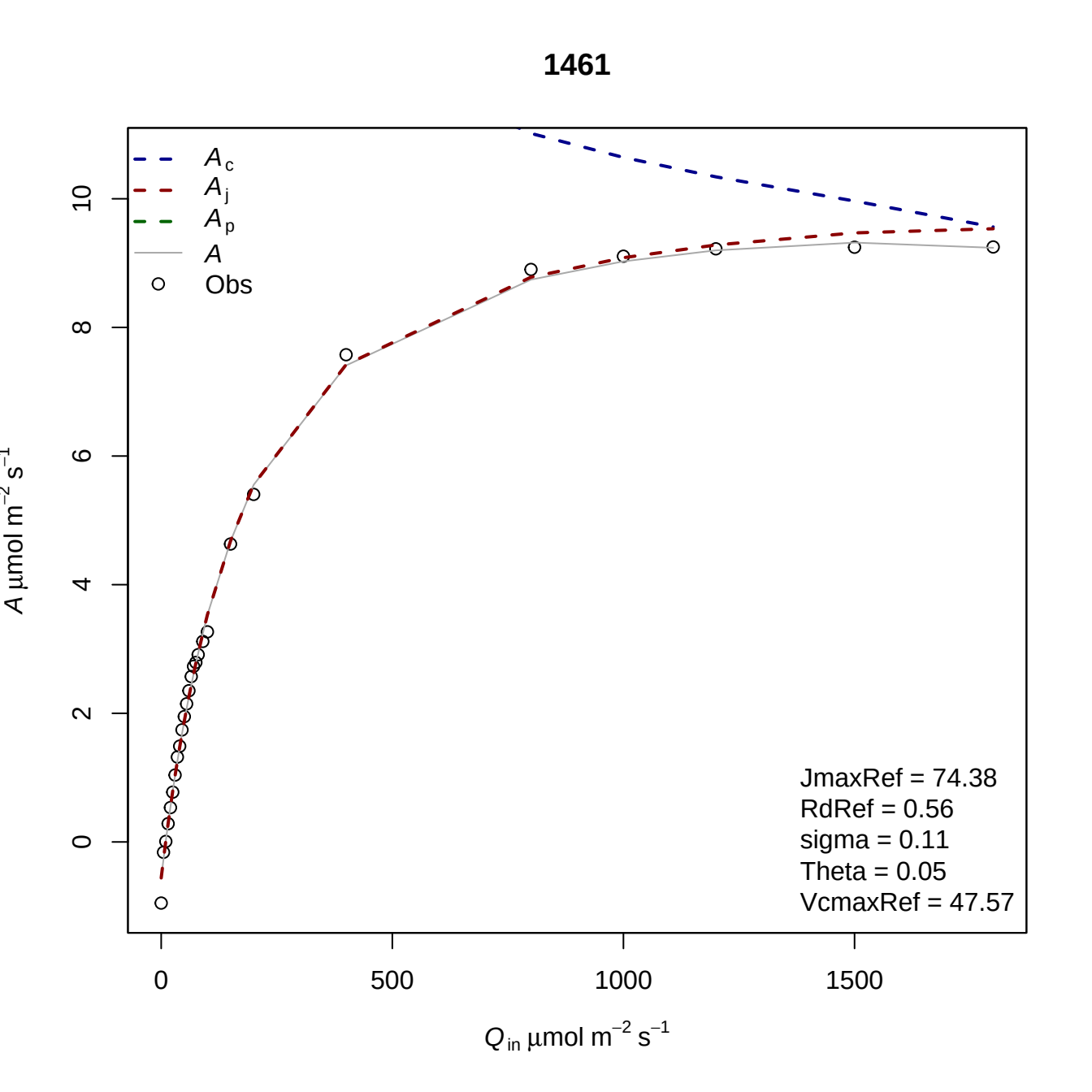

1456
A_c
A_j
A_p
A
Obs
JmaxRef = 12677.85
RdRef = 2.17
sigma = 0.13
Theta = −432.55
VcmaxRef = 97.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

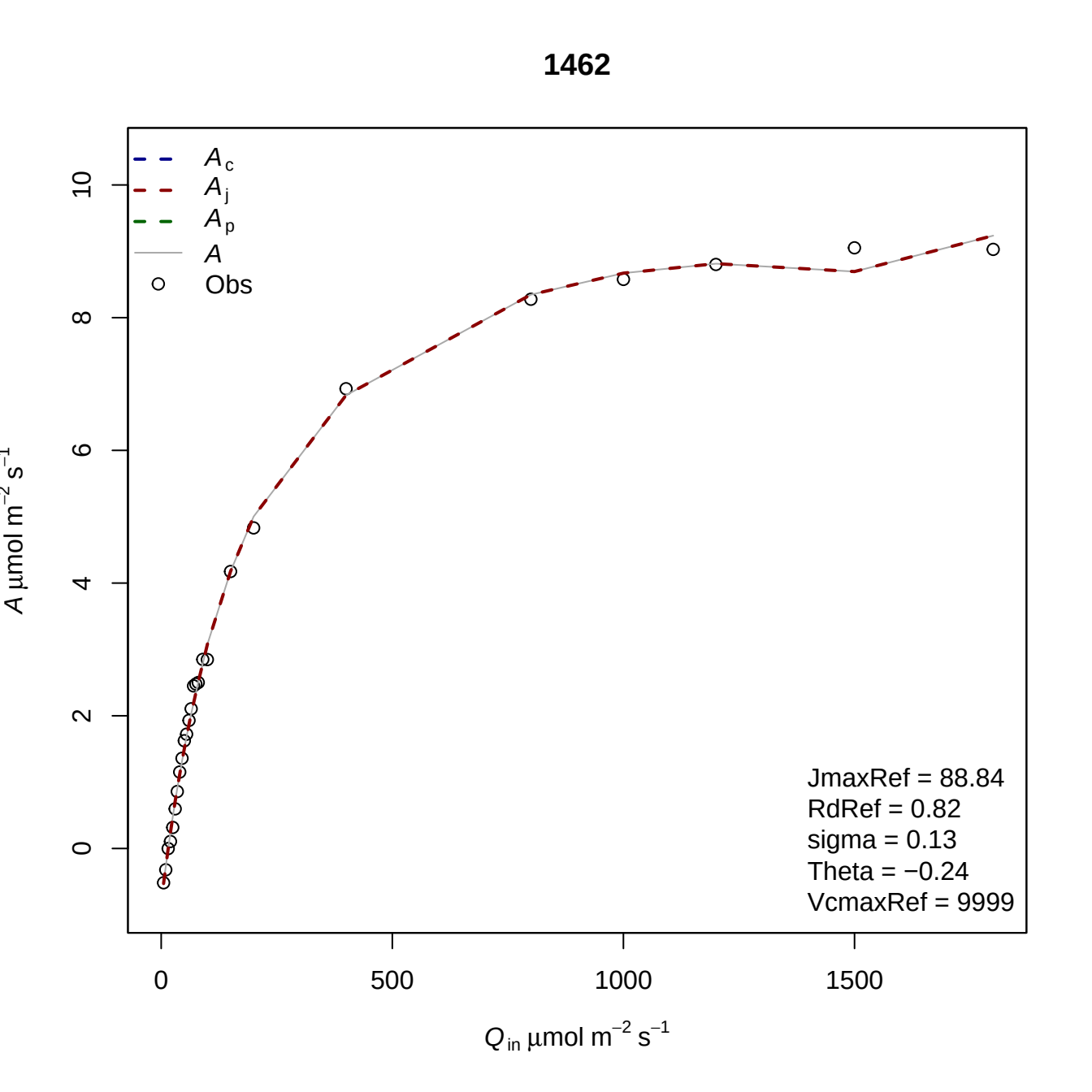

1457
A_c
A_j
A_p
A
Obs
JmaxRef = 120.94
RdRef = 0.89
sigma = 0.19
Theta = 0.15
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200

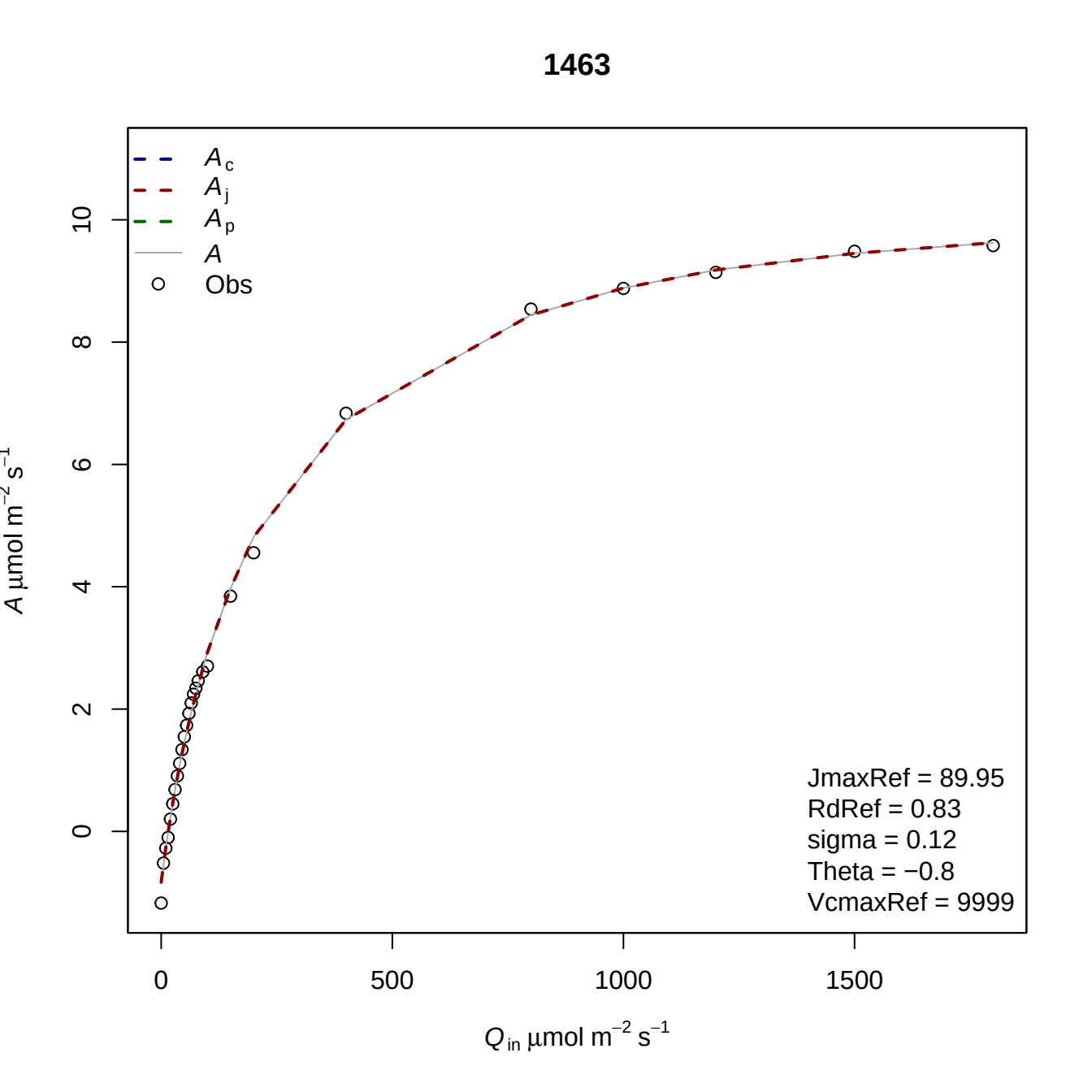

1458
A_c
A_j
A_p
A
Obs
JmaxRef = 106.83
RdRef = 0.9
sigma = 0.09
Theta = −0.44
VcmaxRef = 57.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

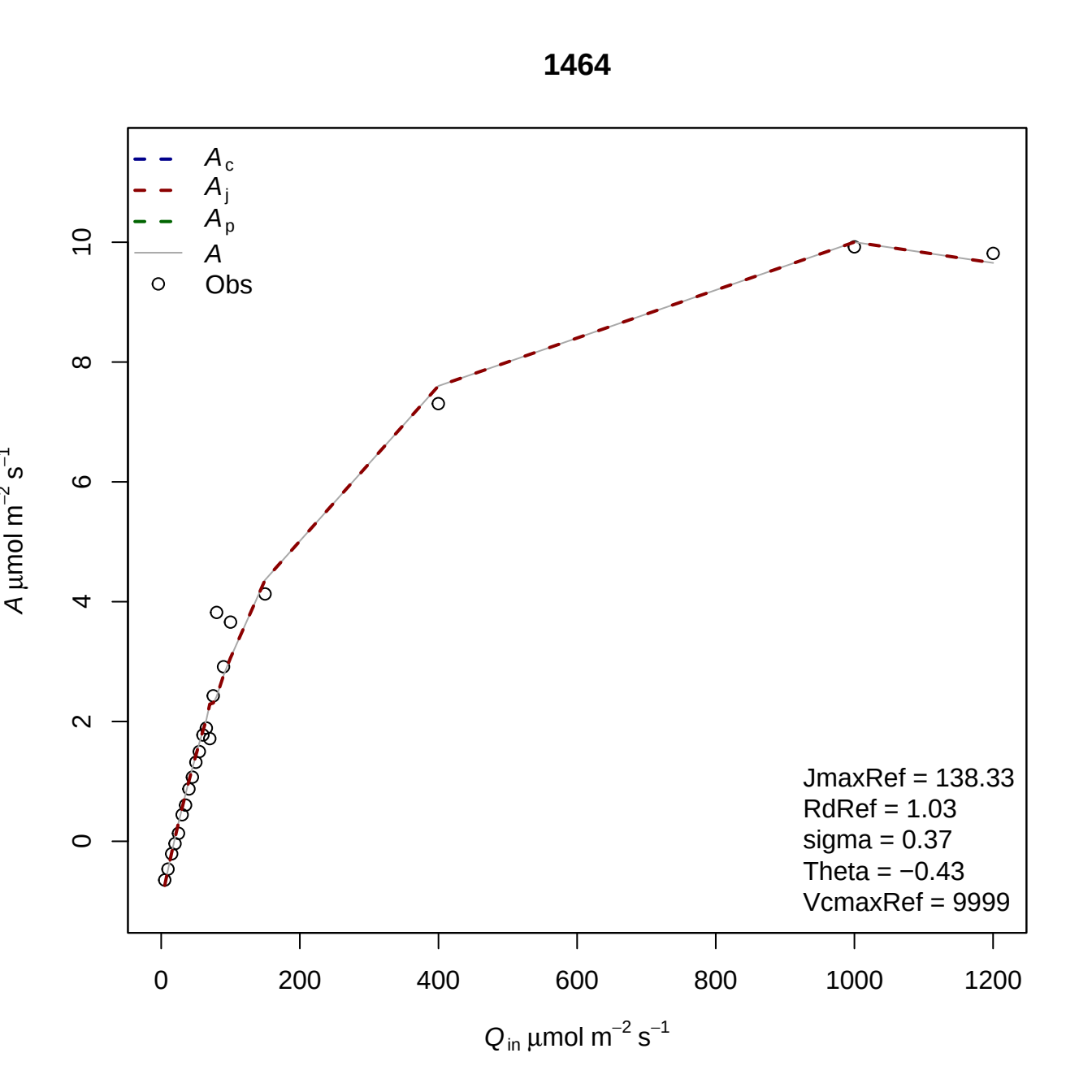

1459
A_c
A_j
A_p
A
Obs
JmaxRef = 146.07
RdRef = 1.48
sigma = 0.09
Theta = −0.62
VcmaxRef = 88.34
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

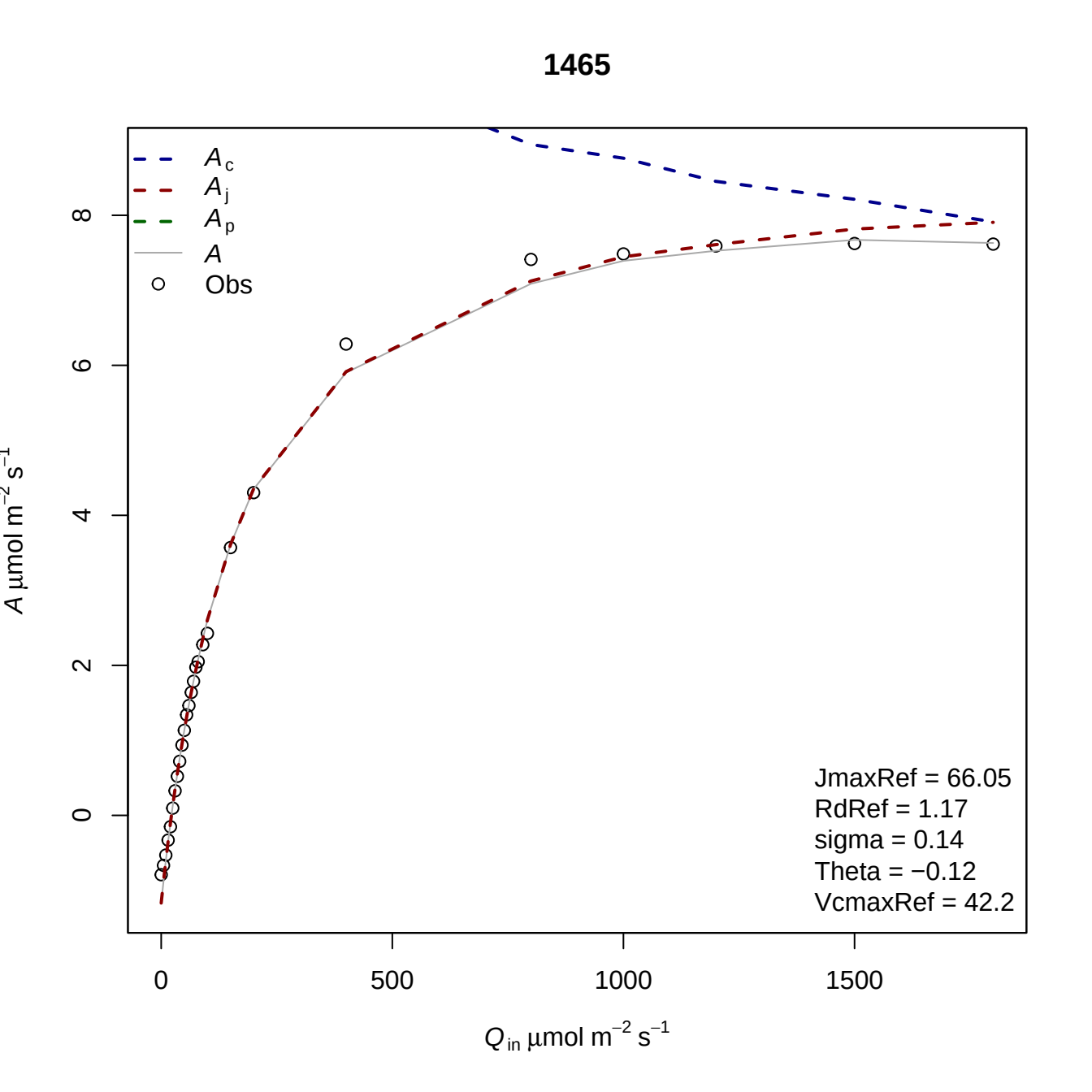

1460
A_c
A_j
A_p
A
Obs
JmaxRef = 123.72
RdRef = 1.77
sigma = 0.11
Theta = −0.64
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

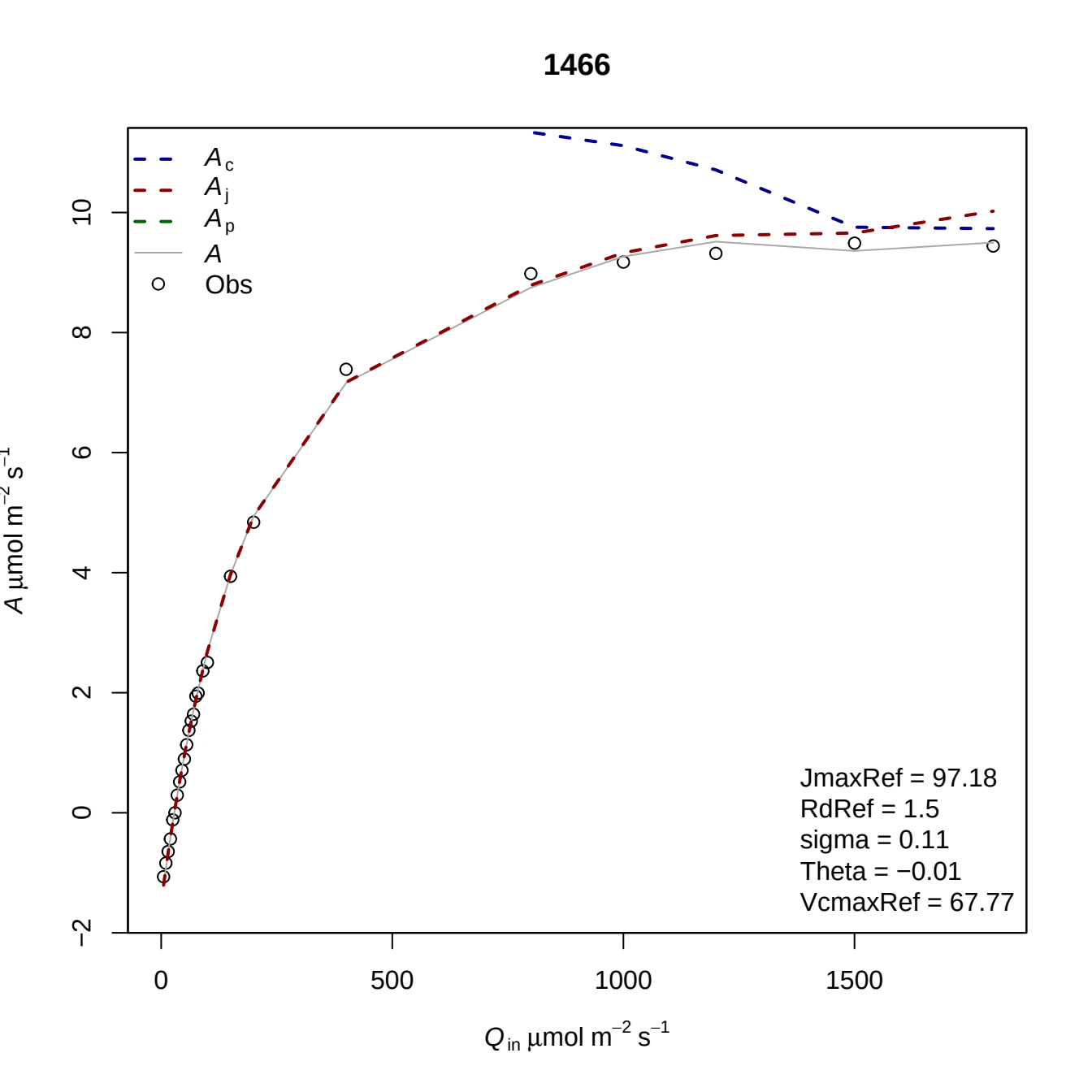

1461
A_c
A_j
A_p
A
Obs
JmaxRef = 74.38
RdRef = 0.56
sigma = 0.11
Theta = 0.05
VcmaxRef = 47.57
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

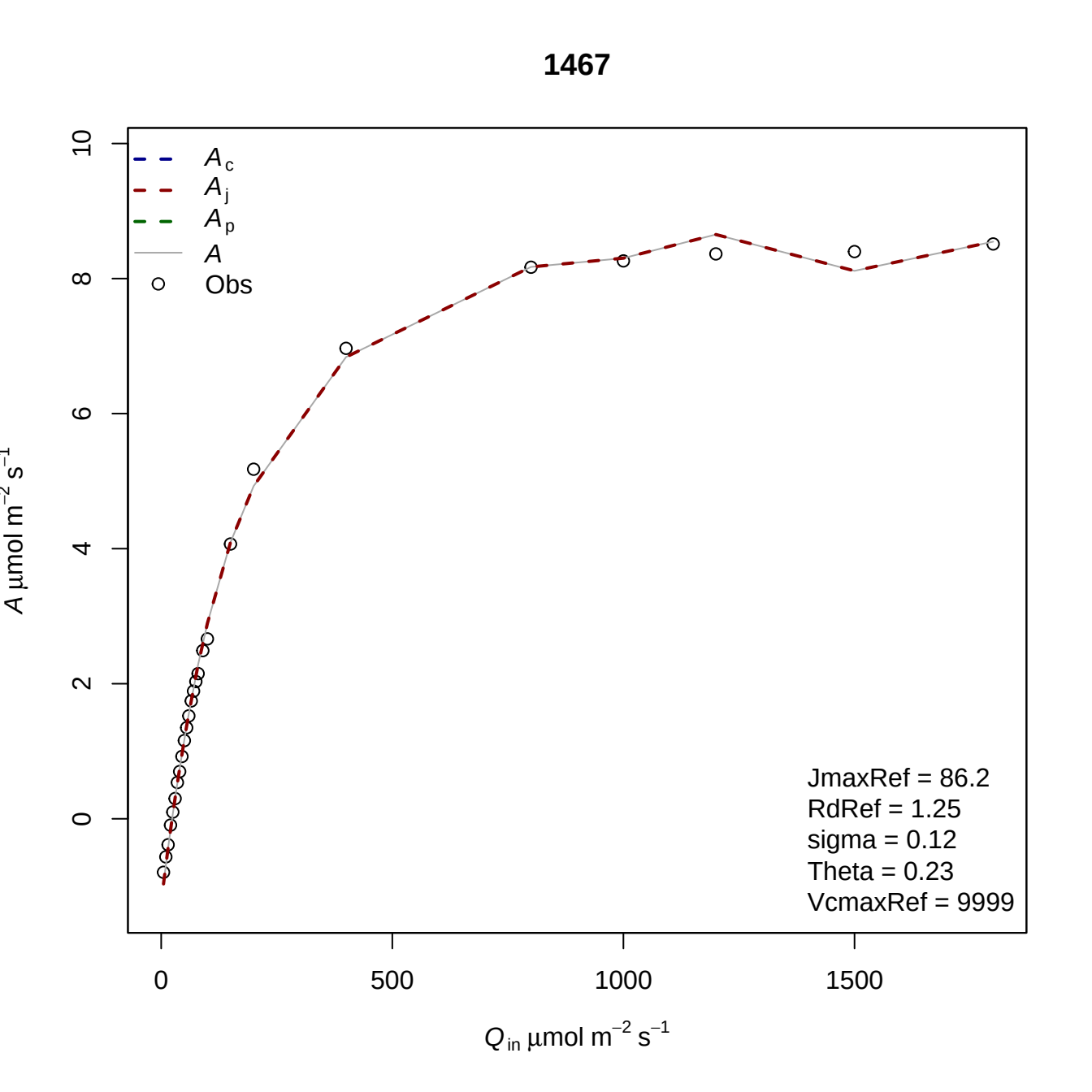

1462
A_c
A_j
A_p
A
Obs
JmaxRef = 88.84
RdRef = 0.82
sigma = 0.13
Theta = −0.24
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

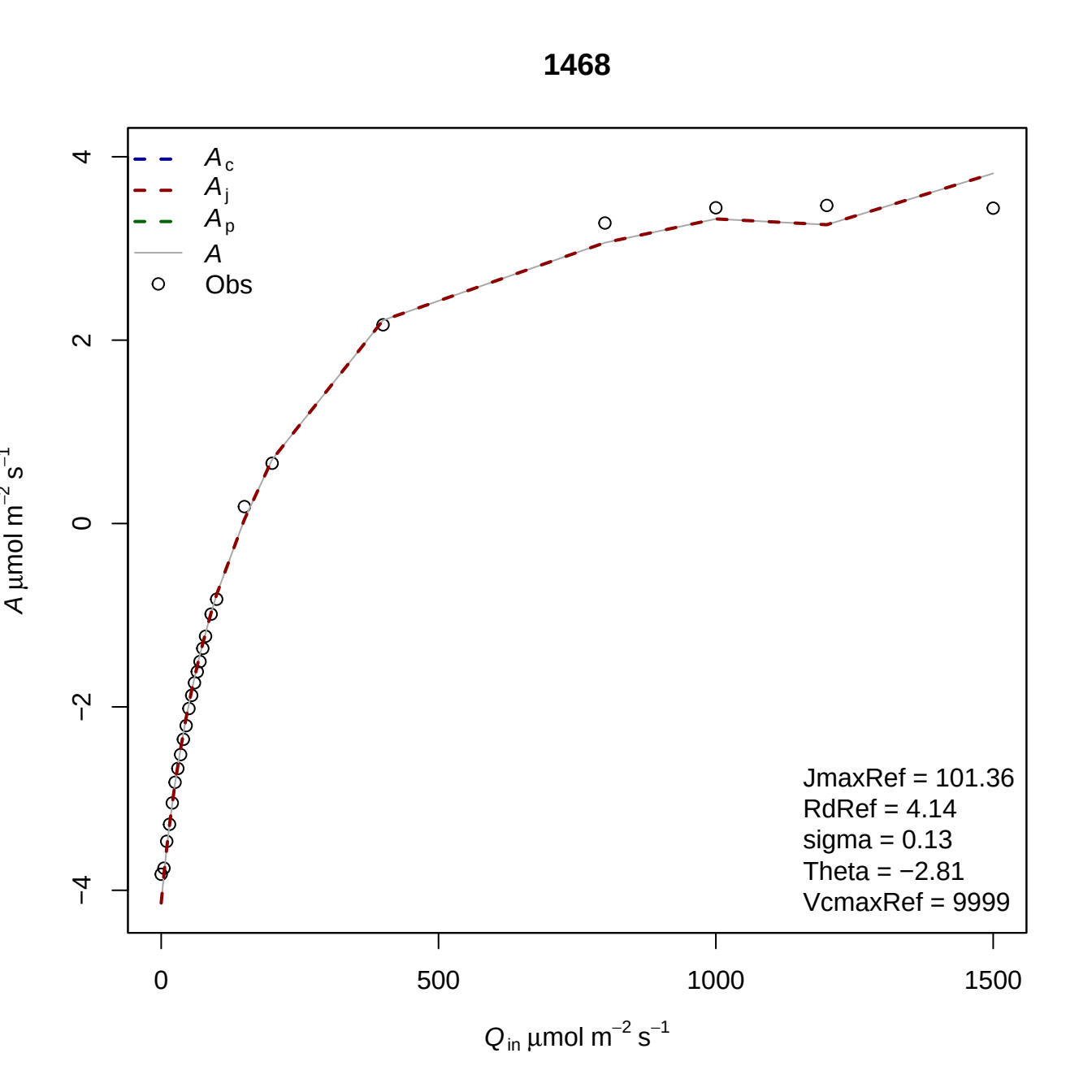

1463
A_c
A_j
A_p
A
Obs
JmaxRef = 89.95
RdRef = 0.83
sigma = 0.12
Theta = −0.8
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

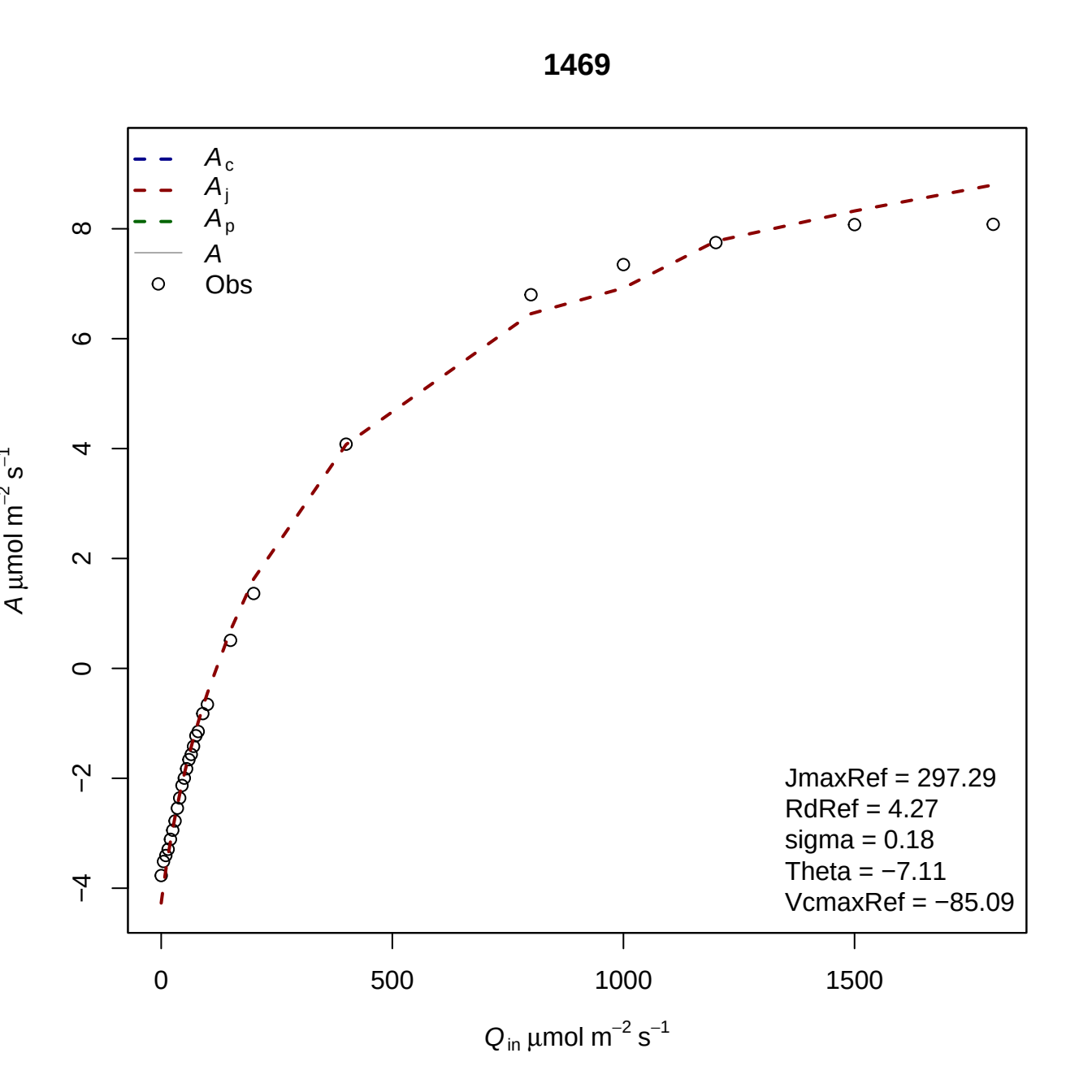

1464
A_c
A_j
A_p
A
Obs
JmaxRef = 138.33
RdRef = 1.03
sigma = 0.37
Theta = −0.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200

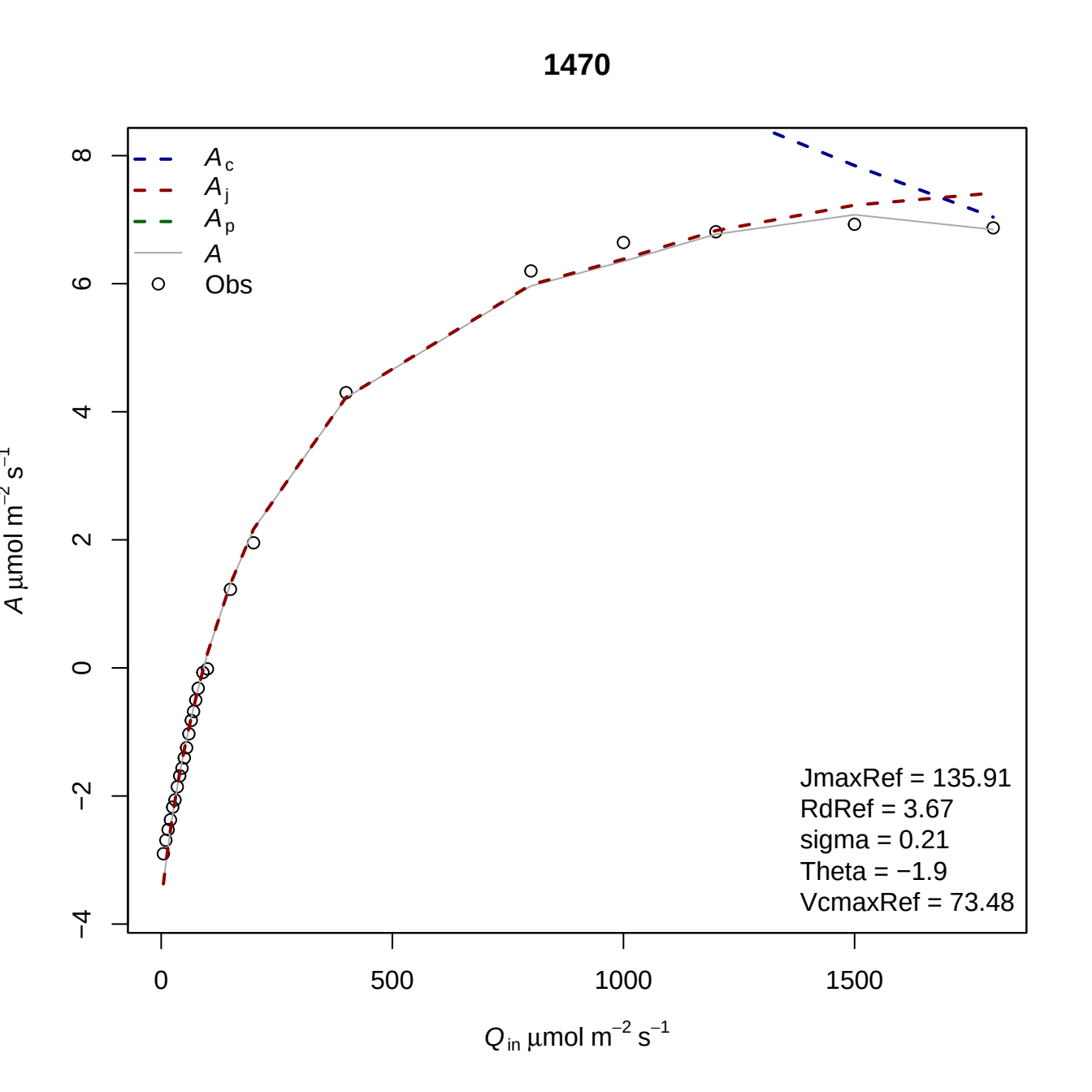

1465
A_c
A_j
A_p
A
Obs
JmaxRef = 66.05
RdRef = 1.17
sigma = 0.14
Theta = −0.12
VcmaxRef = 42.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

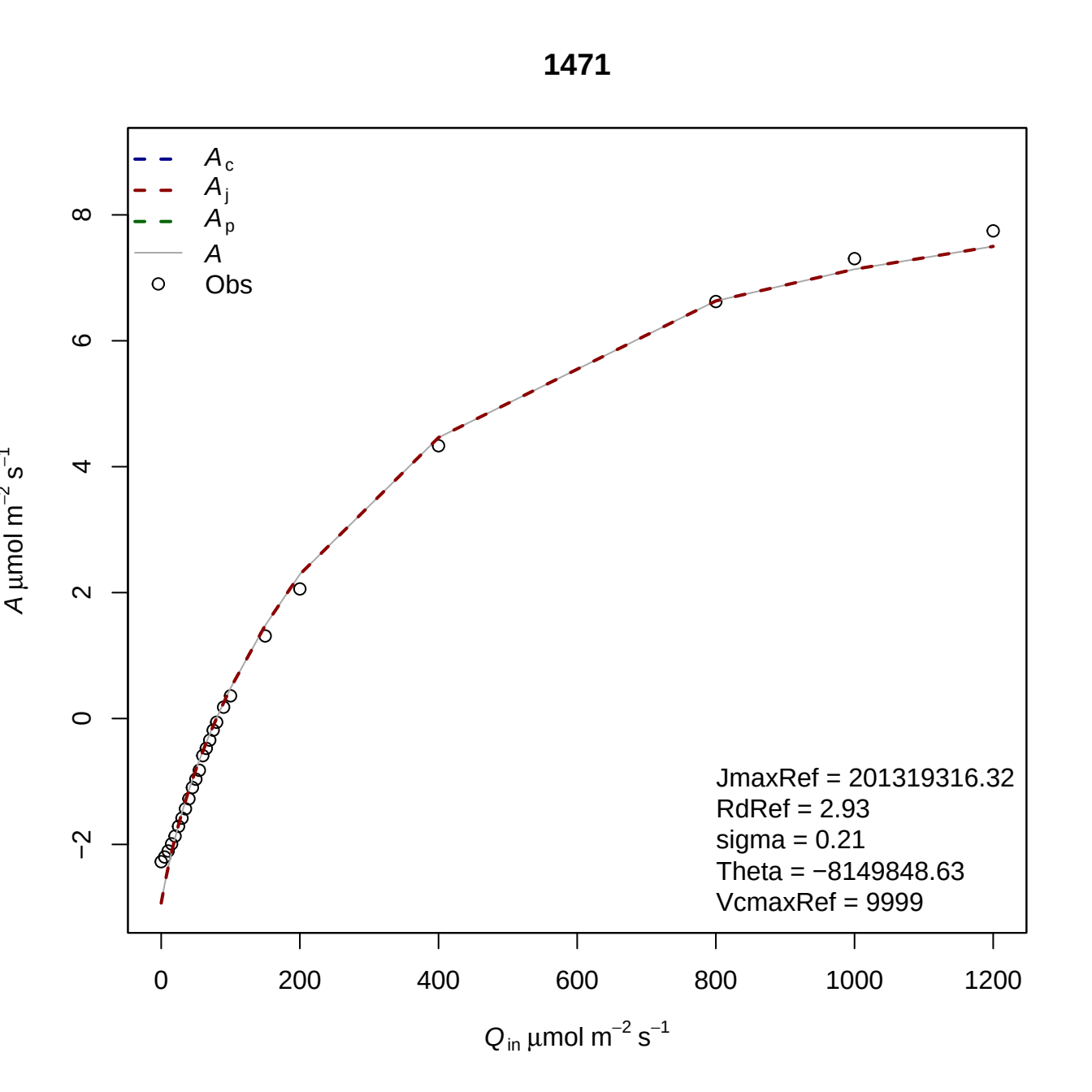

1466
A_c
A_j
A_p
A
Obs
JmaxRef = 97.18
RdRef = 1.5
sigma = 0.11
Theta = −0.01
VcmaxRef = 67.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

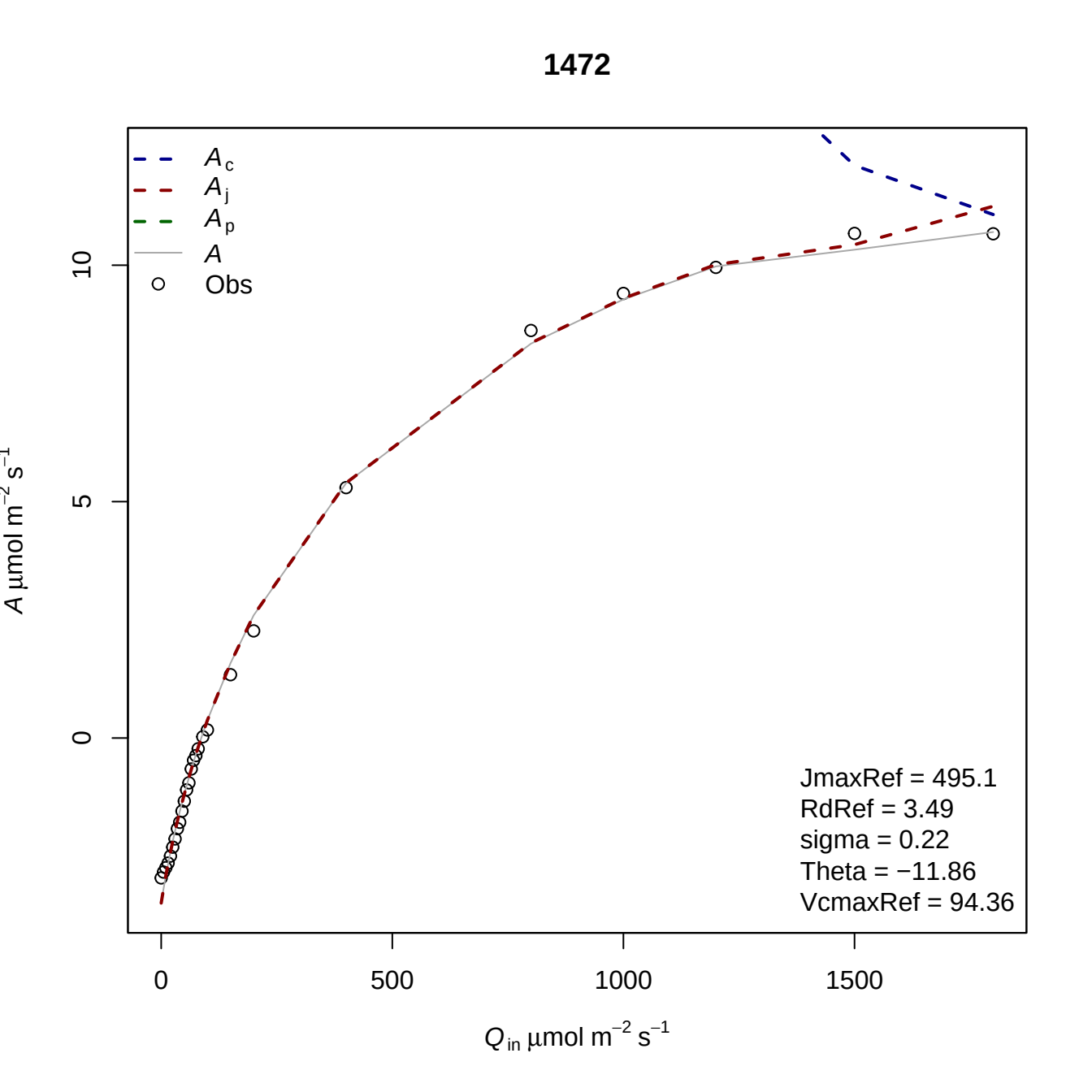

1467
A_c
A_j
A_p
A
Obs
JmaxRef = 86.2
RdRef = 1.25
sigma = 0.12
Theta = 0.23
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

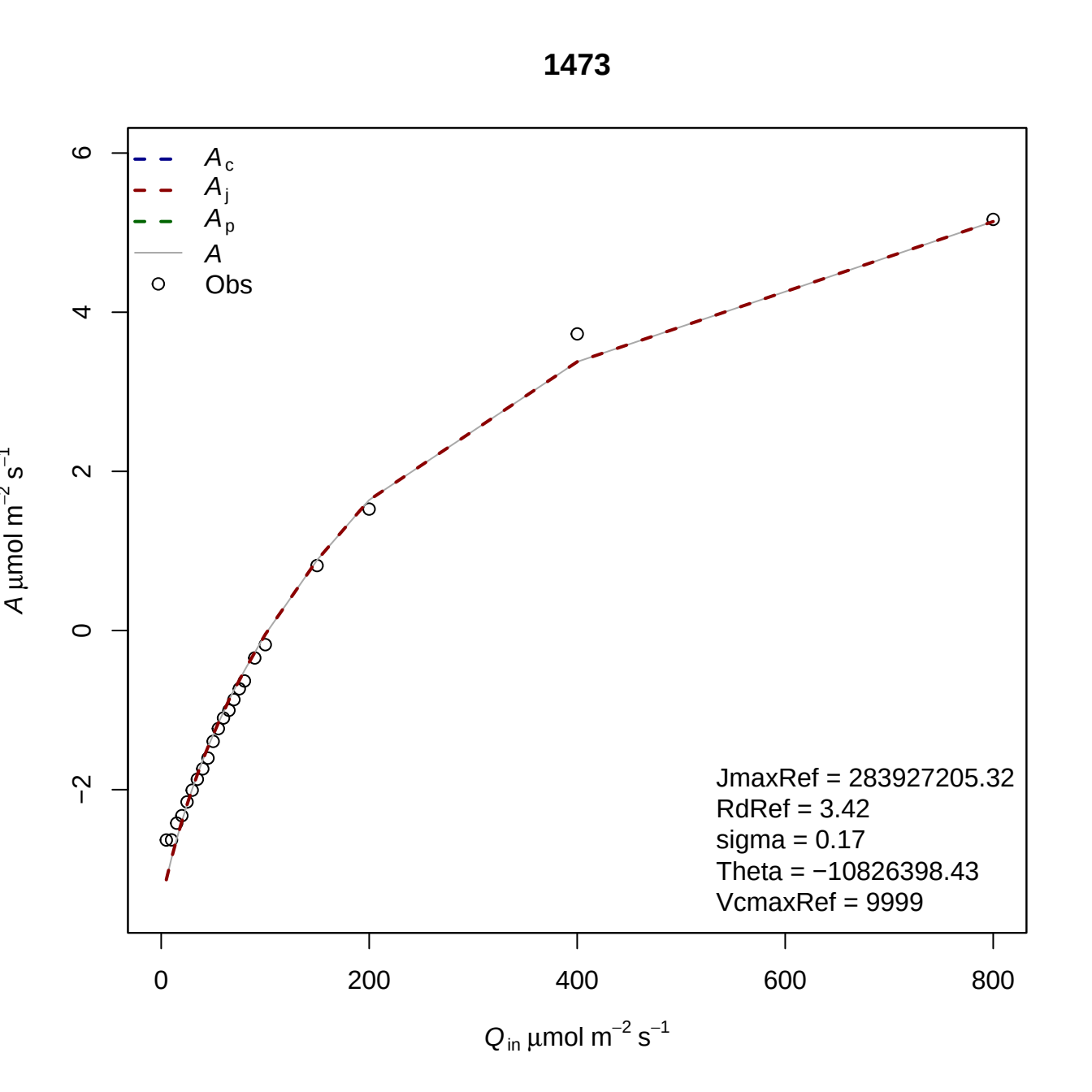

1468
A_c
A_j
A_p
A
Obs
JmaxRef = 101.36
RdRef = 4.14
sigma = 0.13
Theta = −2.81
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

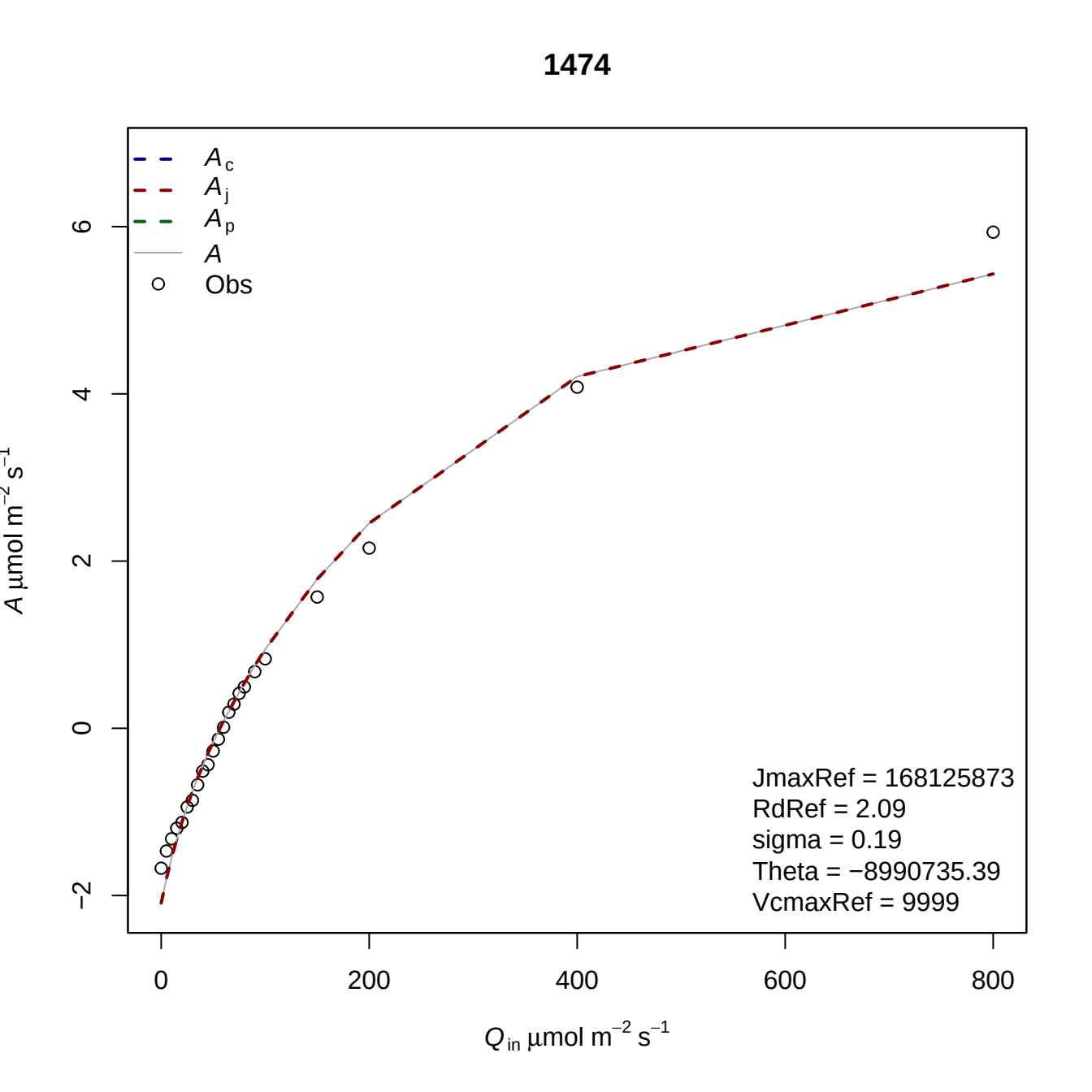

1469
A_c
A_j
A_p
A
Obs
JmaxRef = 297.29
RdRef = 4.27
sigma = 0.18
Theta = −7.11
VcmaxRef = −85.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

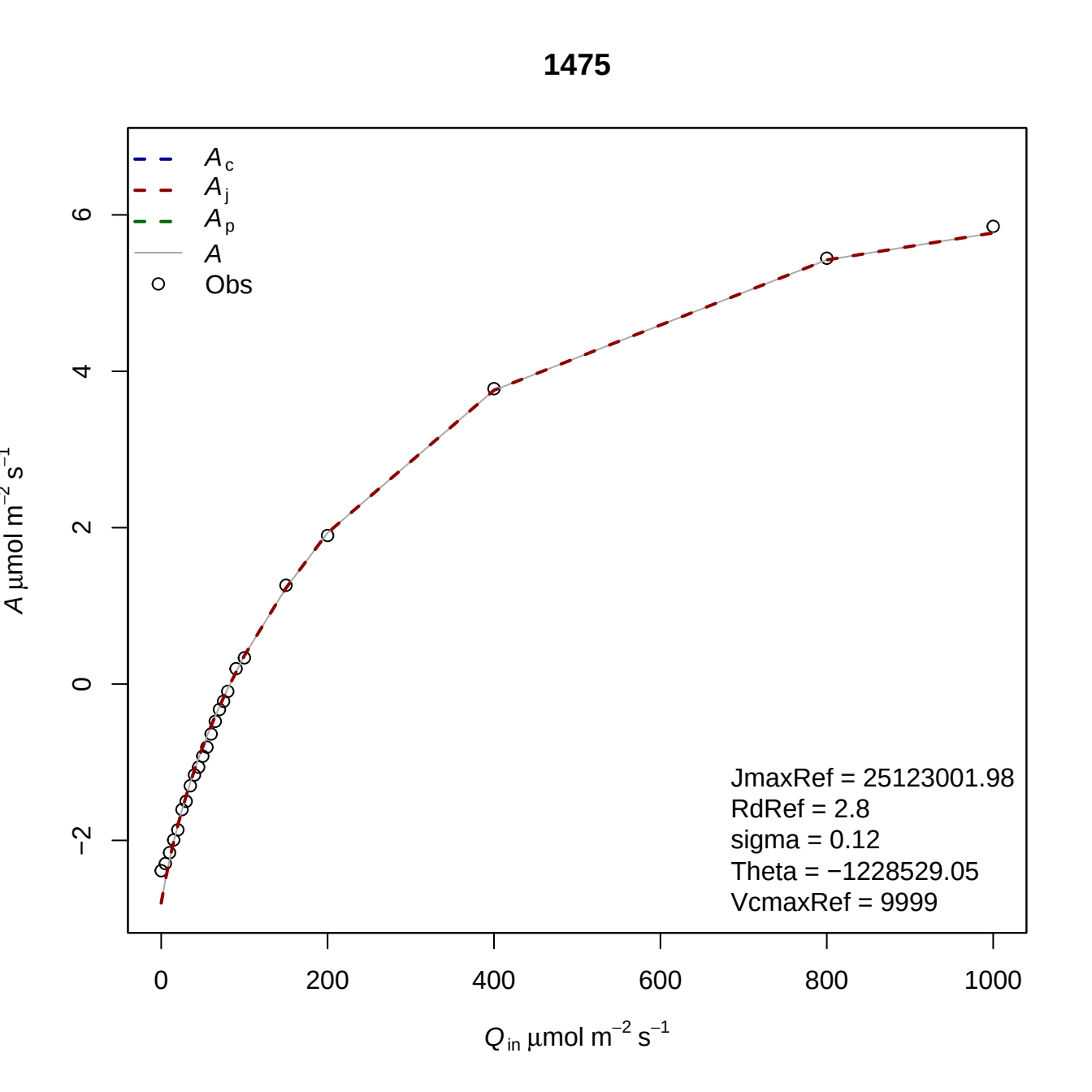

1470
A_c
A_j
A_p
A
Obs
JmaxRef = 135.91
RdRef = 3.67
sigma = 0.21
Theta = −1.9
VcmaxRef = 73.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

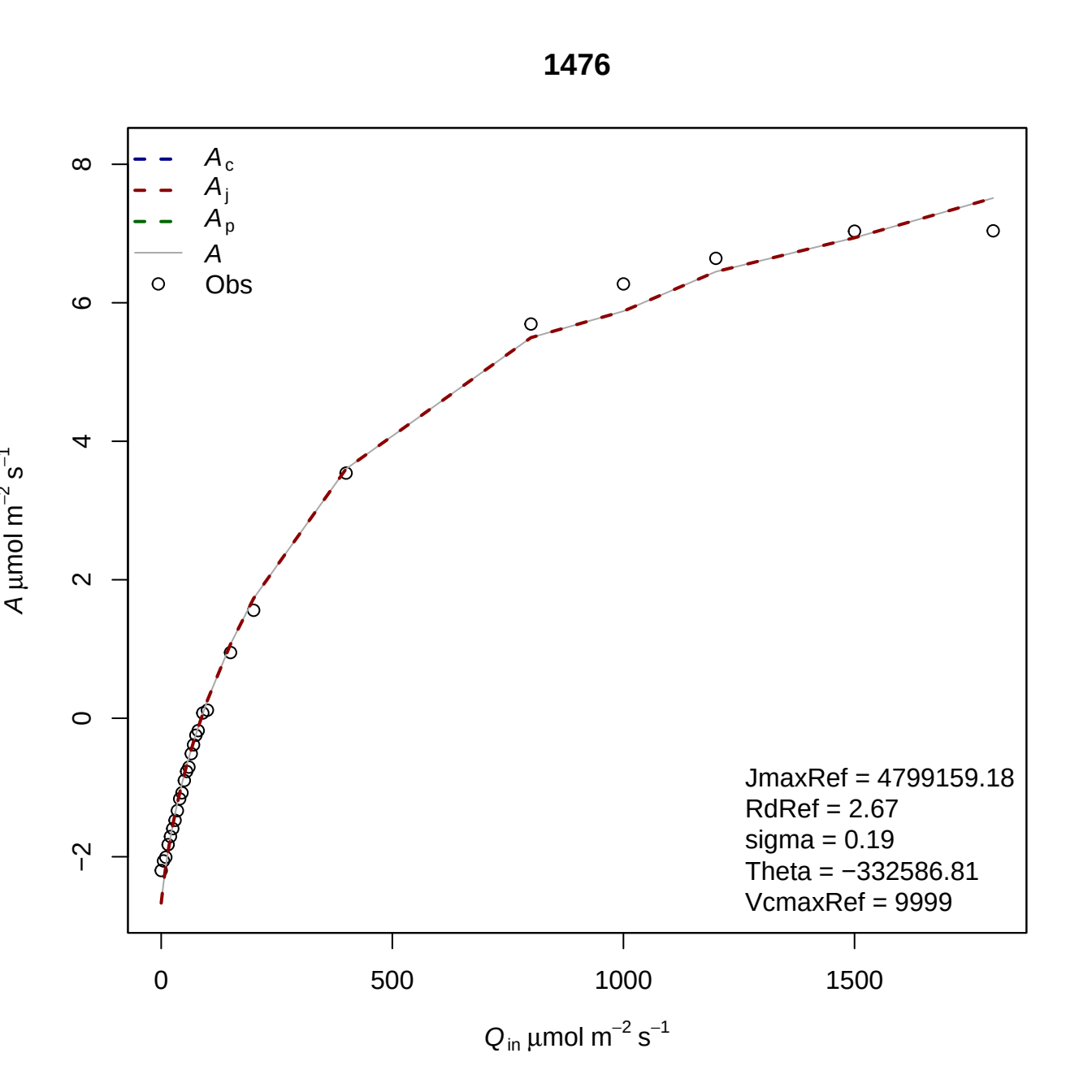

1471
A_c
A_j
A_p
A
Obs
JmaxRef = 201319316.32
RdRef = 2.93
sigma = 0.21
Theta = −8149848.63
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200

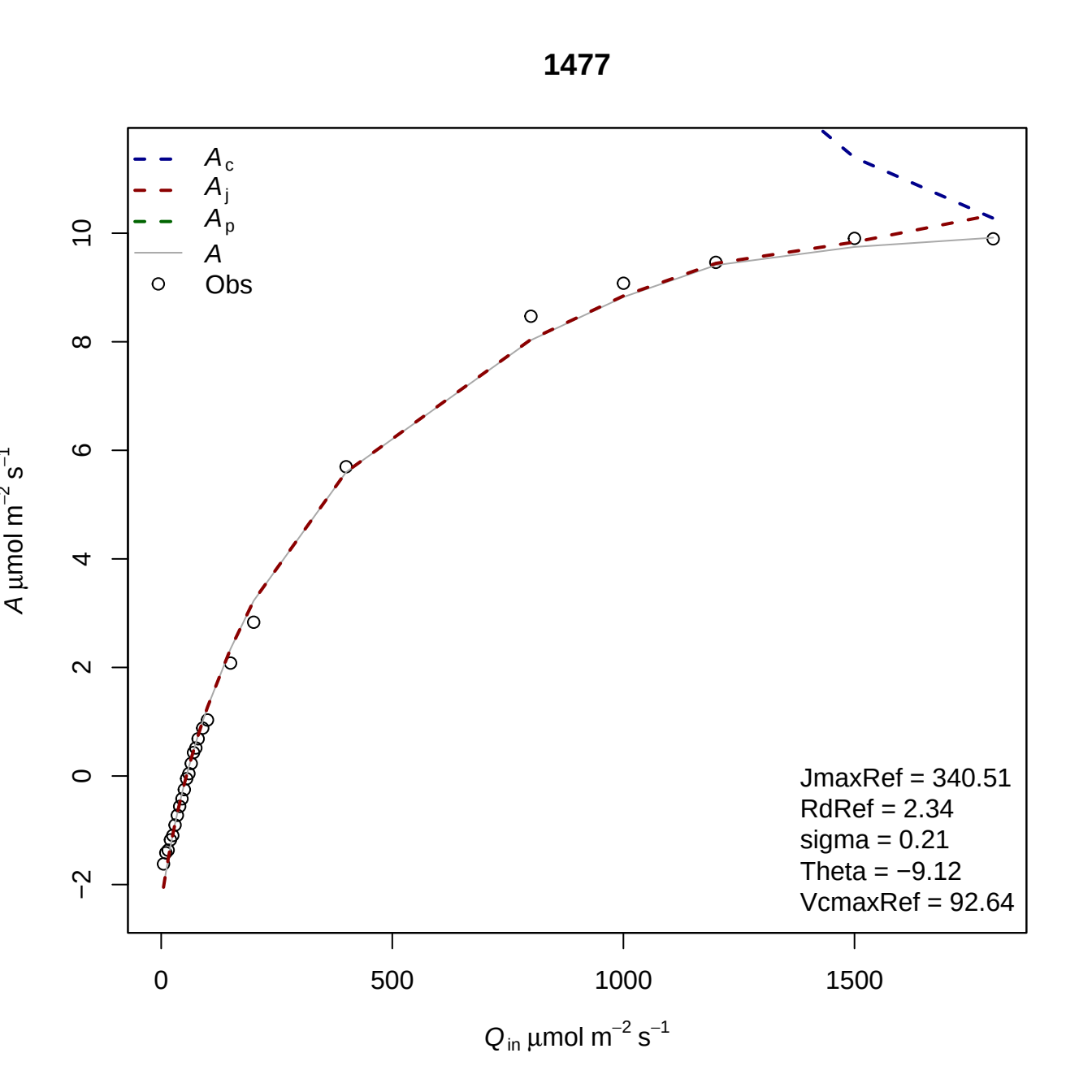

1472
A_c
A_j
A_p
A
Obs
JmaxRef = 495.1
RdRef = 3.49
sigma = 0.22
Theta = −11.86
VcmaxRef = 94.36
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

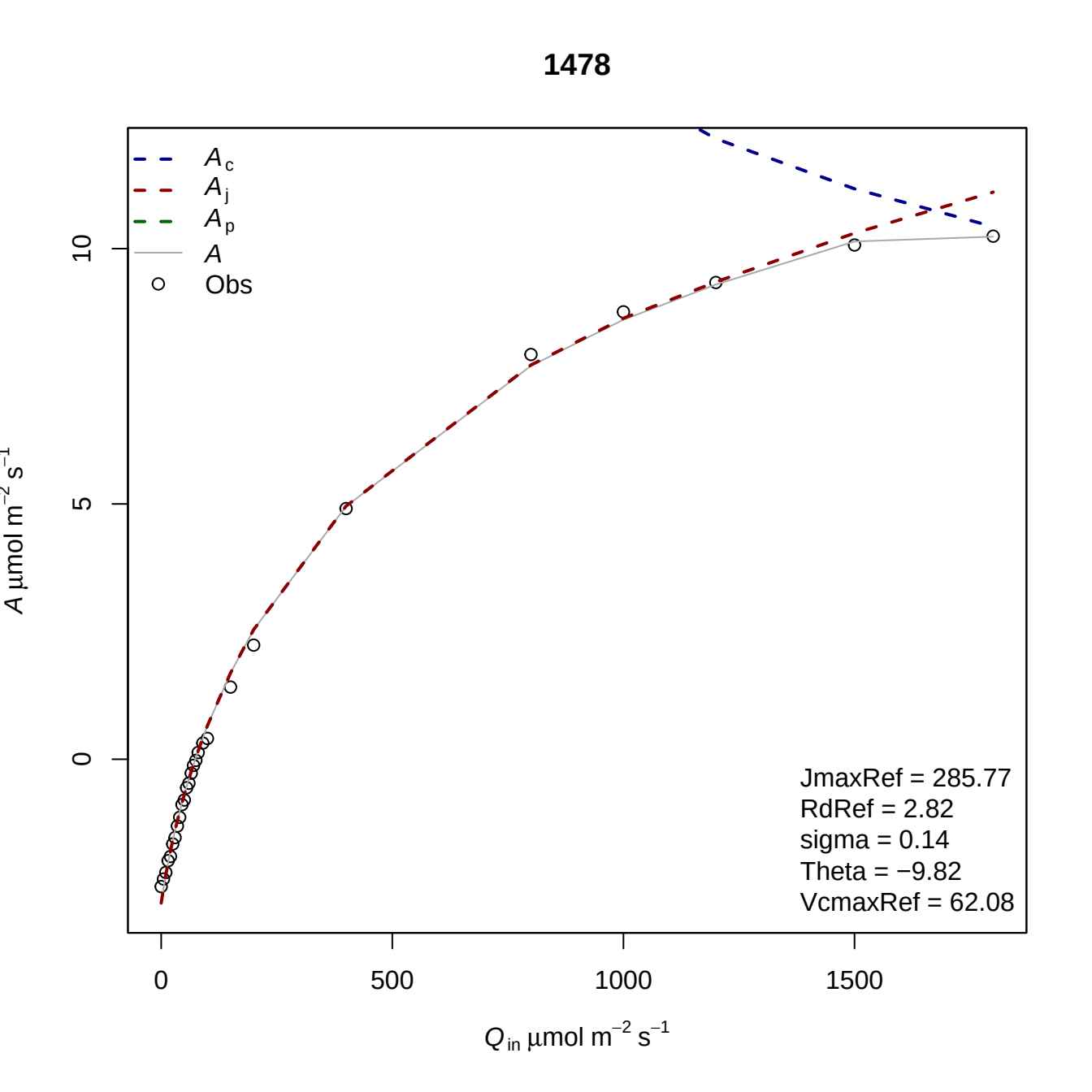

1473
A_c
A_j
A_p
A
Obs
JmaxRef = 283927205.32
RdRef = 3.42
sigma = 0.17
Theta = −10826398.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800

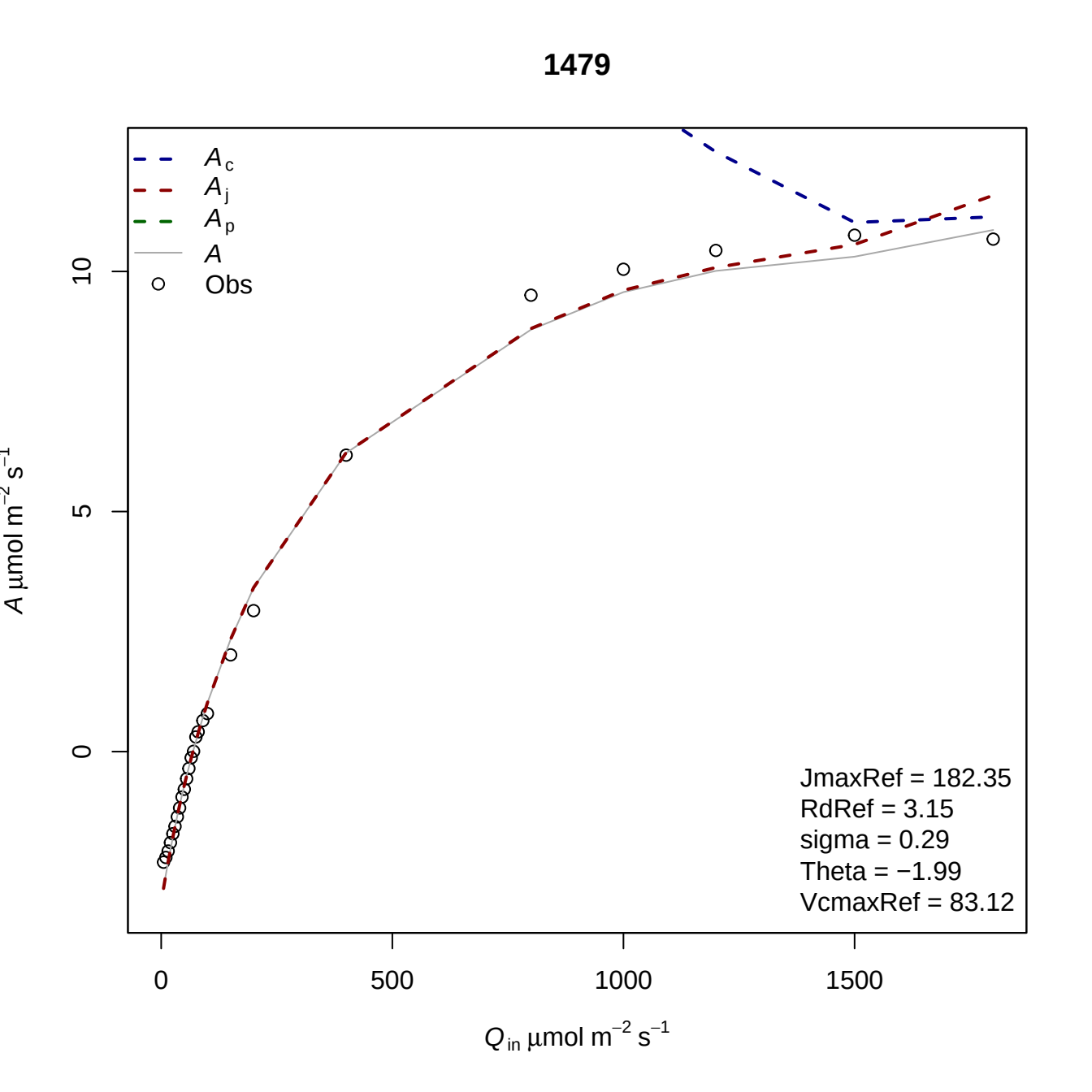

1474
A_c
A_j
A_p
A
Obs
JmaxRef = 168125873
RdRef = 2.09
sigma = 0.19
Theta = −8990735.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800

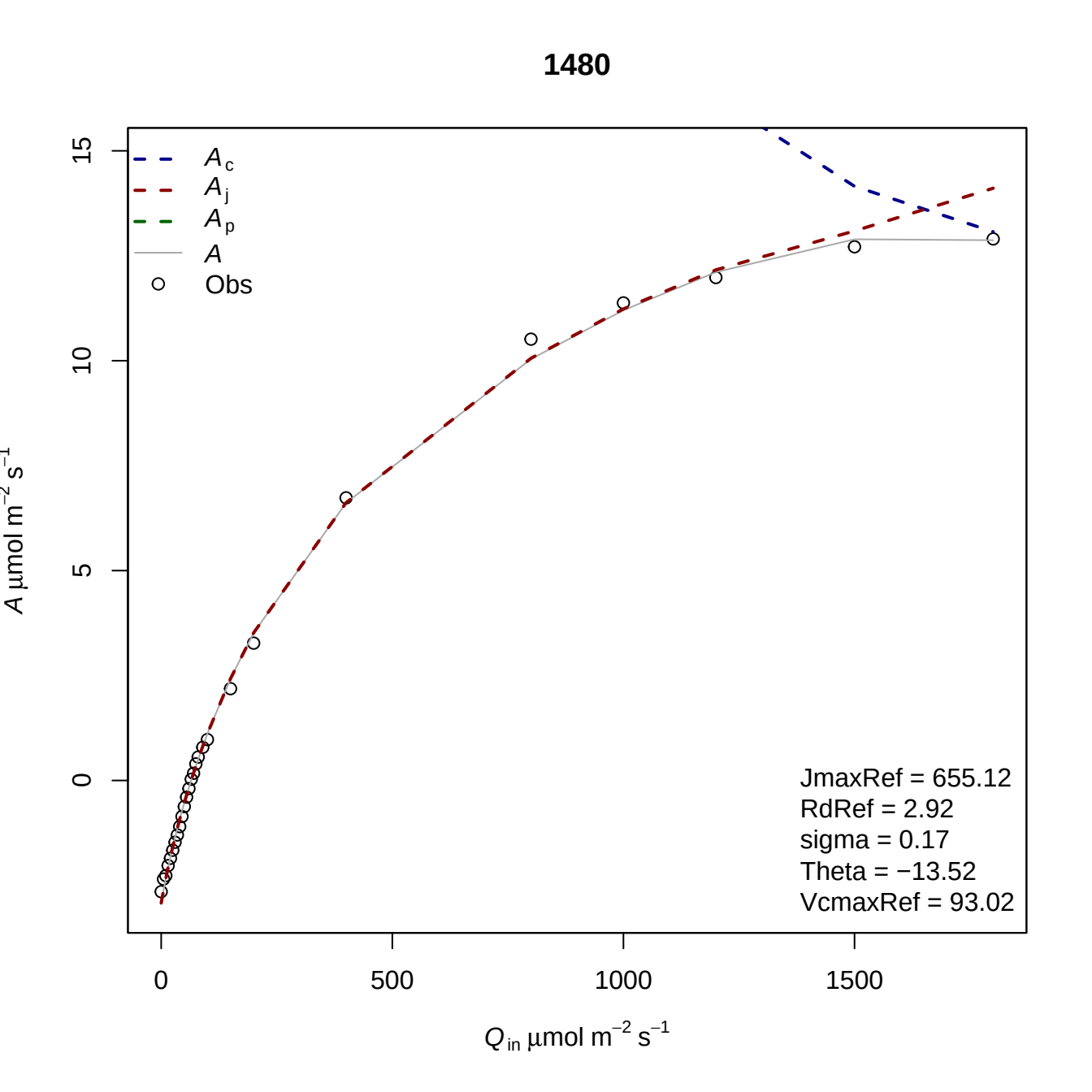

1475
A_c
A_j
A_p
A
Obs
JmaxRef = 25123001.98
RdRef = 2.8
sigma = 0.12
Theta = −1228529.05
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

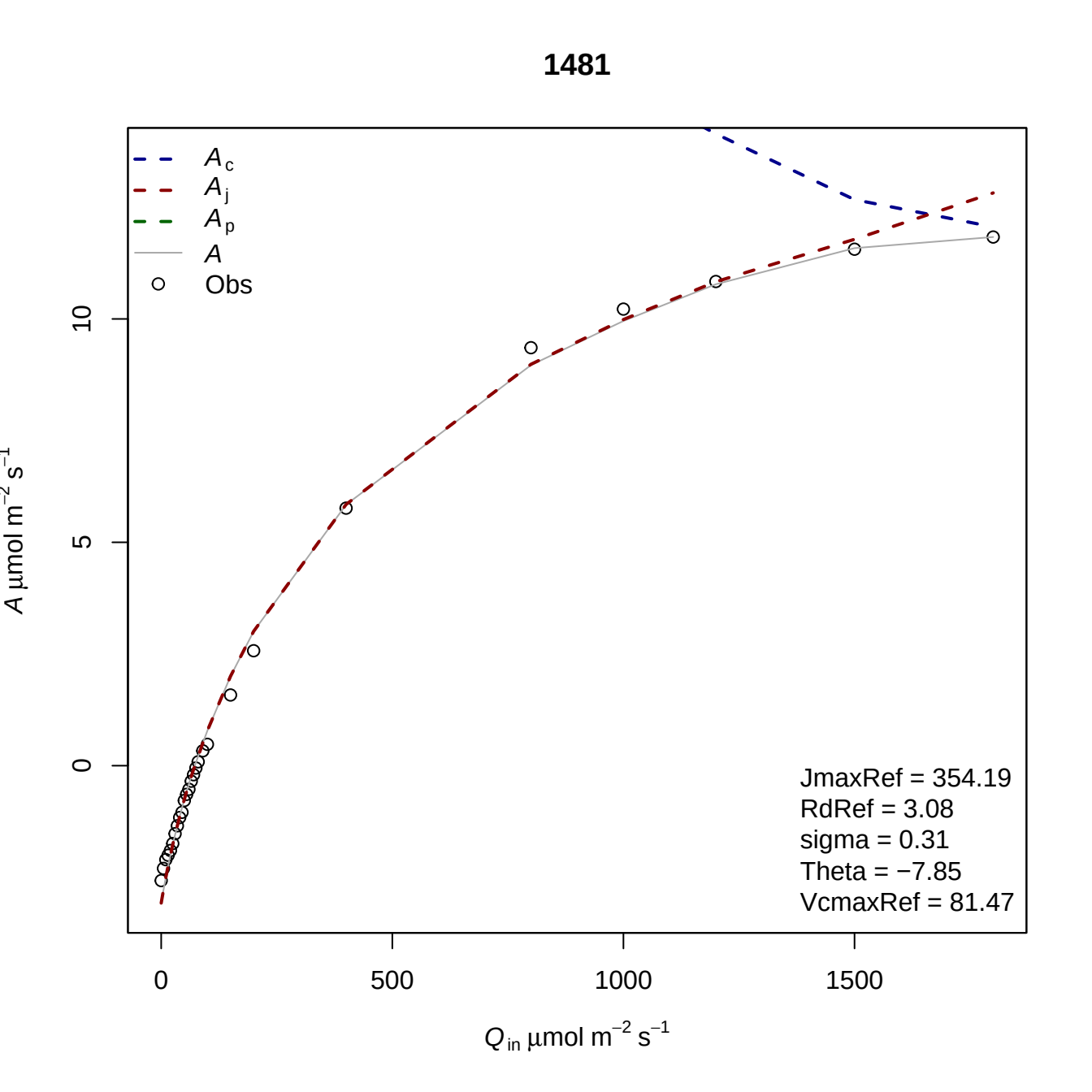

1476
A_c
A_j
A_p
A
Obs
JmaxRef = 4799159.18
RdRef = 2.67
sigma = 0.19
Theta = −332586.81
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

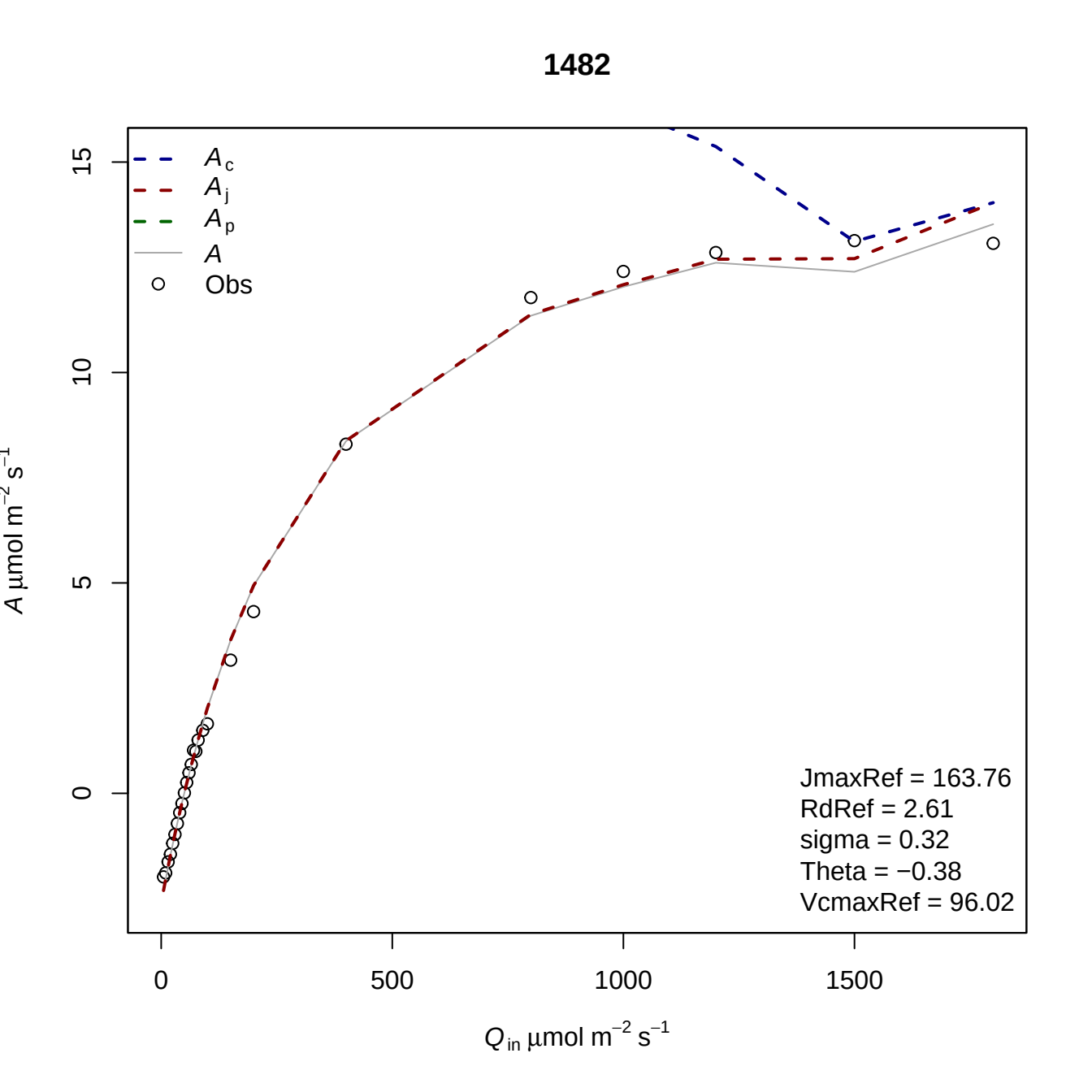

1477
A_c
A_j
A_p
A
Obs
JmaxRef = 340.51
RdRef = 2.34
sigma = 0.21
Theta = −9.12
VcmaxRef = 92.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

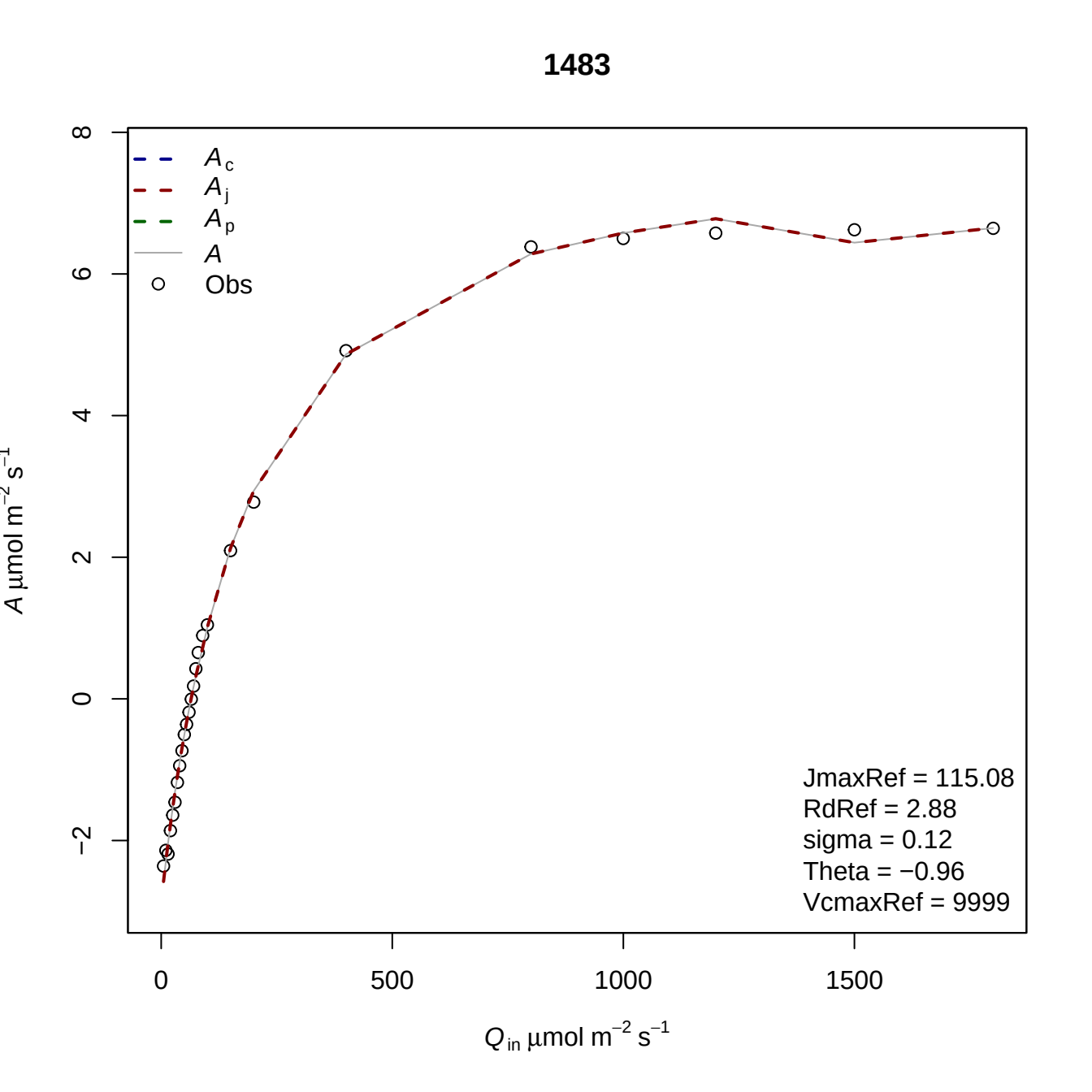

1478
A_c
A_j
A_p
A
Obs
JmaxRef = 285.77
RdRef = 2.82
sigma = 0.14
Theta = −9.82
VcmaxRef = 62.08
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

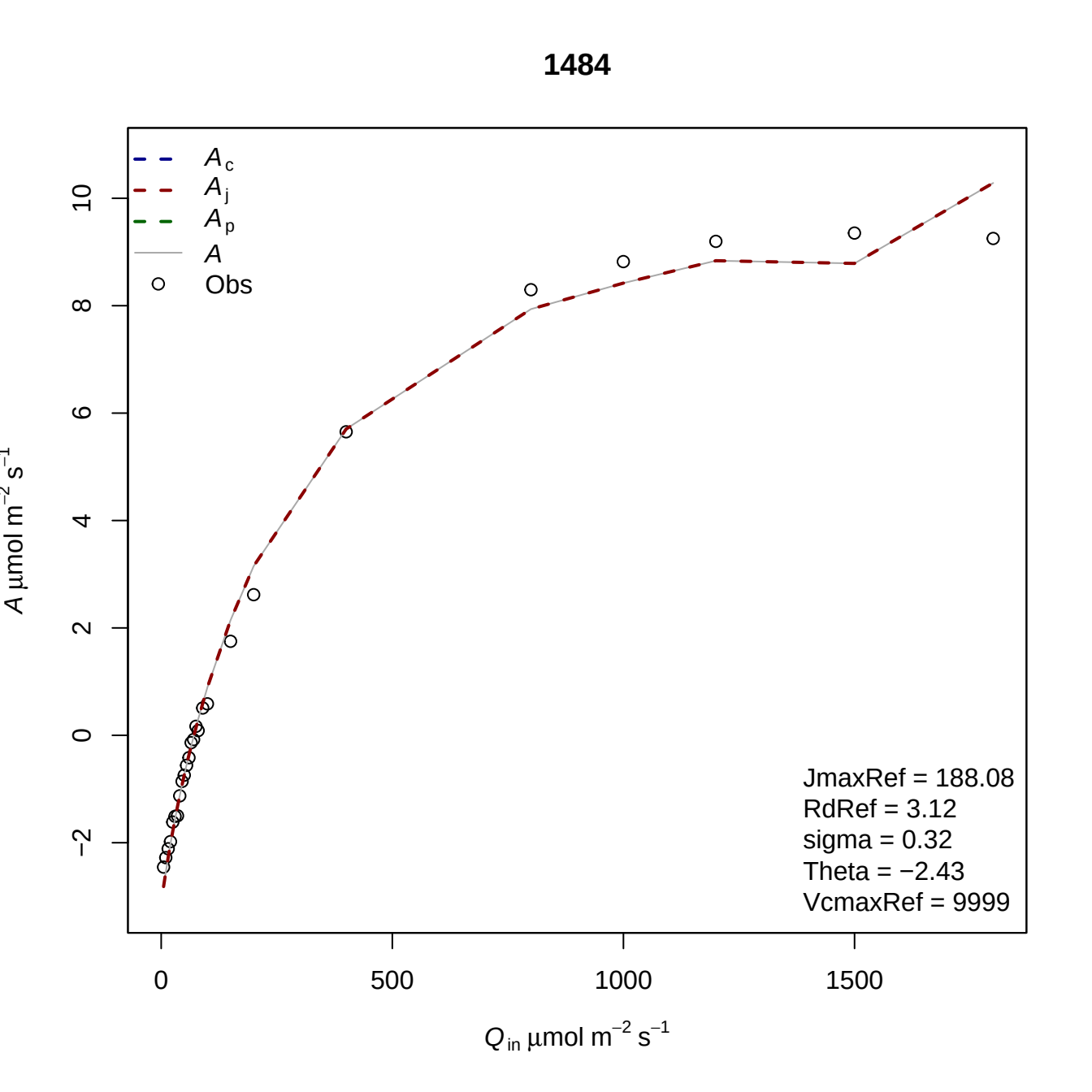

1479
A_c
A_j
A_p
A
Obs
JmaxRef = 182.35
RdRef = 3.15
sigma = 0.29
Theta = −1.99
VcmaxRef = 83.12
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

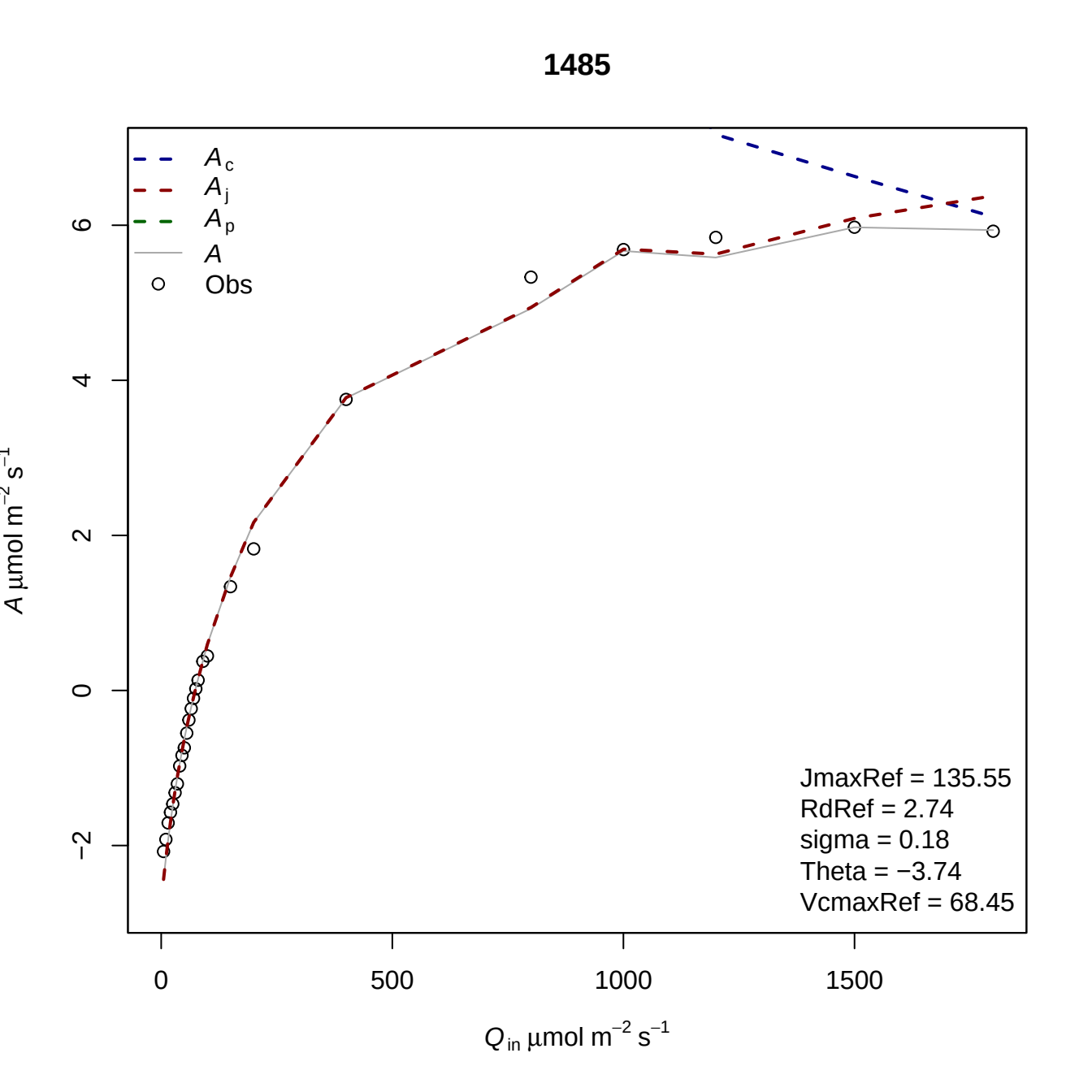

1480
A_c
A_j
A_p
A
Obs
JmaxRef = 655.12
RdRef = 2.92
sigma = 0.17
Theta = −13.52
VcmaxRef = 93.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

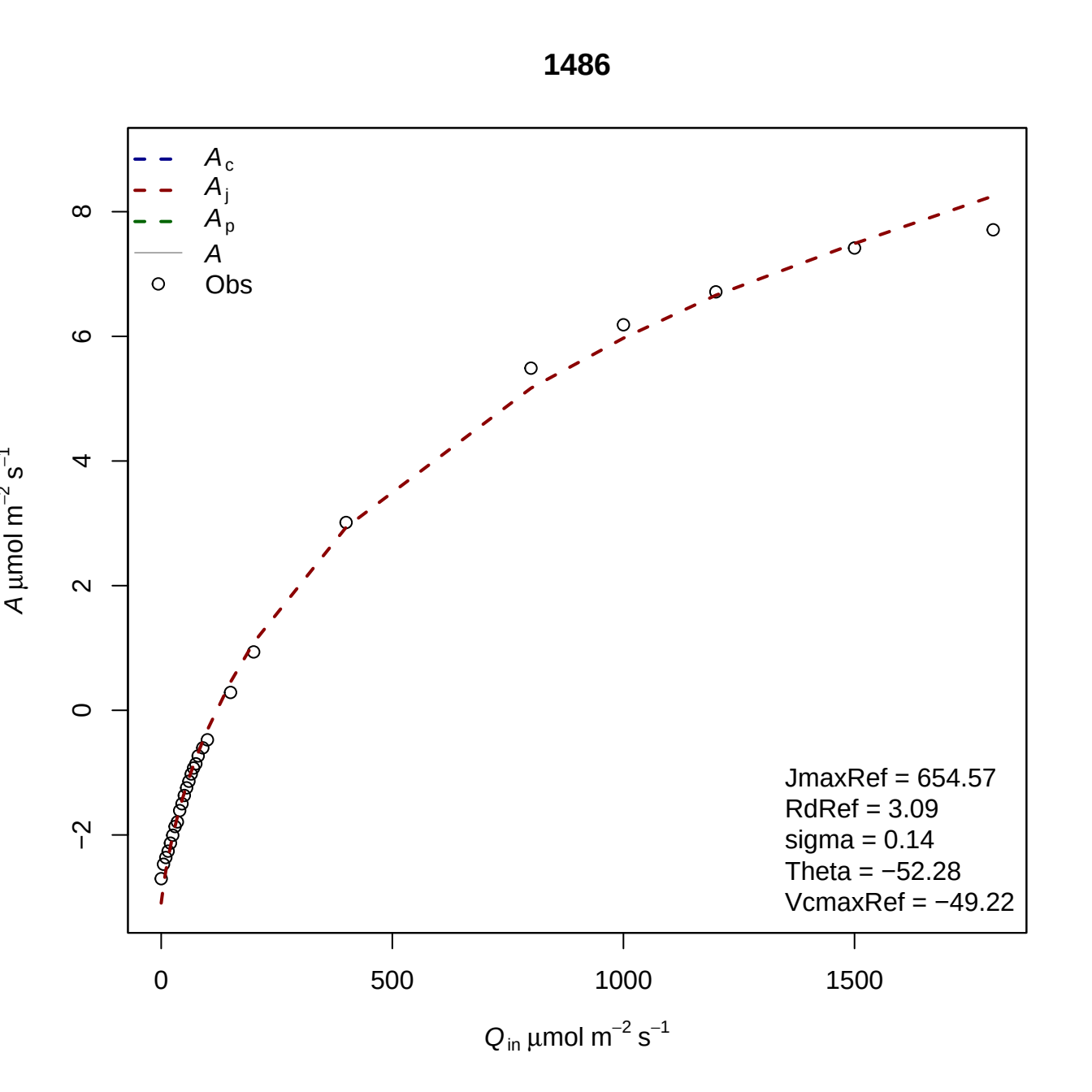

1481
A_c
A_j
A_p
A
Obs
JmaxRef = 354.19
RdRef = 3.08
sigma = 0.31
Theta = −7.85
VcmaxRef = 81.47
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

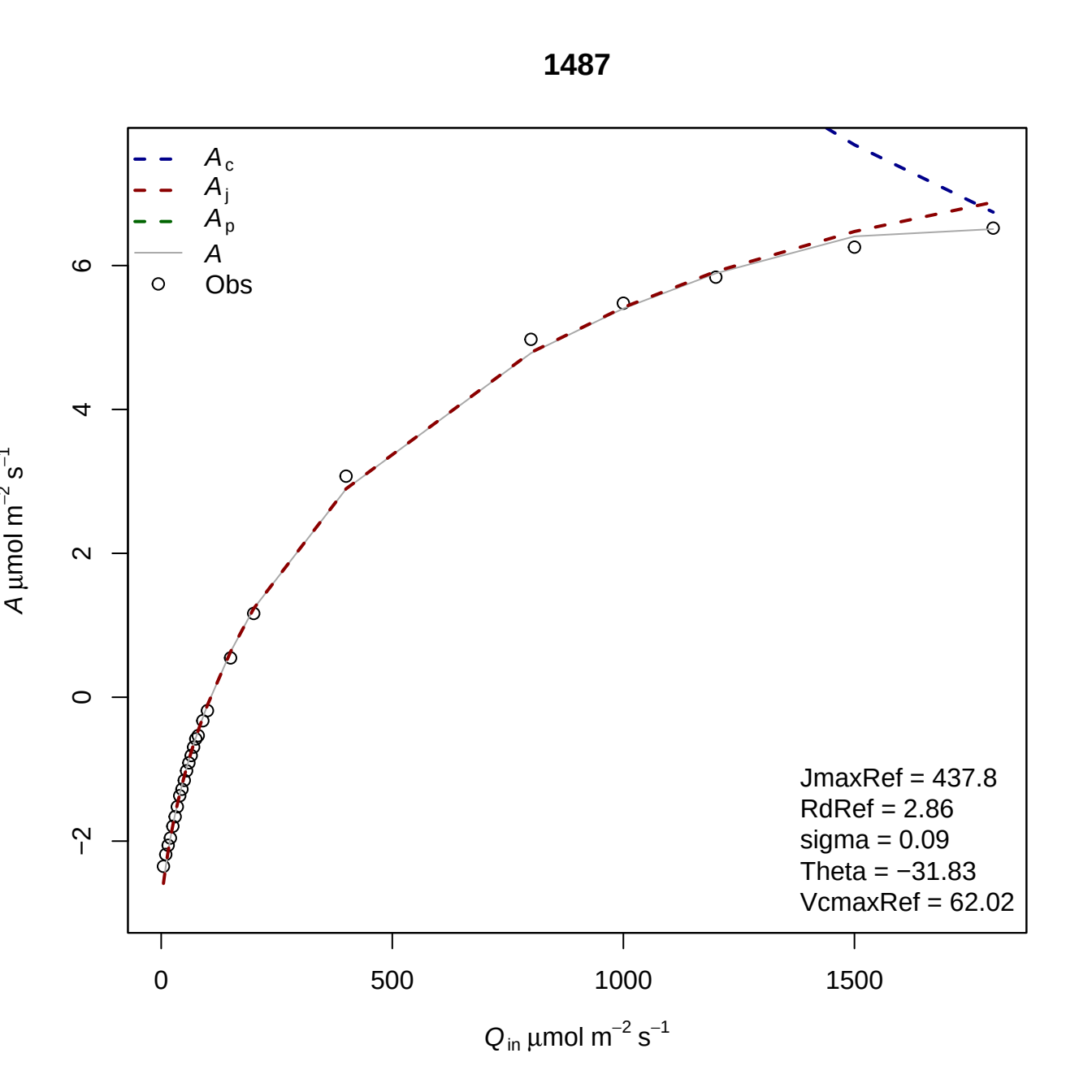

1482
A_c
A_j
A_p
A
Obs
JmaxRef = 163.76
RdRef = 2.61
sigma = 0.32
Theta = −0.38
VcmaxRef = 96.02
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

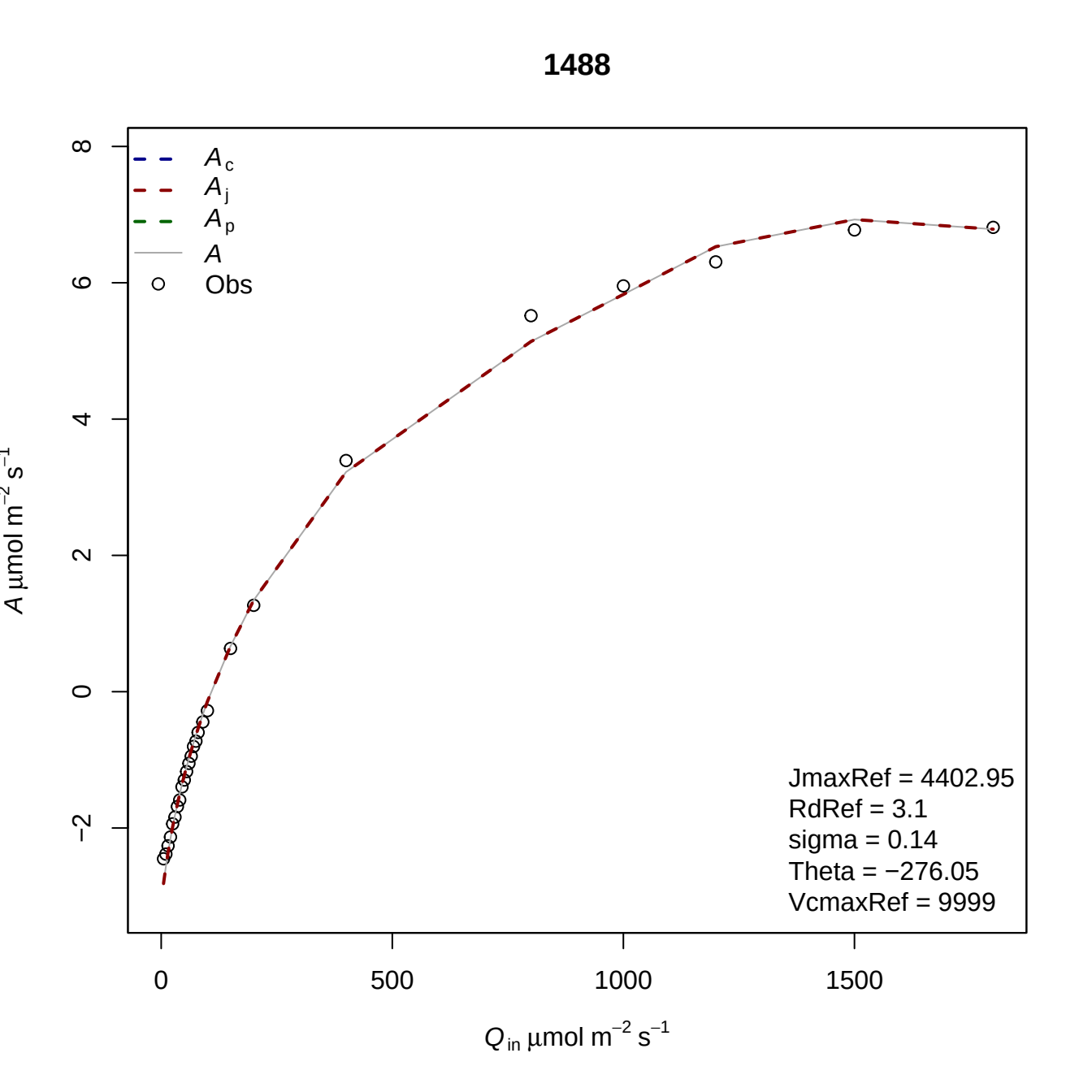

1483
A_c
A_j
A_p
A
Obs
JmaxRef = 115.08
RdRef = 2.88
sigma = 0.12
Theta = −0.96
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

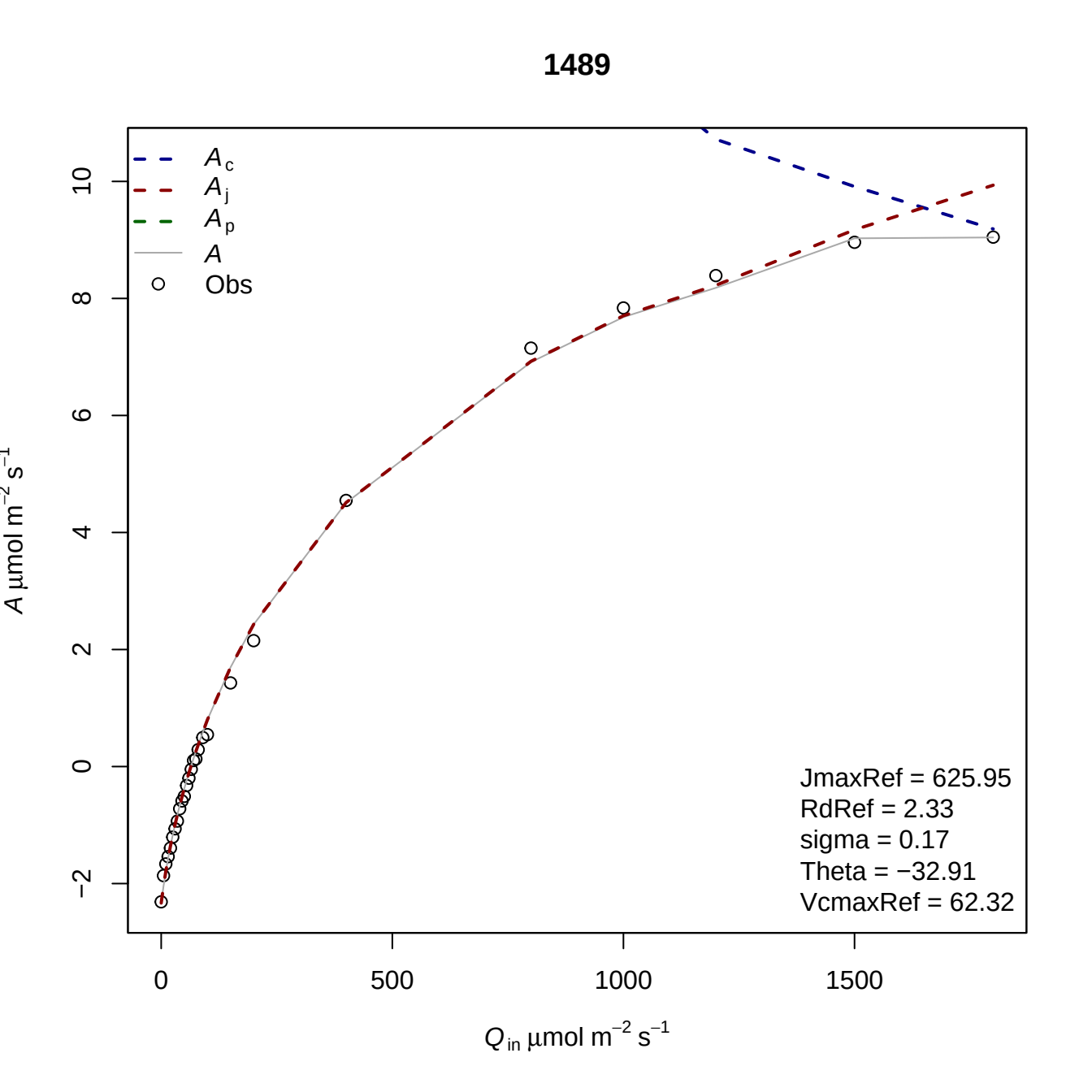

1484
A_c
A_j
A_p
A
Obs
JmaxRef = 188.08
RdRef = 3.12
sigma = 0.32
Theta = −2.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

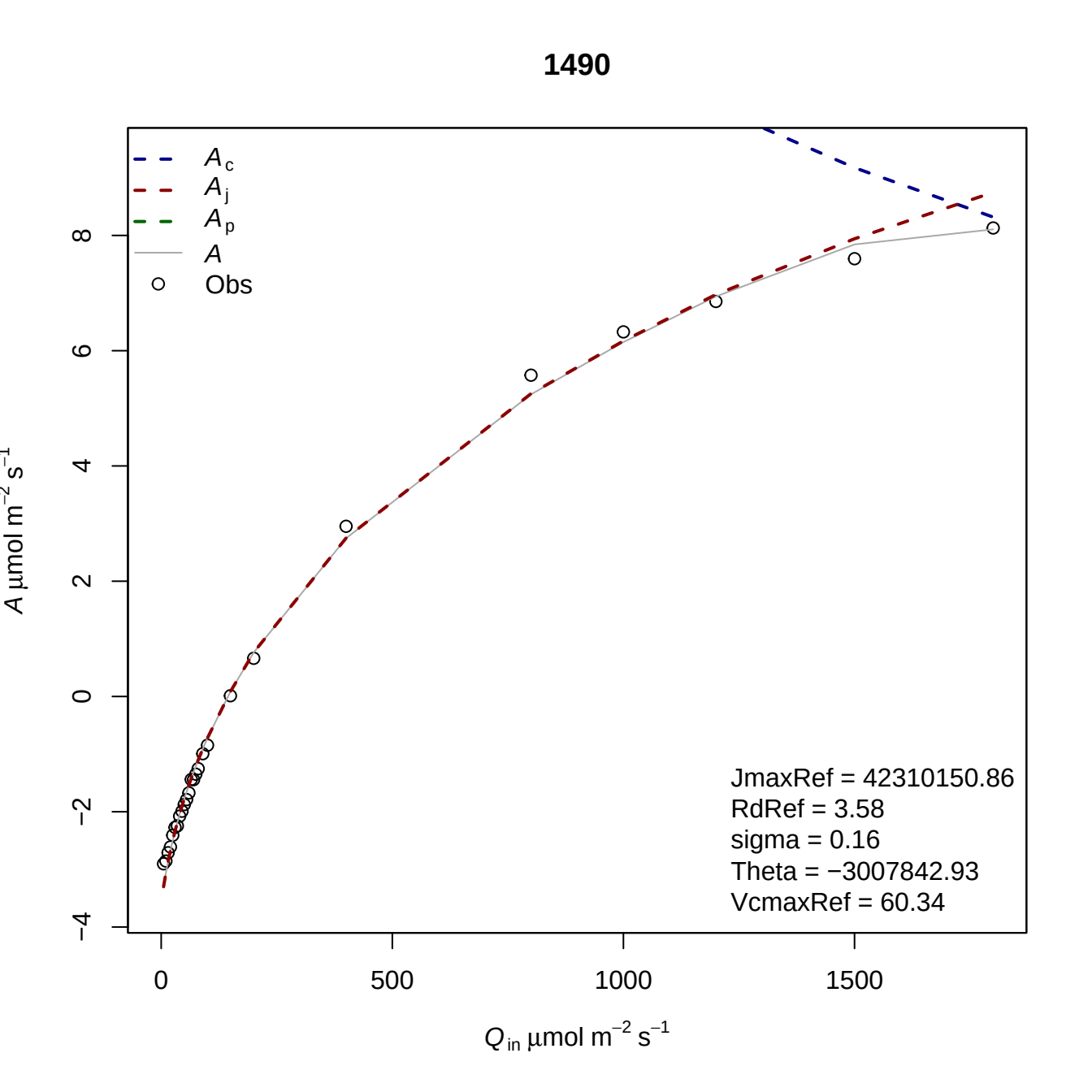

1485
A_c
A_j
A_p
A
Obs
JmaxRef = 135.55
RdRef = 2.74
sigma = 0.18
Theta = −3.74
VcmaxRef = 68.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

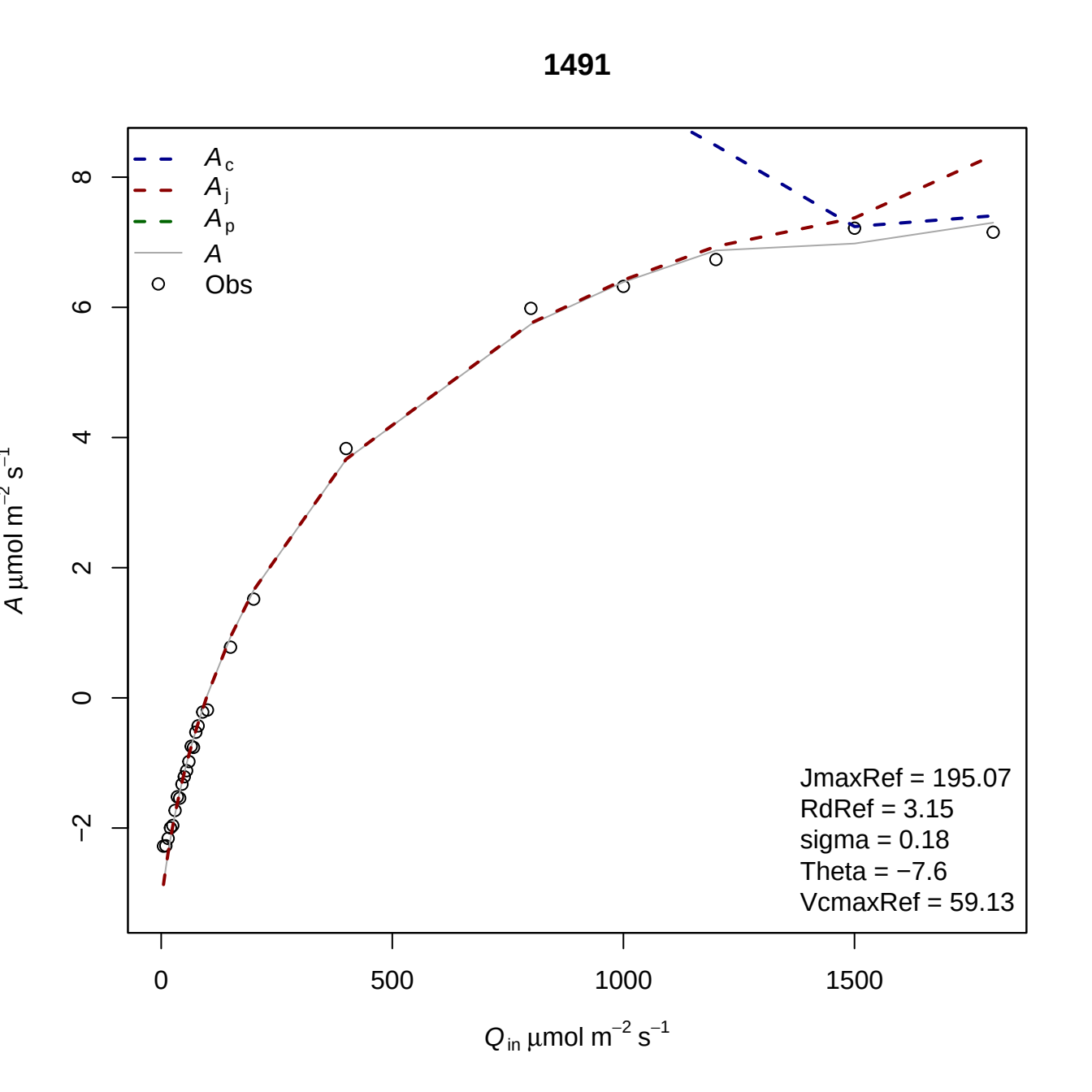

1486
A_c
A_j
A_p
A
Obs
JmaxRef = 654.57
RdRef = 3.09
sigma = 0.14
Theta = −52.28
VcmaxRef = −49.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

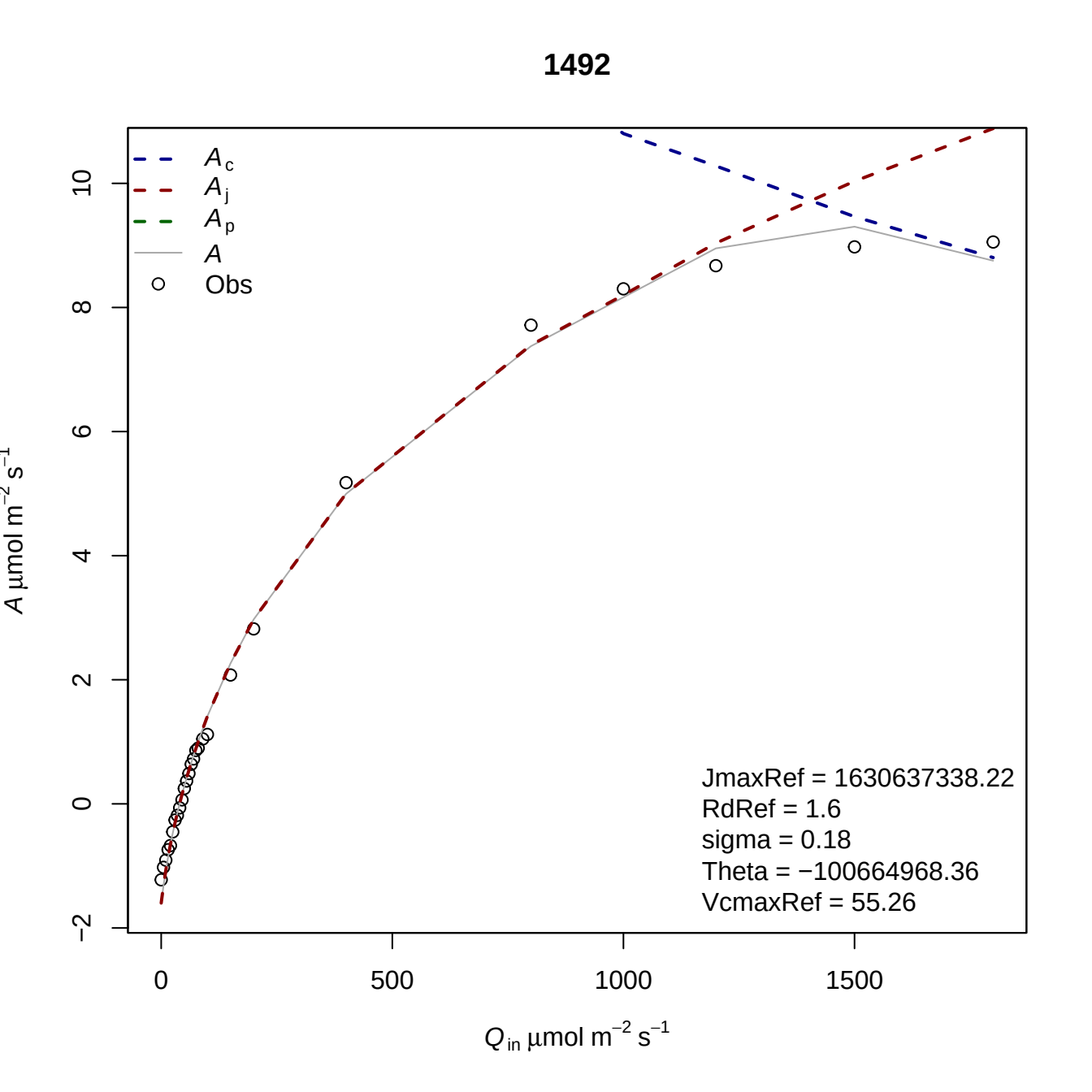

1487
A_c
A_j
A_p
A
Obs
JmaxRef = 437.8
RdRef = 2.86
sigma = 0.09
Theta = −31.83
VcmaxRef = 62.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

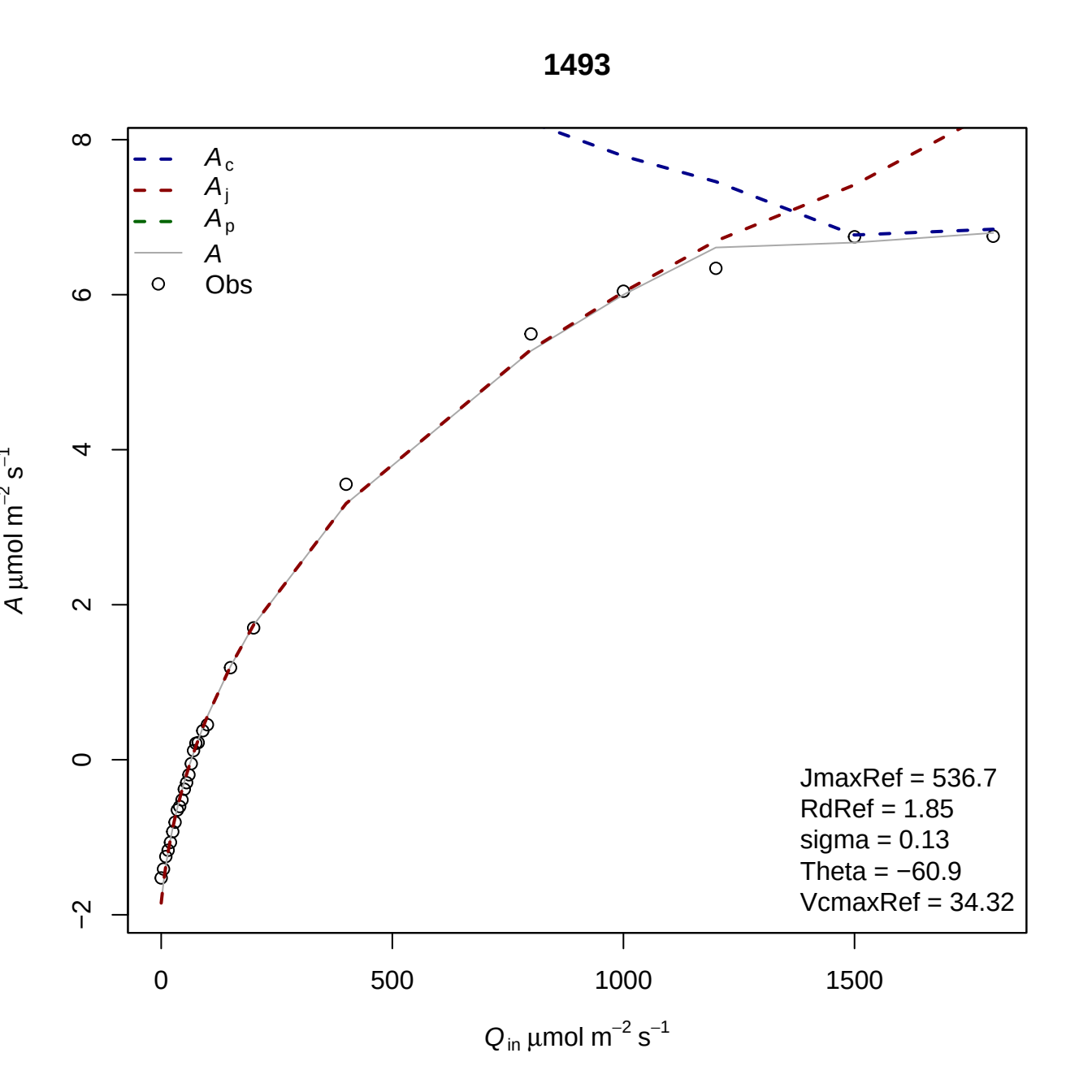

1488
A_c
A_j
A_p
A
Obs
JmaxRef = 4402.95
RdRef = 3.1
sigma = 0.14
Theta = −276.05
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

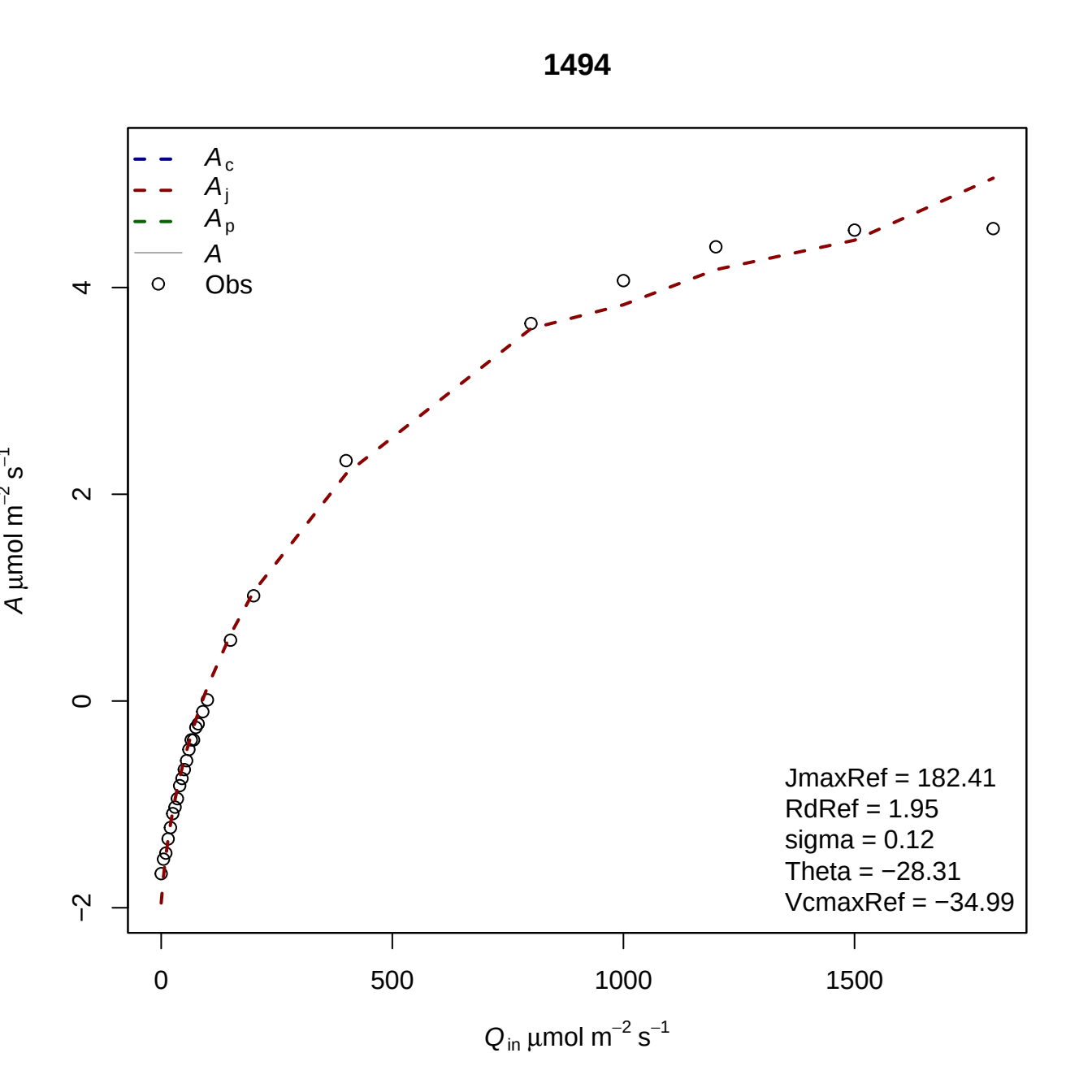

1489
A_c
A_j
A_p
A
Obs
JmaxRef = 625.95
RdRef = 2.33
sigma = 0.17
Theta = −32.91
VcmaxRef = 62.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

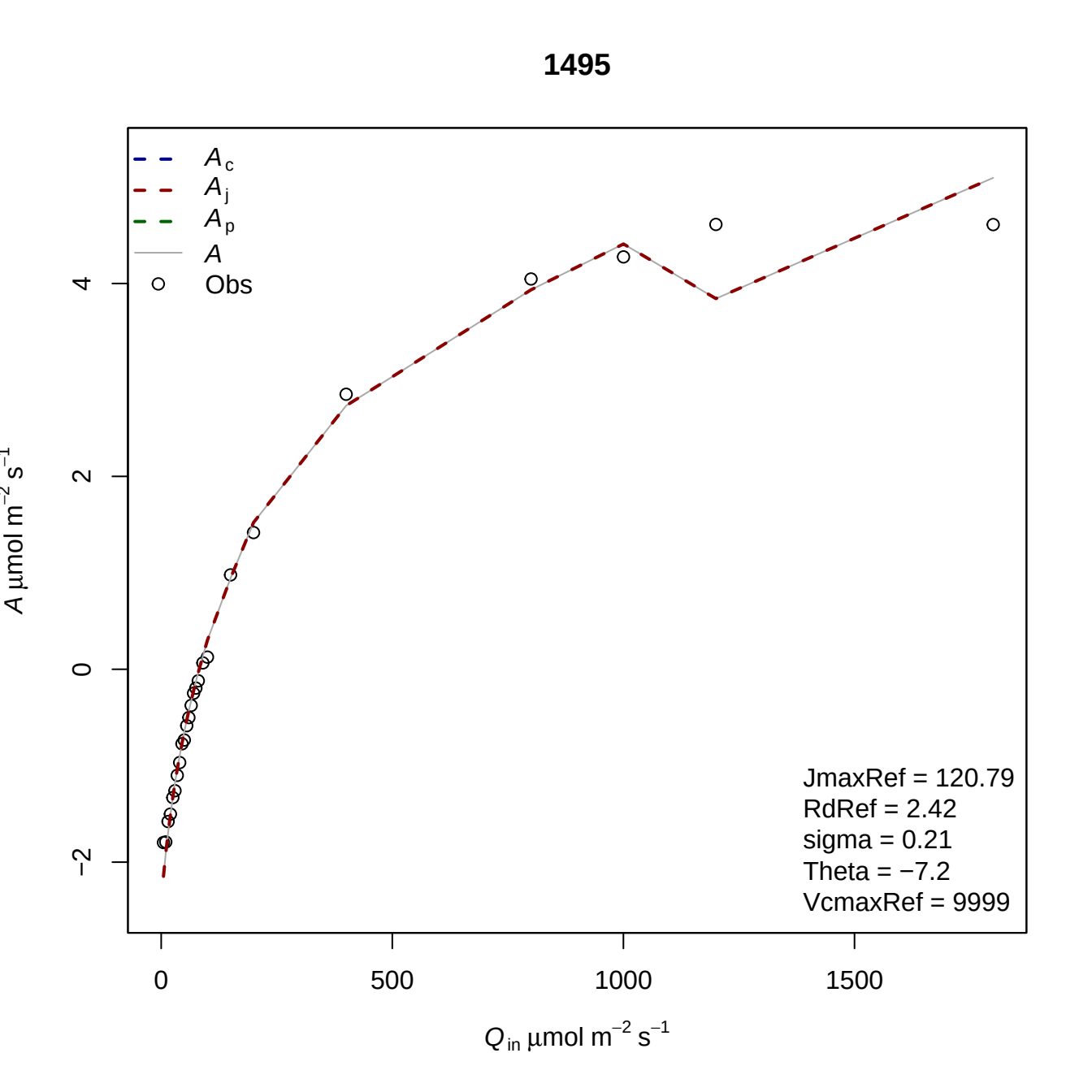

1490
A_c
A_j
A_p
A
Obs
JmaxRef = 42310150.86
RdRef = 3.58
sigma = 0.16
Theta = −3007842.93
VcmaxRef = 60.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

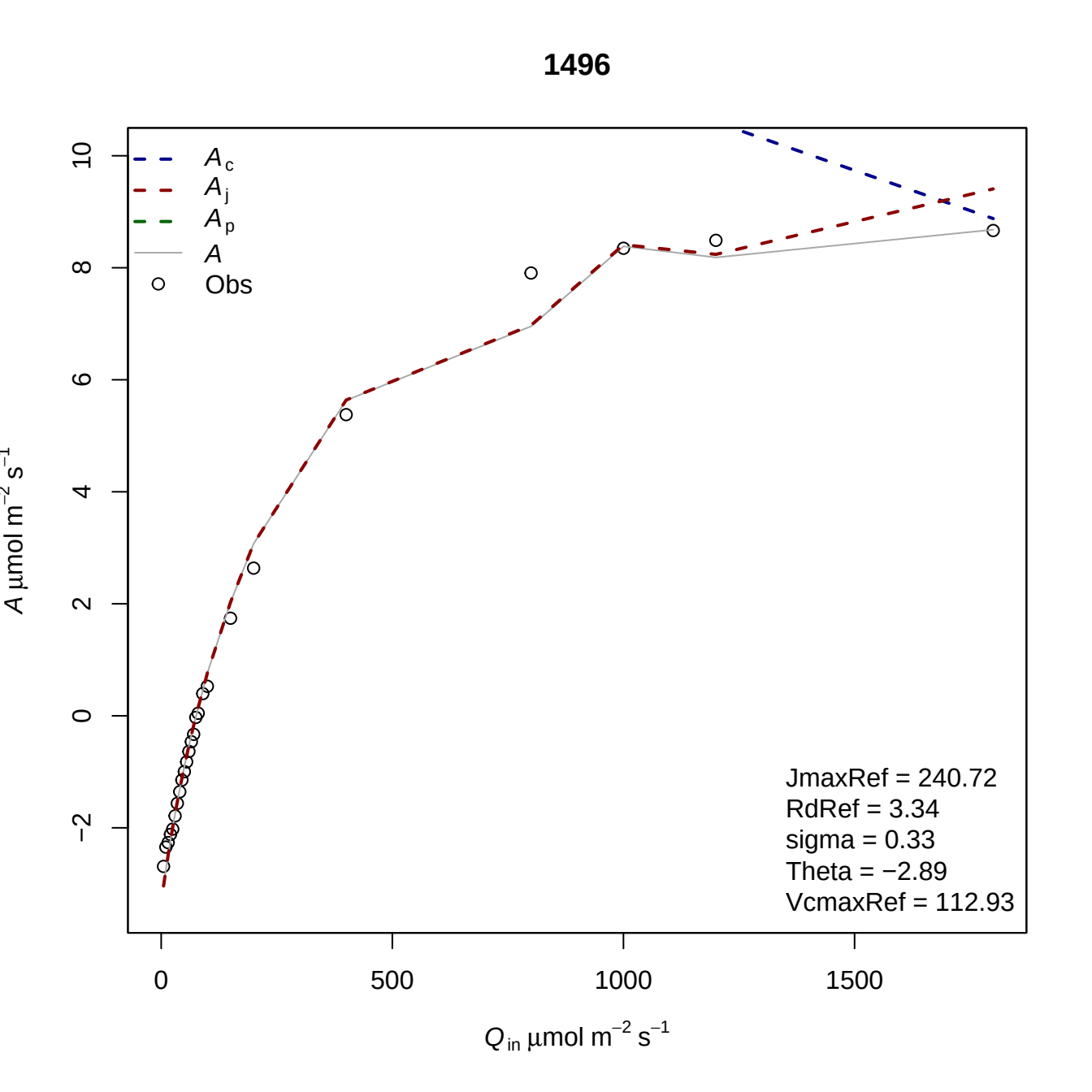

1491
A_c
A_j
A_p
A
Obs
JmaxRef = 195.07
RdRef = 3.15
sigma = 0.18
Theta = −7.6
VcmaxRef = 59.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

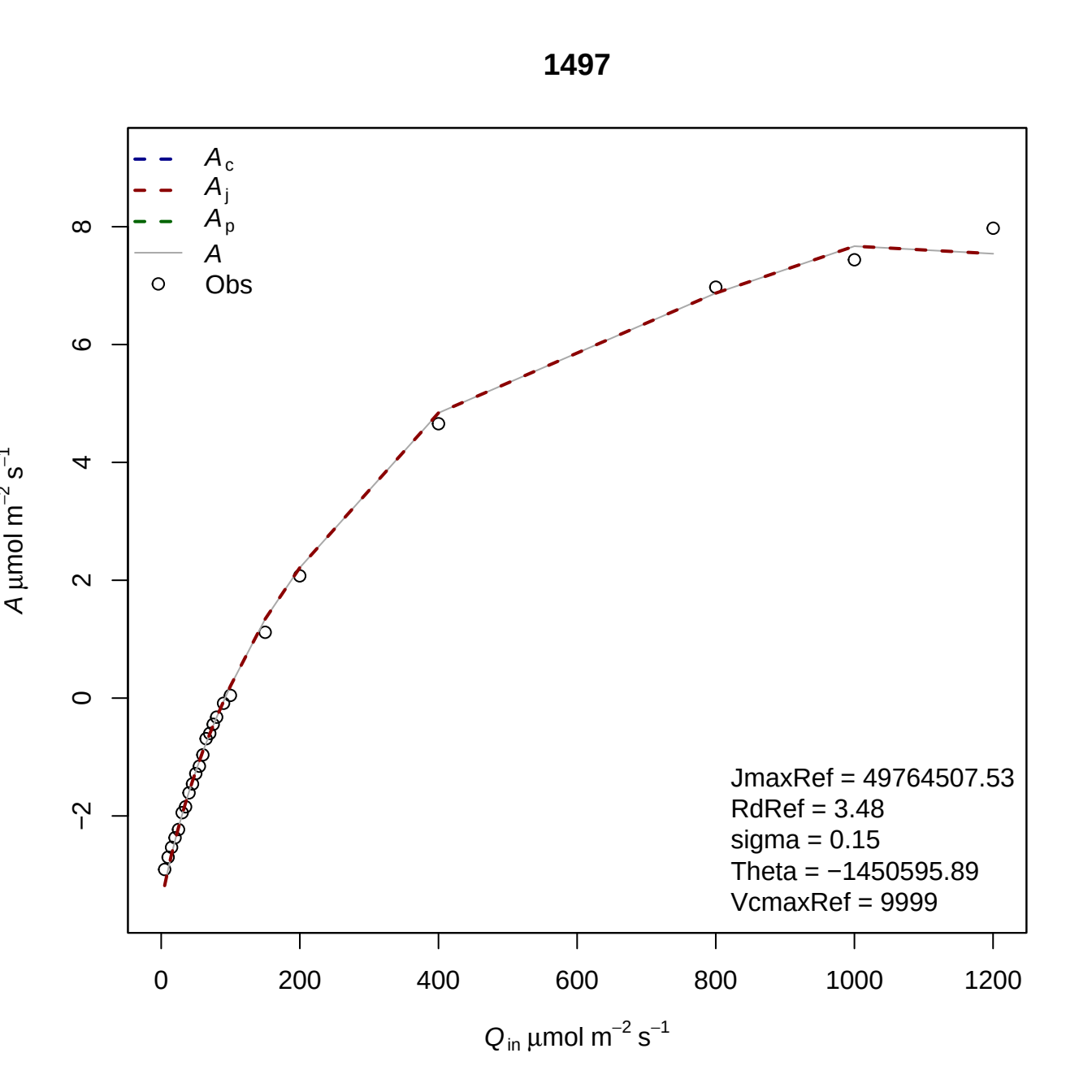

1492
A_c
A_j
A_p
A
Obs
JmaxRef = 1630637338.22
RdRef = 1.6
sigma = 0.18
Theta = −100664968.36
VcmaxRef = 55.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

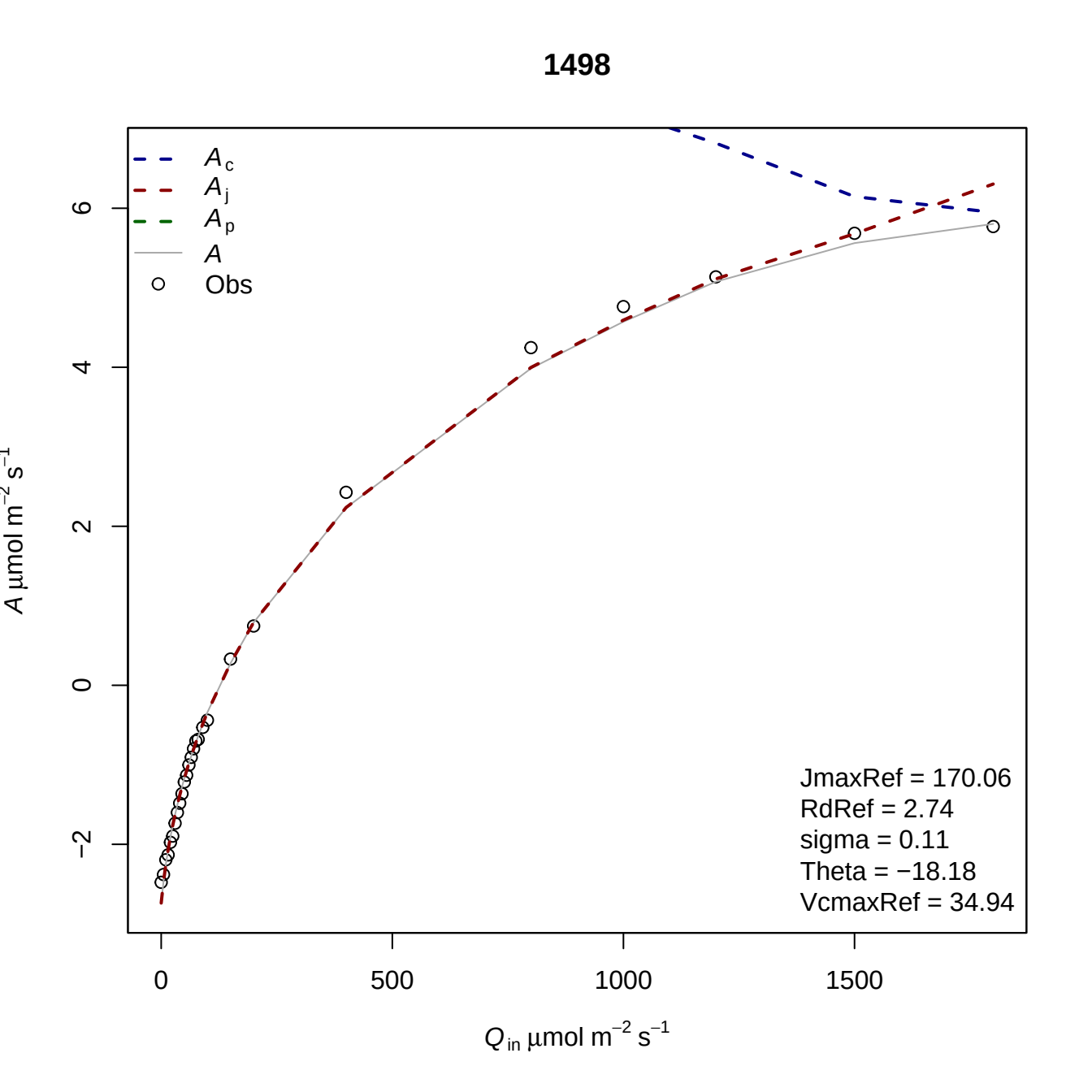

1493
A_c
A_j
A_p
A
Obs
JmaxRef = 536.7
RdRef = 1.85
sigma = 0.13
Theta = −60.9
VcmaxRef = 34.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

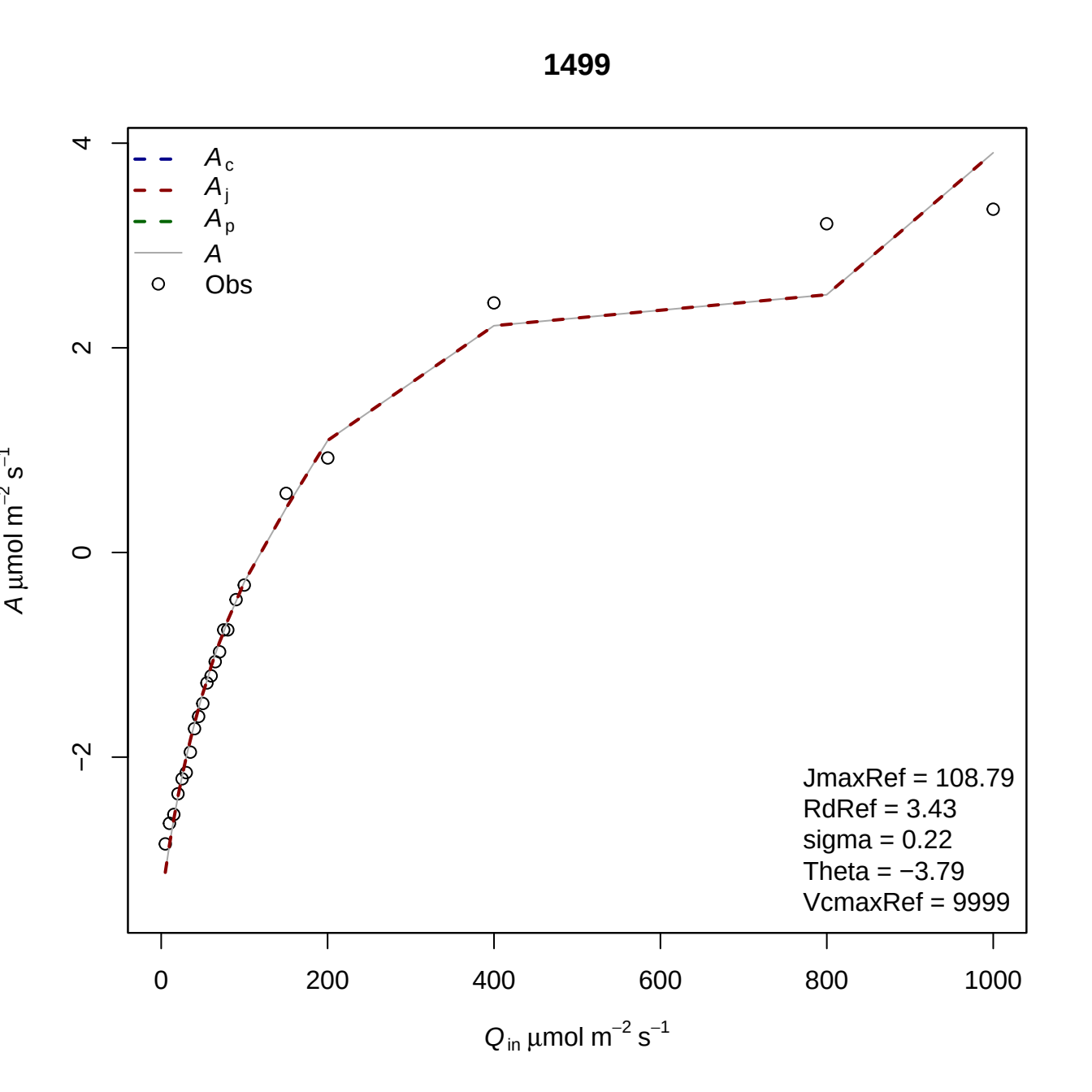

1494
A_c
A_j
A_p
A
Obs
JmaxRef = 182.41
RdRef = 1.95
sigma = 0.12
Theta = −28.31
VcmaxRef = −34.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

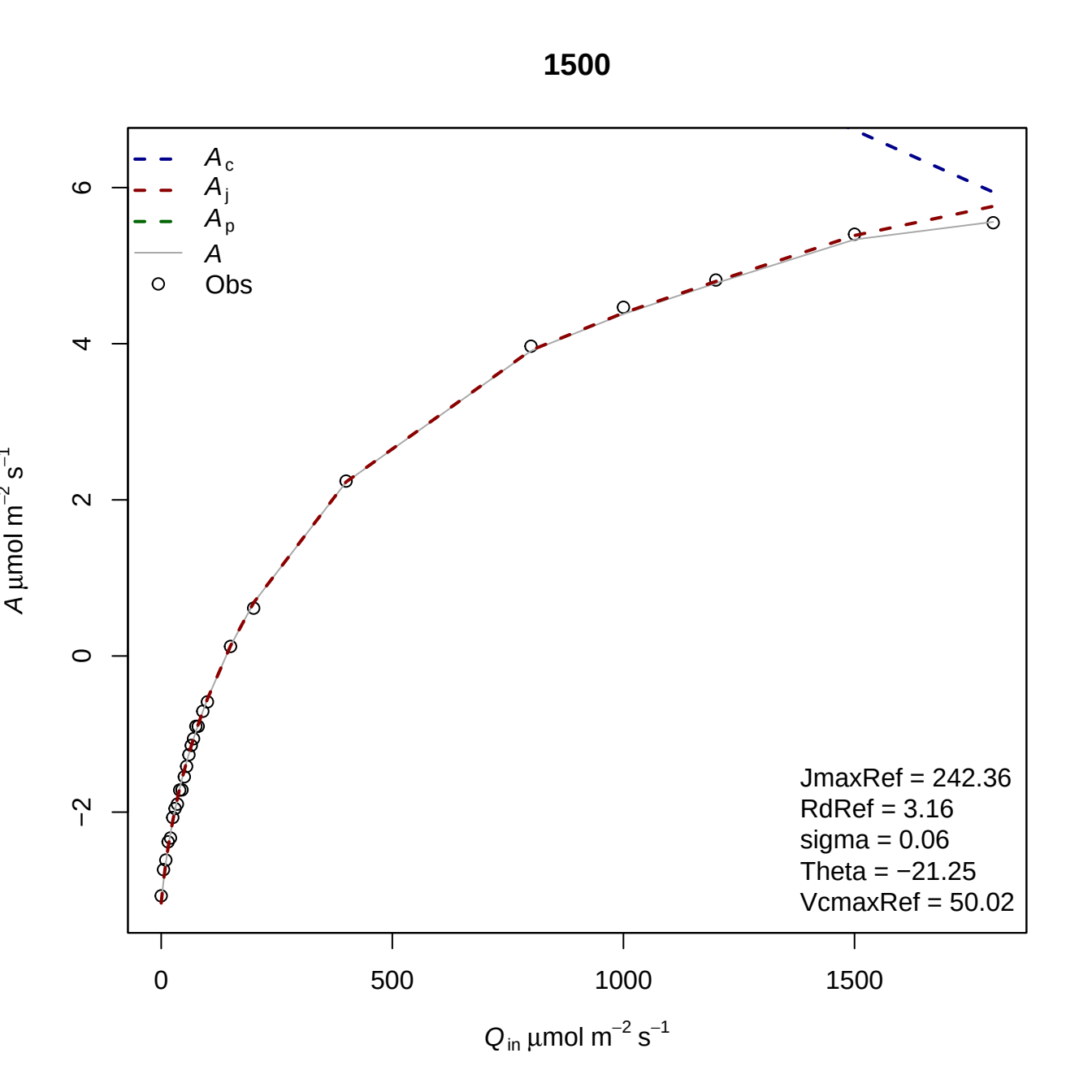

1495
A_c
A_j
A_p
A
Obs
JmaxRef = 120.79
RdRef = 2.42
sigma = 0.21
Theta = −7.2
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

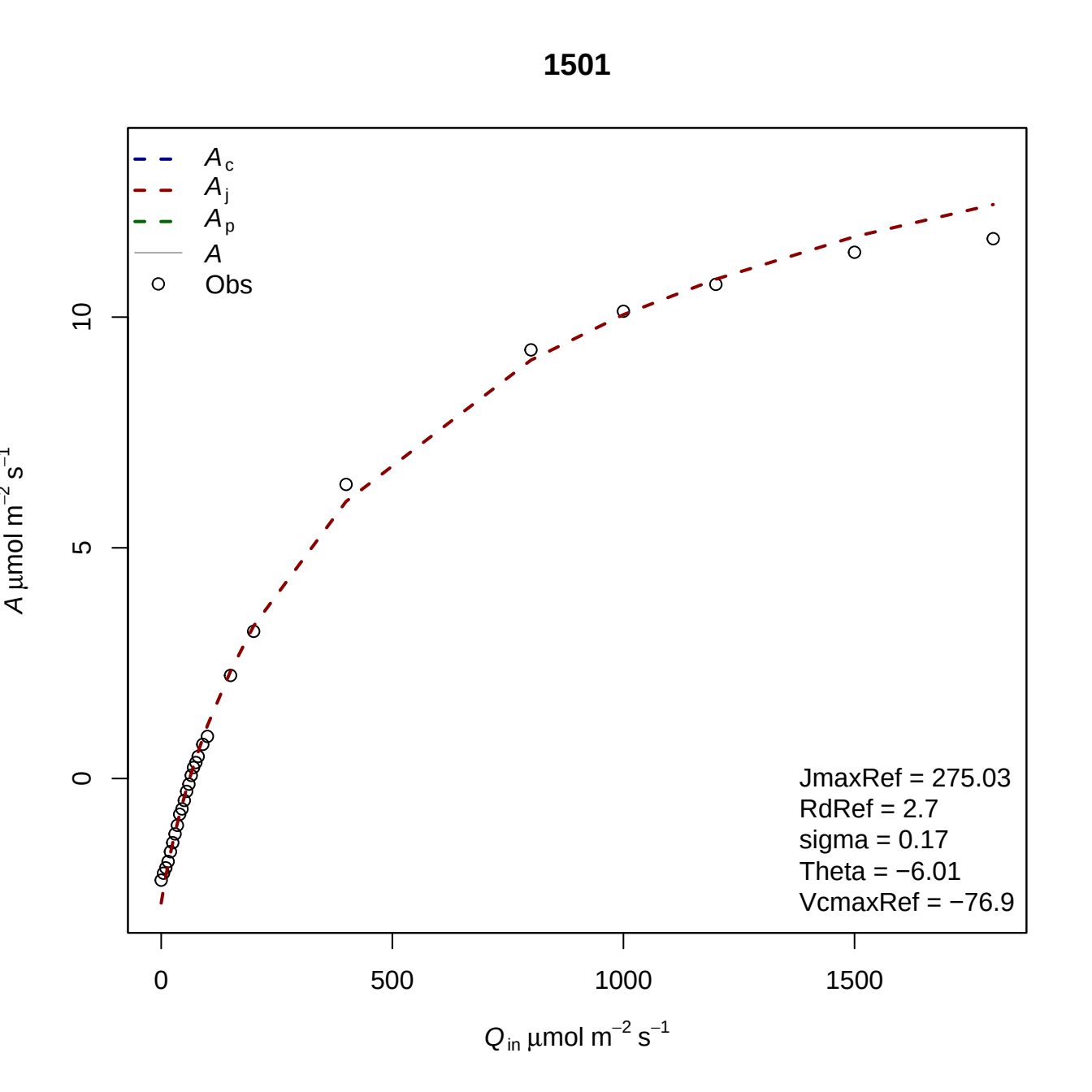

1496
A_c
A_j
A_p
A
Obs
JmaxRef = 240.72
RdRef = 3.34
sigma = 0.33
Theta = −2.89
VcmaxRef = 112.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

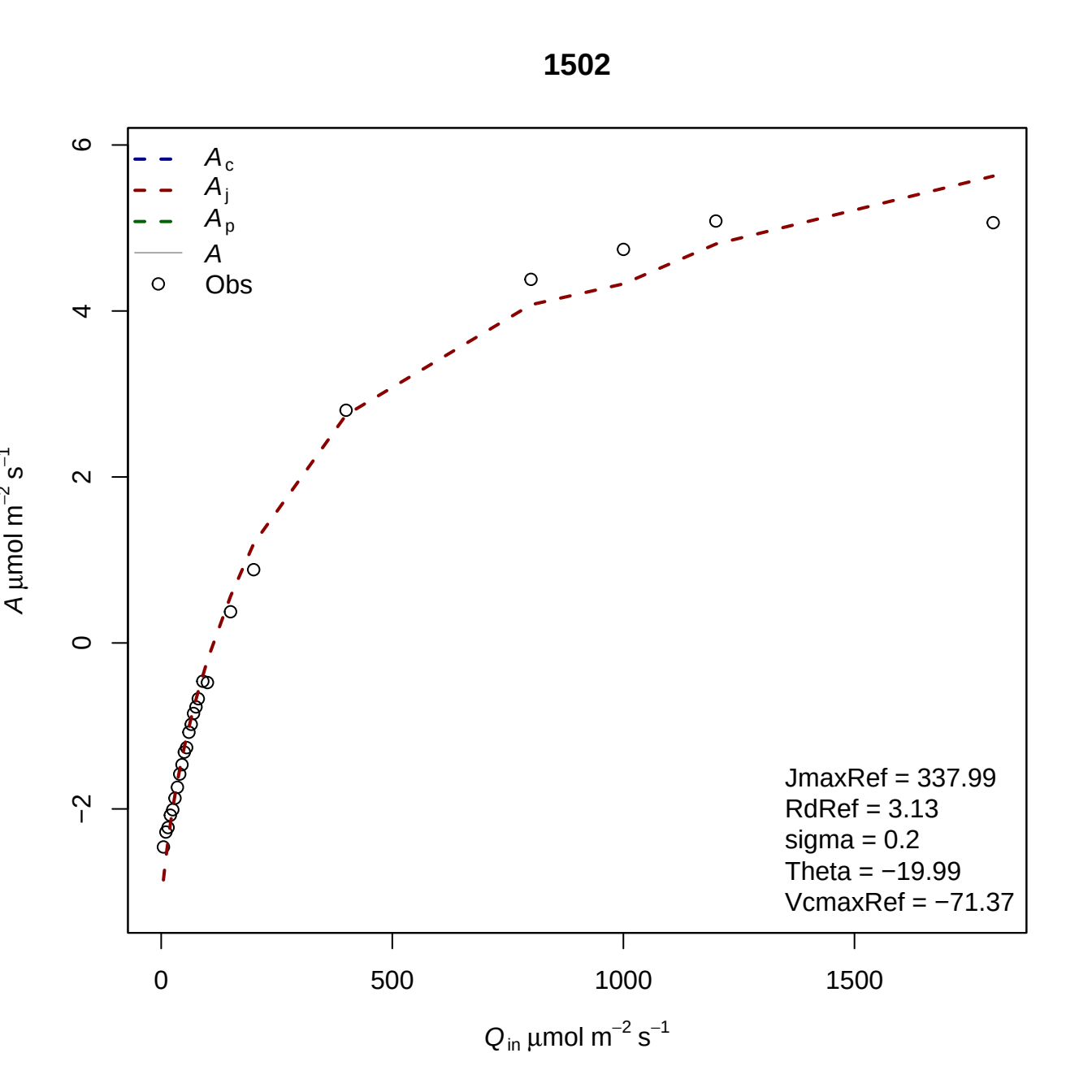

1497
A_c
A_j
A_p
A
Obs
JmaxRef = 49764507.53
RdRef = 3.48
sigma = 0.15
Theta = −1450595.89
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200

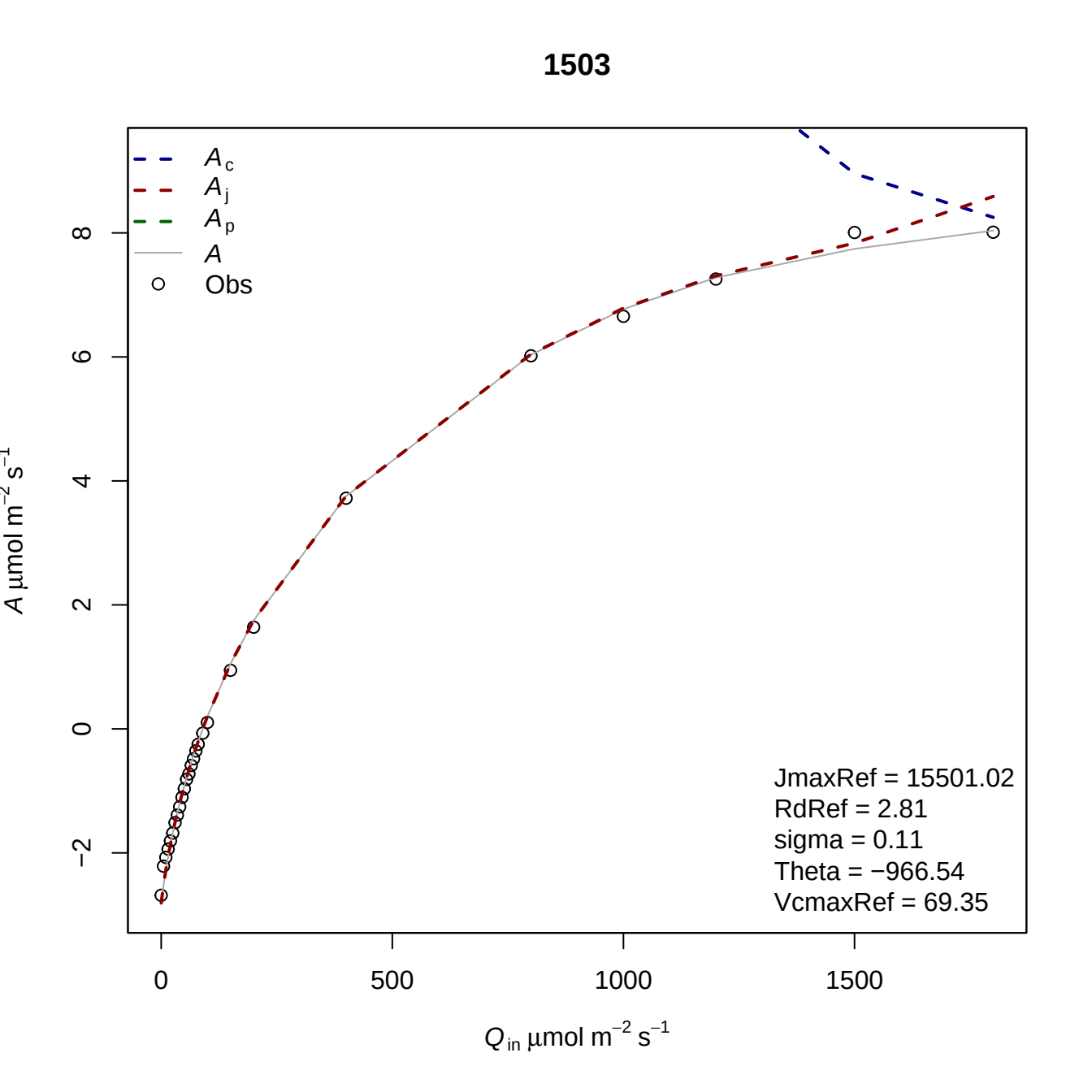

1498
A_c
A_j
A_p
A
Obs
JmaxRef = 170.06
RdRef = 2.74
sigma = 0.11
Theta = −18.18
VcmaxRef = 34.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

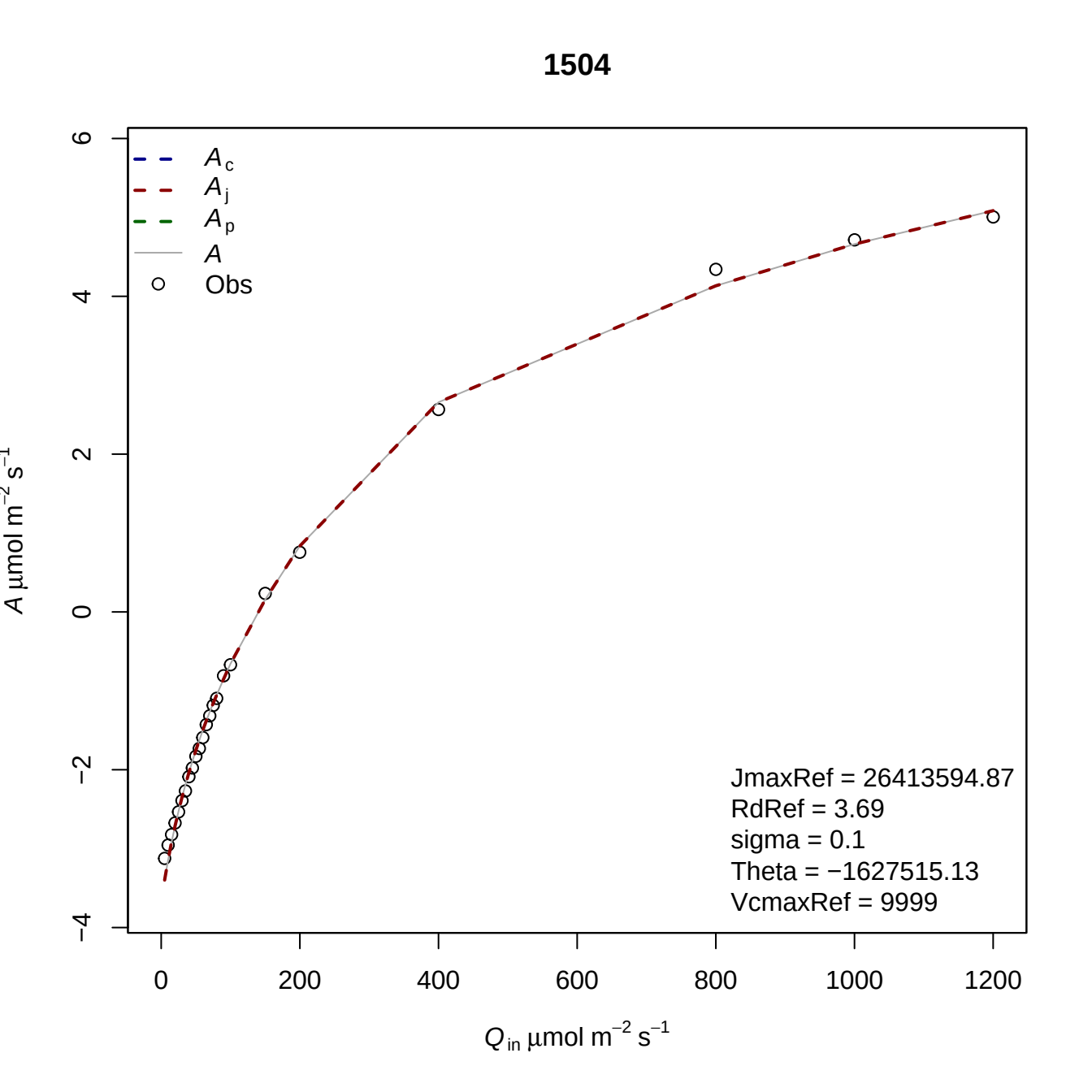

1499
A_c
A_j
A_p
A
Obs
JmaxRef = 108.79
RdRef = 3.43
sigma = 0.22
Theta = −3.79
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
0
200
400
600
800
1000

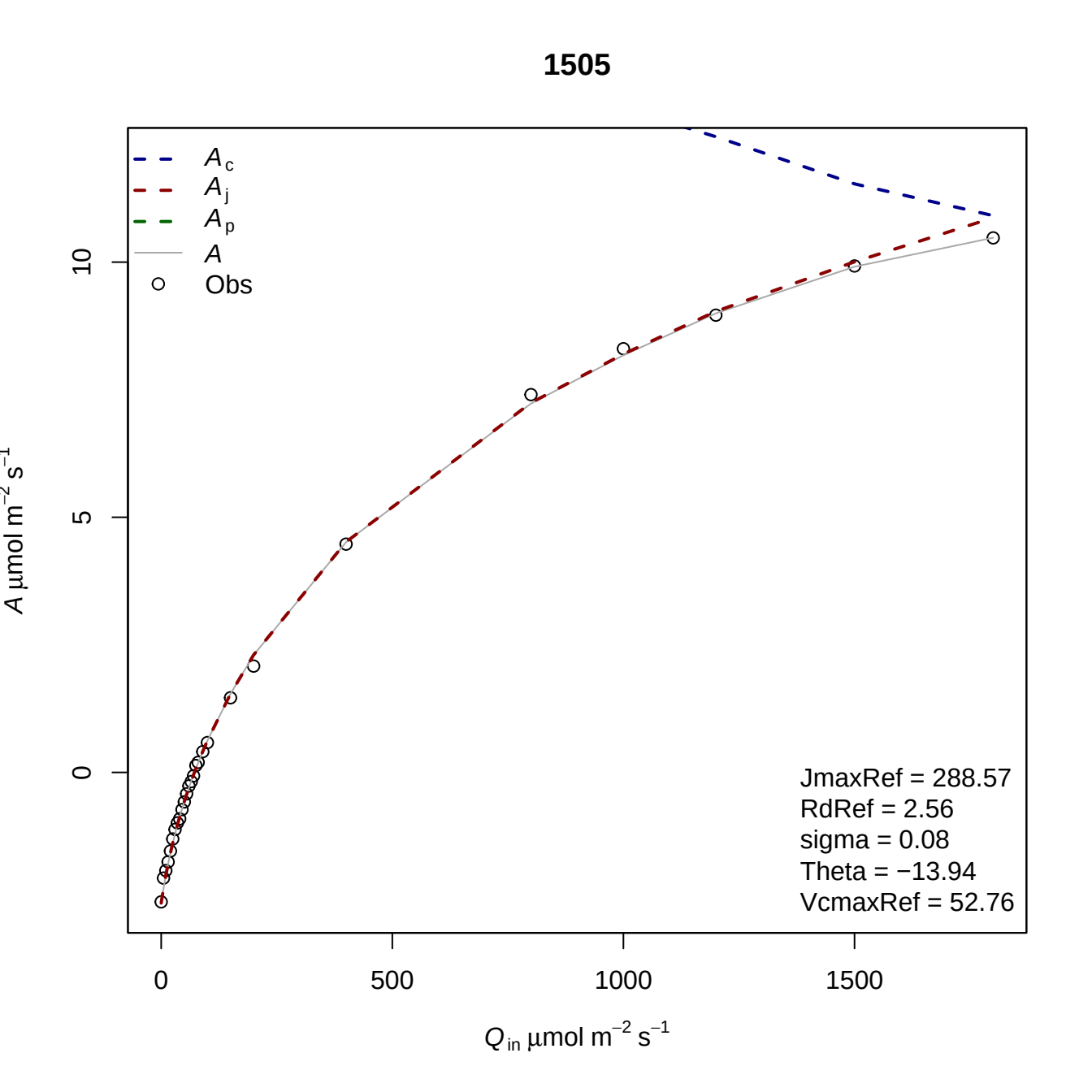

1500
A_c
A_j
A_p
A
Obs
JmaxRef = 242.36
RdRef = 3.16
sigma = 0.06
Theta = −21.25
VcmaxRef = 50.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

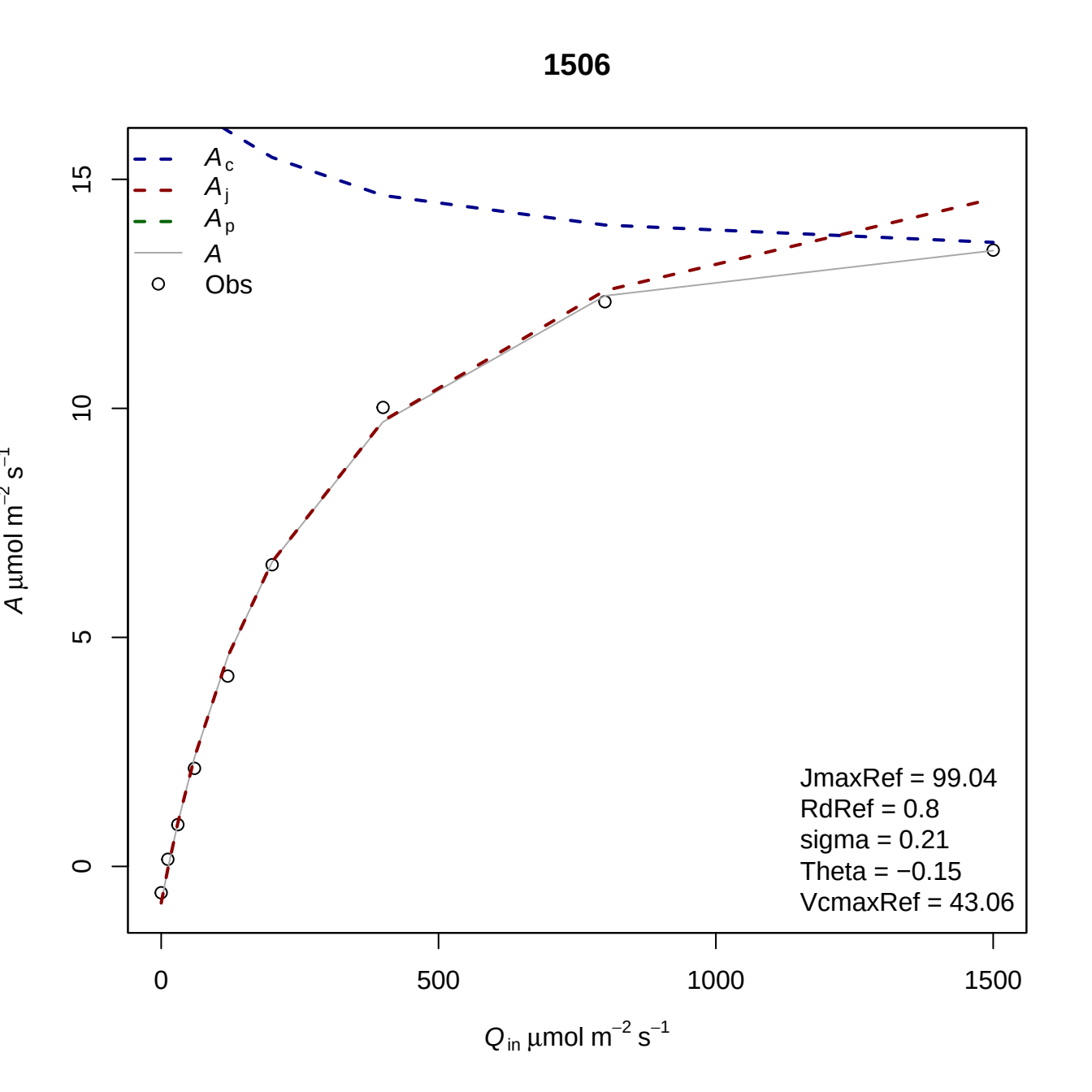

1501
A_c
A_j
A_p
A
Obs
JmaxRef = 275.03
RdRef = 2.7
sigma = 0.17
Theta = −6.01
VcmaxRef = −76.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

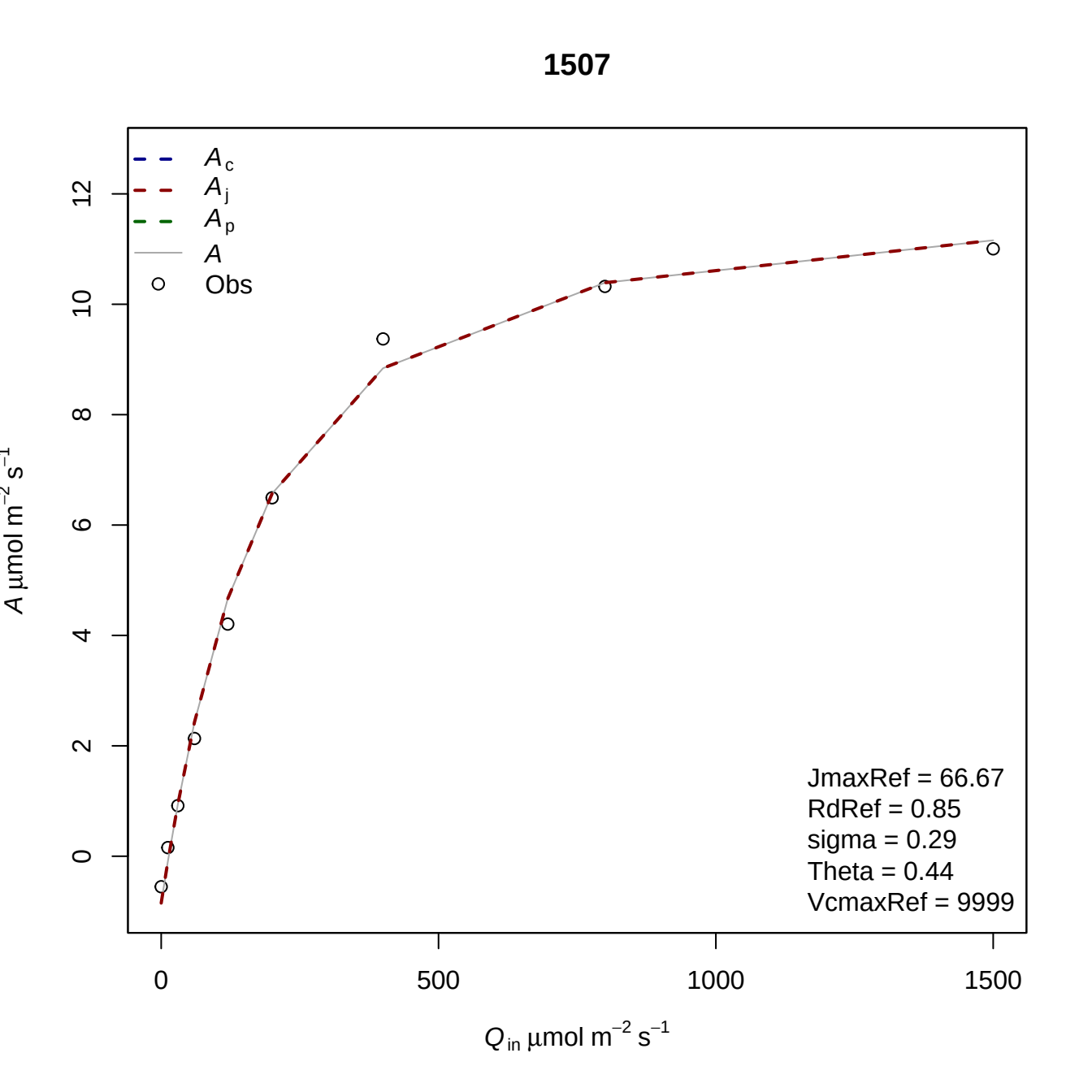

1502
A_c
A_j
A_p
A
Obs
JmaxRef = 337.99
RdRef = 3.13
sigma = 0.2
Theta = −19.99
VcmaxRef = −71.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

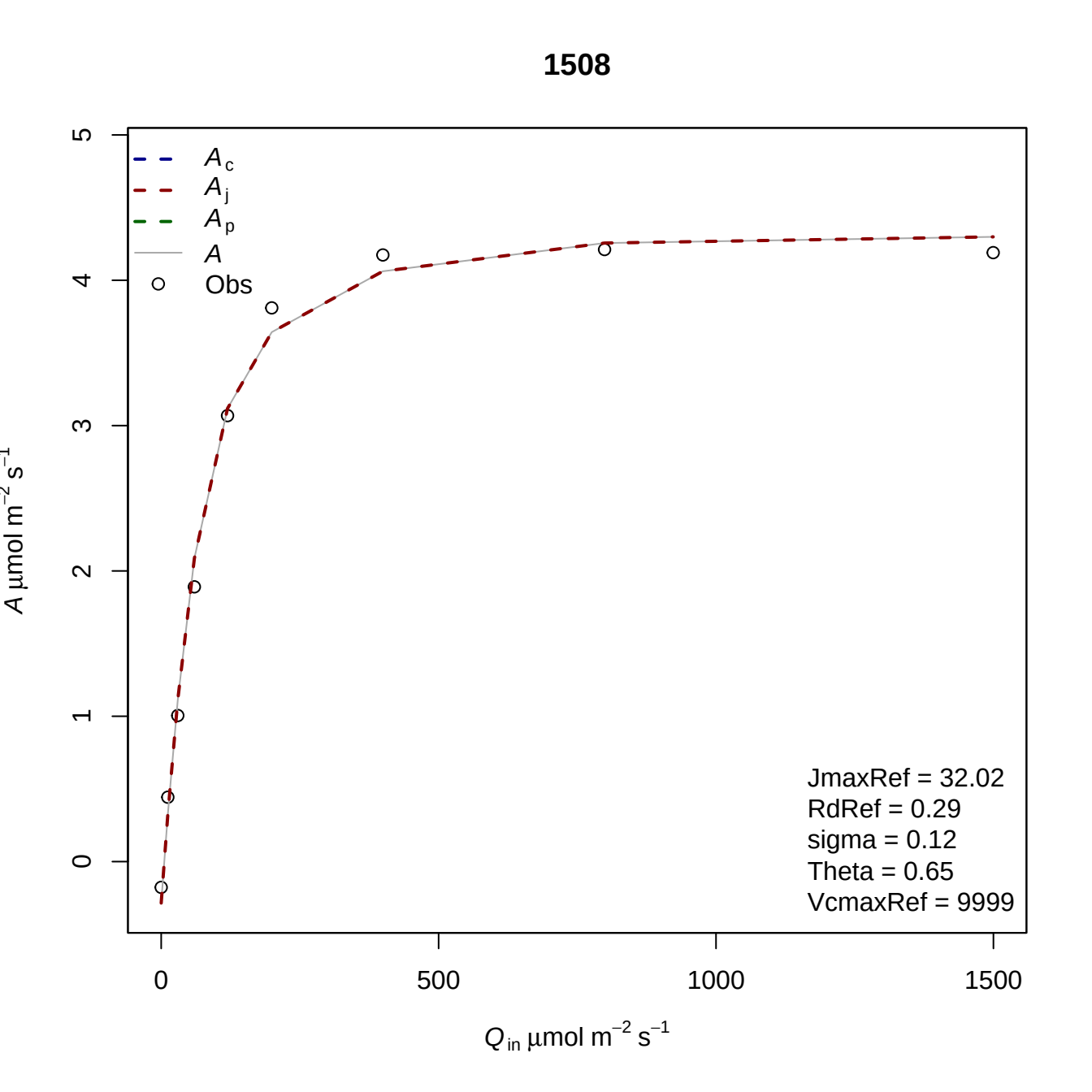

1503
A_c
A_j
A_p
A
Obs
JmaxRef = 15501.02
RdRef = 2.81
sigma = 0.11
Theta = −966.54
VcmaxRef = 69.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

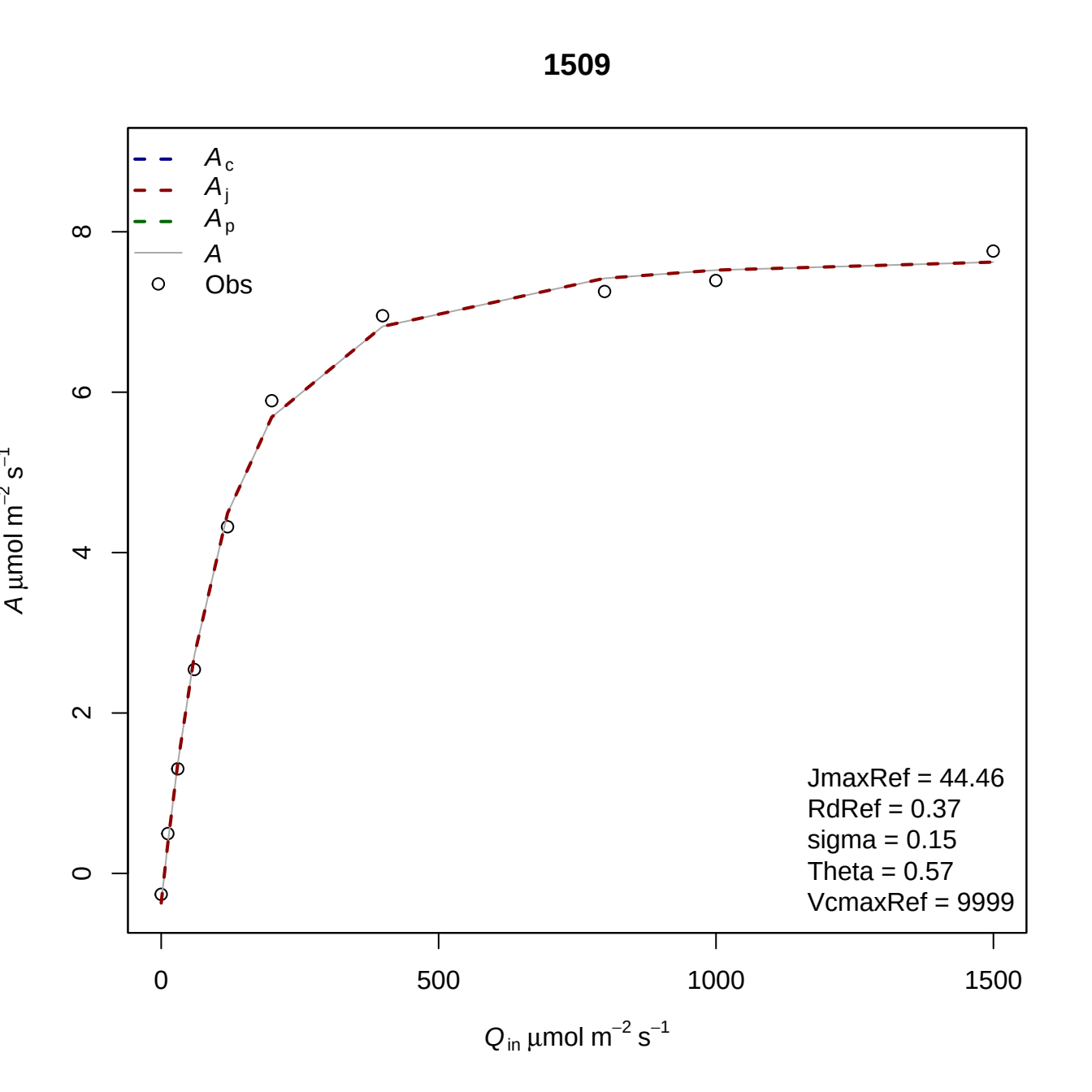

1504
A_c
A_j
A_p
A
Obs
JmaxRef = 26413594.87
RdRef = 3.69
sigma = 0.1
Theta = −1627515.13
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

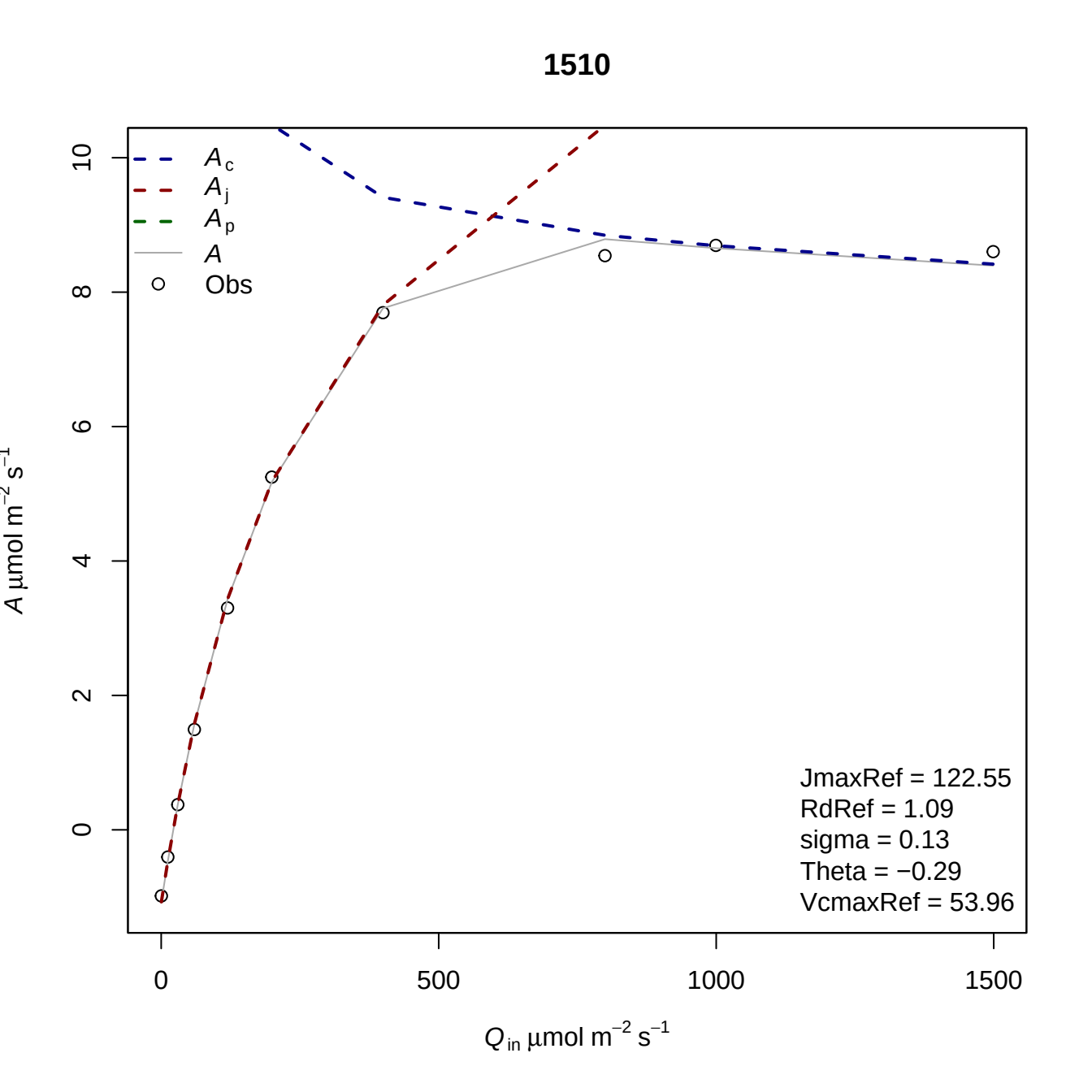

1505
A_c
A_j
A_p
A
Obs
JmaxRef = 288.57
RdRef = 2.56
sigma = 0.08
Theta = −13.94
VcmaxRef = 52.76
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

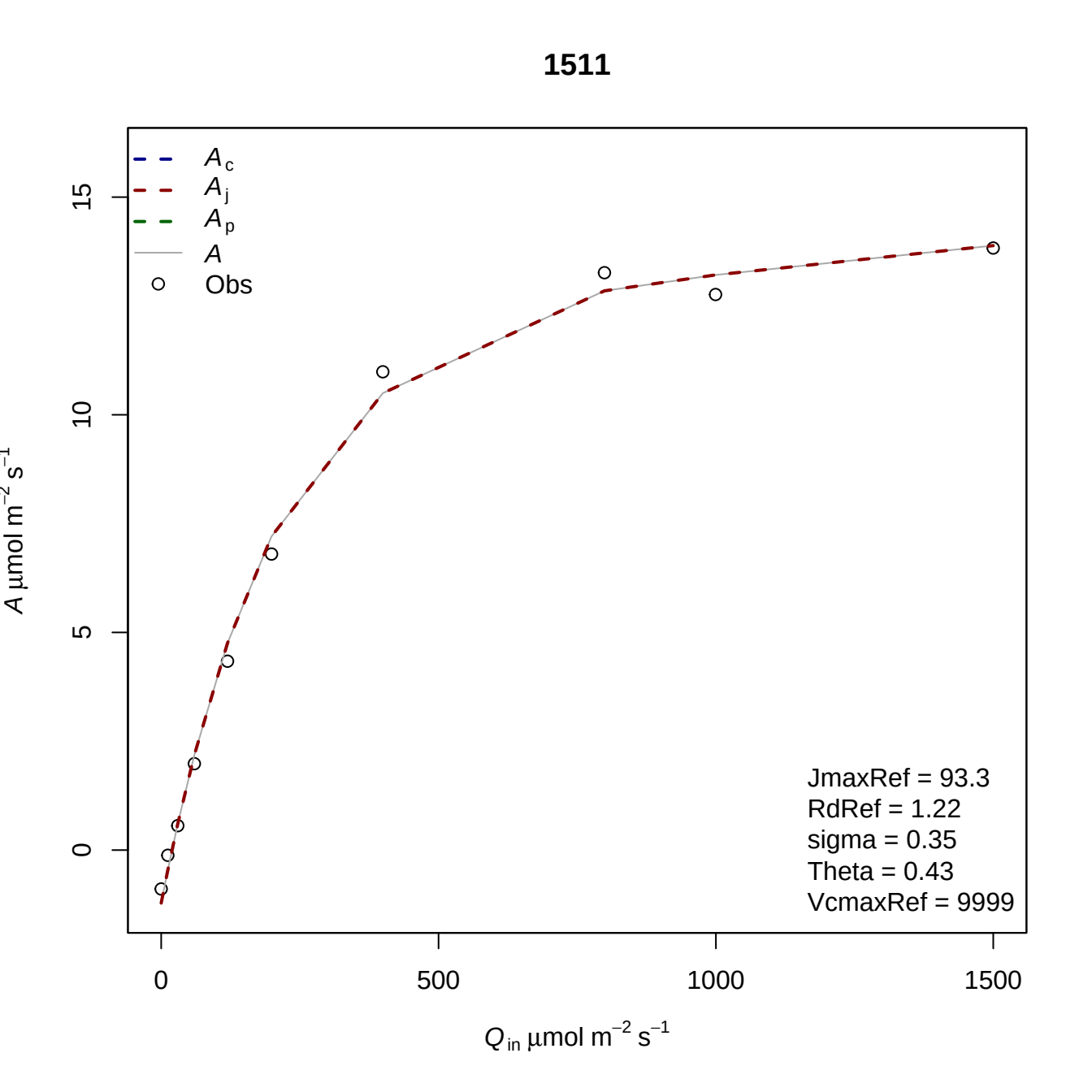

1506
A_c
A_j
A_p
A
Obs
JmaxRef = 99.04
RdRef = 0.8
sigma = 0.21
Theta = −0.15
VcmaxRef = 43.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

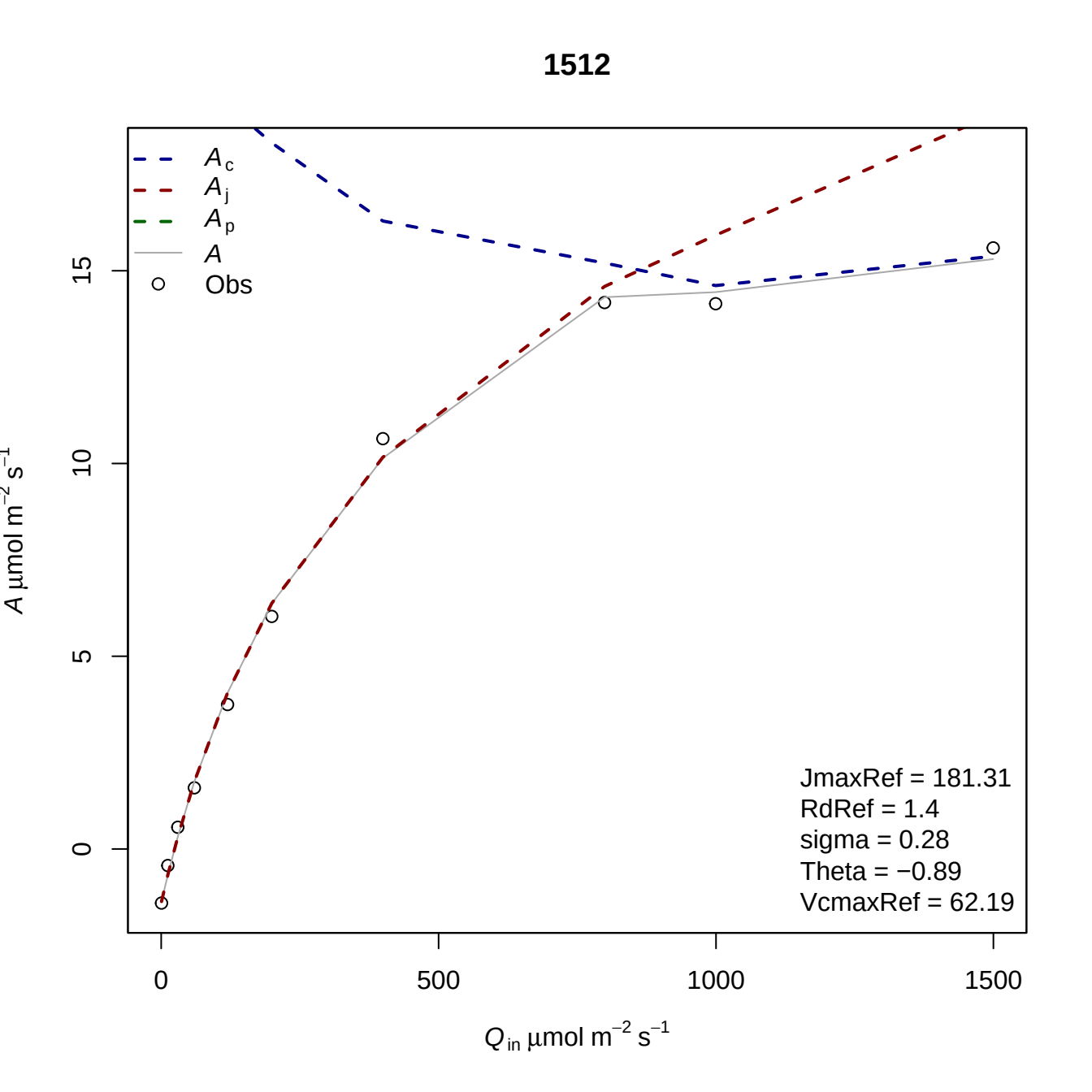

1507
A_c
A_j
A_p
A
Obs
JmaxRef = 66.67
RdRef = 0.85
sigma = 0.29
Theta = 0.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

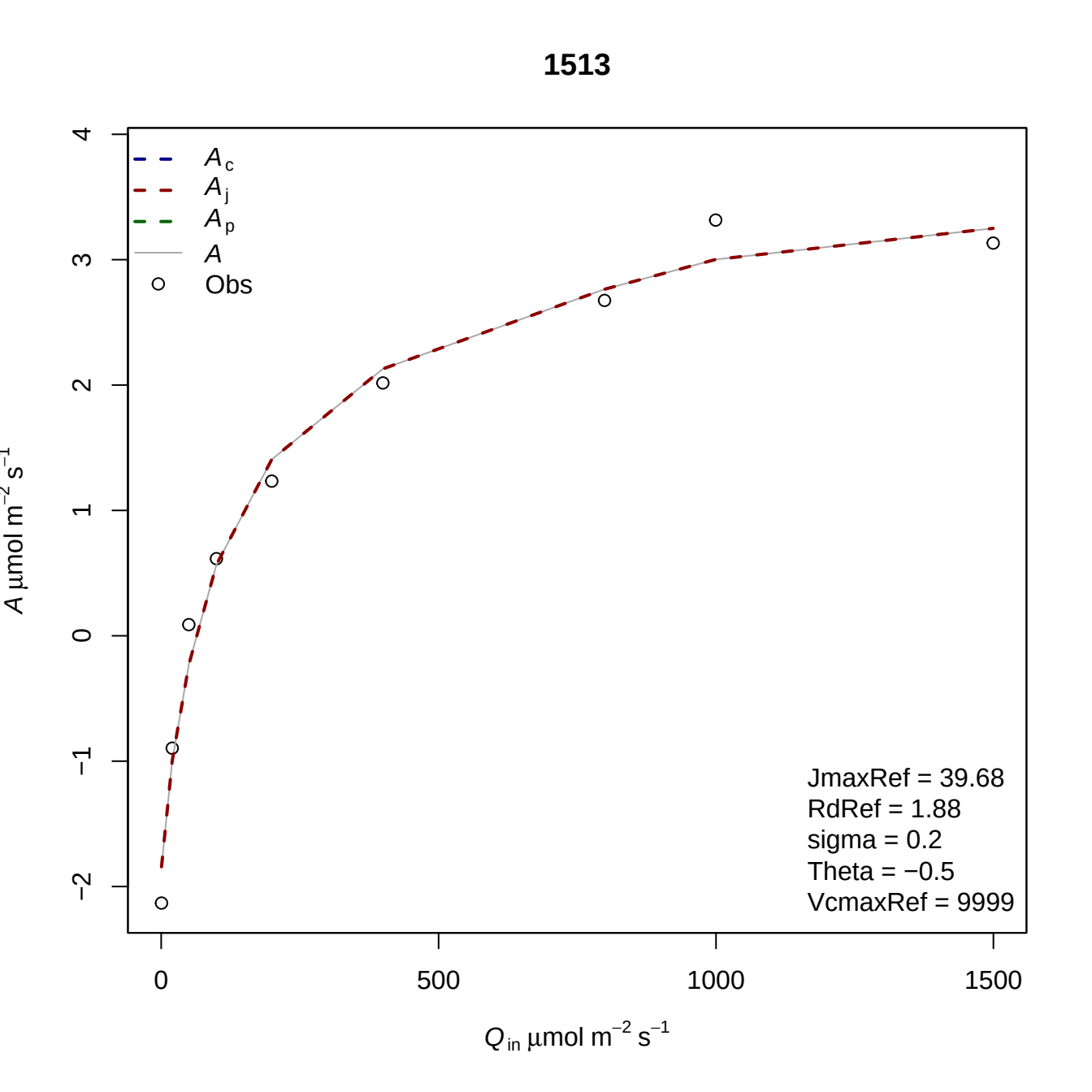

1508
A_c
A_j
A_p
A
Obs
JmaxRef = 32.02
RdRef = 0.29
sigma = 0.12
Theta = 0.65
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

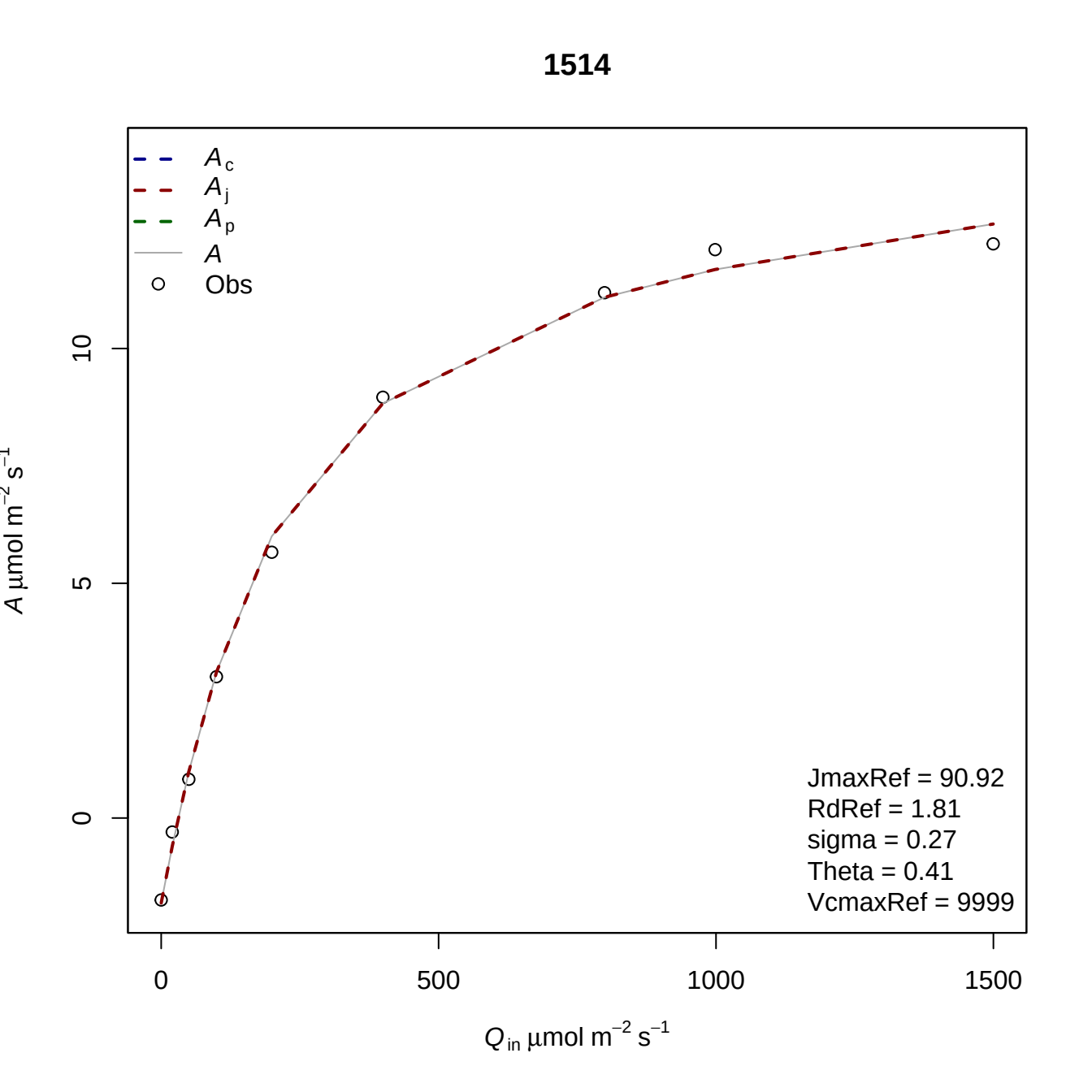

1509
A_c
A_j
A_p
A
Obs
JmaxRef = 44.46
RdRef = 0.37
sigma = 0.15
Theta = 0.57
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

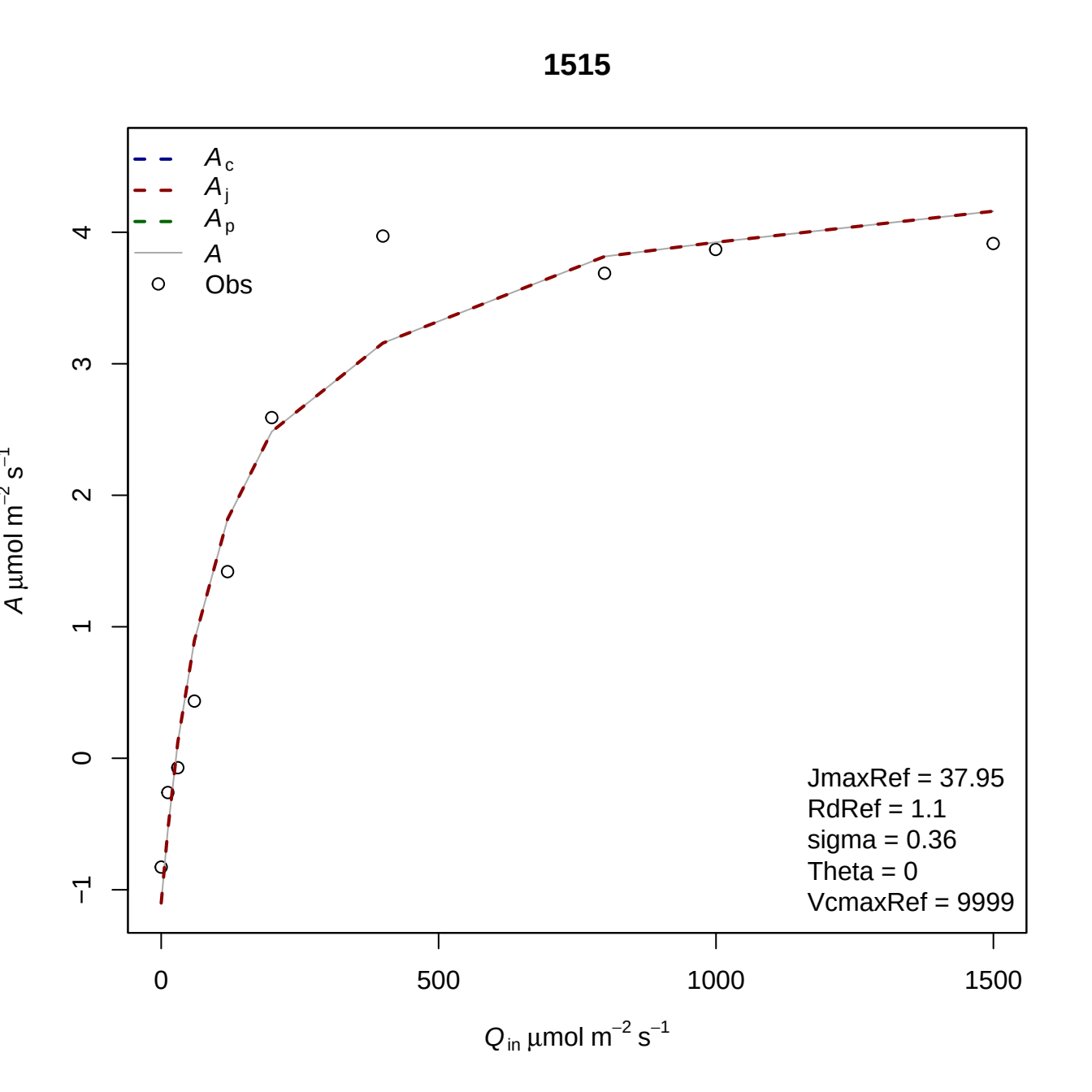

1510
A_c
A_j
A_p
A
Obs
JmaxRef = 122.55
RdRef = 1.09
sigma = 0.13
Theta = −0.29
VcmaxRef = 53.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

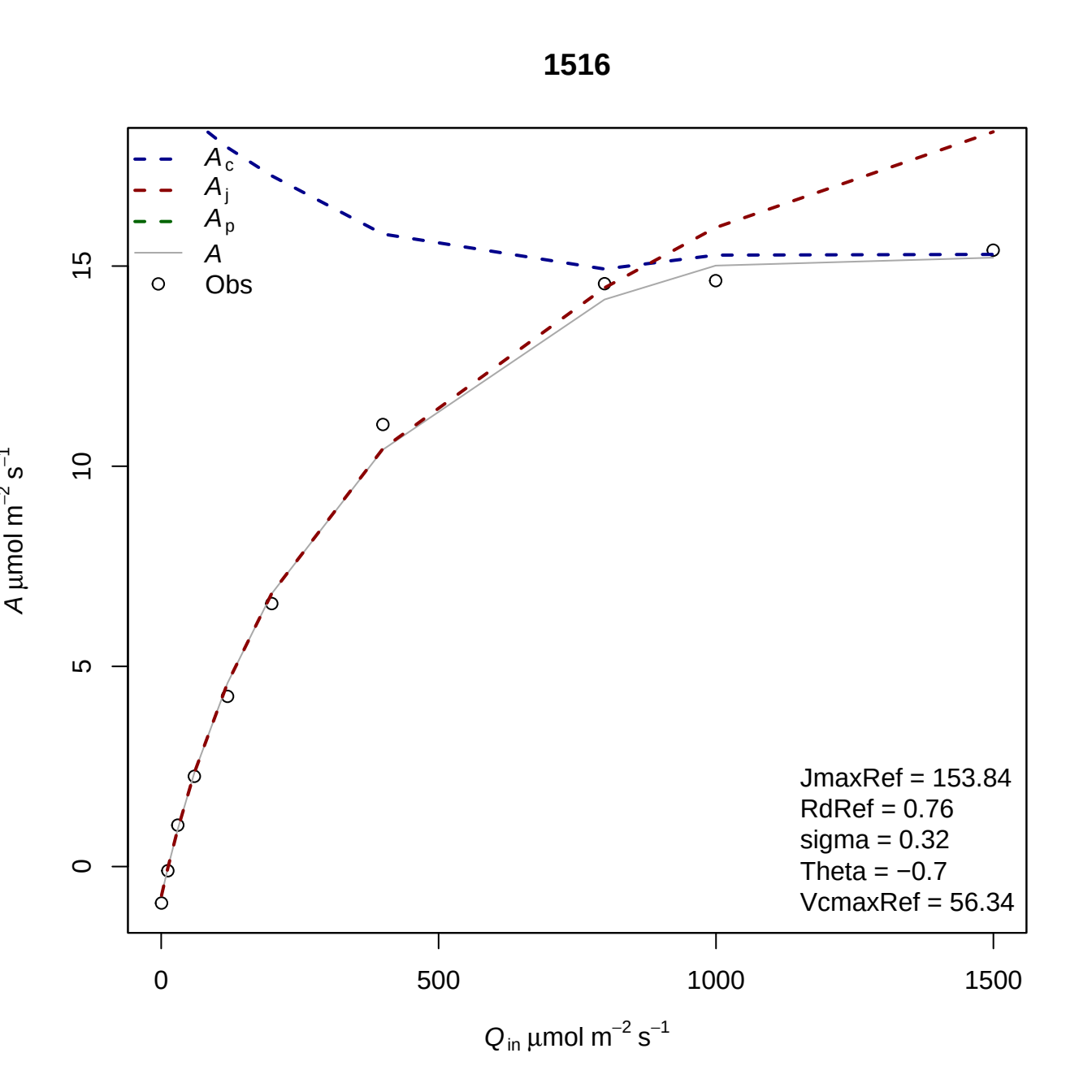

1511
A_c
A_j
A_p
A
Obs
JmaxRef = 93.3
RdRef = 1.22
sigma = 0.35
Theta = 0.43
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

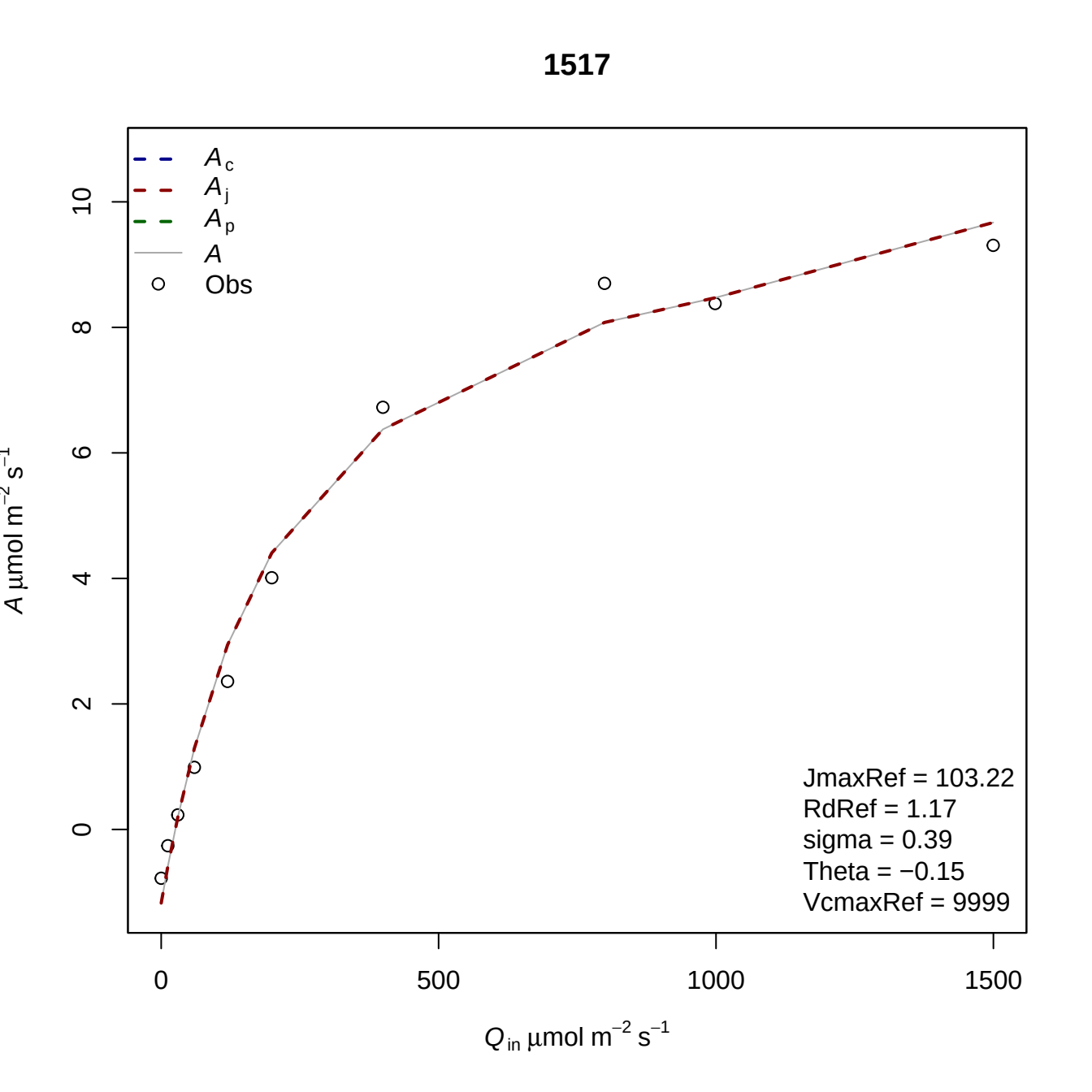

1512
A_c
A_j
A_p
A
Obs
JmaxRef = 181.31
RdRef = 1.4
sigma = 0.28
Theta = −0.89
VcmaxRef = 62.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

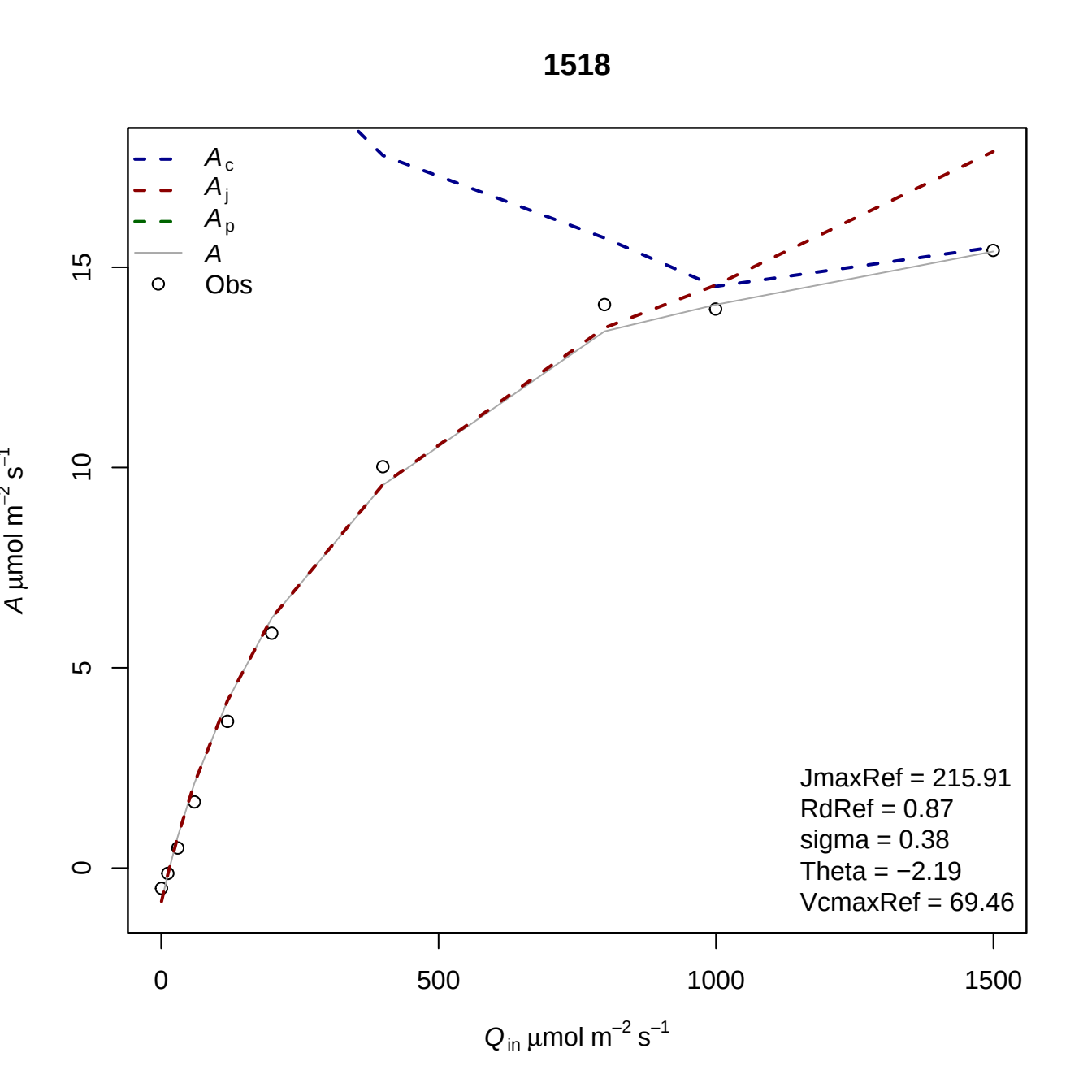

1513
A_c
A_j
A_p
A
Obs
JmaxRef = 39.68
RdRef = 1.88
sigma = 0.2
Theta = −0.5
VcmaxRef = 9999
4
3
2
1
0
−1
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

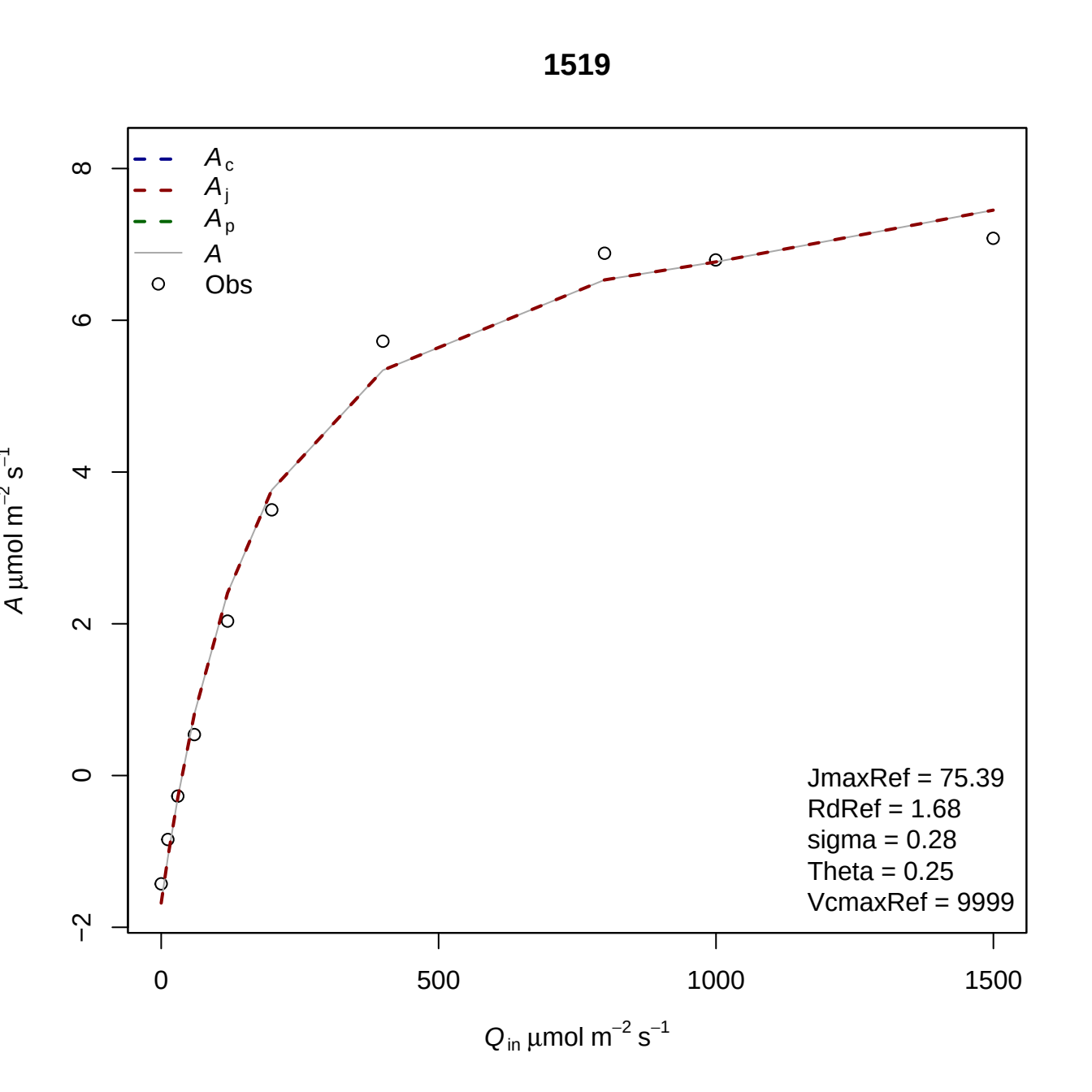

1514
A_c
A_j
A_p
A
Obs
JmaxRef = 90.92
RdRef = 1.81
sigma = 0.27
Theta = 0.41
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

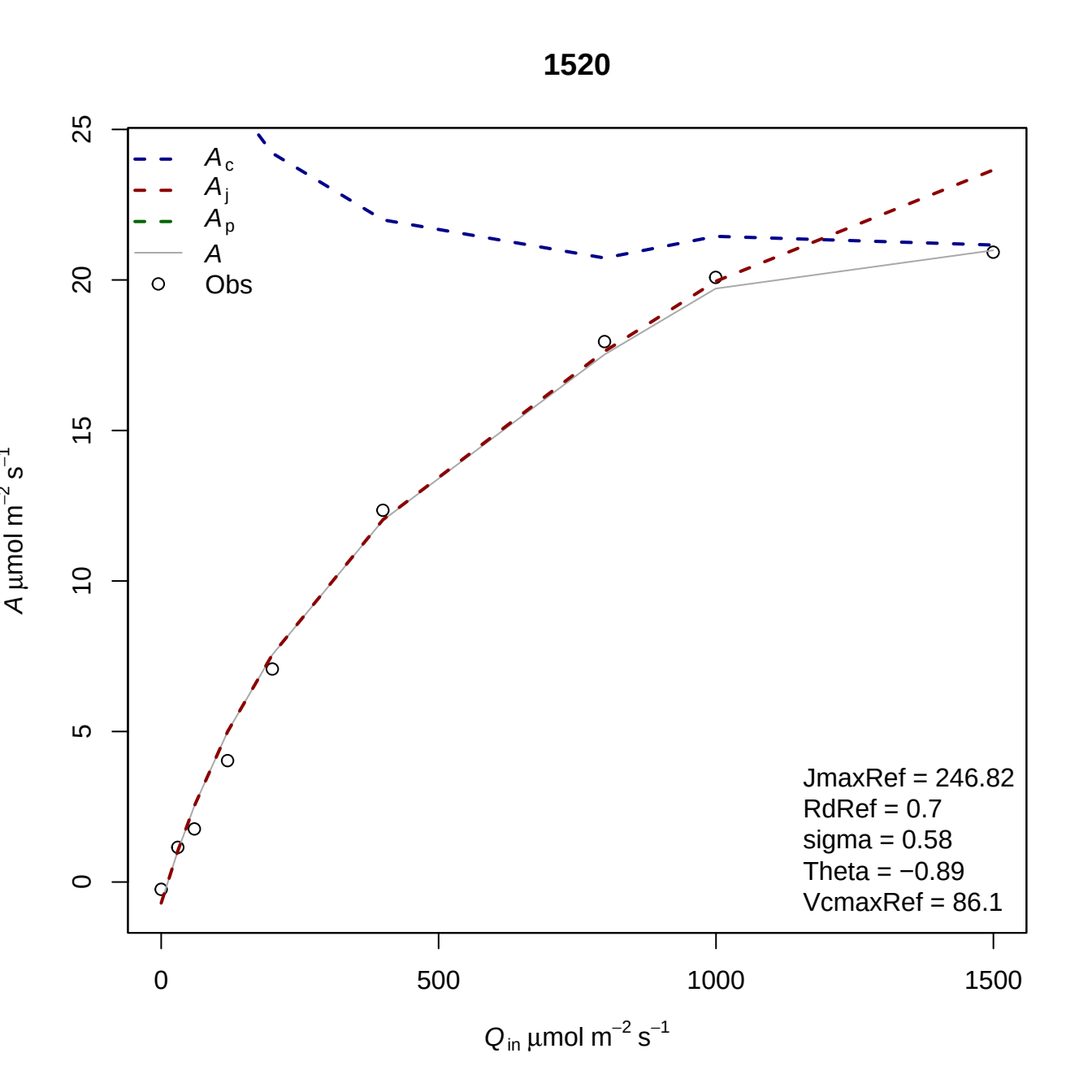

1515
A_c
A_j
A_p
A
Obs
JmaxRef = 37.95
RdRef = 1.1
sigma = 0.36
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

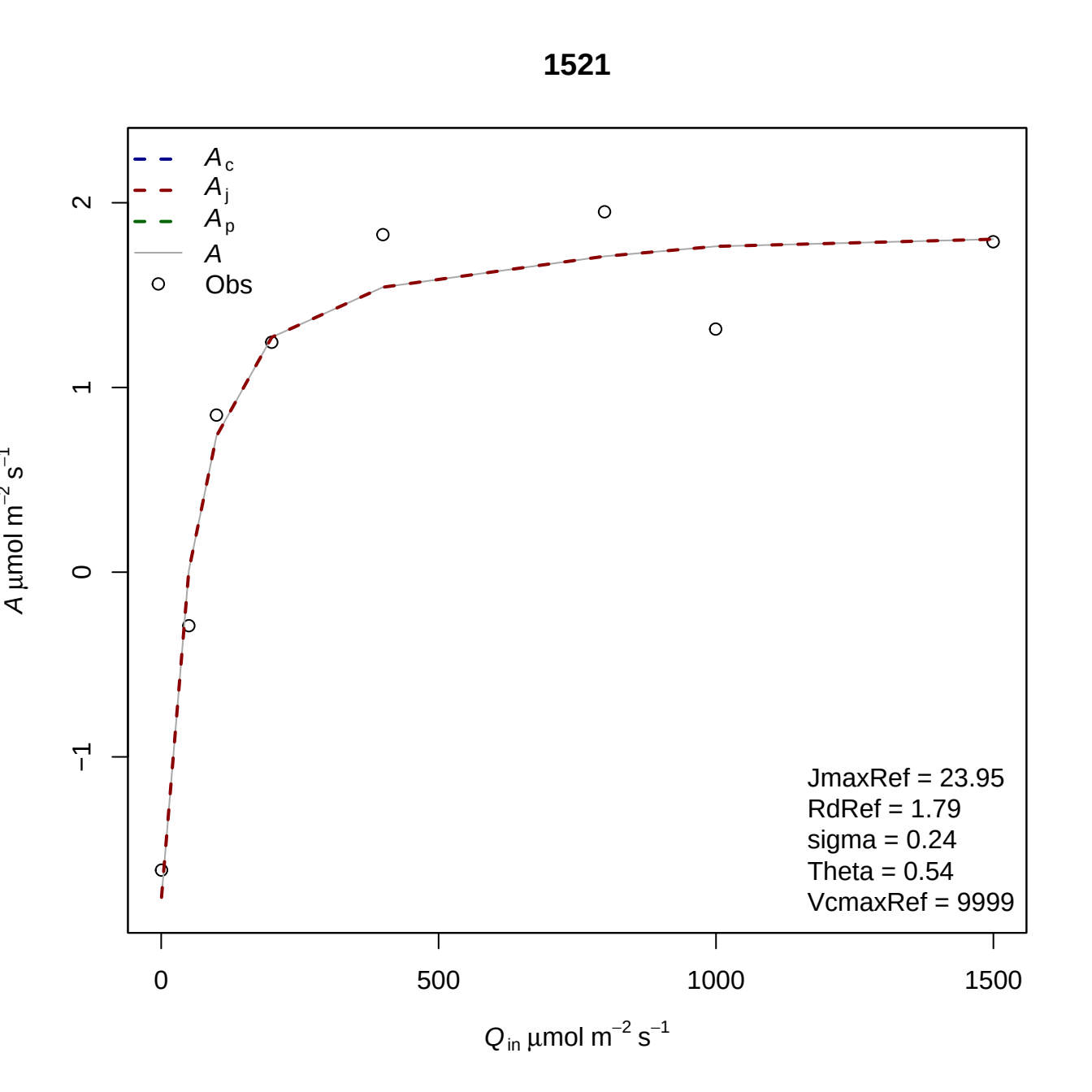

1516
A_c
A_j
A_p
A
Obs
JmaxRef = 153.84
RdRef = 0.76
sigma = 0.32
Theta = −0.7
VcmaxRef = 56.34
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

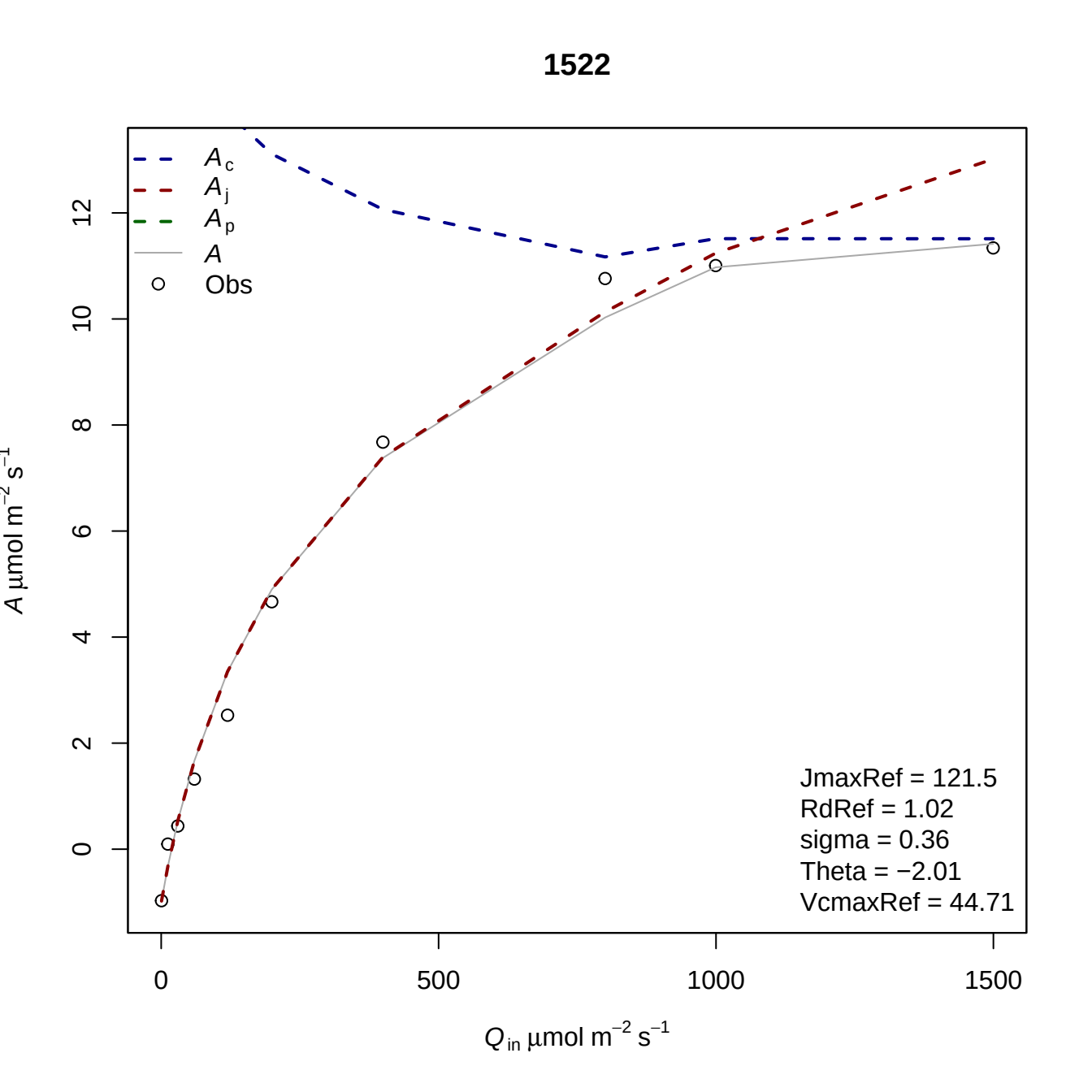

1517
A_c
A_j
A_p
A
Obs
JmaxRef = 103.22
RdRef = 1.17
sigma = 0.39
Theta = −0.15
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

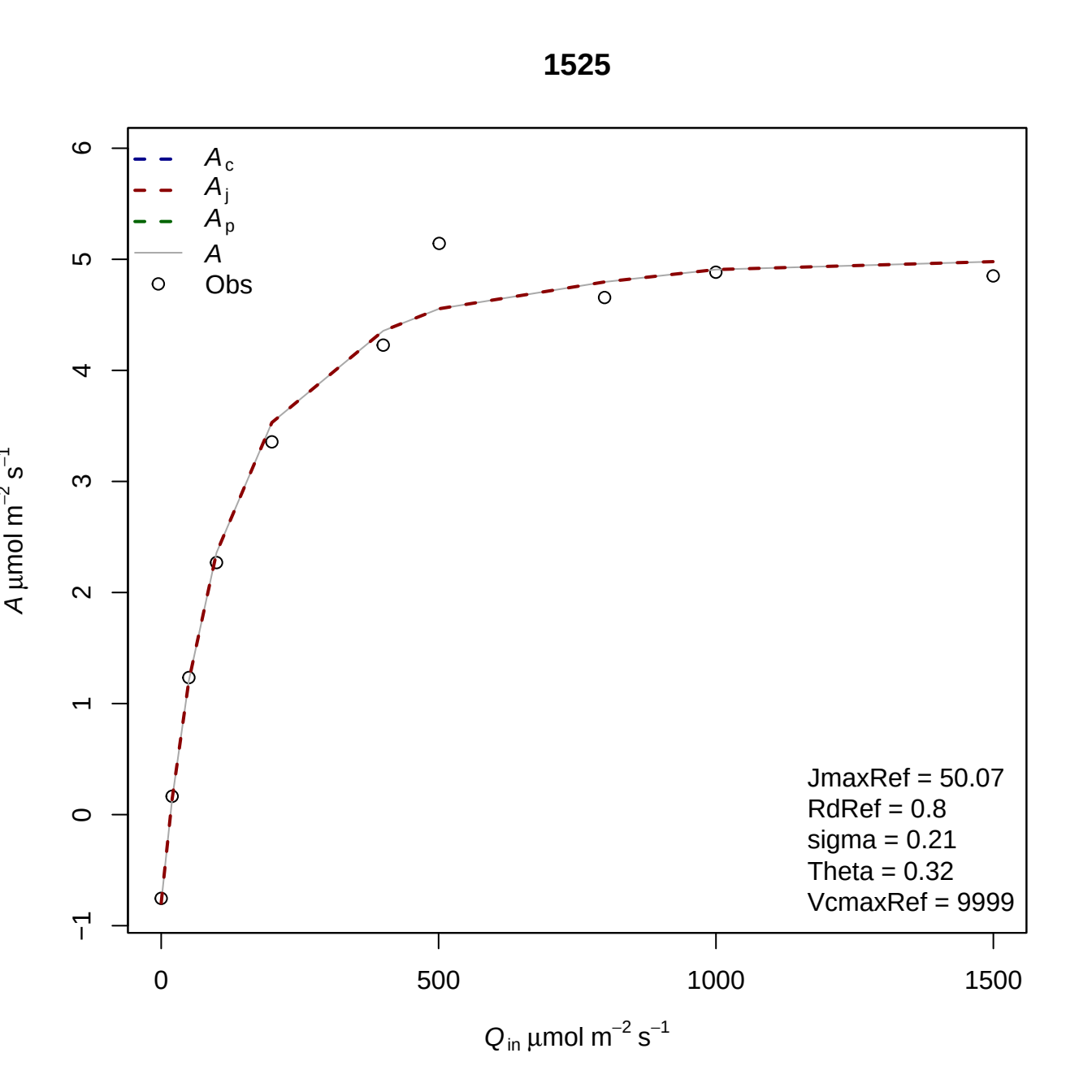

1518
A_c
A_j
A_p
A
Obs
JmaxRef = 215.91
RdRef = 0.87
sigma = 0.38
Theta = −2.19
VcmaxRef = 69.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

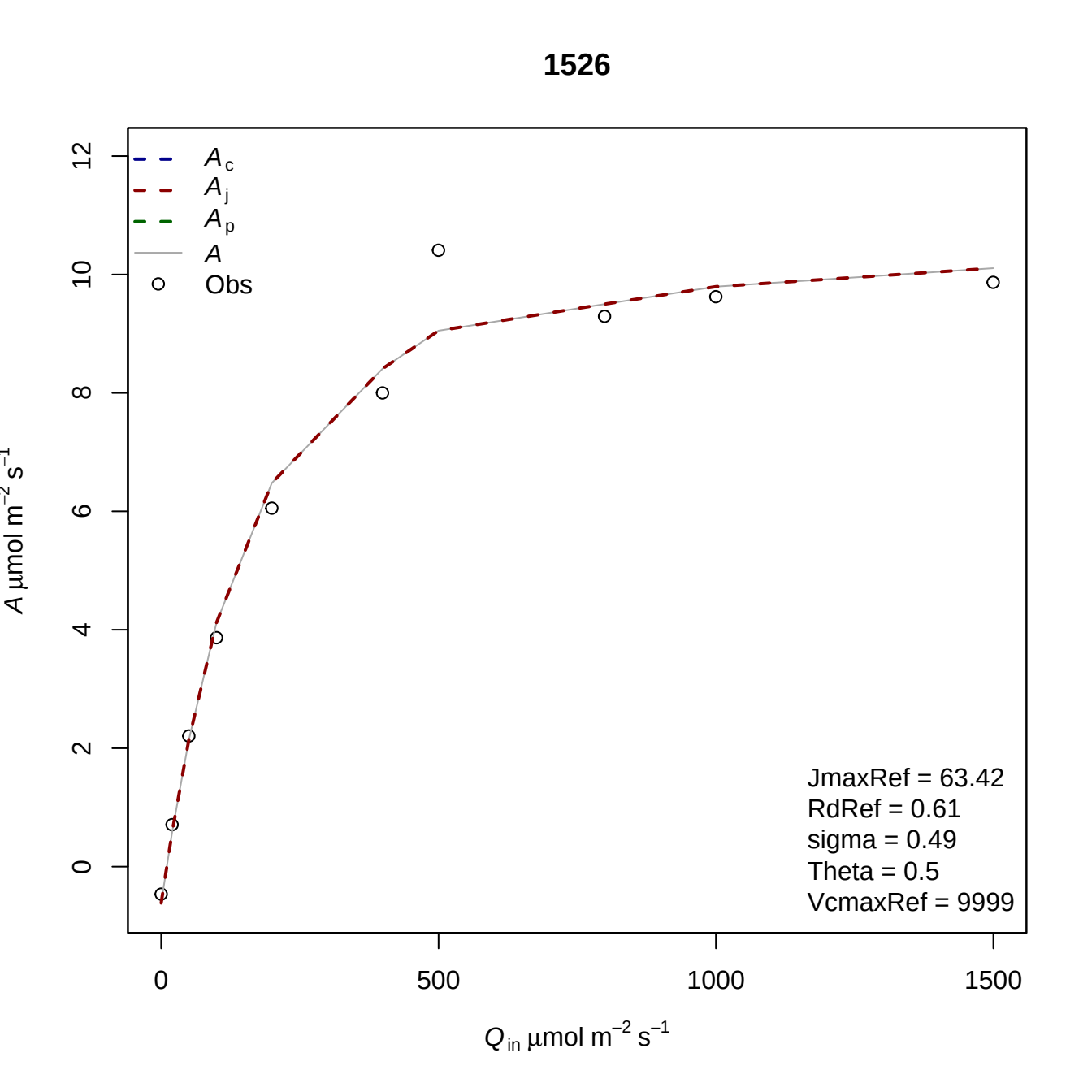

1519
A_c
A_j
A_p
A
Obs
JmaxRef = 75.39
RdRef = 1.68
sigma = 0.28
Theta = 0.25
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

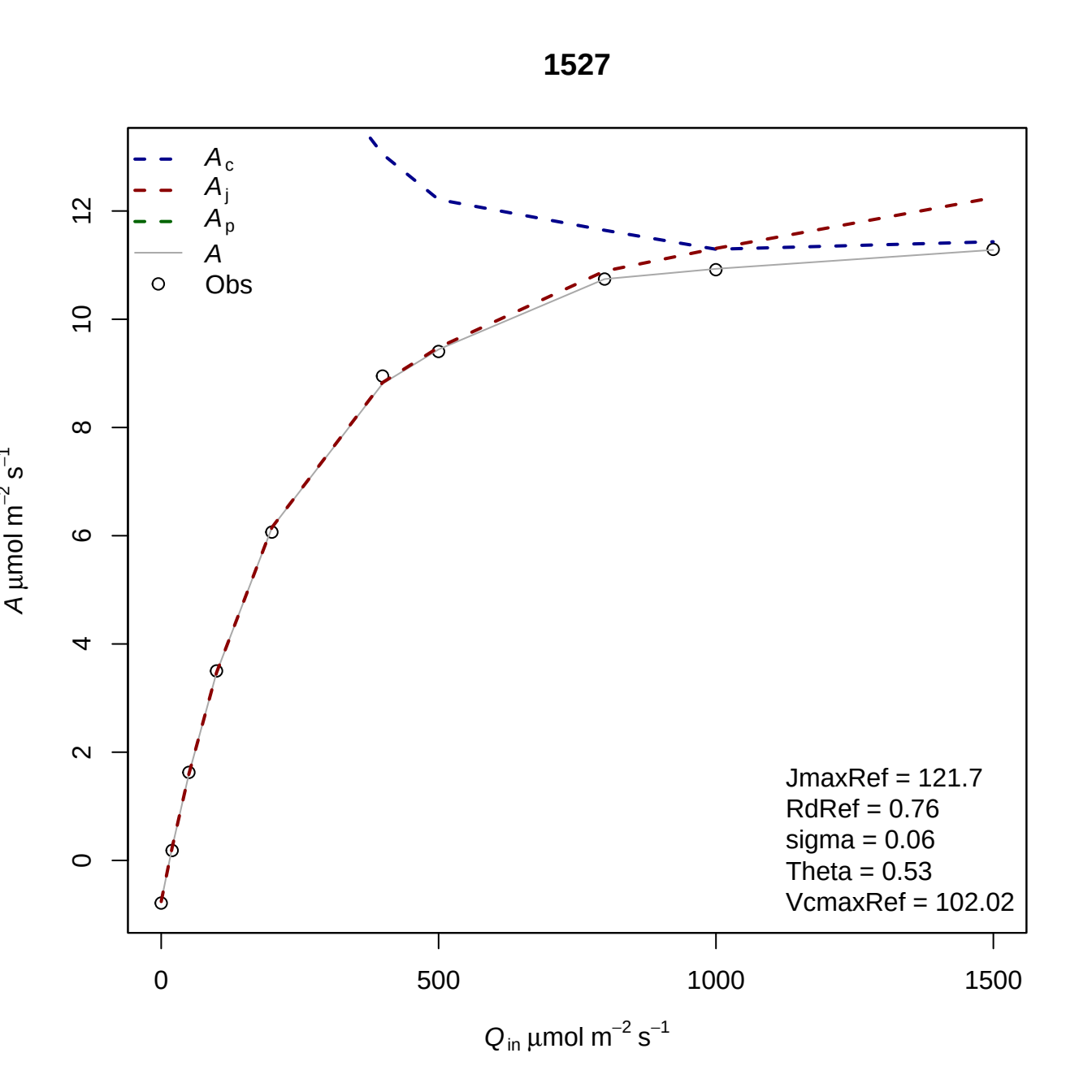

1520
A_c
A_j
A_p
A
Obs
JmaxRef = 246.82
RdRef = 0.7
sigma = 0.58
Theta = −0.89
VcmaxRef = 86.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

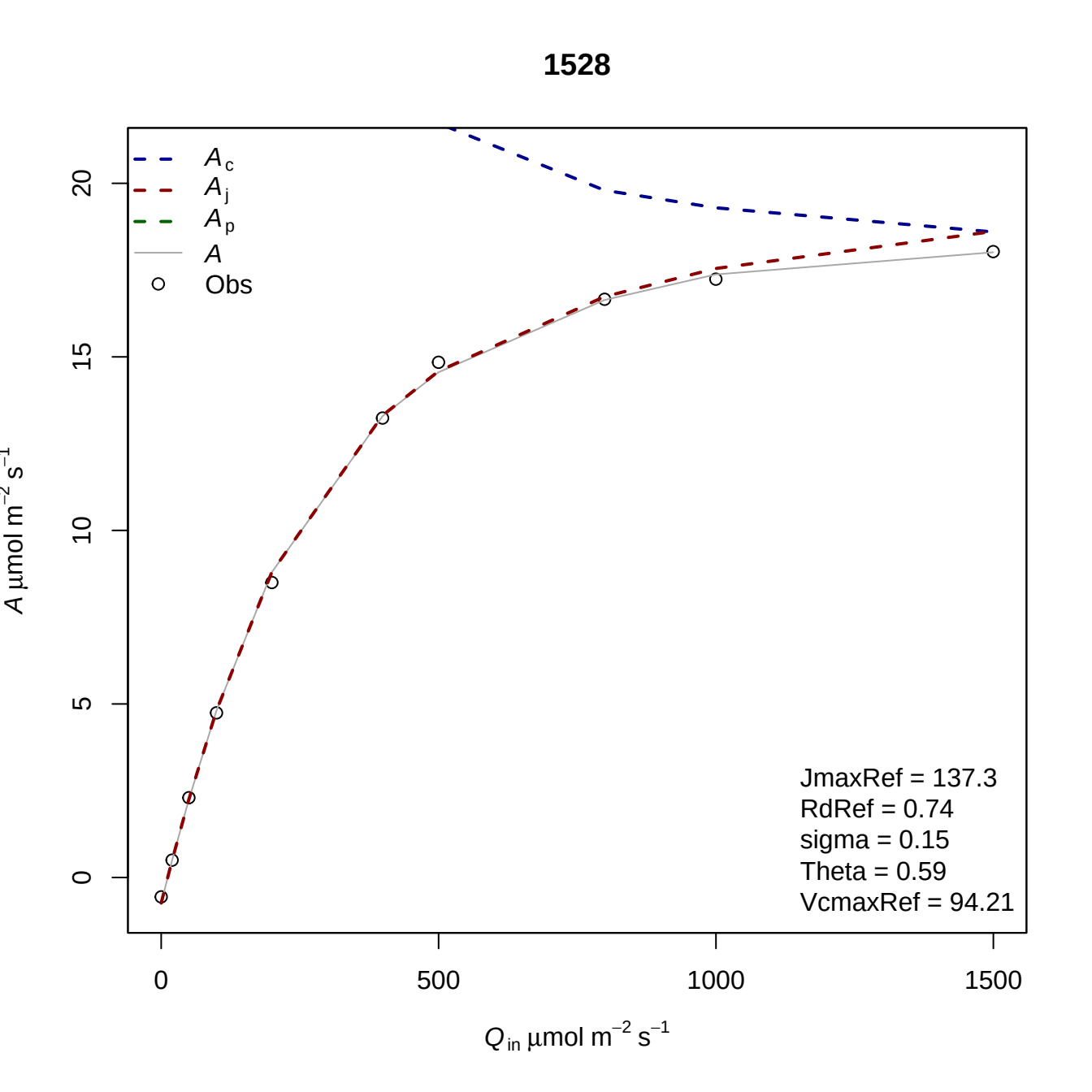

1521
A_c
A_j
A_p
A
Obs
JmaxRef = 23.95
RdRef = 1.79
sigma = 0.24
Theta = 0.54
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
0
500
1000
1500

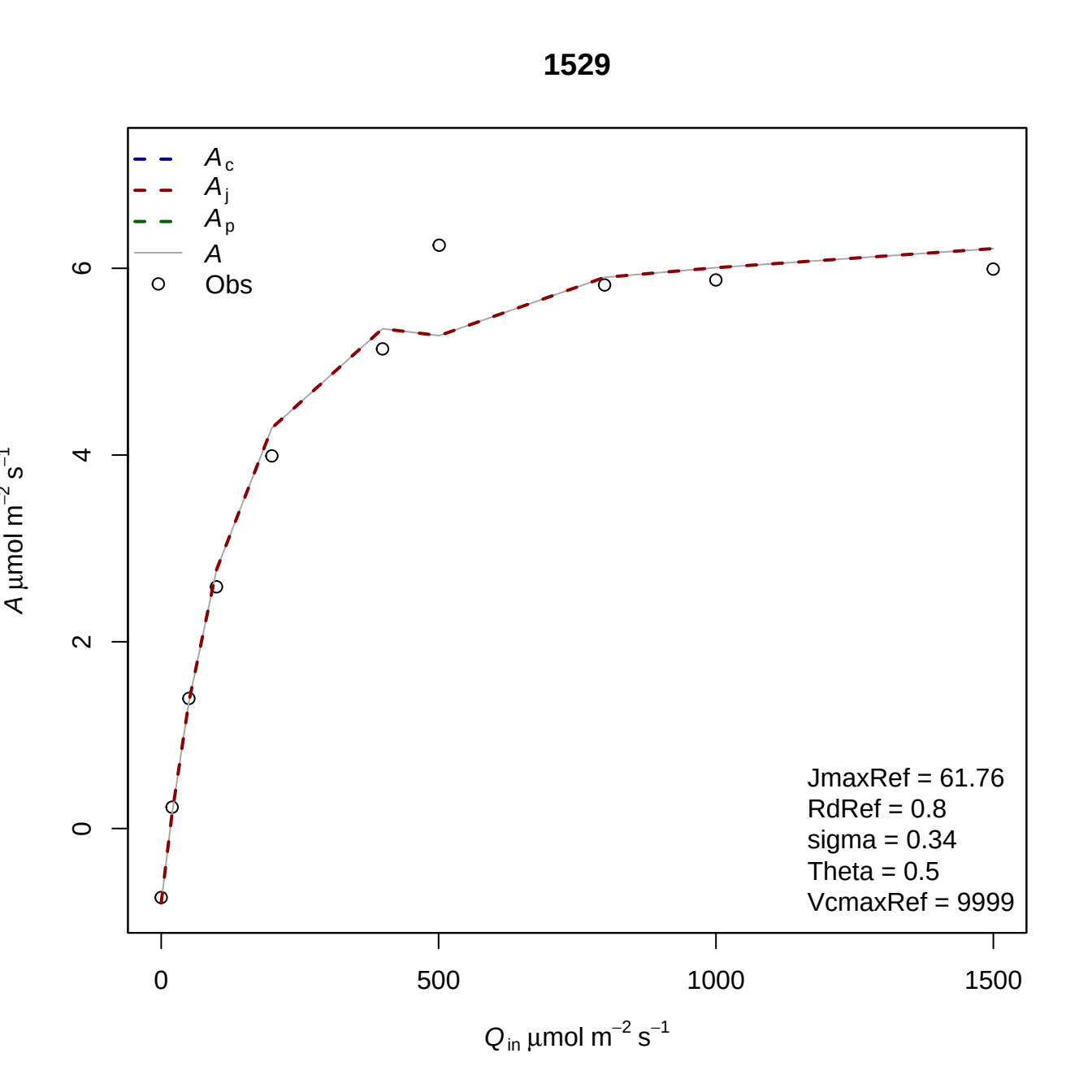

1522
A_c
A_j
A_p
A
Obs
JmaxRef = 121.5
RdRef = 1.02
sigma = 0.36
Theta = −2.01
VcmaxRef = 44.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

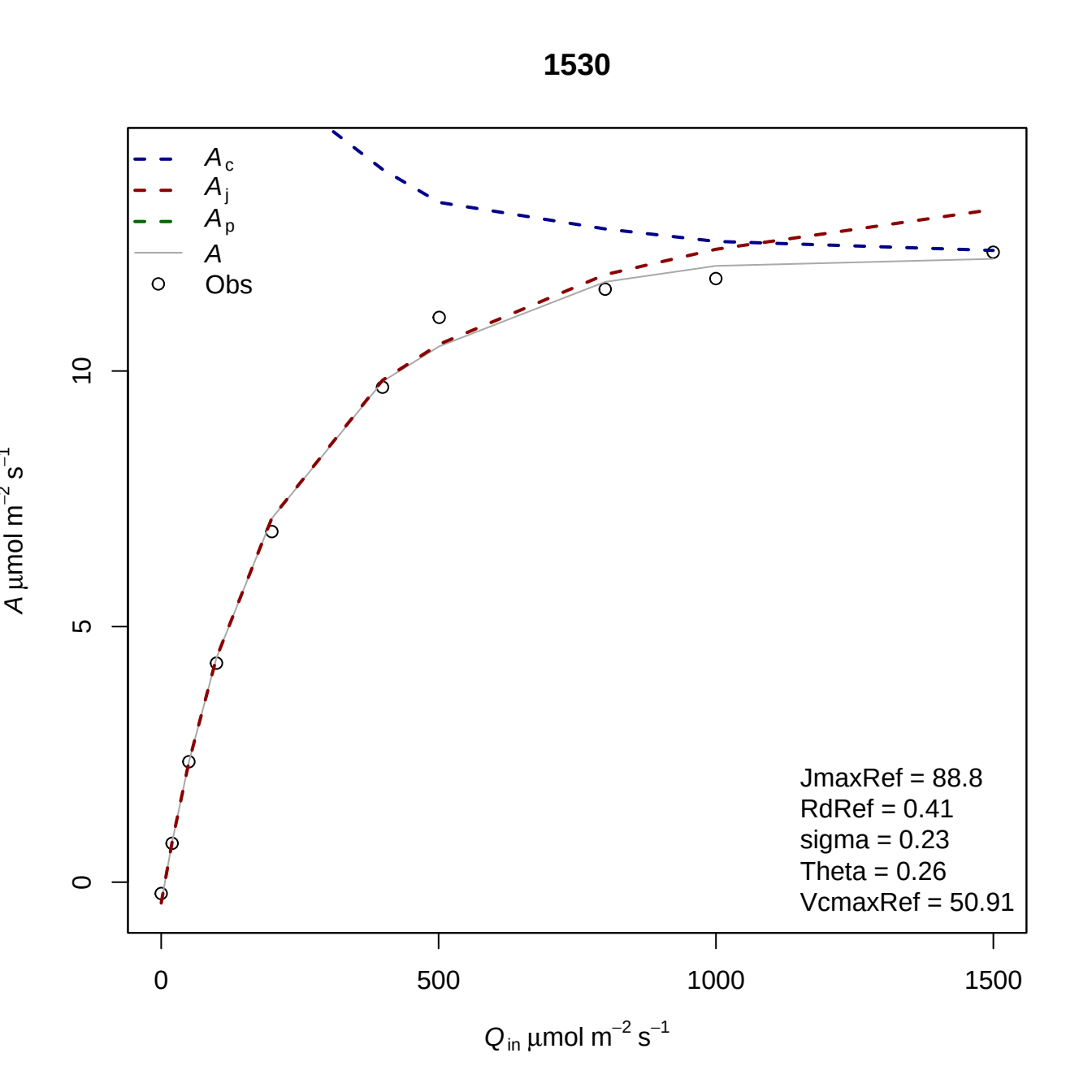

1525
A_c
A_j
A_p
A
Obs
JmaxRef = 50.07
RdRef = 0.8
sigma = 0.21
Theta = 0.32
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

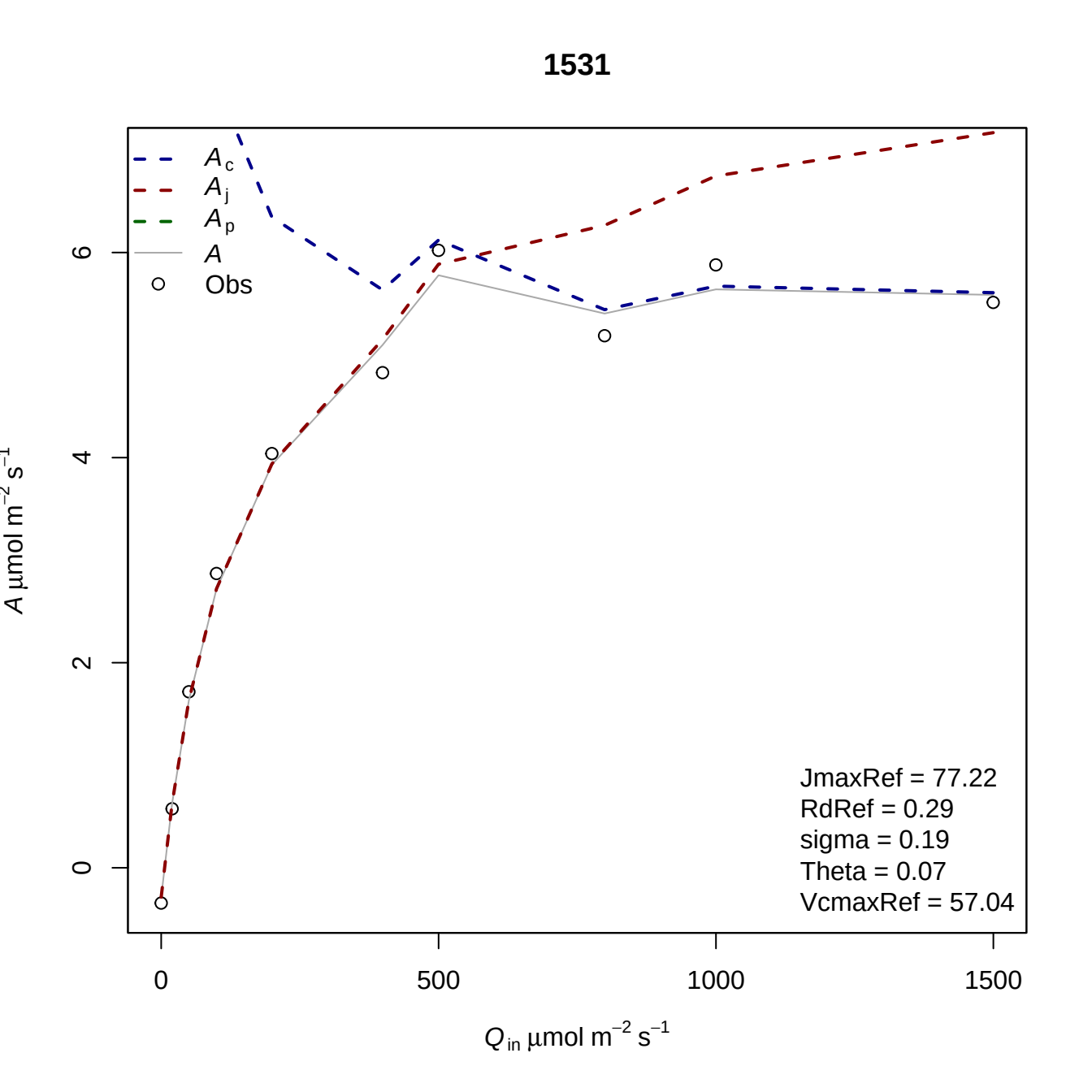

1526
A_c
A_j
A_p
A
Obs
JmaxRef = 63.42
RdRef = 0.61
sigma = 0.49
Theta = 0.5
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

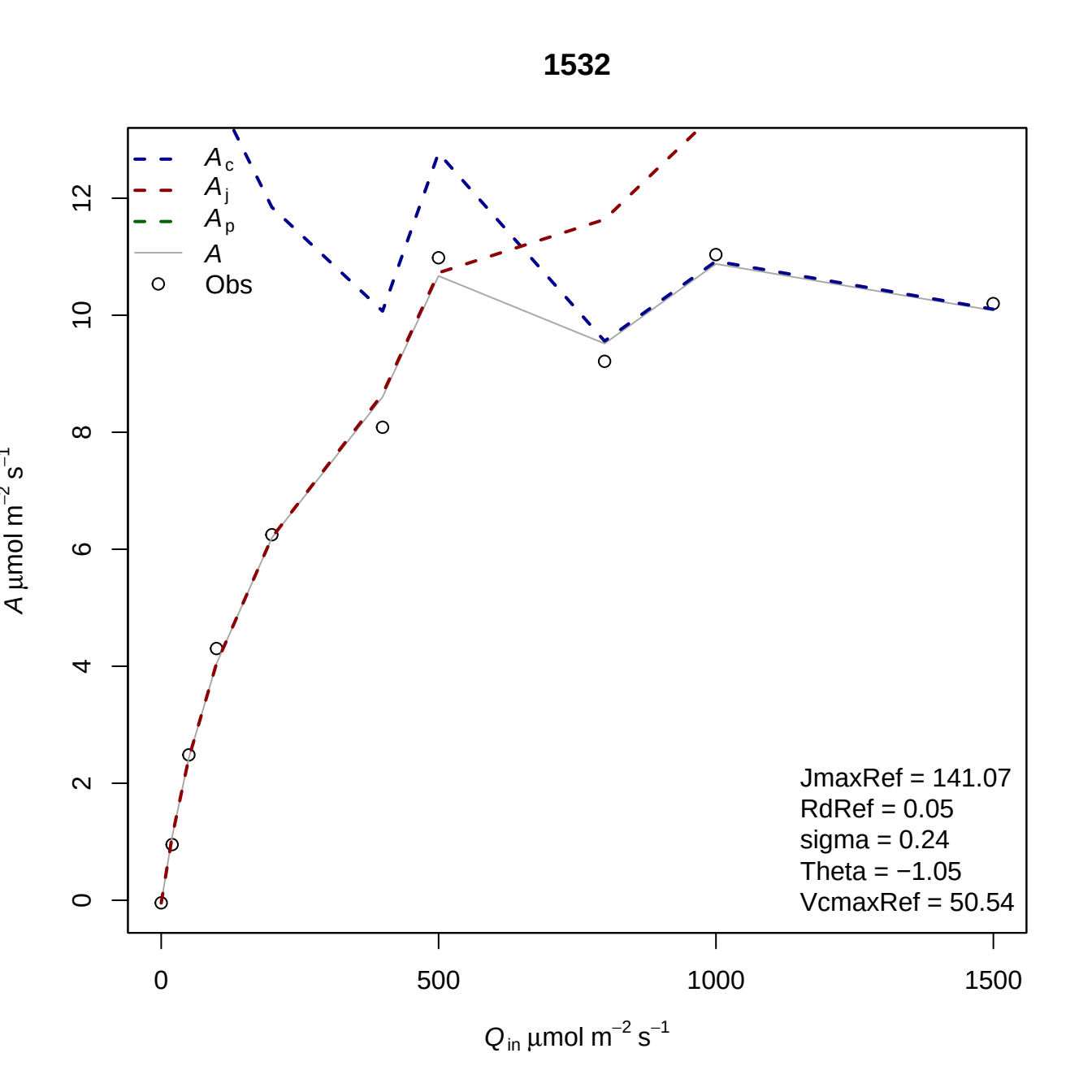

1527
A_c
A_j
A_p
A
Obs
JmaxRef = 121.7
RdRef = 0.76
sigma = 0.06
Theta = 0.53
VcmaxRef = 102.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

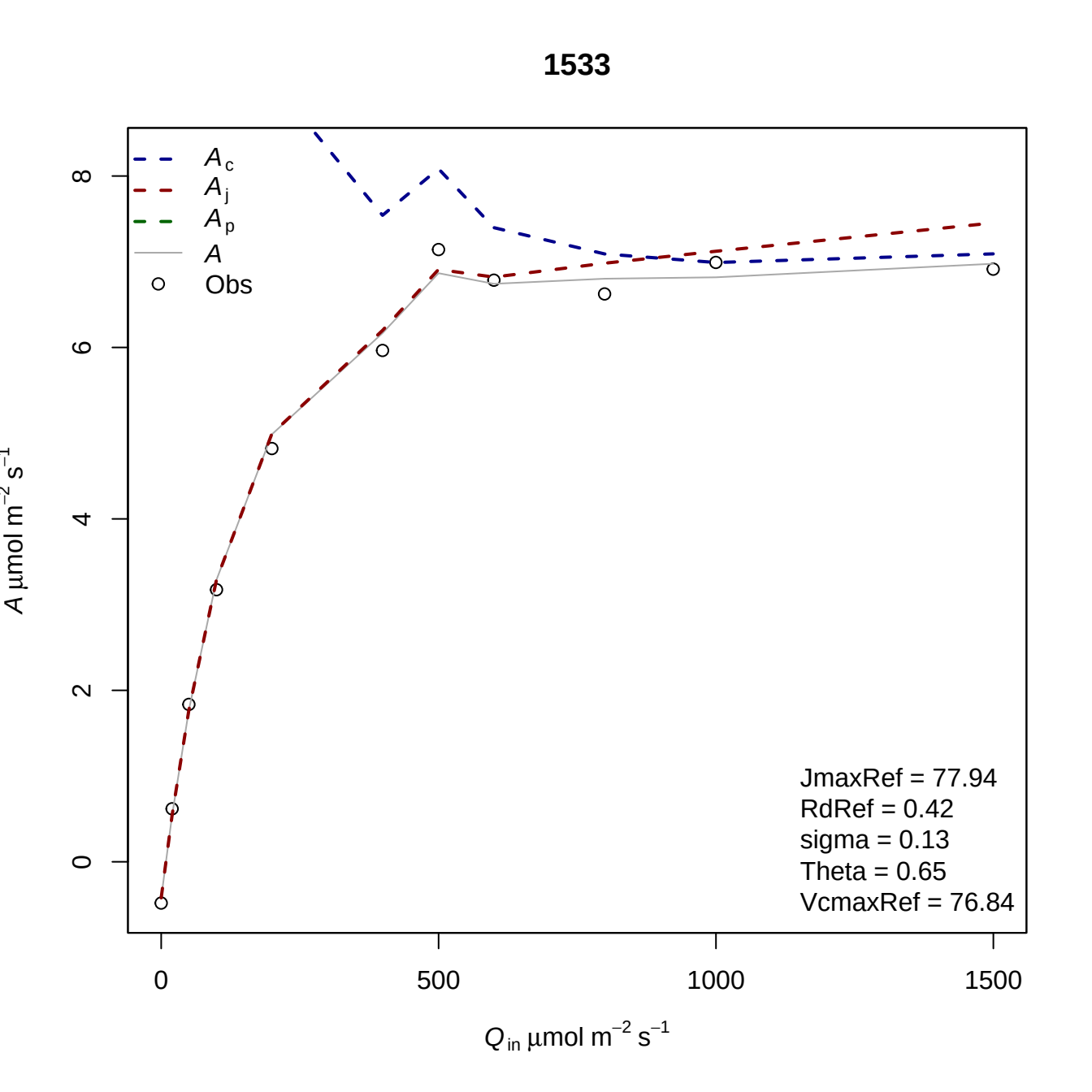

1528
A_c
A_j
A_p
A
Obs
JmaxRef = 137.3
RdRef = 0.74
sigma = 0.15
Theta = 0.59
VcmaxRef = 94.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

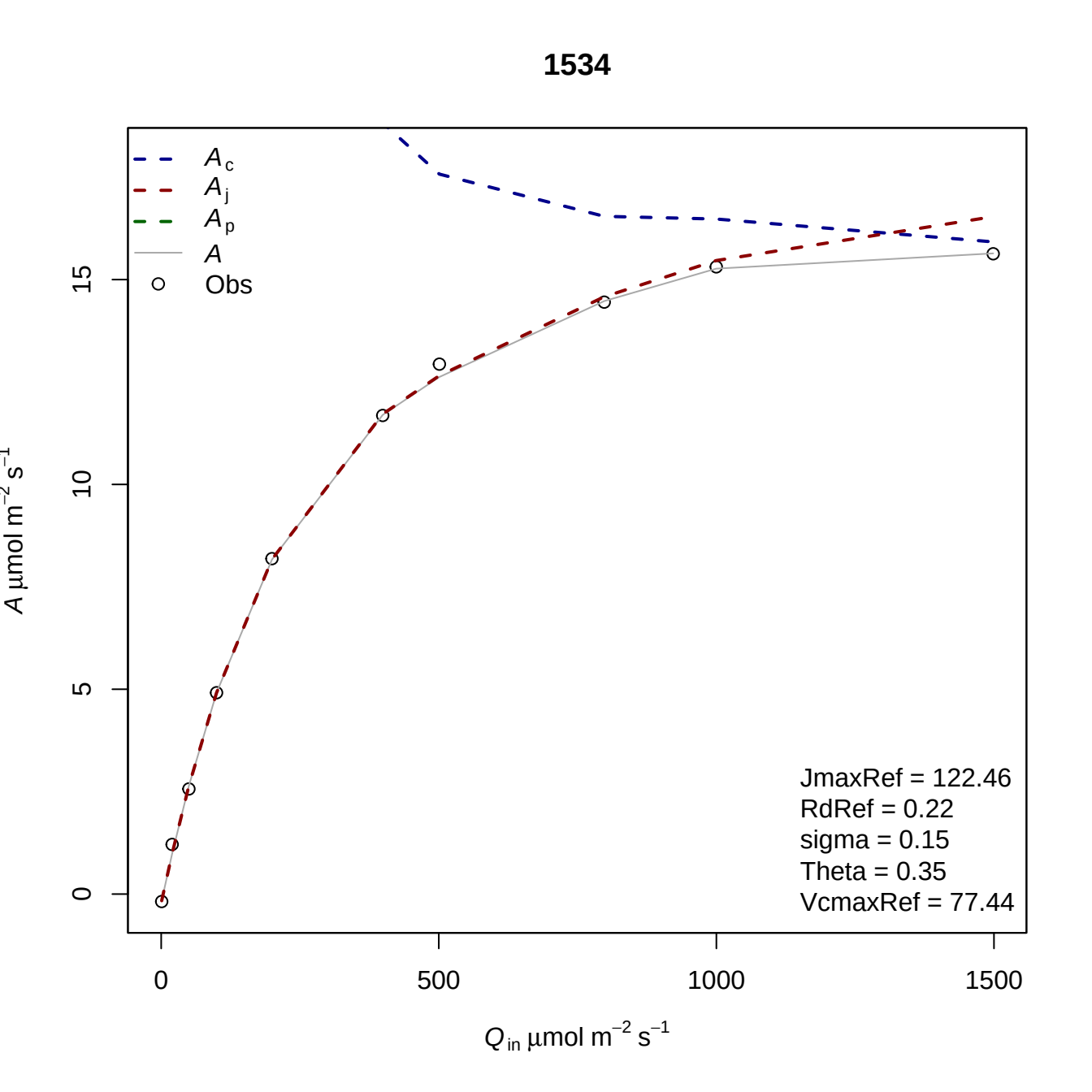

1529
A_c
A_j
A_p
A
Obs
JmaxRef = 61.76
RdRef = 0.8
sigma = 0.34
Theta = 0.5
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

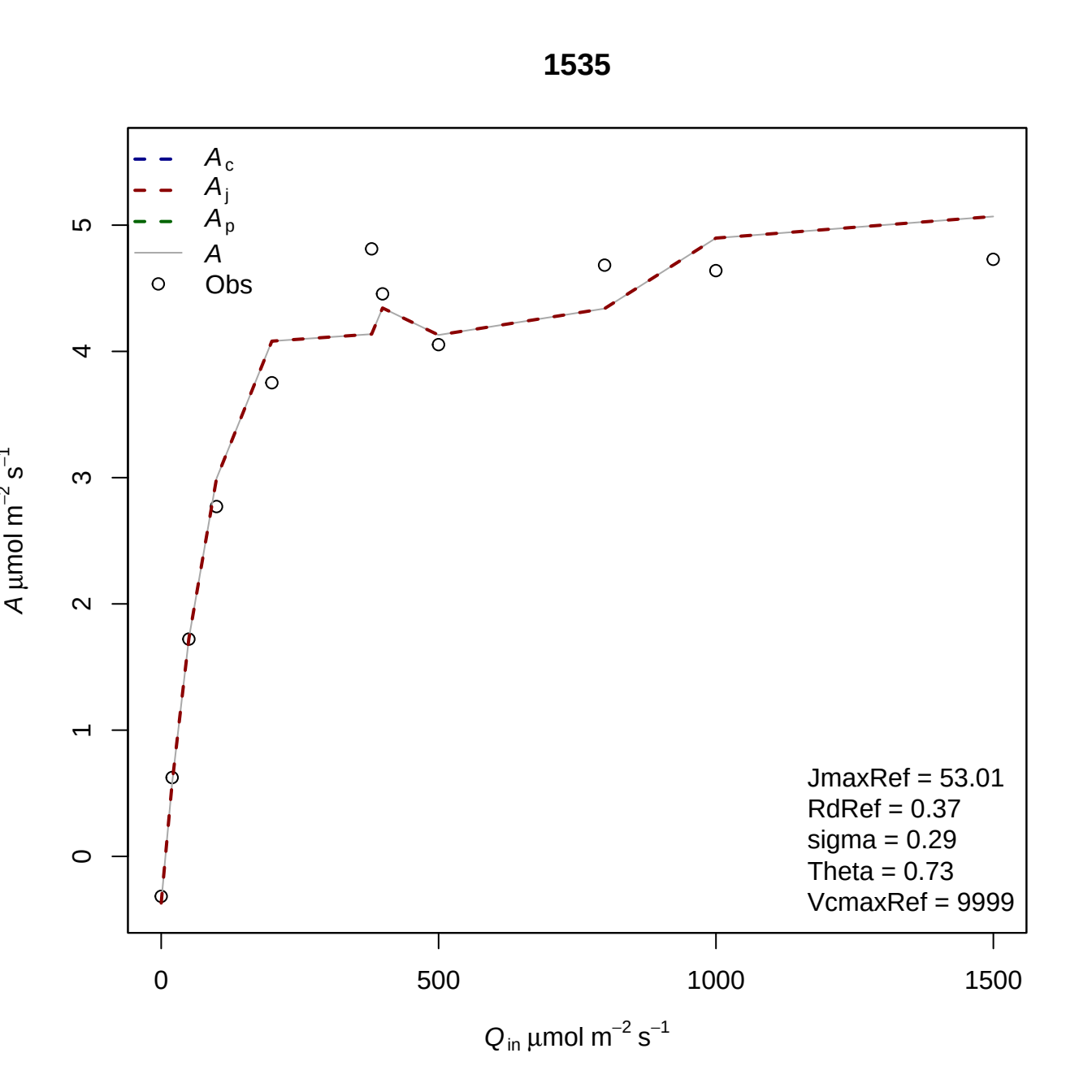

1530
A_c
A_j
A_p
A
Obs
JmaxRef = 88.8
RdRef = 0.41
sigma = 0.23
Theta = 0.26
VcmaxRef = 50.91
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

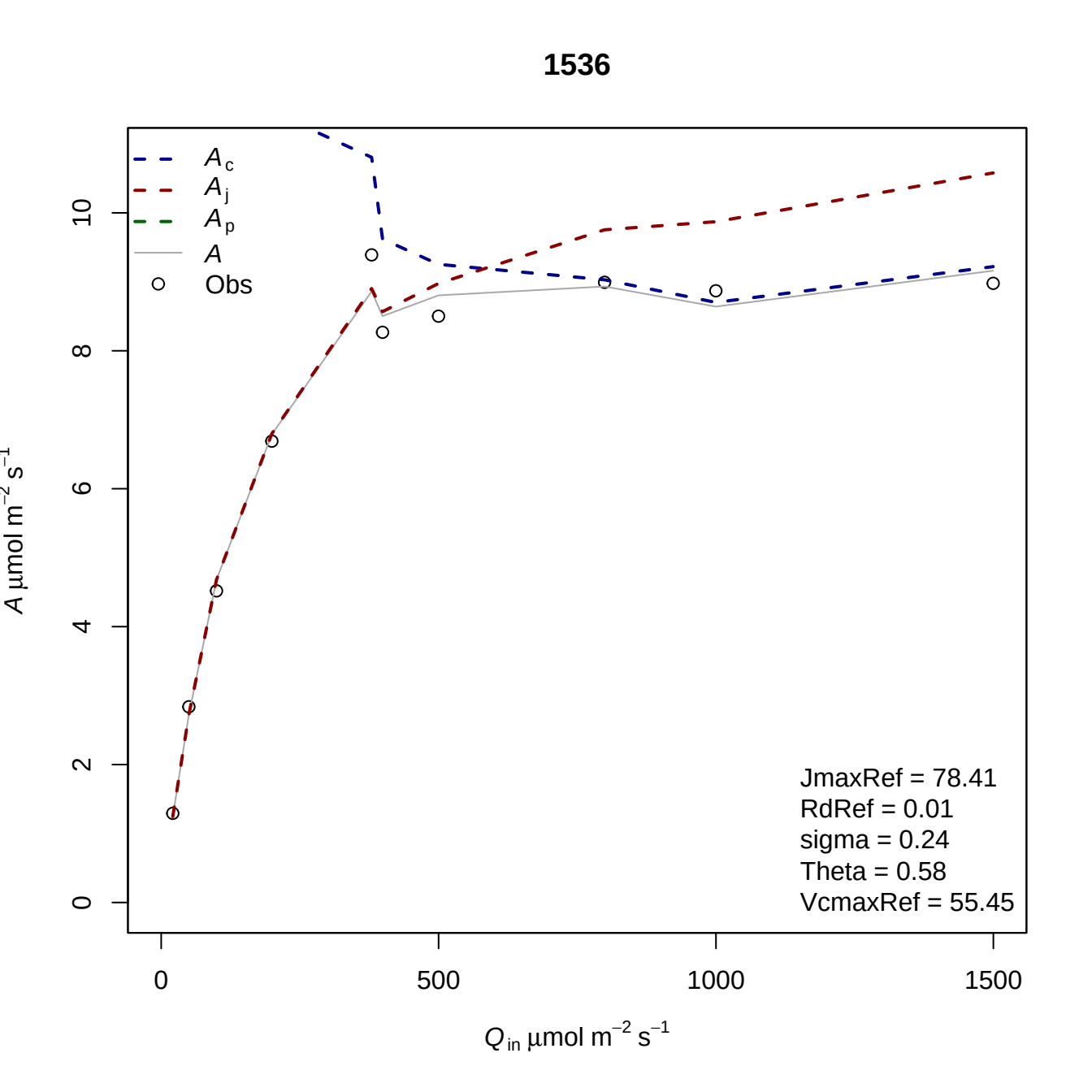

1531
A_c
A_j
A_p
A
Obs
JmaxRef = 77.22
RdRef = 0.29
sigma = 0.19
Theta = 0.07
VcmaxRef = 57.04
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

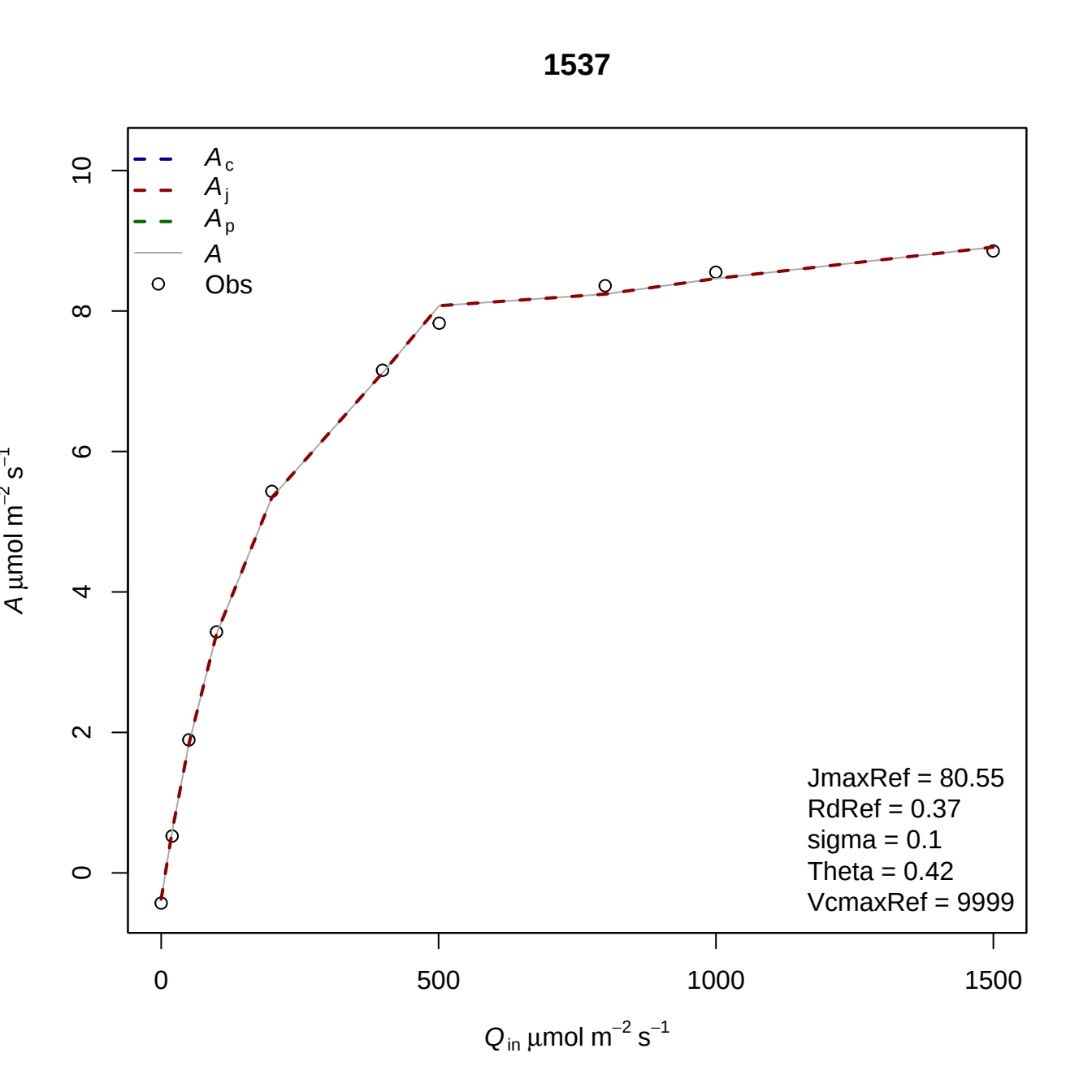

1532
A_c
A_j
A_p
A
Obs
JmaxRef = 141.07
RdRef = 0.05
sigma = 0.24
Theta = −1.05
VcmaxRef = 50.54
0
2
4
6
8
10
12
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

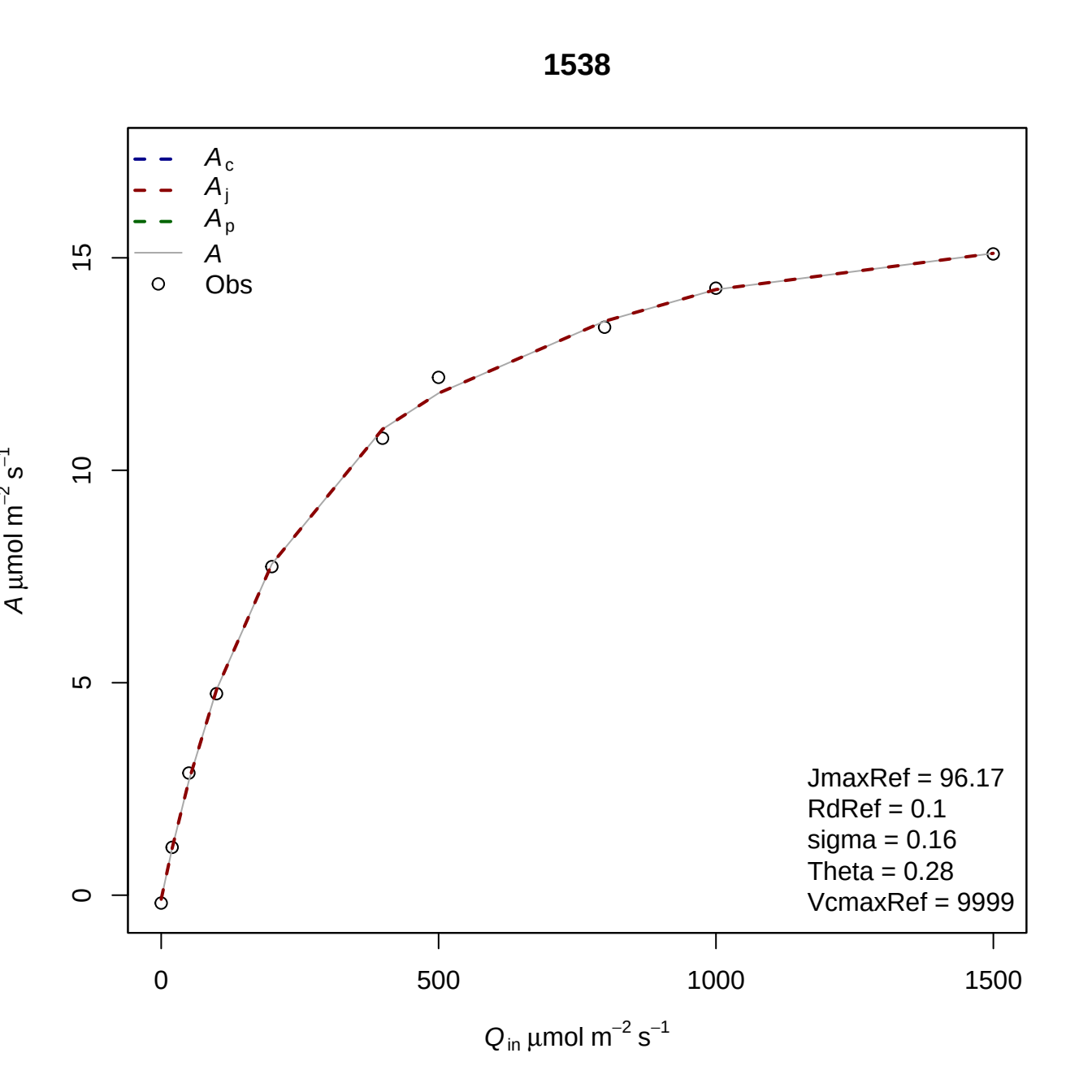

1533
A_c
A_j
A_p
A
Obs
JmaxRef = 77.94
RdRef = 0.42
sigma = 0.13
Theta = 0.65
VcmaxRef = 76.84
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

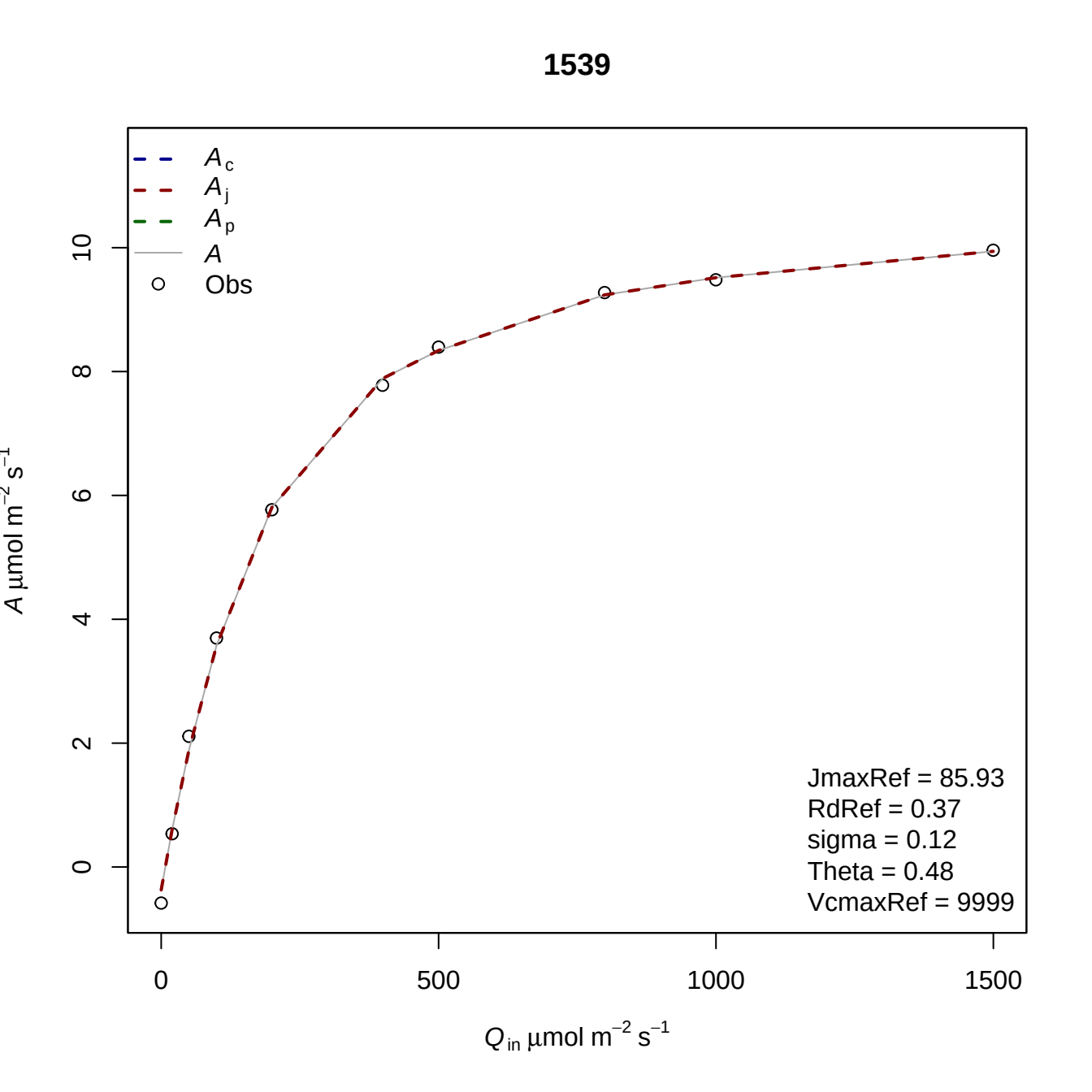

1534
A_c
A_j
A_p
A
Obs
JmaxRef = 122.46
RdRef = 0.22
sigma = 0.15
Theta = 0.35
VcmaxRef = 77.44
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

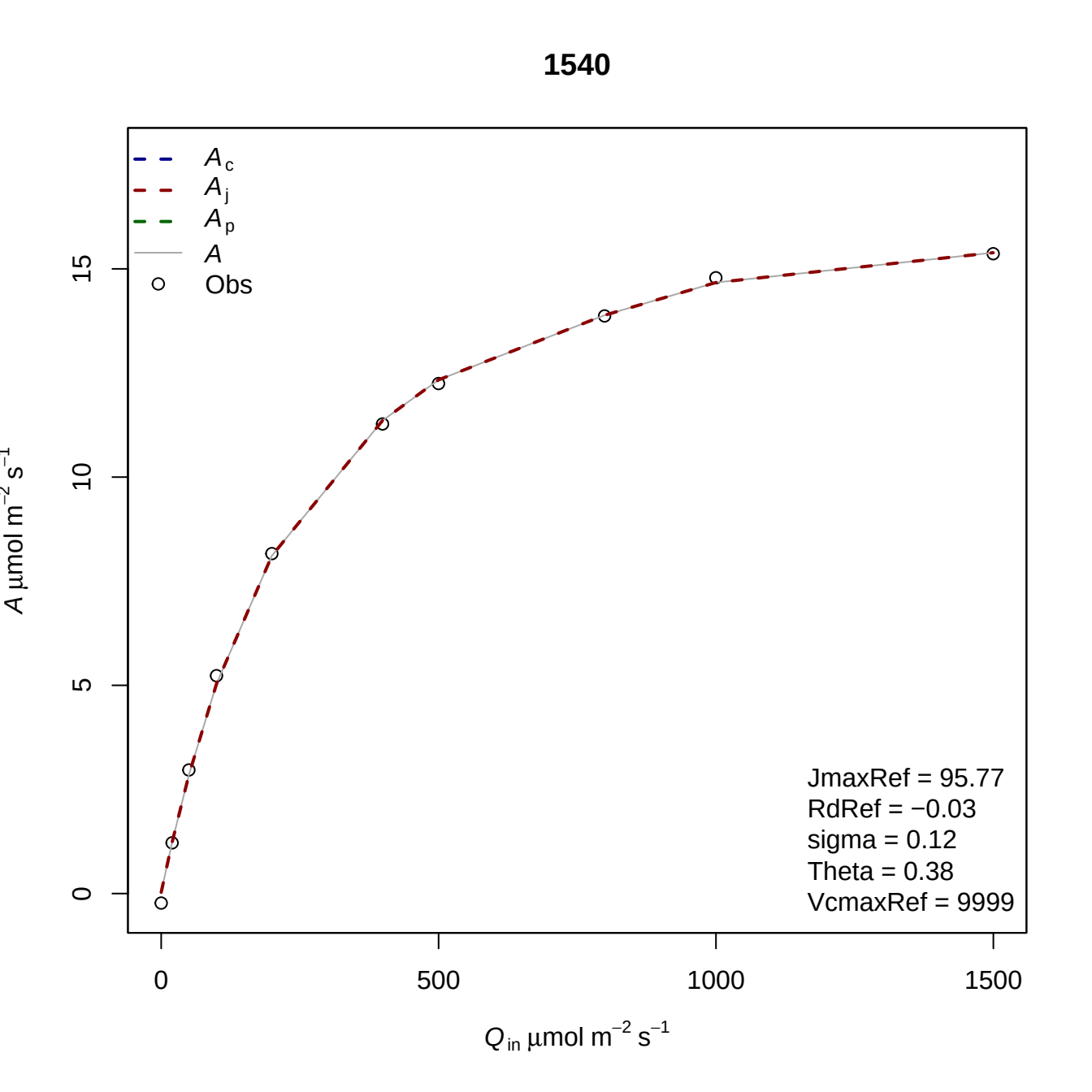

1535
A_c
A_j
A_p
A
Obs
JmaxRef = 53.01
RdRef = 0.37
sigma = 0.29
Theta = 0.73
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

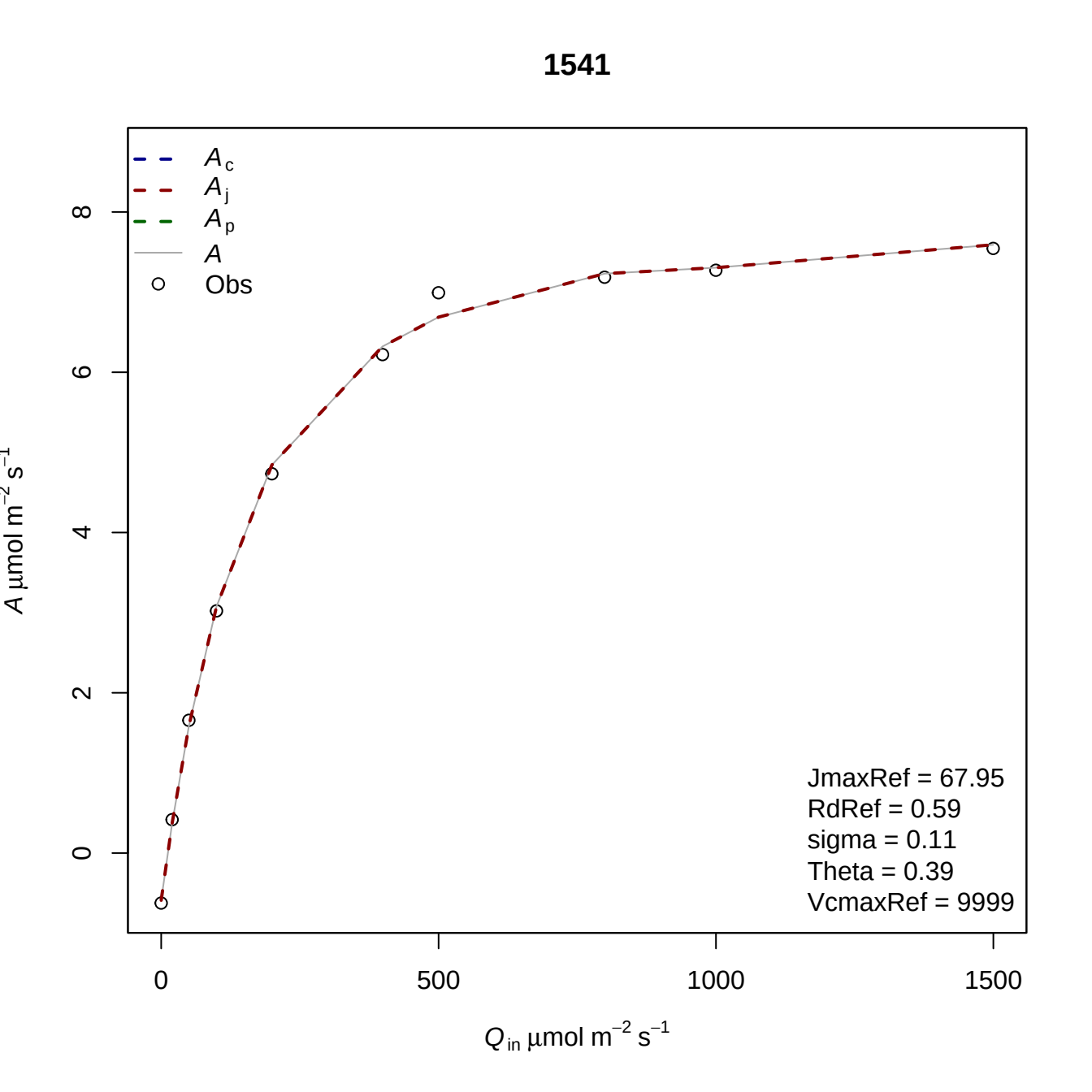

1536
A_c
A_j
A_p
A
Obs
JmaxRef = 78.41
RdRef = 0.01
sigma = 0.24
Theta = 0.58
VcmaxRef = 55.45
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

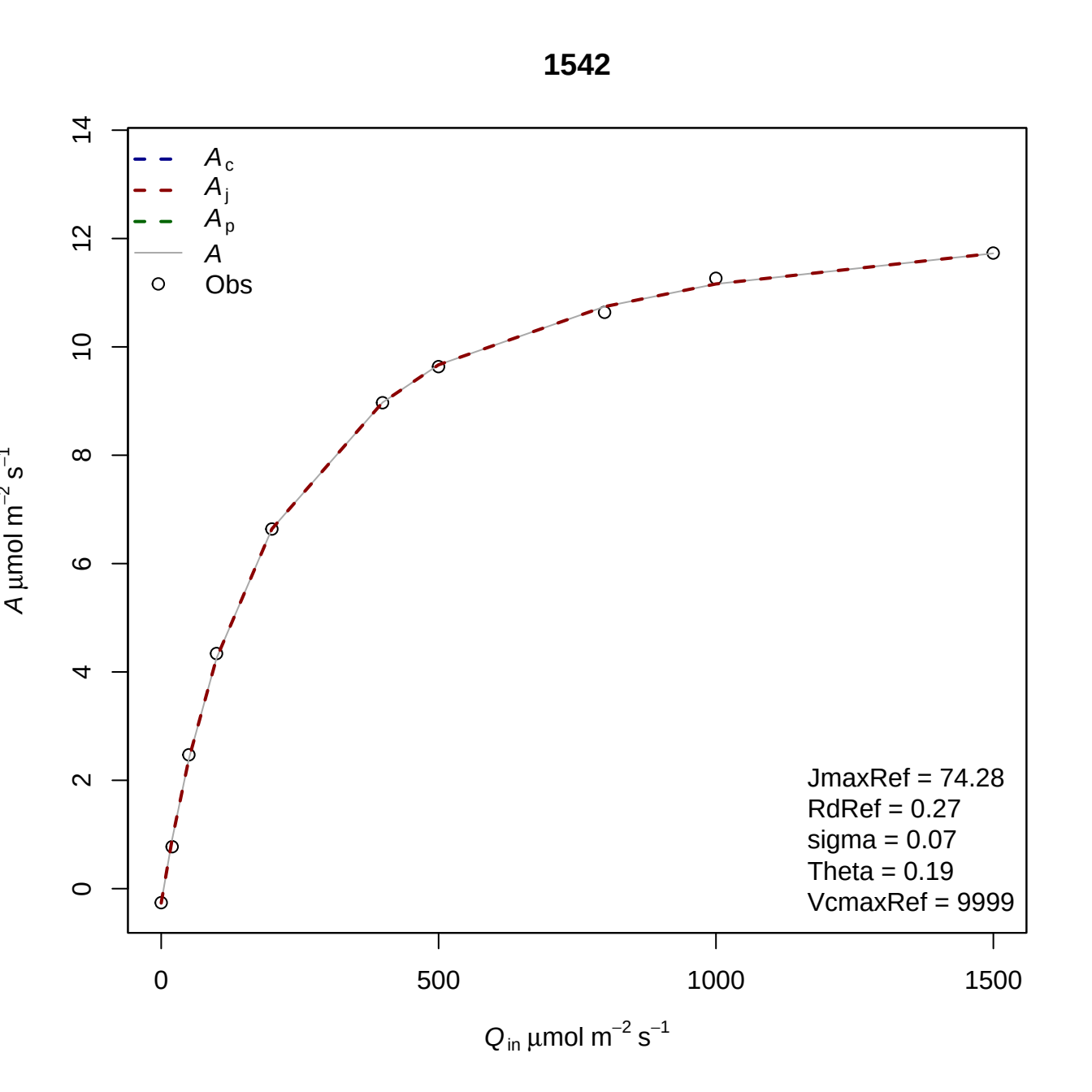

1537
A_c
A_j
A_p
A
Obs
JmaxRef = 80.55
RdRef = 0.37
sigma = 0.1
Theta = 0.42
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

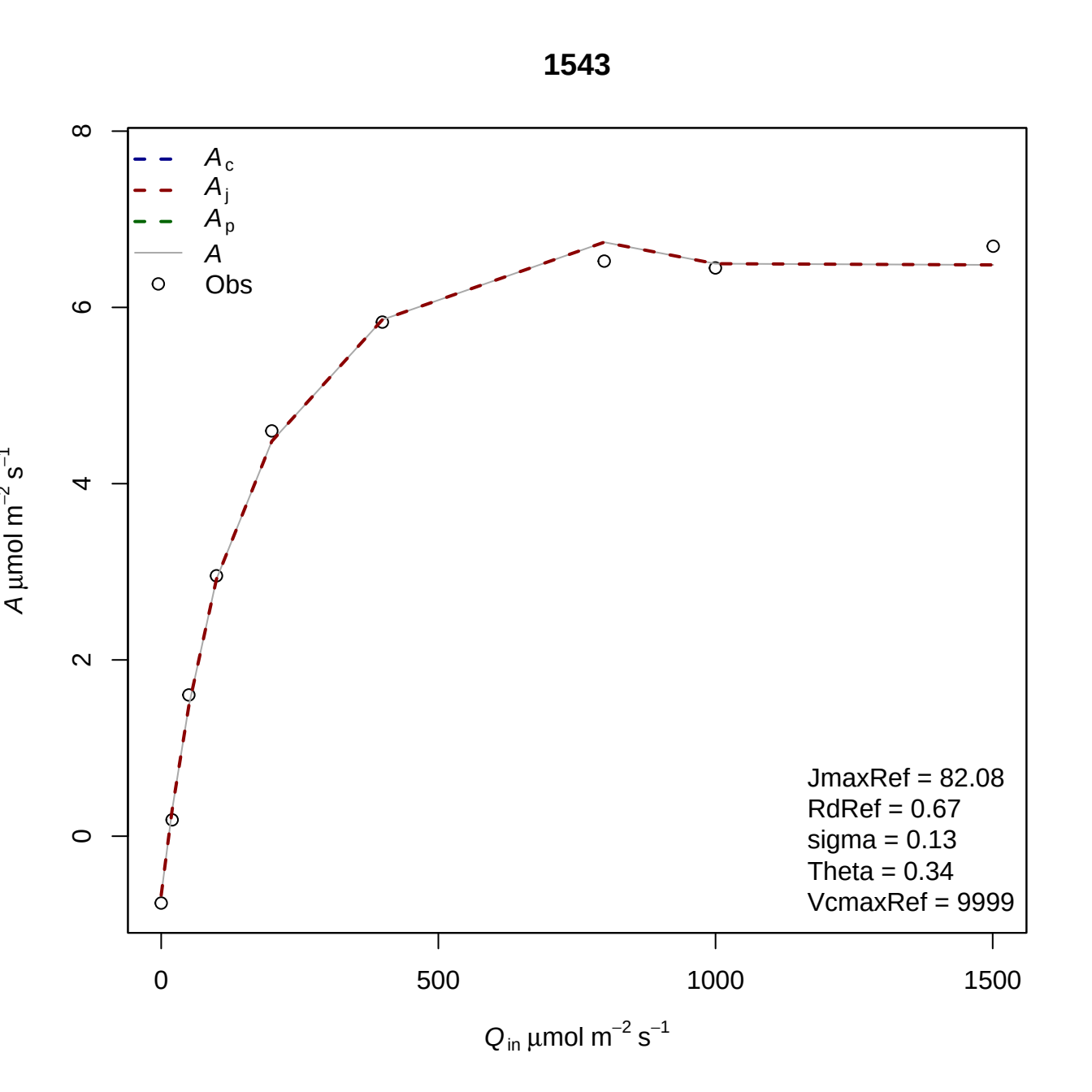

1538
A_c
A_j
A_p
A
Obs
JmaxRef = 96.17
RdRef = 0.1
sigma = 0.16
Theta = 0.28
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

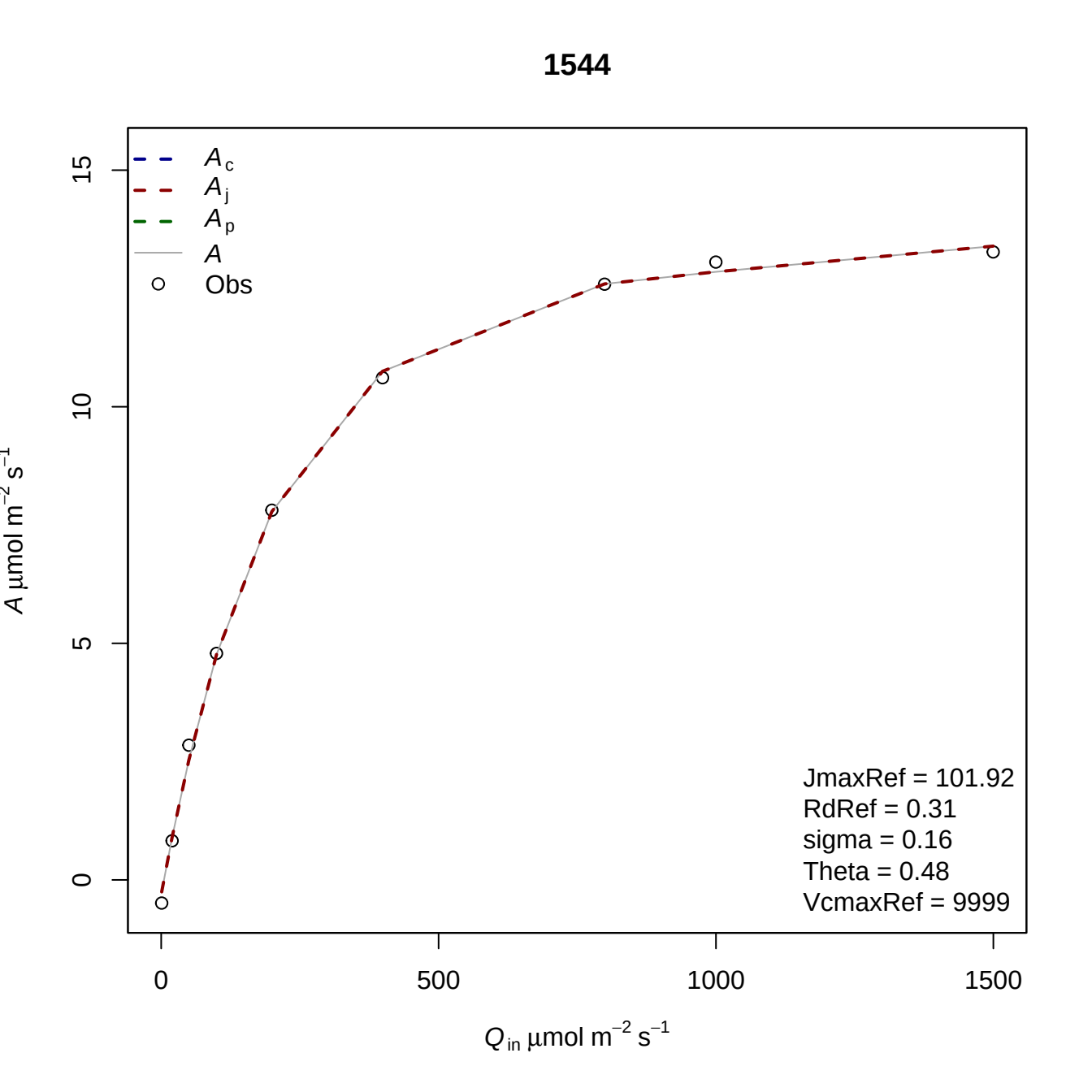

1539
A_c
A_j
A_p
A
Obs
JmaxRef = 85.93
RdRef = 0.37
sigma = 0.12
Theta = 0.48
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

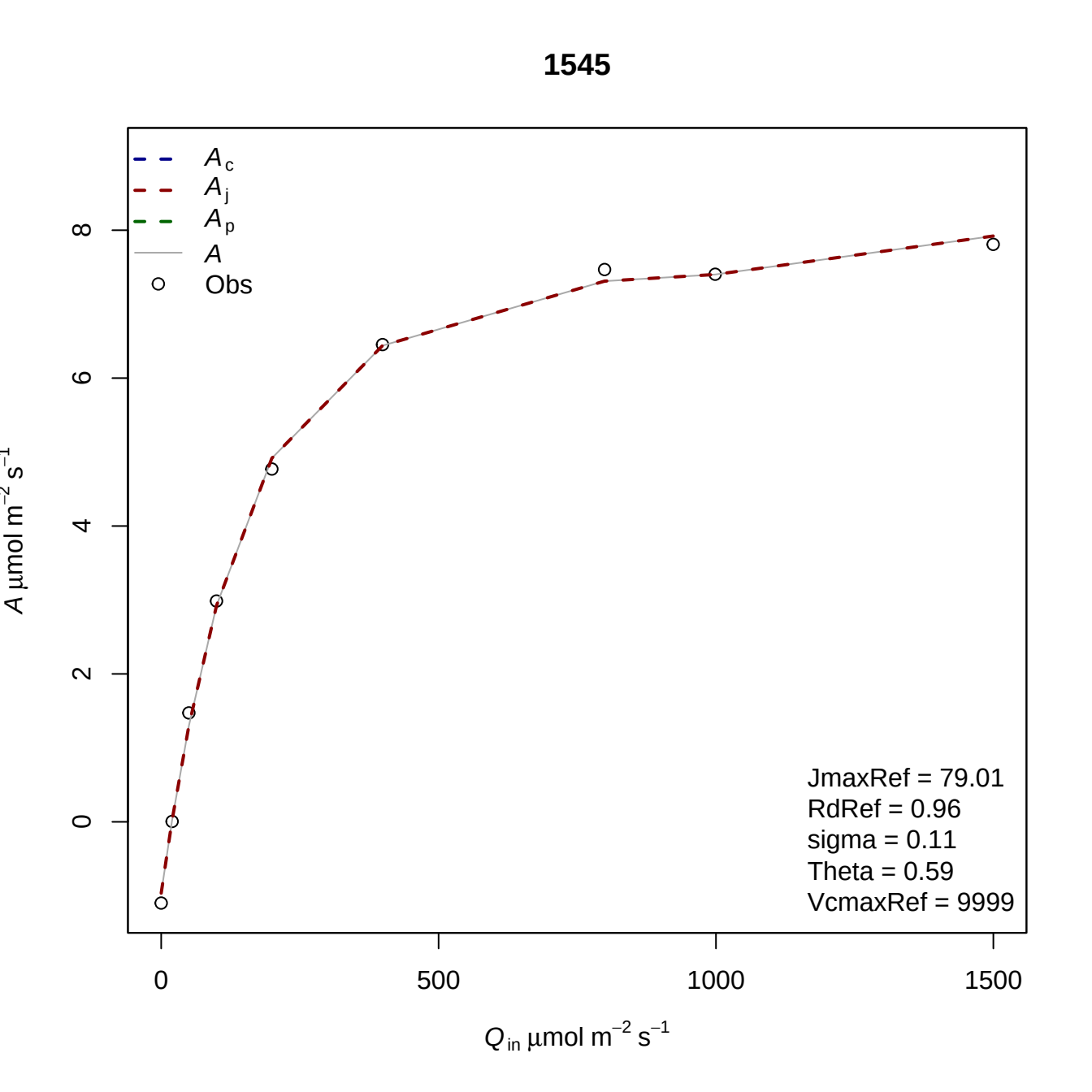

1540
A_c
A_j
A_p
A
Obs
JmaxRef = 95.77
RdRef = −0.03
sigma = 0.12
Theta = 0.38
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

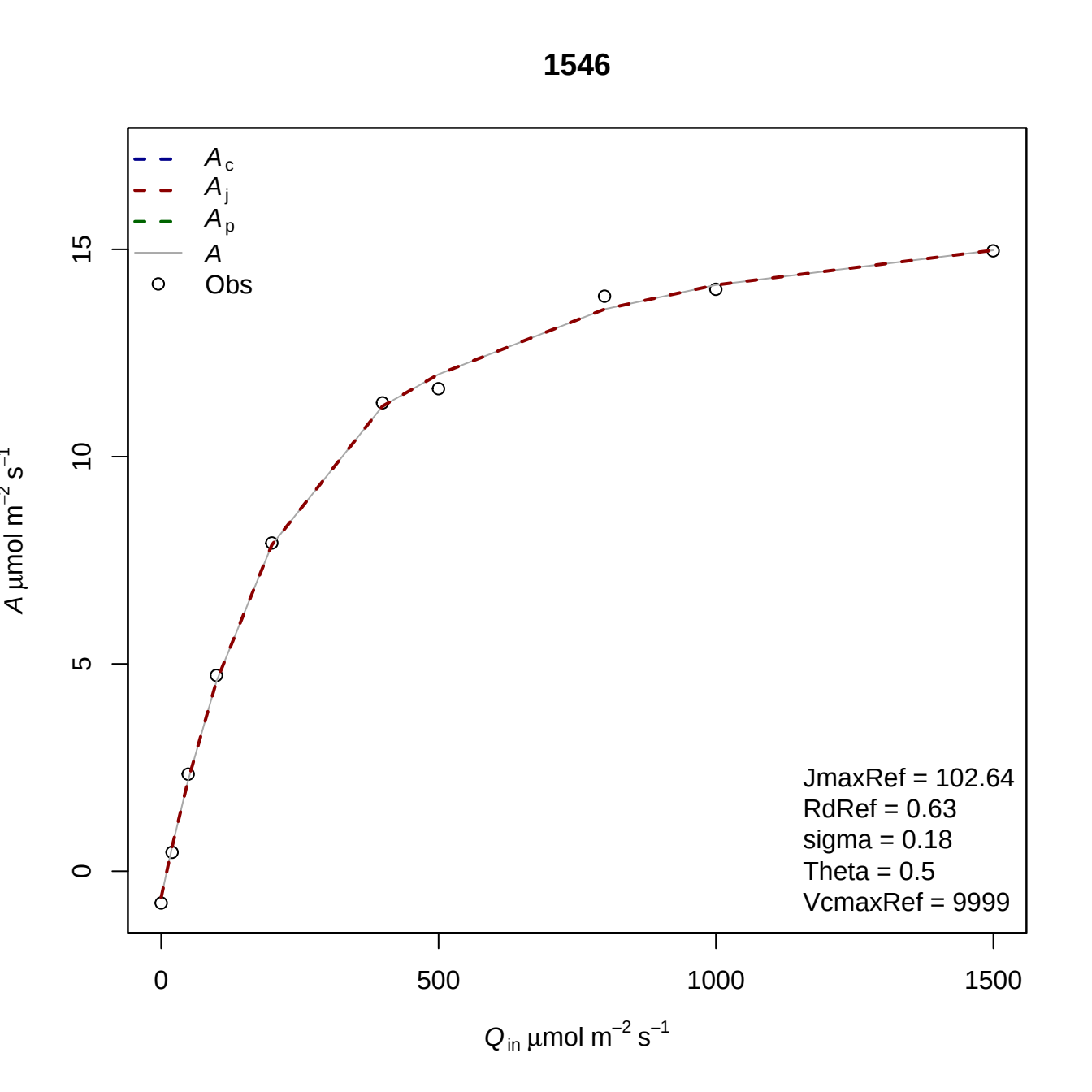

1541
A_c
A_j
A_p
A
Obs
JmaxRef = 67.95
RdRef = 0.59
sigma = 0.11
Theta = 0.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

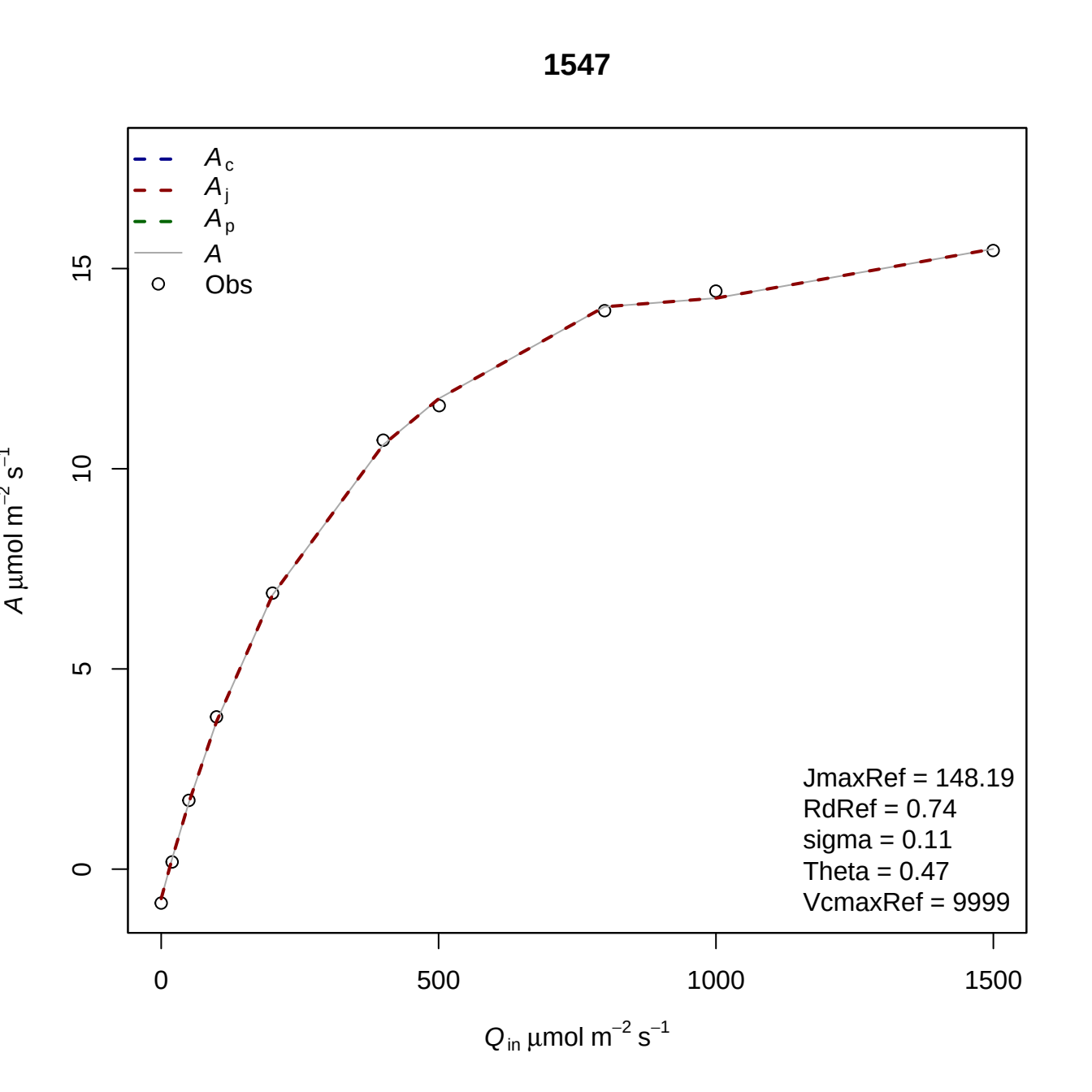

1542
A_c
A_j
A_p
A
Obs
JmaxRef = 74.28
RdRef = 0.27
sigma = 0.07
Theta = 0.19
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

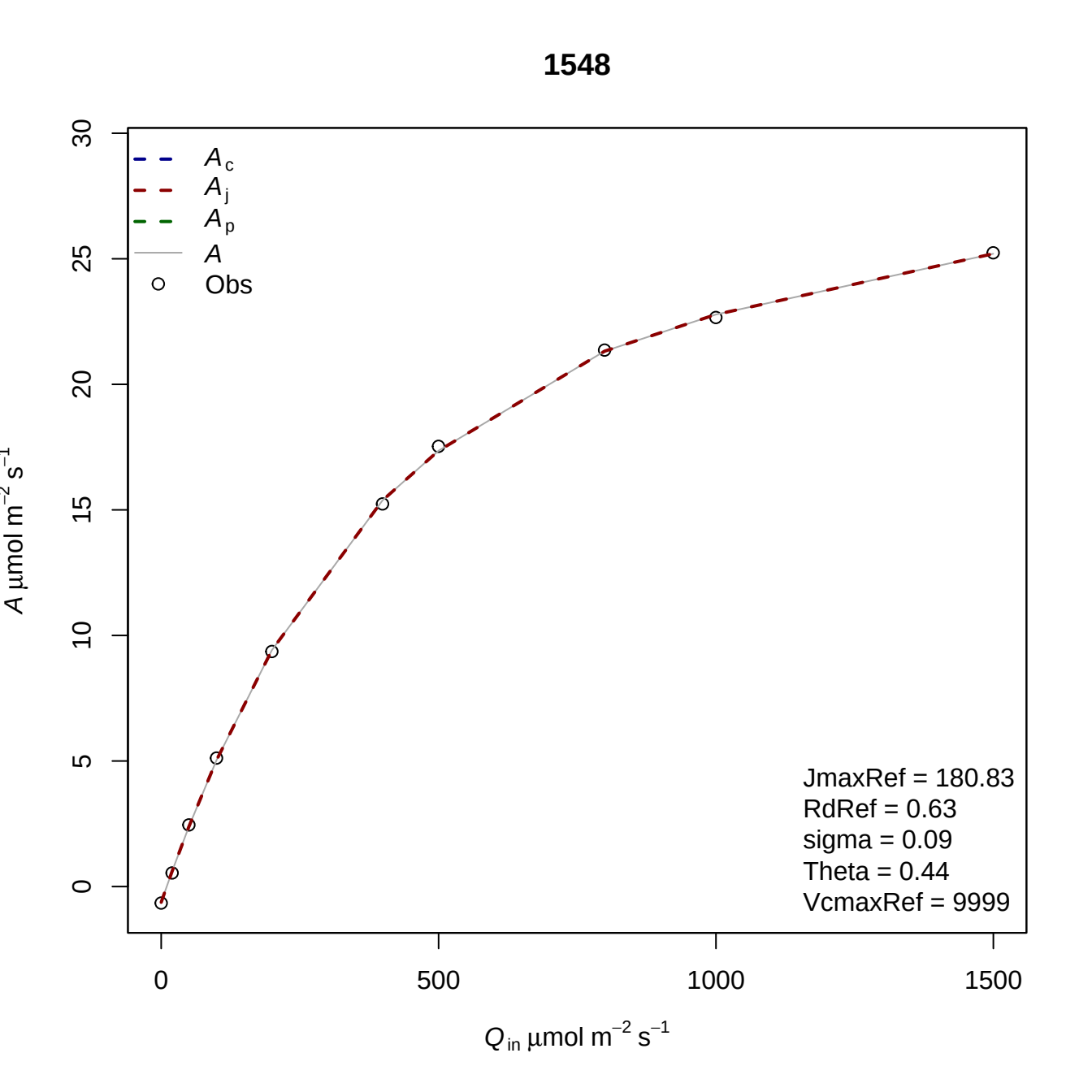

1543
A_c
A_j
A_p
A
Obs
JmaxRef = 82.08
RdRef = 0.67
sigma = 0.13
Theta = 0.34
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

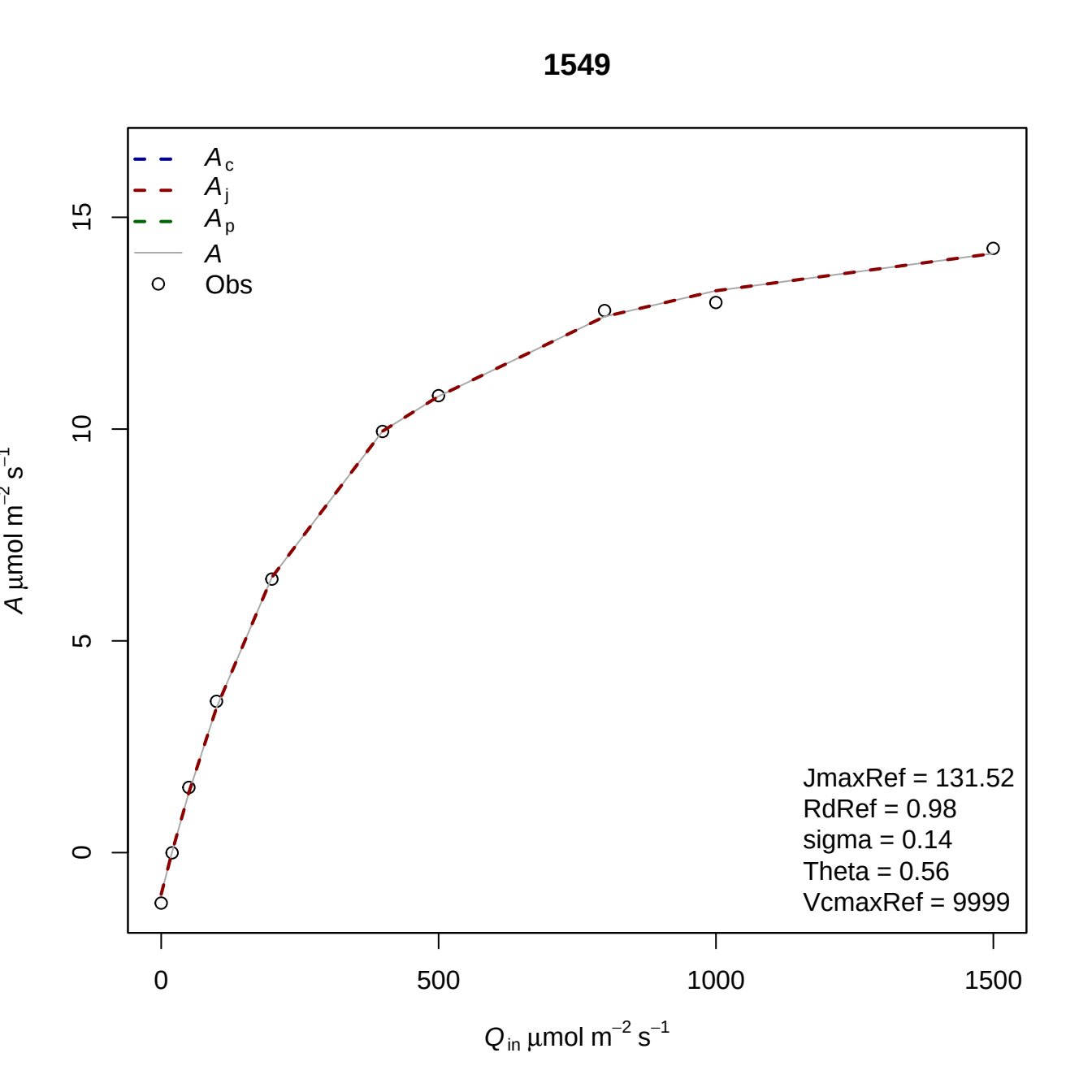

1544
A_c
A_j
A_p
A
Obs
JmaxRef = 101.92
RdRef = 0.31
sigma = 0.16
Theta = 0.48
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

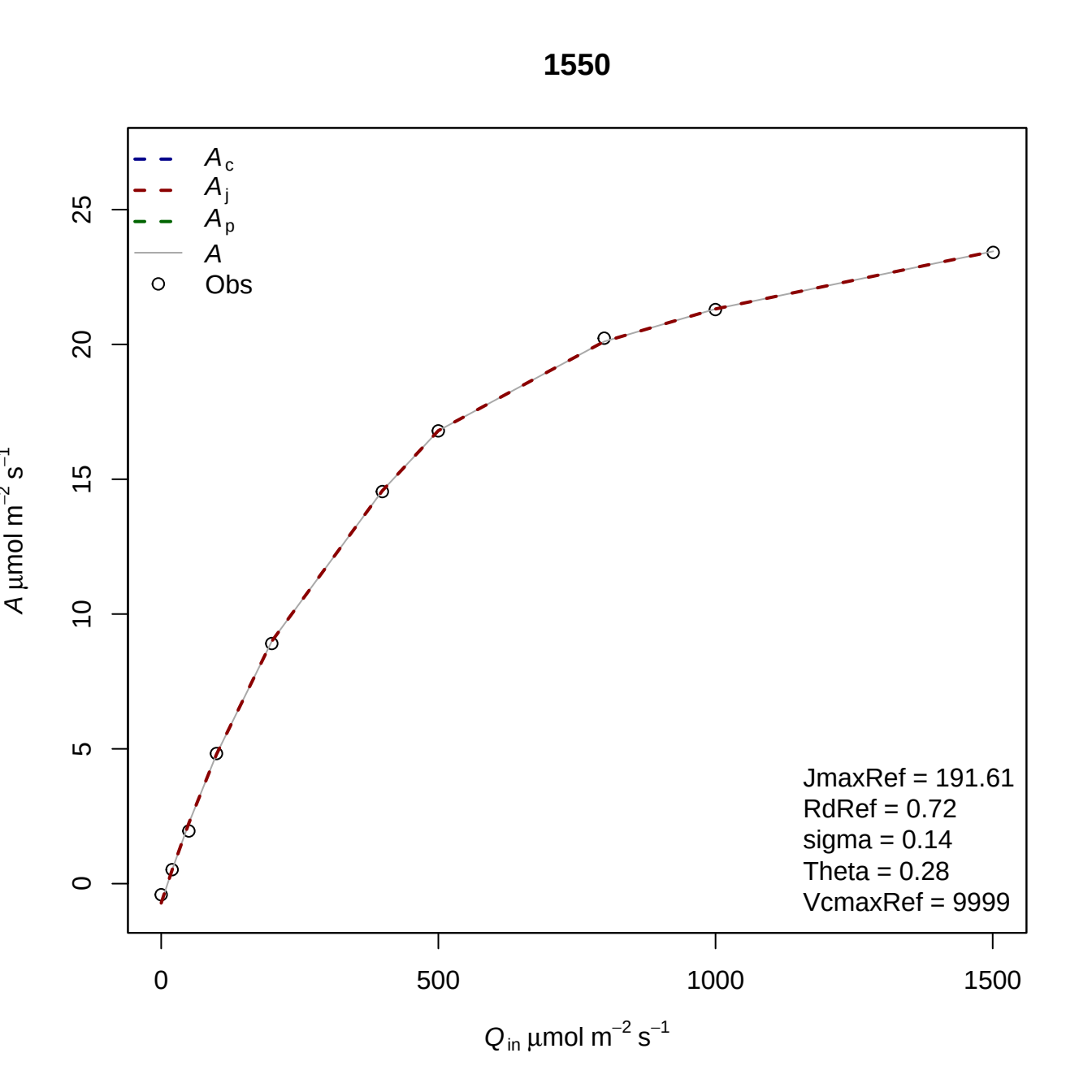

1545
A_c
A_j
A_p
A
Obs
JmaxRef = 79.01
RdRef = 0.96
sigma = 0.11
Theta = 0.59
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

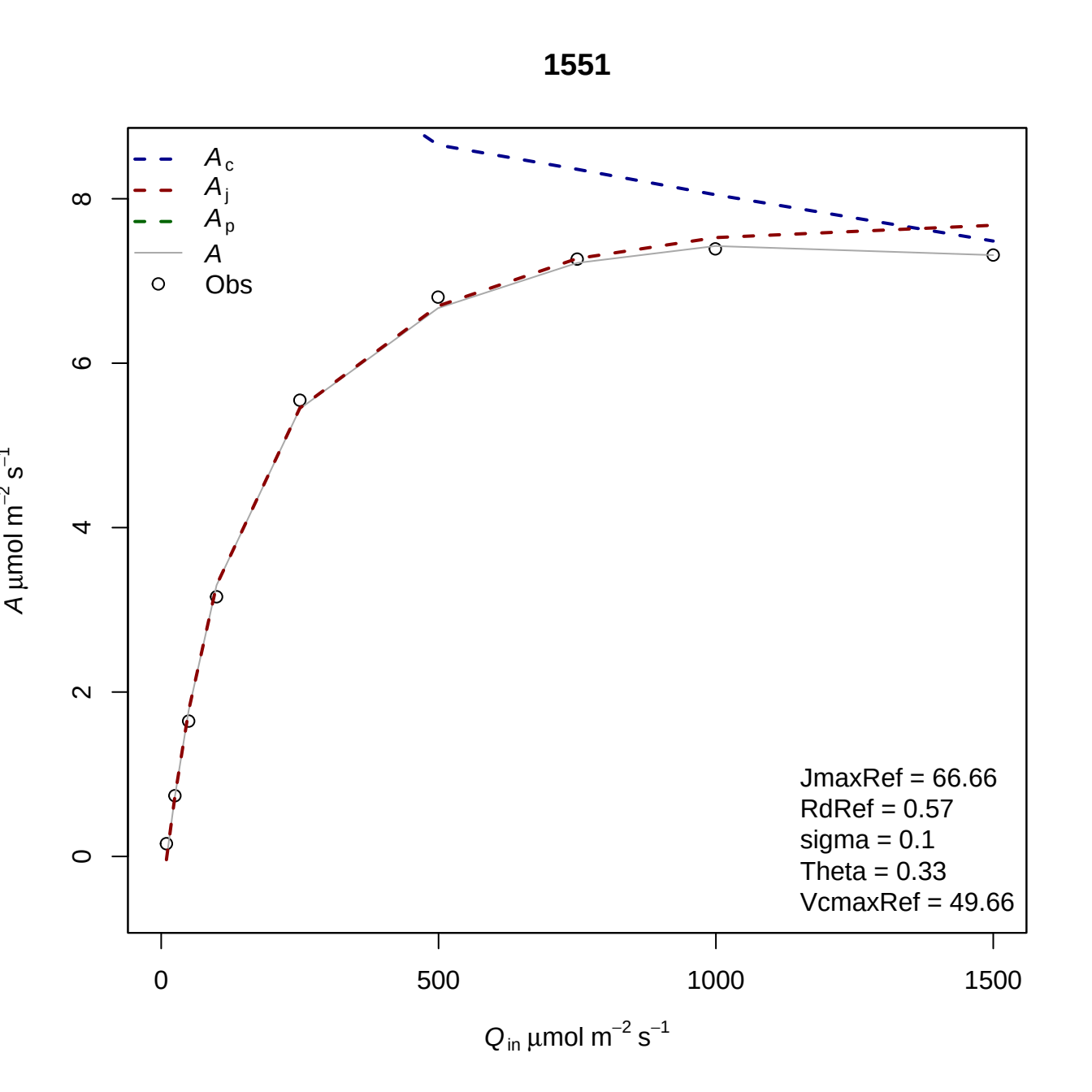

1546
A_c
A_j
A_p
A
Obs
JmaxRef = 102.64
RdRef = 0.63
sigma = 0.18
Theta = 0.5
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

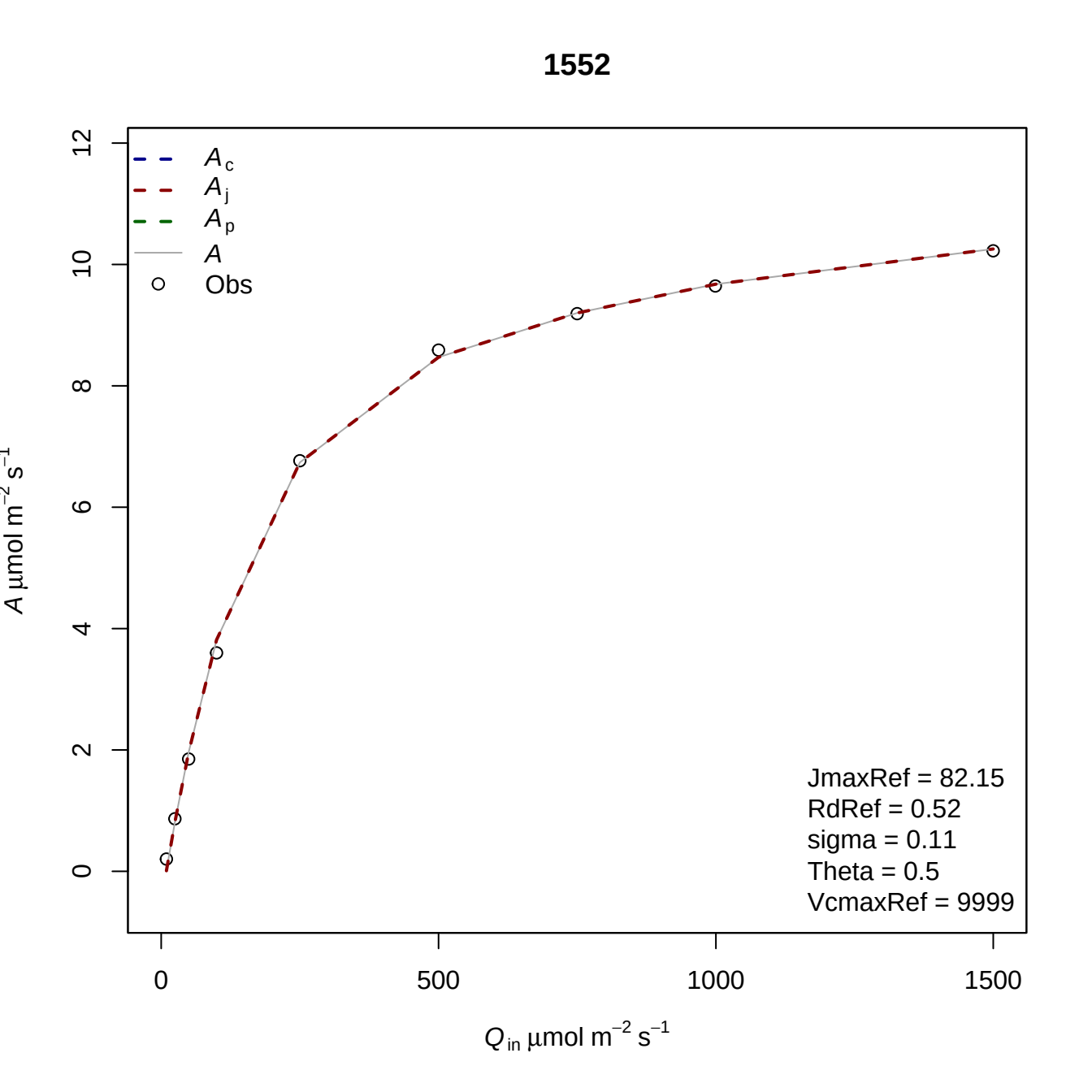

1547
A_c
A_j
A_p
A
Obs
JmaxRef = 148.19
RdRef = 0.74
sigma = 0.11
Theta = 0.47
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

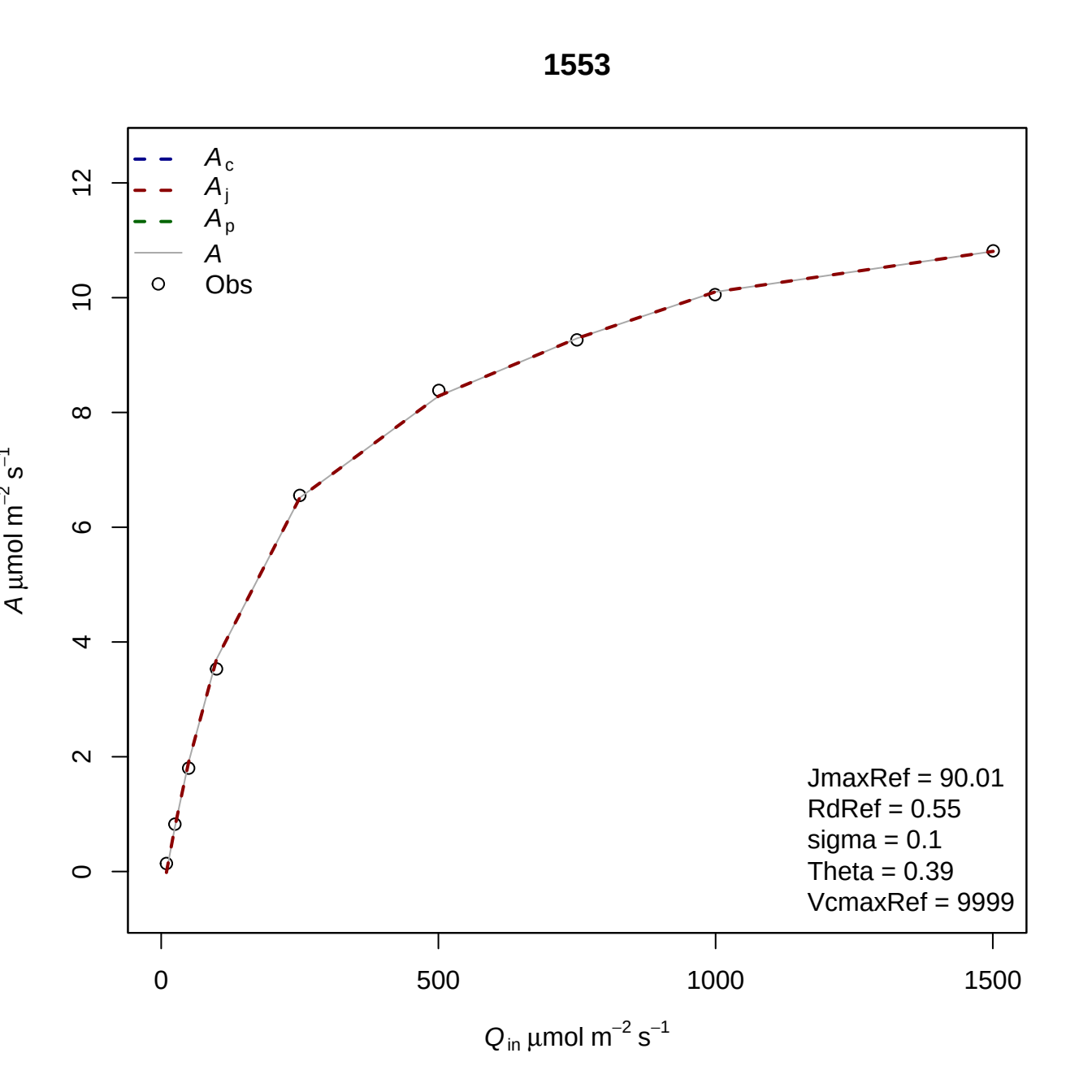

1548
A_c
A_j
A_p
A
Obs
JmaxRef = 180.83
RdRef = 0.63
sigma = 0.09
Theta = 0.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500

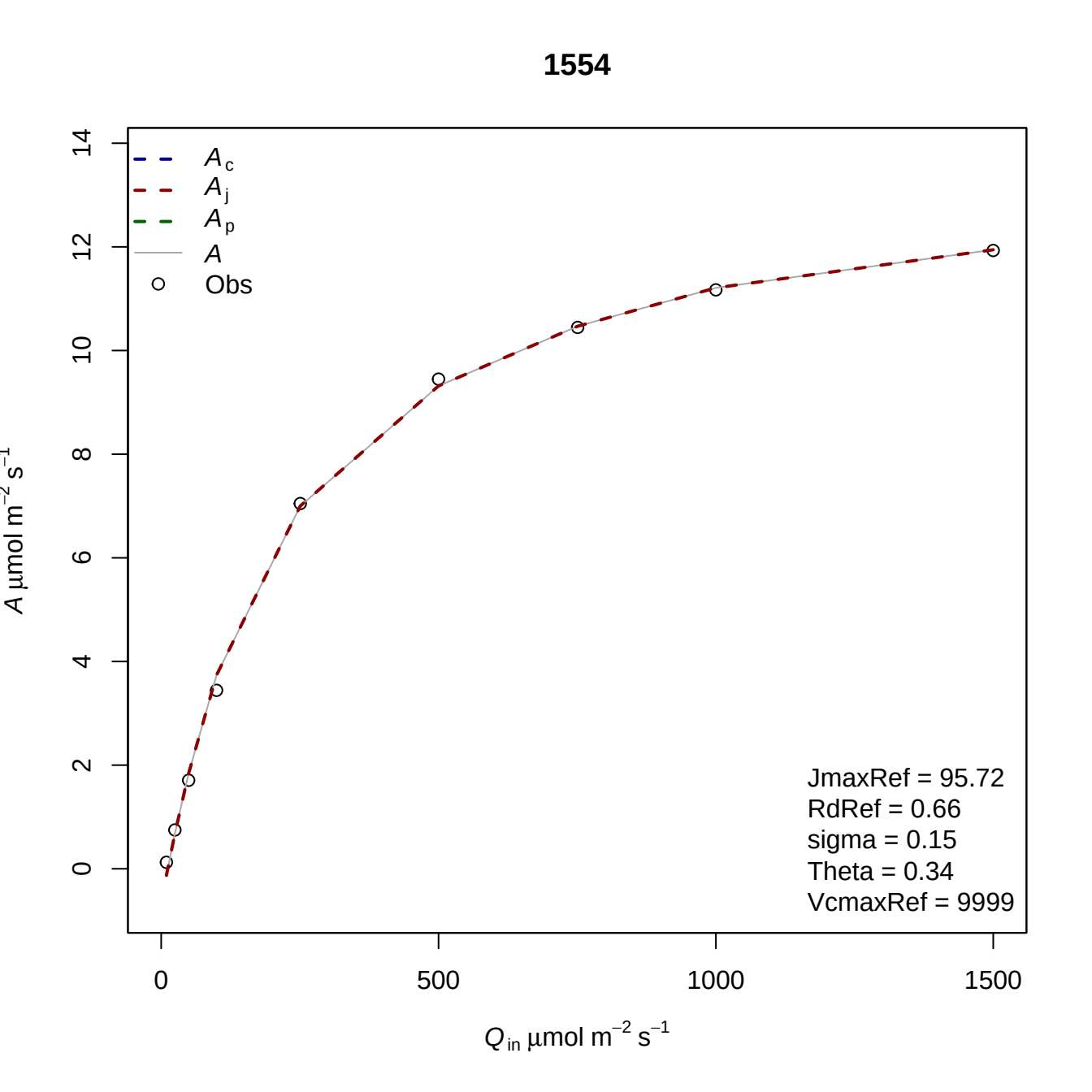

1549
A_c
A_j
A_p
A
Obs
JmaxRef = 131.52
RdRef = 0.98
sigma = 0.14
Theta = 0.56
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

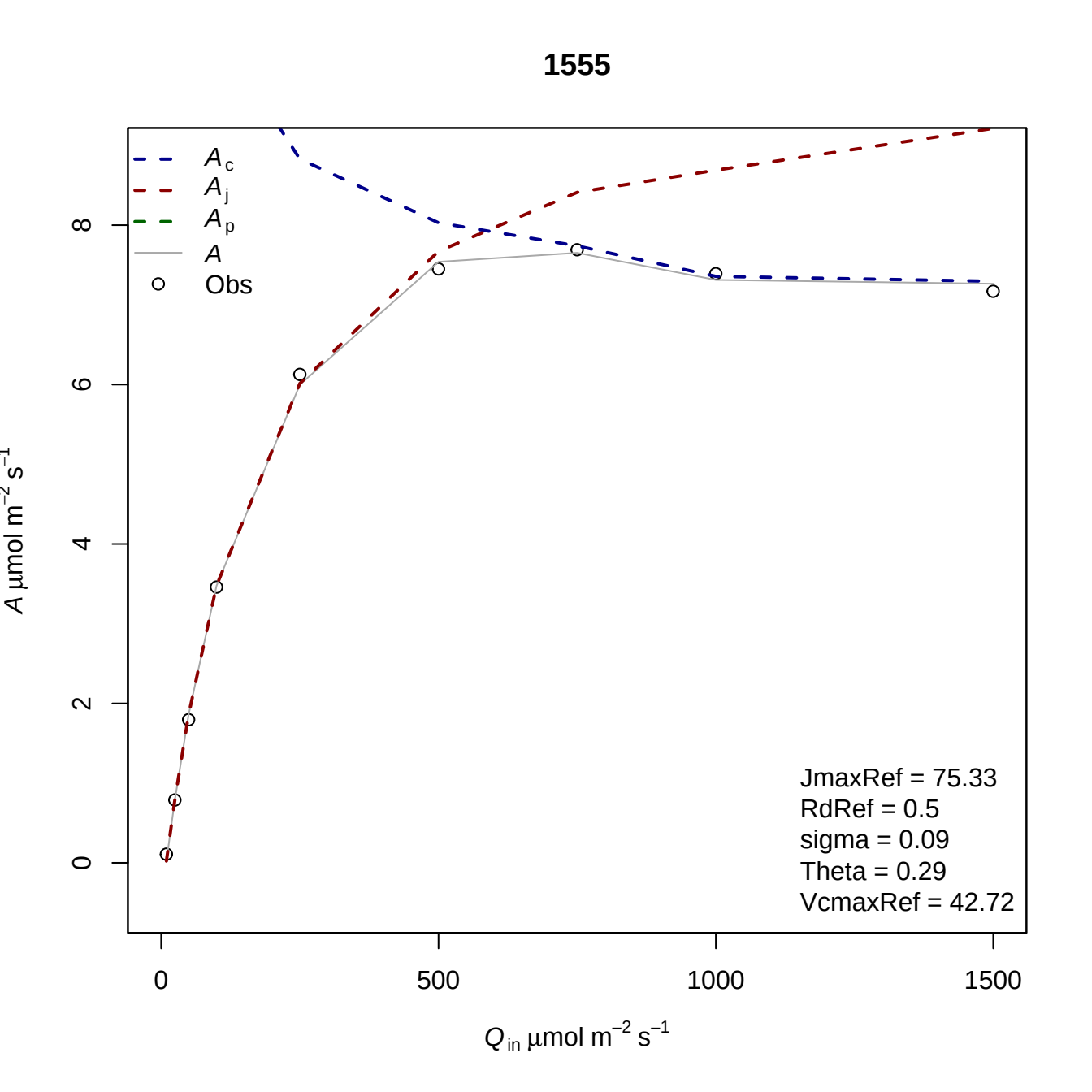

1550
A_c
A_j
A_p
A
Obs
JmaxRef = 191.61
RdRef = 0.72
sigma = 0.14
Theta = 0.28
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

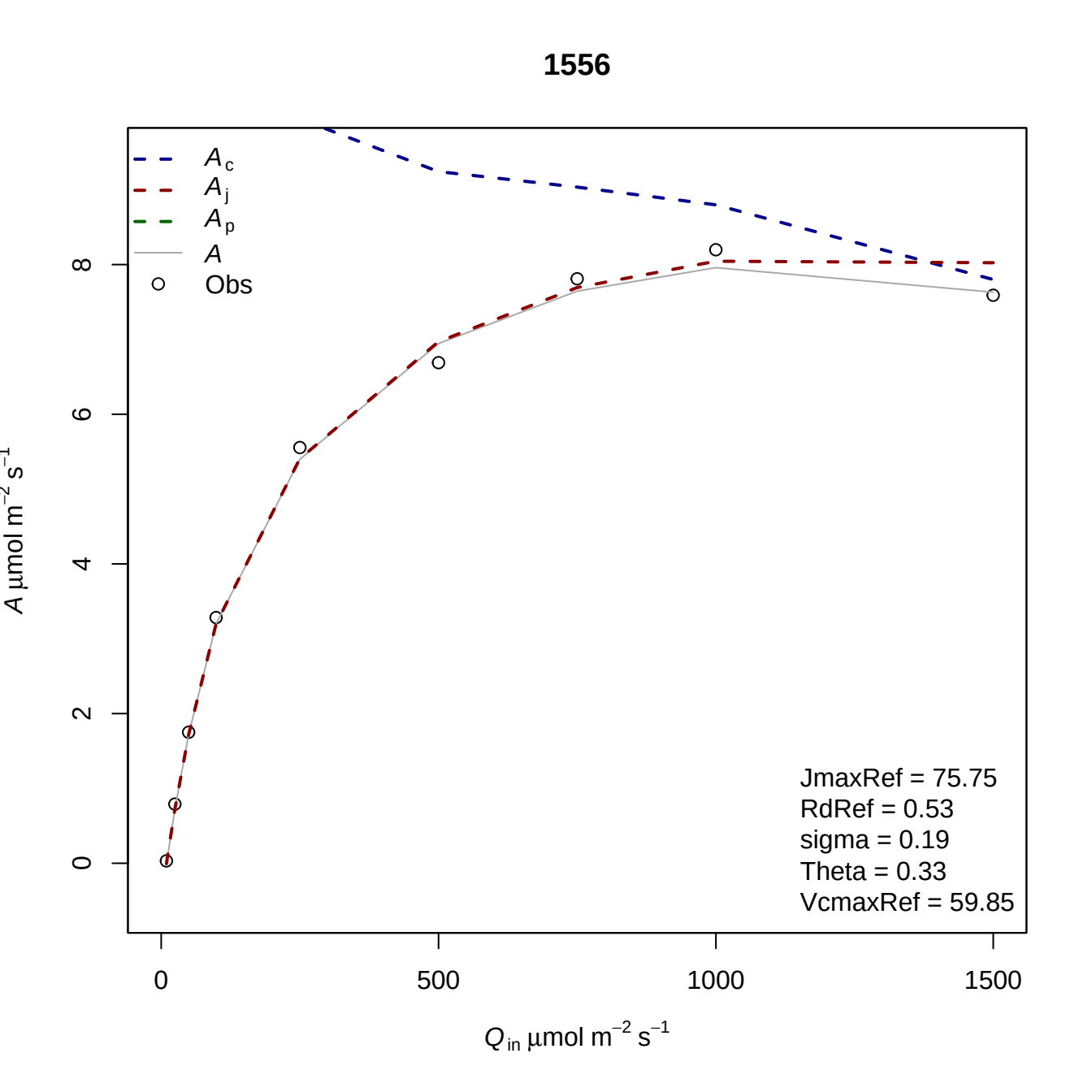

1551
A_c
A_j
A_p
A
Obs
JmaxRef = 66.66
RdRef = 0.57
sigma = 0.1
Theta = 0.33
VcmaxRef = 49.66
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

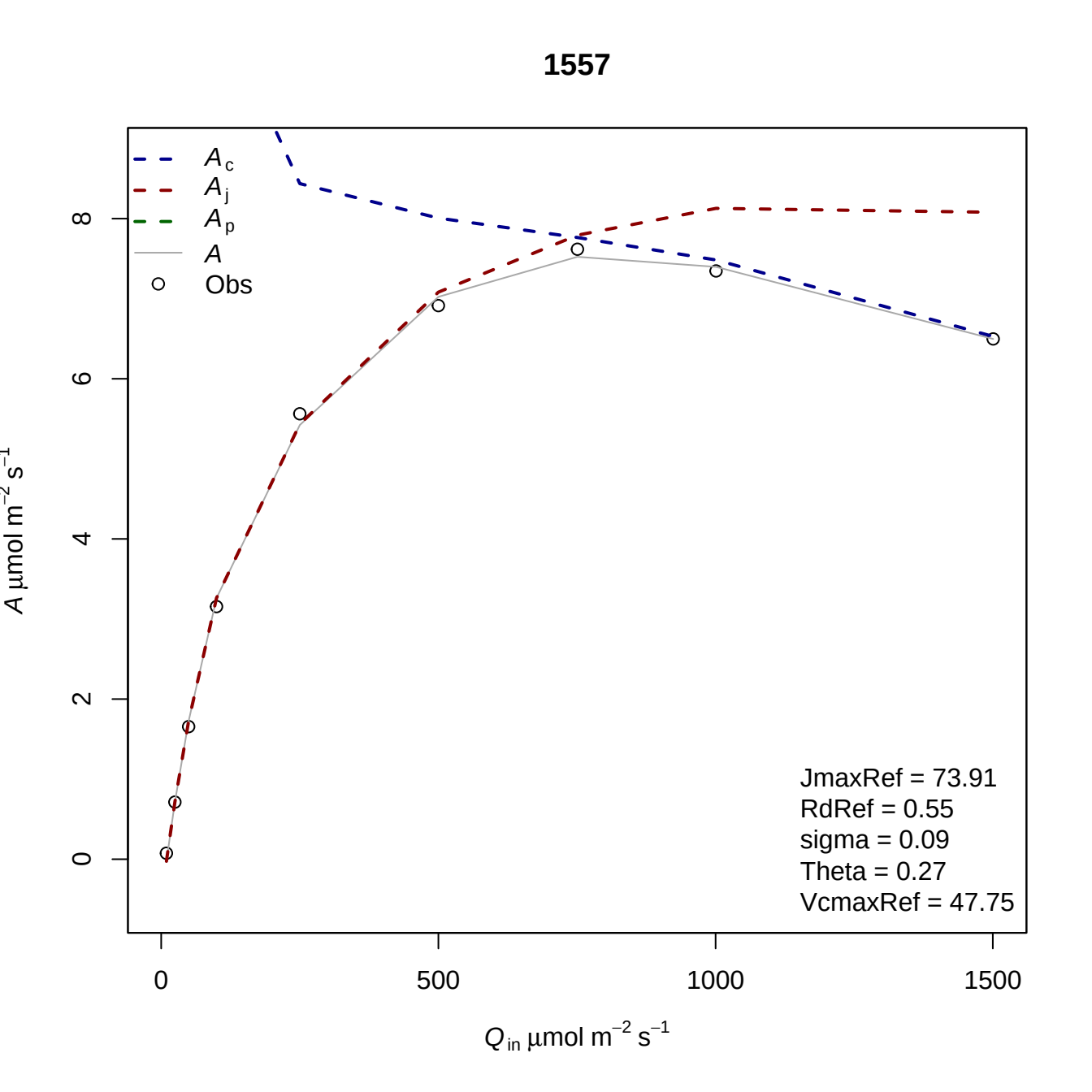

1552
A_c
A_j
A_p
A
Obs
JmaxRef = 82.15
RdRef = 0.52
sigma = 0.11
Theta = 0.5
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

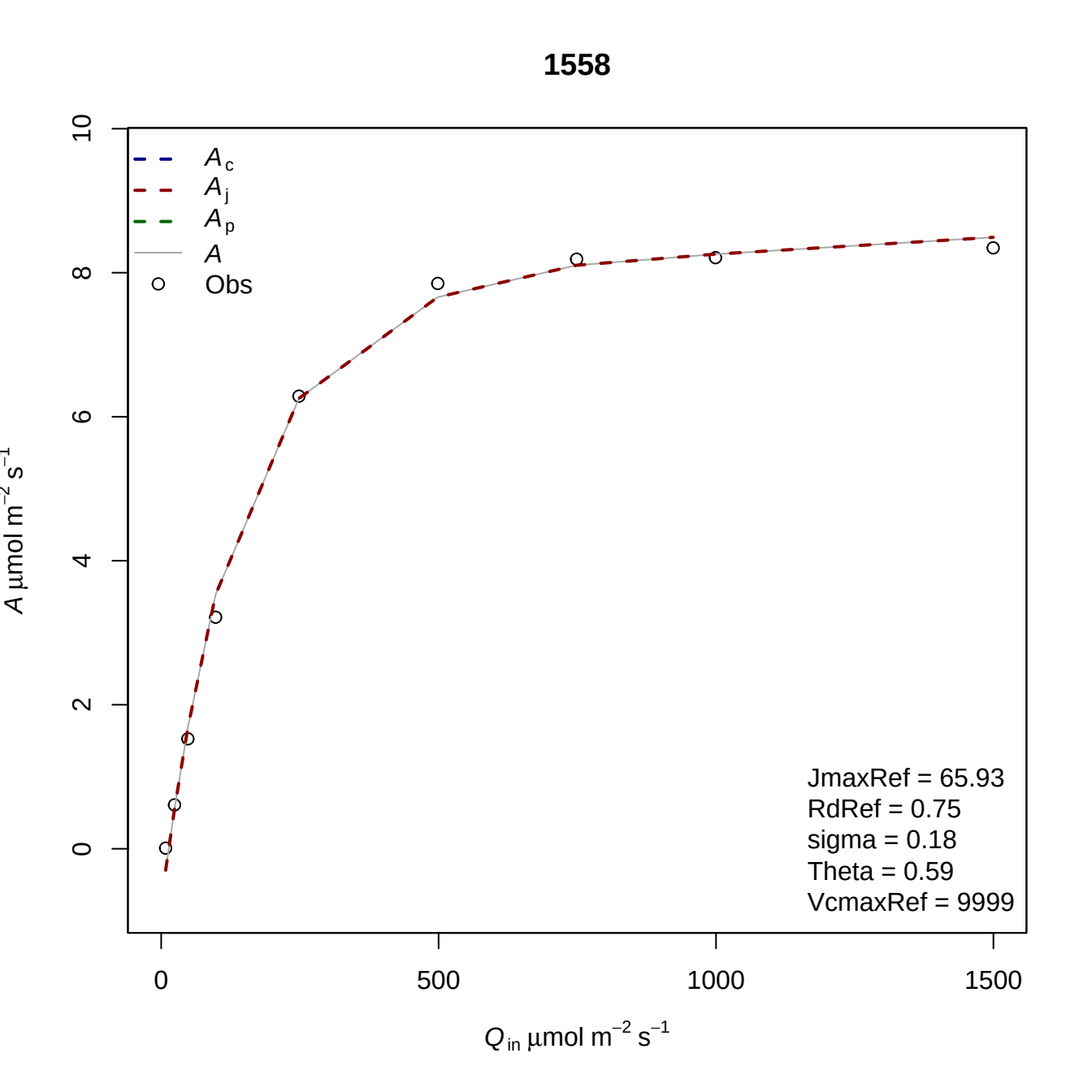

1553
A_c
A_j
A_p
A
Obs
JmaxRef = 90.01
RdRef = 0.55
sigma = 0.1
Theta = 0.39
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

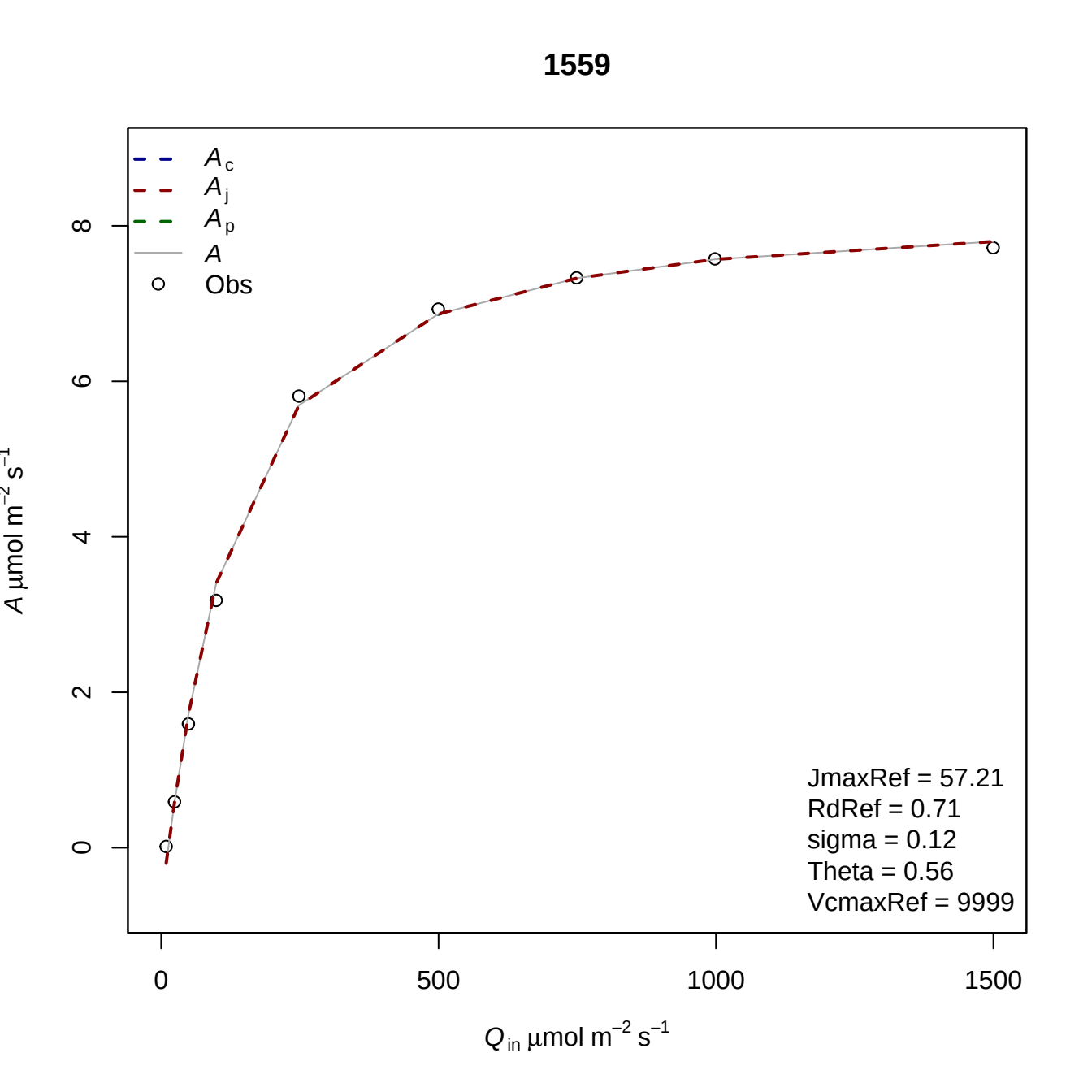

1554
A_c
A_j
A_p
A
Obs
JmaxRef = 95.72
RdRef = 0.66
sigma = 0.15
Theta = 0.34
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

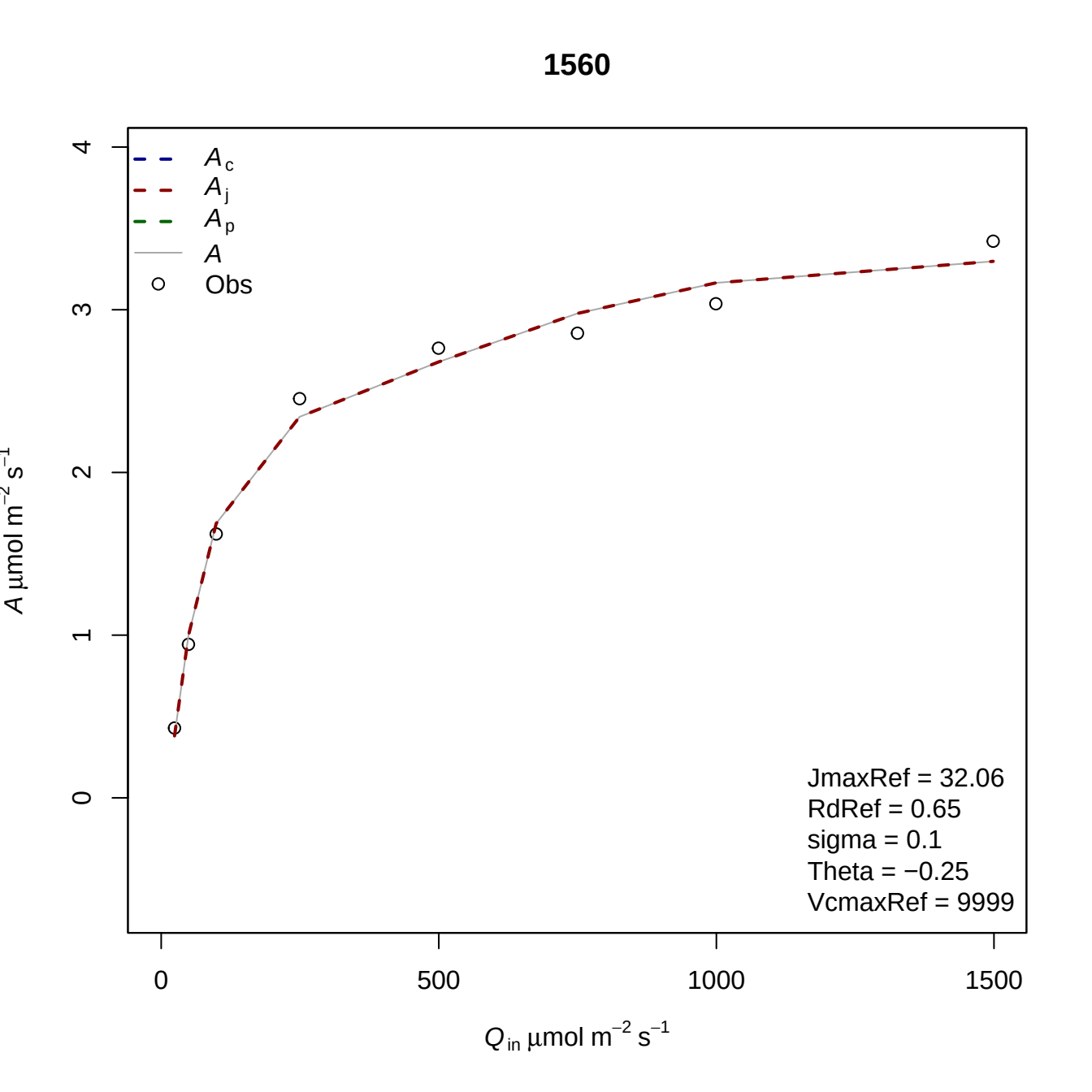

1555
A_c
A_j
A_p
A
Obs
JmaxRef = 75.33
RdRef = 0.5
sigma = 0.09
Theta = 0.29
VcmaxRef = 42.72
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

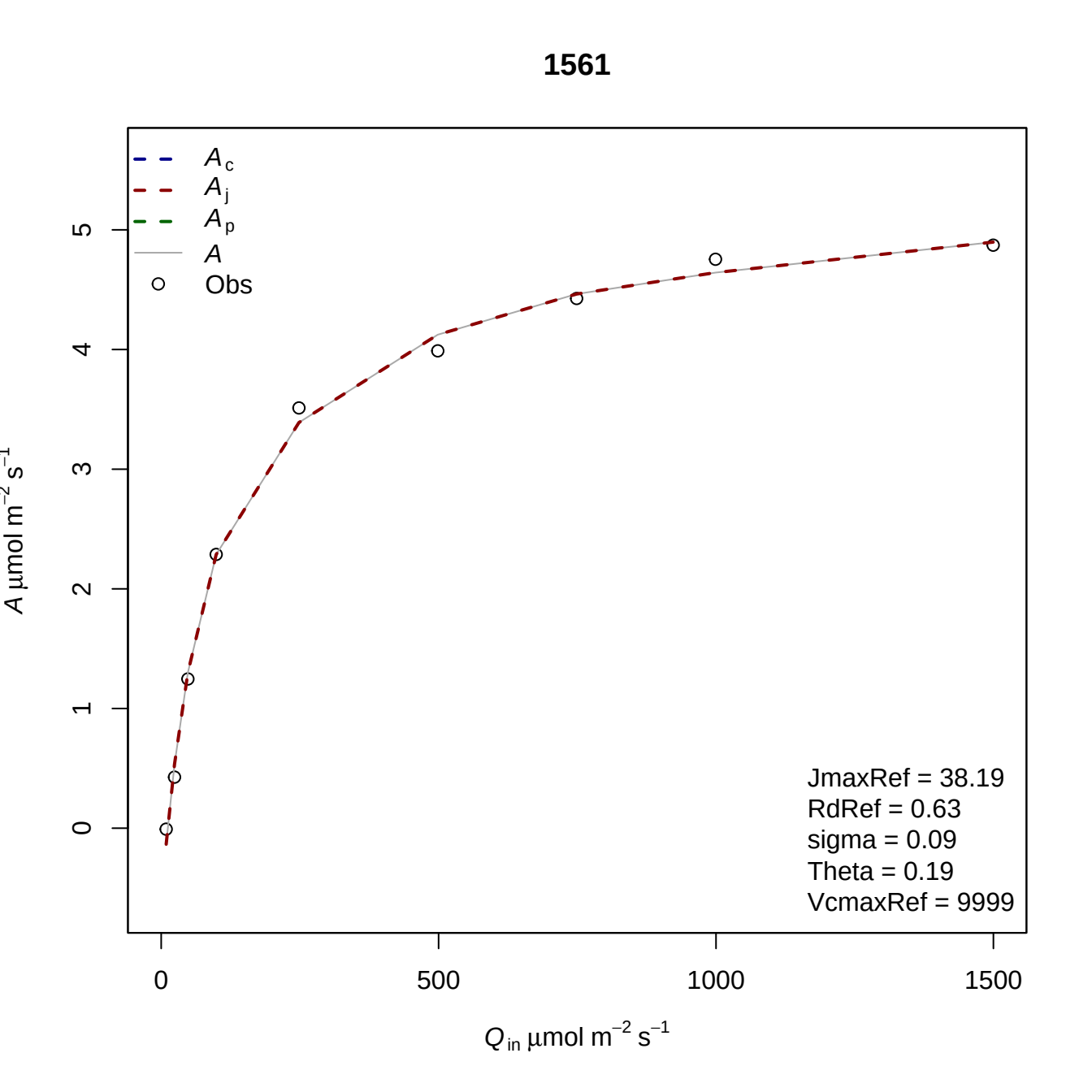

1556
A_c
A_j
A_p
A
Obs
JmaxRef = 75.75
RdRef = 0.53
sigma = 0.19
Theta = 0.33
VcmaxRef = 59.85
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

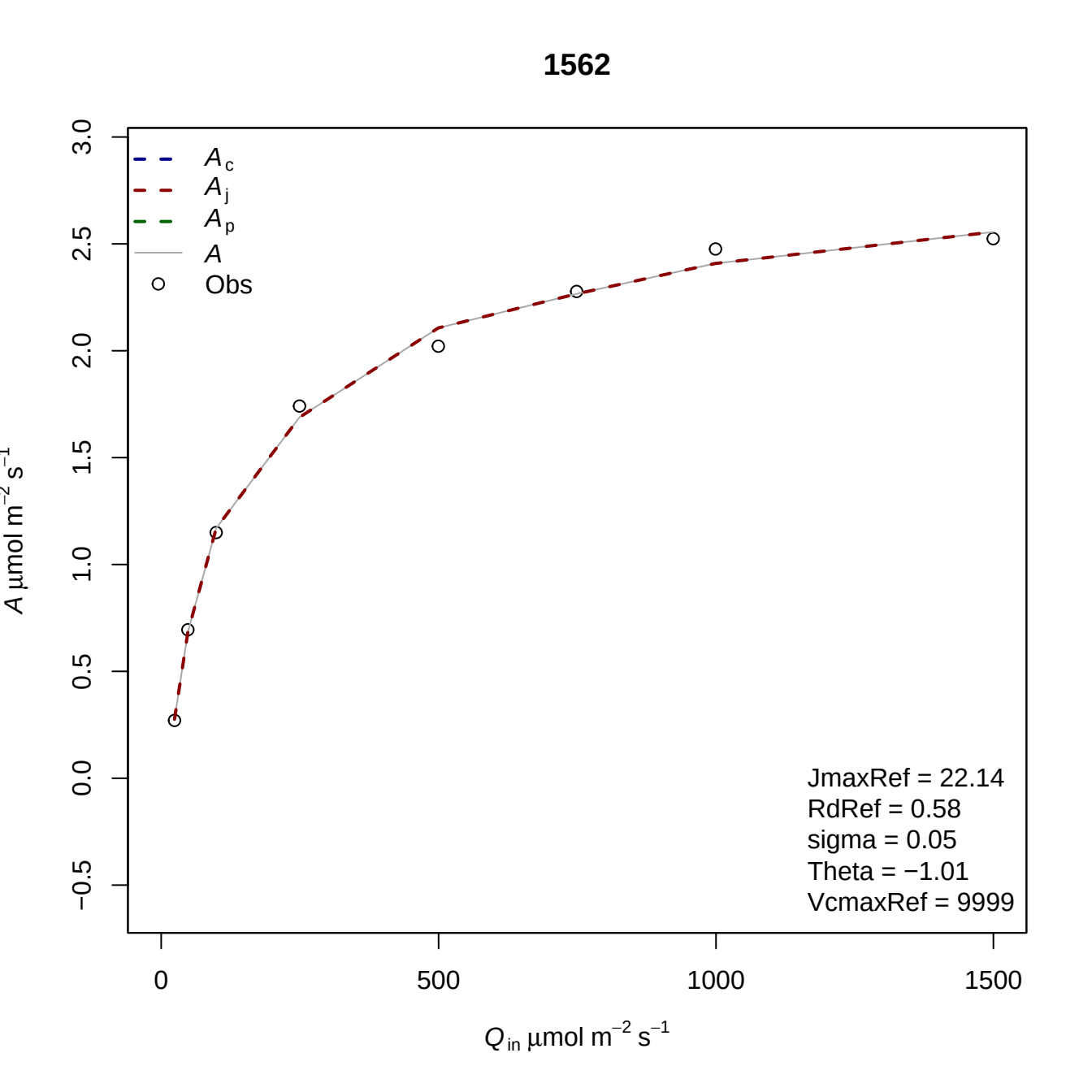

1557
A_c
A_j
A_p
A
Obs
JmaxRef = 73.91
RdRef = 0.55
sigma = 0.09
Theta = 0.27
VcmaxRef = 47.75
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

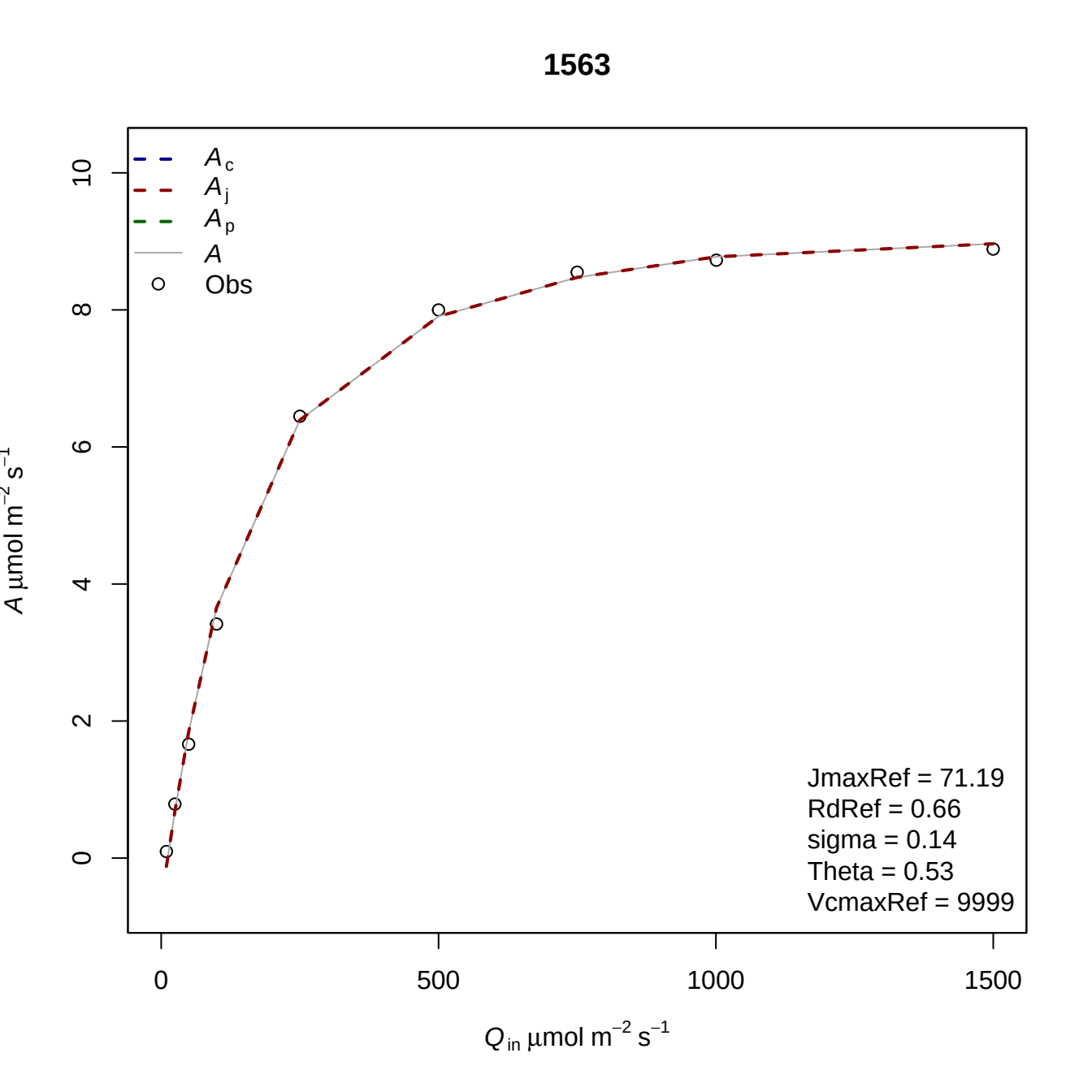

1558
A_c
A_j
A_p
A
Obs
JmaxRef = 65.93
RdRef = 0.75
sigma = 0.18
Theta = 0.59
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

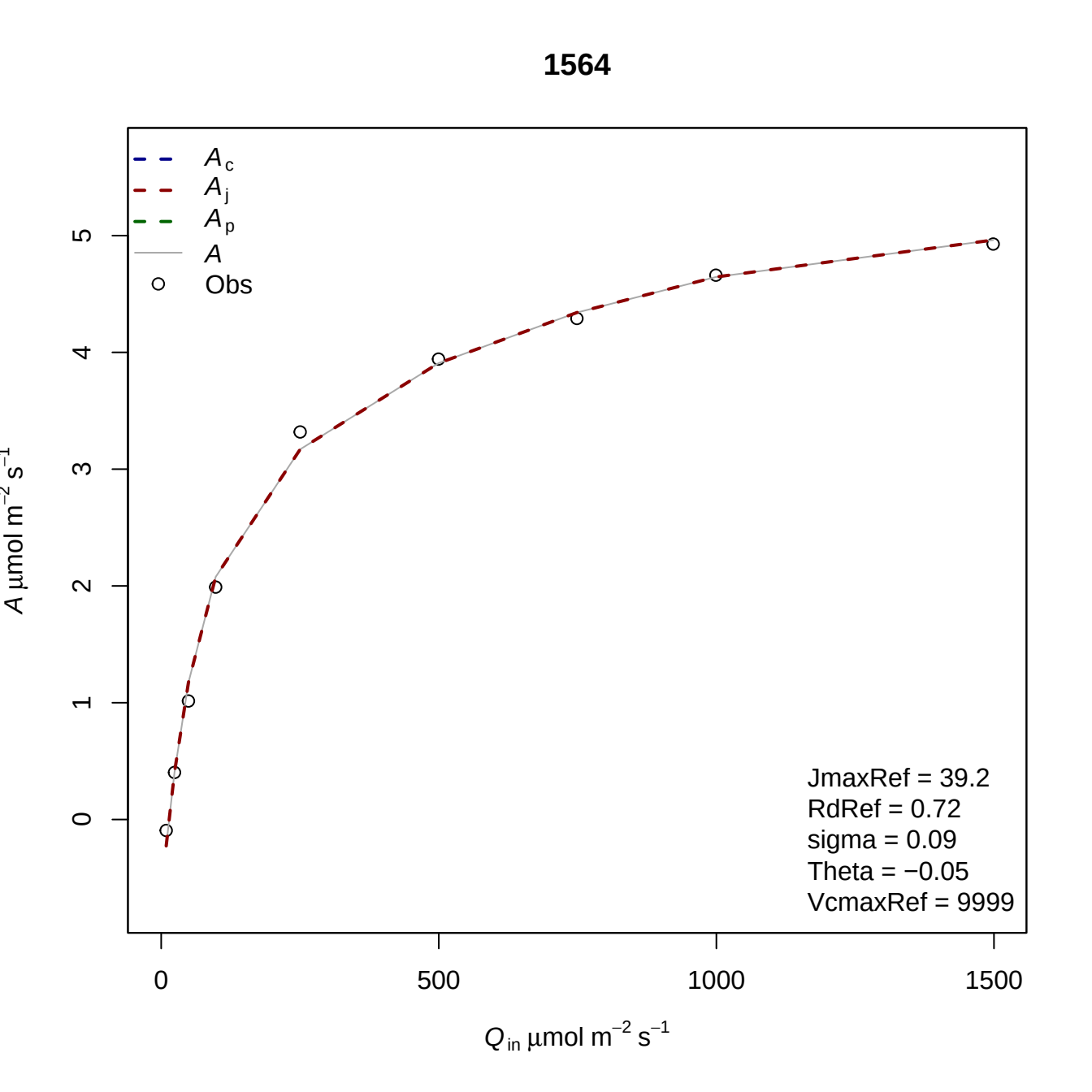

1559
A_c
A_j
A_p
A
Obs
JmaxRef = 57.21
RdRef = 0.71
sigma = 0.12
Theta = 0.56
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

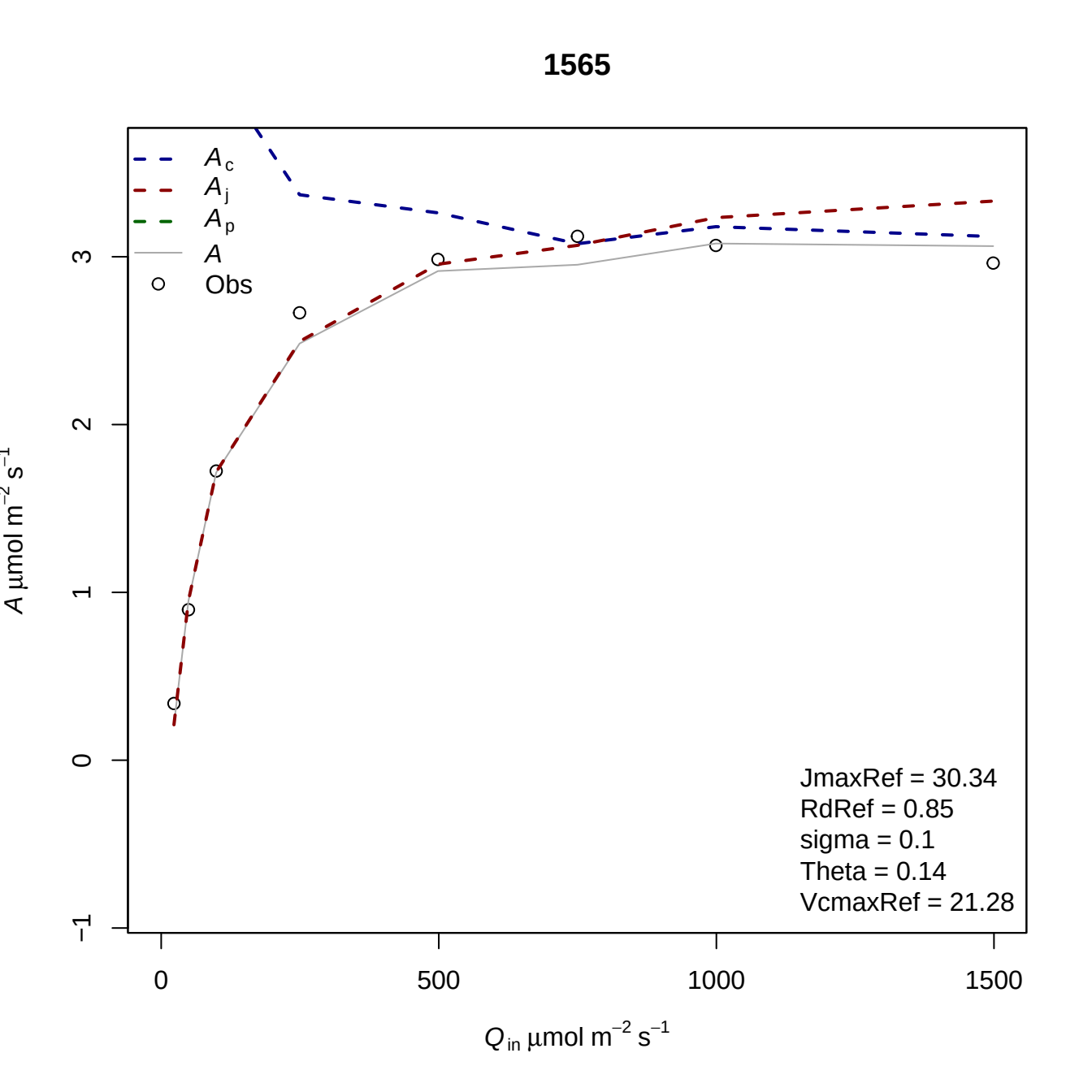

1560
A_c
A_j
A_p
A
Obs
JmaxRef = 32.06
RdRef = 0.65
sigma = 0.1
Theta = −0.25
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

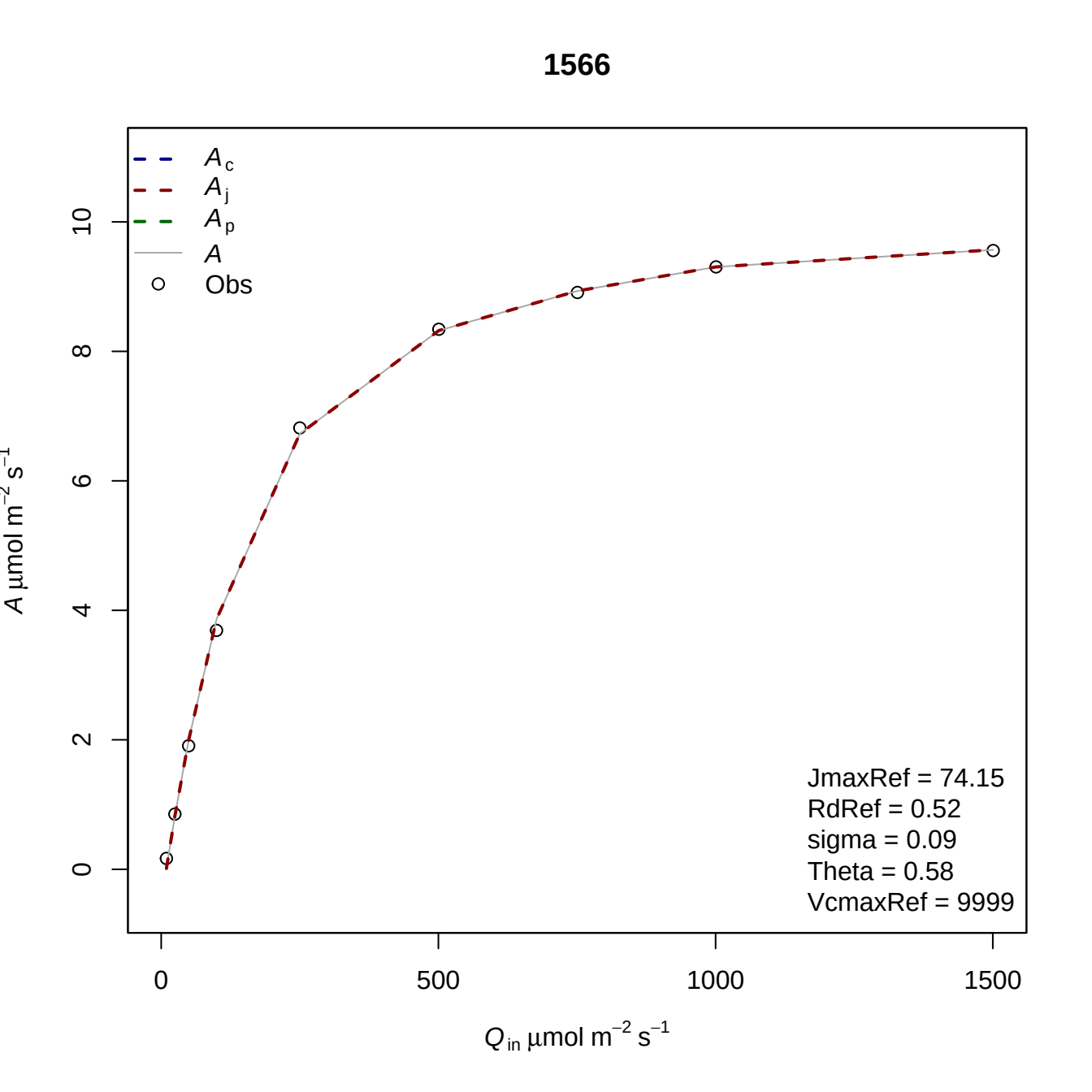

1561
A_c
A_j
A_p
A
Obs
JmaxRef = 38.19
RdRef = 0.63
sigma = 0.09
Theta = 0.19
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

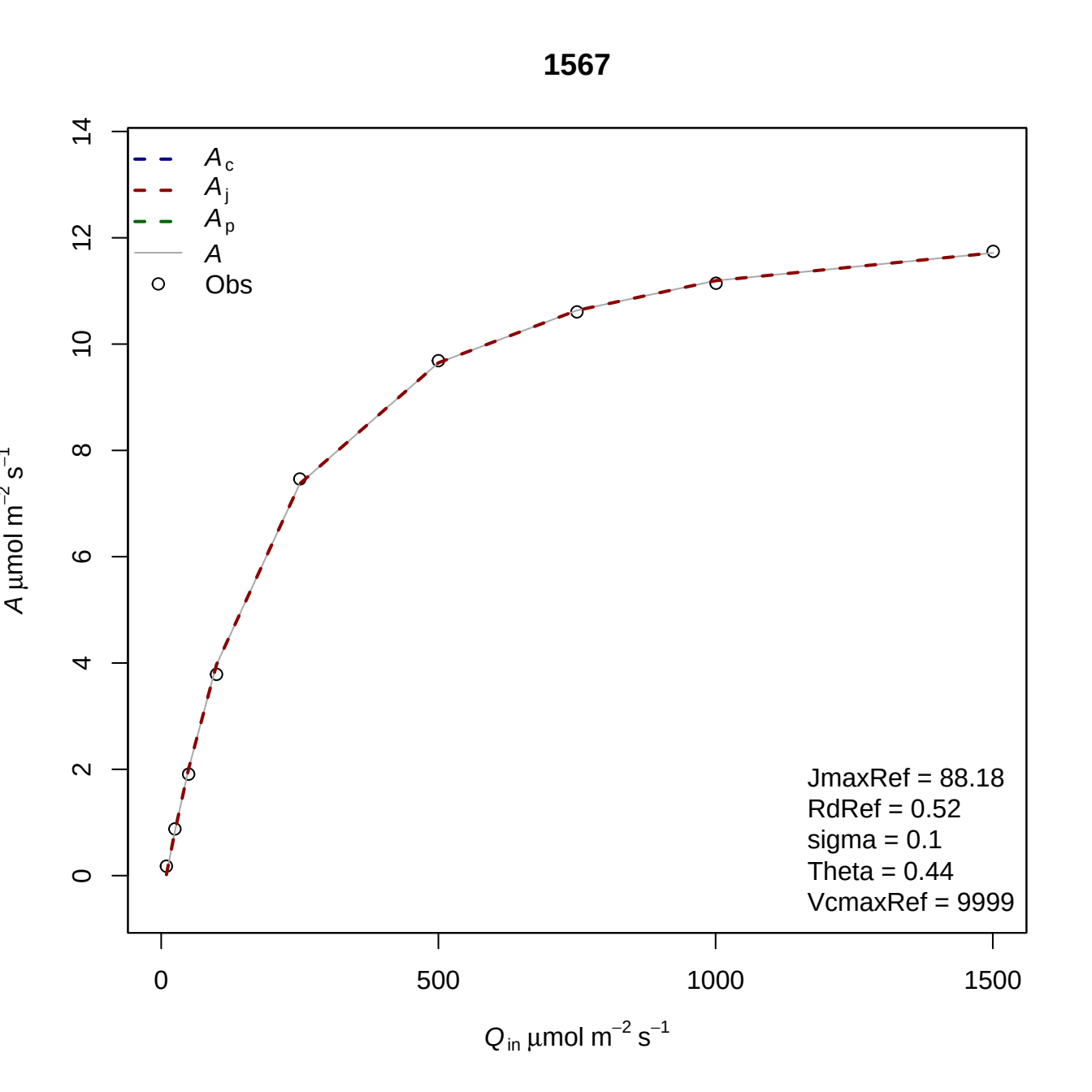

1562
A_c
A_j
A_p
A
Obs
JmaxRef = 22.14
RdRef = 0.58
sigma = 0.05
Theta = −1.01
VcmaxRef = 9999
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

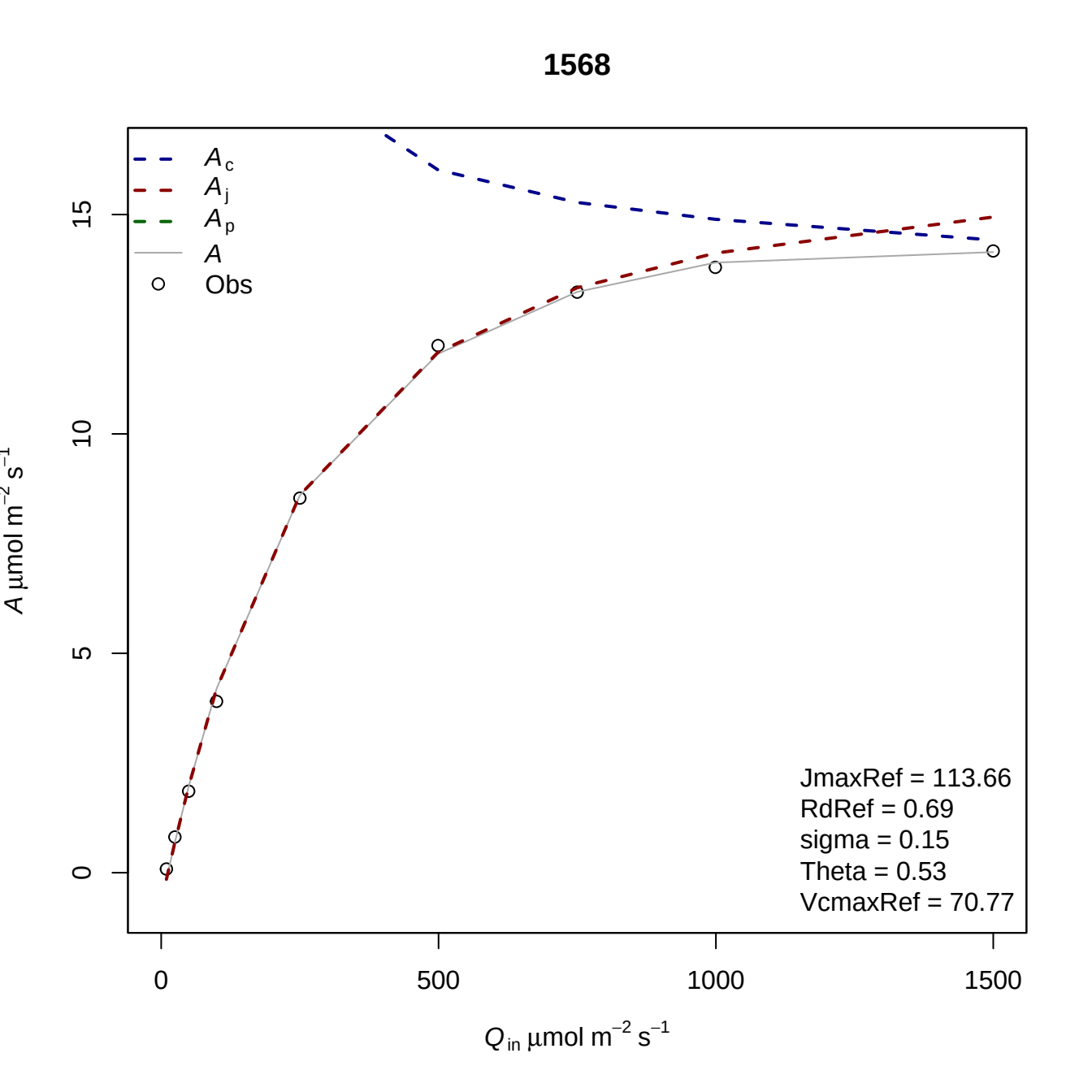

1563
A_c
A_j
A_p
A
Obs
JmaxRef = 71.19
RdRef = 0.66
sigma = 0.14
Theta = 0.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

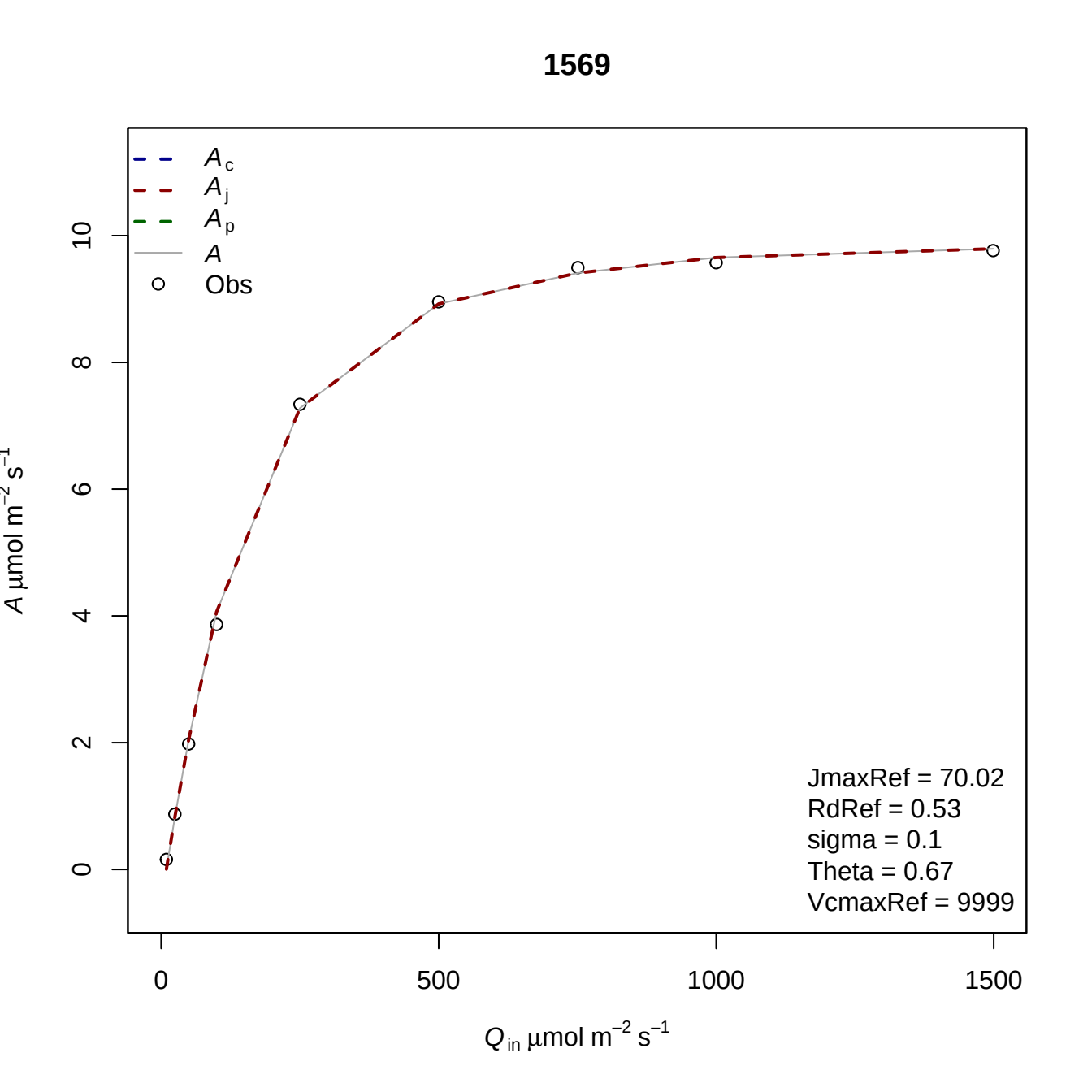

1564
A_c
A_j
A_p
A
Obs
JmaxRef = 39.2
RdRef = 0.72
sigma = 0.09
Theta = −0.05
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

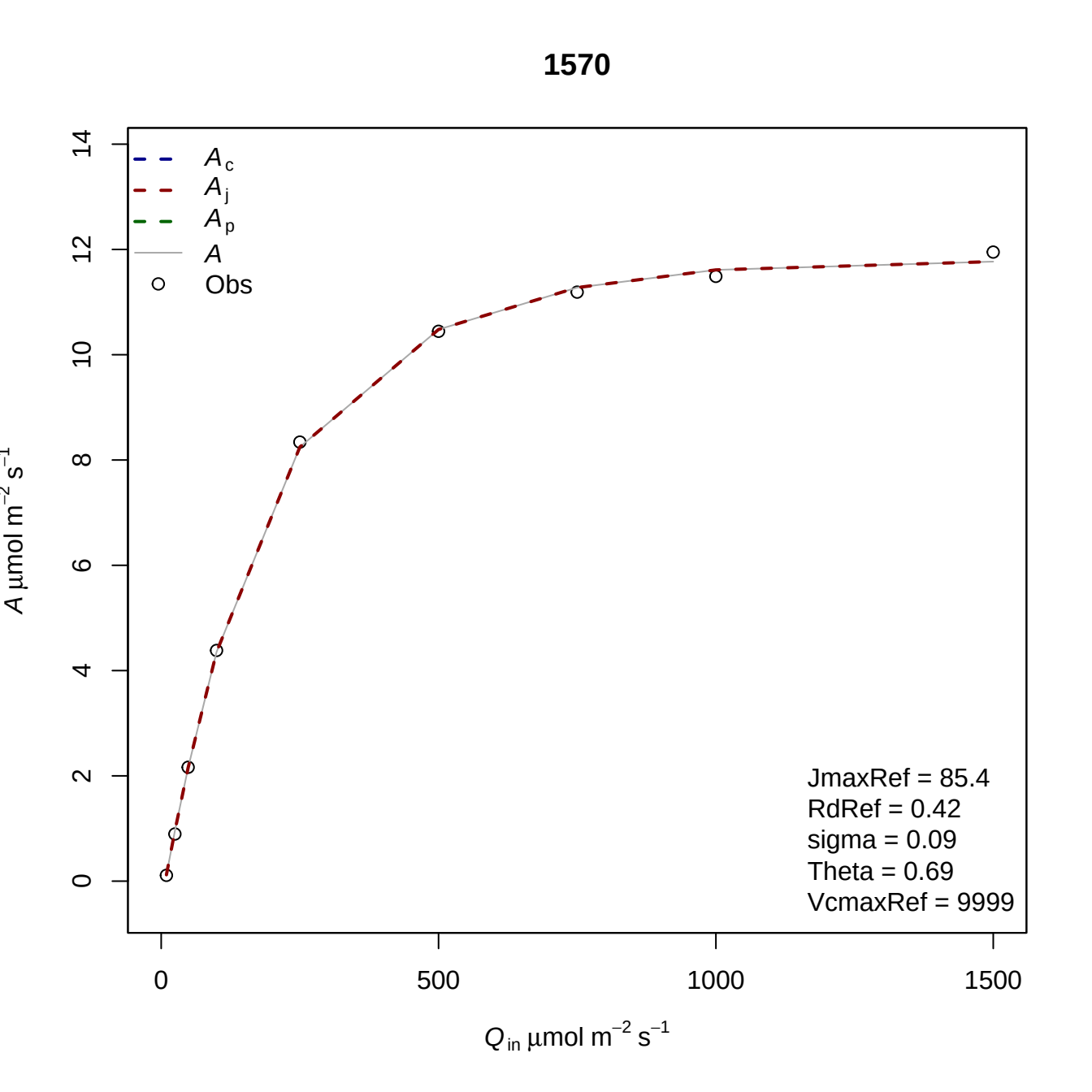

1565
A_c
A_j
A_p
A
Obs
JmaxRef = 30.34
RdRef = 0.85
sigma = 0.1
Theta = 0.14
VcmaxRef = 21.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

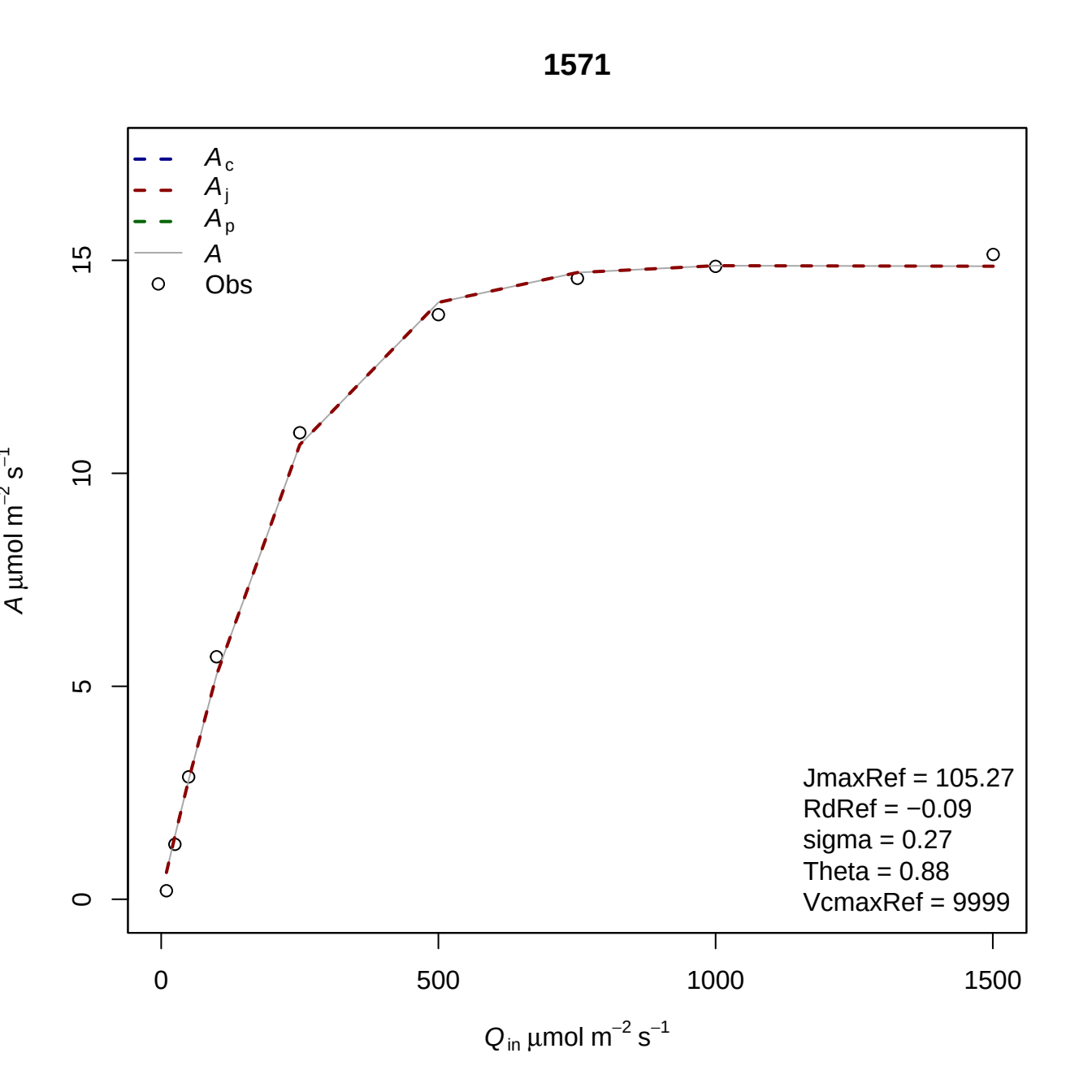

1566
A_c
A_j
A_p
A
Obs
JmaxRef = 74.15
RdRef = 0.52
sigma = 0.09
Theta = 0.58
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

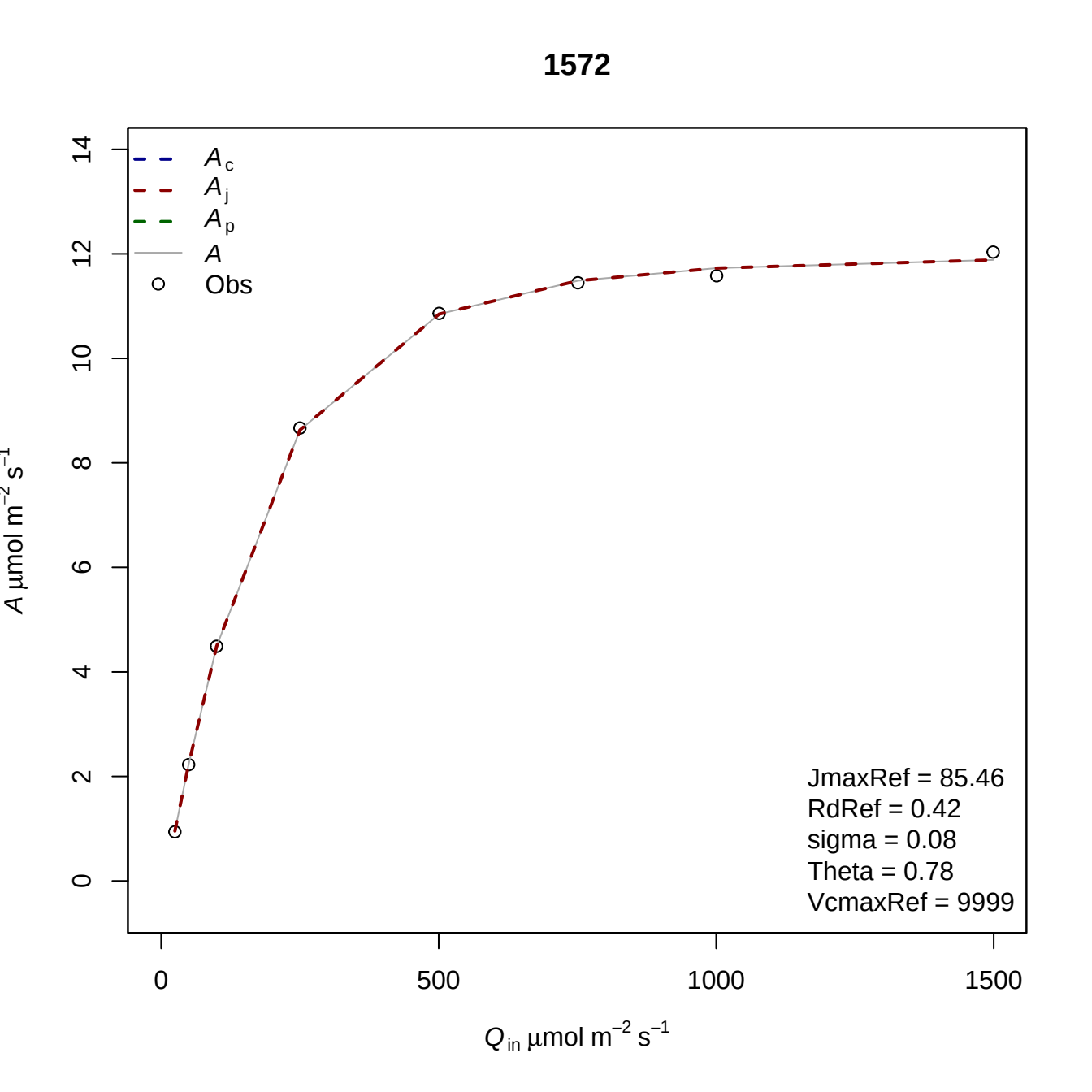

1567
A_c
A_j
A_p
A
Obs
JmaxRef = 88.18
RdRef = 0.52
sigma = 0.1
Theta = 0.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

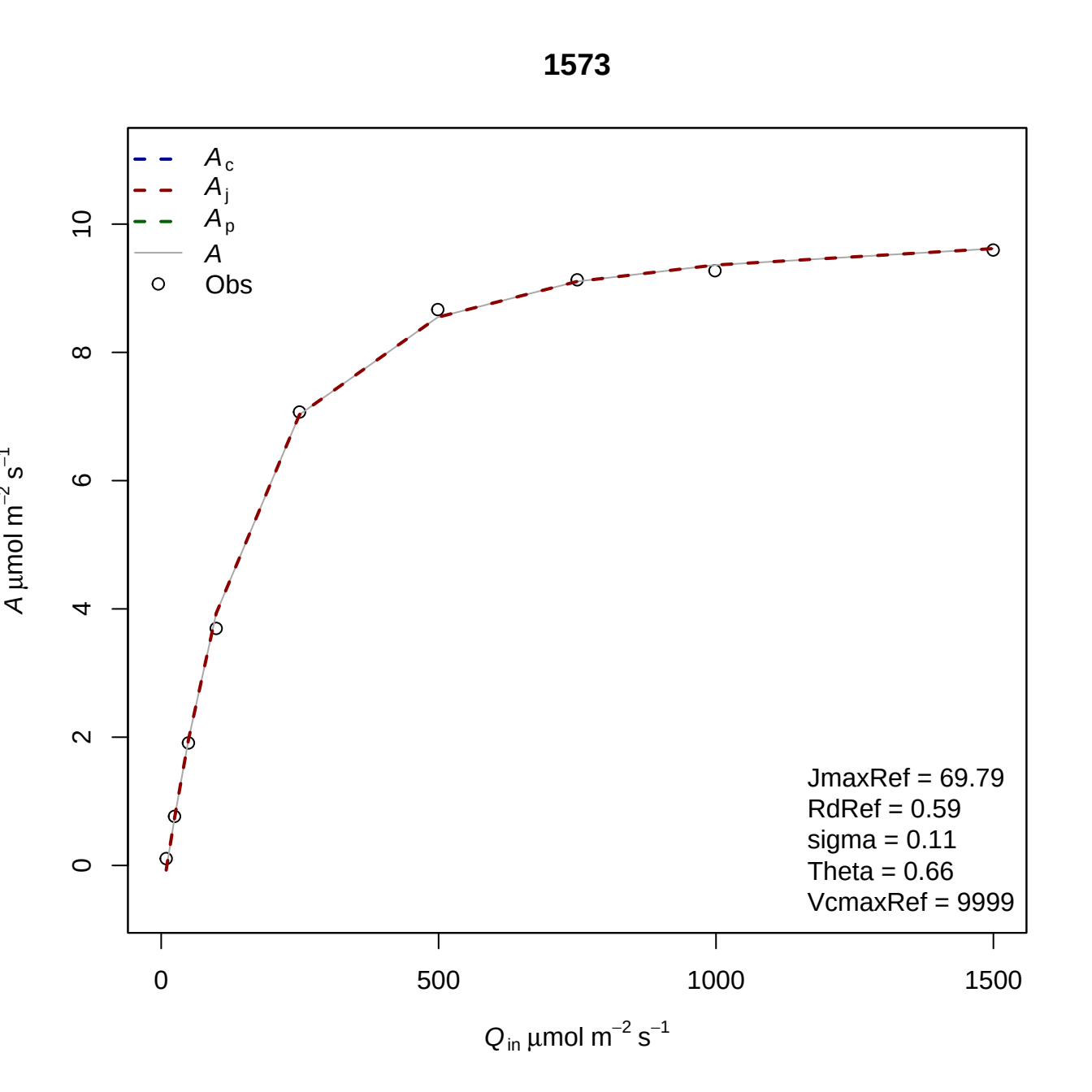

1568
A_c
A_j
A_p
A
Obs
JmaxRef = 113.66
RdRef = 0.69
sigma = 0.15
Theta = 0.53
VcmaxRef = 70.77
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

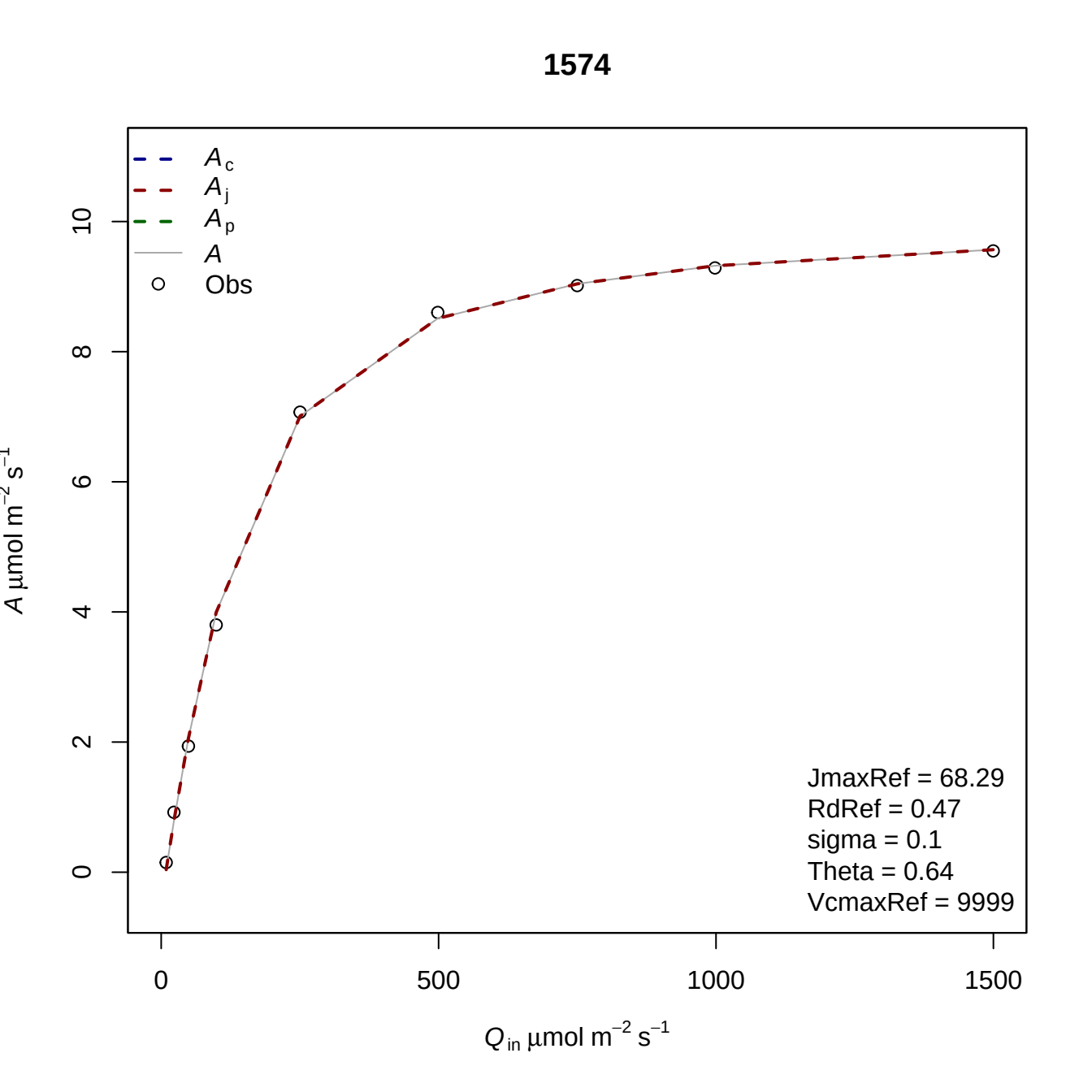

1569
A_c
A_j
A_p
A
Obs
JmaxRef = 70.02
RdRef = 0.53
sigma = 0.1
Theta = 0.67
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

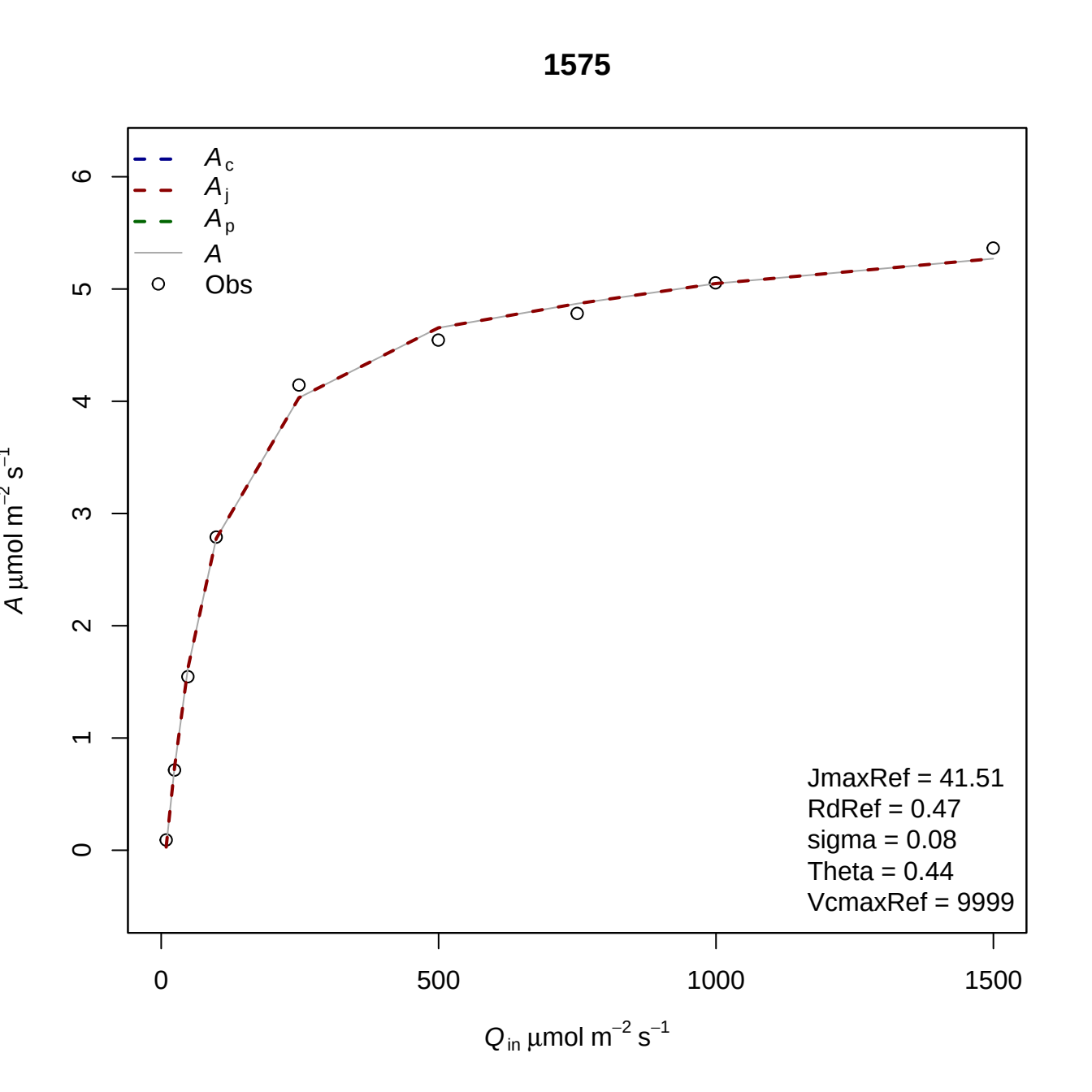

1570
A_c
A_j
A_p
A
Obs
JmaxRef = 85.4
RdRef = 0.42
sigma = 0.09
Theta = 0.69
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

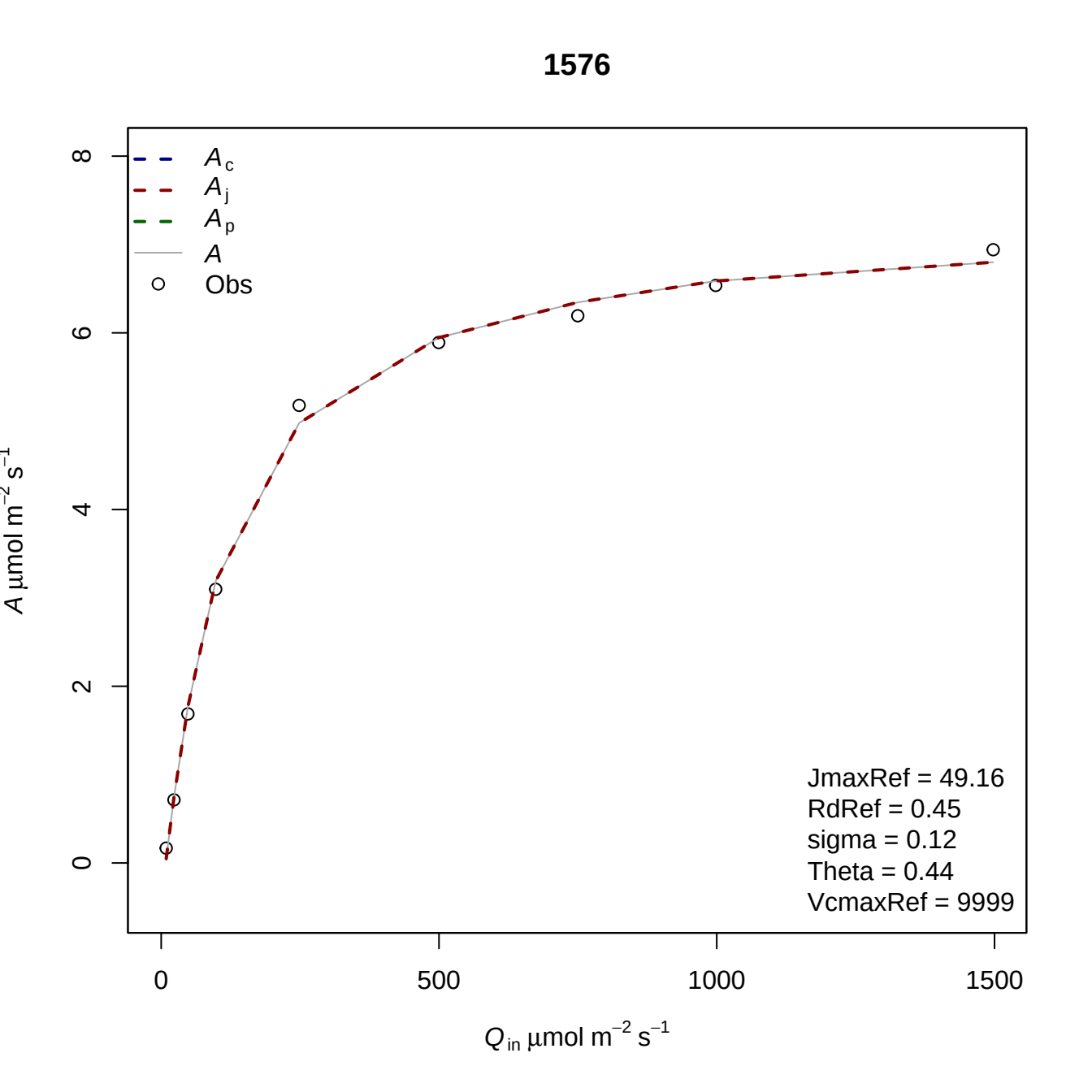

1571
A_c
A_j
A_p
A
Obs
JmaxRef = 105.27
RdRef = −0.09
sigma = 0.27
Theta = 0.88
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

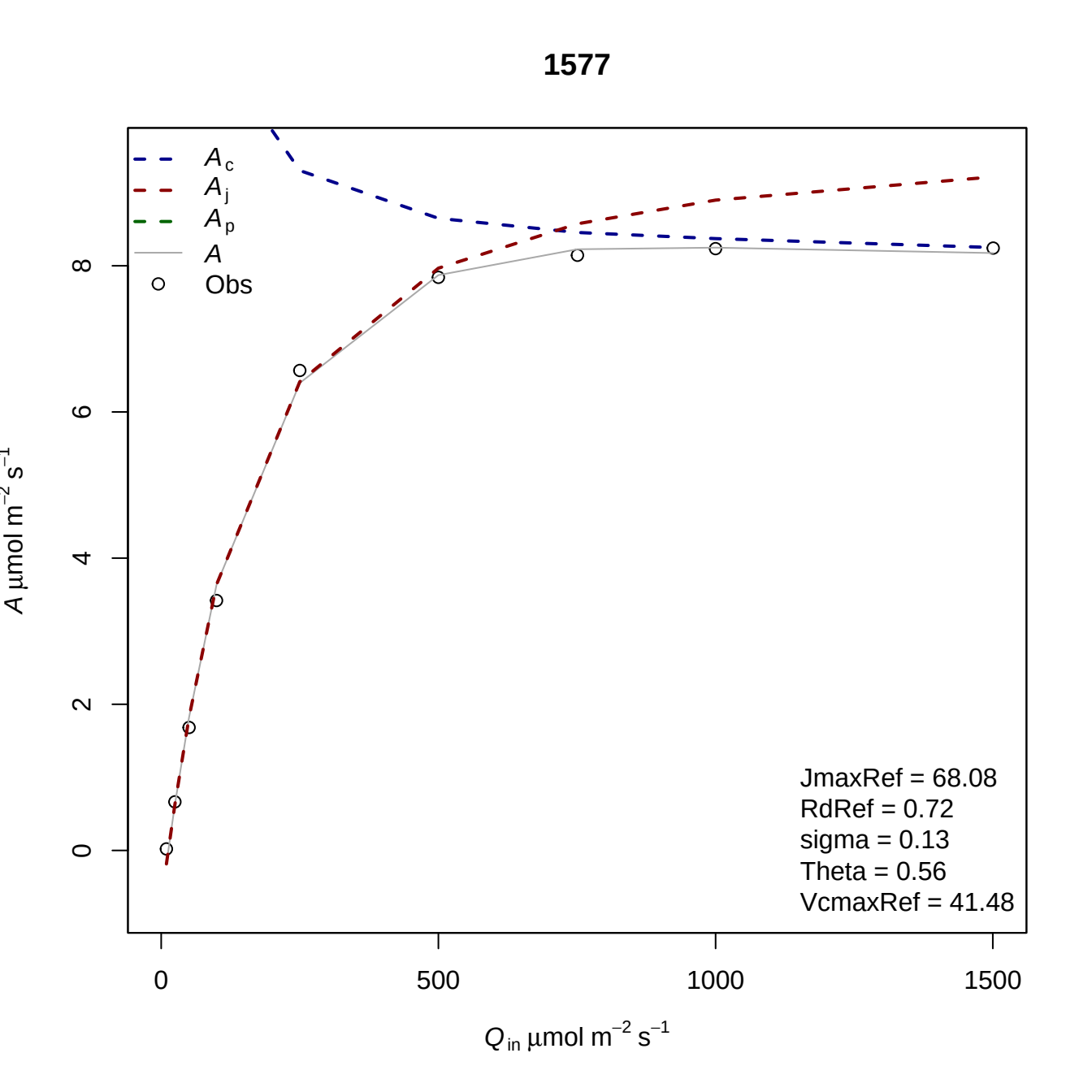

1572
A_c
A_j
A_p
A
Obs
JmaxRef = 85.46
RdRef = 0.42
sigma = 0.08
Theta = 0.78
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

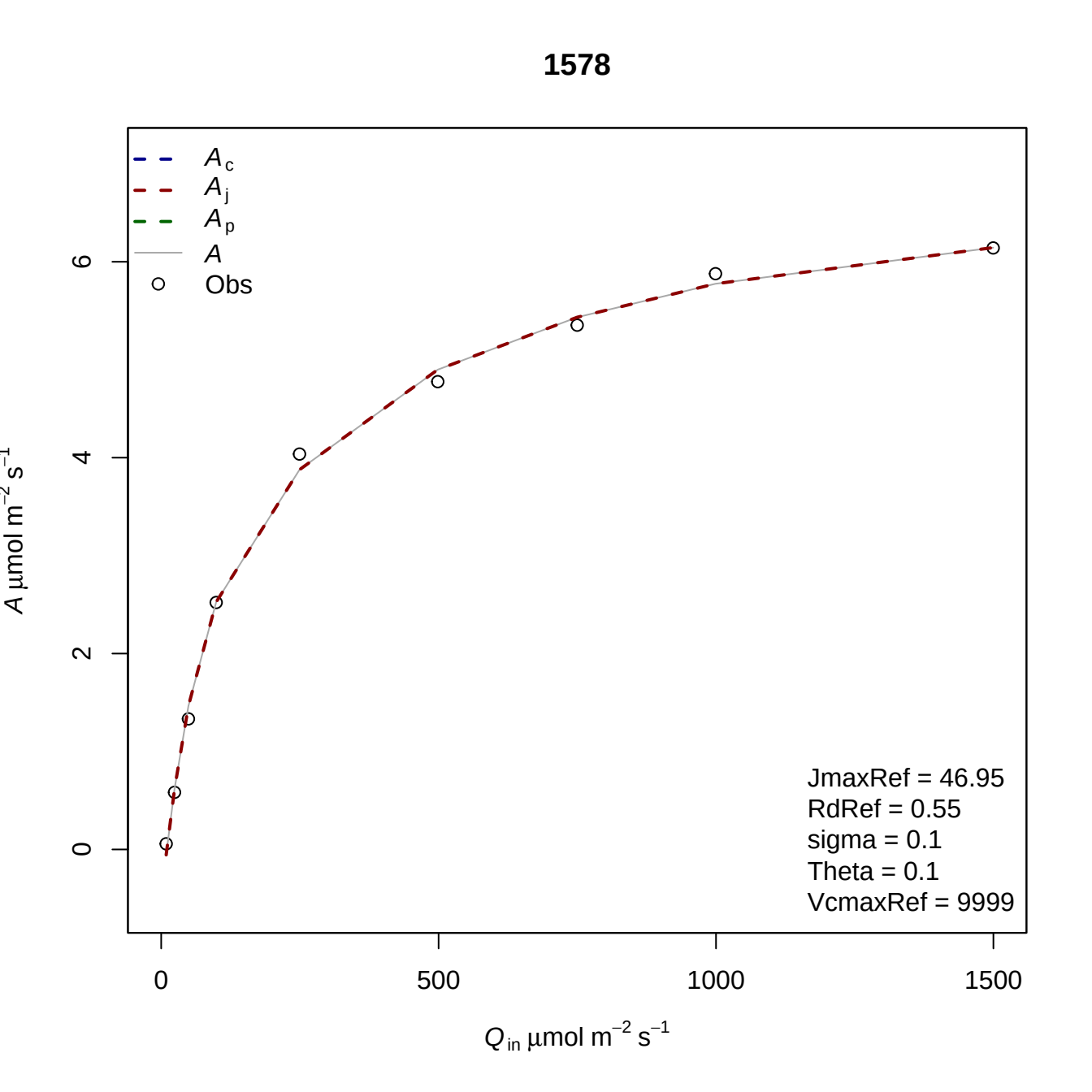

1573
A_c
A_j
A_p
A
Obs
JmaxRef = 69.79
RdRef = 0.59
sigma = 0.11
Theta = 0.66
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

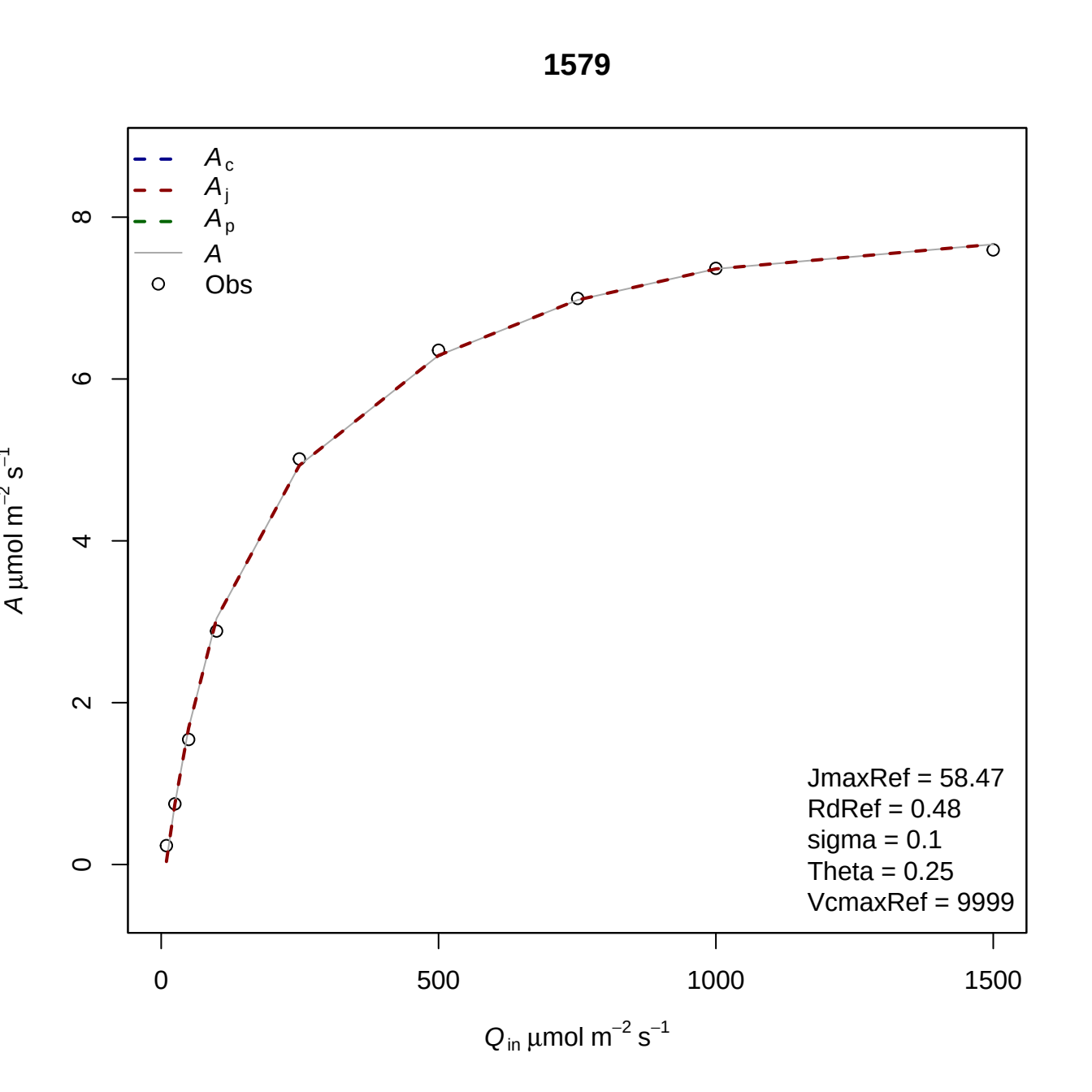

1574
A_c
A_j
A_p
A
Obs
JmaxRef = 68.29
RdRef = 0.47
sigma = 0.1
Theta = 0.64
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

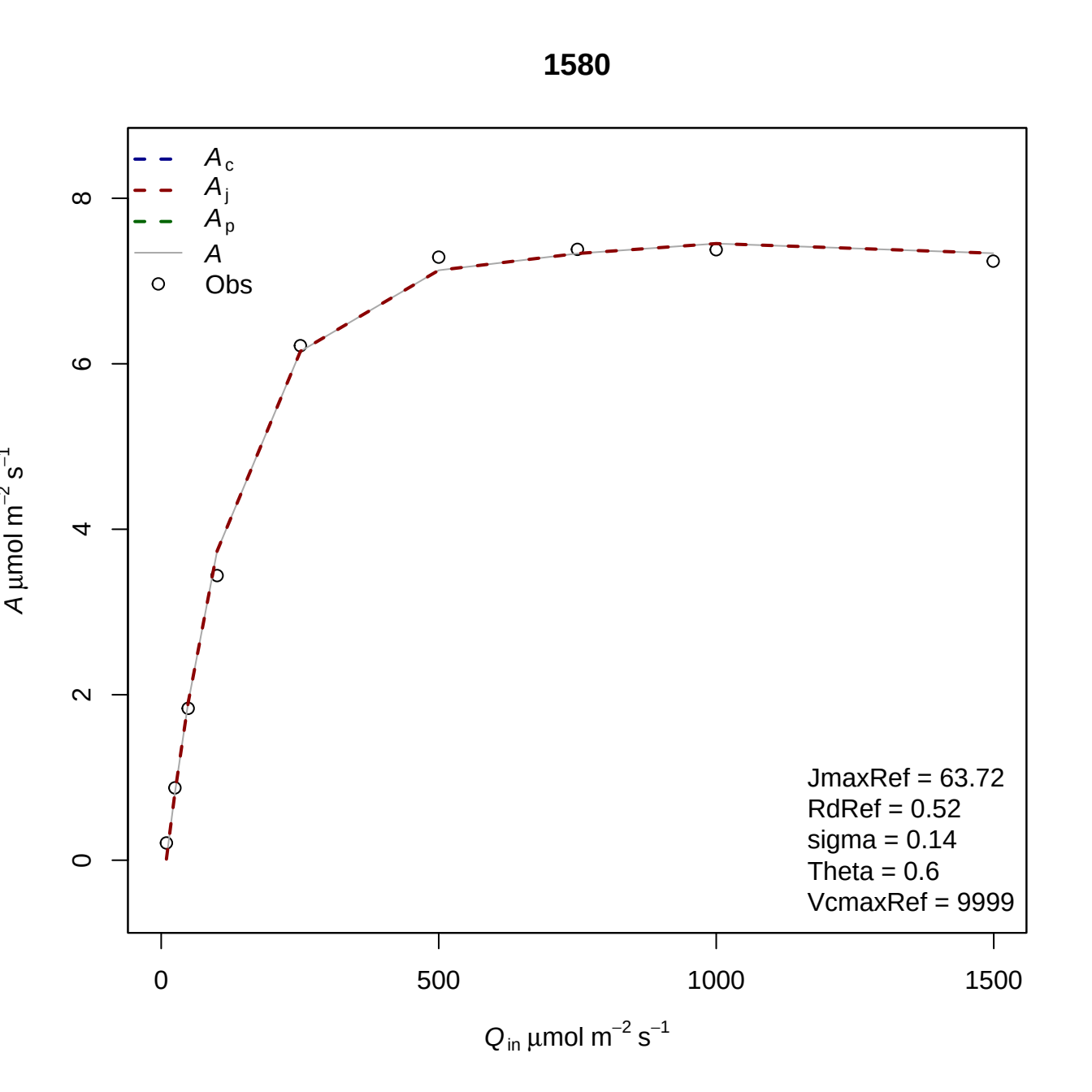

1575
A_c
A_j
A_p
A
Obs
JmaxRef = 41.51
RdRef = 0.47
sigma = 0.08
Theta = 0.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

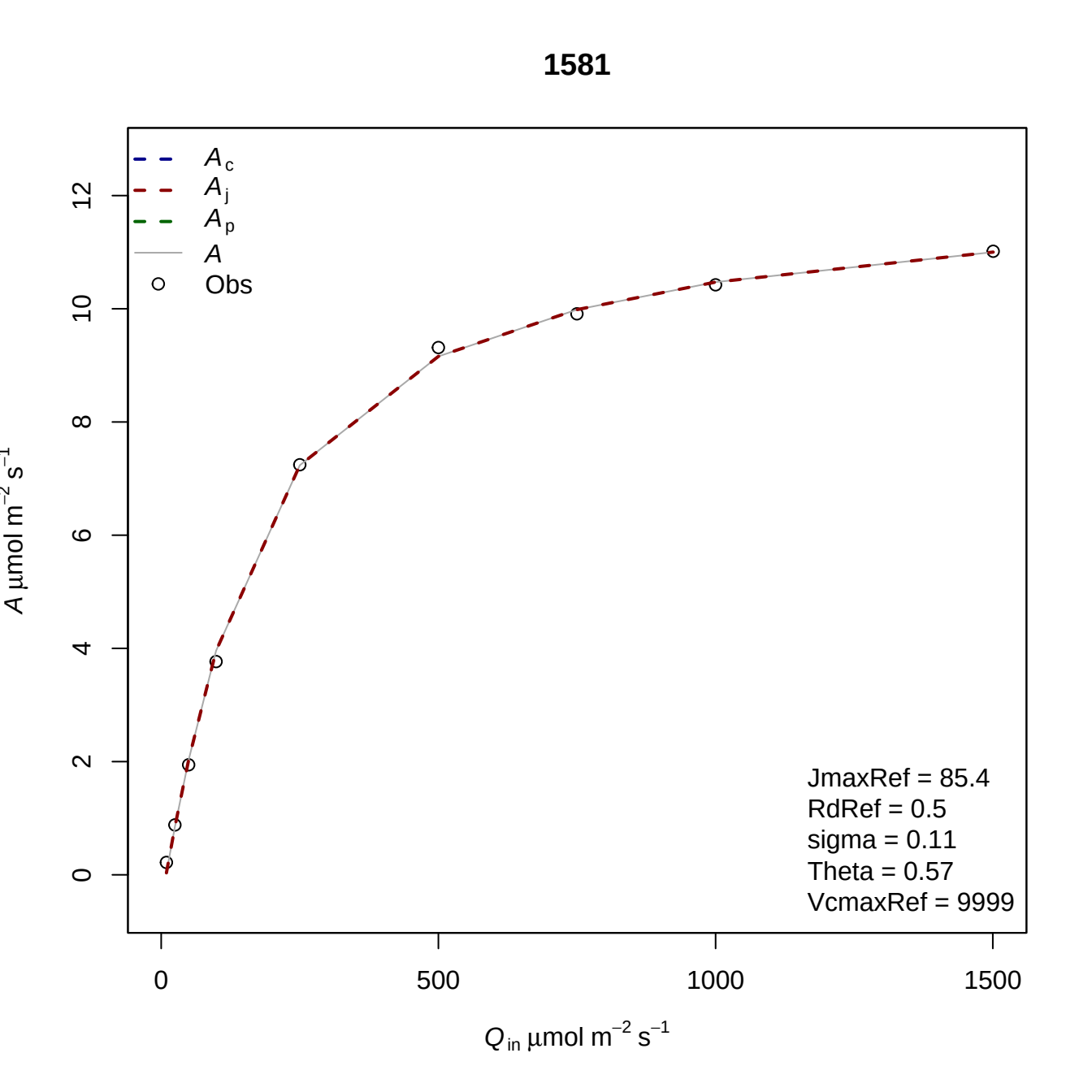

1576
A_c
A_j
A_p
A
Obs
JmaxRef = 49.16
RdRef = 0.45
sigma = 0.12
Theta = 0.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

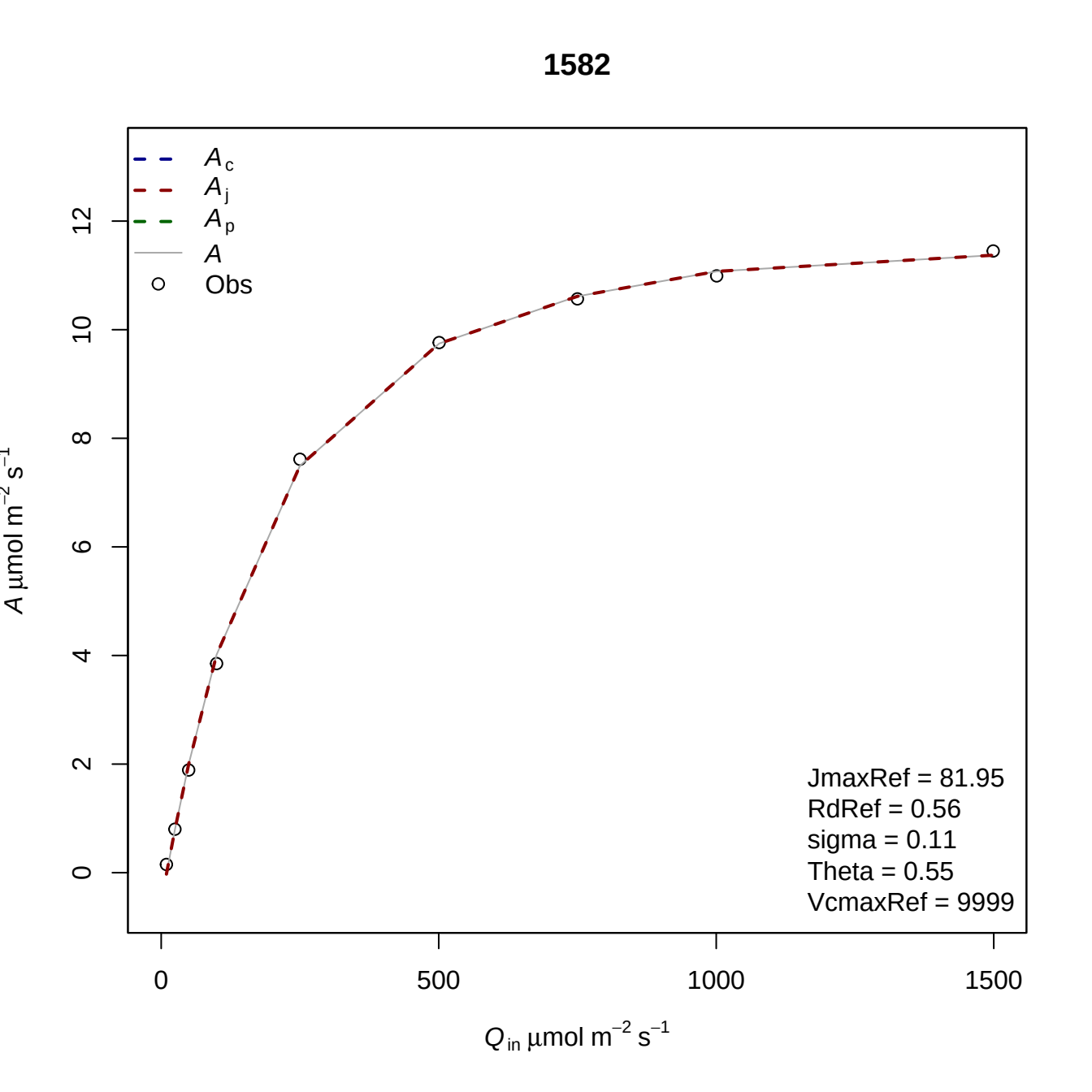

1577
A_c
A_j
A_p
A
Obs
JmaxRef = 68.08
RdRef = 0.72
sigma = 0.13
Theta = 0.56
VcmaxRef = 41.48
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

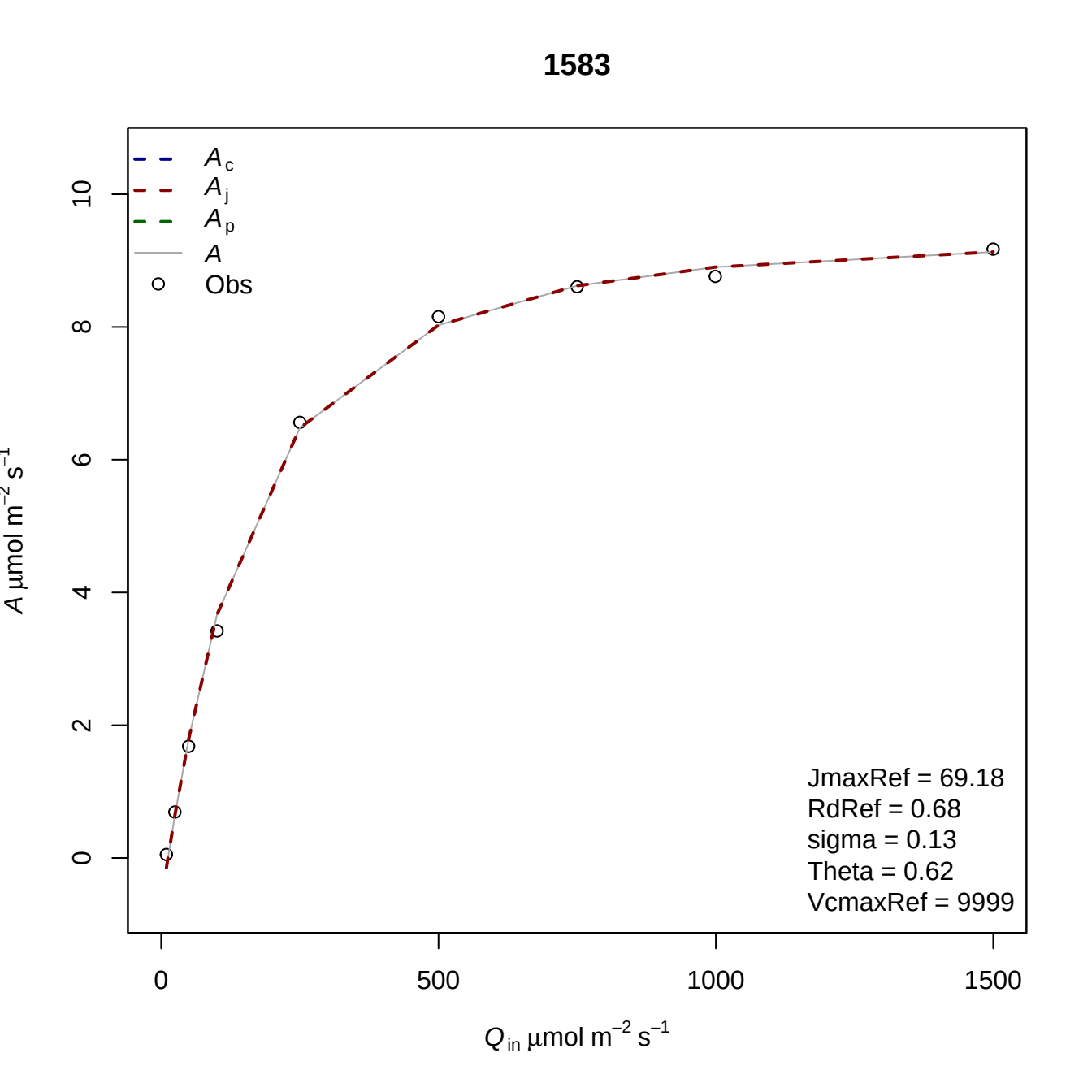

1578
A_c
A_j
A_p
A
Obs
JmaxRef = 46.95
RdRef = 0.55
sigma = 0.1
Theta = 0.1
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

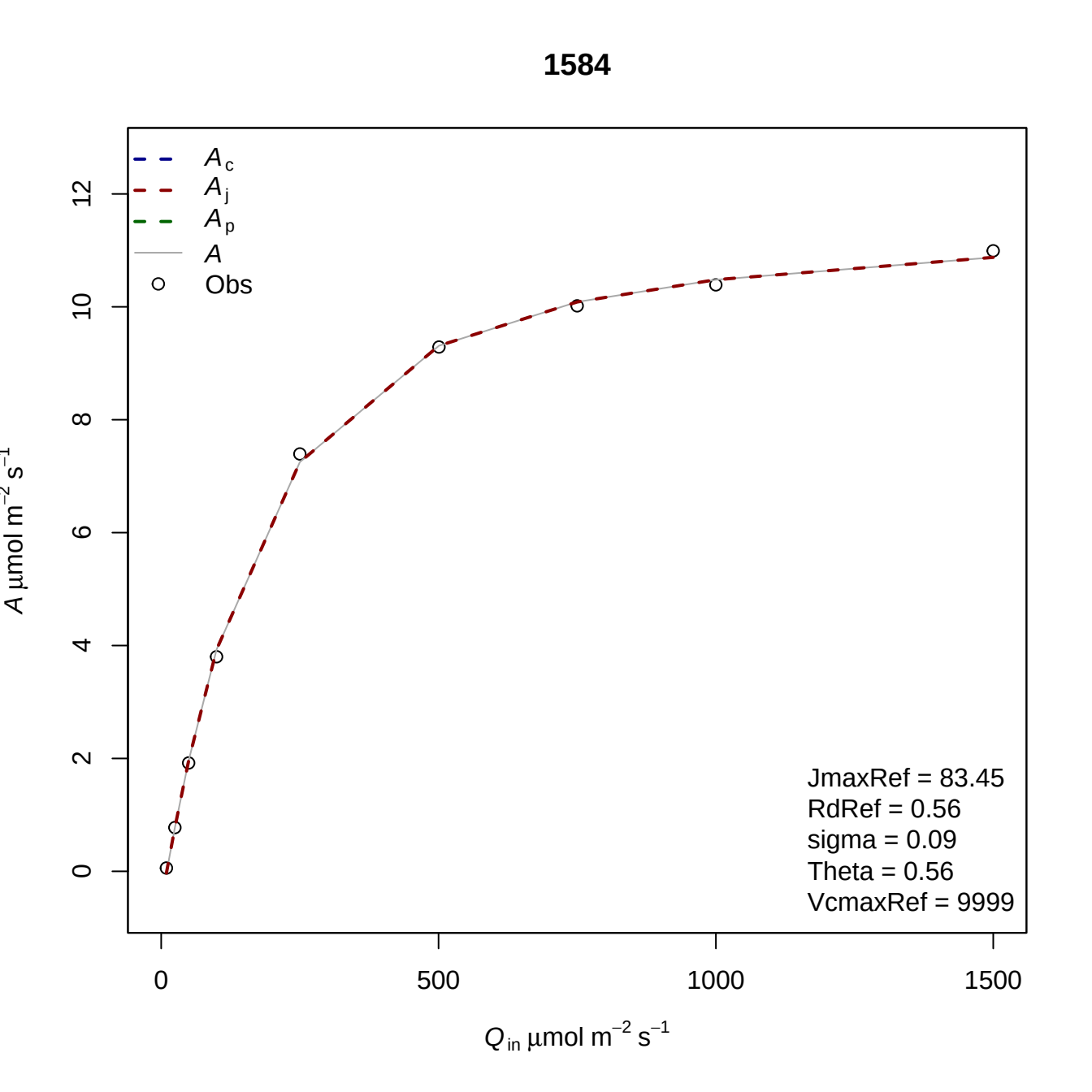

1579
A_c
A_j
A_p
A
Obs
JmaxRef = 58.47
RdRef = 0.48
sigma = 0.1
Theta = 0.25
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

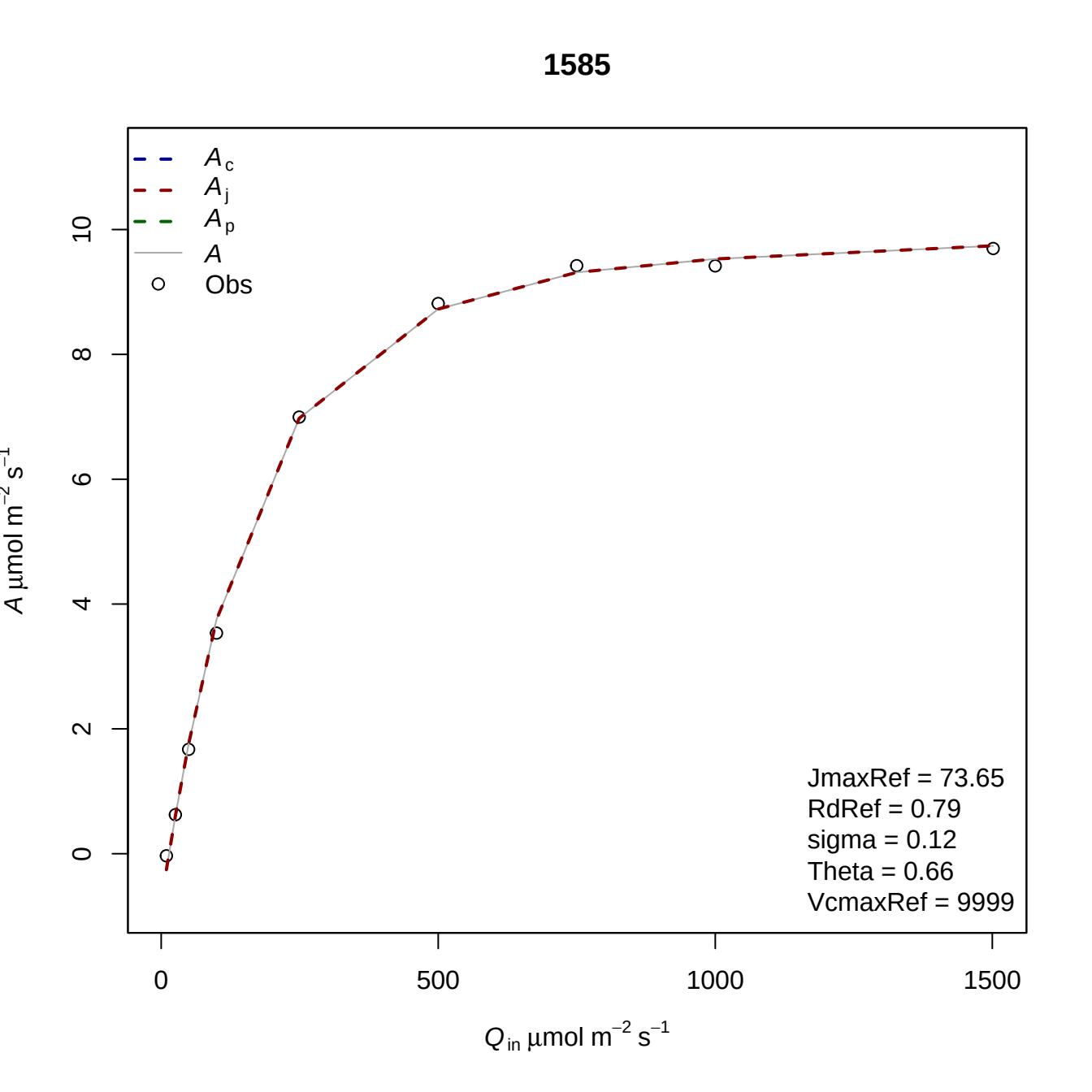

1580
A_c
A_j
A_p
A
Obs
JmaxRef = 63.72
RdRef = 0.52
sigma = 0.14
Theta = 0.6
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

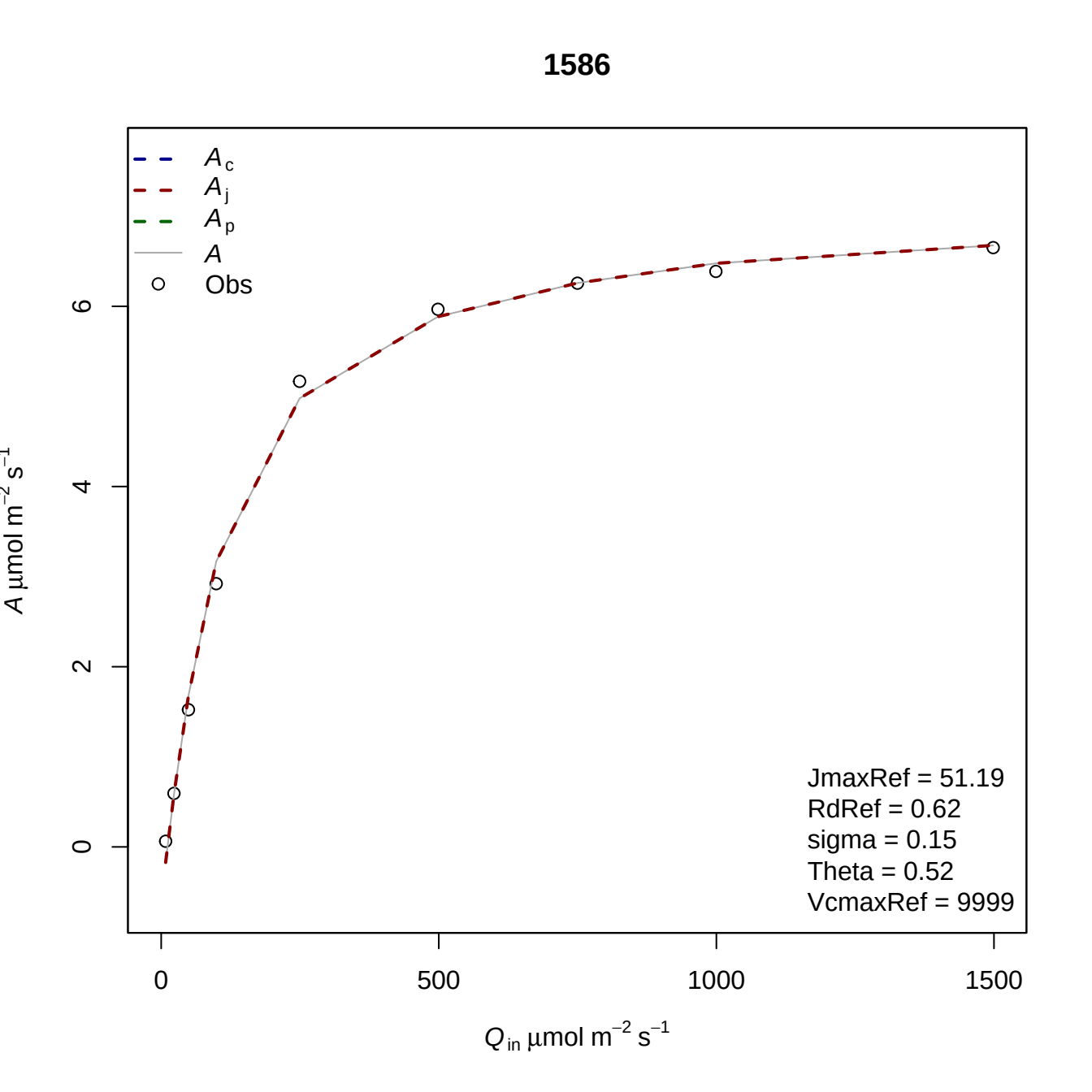

1581
A_c
A_j
A_p
A
Obs
JmaxRef = 85.4
RdRef = 0.5
sigma = 0.11
Theta = 0.57
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

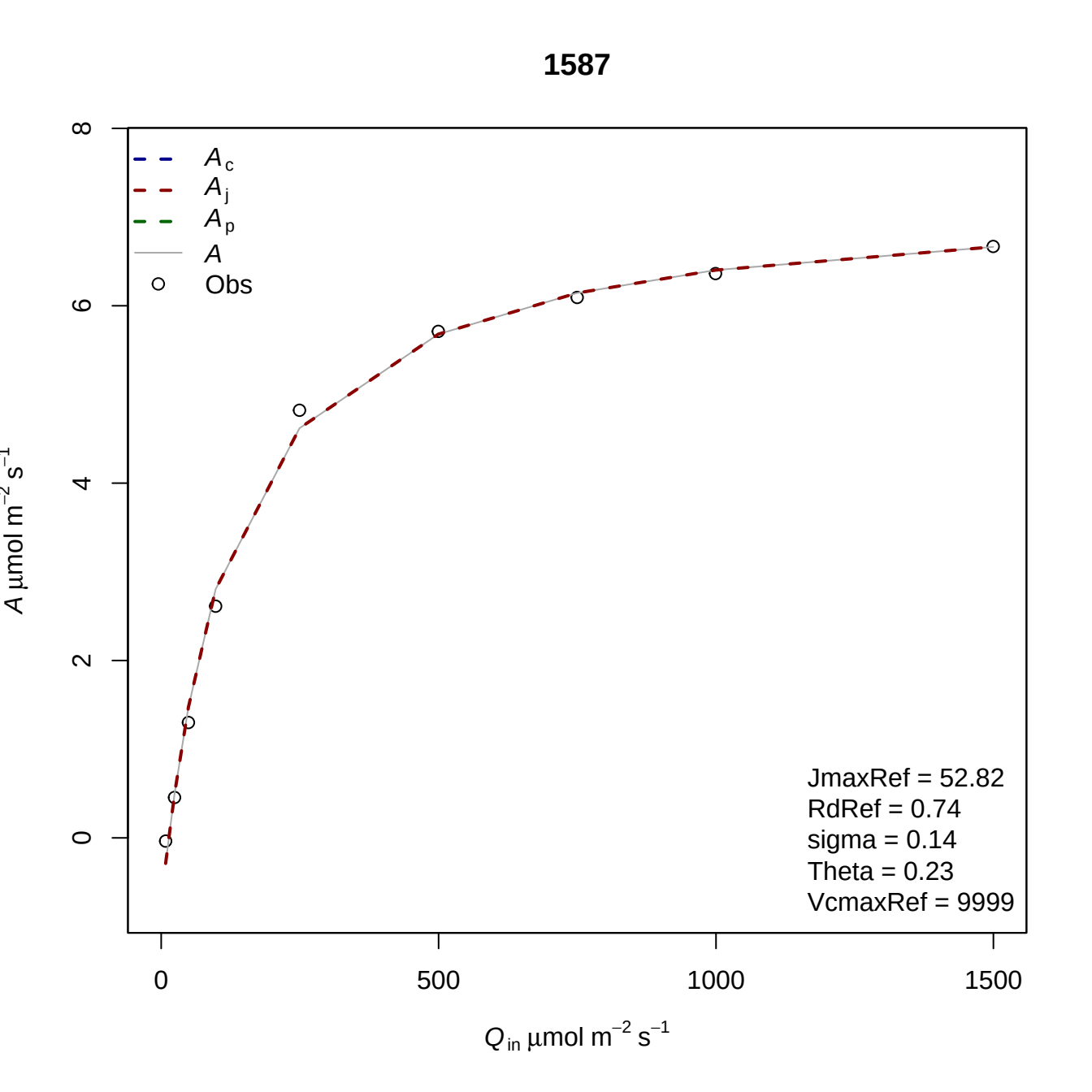

1582
A_c
A_j
A_p
A
Obs
JmaxRef = 81.95
RdRef = 0.56
sigma = 0.11
Theta = 0.55
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

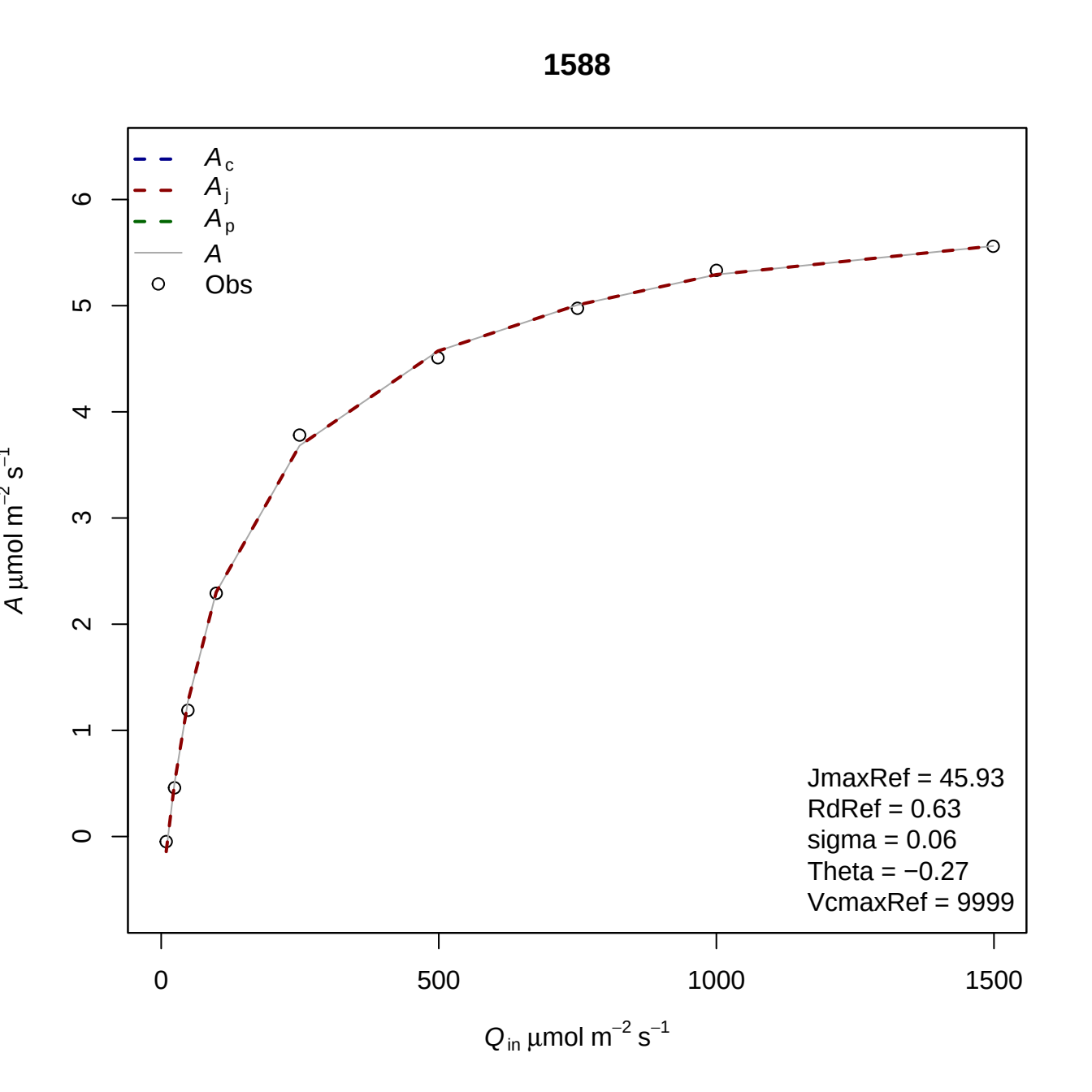

1583
A_c
A_j
A_p
A
Obs
JmaxRef = 69.18
RdRef = 0.68
sigma = 0.13
Theta = 0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

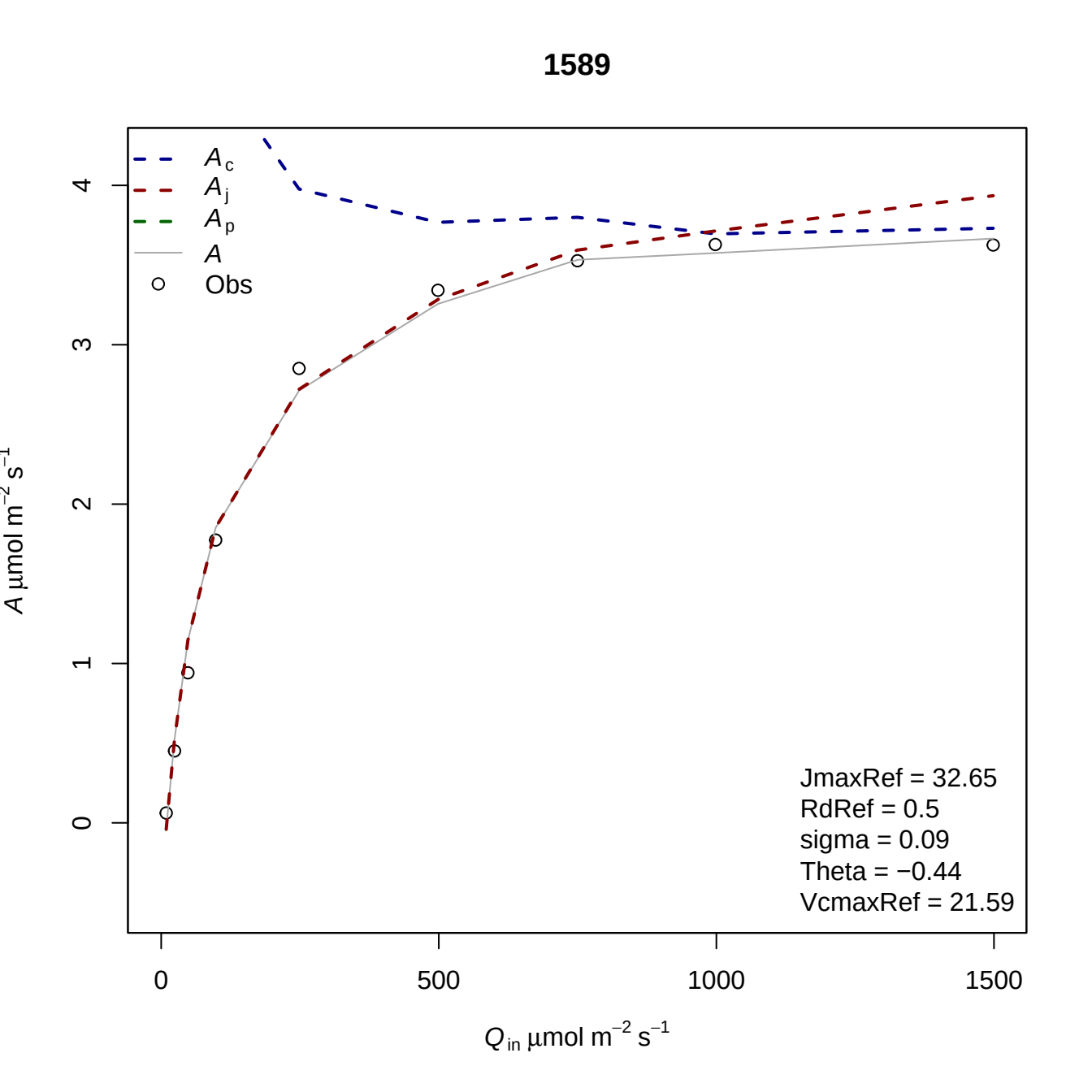

1584
A_c
A_j
A_p
A
Obs
JmaxRef = 83.45
RdRef = 0.56
sigma = 0.09
Theta = 0.56
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

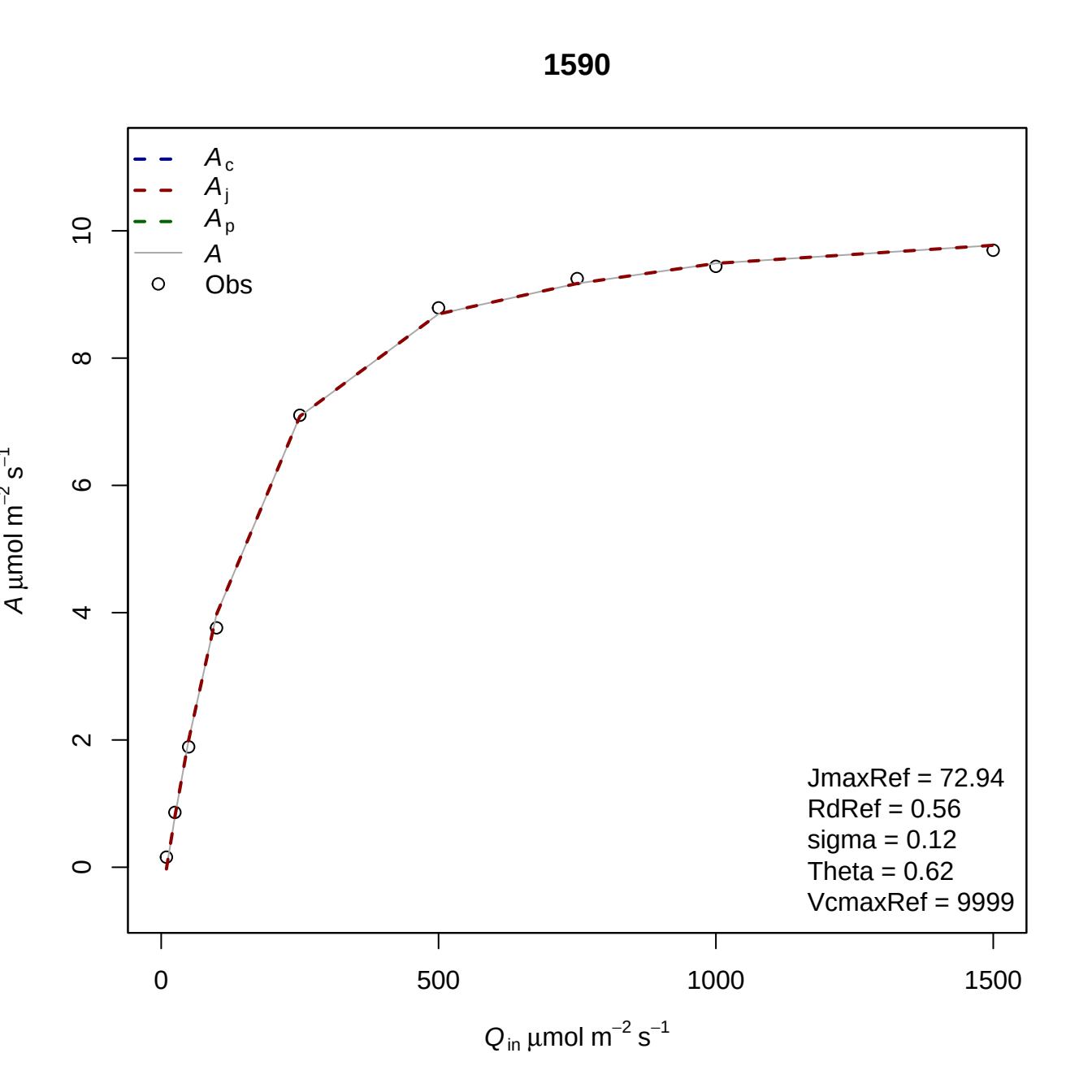

1585
A_c
A_j
A_p
A
Obs
JmaxRef = 73.65
RdRef = 0.79
sigma = 0.12
Theta = 0.66
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

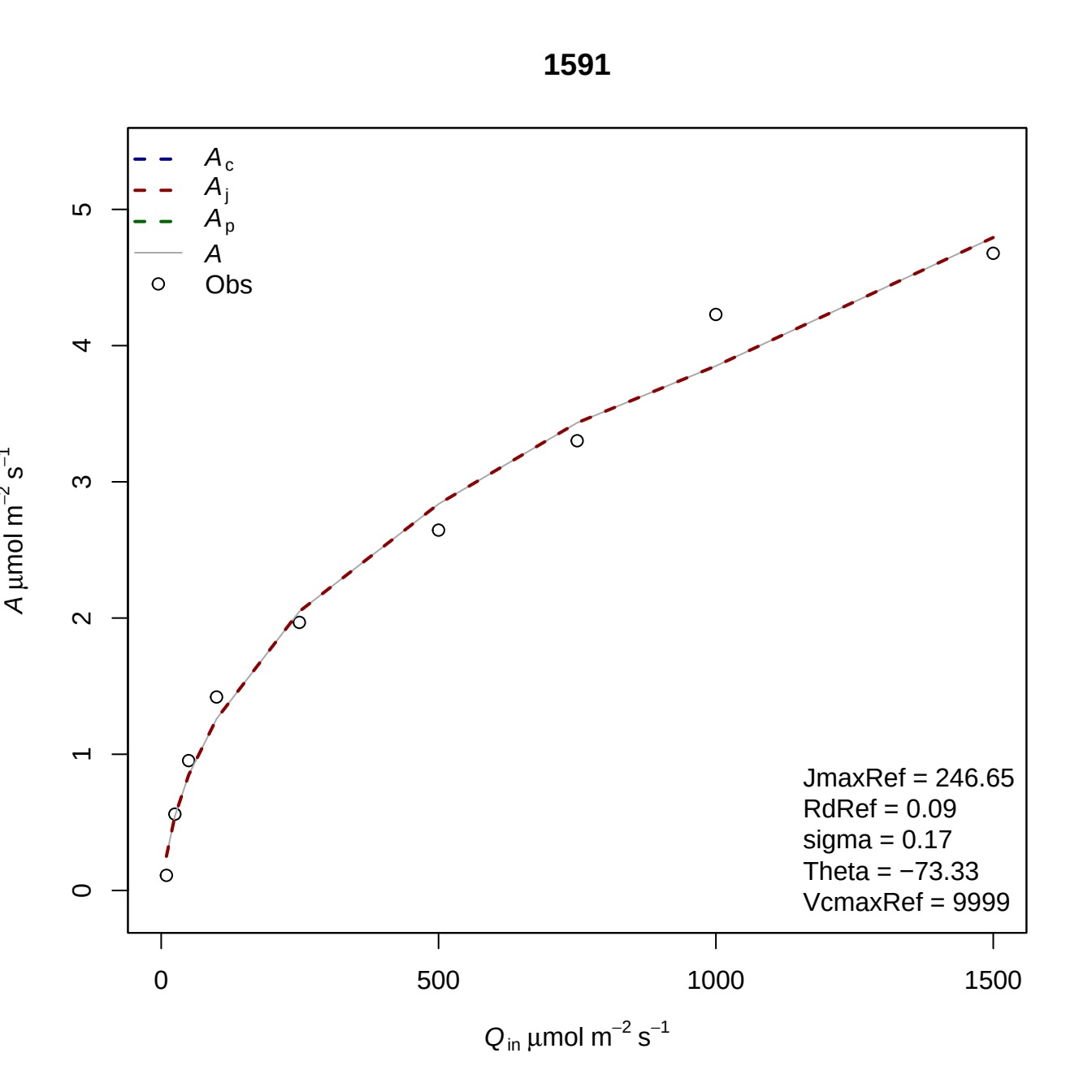

1586
A_c
A_j
A_p
A
Obs
JmaxRef = 51.19
RdRef = 0.62
sigma = 0.15
Theta = 0.52
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

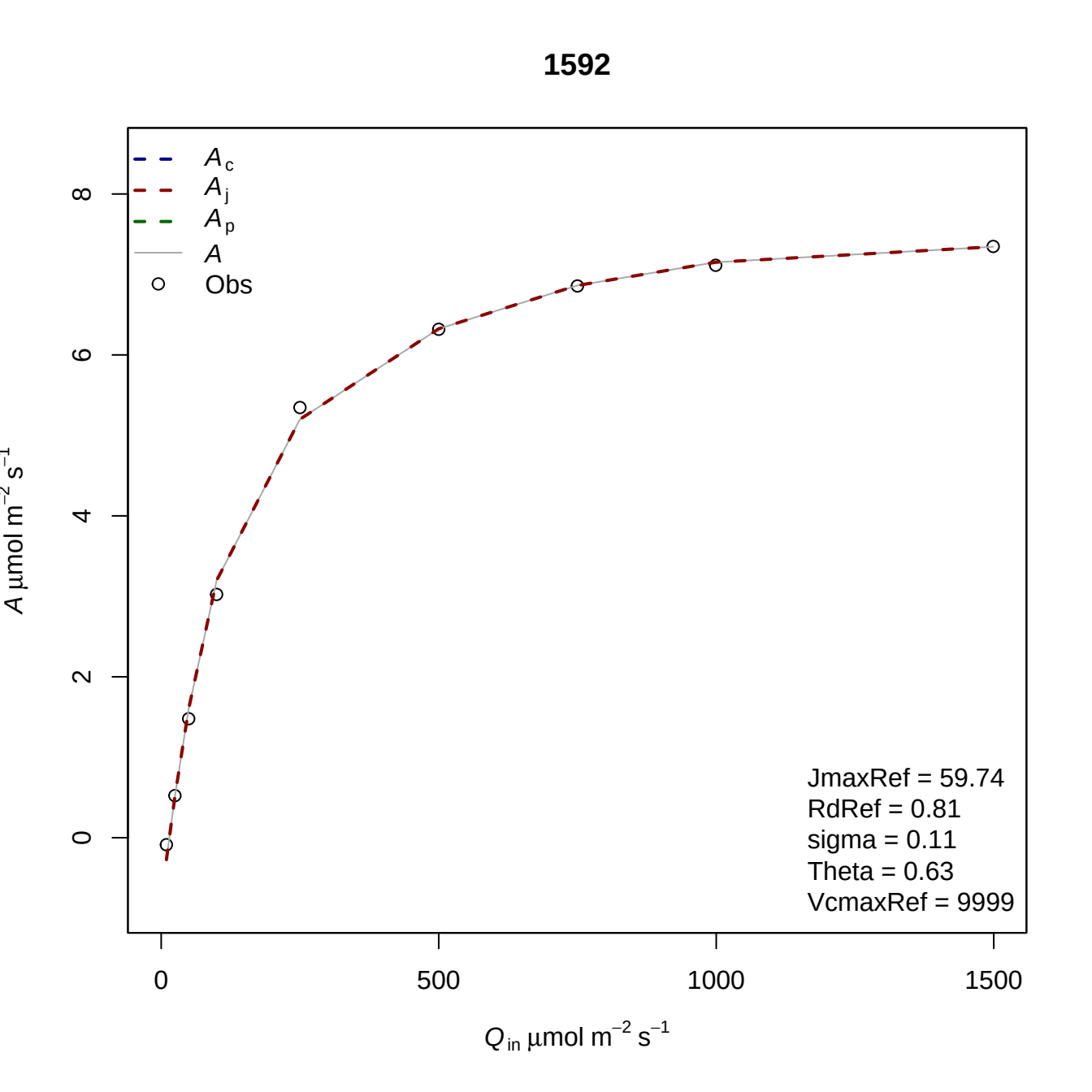

1587
A_c
A_j
A_p
A
Obs
JmaxRef = 52.82
RdRef = 0.74
sigma = 0.14
Theta = 0.23
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

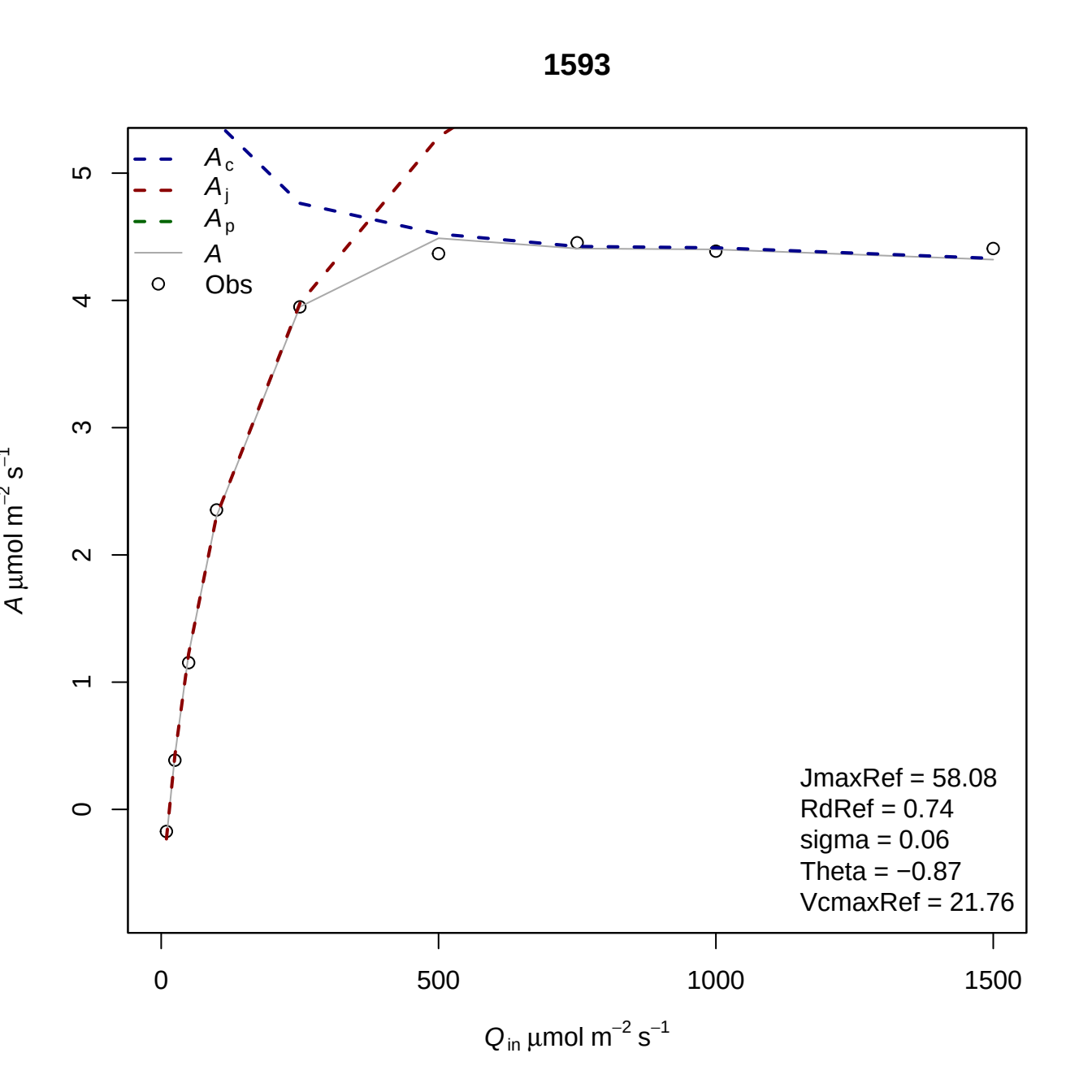

1588
A_c
A_j
A_p
A
Obs
JmaxRef = 45.93
RdRef = 0.63
sigma = 0.06
Theta = −0.27
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

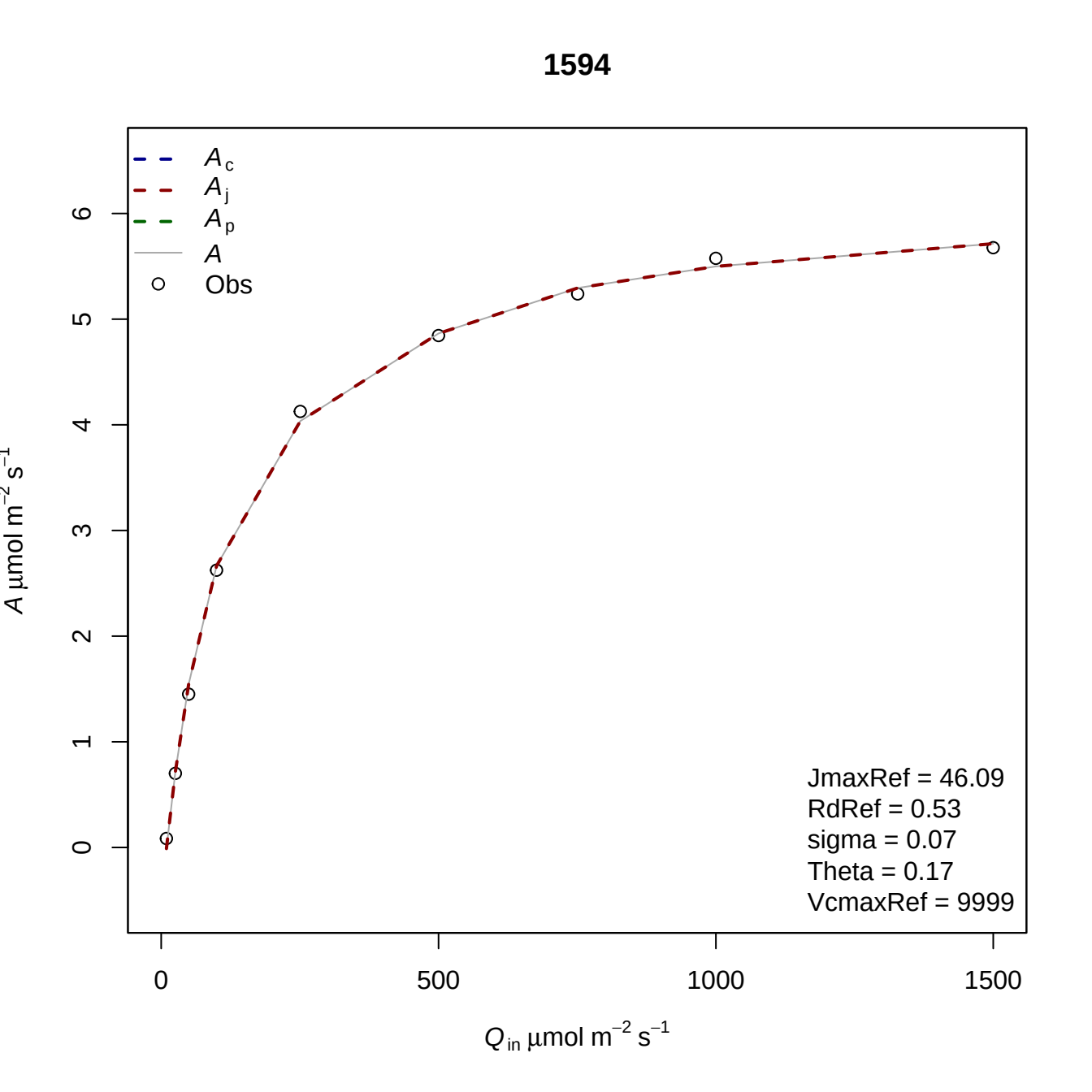

1589
A_c
A_j
A_p
A
Obs
JmaxRef = 32.65
RdRef = 0.5
sigma = 0.09
Theta = −0.44
VcmaxRef = 21.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

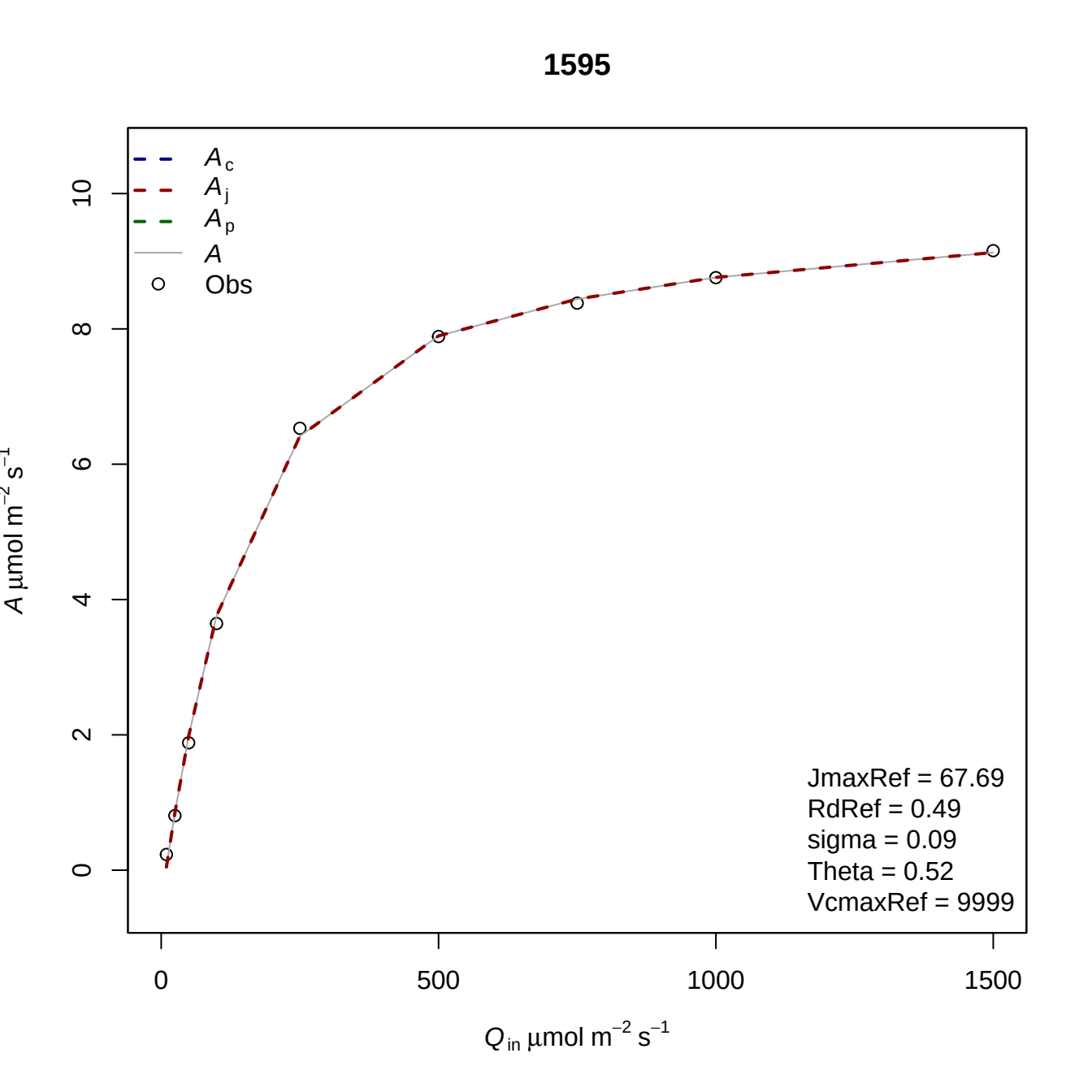

1590
A_c
A_j
A_p
A
Obs
JmaxRef = 72.94
RdRef = 0.56
sigma = 0.12
Theta = 0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

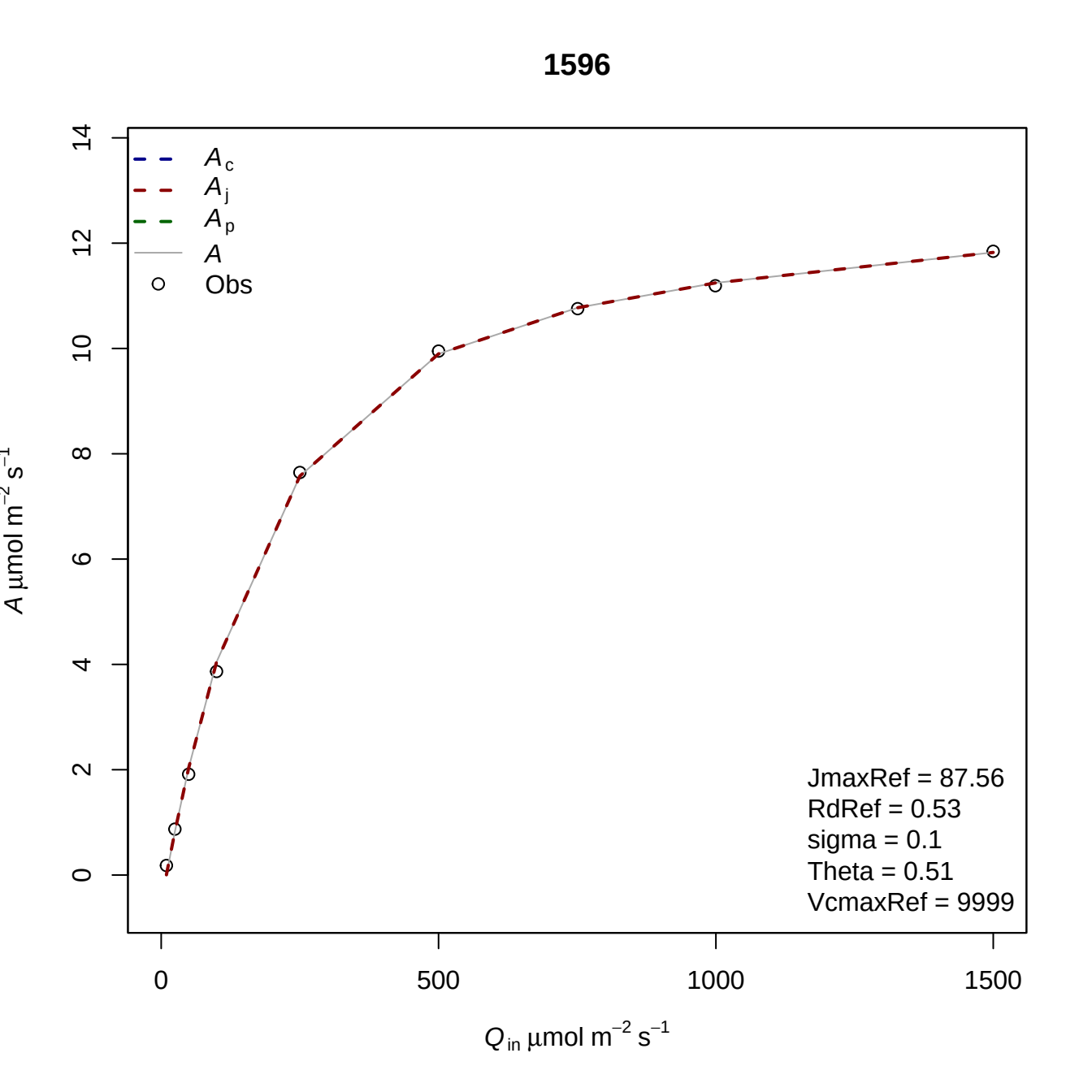

1591
A_c
A_j
A_p
A
Obs
JmaxRef = 246.65
RdRef = 0.09
sigma = 0.17
Theta = −73.33
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

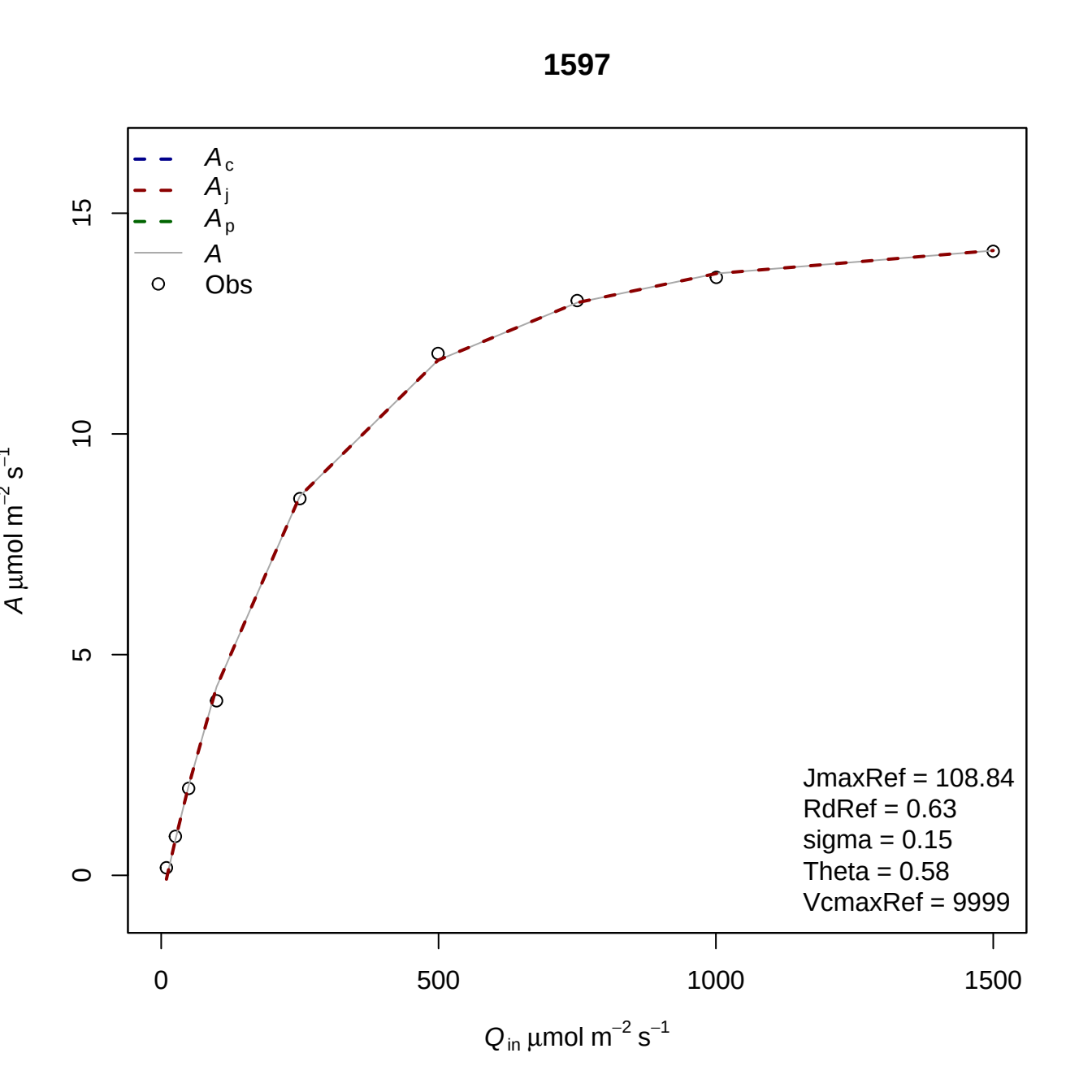

1592
A_c
A_j
A_p
A
Obs
JmaxRef = 59.74
RdRef = 0.81
sigma = 0.11
Theta = 0.63
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

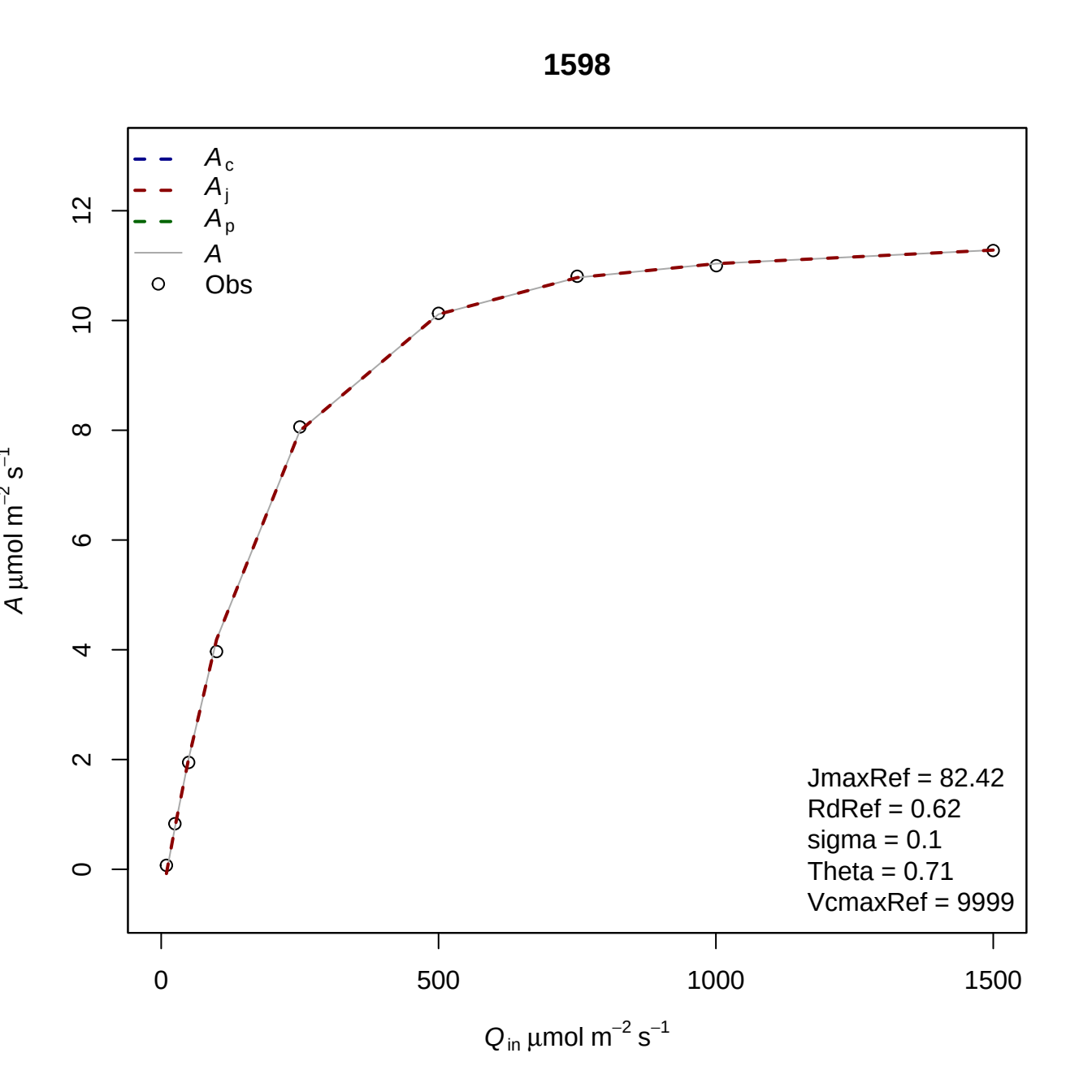

1593
A_c
A_j
A_p
A
Obs
JmaxRef = 58.08
RdRef = 0.74
sigma = 0.06
Theta = −0.87
VcmaxRef = 21.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

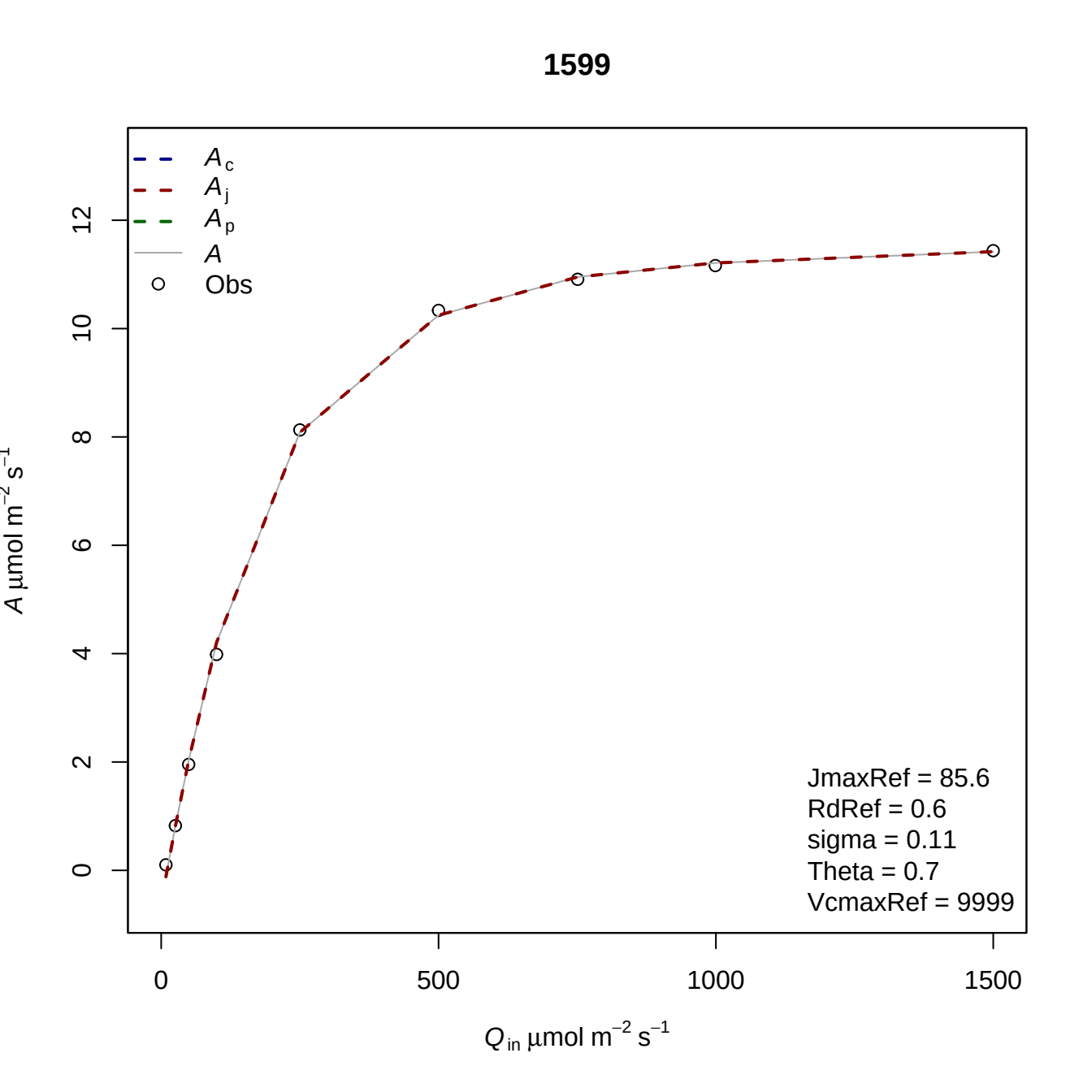

1594
A_c
A_j
A_p
A
Obs
JmaxRef = 46.09
RdRef = 0.53
sigma = 0.07
Theta = 0.17
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

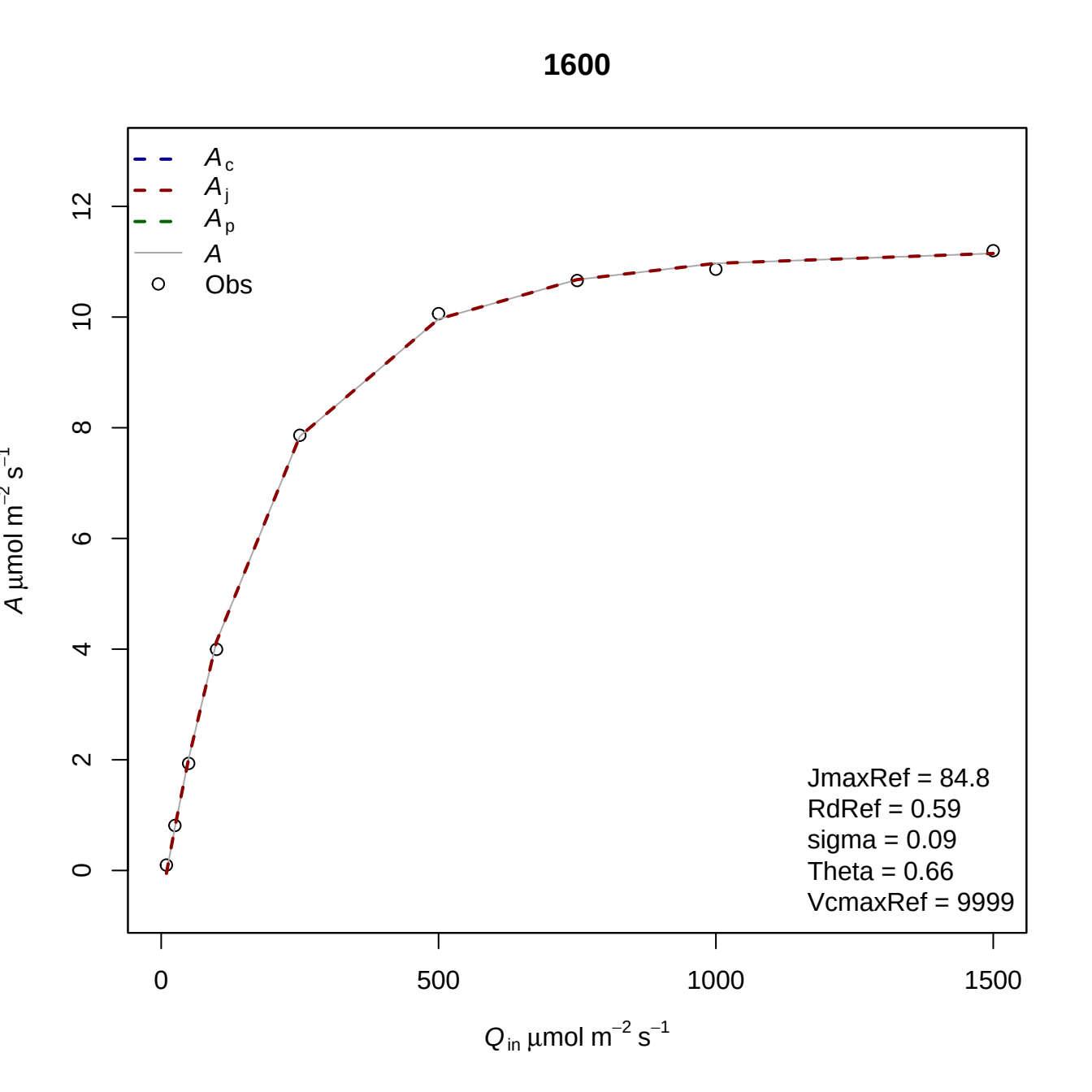

1595
A_c
A_j
A_p
A
Obs
JmaxRef = 67.69
RdRef = 0.49
sigma = 0.09
Theta = 0.52
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

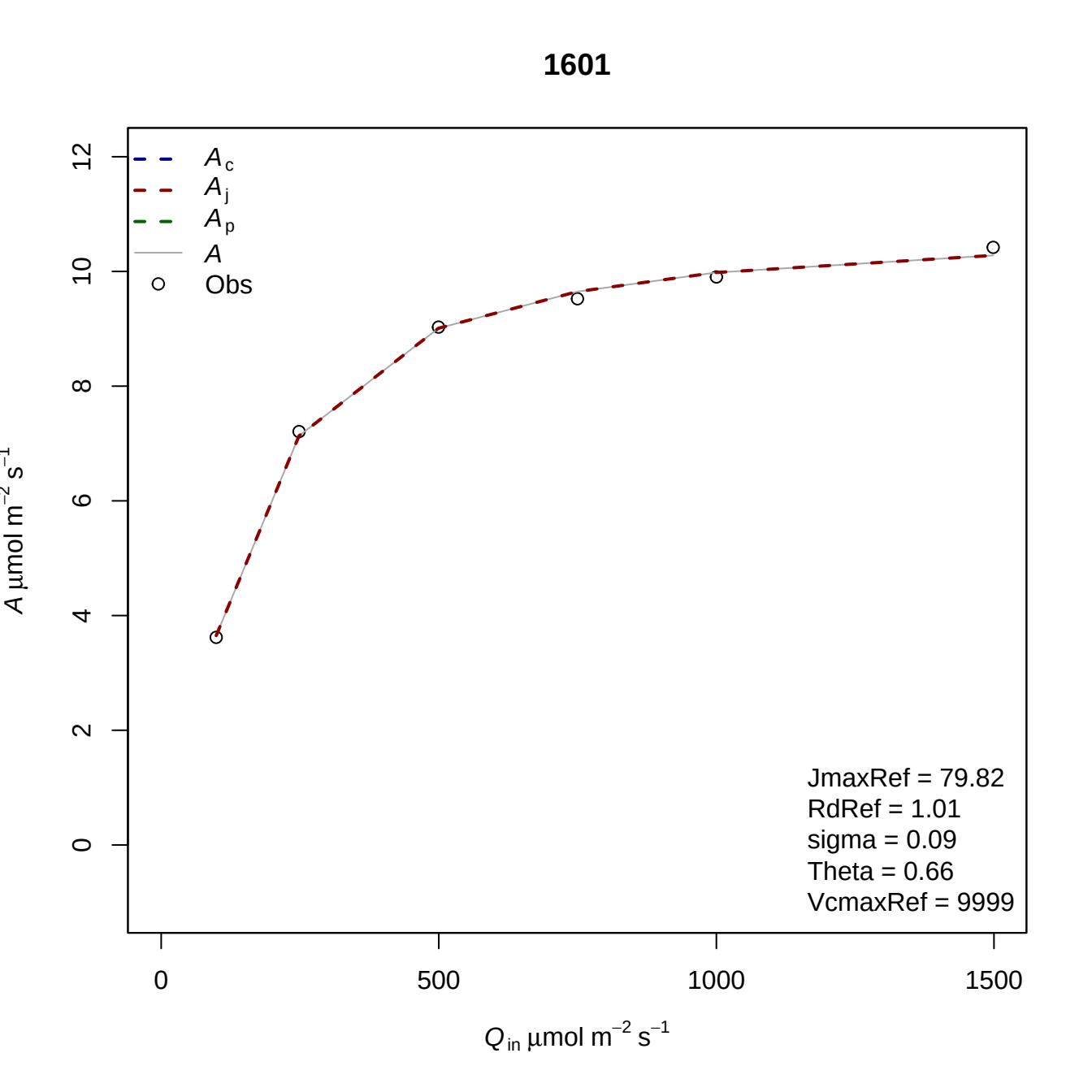

1596
A_c
A_j
A_p
A
Obs
JmaxRef = 87.56
RdRef = 0.53
sigma = 0.1
Theta = 0.51
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

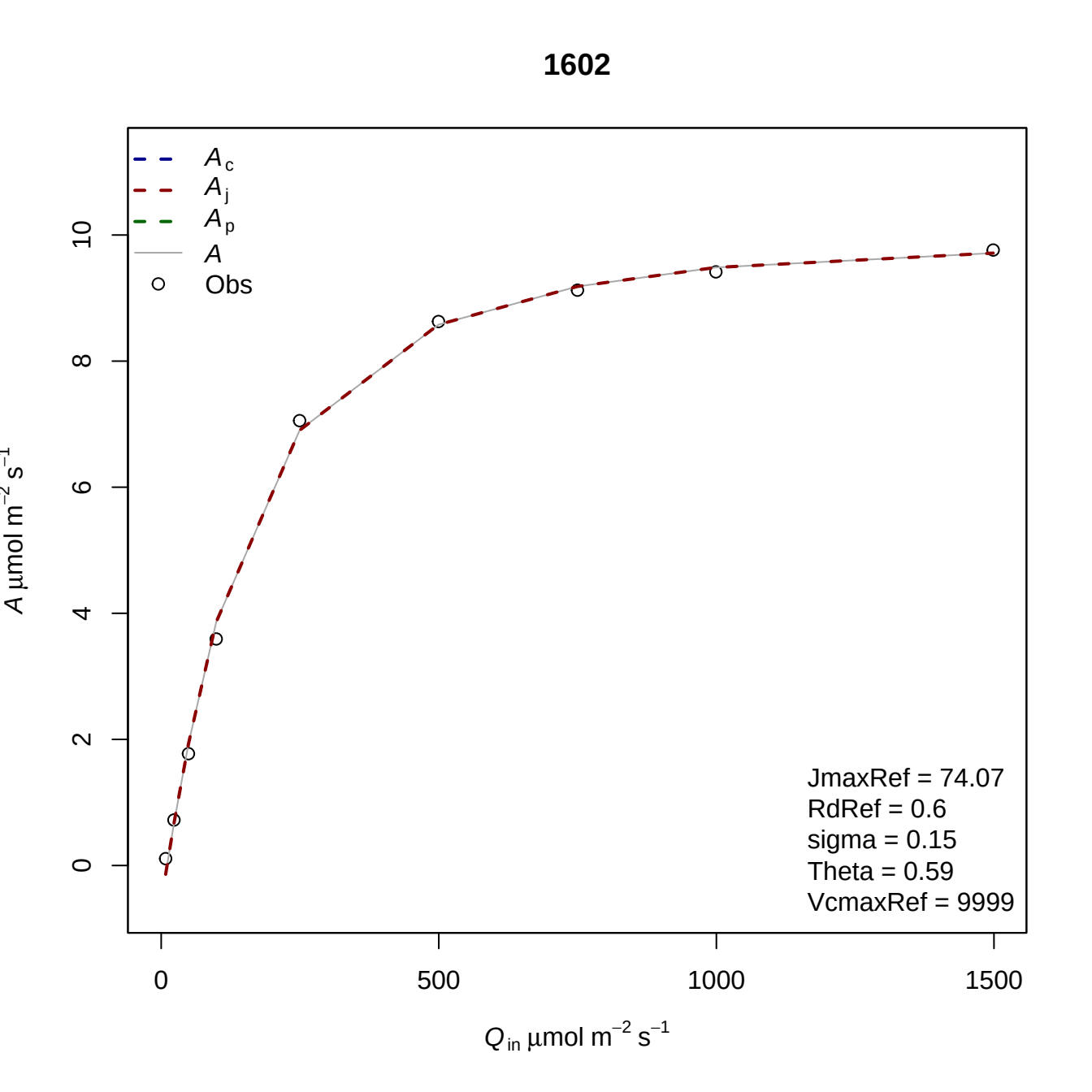

1597
A_c
A_j
A_p
A
Obs
JmaxRef = 108.84
RdRef = 0.63
sigma = 0.15
Theta = 0.58
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

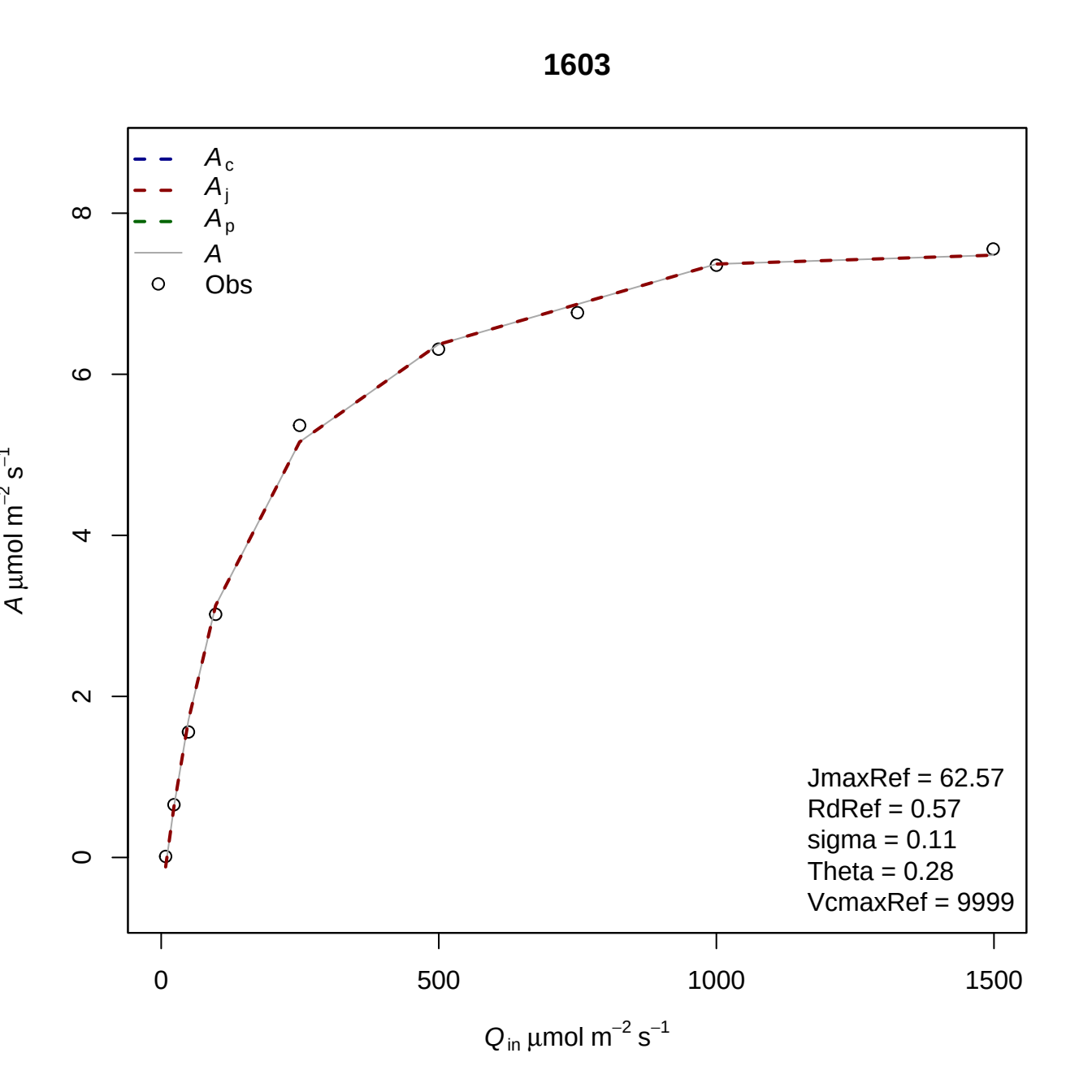

1598
A_c
A_j
A_p
A
Obs
JmaxRef = 82.42
RdRef = 0.62
sigma = 0.1
Theta = 0.71
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

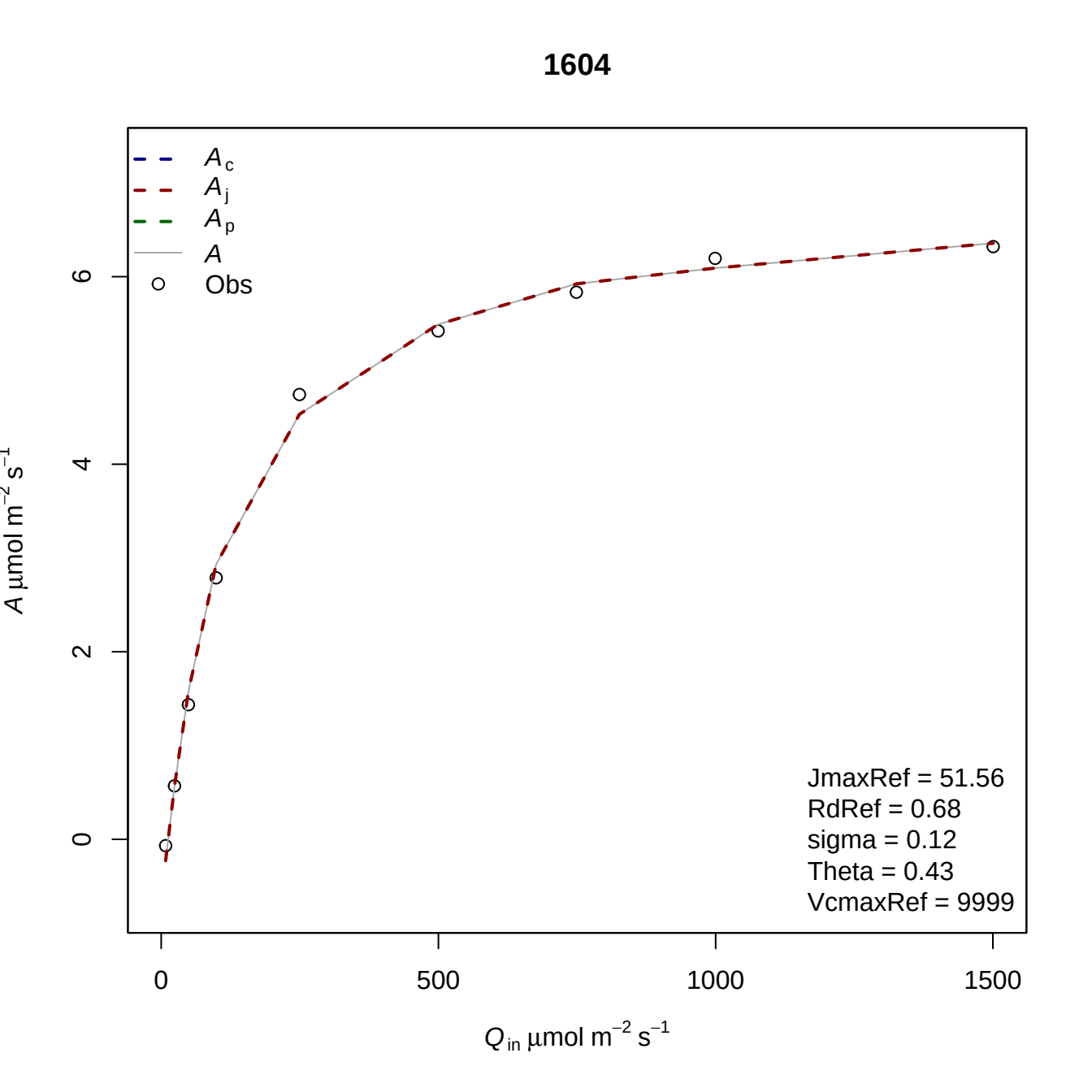

1599
A_c
A_j
A_p
A
Obs
JmaxRef = 85.6
RdRef = 0.6
sigma = 0.11
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

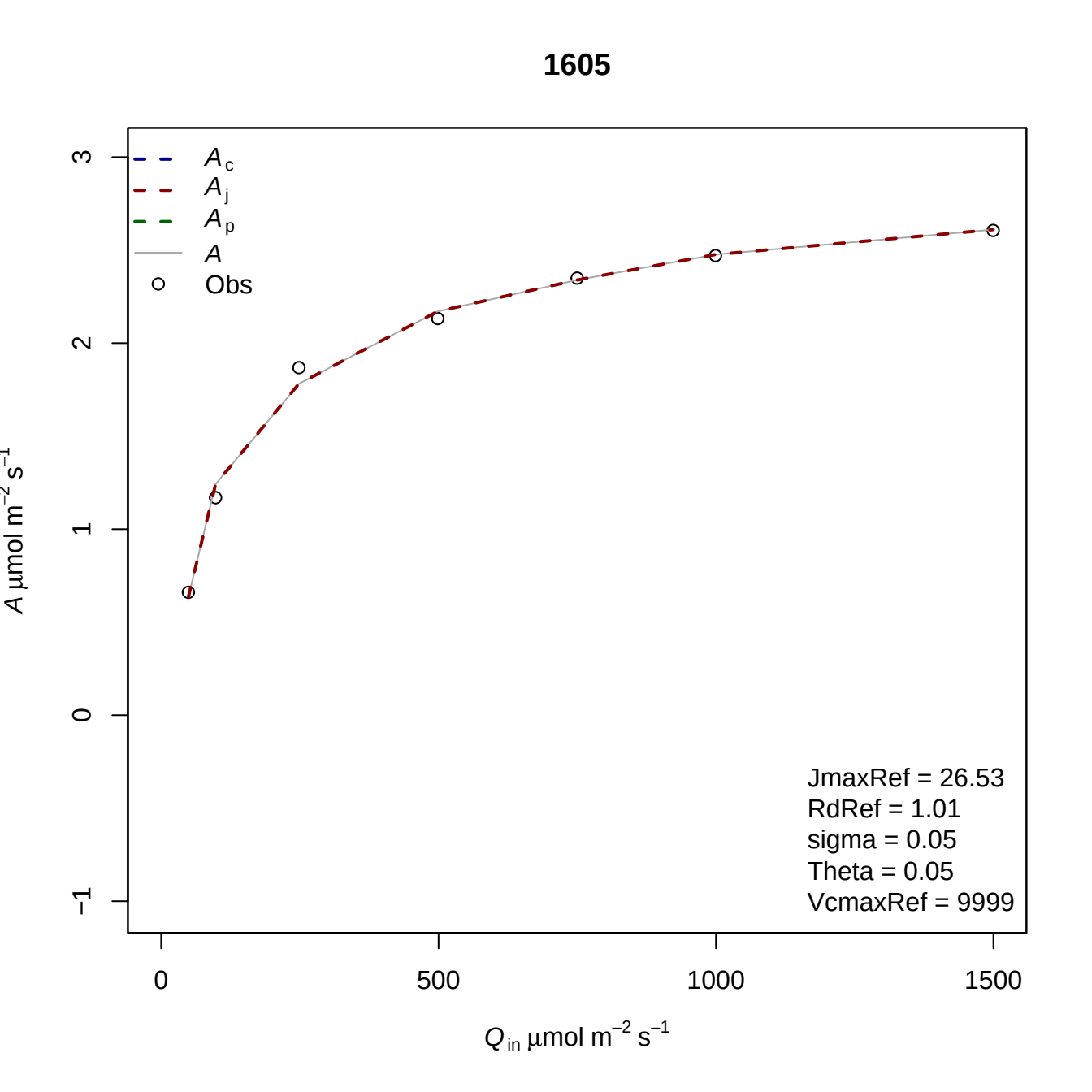

1600
A_c
A_j
A_p
A
Obs
JmaxRef = 84.8
RdRef = 0.59
sigma = 0.09
Theta = 0.66
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

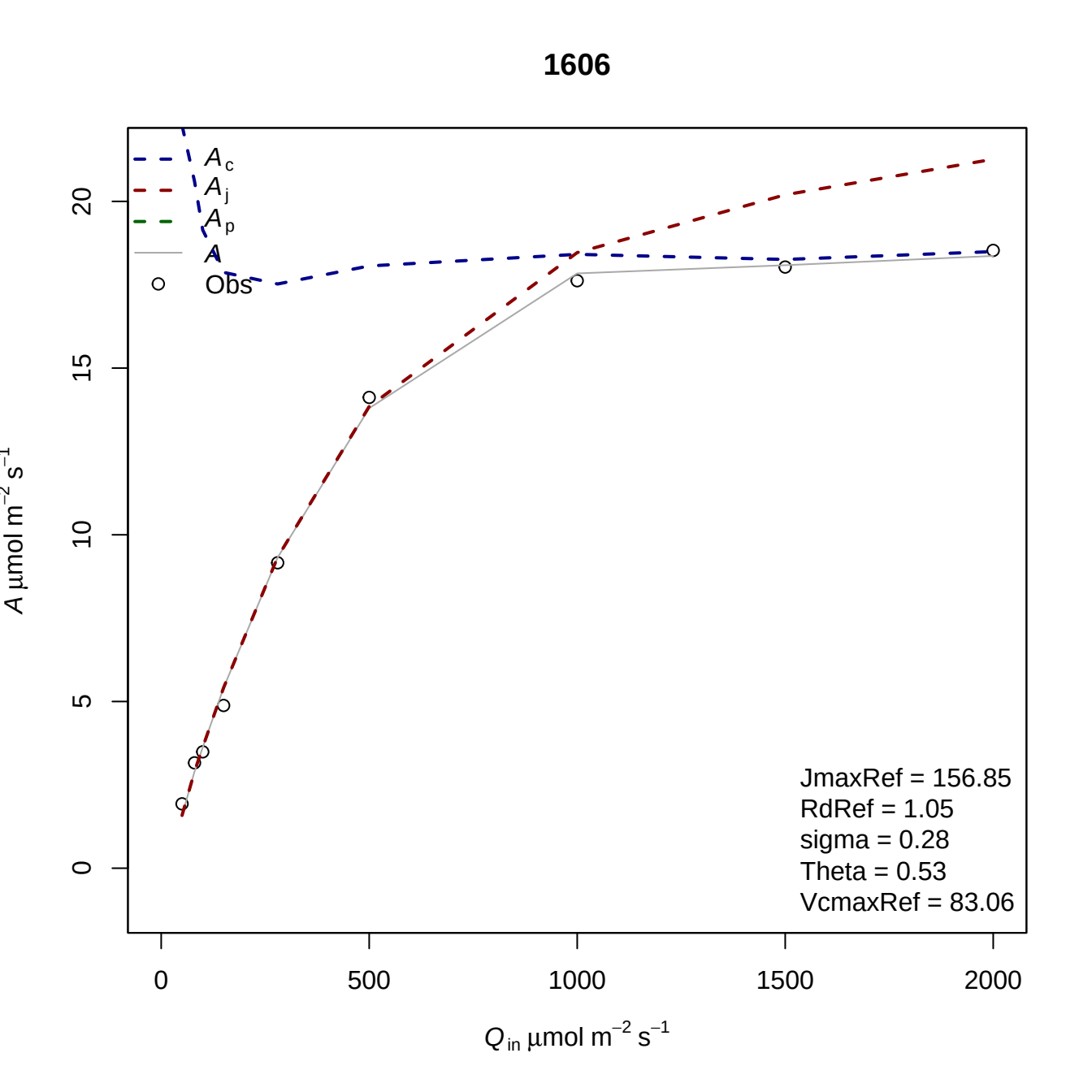

1601
A_c
A_j
A_p
A
Obs
JmaxRef = 79.82
RdRef = 1.01
sigma = 0.09
Theta = 0.66
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

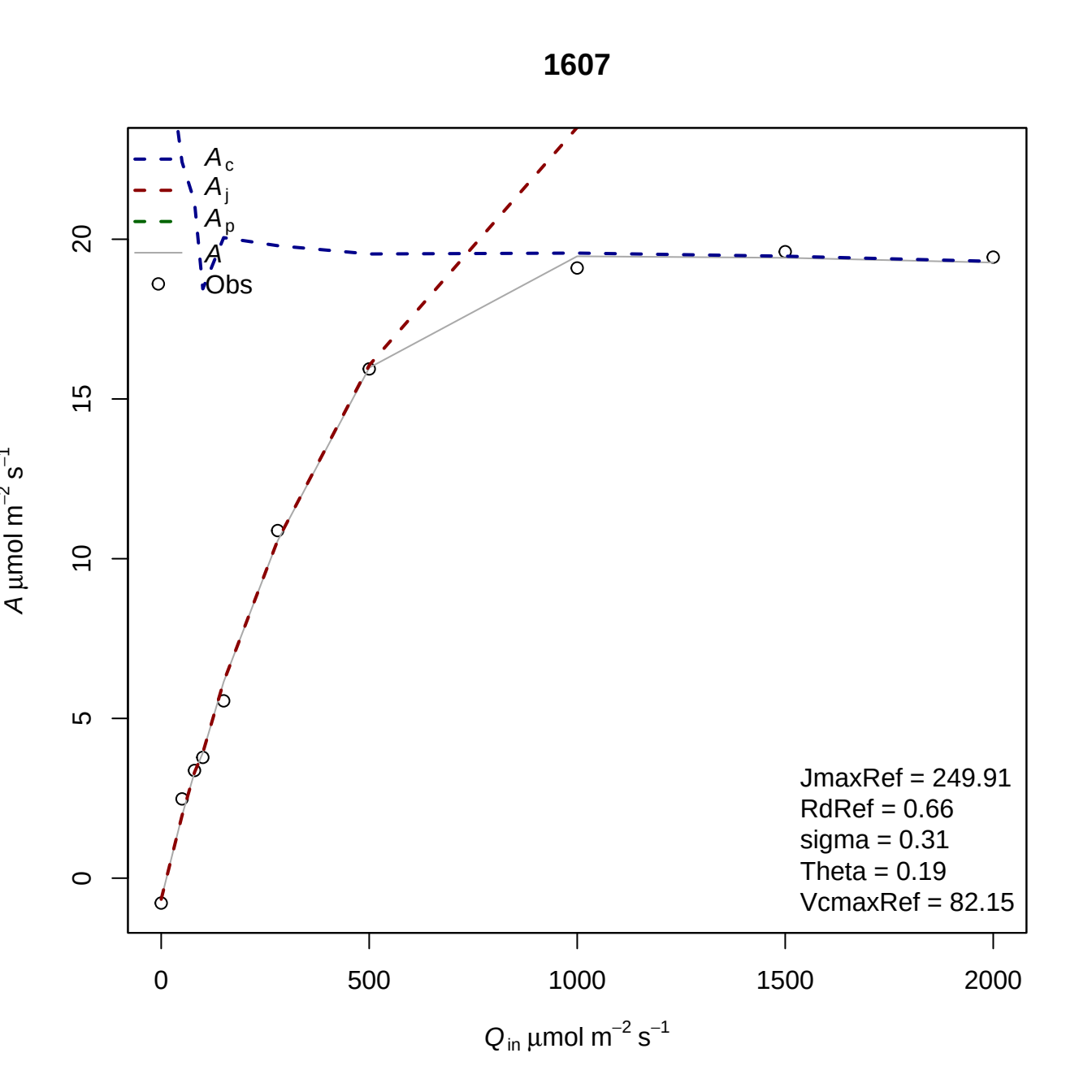

1602
A_c
A_j
A_p
A
Obs
JmaxRef = 74.07
RdRef = 0.6
sigma = 0.15
Theta = 0.59
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

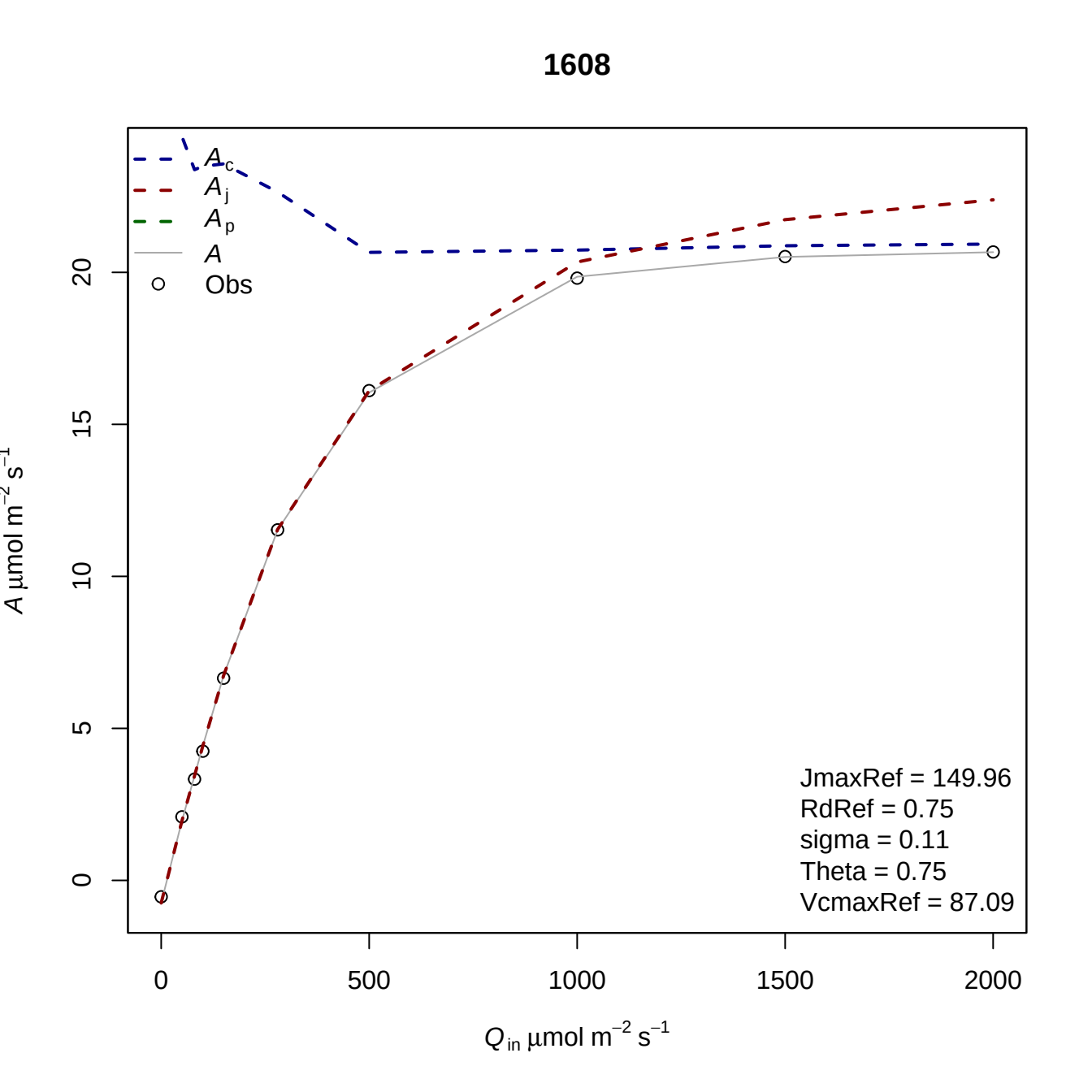

1603
A_c
A_j
A_p
A
Obs
JmaxRef = 62.57
RdRef = 0.57
sigma = 0.11
Theta = 0.28
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

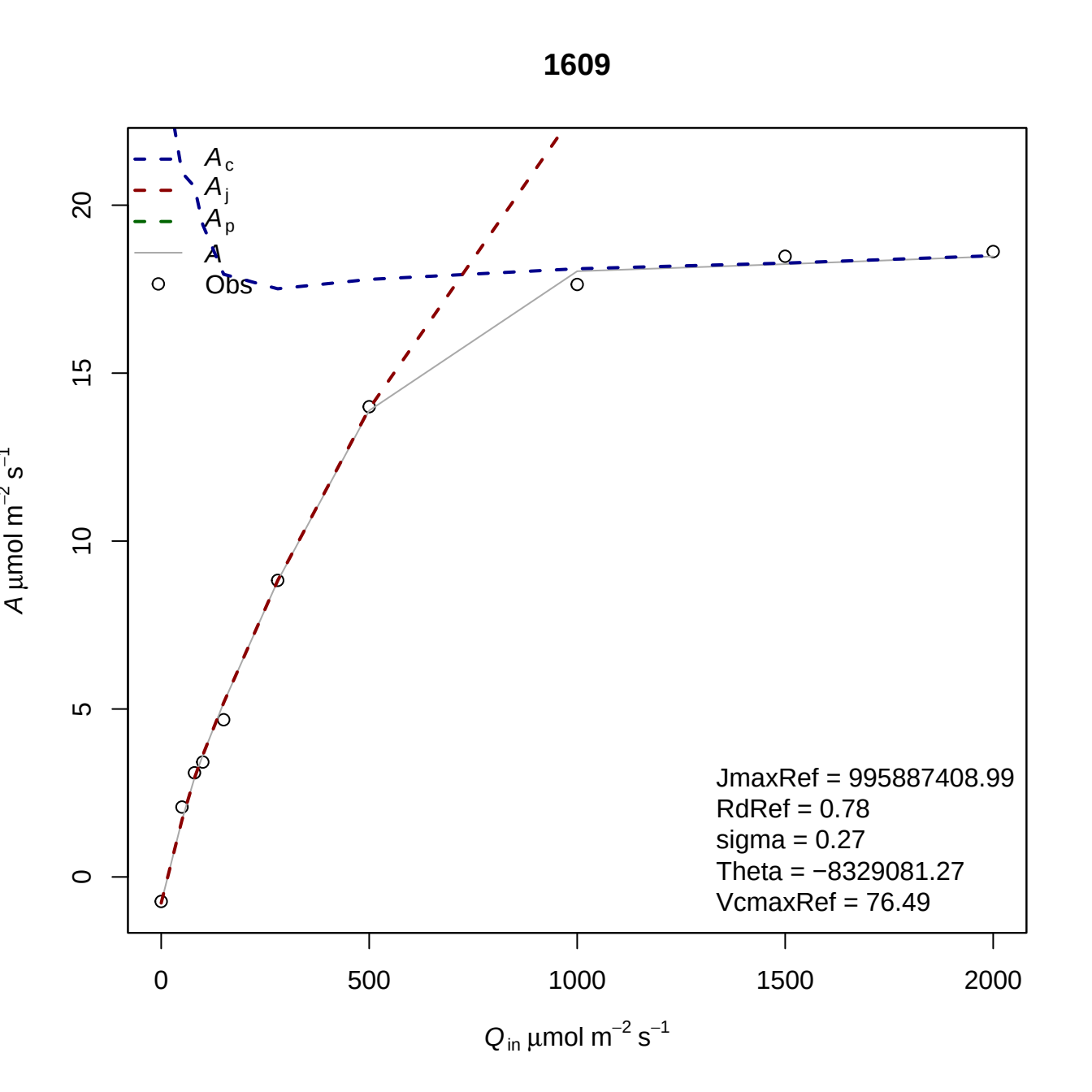

1604
A_c
A_j
A_p
A
Obs
JmaxRef = 51.56
RdRef = 0.68
sigma = 0.12
Theta = 0.43
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

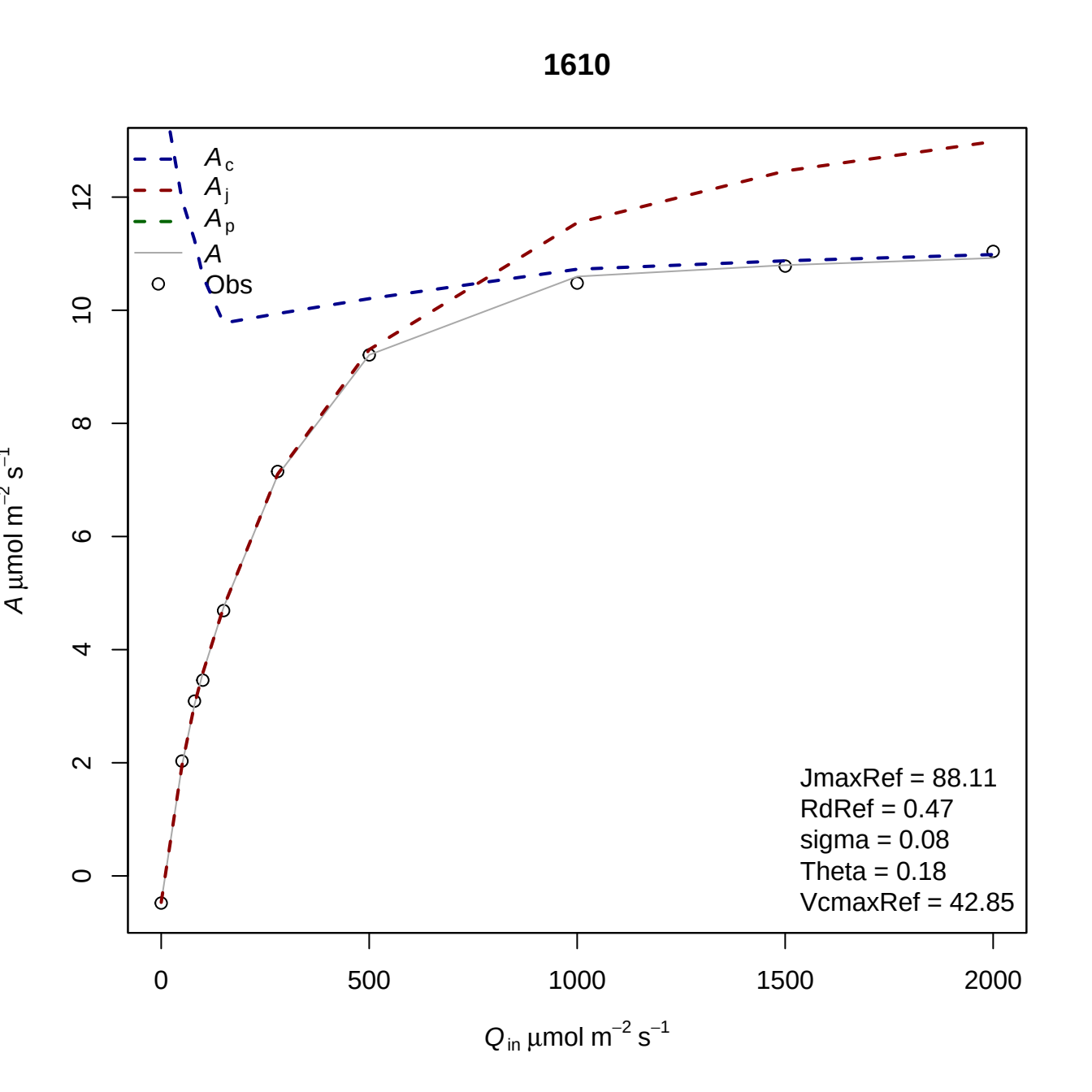

1605
A_c
A_j
A_p
A
Obs
JmaxRef = 26.53
RdRef = 1.01
sigma = 0.05
Theta = 0.05
VcmaxRef = 9999
3
2
1
0
−1
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

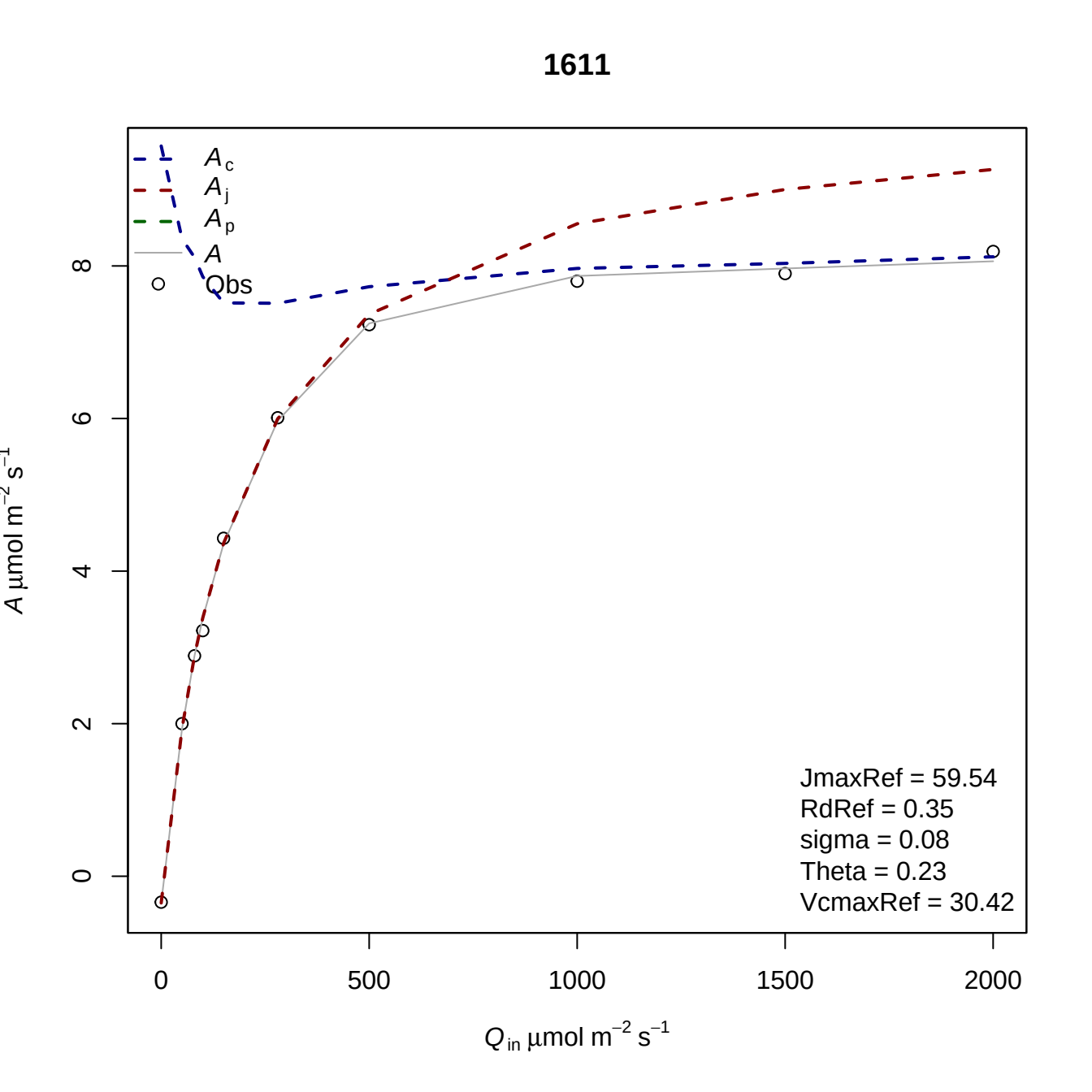

1606
A_c
A_j
A_p
A
Obs
JmaxRef = 156.85
RdRef = 1.05
sigma = 0.28
Theta = 0.53
VcmaxRef = 83.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

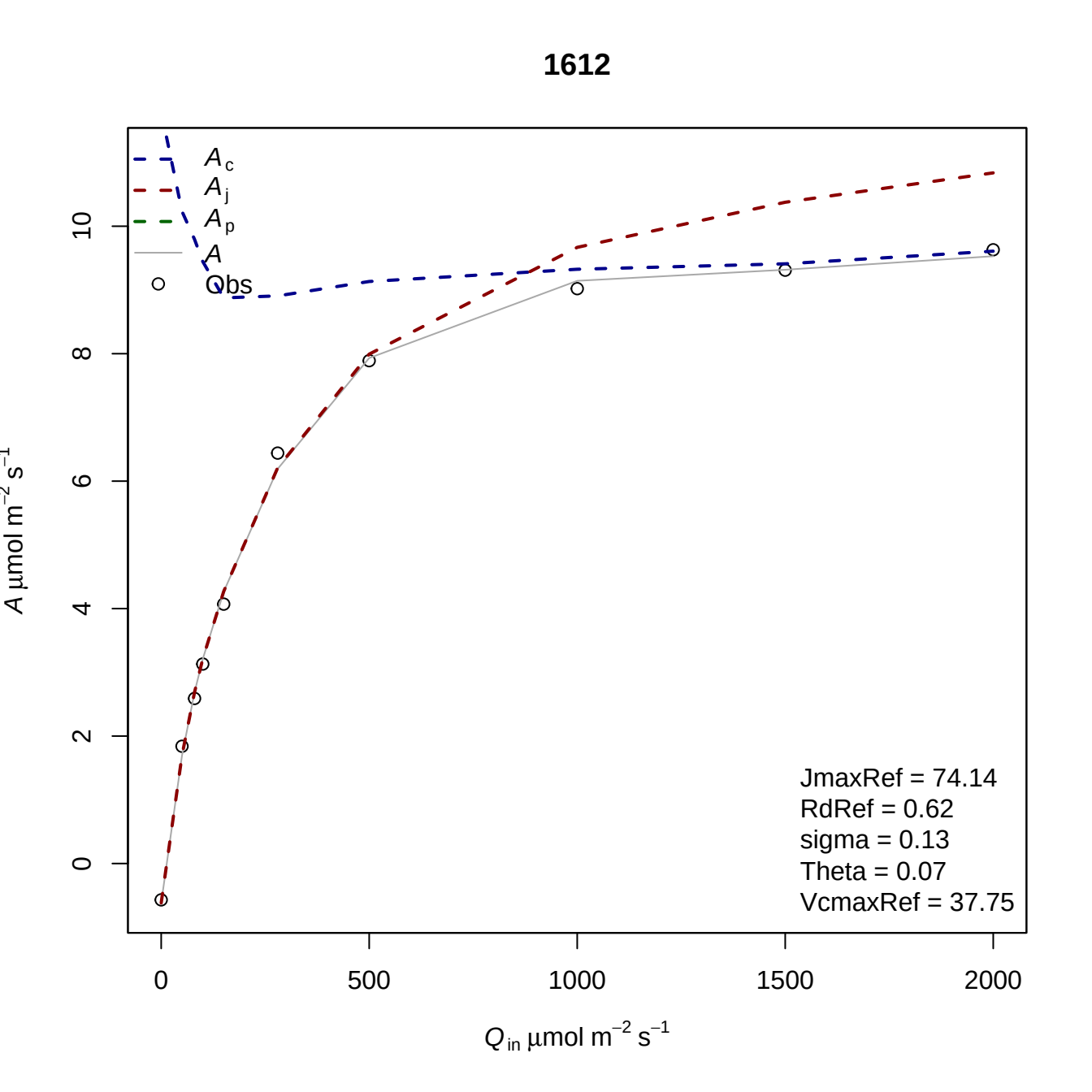

1607
A_c
A_j
A_p
A
Obs
JmaxRef = 249.91
RdRef = 0.66
sigma = 0.31
Theta = 0.19
VcmaxRef = 82.15
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

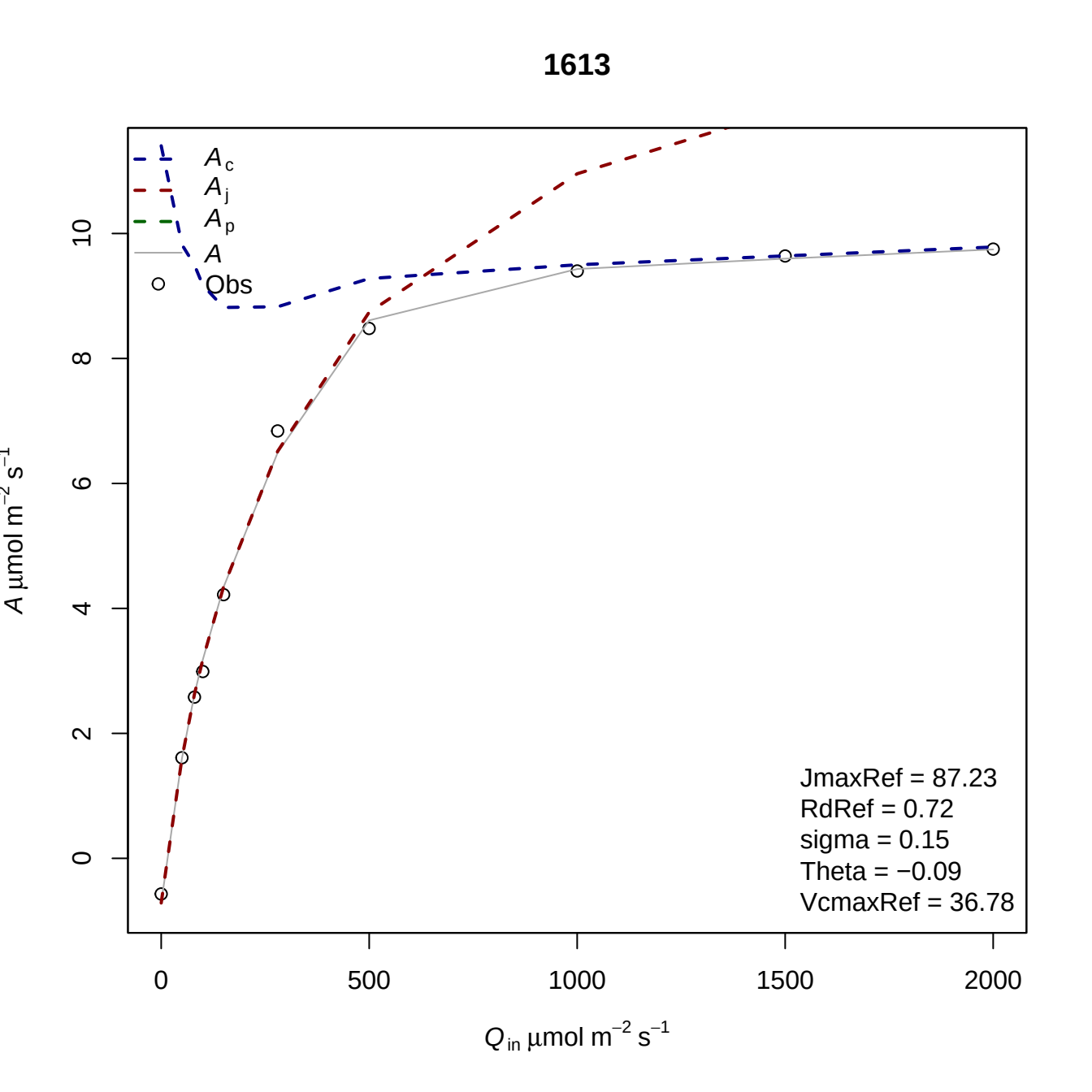

1608
A_c
A_j
A_p
A
Obs
JmaxRef = 149.96
RdRef = 0.75
sigma = 0.11
Theta = 0.75
VcmaxRef = 87.09
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

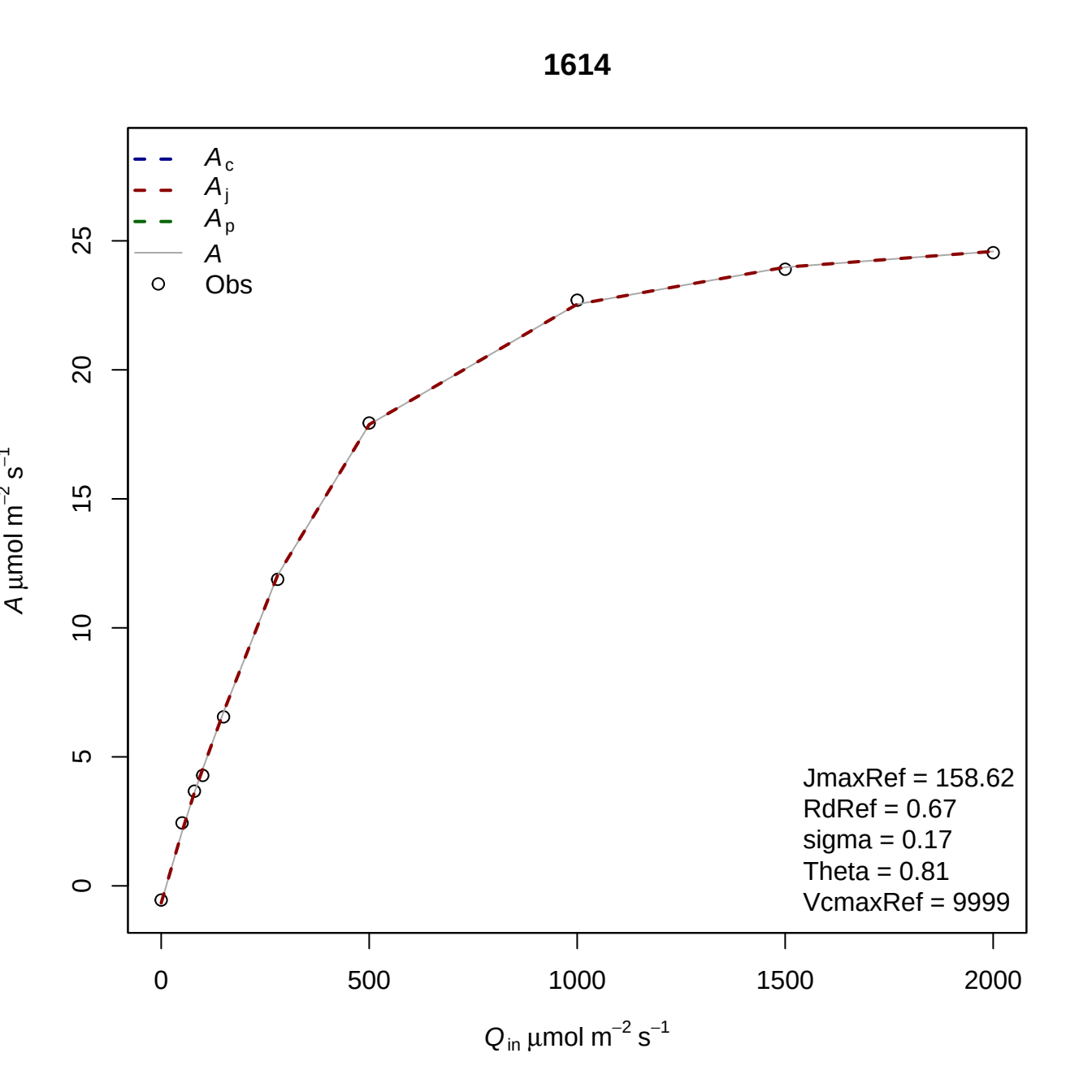

1609
A_c
A_j
A_p
A
Obs
JmaxRef = 995887408.99
RdRef = 0.78
sigma = 0.27
Theta = −8329081.27
VcmaxRef = 76.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

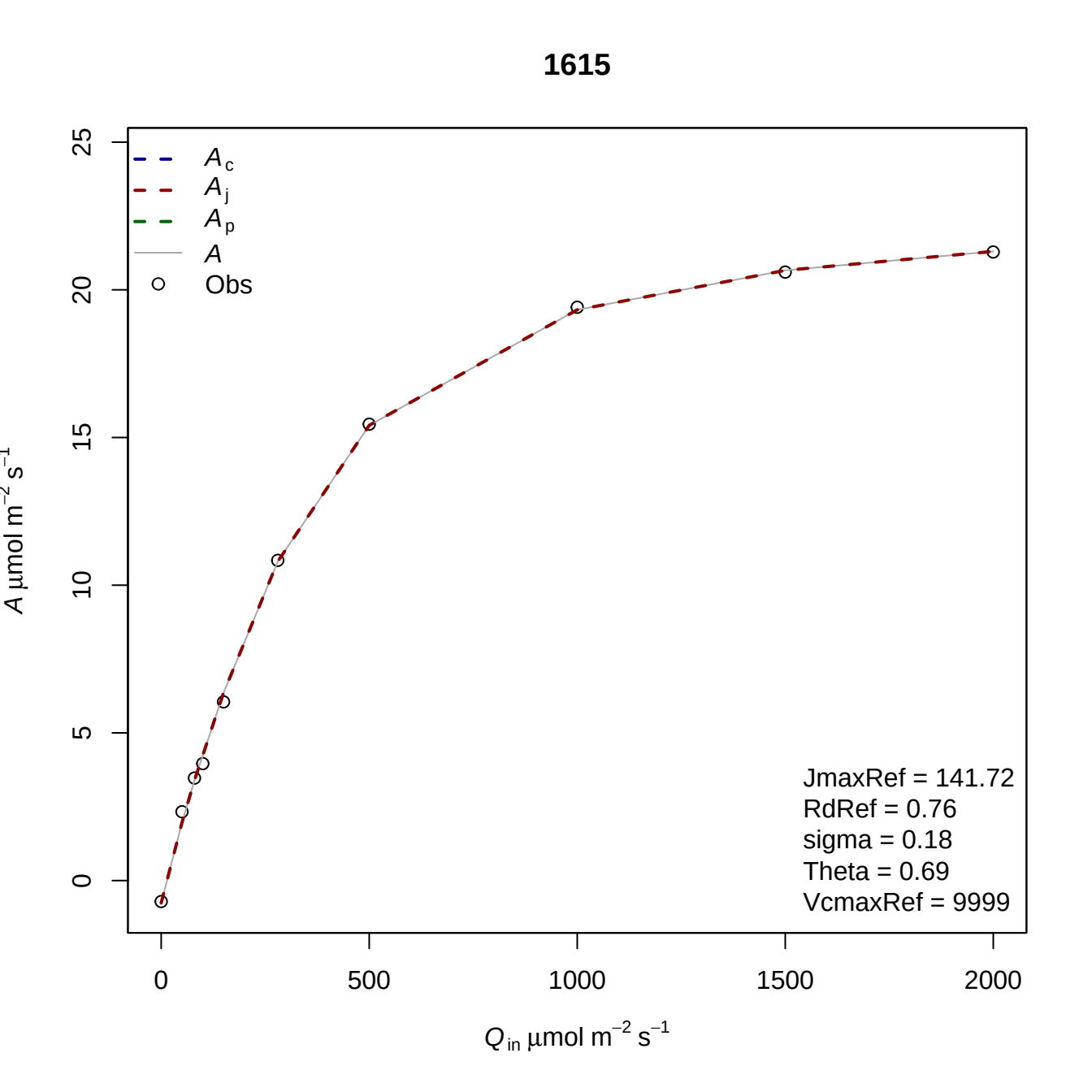

1610
A_c
A_j
A_p
A
Obs
JmaxRef = 88.11
RdRef = 0.47
sigma = 0.08
Theta = 0.18
VcmaxRef = 42.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

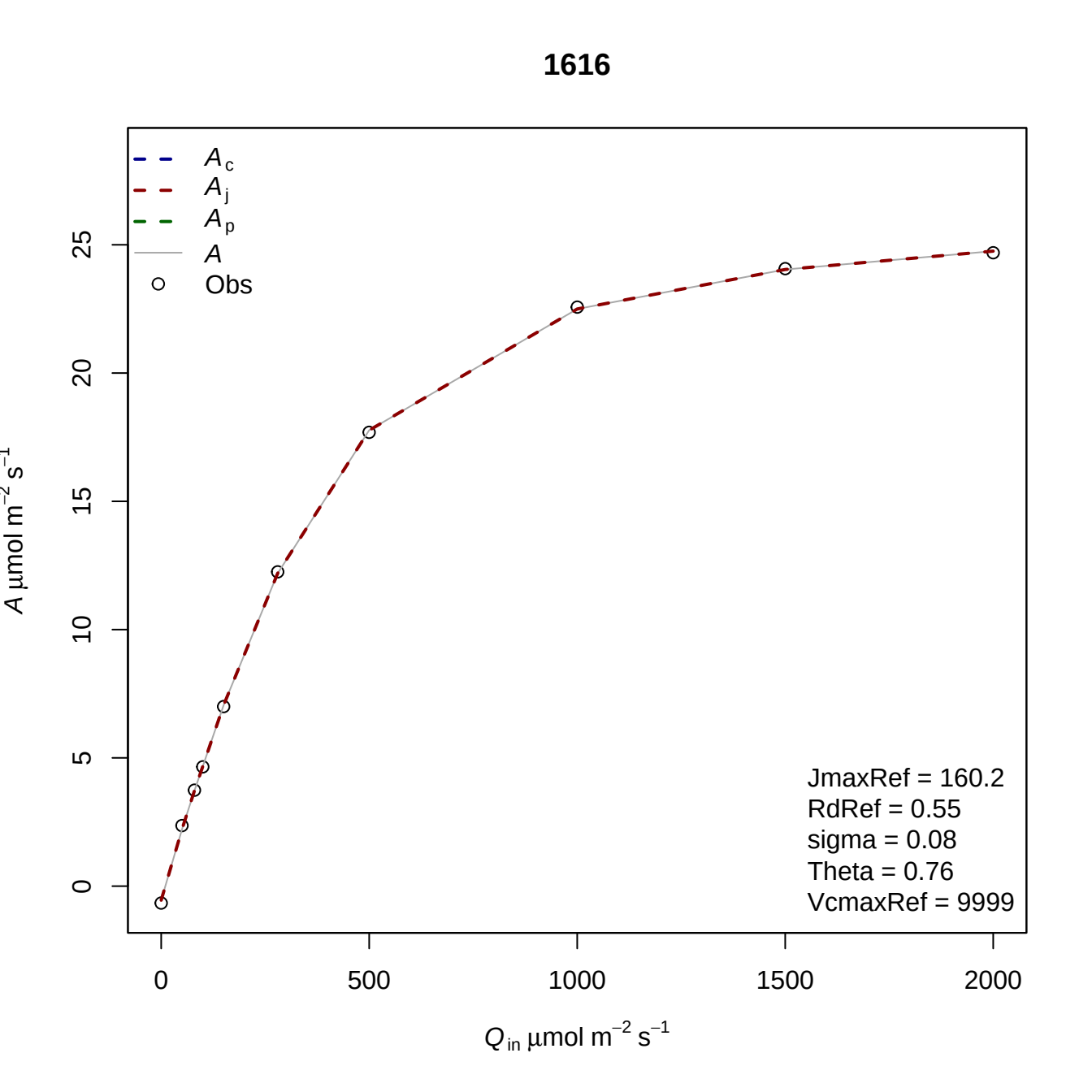

1611
A_c
A_j
A_p
A
Qbs
JmaxRef = 59.54
RdRef = 0.35
sigma = 0.08
Theta = 0.23
VcmaxRef = 30.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

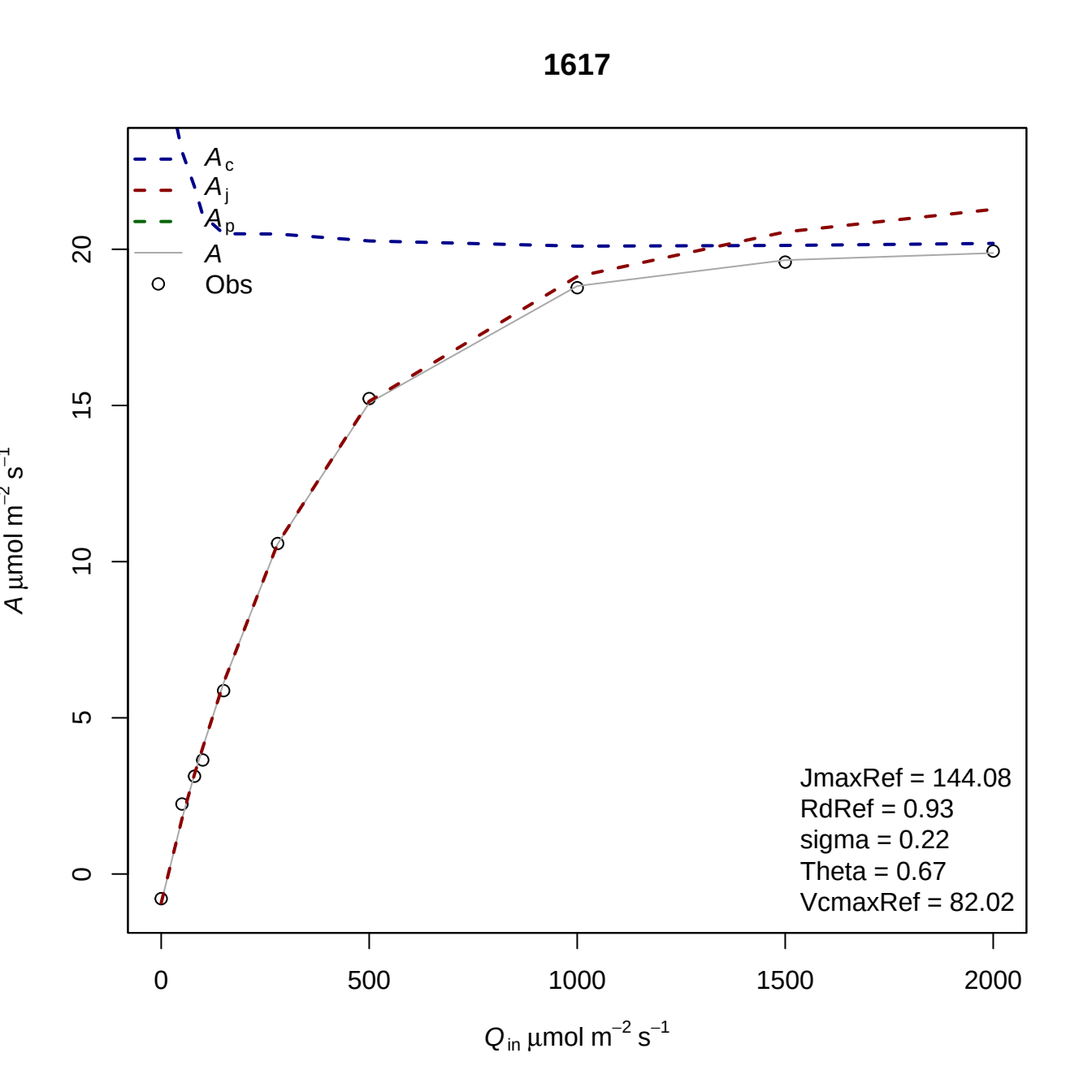

1612
A_c
A_j
A_p
A
Obs
JmaxRef = 74.14
RdRef = 0.62
sigma = 0.13
Theta = 0.07
VcmaxRef = 37.75
0
2
4
6
8
10
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

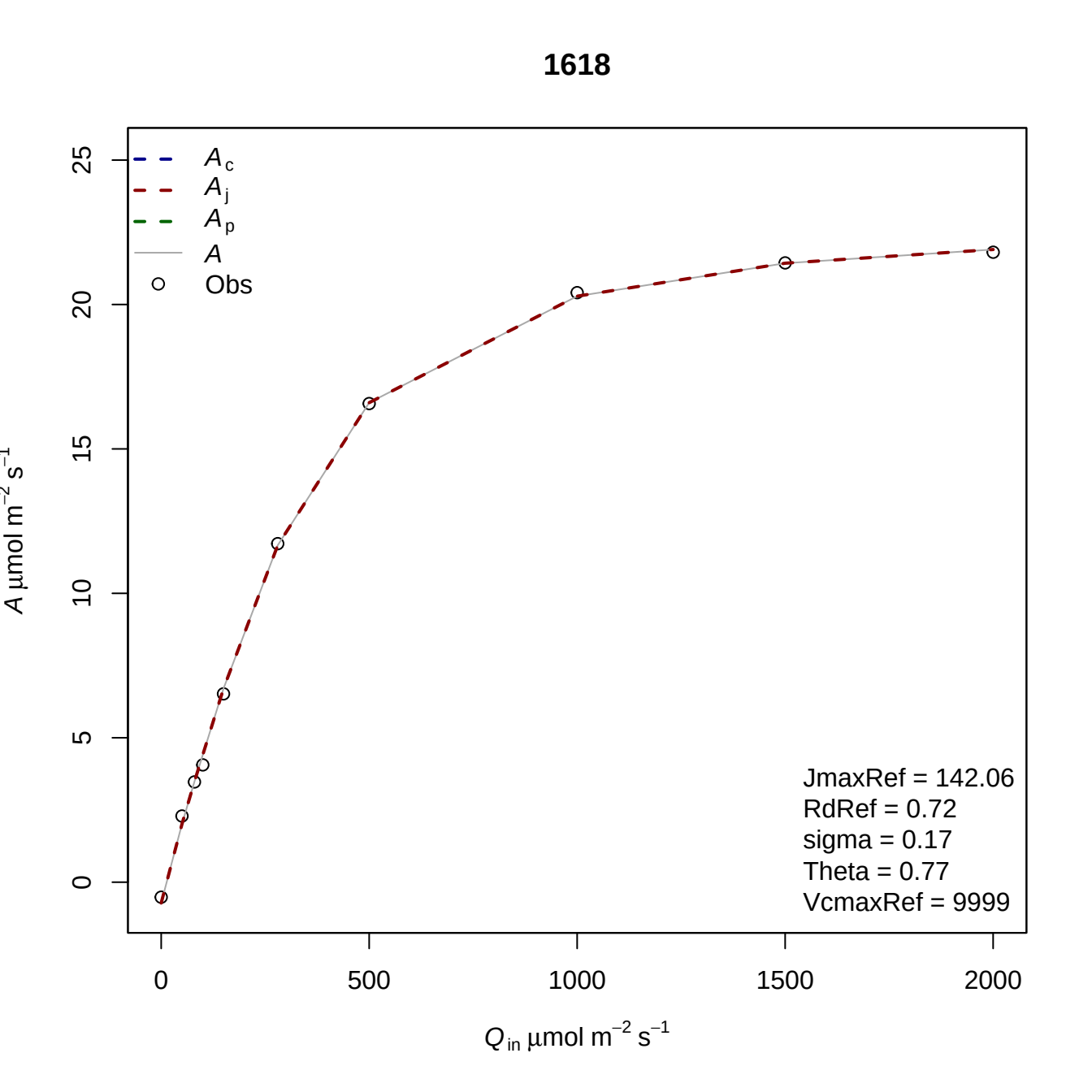

1613
A_c
A_j
A_p
A
Obs
JmaxRef = 87.23
RdRef = 0.72
sigma = 0.15
Theta = −0.09
VcmaxRef = 36.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

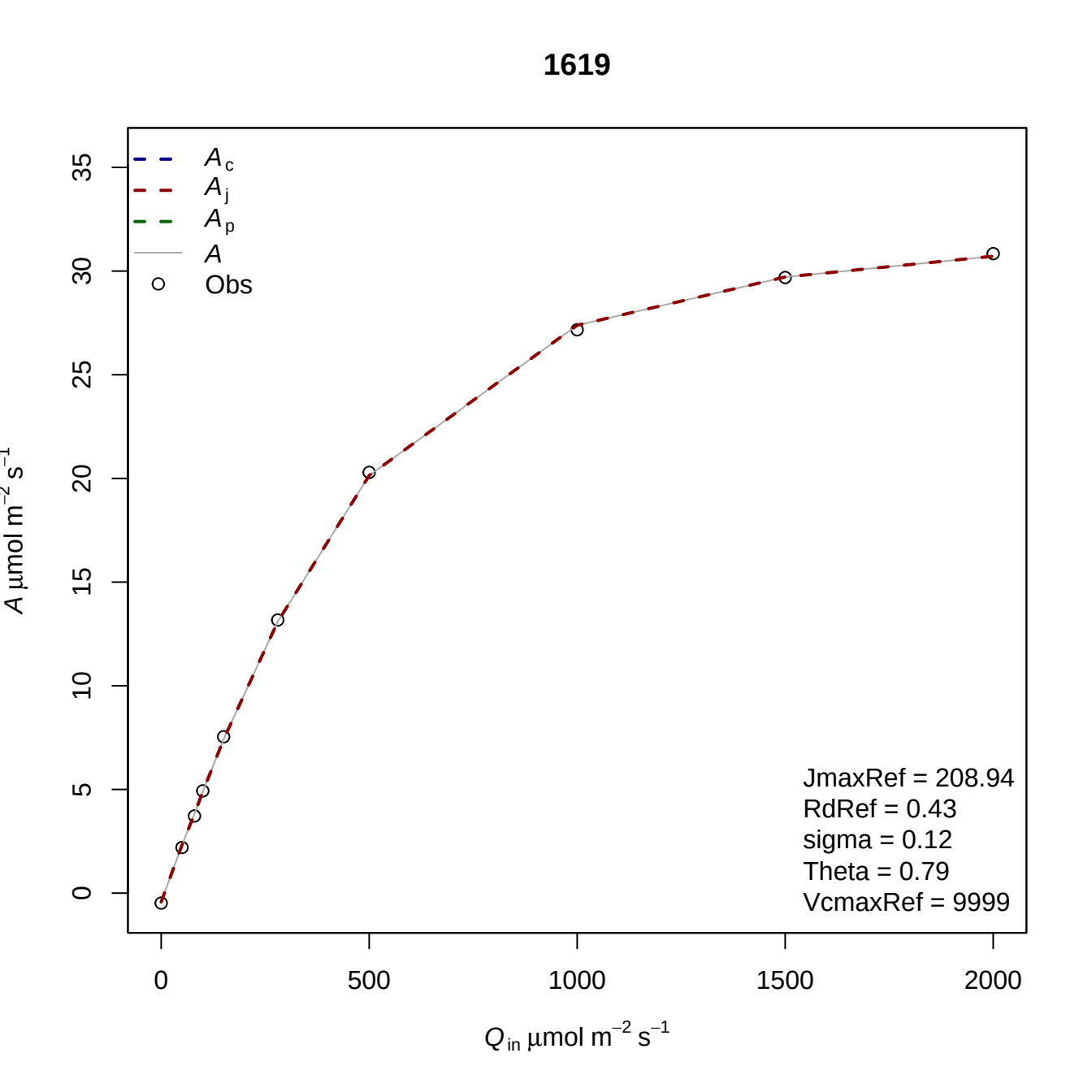

1614
A_c
A_j
A_p
A
Obs
JmaxRef = 158.62
RdRef = 0.67
sigma = 0.17
Theta = 0.81
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

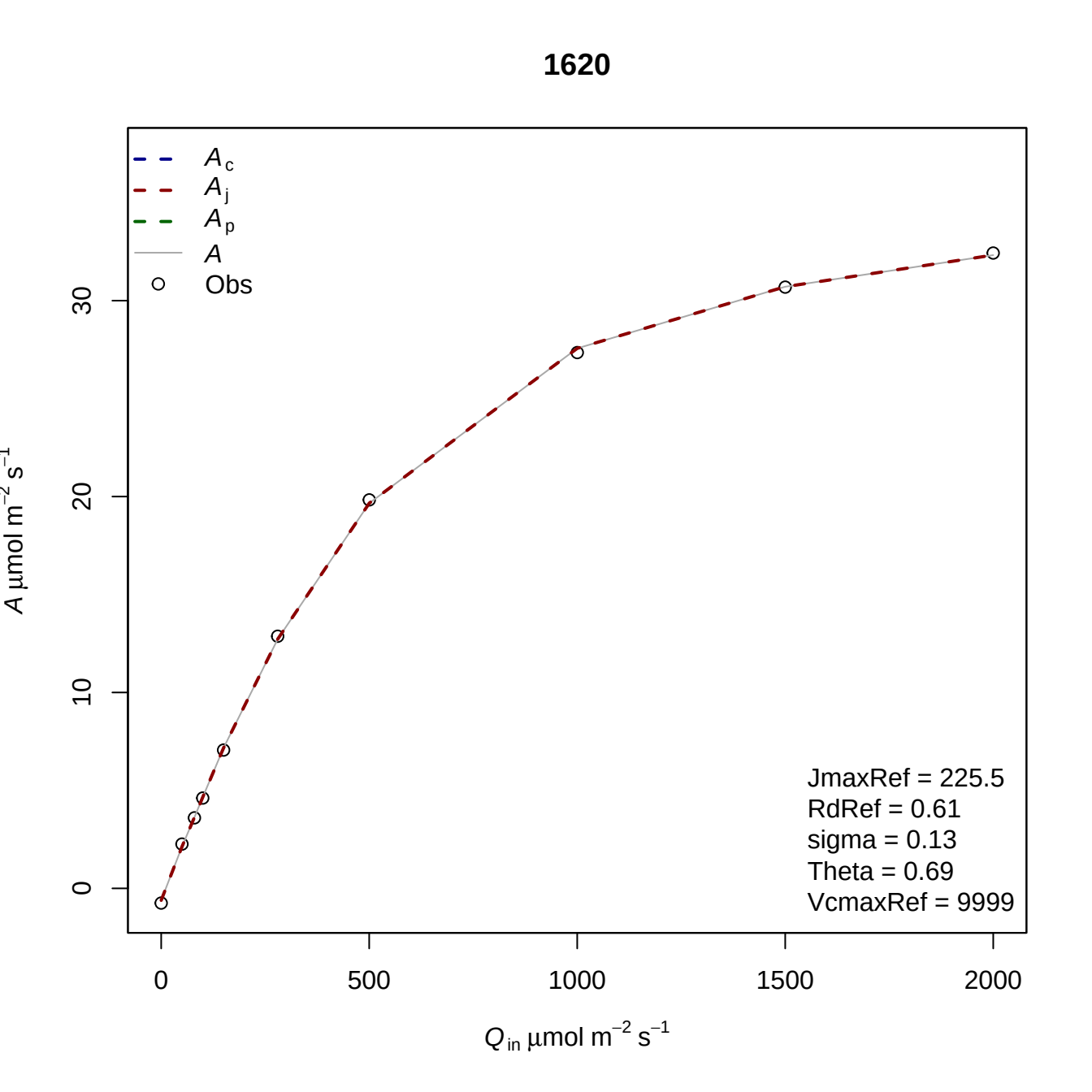

1615
A_c
A_j
A_p
A
Obs
JmaxRef = 141.72
RdRef = 0.76
sigma = 0.18
Theta = 0.69
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

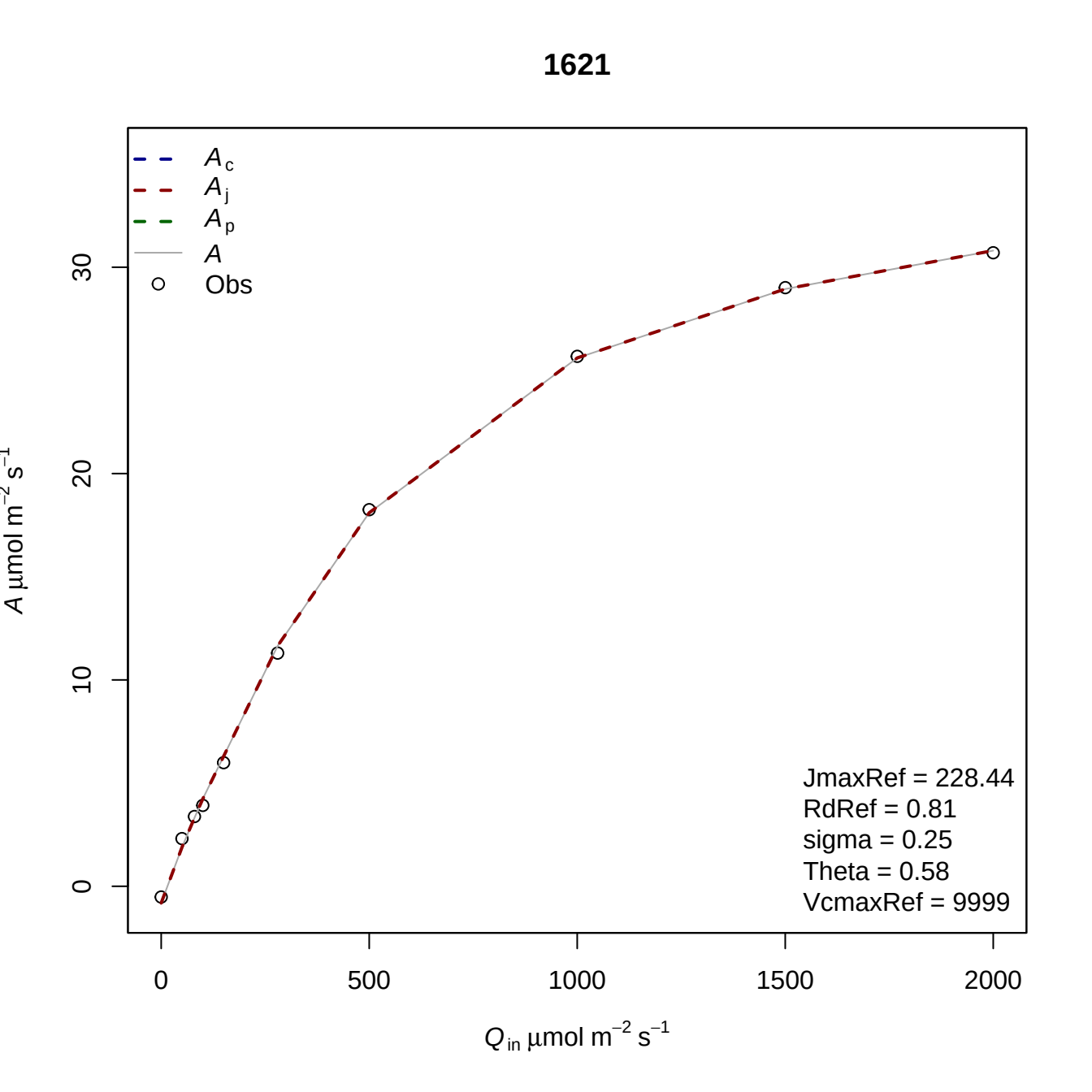

1616
A_c
A_j
A_p
A
Obs
JmaxRef = 160.2
RdRef = 0.55
sigma = 0.08
Theta = 0.76
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

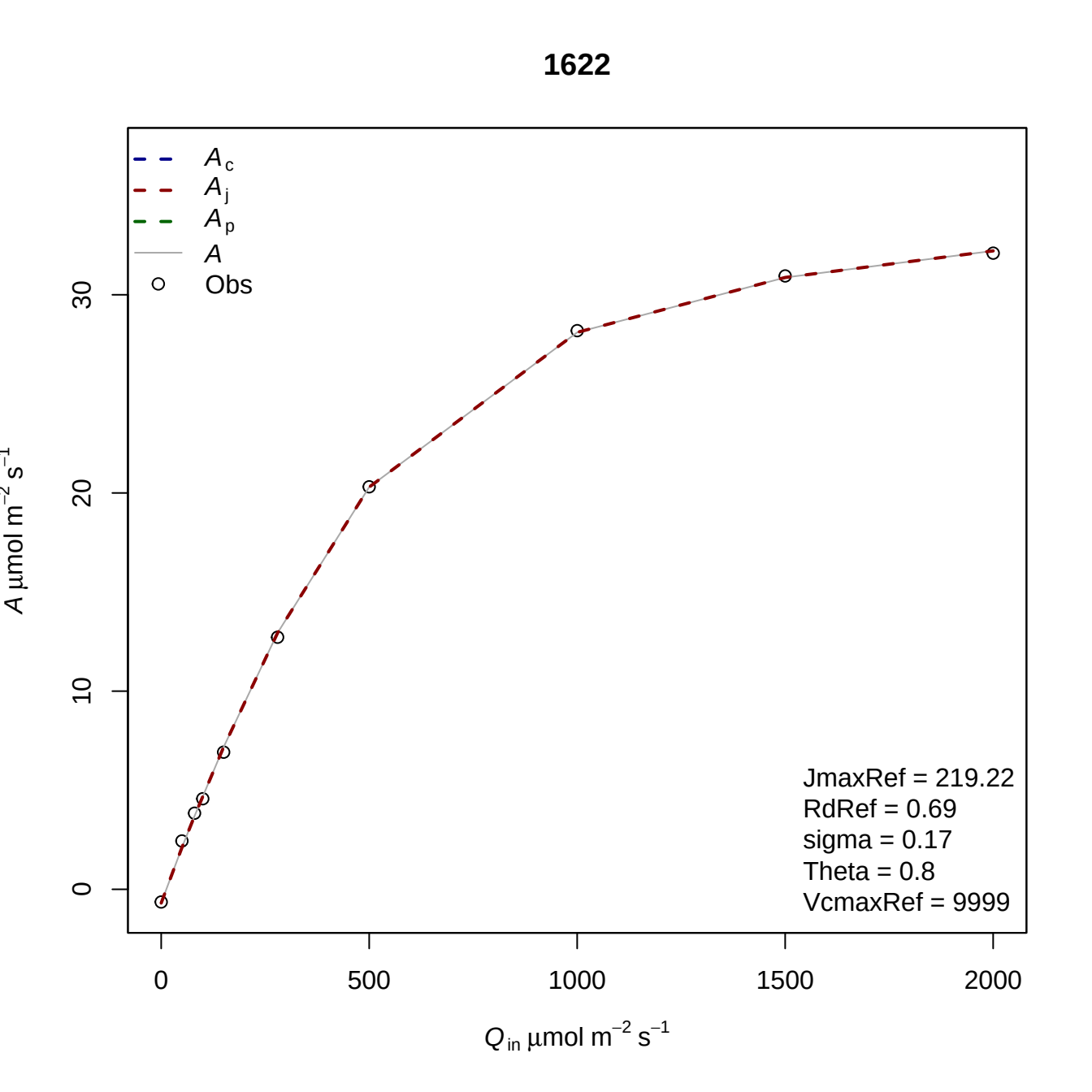

1617
A_c
A_j
A_p
A
Obs
JmaxRef = 144.08
RdRef = 0.93
sigma = 0.22
Theta = 0.67
VcmaxRef = 82.02
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

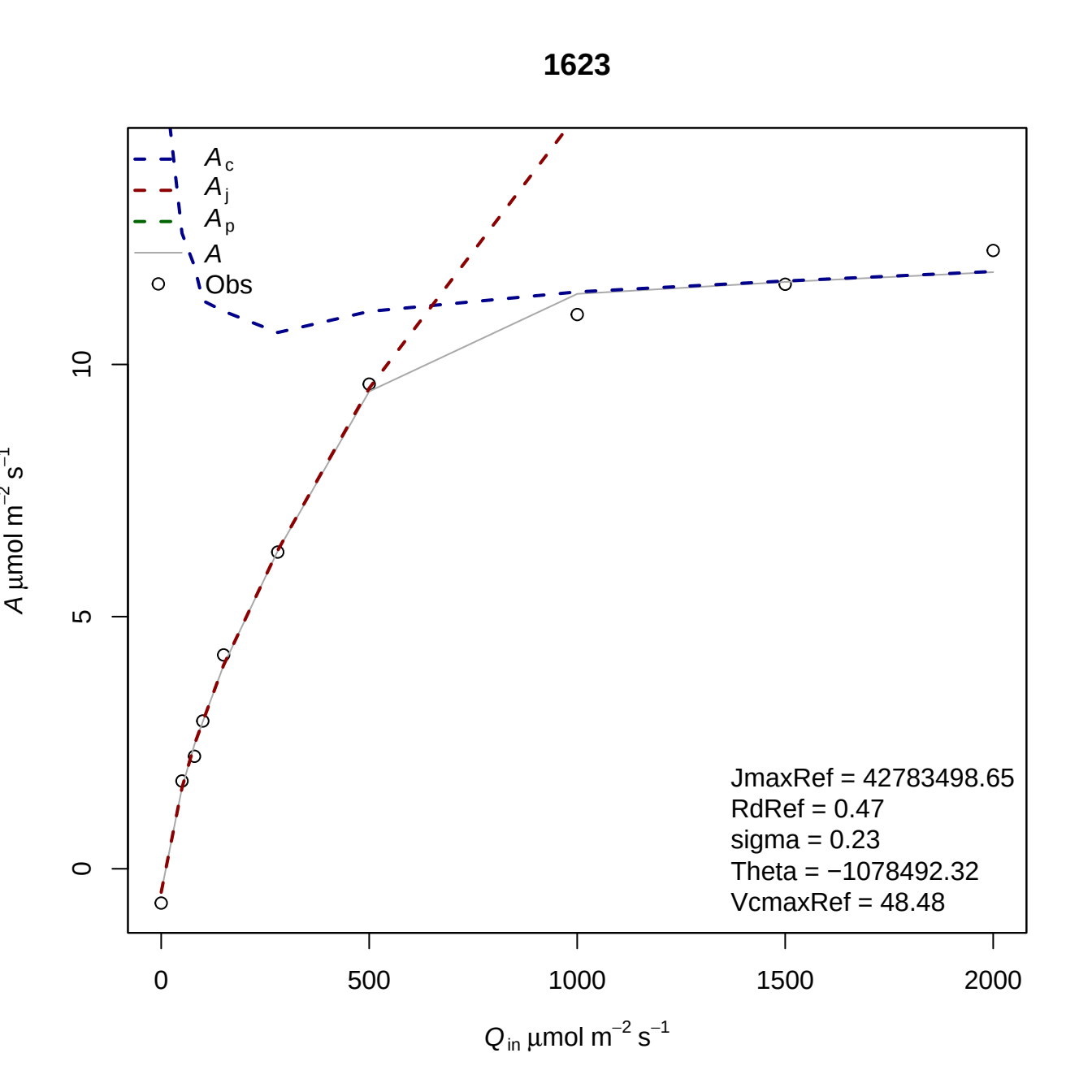

1618
A_c
A_j
A_p
A
Obs
JmaxRef = 142.06
RdRef = 0.72
sigma = 0.17
Theta = 0.77
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

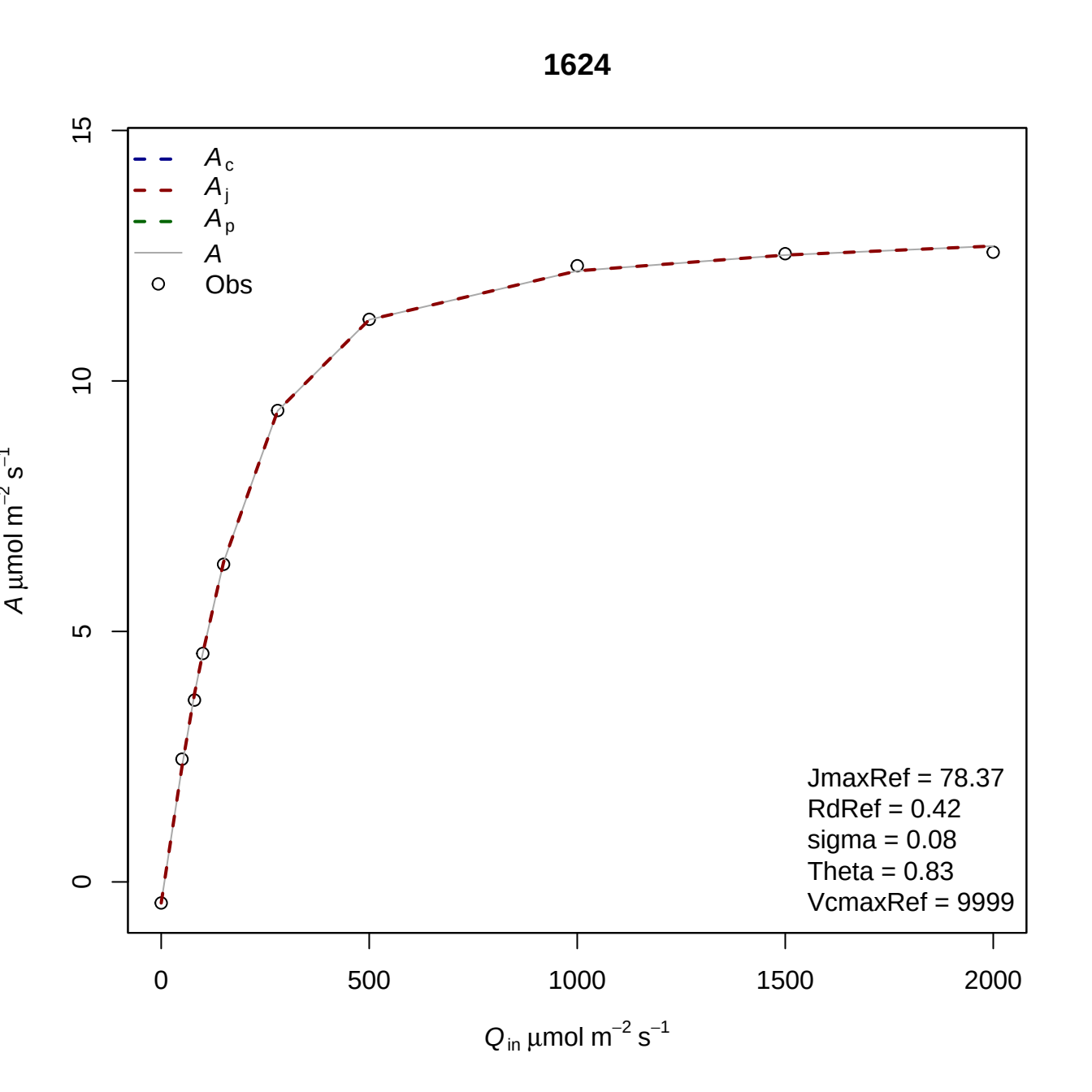

1619
A_c
A_j
A_p
A
Obs
JmaxRef = 208.94
RdRef = 0.43
sigma = 0.12
Theta = 0.79
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

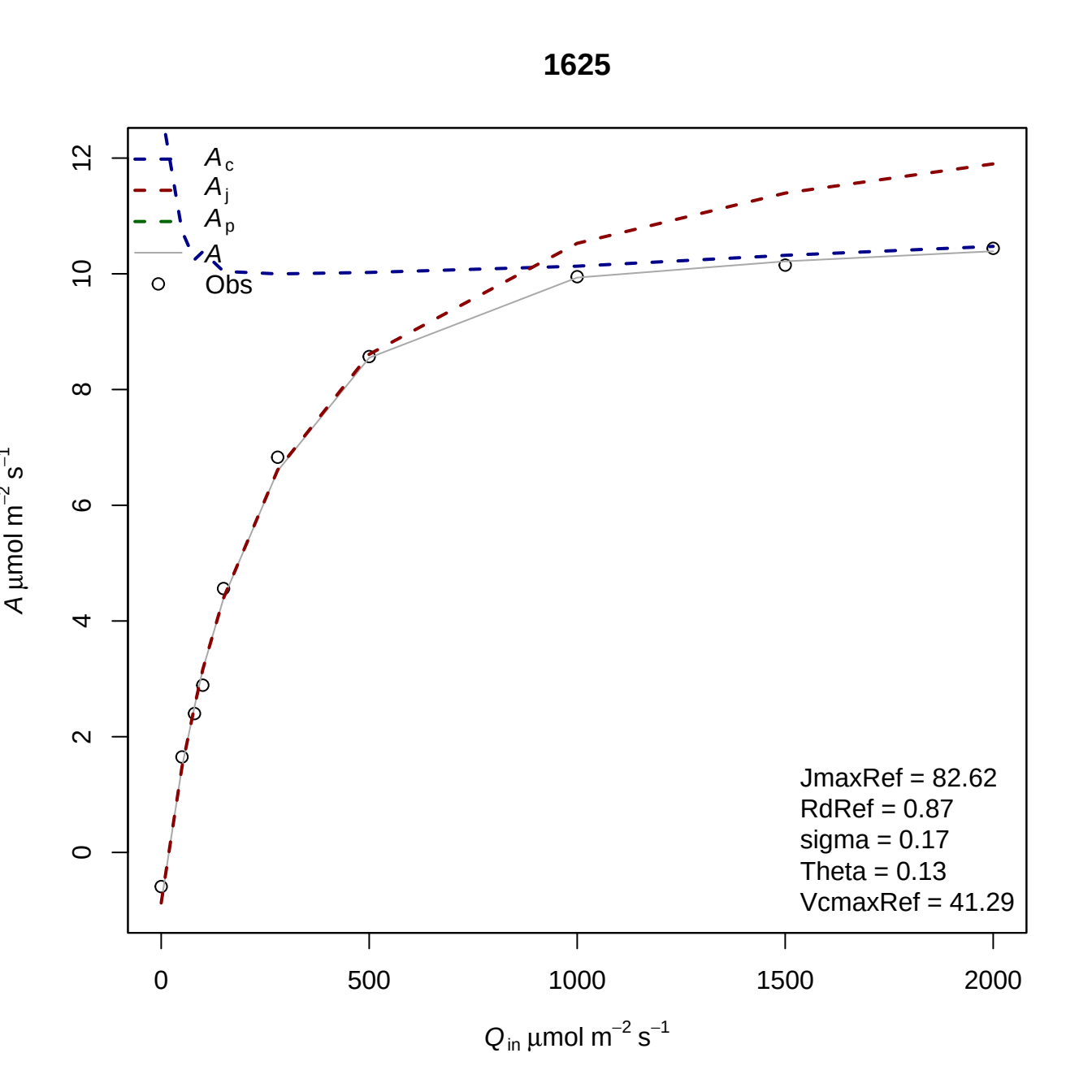

1620
A_c
A_j
A_p
A
Obs
JmaxRef = 225.5
RdRef = 0.61
sigma = 0.13
Theta = 0.69
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

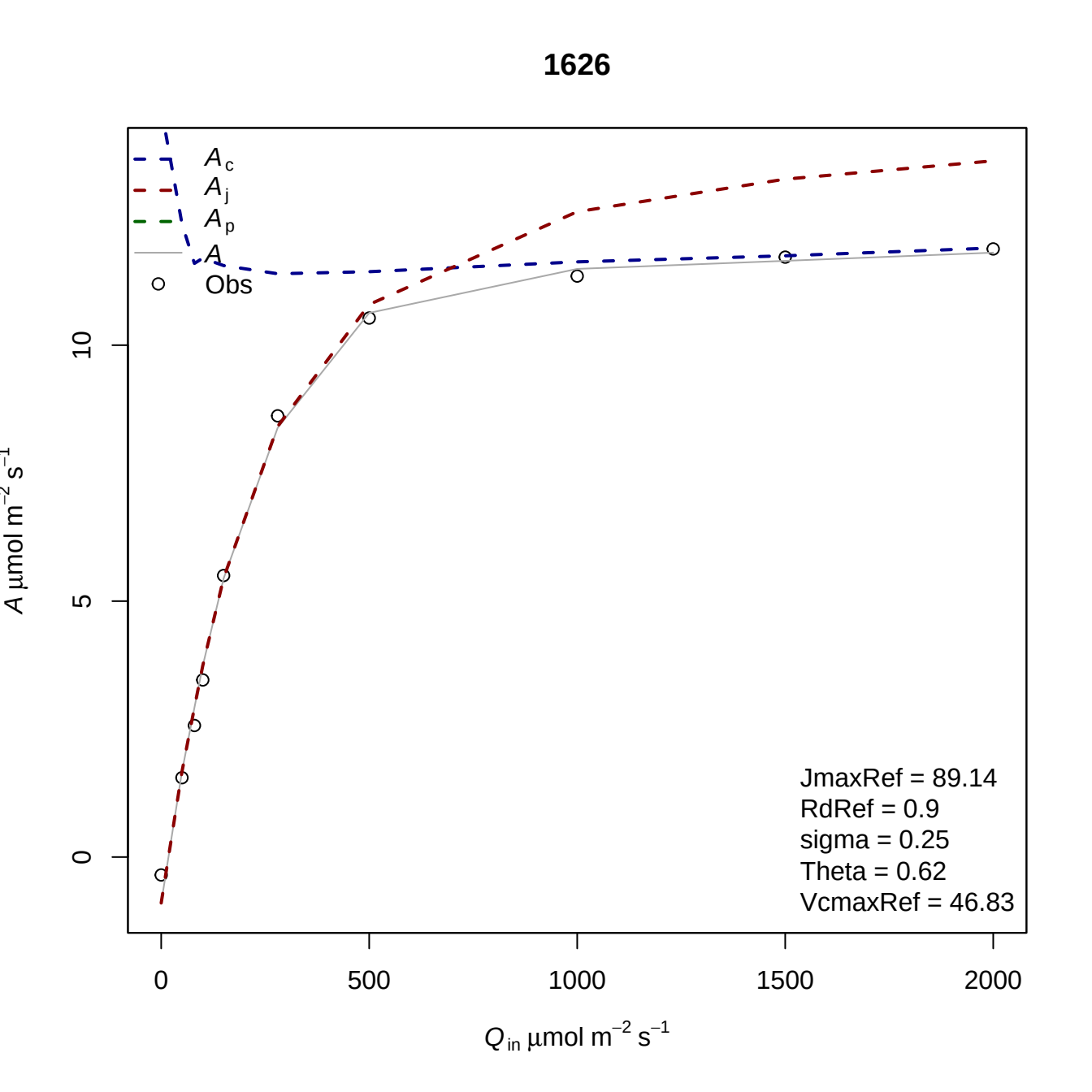

1621
A_c
A_j
A_p
A
Obs
JmaxRef = 228.44
RdRef = 0.81
sigma = 0.25
Theta = 0.58
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

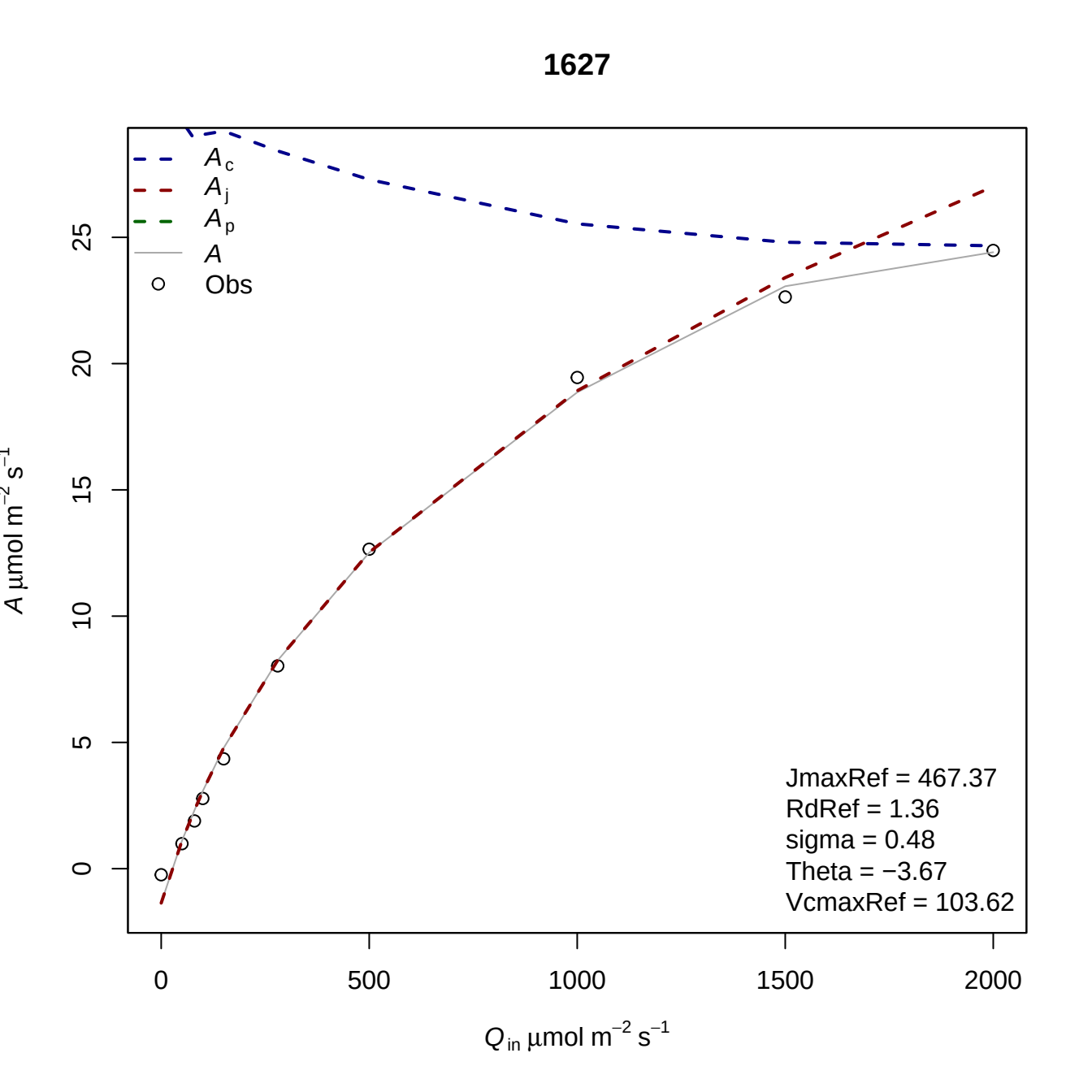

1622
A_c
A_j
A_p
A
Obs
JmaxRef = 219.22
RdRef = 0.69
sigma = 0.17
Theta = 0.8
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

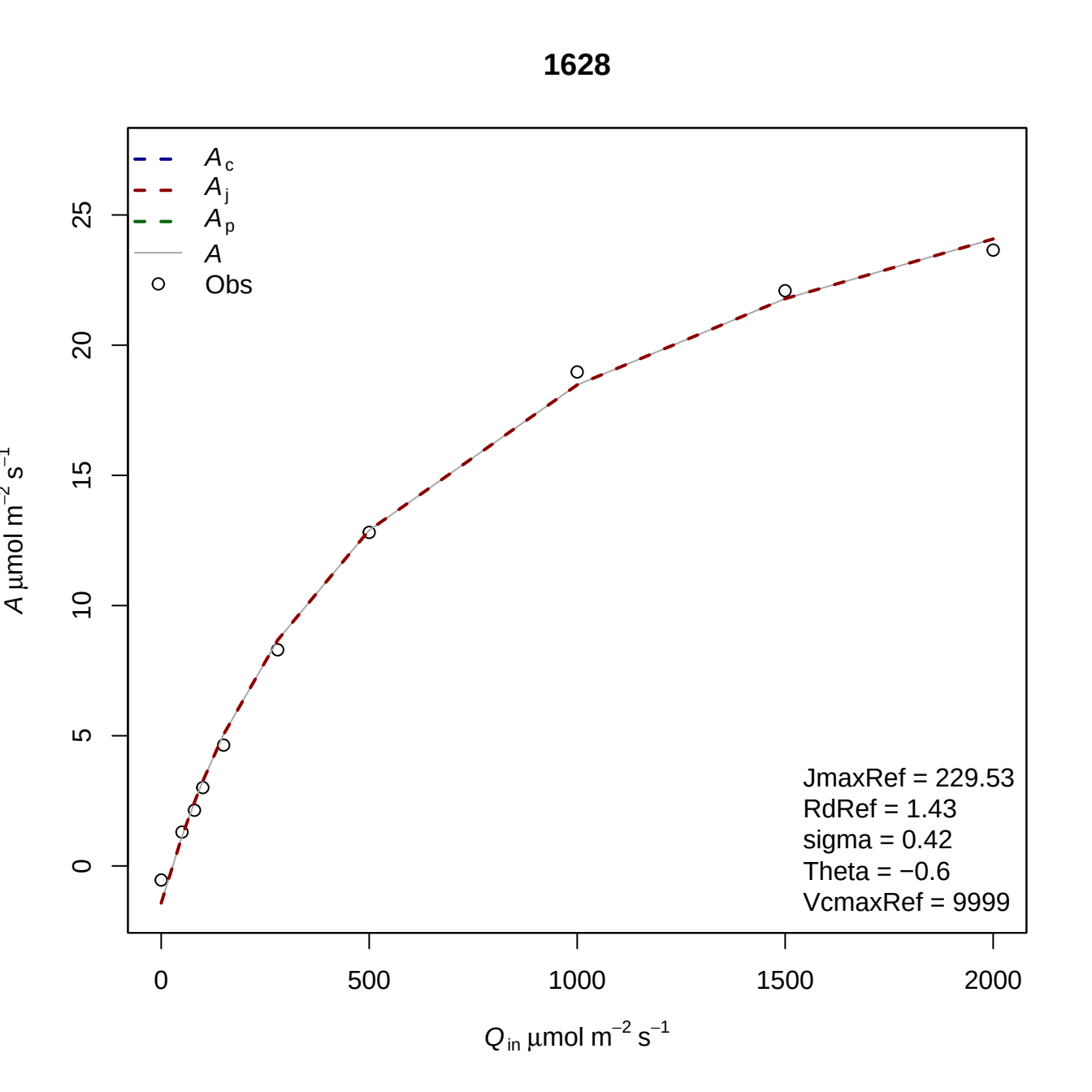

1623
A_c
A_j
A_p
A
Obs
JmaxRef = 42783498.65
RdRef = 0.47
sigma = 0.23
Theta = −1078492.32
VcmaxRef = 48.48
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

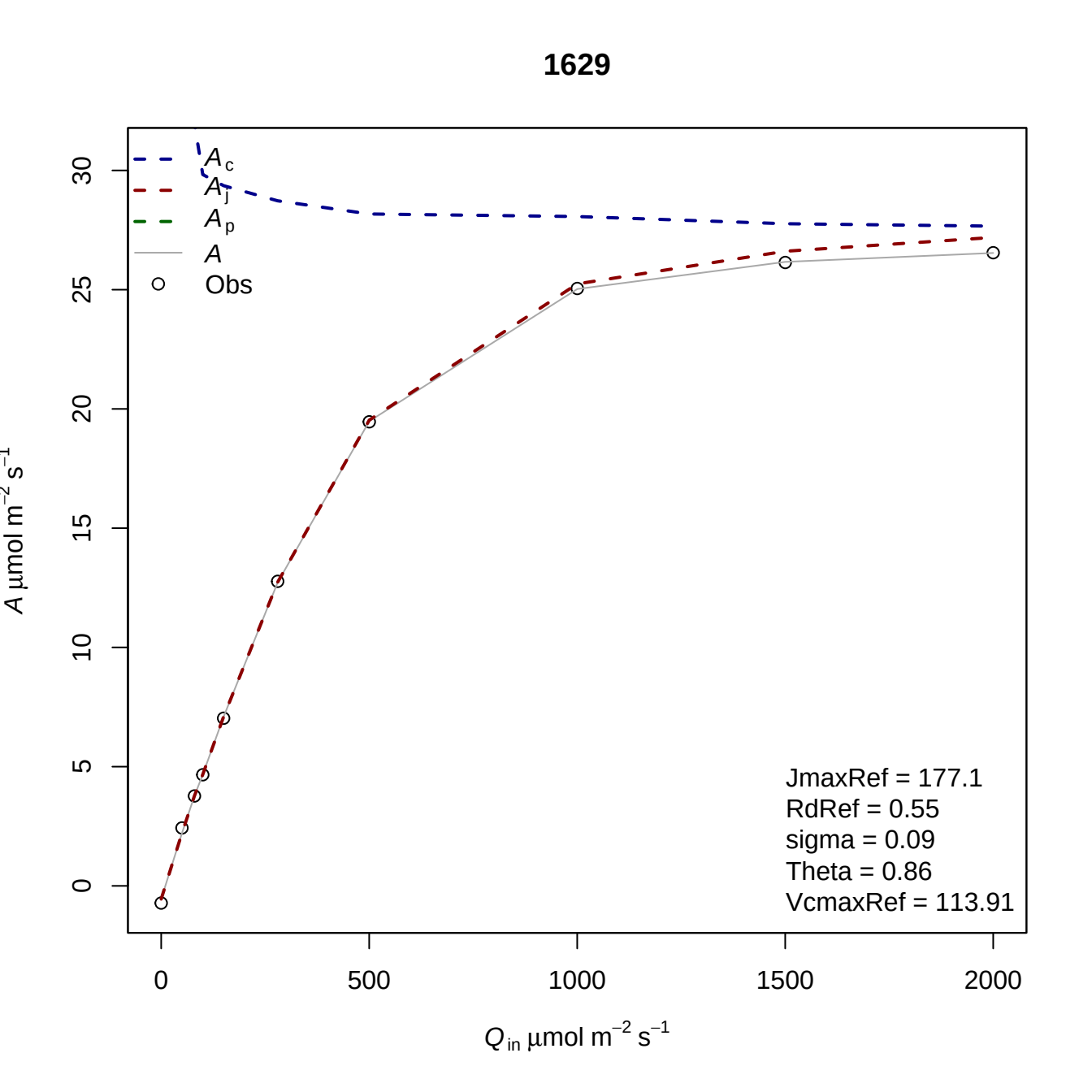

1624
A_c
A_j
A_p
A
Obs
JmaxRef = 78.37
RdRef = 0.42
sigma = 0.08
Theta = 0.83
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

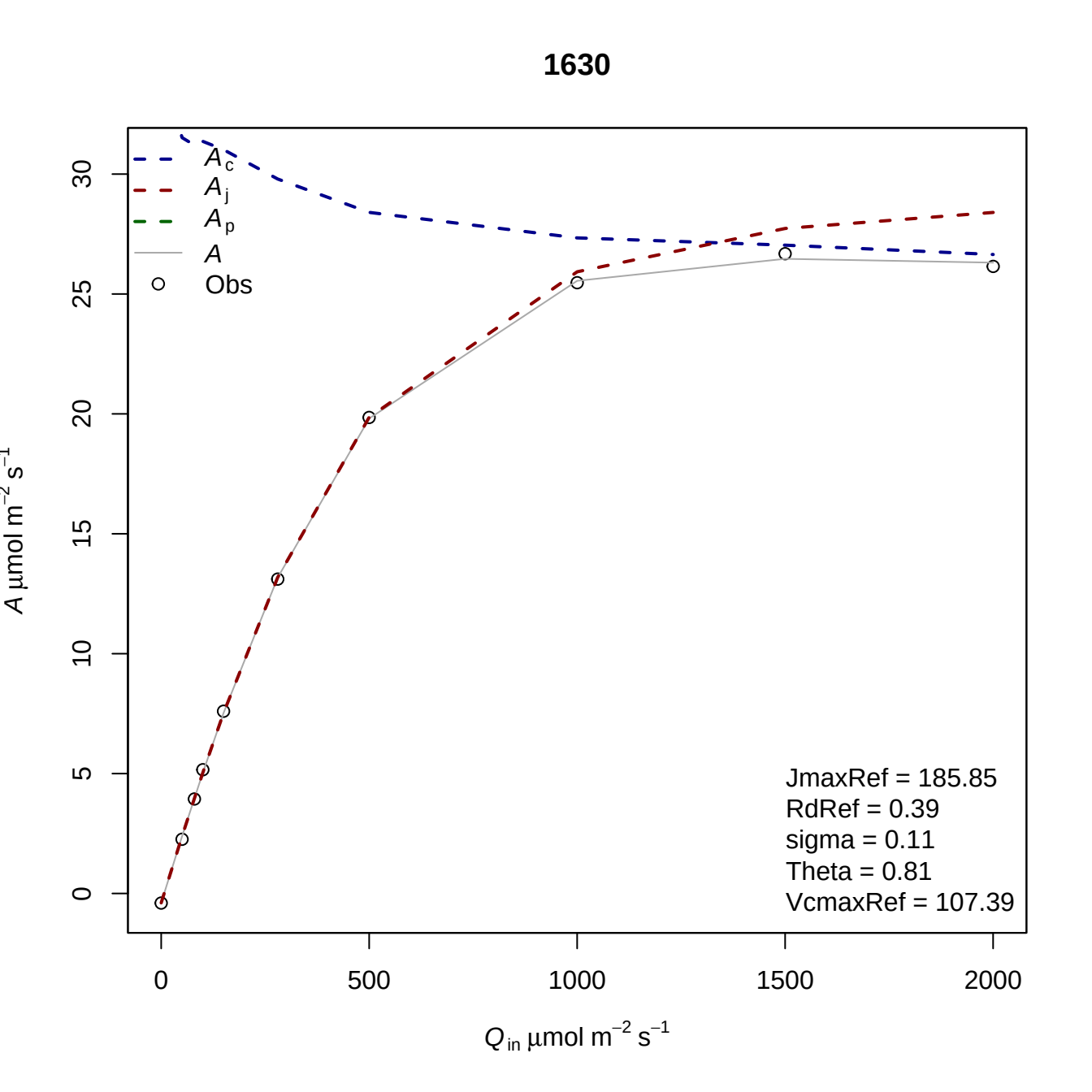

1625
A_c
A_j
A_p
A
Obs
JmaxRef = 82.62
RdRef = 0.87
sigma = 0.17
Theta = 0.13
VcmaxRef = 41.29
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

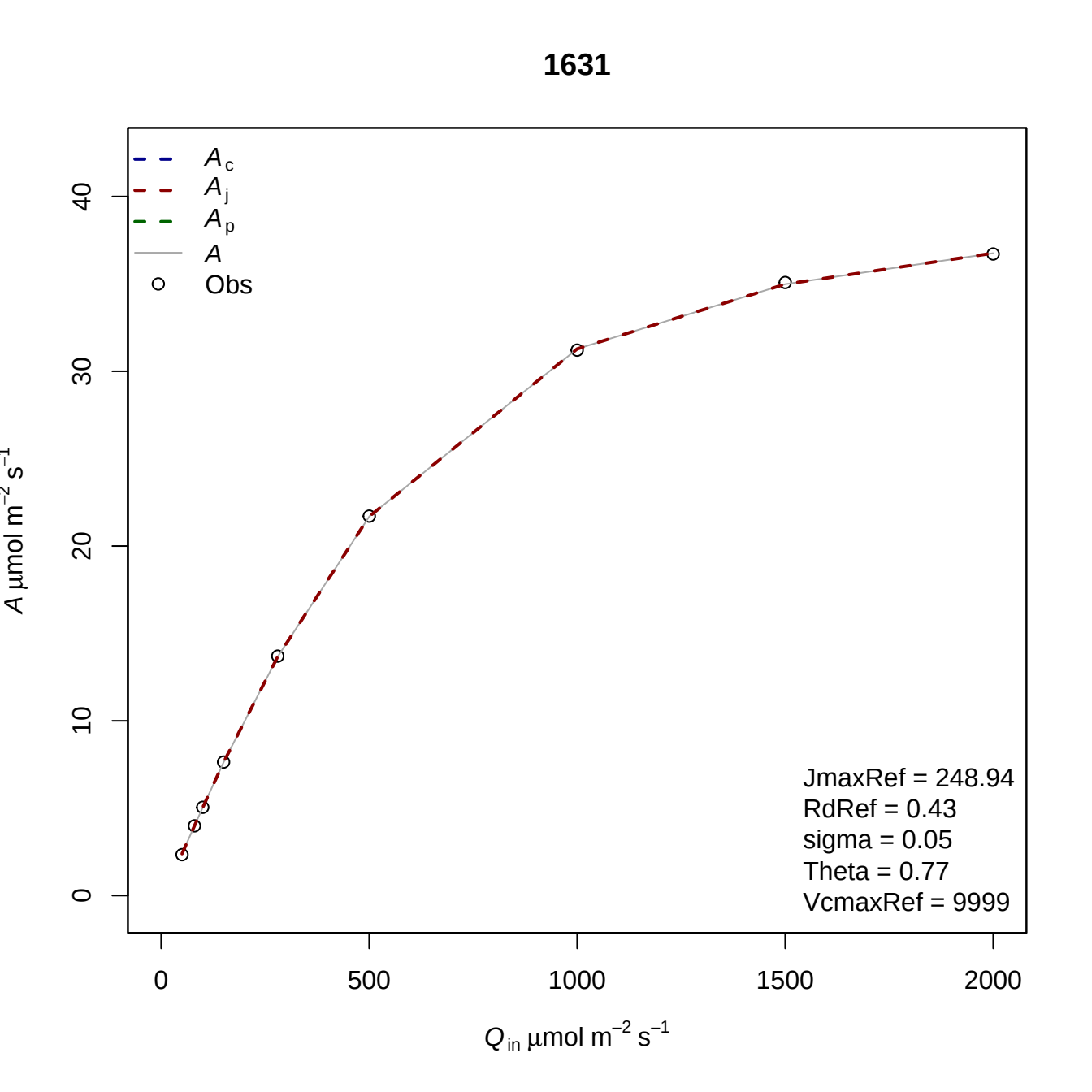

1626
A_c
A_j
A_p
A
Obs
JmaxRef = 89.14
RdRef = 0.9
sigma = 0.25
Theta = 0.62
VcmaxRef = 46.83
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

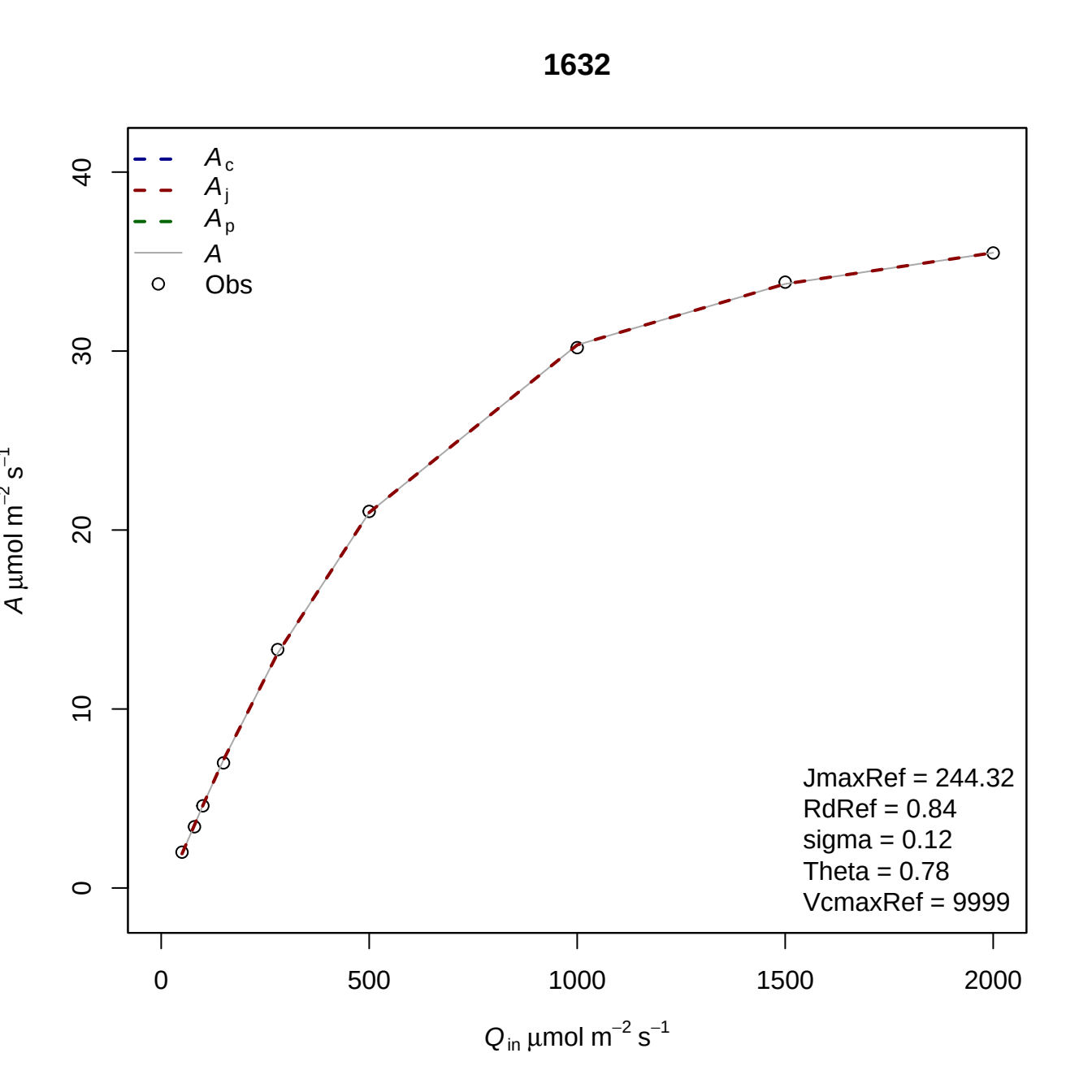

1627
A_c
A_j
A_p
A
Obs
JmaxRef = 467.37
RdRef = 1.36
sigma = 0.48
Theta = −3.67
VcmaxRef = 103.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

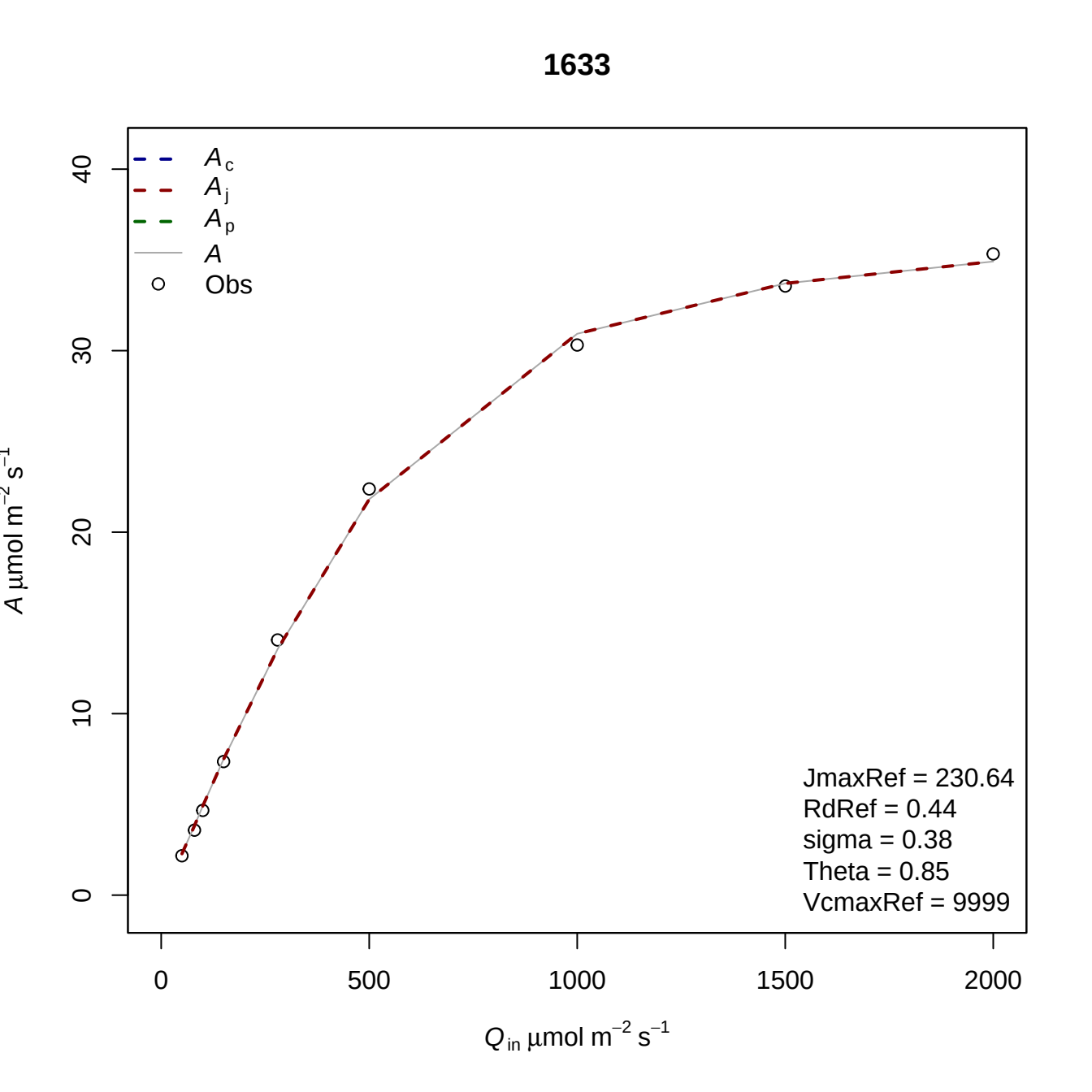

1628
A_c
A_j
A_p
A
Obs
JmaxRef = 229.53
RdRef = 1.43
sigma = 0.42
Theta = −0.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

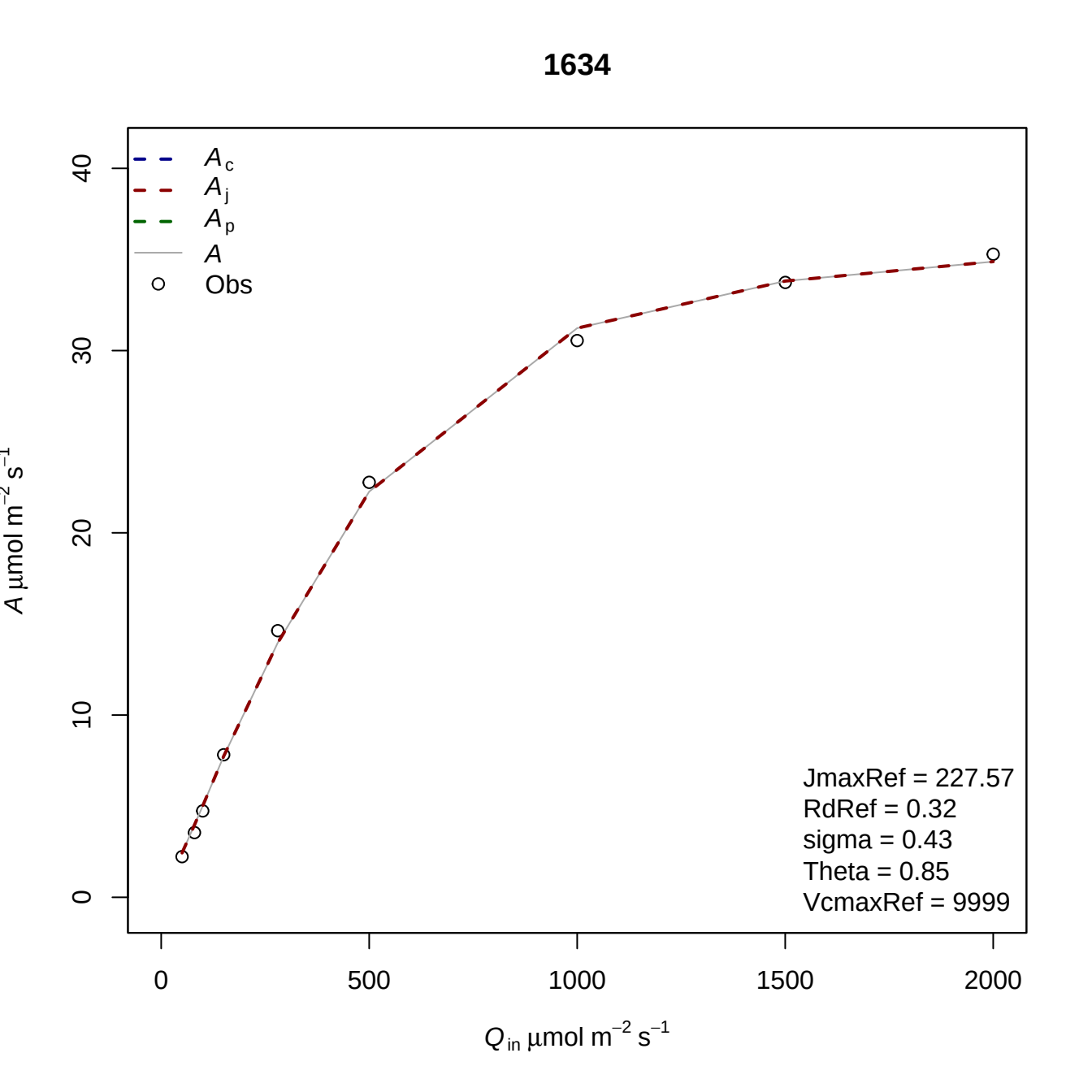

1629
A_c
A_J
A_p
A
Obs
JmaxRef = 177.1
RdRef = 0.55
sigma = 0.09
Theta = 0.86
VcmaxRef = 113.91
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

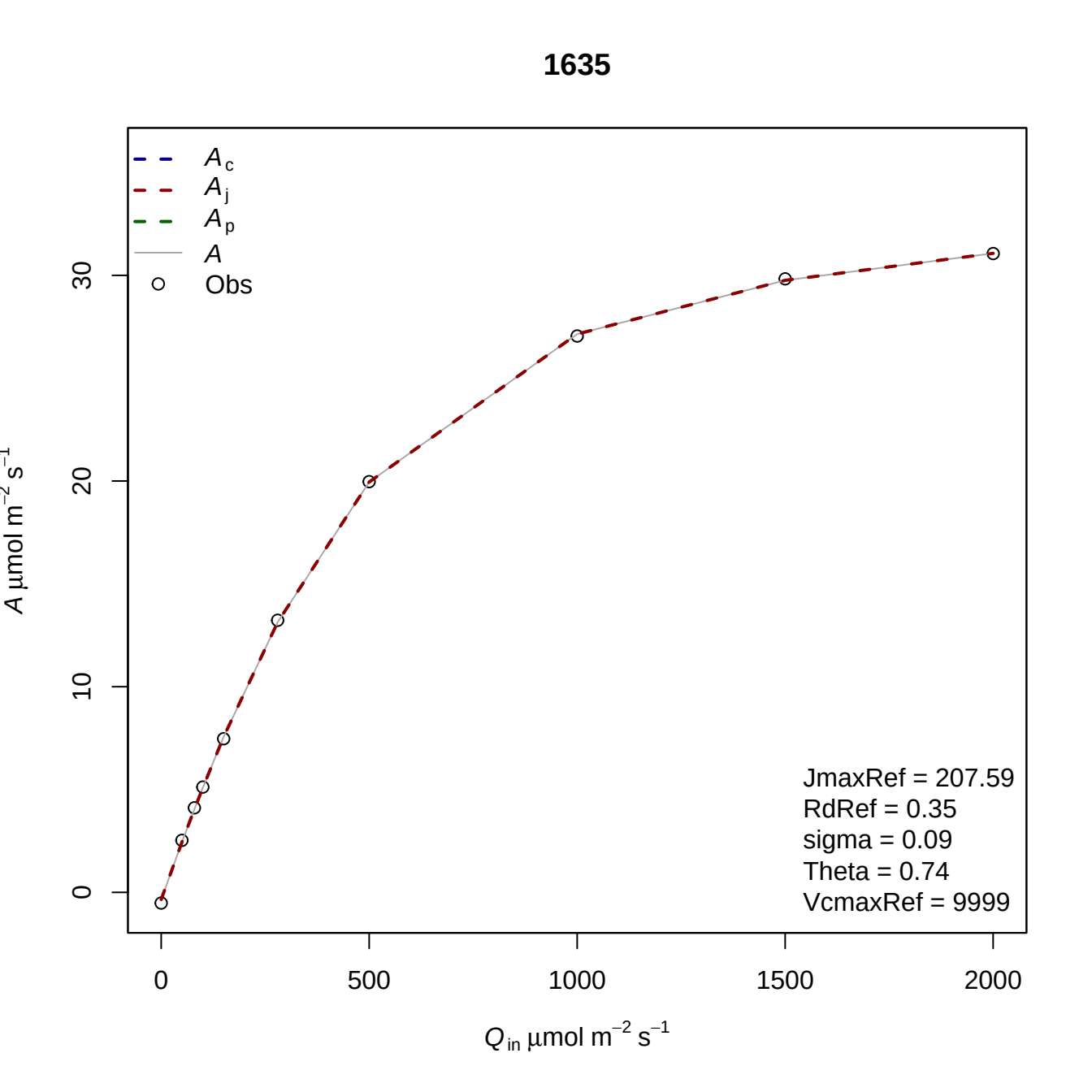

1630
A_c
A_j
A_p
A
Obs
JmaxRef = 185.85
RdRef = 0.39
sigma = 0.11
Theta = 0.81
VcmaxRef = 107.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

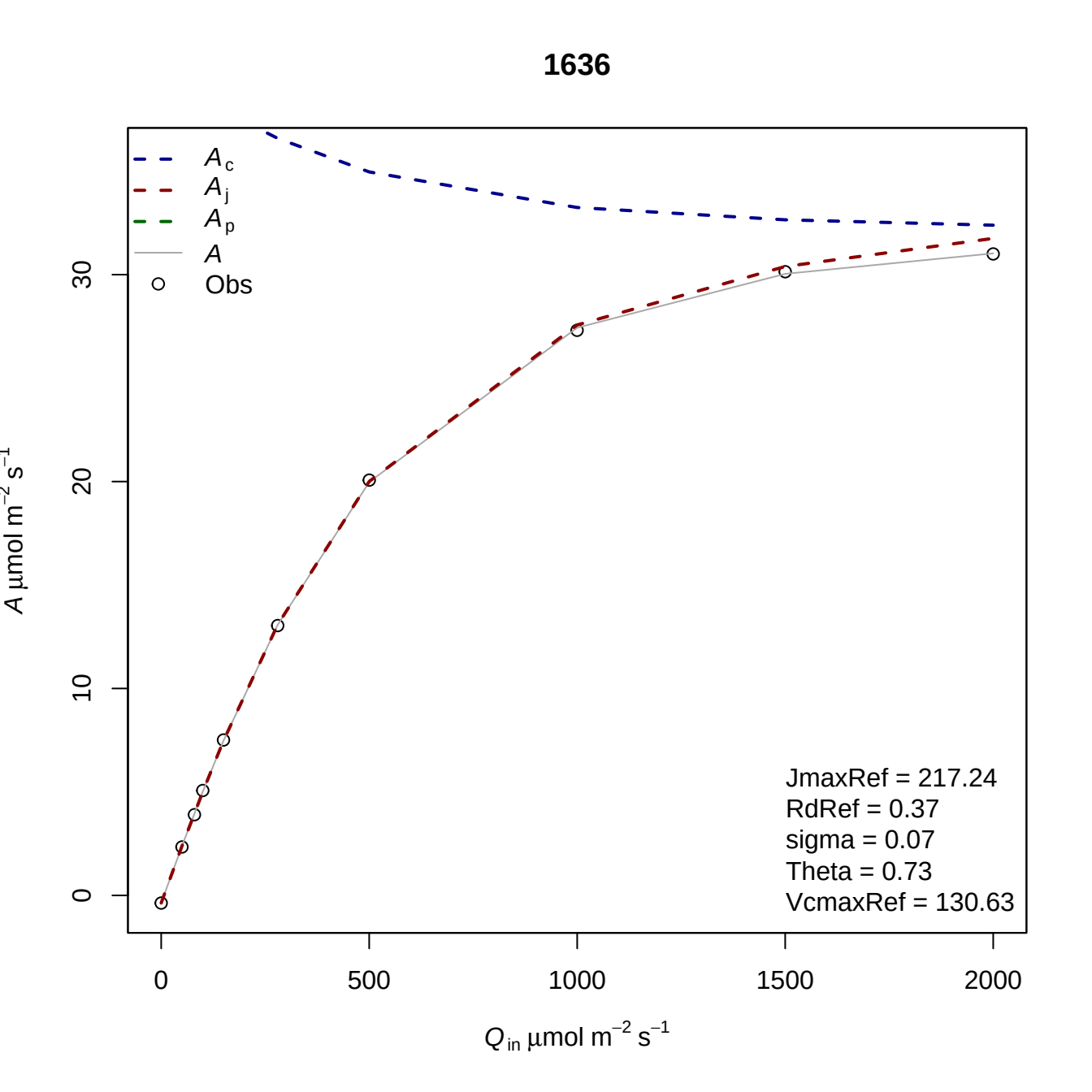

1631
A_c
A_j
A_p
A
Obs
JmaxRef = 248.94
RdRef = 0.43
sigma = 0.05
Theta = 0.77
VcmaxRef = 9999
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

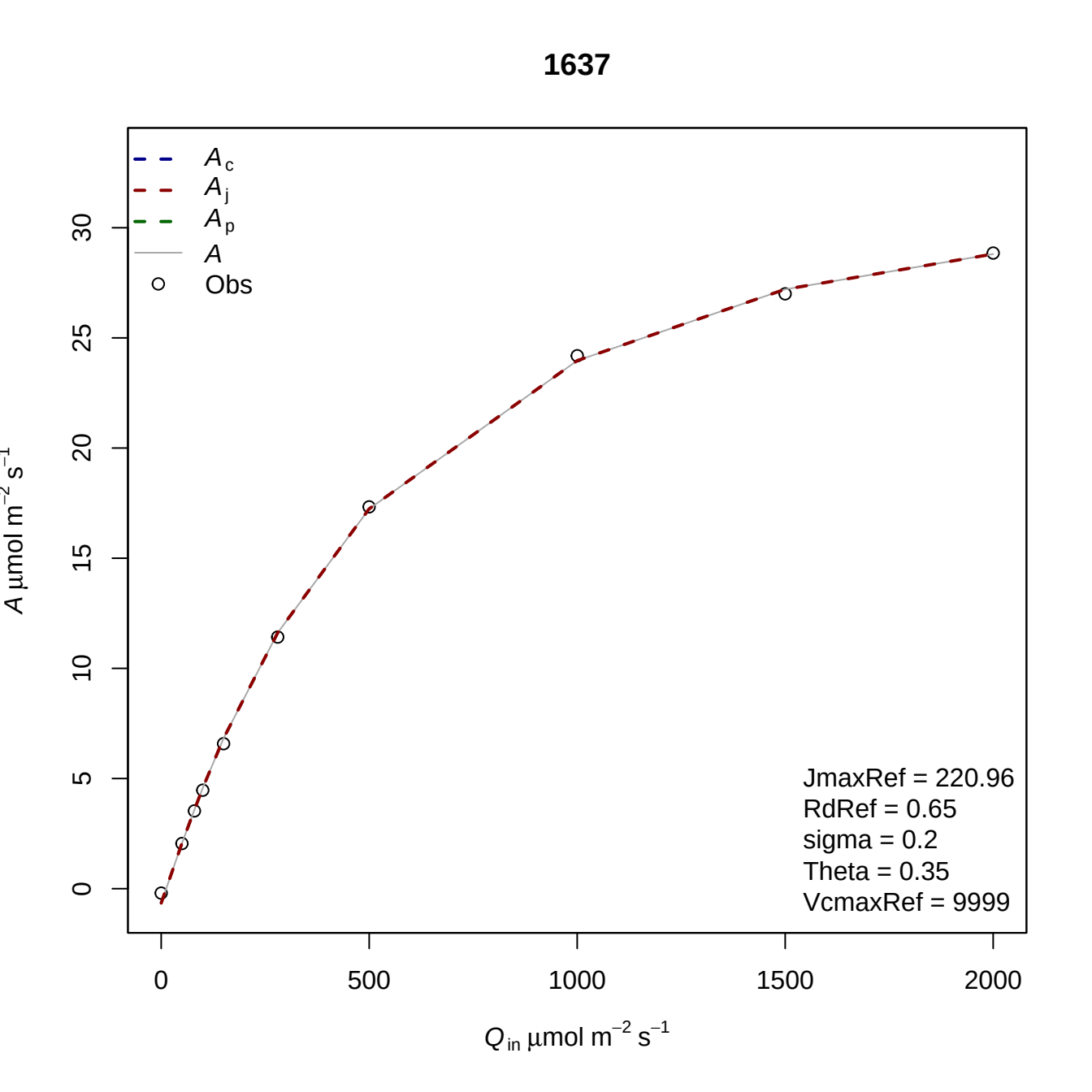

1632
A_c
A_j
A_p
A
Obs
JmaxRef = 244.32
RdRef = 0.84
sigma = 0.12
Theta = 0.78
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

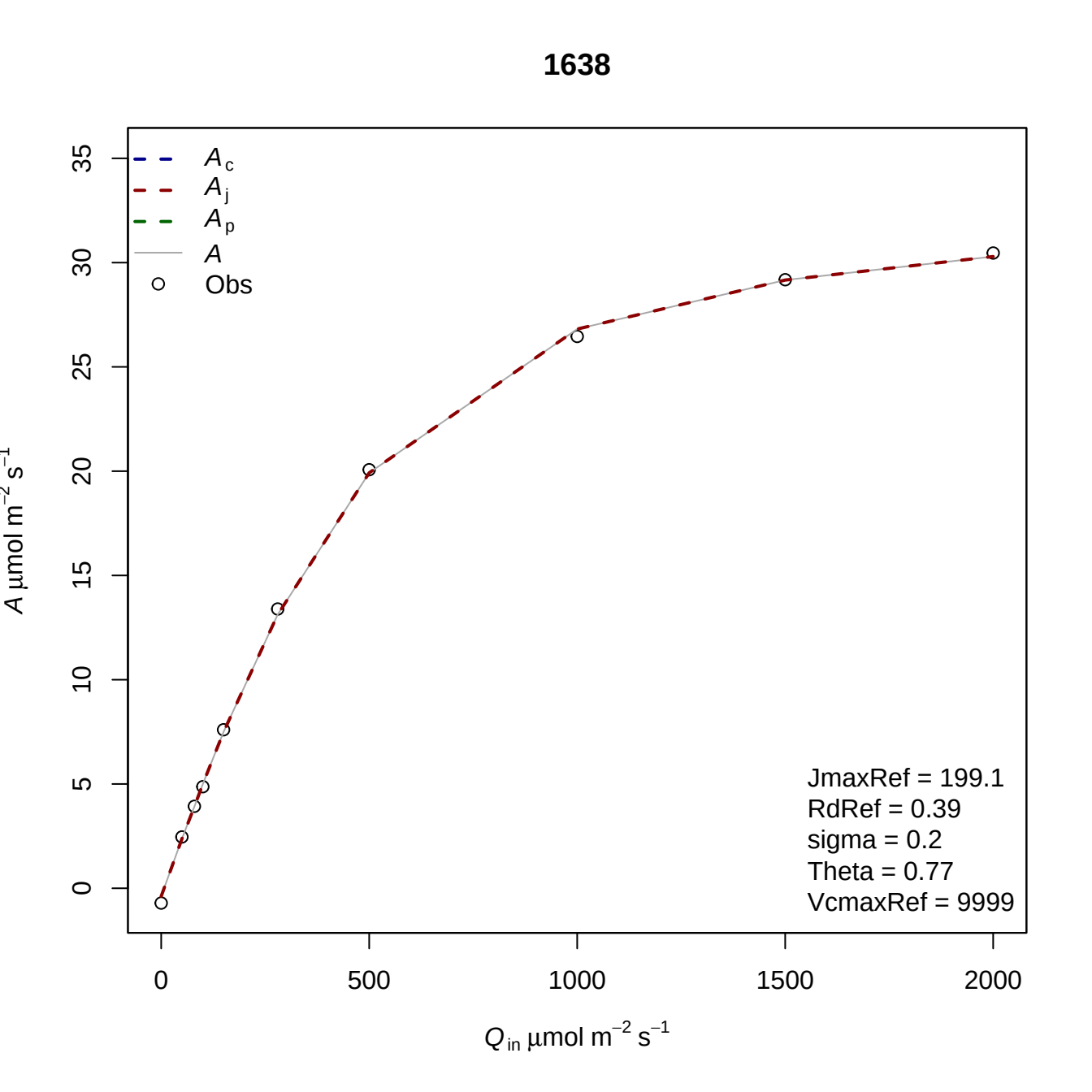

1633
A_c
A_j
A_p
A
Obs
JmaxRef = 230.64
RdRef = 0.44
sigma = 0.38
Theta = 0.85
VcmaxRef = 9999
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

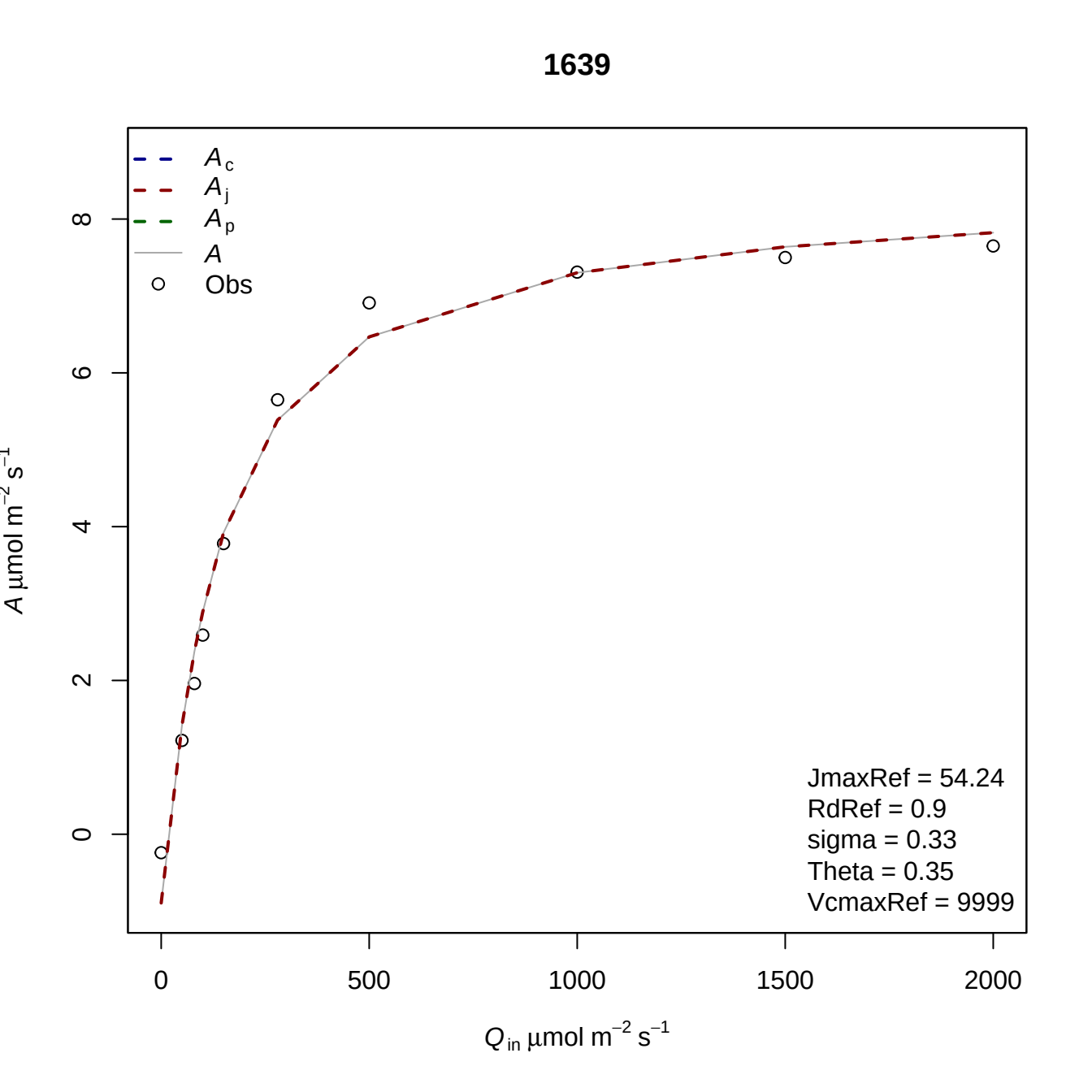

1634
A_c
A_j
A_p
A
Obs
JmaxRef = 227.57
RdRef = 0.32
sigma = 0.43
Theta = 0.85
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

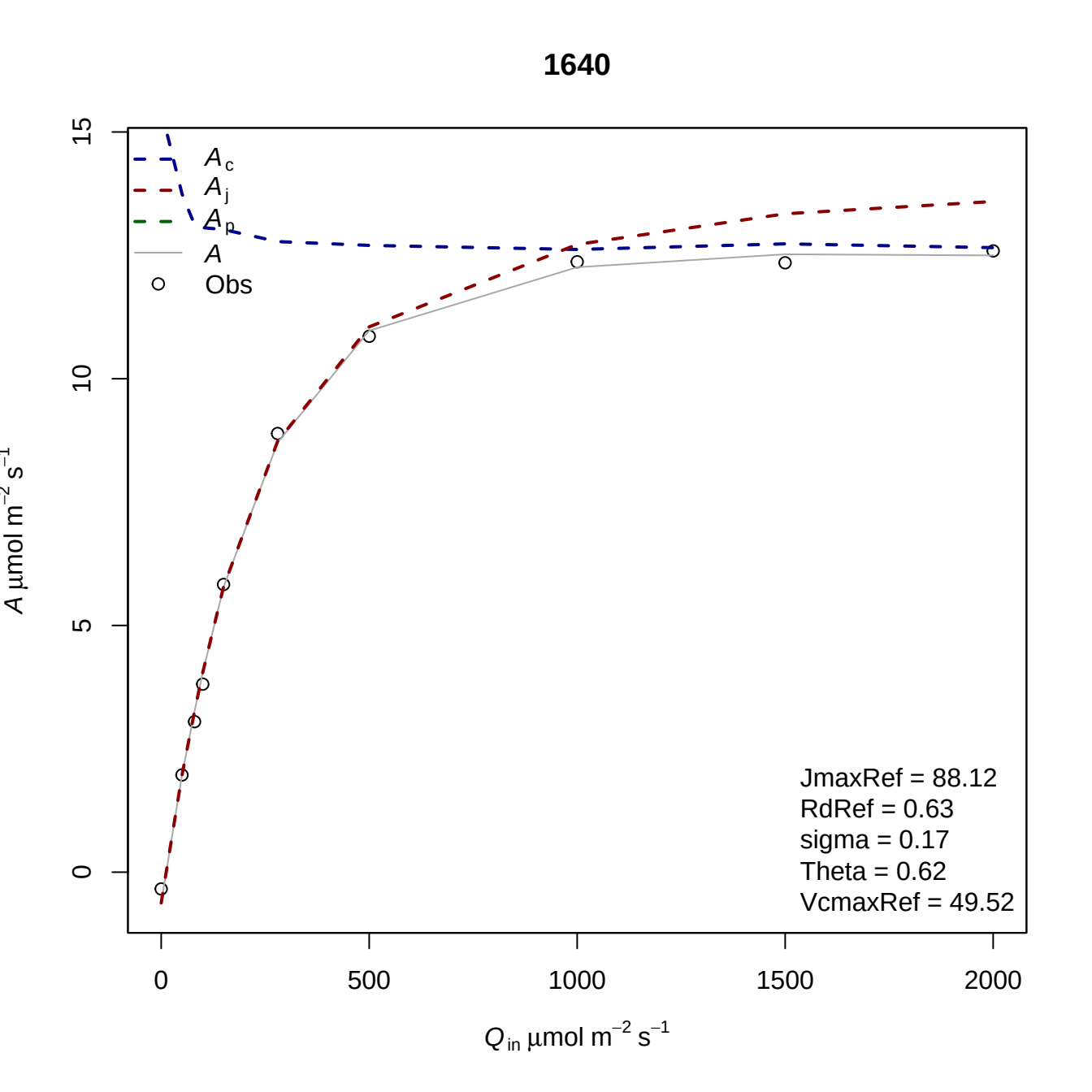

1635
A_c
A_j
A_p
A
Obs
JmaxRef = 207.59
RdRef = 0.35
sigma = 0.09
Theta = 0.74
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

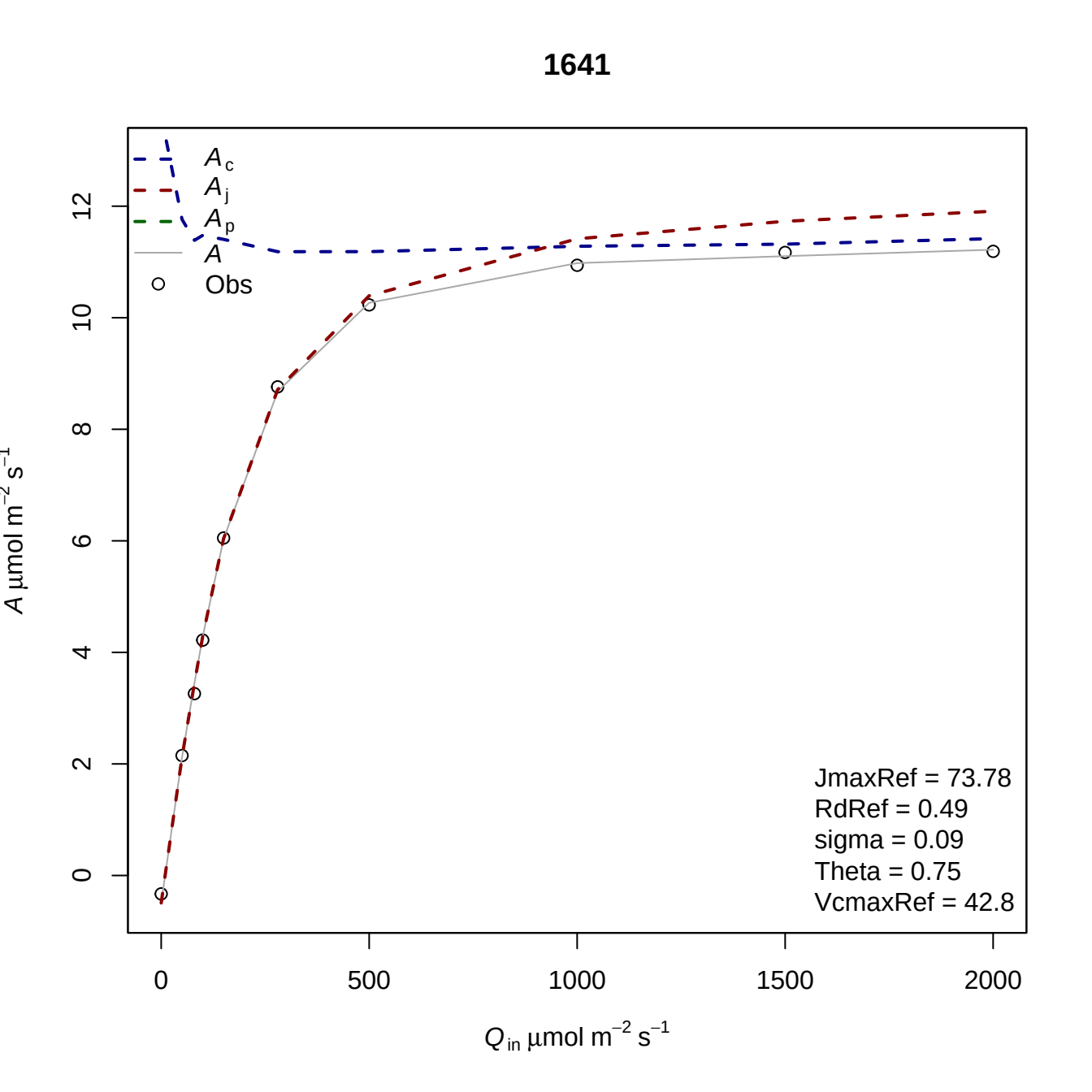

1636
A_c
A_j
A_p
A
Obs
JmaxRef = 217.24
RdRef = 0.37
sigma = 0.07
Theta = 0.73
VcmaxRef = 130.63
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

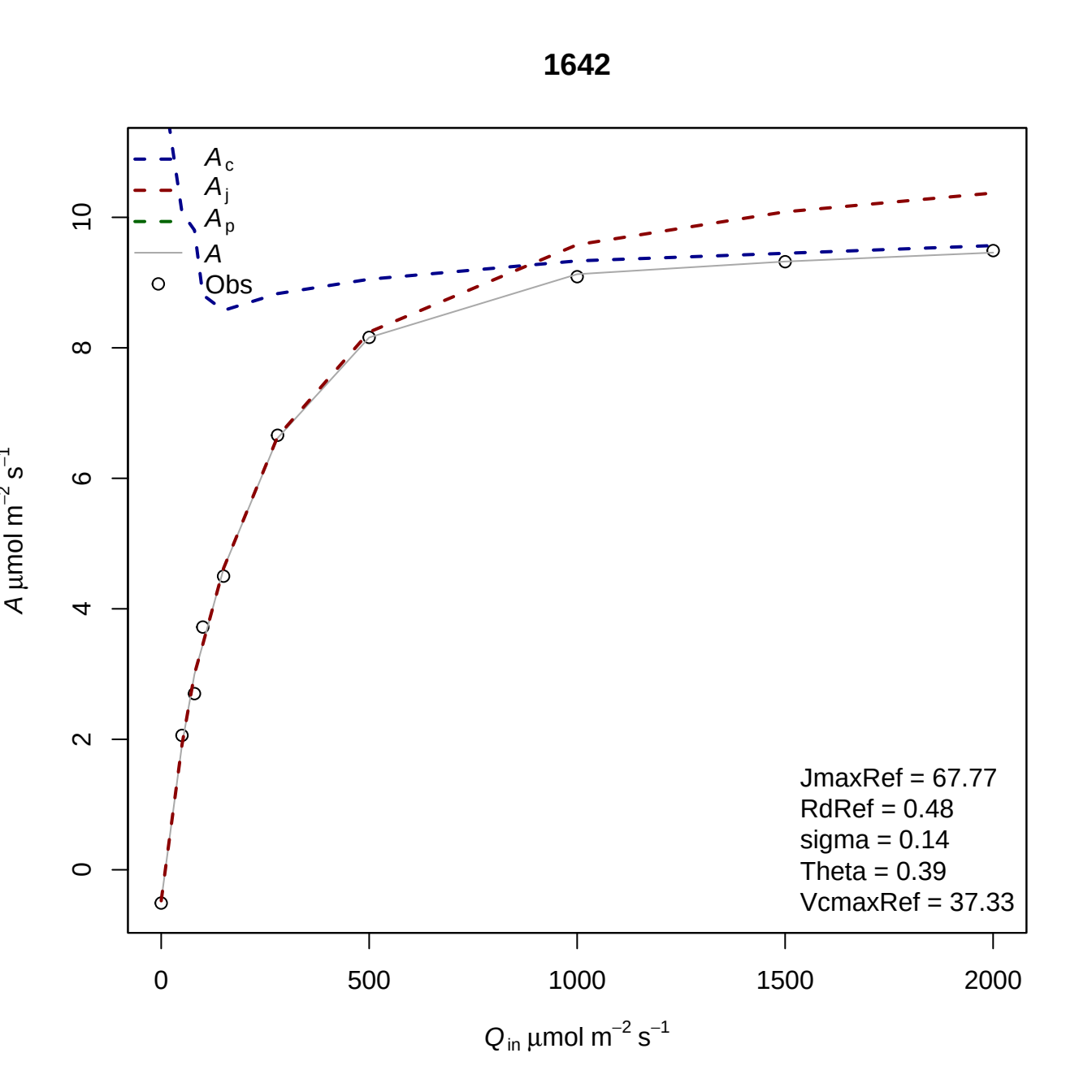

1637
A_c
A_j
A_p
A
Obs
JmaxRef = 220.96
RdRef = 0.65
sigma = 0.2
Theta = 0.35
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

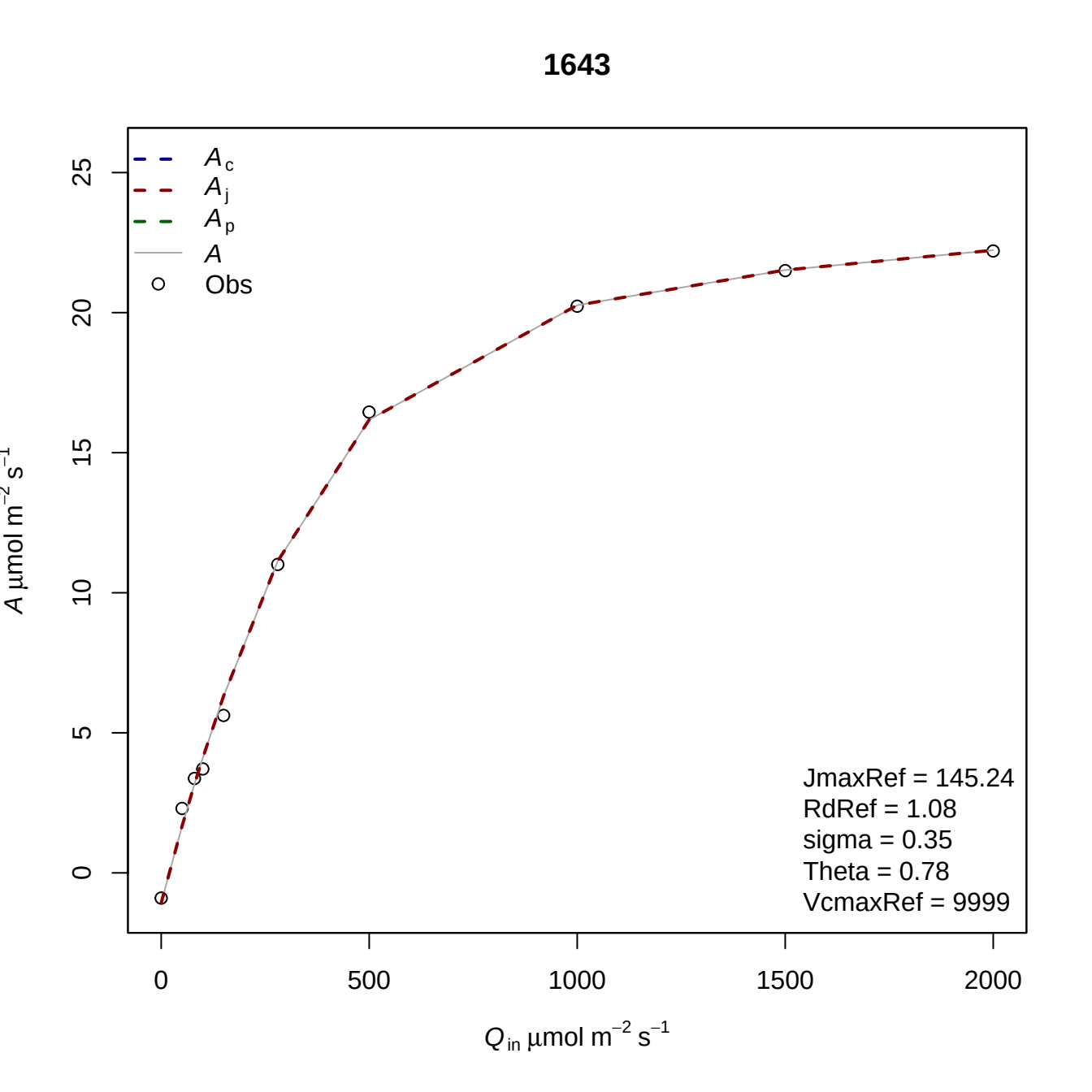

1638
A_c
A_j
A_p
A
Obs
JmaxRef = 199.1
RdRef = 0.39
sigma = 0.2
Theta = 0.77
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

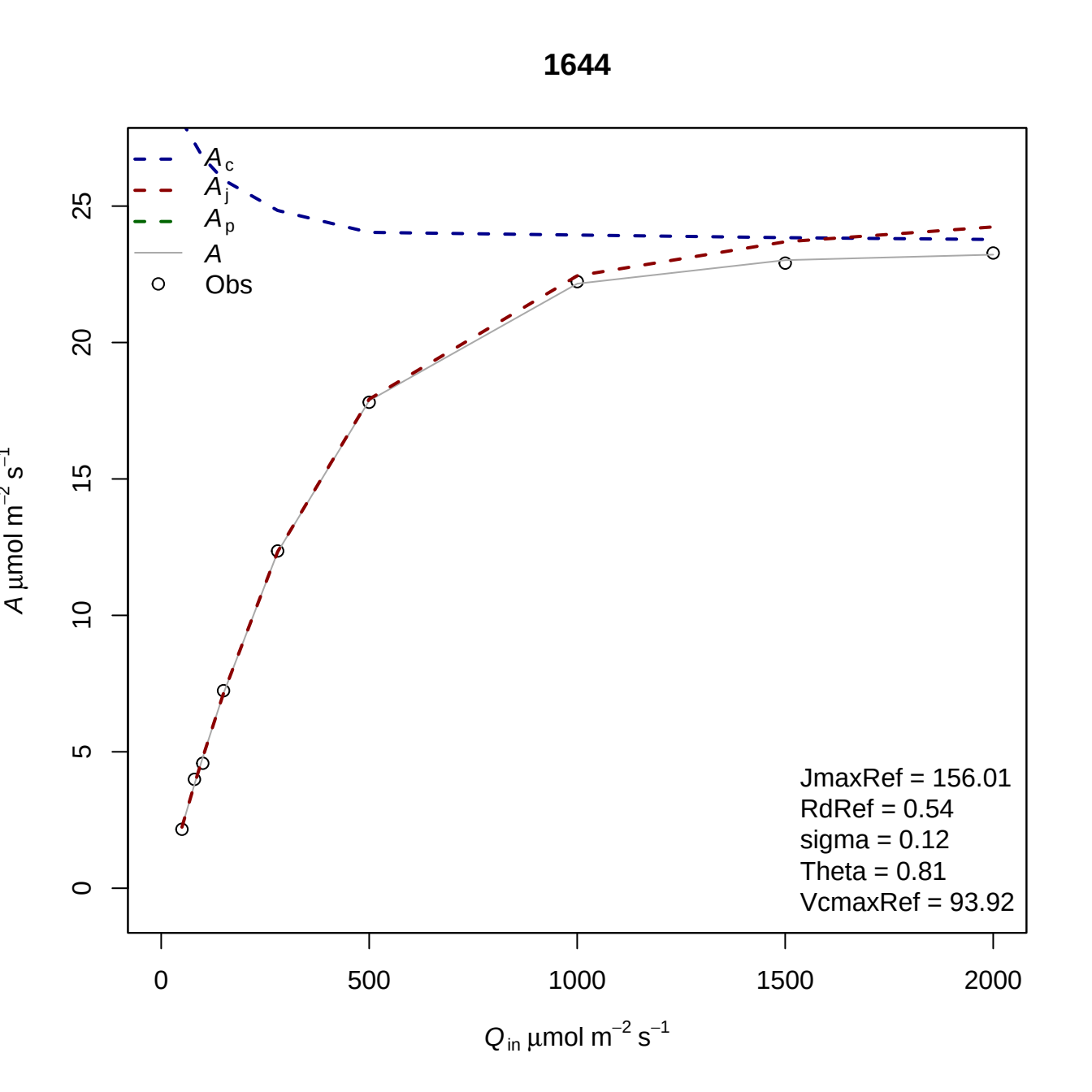

1639
A_c
A_j
A_p
A
Obs
JmaxRef = 54.24
RdRef = 0.9
sigma = 0.33
Theta = 0.35
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

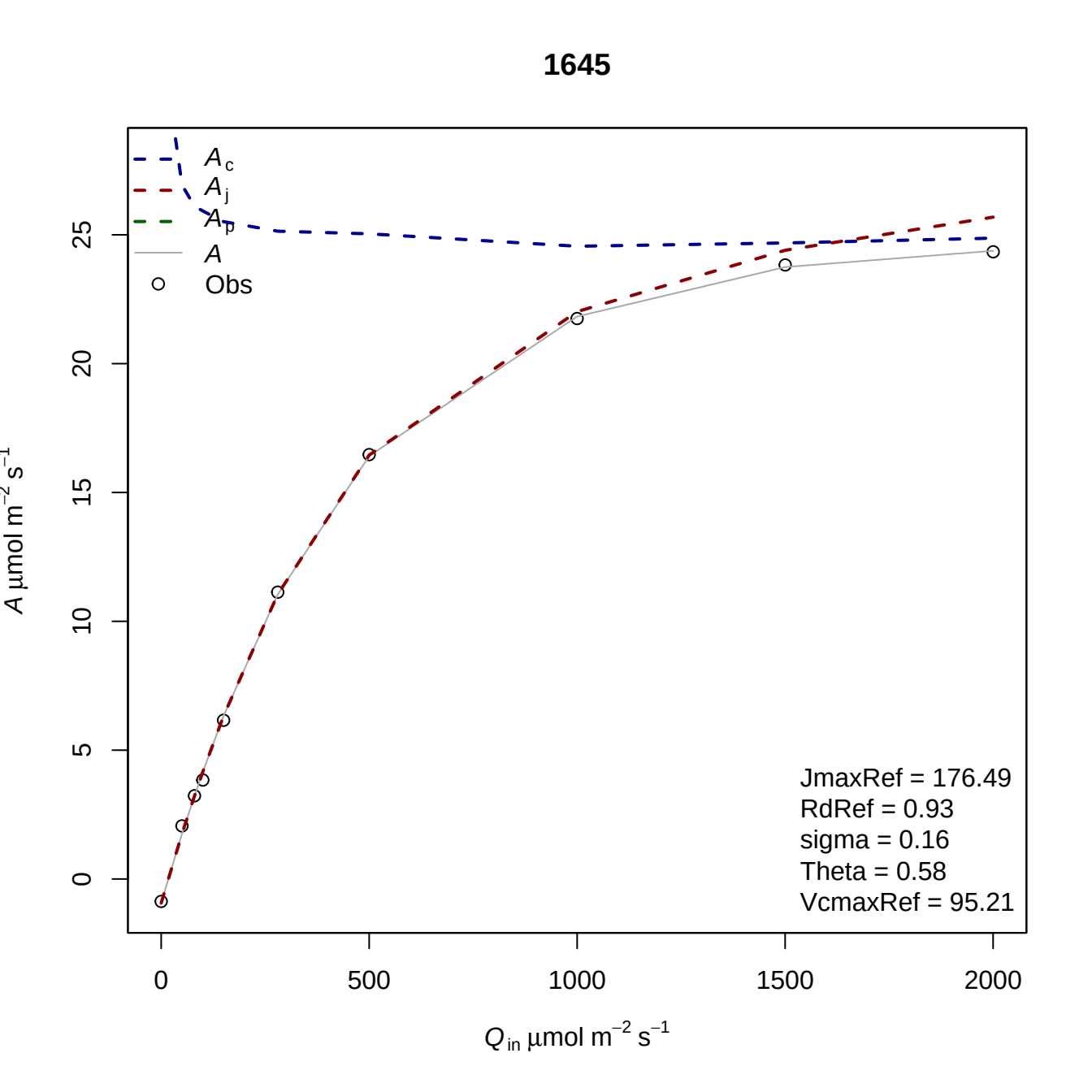

1640
A_c
A_j
A_p
A
Obs
JmaxRef = 88.12
RdRef = 0.63
sigma = 0.17
Theta = 0.62
VcmaxRef = 49.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

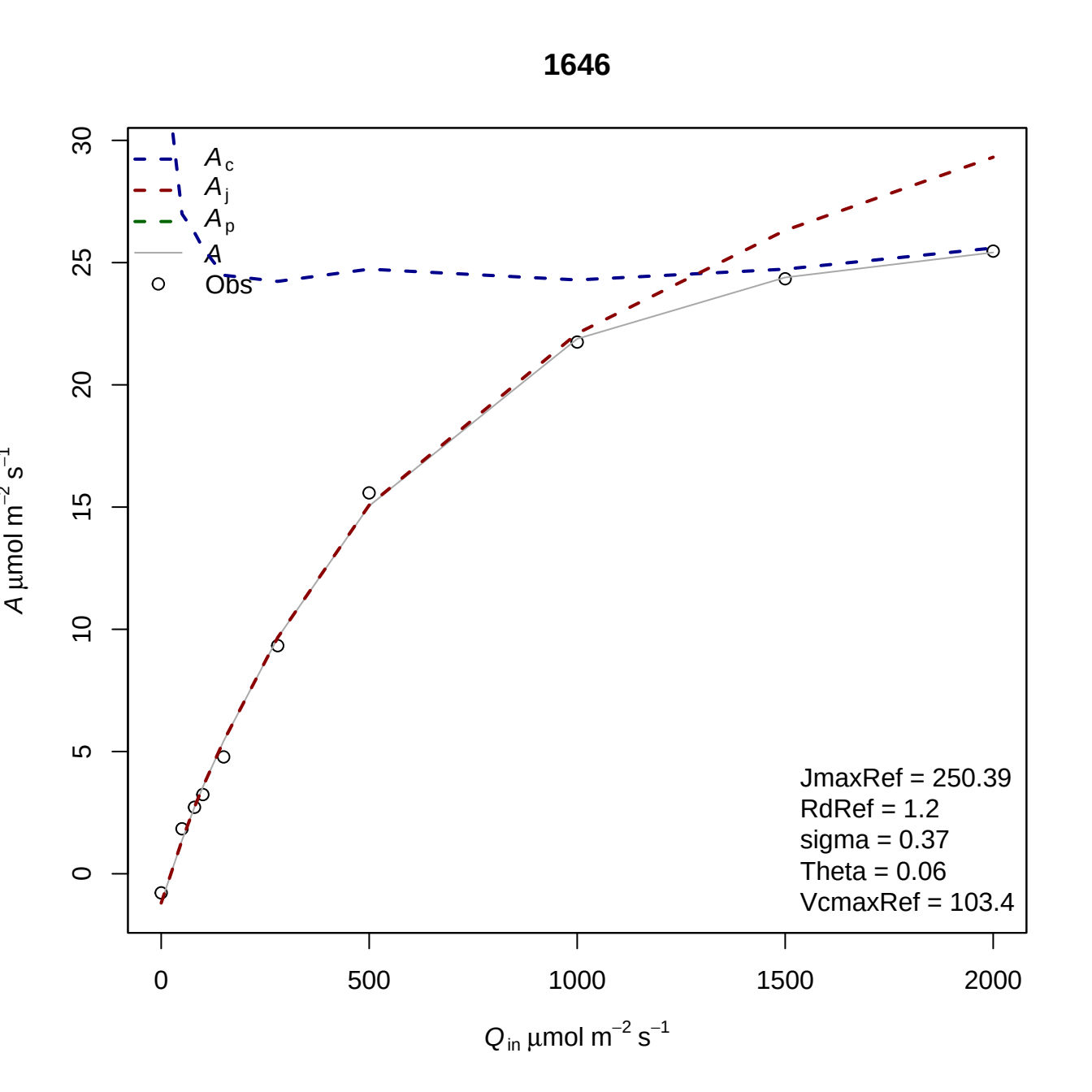

1641
A_c
A_j
A_p
A
Obs
JmaxRef = 73.78
RdRef = 0.49
sigma = 0.09
Theta = 0.75
VcmaxRef = 42.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

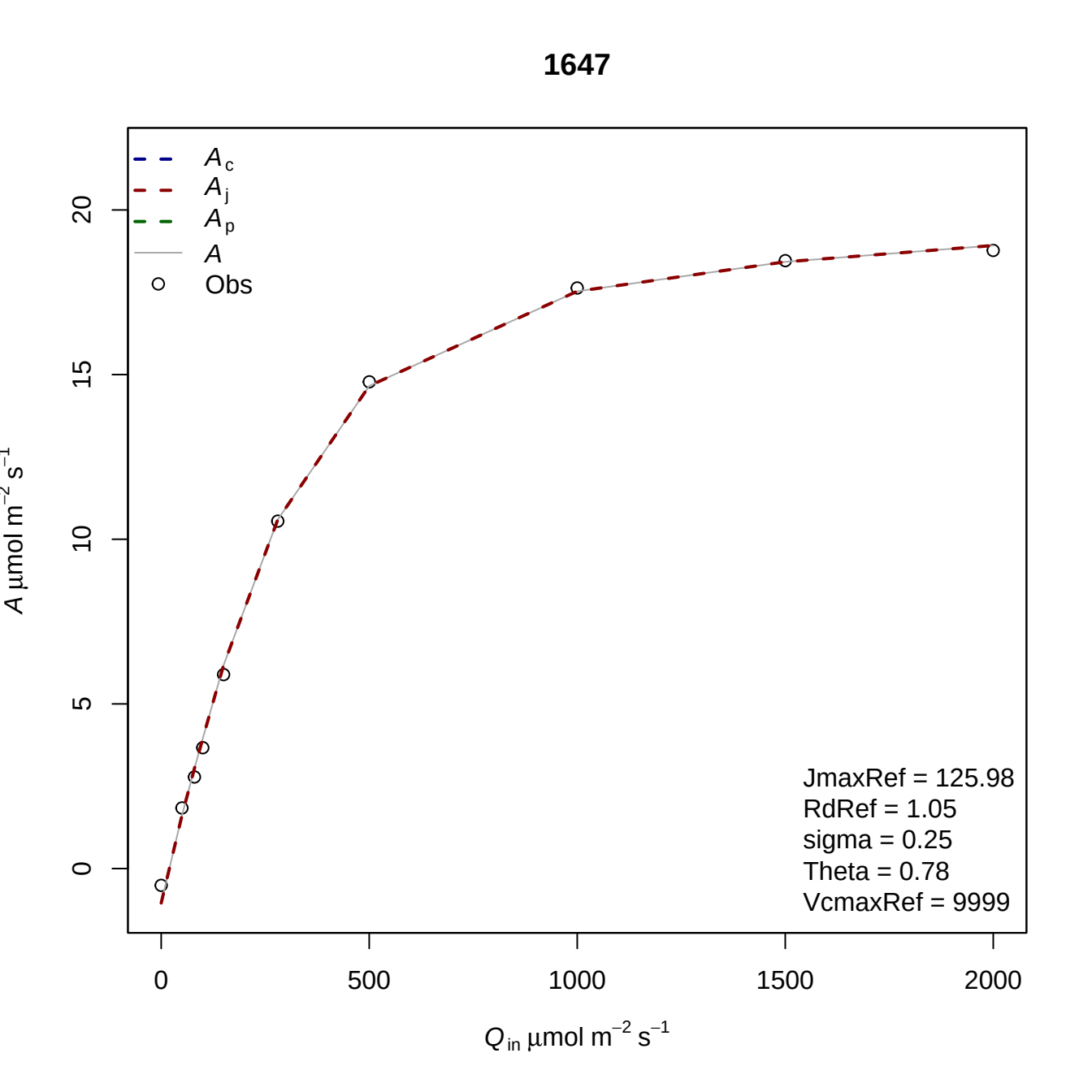

1642
A_c
A_j
A_p
A
Obs
JmaxRef = 67.77
RdRef = 0.48
sigma = 0.14
Theta = 0.39
VcmaxRef = 37.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

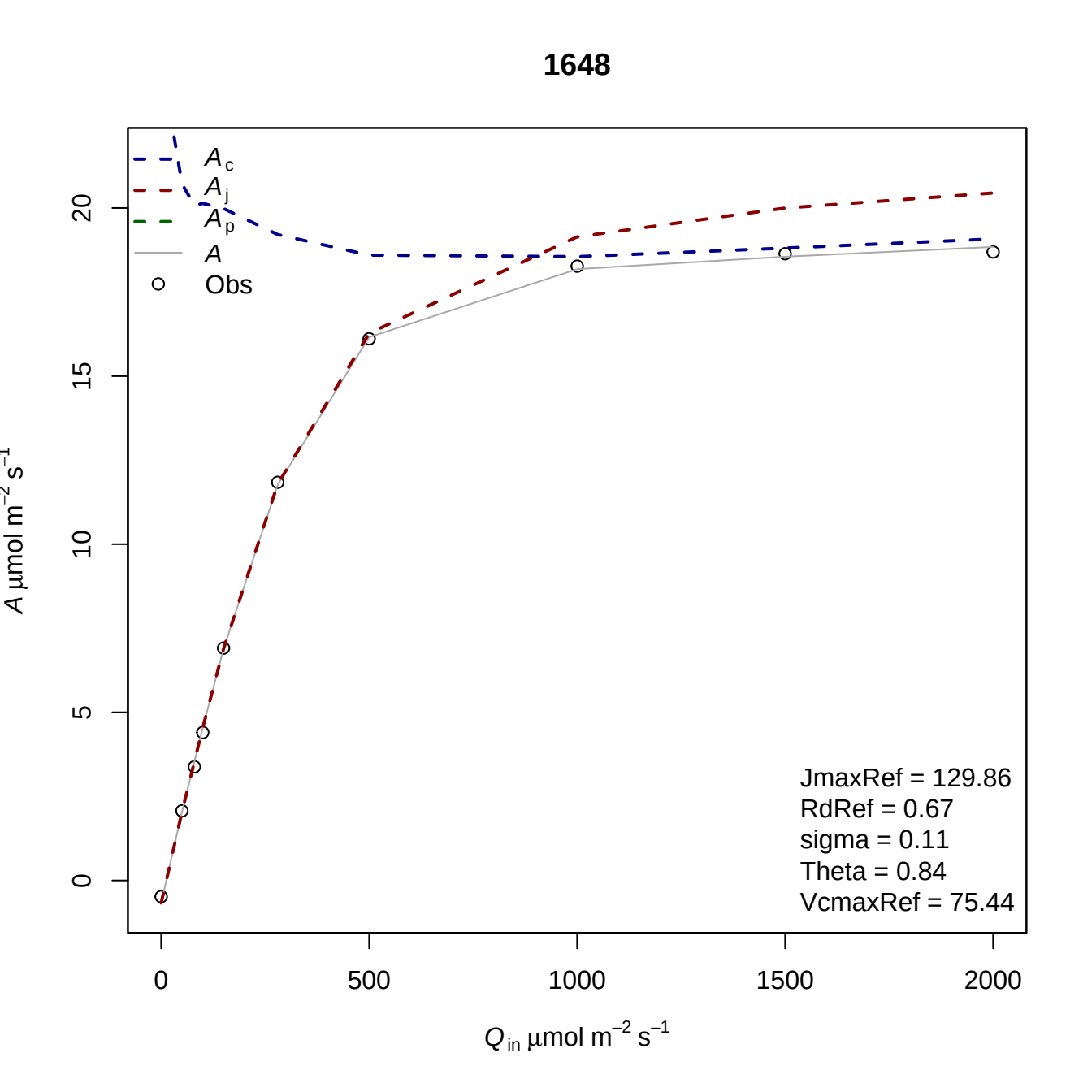

1643
A_c
A_j
A_p
A
Obs
JmaxRef = 145.24
RdRef = 1.08
sigma = 0.35
Theta = 0.78
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

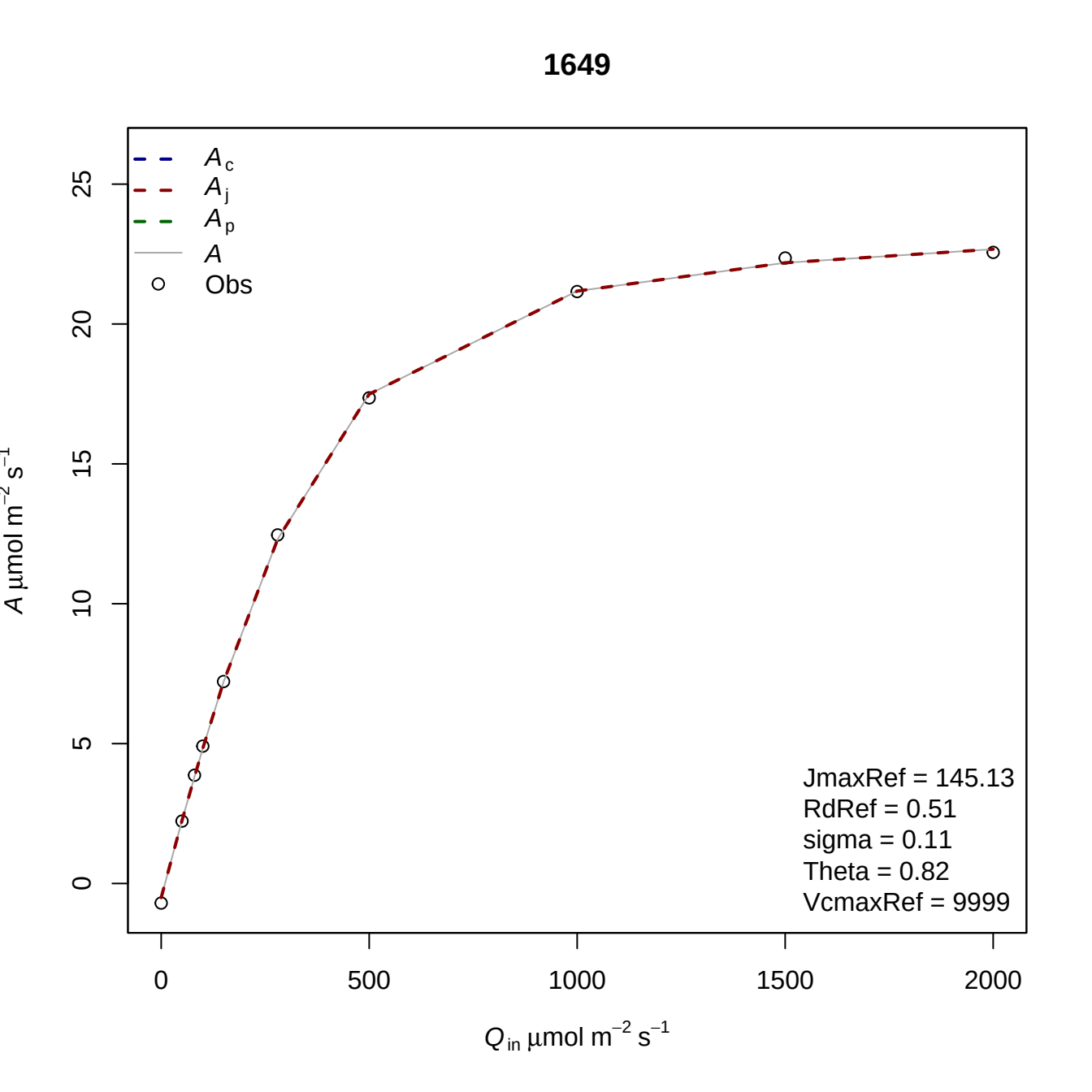

1644
A_c
A_j
A_p
A
Obs
JmaxRef = 156.01
RdRef = 0.54
sigma = 0.12
Theta = 0.81
VcmaxRef = 93.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

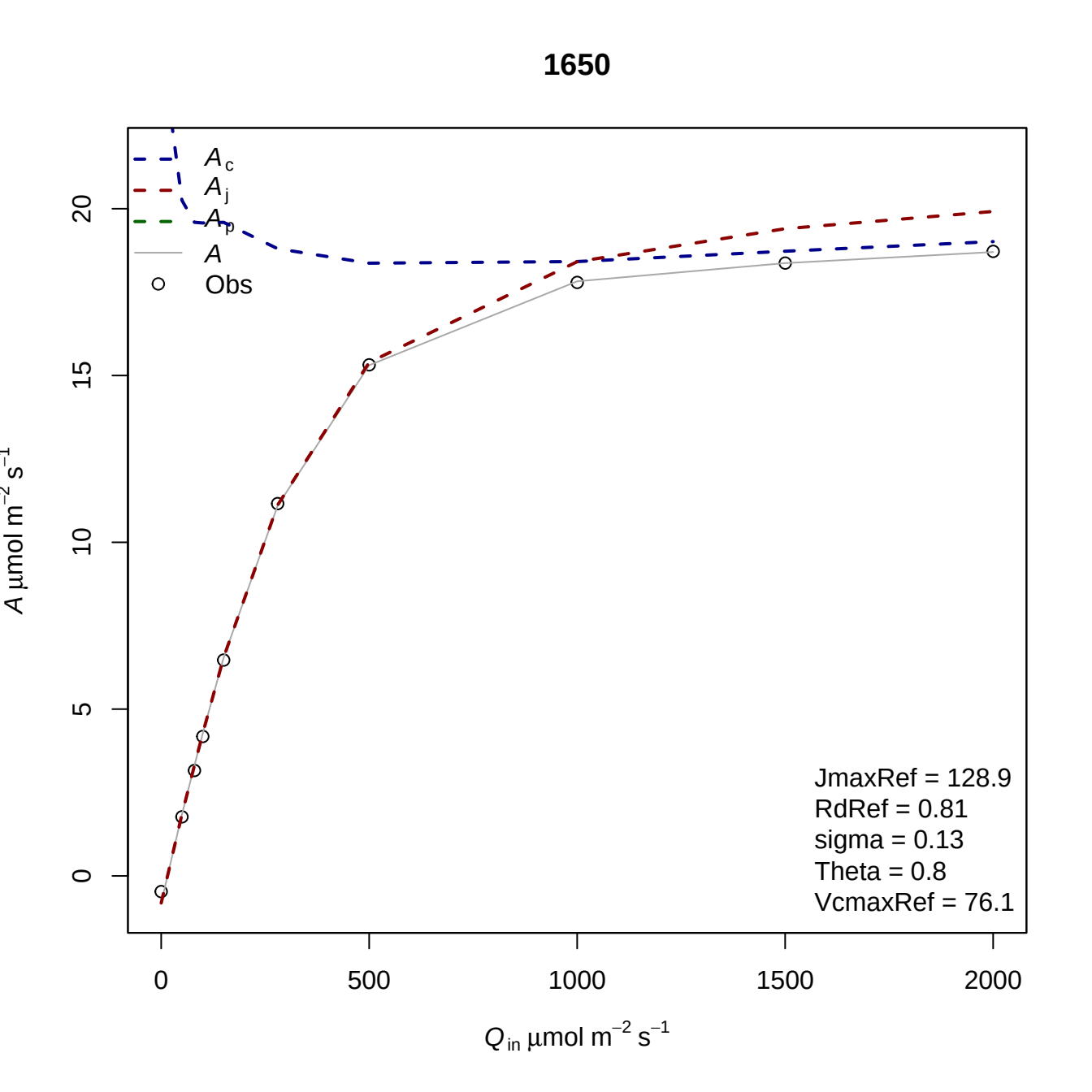

1645
A_c
A_j
A_p
A
Obs
JmaxRef = 176.49
RdRef = 0.93
sigma = 0.16
Theta = 0.58
VcmaxRef = 95.21
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

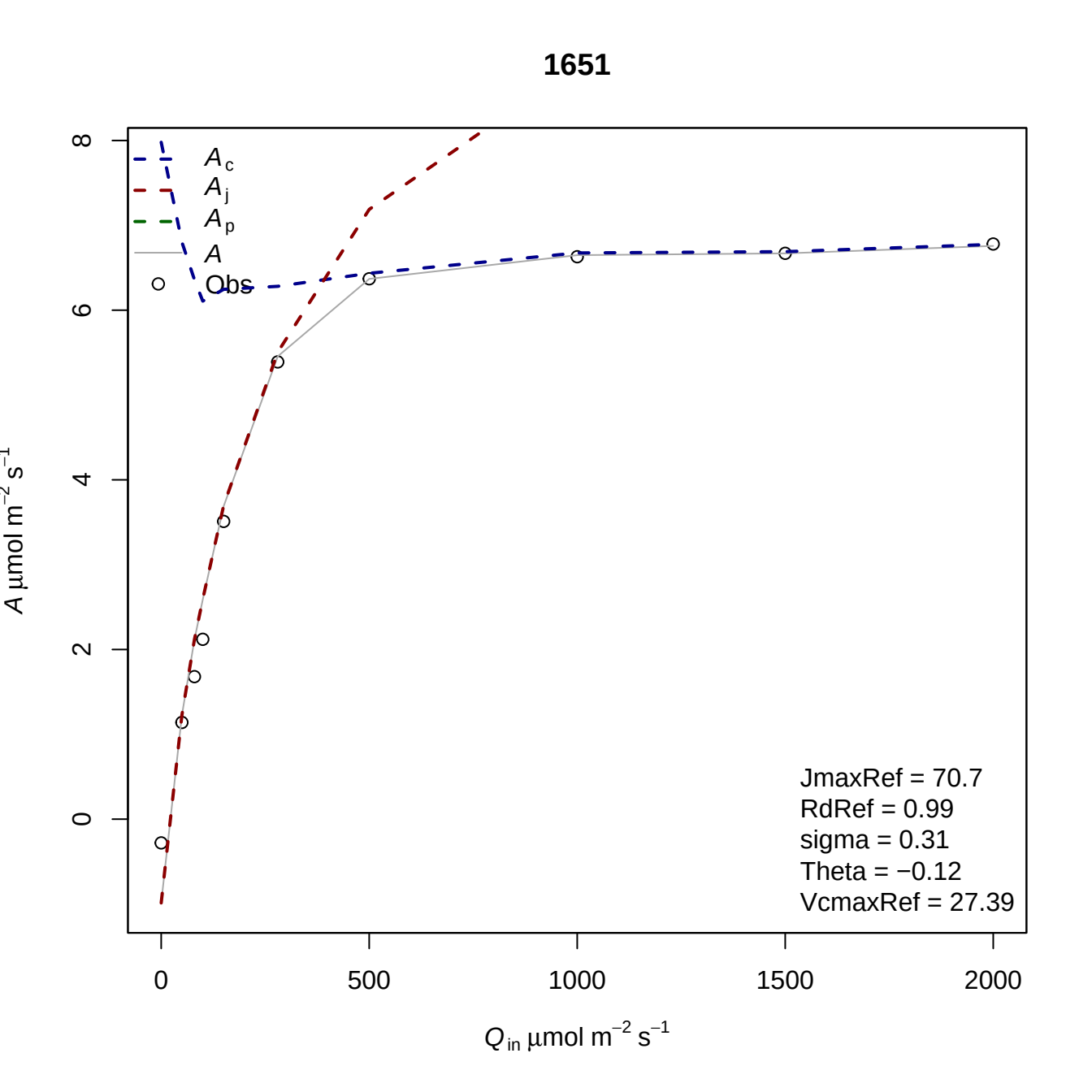

1646
A_c
A_j
A_p
A
Obs
JmaxRef = 250.39
RdRef = 1.2
sigma = 0.37
Theta = 0.06
VcmaxRef = 103.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

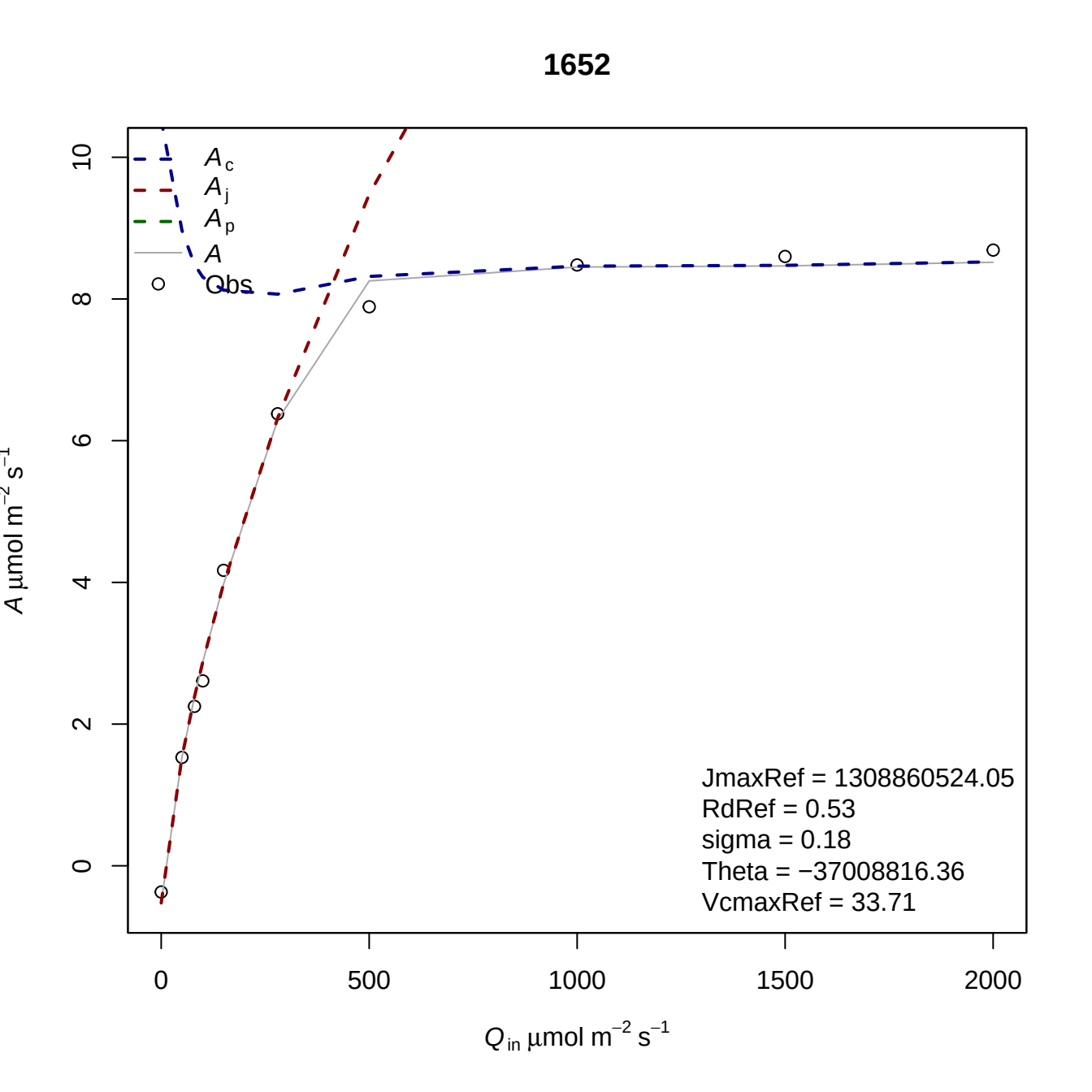

1647
A_c
A_j
A_p
A
Obs
JmaxRef = 125.98
RdRef = 1.05
sigma = 0.25
Theta = 0.78
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

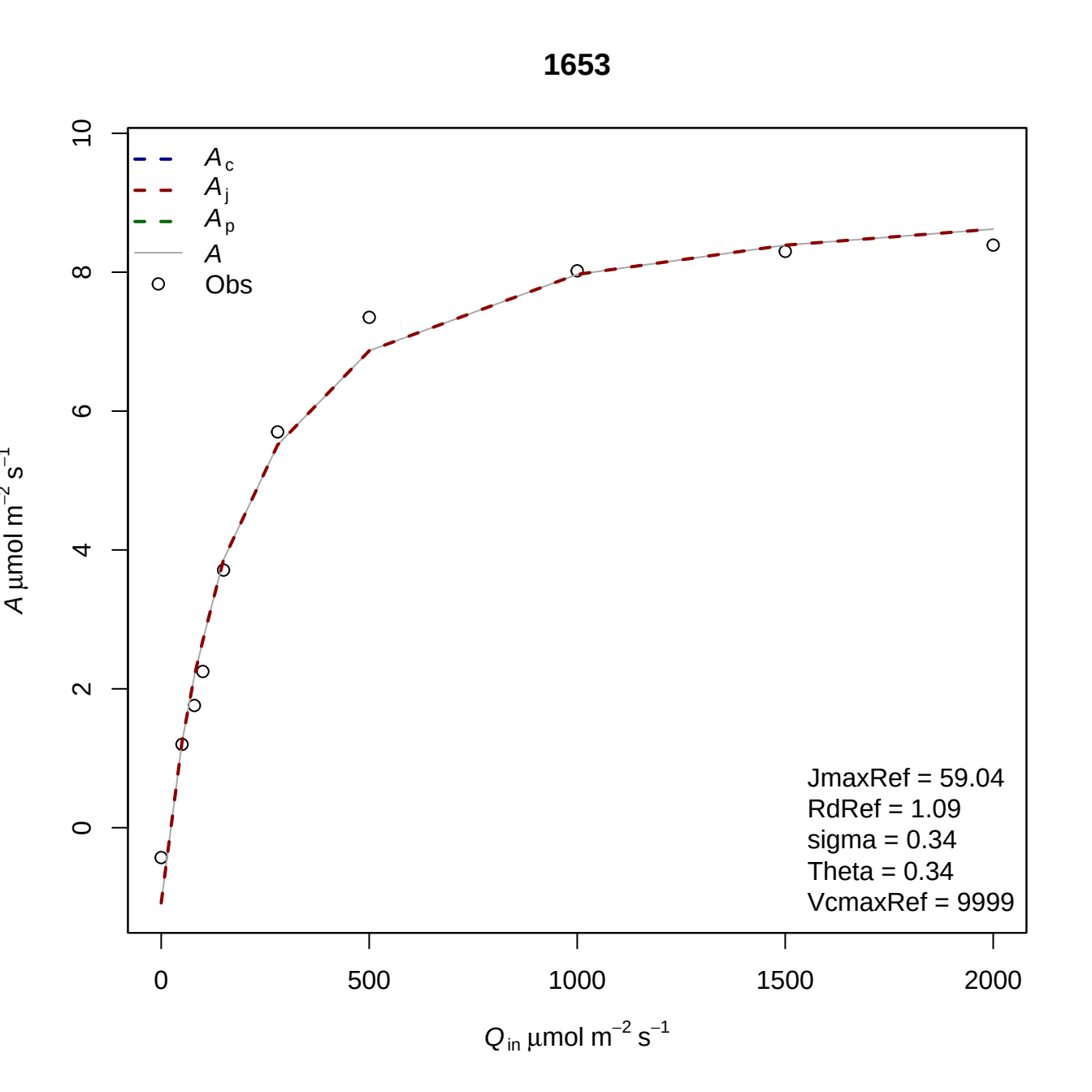

1648
A_c
A_j
A_p
A
Obs
JmaxRef = 129.86
RdRef = 0.67
sigma = 0.11
Theta = 0.84
VcmaxRef = 75.44
0
5
10
15
20
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

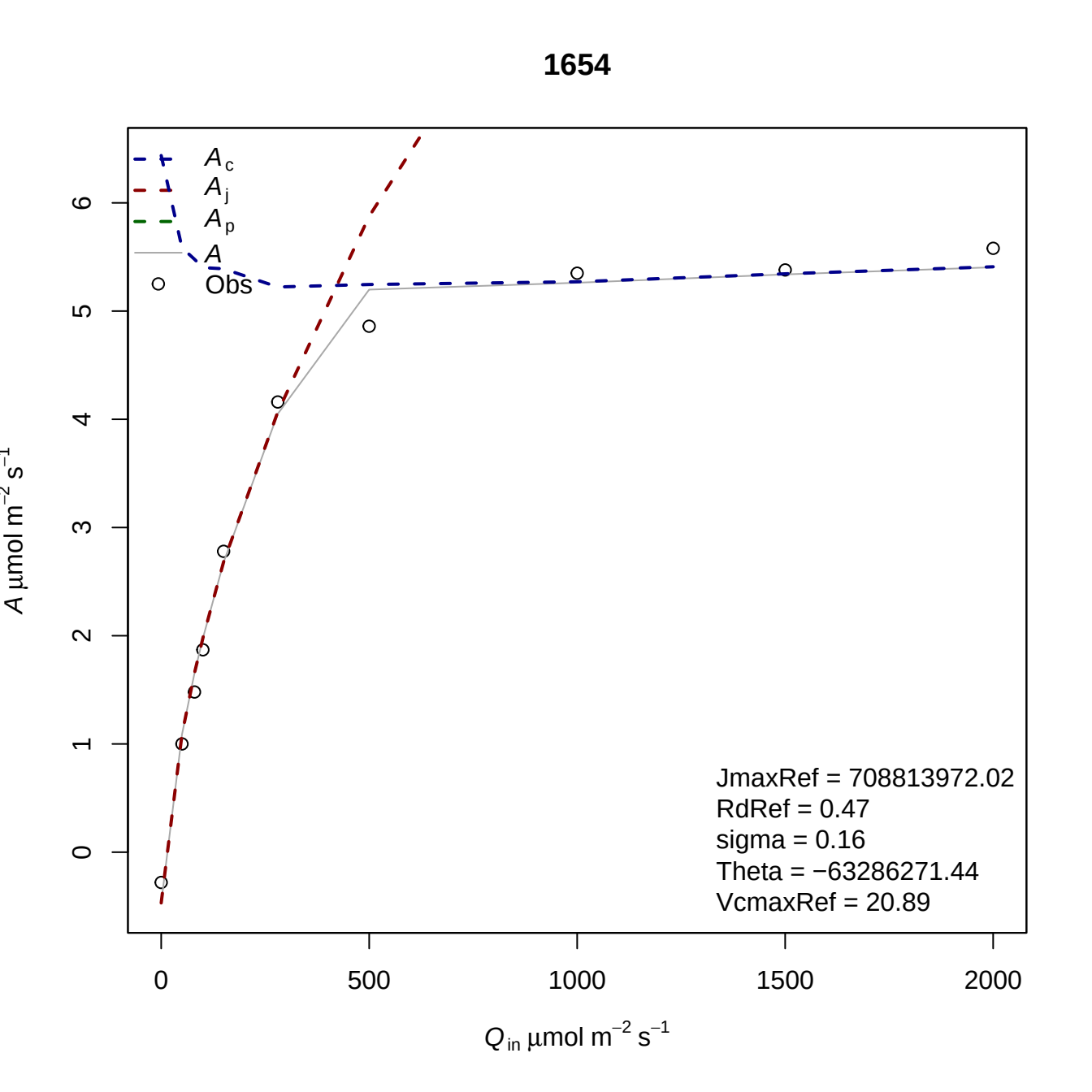

1649
A_c
A_j
A_p
A
Obs
JmaxRef = 145.13
RdRef = 0.51
sigma = 0.11
Theta = 0.82
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

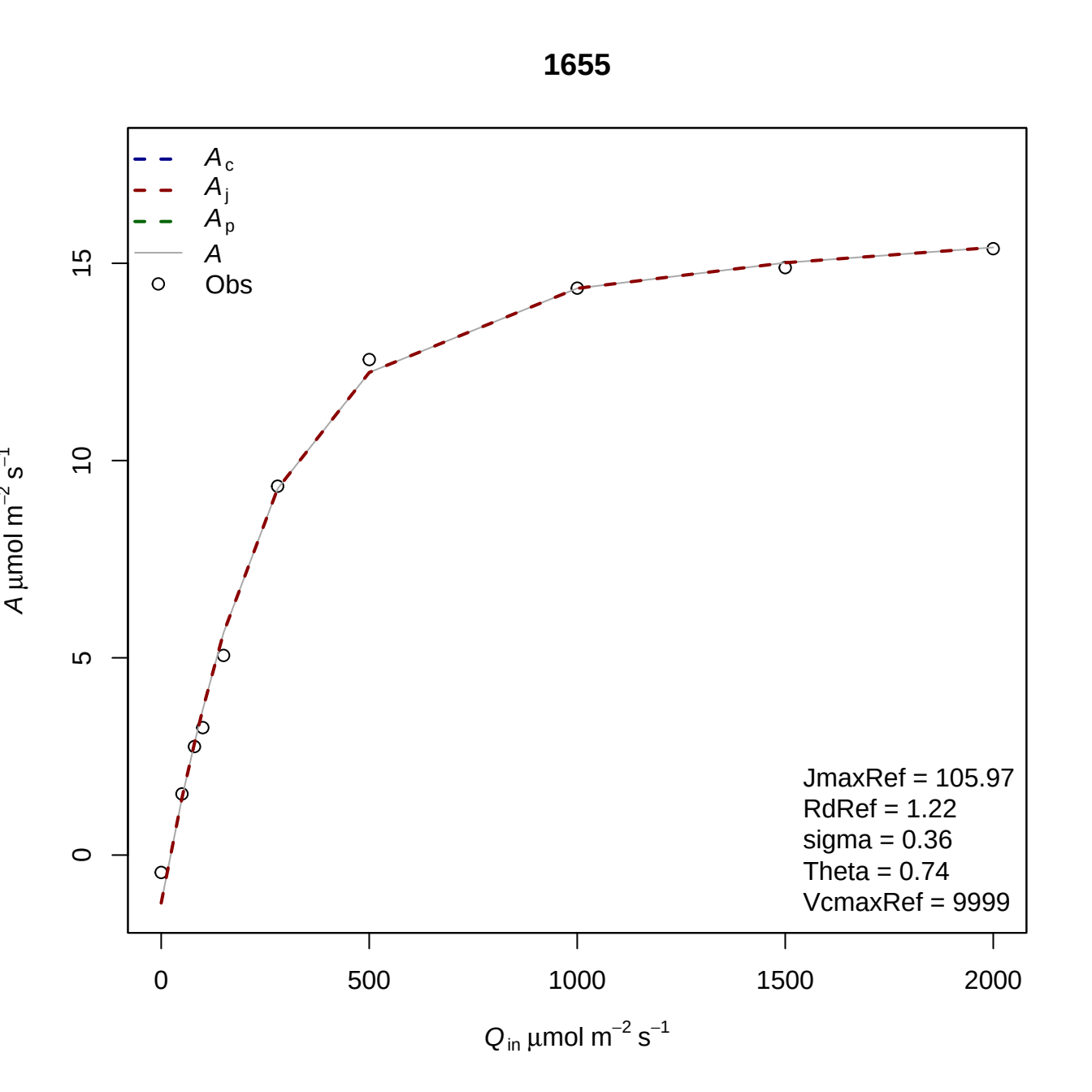

1650
A_c
A_j
A_p
A
Obs
JmaxRef = 128.9
RdRef = 0.81
sigma = 0.13
Theta = 0.8
VcmaxRef = 76.1
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

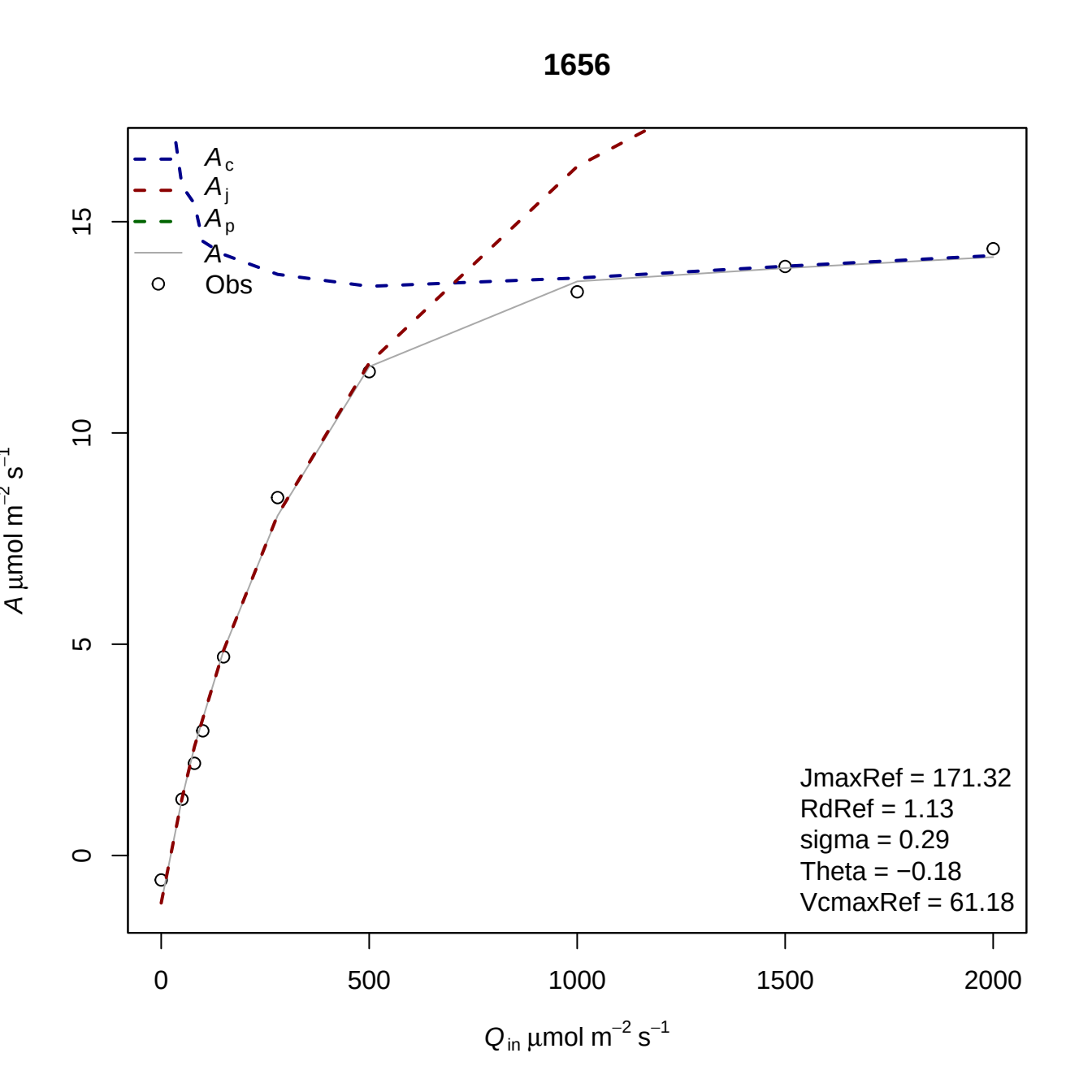

1651
A_c
A_j
A_p
A
Obs
JmaxRef = 70.7
RdRef = 0.99
sigma = 0.31
Theta = −0.12
VcmaxRef = 27.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

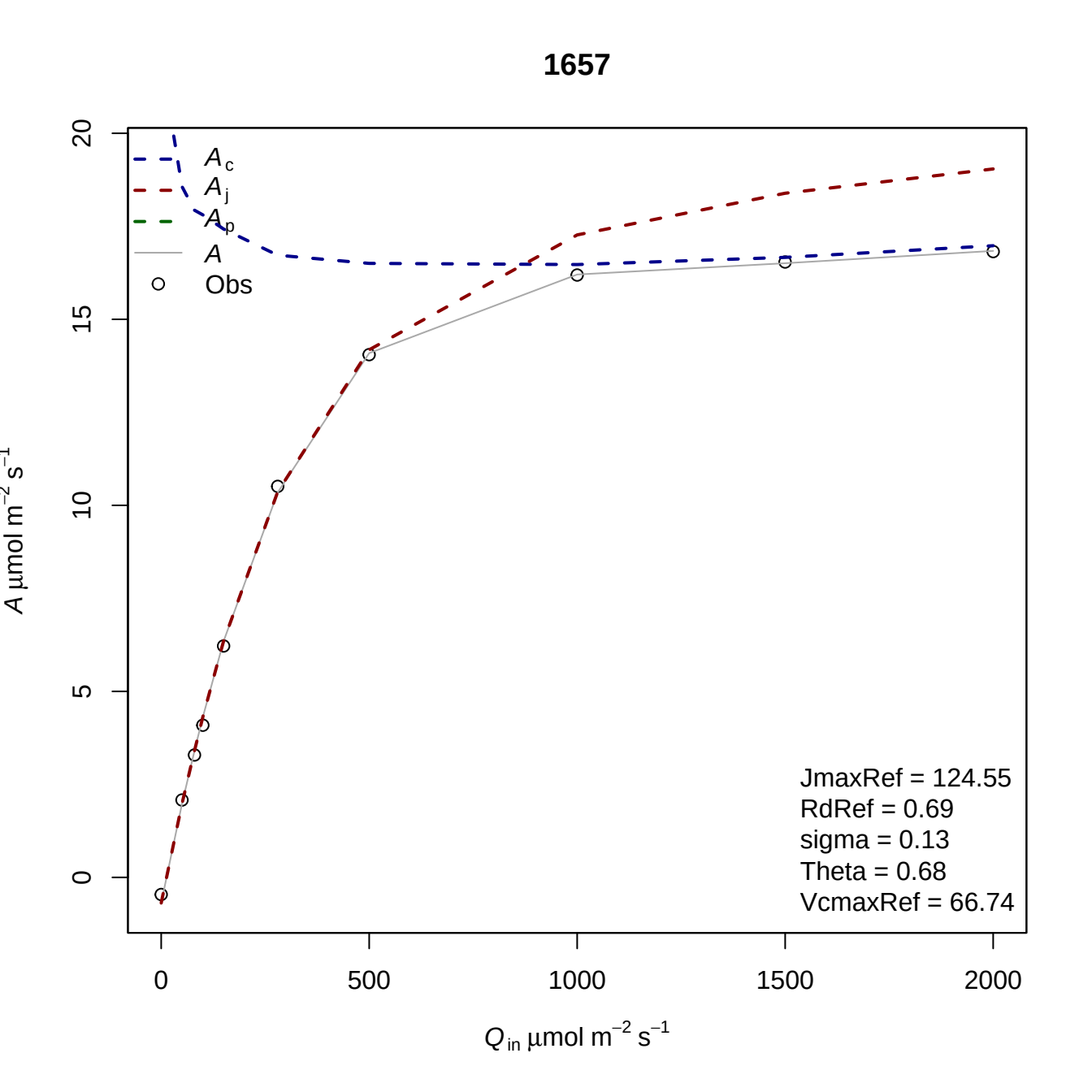

1652
A_c
A_j
A_p
A
Qbs
JmaxRef = 1308860524.05
RdRef = 0.53
sigma = 0.18
Theta = −37008816.36
VcmaxRef = 33.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

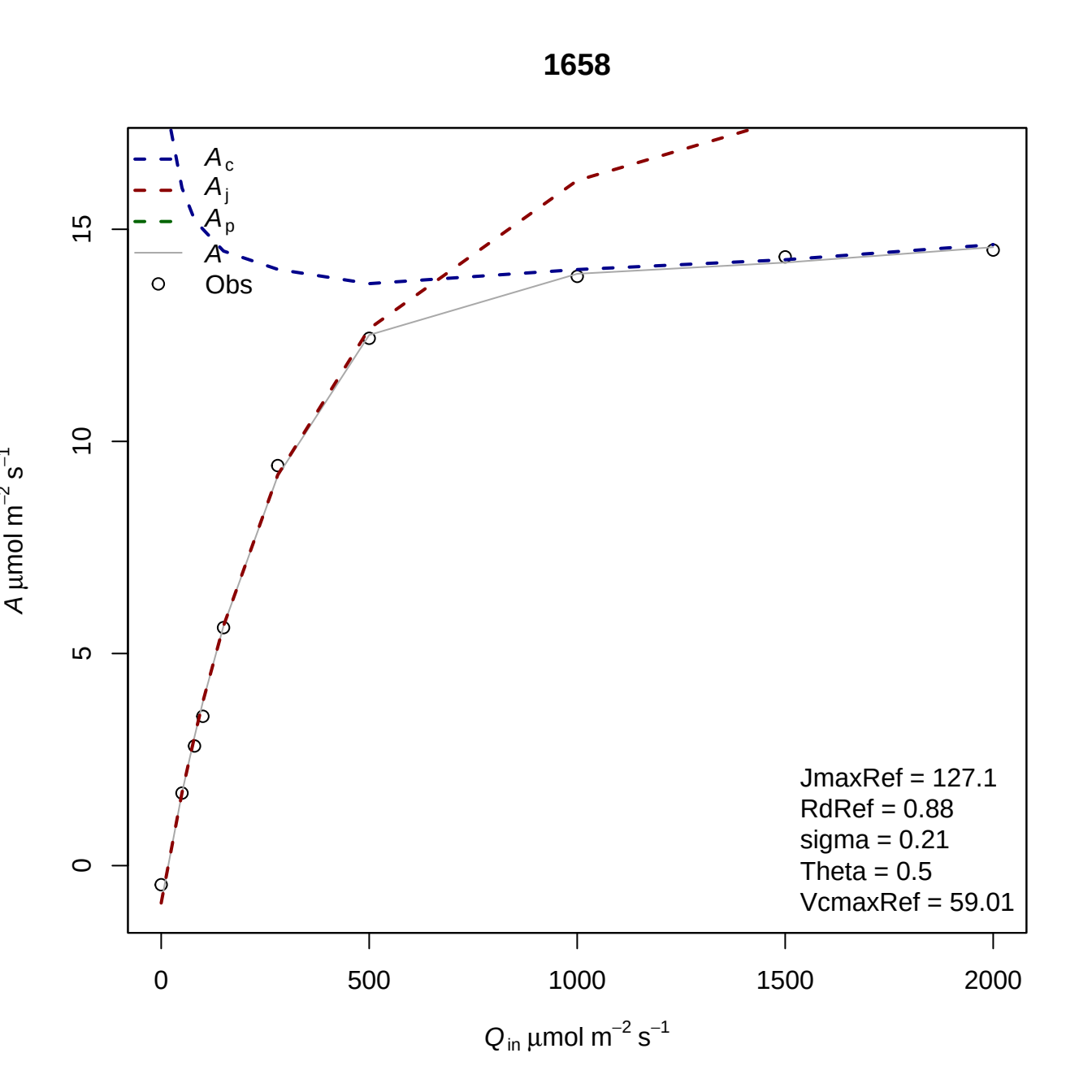

1653
A_c
A_j
A_p
A
Obs
JmaxRef = 59.04
RdRef = 1.09
sigma = 0.34
Theta = 0.34
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

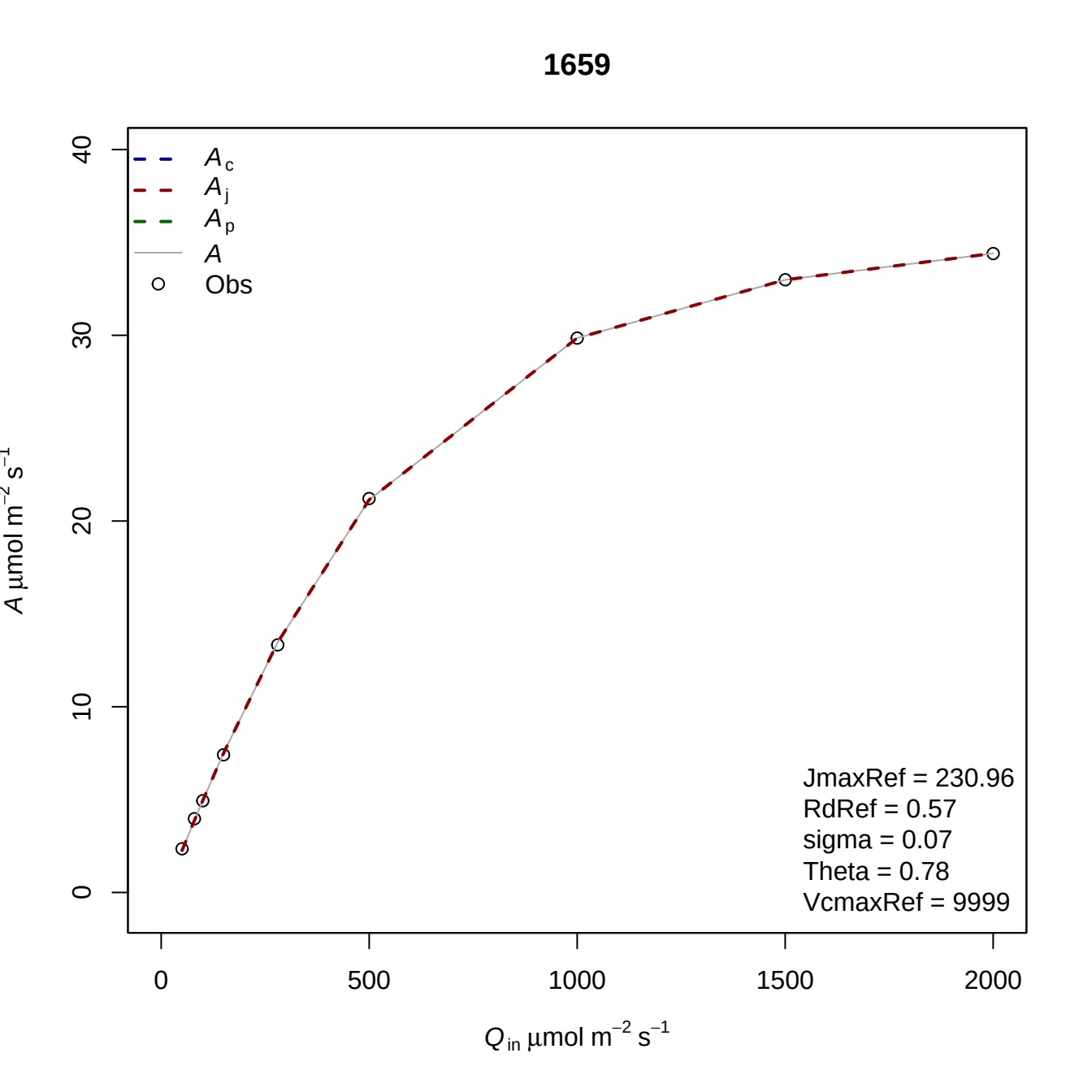

1654
A_c
A_j
A_p
A
Obs
JmaxRef = 708813972.02
RdRef = 0.47
sigma = 0.16
Theta = −63286271.44
VcmaxRef = 20.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500
2000

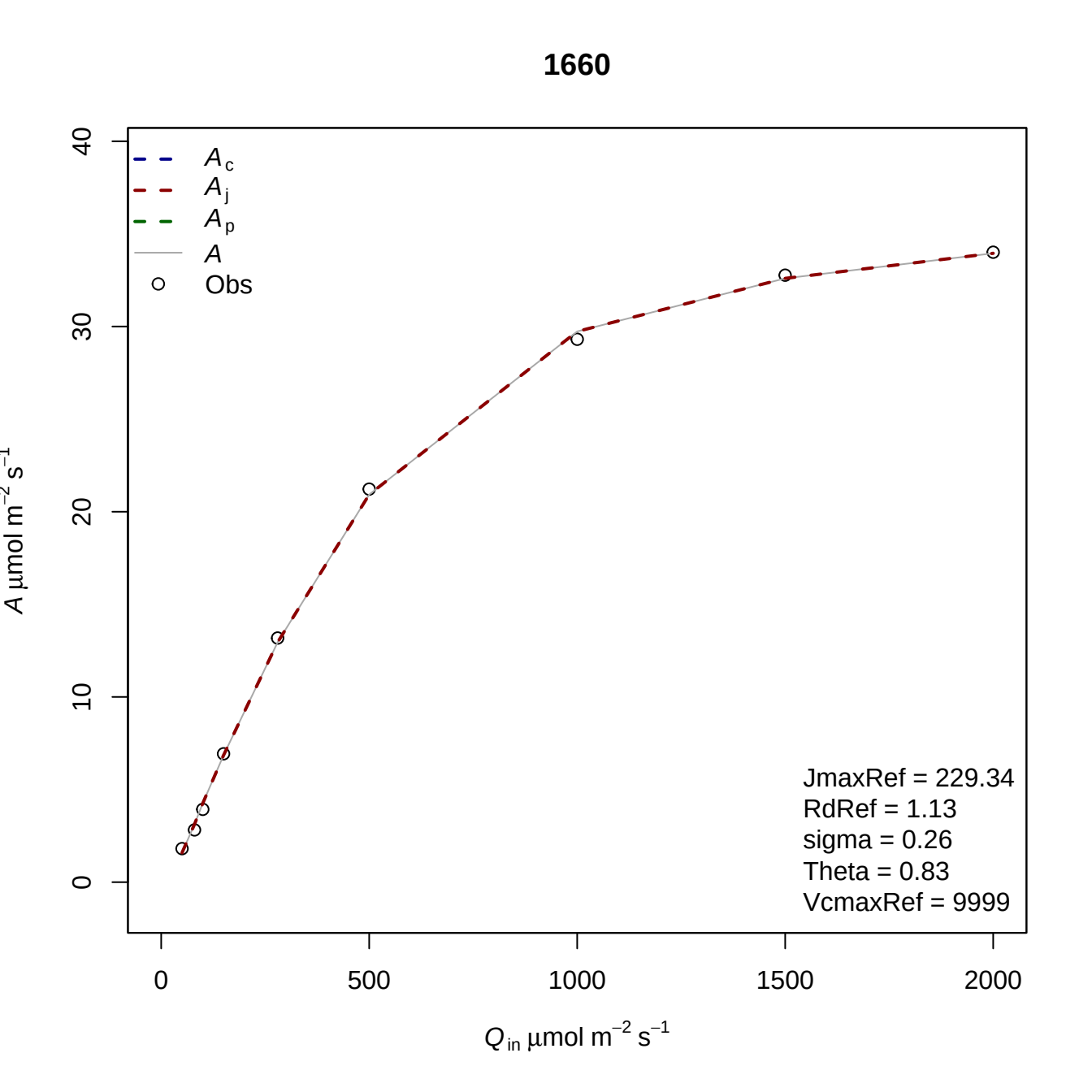

1655
A_c
A_j
A_p
A
Obs
JmaxRef = 105.97
RdRef = 1.22
sigma = 0.36
Theta = 0.74
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

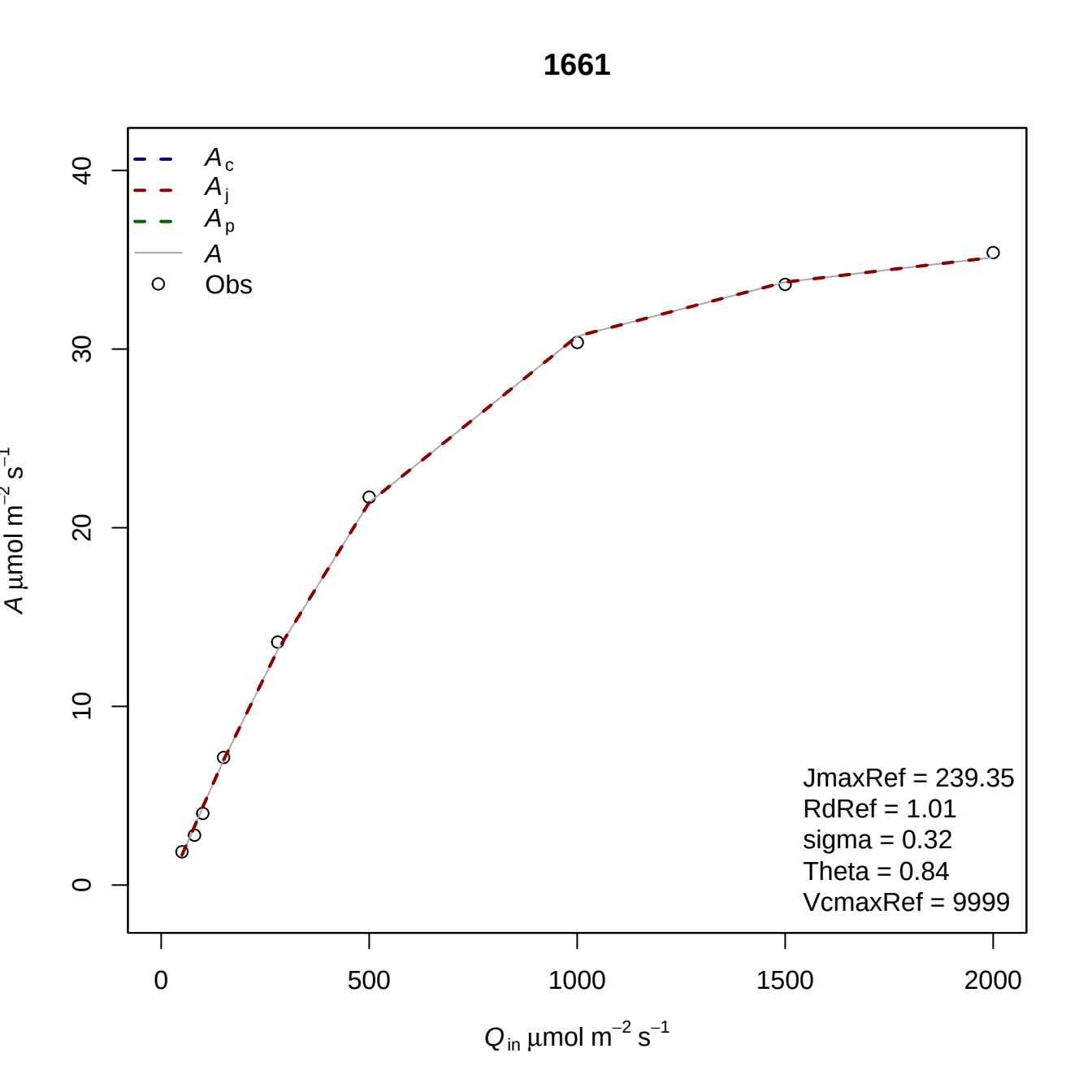

1656
A_c
A_j
A_p
A
Obs
JmaxRef = 171.32
RdRef = 1.13
sigma = 0.29
Theta = −0.18
VcmaxRef = 61.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

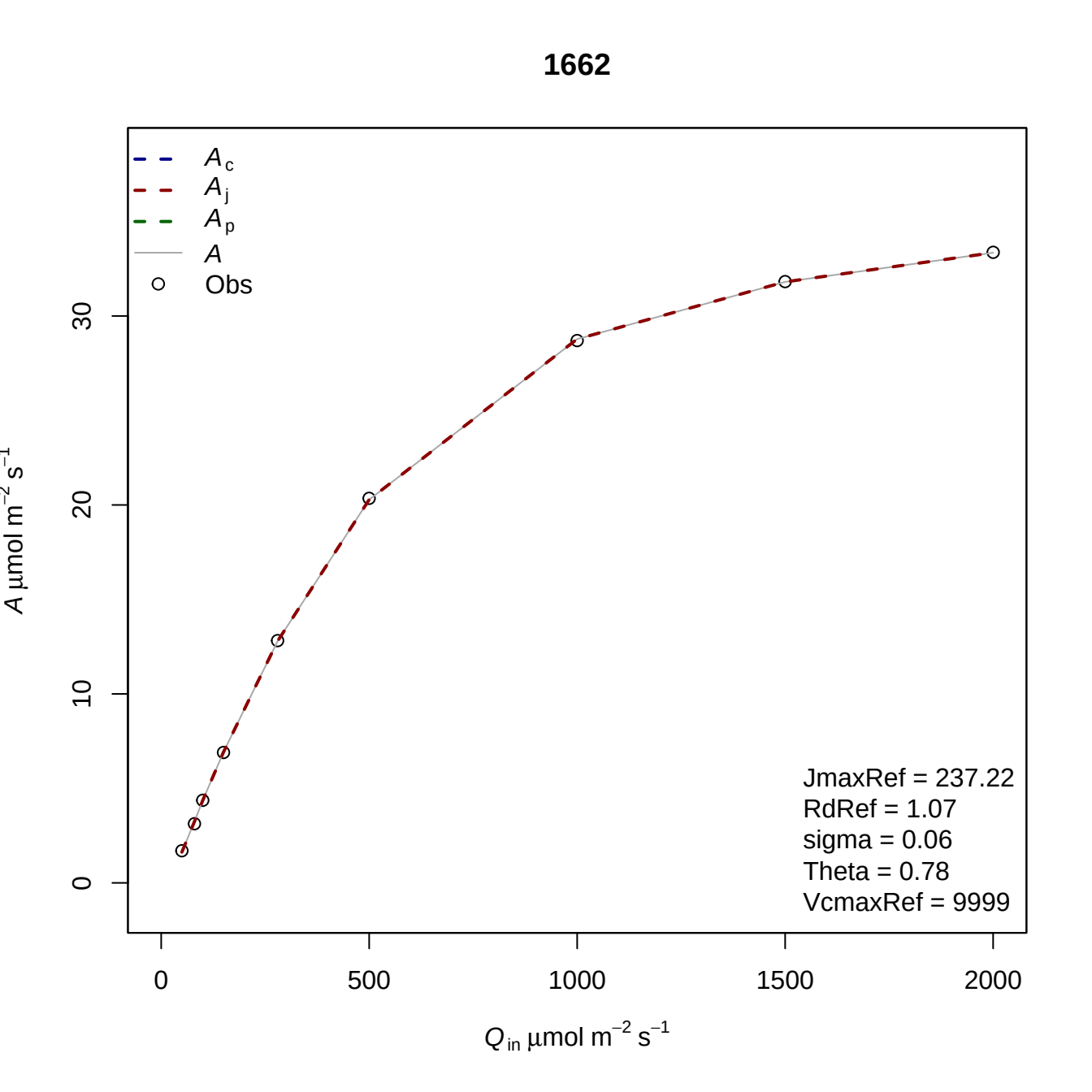

1657
A_c
A_j
A_p
A
Obs
JmaxRef = 124.55
RdRef = 0.69
sigma = 0.13
Theta = 0.68
VcmaxRef = 66.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000


1658
A_c
A_j
A_p
A
Obs
JmaxRef = 127.1
RdRef = 0.88
sigma = 0.21
Theta = 0.5
VcmaxRef = 59.01
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}


1659
A_c
A_j
A_p
A
Obs
JmaxRef = 230.96
RdRef = 0.57
sigma = 0.07
Theta = 0.78
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000


1660
A_c
A_j
A_p
A
Obs
JmaxRef = 229.34
RdRef = 1.13
sigma = 0.26
Theta = 0.83
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000


1661
A_c
A_j
A_p
A
Obs
JmaxRef = 239.35
RdRef = 1.01
sigma = 0.32
Theta = 0.84
VcmaxRef = 9999
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}


1662
A_c
A_j
A_p
A
Obs
JmaxRef = 237.22
RdRef = 1.07
sigma = 0.06
Theta = 0.78
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}